

100 Full-Length Mock Tests

Artificial Intelligence: Search · Logic · Reasoning under Uncertainty

Every set follows the GATE DA question pattern: **25 questions, 40 marks** ($10 \times 1\text{-mark} + 15 \times 2\text{-mark}$), with Multiple-Choice (MCQ), Multiple-Select (MSQ) and Numerical-Answer (NAT) questions and GATE negative marking. Each set ends with an answer key and **step-by-step worked solutions**: full search traces, annotated game trees, truth tables, resolution derivations, enumeration tables, variable-elimination factor tables and sampling computations.

Syllabus unit (GATE DA 2027, Section 7)	Covered by
Search — uninformed	BFS / DFS traces, UCS, IDS / DLS / BFS node counts, concept questions
Search — informed	A*, greedy best-first, admissibility & consistency, effective branching factor
Search — adversarial	Minimax, alpha-beta pruning (leaf counting), expectimax, move ordering
Logic — propositional	Model counting, validity, equivalence, entailment, resolution, forward chaining
Logic — predicate	Finite-model semantics, quantifier order, counting interpretations, unification
Conditional independence representation	Bayesian networks, d-separation, Markov blanket, parameters
Exact inference — variable elimination	Enumeration, factor scopes and sizes, elimination orderings
Approximate inference — sampling	Prior, rejection, likelihood weighting, Gibbs sampling

Diagrams (graphs, game trees, Bayesian networks, relation graphs) are drawn as vector graphics. All numerical questions were generated and solved by exact computation, so the answer keys are reproducible.

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Mock Test 051	665
Mock Test 052	678
Mock Test 053	692
Mock Test 054	704
Mock Test 055	717
Mock Test 056	730
Mock Test 057	743
Mock Test 058	755
Mock Test 059	768
Mock Test 060	781
Mock Test 061	794
Mock Test 062	807
Mock Test 063	820
Mock Test 064	833
Mock Test 065	846
Mock Test 066	860
Mock Test 067	873
Mock Test 068	886
Mock Test 069	899
Mock Test 070	912
Mock Test 071	925
Mock Test 072	938
Mock Test 073	950
Mock Test 074	963
Mock Test 075	976
Mock Test 076	989
Mock Test 077	1002
Mock Test 078	1015
Mock Test 079	1028
Mock Test 080	1041
Mock Test 081	1054
Mock Test 082	1067

Mock Test 083	1079
Mock Test 084	1092
Mock Test 085	1105
Mock Test 086	1118
Mock Test 087	1130
Mock Test 088	1143
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How to Use this Book

This book has 100 independent mock tests for **Section 7 (Artificial Intelligence)** of the GATE Data Science & AI (DA) syllabus. Each test is timed for 75 minutes and mirrors the GATE question styles:

Type	What you do	Marking
MCQ	Choose exactly one of four options	+1 / -1/3 (1-mark); +2 / -2/3 (2-mark)
MSQ	Choose ALL correct options (one or more)	Full marks only for exactly the correct set; no negative
NAT	Type a number (range accepted where stated)	No negative marking

Suggested routine. Attempt a test in one sitting without notes, mark your answers, then use the answer key on the last pages of the test. Read the detailed solution for every question, including the ones you got right: the traces (frontier contents, α/β values, factor scopes, sample weights) show exactly the steps an examiner expects. Re-attempt tests where you score below 60%.

Conventions used throughout. Unless a question says otherwise: ties are broken alphabetically; graph search keeps an explored set; UCS and A* apply the goal test when a node is selected for expansion; game trees are explored left to right; Bayesian-network variables are Boolean and CPTs list $P(X = T \mid \text{parents})$.

Revision Notes 1 — Search

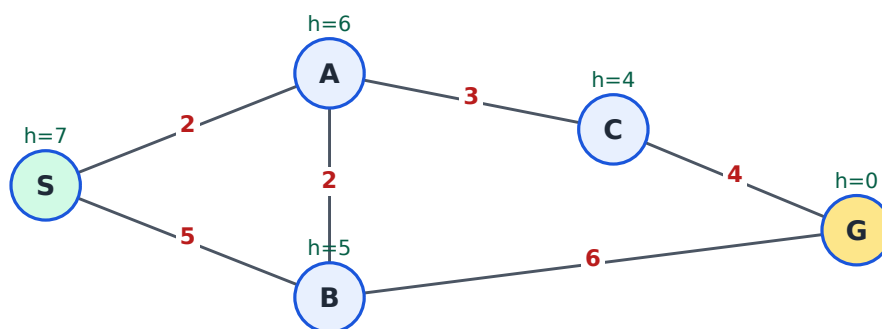
A search problem consists of an initial state, the actions available in a state, a transition model, a goal test and a step-cost function. b = branching factor, d = depth of the shallowest goal, m = maximum depth, C^* = optimal cost, ϵ = minimum step cost, ℓ = depth limit.

Algorithm	Frontier / priority	Complete?	Optimal?	Time	Space
BFS	FIFO queue	Yes (finite b)	Yes if unit costs	$O(b^d)$	$O(b^d)$
Uniform-cost	$g(n)$	Yes if $\epsilon > 0$	Yes	$O(b^{(1+\lceil C^*/\epsilon \rceil)})$	$O(b^{(1+\lceil C^*/\epsilon \rceil)})$
DFS	LIFO stack	No (infinite depth / tree search)	No	$O(b^m)$	$O(bm)$
Depth-limited	LIFO, depth $\leq \ell$	No (if $\ell < d$)	No	$O(b^\ell)$	$O(b\ell)$
Iterative deepening	DLS for $\ell = 0, 1, 2, \dots$	Yes	Yes if unit costs	$O(b^d)$	$O(bd)$
Bidirectional	two BFS frontiers	Yes	Yes if unit costs	$O(b^{(d/2)})$	$O(b^{(d/2)})$
Greedy best-first	$h(n)$	No (tree); yes finite graph	No	$O(b^m)$	$O(b^m)$
A*	$f(n) = g(n) + h(n)$	Yes	Yes (admissible tree / consistent graph)	exp.	exp.
IDA*	f-cost contour limit	Yes	Yes (admissible)	exp.	$O(bd)$

Heuristics

- **Admissible:** $h(n) \leq h^*(n)$ for all n (never overestimates).
- **Consistent (monotone):** $h(n) \leq c(n, a, n') + h(n')$ for every edge and $h(\text{goal}) = 0$. Consistent $\Rightarrow$ admissible; with a consistent h , f is non-decreasing along any path and A* graph search never re-opens a node.
- **Dominance:** if $h_2(n) \geq h_1(n)$ for all n (both admissible), A* with h_2 expands no more nodes than with h_1 . $\max(h_1, h_2)$ of admissible heuristics is admissible and dominates both.
- **Relaxed problems** give admissible heuristics (8-puzzle: misplaced tiles h_1 , Manhattan distance $h_2 \geq h_1$).
- **Effective branching factor** b^* : $N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^d$.
- A* expands all nodes with $f(n) < C^*$, some with $f(n) = C^*$, none with $f(n) > C^*$.

Worked example — A* graph search



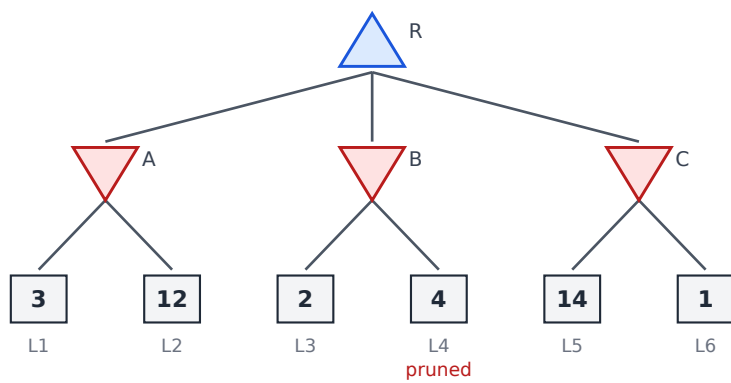
$f(S)=7 \rightarrow$ expand S: A ($g=2, f=8$), B ($g=5, f=10$). Expand A ($f=8$): C ($g=5, f=9$), B improves to $g=4, f=9$. Ties at $f=9$ are broken alphabetically: expand B: G ($g=10, f=10$). Expand C ($f=9$): G improves to $g=9, f=9$. Expand G ($f=9$) $\Rightarrow$ goal. Path $S \rightarrow A \rightarrow C \rightarrow G$, cost 9. h is consistent here, e.g. $h(S)=7 \leq 2+h(A)=8$.

Adversarial search

- **Minimax:** MAX picks the child of maximum value, MIN the minimum; complete for finite trees, optimal against an optimal opponent; time $O(b^m)$, space $O(bm)$.
- **Alpha-beta:** α = best value MAX can guarantee so far on the path; β = best value for MIN. At a MIN node stop when $v \leq \alpha$; at a MAX node stop when $v \geq \beta$. The root value is never changed by pruning.
- Perfect move ordering examines $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$ leaves ($\approx O(b^{(d/2)})$), random ordering $\approx O(b^{(3d/4)})$, worst ordering $O(b^d)$.
- **Expectimax** replaces MIN by chance nodes (probability-weighted average). Its decisions are invariant only to positive linear transformations of the utilities (minimax: any monotonic transformation).

Alpha-beta style pruning needs bounded utilities at chance nodes.

Worked example — alpha-beta



$\triangle$ MAX node ∇ MIN node $\square$ terminal utility

$A = \min(3, 12) = 3 \Rightarrow \alpha = 3$ at the root. At B the first leaf gives $2 \leq \alpha = 3 \Rightarrow L4$ is pruned ($B \leq 2$). C: $14 > 3$, then $1 \leq 3 \Rightarrow C = 1$. Root = $\max(3, \leq 2, 1) = 3$. One leaf is pruned. With C's children reversed (1 first), L5 would be pruned as well — move ordering matters.

Revision Notes 2 — Logic

A model is a truth assignment (propositional) or an interpretation (first-order). $\alpha \models \beta$ iff every model of α is a model of β . α is **valid** iff true in all models; **satisfiable** iff true in some model. $\alpha \models \beta$ iff $(\alpha \rightarrow \beta)$ is valid iff $(\alpha \wedge \neg\beta)$ is unsatisfiable (the basis of proof by refutation).

Equivalence	Law
$\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta \equiv \neg\beta \rightarrow \neg\alpha$	implication elimination, contraposition
$\alpha \leftrightarrow \beta \equiv (\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha) \equiv (\alpha \wedge \beta) \vee (\neg\alpha \wedge \neg\beta)$	biconditional elimination
$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$; $\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$	De Morgan
$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$; $\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$	distributivity
$(\alpha \wedge \beta) \rightarrow \gamma \equiv \alpha \rightarrow (\beta \rightarrow \gamma)$	exportation
$\neg\forall x P(x) \equiv \exists x \neg P(x)$; $\neg\exists x P(x) \equiv \forall x \neg P(x)$	quantifier negation
$\forall x (P \wedge Q) \equiv \forall x P \wedge \forall x Q$; $\exists x (P \vee Q) \equiv \exists x P \vee \exists x Q$	distribution of quantifiers
$\exists x \forall y R(x,y) \models \forall y \exists x R(x,y)$ (converse fails)	quantifier order

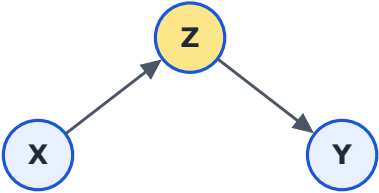
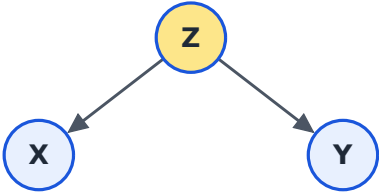
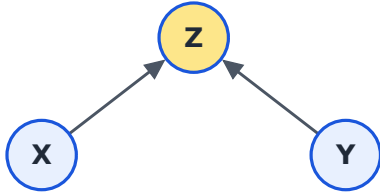
Inference rule	Form	Sound?
Modus ponens	$\alpha \rightarrow \beta, \alpha \vdash \beta$	yes
Modus tollens	$\alpha \rightarrow \beta, \neg\beta \vdash \neg\alpha$	yes
Resolution	$(\alpha \vee \ell), (\neg\ell \vee \beta) \vdash (\alpha \vee \beta)$	yes; refutation-complete
Affirming the consequent	$\alpha \rightarrow \beta, \beta \vdash \alpha$	NO (fallacy)
Denying the antecedent	$\alpha \rightarrow \beta, \neg\alpha \vdash \neg\beta$	NO (fallacy)

- **CNF conversion:** eliminate $\leftrightarrow$, eliminate $\rightarrow$, push $\neg$ inwards (De Morgan, double negation), distribute $\vee$ over $\wedge$. A formula on n variables with k models has $2^n - k$ maxterms in canonical CNF.
- **Horn clause:** at most one positive literal. **Definite clause:** exactly one. Forward chaining (data-driven) and backward chaining (goal-driven) are sound and complete for definite clauses and run in linear time in the size of the KB.
- **Resolution:** resolve on one complementary pair at a time; deriving $\square$ proves unsatisfiability. Resolution is not complete for deriving every entailed clause (e.g. weakenings), only for refutation.
- **Translation patterns:** “All P are Q”: $\forall x (P(x) \rightarrow Q(x))$. “Some P is Q”: $\exists x (P(x) \wedge Q(x))$. The mistakes $\forall x (P(x) \wedge Q(x))$ and $\exists x (P(x) \rightarrow Q(x))$ are classic distractors.
- **Unification:** the MGU makes two atoms identical with the most general substitution; the occurs check forbids $x/f(x)$. **Skolemisation:** $\exists$ inside $\forall x_1 \dots \forall x_k$ becomes a Skolem function $f(x_1, \dots, x_k)$; an outermost $\exists$ becomes a Skolem constant.
- **Counting interpretations:** a binary relation on n elements is an $n \times n$ 0/1 matrix ($2^{(n^2)}$ choices); reflexive $2^{(n^2-n)}$, symmetric $2^{(n(n+1)/2)}$, antisymmetric $2^n \cdot 3^{(n(n-1)/2)}$, asymmetric $3^{(n(n-1)/2)}$.

Revision Notes 3 — Reasoning under Uncertainty

- Product rule $P(a \wedge b) = P(a | b) P(b)$; Bayes' rule $P(a | b) = P(b | a) P(a) / P(b)$; normalisation $P(X | e) = \alpha P(X, e) = \alpha \sum_y P(X, e, y)$.
- $X \perp Y | Z$ iff $P(X, Y | Z) = P(X | Z) P(Y | Z)$ iff $P(X | Y, Z) = P(X | Z)$.
- **Bayesian network semantics:** $P(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$. Each node is conditionally independent of its non-descendants given its parents, and of everything else given its Markov blanket (parents, children, children's other parents).
- **Parameter count:** $\sum (|X_i| - 1) \prod_{U \in \text{Pa}(X_i)} |U|$ (versus $\prod |X_i| - 1$ for the full joint).

d-separation: the three canonical connections

	Chain $X \rightarrow Z \rightarrow Y$	Fork $X \leftarrow Z \rightarrow Y$	Collider $X \rightarrow Z \leftarrow Y$
			
Structure	Z unobserved	Z observed (or, for a collider, a descendant of Z)	
Chain $X \rightarrow Z \rightarrow Y$	active (X, Y dependent)	blocked: $X \perp Y Z$	
Fork $X \leftarrow Z \rightarrow Y$	active	blocked: $X \perp Y Z$ (common cause)	
Collider $X \rightarrow Z \leftarrow Y$	blocked: $X \perp Y$	active — “explaining away”	

Exact inference by variable elimination

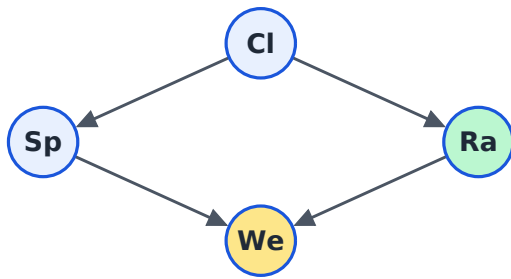
- Start with one factor per CPT, instantiating the evidence. For each hidden variable Z (in the chosen order): multiply all factors mentioning Z (pointwise product), then sum Z out. Multiply the remaining factors and normalise.
- Cost is dominated by the largest factor created — exponential in its number of variables. On polytrees (singly connected networks) VE is linear in network size; in general exact inference is NP-hard (#P-hard) and the best ordering is itself NP-hard to find.
- Every variable that is not an ancestor of the query or evidence variables is irrelevant and can be removed before elimination (it sums out to 1).

Approximate inference by sampling

Method	How a sample is produced	Estimate of $P(X e)$	Notes
Prior (direct)	sample every variable in topological order from $P(X_i \text{parents})$	fraction of samples (no evidence)	$S_{PS}(x) = P(x)$; consistent
Rejection	prior sample; discard samples inconsistent with e	$N(x, e) / N(e)$	wastes samples when $P(e)$ is small
Likelihood weighting	fix evidence; sample the rest; weight $w = \prod P(e_i \text{parents})$	$\sum w$ over samples with $X=x$ / $\sum w$	consistent; degrades with downstream evidence
Gibbs (MCMC)	start anywhere; repeatedly resample one non-evidence variable from $P(X_i \text{mb}(X_i))$	fraction of states visited	converges to the posterior (stationary distribution)

Gibbs conditional: $P(x_i | \text{mb}(X_i)) = \alpha P(x_i | \text{parents}(X_i)) \prod_{Y \in \text{children}(X_i)} P(y | \text{parents}(Y))$. Inverse-CDF rule for a Boolean variable: $X = T$ iff $u < P(X = T | \text{parents})$.

Worked example — enumeration on the sprinkler network



With $P(c)=0.5$, $P(s|c)=0.1$, $P(s|\neg c)=0.5$, $P(r|c)=0.8$, $P(r|\neg c)=0.2$, $P(w|s,r)=0.99$, $P(w|s,\neg r)=0.90$, $P(w|\neg s,r)=0.90$, $P(w|\neg s,\neg r)=0$: $P(r, w) = \sum_{c,s} P(c)P(s|c)P(r|c)P(w|s,r) = 0.4581$ and $P(w) = 0.6471$, so $P(r | w) = 0.4581 / 0.6471 \approx 0.708$.

Mock Test 001

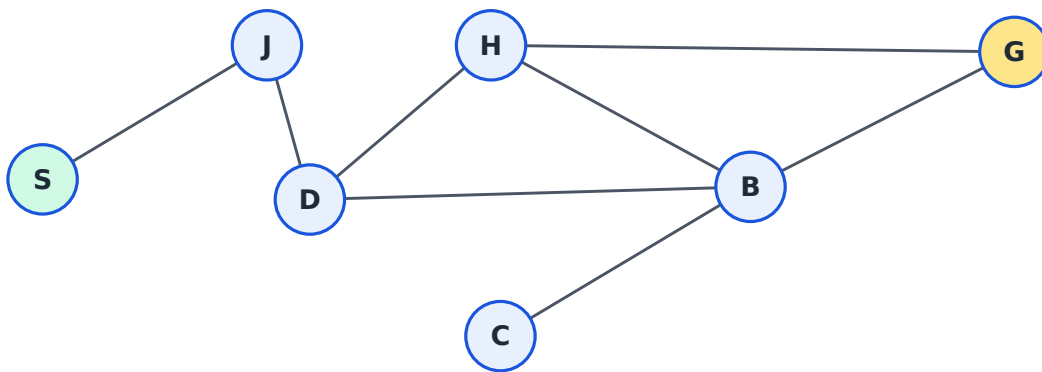
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 7 / 11	15 / 12 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

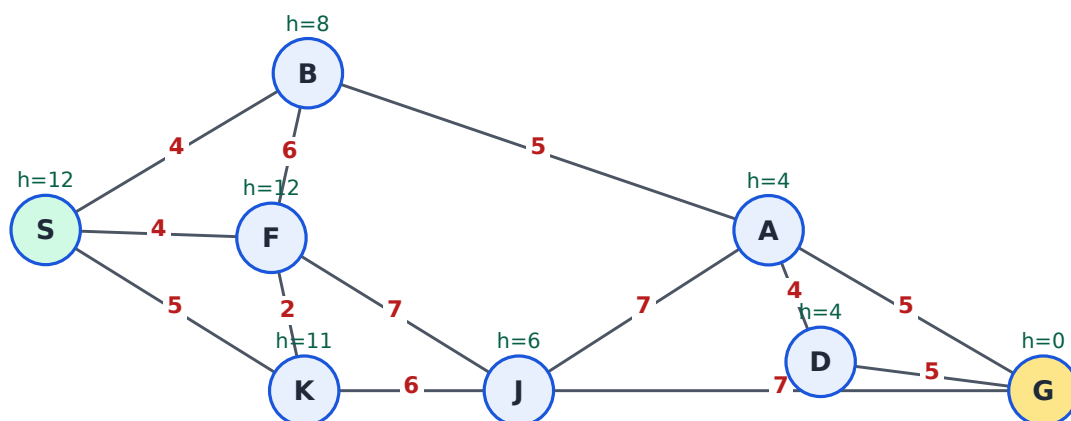
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, J, D, B, H, C, G
- ☐ (B) S, J, D, B, C, G
- ☐ (C) S, J, D, H, B, G
- ☐ (D) S, J, D, B, C, H, G

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

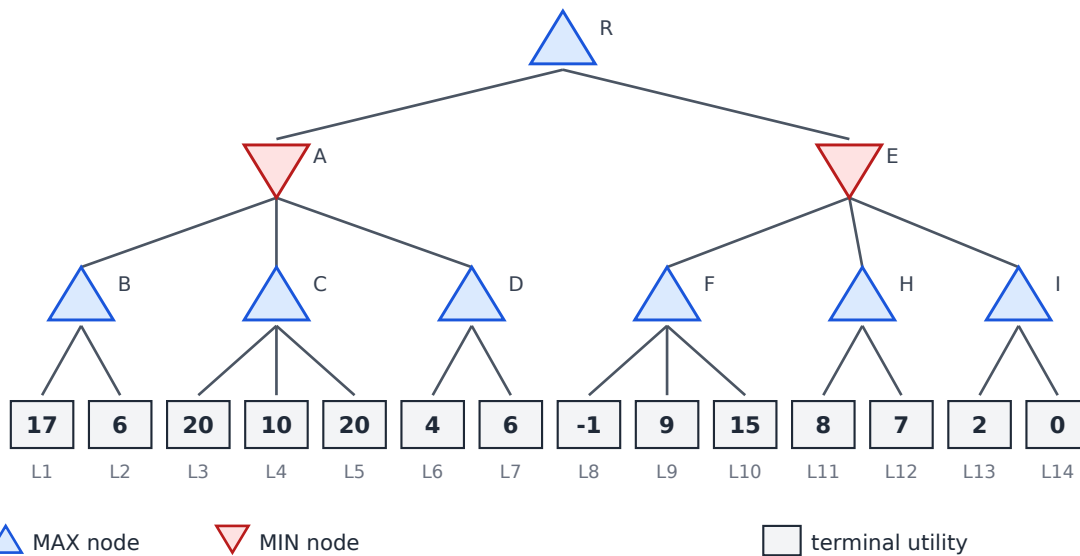
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to A; root value 8
☐ (B) Move to A; root value 20
☐ (C) Move to A; root value 6
☐ (D) Move to E; root value 2

Q.4 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following statements about forward and backward chaining over Horn knowledge bases is correct?

- ☐ (A) Forward chaining is data-driven while backward chaining is goal-driven
☐ (B) Forward chaining is goal-driven while backward chaining is data-driven
☐ (C) Both are complete for arbitrary (non-Horn) propositional knowledge bases
☐ (D) Backward chaining can derive facts that forward chaining cannot

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (R \vee Q) \wedge (P \leftrightarrow (Q \rightarrow R)) ?$$

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following is logically equivalent to $p \rightarrow q$?

- ☐ (A) $\neg q \rightarrow \neg p$
☐ (B) $q \rightarrow p$
☐ (C) $\neg p \rightarrow \neg q$
☐ (D) $p \wedge \neg q$

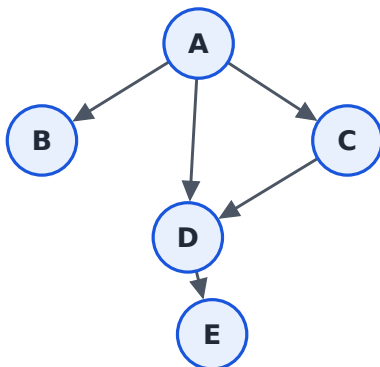
Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 2 elements, so there are $4^2 = 16$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D, E. What is the probability that a single run produces the sample **(A=T, B=F, C=T, D=T, E=T)**? (round off to 4 decimal places)



P(A=T)
0.25

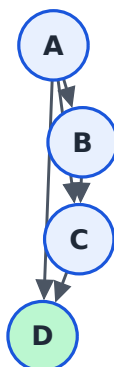
A	P(B=T ...)	A	P(C=T ...)
T	0.3	T	0.95
F	0.2	F	0.75

A	C	P(D=T ...)	D	P(E=T ...)
T	T	0.9	T	0.1
T	F	0.95	F	0.9
F	T	0.3		
F	F	0.95		

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability **$P(D = T)$** (round off to 3 decimal places).



P(A=T)
0.7

A	P(B=T ...)	A	B	P(C=T ...)
T	0.9	T	T	0.7
F	0.1	T	F	0.05
		F	T	0.9
		F	F	0.25

A	C	P(D=T ...)
T	T	0.25
T	F	0.7
F	T	0.95
F	F	0.05

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

A Boolean node has k Boolean parents. Its CPT is specified using a noisy-OR model (no leak). How many parameters are needed, versus a general CPT?

- ☐ (A) k for noisy-OR versus 2^k for a general CPT
☐ (B) 2^k for both
☐ (C) k^2 for noisy-OR versus 2^k for a general CPT
☐ (D) $2k$ for noisy-OR versus k^2 for a general CPT

Section B — 2-mark questions

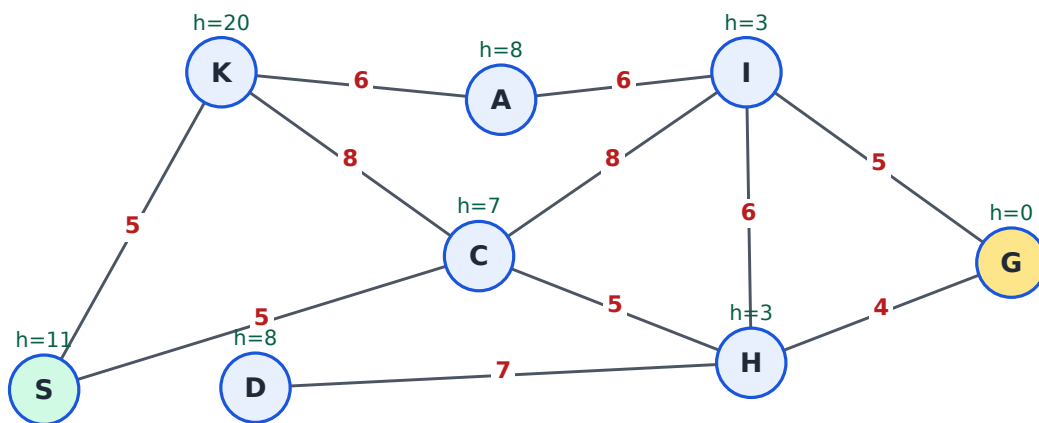
Q.11 [MSQ · 2 marks · no negative marking] Adversarial Search

Minimax search (depth-first) runs on a finite game tree with branching factor b and maximum depth m . Which of the following are TRUE?

- ☐ (A) Its time complexity is $O(b^m)$
- ☐ (B) Its space complexity is $O(bm)$ when all successors are generated at once
- ☐ (C) Against an optimal opponent, the minimax strategy achieves the minimax value of the root
- ☐ (D) It needs a heuristic evaluation function even when every path can be searched to a terminal state

Q.12 [MSQ · 2 marks · no negative marking] Informed Search - Heuristics

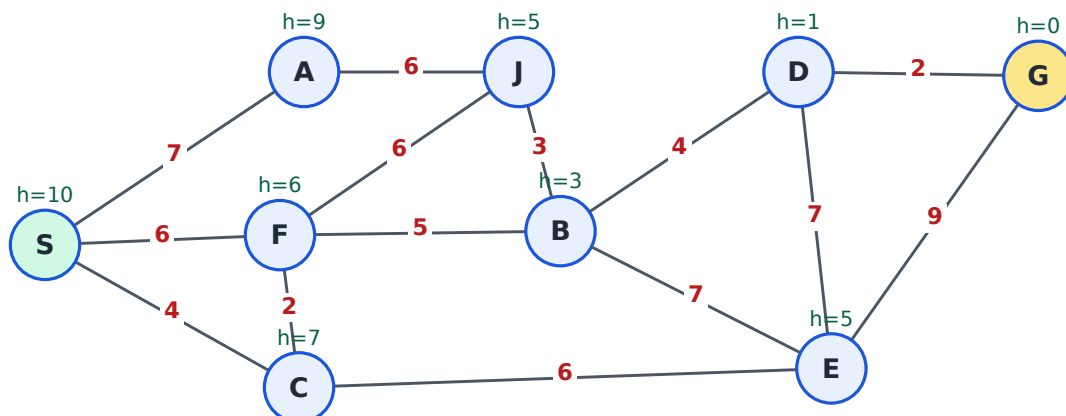
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
- ☐ (B) $h(C)$ overestimates the true cost from C to G .
- ☐ (C) The edge $K-C$ violates the consistency condition $h(K) \leq c(K,C) + h(C)$.
- ☐ (D) h is consistent (monotone).

Q.13 [NAT · 2 marks · no negative marking] Informed Search - A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both **S** and **G**?



Answer (numerical): _____

Q.14 [MCQ · 2 marks · negative: -2/3] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). The heuristic is $h(S)=7$, $h(A)=6$, $h(B)=2$, $h(C)=1$, $h(G)=0$. A* **graph search** is run from S. It keeps an explored set, never re-opens a state that has already been expanded, and goal-tests on expansion. What is the cost of the solution it returns?

- ☐ (A) 7
☐ (B) 8
☐ (C) 9
☐ (D) 10

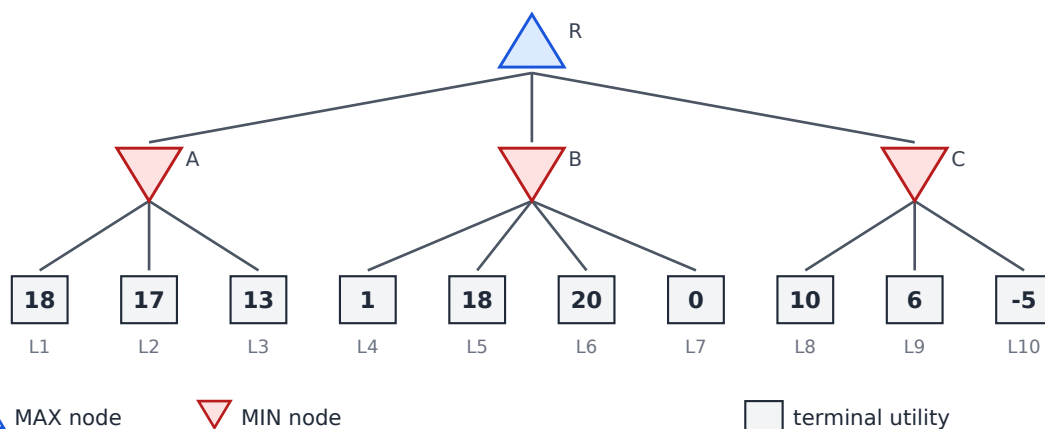
Q.15 [MCQ · 2 marks · negative: -2/3] Local Search

Steepest-ascent hill climbing solves a random 8-queens instance with probability $p \approx 0.14$. Random-restart hill climbing repeats independent trials until one succeeds. What is the expected number of trials?

- ☐ (A) About 1.14
☐ (B) About 3
☐ (C) About 7
☐ (D) About 14

Q.16 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



Answer (numerical): _____

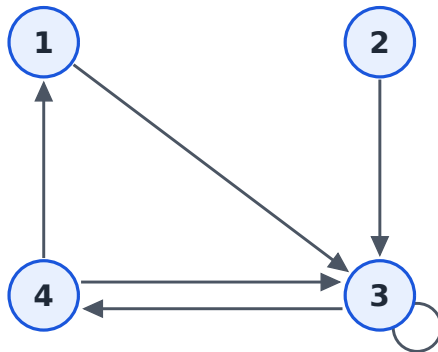
Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$
☐ (B) $\neg(R \wedge P) \leftrightarrow (\neg R \vee \neg P)$
☐ (C) $((Q \rightarrow R) \wedge (P \rightarrow R)) \rightarrow (Q \wedge P)$
☐ (D) $((P \vee R) \wedge (\neg P \vee Q)) \rightarrow (R \vee Q)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (B) $\exists y \forall x R(x, y)$
☐ (C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic

The canonical CNF (product of maxterms, each clause containing all variables) of $p \oplus q \oplus r$ has how many clauses?

Answer (numerical): _____

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

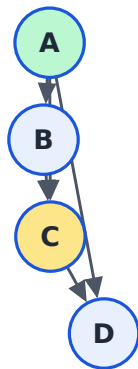
$C1: (\neg P \vee R \vee \neg S), C2: (\neg P \vee R), C3: (P \vee R \vee S), C4: (\neg P \vee \neg Q \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P \vee \neg Q \vee R)$
☐ (B) $(R \vee \neg S)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(R \vee S)$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(A = T \mid C = T)$ (round off to 3 decimal places).



A	$P(A=T)$
T	0.95

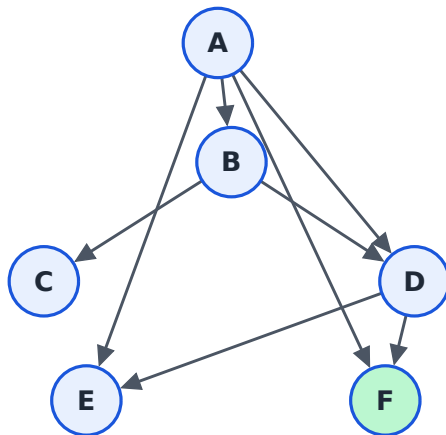
A	$P(B=T \mid \cdot)$	A	B	$P(C=T \mid \dots)$
T	0.95	T	T	0.05
F	0.1	T	F	0.95
		F	T	0.7
		F	F	0.9

A	C	$P(D=T \mid \dots)$
T	T	0.25
T	F	0.8
F	T	0.7
F	F	0.4

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(F)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, B, E, A, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

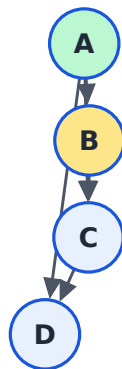
Q.23 [MSQ · 2 marks · no negative marking] Sampling

Which of the following statements about approximate inference in Bayesian networks are correct?

- ☐ (A) Likelihood weighting produces consistent estimates
- ☐ (B) Likelihood weighting performs poorly when evidence variables are downstream (e.g. leaves), since samples of upstream variables ignore that evidence
- ☐ (C) Gibbs sampling resamples each non-evidence variable from its distribution conditioned on the current values of its Markov blanket
- ☐ (D) In rejection sampling, the fraction of accepted samples grows as more evidence variables are added

Q.24 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence **B=T**. The current state of the chain is **(A=T, B=T, C=F, D=T)**. The next step resamples **A**. With what probability is A set to T? (round off to 3 decimal places)



A	P(A=T)
T	0.7

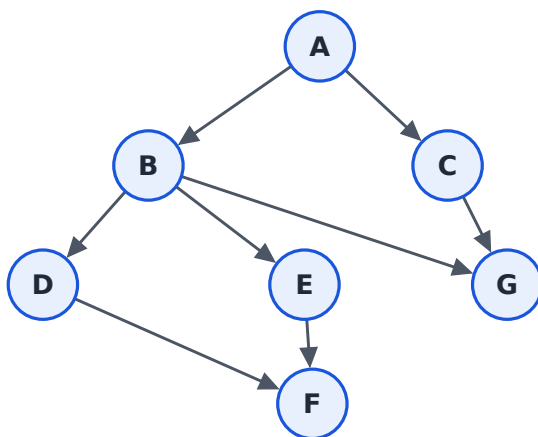
A	P(B=T .)	A	B	P(C=T ...)
T	0.6	T	T	0.25
F	0.1	T	F	0.5
		F	T	0.2
		F	F	0.7

A	C	P(D=T ...)
T	T	0.7
T	F	0.1
F	T	0.95
F	F	0.95

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp C \mid E$
☐ (B) $D \perp C$
☐ (C) $G \perp D \mid E$
☐ (D) $A \perp E \mid B, D$

Answer Key — Mock Test 001

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	MCQ	2	Informed Search	B
2	NAT	1	Informed Search	14	15	MCQ	2	Local Search	C
3	MCQ	1	Adversarial Search	C	16	NAT	2	Adversarial Search	5
4	MCQ	1	Propositional Logic	A	17	MSQ	2	Propositional Logic	B, D
5	NAT	1	Propositional Logic	3	18	MSQ	2	Predicate Logic	A, B, D
6	MCQ	1	Propositional Logic	A	19	NAT	2	Propositional Logic	4
7	NAT	1	Predicate Logic	4	20	MSQ	2	Propositional Logic	A, D
8	NAT	1	Sampling	0.0148 to 0.0152	21	NAT	2	Exact Inference	0.67 to 0.674
9	NAT	1	Exact Inference	0.388 to 0.392	22	NAT	2	Exact Inference	2
10	MCQ	1	Bayesian Networks	A	23	MSQ	2	Sampling	A, B, C
11	MSQ	2	Adversarial Search	A, B, C	24	NAT	2	Sampling	0.578 to 0.582
12	MSQ	2	Informed Search	C	25	MSQ	2	Conditional Independence	D
13	NAT	2	Informed Search	8					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: A (A: S, J, D, B, H, C, G)

Adjacency lists (alphabetical): B: {C, D, G, H}; C: {B}; D: {B, H, J}; G: {B, H}; H: {B, D, G}; J: {D, S}; S: {J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[J]
2	J	[D]
3	D	[B, H]
4	B	[H, C, G]
5	H	[C, G]
6	C	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, J, D, B, H, C, G**. The path returned is $S \rightarrow J \rightarrow D \rightarrow B \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Informed Search] Answer: 14

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

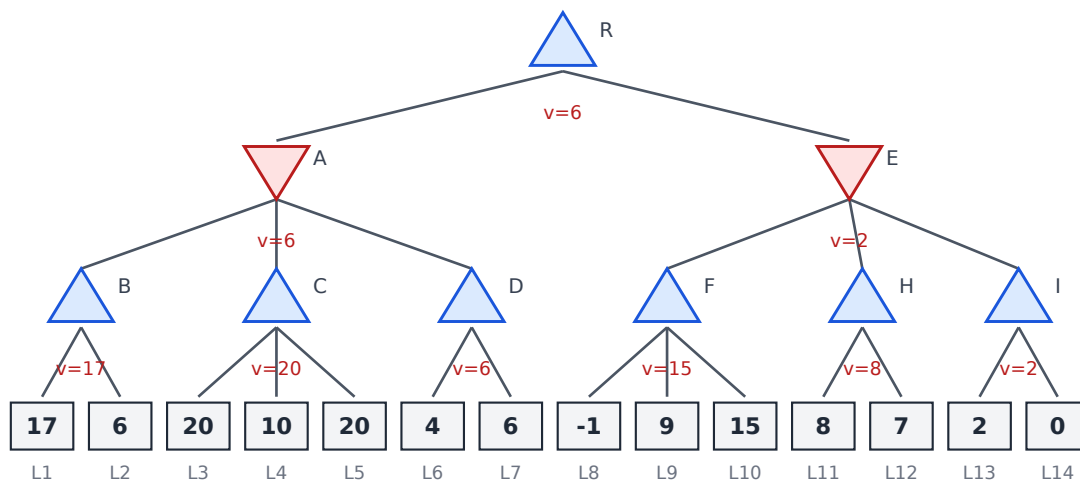
Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (12)	B(8), K(11), F(12)
2	B (8)	A(4), K(11), F(12)
3	A (4)	G(0), D(4), J(6), K(11), F(12)
4	G (0)	D(4), J(6), K(11), F(12) ← goal, stop

Path returned: $S \rightarrow B \rightarrow A \rightarrow G$, cost = $4 + 5 + 5 = 14$.

The optimal cost is $h^*(S) = 14$ — greedy happens to be optimal here, but in general it is not.

Q.3 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to A; root value 6)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 17$, $C = 20$, $D = 6$, $A = 6$, $F = 15$, $H = 8$, $I = 2$, $E = 2$. Root value = **6**; MAX's optimal first move is towards A.

Q.4 [MCQ, 1M · Propositional Logic] **Answer: A** (A: Forward chaining is data-driven while backward chaining is goal-driven)

Forward chaining starts from known facts and fires rules whose premises are satisfied (data-driven). Backward chaining starts from the query and works back through rules whose conclusion matches (goal-driven). Both are complete only for Horn (definite clause) KBs, not arbitrary ones. For Horn KBs both derive exactly the entailed atoms, so neither derives anything the other cannot.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 3**

Build the truth table (sub-formulas shown first):

P	Q	R	$R \vee Q$	$P \leftrightarrow (Q \rightarrow R)$	$(R \vee Q) \wedge (P \leftrightarrow (Q \rightarrow R))$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	F	T	F
F	T	T	T	F	F
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	F	F	F

Rows where $F = T$: **3**.

Q.6 [MCQ, 1M · Propositional Logic] **Answer: A** ($A: \neg q \rightarrow \neg p$)

The contrapositive $\neg q \rightarrow \neg p \equiv q \vee \neg p \equiv p \rightarrow q$. The converse $q \rightarrow p$ and the inverse $\neg p \rightarrow \neg q$ are equivalent to each other but differ from $p \rightarrow q$ at $p=F, q=T$ ($p \rightarrow q$ is T, $q \rightarrow p$ is F). $p \wedge \neg q$ is the negation of $p \rightarrow q$.

Q.7 [NAT, 1M · Predicate Logic] **Answer: 4**

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 2$ this gives **4** (verified by enumerating all interpretations).

Q.8 [NAT, 1M · Sampling] **Answer: 0.0148 to 0.0152**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(A=T) \times P(B=F|A=T) \times P(C=T|A=T) \times P(D=T|A=T, C=T) \times P(E=T|D=T)$
 $= 0.25 \times 0.7 \times 0.95 \times 0.9 \times 0.1 = \mathbf{0.014962} \approx 0.0150$.

Q.9 [NAT, 1M · Exact Inference] **Answer: 0.388 to 0.392**

Only the ancestors of D (A, B, C) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
$A=T, B=T, C=T$	0.11025

Assignment to other variables	joint $P(\dots, D=T)$
A=T, B=T, C=F	0.1323
A=T, B=F, C=T	0.00087
A=T, B=F, C=F	0.04655
A=F, B=T, C=T	0.02565
A=F, B=T, C=F	0.00015
A=F, B=F, C=T	0.06413
A=F, B=F, C=F	0.01013

Sum = **0.39** $\approx$ 0.390.

Q.10 [MCQ, 1M · Bayesian Networks] **Answer: A** (A: k for noisy-OR versus 2^k for a general CPT)

Noisy-OR needs one inhibition probability per parent, i.e. k numbers, and the full table is derived as $P(\neg x \mid \text{parents}) = \prod q_j$ over true parents. A general Boolean CPT has one free number for each of the 2^k parent configurations. This linear-vs-exponential saving is why canonical distributions are used.

Q.11 [MSQ, 2M · Adversarial Search] **Answer: A, B, C**

(A) and (B) are true: minimax is a complete depth-first exploration of the tree. (C) is true: minimax is optimal against an optimal adversary and guarantees the root's minimax value. (D) is false: when the search reaches terminal states, the exact utilities are used. An evaluation function is needed only when the search is cut off early.

Q.12 [MSQ, 2M · Informed Search] **Answer: C** (C: The edge K-C violates the consistency condition $h(K) \leq c(K,C) + h(C)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	C	D	G	H	I	K	S
$h(n)$	8	7	8	0	3	3	20	11
$h^*(n)$	11	9	11	0	4	5	17	14
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✗	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **no** (overestimates at K).

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $K \rightarrow A$ ($20 > 6 + 8$), $K \rightarrow C$ ($20 > 8 + 7$), $K \rightarrow S$ ($20 > 5 + 11$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Informed Search] **Answer: 8**

Heuristic values: $h(A)=9$, $h(B)=3$, $h(C)=7$, $h(D)=1$, $h(E)=5$, $h(F)=6$, $h(G)=0$, $h(J)=5$, $h(S)=10$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (10)	C(11), F(12), A(16)
2	C (11)	F(12), E(15), A(16)
3	F (12)	B(14), E(15), A(16), J(17)
4	B (14)	E(15), A(16), D(16), J(17)
5	E (15)	A(16), D(16), J(17), G(19)
6	A (16)	D(16), J(17), G(19)
7	D (16)	G(17), J(17)
8	G (17)	J(17) $\leftarrow$ goal, stop

Path returned: **S $\rightarrow$ F $\rightarrow$ B $\rightarrow$ D $\rightarrow$ G** with cost **17**; nodes expanded = **8** (S, C, F, B, E, A, D, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 17$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.14 [MCQ, 2M · Informed Search] **Answer: B** (B: 8)

Expand S: push A ($g=1$, $f=7$) and B ($g=4$, $f=6$). Expand B ($f=6$): push C ($g=5$, $f=6$) and G ($g=10$, $f=10$).

Expand C ($f=6$): push G ($g=8$, $f=8$). Expand A ($f=7$): its successors B ($g=3$) and C ($g=6$) are already explored and are discarded. Expand G ($f=8$) and return cost **8** (S $\rightarrow$ B $\rightarrow$ C $\rightarrow$ G). The optimal cost is 7 (S $\rightarrow$ A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ G).

Optimality fails because h is admissible but not consistent ($h(A)=6 > c(A,B)+h(B)=4$), and the cheaper path to B is discovered only after B is closed.

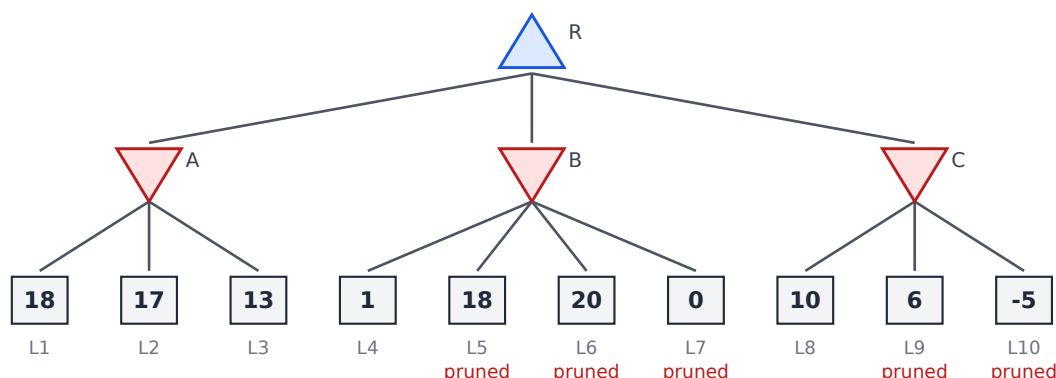
Q.15 [MCQ, 2M · Local Search] Answer: C (C: About 7)

The number of trials until the first success is geometrically distributed with mean $1/p = 1/0.14 \approx 7$. That means about 6 failures and 1 success, matching AIMA's analysis. (A) confuses $1/p$ with $1 + p$. (D) confuses $1/p$ with $100 \cdot p$.

Q.16 [NAT, 2M · Adversarial Search] Answer: 5

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $1 \leq \alpha = 13 \Rightarrow$ prune remaining children (leaves L5, L6, L7).
- At MIN node C: value $10 \leq \alpha = 13 \Rightarrow$ prune remaining children (leaves L9, L10).



Leaves evaluated (5): L1, L2, L3, L4, L8. Leaves pruned (5): L5, L6, L7, L9, L10. Root value = 13 (same as plain minimax, 13 — pruning never changes the root value).

Q.17 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $\neg(R \wedge P) \leftrightarrow (\neg R \vee \neg P)$; D: $((P \vee R) \wedge (\neg P \vee Q)) \rightarrow (R \vee Q)$)

(A) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$: **not valid** — counter-model $Q=F, R=T$.

(B) $\neg(R \wedge P) \leftrightarrow (\neg R \vee \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $((Q \rightarrow R) \wedge (P \rightarrow R)) \rightarrow (Q \wedge P)$: **not valid** — counter-model $P=T, Q=F, R=T$.

(D) $((P \vee R) \wedge (\neg P \vee Q)) \rightarrow (R \vee Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Predicate Logic] Answer: A, B, D (A: $\forall x \exists y (x \neq y \wedge R(x, y))$; B: $\exists y \forall x R(x, y)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,3), (2,3), (3,3), (3,4), (4,1), (4,3)\}$.

(A) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(B) $\exists y \forall x R(x, y)$: **TRUE** — $y = 3$ receives an edge from every x (incl. itself).

(C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(3,4)$ and $R(4,3)$ with $3 \neq 4$.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 2 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [NAT, 2M · Propositional Logic] Answer: 4

In the canonical CNF there is one maxterm (clause) for every falsifying assignment. $p \oplus q \oplus r$ is false exactly when an even number of variables is true: FFF, TTF, TFT, FTT. That gives 4 falsifying rows and hence 4 clauses. It is also known that no shorter CNF exists for parity, since every clause must contain all 3 variables.

Q.20 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $(\neg P \vee \neg Q \vee R)$; D: $(R \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee \neg Q \vee R)$ is derivable:

1. resolve $(\neg P \vee R \vee \neg S)$ and $(\neg P \vee \neg Q \vee S)$ on $S \Rightarrow (\neg P \vee \neg Q \vee R)$

(B) $(R \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(D) $(R \vee S)$ is derivable:

1. resolve $(\neg P \vee R)$ and $(P \vee R \vee S)$ on $P \Rightarrow (R \vee S)$

Q.21 [NAT, 2M · Exact Inference] Answer: 0.67 to 0.674

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B | A) \cdot P(C | A, B) \cdot P(D | A, C)$.

Inference by enumeration: $P(A | e) = \alpha \sum_{\text{hidden}} P(A, e, \text{hidden})$.

Hidden: B, D	A	product of CPT entries (A · B · C · D)	value
T, T	T	$0.95 \times 0.95 \times 0.05 \times 0.25$	0.01128
T, F	T	$0.95 \times 0.95 \times 0.05 \times 0.75$	0.03384
F, T	T	$0.95 \times 0.05 \times 0.95 \times 0.25$	0.01128
F, F	T	$0.95 \times 0.05 \times 0.95 \times 0.75$	0.03384
T, T	F	$0.05 \times 0.1 \times 0.7 \times 0.7$	0.00245
T, F	F	$0.05 \times 0.1 \times 0.7 \times 0.3$	0.00105
F, T	F	$0.05 \times 0.9 \times 0.9 \times 0.7$	0.02835
F, F	F	$0.05 \times 0.9 \times 0.9 \times 0.3$	0.01215

$P(A=T, e) = 0.09025$; $P(A=F, e) = 0.044$; $P(e) = 0.13425$.

$P(A=T | e) = 0.09025 / 0.13425 = \mathbf{0.6723} \approx 0.672$.

Q.22 [NAT, 2M · Exact Inference] Answer: 2

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(A, D, E)$, $f(A, D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C)$	$\{B, C\}$	$f(B)$
B	$f(A, B)$, $f(A, B, D)$, $f(B)$	$\{A, B, D\}$	$f(A, D)$
E	$f(A, D, E)$	$\{A, D, E\}$	$f(A, D)$
A	$f(A)$, $f(A, D, F)$, $f(A, D)$, $f(A, D)$	$\{A, D, F\}$	$f(D, F)$
D	$f(D, F)$	$\{D, F\}$	$f(F)$

The largest factor produced has **2** variable(s) (created when eliminating B); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over F and normalise.

Q.23 [MSQ, 2M · Sampling] Answer: A, B, C

(A) AIMA shows $S_{WS}(z, e) w(z, e) = P(z, e)$, so weighted estimates are consistent. (B) Non-evidence variables are sampled from their parents only, so evidence at leaves does not influence them; many samples get tiny weights. (C) This is exactly the Gibbs step: $P(x_i | mb(X_i))$. (D) is false: each extra evidence variable generally reduces $P(e)$, so fewer samples are accepted.

Q.24 [NAT, 2M · Sampling] Answer: 0.578 to 0.582

Gibbs sampling draws A from $P(A | mb(A))$, where $mb(A) = \{B, C, D\}$:

$P(A | mb) \propto P(A | \text{parents}(A)) \times \prod_{\text{children } Y} P(y | \text{parents}(Y))$.

$A=T$: $P(A=T) \times P(B=T|A=T) \times P(C=F|A=T, B=T) \times P(D=T|A=T, C=F) = 0.7 \times 0.6 \times 0.75 \times 0.1 = 0.0315$

$A=F$: $P(A=F) \times P(B=T|A=F) \times P(C=F|A=F, B=T) \times P(D=T|A=F, C=F) = 0.3 \times 0.1 \times 0.8 \times 0.95 = 0.0228$

Normalise: $0.0315 / (0.0315 + 0.0228) = \mathbf{0.5801}$.

Q.25 [MSQ, 2M · Conditional Independence] Answer: D ($D: A \perp E | B, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $D \perp C | E$: **FALSE** — the path $D \leftarrow B \leftarrow A \rightarrow C$ is active given $\{E\}$.

(B) $D \perp C$: **FALSE** — the path $D \leftarrow B \leftarrow A \rightarrow C$ is active given $\{\}$.

(C) $G \perp D | E$: **FALSE** — the path $G \leftarrow B \rightarrow D$ is active given $\{E\}$.

(D) $A \perp E | B, D$: **TRUE** — $A \rightarrow B \rightarrow D \rightarrow F \leftarrow E$ (blocked: observed chain node B); $A \rightarrow B \rightarrow E$ (blocked: observed chain node B); $A \rightarrow C \rightarrow G \leftarrow B \rightarrow D \rightarrow F \leftarrow E$ (blocked: collider G with neither it nor a descendant observed); $A \rightarrow C \rightarrow G \leftarrow B \rightarrow E$ (blocked: collider G with neither it nor a descendant observed).

Mock Test 002

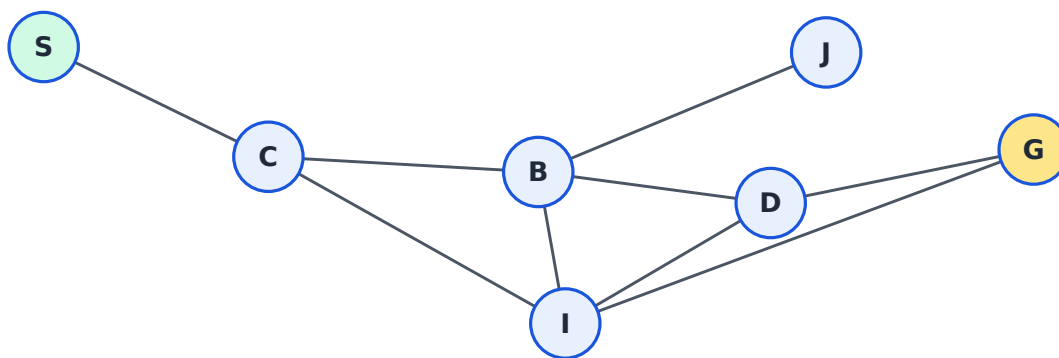
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 10 / 11	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

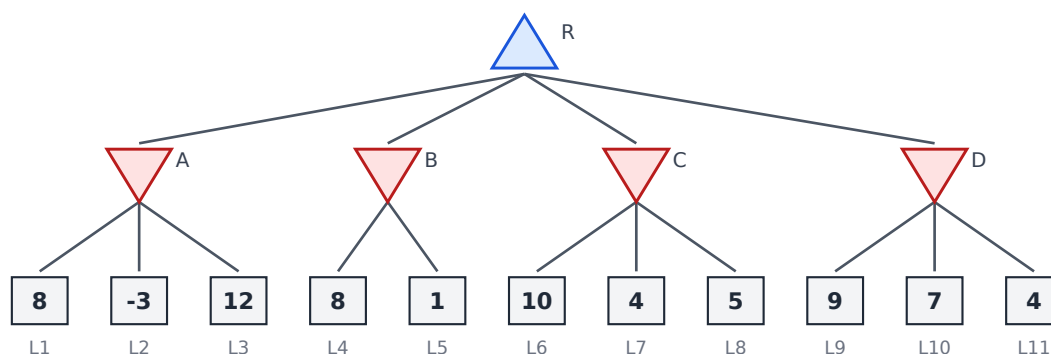
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, C, I, G
- ☐ (B) S, B, C, D, G
- ☐ (C) S, C, B, D, G
- ☐ (D) S, C, B, I, D, J, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



MAX node MIN node

terminal utility

Answer (numerical): _____

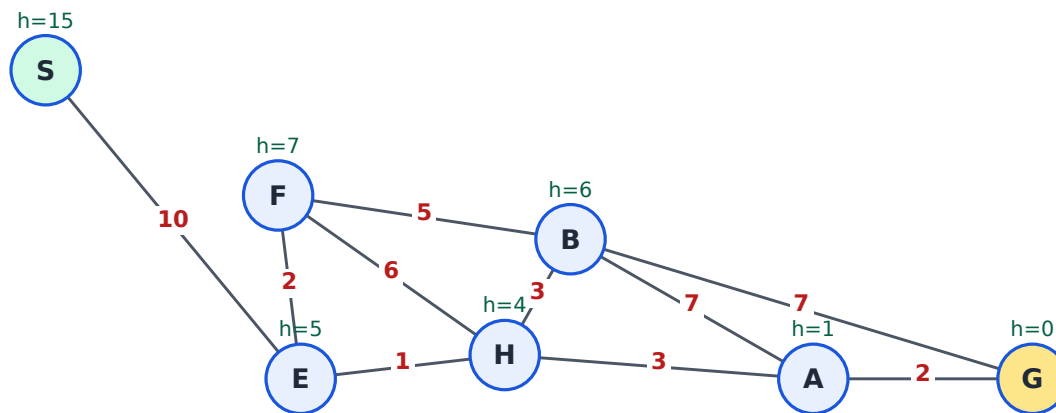
Q.3 [NAT · 1 mark · no negative marking] Informed Search

A* generates $N = 84$ nodes (not counting the root) to find a solution at depth $d = 3$. Using AIMA's definition $N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^d$, what is the effective branching factor b^* ?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over a, b, c does the CNF $(a \vee b \vee c) \wedge (\neg a \vee \neg b) \wedge (\neg b \vee \neg c) \wedge (\neg a \vee \neg c)$ have?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((P \vee R) \wedge Q) \vee \neg(P \rightarrow \neg Q) ?$$

Answer (numerical): _____

Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following inference rules are sound in propositional logic?

- ☐ (A) Modus ponens: from $\alpha \rightarrow \beta$ and α infer β
- ☐ (B) And-elimination: from $\alpha \wedge \beta$ infer α
- ☐ (C) Abduction: from $\alpha \rightarrow \beta$ and β infer α
- ☐ (D) Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$

Q.8 [NAT · 1 mark · no negative marking] Bayesian Networks

A naive Bayes classifier has a class variable with 3 values and 10 Boolean features. How many independent parameters does the Bayesian network have?

Answer (numerical): _____

Q.9 [MSQ · 1 mark · no negative marking] Sampling

Which statements correctly describe how sampling algorithms treat the evidence e ?

- ☐ (A) Likelihood weighting fixes evidence variables to their observed values and samples only non-evidence variables
- ☐ (B) Prior sampling ignores evidence while generating samples
- ☐ (C) Gibbs sampling changes the values of evidence variables during the Markov chain
- ☐ (D) Rejection sampling discards samples that are inconsistent with e

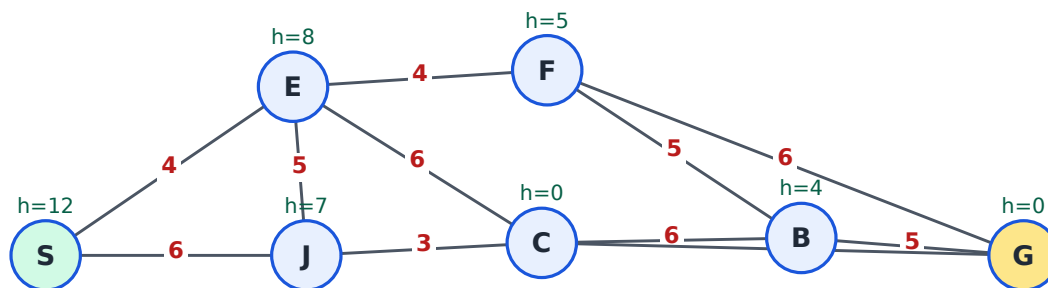
Q.10 [MCQ · 1 mark · negative: -1/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Likelihood weighting is used with evidence $C = \text{true}$ only (a root node), to estimate $P(R | c)$. What is the weight of every sample?

- ☐ (A) 0.5
☐ (B) 1.0
☐ (C) It depends on the sampled value of R
☐ (D) 0.8

Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

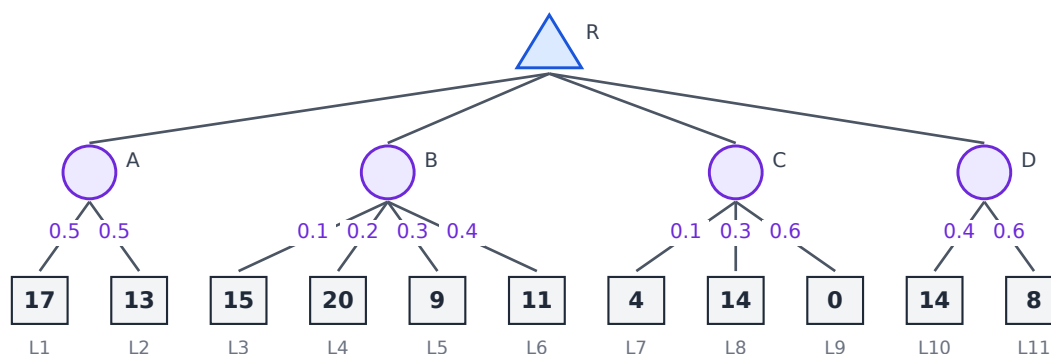
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(C)$ overestimates the true cost from C to G.
☐ (B) h is admissible.
☐ (C) h is consistent (monotone).
☐ (D) The edge J-C violates the consistency condition $h(J) \leq c(J,C) + h(C)$.

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)

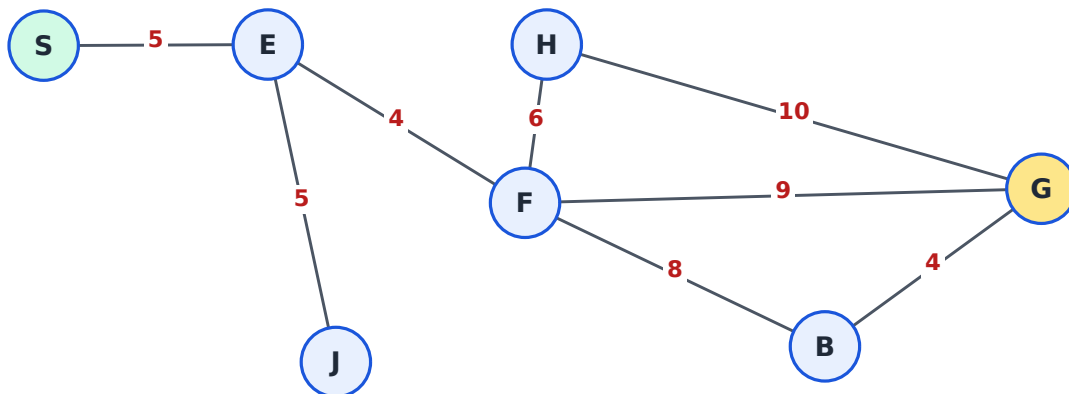


MAX node CHANCE node (edge = probability) terminal utility

Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Uninformed Search

Which of the following statements about tree search versus graph search are TRUE?

- ☐ (A) Graph search keeps an explored (reached) set so that no state is expanded more than once
- ☐ (B) On a finite state space with reversible actions, tree search can face an infinite search tree because of loopy paths
- ☐ (C) Depth-first graph search is complete on infinite state spaces
- ☐ (D) The memory used by depth-first graph search can grow in proportion to the number of reachable states

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \vee P) \wedge (\neg R \vee Q)) \rightarrow (P \vee Q)$
- ☐ (B) $(\neg Q \rightarrow (P \rightarrow R)) \leftrightarrow ((\neg Q \wedge P) \rightarrow R)$
- ☐ (C) $((P \rightarrow R) \wedge R) \rightarrow P$
- ☐ (D) $((P \rightarrow Q) \wedge \neg P) \rightarrow \neg Q$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

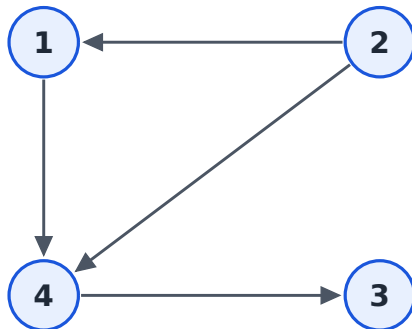
- (1) $(S \leftrightarrow \neg S) \leftrightarrow (\neg Q \leftrightarrow S)$
- (2) $(R \wedge S) \leftrightarrow (Q \leftrightarrow S)$
- (3) $\neg(\neg S \wedge \neg Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg(Q \leftrightarrow Q)$
- ☐ (B) $\neg Q \wedge R$
- ☐ (C) $\neg Q \rightarrow (P \leftrightarrow S)$
- ☐ (D) $\neg S \leftrightarrow Q$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.18 [NAT · 2 marks · no negative marking] Propositional Logic

Consider all propositional formulas over the symbols p, q, r , identified up to logical equivalence. How many of these equivalence classes ϕ satisfy $(p \wedge q) \models \phi$?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **D, C** and rules:

- R1: $D \rightarrow H$
 R2: $H \wedge D \wedge B \rightarrow C$
 R3: $B \wedge D \rightarrow H$
 R4: $A \wedge C \rightarrow F$
 R5: $K \wedge B \wedge E \rightarrow H$
 R6: $F \wedge A \rightarrow J$
 R7: $C \wedge H \rightarrow J$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) E is entailed
☐ (B) H is entailed
☐ (C) J is entailed
☐ (D) K is entailed

Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following statements about first-order logic are TRUE?

- ☐ (A) Entailment in first-order logic is semidecidable
☐ (B) Resolution (with factoring and unification) is refutation-complete for first-order logic
☐ (C) Validity of first-order sentences is decidable
☐ (D) Forward chaining on a Datalog KB (definite clauses without function symbols) always terminates

Q.21 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(A = T \mid C = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	T	F	F
2	F	T	F	F
3	F	T	T	T
4	F	T	F	F
5	T	T	F	F
6	F	T	F	T
7	F	T	F	T
8	F	T	F	T
9	F	T	F	F
10	T	F	F	F
11	T	T	T	T
12	T	T	F	F
13	F	T	F	F
14	T	T	F	F
15	F	T	F	F
16	F	T	F	T
17	F	T	F	T
18	T	F	T	T

Answer (numerical): _____

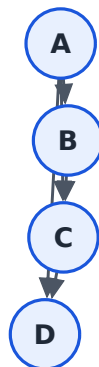
Q.22 [MCQ · 2 marks · negative: -2/3] Conditional Independence

Consider the Bayesian network (DAG) with edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$, $B \rightarrow F$. Which of the following pairs is d-separated by the given set?

- ☐ (A) A and B given {E}
- ☐ (B) A and F given {D}
- ☐ (C) A and E given {D}
- ☐ (D) E and F given $\emptyset$

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.79}, \mathbf{0.06}, \mathbf{0.52}, \mathbf{0.99} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



A	P(A=T)
T	0.7
F	0.3

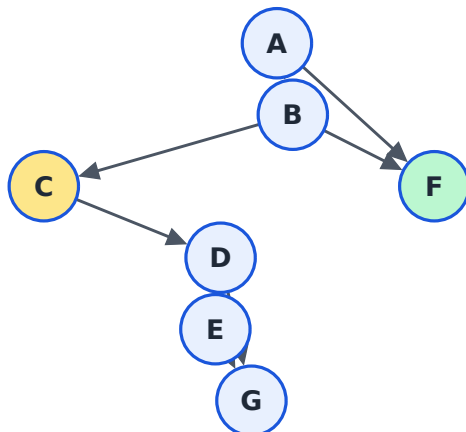
A	P(B=T .)	A	B	P(C=T ...)
T	0.4	T	T	0.8
F	0.3	T	F	0.2
		F	T	0.05
		F	F	0.75

A	C	P(D=T ...)
T	T	0.75
T	F	0.95
F	T	0.3
F	F	0.1

- ☐ (A) A=T, B=F, C=T, D=T
☐ (B) A=T, B=T, C=F, D=F
☐ (C) A=F, B=T, C=F, D=F
☐ (D) A=F, B=F, C=F, D=F

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

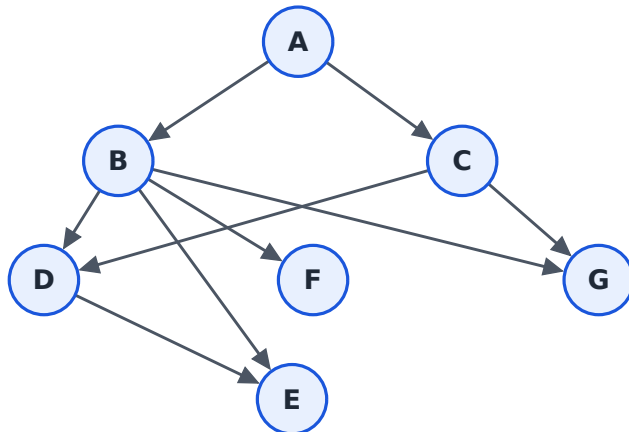
Variable elimination is used to compute $\mathbf{P(F \mid C = T)}$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, B, G, D, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $F \perp C \mid B$
☐ (B) $D \perp A \mid C$
☐ (C) $G \perp C \mid E, F$
☐ (D) $E \perp F \mid A, B$

Answer Key — Mock Test 002

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	C	14	MSQ	2	Uninformed Search	A, B, D
2	NAT	1	Adversarial Search	4	15	MSQ	2	Propositional Logic	A, B
3	NAT	1	Informed Search	4	16	MSQ	2	Propositional Logic	C
4	NAT	1	Informed Search	16	17	MSQ	2	Predicate Logic	A, C, D
5	NAT	1	Propositional Logic	3	18	NAT	2	Propositional Logic	64
6	NAT	1	Propositional Logic	3	19	MSQ	2	Propositional Logic	B, C
7	MSQ	1	Propositional Logic	A, B, D	20	MSQ	2	Predicate Logic	A, B, D
8	NAT	1	Bayesian Networks	32	21	NAT	2	Sampling	0.26 to 0.28
9	MSQ	1	Sampling	A, B, D	22	MCQ	2	Conditional Independence	C
10	MCQ	1	Sampling	A	23	MCQ	2	Sampling	C
11	MSQ	2	Informed Search	B, D	24	NAT	2	Exact Inference	2
12	NAT	2	Adversarial Search	14.99 to 15.01	25	MSQ	2	Conditional Independence	A, D
13	NAT	2	Uninformed Search	7					

Detailed Solutions

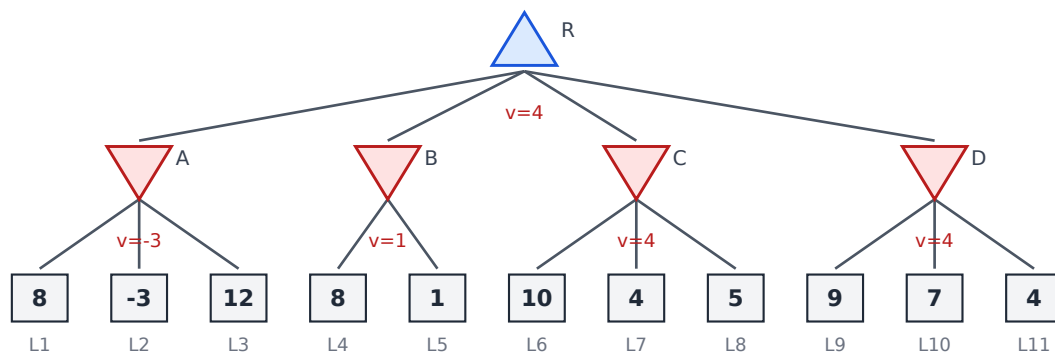
Q.1 [MCQ, 1M · Uninformed Search] Answer: C (C: S, C, B, D, G)

Adjacency lists (alphabetical): B: {C, D, I, J}; C: {B, I, S}; D: {B, G, I}; G: {D, I}; I: {B, C, D, G}; J: {B}; S: {C}. DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, C, B, D, G**; the path found is $S \rightarrow C \rightarrow B \rightarrow D \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] Answer: 4

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = -3, B = 1, C = 4, D = 4. Root value = 4; MAX's optimal first move is towards C, D.

Q.3 [NAT, 1M · Informed Search] Answer: 4

Solve $1 + b^* + b^{*2} + b^{*3} = 85$, which means $b^* + b^{*2} + b^{*3} = 84$. Try $b^* = 4$: $4 + 16 + 64 = 84$. So $b^* = 4$.

Q.4 [NAT, 1M · Informed Search] Answer: 16

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (15)	E(5)
2	E (5)	H(4), F(7)
3	H (4)	A(1), B(6), F(7)
4	A (1)	G(0), B(6), F(7)
5	G (0)	B(6), F(7) ← goal, stop

Path returned: **S** → **E** → **H** → **A** → **G**, cost = 10 + 1 + 3 + 2 = **16**.

The optimal cost is $h^*(S) = 16$ — greedy happens to be optimal here, but in general it is not.

Q.5 [NAT, 1M · Propositional Logic] Answer: 3

The first clause says at least one of a, b, c is true. The three binary clauses say no two are simultaneously true. Together they encode 'exactly one of a, b, c is true', which has 3 models: (T,F,F), (F,T,F), (F,F,T).

Q.6 [NAT, 1M · Propositional Logic] Answer: 3

Build the truth table (sub-formulas shown first):

P	Q	R	$(P \vee R) \wedge Q$	$\neg(P \rightarrow \neg Q)$	$((P \vee R) \wedge Q) \vee \neg(P \rightarrow \neg Q)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	F	F	F
F	T	T	T	F	T
F	T	F	F	F	F
F	F	T	F	F	F
F	F	F	F	F	F

Rows where F = T: **3**.

Q.7 [MSQ, 1M · Propositional Logic] Answer: A, B, D (A: Modus ponens: from $\alpha \rightarrow \beta$ and α infer β ; B:

And-elimination: from $\alpha \wedge \beta$ infer α ; D: Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$)

Modus ponens, and-elimination and resolution are truth-preserving: whenever their premises are true, their conclusion is true. For resolution, if β is true then γ must be true, else α must be true. Abduction is unsound: $\alpha=F, \beta=T$ makes both premises true and the conclusion false.

Q.8 [NAT, 1M · Bayesian Networks] Answer: 32

$P(C)$ needs $3 - 1 = 2$ numbers. Each feature's CPT $P(F_i | C)$ has 3 rows with 1 free number each, i.e. 3 numbers. Total = $2 + 10 \times 3 = 32$ (versus $3 \times 2^{10} - 1 = 3071$ for the full joint).

Q.9 [MSQ, 1M · Sampling] Answer: A, B, D

(A) LW clamps evidence and weights each sample by $\prod P(e_i | \text{parents})$. (B) Prior sampling generates from the prior; rejection sampling then filters these samples (D). (C) is false: in Gibbs sampling evidence variables stay fixed at their observed values; only non-evidence variables are resampled.

Q.10 [MCQ, 1M · Sampling] Answer: A (A: 0.5)

The weight is the product of $P(e_i | \text{parents})$ over evidence variables; here only C, with no parents, so $w = P(c) = 0.5$ for every sample. Since all weights are equal, LW reduces to sampling R from $P(R | c)$, which is the exact posterior. The weight does not depend on R, and 0.8 is $P(r | c)$, a sampling probability, not a weight.

Q.11 [MSQ, 2M · Informed Search] Answer: B, D (B: h is admissible.; D: The edge J-C violates the consistency condition $h(J) \leq c(J,C) + h(C)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	E	F	G	J	S
$h(n)$	4	0	8	5	0	7	12
$h^*(n)$	5	6	10	6	0	9	14
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $J \rightarrow C$ ($7 > 3 + 0$), $E \rightarrow C$ ($8 > 6 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [NAT, 2M · Adversarial Search] Answer: 14.99 to 15.01

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.5 \times 17 + 0.5 \times 13 = 15$
- $B = 0.1 \times 15 + 0.2 \times 20 + 0.3 \times 9 + 0.4 \times 11 = 12.6$
- $C = 0.1 \times 4 + 0.3 \times 14 + 0.6 \times 0 = 4.6$
- $D = 0.4 \times 14 + 0.6 \times 8 = 10.4$

Root = $\max(15, 12.6, 4.6, 10.4) = 15$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.13 [NAT, 2M · Uninformed Search] Answer: 7

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	E(5)
2	E (5)	F(9), J(10)
3	F (9)	J(10), H(15), B(17), G(18)
4	J (10)	H(15), B(17), G(18)
5	H (15)	B(17), G(18)
6	B (17)	G(18)
7	G (18)	← goal, stop

Path returned: **S** $\rightarrow$ **E** $\rightarrow$ **F** $\rightarrow$ **G**, cost = $5 + 4 + 9 = 18$. Nodes expanded (including S and G): **7** (S, E, F, J, H, B, G).

Q.14 [MSQ, 2M · Uninformed Search] Answer: A, B, D

(A) is true: the explored set is what defines graph search. (B) is true: with reversible actions, a path such as $A \rightarrow B \rightarrow A \rightarrow B \dots$ can be extended forever, so even a finite state space produces an infinite tree. (C) is false: in an infinite state space, DFS can follow an infinite branch forever. (D) is true: the explored set can hold every reachable state, which removes DFS's linear-space advantage.

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, B (A: $((R \vee P) \wedge (\neg R \vee Q)) \rightarrow (P \vee Q)$; B: $(\neg Q \rightarrow (P \rightarrow R)) \leftrightarrow ((\neg Q \wedge P) \rightarrow R)$)

(A) $((R \vee P) \wedge (\neg R \vee Q)) \rightarrow (P \vee Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(B) $(\neg Q \rightarrow (P \rightarrow R)) \leftrightarrow ((\neg Q \wedge P) \rightarrow R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $((P \rightarrow R) \wedge R) \rightarrow P$: **not valid** — counter-model $P=F, R=T$.

(D) $((P \rightarrow Q) \wedge \neg P) \rightarrow \neg Q$: **not valid** — counter-model $P=F, Q=T$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.16 [MSQ, 2M · Propositional Logic] Answer: C (C: $\neg Q \rightarrow (P \leftrightarrow S)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
T	T	T	T	T
F	T	T	T	T

(A) $\neg(Q \leftrightarrow Q)$ is false in the KB-model $P=T, Q=T, R=T, S=T \Rightarrow$ not entailed.

(B) $\neg Q \wedge R$ is false in the KB-model $P=T, Q=T, R=T, S=T \Rightarrow$ not entailed.

(C) $\neg Q \rightarrow (P \leftrightarrow S)$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $\neg S \leftrightarrow Q$ is false in the KB-model $P=T, Q=T, R=T, S=T \Rightarrow$ not entailed.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, C, D (A: $\exists x \forall y \neg R(x, y)$; C: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,4), (2,1), (2,4), (4,3)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 3 has no outgoing edge.

(B) $\forall y \exists x R(x, y)$: **FALSE** — 2 has no incoming edge.

(C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 2 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [NAT, 2M · Propositional Logic] Answer: 64

Up to equivalence a formula is determined by its set of models, a subset of the 8 assignments. $(p \wedge q) \models \phi$ iff $M(p \wedge q) \subseteq M(\phi)$. Over p, q, r , $M(p \wedge q) = \{TTT, TTF\}$ has 2 elements. So $M(\phi)$ must contain these 2 assignments and may contain any subset of the other 6: $2^6 = 64$.

Q.19 [MSQ, 2M · Propositional Logic] Answer: B, C (B: H is entailed; C: J is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \rightarrow H$; add H.

Round 2: fire $C \wedge H \rightarrow J$; add J.

Fixed point: $\{C, D, H, J\}$ — 4 atoms are entailed (facts: C, D; newly derived: H, J). Atoms never derived: A, B, E, F, K.

Q.20 [MSQ, 2M · Predicate Logic] Answer: A, B, D

(A) True: an algorithm can confirm every entailed sentence but may run forever on non-entailed ones

(Turing/Church). (B) True: Robinson's resolution theorem. (C) False: FOL validity is undecidable. (D) True:

with no function symbols only finitely many ground atoms exist, so a fixed point is reached in finitely many steps.

Q.21 [NAT, 2M · Sampling] Answer: 0.26 to 0.28

Rejection sampling keeps only samples consistent with the evidence $C = F$: samples 1, 2, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17 (15 accepted, 3 rejected).

Among the accepted samples, $A = T$ in 4. Estimate = $4/15 = \mathbf{0.2667}$.

(For this network the exact value is 0.4451; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.22 [MCQ, 2M · Conditional Independence] Answer: C (C: A and E given $\{D\}$)

(C): every path from A to E goes $A \rightarrow C \rightarrow D \rightarrow E$; D is a non-collider on it and is observed, so it is blocked:

d-separated. (A): the path $A \rightarrow C \leftarrow B$ has collider C whose descendant E is observed, so it is active. (B): the

path $A \rightarrow C \leftarrow B \rightarrow F$ has collider C with observed descendant D, and non-collider B unobserved, so it is active.

(D): $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ has no colliders and nothing observed, so it is active.

Q.23 [MCQ, 2M · Sampling] Answer: C (C: A=F, B=T, C=F, D=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.7$; $u = 0.79 \geq 0.7 \Rightarrow A = F$
- B: $P(B=T \mid A=F) = 0.3$; $u = 0.06 < 0.3 \Rightarrow B = T$
- C: $P(C=T \mid A=F, B=T) = 0.05$; $u = 0.52 \geq 0.05 \Rightarrow C = F$
- D: $P(D=T \mid A=F, C=F) = 0.1$; $u = 0.99 \geq 0.1 \Rightarrow D = F$

Sample: **(A=F, B=T, C=F, D=F)**.

Q.24 [NAT, 2M · Exact Inference] Answer: 2

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B)$, $f(D)$, $f(D, E)$, $f(A, B, F)$, $f(D, E, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, B)$, $f(A, B, F)$	$\{A, B, F\}$	$f(B, F)$
B	$f(B)$, $f(B, F)$	$\{B, F\}$	$f(F)$
G	$f(D, E, G)$	$\{D, E, G\}$	$f(D, E)$
D	$f(D)$, $f(D, E)$, $f(D, E)$	$\{D, E\}$	$f(E)$
E	$f(E)$	$\{E\}$	$f()$

The largest factor produced has **2** variable(s) (created when eliminating A); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over F and normalise.

Q.25 [MSQ, 2M · Conditional Independence] Answer: A, D (A: $F \perp C \mid B$; D: $E \perp F \mid A, B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $F \perp C \mid B$: **TRUE** — $F \leftarrow B \leftarrow A \rightarrow C$ (blocked: observed chain node B); $F \leftarrow B \rightarrow D \leftarrow C$ (blocked: observed fork node B); $F \leftarrow B \rightarrow E \leftarrow D \leftarrow C$ (blocked: observed fork node B); $F \leftarrow B \rightarrow G \leftarrow C$ (blocked: observed fork node B).

(B) $D \perp A \mid C$: **FALSE** — the path $D \leftarrow B \leftarrow A$ is active given $\{C\}$.

(C) $G \perp C \mid E, F$: **FALSE** — the path $G \leftarrow B \leftarrow A \rightarrow C$ is active given $\{E, F\}$.

(D) $E \perp F \mid A, B$: **TRUE** — $E \leftarrow B \rightarrow F$ (blocked: observed fork node B); $E \leftarrow D \leftarrow B \rightarrow F$ (blocked: observed fork node B); $E \leftarrow D \leftarrow C \leftarrow A \rightarrow B \rightarrow F$ (blocked: observed fork node A); $E \leftarrow D \leftarrow C \rightarrow G \leftarrow B \rightarrow F$ (blocked: collider G with neither it nor a descendant observed).

Mock Test 003

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	3 / 9 / 13	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

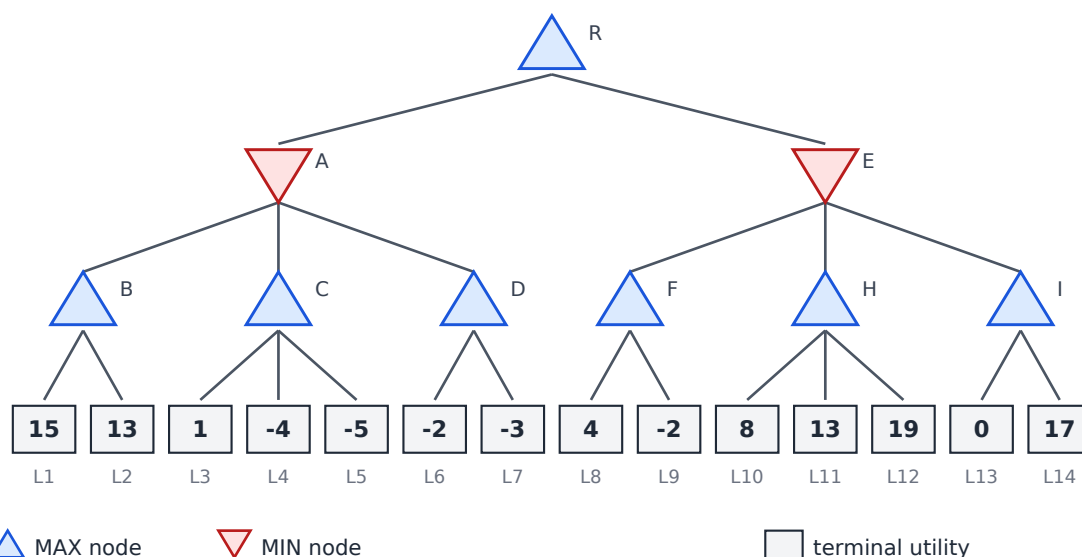
Q.1 [NAT · 1 mark · no negative marking] Uninformed Search

Bidirectional BFS runs on a problem with uniform branching factor $b = 10$ in both directions, and the solution has depth $d = 6$. Each search goes to depth $d/2 = 3$, and the two frontiers meet. Assume each direction generates every node down to depth 3, excluding its own root. How many nodes are generated in total by the two searches?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

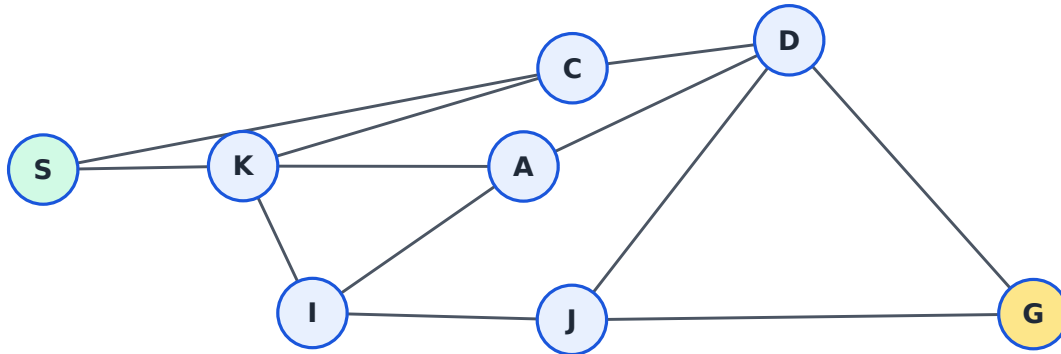
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, K, C, I, A, D, J, G
☐ (B) S, C, D, A, I, J, G
☐ (C) S, C, K, D, A, I, G
☐ (D) S, C, K, D, I, A, G

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 2$ (so $b^d = 4$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (P \rightarrow (R \leftrightarrow \neg R)) \vee ((P \leftrightarrow Q) \leftrightarrow \neg R) ?$$

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following are tautologies?

- ☐ (A) $p \vee \neg p$
☐ (B) $\neg(p \wedge \neg p)$
☐ (C) $p \rightarrow (p \wedge q)$
☐ (D) $(p \wedge (p \rightarrow q)) \rightarrow q$

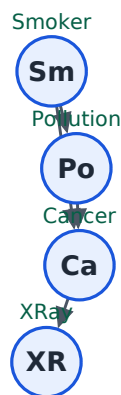
Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following statements are TRUE?

- ☐ (A) Propositional resolution is refutation-complete
☐ (B) Modus ponens alone is complete for arbitrary propositional knowledge bases
☐ (C) Forward chaining is sound and complete for knowledge bases of definite clauses
☐ (D) A sound inference procedure derives only sentences that are entailed by the KB

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) What is the probability that a single run produces the sample **(Sm=T, Po=F, Ca=F, XR=T)**? (round off to 4 decimal places)



Sm	P(Sm=T)
T	0.95
F	

Sm	P(Po=T ..)	Sm	Po	P(Ca=T ...)
T	0.25	T	T	0.4
F	0.7	T	F	0.2
		F	T	0.4
		F	F	0.05

Ca	P(XR=T ...)
T	0.6
F	0.75

Answer (numerical): _____

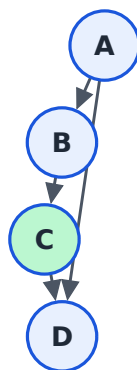
Q.9 [MSQ · 1 mark · no negative marking] Probability for AI

Let A and B be Boolean random variables with $P(b) > 0$ and $P(\neg b) > 0$. Which of the following identities hold for **every** such distribution?

- ☐ (A) $P(a \wedge b) = P(a | b) P(b)$
☐ (B) $P(a | b) = P(b | a) P(a) / P(b)$
☐ (C) $P(a | b) + P(a | \neg b) = 1$
☐ (D) $P(a) = P(a | b) P(b) + P(a | \neg b) P(\neg b)$

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T | \text{parents})$), compute the marginal probability **$P(C = T)$** (round off to 3 decimal places).



A	P(A=T)
T	0.5
F	

A	P(B=T ...)	B	P(C=T ...)
T	0.7	T	0.75
F	0.4	F	0.75

A	C	P(D=T ...)
T	T	0.5
T	F	0.25
F	T	0.5
F	F	0.9

Answer (numerical): _____

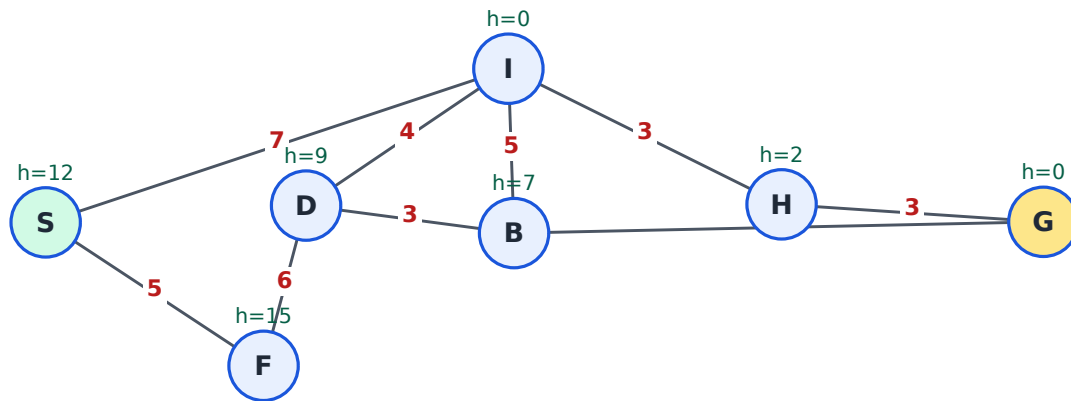
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search

A three-ply binary game tree has levels MAX (root), MIN, MAX, and then leaves. The eight leaf values, left to right, are 5, 6, 7, 4, 5, 3, 6, 7. Each MAX node at the third level owns two consecutive leaves. Alpha-beta pruning processes children left to right. How many leaves are **evaluated**?

Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

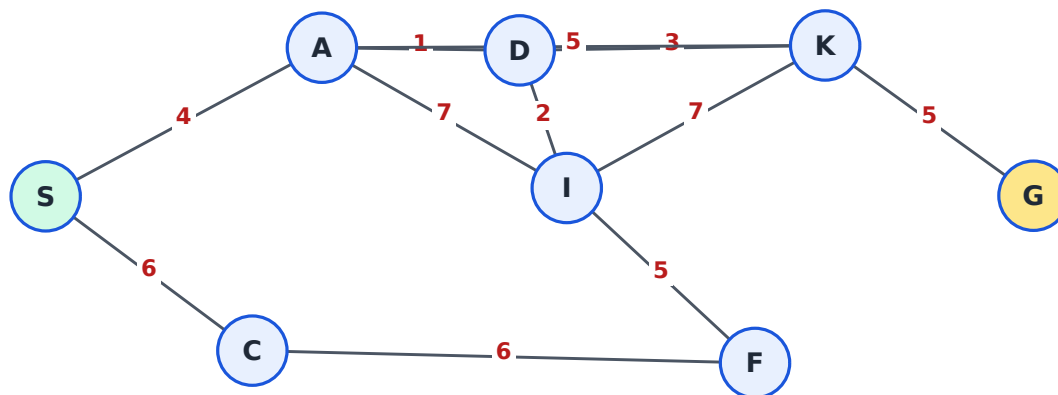
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
☐ (B) h is consistent (monotone).
☐ (C) $h(H)$ overestimates the true cost from H to G .
☐ (D) The edge $S-I$ violates the consistency condition $h(S) \leq c(S,I) + h(I)$.

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

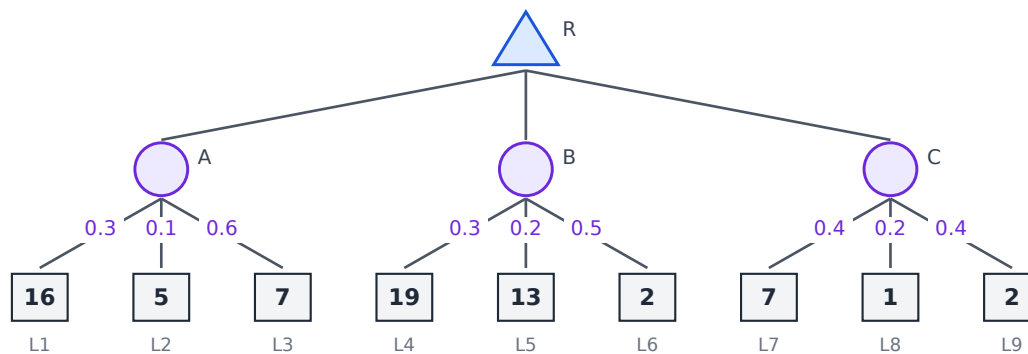
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)

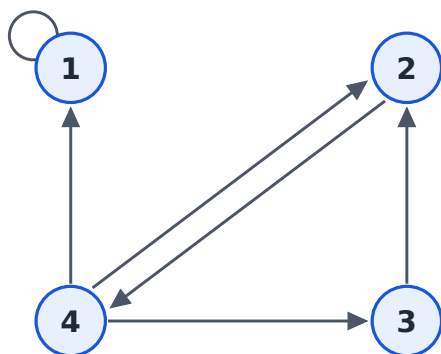


△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
☐ (B) $\exists x R(x, x)$
☐ (C) $\forall y \exists x R(x, y)$
☐ (D) $\exists y \forall x R(x, y)$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(P \leftrightarrow R) \wedge (\neg R \wedge Q)$
 (2) $R \rightarrow (Q \vee Q)$
 (3) $(P \vee R) \vee (S \wedge Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $(S \rightarrow \neg R) \wedge (P \wedge S)$
☐ (B) $R \rightarrow S$
☐ (C) $\neg R \leftrightarrow R$
☐ (D) $\neg(\neg R \vee P)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \rightarrow P) \rightarrow Q) \leftrightarrow (R \rightarrow (P \rightarrow Q))$
- ☐ (B) $((Q \rightarrow R) \wedge \neg R) \rightarrow \neg Q$
- ☐ (C) $(R \rightarrow (P \rightarrow Q)) \leftrightarrow ((R \wedge P) \rightarrow Q)$
- ☐ (D) $(R \rightarrow Q) \rightarrow (Q \rightarrow R)$

Q.18 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Consider the clauses $C_1 = p \vee q \vee \neg r$ and $C_2 = \neg p \vee r \vee s$. Which one of the following statements is correct?

- ☐ (A) $q \vee s$ is a valid resolvent of C_1 and C_2
- ☐ (B) Every resolvent of C_1 and C_2 is a tautology
- ☐ (C) C_1 and C_2 have no resolvent
- ☐ (D) The empty clause can be derived from C_1 and C_2

Q.19 [MCQ · 2 marks · negative: -2/3] Predicate Logic

A KB consists of the definite clauses $\text{Nat}(0)$ and $\text{Nat}(x) \rightarrow \text{Nat}(S(x))$. Forward chaining is used to answer the query $\text{Nat}(\text{Foo})$, where Foo is a constant different from 0. Which one of the following is correct?

- ☐ (A) Forward chaining never terminates, since it keeps adding $\text{Nat}(S(\dots S(0)\dots))$
- ☐ (B) Forward chaining terminates and answers 'not entailed'
- ☐ (C) Forward chaining derives $\text{Nat}(\text{Foo})$
- ☐ (D) Forward chaining fails immediately since the KB is not in Datalog form and therefore cannot be processed

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **D, C, A** and rules:

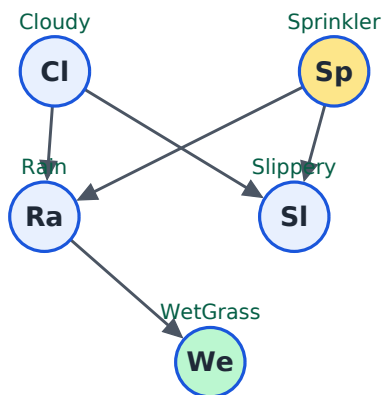
- R1: $J \rightarrow H$
- R2: $A \wedge C \rightarrow J$
- R3: $E \wedge D \rightarrow B$
- R4: $A \wedge E \wedge K \rightarrow C$
- R5: $J \wedge F \rightarrow E$
- R6: $H \rightarrow J$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) B is entailed
- ☐ (B) E is entailed
- ☐ (C) H is entailed
- ☐ (D) J is entailed

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, Sl = Slippery.) Compute $P(\text{WetGrass} = T \mid \text{Sprinkler} = T)$ (round off to 3 decimal places).



$P(Cl=T)$
0.6

$P(Sp=T)$
0.05

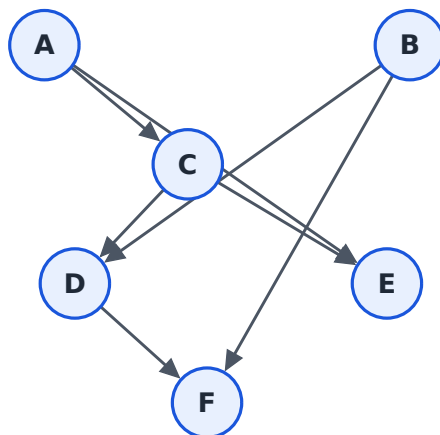
Cl	Sp	$P(Ra=T \dots)$
T	T	0.25
T	F	0.2
F	T	0.3
F	F	0.75

Ra	$P(We=T Ra)$	Cl	Sp	$P(Sl=T \dots)$
T	0.4	T	T	0.75
F	0.25	T	F	0.7
		F	T	0.9
		F	F	0.3

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

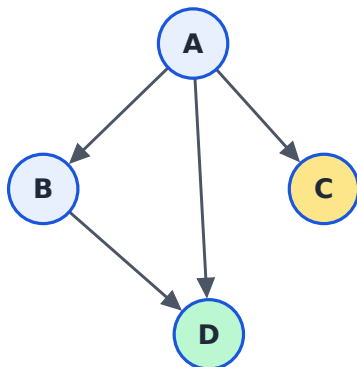
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp A \mid C$
☐ (B) $A \perp B$
☐ (C) $D \perp A$
☐ (D) $D \perp B \mid E$

Q.23 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{C=F}$. The current state of the chain is $(\mathbf{A=F, B=F, C=F, D=F})$. The next step resamples $\mathbf{D}$. With what probability is D set to T? (round off to 3 decimal places)



P(A=T)
0.3

A	P(B=T ...)	A	P(C=T ...)
T	0.4	T	0.75
F	0.2	F	0.95

A	B	P(D=T ...)
T	T	0.4
T	F	0.75
F	T	0.1
F	F	0.75

Answer (numerical): _____

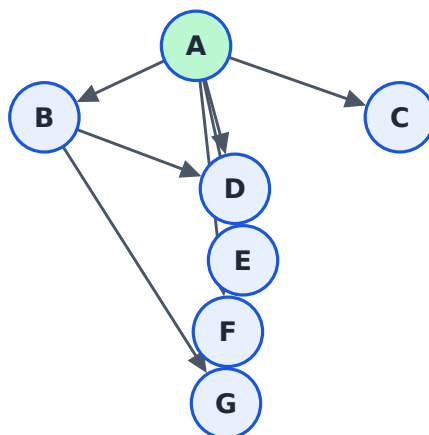
Q.24 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Gibbs sampling with evidence $S = \text{true}$, $W = \text{true}$ starts in state $C = \text{true}$, $R = \text{false}$. One sweep first resamples C (given its Markov blanket), then resamples R (given its Markov blanket, using the new C). What is the probability that $R = \text{true}$ at the end of the sweep? (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(A)}$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{D, F, B, E, C, G}$. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 003

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	2220	14	NAT	2	Adversarial Search	9.49 to 9.51
2	NAT	1	Adversarial Search	4	15	MSQ	2	Predicate Logic	B, C
3	MCQ	1	Uninformed Search	C	16	MSQ	2	Propositional Logic	B
4	NAT	1	Adversarial Search	3	17	MSQ	2	Propositional Logic	B, C
5	NAT	1	Propositional Logic	6	18	MCQ	2	Propositional Logic	B
6	MSQ	1	Propositional Logic	A, B, D	19	MCQ	2	Predicate Logic	A
7	MSQ	1	Propositional Logic	A, C, D	20	MSQ	2	Propositional Logic	C, D
8	NAT	1	Sampling	0.4273 to 0.4277	21	NAT	2	Exact Inference	0.288 to 0.292
9	MSQ	1	Probability for AI	A, B, D	22	MSQ	2	Conditional Independence	A, B
10	NAT	1	Exact Inference	0.748 to 0.752	23	NAT	2	Sampling	0.748 to 0.752
11	NAT	2	Adversarial Search	5	24	NAT	2	Sampling	0.244 to 0.245
12	MSQ	2	Informed Search	A, D	25	NAT	2	Exact Inference	4
13	NAT	2	Uninformed Search	13					

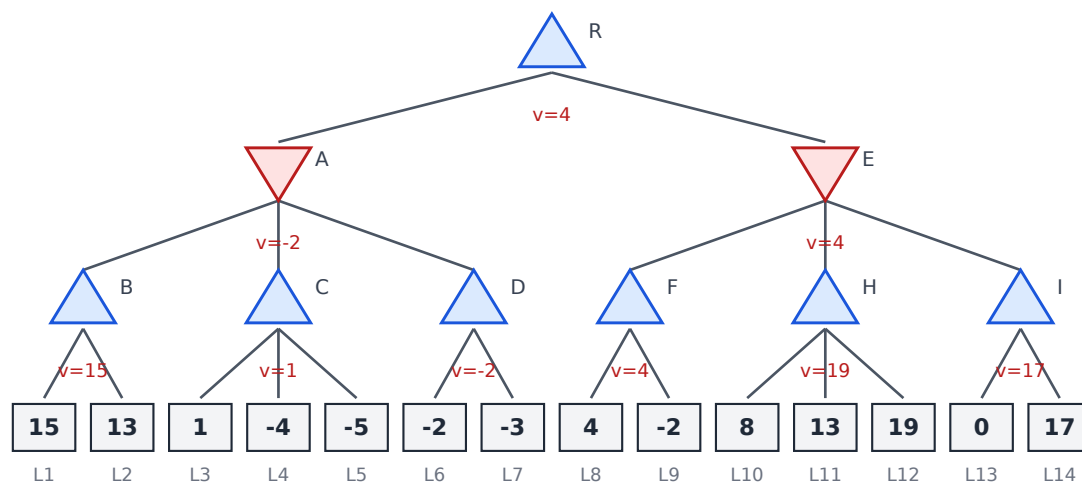
Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] Answer: 2220

Each direction generates $b + b^2 + b^3 = 10 + 100 + 1000 = 1110$ nodes. Two directions give $2 \times 1110 = 2220$. A single BFS to depth 6 would generate about 1.11×10^6 nodes. That is the $O(b^{d/2})$ versus $O(b^d)$ advantage.

Q.2 [NAT, 1M · Adversarial Search] Answer: 4

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 15$, $C = 1$, $D = -2$, $A = -2$, $F = 4$, $H = 19$, $I = 17$, $E = 4$. Root value = **4**; MAX's optimal first move is towards E.

Q.3 [MCQ, 1M · Uninformed Search] Answer: C (C: S, C, K, D, A, I, G)

Adjacency lists (alphabetical): A: {D, I, K}; C: {D, K, S}; D: {A, C, G, J}; G: {D, J}; I: {A, J, K}; J: {D, G, I}; K: {A, C, I, S}; S: {C, K}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[C, K]
2	C	[K, D]
3	K	[D, A, I]
4	D	[A, I, G, J]
5	A	[I, G, J]

Step	Expanded	Frontier (FIFO) after expansion
6	I	[G, J]
7	G	[J] ← goal popped, stop

Expansion order: **S, C, K, D, A, I, G**. The path returned is $S \rightarrow C \rightarrow D \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [NAT, 1M · Adversarial Search] Answer: 3

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 2^{\lceil 1 \rceil} + 2^{\lfloor 1 \rfloor} - 1 = 2 + 2 - 1 = 3$, versus 4 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.5 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$P \rightarrow (R \leftrightarrow \neg R)$	$(P \leftrightarrow Q) \leftrightarrow \neg R$	$(P \rightarrow (R \leftrightarrow \neg R)) \vee ((P \leftrightarrow Q) \leftrightarrow \neg R)$
T	T	T	F	F	F
T	T	F	F	T	T
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	F	T
F	F	F	T	T	T

Rows where F = T: **6**.

Q.6 [MSQ, 1M · Propositional Logic] Answer: A, B, D (A: $p \vee \neg p$; B: $\neg(p \wedge \neg p)$; D: $(p \wedge (p \rightarrow q)) \rightarrow q$)

$p \vee \neg p$ (excluded middle) and $\neg(p \wedge \neg p)$ (non-contradiction) are valid. $(p \wedge (p \rightarrow q)) \rightarrow q$ is the modus ponens tautology. $p \rightarrow (p \wedge q)$ is false at $p=T, q=F$, so it is not valid.

Q.7 [MSQ, 1M · Propositional Logic] Answer: A, C, D

(A) True: if $KB \wedge \neg \alpha$ is unsatisfiable, resolution derives the empty clause. (B) False: from $p \vee q, p \rightarrow r, q \rightarrow r$, modus ponens cannot derive r , although r is entailed. (C) True for definite clauses (Horn KBs with exactly one positive literal per clause). (D) This is the definition of soundness (truth-preserving).

Q.8 [NAT, 1M · Sampling] Answer: 0.4273 to 0.4277

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Sm}=T) \times P(\text{Po}=F|\text{Sm}=T) \times P(\text{Ca}=F|\text{Sm}=T, \text{Po}=F) \times P(\text{XR}=T|\text{Ca}=F)$
 $= 0.95 \times 0.75 \times 0.8 \times 0.75 = \mathbf{0.4275} \approx 0.4275$.

Q.9 [MSQ, 1M · Probability for AI] Answer: A, B, D (A: $P(a \wedge b) = P(a | b) P(b)$; B: $P(a | b) = P(b | a) P(a) / P(b)$; D:

$P(a) = P(a | b) P(b) + P(a | \neg b) P(\neg b)$)

(A) is the product rule. (B) is Bayes rule, derived from writing $P(a \wedge b)$ in two ways via the product rule. (D) is the conditioning rule (marginalization of $P(a, B)$ with the product rule). (C) is false: conditional distributions sum to 1 over the values of the conditioned variable ($P(a | b) + P(\neg a | b) = 1$), not over the conditioning variable; e.g. if $A \perp B$ with $P(a) = 0.9$, then $P(a | b) + P(a | \neg b) = 0.9 + 0.9 = 1.8$.

Q.10 [NAT, 1M · Exact Inference] Answer: 0.748 to 0.752

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T, D=T	0.13125
A=T, B=T, D=F	0.13125
A=T, B=F, D=T	0.05625
A=T, B=F, D=F	0.05625
A=F, B=T, D=T	0.075

Assignment to other variables	joint $P(\dots, C=T)$
$A=F, B=T, D=F$	0.075
$A=F, B=F, D=T$	0.1125
$A=F, B=F, D=F$	0.1125

Sum = **0.75** $\approx$ 0.750.

Q.11 [NAT, 2M · Adversarial Search] **Answer: 5**

The leftmost MAX node sees (5, 6) and returns 6, so its MIN parent has $\beta = 6$. The next MAX node sees $7 \geq \beta$ and prunes leaf 4. The left MIN node therefore has value 6, and the root gets $\alpha = 6$. In the right MIN subtree, the first MAX node sees 5 and then 3 and returns 5. The MIN node's value is now $\leq 5 \leq \alpha$, so the whole MAX subtree with leaves (6, 7) is pruned. Evaluated leaves: 5, 6, 7, 5, 3 = **5**. The root value is 6.

Q.12 [MSQ, 2M · Informed Search] **Answer: A, D** (A: h is admissible.; D: The edge S-I violates the consistency condition $h(S) \leq c(S,I) + h(I)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	D	F	G	H	I	S
$h(n)$	7	9	15	0	2	0	12
$h^*(n)$	8	10	16	0	3	6	13
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $B \rightarrow I$ ($7 > 5 + 0$), $D \rightarrow I$ ($9 > 4 + 0$), $S \rightarrow I$ ($12 > 7 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Uninformed Search] **Answer: 13**

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	A(4), C(6)
2	A (4)	D(5), C(6), K(9), I(11)
3	D (5)	C(6), I(7), K(8)
4	C (6)	I(7), K(8), F(12)
5	I (7)	K(8), F(12)
6	K (8)	F(12), G(13)
7	F (12)	G(13)
8	G (13)	← goal, stop

Path returned: **S** $\rightarrow$ **A** $\rightarrow$ **D** $\rightarrow$ **K** $\rightarrow$ **G**, cost = 4 + 1 + 3 + 5 = **13**. Nodes expanded (including S and G): **8** (S, A, D, C, I, K, F, G).

Q.14 [NAT, 2M · Adversarial Search] **Answer: 9.49 to 9.51**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.3 \times 16 + 0.1 \times 5 + 0.6 \times 7 = 9.5$
- $B = 0.3 \times 19 + 0.2 \times 13 + 0.5 \times 2 = 9.3$
- $C = 0.4 \times 7 + 0.2 \times 1 + 0.4 \times 2 = 3.8$

Root = $\max(9.5, 9.3, 3.8) =$ **9.5**.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [MSQ, 2M · Predicate Logic] **Answer: B, C** (B: $\exists x R(x, x)$; C: $\forall y \exists x R(x, y)$)

$R = \{(1,1), (2,4), (3,2), (4,1), (4,2), (4,3)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

(C) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(D) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

Remember: $\exists y \forall x R(x,y) \Rightarrow \forall x \exists y R(x,y)$ but not conversely; quantifier order matters.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: B** (B: $R \rightarrow S$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 1 of them:

P	Q	R	S	KB
F	T	F	T	T

(A) $(S \rightarrow \neg R) \wedge (P \wedge S)$ is false in the KB-model $P=F, Q=T, R=F, S=T \Rightarrow$ not entailed.

(B) $R \rightarrow S$ is true in all models of KB $\Rightarrow$ **entailed**.

(C) $\neg R \leftrightarrow R$ is false in the KB-model $P=F, Q=T, R=F, S=T \Rightarrow$ not entailed.

(D) $\neg(\neg R \vee P)$ is false in the KB-model $P=F, Q=T, R=F, S=T \Rightarrow$ not entailed.

Q.17 [MSQ, 2M · Propositional Logic] Answer: B, C (B: $((Q \rightarrow R) \wedge \neg R) \rightarrow \neg Q$; C: $(R \rightarrow (P \rightarrow Q)) \leftrightarrow ((R \wedge P) \rightarrow Q)$)

(A) $((R \rightarrow P) \rightarrow Q) \leftrightarrow (R \rightarrow (P \rightarrow Q))$: **not valid** — counter-model $P=T, Q=F, R=F$.

(B) $((Q \rightarrow R) \wedge \neg R) \rightarrow \neg Q$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $(R \rightarrow (P \rightarrow Q)) \leftrightarrow ((R \wedge P) \rightarrow Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $(R \rightarrow Q) \rightarrow (Q \rightarrow R)$: **not valid** — counter-model $Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MCQ, 2M · Propositional Logic] Answer: B (B: Every resolvent of C_1 and C_2 is a tautology)

C_1 and C_2 clash on two pairs, $p/\neg p$ and $\neg r/r$. Resolution removes exactly one complementary pair at a time. Resolving on p gives $q \vee \neg r \vee r \vee s$; resolving on r gives $p \vee \neg p \vee q \vee s$. Both contain a complementary pair, so both are tautologies. Removing both pairs at once to get $q \vee s$ is unsound: $p=T, q=F, r=T, s=F$ satisfies C_1 and C_2 but falsifies $q \vee s$. The pair is satisfiable, so $\square$ cannot be derived.

Q.19 [MCQ, 2M · Predicate Logic] Answer: A (A: Forward chaining never terminates, since it keeps adding $\text{Nat}(S(\dots S(0)\dots))$)

$\text{Nat}(\text{Foo})$ is not entailed (there is a model where Foo is not a natural number). Forward chaining derives $\text{Nat}(S(0))$, $\text{Nat}(S(S(0)))$, ... forever, because the function symbol S yields infinitely many ground terms. This illustrates that FOL entailment with definite clauses is only semidecidable: entailed queries are eventually found, non-entailed ones may run forever. Forward chaining still applies to non-Datalog KBs; it just lacks a termination guarantee.

Q.20 [MSQ, 2M · Propositional Logic] Answer: C, D (C: H is entailed; D: J is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $A \wedge C \rightarrow J$; add J .

Round 2: fire $J \rightarrow H$; add H .

Fixed point: $\{A, C, D, H, J\}$ — 5 atoms are entailed (facts: A, C, D ; newly derived: H, J). Atoms never derived: B, E, F, K .

Q.21 [NAT, 2M · Exact Inference] Answer: 0.288 to 0.292

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}, \text{Slippery}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler}) \cdot P(\text{Rain} \mid \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Rain}) \cdot P(\text{Slippery} \mid \text{Cloudy}, \text{Sprinkler})$.

Inference by enumeration: $P(\text{WetGrass} \mid e) = \alpha \sum_{\text{hidden}} P(\text{WetGrass}, e, \text{hidden})$.

Hidden: Cl, Ra, Sl	We	product of CPT entries (Cl · Sp · Ra · We · Sl)	value
T, T, T	T	$0.6 \times 0.05 \times 0.25 \times 0.4 \times 0.75$	0.00225
T, T, F	T	$0.6 \times 0.05 \times 0.25 \times 0.4 \times 0.25$	0.00075
T, F, T	T	$0.6 \times 0.05 \times 0.75 \times 0.25 \times 0.75$	0.00422
T, F, F	T	$0.6 \times 0.05 \times 0.75 \times 0.25 \times 0.25$	0.00141
F, T, T	T	$0.4 \times 0.05 \times 0.3 \times 0.4 \times 0.9$	0.00216
F, T, F	T	$0.4 \times 0.05 \times 0.3 \times 0.4 \times 0.1$	0.00024
F, F, T	T	$0.4 \times 0.05 \times 0.7 \times 0.25 \times 0.9$	0.00315
F, F, F	T	$0.4 \times 0.05 \times 0.7 \times 0.25 \times 0.1$	0.00035
T, T, T	F	$0.6 \times 0.05 \times 0.25 \times 0.6 \times 0.75$	0.00337
T, T, F	F	$0.6 \times 0.05 \times 0.25 \times 0.6 \times 0.25$	0.00112
T, F, T	F	$0.6 \times 0.05 \times 0.75 \times 0.75 \times 0.75$	0.01266
T, F, F	F	$0.6 \times 0.05 \times 0.75 \times 0.75 \times 0.25$	0.00422
F, T, T	F	$0.4 \times 0.05 \times 0.3 \times 0.6 \times 0.9$	0.00324
F, T, F	F	$0.4 \times 0.05 \times 0.3 \times 0.6 \times 0.1$	0.00036
F, F, T	F	$0.4 \times 0.05 \times 0.7 \times 0.75 \times 0.9$	0.00945

Hidden: Cl, Ra, Sl	We	product of CPT entries (Cl · Sp · Ra · We · Sl)	value
F, F, F	F	$0.4 \times 0.05 \times 0.7 \times 0.75 \times 0.1$	0.00105

$P(\text{WetGrass}=T, e) = 0.01453$; $P(\text{WetGrass}=F, e) = 0.03548$; $P(e) = 0.05$.

$P(\text{WetGrass}=T | e) = 0.01453 / 0.05 = \mathbf{0.2905} \approx 0.290$.

Q.22 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $D \perp A | C$; B: $A \perp B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $D \perp A | C$: **TRUE** — $D \leftarrow C \leftarrow A$ (blocked: observed chain node C); $D \leftarrow C \rightarrow E \leftarrow A$ (blocked: observed fork node C).

(B) $A \perp B$: **TRUE** — $A \rightarrow C \rightarrow D \leftarrow B$ (blocked: collider D with neither it nor a descendant observed); $A \rightarrow C \rightarrow D \rightarrow F \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow E \leftarrow C \rightarrow D \leftarrow B$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow E \leftarrow C \rightarrow D \rightarrow F \leftarrow B$ (blocked: collider E with neither it nor a descendant observed).

(C) $D \perp A$: **FALSE** — the path $D \leftarrow C \leftarrow A$ is active given $\{\}$.

(D) $D \perp B | E$: **FALSE** — the path $D \leftarrow B$ is active given $\{E\}$.

Q.23 [NAT, 2M · Sampling] **Answer: 0.748 to 0.752**

Gibbs sampling draws D from $P(D | \text{mb}(D))$, where $\text{mb}(D) = \{A, B\}$:

$P(D | \text{mb}) \propto P(D | \text{parents}(D)) \times \prod_{\text{children } Y} P(Y | \text{parents}(Y))$.

$D=T$: $P(D=T|A=F, B=F) = 0.75 = 0.75$

$D=F$: $P(D=F|A=F, B=F) = 0.25 = 0.25$

Normalise: $0.75 / (0.75 + 0.25) = \mathbf{0.75}$.

Q.24 [NAT, 2M · Sampling] **Answer: 0.244 to 0.245**

Step 1: $P(c | s, \neg r) = \alpha P(c)P(s | c)P(\neg r | c)$: c : $0.5 \times 0.1 \times 0.2 = 0.01$; $\neg c$: $0.5 \times 0.5 \times 0.8 = 0.2 \rightarrow P(c) = 0.01/0.21 \approx 0.04762$. Step 2: $P(r | c, s, w) = 0.792/0.972 \approx 0.81481$; $P(r | \neg c, s, w) = (0.2 \times 0.99)/(0.2 \times 0.99 + 0.8 \times 0.90) = 0.198/0.918 \approx 0.21569$. Total: $0.04762 \times 0.81481 + 0.95238 \times 0.21569 \approx 0.03880 + 0.20542 = 0.2442 \approx 0.244$.

Q.25 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, C)$, $f(A, B, D)$, $f(A, D, E)$, $f(A, E, F)$, $f(B, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(A, B, D)$, $f(A, D, E)$	$\{A, B, D, E\}$	$f(A, B, E)$
F	$f(A, E, F)$, $f(B, F, G)$	$\{A, B, E, F, G\}$	$f(A, B, E, G)$
B	$f(A, B)$, $f(A, B, E)$, $f(A, B, E, G)$	$\{A, B, E, G\}$	$f(A, E, G)$
E	$f(A, E, G)$	$\{A, E, G\}$	$f(A, G)$
C	$f(A, C)$	$\{A, C\}$	$f(A)$
G	$f(A, G)$	$\{A, G\}$	$f(A)$

The largest factor produced has **4** variable(s) (created when eliminating F); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over A and normalise.

Mock Test 004

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 8 / 11	14 / 11 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

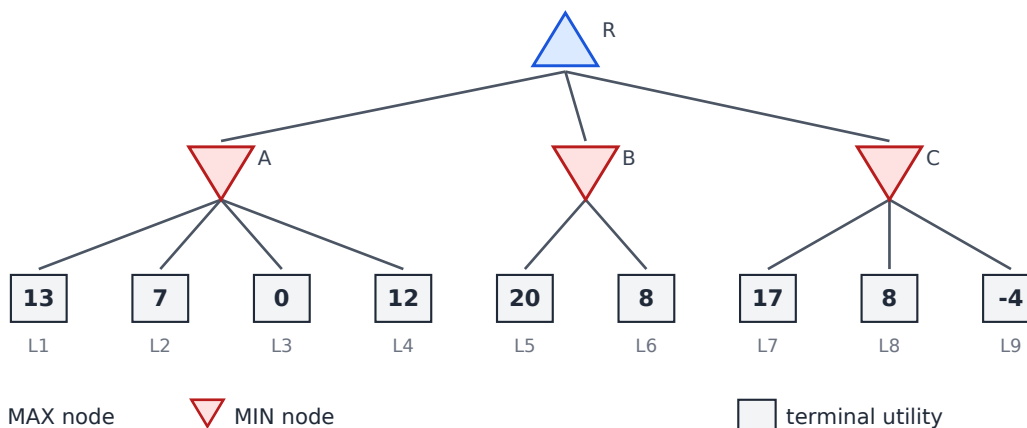
Q.1 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

A search tree has uniform branching factor $b = 3$; the shallowest goal is at depth $d = 4$ and happens to be the **last** node generated at that depth. BFS applies the goal test when a node is **generated**. How many nodes are generated in total (do **not** count the root)?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

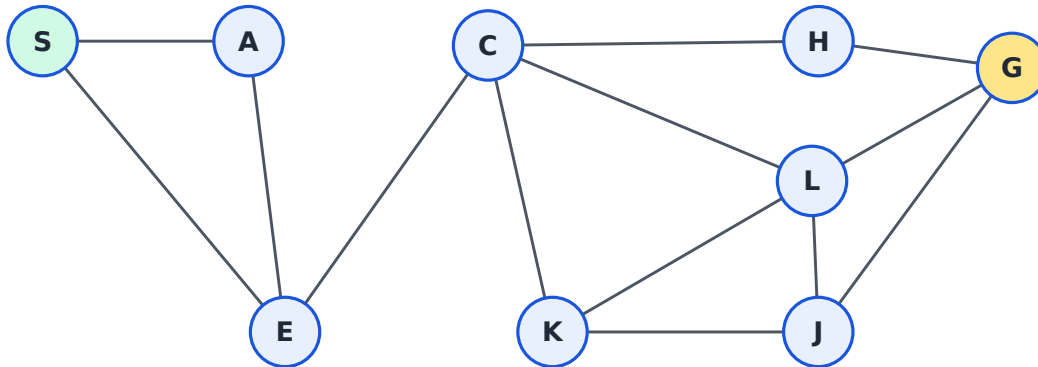
Q.3 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search

For depth-first tree search with branching factor b and maximum depth m , the space complexity is

- ☐ (A) $O(b^m)$
- ☐ (B) $O(bm)$
- ☐ (C) $O(b^d)$
- ☐ (D) $O(m^b)$

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, A, E, C, H, G
☐ (B) S, E, A, C, L, K, H, J, G
☐ (C) S, A, E, C, H, K, L, G
☐ (D) S, A, E, H, C, K, L, G

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **C, D, E** and rules:

- R1: $E \wedge H \rightarrow A$
 R2: $J \wedge B \rightarrow K$
 R3: $D \wedge K \rightarrow F$
 R4: $D \rightarrow E$
 R5: $E \rightarrow H$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following clauses are Horn clauses?

- ☐ (A) $\neg p \vee \neg q \vee r$
☐ (B) $p \vee q$
☐ (C) $\neg p \vee \neg q$
☐ (D) r

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

In how many rows of the truth table over p, q is $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow p)$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = \neg(P \leftrightarrow \neg Q) \wedge ((R \wedge P) \vee \neg P) ?$$

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$ and let f be a unary function symbol. How many interpretations of f satisfy $\forall x f(f(x)) = x$?

Answer (numerical): _____

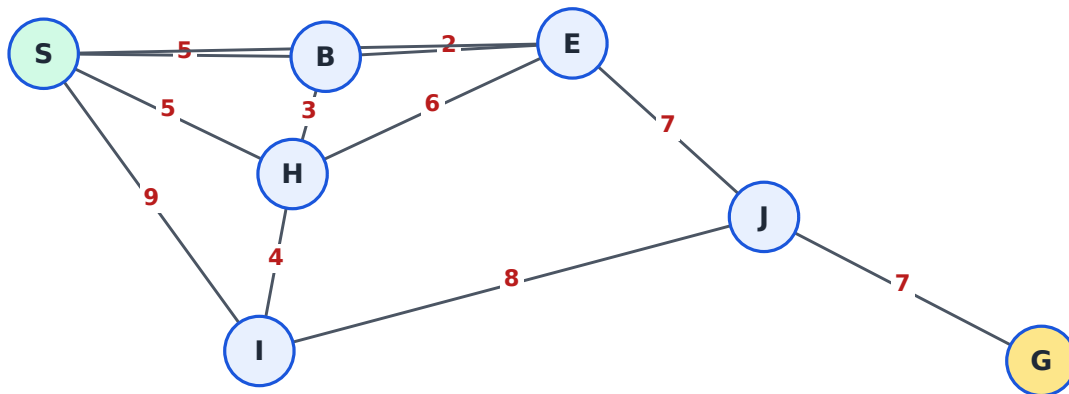
Q.10 [MCQ · 1 mark · negative: -1/3] Variable Elimination

A Bayesian network is called a polytree (singly connected) if

- ☐ (A) there is at most one undirected path between any two nodes
- ☐ (B) every node has at most one parent
- ☐ (C) there is at most one directed path between any two nodes
- ☐ (D) it has exactly one root node

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search – UCS

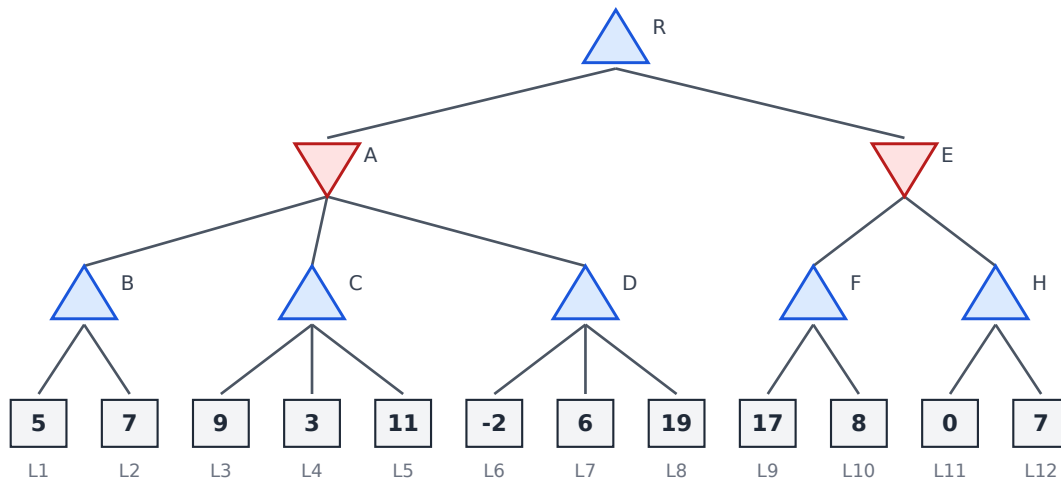
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



- ☐ (A) Leaf L1 is pruned.
☐ (B) Leaf L4 is pruned.
☐ (C) Leaf L8 is pruned.
☐ (D) Leaf L12 is pruned.

Q.13 [MSQ · 2 marks · no negative marking] Informed Search

Which of the following statements about greedy best-first search are TRUE?

- ☐ (A) The graph-search version is complete in finite state spaces
☐ (B) It is optimal whenever the heuristic is admissible
☐ (C) Its worst-case time and space complexity is $O(b^m)$, where m is the maximum depth
☐ (D) If $h(n) = h^*(n)$ exactly, greedy best-first search always returns an optimal solution

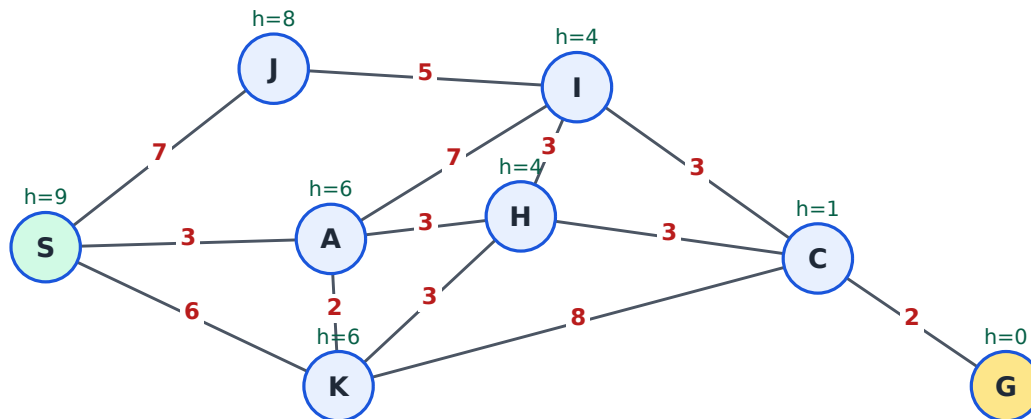
Q.14 [NAT · 2 marks · no negative marking] Informed Search

A search algorithm generates $N = 52$ nodes (not counting the root) to find a solution at depth $d = 5$. Using $N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^5$, find the effective branching factor b^* , correct to two decimal places.

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

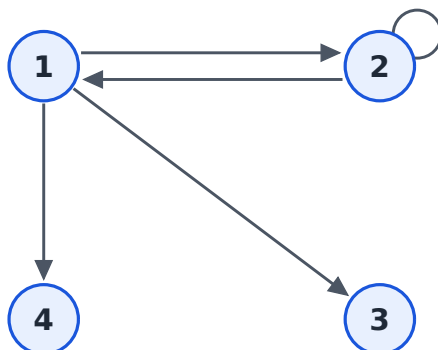
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge C-I violates the consistency condition (in at least one direction).
- ☐ (B) h is admissible.
- ☐ (C) h is consistent (monotone).
- ☐ (D) $h(J)$ overestimates the true cost from J to G.

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
- ☐ (B) $\exists x R(x, x)$
- ☐ (C) $\exists y \forall x R(x, y)$
- ☐ (D) $\forall y \exists x R(x, y)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = P \vee \neg \neg Q$?

- ☐ (A) $\neg(\neg P \vee \neg \neg Q)$
- ☐ (B) $P \wedge \neg \neg Q$
- ☐ (C) $\neg P \vee \neg \neg Q$
- ☐ (D) $\neg \neg Q \vee P$

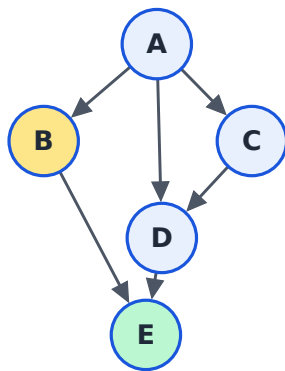
Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Skolemizing $\exists x \forall y \exists z R(x, y, z)$ gives (A is a new constant, F, G new function symbols)

- (A) $\forall y R(A, y, F(y))$
 ○ (B) $\forall y R(A, y, F(A))$
 ○ (C) $\forall y R(G(y), y, F(y))$
 ○ (D) $\forall y R(A, y, B)$, with B a new constant

Q.19 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{B} = \mathbf{F}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(E = T \mid B = F)}$ (round off to 3 decimal places).



P(A=T)
0.9

A	P(B=T ...)	A	P(C=T ...)
T	0.9	T	0.7
F	0.95	F	0.9

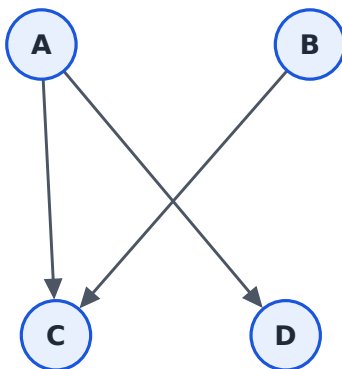
A	C	P(D=T ...)	B	D	P(E=T ...)
T	T	0.1	T	T	0.8
T	F	0.95	T	F	0.05
F	T	0.3	F	T	0.05
F	F	0.05	F	F	0.5

#	A	C	D	E
1	T	T	F	F
2	T	T	F	T
3	T	T	T	F
4	T	T	F	T
5	T	F	T	F

Answer (numerical): _____

Q.20 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.01}, \mathbf{0.52}, \mathbf{0.45}, \mathbf{0.64} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.95

P(B=T)	A	B	P(C=T ...)
0.75	T	T	0.25
	T	F	0.75
	F	T	0.75
	F	F	0.95

A	P(D=T ...)
T	0.4
F	0.25

- (A) A=T, B=F, C=F, D=F
 ○ (B) A=T, B=T, C=F, D=F
 ○ (C) A=F, B=T, C=F, D=F
 ○ (D) A=F, B=F, C=F, D=T

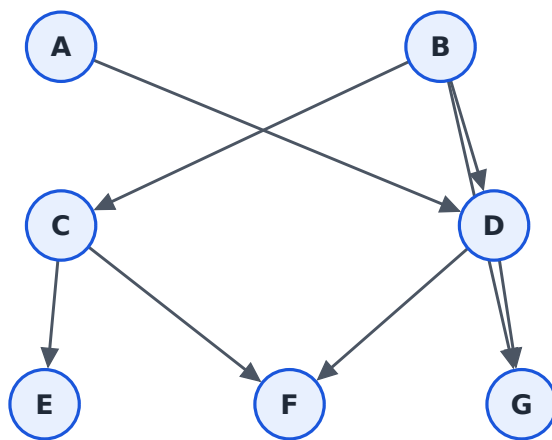
Q.21 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Which of the following are correct (values rounded to 3 decimal places)?

- ☐ (A) $P(s | w, r) \approx 0.194$
☐ (B) $P(s | w) \approx 0.430$
☐ (C) $P(s | w, r) > P(s | w)$
☐ (D) $P(w) \approx 0.647$

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

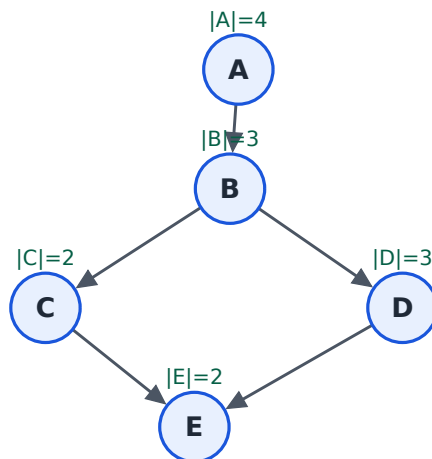
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp A | B, E$
☐ (B) $F \perp G | A, E$
☐ (C) $C \perp A | F$
☐ (D) $D \perp E | A, B$

Q.23 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

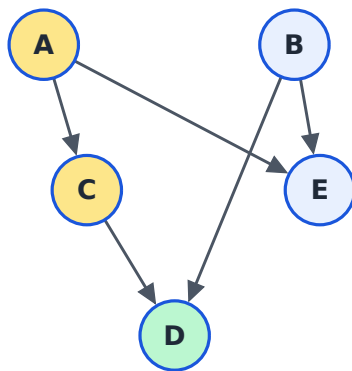
The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(D = T \mid A = T, C = T)$ (round off to 3 decimal places).



$P(A=T)$
0.4

$P(B=T)$
0.3

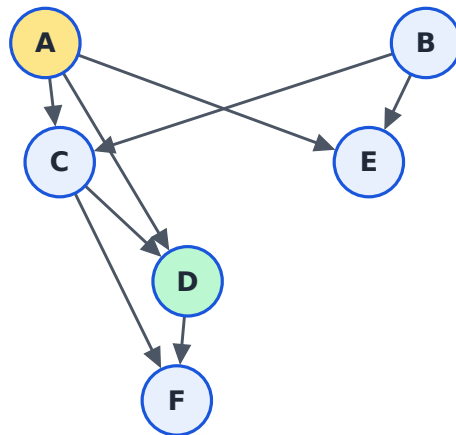
A	$P(C=T \dots)$
T	0.7
F	0.5

B	C	$P(D=T \dots)$	A	B	$P(E=T \dots)$
T	T	0.7	T	T	0.9
T	F	0.2	T	F	0.4
F	T	0.3	F	T	0.4
F	F	0.2	F	F	0.3

Answer (numerical): _____

Q.25 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(D \mid A = T)$ in the Bayesian network below (Boolean variables; evidence $A = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, E, B, F**. What is the scope of the new factor produced when **B** is summed out?



- ☐ (A) $f(B, D, F)$
☐ (B) $f()$
☐ (C) $f(F)$
☐ (D) $f(D, F)$

Answer Key — Mock Test 004

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	120	14	NAT	2	Informed Search	1.9 to 1.93
2	NAT	1	Adversarial Search	8	15	MSQ	2	Informed Search	B, C
3	MCQ	1	Uninformed Search	B	16	MSQ	2	Predicate Logic	A, B, D
4	MCQ	1	Uninformed Search	C	17	MSQ	2	Propositional Logic	D
5	NAT	1	Propositional Logic	5	18	MCQ	2	Predicate Logic	A
6	MSQ	1	Propositional Logic	A, C, D	19	NAT	2	Sampling	0.398 to 0.402
7	NAT	1	Propositional Logic	2	20	MCQ	2	Sampling	B
8	NAT	1	Propositional Logic	3	21	MSQ	2	Bayesian Networks	A, B, D
9	NAT	1	Predicate Logic	4	22	MSQ	2	Conditional Independence	A, D
10	MCQ	1	Variable Elimination	A	23	NAT	2	Bayesian Networks	117
11	NAT	2	Uninformed Search	7	24	NAT	2	Exact Inference	0.418 to 0.422
12	MSQ	2	Adversarial Search	B	25	MCQ	2	Exact Inference	D
13	MSQ	2	Informed Search	A, C					

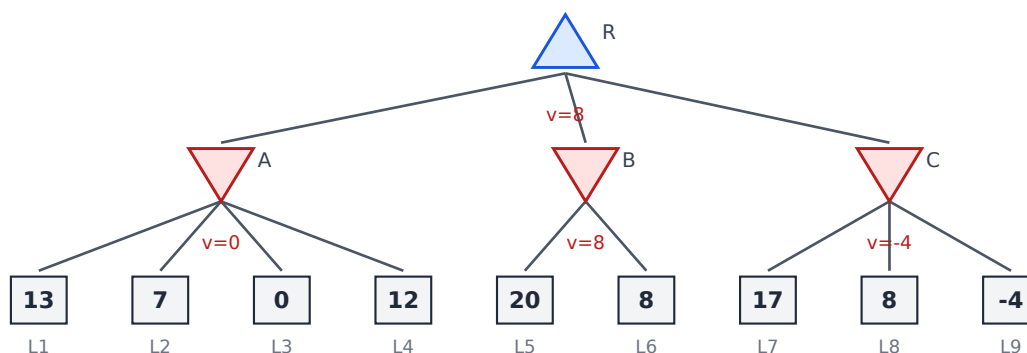
Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] Answer: 120

With goal test at generation, BFS generates every node at depths 1 ... d before (and including) the goal (the worst case). $N = b + b^2 + \dots + b^d = 3 + 9 + 27 + 81 = 120$. This is the $O(b^d)$ of AIMA.

Q.2 [NAT, 1M · Adversarial Search] Answer: 8

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 0$, $B = 8$, $C = -4$. Root value = **8**; MAX's optimal first move is towards B.

Q.3 [MCQ, 1M · Uninformed Search] Answer: B (B: $O(bm)$)

DFS tree search stores only the current path from the root plus the unexpanded siblings of each node on that path. That is at most about b nodes per level over m levels, so the space is **$O(bm)$** . $O(b^m)$ is its worst-case time, not its space. $O(b^d)$ is the space of BFS. $O(m^b)$ does not arise for any standard algorithm.

Q.4 [MCQ, 1M · Uninformed Search] Answer: C (C: S, A, E, C, H, K, L, G)

Adjacency lists (alphabetical): A: {E, S}; C: {E, H, K, L}; E: {A, C, S}; G: {H, J, L}; H: {C, G}; J: {G, K, L}; K: {C, J, L}; L: {C, G, J, K}; S: {A, E}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[A, E]
2	A	[E]
3	E	[C]
4	C	[H, K, L]
5	H	[K, L, G]
6	K	[L, G, J]
7	L	[G, J]

Step	Expanded	Frontier (FIFO) after expansion
8	G	[J] ← goal popped, stop

Expansion order: **S, A, E, C, H, K, L, G**. The path returned is $S \rightarrow E \rightarrow C \rightarrow H \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 5**

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $E \rightarrow H$; add H.

Round 2: fire $E \wedge H \rightarrow A$; add A.

Fixed point: {A, C, D, E, H} — 5 atoms are entailed (facts: C, D, E; newly derived: A, H). Atoms never derived: B, F, J, K.

Q.6 [MSQ, 1M · Propositional Logic] **Answer: A, C, D** (A: $\neg p \vee \neg q \vee r$; C: $\neg p \vee \neg q$; D: r)

A Horn clause is a disjunction of literals with at most one positive literal. $\neg p \vee \neg q \vee r$ has one positive literal (a definite clause, i.e. $p \wedge q \rightarrow r$). $\neg p \vee \neg q$ has zero positive literals (a goal clause). r is a single positive literal (a fact). $p \vee q$ has two positive literals, so it is not Horn.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 2**

$p \rightarrow q \equiv \neg p \vee q$ and $\neg q \rightarrow p \equiv p \vee q$. Row TT: T, $T \rightarrow T$. Row TF: F, $T \rightarrow F$. Row FT: T, $T \rightarrow T$. Row FF: T, $F \rightarrow F$. So the biconditional is true in 2 rows.

Q.8 [NAT, 1M · Propositional Logic] **Answer: 3**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg(P \leftrightarrow \neg Q)$	$(R \wedge P) \vee \neg P$	$\neg(P \leftrightarrow \neg Q) \wedge ((R \wedge P) \vee \neg P)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	F	F
F	T	T	F	T	F
F	T	F	F	T	F
F	F	T	T	T	T
F	F	F	T	T	T

Rows where F = T: **3**.

Q.9 [NAT, 1M · Predicate Logic] **Answer: 4**

$f(f(x)) = x$ for all x means f is an involution: it is a bijection whose cycles have length 1 or 2. On 3 elements: the identity (1 way) or exactly one transposition with one fixed point (3 ways). Total = 4.

Q.10 [MCQ, 1M · Variable Elimination] **Answer: A** (A: there is at most one undirected path between any two nodes)

AIMA defines singly connected networks (polytrees) as those with at most one undirected path between any pair of nodes. Nodes may have several parents (e.g. the alarm network is a polytree although A has two parents). Requiring at most one directed path is weaker: $A \rightarrow B, A \rightarrow C, B \rightarrow C$ has two directed paths A to C, but $A \rightarrow B, C \rightarrow B, A \rightarrow D, C \rightarrow D$ has unique directed paths yet the undirected cycle A-B-C-D-A. A polytree may also have several roots.

Q.11 [NAT, 2M · Uninformed Search] **Answer: 7**

UCS always expands the frontier node with the smallest path cost g(n); if a cheaper path to a frontier node is found its entry is replaced.

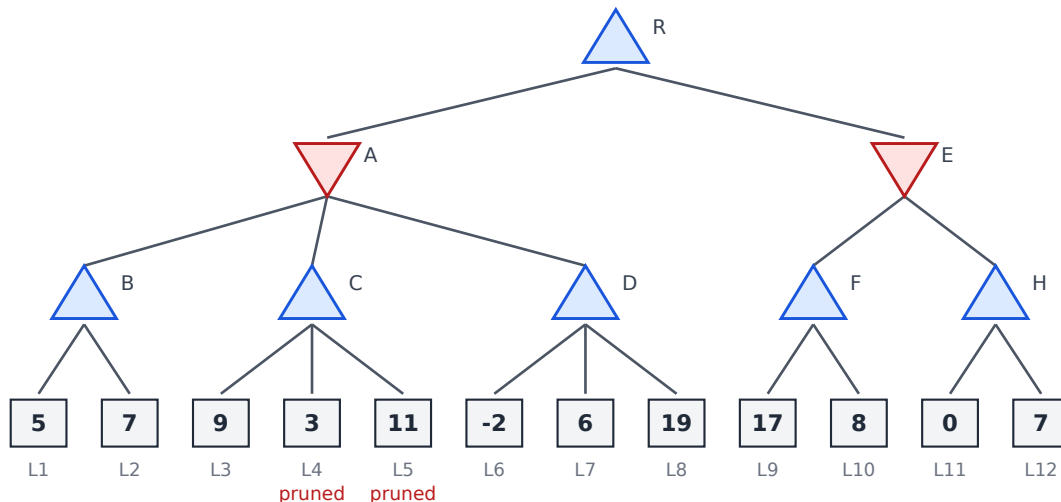
Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(5), H(5), E(6), I(9)
2	B (5)	H(5), E(6), I(9)
3	H (5)	E(6), I(9)
4	E (6)	I(9), J(13)
5	I (9)	J(13)
6	J (13)	G(20)
7	G (20)	← goal, stop

Path returned: **S → E → J → G**, cost = 6 + 7 + 7 = **20**. Nodes expanded (including S and G): **7** (S, B, H, E, I, J, G).

Q.12 [MSQ, 2M · Adversarial Search] Answer: B (B: Leaf L4 is pruned.)

Run alpha-beta depth-first, left to right, passing (α , β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value 9 $\geq \beta = 7 \Rightarrow$ prune remaining children (leaves L4, L5).



Leaves evaluated (10): L1, L2, L3, L6, L7, L8, L9, L10, L11, L12. Leaves pruned (2): L4, L5. Root value = 7 (same as plain minimax, 7 — pruning never changes the root value).

Q.13 [MSQ, 2M · Informed Search] Answer: A, C (A: The graph-search version is complete in finite state spaces; C: Its worst-case time and space complexity is $O(b^m)$, where m is the maximum depth)

(A) is true: in a finite space with an explored set, it eventually expands every state. (B) is false: greedy search ignores g , so admissibility does not help. (C) is true, as stated in AIMA. (D) is false. Counterexample: $S \rightarrow A$ costs 100 and $A \rightarrow G$ costs 1, while $S \rightarrow B$ costs 1 and $B \rightarrow G$ costs 2. Then $h^*(A)=1 < h^*(B)=2$, so greedy expands A and returns cost 101 instead of 3. A perfect heuristic makes A* optimal, not greedy search.

Q.14 [NAT, 2M · Informed Search] Answer: 1.9 to 1.93

Solve $b^* + b^{*2} + b^{*3} + b^{*4} + b^{*5} = 52$. For $b^*=1.9$: $1.9 + 3.61 + 6.859 + 13.03 + 24.76 \approx 50.2$, which is too small. For $b^*=1.95$: about 54.0, which is too large. Bisection gives **$b^* \approx 1.92$** , the AIMA textbook example. A b^* close to 1 indicates a very good heuristic.

Q.15 [MSQ, 2M · Informed Search] Answer: B, C (B: h is admissible.; C: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	C	G	H	I	J	K	S
$h(n)$	6	1	0	4	4	8	6	9
$h^*(n)$	8	2	0	5	5	10	8	11
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.16 [MSQ, 2M · Predicate Logic] Answer: A, B, D (A: $\exists x \forall y \neg R(x, y)$; B: $\exists x R(x, x)$; D: $\forall y \exists x R(x, y)$)

$R = \{(1,2), (1,3), (1,4), (2,1), (2,2)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 3 has no outgoing edge.

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] Answer: D (D: $\neg\neg Q \vee P$)

Compare truth tables with $F = P \vee \neg\neg Q$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

- (A) $\neg(\neg P \vee \neg\neg Q)$ — differs at $P=T, Q=F$ ($F = T$, option = F) $\Rightarrow$ not equivalent.
 (B) $P \wedge \neg\neg Q$ — differs at $P=T, Q=F$ ($F = T$, option = F) $\Rightarrow$ not equivalent.
 (C) $\neg P \vee \neg\neg Q$ — differs at $P=T, Q=F$ ($F = T$, option = F) $\Rightarrow$ not equivalent.
 (D) $\neg\neg Q \vee P$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.18 [MCQ, 2M · Predicate Logic] Answer: A (A: $\forall y R(A, y, F(y))$)

x is existential and not inside any universal quantifier, so it becomes a constant A . z is inside the scope of $\forall y$, so it becomes $F(y)$ (arguments are the enclosing universally quantified variables). $F(A)$ ignores the dependence on y and is an unwarranted strengthening; $G(y)$ for x is wrong because x precedes $\forall y$; a constant for z also loses the dependence on y .

Q.19 [NAT, 2M · Sampling] Answer: 0.398 to 0.402

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	E
1	$P(B=F A=T)=0.1$	0.1	F
2	$P(B=F A=T)=0.1$	0.1	T
3	$P(B=F A=T)=0.1$	0.1	F
4	$P(B=F A=T)=0.1$	0.1	T
5	$P(B=F A=T)=0.1$	0.1	F

$\sum w$ (all) = 0.5; $\sum w$ ($E = T$) = 0.2.

Estimate $P(E=T \mid B = F) = 0.2 / 0.5 = \mathbf{0.4}$.

Q.20 [MCQ, 2M · Sampling] Answer: B (B: $A=T, B=T, C=F, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \neg) = 0.95$; $u = 0.01 < 0.95 \Rightarrow A = T$
- B: $P(B=T \mid \neg) = 0.75$; $u = 0.52 < 0.75 \Rightarrow B = T$
- C: $P(C=T \mid A=T, B=T) = 0.25$; $u = 0.45 \geq 0.25 \Rightarrow C = F$
- D: $P(D=T \mid A=T) = 0.4$; $u = 0.64 \geq 0.4 \Rightarrow D = F$

Sample: **$\{A=T, B=T, C=F, D=F\}$** .

Q.21 [MSQ, 2M · Bayesian Networks] Answer: A, B, D (A: $P(s \mid w, r) \approx 0.194$; B: $P(s \mid w) \approx 0.430$; D: $P(w) \approx 0.647$)

$P(s, w) = c: 0.5 \times 0.1 \times (0.8 \times 0.99 + 0.2 \times 0.90) = 0.0486$; $\neg c: 0.5 \times 0.5 \times (0.2 \times 0.99 + 0.8 \times 0.90) = 0.2295$; total 0.2781. $P(\neg s, w) = c: 0.5 \times 0.9 \times 0.8 \times 0.90 = 0.324$; $\neg c: 0.5 \times 0.5 \times 0.2 \times 0.90 = 0.045$; total 0.369. $P(w) = 0.6471$ (D true) and $P(s \mid w) = 0.2781/0.6471 \approx 0.430$ (B true). $P(s, w, r) = (0.5 \times 0.1 \times 0.8 + 0.5 \times 0.5 \times 0.2) \times 0.99 = 0.0891$; $P(\neg s, w, r) = (0.36 + 0.05) \times 0.90 = 0.369$; $P(s \mid w, r) = 0.0891/0.4581 \approx 0.194$ (A true). Rain explains away the wet grass, so $P(s \mid w, r) < P(s \mid w)$: (C) is false.

Q.22 [MSQ, 2M · Conditional Independence] Answer: A, D (A: $C \perp A \mid B, E$; D: $D \perp E \mid A, B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

- (A) $C \perp A \mid B, E$: **TRUE** — $C \leftarrow B \rightarrow D \leftarrow A$ (blocked: observed fork node B); $C \leftarrow B \rightarrow G \leftarrow D \leftarrow A$ (blocked: observed fork node B); $C \rightarrow F \leftarrow D \leftarrow A$ (blocked: collider F with neither it nor a descendant observed).
 (B) $F \perp G \mid A, E$: **FALSE** — the path $F \leftarrow C \leftarrow B \rightarrow D \rightarrow G$ is active given $\{A, E\}$.
 (C) $C \perp A \mid F$: **FALSE** — the path $C \leftarrow B \rightarrow D \leftarrow A$ is active given $\{F\}$.
 (D) $D \perp E \mid A, B$: **TRUE** — $D \leftarrow B \rightarrow C \rightarrow E$ (blocked: observed fork node B); $D \rightarrow F \leftarrow C \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $D \rightarrow G \leftarrow B \rightarrow C \rightarrow E$ (blocked: collider G with neither it nor a descendant observed).

Q.23 [NAT, 2M · Bayesian Networks] Answer: 117

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (4-1) + B: (3-1) \times 4 + C: (2-1) \times 3 + D: (3-1) \times 3 + E: (2-1) \times 2 \times 3 = \mathbf{26}$.

Full joint table: $\prod |X| - 1 = 144 - 1 = 143$; saving = 117.

Q.24 [NAT, 2M · Exact Inference] Answer: 0.418 to 0.422

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B) \cdot P(C | A) \cdot P(D | B, C) \cdot P(E | A, B)$.

Inference by enumeration: $P(D | e) = \alpha \sum_{\text{hidden}} P(D, e, \text{hidden})$.

Hidden: B, E	D	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T	T	$0.4 \times 0.3 \times 0.7 \times 0.7 \times 0.9$	0.05292
T, F	T	$0.4 \times 0.3 \times 0.7 \times 0.7 \times 0.1$	0.00588
F, T	T	$0.4 \times 0.7 \times 0.7 \times 0.3 \times 0.4$	0.02352
F, F	T	$0.4 \times 0.7 \times 0.7 \times 0.3 \times 0.6$	0.03528
T, T	F	$0.4 \times 0.3 \times 0.7 \times 0.3 \times 0.9$	0.02268
T, F	F	$0.4 \times 0.3 \times 0.7 \times 0.3 \times 0.1$	0.00252
F, T	F	$0.4 \times 0.7 \times 0.7 \times 0.7 \times 0.4$	0.05488
F, F	F	$0.4 \times 0.7 \times 0.7 \times 0.7 \times 0.6$	0.08232

$P(D=T, e) = 0.1176$; $P(D=F, e) = 0.1624$; $P(e) = 0.28$.

$P(D=T | e) = 0.1176 / 0.28 = \mathbf{0.42} \approx 0.420$.

Q.25 [MCQ, 2M · Exact Inference] Answer: D (D: f(D, F))

Initial factors (one per CPT, evidence removed from scopes): $f()$, $f(B)$, $f(B, C)$, $f(C, D)$, $f(B, E)$, $f(C, D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C)$, $f(C, D)$, $f(C, D, F)$	$\{B, C, D, F\}$	$f(B, D, F)$
E	$f(B, E)$	$\{B, E\}$	$f(B)$
B	$f(B)$, $f(B, D, F)$, $f(B)$	$\{B, D, F\}$	$f(D, F)$
F	$f(D, F)$	$\{D, F\}$	$f(D)$

The largest factor produced has **3** variable(s) (created when eliminating C); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over D and normalise.

Mock Test 005

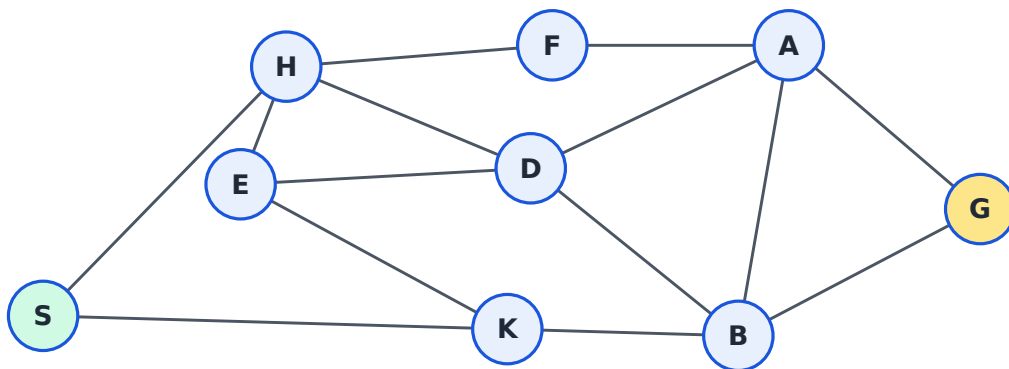
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 7 / 14	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

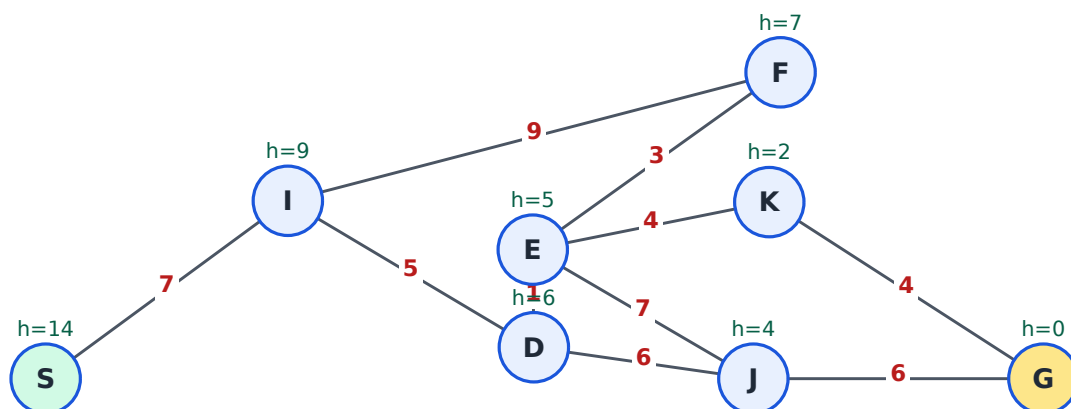
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, H, K, D, E, F, B, A, G
- ☐ (B) S, H, K, D, E, F, A, B, G
- ☐ (C) S, H, D, A, B, G
- ☐ (D) S, K, H, E, B, F, D, G

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

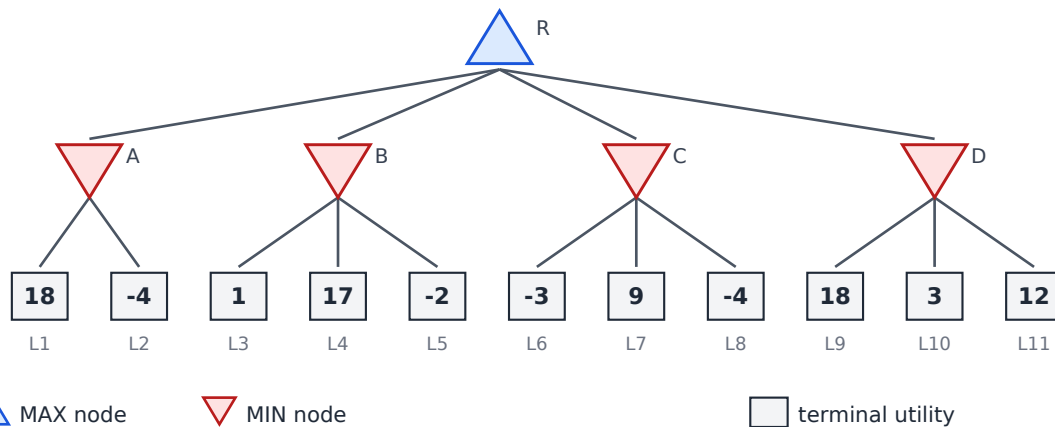
Q.3 [NAT · 1 mark · no negative marking] Adversarial Search

A two-ply game tree has a MAX root with four MIN children. Their leaf values, left to right, are: MIN_1 : (10, 9, 14); MIN_2 : (11, 8, 20); MIN_3 : (2, 15, 30); MIN_4 : (12, 13, 7). What is the minimax value of the root?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 4 elements and R a binary predicate symbol. In how many of the 2^{16} possible interpretations of R over D is the sentence $\forall x R(x, x) \wedge \forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (R \vee (\neg P \leftrightarrow R)) \leftrightarrow ((Q \wedge \neg P) \rightarrow P) ?$$

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic

The KB contains the clauses $\neg \text{Man}(x) \vee \text{Mortal}(x)$ and $\text{Man}(\text{Socrates})$. To prove $\text{Mortal}(\text{Socrates})$ by resolution refutation, $\neg \text{Mortal}(\text{Socrates})$ is added. What is the minimum number of resolution steps needed to derive the empty clause?

Answer (numerical): _____

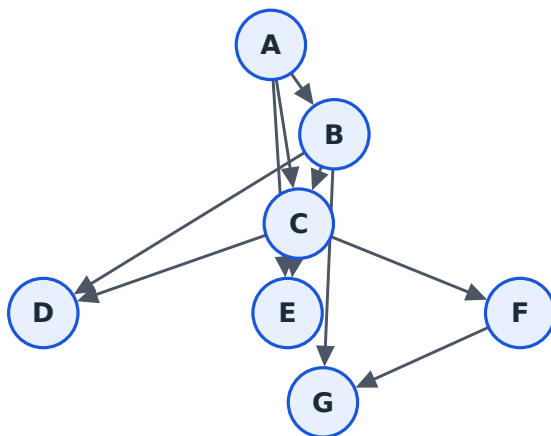
Q.8 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following propositional formulas is a tautology?

- ☐ (A) $(p \rightarrow q) \vee (q \rightarrow p)$
- ☐ (B) $(p \rightarrow q) \rightarrow (q \rightarrow p)$
- ☐ (C) $(p \wedge \neg q) \rightarrow q$
- ☐ (D) $(p \vee q) \rightarrow (p \wedge q)$

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **C** in the Bayesian network below?



Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Sampling

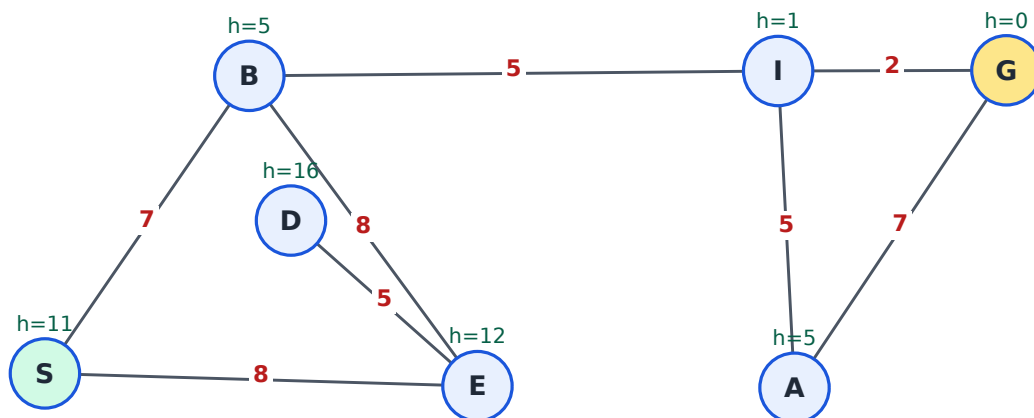
What is the main drawback of rejection sampling for computing $P(X | e)$?

- ☐ (A) It rejects most samples when $P(e)$ is small, and the fraction accepted drops exponentially with the number of evidence variables
- ☐ (B) It is not consistent: estimates do not converge even with infinitely many samples
- ☐ (C) It requires the Markov blanket of every variable
- ☐ (D) It can only be used when the evidence variables are root nodes

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

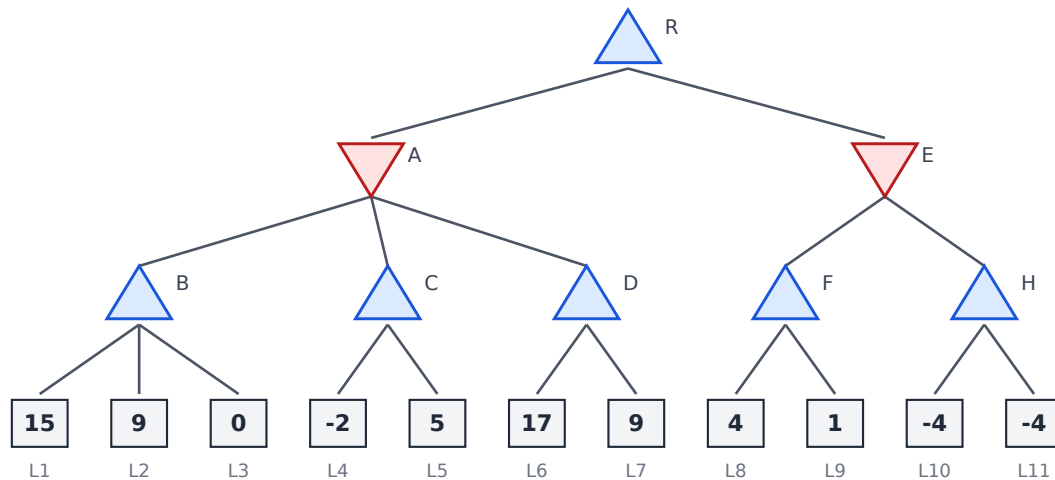
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L1 is pruned.
☐ (B) Leaf L7 is pruned.
☐ (C) Leaf L9 is pruned.
☐ (D) Leaf L11 is pruned.

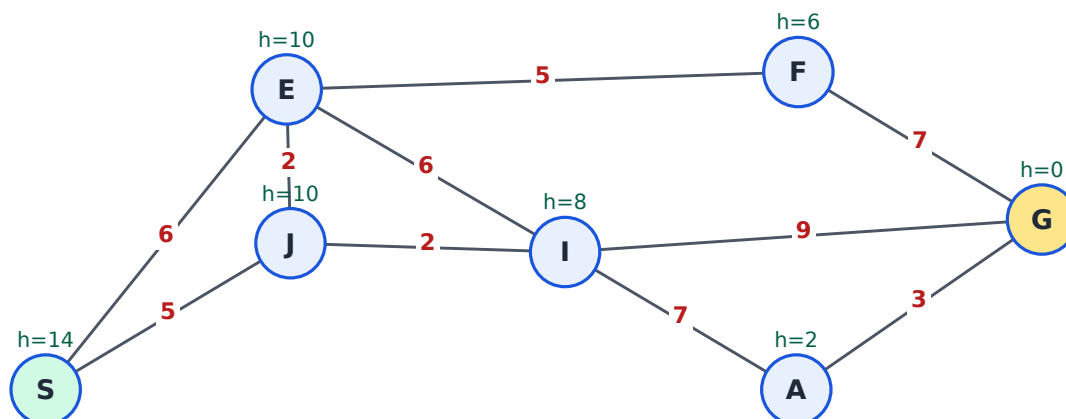
Q.13 [NAT · 2 marks · no negative marking] Problem Formulation

In the missionaries-and-cannibals problem (3 of each, a boat with capacity 2), a state is (m, c, b) , where m and c are the numbers of missionaries and cannibals on the left bank (0 to 3 each) and $b \in \{\text{left, right}\}$ is the boat's side. A state is safe if, on each bank, the missionaries are either zero or at least as many as the cannibals. How many safe states are there, counting both boat positions and ignoring reachability?

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following sentences (with equality) are true in exactly those interpretations where exactly one element satisfies P?

- ☐ (A) $\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x))$
☐ (B) $\exists x \forall y (P(y) \leftrightarrow y = x)$
☐ (C) $\exists x \forall y ((P(x) \wedge P(y)) \rightarrow x = y)$
☐ (D) $\exists x (P(x) \wedge \forall y (y = x \rightarrow P(y)))$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg(R \rightarrow \neg Q)$
 (2) $\neg P \leftrightarrow Q$
 (3) $R \leftrightarrow Q$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg(R \wedge Q)$
☐ (B) $Q \rightarrow \neg Q$
☐ (C) $(\neg R \vee \neg R) \vee R$
☐ (D) $P \vee Q$

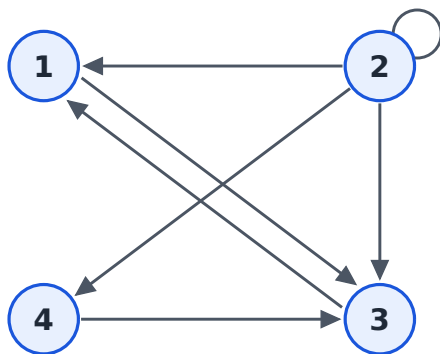
Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following formulas are valid?

- ☐ (A) $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
☐ (B) $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$
☐ (C) $((p \rightarrow q) \rightarrow r) \leftrightarrow (p \rightarrow (q \rightarrow r))$
☐ (D) $(\neg p \rightarrow \perp) \rightarrow p$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
☐ (B) $\exists x \forall y R(x, y)$
☐ (C) $\exists x R(x, x)$
☐ (D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(P \rightarrow R) \rightarrow (R \rightarrow P)$
☐ (B) $(P \rightarrow Q) \rightarrow ((P \vee R) \rightarrow Q)$
☐ (C) $(R \rightarrow Q) \vee (Q \rightarrow R)$
☐ (D) $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$

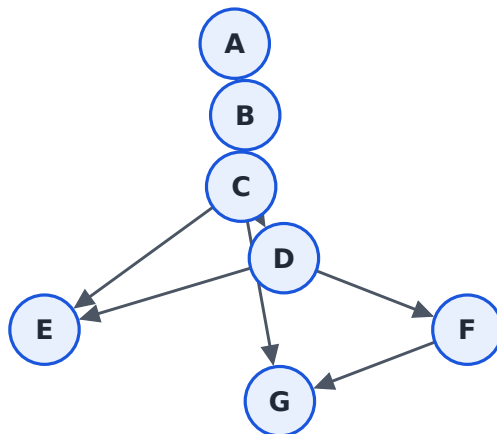
Q.20 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Rejection sampling is used with evidence $S = \text{true}$, $W = \text{true}$. What fraction of the prior samples is expected to be rejected? (Round to 3 decimal places.)

Answer (numerical): _____

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

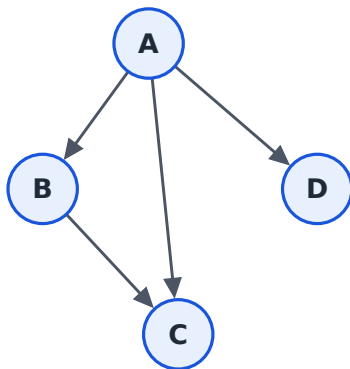
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp B$
☐ (B) $C \perp B | D$
☐ (C) $G \perp C | A, D$
☐ (D) $G \perp E | C, D$

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.95}, \mathbf{0.64}, \mathbf{0.41}, \mathbf{0.11} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



A	P(A=T)
T	0.8
F	

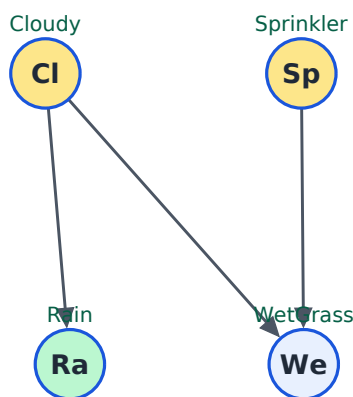
A	P(B=T ...)	A	B	P(C=T ...)
T	0.3	T	T	0.4
F	0.2	T	F	0.1
		F	T	0.6
		F	F	0.9

A	P(D=T ...)
T	0.7
F	0.05

- ☐ (A) A=F, B=T, C=T, D=F
☐ (B) A=F, B=F, C=T, D=F
☐ (C) A=T, B=F, C=T, D=F
☐ (D) A=T, B=T, C=T, D=F

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $\mathbf{P(Rain = T \mid Cloudy = T, Sprinkler = T)}$ (round off to 3 decimal places).



Cl	P(Cl=T)
T	0.8
F	

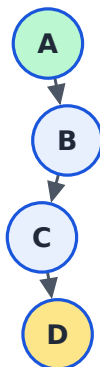
Sp	P(Sp=T)	Cl	P(Ra=T ...)
T	0.8	T	0.4
F		F	0.6

Cl	Sp	P(We=T ...)
T	T	0.05
T	F	0.25
F	T	0.1
F	F	0.6

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{D} = \mathbf{F}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(A = T \mid D = F)}$ (round off to 3 decimal places).



P(A=T)
0.4

A	P(B=T ...)	B	P(C=T ...)
T	0.05	T	0.2
F	0.7	F	0.4

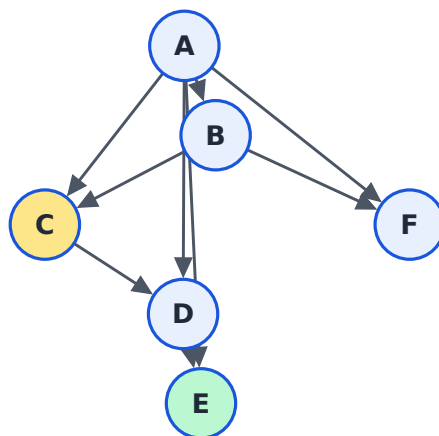
C	P(D=T ...)
T	0.1
F	0.25

#	A	B	C
1	T	F	T
2	T	F	T
3	F	T	T
4	F	T	F

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(E \mid C = T)}$ in the Bayesian network below (Boolean variables; evidence $\mathbf{C = T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{F, D, B, A}$. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 005

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	NAT	2	Informed Search	6
2	NAT	1	Informed Search	24	15	MSQ	2	Predicate Logic	A, B
3	NAT	1	Adversarial Search	9	16	MSQ	2	Propositional Logic	C, D
4	NAT	1	Adversarial Search	3	17	MSQ	2	Propositional Logic	A, B, D
5	NAT	1	Predicate Logic	64	18	MSQ	2	Predicate Logic	A, B, C
6	NAT	1	Propositional Logic	6	19	MSQ	2	Propositional Logic	C, D
7	NAT	1	Predicate Logic	2	20	NAT	2	Sampling	0.721 to 0.723
8	MCQ	1	Propositional Logic	A	21	MSQ	2	Conditional Independence	D
9	NAT	1	Conditional Independence	5	22	MCQ	2	Sampling	B
10	MCQ	1	Sampling	A	23	NAT	2	Exact Inference	0.398 to 0.402
11	NAT	2	Informed Search	14	24	NAT	2	Sampling	0.52 to 0.524
12	MSQ	2	Adversarial Search	B, D	25	NAT	2	Exact Inference	2
13	NAT	2	Problem Formulation	20					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: A (A: S, H, K, D, E, F, B, A, G)

Adjacency lists (alphabetical): A: {B, D, F, G}; B: {A, D, G, K}; D: {A, B, E, H}; E: {D, H, K}; F: {A, H}; G: {A, B}; H: {D, E, F, S}; K: {B, E, S}; S: {H, K}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[H, K]
2	H	[K, D, E, F]
3	K	[D, E, F, B]
4	D	[E, F, B, A]
5	E	[F, B, A]
6	F	[B, A]
7	B	[A, G]
8	A	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, H, K, D, E, F, B, A, G**. The path returned is $S \rightarrow K \rightarrow B \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Informed Search] Answer: 24

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (14)	I(9)
2	I (9)	D(6), F(7)
3	D (6)	J(4), E(5), F(7)
4	J (4)	G(0), E(5), F(7)
5	G (0)	E(5), F(7) ← goal, stop

Path returned: **S → I → D → J → G**, cost = 7 + 5 + 6 + 6 = **24**.

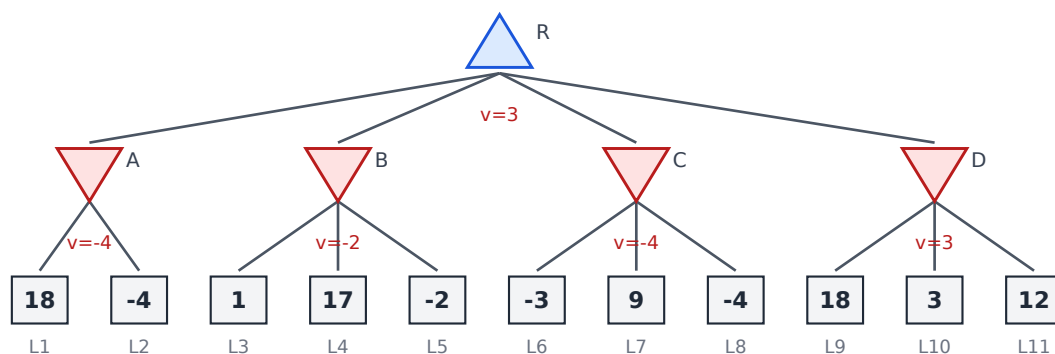
The optimal cost is $h^*(S) = 21$ — greedy search is not optimal here.

Q.3 [NAT, 1M · Adversarial Search] Answer: 9

Each MIN node takes the minimum of its leaves: $\text{MIN}_1 = 9$, $\text{MIN}_2 = 8$, $\text{MIN}_3 = 2$, $\text{MIN}_4 = 7$. The root takes the maximum: $\max(9, 8, 2, 7) = \mathbf{9}$, so MAX plays the first move.

Q.4 [NAT, 1M · Adversarial Search] Answer: 3

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -4$, $B = -2$, $C = -4$, $D = 3$. Root value = **3**; MAX's optimal first move is towards D.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 64**

View R as an 4×4 0/1 matrix. The sentence says R is reflexive and symmetric: only the $n(n-1)/2$ unordered off-diagonal pairs are free $\Rightarrow 2^{n(n-1)/2}$. With $n = 4$: **64**.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	$R \vee (\neg P \leftrightarrow R)$	$(Q \wedge \neg P) \rightarrow P$	$(R \vee (\neg P \leftrightarrow R)) \leftrightarrow ((Q \wedge \neg P) \rightarrow P)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	F	T	F

Rows where F = T: **6**.

Q.7 [NAT, 1M · Predicate Logic] **Answer: 2**

Step 1: resolve $\neg \text{Mortal}(\text{Socrates})$ with $\neg \text{Man}(x) \vee \text{Mortal}(x)$ using $\{x/\text{Socrates}\}$, giving $\neg \text{Man}(\text{Socrates})$. Step 2: resolve $\neg \text{Man}(\text{Socrates})$ with $\text{Man}(\text{Socrates})$, giving $\square$. One step cannot suffice since no two initial clauses are complementary units. So the answer is 2.

Q.8 [MCQ, 1M · Propositional Logic] **Answer: A** ($A: (p \rightarrow q) \vee (q \rightarrow p)$)

$(p \rightarrow q) \vee (q \rightarrow p) \equiv \neg p \vee q \vee \neg q \vee p$, which contains $q \vee \neg q$, so it is true in all 4 rows. $(p \rightarrow q) \rightarrow (q \rightarrow p)$ is false at $p=F, q=T$ (antecedent T, consequent F). $(p \wedge \neg q) \rightarrow q$ is false at $p=T, q=F$. $(p \vee q) \rightarrow (p \wedge q)$ is false whenever exactly one of p, q is true. Hence only the first is valid.

Q.9 [NAT, 1M · Conditional Independence] **Answer: 5**

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of C: {A, B}; children: {D, E, F}; co-parents: $\{\emptyset\}$.

$\text{MB}(C) = \{A, B, D, E, F\}$ — **5** nodes. Given its Markov blanket, C is conditionally independent of every other node in the network.

Q.10 [MCQ, 1M · Sampling] **Answer: A** (A: It rejects most samples when $P(e)$ is small, and the fraction accepted drops exponentially with the number of evidence variables)

Rejection sampling keeps only samples consistent with e , so about $N \cdot P(e)$ samples are useful; $P(e)$ typically shrinks exponentially as evidence grows. It is consistent (B is false). Markov blankets are used by Gibbs sampling, not rejection sampling (C false). Evidence may be anywhere in the network (D false).

Q.11 [NAT, 2M · Informed Search] **Answer: 14**

Heuristic values: $h(A)=5$, $h(B)=5$, $h(D)=16$, $h(E)=12$, $h(G)=0$, $h(I)=1$, $h(S)=11$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	B(12), E(20)

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
2	B (12)	I(13), E(20)
3	I (13)	G(14), E(20), A(22)
4	G (14)	E(20), A(22) ← goal, stop

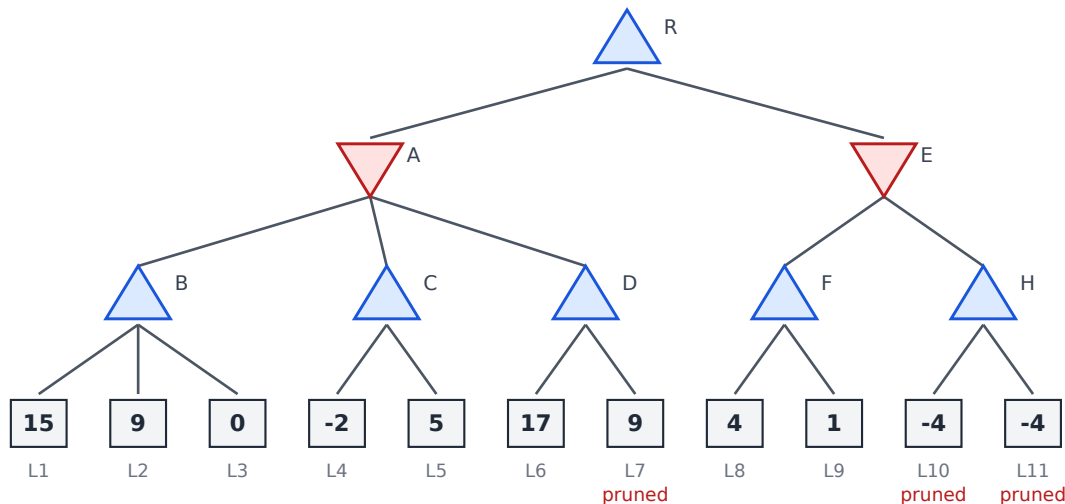
Path returned: **S** → **B** → **I** → **G** with cost **14**; nodes expanded = **4** (S, B, I, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 6 nodes.

Q.12 [MSQ, 2M · Adversarial Search] **Answer: B, D** (B: Leaf L7 is pruned.; D: Leaf L11 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node D: value 17 $\geq \beta = 5 \Rightarrow$ prune remaining children (leaves L7).
- At MIN node E: value 4 $\leq \alpha = 5 \Rightarrow$ prune remaining children (leaves L10, L11).



Leaves evaluated (8): L1, L2, L3, L4, L5, L6, L8, L9. Leaves pruned (3): L7, L10, L11. Root value = 5 (same as plain minimax, 5 — pruning never changes the root value).

Q.13 [NAT, 2M · Problem Formulation] **Answer: 20**

Count the safe (m, c) pairs. $m = 0$: the right bank has 3 missionaries, so every $c = 0..3$ is safe, giving 4 pairs. $m = 3$: the left bank needs $c \leq 3$ and the right bank has no missionaries, giving 4 pairs. $m = 1$: the left bank needs $c \leq 1$ and the right bank (2 missionaries, $3-c$ cannibals) needs $3-c \leq 2$, so $c = 1$: 1 pair. $m = 2$: $c \leq 2$ and $3-c \leq 1$, so $c = 2$: 1 pair. That gives 10 pairs, and 2 boat positions give **20** states. (Only 16 of them are reachable from (3,3,left).)

Q.14 [NAT, 2M · Informed Search] **Answer: 6**

Heuristic values: $h(A)=2$, $h(E)=10$, $h(F)=6$, $h(G)=0$, $h(I)=8$, $h(J)=10$, $h(S)=14$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (14)	J(15), E(16)
2	J (15)	I(15), E(16)
3	I (15)	A(16), E(16), G(16)
4	A (16)	E(16), G(16)
5	E (16)	G(16), F(17)
6	G (16)	F(17) ← goal, stop

Path returned: **S** → **J** → **I** → **G** with cost **16**; nodes expanded = **6** (S, J, I, A, E, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [MSQ, 2M · Predicate Logic] **Answer: A, B** (A: $\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x))$; B: $\exists x \forall y (P(y) \leftrightarrow y = x)$)

(A) Standard 'exactly one': some x is P and every P is x . (B) For some x , $P(y)$ holds iff $y = x$ — also exactly one; A and B are equivalent. (C) Is true when no element satisfies P ($P(x)$ false makes the implication vacuous), so it expresses 'at most one'. (D) The inner part is trivially implied by $P(x)$, so it is just $\exists x P(x)$ ('at least one'). Brute force on domains of size 1–4 confirms that only A and B match 'exactly one'.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $(\neg R \vee \neg R) \vee R$; D: $P \vee Q$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
F	T	T	T

(A) $\neg(R \wedge Q)$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

(B) $Q \rightarrow \neg Q$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

(C) $(\neg R \vee \neg R) \vee R$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $P \vee Q$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, B, D** (A: $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$; B: $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$; D: $(\neg p \rightarrow \perp) \rightarrow p$)

(A) Transitivity of implication; a truth table shows no falsifying row. (B) Exportation: both sides are equivalent to $\neg p \vee \neg q \vee r$. (C) Not valid: at $p=F, q=F, r=F$ the left side $(p \rightarrow q) \rightarrow r$ is $T \rightarrow F = F$, while the right side is T. Implication is not associative. (D) $\neg p \rightarrow \perp \equiv \neg \neg p \equiv p$, so the formula is $p \rightarrow p$, which is valid (this is the basis of proof by contradiction).

Q.18 [MSQ, 2M · Predicate Logic] **Answer: A, B, C** (A: $\forall x \exists y R(x, y)$; B: $\exists x \forall y R(x, y)$; C: $\exists x R(x, x)$)

$R = \{(1,3), (2,1), (2,2), (2,3), (2,4), (3,1), (4,3)\}$.

(A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(B) $\exists x \forall y R(x, y)$: **TRUE** — $x = 2$ has edges to every y .

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

(D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,4)$ holds but $R(4,2)$ does not.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $(R \rightarrow Q) \vee (Q \rightarrow R)$; D: $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$)

(A) $(P \rightarrow R) \rightarrow (R \rightarrow P)$: **not valid** — counter-model $P=F, R=T$.

(B) $(P \rightarrow Q) \rightarrow ((P \vee R) \rightarrow Q)$: **not valid** — counter-model $P=F, Q=F, R=T$.

(C) $(R \rightarrow Q) \vee (Q \rightarrow R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.20 [NAT, 2M · Sampling] **Answer: 0.721 to 0.723**

A sample is kept with probability $P(s, w)$. $P(s, w) = \sum_{c,r} P(c)P(s | c)P(r | c)P(w | s, r)$. c : $0.5 \times 0.1 \times (0.8 \times 0.99 + 0.2 \times 0.90) = 0.05 \times 0.972 = 0.0486$. $\neg c$: $0.5 \times 0.5 \times (0.2 \times 0.99 + 0.8 \times 0.90) = 0.25 \times 0.918 = 0.2295$. $P(s, w) = 0.2781$. Expected rejected fraction = $1 - 0.2781 = 0.7219 \approx 0.722$.

Q.21 [MSQ, 2M · Conditional Independence] **Answer: D** (D: $G \perp E | C, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $C \perp B$: **FALSE** — the path $C \leftarrow A \rightarrow B$ is active given $\{\}$.

(B) $C \perp B | D$: **FALSE** — the path $C \leftarrow A \rightarrow B$ is active given $\{D\}$.

(C) $G \perp C | A, D$: **FALSE** — the path $G \leftarrow C$ is active given $\{A, D\}$.

(D) $G \perp E | C, D$: **TRUE** — $G \leftarrow C \rightarrow D \rightarrow E$ (blocked: observed fork node C); $G \leftarrow C \rightarrow E$ (blocked: observed fork node C); $G \leftarrow F \leftarrow D \leftarrow C \rightarrow E$ (blocked: observed chain node D); $G \leftarrow F \leftarrow D \rightarrow E$ (blocked: observed fork node D).

Q.22 [MCQ, 2M · Sampling] **Answer: B** (B: $A=F, B=F, C=T, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \neg) = 0.8$; $u = 0.95 \geq 0.8 \Rightarrow A = F$
- B: $P(B=T | A=F) = 0.2$; $u = 0.64 \geq 0.2 \Rightarrow B = F$
- C: $P(C=T | A=F, B=F) = 0.9$; $u = 0.41 < 0.9 \Rightarrow C = T$
- D: $P(D=T | A=F) = 0.05$; $u = 0.11 \geq 0.05 \Rightarrow D = F$

Sample: **$\{A=F, B=F, C=T, D=F\}$** .

Q.23 [NAT, 2M · Exact Inference] **Answer: 0.398 to 0.402**

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler}) \cdot P(\text{Rain} | \text{Cloudy}) \cdot P(\text{WetGrass} | \text{Cloudy}, \text{Sprinkler})$.

Inference by enumeration: $P(\text{Rain} | e) = \alpha \sum_{\text{hidden}} P(\text{Rain}, e, \text{hidden})$.

Hidden: We	Ra	product of CPT entries (Cl · Sp · Ra · We)	value
T	T	$0.8 \times 0.8 \times 0.4 \times 0.05$	0.0128
F	T	$0.8 \times 0.8 \times 0.4 \times 0.95$	0.2432
T	F	$0.8 \times 0.8 \times 0.6 \times 0.05$	0.0192
F	F	$0.8 \times 0.8 \times 0.6 \times 0.95$	0.3648

$P(\text{Rain}=T, e) = 0.256$; $P(\text{Rain}=F, e) = 0.384$; $P(e) = 0.64$.

$P(\text{Rain}=T | e) = 0.256 / 0.64 = \mathbf{0.4} \approx 0.400$.

Q.24 [NAT, 2M · Sampling] **Answer: 0.52 to 0.524**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	A
1	$P(D=F C=T)=0.9$	0.9	T
2	$P(D=F C=T)=0.9$	0.9	T
3	$P(D=F C=T)=0.9$	0.9	F
4	$P(D=F C=F)=0.75$	0.75	F

$\sum w (\text{all}) = 3.45$; $\sum w (A = T) = 1.8$.

Estimate $P(A=T | D = F) = 1.8 / 3.45 = \mathbf{0.5217}$.

Q.25 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B)$, $f(A, D)$, $f(A, D, E)$, $f(A, B, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(A, B, F)$	$\{A, B, F\}$	$f(A, B)$
D	$f(A, D)$, $f(A, D, E)$	$\{A, D, E\}$	$f(A, E)$
B	$f(A, B)$, $f(A, B)$, $f(A, B)$	$\{A, B\}$	$f(A)$
A	$f(A)$, $f(A, E)$, $f(A)$	$\{A, E\}$	$f(E)$

The largest factor produced has **2** variable(s) (created when eliminating F); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over E and normalise.

Mock Test 006

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 9 / 10	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

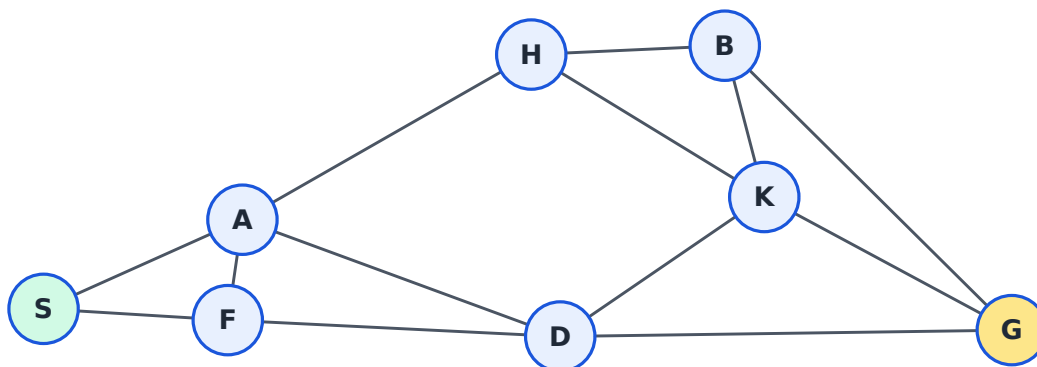
Q.1 [MSQ · 1 mark · no negative marking] Local Search

Steepest-ascent hill climbing (maximisation, moves only to a strictly better neighbour) can halt without reaching a global maximum because of which of the following?

- ☐ (A) A local maximum
- ☐ (B) A plateau or shoulder (a flat region where all neighbours have equal value)
- ☐ (C) A ridge (a sequence of local maxima that single moves cannot climb)
- ☐ (D) A state where the neighbour with the highest value is better than the current state

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

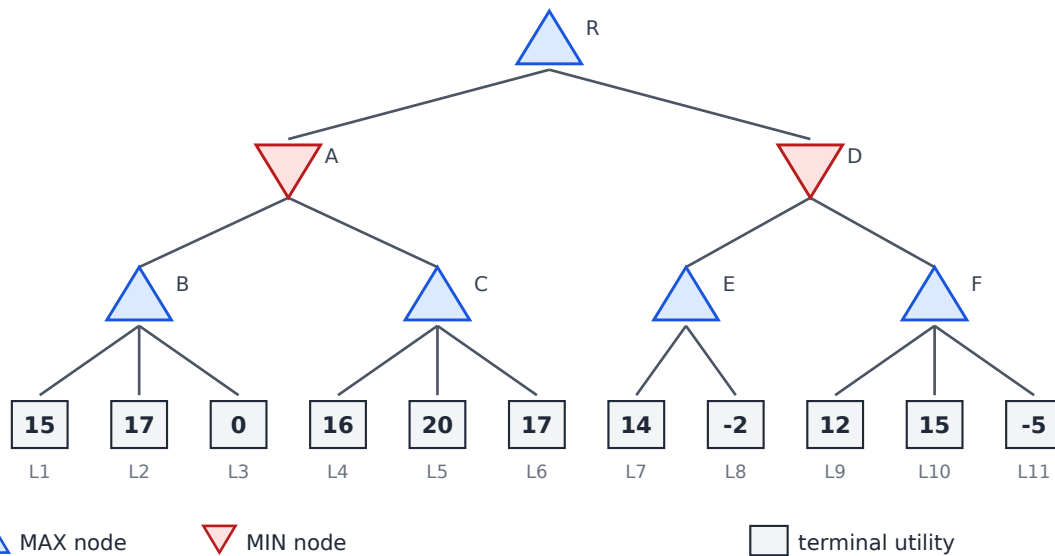
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, A, F, H, D, G
- ☐ (B) S, A, F, D, H, G
- ☐ (C) S, F, A, D, H, K, G
- ☐ (D) S, A, D, F, G

Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to A; root value 20
☐ (B) Move to A; root value 18
☐ (C) Move to A; root value 17
☐ (D) Move to D; root value 14

Q.4 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Depth-limited search with limit $l = 3$ is run on a tree with branching factor $b = 4$. No goal exists within depth 3. How many nodes (including the root) does it generate before returning cutoff?

Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Resolving the first-order clauses $P(x) \vee Q(f(x))$ and $\neg P(g(y)) \vee R(y)$ (variables already standardized apart) on P gives

- ☐ (A) $Q(f(g(y))) \vee R(y)$
☐ (B) $Q(f(x)) \vee R(y)$
☐ (C) $Q(g(f(y))) \vee R(y)$
☐ (D) $Q(f(y)) \vee R(g(y))$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((R \rightarrow Q) \rightarrow Q) \rightarrow (R \leftrightarrow (\neg Q \vee P)) ?$$

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

How many of the 8 truth assignments to p, q, r are models of $(p \vee q) \wedge (\neg p \vee r)$?

Answer (numerical): _____

Q.8 [MCQ · 1 mark · negative: -1/3] Probability for AI

In AIMA's enumeration-style inference, $P(X | e)$ is computed as $\alpha \sum P(X, e)$, where the sum over hidden variables gives $P(X, e)$. What is the value of α ?

- (A) $1/P(e)$
 ○ (B) $P(e)$
 ○ (C) $1/P(X)$
 ○ (D) $1/|\text{domain}(X)|$

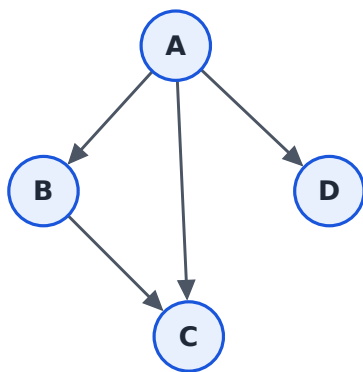
Q.9 [NAT · 1 mark · no negative marking] Variable Elimination

Factors over Boolean variables: $f(A, B)$ with $f(a, b) = 0.4$, $f(a, \neg b) = 0.6$, $f(\neg a, b) = 0.8$, $f(\neg a, \neg b) = 0.2$; $g(B, C)$ with $g(b, c) = 0.3$, $g(b, \neg c) = 0.7$, $g(\neg b, c) = 0.5$, $g(\neg b, \neg c) = 0.5$. Let $h(A, C) = \sum_B f(A, B) \times g(B, C)$. Compute $h(a, c)$. (Round to 2 decimal places.)

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=F, B=T, C=T, D=F)**? (round off to 4 decimal places)



A	P(A=T)
T	0.1
F	

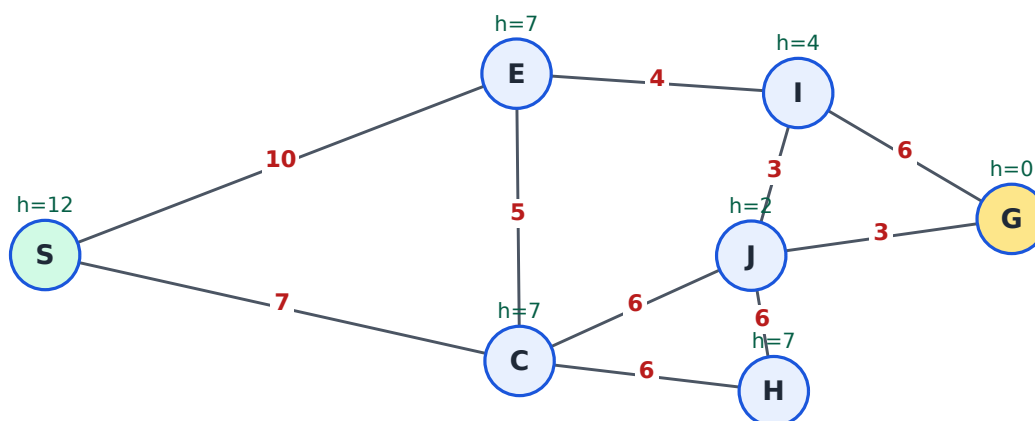
A	P(B=T ...)	A	B	P(C=T ...)
T	0.1	T	T	0.5
F	0.75	T	F	0.95
		F	T	0.9
		F	F	0.4

A	P(D=T ...)
T	0.9
F	0.75

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Informed Search – A*

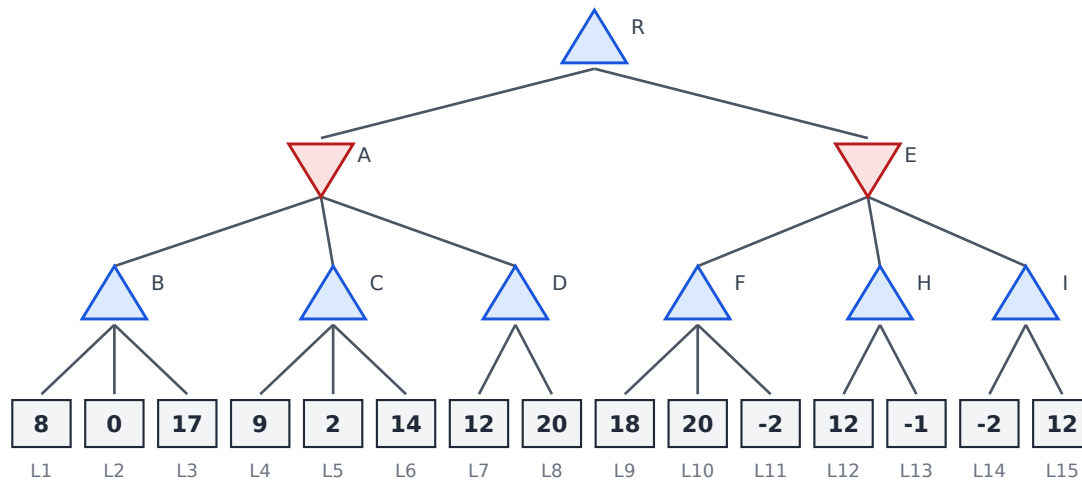
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



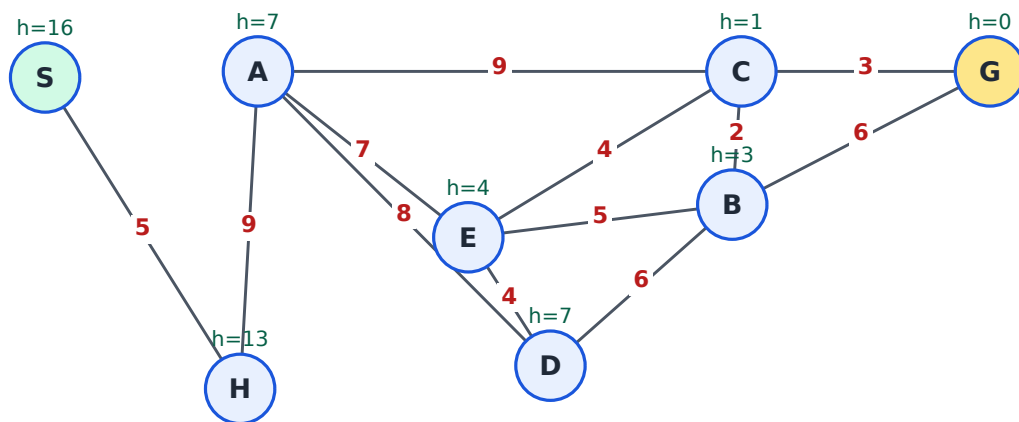
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node E is expanded.
- ☐ (B) Node C is expanded.
- ☐ (C) Node A is expanded.
- ☐ (D) The path returned has cost 28.

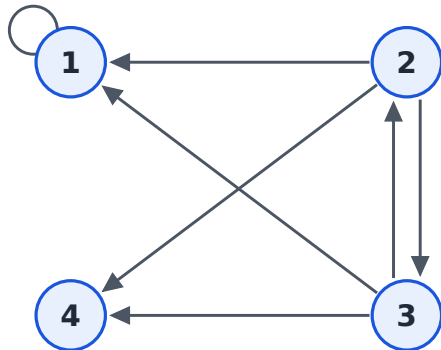
Q.14 [MSQ · 2 marks · no negative marking] Adversarial Search

Which of the following statements about depth-limited (cutoff) game search with heuristic evaluation functions are TRUE?

- ☐ (A) The horizon effect occurs when an unavoidable damaging event is pushed beyond the search depth by delaying moves
- ☐ (B) Quiescence search extends the search at non-quiet positions, such as pending captures, before applying the evaluation function
- ☐ (C) A good evaluation function should be strongly correlated with the actual chance of winning
- ☐ (D) Increasing the cutoff depth by a fixed amount always removes the horizon effect

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y R(x, y)$
☐ (B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\exists x R(x, x)$

Q.16 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following sets of connectives is NOT functionally complete?

- ☐ (A) $\{\neg, \wedge\}$
☐ (B) $\{\text{NAND}\}$
☐ (C) $\{\rightarrow, \perp\}$
☐ (D) $\{\wedge, \vee, \leftrightarrow\}$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $\mathbf{F} = (\mathbf{P} \rightarrow \neg \mathbf{Q}) \vee (\mathbf{P} \leftrightarrow \mathbf{P})$?

- ☐ (A) $\neg(\neg(\mathbf{P} \rightarrow \neg \mathbf{Q}) \vee \neg(\mathbf{P} \leftrightarrow \mathbf{P}))$
☐ (B) $\neg(\neg(\mathbf{P} \rightarrow \neg \mathbf{Q}) \wedge \neg(\mathbf{P} \leftrightarrow \mathbf{P}))$
☐ (C) $(\mathbf{P} \rightarrow \neg \mathbf{Q}) \wedge (\mathbf{P} \leftrightarrow \mathbf{P})$
☐ (D) $\neg(\mathbf{P} \rightarrow \neg \mathbf{Q}) \vee (\mathbf{P} \leftrightarrow \mathbf{P})$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (\neg P \vee Q \vee \neg S)$, $C2: (P \vee Q)$, $C3: (P \vee \neg R \vee \neg S)$, $C4: (\neg P)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(Q \vee \neg R)$
☐ (B) (Q)
☐ (C) $(Q \vee \neg S)$
☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **C, A** and rules:

R1: $H \wedge A \wedge F \rightarrow C$

R2: $F \rightarrow J$

R3: $D \wedge E \rightarrow J$

R4: $C \wedge A \rightarrow H$

R5: $D \wedge K \wedge C \rightarrow E$

R6: $C \rightarrow F$

R7: $E \wedge J \wedge D \rightarrow K$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) B is entailed
- ☐ (B) E is entailed
- ☐ (C) F is entailed
- ☐ (D) K is entailed

Q.20 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Converting $\forall x ((\exists y P(x, y)) \rightarrow (\exists z Q(x, z)))$ to clausal form (with F a new Skolem function) gives

- ☐ (A) $\neg P(x, y) \vee Q(x, F(x))$
- ☐ (B) $\neg P(x, G(x)) \vee Q(x, F(x))$
- ☐ (C) $\neg P(x, y) \vee Q(x, C)$, with C a new constant
- ☐ (D) $\neg P(x, y) \wedge Q(x, F(x))$

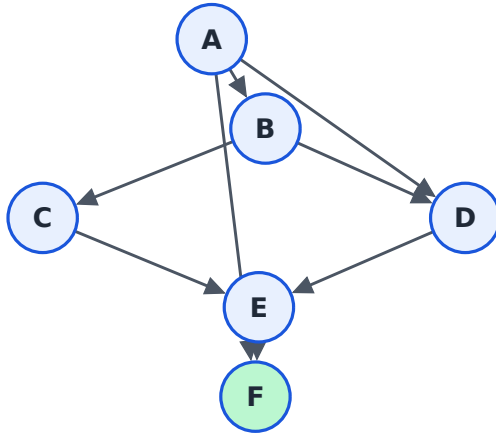
Q.21 [MSQ · 2 marks · no negative marking] Probability for AI

A full joint distribution over Boolean variables X, Y, Z is: $P(x, y, z) = 0.10$, $P(x, y, \neg z) = 0.05$, $P(x, \neg y, z) = 0.15$, $P(x, \neg y, \neg z) = 0.10$, $P(\neg x, y, z) = 0.05$, $P(\neg x, y, \neg z) = 0.20$, $P(\neg x, \neg y, z) = 0.15$, $P(\neg x, \neg y, \neg z) = 0.20$. Which of the following statements are true?

- ☐ (A) $P(x) = 0.4$
- ☐ (B) $X \perp Y$ (X and Y are marginally independent)
- ☐ (C) $X \perp Y \mid Z$
- ☐ (D) $P(y \mid \neg z) \approx 0.455$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

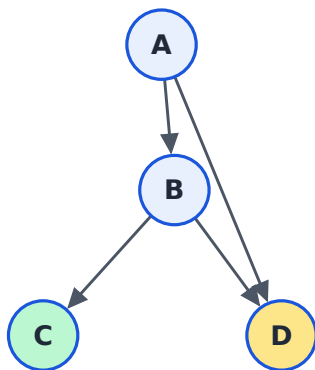
Variable elimination is used to compute $P(\mathbf{F})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, D, C, B, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(\mathbf{C} = \mathbf{T} \mid \mathbf{D} = \mathbf{T})$ (round off to 3 decimal places).



$P(A=T)$
0.1

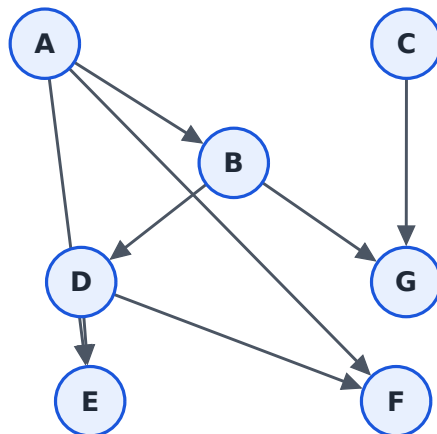
A	$P(B=T \dots)$	B	$P(C=T \dots)$
T	0.5	T	0.75
F	0.75	F	0.6

A	B	$P(D=T \dots)$
T	T	0.5
T	F	0.1
F	T	0.1
F	F	0.5

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp A \mid D$
☐ (B) $B \perp G$
☐ (C) $B \perp G \mid E, F$
☐ (D) $E \perp F \mid A, D$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid A = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	T	F	T	F
2	T	F	F	T
3	F	F	F	T
4	T	F	F	T
5	T	F	F	T
6	T	F	T	T
7	T	T	F	F
8	F	F	F	T
9	T	T	T	F
10	T	F	T	T
11	T	F	F	F
12	T	T	T	T
13	F	F	F	T
14	T	F	T	F
15	T	T	F	T

Answer (numerical): _____

Answer Key — Mock Test 006

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MSQ	1	Local Search	A, B, C	14	MSQ	2	Adversarial Search	A, B, C
2	MCQ	1	Uninformed Search	B	15	MSQ	2	Predicate Logic	D
3	MCQ	1	Adversarial Search	C	16	MCQ	2	Propositional Logic	D
4	NAT	1	Uninformed Search	85	17	MSQ	2	Propositional Logic	B, D
5	MCQ	1	Predicate Logic	A	18	MSQ	2	Propositional Logic	B, C
6	NAT	1	Propositional Logic	6	19	MSQ	2	Propositional Logic	C
7	NAT	1	Propositional Logic	4	20	MCQ	2	Predicate Logic	A
8	MCQ	1	Probability for AI	A	21	MSQ	2	Probability for AI	A, D
9	NAT	1	Variable Elimination	0.415 to 0.425	22	NAT	2	Exact Inference	4
10	NAT	1	Sampling	0.1517 to 0.1521	23	NAT	2	Exact Inference	0.664 to 0.668
11	NAT	2	Informed Search	4	24	MSQ	2	Conditional Independence	A, D
12	NAT	2	Adversarial Search	2	25	NAT	2	Sampling	0.57 to 0.59
13	MSQ	2	Informed Search	A, B, C					

Detailed Solutions

Q.1 [MSQ, 1M · Local Search] Answer: A, B, C (A: A local maximum; B: A plateau or shoulder (a flat region where all neighbours have equal value); C: A ridge (a sequence of local maxima that single moves cannot climb))

(A), (B) and (C) are the three classic traps listed in AIMA. At a local maximum no neighbour is better. On a plateau no neighbour is strictly better. On a ridge every single-move neighbour is worse even though the ridge rises. (D) is false: if the best neighbour is better, hill climbing moves there and does not halt.

Q.2 [MCQ, 1M · Uninformed Search] Answer: B (B: S, A, F, D, H, G)

Adjacency lists (alphabetical): A: {D, F, H, S}; B: {G, H, K}; D: {A, F, G, K}; F: {A, D, S}; G: {B, D, K}; H: {A, B, K}; K: {B, D, G, H}; S: {A, F}.

BFS trace (queue shown front → back):

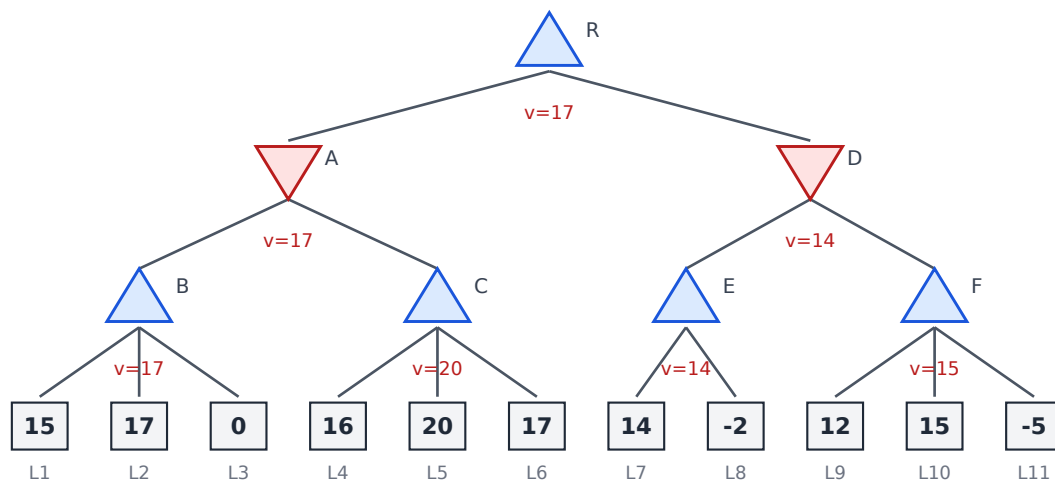
Step	Expanded	Frontier (FIFO) after expansion
1	S	[A, F]
2	A	[F, D, H]
3	F	[D, H]
4	D	[H, G, K]
5	H	[G, K, B]
6	G	[K, B] ← goal popped, stop

Expansion order: **S, A, F, D, H, G**. The path returned is $S \rightarrow A \rightarrow D \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to A; root value 17)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 17$, $C = 20$, $A = 17$, $E = 14$, $F = 15$, $D = 14$. Root value = **17**; MAX's optimal first move is towards A.

Q.4 [NAT, 1M · Uninformed Search] **Answer: 85**

DLS explores the complete tree down to depth ℓ : $1 + b + \dots + b^\ell = 1 + 4 + 16 + 64 = \mathbf{85} = (b^{\ell+1} - 1)/(b - 1)$.

Q.5 [MCQ, 1M · Predicate Logic] **Answer: A** ($A: Q(f(g(y))) \vee R(y)$)

Unify $P(x)$ with $P(g(y))$: MGU $\theta = \{x/g(y)\}$. The resolvent is $(Q(f(x)) \vee R(y))\theta = Q(f(g(y))) \vee R(y)$. Option B forgets to apply θ . Options C and D apply the substitution incorrectly.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \rightarrow Q) \rightarrow Q$	$R \leftrightarrow (\neg Q \vee P)$	$((R \rightarrow Q) \rightarrow Q) \rightarrow (R \leftrightarrow (\neg Q \vee P))$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	F	F	T
F	T	T	T	F	F
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	F	F	T

Rows where F = T: **6**.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 4**

Case $p=T$: the first clause is satisfied and the second requires $r=T$; q is free, giving 2 models. Case $p=F$: the second clause is satisfied and the first requires $q=T$; r is free, giving 2 models. Total = $2 + 2 = 4$.

Q.8 [MCQ, 1M · Probability for AI] **Answer: A** ($A: 1/P(e)$)

By definition $P(X | e) = P(X, e)/P(e)$, so $\alpha = 1/P(e)$. In practice α is found by normalizing the vector $P(X, e)$ to sum to 1, because $\sum_x P(x, e) = P(e)$. The other options do not make the entries sum to 1 in general.

Q.9 [NAT, 1M · Variable Elimination] **Answer: 0.415 to 0.425**

$h(a, c) = f(a, b) g(b, c) + f(a, \neg b) g(\neg b, c) = 0.4 \times 0.3 + 0.6 \times 0.5 = 0.12 + 0.30 = 0.42$. First the pointwise product over $\{A, B, C\}$ is formed, then B is summed out, leaving a factor over $\{A, C\}$.

Q.10 [NAT, 1M · Sampling] **Answer: 0.1517 to 0.1521**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(A=F) \times P(B=T|A=F) \times P(C=T|A=F, B=T) \times P(D=F|A=F)$
 $= 0.9 \times 0.75 \times 0.9 \times 0.25 = \mathbf{0.151875} \approx 0.1519$.

Q.11 [NAT, 2M · Informed Search] Answer: 4

Heuristic values: $h(C)=7$, $h(E)=7$, $h(G)=0$, $h(H)=7$, $h(I)=4$, $h(J)=2$, $h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	C(14), E(17)
2	C (14)	J(15), E(17), H(20)
3	J (15)	G(16), E(17), H(20), I(20)
4	G (16)	E(17), H(20), I(20) ← goal, stop

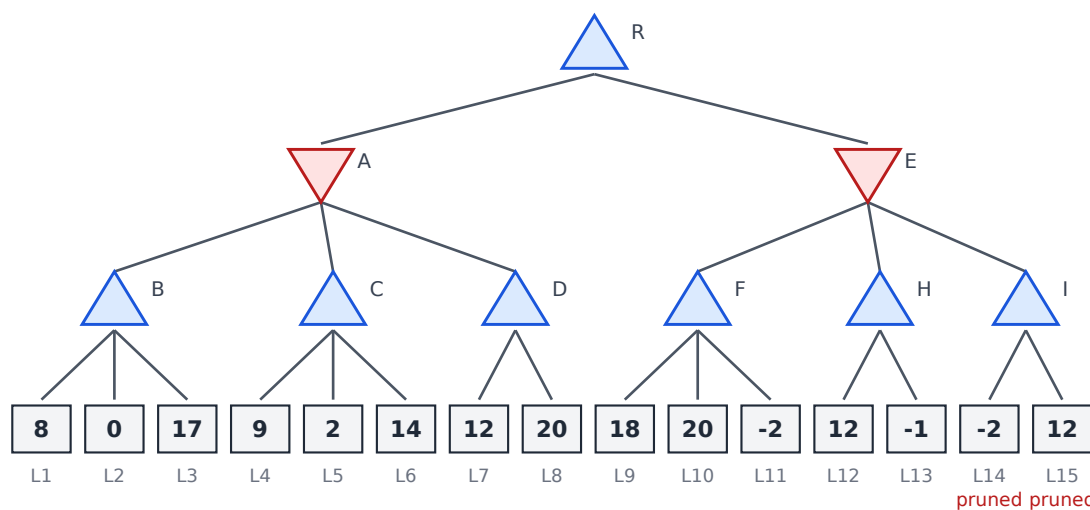
Path returned: **S** → **C** → **J** → **G** with cost **16**; nodes expanded = **4** (S, C, J, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.12 [NAT, 2M · Adversarial Search] Answer: 2

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node E: value $12 \leq \alpha = 14 \Rightarrow$ prune remaining children (leaves L14, L15).



Leaves evaluated (13): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10, L11, L12, L13. Leaves pruned (2): L14, L15. Root value = 14 (same as plain minimax, 14 — pruning never changes the root value).

Q.13 [MSQ, 2M · Informed Search] Answer: A, B, C (A: Node E is expanded.; B: Node C is expanded.; C: Node A is expanded.)

Heuristic values: $h(A)=7$, $h(B)=3$, $h(C)=1$, $h(D)=7$, $h(E)=4$, $h(G)=0$, $h(H)=13$, $h(S)=16$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (16)	H(18)
2	H (18)	A(21)
3	A (21)	C(24), E(25), D(29)
4	C (24)	E(25), G(26), B(28), D(29)
5	E (25)	G(26), B(28), D(29)
6	G (26)	B(28), D(29) ← goal, stop

Path returned: **S** → **H** → **A** → **C** → **G** with cost **26**; nodes expanded = **6** (S, H, A, C, E, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 26$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.14 [MSQ, 2M · Adversarial Search] Answer: A, B, C

(A) defines the horizon effect. (B) defines quiescence search, which applies the evaluation function only at quiet positions. (C) is the AIMA requirement for evaluation functions. (D) is false: at any fixed depth, a delaying tactic can push the bad event just beyond the new horizon.

Q.15 [MSQ, 2M · Predicate Logic] Answer: D (D: $\exists x R(x, x)$)

$R = \{(1,1), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4)\}$.

(A) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(2,3) \wedge R(3,2)$ but $\neg R(2,2)$.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 1 has no edge to a different element.

(D) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.16 [MCQ, 2M · Propositional Logic] Answer: D (D: $\{\wedge, \vee, \leftrightarrow\}$)

$\wedge, \vee$ and $\leftrightarrow$ all output T when every input is T ($T \wedge T = T, T \vee T = T, T \leftrightarrow T = T$). Hence every formula built from them is T under the all-true assignment, so $\neg p$ cannot be expressed. $\{\neg, \wedge\}$ is complete by De Morgan. NAND alone is complete ($\neg p = p \text{ NAND } p$). With $\rightarrow$ and $\perp$ we get $\neg p \equiv p \rightarrow \perp$ and $p \vee q \equiv \neg p \rightarrow q$, so $\{\rightarrow, \perp\}$ is complete.

Q.17 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $\neg(\neg(P \rightarrow \neg Q) \wedge \neg(P \leftrightarrow P))$; D: $\neg(P \rightarrow \neg Q) \vee (P \leftrightarrow P)$)

Compare truth tables with $F = (P \rightarrow \neg Q) \vee (P \leftrightarrow P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg(P \rightarrow \neg Q) \vee \neg(P \leftrightarrow P))$ — differs at $P=T, Q=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(B) $\neg(\neg(P \rightarrow \neg Q) \wedge \neg(P \leftrightarrow P))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $(P \rightarrow \neg Q) \wedge (P \leftrightarrow P)$ — differs at $P=T, Q=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(D) $\neg(P \rightarrow \neg Q) \vee (P \leftrightarrow P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, C (B: (Q) ; C: $(Q \vee \neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(Q \vee \neg R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) (Q) is derivable:

1. resolve $(P \vee Q)$ and $(\neg P)$ on $P \Rightarrow (Q)$

(C) $(Q \vee \neg S)$ is derivable:

1. resolve $(\neg P \vee Q \vee \neg S)$ and $(P \vee Q)$ on $P \Rightarrow (Q \vee \neg S)$

(D) The resolution closure contains 8 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

Q.19 [MSQ, 2M · Propositional Logic] Answer: C (C: F is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $C \wedge A \rightarrow H, C \rightarrow F$; add F, H.

Round 2: fire $F \rightarrow J$; add J.

Fixed point: $\{A, C, F, H, J\}$ — 5 atoms are entailed (facts: A, C; newly derived: F, H, J). Atoms never derived: B, D, E, K.

Q.20 [MCQ, 2M · Predicate Logic] Answer: A (A: $\neg P(x, y) \vee Q(x, F(x))$)

Eliminate $\rightarrow$: $\forall x (\neg \exists y P(x, y) \vee \exists z Q(x, z)) \equiv \forall x (\forall y \neg P(x, y) \vee \exists z Q(x, z))$. y is now universal, so it stays a variable. z lies only within the scope of $\forall x$, so it becomes $F(x)$. Dropping universals gives the clause $\neg P(x, y) \vee Q(x, F(x))$. Option B wrongly Skolemizes the universal y ; option C ignores the dependence of z on x ; option D is not a clause and changes the meaning.

Q.21 [MSQ, 2M · Probability for AI] Answer: A, D (A: $P(x) = 0.4$; D: $P(y \mid \neg z) \approx 0.455$)

$P(x) = 0.10 + 0.05 + 0.15 + 0.10 = 0.40$, so (A) is true. $P(y) = 0.10 + 0.05 + 0.05 + 0.20 = 0.40$ and $P(x, y) = 0.15 \neq 0.4 \times 0.4 = 0.16$, so (B) is false. Given z : $P(z) = 0.45$, $P(x, y \mid z) = 0.10/0.45 \approx 0.222$, while $P(x \mid z)P(y \mid z) = (0.25/0.45)(0.15/0.45) \approx 0.556 \times 0.333 \approx 0.185$, so (C) is false. $P(\neg z) = 0.55$ and $P(y, \neg z) = 0.05 + 0.20 = 0.25$, so $P(y \mid \neg z) = 0.25/0.55 \approx 0.4545$, making (D) true.

Q.22 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(B, C), f(A, B, D), f(C, D, E), f(A, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A), f(A, B), f(A, B, D), f(A, E, F)$	$\{A, B, D, E, F\}$	$f(B, D, E, F)$
D	$f(C, D, E), f(B, D, E, F)$	$\{B, C, D, E, F\}$	$f(B, C, E, F)$

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C), f(B, C, E, F)$	$\{B, C, E, F\}$	$f(B, E, F)$
B	$f(B, E, F)$	$\{B, E, F\}$	$f(E, F)$
E	$f(E, F)$	$\{E, F\}$	$f(F)$

The largest factor produced has **4** variable(s) (created when eliminating A); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over F and normalise.

Q.23 [NAT, 2M · Exact Inference] Answer: 0.664 to 0.668

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B | A) \cdot P(C | B) \cdot P(D | A, B)$.

Inference by enumeration: $P(C | e) = \alpha \sum_{\text{hidden}} P(C, e, \text{hidden})$.

Hidden: A, B	C	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.1 \times 0.5 \times 0.75 \times 0.5$	0.01875
T, F	T	$0.1 \times 0.5 \times 0.6 \times 0.1$	0.003
F, T	T	$0.9 \times 0.75 \times 0.75 \times 0.1$	0.05063
F, F	T	$0.9 \times 0.25 \times 0.6 \times 0.5$	0.0675
T, T	F	$0.1 \times 0.5 \times 0.25 \times 0.5$	0.00625
T, F	F	$0.1 \times 0.5 \times 0.4 \times 0.1$	0.002
F, T	F	$0.9 \times 0.75 \times 0.25 \times 0.1$	0.01688
F, F	F	$0.9 \times 0.25 \times 0.4 \times 0.5$	0.045

$P(C=T, e) = 0.13988$; $P(C=F, e) = 0.07013$; $P(e) = 0.21$.

$P(C=T | e) = 0.13988 / 0.21 = \mathbf{0.6661} \approx 0.666$.

Q.24 [MSQ, 2M · Conditional Independence] Answer: A, D ($A: C \perp A | D$; $D: E \perp F | A, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $C \perp A | D$: **TRUE** — $C \rightarrow G \leftarrow B \leftarrow A$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow G \leftarrow B \rightarrow D \rightarrow E \leftarrow A$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow G \leftarrow B \rightarrow D \rightarrow F \leftarrow A$ (blocked: collider G with neither it nor a descendant observed).

(B) $B \perp G$: **FALSE** — the path $B \rightarrow G$ is active given $\{\}$.

(C) $B \perp G | E, F$: **FALSE** — the path $B \rightarrow G$ is active given $\{E, F\}$.

(D) $E \perp F | A, D$: **TRUE** — $E \leftarrow A \rightarrow B \rightarrow D \rightarrow F$ (blocked: observed fork node A); $E \leftarrow A \rightarrow F$ (blocked: observed fork node A); $E \leftarrow D \leftarrow B \leftarrow A \rightarrow F$ (blocked: observed chain node D); $E \leftarrow D \rightarrow F$ (blocked: observed fork node D).

Q.25 [NAT, 2M · Sampling] Answer: 0.57 to 0.59

Rejection sampling keeps only samples consistent with the evidence $A = T$: samples 1, 2, 4, 5, 6, 7, 9, 10, 11, 12, 14, 15 (12 accepted, 3 rejected).

Among the accepted samples, $D = T$ in 7. Estimate = $7/12 = \mathbf{0.5833}$.

(For this network the exact value is 0.459; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Mock Test 007

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	2 / 9 / 14	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

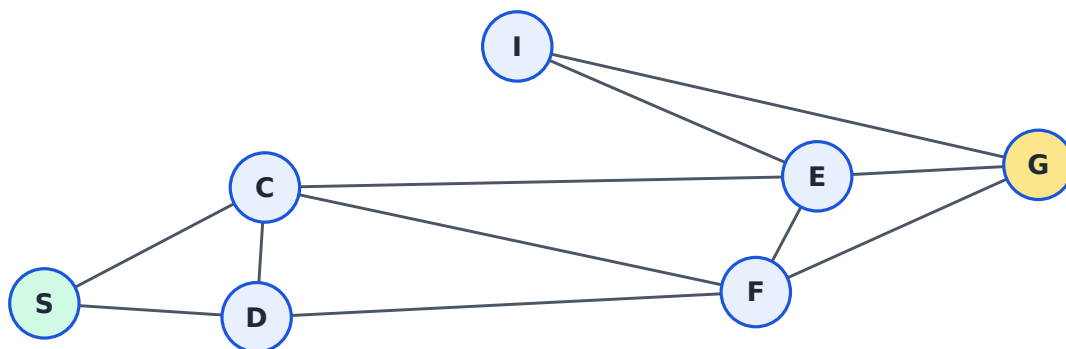
Q.1 [NAT · 1 mark · no negative marking] Problem Formulation

The 8-puzzle has $9!/2$ states reachable from any given start state. Compute this number.

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

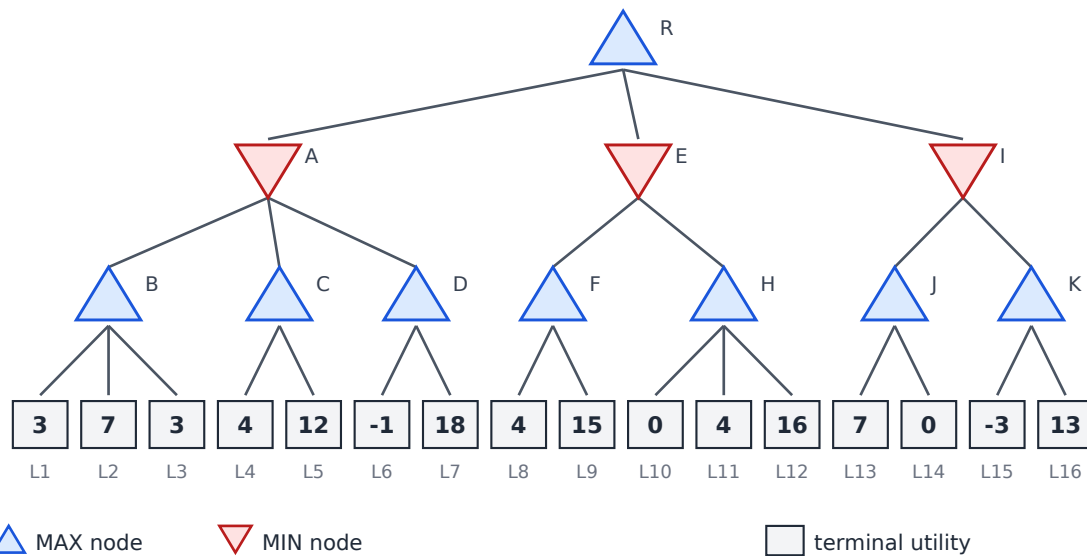
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, C, D, F, E, G
- ☐ (B) S, C, D, E, F, G
- ☐ (C) S, D, C, E, F, G
- ☐ (D) S, D, C, F, E, G

Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

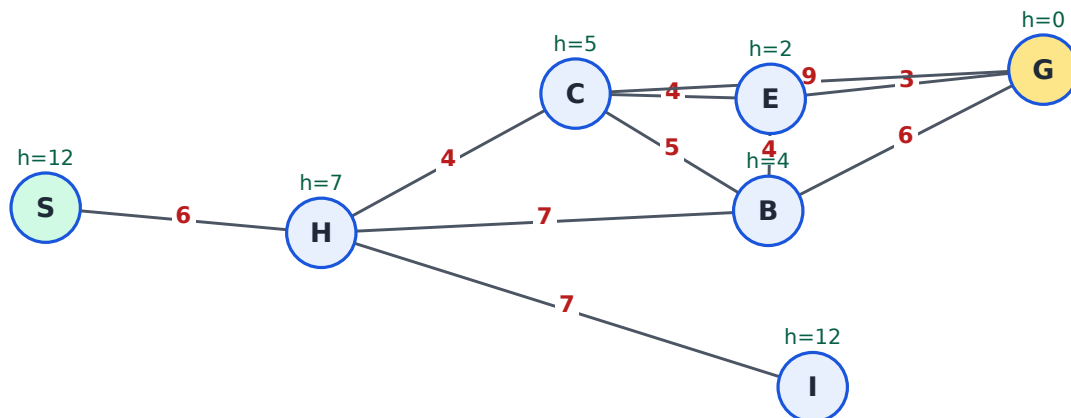
In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to A; root value 7
☐ (B) Move to E; root value 18
☐ (C) Move to E; root value 15
☐ (D) Move to I; root value 7

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following equivalences are valid in first-order logic?

- ☐ (A) $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
☐ (B) $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
☐ (C) $\forall x \exists y P(x, y) \equiv \exists y \forall x P(x, y)$
☐ (D) $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

The clauses $A \vee B \vee \neg C$ and $\neg A \vee B \vee D$ are resolved on the symbol A. How many literals are in the resolvent (after merging duplicate literals)?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

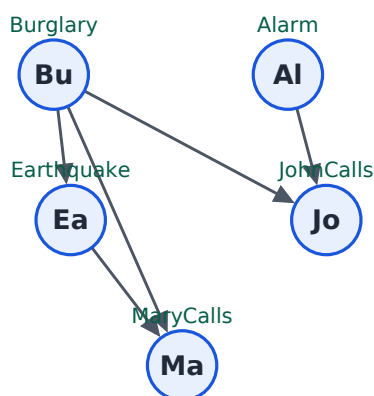
Q.8 [MSQ · 1 mark · no negative marking] Probability for AI

A and B are independent Boolean variables with $P(a) = 0.4$ and $P(b) = 0.5$. Which statements are true?

- ☐ (A) $P(a \vee b) = 0.7$
☐ (B) $P(a \mid b) = 0.4$
☐ (C) $P(\neg a \wedge \neg b) = 0.3$
☐ (D) $P(a \wedge \neg b) = 0.25$

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Bu, Ea, Al, Jo, Ma. (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls, Ma = MaryCalls.) What is the probability that a single run produces the sample **(Bu=F, Ea=T, Al=F, Jo=T, Ma=F)**? (round off to 4 decimal places)



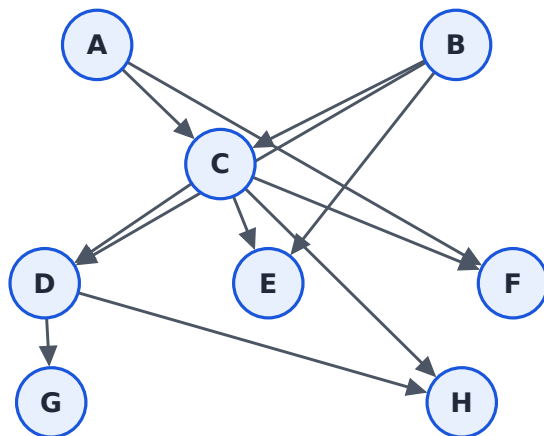
P(Bu=T)	Bu	P(Ea=T ...)	P(Al=T)
0.4	T	0.1	0.75
	F	0.6	

Bu	Al	P(Jo=T ...)	Bu	Ea	P(Ma=T ...)
T	T	0.75	T	T	0.5
T	F	0.7	T	F	0.95
F	T	0.2	F	T	0.25
F	F	0.95	F	F	0.5

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **D** in the Bayesian network below?



Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search

Which of the following statements about IDA* are TRUE?

- ☐ (A) Its memory requirement is linear in the depth of the solution
- ☐ (B) With an admissible heuristic, it returns an optimal solution
- ☐ (C) It keeps every previously visited state in memory to avoid repeated states
- ☐ (D) When step costs are real-valued and almost all distinct, each iteration may add only one new node, which leads to many iterations

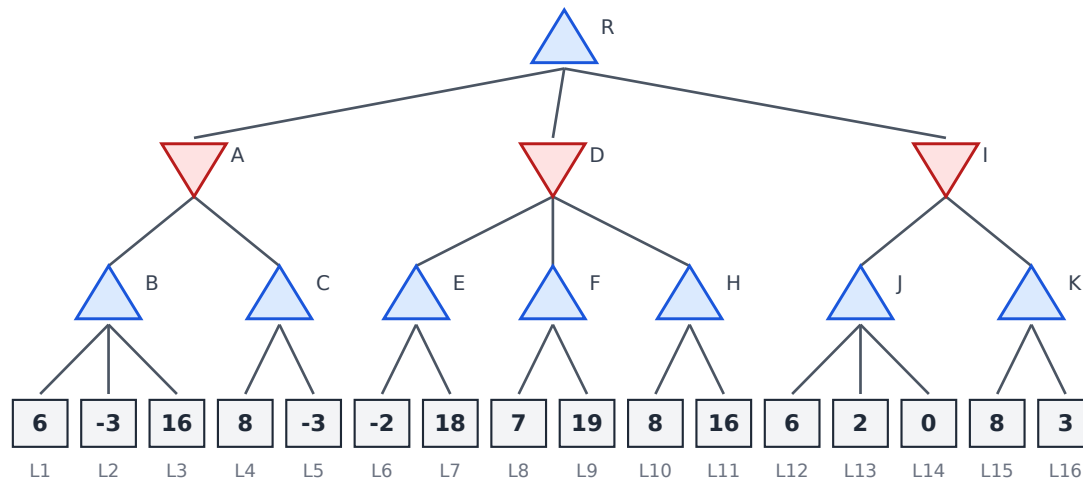
Q.12 [NAT · 2 marks · no negative marking] Informed Search

In the 8-puzzle, the goal configuration is 1 2 3 / 4 5 6 / 7 8 _ and the current state is 8 1 3 / 4 _ 2 / 7 6 5 (rows top to bottom, _ = blank). What is h_2 , the sum of the Manhattan distances of the tiles from their goal positions (excluding the blank)?

Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

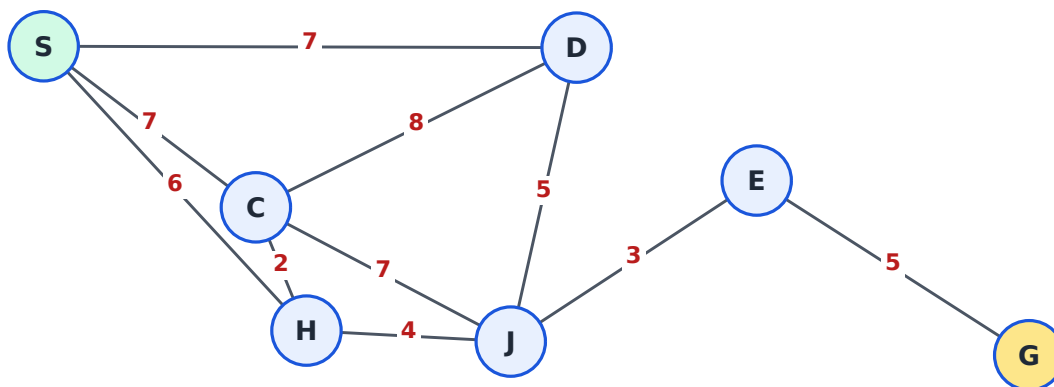
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



- ☐ (A) Leaf L2 is pruned.
☐ (B) Leaf L9 is pruned.
☐ (C) Leaf L14 is pruned.
☐ (D) Leaf L16 is pruned.

Q.14 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

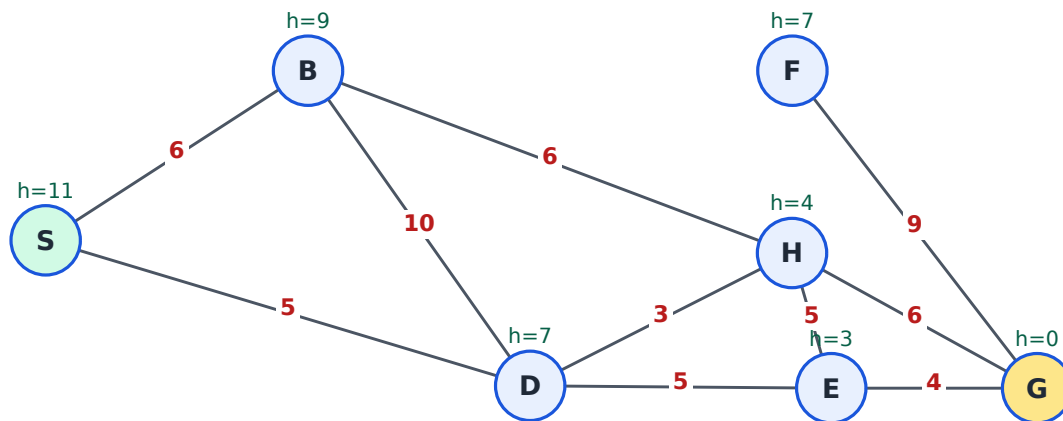
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) The edge S-B violates the consistency condition (in at least one direction).
- ☐ (C) $h(E)$ overestimates the true cost from E to G.
- ☐ (D) h is admissible.

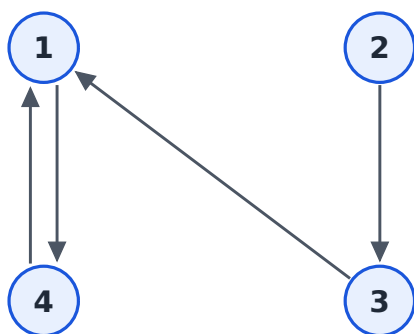
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $\neg(Q \wedge R) \leftrightarrow (\neg Q \wedge \neg R)$
- ☐ (B) $(Q \rightarrow R) \rightarrow (R \rightarrow Q)$
- ☐ (C) $(Q \vee R) \rightarrow (Q \wedge R)$
- ☐ (D) $Q \rightarrow (P \rightarrow Q)$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y (x \neq y \wedge R(x, y))$
- ☐ (B) $\exists x \forall y \neg R(y, x)$
- ☐ (C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
- ☐ (D) $\forall x \exists y R(x, y)$

Q.18 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$ and let f be a unary function symbol. How many interpretations of f (functions $D \rightarrow D$) satisfy $\forall x f(f(x)) = f(x)$?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (\neg R), C2: (Q \vee R), C3: (\neg P \vee \neg Q \vee S), C4: (P \vee \neg Q), C5: (\neg P \vee \neg Q)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(R \vee \neg S)$
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) $(\neg Q)$
☐ (D) $(\neg Q \vee R)$

Q.20 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

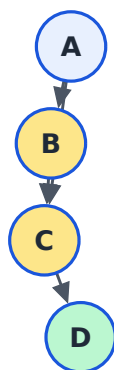
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$F = ((Q \wedge Q) \rightarrow (\neg Q \wedge S)) \vee ((\neg P \rightarrow Q) \leftrightarrow (Q \wedge R))$?

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(D = T \mid C = T, B = T)$ (round off to 3 decimal places).



A	P(A=T)
T	0.6

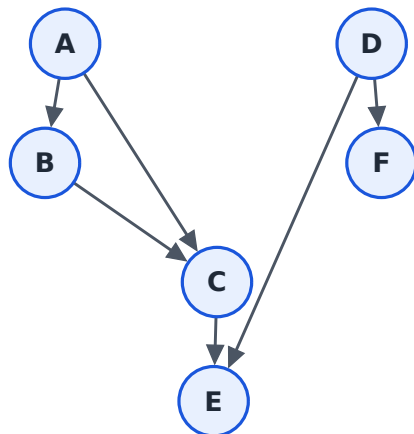
A	P(B=T ...)	A	B	P(C=T ...)
T	0.95	T	T	0.3
F	0.6	T	F	0.75
		F	T	0.05
		F	F	0.9

C	P(D=T ...)
T	0.75
F	0.9

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

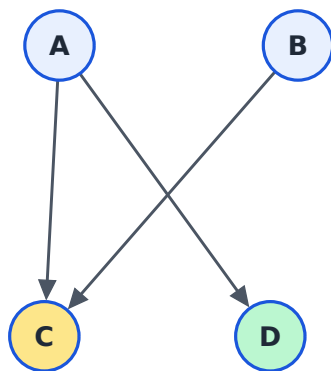
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $F \perp C$
☐ (B) $E \perp B \mid C$
☐ (C) $E \perp F \mid B$
☐ (D) $B \perp D \mid A, E$

Q.23 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $C = T$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $P(D = T \mid C = T)$ (round off to 3 decimal places).



P(A=T)
0.05

P(B=T)	A	B	P(C=T ...)
0.2	T	T	0.9
	T	F	0.75
	F	T	0.75
	F	F	0.95

A	P(D=T ...)
T	0.4
F	0.9

#	A	B	D
1	F	F	T
2	F	F	T
3	F	F	T
4	F	F	T
5	F	F	F

Answer (numerical): _____

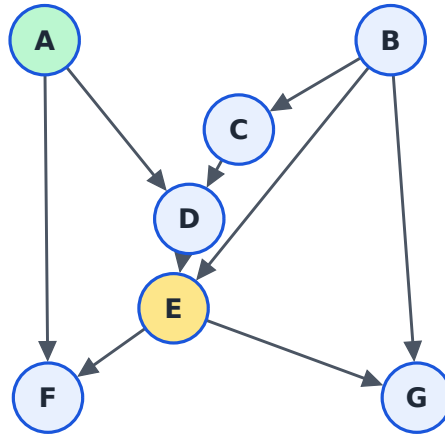
Q.24 [NAT · 2 marks · no negative marking] Bayesian Networks

Fever has three possible causes Cold, Flu and Malaria, modeled with a noisy-OR CPT with no leak. The inhibition probabilities are $q_{\text{cold}} = 0.6$, $q_{\text{flu}} = 0.2$, $q_{\text{malaria}} = 0.1$ (q_i = probability that cause i , when present alone, fails to produce fever). Compute $P(\text{fever} \mid \text{cold}, \text{flu}, \neg \text{malaria})$. (Round to 2 decimal places.)

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(A \mid E = T)$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, F, B, G, C**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 007

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Problem Formulation	181440	14	NAT	2	Uninformed Search	18
2	MCQ	1	Uninformed Search	B	15	MSQ	2	Informed Search	A, D
3	MCQ	1	Adversarial Search	C	16	MSQ	2	Propositional Logic	D
4	NAT	1	Informed Search	19	17	MSQ	2	Predicate Logic	A, B, D
5	MSQ	1	Predicate Logic	A, B, D	18	NAT	2	Predicate Logic	10
6	NAT	1	Propositional Logic	3	19	MSQ	2	Propositional Logic	B, C, D
7	NAT	1	Predicate Logic	19	20	NAT	2	Propositional Logic	12
8	MSQ	1	Probability for AI	A, B, C	21	NAT	2	Exact Inference	0.748 to 0.752
9	NAT	1	Sampling	0.0639 to 0.0643	22	MSQ	2	Conditional Independence	A, B
10	NAT	1	Conditional Independence	4	23	NAT	2	Sampling	0.798 to 0.802
11	MSQ	2	Informed Search	A, B, D	24	NAT	2	Bayesian Networks	0.875 to 0.885
12	NAT	2	Informed Search	10	25	NAT	2	Exact Inference	3
13	MSQ	2	Adversarial Search	D					

Detailed Solutions

Q.1 [NAT, 1M · Problem Formulation] Answer: 181440

$9! = 362880$. The parity of a permutation splits the states into two disjoint halves, and no sequence of moves crosses from one half to the other. So $362880/2 = 181440$ states are reachable.

Q.2 [MCQ, 1M · Uninformed Search] Answer: B (B: S, C, D, E, F, G)

Adjacency lists (alphabetical): C: {D, E, F, S}; D: {C, F, S}; E: {C, F, G, I}; F: {C, D, E, G}; G: {E, F, I}; I: {E, G}; S: {C, D}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[C, D]

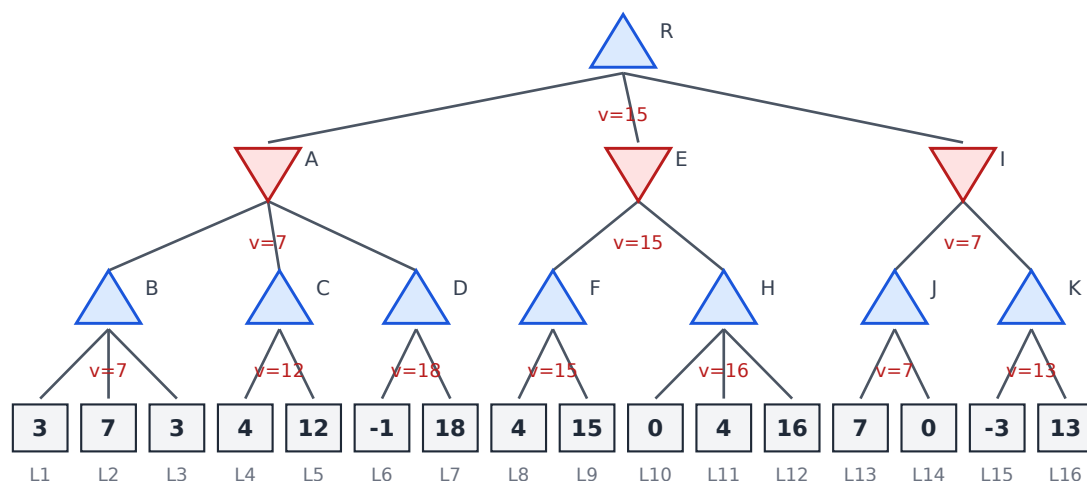
Step	Expanded	Frontier (FIFO) after expansion
2	C	[D, E, F]
3	D	[E, F]
4	E	[F, G, I]
5	F	[G, I]
6	G	[I] ← goal popped, stop

Expansion order: **S, C, D, E, F, G**. The path returned is $S \rightarrow C \rightarrow E \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to E; root value 15)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 7, C = 12, D = 18, A = 7, F = 15, H = 16, E = 15, J = 7, K = 13, I = 7$. Root value = **15**; MAX's optimal first move is towards E.

Q.4 [NAT, 1M · Informed Search] Answer: 19

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (12)	H(7)
2	H (7)	B(4), C(5), I(12)
3	B (4)	G(0), E(2), C(5), I(12)
4	G (0)	E(2), C(5), I(12) ← goal, stop

Path returned: $S \rightarrow H \rightarrow B \rightarrow G$, cost = $6 + 7 + 6 = 19$.

The optimal cost is $h^*(S) = 17$ — greedy search is not optimal here.

Q.5 [MSQ, 1M · Predicate Logic] Answer: A, B, D (A: $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$; B: $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$; D: $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$)

Quantifiers of the same kind commute (A, B). $\exists$ distributes over $\vee$ (D), just as $\forall$ distributes over $\wedge$. (C) Mixed quantifiers do not commute: with $P(x, y) \iff x = y$ on $\{1, 2\}$, $\forall x \exists y P$ is true but $\exists y \forall x P$ is false.

Q.6 [NAT, 1M · Propositional Logic] Answer: 3

Remove A and $\neg A$ and take the union of the remaining literals: $\{B, \neg C\} \cup \{B, D\} = \{B, \neg C, D\}$. The duplicate B is merged (factoring), so the resolvent $B \vee \neg C \vee D$ has 3 literals.

Q.7 [NAT, 1M · Predicate Logic] Answer: 19

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 3$ this gives **19** (verified by enumerating all interpretations).

Q.8 [MSQ, 1M · Probability for AI] Answer: A, B, C (A: $P(a \vee b) = 0.7$; B: $P(a | b) = 0.4$; C: $P(\neg a \wedge \neg b) = 0.3$)

Independence gives $P(a \wedge b) = 0.4 \times 0.5 = 0.2$, so $P(a \vee b) = 0.4 + 0.5 - 0.2 = 0.7$ (A true). Independence means $P(a | b) = P(a) = 0.4$ (B true). $P(\neg a \wedge \neg b) = 0.6 \times 0.5 = 0.3$ (C true). $P(a \wedge \neg b) = 0.4 \times 0.5 = 0.2$, not 0.25 (D false).

Q.9 [NAT, 1M · Sampling] Answer: 0.0639 to 0.0643

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(\text{Bu}=\text{F}) \times P(\text{Ea}=\text{T}|\text{Bu}=\text{F}) \times P(\text{Al}=\text{F}) \times P(\text{Jo}=\text{T}|\text{Bu}=\text{F}, \text{Al}=\text{F}) \times P(\text{Ma}=\text{F}|\text{Bu}=\text{F}, \text{Ea}=\text{T}) \\ = 0.6 \times 0.6 \times 0.25 \times 0.95 \times 0.75 = \mathbf{0.064125} \approx 0.0641.$$

Q.10 [NAT, 1M · Conditional Independence] Answer: 4

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of D: {B, C}; children: {G, H}; co-parents: { $\emptyset$ }.

MB(D) = {B, C, G, H} — **4** nodes. Given its Markov blanket, D is conditionally independent of every other node in the network.

Q.11 [MSQ, 2M · Informed Search] Answer: A, B, D

(A) is true: each iteration is a depth-first search bounded by the f-cutoff. (B) is true: thresholds rise to the smallest f that exceeded the old one, so the first goal found is optimal. (C) is false: IDA* does not keep an explored set, which is exactly why it uses little memory and why it can re-expand states. (D) is true: this is the well-known weakness, which motivates RBFS and SMA*.

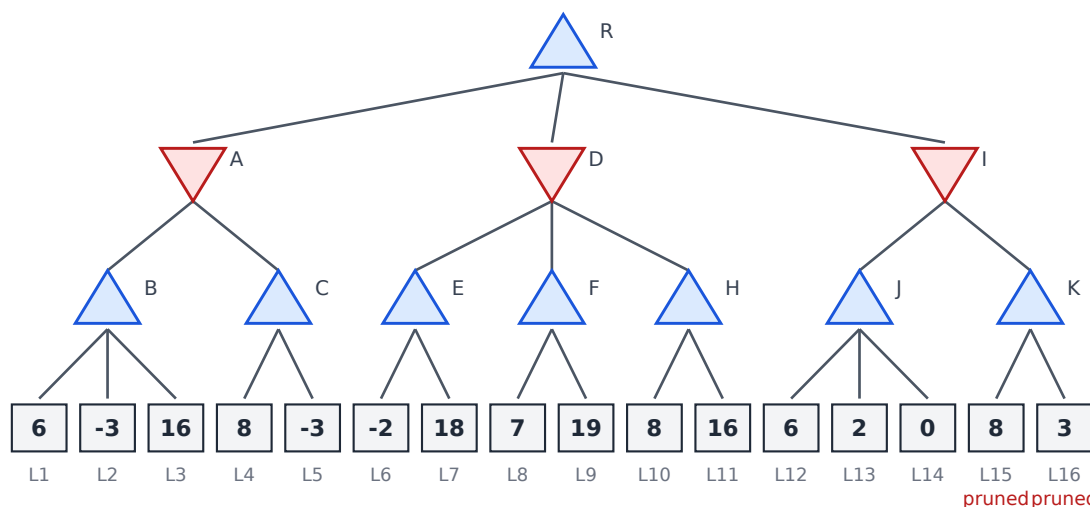
Q.12 [NAT, 2M · Informed Search] Answer: 10

Use (row, column) coordinates from (0,0). Tile 8 is at (0,0) and belongs at (2,1): distance 3. Tile 1 is at (0,1) and belongs at (0,0): 1. Tile 3: 0. Tile 4: 0. Tile 2 is at (1,2) and belongs at (0,1): 2. Tile 7: 0. Tile 6 is at (2,1) and belongs at (1,2): 2. Tile 5 is at (2,2) and belongs at (1,1): 2. Sum = 3+1+2+2+2 = **10**.

Q.13 [MSQ, 2M · Adversarial Search] Answer: D (D: Leaf L16 is pruned.)

Run alpha-beta depth-first, left to right, passing (α , β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node I: value $6 \leq \alpha = 16 \Rightarrow$ prune remaining children (leaves L15, L16).



Leaves evaluated (14): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10, L11, L12, L13, L14. Leaves pruned (2): L15, L16. Root value = 16 (same as plain minimax, 16 — pruning never changes the root value).

Q.14 [NAT, 2M · Uninformed Search] Answer: 18

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	H(6), C(7), D(7)
2	H (6)	C(7), D(7), J(10)
3	C (7)	D(7), J(10)
4	D (7)	J(10)
5	J (10)	E(13)
6	E (13)	G(18)
7	G (18)	← goal, stop

Path returned: **S** → **H** → **J** → **E** → **G**, cost = 6 + 4 + 3 + 5 = **18**. Nodes expanded (including S and G): **7** (S, H, C, D, J, E, G).

Q.15 [MSQ, 2M · Informed Search] **Answer: A, D** (A: h is consistent (monotone).; D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	D	E	F	G	H	S
$h(n)$	9	7	3	7	0	4	11
$h^*(n)$	12	9	4	9	0	6	14
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: D** (D: $Q \rightarrow (P \rightarrow Q)$)

(A) $\neg(Q \wedge R) \Leftrightarrow (\neg Q \wedge \neg R)$: **not valid** — counter-model $Q=T, R=F$.

(B) $(Q \rightarrow R) \rightarrow (R \rightarrow Q)$: **not valid** — counter-model $Q=F, R=T$.

(C) $(Q \vee R) \rightarrow (Q \wedge R)$: **not valid** — counter-model $Q=T, R=F$.

(D) $Q \rightarrow (P \rightarrow Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [MSQ, 2M · Predicate Logic] **Answer: A, B, D** (A: $\forall x \exists y (x \neq y \wedge R(x, y))$; B: $\exists x \forall y \neg R(y, x)$; D: $\forall x \exists y R(x, y)$)

$R = \{(1,4), (2,3), (3,1), (4,1)\}$.

(A) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(B) $\exists x \forall y \neg R(y, x)$: **TRUE** — 2 has no incoming edge.

(C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(4,1)$ and $R(1,4)$ with $4 \neq 1$.

(D) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x,y) \Rightarrow \forall x \exists y R(x,y)$ but not conversely; quantifier order matters.

Q.18 [NAT, 2M · Predicate Logic] **Answer: 10**

The condition says f is idempotent: f fixes every element of its image. Choose the image S (size k) — each element of S maps to itself — and map every element outside S anywhere into S : k^{3-k} ways. Sum over k : $k=1: 3 \times 1^2 = 3$; $k=2: 3 \times 2^1 = 6$; $k=3: 1 \times 3^0 = 1$. Total = 10.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: B, C, D** (B: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable); C: $(\neg Q)$; D: $(\neg Q \vee R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(R \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $\square$ is derivable:

1. resolve $(\neg R)$ and $(Q \vee R)$ on $R \Rightarrow (Q)$
2. resolve $(P \vee \neg Q)$ and $(\neg P \vee \neg Q)$ on $P \Rightarrow (\neg Q)$
3. resolve (Q) and $(\neg Q)$ on $Q \Rightarrow \square$ (empty clause)

(C) $(\neg Q)$ is derivable:

1. resolve $(P \vee \neg Q)$ and $(\neg P \vee \neg Q)$ on $P \Rightarrow (\neg Q)$

(D) $(\neg Q \vee R)$ is derivable:

1. resolve $(Q \vee R)$ and $(\neg P \vee \neg Q)$ on $Q \Rightarrow (\neg P \vee R)$
2. resolve $(P \vee \neg Q)$ and $(\neg P \vee R)$ on $P \Rightarrow (\neg Q \vee R)$

Q.20 [NAT, 2M · Propositional Logic] **Answer: 12**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(Q \wedge Q) \rightarrow (\neg Q \wedge S)$	$(\neg P \rightarrow Q) \Leftrightarrow (Q \wedge R)$	$((Q \wedge Q) \rightarrow (\neg Q \wedge S)) \vee ((\neg P \rightarrow Q) \Leftrightarrow (Q \wedge R))$
T	T	T	T	F	T	T
T	T	T	F	F	T	T
T	T	F	T	F	F	F
T	T	F	F	F	F	F
T	F	T	T	T	F	T
T	F	T	F	T	F	T
T	F	F	T	T	F	T

P	Q	R	S	$(Q \wedge Q) \rightarrow (\neg Q \wedge S)$	$(\neg P \rightarrow Q) \leftrightarrow (Q \wedge R)$	$((Q \wedge Q) \rightarrow (\neg Q \wedge S)) \vee ((\neg P \rightarrow Q) \leftrightarrow (Q \wedge R))$
T	F	F	F	T	F	T
F	T	T	T	F	T	T
F	T	T	F	F	T	T
F	T	F	T	F	F	F
F	T	F	F	F	F	F
F	F	T	T	T	T	T
F	F	T	F	T	T	T
F	F	F	T	T	T	T
F	F	F	F	T	T	T

Rows where F = T: **12**.

Q.21 [NAT, 2M · Exact Inference] **Answer: 0.748 to 0.752**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B | A) \cdot P(C | A, B) \cdot P(D | C)$.

Inference by enumeration: $P(D | e) = \alpha \sum_{\text{hidden}} P(D, e, \text{hidden})$.

Hidden: A	D	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T	T	$0.6 \times 0.95 \times 0.3 \times 0.75$	0.12825
F	T	$0.4 \times 0.6 \times 0.05 \times 0.75$	0.009
T	F	$0.6 \times 0.95 \times 0.3 \times 0.25$	0.04275
F	F	$0.4 \times 0.6 \times 0.05 \times 0.25$	0.003

$P(D=T, e) = 0.13725$; $P(D=F, e) = 0.04575$; $P(e) = 0.183$.

$P(D=T | e) = 0.13725 / 0.183 = \mathbf{0.75} \approx 0.750$.

Q.22 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $F \perp C$; B: $E \perp B | C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $F \perp C$: **TRUE** — $F \leftarrow D \rightarrow E \leftarrow C$ (blocked: collider E with neither it nor a descendant observed).

(B) $E \perp B | C$: **TRUE** — $E \leftarrow C \leftarrow A \rightarrow B$ (blocked: observed chain node C); $E \leftarrow C \leftarrow B$ (blocked: observed chain node C).

(C) $E \perp F | B$: **FALSE** — the path $E \leftarrow D \rightarrow F$ is active given $\{B\}$.

(D) $B \perp D | A, E$: **FALSE** — the path $B \rightarrow C \rightarrow E \leftarrow D$ is active given $\{A, E\}$.

Q.23 [NAT, 2M · Sampling] **Answer: 0.798 to 0.802**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	D
1	$P(C=T A=F, B=F)=0.95$	0.95	T
2	$P(C=T A=F, B=F)=0.95$	0.95	T
3	$P(C=T A=F, B=F)=0.95$	0.95	T
4	$P(C=T A=F, B=F)=0.95$	0.95	T
5	$P(C=T A=F, B=F)=0.95$	0.95	F

$\sum w$ (all) = 4.75; $\sum w$ (D = T) = 3.8.

Estimate $P(D=T | C = T) = 3.8 / 4.75 = \mathbf{0.8}$.

Q.24 [NAT, 2M · Bayesian Networks] **Answer: 0.875 to 0.885**

In noisy-OR, $P(\neg \text{fever} | \text{causes}) = \text{product of } q_i \text{ over the causes that are present}$. Present causes: cold and flu, so $P(\neg \text{fever}) = 0.6 \times 0.2 = 0.12$. $P(\text{fever} | \text{cold, flu, } \neg \text{malaria}) = 1 - 0.12 = 0.88$. Malaria is absent and contributes no factor.

Q.25 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(B, C)$, $f(A, C, D)$, $f(B, D)$, $f(A, F)$, $f(B, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after $\sum$)
D	$f(A, C, D), f(B, D)$	$\{A, B, C, D\}$	$f(A, B, C)$
F	$f(A, F)$	$\{A, F\}$	$f(A)$
B	$f(B), f(B, C), f(B, G), f(A, B, C)$	$\{A, B, C, G\}$	$f(A, C, G)$

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(A, C, G)$	$\{A, C, G\}$	$f(A, C)$
C	$f(A, C)$	$\{A, C\}$	$f(A)$

The largest factor produced has **3** variable(s) (created when eliminating D); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over A and normalise.

Mock Test 008

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	3 / 7 / 15	15 / 12 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

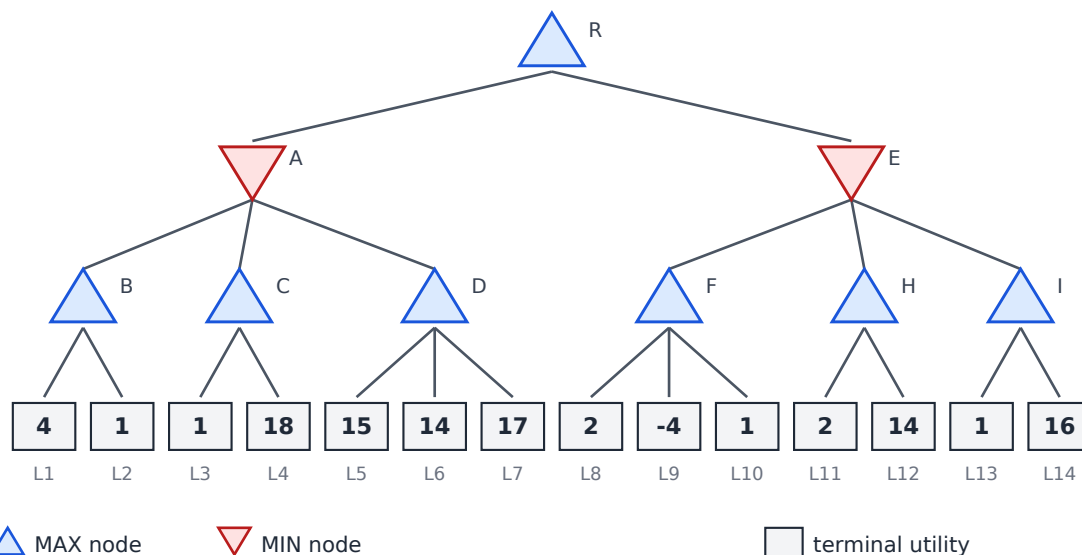
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 4$ (so $b^d = 81$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

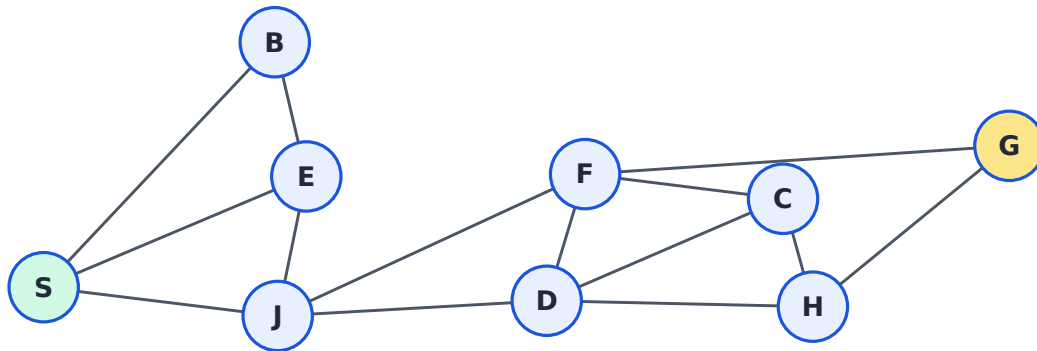
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, E, J, D, F, C, H, G
- ☐ (B) S, B, E, J, D, C, F, G
- ☐ (C) S, B, E, J, D, F, H, C, G
- ☐ (D) S, J, E, B, F, D, G

Q.4 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$ true?

Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following correctly express 'Not every graph is connected'?

- ☐ (A) $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$
- ☐ (B) $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$
- ☐ (C) $\forall x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$
- ☐ (D) $\exists x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = \neg Q \leftrightarrow (P \rightarrow (\neg R \wedge \neg P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

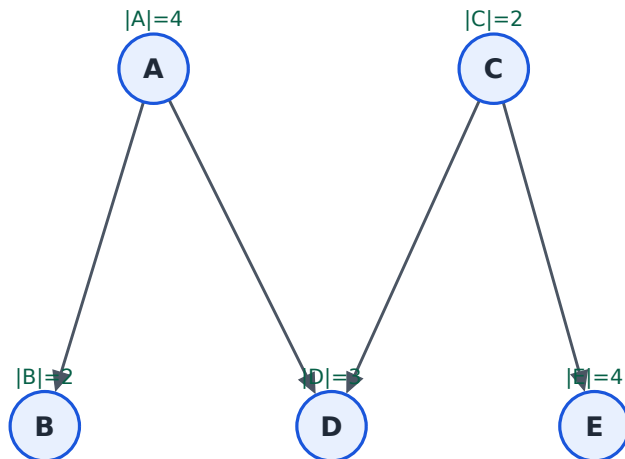
Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

The formula $(A \wedge B) \vee (C \wedge D) \vee (E \wedge F)$ is converted to CNF by repeatedly distributing $\vee$ over $\wedge$ (no simplification). How many clauses does the result have?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

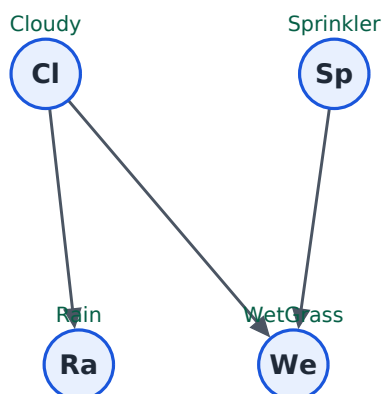
The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample **Cl=T, Sp=T, Ra=T, We=F**? (round off to 4 decimal places)



P(Cl=T)
0.3

P(Sp=T)	Cl	P(Ra=T ...)
0.75	T	0.95
	F	0.25

Cl	Sp	P(We=T ...)
T	T	0.2
T	F	0.2
F	T	0.2
F	F	0.2

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks

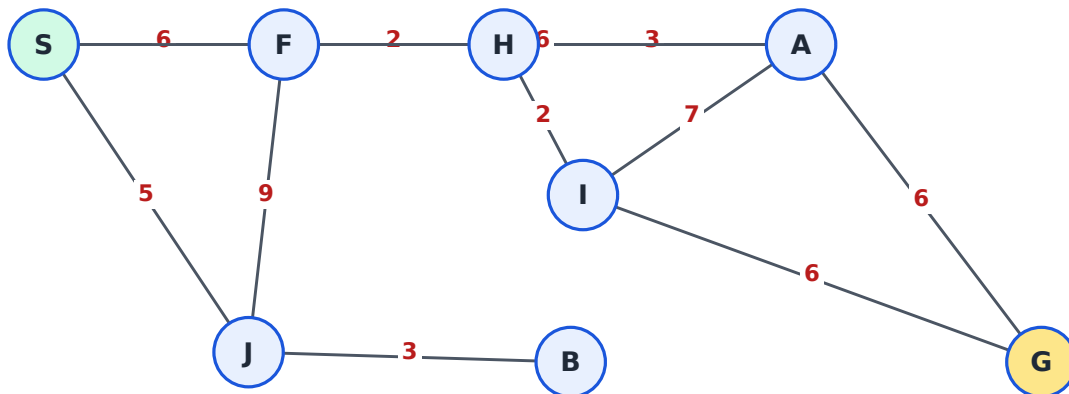
A Bayesian network over Boolean variables has edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $C \rightarrow E$. How many independent parameters do its CPTs contain?

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

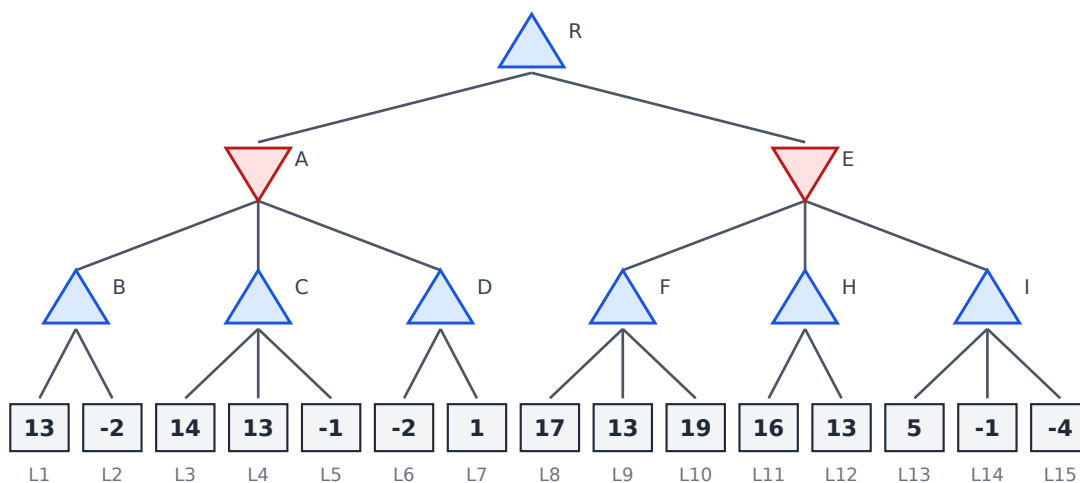
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



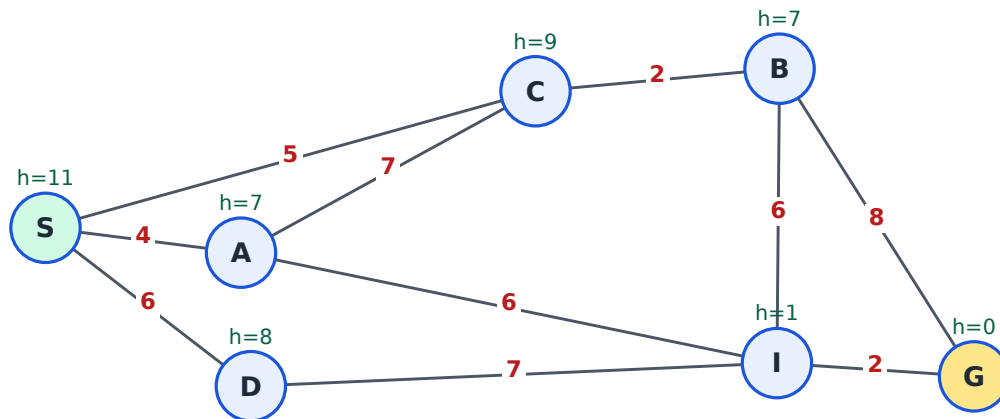
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L1 is pruned.
- ☐ (B) Leaf L4 is pruned.
- ☐ (C) Leaf L9 is pruned.
- ☐ (D) Leaf L12 is pruned.

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
☐ (B) The edge S-C violates the consistency condition (in at least one direction).
☐ (C) $h(C)$ overestimates the true cost from C to G.
☐ (D) h is consistent (monotone).

Q.14 [NAT · 2 marks · no negative marking] Problem Formulation

The 4-queens problem uses the improved incremental formulation. A state places k queens ($0 \leq k \leq 4$), one in each of the leftmost k columns, with no two queens attacking each other. Including the empty board ($k = 0$), how many states are in this state space?

Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Depth-first search pushes the right child before the left child, so the left child is explored first, and it goal-tests a node when the node is removed from the frontier. The goal is node 12. How many nodes are removed from the frontier, including node 12?

Answer (numerical): _____

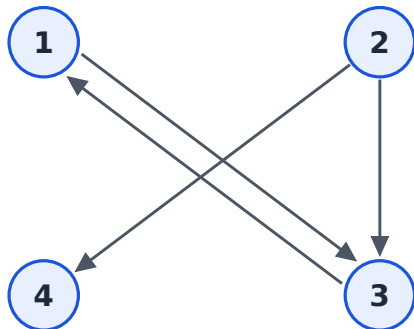
Q.16 [NAT · 2 marks · no negative marking] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Iterative deepening search uses depth limits 0, 1, 2, 3, ... in turn, expands left children before right children, and goal-tests a node when it is visited (popped). The goal is node 11. Over all iterations, counting repeated visits, how many node visits happen up to and including the visit at which node 11 is found?

Answer (numerical): _____

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (B) $\exists x \forall y \neg R(y, x)$
☐ (C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (D) $\forall y \exists x R(x, y)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (R \vee S), C2: (\neg Q \vee \neg R \vee \neg S), C3: (\neg P \vee Q), C4: (Q \vee R \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) $(P \vee Q)$
☐ (C) $(\neg P \vee Q \vee R)$
☐ (D) $(\neg Q \vee R)$

Q.19 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Given the definite clauses $\text{King}(x) \wedge \text{Greedy}(x) \rightarrow \text{Evil}(x)$, $\text{King}(\text{John})$, and $\text{Greedy}(y)$, Generalized Modus Ponens derives $\text{Evil}(\text{John})$ using which substitution?

- ☐ (A) $\{x/\text{John}, y/\text{John}\}$
☐ (B) $\{x/y\}$
☐ (C) $\{x/\text{John}\}$
☐ (D) $\{y/x, x/\text{Evil}\}$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (\neg Q \wedge P) \vee R$?

- ☐ (A) $R \vee (\neg Q \wedge P)$
☐ (B) $\neg(\neg Q \wedge P) \vee R$
☐ (C) $\neg(\neg(\neg Q \wedge P) \wedge \neg R)$
☐ (D) $\neg(\neg(\neg Q \wedge P) \vee \neg R)$

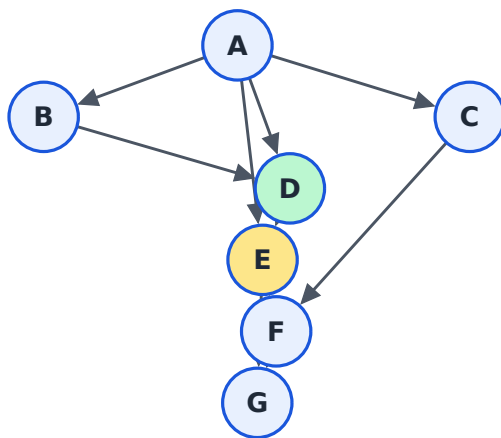
Q.21 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Likelihood weighting with evidence $S = \text{true}$, $W = \text{true}$ produces exactly four samples of (C, R): (c, r), ($\neg c$, r), ($\neg c$, $\neg r$), (c, $\neg r$). Using the correct weights, compute the likelihood-weighting estimate of $P(r | s, w)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.22 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

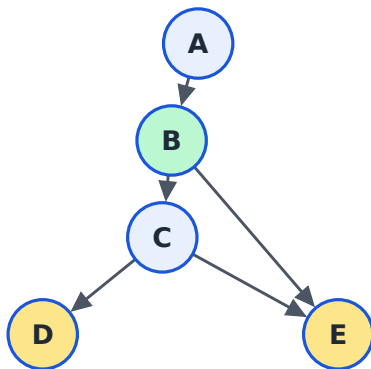
Variable elimination is used to compute $P(D | E = T)$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **G**, **F**, **B**, **A**, **C**. What is the scope of the new factor produced when **B** is summed out?



- ☐ (A) $f(A, B, D)$
- ☐ (B) $f(A, D)$
- ☐ (C) $f(A, B, C, D, F, G)$
- ☐ (D) $f(A)$

Q.23 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{D = T, E = T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(B = T \mid D = T, E = T)}$ (round off to 3 decimal places).



P(A=T)	A	P(B=T ...)	B	P(C=T ...)
0.9	T	0.4	T	0.7
	F	0.75	F	0.1

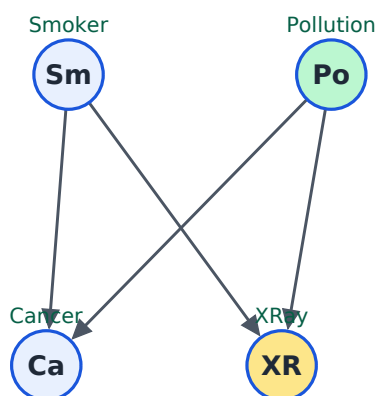
C	P(D=T ...)	B	C	P(E=T ...)
T	0.1	T	T	0.1
F	0.95	T	F	0.1
		F	T	0.75
		F	F	0.75

#	A	B	C
1	T	T	F
2	T	F	T
3	T	F	F
4	T	F	F
5	T	F	F
6	T	F	F

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $\mathbf{P(Pollution = T \mid XRay = T)}$ (round off to 3 decimal places).



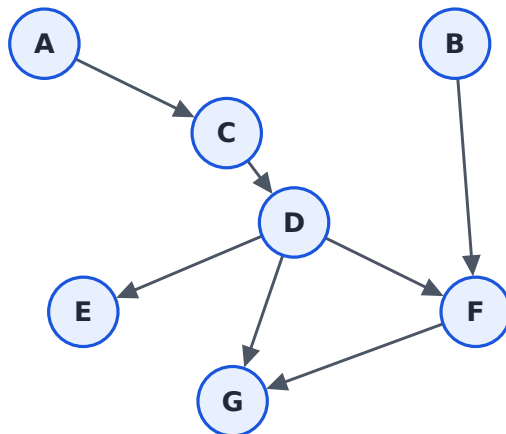
P(Sm=T)	P(Po=T)	Sm	Po	P(Ca=T ...)
0.8	0.9	T	T	0.6
		T	F	0.8
		F	T	0.05
		F	F	0.3

Sm	Po	P(XR=T ...)
T	T	0.4
T	F	0.7
F	T	0.25
F	F	0.05

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp F$
☐ (B) $A \perp G \mid F$
☐ (C) $G \perp A \mid E$
☐ (D) $C \perp G \mid B, D$

Answer Key — Mock Test 008

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	17	14	NAT	2	Problem Formulation	17
2	NAT	1	Adversarial Search	4	15	NAT	2	Uninformed Search	11
3	MCQ	1	Uninformed Search	A	16	NAT	2	Uninformed Search	19
4	NAT	1	Predicate Logic	216	17	MSQ	2	Predicate Logic	B
5	MSQ	1	Predicate Logic	A, B	18	MSQ	2	Propositional Logic	C
6	NAT	1	Propositional Logic	4	19	MCQ	2	Predicate Logic	A
7	NAT	1	Propositional Logic	8	20	MSQ	2	Propositional Logic	A, C
8	NAT	1	Bayesian Networks	30	21	NAT	2	Sampling	0.523 to 0.524
9	NAT	1	Sampling	0.1708 to 0.1712	22	MCQ	2	Exact Inference	B
10	NAT	1	Bayesian Networks	10	23	NAT	2	Sampling	0.029 to 0.033
11	NAT	2	Uninformed Search	15	24	NAT	2	Exact Inference	0.852 to 0.856
12	MSQ	2	Adversarial Search	B	25	MSQ	2	Conditional Independence	D
13	MSQ	2	Informed Search	A, D					

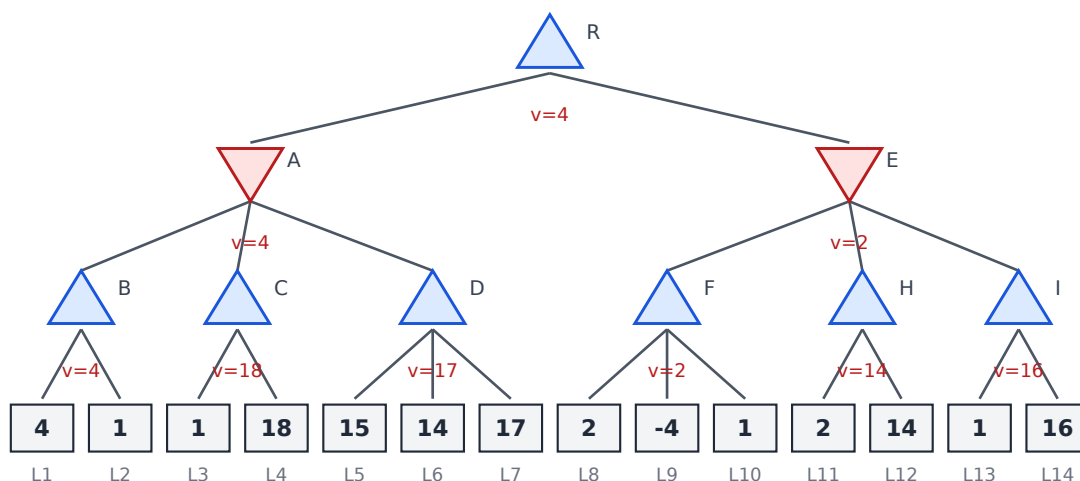
Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 17**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 3^{\lceil 2 \rceil} + 3^{\lfloor 2 \rfloor} - 1 = 9 + 9 - 1 = 17$, versus 81 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.2 [NAT, 1M · Adversarial Search] **Answer: 4**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 4$, $C = 18$, $D = 17$, $A = 4$, $F = 2$, $H = 14$, $I = 16$, $E = 2$. Root value = **4**; MAX's optimal first move is towards A.

Q.3 [MCQ, 1M · Uninformed Search] **Answer: A** (A: S, B, E, J, D, F, C, H, G)

Adjacency lists (alphabetical): B: {E, S}; C: {D, F, H}; D: {C, F, H, J}; E: {B, J, S}; F: {C, D, G, J}; G: {F, H}; H: {C, D, G}; J: {D, E, F, S}; S: {B, E, J}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B, E, J]
2	B	[E, J]
3	E	[J]
4	J	[D, F]
5	D	[F, C, H]
6	F	[C, H, G]
7	C	[H, G]
8	H	[G]
9	G	[] $\leftarrow$ goal popped, stop

Expansion order: **S, B, E, J, D, F, C, H, G**. The path returned is $S \rightarrow J \rightarrow F \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [NAT, 1M · Predicate Logic] **Answer: 216**

View R as an 3×3 0/1 matrix. The sentence says R is antisymmetric: diagonal free (2^n); each unordered pair has 3 options (neither, one way, other way) $\Rightarrow 2^n \cdot 3^{(n(n-1)/2)}$. With $n = 3$: **216**.

Q.5 [MSQ, 1M · Predicate Logic] **Answer: A, B** (A: $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$; B: $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$)

Option A is the direct translation, and pushing the negation inward gives $\exists x \neg (\text{Graph}(x) \rightarrow \text{Connected}(x)) \equiv \exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$ — option B. Option C says no graph is connected, which is stronger. Option D is true whenever any non-graph exists (vacuous implication), so it does not express the statement.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 4**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg Q$	$P \rightarrow (\neg R \wedge \neg P)$	$\neg Q \leftrightarrow (P \rightarrow (\neg R \wedge \neg P))$
T	T	T	F	F	T
T	T	F	F	F	T
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	F	T	F
F	T	F	F	T	F
F	F	T	T	T	T
F	F	F	T	T	T

F has 4 models, so it is false in $8 - 4 = 4$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.7 [NAT, 1M · Propositional Logic] Answer: 8

Distribution produces one clause per way of choosing one literal from each conjunction: $\{A, B\} \times \{C, D\} \times \{E, F\}$, i.e. $2 \times 2 \times 2 = 8$ clauses such as $A \vee C \vee E$, $A \vee C \vee F$, ..., $B \vee D \vee F$. No clause is a tautology or a duplicate because all six symbols are distinct. This illustrates the exponential blow-up of naive CNF conversion.

Q.8 [NAT, 1M · Bayesian Networks] Answer: 30

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (4-1) + B: (2-1) \times 4 + C: (2-1) + D: (3-1) \times 4 \times 2 + E: (4-1) \times 2 = 30$.

Full joint table: $\prod |X| - 1 = 192 - 1 = 191$; saving = 161.

Q.9 [NAT, 1M · Sampling] Answer: 0.1708 to 0.1712

Prior sampling generates each complete event with exactly its joint probability $S_{\text{PS}}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Cl}=T) \times P(\text{Sp}=T) \times P(\text{Ra}=T|\text{Cl}=T) \times P(\text{We}=F|\text{Cl}=T, \text{Sp}=T)$

$= 0.3 \times 0.75 \times 0.95 \times 0.8 = 0.171 \approx 0.1710$.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 10

A Boolean node with k Boolean parents needs 2^k numbers. A: 1, B: 1, C (2 parents): 4, D (1 parent): 2, E (1 parent): 2. Total = 10, compared with $2^5 - 1 = 31$ for the full joint.

Q.11 [NAT, 2M · Uninformed Search] Answer: 15

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

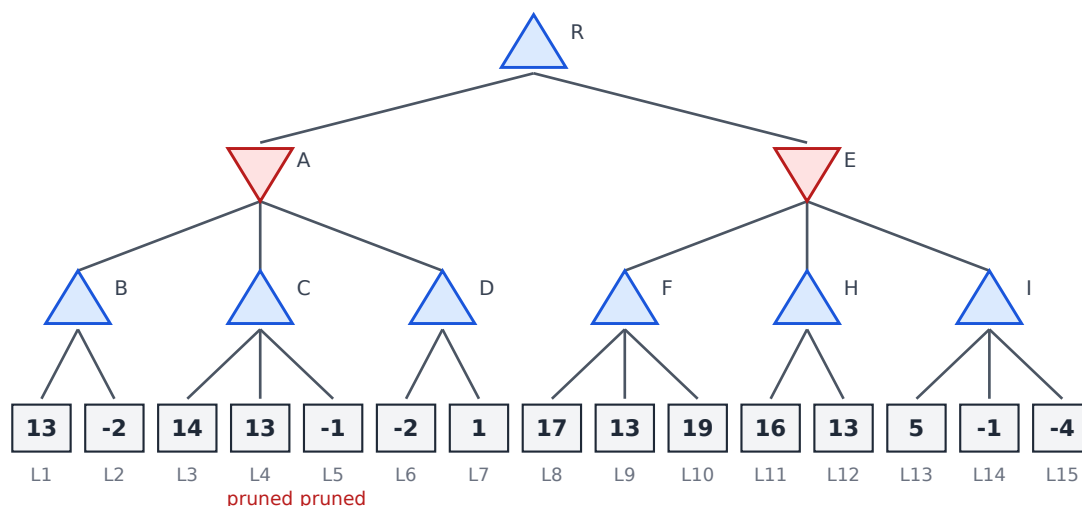
Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(5), F(6), H(7)
2	J (5)	F(6), H(7), B(8)
3	F (6)	H(7), B(8), A(12)
4	H (7)	B(8), I(9), A(10)
5	B (8)	I(9), A(10)
6	I (9)	A(10), G(15)
7	A (10)	G(15)
8	G (15)	← goal, stop

Path returned: **S** → **H** → **I** → **G**, cost = $7 + 2 + 6 = 15$. Nodes expanded (including S and G): **8** (S, J, F, H, B, I, A, G).

Q.12 [MSQ, 2M · Adversarial Search] Answer: B (B: Leaf L4 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MAX node C: value $14 \geq \beta = 13 \Rightarrow$ prune remaining children (leaves L4, L5).



Leaves evaluated (13): L1, L2, L3, L6, L7, L8, L9, L10, L11, L12, L13, L14, L15. Leaves pruned (2): L4, L5.
Root value = 5 (same as plain minimax, 5 — pruning never changes the root value).

Q.13 [MSQ, 2M · Informed Search] Answer: A, D (A: h is admissible.; D: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	C	D	G	I	S
$h(n)$	7	7	9	8	0	1	11
$h^*(n)$	8	8	10	9	0	2	12
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [NAT, 2M · Problem Formulation] Answer: 17

$k = 0$: 1 state. $k = 1$: 4 states. $k = 2$: the pairs of rows (r_1, r_2) with $|r_1 - r_2| \geq 2$ are (1,3), (1,4), (2,4), (3,1), (4,1), (4,2), so 6 states. $k = 3$: extending these without attack gives (1,4,2), (2,4,1), (3,1,4), (4,1,3), so 4 states. $k = 4$: the two solutions (2,4,1,3) and (3,1,4,2). Total = $1+4+6+4+2 = 17$. For 8-queens, the same formulation gives AIMA's 2057 states.

Q.15 [NAT, 2M · Uninformed Search] Answer: 11

DFS visits nodes in preorder: 1, 2, 4, 8, 9, 5, 10, 11, 3, 6, 12, ... Node 12 is the **11th** node removed. For comparison, BFS would remove nodes in label order and find node 12 at the 12th removal.

Q.16 [NAT, 2M · Uninformed Search] Answer: 19

Node 11 lies at depth 3 (path $1 \rightarrow 2 \rightarrow 5 \rightarrow 11$). Limit 0 visits {1}: 1 visit. Limit 1 visits {1,2,3}: 3 visits. Limit 2 visits the whole depth-2 tree {1,2,4,5,3,6,7}: 7 visits. Limit 3 visits 1,2,4,8,9,5,10,11 in preorder and stops at 11: 8 visits. Total = $1 + 3 + 7 + 8 = 19$.

Q.17 [MSQ, 2M · Predicate Logic] Answer: B (B: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,3), (2,3), (2,4), (3,1)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,3)$ holds but $R(3,2)$ does not.

(B) $\exists x \forall y \neg R(y, x)$: **TRUE** — 2 has no incoming edge.

(C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,2)$ nor $R(2,1)$.

(D) $\forall y \exists x R(x, y)$: **FALSE** — 2 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: C (C: $(\neg P \vee Q \vee R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 8 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(P \vee Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $(\neg P \vee Q \vee R)$ is derivable:

1. resolve $(\neg Q \vee \neg R \vee \neg S)$ and $(\neg P \vee Q)$ on $Q \Rightarrow (\neg P \vee \neg R \vee \neg S)$
2. resolve $(Q \vee R \vee \neg S)$ and $(\neg P \vee \neg R \vee \neg S)$ on $R \Rightarrow (\neg P \vee Q \vee \neg S)$
3. resolve $(R \vee S)$ and $(\neg P \vee Q \vee \neg S)$ on $S \Rightarrow (\neg P \vee Q \vee R)$

(D) $(\neg Q \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.19 [MCQ, 2M · Predicate Logic] Answer: A (A: $\{x/\text{John}, y/\text{John}\}$)

GMP needs one θ with $\text{King}(x)\theta = \text{King}(\text{John})\theta$ and $\text{Greedy}(x)\theta = \text{Greedy}(y)\theta$. x/John matches the first premise; y/John then makes $\text{Greedy}(y)$ match $\text{Greedy}(\text{John})$. With only $\{x/\text{John}\}$, $\text{Greedy}(\text{John})$ and $\text{Greedy}(y)$ are not syntactically identical, so the GMP side condition fails; $\{x/y\}$ does not match $\text{King}(\text{John})$. The conclusion is $\text{Evil}(x)\theta = \text{Evil}(\text{John})$.

Q.20 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $R \vee (\neg Q \wedge P)$; C: $\neg(\neg(\neg Q \wedge P) \wedge \neg R)$)

Compare truth tables with $F = (\neg Q \wedge P) \vee R$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $R \vee (\neg Q \wedge P)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(B) $\neg(\neg Q \wedge P) \vee R$ — differs at $P=T, Q=T, R=F$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(C) $\neg(\neg(\neg Q \wedge P) \wedge \neg R)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(D) $\neg(\neg(\neg Q \wedge P) \vee \neg R)$ — differs at $P=T, Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

Q.21 [NAT, 2M · Sampling] Answer: 0.523 to 0.524

Weights $w = P(s | C) \times P(w | s, R)$: (c, r): $0.1 \times 0.99 = 0.099$; ($\neg c$, r): $0.5 \times 0.99 = 0.495$; ($\neg c$, $\neg r$): $0.5 \times 0.90 = 0.45$; (c, $\neg r$): $0.1 \times 0.90 = 0.09$. Total weight = $0.099 + 0.495 + 0.45 + 0.09 = 1.134$; weight with r = $0.099 + 0.495 = 0.594$. Estimate = $0.594/1.134 \approx 0.5238 \approx 0.524$. (The exact $P(r | s, w) \approx 0.320$; with only four samples the estimate is far off, but LW is consistent as $N \rightarrow \infty$.)

Q.22 [MCQ, 2M · Exact Inference] Answer: B (B: $f(A, D)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, C)$, $f(A, B, D)$, $f(A, D)$, $f(C, F)$, $f(F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(F, G)$	$\{F, G\}$	$f(F)$
F	$f(C, F)$, $f(F)$	$\{C, F\}$	$f(C)$
B	$f(A, B)$, $f(A, B, D)$	$\{A, B, D\}$	$f(A, D)$
A	$f(A)$, $f(A, C)$, $f(A, D)$, $f(A, D)$	$\{A, C, D\}$	$f(C, D)$
C	$f(C)$, $f(C, D)$	$\{C, D\}$	$f(D)$

The largest factor produced has **2** variable(s) (created when eliminating B); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over D and normalise.

Q.23 [NAT, 2M · Sampling] Answer: 0.029 to 0.033

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	B
1	$P(D=T C=F)=0.95 \times P(E=T B=T, C=F)=0.1$	0.095	T
2	$P(D=T C=T)=0.1 \times P(E=T B=F, C=T)=0.75$	0.075	F
3	$P(D=T C=F)=0.95 \times P(E=T B=F, C=F)=0.75$	0.7125	F
4	$P(D=T C=F)=0.95 \times P(E=T B=F, C=F)=0.75$	0.7125	F
5	$P(D=T C=F)=0.95 \times P(E=T B=F, C=F)=0.75$	0.7125	F
6	$P(D=T C=F)=0.95 \times P(E=T B=F, C=F)=0.75$	0.7125	F

Σw (all) = 3.02; Σw (B = T) = 0.095.

Estimate $P(B=T | D = T, E = T) = 0.095 / 3.02 = \mathbf{0.0315}$.

Q.24 [NAT, 2M · Exact Inference] Answer: 0.852 to 0.856

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution}) \cdot P(\text{Cancer} | \text{Smoker}, \text{Pollution}) \cdot P(\text{XRay} | \text{Smoker}, \text{Pollution})$.

Inference by enumeration: $P(\text{Pollution} | e) = \alpha \sum_{\text{hidden}} P(\text{Pollution}, e, \text{hidden})$.

Hidden: Sm, Ca	Po	product of CPT entries (Sm · Po · Ca · XR)	value
T, T	T	$0.8 \times 0.9 \times 0.6 \times 0.4$	0.1728
T, F	T	$0.8 \times 0.9 \times 0.4 \times 0.4$	0.1152
F, T	T	$0.2 \times 0.9 \times 0.05 \times 0.25$	0.00225
F, F	T	$0.2 \times 0.9 \times 0.95 \times 0.25$	0.04275
T, T	F	$0.8 \times 0.1 \times 0.8 \times 0.7$	0.0448
T, F	F	$0.8 \times 0.1 \times 0.2 \times 0.7$	0.0112
F, T	F	$0.2 \times 0.1 \times 0.3 \times 0.05$	0.0003
F, F	F	$0.2 \times 0.1 \times 0.7 \times 0.05$	0.0007

$P(\text{Pollution}=T, e) = 0.333$; $P(\text{Pollution}=F, e) = 0.057$; $P(e) = 0.39$.

$P(\text{Pollution}=T | e) = 0.333 / 0.39 = \mathbf{0.8538} \approx 0.854$.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: D** (D: $C \perp G \mid B, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp F$: **FALSE** — the path $C \rightarrow D \rightarrow F$ is active given $\{\}$.

(B) $A \perp G \mid F$: **FALSE** — the path $A \rightarrow C \rightarrow D \rightarrow G$ is active given $\{F\}$.

(C) $G \perp A \mid E$: **FALSE** — the path $G \leftarrow D \leftarrow C \leftarrow A$ is active given $\{E\}$.

(D) $C \perp G \mid B, D$: **TRUE** — $C \rightarrow D \rightarrow F \rightarrow G$ (blocked: observed chain node D); $C \rightarrow D \rightarrow G$ (blocked: observed chain node D).

Mock Test 009

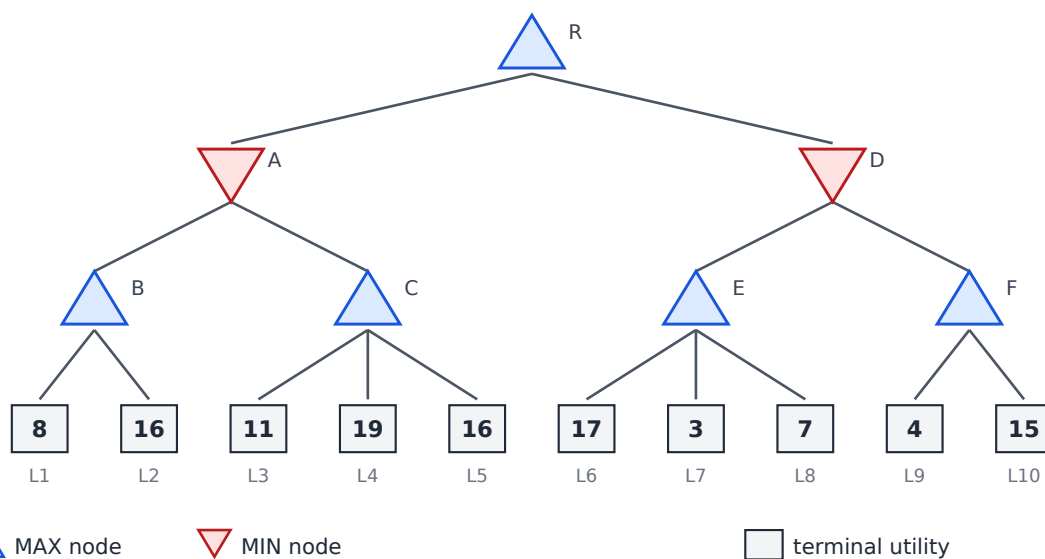
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	3 / 8 / 14	13 / 12 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

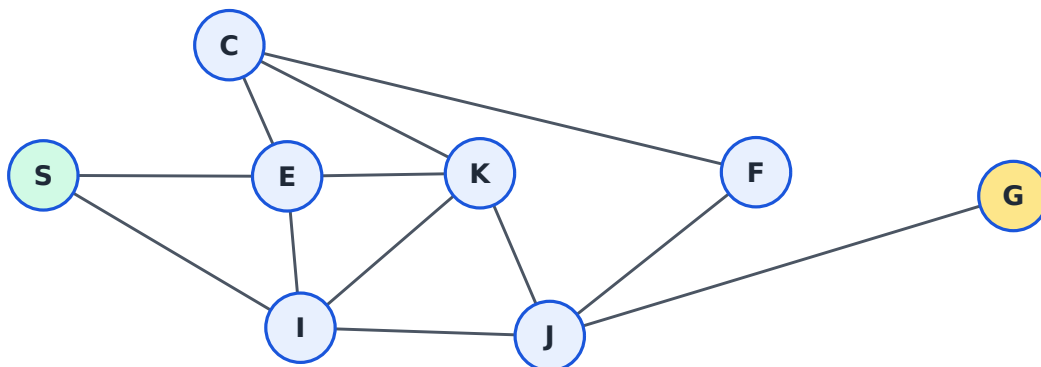
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, I, E, K, J, C, G
- ☐ (B) S, E, I, K, C, J, F, G
- ☐ (C) S, E, I, C, K, J, F, G
- ☐ (D) S, E, C, F, J, G

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Effective branching factor

A* finds a solution at depth $d = 4$ after generating $N = 12$ nodes (root excluded). The effective branching factor b^* satisfies $N = b^* + (b^*)^2 + \dots + (b^*)^d$. Find b^* (round off to 2 decimal places).

Answer (numerical): _____

Q.4 [MSQ · 1 mark · no negative marking] Predicate Logic

Without standardizing variables apart, $\text{Unify}(\text{Knows}(\text{John}, x), q)$ succeeds for which of the following atoms q ?

- ☐ (A) $\text{Knows}(\text{John}, \text{Jane})$
- ☐ (B) $\text{Knows}(y, \text{Bill})$
- ☐ (C) $\text{Knows}(y, \text{Mother}(y))$
- ☐ (D) $\text{Knows}(x, \text{Elizabeth})$

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = \neg(R \leftrightarrow R) \leftrightarrow \neg(\neg Q \vee P) ?$$

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

Interpret $\text{Div}(x, y)$ as 'x divides y' over the domain $D = \{1, 2, \dots, 12\}$. For how many $x \in D$ is the formula $\neg(x = 1) \wedge \forall y (\text{Div}(y, x) \rightarrow (y = 1 \vee y = x))$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

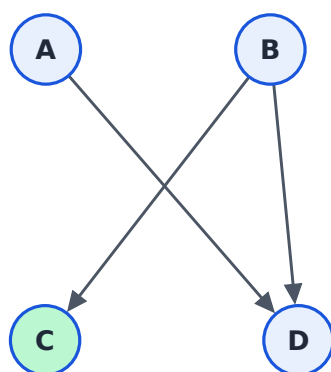
Q.8 [MSQ · 1 mark · no negative marking] Conditional Independence

Which of the following hold for **every** probability distribution over variables A, B, C, D ?

- ☐ (A) $A \perp B \mid C$ implies $B \perp A \mid C$
- ☐ (B) $A \perp \{B, C\} \mid D$ implies $A \perp B \mid D$
- ☐ (C) $A \perp B$ implies $A \perp B \mid C$
- ☐ (D) $A \perp B \mid C$ implies $A \perp B$

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



$P(A=T)$
0.8

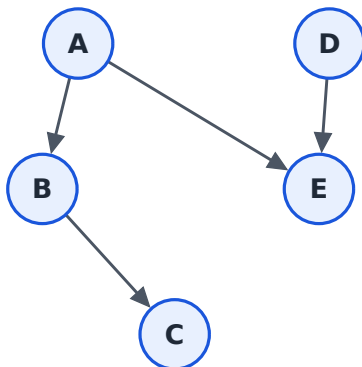
$P(B=T)$	B	$P(C=T \dots)$
0.3	T	0.9
	F	0.7

A	B	$P(D=T \dots)$
T	T	0.3
T	F	0.8
F	T	0.2
F	F	0.1

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D, E. What is the probability that a single run produces the sample **(A=F, B=F, C=F, D=T, E=F)**? (round off to 4 decimal places)



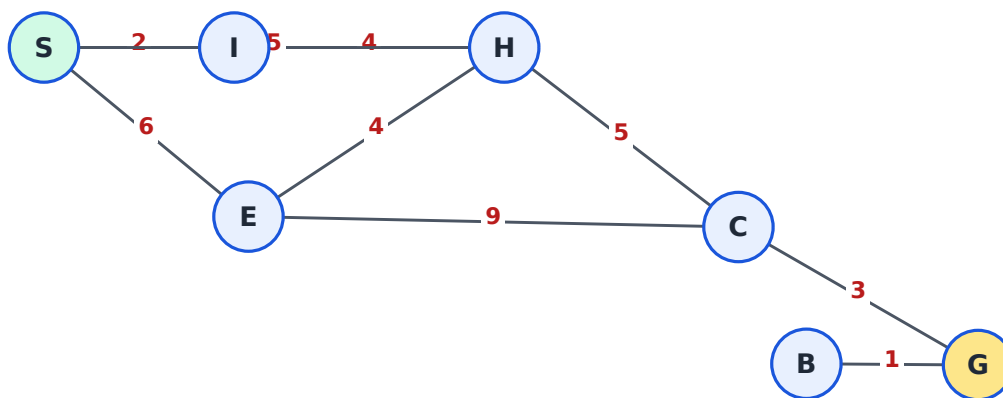
P(A=T)	A	P(B=T ...)	B	P(C=T ...)
0.05	T	0.05	T	0.8
	F	0.2	F	0.4

P(D=T)	A	D	P(E=T ...)
0.7	T	T	0.5
	T	F	0.4
	F	T	0.25
	F	F	0.1

Answer (numerical): _____

Section B – 2-mark questions**Q.11** [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

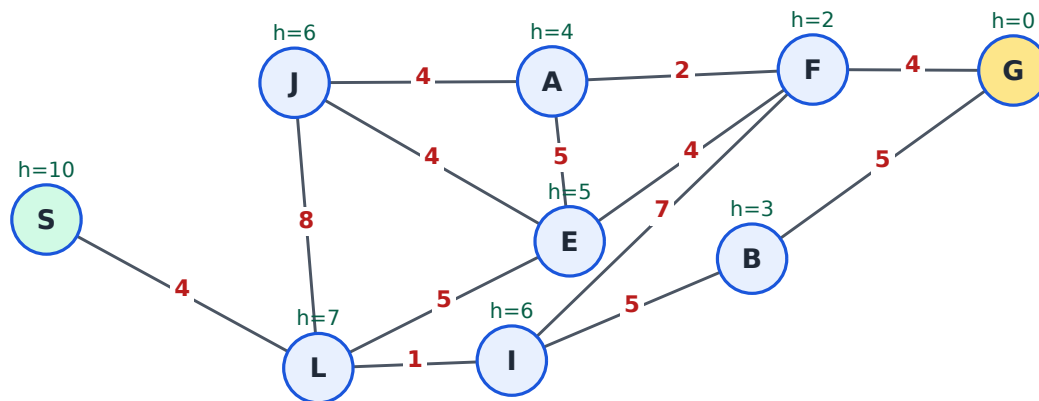
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) S → E → H → C → G
☐ (B) S → I → H → C → G
☐ (C) S → H → C → G
☐ (D) S → E → C → G

Q.12 [NAT · 2 marks · no negative marking] Informed Search - A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.13 [MCQ · 2 marks · negative: -2/3] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (2), $S \rightarrow B$ (4), $A \rightarrow C$ (2), $A \rightarrow D$ (4), $B \rightarrow C$ (1), $C \rightarrow D$ (1), $C \rightarrow G$ (5), $D \rightarrow G$ (2). The heuristic is $h(S)=5$, $h(A)=4$, $h(B)=4$, $h(C)=3$, $h(D)=2$, $h(G)=0$. What is the cost of the optimal path from S to G, and is h consistent?

- ☐ (A) Cost 7; h is consistent
- ☐ (B) Cost 7; h is not consistent
- ☐ (C) Cost 8; h is consistent
- ☐ (D) Cost 9; h is not consistent

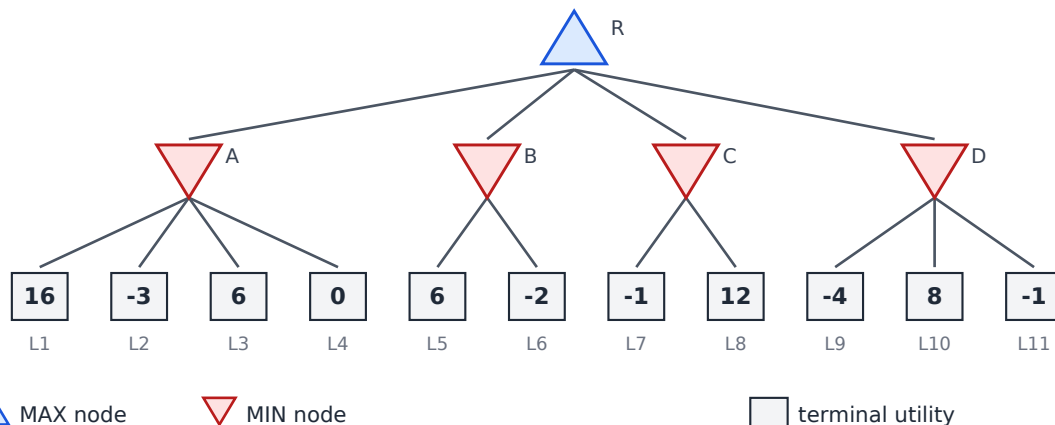
Q.14 [MSQ · 2 marks · no negative marking] Adversarial Search

Which of the following statements about expectimax / expectiminimax are TRUE?

- ☐ (A) The value of a chance node is the probability-weighted average of its children's values
- ☐ (B) With n distinct chance outcomes per chance layer, the time complexity becomes $O(b^m n^m)$
- ☐ (C) Pruning at chance nodes is impossible even when leaf values are known to lie in a bounded range
- ☐ (D) Expectimax is appropriate when the opponent's or environment's moves are random rather than optimally adversarial

Q.15 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(Q \vee R) \vee Q$
- (2) $R \vee R$
- (3) $\neg(R \rightarrow Q)$

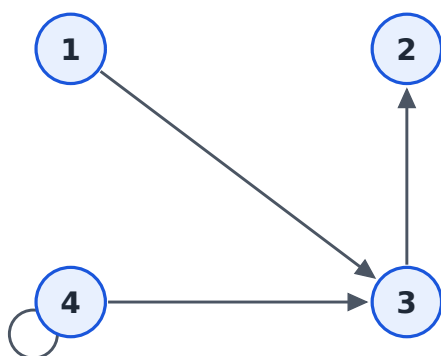
Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg P$
- ☐ (B) $\neg Q \vee R$
- ☐ (C) $P \wedge P$
- ☐ (D) $\neg P \vee R$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true).

Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists y \forall x R(x, y)$
- ☐ (B) $\forall x \exists y R(x, y)$
- ☐ (C) $\exists x \forall y \neg R(x, y)$
- ☐ (D) $\exists x \forall y \neg R(y, x)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = P \wedge (\neg Q \wedge R)$?

- ☐ (A) $\neg(\neg P \wedge \neg(\neg Q \wedge R))$
☐ (B) $\neg P \wedge (\neg Q \wedge R)$
☐ (C) $(\neg Q \wedge R) \wedge P$
☐ (D) $P \vee (\neg Q \wedge R)$

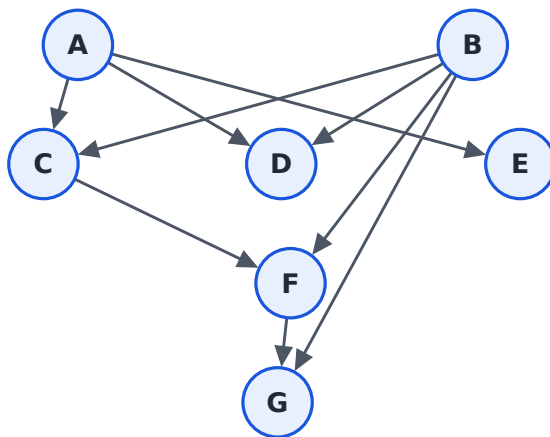
Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic

Let α and β be propositional sentences. Which of the following statements are ALWAYS true?

- ☐ (A) If α is satisfiable then $\neg\alpha$ is unsatisfiable
☐ (B) If α is valid then $\neg\alpha$ is unsatisfiable
☐ (C) $\alpha \vee \beta$ is satisfiable if and only if α is satisfiable or β is satisfiable
☐ (D) $\alpha \wedge \beta$ is satisfiable if and only if α is satisfiable and β is satisfiable

Q.20 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp E$
☐ (B) $D \perp A \mid E, F$
☐ (C) $E \perp B$
☐ (D) $D \perp G \mid B$

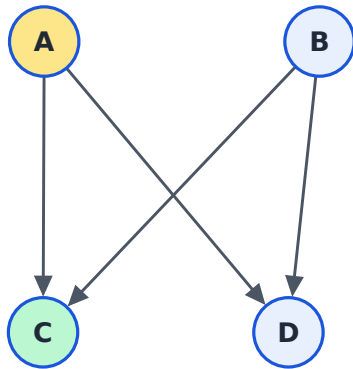
Q.21 [NAT · 2 marks · no negative marking] Variable Elimination

A Bayesian network over Boolean variables has edges $H \rightarrow X_1, H \rightarrow X_2, H \rightarrow X_3, H \rightarrow X_4$. To compute $P(X_1)$ by variable elimination, H is eliminated **first** (then X_2, X_3, X_4). How many entries does the new factor produced by summing out H contain?

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{A} = \mathbf{F}$. The values of the non-evidence variables in the generated samples are listed in the table. What is the weight of sample #5? (round off to 4 decimal places)



$P(A=T)$
0.6

$P(B=T)$	A	B	$P(C=T ...)$
0.2	T	T	0.5
	T	F	0.1
	F	T	0.5
	F	F	0.2

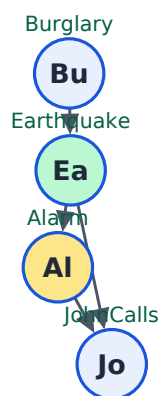
A	B	$P(D=T ...)$
T	T	0.7
T	F	0.3
F	T	0.75
F	F	0.3

#	B	C	D
1	F	T	F
2	F	T	F
3	F	F	F
4	F	F	F
5	F	T	F
6	F	F	F

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T | \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute $\mathbf{P(\text{Earthquake} = T | \text{Alarm} = F)}$ (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.95

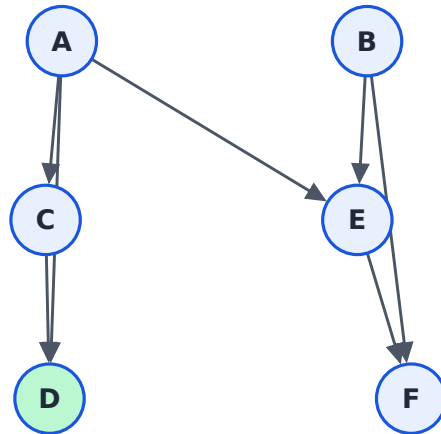
Bu	$P(\text{Ea}=T ..)$	Ea	$P(\text{Al}=T ...)$
T	0.6	T	0.8
F	0.3	F	0.4

Ea	Al	$P(\text{Jo}=T ...)$
T	T	0.4
T	F	0.25
F	T	0.3
F	F	0.4

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(D)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, F, C, B, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Variable Elimination

A Bayesian network is the chain $A \rightarrow B \rightarrow C \rightarrow D$ over Boolean variables with $P(a) = 0.2$; $P(b | a) = 0.9$, $P(b | \neg a) = 0.3$; $P(c | b) = 0.6$, $P(c | \neg b) = 0.1$; $P(d | c) = 0.7$, $P(d | \neg c) = 0.2$. Compute $P(d)$ by eliminating A, B, C in that order. (Round to 3 decimal places.)

Answer (numerical): _____

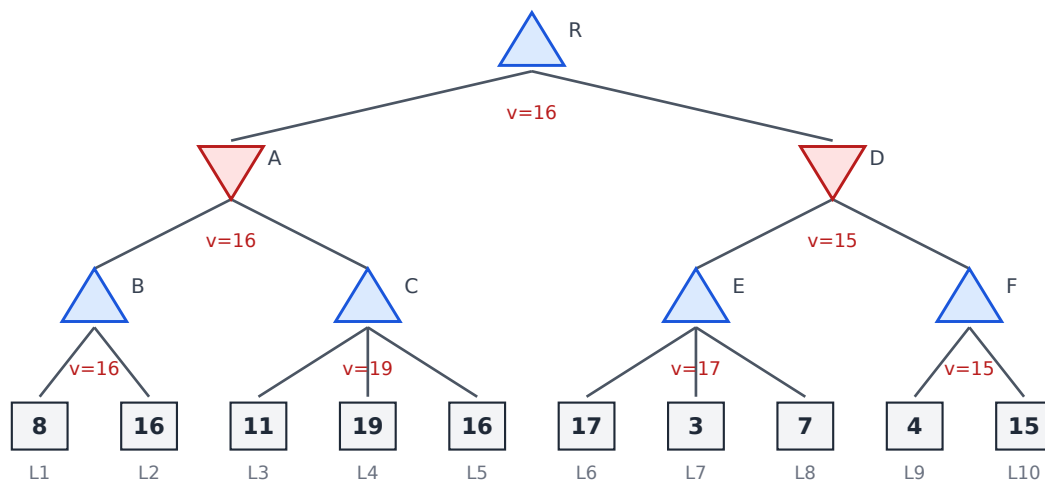
Answer Key — Mock Test 009

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	16	14	MSQ	2	Adversarial Search	A, B, D
2	MCQ	1	Uninformed Search	C	15	NAT	2	Adversarial Search	9
3	NAT	1	Informed Search	1.47 to 1.51	16	MSQ	2	Propositional Logic	B, D
4	MSQ	1	Predicate Logic	A, B, C	17	MSQ	2	Predicate Logic	C, D
5	NAT	1	Propositional Logic	6	18	MSQ	2	Propositional Logic	C
6	NAT	1	Predicate Logic	5	19	MSQ	2	Propositional Logic	B, C
7	NAT	1	Predicate Logic	19	20	MSQ	2	Conditional Independence	A, C
8	MSQ	1	Conditional Independence	A, B	21	NAT	2	Variable Elimination	16
9	NAT	1	Exact Inference	0.758 to 0.762	22	NAT	2	Sampling	0.3998 to 0.4002
10	NAT	1	Sampling	0.2392 to 0.2396	23	NAT	2	Exact Inference	0.318 to 0.322
11	MCQ	2	Uninformed Search	C	24	NAT	2	Exact Inference	4
12	NAT	2	Informed Search	7	25	NAT	2	Variable Elimination	0.354 to 0.356
13	MCQ	2	Informed Search	A					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] **Answer: 16**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 16, C = 19, A = 16, E = 17, F = 15, D = 15. Root value = **16**; MAX's optimal first move is towards A.

Q.2 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, E, I, C, K, J, F, G)

Adjacency lists (alphabetical): C: {E, F, K}; E: {C, I, K, S}; F: {C, J}; G: {}; I: {E, J, K, S}; J: {F, G, I, K}; K: {C, E, I, J}; S: {E, I}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[E, I]
2	E	[I, C, K]
3	I	[C, K, J]
4	C	[K, J, F]
5	K	[J, F]
6	J	[F, G]
7	F	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, E, I, C, K, J, F, G**. The path returned is $S \rightarrow I \rightarrow J \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [NAT, 1M · Informed Search] **Answer: 1.47 to 1.51**

Solve $b + b^2 + \dots + b^4 = 12$ numerically (bisection): $b^* \approx \mathbf{1.49}$.

Check: $1.49^1 + 1.49^2 + 1.49^3 + 1.49^4 \approx 11.95 \approx 12$.

Q.4 [MSQ, 1M · Predicate Logic] **Answer: A, B, C** (A: Knows(John, Jane); B: Knows(y, Bill); C: Knows(y, Mother(y)))

(A) {x/Jane}. (B) {x/Bill, y/John}. (C) {y/John, x/Mother(John)}. (D) fails: x would have to be both John and Elizabeth. Renaming the second x (standardizing apart) would make it succeed, which is why AIMA insists on standardizing apart.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg(R \leftrightarrow R)$	$\neg(\neg Q \vee P)$	$\neg(R \leftrightarrow R) \leftrightarrow \neg(\neg Q \vee P)$
T	T	T	F	F	T
T	T	F	F	F	T
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	F	T	F
F	T	F	F	T	F
F	F	T	F	F	T
F	F	F	F	F	T

Rows where F = T: **6**.

Q.6 [NAT, 1M · Predicate Logic] Answer: 5

The formula says $x \neq 1$ and the only divisors of x in D are 1 and x itself — i.e. x is prime. (Every divisor of $x \leq 12$ lies in D , so the restriction to D loses nothing.) The primes in $\{1, \dots, 12\}$ are 2, 3, 5, 7, 11, so the answer is 5.

Q.7 [NAT, 1M · Predicate Logic] Answer: 19

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 3$ this gives **19** (verified by enumerating all interpretations).

Q.8 [MSQ, 1M · Conditional Independence] Answer: A, B ($A: A \perp B \mid C$ implies $B \perp A \mid C$; $B: A \perp \{B, C\} \mid D$ implies $A \perp B \mid D$)

(A) Symmetry: $P(A, B \mid C) = P(A \mid C)P(B \mid C)$ is symmetric in A and B . (B) Decomposition: summing $P(A, B, C \mid D) = P(A \mid D)P(B, C \mid D)$ over C gives $P(A, B \mid D) = P(A \mid D)P(B \mid D)$. (C) is false: in a v-structure $A \rightarrow C \leftarrow B$, $A \perp B$ but conditioning on C makes them dependent (explaining away). (D) is false: in a fork $A \leftarrow C \rightarrow B$, $A \perp B \mid C$ but A and B are generally dependent.

Q.9 [NAT, 1M · Exact Inference] Answer: 0.758 to 0.762

Only the ancestors of C (B) matter: every other variable sums out to 1.

$P(C=T) = \sum$ over its ancestors of $P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
$A=T, B=T, D=T$	0.0648
$A=T, B=T, D=F$	0.1512
$A=T, B=F, D=T$	0.3136
$A=T, B=F, D=F$	0.0784
$A=F, B=T, D=T$	0.0108
$A=F, B=T, D=F$	0.0432
$A=F, B=F, D=T$	0.0098
$A=F, B=F, D=F$	0.0882

Sum = **0.76** ≈ 0.760 .

Q.10 [NAT, 1M · Sampling] Answer: 0.2392 to 0.2396

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(A=F) \times P(B=F \mid A=F) \times P(C=F \mid B=F) \times P(D=T) \times P(E=F \mid A=F, D=T)$
 $= 0.95 \times 0.8 \times 0.6 \times 0.7 \times 0.75 = \mathbf{0.2394} \approx 0.2394$.

Q.11 [MCQ, 2M · Uninformed Search] Answer: C ($C: S \rightarrow H \rightarrow C \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	I(2), H(5), E(6)
2	I (2)	H(5), E(6)
3	H (5)	E(6), C(10)
4	E (6)	C(10)
5	C (10)	G(13)
6	G (13)	← goal, stop

Path returned: **S** $\rightarrow$ **H** $\rightarrow$ **C** $\rightarrow$ **G**, cost = 5 + 5 + 3 = **13**. Nodes expanded (including S and G): **6** (S, I, H, E, C, G).

Q.12 [NAT, 2M · Informed Search] Answer: 7

Heuristic values: $h(A)=4, h(B)=3, h(E)=5, h(F)=2, h(G)=0, h(I)=6, h(J)=6, h(L)=7, h(S)=10$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (10)	L(11)
2	L (11)	I(11), E(14), J(18)
3	I (11)	B(13), E(14), F(14), J(18)

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
4	B (13)	E(14), F(14), G(15), J(18)
5	E (14)	F(14), G(15), A(18), J(18)
6	F (14)	G(15), A(18), J(18)
7	G (15)	A(18), J(18) ← goal, stop

Path returned: **S** → **L** → **I** → **B** → **G** with cost **15**; nodes expanded = **7** (S, L, I, B, E, F, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.13 [MCQ, 2M · Informed Search] **Answer: A** (A: Cost 7; h is consistent)

Path costs: $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 2+2+1+2 = 7$; $S \rightarrow A \rightarrow C \rightarrow G = 9$; $S \rightarrow A \rightarrow D \rightarrow G = 8$; $S \rightarrow B \rightarrow C \rightarrow D \rightarrow G = 8$; $S \rightarrow B \rightarrow C \rightarrow G = 10$. The optimum is **7**. Consistency, edge by edge: $S \rightarrow A$: $5 \leq 2+4$. $S \rightarrow B$: $5 \leq 4+4$. $A \rightarrow C$: $4 \leq 2+3$. $A \rightarrow D$: $4 \leq 4+2$. $B \rightarrow C$: $4 \leq 1+3$. $C \rightarrow D$: $3 \leq 1+2$. $C \rightarrow G$: $3 \leq 5+0$. $D \rightarrow G$: $2 \leq 2+0$. All hold, so **h is consistent**.

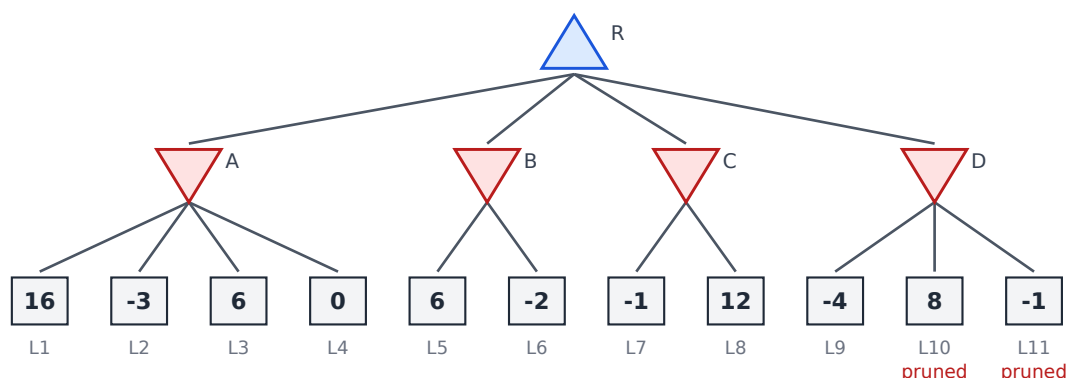
Q.14 [MSQ, 2M · Adversarial Search] **Answer: A, B, D**

(A) is the definition of a chance node's value. (B) is the AIMA bound for expectiminimax. (C) is false: if utilities are bounded, a chance node's average can be bounded before all of its children are seen, so some pruning is possible. (D) is true: expectimax models stochastic behaviour, while minimax models a worst-case adversary.

Q.15 [NAT, 2M · Adversarial Search] **Answer: 9**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $-4 \leq \alpha = -1 \Rightarrow$ prune remaining children (leaves L10, L11).



Leaves evaluated (9): L1, L2, L3, L4, L5, L6, L7, L8, L9. Leaves pruned (2): L10, L11. Root value = -1 (same as plain minimax, -1 — pruning never changes the root value).

Q.16 [MSQ, 2M · Propositional Logic] **Answer: B, D** (B: $\neg Q \vee R$; D: $\neg P \vee R$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	F	T	T
F	F	T	T

- (A) $\neg P$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.
 (B) $\neg Q \vee R$ is true in all models of KB $\Rightarrow$ **entailed**.
 (C) $P \wedge P$ is false in the KB-model $P=F, Q=F, R=T \Rightarrow$ not entailed.
 (D) $\neg P \vee R$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.17 [MSQ, 2M · Predicate Logic] **Answer: C, D** (C: $\exists x \forall y \neg R(x, y)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,3), (3,2), (4,3), (4,4)\}$.

- (A) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).
 (B) $\forall x \exists y R(x, y)$: **FALSE** — 2 has no outgoing edge.
 (C) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.
 (D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 1 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: C ($C: (\neg Q \wedge R) \wedge P$)

Compare truth tables with $F = P \wedge (\neg Q \wedge R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg P \wedge \neg(\neg Q \wedge R))$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(B) $\neg P \wedge (\neg Q \wedge R)$ — differs at $P=T, Q=F, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(C) $(\neg Q \wedge R) \wedge P$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(D) $P \vee (\neg Q \wedge R)$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

Q.19 [MSQ, 2M · Propositional Logic] Answer: B, C (B: If α is valid then $\neg \alpha$ is unsatisfiable; C: $\alpha \vee \beta$ is satisfiable if and only if α is satisfiable or β is satisfiable)

(A) False: p is satisfiable and so is $\neg p$. (B) True: valid means true in all models, so $\neg \alpha$ is true in none. (C)

True: a model of $\alpha \vee \beta$ is a model of α or of β , and conversely any model of α (or of β) is a model of $\alpha \vee \beta$. (D)

False: $\alpha = p$ and $\beta = \neg p$ are each satisfiable but $p \wedge \neg p$ is not (the 'if' direction fails).

Q.20 [MSQ, 2M · Conditional Independence] Answer: A, C (A: $B \perp E$; C: $E \perp B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp E$: **TRUE** — $B \rightarrow C \leftarrow A \rightarrow E$ (blocked: collider C with neither it nor a descendant observed); $B \rightarrow D \leftarrow A \rightarrow E$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow F \leftarrow C \leftarrow A \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow G \leftarrow F \leftarrow C \leftarrow A \rightarrow E$ (blocked: collider G with neither it nor a descendant observed).

(B) $D \perp A \mid E, F$: **FALSE** — the path $D \leftarrow A$ is active given $\{E, F\}$.

(C) $E \perp B$: **TRUE** — $E \leftarrow A \rightarrow C \leftarrow B$ (blocked: collider C with neither it nor a descendant observed); $E \leftarrow A \rightarrow C \rightarrow F \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $E \leftarrow A \rightarrow C \rightarrow F \rightarrow G \leftarrow B$ (blocked: collider G with neither it nor a descendant observed); $E \leftarrow A \rightarrow D \leftarrow B$ (blocked: collider D with neither it nor a descendant observed).

(D) $D \perp G \mid B$: **FALSE** — the path $D \leftarrow A \rightarrow C \rightarrow F \rightarrow G$ is active given $\{B\}$.

Q.21 [NAT, 2M · Variable Elimination] Answer: 16

Eliminating H multiplies all factors that mention H : $P(H)$, $P(X_1 \mid H)$, ..., $P(X_4 \mid H)$. Summing out H leaves a factor over $\{X_1, X_2, X_3, X_4\}$, with $2^4 = 16$ entries. By contrast, eliminating X_2, X_3, X_4 first gives trivial factors over H (each $\sum_x P(x \mid H) = 1$) and the largest new factor has just 1 variable.

Q.22 [NAT, 2M · Sampling] Answer: 0.3998 to 0.4002

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	C
1	$P(A=F)=0.4$	0.4	T
2	$P(A=F)=0.4$	0.4	T
3	$P(A=F)=0.4$	0.4	F
4	$P(A=F)=0.4$	0.4	F
5	$P(A=F)=0.4$	0.4	T
6	$P(A=F)=0.4$	0.4	F

$\sum w$ (all) = 2.4; $\sum w$ ($C = T$) = 1.2.

Estimate $P(C=T \mid A = F) = 1.2 / 2.4 = \mathbf{0.5}$.

Q.23 [NAT, 2M · Exact Inference] Answer: 0.318 to 0.322

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm} \mid \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Earthquake}, \text{Alarm})$.

Inference by enumeration: $P(\text{Earthquake} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Earthquake}, e, \text{hidden})$.

Hidden: Bu, Jo	Ea	product of CPT entries ($Bu \cdot Ea \cdot Al \cdot Jo$)	value
T, T	T	$0.95 \times 0.6 \times 0.2 \times 0.25$	0.0285
T, F	T	$0.95 \times 0.6 \times 0.2 \times 0.75$	0.0855
F, T	T	$0.05 \times 0.3 \times 0.2 \times 0.25$	0.00075
F, F	T	$0.05 \times 0.3 \times 0.2 \times 0.75$	0.00225
T, T	F	$0.95 \times 0.4 \times 0.6 \times 0.4$	0.0912
T, F	F	$0.95 \times 0.4 \times 0.6 \times 0.6$	0.1368
F, T	F	$0.05 \times 0.7 \times 0.6 \times 0.4$	0.0084

Hidden: Bu, Jo	Ea	product of CPT entries (Bu · Ea · Al · Jo)	value
F, F	F	$0.05 \times 0.7 \times 0.6 \times 0.6$	0.0126

$P(\text{Earthquake}=T, e) = 0.117$; $P(\text{Earthquake}=F, e) = 0.249$; $P(e) = 0.366$.

$P(\text{Earthquake}=T | e) = 0.117 / 0.366 = \mathbf{0.3197} \approx 0.320$.

Q.24 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, C)$, $f(A, C, D)$, $f(A, B, E)$, $f(B, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, C)$, $f(A, C, D)$, $f(A, B, E)$	$\{A, B, C, D, E\}$	$f(B, C, D, E)$
F	$f(B, E, F)$	$\{B, E, F\}$	$f(B, E)$
C	$f(B, C, D, E)$	$\{B, C, D, E\}$	$f(B, D, E)$
B	$f(B)$, $f(B, E)$, $f(B, D, E)$	$\{B, D, E\}$	$f(D, E)$
E	$f(D, E)$	$\{D, E\}$	$f(D)$

The largest factor produced has **4** variable(s) (created when eliminating A); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over D and normalise.

Q.25 [NAT, 2M · Variable Elimination] **Answer: 0.354 to 0.356**

Eliminate A: $f_A(b) = 0.2 \times 0.9 + 0.8 \times 0.3 = 0.18 + 0.24 = 0.42$, $f_A(\neg b) = 0.58$. Eliminate B: $f_B(c) = 0.42 \times 0.6 + 0.58 \times 0.1 = 0.252 + 0.058 = 0.31$, $f_B(\neg c) = 0.69$. Eliminate C: $P(d) = 0.31 \times 0.7 + 0.69 \times 0.2 = 0.217 + 0.138 = 0.355$. Each step creates a factor over a single variable, so cost is linear in the chain length.

Mock Test 010

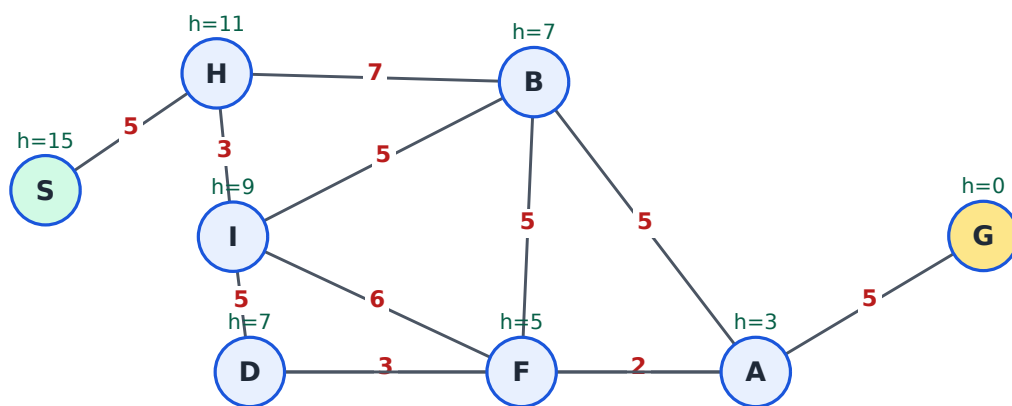
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	9 / 7 / 9	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

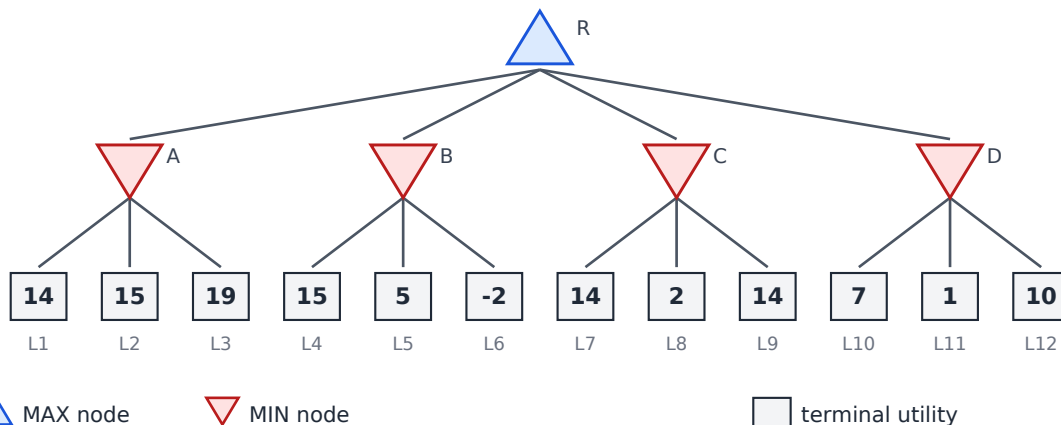
Q.2 [MCQ · 1 mark · negative: $-1/3$] Informed Search

Heuristics are often derived from relaxed problems, in which some constraints on the actions are removed. The cost of an optimal solution to a relaxed problem is

- ☐ (A) an admissible (and in fact consistent) heuristic for the original problem
- ☐ (B) always an overestimate of the original problem's optimal cost
- ☐ (C) always equal to the original problem's optimal cost
- ☐ (D) inadmissible, but useful for greedy search only

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

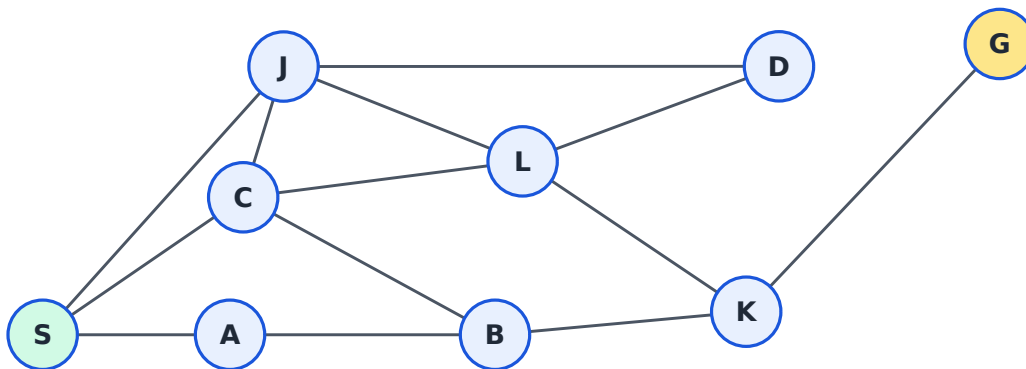
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, A, B, C, J, D, L, K, G
- ☐ (B) S, A, C, J, B, L, D, K, G
- ☐ (C) S, J, C, A, L, D, B, K, G
- ☐ (D) S, A, C, J, B, D, L, K, G

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following is the correct first-order translation of 'Every student likes some course'?

- ☐ (A) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
- ☐ (B) $\forall x (\text{Student}(x) \wedge \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
- ☐ (C) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \rightarrow \text{Likes}(x, y)))$
- ☐ (D) $\exists y \forall x (\text{Student}(x) \wedge \text{Course}(y) \wedge \text{Likes}(x, y))$

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Backward chaining over a propositional definite-clause KB is used to prove the query Q. Which clause is used FIRST?

- ☐ (A) A clause whose head (conclusion) is Q
- ☐ (B) A clause in which Q appears in the body
- ☐ (C) Any fact in the KB, chosen arbitrarily
- ☐ (D) The negated query $\neg Q$ added to the KB

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 2 elements, so there are $4^2 = 16$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

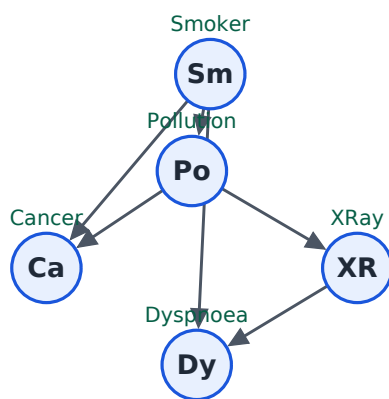
Q.8 [MCQ · 1 mark · negative: -1/3] Variable Elimination

What is the computational complexity of exact inference in general (multiply connected) Bayesian networks?

- ☐ (A) NP-hard (in fact #P-hard)
- ☐ (B) Polynomial time
- ☐ (C) Linear time
- ☐ (D) Undecidable

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR, Dy. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay, Dy = Dyspnoea.) What is the probability that a single run produces the sample **(Sm=T, Po=T, Ca=T, XR=T, Dy=T)**? (round off to 4 decimal places)



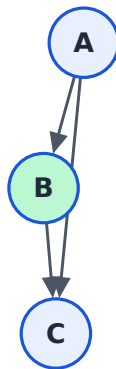
P(Sm=T)	Sm	P(Po=T ...)	Sm	Po	P(Ca=T ...)
0.8	T	0.7	T	T	0.9
	F	0.8	T	F	0.4
			F	T	0.9
			F	F	0.75

Po	P(XR=T ...)	Sm	XR	P(Dy=T ...)
T	0.9	T	T	0.3
F	0.95	T	F	0.75
		F	T	0.05
		F	F	0.2

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(B = T)$ (round off to 3 decimal places).



$P(A=T)$	A	$P(B=T \dots)$	A	B	$P(C=T \dots)$
0.05	T	0.1	T	T	0.5
	F		T	F	0.1
		0.05	F	T	0.05
			F	F	0.25

Answer (numerical): _____

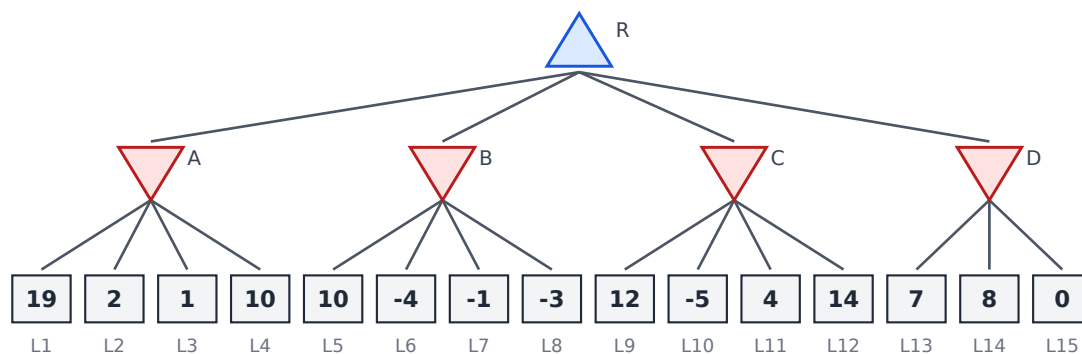
Section B — 2-mark questions**Q.11** [MCQ · 2 marks · negative: -2/3] Problem Formulation

For 8-queens, the complete-state formulation places exactly one queen in each column, in any row, with attacks allowed. How many states does this formulation have?

- ☐ (A) $8^8 = 16,777,216$
☐ (B) 64^8
☐ (C) $C(64, 8) = 4,426,165,368$
☐ (D) $8! = 40,320$

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



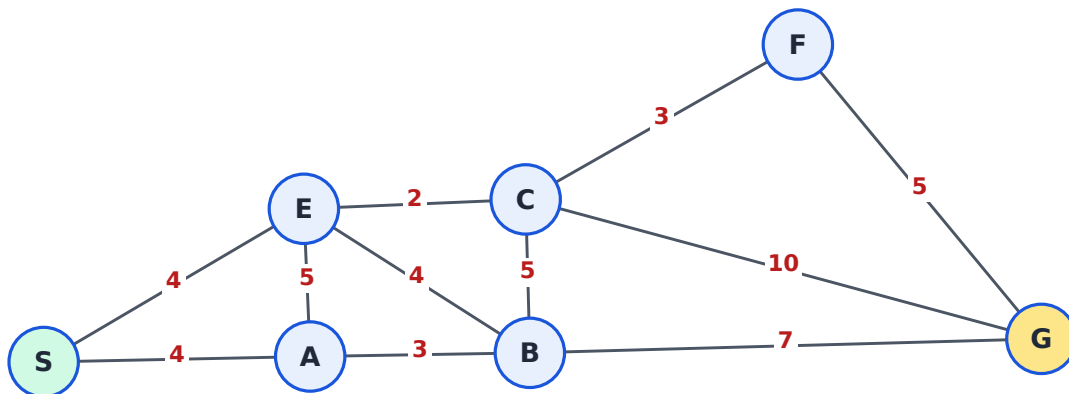
MAX node MIN node

terminal utility

- ☐ (A) Leaf L1 is pruned.
☐ (B) Leaf L2 is pruned.
☐ (C) Leaf L7 is pruned.
☐ (D) Leaf L12 is pruned.

Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

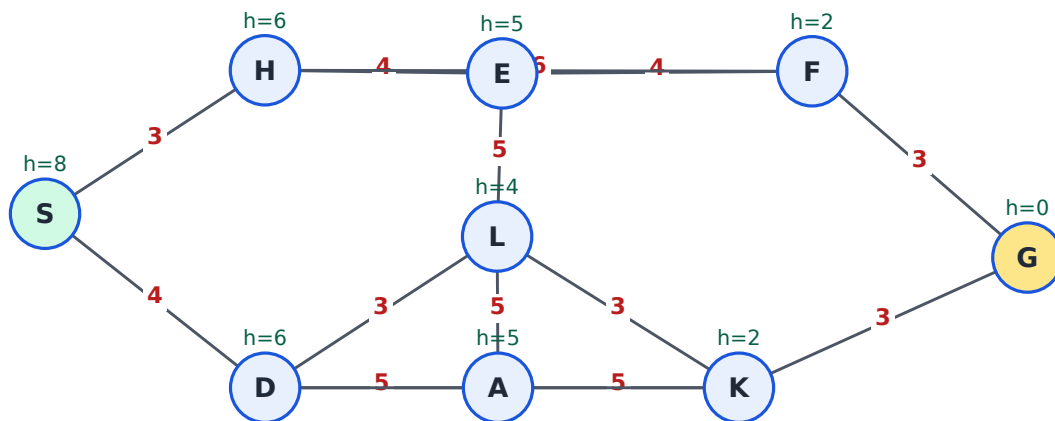
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow E \rightarrow B \rightarrow G$
☐ (B) $S \rightarrow E \rightarrow C \rightarrow G$
☐ (C) $S \rightarrow A \rightarrow B \rightarrow G$
☐ (D) $S \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.15 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following statements about the computational complexity of propositional reasoning is correct (assuming $P \neq NP$)?

- ☐ (A) Satisfiability of 3-CNF formulas can be decided in polynomial time
☐ (B) Deciding whether a propositional formula is valid is co-NP-complete
☐ (C) Satisfiability of Horn formulas is NP-complete
☐ (D) Satisfiability of 2-CNF formulas is NP-complete

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic

Consider the interpretation with domain $D = \{1, 2, 3\}$ in which $L(x, y)$ holds iff $x < y$. Which of the following sentences are TRUE in this interpretation?

- ☐ (A) $\forall x \exists y L(x, y)$
☐ (B) $\exists x \forall y \neg L(y, x)$
☐ (C) $\forall x \forall y (L(x, y) \rightarrow \neg L(y, x))$
☐ (D) $\forall x \forall y (L(x, y) \rightarrow \exists z (L(x, z) \wedge L(z, y)))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (Q \vee R), C2: (\neg Q \vee \neg S), C3: (P \vee \neg R), C4: (P)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) $(\neg P \vee \neg Q)$
☐ (C) $(Q \vee S)$
☐ (D) $(R \vee \neg S)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \rightarrow Q) \rightarrow R) \rightarrow R$
☐ (B) $(\neg Q \rightarrow (P \rightarrow R)) \leftrightarrow ((\neg Q \wedge P) \rightarrow R)$
☐ (C) $((P \rightarrow Q) \wedge (R \rightarrow Q)) \rightarrow (P \wedge R)$
☐ (D) $(R \vee Q) \rightarrow (R \wedge Q)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

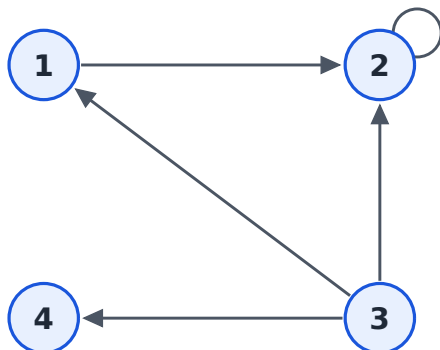
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = (\neg R \wedge S) \rightarrow ((R \vee \neg Q) \wedge (Q \wedge P)) ?$$

Answer (numerical): _____

Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

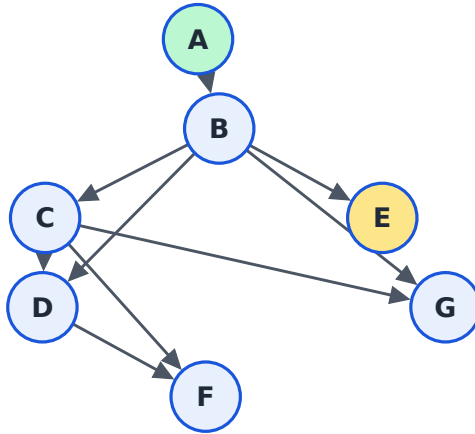
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(A \mid E = T)$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **G, B, D, F, C**. What is the scope of the new factor produced when **C** is summed out?



- ☐ (A) $f(C)$
☐ (B) $f(A)$
☐ (C) $f(A, C)$
☐ (D) $f()$

Q.22 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid A = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	T	F	T	T
2	F	F	F	F
3	F	F	F	F
4	T	F	T	T
5	F	T	T	T
6	F	F	F	T
7	F	F	F	T
8	F	F	F	F
9	F	F	F	F
10	F	F	T	T
11	T	F	F	F
12	T	F	T	F
13	F	F	F	T
14	F	F	T	T
15	F	F	F	F
16	F	F	F	F
17	F	F	T	F
18	F	F	F	F
19	F	T	F	T
20	F	F	F	T

Answer (numerical): _____

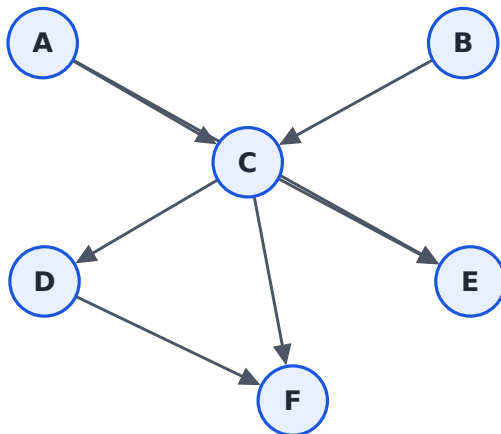
Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence

A Bayesian network over A, B, C, D encodes the factorization $P(A) P(B | A) P(C | A) P(D | B, C)$. Which independences are guaranteed to hold in every distribution with this factorization?

- ☐ (A) $B \perp C \mid \{A\}$
☐ (B) $B \perp C \mid \{A, D\}$
☐ (C) $A \perp D \mid \{B, C\}$
☐ (D) $A \perp D$

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

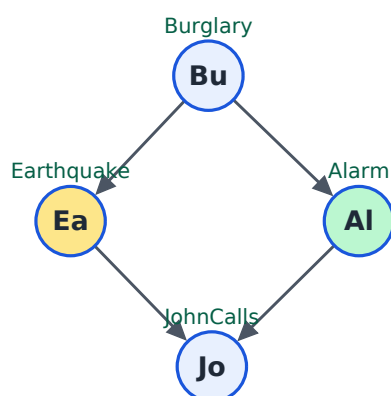
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp F \mid C, E$
☐ (B) $A \perp B$
☐ (C) $B \perp A \mid C, E$
☐ (D) $F \perp E \mid B$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute $P(\text{Alarm} = T \mid \text{Earthquake} = T)$ (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.4

Bu	$P(\text{Ea}=T \mid \dots)$	Bu	$P(\text{Al}=T \mid \dots)$
T	0.3	T	0.9
F	0.5	F	0.6

Ea	Al	$P(\text{Jo}=T \mid \dots)$
T	T	0.7
T	F	0.05
F	T	0.25
F	F	0.5

Answer (numerical): _____

Answer Key — Mock Test 010

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Informed Search	22	14	NAT	2	Informed Search	12
2	MCQ	1	Informed Search	A	15	MCQ	2	Propositional Logic	B
3	NAT	1	Adversarial Search	14	16	MSQ	2	Predicate Logic	B, C
4	MCQ	1	Uninformed Search	B	17	MSQ	2	Propositional Logic	D
5	MCQ	1	Predicate Logic	A	18	MSQ	2	Propositional Logic	A, B
6	MCQ	1	Propositional Logic	A	19	NAT	2	Propositional Logic	12
7	NAT	1	Predicate Logic	4	20	MSQ	2	Predicate Logic	A, C
8	MCQ	1	Variable Elimination	A	21	MCQ	2	Exact Inference	B
9	NAT	1	Sampling	0.1359 to 0.1363	22	NAT	2	Sampling	0.49 to 0.51
10	NAT	1	Exact Inference	0.05 to 0.054	23	MSQ	2	Conditional Independence	A, C
11	MCQ	2	Problem Formulation	A	24	MSQ	2	Conditional Independence	A, B
12	MSQ	2	Adversarial Search	C, D	25	NAT	2	Exact Inference	0.684 to 0.688
13	MCQ	2	Uninformed Search	C					

Detailed Solutions

Q.1 [NAT, 1M · Informed Search] Answer: 22

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (15)	H(11)
2	H (11)	B(7), I(9)
3	B (7)	A(3), F(5), I(9)
4	A (3)	G(0), F(5), D(7), I(9)
5	G (0)	F(5), D(7), I(9) ← goal, stop

Path returned: **S** → **H** → **B** → **A** → **G**, cost = 5 + 7 + 5 + 5 = **22**.

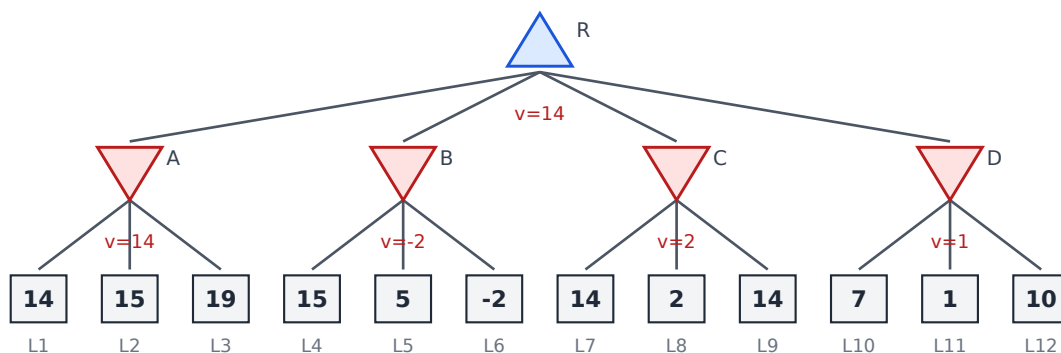
The optimal cost is $h^*(S) = 21$ — greedy search is not optimal here.

Q.2 [MCQ, 1M · Informed Search] Answer: A (A: an admissible (and in fact consistent) heuristic for the original problem)

The relaxed problem's state-space graph is a supergraph of the original, because it has extra edges. So every original solution is also a relaxed solution, and the relaxed optimum is $\leq$ **the true cost: admissible**. It is also an exact cost in a real problem, so it satisfies the triangle inequality and is consistent. Hence (B) and (D) are wrong. (C) holds only when the relaxation happens to be tight.

Q.3 [NAT, 1M · Adversarial Search] Answer: 14

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = 14, B = -2, C = 2, D = 1. Root value = **14**; MAX's optimal first move is towards A.

Q.4 [MCQ, 1M · Uninformed Search] Answer: B (B: S, A, C, J, B, L, D, K, G)

Adjacency lists (alphabetical): A: {B, S}; B: {A, C, K}; C: {B, J, L, S}; D: {J, L}; G: {K}; J: {C, D, L, S}; K: {B, G, L}; L: {C, D, J, K}; S: {A, C, J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[A, C, J]
2	A	[C, J, B]
3	C	[J, B, L]
4	J	[B, L, D]
5	B	[L, D, K]
6	L	[D, K]
7	D	[K]
8	K	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, A, C, J, B, L, D, K, G**. The path returned is $S \rightarrow A \rightarrow B \rightarrow K \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$)

Universal quantifiers pair naturally with $\rightarrow$, existential with $\wedge$. Option B claims everything in the domain is a student. Option C is satisfied by any non-course y (making $\text{Course}(y) \rightarrow \dots$ vacuously true), so it does not require liking a course. Option D says there is a single y that is a course liked by everything, and everything is a student — far too strong.

Q.6 [MCQ, 1M · Propositional Logic] Answer: A (A: A clause whose head (conclusion) is Q)

Backward chaining works from the goal: it looks for rules (or facts) whose conclusion is Q and recursively tries to prove their premises. Using clauses with Q in the body is forward reasoning. Starting from facts is forward chaining. Adding $\neg Q$ is the resolution-refutation approach, not backward chaining.

Q.7 [NAT, 1M · Predicate Logic] Answer: 4

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 2$ this gives **4** (verified by enumerating all interpretations).

Q.8 [MCQ, 1M · Variable Elimination] Answer: A (A: NP-hard (in fact #P-hard))

AIMA shows that exact inference includes propositional satisfiability as a special case (3-SAT can be encoded), so it is NP-hard; computing the probability is #P-hard. It is linear only for polytrees and is certainly decidable (enumeration always terminates).

Q.9 [NAT, 1M · Sampling] Answer: 0.1359 to 0.1363

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Sm}=\text{T}) \times P(\text{Po}=\text{T}|\text{Sm}=\text{T}) \times P(\text{Ca}=\text{T}|\text{Sm}=\text{T}, \text{Po}=\text{T}) \times P(\text{XR}=\text{T}|\text{Po}=\text{T}) \times P(\text{Dy}=\text{T}|\text{Sm}=\text{T}, \text{XR}=\text{T})$
 $= 0.8 \times 0.7 \times 0.9 \times 0.9 \times 0.3 = \mathbf{0.13608} \approx 0.1361$.

Q.10 [NAT, 1M · Exact Inference] Answer: 0.05 to 0.054

Only the ancestors of B (A) matter: every other variable sums out to 1.

$P(B=\text{T}) = \sum \text{over its ancestors of } P(B=\text{T} | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, B=\text{T})$
A=T, C=T	0.0025
A=T, C=F	0.0025
A=F, C=T	0.00238
A=F, C=F	0.04512

Sum = **0.0525** ≈ 0.052 .

Q.11 [MCQ, 2M · Problem Formulation] Answer: A (A: $8^8 = 16,777,216$)

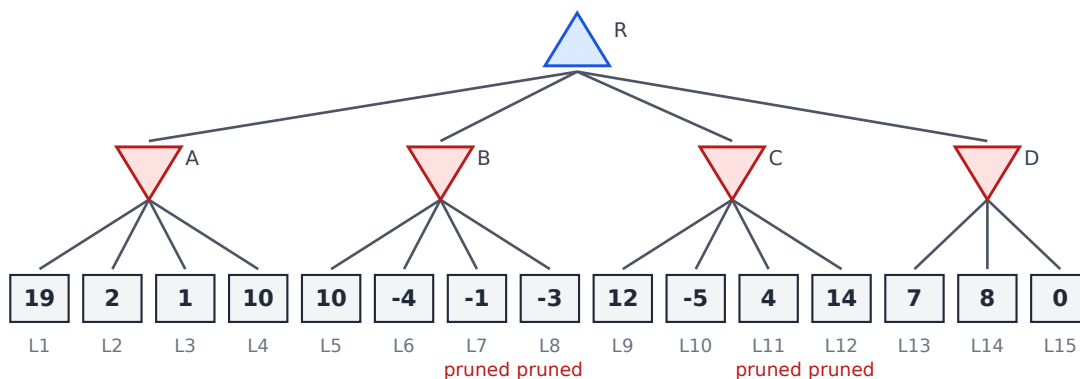
Each of the 8 columns independently holds its queen in one of 8 rows, giving **$8^8 = 16,777,216$** states.

C(64,8) (C) counts any 8 squares without the one-per-column constraint. 64^8 (B) counts ordered placements of distinguishable queens. $8!$ (D) also requires distinct rows.

Q.12 [MSQ, 2M · Adversarial Search] Answer: C, D (C: Leaf L7 is pruned.; D: Leaf L12 is pruned.)

Run alpha-beta depth-first, left to right, passing (α , β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-4 \leq \alpha = 1 \Rightarrow$ prune remaining children (leaves L7, L8).
- At MIN node C: value $-5 \leq \alpha = 1 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (11): L1, L2, L3, L4, L5, L6, L9, L10, L13, L14, L15. Leaves pruned (4): L7, L8, L11, L12. Root value = 1 (same as plain minimax, 1 — pruning never changes the root value).

Q.13 [MCQ, 2M · Uninformed Search] Answer: C (C: $S \rightarrow A \rightarrow B \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	A(4), E(4)
2	A (4)	E(4), B(7)
3	E (4)	C(6), B(7)
4	C (6)	B(7), F(9), G(16)
5	B (7)	F(9), G(14)
6	F (9)	G(14)
7	G (14)	← goal, stop

Path returned: **S** $\rightarrow$ **A** $\rightarrow$ **B** $\rightarrow$ **G**, cost = 4 + 3 + 7 = **14**. Nodes expanded (including S and G): **7** (S, A, E, C, B, F, G).

Q.14 [NAT, 2M · Informed Search] Answer: 12

Heuristic values: $h(A)=5$, $h(D)=6$, $h(E)=5$, $h(F)=2$, $h(G)=0$, $h(H)=6$, $h(K)=2$, $h(L)=4$, $h(S)=8$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (8)	H(9), D(10)
2	H (9)	D(10), F(11), E(12)
3	D (10)	F(11), L(11), E(12), A(14), K(15)
4	F (11)	L(11), E(12), G(12), A(14), K(15)
5	L (11)	E(12), G(12), K(12), A(14)
6	E (12)	G(12), K(12), A(14)
7	G (12)	K(12), A(14) ← goal, stop

Path returned: **S** $\rightarrow$ **H** $\rightarrow$ **F** $\rightarrow$ **G** with cost **12**; nodes expanded = **7** (S, H, D, F, L, E, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 12$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.15 [MCQ, 2M · Propositional Logic] Answer: B (B: Deciding whether a propositional formula is valid is co-NP-complete)

α is valid iff $\neg\alpha$ is unsatisfiable, and SAT is NP-complete, so VALIDITY (TAUT) is co-NP-complete. 3-SAT is NP-complete, so not in P under $P \neq NP$. Horn-SAT is decidable in linear time (unit propagation / forward chaining). 2-SAT is decidable in linear time via the implication graph and strongly connected components.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: B, C** (B: $\exists x \forall y \neg L(y, x)$; C: $\forall x \forall y (L(x, y) \rightarrow \neg L(y, x))$)

(A) False: for $x = 3$ no y in D is larger. (B) True: $x = 1$ has no y with $y < 1$. (C) True: $<$ is asymmetric. (D) False: the order on $\{1, 2, 3\}$ is not dense; for $x = 1, y = 2$ there is no z with $1 < z < 2$. Evaluating all cases by enumeration confirms B and C.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: D** (D: $(R \vee \neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(\neg P \vee \neg Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $(Q \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(R \vee \neg S)$ is derivable:

1. resolve $(Q \vee R)$ and $(\neg Q \vee \neg S)$ on $Q \Rightarrow (R \vee \neg S)$

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $((R \rightarrow Q) \rightarrow R) \rightarrow R$; B: $(\neg Q \rightarrow (P \rightarrow R)) \Leftrightarrow ((\neg Q \wedge P) \rightarrow R)$)

(A) $((R \rightarrow Q) \rightarrow R) \rightarrow R$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $(\neg Q \rightarrow (P \rightarrow R)) \Leftrightarrow ((\neg Q \wedge P) \rightarrow R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $((P \rightarrow Q) \wedge (R \rightarrow Q)) \rightarrow (P \wedge R)$: **not valid** — counter-model $P=T, Q=T, R=F$.

(D) $(R \vee Q) \rightarrow (R \wedge Q)$: **not valid** — counter-model $Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.19 [NAT, 2M · Propositional Logic] **Answer: 12**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$\neg R \wedge S$	$(R \vee \neg Q) \wedge (Q \wedge P)$	$(\neg R \wedge S) \rightarrow ((R \vee \neg Q) \wedge (Q \wedge P))$
T	T	T	T	F	T	T
T	T	T	F	F	T	T
T	T	F	T	T	F	F
T	T	F	F	F	F	T
T	F	T	T	F	F	T
T	F	T	F	F	F	T
T	F	F	T	T	F	F
T	F	F	F	F	F	T
F	T	T	T	F	F	T
F	T	T	F	F	F	T
F	T	F	T	T	F	F
F	T	F	F	F	F	T
F	F	T	T	F	F	T
F	F	T	F	F	F	T
F	F	F	T	T	F	F
F	F	F	F	F	F	T

Rows where F = T: **12**.

Q.20 [MSQ, 2M · Predicate Logic] **Answer: A, C** (A: $\exists x \forall y \neg R(x, y)$; C: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$)

$R = \{(1,2), (2,2), (3,1), (3,2), (3,4)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 4 has no outgoing edge.

(B) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 2 has no edge to a different element.

(C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,4)$ nor $R(4,1)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.21 [MCQ, 2M · Exact Inference] **Answer: B** (B: $f(A)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(B, C), f(B, C, D), f(B), f(C, D, F), f(B, C, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(B, C, G)$	$\{B, C, G\}$	$f(B, C)$
B	$f(A, B), f(B, C), f(B, C, D), f(B), f(B, C)$	$\{A, B, C, D\}$	$f(A, C, D)$
D	$f(C, D, F), f(A, C, D)$	$\{A, C, D, F\}$	$f(A, C, F)$
F	$f(A, C, F)$	$\{A, C, F\}$	$f(A, C)$
C	$f(A, C)$	$\{A, C\}$	$f(A)$

The largest factor produced has **3** variable(s) (created when eliminating B); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over A and normalise.

Q.22 [NAT, 2M · Sampling] Answer: 0.49 to 0.51

Rejection sampling keeps only samples consistent with the evidence $A = F$: samples 2, 3, 5, 6, 7, 8, 9, 10, 13, 14, 15, 16, 17, 18, 19, 20 (16 accepted, 4 rejected).

Among the accepted samples, $D = T$ in 8. Estimate = $8/16 = \mathbf{0.5}$.

(For this network the exact value is 0.4487; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.23 [MSQ, 2M · Conditional Independence] Answer: A, C ($A: B \perp C \mid \{A\}$; $C: A \perp D \mid \{B, C\}$)

The DAG is $A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D$. (A) B-C paths: $B \leftarrow A \rightarrow C$ (fork, A observed: blocked) and $B \rightarrow D \leftarrow C$ (collider D unobserved, no descendants: blocked): d-separated, true. (B) Observing D activates the collider path $B \rightarrow D \leftarrow C$: false. (C) Both A-D paths $A \rightarrow B \rightarrow D$ and $A \rightarrow C \rightarrow D$ pass through observed non-colliders B and C: d-separated, true. (D) With nothing observed, $A \rightarrow B \rightarrow D$ is active: false.

Q.24 [MSQ, 2M · Conditional Independence] Answer: A, B ($A: B \perp F \mid C, E$; $B: A \perp B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp F \mid C, E$: **TRUE** — $B \rightarrow C \rightarrow D \rightarrow F$ (blocked: observed chain node C); $B \rightarrow C \rightarrow F$ (blocked: observed chain node C).

(B) $A \perp B$: **TRUE** — $A \rightarrow C \leftarrow B$ (blocked: collider C with neither it nor a descendant observed); $A \rightarrow E \leftarrow C \leftarrow B$ (blocked: collider E with neither it nor a descendant observed).

(C) $B \perp A \mid C, E$: **FALSE** — the path $B \rightarrow C \leftarrow A$ is active given $\{C, E\}$.

(D) $F \perp E \mid B$: **FALSE** — the path $F \leftarrow C \leftarrow A \rightarrow E$ is active given $\{B\}$.

Q.25 [NAT, 2M · Exact Inference] Answer: 0.684 to 0.688

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm} \mid \text{Burglary}) \cdot P(\text{JohnCalls} \mid \text{Earthquake}, \text{Alarm})$.

Inference by enumeration: $P(\text{Alarm} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Alarm}, e, \text{hidden})$.

Hidden: Bu, Jo	Al	product of CPT entries (Bu · Ea · Al · Jo)	value
T, T	T	$0.4 \times 0.3 \times 0.9 \times 0.7$	0.0756
T, F	T	$0.4 \times 0.3 \times 0.9 \times 0.3$	0.0324
F, T	T	$0.6 \times 0.5 \times 0.6 \times 0.7$	0.126
F, F	T	$0.6 \times 0.5 \times 0.6 \times 0.3$	0.054
T, T	F	$0.4 \times 0.3 \times 0.1 \times 0.05$	0.0006
T, F	F	$0.4 \times 0.3 \times 0.1 \times 0.95$	0.0114
F, T	F	$0.6 \times 0.5 \times 0.4 \times 0.05$	0.006
F, F	F	$0.6 \times 0.5 \times 0.4 \times 0.95$	0.114

$P(\text{Alarm}=T, e) = 0.288$; $P(\text{Alarm}=F, e) = 0.132$; $P(e) = 0.42$.

$P(\text{Alarm}=T \mid e) = 0.288 / 0.42 = \mathbf{0.6857} \approx 0.686$.

Mock Test 011

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 6 / 14	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

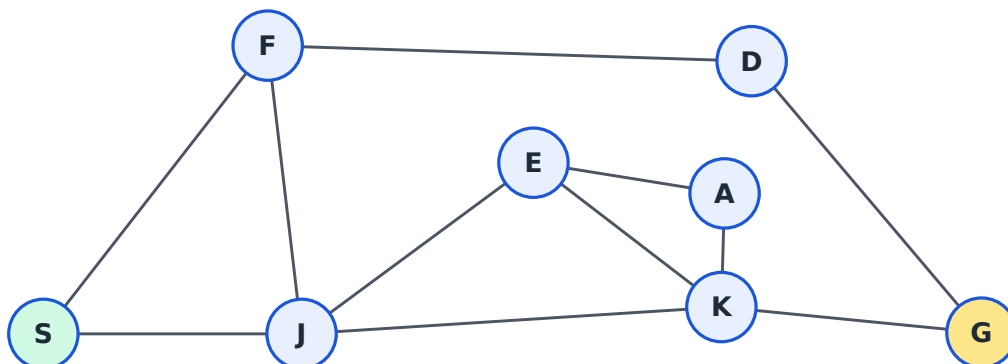
Q.1 [MCQ · 1 mark · negative: $-1/3$] Informed Search

Iterative deepening A* (IDA*) uses a cutoff on $f = g + h$. After an iteration fails, the cutoff for the next iteration is set to

- ☐ (A) the previous cutoff plus 1
- ☐ (B) the smallest f-value of any node that exceeded the cutoff in the previous iteration
- ☐ (C) the largest f-value seen in the previous iteration
- ☐ (D) twice the previous cutoff

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

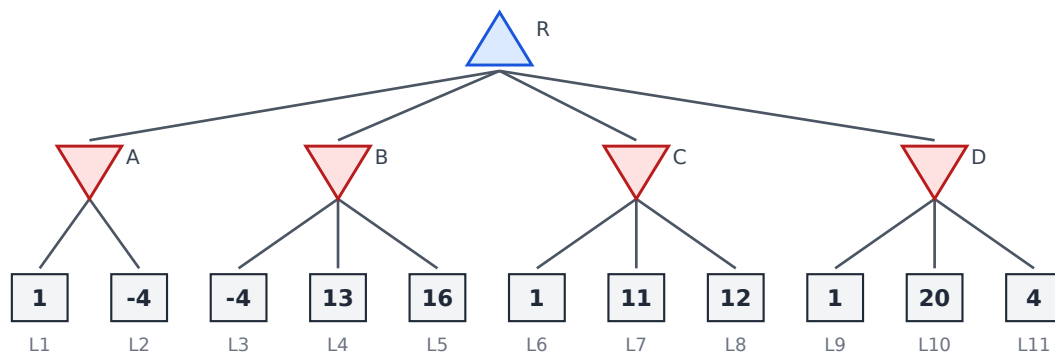
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, F, D, G
- ☐ (B) S, D, F, G
- ☐ (C) S, F, J, D, E, K, G
- ☐ (D) S, J, K, G

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

For the game tree below, what is the minimax value of the root?



△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 5$ and depth $d = 3$ (so $b^d = 125$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Let $KB = \{P \rightarrow Q, Q \rightarrow R, \neg R\}$. Which one of the following is entailed by KB?

- ☐ (A) $\neg P \wedge \neg Q$
☐ (B) $P \vee Q$
☐ (C) $\neg P \wedge Q$
☐ (D) R

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3, 4\}$. How many interpretations of a unary predicate P over D make $\exists x P(x) \wedge \neg \forall x P(x)$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = \neg R \leftrightarrow ((P \rightarrow Q) \leftrightarrow Q) ?$$

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ are both reflexive and symmetric?

Answer (numerical): _____

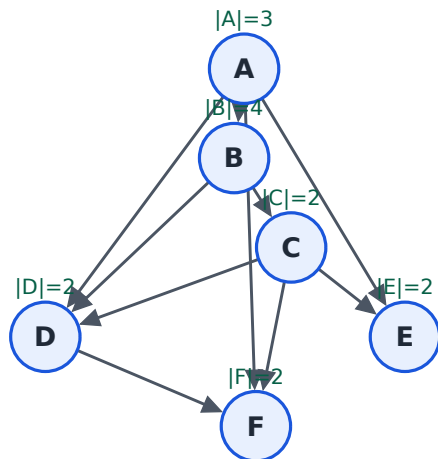
Q.9 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

A Bayesian network has edges $A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D$. Which is the joint distribution it represents?

- ☐ (A) $P(A) P(B | A) P(C | A) P(D | B, C)$
☐ (B) $P(A) P(B | A) P(C | B) P(D | C)$
☐ (C) $P(A) P(B) P(C) P(D | B, C)$
☐ (D) $P(D) P(B | D) P(C | D) P(A | B, C)$

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

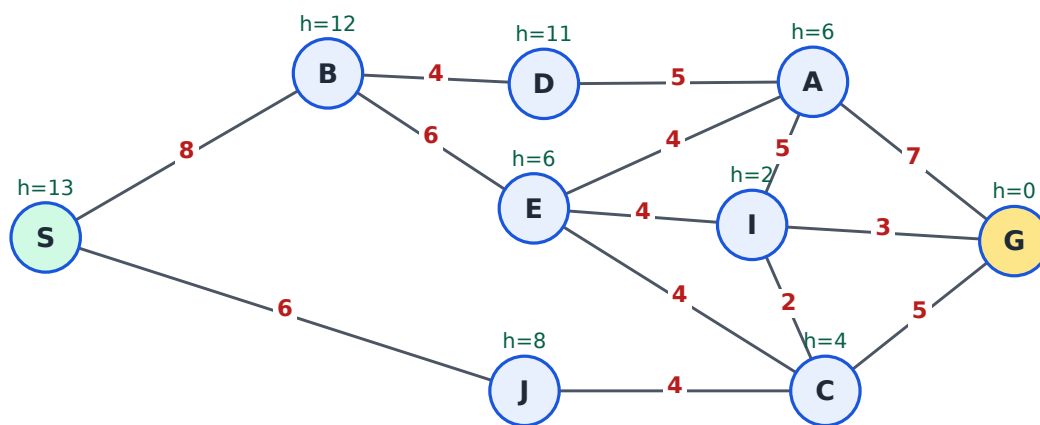
The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Informed Search – A*

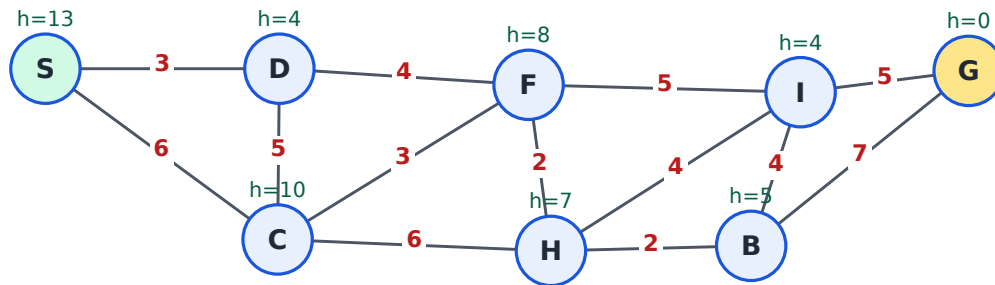
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

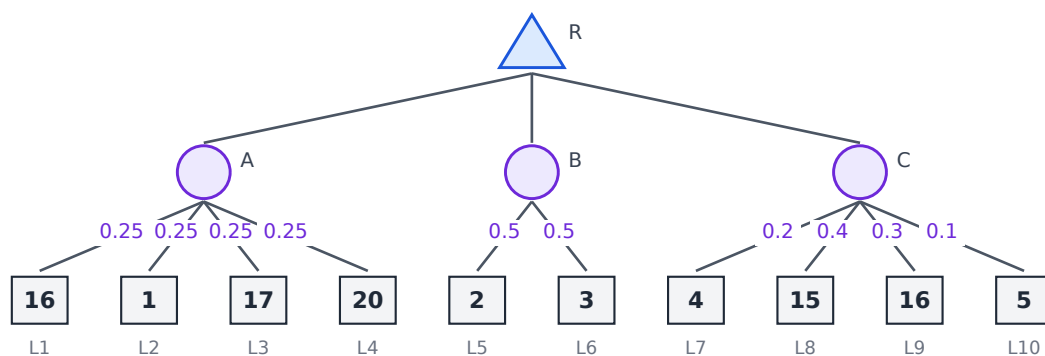
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(I)$ overestimates the true cost from I to G.
☐ (B) h is admissible.
☐ (C) The edge C-D violates the consistency condition $h(C) \leq c(C,D) + h(D)$.
☐ (D) h is consistent (monotone).

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node

○ CHANCE node (edge = probability)

□ terminal utility

Answer (numerical): _____

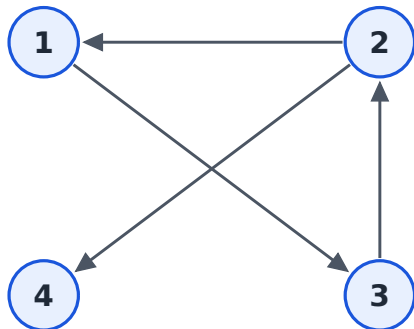
Q.14 [MSQ · 2 marks · no negative marking] Uninformed Search

Iterative deepening search (IDS) runs on a problem with finite branching factor b and shallowest goal depth d . Which of the following statements are TRUE?

- ☐ (A) IDS is complete
☐ (B) IDS is optimal when all step costs are equal
☐ (C) The space complexity of IDS is $O(bd)$
☐ (D) The time complexity of IDS is asymptotically worse than BFS by a factor of d , namely $O(d \cdot b^d)$

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall y \exists x R(x, y)$
☐ (B) $\exists x \forall y \neg R(y, x)$
☐ (C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (D) $\exists x \forall y R(x, y)$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (\neg Q)$, $C2: (\neg P \vee R)$, $C3: (\neg P)$, $C4: (\neg P \vee \neg S)$, $C5: (Q \vee \neg R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg Q \vee S)$
☐ (B) $(\neg P \vee Q)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(\neg R)$

Q.17 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \forall y (R(x, y) \rightarrow \neg R(y, x))$?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **A, C, E** and rules:

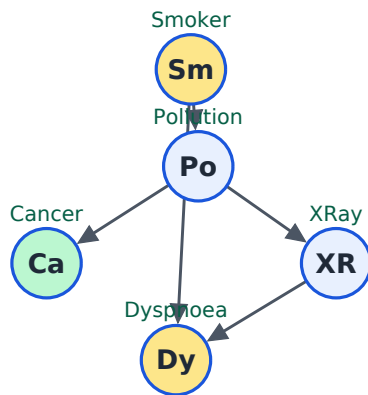
- $R1: D \rightarrow J$
 $R2: J \wedge D \rightarrow C$
 $R3: E \rightarrow H$
 $R4: C \rightarrow D$
 $R5: C \wedge A \wedge B \rightarrow E$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) B is entailed
☐ (B) D is entailed
☐ (C) F is entailed
☐ (D) H is entailed

Q.19 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay, Dy = Dyspnoea.) Compute $P(\text{Cancer} = T \mid \text{Dyspnoea} = T, \text{Smoker} = T)$ (round off to 3 decimal places).



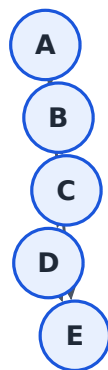
$P(\text{Sm}=T)$	Sm	$P(\text{Po}=T \dots)$	Po	$P(\text{Ca}=T \dots)$
0.05	T	0.75	T	0.25
	F	0.75	F	0.25

Po	$P(\text{XR}=T \dots)$	Sm	XR	$P(\text{Dy}=T \dots)$
T	0.95	T	T	0.1
F	0.4	T	F	0.25
		F	T	0.1
		F	F	0.3

Answer (numerical): _____

Q.20 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.44, 0.12, 0.53, 0.33, 0.4)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampling parent values})$. Which sample is generated?



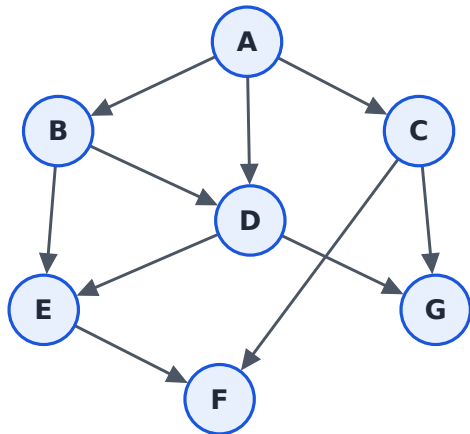
$P(A=T)$	A	$P(B=T \dots)$	A	B	$P(C=T \dots)$
0.25	T	0.75	T	T	0.25
	F	0.25	T	F	0.3
			F	T	0.2
			F	F	0.6

C	$P(D=T \dots)$	A	D	$P(E=T \dots)$
T	0.05	T	T	0.4
F	0.8	T	F	0.9
		F	T	0.05
		F	F	0.3

- ☐ (A) A=F, B=T, C=F, D=T, E=F
☐ (B) A=T, B=T, C=F, D=T, E=F
☐ (C) A=T, B=F, C=T, D=T, E=T
☐ (D) A=F, B=F, C=F, D=T, E=F

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

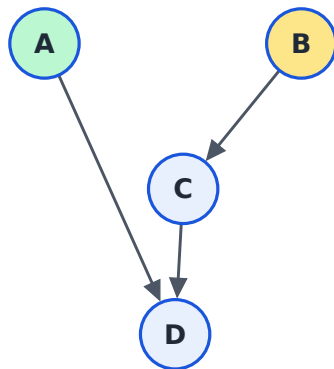
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp E \mid A, D$
☐ (B) $F \perp D \mid G$
☐ (C) $F \perp D \mid A, E$
☐ (D) $A \perp C \mid G$

Q.22 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{B} = \mathbf{T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(A = T \mid B = T)}$ (round off to 3 decimal places).



P(A=T)
0.4

P(B=T)	B	P(C=T ...)
0.75	T	0.95
	F	0.8

A	C	P(D=T ...)
T	T	0.6
T	F	0.4
F	T	0.6
F	F	0.25

#	A	C	D
1	T	T	T
2	T	T	F
3	T	T	T
4	F	T	F

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $C \rightarrow A$ and $C \rightarrow B$ (Boolean variables) with $P(c) = 0.4$, $P(a \mid c) = 0.9$, $P(a \mid \neg c) = 0.2$, $P(b \mid c) = 0.7$, $P(b \mid \neg c) = 0.1$. Compute $P(a, b)$. (Round to 3 decimal places.)

Answer (numerical): _____

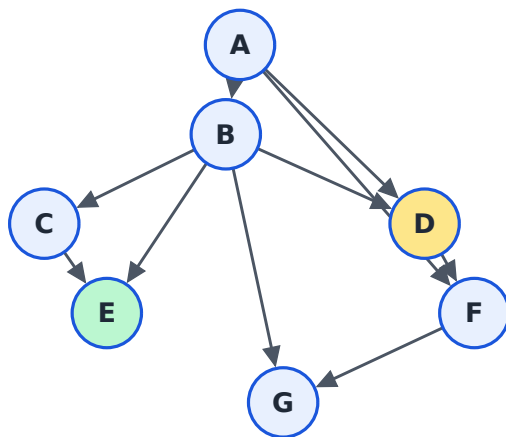
Q.24 [NAT · 2 marks · no negative marking] Probability for AI

A disease D has prior $P(d) = 0.01$. A test has $P(+ | d) = 0.9$ and $P(+ | \neg d) = 0.05$. The test is performed twice; the two results are conditionally independent given D . Both tests come out positive. Compute $P(d | +, +)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(E | D = T)$ in the Bayesian network below (Boolean variables; evidence $D = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, G, F, B, C**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 011

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Informed Search	B	14	MSQ	2	Uninformed Search	A, B, C
2	MCQ	1	Uninformed Search	A	15	MSQ	2	Predicate Logic	A, C
3	NAT	1	Adversarial Search	1	16	MSQ	2	Propositional Logic	B, D
4	NAT	1	Adversarial Search	29	17	NAT	2	Predicate Logic	27
5	MCQ	1	Propositional Logic	A	18	MSQ	2	Propositional Logic	B, D
6	NAT	1	Predicate Logic	14	19	NAT	2	Exact Inference	0.248 to 0.252
7	NAT	1	Propositional Logic	4	20	MCQ	2	Sampling	A
8	NAT	1	Predicate Logic	8	21	MSQ	2	Conditional Independence	A, C
9	MCQ	1	Bayesian Networks	A	22	NAT	2	Sampling	0.748 to 0.752
10	NAT	1	Bayesian Networks	57	23	NAT	2	Conditional Independence	0.263 to 0.265
11	NAT	2	Informed Search	5	24	NAT	2	Probability for AI	0.765 to 0.767
12	MSQ	2	Informed Search	B, C	25	NAT	2	Exact Inference	2
13	NAT	2	Adversarial Search	13.49 to 13.51					

Detailed Solutions

Q.1 [MCQ, 1M · Informed Search] Answer: B (B: the smallest f-value of any node that exceeded the cutoff in the previous iteration)

IDA* raises the threshold to the **smallest f-cost that exceeded the old threshold**. This keeps it optimal with an admissible h, because no solution with a smaller f is skipped. Increment-by-1 (A) assumes integer costs and can waste iterations. The largest f (C) and doubling (D) can jump past the optimal cost and return a suboptimal solution.

Q.2 [MCQ, 1M · Uninformed Search] Answer: A (A: S, F, D, G)

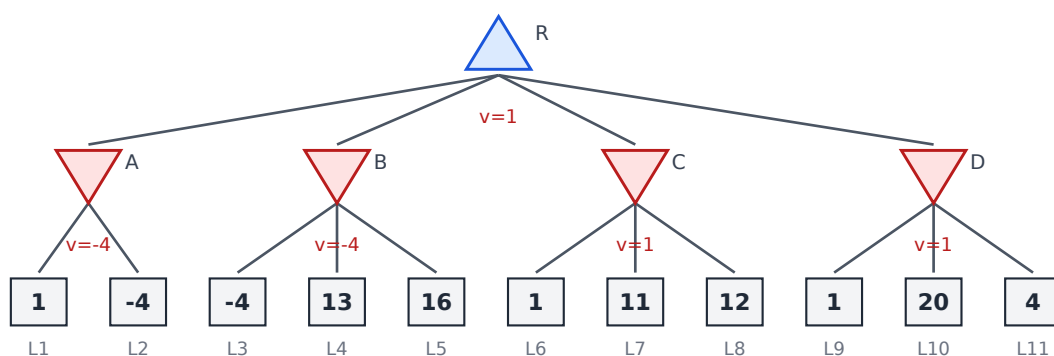
Adjacency lists (alphabetical): A: {E, K}; D: {F, G}; E: {A, J, K}; F: {D, J, S}; G: {D, K}; J: {E, F, K, S}; K: {A, E, G, J}; S: {F, J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, F, D, G**; the path found is $S \rightarrow F \rightarrow D \rightarrow G$ (3 edges). Note that DFS need not find the shortest path.

Q.3 [NAT, 1M · Adversarial Search] Answer: 1

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -4$, $B = -4$, $C = 1$, $D = 1$. Root value = **1**; MAX's optimal first move is towards C, D.

Q.4 [NAT, 1M · Adversarial Search] Answer: 29

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 5^2 + 5^1 - 1 = 25 + 5 - 1 = \mathbf{29}$, versus 125 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.5 [MCQ, 1M · Propositional Logic] Answer: A (A: $\neg P \wedge \neg Q$)

From $\neg R$ and $Q \rightarrow R$, modus tollens gives $\neg Q$; from $\neg Q$ and $P \rightarrow Q$ it gives $\neg P$. The only model of KB (over P, Q, R) is $P=F$, $Q=F$, $R=F$, in which $\neg P \wedge \neg Q$ is true. $P \vee Q$, $\neg P \wedge Q$ and R are all false in that model, so they are not entailed.

Q.6 [NAT, 1M · Predicate Logic] Answer: 14

An interpretation of P is a subset of D: $2^4 = 16$ choices. $\exists x P(x)$ excludes the empty set, $\neg \forall x P(x)$ excludes D itself. Hence $16 - 2 = 14$.

Q.7 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg R$	$(P \rightarrow Q) \leftrightarrow Q$	$\neg R \leftrightarrow ((P \rightarrow Q) \leftrightarrow Q)$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	F

Rows where F = T: **4**.

Q.8 [NAT, 1M · Predicate Logic] Answer: 8

Reflexivity fixes all 3 diagonal pairs to be present. Symmetry ties (x, y) and (y, x) together for $x \neq y$, leaving 3 independent unordered pairs, each in or out. Total = $2^3 = 8$.

Q.9 [MCQ, 1M · Bayesian Networks] Answer: A (A: $P(A)P(B|A)P(C|A)P(D|B,C)$)

A BN represents $\prod_i P(X_i | \text{Parents}(X_i))$. Parents: A: none; B: A; C: A; D: B, C. This gives (A). (B) corresponds to a chain $A \rightarrow B \rightarrow C \rightarrow D$; (C) drops edges from A; (D) reverses every edge.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 57

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (4-1) \times 3 + C: (2-1) \times 4 + D: (2-1) \times 3 \times 4 \times 2 + E: (2-1) \times 3 \times 2 + F: (2-1) \times 3 \times 2 \times 2 = 57$.

Full joint table: $\prod |X| - 1 = 192 - 1 = 191$; saving = 134.

Q.11 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(A)=6$, $h(B)=12$, $h(C)=4$, $h(D)=11$, $h(E)=6$, $h(G)=0$, $h(I)=2$, $h(J)=8$, $h(S)=13$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	J(14), B(20)
2	J (14)	C(14), B(20)
3	C (14)	I(14), G(15), B(20), E(20)
4	I (14)	G(15), B(20), E(20), A(23)
5	G (15)	B(20), E(20), A(23) ← goal, stop

Path returned: **S** → **J** → **C** → **G** with cost **15**; nodes expanded = **5** (S, J, C, I, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.12 [MSQ, 2M · Informed Search] Answer: B, C (B: h is admissible.; C: The edge C-D violates the consistency condition $h(C) \leq c(C,D) + h(D)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	D	F	G	H	I	S
$h(n)$	5	10	4	8	0	7	4	13
$h^*(n)$	7	13	14	10	0	9	5	17
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $S \rightarrow D$ ($13 > 3 + 4$), $C \rightarrow D$ ($10 > 5 + 4$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Adversarial Search] Answer: 13.49 to 13.51

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

• $A = 0.25 \times 16 + 0.25 \times 1 + 0.25 \times 17 + 0.25 \times 20 = 13.5$

• $B = 0.5 \times 2 + 0.5 \times 3 = 2.5$

• $C = 0.2 \times 4 + 0.4 \times 15 + 0.3 \times 16 + 0.1 \times 5 = 12.1$

Root = $\max(13.5, 2.5, 12.1) = 13.5$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.14 [MSQ, 2M · Uninformed Search] Answer: A, B, C (A: IDS is complete; B: IDS is optimal when all step costs are equal; C: The space complexity of IDS is $O(bd)$)

IDS combines DFS's memory use with BFS's level-by-level completeness. (A) is true: with finite b it eventually reaches limit d and finds the goal. (B) is true: like BFS, it finds the shallowest goal first, which is optimal for unit costs. (C) is true: each iteration is a DFS of depth at most d , so the space is $O(bd)$. (D) is false: the total $d \cdot b + (d-1)b^2 + \dots + b^d$ is still $O(b^d)$, because the deepest level dominates and the repeated shallow levels add only a constant factor of about $b/(b-1)$.

Q.15 [MSQ, 2M · Predicate Logic] Answer: A, C (A: $\forall y \exists x R(x, y)$; C: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$)

$R = \{(1,3), (2,1), (2,4), (3,2)\}$.

(A) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(B) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.

(C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(D) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $(\neg P \vee Q)$; D: $(\neg R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg Q \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(\neg P \vee Q)$ is derivable:

1. resolve $(\neg P \vee R)$ and $(Q \vee \neg R)$ on $R \Rightarrow (\neg P \vee Q)$

(C) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(D) $(\neg R)$ is derivable:

1. resolve $(\neg Q)$ and $(Q \vee \neg R)$ on $Q \Rightarrow (\neg R)$

Q.17 [NAT, 2M · Predicate Logic] Answer: 27

Taking $y = x$ gives $R(x, x) \rightarrow \neg R(x, x)$, so no diagonal pair may be in R (R is irreflexive). For each of the 3 unordered pairs $\{x, y\}$ with $x \neq y$, at most one of (x, y) , (y, x) is in R : 3 choices (neither, one, the other).

Total = $1 \times 3^3 = 27$ (asymmetric relations).

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, D (B: D is entailed; D: H is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $E \rightarrow H$, $C \rightarrow D$; add D , H .

Round 2: fire $D \rightarrow J$; add J .

Fixed point: $\{A, C, D, E, H, J\}$ — 6 atoms are entailed (facts: A, C, E ; newly derived: D, H, J). Atoms never derived: B, F, K .

Q.19 [NAT, 2M · Exact Inference] Answer: 0.248 to 0.252

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}, \text{Dyspnoea}) = P(\text{Smoker}) \cdot P(\text{Pollution} \mid \text{Smoker}) \cdot P(\text{Cancer} \mid \text{Pollution}) \cdot P(\text{XRay} \mid \text{Pollution}) \cdot P(\text{Dyspnoea} \mid \text{Smoker}, \text{XRay})$.

Inference by enumeration: $P(\text{Cancer} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cancer}, e, \text{hidden})$.

Hidden: Po, XR	Ca	product of CPT entries ($Sm \cdot Po \cdot Ca \cdot XR \cdot Dy$)	value
T, T	T	$0.05 \times 0.75 \times 0.25 \times 0.95 \times 0.1$	0.00089
T, F	T	$0.05 \times 0.75 \times 0.25 \times 0.05 \times 0.25$	0.00012
F, T	T	$0.05 \times 0.25 \times 0.25 \times 0.4 \times 0.1$	0.00013
F, F	T	$0.05 \times 0.25 \times 0.25 \times 0.6 \times 0.25$	0.00047
T, T	F	$0.05 \times 0.75 \times 0.75 \times 0.95 \times 0.1$	0.00267
T, F	F	$0.05 \times 0.75 \times 0.75 \times 0.05 \times 0.25$	0.00035
F, T	F	$0.05 \times 0.25 \times 0.75 \times 0.4 \times 0.1$	0.00038
F, F	F	$0.05 \times 0.25 \times 0.75 \times 0.6 \times 0.25$	0.00141

$P(\text{Cancer}=T, e) = 0.0016$; $P(\text{Cancer}=F, e) = 0.0048$; $P(e) = 0.00641$.

$P(\text{Cancer}=T \mid e) = 0.0016 / 0.00641 = \mathbf{0.25} \approx 0.250$.

Q.20 [MCQ, 2M · Sampling] Answer: A (A: $A=F$, $B=T$, $C=F$, $D=T$, $E=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.25$; $u = 0.44 \geq 0.25 \Rightarrow A = F$
- B: $P(B=T \mid A=F) = 0.25$; $u = 0.12 < 0.25 \Rightarrow B = T$
- C: $P(C=T \mid A=F, B=T) = 0.2$; $u = 0.53 \geq 0.2 \Rightarrow C = F$
- D: $P(D=T \mid C=F) = 0.8$; $u = 0.33 < 0.8 \Rightarrow D = T$
- E: $P(E=T \mid A=F, D=T) = 0.05$; $u = 0.4 \geq 0.05 \Rightarrow E = F$

Sample: (**A=F, B=T, C=F, D=T, E=F**).

Q.21 [MSQ, 2M · Conditional Independence] Answer: A, C (A: $G \perp E \mid A, D$; C: $F \perp D \mid A, E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp E \mid A, D$: **TRUE** — $G \leftarrow C \leftarrow A \rightarrow B \rightarrow D \rightarrow E$ (blocked: observed fork node A); $G \leftarrow C \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed fork node A); $G \leftarrow C \leftarrow A \rightarrow D \leftarrow B \rightarrow E$ (blocked: observed fork node A); $G \leftarrow C \leftarrow A \rightarrow D \rightarrow E$ (blocked: observed fork node A); $G \leftarrow C \rightarrow F \leftarrow E$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow D \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed chain node D) (+4 more paths, all blocked).

(B) $F \perp D \mid G$: **FALSE** — the path $F \leftarrow C \leftarrow A \rightarrow B \rightarrow D$ is active given $\{G\}$.

(C) $F \perp D \mid A, E$: **TRUE** — $F \leftarrow C \leftarrow A \rightarrow B \rightarrow D$ (blocked: observed fork node A); $F \leftarrow C \leftarrow A \rightarrow B \rightarrow E \leftarrow D$ (blocked: observed fork node A); $F \leftarrow C \leftarrow A \rightarrow D$ (blocked: observed fork node A); $F \leftarrow C \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed); $F \leftarrow E \leftarrow B \leftarrow A \rightarrow C \rightarrow G \leftarrow D$ (blocked: observed chain node E); $F \leftarrow E \leftarrow B \leftarrow A \rightarrow D$ (blocked: observed chain node E) (+2 more paths, all blocked).

(D) $A \perp C \mid G$: **FALSE** — the path $A \rightarrow B \rightarrow D \rightarrow G \leftarrow C$ is active given $\{G\}$.

Q.22 [NAT, 2M · Sampling] Answer: 0.748 to 0.752

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(B=T)=0.75$	0.75	T
2	$P(B=T)=0.75$	0.75	T
3	$P(B=T)=0.75$	0.75	T
4	$P(B=T)=0.75$	0.75	F

$\sum w (\text{all}) = 3$; $\sum w (A = T) = 2.25$.

Estimate $P(A=T \mid B = T) = 2.25 / 3 = \mathbf{0.75}$.

Q.23 [NAT, 2M · Conditional Independence] Answer: 0.263 to 0.265

$A \perp B \mid C$, so $P(a, b) = \sum_c P(c) P(a \mid c) P(b \mid c)$. c : $0.4 \times 0.9 \times 0.7 = 0.252$. $\neg c$: $0.6 \times 0.2 \times 0.1 = 0.012$. Sum = 0.264. Note $P(a) = 0.36 + 0.12 = 0.48$ and $P(b) = 0.28 + 0.06 = 0.34$, so $P(a)P(b) = 0.1632 \neq 0.264$: A and B are marginally dependent.

Q.24 [NAT, 2M · Probability for AI] Answer: 0.765 to 0.767

By Bayes rule with conditional independence, $P(d \mid +, +) = \alpha P(+ \mid d)^2 P(d)$. Unnormalized for d: $0.9^2 \times 0.01 = 0.81 \times 0.01 = 0.0081$. For $\neg d$: $0.05^2 \times 0.99 = 0.0025 \times 0.99 = 0.002475$. Normalizing: $0.0081 / (0.0081 + 0.002475) = 0.0081 / 0.010575 \approx 0.766$. (For comparison, a single positive test gives only $0.009/0.0585 \approx 0.154$.)

Q.25 [NAT, 2M · Exact Inference] Answer: 2

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, B)$, $f(B, C, E)$, $f(A, F)$, $f(B, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, B)$, $f(A, B)$, $f(A, F)$	$\{A, B, F\}$	$f(B, F)$
G	$f(B, F, G)$	$\{B, F, G\}$	$f(B, F)$
F	$f(B, F)$, $f(B, F)$	$\{B, F\}$	$f(B)$
B	$f(B, C)$, $f(B, C, E)$, $f(B)$	$\{B, C, E\}$	$f(C, E)$
C	$f(C, E)$	$\{C, E\}$	$f(E)$

The largest factor produced has **2** variable(s) (created when eliminating A); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over E and normalise.

Mock Test 012

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	2 / 9 / 14	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

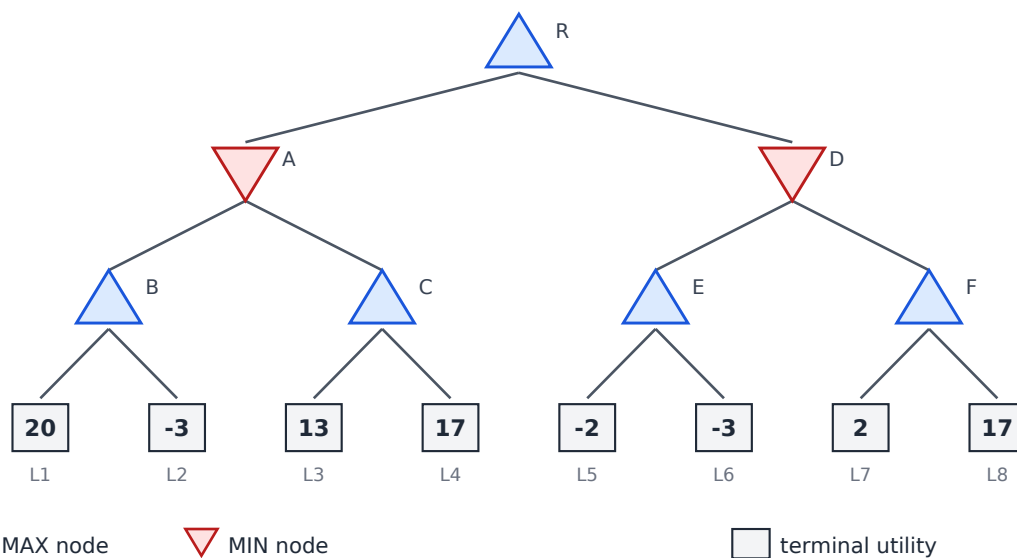
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search

Tic-tac-toe has 9 squares. If early wins are ignored and every game continues until all 9 squares are filled, how many distinct move sequences (orderings of the 9 moves) are there?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

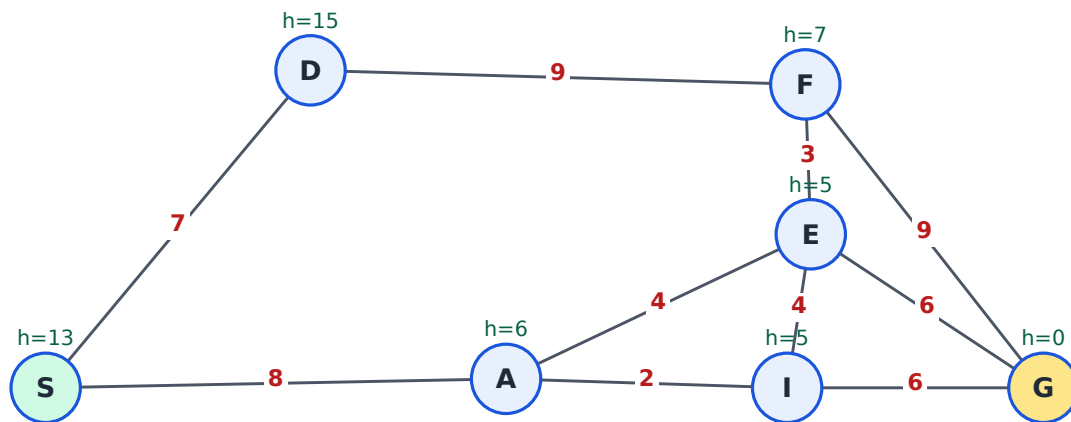
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

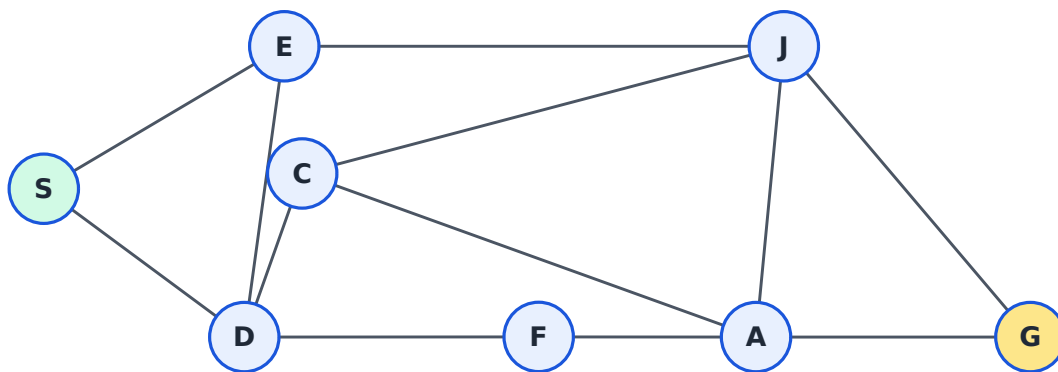
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, D, E, C, F, J, A, G
☐ (B) S, D, C, E, F, J, A, G
☐ (C) S, D, C, A, F, G
☐ (D) S, E, D, J, F, C, G

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are in conjunctive normal form (CNF)?

- ☐ (A) $(p \vee \neg q) \wedge r$
☐ (B) $\neg(p \wedge q)$
☐ (C) $p \vee \neg q \vee r$
☐ (D) $(p \wedge q) \vee r$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((R \wedge R) \leftrightarrow (Q \vee P)) \wedge (\neg R \wedge (R \wedge R)) ?$$

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In Existential Instantiation, the sentence $\exists x \text{Crown}(x) \wedge \text{OnHead}(x, \text{John})$ is replaced by $\text{Crown}(C_1) \wedge \text{OnHead}(C_1, \text{John})$. Which condition must C_1 satisfy?

- ☐ (A) C_1 must be a new constant symbol not appearing anywhere else in the KB
- ☐ (B) C_1 must be John
- ☐ (C) C_1 may be any ground term already in the KB
- ☐ (D) C_1 must be a variable

Q.8 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

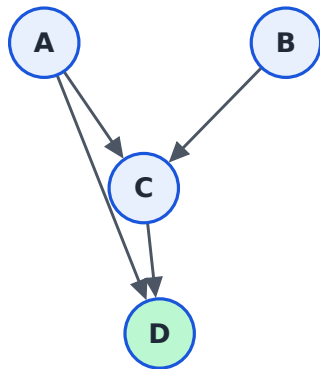
The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(C = T \mid D = F)$, how many of these samples are **rejected**?

#	A	B	C	D
1	F	F	F	F
2	F	F	T	T
3	F	F	F	F
4	F	F	F	F
5	F	T	T	T
6	F	F	F	F
7	F	F	F	F
8	F	F	F	T
9	F	F	F	F
10	T	T	F	T
11	F	T	T	F
12	F	F	F	F
13	T	F	F	F
14	T	F	T	F
15	F	F	F	T
16	F	F	F	F
17	F	F	F	F
18	T	F	T	T
19	F	T	T	F
20	T	F	T	T

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(D = T)$ (round off to 3 decimal places).



$P(A=T)$
0.9

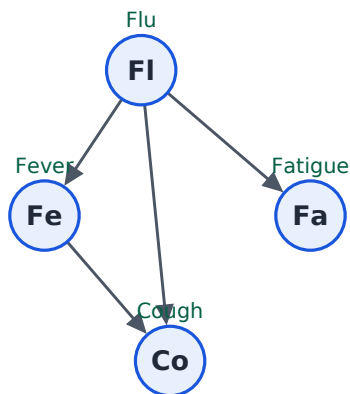
$P(B=T)$	A	B	$P(C=T \dots)$
0.5	T	T	0.6
	T	F	0.75
	F	T	0.25
	F	F	0.1

A	C	$P(D=T \dots)$
T	T	0.95
T	F	0.8
F	T	0.1
F	F	0.2

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order FI, Fe, Co, Fa. (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) What is the probability that a single run produces the sample **(FI=F, Fe=F, Co=T, Fa=F)**? (round off to 4 decimal places)



$P(FI=T)$
0.6

FI	$P(Fe=T ..)$	FI	Fe	$P(Co=T \dots)$
T	0.3	T	T	0.6
F	0.25	T	F	0.6
		F	T	0.5
		F	F	0.7

FI	$P(Fa=T \dots)$
T	0.95
F	0.4

Answer (numerical): _____

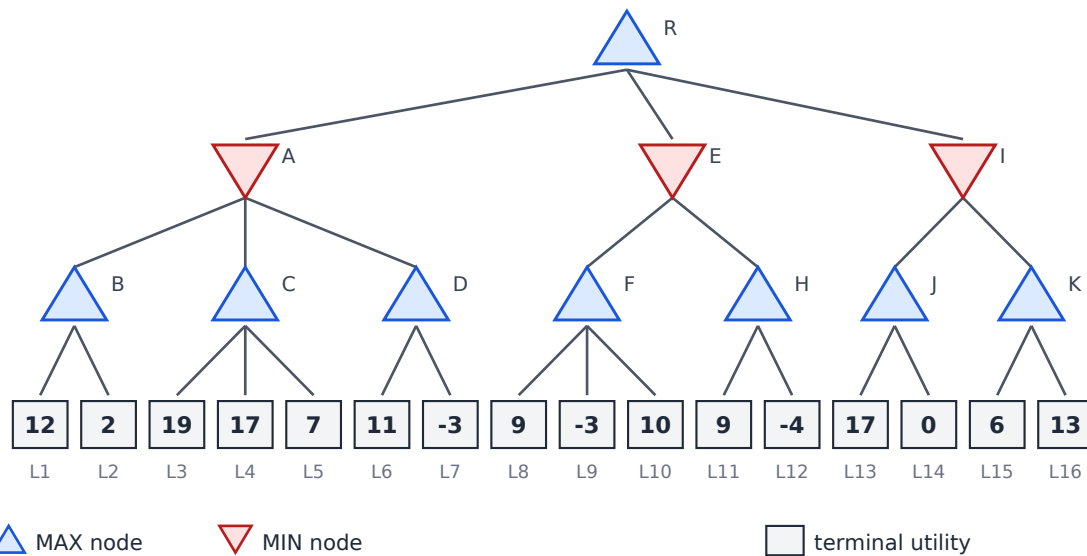
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search

Iterative deepening search runs on a complete binary tree of depth 3 (15 nodes) that contains **no** goal. It uses limits 0, 1, 2, 3 and then stops. A node visit is one goal test on one node, and repeated visits across iterations count separately. What is the total number of node visits?

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

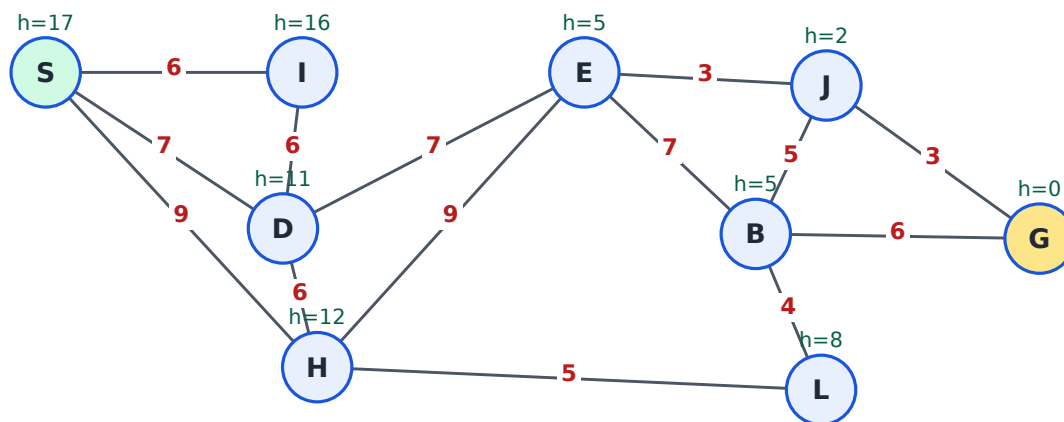
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – A*

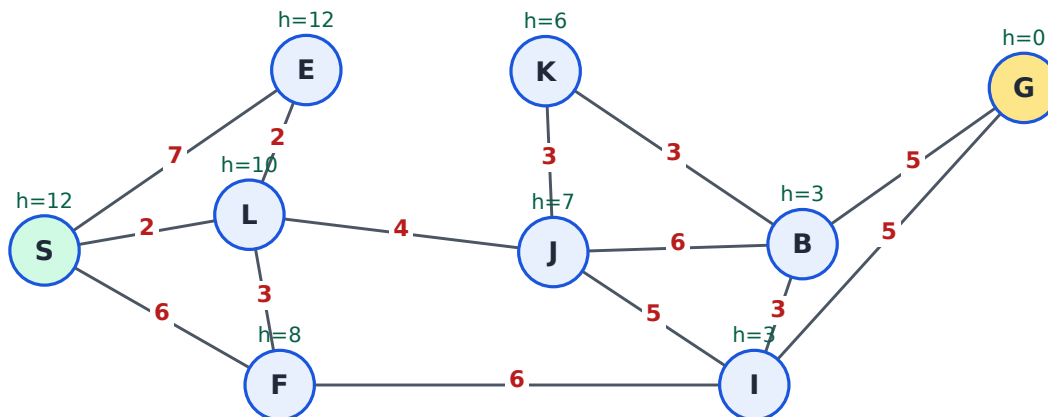
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node D is expanded.
- ☐ (B) Node H is expanded.
- ☐ (C) The path returned has cost 20.
- ☐ (D) Node J is expanded.

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node J is expanded.
- ☐ (B) Node L is expanded.
- ☐ (C) The path returned has cost 15.
- ☐ (D) Node E is expanded.

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \rightarrow R) \wedge (P \rightarrow R)) \rightarrow (Q \wedge P)$
- ☐ (B) $(R \rightarrow Q) \rightarrow ((R \vee P) \rightarrow Q)$
- ☐ (C) $((P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (P \rightarrow Q)$
- ☐ (D) $(R \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg R)$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **D**, **C** and rules:

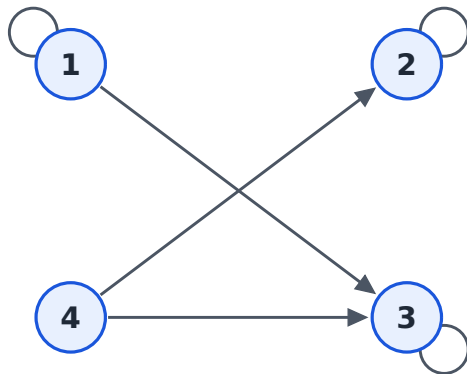
- R1: $B \rightarrow J$
- R2: $D \rightarrow J$
- R3: $J \rightarrow H$
- R4: $K \wedge F \wedge J \rightarrow C$
- R5: $B \rightarrow F$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) A is entailed
- ☐ (B) B is entailed
- ☐ (C) F is entailed
- ☐ (D) J is entailed

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
☐ (B) $\forall x R(x, x)$
☐ (C) $\exists y \forall x R(x, y)$
☐ (D) $\exists x \forall y R(x, y)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (\neg R \vee S)$, $C2: (\neg P \vee R)$, $C3: (\neg P \vee Q \vee S)$, $C4: (\neg P \vee \neg Q \vee \neg R)$, $C5: (\neg P \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P \vee S)$
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) (P)
☐ (D) $(\neg P)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic

To prove R from $KB = \{P \vee Q, P \rightarrow R, Q \rightarrow R\}$ by resolution refutation, the clause $\neg R$ is added to the clauses of KB . What is the minimum number of resolution steps needed to derive the empty clause? (Each step resolves two clauses on one complementary pair; duplicate literals are merged.)

Answer (numerical): _____

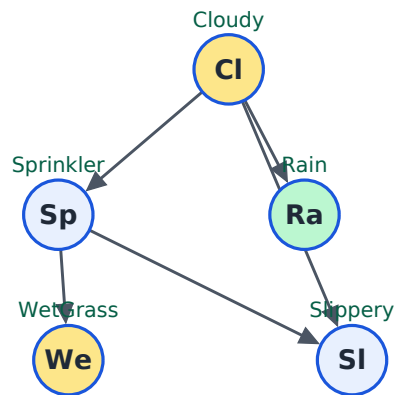
Q.20 [NAT · 2 marks · no negative marking] Probability for AI

A full joint distribution over Boolean variables X, Y, Z is: $P(x, y, z) = 0.10$, $P(x, y, \neg z) = 0.05$, $P(x, \neg y, z) = 0.15$, $P(x, \neg y, \neg z) = 0.10$, $P(\neg x, y, z) = 0.05$, $P(\neg x, y, \neg z) = 0.20$, $P(\neg x, \neg y, z) = 0.15$, $P(\neg x, \neg y, \neg z) = 0.20$. Compute $P(x \mid y \vee z)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: CI = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, SI = Slippery.) Compute $P(\text{Rain} = T \mid \text{WetGrass} = T, \text{Cloudy} = F)$ (round off to 3 decimal places).



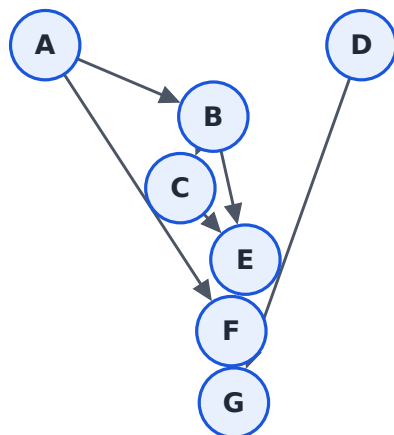
$P(\text{CI}=T)$	CI	$P(\text{Sp}=T \dots)$	CI	$P(\text{Ra}=T \dots)$
0.05	T	0.95	T	0.25
	F	0.05	F	0.25

Sp	$P(\text{We}=T \dots)$	CI	Sp	$P(\text{SI}=T \dots)$
T	0.5	T	T	0.5
F	0.3	T	F	0.5
		F	T	0.4
		F	F	0.9

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

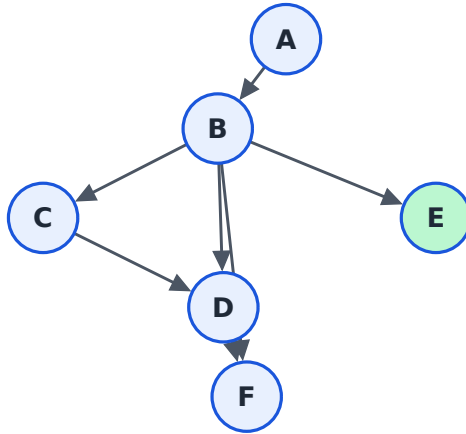
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp E$
☐ (B) $C \perp D$
☐ (C) $G \perp A \mid F$
☐ (D) $C \perp D \mid A$

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{E})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F, D, B, C, A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Bayesian Networks

A naive Bayes spam filter has $P(\text{spam}) = 0.3$ and three Boolean features with $P(f_1 | \text{spam}) = 0.8$, $P(f_1 | \text{ham}) = 0.1$; $P(f_2 | \text{spam}) = 0.6$, $P(f_2 | \text{ham}) = 0.3$; $P(f_3 | \text{spam}) = 0.2$, $P(f_3 | \text{ham}) = 0.5$. An email has $f_1 = \text{true}$, $f_2 = \text{false}$, $f_3 = \text{true}$. Compute $P(\text{spam} | f_1, \neg f_2, f_3)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network (DAG) with edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$, $E \rightarrow D$, $D \rightarrow F$, $C \rightarrow G$, $H \rightarrow G$. Which of the following nodes belong to the Markov blanket of C?

- ☐ (A) B
- ☐ (B) F
- ☐ (C) H
- ☐ (D) A

Answer Key — Mock Test 012

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	362880	14	MSQ	2	Informed Search	A, B, D
2	NAT	1	Adversarial Search	17	15	MSQ	2	Propositional Logic	C, D
3	NAT	1	Informed Search	18	16	MSQ	2	Propositional Logic	D
4	MCQ	1	Uninformed Search	A	17	MSQ	2	Predicate Logic	A
5	MSQ	1	Propositional Logic	A, C	18	MSQ	2	Propositional Logic	A, D
6	NAT	1	Propositional Logic	3	19	NAT	2	Propositional Logic	3
7	MCQ	1	Predicate Logic	A	20	NAT	2	Probability for AI	0.428 to 0.429
8	NAT	1	Sampling	7	21	NAT	2	Exact Inference	0.248 to 0.252
9	NAT	1	Exact Inference	0.827 to 0.831	22	MSQ	2	Conditional Independence	B, C, D
10	NAT	1	Sampling	0.1258 to 0.1262	23	NAT	2	Exact Inference	3
11	NAT	2	Uninformed Search	26	24	NAT	2	Bayesian Networks	0.439 to 0.44
12	NAT	2	Adversarial Search	12	25	MSQ	2	Bayesian Networks	A, C, D
13	MSQ	2	Informed Search	A, C, D					

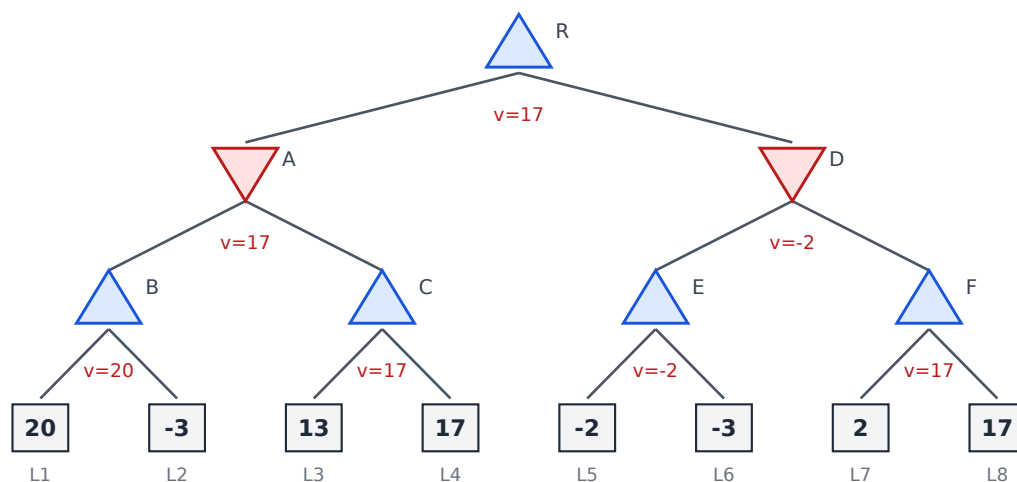
Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 362880

The first move has 9 choices, the second 8, and so on, giving $9! = 362880$ sequences. This is AIMA's upper bound on the size of the tic-tac-toe game tree. Games that end early make the actual tree smaller.

Q.2 [NAT, 1M · Adversarial Search] Answer: 17

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 20$, $C = 17$, $A = 17$, $E = -2$, $F = 17$, $D = -2$. Root value = **17**; MAX's optimal first move is towards A.

Q.3 [NAT, 1M · Informed Search] Answer: 18

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (13)	A(6), D(15)
2	A (6)	E(5), I(5), D(15)
3	E (5)	G(0), I(5), F(7), D(15)
4	G (0)	I(5), F(7), D(15) ← goal, stop

Path returned: **S** → **A** → **E** → **G**, cost = $8 + 4 + 6 = 18$.

The optimal cost is $h^*(S) = 16$ — greedy search is not optimal here.

Q.4 [MCQ, 1M · Uninformed Search] Answer: A (A: S, D, E, C, F, J, A, G)

Adjacency lists (alphabetical): A: {C, F, G, J}; C: {A, D, J}; D: {C, E, F, S}; E: {D, J, S}; F: {A, D}; G: {A, J}; J: {A, C, E, G}; S: {D, E}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[D, E]
2	D	[E, C, F]
3	E	[C, F, J]
4	C	[F, J, A]
5	F	[J, A]
6	J	[A, G]
7	A	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, D, E, C, F, J, A, G**. The path returned is $S \rightarrow E \rightarrow J \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [MSQ, 1M · Propositional Logic] Answer: A, C (A: $(p \vee \neg q) \wedge r$; C: $p \vee \neg q \vee r$)

A CNF is a conjunction of clauses, each a disjunction of literals (negation only on atoms). $(p \vee \neg q) \wedge r$ is two clauses. $p \vee \neg q \vee r$ is a single clause, which is a (one-clause) CNF. $\neg(p \wedge q)$ has negation applied to a compound formula, so it is not in CNF (its CNF is $\neg p \vee \neg q$). $(p \wedge q) \vee r$ has a conjunction inside a disjunction; its CNF is $(p \vee r) \wedge (q \vee r)$.

Q.6 [NAT, 1M · Propositional Logic] Answer: 3

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \wedge R) \leftrightarrow (Q \vee P)$	$\neg \neg R \wedge (R \wedge R)$	$((R \wedge R) \leftrightarrow (Q \vee P)) \wedge (\neg \neg R \wedge (R \wedge R))$
T	T	T	T	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	T	F	F	F	F
F	F	T	F	T	F
F	F	F	T	F	F

Rows where F = T: **3**.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A (A: C_1 must be a new constant symbol not appearing anywhere else in the KB)

Existential Instantiation introduces a fresh Skolem constant naming the (unknown) witness. Using an existing term such as John would assert, without justification, that John is the witness. The result is inferentially equivalent (satisfiable iff the original), not logically equivalent. A variable would make it a non-ground formula.

Q.8 [NAT, 1M · Sampling] Answer: 7

Rejection sampling keeps only samples consistent with the evidence $D = F$: samples 1, 3, 4, 6, 7, 9, 11, 12, 13, 14, 16, 17, 19 (13 accepted, 7 rejected).

Among the accepted samples, $C = T$ in 3. Estimate = $3/13 = \mathbf{0.2308}$.

(For this network the exact value is 0.3177; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.9 [NAT, 1M · Exact Inference] Answer: 0.827 to 0.831

Only the ancestors of D (A, B, C) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
A=T, B=T, C=T	0.2565
A=T, B=T, C=F	0.144

Assignment to other variables	joint $P(\dots, D=T)$
A=T, B=F, C=T	0.32062
A=T, B=F, C=F	0.09
A=F, B=T, C=T	0.00125
A=F, B=T, C=F	0.0075
A=F, B=F, C=T	0.0005
A=F, B=F, C=F	0.009

Sum = **0.8294** $\approx$ 0.829.

Q.10 [NAT, 1M · Sampling] **Answer: 0.1258 to 0.1262**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(FI=F) \times P(Fe=F|FI=F) \times P(Co=T|FI=F, Fe=F) \times P(Fa=F|FI=F) \\ = 0.4 \times 0.75 \times 0.7 \times 0.6 = \mathbf{0.126} \approx 0.1260.$$

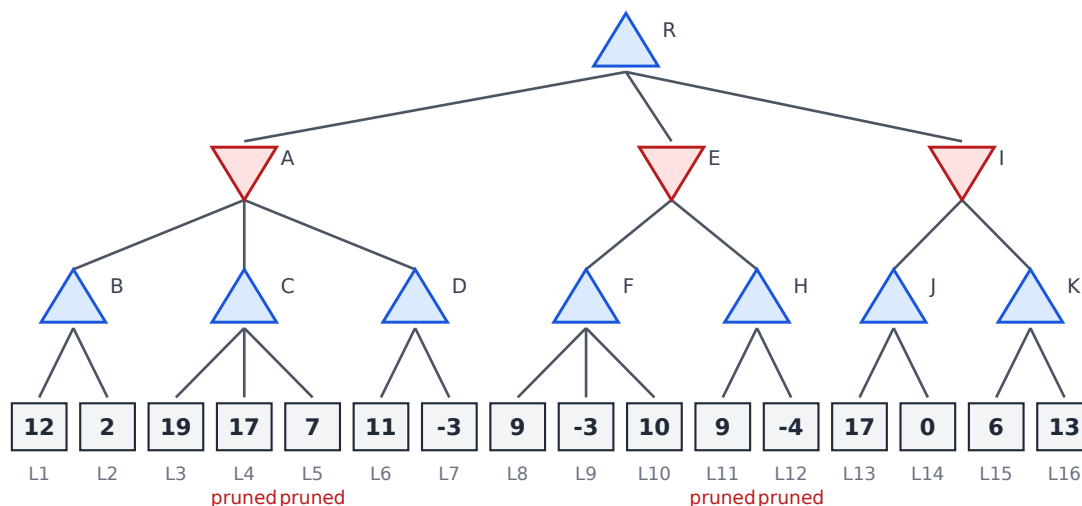
Q.11 [NAT, 2M · Uninformed Search] **Answer: 26**

The iteration with limit L visits the whole tree of depth L, which has $2^{L+1} - 1$ nodes. Limit 0: 1. Limit 1: 3. Limit 2: 7. Limit 3: 15. Total = $1 + 3 + 7 + 15 = \mathbf{26}$. For $b = 2$ the repetition overhead is largest, here 26 versus 15 nodes, which is still less than double.

Q.12 [NAT, 2M · Adversarial Search] **Answer: 12**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $19 \geq \beta = 12 \Rightarrow$ prune remaining children (leaves L4, L5).
- At MIN node E: value $10 \leq \alpha = 11 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (12): L1, L2, L3, L6, L7, L8, L9, L10, L13, L14, L15, L16. Leaves pruned (4): L4, L5, L11, L12. Root value = 13 (same as plain minimax, 13 — pruning never changes the root value).

Q.13 [MSQ, 2M · Informed Search] **Answer: A, C, D** (A: Node D is expanded.; C: The path returned has cost 20.; D: Node J is expanded.)

Heuristic values: $h(B)=5$, $h(D)=11$, $h(E)=5$, $h(G)=0$, $h(H)=12$, $h(I)=16$, $h(J)=2$, $h(L)=8$, $h(S)=17$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (17)	D(18), H(21), I(22)
2	D (18)	E(19), H(21), I(22)
3	E (19)	J(19), H(21), I(22), B(26)
4	J (19)	G(20), H(21), I(22), B(26)
5	G (20)	H(21), I(22), B(26) $\leftarrow$ goal, stop

Path returned: **S $\rightarrow$ D $\rightarrow$ E $\rightarrow$ J $\rightarrow$ G** with cost **20**; nodes expanded = **5** (S, D, E, J, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 20$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.14 [MSQ, 2M · Informed Search] **Answer: A, B, D** (A: Node J is expanded.; B: Node L is expanded.; D: Node E is expanded.)

Heuristic values: $h(B)=3$, $h(E)=12$, $h(F)=8$, $h(G)=0$, $h(I)=3$, $h(J)=7$, $h(K)=6$, $h(L)=10$, $h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	L(12), F(14), E(19)
2	L (12)	F(13), J(13), E(16)
3	F (13)	J(13), I(14), E(16)
4	J (13)	I(14), B(15), K(15), E(16)
5	I (14)	B(15), K(15), E(16), G(16)
6	B (15)	K(15), E(16), G(16)
7	K (15)	E(16), G(16)
8	E (16)	G(16)
9	G (16)	← goal, stop

Path returned: **S** → **L** → **F** → **I** → **G** with cost **16**; nodes expanded = **9** (S, L, F, J, I, B, K, E, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $((P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (P \rightarrow Q)$; D: $(R \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg R)$)

(A) $((Q \rightarrow R) \wedge (P \rightarrow R)) \rightarrow (Q \wedge P)$: **not valid** — counter-model $P=T, Q=F, R=T$.

(B) $(R \rightarrow Q) \rightarrow ((R \vee P) \rightarrow Q)$: **not valid** — counter-model $P=T, Q=F, R=F$.

(C) $((P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (P \rightarrow Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $(R \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg R)$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: D** (D: J is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \rightarrow J$; add J.

Round 2: fire $J \rightarrow H$; add H.

Fixed point: {C, D, H, J} — 4 atoms are entailed (facts: C, D; newly derived: H, J). Atoms never derived: A, B, E, F, K.

Q.17 [MSQ, 2M · Predicate Logic] **Answer: A** (A: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,1), (1,3), (2,2), (3,3), (4,2), (4,3)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

(B) $\forall x R(x, x)$: **FALSE** — no self-loop at 4.

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(\neg P \vee S)$; D: $(\neg P)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee S)$ is derivable:

1. resolve $(\neg R \vee S)$ and $(\neg P \vee R)$ on $R \Rightarrow (\neg P \vee S)$

(B) The resolution closure contains 11 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) (P) never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg P)$ is derivable:

1. resolve $(\neg R \vee S)$ and $(\neg P \vee \neg S)$ on $S \Rightarrow (\neg P \vee \neg R)$

2. resolve $(\neg P \vee R)$ and $(\neg P \vee \neg R)$ on $R \Rightarrow (\neg P)$

Q.19 [NAT, 2M · Propositional Logic] Answer: 3

Clauses: $P \vee Q, \neg P \vee R, \neg Q \vee R, \neg R$. A 3-step refutation: $(P \vee Q, \neg P \vee R) \Rightarrow Q \vee R$; $(Q \vee R, \neg Q \vee R) \Rightarrow R$; $(R, \neg R) \Rightarrow \square$. 2 steps are impossible: the last step must resolve two complementary unit clauses, and the only unit initially is $\neg R$. So R (a unit) must be produced in a single step, but every one-step resolvent is $Q \vee R, P \vee R, \neg P, \neg Q$ or a tautology — never R . Hence the minimum is 3 (a breadth-first search over resolution derivations confirms this).

Q.20 [NAT, 2M · Probability for AI] Answer: 0.428 to 0.429

$P(y \vee z) = 1 - P(\neg y \wedge \neg z) = 1 - [P(x, \neg y, \neg z) + P(\neg x, \neg y, \neg z)] = 1 - (0.10 + 0.20) = 0.70$. $P(x \wedge (y \vee z)) = P(x, y, z) + P(x, y, \neg z) + P(x, \neg y, z) = 0.10 + 0.05 + 0.15 = 0.30$. Therefore $P(x | y \vee z) = 0.30 / 0.70 \approx 0.4286 \approx 0.429$.

Q.21 [NAT, 2M · Exact Inference] Answer: 0.248 to 0.252

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}, \text{Slippery}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} | \text{Cloudy}) \cdot P(\text{Rain} | \text{Cloudy}) \cdot P(\text{WetGrass} | \text{Sprinkler}) \cdot P(\text{Slippery} | \text{Cloudy}, \text{Sprinkler})$.

Inference by enumeration: $P(\text{Rain} | e) = \alpha \sum_{\text{hidden}} P(\text{Rain}, e, \text{hidden})$.

Hidden: Sp, Sl	Ra	product of CPT entries (Cl · Sp · Ra · We · Sl)	value
T, T	T	$0.95 \times 0.05 \times 0.25 \times 0.5 \times 0.4$	0.00238
T, F	T	$0.95 \times 0.05 \times 0.25 \times 0.5 \times 0.6$	0.00356
F, T	T	$0.95 \times 0.95 \times 0.25 \times 0.3 \times 0.9$	0.06092
F, F	T	$0.95 \times 0.95 \times 0.25 \times 0.3 \times 0.1$	0.00677
T, T	F	$0.95 \times 0.05 \times 0.75 \times 0.5 \times 0.4$	0.00713
T, F	F	$0.95 \times 0.05 \times 0.75 \times 0.5 \times 0.6$	0.01069
F, T	F	$0.95 \times 0.95 \times 0.75 \times 0.3 \times 0.9$	0.18276
F, F	F	$0.95 \times 0.95 \times 0.75 \times 0.3 \times 0.1$	0.02031

$P(\text{Rain}=\text{T}, e) = 0.07363$; $P(\text{Rain}=\text{F}, e) = 0.22088$; $P(e) = 0.2945$.

$P(\text{Rain}=\text{T} | e) = 0.07363 / 0.2945 = \mathbf{0.25} \approx 0.250$.

Q.22 [MSQ, 2M · Conditional Independence] Answer: B, C, D (B: $C \perp D$; C: $G \perp A | F$; D: $C \perp D | A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $G \perp E$: **FALSE** — the path $G \leftarrow F \leftarrow A \rightarrow B \rightarrow C \rightarrow E$ is active given $\{ \}$.

(B) $C \perp D$: **TRUE** — $C \leftarrow B \leftarrow A \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed); $C \leftarrow B \rightarrow E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow E \leftarrow B \leftarrow A \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider E with neither it nor a descendant observed); $C \rightarrow E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed).

(C) $G \perp A | F$: **TRUE** — $G \leftarrow F \leftarrow A$ (blocked: observed chain node F); $G \leftarrow F \leftarrow E \leftarrow B \leftarrow A$ (blocked: observed chain node F); $G \leftarrow F \leftarrow E \leftarrow C \leftarrow B \leftarrow A$ (blocked: observed chain node F).

(D) $C \perp D | A$: **TRUE** — $C \leftarrow B \leftarrow A \rightarrow F \rightarrow G \leftarrow D$ (blocked: observed fork node A); $C \leftarrow B \rightarrow E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow E \leftarrow B \leftarrow A \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider E with neither it nor a descendant observed); $C \rightarrow E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed).

Q.23 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(B, C), f(B, C, D), f(B, E), f(B, D, F)$.

Eliminating Z : multiply all factors whose scope contains Z , then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(B, D, F)$	$\{B, D, F\}$	$f(B, D)$
D	$f(B, C, D), f(B, D)$	$\{B, C, D\}$	$f(B, C)$
B	$f(A, B), f(B, C), f(B, E), f(B, C)$	$\{A, B, C, E\}$	$f(A, C, E)$
C	$f(A, C, E)$	$\{A, C, E\}$	$f(A, E)$
A	$f(A), f(A, E)$	$\{A, E\}$	$f(E)$

The largest factor produced has **3** variable(s) (created when eliminating B); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over E and normalise.

Q.24 [NAT, 2M · Bayesian Networks] **Answer: 0.439 to 0.44**

Spam: $0.3 \times 0.8 \times (1 - 0.6) \times 0.2 = 0.3 \times 0.8 \times 0.4 \times 0.2 = 0.0192$. Ham: $0.7 \times 0.1 \times (1 - 0.3) \times 0.5 = 0.7 \times 0.1 \times 0.7 \times 0.5 = 0.0245$. Normalize: $0.0192 / (0.0192 + 0.0245) = 0.0192 / 0.0437 \approx 0.4394 \approx 0.439$. Note $P(\neg f_2 \mid \text{class}) = 1 - P(f_2 \mid \text{class})$ must be used for the false feature.

Q.25 [MSQ, 2M · Bayesian Networks] **Answer: A, C, D** (A: B; C: H; D: A)

Parents of C: {A}. Children of C: {D, G}. Other parents of the children: D has parents B, C, E and G has parents C, H, giving {B, E, H}. Markov blanket of C = {A, B, D, E, G, H}. So A (parent), B (co-parent via D) and H (co-parent via G) belong to it. F is a child of D, i.e. a grandchild of C, and is not in the blanket: given D, F is independent of C.

Mock Test 013

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 8 / 10	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

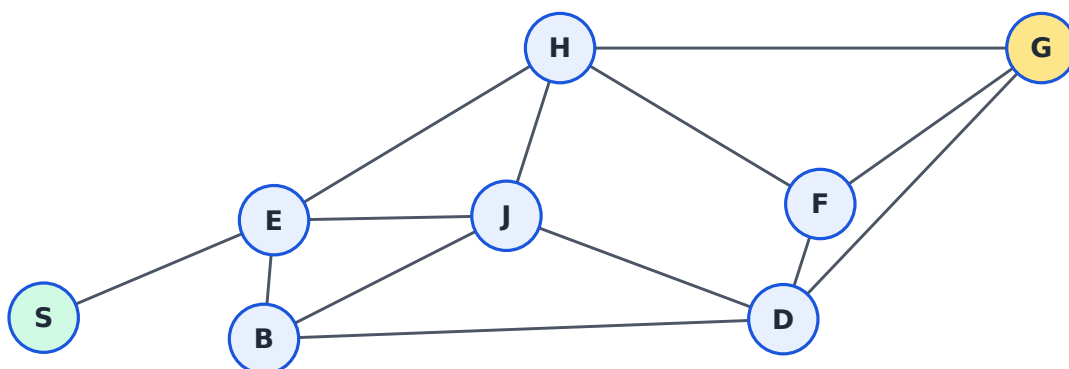
Q.1 [MSQ · 1 mark · no negative marking] Uninformed Search

Every action has the same positive cost. Which of the following algorithms are guaranteed to return an optimal solution on a problem with a finite branching factor that has a solution?

- ☐ (A) Breadth-first search
- ☐ (B) Iterative deepening search
- ☐ (C) Uniform-cost search
- ☐ (D) Depth-first search (graph search)

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, E, B, D, F, G
- ☐ (B) S, E, D, B, F, G
- ☐ (C) S, E, B, H, J, D, F, G
- ☐ (D) S, E, J, H, G

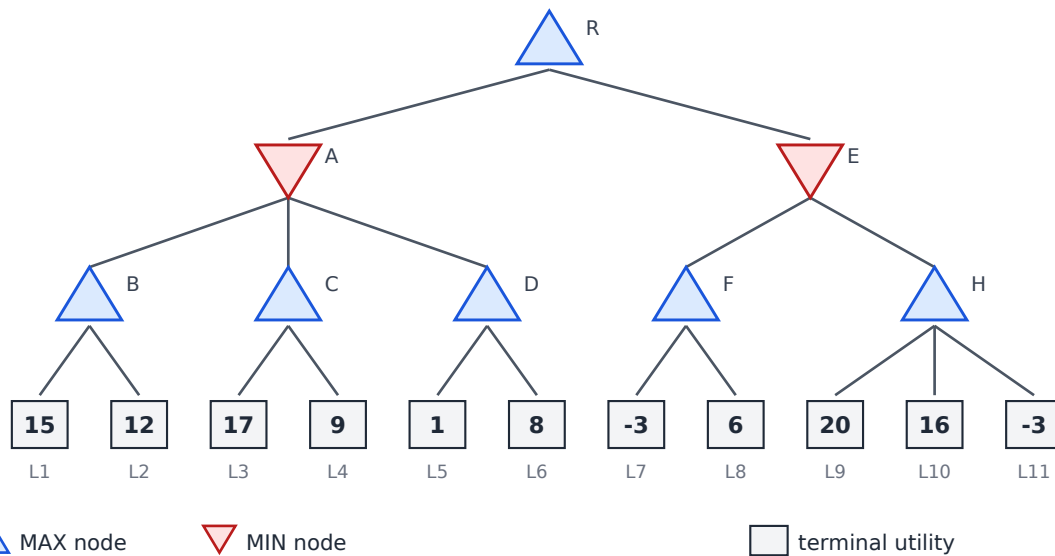
Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 4$ (so $b^d = 256$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to A; root value 8
☐ (B) Move to A; root value 20
☐ (C) Move to A; root value 9
☐ (D) Move to E; root value 6

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q). For how many of them is $\exists x P(x) \wedge \exists x Q(x)$ true?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In the formula $\forall x (P(x, y) \rightarrow \exists y Q(x, y, z))$, which variables have free occurrences?

- ☐ (A) y and z
☐ (B) z only
☐ (C) x, y and z
☐ (D) x and z

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (\neg P \wedge P) \vee ((P \wedge R) \vee Q) ?$$

Answer (numerical): _____

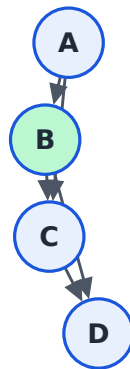
Q.8 [MSQ · 1 mark · no negative marking] Propositional Logic

In the DPLL algorithm, a pure symbol is one that appears with the same sign in every clause in which it occurs. For the clause set $\{A \vee \neg B, \neg B \vee \neg C, C \vee A\}$, which symbols are pure?

- ☐ (A) A
☐ (B) B
☐ (C) C
☐ (D) None of the symbols is pure

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(B = T)$ (round off to 3 decimal places).



A	P(A=T)
T	0.75

A	P(B=T .)	A	B	P(C=T ...)
T	0.2	T	T	0.9
F	0.05	T	F	0.7
		F	T	0.05
		F	F	0.2

B	C	P(D=T ...)
T	T	0.05
T	F	0.25
F	T	0.3
F	F	0.3

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

Which statement is the local Markov property (local semantics) of a Bayesian network?

- ☐ (A) Each node is conditionally independent of its non-descendants given its parents
- ☐ (B) Each node is conditionally independent of its descendants given its parents
- ☐ (C) Each node is marginally independent of all nodes that are not its parents
- ☐ (D) Each node is conditionally independent of its parents given its children

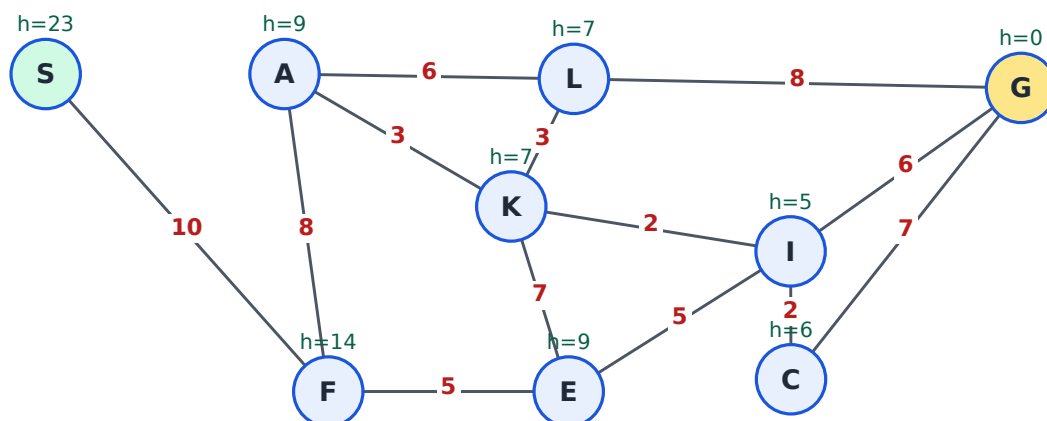
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Local Search

In simulated annealing, a randomly chosen move worsens the objective (which is being maximised) by $\Delta E = -2$ at temperature $T = 4$. With what probability is this move accepted? Give the answer to two decimal places.

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

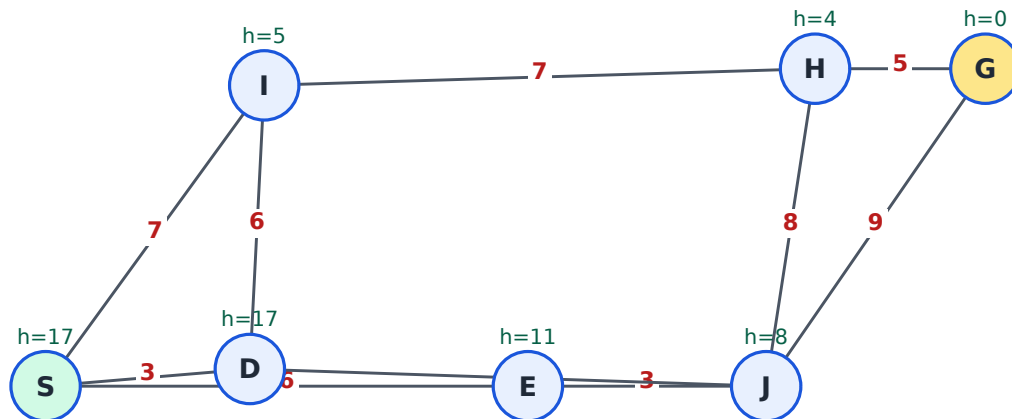
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

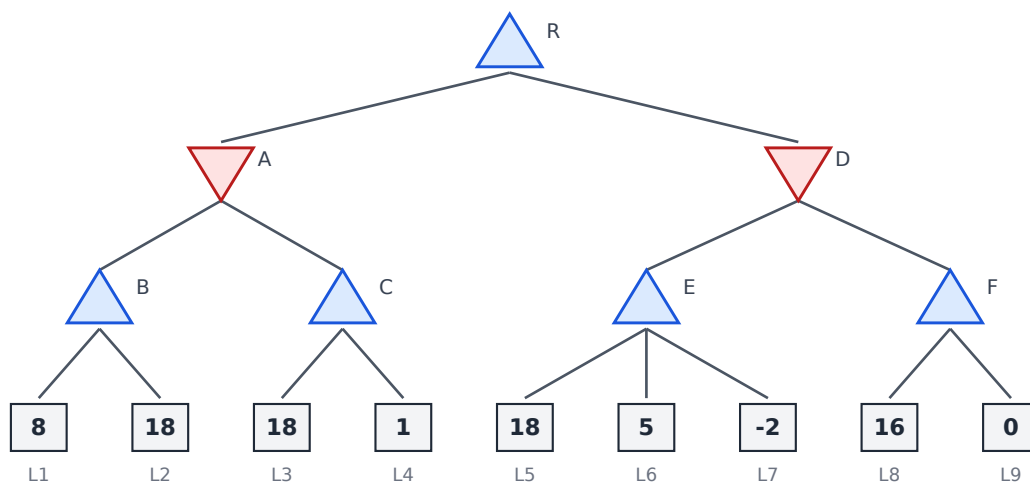
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(J)$ overestimates the true cost from J to G.
☐ (B) h is consistent (monotone).
☐ (C) h is admissible.
☐ (D) The edge S-I violates the consistency condition $h(S) \leq c(S, I) + h(I)$.

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

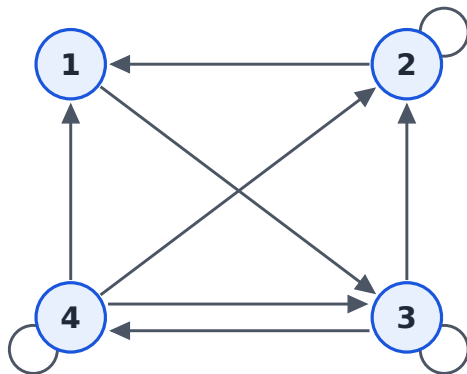
- (1) $(Q \vee R) \wedge (P \wedge Q)$
- (2) $(P \vee R) \wedge R$
- (3) $\neg(\neg Q \vee \neg P)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $(\neg Q \vee \neg P) \vee \neg Q$
☐ (B) $(Q \wedge R) \vee P$
☐ (C) $\neg R$
☐ (D) $\neg R \vee (R \leftrightarrow Q)$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
☐ (B) $\exists x \forall y R(x, y)$
☐ (C) $\exists y \forall x R(x, y)$
☐ (D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$
☐ (B) $\neg P \rightarrow (Q \rightarrow \neg P)$
☐ (C) $(\neg Q \rightarrow R) \leftrightarrow (\neg R \rightarrow \neg \neg Q)$
☐ (D) $(Q \rightarrow P) \rightarrow (P \rightarrow Q)$

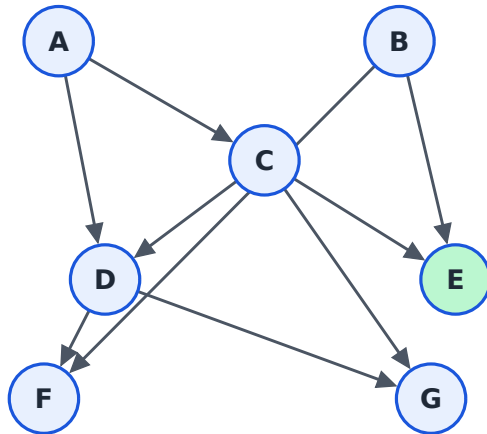
Q.18 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ satisfy all three of $\forall x R(x, x)$, $\forall x \forall y (R(x, y) \rightarrow R(y, x))$ and $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$?

Answer (numerical): _____

Q.19 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

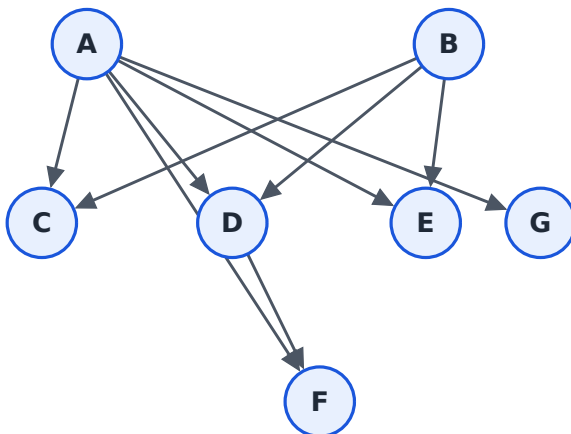
Variable elimination is used to compute $P(\mathbf{E})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, C, F, D, G, A**. What is the scope of the new factor produced when **C** is summed out?



- ☐ (A) $f(A, D, F, G)$
- ☐ (B) $f(D, E, F, G)$
- ☐ (C) $f(A, C, D, E, F, G)$
- ☐ (D) $f(A, D, E, F, G)$

Q.20 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

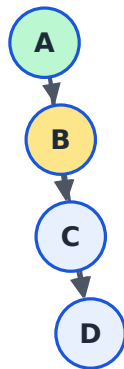
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp E \mid A, B$
- ☐ (B) $B \perp C \mid A, F$
- ☐ (C) $D \perp F \mid E$
- ☐ (D) $A \perp G$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(A = T \mid B = F)$ (round off to 3 decimal places).



A	P(A=T)
T	0.4

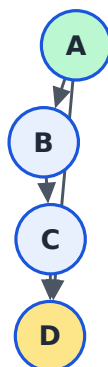
A	P(B=T .)	A	B	P(C=T ...)
T	0.6	T	T	0.3
F	0.75	T	F	0.05
		F	T	0.1
		F	F	0.2

A	C	P(D=T ...)
T	T	0.9
T	F	0.05
F	T	0.6
F	F	0.95

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $D = F$. The values of the non-evidence variables in the generated samples are listed in the table. What is the weight of sample #4? (round off to 4 decimal places)



A	P(A=T)
T	0.75

A	P(B=T ...)	B	P(C=T ...)
T	0.2	T	0.1
F	0.25	F	0.05

A	C	P(D=T ...)
T	T	0.7
T	F	0.75
F	T	0.7
F	F	0.2

#	A	B	C
1	T	T	F
2	T	T	F
3	T	F	F
4	T	T	F
5	T	T	F
6	F	F	F

Answer (numerical): _____

Q.23 [MCQ · 2 marks · negative: -2/3] Conditional Independence

Consider the Bayesian network (DAG) with edges $P \rightarrow Q$, $P \rightarrow R$, $Q \rightarrow S$, $R \rightarrow S$, $S \rightarrow T$, $U \rightarrow R$. Which pair of variables is **NOT** d-separated by the given set?

- ☐ (A) T and P given {Q, R}
☐ (B) U and Q given {P}
☐ (C) U and P given {T}
☐ (D) Q and R given {P}

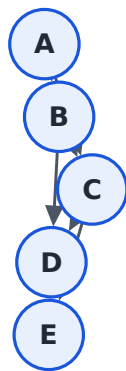
Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence

X and Y are independent fair coin flips (Boolean) and $Z = X \text{ XOR } Y$. Which statements are true?

- ☐ (A) $X \perp Z$
☐ (B) $Y \perp Z$
☐ (C) $X \perp Y \mid Z$
☐ (D) $X \perp \{Y, Z\}$

Q.25 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = \{0.49, 0.88, 0.59, 0.81, 0.66\}$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.75

A	P(B=T ...)	A	B	P(C=T ...)
T	0.7	T	T	0.95
F	0.5	T	F	0.7
		F	T	0.05
		F	F	0.95

B	C	P(D=T ...)	D	P(E=T ...)
T	T	0.9	T	0.9
T	F	0.7	F	0.3
F	T	0.8	T	0.1
F	F	0.3	F	0.8

- ☐ (A) A=T, B=F, C=T, D=F, E=F
☐ (B) A=F, B=F, C=T, D=F, E=F
☐ (C) A=F, B=T, C=T, D=F, E=T
☐ (D) A=T, B=T, C=T, D=F, E=F

Answer Key — Mock Test 013

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MSQ	1	Uninformed Search	A, B, C	14	NAT	2	Adversarial Search	6
2	MCQ	1	Uninformed Search	A	15	MSQ	2	Propositional Logic	B, D
3	NAT	1	Adversarial Search	31	16	MSQ	2	Predicate Logic	A, B
4	MCQ	1	Adversarial Search	A	17	MSQ	2	Propositional Logic	B, C
5	NAT	1	Predicate Logic	225	18	NAT	2	Predicate Logic	5
6	MCQ	1	Predicate Logic	A	19	MCQ	2	Exact Inference	D
7	NAT	1	Propositional Logic	5	20	MSQ	2	Conditional Independence	A
8	MSQ	1	Propositional Logic	A, B	21	NAT	2	Exact Inference	0.514 to 0.518
9	NAT	1	Exact Inference	0.161 to 0.165	22	NAT	2	Sampling	0.2498 to 0.2502
10	MCQ	1	Bayesian Networks	A	23	MCQ	2	Conditional Independence	C
11	NAT	2	Local Search	0.6 to 0.61	24	MSQ	2	Conditional Independence	A, B
12	NAT	2	Informed Search	26	25	MCQ	2	Sampling	A
13	MSQ	2	Informed Search	C, D					

Detailed Solutions

Q.1 [MSQ, 1M · Uninformed Search] Answer: A, B, C (A: Breadth-first search; B: Iterative deepening search; C: Uniform-cost search)

With equal step costs, the cheapest solution is the shallowest one. BFS (A) and IDS (B) find the shallowest goal first. UCS (C) is optimal for any positive step costs. DFS (D) can follow a deep branch and return a deep goal before discovering a shallower one, so it is not optimal.

Q.2 [MCQ, 1M · Uninformed Search] Answer: A (A: S, E, B, D, F, G)

Adjacency lists (alphabetical): B: {D, E, J}; D: {B, F, G, J}; E: {B, H, J, S}; F: {D, G, H}; G: {D, F, H}; H: {E, F, G, J}; J: {B, D, E, H}; S: {E}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, E, B, D, F, G**; the path found is $S \rightarrow E \rightarrow B \rightarrow D \rightarrow F \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

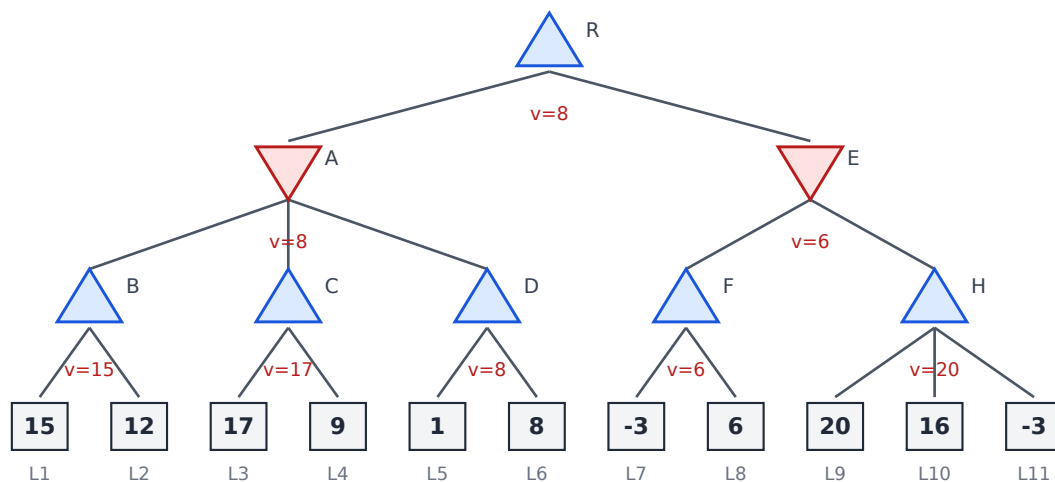
Q.3 [NAT, 1M · Adversarial Search] Answer: 31

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 4^2 + 4^2 - 1 = 16 + 16 - 1 = 31$, versus 256 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.4 [MCQ, 1M · Adversarial Search] Answer: A (A: Move to A; root value 8)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 15, C = 17, D = 8, A = 8, F = 6, H = 20, E = 6. Root value = **8**; MAX's optimal first move is towards A.

Q.5 [NAT, 1M · Predicate Logic] Answer: 225

Interpret P and Q as subsets of the domain; each of P, Q must be non-empty $\Rightarrow (2^n - 1)^2$. With $n = 4$ this gives **225** (verified by enumerating all interpretations).

Q.6 [MCQ, 1M · Predicate Logic] Answer: A (A: y and z)

x is bound by $\forall x$ throughout. The occurrence of y in $P(x, y)$ is not in the scope of any $\exists y/\forall y$, so it is free. The y in $Q(x, y, z)$ is bound by $\exists y$. z is never quantified, so it is free. Hence the variables with free occurrences are y and z.

Q.7 [NAT, 1M · Propositional Logic] Answer: 5

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg P \wedge P$	$(P \wedge R) \vee Q$	$(\neg P \wedge P) \vee ((P \wedge R) \vee Q)$
T	T	T	F	T	T
T	T	F	F	T	T
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	F	T	T
F	T	F	F	T	T
F	F	T	F	F	F

P	Q	R	$\neg P \wedge P$	$(P \wedge R) \vee Q$	$(\neg P \wedge P) \vee ((P \wedge R) \vee Q)$
F	F	F	F	F	F

Rows where F = T: **5**.

Q.8 [MSQ, 1M · Propositional Logic] **Answer: A, B** (A: A; B: B)

A occurs only positively (in $A \vee \neg B$ and $C \vee A$), so it is pure; DPLL can set $A = T$. B occurs only negatively ($\neg B$ twice), so it is pure; DPLL can set $B = F$. C occurs negatively in $\neg B \vee \neg C$ and positively in $C \vee A$, so it is not pure.

Q.9 [NAT, 1M · Exact Inference] **Answer: 0.161 to 0.165**

Only the ancestors of B (A) matter: every other variable sums out to 1.

$P(B=T) = \sum \text{over its ancestors of } P(B=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, B=T)$
A=T, C=T, D=T	0.00675
A=T, C=T, D=F	0.12825
A=T, C=F, D=T	0.00375
A=T, C=F, D=F	0.01125
A=F, C=T, D=T	0.00003
A=F, C=T, D=F	0.00059
A=F, C=F, D=T	0.00297
A=F, C=F, D=F	0.00891

Sum = **0.1625** $\approx$ 0.163.

Q.10 [MCQ, 1M · Bayesian Networks] **Answer: A** (A: Each node is conditionally independent of its non-descendants given its parents)

AIMA: a node is conditionally independent of its non-descendants given its parents; this is equivalent to the global factorization semantics. (B) is false because descendants (e.g. children) carry information about the node even given the parents. (C) ignores dependence through common ancestors, and (D) has no basis.

Q.11 [NAT, 2M · Local Search] **Answer: 0.6 to 0.61**

A worsening move is accepted with probability $e^{\Delta E/T} = e^{-2/4} = e^{-0.5} \approx$ **0.6065**, which rounds to 0.61. Lower temperatures or larger losses make acceptance less likely. Improving moves ($\Delta E > 0$) are always accepted.

Q.12 [NAT, 2M · Informed Search] **Answer: 26**

Heuristic values: $h(A)=9$, $h(C)=6$, $h(E)=9$, $h(F)=14$, $h(G)=0$, $h(I)=5$, $h(K)=7$, $h(L)=7$, $h(S)=23$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (23)	F(24)
2	F (24)	E(24), A(27)
3	E (24)	I(25), A(27), K(29)
4	I (25)	G(26), A(27), C(28), K(29)
5	G (26)	A(27), C(28), K(29) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **F** $\rightarrow$ **E** $\rightarrow$ **I** $\rightarrow$ **G** with cost **26**; nodes expanded = **5** (S, F, E, I, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 26$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.13 [MSQ, 2M · Informed Search] **Answer: C, D** (C: h is admissible.; D: The edge S-I violates the consistency condition $h(S) \leq c(S,I) + h(I)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	D	E	G	H	I	J	S
$h(n)$	17	11	0	4	5	8	17
$h^*(n)$	18	12	0	5	12	9	18
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

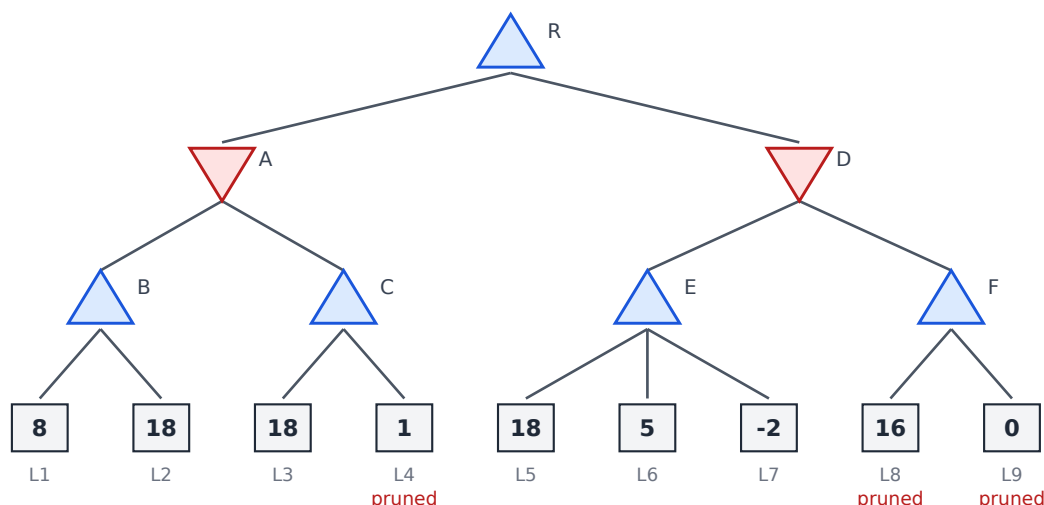
Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $D \rightarrow I$ ($17 > 6 + 5$), $S \rightarrow I$ ($17 > 7 + 5$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [NAT, 2M · Adversarial Search] Answer: 6

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $18 \geq \beta = 18 \Rightarrow$ prune remaining children (leaves L4).
- At MIN node D: value $18 \leq \alpha = 18 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (6): L1, L2, L3, L5, L6, L7. Leaves pruned (3): L4, L8, L9. Root value = 18 (same as plain minimax, 18 — pruning never changes the root value).

Q.15 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $(Q \wedge R) \vee P$; D: $\neg R \vee (R \leftrightarrow Q)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
T	T	T	T

(A) $(\neg Q \vee \neg P) \vee \neg Q$ is false in the KB-model $P=T, Q=T, R=T \Rightarrow$ not entailed.

(B) $(Q \wedge R) \vee P$ is true in all models of KB $\Rightarrow$ **entailed**.

(C) $\neg R$ is false in the KB-model $P=T, Q=T, R=T \Rightarrow$ not entailed.

(D) $\neg R \vee (R \leftrightarrow Q)$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.16 [MSQ, 2M · Predicate Logic] Answer: A, B (A: $\forall x \exists y R(x, y)$; B: $\exists x \forall y R(x, y)$)

$R = \{(1,3), (2,1), (2,2), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4)\}$.

(A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(B) $\exists x \forall y R(x, y)$: **TRUE** — $x = 4$ has edges to every y .

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(2,1) \wedge R(1,3)$ but $\neg R(2,3)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] Answer: B, C (B: $\neg P \rightarrow (Q \rightarrow \neg P)$; C: $(\neg Q \rightarrow R) \leftrightarrow (\neg R \rightarrow \neg \neg Q)$)

(A) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$: **not valid** — counter-model $P=T, Q=F, R=T$.

(B) $\neg P \rightarrow (Q \rightarrow \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $(\neg Q \rightarrow R) \leftrightarrow (\neg R \rightarrow \neg \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $(Q \rightarrow P) \rightarrow (P \rightarrow Q)$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [NAT, 2M · Predicate Logic] Answer: 5

These are exactly the equivalence relations on a 3-element set, which correspond one-to-one with its partitions. The number of partitions of a 3-set is the Bell number $B_3 = 5$: $\{123\}, \{1|23\}, \{2|13\}, \{3|12\}, \{1|2|3\}$. Enumeration of all 512 relations confirms 5.

Q.19 [MCQ, 2M · Exact Inference] Answer: D (D: $f(A, D, E, F, G)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(B), f(A, C), f(A, C, D), f(B, C, E), f(B, D, F), f(C, D, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(B), f(B, C, E), f(B, D, F)$	$\{B, C, D, E, F\}$	$f(C, D, E, F)$
C	$f(A, C), f(A, C, D), f(C, D, G), f(C, D, E, F)$	$\{A, C, D, E, F, G\}$	$f(A, D, E, F, G)$
F	$f(A, D, E, F, G)$	$\{A, D, E, F, G\}$	$f(A, D, E, G)$
D	$f(A, D, E, G)$	$\{A, D, E, G\}$	$f(A, E, G)$
G	$f(A, E, G)$	$\{A, E, G\}$	$f(A, E)$
A	$f(A), f(A, E)$	$\{A, E\}$	$f(E)$

The largest factor produced has **5** variable(s) (created when eliminating C); with Boolean variables it has $2^5 = 32$ entries. Finally, multiply the remaining factors over E and normalise.

Q.20 [MSQ, 2M · Conditional Independence] **Answer: A** ($A: C \perp E \mid A, B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp E \mid A, B$: **TRUE** — $C \leftarrow A \rightarrow D \leftarrow B \rightarrow E$ (blocked: observed fork node A); $C \leftarrow A \rightarrow E$ (blocked: observed fork node A); $C \leftarrow A \rightarrow F \leftarrow D \leftarrow B \rightarrow E$ (blocked: observed fork node A); $C \leftarrow B \rightarrow D \leftarrow A \rightarrow E$ (blocked: observed fork node B); $C \leftarrow B \rightarrow D \rightarrow F \leftarrow A \rightarrow E$ (blocked: observed fork node B); $C \leftarrow B \rightarrow E$ (blocked: observed fork node B).

(B) $B \perp C \mid A, F$: **FALSE** — the path $B \rightarrow C$ is active given $\{A, F\}$.

(C) $D \perp F \mid E$: **FALSE** — the path $D \leftarrow A \rightarrow F$ is active given $\{E\}$.

(D) $A \perp G$: **FALSE** — the path $A \rightarrow G$ is active given $\{\}$.

Q.21 [NAT, 2M · Exact Inference] **Answer: 0.514 to 0.518**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B \mid A) \cdot P(C \mid A, B) \cdot P(D \mid A, C)$.

Inference by enumeration: $P(A \mid e) = \alpha \sum_{\text{hidden}} P(A, e, \text{hidden})$.

Hidden: C, D	A	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.4 \times 0.4 \times 0.05 \times 0.9$	0.0072
T, F	T	$0.4 \times 0.4 \times 0.05 \times 0.1$	0.0008
F, T	T	$0.4 \times 0.4 \times 0.95 \times 0.05$	0.0076
F, F	T	$0.4 \times 0.4 \times 0.95 \times 0.95$	0.1444
T, T	F	$0.6 \times 0.25 \times 0.2 \times 0.6$	0.018
T, F	F	$0.6 \times 0.25 \times 0.2 \times 0.4$	0.012
F, T	F	$0.6 \times 0.25 \times 0.8 \times 0.95$	0.114
F, F	F	$0.6 \times 0.25 \times 0.8 \times 0.05$	0.006

$P(A=T, e) = 0.16$; $P(A=F, e) = 0.15$; $P(e) = 0.31$.

$P(A=T \mid e) = 0.16 / 0.31 = \mathbf{0.5161} \approx 0.516$.

Q.22 [NAT, 2M · Sampling] **Answer: 0.2498 to 0.2502**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(D=F \mid A=T, C=F) = 0.25$	0.25	T
2	$P(D=F \mid A=T, C=F) = 0.25$	0.25	T
3	$P(D=F \mid A=T, C=F) = 0.25$	0.25	T
4	$P(D=F \mid A=T, C=F) = 0.25$	0.25	T
5	$P(D=F \mid A=T, C=F) = 0.25$	0.25	T
6	$P(D=F \mid A=F, C=F) = 0.8$	0.8	F

$\sum w (\text{all}) = 2.05$; $\sum w (A = T) = 1.25$.

Estimate $P(A=T \mid D = F) = 1.25 / 2.05 = \mathbf{0.6098}$.

Q.23 [MCQ, 2M · Conditional Independence] **Answer: C** ($C: U$ and P given $\{T\}$)

(C): T is a descendant of collider R on the path $U \rightarrow R \leftarrow P$ ($R \rightarrow S \rightarrow T$), so observing T activates it: not d-separated. (A): every path from T to P goes through Q or R as non-colliders ($T \leftarrow S \leftarrow Q \leftarrow P$, $T \leftarrow S \leftarrow R \leftarrow P$), both observed: blocked. (B): $U \rightarrow R \leftarrow P \rightarrow Q$ is blocked at observed P and at unobserved collider R; $U \rightarrow R \rightarrow S \leftarrow Q$ is blocked at unobserved collider S. (D): fork P is observed and collider S is unobserved with no observed descendants: blocked.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $X \perp Z$; B: $Y \perp Z$)

$P(z) = 0.5$ and $P(x, z) = P(x, \neg y) = 0.25 = P(x)P(z)$; every entry factorizes, so $X \perp Z$ (A true) and by symmetry $Y \perp Z$ (B true). Given Z , X determines Y : $P(x | y, z) = 0$ while $P(x | z) = 0.5$, so X and Y are dependent given Z (C false). Since $P(x | y, z) = 0 \neq P(x) = 0.5$, X is not independent of the pair $\{Y, Z\}$ (D false). Thus pairwise independence does not imply joint independence.

Q.25 [MCQ, 2M · Sampling] **Answer: A** (A: $A=T, B=F, C=T, D=F, E=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.75$; $u = 0.49 < 0.75 \Rightarrow A = T$
- B: $P(B=T | A=T) = 0.7$; $u = 0.88 \geq 0.7 \Rightarrow B = F$
- C: $P(C=T | A=T, B=F) = 0.7$; $u = 0.59 < 0.7 \Rightarrow C = T$
- D: $P(D=T | B=F, C=T) = 0.8$; $u = 0.81 \geq 0.8 \Rightarrow D = F$
- E: $P(E=T | C=T, D=F) = 0.3$; $u = 0.66 \geq 0.3 \Rightarrow E = F$

Sample: **(A=T, B=F, C=T, D=F, E=F)**.

Mock Test 014

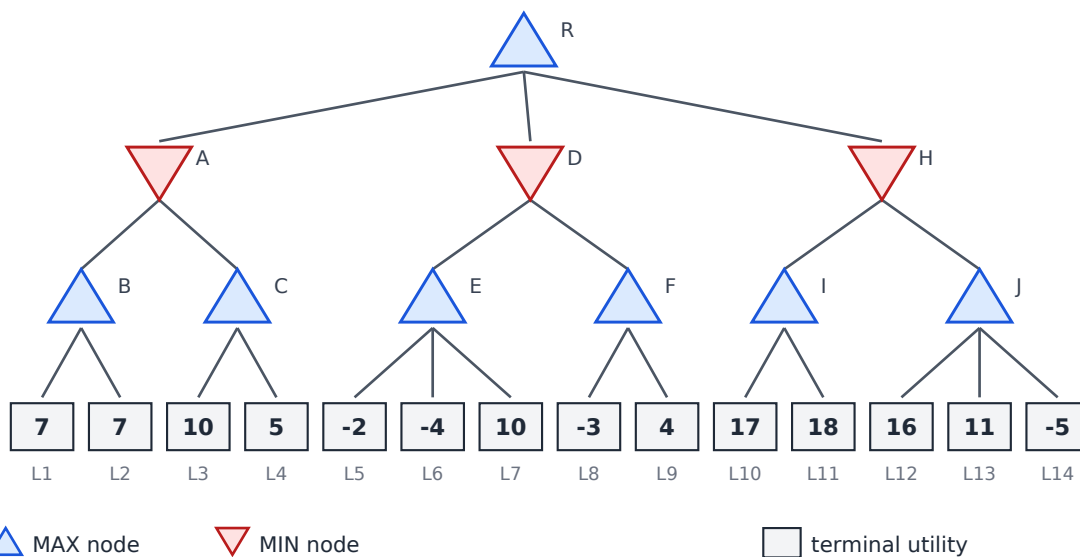
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 8 / 13	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to H; root value 19
☐ (B) Move to A; root value 7
☐ (C) Move to D; root value 4
☐ (D) Move to H; root value 16

Q.2 [NAT · 1 mark · no negative marking] Problem Formulation

The Tower of Hanoi has 3 pegs and 4 disks of distinct sizes, and a larger disk may never sit on a smaller one. How many distinct legal states are there?

Answer (numerical): _____

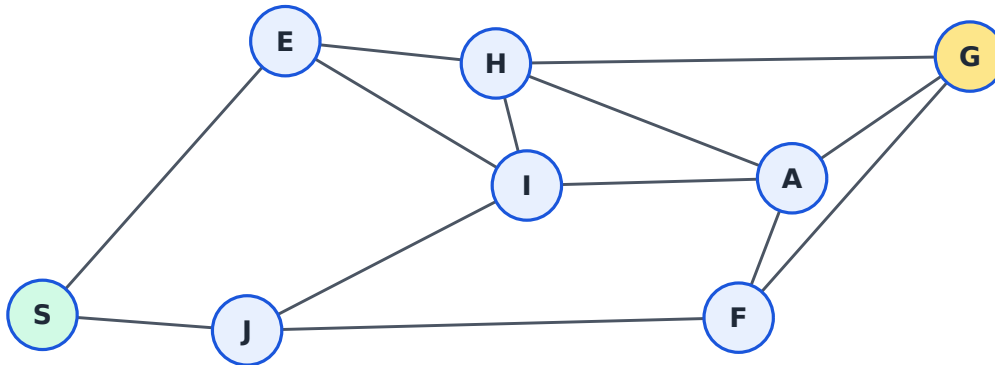
Q.3 [NAT · 1 mark · no negative marking] Informed Search – Effective branching factor

A* finds a solution at depth $d = 4$ after generating $N = 120$ nodes (root excluded). The effective branching factor b^* satisfies $N = b^* + (b^*)^2 + \dots + (b^*)^d$. Find b^* (round off to 2 decimal places).

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, E, J, I, H, F, A, G
- ☐ (B) S, J, E, I, F, H, A, G
- ☐ (C) S, E, J, H, I, F, A, G
- ☐ (D) S, E, H, A, F, G

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are logically equivalent to p ?

- ☐ (A) $p \vee (p \wedge q)$
- ☐ (B) $p \wedge (p \vee q)$
- ☐ (C) $(p \vee q) \wedge (p \vee \neg q)$
- ☐ (D) $(p \rightarrow q) \rightarrow q$

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Skolemizing the sentence $\forall x \exists y P(x, y)$ yields

- ☐ (A) $\forall x P(x, F(x))$, where F is a new function symbol
- ☐ (B) $\forall x P(x, A)$, where A is a new constant
- ☐ (C) $\exists y P(y, y)$
- ☐ (D) $\forall x \forall y P(x, y)$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

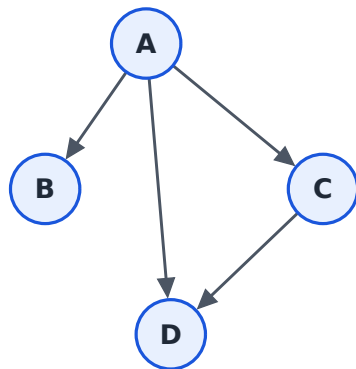
How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (P \leftrightarrow (Q \vee Q)) \rightarrow ((Q \vee Q) \leftrightarrow R) ?$$

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=T, B=F, C=T, D=T)**? (round off to 4 decimal places)



P(A=T)
0.7

A	P(B=T ...)	A	P(C=T ...)
T	0.05	T	0.75
F	0.6	F	0.4

A	C	P(D=T ...)
T	T	0.3
T	F	0.95
F	T	0.75
F	F	0.8

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

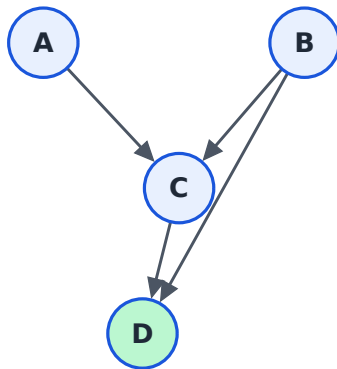
The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(D = T \mid A = F)$, how many of these samples are **rejected**?

#	A	B	C	D
1	T	T	T	T
2	F	T	F	F
3	F	T	F	T
4	F	F	T	F
5	F	T	F	F
6	F	T	T	T
7	F	T	F	T
8	F	T	F	T
9	F	F	T	F
10	F	T	T	F
11	T	T	T	F
12	F	T	T	F
13	F	T	F	T
14	F	T	T	F
15	T	T	T	F
16	F	T	F	F
17	F	T	F	T
18	F	T	T	F
19	F	F	T	F
20	T	T	T	F

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(D = T)$ (round off to 3 decimal places).



$P(A=T)$
0.75

$P(B=T)$	A	B	$P(C=T \dots)$
0.05	T	T	0.5
	T	F	0.2
	F	T	0.25
	F	F	0.6

B	C	$P(D=T \dots)$
T	T	0.75
T	F	0.7
F	T	0.9
F	F	0.2

Answer (numerical): _____

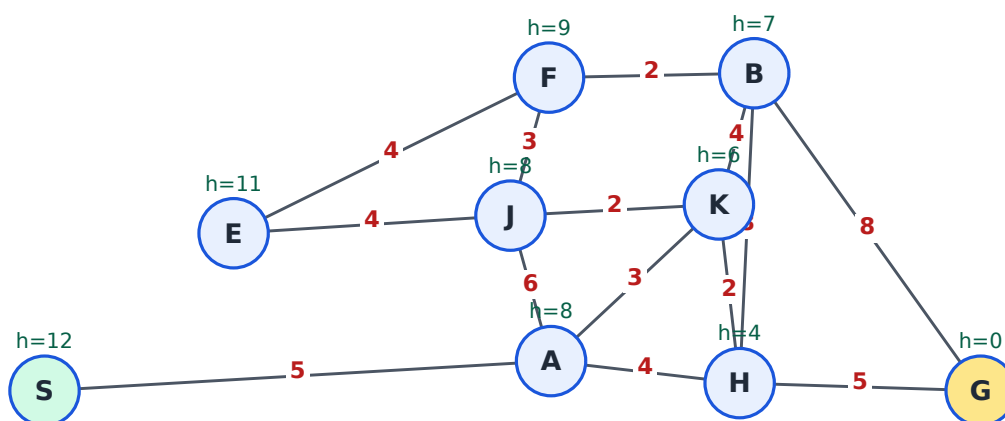
Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search

The 8-puzzle rule is: a tile can move from square X to square Y if X is adjacent to Y and Y is blank. Which of the following statements about relaxations are TRUE?

- ☐ (A) Dropping both conditions (a tile can move anywhere in one step) gives the misplaced-tiles heuristic h_1
- ☐ (B) Dropping only the 'Y is blank' condition gives the Manhattan-distance heuristic h_2
- ☐ (C) Dropping only the adjacency condition gives the Manhattan-distance heuristic h_2
- ☐ (D) The heuristics obtained from these relaxations are admissible for the original 8-puzzle

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

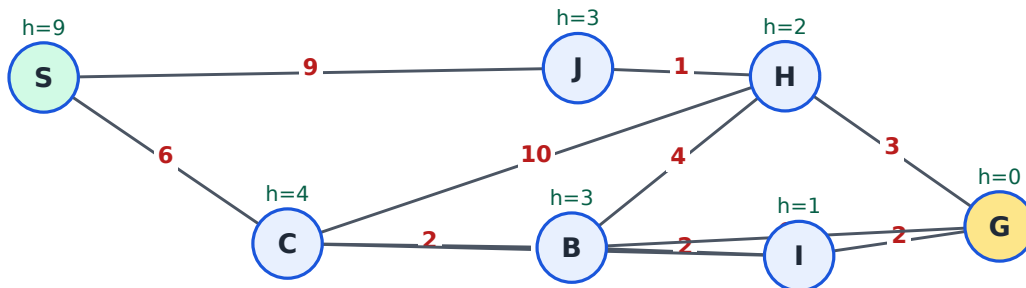
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from S to G on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

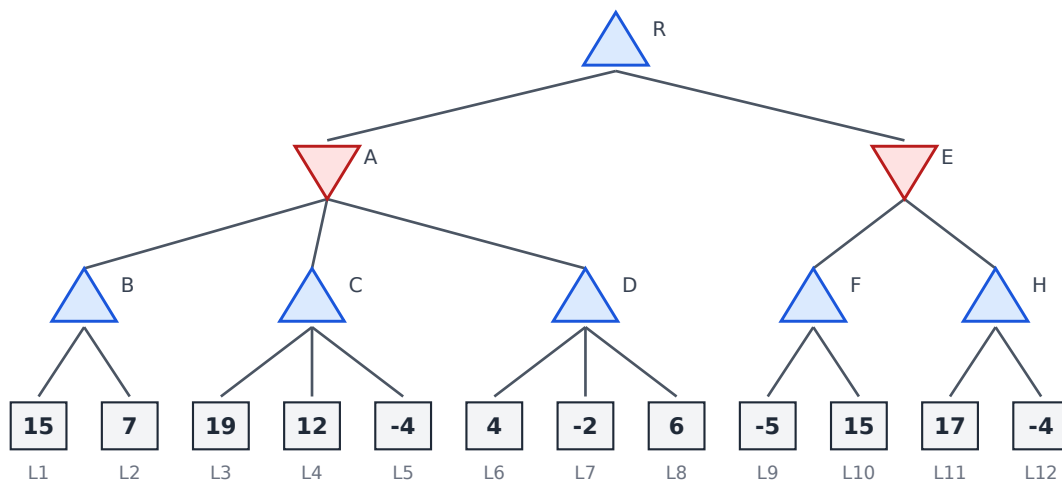
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (Q \wedge P) \rightarrow \neg R$?

- ☐ (A) $\neg(Q \wedge P) \vee \neg R$
☐ (B) $\neg((Q \wedge P) \wedge \neg \neg R)$
☐ (C) $\neg(Q \wedge P) \rightarrow \neg \neg R$
☐ (D) $(Q \wedge P) \vee \neg \neg R$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **D, C, E** and rules:

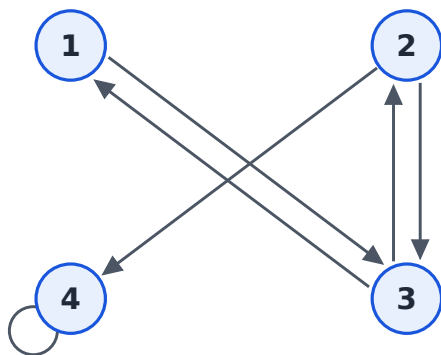
- R1: $E \wedge C \rightarrow H$
 R2: $D \wedge J \rightarrow A$
 R3: $E \rightarrow D$
 R4: $D \wedge E \rightarrow J$
 R5: $A \wedge F \rightarrow D$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) A is entailed
☐ (B) B is entailed
☐ (C) H is entailed
☐ (D) J is entailed

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall y \exists x R(x, y)$
☐ (B) $\exists x R(x, x)$
☐ (C) $\exists x \forall y \neg R(x, y)$
☐ (D) $\forall x \exists y R(x, y)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic

Let KB be a propositional knowledge base and α a sentence, and let $M(\cdot)$ denote the set of models. Which of the following are equivalent to $KB \models \alpha$?

- ☐ (A) $M(KB) \subseteq M(\alpha)$
☐ (B) $KB \rightarrow \alpha$ is valid
☐ (C) $KB \wedge \neg \alpha$ is unsatisfiable
☐ (D) $KB \wedge \alpha$ is satisfiable

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(S \rightarrow P) \wedge P$
- (2) $(S \wedge Q) \wedge P$
- (3) $Q \wedge R$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg\neg P \wedge (P \vee \neg P)$
- ☐ (B) $P \rightarrow \neg P$
- ☐ (C) $(\neg S \leftrightarrow P) \wedge (\neg R \rightarrow Q)$
- ☐ (D) $\neg\neg R$

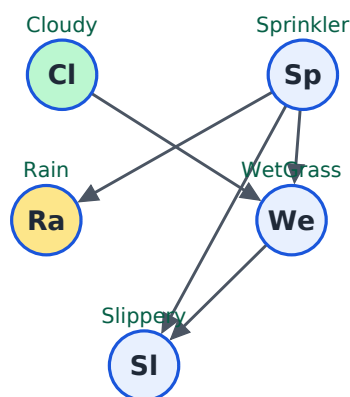
Q.20 [MCQ · 2 marks · negative: -2/3] Predicate Logic

What is the most general unifier of $P(f(x), g(y), y)$ and $P(f(g(z)), w, z)$, where x, y, z, w are variables?

- ☐ (A) $\{x/g(z), w/g(z), y/z\}$
- ☐ (B) $\{x/g(z), w/g(y)\}$
- ☐ (C) $\{x/g(a), w/g(a), y/a, z/a\}$
- ☐ (D) The atoms are not unifiable

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, Sl = Slippery.) Compute $P(\text{Cloudy} = T \mid \text{Rain} = F)$ (round off to 3 decimal places).



$P(Cl=T)$
0.05

$P(Sp=T)$
0.4

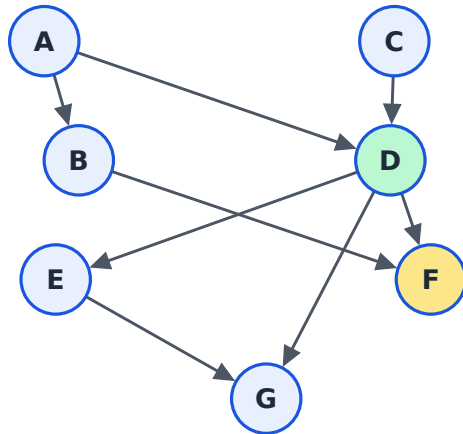
Sp	$P(Ra=T \dots)$
T	0.25
F	0.2

Cl	Sp	$P(We=T \dots)$	Sp	We	$P(Sl=T \dots)$
T	T	0.05	T	T	0.2
T	F	0.05	T	F	0.2
F	T	0.05	F	T	0.25
F	F	0.75	F	F	0.75

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

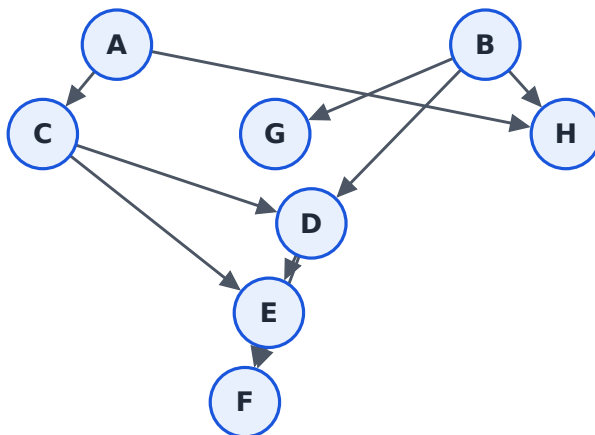
Variable elimination is used to compute $P(D \mid F = T)$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E**, **G**, **B**, **A**, **C**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp B \mid H$
- ☐ (B) $A \perp B \mid C, E$
- ☐ (C) $C \perp B \mid A$
- ☐ (D) $H \perp F \mid C, E$

Q.24 [NAT · 2 marks · no negative marking] Probability for AI

$P(\text{cavity}) = 0.2$. Toothache and Catch are conditionally independent given Cavity with $P(\text{toothache} \mid \text{cavity}) = 0.6$, $P(\text{toothache} \mid \neg \text{cavity}) = 0.1$, $P(\text{catch} \mid \text{cavity}) = 0.9$, $P(\text{catch} \mid \neg \text{cavity}) = 0.2$. Compute $P(\text{cavity} \mid \text{toothache}, \text{catch})$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $A \rightarrow C$ and $B \rightarrow C$ (Boolean variables) with $P(a) = 0.1$, $P(b) = 0.2$, $P(c | a, b) = 0.95$, $P(c | a, \neg b) = 0.90$, $P(c | \neg a, b) = 0.80$, $P(c | \neg a, \neg b) = 0.05$. Compute $P(a | c, b)$. (Round to 3 decimal places.)

Answer (numerical): _____

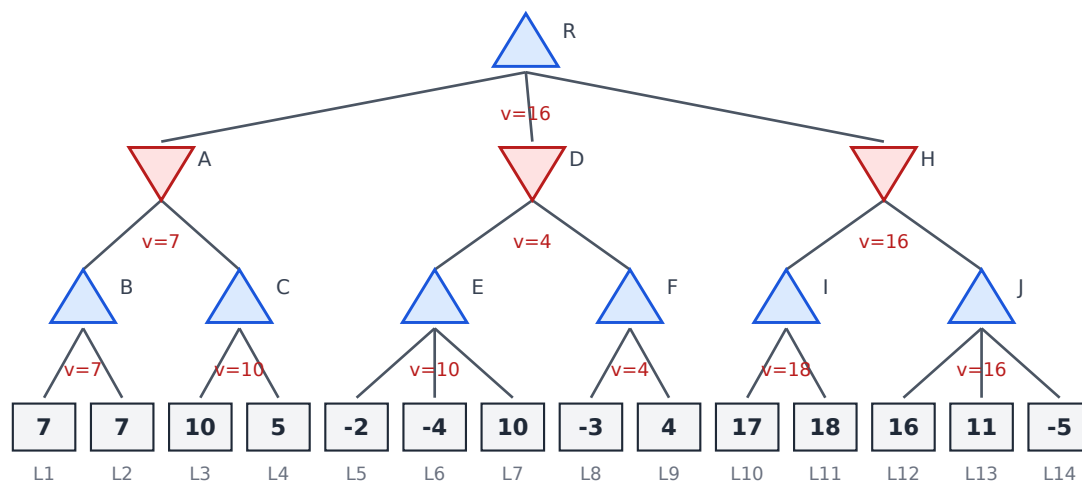
Answer Key — Mock Test 014

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	D	14	NAT	2	Adversarial Search	9
2	NAT	1	Problem Formulation	81	15	MSQ	2	Propositional Logic	A, B
3	NAT	1	Informed Search	2.98 to 3.02	16	MSQ	2	Propositional Logic	A, C, D
4	MCQ	1	Uninformed Search	C	17	MSQ	2	Predicate Logic	A, B, D
5	MSQ	1	Propositional Logic	A, B, C	18	MSQ	2	Propositional Logic	A, B, C
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Propositional Logic	A, D
7	NAT	1	Propositional Logic	6	20	MCQ	2	Predicate Logic	A
8	NAT	1	Sampling	0.1494 to 0.1498	21	NAT	2	Exact Inference	0.048 to 0.052
9	NAT	1	Sampling	4	22	NAT	2	Exact Inference	2
10	NAT	1	Exact Inference	0.424 to 0.428	23	MSQ	2	Conditional Independence	B, C
11	MSQ	2	Informed Search	A, B, D	24	NAT	2	Probability for AI	0.87 to 0.872
12	NAT	2	Informed Search	4	25	NAT	2	Conditional Independence	0.116 to 0.117
13	NAT	2	Informed Search	5					

Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] **Answer: D** (D: Move to H; root value 16)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 7$, $C = 10$, $A = 7$, $E = 10$, $F = 4$, $D = 4$, $I = 18$, $J = 16$, $H = 16$. Root value = **16**; MAX's optimal first move is towards H.

Q.2 [NAT, 1M · Problem Formulation] **Answer: 81**

Each disk can be on any of the 3 pegs. For a given assignment of disks to pegs, the size rule allows exactly one legal stacking order on each peg. So the number of states is $3^4 = 81$.

Q.3 [NAT, 1M · Informed Search] **Answer: 2.98 to 3.02**

Solve $b + b^2 + \dots + b^4 = 120$ numerically (bisection): $b^* \approx 3.0$.

Check: $3.0^1 + 3.0^2 + 3.0^3 + 3.0^4 \approx 120 \approx 120$.

Q.4 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, E, J, H, I, F, A, G)

Adjacency lists (alphabetical): A: {F, G, H, I}; E: {H, I, S}; F: {A, G, J}; G: {A, F, H}; H: {A, E, G, I}; I: {A, E, H, J}; J: {F, I, S}; S: {E, J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[E, J]
2	E	[J, H, I]
3	J	[H, I, F]
4	H	[I, F, A, G]
5	I	[F, A, G]
6	F	[A, G]
7	A	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, E, J, H, I, F, A, G**. The path returned is $S \rightarrow E \rightarrow H \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [MSQ, 1M · Propositional Logic] **Answer: A, B, C** (A: $p \vee (p \wedge q)$; B: $p \wedge (p \vee q)$; C: $(p \vee q) \wedge (p \vee \neg q)$)

(A) and (B) are the absorption laws. (C) By distribution, $(p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q) \equiv p$. (D) $(p \rightarrow q) \rightarrow q \equiv \neg(\neg p \vee q) \vee q \equiv (p \wedge \neg q) \vee q \equiv p \vee q$, which differs from p at $p=F, q=T$.

Q.6 [MCQ, 1M · Predicate Logic] **Answer: A** (A: $\forall x P(x, F(x))$, where F is a new function symbol)

Since the existential $\exists y$ lies within the scope of $\forall x$, the chosen y may depend on x , so it is replaced by a Skolem function $F(x)$. A Skolem constant would wrongly assert one y works for all x . The result is satisfiable iff the original is (though not logically equivalent). $\forall x \forall y P(x, y)$ is much stronger.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	$P \leftrightarrow (Q \vee Q)$	$(Q \vee Q) \leftrightarrow R$	$(P \leftrightarrow (Q \vee Q)) \rightarrow ((Q \vee Q) \leftrightarrow R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	F	T
T	F	F	F	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	T	T	T

Rows where $F = T$: **6**.

Q.8 [NAT, 1M · Sampling] **Answer: 0.1494 to 0.1498**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(A=T) \times P(B=F|A=T) \times P(C=T|A=T) \times P(D=T|A=T, C=T)$
 $= 0.7 \times 0.95 \times 0.75 \times 0.3 = \mathbf{0.149625} \approx 0.1496$.

Q.9 [NAT, 1M · Sampling] **Answer: 4**

Rejection sampling keeps only samples consistent with the evidence $A = F$: samples 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 16, 17, 18, 19 (16 accepted, 4 rejected).

Among the accepted samples, $D = T$ in 6. Estimate = $6/16 = \mathbf{0.375}$.

(For this network the exact value is 0.425; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.10 [NAT, 1M · Exact Inference] **Answer: 0.424 to 0.428**

Only the ancestors of D (A, B, C) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
$A=T, B=T, C=T$	0.01406
$A=T, B=T, C=F$	0.01313
$A=T, B=F, C=T$	0.12825
$A=T, B=F, C=F$	0.114
$A=F, B=T, C=T$	0.00234
$A=F, B=T, C=F$	0.00656
$A=F, B=F, C=T$	0.12825
$A=F, B=F, C=F$	0.019

Sum = **0.4256** $\approx$ 0.426.

Q.11 [MSQ, 2M · Informed Search] **Answer: A, B, D**

(A) is true: if a tile can jump anywhere, each misplaced tile needs exactly 1 move. (B) is true: if tiles move to adjacent squares regardless of the blank, each tile needs exactly its Manhattan distance. (C) is false: dropping adjacency but keeping 'Y is blank' gives Gaschnig's heuristic, not Manhattan distance. (D) is true: optimal costs of relaxed problems are admissible.

Q.12 [NAT, 2M · Informed Search] **Answer: 4**

Heuristic values: $h(A)=8$, $h(B)=7$, $h(E)=11$, $h(F)=9$, $h(G)=0$, $h(H)=4$, $h(J)=8$, $h(K)=6$, $h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	A(13)
2	A (13)	H(13), K(14), J(19)
3	H (13)	G(14), K(14), J(19), B(24)
4	G (14)	K(14), J(19), B(24) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **A** $\rightarrow$ **H** $\rightarrow$ **G** with cost **14**; nodes expanded = **4** (S, A, H, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.13 [NAT, 2M · Informed Search] **Answer: 5**

Heuristic values: $h(B)=3$, $h(C)=4$, $h(G)=0$, $h(H)=2$, $h(I)=1$, $h(J)=3$, $h(S)=9$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	C(10), J(12)
2	C (10)	B(11), J(12), I(13), H(18)
3	B (11)	I(11), J(12), G(14), H(14)
4	I (11)	G(12), J(12), H(14)
5	G (12)	J(12), H(14) $\leftarrow$ goal, stop

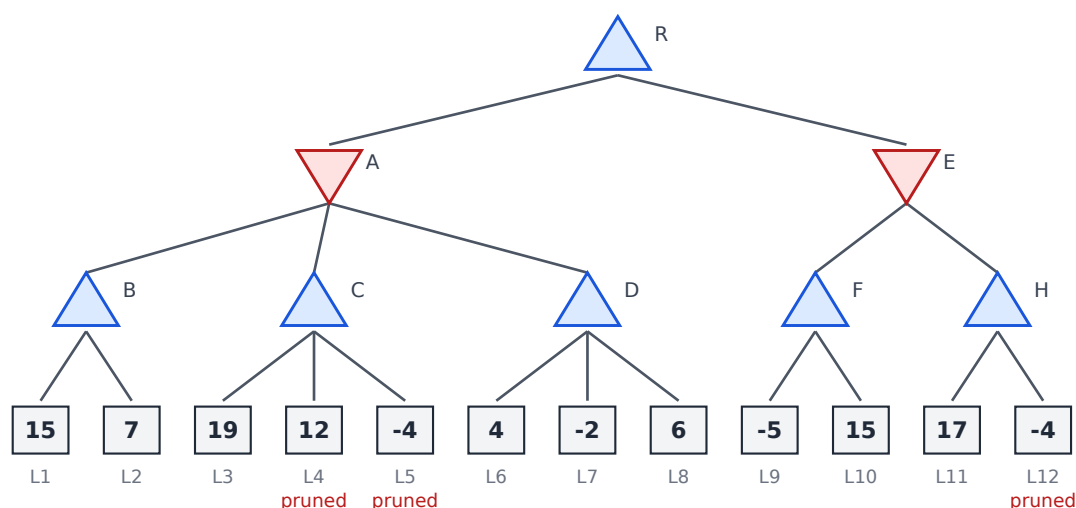
Path returned: **S** $\rightarrow$ **C** $\rightarrow$ **B** $\rightarrow$ **I** $\rightarrow$ **G** with cost **12**; nodes expanded = **5** (S, C, B, I, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 12$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.14 [NAT, 2M · Adversarial Search] **Answer: 9**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $19 \geq \beta = 15 \Rightarrow$ prune remaining children (leaves L4, L5).
- At MAX node H: value $17 \geq \beta = 15 \Rightarrow$ prune remaining children (leaves L12).



Leaves evaluated (9): L1, L2, L3, L6, L7, L8, L9, L10, L11. Leaves pruned (3): L4, L5, L12. Root value = 15 (same as plain minimax, 15 — pruning never changes the root value).

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, B (A: $\neg(Q \wedge P) \vee \neg R$; B: $\neg((Q \wedge P) \wedge \neg \neg R)$)

Compare truth tables with $F = (Q \wedge P) \rightarrow \neg R$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(Q \wedge P) \vee \neg R$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(B) $\neg((Q \wedge P) \wedge \neg \neg R)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(C) $\neg(Q \wedge P) \rightarrow \neg \neg R$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(D) $(Q \wedge P) \vee \neg \neg R$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A, C, D (A: A is entailed; C: H is entailed; D: J is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $E \wedge C \rightarrow H$, $D \wedge E \rightarrow J$; add H, J.

Round 2: fire $D \wedge J \rightarrow A$; add A.

Fixed point: {A, C, D, E, H, J} — 6 atoms are entailed (facts: C, D, E; newly derived: A, H, J). Atoms never derived: B, F, K.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, B, D (A: $\forall y \exists x R(x, y)$; B: $\exists x R(x, x)$; D: $\forall x \exists y R(x, y)$)

$R = \{(1,3), (2,3), (2,4), (3,1), (3,2), (4,4)\}$.

(A) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 4.

(C) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

(D) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, B, C (A: $M(KB) \subseteq M(\alpha)$; B: $KB \rightarrow \alpha$ is valid; C: $KB \wedge \neg \alpha$ is unsatisfiable)

(A) is the definition of entailment: α is true in every model of KB. (B) is the deduction theorem. (C) is the basis of proof by refutation (resolution). (D) is weaker: $KB = T$, $\alpha = p$ gives $KB \wedge \alpha$ satisfiable, but $KB \not\models \alpha$.

Q.19 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $\neg \neg P \wedge (P \vee \neg P)$; D: $\neg \neg R$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 1 of them:

P	Q	R	S	KB
T	T	T	T	T

(A) $\neg \neg P \wedge (P \vee \neg P)$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $P \rightarrow \neg P$ is false in the KB-model $P=T, Q=T, R=T, S=T \Rightarrow$ not entailed.

(C) $(\neg S \leftrightarrow P) \wedge (\neg R \rightarrow Q)$ is false in the KB-model $P=T, Q=T, R=T, S=T \Rightarrow$ not entailed.

(D) $\neg \neg R$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.20 [MCQ, 2M · Predicate Logic] Answer: A ($A: \{x/g(z), w/g(z), y/z\}$)

$f(x)$ vs $f(g(z))$ gives $x/g(z)$. $g(y)$ vs w gives $w/g(y)$. y vs z gives y/z , and composing updates $w/g(y)$ to $w/g(z)$. Both atoms become $P(f(g(z)), g(z), z)$. Option B is not a unifier because y and z remain different. Option C is a unifier but not most general (it is an instance of A with z/a). No occurs-check failure arises, so D is wrong.

Q.21 [NAT, 2M · Exact Inference] Answer: 0.048 to 0.052

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}, \text{Slippery}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler}) \cdot P(\text{Rain} \mid \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{Slippery} \mid \text{Sprinkler}, \text{WetGrass})$.

Inference by enumeration: $P(\text{Cloudy} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cloudy}, e, \text{hidden})$.

Hidden: Sp, We, SI	CI	product of CPT entries (CI · Sp · Ra · We · SI)	value
T, T, T	T	$0.05 \times 0.4 \times 0.75 \times 0.05 \times 0.2$	0.00015
T, T, F	T	$0.05 \times 0.4 \times 0.75 \times 0.05 \times 0.8$	0.0006
T, F, T	T	$0.05 \times 0.4 \times 0.75 \times 0.95 \times 0.2$	0.00285
T, F, F	T	$0.05 \times 0.4 \times 0.75 \times 0.95 \times 0.8$	0.0114
F, T, T	T	$0.05 \times 0.6 \times 0.8 \times 0.05 \times 0.25$	0.0003
F, T, F	T	$0.05 \times 0.6 \times 0.8 \times 0.05 \times 0.75$	0.0009
F, F, T	T	$0.05 \times 0.6 \times 0.8 \times 0.95 \times 0.75$	0.0171
F, F, F	T	$0.05 \times 0.6 \times 0.8 \times 0.95 \times 0.25$	0.0057
T, T, T	F	$0.95 \times 0.4 \times 0.75 \times 0.05 \times 0.2$	0.00285
T, T, F	F	$0.95 \times 0.4 \times 0.75 \times 0.05 \times 0.8$	0.0114
T, F, T	F	$0.95 \times 0.4 \times 0.75 \times 0.95 \times 0.2$	0.05415
T, F, F	F	$0.95 \times 0.4 \times 0.75 \times 0.95 \times 0.8$	0.2166
F, T, T	F	$0.95 \times 0.6 \times 0.8 \times 0.75 \times 0.25$	0.0855
F, T, F	F	$0.95 \times 0.6 \times 0.8 \times 0.75 \times 0.75$	0.2565
F, F, T	F	$0.95 \times 0.6 \times 0.8 \times 0.25 \times 0.75$	0.0855
F, F, F	F	$0.95 \times 0.6 \times 0.8 \times 0.25 \times 0.25$	0.0285

$P(\text{Cloudy}=\text{T}, e) = 0.039$; $P(\text{Cloudy}=\text{F}, e) = 0.741$; $P(e) = 0.78$.

$P(\text{Cloudy}=\text{T} \mid e) = 0.039 / 0.78 = \mathbf{0.05} \approx 0.050$.

Q.22 [NAT, 2M · Exact Inference] Answer: 2

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(C)$, $f(A, C, D)$, $f(D, E)$, $f(B, D)$, $f(D, E, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(D, E)$, $f(D, E, G)$	$\{D, E, G\}$	$f(D, G)$
G	$f(D, G)$	$\{D, G\}$	$f(D)$
B	$f(A, B)$, $f(B, D)$	$\{A, B, D\}$	$f(A, D)$
A	$f(A)$, $f(A, C, D)$, $f(A, D)$	$\{A, C, D\}$	$f(C, D)$
C	$f(C)$, $f(C, D)$	$\{C, D\}$	$f(D)$

The largest factor produced has **2** variable(s) (created when eliminating E); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over D and normalise.

Q.23 [MSQ, 2M · Conditional Independence] Answer: B, C ($B: A \perp B \mid C, E$; $C: C \perp B \mid A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp B \mid H$: **FALSE** — the path $E \leftarrow C \leftarrow A \rightarrow H \leftarrow B$ is active given $\{H\}$.

(B) $A \perp B \mid C, E$: **TRUE** — $A \rightarrow C \rightarrow D \leftarrow B$ (blocked: observed chain node C); $A \rightarrow C \rightarrow E \leftarrow D \leftarrow B$ (blocked: observed chain node C); $A \rightarrow C \rightarrow E \rightarrow F \leftarrow D \leftarrow B$ (blocked: observed chain node C); $A \rightarrow H \leftarrow B$ (blocked: collider H with neither it nor a descendant observed).

(C) $C \perp B \mid A$: **TRUE** — $C \leftarrow A \rightarrow H \leftarrow B$ (blocked: observed fork node A); $C \rightarrow D \leftarrow B$ (blocked: collider D with neither it nor a descendant observed); $C \rightarrow E \leftarrow D \leftarrow B$ (blocked: collider E with neither it nor a descendant observed); $C \rightarrow E \rightarrow F \leftarrow D \leftarrow B$ (blocked: collider F with neither it nor a descendant observed).

(D) $H \perp F \mid C, E$: **FALSE** — the path $H \leftarrow B \rightarrow D \rightarrow F$ is active given $\{C, E\}$.

Q.24 [NAT, 2M · Probability for AI] **Answer: 0.87 to 0.872**

Using conditional independence, $P(\text{Cavity} \mid \text{toothache}, \text{catch}) = \alpha P(\text{toothache} \mid \text{Cavity}) P(\text{catch} \mid \text{Cavity}) P(\text{Cavity})$. For cavity: $0.6 \times 0.9 \times 0.2 = 0.108$. For $\neg\text{cavity}$: $0.1 \times 0.2 \times 0.8 = 0.016$. $\alpha = 1/(0.108 + 0.016) = 1/0.124$, so $P(\text{cavity} \mid \text{toothache}, \text{catch}) = 0.108/0.124 \approx 0.871$.

Q.25 [NAT, 2M · Conditional Independence] **Answer: 0.116 to 0.117**

Since $A \perp B$, $P(A \mid c, b) = \alpha P(A) P(c \mid A, b)$ ($P(b)$ cancels). For a : $0.1 \times 0.95 = 0.095$. For $\neg a$: $0.9 \times 0.80 = 0.72$. $P(a \mid c, b) = 0.095/(0.095 + 0.72) = 0.095/0.815 \approx 0.1166 \approx 0.117$. B already explains C , so A remains near its prior 0.1 .

Mock Test 015

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 9 / 10	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

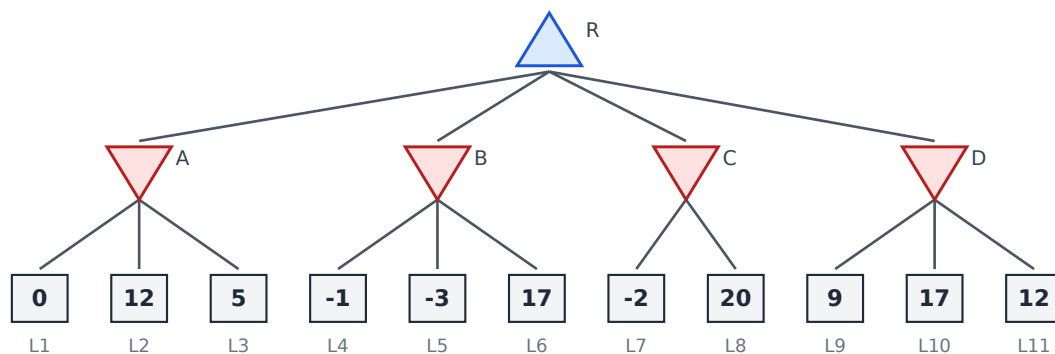
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 2$ (so $b^d = 4$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- △ MAX node ▽ MIN node terminal utility
- ☐ (A) Move to D; root value 9
 - ☐ (B) Move to B; root value -3
 - ☐ (C) Move to C; root value -2
 - ☐ (D) Move to A; root value 0

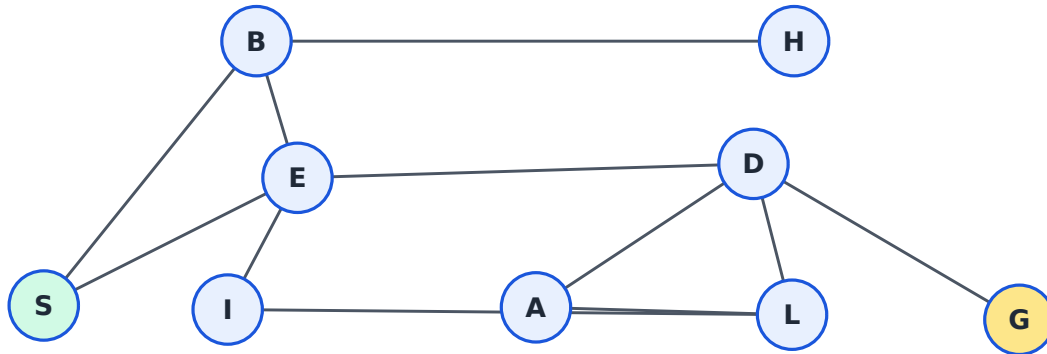
Q.3 [MCQ · 1 mark · negative: $-1/3$] Problem Formulation

In AIMA, a search problem is formally defined by

- ☐ (A) initial state, actions, transition model, goal test, and path (action) cost function
- ☐ (B) initial state and goal state only
- ☐ (C) a heuristic function and an evaluation function
- ☐ (D) a frontier and an explored set

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, H, E, D, I, A, G
☐ (B) S, E, B, I, D, H, L, G
☐ (C) S, B, E, D, A, L, I, G
☐ (D) S, B, E, H, D, I, A, G

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = (Q \rightarrow (P \vee R)) \rightarrow ((\neg Q \rightarrow \neg P) \wedge Q)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\exists x \forall y R(x, y)$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over the symbols p, q, r does $(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)$ have?

Answer (numerical): _____

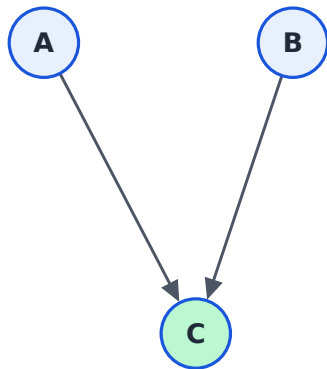
Q.8 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following are sentences, i.e. formulas with no free variables (a is a constant)?

- ☐ (A) $(\forall x P(x)) \rightarrow Q(x)$
☐ (B) $\forall x (P(x) \rightarrow \exists y R(x, y))$
☐ (C) $\exists x R(x, y)$
☐ (D) $\forall x \exists y (R(x, y) \wedge P(a))$

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



$P(A=T)$	$P(B=T)$	A	B	$P(C=T \dots)$
0.9	0.9	T	T	0.6
		T	F	0.25
		F	T	0.6
		F	F	0.95

Answer (numerical): _____

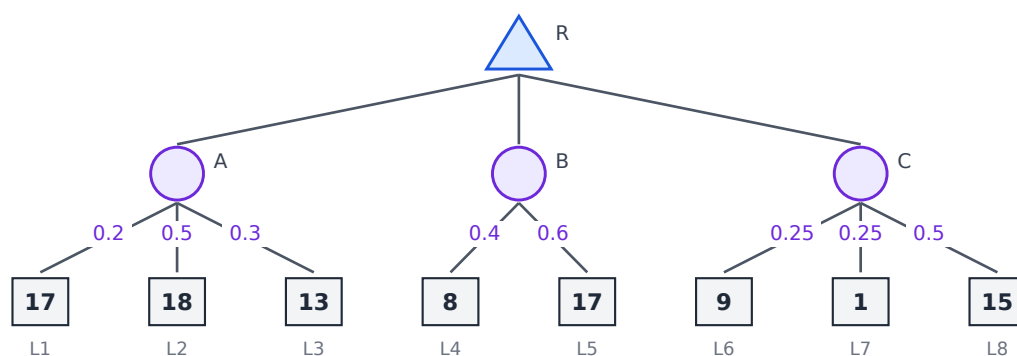
Q.10 [MCQ · 1 mark · negative: -1/3] Conditional Independence

Burglary (B) and Earthquake (E) are independent causes of Alarm (A). Given that the alarm is ringing, learning that an earthquake occurred lowers the probability of a burglary. This phenomenon is called

- ☐ (A) explaining away (intercausal reasoning)
- ☐ (B) the local Markov property
- ☐ (C) the naive Bayes assumption
- ☐ (D) marginalization

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



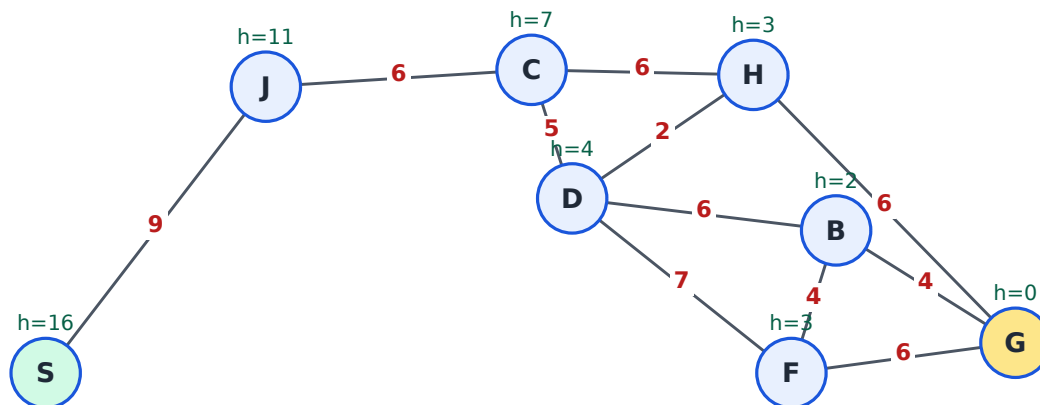
△ MAX node ○ CHANCE node (edge = probability)

□ terminal utility

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

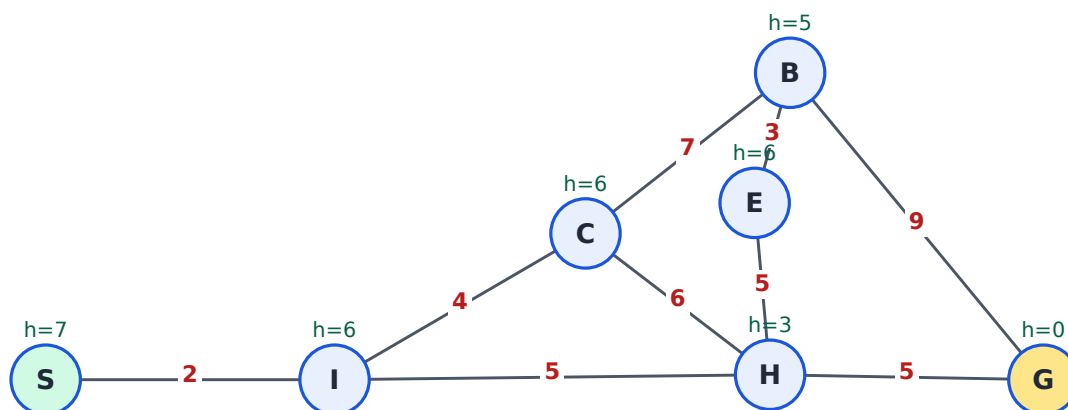
Q.13 [NAT · 2 marks · no negative marking] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (2), $S \rightarrow B$ (4), $A \rightarrow C$ (2), $A \rightarrow D$ (4), $B \rightarrow C$ (1), $C \rightarrow D$ (1), $C \rightarrow G$ (5), $D \rightarrow G$ (2). The heuristic is $h(S)=5$, $h(A)=4$, $h(B)=4$, $h(C)=3$, $h(D)=2$, $h(G)=0$. A* graph search is run from S. It goal-tests a node when the node is selected from the frontier, and it keeps the cheapest entry for each state. How many nodes are selected from the frontier for expansion, including the selection of G that ends the search?

Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node C is expanded.
- ☐ (B) Node E is expanded.
- ☐ (C) The path returned has cost 14.
- ☐ (D) Node B is expanded.

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((\neg R \vee Q) \wedge (\neg \neg R \vee P)) \rightarrow (Q \vee P)$
☐ (B) $((\neg Q \rightarrow R) \wedge \neg R) \rightarrow \neg \neg Q$
☐ (C) $P \rightarrow (Q \rightarrow P)$
☐ (D) $((P \rightarrow Q) \wedge \neg P) \rightarrow \neg Q$

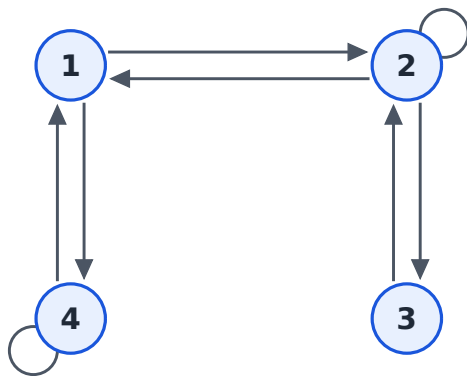
Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following first-order sentences are valid?

- ☐ (A) $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$
☐ (B) $\forall y \exists x P(x, y) \rightarrow \exists x \forall y P(x, y)$
☐ (C) $\forall x (P(x) \vee Q(x)) \rightarrow (\forall x P(x) \vee \forall x Q(x))$
☐ (D) $\exists x (P(x) \wedge Q(x)) \rightarrow (\exists x P(x) \wedge \exists x Q(x))$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\exists y \forall x R(x, y)$
☐ (D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(Q \leftrightarrow R) \leftrightarrow (\neg R \leftrightarrow R)$
 (2) $\neg(P \wedge \neg P)$
 (3) $(\neg R \wedge Q) \wedge (Q \wedge Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg Q \rightarrow R$
☐ (B) $\neg(P \wedge Q)$
☐ (C) $(Q \wedge Q) \rightarrow R$
☐ (D) $\neg R \vee R$

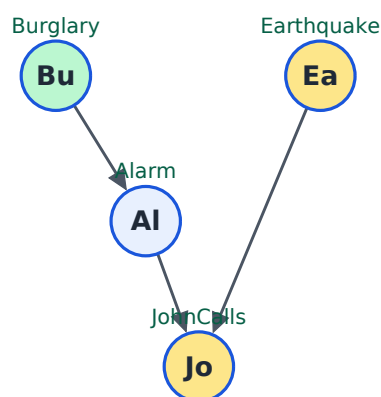
Q.19 [MSQ · 2 marks · no negative marking] Bayesian Networks

Which of the following statements about I-maps of a distribution P are correct? (G is an I-map of P if every independence implied by G by d-separation holds in P .)

- ☐ (A) A complete (fully connected) DAG over the variables is an I-map of every distribution
- ☐ (B) Adding nodes in a causal order (causes before effects) usually yields a sparser minimal I-map
- ☐ (C) Adding an arc (without creating a cycle) to an I-map of P can make it fail to be an I-map of P
- ☐ (D) The minimal I-map of P is the same DAG for every variable ordering

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute **$P(\text{Burglary} = T \mid \text{Earthquake} = T, \text{JohnCalls} = T)$** (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.4

$P(\text{Ea}=T)$	Bu	$P(\text{Al}=T \dots)$
0.05	T	0.9
	F	0.8

Ea	Al	$P(\text{Jo}=T \dots)$
T	T	0.2
T	F	0.8
F	T	0.6
F	F	0.1

Answer (numerical): _____

Q.21 [MSQ · 2 marks · no negative marking] Variable Elimination

Which statements about variable elimination (VE) are correct?

- ☐ (A) Every elimination ordering yields the same final answer
- ☐ (B) Finding an optimal elimination ordering is NP-hard in general
- ☐ (C) VE runs in time polynomial in the number of variables for every network and ordering
- ☐ (D) VE avoids the repeated computation of identical subexpressions that occurs in enumeration

Q.22 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

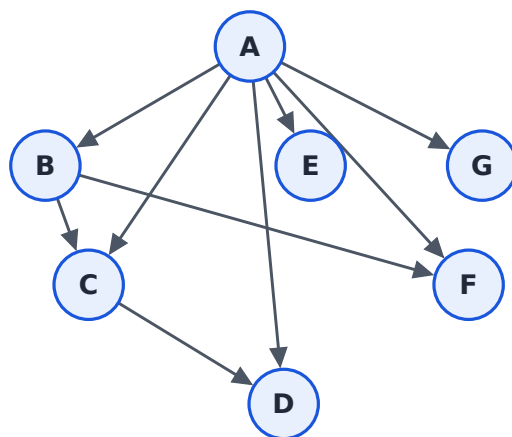
The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D, E. Using **rejection sampling**, estimate $P(D = T \mid E = T)$ (round off to 2 decimal places).

#	A	B	C	D	E
1	T	F	T	T	F
2	T	F	F	T	F
3	T	F	T	T	F
4	T	T	F	T	T
5	T	T	T	T	F
6	F	F	F	F	F
7	T	F	T	F	T
8	F	T	F	F	T
9	T	T	F	T	T
10	T	F	T	T	F
11	T	F	T	F	F
12	T	F	T	F	T
13	T	F	T	F	T
14	T	F	T	F	T
15	T	T	F	T	T
16	T	F	T	T	F
17	F	F	F	F	F
18	T	F	T	F	T
19	T	F	F	T	F
20	T	T	F	T	T

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

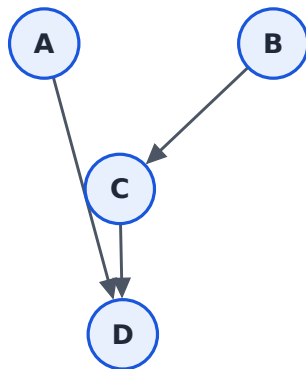
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp A \mid D$
- ☐ (B) $F \perp G \mid A, D$
- ☐ (C) $C \perp B \mid A$
- ☐ (D) $E \perp G \mid A, C$

Q.24 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.01, 0.74, 0.33, 0.55)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.8

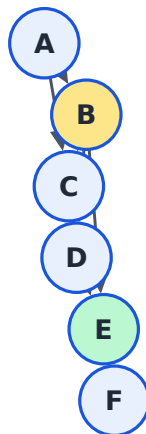
P(B=T)	B	P(C=T ...)
0.8	T	0.5
	F	0.25

A	C	P(D=T ...)
T	T	0.1
T	F	0.25
F	T	0.25
F	F	0.3

- ☐ (A) A=F, B=F, C=T, D=T
☐ (B) A=T, B=F, C=T, D=F
☐ (C) A=F, B=T, C=T, D=F
☐ (D) A=T, B=T, C=T, D=F

Q.25 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(E \mid B = T)}$ in the Bayesian network below (Boolean variables; evidence B = T is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, F, A, D**. What is the scope of the new factor produced when **C** is summed out?



- ☐ (A) f(A)
☐ (B) f(A, D, E)
☐ (C) f(A, D)
☐ (D) f(A, C, D)

Answer Key — Mock Test 015

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	3	14	MSQ	2	Informed Search	A
2	MCQ	1	Adversarial Search	A	15	MSQ	2	Propositional Logic	A, B, C
3	MCQ	1	Problem Formulation	A	16	MSQ	2	Predicate Logic	A, D
4	MCQ	1	Uninformed Search	D	17	MSQ	2	Predicate Logic	B
5	NAT	1	Propositional Logic	4	18	MSQ	2	Propositional Logic	A, D
6	NAT	1	Predicate Logic	169	19	MSQ	2	Bayesian Networks	A, B
7	NAT	1	Propositional Logic	2	20	NAT	2	Exact Inference	0.349 to 0.353
8	MSQ	1	Predicate Logic	B, D	21	MSQ	2	Variable Elimination	A, B, D
9	NAT	1	Exact Inference	0.57 to 0.574	22	NAT	2	Sampling	0.39 to 0.41
10	MCQ	1	Conditional Independence	A	23	MSQ	2	Conditional Independence	B, D
11	NAT	2	Adversarial Search	16.29 to 16.31	24	MCQ	2	Sampling	D
12	NAT	2	Informed Search	6	25	MCQ	2	Exact Inference	C
13	NAT	2	Informed Search	5					

Detailed Solutions

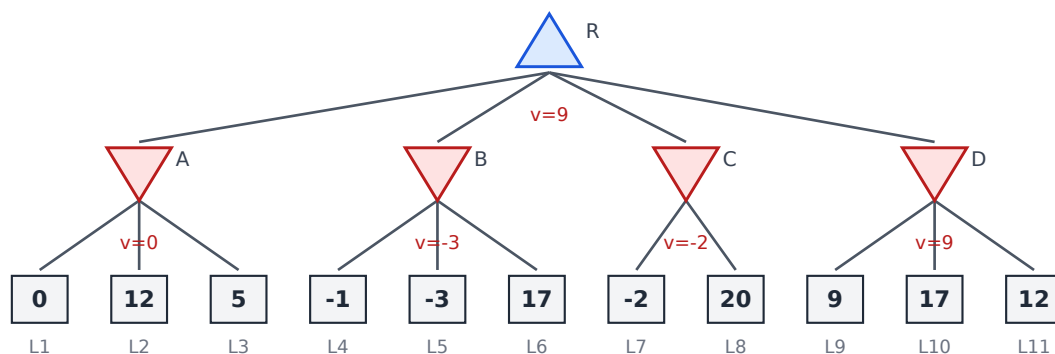
Q.1 [NAT, 1M · Adversarial Search] Answer: 3

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 2^{\lceil 1 \rceil} + 2^{\lfloor 1 \rfloor} - 1 = 2 + 2 - 1 = 3$, versus 4 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.2 [MCQ, 1M · Adversarial Search] Answer: A (A: Move to D; root value 9)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = 0, B = -3, C = -2, D = 9. Root value = **9**; MAX's optimal first move is towards D.

Q.3 [MCQ, 1M · Problem Formulation] Answer: A (A: initial state, actions, transition model, goal test, and path (action) cost function)

A problem consists of the **initial state, the actions available in each state, the transition model** **RESULT(s,a), the goal test, and the action-cost / path-cost function**. Together these define the state space implicitly. (B) leaves out actions and costs. (C) belongs to particular algorithms, not to the problem. (D) is the data structure of a search algorithm.

Q.4 [MCQ, 1M · Uninformed Search] Answer: D (D: S, B, E, H, D, I, A, G)

Adjacency lists (alphabetical): A: {D, L}; B: {E, H, S}; D: {A, E, G, L}; E: {B, D, I, S}; G: {D}; H: {B}; I: {E, L}; L: {A, D, I}; S: {B, E}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B, E]
2	B	[E, H]
3	E	[H, D, I]

Step	Expanded	Frontier (FIFO) after expansion
4	H	[D, I]
5	D	[I, A, G, L]
6	I	[A, G, L]
7	A	[G, L]
8	G	[L] ← goal popped, stop

Expansion order: **S, B, E, H, D, I, A, G**. The path returned is $S \rightarrow E \rightarrow D \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 4**

Build the truth table (sub-formulas shown first):

P	Q	R	$Q \rightarrow (P \vee R)$	$(\neg Q \rightarrow \neg P) \wedge Q$	$(Q \rightarrow (P \vee R)) \rightarrow ((\neg Q \rightarrow \neg P) \wedge Q)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	F	T	T
F	F	T	T	F	F
F	F	F	T	F	F

F has 4 models, so it is false in $8 - 4 = 4$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 169**

View R as an 3×3 0/1 matrix. The sentence says R is complement of 'every row has at least one 0': $2^{(n^2)} - (2^n - 1)^n$. With $n = 3$: **169**.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 2**

The cycle of implications $p \rightarrow q \rightarrow r \rightarrow p$ forces p, q, r to have the same truth value: if any one is T, all following ones are T, and the cycle returns. So the only models are all-T and all-F, i.e. 2 models.

Q.8 [MSQ, 1M · Predicate Logic] **Answer: B, D** (B: $\forall x (P(x) \rightarrow \exists y R(x, y))$; D: $\forall x \exists y (R(x, y) \wedge P(a))$)

(A) The scope of $\forall x$ ends before $\rightarrow$, so x in $Q(x)$ is free. (B) x and y are bound everywhere. (C) y is free. (D) x and y are bound; a is a constant, not a variable. Hence B and D are sentences.

Q.9 [NAT, 1M · Exact Inference] **Answer: 0.57 to 0.574**

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T	0.486
A=T, B=F	0.0225
A=F, B=T	0.054
A=F, B=F	0.0095

Sum = **0.572** $\approx$ 0.572.

Q.10 [MCQ, 1M · Conditional Independence] **Answer: A** (A: explaining away (intercausal reasoning))

In the v-structure $B \rightarrow A \leftarrow E$ the causes are marginally independent but become dependent once the common effect is observed. One cause 'explains away' the effect and reduces the posterior of the other cause. The local Markov property concerns independence of a node from non-descendants given parents; naive Bayes is a class-feature model; marginalization is summing out variables.

Q.11 [NAT, 2M · Adversarial Search] Answer: 16.29 to 16.31

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.2 \times 17 + 0.5 \times 18 + 0.3 \times 13 = 16.3$
- $B = 0.4 \times 8 + 0.6 \times 17 = 13.4$
- $C = 0.25 \times 9 + 0.25 \times 1 + 0.5 \times 15 = 10$

Root = $\max(16.3, 13.4, 10) = 16.3$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.12 [NAT, 2M · Informed Search] Answer: 6

Heuristic values: $h(B)=2$, $h(C)=7$, $h(D)=4$, $h(F)=3$, $h(G)=0$, $h(H)=3$, $h(J)=11$, $h(S)=16$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (16)	J(20)
2	J (20)	C(22)
3	C (22)	D(24), H(24)
4	D (24)	H(24), B(28), F(30)
5	H (24)	G(27), B(28), F(30)
6	G (27)	B(28), F(30) ← goal, stop

Path returned: **S** → **J** → **C** → **H** → **G** with cost **27**; nodes expanded = **6** (S, J, C, D, H, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 27$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.13 [NAT, 2M · Informed Search] Answer: 5

$f(S)=5$. Expand S: $A(g=2, f=6)$, $B(g=4, f=8)$. Expand A ($f=6$): $C(g=4, f=7)$, $D(g=6, f=8)$. Expand C ($f=7$): D improves to $g=5, f=7$, and G is added with $g=9, f=9$. Expand D ($f=7$): G improves to $g=7, f=7$. Select G ($f=7$) and stop with cost 7 via $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$. Selections: S, A, C, D, G = **5**. B ($f=8$) is never expanded.

Q.14 [MSQ, 2M · Informed Search] Answer: A (A: Node C is expanded.)

Heuristic values: $h(B)=5$, $h(C)=6$, $h(E)=6$, $h(G)=0$, $h(H)=3$, $h(I)=6$, $h(S)=7$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (7)	I(8)
2	I (8)	H(10), C(12)
3	H (10)	C(12), G(12), E(18)
4	C (12)	G(12), B(18), E(18)
5	G (12)	B(18), E(18) ← goal, stop

Path returned: **S** → **I** → **H** → **G** with cost **12**; nodes expanded = **5** (S, I, H, C, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 12$. For comparison UCS ($h = 0$) would expand 6 nodes.

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, B, C (A: $((\neg R \vee Q) \wedge (\neg \neg R \vee P)) \rightarrow (Q \vee P)$; B: $((\neg Q \rightarrow R) \wedge \neg R) \rightarrow \neg \neg Q$; C: $P \rightarrow (Q \rightarrow P)$)

(A) $((\neg R \vee Q) \wedge (\neg \neg R \vee P)) \rightarrow (Q \vee P)$: true in all 8 assignments ⇒ **valid**.

(B) $((\neg Q \rightarrow R) \wedge \neg R) \rightarrow \neg \neg Q$: true in all 4 assignments ⇒ **valid**.

(C) $P \rightarrow (Q \rightarrow P)$: true in all 4 assignments ⇒ **valid**.

(D) $((P \rightarrow Q) \wedge \neg P) \rightarrow \neg Q$: **not valid** — counter-model $P=F, Q=T$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.16 [MSQ, 2M · Predicate Logic] Answer: A, D (A: $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$; D: $\exists x (P(x) \wedge Q(x)) \rightarrow (\exists x P(x) \wedge \exists x Q(x))$)

(A) If some a satisfies $P(a, y)$ for all y , then for each y that same a is a witness — valid. (B) Fails on domain $\{1, 2\}$ with $P(x, y) \iff x = y$: each y has a witness, but no single x works for all y . (C) Fails on domain $\{1, 2\}$ with $P = \{1\}$, $Q = \{2\}$: every element is P or Q , but neither P nor Q holds everywhere. (D) A witness a for $P(a) \wedge Q(a)$ witnesses both $\exists x P(x)$ and $\exists x Q(x)$ — valid.

Q.17 [MSQ, 2M · Predicate Logic] Answer: B (B: $\forall y \exists x R(x, y)$)

$R = \{(1,2), (1,4), (2,1), (2,2), (2,3), (3,2), (4,1), (4,4)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(1,2) \wedge R(2,1)$ but $\neg R(1,1)$.

(B) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(1,2)$ and $R(2,1)$ with $1 \neq 2$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $\neg Q \rightarrow R$; D: $\neg R \vee R$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	T	F	T
F	T	F	T

(A) $\neg Q \rightarrow R$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $\neg(P \wedge Q)$ is false in the KB-model $P=T, Q=T, R=F \Rightarrow$ not entailed.

(C) $(Q \wedge Q) \rightarrow R$ is false in the KB-model $P=T, Q=T, R=F \Rightarrow$ not entailed.

(D) $\neg R \vee R$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.19 [MSQ, 2M · Bayesian Networks] Answer: A, B

(A) A complete DAG implies no independences, so it is trivially an I-map. (B) AIMA shows causal orderings (e.g. B, E, A, J, M) give compact networks while diagnostic orderings (M, J, E, B, A) can be fully connected. (C) is false: adding an arc can only remove d-separation statements, so the set of implied independences shrinks and remains true in P. (D) is false: for the alarm example, order B, E, A, J, M gives 10 parameters, order M, J, A, B, E gives 13, and M, J, E, B, A gives 31.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.349 to 0.353

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot P(\text{Alarm} \mid \text{Burglary}) \cdot P(\text{JohnCalls} \mid \text{Earthquake}, \text{Alarm})$.

Inference by enumeration: $P(\text{Burglary} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Burglary}, e, \text{hidden})$.

Hidden: Al	Bu	product of CPT entries ($Bu \cdot Ea \cdot Al \cdot Jo$)	value
T	T	$0.4 \times 0.05 \times 0.9 \times 0.2$	0.0036
F	T	$0.4 \times 0.05 \times 0.1 \times 0.8$	0.0016
T	F	$0.6 \times 0.05 \times 0.8 \times 0.2$	0.0048
F	F	$0.6 \times 0.05 \times 0.2 \times 0.8$	0.0048

$P(\text{Burglary}=T, e) = 0.0052$; $P(\text{Burglary}=F, e) = 0.0096$; $P(e) = 0.0148$.

$P(\text{Burglary}=T \mid e) = 0.0052 / 0.0148 = \mathbf{0.3514} \approx 0.351$.

Q.21 [MSQ, 2M · Variable Elimination] Answer: A, B, D

(A) Sums of products can be reordered (distributive law), so every ordering computes the same exact result; only cost differs. (B) AIMA notes finding the optimal ordering is intractable (NP-hard), so heuristics such as min-fill or min-degree are used. (D) VE stores intermediate results as factors so each is computed once. (C) is false: the cost is exponential in the size of the largest factor, which can be large for multiply connected networks or bad orderings.

Q.22 [NAT, 2M · Sampling] Answer: 0.39 to 0.41

Rejection sampling keeps only samples consistent with the evidence $E = T$: samples 4, 7, 8, 9, 12, 13, 14, 15, 18, 20 (10 accepted, 10 rejected).

Among the accepted samples, $D = T$ in 4. Estimate = $4/10 = \mathbf{0.4}$.

(For this network the exact value is 0.5272; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.23 [MSQ, 2M · Conditional Independence] **Answer: B, D** (B: $F \perp G \mid A, D$; D: $E \perp G \mid A, C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp A \mid D$: **FALSE** — the path $G \leftarrow A$ is active given $\{D\}$.

(B) $F \perp G \mid A, D$: **TRUE** — $F \leftarrow A \rightarrow G$ (blocked: observed fork node A); $F \leftarrow B \leftarrow A \rightarrow G$ (blocked: observed fork node A); $F \leftarrow B \rightarrow C \leftarrow A \rightarrow G$ (blocked: observed fork node A); $F \leftarrow B \rightarrow C \rightarrow D \leftarrow A \rightarrow G$ (blocked: observed fork node A).

(C) $C \perp B \mid A$: **FALSE** — the path $C \leftarrow B$ is active given $\{A\}$.

(D) $E \perp G \mid A, C$: **TRUE** — $E \leftarrow A \rightarrow G$ (blocked: observed fork node A).

Q.24 [MCQ, 2M · Sampling] **Answer: D** (D: $A=T, B=T, C=T, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.8$; $u = 0.01 < 0.8 \Rightarrow A = T$
- B: $P(B=T \mid \text{—}) = 0.8$; $u = 0.74 < 0.8 \Rightarrow B = T$
- C: $P(C=T \mid B=T) = 0.5$; $u = 0.33 < 0.5 \Rightarrow C = T$
- D: $P(D=T \mid A=T, C=T) = 0.1$; $u = 0.55 \geq 0.1 \Rightarrow D = F$

Sample: **(A=T, B=T, C=T, D=F)**.

Q.25 [MCQ, 2M · Exact Inference] **Answer: C** (C: $f(A, D)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A)$, $f(A, C)$, $f(C, D)$, $f(D, E)$, $f(E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(A, C)$, $f(C, D)$	$\{A, C, D\}$	$f(A, D)$
F	$f(E, F)$	$\{E, F\}$	$f(E)$
A	$f(A)$, $f(A)$, $f(A, D)$	$\{A, D\}$	$f(D)$
D	$f(D, E)$, $f(D)$	$\{D, E\}$	$f(E)$

The largest factor produced has **2** variable(s) (created when eliminating C); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over E and normalise.

Mock Test 016

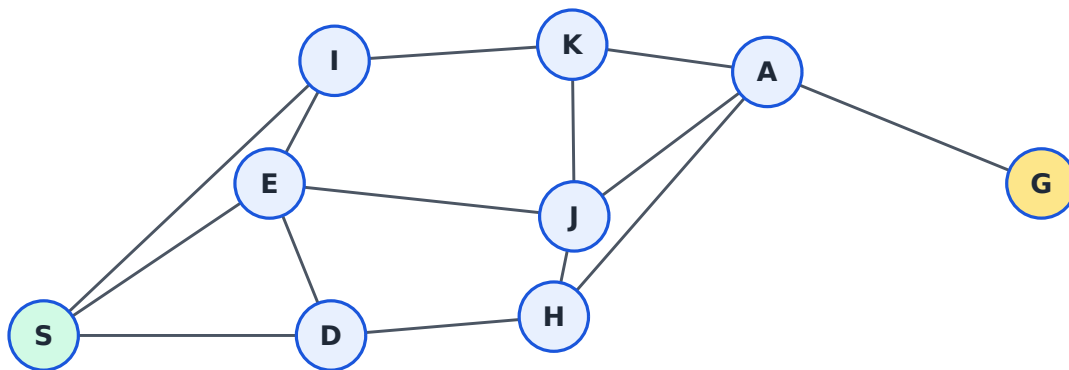
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 6 / 14	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, E, I, K, A, G
- ☐ (B) S, I, K, J, H, D, E, A, G
- ☐ (C) S, D, E, I, H, J, K, A, G
- ☐ (D) S, D, E, I, A, K, G

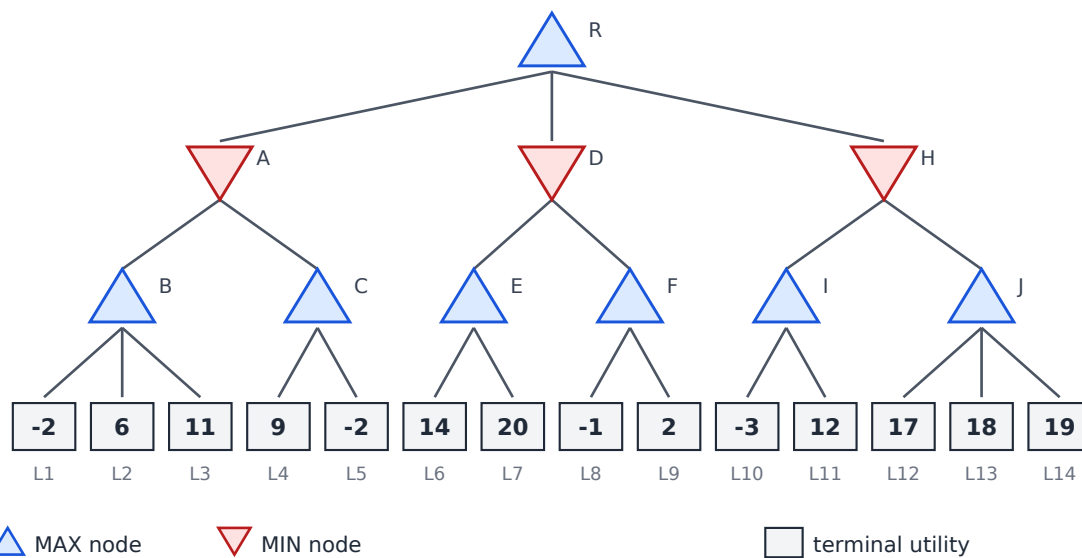
Q.2 [NAT · 1 mark · no negative marking] Informed Search – Effective branching factor

A* finds a solution at depth $d = 4$ after generating $N = 41$ nodes (root excluded). The effective branching factor b^* satisfies $N = b^* + (b^*)^2 + \dots + (b^*)^d$. Find b^* (round off to 2 decimal places).

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search

Chess has an average branching factor of about $b = 36$. With perfect move ordering, alpha-beta reduces the effective branching factor from b to $\sqrt{b}$. What is the effective branching factor?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, D, E** and rules:

- R1: $D \wedge J \rightarrow B$
- R2: $A \wedge F \rightarrow C$
- R3: $E \rightarrow J$
- R4: $C \wedge D \wedge J \rightarrow A$
- R5: $K \wedge E \rightarrow J$
- R6: $K \rightarrow E$
- R7: $D \wedge J \wedge B \rightarrow F$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following expresses 'None of my friends are perfect' ($F(x)$: x is my friend, $P(x)$: x is perfect)?

- ☐ (A) $\forall x (F(x) \rightarrow \neg P(x))$
- ☐ (B) $\exists x (F(x) \wedge \neg P(x))$
- ☐ (C) $\neg \forall x (F(x) \rightarrow P(x))$
- ☐ (D) $\forall x (\neg F(x) \rightarrow P(x))$

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

$\neg(p \rightarrow q)$ is logically equivalent to

- ☐ (A) $p \wedge \neg q$
☐ (B) $\neg p \wedge q$
☐ (C) $\neg p \rightarrow \neg q$
☐ (D) $\neg p \vee q$

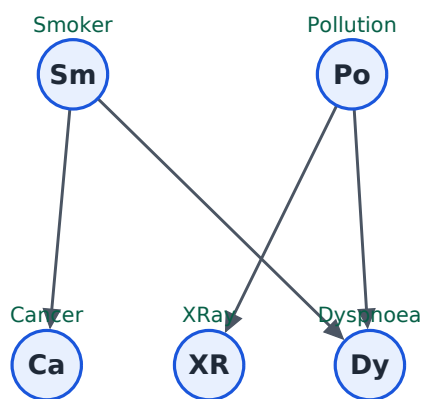
Q.8 [NAT · 1 mark · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. Compute the joint probability $P(j, m, a, \neg b, \neg e)$. (Round to 4 decimal places.)

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR, Dy. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay, Dy = Dyspnoea.) What is the probability that a single run produces the sample **(Sm=T, Po=T, Ca=F, XR=F, Dy=T)**? (round off to 4 decimal places)



P(Sm=T)
0.2

P(Po=T)
0.4

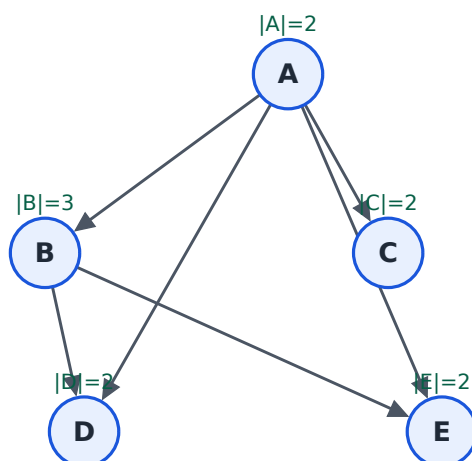
Sm	P(Ca=T ...)
T	0.6
F	0.25

Po	P(XR=T ..)	Sm	Po	P(Dy=T ...)
T	0.7	T	T	0.2
F	0.95	T	F	0.9
		F	T	0.25
		F	F	0.75

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Section B — 2-mark questions

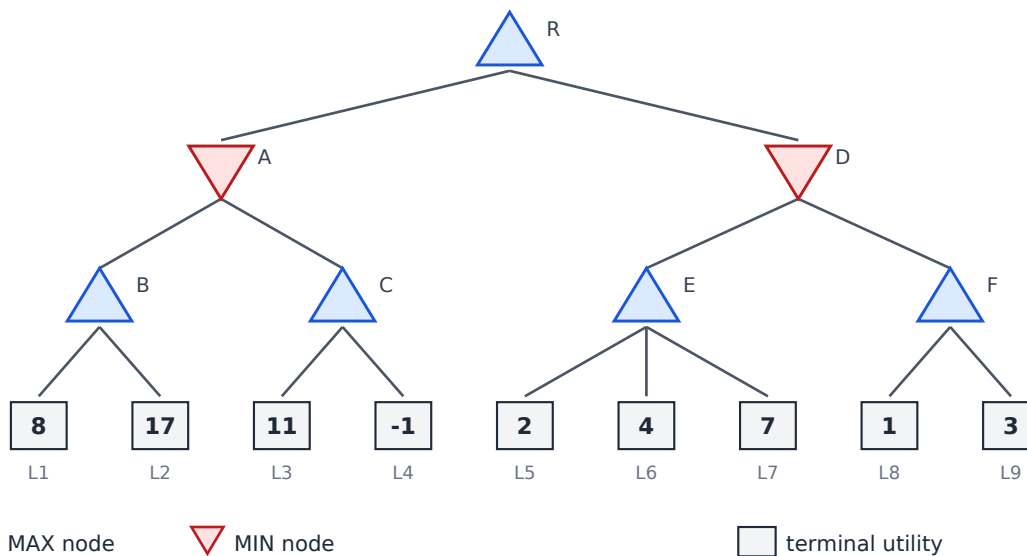
Q.11 [MSQ · 2 marks · no negative marking] Local Search

Which of the following statements about simulated annealing (maximisation) are TRUE?

- ☐ (A) If the temperature is lowered slowly enough, it finds a global optimum with probability approaching 1
- ☐ (B) It always accepts a move that improves the objective
- ☐ (C) At very high temperature, it behaves almost like a random walk
- ☐ (D) At $T = 0$, it accepts worsening moves with probability 1

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

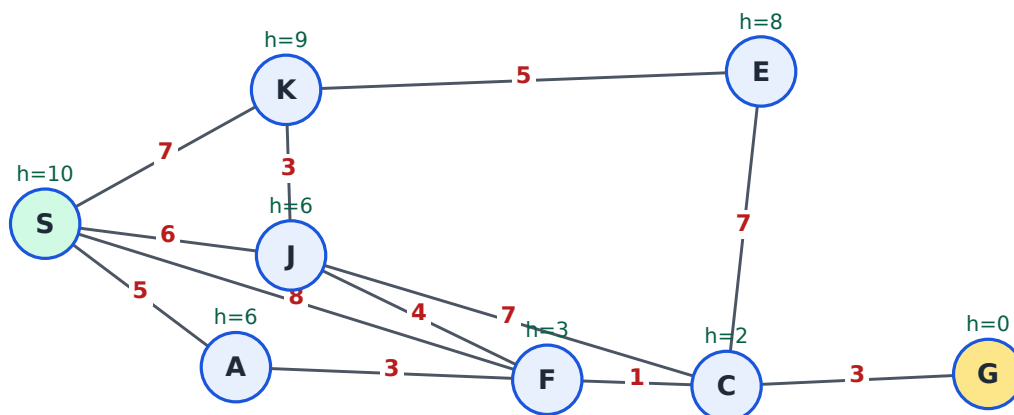
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – A*

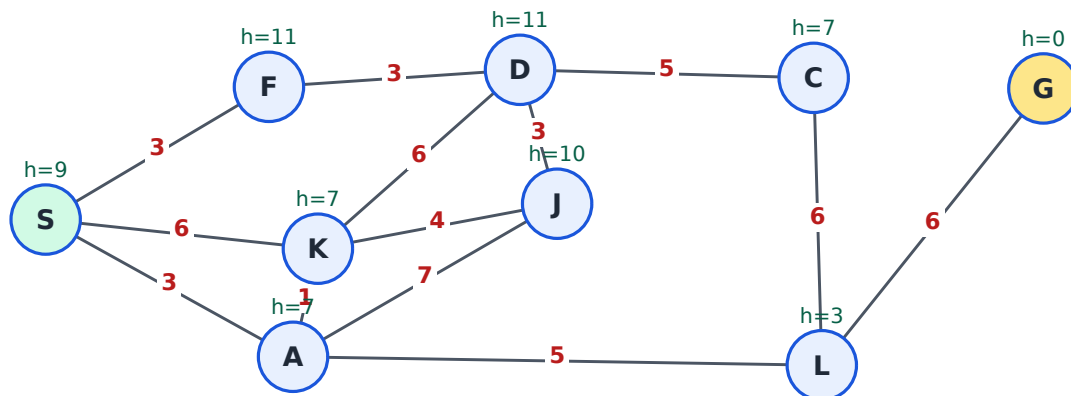
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node E is expanded.
- ☐ (B) Node A is expanded.
- ☐ (C) The path returned has cost 12.
- ☐ (D) Node C is expanded.

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Adversarial Search

A game tree has a MAX root with three MIN children. Each MIN child has two MAX children, and each of those has two leaves, for 12 leaves in total. The leaves, left to right, are 4, 8, 9, 3, 2, 1, 7, 6, 5, 10, 2, 11. Alpha-beta pruning processes children left to right. How many leaves are **evaluated**?

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg P \vee \neg \neg R$
- (2) $(R \rightarrow \neg R) \wedge (R \rightarrow P)$
- (3) $(Q \vee \neg Q) \rightarrow (\neg R \leftrightarrow Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg R$
- ☐ (B) $Q \rightarrow P$
- ☐ (C) $R \wedge R$
- ☐ (D) $Q \rightarrow (Q \leftrightarrow P)$

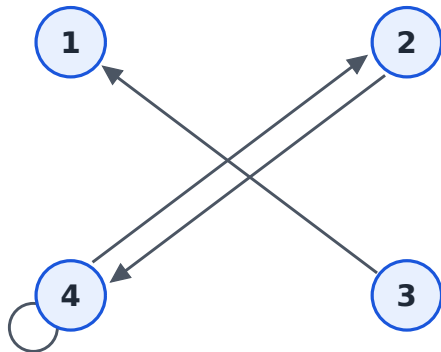
Q.17 [MCQ · 2 marks · negative: -2/3] Predicate Logic

The negation of $\forall x \exists y (P(x, y) \rightarrow Q(y))$ is logically equivalent to

- ☐ (A) $\exists x \forall y (P(x, y) \wedge \neg Q(y))$
- ☐ (B) $\exists x \forall y (\neg P(x, y) \rightarrow Q(y))$
- ☐ (C) $\forall x \exists y (P(x, y) \wedge \neg Q(y))$
- ☐ (D) $\exists x \exists y (P(x, y) \wedge \neg Q(y))$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\exists y \forall x R(x, y)$
☐ (C) $\exists x R(x, x)$
☐ (D) $\exists x \forall y \neg R(x, y)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$\mathbf{F} = ((Q \wedge \neg P) \leftrightarrow \neg R) \rightarrow (S \leftrightarrow \neg S) ?$$

Answer (numerical): _____

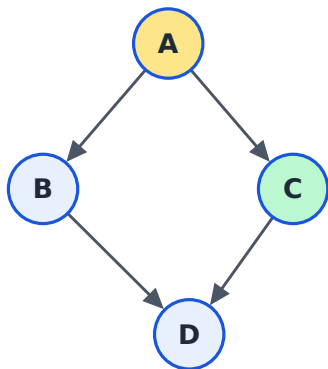
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \rightarrow Q) \wedge Q) \rightarrow R$
☐ (B) $(Q \rightarrow (R \rightarrow P)) \leftrightarrow ((Q \wedge R) \rightarrow P)$
☐ (C) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$
☐ (D) $\neg(\neg P \wedge Q) \leftrightarrow (\neg\neg P \vee \neg Q)$

Q.21 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{A} = \mathbf{T}$. The values of the non-evidence variables in the generated samples are listed in the table. What is the weight of sample #1? (round off to 4 decimal places)



P(A=T)
0.95

A	P(B=T ...)	A	P(C=T ...)
T	0.1	T	0.6
F	0.25	F	0.2

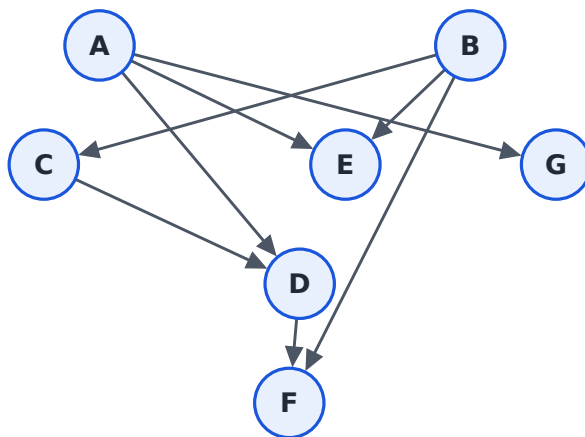
B	C	P(D=T ...)
T	T	0.5
T	F	0.1
F	T	0.5
F	F	0.6

#	B	C	D
1	F	T	T
2	F	T	T
3	F	T	F
4	T	F	F

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp G$
☐ (B) $E \perp F \mid A$
☐ (C) $A \perp C \mid B$
☐ (D) $C \perp G \mid A$

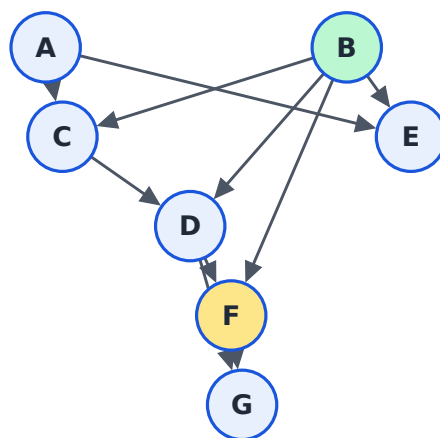
Q.23 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Gibbs sampling is run with evidence $S = \text{true}$, $W = \text{true}$. In the current state $C = \text{true}$. Compute the probability with which R is resampled to true, i.e. $P(r | c, s, w)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.24 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

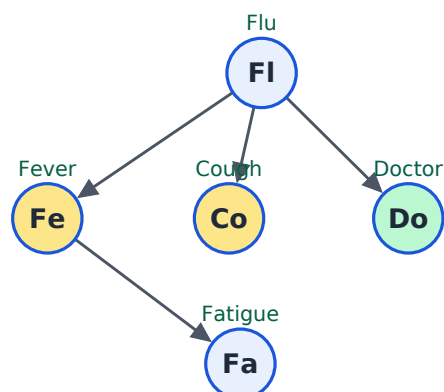
Variable elimination is used to compute $\mathbf{P(B | F = T)}$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, D, G, C, E**. What is the scope of the new factor produced when **A** is summed out?



- ☐ (A) $f(A, B, C, E)$
☐ (B) $f(B, C, E)$
☐ (C) $f(C, E)$
☐ (D) $f(B, C)$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T | \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) Compute $\mathbf{P(\text{Doctor} = T | \text{Cough} = T, \text{Fever} = T)}$ (round off to 3 decimal places).



P(FI=T)					
0.3		FI	P(Fe=T ...)	FI	P(Co=T ...)
		T	0.8	T	0.1
		F	0.5	F	0.2

Fe	P(Fa=T ...)	FI	P(Do=T ...)
T	0.5	T	0.6
F	0.7	F	0.3

Answer (numerical): _____

Answer Key — Mock Test 016

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	NAT	2	Informed Search	6
2	NAT	1	Informed Search	2.18 to 2.22	15	NAT	2	Adversarial Search	9
3	NAT	1	Adversarial Search	12	16	MSQ	2	Propositional Logic	A
4	NAT	1	Adversarial Search	6	17	MCQ	2	Predicate Logic	A
5	NAT	1	Propositional Logic	5	18	MSQ	2	Predicate Logic	C, D
6	MCQ	1	Predicate Logic	A	19	NAT	2	Propositional Logic	8
7	MCQ	1	Propositional Logic	A	20	MSQ	2	Propositional Logic	B, D
8	NAT	1	Bayesian Networks	0.0054 to 0.0055	21	NAT	2	Sampling	0.9498 to 0.9502
9	NAT	1	Sampling	0.0017 to 0.0021	22	MSQ	2	Conditional Independence	A, C, D
10	NAT	1	Bayesian Networks	19	23	NAT	2	Sampling	0.814 to 0.815
11	MSQ	2	Local Search	A, B, C	24	MCQ	2	Exact Inference	B
12	NAT	2	Adversarial Search	2	25	NAT	2	Exact Inference	0.375 to 0.379
13	MSQ	2	Informed Search	B, C, D					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: A (A: S, D, E, I, K, A, G)

Adjacency lists (alphabetical): A: {G, H, J, K}; D: {E, H, S}; E: {D, I, J, S}; G: {A}; H: {A, D, J}; I: {E, K, S}; J: {A, E, H, K}; K: {A, I, J}; S: {D, E, I}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, E, I, K, A, G**; the path found is $S \rightarrow D \rightarrow E \rightarrow I \rightarrow K \rightarrow A \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

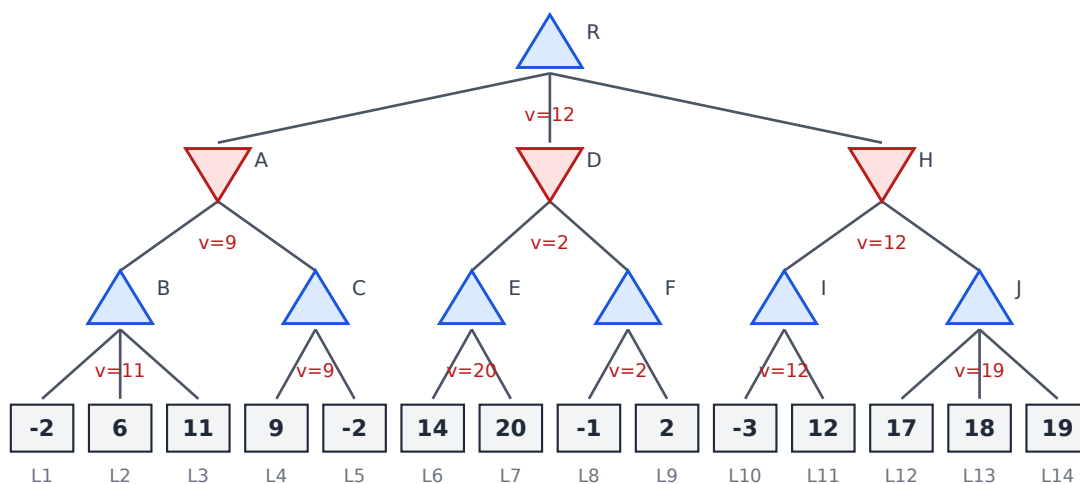
Q.2 [NAT, 1M · Informed Search] Answer: 2.18 to 2.22

Solve $b + b^2 + \dots + b^4 = 41$ numerically (bisection): $b^* \approx 2.2$.

Check: $2.2^1 + 2.2^2 + 2.2^3 + 2.2^4 \approx 41.11 \approx 41$.

Q.3 [NAT, 1M · Adversarial Search] Answer: 12

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 11$, $C = 9$, $A = 9$, $E = 20$, $F = 2$, $D = 2$, $I = 12$, $J = 19$, $H = 12$. Root value = **12**; MAX's optimal first move is towards H.

Q.4 [NAT, 1M · Adversarial Search] Answer: 6

Perfect ordering gives $O(b^{m/2}) = O((\sqrt{b})^m)$. The effective branching factor is $\sqrt{36} = 6$. Equivalently, alpha-beta can search roughly twice as deep as minimax in the same time.

Q.5 [NAT, 1M · Propositional Logic] Answer: 5

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $E \rightarrow J$; add J.

Round 2: fire $D \wedge J \wedge B \rightarrow F$; add F.

Fixed point: {B, D, E, F, J} — 5 atoms are entailed (facts: B, D, E; newly derived: F, J). Atoms never derived: A, C, H, K.

Q.6 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x (F(x) \rightarrow \neg P(x))$)

'None of my friends are perfect' $\equiv \neg \exists x (F(x) \wedge P(x)) \equiv \forall x \neg (F(x) \wedge P(x)) \equiv \forall x (F(x) \rightarrow \neg P(x))$. Options B and C (which are equivalent to each other) only say some friend is not perfect. Option D speaks about non-friends.

Q.7 [MCQ, 1M · Propositional Logic] Answer: A (A: $p \wedge \neg q$)

$p \rightarrow q \equiv \neg p \vee q$, so $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q$ by De Morgan's law. It is true only in the row $p=T, q=F$, which is exactly the row where $p \rightarrow q$ is false. $\neg p \wedge q$ is true only at $p=F, q=T$; $\neg p \rightarrow \neg q$ is the inverse; $\neg p \vee q$ is $p \rightarrow q$ itself.

Q.8 [NAT, 1M · Bayesian Networks] Answer: 0.0054 to 0.0055

By the BN factorization, $P(j, m, a, \neg b, \neg e) = P(j | a) P(m | a) P(a | \neg b, \neg e) P(\neg b) P(\neg e) = 0.8 \times 0.7 \times 0.01 \times 0.98 \times 0.99 = 0.56 \times 0.01 \times 0.9702 = 0.005433 \approx 0.0054$.

Q.9 [NAT, 1M · Sampling] Answer: 0.0017 to 0.0021

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Sm}=T) \times P(\text{Po}=T) \times P(\text{Ca}=F | \text{Sm}=T) \times P(\text{XR}=F | \text{Po}=T) \times P(\text{Dy}=T | \text{Sm}=T, \text{Po}=T)$
 $= 0.2 \times 0.4 \times 0.4 \times 0.3 \times 0.2 = \mathbf{0.00192} \approx 0.0019$.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 19

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (3-1) \times 2 + C: (2-1) \times 2 + D: (2-1) \times 2 \times 3 + E: (2-1) \times 2 \times 3 = \mathbf{19}$.

Full joint table: $\prod |X| - 1 = 48 - 1 = 47$; saving = 28.

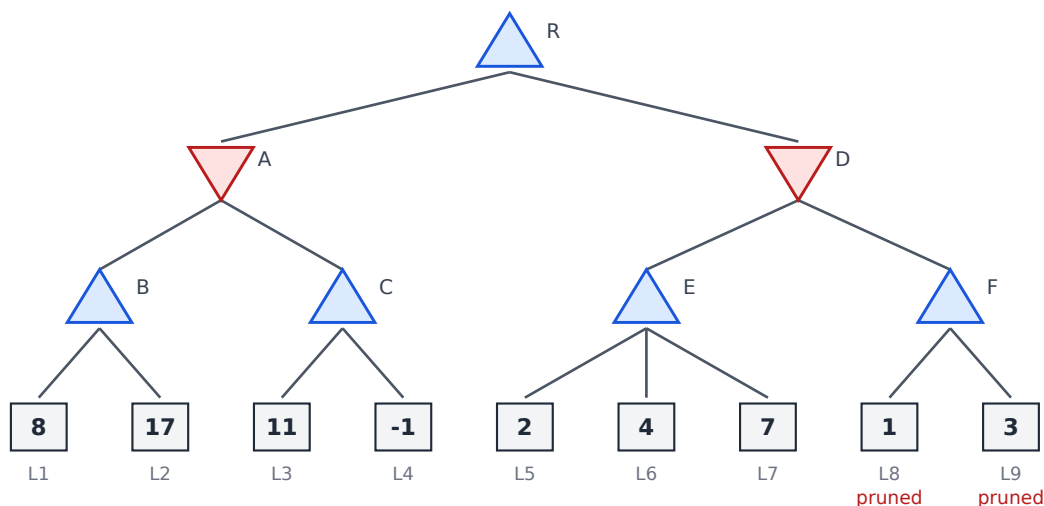
Q.11 [MSQ, 2M · Local Search] Answer: A, B, C

(A) is the classical convergence result for sufficiently slow cooling. (B) is true: moves with $\Delta E > 0$ are always taken. (C) is true: $e^{\Delta E/T} \rightarrow 1$ as $T \rightarrow \infty$, so almost every move is accepted. (D) is false: as $T \rightarrow 0$, $e^{\Delta E/T} \rightarrow 0$ for $\Delta E < 0$, so worsening moves are rejected and the algorithm becomes hill climbing.

Q.12 [NAT, 2M · Adversarial Search] Answer: 2

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node D: value $7 \leq \alpha = 11 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (7): L1, L2, L3, L4, L5, L6, L7. Leaves pruned (2): L8, L9. Root value = 11 (same as plain minimax, 11 — pruning never changes the root value).

Q.13 [MSQ, 2M · Informed Search] Answer: B, C, D (B: Node A is expanded.; C: The path returned has cost 12.; D: Node C is expanded.)

Heuristic values: $h(A)=6$, $h(C)=2$, $h(E)=8$, $h(F)=3$, $h(G)=0$, $h(J)=6$, $h(K)=9$, $h(S)=10$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (10)	A(11), F(11), J(12), K(16)
2	A (11)	F(11), J(12), K(16)
3	F (11)	C(11), J(12), K(16)
4	C (11)	G(12), J(12), K(16), E(24)
5	G (12)	J(12), K(16), E(24) ← goal, stop

Path returned: **S → F → C → G** with cost **12**; nodes expanded = **5** (S, A, F, C, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 12$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.14 [NAT, 2M · Informed Search] Answer: 6

Heuristic values: $h(A)=7$, $h(C)=7$, $h(D)=11$, $h(F)=11$, $h(G)=0$, $h(J)=10$, $h(K)=7$, $h(L)=3$, $h(S)=9$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	A(10), K(13), F(14)
2	A (10)	K(11), L(11), F(14), J(20)
3	K (11)	L(11), F(14), J(18), D(21)
4	L (11)	F(14), G(14), J(18), C(21), D(21)
5	F (14)	G(14), D(17), J(18), C(21)
6	G (14)	D(17), J(18), C(21) ← goal, stop

Path returned: **S → A → L → G** with cost **14**; nodes expanded = **6** (S, A, K, L, F, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.15 [NAT, 2M · Adversarial Search] Answer: 9

MIN_1 : $\text{MAX}(4, 8) = 8$, so $\beta = 8$. The next MAX node sees $9 \geq 8$ and prunes leaf 3. $\text{MIN}_1 = 8$, so root $\alpha = 8$.
 MIN_2 : $\text{MAX}(2, 1) = 2 \leq \alpha$, so the MAX node with leaves (7, 6) is pruned. MIN_3 : $\text{MAX}(5, 10) = 10$, so $\beta = 10$.
 $\text{MAX}(2, 11)$: leaf 2 does not cut and $11 \geq 10$ cuts, but no siblings remain, so both are evaluated. $\text{MIN}_3 = 10$, and the root value is 10. Pruned: 3, 7, 6. Evaluated: $12 - 3 = 9$.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A (A: $\neg R$)

$\text{KB} \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
F	T	F	T

(A) $\neg R$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $Q \rightarrow P$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(C) $R \wedge R$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(D) $Q \rightarrow (Q \leftrightarrow P)$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

Q.17 [MCQ, 2M · Predicate Logic] Answer: A (A: $\exists x \forall y (P(x, y) \wedge \neg Q(y))$)

Push $\neg$ inward: $\neg \forall x \exists y \phi \equiv \exists x \neg \exists y \phi \equiv \exists x \forall y \neg \phi$. Then $\neg(P \rightarrow Q) \equiv P \wedge \neg Q$, giving $\exists x \forall y (P(x, y) \wedge \neg Q(y))$.

Option B uses $\neg P \rightarrow Q \equiv P \vee Q$, which is not $\neg(P \rightarrow Q)$. Option C does not flip the quantifiers. Option D flips only one quantifier correctly ($\exists y$ should become $\forall y$).

Q.18 [MSQ, 2M · Predicate Logic] Answer: C, D (C: $\exists x R(x, x)$; D: $\exists x \forall y \neg R(x, y)$)

$R = \{(2,4), (3,1), (4,2), (4,4)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(2,4) \wedge R(4,2)$ but $\neg R(2,2)$.

(B) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 4.

(D) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [NAT, 2M · Propositional Logic] Answer: 8

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(Q \wedge \neg P) \leftrightarrow \neg R$	$S \leftrightarrow \neg S$	$((Q \wedge \neg P) \leftrightarrow \neg R) \rightarrow (S \leftrightarrow \neg S)$
T	T	T	T	T	F	F
T	T	T	F	T	F	F
T	T	F	T	F	F	T
T	T	F	F	F	F	T
T	F	T	T	T	F	F
T	F	T	F	T	F	F
T	F	F	T	F	F	T
T	F	F	F	F	F	T
F	T	T	T	F	F	T
F	T	T	F	F	F	T
F	T	F	T	T	F	F
F	T	F	F	T	F	F
F	F	T	T	T	F	F
F	F	T	F	T	F	F
F	F	F	T	F	F	T
F	F	F	F	F	F	T

Rows where F = T: **8**.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: B, D** (B: $(Q \rightarrow (R \rightarrow P)) \leftrightarrow ((Q \wedge R) \rightarrow P)$; D: $\neg(\neg P \wedge Q) \leftrightarrow (\neg\neg P \vee \neg Q)$)

(A) $((R \rightarrow Q) \wedge Q) \rightarrow R$: **not valid** — counter-model $Q=T, R=F$.

(B) $(Q \rightarrow (R \rightarrow P)) \leftrightarrow ((Q \wedge R) \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$: **not valid** — counter-model $P=T, R=F$.

(D) $\neg(\neg P \wedge Q) \leftrightarrow (\neg\neg P \vee \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.21 [NAT, 2M · Sampling] **Answer: 0.9498 to 0.9502**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	C
1	$P(A=T)=0.95$	0.95	T
2	$P(A=T)=0.95$	0.95	T
3	$P(A=T)=0.95$	0.95	T
4	$P(A=T)=0.95$	0.95	F

$\sum w$ (all) = 3.8; $\sum w$ (C = T) = 2.85.

Estimate $P(C=T | A = T) = 2.85 / 3.8 = \mathbf{0.75}$.

Q.22 [MSQ, 2M · Conditional Independence] **Answer: A, C, D** (A: $C \perp G$; C: $A \perp C | B$; D: $C \perp G | A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $C \perp G$: **TRUE** — $C \leftarrow B \rightarrow E \leftarrow A \rightarrow G$ (blocked: collider E with neither it nor a descendant observed); $C \leftarrow B \rightarrow F \leftarrow D \leftarrow A \rightarrow G$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow D \leftarrow A \rightarrow G$ (blocked: collider D with neither it nor a descendant observed); $C \rightarrow D \rightarrow F \leftarrow B \rightarrow E \leftarrow A \rightarrow G$ (blocked: collider F with neither it nor a descendant observed).

(B) $E \perp F | A$: **FALSE** — the path $E \leftarrow B \rightarrow C \rightarrow D \rightarrow F$ is active given $\{A\}$.

(C) $A \perp C | B$: **TRUE** — $A \rightarrow D \leftarrow C$ (blocked: collider D with neither it nor a descendant observed); $A \rightarrow D \rightarrow F \leftarrow B \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow E \leftarrow B \rightarrow C$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow E \leftarrow B \rightarrow F \leftarrow D \leftarrow C$ (blocked: collider E with neither it nor a descendant observed).

(D) $C \perp G | A$: **TRUE** — $C \leftarrow B \rightarrow E \leftarrow A \rightarrow G$ (blocked: collider E with neither it nor a descendant observed); $C \leftarrow B \rightarrow F \leftarrow D \leftarrow A \rightarrow G$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow D \leftarrow A \rightarrow G$ (blocked: collider D with neither it nor a descendant observed); $C \rightarrow D \rightarrow F \leftarrow B \rightarrow E \leftarrow A \rightarrow G$ (blocked: collider F with neither it nor a descendant observed).

Q.23 [NAT, 2M · Sampling] Answer: 0.814 to 0.815

The Markov blanket of R is {C, S, W}. $P(R | mb) = \alpha P(R | c) P(w | s, R)$. For r: $0.8 \times 0.99 = 0.792$. For $\neg r$: $0.2 \times 0.90 = 0.18$. $P(r | c, s, w) = 0.792 / (0.792 + 0.18) = 0.792 / 0.972 \approx 0.8148 \approx 0.815$.

Q.24 [MCQ, 2M · Exact Inference] Answer: B (B: f(B, C, E))

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, B, C)$, $f(B, C, D)$, $f(A, B, E)$, $f(B, D)$, $f(D, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, B, C)$, $f(A, B, E)$	{A, B, C, E}	$f(B, C, E)$
D	$f(B, C, D)$, $f(B, D)$, $f(D, G)$	{B, C, D, G}	$f(B, C, G)$
G	$f(B, C, G)$	{B, C, G}	$f(B, C)$
C	$f(B, C, E)$, $f(B, C)$	{B, C, E}	$f(B, E)$
E	$f(B, E)$	{B, E}	$f(B)$

The largest factor produced has **3** variable(s) (created when eliminating A); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over B and normalise.

Q.25 [NAT, 2M · Exact Inference] Answer: 0.375 to 0.379

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}, \text{Doctor}) = P(\text{Flu}) \cdot P(\text{Fever} | \text{Flu}) \cdot P(\text{Cough} | \text{Flu}) \cdot P(\text{Fatigue} | \text{Fever}) \cdot P(\text{Doctor} | \text{Flu})$.

Inference by enumeration: $P(\text{Doctor} | e) = \alpha \sum_{\text{hidden}} P(\text{Doctor}, e, \text{hidden})$.

Hidden: Fl, Fa	Do	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, T	T	$0.3 \times 0.8 \times 0.1 \times 0.5 \times 0.6$	0.0072
T, F	T	$0.3 \times 0.8 \times 0.1 \times 0.5 \times 0.6$	0.0072
F, T	T	$0.7 \times 0.5 \times 0.2 \times 0.5 \times 0.3$	0.0105
F, F	T	$0.7 \times 0.5 \times 0.2 \times 0.5 \times 0.3$	0.0105
T, T	F	$0.3 \times 0.8 \times 0.1 \times 0.5 \times 0.4$	0.0048
T, F	F	$0.3 \times 0.8 \times 0.1 \times 0.5 \times 0.4$	0.0048
F, T	F	$0.7 \times 0.5 \times 0.2 \times 0.5 \times 0.7$	0.0245
F, F	F	$0.7 \times 0.5 \times 0.2 \times 0.5 \times 0.7$	0.0245

$P(\text{Doctor}=T, e) = 0.0354$; $P(\text{Doctor}=F, e) = 0.0586$; $P(e) = 0.094$.

$P(\text{Doctor}=T | e) = 0.0354 / 0.094 = \mathbf{0.3766} \approx 0.377$.

Mock Test 017

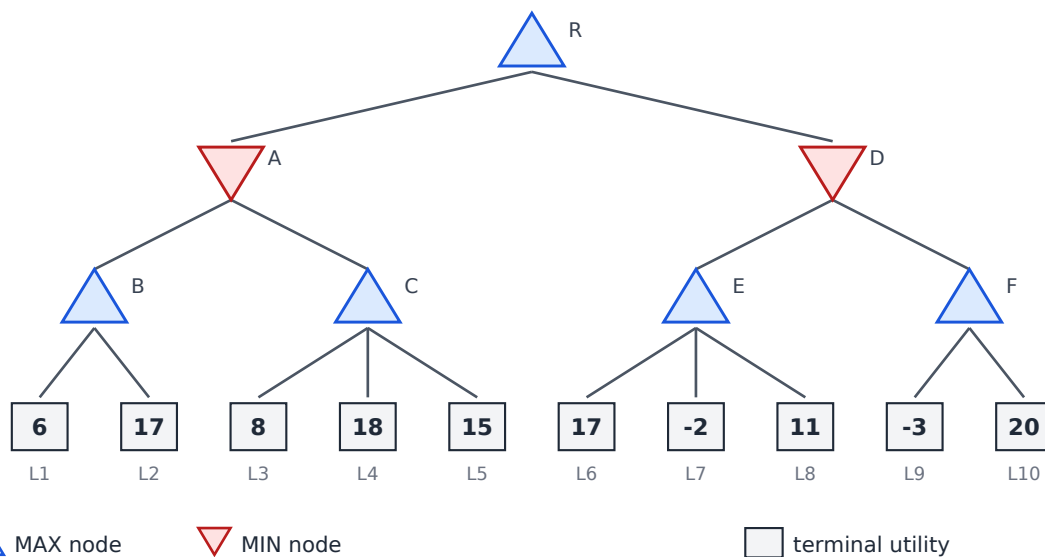
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	8 / 6 / 11	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

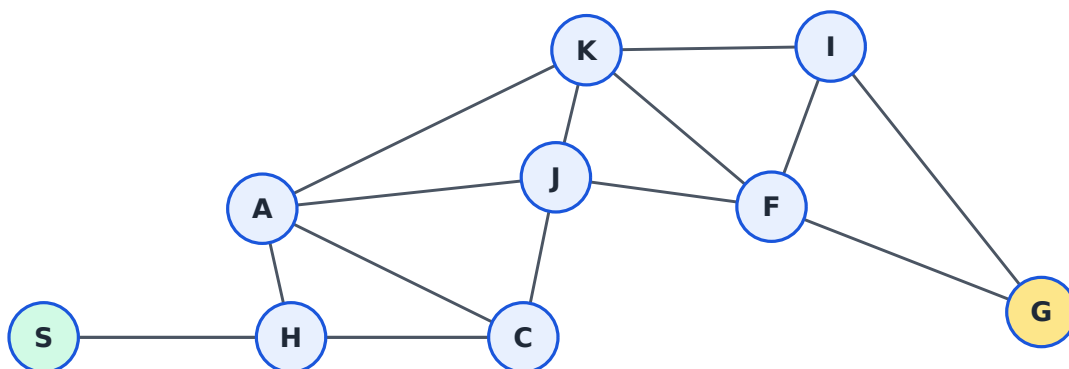
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

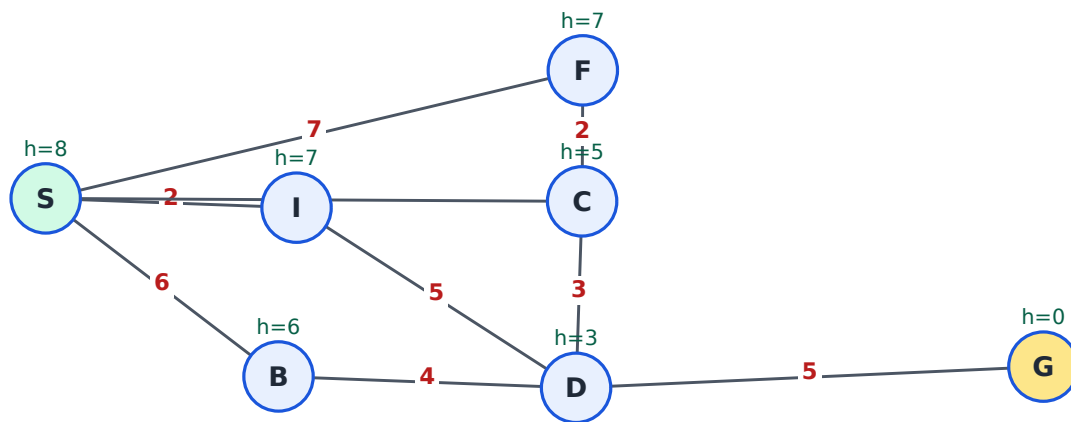
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, C, A, H, J, F, G
- ☐ (B) S, H, C, J, K, I, G
- ☐ (C) S, C, H, A, J, K, F, I, G
- ☐ (D) S, A, C, H, J, F, G

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Informed Search

A* search with $h(n) = 0$ for every node n is equivalent to

- ☐ (A) breadth-first search
- ☐ (B) depth-first search
- ☐ (C) uniform-cost search
- ☐ (D) greedy best-first search

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Over the finite domain $\{a, b\}$, the sentence $\forall x \exists y R(x, y)$ is equivalent to which ground formula?

- ☐ (A) $(R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$
- ☐ (B) $(R(a, a) \wedge R(a, b)) \vee (R(b, a) \wedge R(b, b))$
- ☐ (C) $(R(a, a) \vee R(b, a)) \wedge (R(a, b) \vee R(b, b))$
- ☐ (D) $R(a, a) \wedge R(b, b)$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = (Q \vee (\neg P \wedge P)) \rightarrow ((P \leftrightarrow Q) \wedge (R \leftrightarrow P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

What is the most general unifier (MGU) of $\text{Knows}(\text{John}, x)$ and $\text{Knows}(y, \text{Mother}(y))$?

- ☐ (A) $\{y/\text{John}, x/\text{Mother}(\text{John})\}$
- ☐ (B) $\{x/\text{John}, y/\text{Mother}(\text{John})\}$
- ☐ (C) $\{y/\text{John}, x/\text{Mother}(y)\}$
- ☐ (D) The two atoms are not unifiable

Q.8 [NAT · 1 mark · no negative marking] Predicate Logic

Let the domain be $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \forall y (R(x, y) \rightarrow R(y, x))$?

Answer (numerical): _____

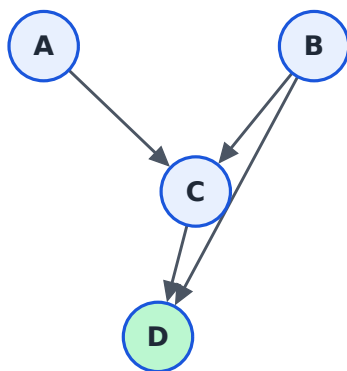
Q.9 [MCQ · 1 mark · negative: -1/3] Probability for AI

If n Boolean random variables are mutually (absolutely) independent, how many independent numbers suffice to specify their full joint distribution?

- (A) n
 ○ (B) $2^n - 1$
 ○ (C) $2n$
 ○ (D) n^2

Q.10 [NAT · 1 mark · no negative marking] Exact Inference - Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(D = T)$ (round off to 3 decimal places).



$P(A=T)$
0.4

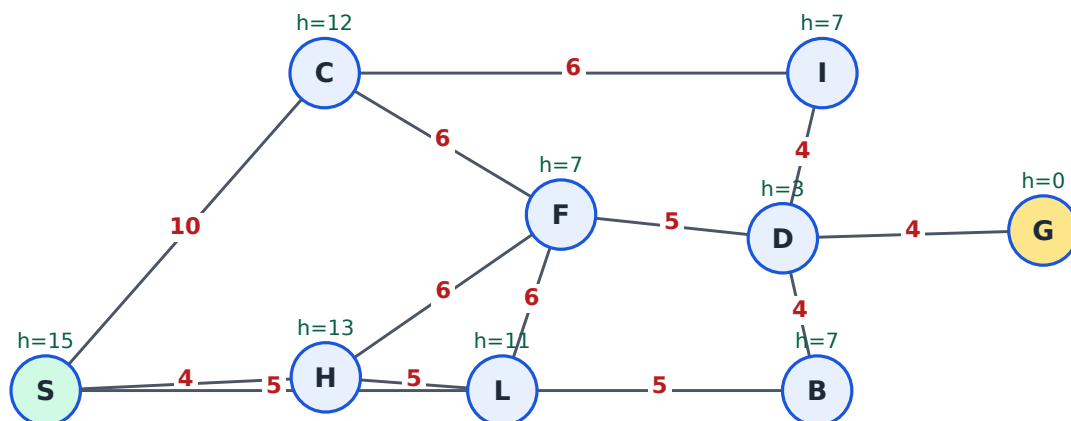
$P(B=T)$	A	B	$P(C=T \dots)$
0.4	T	T	0.25
	T	F	0.5
	F	T	0.1
	F	F	0.5

B	C	$P(D=T \dots)$
T	T	0.2
T	F	0.3
F	T	0.3
F	F	0.05

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Informed Search - A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Breadth-first search goal-tests a node when it is generated, generating the left child before the right, and the goal is node 13. Depth-first search goal-tests on removal from the frontier, exploring the left child first. Which is correct about the number of nodes each algorithm goal-tests up to and including node 13 (counting the root)?

- ☐ (A) BFS: 13, DFS: 12
☐ (B) BFS: 13, DFS: 11
☐ (C) BFS: 12, DFS: 12
☐ (D) BFS: 7, DFS: 12

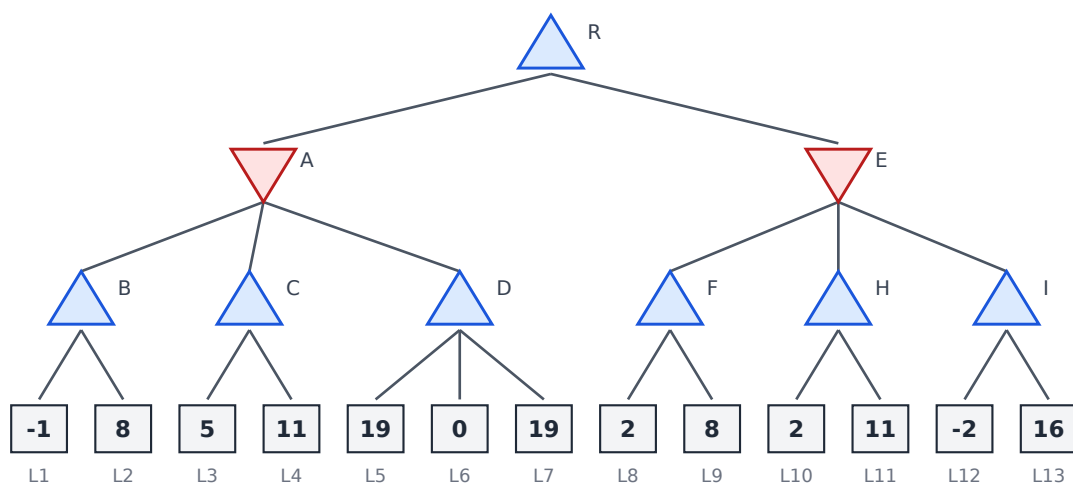
Q.13 [NAT · 2 marks · no negative marking] Informed Search

In the 8-puzzle, the goal is 1 2 3 / 8 _ 4 / 7 6 5 (rows top to bottom, _ = blank), and the current state is 2 8 3 / 1 6 4 / 7 _ 5. Let h_1 be the number of misplaced tiles and h_2 the total Manhattan distance. What is $h_1 + h_2$?

Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



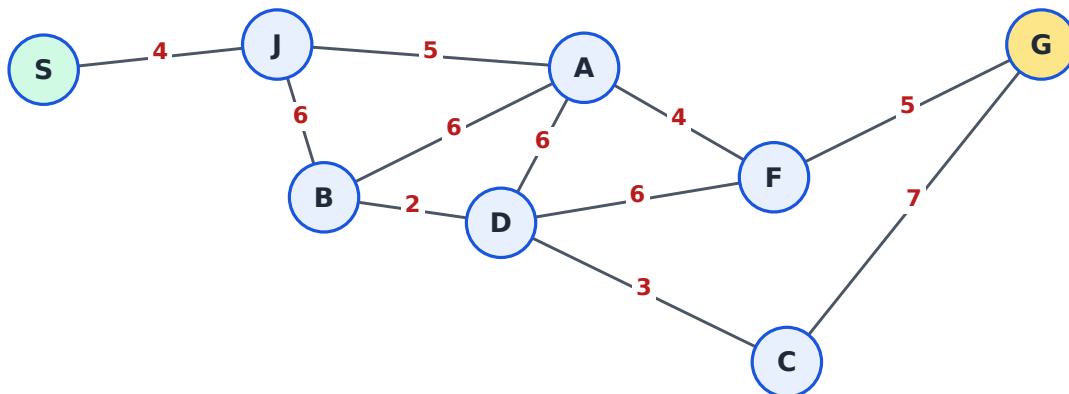
MAX node MIN node

terminal utility

- ☐ (A) Leaf L1 is pruned.
☐ (B) Leaf L2 is pruned.
☐ (C) Leaf L6 is pruned.
☐ (D) Leaf L10 is pruned.

Q.15 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow J \rightarrow A \rightarrow F \rightarrow G$
☐ (B) $S \rightarrow J \rightarrow B \rightarrow D \rightarrow C \rightarrow G$
☐ (C) $S \rightarrow J \rightarrow B \rightarrow D \rightarrow F \rightarrow G$
☐ (D) $S \rightarrow J \rightarrow A \rightarrow D \rightarrow C \rightarrow G$

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic

A propositional Horn knowledge base contains the facts A and B and the definite clauses $A \wedge B \rightarrow C$, $C \rightarrow D$, $D \wedge E \rightarrow F$, $B \rightarrow E$, $F \wedge G \rightarrow H$, $H \rightarrow A$. Forward chaining is run to completion. How many distinct proposition symbols are inferred to be true (count the initial facts too)?

Answer (numerical): _____

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = P \rightarrow (P \wedge R)$?

- ☐ (A) $(P \wedge R) \rightarrow P$
☐ (B) $\neg P \wedge (P \wedge R)$
☐ (C) $\neg(P \wedge \neg(P \wedge R))$
☐ (D) $P \vee \neg(P \wedge R)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

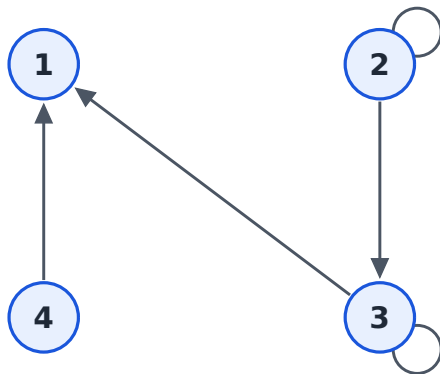
C1: $(Q \vee \neg R)$, C2: $(P \vee \neg Q \vee \neg R)$, C3: $(\neg P \vee \neg R \vee S)$, C4: $(\neg P)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg Q \vee \neg R)$
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) $(Q \vee S)$
☐ (D) $(P \vee \neg R)$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

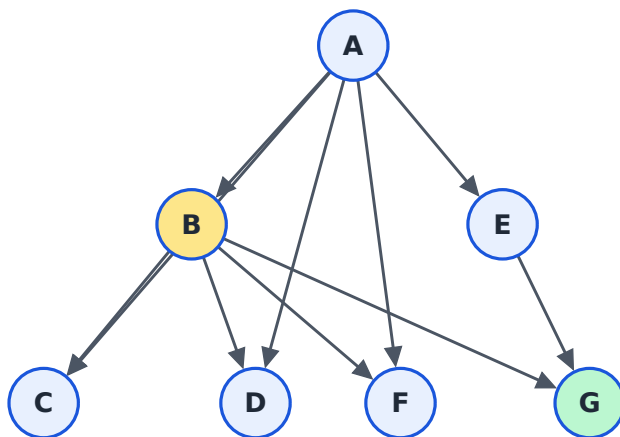
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x R(x, x)$
☐ (B) $\exists x \forall y R(x, y)$
☐ (C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (D) $\exists x \forall y \neg R(x, y)$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

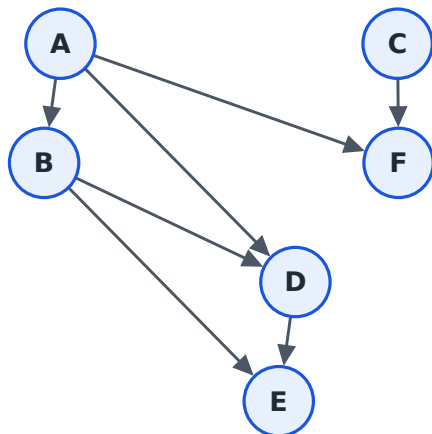
Variable elimination is used to compute $P(\mathbf{G} \mid \mathbf{B} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{B} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, E, A, F, C**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp E \mid C, F$
☐ (B) $C \perp D$
☐ (C) $D \perp F$
☐ (D) $D \perp C$

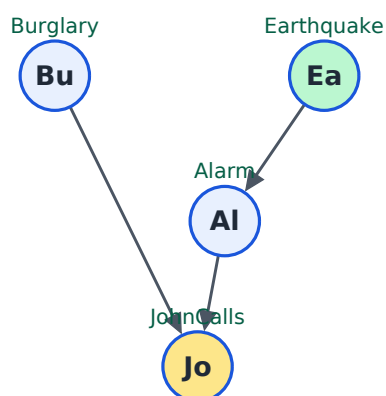
Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $C \rightarrow A$ and $C \rightarrow B$ (Boolean variables) with $P(c) = 0.4$, $P(a \mid c) = 0.9$, $P(a \mid \neg c) = 0.2$, $P(b \mid c) = 0.7$, $P(b \mid \neg c) = 0.1$. Which of the following are true?

- ☐ (A) $P(a) = 0.48$
☐ (B) $A \perp B$ (marginally)
☐ (C) $P(a \mid b, c) = 0.9$
☐ (D) $P(a \mid b) \approx 0.776$

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute **$P(\text{Earthquake} = T \mid \text{JohnCalls} = T)$** (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.2

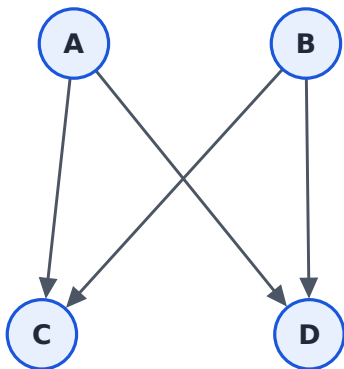
$P(\text{Ea}=T)$	Ea	$P(\text{Al}=T \dots)$
0.05	T	0.8
	F	0.75

Bu	Al	$P(\text{Jo}=T \dots)$
T	T	0.6
T	F	0.3
F	T	0.4
F	F	0.7

Answer (numerical): _____

Q.24 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.57}, \mathbf{0.69}, \mathbf{0.97}, \mathbf{0.1} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.7

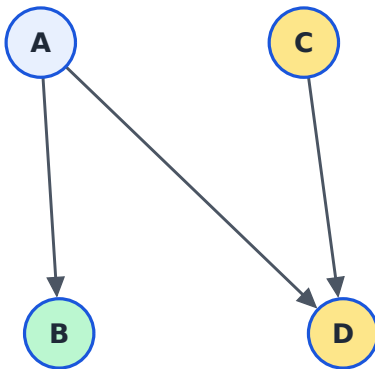
P(B=T)	A	B	P(C=T ...)
0.05	T	T	0.25
	T	F	0.8
	F	T	0.25
	F	F	0.25

A	B	P(D=T ...)
T	T	0.7
T	F	0.6
F	T	0.7
F	F	0.75

- ☐ (A) A=T, B=F, C=F, D=T
☐ (B) A=F, B=F, C=F, D=T
☐ (C) A=F, B=T, C=T, D=F
☐ (D) A=T, B=T, C=F, D=T

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{C} = \mathbf{T}$, $\mathbf{D} = \mathbf{F}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(B} = \mathbf{T} \mid \mathbf{C} = \mathbf{T}, \mathbf{D} = \mathbf{F})}$ (round off to 3 decimal places).



P(A=T)
0.75

A	P(B=T ...)	P(C=T)
T	0.7	0.7
F	0.75	

A	C	P(D=T ...)
T	T	0.25
T	F	0.3
F	T	0.95
F	F	0.1

#	A	B
1	T	F
2	T	F
3	T	T
4	T	T
5	T	F
6	F	T

Answer (numerical): _____

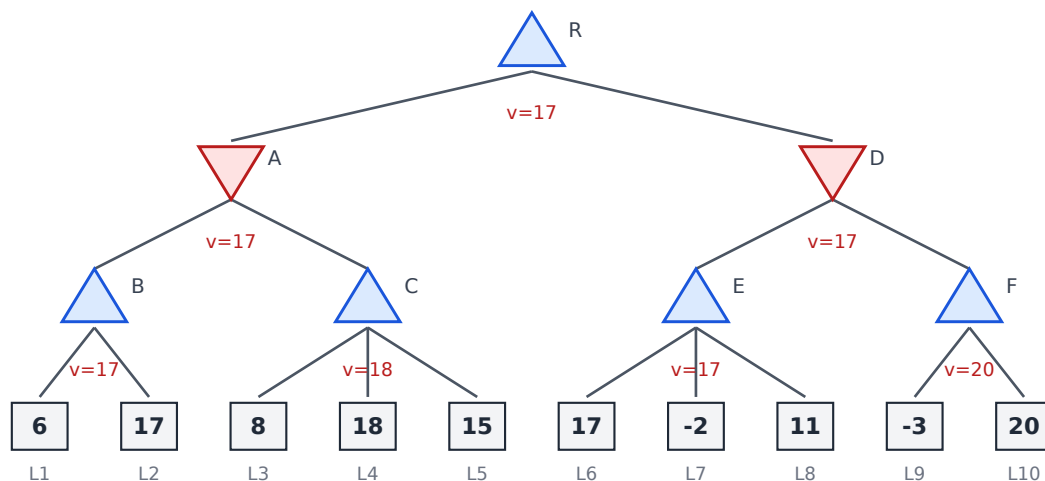
Answer Key — Mock Test 017

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	17	14	MSQ	2	Adversarial Search	C, D
2	MCQ	1	Uninformed Search	A	15	MCQ	2	Uninformed Search	A
3	NAT	1	Informed Search	17	16	NAT	2	Propositional Logic	6
4	MCQ	1	Informed Search	C	17	MSQ	2	Propositional Logic	C
5	MCQ	1	Predicate Logic	A	18	MSQ	2	Propositional Logic	A, D
6	NAT	1	Propositional Logic	3	19	MSQ	2	Predicate Logic	A, C, D
7	MCQ	1	Predicate Logic	A	20	NAT	2	Exact Inference	3
8	NAT	1	Predicate Logic	64	21	MSQ	2	Conditional Independence	B, D
9	MCQ	1	Probability for AI	A	22	MSQ	2	Conditional Independence	A, C, D
10	NAT	1	Exact Inference	0.217 to 0.221	23	NAT	2	Exact Inference	0.047 to 0.051
11	NAT	2	Informed Search	7	24	MCQ	2	Sampling	A
12	MCQ	2	Uninformed Search	A	25	NAT	2	Sampling	0.406 to 0.41
13	NAT	2	Informed Search	9					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 17

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 17, C = 18, A = 17, E = 17, F = 20, D = 17. Root value = **17**; MAX's optimal first move is towards A, D.

Q.2 [MCQ, 1M · Uninformed Search] Answer: A (A: S, C, A, H, J, F, G)

Adjacency lists (alphabetical): A: {C, H, J, K}; C: {A, H, J, S}; F: {G, I, J, K}; G: {F, I}; H: {A, C, S}; I: {F, G, K}; J: {A, C, F, K}; K: {A, F, I, J}; S: {C, H}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, C, A, H, J, F, G**; the path found is S → C → A → J → F → G (5 edges). Note that DFS need not find the shortest path.

Q.3 [NAT, 1M · Informed Search] Answer: 17

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (8)	C(5), B(6), F(7), I(7)
2	C (5)	D(3), B(6), F(7), I(7)
3	D (3)	G(0), B(6), F(7), I(7)
4	G (0)	B(6), F(7), I(7) ← goal, stop

Path returned: **S → C → D → G**, cost = 9 + 3 + 5 = **17**.

The optimal cost is $h^*(S) = 12$ — greedy search is not optimal here.

Q.4 [MCQ, 1M · Informed Search] Answer: C (C: uniform-cost search)

With $h = 0$, $f(n) = g(n)$, so A* expands nodes in order of path cost, which is **uniform-cost search**. It equals BFS (A) only when all step costs are equal. DFS (D) uses a LIFO order. Greedy search with $h \equiv 0$ has every node tied, so its behaviour depends entirely on tie-breaking.

Q.5 [MCQ, 1M · Predicate Logic] Answer: A (A: $(R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$)

Over a finite domain with every element named, $\forall$ becomes a conjunction over x and $\exists$ a disjunction over y . For $x = a$: $R(a, a) \vee R(a, b)$; for $x = b$: $R(b, a) \vee R(b, b)$. Their conjunction is option A. Option B is $\exists x \forall y R(x, y)$. Option C is $\forall y \exists x R(x, y)$ (every element has an R-predecessor). Option D is reflexivity.

Q.6 [NAT, 1M · Propositional Logic] Answer: 3

Build the truth table (sub-formulas shown first):

P	Q	R	$Q \vee (\neg P \wedge P)$	$(P \leftrightarrow Q) \wedge (R \leftrightarrow P)$	$(Q \vee (\neg P \wedge P)) \rightarrow ((P \leftrightarrow Q) \wedge (R \leftrightarrow P))$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	F	F	T
F	F	F	F	T	T

F has 5 models, so it is false in $8 - 5 = 3$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A (A: $\{y/\text{John}, x/\text{Mother}(\text{John})\}$)

Unify arguments left to right. John vs y gives y/John . Then x vs $\text{Mother}(y)$ becomes x vs $\text{Mother}(\text{John})$, giving $x/\text{Mother}(\text{John})$. Applying $\{y/\text{John}, x/\text{Mother}(\text{John})\}$ to both atoms yields $\text{Knows}(\text{John}, \text{Mother}(\text{John}))$. Option B binds the wrong variables. Option C leaves y in the binding of x , so it is not idempotent and does not make the atoms identical in one application.

Q.8 [NAT, 1M · Predicate Logic] Answer: 64

An interpretation of R is a subset of $D \times D$ (9 pairs). The sentence says R is symmetric. The 3 diagonal pairs (x, x) can be chosen freely; the 3 unordered pairs $\{x, y\}$ with $x \neq y$ must be included or excluded as a whole. That is $2^3 \times 2^3 = 64$ relations.

Q.9 [MCQ, 1M · Probability for AI] Answer: A (A: n)

Under mutual independence $P(x_1, \dots, x_n) = \prod P(x_i)$, and each Boolean marginal needs one number $P(x_i = \text{true})$. Hence n numbers suffice. $2^n - 1$ is the count with no independence; $2n$ double-counts $P(\neg x_i) = 1 - P(x_i)$; n^2 has no basis.

Q.10 [NAT, 1M · Exact Inference] Answer: 0.217 to 0.221

Only the ancestors of D (A, B, C) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
$A=T, B=T, C=T$	0.008
$A=T, B=T, C=F$	0.036
$A=T, B=F, C=T$	0.036
$A=T, B=F, C=F$	0.006
$A=F, B=T, C=T$	0.0048
$A=F, B=T, C=F$	0.0648
$A=F, B=F, C=T$	0.054
$A=F, B=F, C=F$	0.009

Sum = **0.2186** ≈ 0.219 .

Q.11 [NAT, 2M · Informed Search] Answer: 7

Heuristic values: $h(B)=7$, $h(C)=12$, $h(D)=3$, $h(F)=7$, $h(G)=0$, $h(H)=13$, $h(I)=7$, $h(L)=11$, $h(S)=15$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	L(16), H(17), C(22)
2	L (16)	B(17), H(17), F(18), C(22)
3	B (17)	D(17), H(17), F(18), C(22)
4	D (17)	H(17), F(18), G(18), C(22), I(25)
5	H (17)	F(17), G(18), C(22), I(25)
6	F (17)	G(18), C(22), I(25)
7	G (18)	C(22), I(25) ← goal, stop

Path returned: **S** → **L** → **B** → **D** → **G** with cost **18**; nodes expanded = **7** (S, L, B, D, H, F, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 18$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.12 [MCQ, 2M · Uninformed Search] Answer: A (A: BFS: 13, DFS: 12)

BFS generates nodes in label order (root, then 2, 3, 4, ...), so node 13 is the **13th** node tested. DFS preorder is 1, 2, 4, 8, 9, 5, 10, 11, 3, 6, 12, 13, ..., so node 13 is the **12th**. The other options miscount one or both orders.

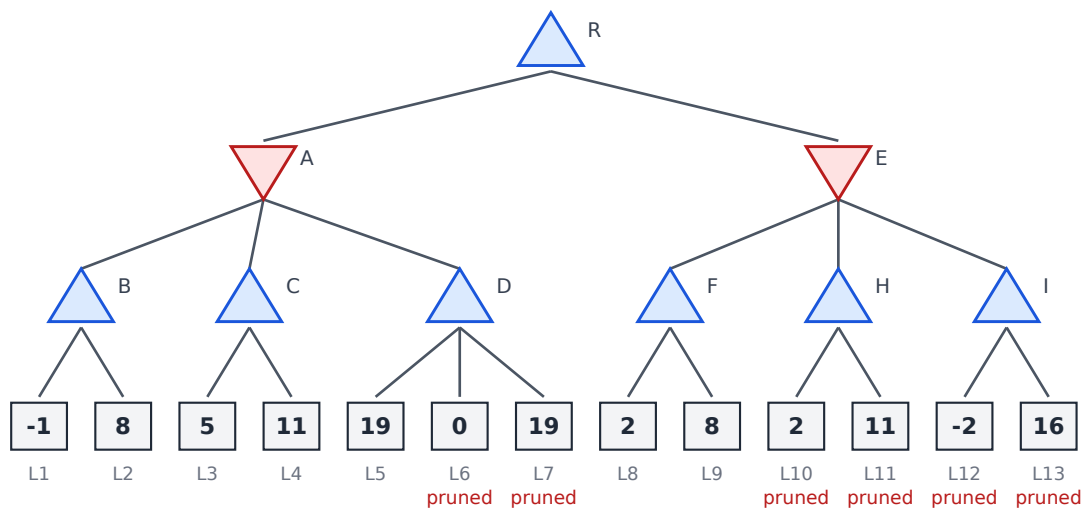
Q.13 [NAT, 2M · Informed Search] Answer: 9

Misplaced tiles: 2 (at (0,0), belongs at (0,1)), 8 (at (0,1), belongs at (1,0)), 1 (at (1,0), belongs at (0,0)), and 6 (at (1,1), belongs at (2,1)). Tiles 3, 4, 7 and 5 are in place, so $h_1 = 4$. Manhattan distances: $2 \rightarrow 1$, $8 \rightarrow 1+1 = 2$, $1 \rightarrow 1$, $6 \rightarrow 1$, so $h_2 = 5$. $h_1 + h_2 = 9$. (The sum is not necessarily admissible. This question is purely arithmetic.)

Q.14 [MSQ, 2M · Adversarial Search] Answer: C, D (C: Leaf L6 is pruned.; D: Leaf L10 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node D: value $19 \geq \beta = 8 \Rightarrow$ prune remaining children (leaves L6, L7).
- At MIN node E: value $8 \leq \alpha = 8 \Rightarrow$ prune remaining children (leaves L10, L11, L12, L13).



Leaves evaluated (7): L1, L2, L3, L4, L5, L8, L9. Leaves pruned (6): L6, L7, L10, L11, L12, L13. Root value = 8 (same as plain minimax, 8 — pruning never changes the root value).

Q.15 [MCQ, 2M · Uninformed Search] Answer: A (A: $S \rightarrow J \rightarrow A \rightarrow F \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(4)
2	J (4)	A(9), B(10)
3	A (9)	B(10), F(13), D(15)
4	B (10)	D(12), F(13)
5	D (12)	F(13), C(15)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
6	F (13)	C(15), G(18)
7	C (15)	G(18)
8	G (18)	← goal, stop

Path returned: **S** → **J** → **A** → **F** → **G**, cost = 4 + 5 + 4 + 5 = **18**. Nodes expanded (including S and G): **8** (S, J, A, B, D, F, C, G).

Q.16 [NAT, 2M · Propositional Logic] **Answer: 6**

Start with {A, B}. $A \wedge B \rightarrow C$ adds C; $C \rightarrow D$ adds D; $B \rightarrow E$ adds E; $D \wedge E \rightarrow F$ adds F. $F \wedge G \rightarrow H$ never fires because G is never derived, and $H \rightarrow A$ adds nothing new. The agenda empties with {A, B, C, D, E, F}, i.e. 6 symbols. Because forward chaining is sound and complete for Horn KBs, these are exactly the entailed atoms.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: C** ($C: \neg(P \wedge \neg(P \wedge R))$)

Compare truth tables with $F = P \rightarrow (P \wedge R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(P \wedge R) \rightarrow P$ — differs at $P=T, R=F$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(B) $\neg P \wedge (P \wedge R)$ — differs at $P=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(C) $\neg(P \wedge \neg(P \wedge R))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(D) $P \vee \neg(P \wedge R)$ — differs at $P=T, R=F$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, D** ($A: (\neg Q \vee \neg R)$; $D: (P \vee \neg R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg Q \vee \neg R)$ is derivable:

1. resolve $(P \vee \neg Q \vee \neg R)$ and $(\neg P)$ on P $\Rightarrow (\neg Q \vee \neg R)$

(B) The resolution closure contains 9 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(Q \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(P \vee \neg R)$ is derivable:

1. resolve $(Q \vee \neg R)$ and $(P \vee \neg Q \vee \neg R)$ on Q $\Rightarrow (P \vee \neg R)$

Q.19 [MSQ, 2M · Predicate Logic] **Answer: A, C, D** ($A: \exists x R(x, x)$; $C: \forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$; $D: \exists x \forall y \neg R(x, y)$)

$R = \{(2,2), (2,3), (3,1), (3,3), (4,1)\}$.

(A) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

(B) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(D) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A)$, $f(A, C)$, $f(A, D)$, $f(A, E)$, $f(A, F)$, $f(E, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(A, D)$	{A, D}	$f(A)$
E	$f(A, E)$, $f(E, G)$	{A, E, G}	$f(A, G)$
A	$f(A)$, $f(A)$, $f(A, C)$, $f(A, F)$, $f(A)$, $f(A, G)$	{A, C, F, G}	$f(C, F, G)$
F	$f(C, F, G)$	{C, F, G}	$f(C, G)$
C	$f(C, G)$	{C, G}	$f(G)$

The largest factor produced has **3** variable(s) (created when eliminating A); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over G and normalise.

Q.21 [MSQ, 2M · Conditional Independence] Answer: B, D ($B: C \perp D$; $D: D \perp C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp E \mid C, F$: **FALSE** — the path $A \rightarrow B \rightarrow D \rightarrow E$ is active given $\{C, F\}$.

(B) $C \perp D$: **TRUE** — $C \rightarrow F \leftarrow A \rightarrow B \rightarrow D$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow F \leftarrow A \rightarrow B \rightarrow E \leftarrow D$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow F \leftarrow A \rightarrow D$ (blocked: collider F with neither it nor a descendant observed).

(C) $D \perp F$: **FALSE** — the path $D \leftarrow A \rightarrow F$ is active given $\{\}$.

(D) $D \perp C$: **TRUE** — $D \leftarrow A \rightarrow F \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $D \leftarrow B \leftarrow A \rightarrow F \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $D \rightarrow E \leftarrow B \leftarrow A \rightarrow F \leftarrow C$ (blocked: collider E with neither it nor a descendant observed).

Q.22 [MSQ, 2M · Conditional Independence] Answer: A, C, D ($A: P(a) = 0.48$; $C: P(a \mid b, c) = 0.9$; $D: P(a \mid b) \approx 0.776$)

$P(a) = 0.4 \times 0.9 + 0.6 \times 0.2 = 0.36 + 0.12 = 0.48$ (A true). $P(a, b) = 0.252 + 0.012 = 0.264$ while $P(a)P(b) = 0.48 \times 0.34 = 0.1632$, so A and B are dependent (B false). In the fork $C \rightarrow A$, $C \rightarrow B$, $A \perp B \mid C$, so $P(a \mid b, c) = P(a \mid c) = 0.9$ (C true). $P(a \mid b) = 0.264/0.34 \approx 0.7765$ (D true): observing b raises belief in c and hence in a.

Q.23 [NAT, 2M · Exact Inference] Answer: 0.047 to 0.051

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot P(\text{Alarm} \mid \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Alarm})$.

Inference by enumeration: $P(\text{Earthquake} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Earthquake}, e, \text{hidden})$.

Hidden: Bu, Al	Ea	product of CPT entries (Bu · Ea · Al · Jo)	value
T, T	T	$0.2 \times 0.05 \times 0.8 \times 0.6$	0.0048
T, F	T	$0.2 \times 0.05 \times 0.2 \times 0.3$	0.0006
F, T	T	$0.8 \times 0.05 \times 0.8 \times 0.4$	0.0128
F, F	T	$0.8 \times 0.05 \times 0.2 \times 0.7$	0.0056
T, T	F	$0.2 \times 0.95 \times 0.75 \times 0.6$	0.0855
T, F	F	$0.2 \times 0.95 \times 0.25 \times 0.3$	0.01425
F, T	F	$0.8 \times 0.95 \times 0.75 \times 0.4$	0.228
F, F	F	$0.8 \times 0.95 \times 0.25 \times 0.7$	0.133

$P(\text{Earthquake}=T, e) = 0.0238$; $P(\text{Earthquake}=F, e) = 0.46075$; $P(e) = 0.48455$.

$P(\text{Earthquake}=T \mid e) = 0.0238 / 0.48455 = \mathbf{0.0491} \approx 0.049$.

Q.24 [MCQ, 2M · Sampling] Answer: A ($A: A=T, B=F, C=F, D=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.7$; $u = 0.57 < 0.7 \Rightarrow A = T$
- B: $P(B=T \mid \text{—}) = 0.05$; $u = 0.69 \geq 0.05 \Rightarrow B = F$
- C: $P(C=T \mid A=T, B=F) = 0.8$; $u = 0.97 \geq 0.8 \Rightarrow C = F$
- D: $P(D=T \mid A=T, B=F) = 0.6$; $u = 0.1 < 0.6 \Rightarrow D = T$

Sample: **(A=T, B=F, C=F, D=T)**.

Q.25 [NAT, 2M · Sampling] Answer: 0.406 to 0.41

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	B
1	$P(C=T)=0.7 \times P(D=F \mid A=T, C=T)=0.75$	0.525	F
2	$P(C=T)=0.7 \times P(D=F \mid A=T, C=T)=0.75$	0.525	F
3	$P(C=T)=0.7 \times P(D=F \mid A=T, C=T)=0.75$	0.525	T
4	$P(C=T)=0.7 \times P(D=F \mid A=T, C=T)=0.75$	0.525	T
5	$P(C=T)=0.7 \times P(D=F \mid A=T, C=T)=0.75$	0.525	F
6	$P(C=T)=0.7 \times P(D=F \mid A=F, C=T)=0.05$	0.035	T

$\sum w (\text{all}) = 2.66$; $\sum w (B = T) = 1.085$.

Estimate $P(B=T \mid C = T, D = F) = 1.085 / 2.66 = \mathbf{0.4079}$.

Mock Test 018

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 6 / 13	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search

In alpha-beta search, α denotes

- ☐ (A) the value of the best choice (highest value) found so far for MAX along the current path
- ☐ (B) the value of the best choice (lowest value) found so far for MIN along the current path
- ☐ (C) the depth limit
- ☐ (D) the number of pruned nodes

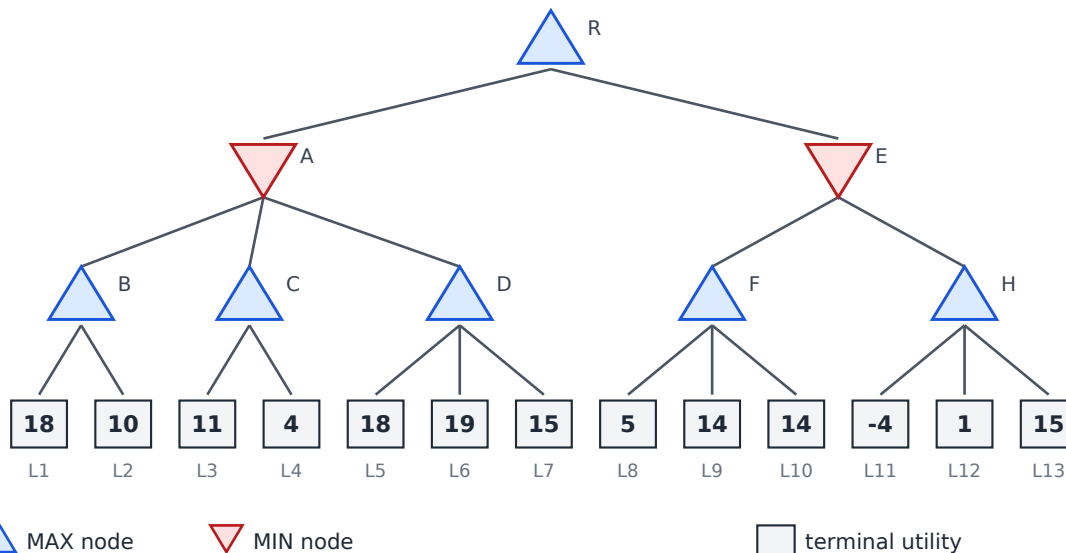
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 4$ (so $b^d = 81$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

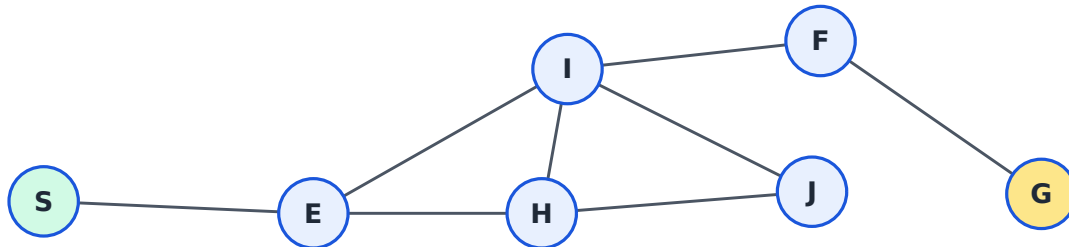
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, E, H, I, J, F, G
☐ (B) S, E, I, H, F, G
☐ (C) S, E, I, J, H, F, G
☐ (D) S, E, H, I, F, G

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((R \vee \neg P) \vee Q) \leftrightarrow ((\neg R \rightarrow R) \rightarrow (R \vee R))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following formulas is satisfiable but NOT valid?

- ☐ (A) $p \rightarrow (q \rightarrow p)$
☐ (B) $p \wedge \neg p$
☐ (C) $(p \wedge q) \rightarrow (p \vee q)$
☐ (D) $(p \rightarrow q) \rightarrow p$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

Of the 16 possible binary Boolean connectives $\circ$ (functions $\{T, F\}^2 \rightarrow \{T, F\}$), how many are commutative, i.e. satisfy $p \circ q \equiv q \circ p$?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **D**, **C** and rules:

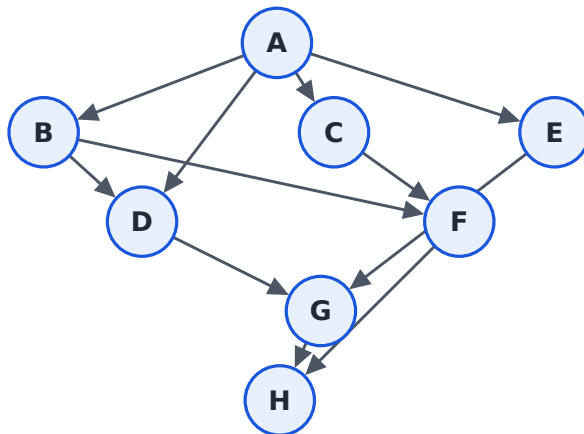
- R1: $H \wedge B \wedge D \rightarrow A$
 R2: $C \wedge B \rightarrow A$
 R3: $H \rightarrow J$
 R4: $B \wedge K \rightarrow J$
 R5: $F \wedge E \rightarrow J$
 R6: $C \rightarrow B$
 R7: $F \wedge K \rightarrow A$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **B** in the Bayesian network below?



Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. When Gibbs sampling resamples Rain (R), the values of which variables must be consulted?

- ☐ (A) C, S and W
- ☐ (B) C only
- ☐ (C) W only
- ☐ (D) C and W only

Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Problem Formulation

In a grid route-finding problem, the agent moves N, S, E or W at cost 1 per move. The start state is (0,0), the only goal is (3,2), and there are no obstacles in an unbounded grid. Which of the following are TRUE?

- ☐ (A) The optimal solution cost is 5
- ☐ (B) BFS with graph search is optimal for this problem
- ☐ (C) Manhattan distance is a consistent heuristic for this problem
- ☐ (D) The branching factor b is 8

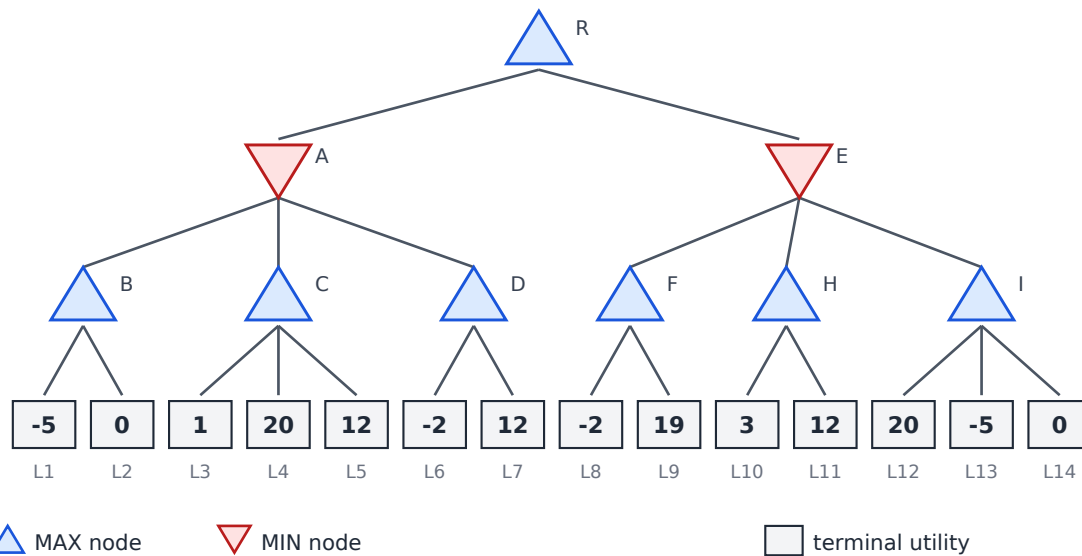
Q.12 [MSQ · 2 marks · no negative marking] Uninformed Search

Uniform-cost search runs with every step cost at least some $\epsilon > 0$. Which of the following are TRUE?

- ☐ (A) UCS is optimal
- ☐ (B) If every step cost equals the same constant, UCS behaves essentially like BFS (expanding level by level), but because it goal-tests on expansion it may generate a whole extra level of nodes
- ☐ (C) UCS is complete when the branching factor is finite
- ☐ (D) UCS remains complete even when the graph contains a cycle of zero-cost actions

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

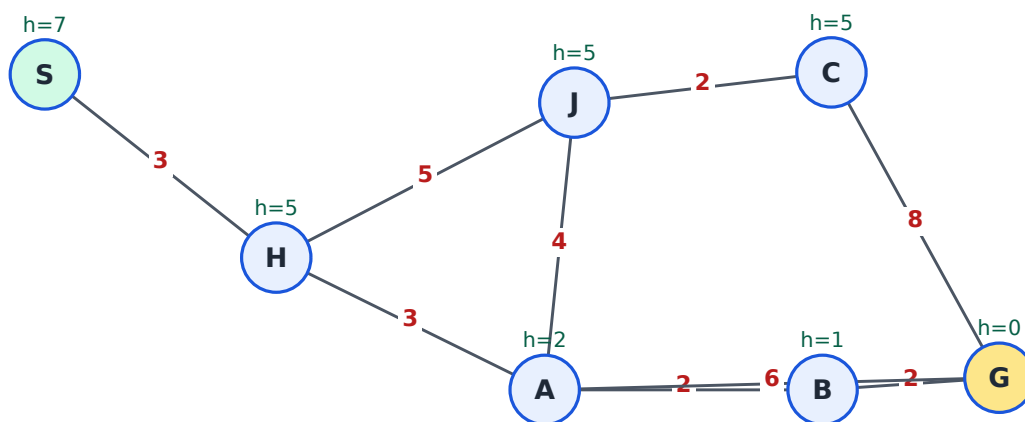
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

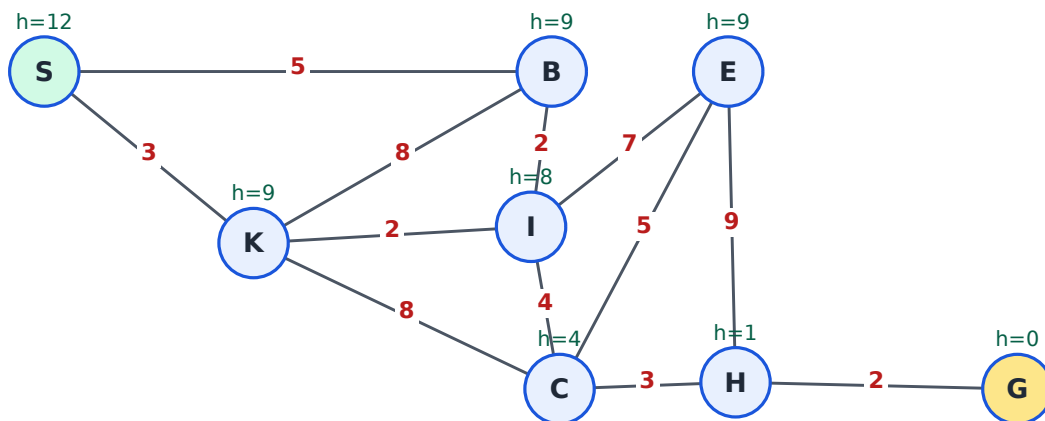
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(Q \vee \neg R \vee S)$, C2: $(\neg P)$, C3: $(\neg Q \vee \neg R)$, C4: $(Q \vee S)$, C5: $(\neg Q \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg R \vee \neg S)$
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) (S)
☐ (D) $(P \vee S)$

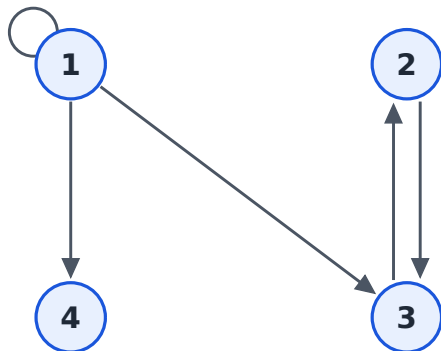
Q.17 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \exists y R(x, y) \wedge \exists y \forall x R(x, y)$?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (C) $\forall y \exists x R(x, y)$
☐ (D) $\forall x R(x, x)$

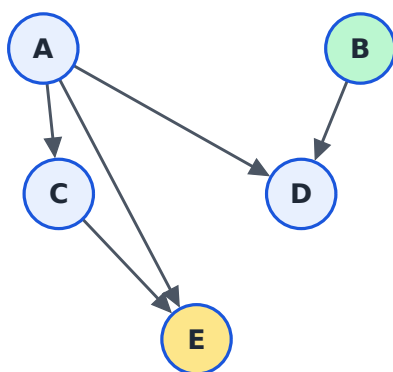
Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$
☐ (B) $(\neg R \rightarrow P) \vee (P \rightarrow \neg R)$
☐ (C) $((Q \rightarrow P) \wedge \neg Q) \rightarrow \neg P$
☐ (D) $((Q \rightarrow P) \wedge (R \rightarrow P)) \rightarrow (Q \wedge R)$

Q.20 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{E=T}$. The current state of the chain is $(\mathbf{A=F}, \mathbf{B=T}, \mathbf{C=F}, \mathbf{D=T}, \mathbf{E=T})$. The next step resamples $\mathbf{B}$. With what probability is $\mathbf{B}$ set to T? (round off to 3 decimal places)



$P(A=T)$
0.9

$P(B=T)$
0.1

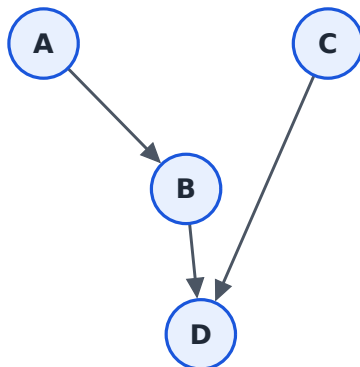
A	$P(C=T \dots)$
T	0.9
F	0.95

A	B	$P(D=T A, C)$	A	C	$P(E=T \dots)$
T	T	0.1	T	T	0.9
T	F	0.05	T	F	0.3
F	T	0.25	F	T	0.2
F	F	0.05	F	F	0.75

Answer (numerical): _____

Q.21 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.6}, \mathbf{0.38}, \mathbf{0.61}, \mathbf{0.48} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.7

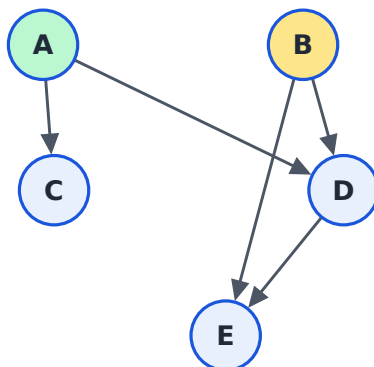
A	P(B=T ...)	P(C=T)
T	0.4	0.75
F	0.3	

B	C	P(D=T ...)
T	T	0.8
T	F	0.95
F	T	0.95
F	F	0.9

- ☐ (A) A=F, B=T, C=F, D=F
☐ (B) A=F, B=T, C=T, D=T
☐ (C) A=T, B=T, C=T, D=T
☐ (D) A=T, B=F, C=T, D=T

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $\mathbf{P(A = T \mid B = T)}$ (round off to 3 decimal places).



P(A=T)
0.25

P(B=T)
0.3

A	P(C=T ...)
T	0.05
F	0.2

A	B	P(D=T B, D)	P(E=T ...)
T	T	0.9	0.8
T	F	0.95	0.4
F	T	0.4	0.05
F	F	0.5	0.5

Answer (numerical): _____

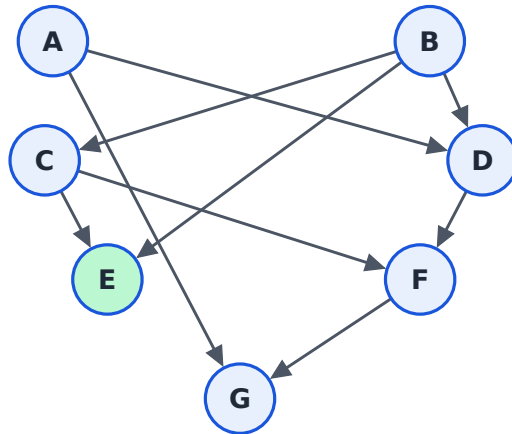
Q.23 [NAT · 2 marks · no negative marking] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. In computing $P(B \mid j, m)$ by variable elimination, A is summed out first, producing a factor $f(B, E) = \sum_a P(a \mid B, E) P(j \mid a) P(m \mid a)$. Compute $f(b, e)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.24 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

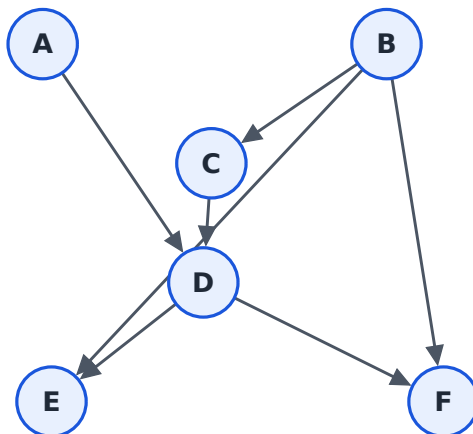
Variable elimination is used to compute $P(\mathbf{E})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{G, D, B, F, C, A}$. What is the scope of the new factor produced when $\mathbf{F}$ is summed out?



- ☐ (A) $f(A, C, E, F)$
- ☐ (B) $f(C, D)$
- ☐ (C) $f(A, C, E)$
- ☐ (D) $f(A, C)$

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp A \mid F$
- ☐ (B) $B \perp C$
- ☐ (C) $E \perp D \mid B$
- ☐ (D) $D \perp B \mid A, C$

Answer Key — Mock Test 018

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	A	14	NAT	2	Informed Search	5
2	NAT	1	Adversarial Search	17	15	NAT	2	Informed Search	7
3	NAT	1	Adversarial Search	14	16	MSQ	2	Propositional Logic	C
4	MCQ	1	Uninformed Search	D	17	NAT	2	Predicate Logic	169
5	NAT	1	Propositional Logic	1	18	MSQ	2	Predicate Logic	A, C
6	MCQ	1	Propositional Logic	D	19	MSQ	2	Propositional Logic	A, B
7	NAT	1	Propositional Logic	8	20	NAT	2	Sampling	0.355 to 0.359
8	NAT	1	Propositional Logic	4	21	MCQ	2	Sampling	C
9	NAT	1	Conditional Independence	4	22	NAT	2	Exact Inference	0.248 to 0.252
10	MCQ	1	Sampling	A	23	NAT	2	Variable Elimination	0.531 to 0.533
11	MSQ	2	Problem Formulation	A, B, C	24	MCQ	2	Exact Inference	C
12	MSQ	2	Uninformed Search	A, B, C	25	MSQ	2	Conditional Independence	D
13	NAT	2	Adversarial Search	10					

Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] Answer: A (A: the value of the best choice (highest value) found so far for MAX along the current path)

α is **MAX's best guaranteed value so far**, a lower bound, along the path to the root. β is MIN's best value so far, an upper bound (B). A branch is cut when the bounds cross, i.e. when $\alpha \geq \beta$. (C) and (D) are not what α means.

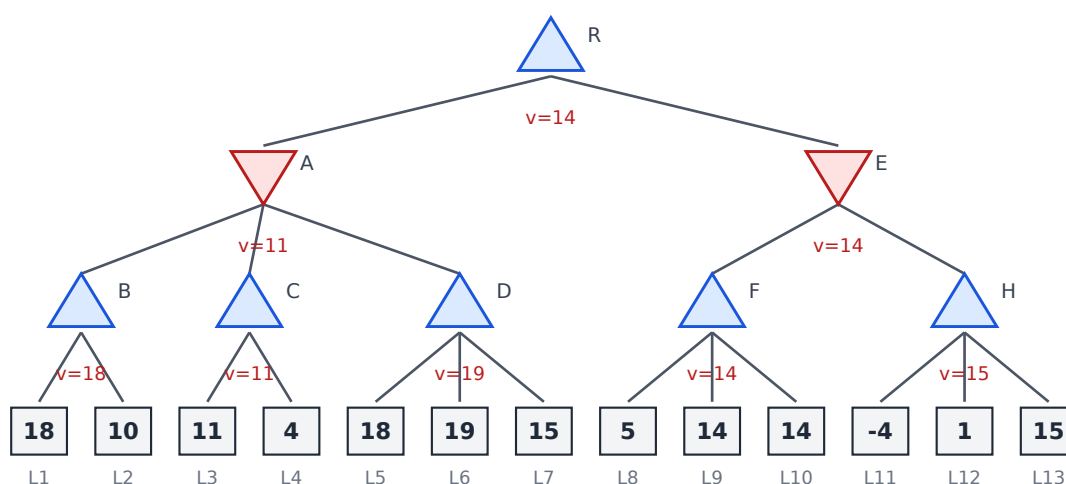
Q.2 [NAT, 1M · Adversarial Search] Answer: 17

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 3^2 + 3^2 - 1 = 9 + 9 - 1 = 17$, versus 81 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [NAT, 1M · Adversarial Search] Answer: 14

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 18, C = 11, D = 19, A = 11, F = 14, H = 15, E = 14. Root value = **14**; MAX's optimal first move is towards E.

Q.4 [MCQ, 1M · Uninformed Search] Answer: D (D: S, E, H, I, F, G)

Adjacency lists (alphabetical): E: {H, I, S}; F: {G, I}; G: {F}; H: {E, I, J}; I: {E, F, H, J}; J: {H, I}; S: {E}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, E, H, I, F, G**; the path found is $S \rightarrow E \rightarrow H \rightarrow I \rightarrow F \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.5 [NAT, 1M · Propositional Logic] Answer: 1

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \vee \neg P) \vee Q$	$(\neg R \rightarrow R) \rightarrow (R \vee R)$	$((R \vee \neg P) \vee Q) \leftrightarrow ((\neg R \rightarrow R) \rightarrow (R \vee R))$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

F has 7 models, so it is false in $8 - 7 = 1$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.6 [MCQ, 1M · Propositional Logic] Answer: D (D: $(p \rightarrow q) \rightarrow p$)

$(p \rightarrow q) \rightarrow p$: when $p=T$ it is true (consequent T); when $p=F$, $p \rightarrow q$ is T and the consequent is F, so it is false. Hence it is satisfiable but not valid. $p \rightarrow (q \rightarrow p)$ is valid (if p is T the consequent is T). $p \wedge \neg p$ is unsatisfiable. $(p \wedge q) \rightarrow (p \vee q)$ is valid because $p \wedge q \models p \vee q$.

Q.7 [NAT, 1M · Propositional Logic] Answer: 8

A binary connective is fixed by its outputs on TT, TF, FT, FF. Commutativity requires $\text{output}(TF) = \text{output}(FT)$. So we may freely choose the outputs on TT, FF and the common value on TF/FT: $2^3 = 8$. Examples: $\wedge$, $\vee$, $\leftrightarrow$, $\oplus$, NAND, NOR, constant T, constant F.

Q.8 [NAT, 1M · Propositional Logic] Answer: 4

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $C \rightarrow B$; add B.

Round 2: fire $C \wedge B \rightarrow A$; add A.

Fixed point: {A, B, C, D} — 4 atoms are entailed (facts: C, D; newly derived: A, B). Atoms never derived: E, F, H, J, K.

Q.9 [NAT, 1M · Conditional Independence] Answer: 4

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of B: {A}; children: {D, F}; co-parents: {C}.

$\text{MB}(B) = \{A, C, D, F\}$ — 4 nodes. Given its Markov blanket, B is conditionally independent of every other node in the network.

Q.10 [MCQ, 1M · Sampling] Answer: A (A: C, S and W)

Gibbs samples R from $P(R \mid \text{mb}(R))$. $\text{mb}(R) = \text{parents } \{C\} \cup \text{children } \{W\} \cup \text{children's other parents } \{S\} = \{C, S, W\}$. S is needed because $P(w \mid S, R)$ depends on it (explaining away). The other options omit blanket members.

Q.11 [MSQ, 2M · Problem Formulation] Answer: A, B, C (A: The optimal solution cost is 5; B: BFS with graph search is optimal for this problem; C: Manhattan distance is a consistent heuristic for this problem)

(A) is true: the shortest path needs $|3| + |2| = 5$ unit moves. (B) is true: costs are uniform, so the shallowest goal is cheapest. (C) is true: one move changes the Manhattan distance by exactly 1 = the step cost, so $h(n) \leq 1 + h(n')$. (D) is false: there are 4 actions, so $b = 4$. $b = 8$ would need diagonal moves.

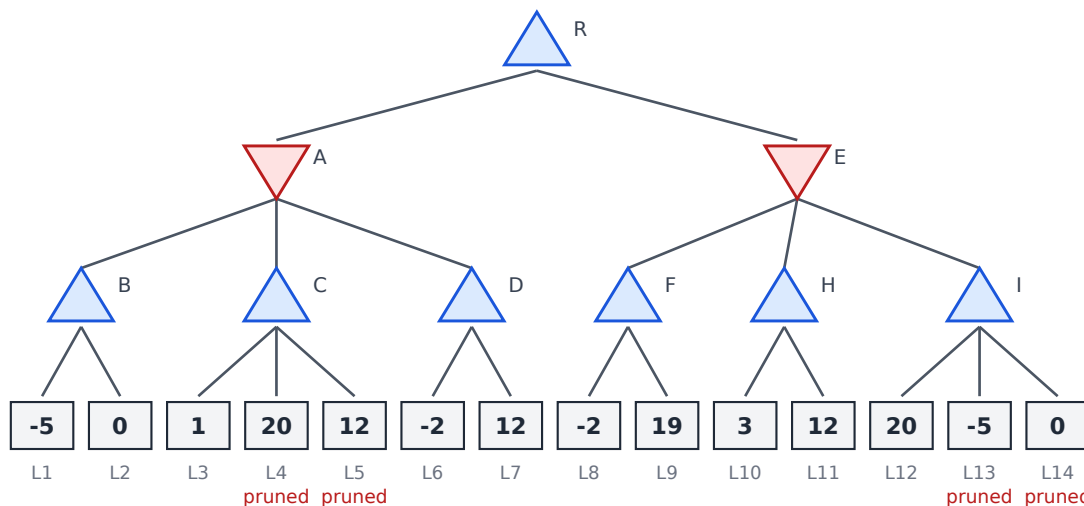
Q.12 [MSQ, 2M · Uninformed Search] Answer: A, B, C

(A) is true: UCS expands nodes in order of g , so the first goal it selects is the cheapest. (B) is true: equal costs make g proportional to depth, but the goal test at expansion means the children at depth $d+1$ may already have been generated. (C) is true: with costs $\geq \epsilon$ and finite b , only finitely many nodes have $g \leq C^*$. (D) is false: a zero-cost cycle lets g stay constant while the path grows forever, so UCS may never terminate. This is why the $\epsilon > 0$ condition is needed.

Q.13 [NAT, 2M · Adversarial Search] Answer: 10

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $1 \geq \beta = 0 \Rightarrow$ prune remaining children (leaves L4, L5).
- At MAX node I: value $20 \geq \beta = 12 \Rightarrow$ prune remaining children (leaves L13, L14).



Leaves evaluated (10): L1, L2, L3, L6, L7, L8, L9, L10, L11, L12. Leaves pruned (4): L4, L5, L13, L14. Root value = 12 (same as plain minimax, 12 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(A)=2$, $h(B)=1$, $h(C)=5$, $h(G)=0$, $h(H)=5$, $h(J)=5$, $h(S)=7$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (7)	H(8)
2	H (8)	A(8), J(13)
3	A (8)	B(9), G(12), J(13)
4	B (9)	G(10), J(13)
5	G (10)	J(13) ← goal, stop

Path returned: $S \rightarrow H \rightarrow A \rightarrow B \rightarrow G$ with cost **10**; nodes expanded = **5** (S, H, A, B, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 10$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [NAT, 2M · Informed Search] Answer: 7

Heuristic values: $h(B)=9$, $h(C)=4$, $h(E)=9$, $h(G)=0$, $h(H)=1$, $h(I)=8$, $h(K)=9$, $h(S)=12$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	K(12), B(14)
2	K (12)	I(13), B(14), C(15)
3	I (13)	C(13), B(14), E(21)
4	C (13)	H(13), B(14), E(21)
5	H (13)	B(14), G(14), E(21)
6	B (14)	G(14), E(21)
7	G (14)	E(21) ← goal, stop

Path returned: $S \rightarrow K \rightarrow I \rightarrow C \rightarrow H \rightarrow G$ with cost **14**; nodes expanded = **7** (S, K, I, C, H, B, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: C** (C: (S))

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg R \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) (S) is derivable:

1. resolve $(Q \vee S)$ and $(\neg Q \vee S)$ on $Q \Rightarrow (S)$

(D) $(P \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.17 [NAT, 2M · Predicate Logic] **Answer: 169**

$\exists y \forall x R(x, y)$ says some column of the 3×3 matrix is all T. This already implies $\forall x \exists y R(x, y)$ (that column is a witness for every row), so we only count relations with at least one full column. Relations with no full column: each column has 7 non-full choices $\rightarrow 7^3 = 343$. Answer = $512 - 343 = 169$.

Q.18 [MSQ, 2M · Predicate Logic] **Answer: A, C** (A: $\exists x \forall y \neg R(x, y)$; C: $\forall y \exists x R(x, y)$)

$R = \{(1,1), (1,3), (1,4), (2,3), (3,2)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 4 has no outgoing edge.

(B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(1,4)$ holds but $R(4,1)$ does not.

(C) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(D) $\forall x R(x, x)$: **FALSE** — no self-loop at 2.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$; B: $(\neg R \rightarrow P) \vee (P \rightarrow \neg R)$)

(A) $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

(B) $(\neg R \rightarrow P) \vee (P \rightarrow \neg R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $((Q \rightarrow P) \wedge \neg Q) \rightarrow \neg P$: **not valid** — counter-model $P=T, Q=F$.

(D) $((Q \rightarrow P) \wedge (R \rightarrow P)) \rightarrow (Q \wedge R)$: **not valid** — counter-model $P=T, Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.20 [NAT, 2M · Sampling] **Answer: 0.355 to 0.359**

Gibbs sampling draws B from $P(B \mid \text{mb}(B))$, where $\text{mb}(B) = \{A, D\}$:

$P(B \mid \text{mb}) \propto P(B \mid \text{parents}(B)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$B=T$: $P(B=T) \times P(D=T \mid A=F, B=T) = 0.1 \times 0.25 = 0.025$

$B=F$: $P(B=F) \times P(D=T \mid A=F, B=F) = 0.9 \times 0.05 = 0.045$

Normalise: $0.025 / (0.025 + 0.045) = \mathbf{0.3571}$.

Q.21 [MCQ, 2M · Sampling] **Answer: C** (C: $A=T, B=T, C=T, D=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.7$; $u = 0.6 < 0.7 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.4$; $u = 0.38 < 0.4 \Rightarrow B = T$
- C: $P(C=T \mid \text{—}) = 0.75$; $u = 0.61 < 0.75 \Rightarrow C = T$
- D: $P(D=T \mid B=T, C=T) = 0.8$; $u = 0.48 < 0.8 \Rightarrow D = T$

Sample: **$A=T, B=T, C=T, D=T$** .

Q.22 [NAT, 2M · Exact Inference] **Answer: 0.248 to 0.252**

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B) \cdot P(C \mid A) \cdot P(D \mid A, B) \cdot P(E \mid B, D)$.

Inference by enumeration: $P(A \mid e) = \alpha \sum_{\text{hidden}} P(A, e, \text{hidden})$.

Hidden: C, D, E	A	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T, T	T	$0.25 \times 0.3 \times 0.05 \times 0.9 \times 0.8$	0.0027
T, T, F	T	$0.25 \times 0.3 \times 0.05 \times 0.9 \times 0.2$	0.00067
T, F, T	T	$0.25 \times 0.3 \times 0.05 \times 0.1 \times 0.4$	0.00015
T, F, F	T	$0.25 \times 0.3 \times 0.05 \times 0.1 \times 0.6$	0.00022
F, T, T	T	$0.25 \times 0.3 \times 0.95 \times 0.9 \times 0.8$	0.0513
F, T, F	T	$0.25 \times 0.3 \times 0.95 \times 0.9 \times 0.2$	0.01282
F, F, T	T	$0.25 \times 0.3 \times 0.95 \times 0.1 \times 0.4$	0.00285

Hidden: C, D, E	A	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
F, F, F	T	$0.25 \times 0.3 \times 0.95 \times 0.1 \times 0.6$	0.00427
T, T, T	F	$0.75 \times 0.3 \times 0.2 \times 0.4 \times 0.8$	0.0144
T, T, F	F	$0.75 \times 0.3 \times 0.2 \times 0.4 \times 0.2$	0.0036
T, F, T	F	$0.75 \times 0.3 \times 0.2 \times 0.6 \times 0.4$	0.0108
T, F, F	F	$0.75 \times 0.3 \times 0.2 \times 0.6 \times 0.6$	0.0162
F, T, T	F	$0.75 \times 0.3 \times 0.8 \times 0.4 \times 0.8$	0.0576
F, T, F	F	$0.75 \times 0.3 \times 0.8 \times 0.4 \times 0.2$	0.0144
F, F, T	F	$0.75 \times 0.3 \times 0.8 \times 0.6 \times 0.4$	0.0432
F, F, F	F	$0.75 \times 0.3 \times 0.8 \times 0.6 \times 0.6$	0.0648

$P(A=T, e) = 0.075$; $P(A=F, e) = 0.225$; $P(e) = 0.3$.

$P(A=T | e) = 0.075 / 0.3 = \mathbf{0.25} \approx 0.250$.

Q.23 [NAT, 2M · Variable Elimination] **Answer: 0.531 to 0.533**

$f(b, e) = P(a | b, e) P(j | a) P(m | a) + P(\neg a | b, e) P(j | \neg a) P(m | \neg a) = 0.95 \times 0.8 \times 0.7 + 0.05 \times 0.05 \times 0.02 = 0.532 + 0.00005 = 0.53205 \approx 0.532$. This is not a probability distribution over B, E; it is an intermediate factor.

Q.24 [MCQ, 2M · Exact Inference] **Answer: C** (C: $f(A, C, E)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(B, C)$, $f(A, B, D)$, $f(B, C, E)$, $f(C, D, F)$, $f(A, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(A, F, G)$	$\{A, F, G\}$	$f(A, F)$
D	$f(A, B, D)$, $f(C, D, F)$	$\{A, B, C, D, F\}$	$f(A, B, C, F)$
B	$f(B)$, $f(B, C)$, $f(B, C, E)$, $f(A, B, C, F)$	$\{A, B, C, E, F\}$	$f(A, C, E, F)$
F	$f(A, F)$, $f(A, C, E, F)$	$\{A, C, E, F\}$	$f(A, C, E)$
C	$f(A, C, E)$	$\{A, C, E\}$	$f(A, E)$
A	$f(A)$, $f(A, E)$	$\{A, E\}$	$f(E)$

The largest factor produced has **4** variable(s) (created when eliminating D); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over E and normalise.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: D** ($D: D \perp B | A, C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $D \perp A | F$: **FALSE** — the path $D \leftarrow A$ is active given $\{F\}$.

(B) $B \perp C$: **FALSE** — the path $B \rightarrow C$ is active given $\{\}$.

(C) $E \perp D | B$: **FALSE** — the path $E \leftarrow D$ is active given $\{B\}$.

(D) $D \perp B | A, C$: **TRUE** — $D \leftarrow C \leftarrow B$ (blocked: observed chain node C); $D \rightarrow E \leftarrow B$ (blocked: collider E with neither it nor a descendant observed); $D \rightarrow F \leftarrow B$ (blocked: collider F with neither it nor a descendant observed).

Mock Test 019

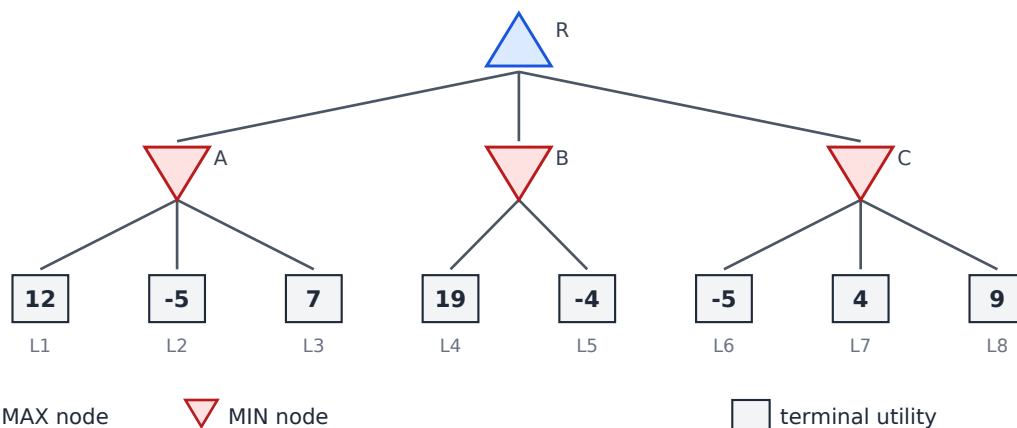
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 6 / 13	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

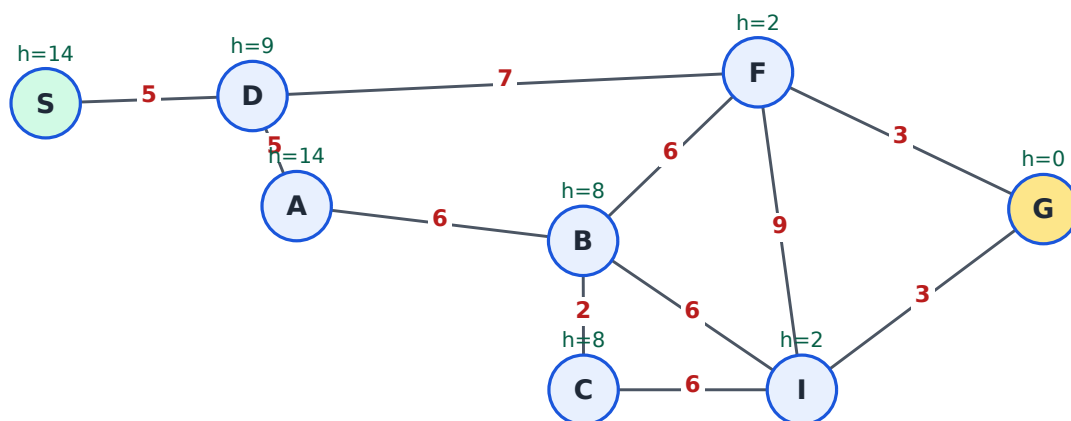
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

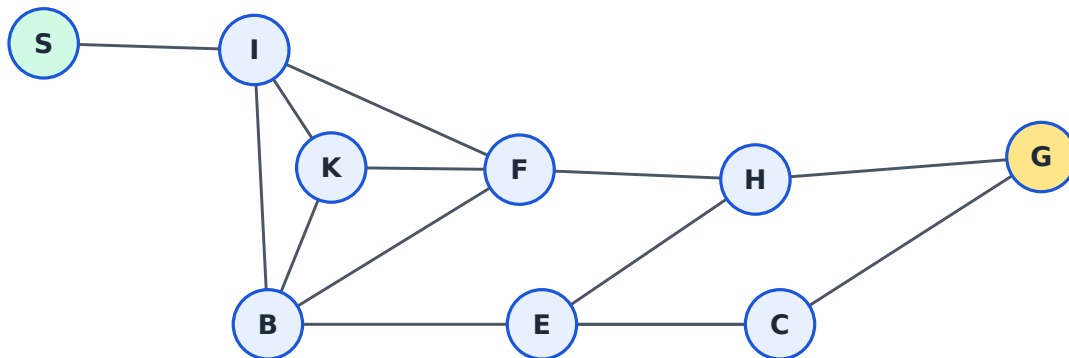
Q.3 [NAT · 1 mark · no negative marking] Problem Formulation

In the vacuum world with $n = 3$ squares in a row, a state specifies the agent's location and whether each square is dirty or clean. How many states are there?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, I, K, F, H, G
- ☐ (B) S, I, B, C, E, H, F, K, G
- ☐ (C) S, I, B, F, K, C, H, E, G
- ☐ (D) S, B, I, C, E, H, F, K, G

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

For propositional sentences α and β , $\alpha \models \beta$ holds if and only if

- ☐ (A) $\alpha \wedge \neg\beta$ is unsatisfiable
- ☐ (B) $\alpha \wedge \beta$ is satisfiable
- ☐ (C) $\neg\alpha \vee \beta$ is satisfiable
- ☐ (D) $\alpha \vee \neg\beta$ is valid

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$\mathbf{F} = (\neg Q \vee (\neg R \rightarrow \neg Q)) \rightarrow (P \vee \neg R) ?$$

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In Existential Instantiation, the sentence $\exists x \text{Crown}(x) \wedge \text{OnHead}(x, \text{John})$ is replaced by $\text{Crown}(C_1) \wedge \text{OnHead}(C_1, \text{John})$. Which condition must C_1 satisfy?

- ☐ (A) C_1 must be a new constant symbol not appearing anywhere else in the KB
- ☐ (B) C_1 must be John
- ☐ (C) C_1 may be any ground term already in the KB
- ☐ (D) C_1 must be a variable

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, D, C** and rules:

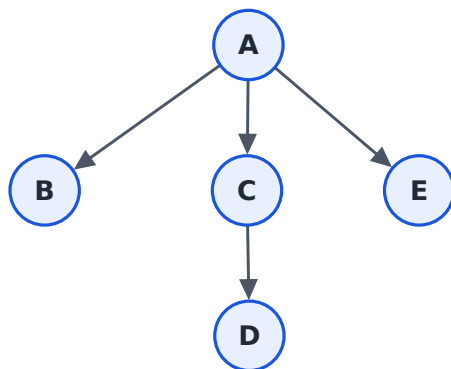
- R1: $K \wedge F \rightarrow A$
 R2: $C \rightarrow F$
 R3: $A \rightarrow E$
 R4: $C \wedge D \rightarrow A$
 R5: $A \wedge F \wedge C \rightarrow D$
 R6: $H \wedge D \rightarrow B$
 R7: $D \wedge E \rightarrow A$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D, E. What is the probability that a single run produces the sample **(A=T, B=F, C=F, D=F, E=F)**? (round off to 4 decimal places)



P(A=T)
0.75

A	P(B=T ...)	A	P(C=T ...)
T	0.25	T	0.1
F	0.4	F	0.1

C	P(D=T ...)	A	P(E=T ...)
T	0.75	T	0.25
F	0.5	F	0.4

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling

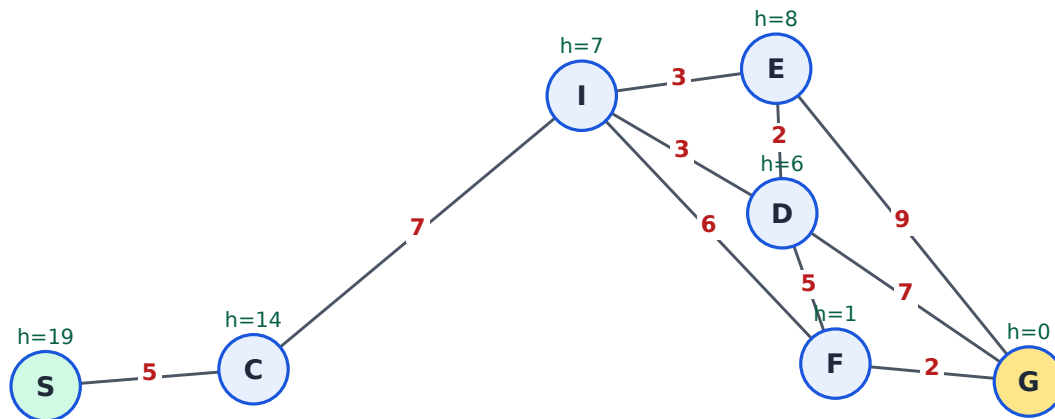
A continuous variable X has exponential density $f(x) = 2e^{-2x}$ for $x \geq 0$. Using inverse-CDF sampling with uniform random number $u = 0.75$, what value of X is generated? (Round to 3 decimal places.)

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge E-G violates the consistency condition (in at least one direction).
- ☐ (B) $h(D)$ overestimates the true cost from D to G.
- ☐ (C) h is admissible.
- ☐ (D) h is consistent (monotone).

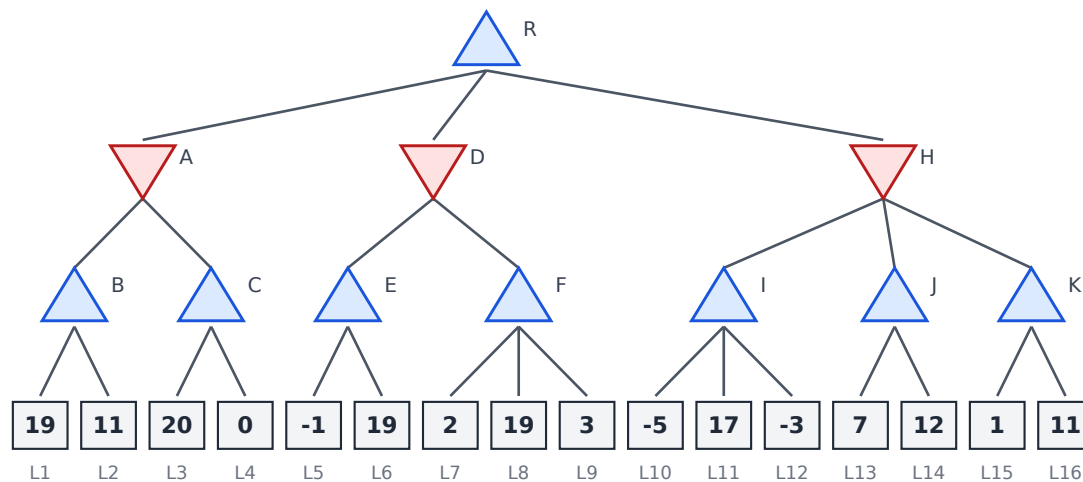
Q.12 [MCQ · 2 marks · negative: -2/3] Adversarial Search

A three-ply binary game tree has levels MAX (root), MIN, MAX, and then leaves. The leaves, left to right, are 3, 4, 2, 9, 7, 1, 8, 6. Each bottom MAX node owns two consecutive leaves. Alpha-beta processes children left to right. Which statement is correct?

- ☐ (A) Root value 7; only the last leaf (6) is pruned
- ☐ (B) Root value 4; leaves 2 and 9 are pruned
- ☐ (C) Root value 7; leaves 1 and 6 are pruned
- ☐ (D) Root value 8; no leaf is pruned

Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

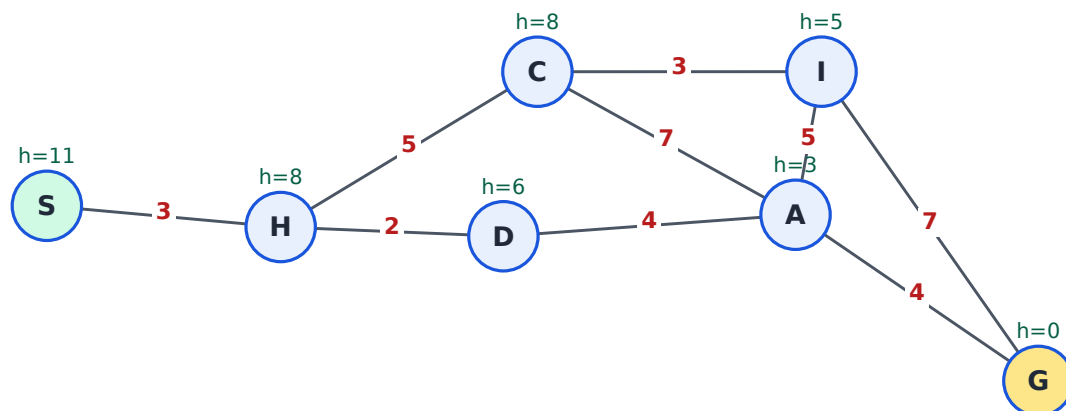
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



- ☐ (A) Leaf L5 is pruned.
☐ (B) Leaf L7 is pruned.
☐ (C) Leaf L9 is pruned.
☐ (D) Leaf L12 is pruned.

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

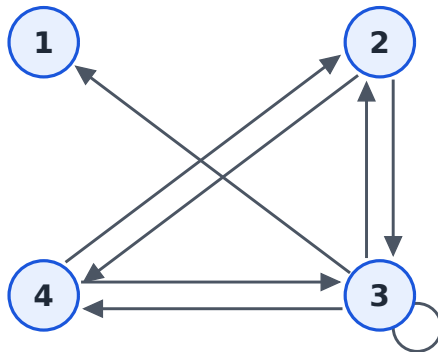
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\exists y \forall x R(x, y)$
☐ (C) $\exists x R(x, x)$
☐ (D) $\exists x \forall y R(x, y)$

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic

Of the 16 binary connectives $\circ$, for how many is the formula $(p \circ q) \rightarrow p$ a tautology?

Answer (numerical): _____

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(P \rightarrow Q) \vee (\neg Q \vee R)$
- (2) $(\neg Q \leftrightarrow R) \wedge (Q \rightarrow R)$
- (3) $Q \rightarrow (R \leftrightarrow Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $(Q \vee P) \leftrightarrow (R \wedge \neg P)$
☐ (B) $\neg R \rightarrow R$
☐ (C) $P \vee \neg R$
☐ (D) $\neg(Q \vee P)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(P \rightarrow Q) \vee (Q \rightarrow P)$
☐ (B) $((P \rightarrow Q) \wedge (R \rightarrow Q)) \rightarrow (P \wedge R)$
☐ (C) $(\neg Q \rightarrow (R \rightarrow P)) \leftrightarrow ((\neg Q \wedge R) \rightarrow P)$
☐ (D) $((Q \rightarrow P) \wedge P) \rightarrow Q$

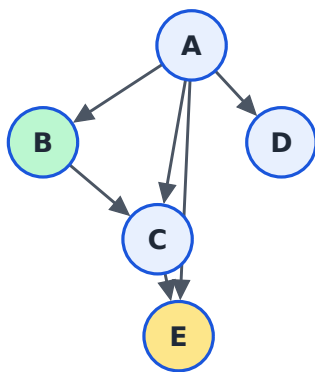
Q.19 [MCQ · 2 marks · negative: -2/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Gibbs sampling with evidence $S = \text{true}$, $W = \text{true}$ is in the state where $R = \text{true}$. What is the probability that C is resampled to true, $P(c | s, r)$?

- (A) 0.444
○ (B) 0.040
○ (C) 0.500
○ (D) 0.878

Q.20 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{E=F}$. The current state of the chain is $(\mathbf{A=F}, \mathbf{B=F}, \mathbf{C=T}, \mathbf{D=T}, \mathbf{E=F})$. The next step resamples **B**. With what probability is B set to T? (round off to 3 decimal places)



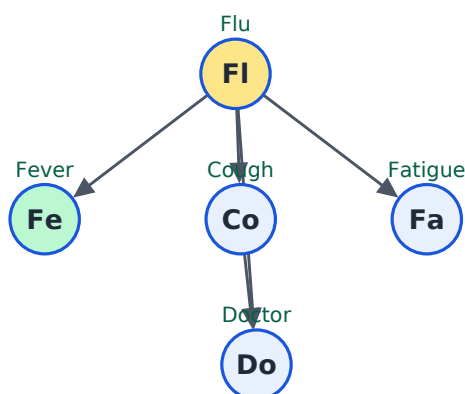
P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.4	T	0.25	T	T	0.25
	F	0.5	T	F	0.9
			F	T	0.05
			F	F	0.1

A	P(D=T ..)	A	C	P(E=T ...)
T	0.1	T	T	0.5
F	0.4	T	F	0.3
		F	T	0.9
		F	F	0.75

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T | \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) Compute $\mathbf{P(\text{Fever} = T | \text{Flu} = T)}$ (round off to 3 decimal places).



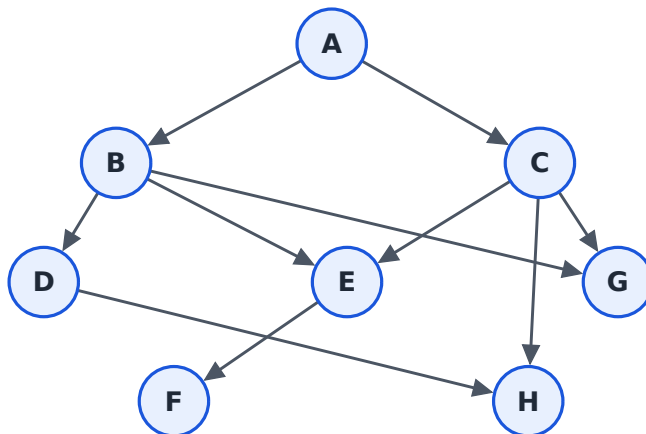
P(FI=T)	FI	P(Fe=T ...)	FI	P(Co=T ...)
0.5	T	0.7	T	0.4
	F	0.3	F	0.7

FI	P(Fa=T ..)	FI	Co	P(Do=T ...)
T	0.95	T	T	0.2
F	0.7	T	F	0.8
		F	T	0.5
		F	F	0.6

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp B \mid A$
☐ (B) $A \perp G \mid B, C$
☐ (C) $D \perp C \mid A, B$
☐ (D) $H \perp F \mid D$

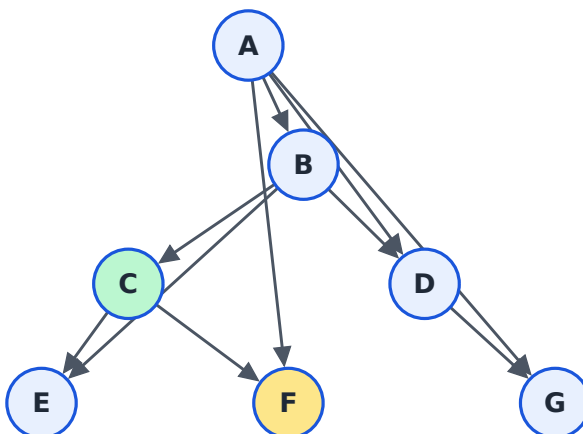
Q.23 [MCQ · 2 marks · negative: -2/3] Variable Elimination

For a Bayesian network with n Boolean variables, the depth-first ENUMERATION-ASK algorithm (inference by enumeration) has worst-case

- ☐ (A) $O(2^n)$ time and $O(n)$ space
☐ (B) $O(n)$ time and $O(2^n)$ space
☐ (C) $O(2^n)$ time and $O(2^n)$ space
☐ (D) $O(n^2)$ time and $O(n)$ space

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

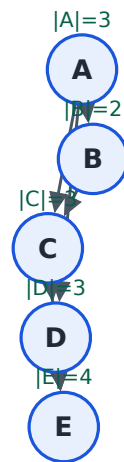
Variable elimination is used to compute $P(C \mid F = T)$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **G, B, A, E, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

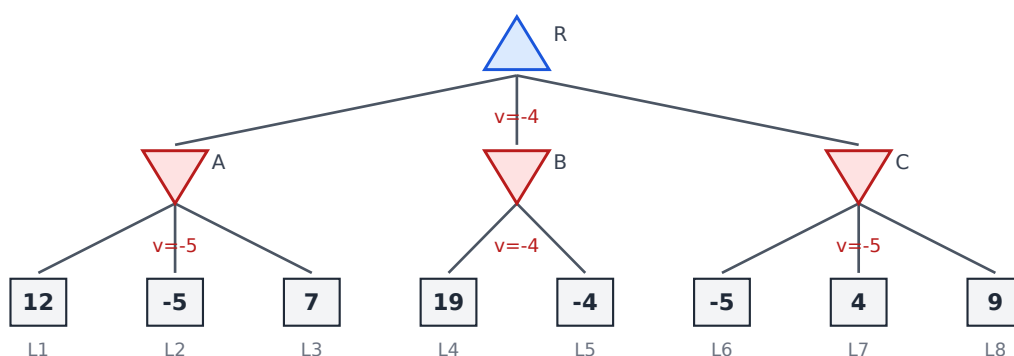
Answer Key — Mock Test 019

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	-4	14	NAT	2	Informed Search	13
2	NAT	1	Informed Search	15	15	MSQ	2	Predicate Logic	A, C, D
3	NAT	1	Problem Formulation	24	16	NAT	2	Propositional Logic	4
4	MCQ	1	Uninformed Search	B	17	MSQ	2	Propositional Logic	B
5	MCQ	1	Propositional Logic	A	18	MSQ	2	Propositional Logic	A, C
6	NAT	1	Propositional Logic	6	19	MCQ	2	Sampling	A
7	MCQ	1	Predicate Logic	A	20	NAT	2	Sampling	0.331 to 0.335
8	NAT	1	Propositional Logic	6	21	NAT	2	Exact Inference	0.698 to 0.702
9	NAT	1	Sampling	0.1896 to 0.19	22	MSQ	2	Conditional Independence	A, B, C
10	NAT	1	Sampling	0.692 to 0.694	23	MCQ	2	Variable Elimination	A
11	MSQ	2	Informed Search	C, D	24	NAT	2	Exact Inference	4
12	MCQ	2	Adversarial Search	A	25	NAT	2	Bayesian Networks	153
13	MSQ	2	Adversarial Search	B, C					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] **Answer: -4**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = -5, B = -4, C = -5. Root value = **-4**; MAX's optimal first move is towards B.

Q.2 [NAT, 1M · Informed Search] Answer: 15

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (14)	D(9)
2	D (9)	F(2), A(14)
3	F (2)	G(0), I(2), B(8), A(14)
4	G (0)	I(2), B(8), A(14) ← goal, stop

Path returned: **S → D → F → G**, cost = 5 + 7 + 3 = **15**.

The optimal cost is $h^*(S) = 15$ — greedy happens to be optimal here, but in general it is not.

Q.3 [NAT, 1M · Problem Formulation] Answer: 24

The agent can be in any of 3 squares, and each of the 3 squares is dirty or clean, giving 2^3 combinations. States = $n \cdot 2^n = 3 \times 8 = \mathbf{24}$. The 2-square world has $2 \cdot 2^2 = 8$ states.

Q.4 [MCQ, 1M · Uninformed Search] Answer: B (B: S, I, B, C, E, H, F, K, G)

Adjacency lists (alphabetical): B: {C, F, I, K}; C: {B, E, G}; E: {C, H}; F: {B, H, I, K}; G: {C, H}; H: {E, F, G}; I: {B, F, K, S}; K: {B, F, I}; S: {I}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, I, B, C, E, H, F, K, G**; the path found is $S \rightarrow I \rightarrow B \rightarrow C \rightarrow E \rightarrow H \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.5 [MCQ, 1M · Propositional Logic] Answer: A ($A: \alpha \wedge \neg\beta$ is unsatisfiable)

$\alpha \models \beta$ means every model of α is a model of β , i.e. no model makes α true and β false, i.e. $\alpha \wedge \neg\beta$ is unsatisfiable. Equivalently $\alpha \rightarrow \beta$ is valid. Satisfiability of $\neg\alpha \vee \beta$ or $\alpha \wedge \beta$ is too weak (one model is enough). $\alpha \vee \neg\beta$ being valid would mean $\beta \models \alpha$, the reverse direction.

Q.6 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg Q \vee (\neg R \rightarrow \neg Q)$	$P \vee \neg R$	$(\neg Q \vee (\neg R \rightarrow \neg Q)) \rightarrow (P \vee \neg R)$
T	T	T	T	T	T
T	T	F	F	T	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	F	T	T
F	F	T	T	F	F
F	F	F	T	T	T

Rows where F = T: **6**.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A ($A: C_1$ must be a new constant symbol not appearing anywhere else in the KB)

Existential Instantiation introduces a fresh Skolem constant naming the (unknown) witness. Using an existing term such as John would assert, without justification, that John is the witness. The result is inferentially equivalent (satisfiable iff the original), not logically equivalent. A variable would make it a non-ground formula.

Q.8 [NAT, 1M · Propositional Logic] Answer: 6

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $C \rightarrow F$, $C \wedge D \rightarrow A$; add A, F.

Round 2: fire $A \rightarrow E$; add E.

Fixed point: {A, B, C, D, E, F} — 6 atoms are entailed (facts: B, C, D; newly derived: A, E, F). Atoms never derived: H, J, K.

Q.9 [NAT, 1M · Sampling] Answer: 0.1896 to 0.19

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(A=T) \times P(B=F|A=T) \times P(C=F|A=T) \times P(D=F|C=F) \times P(E=F|A=T) \\ = 0.75 \times 0.75 \times 0.9 \times 0.5 \times 0.75 = \mathbf{0.189844} \approx 0.1898.$$

Q.10 [NAT, 1M · Sampling] Answer: 0.692 to 0.694

The CDF is $F(x) = 1 - e^{-2x}$. Solving $F(x) = u$ gives $x = -\ln(1 - u)/2$. With $u = 0.75$: $x = -\ln(0.25)/2 = \ln 4/2 = 1.3863/2 \approx 0.693$.

Q.11 [MSQ, 2M · Informed Search] Answer: C, D (C: h is admissible.; D: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	D	E	F	G	I	S
$h(n)$	14	6	8	1	0	7	19
$h^*(n)$	15	7	9	2	0	8	20
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

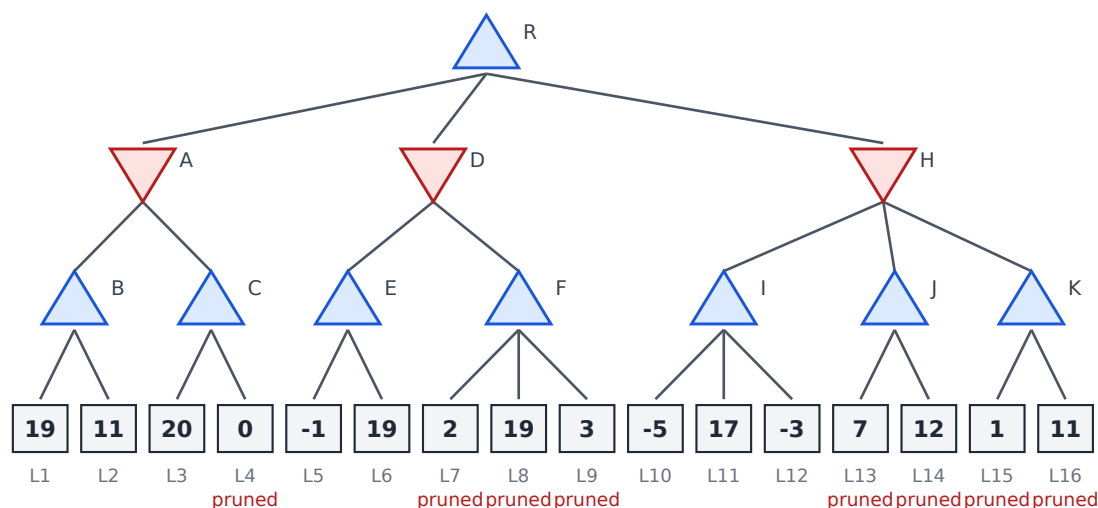
Q.12 [MCQ, 2M · Adversarial Search] Answer: A (A: Root value 7; only the last leaf (6) is pruned)

Left MIN: $\text{MAX}(3,4) = 4$, so $\beta = 4$. $\text{MAX}(2,9)$: leaf 2 does not cut, and $9 \geq 4$ cuts, but it is the last child, so both leaves are evaluated. Left MIN = 4 and root $\alpha = 4$. Right MIN: $\text{MAX}(7,1) = 7$ ($7 > \alpha$, so no cut), so $\beta = 7$. $\text{MAX}(8,6)$: $8 \geq \beta = 7$, so leaf 6 is pruned. Right MIN = 7, and the root is $\max(4, 7) = \mathbf{7}$. Only leaf 6 is pruned, so (A) is correct. Leaf 1 must be evaluated because $7 > \alpha$ at that point.

Q.13 [MSQ, 2M · Adversarial Search] Answer: B, C (B: Leaf L7 is pruned.; C: Leaf L9 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $20 \geq \beta = 19 \Rightarrow$ prune remaining children (leaves L4).
- At MIN node D: value $19 \leq \alpha = 19 \Rightarrow$ prune remaining children (leaves L7, L8, L9).
- At MIN node H: value $17 \leq \alpha = 19 \Rightarrow$ prune remaining children (leaves L13, L14, L15, L16).



Leaves evaluated (8): L1, L2, L3, L5, L6, L10, L11, L12. Leaves pruned (8): L4, L7, L8, L9, L13, L14, L15, L16. Root value = 19 (same as plain minimax, 19 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] Answer: 13

Heuristic values: $h(A)=3$, $h(C)=8$, $h(D)=6$, $h(G)=0$, $h(H)=8$, $h(I)=5$, $h(S)=11$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	H(11)
2	H (11)	D(11), C(16)
3	D (11)	A(12), C(16)

Step	Expanded (f = g + h)	Frontier after expansion (sorted by priority)
4	A (12)	G(13), C(16), I(19)
5	G (13)	C(16), I(19) ← goal, stop

Path returned: **S** → **H** → **D** → **A** → **G** with cost **13**; nodes expanded = **5** (S, H, D, A, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 13$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [MSQ, 2M · Predicate Logic] **Answer: A, C, D** (A: $\exists x \forall y \neg R(x, y)$; C: $\exists x R(x, x)$; D: $\exists x \forall y R(x, y)$)

$R = \{(2,3), (2,4), (3,1), (3,2), (3,3), (3,4), (4,2), (4,3)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

(B) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 3.

(D) $\exists x \forall y R(x, y)$: **TRUE** — $x = 3$ has edges to every y .

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.16 [NAT, 2M · Propositional Logic] **Answer: 4**

$(p \circ q) \rightarrow p$ can fail only in rows where $p = F$, and it fails there iff $p \circ q = T$. So we need $F \circ T = F$ and $F \circ F = F$, while $T \circ T$ and $T \circ F$ are free. That gives $2^2 = 4$ connectives: constant F , $p \wedge q$, $p \wedge \neg q$, and the projection p .

Q.17 [MSQ, 2M · Propositional Logic] **Answer: B** (B: $\neg R \rightarrow R$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	F	T	T
F	F	T	T

(A) $(Q \vee P) \leftrightarrow (R \wedge \neg P)$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.

(B) $\neg R \rightarrow R$ is true in all models of $KB \Rightarrow$ **entailed**.

(C) $P \vee \neg R$ is false in the KB-model $P=F, Q=F, R=T \Rightarrow$ not entailed.

(D) $\neg(Q \vee P)$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: $(P \rightarrow Q) \vee (Q \rightarrow P)$; C: $(\neg Q \rightarrow (R \rightarrow P)) \leftrightarrow ((\neg Q \wedge R) \rightarrow P)$)

(A) $(P \rightarrow Q) \vee (Q \rightarrow P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $((P \rightarrow Q) \wedge (R \rightarrow Q)) \rightarrow (P \wedge R)$: **not valid** — counter-model $P=T, Q=T, R=F$.

(C) $(\neg Q \rightarrow (R \rightarrow P)) \leftrightarrow ((\neg Q \wedge R) \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow P) \wedge P) \rightarrow Q$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.19 [MCQ, 2M · Sampling] **Answer: A** (A: 0.444)

$MB(C) = \{S, R\}$ (its children; they have no other parents). $P(C | s, r) = \alpha P(C) P(s | C) P(r | C)$. $c: 0.5 \times 0.1 \times 0.8 = 0.04$. $\neg c: 0.5 \times 0.5 \times 0.2 = 0.05$. $P(c | s, r) = 0.04/0.09 \approx 0.444$. 0.040 is the unnormalized value, 0.500 is the prior, and 0.878 is $P(c | \neg s, r)$.

Q.20 [NAT, 2M · Sampling] **Answer: 0.331 to 0.335**

Gibbs sampling draws B from $P(B | mb(B))$, where $mb(B) = \{A, C\}$:

$P(B | mb) \propto P(B | \text{parents}(B)) \times \prod_{\text{children } Y} P(Y | \text{parents}(Y))$.

$B=T: P(B=T|A=F) \times P(C=T|A=F, B=T) = 0.5 \times 0.05 = 0.025$

$B=F: P(B=F|A=F) \times P(C=T|A=F, B=F) = 0.5 \times 0.1 = 0.05$

Normalise: $0.025 / (0.025 + 0.05) = \mathbf{0.3333}$.

Q.21 [NAT, 2M · Exact Inference] **Answer: 0.698 to 0.702**

Factorisation: $P(\text{Flu, Fever, Cough, Fatigue, Doctor}) = P(\text{Flu}) \cdot P(\text{Fever} | \text{Flu}) \cdot P(\text{Cough} | \text{Flu}) \cdot P(\text{Fatigue} | \text{Flu}) \cdot P(\text{Doctor} | \text{Flu, Cough})$.

Inference by enumeration: $P(\text{Fever} | e) = \alpha \sum_{\text{hidden}} P(\text{Fever, e, hidden})$.

Hidden: Co, Fa, Do	Fe	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, T, T	T	$0.5 \times 0.7 \times 0.4 \times 0.95 \times 0.2$	0.0266
T, T, F	T	$0.5 \times 0.7 \times 0.4 \times 0.95 \times 0.8$	0.1064

Hidden: Co, Fa, Do	Fe	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, F, T	T	$0.5 \times 0.7 \times 0.4 \times 0.05 \times 0.2$	0.0014
T, F, F	T	$0.5 \times 0.7 \times 0.4 \times 0.05 \times 0.8$	0.0056
F, T, T	T	$0.5 \times 0.7 \times 0.6 \times 0.95 \times 0.8$	0.1596
F, T, F	T	$0.5 \times 0.7 \times 0.6 \times 0.95 \times 0.2$	0.0399
F, F, T	T	$0.5 \times 0.7 \times 0.6 \times 0.05 \times 0.8$	0.0084
F, F, F	T	$0.5 \times 0.7 \times 0.6 \times 0.05 \times 0.2$	0.0021
T, T, T	F	$0.5 \times 0.3 \times 0.4 \times 0.95 \times 0.2$	0.0114
T, T, F	F	$0.5 \times 0.3 \times 0.4 \times 0.95 \times 0.8$	0.0456
T, F, T	F	$0.5 \times 0.3 \times 0.4 \times 0.05 \times 0.2$	0.0006
T, F, F	F	$0.5 \times 0.3 \times 0.4 \times 0.05 \times 0.8$	0.0024
F, T, T	F	$0.5 \times 0.3 \times 0.6 \times 0.95 \times 0.8$	0.0684
F, T, F	F	$0.5 \times 0.3 \times 0.6 \times 0.95 \times 0.2$	0.0171
F, F, T	F	$0.5 \times 0.3 \times 0.6 \times 0.05 \times 0.8$	0.0036
F, F, F	F	$0.5 \times 0.3 \times 0.6 \times 0.05 \times 0.2$	0.0009

$P(\text{Fever}=T, e) = 0.35$; $P(\text{Fever}=F, e) = 0.15$; $P(e) = 0.5$.

$P(\text{Fever}=T | e) = 0.35 / 0.5 = \mathbf{0.7} \approx 0.700$.

Q.22 [MSQ, 2M · Conditional Independence] Answer: A, B, C (A: $C \perp B | A$; B: $A \perp G | B, C$; C: $D \perp C | A, B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $C \perp B | A$: **TRUE** — $C \leftarrow A \rightarrow B$ (blocked: observed fork node A); $C \rightarrow E \leftarrow B$ (blocked: collider E with neither it nor a descendant observed); $C \rightarrow G \leftarrow B$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow H \leftarrow D \leftarrow B$ (blocked: collider H with neither it nor a descendant observed).

(B) $A \perp G | B, C$: **TRUE** — $A \rightarrow B \rightarrow D \rightarrow H \leftarrow C \rightarrow G$ (blocked: observed chain node B); $A \rightarrow B \rightarrow E \leftarrow C \rightarrow G$ (blocked: observed chain node B); $A \rightarrow B \rightarrow G$ (blocked: observed chain node B); $A \rightarrow C \rightarrow E \leftarrow B \rightarrow G$ (blocked: observed chain node C); $A \rightarrow C \rightarrow G$ (blocked: observed chain node C); $A \rightarrow C \rightarrow H \leftarrow D \leftarrow B \rightarrow G$ (blocked: observed chain node C).

(C) $D \perp C | A, B$: **TRUE** — $D \leftarrow B \leftarrow A \rightarrow C$ (blocked: observed chain node B); $D \leftarrow B \rightarrow E \leftarrow C$ (blocked: observed fork node B); $D \leftarrow B \rightarrow G \leftarrow C$ (blocked: observed fork node B); $D \rightarrow H \leftarrow C$ (blocked: collider H with neither it nor a descendant observed).

(D) $H \perp F | D$: **FALSE** — the path $H \leftarrow C \leftarrow A \rightarrow B \rightarrow E \rightarrow F$ is active given $\{D\}$.

Q.23 [MCQ, 2M · Variable Elimination] Answer: A (A: $O(2^n)$ time and $O(n)$ space)

Enumeration evaluates a depth-first expression tree that branches over every hidden variable, giving $O(2^n)$ time. Being depth-first it stores only the current path, so space is $O(n)$. It never builds the full joint table, so it does not need $O(2^n)$ space; polynomial time is impossible in general since inference is NP-hard. Variable elimination improves time by caching repeated subexpressions as factors.

Q.24 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(B, C, E)$, $f(A, C)$, $f(A, D, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(A, D, G)$	$\{A, D, G\}$	$f(A, D)$
B	$f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(B, C, E)$	$\{A, B, C, D, E\}$	$f(A, C, D, E)$
A	$f(A)$, $f(A, C)$, $f(A, D)$, $f(A, C, D, E)$	$\{A, C, D, E\}$	$f(C, D, E)$
E	$f(C, D, E)$	$\{C, D, E\}$	$f(C, D)$
D	$f(C, D)$	$\{C, D\}$	$f(C)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over C and normalise.

Q.25 [NAT, 2M · Bayesian Networks] Answer: 153

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (2-1) \times 3 + C: (3-1) \times 3 \times 2 + D: (3-1) \times 3 \times 3 + E: (4-1) \times 3 \times 3 = \mathbf{62}$.

Full joint table: $\prod |X| - 1 = 216 - 1 = 215$; saving = 153.

Mock Test 020

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 9 / 10	13 / 14 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

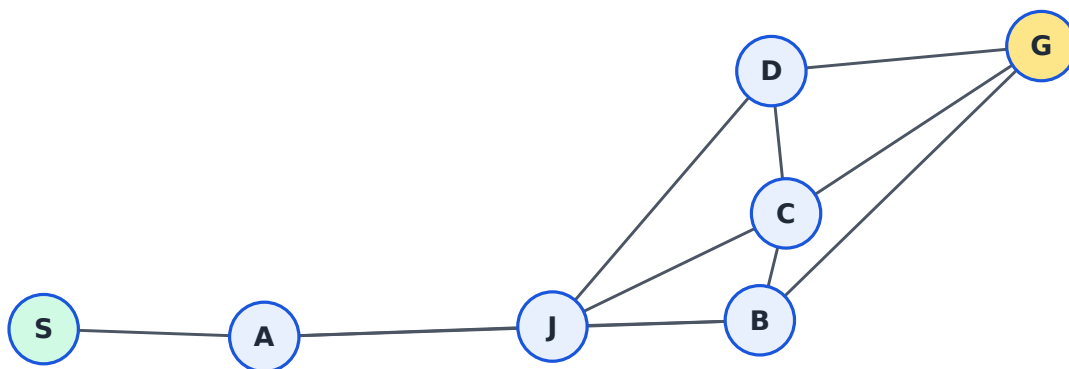
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 4$ (so $b^d = 256$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, A, B, C, D, G
- ☐ (B) S, A, J, B, D, C, G
- ☐ (C) S, A, B, J, C, G
- ☐ (D) S, A, B, C, J, G

Q.3 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search

The frontier of depth-first search is implemented as a

- ☐ (A) FIFO queue
- ☐ (B) LIFO stack
- ☐ (C) priority queue ordered by $g(n)$
- ☐ (D) priority queue ordered by $h(n)$

Q.4 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **D, C** and rules:

- R1: $A \wedge H \wedge J \rightarrow K$
 R2: $E \wedge A \rightarrow K$
 R3: $J \rightarrow B$
 R4: $J \wedge D \wedge E \rightarrow B$
 R5: $D \rightarrow J$
 R6: $C \rightarrow D$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following statements are TRUE?

- ☐ (A) Propositional resolution is refutation-complete
☐ (B) Modus ponens alone is complete for arbitrary propositional knowledge bases
☐ (C) Forward chaining is sound and complete for knowledge bases of definite clauses
☐ (D) A sound inference procedure derives only sentences that are entailed by the KB

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((P \vee Q) \vee (R \leftrightarrow R)) \leftrightarrow ((R \vee \neg Q) \wedge (R \wedge \neg Q)) ?$$

Answer (numerical): _____

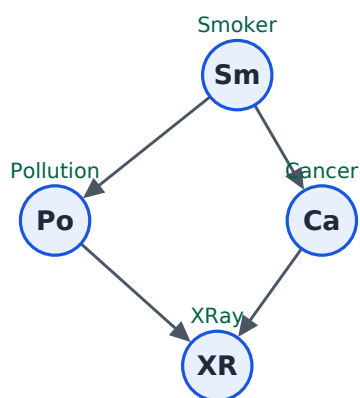
Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Skolemizing the sentence $\forall x \exists y P(x, y)$ yields

- ☐ (A) $\forall x P(x, F(x))$, where F is a new function symbol
☐ (B) $\forall x P(x, A)$, where A is a new constant
☐ (C) $\exists y P(y, y)$
☐ (D) $\forall x \forall y P(x, y)$

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) What is the probability that a single run produces the sample **(Sm=F, Po=T, Ca=T, XR=T)**? (round off to 4 decimal places)



P(Sm=T)
0.5

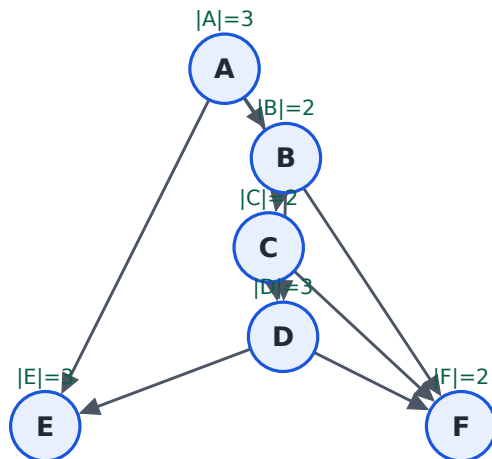
Sm	P(Po=T ..)	Sm	P(Ca=T ...)
T	0.1	T	0.8
F	0.9	F	0.75

Po	Ca	P(XR=T ...)
T	T	0.1
T	F	0.9
F	T	0.9
F	F	0.75

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D, E. If rejection sampling is used to estimate $P(B = T \mid D = F)$, how many of these samples are **rejected**?

#	A	B	C	D	E
1	F	F	F	F	T
2	F	F	T	T	F
3	F	T	T	T	T
4	F	F	F	F	F
5	T	T	F	T	T
6	F	F	T	F	F
7	F	T	F	T	T
8	F	F	F	T	T
9	T	T	T	F	T
10	F	F	F	T	T
11	F	F	F	F	T
12	F	F	T	T	T
13	F	T	T	F	F
14	F	F	T	F	F
15	F	T	F	T	T
16	F	F	T	T	T
17	T	T	F	F	T
18	T	T	F	F	F
19	T	T	F	T	T
20	F	T	F	F	F

Answer (numerical): _____

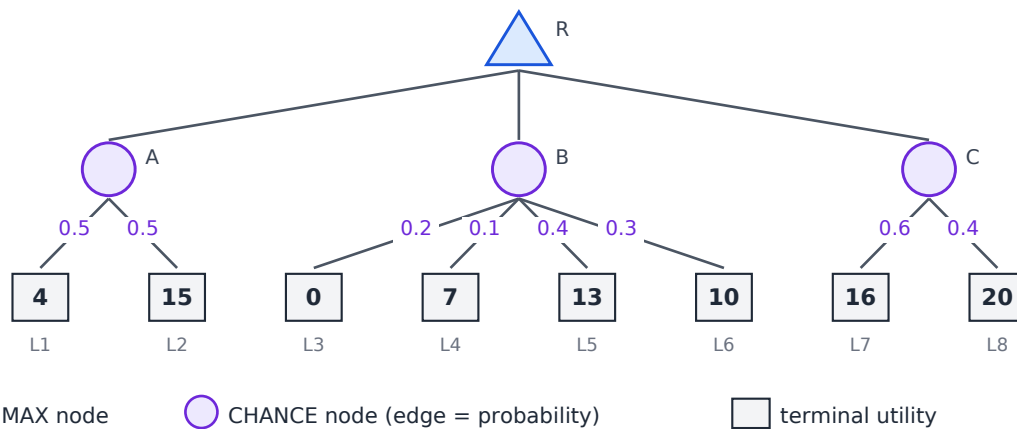
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). Uniform-cost search (graph search with an explored set, goal test when a node is selected for expansion) is run from S. How many nodes are expanded before the goal G is selected? Count S, and do not count G.

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



Answer (numerical): _____

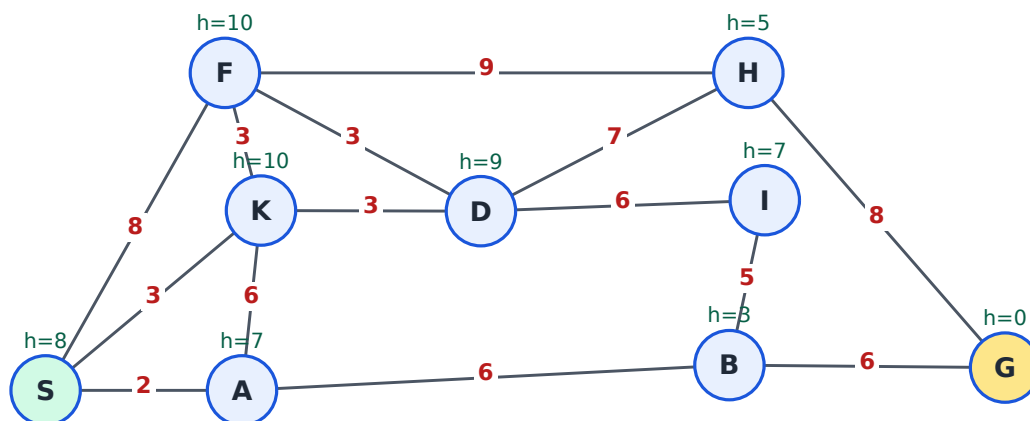
Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search

Which of the following statements about alpha-beta pruning are TRUE?

- ☐ (A) It returns the same root value as minimax
- ☐ (B) A pruned subtree cannot change the root's minimax decision
- ☐ (C) Move ordering affects how much is pruned, but not the root value
- ☐ (D) It can be applied unchanged to expectimax trees containing chance nodes, with the same pruning conditions

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

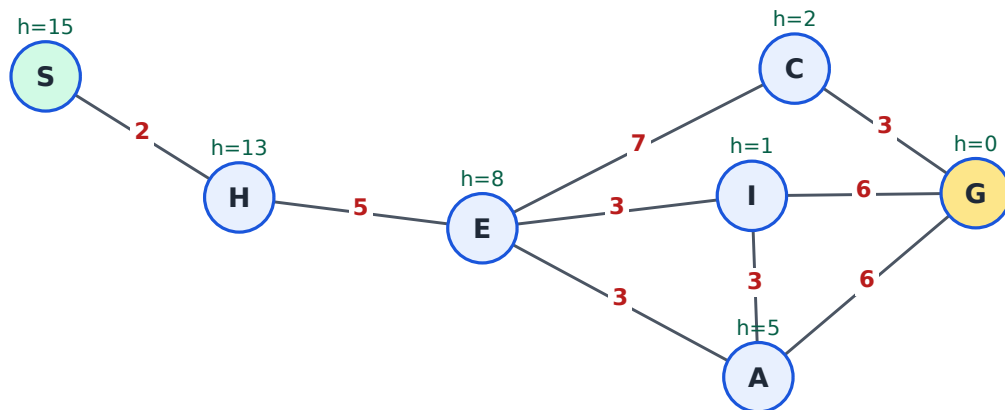
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

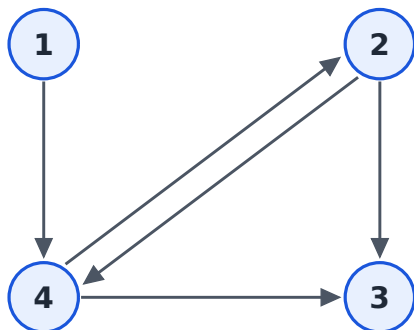
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(H)$ overestimates the true cost from H to G.
- ☐ (B) h is consistent (monotone).
- ☐ (C) h is admissible.
- ☐ (D) The edge C-E violates the consistency condition (in at least one direction).

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y (x \neq y \wedge R(x, y))$
- ☐ (B) $\exists x \forall y \neg R(x, y)$
- ☐ (C) $\exists x \forall y \neg R(y, x)$
- ☐ (D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$

Q.17 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following statements about resolution is correct?

- ☐ (A) Resolution is complete for deriving every entailed clause directly, without refutation
- ☐ (B) If a clause set is unsatisfiable, the resolution closure of the set contains the empty clause
- ☐ (C) If the resolution closure of a clause set does not contain the empty clause, the set may still be unsatisfiable
- ☐ (D) Resolution can derive the empty clause from a satisfiable set of clauses

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (P \rightarrow P) \wedge (Q \wedge P)$?

- ☐ (A) $\neg(\neg(P \rightarrow P) \vee \neg(Q \wedge P))$
☐ (B) $\neg(\neg(P \rightarrow P) \wedge \neg(Q \wedge P))$
☐ (C) $(Q \wedge P) \wedge (P \rightarrow P)$
☐ (D) $(P \rightarrow P) \vee (Q \wedge P)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following statements about Horn clauses and Horn-form knowledge bases are TRUE?

- ☐ (A) Entailment of an atom from a Horn KB can be decided in time linear in the size of the KB
☐ (B) Every Horn clause has at most one positive literal
☐ (C) The clause $\neg p \vee q \vee r$ is a Horn clause
☐ (D) The resolvent of two Horn clauses is again a Horn clause

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

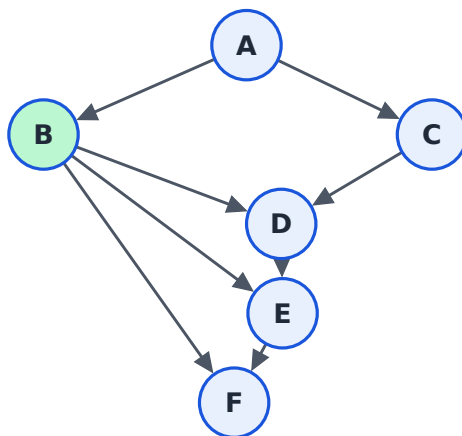
C1: $(P \vee S)$, C2: $(\neg P \vee \neg Q \vee \neg R)$, C3: $(\neg P \vee \neg Q \vee \neg S)$, C4: $(Q \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg R \vee S)$
☐ (B) $(\neg Q)$
☐ (C) $(\neg P \vee \neg R \vee S)$
☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{B})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E, D, A, F, C**. What is the scope of the new factor produced when **F** is summed out?



- ☐ (A) $f(C)$
☐ (B) $f(B, C)$
☐ (C) $f(B, C, F)$
☐ (D) $f(B, E)$

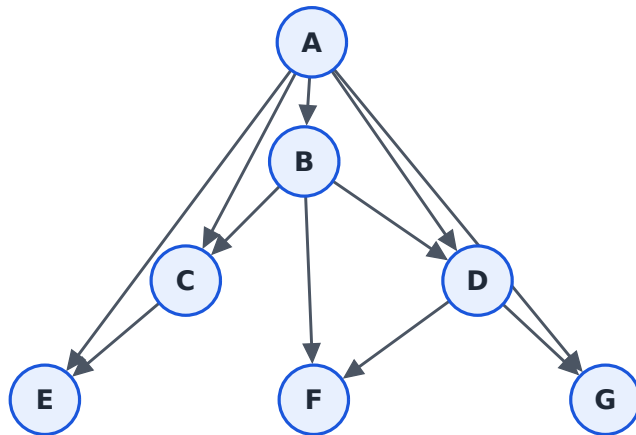
Q.22 [MSQ · 2 marks · no negative marking] Variable Elimination

Which of the following Bayesian networks (edges listed) are polytrees?

- ☐ (A) $A \rightarrow C, B \rightarrow C, C \rightarrow D, C \rightarrow E$
☐ (B) $A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D$
☐ (C) $A \rightarrow B, B \rightarrow C, D \rightarrow C, D \rightarrow E$
☐ (D) $A \rightarrow B, A \rightarrow C, B \rightarrow C$

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp E \mid A, B$
☐ (B) $G \perp C \mid B$
☐ (C) $E \perp D \mid A, B$
☐ (D) $G \perp C \mid F$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $P(\mathbf{XRay} = \mathbf{T} \mid \mathbf{Cancer} = \mathbf{T})$ (round off to 3 decimal places).



P(Sm=T)					
0.2		Sm	P(Po=T ..)	Sm	Po
T	0.8	T	T	0.6	
F	0.25	T	F	0.25	
		F	T	0.4	
		F	F	0.2	

Ca	P(XR=T ...)
T	0.05
F	0.95

Answer (numerical): _____

Q.25 [MCQ · 2 marks · negative: -2/3] Variable Elimination

A Bayesian network over Boolean variables has edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$, $D \rightarrow E$. $P(E)$ is computed by variable elimination with no evidence. For which elimination ordering does **every** new factor created by summing out a variable contain at most two variables?

- (A) A, B, C, D
 ○ (B) D, C, B, A
 ○ (C) B, C, A, D
 ○ (D) C, B, A, D

Answer Key — Mock Test 020

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	31	14	NAT	2	Informed Search	5
2	MCQ	1	Uninformed Search	C	15	MSQ	2	Informed Search	C
3	MCQ	1	Uninformed Search	B	16	MSQ	2	Predicate Logic	B, C
4	NAT	1	Propositional Logic	4	17	MCQ	2	Propositional Logic	B
5	MSQ	1	Propositional Logic	A, C, D	18	MSQ	2	Propositional Logic	A, C
6	NAT	1	Propositional Logic	2	19	MSQ	2	Propositional Logic	A, B, D
7	MCQ	1	Predicate Logic	A	20	MSQ	2	Propositional Logic	A, C
8	NAT	1	Sampling	0.0336 to 0.034	21	MCQ	2	Exact Inference	B
9	NAT	1	Bayesian Networks	51	22	MSQ	2	Variable Elimination	A, C
10	NAT	1	Sampling	10	23	MSQ	2	Conditional Independence	A, C
11	NAT	2	Uninformed Search	4	24	NAT	2	Exact Inference	0.048 to 0.052
12	NAT	2	Adversarial Search	17.59 to 17.61	25	MCQ	2	Variable Elimination	A
13	MSQ	2	Adversarial Search	A, B, C					

Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 31**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 4^2 + 4^2 - 1 = 16 + 16 - 1 = 31$, versus 256 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.2 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, A, B, J, C, G)

Adjacency lists (alphabetical): A: {B, J, S}; B: {A, C, G, J}; C: {B, D, G, J}; D: {C, G, J}; G: {B, C, D}; J: {A, B, C, D}; S: {A}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[A]
2	A	[B, J]
3	B	[J, C, G]
4	J	[C, G, D]
5	C	[G, D]
6	G	[D] $\leftarrow$ goal popped, stop

Expansion order: **S, A, B, J, C, G**. The path returned is $S \rightarrow A \rightarrow B \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [MCQ, 1M · Uninformed Search] **Answer: B** (B: LIFO stack)

DFS always expands the most recently generated (deepest) node, which is **LIFO** behaviour. A FIFO queue gives BFS (A). A priority queue on $g(n)$ gives UCS (C). A priority queue on $h(n)$ gives greedy best-first search (D).

Q.4 [NAT, 1M · Propositional Logic] Answer: 4

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \rightarrow J$; add J.

Round 2: fire $J \rightarrow B$; add B.

Fixed point: $\{B, C, D, J\}$ — 4 atoms are entailed (facts: C, D; newly derived: B, J). Atoms never derived: A, E, F, H, K.

Q.5 [MSQ, 1M · Propositional Logic] Answer: A, C, D

(A) True: if $KB \wedge \neg \alpha$ is unsatisfiable, resolution derives the empty clause. (B) False: from $p \vee q$, $p \rightarrow r$, $q \rightarrow r$, modus ponens cannot derive r, although r is entailed. (C) True for definite clauses (Horn KBs with exactly one positive literal per clause). (D) This is the definition of soundness (truth-preserving).

Q.6 [NAT, 1M · Propositional Logic] Answer: 2

Build the truth table (sub-formulas shown first):

P	Q	R	$(P \vee Q) \vee (R \leftrightarrow R)$	$(R \vee \neg Q) \wedge (R \wedge \neg Q)$	$((P \vee Q) \vee (R \leftrightarrow R)) \leftrightarrow ((R \vee \neg Q) \wedge (R \wedge \neg Q))$
T	T	T	T	F	F
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	F	F

Rows where F = T: **2**.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x P(x, F(x))$, where F is a new function symbol)

Since the existential $\exists y$ lies within the scope of $\forall x$, the chosen y may depend on x, so it is replaced by a Skolem function $F(x)$. A Skolem constant would wrongly assert one y works for all x. The result is satisfiable iff the original is (though not logically equivalent). $\forall x \forall y P(x, y)$ is much stronger.

Q.8 [NAT, 1M · Sampling] Answer: 0.0336 to 0.034

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Sm}=F) \times P(\text{Po}=T|\text{Sm}=F) \times P(\text{Ca}=T|\text{Sm}=F) \times P(\text{XR}=T|\text{Po}=T, \text{Ca}=T)$

$= 0.5 \times 0.9 \times 0.75 \times 0.1 = \mathbf{0.03375} \approx 0.0338$.

Q.9 [NAT, 1M · Bayesian Networks] Answer: 51

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (2-1) \times 3 + C: (2-1) \times 2 + D: (3-1) \times 2 \times 2 + E: (3-1) \times 3 \times 3 + F: (2-1) \times 3 \times 2 \times 3 = \mathbf{51}$.

Full joint table: $\prod |X| - 1 = 216 - 1 = 215$; saving = 164.

Q.10 [NAT, 1M · Sampling] Answer: 10

Rejection sampling keeps only samples consistent with the evidence $D = F$: samples 1, 4, 6, 9, 11, 13, 14, 17, 18, 20 (10 accepted, 10 rejected).

Among the accepted samples, $B = T$ in 5. Estimate = $5/10 = \mathbf{0.5}$.

(For this network the exact value is 0.5192; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.11 [NAT, 2M · Uninformed Search] Answer: 4

The expansion order by g-value is S(0), A(1), B(3) (reached via A), C(4) (reached via B). After that, the cheapest entry in the frontier is G(7), which is selected and terminates the search. The stale entries B(4) and C(6) are never expanded. Nodes expanded = $\{S, A, B, C\} = \mathbf{4}$.

Q.12 [NAT, 2M · Adversarial Search] Answer: 17.59 to 17.61

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.5 \times 4 + 0.5 \times 15 = 9.5$
- $B = 0.2 \times 0 + 0.1 \times 7 + 0.4 \times 13 + 0.3 \times 10 = 8.9$
- $C = 0.6 \times 16 + 0.4 \times 20 = 17.6$

Root = $\max(9.5, 8.9, 17.6) = 17.6$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.13 [MSQ, 2M · Adversarial Search] Answer: A, B, C

(A), (B) and (C) are the defining properties of alpha-beta: it prunes only subtrees that provably cannot affect the decision, so the result matches minimax for any ordering, and only the amount of pruning changes. (D) is false: the value of a chance node is an average, so one child does not bound it. Pruning there needs extra bounds on the leaf values and modified conditions.

Q.14 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(A)=7$, $h(B)=3$, $h(D)=9$, $h(F)=10$, $h(G)=0$, $h(H)=5$, $h(I)=7$, $h(K)=10$, $h(S)=8$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (8)	A(9), K(13), F(18)
2	A (9)	B(11), K(13), F(18)
3	B (11)	K(13), G(14), F(18), I(20)
4	K (13)	G(14), D(15), F(16), I(20)
5	G (14)	D(15), F(16), I(20) ← goal, stop

Path returned: **S** → **A** → **B** → **G** with cost **14**; nodes expanded = **5** (S, A, B, K, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.15 [MSQ, 2M · Informed Search] Answer: C (C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	C	E	G	H	I	S
$h(n)$	5	2	8	0	13	1	15
$h^*(n)$	6	3	9	0	14	6	16
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $E \rightarrow I$ ($8 > 3 + 1$), $A \rightarrow I$ ($5 > 3 + 1$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.16 [MSQ, 2M · Predicate Logic] Answer: B, C (B: $\exists x \forall y \neg R(x, y)$; C: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,4), (2,3), (2,4), (4,2), (4,3)\}$.

(A) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 3 has no edge to a different element.

(B) $\exists x \forall y \neg R(x, y)$: **TRUE** — 3 has no outgoing edge.

(C) $\exists x \forall y \neg R(y, x)$: **TRUE** — 1 has no incoming edge.

(D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(4,3)$ holds but $R(3,4)$ does not.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MCQ, 2M · Propositional Logic] Answer: B (B: If a clause set is unsatisfiable, the resolution closure of the set contains the empty clause)

The ground resolution theorem: a clause set is unsatisfiable iff its resolution closure contains $\square$. So B is correct and C is false. D would contradict soundness. A is false: resolution is only refutation-complete; e.g. from $\{p\}$ it cannot derive the entailed clause $p \vee q$ directly.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $\neg(\neg(P \rightarrow P) \vee \neg(Q \wedge P))$; C: $(Q \wedge P) \wedge (P \rightarrow P)$)

Compare truth tables with $F = (P \rightarrow P) \wedge (Q \wedge P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg(P \rightarrow P) \vee \neg(Q \wedge P))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $\neg(\neg(P \rightarrow P) \wedge \neg(Q \wedge P))$ — differs at $P=T, Q=F$ ($F = F$, option $= T$) $\Rightarrow$ not equivalent.

(C) $(Q \wedge P) \wedge (P \rightarrow P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(D) $(P \rightarrow P) \vee (Q \wedge P)$ — differs at $P=T, Q=F$ ($F = F$, option $= T$) $\Rightarrow$ not equivalent.

Q.19 [MSQ, 2M · Propositional Logic] Answer: A, B, D

(A) True: forward chaining with premise counters runs in linear time. (B) True: this is the definition. (C) False: q and r are both positive. (D) True: if the clashing positive literal comes from C_1 (≤ 1 positive) and the negative from C_2 , the resolvent contains C_1 's other literals (all negative) plus C_2 's remaining literals (≤ 1 positive), so it has at most one positive literal.

Q.20 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $(\neg R \vee S)$; C: $(\neg P \vee \neg R \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg R \vee S)$ is derivable:

1. resolve $(\neg P \vee \neg Q \vee \neg R)$ and $(Q \vee S)$ on $Q \Rightarrow (\neg P \vee \neg R \vee S)$

2. resolve $(P \vee S)$ and $(\neg P \vee \neg R \vee S)$ on $P \Rightarrow (\neg R \vee S)$

(B) $(\neg Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $(\neg P \vee \neg R \vee S)$ is derivable:

1. resolve $(\neg P \vee \neg Q \vee \neg R)$ and $(Q \vee S)$ on $Q \Rightarrow (\neg P \vee \neg R \vee S)$

(D) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

Q.21 [MCQ, 2M · Exact Inference] Answer: B (B: $f(B, C)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, C)$, $f(B, C, D)$, $f(B, D, E)$, $f(B, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(B, D, E)$, $f(B, E, F)$	$\{B, D, E, F\}$	$f(B, D, F)$
D	$f(B, C, D)$, $f(B, D, F)$	$\{B, C, D, F\}$	$f(B, C, F)$
A	$f(A)$, $f(A, B)$, $f(A, C)$	$\{A, B, C\}$	$f(B, C)$
F	$f(B, C, F)$	$\{B, C, F\}$	$f(B, C)$
C	$f(B, C)$, $f(B, C)$	$\{B, C\}$	$f(B)$

The largest factor produced has **3** variable(s) (created when eliminating E); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over B and normalise.

Q.22 [MSQ, 2M · Variable Elimination] Answer: A, C (A: $A \rightarrow C, B \rightarrow C, C \rightarrow D, C \rightarrow E$; C: $A \rightarrow B, B \rightarrow C, D \rightarrow C, D \rightarrow E$)

A polytree has no undirected cycle. (A) Underlying undirected graph A-C, B-C, C-D, C-E is a tree: polytree.

(B) A-B-D-C-A is an undirected cycle: multiply connected. (C) A-B-C-D-E is a simple path, no cycle: polytree

(C has two parents, which is allowed). (D) A-B-C-A forms a triangle: not a polytree.

Q.23 [MSQ, 2M · Conditional Independence] Answer: A, C (A: $D \perp E \mid A, B$; C: $E \perp D \mid A, B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $D \perp E \mid A, B$: **TRUE** — $D \leftarrow A \rightarrow B \rightarrow C \rightarrow E$ (blocked: observed fork node A); $D \leftarrow A \rightarrow C \rightarrow E$ (blocked: observed fork node A); $D \leftarrow A \rightarrow E$ (blocked: observed fork node A); $D \leftarrow B \leftarrow A \rightarrow C \rightarrow E$ (blocked: observed chain node B); $D \leftarrow B \leftarrow A \rightarrow E$ (blocked: observed chain node B); $D \leftarrow B \rightarrow C \leftarrow A \rightarrow E$ (blocked: observed fork node B) (+8 more paths, all blocked).

(B) $G \perp C \mid B$: **FALSE** — the path $G \leftarrow A \rightarrow C$ is active given $\{B\}$.

(C) $E \perp D \mid A, B$: **TRUE** — $E \leftarrow A \rightarrow B \rightarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow B \rightarrow F \leftarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow C \leftarrow B \rightarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow C \leftarrow B \rightarrow F \leftarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow G \leftarrow D$ (blocked: observed fork node A) (+8 more paths, all blocked).

(D) $G \perp C \mid F$: **FALSE** — the path $G \leftarrow A \rightarrow B \rightarrow C$ is active given $\{F\}$.

Q.24 [NAT, 2M · Exact Inference] **Answer: 0.048 to 0.052**

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution} \mid \text{Smoker}) \cdot P(\text{Cancer} \mid \text{Smoker}, \text{Pollution}) \cdot P(\text{XRay} \mid \text{Cancer})$.

Inference by enumeration: $P(\text{XRay} \mid e) = \alpha \sum_{\text{hidden}} P(\text{XRay}, e, \text{hidden})$.

Hidden: Sm, Po	XR	product of CPT entries (Sm · Po · Ca · XR)	value
T, T	T	$0.2 \times 0.8 \times 0.6 \times 0.05$	0.0048
T, F	T	$0.2 \times 0.2 \times 0.25 \times 0.05$	0.0005
F, T	T	$0.8 \times 0.25 \times 0.4 \times 0.05$	0.004
F, F	T	$0.8 \times 0.75 \times 0.2 \times 0.05$	0.006
T, T	F	$0.2 \times 0.8 \times 0.6 \times 0.95$	0.0912
T, F	F	$0.2 \times 0.2 \times 0.25 \times 0.95$	0.0095
F, T	F	$0.8 \times 0.25 \times 0.4 \times 0.95$	0.076
F, F	F	$0.8 \times 0.75 \times 0.2 \times 0.95$	0.114

$P(\text{XRay}=\text{T}, e) = 0.0153$; $P(\text{XRay}=\text{F}, e) = 0.2907$; $P(e) = 0.306$.

$P(\text{XRay}=\text{T} \mid e) = 0.0153 / 0.306 = \mathbf{0.05} \approx 0.050$.

Q.25 [MCQ, 2M · Variable Elimination] **Answer: A** (A: A, B, C, D)

Order A, B, C, D: eliminating A combines $P(A)$, $P(B \mid A)$, $P(C \mid A) \rightarrow f(B, C)$; B combines $f(B, C)$, $P(D \mid B, C) \rightarrow f(C, D)$; C $\rightarrow f(D)$; D with $P(E \mid D) \rightarrow f(E)$. Max 2 variables. D first: $P(D \mid B, C)$ and $P(E \mid D) \rightarrow f(B, C, E)$ (3 variables). B first: $P(B \mid A)$, $P(D \mid B, C) \rightarrow f(A, C, D)$ (3 variables); C first similarly gives $f(A, B, D)$. So only (A) keeps all factors at two variables or fewer.

Mock Test 021

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 6 / 14	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

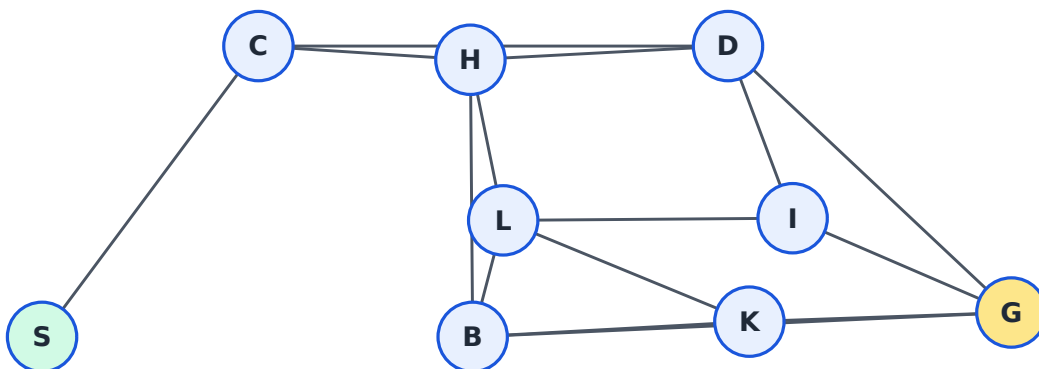
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 2$ (so $b^d = 4$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, C, H, D, L, B, I, G
- ☐ (B) S, C, H, D, G
- ☐ (C) S, C, D, G
- ☐ (D) S, C, D, H, G

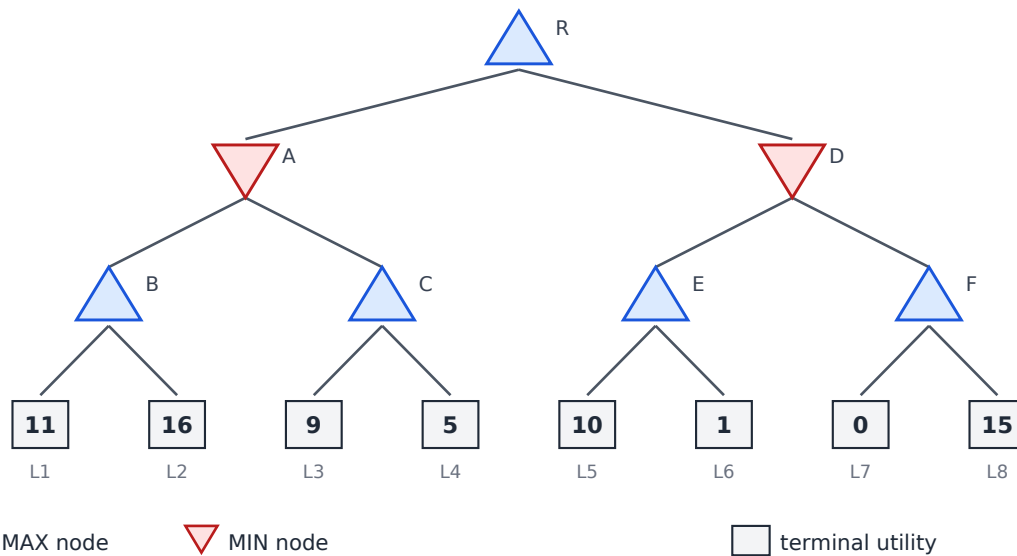
Q.3 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search

With perfect move ordering, the time complexity of alpha-beta search to depth m with branching factor b is

- ☐ (A) $O(b^m)$
- ☐ (B) $O(b^{m/2})$
- ☐ (C) $O(b^{3m/4})$
- ☐ (D) $O(bm)$

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

$\neg(p \rightarrow q)$ is logically equivalent to

- ☐ (A) $p \wedge \neg q$
☐ (B) $\neg p \wedge q$
☐ (C) $\neg p \rightarrow \neg q$
☐ (D) $\neg p \vee q$

Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following inference rules are sound in propositional logic?

- ☐ (A) Modus ponens: from $\alpha \rightarrow \beta$ and α infer β
☐ (B) And-elimination: from $\alpha \wedge \beta$ infer α
☐ (C) Abduction: from $\alpha \rightarrow \beta$ and β infer α
☐ (D) Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((R \wedge R) \vee (R \leftrightarrow \neg Q)) \wedge (\neg R \rightarrow (P \vee P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$ and let f be a unary function symbol. How many interpretations of f satisfy $\forall x f(f(x)) = x$?

Answer (numerical): _____

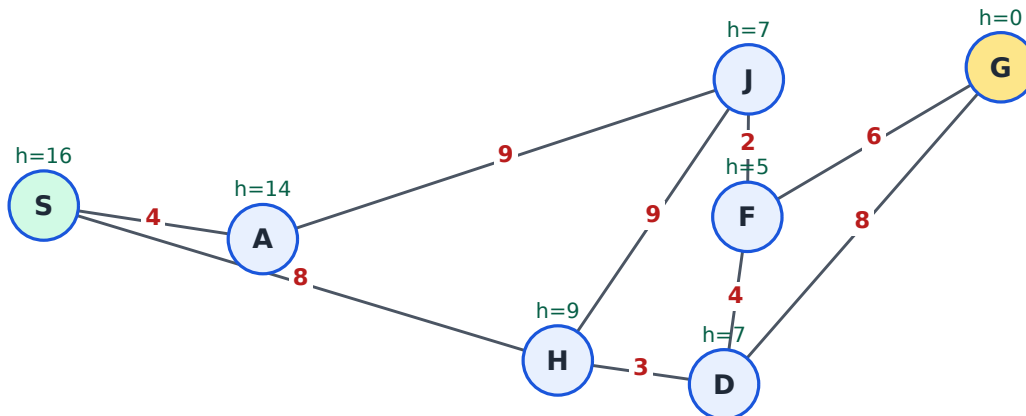
Q.10 [NAT · 1 mark · no negative marking] Probability for AI

A domain has four Boolean random variables and one random variable with 3 possible values. How many independent numbers are needed to specify an arbitrary full joint distribution over these five variables?

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(D)$ overestimates the true cost from D to G.
- ☐ (B) h is admissible.
- ☐ (C) h is consistent (monotone).
- ☐ (D) The edge D-G violates the consistency condition (in at least one direction).

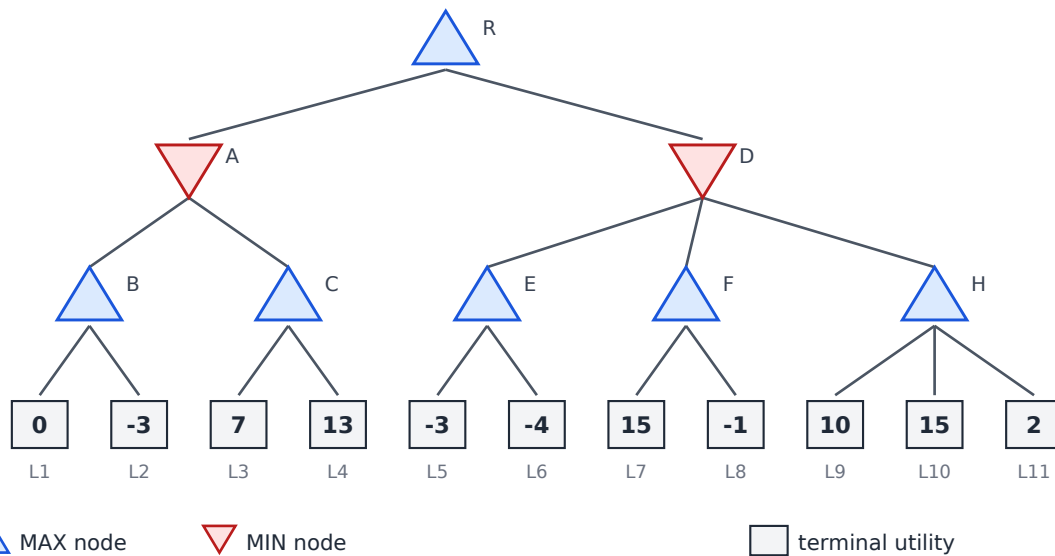
Q.12 [MCQ · 2 marks · negative: -2/3] Informed Search

$h_1, \dots, h_m$ are admissible heuristics for a problem. Which composite heuristic is guaranteed to be admissible and to dominate each h_i ?

- ☐ (A) $h(n) = h_1(n) + \dots + h_m(n)$
- ☐ (B) $h(n) = \max(h_1(n), \dots, h_m(n))$
- ☐ (C) $h(n) = \min(h_1(n), \dots, h_m(n))$
- ☐ (D) $h(n) = m \cdot h_1(n)$

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

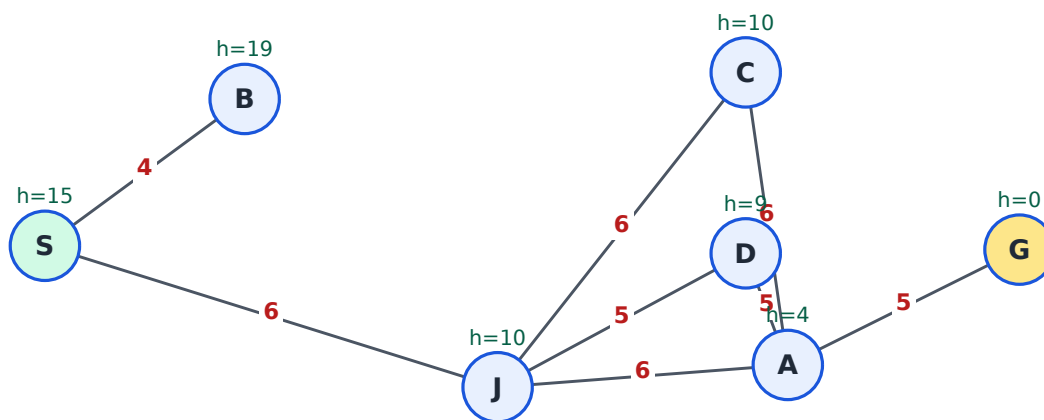
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node B is expanded.
- ☐ (B) Node D is expanded.
- ☐ (C) The path returned has cost 17.
- ☐ (D) Node C is expanded.

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $Q \wedge \neg S$
- (2) $(Q \vee \neg S) \wedge \neg S$
- (3) $R \wedge P$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg S \vee (S \rightarrow R)$
- ☐ (B) $\neg\neg S \wedge \neg Q$
- ☐ (C) $P \vee Q$
- ☐ (D) $\neg P \wedge Q$

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic

How many truth assignments to $x_1, \dots, x_5$ satisfy the 2-CNF $(x_1 \vee x_2) \wedge (x_2 \vee x_3) \wedge (x_3 \vee x_4) \wedge (x_4 \vee x_5)$?

Answer (numerical): _____

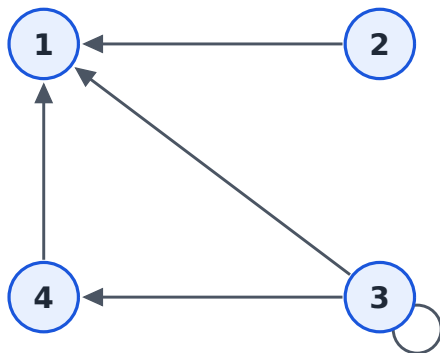
Q.17 [NAT · 2 marks · no negative marking] Propositional Logic

How many of the 16 truth assignments to p, q, r, s satisfy exactly two of the four clauses $p \vee q, \neg p \vee r, \neg q \vee s, \neg r \vee \neg s$? (Enter the number of assignments, not clauses.)

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

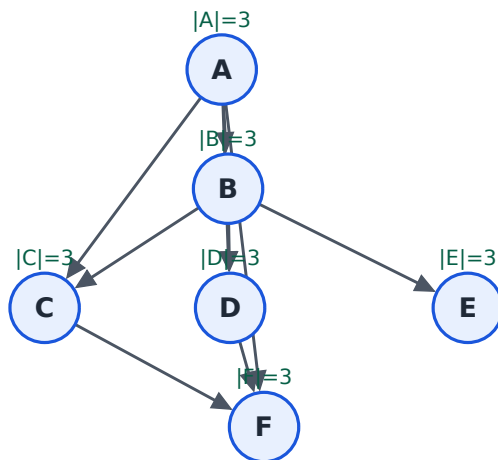
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
- ☐ (B) $\exists x R(x, x)$
- ☐ (C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
- ☐ (D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$

Q.19 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

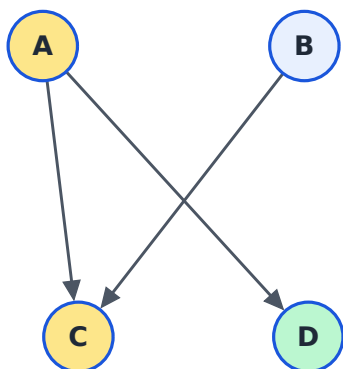
The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(D = T \mid A = T, C = F)$ (round off to 3 decimal places).



P(A=T)
0.2

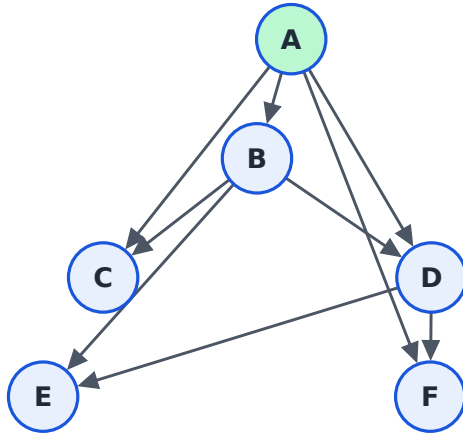
P(B=T)	A	B	P(C=T ...)
0.8	T	T	0.2
	T	F	0.9
	F	T	0.2
	F	F	0.9

A	P(D=T ...)
T	0.1
F	0.95

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{A})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, C, E, F, B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

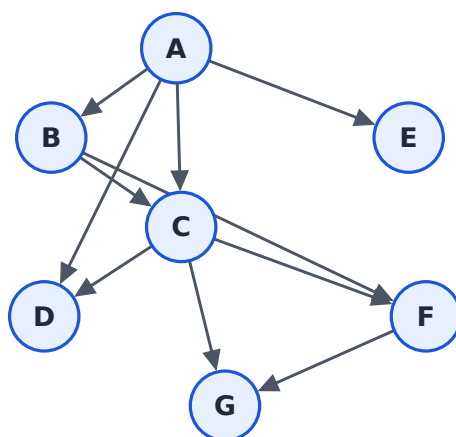
Q.22 [NAT · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. Compute $P(b | j, m)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

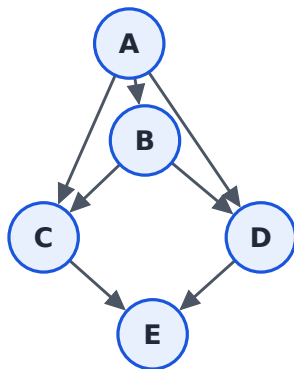
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp B \mid E$
- ☐ (B) $E \perp F$
- ☐ (C) $C \perp E \mid A, G$
- ☐ (D) $G \perp E \mid A$

Q.24 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.19, 0.83, 0.06, 0.98, 0.73)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



A	P(A=T)
T	0.4
F	

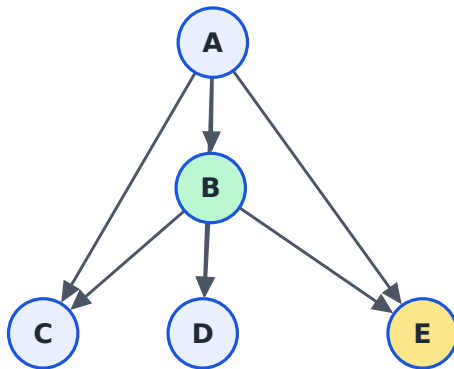
A	P(B=T ...)	A	B	P(C=T ...)
T	0.9	T	T	0.5
F	0.25	T	F	0.6
		F	T	0.1
		F	F	0.6

A	B	P(D=T ...)	C	D	P(E=T ...)
T	T	0.3	T	T	0.8
T	F	0.25	T	F	0.1
F	T	0.9	F	T	0.3
F	F	0.75	F	F	0.4

- ☐ (A) A=T, B=T, C=T, D=F, E=F
☐ (B) A=F, B=T, C=T, D=F, E=F
☐ (C) A=F, B=T, C=F, D=T, E=T
☐ (D) A=T, B=F, C=T, D=F, E=F

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{E} = \mathbf{T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(B = T \mid E = T)}$ (round off to 3 decimal places).



A	P(A=T)
T	0.4
F	

A	P(B=T ...)	A	B	P(C=T ...)
T	0.8	T	T	0.2
F	0.25	T	F	0.9
		F	T	0.25
		F	F	0.25

A	B	P(D=T ...)	A	B	P(E=T ...)
T	T	0.3	T	T	0.4
T	F	0.95	T	F	0.4
F	T	0.1	F	T	0.05
F	F	0.75	F	F	0.8

#	A	B	C	D
1	T	T	F	F
2	T	T	F	F
3	F	F	F	T
4	T	T	F	F
5	T	T	F	T
6	F	F	T	T

Answer (numerical): _____

Answer Key — Mock Test 021

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	3	14	MSQ	2	Informed Search	C
2	MCQ	1	Uninformed Search	D	15	MSQ	2	Propositional Logic	A, C
3	MCQ	1	Adversarial Search	B	16	NAT	2	Propositional Logic	13
4	NAT	1	Adversarial Search	10	17	NAT	2	Propositional Logic	2
5	NAT	1	Predicate Logic	65	18	MSQ	2	Predicate Logic	A, B, D
6	MCQ	1	Propositional Logic	A	19	NAT	2	Bayesian Networks	624
7	MSQ	1	Propositional Logic	A, B, D	20	NAT	2	Exact Inference	0.098 to 0.102
8	NAT	1	Propositional Logic	3	21	NAT	2	Exact Inference	4
9	NAT	1	Predicate Logic	4	22	NAT	2	Bayesian Networks	0.555 to 0.557
10	NAT	1	Probability for AI	47	23	MSQ	2	Conditional Independence	C, D
11	MSQ	2	Informed Search	B, C	24	MCQ	2	Sampling	A
12	MCQ	2	Informed Search	B	25	NAT	2	Sampling	0.498 to 0.502
13	NAT	2	Adversarial Search	5					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 3

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 2^{\lceil 1 \rceil} + 2^{\lfloor 1 \rfloor} - 1 = 2 + 2 - 1 = 3$, versus 4 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.2 [MCQ, 1M · Uninformed Search] Answer: D (D: S, C, D, H, G)

Adjacency lists (alphabetical): B: {G, H, K, L}; C: {D, H, S}; D: {C, G, H, I}; G: {B, D, I, K}; H: {B, C, D, L}; I: {D, G, L}; K: {B, G, L}; L: {B, H, I, K}; S: {C}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[C]
2	C	[D, H]
3	D	[H, G, I]
4	H	[G, I, B, L]
5	G	[I, B, L] ← goal popped, stop

Expansion order: **S, C, D, H, G**. The path returned is $S \rightarrow C \rightarrow D \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

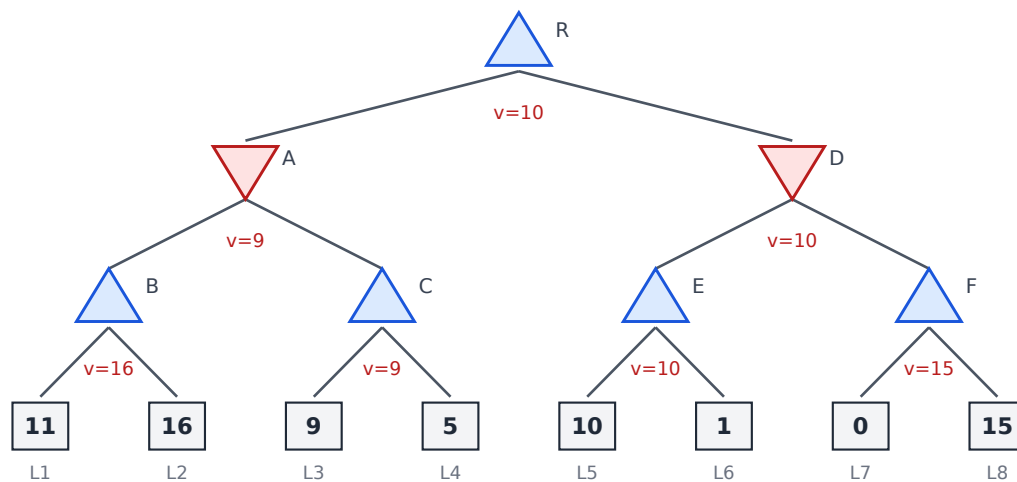
Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [MCQ, 1M · Adversarial Search] Answer: B (B: $O(b^{m/2})$)

Perfect ordering examines about $b^{\lceil m/2 \rceil} + b^{\lfloor m/2 \rfloor} - 1$ leaves, which is **$O(b^{m/2})$** . $O(b^m)$ is minimax, or the worst ordering (A). $O(b^{3m/4})$ is the random-ordering estimate (C). $O(bm)$ is minimax's space, not its time (D).

Q.4 [NAT, 1M · Adversarial Search] Answer: 10

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 16$, $C = 9$, $A = 9$, $E = 10$, $F = 15$, $D = 10$. Root value = **10**; MAX's optimal first move is towards D.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 65**

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 4$ this gives **65** (verified by enumerating all interpretations).

Q.6 [MCQ, 1M · Propositional Logic] **Answer: A** ($A: p \wedge \neg q$)

$p \rightarrow q \equiv \neg p \vee q$, so $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q$ by De Morgan's law. It is true only in the row $p=T, q=F$, which is exactly the row where $p \rightarrow q$ is false. $\neg p \wedge q$ is true only at $p=F, q=T$; $\neg p \rightarrow \neg q$ is the inverse; $\neg p \vee q$ is $p \rightarrow q$ itself.

Q.7 [MSQ, 1M · Propositional Logic] **Answer: A, B, D** (A : Modus ponens: from $\alpha \rightarrow \beta$ and α infer β ; B :

And-elimination: from $\alpha \wedge \beta$ infer α ; D : Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$)

Modus ponens, and-elimination and resolution are truth-preserving: whenever their premises are true, their conclusion is true. For resolution, if β is true then γ must be true, else α must be true. Abduction is unsound: $\alpha=F, \beta=T$ makes both premises true and the conclusion false.

Q.8 [NAT, 1M · Propositional Logic] **Answer: 3**

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \wedge R) \vee (R \leftrightarrow \neg Q)$	$\neg R \rightarrow (P \vee P)$	$((R \wedge R) \vee (R \leftrightarrow \neg Q)) \wedge (\neg R \rightarrow (P \vee P))$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	F	F	F

F has 5 models, so it is false in $8 - 5 = 3$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.9 [NAT, 1M · Predicate Logic] **Answer: 4**

$f(f(x)) = x$ for all x means f is an involution: it is a bijection whose cycles have length 1 or 2. On 3 elements: the identity (1 way) or exactly one transposition with one fixed point (3 ways). Total = 4.

Q.10 [NAT, 1M · Probability for AI] **Answer: 47**

The full joint table has $2^4 \times 3 = 48$ entries. Because the entries must sum to 1, one of them is determined by the rest, giving $48 - 1 = 47$ independent numbers. No independence assumptions are made, so no further reduction is possible.

Q.11 [MSQ, 2M · Informed Search] Answer: B, C (B: h is admissible.; C: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	D	F	G	H	J	S
$h(n)$	14	7	5	0	9	7	16
$h^*(n)$	17	8	6	0	11	8	19
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

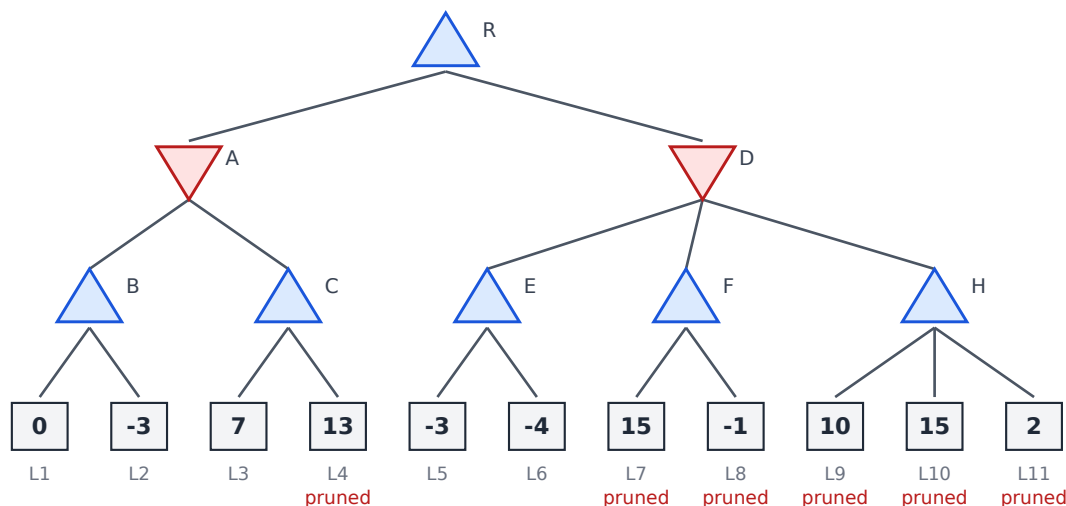
Q.12 [MCQ, 2M · Informed Search] Answer: B (B: $h(n) = \max(h_1(n), \dots, h_m(n))$)

Each $h_i(n) \leq h^*(n)$, so their maximum is also $\leq h^*(n)$, and it is $\geq$ each component by construction. So the **max** is admissible and dominates every h_i . The sum (A) and the multiple (D) can exceed h^* . The min (C) is admissible but is dominated by every h_i , not dominating.

Q.13 [NAT, 2M · Adversarial Search] Answer: 5

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $7 \geq \beta = 0 \Rightarrow$ prune remaining children (leaves L4).
- At MIN node D: value $-3 \leq \alpha = 0 \Rightarrow$ prune remaining children (leaves L7, L8, L9, L10, L11).



Leaves evaluated (5): L1, L2, L3, L5, L6. Leaves pruned (6): L4, L7, L8, L9, L10, L11. Root value = 0 (same as plain minimax, 0 — pruning never changes the root value).

Q.14 [MSQ, 2M · Informed Search] Answer: C (C: The path returned has cost 17.)

Heuristic values: $h(A)=4$, $h(B)=19$, $h(C)=10$, $h(D)=9$, $h(G)=0$, $h(J)=10$, $h(S)=15$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	J(16), B(23)
2	J (16)	A(16), D(20), C(22), B(23)
3	A (16)	G(17), D(20), C(22), B(23)
4	G (17)	D(20), C(22), B(23) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **J** $\rightarrow$ **A** $\rightarrow$ **G** with cost **17**; nodes expanded = **4** (S, J, A, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 17$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $\neg S \vee (S \rightarrow R)$; C: $P \vee Q$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 16 assignments, KB is true in exactly 1 of them:

P	Q	R	S	KB
T	T	T	F	T

(A) $\neg S \vee (S \rightarrow R)$ is true in all models of $KB \Rightarrow$ **entailed**.

(B) $\neg \neg S \wedge \neg Q$ is false in the KB -model $P=T, Q=T, R=T, S=F \Rightarrow$ not entailed.

(C) $P \vee Q$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $\neg P \wedge Q$ is false in the KB-model $P=T, Q=T, R=T, S=F \Rightarrow$ not entailed.

Q.16 [NAT, 2M · Propositional Logic] **Answer: 13**

The formula says no two consecutive variables are both F. Let a_n be the number of such strings of length n . A valid string either ends in T (preceded by any valid string of length $n-1$) or ends in F, in which case the previous symbol must be T (preceded by any valid string of length $n-2$). So $a_n = a_{n-1} + a_{n-2}$ with $a_1 = 2, a_2 = 3$. Then $a_3 = 5, a_4 = 8, a_5 = 13$. Enumeration of all 32 assignments confirms 13.

Q.17 [NAT, 2M · Propositional Logic] **Answer: 2**

Each clause, being a disjunction, is false only in specific situations: C1 false iff $p=q=F$; C2 iff $p=T, r=F$; C3 iff $q=T, s=F$; C4 iff $r=s=T$. With 4 clauses, satisfying exactly two means falsifying exactly two. C1 is incompatible with C2 and C3 (they need $p=T$ or $q=T$). C1 & C4: $p=q=F, r=s=T$ — then C2, C3 hold $\rightarrow$ 1 assignment. C2 & C3: $p=T, r=F, q=T, s=F$ — C4 holds, C1 holds $\rightarrow$ 1 assignment. C2 & C4 needs $r=F$ and $r=T$; C3 & C4 needs $s=F$ and $s=T$ — impossible. Total = 2 (confirmed by enumeration).

Q.18 [MSQ, 2M · Predicate Logic] **Answer: A, B, D** (A: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$; B: $\exists x R(x, x)$; D: $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$)

$R = \{(2,1), (3,1), (3,3), (3,4), (4,1)\}$.

(A) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 3.

(C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,1)$ holds but $R(1,2)$ does not.

(D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **TRUE** — all two-step paths are short-cut.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [NAT, 2M · Bayesian Networks] **Answer: 624**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (3-1) \times 3 + C: (3-1) \times 3 \times 3 + D: (3-1) \times 3 \times 3 + E: (3-1) \times 3 + F: (3-1) \times 3 \times 3 \times 3 = 104$.

Full joint table: $\prod |X| - 1 = 729 - 1 = 728$; saving = 624.

Q.20 [NAT, 2M · Exact Inference] **Answer: 0.098 to 0.102**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B) \cdot P(C | A, B) \cdot P(D | A)$.

Inference by enumeration: $P(D | e) = \alpha \sum_{\text{hidden}} P(D, e, \text{hidden})$.

Hidden: B	D	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T	T	$0.2 \times 0.8 \times 0.8 \times 0.1$	0.0128
F	T	$0.2 \times 0.2 \times 0.1 \times 0.1$	0.0004
T	F	$0.2 \times 0.8 \times 0.8 \times 0.9$	0.1152
F	F	$0.2 \times 0.2 \times 0.1 \times 0.9$	0.0036

$P(D=T, e) = 0.0132$; $P(D=F, e) = 0.1188$; $P(e) = 0.132$.

$P(D=T | e) = 0.0132 / 0.132 = 0.1 \approx 0.100$.

Q.21 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(A, B, C), f(A, B, D), f(B, D, E), f(A, D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(A, B, D), f(B, D, E), f(A, D, F)$	$\{A, B, D, E, F\}$	$f(A, B, E, F)$
C	$f(A, B, C)$	$\{A, B, C\}$	$f(A, B)$
E	$f(A, B, E, F)$	$\{A, B, E, F\}$	$f(A, B, F)$
F	$f(A, B, F)$	$\{A, B, F\}$	$f(A, B)$
B	$f(A, B), f(A, B), f(A, B)$	$\{A, B\}$	$f(A)$

The largest factor produced has **4** variable(s) (created when eliminating D); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over A and normalise.

Q.22 [NAT, 2M · Bayesian Networks] Answer: 0.555 to 0.557

$P(B | j, m) = \alpha P(B) \sum_e P(e) \sum_a P(a | B, e) P(j | a) P(m | a)$. Let $g(a) = P(j | a)P(m | a)$: $g(a) = 0.56$, $g(\neg a) = 0.001$. For b : e : $0.95 \times 0.56 + 0.05 \times 0.001 = 0.53205$; $\neg e$: $0.90 \times 0.56 + 0.10 \times 0.001 = 0.5041$; weighted: $0.01 \times 0.53205 + 0.99 \times 0.5041 = 0.504381$; times $P(b) = 0.02 \rightarrow 0.0100876$. For $\neg b$: e : $0.30 \times 0.56 + 0.70 \times 0.001 = 0.1687$; $\neg e$: $0.01 \times 0.56 + 0.99 \times 0.001 = 0.006589$; weighted: $0.01 \times 0.1687 + 0.99 \times 0.006589 = 0.0082101$; times $0.98 \rightarrow 0.0080459$. $P(b | j, m) = 0.0100876 / (0.0100876 + 0.0080459) \approx 0.556$.

Q.23 [MSQ, 2M · Conditional Independence] Answer: C, D (C: $C \perp E | A, G$; D: $G \perp E | A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $D \perp B | E$: **FALSE** — the path $D \leftarrow A \rightarrow B$ is active given $\{E\}$.

(B) $E \perp F$: **FALSE** — the path $E \leftarrow A \rightarrow B \rightarrow C \rightarrow F$ is active given $\{\}$.

(C) $C \perp E | A, G$: **TRUE** — $C \leftarrow A \rightarrow E$ (blocked: observed fork node A); $C \leftarrow B \leftarrow A \rightarrow E$ (blocked: observed fork node A); $C \rightarrow D \leftarrow A \rightarrow E$ (blocked: collider D with neither it nor a descendant observed); $C \rightarrow F \leftarrow B \leftarrow A \rightarrow E$ (blocked: observed fork node A); $C \rightarrow G \leftarrow F \leftarrow B \leftarrow A \rightarrow E$ (blocked: observed fork node A).

(D) $G \perp E | A$: **TRUE** — $G \leftarrow C \leftarrow A \rightarrow E$ (blocked: observed fork node A); $G \leftarrow C \leftarrow B \leftarrow A \rightarrow E$ (blocked: observed fork node A); $G \leftarrow C \rightarrow D \leftarrow A \rightarrow E$ (blocked: collider D with neither it nor a descendant observed); $G \leftarrow C \rightarrow F \leftarrow B \leftarrow A \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow F \leftarrow B \leftarrow A \rightarrow E$ (blocked: observed fork node A); $G \leftarrow F \leftarrow B \rightarrow C \leftarrow A \rightarrow E$ (blocked: collider C with neither it nor a descendant observed) (+4 more paths, all blocked).

Q.24 [MCQ, 2M · Sampling] Answer: A (A: $A=T$, $B=T$, $C=T$, $D=F$, $E=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.4$; $u = 0.19 < 0.4 \Rightarrow A = T$
- B: $P(B=T | A=T) = 0.9$; $u = 0.83 < 0.9 \Rightarrow B = T$
- C: $P(C=T | A=T, B=T) = 0.5$; $u = 0.06 < 0.5 \Rightarrow C = T$
- D: $P(D=T | A=T, B=T) = 0.3$; $u = 0.98 \geq 0.3 \Rightarrow D = F$
- E: $P(E=T | C=T, D=F) = 0.1$; $u = 0.73 \geq 0.1 \Rightarrow E = F$

Sample: (**A=T, B=T, C=T, D=F, E=F**).

Q.25 [NAT, 2M · Sampling] Answer: 0.498 to 0.502

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	B
1	$P(E=T A=T, B=T) = 0.4$	0.4	T
2	$P(E=T A=T, B=T) = 0.4$	0.4	T
3	$P(E=T A=F, B=F) = 0.8$	0.8	F
4	$P(E=T A=T, B=T) = 0.4$	0.4	T
5	$P(E=T A=T, B=T) = 0.4$	0.4	T
6	$P(E=T A=F, B=F) = 0.8$	0.8	F

$\sum w (\text{all}) = 3.2$; $\sum w (B = T) = 1.6$.

Estimate $P(B=T | E = T) = 1.6 / 3.2 = \mathbf{0.5}$.

Mock Test 022

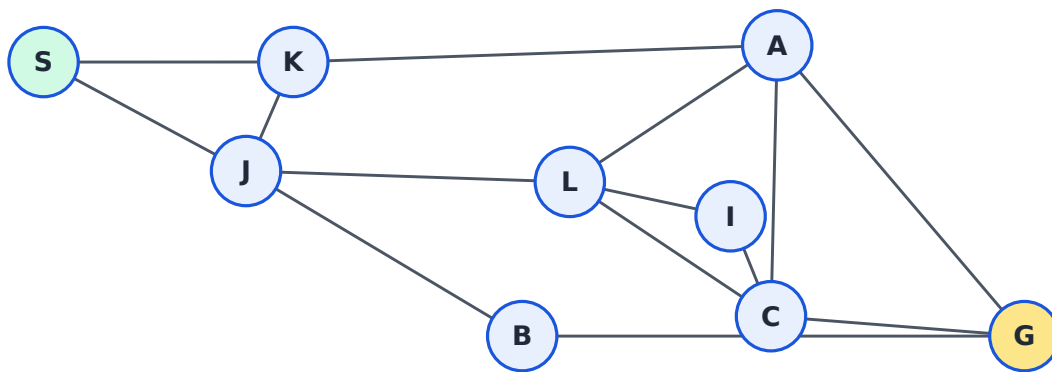
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 8 / 10	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

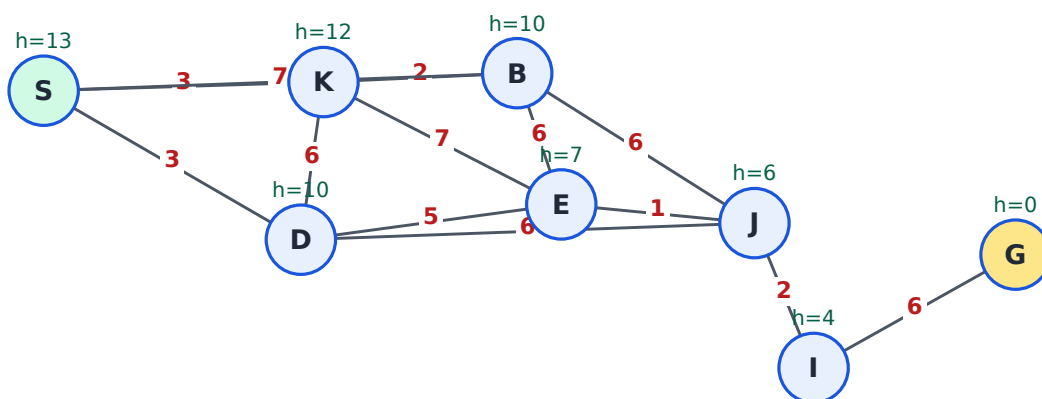
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, J, B, G
- ☐ (B) S, K, J, A, L, B, G
- ☐ (C) S, J, K, B, L, A, G
- ☐ (D) S, J, B, K, L, A, G

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

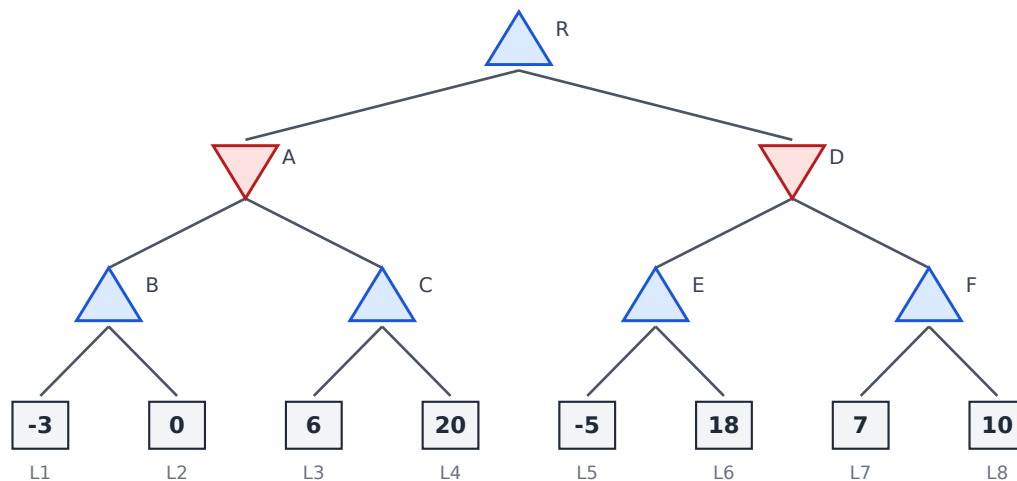
Q.3 [MCQ · 1 mark · negative: -1/3] Informed Search

h_1 and h_2 are both admissible, and $h_2(n) \geq h_1(n)$ for every n . Which statement is correct?

- ☐ (A) h_1 dominates h_2 , so A^* with h_1 is preferable
- ☐ (B) h_2 dominates h_1 , and A^* with h_2 never expands more nodes than A^* with h_1 (apart from ties at $f = C^*$)
- ☐ (C) A^* with h_2 may return a more expensive solution than A^* with h_1
- ☐ (D) Dominance requires $h_2(n) = h_1(n) + c$ for a constant c

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



MAX node MIN node

terminal utility

- ☐ (A) Move to D; root value 20
- ☐ (B) Move to D; root value 13
- ☐ (C) Move to A; root value 0
- ☐ (D) Move to D; root value 10

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following correctly express 'Not every graph is connected'?

- ☐ (A) $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$
- ☐ (B) $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$
- ☐ (C) $\forall x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$
- ☐ (D) $\exists x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over a, b, c does the CNF $(a \vee b \vee c) \wedge (\neg a \vee \neg b) \wedge (\neg b \vee \neg c) \wedge (\neg a \vee \neg c)$ have?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

F = $((R \rightarrow R) \wedge \neg Q) \wedge \neg \neg P$?

Answer (numerical): _____

Q.8 [MCQ · 1 mark · negative: -1/3] Sampling

A discrete variable Weather has distribution (sunny: 0.2, cloudy: 0.5, rainy: 0.3) (in that order).

Using inverse-CDF sampling with cumulative intervals in the same order, which value is produced by the uniform random number $u = 0.65$?

- ☐ (A) cloudy
☐ (B) sunny
☐ (C) rainy
☐ (D) Cannot be determined from u alone

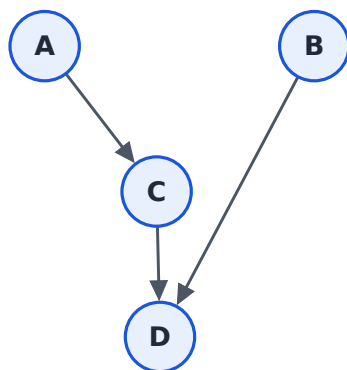
Q.9 [NAT · 1 mark · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Likelihood weighting is used with evidence $S = \text{true}$, $W = \text{true}$. A sample is generated with $C = \text{false}$ and $R = \text{false}$. What is the weight of this sample? (Round to 2 decimal places.)

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=F, B=T, C=T, D=F)**? (round off to 4 decimal places)



P(A=T)
0.3

P(B=T)	A	P(C=T ...)
0.25	T	0.3
	F	0.7

B	C	P(D=T ...)
T	T	0.05
T	F	0.8
F	T	0.8
F	F	0.5

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search

A two-ply game tree has a MAX root with four MIN children, whose leaves are listed left to right: $\text{MIN}_1: (10, 9, 14)$; $\text{MIN}_2: (11, 8, 20)$; $\text{MIN}_3: (2, 15, 30)$; $\text{MIN}_4: (12, 13, 7)$. Alpha-beta pruning visits children left to right. How many leaves are **pruned** (never evaluated)?

Answer (numerical): _____

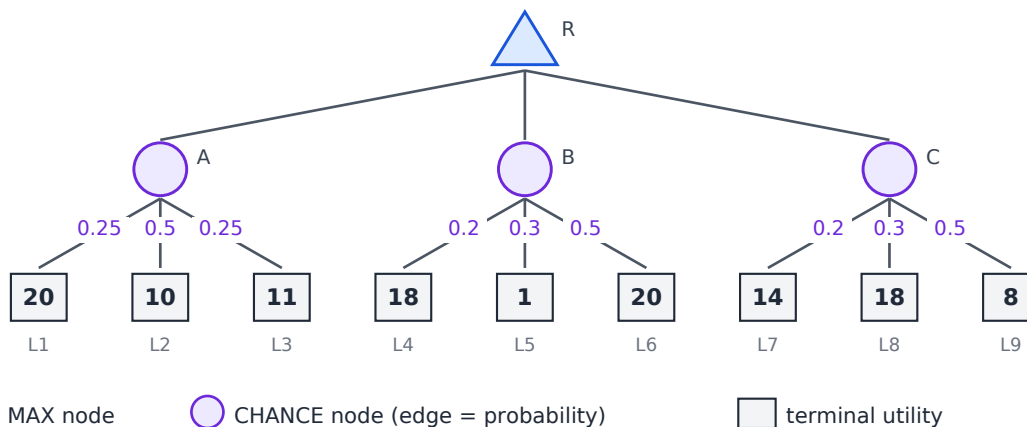
Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). Uniform-cost search (graph search, goal test when a node is selected for expansion) is run from S. Which path does it return?

- ☐ (A) $S \rightarrow B \rightarrow G$ (cost 10)
☐ (B) $S \rightarrow A \rightarrow C \rightarrow G$ (cost 9)
☐ (C) $S \rightarrow B \rightarrow C \rightarrow G$ (cost 8)
☐ (D) $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$ (cost 7)

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

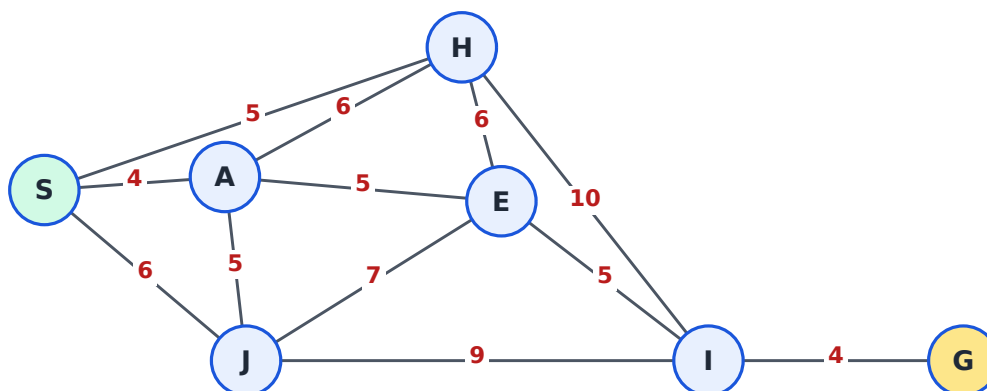
In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



Answer (numerical): _____

Q.14 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

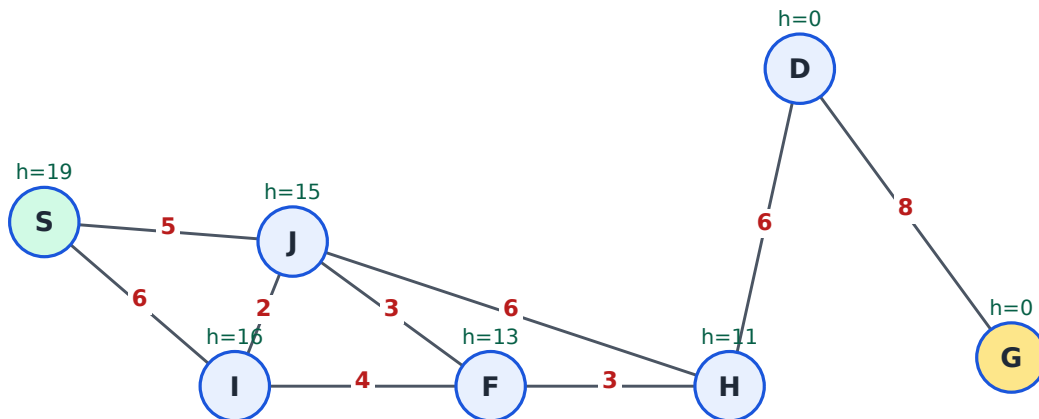
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow H \rightarrow I \rightarrow G$
☐ (B) $S \rightarrow H \rightarrow E \rightarrow I \rightarrow G$
☐ (C) $S \rightarrow J \rightarrow I \rightarrow G$
☐ (D) $S \rightarrow A \rightarrow E \rightarrow I \rightarrow G$

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

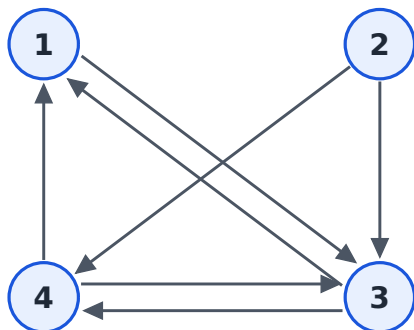
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(F)$ overestimates the true cost from F to G.
☐ (B) The edge J-H violates the consistency condition (in at least one direction).
☐ (C) h is consistent (monotone).
☐ (D) h is admissible.

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x R(x, x)$
☐ (B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **E, C, B** and rules:

R1: $F \wedge K \wedge B \rightarrow D$

R2: $J \wedge K \wedge A \rightarrow H$

R3: $E \wedge K \rightarrow B$

R4: $B \rightarrow F$

R5: $B \wedge A \rightarrow H$

R6: $F \wedge B \rightarrow A$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) A is entailed
- ☐ (B) D is entailed
- ☐ (C) J is entailed
- ☐ (D) K is entailed

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$
- ☐ (B) $(\neg R \rightarrow Q) \vee (Q \rightarrow \neg R)$
- ☐ (C) $P \rightarrow (R \rightarrow P)$
- ☐ (D) $((Q \rightarrow P) \wedge P) \rightarrow Q$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg P \vee \neg Q \vee S)$, C2: (R) , C3: $(P \vee S)$, C4: $(P \vee R)$, C5: (P)

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
- ☐ (B) $(\neg Q \vee S)$
- ☐ (C) $(\neg R)$
- ☐ (D) $(\neg R \vee \neg S)$

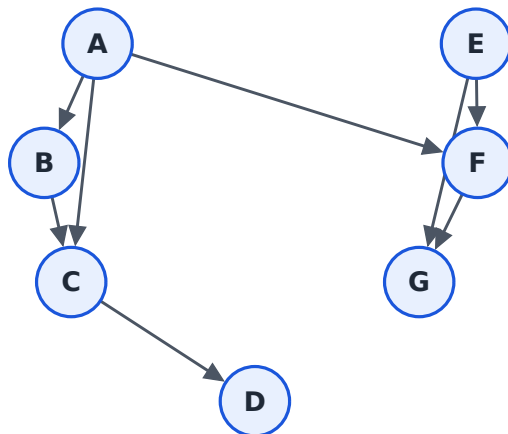
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following sets of propositional formulas are unsatisfiable?

- ☐ (A) $\{p \vee q, \neg p \vee q, p \vee \neg q, \neg p \vee \neg q\}$
- ☐ (B) $\{p \rightarrow q, q \rightarrow r, r \rightarrow \neg p, p\}$
- ☐ (C) $\{p \leftrightarrow \neg q, q \leftrightarrow \neg r, r \leftrightarrow \neg p\}$
- ☐ (D) $\{p \vee q \vee r, \neg p \vee \neg q, \neg q \vee \neg r, \neg p \vee \neg r\}$

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

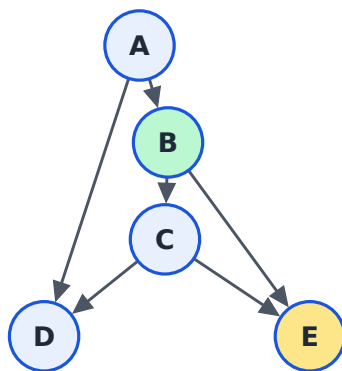
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp G$
☐ (B) $F \perp C \mid A, G$
☐ (C) $E \perp D$
☐ (D) $G \perp A \mid E, F$

Q.22 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $E=T$. The current state of the chain is $(A=F, B=T, C=F, D=F, E=T)$. The next step resamples **B**. With what probability is B set to T? (round off to 3 decimal places)



P(A=T)
0.7

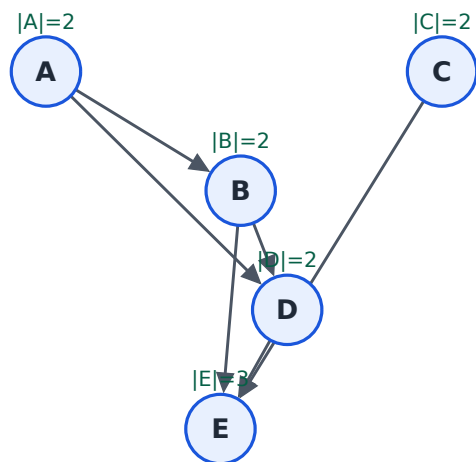
A	P(B=T ...)	B	P(C=T ...)
T	0.4	T	0.1
F	0.95	F	0.95

A	C	P(D=T ...)	B	C	P(E=T ...)
T	T	0.25	T	T	0.3
T	F	0.3	T	F	0.6
F	T	0.75	F	T	0.95
F	F	0.8	F	F	0.05

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

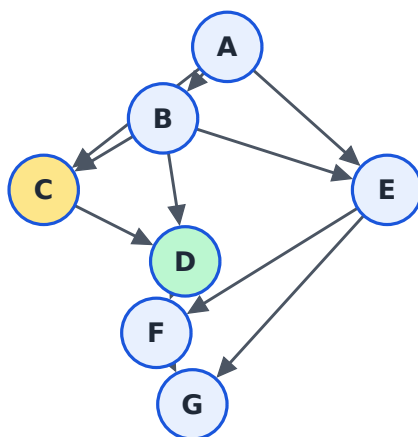
Q.24 [MCQ · 2 marks · negative: -2/3] Probability for AI

The prior odds of hypothesis h against $\neg h$ are 1 : 4. Evidence e has likelihood ratio $P(e | h)/P(e | \neg h) = 6$. What is $P(h | e)$?

- ☐ (A) 0.600
☐ (B) 0.857
☐ (C) 0.250
☐ (D) 0.200

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(D | C = T)$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, G, A, F, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 022

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	C	14	MCQ	2	Uninformed Search	D
2	NAT	1	Informed Search	21	15	MSQ	2	Informed Search	D
3	MCQ	1	Informed Search	B	16	MSQ	2	Predicate Logic	C, D
4	MCQ	1	Adversarial Search	D	17	MSQ	2	Propositional Logic	A
5	MSQ	1	Predicate Logic	A, B	18	MSQ	2	Propositional Logic	B, C
6	NAT	1	Propositional Logic	3	19	MSQ	2	Propositional Logic	B
7	NAT	1	Propositional Logic	2	20	MSQ	2	Propositional Logic	A, B, C
8	MCQ	1	Sampling	A	21	MSQ	2	Conditional Independence	B, C, D
9	NAT	1	Sampling	0.445 to 0.455	22	NAT	2	Sampling	0.998 to 1.002
10	NAT	1	Sampling	0.1162 to 0.1166	23	NAT	2	Bayesian Networks	23
11	NAT	2	Adversarial Search	3	24	MCQ	2	Probability for AI	A
12	MCQ	2	Uninformed Search	D	25	NAT	2	Exact Inference	3
13	NAT	2	Adversarial Search	13.89 to 13.91					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: C (C: S, J, K, B, L, A, G)

Adjacency lists (alphabetical): A: {C, G, K, L}; B: {G, J}; C: {A, G, I, L}; G: {A, B, C}; I: {C, L}; J: {B, K, L, S}; K: {A, J, S}; L: {A, C, I, J}; S: {J, K}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[J, K]
2	J	[K, B, L]
3	K	[B, L, A]
4	B	[L, A, G]
5	L	[A, G, C, I]
6	A	[G, C, I]
7	G	[C, I] ← goal popped, stop

Expansion order: **S, J, K, B, L, A, G**. The path returned is $S \rightarrow J \rightarrow B \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Informed Search] Answer: 21

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (13)	B(10), D(10), K(12)
2	B (10)	J(6), E(7), D(10), K(12)
3	J (6)	I(4), E(7), D(10), K(12)
4	I (4)	G(0), E(7), D(10), K(12)
5	G (0)	E(7), D(10), K(12) ← goal, stop

Path returned: **S → B → J → I → G**, cost = $7 + 6 + 2 + 6 = 21$.

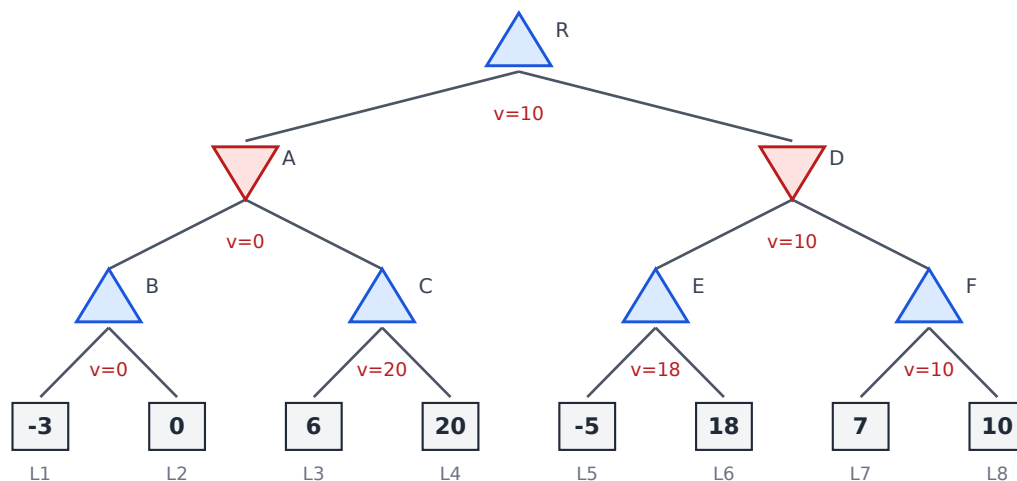
The optimal cost is $h^*(S) = 17$ — greedy search is not optimal here.

Q.3 [MCQ, 1M · Informed Search] Answer: B ($B: h_2$ dominates h_1 , and A^* with h_2 never expands more nodes than A^* with h_1 (apart from ties at $f = C^*$))

h_2 dominates h_1 . With consistent heuristics, A^* expands every node with $f(n) < C^*$, which means $h(n) < C^* - g(n)$. A larger h satisfies this inequality at fewer nodes, so A^* with h_2 expands a subset. (A) reverses the relation. (C) is wrong because both heuristics are admissible, so both runs are optimal. (D) is not the definition of dominance.

Q.4 [MCQ, 1M · Adversarial Search] Answer: D (D: Move to D; root value 10)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 0, C = 20, A = 0, E = 18, F = 10, D = 10. Root value = **10**; MAX's optimal first move is towards D.

Q.5 [MSQ, 1M · Predicate Logic] Answer: A, B (A: $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$; B: $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$)

Option A is the direct translation, and pushing the negation inward gives $\exists x \neg(\text{Graph}(x) \rightarrow \text{Connected}(x)) \equiv \exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$ — option B. Option C says no graph is connected, which is stronger. Option D is true whenever any non-graph exists (vacuous implication), so it does not express the statement.

Q.6 [NAT, 1M · Propositional Logic] Answer: 3

The first clause says at least one of a, b, c is true. The three binary clauses say no two are simultaneously true. Together they encode 'exactly one of a, b, c is true', which has 3 models: (T,F,F), (F,T,F), (F,F,T).

Q.7 [NAT, 1M · Propositional Logic] Answer: 2

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \rightarrow R) \wedge \neg Q$	$\neg \neg P$	$((R \rightarrow R) \wedge \neg Q) \wedge \neg \neg P$
T	T	T	F	T	F
T	T	F	F	T	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	F	F
F	T	F	F	F	F
F	F	T	T	F	F
F	F	F	T	F	F

Rows where F = T: **2**.

Q.8 [MCQ, 1M · Sampling] Answer: A (A: cloudy)

Cumulative distribution: sunny [0, 0.2), cloudy [0.2, 0.7), rainy [0.7, 1.0). $u = 0.65$ lies in [0.2, 0.7), so the sample is cloudy. Inverse-CDF sampling is deterministic given u and the ordering, so (D) is wrong.

Q.9 [NAT, 1M · Sampling] Answer: 0.445 to 0.455

In likelihood weighting the weight is the product of the likelihoods of the evidence variables given their parents in the sample. $w = P(s \mid \neg c) \times P(w \mid s, \neg r) = 0.5 \times 0.90 = 0.45$. The probabilities of the sampled non-evidence values ($P(\neg c)$, $P(\neg r \mid \neg c)$) are not part of the weight.

Q.10 [NAT, 1M · Sampling] Answer: 0.1162 to 0.1166

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(A=F) \times P(B=T) \times P(C=T \mid A=F) \times P(D=F \mid B=T, C=T)$
 $= 0.7 \times 0.25 \times 0.7 \times 0.95 = \mathbf{0.116375} \approx 0.1164$.

Q.11 [NAT, 2M · Adversarial Search] Answer: 3

MIN_1 : evaluate 10, 9, 14, giving 9, so $\alpha = 9$ at the root. MIN_2 : 11, then $8 \leq \alpha$, so prune 20. MIN_3 : $2 \leq \alpha$, so prune 15 and 30. MIN_4 : 12, 13, 7 are all evaluated, because the running minimum stays above 9 until the last leaf. Evaluated leaves: 9 of 12. Pruned: 20, 15, 30 = **3**.

Q.12 [MCQ, 2M · Uninformed Search] Answer: D (D: $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$ (cost 7))

UCS expands nodes in order of g. Pop S(0): push A(1), B(4). Pop A(1): push B(3), C(6). Pop B(3): push C(4), G(9). Pop C(4): push G(7). The stale entries B(4) and C(6) are skipped. Pop G(7) and return **$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$ with cost 7**. The other paths are real paths in the graph, but each costs more than 7. UCS returns the cheapest path because it tests the goal on expansion.

Q.13 [NAT, 2M · Adversarial Search] Answer: 13.89 to 13.91

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.25 \times 20 + 0.5 \times 10 + 0.25 \times 11 = 12.75$
- $B = 0.2 \times 18 + 0.3 \times 1 + 0.5 \times 20 = 13.9$
- $C = 0.2 \times 14 + 0.3 \times 18 + 0.5 \times 8 = 12.2$

Root = $\max(12.75, 13.9, 12.2) = \mathbf{13.9}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.14 [MCQ, 2M · Uninformed Search] Answer: D (D: $S \rightarrow A \rightarrow E \rightarrow I \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	A(4), H(5), J(6)
2	A (4)	H(5), J(6), E(9)
3	H (5)	J(6), E(9), I(15)
4	J (6)	E(9), I(15)
5	E (9)	I(14)
6	I (14)	G(18)
7	G (18)	← goal, stop

Path returned: **$S \rightarrow A \rightarrow E \rightarrow I \rightarrow G$** , cost = 4 + 5 + 5 + 4 = **18**. Nodes expanded (including S and G): **7** (S, A, H, J, E, I, G).

Q.15 [MSQ, 2M · Informed Search] Answer: D (D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	D	F	G	H	I	J	S
$h(n)$	0	13	0	11	16	15	19
$h^*(n)$	8	17	0	14	21	20	25
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges:

$H \rightarrow D$ ($11 > 6 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.16 [MSQ, 2M · Predicate Logic] Answer: C, D (C: $\forall x \exists y (x \neq y \wedge R(x, y))$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,3), (2,3), (2,4), (3,1), (3,4), (4,1), (4,3)\}$.

(A) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(3,4)$ and $R(4,3)$ with $3 \neq 4$.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 2 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A (A: A is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $B \rightarrow F$; add F.

Round 2: fire $F \wedge B \rightarrow A$; add A.

Round 3: fire $B \wedge A \rightarrow H$; add H.

Fixed point: $\{A, B, C, E, F, H\}$ — 6 atoms are entailed (facts: B, C, E; newly derived: A, F, H). Atoms never derived: D, J, K.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, C ($B: (\neg R \rightarrow Q) \vee (Q \rightarrow \neg R)$; $C: P \rightarrow (R \rightarrow P)$)

(A) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$: **not valid** — counter-model $Q=F, R=T$.

(B) $(\neg R \rightarrow Q) \vee (Q \rightarrow \neg R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $P \rightarrow (R \rightarrow P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow P) \wedge P) \rightarrow Q$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.19 [MSQ, 2M · Propositional Logic] Answer: B ($B: (\neg Q \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(\neg Q \vee S)$ is derivable:

1. resolve $(\neg P \vee \neg Q \vee S)$ and $(P \vee S)$ on $P \Rightarrow (\neg Q \vee S)$

(C) $(\neg R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg R \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.20 [MSQ, 2M · Propositional Logic] Answer: A, B, C ($A: \{p \vee q, \neg p \vee q, p \vee \neg q, \neg p \vee \neg q\}$; $B: \{p \rightarrow q, q \rightarrow r, r \rightarrow \neg p, p\}$; $C: \{p \leftrightarrow \neg q, q \leftrightarrow \neg r, r \leftrightarrow \neg p\}$)

(A) The four clauses exclude all four (p, q) assignments, one each. (B) From p we get q , then r , then $\neg p$ — a contradiction. (C) $p = \neg q$ and $q = \neg r$ give $p = r$, but $r \leftrightarrow \neg p$ demands $r = \neg p$ — contradiction (an odd cycle of negations). (D) is satisfiable, e.g. $p=T, q=F, r=F$ (exactly one true).

Q.21 [MSQ, 2M · Conditional Independence] Answer: B, C, D ($B: F \perp C \mid A, G$; $C: E \perp D$; $D: G \perp A \mid E, F$)

Rules: a chain $(\rightarrow B \rightarrow)$ or fork $(\leftarrow B \leftarrow)$ is blocked iff B is observed; a collider $(\rightarrow B \leftarrow)$ is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $D \perp G$: **FALSE** — the path $D \leftarrow C \leftarrow A \rightarrow F \rightarrow G$ is active given $\{\}$.

(B) $F \perp C \mid A, G$: **TRUE** — $F \leftarrow A \rightarrow B \rightarrow C$ (blocked: observed fork node A); $F \leftarrow A \rightarrow C$ (blocked: observed fork node A).

(C) $E \perp D$: **TRUE** — $E \rightarrow F \leftarrow A \rightarrow B \rightarrow C \rightarrow D$ (blocked: collider F with neither it nor a descendant observed); $E \rightarrow F \leftarrow A \rightarrow C \rightarrow D$ (blocked: collider F with neither it nor a descendant observed); $E \rightarrow G \leftarrow F \leftarrow A \rightarrow B \rightarrow C \rightarrow D$ (blocked: collider G with neither it nor a descendant observed); $E \rightarrow G \leftarrow F \leftarrow A \rightarrow C \rightarrow D$ (blocked: collider G with neither it nor a descendant observed).

(D) $G \perp A \mid E, F$: **TRUE** — $G \leftarrow E \rightarrow F \leftarrow A$ (blocked: observed fork node E); $G \leftarrow F \leftarrow A$ (blocked: observed chain node F).

Q.22 [NAT, 2M · Sampling] Answer: 0.998 to 1.002

Gibbs sampling draws B from $P(B \mid \text{mb}(B))$, where $\text{mb}(B) = \{A, C, E\}$:

$P(B \mid \text{mb}) \propto P(B \mid \text{parents}(B)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$B=T$: $P(B=T \mid A=F) \times P(C=F \mid B=T) \times P(E=T \mid B=T, C=F) = 0.95 \times 0.9 \times 0.6 = 0.513$

$B=F$: $P(B=F \mid A=F) \times P(C=F \mid B=F) \times P(E=T \mid B=F, C=F) = 0.05 \times 0.05 \times 0.05 = 0.00013$

Normalise: $0.513 / (0.513 + 0.00013) = \mathbf{0.9998}$.

Q.23 [NAT, 2M · Bayesian Networks] Answer: 23

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (2-1) \times 2 + C: (2-1) + D: (2-1) \times 2 \times 2 + E: (3-1) \times 2 \times 2 \times 2 = \mathbf{24}$.

Full joint table: $\prod |X| - 1 = 48 - 1 = 47$; saving = 23.

Q.24 [MCQ, 2M · Probability for AI] **Answer: A** (A: 0.600)

Bayes rule in odds form: posterior odds = likelihood ratio $\times$ prior odds = $6 \times (1/4) = 1.5$. Then $P(h | e) = 1.5/(1 + 1.5) = 0.6$. Check directly: $P(h) = 0.2$; with $P(e | h) = 6k$ and $P(e | \neg h) = k$, $P(h | e) = 1.2k/(1.2k + 0.8k) = 0.6$. $0.857 = 6/7$ ignores the prior, 0.250 reads the prior odds $1 : 4$ as a probability, and 0.200 is the prior $P(h) = 1/5$, ignoring the evidence.

Q.25 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B)$, $f(B, D)$, $f(A, B, E)$, $f(D, E, F)$, $f(E, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B)$, $f(A, B)$, $f(B, D)$, $f(A, B, E)$	$\{A, B, D, E\}$	$f(A, D, E)$
G	$f(E, F, G)$	$\{E, F, G\}$	$f(E, F)$
A	$f(A)$, $f(A, D, E)$	$\{A, D, E\}$	$f(D, E)$
F	$f(D, E, F)$, $f(E, F)$	$\{D, E, F\}$	$f(D, E)$
E	$f(D, E)$, $f(D, E)$	$\{D, E\}$	$f(D)$

The largest factor produced has **3** variable(s) (created when eliminating B); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over D and normalise.

Mock Test 023

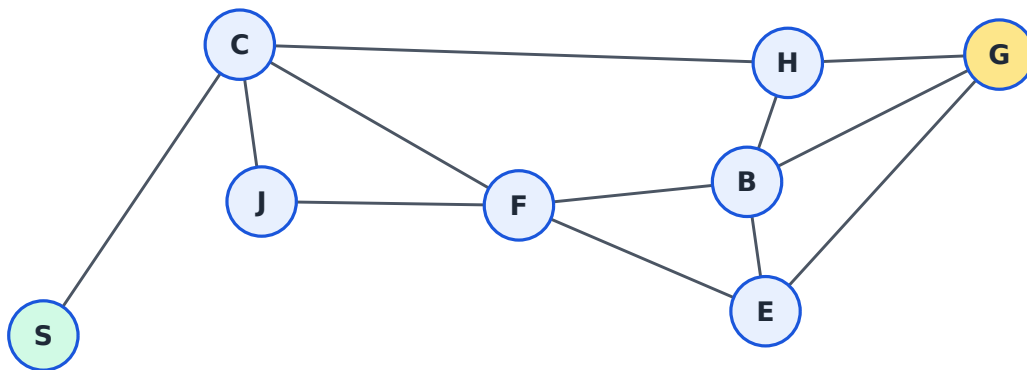
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 8 / 12	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, C, F, H, J, B, E, G
- ☐ (B) S, C, J, F, E, G
- ☐ (C) S, C, B, F, E, G
- ☐ (D) S, C, F, B, E, G

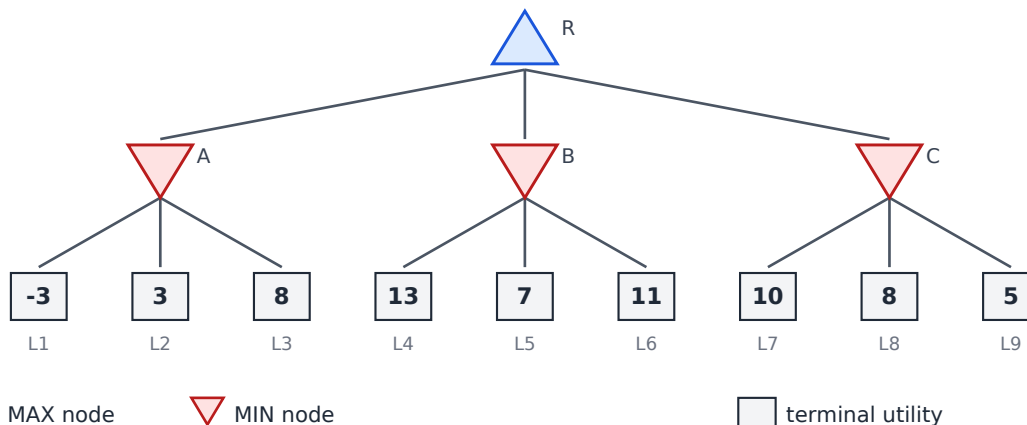
Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search

In alpha-beta search, while a MIN node is being evaluated with current bounds (α, β) , the remaining children of the MIN node can be pruned as soon as its running value v satisfies

- ☐ (A) $v \geq \beta$
- ☐ (B) $v \leq \alpha$
- ☐ (C) $v \geq \alpha$
- ☐ (D) $v \leq \beta$

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 2$ (so $b^d = 9$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over the symbols p, q, r does $(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)$ have?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following are tautologies?

- ☐ (A) $p \vee \neg p$
☐ (B) $\neg(p \wedge \neg p)$
☐ (C) $p \rightarrow (p \wedge q)$
☐ (D) $(p \wedge (p \rightarrow q)) \rightarrow q$

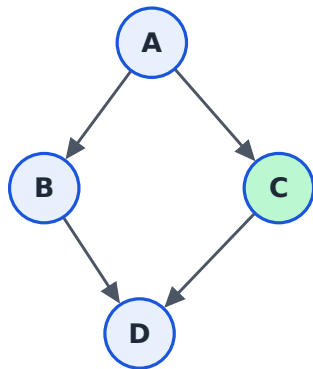
Q.8 [MSQ · 1 mark · no negative marking] Bayesian Networks

Which of the following are correct about Bayesian networks with Boolean variables?

- ☐ (A) The graph must be a directed acyclic graph
☐ (B) The CPT of a node with k Boolean parents has 2^k independent numbers
☐ (C) Each row of a CPT (one parent configuration) sums to 1 over the node's values
☐ (D) If there is no arc between X and Y , then $X \perp Y$ marginally

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(\mathbf{C} = \mathbf{T})$ (round off to 3 decimal places).



$P(A=T)$
0.5

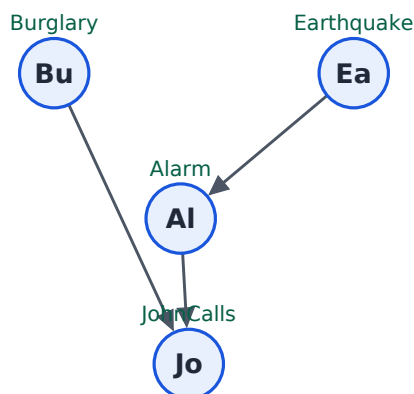
A	$P(B=T \dots)$	A	$P(C=T \dots)$
T	0.8	T	0.25
F	0.3	F	0.5

B	C	$P(D=T \dots)$
T	T	0.4
T	F	0.05
F	T	0.75
F	F	0.9

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Bu, Ea, Al, Jo. (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) What is the probability that a single run produces the sample **(Bu=F, Ea=T, Al=F, Jo=F)**? (round off to 4 decimal places)



$P(\text{Bu}=T)$
0.05

$P(\text{Ea}=T)$	Ea	$P(\text{Al}=T \dots)$
0.4	T	0.2
	F	0.7

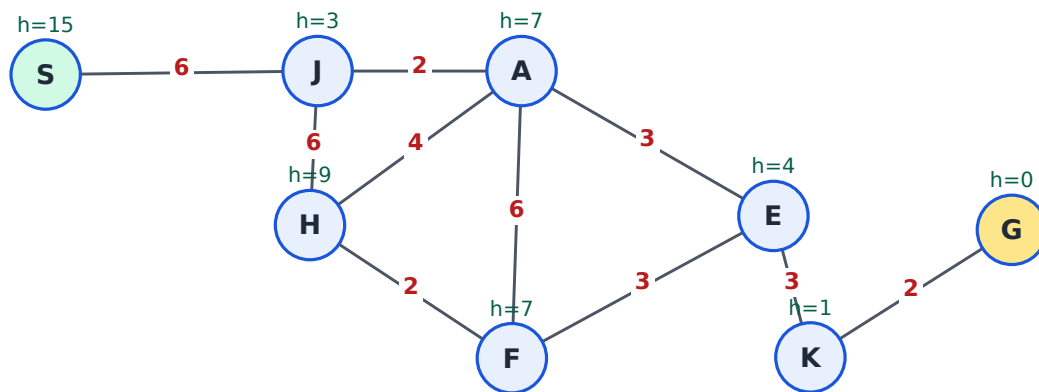
Bu	Al	$P(\text{Jo}=T \dots)$
T	T	0.95
T	F	0.25
F	T	0.95
F	F	0.7

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

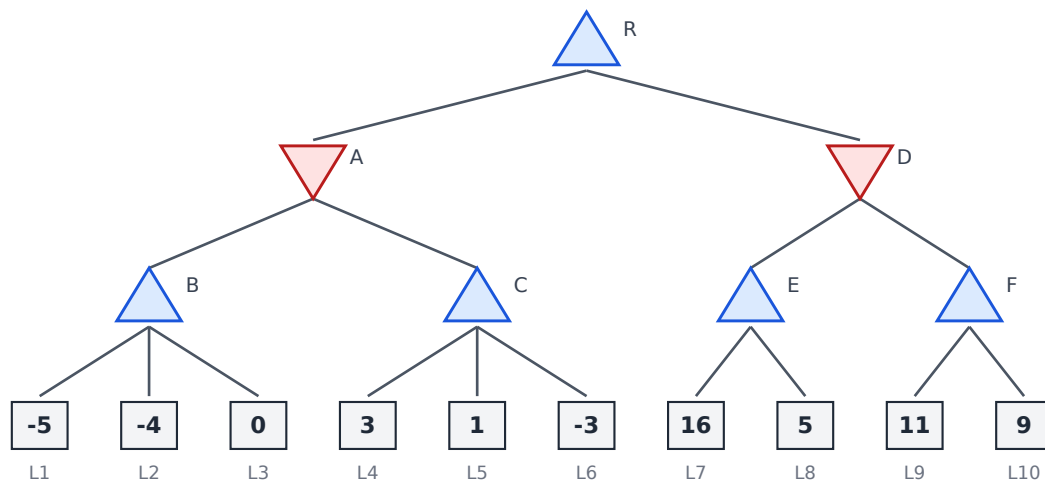
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge A-J violates the consistency condition $h(A) \leq c(A,J) + h(J)$.
☐ (B) $h(H)$ overestimates the true cost from H to G.
☐ (C) h is consistent (monotone).
☐ (D) h is admissible.

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



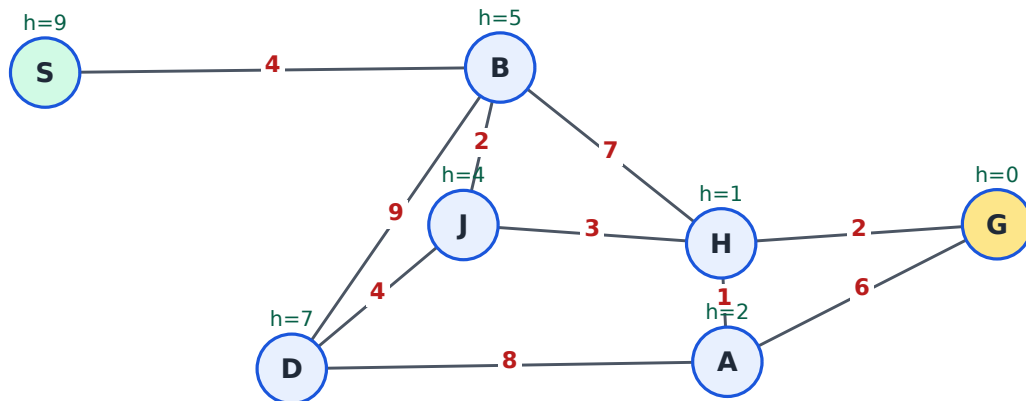
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.14 [MCQ · 2 marks · negative: -2/3] Uninformed Search

BFS runs on a problem with $b = 10$ and solution depth $d = 8$. It stores every generated node, and each node needs 1 KB. Approximately how much memory is required? (1 GB = 10^6 KB.)

- ☐ (A) About 100 MB
☐ (B) About 1 GB
☐ (C) About 100 GB
☐ (D) About 10 TB

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: (Q), C2: ($\neg Q$), C3: ($P \vee R \vee \neg S$), C4: ($Q \vee \neg S$), C5: ($Q \vee \neg R$)

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) ($\neg P \vee R$)
☐ (B) ($P \vee Q \vee \neg S$)
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) ($\neg R$)

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

F = $((S \rightarrow P) \rightarrow (P \wedge P)) \leftrightarrow (R \leftrightarrow (S \rightarrow Q))$?

Answer (numerical): _____

Q.17 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following formulas is in disjunctive normal form and is equivalent to $(p \vee q) \wedge \neg(p \wedge q \wedge r)$?

- ☐ (A) $(p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg r)$
☐ (B) $(p \wedge \neg q) \vee (\neg p \wedge q)$
☐ (C) $(p \wedge \neg r) \vee (q \wedge \neg r)$
☐ (D) $(p \vee q) \wedge (\neg p \vee \neg q \vee \neg r)$

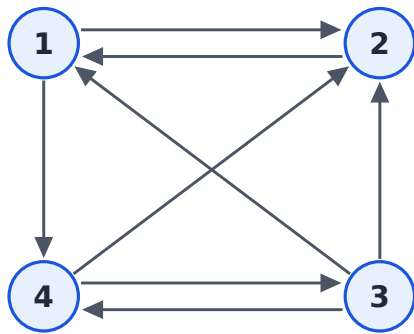
Q.18 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2\}$. How many interpretations of a binary predicate R over D make the sentence $\forall x \exists y R(x, y) \rightarrow \exists y \forall x R(x, y)$ true?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
- ☐ (B) $\forall x \exists y (x \neq y \wedge R(x, y))$
- ☐ (C) $\exists x \forall y R(x, y)$
- ☐ (D) $\exists x \forall y \neg R(y, x)$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((\neg R \rightarrow P) \wedge (P \rightarrow Q)) \rightarrow (\neg R \rightarrow Q)$
- ☐ (B) $(R \rightarrow P) \rightarrow (P \rightarrow R)$
- ☐ (C) $\neg(P \wedge R) \leftrightarrow (\neg P \wedge \neg R)$
- ☐ (D) $(\neg Q \rightarrow P) \vee (P \rightarrow \neg Q)$

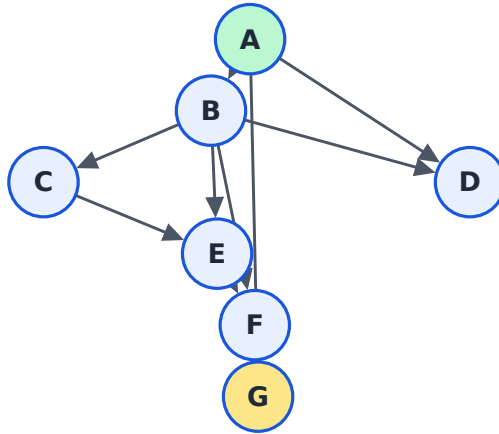
Q.21 [MSQ · 2 marks · no negative marking] Probability for AI

For arbitrary random variables X_1, X_2, X_3, X_4 (no independence assumed), which of the following expressions are always equal to $P(x_1, x_2, x_3, x_4)$?

- ☐ (A) $P(x_4 | x_3) P(x_3 | x_2) P(x_2 | x_1) P(x_1)$
- ☐ (B) $P(x_4 | x_1, x_2, x_3) P(x_3 | x_1, x_2) P(x_2 | x_1) P(x_1)$
- ☐ (C) $P(x_1 | x_2, x_3, x_4) P(x_2 | x_3, x_4) P(x_3 | x_4) P(x_4)$
- ☐ (D) $P(x_4 | x_1, x_2, x_3) P(x_3 | x_2) P(x_2 | x_1) P(x_1)$

Q.22 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

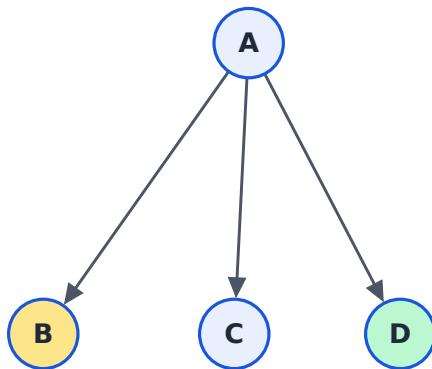
Variable elimination is used to compute $P(A \mid G = T)$ in the Bayesian network below (Boolean variables; evidence $G = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, D, E, F, B**. What is the scope of the new factor produced when **E** is summed out?



- ☐ (A) $f(A, B, F)$
☐ (B) $f(B, F)$
☐ (C) $f(B, E, F)$
☐ (D) $f(B)$

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(D = T \mid B = T)$ (round off to 3 decimal places).



$P(A=T)$
0.4

A	$P(B=T \dots)$	A	$P(C=T \dots)$
T	0.3	T	0.8
F	0.1	F	0.4

A	$P(D=T \dots)$
T	0.9
F	0.5

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

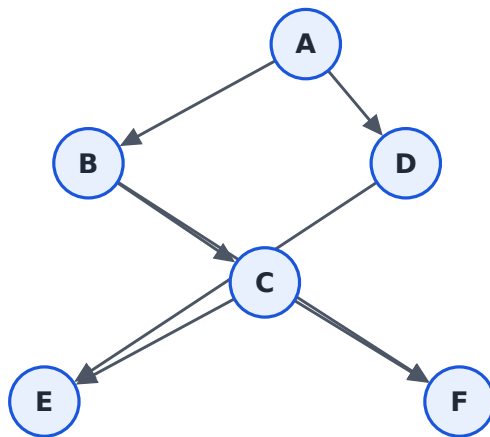
The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid C = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	F	T	T
2	T	F	T	F
3	F	T	F	F
4	F	F	T	T
5	F	F	T	F
6	F	F	T	T
7	T	F	F	F
8	T	F	F	F
9	F	T	F	F
10	F	F	T	T
11	T	F	F	F
12	F	F	T	T
13	F	T	T	F
14	T	F	F	F
15	F	F	T	T
16	F	T	F	T
17	F	F	T	T
18	T	F	F	T

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp A \mid F$
☐ (B) $E \perp B \mid C, D$
☐ (C) $A \perp D$
☐ (D) $F \perp B \mid D$

Answer Key — Mock Test 023

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	MCQ	2	Uninformed Search	C
2	MCQ	1	Adversarial Search	B	15	MSQ	2	Propositional Logic	B, C, D
3	NAT	1	Adversarial Search	7	16	NAT	2	Propositional Logic	8
4	NAT	1	Adversarial Search	5	17	MCQ	2	Propositional Logic	A
5	NAT	1	Propositional Logic	2	18	NAT	2	Predicate Logic	14
6	NAT	1	Predicate Logic	16	19	MSQ	2	Predicate Logic	A, B
7	MSQ	1	Propositional Logic	A, B, D	20	MSQ	2	Propositional Logic	A, D
8	MSQ	1	Bayesian Networks	A, B, C	21	MSQ	2	Probability for AI	B, C
9	NAT	1	Exact Inference	0.373 to 0.377	22	MCQ	2	Exact Inference	B
10	NAT	1	Sampling	0.091 to 0.0914	23	NAT	2	Exact Inference	0.765 to 0.769
11	MSQ	2	Informed Search	A, D	24	NAT	2	Sampling	0.24 to 0.26
12	NAT	2	Adversarial Search	2	25	MSQ	2	Conditional Independence	B
13	NAT	2	Informed Search	11					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: D (D: S, C, F, B, E, G)

Adjacency lists (alphabetical): B: {E, F, G, H}; C: {F, H, J, S}; E: {B, F, G}; F: {B, C, E, J}; G: {B, E, H}; H: {B, C, G}; J: {C, F}; S: {C}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

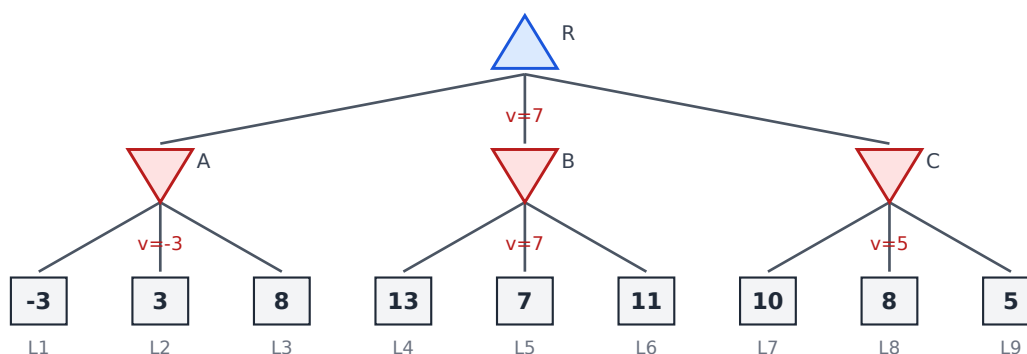
Visit order: **S, C, F, B, E, G**; the path found is $S \rightarrow C \rightarrow F \rightarrow B \rightarrow E \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.2 [MCQ, 1M · Adversarial Search] Answer: B ($B: v \leq \alpha$)

A MIN node's value can only decrease as more children are seen. Once $v \leq \alpha$, MAX already has an alternative worth α higher up the tree and will never let play reach this node. $v \geq \beta$ (A) is the cutoff test at MAX nodes. (C) and (D) do not justify pruning.

Q.3 [NAT, 1M · Adversarial Search] Answer: 7

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -3$, $B = 7$, $C = 5$. Root value = **7**; MAX's optimal first move is towards B.

Q.4 [NAT, 1M · Adversarial Search] Answer: 5

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 3^{\lceil 1 \rceil} + 3^{\lfloor 1 \rfloor} - 1 = 3 + 3 - 1 = 5$, versus 9 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.5 [NAT, 1M · Propositional Logic] Answer: 2

The cycle of implications $p \rightarrow q \rightarrow r \rightarrow p$ forces p, q, r to have the same truth value: if any one is T, all following ones are T, and the cycle returns. So the only models are all-T and all-F, i.e. 2 models.

Q.6 [NAT, 1M · Predicate Logic] Answer: 16

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 4$ this gives **16** (verified by enumerating all interpretations).

Q.7 [MSQ, 1M · Propositional Logic] Answer: A, B, D (A: $p \vee \neg p$; B: $\neg(p \wedge \neg p)$; D: $(p \wedge (p \rightarrow q)) \rightarrow q$)

$p \vee \neg p$ (excluded middle) and $\neg(p \wedge \neg p)$ (non-contradiction) are valid. $(p \wedge (p \rightarrow q)) \rightarrow q$ is the modus ponens tautology. $p \rightarrow (p \wedge q)$ is false at $p=T, q=F$, so it is not valid.

Q.8 [MSQ, 1M · Bayesian Networks] Answer: A, B, C

(A) BNs are DAGs by definition. (B) There are 2^k rows, each needing one number $P(X = \text{true} \mid \text{row})$. (C) Each row is a conditional distribution, so it sums to 1. (D) is false: in $A \rightarrow B \rightarrow C$ there is no arc between A and C, but they are dependent through B.

Q.9 [NAT, 1M · Exact Inference] Answer: 0.373 to 0.377

Only the ancestors of C (A) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T, D=T	0.04
A=T, B=T, D=F	0.06
A=T, B=F, D=T	0.01875
A=T, B=F, D=F	0.00625
A=F, B=T, D=T	0.03
A=F, B=T, D=F	0.045
A=F, B=F, D=T	0.13125
A=F, B=F, D=F	0.04375

Sum = **0.375** ≈ 0.375 .

Q.10 [NAT, 1M · Sampling] Answer: 0.091 to 0.0914

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(Bu=F) \times P(Ea=T) \times P(Al=F \mid Ea=T) \times P(Jo=F \mid Bu=F, Al=F)$
 $= 0.95 \times 0.4 \times 0.8 \times 0.3 = \mathbf{0.0912} \approx 0.0912$.

Q.11 [MSQ, 2M · Informed Search] Answer: A, D (A: The edge A-J violates the consistency condition $h(A) \leq c(A,J) + h(J)$.; D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	E	F	G	H	J	K	S
$h(n)$	7	4	7	0	9	3	1	15
$h^*(n)$	8	5	8	0	10	10	2	16
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

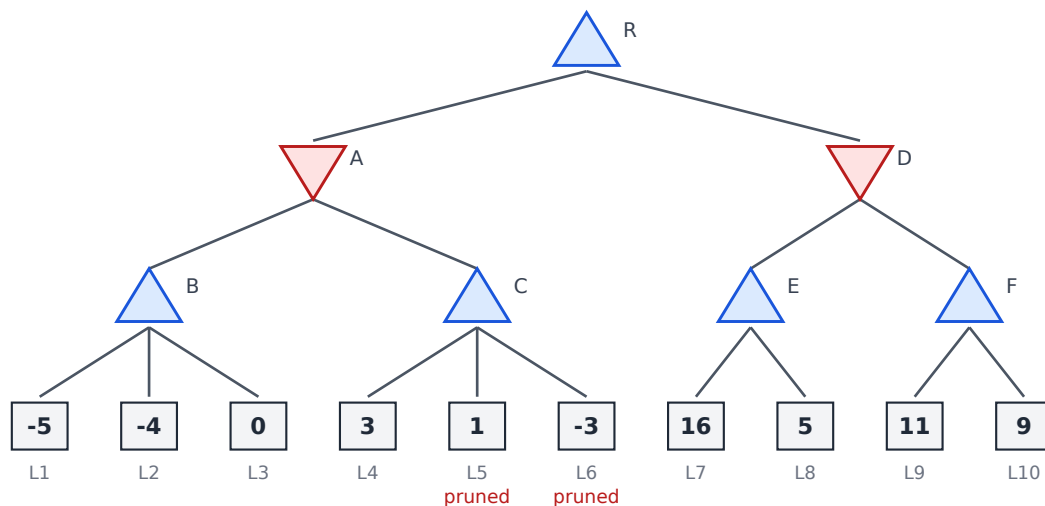
Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $A \rightarrow J$ ($7 > 2 + 3$), $S \rightarrow J$ ($15 > 6 + 3$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [NAT, 2M · Adversarial Search] Answer: 2

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MAX node C: value $3 \geq \beta = 0 \Rightarrow$ prune remaining children (leaves L5, L6).



Leaves evaluated (8): L1, L2, L3, L4, L7, L8, L9, L10. Leaves pruned (2): L5, L6. Root value = 11 (same as plain minimax, 11 — pruning never changes the root value).

Q.13 [NAT, 2M · Informed Search] Answer: 11

Heuristic values: $h(A)=2$, $h(B)=5$, $h(D)=7$, $h(G)=0$, $h(H)=1$, $h(J)=4$, $h(S)=9$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	B(9)
2	B (9)	J(10), H(12), D(20)
3	J (10)	H(10), D(17)
4	H (10)	G(11), A(12), D(17)
5	G (11)	A(12), D(17) ← goal, stop

Path returned: **S** → **B** → **J** → **H** → **G** with cost **11**; nodes expanded = **5** (S, B, J, H, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 11$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.14 [MCQ, 2M · Uninformed Search] Answer: C (C: About 100 GB)

The nodes generated are $b + \dots + b^8 \approx 1.11 \times 10^8$. At 1 KB each, that is about 1.11×10^8 KB $\approx$ **111 GB, i.e. about 100 GB**. This is AIMA's point that memory, not time, is the binding constraint for BFS. 100 MB (A) and 1 GB (B) are 2 to 3 orders of magnitude too small, and 10 TB (D) is 100 times too large.

Q.15 [MSQ, 2M · Propositional Logic] Answer: B, C, D (B: $(P \vee Q \vee \neg S)$; C: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable); D: $(\neg R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(P \vee Q \vee \neg S)$ is derivable:

1. resolve $(P \vee R \vee \neg S)$ and $(Q \vee \neg R)$ on $R \Rightarrow (P \vee Q \vee \neg S)$

(C) $\square$ is derivable:

1. resolve (Q) and $(\neg Q)$ on $Q \Rightarrow \square$ (empty clause)

(D) $(\neg R)$ is derivable:

1. resolve $(\neg Q)$ and $(Q \vee \neg R)$ on $Q \Rightarrow (\neg R)$

Q.16 [NAT, 2M · Propositional Logic] Answer: 8

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(S \rightarrow P) \rightarrow (P \wedge P)$	$R \leftrightarrow (S \rightarrow Q)$	$((S \rightarrow P) \rightarrow (P \wedge P)) \leftrightarrow (R \leftrightarrow (S \rightarrow Q))$
T	T	T	T	T	T	T
T	T	T	F	T	T	T
T	T	F	T	T	F	F
T	T	F	F	T	F	F
T	F	T	T	T	F	F
T	F	T	F	T	T	T

P	Q	R	S	$(S \rightarrow P) \rightarrow (P \wedge P)$	$R \leftrightarrow (S \rightarrow Q)$	$((S \rightarrow P) \rightarrow (P \wedge P)) \leftrightarrow (R \leftrightarrow (S \rightarrow Q))$
T	F	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	T	F	F	T	F
F	T	F	T	T	F	F
F	T	F	F	F	F	T
F	F	T	T	T	F	F
F	F	T	F	F	T	F
F	F	F	T	T	T	T
F	F	F	F	F	F	T

Rows where F = T: **8**.

Q.17 [MCQ, 2M · Propositional Logic] Answer: A $(A: (p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg r))$

Distribute: $(p \vee q) \wedge (\neg p \vee \neg q \vee \neg r) = (p \wedge \neg p) \vee (p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg q) \vee (q \wedge \neg r)$; dropping the contradictory terms gives option A. Option B $(p \oplus q)$ misses the model $p=q=T, r=F$. Option C misses $p=T, q=F, r=T$. Option D is equivalent but is a CNF, not a DNF. Truth tables confirm A has exactly the 5 models of the original.

Q.18 [NAT, 2M · Predicate Logic] Answer: 14

There are $2^4 = 16$ relations. The sentence is false iff every row of the 2×2 matrix is non-empty but no column is full. A column is full if both rows contain that column; so each row must be non-empty and the rows must have no common column, forcing each row to be a single, different column: $R = \{(1,1), (2,2)\}$ or $R = \{(1,2), (2,1)\}$. So 2 interpretations falsify it and $16 - 2 = 14$ satisfy it.

Q.19 [MSQ, 2M · Predicate Logic] Answer: A, B $(A: \forall x \exists y R(x, y); B: \forall x \exists y (x \neq y \wedge R(x, y)))$

$R = \{(1,2), (1,4), (2,1), (3,1), (3,2), (3,4), (4,2), (4,3)\}$.

(A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(B) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(C) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(D) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Propositional Logic] Answer: A, D $(A: ((\neg R \rightarrow P) \wedge (P \rightarrow Q)) \rightarrow (\neg R \rightarrow Q); D: (\neg Q \rightarrow P) \vee (P \rightarrow \neg Q))$

(A) $((\neg R \rightarrow P) \wedge (P \rightarrow Q)) \rightarrow (\neg R \rightarrow Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(B) $(R \rightarrow P) \rightarrow (P \rightarrow R)$: **not valid** — counter-model $P=T, R=F$.

(C) $\neg(P \wedge R) \leftrightarrow (\neg P \wedge \neg R)$: **not valid** — counter-model $P=T, R=F$.

(D) $(\neg Q \rightarrow P) \vee (P \rightarrow \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.21 [MSQ, 2M · Probability for AI] Answer: B, C

The chain rule $P(x_1, \dots, x_n) = \prod_i P(x_i | x_1, \dots, x_{i-1})$ holds for any ordering of the variables. (B) is the chain rule for order X_1, X_2, X_3, X_4 ; (C) is the chain rule for the reverse order X_4, X_3, X_2, X_1 . (A) drops conditioning variables, which is valid only under Markov-chain independence assumptions. (D) replaces $P(x_3 | x_1, x_2)$ by $P(x_3 | x_2)$, which needs $X_3 \perp X_1 | X_2$, not generally true.

Q.22 [MCQ, 2M · Exact Inference] Answer: B $(B: f(B, F))$

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(B, C), f(A, B, D), f(B, C, E), f(B, E), f(A, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C), f(B, C, E)$	$\{B, C, E\}$	$f(B, E)$
D	$f(A, B, D)$	$\{A, B, D\}$	$f(A, B)$
E	$f(B, E, F), f(B, E)$	$\{B, E, F\}$	$f(B, F)$
F	$f(A, F), f(B, F)$	$\{A, B, F\}$	$f(A, B)$
B	$f(A, B), f(A, B), f(A, B)$	$\{A, B\}$	$f(A)$

The largest factor produced has **2** variable(s) (created when eliminating C); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.23 [NAT, 2M · Exact Inference] **Answer: 0.765 to 0.769**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B | A) \cdot P(C | A) \cdot P(D | A)$.

Inference by enumeration: $P(D | e) = \alpha \sum_{\text{hidden}} P(D, e, \text{hidden})$.

Hidden: A, C	D	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.4 \times 0.3 \times 0.8 \times 0.9$	0.0864
T, F	T	$0.4 \times 0.3 \times 0.2 \times 0.9$	0.0216
F, T	T	$0.6 \times 0.1 \times 0.4 \times 0.5$	0.012
F, F	T	$0.6 \times 0.1 \times 0.6 \times 0.5$	0.018
T, T	F	$0.4 \times 0.3 \times 0.8 \times 0.1$	0.0096
T, F	F	$0.4 \times 0.3 \times 0.2 \times 0.1$	0.0024
F, T	F	$0.6 \times 0.1 \times 0.4 \times 0.5$	0.012
F, F	F	$0.6 \times 0.1 \times 0.6 \times 0.5$	0.018

$P(D=T, e) = 0.138$; $P(D=F, e) = 0.042$; $P(e) = 0.18$.

$P(D=T | e) = 0.138 / 0.18 = \mathbf{0.7667} \approx 0.767$.

Q.24 [NAT, 2M · Sampling] **Answer: 0.24 to 0.26**

Rejection sampling keeps only samples consistent with the evidence $C = F$: samples 3, 7, 8, 9, 11, 14, 16, 18 (8 accepted, 10 rejected).

Among the accepted samples, $D = T$ in 2. Estimate = $2/8 = \mathbf{0.25}$.

(For this network the exact value is 0.2143; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.25 [MSQ, 2M · Conditional Independence] **Answer: B** ($B: E \perp B | C, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $B \perp A | F$: **FALSE** — the path $B \leftarrow A$ is active given $\{F\}$.

(B) $E \perp B | C, D$: **TRUE** — $E \leftarrow C \leftarrow B$ (blocked: observed chain node C); $E \leftarrow C \rightarrow F \leftarrow B$ (blocked: observed fork node C); $E \leftarrow D \leftarrow A \rightarrow B$ (blocked: observed chain node D).

(C) $A \perp D$: **FALSE** — the path $A \rightarrow D$ is active given $\{\}$.

(D) $F \perp B | D$: **FALSE** — the path $F \leftarrow B$ is active given $\{D\}$.

Mock Test 024

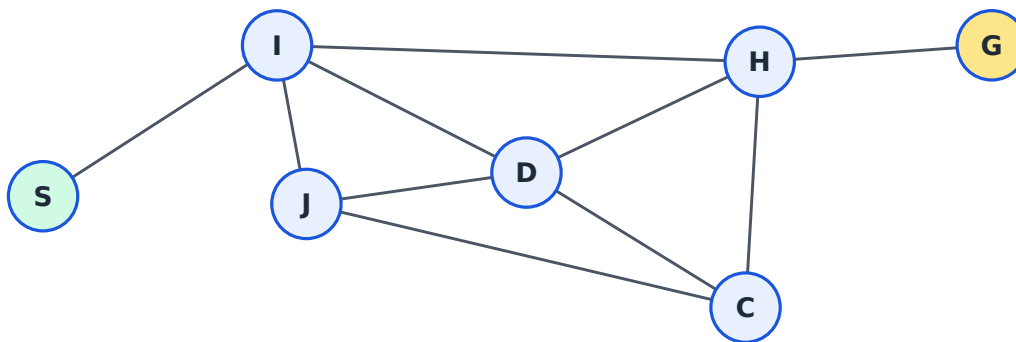
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 7 / 11	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, I, J, H, D, C, G
- ☐ (B) S, I, D, H, J, C, G
- ☐ (C) S, I, D, C, H, G
- ☐ (D) S, I, H, D, J, C, G

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search

Which statement about depth-first search on a **finite** state space is correct?

- ☐ (A) The graph-search version is complete, but the tree-search version can loop forever on cycles
- ☐ (B) Both versions are complete and optimal
- ☐ (C) The tree-search version is complete, but the graph-search version is not
- ☐ (D) Neither version is complete

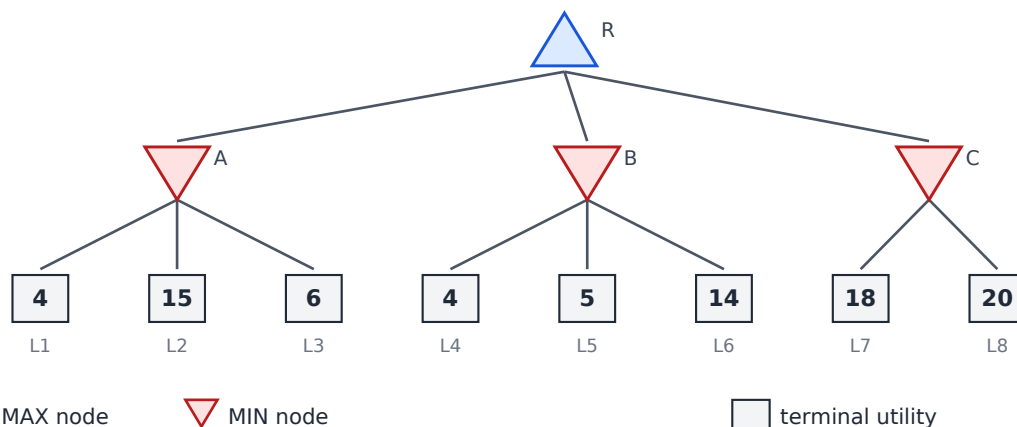
Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 4$ (so $b^d = 81$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic

Let the domain be $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \forall y (R(x, y) \rightarrow R(y, x))$?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((P \wedge Q) \leftrightarrow \neg \neg P) \rightarrow ((P \rightarrow \neg R) \wedge (\neg P \wedge R)) ?$$

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Over the finite domain $\{a, b\}$, the sentence $\forall x \exists y R(x, y)$ is equivalent to which ground formula?

- ☐ (A) $(R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$
☐ (B) $(R(a, a) \wedge R(a, b)) \vee (R(b, a) \wedge R(b, b))$
☐ (C) $(R(a, a) \vee R(b, a)) \wedge (R(a, b) \vee R(b, b))$
☐ (D) $R(a, a) \wedge R(b, b)$

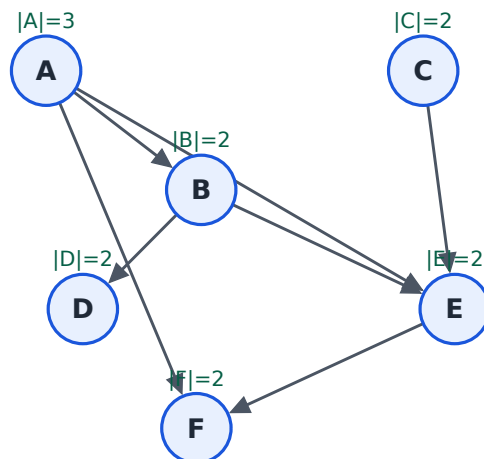
Q.8 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 4 elements and R a binary predicate symbol. In how many of the 2^{16} possible interpretations of R over D is the sentence $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$ true?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

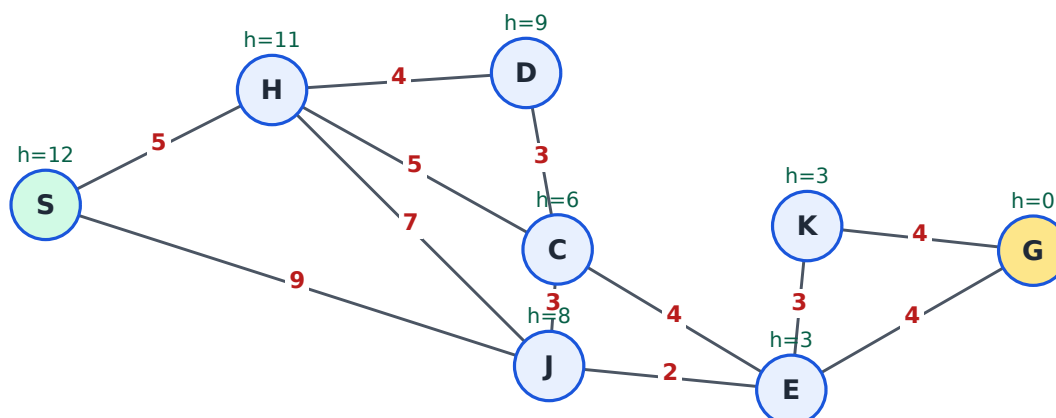
Q.10 [MCQ · 1 mark · negative: -1/3] Variable Elimination

In VE for a query with evidence $J = \text{true}$, the CPT $P(J | A)$ (A and J Boolean) is converted into which factor?

- ☐ (A) A factor over A with 2 entries: $P(j | a)$, $P(j | \neg a)$
- ☐ (B) A factor over $\{A, J\}$ with 4 entries
- ☐ (C) A factor over J with 2 entries
- ☐ (D) A scalar equal to $P(j)$

Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

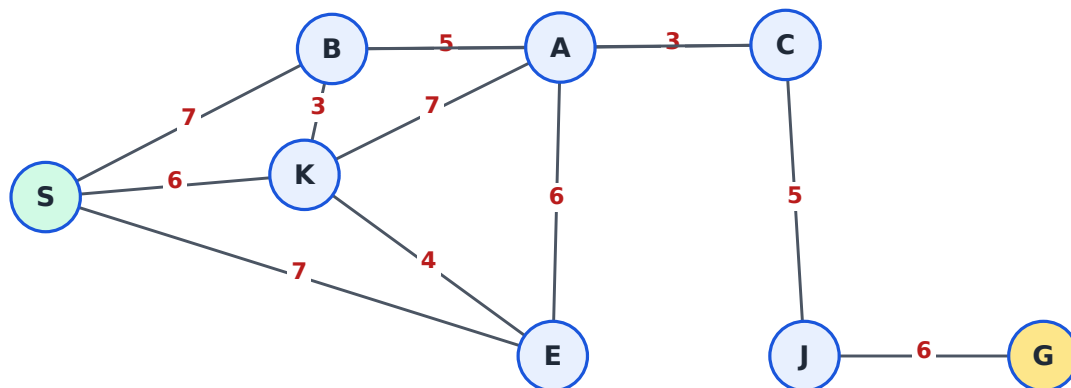
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) The edge J - E violates the consistency condition $h(J) \leq c(J, E) + h(E)$.
- ☐ (C) $h(J)$ overestimates the true cost from J to G .
- ☐ (D) h is admissible.

Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow K \rightarrow B \rightarrow C \rightarrow J \rightarrow G$
☐ (B) $S \rightarrow B \rightarrow C \rightarrow J \rightarrow G$
☐ (C) $S \rightarrow E \rightarrow A \rightarrow C \rightarrow J \rightarrow G$
☐ (D) $S \rightarrow B \rightarrow A \rightarrow C \rightarrow J \rightarrow G$

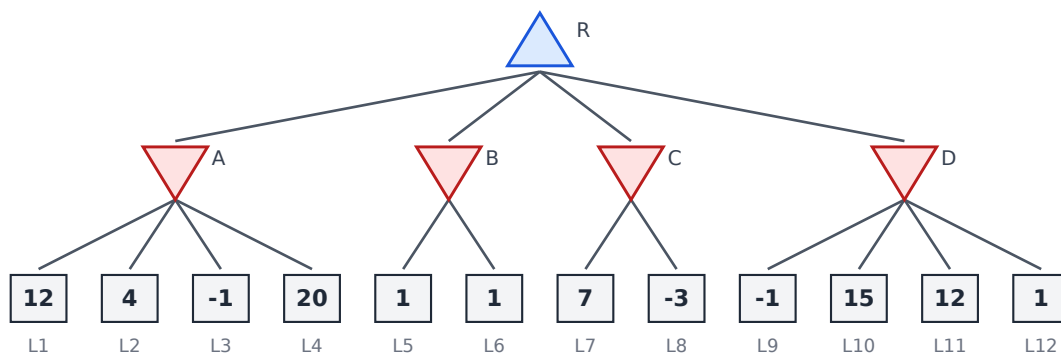
Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Suppose breadth-first search were changed to apply the goal test when a node is selected for expansion instead of when it is generated. The branching factor is b and the shallowest goal is at depth d . In the worst case, the number of nodes generated becomes

- ☐ (A) $O(b^d)$
☐ (B) $O(b^{d+1})$
☐ (C) $O(b^{d/2})$
☐ (D) $O(bd)$

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $P \leftrightarrow (Q \leftrightarrow P)$
- (2) $\neg\neg R \wedge (P \wedge P)$
- (3) $S \vee R$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $S \vee S$
- ☐ (B) $\neg P \wedge S$
- ☐ (C) $P \wedge P$
- ☐ (D) $\neg Q \vee Q$

Q.16 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ are partial orders, i.e. satisfy reflexivity $\forall x R(x, x)$, antisymmetry $\forall x \forall y (R(x, y) \wedge R(y, x) \rightarrow x = y)$, and transitivity?

Answer (numerical): _____

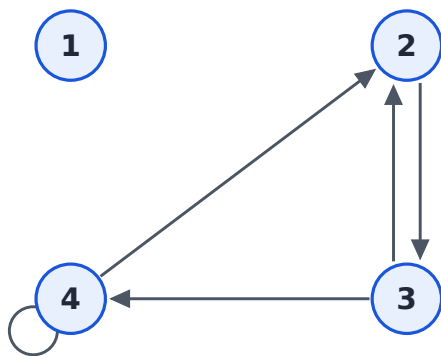
Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((P \rightarrow Q) \rightarrow R) \leftrightarrow (P \rightarrow (Q \rightarrow R))$
- ☐ (B) $\neg(P \wedge Q) \leftrightarrow (\neg P \wedge \neg Q)$
- ☐ (C) $((Q \vee P) \wedge (\neg Q \vee R)) \rightarrow (P \vee R)$
- ☐ (D) $((R \rightarrow P) \wedge (P \rightarrow Q)) \rightarrow (R \rightarrow Q)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x R(x, x)$
- ☐ (B) $\exists x \forall y R(x, y)$
- ☐ (C) $\exists y \forall x R(x, y)$
- ☐ (D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic

Let $\phi = \forall x \exists y (R(x, y) \wedge \neg R(y, x))$. Which of the following statements are TRUE?

- ☐ (A) ϕ has no model with a domain of size 1
- ☐ (B) ϕ has a model with domain $\{1, 2\}$
- ☐ (C) ϕ has a model with domain $\{1, 2, 3\}$
- ☐ (D) ϕ has a model with domain $\mathbb{N}$

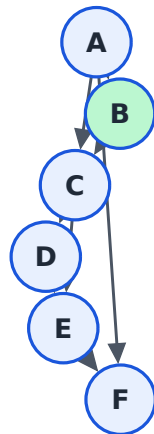
Q.20 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. Which of the following are correct (values rounded)?

- ☐ (A) $P(a) \approx 0.0307$
☐ (B) $P(b | a) \approx 0.588$
☐ (C) $P(b | a, e) > P(b | a)$
☐ (D) $P(j | b) \approx 0.725$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

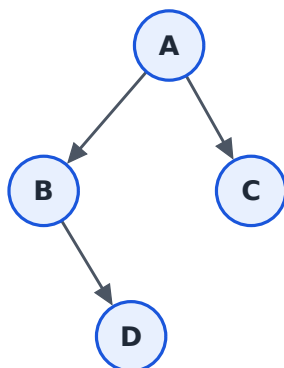
Variable elimination is used to compute $\mathbf{P(B)}$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E, A, C, D, F**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.69}, \mathbf{0.11}, \mathbf{0.75}, \mathbf{0.62} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T | \text{sampled parent values})$. Which sample is generated?



A	P(A=T)
T	0.5
F	0.5

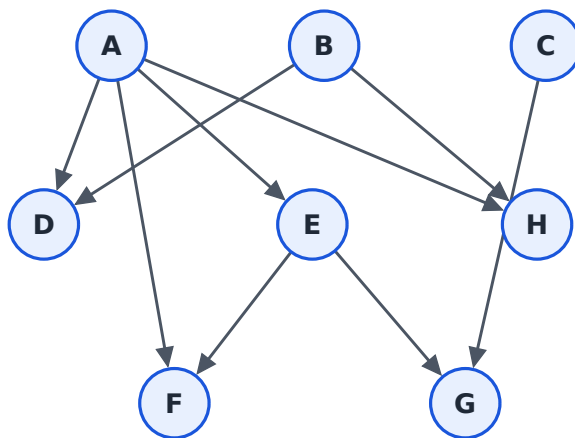
A	P(B=T ...)	A	P(C=T ...)
T	0.5	T	0.9
F	0.5	F	0.05

B	P(D=T ...)
T	0.7
F	0.25

- ☐ (A) A=F, B=T, C=F, D=T
☐ (B) A=T, B=F, C=F, D=T
☐ (C) A=T, B=T, C=F, D=T
☐ (D) A=F, B=F, C=F, D=T

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

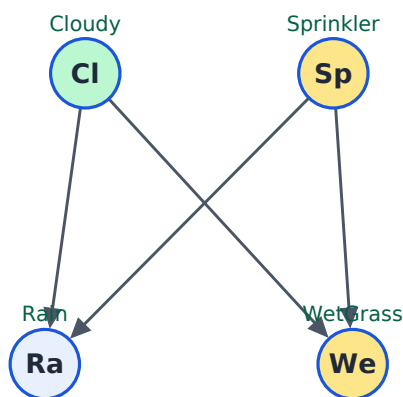
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp B \mid H$
☐ (B) $E \perp H \mid A$
☐ (C) $B \perp G \mid D$
☐ (D) $A \perp B$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\text{Cloudy} = T \mid \text{WetGrass} = T, \text{Sprinkler} = T)$ (round off to 3 decimal places).



$P(Cl=T)$
0.5

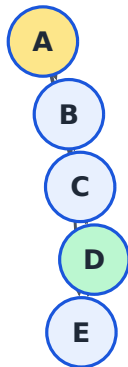
$P(Sp=T)$	Cl	Sp	$P(Ra=T \dots)$
0.75	T	T	0.25
	T	F	0.2
	F	T	0.4
	F	F	0.75

Cl	Sp	$P(We=T \dots)$
T	T	0.75
T	F	0.8
F	T	0.5
F	F	0.95

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{A} = \mathbf{T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(D = T \mid A = T)}$ (round off to 3 decimal places).



P(A=T)
0.05

A	P(B=T ...)	A	B	P(C=T ...)
T	0.8	T	T	0.95
F	0.25	T	F	0.1
		F	T	0.95
		F	F	0.3

A	C	P(D=T ...)	B	D	P(E=T ...)
T	T	0.7	T	T	0.8
T	F	0.1	T	F	0.75
F	T	0.5	F	T	0.2
F	F	0.2	F	F	0.3

#	B	C	D	E
1	T	T	T	T
2	T	T	T	T
3	T	T	F	T
4	T	T	T	T

Answer (numerical): _____

Answer Key — Mock Test 024

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	B	14	NAT	2	Adversarial Search	3
2	MCQ	1	Uninformed Search	A	15	MSQ	2	Propositional Logic	C, D
3	NAT	1	Adversarial Search	17	16	NAT	2	Predicate Logic	19
4	NAT	1	Adversarial Search	18	17	MSQ	2	Propositional Logic	C, D
5	NAT	1	Predicate Logic	64	18	MSQ	2	Predicate Logic	A
6	NAT	1	Propositional Logic	4	19	MSQ	2	Predicate Logic	A, C, D
7	MCQ	1	Predicate Logic	A	20	MSQ	2	Bayesian Networks	A, B, D
8	NAT	1	Predicate Logic	11664	21	NAT	2	Exact Inference	4
9	NAT	1	Bayesian Networks	26	22	MCQ	2	Sampling	A
10	MCQ	1	Variable Elimination	A	23	MSQ	2	Conditional Independence	B, D
11	MSQ	2	Informed Search	B, C	24	NAT	2	Exact Inference	0.598 to 0.602
12	MCQ	2	Uninformed Search	B	25	NAT	2	Sampling	0.748 to 0.752
13	MCQ	2	Uninformed Search	B					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: B** (B: S, I, D, H, J, C, G)

Adjacency lists (alphabetical): C: {D, H, J}; D: {C, H, I, J}; G: {H}; H: {C, D, G, I}; I: {D, H, J, S}; J: {C, D, I}; S: {I}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[I]
2	I	[D, H, J]
3	D	[H, J, C]
4	H	[J, C, G]
5	J	[C, G]

Step	Expanded	Frontier (FIFO) after expansion
6	C	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, I, D, H, J, C, G**. The path returned is $S \rightarrow I \rightarrow H \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [MCQ, 1M · Uninformed Search] Answer: A (A: The graph-search version is complete, but the tree-search version can loop forever on cycles)

Graph-search DFS never re-expands a state, so in a finite space it eventually expands every reachable state and is complete. Tree-search DFS keeps no explored set, so it can follow a cycle such as $A \rightarrow B \rightarrow A \rightarrow \dots$ forever and is incomplete. Neither version is optimal: DFS can return a deep, expensive goal before a shallow one. So (B), (C) and (D) are all wrong.

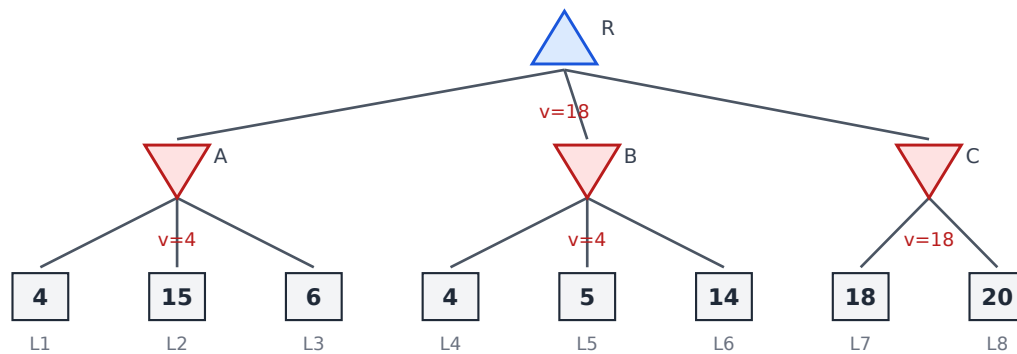
Q.3 [NAT, 1M · Adversarial Search] Answer: 17

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 3^2 + 3^2 - 1 = 9 + 9 - 1 = 17$, versus 81 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.4 [NAT, 1M · Adversarial Search] Answer: 18

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 4$, $B = 4$, $C = 18$. Root value = **18**; MAX's optimal first move is towards C.

Q.5 [NAT, 1M · Predicate Logic] Answer: 64

An interpretation of R is a subset of $D \times D$ (9 pairs). The sentence says R is symmetric. The 3 diagonal pairs (x, x) can be chosen freely; the 3 unordered pairs $\{x, y\}$ with $x \neq y$ must be included or excluded as a whole. That is $2^3 \times 2^3 = 64$ relations.

Q.6 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$(P \wedge Q) \leftrightarrow \neg \neg P$	$(P \rightarrow \neg R) \wedge (\neg P \wedge R)$	$((P \wedge Q) \leftrightarrow \neg \neg P) \rightarrow ((P \rightarrow \neg R) \wedge (\neg P \wedge R))$
T	T	T	T	F	F
T	T	F	T	F	F
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	F	F

Rows where $F = T$: **4**.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A ($A: (R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$)

Over a finite domain with every element named, $\forall$ becomes a conjunction over x and $\exists$ a disjunction over y . For $x = a$: $R(a, a) \vee R(a, b)$; for $x = b$: $R(b, a) \vee R(b, b)$. Their conjunction is option A. Option B is $\exists x \forall y R(x, y)$. Option C is $\forall y \exists x R(x, y)$ (every element has an R -predecessor). Option D is reflexivity.

Q.8 [NAT, 1M · Predicate Logic] Answer: 11664

View R as an 4×4 0/1 matrix. The sentence says R is antisymmetric: diagonal free (2^n); each unordered pair has 3 options (neither, one way, other way) $\Rightarrow 2^n \cdot 3^{(n-1)/2}$. With $n = 4$: **11664**.

Q.9 [NAT, 1M · Bayesian Networks] Answer: 26

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (2-1) \times 3 + C: (2-1) + D: (2-1) \times 2 + E: (2-1) \times 3 \times 2 \times 2 + F: (2-1) \times 3 \times 2 = 26$.

Full joint table: $\prod |X| - 1 = 96 - 1 = 95$; saving = 69.

Q.10 [MCQ, 1M · Variable Elimination] Answer: A (A: A factor over A with 2 entries: $P(j | a)$, $P(j | \neg a)$)

Evidence variables are instantiated, so only the column $J = \text{true}$ is kept, giving $f(A) = \langle P(j | a), P(j | \neg a) \rangle$. J no longer appears as a dimension. $P(j)$ would require summing over A weighted by $P(A)$, which VE does only later and only if needed.

Q.11 [MSQ, 2M · Informed Search] Answer: B, C (B: The edge J-E violates the consistency condition $h(J) \leq c(J,E) + h(E)$.; C: $h(J)$ overestimates the true cost from J to G.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	D	E	G	H	J	K	S
$h(n)$	6	9	3	0	11	8	3	12
$h^*(n)$	8	11	4	0	13	6	4	15
$h \leq h^*$?	✓	✓	✓	✓	✓	✗	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **no** (overestimates at J).

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $J \rightarrow E$ ($8 > 2 + 3$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [MCQ, 2M · Uninformed Search] Answer: B (B: $S \rightarrow B \rightarrow C \rightarrow J \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	K(6), B(7), E(7)
2	K (6)	B(7), E(7), A(13)
3	B (7)	E(7), A(12), C(12)
4	E (7)	A(12), C(12)
5	A (12)	C(12)
6	C (12)	J(17)
7	J (17)	G(23)
8	G (23)	← goal, stop

Path returned: **S** $\rightarrow$ **B** $\rightarrow$ **C** $\rightarrow$ **J** $\rightarrow$ **G**, cost = 7 + 5 + 5 + 6 = **23**. Nodes expanded (including S and G): **8** (S, K, B, E, A, C, J, G).

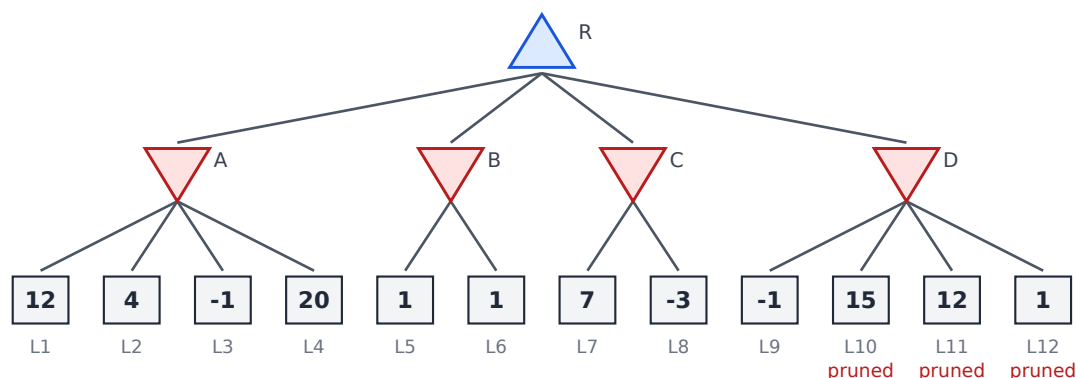
Q.13 [MCQ, 2M · Uninformed Search] Answer: B (B: $O(b^{d+1})$)

With a late goal test, BFS expands every node at depth d that precedes the goal and generates their children at depth $d+1$. That adds up to $b^{d+1} - b$ nodes, so the bound is **$O(b^{d+1})$** . Early testing (at generation) gives $O(b^d)$ (A). $O(b^{d/2})$ is the bidirectional bound (C). $O(bd)$ is a space bound for IDS (D).

Q.14 [NAT, 2M · Adversarial Search] Answer: 3

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node D: value $-1 \leq \alpha = 1 \Rightarrow$ prune remaining children (leaves L10, L11, L12).



Leaves evaluated (9): L1, L2, L3, L4, L5, L6, L7, L8, L9. Leaves pruned (3): L10, L11, L12. Root value = 1 (same as plain minimax, 1 — pruning never changes the root value).

Q.15 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $P \wedge P$; D: $\neg Q \vee Q$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
T	T	T	T	T
T	T	T	F	T

(A) $S \vee S$ is false in the KB-model $P=T, Q=T, R=T, S=F \Rightarrow$ not entailed.

(B) $\neg P \wedge S$ is false in the KB-model $P=T, Q=T, R=T, S=T \Rightarrow$ not entailed.

(C) $P \wedge P$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $\neg Q \vee Q$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.16 [NAT, 2M · Predicate Logic] **Answer: 19**

Count labelled posets on 3 elements by shape: the antichain (1); one comparable pair plus an isolated point ($3 \times 2 = 6$); a chain ($3! = 6$); one element below two incomparable ones (3); one element above two incomparable ones (3). Total = $1 + 6 + 6 + 3 + 3 = 19$, which matches brute-force enumeration of all 512 relations.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $((Q \vee P) \wedge (\neg Q \vee R)) \rightarrow (P \vee R)$; D: $((R \rightarrow P) \wedge (P \rightarrow Q)) \rightarrow (R \rightarrow Q)$)

(A) $((P \rightarrow Q) \rightarrow R) \leftrightarrow (P \rightarrow (Q \rightarrow R))$: **not valid** — counter-model $P=F, Q=T, R=F$.

(B) $\neg(P \wedge Q) \leftrightarrow (\neg P \wedge \neg Q)$: **not valid** — counter-model $P=T, Q=F$.

(C) $((Q \vee P) \wedge (\neg Q \vee R)) \rightarrow (P \vee R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $((R \rightarrow P) \wedge (P \rightarrow Q)) \rightarrow (R \rightarrow Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Predicate Logic] **Answer: A** (A: $\exists x R(x, x)$)

$R = \{(2,3), (3,2), (3,4), (4,2), (4,4)\}$.

(A) $\exists x R(x, x)$: **TRUE** — self-loop at 4.

(B) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(3,4)$ holds but $R(4,3)$ does not.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: A, C, D** (A: ϕ has no model with a domain of size 1; C: ϕ has a model with domain $\{1, 2, 3\}$; D: ϕ has a model with domain $\mathbb{N}$)

On $\{1\}$: y must be 1, and $R(1, 1) \wedge \neg R(1, 1)$ is impossible, so no model. On $\{1, 2\}$: element 1 needs y with $R(1, y) \wedge \neg R(y, 1)$; $y = 1$ is impossible, so $y = 2$: $R(1, 2), \neg R(2, 1)$. Element 2 similarly needs $R(2, 1) \wedge \neg R(1, 2)$ — contradiction; brute force over all 16 relations finds no model. On $\{1, 2, 3\}$ the 3-cycle $R = \{(1, 2), (2, 3), (3, 1)\}$ works, and on $\mathbb{N}$ $R(x, y) \Leftrightarrow x < y$ works.

Q.20 [MSQ, 2M · Bayesian Networks] Answer: A, B, D (A: $P(a) \approx 0.0307$; B: $P(b | a) \approx 0.588$; D: $P(j | b) \approx 0.725$)
 $P(a) = 0.02 \times 0.01 \times 0.95 + 0.02 \times 0.99 \times 0.90 + 0.98 \times 0.01 \times 0.30 + 0.98 \times 0.99 \times 0.01 = 0.00019 + 0.01782 + 0.00294 + 0.009702 = 0.030652$ (A true). $P(b, a) = 0.00019 + 0.01782 = 0.01801$, so $P(b | a) = 0.01801/0.030652 \approx 0.588$ (B true). $P(b | a, e) = 0.019/0.313 \approx 0.061 < 0.588$ (explaining away), so (C) is false. $P(a | b) = 0.01 \times 0.95 + 0.99 \times 0.90 = 0.9005$, so $P(j | b) = 0.9005 \times 0.8 + 0.0995 \times 0.05 = 0.7204 + 0.004975 \approx 0.725$ (D true).

Q.21 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(C, D)$, $f(C, D, E)$, $f(A, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(C, D, E)$, $f(A, E, F)$	$\{A, C, D, E, F\}$	$f(A, C, D, F)$
A	$f(A)$, $f(A, B)$, $f(A, B, C)$, $f(A, C, D, F)$	$\{A, B, C, D, F\}$	$f(B, C, D, F)$
C	$f(C, D)$, $f(B, C, D, F)$	$\{B, C, D, F\}$	$f(B, D, F)$
D	$f(B, D, F)$	$\{B, D, F\}$	$f(B, F)$
F	$f(B, F)$	$\{B, F\}$	$f(B)$

The largest factor produced has **4** variable(s) (created when eliminating E); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over B and normalise.

Q.22 [MCQ, 2M · Sampling] Answer: A (A: $A=F$, B=T, C=F, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.5$; $u = 0.69 \geq 0.5 \Rightarrow A = F$
- B: $P(B=T | A=F) = 0.5$; $u = 0.11 < 0.5 \Rightarrow B = T$
- C: $P(C=T | A=F) = 0.05$; $u = 0.75 \geq 0.05 \Rightarrow C = F$
- D: $P(D=T | B=T) = 0.7$; $u = 0.62 < 0.7 \Rightarrow D = T$

Sample: **(A=F, B=T, C=F, D=T)**.

Q.23 [MSQ, 2M · Conditional Independence] Answer: B, D (B: $E \perp H | A$; D: $A \perp B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $G \perp B | H$: **FALSE** — the path $G \leftarrow E \leftarrow A \rightarrow H \leftarrow B$ is active given $\{H\}$.

(B) $E \perp H | A$: **TRUE** — $E \leftarrow A \rightarrow D \leftarrow B \rightarrow H$ (blocked: observed fork node A); $E \leftarrow A \rightarrow H$ (blocked: observed fork node A); $E \rightarrow F \leftarrow A \rightarrow D \leftarrow B \rightarrow H$ (blocked: collider F with neither it nor a descendant observed); $E \rightarrow F \leftarrow A \rightarrow H$ (blocked: collider F with neither it nor a descendant observed).

(C) $B \perp G | D$: **FALSE** — the path $B \rightarrow D \leftarrow A \rightarrow E \rightarrow G$ is active given $\{D\}$.

(D) $A \perp B$: **TRUE** — $A \rightarrow D \leftarrow B$ (blocked: collider D with neither it nor a descendant observed); $A \rightarrow H \leftarrow B$ (blocked: collider H with neither it nor a descendant observed).

Q.24 [NAT, 2M · Exact Inference] Answer: 0.598 to 0.602

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler}) \cdot P(\text{Rain} | \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{WetGrass} | \text{Cloudy}, \text{Sprinkler})$.

Inference by enumeration: $P(\text{Cloudy} | e) = \alpha \sum_{\text{hidden}} P(\text{Cloudy}, e, \text{hidden})$.

Hidden: Ra	Cl	product of CPT entries (Cl · Sp · Ra · We)	value
T	T	$0.5 \times 0.75 \times 0.25 \times 0.75$	0.07031
F	T	$0.5 \times 0.75 \times 0.75 \times 0.75$	0.21094
T	F	$0.5 \times 0.75 \times 0.4 \times 0.5$	0.075
F	F	$0.5 \times 0.75 \times 0.6 \times 0.5$	0.1125

$P(\text{Cloudy}=T, e) = 0.28125$; $P(\text{Cloudy}=F, e) = 0.1875$; $P(e) = 0.46875$.

$P(\text{Cloudy}=T | e) = 0.28125 / 0.46875 = \mathbf{0.6} \approx 0.600$.

Q.25 [NAT, 2M · Sampling] Answer: 0.748 to 0.752

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	D
1	$P(A=T)=0.05$	0.05	T
2	$P(A=T)=0.05$	0.05	T
3	$P(A=T)=0.05$	0.05	F

#	weight $w = \prod P(e \mid \text{parents})$	w	D
4	$P(A=T)=0.05$	0.05	T

$\sum w (\text{all}) = 0.2$; $\sum w (D = T) = 0.15$.

Estimate $P(D=T \mid A = T) = 0.15 / 0.2 = \mathbf{0.75}$.

Mock Test 025

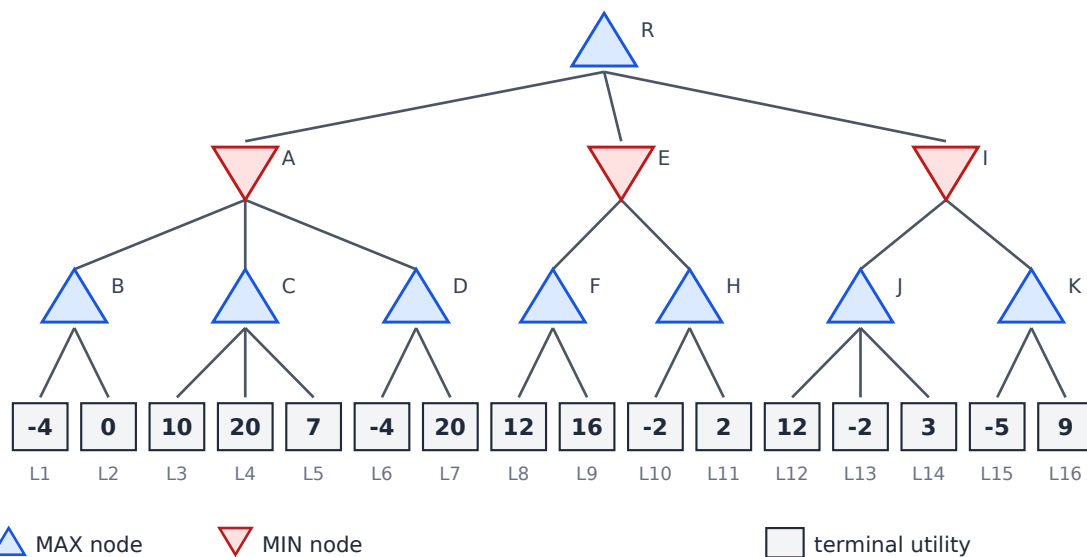
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 7 / 12	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

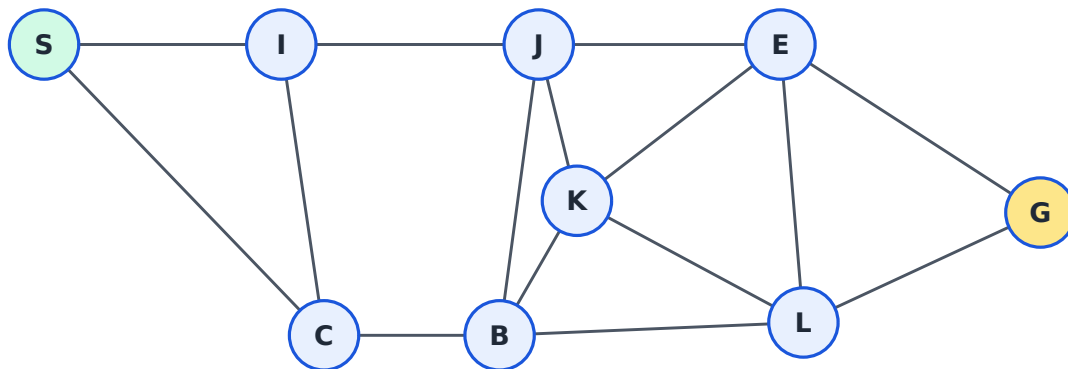
In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to E; root value 2
- ☐ (B) Move to I; root value 9
- ☐ (C) Move to A; root value 0
- ☐ (D) Move to I; root value 10

Q.2 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, C, B, J, E, I, G
- ☐ (B) S, C, I, J, B, E, K, L, G
- ☐ (C) S, J, K, L, G
- ☐ (D) S, C, B, J, I, E, G

Q.3 [MCQ · 1 mark · negative: -1/3] Informed Search

In route finding on a road map, why is the straight-line (Euclidean) distance to the destination an admissible heuristic?

- ☐ (A) Because it is easy to compute
- ☐ (B) Because the shortest route between two points can never be shorter than the straight line between them
- ☐ (C) Because roads are always straight
- ☐ (D) Because it equals the true road distance

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Effective branching factor

A* finds a solution at depth $d = 5$ after generating $N = 20$ nodes (root excluded). The effective branching factor b^* satisfies $N = b^* + (b^*)^2 + \dots + (b^*)^d$. Find b^* (round off to 2 decimal places).

Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following equivalences are valid in first-order logic?

- ☐ (A) $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
- ☐ (B) $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
- ☐ (C) $\forall x \exists y P(x, y) \equiv \exists y \forall x P(x, y)$
- ☐ (D) $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

The KB contains the clauses $\neg \text{Man}(x) \vee \text{Mortal}(x)$ and $\text{Man}(\text{Socrates})$. To prove $\text{Mortal}(\text{Socrates})$ by resolution refutation, $\neg \text{Mortal}(\text{Socrates})$ is added. What is the minimum number of resolution steps needed to derive the empty clause?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x R(x, x) \wedge \forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = (R \rightarrow (P \rightarrow \neg R)) \rightarrow (R \rightarrow \neg Q)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

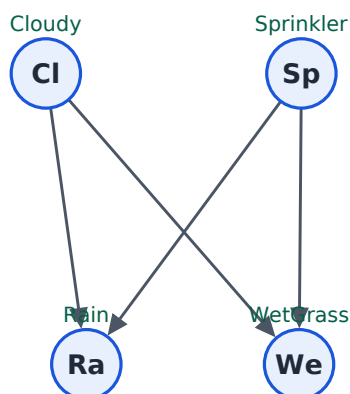
Q.9 [MCQ · 1 mark · negative: -1/3] Probability for AI

For random variables Y and Z , which expression is the conditioning rule for computing the marginal $P(Y)$?

- ☐ (A) $P(Y) = \sum_z P(Y | z) P(z)$
☐ (B) $P(Y) = \sum_z P(z | Y) P(Y)$
☐ (C) $P(Y) = \prod_z P(Y | z)$
☐ (D) $P(Y) = \sum_z P(Y | z)$

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We . (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample $\langle Cl=F, Sp=T, Ra=F, We=T \rangle$? (round off to 4 decimal places)



$P(Cl=T)$
0.3

$P(Sp=T)$	Cl	Sp	$P(Ra=T ...)$
0.2	T	T	0.3
	T	F	0.7
	F	T	0.4
	F	F	0.1

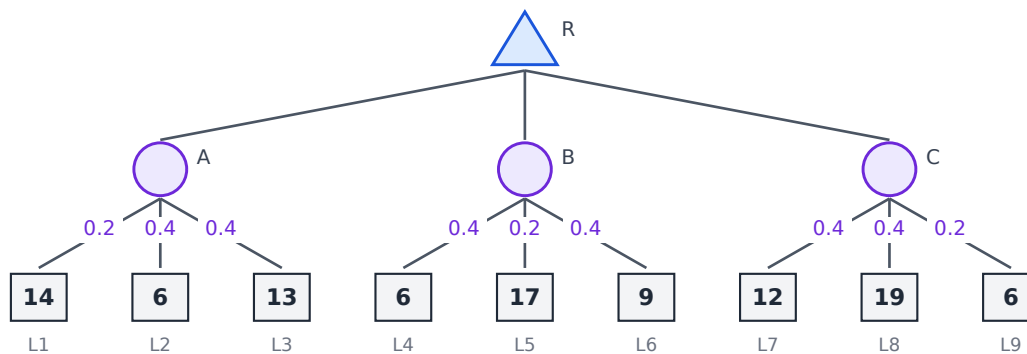
Cl	Sp	$P(We=T ...)$
T	T	0.5
T	F	0.1
F	T	0.6
F	F	0.8

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)

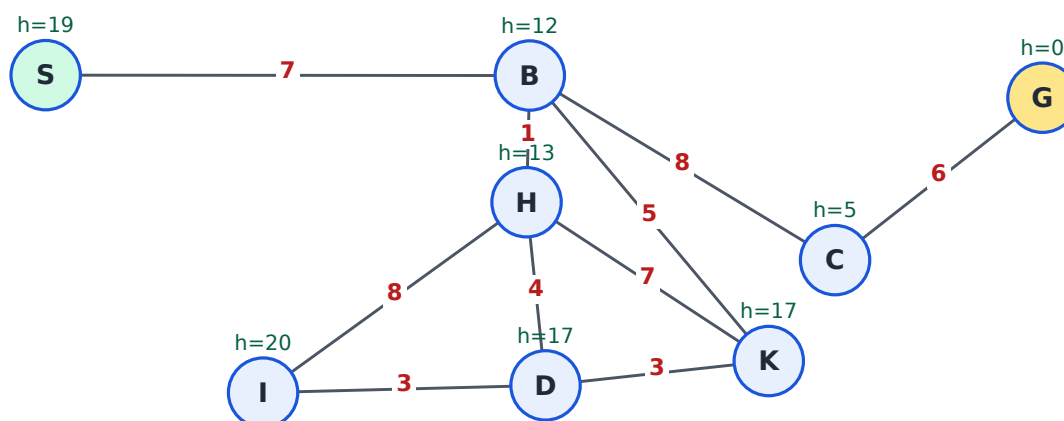


△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

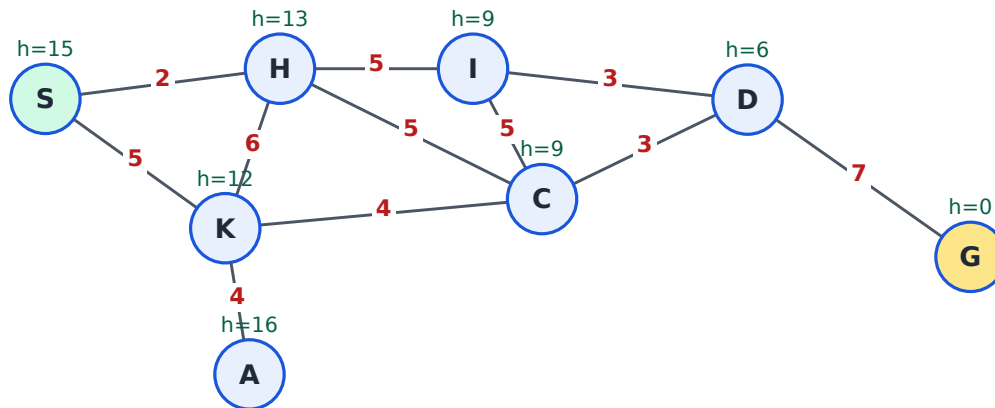
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge C-K violates the consistency condition (in at least one direction).
- ☐ (B) h is consistent (monotone).
- ☐ (C) h is admissible.
- ☐ (D) $h(K)$ overestimates the true cost from K to G.

Q.14 [MCQ · 2 marks · negative: -2/3] Adversarial Search

MAX has two moves, each leading to a chance node with two equally likely outcomes. Move A has leaf values (0, 10) and move B has leaf values (4, 5). Every leaf value is then replaced by its square root. Which move does expectimax choose before and after the transformation?

- ☐ (A) A before, A after
- ☐ (B) A before, B after
- ☐ (C) B before, A after
- ☐ (D) B before, B after

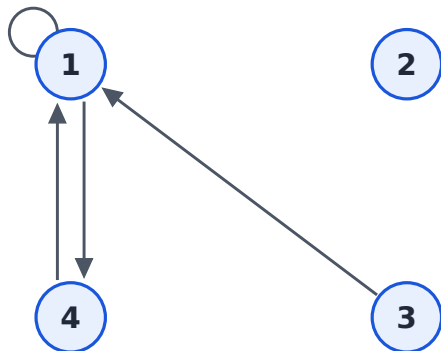
Q.15 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of two unary predicates P and Q over D (i.e. ordered pairs of subsets) satisfy $\forall x (P(x) \rightarrow Q(x)) \wedge \exists x (Q(x) \wedge \neg P(x))$?

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall y \exists x R(x, y)$
☐ (B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (D) $\exists x R(x, x)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((P \rightarrow Q) \wedge (R \rightarrow Q)) \rightarrow (P \wedge R)$
☐ (B) $(R \wedge (R \rightarrow P)) \rightarrow P$
☐ (C) $\neg(R \wedge Q) \Leftrightarrow (\neg R \vee \neg Q)$
☐ (D) $\neg(P \wedge Q) \Leftrightarrow (\neg P \wedge \neg Q)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

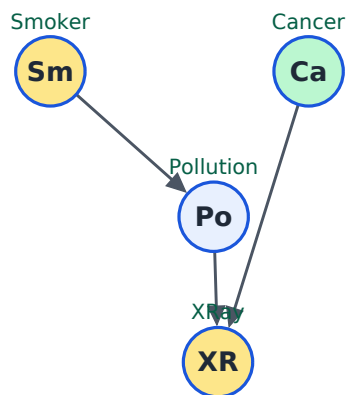
$C1: (P \vee R \vee \neg S), C2: (P \vee Q \vee \neg R), C3: (S), C4: (P \vee \neg R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(P \vee R)$
☐ (B) $(\neg P \vee S)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(P \vee Q \vee \neg S)$

Q.19 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $P(\text{Cancer} = T \mid \text{XRay} = T, \text{Smoker} = F)$ (round off to 3 decimal places).



$P(\text{Sm}=T)$
0.1

Sm	$P(\text{Po}=T \dots)$	$P(\text{Ca}=T)$
T	0.1	0.5
F	0.1	

Po	Ca	$P(\text{XR}=T \dots)$
T	T	0.5
T	F	0.3
F	T	0.25
F	F	0.95

Answer (numerical): _____

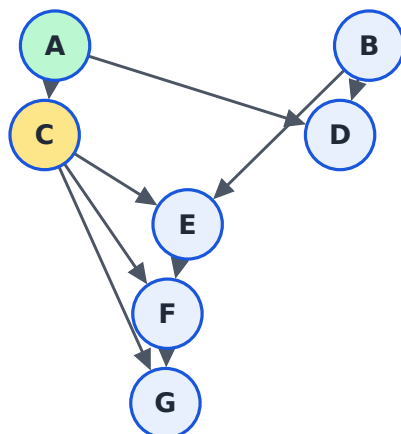
Q.20 [MCQ · 2 marks · negative: -2/3] Bayesian Networks

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. What is $P(r \mid w)$ (rounded to 3 decimal places)?

- ☐ (A) 0.708
☐ (B) 0.647
☐ (C) 0.430
☐ (D) 0.800

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(A \mid C = T)$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **G, F, E, B, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

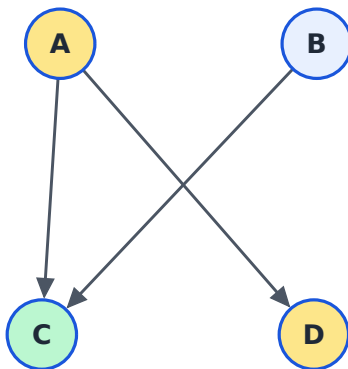
Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence

Consider the Bayesian network (DAG) with edges $P \rightarrow Q$, $P \rightarrow R$, $Q \rightarrow S$, $R \rightarrow S$, $S \rightarrow T$, $U \rightarrow R$. Which of the following independences are guaranteed by d-separation (i.e., hold in every distribution that factorizes over this DAG)?

- ☐ (A) $Q \perp R \mid \{P\}$
☐ (B) $Q \perp R \mid \{P, T\}$
☐ (C) $U \perp P$
☐ (D) $U \perp Q \mid \{P, S\}$

Q.23 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{A} = \mathbf{T}$, $\mathbf{D} = \mathbf{T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P}(\mathbf{C} = \mathbf{T} \mid \mathbf{A} = \mathbf{T}, \mathbf{D} = \mathbf{T})$ (round off to 3 decimal places).



P(A=T)
0.8

P(B=T)	A	B	P(C=T ...)
0.4	T	T	0.9
	T	F	0.2
	F	T	0.9
	F	F	0.5

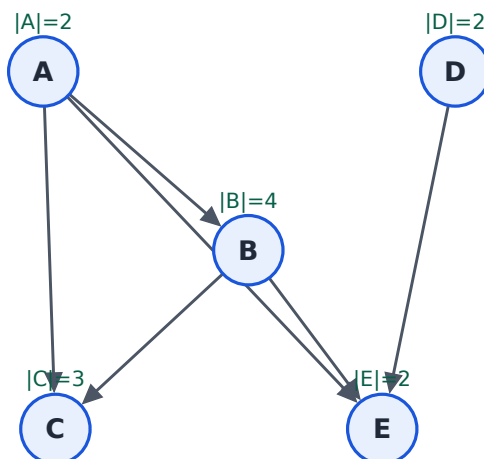
A	P(D=T ...)
T	0.2
F	0.95

#	B	C
1	T	T
2	T	T
3	T	T
4	F	F

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

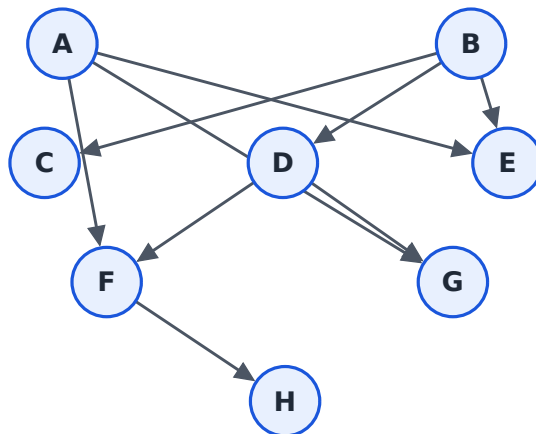
The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp B$
☐ (B) $B \perp H \mid D, F$
☐ (C) $A \perp C \mid B, D$
☐ (D) $H \perp C \mid E$

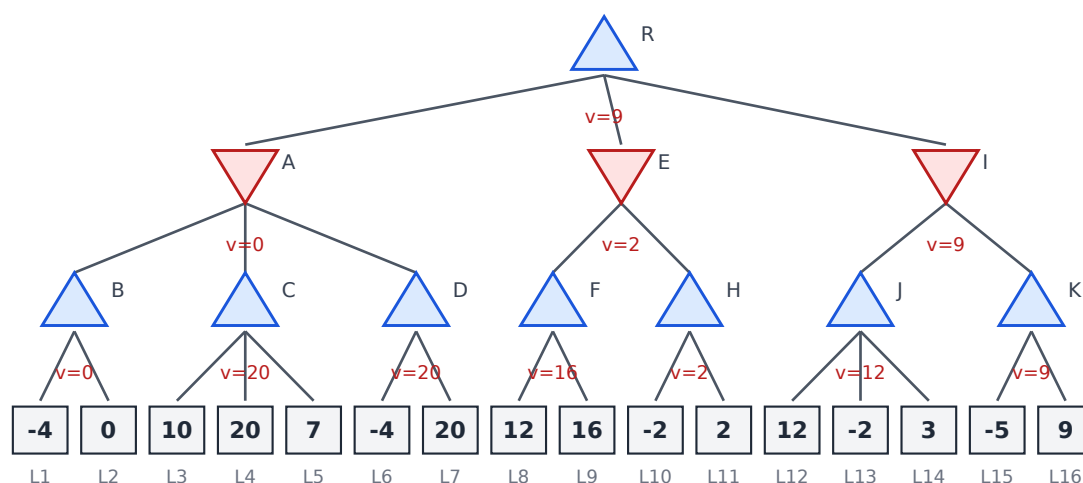
Answer Key — Mock Test 025

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	B	14	MCQ	2	Adversarial Search	B
2	MCQ	1	Uninformed Search	D	15	NAT	2	Predicate Logic	19
3	MCQ	1	Informed Search	B	16	MSQ	2	Predicate Logic	D
4	NAT	1	Informed Search	1.48 to 1.52	17	MSQ	2	Propositional Logic	B, C
5	MSQ	1	Predicate Logic	A, B, D	18	MSQ	2	Propositional Logic	A, D
6	NAT	1	Predicate Logic	2	19	NAT	2	Exact Inference	0.235 to 0.239
7	NAT	1	Predicate Logic	8	20	MCQ	2	Bayesian Networks	A
8	NAT	1	Propositional Logic	1	21	NAT	2	Exact Inference	2
9	MCQ	1	Probability for AI	A	22	MSQ	2	Conditional Independence	A, C
10	NAT	1	Sampling	0.0502 to 0.0506	23	NAT	2	Sampling	0.748 to 0.752
11	NAT	2	Adversarial Search	13.59 to 13.61	24	NAT	2	Bayesian Networks	55
12	NAT	2	Informed Search	4	25	MSQ	2	Conditional Independence	B, C
13	MSQ	2	Informed Search	B, C					

Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] **Answer: B** (B: Move to I; root value 9)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 0$, $C = 20$, $D = 20$, $A = 0$, $F = 16$, $H = 2$, $E = 2$, $J = 12$, $K = 9$, $I = 9$. Root value = **9**; MAX's optimal first move is towards I.

Q.2 [MCQ, 1M · Uninformed Search] Answer: D (D: S, C, B, J, I, E, G)

Adjacency lists (alphabetical): B: {C, J, K, L}; C: {B, I, S}; E: {G, I, K, L}; G: {E, L}; I: {C, E, J, S}; J: {B, I, K, S}; K: {B, E, J, L}; L: {B, E, G, K}; S: {C, I, J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, C, B, J, I, E, G**; the path found is $S \rightarrow C \rightarrow B \rightarrow J \rightarrow I \rightarrow E \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.3 [MCQ, 1M · Informed Search] Answer: B (B: Because the shortest route between two points can never be shorter than the straight line between them)

Admissibility needs $h(n) \leq h^*(n)$. Any road route is a path in the plane, and no path is **shorter than the straight line** between its endpoints, so the heuristic never overestimates. Being cheap to compute (A) is useful but irrelevant to admissibility. Roads are generally not straight (C), and the heuristic generally underestimates rather than equals the road distance (D).

Q.4 [NAT, 1M · Informed Search] Answer: 1.48 to 1.52

Solve $b + b^2 + \dots + b^5 = 20$ numerically (bisection): $b^* \approx 1.5$.

Check: $1.5^1 + 1.5^2 + 1.5^3 + 1.5^4 + 1.5^5 \approx 19.78 \approx 20$.

Q.5 [MSQ, 1M · Predicate Logic] Answer: A, B, D (A: $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$; B: $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$; D: $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$)

Quantifiers of the same kind commute (A, B). $\exists$ distributes over $\vee$ (D), just as $\forall$ distributes over $\wedge$. (C) Mixed quantifiers do not commute: with $P(x, y) \iff x = y$ on $\{1, 2\}$, $\forall x \exists y P$ is true but $\exists y \forall x P$ is false.

Q.6 [NAT, 1M · Predicate Logic] Answer: 2

Step 1: resolve $\neg \text{Mortal}(\text{Socrates})$ with $\neg \text{Man}(x) \vee \text{Mortal}(x)$ using $\{x/\text{Socrates}\}$, giving $\neg \text{Man}(\text{Socrates})$. Step 2: resolve $\neg \text{Man}(\text{Socrates})$ with $\text{Man}(\text{Socrates})$, giving $\square$. One step cannot suffice since no two initial clauses are complementary units. So the answer is 2.

Q.7 [NAT, 1M · Predicate Logic] Answer: 8

View R as an 3×3 0/1 matrix. The sentence says R is reflexive and symmetric: only the $n(n-1)/2$ unordered off-diagonal pairs are free $\Rightarrow 2^{n(n-1)/2}$. With $n = 3$: **8**.

Q.8 [NAT, 1M · Propositional Logic] Answer: 1

Build the truth table (sub-formulas shown first):

P	Q	R	$R \rightarrow (P \rightarrow \neg R)$	$R \rightarrow \neg Q$	$(R \rightarrow (P \rightarrow \neg R)) \rightarrow (R \rightarrow \neg Q)$
T	T	T	F	F	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	T	T	T
F	T	T	T	F	F

P	Q	R	$R \rightarrow (P \rightarrow \neg R)$	$R \rightarrow \neg Q$	$(R \rightarrow (P \rightarrow \neg R)) \rightarrow (R \rightarrow \neg Q)$
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

F has 7 models, so it is false in $8 - 7 = 1$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.9 [MCQ, 1M · Probability for AI] Answer: A (A: $P(Y) = \sum_z P(Y | z) P(z)$)

Marginalization gives $P(Y) = \sum_z P(Y, z)$, and the product rule rewrites $P(Y, z) = P(Y | z)P(z)$; together this is the conditioning rule (A). (B) equals $P(Y) \sum_z P(z | Y) = P(Y)$, which is circular since it needs $P(Y)$. (C) multiplies probabilities and is not a marginal. (D) omits the weights $P(z)$ and need not even be ≤ 1 .

Q.10 [NAT, 1M · Sampling] Answer: 0.0502 to 0.0506

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(CI=F) \times P(Sp=T) \times P(Ra=F|CI=F, Sp=T) \times P(We=T|CI=F, Sp=T)$
 $= 0.7 \times 0.2 \times 0.6 \times 0.6 = \mathbf{0.0504} \approx 0.0504$.

Q.11 [NAT, 2M · Adversarial Search] Answer: 13.59 to 13.61

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.2 \times 14 + 0.4 \times 6 + 0.4 \times 13 = 10.4$
- $B = 0.4 \times 6 + 0.2 \times 17 + 0.4 \times 9 = 9.4$
- $C = 0.4 \times 12 + 0.4 \times 19 + 0.2 \times 6 = 13.6$

Root = $\max(10.4, 9.4, 13.6) = \mathbf{13.6}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.12 [NAT, 2M · Informed Search] Answer: 4

Heuristic values: $h(B)=12$, $h(C)=5$, $h(D)=17$, $h(G)=0$, $h(H)=13$, $h(I)=20$, $h(K)=17$, $h(S)=19$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (19)	B(19)
2	B (19)	C(20), H(21), K(29)
3	C (20)	G(21), H(21), K(29)
4	G (21)	H(21), K(29) ← goal, stop

Path returned: **S** → **B** → **C** → **G** with cost **21**; nodes expanded = **4** (S, B, C, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 21$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.13 [MSQ, 2M · Informed Search] Answer: B, C (B: h is consistent (monotone).; C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	C	D	G	H	I	K	S
$h(n)$	16	9	6	0	13	9	12	15
$h^*(n)$	18	10	7	0	15	10	14	17
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [MCQ, 2M · Adversarial Search] Answer: B (B: A before, B after)

Before: $E[A] = (0+10)/2 = 5$ and $E[B] = 4.5$, so A is chosen. After: $E[A] = (0 + 3.162)/2 \approx 1.58$ and $E[B] = (2 + 2.236)/2 \approx 2.12$, so B is chosen. Expectimax decisions are preserved only by positive linear transformations of the evaluation function. A merely order-preserving map such as $\sqrt{x}$ can change them. Minimax decisions are invariant under any order-preserving map.

Q.15 [NAT, 2M · Predicate Logic] **Answer: 19**

$\forall x (P(x) \rightarrow Q(x))$ leaves each element 3 choices of (P, Q): (F,F), (F,T), (T,T); so $3^3 = 27$ pairs. The second conjunct requires at least one element with (F,T). Pairs with no (F,T) element: $2^3 = 8$. Answer = $27 - 8 = 19$ (P is a proper subset of Q).

Q.16 [MSQ, 2M · Predicate Logic] **Answer: D** (D: $\exists x R(x, x)$)

$R = \{(1,1), (1,4), (3,1), (4,1)\}$.

(A) $\forall y \exists x R(x, y)$: **FALSE** — 2 has no incoming edge.

(B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,2)$ nor $R(2,1)$.

(C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(3,1)$ holds but $R(1,3)$ does not.

(D) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: B, C** (B: $(R \wedge (R \rightarrow P)) \rightarrow P$; C: $\neg(R \wedge Q) \leftrightarrow (\neg R \vee \neg Q)$)

(A) $((P \rightarrow Q) \wedge (R \rightarrow Q)) \rightarrow (P \wedge R)$: **not valid** — counter-model $P=T, Q=T, R=F$.

(B) $(R \wedge (R \rightarrow P)) \rightarrow P$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $\neg(R \wedge Q) \leftrightarrow (\neg R \vee \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $\neg(P \wedge Q) \leftrightarrow (\neg P \wedge \neg Q)$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(P \vee R)$; D: $(P \vee Q \vee \neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(P \vee R)$ is derivable:

1. resolve $(P \vee R \vee \neg S)$ and (S) on S $\Rightarrow (P \vee R)$

(B) $(\neg P \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) The resolution closure contains 9 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(D) $(P \vee Q \vee \neg S)$ is derivable:

1. resolve $(P \vee R \vee \neg S)$ and $(P \vee Q \vee \neg R)$ on R $\Rightarrow (P \vee Q \vee \neg S)$

Q.19 [NAT, 2M · Exact Inference] **Answer: 0.235 to 0.239**

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution} | \text{Smoker}) \cdot P(\text{Cancer}) \cdot P(\text{XRay} | \text{Pollution}, \text{Cancer})$.

Inference by enumeration: $P(\text{Cancer} | e) = \alpha \sum_{\text{hidden}} P(\text{Cancer}, e, \text{hidden})$.

Hidden: Po	Ca	product of CPT entries (Sm · Po · Ca · XR)	value
T	T	$0.9 \times 0.1 \times 0.5 \times 0.5$	0.0225
F	T	$0.9 \times 0.9 \times 0.5 \times 0.25$	0.10125
T	F	$0.9 \times 0.1 \times 0.5 \times 0.3$	0.0135
F	F	$0.9 \times 0.9 \times 0.5 \times 0.95$	0.38475

$P(\text{Cancer}=T, e) = 0.12375$; $P(\text{Cancer}=F, e) = 0.39825$; $P(e) = 0.522$.

$P(\text{Cancer}=T | e) = 0.12375 / 0.522 = \mathbf{0.2371} \approx 0.237$.

Q.20 [MCQ, 2M · Bayesian Networks] **Answer: A** (A: 0.708)

Compute $P(r, w) = \sum_{c,s} P(c)P(s | c)P(r | c)P(w | s, r)$. With c: $P(r | c) = 0.8$ and $P(w | r)$ averaged over S:

$0.1 \times 0.99 + 0.9 \times 0.90 = 0.909 \rightarrow 0.5 \times 0.8 \times 0.909 = 0.3636$. With $\neg c$: $P(r | \neg c) = 0.2$ and $0.5 \times 0.99 + 0.5 \times 0.90 = 0.945 \rightarrow 0.5 \times 0.2 \times 0.945 = 0.0945$. So $P(r, w) = 0.4581$. $P(\neg r, w)$: c: $0.5 \times 0.2 \times (0.1 \times 0.9 + 0.9 \times 0) = 0.009$; $\neg c$: $0.5 \times 0.8 \times (0.5 \times 0.9) = 0.18$; total 0.189. $P(w) = 0.6471$. $P(r | w) = 0.4581 / 0.6471 \approx 0.708$. 0.647 is $P(w)$, 0.430 is $P(s | w)$, 0.800 is $P(r | c)$.

Q.21 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A)$, $f(A, B, D)$, $f(B, E)$, $f(E, F)$, $f(F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(F, G)$	$\{F, G\}$	$f(F)$
F	$f(E, F)$, $f(F)$	$\{E, F\}$	$f(E)$
E	$f(B, E)$, $f(E)$	$\{B, E\}$	$f(B)$

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(B), f(A, B, D), f(B)$	$\{A, B, D\}$	$f(A, D)$
D	$f(A, D)$	$\{A, D\}$	$f(A)$

The largest factor produced has **2** variable(s) (created when eliminating B); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.22 [MSQ, 2M · Conditional Independence] **Answer: A, C** (A: $Q \perp R \mid \{P\}$; C: $U \perp P$)

Paths between Q and R: $Q \leftarrow P \rightarrow R$ (fork at P) and $Q \rightarrow S \leftarrow R$ (collider S). (A) Given P, the fork is blocked and the collider S has no observed descendant: d-separated, true. (B) T is a descendant of S, so observing T activates $Q \rightarrow S \leftarrow R$: false. (C) Paths from U to P all pass through R as a collider ($U \rightarrow R \leftarrow P$, or $U \rightarrow R \rightarrow S \leftarrow P \leftarrow P$ with collider S); with nothing observed they are blocked: true. (D) Given S, the path $U \rightarrow R \rightarrow S \leftarrow Q$ has observed collider S and unobserved non-collider R, so it is active: false.

Q.23 [NAT, 2M · Sampling] **Answer: 0.748 to 0.752**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	C
1	$P(A=T)=0.8 \times P(D=T A=T)=0.2$	0.16	T
2	$P(A=T)=0.8 \times P(D=T A=T)=0.2$	0.16	T
3	$P(A=T)=0.8 \times P(D=T A=T)=0.2$	0.16	T
4	$P(A=T)=0.8 \times P(D=T A=T)=0.2$	0.16	F

Σw (all) = 0.64; Σw (C = T) = 0.48.

Estimate $P(C=T \mid A = T, D = T) = 0.48 / 0.64 = \mathbf{0.75}$.

Q.24 [NAT, 2M · Bayesian Networks] **Answer: 55**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (4-1) \times 2 + C: (3-1) \times 2 \times 4 + D: (2-1) + E: (2-1) \times 2 \times 4 \times 2 = \mathbf{40}$.

Full joint table: $\prod |X| - 1 = 96 - 1 = 95$; saving = 55.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: B, C** (B: $B \perp H \mid D, F$; C: $A \perp C \mid B, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp B$: **FALSE** — the path $C \leftarrow B$ is active given $\{\}$.

(B) $B \perp H \mid D, F$: **TRUE** — $B \rightarrow D \rightarrow F \rightarrow H$ (blocked: observed chain node D); $B \rightarrow D \rightarrow G \leftarrow A \rightarrow F \rightarrow H$ (blocked: observed chain node D); $B \rightarrow E \leftarrow A \rightarrow F \rightarrow H$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow E \leftarrow A \rightarrow G \leftarrow D \rightarrow F \rightarrow H$ (blocked: collider E with neither it nor a descendant observed).

(C) $A \perp C \mid B, D$: **TRUE** — $A \rightarrow E \leftarrow B \rightarrow C$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow F \leftarrow D \leftarrow B \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow G \leftarrow D \leftarrow B \rightarrow C$ (blocked: collider G with neither it nor a descendant observed).

(D) $H \perp C \mid E$: **FALSE** — the path $H \leftarrow F \leftarrow A \rightarrow E \leftarrow B \rightarrow C$ is active given $\{E\}$.

Mock Test 026

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	10 / 6 / 9	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

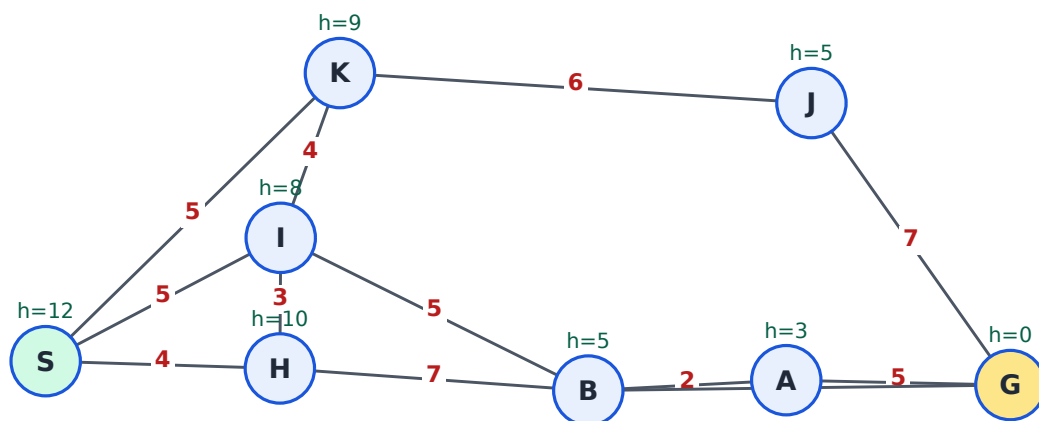
Q.1 [MCQ · 1 mark · negative: $-1/3$] Problem Formulation

In a search problem, the state space is defined implicitly by

- ☐ (A) the initial state, the actions, and the transition model
- ☐ (B) the goal test only
- ☐ (C) the heuristic function
- ☐ (D) the path cost function only

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

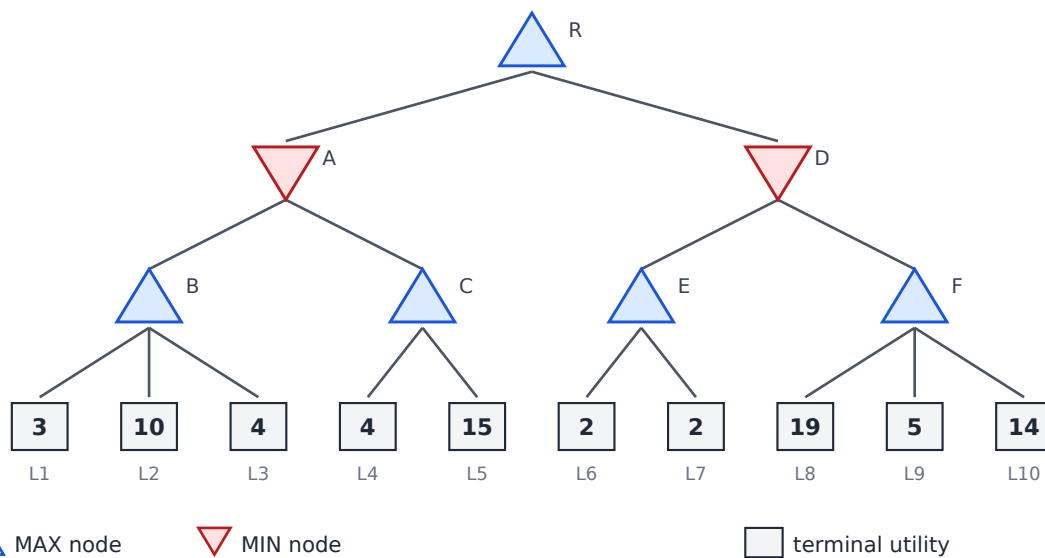
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

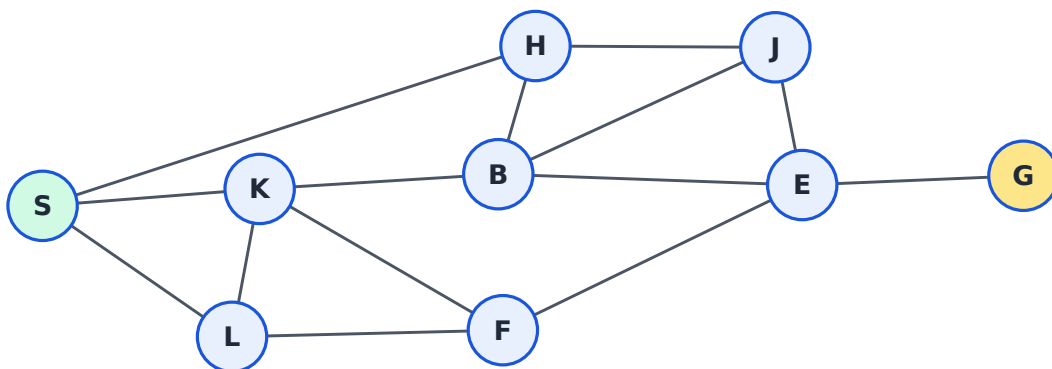
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, H, K, L, B, J, F, E, G
- ☐ (B) S, L, K, H, F, B, J, E, G
- ☐ (C) S, H, B, E, F, K, L, G
- ☐ (D) S, H, L, K, B, J, F, E, G

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following is logically equivalent to $p \rightarrow q$?

- ☐ (A) $\neg q \rightarrow \neg p$
- ☐ (B) $q \rightarrow p$
- ☐ (C) $\neg p \rightarrow \neg q$
- ☐ (D) $p \wedge \neg q$

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following formulas is satisfiable but NOT valid?

- ☐ (A) $p \rightarrow (q \rightarrow p)$
- ☐ (B) $p \wedge \neg p$
- ☐ (C) $(p \wedge q) \rightarrow (p \vee q)$
- ☐ (D) $(p \rightarrow q) \rightarrow p$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic - Model counting

The formula $F = ((P \wedge Q) \rightarrow (R \leftrightarrow R)) \wedge (\neg P \rightarrow (\neg Q \leftrightarrow P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

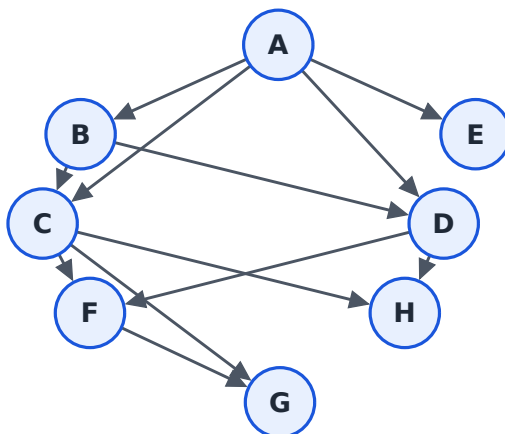
Q.8 [MCQ · 1 mark · negative: -1/3] Probability for AI

Given $P(b | a) = 0.8$, $P(a) = 0.25$ and $P(b) = 0.4$, what is $P(a | b)$?

- ☐ (A) 0.50
- ☐ (B) 0.32
- ☐ (C) 0.20
- ☐ (D) 0.80

Q.9 [MCQ · 1 mark · negative: -1/3] Conditional Independence - Markov blanket

Which set is the Markov blanket of **B** in the Bayesian network below?



- ☐ (A) $\{A, C, D, E\}$
- ☐ (B) $\{A, C, D\}$
- ☐ (C) $\{A, C, D, F, G, H\}$
- ☐ (D) $\{C, D\}$

Q.10 [MCQ · 1 mark · negative: -1/3] Variable Elimination

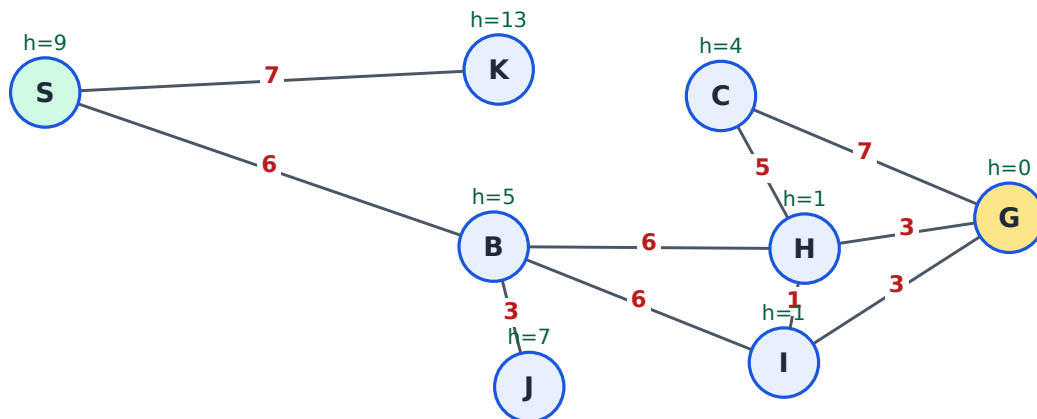
In variable elimination, $f_1(A, B)$ and $f_2(B, C)$ are factors over Boolean variables. Their pointwise product $f_1 \times f_2$ is

- ☐ (A) a factor over $\{A, B, C\}$ with 8 entries
- ☐ (B) a factor over $\{A, C\}$ with 4 entries
- ☐ (C) a factor over $\{B\}$ with 2 entries
- ☐ (D) a factor over $\{A, B, B, C\}$ with 16 entries

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – A*

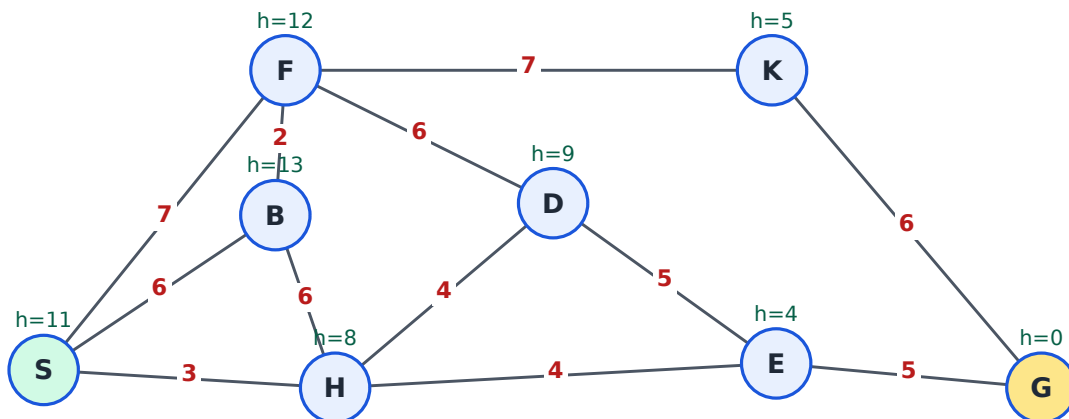
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) The path returned has cost 15.
☐ (B) Node I is expanded.
☐ (C) Node C is expanded.
☐ (D) Node H is expanded.

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

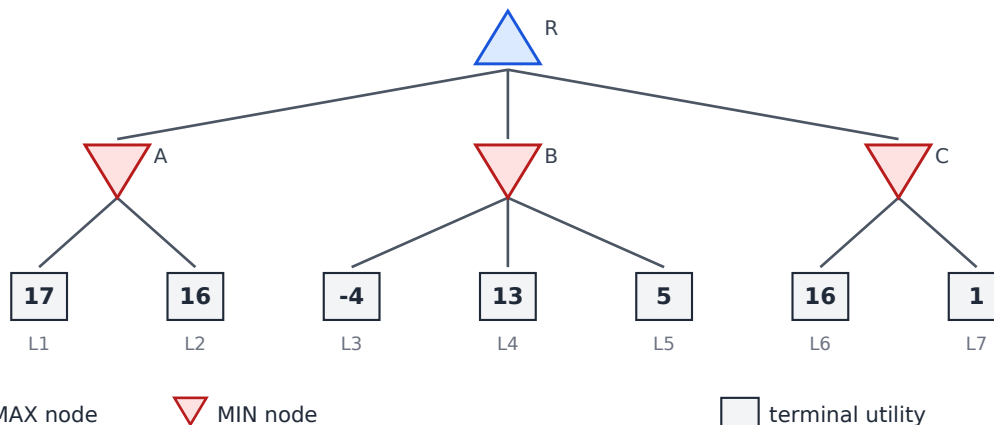
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



Answer (numerical): _____

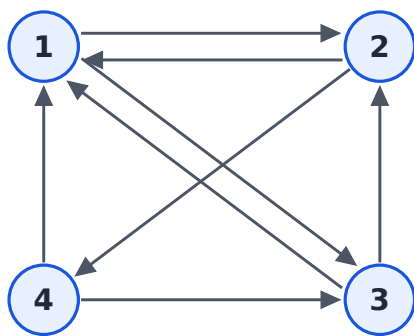
Q.14 [MSQ · 2 marks · no negative marking] Informed Search

Assume $h(\text{goal}) = 0$ and all step costs are positive. Which of the following are TRUE?

- ☐ (A) Every consistent heuristic is admissible
- ☐ (B) Every admissible heuristic is consistent
- ☐ (C) If h is consistent, the values of $f(n) = g(n) + h(n)$ along any path are non-decreasing
- ☐ (D) A* tree search with an admissible heuristic is optimal

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
- ☐ (B) $\forall x R(x, x)$
- ☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
- ☐ (D) $\forall x \exists y R(x, y)$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, D, A** and rules:

R1: $D \wedge F \wedge E \rightarrow C$

R2: $C \rightarrow H$

R3: $D \wedge A \rightarrow H$

R4: $A \wedge H \rightarrow E$

R5: $H \rightarrow A$

R6: $E \rightarrow H$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) C is entailed
- ☐ (B) F is entailed
- ☐ (C) H is entailed
- ☐ (D) K is entailed

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

(1) $(P \vee \neg P) \wedge (\neg Q \leftrightarrow P)$

(2) $R \wedge R$

(3) $\neg Q \vee \neg S$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $R \vee P$
- ☐ (B) $\neg Q \vee \neg S$
- ☐ (C) $\neg R \leftrightarrow R$
- ☐ (D) $\neg Q \wedge (\neg R \rightarrow Q)$

Q.18 [NAT · 2 marks · no negative marking] Propositional Logic

How many of the 32 truth assignments to $x_1, \dots, x_5$ satisfy $(x_1 \rightarrow x_2) \wedge (x_2 \rightarrow x_3) \wedge (x_3 \rightarrow x_4) \wedge (x_4 \rightarrow x_5)$?

Answer (numerical): _____

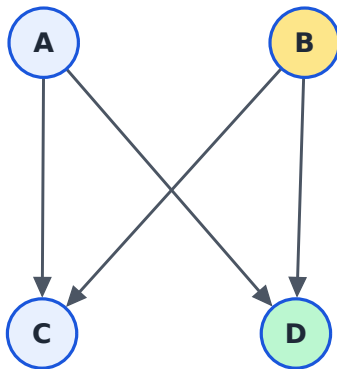
Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$
- ☐ (B) $(P \rightarrow Q) \rightarrow (Q \rightarrow P)$
- ☐ (C) $((R \rightarrow Q) \wedge \neg Q) \rightarrow \neg R$
- ☐ (D) $((Q \rightarrow P) \wedge (P \rightarrow R)) \rightarrow (Q \rightarrow R)$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(D = T \mid B = F)$ (round off to 3 decimal places).



A	P(A=T)
T	0.5

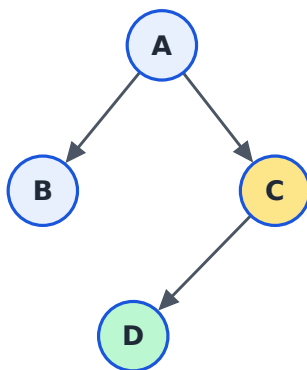
P(B=T)	A	B	P(C=T ...)
0.9	T	T	0.9
	T	F	0.75
	F	T	0.25
	F	F	0.8

A	B	P(D=T ...)
T	T	0.3
T	F	0.05
F	T	0.9
F	F	0.6

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $C=F$. The current state of the chain is $(A=T, B=F, C=F, D=T)$. The next step resamples D . With what probability is D set to T ? (round off to 3 decimal places)



A	P(A=T)
T	0.9

A	P(B=T ...)	A	P(C=T ...)
T	0.05	T	0.05
F	0.4	F	0.9

C	P(D=T ...)
T	0.05
F	0.25

Answer (numerical): _____

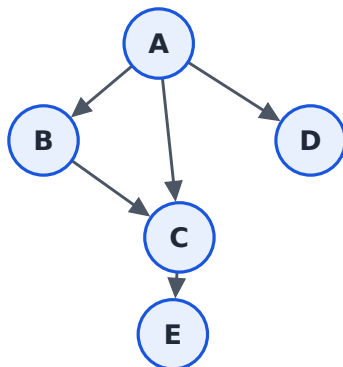
Q.22 [NAT · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. Suppose a new network is built for the same distribution by adding nodes in the order M, J, A, B, E, choosing for each node a minimal set of parents among the previously added nodes such that the node is conditionally independent of its other predecessors given its parents. How many independent CPT parameters does this new network have (all variables Boolean)? Assume no independences beyond those implied by the original structure.

Answer (numerical): _____

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.84, 0.57, 0.91, 0.76, 0.2)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



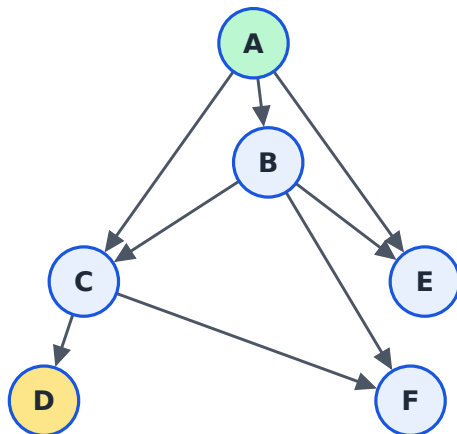
P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.6	T	0.75	T	T	0.1
	F	0.25	T	F	0.1
			F	T	0.6
			F	F	0.05

A	P(D=T ...)	C	P(E=T ...)
T	0.75	T	0.4
F	0.7	F	0.7

- ☐ (A) A=F, B=T, C=F, D=F, E=T
☐ (B) A=F, B=F, C=F, D=F, E=T
☐ (C) A=T, B=F, C=F, D=F, E=T
☐ (D) A=T, B=F, C=T, D=T, E=F

Q.24 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(A \mid D = T)}$ in the Bayesian network below (Boolean variables; evidence $D = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F**, **C**, **E**, **B**. What is the scope of the new factor produced when **F** is summed out?



- ☐ (A) $f(B, C)$
☐ (B) $f(A, B, C)$
☐ (C) $f(C)$
☐ (D) $f(B, C, F)$

Q.25 [MCQ · 2 marks · negative: -2/3] Sampling

In likelihood weighting with evidence e and non-evidence variables z , let $S_{WS}(z, e)$ be the sampling probability and $w(z, e)$ the weight. Which statement is correct?

- (A) $S_{WS}(z, e) = \prod P(z_i | \text{parents}(Z_i))$, $w(z, e) = \prod P(e_j | \text{parents}(E_j))$, and $S_{WS}(z, e) \cdot w(z, e) = P(z, e)$
 ○ (B) $S_{WS}(z, e) = P(z | e)$, so all weights equal 1
 ○ (C) $S_{WS}(z, e) = P(z, e)$ and $w(z, e) = P(e)$
 ○ (D) $S_{WS}(z, e) = \prod P(e_j | \text{parents}(E_j))$, $w(z, e) = \prod P(z_i | \text{parents}(Z_i))$

Answer Key — Mock Test 026

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Problem Formulation	A	14	MSQ	2	Informed Search	A, C, D
2	NAT	1	Informed Search	18	15	MSQ	2	Predicate Logic	C, D
3	NAT	1	Adversarial Search	10	16	MSQ	2	Propositional Logic	C
4	MCQ	1	Uninformed Search	A	17	MSQ	2	Propositional Logic	A, B
5	MCQ	1	Propositional Logic	A	18	NAT	2	Propositional Logic	6
6	MCQ	1	Propositional Logic	D	19	MSQ	2	Propositional Logic	C, D
7	NAT	1	Propositional Logic	2	20	NAT	2	Exact Inference	0.323 to 0.327
8	MCQ	1	Probability for AI	A	21	NAT	2	Sampling	0.248 to 0.252
9	MCQ	1	Conditional Independence	B	22	NAT	2	Bayesian Networks	13
10	MCQ	1	Variable Elimination	A	23	MCQ	2	Sampling	B
11	MSQ	2	Informed Search	A, B, D	24	MCQ	2	Exact Inference	A
12	NAT	2	Informed Search	4	25	MCQ	2	Sampling	A
13	NAT	2	Adversarial Search	3					

Detailed Solutions**Q.1** [MCQ, 1M · Problem Formulation] **Answer: A** (A: the initial state, the actions, and the transition model)

Starting from the **initial state** and applying the **actions** through the **transition model** generates every reachable state and the edges between them, which is the state-space graph. The goal test (B) only identifies goal states. The heuristic (C) guides search but does not define states. The path cost (D) only labels edges.

Q.2 [NAT, 1M · Informed Search] **Answer: 18**

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

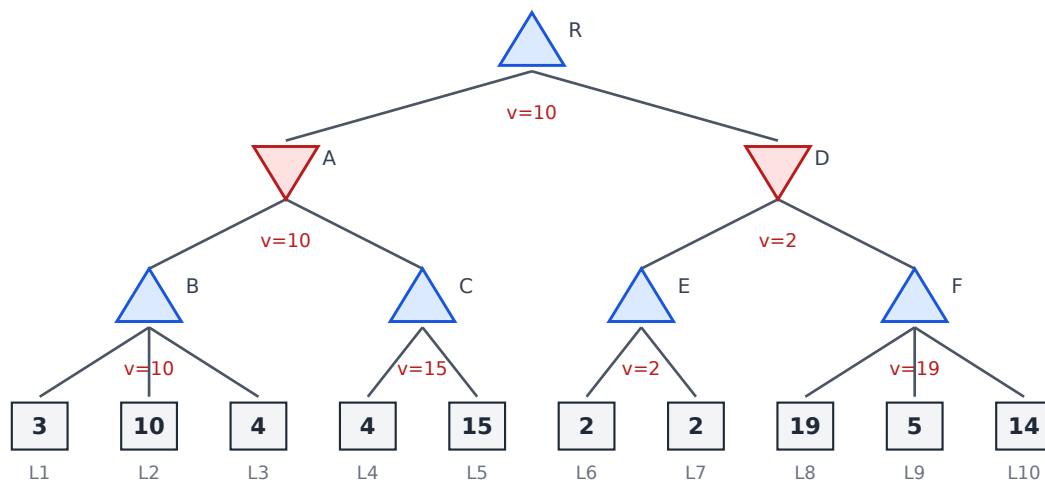
Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (12)	I(8), K(9), H(10)
2	I (8)	B(5), K(9), H(10)
3	B (5)	G(0), A(3), K(9), H(10)
4	G (0)	A(3), K(9), H(10) ← goal, stop

Path returned: **S** → **I** → **B** → **G**, cost = 5 + 5 + 8 = **18**.

The optimal cost is $h^*(S) = 17$ — greedy search is not optimal here.

Q.3 [NAT, 1M · Adversarial Search] **Answer: 10**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 10, C = 15, A = 10, E = 2, F = 19, D = 2. Root value = **10**; MAX's optimal first move is towards A.

Q.4 [MCQ, 1M · Uninformed Search] Answer: A (A: S, H, K, L, B, J, F, E, G)

Adjacency lists (alphabetical): B: {E, H, J, K}; E: {B, F, G, J}; F: {E, K, L}; G: {E}; H: {B, J, S}; J: {B, E, H}; K: {B, F, L, S}; L: {F, K, S}; S: {H, K, L}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[H, K, L]
2	H	[K, L, B, J]
3	K	[L, B, J, F]
4	L	[B, J, F]
5	B	[J, F, E]
6	J	[F, E]
7	F	[E]
8	E	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, H, K, L, B, J, F, E, G**. The path returned is S → H → B → E → G with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [MCQ, 1M · Propositional Logic] Answer: A (A: $\neg q \rightarrow \neg p$)

The contrapositive $\neg q \rightarrow \neg p \equiv q \vee \neg p \equiv p \rightarrow q$. The converse $q \rightarrow p$ and the inverse $\neg p \rightarrow \neg q$ are equivalent to each other but differ from $p \rightarrow q$ at $p=F, q=T$ ($p \rightarrow q$ is T, $q \rightarrow p$ is F). $p \wedge \neg q$ is the negation of $p \rightarrow q$.

Q.6 [MCQ, 1M · Propositional Logic] Answer: D (D: $(p \rightarrow q) \rightarrow p$)

$(p \rightarrow q) \rightarrow p$: when $p=T$ it is true (consequent T); when $p=F$, $p \rightarrow q$ is T and the consequent is F, so it is false. Hence it is satisfiable but not valid. $p \rightarrow (q \rightarrow p)$ is valid (if p is T the consequent is T). $p \wedge \neg p$ is unsatisfiable. $(p \wedge q) \rightarrow (p \vee q)$ is valid because $p \wedge q \models p \vee q$.

Q.7 [NAT, 1M · Propositional Logic] Answer: 2

Build the truth table (sub-formulas shown first):

P	Q	R	$(P \wedge Q) \rightarrow (R \leftrightarrow R)$	$\neg P \rightarrow (\neg Q \leftrightarrow P)$	$((P \wedge Q) \rightarrow (R \leftrightarrow R)) \wedge (\neg P \rightarrow (\neg Q \leftrightarrow P))$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	T	F	F

F has 6 models, so it is false in $8 - 6 = 2$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.8 [MCQ, 1M · Probability for AI] Answer: A (A: 0.50)

By Bayes rule, $P(a | b) = P(b | a)P(a)/P(b) = 0.8 \times 0.25 / 0.4 = 0.2 / 0.4 = 0.5$. $0.32 = 0.8 \times 0.4$ multiplies by $P(b)$ instead of dividing; $0.20 = P(a \wedge b)$ is the joint, not the conditional; 0.80 confuses $P(a | b)$ with $P(b | a)$.

Q.9 [MCQ, 1M · Conditional Independence] Answer: B (B: {A, C, D})

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of B: {A}; children: {C, D}; co-parents: $\{\emptyset\}$.

$MB(B) = \{A, C, D\}$ — 3 nodes. Given its Markov blanket, B is conditionally independent of every other node in the network.

Q.10 [MCQ, 1M · Variable Elimination] Answer: A (A: a factor over {A, B, C} with 8 entries)

The pointwise product is defined on the union of the variables: $f(a, b, c) = f_1(a, b) \times f_2(b, c)$. The union {A, B, C} has $2^3 = 8$ entries. {A, C} would result only after summing out B; shared variable B appears once, not twice.

Q.11 [MSQ, 2M · Informed Search] Answer: A, B, D (A: The path returned has cost 15.; B: Node I is expanded.; D: Node H is expanded.)

Heuristic values: $h(B)=5$, $h(C)=4$, $h(G)=0$, $h(H)=1$, $h(I)=1$, $h(J)=7$, $h(K)=13$, $h(S)=9$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	B(11), K(20)
2	B (11)	H(13), I(13), J(16), K(20)
3	H (13)	I(13), G(15), J(16), K(20), C(21)
4	I (13)	G(15), J(16), K(20), C(21)
5	G (15)	J(16), K(20), C(21) ← goal, stop

Path returned: **S** → **B** → **H** → **G** with cost **15**; nodes expanded = **5** (S, B, H, I, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.12 [NAT, 2M · Informed Search] Answer: 4

Heuristic values: $h(B)=13$, $h(D)=9$, $h(E)=4$, $h(F)=12$, $h(G)=0$, $h(H)=8$, $h(K)=5$, $h(S)=11$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	H(11), B(19), F(19)
2	H (11)	E(11), D(16), B(19), F(19)
3	E (11)	G(12), D(16), B(19), F(19)
4	G (12)	D(16), B(19), F(19) ← goal, stop

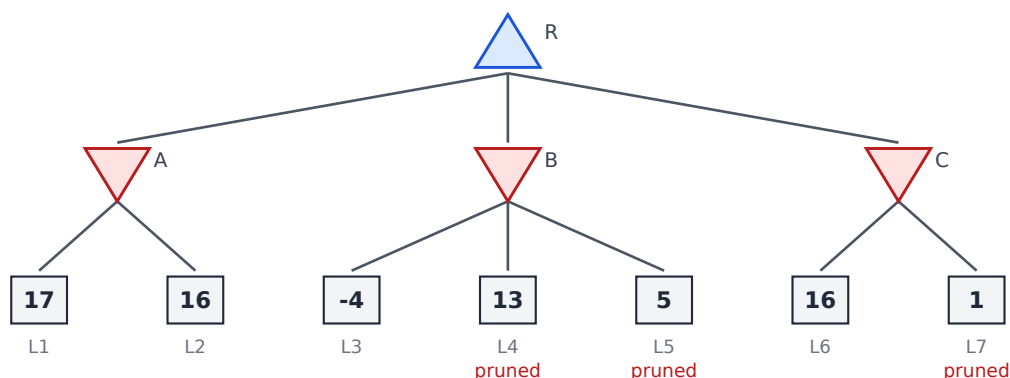
Path returned: **S** → **H** → **E** → **G** with cost **12**; nodes expanded = **4** (S, H, E, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 12$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.13 [NAT, 2M · Adversarial Search] Answer: 3

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-4 \leq \alpha = 16 \Rightarrow$ prune remaining children (leaves L4, L5).
- At MIN node C: value $16 \leq \alpha = 16 \Rightarrow$ prune remaining children (leaves L7).



Leaves evaluated (4): L1, L2, L3, L6. Leaves pruned (3): L4, L5, L7. Root value = 16 (same as plain minimax, 16 — pruning never changes the root value).

Q.14 [MSQ, 2M · Informed Search] **Answer: A, C, D**

(A) is true: induct backward along an optimal path from the goal, using $h(n) \leq c + h(n') \leq c + h^*(n') = h^*(n)$. (B) is false: admissible but inconsistent heuristics exist, for example $h(A)=6$ and $h(B)=2$ with $c(A,B)=2$. (C) is true: $f(n') = g(n) + c + h(n') \geq g(n) + h(n) = f(n)$. (D) is true: for tree search, admissibility is enough for optimality. Consistency is needed for graph search without reopening.

Q.15 [MSQ, 2M · Predicate Logic] **Answer: C, D** (C: $\forall x \exists y (x \neq y \wedge R(x, y))$; D: $\forall x \exists y R(x, y)$)

$R = \{(1,2), (1,3), (2,1), (2,4), (3,1), (3,2), (4,1), (4,3)\}$.

(A) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(1,2)$ and $R(2,1)$ with $1 \neq 2$.

(B) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(D) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: C** (C: H is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \wedge A \rightarrow H$; add H.

Round 2: fire $A \wedge H \rightarrow E$; add E.

Fixed point: $\{A, B, D, E, H\}$ — 5 atoms are entailed (facts: A, B, D; newly derived: E, H). Atoms never derived: C, F, J, K.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $R \vee P$; B: $\neg Q \vee \neg S$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 3 of them:

P	Q	R	S	KB
T	F	T	T	T
T	F	T	F	T
F	T	T	F	T

(A) $R \vee P$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $\neg Q \vee \neg S$ is true in all models of KB $\Rightarrow$ **entailed**.

(C) $\neg R \leftrightarrow R$ is false in the KB-model $P=T, Q=F, R=T, S=T \Rightarrow$ not entailed.

(D) $\neg Q \wedge (\neg R \rightarrow Q)$ is false in the KB-model $P=F, Q=T, R=T, S=F \Rightarrow$ not entailed.

Q.18 [NAT, 2M · Propositional Logic] **Answer: 6**

The constraints say once some x_i is T, every later x_j ($j > i$) must be T. So a model is a string of F's followed by a string of T's, e.g. FFTTT. The position where T's begin can be 1, 2, 3, 4, 5 or 'never' (all F), giving 6 models. Brute-force over all 32 assignments confirms 6.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $((R \rightarrow Q) \wedge \neg Q) \rightarrow \neg R$; D: $((Q \rightarrow P) \wedge (P \rightarrow R)) \rightarrow (Q \rightarrow R)$)

(A) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$: **not valid** — counter-model $Q=F, R=T$.

(B) $(P \rightarrow Q) \rightarrow (Q \rightarrow P)$: **not valid** — counter-model $P=F, Q=T$.

(C) $((R \rightarrow Q) \wedge \neg Q) \rightarrow \neg R$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow P) \wedge (P \rightarrow R)) \rightarrow (Q \rightarrow R)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.20 [NAT, 2M · Exact Inference] **Answer: 0.323 to 0.327**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B) \cdot P(C | A, B) \cdot P(D | A, B)$.

Inference by enumeration: $P(D | e) = \alpha \sum_{\text{hidden}} P(D, e, \text{hidden})$.

Hidden: A, C	D	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.5 \times 0.1 \times 0.75 \times 0.05$	0.00187
T, F	T	$0.5 \times 0.1 \times 0.25 \times 0.05$	0.00062
F, T	T	$0.5 \times 0.1 \times 0.8 \times 0.6$	0.024
F, F	T	$0.5 \times 0.1 \times 0.2 \times 0.6$	0.006
T, T	F	$0.5 \times 0.1 \times 0.75 \times 0.95$	0.03562
T, F	F	$0.5 \times 0.1 \times 0.25 \times 0.95$	0.01187
F, T	F	$0.5 \times 0.1 \times 0.8 \times 0.4$	0.016
F, F	F	$0.5 \times 0.1 \times 0.2 \times 0.4$	0.004

$P(D=T, e) = 0.0325$; $P(D=F, e) = 0.0675$; $P(e) = 0.1$.

$P(D=T | e) = 0.0325 / 0.1 = \mathbf{0.325} \approx 0.325$.

Q.21 [NAT, 2M · Sampling] **Answer: 0.248 to 0.252**

Gibbs sampling draws D from $P(D | \text{mb}(D))$, where $\text{mb}(D) = \{C\}$:

$P(D | \text{mb}) \propto P(D | \text{parents}(D)) \times \prod_{\text{children } Y} P(Y | \text{parents}(Y))$.

$D=T$: $P(D=T|C=F) = 0.25 = 0.25$

$D=F$: $P(D=F|C=F) = 0.75 = 0.75$

Normalise: $0.25 / (0.25 + 0.75) = \mathbf{0.25}$.

Q.22 [NAT, 2M · Bayesian Networks] **Answer: 13**

M: no predecessors $\rightarrow 1$ parameter. J: not independent of M (both depend on A) $\rightarrow$ parent M $\rightarrow 2$. A: depends on both M and J and neither screens the other $\rightarrow$ parents $\{M, J\} \rightarrow 4$. B: given A, B is independent of M and J $\rightarrow$ parent $\{A\} \rightarrow 2$. E: given A alone, E still depends on B (explaining away), but given $\{A, B\}$ it is independent of M, J $\rightarrow$ parents $\{A, B\} \rightarrow 4$. Total = $1 + 2 + 4 + 2 + 4 = 13$, versus 10 for the causal ordering B, E, A, J, M.

Q.23 [MCQ, 2M · Sampling] **Answer: B** (B: $A=F, B=F, C=F, D=F, E=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.6$; $u = 0.84 \geq 0.6 \Rightarrow A = F$
- B: $P(B=T | A=F) = 0.25$; $u = 0.57 \geq 0.25 \Rightarrow B = F$
- C: $P(C=T | A=F, B=F) = 0.05$; $u = 0.91 \geq 0.05 \Rightarrow C = F$
- D: $P(D=T | A=F) = 0.7$; $u = 0.76 \geq 0.7 \Rightarrow D = F$
- E: $P(E=T | C=F) = 0.7$; $u = 0.2 < 0.7 \Rightarrow E = T$

Sample: **$\{A=F, B=F, C=F, D=F, E=T\}$** .

Q.24 [MCQ, 2M · Exact Inference] **Answer: A** (A: $f(B, C)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(A, B, C), f(C), f(A, B, E), f(B, C, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(B, C, F)$	$\{B, C, F\}$	$f(B, C)$
C	$f(A, B, C), f(C), f(B, C)$	$\{A, B, C\}$	$f(A, B)$
E	$f(A, B, E)$	$\{A, B, E\}$	$f(A, B)$
B	$f(A, B), f(A, B), f(A, B)$	$\{A, B\}$	$f(A)$

The largest factor produced has **2** variable(s) (created when eliminating F); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.25 [MCQ, 2M · Sampling] **Answer: A**

Each non-evidence variable is sampled from its CPT given sampled/fixed parents, giving $S_{WS} = \prod P(z_i \mid \text{parents})$. The weight multiplies the evidence likelihoods. The product of these two is exactly the BN factorization of $P(z, e)$, which proves consistency. (B) is false: evidence at descendants is not taken into account while sampling, so $S_{WS} \neq P(z \mid e)$ in general. (C) and (D) misstate or swap the two products.

Mock Test 027

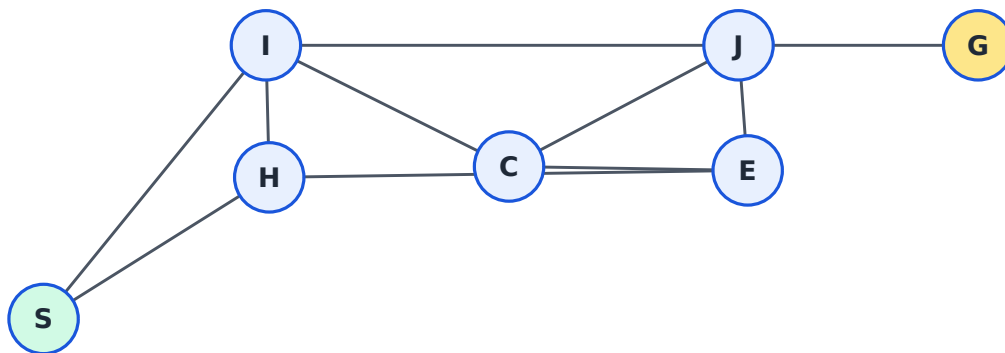
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	9 / 7 / 9	13 / 12 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, I, H, J, C, E, G
- ☐ (B) S, I, H, E, C, J, G
- ☐ (C) S, H, I, E, C, J, G
- ☐ (D) S, H, E, C, I, J, G

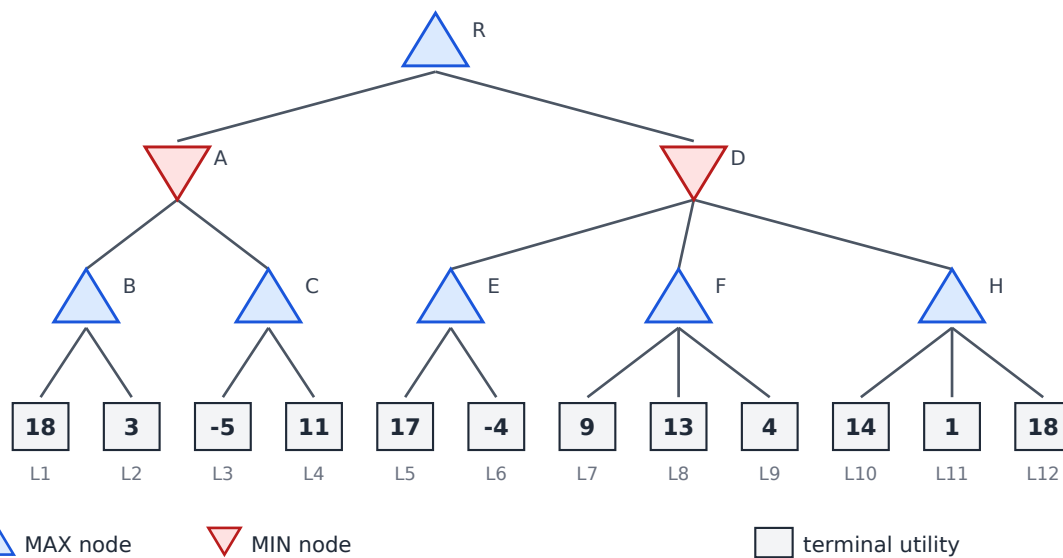
Q.2 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

How many nodes are there in a complete search tree of depth 5 (root at depth 0) with branching factor 5?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to D; root value 18
☐ (B) Move to A; root value 11
☐ (C) Move to D; root value 13
☐ (D) Move to D; root value 16

Q.4 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = (\neg R \wedge (Q \vee R)) \leftrightarrow ((\neg R \rightarrow Q) \vee (\neg P \leftrightarrow \neg P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Backward chaining over a propositional definite-clause KB is used to prove the query Q. Which clause is used FIRST?

- ☐ (A) A clause whose head (conclusion) is Q
☐ (B) A clause in which Q appears in the body
☐ (C) Any fact in the KB, chosen arbitrarily
☐ (D) The negated query $\neg Q$ added to the KB

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ are both reflexive and symmetric?

Answer (numerical): _____

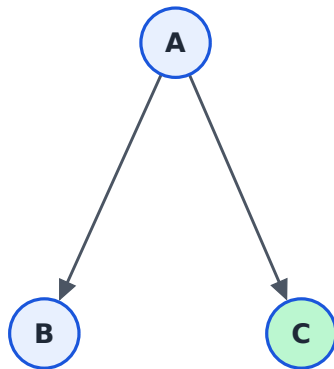
Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



$P(A=T)$	A	$P(B=T \dots)$	A	$P(C=T \dots)$
0.9	T	0.4	T	0.5
	F	0.95	F	0.75

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(B = T \mid D = T)$, how many of these samples are **rejected**?

#	A	B	C	D
1	T	T	T	T
2	F	T	T	F
3	T	T	F	T
4	F	T	T	T
5	T	T	T	F
6	T	F	F	F
7	F	T	T	T
8	F	T	T	F
9	T	T	T	F
10	T	T	T	F
11	T	T	T	F
12	F	T	T	F
13	T	F	F	T
14	F	T	T	F
15	T	T	T	T
16	F	T	T	F
17	T	F	T	T
18	T	T	F	T
19	T	T	F	T
20	T	T	T	F

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Sampling

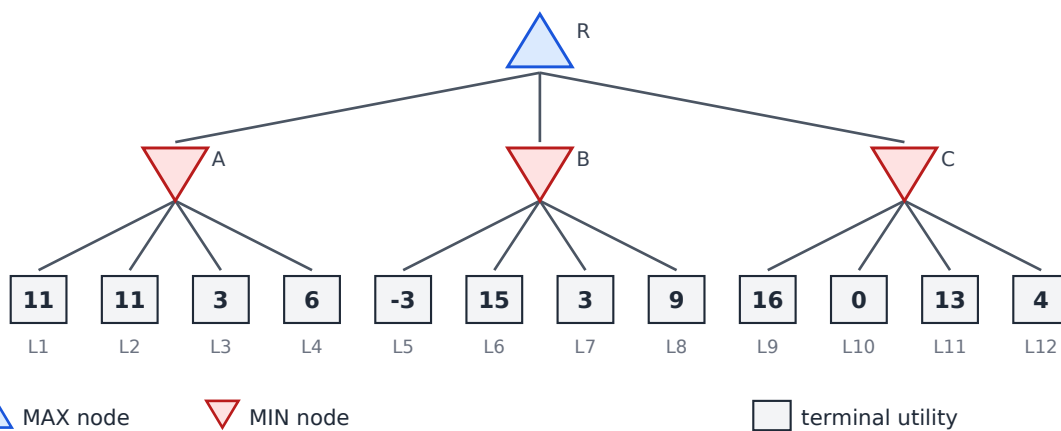
For sampling-based estimates of a probability, the standard deviation of the estimate decreases as $O(1/\sqrt{N})$ in the number of samples N . Roughly how many samples are needed to halve the error obtained with N samples?

- ☐ (A) $4N$
☐ (B) $2N$
☐ (C) $N\sqrt{2}$
☐ (D) N^2

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

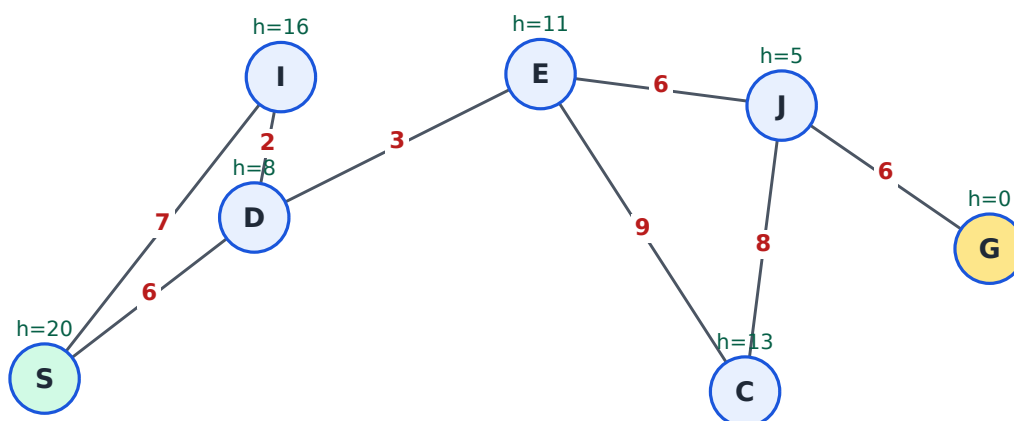
Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Depth-limited search (DLS) with limit l is applied to a problem with unit step costs whose shallowest goal lies at depth d . Which statement is correct?

- ☐ (A) If $l < d$, DLS is incomplete. If $l > d$, it finds a solution but not necessarily the shallowest (optimal) one
- ☐ (B) If $l < d$, DLS still finds the goal because it backtracks
- ☐ (C) If $l > d$, DLS is always optimal because it searches deeper than needed
- ☐ (D) DLS is complete and optimal for every $l \geq 0$

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(J)$ overestimates the true cost from J to G.
- ☐ (B) The edge S-D violates the consistency condition $h(S) \leq c(S,D) + h(D)$.
- ☐ (C) h is admissible.
- ☐ (D) h is consistent (monotone).

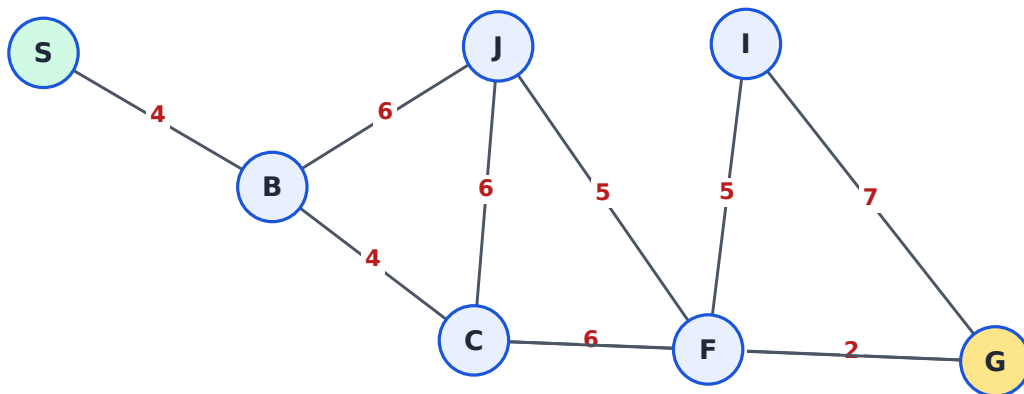
Q.14 [MCQ · 2 marks · negative: -2/3] Informed Search

A* tree search uses an admissible h , the optimal goal is G_1 with cost C^* , and G_2 is a suboptimal goal with $g(G_2) > C^*$. Why can G_2 never be selected for expansion before G_1 ?

- ☐ (A) Because $h(G_2) > h(G_1)$
- ☐ (B) Because some node n on the optimal path is always in the frontier with $f(n) \leq C^* < g(G_2) = f(G_2)$
- ☐ (C) Because A* expands nodes in order of depth
- ☐ (D) Because G_2 is never generated

Q.15 [MCQ · 2 marks · negative: -2/3] Uninformed Search - UCS

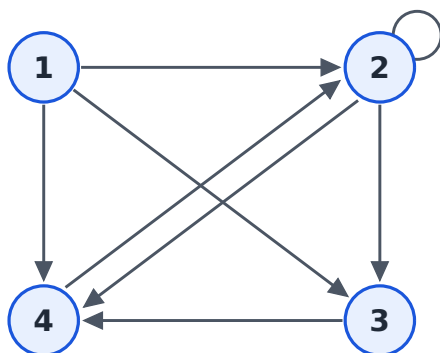
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow B \rightarrow C \rightarrow G$
- ☐ (B) $S \rightarrow B \rightarrow C \rightarrow J \rightarrow F \rightarrow G$
- ☐ (C) $S \rightarrow B \rightarrow C \rightarrow F \rightarrow G$
- ☐ (D) $S \rightarrow B \rightarrow J \rightarrow F \rightarrow G$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic - FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
- ☐ (B) $\forall x \exists y (x \neq y \wedge R(x, y))$
- ☐ (C) $\forall x \exists y R(x, y)$
- ☐ (D) $\exists x \forall y \neg R(x, y)$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic

On the domain $D = \{1, 2, 3\}$, let $R = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$. Which of the following sentences are TRUE in this interpretation?

- ☐ (A) $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$
☐ (B) $\forall x R(x, x)$
☐ (C) $\exists x \forall y R(x, y)$
☐ (D) $\forall y \exists x R(x, y)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = R \vee \neg\neg P$?

- ☐ (A) $\neg\neg P \vee R$
☐ (B) $\neg(\neg R \vee \neg\neg\neg P)$
☐ (C) $\neg(\neg R \wedge \neg\neg\neg P)$
☐ (D) $\neg R \vee \neg\neg P$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

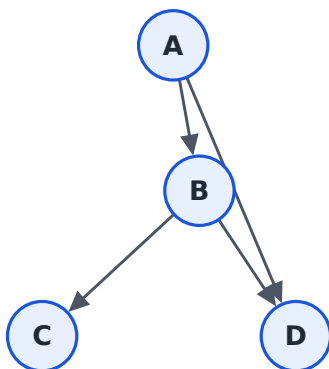
$C1: (\neg R), C2: (\neg P), C3: (\neg P \vee \neg R), C4: (\neg P \vee \neg Q), C5: (P \vee \neg R \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) $(\neg Q \vee \neg R \vee S)$
☐ (C) $(P \vee \neg S)$
☐ (D) $(\neg Q \vee \neg S)$

Q.20 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.11, 0.32, 0.65, 0.35)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.9

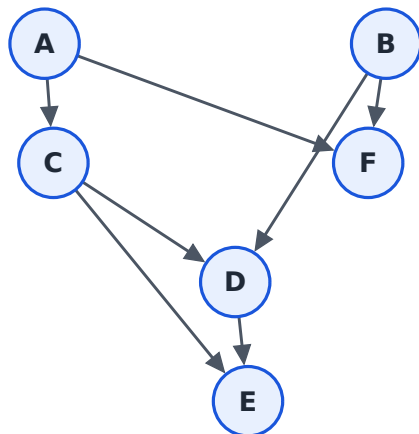
A	P(B=T ...)	B	P(C=T ...)
T	0.5	T	0.3
F	0.4	F	0.05

A	B	P(D=T ...)
T	T	0.9
T	F	0.9
F	T	0.2
F	F	0.2

- ☐ (A) $A=F, B=T, C=F, D=T$
☐ (B) $A=F, B=F, C=T, D=T$
☐ (C) $A=T, B=T, C=F, D=T$
☐ (D) $A=T, B=F, C=F, D=T$

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp C$
- ☐ (B) $A \perp E \mid D, F$
- ☐ (C) $A \perp B \mid C$
- ☐ (D) $F \perp C \mid A, E$

Q.22 [MCQ · 2 marks · negative: -2/3] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. For the query $P(j \mid b)$ (J given Burglary = true), which variables are irrelevant to the query and can be removed before running variable elimination?

- ☐ (A) {M}
- ☐ (B) {E, M}
- ☐ (C) {E}
- ☐ (D) No variable is irrelevant

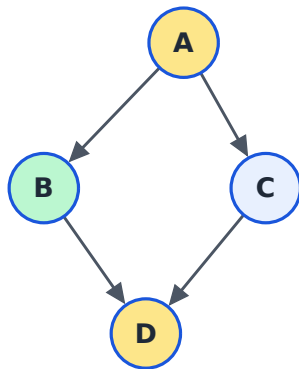
Q.23 [MSQ · 2 marks · no negative marking] Bayesian Networks

A Bayesian network is the chain $A \rightarrow B \rightarrow C$ over Boolean variables with $P(a) = 0.6$, $P(b \mid a) = 0.7$, $P(b \mid \neg a) = 0.2$, $P(c \mid b) = 0.9$, $P(c \mid \neg b) = 0.3$. Which of the following are correct?

- ☐ (A) $P(b) = 0.5$
- ☐ (B) $P(c) = 0.6$
- ☐ (C) $P(a \mid c) = 0.72$
- ☐ (D) $P(a \mid c) = P(a)$, since A and C are not adjacent

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(\mathbf{B} = \mathbf{T} \mid \mathbf{A} = \mathbf{T}, \mathbf{D} = \mathbf{T})$ (round off to 3 decimal places).



$P(A=T)$
0.7

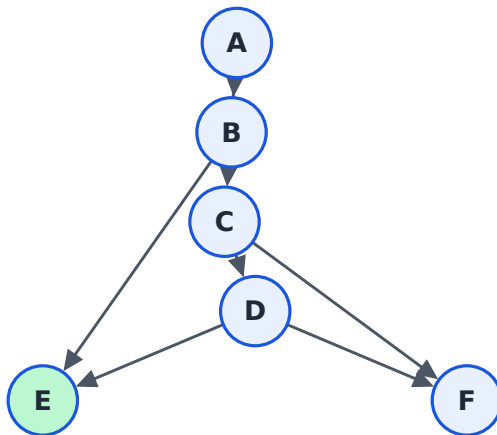
A	$P(B=T \dots)$	A	$P(C=T \dots)$
T	0.95	T	0.2
F	0.05	F	0.9

B	C	$P(D=T \dots)$
T	T	0.4
T	F	0.05
F	T	0.7
F	F	0.4

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{E})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, B, C, A, F**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 027

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	C	14	MCQ	2	Informed Search	B
2	NAT	1	Uninformed Search	3906	15	MCQ	2	Uninformed Search	C
3	MCQ	1	Adversarial Search	C	16	MSQ	2	Predicate Logic	A, B, C
4	NAT	1	Propositional Logic	6	17	MSQ	2	Predicate Logic	A, C, D
5	MCQ	1	Propositional Logic	A	18	MSQ	2	Propositional Logic	A, C
6	NAT	1	Predicate Logic	8	19	MSQ	2	Propositional Logic	B
7	NAT	1	Predicate Logic	8	20	MCQ	2	Sampling	C
8	NAT	1	Exact Inference	0.523 to 0.527	21	MSQ	2	Conditional Independence	A, C
9	NAT	1	Sampling	11	22	MCQ	2	Variable Elimination	A
10	MCQ	1	Sampling	A	23	MSQ	2	Bayesian Networks	A, B, C
11	NAT	2	Adversarial Search	7	24	NAT	2	Exact Inference	0.83 to 0.834
12	MCQ	2	Uninformed Search	A	25	NAT	2	Exact Inference	4
13	MSQ	2	Informed Search	B, C					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: C (C: S, H, I, E, C, J, G)

Adjacency lists (alphabetical): C: {E, I, J}; E: {C, H, J}; G: {J}; H: {E, I, S}; I: {C, H, J, S}; J: {C, E, G, I}; S: {H, I}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[H, I]
2	H	[I, E]
3	I	[E, C, J]
4	E	[C, J]
5	C	[J]
6	J	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, H, I, E, C, J, G**. The path returned is **S → I → J → G** with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

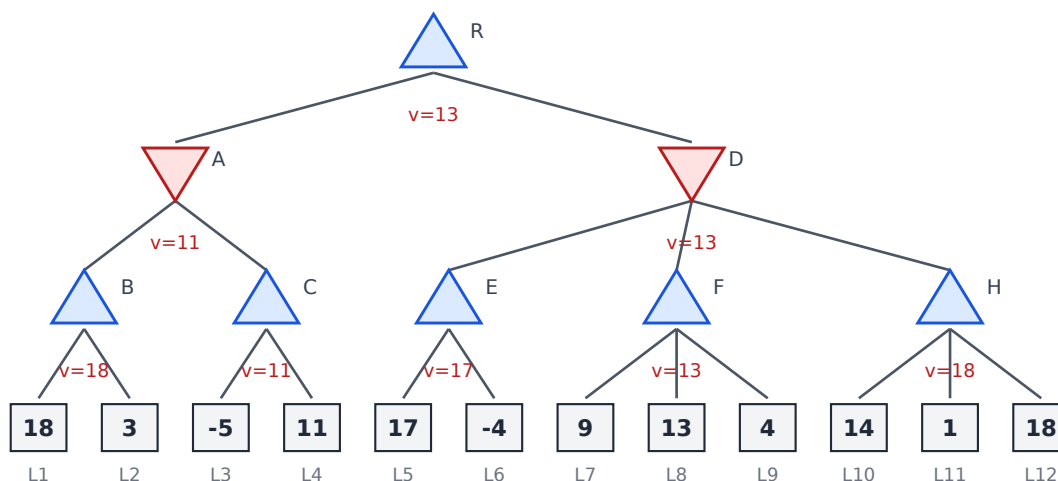
Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Uninformed Search] Answer: 3906

$$\sum_{i=0..5} 5^i = (5^6 - 1)/(5 - 1) = \mathbf{3906}.$$

Q.3 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to D; root value 13)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 18, C = 11, A = 11, E = 17, F = 13, H = 18, D = 13. Root value = **13**; MAX's optimal first move is towards D.

Q.4 [NAT, 1M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg R \wedge (Q \vee R)$	$(\neg R \rightarrow Q) \vee (\neg P \leftrightarrow \neg P)$	$(\neg R \wedge (Q \vee R)) \leftrightarrow ((\neg R \rightarrow Q) \vee (\neg P \leftrightarrow \neg P))$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	T	F
F	F	F	F	T	F

F has 2 models, so it is false in $8 - 2 = 6$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.5 [MCQ, 1M · Propositional Logic] **Answer: A** (A: A clause whose head (conclusion) is Q)

Backward chaining works from the goal: it looks for rules (or facts) whose conclusion is Q and recursively tries to prove their premises. Using clauses with Q in the body is forward reasoning. Starting from facts is forward chaining. Adding $\neg Q$ is the resolution-refutation approach, not backward chaining.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 8**

Reflexivity fixes all 3 diagonal pairs to be present. Symmetry ties (x, y) and (y, x) together for $x \neq y$, leaving 3 independent unordered pairs, each in or out. Total = $2^3 = 8$.

Q.7 [NAT, 1M · Predicate Logic] **Answer: 8**

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 3$ this gives **8** (verified by enumerating all interpretations).

Q.8 [NAT, 1M · Exact Inference] **Answer: 0.523 to 0.527**

Only the ancestors of C (A) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T	0.18
A=T, B=F	0.27
A=F, B=T	0.07125
A=F, B=F	0.00375

Sum = **0.525** ≈ 0.525 .

Q.9 [NAT, 1M · Sampling] **Answer: 11**

Rejection sampling keeps only samples consistent with the evidence $D = T$: samples 1, 3, 4, 7, 13, 15, 17, 18, 19 (9 accepted, 11 rejected).

Among the accepted samples, $B = T$ in 7. Estimate = $7/9 = 0.7778$.

(For this network the exact value is 0.6833; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

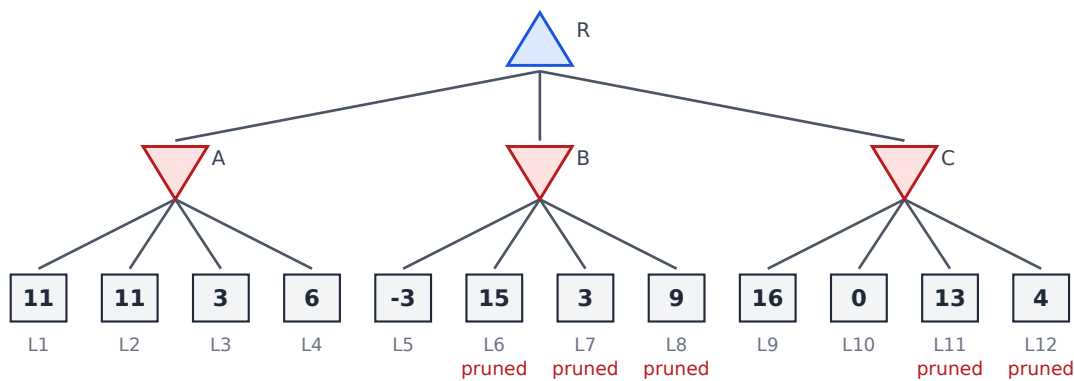
Q.10 [MCQ, 1M · Sampling] **Answer: A** (A: $4N$)

Error $\propto 1/\sqrt{N}$, so halving the error requires $\sqrt{N'} = 2\sqrt{N}$, i.e. $N' = 4N$. $2N$ would reduce error only by a factor $\sqrt{2} \approx 1.41$; $N/2$ by $2^{1/4}$; N^2 is excessive and depends on N .

Q.11 [NAT, 2M · Adversarial Search] **Answer: 7**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-3 \leq \alpha = 3 \Rightarrow$ prune remaining children (leaves L6, L7, L8).
- At MIN node C: value $0 \leq \alpha = 3 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (7): L1, L2, L3, L4, L5, L9, L10. Leaves pruned (5): L6, L7, L8, L11, L12. Root value = 3 (same as plain minimax, 3 — pruning never changes the root value).

Q.12 [MCQ, 2M · Uninformed Search] Answer: A (A: If $l < d$, DLS is incomplete. If $l > d$, it finds a solution but not necessarily the shallowest (optimal) one)

DLS treats nodes at depth l as having no successors. If $l < d$, the goal can never be reached, so DLS is incomplete; backtracking does not help, so (B) is wrong. If $l > d$, a goal is reachable, but DLS explores depth-first and may meet a deeper goal (depth between d and l) before the shallowest one, so it is not optimal and (C) is wrong. (D) is wrong for both reasons.

Q.13 [MSQ, 2M · Informed Search] Answer: B, C (B: The edge S-D violates the consistency condition $h(S) \leq c(S,D) + h(D)$.; C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	D	E	G	I	J	S
$h(n)$	13	8	11	0	16	5	20
$h^*(n)$	14	15	12	0	17	6	21
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $I \rightarrow D$ ($16 > 2 + 8$), $S \rightarrow D$ ($20 > 6 + 8$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [MCQ, 2M · Informed Search] Answer: B (B: Because some node n on the optimal path is always in the frontier with $f(n) \leq C^* < g(G_2) = f(G_2)$)

This is the standard optimality proof. Until G_1 is selected, some node n on the optimal path is still in the frontier, and admissibility gives $f(n) = g(n) + h(n) \leq C^*$. Also $f(G_2) = g(G_2) > C^*$ because $h(\text{goal}) = 0$. So n , and eventually G_1 , are selected before G_2 . (A) is wrong because $h = 0$ at every goal. (C) is wrong because A^* orders by f , not depth. (D) is wrong because G_2 may well be generated, just not selected.

Q.15 [MCQ, 2M · Uninformed Search] Answer: C (C: $S \rightarrow B \rightarrow C \rightarrow F \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(4)
2	B (4)	C(8), J(10)
3	C (8)	J(10), F(14), G(17)
4	J (10)	F(14), G(17)
5	F (14)	G(16), I(19)
6	G (16)	I(19) ← goal, stop

Path returned: **S** $\rightarrow$ **B** $\rightarrow$ **C** $\rightarrow$ **F** $\rightarrow$ **G**, cost = 4 + 4 + 6 + 2 = **16**. Nodes expanded (including S and G): **6** (S, B, C, J, F, G).

Q.16 [MSQ, 2M · Predicate Logic] Answer: A, B, C (A: $\exists x \forall y \neg R(y, x)$; B: $\forall x \exists y (x \neq y \wedge R(x, y))$; C: $\forall x \exists y R(x, y)$)

$R = \{(1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,4), (4,2)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **TRUE** — 1 has no incoming edge.

(B) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(C) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(D) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, C, D (A: $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$; C: $\exists x \forall y R(x, y)$; D: $\forall y \exists x R(x, y)$)

(A) The composable pairs are $(1,1)(1,y) \rightarrow (1,y) \in R$, and $(1,2)(2,3) \rightarrow (1,3) \in R$; so R is transitive. (B) False: $(2,2) \notin R$. (C) True with $x = 1$: $(1,1), (1,2), (1,3) \in R$. (D) True: $x = 1$ works for every y . Enumeration over all triples confirms A, C, D.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $\neg \neg P \vee R$; C: $\neg(\neg R \wedge \neg \neg P)$)

Compare truth tables with $F = R \vee \neg \neg P$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg \neg P \vee R$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $\neg(\neg R \vee \neg \neg P)$ — differs at $P=T, R=F$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(C) $\neg(\neg R \wedge \neg \neg P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(D) $\neg R \vee \neg \neg P$ — differs at $P=F, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

Q.19 [MSQ, 2M · Propositional Logic] Answer: B (B: $(\neg Q \vee \neg R \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(\neg Q \vee \neg R \vee S)$ is derivable:

1. resolve $(\neg P \vee \neg Q)$ and $(P \vee \neg R \vee S)$ on $P \Rightarrow (\neg Q \vee \neg R \vee S)$

(C) $(P \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg Q \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.20 [MCQ, 2M · Sampling] Answer: C (C: $A=T, B=T, C=F, D=T$)

Sample each variable after its parents, using the parent values already drawn:

• A: $P(A=T \mid _) = 0.9$; $u = 0.11 < 0.9 \Rightarrow A = T$

• B: $P(B=T \mid A=T) = 0.5$; $u = 0.32 < 0.5 \Rightarrow B = T$

• C: $P(C=T \mid B=T) = 0.3$; $u = 0.65 \geq 0.3 \Rightarrow C = F$

• D: $P(D=T \mid A=T, B=T) = 0.9$; $u = 0.35 < 0.9 \Rightarrow D = T$

Sample: **$\{A=T, B=T, C=F, D=T\}$** .

Q.21 [MSQ, 2M · Conditional Independence] Answer: A, C (A: $B \perp C$; C: $A \perp B \mid C$)

Rules: a chain $(\rightarrow B \rightarrow)$ or fork $(\leftarrow B \leftarrow)$ is blocked iff B is observed; a collider $(\rightarrow B \leftarrow)$ is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp C$: **TRUE** — $B \rightarrow D \leftarrow C$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow D \rightarrow E \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow F \leftarrow A \rightarrow C$ (blocked: collider F with neither it nor a descendant observed).

(B) $A \perp E \mid D, F$: **FALSE** — the path $A \rightarrow C \rightarrow E$ is active given $\{D, F\}$.

(C) $A \perp B \mid C$: **TRUE** — $A \rightarrow C \rightarrow D \leftarrow B$ (blocked: observed chain node C); $A \rightarrow C \rightarrow E \leftarrow D \leftarrow B$ (blocked: observed chain node C); $A \rightarrow F \leftarrow B$ (blocked: collider F with neither it nor a descendant observed).

(D) $F \perp C \mid A, E$: **FALSE** — the path $F \leftarrow B \rightarrow D \leftarrow C$ is active given $\{A, E\}$.

Q.22 [MCQ, 2M · Variable Elimination] Answer: A (A: $\{M\}$)

AIMA: every variable that is not an ancestor of a query variable or an evidence variable is irrelevant.

Ancestors of J and B : $\{A, B, E\}$. M is neither, so it is irrelevant: $\sum_m P(m \mid a) = 1$. E is an ancestor of J ($E \rightarrow A \rightarrow J$), and $P(a \mid b) = \sum_e P(e) P(a \mid b, e)$ genuinely depends on E 's CPT, so E cannot be dropped.

Q.23 [MSQ, 2M · Bayesian Networks] **Answer: A, B, C** (A: $P(b) = 0.5$; B: $P(c) = 0.6$; C: $P(a | c) = 0.72$)

$P(b) = 0.6 \times 0.7 + 0.4 \times 0.2 = 0.42 + 0.08 = 0.5$ (A true). $P(c) = 0.5 \times 0.9 + 0.5 \times 0.3 = 0.45 + 0.15 = 0.6$ (B true). $P(a, c) = 0.6 \times (0.7 \times 0.9 + 0.3 \times 0.3) = 0.6 \times 0.72 = 0.432$, so $P(a | c) = 0.432 / 0.6 = 0.72$ (C true). (D) is false: A and C are connected by the active path $A \rightarrow B \rightarrow C$ (B unobserved), so observing c raises $P(a)$ from 0.6 to 0.72.

Q.24 [NAT, 2M · Exact Inference] **Answer: 0.83 to 0.834**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B | A) \cdot P(C | A) \cdot P(D | B, C)$.

Inference by enumeration: $P(B | e) = \alpha \sum_{\text{hidden}} P(B, e, \text{hidden})$.

Hidden: C	B	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T	T	$0.7 \times 0.95 \times 0.2 \times 0.4$	0.0532
F	T	$0.7 \times 0.95 \times 0.8 \times 0.05$	0.0266
T	F	$0.7 \times 0.05 \times 0.2 \times 0.7$	0.0049
F	F	$0.7 \times 0.05 \times 0.8 \times 0.4$	0.0112

$P(B=T, e) = 0.0798$; $P(B=F, e) = 0.0161$; $P(e) = 0.0959$.

$P(B=T | e) = 0.0798 / 0.0959 = \mathbf{0.8321} \approx 0.832$.

Q.25 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(C, D)$, $f(B, D, E)$, $f(C, D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(C, D)$, $f(B, D, E)$, $f(C, D, F)$	$\{B, C, D, E, F\}$	$f(B, C, E, F)$
B	$f(A, B)$, $f(A, B, C)$, $f(B, C, E, F)$	$\{A, B, C, E, F\}$	$f(A, C, E, F)$
C	$f(A, C, E, F)$	$\{A, C, E, F\}$	$f(A, E, F)$
A	$f(A)$, $f(A, E, F)$	$\{A, E, F\}$	$f(E, F)$
F	$f(E, F)$	$\{E, F\}$	$f(E)$

The largest factor produced has **4** variable(s) (created when eliminating D); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over E and normalise.

Mock Test 028

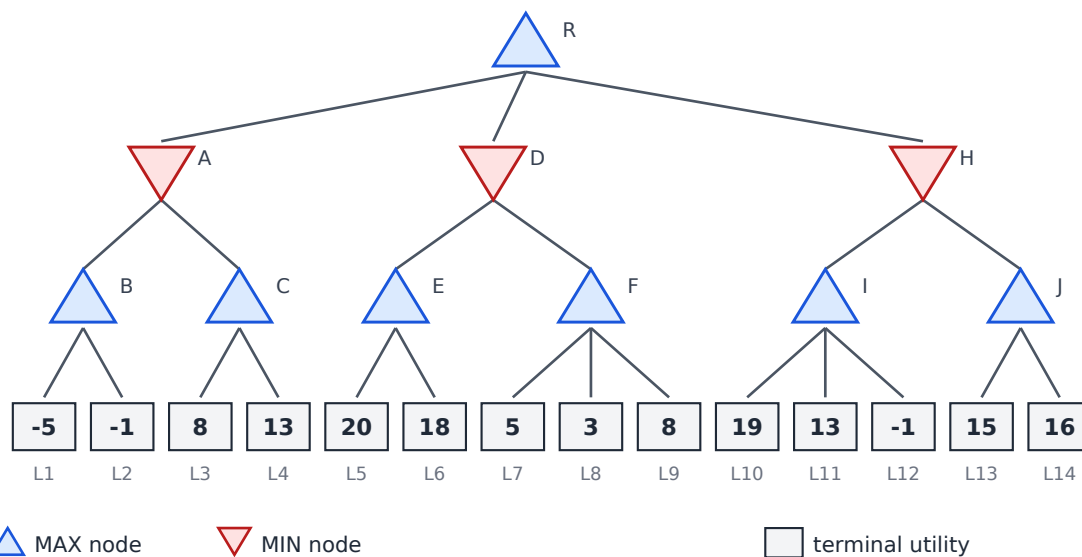
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 8 / 13	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

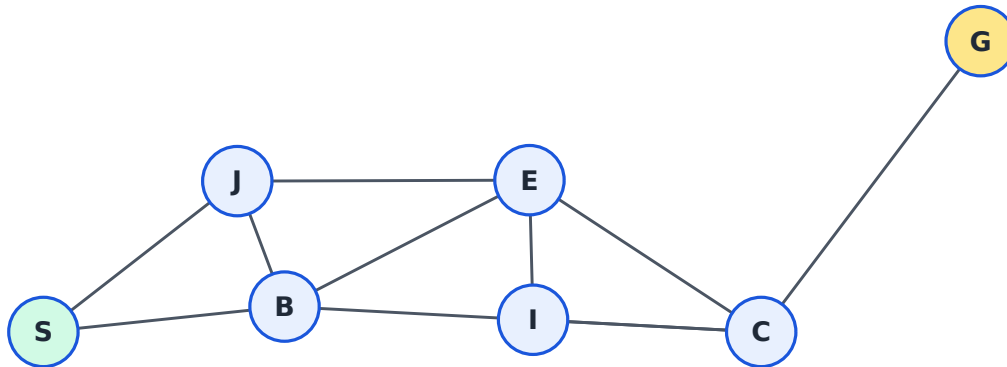
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 3$ (so $b^d = 64$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, C, E, I, J, G
☐ (B) S, J, B, E, C, I, G
☐ (C) S, B, J, C, E, G
☐ (D) S, B, J, E, C, G

Q.4 [NAT · 1 mark · no negative marking] Informed Search

In the 8-puzzle, the goal configuration is (rows top to bottom) 1 2 3 / 4 5 6 / 7 8 _, where _ is the blank. The current state is 8 1 3 / 4 _ 2 / 7 6 5. What is the value of h_1 , the number of misplaced tiles (do not count the blank)?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((\neg R \leftrightarrow Q) \rightarrow (R \rightarrow \neg P)) \rightarrow (Q \rightarrow P) ?$$

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

How many of the 8 truth assignments to p, q, r are models of $(p \vee q) \wedge (\neg p \vee r)$?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following propositional formulas is a tautology?

- ☐ (A) $(p \rightarrow q) \vee (q \rightarrow p)$
☐ (B) $(p \rightarrow q) \rightarrow (q \rightarrow p)$
☐ (C) $(p \wedge \neg q) \rightarrow q$
☐ (D) $(p \vee q) \rightarrow (p \wedge q)$

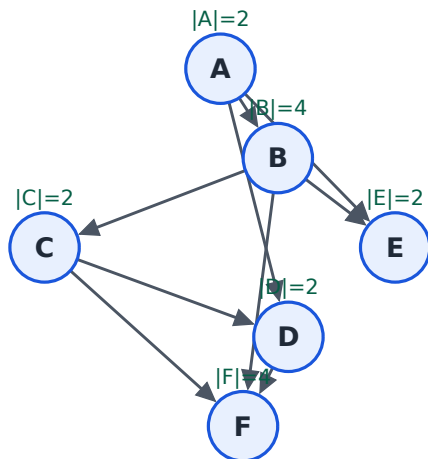
Q.8 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \exists y R(x, y)$ true?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(A = T \mid D = T)$, how many of these samples are **rejected**?

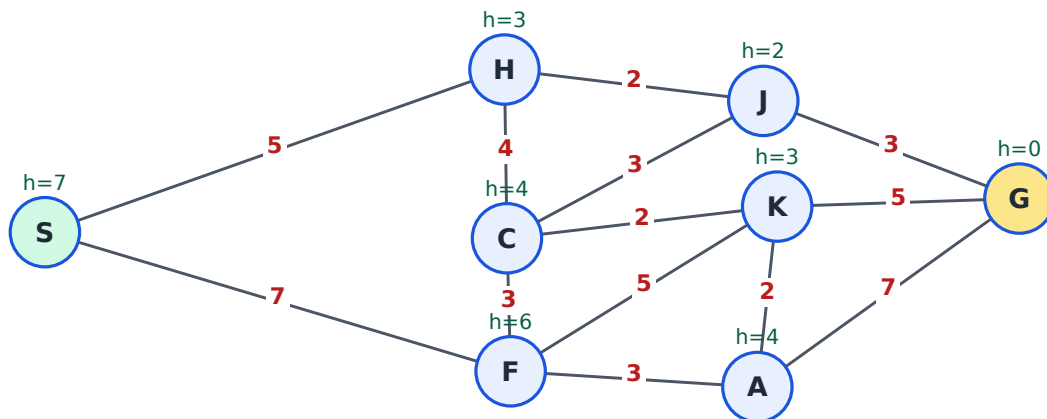
#	A	B	C	D
1	F	F	T	F
2	F	T	F	T
3	F	T	T	F
4	F	F	F	T
5	F	T	T	F
6	F	F	T	F
7	F	T	T	F
8	T	T	F	F
9	F	T	T	F
10	T	F	T	T
11	F	T	F	T
12	F	T	T	F
13	T	T	F	F
14	T	F	F	F
15	F	F	T	F

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Problem Formulation

Which of the following statements match standard (AIMA) definitions of the terms used to analyse search algorithms?

- ☐ (A) Completeness: the algorithm is guaranteed to find a solution when one exists
- ☐ (B) Cost-optimality: the algorithm finds a solution with the lowest path cost among all solutions
- ☐ (C) b , the branching factor, is the maximum number of successors of any node
- ☐ (D) d is the depth of the deepest node in the state space

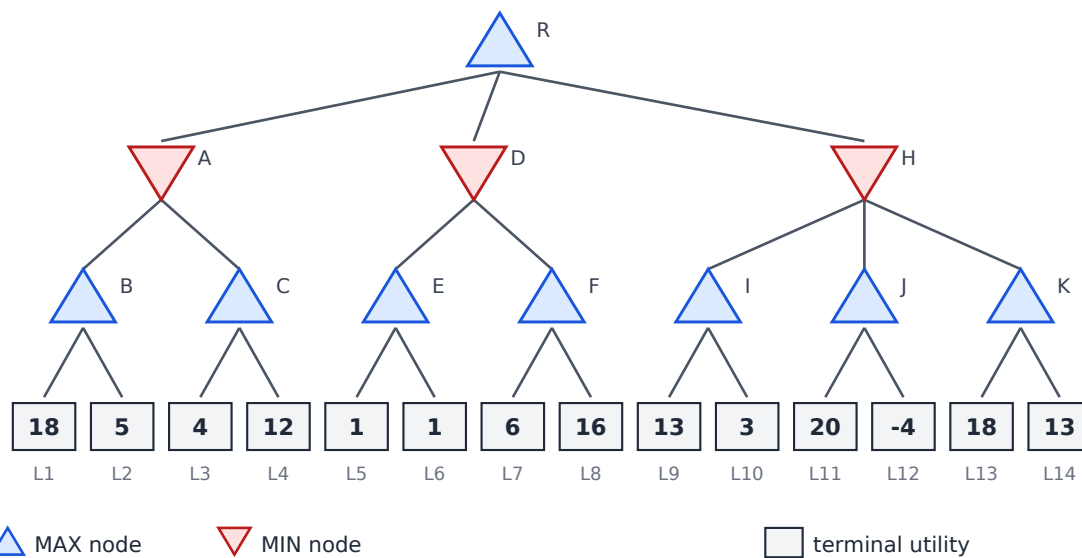
Q.13 [MCQ · 2 marks · negative: -2/3] Adversarial Search

To get the maximum pruning from alpha-beta search, how should children be ordered?

- ☐ (A) At MAX nodes in decreasing order of value, at MIN nodes in increasing order of value
- ☐ (B) At MAX nodes in increasing order of value, at MIN nodes in decreasing order of value
- ☐ (C) In increasing order at both node types
- ☐ (D) Ordering does not affect pruning

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

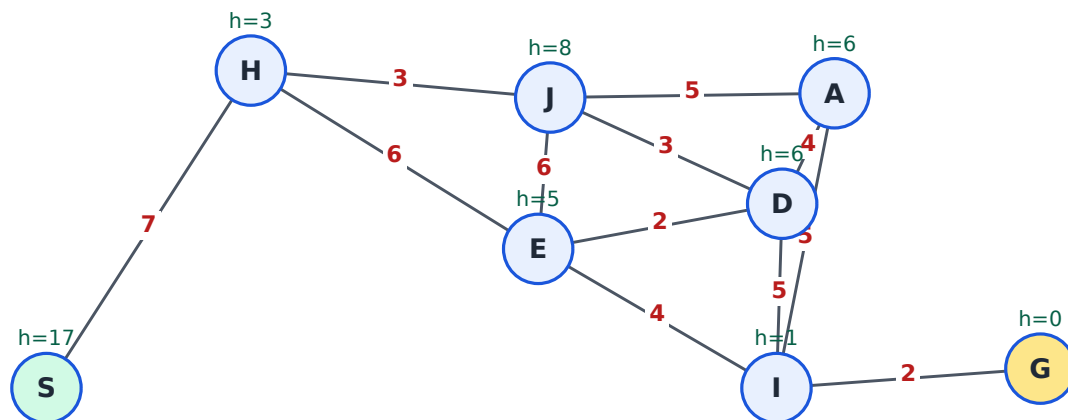
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) The edge D-I violates the consistency condition (in at least one direction).
- ☐ (C) h is admissible.
- ☐ (D) $h(J)$ overestimates the true cost from J to G.

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following entailments hold?

- ☐ (A) $\{p \rightarrow q, q \rightarrow r\} \models p \rightarrow r$
- ☐ (B) $\{p \vee q, \neg p \vee r\} \models q \vee r$
- ☐ (C) $\{p \rightarrow q, q\} \models p$
- ☐ (D) $\{p \rightarrow q, \neg q\} \models \neg p$

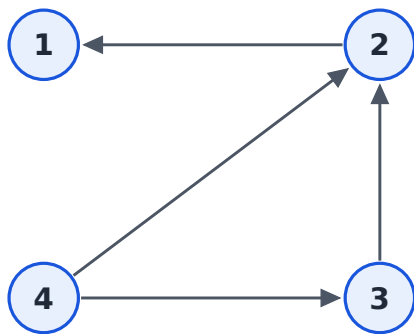
Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(Q \rightarrow R) \vee (R \rightarrow Q)$
☐ (B) $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$
☐ (C) $(P \rightarrow R) \leftrightarrow (\neg R \rightarrow \neg P)$
☐ (D) $(R \vee Q) \rightarrow (R \wedge Q)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\exists y \forall x R(x, y)$
☐ (C) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

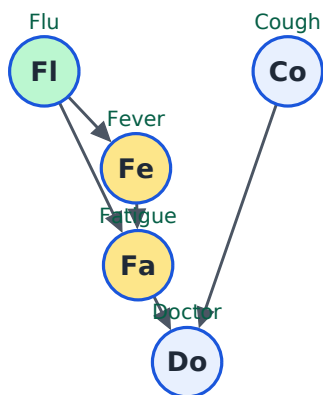
$C1: (\neg P \vee \neg R \vee S)$, $C2: (P \vee \neg R)$, $C3: (\neg Q \vee \neg R)$, $C4: (\neg P \vee \neg S)$, $C5: (Q \vee R \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(P \vee Q)$
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) $(\neg P)$
☐ (D) $(P \vee \neg R \vee S)$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) Compute $P(\text{Flu} = T \mid \text{Fatigue} = T, \text{Fever} = F)$ (round off to 3 decimal places).



$P(\text{Fl}=T)$
0.1

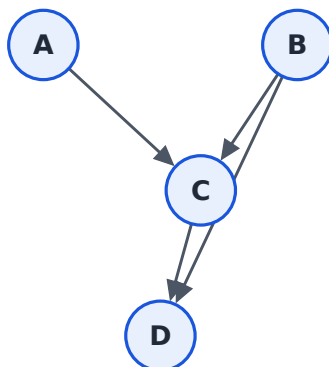
Fl	$P(\text{Fe}=T \dots)$	$P(\text{Co}=T)$
T	0.1	0.1
F	0.3	

Fl	Fe	$P(\text{Fa}=T)$	Co	Fa	$P(\text{Do}=T \dots)$
T	T	0.3	T	T	0.3
T	F	0.3	T	F	0.25
F	T	0.2	F	T	0.6
F	F	0.9	F	F	0.95

Answer (numerical): _____

Q.21 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.03, 0.19, 0.24, 0.24)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.6

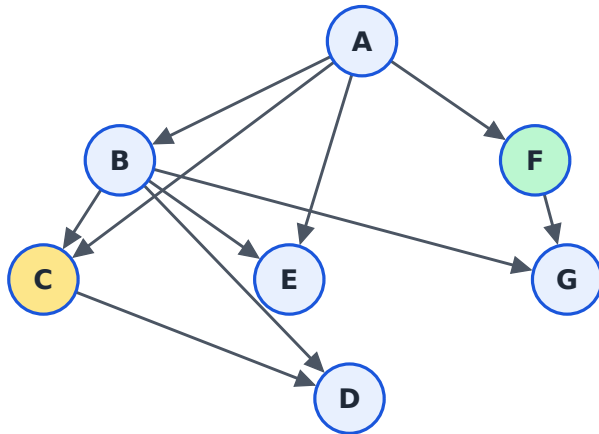
$P(B=T)$	A	B	$P(C=T \dots)$
0.75	T	T	0.2
	T	F	0.25
	F	T	0.5
	F	F	0.4

B	C	$P(D=T \dots)$
T	T	0.05
T	F	0.1
F	T	0.6
F	F	0.9

- ☐ (A) A=F, B=T, C=F, D=F
☐ (B) A=T, B=T, C=F, D=F
☐ (C) A=F, B=F, C=F, D=F
☐ (D) A=T, B=F, C=F, D=F

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

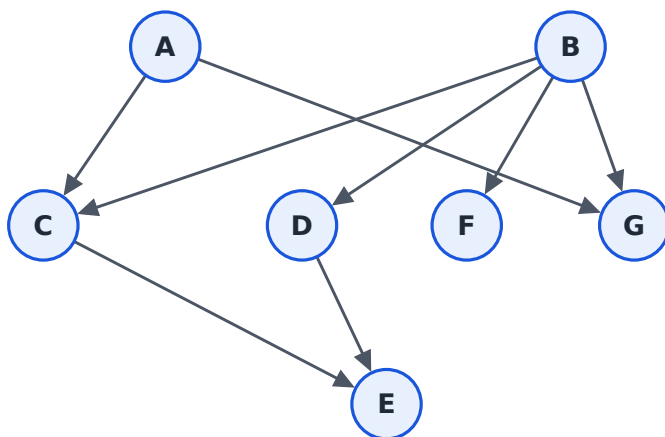
Variable elimination is used to compute $P(F \mid C = T)$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, G, E, B, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp F \mid B, D$
- ☐ (B) $F \perp A \mid B$
- ☐ (C) $D \perp E \mid G$
- ☐ (D) $C \perp D \mid B$

Q.24 [MSQ · 2 marks · no negative marking] Variable Elimination

A Bayesian network has edges $A \rightarrow B$ and $A \rightarrow C$ over Boolean variables with $P(a) = 0.3$, $P(b | a) = 0.8$, $P(b | \neg a) = 0.4$, $P(c | a) = 0.5$, $P(c | \neg a) = 0.1$. $P(A | b, c)$ is computed by variable elimination, multiplying the evidence-instantiated factors $f_1(A) = P(A)$, $f_2(A) = P(b | A)$, $f_3(A) = P(c | A)$ into an unnormalized factor $f(A)$. Which are correct?

- ☐ (A) $f(a) = 0.12$
☐ (B) $f(\neg a) = 0.028$
☐ (C) $P(a | b, c) \approx 0.811$
☐ (D) $P(b, c) = 0.12$

Q.25 [NAT · 2 marks · no negative marking] Bayesian Networks

A Bayesian network has: A with 3 values (no parents); B Boolean (no parents); C with 4 values and parents A, B; D Boolean with parent C; E with 3 values and parents C, D. How many independent parameters do the CPTs require?

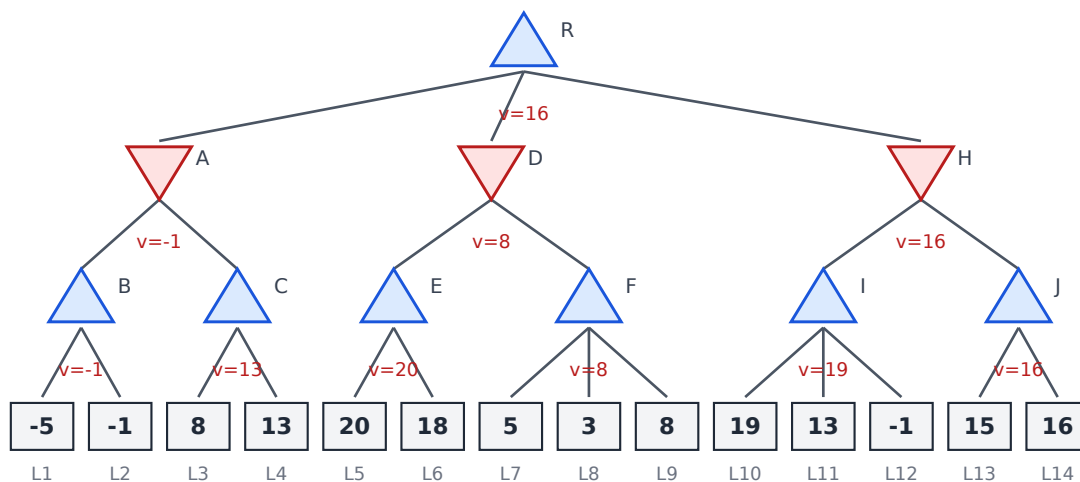
Answer (numerical): _____

Answer Key — Mock Test 028

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	16	14	NAT	2	Adversarial Search	10
2	NAT	1	Adversarial Search	19	15	MSQ	2	Informed Search	C
3	MCQ	1	Uninformed Search	C	16	MSQ	2	Propositional Logic	A, B, D
4	NAT	1	Informed Search	5	17	MSQ	2	Propositional Logic	A, B, C
5	NAT	1	Propositional Logic	6	18	MSQ	2	Predicate Logic	A, D
6	NAT	1	Propositional Logic	4	19	MSQ	2	Propositional Logic	D
7	MCQ	1	Propositional Logic	A	20	NAT	2	Exact Inference	0.043 to 0.047
8	NAT	1	Predicate Logic	343	21	MCQ	2	Sampling	B
9	NAT	1	Bayesian Networks	71	22	NAT	2	Exact Inference	3
10	NAT	1	Sampling	11	23	MSQ	2	Conditional Independence	A, B, D
11	NAT	2	Informed Search	10	24	MSQ	2	Variable Elimination	A, B, C
12	MSQ	2	Problem Formulation	A, B, C	25	NAT	2	Bayesian Networks	41
13	MCQ	2	Adversarial Search	A					

Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 16**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = -1, C = 13, A = -1, E = 20, F = 8, D = 8, I = 19, J = 16, H = 16. Root value = **16**; MAX's optimal first move is towards H.

Q.2 [NAT, 1M · Adversarial Search] Answer: 19

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 4^2 + 4^1 - 1 = 16 + 4 - 1 = 19$, versus 64 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [MCQ, 1M · Uninformed Search] Answer: C (C: S, B, J, C, E, G)

Adjacency lists (alphabetical): B: {C, E, J, S}; C: {B, E, G, I}; E: {B, C, I, J}; G: {C}; I: {C, E}; J: {B, E, S}; S: {B, J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B, J]
2	B	[J, C, E]
3	J	[C, E]
4	C	[E, G, I]
5	E	[G, I]
6	G	[I] ← goal popped, stop

Expansion order: **S, B, J, C, E, G**. The path returned is $S \rightarrow B \rightarrow C \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [NAT, 1M · Informed Search] Answer: 5

Compare position by position. Tiles 3, 4 and 7 are already in place. Tiles 8, 1, 2, 6 and 5 are not. $h_1 = 5$.

Q.5 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg R \leftrightarrow Q) \rightarrow (R \rightarrow \neg P)$	$Q \rightarrow P$	$((\neg R \leftrightarrow Q) \rightarrow (R \rightarrow \neg P)) \rightarrow (Q \rightarrow P)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

Rows where F = T: **6**.

Q.6 [NAT, 1M · Propositional Logic] Answer: 4

Case $p=T$: the first clause is satisfied and the second requires $r=T$; q is free, giving 2 models. Case $p=F$: the second clause is satisfied and the first requires $q=T$; r is free, giving 2 models. Total = $2 + 2 = 4$.

Q.7 [MCQ, 1M · Propositional Logic] Answer: A ($A: (p \rightarrow q) \vee (q \rightarrow p)$)

$(p \rightarrow q) \vee (q \rightarrow p) \equiv \neg p \vee q \vee \neg q \vee p$, which contains $q \vee \neg q$, so it is true in all 4 rows. $(p \rightarrow q) \rightarrow (q \rightarrow p)$ is false at $p=F, q=T$ (antecedent T, consequent F). $(p \wedge \neg q) \rightarrow q$ is false at $p=T, q=F$. $(p \vee q) \rightarrow (p \wedge q)$ is false whenever exactly one of p, q is true. Hence only the first is valid.

Q.8 [NAT, 1M · Predicate Logic] Answer: 343

View R as an 3×3 0/1 matrix. The sentence says R is every row of the relation matrix must be non-empty: $(2^n - 1)$ choices per row $\Rightarrow (2^n - 1)^n$. With $n = 3$: **343**.

Q.9 [NAT, 1M · Bayesian Networks] Answer: 71

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (4-1) \times 2 + C: (2-1) \times 4 + D: (2-1) \times 2 \times 2 + E: (2-1) \times 2 \times 4 + F: (4-1) \times 4 \times 2 \times 2 = 71$.

Full joint table: $\prod |X| - 1 = 256 - 1 = 255$; saving = 184.

Q.10 [NAT, 1M · Sampling] Answer: 11

Rejection sampling keeps only samples consistent with the evidence $D = T$: samples 2, 4, 10, 11 (4 accepted, 11 rejected).

Among the accepted samples, $A = T$ in 1. Estimate = $1/4 = 0.25$.

(For this network the exact value is 0.4023; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.11 [NAT, 2M · Informed Search] Answer: 10

Heuristic values: $h(A)=4$, $h(C)=4$, $h(F)=6$, $h(G)=0$, $h(H)=3$, $h(J)=2$, $h(K)=3$, $h(S)=7$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (7)	H(8), F(13)
2	H (8)	J(9), C(13), F(13)
3	J (9)	G(10), C(13), F(13)
4	G (10)	C(13), F(13) ← goal, stop

Path returned: **S** → **H** → **J** → **G** with cost **10**; nodes expanded = **4** (S, H, J, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 10$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.12 [MSQ, 2M · Problem Formulation] Answer: A, B, C

(A), (B) and (C) are the textbook definitions. (D) is false: d is the depth of the **shallowest** goal node. The maximum length of any path in the state space is m .

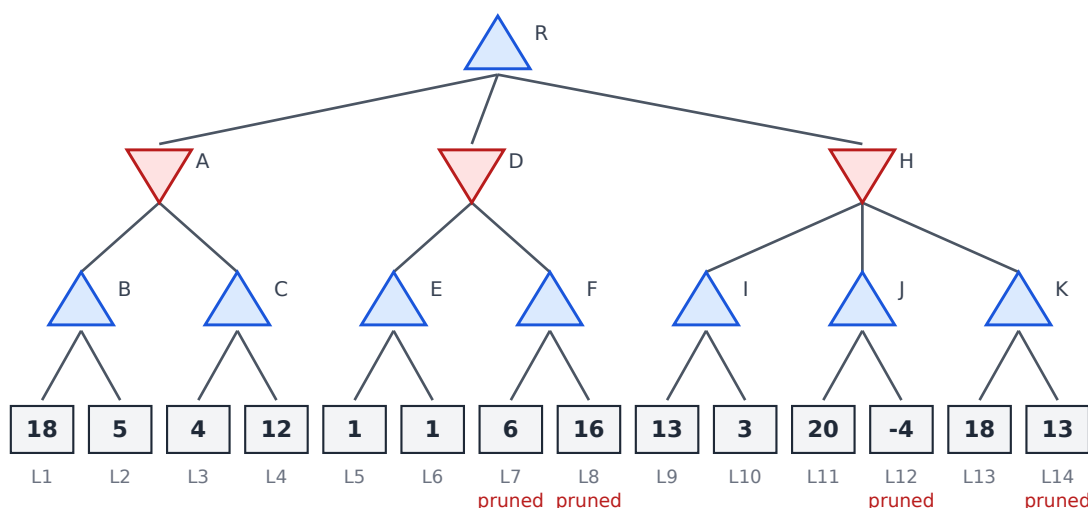
Q.13 [MCQ, 2M · Adversarial Search] Answer: A (A: At MAX nodes in decreasing order of value, at MIN nodes in increasing order of value)

Pruning is greatest when the best move is examined first: the highest value for MAX and the lowest for MIN. That gives the tightest α (or β) as early as possible, so later siblings are cut sooner. (B) is the worst ordering, under which alpha-beta degenerates to $O(b^m)$. (C) is best for only one of the two players. (D) is false: ordering decides whether the cost is $O(b^{m/2})$ or $O(b^m)$.

Q.14 [NAT, 2M · Adversarial Search] Answer: 10

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $1 \leq \alpha = 12 \Rightarrow$ prune remaining children (leaves L7, L8).
- At MAX node J: value $20 \geq \beta = 13 \Rightarrow$ prune remaining children (leaves L12).
- At MAX node K: value $18 \geq \beta = 13 \Rightarrow$ prune remaining children (leaves L14).



Leaves evaluated (10): L1, L2, L3, L4, L5, L6, L9, L10, L11, L13. Leaves pruned (4): L7, L8, L12, L14. Root value = 13 (same as plain minimax, 13 — pruning never changes the root value).

Q.15 [MSQ, 2M · Informed Search] Answer: C (C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	D	E	G	H	I	J	S
$h(n)$	6	6	5	0	3	1	8	17

$h^*(n)$	7	7	6	0	12	2	10	19
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $J \rightarrow H$ ($8 > 3 + 3$), $S \rightarrow H$ ($17 > 7 + 3$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A, B, D (A: $\{p \rightarrow q, q \rightarrow r\} \models p \rightarrow r$; B: $\{p \vee q, \neg p \vee r\} \models q \vee r$; D: $\{p \rightarrow q, \neg q\} \models \neg p$)

(A) Hypothetical syllogism: every model of both premises with $p=T$ has $q=T$ and hence $r=T$. (B) This is exactly the resolution rule on p : if $p=T$ then r must be T ; if $p=F$ then q must be T . (C) Fails: $p=F, q=T$ satisfies both premises but falsifies p (affirming the consequent). (D) Modus tollens: if p were T then q would be T , contradicting $\neg q$. Truth tables confirm A, B, D.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A, B, C (A: $(Q \rightarrow R) \vee (R \rightarrow Q)$; B: $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$; C: $(P \rightarrow R) \Leftrightarrow (\neg R \rightarrow \neg P)$)

(A) $(Q \rightarrow R) \vee (R \rightarrow Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $(P \rightarrow R) \Leftrightarrow (\neg R \rightarrow \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $(R \vee Q) \rightarrow (R \wedge Q)$: **not valid** — counter-model $Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Predicate Logic] Answer: A, D (A: $\exists x \forall y \neg R(x, y)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(2,1), (3,2), (4,2), (4,3)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

(B) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(C) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(3,2) \wedge R(2,1)$ but $\neg R(3,1)$.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Propositional Logic] Answer: D (D: $(P \vee \neg R \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(P \vee Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) The resolution closure contains 22 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(\neg P)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(P \vee \neg R \vee S)$ is derivable:

1. resolve $(P \vee \neg R)$ and $(Q \vee R \vee S)$ on $R \Rightarrow (P \vee Q \vee S)$
2. resolve $(\neg Q \vee \neg R)$ and $(P \vee Q \vee S)$ on $Q \Rightarrow (P \vee \neg R \vee S)$

Q.20 [NAT, 2M · Exact Inference] Answer: 0.043 to 0.047

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}, \text{Doctor}) = P(\text{Flu}) \cdot P(\text{Fever} \mid \text{Flu}) \cdot P(\text{Cough}) \cdot P(\text{Fatigue} \mid \text{Flu}, \text{Fever}) \cdot P(\text{Doctor} \mid \text{Cough}, \text{Fatigue})$.

Inference by enumeration: $P(\text{Flu} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Flu}, e, \text{hidden})$.

Hidden: Co, Do	Fl	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, T	T	$0.1 \times 0.9 \times 0.1 \times 0.3 \times 0.3$	0.00081
T, F	T	$0.1 \times 0.9 \times 0.1 \times 0.3 \times 0.7$	0.00189
F, T	T	$0.1 \times 0.9 \times 0.9 \times 0.3 \times 0.6$	0.01458
F, F	T	$0.1 \times 0.9 \times 0.9 \times 0.3 \times 0.4$	0.00972
T, T	F	$0.9 \times 0.7 \times 0.1 \times 0.9 \times 0.3$	0.01701
T, F	F	$0.9 \times 0.7 \times 0.1 \times 0.9 \times 0.7$	0.03969
F, T	F	$0.9 \times 0.7 \times 0.9 \times 0.9 \times 0.6$	0.30618
F, F	F	$0.9 \times 0.7 \times 0.9 \times 0.9 \times 0.4$	0.20412

$P(\text{Flu}=T, e) = 0.027$; $P(\text{Flu}=F, e) = 0.567$; $P(e) = 0.594$.

$P(\text{Flu}=T \mid e) = 0.027 / 0.594 = \mathbf{0.0455} \approx 0.045$.

Q.21 [MCQ, 2M · Sampling] **Answer: B** (B: A=T, B=T, C=F, D=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.6$; $u = 0.03 < 0.6 \Rightarrow A = T$
- B: $P(B=T \mid \text{—}) = 0.75$; $u = 0.19 < 0.75 \Rightarrow B = T$
- C: $P(C=T \mid A=T, B=T) = 0.2$; $u = 0.24 \geq 0.2 \Rightarrow C = F$
- D: $P(D=T \mid B=T, C=F) = 0.1$; $u = 0.24 \geq 0.1 \Rightarrow D = F$

Sample: **(A=T, B=T, C=F, D=F)**.

Q.22 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B)$, $f(B, D)$, $f(A, B, E)$, $f(A, F)$, $f(B, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, B)$, $f(A, B)$, $f(A, B, E)$, $f(A, F)$	$\{A, B, E, F\}$	$f(B, E, F)$
G	$f(B, F, G)$	$\{B, F, G\}$	$f(B, F)$
E	$f(B, E, F)$	$\{B, E, F\}$	$f(B, F)$
B	$f(B, D)$, $f(B, F)$, $f(B, F)$	$\{B, D, F\}$	$f(D, F)$
D	$f(D, F)$	$\{D, F\}$	$f(F)$

The largest factor produced has **3** variable(s) (created when eliminating A); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over F and normalise.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: A, B, D** (A: $G \perp F \mid B, D$; B: $F \perp A \mid B$; D: $C \perp D \mid B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp F \mid B, D$: **TRUE** — $G \leftarrow A \rightarrow C \leftarrow B \rightarrow F$ (blocked: collider C with neither it nor a descendant observed); $G \leftarrow A \rightarrow C \rightarrow E \leftarrow D \leftarrow B \rightarrow F$ (blocked: collider E with neither it nor a descendant observed); $G \leftarrow B \rightarrow F$ (blocked: observed fork node B).

(B) $F \perp A \mid B$: **TRUE** — $F \leftarrow B \rightarrow C \leftarrow A$ (blocked: observed fork node B); $F \leftarrow B \rightarrow D \rightarrow E \leftarrow C \leftarrow A$ (blocked: observed fork node B); $F \leftarrow B \rightarrow G \leftarrow A$ (blocked: observed fork node B).

(C) $D \perp E \mid G$: **FALSE** — the path $D \leftarrow B \rightarrow C \rightarrow E$ is active given $\{G\}$.

(D) $C \perp D \mid B$: **TRUE** — $C \leftarrow A \rightarrow G \leftarrow B \rightarrow D$ (blocked: collider G with neither it nor a descendant observed); $C \leftarrow B \rightarrow D$ (blocked: observed fork node B); $C \rightarrow E \leftarrow D$ (blocked: collider E with neither it nor a descendant observed).

Q.24 [MSQ, 2M · Variable Elimination] **Answer: A, B, C** (A: $f(a) = 0.12$; B: $f(\neg a) = 0.028$; C: $P(a \mid b, c) \approx 0.811$)

$f(a) = 0.3 \times 0.8 \times 0.5 = 0.12$ (A true). $f(\neg a) = 0.7 \times 0.4 \times 0.1 = 0.028$ (B true). Normalizing, $P(a \mid b, c) = 0.12 / (0.12 + 0.028) = 0.12 / 0.148 \approx 0.811$ (C true). The normalizer is $P(b, c) = f(a) + f(\neg a) = 0.148$, not 0.12 (which is $P(a, b, c)$), so (D) is false.

Q.25 [NAT, 2M · Bayesian Networks] **Answer: 41**

Each CPT needs (number of parent configurations) $\times$ (arity $- 1$). A: $1 \times 2 = 2$. B: $1 \times 1 = 1$. C: $(3 \times 2) \times 3 = 18$. D: $4 \times 1 = 4$. E: $(4 \times 2) \times 2 = 16$. Total = $2 + 1 + 18 + 4 + 16 = 41$, versus $3 \times 2 \times 4 \times 2 \times 3 - 1 = 143$ for the full joint.

Mock Test 029

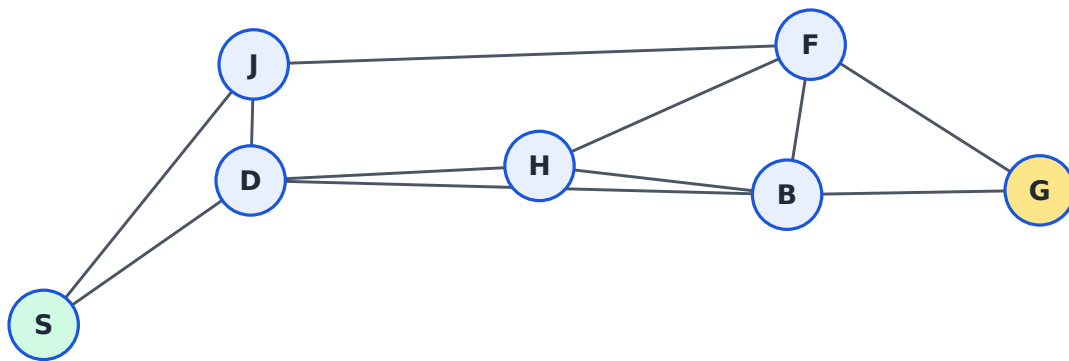
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 7 / 11	13 / 12 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

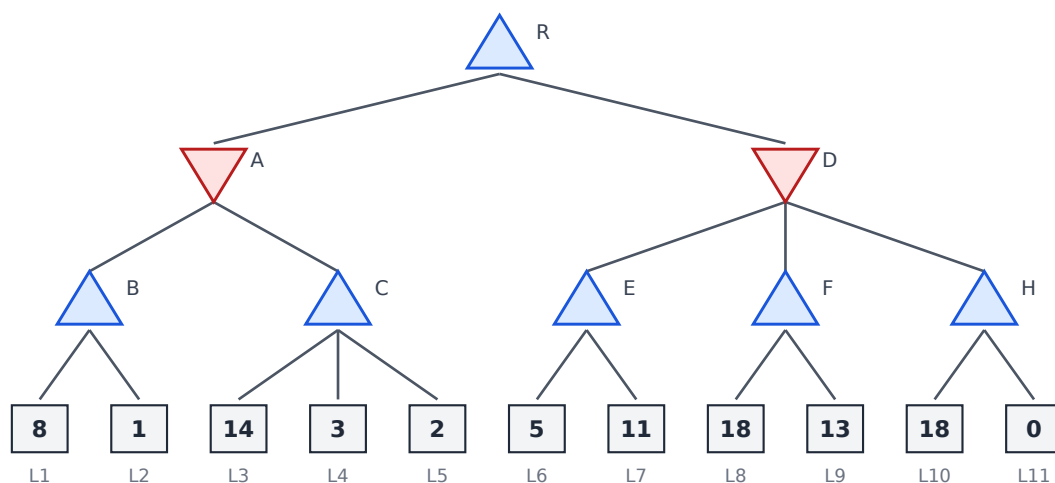
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, J, B, H, F, G
- ☐ (B) S, B, D, F, G
- ☐ (C) S, J, F, H, D, B, G
- ☐ (D) S, D, B, F, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



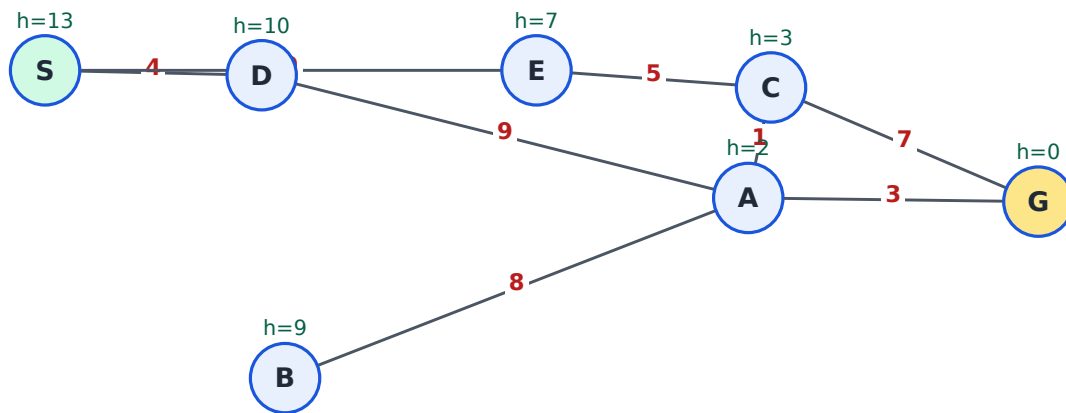
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (\neg\neg Q \vee (\neg Q \leftrightarrow R)) \wedge ((\neg R \rightarrow \neg P) \wedge (\neg P \rightarrow P)) ?$$

Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following statements about forward and backward chaining over Horn knowledge bases is correct?

- ☐ (A) Forward chaining is data-driven while backward chaining is goal-driven
- ☐ (B) Forward chaining is goal-driven while backward chaining is data-driven
- ☐ (C) Both are complete for arbitrary (non-Horn) propositional knowledge bases
- ☐ (D) Backward chaining can derive facts that forward chaining cannot

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following clauses are Horn clauses?

- ☐ (A) $\neg p \vee \neg q \vee r$
- ☐ (B) $p \vee q$
- ☐ (C) $\neg p \vee \neg q$
- ☐ (D) r

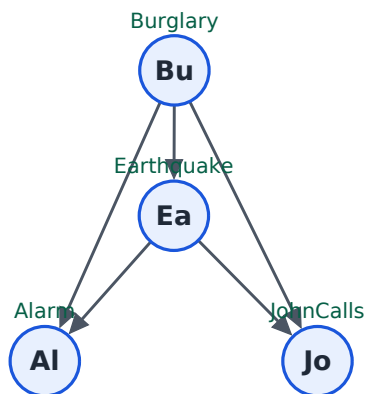
Q.8 [MCQ · 1 mark · negative: -1/3] Conditional Independence

Which three-node structure is such that X and Z are d-separated by the empty set but **not** d-separated given Y ?

- ☐ (A) $X \rightarrow Y \rightarrow Z$
- ☐ (B) $X \leftarrow Y \rightarrow Z$
- ☐ (C) $X \rightarrow Y \leftarrow Z$
- ☐ (D) $X \leftarrow Y \leftarrow Z$

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Bu, Ea, Al, Jo. (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) What is the probability that a single run produces the sample (**Bu=T, Ea=F, Al=F, Jo=F**)? (round off to 4 decimal places)



P(Bu=T)
0.95

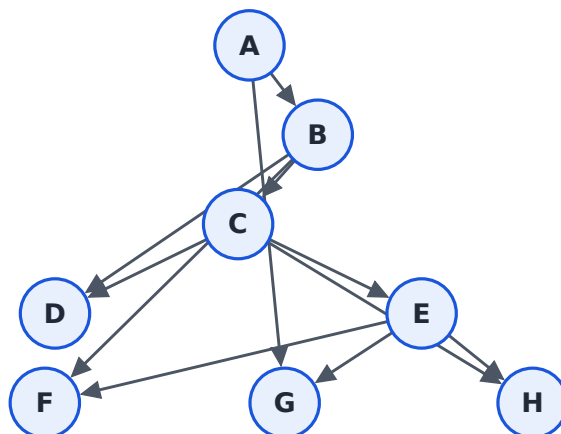
Bu	P(Ea=T Bu)	Bu	Ea	P(Al=T ...)
T	0.2	T	T	0.25
F	0.2	T	F	0.2
		F	T	0.95
		F	F	0.8

Bu	Ea	P(Jo=T ...)
T	T	0.5
T	F	0.05
F	T	0.7
F	F	0.5

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Conditional Independence – Markov blanket

Which set is the Markov blanket of **B** in the Bayesian network below?

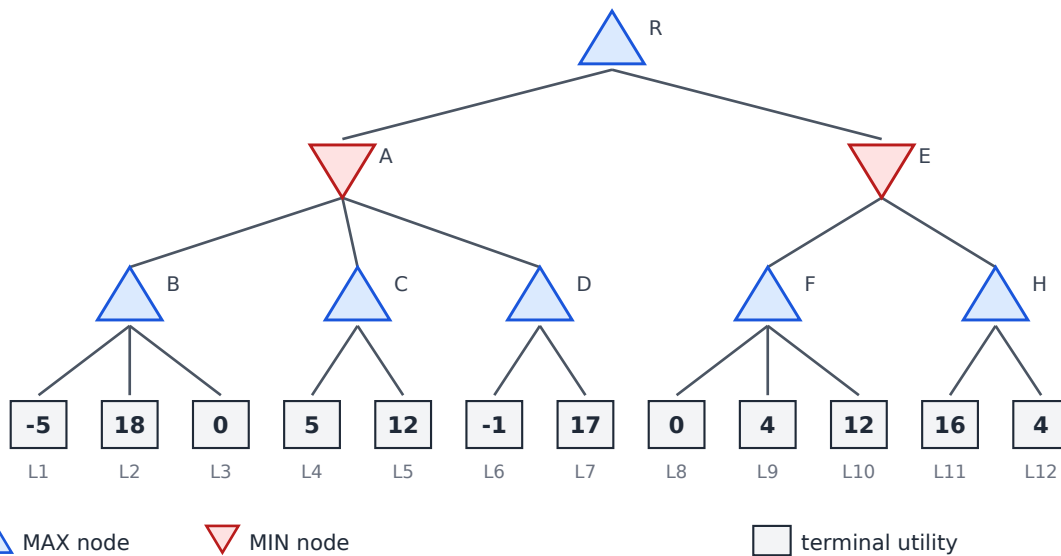


- ☐ (A) {A, C, D, E, F}
☐ (B) {A, C, D, E, F, G, H}
☐ (C) {A, C, D, E, F, G}
☐ (D) {A, C, D, F}

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

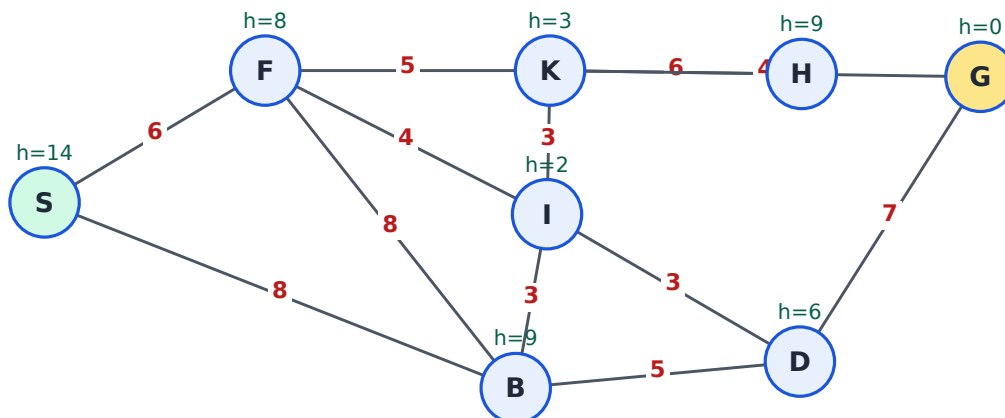
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) h is admissible.
- ☐ (C) $h(B)$ overestimates the true cost from B to G.
- ☐ (D) The edge F-I violates the consistency condition $h(F) \leq c(F, I) + h(I)$.

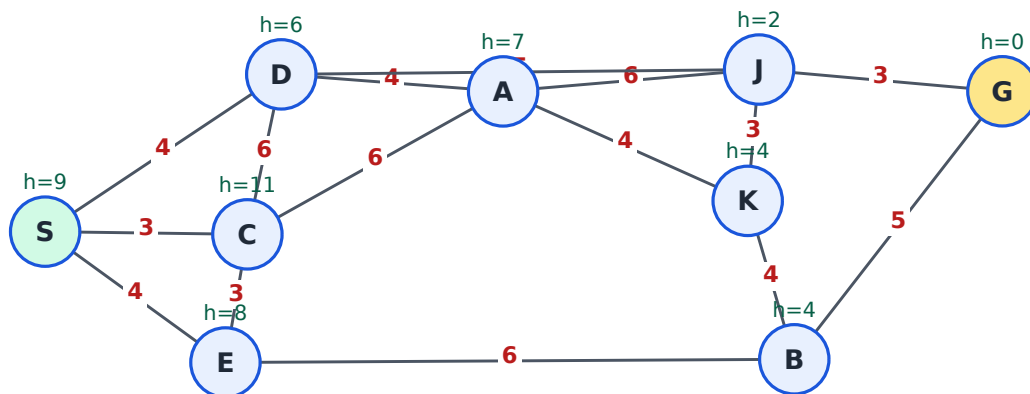
Q.13 [NAT · 2 marks · no negative marking] Adversarial Search

A MAX root has two chance-node children. Chance node C_1 leads with probability 0.4 to a MIN node with leaves (6, 10) and with probability 0.6 to a MIN node with leaves (3, 9). Chance node C_2 leads with probability 0.5 to a leaf of value 8 and with probability 0.5 to a leaf of value 1. What is the expectiminimax value of the root?

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

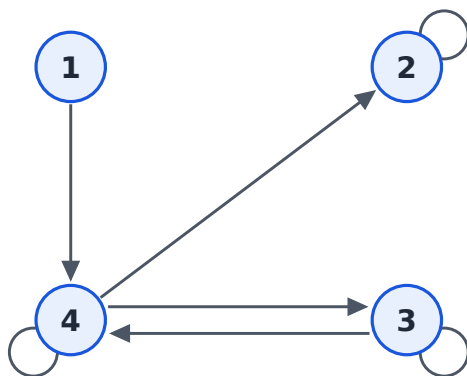
Q.15 [MCQ · 2 marks · negative: -2/3] Informed Search

Weighted A* uses $f(n) = g(n) + W \cdot h(n)$. As the weight $W \rightarrow \infty$, the order in which it expands nodes approaches that of

- ☐ (A) uniform-cost search
- ☐ (B) A* search
- ☐ (C) greedy best-first search
- ☐ (D) breadth-first search

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
- ☐ (B) $\exists x \forall y \neg R(y, x)$
- ☐ (C) $\forall x R(x, x)$
- ☐ (D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(\neg R \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg \neg R)$
☐ (B) $((P \rightarrow Q) \rightarrow R) \leftrightarrow (P \rightarrow (Q \rightarrow R))$
☐ (C) $(Q \vee P) \rightarrow (Q \wedge P)$
☐ (D) $((R \rightarrow Q) \rightarrow R) \rightarrow R$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(P \wedge Q) \rightarrow (R \vee \neg Q)$
 (2) $P \rightarrow (P \vee P)$
 (3) $\neg(\neg P \vee R)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $S \wedge S$
☐ (B) $\neg \neg Q \wedge (P \vee R)$
☐ (C) $(P \rightarrow \neg S) \rightarrow (\neg Q \wedge P)$
☐ (D) $R \rightarrow (Q \wedge \neg P)$

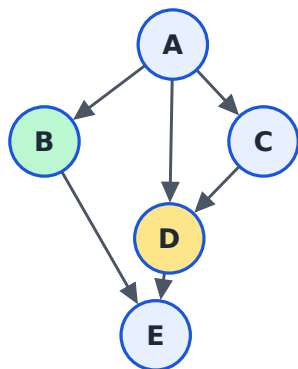
Q.19 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following is a prenex normal form equivalent to $(\forall x P(x)) \rightarrow (\exists y Q(y))$?

- ☐ (A) $\exists x \exists y (P(x) \rightarrow Q(y))$
☐ (B) $\forall x \exists y (P(x) \rightarrow Q(y))$
☐ (C) $\forall x \forall y (P(x) \rightarrow Q(y))$
☐ (D) $\exists y \forall x (P(x) \rightarrow Q(y))$

Q.20 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence **D=F**. The current state of the chain is **(A=F, B=T, C=F, D=F, E=F)**. The next step resamples **B**. With what probability is B set to T? (round off to 3 decimal places)



P(A=T)
0.05

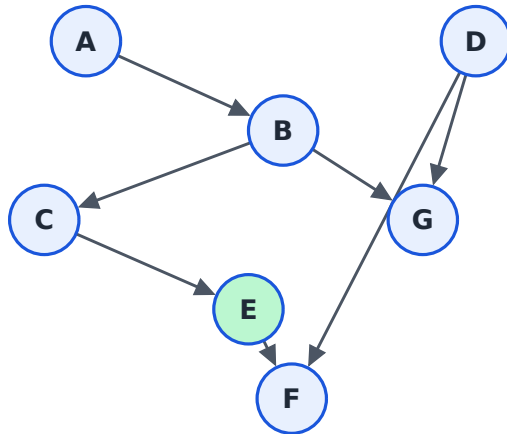
A	P(B=T ...)	A	P(C=T ...)
T	0.6	T	0.8
F	0.9	F	0.6

A	C	P(D=T ...)	B	D	P(E=T ...)
T	T	0.25	T	T	0.7
T	F	0.8	T	F	0.2
F	T	0.4	F	T	0.2
F	F	0.9	F	F	0.5

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{E})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, B, G, D, F, A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

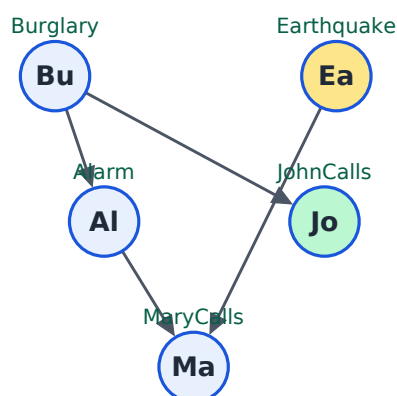
Q.22 [MCQ · 2 marks · negative: -2/3] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. For $P(B | j, m)$, the elimination order A then E is used. After summing out A, which factors remain to be multiplied when E is eliminated?

- ☐ (A) $P(E)$ and $f_A(B, E)$
☐ (B) $P(E)$, $P(B)$ and $f_A(B, E)$
☐ (C) $P(E)$ and $f_A(B, E, J, M)$
☐ (D) $P(E)$, $P(A | B, E)$ and $f_A(B)$

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T | \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls, Ma = MaryCalls.) Compute $P(\text{JohnCalls} = T | \text{Earthquake} = T)$ (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.7

$P(\text{Ea}=T)$
0.9

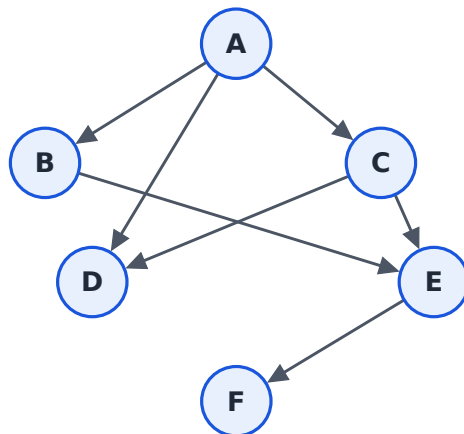
Bu	$P(\text{Al}=T \dots)$
T	0.95
F	0.5

Bu	$P(\text{Jo}=T \dots)$	Ea	Al	$P(\text{Ma}=T \dots)$
T	0.8	T	T	0.6
F	0.9	T	F	0.5
		F	T	0.95
		F	F	0.6

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp D \mid A$
☐ (B) $A \perp E \mid B, C$
☐ (C) $F \perp A \mid E$
☐ (D) $B \perp E \mid F$

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence

Consider the Bayesian network (DAG) with edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$, $B \rightarrow F$. Which of the following d-separation statements are true?

- ☐ (A) A and B are d-separated given $\emptyset$
☐ (B) A and F are d-separated given $\{B\}$
☐ (C) E and F are d-separated given $\{C\}$
☐ (D) E and F are d-separated given $\emptyset$

Answer Key — Mock Test 029

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	NAT	2	Informed Search	5
2	NAT	1	Adversarial Search	11	15	MCQ	2	Informed Search	C
3	NAT	1	Informed Search	21	16	MSQ	2	Predicate Logic	B
4	NAT	1	Propositional Logic	2	17	MSQ	2	Propositional Logic	A, D
5	MCQ	1	Propositional Logic	A	18	MSQ	2	Propositional Logic	C, D
6	NAT	1	Predicate Logic	65	19	MCQ	2	Predicate Logic	A
7	MSQ	1	Propositional Logic	A, C, D	20	NAT	2	Sampling	0.933 to 0.937
8	MCQ	1	Conditional Independence	C	21	NAT	2	Exact Inference	4
9	NAT	1	Sampling	0.5774 to 0.5778	22	MCQ	2	Variable Elimination	A
10	MCQ	1	Conditional Independence	A	23	NAT	2	Exact Inference	0.828 to 0.832
11	NAT	2	Adversarial Search	2	24	MSQ	2	Conditional Independence	A, B, C
12	MSQ	2	Informed Search	B, D	25	MSQ	2	Conditional Independence	A, B, C
13	NAT	2	Adversarial Search	4.5					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: D (D: S, D, B, F, G)

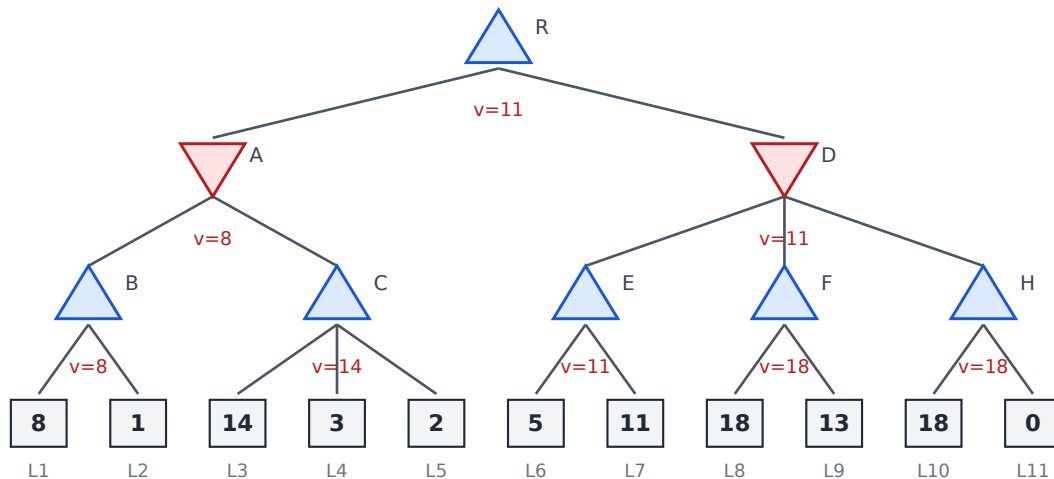
Adjacency lists (alphabetical): B: {D, F, G, H}; D: {B, H, J, S}; F: {B, G, H, J}; G: {B, F}; H: {B, D, F}; J: {D, F, S}; S: {D, J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, B, F, G**; the path found is $S \rightarrow D \rightarrow B \rightarrow F \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] Answer: 11

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 8, C = 14, A = 8, E = 11, F = 18, H = 18, D = 11. Root value = **11**; MAX's optimal first move is towards D.

Q.3 [NAT, 1M · Informed Search] Answer: 21

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (13)	E(7), D(10)
2	E (7)	C(3), D(10)
3	C (3)	G(0), A(2), D(10)
4	G (0)	A(2), D(10) ← goal, stop

Path returned: **S** → **E** → **C** → **G**, cost = 9 + 5 + 7 = **21**.

The optimal cost is $h^*(S) = 16$ — greedy search is not optimal here.

Q.4 [NAT, 1M · Propositional Logic] Answer: 2

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg\neg Q \vee (\neg Q \leftrightarrow R)$	$(\neg R \rightarrow \neg P) \wedge (\neg P \rightarrow P)$	$(\neg\neg Q \vee (\neg Q \leftrightarrow R)) \wedge ((\neg R \rightarrow \neg P) \wedge (\neg P \rightarrow P))$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	F	F	F
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	T	F	F
F	F	F	F	F	F

Rows where F = T: **2**.

Q.5 [MCQ, 1M · Propositional Logic] Answer: A (A: Forward chaining is data-driven while backward chaining is goal-driven)

Forward chaining starts from known facts and fires rules whose premises are satisfied (data-driven).

Backward chaining starts from the query and works back through rules whose conclusion matches (goal-driven). Both are complete only for Horn (definite clause) KBs, not arbitrary ones. For Horn KBs both derive exactly the entailed atoms, so neither derives anything the other cannot.

Q.6 [NAT, 1M · Predicate Logic] Answer: 65

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 4$ this gives **65** (verified by enumerating all interpretations).

Q.7 [MSQ, 1M · Propositional Logic] Answer: A, C, D (A: $\neg p \vee \neg q \vee r$; C: $\neg p \vee \neg q$; D: r)

A Horn clause is a disjunction of literals with at most one positive literal. $\neg p \vee \neg q \vee r$ has one positive literal (a definite clause, i.e. $p \wedge q \rightarrow r$). $\neg p \vee \neg q$ has zero positive literals (a goal clause). r is a single positive literal (a fact). $p \vee q$ has two positive literals, so it is not Horn.

Q.8 [MCQ, 1M · Conditional Independence] Answer: C (C: $X \rightarrow Y \leftarrow Z$)

In a chain (A, D) or fork (B), the path $X-Y-Z$ is active when Y is unobserved and blocked when Y is observed. In the collider (v-structure) $X \rightarrow Y \leftarrow Z$ the path is blocked when neither Y nor a descendant is observed, so X and Z are d-separated by $\emptyset$, and it becomes active when Y is observed. Hence only (C) has the stated behaviour.

Q.9 [NAT, 1M · Sampling] Answer: 0.5774 to 0.5778

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(\text{Bu}=\text{T}) \times P(\text{Ea}=\text{F}|\text{Bu}=\text{T}) \times P(\text{Al}=\text{F}|\text{Bu}=\text{T}, \text{Ea}=\text{F}) \times P(\text{Jo}=\text{F}|\text{Bu}=\text{T}, \text{Ea}=\text{F}) \\ = 0.95 \times 0.8 \times 0.8 \times 0.95 = \mathbf{0.5776} \approx 0.5776.$$

Q.10 [MCQ, 1M · Conditional Independence] Answer: A (A: {A, C, D, E, F})

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

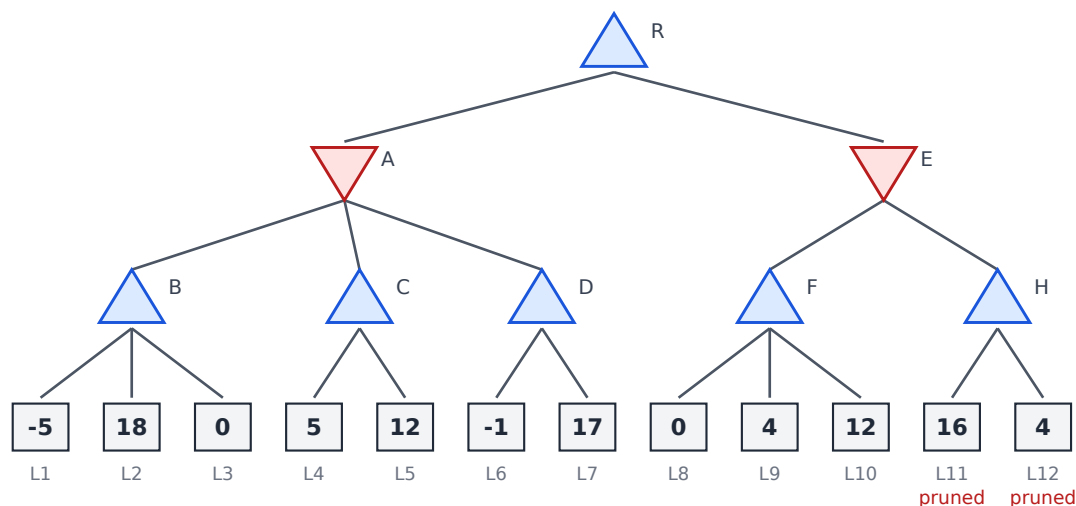
Parents of B: {A}; children: {C, D, F}; co-parents: {E}.

MB(B) = {A, C, D, E, F} — **5** nodes. Given its Markov blanket, B is conditionally independent of every other node in the network.

Q.11 [NAT, 2M · Adversarial Search] Answer: 2

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node E: value $12 \leq \alpha = 12 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (10): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10. Leaves pruned (2): L11, L12. Root value = 12 (same as plain minimax, 12 — pruning never changes the root value).

Q.12 [MSQ, 2M · Informed Search] Answer: B, D (B: h is admissible.; D: The edge F-I violates the consistency condition $h(F) \leq c(F, I) + h(I)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	D	F	G	H	I	K	S
$h(n)$	9	6	8	0	9	2	3	14
$h^*(n)$	10	7	9	0	10	7	4	15
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $B \rightarrow I$ ($9 > 3 + 2$), $D \rightarrow I$ ($6 > 3 + 2$), $F \rightarrow I$ ($8 > 4 + 2$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Adversarial Search] **Answer: 4.5**

The MIN nodes under C_1 have values $\min(6,10) = 6$ and $\min(3,9) = 3$. $C_1 = 0.4 \cdot 6 + 0.6 \cdot 3 = 2.4 + 1.8 = 4.2$. $C_2 = 0.5 \cdot 8 + 0.5 \cdot 1 = 4.5$. Root = $\max(4.2, 4.5) = 4.5$, so MAX chooses C_2 .

Q.14 [NAT, 2M · Informed Search] **Answer: 5**

Heuristic values: $h(A)=7$, $h(B)=4$, $h(C)=11$, $h(D)=6$, $h(E)=8$, $h(G)=0$, $h(J)=2$, $h(K)=4$, $h(S)=9$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	D(10), E(12), C(14)
2	D (10)	J(11), E(12), C(14), A(15)
3	J (11)	E(12), G(12), C(14), A(15), K(16)
4	E (12)	G(12), B(14), C(14), A(15), K(16)
5	G (12)	B(14), C(14), A(15), K(16) $\leftarrow$ goal, stop

Path returned: **S $\rightarrow$ D $\rightarrow$ J $\rightarrow$ G** with cost **12**; nodes expanded = **5** (S, D, J, E, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 12$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [MCQ, 2M · Informed Search] **Answer: C** (C: greedy best-first search)

For very large W , the term $W \cdot h$ dominates g , so nodes are effectively ordered by h alone, which is **greedy best-first search**. $W = 1$ gives A^* (B). $W = 0$ gives UCS (A). BFS (D) would need ordering by depth.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: B** (B: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,4), (2,2), (3,3), (3,4), (4,2), (4,3), (4,4)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(4,2)$ holds but $R(2,4)$ does not.

(B) $\exists x \forall y \neg R(y, x)$: **TRUE** — 1 has no incoming edge.

(C) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,2)$ nor $R(2,1)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(\neg R \rightarrow P) \Leftrightarrow (\neg P \rightarrow \neg \neg R)$; D: $((R \rightarrow Q) \rightarrow R) \rightarrow R$)

(A) $(\neg R \rightarrow P) \Leftrightarrow (\neg P \rightarrow \neg \neg R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $((P \rightarrow Q) \rightarrow R) \Leftrightarrow (P \rightarrow (Q \rightarrow R))$: **not valid** — counter-model $P=F, Q=T, R=F$.

(C) $(Q \vee P) \rightarrow (Q \wedge P)$: **not valid** — counter-model $P=T, Q=F$.

(D) $((R \rightarrow Q) \rightarrow R) \rightarrow R$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $(P \rightarrow \neg S) \rightarrow (\neg Q \wedge P)$; D: $R \rightarrow (Q \wedge \neg P)$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
T	F	F	T	T
T	F	F	F	T

(A) $S \wedge S$ is false in the KB-model $P=T, Q=F, R=F, S=F \Rightarrow$ not entailed.

(B) $\neg \neg Q \wedge (P \vee R)$ is false in the KB-model $P=T, Q=F, R=F, S=T \Rightarrow$ not entailed.

(C) $(P \rightarrow \neg S) \rightarrow (\neg Q \wedge P)$ is true in all models of $KB \Rightarrow$ **entailed**.

(D) $R \rightarrow (Q \wedge \neg P)$ is true in all models of $KB \Rightarrow$ **entailed**.

Q.19 [MCQ, 2M · Predicate Logic] **Answer: A** (A: $\exists x \exists y (P(x) \rightarrow Q(y))$)

A quantifier pulled out of the antecedent of $\rightarrow$ flips: $(\forall x P(x)) \rightarrow \psi \equiv \exists x (P(x) \rightarrow \psi)$. From the consequent it does not flip: $\phi \rightarrow \exists y Q(y) \equiv \exists y (\phi \rightarrow Q(y))$. So we get $\exists x \exists y (P(x) \rightarrow Q(y))$. Check option B on domain $\{1, 2\}$ with $P = \{1\}$, $Q = \emptyset$: the original is true ($\forall x P(x)$ is false), but at $x=1$ no y makes $P(1) \rightarrow Q(y)$ true, so B is false. C and D are false on the same interpretation.

Q.20 [NAT, 2M · Sampling] Answer: 0.933 to 0.937

Gibbs sampling draws B from $P(B \mid \text{mb}(B))$, where $\text{mb}(B) = \{A, D, E\}$:

$$P(B \mid \text{mb}) \propto P(B \mid \text{parents}(B)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y)).$$

$$B=T: P(B=T \mid A=F) \times P(E=F \mid B=T, D=F) = 0.9 \times 0.8 = 0.72$$

$$B=F: P(B=F \mid A=F) \times P(E=F \mid B=F, D=F) = 0.1 \times 0.5 = 0.05$$

$$\text{Normalise: } 0.72 / (0.72 + 0.05) = \mathbf{0.9351}.$$

Q.21 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(D)$, $f(C, E)$, $f(D, E, F)$, $f(B, D, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C)$, $f(C, E)$	$\{B, C, E\}$	$f(B, E)$
B	$f(A, B)$, $f(B, D, G)$, $f(B, E)$	$\{A, B, D, E, G\}$	$f(A, D, E, G)$
G	$f(A, D, E, G)$	$\{A, D, E, G\}$	$f(A, D, E)$
D	$f(D)$, $f(D, E, F)$, $f(A, D, E)$	$\{A, D, E, F\}$	$f(A, E, F)$
F	$f(A, E, F)$	$\{A, E, F\}$	$f(A, E)$
A	$f(A)$, $f(A, E)$	$\{A, E\}$	$f(E)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over E and normalise.

Q.22 [MCQ, 2M · Variable Elimination] Answer: A (A: $P(E)$ and $f_A(B, E)$)

Initially the factors are $P(B)$, $P(E)$, $P(A \mid B, E)$, $f(A) = P(j \mid A)$, $g(A) = P(m \mid A)$. Summing out A multiplies the three factors containing A and yields $f_A(B, E)$. Eliminating E then multiplies only the factors mentioning E: $P(E)$ and $f_A(B, E)$, giving $f_E(B)$. $P(B)$ does not mention E so it is left aside until the end; J and M are evidence, not dimensions; $P(A \mid B, E)$ was already consumed.

Q.23 [NAT, 2M · Exact Inference] Answer: 0.828 to 0.832

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot P(\text{Alarm} \mid \text{Burglary}) \cdot P(\text{JohnCalls} \mid \text{Burglary}) \cdot P(\text{MaryCalls} \mid \text{Earthquake}, \text{Alarm})$.

Inference by enumeration: $P(\text{JohnCalls} \mid e) = \alpha \sum_{\text{hidden}} P(\text{JohnCalls}, e, \text{hidden})$.

Hidden: Bu, Al, Ma	Jo	product of CPT entries (Bu · Ea · Al · Jo · Ma)	value
T, T, T	T	$0.7 \times 0.9 \times 0.95 \times 0.8 \times 0.6$	0.28728
T, T, F	T	$0.7 \times 0.9 \times 0.95 \times 0.8 \times 0.4$	0.19152
T, F, T	T	$0.7 \times 0.9 \times 0.05 \times 0.8 \times 0.5$	0.0126
T, F, F	T	$0.7 \times 0.9 \times 0.05 \times 0.8 \times 0.5$	0.0126
F, T, T	T	$0.3 \times 0.9 \times 0.5 \times 0.9 \times 0.6$	0.0729
F, T, F	T	$0.3 \times 0.9 \times 0.5 \times 0.9 \times 0.4$	0.0486
F, F, T	T	$0.3 \times 0.9 \times 0.5 \times 0.9 \times 0.5$	0.06075
F, F, F	T	$0.3 \times 0.9 \times 0.5 \times 0.9 \times 0.5$	0.06075
T, T, T	F	$0.7 \times 0.9 \times 0.95 \times 0.2 \times 0.6$	0.07182
T, T, F	F	$0.7 \times 0.9 \times 0.95 \times 0.2 \times 0.4$	0.04788
T, F, T	F	$0.7 \times 0.9 \times 0.05 \times 0.2 \times 0.5$	0.00315
T, F, F	F	$0.7 \times 0.9 \times 0.05 \times 0.2 \times 0.5$	0.00315
F, T, T	F	$0.3 \times 0.9 \times 0.5 \times 0.1 \times 0.6$	0.0081
F, T, F	F	$0.3 \times 0.9 \times 0.5 \times 0.1 \times 0.4$	0.0054
F, F, T	F	$0.3 \times 0.9 \times 0.5 \times 0.1 \times 0.5$	0.00675
F, F, F	F	$0.3 \times 0.9 \times 0.5 \times 0.1 \times 0.5$	0.00675

$$P(\text{JohnCalls}=T, e) = 0.747; P(\text{JohnCalls}=F, e) = 0.153; P(e) = 0.9.$$

$$P(\text{JohnCalls}=T \mid e) = 0.747 / 0.9 = \mathbf{0.83} \approx 0.830.$$

Q.24 [MSQ, 2M · Conditional Independence] **Answer: A, B, C** (A: $B \perp D \mid A$; B: $A \perp E \mid B, C$; C: $F \perp A \mid E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp D \mid A$: **TRUE** — $B \leftarrow A \rightarrow C \rightarrow D$ (blocked: observed fork node A); $B \leftarrow A \rightarrow D$ (blocked: observed fork node A); $B \rightarrow E \leftarrow C \leftarrow A \rightarrow D$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow E \leftarrow C \rightarrow D$ (blocked: collider E with neither it nor a descendant observed).

(B) $A \perp E \mid B, C$: **TRUE** — $A \rightarrow B \rightarrow E$ (blocked: observed chain node B); $A \rightarrow C \rightarrow E$ (blocked: observed chain node C); $A \rightarrow D \leftarrow C \rightarrow E$ (blocked: collider D with neither it nor a descendant observed).

(C) $F \perp A \mid E$: **TRUE** — $F \leftarrow E \leftarrow B \leftarrow A$ (blocked: observed chain node E); $F \leftarrow E \leftarrow C \leftarrow A$ (blocked: observed chain node E); $F \leftarrow E \leftarrow C \rightarrow D \leftarrow A$ (blocked: observed chain node E).

(D) $B \perp E \mid F$: **FALSE** — the path $B \leftarrow A \rightarrow C \rightarrow E$ is active given $\{F\}$.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: A, B, C** (A: A and B are d-separated given $\emptyset$; B: A and F are d-separated given $\{B\}$; C: E and F are d-separated given $\{C\}$)

(A) The only path $A \rightarrow C \leftarrow B$ has collider C with no observed descendant: blocked, so true. (B) The only path $A \rightarrow C \leftarrow B \rightarrow F$ is blocked by the unobserved collider C (and also by the observed fork node B): true. (C) The only path $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ contains observed non-collider C: blocked, true. (D) With nothing observed, $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ has no colliders and is active, so false.

Mock Test 030

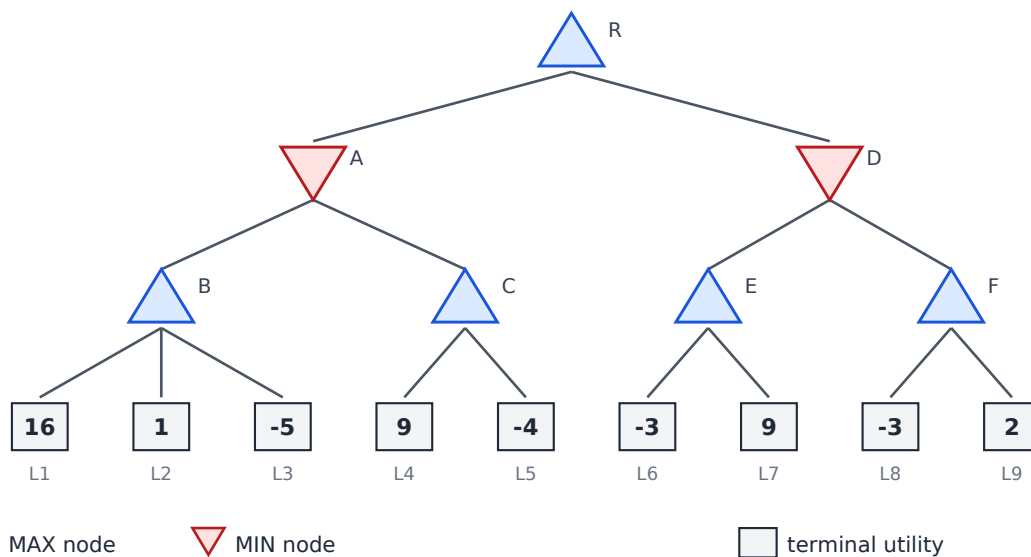
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 7 / 12	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

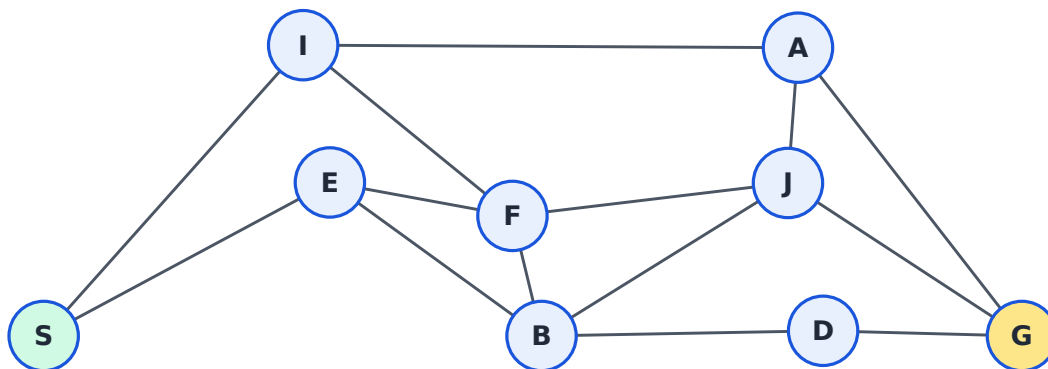
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

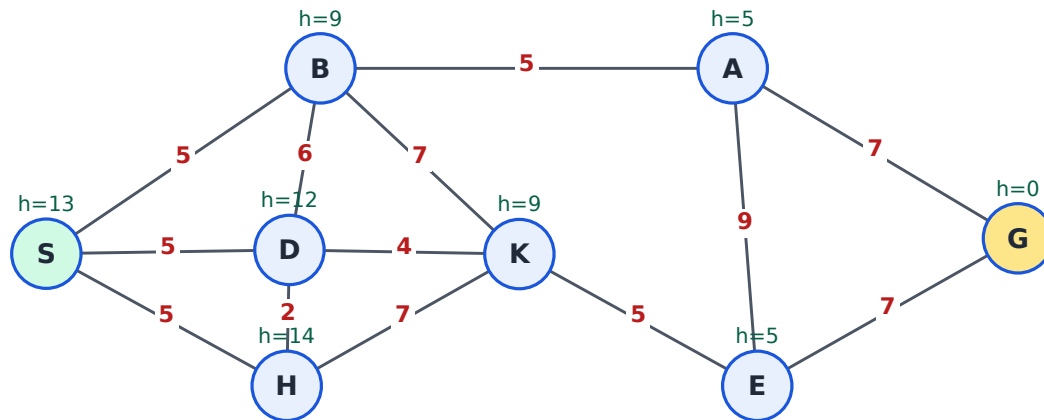
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, E, B, D, G
- ☐ (B) S, E, I, B, A, F, D, J, G
- ☐ (C) S, E, I, B, F, A, D, J, G
- ☐ (D) S, I, E, F, A, B, J, G

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search

Chess has an average branching factor of about 35, and a game lasts about 100 plies. The full game tree therefore has about how many nodes?

- ☐ (A) 10^{40}
☐ (B) 10^{100}
☐ (C) 10^{154}
☐ (D) 10^{400}

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B**, **D** and rules:

- R1: $H \wedge E \wedge C \rightarrow B$
 R2: $A \wedge F \rightarrow K$
 R3: $D \wedge H \wedge E \rightarrow F$
 R4: $F \rightarrow H$
 R5: $D \rightarrow J$
 R6: $A \wedge H \rightarrow E$
 R7: $J \rightarrow C$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Let $KB = \{P \rightarrow Q, Q \rightarrow R, \neg R\}$. Which one of the following is entailed by KB?

- ☐ (A) $\neg P \wedge \neg Q$
☐ (B) $P \vee Q$
☐ (C) $\neg P \wedge Q$
☐ (D) R

Q.7 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following are sentences, i.e. formulas with no free variables (a is a constant)?

- ☐ (A) $(\forall x P(x)) \rightarrow Q(x)$
☐ (B) $\forall x (P(x) \rightarrow \exists y R(x, y))$
☐ (C) $\exists x R(x, y)$
☐ (D) $\forall x \exists y (R(x, y) \wedge P(a))$

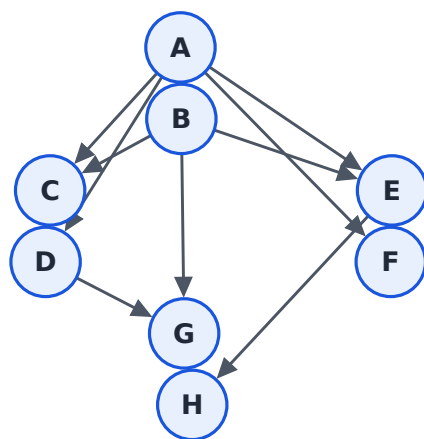
Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = \neg(R \vee R) \rightarrow (\neg P \rightarrow (\neg Q \wedge P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **E** in the Bayesian network below?



Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. With prior sampling, what is the probability that a single generated sample is exactly (c, $\neg s$, r, w)? (Round to 3 decimal places.)

Answer (numerical): _____

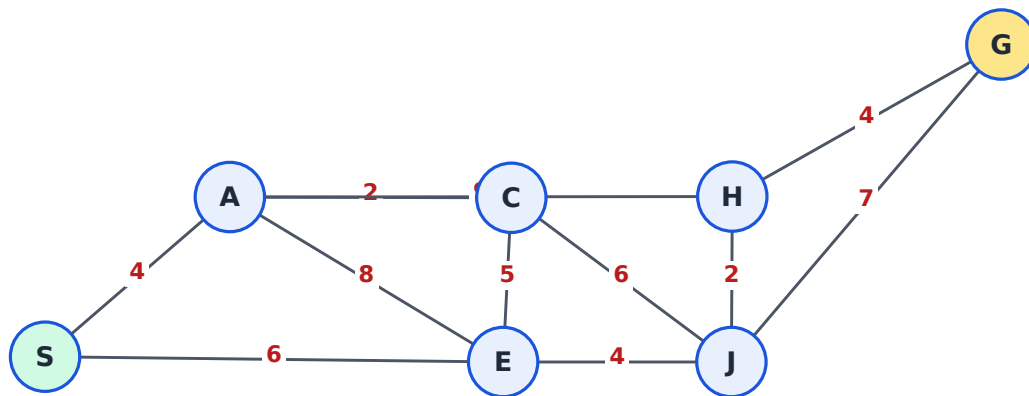
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search

Iterative deepening search (IDS) runs on a uniform tree with $b = 10$ and the shallowest goal at $d = 5$. Assume the worst case: the final iteration (limit 5) generates every node at depth 5. Each time a non-root node is generated, it counts once, so a node regenerated in a later iteration counts again. The root is not counted. What is the total number of node generations?

Answer (numerical): _____

Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

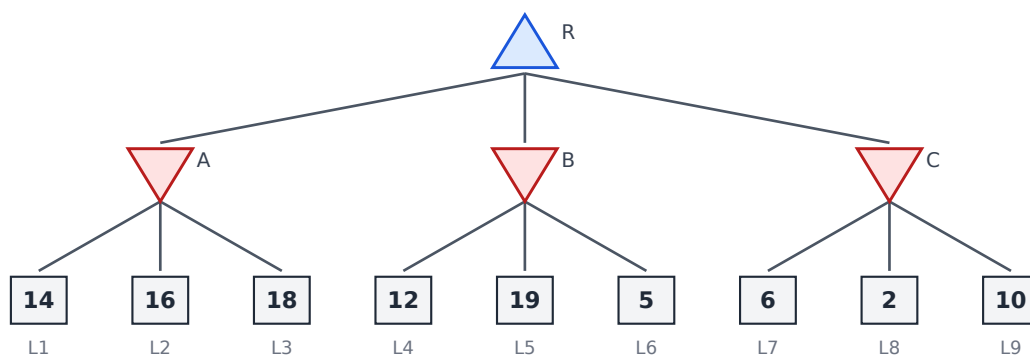
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow E \rightarrow J \rightarrow H \rightarrow G$
☐ (B) $S \rightarrow A \rightarrow H \rightarrow G$
☐ (C) $S \rightarrow E \rightarrow J \rightarrow G$
☐ (D) $S \rightarrow A \rightarrow C \rightarrow J \rightarrow H \rightarrow G$

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



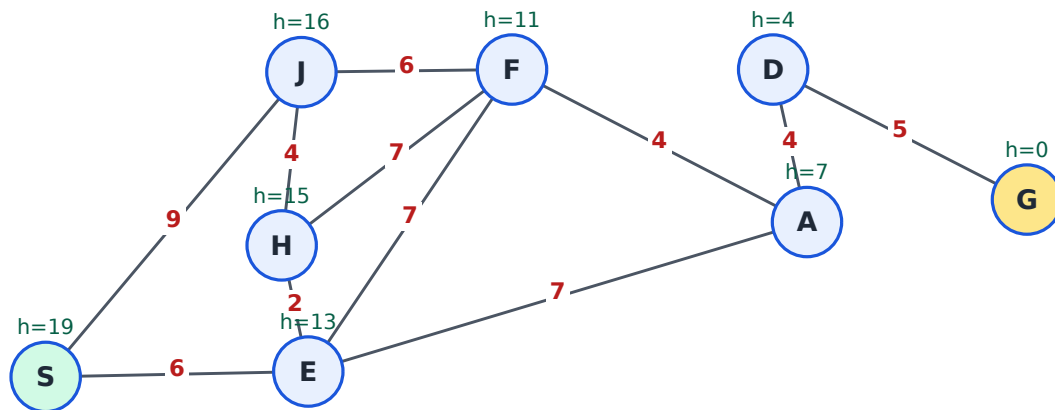
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following correctly express 'Every person who owns a dog is happy'?

- ☐ (A) $\forall x ((\text{Person}(x) \wedge \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y))) \rightarrow \text{Happy}(x))$
- ☐ (B) $\forall x \forall y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$
- ☐ (C) $\forall x \exists y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$
- ☐ (D) $\forall x (\text{Person}(x) \rightarrow \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y) \wedge \text{Happy}(x)))$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(P \rightarrow Q) \rightarrow (Q \rightarrow P)$
- ☐ (B) $(Q \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg Q)$
- ☐ (C) $\neg(P \wedge R) \leftrightarrow (\neg P \wedge \neg R)$
- ☐ (D) $((P \rightarrow R) \rightarrow Q) \leftrightarrow (P \rightarrow (R \rightarrow Q))$

Q.17 [NAT · 2 marks · no negative marking] Predicate Logic

A first-order language has constants a, b and one binary function symbol g . Let $H_0 = \{a, b\}$ and $H_{i+1} = H_i \cup \{g(s, t) : s, t \in H_i\}$. How many ground terms are in H_2 ?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

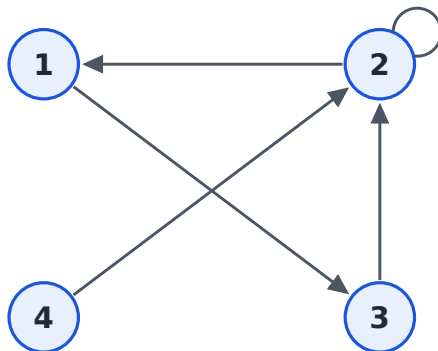
- (1) $\neg(\neg R \wedge R)$
- (2) $(R \rightarrow Q) \leftrightarrow (P \leftrightarrow R)$
- (3) $(P \leftrightarrow \neg Q) \vee (\neg R \wedge R)$

Which of the following sentences is/are entailed by KB ($\text{KB} \models \alpha$)?

- ☐ (A) $(Q \vee Q) \wedge \neg Q$
- ☐ (B) $\neg P \leftrightarrow (R \vee R)$
- ☐ (C) $\neg Q$
- ☐ (D) $(P \leftrightarrow Q) \vee \neg P$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (B) $\forall x R(x, x)$
☐ (C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.20 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

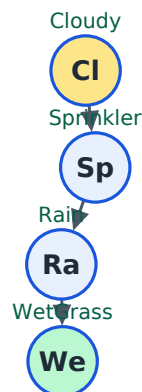
The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D . Using **rejection sampling**, estimate $P(C = T \mid B = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	T	T	F	T
2	T	F	F	T
3	T	T	T	F
4	T	F	T	F
5	T	T	F	F
6	T	F	T	F
7	T	T	T	T
8	T	F	T	F
9	T	T	T	F
10	T	F	F	T
11	T	F	F	T
12	T	T	T	F
13	T	F	T	F
14	T	F	T	F
15	T	F	F	T

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\text{WetGrass} = T \mid \text{Cloudy} = F)$ (round off to 3 decimal places).



Cl	P(Cl=T)
T	0.9
F	

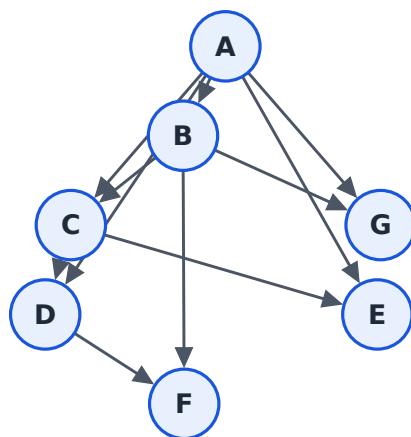
Cl	P(Sp=T ...)	Sp	P(Ra=T ...)
T	0.95	T	0.7
F	0.25	F	0.8

Ra	P(We=T ...)
T	0.7
F	0.2

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

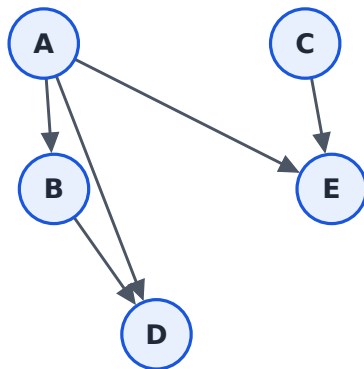
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp G \mid B, E$
☐ (B) $C \perp A \mid D$
☐ (C) $D \perp G \mid A, B$
☐ (D) $F \perp E \mid A, C$

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.2, 0.32, 0.84, 0.2, 0.22)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.05

A	P(B=T ...)
T	0.9
F	0.1

P(C=T)
0.05

A	B	P(D=T ...)	A	C	P(E=T ...)
T	T	0.4	T	T	0.75
T	F	0.4	T	F	0.6
F	T	0.5	F	T	0.75
F	F	0.9	F	F	0.1

- ☐ (A) A=T, B=F, C=T, D=F, E=F
☐ (B) A=F, B=F, C=F, D=T, E=F
☐ (C) A=T, B=F, C=F, D=T, E=F
☐ (D) A=F, B=T, C=F, D=T, E=F

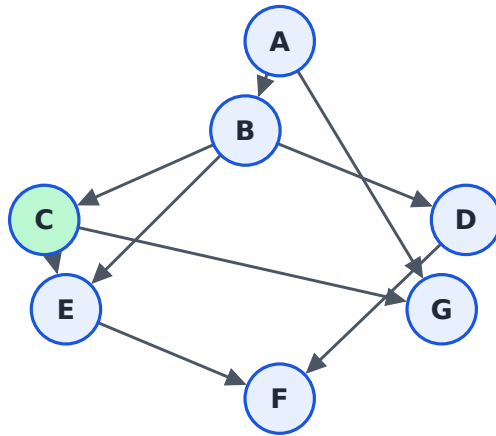
Q.24 [MSQ · 2 marks · no negative marking] Probability for AI

A hidden variable H takes values h_1, h_2, h_3 with prior probabilities 0.5, 0.3, 0.2. For an observation e, $P(e \mid h_1) = 0.1$, $P(e \mid h_2) = 0.4$, $P(e \mid h_3) = 0.7$. Which of the following are correct?

- ☐ (A) $P(e) = 0.31$
☐ (B) $P(h_3 \mid e) \approx 0.452$
☐ (C) $P(h_1 \mid e) \approx 0.161$
☐ (D) The most probable hypothesis after observing e (MAP) is h_2

Q.25 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{C})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{E, G, D, B, F, A}$. What is the scope of the new factor produced when $\mathbf{E}$ is summed out?



- ☐ (A) $f(B, D, F)$
☐ (B) $f(B, C, F)$
☐ (C) $f(B, C, D, F)$
☐ (D) $f(B, C, D, E, F)$

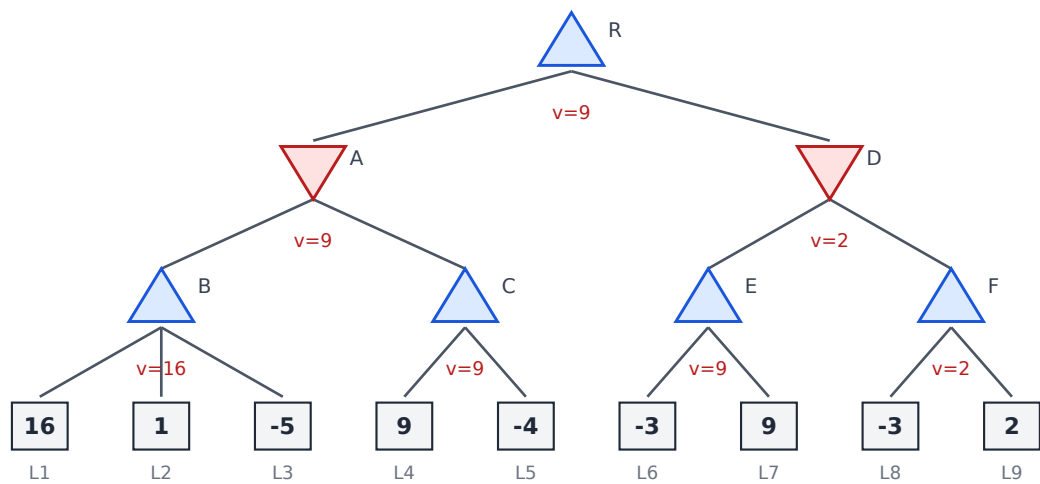
Answer Key — Mock Test 030

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	9	14	NAT	2	Informed Search	5
2	MCQ	1	Uninformed Search	C	15	MSQ	2	Predicate Logic	A, B
3	NAT	1	Informed Search	17	16	MSQ	2	Propositional Logic	B
4	MCQ	1	Adversarial Search	C	17	NAT	2	Predicate Logic	38
5	NAT	1	Propositional Logic	4	18	MSQ	2	Propositional Logic	D
6	MCQ	1	Propositional Logic	A	19	MSQ	2	Predicate Logic	A, C, D
7	MSQ	1	Predicate Logic	B, D	20	NAT	2	Sampling	0.55 to 0.57
8	NAT	1	Propositional Logic	2	21	NAT	2	Exact Inference	0.585 to 0.589
9	NAT	1	Conditional Independence	5	22	MSQ	2	Conditional Independence	C, D
10	NAT	1	Sampling	0.323 to 0.325	23	MCQ	2	Sampling	B
11	NAT	2	Uninformed Search	123450	24	MSQ	2	Probability for AI	A, B, C
12	MCQ	2	Uninformed Search	A	25	MCQ	2	Exact Inference	C
13	NAT	2	Adversarial Search	5					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 9

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 16, C = 9, A = 9, E = 9, F = 2, D = 2. Root value = **9**; MAX's optimal first move is towards A.

Q.2 [MCQ, 1M · Uninformed Search] Answer: C (C: S, E, I, B, F, A, D, J, G)
Adjacency lists (alphabetical): A: {G, I, J}; B: {D, E, F, J}; D: {B, G}; E: {B, F, S}; F: {B, E, I, J}; G: {A, D, J}; I: {A, F, S}; J: {A, B, F, G}; S: {E, I}.
BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[E, I]
2	E	[I, B, F]
3	I	[B, F, A]
4	B	[F, A, D, J]
5	F	[A, D, J]
6	A	[D, J, G]
7	D	[J, G]
8	J	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, E, I, B, F, A, D, J, G**. The path returned is S → I → A → G with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).
Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [NAT, 1M · Informed Search] Answer: 17
Greedy best-first search orders the frontier by h(n) only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (13)	B(9), D(12), H(14)
2	B (9)	A(5), K(9), D(12), H(14)
3	A (5)	G(0), E(5), K(9), D(12), H(14)
4	G (0)	E(5), K(9), D(12), H(14) ← goal, stop

Path returned: **S → B → A → G**, cost = 5 + 5 + 7 = **17**.
The optimal cost is h*(S) = 17 — greedy happens to be optimal here, but in general it is not.

Q.4 [MCQ, 1M · Adversarial Search] Answer: C (C: 10^{154})
 $35^{100} = 10^{100 \cdot \log_{10} 35} \approx 10^{100 \times 1.544} \approx \mathbf{10^{154}}$, the AIMA estimate. 10^{40} (A) is AIMA's estimate of the number of distinct chess positions, not of the tree. (B) and (D) do not match the calculation.

Q.5 [NAT, 1M · Propositional Logic] Answer: 4
Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).
Round 1: fire $D \rightarrow J$; add J.
Round 2: fire $J \rightarrow C$; add C.
Fixed point: {B, C, D, J} — 4 atoms are entailed (facts: B, D; newly derived: C, J). Atoms never derived: A, E, F, H, K.

Q.6 [MCQ, 1M · Propositional Logic] Answer: A ($A: \neg P \wedge \neg Q$)

From $\neg R$ and $Q \rightarrow R$, modus tollens gives $\neg Q$; from $\neg Q$ and $P \rightarrow Q$ it gives $\neg P$. The only model of KB (over P, Q, R) is $P=F, Q=F, R=F$, in which $\neg P \wedge \neg Q$ is true. $P \vee Q, \neg P \wedge Q$ and R are all false in that model, so they are not entailed.

Q.7 [MSQ, 1M · Predicate Logic] Answer: B, D ($B: \forall x (P(x) \rightarrow \exists y R(x, y))$; $D: \forall x \exists y (R(x, y) \wedge P(a))$)

(A) The scope of $\forall x$ ends before $\rightarrow$, so x in $Q(x)$ is free. (B) x and y are bound everywhere. (C) y is free. (D) x and y are bound; a is a constant, not a variable. Hence B and D are sentences.

Q.8 [NAT, 1M · Propositional Logic] Answer: 2

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg(R \vee R)$	$\neg P \rightarrow (\neg Q \wedge P)$	$\neg(R \vee R) \rightarrow (\neg P \rightarrow (\neg Q \wedge P))$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	T	T	T
F	T	T	F	F	T
F	T	F	T	F	F
F	F	T	F	F	T
F	F	F	T	F	F

F has 6 models, so it is false in $8 - 6 = 2$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.9 [NAT, 1M · Conditional Independence] Answer: 5

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of E: {A, B}; children: {F, H}; co-parents: {G}.

$MB(E) = \{A, B, F, G, H\}$ — **5** nodes. Given its Markov blanket, E is conditionally independent of every other node in the network.

Q.10 [NAT, 1M · Sampling] Answer: 0.323 to 0.325

Prior sampling generates each event with probability equal to its joint probability: $S_{PS}(x) = \prod P(x_i | \text{parents})$.
 $= P(c) P(\neg s | c) P(r | c) P(w | \neg s, r) = 0.5 \times 0.9 \times 0.8 \times 0.9 = 0.324$.

Q.11 [NAT, 2M · Uninformed Search] Answer: 123450

Nodes at depth i are generated once in each iteration with limit $\geq i$, so they are generated $(d - i + 1)$ times.
 $N(IDS) = d \cdot b + (d-1) \cdot b^2 + \dots + 1 \cdot b^d = 5 \cdot 10 + 4 \cdot 100 + 3 \cdot 1000 + 2 \cdot 10000 + 1 \cdot 100000 = 50 + 400 + 3000 + 20000 + 100000 = \mathbf{123450}$. BFS generates 111110 nodes on the same problem, so the overhead of repeating the shallow levels is only about 11%.

Q.12 [MCQ, 2M · Uninformed Search] Answer: A ($A: S \rightarrow E \rightarrow J \rightarrow H \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

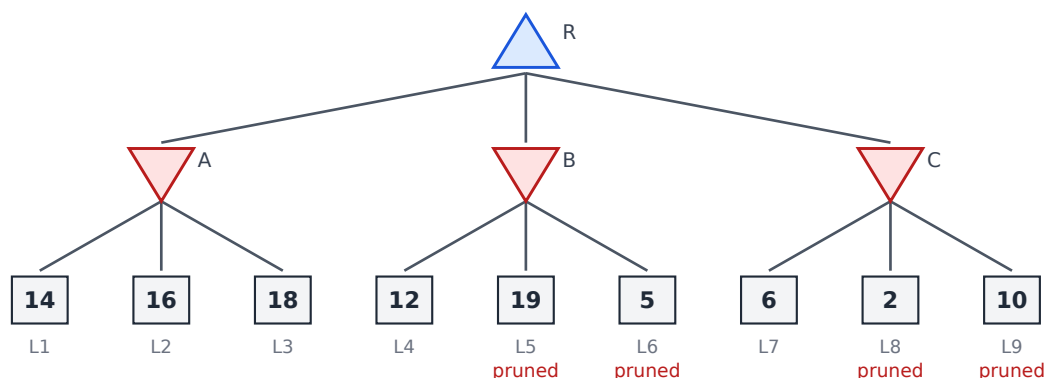
Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	A(4), E(6)
2	A (4)	C(6), E(6), H(13)
3	C (6)	E(6), J(12), H(13)
4	E (6)	J(10), H(13)
5	J (10)	H(12), G(17)
6	H (12)	G(16)
7	G (16)	← goal, stop

Path returned: **S** $\rightarrow$ **E** $\rightarrow$ **J** $\rightarrow$ **H** $\rightarrow$ **G**, cost = $6 + 4 + 2 + 4 = \mathbf{16}$. Nodes expanded (including S and G): **7** (S, A, C, E, J, H, G).

Q.13 [NAT, 2M · Adversarial Search] Answer: 5

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $12 \leq \alpha = 14 \Rightarrow$ prune remaining children (leaves L5, L6).
- At MIN node C: value $6 \leq \alpha = 14 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (5): L1, L2, L3, L4, L7. Leaves pruned (4): L5, L6, L8, L9. Root value = 14 (same as plain minimax, 14 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(A)=7$, $h(D)=4$, $h(E)=13$, $h(F)=11$, $h(G)=0$, $h(H)=15$, $h(J)=16$, $h(S)=19$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (19)	E(19), J(25)
2	E (19)	A(20), H(23), F(24), J(25)
3	A (20)	D(21), H(23), F(24), J(25)
4	D (21)	G(22), H(23), F(24), J(25)
5	G (22)	H(23), F(24), J(25) ← goal, stop

Path returned: **S** → **E** → **A** → **D** → **G** with cost **22**; nodes expanded = **5** (S, E, A, D, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 22$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [MSQ, 2M · Predicate Logic] Answer: A, B (A: $\forall x ((\text{Person}(x) \wedge \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y))) \rightarrow \text{Happy}(x))$; B: $\forall x \forall y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$)

(A) is the direct reading. (B) is equivalent: $(\exists y \phi(y)) \rightarrow \psi \equiv \forall y (\phi(y) \rightarrow \psi)$ when y is not free in ψ . (C) $\exists y (\phi(y) \rightarrow \psi) \equiv (\forall y \phi(y)) \rightarrow \psi$, which is vacuously true for x as soon as some object is not a dog owned by x , so it imposes almost no constraint. (D) asserts every person owns a dog. Brute force on small domains confirms $A \equiv B$ and that C, D differ.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B (B: $(Q \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg Q)$)

(A) $(P \rightarrow Q) \rightarrow (Q \rightarrow P)$: **not valid** — counter-model $P=F, Q=T$.

(B) $(Q \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $\neg(P \wedge R) \leftrightarrow (\neg P \wedge \neg R)$: **not valid** — counter-model $P=T, R=F$.

(D) $((P \rightarrow R) \rightarrow Q) \leftrightarrow (P \rightarrow (R \rightarrow Q))$: **not valid** — counter-model $P=F, Q=F, R=T$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [NAT, 2M · Predicate Logic] Answer: 38

$|H_1| = 2 + 2^2 = 6$: $a, b, g(a,a), g(a,b), g(b,a), g(b,b)$. $H_2 = H_1 \cup \{g(s, t) : s, t \in H_1\}$. The 36 terms $g(s, t)$ are syntactically distinct, and they already include the 4 g -terms of H_1 (since $H_0 \subseteq H_1$); adding a and b gives $|H_2| = 2 + 6^2 = 38$.

Q.18 [MSQ, 2M · Propositional Logic] Answer: D (D: $(P \leftrightarrow Q) \vee \neg P$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
F	T	F	T

(A) $(Q \vee Q) \wedge \neg Q$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(B) $\neg P \leftrightarrow (R \vee R)$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(C) $\neg Q$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(D) $(P \leftrightarrow Q) \vee \neg P$ is true in all models of $KB \Rightarrow$ **entailed**.

Q.19 [MSQ, 2M · Predicate Logic] Answer: A, C, D (A: $\forall x \exists y (x \neq y \wedge R(x, y))$; C: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,3), (2,1), (2,2), (3,2), (4,2)\}$.

(A) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(B) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [NAT, 2M · Sampling] Answer: 0.55 to 0.57

Rejection sampling keeps only samples consistent with the evidence $B = F$: samples 2, 4, 6, 8, 10, 11, 13, 14, 15 (9 accepted, 6 rejected).

Among the accepted samples, $C = T$ in 5. Estimate = $5/9 = 0.5556$.

(For this network the exact value is 0.4; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.21 [NAT, 2M · Exact Inference] Answer: 0.585 to 0.589

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Rain})$.

Inference by enumeration: $P(\text{WetGrass} \mid e) = \alpha \sum_{\text{hidden}} P(\text{WetGrass}, e, \text{hidden})$.

Hidden: Sp, Ra	We	product of CPT entries (Cl · Sp · Ra · We)	value
T, T	T	$0.1 \times 0.25 \times 0.7 \times 0.7$	0.01225
T, F	T	$0.1 \times 0.25 \times 0.3 \times 0.2$	0.0015
F, T	T	$0.1 \times 0.75 \times 0.8 \times 0.7$	0.042
F, F	T	$0.1 \times 0.75 \times 0.2 \times 0.2$	0.003
T, T	F	$0.1 \times 0.25 \times 0.7 \times 0.3$	0.00525
T, F	F	$0.1 \times 0.25 \times 0.3 \times 0.8$	0.006
F, T	F	$0.1 \times 0.75 \times 0.8 \times 0.3$	0.018
F, F	F	$0.1 \times 0.75 \times 0.2 \times 0.8$	0.012

$P(\text{WetGrass}=T, e) = 0.05875$; $P(\text{WetGrass}=F, e) = 0.04125$; $P(e) = 0.1$.

$P(\text{WetGrass}=T \mid e) = 0.05875 / 0.1 = 0.5875 \approx 0.587$.

Q.22 [MSQ, 2M · Conditional Independence] Answer: C, D (C: $D \perp G \mid A, B$; D: $F \perp E \mid A, C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp G \mid B, E$: **FALSE** — the path $A \rightarrow G$ is active given $\{B, E\}$.

(B) $C \perp A \mid D$: **FALSE** — the path $C \leftarrow A$ is active given $\{D\}$.

(C) $D \perp G \mid A, B$: **TRUE** — $D \leftarrow A \rightarrow B \rightarrow G$ (blocked: observed fork node A); $D \leftarrow A \rightarrow C \leftarrow B \rightarrow G$ (blocked: observed fork node A); $D \leftarrow A \rightarrow E \leftarrow C \leftarrow B \rightarrow G$ (blocked: observed fork node A); $D \leftarrow A \rightarrow G$ (blocked: observed fork node A); $D \leftarrow C \leftarrow A \rightarrow B \rightarrow G$ (blocked: observed fork node A); $D \leftarrow C \leftarrow A \rightarrow G$ (blocked: observed fork node A) (+8 more paths, all blocked).

(D) $F \perp E \mid A, C$: **TRUE** — $F \leftarrow B \leftarrow A \rightarrow C \rightarrow E$ (blocked: observed fork node A); $F \leftarrow B \leftarrow A \rightarrow D \leftarrow C \rightarrow E$ (blocked: observed fork node A); $F \leftarrow B \leftarrow A \rightarrow E$ (blocked: observed fork node A); $F \leftarrow B \rightarrow C \leftarrow A \rightarrow E$ (blocked: observed fork node A); $F \leftarrow B \rightarrow C \rightarrow D \leftarrow A \rightarrow E$ (blocked: observed chain node C); $F \leftarrow B \rightarrow C \rightarrow E$ (blocked: observed chain node C) (+11 more paths, all blocked).

Q.23 [MCQ, 2M · Sampling] Answer: B (B: $A=F, B=F, C=F, D=T, E=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.05$; $u = 0.2 \geq 0.05 \Rightarrow A = F$
- B: $P(B=T \mid A=F) = 0.1$; $u = 0.32 \geq 0.1 \Rightarrow B = F$
- C: $P(C=T \mid \text{—}) = 0.05$; $u = 0.84 \geq 0.05 \Rightarrow C = F$
- D: $P(D=T \mid A=F, B=F) = 0.9$; $u = 0.2 < 0.9 \Rightarrow D = T$
- E: $P(E=T \mid A=F, C=F) = 0.1$; $u = 0.22 \geq 0.1 \Rightarrow E = F$

Sample: **(A=F, B=F, C=F, D=T, E=F)**.

Q.24 [MSQ, 2M · Probability for AI] **Answer: A, B, C** (A: $P(e) = 0.31$; B: $P(h_3 | e) \approx 0.452$; C: $P(h_1 | e) \approx 0.161$)

Unnormalized posteriors $P(e | h_i)P(h_i)$: $0.1 \times 0.5 = 0.05$, $0.4 \times 0.3 = 0.12$, $0.7 \times 0.2 = 0.14$. Marginalizing, $P(e) = 0.05 + 0.12 + 0.14 = 0.31$ (A true). $P(h_3 | e) = 0.14/0.31 \approx 0.452$ (B true); $P(h_1 | e) = 0.05/0.31 \approx 0.161$ (C true); $P(h_2 | e) = 0.12/0.31 \approx 0.387$. The MAP hypothesis is h_3 (largest posterior), even though it had the smallest prior, so (D) is false.

Q.25 [MCQ, 2M · Exact Inference] **Answer: C** (C: $f(B, C, D, F)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(B, D)$, $f(B, C, E)$, $f(D, E, F)$, $f(A, C, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(B, C, E)$, $f(D, E, F)$	$\{B, C, D, E, F\}$	$f(B, C, D, F)$
G	$f(A, C, G)$	$\{A, C, G\}$	$f(A, C)$
D	$f(B, D)$, $f(B, C, D, F)$	$\{B, C, D, F\}$	$f(B, C, F)$
B	$f(A, B)$, $f(B, C)$, $f(B, C, F)$	$\{A, B, C, F\}$	$f(A, C, F)$
F	$f(A, C, F)$	$\{A, C, F\}$	$f(A, C)$
A	$f(A)$, $f(A, C)$, $f(A, C)$	$\{A, C\}$	$f(C)$

The largest factor produced has **4** variable(s) (created when eliminating E); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over C and normalise.

Mock Test 031

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 8 / 13	12 / 16 / 12

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

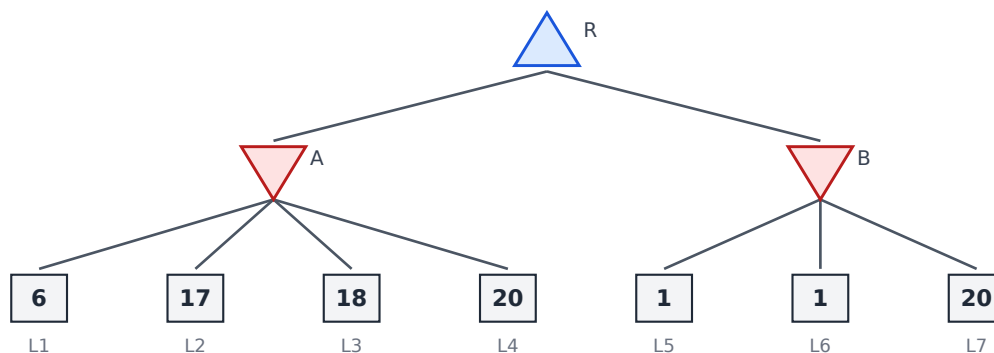
Q.1 [MSQ · 1 mark · no negative marking] Informed Search

h_1 and h_2 are admissible heuristics. Which of the following are **guaranteed** to be admissible?

- ☐ (A) $\max(h_1, h_2)$
- ☐ (B) $(h_1 + h_2) / 2$
- ☐ (C) $h_1 + h_2$
- ☐ (D) $0.5 \cdot h_1$

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



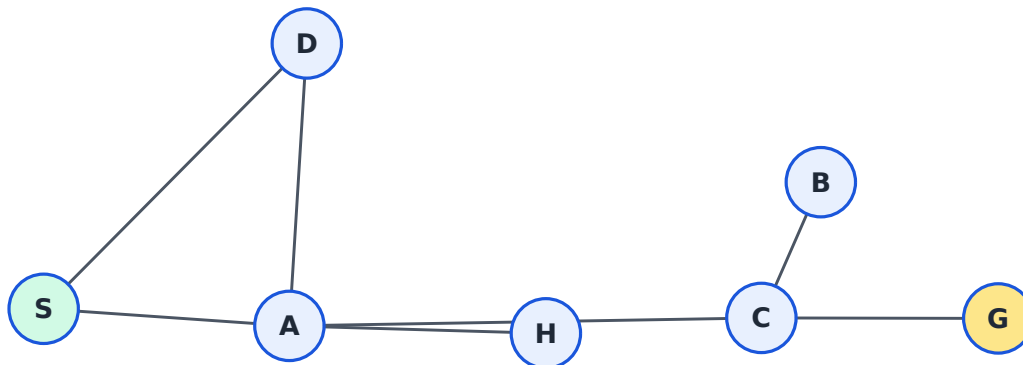
 MAX node  MIN node

 terminal utility

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, A, H, C, G
- ☐ (B) S, A, B, C, G
- ☐ (C) S, A, D, C, H, B, G
- ☐ (D) S, A, C, B, G

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Effective branching factor

A* finds a solution at depth $d = 5$ after generating $N = 161$ nodes (root excluded). The effective branching factor b^* satisfies $N = b^* + (b^*)^2 + \dots + (b^*)^d$. Find b^* (round off to 2 decimal places).

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

The clauses $A \vee B \vee \neg C$ and $\neg A \vee B \vee D$ are resolved on the symbol A. How many literals are in the resolvent (after merging duplicate literals)?

Answer (numerical): _____

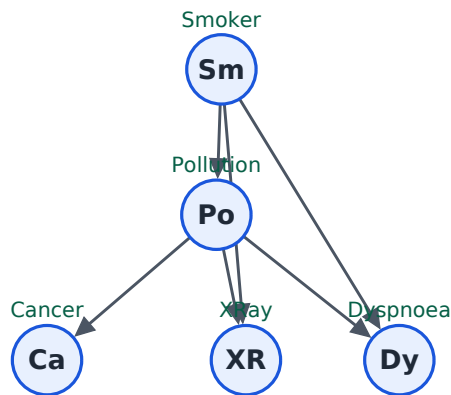
Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Resolving the first-order clauses $P(x) \vee Q(f(x))$ and $\neg P(g(y)) \vee R(y)$ (variables already standardized apart) on P gives

- ☐ (A) $Q(f(g(y))) \vee R(y)$
- ☐ (B) $Q(f(x)) \vee R(y)$
- ☐ (C) $Q(g(f(y))) \vee R(y)$
- ☐ (D) $Q(f(y)) \vee R(g(y))$

Q.7 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR, Dy. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay, Dy = Dyspnoea.) What is the probability that a single run produces the sample **(Sm=T, Po=F, Ca=F, XR=F, Dy=F)**? (round off to 4 decimal places)



P(Sm=T)
0.25

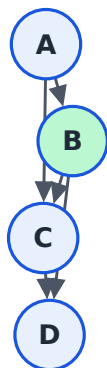
Sm	P(Po=T ...)	Po	P(Ca=T ...)
T	0.2	T	0.2
F	0.7	F	0.6

Sm	Po	P(XR=T ...)	Sm	Po	P(Dy=T ...)
T	T	0.3	T	T	0.5
T	F	0.05	T	F	0.1
F	T	0.7	F	T	0.3
F	F	0.8	F	F	0.8

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability **$P(B = T)$** (round off to 3 decimal places).



P(A=T)
0.1

A	P(B=T .)	A	B	P(C=T ...)
T	0.4	T	T	0.7
F	0.75	T	F	0.3
		F	T	0.9
		F	F	0.1

B	C	P(D=T ...)
T	T	0.05
T	F	0.6
F	T	0.2
F	F	0.95

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(A = T \mid C = T)$, how many of these samples are **rejected**?

#	A	B	C	D
1	T	F	T	F
2	T	F	F	T
3	F	F	F	T
4	T	F	T	F
5	F	F	T	T
6	T	F	T	F
7	F	F	F	T
8	F	F	F	T
9	F	F	F	T
10	F	F	T	T
11	T	F	T	F
12	F	F	F	T
13	T	F	F	F
14	F	F	F	T
15	F	F	T	T
16	F	F	T	T
17	F	T	F	T
18	F	T	T	T
19	F	F	F	T
20	F	F	T	T

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Sampling

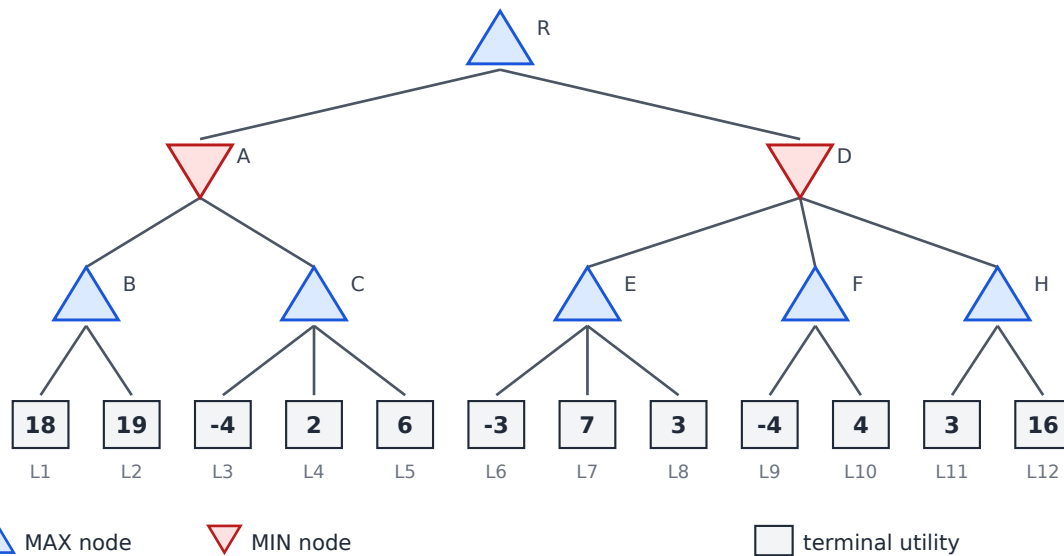
Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. Prior (direct) sampling visits variables in the order C, S, R, W and sets a variable to true if and only if the next uniform random number u satisfies $u < P(\text{variable} = \text{true} \mid \text{sampled parents})$. The random numbers used are 0.42, 0.07, 0.85, 0.30 (in this order). What sample is generated?

- ☐ (A) (c, s, $\neg r$, w)
☐ (B) (c, $\neg s$, r, w)
☐ (C) (c, s, r, w)
☐ (D) ($\neg c$, s, $\neg r$, $\neg w$)

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

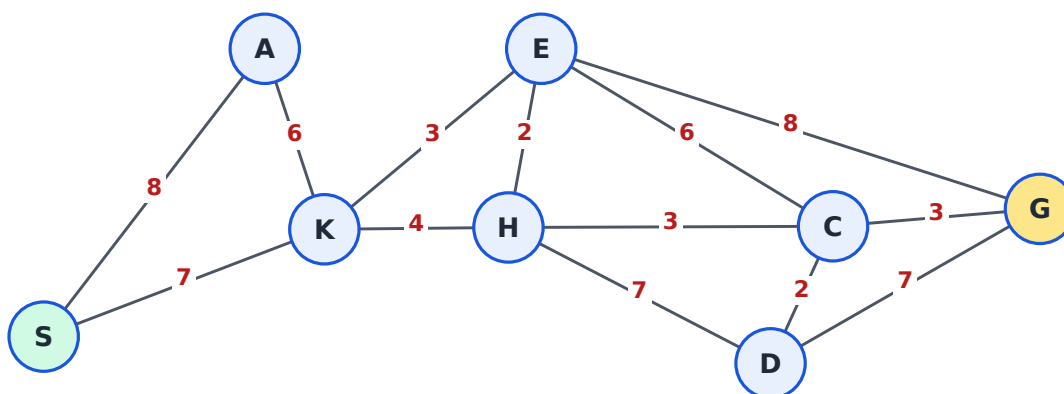
Q.12 [MSQ · 2 marks · no negative marking] Informed Search

In the 8-puzzle, h_1 is the number of misplaced tiles and h_2 is the total Manhattan distance. Which of the following are TRUE?

- ☐ (A) $h_2(n) \geq h_1(n)$ for every state n
- ☐ (B) Both h_1 and h_2 are consistent
- ☐ (C) $h_1 + h_2$ is admissible
- ☐ (D) The number of reachable states from any start state is $9!/2 = 181440$

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

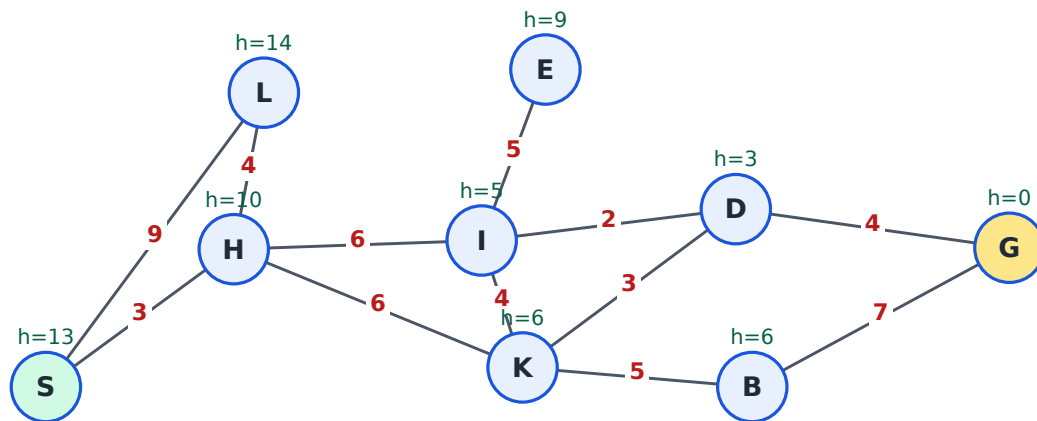
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(P \vee \neg Q)$, C2: $(P \vee R)$, C3: $(P \vee \neg R \vee S)$, C4: $(P \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
- ☐ (B) (P)
- ☐ (C) (Q)
- ☐ (D) $(\neg Q \vee R)$

Q.16 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following is logically equivalent to $\neg \exists x (P(x) \wedge \forall y Q(x, y))$?

- ☐ (A) $\forall x (P(x) \rightarrow \exists y \neg Q(x, y))$
- ☐ (B) $\forall x (\neg P(x) \wedge \exists y \neg Q(x, y))$
- ☐ (C) $\exists x (\neg P(x) \vee \exists y \neg Q(x, y))$
- ☐ (D) $\forall x (P(x) \wedge \exists y \neg Q(x, y))$

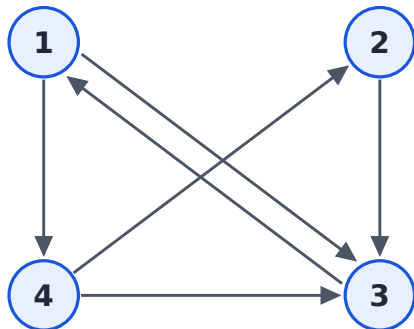
Q.17 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on the set $\{1, 2, 3\}$ are transitive, i.e. satisfy $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\forall x R(x, x)$
☐ (D) $\exists x R(x, x)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = ((Q \wedge R) \wedge (S \rightarrow Q)) \rightarrow (\neg R \vee (\neg S \wedge P)) ?$$

Answer (numerical): _____

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **E, A, C** and rules:

- R1: $J \wedge E \rightarrow K$
 R2: $C \rightarrow J$
 R3: $K \wedge E \wedge B \rightarrow J$
 R4: $H \wedge B \wedge A \rightarrow E$
 R5: $D \wedge H \wedge K \rightarrow B$
 R6: $E \wedge J \rightarrow F$
 R7: $D \wedge H \rightarrow C$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) D is entailed
☐ (B) F is entailed
☐ (C) J is entailed
☐ (D) K is entailed

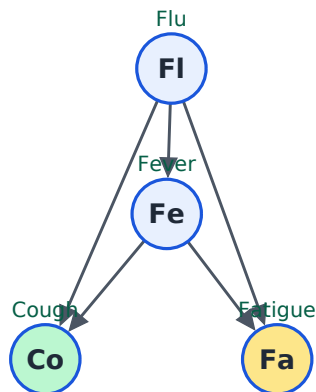
Q.21 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$
☐ (B) $\neg(\neg R \wedge Q) \leftrightarrow (\neg\neg R \vee \neg Q)$
☐ (C) $(P \vee R) \rightarrow (P \wedge R)$
☐ (D) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Cough} = T \mid \text{Fatigue} = F)$ (round off to 3 decimal places).



FI	P(FI=T)
T	0.4

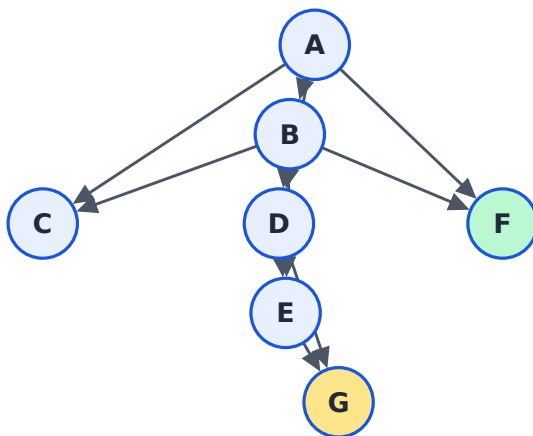
FI	P(Fe=T FI)	FI	Fe	P(Co=T ...)
T	0.5	T	T	0.25
F	0.25	T	F	0.5
		F	T	0.05
		F	F	0.6

FI	Fe	P(Fa=T ...)
T	T	0.8
T	F	0.5
F	T	0.8
F	F	0.9

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(F \mid G = T)$ in the Bayesian network below (Boolean variables; evidence $G = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D**, **A**, **E**, **C**, **B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

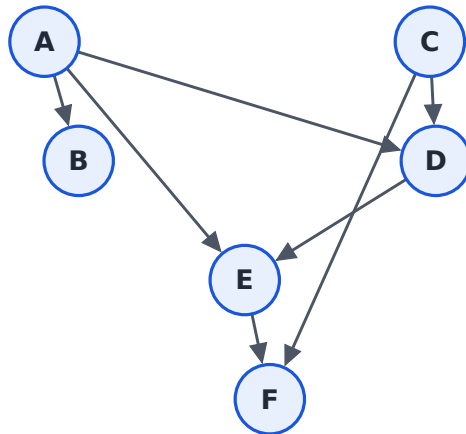
Q.24 [MSQ · 2 marks · no negative marking] Sampling

Which statements about Gibbs sampling (as an MCMC method in Bayesian networks) are correct?

- ☐ (A) Successive samples are generally correlated, not independent
- ☐ (B) With fixed evidence and all CPT entries strictly between 0 and 1, the chain's stationary distribution is $P(X \mid e)$
- ☐ (C) Each state of the Markov chain is a complete assignment to all variables, with evidence variables fixed
- ☐ (D) Each sample must be accepted or rejected depending on whether it agrees with the evidence

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp C \mid E$
☐ (B) $F \perp B$
☐ (C) $B \perp C \mid A$
☐ (D) $B \perp E \mid A, F$

Answer Key — Mock Test 031

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MSQ	1	Informed Search	A, B, D	14	NAT	2	Informed Search	5
2	NAT	1	Adversarial Search	6	15	MSQ	2	Propositional Logic	B
3	MCQ	1	Uninformed Search	D	16	MCQ	2	Predicate Logic	A
4	NAT	1	Informed Search	2.48 to 2.52	17	NAT	2	Predicate Logic	171
5	NAT	1	Propositional Logic	3	18	MSQ	2	Predicate Logic	B
6	MCQ	1	Predicate Logic	A	19	NAT	2	Propositional Logic	13
7	NAT	1	Sampling	0.0682 to 0.0686	20	MSQ	2	Propositional Logic	B, C, D
8	NAT	1	Exact Inference	0.713 to 0.717	21	MSQ	2	Propositional Logic	A, B
9	NAT	1	Sampling	10	22	NAT	2	Exact Inference	0.41 to 0.414
10	MCQ	1	Sampling	A	23	NAT	2	Exact Inference	4
11	NAT	2	Adversarial Search	10	24	MSQ	2	Sampling	A, B, C
12	MSQ	2	Informed Search	A, B, D	25	MSQ	2	Conditional Independence	C, D
13	NAT	2	Uninformed Search	17					

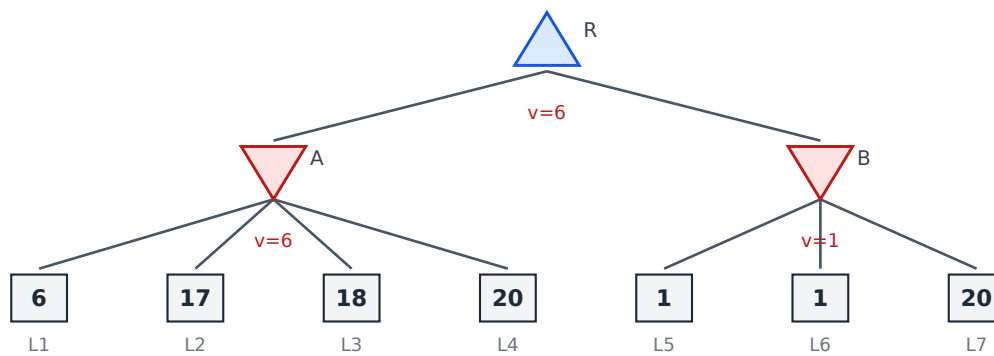
Detailed Solutions

Q.1 [MSQ, 1M · Informed Search] **Answer: A, B, D** (A: $\max(h_1, h_2)$; B: $(h_1 + h_2) / 2$; D: $0.5 \cdot h_1$)

Since $h_1, h_2 \leq h^*$, the max is $\leq h^*$ (A), the average is $\leq h^*$ (B), and any non-negative multiple ≤ 1 such as $0.5 \cdot h_1$ is $\leq h^*$ (D). The sum (C) can reach $2h^*$, for example when both heuristics equal h^* , so it is not guaranteed admissible.

Q.2 [NAT, 1M · Adversarial Search] **Answer: 6**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 6$, $B = 1$. Root value = **6**; MAX's optimal first move is towards A.

Q.3 [MCQ, 1M · Uninformed Search] **Answer: D** (D: S, A, C, B, G)

Adjacency lists (alphabetical): A: {C, D, H, S}; B: {C}; C: {A, B, G}; D: {A, S}; G: {C}; H: {A}; S: {A, D}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, A, C, B, G**; the path found is $S \rightarrow A \rightarrow C \rightarrow G$ (3 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Informed Search] **Answer: 2.48 to 2.52**

Solve $b + b^2 + \dots + b^5 = 161$ numerically (bisection): $b^* \approx 2.5$.

Check: $2.5^1 + 2.5^2 + 2.5^3 + 2.5^4 + 2.5^5 \approx 161.09 \approx 161$.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 3**

Remove A and $\neg A$ and take the union of the remaining literals: $\{B, \neg C\} \cup \{B, D\} = \{B, \neg C, D\}$. The duplicate B is merged (factoring), so the resolvent $B \vee \neg C \vee D$ has 3 literals.

Q.6 [MCQ, 1M · Predicate Logic] **Answer: A** (A: $Q(f(g(y))) \vee R(y)$)

Unify $P(x)$ with $P(g(y))$: MGU $\theta = \{x/g(y)\}$. The resolvent is $(Q(f(x)) \vee R(y))\theta = Q(f(g(y))) \vee R(y)$. Option B forgets to apply θ . Options C and D apply the substitution incorrectly.

Q.7 [NAT, 1M · Sampling] **Answer: 0.0682 to 0.0686**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Sm}=\text{T}) \times P(\text{Po}=\text{F}|\text{Sm}=\text{T}) \times P(\text{Ca}=\text{F}|\text{Po}=\text{F}) \times P(\text{XR}=\text{F}|\text{Sm}=\text{T}, \text{Po}=\text{F}) \times P(\text{Dy}=\text{F}|\text{Sm}=\text{T}, \text{Po}=\text{F})$
 $= 0.25 \times 0.8 \times 0.4 \times 0.95 \times 0.9 = \mathbf{0.0684} \approx 0.0684$.

Q.8 [NAT, 1M · Exact Inference] **Answer: 0.713 to 0.717**

Only the ancestors of B (A) matter: every other variable sums out to 1.

$P(B=\text{T}) = \sum \text{over its ancestors of } P(B=\text{T} | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, B=\text{T})$
A=T, C=T, D=T	0.0014
A=T, C=T, D=F	0.0266
A=T, C=F, D=T	0.0072
A=T, C=F, D=F	0.0048
A=F, C=T, D=T	0.03038
A=F, C=T, D=F	0.57712
A=F, C=F, D=T	0.0405
A=F, C=F, D=F	0.027

Sum = **0.715** ≈ 0.715 .

Q.9 [NAT, 1M · Sampling] **Answer: 10**

Rejection sampling keeps only samples consistent with the evidence $C = \text{T}$: samples 1, 4, 5, 6, 10, 11, 15, 16, 18, 20 (10 accepted, 10 rejected).

Among the accepted samples, $A = \text{T}$ in 4. Estimate = $4/10 = \mathbf{0.4}$.

(For this network the exact value is 0.5455; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

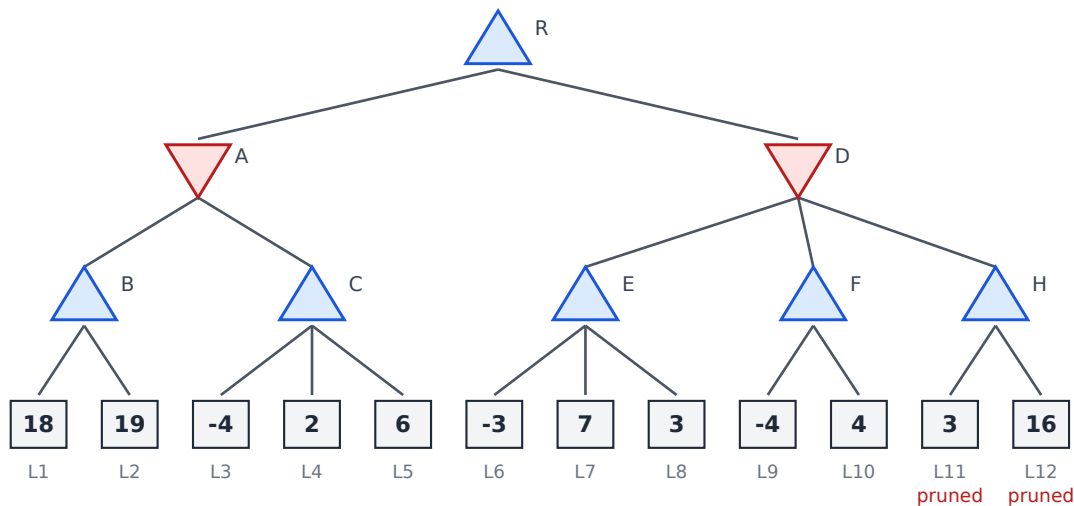
Q.10 [MCQ, 1M · Sampling] Answer: A (A: (c, s, ¬r, w))

C: $0.42 < P(c) = 0.5 \rightarrow c$. S: $P(s | c) = 0.1, 0.07 < 0.1 \rightarrow s$. R: $P(r | c) = 0.8, 0.85 \geq 0.8 \rightarrow \neg r$. W: $P(w | s, \neg r) = 0.90, 0.30 < 0.90 \rightarrow w$. Sample = (c, s, ¬r, w). (B) mis-samples S and R; (C) mis-samples R; (D) mis-samples C.

Q.11 [NAT, 2M · Adversarial Search] Answer: 10

Run alpha-beta depth-first, left to right, passing (α , β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $4 \leq \alpha = 6 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (10): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10. Leaves pruned (2): L11, L12. Root value = 6 (same as plain minimax, 6 — pruning never changes the root value).

Q.12 [MSQ, 2M · Informed Search] Answer: A, B, D

(A) is true: every misplaced tile is at Manhattan distance at least 1, so $h_2 \geq h_1$. (B) is true: one move changes h_1 by at most 1 and h_2 by exactly 1, while the step cost is 1. (C) is false: in a state one move from the goal, $h_1 = h_2 = 1$, so the sum is $2 > h^* = 1$. (D) is true: the parity invariant splits the $9!$ states into two halves that cannot reach each other.

Q.13 [NAT, 2M · Uninformed Search] Answer: 17

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	K(7), A(8)
2	K (7)	A(8), E(10), H(11)
3	A (8)	E(10), H(11)
4	E (10)	H(11), C(16), G(18)
5	H (11)	C(14), D(18), G(18)
6	C (14)	D(16), G(17)
7	D (16)	G(17)
8	G (17)	← goal, stop

Path returned: **S** → **K** → **H** → **C** → **G**, cost = $7 + 4 + 3 + 3 = 17$. Nodes expanded (including S and G): **8** (S, K, A, E, H, C, D, G).

Q.14 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(B)=6, h(D)=3, h(E)=9, h(G)=0, h(H)=10, h(I)=5, h(K)=6, h(L)=14, h(S)=13$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	H(13), L(23)
2	H (13)	I(14), K(15), L(21)
3	I (14)	D(14), K(15), L(21), E(23)
4	D (14)	G(15), K(15), L(21), E(23)
5	G (15)	K(15), L(21), E(23) ← goal, stop

Path returned: **S** → **H** → **I** → **D** → **G** with cost **15**; nodes expanded = **5** (S, H, I, D, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: B** (B: (P))

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) (P) is derivable:

1. resolve $(P \vee \neg R \vee S)$ and $(P \vee \neg S)$ on $S \Rightarrow (P \vee \neg R)$
2. resolve $(P \vee R)$ and $(P \vee \neg R)$ on $R \Rightarrow (P)$

(C) (Q) never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg Q \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.16 [MCQ, 2M · Predicate Logic] **Answer: A** (A: $\forall x (P(x) \rightarrow \exists y \neg Q(x, y))$)

$\neg \exists x (\dots) \equiv \forall x \neg(P(x) \wedge \forall y Q(x, y)) \equiv \forall x (\neg P(x) \vee \exists y \neg Q(x, y)) \equiv \forall x (P(x) \rightarrow \exists y \neg Q(x, y))$. Option B uses $\wedge$ instead of $\vee$ (De Morgan applied wrongly). Option C fails to flip $\exists x$ to $\forall x$. Option D requires every element to satisfy P.

Q.17 [NAT, 2M · Predicate Logic] **Answer: 171**

There is no simple closed formula; we enumerate all $2^9 = 512$ relations and test the sentence for all 27 triples (x, y, z) . The count of transitive relations on a 3-element set is 171 (OEIS A006905: 1, 2, 13, 171, 3994, ...). Sanity check: on a 2-element set the same enumeration gives 13 of 16; the 3 failures contain both (1,2) and (2,1) but miss (1,1) or (2,2).

Q.18 [MSQ, 2M · Predicate Logic] **Answer: B** (B: $\forall y \exists x R(x, y)$)

$R = \{(1,3), (1,4), (2,3), (3,1), (4,2), (4,3)\}$.

(A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,2)$ nor $R(2,1)$.

(B) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(C) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(D) $\exists x R(x, x)$: **FALSE** — there are no self-loops.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [NAT, 2M · Propositional Logic] **Answer: 13**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(Q \wedge R) \wedge (S \rightarrow Q)$	$\neg R \vee (\neg S \wedge P)$	$((Q \wedge R) \wedge (S \rightarrow Q)) \rightarrow (\neg R \vee (\neg S \wedge P))$
T	T	T	T	T	F	F
T	T	T	F	T	T	T
T	T	F	T	F	T	T
T	T	F	F	F	T	T
T	F	T	T	F	F	T
T	F	T	F	F	T	T
T	F	F	T	F	T	T
T	F	F	F	F	T	T
F	T	T	T	T	F	F
F	T	T	F	T	F	F
F	T	F	T	F	T	T
F	T	F	F	F	T	T
F	F	T	T	F	F	T
F	F	T	F	F	F	T
F	F	F	T	F	T	T
F	F	F	F	F	T	T

Rows where F = T: **13**.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: B, C, D** (B: F is entailed; C: J is entailed; D: K is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $C \rightarrow J$; add J.

Round 2: fire $J \wedge E \rightarrow K$, $E \wedge J \rightarrow F$; add F, K.

Fixed point: {A, C, E, F, J, K} — 6 atoms are entailed (facts: A, C, E; newly derived: F, J, K). Atoms never derived: B, D, H.

Q.21 [MSQ, 2M · Propositional Logic] **Answer: A, B** ($A: (P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$; $B: \neg(\neg R \wedge Q) \leftrightarrow (\neg\neg R \vee \neg Q)$)

(A) $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $\neg(\neg R \wedge Q) \leftrightarrow (\neg\neg R \vee \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $(P \vee R) \rightarrow (P \wedge R)$: **not valid** — counter-model $P=T, R=F$.

(D) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$: **not valid** — counter-model $P=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.22 [NAT, 2M · Exact Inference] **Answer: 0.41 to 0.414**

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever} | \text{Flu}) \cdot P(\text{Cough} | \text{Flu}, \text{Fever}) \cdot P(\text{Fatigue} | \text{Flu}, \text{Fever})$.

Inference by enumeration: $P(\text{Cough} | e) = \alpha \sum_{\text{hidden}} P(\text{Cough}, e, \text{hidden})$.

Hidden: Fl, Fe	Co	product of CPT entries (Fl · Fe · Co · Fa)	value
T, T	T	$0.4 \times 0.5 \times 0.25 \times 0.2$	0.01
T, F	T	$0.4 \times 0.5 \times 0.5 \times 0.5$	0.05
F, T	T	$0.6 \times 0.25 \times 0.05 \times 0.2$	0.0015
F, F	T	$0.6 \times 0.75 \times 0.6 \times 0.1$	0.027
T, T	F	$0.4 \times 0.5 \times 0.75 \times 0.2$	0.03
T, F	F	$0.4 \times 0.5 \times 0.5 \times 0.5$	0.05
F, T	F	$0.6 \times 0.25 \times 0.95 \times 0.2$	0.0285
F, F	F	$0.6 \times 0.75 \times 0.4 \times 0.1$	0.018

$P(\text{Cough}=T, e) = 0.0885$; $P(\text{Cough}=F, e) = 0.1265$; $P(e) = 0.215$.

$P(\text{Cough}=T | e) = 0.0885 / 0.215 = \mathbf{0.4116} \approx 0.412$.

Q.23 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(A, B, D)$, $f(B, D, E)$, $f(A, B, F)$, $f(D, E)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(A, B, D)$, $f(B, D, E)$, $f(D, E)$	{A, B, D, E}	$f(A, B, E)$
A	$f(A)$, $f(A, B)$, $f(A, B, C)$, $f(A, B, F)$, $f(A, B, E)$	{A, B, C, E, F}	$f(B, C, E, F)$
E	$f(B, C, E, F)$	{B, C, E, F}	$f(B, C, F)$
C	$f(B, C, F)$	{B, C, F}	$f(B, F)$
B	$f(B, F)$	{B, F}	$f(F)$

The largest factor produced has **4** variable(s) (created when eliminating A); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over F and normalise.

Q.24 [MSQ, 2M · Sampling] **Answer: A, B, C**

(A) Each state is obtained by modifying the previous one, so samples are autocorrelated. (B) Gibbs steps satisfy detailed balance with respect to $P(x | e)$ and the chain is ergodic when there are no deterministic (0/1) entries, so it converges to the posterior. (C) The state is a full assignment with evidence clamped. (D) describes rejection sampling; Gibbs never rejects since evidence is clamped.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: C, D** (C: $B \perp C \mid A$; D: $B \perp E \mid A, F$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp C \mid E$: **FALSE** — the path $B \leftarrow A \rightarrow D \leftarrow C$ is active given $\{E\}$.

(B) $F \perp B$: **FALSE** — the path $F \leftarrow E \leftarrow A \rightarrow B$ is active given $\{\}$.

(C) $B \perp C \mid A$: **TRUE** — $B \leftarrow A \rightarrow D \leftarrow C$ (blocked: observed fork node A); $B \leftarrow A \rightarrow D \rightarrow E \rightarrow F \leftarrow C$ (blocked: observed fork node A); $B \leftarrow A \rightarrow E \leftarrow D \leftarrow C$ (blocked: observed fork node A); $B \leftarrow A \rightarrow E \rightarrow F \leftarrow C$ (blocked: observed fork node A).

(D) $B \perp E \mid A, F$: **TRUE** — $B \leftarrow A \rightarrow D \leftarrow C \rightarrow F \leftarrow E$ (blocked: observed fork node A); $B \leftarrow A \rightarrow D \rightarrow E$ (blocked: observed fork node A); $B \leftarrow A \rightarrow E$ (blocked: observed fork node A).

Mock Test 032

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 7 / 11	11 / 13 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

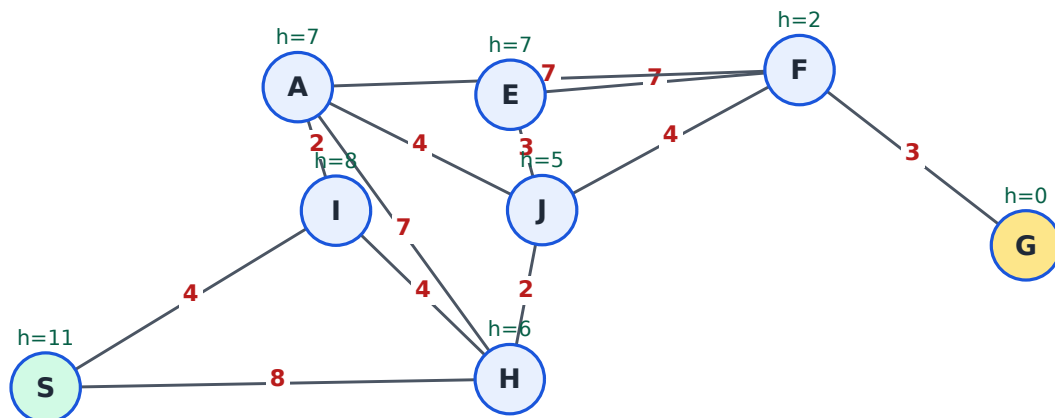
Q.1 [MSQ · 1 mark · no negative marking] Adversarial Search

Which of the following statements about two-player zero-sum games are TRUE?

- ☐ (A) The players' utilities sum to the same constant in every terminal state
- ☐ (B) Chess, with outcomes $+1/0/-1$ or $1/1/2/0$, can be modelled as a zero-sum (constant-sum) game
- ☐ (C) Minimax assumes that the opponent plays optimally
- ☐ (D) In a zero-sum game, both players can strictly gain from the same outcome change

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

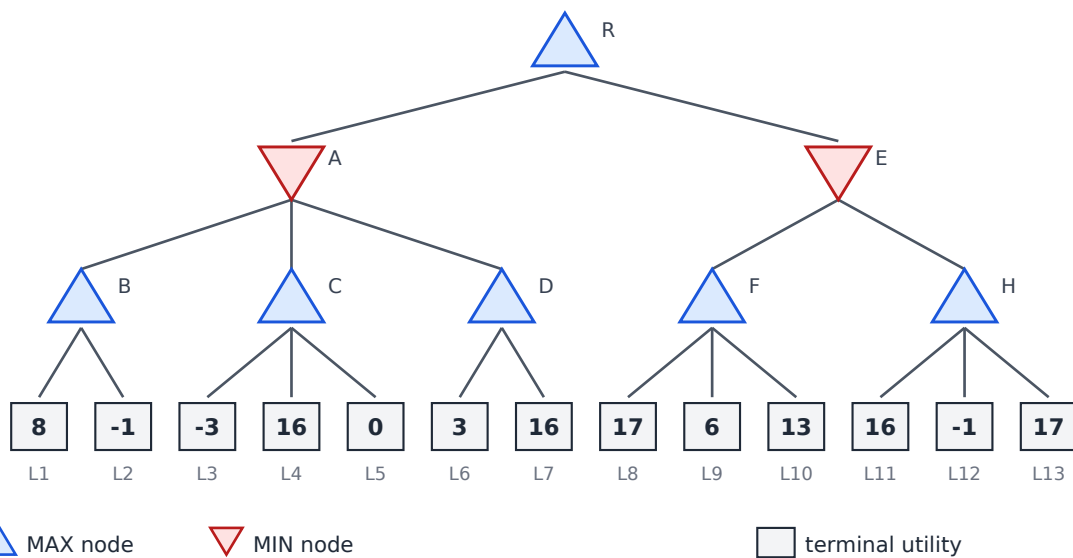
Q.3 [NAT · 1 mark · no negative marking] Informed Search

Weighted A* uses $f(n) = g(n) + W \cdot h(n)$ with $W = 1.5$ and an admissible heuristic h . The optimal solution cost is $C^* = 40$. What is the largest solution cost that weighted A* is guaranteed not to exceed?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

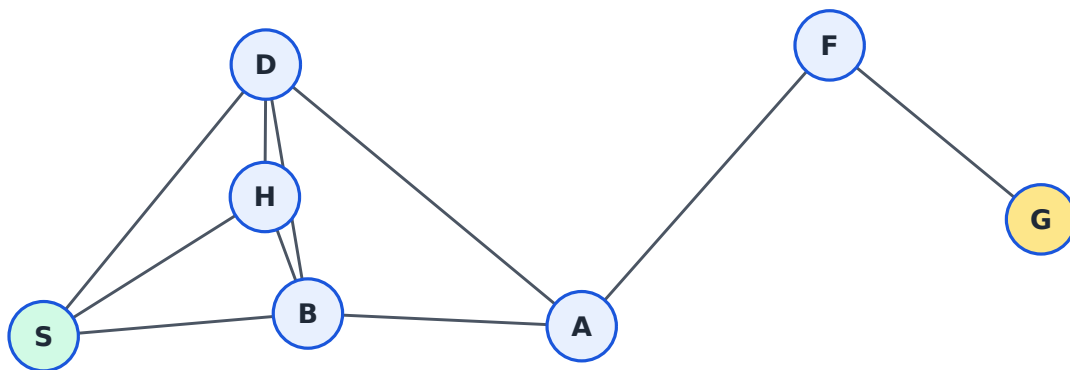
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, B, A, D, H, F, G
☐ (B) S, B, D, H, A, F, G
☐ (C) S, H, D, B, A, F, G
☐ (D) S, A, B, D, H, F, G

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

In how many rows of the truth table over p, q is $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow p)$ true?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following is the correct first-order translation of 'Every student likes some course'?

- ☐ (A) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
☐ (B) $\forall x (\text{Student}(x) \wedge \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
☐ (C) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \rightarrow \text{Likes}(x, y)))$
☐ (D) $\exists y \forall x (\text{Student}(x) \wedge \text{Course}(y) \wedge \text{Likes}(x, y))$

Q.8 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \exists y R(x, y)$ true?

Answer (numerical): _____

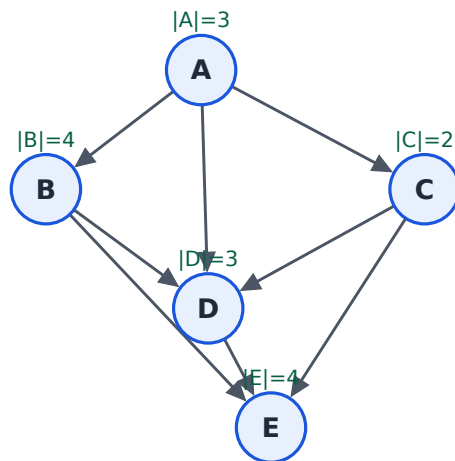
Q.9 [MCQ · 1 mark · negative: -1/3] Variable Elimination

For a polytree (singly connected) Bayesian network, the time and space complexity of exact inference by variable elimination is

- ☐ (A) linear in the size of the network (total CPT entries)
- ☐ (B) exponential in the number of variables in the worst case
- ☐ (C) quadratic in the number of variables, independent of CPT sizes
- ☐ (D) NP-hard

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?

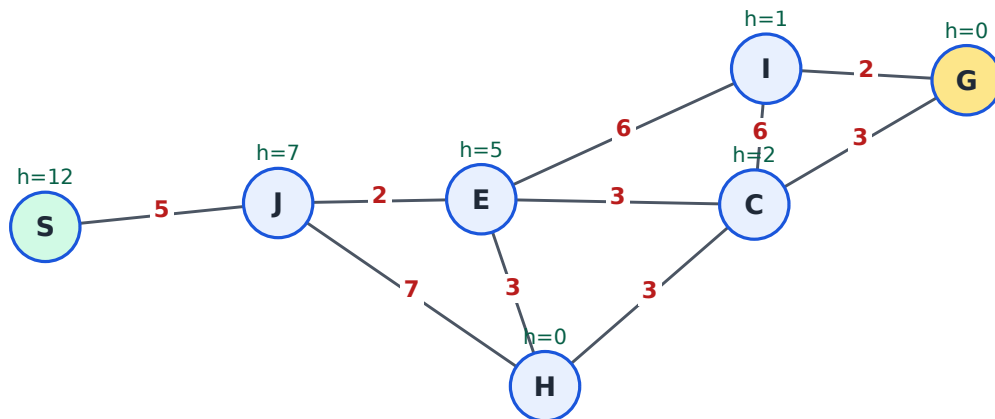


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

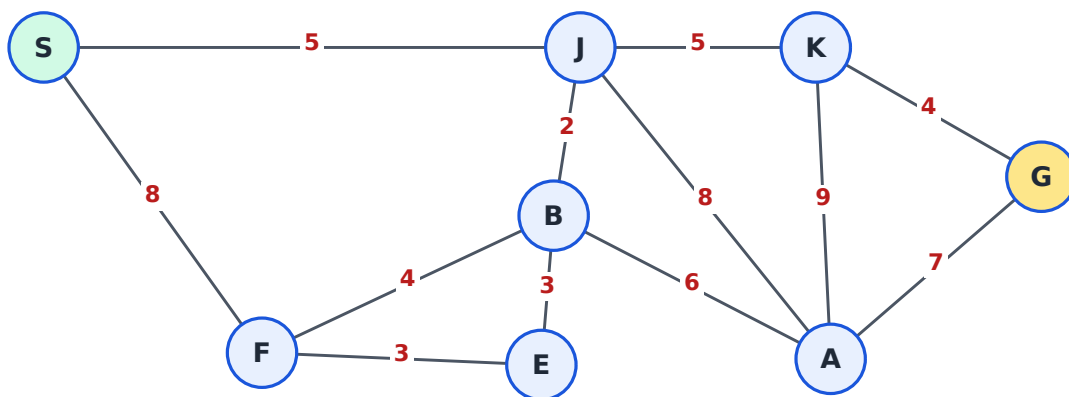
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
☐ (B) h is admissible.
☐ (C) $h(C)$ overestimates the true cost from C to G .
☐ (D) The edge $E-H$ violates the consistency condition $h(E) \leq c(E,H) + h(H)$.

Q.12 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.13 [MCQ · 2 marks · negative: -2/3] Informed Search

A* graph search runs with a consistent heuristic, and C^* is the optimal solution cost. Which statement is correct?

- ☐ (A) Every node reachable from the start with $f(n) < C^*$ is expanded
☐ (B) No node with $f(n) < C^*$ is ever expanded
☐ (C) Nodes with $f(n) > C^*$ may be expanded before the goal is selected
☐ (D) A* expands exactly the nodes on the optimal path and no others

Q.14 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (Q \leftrightarrow P) \vee P$?

- ☐ (A) $\neg(\neg(Q \leftrightarrow P) \vee \neg P)$
☐ (B) $(Q \leftrightarrow P) \wedge P$
☐ (C) $P \vee (Q \leftrightarrow P)$
☐ (D) $\neg(\neg(Q \leftrightarrow P) \wedge \neg P)$

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

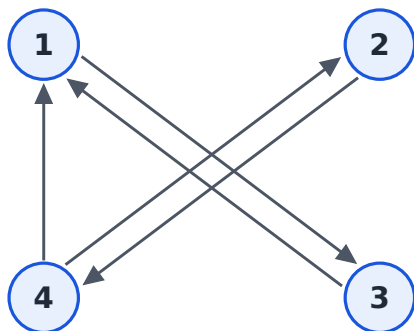
C1: $(P \vee \neg R)$, C2: $(\neg P)$, C3: $(\neg S)$, C4: $(P \vee S)$, C5: $(\neg P \vee \neg Q \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg Q \vee \neg R)$
☐ (B) $(\neg R \vee S)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(\neg R)$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (B) $\forall x \exists y R(x, y)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\forall y \exists x R(x, y)$

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$F = ((R \vee R) \wedge \neg S) \leftrightarrow (\neg P \wedge (S \rightarrow Q))$?

Answer (numerical): _____

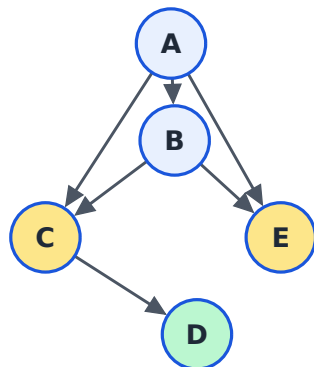
Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following expresses 'No student likes every course' (S : student, C : course, $L(x, y)$: x likes y)?

- ☐ (A) $\forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$
☐ (B) $\forall x (S(x) \rightarrow \exists y (C(y) \rightarrow \neg L(x, y)))$
☐ (C) $\forall x \forall y ((S(x) \wedge C(y)) \rightarrow \neg L(x, y))$
☐ (D) $\exists x (S(x) \wedge \exists y (C(y) \wedge \neg L(x, y)))$

Q.19 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{E = T}$, $\mathbf{C = F}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(D = T \mid E = T, C = F)}$ (round off to 3 decimal places).



P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.25	T	0.95	T	T	0.1
	F	0.9	T	F	0.7
			F	T	0.7
			F	F	0.6

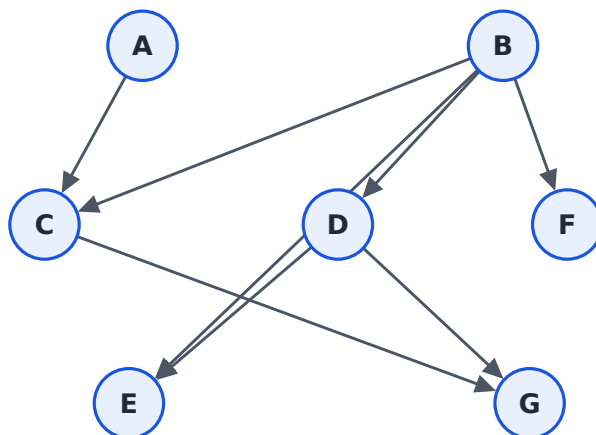
C	P(D=T .)	A	B	P(E=T ...)
T	0.7	T	T	0.1
F	0.5	T	F	0.2
		F	T	0.2
		F	F	0.5

#	A	B	D
1	F	T	T
2	F	T	T
3	T	T	T
4	F	F	F
5	T	T	F
6	F	T	T

Answer (numerical): _____

Q.20 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $F \perp B \mid C, E$
☐ (B) $D \perp C \mid B$
☐ (C) $C \perp G \mid A, E$
☐ (D) $C \perp E$

Q.21 [MCQ · 2 marks · negative: -2/3] Conditional Independence

A Bayesian network has edges $A \rightarrow C$ and $B \rightarrow C$ (Boolean variables) with $P(a) = 0.1$, $P(b) = 0.2$, $P(c | a, b) = 0.95$, $P(c | a, \neg b) = 0.90$, $P(c | \neg a, b) = 0.80$, $P(c | \neg a, \neg b) = 0.05$. Which ordering of the three quantities $P(a)$, $P(a | c)$ and $P(a | c, b)$ is correct?

- ☐ (A) $P(a) < P(a | c, b) < P(a | c)$
☐ (B) $P(a | c, b) < P(a) < P(a | c)$
☐ (C) $P(a) < P(a | c) < P(a | c, b)$
☐ (D) $P(a | c) < P(a) < P(a | c, b)$

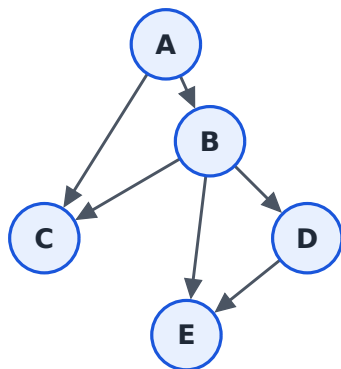
Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence

X and Y are independent fair coin flips (Boolean) and $Z = X \text{ XOR } Y$. Which statements are true?

- ☐ (A) $X \perp Z$
☐ (B) $Y \perp Z$
☐ (C) $X \perp Y | Z$
☐ (D) $X \perp \{Y, Z\}$

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E , using the uniform random numbers $\mathbf{u} = (0.56, 0.49, 0.56, 0.34, 0.59)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T | \text{sampled parent values})$. Which sample is generated?



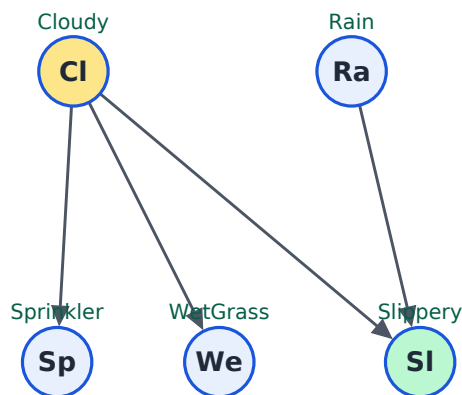
P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.9	T	0.3	T	T	0.95
	F	0.95	T	F	0.6
			F	T	0.9
			F	F	0.6

B	P(D=T .)	B	D	P(E=T ...)
T	0.9	T	T	0.7
F	0.25	T	F	0.1
		F	T	0.5
		F	F	0.5

- ☐ (A) $A=T, B=F, C=T, D=F, E=F$
☐ (B) $A=T, B=T, C=T, D=F, E=F$
☐ (C) $A=F, B=F, C=T, D=F, E=F$
☐ (D) $A=F, B=F, C=F, D=T, E=T$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, Sl = Slippery.) Compute $P(\text{Slippery} = T \mid \text{Cloudy} = T)$ (round off to 3 decimal places).



$P(Cl=T)$
0.75

Cl	$P(Sp=T \dots)$
T	0.95
F	0.75

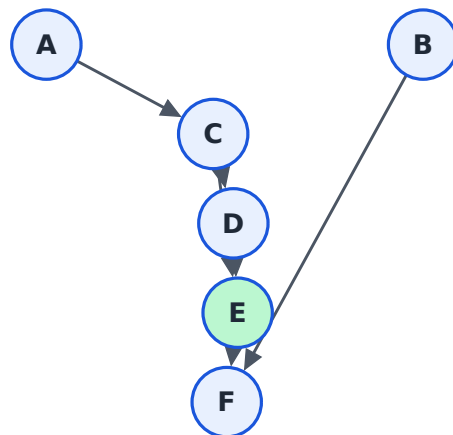
$P(Ra=T)$
0.5

Cl	$P(We=T \dots)$	Cl	Ra	$P(Sl=T \dots)$
T	0.4	T	T	0.05
F	0.5	T	F	0.9
		F	T	0.05
		F	F	0.4

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(E)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F, C, A, B, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 032

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MSQ	1	Adversarial Search	A, B, C	14	MSQ	2	Propositional Logic	C, D
2	NAT	1	Informed Search	17	15	MSQ	2	Propositional Logic	A, C, D
3	NAT	1	Informed Search	60	16	MSQ	2	Predicate Logic	B, C, D
4	NAT	1	Adversarial Search	17	17	NAT	2	Propositional Logic	10
5	MCQ	1	Uninformed Search	A	18	MCQ	2	Predicate Logic	A
6	NAT	1	Propositional Logic	2	19	NAT	2	Sampling	0.48 to 0.484
7	MCQ	1	Predicate Logic	A	20	MSQ	2	Conditional Independence	B
8	NAT	1	Predicate Logic	343	21	MCQ	2	Conditional Independence	A
9	MCQ	1	Variable Elimination	A	22	MSQ	2	Conditional Independence	A, B
10	NAT	1	Bayesian Networks	134	23	MCQ	2	Sampling	A
11	MSQ	2	Informed Search	B, D	24	NAT	2	Exact Inference	0.473 to 0.477
12	NAT	2	Uninformed Search	14	25	NAT	2	Exact Inference	3
13	MCQ	2	Informed Search	A					

Detailed Solutions

Q.1 [MSQ, 1M · Adversarial Search] Answer: A, B, C

(A) is the definition of zero-sum (constant-sum). (B) is true: AIMA notes that chess's payoffs always total a constant. (C) is true: minimax computes the best play against a perfect, adversarial opponent. (D) is false: since the sum is constant, one player's gain is exactly the other's loss.

Q.2 [NAT, 1M · Informed Search] Answer: 17

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (11)	H(6), I(8)
2	H (6)	J(5), A(7), I(8)
3	J (5)	F(2), A(7), E(7), I(8)
4	F (2)	G(0), A(7), E(7), I(8)
5	G (0)	A(7), E(7), I(8) ← goal, stop

Path returned: **S** → **H** → **J** → **F** → **G**, cost = $8 + 2 + 4 + 3 = 17$.

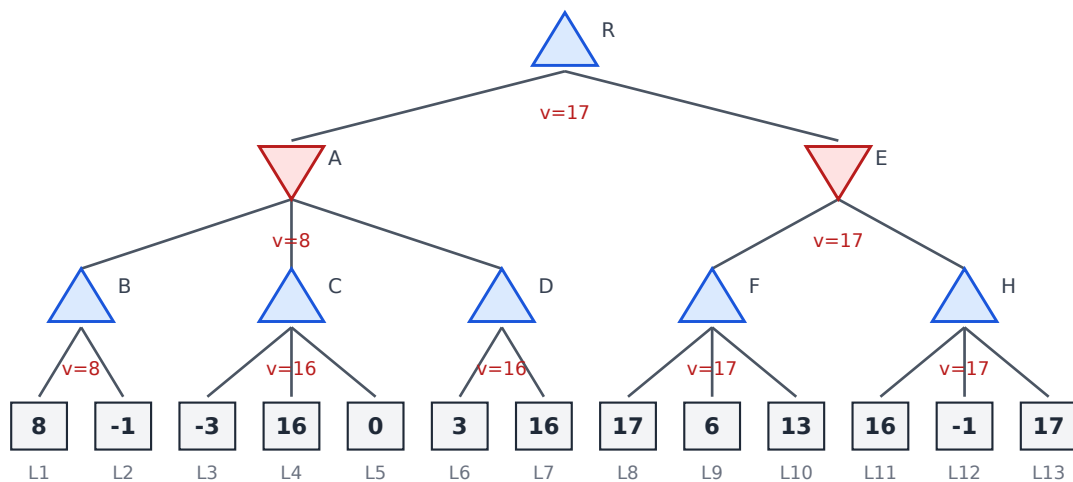
The optimal cost is $h^*(S) = 16$ — greedy search is not optimal here.

Q.3 [NAT, 1M · Informed Search] Answer: 60

With an admissible h , weighted A^* is W -admissible: the cost it returns is $\leq W \cdot C^* = 1.5 \times 40 = 60$. Inflating h makes the search greedier and faster, and the price is this bounded loss of optimality.

Q.4 [NAT, 1M · Adversarial Search] Answer: 17

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 8, C = 16, D = 16, A = 8, F = 17, H = 17, E = 17. Root value = **17**; MAX's optimal first move is towards E.

Q.5 [MCQ, 1M · Uninformed Search] Answer: A (A: S, B, A, D, H, F, G)

Adjacency lists (alphabetical): A: {B, D, F}; B: {A, D, H, S}; D: {A, B, H, S}; F: {A, G}; G: {F}; H: {B, D, S}; S: {B, D, H}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, B, A, D, H, F, G**; the path found is $S \rightarrow B \rightarrow A \rightarrow F \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.6 [NAT, 1M · Propositional Logic] Answer: 2

$p \rightarrow q \equiv \neg p \vee q$ and $\neg q \rightarrow p \equiv p \vee q$. Row TT: T, $T \rightarrow T$. Row TF: F, $T \rightarrow F$. Row FT: T, $T \rightarrow T$. Row FF: T, $F \rightarrow F$. So the biconditional is true in 2 rows.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$)

Universal quantifiers pair naturally with $\rightarrow$, existential with $\wedge$. Option B claims everything in the domain is a student. Option C is satisfied by any non-course y (making $\text{Course}(y) \rightarrow \dots$ vacuously true), so it does not require liking a course. Option D says there is a single y that is a course liked by everything, and everything is a student — far too strong.

Q.8 [NAT, 1M · Predicate Logic] Answer: 343

View R as an 3×3 0/1 matrix. The sentence says R is every row of the relation matrix must be non-empty: $(2^n - 1)$ choices per row $\Rightarrow (2^n - 1)^n$. With $n = 3$: **343**.

Q.9 [MCQ, 1M · Variable Elimination] Answer: A (A: linear in the size of the network (total CPT entries))

AIMA: in polytrees there is at most one undirected path between any two nodes, and VE (with a suitable ordering) runs in time and space linear in the size of the network, measured by CPT entries.

Exponential/NP-hard behaviour arises for multiply connected networks. Quadratic in variables ignores that CPT sizes grow with the number of parents.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 134

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (4-1) \times 3 + C: (2-1) \times 3 + D: (3-1) \times 3 \times 4 \times 2 + E: (4-1) \times 4 \times 2 \times 3 = \mathbf{134}$.

Full joint table: $\prod |X| - 1 = 288 - 1 = 287$; saving = 153.

Q.11 [MSQ, 2M · Informed Search] Answer: B, D (B: h is admissible.; D: The edge E-H violates the consistency condition $h(E) \leq c(E, H) + h(H)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	E	G	H	I	J	S
$h(n)$	2	5	0	0	1	7	12
$h^*(n)$	3	6	0	6	2	8	13
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges:

$E \rightarrow H$ ($5 > 3 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [NAT, 2M · Uninformed Search] **Answer: 14**

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(5), F(8)
2	J (5)	B(7), F(8), K(10), A(13)
3	B (7)	F(8), E(10), K(10), A(13)
4	F (8)	E(10), K(10), A(13)
5	E (10)	K(10), A(13)
6	K (10)	A(13), G(14)
7	A (13)	G(14)
8	G (14)	$\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **J** $\rightarrow$ **K** $\rightarrow$ **G**, cost = $5 + 5 + 4 = 14$. Nodes expanded (including S and G): **8** (S, J, B, F, E, K, A, G).

Q.13 [MCQ, 2M · Informed Search] **Answer: A** (A: Every node reachable from the start with $f(n) < C^*$ is expanded)

With a consistent h , A^* expands nodes in non-decreasing order of f . It therefore expands **all nodes with $f < C^*$** , some nodes with $f = C^*$, and none with $f > C^*$. (B) is the opposite. (C) contradicts that ordering, because the goal with $f = C^*$ would be selected first. (D) is true only for a perfect heuristic with favourable tie-breaking.

Q.14 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $P \vee (Q \leftrightarrow P)$; D: $\neg(\neg(Q \leftrightarrow P) \wedge \neg P)$)

Compare truth tables with $F = (Q \leftrightarrow P) \vee P$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg(Q \leftrightarrow P) \vee \neg P)$ — differs at $P=T, Q=F$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(B) $(Q \leftrightarrow P) \wedge P$ — differs at $P=T, Q=F$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(C) $P \vee (Q \leftrightarrow P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(D) $\neg(\neg(Q \leftrightarrow P) \wedge \neg P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: A, C, D** (A: $(\neg Q \vee \neg R)$; C: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable); D: $(\neg R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg Q \vee \neg R)$ is derivable:

1. resolve $(P \vee \neg R)$ and $(\neg P)$ on $P \Rightarrow (\neg R)$
2. resolve $(\neg P \vee \neg Q \vee R)$ and $(\neg R)$ on $R \Rightarrow (\neg P \vee \neg Q)$
3. resolve $(P \vee \neg R)$ and $(\neg P \vee \neg Q)$ on $P \Rightarrow (\neg Q \vee \neg R)$

(B) $(\neg R \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $\square$ is derivable:

1. resolve $(\neg S)$ and $(P \vee S)$ on $S \Rightarrow (P)$
2. resolve $(\neg P)$ and (P) on $P \Rightarrow \square$ (empty clause)

(D) $(\neg R)$ is derivable:

1. resolve $(P \vee \neg R)$ and $(\neg P)$ on $P \Rightarrow (\neg R)$

Q.16 [MSQ, 2M · Predicate Logic] **Answer: B, C, D** (B: $\forall x \exists y R(x, y)$; C: $\forall x \exists y (x \neq y \wedge R(x, y))$; D: $\forall y \exists x R(x, y)$)

$R = \{(1,3), (2,4), (3,1), (4,1), (4,2)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(4,1)$ holds but $R(1,4)$ does not.

(B) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(D) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [NAT, 2M · Propositional Logic] **Answer: 10**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(R \vee R) \wedge \neg S$	$\neg P \wedge (S \rightarrow Q)$	$((R \vee R) \wedge \neg S) \leftrightarrow (\neg P \wedge (S \rightarrow Q))$
T	T	T	T	F	F	T
T	T	T	F	T	F	F
T	T	F	T	F	F	T
T	T	F	F	F	F	T
T	F	T	T	F	F	T
T	F	T	F	T	F	F
T	F	F	T	F	F	T
T	F	F	F	F	F	T
F	T	T	T	F	T	F
F	T	T	F	T	T	T
F	T	F	T	F	T	F
F	T	F	F	F	T	F
F	F	T	T	F	F	T
F	F	T	F	T	T	T
F	F	F	T	F	F	T
F	F	F	F	F	T	F

Rows where F = T: **10**.

Q.18 [MCQ, 2M · Predicate Logic] **Answer: A** (A: $\forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$)

'No student likes every course' $\equiv \neg \exists x (S(x) \wedge \forall y (C(y) \rightarrow L(x, y))) \equiv \forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$. Option B is satisfied by any non-course y, so it is too weak. Option C says no student likes any course — too strong. Option D only says some student fails to like some course.

Q.19 [NAT, 2M · Sampling] **Answer: 0.48 to 0.484**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	D
1	$P(C=F A=F,B=T)=0.3 \times P(E=T A=F,B=T)=0.2$	0.06	T
2	$P(C=F A=F,B=T)=0.3 \times P(E=T A=F,B=T)=0.2$	0.06	T
3	$P(C=F A=T,B=T)=0.9 \times P(E=T A=T,B=T)=0.1$	0.09	T
4	$P(C=F A=F,B=F)=0.4 \times P(E=T A=F,B=F)=0.5$	0.2	F
5	$P(C=F A=T,B=T)=0.9 \times P(E=T A=T,B=T)=0.1$	0.09	F
6	$P(C=F A=F,B=T)=0.3 \times P(E=T A=F,B=T)=0.2$	0.06	T

$\sum w \text{ (all)} = 0.56$; $\sum w \text{ (D = T)} = 0.27$.

Estimate $P(D=T \mid E = T, C = F) = 0.27 / 0.56 = \mathbf{0.4821}$.

Q.20 [MSQ, 2M · Conditional Independence] **Answer: B** (B: $D \perp C \mid B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $F \perp B \mid C, E$: **FALSE** — the path $F \leftarrow B$ is active given $\{C, E\}$.

(B) $D \perp C \mid B$: **TRUE** — $D \leftarrow B \rightarrow C$ (blocked: observed fork node B); $D \rightarrow E \leftarrow B \rightarrow C$ (blocked: collider E with neither it nor a descendant observed); $D \rightarrow G \leftarrow C$ (blocked: collider G with neither it nor a descendant observed).

(C) $C \perp G \mid A, E$: **FALSE** — the path $C \leftarrow B \rightarrow D \rightarrow G$ is active given $\{A, E\}$.

(D) $C \perp E$: **FALSE** — the path $C \leftarrow B \rightarrow D \rightarrow E$ is active given $\{\}$.

Q.21 [MCQ, 2M · Conditional Independence] **Answer: A** (A: $P(a) < P(a \mid c, b) < P(a \mid c)$)

$P(c) = 0.1 \times 0.2 \times 0.95 + 0.1 \times 0.8 \times 0.90 + 0.9 \times 0.2 \times 0.80 + 0.9 \times 0.8 \times 0.05 = 0.019 + 0.072 + 0.144 + 0.036 = 0.271$, and $P(a, c) = 0.019 + 0.072 = 0.091$, so $P(a \mid c) = 0.091/0.271 \approx 0.336$. $P(a \mid c, b) = 0.095/(0.095 + 0.72) \approx 0.117$. $P(a) = 0.1$. Hence $0.1 < 0.117 < 0.336$: observing c raises P(a); additionally observing b explains c away and lowers it, but not below the prior here.

Q.22 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $X \perp Z$; B: $Y \perp Z$)

$P(z) = 0.5$ and $P(x, z) = P(x, \neg y) = 0.25 = P(x)P(z)$; every entry factorizes, so $X \perp Z$ (A true) and by symmetry $Y \perp Z$ (B true). Given Z, X determines Y: $P(x \mid y, z) = 0$ while $P(x \mid z) = 0.5$, so X and Y are dependent given Z (C false). Since $P(x \mid y, z) = 0 \neq P(x) = 0.5$, X is not independent of the pair $\{Y, Z\}$ (D false). Thus pairwise independence does not imply joint independence.

Q.23 [MCQ, 2M · Sampling] Answer: A (A: A=T, B=F, C=T, D=F, E=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.9$; $u = 0.56 < 0.9 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.3$; $u = 0.49 \geq 0.3 \Rightarrow B = F$
- C: $P(C=T \mid A=T, B=F) = 0.6$; $u = 0.56 < 0.6 \Rightarrow C = T$
- D: $P(D=T \mid B=F) = 0.25$; $u = 0.34 \geq 0.25 \Rightarrow D = F$
- E: $P(E=T \mid B=F, D=F) = 0.5$; $u = 0.59 \geq 0.5 \Rightarrow E = F$

Sample: **(A=T, B=F, C=T, D=F, E=F)**.

Q.24 [NAT, 2M · Exact Inference] Answer: 0.473 to 0.477

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}, \text{Slippery}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain}) \cdot P(\text{WetGrass} \mid \text{Cloudy}) \cdot P(\text{Slippery} \mid \text{Cloudy}, \text{Rain})$.

Inference by enumeration: $P(\text{Slippery} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Slippery}, e, \text{hidden})$.

Hidden: Sp, Ra, We	Sl	product of CPT entries (Cl · Sp · Ra · We · Sl)	value
T, T, T	T	$0.75 \times 0.95 \times 0.5 \times 0.4 \times 0.05$	0.00712
T, T, F	T	$0.75 \times 0.95 \times 0.5 \times 0.6 \times 0.05$	0.01069
T, F, T	T	$0.75 \times 0.95 \times 0.5 \times 0.4 \times 0.9$	0.12825
T, F, F	T	$0.75 \times 0.95 \times 0.5 \times 0.6 \times 0.9$	0.19237
F, T, T	T	$0.75 \times 0.05 \times 0.5 \times 0.4 \times 0.05$	0.00038
F, T, F	T	$0.75 \times 0.05 \times 0.5 \times 0.6 \times 0.05$	0.00056
F, F, T	T	$0.75 \times 0.05 \times 0.5 \times 0.4 \times 0.9$	0.00675
F, F, F	T	$0.75 \times 0.05 \times 0.5 \times 0.6 \times 0.9$	0.01013
T, T, T	F	$0.75 \times 0.95 \times 0.5 \times 0.4 \times 0.95$	0.13537
T, T, F	F	$0.75 \times 0.95 \times 0.5 \times 0.6 \times 0.95$	0.20306
T, F, T	F	$0.75 \times 0.95 \times 0.5 \times 0.4 \times 0.1$	0.01425
T, F, F	F	$0.75 \times 0.95 \times 0.5 \times 0.6 \times 0.1$	0.02137
F, T, T	F	$0.75 \times 0.05 \times 0.5 \times 0.4 \times 0.95$	0.00713
F, T, F	F	$0.75 \times 0.05 \times 0.5 \times 0.6 \times 0.95$	0.01069
F, F, T	F	$0.75 \times 0.05 \times 0.5 \times 0.4 \times 0.1$	0.00075
F, F, F	F	$0.75 \times 0.05 \times 0.5 \times 0.6 \times 0.1$	0.00113

$P(\text{Slippery}=T, e) = 0.35625$; $P(\text{Slippery}=F, e) = 0.39375$; $P(e) = 0.75$.

$P(\text{Slippery}=T \mid e) = 0.35625 / 0.75 = \mathbf{0.475} \approx 0.475$.

Q.25 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, C)$, $f(C, D)$, $f(C, D, E)$, $f(B, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(B, E, F)$	$\{B, E, F\}$	$f(B, E)$
C	$f(A, C)$, $f(C, D)$, $f(C, D, E)$	$\{A, C, D, E\}$	$f(A, D, E)$
A	$f(A)$, $f(A, D, E)$	$\{A, D, E\}$	$f(D, E)$
B	$f(B)$, $f(B, E)$	$\{B, E\}$	$f(E)$
D	$f(D, E)$	$\{D, E\}$	$f(E)$

The largest factor produced has **3** variable(s) (created when eliminating C); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over E and normalise.

Mock Test 033

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 10 / 8	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search

Breadth-first search (graph search) is guaranteed to return an optimal (least-cost) solution whenever a solution exists, provided that

- ☐ (A) all step costs are identical (e.g. every action costs 1)
- ☐ (B) an admissible heuristic is supplied
- ☐ (C) the branching factor is infinite
- ☐ (D) the state space contains no cycles

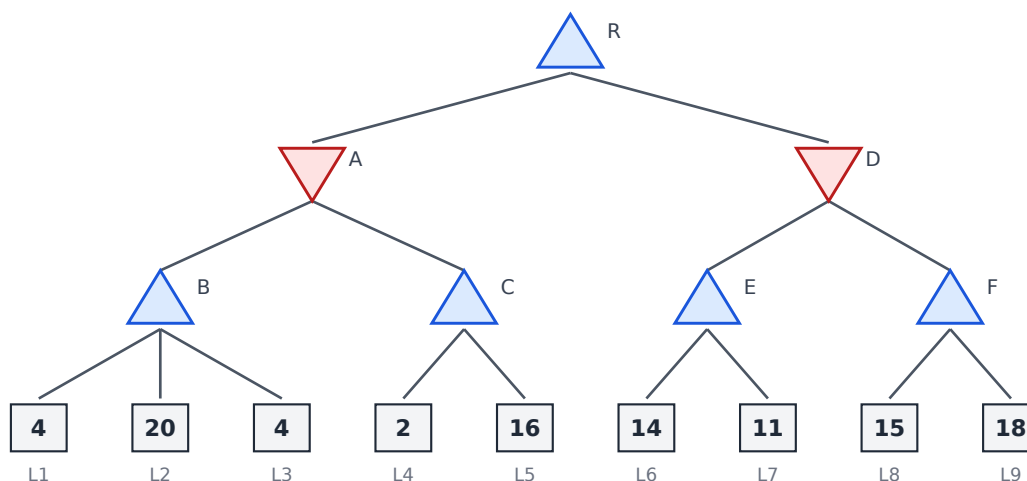
Q.2 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Iterative deepening search is applied to a tree with branching factor $b = 3$ whose shallowest goal is at depth $d = 4$ (goal is the last node at that depth). Using the standard count $N(\text{IDS}) = d \cdot b + (d-1) \cdot b^2 + \dots + 1 \cdot b^d$ (root not counted), how many nodes are generated?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



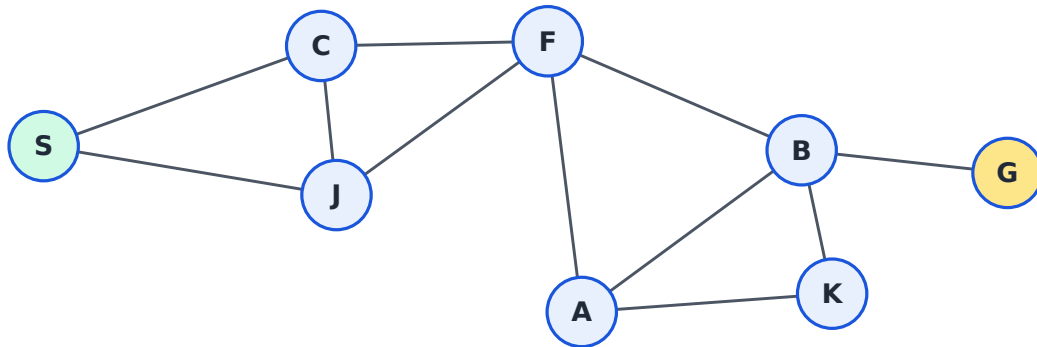
MAX node MIN node

terminal utility

- ☐ (A) Move to A; root value 19
- ☐ (B) Move to A; root value 20
- ☐ (C) Move to A; root value 16
- ☐ (D) Move to D; root value 14

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, C, J, F, A, B, K, G
- ☐ (B) S, C, F, A, B, G
- ☐ (C) S, J, C, F, B, A, K, G
- ☐ (D) S, C, J, F, A, K, B, G

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((Q \vee R) \leftrightarrow R) \wedge (\neg R \vee (R \vee \neg P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are logically equivalent to p?

- ☐ (A) $p \vee (p \wedge q)$
- ☐ (B) $p \wedge (p \vee q)$
- ☐ (C) $(p \vee q) \wedge (p \vee \neg q)$
- ☐ (D) $(p \rightarrow q) \rightarrow q$

Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are in conjunctive normal form (CNF)?

- ☐ (A) $(p \vee \neg q) \wedge r$
- ☐ (B) $\neg(p \wedge q)$
- ☐ (C) $p \vee \neg q \vee r$
- ☐ (D) $(p \wedge q) \vee r$

Q.8 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In the formula $\forall x (P(x, y) \rightarrow \exists y Q(x, y, z))$, which variables have free occurrences?

- ☐ (A) y and z
- ☐ (B) z only
- ☐ (C) x, y and z
- ☐ (D) x and z

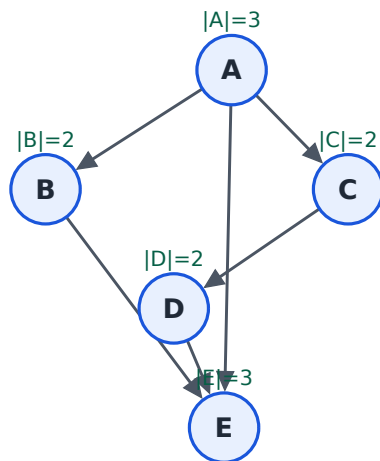
Q.9 [MCQ · 1 mark · negative: -1/3] Conditional Independence

For random variables A, B, C with all conditioning events of positive probability, $A \perp B \mid C$ holds if and only if

- ☐ (A) $P(A \mid B, C) = P(A \mid C)$
- ☐ (B) $P(A, B) = P(A) P(B)$
- ☐ (C) $P(A \mid C) = P(B \mid C)$
- ☐ (D) $P(A, B, C) = P(A) P(B) P(C)$

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

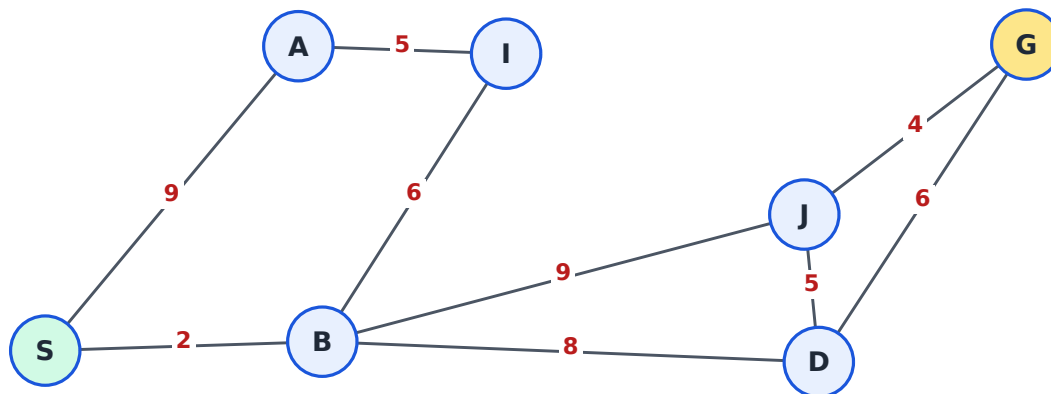
Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). The heuristic is $h(S)=7$, $h(A)=6$, $h(B)=2$, $h(C)=1$, $h(G)=0$. Which of the following statements are TRUE?

- ☐ (A) h is admissible
- ☐ (B) h is consistent
- ☐ (C) A* tree search (no explored set) with this h returns a path of cost 7
- ☐ (D) Greedy best-first search (graph search) with this h returns the path $S \rightarrow B \rightarrow G$

Q.12 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

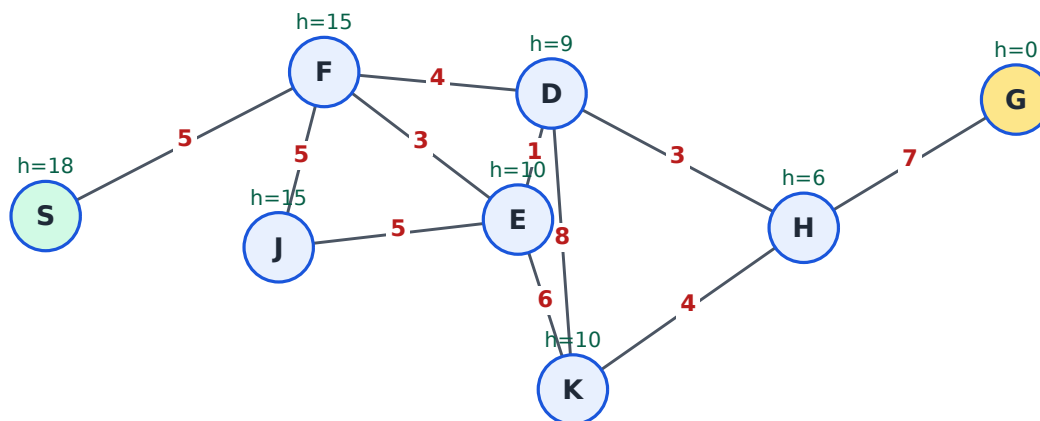
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

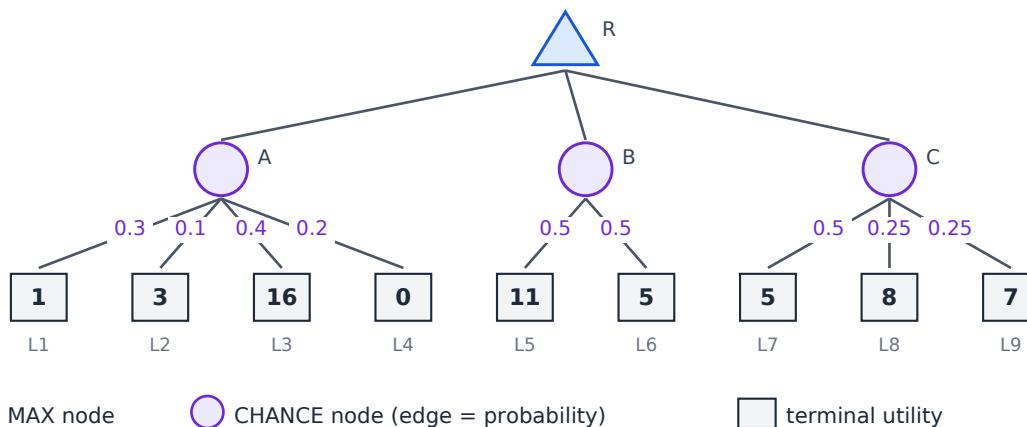
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(J)$ overestimates the true cost from J to G.
- ☐ (B) The edge F-E violates the consistency condition $h(F) \leq c(F,E) + h(E)$.
- ☐ (C) h is admissible.
- ☐ (D) h is consistent (monotone).

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic

Here x and y are variables, and a, b are constants. Which of the following pairs of atoms are unifiable (with the occurs check)?

- ☐ (A) $P(x, f(x))$ and $P(f(y), y)$
☐ (B) $Q(a, x, f(x))$ and $Q(y, b, f(b))$
☐ (C) $R(x, g(x))$ and $R(g(y), g(g(a)))$
☐ (D) $S(f(a), x)$ and $S(x, f(b))$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **B, C, D** and rules:

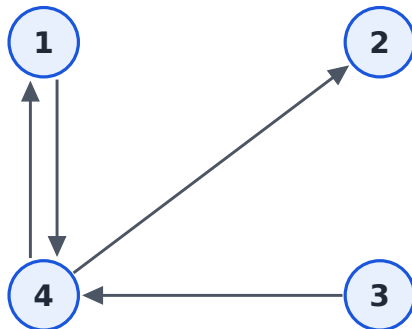
- R1: $H \rightarrow B$
 R2: $K \rightarrow E$
 R3: $C \rightarrow K$
 R4: $J \wedge A \rightarrow F$
 R5: $C \wedge A \rightarrow H$
 R6: $C \rightarrow F$
 R7: $C \wedge F \wedge H \rightarrow A$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) A is entailed
☐ (B) E is entailed
☐ (C) F is entailed
☐ (D) K is entailed

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\forall x \exists y R(x, y)$
☐ (C) $\exists x \forall y \neg R(y, x)$
☐ (D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$

Q.18 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following formulas is in conjunctive normal form AND is logically equivalent to $p \leftrightarrow q$?

- ☐ (A) $(\neg p \vee q) \wedge (p \vee \neg q)$
☐ (B) $(p \wedge q) \vee (\neg p \wedge \neg q)$
☐ (C) $(p \vee q) \wedge (\neg p \vee \neg q)$
☐ (D) $(\neg p \vee q) \wedge (p \vee \neg q) \wedge (p \vee q)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

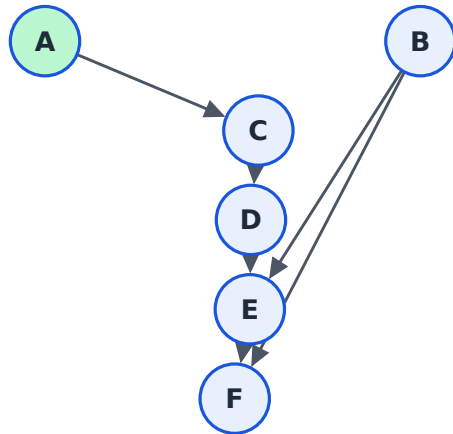
- (1) $(\neg P \rightarrow Q) \leftrightarrow (Q \vee R)$
 (2) $\neg Q$
 (3) $\neg(\neg P \wedge Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg\neg R$
☐ (B) $Q \wedge R$
☐ (C) $\neg R$
☐ (D) $\neg\neg P \vee (Q \vee \neg Q)$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{A})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F, B, D, C, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

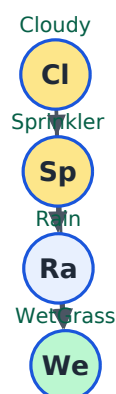
Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence

Consider the Bayesian network (DAG) with edges $P \rightarrow Q$, $P \rightarrow R$, $Q \rightarrow S$, $R \rightarrow S$, $S \rightarrow T$, $U \rightarrow R$. Which of the following independences are guaranteed by d-separation (i.e., hold in every distribution that factorizes over this DAG)?

- ☐ (A) $Q \perp R \mid \{P\}$
☐ (B) $Q \perp R \mid \{P, T\}$
☐ (C) $U \perp P$
☐ (D) $U \perp Q \mid \{P, S\}$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: CI = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\mathbf{WetGrass} = T \mid \mathbf{Cloudy} = T, \mathbf{Sprinkler} = T)$ (round off to 3 decimal places).



P(CI=T)
0.9

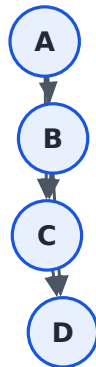
CI	P(Sp=T CI)	CI	Sp	P(Ra=T ...)
T	0.7	T	T	0.95
F	0.1	T	F	0.05
		F	T	0.7
		F	F	0.75

CI	Ra	P(We=T ...)
T	T	0.05
T	F	0.9
F	T	0.1
F	F	0.6

Answer (numerical): _____

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.78, 0.67, 0.09, 0.56)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



A	P(A=T)
T	0.05
F	

A	P(B=T .)	A	B	P(C=T ...)
T	0.75	T	T	0.8
F	0.2	T	F	0.5
		F	T	0.7
		F	F	0.7

A	C	P(D=T ...)
T	T	0.05
T	F	0.5
F	T	0.25
F	F	0.4

- ☐ (A) A=F, B=F, C=T, D=F
☐ (B) A=T, B=F, C=T, D=F
☐ (C) A=F, B=T, C=T, D=F
☐ (D) A=T, B=F, C=F, D=T

Q.24 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

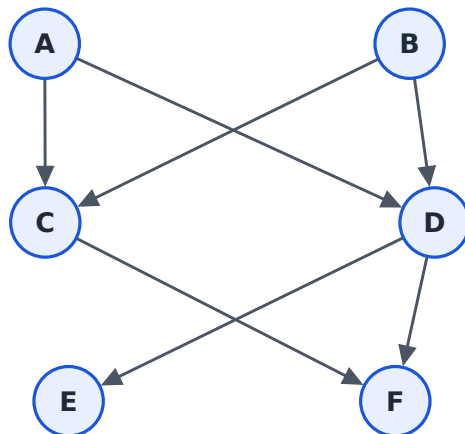
The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid C = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	T	T	T	T
2	T	T	F	T
3	T	T	T	F
4	T	T	T	F
5	F	F	T	T
6	T	T	T	F
7	T	F	T	F
8	T	T	T	T
9	T	T	T	F
10	T	T	T	F
11	T	T	T	T
12	T	T	T	F
13	T	T	T	T
14	T	T	T	F
15	T	T	T	F
16	T	T	T	T
17	T	T	F	T
18	T	T	F	T

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp E \mid B$
☐ (B) $E \perp A \mid D$
☐ (C) $E \perp A$
☐ (D) $C \perp A \mid B, D$

Answer Key — Mock Test 033

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	NAT	2	Adversarial Search	7.99 to 8.01
2	NAT	1	Uninformed Search	174	15	MSQ	2	Predicate Logic	B, C
3	MCQ	1	Adversarial Search	C	16	MSQ	2	Propositional Logic	B, C, D
4	MCQ	1	Uninformed Search	A	17	MSQ	2	Predicate Logic	A, C
5	NAT	1	Propositional Logic	2	18	MCQ	2	Propositional Logic	A
6	MSQ	1	Propositional Logic	A, B, C	19	MSQ	2	Propositional Logic	D
7	MSQ	1	Propositional Logic	A, C	20	NAT	2	Exact Inference	2
8	MCQ	1	Predicate Logic	A	21	MSQ	2	Conditional Independence	A, C
9	MCQ	1	Conditional Independence	A	22	NAT	2	Exact Inference	0.091 to 0.095
10	NAT	1	Bayesian Networks	34	23	MCQ	2	Sampling	A
11	MSQ	2	Informed Search	A, C, D	24	NAT	2	Sampling	0.39 to 0.41
12	NAT	2	Uninformed Search	15	25	MSQ	2	Conditional Independence	B
13	MSQ	2	Informed Search	B					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: A** (A: all step costs are identical (e.g. every action costs 1))

BFS always returns the **shallowest** goal. The shallowest goal is the cheapest one only when path cost is a non-decreasing function of depth, the standard case being identical step costs. BFS uses no heuristic, so (B) has nothing to do with it. With an infinite branching factor BFS cannot even finish a single level, so (C) is wrong. Removing cycles does not help when step costs differ: a deep cheap path can still beat a shallow expensive one, so (D) is wrong.

Q.2 [NAT, 1M · Uninformed Search] **Answer: 174**

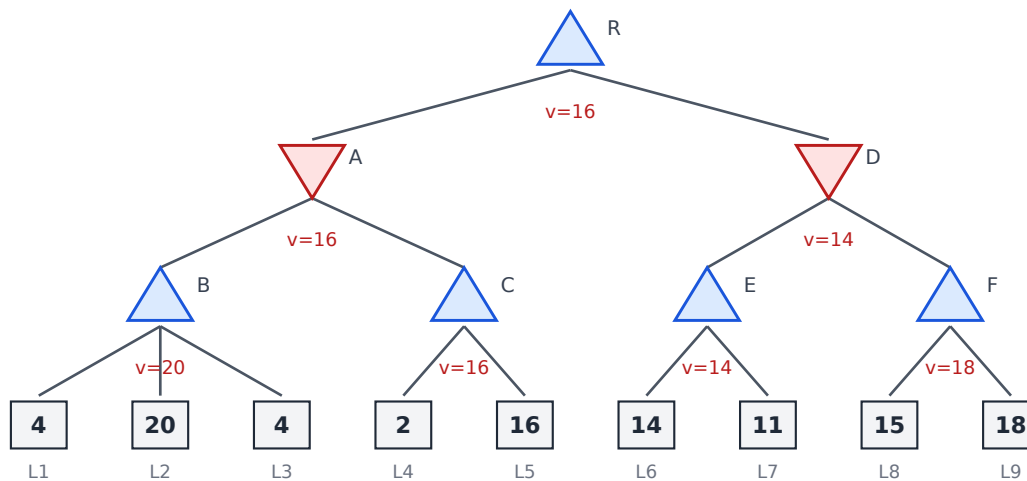
Nodes at depth i are regenerated in every iteration with limit $\geq i$, i.e. $(d + 1 - i)$ times.

$$N = 4 \times 3 + 3 \times 9 + 2 \times 27 + 1 \times 81 = \mathbf{174}.$$

Compare BFS: 120; the overhead of IDS is small for $b \geq 2$.

Q.3 [MCQ, 1M · Adversarial Search] **Answer: C** (C: Move to A; root value 16)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 20, C = 16, A = 16, E = 14, F = 18, D = 14. Root value = **16**; MAX's optimal first move is towards A.

Q.4 [MCQ, 1M · Uninformed Search] **Answer: A** (A: S, C, J, F, A, B, K, G)

Adjacency lists (alphabetical): A: {B, F, K}; B: {A, F, G, K}; C: {F, J, S}; F: {A, B, C, J}; G: {B}; J: {C, F, S}; K: {A, B}; S: {C, J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[C, J]
2	C	[J, F]
3	J	[F]
4	F	[A, B]
5	A	[B, K]
6	B	[K, G]
7	K	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, C, J, F, A, B, K, G**. The path returned is S → C → F → B → G with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 2**

Build the truth table (sub-formulas shown first):

P	Q	R	$(Q \vee R) \leftrightarrow R$	$\neg R \vee (R \vee \neg P)$	$((Q \vee R) \leftrightarrow R) \wedge (\neg R \vee (R \vee \neg P))$
T	T	T	T	T	T
T	T	F	F	T	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	F	T	F
F	F	T	T	T	T
F	F	F	T	T	T

F has 6 models, so it is false in $8 - 6 = 2$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.6 [MSQ, 1M · Propositional Logic] **Answer: A, B, C** (A: $p \vee (p \wedge q)$; B: $p \wedge (p \vee q)$; C: $(p \vee q) \wedge (p \vee \neg q)$)

(A) and (B) are the absorption laws. (C) By distribution, $(p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q) \equiv p$. (D) $(p \rightarrow q) \rightarrow q \equiv \neg(\neg p \vee q) \vee q \equiv (p \wedge \neg q) \vee q \equiv p \vee q$, which differs from p at $p=F, q=T$.

Q.7 [MSQ, 1M · Propositional Logic] Answer: A, C ($A: (p \vee \neg q) \wedge r$; $C: p \vee \neg q \vee r$)

A CNF is a conjunction of clauses, each a disjunction of literals (negation only on atoms). $(p \vee \neg q) \wedge r$ is two clauses. $p \vee \neg q \vee r$ is a single clause, which is a (one-clause) CNF. $\neg(p \wedge q)$ has negation applied to a compound formula, so it is not in CNF (its CNF is $\neg p \vee \neg q$). $(p \wedge q) \vee r$ has a conjunction inside a disjunction; its CNF is $(p \vee r) \wedge (q \vee r)$.

Q.8 [MCQ, 1M · Predicate Logic] Answer: A ($A: y$ and z)

x is bound by $\forall x$ throughout. The occurrence of y in $P(x, y)$ is not in the scope of any $\exists y/\forall y$, so it is free. The y in $Q(x, y, z)$ is bound by $\exists y$. z is never quantified, so it is free. Hence the variables with free occurrences are y and z .

Q.9 [MCQ, 1M · Conditional Independence] Answer: A ($A: P(A | B, C) = P(A | C)$)

Conditional independence of A and B given C means $P(A, B | C) = P(A | C)P(B | C)$, equivalently $P(A | B, C) = P(A | C)$. (B) is marginal independence, which neither implies nor is implied by $A \perp B | C$. (C) compares two different variables' distributions and is meaningless here. (D) is mutual independence, a much stronger condition.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 34

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (2-1) \times 3 + C: (2-1) \times 3 + D: (2-1) \times 2 + E: (3-1) \times 3 \times 2 \times 2 = 34$.

Full joint table: $\prod |X| - 1 = 72 - 1 = 71$; saving = 37.

Q.11 [MSQ, 2M · Informed Search] Answer: A, C, D

The true costs-to-go are $h^*(C)=3$, $h^*(B)=\min(6, 1+3)=4$, $h^*(A)=\min(5+3, 2+4)=6$, and $h^*(S)=\min(1+6, 4+4)=7$. In every case $h \leq h^*$, so (A) is true. (B) is false: on edge $A \rightarrow B$, $h(A)=6 > 2 + h(B) = 4$. (C) is true: A^* tree search with an admissible h is optimal and returns $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$ of cost 7. (D) is true: greedy expands S , then B ($h=2$), which generates C ($h=1$) and G ($h=0$). It then selects G and returns $S \rightarrow B \rightarrow G$ of cost 10.

Q.12 [NAT, 2M · Uninformed Search] Answer: 15

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(2), A(9)
2	B (2)	I(8), A(9), D(10), J(11)
3	I (8)	A(9), D(10), J(11)
4	A (9)	D(10), J(11)
5	D (10)	J(11), G(16)
6	J (11)	G(15)
7	G (15)	← goal, stop

Path returned: **S** $\rightarrow$ **B** $\rightarrow$ **J** $\rightarrow$ **G**, cost = $2 + 9 + 4 = 15$. Nodes expanded (including S and G): **7** (S, B, I, A, D, J, G).

Q.13 [MSQ, 2M · Informed Search] Answer: B (B: The edge $F \rightarrow E$ violates the consistency condition $h(F) \leq c(F, E) + h(E)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G :

n	D	E	F	G	H	J	K	S
$h(n)$	9	10	15	0	6	15	10	18
$h^*(n)$	10	11	14	0	7	16	11	19
$h \leq h^*$?	✓	✓	✗	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **no** (overestimates at F).

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u, v) + h(v)$ for every edge in both directions. Violating directed edges: $F \rightarrow D$ ($15 > 4 + 9$), $F \rightarrow E$ ($15 > 3 + 10$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [NAT, 2M · Adversarial Search] Answer: 7.99 to 8.01

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.3 \times 1 + 0.1 \times 3 + 0.4 \times 16 + 0.2 \times 0 = 7$
- $B = 0.5 \times 11 + 0.5 \times 5 = 8$
- $C = 0.5 \times 5 + 0.25 \times 8 + 0.25 \times 7 = 6.25$

Root = $\max(7, 8, 6.25) = 8$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [MSQ, 2M · Predicate Logic] Answer: B, C (B: $Q(a, x, f(x))$ and $Q(y, b, f(b))$; C: $R(x, g(x))$ and $R(g(y), g(g(a)))$)
 (A) $x/f(y)$; then $f(f(y))$ must unify with y , which fails the occurs check. (B) a/y gives y/a , x/b , then $f(b) = f(b)$: MGU $\{y/a, x/b\}$. (C) $x/g(y)$, then $g(g(y))$ vs $g(g(a))$ gives y/a : MGU $\{x/g(a), y/a\}$. (D) $x/f(a)$, then $f(a)$ must equal $f(b)$, which fails since $a \neq b$ are distinct constants.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, C, D (B: E is entailed; C: F is entailed; D: K is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $C \rightarrow K$, $C \rightarrow F$; add F, K.

Round 2: fire $K \rightarrow E$; add E.

Fixed point: $\{B, C, D, E, F, K\}$ — 6 atoms are entailed (facts: B, C, D; newly derived: E, F, K). Atoms never derived: A, H, J.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, C (A: $\exists x \forall y \neg R(x, y)$; C: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,4), (3,4), (4,1), (4,2)\}$.

- (A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.
- (B) $\forall x \exists y R(x, y)$: **FALSE** — 2 has no outgoing edge.
- (C) $\exists x \forall y \neg R(y, x)$: **TRUE** — 3 has no incoming edge.
- (D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,2)$ nor $R(2,1)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MCQ, 2M · Propositional Logic] Answer: A (A: $(\neg p \vee q) \wedge (p \vee \neg q)$)

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv (\neg p \vee q) \wedge (p \vee \neg q)$, which is a CNF. $(p \wedge q) \vee (\neg p \wedge \neg q)$ is equivalent but is in DNF, not CNF. $(p \vee q) \wedge (\neg p \vee \neg q)$ is true exactly when one of p, q is true, i.e. $p \oplus q$. The last option adds the clause $p \vee q$, which removes the model $p=q=F$, leaving only $p \wedge q$.

Q.19 [MSQ, 2M · Propositional Logic] Answer: D (D: $\neg \neg P \vee (Q \vee \neg Q)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	F	T	T
F	F	F	T

- (A) $\neg \neg R$ is false in the KB-model $P=F, Q=F, R=F \Rightarrow$ not entailed.
- (B) $Q \wedge R$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.
- (C) $\neg R$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.
- (D) $\neg \neg P \vee (Q \vee \neg Q)$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.20 [NAT, 2M · Exact Inference] Answer: 2

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(B), f(A, C), f(C, D), f(B, D, E), f(B, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(B, E, F)$	$\{B, E, F\}$	$f(B, E)$
B	$f(B), f(B, D, E), f(B, E)$	$\{B, D, E\}$	$f(D, E)$
D	$f(C, D), f(D, E)$	$\{C, D, E\}$	$f(C, E)$
C	$f(A, C), f(C, E)$	$\{A, C, E\}$	$f(A, E)$
E	$f(A, E)$	$\{A, E\}$	$f(A)$

The largest factor produced has **2** variable(s) (created when eliminating F); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.21 [MSQ, 2M · Conditional Independence] **Answer: A, C** (A: $Q \perp R \mid \{P\}$; C: $U \perp P$)

Paths between Q and R: $Q \leftarrow P \rightarrow R$ (fork at P) and $Q \rightarrow S \leftarrow R$ (collider S). (A) Given P, the fork is blocked and the collider S has no observed descendant: d-separated, true. (B) T is a descendant of S, so observing T activates $Q \rightarrow S \leftarrow R$: false. (C) Paths from U to P all pass through R as a collider ($U \rightarrow R \leftarrow P$, or $U \rightarrow R \rightarrow S \leftarrow P$ with collider S); with nothing observed they are blocked: true. (D) Given S, the path $U \rightarrow R \rightarrow S \leftarrow Q$ has observed collider S and unobserved non-collider R, so it is active: false.

Q.22 [NAT, 2M · Exact Inference] **Answer: 0.091 to 0.095**

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Cloudy}, \text{Rain})$.

Inference by enumeration: $P(\text{WetGrass} \mid e) = \alpha \sum_{\text{hidden}} P(\text{WetGrass}, e, \text{hidden})$.

Hidden: Ra	We	product of CPT entries (Cl · Sp · Ra · We)	value
T	T	$0.9 \times 0.7 \times 0.95 \times 0.05$	0.02992
F	T	$0.9 \times 0.7 \times 0.05 \times 0.9$	0.02835
T	F	$0.9 \times 0.7 \times 0.95 \times 0.95$	0.56857
F	F	$0.9 \times 0.7 \times 0.05 \times 0.1$	0.00315

$P(\text{WetGrass}=T, e) = 0.05828$; $P(\text{WetGrass}=F, e) = 0.57172$; $P(e) = 0.63$.

$P(\text{WetGrass}=T \mid e) = 0.05828 / 0.63 = \mathbf{0.0925} \approx 0.093$.

Q.23 [MCQ, 2M · Sampling] **Answer: A** (A: $A=F$, B=F, C=T, D=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.05$; $u = 0.78 \geq 0.05 \Rightarrow A = F$
- B: $P(B=T \mid A=F) = 0.2$; $u = 0.67 \geq 0.2 \Rightarrow B = F$
- C: $P(C=T \mid A=F, B=F) = 0.7$; $u = 0.09 < 0.7 \Rightarrow C = T$
- D: $P(D=T \mid A=F, C=T) = 0.25$; $u = 0.56 \geq 0.25 \Rightarrow D = F$

Sample: **(A=F, B=F, C=T, D=F)**.

Q.24 [NAT, 2M · Sampling] **Answer: 0.39 to 0.41**

Rejection sampling keeps only samples consistent with the evidence $C = T$: samples 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (15 accepted, 3 rejected).

Among the accepted samples, $D = T$ in 6. Estimate = $6/15 = \mathbf{0.4}$.

(For this network the exact value is 0.3162; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.25 [MSQ, 2M · Conditional Independence] **Answer: B** (B: $E \perp A \mid D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp E \mid B$: **FALSE** — the path $A \rightarrow D \rightarrow E$ is active given $\{B\}$.

(B) $E \perp A \mid D$: **TRUE** — $E \leftarrow D \leftarrow A$ (blocked: observed chain node D); $E \leftarrow D \leftarrow B \rightarrow C \leftarrow A$ (blocked: observed chain node D); $E \leftarrow D \rightarrow F \leftarrow C \leftarrow A$ (blocked: observed fork node D).

(C) $E \perp A$: **FALSE** — the path $E \leftarrow D \leftarrow A$ is active given $\{\}$.

(D) $C \perp A \mid B, D$: **FALSE** — the path $C \leftarrow A$ is active given $\{B, D\}$.

Mock Test 034

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 8 / 12	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

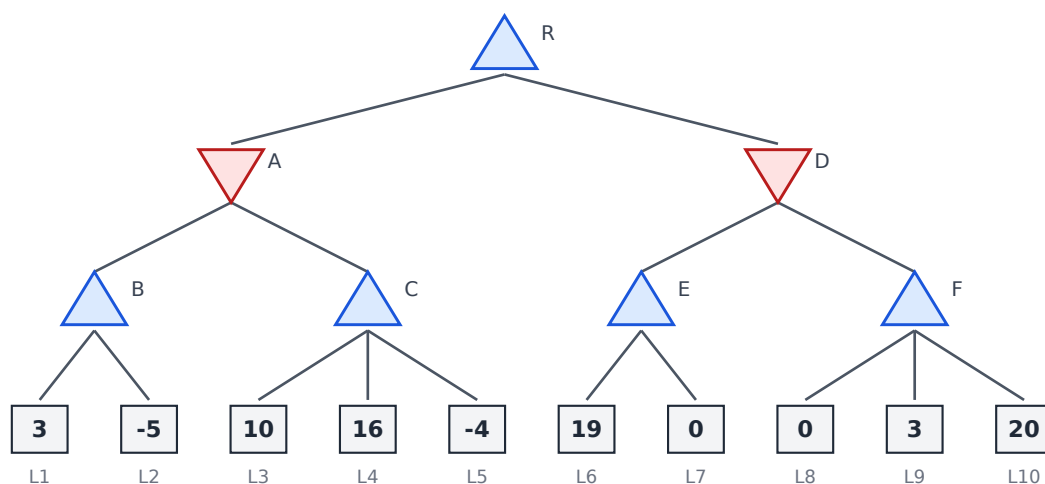
Q.1 [MCQ · 1 mark · negative: $-1/3$] Informed Search

A heuristic h is **consistent** (monotone) if, for every node n , every action a and its successor n' ,

- ☐ (A) $h(n) \leq h^*(n)$
- ☐ (B) $h(n) \leq c(n, a, n') + h(n')$
- ☐ (C) $h(n') \leq h(n)$
- ☐ (D) $h(n) + h(n') \leq c(n, a, n')$

Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



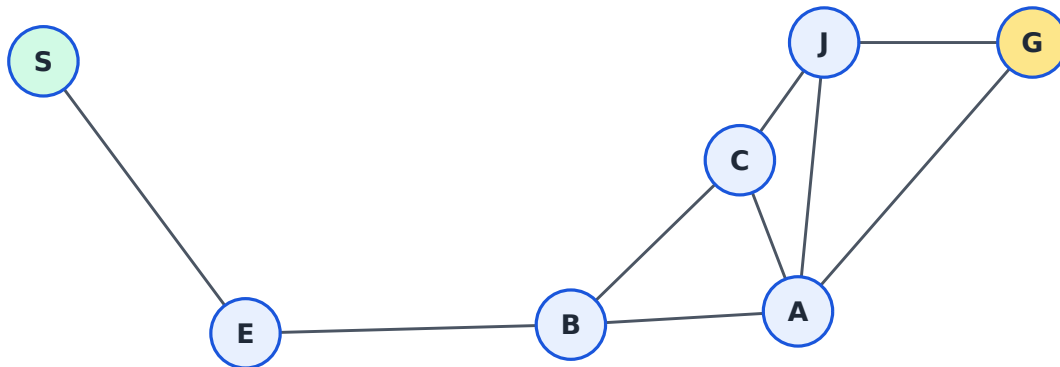
MAX node MIN node

terminal utility

- ☐ (A) Move to A; root value 3
- ☐ (B) Move to D; root value 22
- ☐ (C) Move to D; root value 20
- ☐ (D) Move to D; root value 19

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

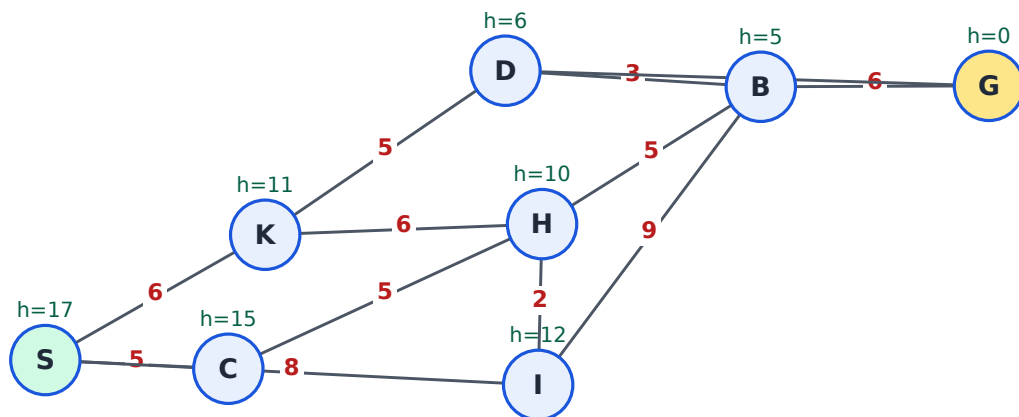
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, E, A, B, C, J, G
☐ (B) S, E, B, A, C, G
☐ (C) S, E, B, C, J, G
☐ (D) S, E, B, A, C, J, G

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3, 4\}$. How many interpretations of a unary predicate P over D make $\exists x P(x) \wedge \neg \forall x P(x)$ true?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

For propositional sentences α and β , $\alpha \models \beta$ holds if and only if

- ☐ (A) $\alpha \wedge \neg\beta$ is unsatisfiable
☐ (B) $\alpha \wedge \beta$ is satisfiable
☐ (C) $\neg\alpha \vee \beta$ is satisfiable
☐ (D) $\alpha \vee \neg\beta$ is valid

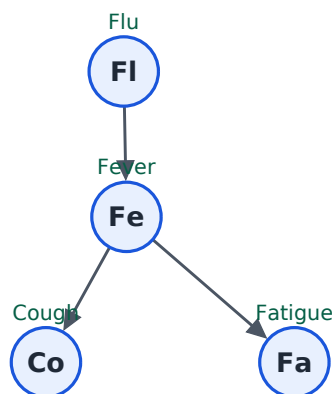
Q.8 [MSQ · 1 mark · no negative marking] Variable Elimination

Which of the following statements about factors in variable elimination are correct (all variables Boolean)?

- ☐ (A) Summing out one variable from a factor over k variables yields a factor with 2^{k-1} entries
☐ (B) The pointwise product of factors is commutative
☐ (C) The entries of every intermediate factor must sum to 1
☐ (D) Evidence variables are fixed to their observed values, so they do not appear as dimensions of factors

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Fl, Fe, Co, Fa. (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) What is the probability that a single run produces the sample **(Fl=T, Fe=F, Co=T, Fa=F)**? (round off to 4 decimal places)



P(Fl=T)
0.5

Fl	P(Fe=T ...)	Fe	P(Co=T ...)
T	0.3	T	0.3
F	0.95	F	0.6

Fe	P(Fa=T ...)
T	0.05
F	0.7

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Conditional Independence

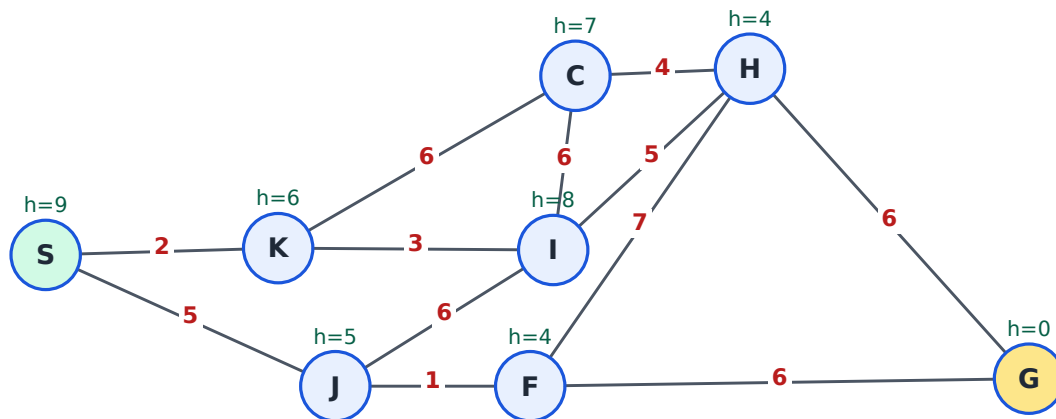
In a Bayesian network, a node is conditionally independent of all other nodes in the network given its Markov blanket. The Markov blanket consists of the node's

- ☐ (A) parents, children, and children's other parents
☐ (B) parents and children only
☐ (C) parents and all ancestors
☐ (D) all non-descendants

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

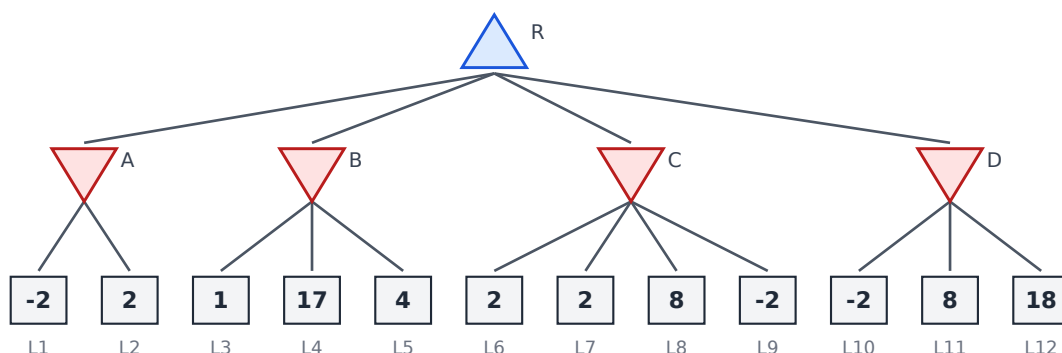
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(H)$ overestimates the true cost from H to G.
☐ (B) h is consistent (monotone).
☐ (C) The edge S-K violates the consistency condition $h(S) \leq c(S,K) + h(K)$.
☐ (D) h is admissible.

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



MAX node MIN node

terminal utility

- ☐ (A) Leaf L3 is pruned.
☐ (B) Leaf L4 is pruned.
☐ (C) Leaf L9 is pruned.
☐ (D) Leaf L12 is pruned.

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). Uniform-cost search (graph search with an explored set, goal test when a node is selected for expansion) is run from S. How many nodes are expanded before the goal G is selected? Count S, and do not count G.

Answer (numerical): _____

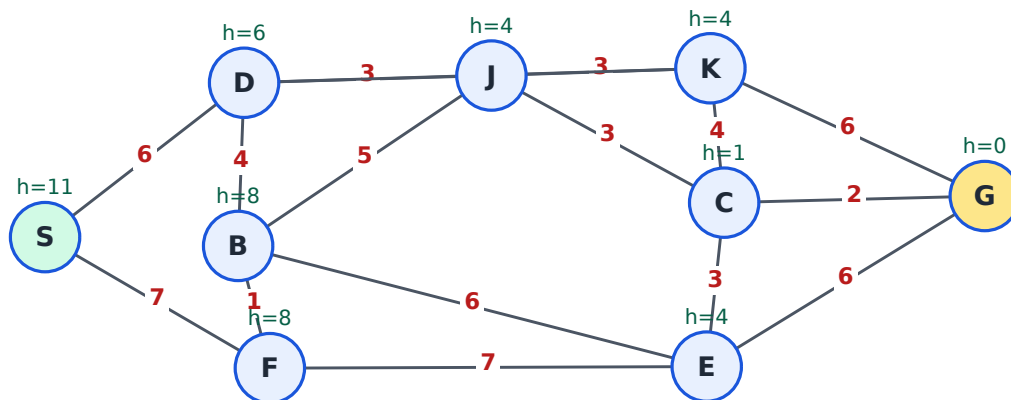
Q.14 [NAT · 2 marks · no negative marking] Uninformed Search

Depth-first tree search generates all $b = 3$ children of a node when it expands that node. The tree has depth $m = 10$, and no goal is found along the leftmost path. Consider the moment just after DFS expands the leftmost node at depth 9, so that its 3 children at depth 10 have been added to the frontier. How many nodes are in the frontier at that moment?

Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Informed Search - A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

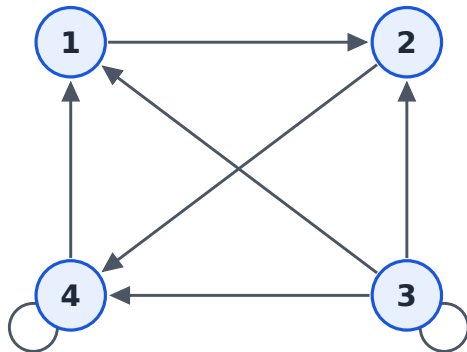
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following are logically equivalent to $p \rightarrow (q \vee r)$?

- ☐ (A) $(p \wedge \neg q) \rightarrow r$
- ☐ (B) $(p \rightarrow q) \vee (p \rightarrow r)$
- ☐ (C) $\neg r \rightarrow (p \rightarrow q)$
- ☐ (D) $(p \rightarrow q) \wedge (p \rightarrow r)$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (B) $\exists x \forall y \neg R(x, y)$
☐ (C) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (D) $\exists x \forall y R(x, y)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(\neg P \rightarrow R) \vee (Q \leftrightarrow P)$
 (2) $(R \leftrightarrow \neg R) \leftrightarrow (\neg Q \rightarrow P)$
 (3) $(P \wedge \neg P) \rightarrow Q$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $Q \wedge P$
☐ (B) $(Q \wedge P) \rightarrow (R \vee \neg R)$
☐ (C) $(\neg P \vee \neg Q) \leftrightarrow (R \leftrightarrow \neg P)$
☐ (D) $(\neg Q \rightarrow Q) \rightarrow (\neg R \wedge \neg Q)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

The formula $F = ((\neg Q \leftrightarrow S) \wedge (S \leftrightarrow Q)) \vee ((Q \vee S) \wedge (P \leftrightarrow R))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R, S . How many maxterms (clauses) does it contain?

Answer (numerical): _____

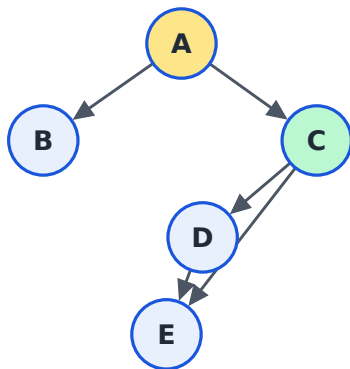
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (\neg R \wedge \neg Q) \wedge (Q \wedge P)$?

- ☐ (A) $\neg(\neg(\neg R \wedge \neg Q) \vee \neg(Q \wedge P))$
☐ (B) $\neg(\neg(\neg R \wedge \neg Q) \wedge \neg(Q \wedge P))$
☐ (C) $(\neg R \wedge \neg Q) \vee (Q \wedge P)$
☐ (D) $(Q \wedge P) \wedge (\neg R \wedge \neg Q)$

Q.21 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence **A=F**. The current state of the chain is **(A=F, B=F, C=F, D=T, E=F)**. The next step resamples **C**. With what probability is C set to T? (round off to 3 decimal places)



P(A=T)	A	P(B=T ...)	A	P(C=T ...)
0.25	T	0.2	T	0.05
	F	0.75	F	0.05

C	P(D=T ...)	C	D	P(E=T ...)
T	0.25	T	T	0.2
F	0.6	T	F	0.25
		F	T	0.4
		F	F	0.7

Answer (numerical): _____

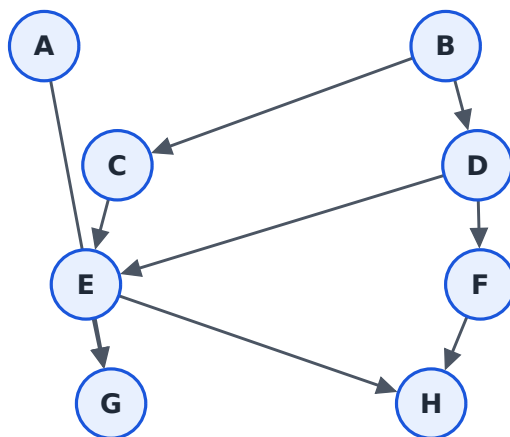
Q.22 [NAT · 2 marks · no negative marking] Bayesian Networks

A Bayesian network has: A with 3 values (no parents); B Boolean (no parents); C with 4 values and parents A, B; D Boolean with parent C; E with 3 values and parents C, D. How many independent parameters do the CPTs require?

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

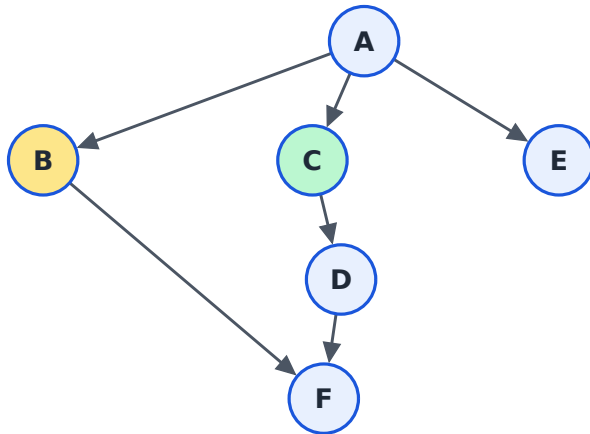
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp D \mid E, G$
☐ (B) $H \perp A$
☐ (C) $H \perp G \mid C$
☐ (D) $G \perp E \mid H$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

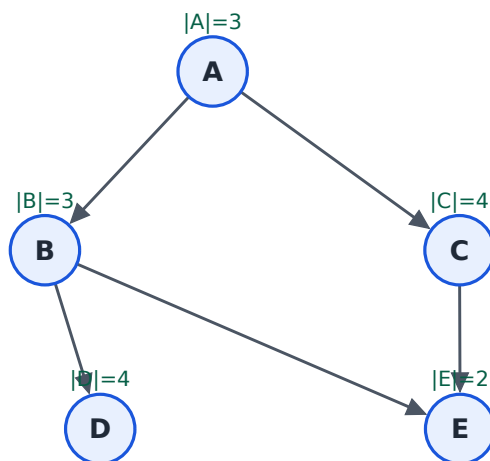
Variable elimination is used to compute $P(C \mid B = T)$ in the Bayesian network below (Boolean variables; evidence $B = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F**, **A**, **E**, **D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Answer Key — Mock Test 034

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Informed Search	B	14	NAT	2	Uninformed Search	21
2	MCQ	1	Adversarial Search	D	15	NAT	2	Informed Search	5
3	MCQ	1	Uninformed Search	D	16	MSQ	2	Propositional Logic	A, B, C
4	NAT	1	Informed Search	18	17	MSQ	2	Predicate Logic	A, D
5	NAT	1	Predicate Logic	14	18	MSQ	2	Propositional Logic	B, D
6	NAT	1	Predicate Logic	8	19	NAT	2	Propositional Logic	10
7	MCQ	1	Propositional Logic	A	20	MSQ	2	Propositional Logic	A, D
8	MSQ	1	Variable Elimination	A, B, D	21	NAT	2	Sampling	0.026 to 0.03
9	NAT	1	Sampling	0.0628 to 0.0632	22	NAT	2	Bayesian Networks	41
10	MCQ	1	Conditional Independence	A	23	MSQ	2	Conditional Independence	A, B
11	MSQ	2	Informed Search	C, D	24	NAT	2	Exact Inference	2
12	MSQ	2	Adversarial Search	D	25	NAT	2	Bayesian Networks	249
13	NAT	2	Uninformed Search	4					

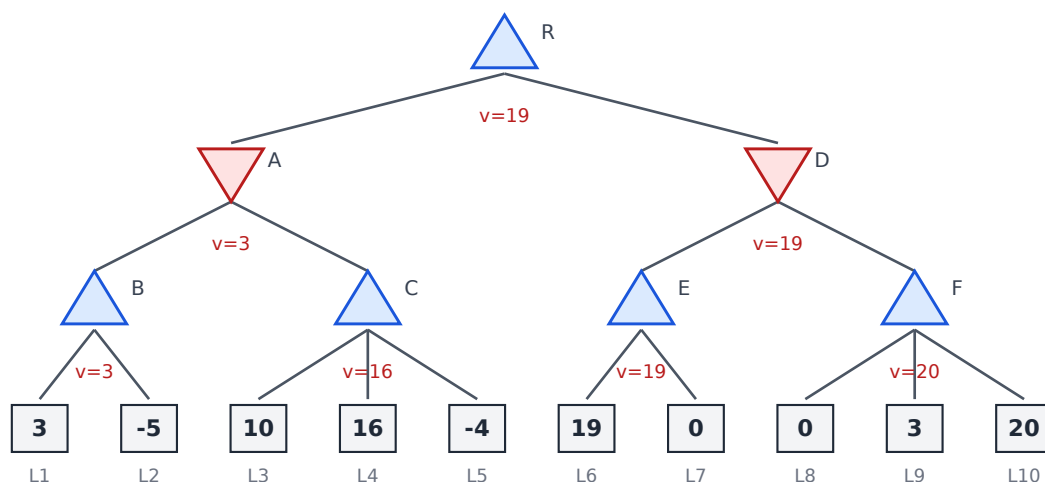
Detailed Solutions

Q.1 [MCQ, 1M · Informed Search] Answer: B ($B: h(n) \leq c(n, a, n') + h(n')$)

Consistency is the triangle inequality $h(n) \leq c(n, a, n') + h(n')$. (A) is admissibility. (C) is not required: a successor can have a larger h than its parent. (D) is meaningless as a condition and would force h to be tiny.

Q.2 [MCQ, 1M · Adversarial Search] Answer: D (D: Move to D; root value 19)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 3$, $C = 16$, $A = 3$, $E = 19$, $F = 20$, $D = 19$. Root value = **19**; MAX's optimal first move is towards D.

Q.3 [MCQ, 1M · Uninformed Search] Answer: D (D: S, E, B, A, C, J, G)

Adjacency lists (alphabetical): A: {B, C, G, J}; B: {A, C, E}; C: {A, B, J}; E: {B, S}; G: {A, J}; J: {A, C, G}; S: {E}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, E, B, A, C, J, G**; the path found is $S \rightarrow E \rightarrow B \rightarrow A \rightarrow C \rightarrow J \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Informed Search] Answer: 18

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (17)	K(11), I(12), C(15)
2	K (11)	D(6), H(10), I(12), C(15)

Step	Expanded (h)	Frontier after expansion (sorted by priority)
3	D (6)	G(0), B(5), H(10), I(12), C(15)
4	G (0)	B(5), H(10), I(12), C(15) ← goal, stop

Path returned: **S** → **K** → **D** → **G**, cost = 6 + 5 + 7 = **18**.

The optimal cost is $h^*(S) = 18$ — greedy happens to be optimal here, but in general it is not.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 14**

An interpretation of P is a subset of D: $2^4 = 16$ choices. $\exists x P(x)$ excludes the empty set, $\neg \forall x P(x)$ excludes D itself. Hence $16 - 2 = 14$.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 8**

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 3$ this gives **8** (verified by enumerating all interpretations).

Q.7 [MCQ, 1M · Propositional Logic] **Answer: A** (A: $\alpha \wedge \neg \beta$ is unsatisfiable)

$\alpha \models \beta$ means every model of α is a model of β , i.e. no model makes α true and β false, i.e. $\alpha \wedge \neg \beta$ is unsatisfiable. Equivalently $\alpha \rightarrow \beta$ is valid. Satisfiability of $\neg \alpha \vee \beta$ or $\alpha \wedge \beta$ is too weak (one model is enough). $\alpha \vee \neg \beta$ being valid would mean $\beta \models \alpha$, the reverse direction.

Q.8 [MSQ, 1M · Variable Elimination] **Answer: A, B, D**

(A) Summing out removes one Boolean dimension, from 2^k to 2^{k-1} entries. (B) Pointwise product is entry-wise multiplication of real numbers, hence commutative (and associative). (D) AIMA instantiates evidence, e.g. $P(j | A)$ becomes a factor over A alone. (C) is false: factors are generally unnormalized (e.g. $P(j | A)$ as a function of A, or intermediate sums), normalization happens only at the end.

Q.9 [NAT, 1M · Sampling] **Answer: 0.0628 to 0.0632**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(FI=T) \times P(Fe=F|FI=T) \times P(Co=T|Fe=F) \times P(Fa=F|Fe=F)$
 $= 0.5 \times 0.7 \times 0.6 \times 0.3 = \mathbf{0.063} \approx 0.0630$.

Q.10 [MCQ, 1M · Conditional Independence] **Answer: A** (A: parents, children, and children's other parents)

AIMA defines the Markov blanket as parents, children and children's parents. Children's other parents are needed because of explaining away: given a child, a co-parent carries information about the node. (B) omits the co-parents; (C) and (D) describe sets relevant to the local Markov property (independence from non-descendants given parents), not the blanket.

Q.11 [MSQ, 2M · Informed Search] **Answer: C, D** (C: The edge S-K violates the consistency condition $h(S) \leq c(S,K) + h(K)$.; D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	F	G	H	I	J	K	S
$h(n)$	7	4	0	4	8	5	6	9
$h^*(n)$	10	6	0	6	11	7	14	12
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

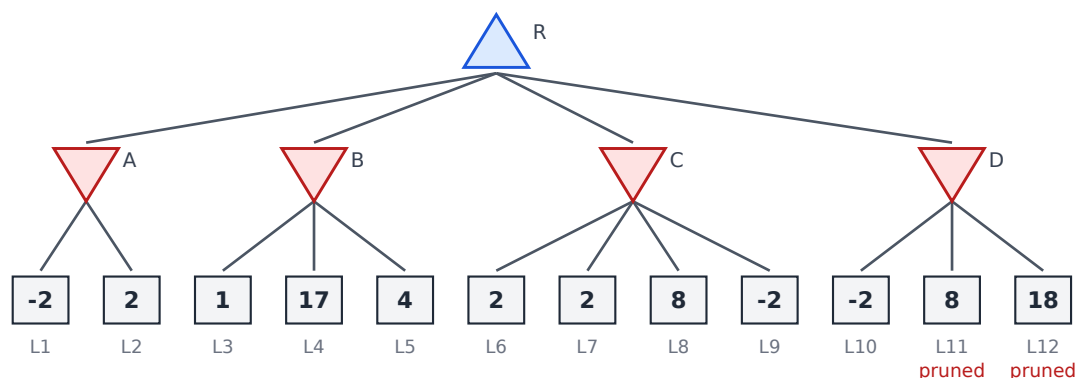
Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $S \rightarrow K$ ($9 > 2 + 6$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [MSQ, 2M · Adversarial Search] **Answer: D** (D: Leaf L12 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node D: value $-2 \leq \alpha = 1 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (10): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10. Leaves pruned (2): L11, L12. Root value = 1 (same as plain minimax, 1 — pruning never changes the root value).

Q.13 [NAT, 2M · Uninformed Search] Answer: 4

The expansion order by g-value is S(0), A(1), B(3) (reached via A), C(4) (reached via B). After that, the cheapest entry in the frontier is G(7), which is selected and terminates the search. The stale entries B(4) and C(6) are never expanded. Nodes expanded = {S, A, B, C} = 4.

Q.14 [NAT, 2M · Uninformed Search] Answer: 21

Along the leftmost path, the nodes at depths 0 to 9 have all been expanded. At each depth from 1 to 9, the $b - 1 = 2$ unexpanded siblings of the path node are still in the frontier: $9 \times 2 = 18$ nodes. The 3 children at depth 10 have just been added. Frontier size = $18 + 3 = 21 = (b-1)m + 1$, which illustrates DFS's $O(bm)$ space.

Q.15 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(B)=8$, $h(C)=1$, $h(D)=6$, $h(E)=4$, $h(F)=8$, $h(G)=0$, $h(J)=4$, $h(K)=4$, $h(S)=11$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	D(12), F(15)
2	D (12)	J(13), F(15), K(17), B(18)
3	J (13)	C(13), F(15), K(16), B(18)
4	C (13)	G(14), F(15), K(16), B(18), E(19)
5	G (14)	F(15), K(16), B(18), E(19) ← goal, stop

Path returned: **S** → **D** → **J** → **C** → **G** with cost **14**; nodes expanded = **5** (S, D, J, C, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A, B, C (A: $(p \wedge \neg q) \rightarrow r$; B: $(p \rightarrow q) \vee (p \rightarrow r)$; C: $\neg r \rightarrow (p \rightarrow q)$)

$p \rightarrow (q \vee r) \equiv \neg p \vee q \vee r$. (A) $(p \wedge \neg q) \rightarrow r \equiv \neg p \vee q \vee r$. (B) $(\neg p \vee q) \vee (\neg p \vee r) \equiv \neg p \vee q \vee r$. (C) $\neg r \rightarrow (\neg p \vee q) \equiv r \vee \neg p \vee q$. (D) $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$, which is false at $p=T, q=T, r=F$ whereas the original is true there.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, D (A: $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$; D: $\exists x \forall y R(x, y)$)

$R = \{(1,2), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,4)\}$.

(A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **TRUE** — every two distinct elements are related in some direction.

(B) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

(C) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(2,4) \wedge R(4,1)$ but $\neg R(2,1)$.

(D) $\exists x \forall y R(x, y)$: **TRUE** — $x = 3$ has edges to every y .

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $(Q \wedge P) \rightarrow (R \vee \neg R)$; D: $(\neg Q \rightarrow Q) \rightarrow (\neg R \wedge \neg Q)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
F	F	T	T
F	F	F	T

(A) $Q \wedge P$ is false in the KB-model $P=F, Q=F, R=T \Rightarrow$ not entailed.

(B) $(Q \wedge P) \rightarrow (R \vee \neg R)$ is true in all models of KB $\Rightarrow$ **entailed**.

(C) $(\neg P \vee \neg Q) \leftrightarrow (R \leftrightarrow \neg P)$ is false in the KB-model $P=F, Q=F, R=F \Rightarrow$ not entailed.

(D) $(\neg Q \rightarrow Q) \rightarrow (\neg R \wedge \neg Q)$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.19 [NAT, 2M · Propositional Logic] **Answer: 10**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(\neg Q \leftrightarrow S) \wedge (S \leftrightarrow Q)$	$(Q \vee S) \wedge (P \leftrightarrow R)$	$((\neg Q \leftrightarrow S) \wedge (S \leftrightarrow Q)) \vee ((Q \vee S) \wedge (P \leftrightarrow R))$
T	T	T	T	F	T	T
T	T	T	F	F	T	T
T	T	F	T	F	F	F
T	T	F	F	F	F	F
T	F	T	T	F	T	T
T	F	T	F	F	F	F
T	F	F	T	F	F	F
T	F	F	F	F	F	F
F	T	T	T	F	F	F
F	T	T	F	F	F	F
F	T	F	T	F	T	T
F	T	F	F	F	T	T
F	F	T	T	F	F	F
F	F	T	F	F	F	F
F	F	F	T	F	T	T
F	F	F	F	F	F	F

F has 6 models, so it is false in $16 - 6 = 10$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $\neg(\neg(\neg R \wedge \neg Q) \vee \neg(Q \wedge P))$; D: $(Q \wedge P) \wedge (\neg R \wedge \neg Q)$)

Compare truth tables with $F = (\neg R \wedge \neg Q) \wedge (Q \wedge P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg(\neg R \wedge \neg Q) \vee \neg(Q \wedge P))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(B) $\neg(\neg(\neg R \wedge \neg Q) \wedge \neg(Q \wedge P))$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(C) $(\neg R \wedge \neg Q) \vee (Q \wedge P)$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(D) $(Q \wedge P) \wedge (\neg R \wedge \neg Q)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

Q.21 [NAT, 2M · Sampling] **Answer: 0.026 to 0.03**

Gibbs sampling draws C from $P(C \mid \text{mb}(C))$, where $\text{mb}(C) = \{A, D, E\}$:

$P(C \mid \text{mb}) \propto P(C \mid \text{parents}(C)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$C=T$: $P(C=T \mid A=F) \times P(D=T \mid C=T) \times P(E=F \mid C=T, D=T) = 0.05 \times 0.25 \times 0.8 = 0.01$

$C=F$: $P(C=F \mid A=F) \times P(D=T \mid C=F) \times P(E=F \mid C=F, D=T) = 0.95 \times 0.6 \times 0.6 = 0.342$

Normalise: $0.01 / (0.01 + 0.342) = 0.0284$.

Q.22 [NAT, 2M · Bayesian Networks] **Answer: 41**

Each CPT needs (number of parent configurations) $\times$ (arity - 1). A: $1 \times 2 = 2$. B: $1 \times 1 = 1$. C: $(3 \times 2) \times 3 = 18$. D: $4 \times 1 = 4$. E: $(4 \times 2) \times 2 = 16$. Total = $2 + 1 + 18 + 4 + 16 = 41$, versus $3 \times 2 \times 4 \times 2 \times 3 - 1 = 143$ for the full joint.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $A \perp D \mid E, G$; B: $H \perp A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp D \mid E, G$: **TRUE** — $A \rightarrow G \leftarrow E \leftarrow C \leftarrow B \rightarrow D$ (blocked: observed chain node E); $A \rightarrow G \leftarrow E \leftarrow D$ (blocked: observed chain node E); $A \rightarrow G \leftarrow E \rightarrow H \leftarrow F \leftarrow D$ (blocked: observed fork node E).

(B) $H \perp A$: **TRUE** — $H \leftarrow E \rightarrow G \leftarrow A$ (blocked: collider G with neither it nor a descendant observed); $H \leftarrow F \leftarrow D \leftarrow B \rightarrow C \rightarrow E \rightarrow G \leftarrow A$ (blocked: collider G with neither it nor a descendant observed); $H \leftarrow F \leftarrow D \rightarrow E \rightarrow G \leftarrow A$ (blocked: collider G with neither it nor a descendant observed).

(C) $H \perp G \mid C$: **FALSE** — the path $H \leftarrow E \rightarrow G$ is active given $\{C\}$.

(D) $G \perp E \mid H$: **FALSE** — the path $G \leftarrow E$ is active given $\{H\}$.

Q.24 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A), f(A, C), f(C, D), f(A, E), f(D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(D, F)$	$\{D, F\}$	$f(D)$
A	$f(A), f(A), f(A, C), f(A, E)$	$\{A, C, E\}$	$f(C, E)$
E	$f(C, E)$	$\{C, E\}$	$f(C)$
D	$f(C, D), f(D)$	$\{C, D\}$	$f(C)$

The largest factor produced has **2** variable(s) (created when eliminating A); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over C and normalise.

Q.25 [NAT, 2M · Bayesian Networks] **Answer: 249**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (3-1) \times 3 + C: (4-1) \times 3 + D: (4-1) \times 3 + E: (2-1) \times 3 \times 4 = \mathbf{38}$.

Full joint table: $\prod |X| - 1 = 288 - 1 = 287$; saving = 249.

Mock Test 035

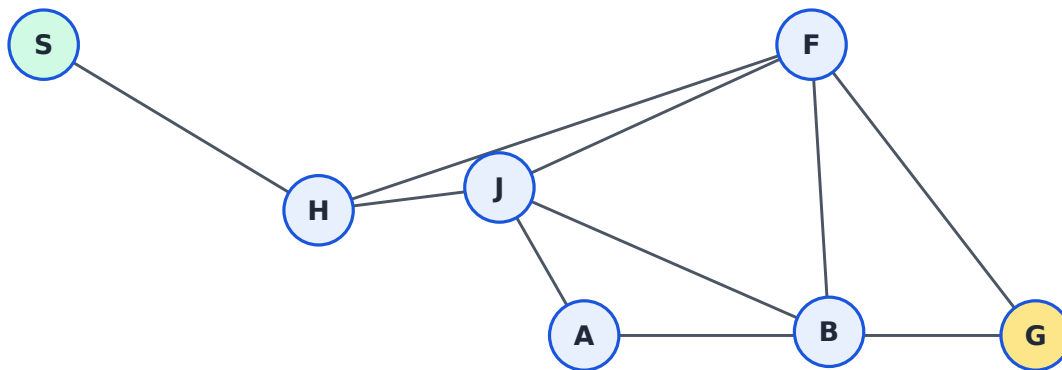
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 6 / 13	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, H, F, J, B, G
- ☐ (B) S, H, J, F, G
- ☐ (C) S, H, F, B, A, J, G
- ☐ (D) S, H, F, B, J, A, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 5$ (so $b^d = 243$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

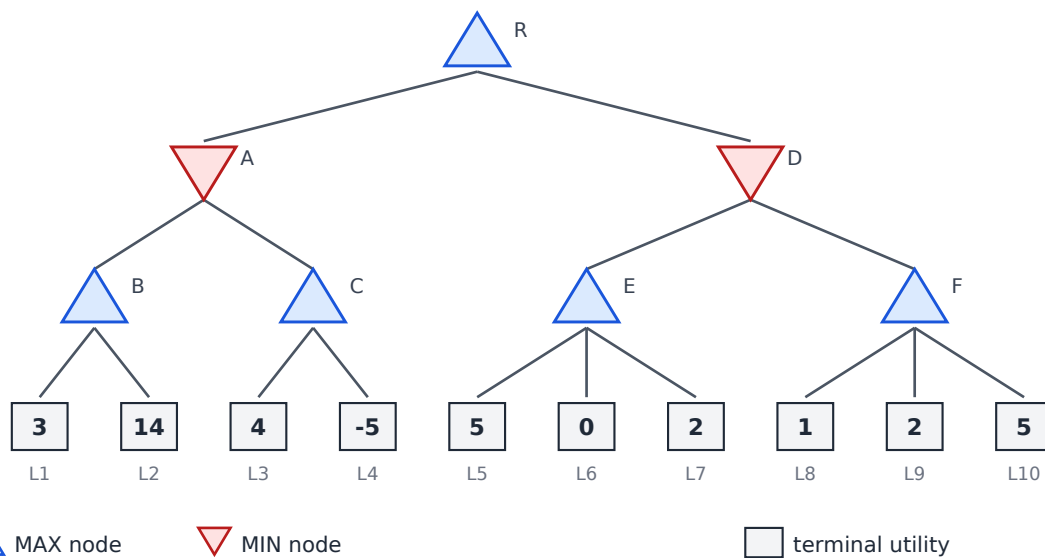
Q.3 [NAT · 1 mark · no negative marking] Uninformed Search

A uniform tree has branching factor $b = 10$, and the only goal is at depth $d = 5$. Breadth-first search applies the goal test when a node is generated. In the worst case the goal is the last node generated at depth 5. How many nodes are generated in total, not counting the root?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((Q \wedge R) \wedge (P \rightarrow P)) \wedge ((\neg P \leftrightarrow \neg Q) \wedge (Q \leftrightarrow P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

Of the 16 possible binary Boolean connectives $\circ$ (functions $\{T, F\}^2 \rightarrow \{T, F\}$), how many are commutative, i.e. satisfy $p \circ q \equiv q \circ p$?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

Q.8 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following expresses 'None of my friends are perfect' (F(x): x is my friend, P(x): x is perfect)?

- ☐ (A) $\forall x (F(x) \rightarrow \neg P(x))$
☐ (B) $\exists x (F(x) \wedge \neg P(x))$
☐ (C) $\neg \forall x (F(x) \rightarrow P(x))$
☐ (D) $\forall x (\neg F(x) \rightarrow P(x))$

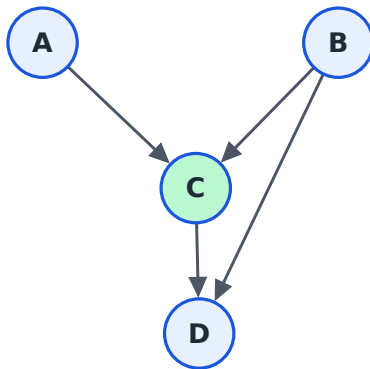
Q.9 [NAT · 1 mark · no negative marking] Sampling

To estimate $P(\text{Rain} \mid \text{sprinkler})$ by rejection sampling, 500 samples are generated by prior sampling. 140 of them have Sprinkler = true, and of these 35 have Rain = true. What is the rejection-sampling estimate of $P(\text{rain} \mid \text{sprinkler})$? (Round to 2 decimal places.)

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



P(A=T)
0.95

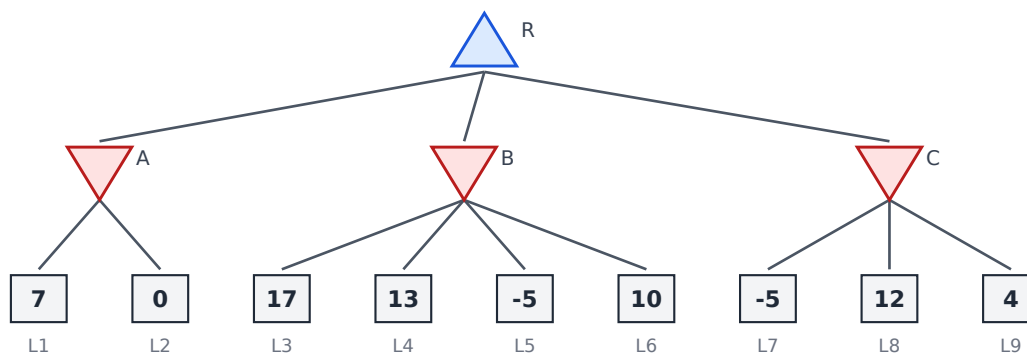
P(B=T)	A	B	P(C=T ...)
0.3	T	T	0.25
	T	F	0.25
	F	T	0.2
	F	F	0.2

B	C	P(D=T ...)
T	T	0.25
T	F	0.3
F	T	0.7
F	F	0.1

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



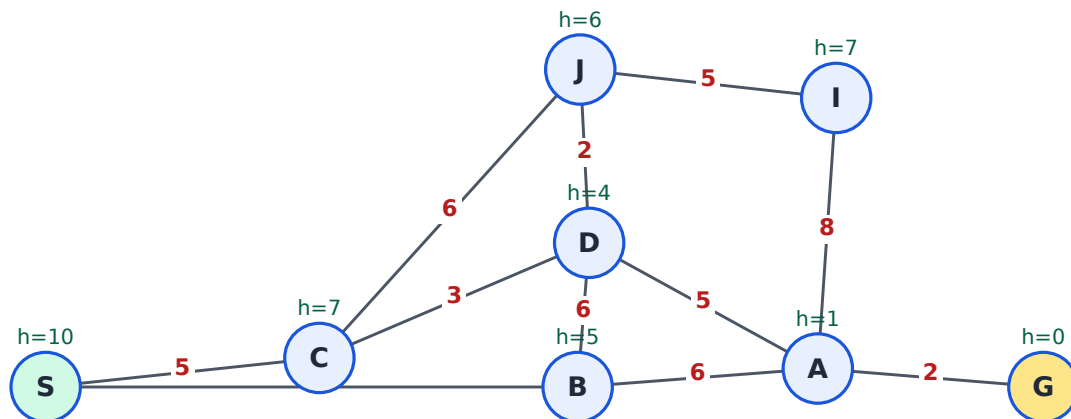
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

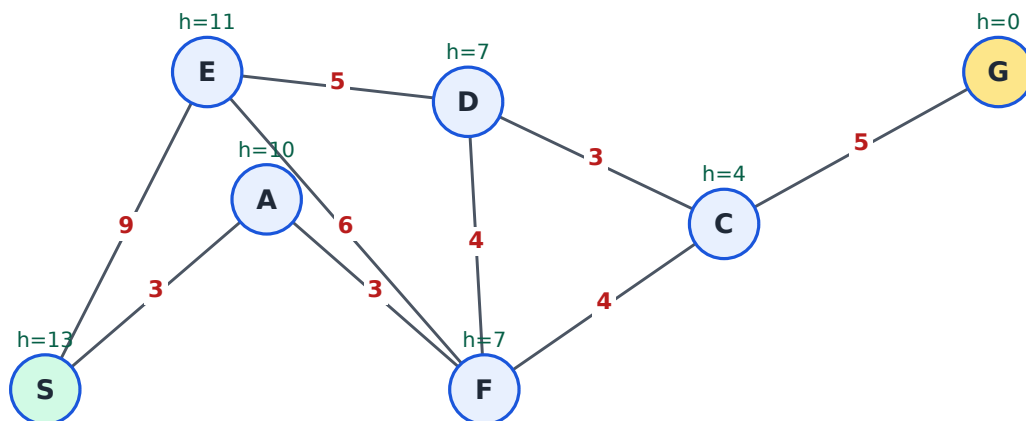
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both **S** and **G**?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge F-C violates the consistency condition (in at least one direction).
- ☐ (B) h is admissible.
- ☐ (C) h is consistent (monotone).
- ☐ (D) $h(C)$ overestimates the true cost from **C** to **G**.

Q.14 [MCQ · 2 marks · negative: -2/3] Problem Formulation

For 8-queens, the complete-state formulation places exactly one queen in each column, in any row, with attacks allowed. How many states does this formulation have?

- ☐ (A) $8^8 = 16,777,216$
- ☐ (B) 64^8
- ☐ (C) $C(64, 8) = 4,426,165,368$
- ☐ (D) $8! = 40,320$

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (P \leftrightarrow \neg P) \rightarrow \neg Q$?

- ☐ (A) $\neg\neg Q \rightarrow \neg(P \leftrightarrow \neg P)$
☐ (B) $\neg(P \leftrightarrow \neg P) \vee \neg Q$
☐ (C) $\neg Q \rightarrow (P \leftrightarrow \neg P)$
☐ (D) $\neg((P \leftrightarrow \neg P) \wedge \neg\neg Q)$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(\neg Q \wedge \neg R) \vee \neg P$
 (2) $(S \vee P) \wedge (Q \vee Q)$
 (3) $\neg(\neg Q \leftrightarrow Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg P \leftrightarrow R$
☐ (B) $P \rightarrow \neg P$
☐ (C) $\neg P$
☐ (D) $\neg Q$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following formulas are contingent (satisfiable but not valid)?

- ☐ (A) $(p \rightarrow q) \wedge (p \rightarrow \neg q)$
☐ (B) $(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$
☐ (C) $(p \leftrightarrow q) \wedge (p \oplus q)$
☐ (D) $(p \vee q) \rightarrow (p \wedge \neg q)$

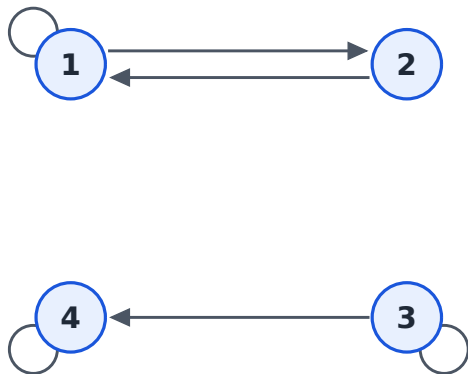
Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following is a prenex normal form equivalent to $(\forall x P(x)) \rightarrow (\exists y Q(y))$?

- ☐ (A) $\exists x \exists y (P(x) \rightarrow Q(y))$
☐ (B) $\forall x \exists y (P(x) \rightarrow Q(y))$
☐ (C) $\forall x \forall y (P(x) \rightarrow Q(y))$
☐ (D) $\exists y \forall x (P(x) \rightarrow Q(y))$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

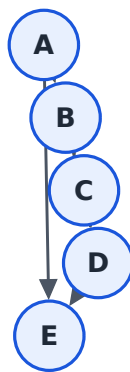
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x R(x, x)$
☐ (B) $\exists x \forall y \neg R(y, x)$
☐ (C) $\forall x \exists y R(x, y)$
☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.20 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.29, 0.1, 0.38, 0.02, 0.96)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.7

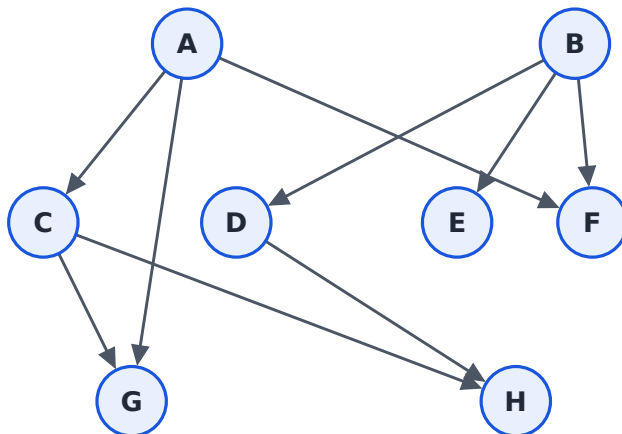
A	$P(B=T \dots)$	B	$P(C=T \dots)$
T	0.25	T	0.75
F	0.25	F	0.95

B	C	$P(D=T \dots)$	A	D	$P(E=T \dots)$
T	T	0.25	T	T	0.25
T	F	0.4	T	F	0.25
F	T	0.9	F	T	0.4
F	F	0.95	F	F	0.6

- ☐ (A) $A=T, B=T, C=T, D=T, E=F$
☐ (B) $A=F, B=T, C=T, D=T, E=F$
☐ (C) $A=F, B=F, C=F, D=F, E=T$
☐ (D) $A=T, B=F, C=T, D=T, E=F$

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp B \mid E$
☐ (B) $F \perp C \mid E$
☐ (C) $C \perp B \mid D$
☐ (D) $F \perp D$

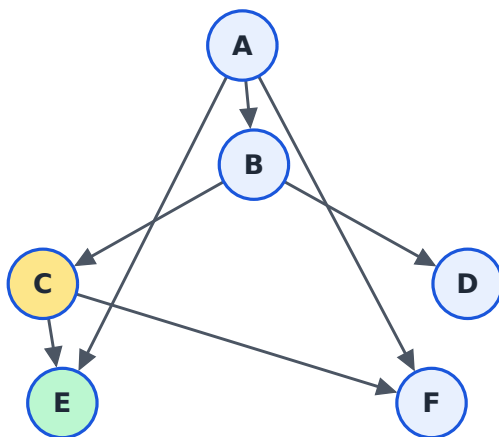
Q.22 [MCQ · 2 marks · negative: -2/3] Conditional Independence

Consider the Bayesian network (DAG) with edges $P \rightarrow Q$, $P \rightarrow R$, $Q \rightarrow S$, $R \rightarrow S$, $S \rightarrow T$, $U \rightarrow R$. Which pair of variables is **NOT** d-separated by the given set?

- ☐ (A) T and P given {Q, R}
☐ (B) U and Q given {P}
☐ (C) U and P given {T}
☐ (D) Q and R given {P}

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

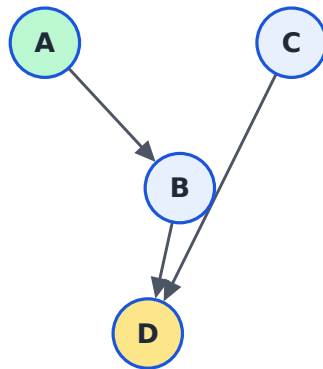
Variable elimination is used to compute $P(E \mid C = T)$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F**, **B**, **D**, **A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(A = T \mid D = F)$ (round off to 3 decimal places).



$P(A=T)$
0.7

A	$P(B=T \dots)$	$P(C=T)$
T	0.8	0.95
F	0.25	

B	C	$P(D=T \dots)$
T	T	0.5
T	F	0.2
F	T	0.6
F	F	0.2

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid B = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	T	F	F	F
2	T	T	T	T
3	F	F	F	T
4	T	T	T	F
5	F	F	F	T
6	F	T	F	T
7	F	T	T	F
8	T	T	T	T
9	F	F	F	T
10	F	F	F	T
11	F	T	F	F
12	F	F	F	T
13	F	T	F	F
14	F	F	F	T
15	F	T	F	F

Answer (numerical): _____

Answer Key — Mock Test 035

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	MCQ	2	Problem Formulation	A
2	NAT	1	Adversarial Search	35	15	MSQ	2	Propositional Logic	A, B, D
3	NAT	1	Uninformed Search	111110	16	MSQ	2	Propositional Logic	B, C
4	NAT	1	Adversarial Search	5	17	MSQ	2	Propositional Logic	A, D
5	NAT	1	Propositional Logic	7	18	MCQ	2	Predicate Logic	A
6	NAT	1	Propositional Logic	8	19	MSQ	2	Predicate Logic	A, C
7	NAT	1	Predicate Logic	64	20	MCQ	2	Sampling	A
8	MCQ	1	Predicate Logic	A	21	MSQ	2	Conditional Independence	C
9	NAT	1	Sampling	0.245 to 0.255	22	MCQ	2	Conditional Independence	C
10	NAT	1	Exact Inference	0.245 to 0.249	23	NAT	2	Exact Inference	2
11	NAT	2	Adversarial Search	6	24	NAT	2	Exact Inference	0.721 to 0.725
12	NAT	2	Informed Search	6	25	NAT	2	Sampling	0.36 to 0.39
13	MSQ	2	Informed Search	B, C					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: D (D: S, H, F, B, J, A, G)

Adjacency lists (alphabetical): A: {G, J}; B: {F, J}; F: {B, G, H, J}; G: {A, F}; H: {F, J, S}; J: {A, B, F, H}; S: {H}. DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, H, F, B, J, A, G**; the path found is $S \rightarrow H \rightarrow F \rightarrow B \rightarrow J \rightarrow A \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] Answer: 35

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

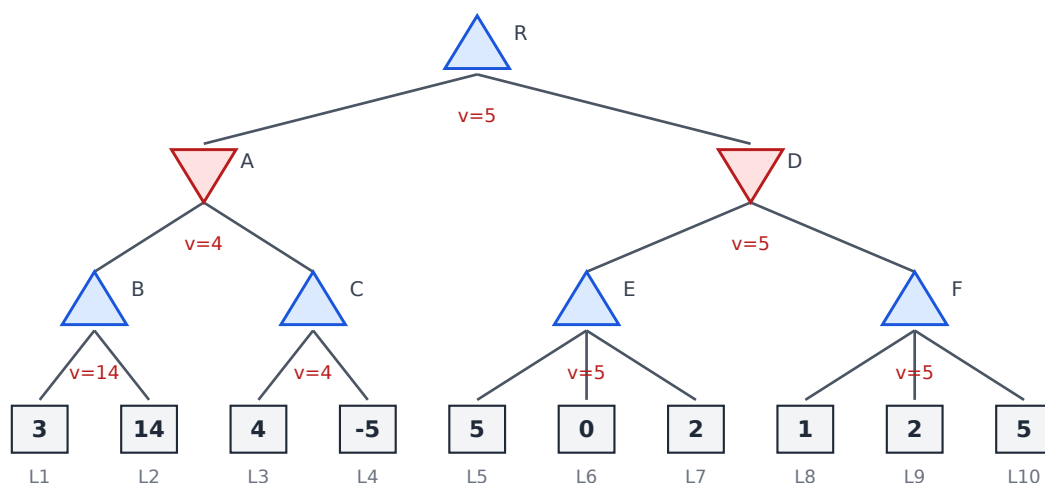
$= 3^3 + 3^2 - 1 = 27 + 9 - 1 = \mathbf{35}$, versus 243 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [NAT, 1M · Uninformed Search] Answer: 111110

With the goal test at generation, BFS stops as soon as the goal is generated, so it generates at most every node down to depth d. Nodes generated $= b + b^2 + b^3 + b^4 + b^5 = 10 + 100 + 1000 + 10000 + 100000 = \mathbf{111110}$. This is the AIMA figure behind BFS's $O(b^d)$ time bound.

Q.4 [NAT, 1M · Adversarial Search] Answer: 5

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 14, C = 4, A = 4, E = 5, F = 5, D = 5. Root value = **5**; MAX's optimal first move is towards D.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 7**

Build the truth table (sub-formulas shown first):

P	Q	R	$(Q \wedge R) \wedge (P \rightarrow P)$	$(\neg P \leftrightarrow \neg Q) \wedge (Q \leftrightarrow P)$	$((Q \wedge R) \wedge (P \rightarrow P)) \wedge ((\neg P \leftrightarrow \neg Q) \wedge (Q \leftrightarrow P))$
T	T	T	T	T	T
T	T	F	F	T	F
T	F	T	F	F	F
T	F	F	F	F	F
F	T	T	T	F	F
F	T	F	F	F	F
F	F	T	F	T	F
F	F	F	F	T	F

F has 1 models, so it is false in $8 - 1 = 7$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 8**

A binary connective is fixed by its outputs on TT, TF, FT, FF. Commutativity requires $\text{output}(TF) = \text{output}(FT)$. So we may freely choose the outputs on TT, FF and the common value on TF/FT: $2^3 = 8$. Examples: $\wedge$, $\vee$, $\leftrightarrow$, $\oplus$, NAND, NOR, constant T, constant F.

Q.7 [NAT, 1M · Predicate Logic] **Answer: 64**

View R as an 3×3 0/1 matrix. The sentence says R is symmetric: choose freely the n diagonal pairs and one decision per unordered pair $\{x, y\} \Rightarrow 2^{n(n+1)/2}$. With $n = 3$: **64**.

Q.8 [MCQ, 1M · Predicate Logic] **Answer: A** (A: $\forall x (F(x) \rightarrow \neg P(x))$)

'None of my friends are perfect' $\equiv \neg \exists x (F(x) \wedge P(x)) \equiv \forall x \neg (F(x) \wedge P(x)) \equiv \forall x (F(x) \rightarrow \neg P(x))$. Options B and C (which are equivalent to each other) only say some friend is not perfect. Option D speaks about non-friends.

Q.9 [NAT, 1M · Sampling] **Answer: 0.245 to 0.255**

Rejection sampling discards the $500 - 140 = 360$ samples inconsistent with the evidence. Among the 140 retained samples, the fraction with Rain = true is $35/140 = 0.25$. Dividing by 500 would be wrong because it estimates $P(\text{rain}, \text{sprinkler})$, not the conditional.

Q.10 [NAT, 1M · Exact Inference] **Answer: 0.245 to 0.249**

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

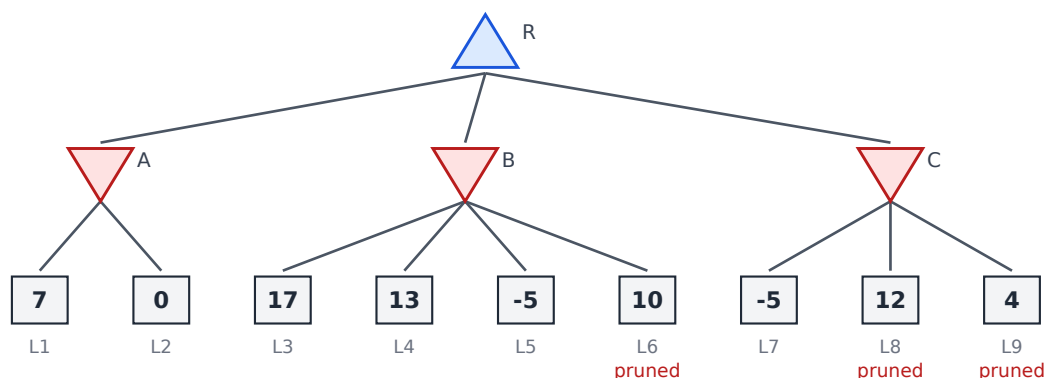
Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T, D=T	0.01781
A=T, B=T, D=F	0.05344
A=T, B=F, D=T	0.11637
A=T, B=F, D=F	0.04988
A=F, B=T, D=T	0.00075
A=F, B=T, D=F	0.00225
A=F, B=F, D=T	0.0049
A=F, B=F, D=F	0.0021

Sum = **0.2475** ≈ 0.247 .

Q.11 [NAT, 2M · Adversarial Search] **Answer: 6**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-5 \leq \alpha = 0 \Rightarrow$ prune remaining children (leaves L6).
- At MIN node C: value $-5 \leq \alpha = 0 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (6): L1, L2, L3, L4, L5, L7. Leaves pruned (3): L6, L8, L9. Root value = 0 (same as plain minimax, 0 — pruning never changes the root value).

Q.12 [NAT, 2M · Informed Search] Answer: 6

Heuristic values: $h(A)=1$, $h(B)=5$, $h(C)=7$, $h(D)=4$, $h(G)=0$, $h(I)=7$, $h(J)=6$, $h(S)=10$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (10)	C(12), B(13)
2	C (12)	D(12), B(13), J(17)
3	D (12)	B(13), A(14), J(16)
4	B (13)	A(14), J(16)
5	A (14)	G(15), J(16), I(28)
6	G (15)	J(16), I(28) ← goal, stop

Path returned: **S** → **C** → **D** → **A** → **G** with cost **15**; nodes expanded = **6** (S, C, D, B, A, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.13 [MSQ, 2M · Informed Search] Answer: B, C (B: h is admissible.; C: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	C	D	E	F	G	S
$h(n)$	10	4	7	11	7	0	13
$h^*(n)$	12	5	8	13	9	0	15
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [MCQ, 2M · Problem Formulation] Answer: A ($A: 8^8 = 16,777,216$)

Each of the 8 columns independently holds its queen in one of 8 rows, giving **$8^8 = 16,777,216$** states.

$C(64,8)$ (C) counts any 8 squares without the one-per-column constraint. 64^8 (B) counts ordered placements of distinguishable queens. $8!$ (D) also requires distinct rows.

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, B, D ($A: \neg\neg Q \rightarrow \neg(P \leftrightarrow \neg P)$; $B: \neg(P \leftrightarrow \neg P) \vee \neg Q$; $D: \neg((P \leftrightarrow \neg P) \wedge \neg\neg Q)$)

Compare truth tables with $F = (P \leftrightarrow \neg P) \rightarrow \neg Q$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg\neg Q \rightarrow \neg(P \leftrightarrow \neg P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $\neg(P \leftrightarrow \neg P) \vee \neg Q$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $\neg Q \rightarrow (P \leftrightarrow \neg P)$ — differs at $P=T, Q=F$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(D) $\neg((P \leftrightarrow \neg P) \wedge \neg\neg Q)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, C (B: $P \rightarrow \neg P$; C: $\neg P$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
F	T	T	T	T
F	T	F	T	T

- (A) $\neg P \leftrightarrow R$ is false in the KB-model $P=F, Q=T, R=F, S=T \Rightarrow$ not entailed.
 (B) $P \rightarrow \neg P$ is true in all models of KB $\Rightarrow$ **entailed**.
 (C) $\neg P$ is true in all models of KB $\Rightarrow$ **entailed**.
 (D) $\neg Q$ is false in the KB-model $P=F, Q=T, R=T, S=T \Rightarrow$ not entailed.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(p \rightarrow q) \wedge (p \rightarrow \neg q)$; D: $(p \vee q) \rightarrow (p \wedge \neg q)$)

(A) $\equiv p \rightarrow (q \wedge \neg q) \equiv \neg p$: true at $p=F$, false at $p=T$, so contingent. (B) An implication and its contrapositive are equivalent, so this is valid. (C) $p \leftrightarrow q$ and $p \oplus q$ are negations of each other, so their conjunction is unsatisfiable. (D) True at $p=q=F$ (antecedent false), false at $p=F, q=T$, so contingent.

Q.18 [MCQ, 2M · Predicate Logic] **Answer: A** (A: $\exists x \exists y (P(x) \rightarrow Q(y))$)

A quantifier pulled out of the antecedent of $\rightarrow$ flips: $(\forall x P(x)) \rightarrow \psi \equiv \exists x (P(x) \rightarrow \psi)$. From the consequent it does not flip: $\phi \rightarrow \exists y Q(y) \equiv \exists y (\phi \rightarrow Q(y))$. So we get $\exists x \exists y (P(x) \rightarrow Q(y))$. Check option B on domain $\{1, 2\}$ with $P = \{1\}$, $Q = \emptyset$: the original is true ($\forall x P(x)$ is false), but at $x=1$ no y makes $P(1) \rightarrow Q(y)$ true, so B is false. C and D are false on the same interpretation.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: A, C** (A: $\exists x R(x, x)$; C: $\forall x \exists y R(x, y)$)

$R = \{(1,1), (1,2), (2,1), (3,3), (3,4), (4,4)\}$.

- (A) $\exists x R(x, x)$: **TRUE** — self-loop at 1.
 (B) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.
 (C) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.
 (D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 4 has no edge to a different element.
 Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MCQ, 2M · Sampling] **Answer: A** (A: $A=T, B=T, C=T, D=T, E=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \neg) = 0.7$; $u = 0.29 < 0.7 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.25$; $u = 0.1 < 0.25 \Rightarrow B = T$
- C: $P(C=T \mid B=T) = 0.75$; $u = 0.38 < 0.75 \Rightarrow C = T$
- D: $P(D=T \mid B=T, C=T) = 0.25$; $u = 0.02 < 0.25 \Rightarrow D = T$
- E: $P(E=T \mid A=T, D=T) = 0.25$; $u = 0.96 \geq 0.25 \Rightarrow E = F$

Sample: **(A=T, B=T, C=T, D=T, E=F)**.

Q.21 [MSQ, 2M · Conditional Independence] **Answer: C** (C: $C \perp B \mid D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

- (A) $D \perp B \mid E$: **FALSE** — the path $D \leftarrow B$ is active given $\{E\}$.
 (B) $F \perp C \mid E$: **FALSE** — the path $F \leftarrow A \rightarrow C$ is active given $\{E\}$.
 (C) $C \perp B \mid D$: **TRUE** — $C \leftarrow A \rightarrow F \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow G \leftarrow A \rightarrow F \leftarrow B$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow H \leftarrow D \leftarrow B$ (blocked: collider H with neither it nor a descendant observed).
 (D) $F \perp D$: **FALSE** — the path $F \leftarrow B \rightarrow D$ is active given $\{\}$.

Q.22 [MCQ, 2M · Conditional Independence] **Answer: C** (C: U and P given $\{T\}$)

(C): T is a descendant of collider R on the path $U \rightarrow R \leftarrow P$ ($R \rightarrow S \rightarrow T$), so observing T activates it: not d-separated. (A): every path from T to P goes through Q or R as non-colliders ($T \leftarrow S \leftarrow Q \leftarrow P$, $T \leftarrow S \leftarrow R \leftarrow P$), both observed: blocked. (B): $U \rightarrow R \leftarrow P \rightarrow Q$ is blocked at observed P and at unobserved collider R; $U \rightarrow R \rightarrow S \leftarrow Q$ is blocked at unobserved collider S. (D): fork P is observed and collider S is unobserved with no observed descendants: blocked.

Q.23 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(B), f(B, D), f(A, E), f(A, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(A, F)$	$\{A, F\}$	$f(A)$
B	$f(A, B), f(B), f(B, D)$	$\{A, B, D\}$	$f(A, D)$
D	$f(A, D)$	$\{A, D\}$	$f(A)$
A	$f(A), f(A, E), f(A), f(A)$	$\{A, E\}$	$f(E)$

The largest factor produced has **2** variable(s) (created when eliminating B); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over E and normalise.

Q.24 [NAT, 2M · Exact Inference] Answer: 0.721 to 0.725

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B | A) \cdot P(C) \cdot P(D | B, C)$.

Inference by enumeration: $P(A | e) = \alpha \sum_{\text{hidden}} P(A, e, \text{hidden})$.

Hidden: B, C	A	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.7 \times 0.8 \times 0.95 \times 0.5$	0.266
T, F	T	$0.7 \times 0.8 \times 0.05 \times 0.8$	0.0224
F, T	T	$0.7 \times 0.2 \times 0.95 \times 0.4$	0.0532
F, F	T	$0.7 \times 0.2 \times 0.05 \times 0.8$	0.0056
T, T	F	$0.3 \times 0.25 \times 0.95 \times 0.5$	0.03563
T, F	F	$0.3 \times 0.25 \times 0.05 \times 0.8$	0.003
F, T	F	$0.3 \times 0.75 \times 0.95 \times 0.4$	0.0855
F, F	F	$0.3 \times 0.75 \times 0.05 \times 0.8$	0.009

$P(A=T, e) = 0.3472$; $P(A=F, e) = 0.13313$; $P(e) = 0.48033$.

$P(A=T | e) = 0.3472 / 0.48033 = \mathbf{0.7228} \approx 0.723$.

Q.25 [NAT, 2M · Sampling] Answer: 0.36 to 0.39

Rejection sampling keeps only samples consistent with the evidence $B = T$: samples 2, 4, 6, 7, 8, 11, 13, 15 (8 accepted, 7 rejected).

Among the accepted samples, $D = T$ in 3. Estimate = $3/8 = \mathbf{0.375}$.

(For this network the exact value is 0.4945; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Mock Test 036

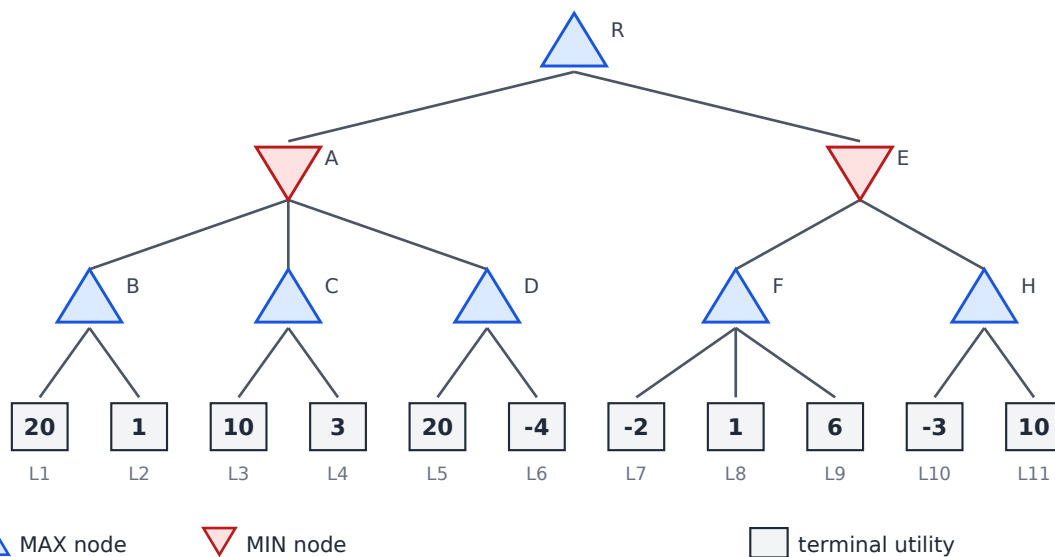
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 6 / 12	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

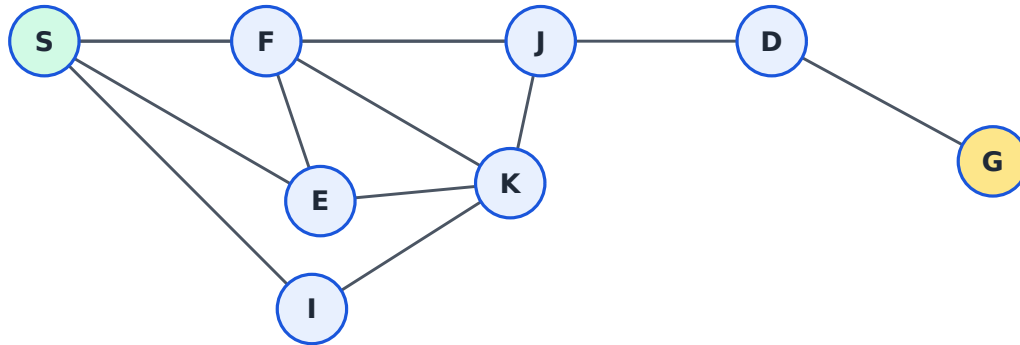
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 5$ (so $b^d = 32$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, E, F, J, I, K, D, G
- ☐ (B) S, J, I, F, E, K, D, G
- ☐ (C) S, E, F, I, J, K, D, G
- ☐ (D) S, E, F, J, D, G

Q.4 [MCQ · 1 mark · negative: -1/3] Informed Search

Greedy best-first search orders its frontier by

- ☐ (A) $g(n)$
- ☐ (B) $h(n)$
- ☐ (C) $g(n) + h(n)$
- ☐ (D) $\text{depth}(n)$

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic

Interpret $\text{Div}(x, y)$ as 'x divides y' over the domain $D = \{1, 2, \dots, 12\}$. For how many $x \in D$ is the formula $\neg(x = 1) \wedge \forall y (\text{Div}(y, x) \rightarrow (y = 1 \vee y = x))$ true?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

The formula $(A \wedge B) \vee (C \wedge D) \vee (E \wedge F)$ is converted to CNF by repeatedly distributing $\vee$ over $\wedge$ (no simplification). How many clauses does the result have?

Answer (numerical): _____

Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

In the DPLL algorithm, a pure symbol is one that appears with the same sign in every clause in which it occurs. For the clause set $\{A \vee \neg B, \neg B \vee \neg C, C \vee A\}$, which symbols are pure?

- ☐ (A) A
- ☐ (B) B
- ☐ (C) C
- ☐ (D) None of the symbols is pure

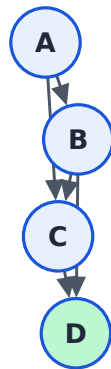
Q.8 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x R(x, x) \wedge \forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(D = T)$ (round off to 3 decimal places).



$P(A=T)$
0.8

A	$P(B=T \mid \cdot)$	A	B	$P(C=T \mid \dots)$
T	0.6	T	T	0.75
F	0.6	T	F	0.4
		F	T	0.7
		F	F	0.5

B	C	$P(D=T \mid \dots)$
T	T	0.4
T	F	0.95
F	T	0.9
F	F	0.3

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Probability for AI

While answering a query $P(\text{Cavity} \mid \text{toothache})$, the unnormalized vector obtained for $\text{Cavity} = \langle \text{true}, \text{false} \rangle$ is $\langle 0.12, 0.08 \rangle$. Which option gives the normalization constant α and the resulting posterior distribution?

- ☐ (A) $\alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.6, 0.4 \rangle$
- ☐ (B) $\alpha = 0.2$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.024, 0.016 \rangle$
- ☐ (C) $\alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.4, 0.6 \rangle$
- ☐ (D) $\alpha = 2$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.24, 0.16 \rangle$

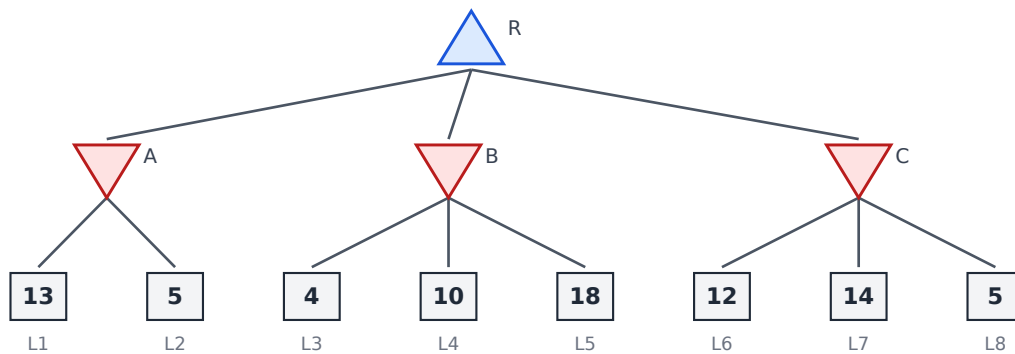
Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Problem Formulation

Which of the following statements match standard (AIMA) definitions of the terms used to analyse search algorithms?

- ☐ (A) Completeness: the algorithm is guaranteed to find a solution when one exists
- ☐ (B) Cost-optimality: the algorithm finds a solution with the lowest path cost among all solutions
- ☐ (C) b , the branching factor, is the maximum number of successors of any node
- ☐ (D) d is the depth of the deepest node in the state space

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



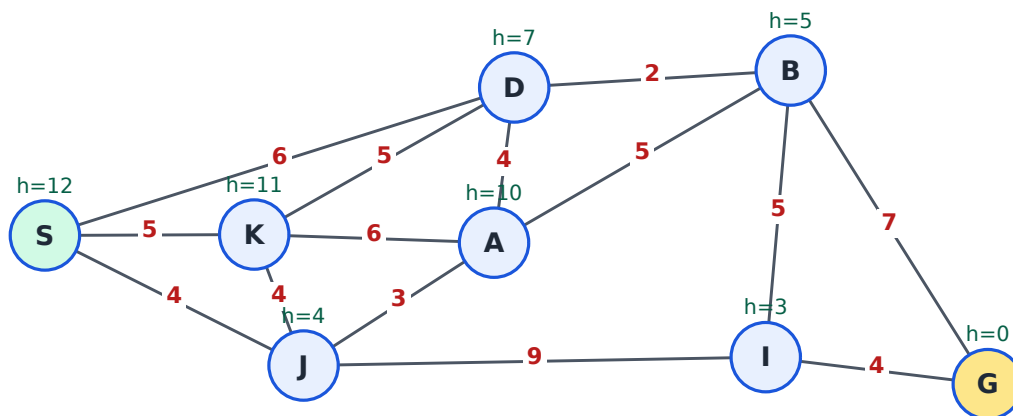
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

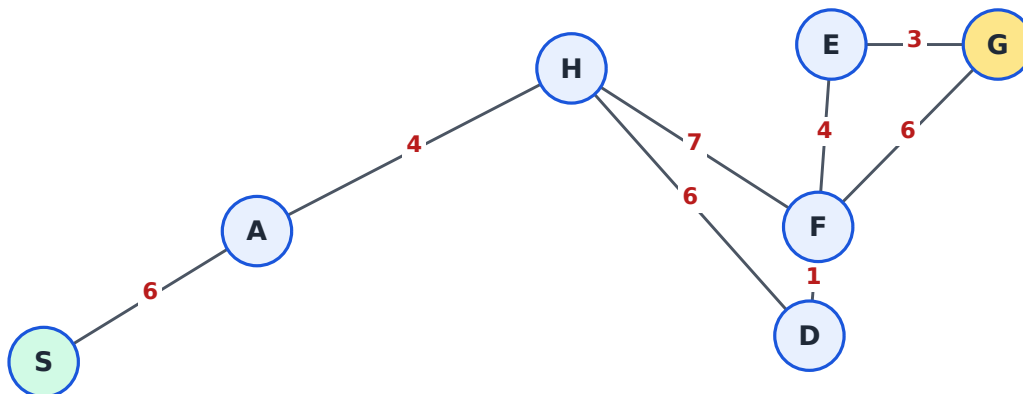
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) The edge K-J violates the consistency condition $h(K) \leq c(K,J) + h(J)$.
- ☐ (C) $h(B)$ overestimates the true cost from B to G.
- ☐ (D) h is admissible.

Q.14 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow A \rightarrow H \rightarrow F \rightarrow E \rightarrow G$
- ☐ (B) $S \rightarrow A \rightarrow H \rightarrow F \rightarrow G$
- ☐ (C) $S \rightarrow A \rightarrow H \rightarrow D \rightarrow F \rightarrow G$
- ☐ (D) $S \rightarrow A \rightarrow H \rightarrow D \rightarrow F \rightarrow E \rightarrow G$

Q.15 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following statements about the computational complexity of propositional reasoning is correct (assuming $P \neq NP$)?

- ☐ (A) Satisfiability of 3-CNF formulas can be decided in polynomial time
- ☐ (B) Deciding whether a propositional formula is valid is co-NP-complete
- ☐ (C) Satisfiability of Horn formulas is NP-complete
- ☐ (D) Satisfiability of 2-CNF formulas is NP-complete

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

The formula $F = ((R \leftrightarrow S) \vee (Q \leftrightarrow \neg Q)) \wedge ((S \vee \neg Q) \wedge \neg P)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R, S . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \rightarrow P) \wedge \neg Q) \rightarrow \neg P$
- ☐ (B) $(R \vee Q) \rightarrow (R \wedge Q)$
- ☐ (C) $(R \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg R)$
- ☐ (D) $((Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow (Q \rightarrow P)$

Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Skolemizing $\exists x \forall y \exists z R(x, y, z)$ gives (A is a new constant, F, G new function symbols)

- ☐ (A) $\forall y R(A, y, F(y))$
- ☐ (B) $\forall y R(A, y, F(A))$
- ☐ (C) $\forall y R(G(y), y, F(y))$
- ☐ (D) $\forall y R(A, y, B)$, with B a new constant

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

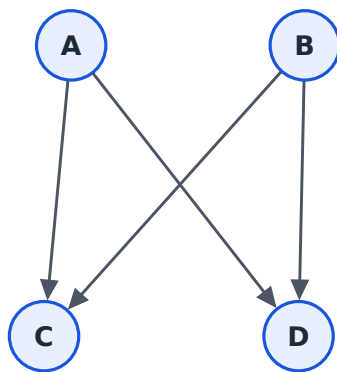
- (1) $(Q \vee P) \wedge (P \leftrightarrow \neg Q)$
- (2) $\neg P \leftrightarrow \neg \neg Q$
- (3) $\neg P \vee (\neg Q \rightarrow Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg(Q \wedge \neg Q)$
- ☐ (B) $(\neg Q \wedge R) \wedge \neg Q$
- ☐ (C) $\neg(\neg Q \leftrightarrow R)$
- ☐ (D) $\neg \neg Q$

Q.20 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.07}, \mathbf{0.41}, \mathbf{0.08}, \mathbf{0.45} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.2

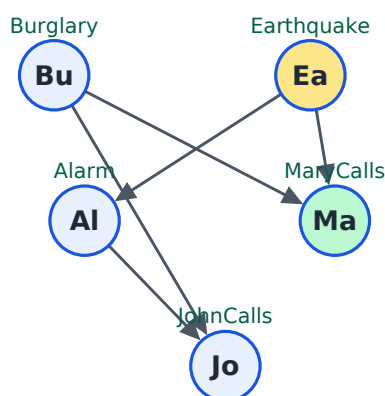
P(B=T)	A	B	P(C=T ...)
0.05	T	T	0.75
	T	F	0.4
	F	T	0.1
	F	F	0.9

A	B	P(D=T ...)
T	T	0.7
T	F	0.95
F	T	0.2
F	F	0.5

- ☐ (A) A=T, B=F, C=T, D=T
- ☐ (B) A=F, B=T, C=F, D=T
- ☐ (C) A=T, B=T, C=T, D=T
- ☐ (D) A=F, B=F, C=T, D=T

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls, Ma = MaryCalls.) Compute $\mathbf{P(\text{MaryCalls} = T \mid \text{Earthquake} = T)}$ (round off to 3 decimal places).



P(Bu=T)
0.75

P(Ea=T)
0.9

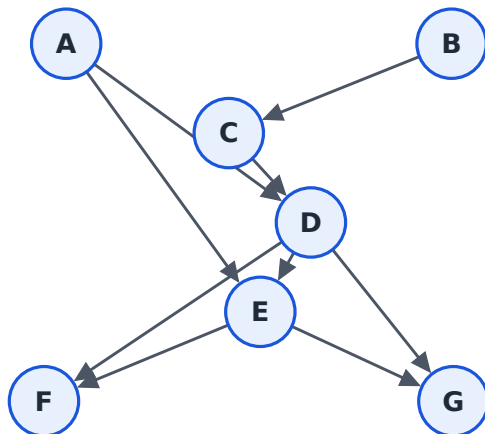
Ea	P(Al=T ...)
T	0.1
F	0.7

Bu	Al	P(Jo=T ...)	Bu	Ea	P(Ma=T ...)
T	T	0.95	T	T	0.25
T	F	0.95	T	F	0.7
F	T	0.8	F	T	0.7
F	F	0.8	F	F	0.6

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

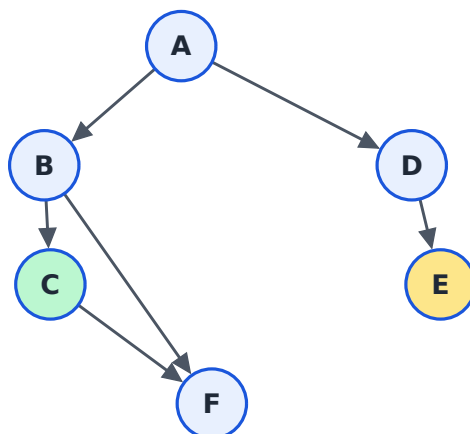
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp A \mid E, G$
☐ (B) $B \perp G \mid C$
☐ (C) $E \perp F \mid D$
☐ (D) $B \perp C \mid E, G$

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(C \mid E = T)$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, D, B, F**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid B = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	T	T	F
2	F	F	F	T
3	F	T	F	T
4	T	F	T	T
5	F	F	F	T
6	F	F	F	T
7	F	T	F	T
8	F	T	F	T
9	F	T	F	T
10	T	T	F	T
11	F	F	F	T
12	F	T	T	F
13	F	T	F	F
14	F	T	F	T
15	T	T	F	T
16	F	F	F	T
17	F	F	F	T
18	F	F	F	T
19	F	T	F	T
20	F	T	T	T

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Probability for AI

A full joint distribution over Boolean variables X, Y, Z is: $P(x, y, z) = 0.10$, $P(x, y, \neg z) = 0.05$, $P(x, \neg y, z) = 0.15$, $P(x, \neg y, \neg z) = 0.10$, $P(\neg x, y, z) = 0.05$, $P(\neg x, y, \neg z) = 0.20$, $P(\neg x, \neg y, z) = 0.15$, $P(\neg x, \neg y, \neg z) = 0.20$. Compute $P(x \mid y \vee z)$. (Round to 3 decimal places.)

Answer (numerical): _____

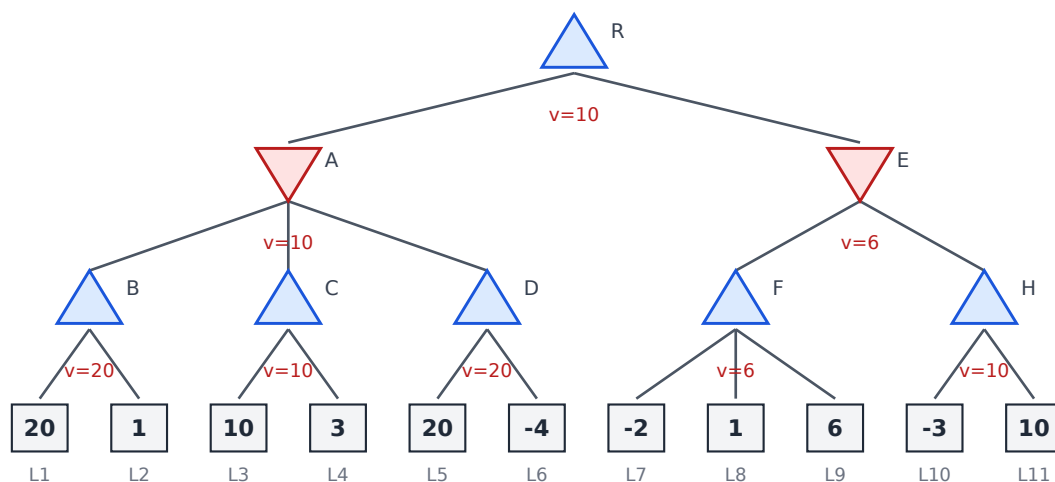
Answer Key — Mock Test 036

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	10	14	MCQ	2	Uninformed Search	B
2	NAT	1	Adversarial Search	11	15	MCQ	2	Propositional Logic	B
3	MCQ	1	Uninformed Search	C	16	NAT	2	Propositional Logic	13
4	MCQ	1	Informed Search	B	17	MSQ	2	Propositional Logic	C, D
5	NAT	1	Predicate Logic	5	18	MCQ	2	Predicate Logic	A
6	NAT	1	Propositional Logic	8	19	MSQ	2	Propositional Logic	A, D
7	MSQ	1	Propositional Logic	A, B	20	MCQ	2	Sampling	A
8	NAT	1	Predicate Logic	8	21	NAT	2	Exact Inference	0.361 to 0.365
9	NAT	1	Exact Inference	0.545 to 0.549	22	MSQ	2	Conditional Independence	B
10	MCQ	1	Probability for AI	A	23	NAT	2	Exact Inference	2
11	MSQ	2	Problem Formulation	A, B, C	24	NAT	2	Sampling	0.74 to 0.76
12	NAT	2	Adversarial Search	2	25	NAT	2	Probability for AI	0.428 to 0.429
13	MSQ	2	Informed Search	B, D					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 10

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 20, C = 10, D = 20, A = 10, F = 6, H = 10, E = 6. Root value = **10**; MAX's optimal first move is towards A.

Q.2 [NAT, 1M · Adversarial Search] Answer: 11

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 2^3 + 2^2 - 1 = 8 + 4 - 1 = \mathbf{11}$, versus 32 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [MCQ, 1M · Uninformed Search] Answer: C (C: S, E, F, I, J, K, D, G)

Adjacency lists (alphabetical): D: {G, J}; E: {F, K, S}; F: {E, J, K, S}; G: {D}; I: {K, S}; J: {D, F, K, S}; K: {E, F, I, J}; S: {E, F, I, J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[E, F, I, J]
2	E	[F, I, J, K]
3	F	[I, J, K]
4	I	[J, K]
5	J	[K, D]
6	K	[D]
7	D	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, E, F, I, J, K, D, G**. The path returned is S → J → D → G with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [MCQ, 1M · Informed Search] Answer: B (B: $h(n)$)

Greedy best-first search always expands the node that looks closest to the goal, so it uses $f(n) = \mathbf{h(n)}$. $g(n)$ gives UCS (A). $g + h$ gives A* (C). Depth gives BFS when the shallowest node is expanded first, or DFS when the deepest is (D).

Q.5 [NAT, 1M · Predicate Logic] Answer: 5

The formula says $x \neq 1$ and the only divisors of x in D are 1 and x itself — i.e. x is prime. (Every divisor of $x \leq 12$ lies in D , so the restriction to D loses nothing.) The primes in $\{1, \dots, 12\}$ are 2, 3, 5, 7, 11, so the answer is 5.

Q.6 [NAT, 1M · Propositional Logic] Answer: 8

Distribution produces one clause per way of choosing one literal from each conjunction: $\{A, B\} \times \{C, D\} \times \{E, F\}$, i.e. $2 \times 2 \times 2 = 8$ clauses such as $A \vee C \vee E$, $A \vee C \vee F$, ..., $B \vee D \vee F$. No clause is a tautology or a duplicate because all six symbols are distinct. This illustrates the exponential blow-up of naive CNF conversion.

Q.7 [MSQ, 1M · Propositional Logic] Answer: A, B (A: A; B: B)

A occurs only positively (in $A \vee \neg B$ and $C \vee A$), so it is pure; DPLL can set $A = T$. B occurs only negatively ($\neg B$ twice), so it is pure; DPLL can set $B = F$. C occurs negatively in $\neg B \vee \neg C$ and positively in $C \vee A$, so it is not pure.

Q.8 [NAT, 1M · Predicate Logic] Answer: 8

View R as an 3×3 0/1 matrix. The sentence says R is reflexive and symmetric: only the $n(n-1)/2$ unordered off-diagonal pairs are free $\Rightarrow 2^{(n(n-1)/2)}$. With $n = 3$: **8**.

Q.9 [NAT, 1M · Exact Inference] Answer: 0.545 to 0.549

Only the ancestors of D (A, B, C) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
A=T, B=T, C=T	0.144
A=T, B=T, C=F	0.114
A=T, B=F, C=T	0.1152
A=T, B=F, C=F	0.0576
A=F, B=T, C=T	0.0336
A=F, B=T, C=F	0.0342
A=F, B=F, C=T	0.036
A=F, B=F, C=F	0.012

Sum = **0.5466** ≈ 0.547 .

Q.10 [MCQ, 1M · Probability for AI] Answer: A (A: $\alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.6, 0.4 \rangle$)

The entries must sum to 1 after scaling, so $\alpha = 1/(0.12 + 0.08) = 1/0.20 = 5$. Then $\alpha \langle 0.12, 0.08 \rangle = \langle 0.6, 0.4 \rangle$. $\alpha = 0.2$ is the sum itself (multiplying by it does not normalize), $\langle 0.4, 0.6 \rangle$ swaps the order of the values, and $\alpha = 2$ gives entries summing to 0.4, not 1.

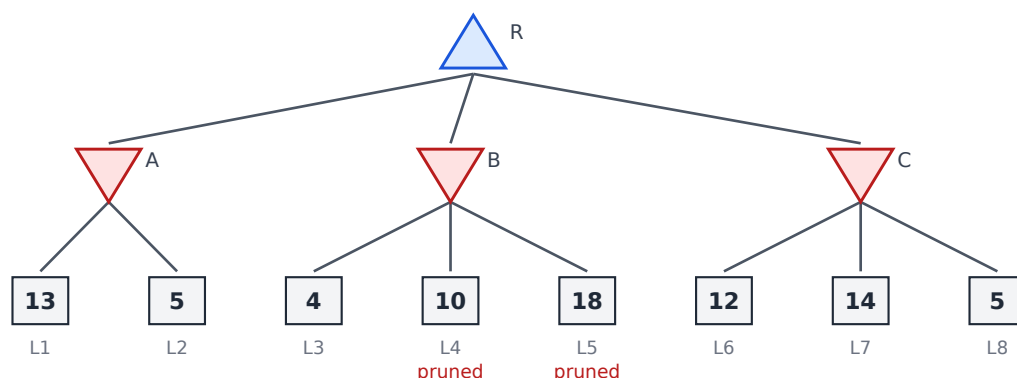
Q.11 [MSQ, 2M · Problem Formulation] Answer: A, B, C

(A), (B) and (C) are the textbook definitions. (D) is false: d is the depth of the **shallowest** goal node. The maximum length of any path in the state space is m.

Q.12 [NAT, 2M · Adversarial Search] Answer: 2

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $4 \leq \alpha = 5 \Rightarrow$ prune remaining children (leaves L4, L5).



Leaves evaluated (6): L1, L2, L3, L6, L7, L8. Leaves pruned (2): L4, L5. Root value = 5 (same as plain minimax, 5 — pruning never changes the root value).

Q.13 [MSQ, 2M · Informed Search] Answer: B, D (B: The edge K-J violates the consistency condition $h(K) \leq c(K,J) + h(J)$.; D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	D	G	I	J	K	S
$h(n)$	10	5	7	0	3	4	11	12
$h^*(n)$	12	7	9	0	4	13	14	15
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $K \rightarrow J$ ($11 > 4 + 4$), $A \rightarrow J$ ($10 > 3 + 4$), $S \rightarrow J$ ($12 > 4 + 4$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [MCQ, 2M · Uninformed Search] **Answer: B** (B: $S \rightarrow A \rightarrow H \rightarrow F \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	A(6)
2	A (6)	H(10)
3	H (10)	D(16), F(17)
4	D (16)	F(17)
5	F (17)	E(21), G(23)
6	E (21)	G(23)
7	G (23)	$\leftarrow$ goal, stop

Path returned: **S $\rightarrow$ A $\rightarrow$ H $\rightarrow$ F $\rightarrow$ G**, cost = $6 + 4 + 7 + 6 = 23$. Nodes expanded (including S and G): **7** (S, A, H, D, F, E, G).

Q.15 [MCQ, 2M · Propositional Logic] **Answer: B** (B: Deciding whether a propositional formula is valid is co-NP-complete)

α is valid iff $\neg\alpha$ is unsatisfiable, and SAT is NP-complete, so VALIDITY (TAUT) is co-NP-complete. 3-SAT is NP-complete, so not in P under $P \neq NP$. Horn-SAT is decidable in linear time (unit propagation / forward chaining). 2-SAT is decidable in linear time via the implication graph and strongly connected components.

Q.16 [NAT, 2M · Propositional Logic] **Answer: 13**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(R \leftrightarrow S) \vee (Q \leftrightarrow \neg Q)$	$(S \vee \neg Q) \wedge \neg P$	$((R \leftrightarrow S) \vee (Q \leftrightarrow \neg Q)) \wedge ((S \vee \neg Q) \wedge \neg P)$
T	T	T	T	T	F	F
T	T	T	F	F	F	F
T	T	F	T	F	F	F
T	T	F	F	T	F	F
T	F	T	T	T	F	F
T	F	T	F	F	F	F
T	F	F	T	F	F	F
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	T	F	F	F	F
F	T	F	T	F	T	F
F	T	F	F	T	F	F
F	F	T	T	T	T	T
F	F	T	F	F	T	F
F	F	F	T	F	T	F
F	F	F	F	T	T	T

F has 3 models, so it is false in $16 - 3 = 13$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $(R \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg R)$; D: $((Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow (Q \rightarrow P)$)

(A) $((Q \rightarrow P) \wedge \neg Q) \rightarrow \neg P$: **not valid** — counter-model $P=T, Q=F$.

(B) $(R \vee Q) \rightarrow (R \wedge Q)$: **not valid** — counter-model $Q=T, R=F$.

(C) $(R \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow (Q \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MCQ, 2M · Predicate Logic] Answer: A ($A: \forall y R(A, y, F(y))$)

x is existential and not inside any universal quantifier, so it becomes a constant A . z is inside the scope of $\forall y$, so it becomes $F(y)$ (arguments are the enclosing universally quantified variables). $F(A)$ ignores the dependence on y and is an unwarranted strengthening; $G(y)$ for x is wrong because x precedes $\forall y$; a constant for z also loses the dependence on y .

Q.19 [MSQ, 2M · Propositional Logic] Answer: A, D ($A: \neg(Q \wedge \neg Q)$; $D: \neg\neg Q$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
F	T	T	T
F	T	F	T

(A) $\neg(Q \wedge \neg Q)$ is true in all models of $KB \Rightarrow$ **entailed**.

(B) $(\neg Q \wedge R) \wedge \neg Q$ is false in the KB -model $P=F, Q=T, R=T \Rightarrow$ not entailed.

(C) $\neg(\neg Q \leftrightarrow R)$ is false in the KB -model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(D) $\neg\neg Q$ is true in all models of $KB \Rightarrow$ **entailed**.

Q.20 [MCQ, 2M · Sampling] Answer: A ($A: A=T, B=F, C=T, D=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \neg) = 0.2$; $u = 0.07 < 0.2 \Rightarrow A = T$
- B: $P(B=T \mid \neg) = 0.05$; $u = 0.41 \geq 0.05 \Rightarrow B = F$
- C: $P(C=T \mid A=T, B=F) = 0.4$; $u = 0.08 < 0.4 \Rightarrow C = T$
- D: $P(D=T \mid A=T, B=F) = 0.95$; $u = 0.45 < 0.95 \Rightarrow D = T$

Sample: **(A=T, B=F, C=T, D=T)**.

Q.21 [NAT, 2M · Exact Inference] Answer: 0.361 to 0.365

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot P(\text{Alarm} \mid \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Alarm}) \cdot P(\text{MaryCalls} \mid \text{Burglary}, \text{Earthquake})$.

Inference by enumeration: $P(\text{MaryCalls} \mid e) = \alpha \sum_{\text{hidden}} P(\text{MaryCalls}, e, \text{hidden})$.

Hidden: Bu, Al, Jo	Ma	product of CPT entries (Bu · Ea · Al · Jo · Ma)	value
T, T, T	T	$0.75 \times 0.9 \times 0.1 \times 0.95 \times 0.25$	0.01603
T, T, F	T	$0.75 \times 0.9 \times 0.1 \times 0.05 \times 0.25$	0.00084
T, F, T	T	$0.75 \times 0.9 \times 0.9 \times 0.95 \times 0.25$	0.14428
T, F, F	T	$0.75 \times 0.9 \times 0.9 \times 0.05 \times 0.25$	0.00759
F, T, T	T	$0.25 \times 0.9 \times 0.1 \times 0.8 \times 0.7$	0.0126
F, T, F	T	$0.25 \times 0.9 \times 0.1 \times 0.2 \times 0.7$	0.00315
F, F, T	T	$0.25 \times 0.9 \times 0.9 \times 0.8 \times 0.7$	0.1134
F, F, F	T	$0.25 \times 0.9 \times 0.9 \times 0.2 \times 0.7$	0.02835
T, T, T	F	$0.75 \times 0.9 \times 0.1 \times 0.95 \times 0.75$	0.04809
T, T, F	F	$0.75 \times 0.9 \times 0.1 \times 0.05 \times 0.75$	0.00253
T, F, T	F	$0.75 \times 0.9 \times 0.9 \times 0.95 \times 0.75$	0.43284
T, F, F	F	$0.75 \times 0.9 \times 0.9 \times 0.05 \times 0.75$	0.02278
F, T, T	F	$0.25 \times 0.9 \times 0.1 \times 0.8 \times 0.3$	0.0054
F, T, F	F	$0.25 \times 0.9 \times 0.1 \times 0.2 \times 0.3$	0.00135
F, F, T	F	$0.25 \times 0.9 \times 0.9 \times 0.8 \times 0.3$	0.0486
F, F, F	F	$0.25 \times 0.9 \times 0.9 \times 0.2 \times 0.3$	0.01215

$P(\text{MaryCalls}=T, e) = 0.32625$; $P(\text{MaryCalls}=F, e) = 0.57375$; $P(e) = 0.9$.

$P(\text{MaryCalls}=T \mid e) = 0.32625 / 0.9 = \mathbf{0.3625} \approx 0.363$.

Q.22 [MSQ, 2M · Conditional Independence] Answer: B ($B: B \perp G \mid C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp A \mid E, G$: **FALSE** — the path $B \rightarrow C \rightarrow D \leftarrow A$ is active given $\{E, G\}$.

(B) $B \perp G \mid C$: **TRUE** — $B \rightarrow C \rightarrow D \leftarrow A \rightarrow E \rightarrow G$ (blocked: observed chain node C); $B \rightarrow C \rightarrow D \rightarrow E \rightarrow G$ (blocked: observed chain node C); $B \rightarrow C \rightarrow D \rightarrow F \leftarrow E \rightarrow G$ (blocked: observed chain node C); $B \rightarrow C \rightarrow D \rightarrow G$ (blocked: observed chain node C).

(C) $E \perp F \mid D$: **FALSE** — the path $E \rightarrow F$ is active given $\{D\}$.

(D) $B \perp C \mid E, G$: **FALSE** — the path $B \rightarrow C$ is active given $\{E, G\}$.

Q.23 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, D)$, $f(D)$, $f(B, C, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, B)$, $f(A, D)$	$\{A, B, D\}$	$f(B, D)$
D	$f(D)$, $f(B, D)$	$\{B, D\}$	$f(B)$
B	$f(B, C)$, $f(B, C, F)$, $f(B)$	$\{B, C, F\}$	$f(C, F)$
F	$f(C, F)$	$\{C, F\}$	$f(C)$

The largest factor produced has **2** variable(s) (created when eliminating A); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over C and normalise.

Q.24 [NAT, 2M · Sampling] **Answer: 0.74 to 0.76**

Rejection sampling keeps only samples consistent with the evidence $B = T$: samples 1, 3, 7, 8, 9, 10, 12, 13, 14, 15, 19, 20 (12 accepted, 8 rejected).

Among the accepted samples, $D = T$ in 9. Estimate = $9/12 = \mathbf{0.75}$.

(For this network the exact value is 0.69; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.25 [NAT, 2M · Probability for AI] **Answer: 0.428 to 0.429**

$P(y \vee z) = 1 - P(\neg y \wedge \neg z) = 1 - [P(x, \neg y, \neg z) + P(\neg x, \neg y, \neg z)] = 1 - (0.10 + 0.20) = 0.70$. $P(x \wedge (y \vee z)) = P(x, y, z) + P(x, y, \neg z) + P(x, \neg y, z) = 0.10 + 0.05 + 0.15 = 0.30$. Therefore $P(x \mid y \vee z) = 0.30 / 0.70 \approx 0.4286 \approx 0.429$.

Mock Test 037

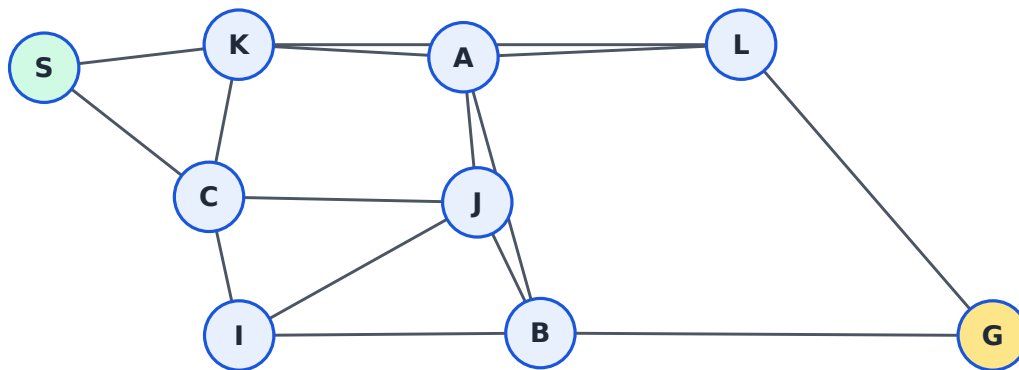
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 9 / 11	13 / 12 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

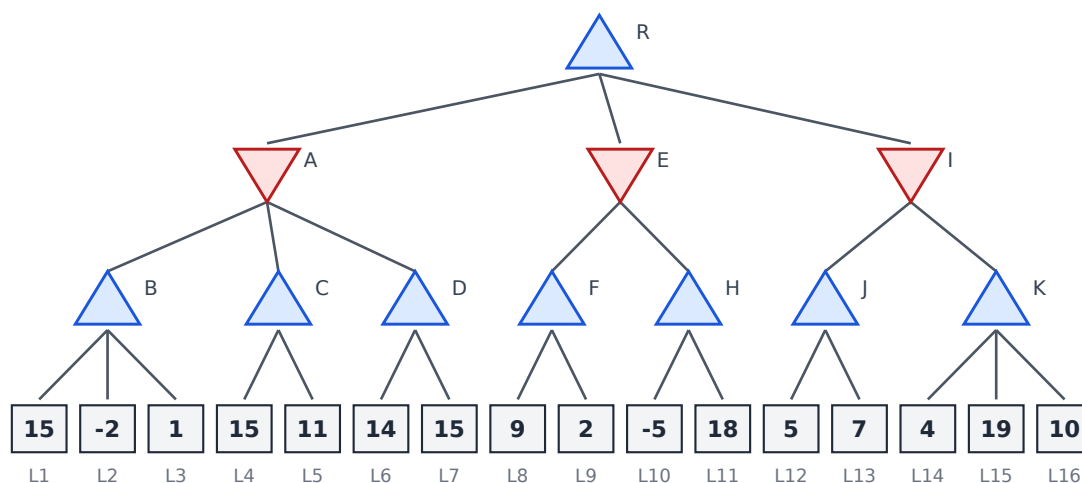
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, C, K, I, J, A, L, B, G
- ☐ (B) S, K, L, G
- ☐ (C) S, C, I, B, A, J, K, L, G
- ☐ (D) S, I, C, B, A, J, K, L, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Iterative deepening search is applied to a tree with branching factor $b = 5$ whose shallowest goal is at depth $d = 4$ (goal is the last node at that depth). Using the standard count $N(IDS) = d \cdot b + (d-1) \cdot b^2 + \dots + 1 \cdot b^d$ (root not counted), how many nodes are generated?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q) . For how many of them is $\forall x P(x) \vee \forall x Q(x)$ true?

Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Without standardizing variables apart, $\text{Unify}(\text{Knows}(\text{John}, x), q)$ succeeds for which of the following atoms q ?

- ☐ (A) $\text{Knows}(\text{John}, \text{Jane})$
☐ (B) $\text{Knows}(y, \text{Bill})$
☐ (C) $\text{Knows}(y, \text{Mother}(y))$
☐ (D) $\text{Knows}(x, \text{Elizabeth})$

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

What is the most general unifier (MGU) of $\text{Knows}(\text{John}, x)$ and $\text{Knows}(y, \text{Mother}(y))$?

- ☐ (A) $\{y/\text{John}, x/\text{Mother}(\text{John})\}$
☐ (B) $\{x/\text{John}, y/\text{Mother}(\text{John})\}$
☐ (C) $\{y/\text{John}, x/\text{Mother}(y)\}$
☐ (D) The two atoms are not unifiable

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

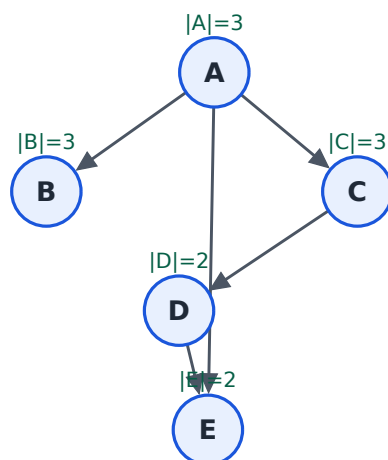
How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = \neg\neg R \leftrightarrow ((\neg R \wedge Q) \wedge P) ?$$

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

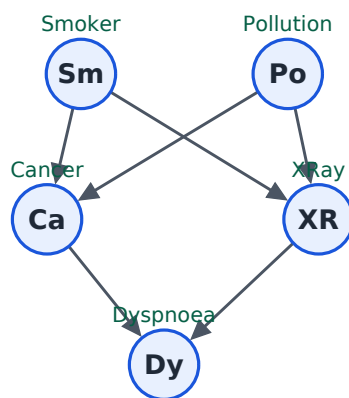
Q.9 [MSQ · 1 mark · no negative marking] Conditional Independence

Which of the following statements about d-separation in three-node structures are correct?

- ☐ (A) In the chain $X \rightarrow Y \rightarrow Z$, X and Z are d-separated given Y
- ☐ (B) In the fork $X \leftarrow Y \rightarrow Z$, X and Z are d-separated given Y
- ☐ (C) In the collider $X \rightarrow Y \leftarrow Z$, X and Z are d-separated given Y
- ☐ (D) In the collider $X \rightarrow Y \leftarrow Z$, X and Z are d-separated given $\emptyset$

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR, Dy . (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay, Dy = Dyspnoea.) What is the probability that a single run produces the sample (**$Sm=F, Po=F, Ca=F, XR=T, Dy=F$**)? (round off to 4 decimal places)



$P(Sm=T)$
0.6

$P(Po=T)$
0.05

Sm	Po	$P(Ca=T ...)$
T	T	0.75
T	F	0.4
F	T	0.75
F	F	0.2

Sm	Po	$P(XR=T Ca, XR)$	$P(Dy=T ...)$
T	T	0.05T	0.95
T	F	0.75T	0.1
F	T	0.8F	0.5
F	F	0.7F	0.8

Answer (numerical): _____

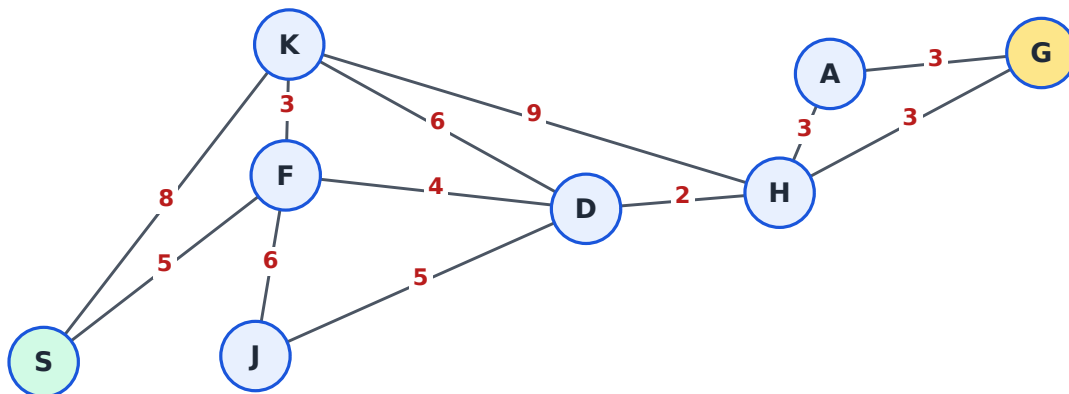
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search

A three-ply binary game tree has levels MAX (root), MIN, MAX, and then leaves. The eight leaf values, left to right, are 5, 6, 7, 4, 5, 3, 6, 7. Each MAX node at the third level owns two consecutive leaves. Alpha-beta pruning processes children left to right. How many leaves are **evaluated**?

Answer (numerical): _____

Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

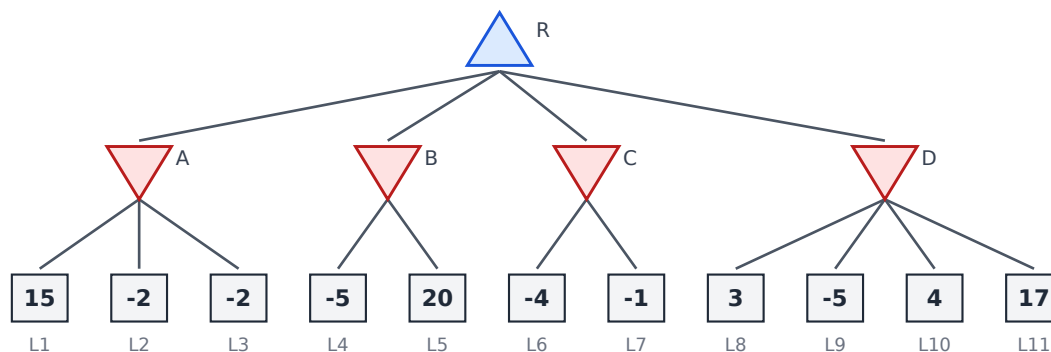
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow F \rightarrow K \rightarrow D \rightarrow H \rightarrow G$
☐ (B) $S \rightarrow F \rightarrow D \rightarrow H \rightarrow G$
☐ (C) $S \rightarrow K \rightarrow D \rightarrow H \rightarrow G$
☐ (D) $S \rightarrow F \rightarrow D \rightarrow H \rightarrow A \rightarrow G$

Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



MAX node MIN node

terminal utility

- ☐ (A) Leaf L2 is pruned.
☐ (B) Leaf L3 is pruned.
☐ (C) Leaf L4 is pruned.
☐ (D) Leaf L5 is pruned.

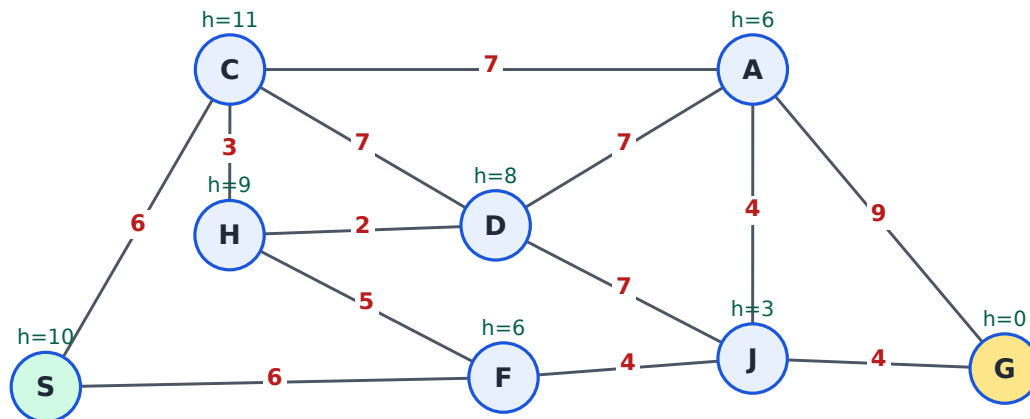
Q.14 [MCQ · 2 marks · negative: -2/3] Informed Search

$h_1, \dots, h_m$ are admissible heuristics for a problem. Which composite heuristic is guaranteed to be admissible and to dominate each h_i ?

- ☐ (A) $h(n) = h_1(n) + \dots + h_m(n)$
☐ (B) $h(n) = \max(h_1(n), \dots, h_m(n))$
☐ (C) $h(n) = \min(h_1(n), \dots, h_m(n))$
☐ (D) $h(n) = m \cdot h_1(n)$

Q.15 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (R \rightarrow \neg R) \wedge (Q \wedge \neg R)$?

- ☐ (A) $\neg(\neg(R \rightarrow \neg R) \vee \neg(Q \wedge \neg R))$
☐ (B) $\neg(\neg(R \rightarrow \neg R) \wedge \neg(Q \wedge \neg R))$
☐ (C) $(R \rightarrow \neg R) \vee (Q \wedge \neg R)$
☐ (D) $\neg(R \rightarrow \neg R) \wedge (Q \wedge \neg R)$

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic

The canonical CNF (product of maxterms, each clause containing all variables) of $p \oplus q \oplus r$ has how many clauses?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

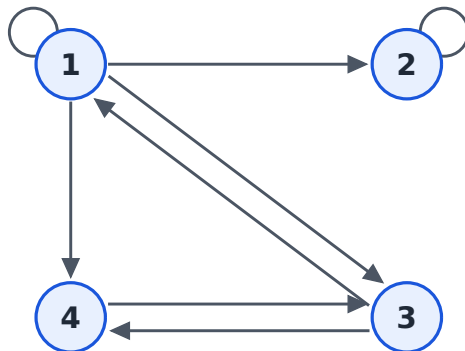
C1: $(P \vee \neg Q)$, C2: $(Q \vee R \vee S)$, C3: $(P \vee \neg R)$, C4: $(Q \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg Q \vee \neg R)$
☐ (B) (P)
☐ (C) $(Q \vee \neg S)$
☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (B) $\exists x \forall y \neg R(x, y)$
☐ (C) $\exists x \forall y R(x, y)$
☐ (D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$

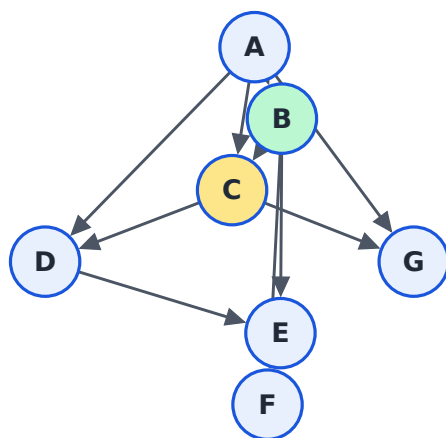
Q.20 [MSQ · 2 marks · no negative marking] Variable Elimination

Which statements about variable elimination (VE) are correct?

- ☐ (A) Every elimination ordering yields the same final answer
☐ (B) Finding an optimal elimination ordering is NP-hard in general
☐ (C) VE runs in time polynomial in the number of variables for every network and ordering
☐ (D) VE avoids the repeated computation of identical subexpressions that occurs in enumeration

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{B} \mid \mathbf{C} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{C} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, A, E, F, G**. What is the scope of the new factor produced when **D** is summed out?



- ☐ (A) $f(A, E)$
☐ (B) $f(A, B, D, E)$
☐ (C) $f(A, B, E)$
☐ (D) $f(A, C)$

Q.22 [MSQ · 2 marks · no negative marking] Variable Elimination

Which of the following Bayesian networks (edges listed) are polytrees?

- ☐ (A) $A \rightarrow C, B \rightarrow C, C \rightarrow D, C \rightarrow E$
☐ (B) $A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D$
☐ (C) $A \rightarrow B, B \rightarrow C, D \rightarrow C, D \rightarrow E$
☐ (D) $A \rightarrow B, A \rightarrow C, B \rightarrow C$

Q.23 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

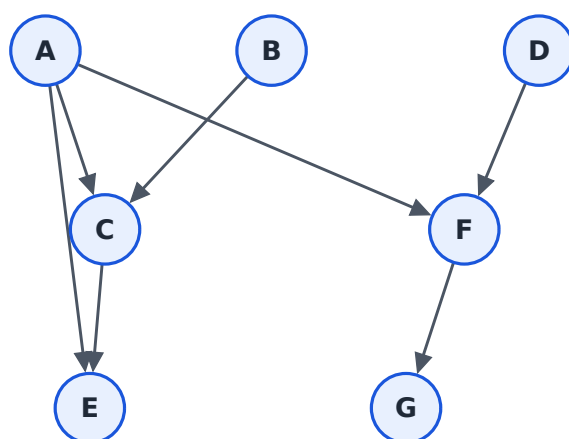
The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid C = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	T	F	T	T
2	T	T	T	F
3	F	T	F	T
4	T	T	T	F
5	F	T	T	F
6	T	T	F	F
7	T	T	F	T
8	T	F	T	F
9	T	T	F	T
10	T	T	T	F
11	T	T	F	F
12	T	T	F	F
13	F	F	T	T
14	F	T	F	F
15	T	F	F	T

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

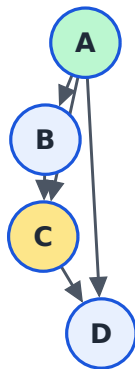
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp D \mid B, F$
☐ (B) $B \perp G$
☐ (C) $D \perp F \mid E$
☐ (D) $G \perp E$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(A = T \mid C = T)$ (round off to 3 decimal places).



A	P(A=T)
T	0.8

A	P(B=T .)	A	B	P(C=T ...)
T	0.6	T	T	0.8
F	0.25	T	F	0.1
		F	T	0.6
		F	F	0.25

A	C	P(D=T ...)
T	T	0.7
T	F	0.9
F	T	0.9
F	F	0.8

Answer (numerical): _____

Answer Key — Mock Test 037

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	C	14	MCQ	2	Informed Search	B
2	NAT	1	Adversarial Search	15	15	NAT	2	Informed Search	4
3	NAT	1	Uninformed Search	970	16	MSQ	2	Propositional Logic	A
4	NAT	1	Predicate Logic	31	17	NAT	2	Propositional Logic	4
5	MSQ	1	Predicate Logic	A, B, C	18	MSQ	2	Propositional Logic	B
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Predicate Logic	C
7	NAT	1	Propositional Logic	3	20	MSQ	2	Variable Elimination	A, B, D
8	NAT	1	Bayesian Networks	23	21	MCQ	2	Exact Inference	C
9	MSQ	1	Conditional Independence	A, B, D	22	MSQ	2	Variable Elimination	A, C
10	NAT	1	Sampling	0.1062 to 0.1066	23	NAT	2	Sampling	0.28 to 0.3
11	NAT	2	Adversarial Search	5	24	MSQ	2	Conditional Independence	A, B
12	MCQ	2	Uninformed Search	B	25	NAT	2	Exact Inference	0.858 to 0.862
13	MSQ	2	Adversarial Search	D					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, C, I, B, A, J, K, L, G)

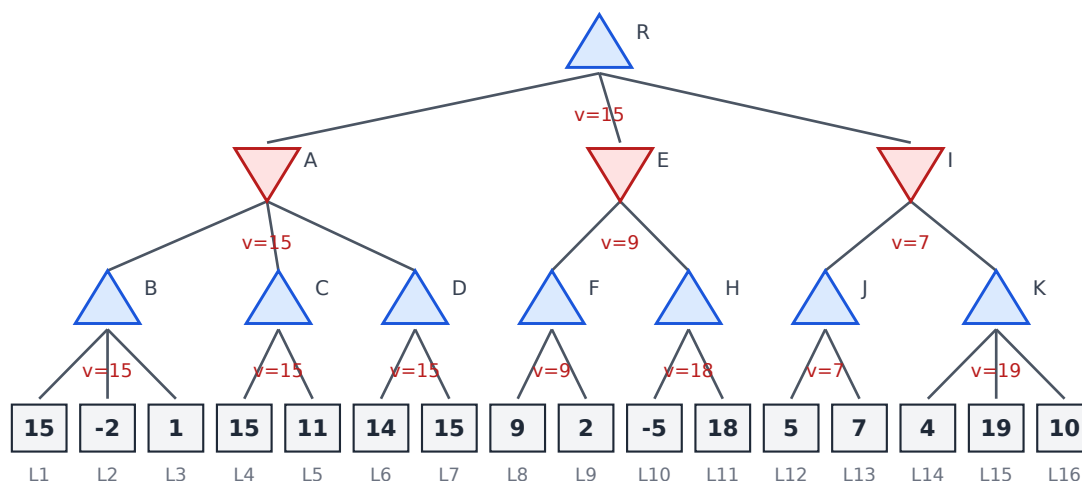
Adjacency lists (alphabetical): A: {B, J, K, L}; B: {A, G, I, J}; C: {I, J, K, S}; G: {B, L}; I: {B, C, J}; J: {A, B, C, I}; K: {A, C, L, S}; L: {A, G, K}; S: {C, K}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, C, I, B, A, J, K, L, G**; the path found is $S \rightarrow C \rightarrow I \rightarrow B \rightarrow A \rightarrow K \rightarrow L \rightarrow G$ (7 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] **Answer: 15**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 15$, $C = 15$, $D = 15$, $A = 15$, $F = 9$, $H = 18$, $E = 9$, $J = 7$, $K = 19$, $I = 7$. Root value = **15**; MAX's optimal first move is towards A.

Q.3 [NAT, 1M · Uninformed Search] **Answer: 970**

Nodes at depth i are regenerated in every iteration with limit $\geq i$, i.e. $(d + 1 - i)$ times.

$N = 4 \times 5 + 3 \times 25 + 2 \times 125 + 1 \times 625 = \mathbf{970}$.

Compare BFS: 780; the overhead of IDS is small for $b \geq 2$.

Q.4 [NAT, 1M · Predicate Logic] **Answer: 31**

Interpret P and Q as subsets of the domain; inclusion-exclusion: $2^n + 2^n - 1$. With $n = 4$ this gives **31** (verified by enumerating all interpretations).

Q.5 [MSQ, 1M · Predicate Logic] **Answer: A, B, C** (A: Knows(John, Jane); B: Knows(y, Bill); C: Knows(y, Mother(y)))

(A) $\{x/\text{Jane}\}$. (B) $\{x/\text{Bill}, y/\text{John}\}$. (C) $\{y/\text{John}, x/\text{Mother}(\text{John})\}$. (D) fails: x would have to be both John and Elizabeth. Renaming the second x (standardizing apart) would make it succeed, which is why AIMA insists on standardizing apart.

Q.6 [MCQ, 1M · Predicate Logic] **Answer: A** (A: $\{y/\text{John}, x/\text{Mother}(\text{John})\}$)

Unify arguments left to right. John vs y gives y/John . Then x vs $\text{Mother}(y)$ becomes x vs $\text{Mother}(\text{John})$, giving $x/\text{Mother}(\text{John})$. Applying $\{y/\text{John}, x/\text{Mother}(\text{John})\}$ to both atoms yields $\text{Knows}(\text{John}, \text{Mother}(\text{John}))$. Option B binds the wrong variables. Option C leaves y in the binding of x , so it is not idempotent and does not make the atoms identical in one application.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 3**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg\neg R$	$(\neg R \wedge Q) \wedge P$	$\neg\neg R \leftrightarrow ((\neg R \wedge Q) \wedge P)$
T	T	T	T	F	F
T	T	F	F	T	F
T	F	T	T	F	F
T	F	F	F	F	T
F	T	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	F	F	T

Rows where $F = T$: **3**.

Q.8 [NAT, 1M · Bayesian Networks] **Answer: 23**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (3-1) \times 3 + C: (3-1) \times 3 + D: (2-1) \times 3 + E: (2-1) \times 3 \times 2 = \mathbf{23}$.

Full joint table: $\prod |X| - 1 = 108 - 1 = 107$; saving = 84.

Q.9 [MSQ, 1M · Conditional Independence] Answer: A, B, D

An observed non-collider blocks a path, so (A) and (B) are true. A collider blocks the path when neither it nor any descendant is observed, so (D) is true. Observing the collider activates the path, so (C) is false.

Q.10 [NAT, 1M · Sampling] Answer: 0.1062 to 0.1066

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(\text{Sm}=F) \times P(\text{Po}=F) \times P(\text{Ca}=F|\text{Sm}=F, \text{Po}=F) \times P(\text{XR}=T|\text{Sm}=F, \text{Po}=F) \times P(\text{Dy}=F|\text{Ca}=F, \text{XR}=T) \\ = 0.4 \times 0.95 \times 0.8 \times 0.7 \times 0.5 = \mathbf{0.1064} \approx 0.1066.$$

Q.11 [NAT, 2M · Adversarial Search] Answer: 5

The leftmost MAX node sees (5, 6) and returns 6, so its MIN parent has $\beta = 6$. The next MAX node sees $7 \geq \beta$ and prunes leaf 4. The left MIN node therefore has value 6, and the root gets $\alpha = 6$. In the right MIN subtree, the first MAX node sees 5 and then 3 and returns 5. The MIN node's value is now $\leq 5 \leq \alpha$, so the whole MAX subtree with leaves (6, 7) is pruned. Evaluated leaves: 5, 6, 7, 5, 3 = **5**. The root value is 6.

Q.12 [MCQ, 2M · Uninformed Search] Answer: B (B: $S \rightarrow F \rightarrow D \rightarrow H \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

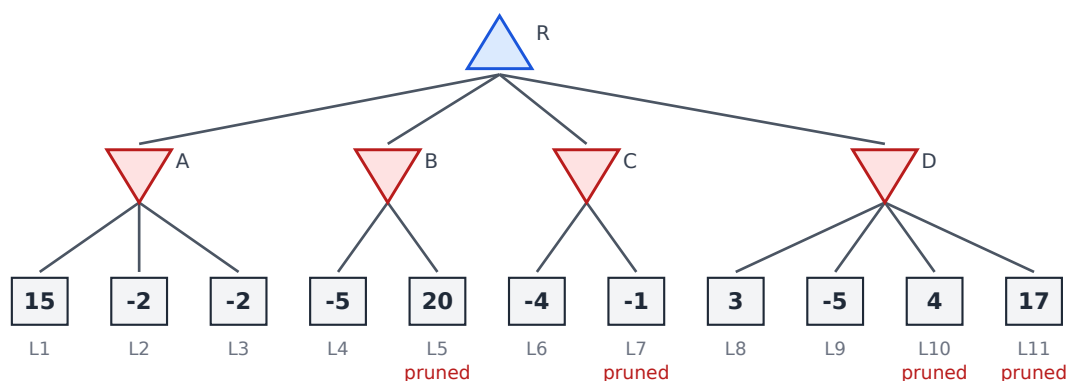
Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	F(5), K(8)
2	F (5)	K(8), D(9), J(11)
3	K (8)	D(9), J(11), H(17)
4	D (9)	H(11), J(11)
5	H (11)	J(11), A(14), G(14)
6	J (11)	A(14), G(14)
7	A (14)	G(14)
8	G (14)	← goal, stop

Path returned: **S** → **F** → **D** → **H** → **G**, cost = 5 + 4 + 2 + 3 = **14**. Nodes expanded (including S and G): **8** (S, F, K, D, H, J, A, G).

Q.13 [MSQ, 2M · Adversarial Search] Answer: D (D: Leaf L5 is pruned.)

Run alpha-beta depth-first, left to right, passing (α , β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-5 \leq \alpha = -2 \Rightarrow$ prune remaining children (leaves L5).
- At MIN node C: value $-4 \leq \alpha = -2 \Rightarrow$ prune remaining children (leaves L7).
- At MIN node D: value $-5 \leq \alpha = -2 \Rightarrow$ prune remaining children (leaves L10, L11).



Leaves evaluated (7): L1, L2, L3, L4, L6, L8, L9. Leaves pruned (4): L5, L7, L10, L11. Root value = -2 (same as plain minimax, -2 — pruning never changes the root value).

Q.14 [MCQ, 2M · Informed Search] Answer: B (B: $h(n) = \max(h_1(n), \dots, h_m(n))$)

Each $h_i(n) \leq h^*(n)$, so their maximum is also $\leq h^*(n)$, and it is $\geq$ each component by construction. So the **max** is admissible and dominates every h_i . The sum (A) and the multiple (D) can exceed h^* . The min (C) is admissible but is dominated by every h_i , not dominating.

Q.15 [NAT, 2M · Informed Search] Answer: 4

Heuristic values: $h(A)=6$, $h(C)=11$, $h(D)=8$, $h(F)=6$, $h(G)=0$, $h(H)=9$, $h(J)=3$, $h(S)=10$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (10)	F(12), C(17)
2	F (12)	J(13), C(17), H(20)
3	J (13)	G(14), C(17), A(20), H(20), D(25)
4	G (14)	C(17), A(20), H(20), D(25) ← goal, stop

Path returned: **S** → **F** → **J** → **G** with cost **14**; nodes expanded = **4** (S, F, J, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A ($A: \neg(\neg(R \rightarrow \neg R) \vee \neg(Q \wedge \neg R))$)

Compare truth tables with $F = (R \rightarrow \neg R) \wedge (Q \wedge \neg R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg(R \rightarrow \neg R) \vee \neg(Q \wedge \neg R))$ — agrees with F on all 4 rows ⇒ **equivalent**.

(B) $\neg(\neg(R \rightarrow \neg R) \wedge \neg(Q \wedge \neg R))$ — differs at $Q=F, R=F$ ($F = F$, option = T) ⇒ not equivalent.

(C) $(R \rightarrow \neg R) \vee (Q \wedge \neg R)$ — differs at $Q=F, R=F$ ($F = F$, option = T) ⇒ not equivalent.

(D) $\neg(R \rightarrow \neg R) \wedge (Q \wedge \neg R)$ — differs at $Q=T, R=F$ ($F = T$, option = F) ⇒ not equivalent.

Q.17 [NAT, 2M · Propositional Logic] Answer: 4

In the canonical CNF there is one maxterm (clause) for every falsifying assignment. $p \oplus q \oplus r$ is false exactly when an even number of variables is true: FFF, TTF, TFT, FTT. That gives 4 falsifying rows and hence 4 clauses. It is also known that no shorter CNF exists for parity, since every clause must contain all 3 variables.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B ($B: (P)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg Q \vee \neg R)$ never appears in the resolution closure ⇒ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) (P) is derivable:

1. resolve $(P \vee \neg R)$ and $(Q \vee R)$ on $R \Rightarrow (P \vee Q)$

2. resolve $(P \vee \neg Q)$ and $(P \vee Q)$ on $Q \Rightarrow (P)$

(C) $(Q \vee \neg S)$ never appears in the resolution closure ⇒ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) The resolution closure contains 10 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) ⇒ not derivable.

Q.19 [MSQ, 2M · Predicate Logic] Answer: C ($C: \exists x \forall y R(x, y)$)

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (3,1), (3,4), (4,3)\}$.

(A) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 2 has no edge to a different element.

(B) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

(C) $\exists x \forall y R(x, y)$: **TRUE** — $x = 1$ has edges to every y .

(D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(3,4) \wedge R(4,3)$ but $\neg R(3,3)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Variable Elimination] Answer: A, B, D

(A) Sums of products can be reordered (distributive law), so every ordering computes the same exact result; only cost differs. (B) AIMA notes finding the optimal ordering is intractable (NP-hard), so heuristics such as min-fill or min-degree are used. (D) VE stores intermediate results as factors so each is computed once. (C) is false: the cost is exponential in the size of the largest factor, which can be large for multiply connected networks or bad orderings.

Q.21 [MCQ, 2M · Exact Inference] Answer: C ($C: f(A, B, E)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B)$, $f(A, D)$, $f(B, D, E)$, $f(B, E, F)$, $f(A, G)$.

Eliminating Z : multiply all factors whose scope contains Z , then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(A, D)$, $f(B, D, E)$	$\{A, B, D, E\}$	$f(A, B, E)$
A	$f(A)$, $f(A, B)$, $f(A, B)$, $f(A, G)$, $f(A, B, E)$	$\{A, B, E, G\}$	$f(B, E, G)$

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(B, E, F), f(B, E, G)$	$\{B, E, F, G\}$	$f(B, F, G)$
F	$f(B, F, G)$	$\{B, F, G\}$	$f(B, G)$
G	$f(B, G)$	$\{B, G\}$	$f(B)$

The largest factor produced has **3** variable(s) (created when eliminating D); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over B and normalise.

Q.22 [MSQ, 2M · Variable Elimination] **Answer: A, C** (A: $A \rightarrow C, B \rightarrow C, C \rightarrow D, C \rightarrow E$; C: $A \rightarrow B, B \rightarrow C, D \rightarrow C, D \rightarrow E$)

A polytree has no undirected cycle. (A) Underlying undirected graph A-C, B-C, C-D, C-E is a tree: polytree. (B) A-B-D-C-A is an undirected cycle: multiply connected. (C) A-B-C-D-E is a simple path, no cycle: polytree (C has two parents, which is allowed). (D) A-B-C-A forms a triangle: not a polytree.

Q.23 [NAT, 2M · Sampling] **Answer: 0.28 to 0.3**

Rejection sampling keeps only samples consistent with the evidence $C = T$: samples 1, 2, 4, 5, 8, 10, 13 (7 accepted, 8 rejected).

Among the accepted samples, $D = T$ in 2. Estimate = $2/7 = \mathbf{0.2857}$.

(For this network the exact value is 0.3; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.24 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $G \perp D \mid B, F$; B: $B \perp G$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp D \mid B, F$: **TRUE** — $G \leftarrow F \leftarrow D$ (blocked: observed chain node F).

(B) $B \perp G$: **TRUE** — $B \rightarrow C \leftarrow A \rightarrow F \rightarrow G$ (blocked: collider C with neither it nor a descendant observed); $B \rightarrow C \rightarrow E \leftarrow A \rightarrow F \rightarrow G$ (blocked: collider E with neither it nor a descendant observed).

(C) $D \perp F \mid E$: **FALSE** — the path $D \rightarrow F$ is active given $\{E\}$.

(D) $G \perp E$: **FALSE** — the path $G \leftarrow F \leftarrow A \rightarrow C \rightarrow E$ is active given $\{\}$.

Q.25 [NAT, 2M · Exact Inference] **Answer: 0.858 to 0.862**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B \mid A) \cdot P(C \mid A, B) \cdot P(D \mid A, C)$.

Inference by enumeration: $P(A \mid e) = \alpha \sum_{\text{hidden}} P(A, e, \text{hidden})$.

Hidden: B, D	A	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.8 \times 0.6 \times 0.8 \times 0.7$	0.2688
T, F	T	$0.8 \times 0.6 \times 0.8 \times 0.3$	0.1152
F, T	T	$0.8 \times 0.4 \times 0.1 \times 0.7$	0.0224
F, F	T	$0.8 \times 0.4 \times 0.1 \times 0.3$	0.0096
T, T	F	$0.2 \times 0.25 \times 0.6 \times 0.9$	0.027
T, F	F	$0.2 \times 0.25 \times 0.6 \times 0.1$	0.003
F, T	F	$0.2 \times 0.75 \times 0.25 \times 0.9$	0.03375
F, F	F	$0.2 \times 0.75 \times 0.25 \times 0.1$	0.00375

$P(A=T, e) = 0.416$; $P(A=F, e) = 0.0675$; $P(e) = 0.4835$.

$P(A=T \mid e) = 0.416 / 0.4835 = \mathbf{0.8604} \approx 0.860$.

Mock Test 038

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 9 / 10	11 / 15 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

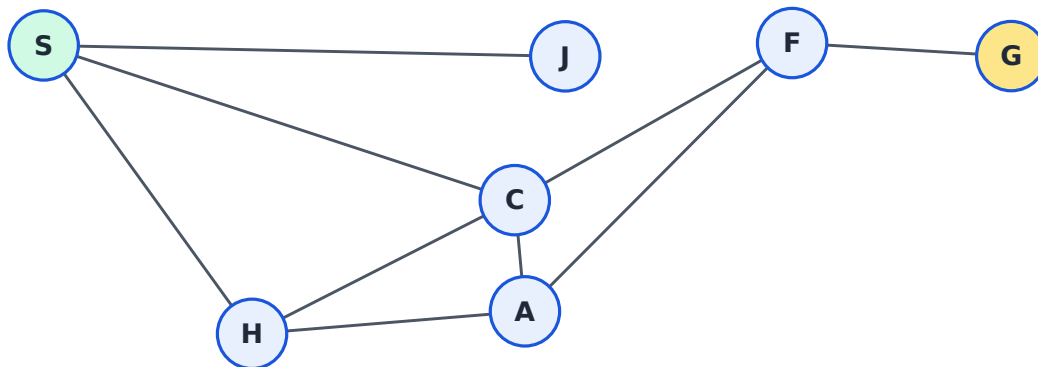
Q.1 [MSQ · 1 mark · no negative marking] Uninformed Search

Which of the following are requirements for applying bidirectional search?

- ☐ (A) Predecessors of a state can be computed, so the problem can be searched backward
- ☐ (B) The goal state is explicitly known (or there are only a few goal states), so the backward search has somewhere to start
- ☐ (C) An admissible heuristic must be available
- ☐ (D) The state space must be a tree

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, C, H, A, J, F, G
- ☐ (B) S, C, A, F, G
- ☐ (C) S, J, H, C, A, F, G
- ☐ (D) S, C, H, J, A, F, G

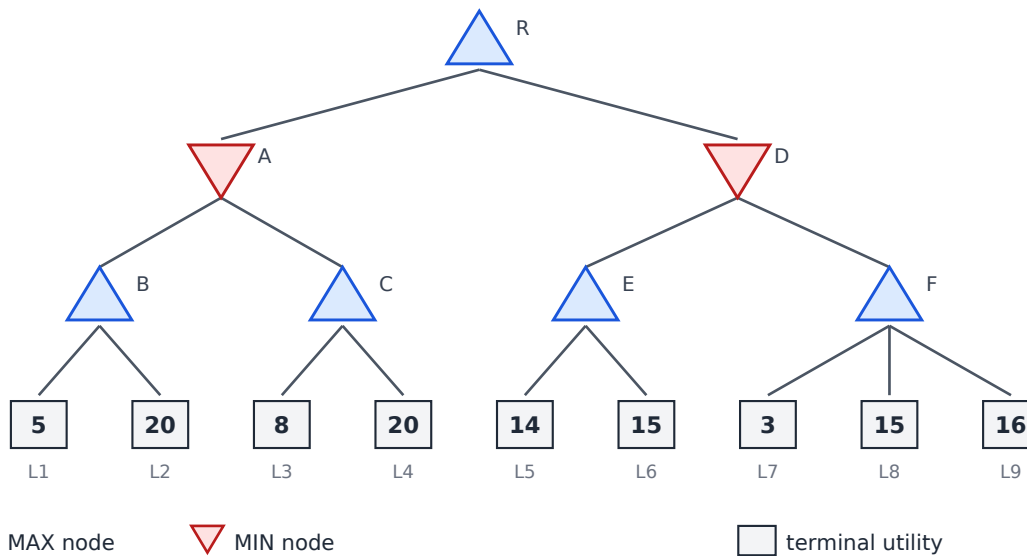
Q.3 [MSQ · 1 mark · no negative marking] Local Search

Which of the following statements about local search are TRUE?

- ☐ (A) Local beam search with k states passes useful information among the k parallel searches
- ☐ (B) Local beam search with $k = 1$ is equivalent to hill climbing
- ☐ (C) Hill climbing stores the whole search tree in memory
- ☐ (D) Typical local-search algorithms use only a constant amount of memory

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

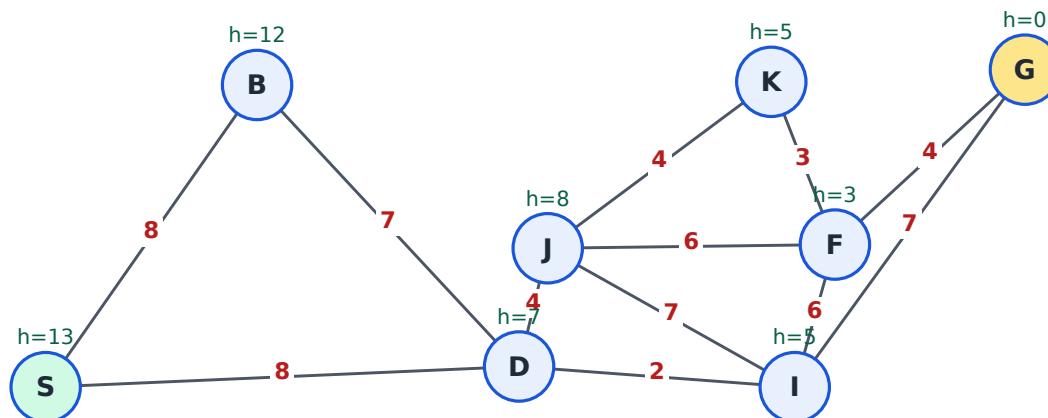
In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to D; root value 15
☐ (B) Move to A; root value 21
☐ (C) Move to D; root value 20
☐ (D) Move to A; root value 20

Q.5 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

$\neg(p \rightarrow q)$ is logically equivalent to

- ☐ (A) $p \wedge \neg q$
☐ (B) $\neg p \wedge q$
☐ (C) $\neg p \rightarrow \neg q$
☐ (D) $\neg p \vee q$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = \neg(R \leftrightarrow R) \vee ((R \wedge P) \leftrightarrow (Q \rightarrow Q))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ are both reflexive and symmetric?

Answer (numerical): _____

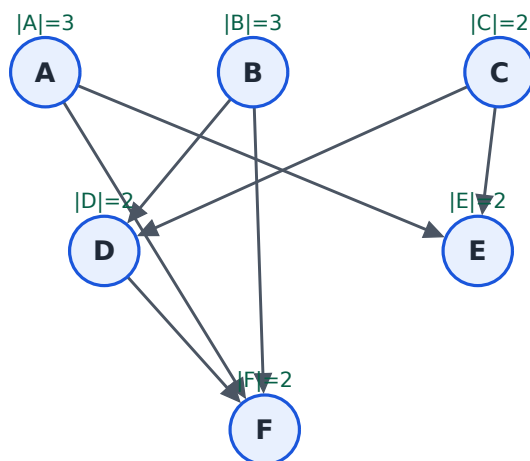
Q.9 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

A naive Bayes model with class variable C and features $F_1, \dots, F_n$, viewed as a Bayesian network, has which structure and assumption?

- ☐ (A) Edges $C \rightarrow F_i$ for each i; features are conditionally independent given C
- ☐ (B) Edges $F_i \rightarrow C$ for each i; features are marginally independent
- ☐ (C) A chain $C \rightarrow F_1 \rightarrow F_2 \rightarrow \dots \rightarrow F_n$
- ☐ (D) Edges $C \rightarrow F_i$ for each i; features are marginally independent

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Section B — 2-mark questions

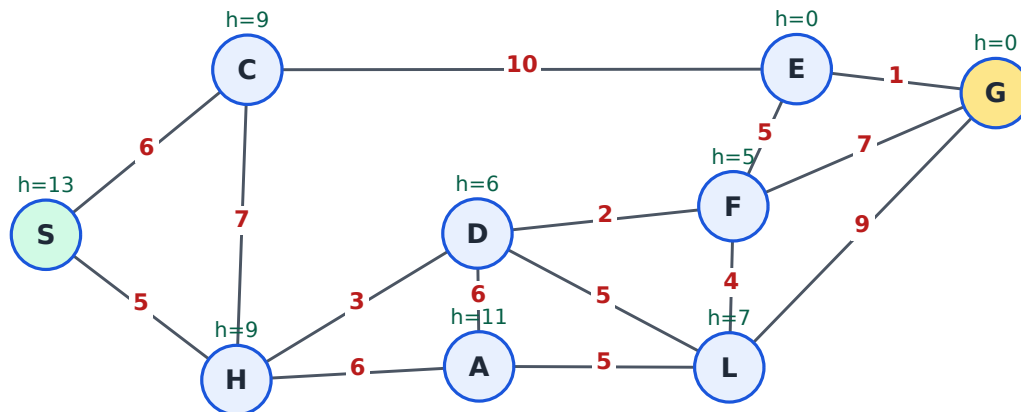
Q.11 [MSQ · 2 marks · no negative marking] Informed Search

Which of the following statements about IDA* are TRUE?

- ☐ (A) Its memory requirement is linear in the depth of the solution
- ☐ (B) With an admissible heuristic, it returns an optimal solution
- ☐ (C) It keeps every previously visited state in memory to avoid repeated states
- ☐ (D) When step costs are real-valued and almost all distinct, each iteration may add only one new node, which leads to many iterations

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

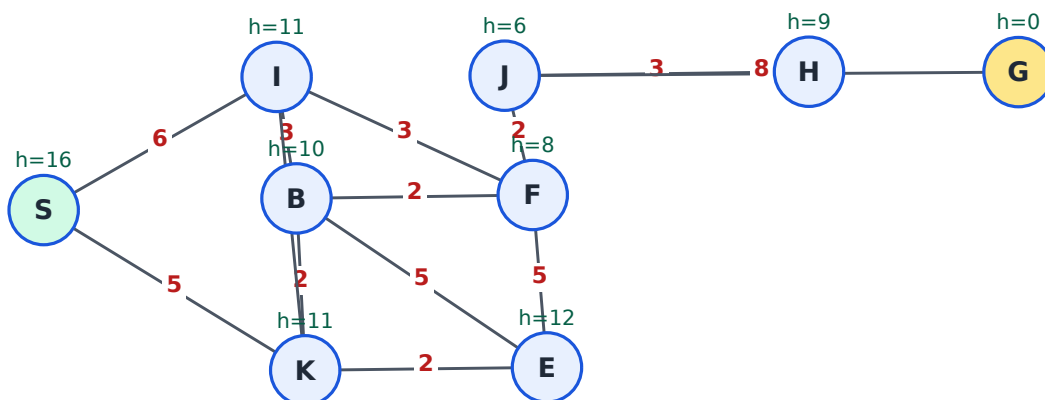
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node E is expanded.
- ☐ (B) Node F is expanded.
- ☐ (C) Node K is expanded.
- ☐ (D) The path returned has cost 21.

Q.14 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2\}$. How many interpretations of a binary predicate R over D make the sentence $\forall x \exists y R(x, y) \rightarrow \exists y \forall x R(x, y)$ true?

Answer (numerical): _____

Q.15 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Consider the clauses $C_1 = p \vee q \vee \neg r$ and $C_2 = \neg p \vee r \vee s$. Which one of the following statements is correct?

- ☐ (A) $q \vee s$ is a valid resolvent of C_1 and C_2
- ☐ (B) Every resolvent of C_1 and C_2 is a tautology
- ☐ (C) C_1 and C_2 have no resolvent
- ☐ (D) The empty clause can be derived from C_1 and C_2

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, D** and rules:

R1: $H \wedge C \wedge B \rightarrow A$

R2: $H \rightarrow C$

R3: $B \wedge D \rightarrow C$

R4: $A \rightarrow H$

R5: $J \wedge E \wedge C \rightarrow B$

R6: $C \wedge H \wedge D \rightarrow B$

R7: $C \rightarrow J$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) C is entailed
- ☐ (B) E is entailed
- ☐ (C) J is entailed
- ☐ (D) K is entailed

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

(1) $(Q \wedge P) \vee (P \wedge S)$

(2) $(Q \rightarrow Q) \rightarrow (\neg S \leftrightarrow \neg Q)$

(3) $(P \wedge R) \vee (Q \wedge R)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $(P \rightarrow Q) \leftrightarrow P$
- ☐ (B) $(S \leftrightarrow \neg S) \vee Q$
- ☐ (C) $\neg S \leftrightarrow S$
- ☐ (D) $\neg(S \rightarrow S)$

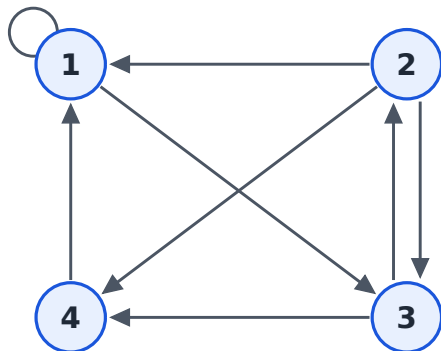
Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(R \rightarrow Q) \vee (Q \rightarrow R)$
- ☐ (B) $\neg(\neg Q \wedge P) \leftrightarrow (\neg\neg Q \vee \neg P)$
- ☐ (C) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$
- ☐ (D) $(R \rightarrow P) \rightarrow ((R \vee Q) \rightarrow P)$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

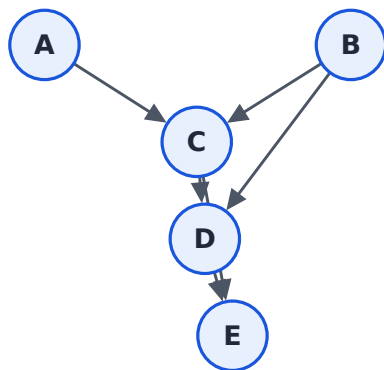
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x R(x, x)$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$

Q.20 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.39, 0.97, 0.73, 0.03, 0.05)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.05

$P(B=T)$
0.2

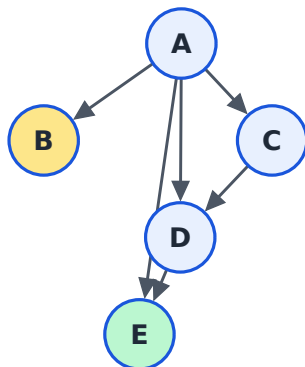
A	B	$P(C=T \dots)$
T	T	0.7
T	F	0.25
F	T	0.75
F	F	0.95

B	C	$P(D=T C, D)$	$P(E=T \dots)$
T	T	0.3	0.9
T	F	0.75	0.95
F	T	0.4	0.5
F	F	0.95	0.95

- ☐ (A) $A=F, B=T, C=T, D=T, E=T$
☐ (B) $A=F, B=F, C=T, D=T, E=T$
☐ (C) $A=T, B=F, C=T, D=T, E=T$
☐ (D) $A=T, B=T, C=T, D=F, E=F$

Q.21 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{B=F}$. The current state of the chain is $(\mathbf{A=F, B=F, C=T, D=T, E=F})$. The next step resamples $\mathbf{E}$. With what probability is E set to T? (round off to 3 decimal places)



P(A=T)
0.8

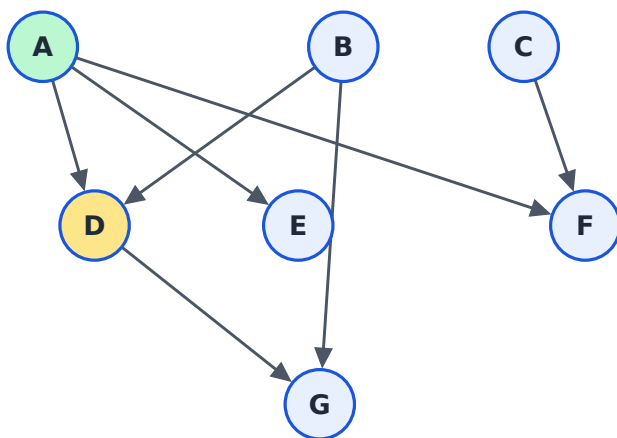
A	P(B=T ...)	A	P(C=T ...)
T	0.2	T	0.5
F	0.4	F	0.8

A	C	P(D=T)	A	D	P(E=T ...)
T	T	0.6	T	T	0.2
T	F	0.6	T	F	0.25
F	T	0.05	F	T	0.05
F	F	0.7	F	F	0.5

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(A \mid D = T)}$ in the Bayesian network below (Boolean variables; evidence $D = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{F, C, E, B, G}$. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

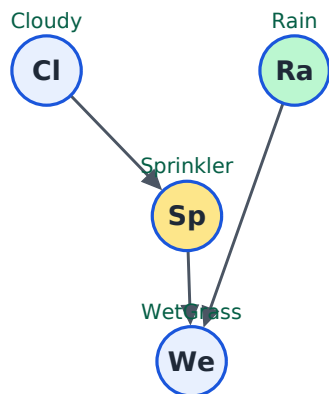
Q.23 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S, C \rightarrow R, S \rightarrow W, R \rightarrow W$ and CPTs: $P(c) = 0.5; P(s \mid c) = 0.1, P(s \mid \neg c) = 0.5; P(r \mid c) = 0.8, P(r \mid \neg c) = 0.2; P(w \mid s, r) = 0.99, P(w \mid s, \neg r) = 0.90, P(w \mid \neg s, r) = 0.90, P(w \mid \neg s, \neg r) = 0.0$. Gibbs sampling is run with evidence $S = \text{true}, W = \text{true}$. In the current state $C = \text{true}$. Compute the probability with which R is resampled to true, i.e. $P(r \mid c, s, w)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: CI = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\text{Rain} = T \mid \text{Sprinkler} = T)$ (round off to 3 decimal places).



$P(\text{CI}=T)$
0.4

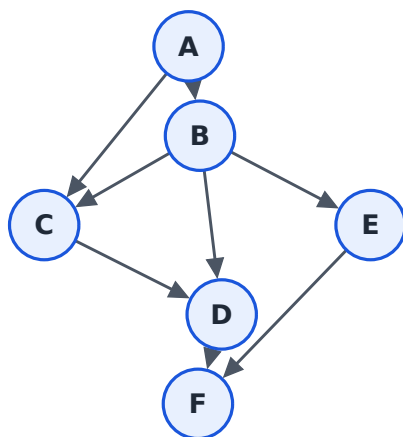
CI	$P(\text{Sp}=T \dots)$	$P(\text{Ra}=T)$
T	0.7	0.8
F	0.2	

Sp	Ra	$P(\text{We}=T \dots)$
T	T	0.25
T	F	0.3
F	T	0.9
F	F	0.2

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp F \mid E$
☐ (B) $B \perp C \mid D, E$
☐ (C) $D \perp F \mid B, C$
☐ (D) $E \perp A \mid B, C$

Answer Key — Mock Test 038

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MSQ	1	Uninformed Search	A, B	14	NAT	2	Predicate Logic	14
2	MCQ	1	Uninformed Search	D	15	MCQ	2	Propositional Logic	B
3	MSQ	1	Local Search	A, B, D	16	MSQ	2	Propositional Logic	A, C
4	MCQ	1	Adversarial Search	D	17	MSQ	2	Propositional Logic	A, B
5	NAT	1	Informed Search	17	18	MSQ	2	Propositional Logic	A, B
6	MCQ	1	Propositional Logic	A	19	MSQ	2	Predicate Logic	A, B, C
7	NAT	1	Propositional Logic	6	20	MCQ	2	Sampling	B
8	NAT	1	Predicate Logic	8	21	NAT	2	Sampling	0.048 to 0.052
9	MCQ	1	Bayesian Networks	A	22	NAT	2	Exact Inference	2
10	NAT	1	Bayesian Networks	35	23	NAT	2	Sampling	0.814 to 0.815
11	MSQ	2	Informed Search	A, B, D	24	NAT	2	Exact Inference	0.798 to 0.802
12	NAT	2	Informed Search	7	25	MSQ	2	Conditional Independence	D
13	MSQ	2	Informed Search	A, B, C					

Detailed Solutions

Q.1 [MSQ, 1M · Uninformed Search] Answer: A, B

The backward search expands states toward their predecessors, so (A) predecessors must be computable and (B) the goal must be explicitly known. An implicit goal test such as 'checkmate' gives backward search no starting state. A heuristic is not needed (C): bidirectional BFS and bidirectional UCS are uninformed. The state space may be any graph (D).

Q.2 [MCQ, 1M · Uninformed Search] Answer: D (D: S, C, H, J, A, F, G)

Adjacency lists (alphabetical): A: {C, F, H}; C: {A, F, H, S}; F: {A, C, G}; G: {F}; H: {A, C, S}; J: {S}; S: {C, H, J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[C, H, J]
2	C	[H, J, A, F]
3	H	[J, A, F]
4	J	[A, F]
5	A	[F]
6	F	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, C, H, J, A, F, G**. The path returned is $S \rightarrow C \rightarrow F \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

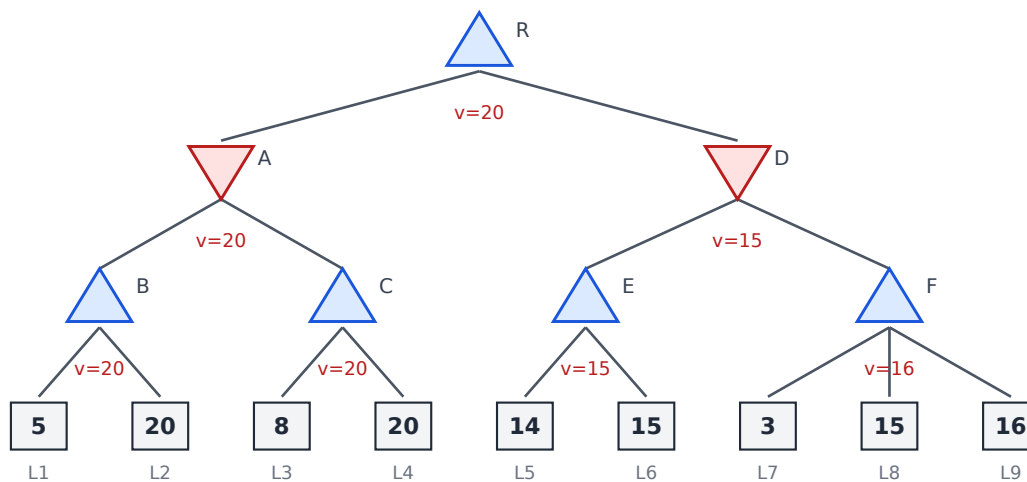
Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [MSQ, 1M · Local Search] Answer: A, B, D

(A) is true: beam search selects the k best successors from the pooled successors of all k states. (B) is true: with one state, keeping the best successor is hill climbing. (C) is false: hill climbing keeps only the current state, not a tree or paths. (D) is true: local search keeps just the current state or states.

Q.4 [MCQ, 1M · Adversarial Search] Answer: D (D: Move to A; root value 20)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 20, C = 20, A = 20, E = 15, F = 16, D = 15. Root value = **20**; MAX's optimal first move is towards A.

Q.5 [NAT, 1M · Informed Search] Answer: 17

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (13)	D(7), B(12)
2	D (7)	I(5), J(8), B(12)
3	I (5)	G(0), F(3), J(8), B(12)
4	G (0)	F(3), J(8), B(12) ← goal, stop

Path returned: **S** → **D** → **I** → **G**, cost = 8 + 2 + 7 = **17**.

The optimal cost is $h^*(S) = 17$ — greedy happens to be optimal here, but in general it is not.

Q.6 [MCQ, 1M · Propositional Logic] Answer: A (A: $p \wedge \neg q$)

$p \rightarrow q \equiv \neg p \vee q$, so $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q$ by De Morgan's law. It is true only in the row $p=T, q=F$, which is exactly the row where $p \rightarrow q$ is false. $\neg p \wedge q$ is true only at $p=F, q=T$; $\neg p \rightarrow \neg q$ is the inverse; $\neg p \vee q$ is $p \rightarrow q$ itself.

Q.7 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg(R \leftrightarrow R)$	$(R \wedge P) \leftrightarrow (Q \rightarrow Q)$	$\neg(R \leftrightarrow R) \vee ((R \wedge P) \leftrightarrow (Q \rightarrow Q))$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	F	F	F
F	T	F	F	F	F
F	F	T	F	F	F
F	F	F	F	F	F

F has 2 models, so it is false in $8 - 2 = 6$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.8 [NAT, 1M · Predicate Logic] Answer: 8

Reflexivity fixes all 3 diagonal pairs to be present. Symmetry ties (x, y) and (y, x) together for $x \neq y$, leaving 3 independent unordered pairs, each in or out. Total = $2^3 = 8$.

Q.9 [MCQ, 1M · Bayesian Networks] Answer: A (A: Edges $C \rightarrow F_i$ for each i ; features are conditionally independent given C)

Naive Bayes uses $P(C, F_1, \dots, F_n) = P(C) \prod P(F_i | C)$, i.e. the class is the single parent of every feature. The DAG implies $F_i \perp F_j | C$. (D) is wrong because the features are generally dependent marginally (common cause C is unobserved). (B) reverses the edges (that would make features marginally independent but dependent given C); (C) is a different model.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 35

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (3-1) + C: (2-1) + D: (2-1) \times 3 \times 2 + E: (2-1) \times 3 \times 2 + F: (2-1) \times 3 \times 3 \times 2 = 35$.

Full joint table: $\prod |X| - 1 = 144 - 1 = 143$; saving = 108.

Q.11 [MSQ, 2M · Informed Search] Answer: A, B, D

(A) is true: each iteration is a depth-first search bounded by the f -cutoff. (B) is true: thresholds rise to the smallest f that exceeded the old one, so the first goal found is optimal. (C) is false: IDA* does not keep an explored set, which is exactly why it uses little memory and why it can re-expand states. (D) is true: this is the well-known weakness, which motivates RBFS and SMA*.

Q.12 [NAT, 2M · Informed Search] Answer: 7

Heuristic values: $h(A)=11$, $h(C)=9$, $h(D)=6$, $h(E)=0$, $h(F)=5$, $h(G)=0$, $h(H)=9$, $h(L)=7$, $h(S)=13$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	H(14), C(15)
2	H (14)	D(14), C(15), A(22)
3	D (14)	C(15), F(15), L(20), A(22)
4	C (15)	F(15), E(16), L(20), A(22)
5	F (15)	E(15), G(17), L(20), A(22)
6	E (15)	G(16), L(20), A(22)
7	G (16)	L(20), A(22) ← goal, stop

Path returned: **S** → **H** → **D** → **F** → **E** → **G** with cost **16**; nodes expanded = **7** (S, H, D, C, F, E, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.13 [MSQ, 2M · Informed Search] Answer: A, B, C (A: Node E is expanded.; B: Node F is expanded.; C: Node K is expanded.)

Heuristic values: $h(B)=10$, $h(E)=12$, $h(F)=8$, $h(G)=0$, $h(H)=9$, $h(I)=11$, $h(J)=6$, $h(K)=11$, $h(S)=16$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (16)	K(16), I(17)
2	K (16)	B(17), I(17), E(19)
3	B (17)	F(17), I(17), E(19)
4	F (17)	I(17), J(17), E(19)
5	I (17)	J(17), E(19)
6	J (17)	E(19), G(19), H(23)
7	E (19)	G(19), H(23)
8	G (19)	H(23) ← goal, stop

Path returned: **S** → **K** → **B** → **F** → **J** → **G** with cost **19**; nodes expanded = **8** (S, K, B, F, I, J, E, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.14 [NAT, 2M · Predicate Logic] Answer: 14

There are $2^4 = 16$ relations. The sentence is false iff every row of the 2×2 matrix is non-empty but no column is full. A column is full if both rows contain that column; so each row must be non-empty and the rows must have no common column, forcing each row to be a single, different column: $R = \{(1,1), (2,2)\}$ or $R = \{(1,2), (2,1)\}$. So 2 interpretations falsify it and $16 - 2 = 14$ satisfy it.

Q.15 [MCQ, 2M · Propositional Logic] Answer: B (B: Every resolvent of C_1 and C_2 is a tautology)

C_1 and C_2 clash on two pairs, $p/\neg p$ and $\neg r/r$. Resolution removes exactly one complementary pair at a time. Resolving on p gives $q \vee \neg r \vee r \vee s$; resolving on r gives $p \vee \neg p \vee q \vee s$. Both contain a complementary pair, so both are tautologies. Removing both pairs at once to get $q \vee s$ is unsound: $p=T, q=F, r=T, s=F$ satisfies C_1 and C_2 but falsifies $q \vee s$. The pair is satisfiable, so $\square$ cannot be derived.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: C is entailed; C: J is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $B \wedge D \rightarrow C$; add C.

Round 2: fire $C \rightarrow J$; add J.

Fixed point: {B, C, D, J} — 4 atoms are entailed (facts: B, D; newly derived: C, J). Atoms never derived: A, E, F, H, K.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $(P \rightarrow Q) \leftrightarrow P$; B: $(S \leftrightarrow \neg S) \vee Q$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 1 of them:

P	Q	R	S	KB
T	T	T	T	T

(A) $(P \rightarrow Q) \leftrightarrow P$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $(S \leftrightarrow \neg S) \vee Q$ is true in all models of KB $\Rightarrow$ **entailed**.

(C) $\neg S \leftrightarrow S$ is false in the KB-model $P=T, Q=T, R=T, S=T \Rightarrow$ not entailed.

(D) $\neg(S \rightarrow S)$ is false in the KB-model $P=T, Q=T, R=T, S=T \Rightarrow$ not entailed.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $(R \rightarrow Q) \vee (Q \rightarrow R)$; B: $\neg(\neg Q \wedge P) \leftrightarrow (\neg\neg Q \vee \neg P)$)

(A) $(R \rightarrow Q) \vee (Q \rightarrow R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $\neg(\neg Q \wedge P) \leftrightarrow (\neg\neg Q \vee \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$: **not valid** — counter-model $P=T, Q=F, R=T$.

(D) $(R \rightarrow P) \rightarrow ((R \vee Q) \rightarrow P)$: **not valid** — counter-model $P=F, Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: A, B, C** (A: $\exists x R(x, x)$; B: $\forall y \exists x R(x, y)$; C: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,1), (1,3), (2,1), (2,3), (2,4), (3,2), (3,4), (4,1)\}$.

(A) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

(B) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(2,3)$ and $R(3,2)$ with $2 \neq 3$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MCQ, 2M · Sampling] **Answer: B** (B: $A=F, B=F, C=T, D=T, E=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \neg) = 0.05$; $u = 0.39 \geq 0.05 \Rightarrow A = F$
- B: $P(B=T \mid \neg) = 0.2$; $u = 0.97 \geq 0.2 \Rightarrow B = F$
- C: $P(C=T \mid A=F, B=F) = 0.95$; $u = 0.73 < 0.95 \Rightarrow C = T$
- D: $P(D=T \mid B=F, C=T) = 0.4$; $u = 0.03 < 0.4 \Rightarrow D = T$
- E: $P(E=T \mid C=T, D=T) = 0.9$; $u = 0.05 < 0.9 \Rightarrow E = T$

Sample: **$\{A=F, B=F, C=T, D=T, E=T\}$** .

Q.21 [NAT, 2M · Sampling] **Answer: 0.048 to 0.052**

Gibbs sampling draws E from $P(E \mid \text{mb}(E))$, where $\text{mb}(E) = \{A, D\}$:

$P(E \mid \text{mb}) \propto P(E \mid \text{parents}(E)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$E=T$: $P(E=T \mid A=F, D=T) = 0.05 = 0.05$

$E=F$: $P(E=F \mid A=F, D=T) = 0.95 = 0.95$

Normalise: $0.05 / (0.05 + 0.95) = \mathbf{0.05}$.

Q.22 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(B), f(C), f(A, B), f(A, E), f(A, C, F), f(B, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(A, C, F)$	$\{A, C, F\}$	$f(A, C)$
C	$f(C), f(A, C)$	$\{A, C\}$	$f(A)$
E	$f(A, E)$	$\{A, E\}$	$f(A)$
B	$f(B), f(A, B), f(B, G)$	$\{A, B, G\}$	$f(A, G)$
G	$f(A, G)$	$\{A, G\}$	$f(A)$

The largest factor produced has **2** variable(s) (created when eliminating F); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.23 [NAT, 2M · Sampling] **Answer: 0.814 to 0.815**

The Markov blanket of R is {C, S, W}. $P(R | mb) = \alpha P(R | c) P(w | s, R)$. For r: $0.8 \times 0.99 = 0.792$. For $\neg r$: $0.2 \times 0.90 = 0.18$. $P(r | c, s, w) = 0.792 / (0.792 + 0.18) = 0.792 / 0.972 \approx 0.8148 \approx 0.815$.

Q.24 [NAT, 2M · Exact Inference] **Answer: 0.798 to 0.802**

Factorisation: $P(\text{Cloudy, Sprinkler, Rain, WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} | \text{Cloudy}) \cdot P(\text{Rain}) \cdot P(\text{WetGrass} | \text{Sprinkler, Rain})$.

Inference by enumeration: $P(\text{Rain} | e) = \alpha \sum_{\text{hidden}} P(\text{Rain}, e, \text{hidden})$.

Hidden: Cl, We	Ra	product of CPT entries (Cl · Sp · Ra · We)	value
T, T	T	$0.4 \times 0.7 \times 0.8 \times 0.25$	0.056
T, F	T	$0.4 \times 0.7 \times 0.8 \times 0.75$	0.168
F, T	T	$0.6 \times 0.2 \times 0.8 \times 0.25$	0.024
F, F	T	$0.6 \times 0.2 \times 0.8 \times 0.75$	0.072
T, T	F	$0.4 \times 0.7 \times 0.2 \times 0.3$	0.0168
T, F	F	$0.4 \times 0.7 \times 0.2 \times 0.7$	0.0392
F, T	F	$0.6 \times 0.2 \times 0.2 \times 0.3$	0.0072
F, F	F	$0.6 \times 0.2 \times 0.2 \times 0.7$	0.0168

$P(\text{Rain}=T, e) = 0.32$; $P(\text{Rain}=F, e) = 0.08$; $P(e) = 0.4$.

$P(\text{Rain}=T | e) = 0.32 / 0.4 = \mathbf{0.8} \approx 0.800$.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: D** ($D: E \perp A | B, C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $C \perp F | E$: **FALSE** — the path $C \leftarrow A \rightarrow B \rightarrow D \rightarrow F$ is active given {E}.

(B) $B \perp C | D, E$: **FALSE** — the path $B \leftarrow A \rightarrow C$ is active given {D, E}.

(C) $D \perp F | B, C$: **FALSE** — the path $D \rightarrow F$ is active given {B, C}.

(D) $E \perp A | B, C$: **TRUE** — $E \leftarrow B \leftarrow A$ (blocked: observed chain node B); $E \leftarrow B \rightarrow C \leftarrow A$ (blocked: observed fork node B); $E \leftarrow B \rightarrow D \leftarrow C \leftarrow A$ (blocked: observed fork node B); $E \rightarrow F \leftarrow D \leftarrow B \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $E \rightarrow F \leftarrow D \leftarrow B \rightarrow C \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $E \rightarrow F \leftarrow D \leftarrow C \leftarrow A$ (blocked: collider F with neither it nor a descendant observed) (+1 more paths, all blocked).

Mock Test 039

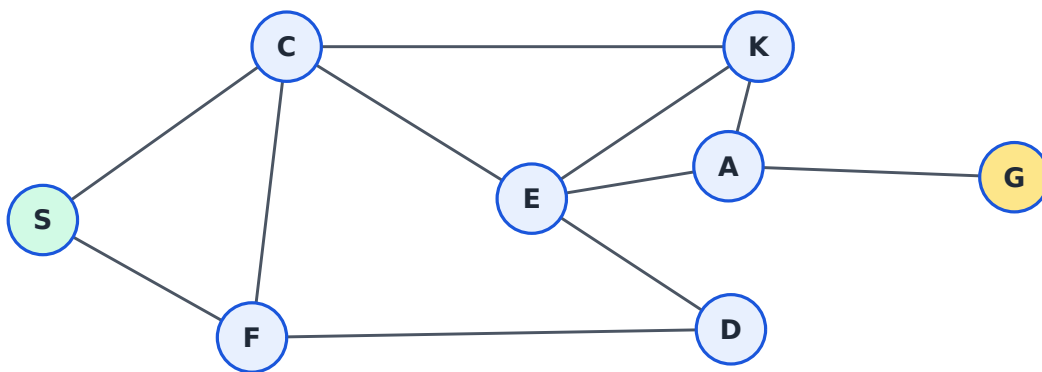
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	3 / 8 / 14	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, E, C, A, G
- ☐ (B) S, C, E, A, G
- ☐ (C) S, C, F, E, K, D, A, G
- ☐ (D) S, F, D, E, K, C, A, G

Q.2 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Iterative deepening search is applied to a tree with branching factor $b = 3$ whose shallowest goal is at depth $d = 5$ (goal is the last node at that depth). Using the standard count $N(IDS) = d \cdot b + (d-1) \cdot b^2 + \dots + 1 \cdot b^d$ (root not counted), how many nodes are generated?

Answer (numerical): _____

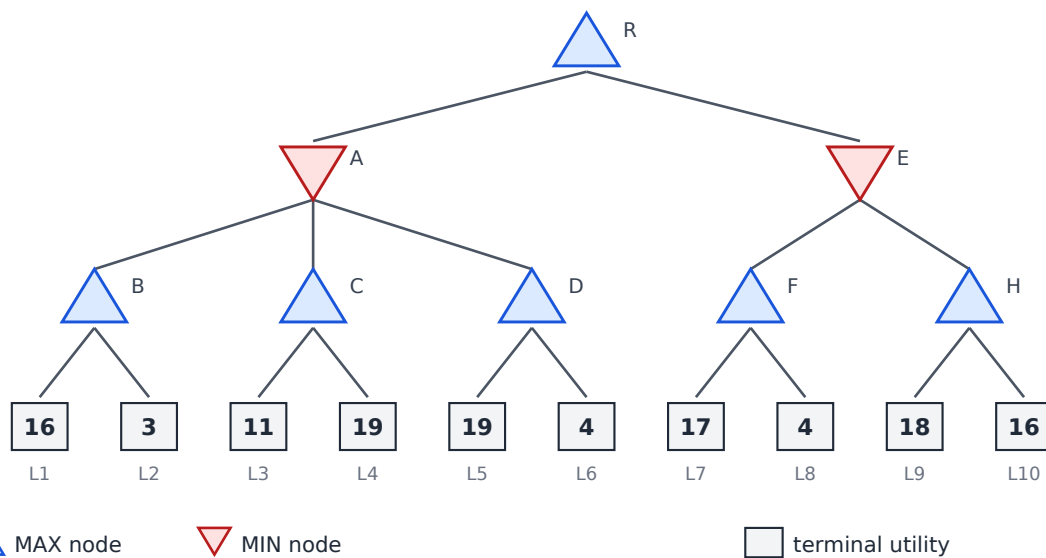
Q.3 [NAT · 1 mark · no negative marking] Local Search

In the 8-queens complete-state formulation (one queen per column), the successors of a state are all states obtained by moving a single queen to a different square in the same column. How many successors does each state have?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following equivalences are valid in first-order logic?

- ☐ (A) $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
☐ (B) $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
☐ (C) $\forall x \exists y P(x, y) \equiv \exists y \forall x P(x, y)$
☐ (D) $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 4 elements and R a binary predicate symbol. In how many of the 2^{16} possible interpretations of R over D is the sentence $\forall x R(x, x) \wedge \forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

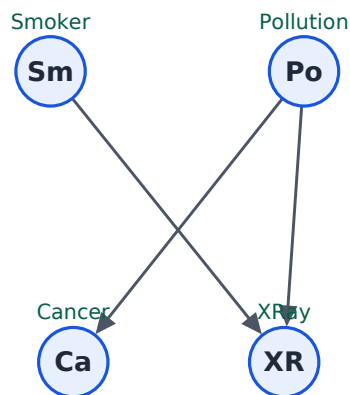
Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following is logically equivalent to $p \rightarrow q$?

- ☐ (A) $\neg q \rightarrow \neg p$
☐ (B) $q \rightarrow p$
☐ (C) $\neg p \rightarrow \neg q$
☐ (D) $p \wedge \neg q$

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) What is the probability that a single run produces the sample **(Sm=T, Po=T, Ca=F, XR=F)**? (round off to 4 decimal places)



P(Sm=T)
0.25

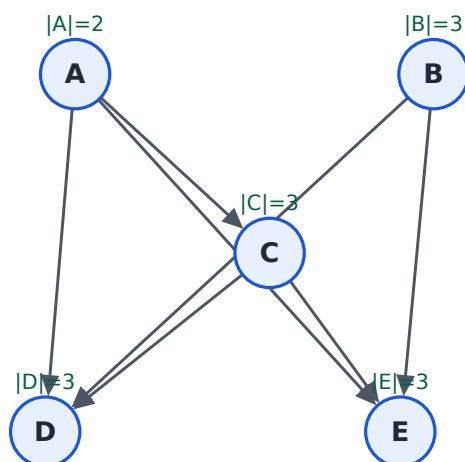
P(Po=T)	Po	P(Ca=T ...)
0.8	T	0.25
	F	0.5

Sm	Po	P(XR=T ...)
T	T	0.3
T	F	0.75
F	T	0.25
F	F	0.7

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Conditional Independence

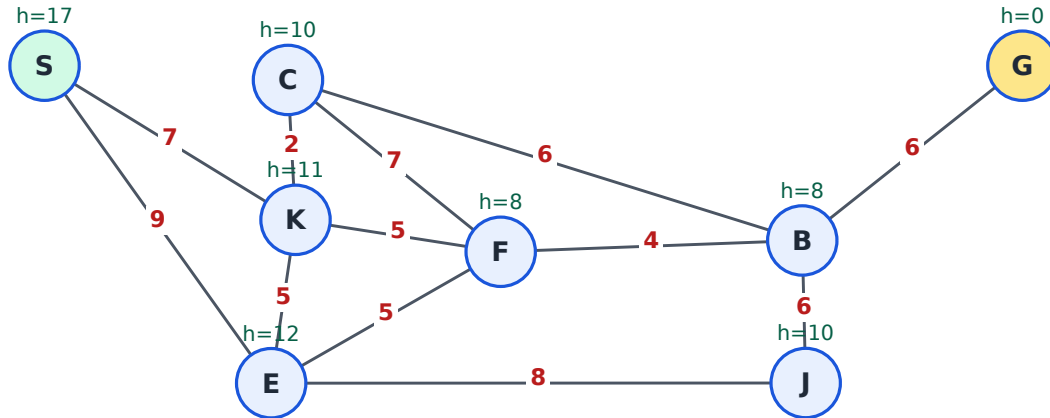
Boolean variables A, B, C satisfy $A \perp B \mid C$, and no other independence is known. What is the minimum number of independent parameters needed to specify $P(A, B, C)$?

- ☐ (A) 5
☐ (B) 7
☐ (C) 6
☐ (D) 4

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
☐ (B) The edge S-E violates the consistency condition (in at least one direction).
☐ (C) h is consistent (monotone).
☐ (D) $h(B)$ overestimates the true cost from B to G.

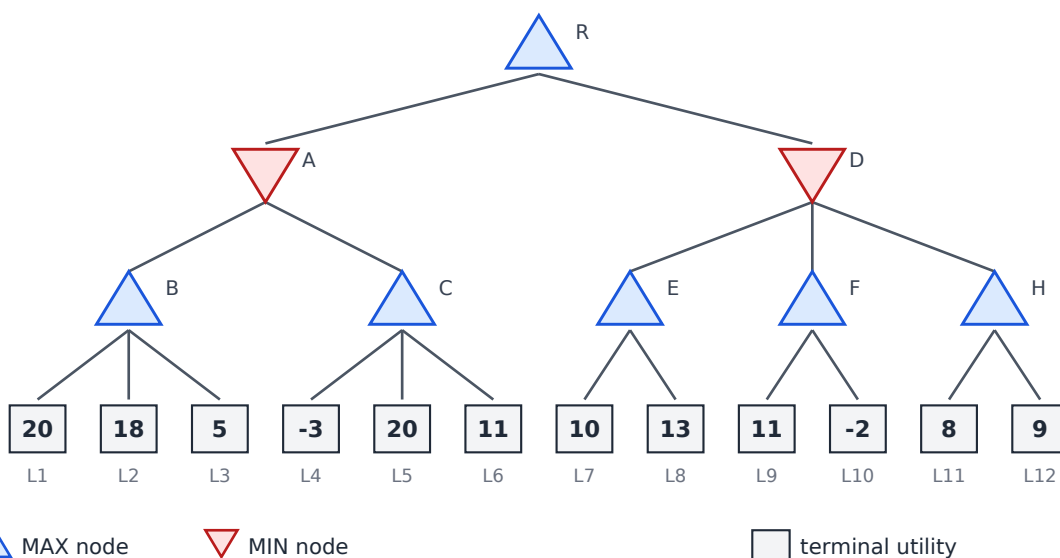
Q.12 [MSQ · 2 marks · no negative marking] Uninformed Search

Iterative deepening search (IDS) runs on a problem with finite branching factor b and shallowest goal depth d . Which of the following statements are TRUE?

- ☐ (A) IDS is complete
☐ (B) IDS is optimal when all step costs are equal
☐ (C) The space complexity of IDS is $O(bd)$
☐ (D) The time complexity of IDS is asymptotically worse than BFS by a factor of d , namely $O(d \cdot b^d)$

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

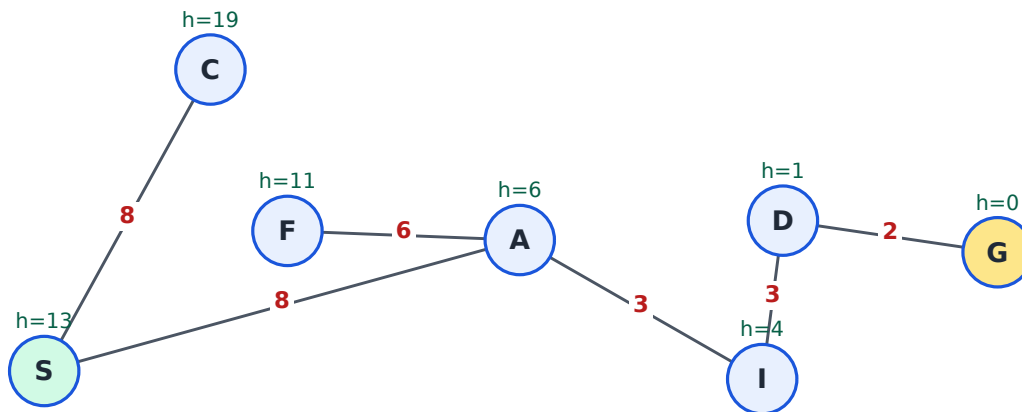
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Uninformed Search

Iterative deepening search (IDS) runs on a uniform tree with $b = 10$ and the shallowest goal at $d = 5$. Assume the worst case: the final iteration (limit 5) generates every node at depth 5. Each time a non-root node is generated, it counts once, so a node regenerated in a later iteration counts again. The root is not counted. What is the total number of node generations?

Answer (numerical): _____

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

The formula $F = ((P \wedge Q) \rightarrow (\neg P \vee \neg R)) \rightarrow \neg(\neg S \rightarrow R)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R, S . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (\neg Q \rightarrow P) \rightarrow \neg P$?

- ☐ (A) $\neg(\neg Q \rightarrow P) \wedge \neg P$
☐ (B) $\neg P \rightarrow (\neg Q \rightarrow P)$
☐ (C) $\neg((\neg Q \rightarrow P) \wedge \neg \neg P)$
☐ (D) $\neg(\neg Q \rightarrow P) \vee \neg P$

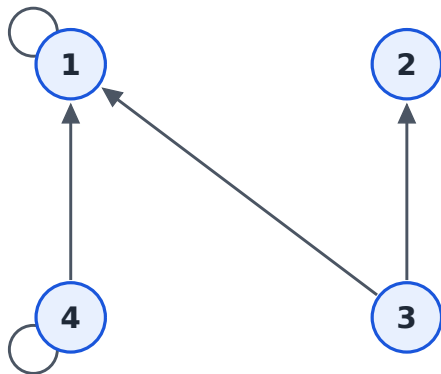
Q.18 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \forall y (R(x, y) \rightarrow \neg R(y, x))$?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y R(x, y)$
☐ (B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (C) $\exists x \forall y \neg R(y, x)$
☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

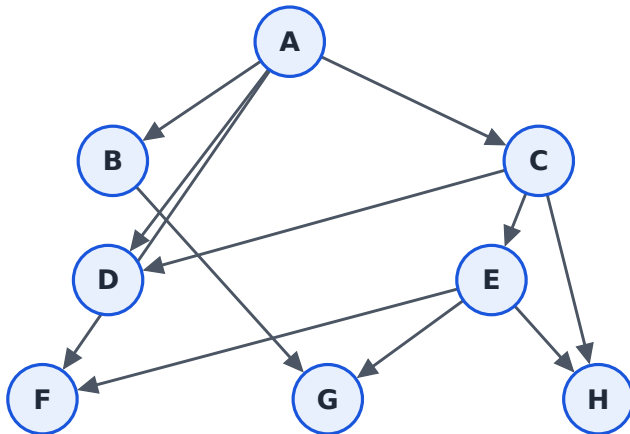
$C1: (P \vee \neg Q \vee \neg S)$, $C2: (\neg Q \vee R)$, $C3: (\neg P \vee \neg Q)$, $C4: (\neg P \vee \neg S)$, $C5: (\neg P \vee \neg Q \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P \vee R)$
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) $(\neg Q \vee \neg S)$
☐ (D) $(\neg Q \vee R \vee \neg S)$

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

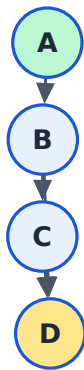
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $F \perp C \mid D$
☐ (B) $D \perp B \mid A, H$
☐ (C) $G \perp F$
☐ (D) $F \perp B \mid E$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(A = T \mid D = F)$ (round off to 3 decimal places).



P(A=T)
0.6

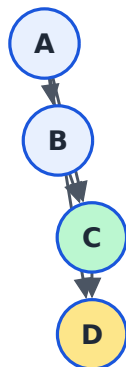
A	P(B=T ...)	B	P(C=T ...)
T	0.2	T	0.2
F	0.7	F	0.9

B	C	P(D=T ...)
T	T	0.4
T	F	0.3
F	T	0.1
F	F	0.5

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{D=F}$. The current state of the chain is $\mathbf{(A=F, B=F, C=F, D=F)}$. The next step resamples $\mathbf{C}$. With what probability is C set to T? (round off to 3 decimal places)



A	P(A=T)
T	0.05

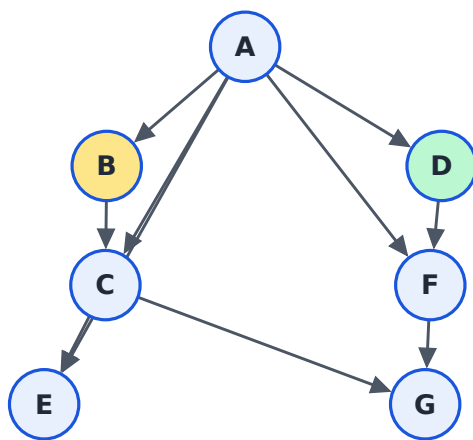
A	P(B=T .)	A	B	P(C=T ...)
T	0.8	T	T	0.25
F	0.95	T	F	0.3
		F	T	0.95
		F	F	0.1

A	C	P(D=T ...)
T	T	0.7
T	F	0.7
F	T	0.3
F	F	0.6

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(D \mid B = T)}$ in the Bayesian network below (Boolean variables; evidence $\mathbf{B = T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{G, C, F, E, A}$. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $\mathbf{B \rightarrow A, E \rightarrow A, A \rightarrow J, A \rightarrow M}$ and CPTs: $\mathbf{P(b) = 0.02, P(e) = 0.01; P(a \mid b, e) = 0.95, P(a \mid b, \neg e) = 0.90, P(a \mid \neg b, e) = 0.30, P(a \mid \neg b, \neg e) = 0.01; P(j \mid a) = 0.8, P(j \mid \neg a) = 0.05; P(m \mid a) = 0.7, P(m \mid \neg a) = 0.02}$. Which of the following are correct (values rounded)?

- ☐ (A) $P(a) \approx 0.0307$
- ☐ (B) $P(b \mid a) \approx 0.588$
- ☐ (C) $P(b \mid a, e) > P(b \mid a)$
- ☐ (D) $P(j \mid b) \approx 0.725$

Answer Key — Mock Test 039

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	B	14	NAT	2	Informed Search	5
2	NAT	1	Uninformed Search	537	15	NAT	2	Uninformed Search	123450
3	NAT	1	Local Search	56	16	NAT	2	Propositional Logic	10
4	NAT	1	Adversarial Search	17	17	MSQ	2	Propositional Logic	C, D
5	MSQ	1	Predicate Logic	A, B, D	18	NAT	2	Predicate Logic	27
6	NAT	1	Predicate Logic	64	19	MSQ	2	Predicate Logic	B, C
7	MCQ	1	Propositional Logic	A	20	MSQ	2	Propositional Logic	C, D
8	NAT	1	Sampling	0.1048 to 0.1052	21	MSQ	2	Conditional Independence	B
9	NAT	1	Bayesian Networks	79	22	NAT	2	Exact Inference	0.625 to 0.629
10	MCQ	1	Conditional Independence	A	23	NAT	2	Sampling	0.161 to 0.165
11	MSQ	2	Informed Search	D	24	NAT	2	Exact Inference	3
12	MSQ	2	Uninformed Search	A, B, C	25	MSQ	2	Bayesian Networks	A, B, D
13	NAT	2	Adversarial Search	7					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: B (B: S, C, E, A, G)

Adjacency lists (alphabetical): A: {E, G, K}; C: {E, F, K, S}; D: {E, F}; E: {A, C, D, K}; F: {C, D, S}; G: {A}; K: {A, C, E}; S: {C, F}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, C, E, A, G**; the path found is $S \rightarrow C \rightarrow E \rightarrow A \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Uninformed Search] Answer: 537

Nodes at depth i are regenerated in every iteration with limit $\geq i$, i.e. $(d + 1 - i)$ times.

$N = 5 \times 3 + 4 \times 9 + 3 \times 27 + 2 \times 81 + 1 \times 243 = 537$.

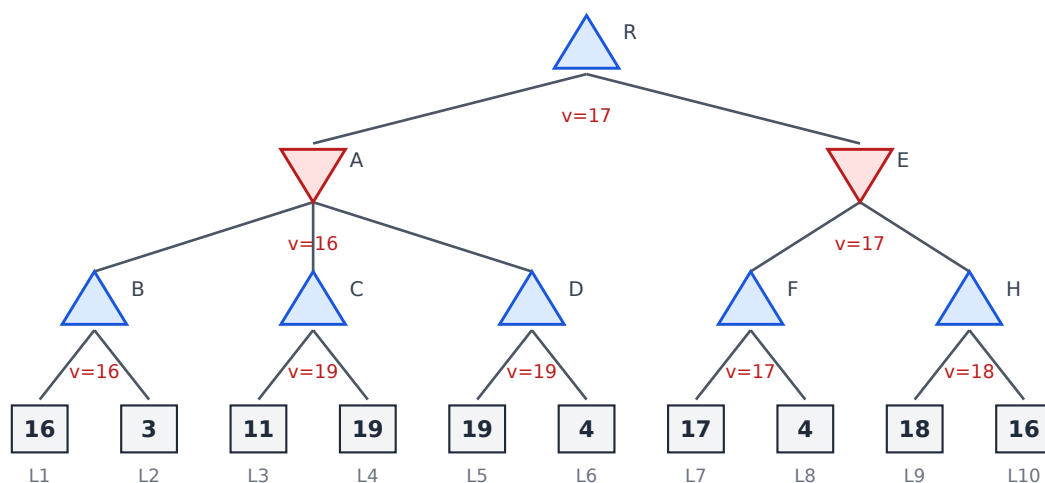
Compare BFS: 363; the overhead of IDS is small for $b \geq 2$.

Q.3 [NAT, 1M · Local Search] Answer: 56

Each of the 8 queens can move to any of the 7 other squares in its column, giving $8 \times 7 = 56$ successors. Steepest-ascent hill climbing evaluates all 56 and picks the best.

Q.4 [NAT, 1M · Adversarial Search] Answer: 17

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 16$, $C = 19$, $D = 19$, $A = 16$, $F = 17$, $H = 18$, $E = 17$. Root value = **17**; MAX's optimal first move is towards E.

Q.5 [MSQ, 1M · Predicate Logic] Answer: A, B, D (A: $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$; B: $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$; D: $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$)

Quantifiers of the same kind commute (A, B). $\exists$ distributes over $\vee$ (D), just as $\forall$ distributes over $\wedge$. (C) Mixed quantifiers do not commute: with $P(x, y) \iff x = y$ on $\{1, 2\}$, $\forall x \exists y P$ is true but $\exists y \forall x P$ is false.

Q.6 [NAT, 1M · Predicate Logic] Answer: 64

View R as an 4×4 0/1 matrix. The sentence says R is reflexive and symmetric: only the $n(n-1)/2$ unordered off-diagonal pairs are free $\Rightarrow 2^{n(n-1)/2}$. With $n = 4$: **64**.

Q.7 [MCQ, 1M · Propositional Logic] Answer: A (A: $\neg q \rightarrow \neg p$)

The contrapositive $\neg q \rightarrow \neg p \equiv q \vee \neg p \equiv p \rightarrow q$. The converse $q \rightarrow p$ and the inverse $\neg p \rightarrow \neg q$ are equivalent to each other but differ from $p \rightarrow q$ at $p=F, q=T$ ($p \rightarrow q$ is T, $q \rightarrow p$ is F). $p \wedge \neg q$ is the negation of $p \rightarrow q$.

Q.8 [NAT, 1M · Sampling] Answer: 0.1048 to 0.1052

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Sm}=T) \times P(\text{Po}=T) \times P(\text{Ca}=F | \text{Po}=T) \times P(\text{XR}=F | \text{Sm}=T, \text{Po}=T)$
 $= 0.25 \times 0.8 \times 0.75 \times 0.7 = \mathbf{0.105} \approx 0.1050$.

Q.9 [NAT, 1M · Bayesian Networks] Answer: 79

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = \text{A: } (2-1) + \text{B: } (3-1) + \text{C: } (3-1) \times 2 + \text{D: } (3-1) \times 2 \times 3 \times 3 + \text{E: } (3-1) \times 2 \times 3 \times 3 = \mathbf{79}$.

Full joint table: $\prod |X| - 1 = 162 - 1 = 161$; saving = 82.

Q.10 [MCQ, 1M · Conditional Independence] Answer: A (A: 5)

Write $P(A, B, C) = P(C) P(A | C) P(B | C)$, using $A \perp B | C$. $P(C)$ needs 1 number, $P(A | C)$ needs 2 (one per value of C), $P(B | C)$ needs 2. Total = 5. $7 = 2^3 - 1$ is the unconstrained full joint; 6 and 4 do not correspond to this factorization.

Q.11 [MSQ, 2M · Informed Search] Answer: D (D: $h(B)$ overestimates the true cost from B to G.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	E	F	G	J	K	S
$h(n)$	8	10	12	8	0	10	11	17
$h^*(n)$	6	12	15	10	0	12	14	21
$h \leq h^*$?	X	✓	✓	✓	✓	✓	✓	✓

Admissible $\iff h(n) \leq h^*(n)$ for every n: **no** (overestimates at B).

Consistent $\iff h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $B \rightarrow G$ ($8 > 6 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

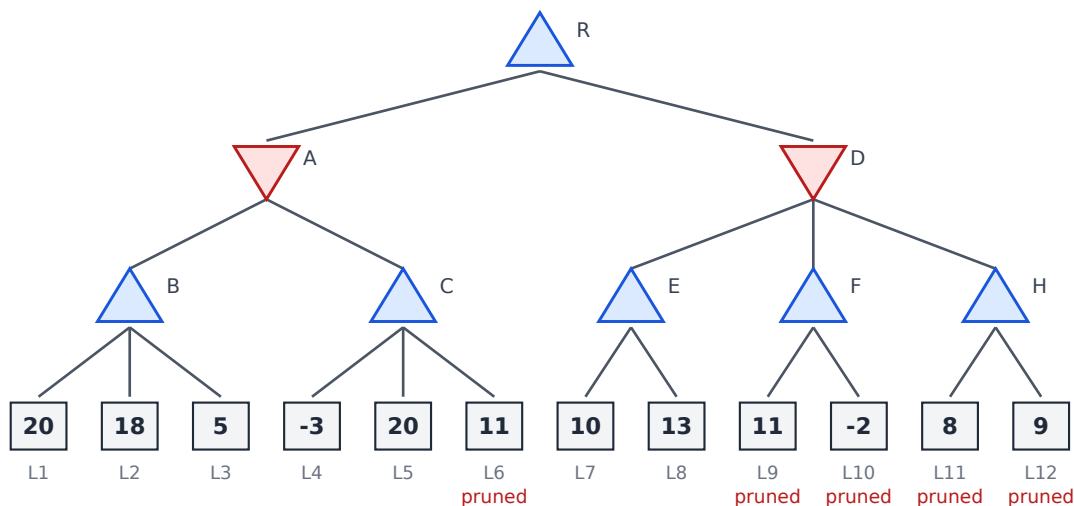
Q.12 [MSQ, 2M · Uninformed Search] Answer: A, B, C (A: IDS is complete; B: IDS is optimal when all step costs are equal; C: The space complexity of IDS is $O(bd)$)

IDS combines DFS's memory use with BFS's level-by-level completeness. (A) is true: with finite b it eventually reaches limit d and finds the goal. (B) is true: like BFS, it finds the shallowest goal first, which is optimal for unit costs. (C) is true: each iteration is a DFS of depth at most d , so the space is $O(bd)$. (D) is false: the total $d \cdot b + (d-1)b^2 + \dots + b^d$ is still $O(b^d)$, because the deepest level dominates and the repeated shallow levels add only a constant factor of about $b/(b-1)$.

Q.13 [NAT, 2M · Adversarial Search] Answer: 7

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $20 \geq \beta = 20 \Rightarrow$ prune remaining children (leaves L6).
- At MIN node D: value $13 \leq \alpha = 20 \Rightarrow$ prune remaining children (leaves L9, L10, L11, L12).



Leaves evaluated (7): L1, L2, L3, L4, L5, L7, L8. Leaves pruned (5): L6, L9, L10, L11, L12. Root value = 20 (same as plain minimax, 20 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] **Answer: 5**

Heuristic values: $h(A)=6$, $h(C)=19$, $h(D)=1$, $h(F)=11$, $h(G)=0$, $h(I)=4$, $h(S)=13$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	A(14), C(27)
2	A (14)	I(15), F(25), C(27)
3	I (15)	D(15), F(25), C(27)
4	D (15)	G(16), F(25), C(27)
5	G (16)	F(25), C(27) ← goal, stop

Path returned: **S** → **A** → **I** → **D** → **G** with cost **16**; nodes expanded = **5** (S, A, I, D, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [NAT, 2M · Uninformed Search] **Answer: 123450**

Nodes at depth i are generated once in each iteration with limit $\geq i$, so they are generated $(d - i + 1)$ times. $N(\text{IDS}) = d \cdot b + (d-1) \cdot b^2 + \dots + 1 \cdot b^d = 5 \cdot 10 + 4 \cdot 100 + 3 \cdot 1000 + 2 \cdot 10000 + 1 \cdot 100000 = 50 + 400 + 3000 + 20000 + 100000 = \mathbf{123450}$. BFS generates 111110 nodes on the same problem, so the overhead of repeating the shallow levels is only about 11%.

Q.16 [NAT, 2M · Propositional Logic] **Answer: 10**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(P \wedge Q) \rightarrow (\neg P \vee \neg R)$	$\neg(\neg S \rightarrow R)$	$((P \wedge Q) \rightarrow (\neg P \vee \neg R)) \rightarrow \neg(\neg S \rightarrow R)$
T	T	T	T	F	F	T
T	T	T	F	F	F	T
T	T	F	T	T	F	F
T	T	F	F	T	T	T
T	F	T	T	T	F	F
T	F	T	F	T	F	F
T	F	F	T	T	F	F
T	F	F	F	T	T	T
F	T	T	T	T	F	F
F	T	T	F	T	F	F
F	T	F	T	T	F	F
F	T	F	F	T	T	T
F	F	T	T	T	F	F
F	F	T	F	T	F	F
F	F	F	T	T	F	F
F	F	F	F	T	T	T

F has 6 models, so it is false in $16 - 6 = 10$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.17 [MSQ, 2M · Propositional Logic] Answer: C, D (C: $\neg((\neg Q \rightarrow P) \wedge \neg\neg P)$; D: $\neg(\neg Q \rightarrow P) \vee \neg P$)

Compare truth tables with $F = (\neg Q \rightarrow P) \rightarrow \neg P$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg Q \rightarrow P) \wedge \neg P$ — differs at $P=F, Q=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(B) $\neg P \rightarrow (\neg Q \rightarrow P)$ — differs at $P=T, Q=T$ ($F=F$, option = T) $\Rightarrow$ not equivalent.

(C) $\neg((\neg Q \rightarrow P) \wedge \neg\neg P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(D) $\neg(\neg Q \rightarrow P) \vee \neg P$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.18 [NAT, 2M · Predicate Logic] Answer: 27

Taking $y = x$ gives $R(x, x) \rightarrow \neg R(x, x)$, so no diagonal pair may be in R (R is irreflexive). For each of the 3 unordered pairs $\{x, y\}$ with $x \neq y$, at most one of (x, y) , (y, x) is in R: 3 choices (neither, one, the other). Total = $1 \times 3^3 = 27$ (asymmetric relations).

Q.19 [MSQ, 2M · Predicate Logic] Answer: B, C (B: $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$; C: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,1), (3,1), (3,2), (4,1), (4,4)\}$.

(A) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **TRUE** — all two-step paths are short-cut.

(C) $\exists x \forall y \neg R(y, x)$: **TRUE** — 3 has no incoming edge.

(D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 1 has no edge to a different element.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Propositional Logic] Answer: C, D (C: $(\neg Q \vee \neg S)$; D: $(\neg Q \vee R \vee \neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(\neg Q \vee \neg S)$ is derivable:

1. resolve $(P \vee \neg Q \vee \neg S)$ and $(\neg P \vee \neg Q)$ on $P \Rightarrow (\neg Q \vee \neg S)$

(D) $(\neg Q \vee R \vee \neg S)$ is derivable:

1. resolve $(P \vee \neg Q \vee \neg S)$ and $(\neg P \vee \neg Q \vee R)$ on $P \Rightarrow (\neg Q \vee R \vee \neg S)$

Q.21 [MSQ, 2M · Conditional Independence] Answer: B (B: $D \perp B \mid A, H$)

Rules: a chain $(\rightarrow B \rightarrow)$ or fork $(\leftarrow B \leftarrow)$ is blocked iff B is observed; a collider $(\rightarrow B \leftarrow)$ is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $F \perp C \mid D$: **FALSE** — the path $F \leftarrow A \rightarrow C$ is active given $\{D\}$.

(B) $D \perp B \mid A, H$: **TRUE** — $D \leftarrow A \rightarrow B$ (blocked: observed fork node A); $D \leftarrow A \rightarrow C \rightarrow E \rightarrow G \leftarrow B$ (blocked: observed fork node A); $D \leftarrow A \rightarrow C \rightarrow H \leftarrow E \rightarrow G \leftarrow B$ (blocked: observed fork node A); $D \leftarrow A \rightarrow F \leftarrow E \rightarrow G \leftarrow B$ (blocked: observed fork node A); $D \leftarrow C \leftarrow A \rightarrow B$ (blocked: observed fork node A); $D \leftarrow C \leftarrow A \rightarrow F \leftarrow E \rightarrow G \leftarrow B$ (blocked: observed fork node A) (+4 more paths, all blocked).

(C) $G \perp F$: **FALSE** — the path $G \leftarrow B \leftarrow A \rightarrow C \rightarrow E \rightarrow F$ is active given $\{\}$.

(D) $F \perp B \mid E$: **FALSE** — the path $F \leftarrow A \rightarrow B$ is active given $\{E\}$.

Q.22 [NAT, 2M · Exact Inference] Answer: 0.625 to 0.629

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B \mid A) \cdot P(C \mid B) \cdot P(D \mid B, C)$.

Inference by enumeration: $P(A \mid e) = \alpha \sum_{\text{hidden}} P(A, e, \text{hidden})$.

Hidden: B, C	A	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.6 \times 0.2 \times 0.2 \times 0.6$	0.0144
T, F	T	$0.6 \times 0.2 \times 0.8 \times 0.7$	0.0672
F, T	T	$0.6 \times 0.8 \times 0.9 \times 0.9$	0.3888
F, F	T	$0.6 \times 0.8 \times 0.1 \times 0.5$	0.024
T, T	F	$0.4 \times 0.7 \times 0.2 \times 0.6$	0.0336
T, F	F	$0.4 \times 0.7 \times 0.8 \times 0.7$	0.1568
F, T	F	$0.4 \times 0.3 \times 0.9 \times 0.9$	0.0972
F, F	F	$0.4 \times 0.3 \times 0.1 \times 0.5$	0.006

$P(A=T, e) = 0.4944$; $P(A=F, e) = 0.2936$; $P(e) = 0.788$.

$$P(A=T | e) = 0.4944 / 0.788 = \mathbf{0.6274} \approx 0.627.$$

Q.23 [NAT, 2M · Sampling] **Answer: 0.161 to 0.165**

Gibbs sampling draws C from $P(C | \text{mb}(C))$, where $\text{mb}(C) = \{A, B, D\}$:

$$P(C | \text{mb}) \propto P(C | \text{parents}(C)) \times \prod_{\text{children } Y} P(Y | \text{parents}(Y)).$$

$$C=T: P(C=T|A=F, B=F) \times P(D=F|A=F, C=T) = 0.1 \times 0.7 = 0.07$$

$$C=F: P(C=F|A=F, B=F) \times P(D=F|A=F, C=F) = 0.9 \times 0.4 = 0.36$$

$$\text{Normalise: } 0.07 / (0.07 + 0.36) = \mathbf{0.1628}.$$

Q.24 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A)$, $f(A, C)$, $f(A, D)$, $f(A, C, E)$, $f(A, D, F)$, $f(C, F, G)$.

Eliminating Z : multiply all factors whose scope contains Z , then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(C, F, G)$	$\{C, F, G\}$	$f(C, F)$
C	$f(A, C)$, $f(A, C, E)$, $f(C, F)$	$\{A, C, E, F\}$	$f(A, E, F)$
F	$f(A, D, F)$, $f(A, E, F)$	$\{A, D, E, F\}$	$f(A, D, E)$
E	$f(A, D, E)$	$\{A, D, E\}$	$f(A, D)$
A	$f(A)$, $f(A)$, $f(A, D)$, $f(A, D)$	$\{A, D\}$	$f(D)$

The largest factor produced has **3** variable(s) (created when eliminating C); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over D and normalise.

Q.25 [MSQ, 2M · Bayesian Networks] **Answer: A, B, D** (A: $P(a) \approx 0.0307$; B: $P(b | a) \approx 0.588$; D: $P(j | b) \approx 0.725$)

$P(a) = 0.02 \times 0.01 \times 0.95 + 0.02 \times 0.99 \times 0.90 + 0.98 \times 0.01 \times 0.30 + 0.98 \times 0.99 \times 0.01 = 0.00019 + 0.01782 + 0.00294 + 0.009702 = 0.030652$ (A true). $P(b, a) = 0.00019 + 0.01782 = 0.01801$, so $P(b | a) = 0.01801 / 0.030652 \approx 0.588$ (B true). $P(b | a, e) = 0.019 / 0.313 \approx 0.061 < 0.588$ (explaining away), so (C) is false. $P(a | b) = 0.01 \times 0.95 + 0.99 \times 0.90 = 0.9005$, so $P(j | b) = 0.9005 \times 0.8 + 0.0995 \times 0.05 = 0.7204 + 0.004975 \approx 0.725$ (D true).

Mock Test 040

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 7 / 13	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

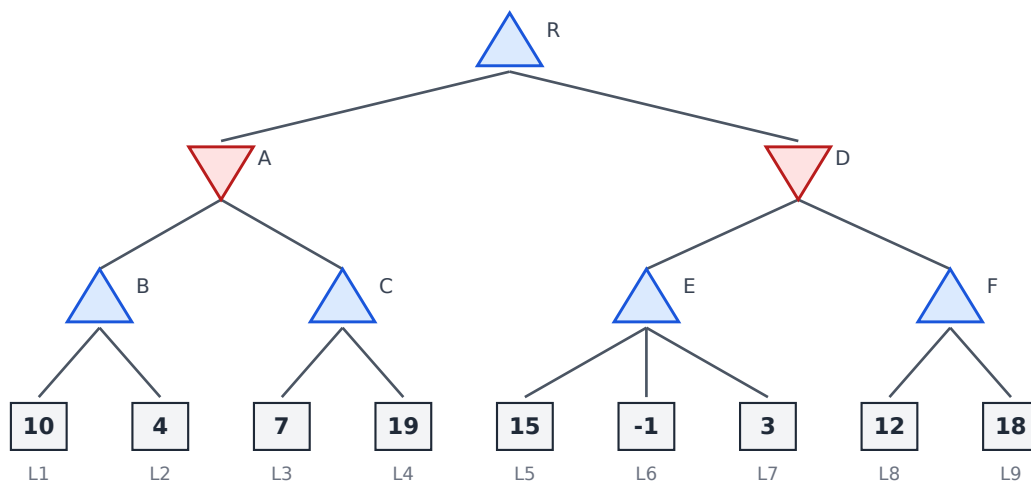
Q.1 [NAT · 1 mark · no negative marking] Uninformed Search

AIMA bounds the time and space of uniform-cost search by $O(b^{1+\lceil C^*/\epsilon \rceil})$, where C^* is the optimal solution cost and ϵ is the minimum step cost. For $C^* = 10$ and $\epsilon = 2$, what is the exponent of b in this bound?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



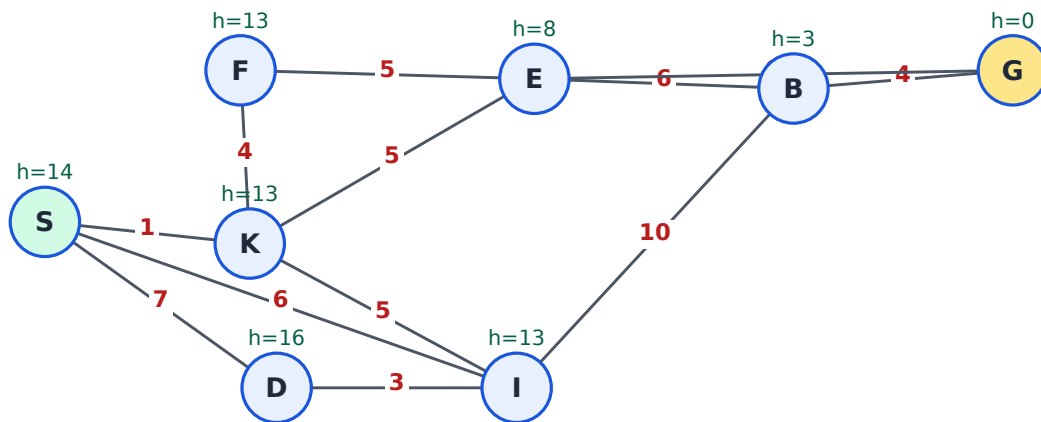
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to D; root value 18
- ☐ (B) Move to A; root value 10
- ☐ (C) Move to D; root value 15
- ☐ (D) Move to D; root value 19

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

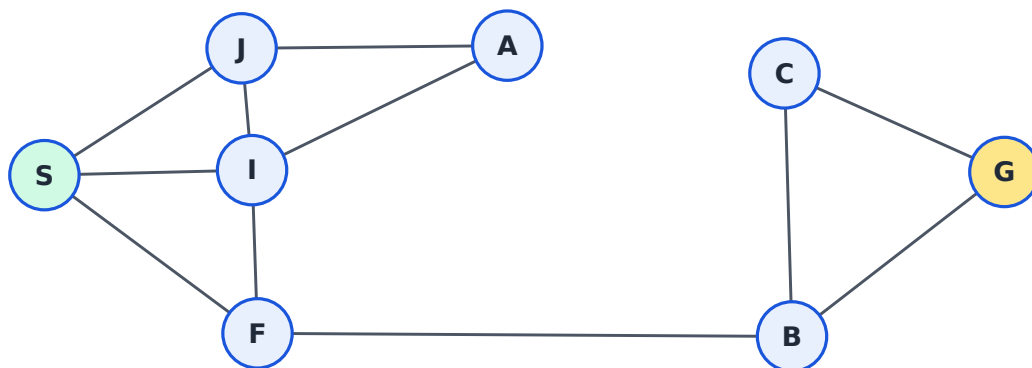
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, B, F, C, G
☐ (B) S, J, I, F, B, G
☐ (C) S, F, I, J, B, A, C, G
☐ (D) S, F, B, C, G

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$\mathbf{F = ((Q \rightarrow P) \wedge (Q \leftrightarrow R)) \rightarrow ((P \rightarrow Q) \wedge R) ?}$$

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

The KB contains the clauses $\neg \text{Man}(x) \vee \text{Mortal}(x)$ and $\text{Man}(\text{Socrates})$. To prove $\text{Mortal}(\text{Socrates})$ by resolution refutation, $\neg \text{Mortal}(\text{Socrates})$ is added. What is the minimum number of resolution steps needed to derive the empty clause?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, D, A** and rules:

R1: $E \wedge A \wedge K \rightarrow F$

R2: $H \wedge F \rightarrow B$

R3: $A \wedge D \rightarrow E$

R4: $F \wedge A \rightarrow B$

R5: $A \rightarrow F$

R6: $D \rightarrow K$

R7: $B \wedge K \rightarrow C$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.8 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are logically equivalent to p?

- ☐ (A) $p \vee (p \wedge q)$
- ☐ (B) $p \wedge (p \vee q)$
- ☐ (C) $(p \vee q) \wedge (p \vee \neg q)$
- ☐ (D) $(p \rightarrow q) \rightarrow q$

Q.9 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over a, b, c does the CNF $(a \vee b \vee c) \wedge (\neg a \vee \neg b) \wedge (\neg b \vee \neg c) \wedge (\neg a \vee \neg c)$ have?

Answer (numerical): _____

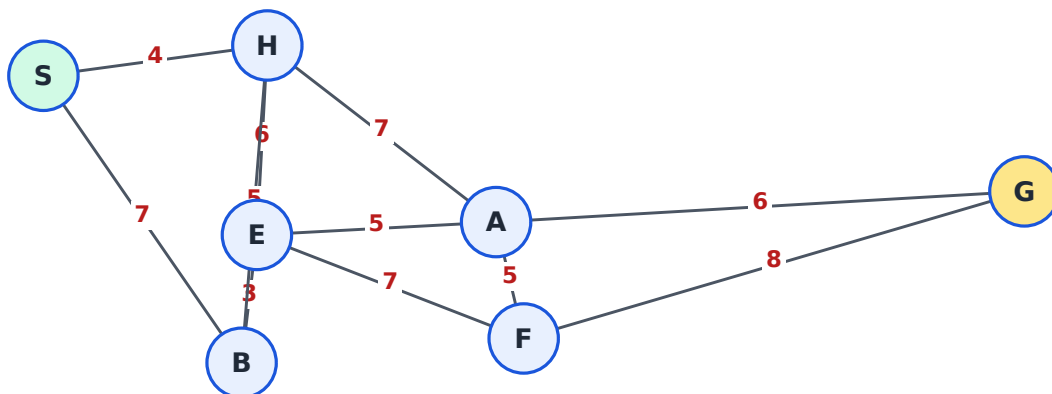
Q.10 [MSQ · 1 mark · no negative marking] Probability for AI

Let A and B be Boolean random variables with $P(b) > 0$ and $P(\neg b) > 0$. Which of the following identities hold for **every** such distribution?

- ☐ (A) $P(a \wedge b) = P(a \mid b) P(b)$
- ☐ (B) $P(a \mid b) = P(b \mid a) P(a) / P(b)$
- ☐ (C) $P(a \mid b) + P(a \mid \neg b) = 1$
- ☐ (D) $P(a) = P(a \mid b) P(b) + P(a \mid \neg b) P(\neg b)$

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search – UCS

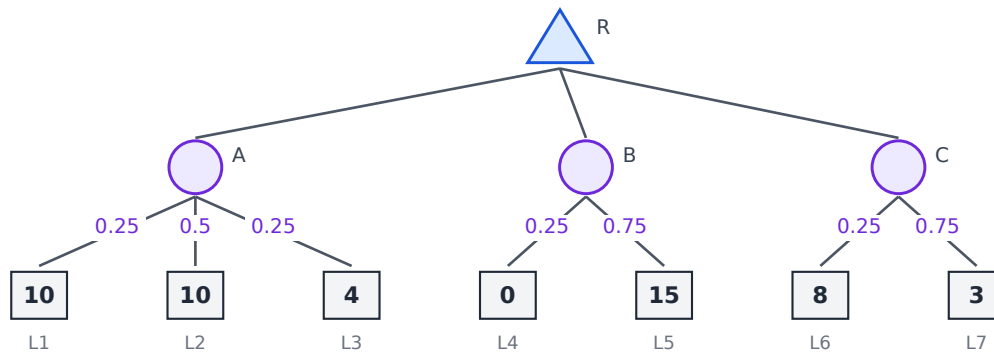
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

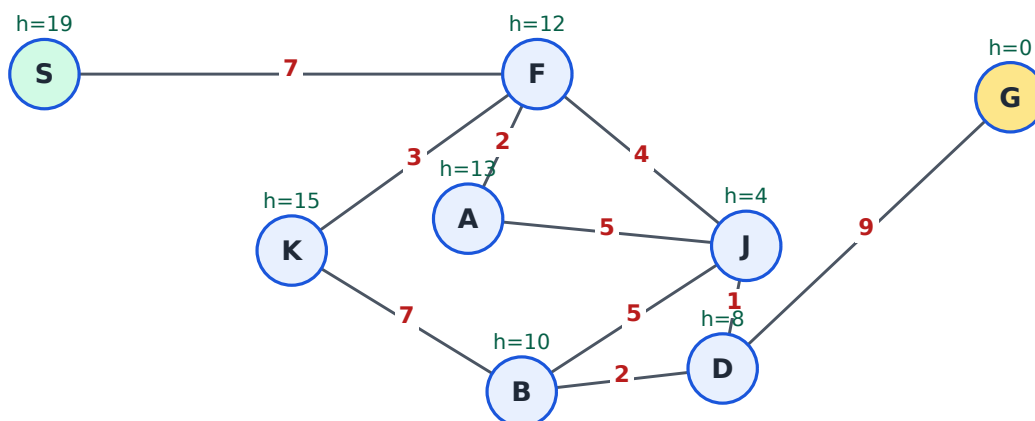
Q.13 [NAT · 2 marks · no negative marking] Local Search

In simulated annealing, a randomly chosen move worsens the objective (which is being maximised) by $\Delta E = -2$ at temperature $T = 4$. With what probability is this move accepted? Give the answer to two decimal places.

Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) $h(B)$ overestimates the true cost from B to G.
- ☐ (C) The edge D-J violates the consistency condition $h(D) \leq c(D,J) + h(J)$.
- ☐ (D) h is admissible.

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: (Q), C2: (P ∨ R ∨ S), C3: (¬Q ∨ R ∨ ¬S), C4: (¬P ∨ ¬Q ∨ R), C5: (P)

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) (¬R ∨ S)
- ☐ (B) The empty clause □ (i.e. the clause set is shown unsatisfiable)
- ☐ (C) (¬S)
- ☐ (D) (R)

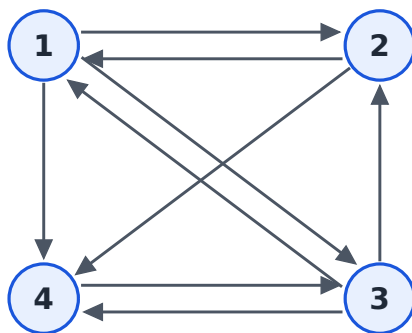
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic

Let KB be a propositional knowledge base and α a sentence, and let $M(\cdot)$ denote the set of models. Which of the following are equivalent to $KB \models \alpha$?

- ☐ (A) $M(KB) \subseteq M(\alpha)$
- ☐ (B) $KB \rightarrow \alpha$ is valid
- ☐ (C) $KB \wedge \neg\alpha$ is unsatisfiable
- ☐ (D) $KB \wedge \alpha$ is satisfiable

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
- ☐ (B) $\exists x \forall y R(x, y)$
- ☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
- ☐ (D) $\exists x \forall y \neg R(x, y)$

Q.18 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following formulas is in conjunctive normal form AND is logically equivalent to $p \leftrightarrow q$?

- ☐ (A) $(\neg p \vee q) \wedge (p \vee \neg q)$
- ☐ (B) $(p \wedge q) \vee (\neg p \wedge \neg q)$
- ☐ (C) $(p \vee q) \wedge (\neg p \vee \neg q)$
- ☐ (D) $(\neg p \vee q) \wedge (p \vee \neg q) \wedge (p \vee q)$

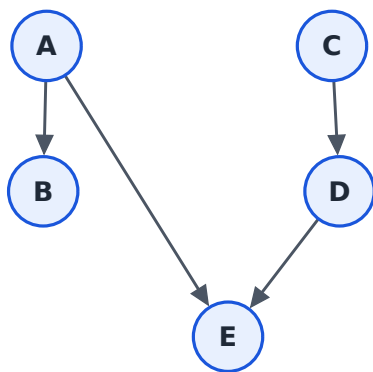
Q.19 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Rejection sampling is used with evidence $S = \text{true}$, $W = \text{true}$. What fraction of the prior samples is expected to be rejected? (Round to 3 decimal places.)

Answer (numerical): _____

Q.20 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = \{0.34, 0.8, 0.98, 0.97, 0.31\}$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T | \text{sampld parent values})$. Which sample is generated?



P(A=T)
0.5

A	P(B=T ...)
T	0.6
F	0.6

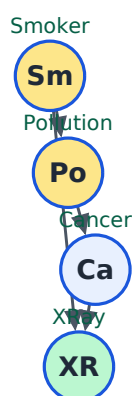
P(C=T)
0.05

C	P(D=T ...)	A	D	P(E=T ...)
T	0.2	T	T	0.9
F	0.2	T	F	0.8
		F	T	0.6
		F	F	0.2

- ☐ (A) A=T, B=T, C=F, D=F, E=T
☐ (B) A=F, B=F, C=F, D=F, E=T
☐ (C) A=T, B=F, C=F, D=F, E=T
☐ (D) A=F, B=T, C=T, D=T, E=F

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T | \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $\mathbf{P(XRay = T | Smoker = T, Pollution = T)}$ (round off to 3 decimal places).



P(Sm=T)
0.2

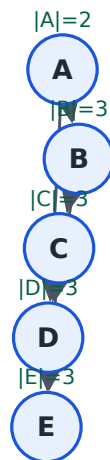
Sm	P(Po=T ..)	Po	P(Ca=T ...)
T	0.3	T	0.4
F	0.25	F	0.5

Sm	Ca	P(XR=T ...)
T	T	0.3
T	F	0.3
F	T	0.7
F	F	0.8

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

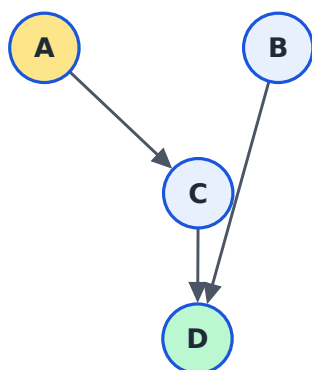
The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{A=F}$. The current state of the chain is $\mathbf{(A=F, B=F, C=T, D=T)}$. The next step resamples $\mathbf{D}$. With what probability is D set to T? (round off to 3 decimal places)



P(A=T)
0.75

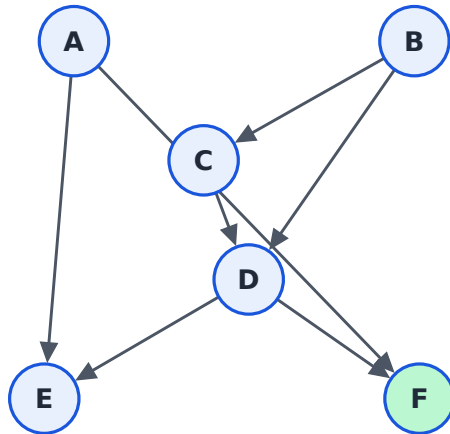
P(B=T)	A	P(C=T ...)
0.95	T	0.7
	F	0.9

B	C	P(D=T ...)
T	T	0.25
T	F	0.1
F	T	0.3
F	F	0.95

Answer (numerical): _____

Q.24 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

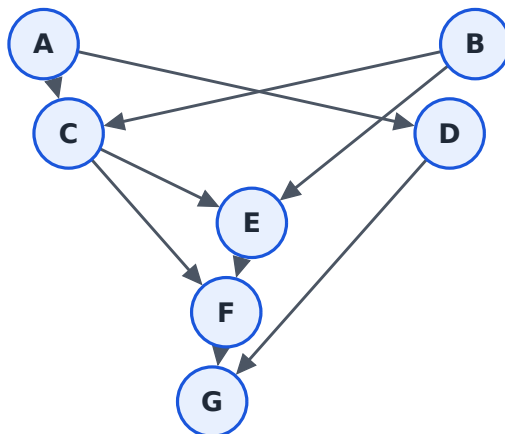
Variable elimination is used to compute $P(\mathbf{F})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, B, E, D, A**. What is the scope of the new factor produced when **E** is summed out?



- ☐ (A) $f(A, D, E)$
☐ (B) $f(A)$
☐ (C) $f(A, D, F)$
☐ (D) $f(A, D)$

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp E \mid A, B$
☐ (B) $G \perp B$
☐ (C) $F \perp A \mid D, E$
☐ (D) $C \perp D \mid A$

Answer Key — Mock Test 040

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	6	14	MSQ	2	Informed Search	C, D
2	MCQ	1	Adversarial Search	C	15	MSQ	2	Propositional Logic	D
3	NAT	1	Informed Search	20	16	MSQ	2	Propositional Logic	A, B, C
4	MCQ	1	Uninformed Search	D	17	MSQ	2	Predicate Logic	A, C
5	NAT	1	Propositional Logic	6	18	MCQ	2	Propositional Logic	A
6	NAT	1	Predicate Logic	2	19	NAT	2	Sampling	0.721 to 0.723
7	NAT	1	Propositional Logic	7	20	MCQ	2	Sampling	C
8	MSQ	1	Propositional Logic	A, B, C	21	NAT	2	Exact Inference	0.298 to 0.302
9	NAT	1	Propositional Logic	3	22	NAT	2	Bayesian Networks	96
10	MSQ	1	Probability for AI	A, B, D	23	NAT	2	Sampling	0.298 to 0.302
11	NAT	2	Uninformed Search	7	24	MCQ	2	Exact Inference	D
12	NAT	2	Adversarial Search	11.24 to 11.26	25	MSQ	2	Conditional Independence	A, D
13	NAT	2	Local Search	0.6 to 0.61					

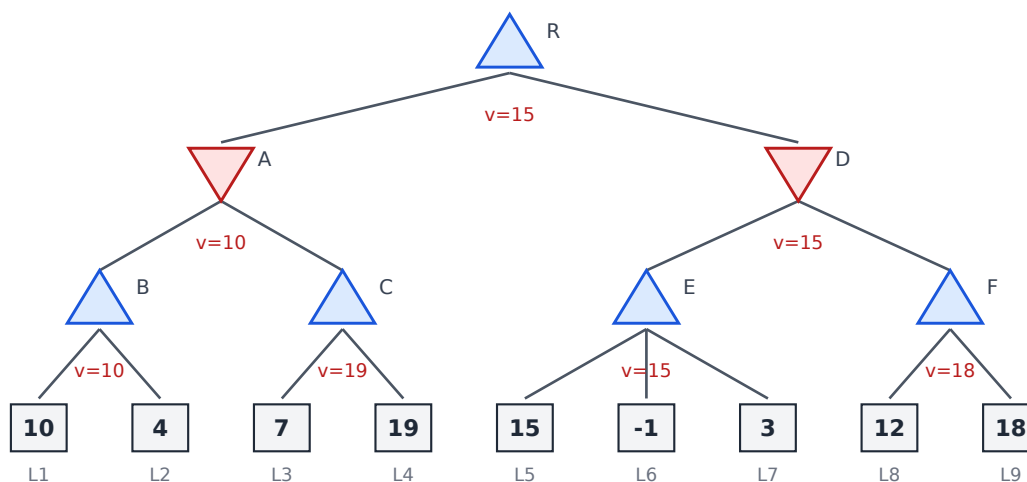
Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] Answer: 6

The exponent is $1 + \lceil C^*/\epsilon \rceil = 1 + \lceil 10/2 \rceil = 1 + 5 = 6$. Each step costs at least ϵ , so an optimal path has at most $C^*/\epsilon = 5$ steps. UCS can then generate one further level when it expands nodes of cost about C^* .

Q.2 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to D; root value 15)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 10$, $C = 19$, $A = 10$, $E = 15$, $F = 18$, $D = 15$. Root value = **15**; MAX's optimal first move is towards D.

Q.3 [NAT, 1M · Informed Search] Answer: 20

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (14)	I(13), K(13), D(16)
2	I (13)	B(3), K(13), D(16)
3	B (3)	G(0), E(8), K(13), D(16)
4	G (0)	E(8), K(13), D(16) ← goal, stop

Path returned: **S** → **I** → **B** → **G**, cost = $6 + 10 + 4 = 20$.

The optimal cost is $h^*(S) = 15$ — greedy search is not optimal here.

Q.4 [MCQ, 1M · Uninformed Search] Answer: D (D: S, F, B, C, G)

Adjacency lists (alphabetical): A: {I, J}; B: {C, F, G}; C: {B, G}; F: {B, I, S}; G: {B, C}; I: {A, F, J, S}; J: {A, I, S}; S: {F, I, J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, F, B, C, G**; the path found is $S \rightarrow F \rightarrow B \rightarrow C \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.5 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$(Q \rightarrow P) \wedge (Q \leftrightarrow R)$	$(P \rightarrow Q) \wedge R$	$((Q \rightarrow P) \wedge (Q \leftrightarrow R)) \rightarrow ((P \rightarrow Q) \wedge R)$
T	T	T	T	T	T
T	T	F	F	F	T
T	F	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	F	T	T
F	F	F	T	F	F

Rows where F = T: **6**.

Q.6 [NAT, 1M · Predicate Logic] Answer: 2

Step 1: resolve $\neg \text{Mortal}(\text{Socrates})$ with $\neg \text{Man}(x) \vee \text{Mortal}(x)$ using $\{x/\text{Socrates}\}$, giving $\neg \text{Man}(\text{Socrates})$. Step 2: resolve $\neg \text{Man}(\text{Socrates})$ with $\text{Man}(\text{Socrates})$, giving $\square$. One step cannot suffice since no two initial clauses are complementary units. So the answer is 2.

Q.7 [NAT, 1M · Propositional Logic] Answer: 7

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $A \wedge D \rightarrow E$, $A \rightarrow F$, $D \rightarrow K$; add E, F, K.

Round 2: fire $B \wedge K \rightarrow C$; add C.

Fixed point: $\{A, B, C, D, E, F, K\}$ — 7 atoms are entailed (facts: A, B, D; newly derived: C, E, F, K). Atoms never derived: H, J.

Q.8 [MSQ, 1M · Propositional Logic] Answer: A, B, C (A: $p \vee (p \wedge q)$; B: $p \wedge (p \vee q)$; C: $(p \vee q) \wedge (p \vee \neg q)$)

(A) and (B) are the absorption laws. (C) By distribution, $(p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q) \equiv p$. (D) $(p \rightarrow q) \rightarrow q \equiv \neg(\neg p \vee q) \vee q \equiv (p \wedge \neg q) \vee q \equiv p \vee q$, which differs from p at $p=F, q=T$.

Q.9 [NAT, 1M · Propositional Logic] Answer: 3

The first clause says at least one of a, b, c is true. The three binary clauses say no two are simultaneously true. Together they encode 'exactly one of a, b, c is true', which has 3 models: (T,F,F), (F,T,F), (F,F,T).

Q.10 [MSQ, 1M · Probability for AI] Answer: A, B, D (A: $P(a \wedge b) = P(a | b) P(b)$; B: $P(a | b) = P(b | a) P(a) / P(b)$; D: $P(a) = P(a | b) P(b) + P(a | \neg b) P(\neg b)$)

(A) is the product rule. (B) is Bayes rule, derived from writing $P(a \wedge b)$ in two ways via the product rule. (D) is the conditioning rule (marginalization of $P(a, B)$ with the product rule). (C) is false: conditional distributions sum to 1 over the values of the conditioned variable ($P(a | b) + P(\neg a | b) = 1$), not over the conditioning variable; e.g. if $A \perp B$ with $P(a) = 0.9$, then $P(a | b) + P(a | \neg b) = 0.9 + 0.9 = 1.8$.

Q.11 [NAT, 2M · Uninformed Search] Answer: 7

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	H(4), B(7)
2	H (4)	B(7), E(10), A(11)
3	B (7)	E(10), A(11)
4	E (10)	A(11), F(17)
5	A (11)	F(16), G(17)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
6	F (16)	G(17)
7	G (17)	← goal, stop

Path returned: **S** → **H** → **A** → **G**, cost = 4 + 7 + 6 = **17**. Nodes expanded (including S and G): **7** (S, H, B, E, A, F, G).

Q.12 [NAT, 2M · Adversarial Search] **Answer: 11.24 to 11.26**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.25 \times 10 + 0.5 \times 10 + 0.25 \times 4 = 8.5$
- $B = 0.25 \times 0 + 0.75 \times 15 = 11.25$
- $C = 0.25 \times 8 + 0.75 \times 3 = 4.25$

Root = $\max(8.5, 11.25, 4.25) = \mathbf{11.25}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.13 [NAT, 2M · Local Search] **Answer: 0.6 to 0.61**

A worsening move is accepted with probability $e^{\Delta E/T} = e^{-2/4} = e^{-0.5} \approx \mathbf{0.6065}$, which rounds to 0.61. Lower temperatures or larger losses make acceptance less likely. Improving moves ($\Delta E > 0$) are always accepted.

Q.14 [MSQ, 2M · Informed Search] **Answer: C, D** (C: The edge D-J violates the consistency condition $h(D) \leq c(D,J) + h(J)$.; D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	D	F	G	J	K	S
$h(n)$	13	10	8	12	0	4	15	19
$h^*(n)$	15	11	9	14	0	10	17	21
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: D→J ($8 > 1 + 4$), A→J ($13 > 5 + 4$), B→J ($10 > 5 + 4$), F→J ($12 > 4 + 4$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: D** (D: (R))

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg R \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) The resolution closure contains 13 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(\neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) (R) is derivable:

1. resolve $(\neg P \vee \neg Q \vee R)$ and (P) on P $\Rightarrow (\neg Q \vee R)$
2. resolve (Q) and $(\neg Q \vee R)$ on Q $\Rightarrow (R)$

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A, B, C** (A: $M(KB) \subseteq M(\alpha)$; B: $KB \rightarrow \alpha$ is valid; C: $KB \wedge \neg \alpha$ is unsatisfiable)

(A) is the definition of entailment: α is true in every model of KB. (B) is the deduction theorem. (C) is the basis of proof by refutation (resolution). (D) is weaker: $KB = \top$, $\alpha = p$ gives $KB \wedge \alpha$ satisfiable, but $KB \not\models \alpha$.

Q.17 [MSQ, 2M · Predicate Logic] **Answer: A, C** (A: $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$; C: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,2), (1,3), (1,4), (2,1), (2,4), (3,1), (3,2), (3,4), (4,3)\}$.

(A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **TRUE** — every two distinct elements are related in some direction.

(B) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(D) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MCQ, 2M · Propositional Logic] **Answer: A** ($A: (\neg p \vee q) \wedge (p \vee \neg q)$)

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv (\neg p \vee q) \wedge (p \vee \neg q)$, which is a CNF. $(p \wedge q) \vee (\neg p \wedge \neg q)$ is equivalent but is in DNF, not CNF. $(p \vee q) \wedge (\neg p \vee \neg q)$ is true exactly when one of p, q is true, i.e. $p \oplus q$. The last option adds the clause $p \vee q$, which removes the model $p=q=F$, leaving only $p \wedge q$.

Q.19 [NAT, 2M · Sampling] **Answer: 0.721 to 0.723**

A sample is kept with probability $P(s, w)$. $P(s, w) = \sum_{c,r} P(c)P(s | c)P(r | c)P(w | s, r)$. $c: 0.5 \times 0.1 \times (0.8 \times 0.99 + 0.2 \times 0.90) = 0.05 \times 0.972 = 0.0486$. $\neg c: 0.5 \times 0.5 \times (0.2 \times 0.99 + 0.8 \times 0.90) = 0.25 \times 0.918 = 0.2295$. $P(s, w) = 0.2781$. Expected rejected fraction = $1 - 0.2781 = 0.7219 \approx 0.722$.

Q.20 [MCQ, 2M · Sampling] **Answer: C** ($C: A=T, B=F, C=F, D=F, E=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.5$; $u = 0.34 < 0.5 \Rightarrow A = T$
- B: $P(B=T | A=T) = 0.6$; $u = 0.8 \geq 0.6 \Rightarrow B = F$
- C: $P(C=T | \text{—}) = 0.05$; $u = 0.98 \geq 0.05 \Rightarrow C = F$
- D: $P(D=T | C=F) = 0.2$; $u = 0.97 \geq 0.2 \Rightarrow D = F$
- E: $P(E=T | A=T, D=F) = 0.8$; $u = 0.31 < 0.8 \Rightarrow E = T$

Sample: **(A=T, B=F, C=F, D=F, E=T)**.

Q.21 [NAT, 2M · Exact Inference] **Answer: 0.298 to 0.302**

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution} | \text{Smoker}) \cdot P(\text{Cancer} | \text{Pollution}) \cdot P(\text{XRay} | \text{Smoker}, \text{Cancer})$.

Inference by enumeration: $P(\text{XRay} | e) = \alpha \sum_{\text{hidden}} P(\text{XRay}, e, \text{hidden})$.

Hidden: Ca	XR	product of CPT entries ($S_m \cdot P_o \cdot C_a \cdot X_R$)	value
T	T	$0.2 \times 0.3 \times 0.4 \times 0.3$	0.0072
F	T	$0.2 \times 0.3 \times 0.6 \times 0.3$	0.0108
T	F	$0.2 \times 0.3 \times 0.4 \times 0.7$	0.0168
F	F	$0.2 \times 0.3 \times 0.6 \times 0.7$	0.0252

$P(\text{XRay}=T, e) = 0.018$; $P(\text{XRay}=F, e) = 0.042$; $P(e) = 0.06$.

$P(\text{XRay}=T | e) = 0.018 / 0.06 = \mathbf{0.3} \approx 0.300$.

Q.22 [NAT, 2M · Bayesian Networks] **Answer: 96**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (3-1) \times 2 + C: (3-1) \times 3 + D: (3-1) \times 2 \times 3 \times 3 + E: (3-1) \times 3 \times 3 = \mathbf{65}$.

Full joint table: $\prod |X| - 1 = 162 - 1 = 161$; saving = 96.

Q.23 [NAT, 2M · Sampling] **Answer: 0.298 to 0.302**

Gibbs sampling draws D from $P(D | \text{mb}(D))$, where $\text{mb}(D) = \{B, C\}$:

$P(D | \text{mb}) \propto P(D | \text{parents}(D)) \times \prod_{\text{children } Y} P(Y | \text{parents}(Y))$.

$D=T: P(D=T | B=F, C=T) = 0.3 = 0.3$

$D=F: P(D=F | B=F, C=T) = 0.7 = 0.7$

Normalise: $0.3 / (0.3 + 0.7) = \mathbf{0.3}$.

Q.24 [MCQ, 2M · Exact Inference] **Answer: D** ($D: f(A, D)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(B), f(B, C), f(B, C, D), f(A, D, E), f(A, D, F)$.

Eliminating Z : multiply all factors whose scope contains Z , then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C), f(B, C, D)$	$\{B, C, D\}$	$f(B, D)$
B	$f(B), f(B, D)$	$\{B, D\}$	$f(D)$
E	$f(A, D, E)$	$\{A, D, E\}$	$f(A, D)$
D	$f(A, D, F), f(D), f(A, D)$	$\{A, D, F\}$	$f(A, F)$
A	$f(A), f(A, F)$	$\{A, F\}$	$f(F)$

The largest factor produced has **2** variable(s) (created when eliminating C); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over F and normalise.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: A, D** (A: $D \perp E \mid A, B$; D: $C \perp D \mid A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $D \perp E \mid A, B$: **TRUE** — $D \leftarrow A \rightarrow C \leftarrow B \rightarrow E$ (blocked: observed fork node A); $D \leftarrow A \rightarrow C \rightarrow E$ (blocked: observed fork node A); $D \leftarrow A \rightarrow C \rightarrow F \leftarrow E$ (blocked: observed fork node A); $D \rightarrow G \leftarrow F \leftarrow C \leftarrow B \rightarrow E$ (blocked: collider G with neither it nor a descendant observed); $D \rightarrow G \leftarrow F \leftarrow C \rightarrow E$ (blocked: collider G with neither it nor a descendant observed); $D \rightarrow G \leftarrow F \leftarrow E$ (blocked: collider G with neither it nor a descendant observed).

(B) $G \perp B$: **FALSE** — the path $G \leftarrow F \leftarrow C \leftarrow B$ is active given $\{\}$.

(C) $F \perp A \mid D, E$: **FALSE** — the path $F \leftarrow C \leftarrow A$ is active given $\{D, E\}$.

(D) $C \perp D \mid A$: **TRUE** — $C \leftarrow A \rightarrow D$ (blocked: observed fork node A); $C \leftarrow B \rightarrow E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed).

Mock Test 041

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 7 / 11	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

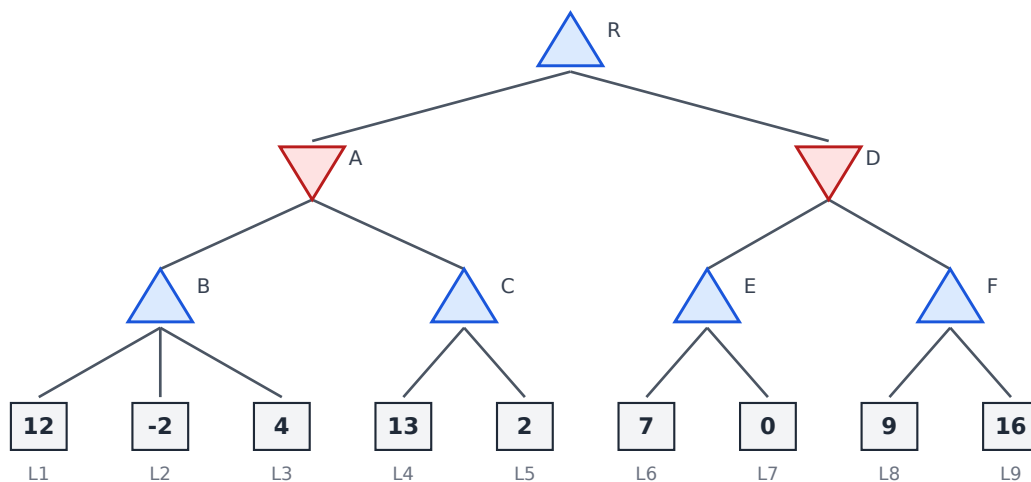
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 5$ (so $b^d = 1024$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



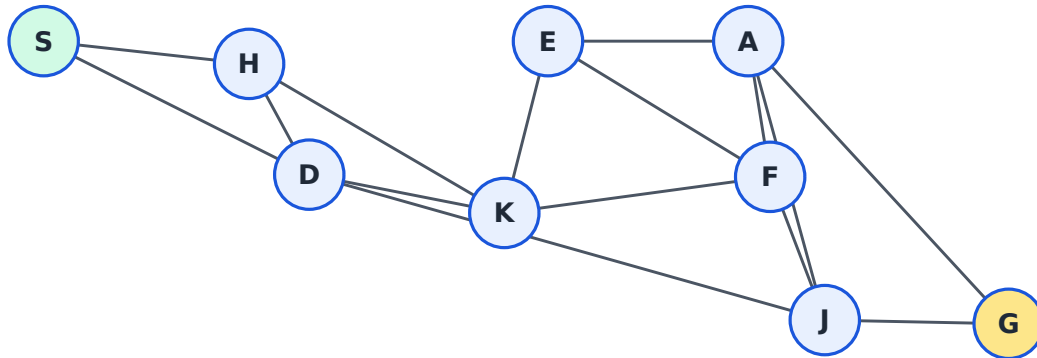
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, D, J, H, K, A, F, G
- ☐ (B) S, H, D, K, J, F, E, G
- ☐ (C) S, D, H, K, E, A, F, J, G
- ☐ (D) S, D, H, J, K, A, F, G

Q.4 [MCQ · 1 mark · negative: -1/3] Informed Search

A heuristic h is **admissible** if, for every node n (where $h^*(n)$ is the true cost of an optimal path from n to a goal),

- ☐ (A) $h(n) \geq h^*(n)$
- ☐ (B) $h(n) \leq h^*(n)$
- ☐ (C) $h(n) \leq c(n, a, n') + h(n')$ for every successor n'
- ☐ (D) $h(n) = 0$

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following inference rules are sound in propositional logic?

- ☐ (A) Modus ponens: from $\alpha \rightarrow \beta$ and α infer β
- ☐ (B) And-elimination: from $\alpha \wedge \beta$ infer α
- ☐ (C) Abduction: from $\alpha \rightarrow \beta$ and β infer α
- ☐ (D) Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((Q \rightarrow R) \leftrightarrow (\neg P \rightarrow \neg Q)) \rightarrow \neg(\neg P \leftrightarrow \neg P)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **A, E** and rules:

- R1: $E \rightarrow J$
 R2: $D \wedge K \wedge A \rightarrow F$
 R3: $C \wedge H \wedge K \rightarrow B$
 R4: $J \wedge A \rightarrow K$
 R5: $E \wedge D \rightarrow C$
 R6: $F \rightarrow J$
 R7: $F \wedge K \rightarrow D$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.8 [MCQ · 1 mark · negative: -1/3] Probability for AI

If n Boolean random variables are mutually (absolutely) independent, how many independent numbers suffice to specify their full joint distribution?

- ☐ (A) n
☐ (B) $2^n - 1$
☐ (C) $2n$
☐ (D) n^2

Q.9 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

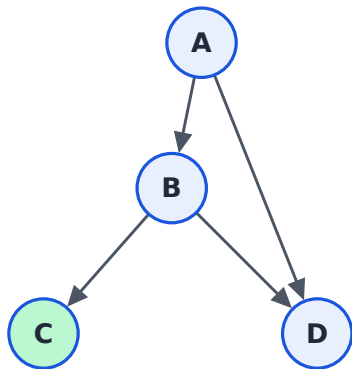
The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(D = T \mid B = T)$, how many of these samples are **rejected**?

#	A	B	C	D
1	T	T	F	T
2	T	F	F	T
3	T	T	F	T
4	T	T	F	T
5	F	T	T	F
6	T	T	F	T
7	T	F	F	T
8	T	T	F	T
9	T	T	F	T
10	T	F	T	F
11	T	T	F	F
12	T	F	T	F
13	T	T	F	T
14	T	F	T	F
15	T	T	F	F

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



$P(A=T)$
0.25

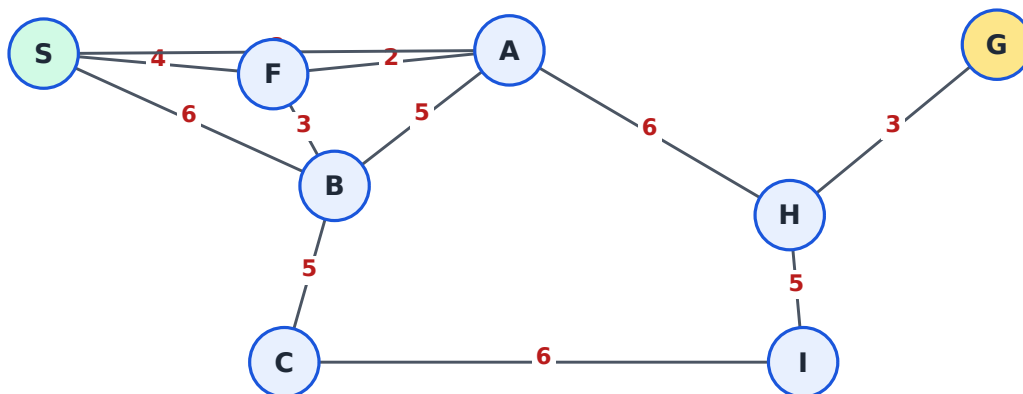
A	$P(B=T \dots)$	B	$P(C=T \dots)$
T	0.8	T	0.05
F	0.6	F	0.2

A	B	$P(D=T \dots)$
T	T	0.8
T	F	0.25
F	T	0.3
F	F	0.2

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

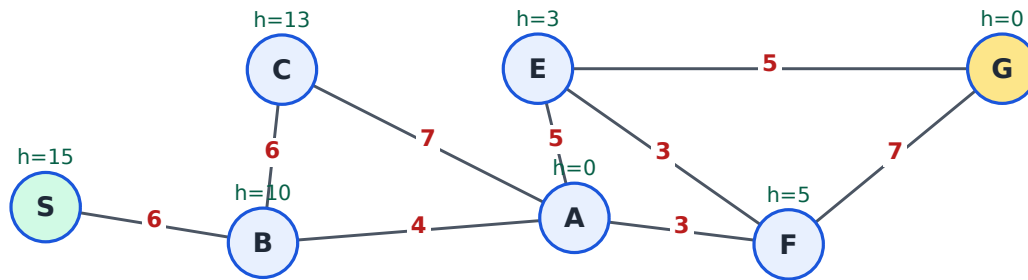
Q.12 [NAT · 2 marks · no negative marking] Problem Formulation

The 4-queens problem uses the improved incremental formulation. A state places k queens ($0 \leq k \leq 4$), one in each of the leftmost k columns, with no two queens attacking each other. Including the empty board ($k = 0$), how many states are in this state space?

Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

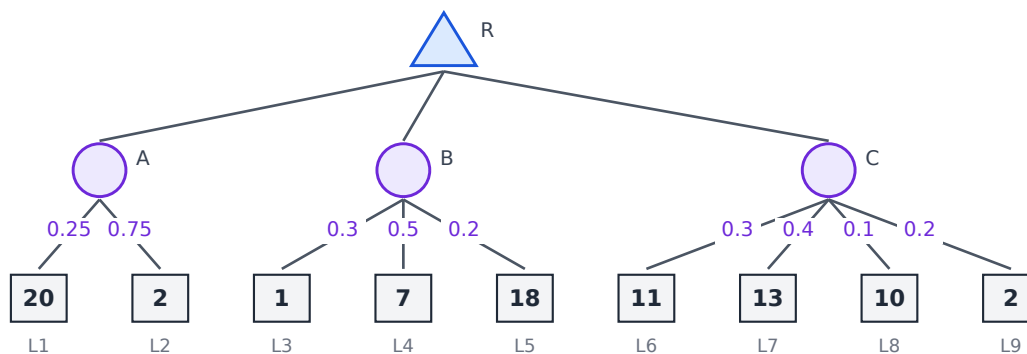
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(F)$ overestimates the true cost from F to G.
☐ (B) The edge C-A violates the consistency condition $h(C) \leq c(C,A) + h(A)$.
☐ (C) h is admissible.
☐ (D) h is consistent (monotone).

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node

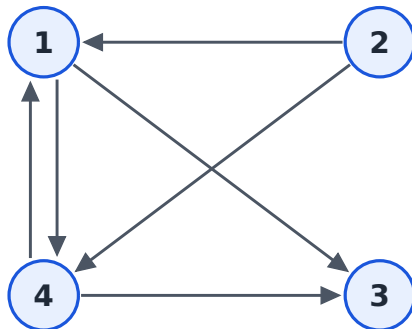
○ CHANCE node (edge = probability)

□ terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\exists x \forall y R(x, y)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$

Q.16 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following sets of connectives is NOT functionally complete?

- ☐ (A) $\{\neg, \wedge\}$
☐ (B) $\{\text{NAND}\}$
☐ (C) $\{\rightarrow, \perp\}$
☐ (D) $\{\wedge, \vee, \leftrightarrow\}$

Q.17 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following statements about resolution is correct?

- ☐ (A) Resolution is complete for deriving every entailed clause directly, without refutation
☐ (B) If a clause set is unsatisfiable, the resolution closure of the set contains the empty clause
☐ (C) If the resolution closure of a clause set does not contain the empty clause, the set may still be unsatisfiable
☐ (D) Resolution can derive the empty clause from a satisfiable set of clauses

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(Q \leftrightarrow P) \leftrightarrow (R \rightarrow Q)$
 (2) $\neg(\neg P \leftrightarrow \neg R)$
 (3) $Q \leftrightarrow (\neg R \vee P)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $(Q \wedge S) \rightarrow \neg S$
☐ (B) $(\neg S \leftrightarrow Q) \rightarrow (Q \rightarrow Q)$
☐ (C) $Q \leftrightarrow P$
☐ (D) $P \leftrightarrow S$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following entailments hold?

- ☐ (A) $\{p \rightarrow q, q \rightarrow r\} \models p \rightarrow r$
☐ (B) $\{p \vee q, \neg p \vee r\} \models q \vee r$
☐ (C) $\{p \rightarrow q, q\} \models p$
☐ (D) $\{p \rightarrow q, \neg q\} \models \neg p$

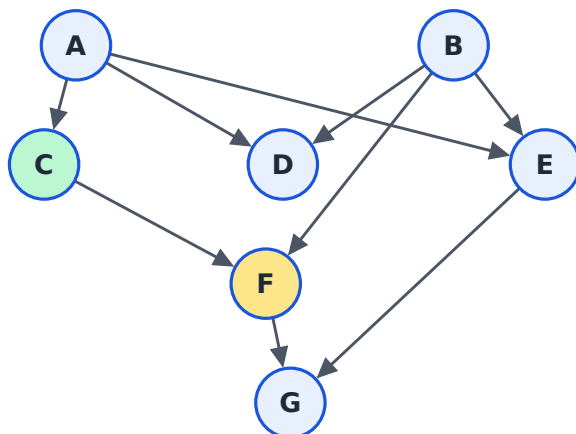
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(P \rightarrow (R \rightarrow Q)) \leftrightarrow ((P \wedge R) \rightarrow Q)$
☐ (B) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$
☐ (C) $((P \rightarrow Q) \rightarrow P) \rightarrow P$
☐ (D) $(R \vee Q) \rightarrow (R \wedge Q)$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

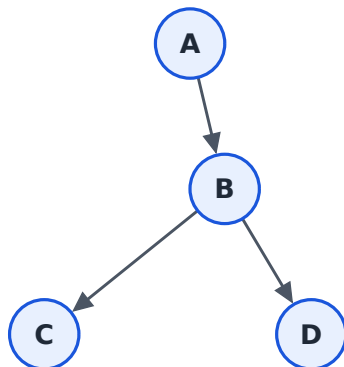
Variable elimination is used to compute $P(\mathbf{C} \mid \mathbf{F} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{F} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, E, B, D, G**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.54}, \mathbf{0.28}, \mathbf{0.45}, \mathbf{0.34} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.6

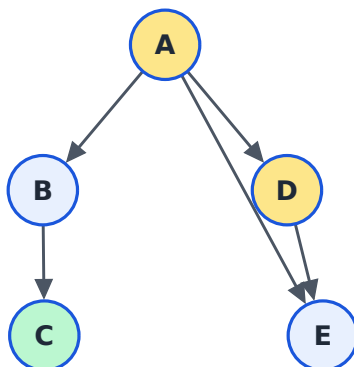
A	P(B=T ...)	B	P(C=T ...)
T	0.7	T	0.95
F	0.1	F	0.2

B	P(D=T ...)
T	0.3
F	0.1

- ☐ (A) A=T, B=F, C=T, D=F
☐ (B) A=T, B=T, C=T, D=F
☐ (C) A=F, B=T, C=T, D=F
☐ (D) A=F, B=T, C=F, D=T

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $\mathbf{P(C = T \mid D = F, A = T)}$ (round off to 3 decimal places).



P(A=T)
0.75

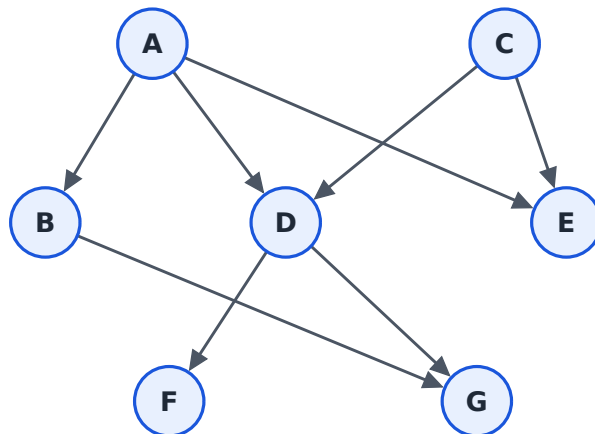
A	P(B=T ...)	B	P(C=T ...)
T	0.3	T	0.9
F	0.5	F	0.2

A	P(D=T ...)	A	D	P(E=T ...)
T	0.4	T	T	0.7
F	0.75	T	F	0.75
		F	T	0.9
		F	F	0.6

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $F \perp A \mid D, E$
☐ (B) $C \perp A \mid D, G$
☐ (C) $B \perp C$
☐ (D) $F \perp A \mid D$

Q.25 [MCQ · 2 marks · negative: -2/3] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. For $P(B \mid j, m)$, the elimination order A then E is used. After summing out A, which factors remain to be multiplied when E is eliminated?

- ☐ (A) $P(E)$ and $f_A(B, E)$
☐ (B) $P(E)$, $P(B)$ and $f_A(B, E)$
☐ (C) $P(E)$ and $f_A(B, E, J, M)$
☐ (D) $P(E)$, $P(A \mid B, E)$ and $f_A(B)$

Answer Key — Mock Test 041

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	79	14	NAT	2	Adversarial Search	9.89 to 9.91
2	NAT	1	Adversarial Search	12	15	MSQ	2	Predicate Logic	A
3	MCQ	1	Uninformed Search	D	16	MCQ	2	Propositional Logic	D
4	MCQ	1	Informed Search	B	17	MCQ	2	Propositional Logic	B
5	MSQ	1	Propositional Logic	A, B, D	18	MSQ	2	Propositional Logic	B, C
6	NAT	1	Propositional Logic	6	19	MSQ	2	Propositional Logic	A, B, D
7	NAT	1	Propositional Logic	4	20	MSQ	2	Propositional Logic	A, C
8	MCQ	1	Probability for AI	A	21	NAT	2	Exact Inference	4
9	NAT	1	Sampling	5	22	MCQ	2	Sampling	B
10	NAT	1	Exact Inference	0.101 to 0.105	23	NAT	2	Exact Inference	0.408 to 0.412
11	NAT	2	Uninformed Search	15	24	MSQ	2	Conditional Independence	A, C, D
12	NAT	2	Problem Formulation	17	25	MCQ	2	Variable Elimination	A
13	MSQ	2	Informed Search	B, C					

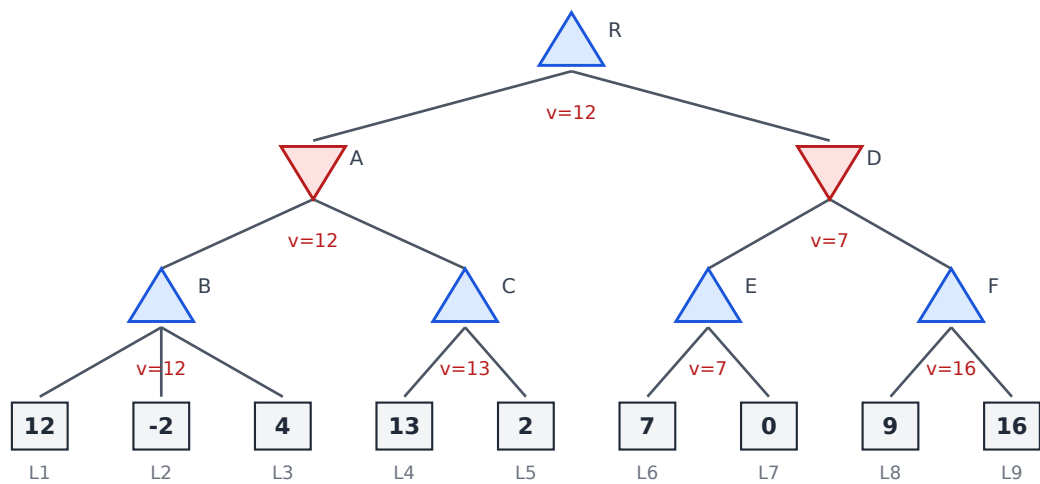
Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 79

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.
 $= 4^3 + 4^2 - 1 = 64 + 16 - 1 = \mathbf{79}$, versus 1024 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.2 [NAT, 1M · Adversarial Search] Answer: 12

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 12, C = 13, A = 12, E = 7, F = 16, D = 7. Root value = **12**; MAX's optimal first move is towards A.

Q.3 [MCQ, 1M · Uninformed Search] Answer: D (D: S, D, H, J, K, A, F, G)

Adjacency lists (alphabetical): A: {E, F, G, J}; D: {H, J, K, S}; E: {A, F, K}; F: {A, E, J, K}; G: {A, J}; H: {D, K, S}; J: {A, D, F, G}; K: {D, E, F, H}; S: {D, H}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[D, H]
2	D	[H, J, K]
3	H	[J, K]
4	J	[K, A, F, G]
5	K	[A, F, G, E]
6	A	[F, G, E]
7	F	[G, E]
8	G	[E] ← goal popped, stop

Expansion order: **S, D, H, J, K, A, F, G**. The path returned is S → D → J → G with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [MCQ, 1M · Informed Search] Answer: B (B: $h(n) \leq h^*(n)$)

An admissible heuristic never overestimates the cost to reach a goal: **$h(n) \leq h^*(n)$** . (A) describes a pessimistic, inadmissible heuristic. (C) is the definition of consistency, which is a stronger condition. (D) is one particular admissible heuristic, not the definition.

Q.5 [MSQ, 1M · Propositional Logic] Answer: A, B, D (A: Modus ponens: from $\alpha \rightarrow \beta$ and α infer β ; B:

And-elimination: from $\alpha \wedge \beta$ infer α ; D: Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$)

Modus ponens, and-elimination and resolution are truth-preserving: whenever their premises are true, their conclusion is true. For resolution, if β is true then γ must be true, else α must be true. Abduction is unsound: $\alpha=F, \beta=T$ makes both premises true and the conclusion false.

Q.6 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$(Q \rightarrow R) \leftrightarrow (\neg P \rightarrow \neg Q)$	$\neg(\neg P \leftrightarrow \neg P)$	$((Q \rightarrow R) \leftrightarrow (\neg P \rightarrow \neg Q)) \rightarrow \neg(\neg P \leftrightarrow \neg P)$
T	T	T	T	F	F
T	T	F	F	F	T
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	F	F	T
F	T	F	T	F	F
F	F	T	T	F	F
F	F	F	T	F	F

F has 2 models, so it is false in $8 - 2 = 6$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 4**

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $E \rightarrow J$; add J.

Round 2: fire $J \wedge A \rightarrow K$; add K.

Fixed point: {A, E, J, K} — 4 atoms are entailed (facts: A, E; newly derived: J, K). Atoms never derived: B, C, D, F, H.

Q.8 [MCQ, 1M · Probability for AI] **Answer: A** (A: n)

Under mutual independence $P(x_1, \dots, x_n) = \prod P(x_i)$, and each Boolean marginal needs one number $P(x_i = \text{true})$. Hence n numbers suffice. $2^n - 1$ is the count with no independence; $2n$ double-counts $P(\neg x_i) = 1 - P(x_i)$; n^2 has no basis.

Q.9 [NAT, 1M · Sampling] **Answer: 5**

Rejection sampling keeps only samples consistent with the evidence $B = T$: samples 1, 3, 4, 5, 6, 8, 9, 11, 13, 15 (10 accepted, 5 rejected).

Among the accepted samples, $D = T$ in 7. Estimate = $7/10 = 0.7$.

(For this network the exact value is 0.6965; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.10 [NAT, 1M · Exact Inference] **Answer: 0.101 to 0.105**

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T, D=T	0.008
A=T, B=T, D=F	0.002
A=T, B=F, D=T	0.0025
A=T, B=F, D=F	0.0075
A=F, B=T, D=T	0.00675
A=F, B=T, D=F	0.01575
A=F, B=F, D=T	0.012
A=F, B=F, D=F	0.048

Sum = **0.1025** ≈ 0.103 .

Q.11 [NAT, 2M · Uninformed Search] **Answer: 15**

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	F(4), B(6), A(8)
2	F (4)	A(6), B(6)
3	A (6)	B(6), H(12)
4	B (6)	C(11), H(12)
5	C (11)	H(12), I(17)
6	H (12)	G(15), I(17)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
7	G (15)	I(17) ← goal, stop

Path returned: **S → F → A → H → G**, cost = 4 + 2 + 6 + 3 = **15**. Nodes expanded (including S and G): **7** (S, F, A, B, C, H, G).

Q.12 [NAT, 2M · Problem Formulation] **Answer: 17**

$k = 0$: 1 state. $k = 1$: 4 states. $k = 2$: the pairs of rows (r_1, r_2) with $|r_1 - r_2| \geq 2$ are (1,3), (1,4), (2,4), (3,1), (4,1), (4,2), so 6 states. $k = 3$: extending these without attack gives (1,4,2), (2,4,1), (3,1,4), (4,1,3), so 4 states. $k = 4$: the two solutions (2,4,1,3) and (3,1,4,2). Total = 1+4+6+4+2 = **17**. For 8-queens, the same formulation gives AIMA's 2057 states.

Q.13 [MSQ, 2M · Informed Search] **Answer: B, C** (B: The edge C-A violates the consistency condition $h(C) \leq c(C,A) + h(A)$.; C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	C	E	F	G	S
$h(n)$	0	10	13	3	5	0	15
$h^*(n)$	10	14	17	5	7	0	20
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $F \rightarrow A$ ($5 > 3 + 0$), $B \rightarrow A$ ($10 > 4 + 0$), $C \rightarrow A$ ($13 > 7 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [NAT, 2M · Adversarial Search] **Answer: 9.89 to 9.91**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.25 \times 20 + 0.75 \times 2 = 6.5$
- $B = 0.3 \times 1 + 0.5 \times 7 + 0.2 \times 18 = 7.4$
- $C = 0.3 \times 11 + 0.4 \times 13 + 0.1 \times 10 + 0.2 \times 2 = 9.9$

Root = $\max(6.5, 7.4, 9.9) = \mathbf{9.9}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [MSQ, 2M · Predicate Logic] **Answer: A** (A: $\exists x \forall y \neg R(x, y)$)

$R = \{(1,3), (1,4), (2,1), (2,4), (4,1), (4,3)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 3 has no outgoing edge.

(B) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 3 has no edge to a different element.

(D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(2,3)$ nor $R(3,2)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.16 [MCQ, 2M · Propositional Logic] **Answer: D** (D: $\{\wedge, \vee, \leftrightarrow\}$)

$\wedge$, $\vee$ and $\leftrightarrow$ all output T when every input is T ($T \wedge T = T$, $T \vee T = T$, $T \leftrightarrow T = T$). Hence every formula built from them is T under the all-true assignment, so $\neg p$ cannot be expressed. $\{\neg, \wedge\}$ is complete by De Morgan. NAND alone is complete ($\neg p = p \text{ NAND } p$). With $\rightarrow$ and $\perp$ we get $\neg p \equiv p \rightarrow \perp$ and $p \vee q \equiv \neg p \rightarrow q$, so $\{\rightarrow, \perp\}$ is complete.

Q.17 [MCQ, 2M · Propositional Logic] **Answer: B** (B: If a clause set is unsatisfiable, the resolution closure of the set contains the empty clause)

The ground resolution theorem: a clause set is unsatisfiable iff its resolution closure contains $\square$. So B is correct and C is false. D would contradict soundness. A is false: resolution is only refutation-complete; e.g. from $\{p\}$ it cannot derive the entailed clause $p \vee q$ directly.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: B, C** (B: $(\neg S \leftrightarrow Q) \rightarrow (Q \rightarrow Q)$; C: $Q \leftrightarrow P$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
T	T	F	T	T
T	T	F	F	T

(A) $(Q \wedge S) \rightarrow \neg S$ is false in the KB-model $P=T, Q=T, R=F, S=T \Rightarrow$ not entailed.

(B) $(\neg S \leftrightarrow Q) \rightarrow (Q \rightarrow Q)$ is true in all models of KB $\Rightarrow$ **entailed**.

(C) $Q \leftrightarrow P$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $P \leftrightarrow S$ is false in the KB-model $P=T, Q=T, R=F, S=F \Rightarrow$ not entailed.

Q.19 [MSQ, 2M · Propositional Logic] Answer: A, B, D (A: $\{p \rightarrow q, q \rightarrow r\} \models p \rightarrow r$; B: $\{p \vee q, \neg p \vee r\} \models q \vee r$; D: $\{p \rightarrow q, \neg q\} \models \neg p$)

(A) Hypothetical syllogism: every model of both premises with $p=T$ has $q=T$ and hence $r=T$. (B) This is exactly the resolution rule on p : if $p=T$ then r must be T ; if $p=F$ then q must be T . (C) Fails: $p=F, q=T$ satisfies both premises but falsifies p (affirming the consequent). (D) Modus tollens: if p were T then q would be T , contradicting $\neg q$. Truth tables confirm A, B, D.

Q.20 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $(P \rightarrow (R \rightarrow Q)) \leftrightarrow ((P \wedge R) \rightarrow Q)$; C: $((P \rightarrow Q) \rightarrow P) \rightarrow P$)

(A) $(P \rightarrow (R \rightarrow Q)) \leftrightarrow ((P \wedge R) \rightarrow Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(B) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$: **not valid** — counter-model $P=T, R=F$.

(C) $((P \rightarrow Q) \rightarrow P) \rightarrow P$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $(R \vee Q) \rightarrow (R \wedge Q)$: **not valid** — counter-model $Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.21 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(B), f(A, C), f(A, B, D), f(A, B, E), f(B, C), f(E, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A), f(A, C), f(A, B, D), f(A, B, E)$	$\{A, B, C, D, E\}$	$f(B, C, D, E)$
E	$f(E, G), f(B, C, D, E)$	$\{B, C, D, E, G\}$	$f(B, C, D, G)$
B	$f(B), f(B, C), f(B, C, D, G)$	$\{B, C, D, G\}$	$f(C, D, G)$
D	$f(C, D, G)$	$\{C, D, G\}$	$f(C, G)$
G	$f(C, G)$	$\{C, G\}$	$f(C)$

The largest factor produced has **4** variable(s) (created when eliminating A); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over C and normalise.

Q.22 [MCQ, 2M · Sampling] Answer: B (B: $A=T, B=T, C=T, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.6$; $u = 0.54 < 0.6 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.7$; $u = 0.28 < 0.7 \Rightarrow B = T$
- C: $P(C=T \mid B=T) = 0.95$; $u = 0.45 < 0.95 \Rightarrow C = T$
- D: $P(D=T \mid B=T) = 0.3$; $u = 0.34 \geq 0.3 \Rightarrow D = F$

Sample: **$\langle A=T, B=T, C=T, D=F \rangle$** .

Q.23 [NAT, 2M · Exact Inference] Answer: 0.408 to 0.412

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B \mid A) \cdot P(C \mid B) \cdot P(D \mid A) \cdot P(E \mid A, D)$.

Inference by enumeration: $P(C \mid e) = \alpha \sum_{\text{hidden}} P(C, e, \text{hidden})$.

Hidden: B, E	C	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T	T	$0.75 \times 0.3 \times 0.9 \times 0.6 \times 0.75$	0.09112
T, F	T	$0.75 \times 0.3 \times 0.9 \times 0.6 \times 0.25$	0.03037
F, T	T	$0.75 \times 0.7 \times 0.2 \times 0.6 \times 0.75$	0.04725
F, F	T	$0.75 \times 0.7 \times 0.2 \times 0.6 \times 0.25$	0.01575
T, T	F	$0.75 \times 0.3 \times 0.1 \times 0.6 \times 0.75$	0.01012
T, F	F	$0.75 \times 0.3 \times 0.1 \times 0.6 \times 0.25$	0.00337
F, T	F	$0.75 \times 0.7 \times 0.8 \times 0.6 \times 0.75$	0.189
F, F	F	$0.75 \times 0.7 \times 0.8 \times 0.6 \times 0.25$	0.063

$P(C=T, e) = 0.1845$; $P(C=F, e) = 0.2655$; $P(e) = 0.45$.

$P(C=T \mid e) = 0.1845 / 0.45 = \mathbf{0.41} \approx 0.410$.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: A, C, D** (A: $F \perp A \mid D, E$; C: $B \perp C$; D: $F \perp A \mid D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $F \perp A \mid D, E$: **TRUE** — $F \leftarrow D \leftarrow A$ (blocked: observed chain node D); $F \leftarrow D \leftarrow C \rightarrow E \leftarrow A$ (blocked: observed chain node D); $F \leftarrow D \rightarrow G \leftarrow B \leftarrow A$ (blocked: observed fork node D).

(B) $C \perp A \mid D, G$: **FALSE** — the path $C \rightarrow D \leftarrow A$ is active given $\{D, G\}$.

(C) $B \perp C$: **TRUE** — $B \leftarrow A \rightarrow D \leftarrow C$ (blocked: collider D with neither it nor a descendant observed); $B \leftarrow A \rightarrow E \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow G \leftarrow D \leftarrow A \rightarrow E \leftarrow C$ (blocked: collider G with neither it nor a descendant observed); $B \rightarrow G \leftarrow D \leftarrow C$ (blocked: collider G with neither it nor a descendant observed).

(D) $F \perp A \mid D$: **TRUE** — $F \leftarrow D \leftarrow A$ (blocked: observed chain node D); $F \leftarrow D \leftarrow C \rightarrow E \leftarrow A$ (blocked: observed chain node D); $F \leftarrow D \rightarrow G \leftarrow B \leftarrow A$ (blocked: observed fork node D).

Q.25 [MCQ, 2M · Variable Elimination] **Answer: A** (A: $P(E)$ and $f_A(B, E)$)

Initially the factors are $P(B)$, $P(E)$, $P(A \mid B, E)$, $f(A) = P(j \mid A)$, $g(A) = P(m \mid A)$. Summing out A multiplies the three factors containing A and yields $f_A(B, E)$. Eliminating E then multiplies only the factors mentioning E: $P(E)$ and $f_A(B, E)$, giving $f_E(B)$. $P(B)$ does not mention E so it is left aside until the end; J and M are evidence, not dimensions; $P(A \mid B, E)$ was already consumed.

Mock Test 042

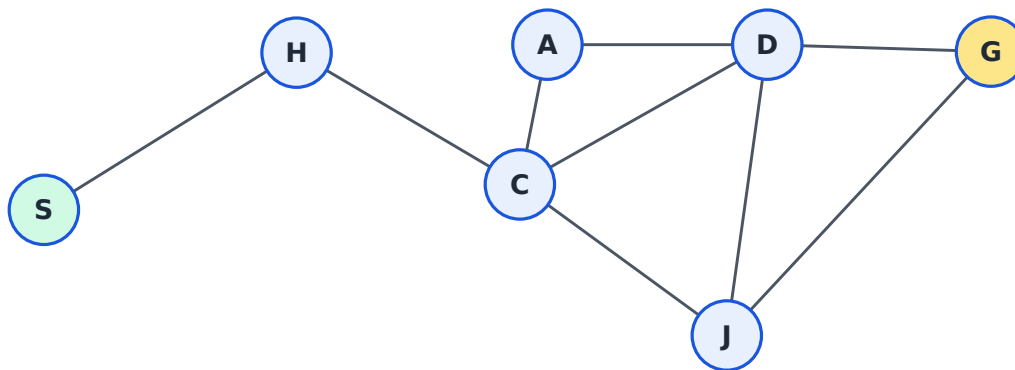
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 6 / 14	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

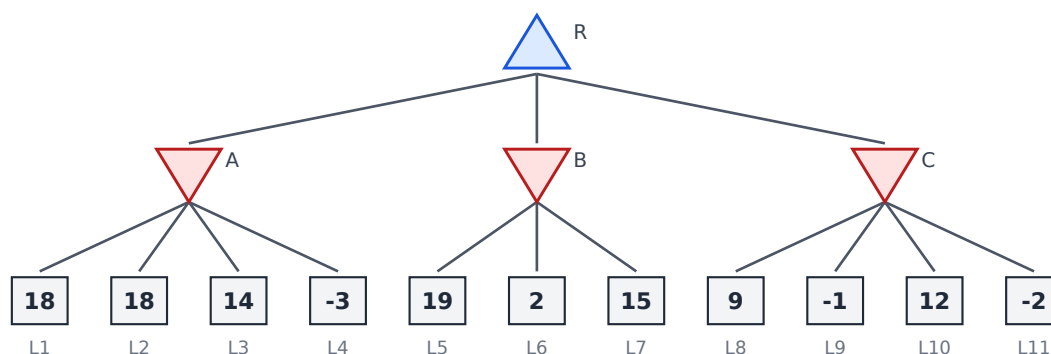
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, H, C, A, D, G
- ☐ (B) S, H, C, A, D, J, G
- ☐ (C) S, H, C, J, G
- ☐ (D) S, C, H, A, D, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search

A chess evaluation function is the weighted linear function $\text{Eval} = 9 \cdot \Delta q + 5 \cdot \Delta r + 3 \cdot \Delta b + 1 \cdot \Delta p$, where each Δ is White's count minus Black's count of queens, rooks, bishops and pawns. In a position, the queens are equal, White has one extra rook, Black has one extra bishop, and White has two extra pawns. What is Eval?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 5$ (so $b^d = 243$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

Of the 16 possible binary Boolean connectives $\circ$ (functions $\{T, F\}^2 \rightarrow \{T, F\}$), how many are commutative, i.e. satisfy $p \circ q \equiv q \circ p$?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((R \rightarrow R) \leftrightarrow (\neg R \leftrightarrow \neg R)) \wedge (\neg R \rightarrow (\neg P \wedge Q)) ?$$

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic

Let the domain be $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \forall y (R(x, y) \rightarrow R(y, x))$?

Answer (numerical): _____

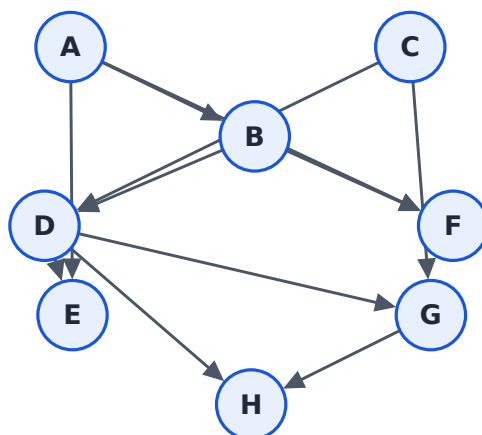
Q.8 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\exists x \forall y R(x, y)$ true?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

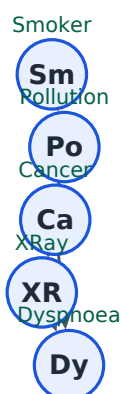
How many nodes are in the Markov blanket of **D** in the Bayesian network below?



Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR, Dy. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay, Dy = Dyspnoea.) What is the probability that a single run produces the sample **(Sm=F, Po=F, Ca=F, XR=F, Dy=F)**? (round off to 4 decimal places)



P(Sm=T)
0.2

Sm	P(Po=T ...)	Po	P(Ca=T ...)
T	0.05	T	0.3
F	0.25	F	0.75

Po	Ca	P(XR=T ...)	Ca	XR	P(Dy=T ...)
T	T	0.7	T	T	0.5
T	F	0.75	T	F	0.3
F	T	0.7	F	T	0.95
F	F	0.8	F	F	0.2

Answer (numerical): _____

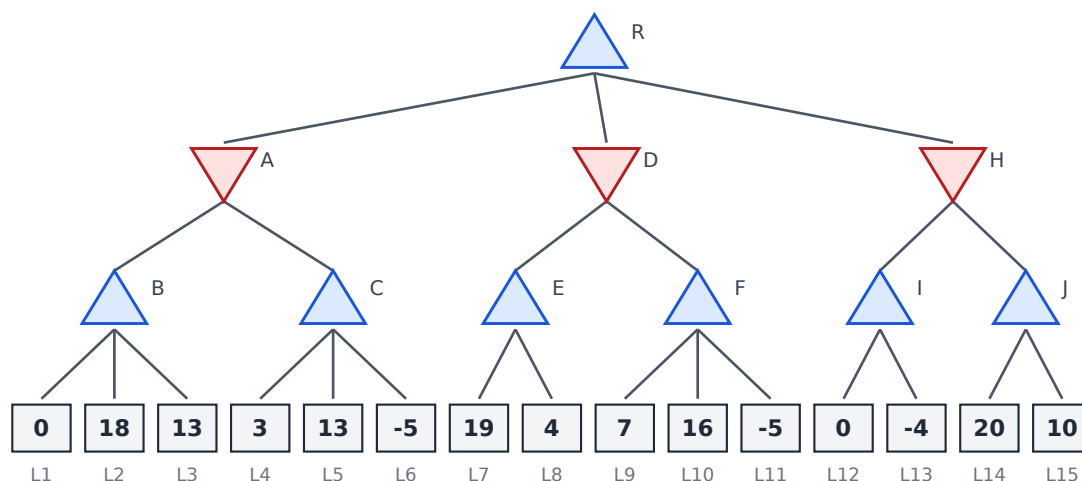
Section B – 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search

Iterative deepening search runs on a complete binary tree of depth 3 (15 nodes) that contains **no** goal. It uses limits 0, 1, 2, 3 and then stops. A node visit is one goal test on one node, and repeated visits across iterations count separately. What is the total number of node visits?

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



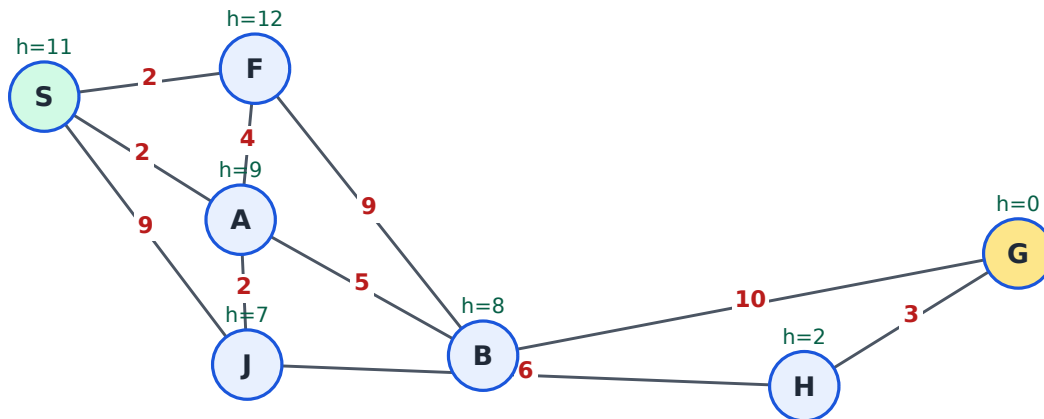
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

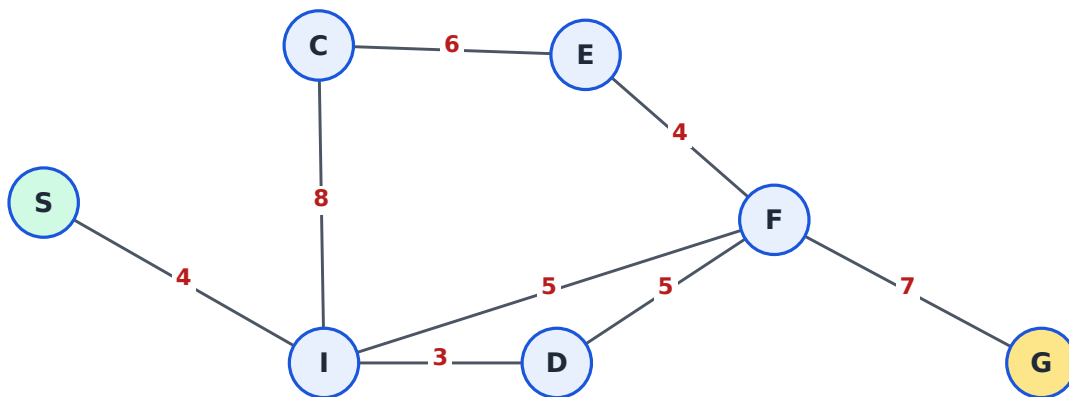
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
☐ (B) The edge H-G violates the consistency condition (in at least one direction).
☐ (C) h is consistent (monotone).
☐ (D) $h(A)$ overestimates the true cost from A to G.

Q.14 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.15 [MCQ · 2 marks · negative: -2/3] Adversarial Search

A three-ply binary game tree has levels MAX (root), MIN, MAX, and then leaves. The leaves, left to right, are 3, 4, 2, 9, 7, 1, 8, 6. Each bottom MAX node owns two consecutive leaves. Alpha-beta processes children left to right. Which statement is correct?

- ☐ (A) Root value 7; only the last leaf (6) is pruned
☐ (B) Root value 4; leaves 2 and 9 are pruned
☐ (C) Root value 7; leaves 1 and 6 are pruned
☐ (D) Root value 8; no leaf is pruned

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg P)$, C2: $(P \vee \neg Q)$, C3: $(P \vee Q \vee \neg S)$, C4: $(P \vee Q \vee S)$, C5: $(Q \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P \vee \neg Q)$
☐ (B) $(\neg S)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(R \vee S)$

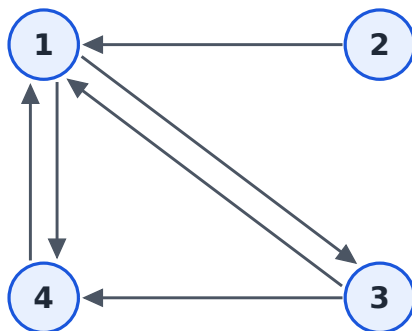
Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (\neg R \vee P) \wedge (Q \rightarrow P)$?

- ☐ (A) $(\neg R \vee P) \vee (Q \rightarrow P)$
☐ (B) $\neg(\neg(\neg R \vee P) \wedge \neg(Q \rightarrow P))$
☐ (C) $\neg(\neg(\neg R \vee P) \vee \neg(Q \rightarrow P))$
☐ (D) $(Q \rightarrow P) \wedge (\neg R \vee P)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
☐ (B) $\exists x R(x, x)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\exists x \forall y \neg R(x, y)$

Q.19 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following expresses 'No student likes every course' (S: student, C: course, $L(x, y)$: x likes y)?

- ☐ (A) $\forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$
☐ (B) $\forall x (S(x) \rightarrow \exists y (C(y) \rightarrow \neg L(x, y)))$
☐ (C) $\forall x \forall y ((S(x) \wedge C(y)) \rightarrow \neg L(x, y))$
☐ (D) $\exists x (S(x) \wedge \exists y (C(y) \wedge \neg L(x, y)))$

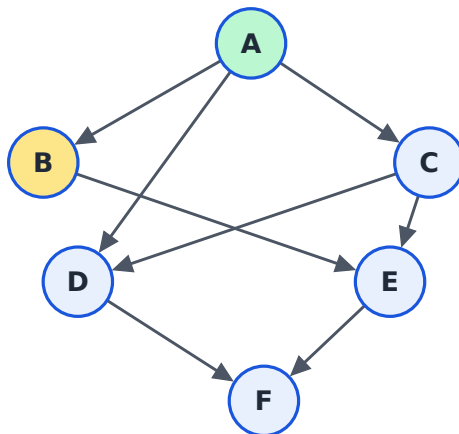
Q.20 [MCQ · 2 marks · negative: -2/3] Variable Elimination

A Bayesian network over Boolean variables has edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$, $D \rightarrow E$. $P(E)$ is computed by variable elimination with no evidence. For which elimination ordering does **every** new factor created by summing out a variable contain at most two variables?

- ☐ (A) A, B, C, D
☐ (B) D, C, B, A
☐ (C) B, C, A, D
☐ (D) C, B, A, D

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

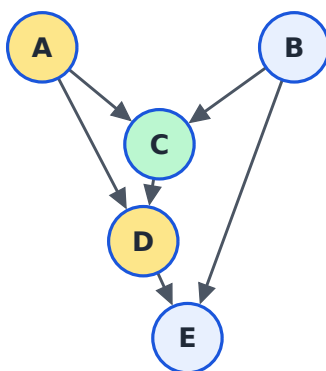
Variable elimination is used to compute $P(A \mid B = T)$ in the Bayesian network below (Boolean variables; evidence $B = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E, F, D, C**. What is the scope of the new factor produced when **C** is summed out?



- ☐ (A) $f(C)$
☐ (B) $f()$
☐ (C) $f(A)$
☐ (D) $f(A, C)$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(C = T \mid A = F, D = F)$ (round off to 3 decimal places).



P(A=T)		P(B=T)		A	B	P(C=T ...)
0.05		0.4		T	T	0.8
				T	F	0.5
				F	T	0.25
				F	F	0.6

A	C	P(D=T B,D)	P(E=T ...)
T	T	0.9	0.4
T	F	0.1	0.95
F	T	0.75	0.6
F	F	0.6	0.5

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Variable Elimination

A Bayesian network has edges $A \rightarrow B$ and $A \rightarrow C$ over Boolean variables with $P(a) = 0.3$, $P(b | a) = 0.8$, $P(b | \neg a) = 0.4$, $P(c | a) = 0.5$, $P(c | \neg a) = 0.1$. $P(A | b, c)$ is computed by variable elimination, multiplying the evidence-instantiated factors $f_1(A) = P(A)$, $f_2(A) = P(b | A)$, $f_3(A) = P(c | A)$ into an unnormalized factor $f(A)$. Which are correct?

- ☐ (A) $f(a) = 0.12$
☐ (B) $f(\neg a) = 0.028$
☐ (C) $P(a | b, c) \approx 0.811$
☐ (D) $P(b, c) = 0.12$

Q.24 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

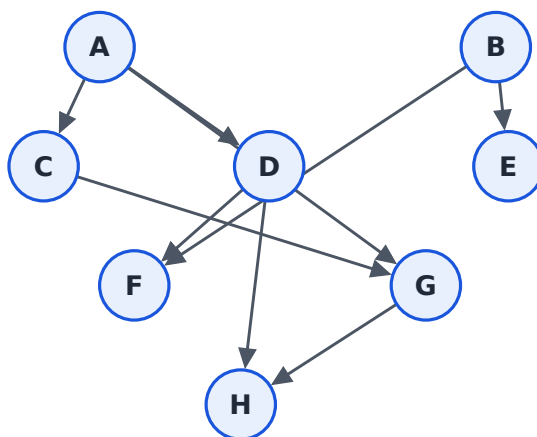
The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D, E. Using **rejection sampling**, estimate $P(C = T | D = F)$ (round off to 2 decimal places).

#	A	B	C	D	E
1	F	T	T	F	F
2	F	F	F	F	T
3	F	T	T	T	F
4	F	T	T	F	T
5	F	F	T	T	T
6	F	T	T	T	F
7	F	T	T	T	T
8	F	F	F	F	T
9	F	F	T	T	T
10	F	F	T	F	T
11	F	F	T	F	T
12	F	F	T	F	T
13	F	F	T	F	T
14	F	T	T	F	F
15	F	T	T	F	F

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp G$
☐ (B) $D \perp H | A, F$
☐ (C) $D \perp G | A$
☐ (D) $E \perp B | D$

Answer Key — Mock Test 042

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	NAT	2	Uninformed Search	7
2	NAT	1	Adversarial Search	2	15	MCQ	2	Adversarial Search	A
3	NAT	1	Adversarial Search	4	16	MSQ	2	Propositional Logic	B, C
4	NAT	1	Adversarial Search	35	17	MSQ	2	Propositional Logic	C, D
5	NAT	1	Propositional Logic	8	18	MSQ	2	Predicate Logic	A, C
6	NAT	1	Propositional Logic	5	19	MCQ	2	Predicate Logic	A
7	NAT	1	Predicate Logic	64	20	MCQ	2	Variable Elimination	A
8	NAT	1	Predicate Logic	169	21	MCQ	2	Exact Inference	C
9	NAT	1	Conditional Independence	6	22	NAT	2	Exact Inference	0.345 to 0.349
10	NAT	1	Sampling	0.0238 to 0.0242	23	MSQ	2	Variable Elimination	A, B, C
11	NAT	2	Uninformed Search	26	24	NAT	2	Sampling	0.79 to 0.81
12	NAT	2	Adversarial Search	2	25	MSQ	2	Conditional Independence	A, C
13	MSQ	2	Informed Search	A, C					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: A (A: S, H, C, A, D, G)

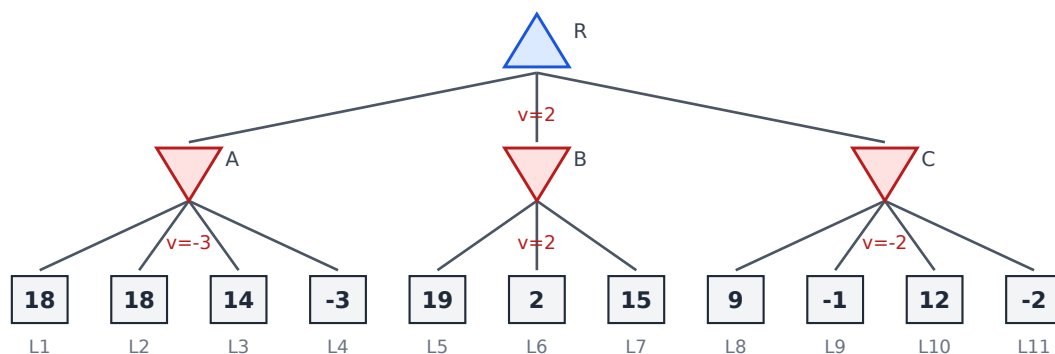
Adjacency lists (alphabetical): A: {C, D}; C: {A, D, H, J}; D: {A, C, G, J}; G: {D, J}; H: {C, S}; J: {C, D, G}; S: {H}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, H, C, A, D, G**; the path found is $S \rightarrow H \rightarrow C \rightarrow A \rightarrow D \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] Answer: 2

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -3$, $B = 2$, $C = -2$. Root value = **2**; MAX's optimal first move is towards B.

Q.3 [NAT, 1M · Adversarial Search] Answer: 4

$\Delta q = 0$, $\Delta r = +1$, $\Delta b = -1$, $\Delta p = +2$. $\text{Eval} = 9 \cdot 0 + 5 \cdot 1 + 3 \cdot (-1) + 1 \cdot 2 = 0 + 5 - 3 + 2 = 4$. A weighted linear evaluation function treats features as independent contributions.

Q.4 [NAT, 1M · Adversarial Search] Answer: 35

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 3^3 + 3^2 - 1 = 27 + 9 - 1 = 35$, versus 243 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.5 [NAT, 1M · Propositional Logic] Answer: 8

A binary connective is fixed by its outputs on TT, TF, FT, FF. Commutativity requires $\text{output}(TF) = \text{output}(FT)$. So we may freely choose the outputs on TT, FF and the common value on TF/FT: $2^3 = 8$. Examples: $\wedge$, $\vee$, $\leftrightarrow$, $\oplus$, NAND, NOR, constant T, constant F.

Q.6 [NAT, 1M · Propositional Logic] Answer: 5

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \rightarrow R) \leftrightarrow (\neg R \leftrightarrow \neg R)$	$\neg R \rightarrow (\neg P \wedge Q)$	$((R \rightarrow R) \leftrightarrow (\neg R \leftrightarrow \neg R)) \wedge (\neg R \rightarrow (\neg P \wedge Q))$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	F	F

Rows where F = T: **5**.

Q.7 [NAT, 1M · Predicate Logic] Answer: 64

An interpretation of R is a subset of $D \times D$ (9 pairs). The sentence says R is symmetric. The 3 diagonal pairs (x, x) can be chosen freely; the 3 unordered pairs $\{x, y\}$ with $x \neq y$ must be included or excluded as a whole. That is $2^3 \times 2^3 = 64$ relations.

Q.8 [NAT, 1M · Predicate Logic] Answer: 169

View R as an 3×3 0/1 matrix. The sentence says R is complement of 'every row has at least one 0': $2^{(n^2)} - (2^n - 1)^n$. With $n = 3$: **169**.

Q.9 [NAT, 1M · Conditional Independence] Answer: 6

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of D: {B, C}; children: {E, G, H}; co-parents: {A}.

$\text{MB}(D) = \{A, B, C, E, G, H\}$ — **6** nodes. Given its Markov blanket, D is conditionally independent of every other node in the network.

Q.10 [NAT, 1M · Sampling] Answer: 0.0238 to 0.0242

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Sm}=F) \times P(\text{Po}=F | \text{Sm}=F) \times P(\text{Ca}=F | \text{Po}=F) \times P(\text{XR}=F | \text{Po}=F, \text{Ca}=F) \times P(\text{Dy}=F | \text{Ca}=F, \text{XR}=F)$
 $= 0.8 \times 0.75 \times 0.25 \times 0.2 \times 0.8 = \mathbf{0.024} \approx 0.0240$.

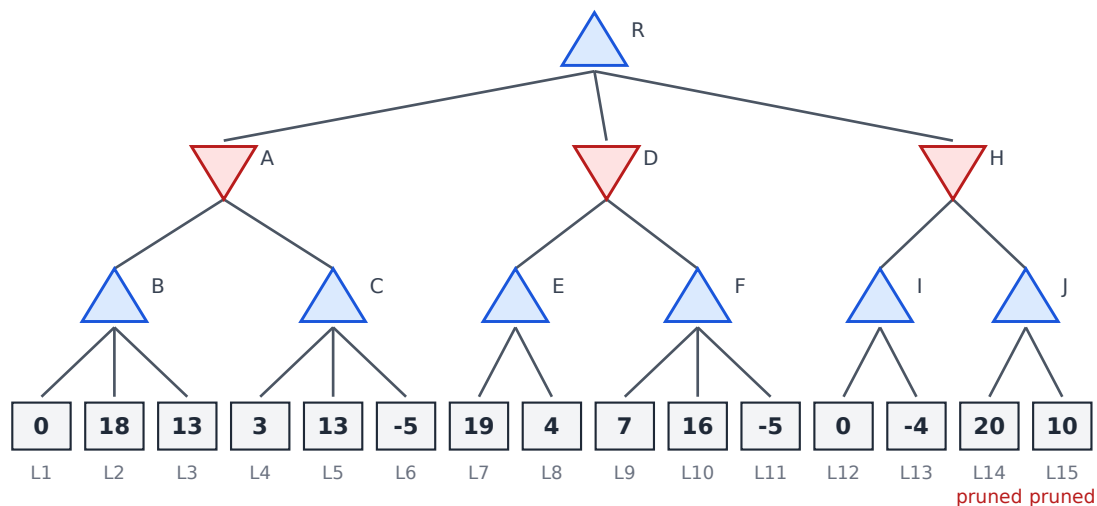
Q.11 [NAT, 2M · Uninformed Search] Answer: 26

The iteration with limit L visits the whole tree of depth L, which has $2^{L+1} - 1$ nodes. Limit 0: 1. Limit 1: 3. Limit 2: 7. Limit 3: 15. Total = $1 + 3 + 7 + 15 = \mathbf{26}$. For $b = 2$ the repetition overhead is largest, here 26 versus 15 nodes, which is still less than double.

Q.12 [NAT, 2M · Adversarial Search] Answer: 2

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node H: value $0 \leq \alpha = 16 \Rightarrow$ prune remaining children (leaves L14, L15).



Leaves evaluated (13): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10, L11, L12, L13. Leaves pruned (2): L14, L15. Root value = 16 (same as plain minimax, 16 — pruning never changes the root value).

Q.13 [MSQ, 2M · Informed Search] Answer: A, C (A: h is admissible.; C: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	F	G	H	J	S
$h(n)$	9	8	12	0	2	7	11
$h^*(n)$	11	10	15	0	3	9	13
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [NAT, 2M · Uninformed Search] Answer: 7

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	I(4)
2	I (4)	D(7), F(9), C(12)
3	D (7)	F(9), C(12)
4	F (9)	C(12), E(13), G(16)
5	C (12)	E(13), G(16)
6	E (13)	G(16)
7	G (16)	← goal, stop

Path returned: **S** $\rightarrow$ **I** $\rightarrow$ **F** $\rightarrow$ **G**, cost = 4 + 5 + 7 = **16**. Nodes expanded (including S and G): **7** (S, I, D, F, C, E, G).

Q.15 [MCQ, 2M · Adversarial Search] Answer: A (A: Root value 7; only the last leaf (6) is pruned)

Left MIN: $\text{MAX}(3,4) = 4$, so $\beta = 4$. $\text{MAX}(2,9)$: leaf 2 does not cut, and $9 \geq 4$ cuts, but it is the last child, so both leaves are evaluated. Left MIN = 4 and root $\alpha = 4$. Right MIN: $\text{MAX}(7,1) = 7$ ($7 > \alpha$, so no cut), so $\beta = 7$. $\text{MAX}(8, 6)$: $8 \geq \beta = 7$, so leaf 6 is pruned. Right MIN = 7, and the root is $\text{max}(4, 7) = 7$. Only leaf 6 is pruned, so (A) is correct. Leaf 1 must be evaluated because $7 > \alpha$ at that point.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, C (B: $(\neg S)$; C: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable))

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee \neg Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(\neg S)$ is derivable:

1. resolve $(P \vee \neg Q)$ and $(P \vee Q \vee \neg S)$ on $Q \Rightarrow (P \vee \neg S)$
2. resolve $(\neg P)$ and $(P \vee \neg S)$ on $P \Rightarrow (\neg S)$

(C) $\square$ is derivable:

1. resolve $(P \vee Q \vee \neg S)$ and $(P \vee Q \vee S)$ on $S \Rightarrow (P \vee Q)$
 2. resolve $(P \vee \neg Q)$ and $(P \vee Q)$ on $Q \Rightarrow (P)$
 3. resolve $(\neg P)$ and (P) on $P \Rightarrow \square$ (empty clause)
- (D) $(R \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.17 [MSQ, 2M · Propositional Logic] Answer: C, D (C: $\neg(\neg(\neg R \vee P) \vee \neg(Q \rightarrow P))$; D: $(Q \rightarrow P) \wedge (\neg R \vee P)$)

Compare truth tables with $F = (\neg R \vee P) \wedge (Q \rightarrow P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

- (A) $(\neg R \vee P) \vee (Q \rightarrow P)$ — differs at $P=F, Q=T, R=F$ ($F=F$, option = T) $\Rightarrow$ not equivalent.
 (B) $\neg(\neg(\neg R \vee P) \wedge \neg(Q \rightarrow P))$ — differs at $P=F, Q=T, R=F$ ($F=F$, option = T) $\Rightarrow$ not equivalent.
 (C) $\neg(\neg(\neg R \vee P) \vee \neg(Q \rightarrow P))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.
 (D) $(Q \rightarrow P) \wedge (\neg R \vee P)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

Q.18 [MSQ, 2M · Predicate Logic] Answer: A, C (A: $\forall x \exists y R(x, y)$; C: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,3), (1,4), (2,1), (3,1), (3,4), (4,1)\}$.

- (A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.
 (B) $\exists x R(x, x)$: **FALSE** — there are no self-loops.
 (C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.
 (D) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.
 Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MCQ, 2M · Predicate Logic] Answer: A (A: $\forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$)

'No student likes every course' $\equiv \neg \exists x (S(x) \wedge \forall y (C(y) \rightarrow L(x, y))) \equiv \forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$. Option B is satisfied by any non-course y , so it is too weak. Option C says no student likes any course — too strong. Option D only says some student fails to like some course.

Q.20 [MCQ, 2M · Variable Elimination] Answer: A (A: A, B, C, D)

Order A, B, C, D: eliminating A combines $P(A), P(B | A), P(C | A) \rightarrow f(B, C)$; B combines $f(B, C), P(D | B, C) \rightarrow f(C, D)$; C $\rightarrow f(D)$; D with $P(E | D) \rightarrow f(E)$. Max 2 variables. D first: $P(D | B, C)$ and $P(E | D) \rightarrow f(B, C, E)$ (3 variables). B first: $P(B | A), P(D | B, C) \rightarrow f(A, C, D)$ (3 variables); C first similarly gives $f(A, B, D)$. So only (A) keeps all factors at two variables or fewer.

Q.21 [MCQ, 2M · Exact Inference] Answer: C (C: $f(A)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A), f(A, C), f(A, C, D), f(C, E), f(D, E, F)$.
 Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(C, E), f(D, E, F)$	$\{C, D, E, F\}$	$f(C, D, F)$
F	$f(C, D, F)$	$\{C, D, F\}$	$f(C, D)$
D	$f(A, C, D), f(C, D)$	$\{A, C, D\}$	$f(A, C)$
C	$f(A, C), f(A, C)$	$\{A, C\}$	$f(A)$

The largest factor produced has **3** variable(s) (created when eliminating E); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over A and normalise.

Q.22 [NAT, 2M · Exact Inference] Answer: 0.345 to 0.349

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B) \cdot P(C | A, B) \cdot P(D | A, C) \cdot P(E | B, D)$.

Inference by enumeration: $P(C | e) = \alpha \sum_{\text{hidden}} P(C, e, \text{hidden})$.

Hidden: B, E	C	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T	T	$0.95 \times 0.4 \times 0.25 \times 0.25 \times 0.95$	0.02256
T, F	T	$0.95 \times 0.4 \times 0.25 \times 0.25 \times 0.05$	0.00119
F, T	T	$0.95 \times 0.6 \times 0.6 \times 0.25 \times 0.5$	0.04275
F, F	T	$0.95 \times 0.6 \times 0.6 \times 0.25 \times 0.5$	0.04275
T, T	F	$0.95 \times 0.4 \times 0.75 \times 0.4 \times 0.95$	0.1083
T, F	F	$0.95 \times 0.4 \times 0.75 \times 0.4 \times 0.05$	0.0057
F, T	F	$0.95 \times 0.6 \times 0.4 \times 0.4 \times 0.5$	0.0456
F, F	F	$0.95 \times 0.6 \times 0.4 \times 0.4 \times 0.5$	0.0456

$P(C=T, e) = 0.10925$; $P(C=F, e) = 0.2052$; $P(e) = 0.31445$.

$P(C=T | e) = 0.10925 / 0.31445 = \mathbf{0.3474} \approx 0.347$.

Q.23 [MSQ, 2M · Variable Elimination] **Answer: A, B, C** (A: $f(a) = 0.12$; B: $f(\neg a) = 0.028$; C: $P(a \mid b, c) \approx 0.811$)
 $f(a) = 0.3 \times 0.8 \times 0.5 = 0.12$ (A true). $f(\neg a) = 0.7 \times 0.4 \times 0.1 = 0.028$ (B true). Normalizing, $P(a \mid b, c) = 0.12 / (0.12 + 0.028) = 0.12 / 0.148 \approx 0.811$ (C true). The normalizer is $P(b, c) = f(a) + f(\neg a) = 0.148$, not 0.12 (which is $P(a, b, c)$), so (D) is false.

Q.24 [NAT, 2M · Sampling] **Answer: 0.79 to 0.81**

Rejection sampling keeps only samples consistent with the evidence $D = F$: samples 1, 2, 4, 8, 10, 11, 12, 13, 14, 15 (10 accepted, 5 rejected).

Among the accepted samples, $C = T$ in 8. Estimate = $8/10 = 0.8$.

(For this network the exact value is 0.5826; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.25 [MSQ, 2M · Conditional Independence] **Answer: A, C** (A: $B \perp G$; C: $D \perp G \mid A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp G$: **TRUE** — $B \rightarrow F \leftarrow D \leftarrow A \rightarrow C \rightarrow G$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow F \leftarrow D \leftarrow A \rightarrow G$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow F \leftarrow D \rightarrow H \leftarrow G$ (blocked: collider F with neither it nor a descendant observed).

(B) $D \perp H \mid A, F$: **FALSE** — the path $D \rightarrow H$ is active given $\{A, F\}$.

(C) $D \perp G \mid A$: **TRUE** — $D \leftarrow A \rightarrow C \rightarrow G$ (blocked: observed fork node A); $D \leftarrow A \rightarrow G$ (blocked: observed fork node A); $D \rightarrow H \leftarrow G$ (blocked: collider H with neither it nor a descendant observed).

(D) $E \perp B \mid D$: **FALSE** — the path $E \leftarrow B$ is active given $\{D\}$.

Mock Test 043

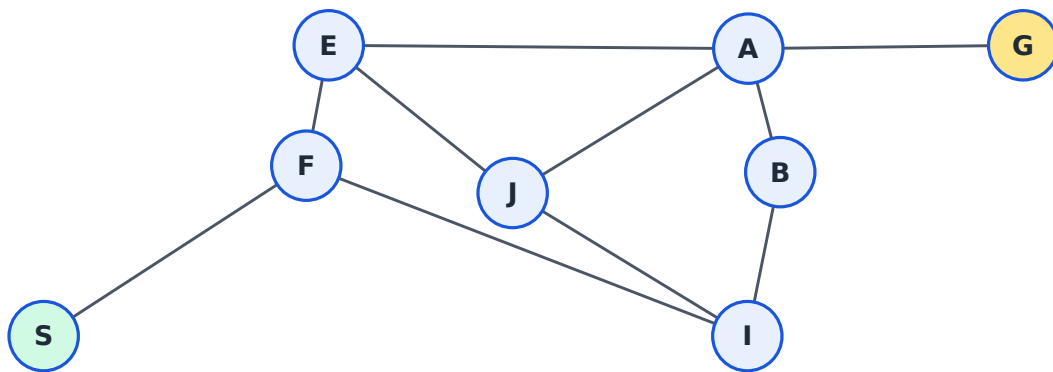
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 7 / 12	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, F, E, I, A, J, B, G
- ☐ (B) S, F, E, I, A, B, J, G
- ☐ (C) S, F, E, A, B, I, J, G
- ☐ (D) S, F, I, E, J, B, A, G

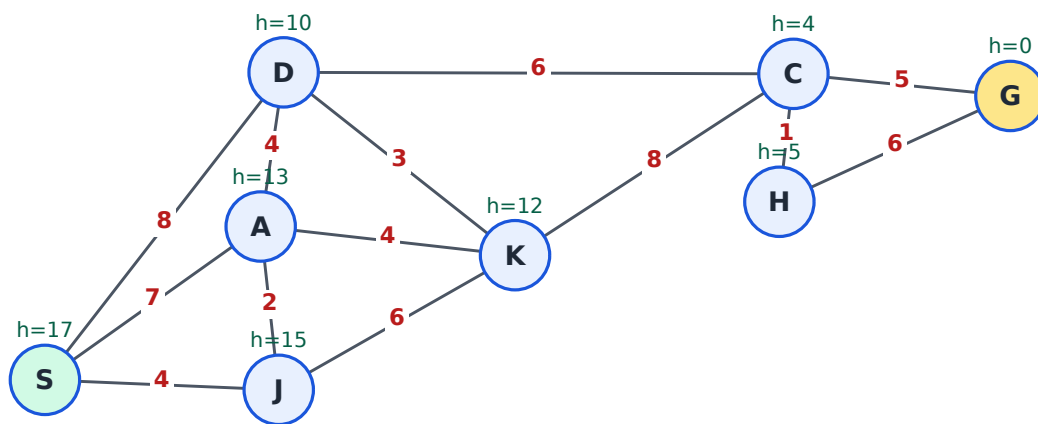
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search

A uniform game tree has branching factor $b = 3$ and depth $d = 4$ plies. Alpha-beta search examines the moves in perfect order. Using the best-case formula $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$, how many leaves are evaluated?

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

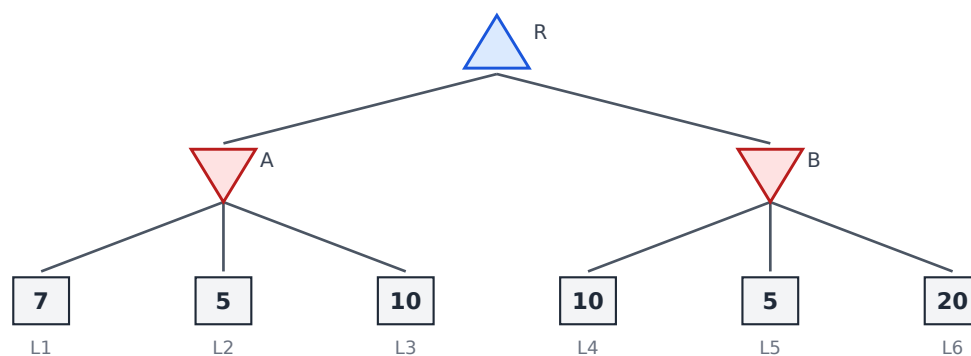
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

For the game tree below, what is the minimax value of the root?



△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following formulas is satisfiable but NOT valid?

- ☐ (A) $p \rightarrow (q \rightarrow p)$
- ☐ (B) $p \wedge \neg p$
- ☐ (C) $(p \wedge q) \rightarrow (p \vee q)$
- ☐ (D) $(p \rightarrow q) \rightarrow p$

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following is the correct first-order translation of 'Every student likes some course'?

- ☐ (A) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
- ☐ (B) $\forall x (\text{Student}(x) \wedge \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
- ☐ (C) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \rightarrow \text{Likes}(x, y)))$
- ☐ (D) $\exists y \forall x (\text{Student}(x) \wedge \text{Course}(y) \wedge \text{Likes}(x, y))$

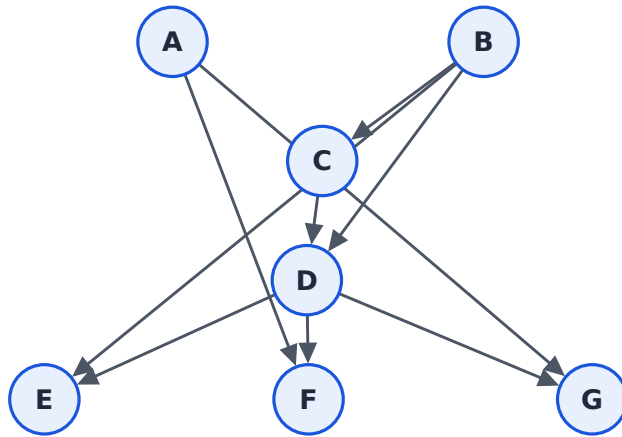
Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q). For how many of them is $\exists x P(x) \wedge \exists x Q(x)$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **D** in the Bayesian network below?



Answer (numerical): _____

Q.9 [MSQ · 1 mark · no negative marking] Sampling

Which statements correctly describe how sampling algorithms treat the evidence e ?

- ☐ (A) Likelihood weighting fixes evidence variables to their observed values and samples only non-evidence variables
- ☐ (B) Prior sampling ignores evidence while generating samples
- ☐ (C) Gibbs sampling changes the values of evidence variables during the Markov chain
- ☐ (D) Rejection sampling discards samples that are inconsistent with e

Q.10 [MSQ · 1 mark · no negative marking] Variable Elimination

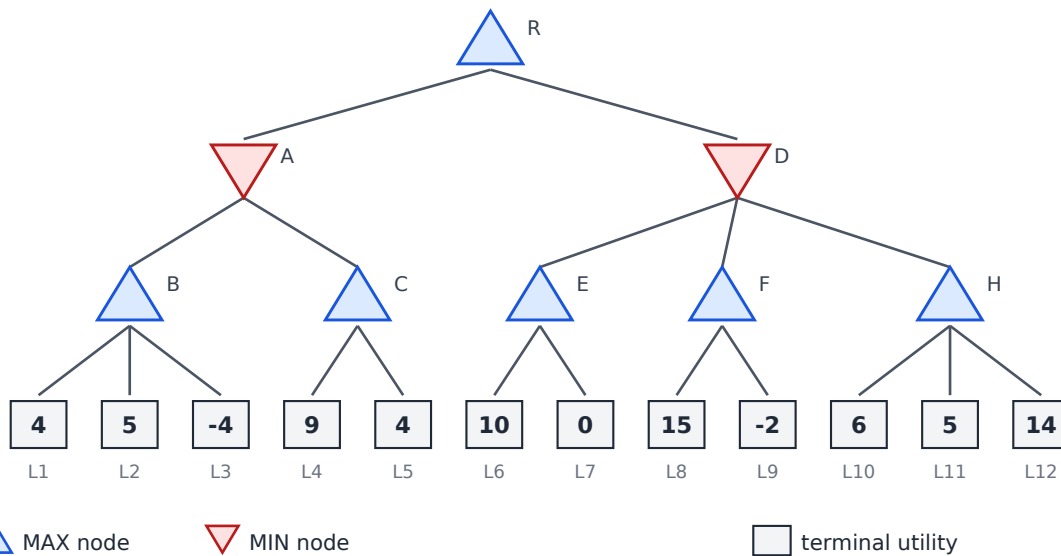
Which of the following statements about factors in variable elimination are correct (all variables Boolean)?

- ☐ (A) Summing out one variable from a factor over k variables yields a factor with 2^{k-1} entries
- ☐ (B) The pointwise product of factors is commutative
- ☐ (C) The entries of every intermediate factor must sum to 1
- ☐ (D) Evidence variables are fixed to their observed values, so they do not appear as dimensions of factors

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

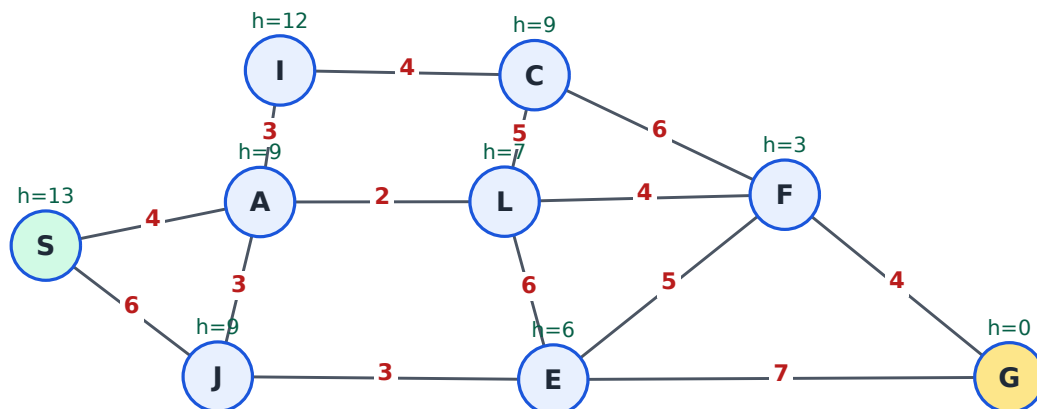
Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Breadth-first search goal-tests a node when it is generated, generating the left child before the right, and the goal is node 13. Depth-first search goal-tests on removal from the frontier, exploring the left child first. Which is correct about the number of nodes each algorithm goal-tests up to and including node 13 (counting the root)?

- ☐ (A) BFS: 13, DFS: 12
☐ (B) BFS: 13, DFS: 11
☐ (C) BFS: 12, DFS: 12
☐ (D) BFS: 7, DFS: 12

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

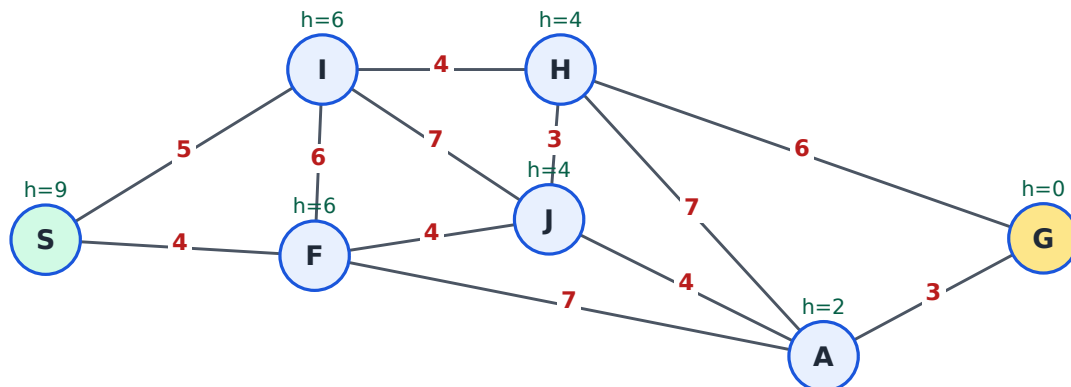
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on the set $\{1, 2, 3\}$ are transitive, i.e. satisfy $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$?

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (R \rightarrow Q) \wedge \neg P$?

- ☐ (A) $\neg P \wedge (R \rightarrow Q)$
☐ (B) $\neg(R \rightarrow Q) \wedge \neg P$
☐ (C) $(R \rightarrow Q) \vee \neg P$
☐ (D) $\neg(\neg(R \rightarrow Q) \wedge \neg\neg P)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(P \vee Q)$, C2: $(\neg P \vee Q \vee \neg R)$, C3: $(\neg P \vee S)$, C4: $(P \vee \neg R \vee S)$, C5: $(\neg P \vee Q)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(Q \vee S)$
☐ (B) $(Q \vee \neg R)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(\neg P \vee R)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic

Let α and β be propositional sentences. Which of the following statements are ALWAYS true?

- ☐ (A) If α is satisfiable then $\neg\alpha$ is unsatisfiable
☐ (B) If α is valid then $\neg\alpha$ is unsatisfiable
☐ (C) $\alpha \vee \beta$ is satisfiable if and only if α is satisfiable or β is satisfiable
☐ (D) $\alpha \wedge \beta$ is satisfiable if and only if α is satisfiable and β is satisfiable

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

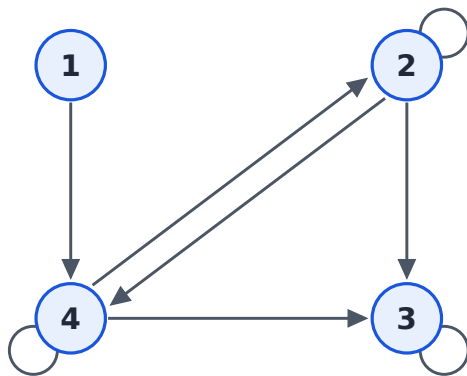
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = ((\neg Q \wedge P) \wedge (Q \wedge P)) \leftrightarrow (R \wedge (S \rightarrow S)) ?$$

Answer (numerical): _____

Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
- ☐ (B) $\forall x \exists y (x \neq y \wedge R(x, y))$
- ☐ (C) $\forall x R(x, x)$
- ☐ (D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$

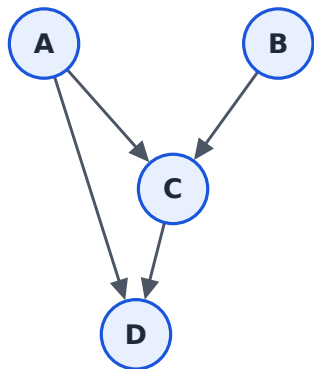
Q.21 [MCQ · 2 marks · negative: -2/3] Conditional Independence

Consider the Bayesian network (DAG) with edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$, $B \rightarrow F$. Which of the following pairs is d-separated by the given set?

- ☐ (A) A and B given $\{E\}$
- ☐ (B) A and F given $\{D\}$
- ☐ (C) A and E given $\{D\}$
- ☐ (D) E and F given $\emptyset$

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.22}, \mathbf{0.42}, \mathbf{0.56}, \mathbf{0.4} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.9

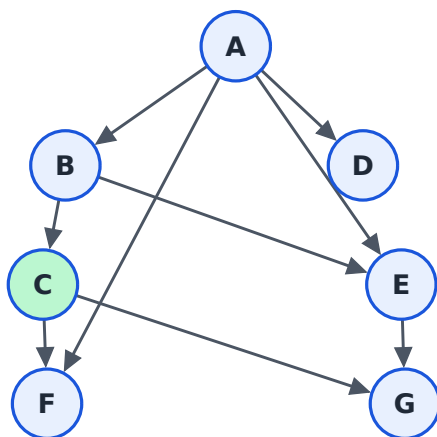
P(B=T)	A	B	P(C=T ...)
0.4	T	T	0.75
	T	F	0.5
	F	T	0.2
	F	F	0.95

A	C	P(D=T ...)
T	T	0.2
T	F	0.5
F	T	0.3
F	F	0.3

- ☐ (A) A=T, B=T, C=F, D=T
☐ (B) A=F, B=T, C=T, D=T
☐ (C) A=T, B=F, C=F, D=T
☐ (D) A=F, B=F, C=F, D=T

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

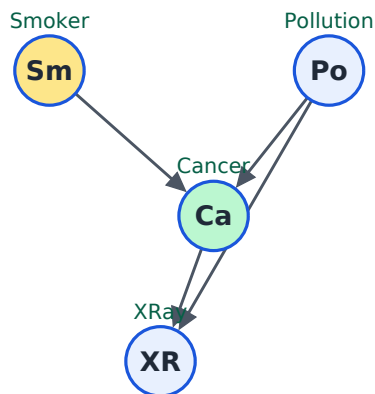
Variable elimination is used to compute $\mathbf{P(C)}$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F, D, G, E, A, B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $P(\text{Cancer} = T \mid \text{Smoker} = T)$ (round off to 3 decimal places).



$P(\text{Sm}=T)$
0.5

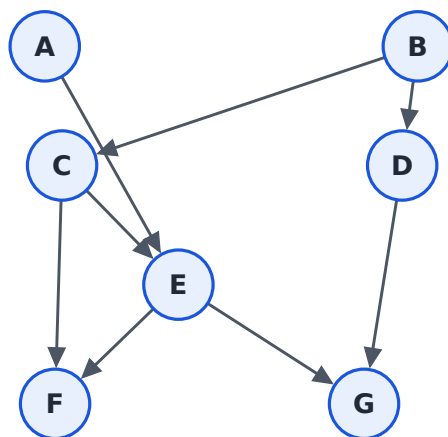
$P(\text{Po}=T)$	Sm	Po	$P(\text{Ca}=T \dots)$
0.4	T	T	0.6
	T	F	0.6
	F	T	0.7
	F	F	0.75

Po	Ca	$P(\text{XR}=T \dots)$
T	T	0.05
T	F	0.25
F	T	0.95
F	F	0.7

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp A$
☐ (B) $C \perp A \mid D$
☐ (C) $B \perp F \mid A, C$
☐ (D) $A \perp D \mid G$

Answer Key — Mock Test 043

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	NAT	2	Informed Search	7
2	NAT	1	Adversarial Search	17	15	NAT	2	Predicate Logic	171
3	NAT	1	Informed Search	19	16	MSQ	2	Propositional Logic	A
4	NAT	1	Adversarial Search	5	17	MSQ	2	Propositional Logic	A, B
5	MCQ	1	Propositional Logic	D	18	MSQ	2	Propositional Logic	B, C
6	MCQ	1	Predicate Logic	A	19	NAT	2	Propositional Logic	8
7	NAT	1	Predicate Logic	225	20	MSQ	2	Predicate Logic	A
8	NAT	1	Conditional Independence	6	21	MCQ	2	Conditional Independence	C
9	MSQ	1	Sampling	A, B, D	22	MCQ	2	Sampling	C
10	MSQ	1	Variable Elimination	A, B, D	23	NAT	2	Exact Inference	3
11	NAT	2	Adversarial Search	10	24	NAT	2	Exact Inference	0.598 to 0.602
12	MCQ	2	Uninformed Search	A	25	MSQ	2	Conditional Independence	A, B, C
13	NAT	2	Informed Search	5					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: A (A: S, F, E, I, A, J, B, G)

Adjacency lists (alphabetical): A: {B, E, G, J}; B: {A, I}; E: {A, F, J}; F: {E, I, S}; G: {A}; I: {B, F, J}; J: {A, E, I}; S: {F}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[F]
2	F	[E, I]
3	E	[I, A, J]
4	I	[A, J, B]
5	A	[J, B, G]
6	J	[B, G]
7	B	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, F, E, I, A, J, B, G**. The path returned is $S \rightarrow F \rightarrow E \rightarrow A \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Adversarial Search] Answer: 17

$3^2 + 3^2 - 1 = 9 + 9 - 1 = 17$. Plain minimax evaluates $3^4 = 81$ leaves, so perfect ordering saves 64 leaf evaluations.

Q.3 [NAT, 1M · Informed Search] Answer: 19

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

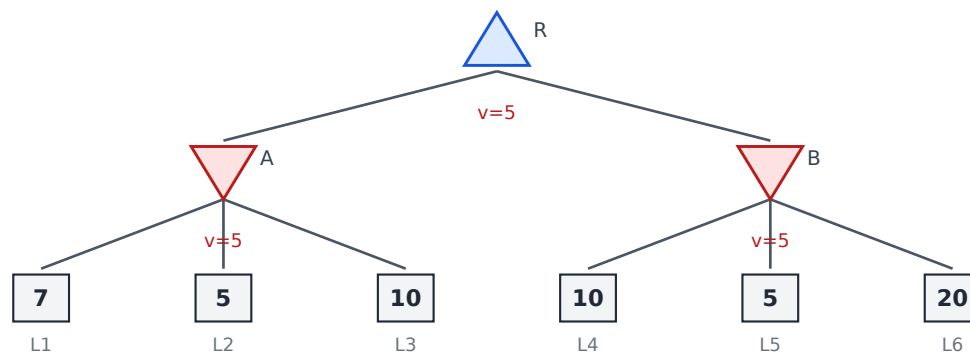
Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (17)	D(10), A(13), J(15)
2	D (10)	C(4), K(12), A(13), J(15)
3	C (4)	G(0), H(5), K(12), A(13), J(15)
4	G (0)	H(5), K(12), A(13), J(15) ← goal, stop

Path returned: **S → D → C → G**, cost = $8 + 6 + 5 = 19$.

The optimal cost is $h^*(S) = 19$ — greedy happens to be optimal here, but in general it is not.

Q.4 [NAT, 1M · Adversarial Search] Answer: 5

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 5$, $B = 5$. Root value = **5**; MAX's optimal first move is towards A, B.

Q.5 [MCQ, 1M · Propositional Logic] **Answer: D** ($D: (p \rightarrow q) \rightarrow p$)

$(p \rightarrow q) \rightarrow p$: when $p=T$ it is true (consequent T); when $p=F$, $p \rightarrow q$ is T and the consequent is F, so it is false. Hence it is satisfiable but not valid. $p \rightarrow (q \rightarrow p)$ is valid (if p is T the consequent is T). $p \wedge \neg p$ is unsatisfiable. $(p \wedge q) \rightarrow (p \vee q)$ is valid because $p \wedge q \models p \vee q$.

Q.6 [MCQ, 1M · Predicate Logic] **Answer: A** ($A: \forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$)

Universal quantifiers pair naturally with $\rightarrow$, existential with $\wedge$. Option B claims everything in the domain is a student. Option C is satisfied by any non-course y (making $\text{Course}(y) \rightarrow \dots$ vacuously true), so it does not require liking a course. Option D says there is a single y that is a course liked by everything, and everything is a student — far too strong.

Q.7 [NAT, 1M · Predicate Logic] **Answer: 225**

Interpret P and Q as subsets of the domain; each of P, Q must be non-empty $\Rightarrow (2^n - 1)^2$. With $n = 4$ this gives **225** (verified by enumerating all interpretations).

Q.8 [NAT, 1M · Conditional Independence] **Answer: 6**

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of D : $\{B, C\}$; children: $\{E, F, G\}$; co-parents: $\{A\}$.

$MB(D) = \{A, B, C, E, F, G\}$ — **6** nodes. Given its Markov blanket, D is conditionally independent of every other node in the network.

Q.9 [MSQ, 1M · Sampling] **Answer: A, B, D**

(A) LW clamps evidence and weights each sample by $\prod P(e_i | \text{parents})$. (B) Prior sampling generates from the prior; rejection sampling then filters these samples (D). (C) is false: in Gibbs sampling evidence variables stay fixed at their observed values; only non-evidence variables are resampled.

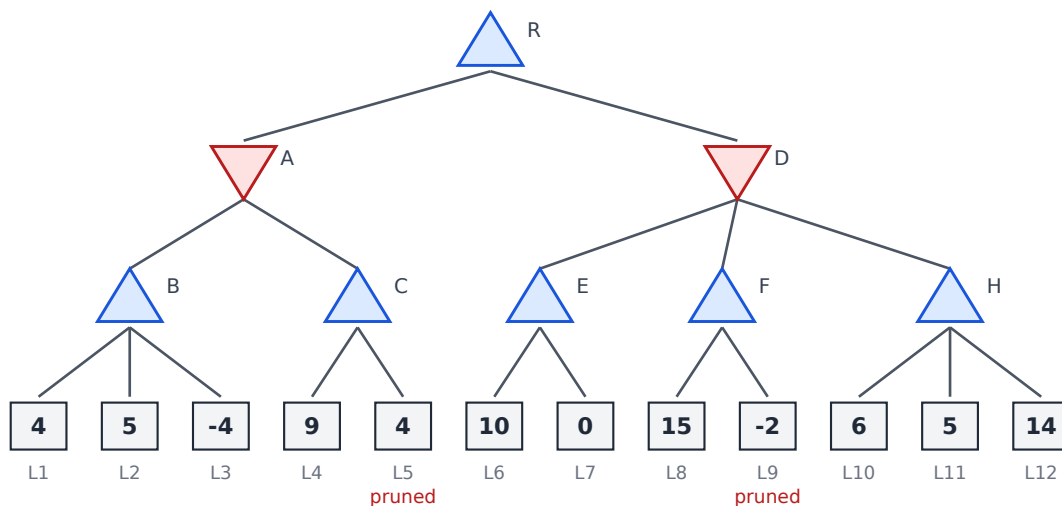
Q.10 [MSQ, 1M · Variable Elimination] **Answer: A, B, D**

(A) Summing out removes one Boolean dimension, from 2^k to 2^{k-1} entries. (B) Pointwise product is entry-wise multiplication of real numbers, hence commutative (and associative). (D) AIMA instantiates evidence, e.g. $P(j | A)$ becomes a factor over A alone. (C) is false: factors are generally unnormalized (e.g. $P(j | A)$ as a function of A , or intermediate sums), normalization happens only at the end.

Q.11 [NAT, 2M · Adversarial Search] **Answer: 10**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $9 \geq \beta = 5 \Rightarrow$ prune remaining children (leaves L5).
- At MAX node F: value $15 \geq \beta = 10 \Rightarrow$ prune remaining children (leaves L9).



Leaves evaluated (10): L1, L2, L3, L4, L6, L7, L8, L10, L11, L12. Leaves pruned (2): L5, L9. Root value = 10 (same as plain minimax, 10 — pruning never changes the root value).

Q.12 [MCQ, 2M · Uninformed Search] **Answer: A** (A: BFS: 13, DFS: 12)

BFS generates nodes in label order (root, then 2, 3, 4, ...), so node 13 is the **13th** node tested. DFS preorder is 1, 2, 4, 8, 9, 5, 10, 11, 3, 6, 12, 13, ..., so node 13 is the **12th**. The other options miscount one or both orders.

Q.13 [NAT, 2M · Informed Search] **Answer: 5**

Heuristic values: $h(A)=9$, $h(C)=9$, $h(E)=6$, $h(F)=3$, $h(G)=0$, $h(I)=12$, $h(J)=9$, $h(L)=7$, $h(S)=13$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	A(13), J(15)
2	A (13)	L(13), J(15), I(19)
3	L (13)	F(13), J(15), E(18), I(19), C(20)
4	F (13)	G(14), J(15), E(18), I(19), C(20)
5	G (14)	J(15), E(18), I(19), C(20) ← goal, stop

Path returned: **S** → **A** → **L** → **F** → **G** with cost **14**; nodes expanded = **5** (S, A, L, F, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.14 [NAT, 2M · Informed Search] **Answer: 7**

Heuristic values: $h(A)=2$, $h(F)=6$, $h(G)=0$, $h(H)=4$, $h(I)=6$, $h(J)=4$, $h(S)=9$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	F(10), I(11)
2	F (10)	I(11), J(12), A(13)
3	I (11)	J(12), A(13), H(13)
4	J (12)	A(13), H(13)
5	A (13)	H(13), G(14)
6	H (13)	G(14)
7	G (14)	← goal, stop

Path returned: **S** → **F** → **A** → **G** with cost **14**; nodes expanded = **7** (S, F, I, J, A, H, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [NAT, 2M · Predicate Logic] **Answer: 171**

There is no simple closed formula; we enumerate all $2^9 = 512$ relations and test the sentence for all 27 triples (x, y, z) . The count of transitive relations on a 3-element set is 171 (OEIS A006905: 1, 2, 13, 171, 3994, ...). Sanity check: on a 2-element set the same enumeration gives 13 of 16; the 3 failures contain both $(1,2)$ and $(2,1)$ but miss $(1,1)$ or $(2,2)$.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A (A: $\neg P \wedge (R \rightarrow Q)$)

Compare truth tables with $F = (R \rightarrow Q) \wedge \neg P$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg P \wedge (R \rightarrow Q)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(B) $\neg(R \rightarrow Q) \wedge \neg P$ — differs at $P=F, Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(C) $(R \rightarrow Q) \vee \neg P$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(D) $\neg(\neg(R \rightarrow Q) \wedge \neg P)$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A, B (A: $(Q \vee S)$; B: $(Q \vee \neg R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(Q \vee S)$ is derivable:

1. resolve $(P \vee Q)$ and $(\neg P \vee S)$ on $P \Rightarrow (Q \vee S)$

(B) $(Q \vee \neg R)$ is derivable:

1. resolve $(P \vee Q)$ and $(\neg P \vee Q \vee \neg R)$ on $P \Rightarrow (Q \vee \neg R)$

(C) The resolution closure contains 10 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(D) $(\neg P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, C (B: If α is valid then $\neg\alpha$ is unsatisfiable; C: $\alpha \vee \beta$ is satisfiable if and only if α is satisfiable or β is satisfiable)

(A) False: p is satisfiable and so is $\neg p$. (B) True: valid means true in all models, so $\neg\alpha$ is true in none. (C)

True: a model of $\alpha \vee \beta$ is a model of α or of β , and conversely any model of α (or of β) is a model of $\alpha \vee \beta$. (D)

False: $\alpha = p$ and $\beta = \neg p$ are each satisfiable but $p \wedge \neg p$ is not (the 'if' direction fails).

Q.19 [NAT, 2M · Propositional Logic] Answer: 8

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(\neg Q \wedge P) \wedge (Q \wedge P)$	$R \wedge (S \rightarrow S)$	$((\neg Q \wedge P) \wedge (Q \wedge P)) \leftrightarrow (R \wedge (S \rightarrow S))$
T	T	T	T	F	T	F
T	T	T	F	F	T	F
T	T	F	T	F	F	T
T	T	F	F	F	F	T
T	F	T	T	F	T	F
T	F	T	F	F	T	F
T	F	F	T	F	F	T
T	F	F	F	F	F	T
F	T	T	T	F	T	F
F	T	T	F	F	T	F
F	T	F	T	F	F	T
F	T	F	F	F	F	T
F	F	T	T	F	T	F
F	F	T	F	F	T	F
F	F	F	T	F	F	T
F	F	F	F	F	F	T

Rows where $F = T$: **8**.

Q.20 [MSQ, 2M · Predicate Logic] Answer: A (A: $\forall x \exists y R(x, y)$)

$R = \{(1,4), (2,2), (2,3), (2,4), (3,3), (4,2), (4,3), (4,4)\}$.

(A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(B) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 3 has no edge to a different element.

(C) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,2)$ nor $R(2,1)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.21 [MCQ, 2M · Conditional Independence] **Answer: C** (C: A and E given {D})

(C): every path from A to E goes $A \rightarrow C \rightarrow D \rightarrow E$; D is a non-collider on it and is observed, so it is blocked: d-separated. (A): the path $A \rightarrow C \leftarrow B$ has collider C whose descendant E is observed, so it is active. (B): the path $A \rightarrow C \leftarrow B \rightarrow F$ has collider C with observed descendant D, and non-collider B unobserved, so it is active. (D): $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ has no colliders and nothing observed, so it is active.

Q.22 [MCQ, 2M · Sampling] **Answer: C** (C: A=T, B=F, C=F, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.9$; $u = 0.22 < 0.9 \Rightarrow A = T$
- B: $P(B=T \mid \text{—}) = 0.4$; $u = 0.42 \geq 0.4 \Rightarrow B = F$
- C: $P(C=T \mid A=T, B=F) = 0.5$; $u = 0.56 \geq 0.5 \Rightarrow C = F$
- D: $P(D=T \mid A=T, C=F) = 0.5$; $u = 0.4 < 0.5 \Rightarrow D = T$

Sample: **(A=T, B=F, C=F, D=T)**.

Q.23 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, D)$, $f(A, B, E)$, $f(A, C, F)$, $f(C, E, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(A, C, F)$	$\{A, C, F\}$	$f(A, C)$
D	$f(A, D)$	$\{A, D\}$	$f(A)$
G	$f(C, E, G)$	$\{C, E, G\}$	$f(C, E)$
E	$f(A, B, E)$, $f(C, E)$	$\{A, B, C, E\}$	$f(A, B, C)$
A	$f(A)$, $f(A, B)$, $f(A, C)$, $f(A)$, $f(A, B, C)$	$\{A, B, C\}$	$f(B, C)$
B	$f(B, C)$, $f(B, C)$	$\{B, C\}$	$f(C)$

The largest factor produced has **3** variable(s) (created when eliminating E); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over C and normalise.

Q.24 [NAT, 2M · Exact Inference] **Answer: 0.598 to 0.602**

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution}) \cdot P(\text{Cancer} \mid \text{Smoker}, \text{Pollution}) \cdot P(\text{XRay} \mid \text{Pollution}, \text{Cancer})$.

Inference by enumeration: $P(\text{Cancer} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cancer}, e, \text{hidden})$.

Hidden: Po, XR	Ca	product of CPT entries ($Sm \cdot Po \cdot Ca \cdot XR$)	value
T, T	T	$0.5 \times 0.4 \times 0.6 \times 0.05$	0.006
T, F	T	$0.5 \times 0.4 \times 0.6 \times 0.95$	0.114
F, T	T	$0.5 \times 0.6 \times 0.6 \times 0.95$	0.171
F, F	T	$0.5 \times 0.6 \times 0.6 \times 0.05$	0.009
T, T	F	$0.5 \times 0.4 \times 0.4 \times 0.25$	0.02
T, F	F	$0.5 \times 0.4 \times 0.4 \times 0.75$	0.06
F, T	F	$0.5 \times 0.6 \times 0.4 \times 0.7$	0.084
F, F	F	$0.5 \times 0.6 \times 0.4 \times 0.3$	0.036

$P(\text{Cancer}=T, e) = 0.3$; $P(\text{Cancer}=F, e) = 0.2$; $P(e) = 0.5$.

$P(\text{Cancer}=T \mid e) = 0.3 / 0.5 = \mathbf{0.6} \approx 0.600$.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: A, B, C** (A: $B \perp A$; B: $C \perp A \mid D$; C: $B \perp F \mid A, C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp A$: **TRUE** — $B \rightarrow C \rightarrow E \leftarrow A$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow C \rightarrow F \leftarrow E \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow D \rightarrow G \leftarrow E \leftarrow A$ (blocked: collider G with neither it nor a descendant observed).

(B) $C \perp A \mid D$: **TRUE** — $C \leftarrow B \rightarrow D \rightarrow G \leftarrow E \leftarrow A$ (blocked: observed chain node D); $C \rightarrow E \leftarrow A$ (blocked: collider E with neither it nor a descendant observed); $C \rightarrow F \leftarrow E \leftarrow A$ (blocked: collider F with neither it nor a descendant observed).

(C) $B \perp F \mid A, C$: **TRUE** — $B \rightarrow C \rightarrow E \rightarrow F$ (blocked: observed chain node C); $B \rightarrow C \rightarrow F$ (blocked: observed chain node C); $B \rightarrow D \rightarrow G \leftarrow E \leftarrow C \rightarrow F$ (blocked: collider G with neither it nor a descendant observed); $B \rightarrow D \rightarrow G \leftarrow E \rightarrow F$ (blocked: collider G with neither it nor a descendant observed).

(D) $A \perp D \mid G$: **FALSE** — the path $A \rightarrow E \leftarrow C \leftarrow B \rightarrow D$ is active given {G}.

Mock Test 044

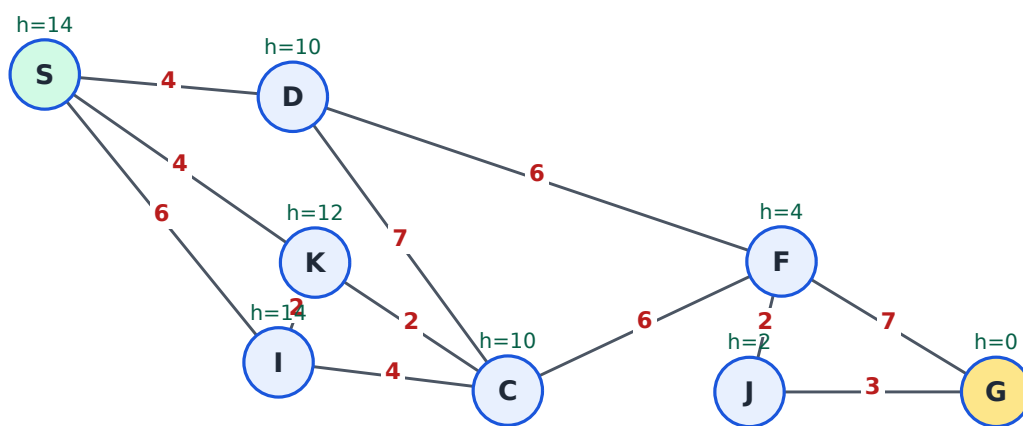
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	8 / 7 / 10	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

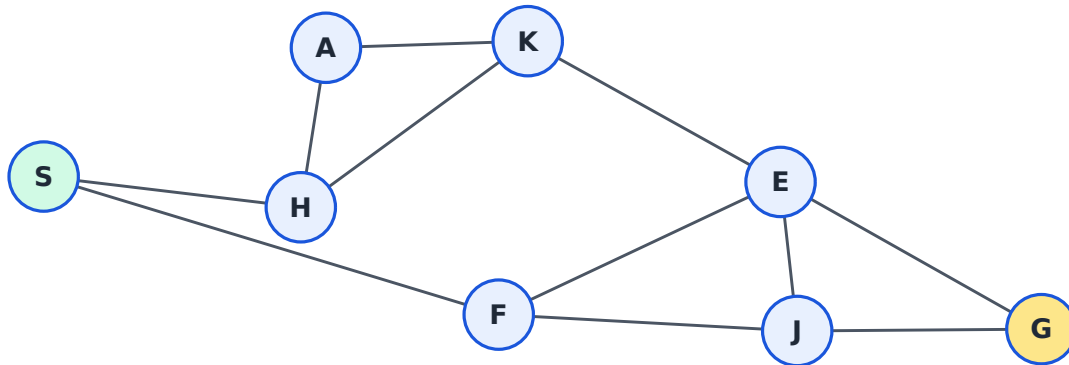
Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search

When successors are examined in **random** order, alpha-beta search is expected to examine roughly

- ☐ (A) $O(b^{m/2})$ nodes
- ☐ (B) $O(b^{3m/4})$ nodes
- ☐ (C) $O(b^m)$ nodes
- ☐ (D) $O(m \cdot b)$ nodes

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

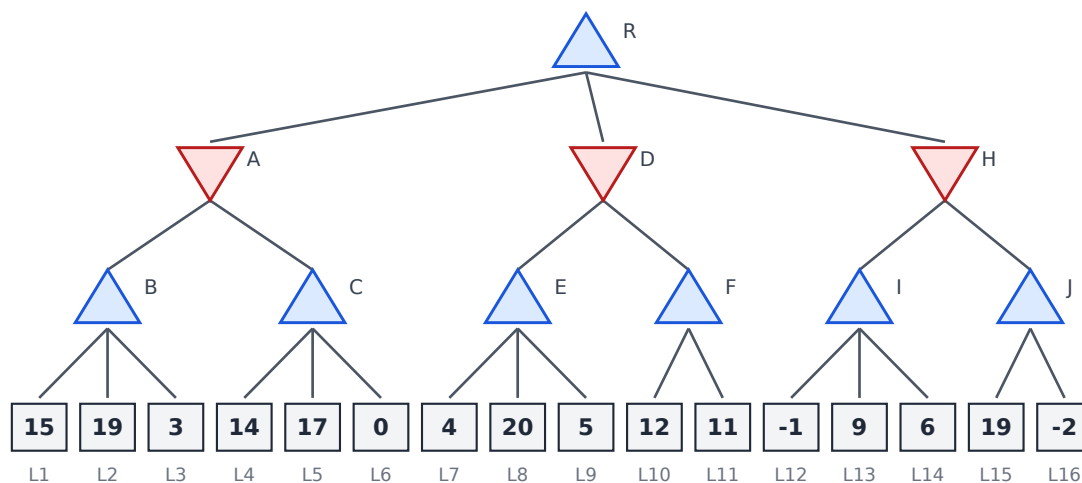
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, F, E, G
☐ (B) S, H, F, K, A, J, E, G
☐ (C) S, F, H, E, J, A, K, G
☐ (D) S, F, H, J, E, A, K, G

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



MAX node MIN node

terminal utility

- ☐ (A) Move to A; root value 17
☐ (B) Move to H; root value 9
☐ (C) Move to A; root value 19
☐ (D) Move to D; root value 12

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \exists y R(x, y)$ true?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are in conjunctive normal form (CNF)?

- ☐ (A) $(p \vee \neg q) \wedge r$
☐ (B) $\neg(p \wedge q)$
☐ (C) $p \vee \neg q \vee r$
☐ (D) $(p \wedge q) \vee r$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (\neg Q \wedge (P \vee R)) \leftrightarrow (\neg R \wedge (\neg P \wedge Q)) ?$$

Answer (numerical): _____

Q.8 [MSQ · 1 mark · no negative marking] Predicate Logic

Without standardizing variables apart, $\text{Unify}(\text{Knows}(\text{John}, x), q)$ succeeds for which of the following atoms q ?

- ☐ (A) $\text{Knows}(\text{John}, \text{Jane})$
☐ (B) $\text{Knows}(y, \text{Bill})$
☐ (C) $\text{Knows}(y, \text{Mother}(y))$
☐ (D) $\text{Knows}(x, \text{Elizabeth})$

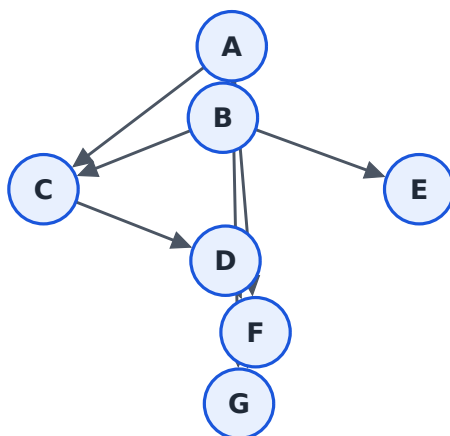
Q.9 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

A Boolean node has k Boolean parents. Its CPT is specified using a noisy-OR model (no leak). How many parameters are needed, versus a general CPT?

- ☐ (A) k for noisy-OR versus 2^k for a general CPT
☐ (B) 2^k for both
☐ (C) k^2 for noisy-OR versus 2^k for a general CPT
☐ (D) $2k$ for noisy-OR versus k^2 for a general CPT

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **B** in the Bayesian network below?

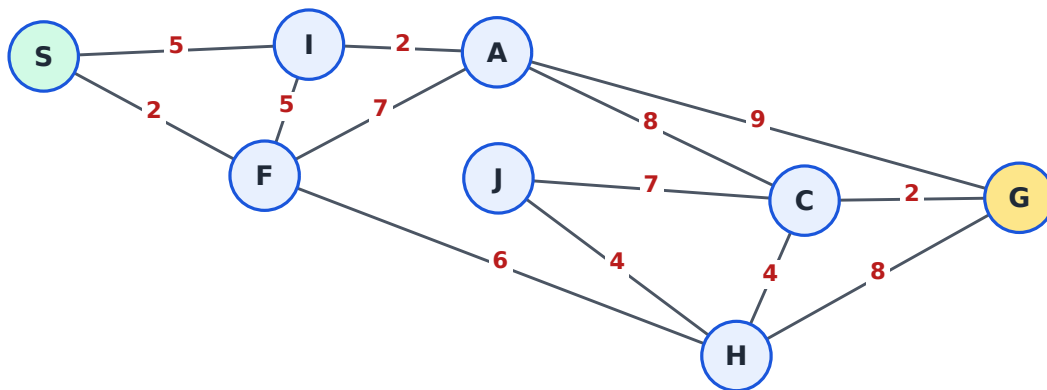


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

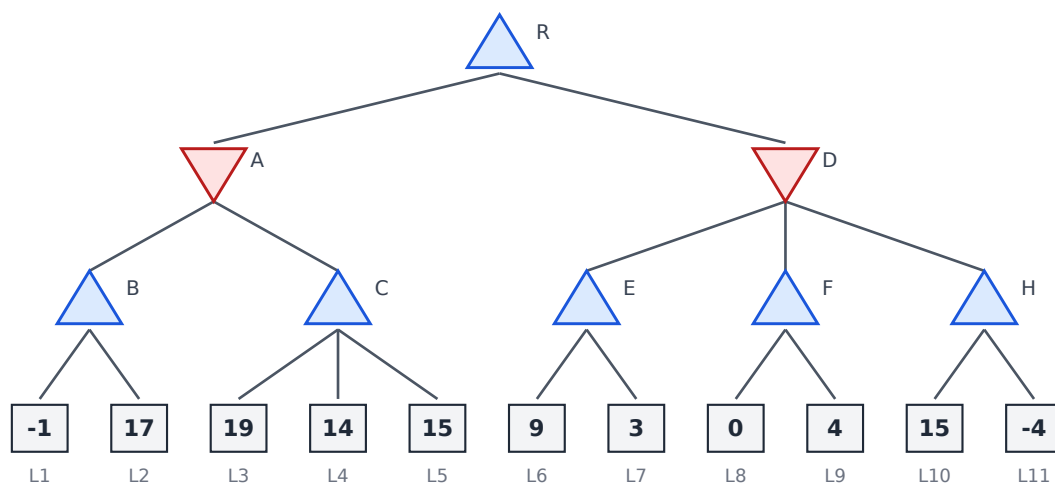
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow F \rightarrow H \rightarrow C \rightarrow G$
☐ (B) $S \rightarrow I \rightarrow A \rightarrow G$
☐ (C) $S \rightarrow I \rightarrow A \rightarrow C \rightarrow G$
☐ (D) $S \rightarrow F \rightarrow H \rightarrow G$

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



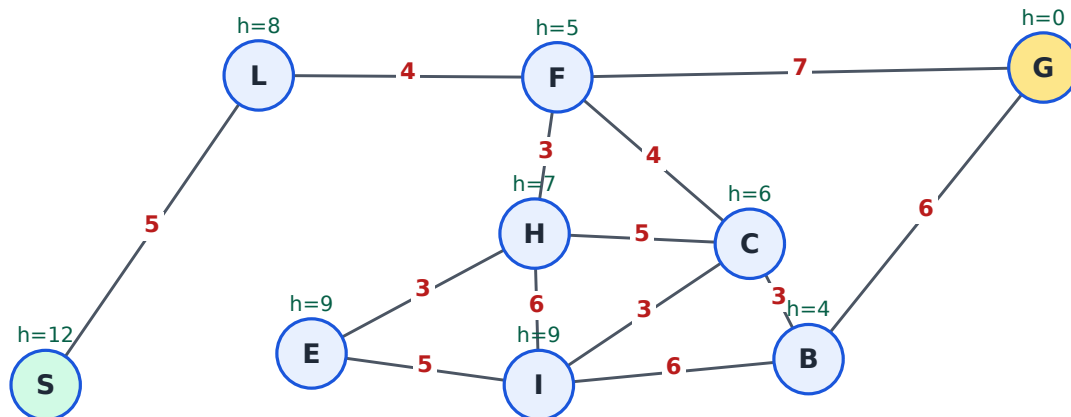
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Adversarial Search

Minimax search (depth-first) runs on a finite game tree with branching factor b and maximum depth m . Which of the following are TRUE?

- ☐ (A) Its time complexity is $O(b^m)$
- ☐ (B) Its space complexity is $O(bm)$ when all successors are generated at once
- ☐ (C) Against an optimal opponent, the minimax strategy achieves the minimax value of the root
- ☐ (D) It needs a heuristic evaluation function even when every path can be searched to a terminal state

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg Q \vee S)$, C2: (S) , C3: $(\neg P \vee \neg R)$, C4: $(\neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(P \vee R)$
- ☐ (B) $(\neg Q)$
- ☐ (C) $(\neg R)$
- ☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

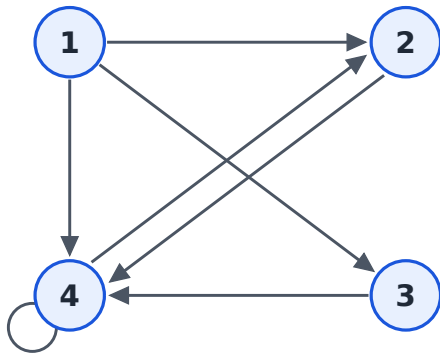
Q.16 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following is logically equivalent to $\neg \exists x (P(x) \wedge \forall y Q(x, y))$?

- ☐ (A) $\forall x (P(x) \rightarrow \exists y \neg Q(x, y))$
- ☐ (B) $\forall x (\neg P(x) \wedge \exists y \neg Q(x, y))$
- ☐ (C) $\exists x (\neg P(x) \vee \exists y \neg Q(x, y))$
- ☐ (D) $\forall x (P(x) \wedge \exists y \neg Q(x, y))$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x R(x, x)$
- ☐ (B) $\exists y \forall x R(x, y)$
- ☐ (C) $\forall y \exists x R(x, y)$
- ☐ (D) $\exists x \forall y \neg R(y, x)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(Q \rightarrow R) \rightarrow (R \rightarrow Q)$
- ☐ (B) $R \rightarrow (Q \rightarrow R)$
- ☐ (C) $\neg(P \wedge Q) \leftrightarrow (\neg P \wedge \neg Q)$
- ☐ (D) $((R \rightarrow P) \wedge P) \rightarrow R$

Q.19 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(A = T \mid B = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	T	T	F
2	F	F	T	F
3	F	F	T	T
4	F	F	T	T
5	F	F	F	F
6	F	F	F	F
7	T	T	T	F
8	F	F	T	T
9	F	F	F	F
10	F	F	F	F
11	F	F	F	T
12	F	F	F	T
13	F	F	F	F
14	T	F	T	F
15	F	F	F	F
16	F	F	T	F
17	T	F	T	F
18	F	F	T	T
19	F	T	F	T
20	T	T	T	T

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. Likelihood weighting with evidence $S = \text{true}$, $W = \text{true}$ produces exactly four samples of (C, R): (c, r), ($\neg c$, r), ($\neg c$, $\neg r$), (c, $\neg r$). Using the correct weights, compute the likelihood-weighting estimate of $P(r \mid s, w)$. (Round to 3 decimal places.)

Answer (numerical): _____

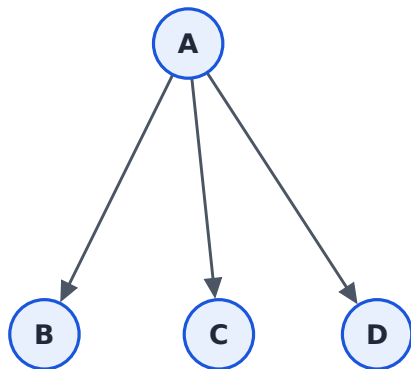
Q.21 [NAT · 2 marks · no negative marking] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. In computing $P(B \mid j, m)$ by variable elimination, A is summed out first, producing a factor $f(B, E) = \sum_a P(a \mid B, E) P(j \mid a) P(m \mid a)$. Compute $f(b, e)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.03, 0.88, 0.3, 0.76)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.7

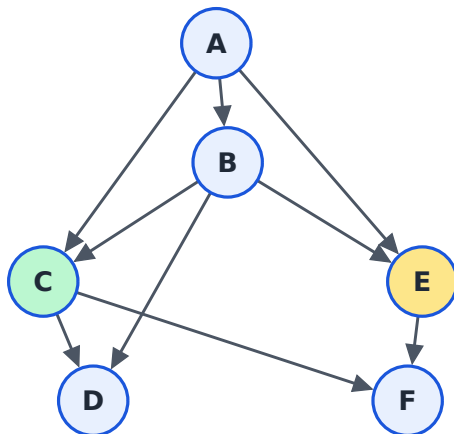
A	P(B=T ...)	A	P(C=T ...)
T	0.6	T	0.2
F	0.75	F	0.25

A	P(D=T ...)
T	0.9
F	0.4

- ☐ (A) A=T, B=T, C=F, D=T
☐ (B) A=F, B=T, C=T, D=T
☐ (C) A=F, B=F, C=F, D=T
☐ (D) A=T, B=F, C=F, D=T

Q.23 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

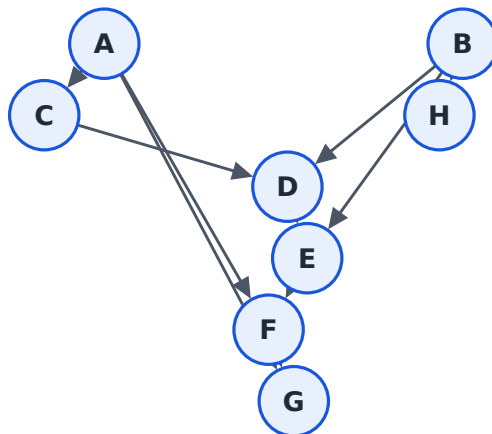
Variable elimination is used to compute $\mathbf{P(C \mid E = T)}$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, D, F, A**. What is the scope of the new factor produced when **D** is summed out?



- ☐ (A) $f(A, C)$
☐ (B) $f(A, C, D)$
☐ (C) $f(C)$
☐ (D) $f(A)$

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

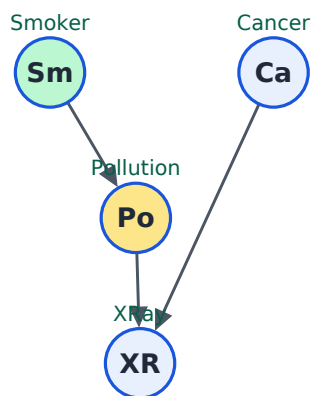
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $H \perp C$
☐ (B) $G \perp C \mid A$
☐ (C) $H \perp G \mid B, E$
☐ (D) $D \perp B \mid E$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute **$P(\text{Smoker} = T \mid \text{Pollution} = F)$** (round off to 3 decimal places).



$P(\text{Sm}=T)$
0.05

Sm	$P(\text{Po}=T \dots)$	$P(\text{Ca}=T)$
T	0.7	0.4
F	0.25	

Po	Ca	$P(\text{XR}=T \dots)$
T	T	0.4
T	F	0.7
F	T	0.4
F	F	0.95

Answer (numerical): _____

Answer Key — Mock Test 044

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Informed Search	17	14	MSQ	2	Adversarial Search	A, B, C
2	MCQ	1	Adversarial Search	B	15	MSQ	2	Propositional Logic	B, D
3	MCQ	1	Uninformed Search	C	16	MCQ	2	Predicate Logic	A
4	MCQ	1	Adversarial Search	A	17	MSQ	2	Predicate Logic	B, D
5	NAT	1	Predicate Logic	343	18	MSQ	2	Propositional Logic	B
6	MSQ	1	Propositional Logic	A, C	19	NAT	2	Sampling	0.49 to 0.51
7	NAT	1	Propositional Logic	4	20	NAT	2	Sampling	0.523 to 0.524
8	MSQ	1	Predicate Logic	A, B, C	21	NAT	2	Variable Elimination	0.531 to 0.533
9	MCQ	1	Bayesian Networks	A	22	MCQ	2	Sampling	D
10	NAT	1	Conditional Independence	3	23	MCQ	2	Exact Inference	A
11	MCQ	2	Uninformed Search	A	24	MSQ	2	Conditional Independence	A, C
12	NAT	2	Adversarial Search	5	25	NAT	2	Exact Inference	0.019 to 0.023
13	NAT	2	Informed Search	16					

Detailed Solutions

Q.1 [NAT, 1M · Informed Search] Answer: 17

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (14)	D(10), K(12), I(14)
2	D (10)	F(4), C(10), K(12), I(14)
3	F (4)	G(0), J(2), C(10), K(12), I(14)
4	G (0)	J(2), C(10), K(12), I(14) ← goal, stop

Path returned: **S** → **D** → **F** → **G**, cost = 4 + 6 + 7 = **17**.

The optimal cost is $h^*(S) = 15$ — greedy search is not optimal here.

Q.2 [MCQ, 1M · Adversarial Search] Answer: B (B: $O(b^{3m/4})$ nodes)

AIMA reports that random move ordering gives about **$O(b^{3m/4})$** for moderate b . $O(b^{m/2})$ (A) needs perfect ordering. $O(b^m)$ (C) is the worst case. $O(mb)$ (D) is a space bound.

Q.3 [MCQ, 1M · Uninformed Search] Answer: C (C: S, F, H, E, J, A, K, G)

Adjacency lists (alphabetical): A: {H, K}; E: {F, G, J, K}; F: {E, J, S}; G: {E, J}; H: {A, K, S}; J: {E, F, G}; K: {A, E, H}; S: {F, H}.

BFS trace (queue shown front → back):

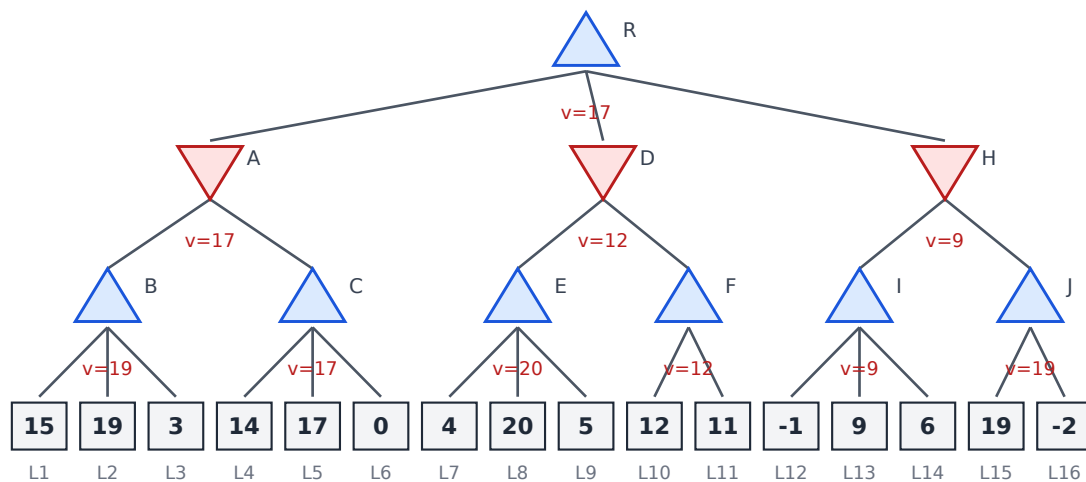
Step	Expanded	Frontier (FIFO) after expansion
1	S	[F, H]
2	F	[H, E, J]
3	H	[E, J, A, K]
4	E	[J, A, K, G]
5	J	[A, K, G]
6	A	[K, G]
7	K	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, F, H, E, J, A, K, G**. The path returned is **S** → **F** → **E** → **G** with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [MCQ, 1M · Adversarial Search] Answer: A (A: Move to A; root value 17)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 19$, $C = 17$, $A = 17$, $E = 20$, $F = 12$, $D = 12$, $I = 9$, $J = 19$, $H = 9$. Root value = **17**; MAX's optimal first move is towards A.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 343**

View R as an 3×3 0/1 matrix. The sentence says R is every row of the relation matrix must be non-empty: $(2^n - 1)$ choices per row $\Rightarrow (2^n - 1)^n$. With $n = 3$: **343**.

Q.6 [MSQ, 1M · Propositional Logic] **Answer: A, C** (A: $(p \vee \neg q) \wedge r$; C: $p \vee \neg q \vee r$)

A CNF is a conjunction of clauses, each a disjunction of literals (negation only on atoms). $(p \vee \neg q) \wedge r$ is two clauses. $p \vee \neg q \vee r$ is a single clause, which is a (one-clause) CNF. $\neg(p \wedge q)$ has negation applied to a compound formula, so it is not in CNF (its CNF is $\neg p \vee \neg q$). $(p \wedge q) \vee r$ has a conjunction inside a disjunction; its CNF is $(p \vee r) \wedge (q \vee r)$.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 4**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg Q \wedge (P \vee R)$	$\neg R \wedge (\neg P \wedge Q)$	$(\neg Q \wedge (P \vee R)) \leftrightarrow (\neg R \wedge (\neg P \wedge Q))$
T	T	T	F	F	T
T	T	F	F	F	T
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	F	F	T
F	T	F	F	T	F
F	F	T	T	F	F
F	F	F	F	F	T

Rows where F = T: **4**.

Q.8 [MSQ, 1M · Predicate Logic] **Answer: A, B, C** (A: Knows(John, Jane); B: Knows(y, Bill); C: Knows(y, Mother(y)))

(A) $\{x/\text{Jane}\}$. (B) $\{x/\text{Bill}, y/\text{John}\}$. (C) $\{y/\text{John}, x/\text{Mother(John)}\}$. (D) fails: x would have to be both John and Elizabeth. Renaming the second x (standardizing apart) would make it succeed, which is why AIMA insists on standardizing apart.

Q.9 [MCQ, 1M · Bayesian Networks] **Answer: A** (A: k for noisy-OR versus 2^k for a general CPT)

Noisy-OR needs one inhibition probability per parent, i.e. k numbers, and the full table is derived as $P(\neg x \mid \text{parents}) = \prod q_i$ over true parents. A general Boolean CPT has one free number for each of the 2^k parent configurations. This linear-vs-exponential saving is why canonical distributions are used.

Q.10 [NAT, 1M · Conditional Independence] **Answer: 3**

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of B: {A}; children: {C, E}; co-parents: $\{\emptyset\}$.

$MB(B) = \{A, C, E\}$ — **3** nodes. Given its Markov blanket, B is conditionally independent of every other node in the network.

Q.11 [MCQ, 2M · Uninformed Search] Answer: A (A: $S \rightarrow F \rightarrow H \rightarrow C \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

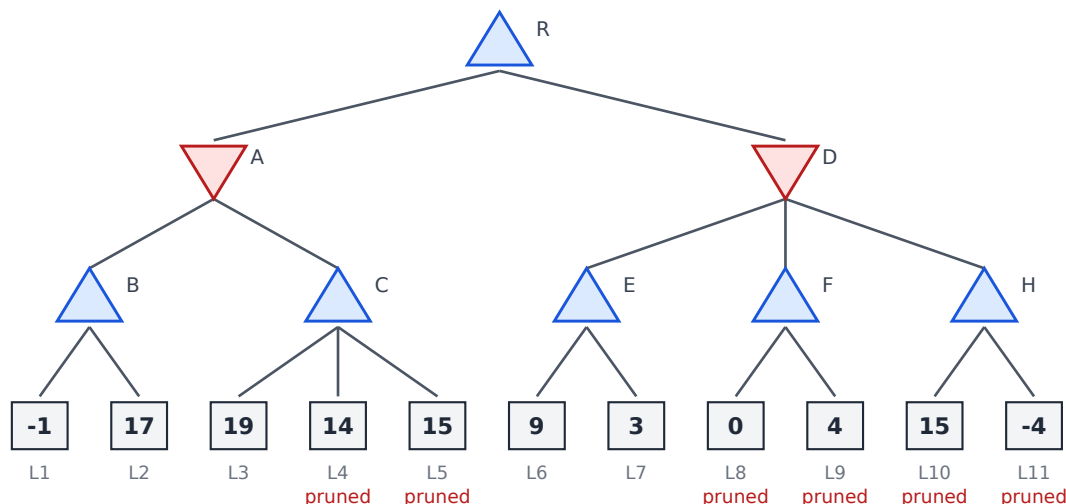
Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	F(2), I(5)
2	F (2)	I(5), H(8), A(9)
3	I (5)	A(7), H(8)
4	A (7)	H(8), C(15), G(16)
5	H (8)	C(12), J(12), G(16)
6	C (12)	J(12), G(14)
7	J (12)	G(14)
8	G (14)	← goal, stop

Path returned: **S** → **F** → **H** → **C** → **G**, cost = 2 + 6 + 4 + 2 = **14**. Nodes expanded (including S and G): **8** (S, F, I, A, H, C, J, G).

Q.12 [NAT, 2M · Adversarial Search] Answer: 5

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value 19 $\geq \beta = 17 \Rightarrow$ prune remaining children (leaves L4, L5).
- At MIN node D: value 9 $\leq \alpha = 17 \Rightarrow$ prune remaining children (leaves L8, L9, L10, L11).



Leaves evaluated (5): L1, L2, L3, L6, L7. Leaves pruned (6): L4, L5, L8, L9, L10, L11. Root value = 17 (same as plain minimax, 17 — pruning never changes the root value).

Q.13 [NAT, 2M · Informed Search] Answer: 16

Heuristic values: $h(B)=4$, $h(C)=6$, $h(E)=9$, $h(F)=5$, $h(G)=0$, $h(H)=7$, $h(I)=9$, $h(L)=8$, $h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	L(13)
2	L (13)	F(14)
3	F (14)	G(16), C(19), H(19)
4	G (16)	C(19), H(19) ← goal, stop

Path returned: **S** → **L** → **F** → **G** with cost **16**; nodes expanded = **4** (S, L, F, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.14 [MSQ, 2M · Adversarial Search] Answer: A, B, C

(A) and (B) are true: minimax is a complete depth-first exploration of the tree. (C) is true: minimax is optimal against an optimal adversary and guarantees the root's minimax value. (D) is false: when the search reaches terminal states, the exact utilities are used. An evaluation function is needed only when the search is cut off early.

Q.15 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $(\neg Q)$; D: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable))

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(\neg Q)$ is derivable:

1. resolve $(\neg Q \vee S)$ and $(\neg S)$ on $S \Rightarrow (\neg Q)$

(C) $(\neg R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $\square$ is derivable:

1. resolve (S) and $(\neg S)$ on $S \Rightarrow \square$ (empty clause)

Q.16 [MCQ, 2M · Predicate Logic] Answer: A (A: $\forall x (P(x) \rightarrow \exists y \neg Q(x, y))$)

$\neg \exists x (\dots) \equiv \forall x \neg(P(x) \wedge \forall y Q(x, y)) \equiv \forall x (\neg P(x) \vee \exists y \neg Q(x, y)) \equiv \forall x (P(x) \rightarrow \exists y \neg Q(x, y))$. Option B uses $\wedge$ instead of $\vee$ (De Morgan applied wrongly). Option C fails to flip $\exists x$ to $\forall x$. Option D requires every element to satisfy P.

Q.17 [MSQ, 2M · Predicate Logic] Answer: B, D (B: $\exists y \forall x R(x, y)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,2), (1,3), (1,4), (2,4), (3,4), (4,2), (4,4)\}$.

(A) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(B) $\exists y \forall x R(x, y)$: **TRUE** — $y = 4$ receives an edge from every x (incl. itself).

(C) $\forall y \exists x R(x, y)$: **FALSE** — 1 has no incoming edge.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 1 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B (B: $R \rightarrow (Q \rightarrow R)$)

(A) $(Q \rightarrow R) \rightarrow (R \rightarrow Q)$: **not valid** — counter-model $Q=F, R=T$.

(B) $R \rightarrow (Q \rightarrow R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $\neg(P \wedge Q) \leftrightarrow (\neg P \wedge \neg Q)$: **not valid** — counter-model $P=T, Q=F$.

(D) $((R \rightarrow P) \wedge P) \rightarrow R$: **not valid** — counter-model $P=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.19 [NAT, 2M · Sampling] Answer: 0.49 to 0.51

Rejection sampling keeps only samples consistent with the evidence $B = T$: samples 1, 7, 19, 20 (4 accepted, 16 rejected).

Among the accepted samples, $A = T$ in 2. Estimate = $2/4 = 0.5$.

(For this network the exact value is 0.8235; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.20 [NAT, 2M · Sampling] Answer: 0.523 to 0.524

Weights $w = P(s | C) \times P(w | s, R)$: (c, r) : $0.1 \times 0.99 = 0.099$; $(\neg c, r)$: $0.5 \times 0.99 = 0.495$; $(\neg c, \neg r)$: $0.5 \times 0.90 = 0.45$; $(c, \neg r)$: $0.1 \times 0.90 = 0.09$. Total weight = $0.099 + 0.495 + 0.45 + 0.09 = 1.134$; weight with $r = 0.099 + 0.495 = 0.594$. Estimate = $0.594/1.134 \approx 0.5238 \approx 0.524$. (The exact $P(r | s, w) \approx 0.320$; with only four samples the estimate is far off, but LW is consistent as $N \rightarrow \infty$.)

Q.21 [NAT, 2M · Variable Elimination] Answer: 0.531 to 0.533

$f(b, e) = P(a | b, e) P(j | a) P(m | a) + P(\neg a | b, e) P(j | \neg a) P(m | \neg a) = 0.95 \times 0.8 \times 0.7 + 0.05 \times 0.05 \times 0.02 = 0.532 + 0.00005 = 0.53205 \approx 0.532$. This is not a probability distribution over B, E ; it is an intermediate factor.

Q.22 [MCQ, 2M · Sampling] Answer: D (D: $A=T, B=F, C=F, D=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.7$; $u = 0.03 < 0.7 \Rightarrow A = T$
- B: $P(B=T | A=T) = 0.6$; $u = 0.88 \geq 0.6 \Rightarrow B = F$
- C: $P(C=T | A=T) = 0.2$; $u = 0.3 \geq 0.2 \Rightarrow C = F$
- D: $P(D=T | A=T) = 0.9$; $u = 0.76 < 0.9 \Rightarrow D = T$

Sample: **$\{A=T, B=F, C=F, D=T\}$** .

Q.23 [MCQ, 2M · Exact Inference] Answer: A (A: f(A, C))

Initial factors (one per CPT, evidence removed from scopes): f(A), f(A, B), f(A, B, C), f(B, C, D), f(A, B), f(C, F).
Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	f(A, B), f(A, B, C), f(B, C, D), f(A, B)	{A, B, C, D}	f(A, C, D)
D	f(A, C, D)	{A, C, D}	f(A, C)
F	f(C, F)	{C, F}	f(C)
A	f(A), f(A, C)	{A, C}	f(C)

The largest factor produced has **3** variable(s) (created when eliminating B); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over C and normalise.

Q.24 [MSQ, 2M · Conditional Independence] Answer: A, C (A: $H \perp C$; C: $H \perp G \mid B, E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $H \perp C$: **TRUE** — $H \leftarrow B \rightarrow D \leftarrow C$ (blocked: collider D with neither it nor a descendant observed); $H \leftarrow B \rightarrow D \rightarrow E \rightarrow F \leftarrow A \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $H \leftarrow B \rightarrow D \rightarrow E \rightarrow F \rightarrow G \leftarrow A \rightarrow C$ (blocked: collider G with neither it nor a descendant observed); $H \leftarrow B \rightarrow E \leftarrow D \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $H \leftarrow B \rightarrow E \rightarrow F \leftarrow A \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $H \leftarrow B \rightarrow E \rightarrow F \rightarrow G \leftarrow A \rightarrow C$ (blocked: collider G with neither it nor a descendant observed).

(B) $G \perp C \mid A$: **FALSE** — the path $G \leftarrow F \leftarrow E \leftarrow D \leftarrow C$ is active given {A}.

(C) $H \perp G \mid B, E$: **TRUE** — $H \leftarrow B \rightarrow D \leftarrow C \leftarrow A \rightarrow F \rightarrow G$ (blocked: observed fork node B); $H \leftarrow B \rightarrow D \leftarrow C \leftarrow A \rightarrow G$ (blocked: observed fork node B); $H \leftarrow B \rightarrow D \rightarrow E \rightarrow F \leftarrow A \rightarrow G$ (blocked: observed fork node B); $H \leftarrow B \rightarrow D \rightarrow E \rightarrow F \rightarrow G$ (blocked: observed fork node B); $H \leftarrow B \rightarrow E \leftarrow D \leftarrow C \leftarrow A \rightarrow F \rightarrow G$ (blocked: observed fork node B); $H \leftarrow B \rightarrow E \leftarrow D \leftarrow C \leftarrow A \rightarrow G$ (blocked: observed fork node B) (+2 more paths, all blocked).

(D) $D \perp B \mid E$: **FALSE** — the path $D \leftarrow B$ is active given {E}.

Q.25 [NAT, 2M · Exact Inference] Answer: 0.019 to 0.023

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution} \mid \text{Smoker}) \cdot P(\text{Cancer}) \cdot P(\text{XRay} \mid \text{Pollution}, \text{Cancer})$.

Inference by enumeration: $P(\text{Smoker} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Smoker}, e, \text{hidden})$.

Hidden: Ca, XR	Sm	product of CPT entries (Sm · Po · Ca · XR)	value
T, T	T	$0.05 \times 0.3 \times 0.4 \times 0.4$	0.0024
T, F	T	$0.05 \times 0.3 \times 0.4 \times 0.6$	0.0036
F, T	T	$0.05 \times 0.3 \times 0.6 \times 0.95$	0.00855
F, F	T	$0.05 \times 0.3 \times 0.6 \times 0.05$	0.00045
T, T	F	$0.95 \times 0.75 \times 0.4 \times 0.4$	0.114
T, F	F	$0.95 \times 0.75 \times 0.4 \times 0.6$	0.171
F, T	F	$0.95 \times 0.75 \times 0.6 \times 0.95$	0.40612
F, F	F	$0.95 \times 0.75 \times 0.6 \times 0.05$	0.02138

$P(\text{Smoker}=T, e) = 0.015$; $P(\text{Smoker}=F, e) = 0.7125$; $P(e) = 0.7275$.

$P(\text{Smoker}=T \mid e) = 0.015 / 0.7275 = \mathbf{0.0206} \approx 0.021$.

Mock Test 045

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 8 / 10	11 / 15 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

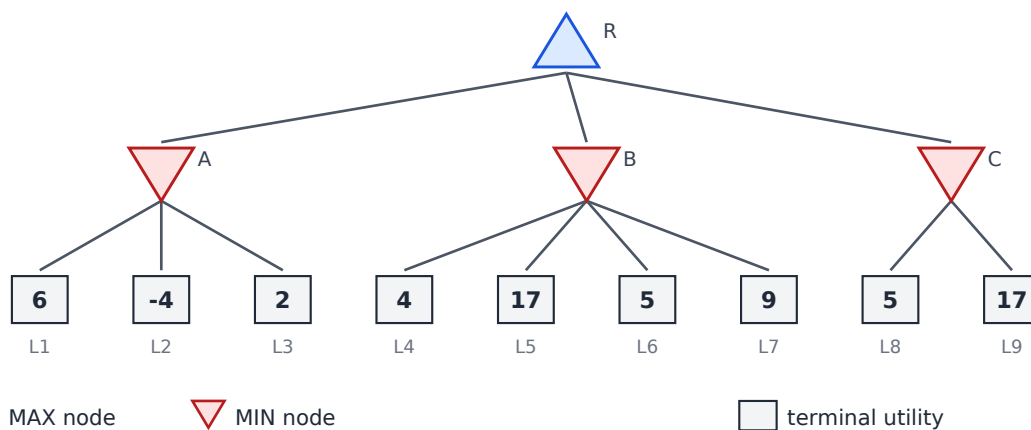
Q.1 [NAT · 1 mark · no negative marking] Informed Search – Effective branching factor

A* finds a solution at depth $d = 3$ after generating $N = 7$ nodes (root excluded). The effective branching factor b^* satisfies $N = b^* + (b^*)^2 + \dots + (b^*)^d$. Find b^* (round off to 2 decimal places).

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

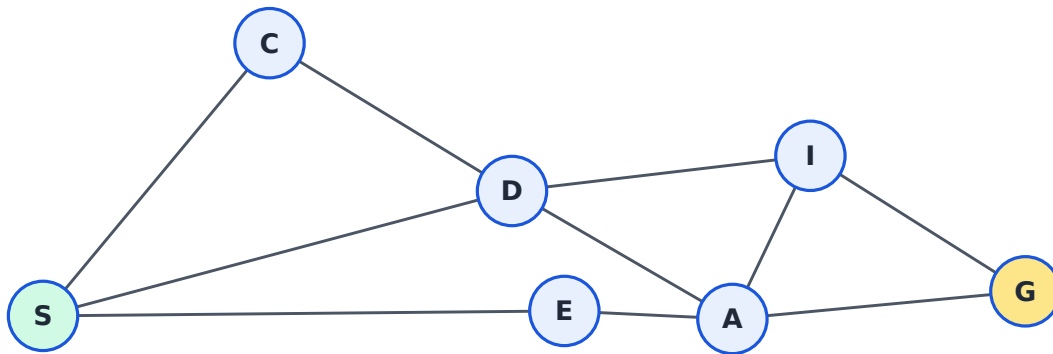
Q.3 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search

With perfect move ordering, the time complexity of alpha-beta search to depth m with branching factor b is

- ☐ (A) $O(b^m)$
- ☐ (B) $O(b^{m/2})$
- ☐ (C) $O(b^{3m/4})$
- ☐ (D) $O(bm)$

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, C, D, E, A, I, G
- ☐ (B) S, E, D, C, A, I, G
- ☐ (C) S, C, D, A, E, G
- ☐ (D) S, C, E, D, A, I, G

Q.5 [NAT · 1 mark · no negative marking] Informed Search

In the 8-puzzle, the goal configuration is (rows top to bottom) 1 2 3 / 4 5 6 / 7 8 _, where _ is the blank. The current state is 8 1 3 / 4 _ 2 / 7 6 5. What is the value of h_1 , the number of misplaced tiles (do not count the blank)?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following statements about forward and backward chaining over Horn knowledge bases is correct?

- ☐ (A) Forward chaining is data-driven while backward chaining is goal-driven
- ☐ (B) Forward chaining is goal-driven while backward chaining is data-driven
- ☐ (C) Both are complete for arbitrary (non-Horn) propositional knowledge bases
- ☐ (D) Backward chaining can derive facts that forward chaining cannot

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Resolving the first-order clauses $P(x) \vee Q(f(x))$ and $\neg P(g(y)) \vee R(y)$ (variables already standardized apart) on P gives

- ☐ (A) $Q(f(g(y))) \vee R(y)$
- ☐ (B) $Q(f(x)) \vee R(y)$
- ☐ (C) $Q(g(f(y))) \vee R(y)$
- ☐ (D) $Q(f(y)) \vee R(g(y))$

Q.8 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q). For how many of them is $\neg \exists x (P(x) \wedge \neg Q(x)) \wedge \exists x P(x)$ true?

Answer (numerical): _____

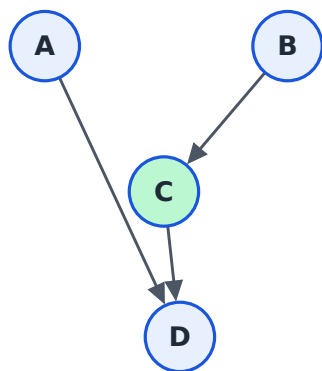
Q.9 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

A Bayesian network has edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$. Which is the joint distribution it represents?

- ☐ (A) $P(A) P(B | A) P(C | A) P(D | B, C)$
☐ (B) $P(A) P(B | A) P(C | B) P(D | C)$
☐ (C) $P(A) P(B) P(C) P(D | B, C)$
☐ (D) $P(D) P(B | D) P(C | D) P(A | B, C)$

Q.10 [NAT · 1 mark · no negative marking] Exact Inference - Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T | \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



$P(A=T)$
0.3

$P(B=T)$	B	$P(C=T ...)$
0.25	T	0.95
	F	0.3

A	C	$P(D=T ...)$
T	T	0.75
T	F	0.3
F	T	0.1
F	F	0.7

Answer (numerical): _____

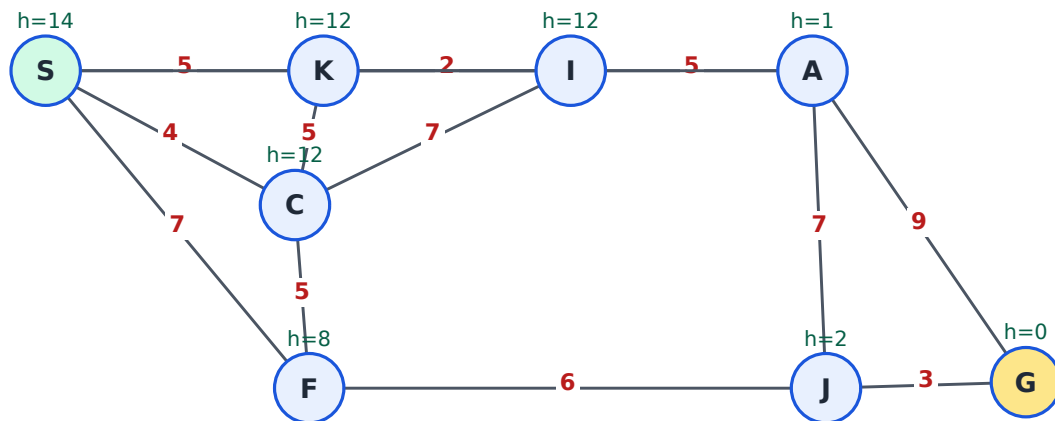
Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search

Assume $h(\text{goal}) = 0$ and all step costs are positive. Which of the following are TRUE?

- ☐ (A) Every consistent heuristic is admissible
☐ (B) Every admissible heuristic is consistent
☐ (C) If h is consistent, the values of $f(n) = g(n) + h(n)$ along any path are non-decreasing
☐ (D) A* tree search with an admissible heuristic is optimal

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

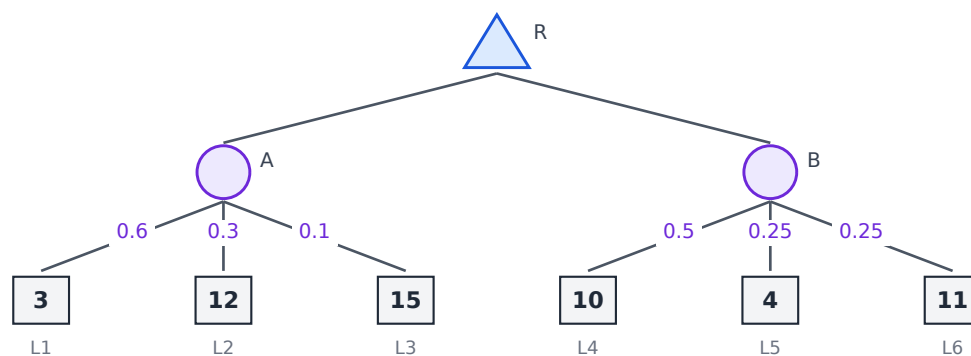
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge K-A violates the consistency condition $h(K) \leq c(K,A) + h(A)$.
☐ (B) h is admissible.
☐ (C) $h(A)$ overestimates the true cost from A to G.
☐ (D) h is consistent (monotone).

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node

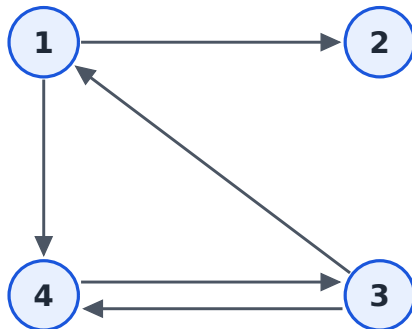
○ CHANCE node (edge = probability)

□ terminal utility

Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (C) $\forall y \exists x R(x, y)$
☐ (D) $\exists x R(x, x)$

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following correctly express 'Every person who owns a dog is happy'?

- ☐ (A) $\forall x ((\text{Person}(x) \wedge \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y))) \rightarrow \text{Happy}(x))$
☐ (B) $\forall x \forall y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$
☐ (C) $\forall x \exists y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$
☐ (D) $\forall x (\text{Person}(x) \rightarrow \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y) \wedge \text{Happy}(x)))$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$
☐ (B) $((\neg R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (\neg R \rightarrow P)$
☐ (C) $((R \rightarrow Q) \wedge Q) \rightarrow R$
☐ (D) $((Q \rightarrow R) \wedge \neg R) \rightarrow \neg Q$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(\neg Q \vee Q) \rightarrow R$
 (2) $(P \rightarrow \neg R) \leftrightarrow (R \rightarrow Q)$
 (3) $\neg R \vee Q$

Which of the following sentences is/are entailed by KB ($\text{KB} \models \alpha$)?

- ☐ (A) $\neg R$
☐ (B) $\neg Q \leftrightarrow Q$
☐ (C) $(P \rightarrow R) \wedge \neg \neg P$
☐ (D) $P \vee R$

Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

A KB consists of the definite clauses $\text{Nat}(0)$ and $\text{Nat}(x) \rightarrow \text{Nat}(S(x))$. Forward chaining is used to answer the query $\text{Nat}(\text{Foo})$, where Foo is a constant different from 0. Which one of the following is correct?

- ☐ (A) Forward chaining never terminates, since it keeps adding $\text{Nat}(S(\dots S(0)\dots))$
- ☐ (B) Forward chaining terminates and answers 'not entailed'
- ☐ (C) Forward chaining derives $\text{Nat}(\text{Foo})$
- ☐ (D) Forward chaining fails immediately since the KB is not in Datalog form and therefore cannot be processed

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

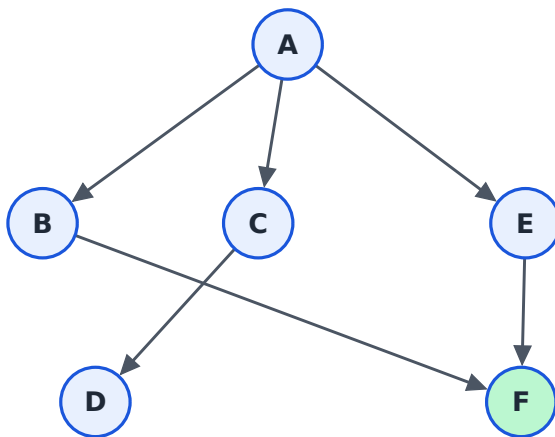
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = \neg(S \rightarrow R) \vee (P \wedge (\neg S \rightarrow \neg Q)) ?$$

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(F)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, C, B, E, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

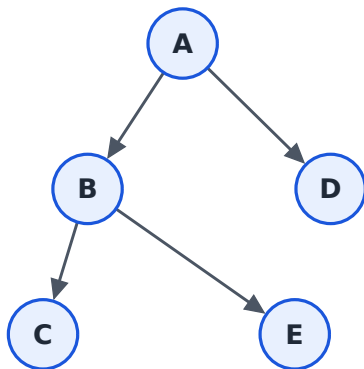
Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence

A Bayesian network over A, B, C, D encodes the factorization $P(A) P(B | A) P(C | A) P(D | B, C)$. Which independences are guaranteed to hold in every distribution with this factorization?

- ☐ (A) $B \perp C \mid \{A\}$
- ☐ (B) $B \perp C \mid \{A, D\}$
- ☐ (C) $A \perp D \mid \{B, C\}$
- ☐ (D) $A \perp D$

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.35, 0.09, 0.57, 0.61, 0.91)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



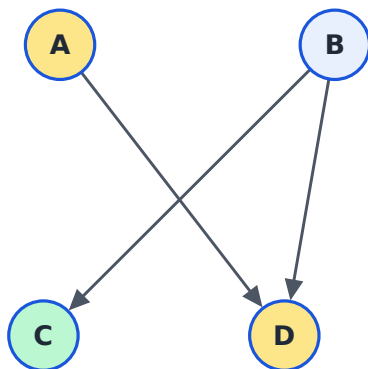
P(A=T)	A	P(B=T ...)	B	P(C=T ...)
0.7	T	0.2	T	0.7
	F	0.05	F	0.25

A	P(D=T ...)	B	P(E=T ...)
T	0.6	T	0.95
F	0.9	F	0.9

- ☐ (A) A=T, B=T, C=T, D=F, E=T
☐ (B) A=T, B=F, C=T, D=F, E=T
☐ (C) A=F, B=T, C=F, D=F, E=F
☐ (D) A=F, B=T, C=T, D=F, E=T

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $\mathbf{P(C = T \mid D = T, A = T)}$ (round off to 3 decimal places).



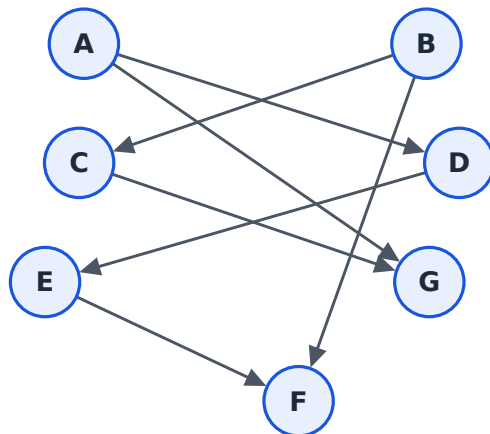
P(A=T)	P(B=T)	B	P(C=T ...)
0.8	0.75	T	0.75
		F	0.05

A	B	P(D=T ...)
T	T	0.95
T	F	0.1
F	T	0.8
F	F	0.1

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

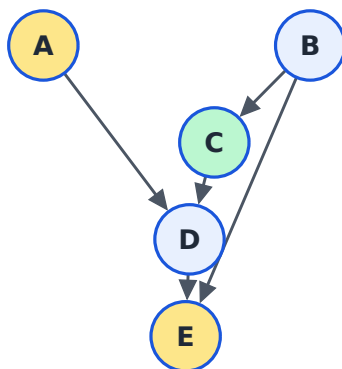
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp D \mid B, F$
☐ (B) $C \perp E$
☐ (C) $G \perp C$
☐ (D) $E \perp B$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $E = T$, $A = T$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $P(C = T \mid E = T, A = T)$ (round off to 3 decimal places).



P(A=T)
0.1

P(B=T)
0.9

B	P(C=T ...)
T	0.75
F	0.25

A	C	P(D=T B, D)	B	D	P(E=T ...)
T	T	0.7	T	T	0.95
T	F	0.4	T	F	0.05
F	T	0.25	F	T	0.8
F	F	0.75	F	F	0.5

#	B	C	D
1	T	T	F
2	T	T	F
3	T	F	T
4	T	T	T
5	T	T	T

Answer (numerical): _____

Answer Key — Mock Test 045

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Informed Search	1.47 to 1.51	14	MSQ	2	Predicate Logic	A, C
2	NAT	1	Adversarial Search	5	15	MSQ	2	Predicate Logic	A, B
3	MCQ	1	Adversarial Search	B	16	MSQ	2	Propositional Logic	B, D
4	MCQ	1	Uninformed Search	A	17	MSQ	2	Propositional Logic	D
5	NAT	1	Informed Search	5	18	MCQ	2	Predicate Logic	A
6	MCQ	1	Propositional Logic	A	19	NAT	2	Propositional Logic	8
7	MCQ	1	Predicate Logic	A	20	NAT	2	Exact Inference	3
8	NAT	1	Predicate Logic	65	21	MSQ	2	Conditional Independence	A, C
9	MCQ	1	Bayesian Networks	A	22	MCQ	2	Sampling	A
10	NAT	1	Exact Inference	0.46 to 0.464	23	NAT	2	Exact Inference	0.724 to 0.728
11	MSQ	2	Informed Search	A, C, D	24	MSQ	2	Conditional Independence	A, B, D
12	MSQ	2	Informed Search	A, B	25	NAT	2	Sampling	0.676 to 0.68
13	NAT	2	Adversarial Search	8.74 to 8.76					

Detailed Solutions

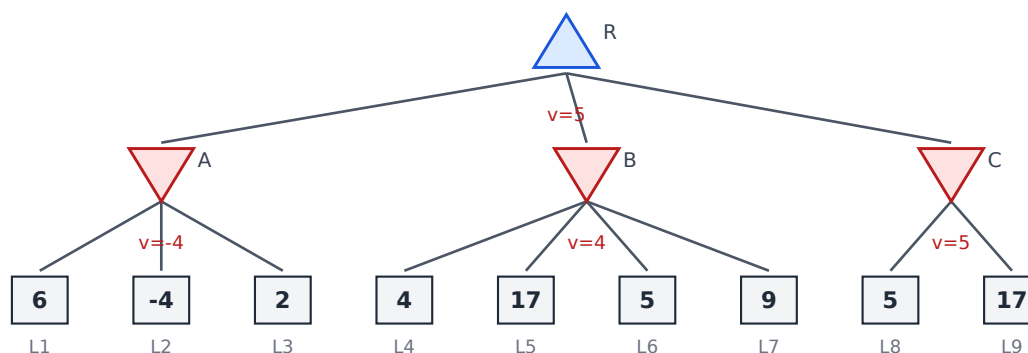
Q.1 [NAT, 1M · Informed Search] **Answer: 1.47 to 1.51**

Solve $b + b^2 + \dots + b^3 = 7$ numerically (bisection): $b^* \approx 1.49$.

Check: $1.49^1 + 1.49^2 + 1.49^3 \approx 7.02 \approx 7$.

Q.2 [NAT, 1M · Adversarial Search] **Answer: 5**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -4$, $B = 4$, $C = 5$. Root value = **5**; MAX's optimal first move is towards C.

Q.3 [MCQ, 1M · Adversarial Search] **Answer: B** ($O(b^{m/2})$)

Perfect ordering examines about $b^{\lceil m/2 \rceil} + b^{\lfloor m/2 \rfloor} - 1$ leaves, which is **$O(b^{m/2})$** . $O(b^m)$ is minimax, or the worst ordering (A). $O(b^{3m/4})$ is the random-ordering estimate (C). $O(bm)$ is minimax's space, not its time (D).

Q.4 [MCQ, 1M · Uninformed Search] **Answer: A** (A: S, C, D, E, A, I, G)

Adjacency lists (alphabetical): A: {D, E, G, I}; C: {D, S}; D: {A, C, I, S}; E: {A, S}; G: {A, I}; I: {A, D, G}; S: {C, D, E}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[C, D, E]
2	C	[D, E]
3	D	[E, A, I]
4	E	[A, I]
5	A	[I, G]
6	I	[G]
7	G	[] $\leftarrow$ goal popped, stop

Expansion order: **S, C, D, E, A, I, G**. The path returned is $S \rightarrow D \rightarrow A \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [NAT, 1M · Informed Search] Answer: 5

Compare position by position. Tiles 3, 4 and 7 are already in place. Tiles 8, 1, 2, 6 and 5 are not. $h_1 = 5$.

Q.6 [MCQ, 1M · Propositional Logic] Answer: A (A: Forward chaining is data-driven while backward chaining is goal-driven)

Forward chaining starts from known facts and fires rules whose premises are satisfied (data-driven).

Backward chaining starts from the query and works back through rules whose conclusion matches (goal-driven). Both are complete only for Horn (definite clause) KBs, not arbitrary ones. For Horn KBs both derive exactly the entailed atoms, so neither derives anything the other cannot.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A (A: $Q(f(g(y))) \vee R(y)$)

Unify $P(x)$ with $P(g(y))$: MGU $\theta = \{x/g(y)\}$. The resolvent is $(Q(f(x)) \vee R(y))\theta = Q(f(g(y))) \vee R(y)$. Option B forgets to apply θ . Options C and D apply the substitution incorrectly.

Q.8 [NAT, 1M · Predicate Logic] Answer: 65

Interpret P and Q as subsets of the domain; $P \subseteq Q$ gives 3^n ; subtract the 2^n with P empty $\Rightarrow 3^n - 2^n$. With $n = 4$ this gives **65** (verified by enumerating all interpretations).

Q.9 [MCQ, 1M · Bayesian Networks] Answer: A (A: $P(A) P(B | A) P(C | A) P(D | B, C)$)

A BN represents $\prod_i P(X_i | \text{Parents}(X_i))$. Parents: A: none; B: A; C: A; D: B, C. This gives (A). (B) corresponds to a chain $A \rightarrow B \rightarrow C \rightarrow D$; (C) drops edges from A; (D) reverses every edge.

Q.10 [NAT, 1M · Exact Inference] Answer: 0.46 to 0.464

Only the ancestors of C (B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T, D=T	0.05344
A=T, B=T, D=F	0.01781
A=T, B=F, D=T	0.05062
A=T, B=F, D=F	0.01687
A=F, B=T, D=T	0.01662
A=F, B=T, D=F	0.14962
A=F, B=F, D=T	0.01575
A=F, B=F, D=F	0.14175

Sum = **0.4625** ≈ 0.462 .

Q.11 [MSQ, 2M · Informed Search] Answer: A, C, D

(A) is true: induct backward along an optimal path from the goal, using $h(n) \leq c + h(n') \leq c + h^*(n') = h^*(n)$.

(B) is false: admissible but inconsistent heuristics exist, for example $h(A)=6$ and $h(B)=2$ with $c(A,B)=2$. (C) is

true: $f(n') = g(n) + c + h(n') \geq g(n) + h(n) = f(n)$. (D) is true: for tree search, admissibility is enough for optimality. Consistency is needed for graph search without reopening.

Q.12 [MSQ, 2M · Informed Search] Answer: A, B (A: The edge $K \rightarrow A$ violates the consistency condition $h(K) \leq c(K,A) + h(A)$.; B: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	C	F	G	I	J	K	S
$h(n)$	1	12	8	0	12	2	12	14
$h^*(n)$	9	14	9	0	14	3	14	16
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $I \rightarrow A$ ($12 > 5 + 1$), $K \rightarrow A$ ($12 > 5 + 1$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Adversarial Search] Answer: 8.74 to 8.76

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

$$\bullet A = 0.6 \times 3 + 0.3 \times 12 + 0.1 \times 15 = 6.9$$

$$\bullet B = 0.5 \times 10 + 0.25 \times 4 + 0.25 \times 11 = 8.75$$

$$\text{Root} = \max(6.9, 8.75) = \mathbf{8.75}.$$

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.14 [MSQ, 2M · Predicate Logic] Answer: A, C (A: $\exists x \forall y \neg R(x, y)$; C: $\forall y \exists x R(x, y)$)

$$R = \{(1,2), (1,4), (3,1), (3,4), (4,3)\}.$$

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.

(B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(2,3)$ nor $R(3,2)$.

(C) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(D) $\exists x R(x, x)$: **FALSE** — there are no self-loops.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.15 [MSQ, 2M · Predicate Logic] Answer: A, B (A: $\forall x ((\text{Person}(x) \wedge \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y))) \rightarrow \text{Happy}(x))$; B: $\forall x \forall y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$)

(A) is the direct reading. (B) is equivalent: $(\exists y \phi(y)) \rightarrow \psi \equiv \forall y (\phi(y) \rightarrow \psi)$ when y is not free in ψ . (C) $\exists y (\phi(y) \rightarrow \psi) \equiv (\forall y \phi(y)) \rightarrow \psi$, which is vacuously true for x as soon as some object is not a dog owned by x , so it imposes almost no constraint. (D) asserts every person owns a dog. Brute force on small domains confirms $A \equiv B$ and that C, D differ.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $((\neg R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (\neg R \rightarrow P)$; D: $((Q \rightarrow R) \wedge \neg R) \rightarrow \neg Q$)

(A) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$: **not valid** — counter-model $P=T, Q=F, R=T$.

(B) $((\neg R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (\neg R \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $((R \rightarrow Q) \wedge Q) \rightarrow R$: **not valid** — counter-model $Q=T, R=F$.

(D) $((Q \rightarrow R) \wedge \neg R) \rightarrow \neg Q$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [MSQ, 2M · Propositional Logic] Answer: D (D: $P \vee R$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
F	T	T	T

(A) $\neg R$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

(B) $\neg Q \leftrightarrow Q$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

(C) $(P \rightarrow R) \wedge \neg P$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

(D) $P \vee R$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.18 [MCQ, 2M · Predicate Logic] Answer: A (A: Forward chaining never terminates, since it keeps adding $\text{Nat}(S(\dots S(0)\dots))$)

$\text{Nat}(\text{Foo})$ is not entailed (there is a model where Foo is not a natural number). Forward chaining derives $\text{Nat}(S(0))$, $\text{Nat}(S(S(0)))$, ... forever, because the function symbol S yields infinitely many ground terms. This illustrates that FOL entailment with definite clauses is only semidecidable: entailed queries are eventually found, non-entailed ones may run forever. Forward chaining still applies to non-Datalog KBs; it just lacks a termination guarantee.

Q.19 [NAT, 2M · Propositional Logic] Answer: 8

Build the truth table (sub-formulas shown first):

P	Q	R	S	$\neg(S \rightarrow R)$	$P \wedge (\neg S \rightarrow \neg Q)$	$\neg(S \rightarrow R) \vee (P \wedge (\neg S \rightarrow \neg Q))$
T	T	T	T	F	T	T
T	T	T	F	F	F	F
T	T	F	T	T	T	T
T	T	F	F	F	F	F
T	F	T	T	F	T	T
T	F	T	F	F	T	T

P	Q	R	S	$\neg(S \rightarrow R)$	$P \wedge (\neg S \rightarrow \neg Q)$	$\neg(S \rightarrow R) \vee (P \wedge (\neg S \rightarrow \neg Q))$
T	F	F	T	T	T	T
T	F	F	F	F	T	T
F	T	T	T	F	F	F
F	T	T	F	F	F	F
F	T	F	T	T	F	T
F	T	F	F	F	F	F
F	F	T	T	F	F	F
F	F	T	F	F	F	F
F	F	F	T	T	F	T
F	F	F	F	F	F	F

Rows where F = T: **8**.

Q.20 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, C)$, $f(C, D)$, $f(A, E)$, $f(B, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, B)$, $f(A, C)$, $f(A, E)$	$\{A, B, C, E\}$	$f(B, C, E)$
C	$f(C, D)$, $f(B, C, E)$	$\{B, C, D, E\}$	$f(B, D, E)$
B	$f(B, E, F)$, $f(B, D, E)$	$\{B, D, E, F\}$	$f(D, E, F)$
E	$f(D, E, F)$	$\{D, E, F\}$	$f(D, F)$
D	$f(D, F)$	$\{D, F\}$	$f(F)$

The largest factor produced has **3** variable(s) (created when eliminating A); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over F and normalise.

Q.21 [MSQ, 2M · Conditional Independence] **Answer: A, C** ($A: B \perp C \mid \{A\}$; $C: A \perp D \mid \{B, C\}$)

The DAG is $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$. (A) B-C paths: $B \leftarrow A \rightarrow C$ (fork, A observed: blocked) and $B \rightarrow D \leftarrow C$ (collider D unobserved, no descendants: blocked): d-separated, true. (B) Observing D activates the collider path $B \rightarrow D \leftarrow C$: false. (C) Both A-D paths $A \rightarrow B \rightarrow D$ and $A \rightarrow C \rightarrow D$ pass through observed non-colliders B and C: d-separated, true. (D) With nothing observed, $A \rightarrow B \rightarrow D$ is active: false.

Q.22 [MCQ, 2M · Sampling] **Answer: A** ($A: A=T, B=T, C=T, D=F, E=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.7$; $u = 0.35 < 0.7 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.2$; $u = 0.09 < 0.2 \Rightarrow B = T$
- C: $P(C=T \mid B=T) = 0.7$; $u = 0.57 < 0.7 \Rightarrow C = T$
- D: $P(D=T \mid A=T) = 0.6$; $u = 0.61 \geq 0.6 \Rightarrow D = F$
- E: $P(E=T \mid B=T) = 0.95$; $u = 0.91 < 0.95 \Rightarrow E = T$

Sample: **$\{A=T, B=T, C=T, D=F, E=T\}$** .

Q.23 [NAT, 2M · Exact Inference] **Answer: 0.724 to 0.728**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B) \cdot P(C \mid B) \cdot P(D \mid A, B)$.

Inference by enumeration: $P(C \mid e) = \alpha \sum_{\text{hidden}} P(C, e, \text{hidden})$.

Hidden: B	C	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T	T	$0.8 \times 0.75 \times 0.75 \times 0.95$	0.4275
F	T	$0.8 \times 0.25 \times 0.05 \times 0.1$	0.001
T	F	$0.8 \times 0.75 \times 0.25 \times 0.95$	0.1425
F	F	$0.8 \times 0.25 \times 0.95 \times 0.1$	0.019

$P(C=T, e) = 0.4285$; $P(C=F, e) = 0.1615$; $P(e) = 0.59$.

$P(C=T \mid e) = 0.4285 / 0.59 = \mathbf{0.7263} \approx 0.726$.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: A, B, D** ($A: C \perp D \mid B, F$; $B: C \perp E$; $D: E \perp B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp D \mid B, F$: **TRUE** — $C \leftarrow B \rightarrow F \leftarrow E \leftarrow D$ (blocked: observed fork node B); $C \rightarrow G \leftarrow A \rightarrow D$ (blocked: collider G with neither it nor a descendant observed).

(B) $C \perp E$: **TRUE** — $C \leftarrow B \rightarrow F \leftarrow E$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow G \leftarrow A \rightarrow D \rightarrow E$ (blocked: collider G with neither it nor a descendant observed).

(C) $G \perp C$: **FALSE** — the path $G \leftarrow C$ is active given $\{\}$.

(D) $E \perp B$: **TRUE** — $E \leftarrow D \leftarrow A \rightarrow G \leftarrow C \leftarrow B$ (blocked: collider G with neither it nor a descendant observed); $E \rightarrow F \leftarrow B$ (blocked: collider F with neither it nor a descendant observed).

Q.25 [NAT, 2M · Sampling] **Answer: 0.676 to 0.68**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	C
1	$P(A=T)=0.1 \times P(E=T \mid B=T, D=F)=0.05$	0.005	T
2	$P(A=T)=0.1 \times P(E=T \mid B=T, D=F)=0.05$	0.005	T
3	$P(A=T)=0.1 \times P(E=T \mid B=T, D=T)=0.95$	0.095	F
4	$P(A=T)=0.1 \times P(E=T \mid B=T, D=T)=0.95$	0.095	T
5	$P(A=T)=0.1 \times P(E=T \mid B=T, D=T)=0.95$	0.095	T

$\sum w (\text{all}) = 0.295$; $\sum w (C = T) = 0.2$.

Estimate $P(C=T \mid E = T, A = T) = 0.2 / 0.295 = \mathbf{0.678}$.

Mock Test 046

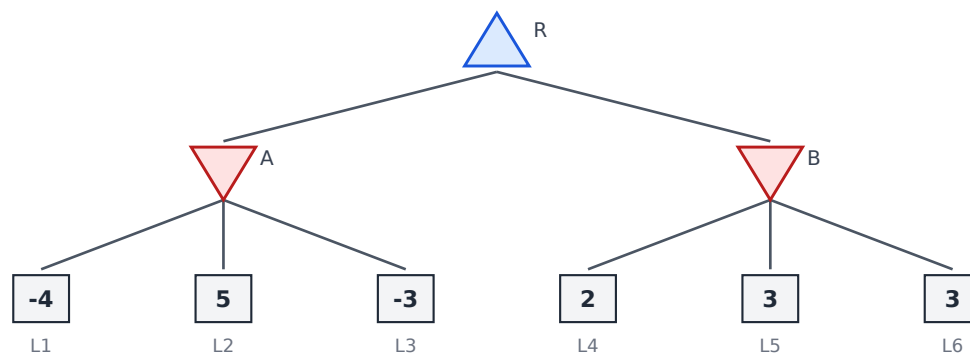
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 10 / 9	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



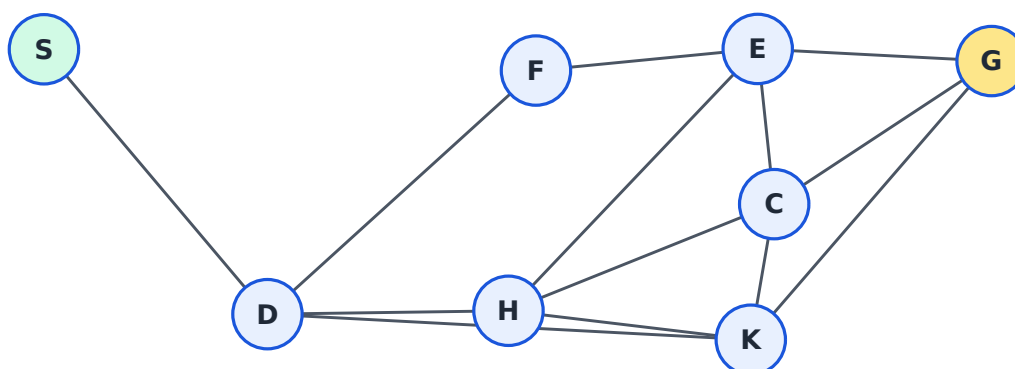
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to B; root value 2
- ☐ (B) Move to B; root value 4
- ☐ (C) Move to B; root value 5
- ☐ (D) Move to A; root value -4

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

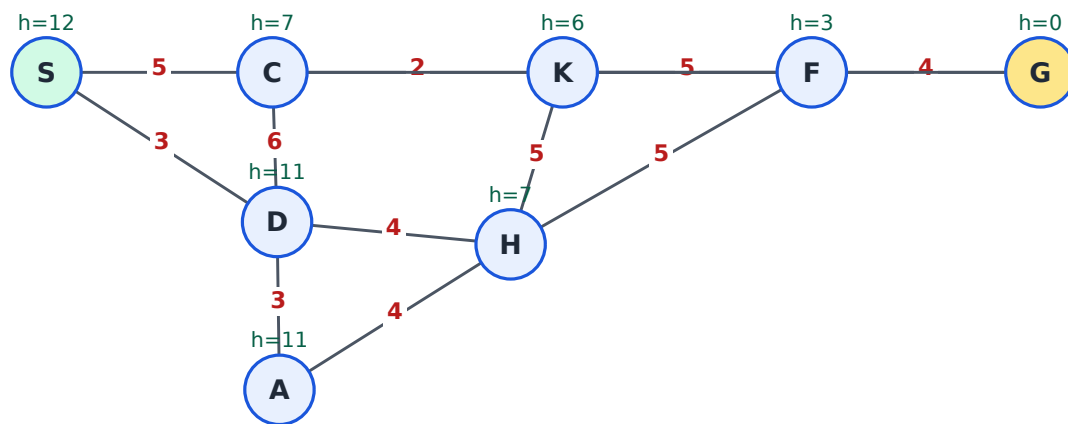
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, K, H, E, G
- ☐ (B) S, F, D, E, C, G
- ☐ (C) S, D, F, H, K, E, C, G
- ☐ (D) S, D, F, E, C, G

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Problem Formulation

In a search problem, the state space is defined implicitly by

- ☐ (A) the initial state, the actions, and the transition model
- ☐ (B) the goal test only
- ☐ (C) the heuristic function
- ☐ (D) the path cost function only

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following are tautologies?

- ☐ (A) $p \vee \neg p$
- ☐ (B) $\neg(p \wedge \neg p)$
- ☐ (C) $p \rightarrow (p \wedge q)$
- ☐ (D) $(p \wedge (p \rightarrow q)) \rightarrow q$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

In how many rows of the truth table over p, q is $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow p)$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((\neg P \wedge \neg P) \wedge (\neg R \vee Q)) \rightarrow (Q \wedge (P \wedge R))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **A, B** and rules:

R1: $A \wedge B \rightarrow H$

R2: $D \wedge B \wedge H \rightarrow J$

R3: $C \wedge J \rightarrow B$

R4: $H \rightarrow K$

R5: $E \wedge H \rightarrow F$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

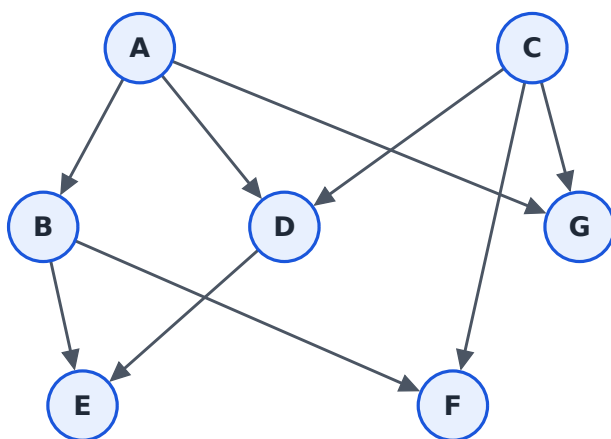
The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(A = T \mid D = F)$, how many of these samples are **rejected**?

#	A	B	C	D
1	F	F	T	T
2	F	F	T	T
3	F	F	F	T
4	F	F	T	T
5	F	F	F	T
6	F	F	T	T
7	F	F	T	T
8	F	F	F	F
9	T	F	T	F
10	F	F	T	T
11	F	F	F	F
12	T	F	T	F
13	T	F	F	F
14	T	F	F	F
15	F	F	T	F
16	F	F	T	T
17	T	F	T	F
18	F	F	F	F

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **D** in the Bayesian network below?

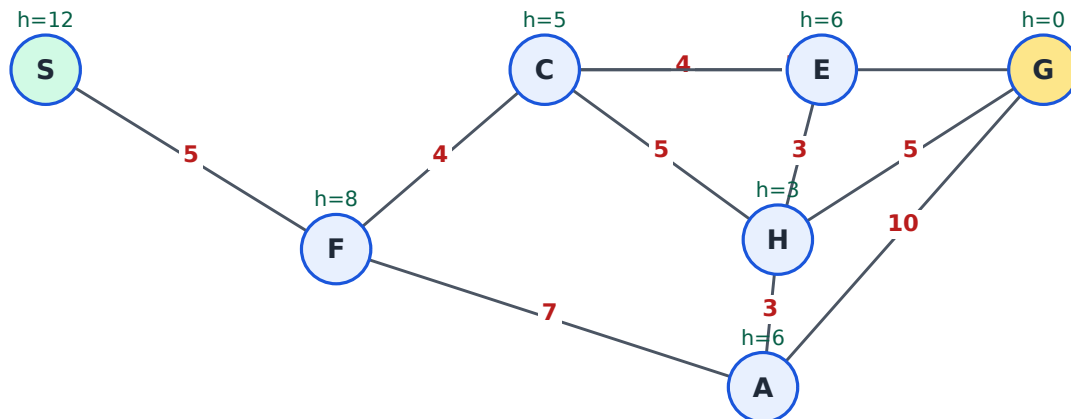


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

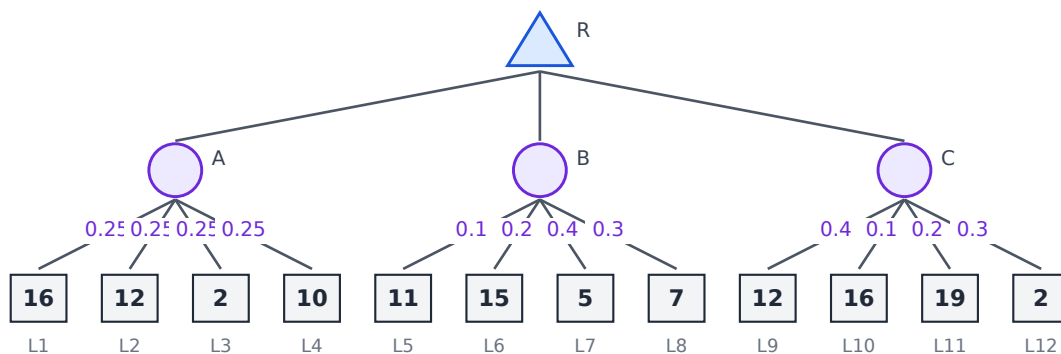
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) The edge S-F violates the consistency condition (in at least one direction).
- ☐ (C) h is admissible.
- ☐ (D) $h(F)$ overestimates the true cost from F to G.

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search

A MAX root has two chance-node children. Chance node C_1 leads with probability 0.4 to a MIN node with leaves (6, 10) and with probability 0.6 to a MIN node with leaves (3, 9). Chance node C_2 leads with probability 0.5 to a leaf of value 8 and with probability 0.5 to a leaf of value 1. What is the expectiminimax value of the root?

Answer (numerical): _____

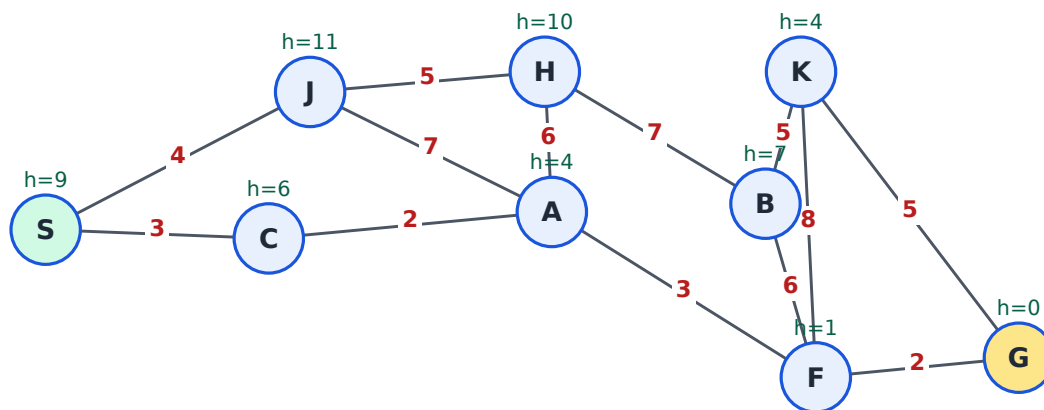
Q.14 [MSQ · 2 marks · no negative marking] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). The heuristic is $h(S)=7$, $h(A)=6$, $h(B)=2$, $h(C)=1$, $h(G)=0$. Which of the following statements are TRUE?

- ☐ (A) h is admissible
☐ (B) h is consistent
☐ (C) A* tree search (no explored set) with this h returns a path of cost 7
☐ (D) Greedy best-first search (graph search) with this h returns the path $S \rightarrow B \rightarrow G$

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from S to G on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) The path returned has cost 10.
☐ (B) Node C is expanded.
☐ (C) Node J is expanded.
☐ (D) Node K is expanded.

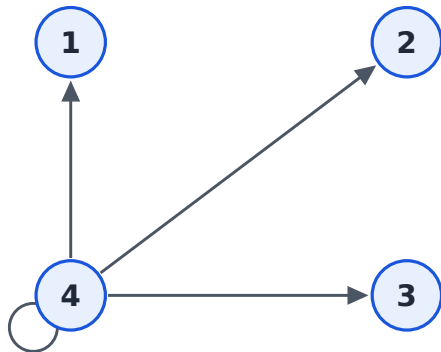
Q.16 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Converting $\forall x ((\exists y P(x, y)) \rightarrow (\exists z Q(x, z)))$ to clausal form (with F a new Skolem function) gives

- ☐ (A) $\neg P(x, y) \vee Q(x, F(x))$
☐ (B) $\neg P(x, G(x)) \vee Q(x, F(x))$
☐ (C) $\neg P(x, y) \vee Q(x, C)$, with C a new constant
☐ (D) $\neg P(x, y) \wedge Q(x, F(x))$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y R(x, y)$
☐ (B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (C) $\forall x \exists y R(x, y)$
☐ (D) $\forall y \exists x R(x, y)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(\neg P \rightarrow (Q \rightarrow R)) \leftrightarrow ((\neg P \wedge Q) \rightarrow R)$
☐ (B) $((R \rightarrow P) \wedge \neg R) \rightarrow \neg P$
☐ (C) $((\neg R \vee P) \wedge (\neg \neg R \vee Q)) \rightarrow (P \vee Q)$
☐ (D) $(P \vee Q) \rightarrow (P \wedge Q)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (\neg R), C2: (Q), C3: (\neg Q \vee S), C4: (\neg P \vee \neg R \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) (S)
☐ (C) $(Q \vee S)$
☐ (D) $(\neg P \vee \neg R)$

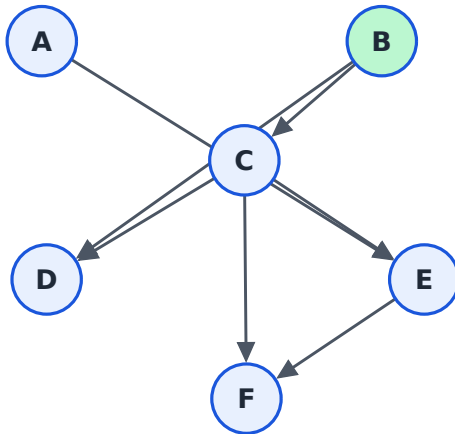
Q.20 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S, C \rightarrow R, S \rightarrow W, R \rightarrow W$ and CPTs: $P(c) = 0.5; P(s | c) = 0.1, P(s | \neg c) = 0.5; P(r | c) = 0.8, P(r | \neg c) = 0.2; P(w | s, r) = 0.99, P(w | s, \neg r) = 0.90, P(w | \neg s, r) = 0.90, P(w | \neg s, \neg r) = 0.0$. Which of the following are correct (values rounded to 3 decimal places)?

- ☐ (A) $P(s | w, r) \approx 0.194$
☐ (B) $P(s | w) \approx 0.430$
☐ (C) $P(s | w, r) > P(s | w)$
☐ (D) $P(w) \approx 0.647$

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

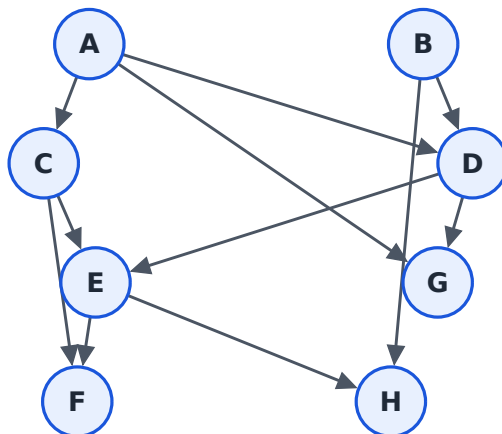
Variable elimination is used to compute $P(\mathbf{B})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, E, F, A, D**. What is the scope of the new factor produced when **C** is summed out?



- ☐ (A) $f(A, B, D, E, F)$
☐ (B) $f(A, D, E, F)$
☐ (C) $f(B, D, E, F)$
☐ (D) $f(A, B, C, D, E, F)$

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

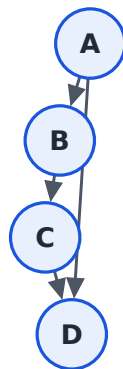
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp H \mid B$
☐ (B) $B \perp D$
☐ (C) $C \perp D \mid A$
☐ (D) $C \perp E \mid B$

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.7}, \mathbf{0.79}, \mathbf{0.59}, \mathbf{0.83} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.75

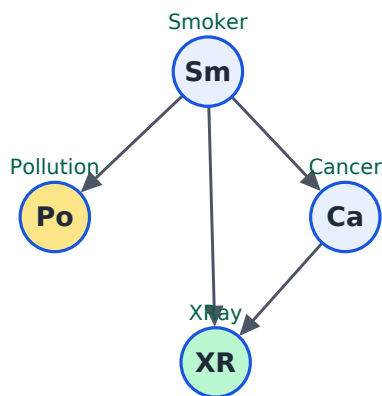
A	P(B=T ...)	B	P(C=T ...)
T	0.8	T	0.75
F	0.95	F	0.05

A	C	P(D=T ...)
T	T	0.2
T	F	0.8
F	T	0.95
F	F	0.1

- ☐ (A) A=T, B=F, C=T, D=F
☐ (B) A=F, B=T, C=T, D=F
☐ (C) A=F, B=F, C=T, D=F
☐ (D) A=T, B=T, C=T, D=F

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $\mathbf{P(XRay = T \mid Pollution = T)}$ (round off to 3 decimal places).



P(Sm=T)
0.6

Sm	P(Po=T .. Sm)	Sm	P(Ca=T ...)
T	0.95	T	0.2
F	0.75	F	0.3

Sm	Ca	P(XR=T ...)
T	T	0.5
T	F	0.5
F	T	0.95
F	F	0.9

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Probability for AI

For arbitrary random variables X_1, X_2, X_3, X_4 (no independence assumed), which of the following expressions are always equal to $P(x_1, x_2, x_3, x_4)$?

- ☐ (A) $P(x_4 \mid x_3) P(x_3 \mid x_2) P(x_2 \mid x_1) P(x_1)$
☐ (B) $P(x_4 \mid x_1, x_2, x_3) P(x_3 \mid x_1, x_2) P(x_2 \mid x_1) P(x_1)$
☐ (C) $P(x_1 \mid x_2, x_3, x_4) P(x_2 \mid x_3, x_4) P(x_3 \mid x_4) P(x_4)$
☐ (D) $P(x_4 \mid x_1, x_2, x_3) P(x_3 \mid x_2) P(x_2 \mid x_1) P(x_1)$

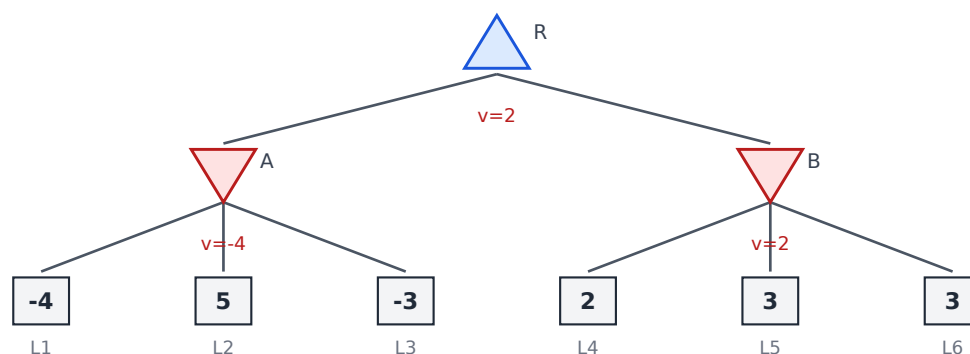
Answer Key — Mock Test 046

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	A	14	MSQ	2	Informed Search	A, C, D
2	MCQ	1	Uninformed Search	D	15	MSQ	2	Informed Search	A, B
3	NAT	1	Informed Search	18	16	MCQ	2	Predicate Logic	A
4	MCQ	1	Problem Formulation	A	17	MSQ	2	Predicate Logic	A, B, D
5	MSQ	1	Propositional Logic	A, B, D	18	MSQ	2	Propositional Logic	A, C
6	NAT	1	Propositional Logic	2	19	MSQ	2	Propositional Logic	B, D
7	NAT	1	Propositional Logic	3	20	MSQ	2	Bayesian Networks	A, B, D
8	NAT	1	Propositional Logic	4	21	MCQ	2	Exact Inference	A
9	NAT	1	Sampling	9	22	MSQ	2	Conditional Independence	C
10	NAT	1	Conditional Independence	4	23	MCQ	2	Sampling	D
11	MSQ	2	Informed Search	A, C	24	NAT	2	Exact Inference	0.641 to 0.645
12	NAT	2	Adversarial Search	10.79 to 10.81	25	MSQ	2	Probability for AI	B, C
13	NAT	2	Adversarial Search	4.5					

Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] Answer: A (A: Move to B; root value 2)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = -4, B = 2. Root value = **2**; MAX's optimal first move is towards B.

Q.2 [MCQ, 1M · Uninformed Search] Answer: D (D: S, D, F, E, C, G)

Adjacency lists (alphabetical): C: {E, G, H, K}; D: {F, H, K, S}; E: {C, F, G, H}; F: {D, E}; G: {C, E, K}; H: {C, D, E, K}; K: {C, D, G, H}; S: {D}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, F, E, C, G**; the path found is S → D → F → E → C → G (5 edges). Note that DFS need not find the shortest path.

Q.3 [NAT, 1M · Informed Search] Answer: 18

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (12)	C(7), D(11)
2	C (7)	F(3), K(6), D(11)
3	F (3)	G(0), K(6), H(7), D(11)
4	G (0)	K(6), H(7), D(11) ← goal, stop

Path returned: **S → C → F → G**, cost = 5 + 9 + 4 = **18**.

The optimal cost is $h^*(S) = 14$ — greedy search is not optimal here.

Q.4 [MCQ, 1M · Problem Formulation] Answer: A (A: the initial state, the actions, and the transition model)

Starting from the **initial state** and applying the **actions** through the **transition model** generates every reachable state and the edges between them, which is the state-space graph. The goal test (B) only identifies goal states. The heuristic (C) guides search but does not define states. The path cost (D) only labels edges.

Q.5 [MSQ, 1M · Propositional Logic] Answer: A, B, D (A: $p \vee \neg p$; B: $\neg(p \wedge \neg p)$; D: $(p \wedge (p \rightarrow q)) \rightarrow q$)

$p \vee \neg p$ (excluded middle) and $\neg(p \wedge \neg p)$ (non-contradiction) are valid. $(p \wedge (p \rightarrow q)) \rightarrow q$ is the modus ponens tautology. $p \rightarrow (p \wedge q)$ is false at $p=T, q=F$, so it is not valid.

Q.6 [NAT, 1M · Propositional Logic] Answer: 2

$p \rightarrow q \equiv \neg p \vee q$ and $\neg q \rightarrow p \equiv p \vee q$. Row TT: T, $T \rightarrow T$. Row TF: F, $T \rightarrow F$. Row FT: T, $T \rightarrow T$. Row FF: T, $F \rightarrow F$. So the biconditional is true in 2 rows.

Q.7 [NAT, 1M · Propositional Logic] Answer: 3

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg P \wedge \neg P) \wedge (\neg R \vee Q)$	$Q \wedge (P \wedge R)$	$((\neg P \wedge \neg P) \wedge (\neg R \vee Q)) \rightarrow (Q \wedge (P \wedge R))$
T	T	T	F	T	T
T	T	F	F	F	T
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	F	F	T
F	F	F	T	F	F

F has 5 models, so it is false in $8 - 5 = 3$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.8 [NAT, 1M · Propositional Logic] Answer: 4

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $A \wedge B \rightarrow H$; add H.

Round 2: fire $H \rightarrow K$; add K.

Fixed point: {A, B, H, K} — 4 atoms are entailed (facts: A, B; newly derived: H, K). Atoms never derived: C, D, E, F, J.

Q.9 [NAT, 1M · Sampling] Answer: 9

Rejection sampling keeps only samples consistent with the evidence $D = F$: samples 8, 9, 11, 12, 13, 14, 15, 17, 18 (9 accepted, 9 rejected).

Among the accepted samples, $A = T$ in 5. Estimate = $5/9 = 0.5556$.

(For this network the exact value is 0.2803; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.10 [NAT, 1M · Conditional Independence] Answer: 4

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of D: {A, C}; children: {E}; co-parents: {B}.

$MB(D) = \{A, B, C, E\}$ — 4 nodes. Given its Markov blanket, D is conditionally independent of every other node in the network.

Q.11 [MSQ, 2M · Informed Search] Answer: A, C (A: h is consistent (monotone).; C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	C	E	F	G	H	S
$h(n)$	6	5	6	8	0	3	12
$h^*(n)$	8	7	8	11	0	5	16
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [NAT, 2M · Adversarial Search] Answer: 10.79 to 10.81

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

$$\bullet A = 0.25 \times 16 + 0.25 \times 12 + 0.25 \times 2 + 0.25 \times 10 = 10$$

$$\bullet B = 0.1 \times 11 + 0.2 \times 15 + 0.4 \times 5 + 0.3 \times 7 = 8.2$$

$$\bullet C = 0.4 \times 12 + 0.1 \times 16 + 0.2 \times 19 + 0.3 \times 2 = 10.8$$

$$\text{Root} = \max(10, 8.2, 10.8) = \mathbf{10.8}.$$

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.13 [NAT, 2M · Adversarial Search] Answer: 4.5

The MIN nodes under C_1 have values $\min(6, 10) = 6$ and $\min(3, 9) = 3$. $C_1 = 0.4 \cdot 6 + 0.6 \cdot 3 = 2.4 + 1.8 = 4.2$. $C_2 = 0.5 \cdot 8 + 0.5 \cdot 1 = 4.5$. $\text{Root} = \max(4.2, 4.5) = \mathbf{4.5}$, so MAX chooses C_2 .

Q.14 [MSQ, 2M · Informed Search] Answer: A, C, D

The true costs-to-go are $h^*(C)=3$, $h^*(B)=\min(6, 1+3)=4$, $h^*(A)=\min(5+3, 2+4)=6$, and $h^*(S)=\min(1+6, 4+4)=7$. In every case $h \leq h^*$, so (A) is true. (B) is false: on edge $A \rightarrow B$, $h(A)=6 > 2 + h(B) = 4$. (C) is true: A^* tree search with an admissible h is optimal and returns $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$ of cost 7. (D) is true: greedy expands S , then B ($h=2$), which generates C ($h=1$) and G ($h=0$). It then selects G and returns $S \rightarrow B \rightarrow G$ of cost 10.

Q.15 [MSQ, 2M · Informed Search] Answer: A, B (A: The path returned has cost 10.; B: Node C is expanded.)

Heuristic values: $h(A)=4$, $h(B)=7$, $h(C)=6$, $h(F)=1$, $h(G)=0$, $h(H)=10$, $h(J)=11$, $h(K)=4$, $h(S)=9$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	C(9), J(15)
2	C (9)	A(9), J(15)
3	A (9)	F(9), J(15), H(21)
4	F (9)	G(10), J(15), K(20), B(21), H(21)
5	G (10)	J(15), K(20), B(21), H(21) $\leftarrow$ goal, stop

Path returned: **S $\rightarrow$ C $\rightarrow$ A $\rightarrow$ F $\rightarrow$ G** with cost **10**; nodes expanded = **5** (S, C, A, F, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 10$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.16 [MCQ, 2M · Predicate Logic] Answer: A (A: $\neg P(x, y) \vee Q(x, F(x))$)

Eliminate $\rightarrow$: $\forall x (\neg \exists y P(x, y) \vee \exists z Q(x, z)) \equiv \forall x (\forall y \neg P(x, y) \vee \exists z Q(x, z))$. y is now universal, so it stays a variable. z lies only within the scope of $\forall x$, so it becomes $F(x)$. Dropping universals gives the clause $\neg P(x, y) \vee Q(x, F(x))$. Option B wrongly Skolemizes the universal y ; option C ignores the dependence of z on x ; option D is not a clause and changes the meaning.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, B, D (A: $\exists x \forall y R(x, y)$; B: $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$; D: $\forall y \exists x R(x, y)$)

$$R = \{(4, 1), (4, 2), (4, 3), (4, 4)\}.$$

(A) $\exists x \forall y R(x, y)$: **TRUE** — $x = 4$ has edges to every y .

(B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **TRUE** — all two-step paths are short-cut.

(C) $\forall x \exists y R(x, y)$: **FALSE** — 1 has no outgoing edge.

(D) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $(\neg P \rightarrow (Q \rightarrow R)) \Leftrightarrow ((\neg P \wedge Q) \rightarrow R)$; C: $((\neg R \vee P) \wedge (\neg \neg R \vee Q)) \rightarrow (P \vee Q)$)

(A) $(\neg P \rightarrow (Q \rightarrow R)) \Leftrightarrow ((\neg P \wedge Q) \rightarrow R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(B) $((R \rightarrow P) \wedge \neg R) \rightarrow \neg P$: **not valid** — counter-model $P=T, R=F$.

(C) $((\neg R \vee P) \wedge (\neg \neg R \vee Q)) \rightarrow (P \vee Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $(P \vee Q) \rightarrow (P \wedge Q)$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.19 [MSQ, 2M · Propositional Logic] Answer: B, D (B: (S); D: $(\neg P \vee \neg R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) (S) is derivable:

1. resolve (Q) and $(\neg Q \vee S)$ on Q $\Rightarrow$ (S)

(C) $(Q \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg P \vee \neg R)$ is derivable:

1. resolve $(\neg Q \vee S)$ and $(\neg P \vee \neg R \vee \neg S)$ on S $\Rightarrow$ $(\neg P \vee \neg R)$
2. resolve (Q) and $(\neg P \vee \neg R)$ on Q $\Rightarrow$ $(\neg P \vee \neg R)$

Q.20 [MSQ, 2M · Bayesian Networks] Answer: A, B, D (A: $P(s | w, r) \approx 0.194$; B: $P(s | w) \approx 0.430$; D: $P(w) \approx 0.647$)

$P(s, w) = c: 0.5 \times 0.1 \times (0.8 \times 0.99 + 0.2 \times 0.90) = 0.0486$; $\neg c: 0.5 \times 0.5 \times (0.2 \times 0.99 + 0.8 \times 0.90) = 0.2295$; total 0.2781. $P(\neg s, w) = c: 0.5 \times 0.9 \times 0.8 \times 0.90 = 0.324$; $\neg c: 0.5 \times 0.5 \times 0.2 \times 0.90 = 0.045$; total 0.369. $P(w) = 0.6471$ (D true) and $P(s | w) = 0.2781/0.6471 \approx 0.430$ (B true). $P(s, w, r) = (0.5 \times 0.1 \times 0.8 + 0.5 \times 0.5 \times 0.2) \times 0.99 = 0.0891$; $P(\neg s, w, r) = (0.36 + 0.05) \times 0.90 = 0.369$; $P(s | w, r) = 0.0891/0.4581 \approx 0.194$ (A true). Rain explains away the wet grass, so $P(s | w, r) < P(s | w)$: (C) is false.

Q.21 [MCQ, 2M · Exact Inference] Answer: A (A: $f(A, B, D, E, F)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(B, C)$, $f(B, C, D)$, $f(A, C, E)$, $f(C, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C)$, $f(B, C, D)$, $f(A, C, E)$, $f(C, E, F)$	$\{A, B, C, D, E, F\}$	$f(A, B, D, E, F)$
E	$f(A, B, D, E, F)$	$\{A, B, D, E, F\}$	$f(A, B, D, F)$
F	$f(A, B, D, F)$	$\{A, B, D, F\}$	$f(A, B, D)$
A	$f(A)$, $f(A, B, D)$	$\{A, B, D\}$	$f(B, D)$
D	$f(B, D)$	$\{B, D\}$	$f(B)$

The largest factor produced has **5** variable(s) (created when eliminating C); with Boolean variables it has $2^5 = 32$ entries. Finally, multiply the remaining factors over B and normalise.

Q.22 [MSQ, 2M · Conditional Independence] Answer: C (C: $C \perp D | A$)

Rules: a chain $(\rightarrow B \rightarrow)$ or fork $(\leftarrow B \leftarrow)$ is blocked iff B is observed; a collider $(\rightarrow B \leftarrow)$ is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $E \perp H | B$: **FALSE** — the path $E \rightarrow H$ is active given $\{B\}$.

(B) $B \perp D$: **FALSE** — the path $B \rightarrow D$ is active given $\{\}$.

(C) $C \perp D | A$: **TRUE** — $C \leftarrow A \rightarrow D$ (blocked: observed fork node A); $C \leftarrow A \rightarrow G \leftarrow D$ (blocked: observed fork node A); $C \rightarrow E \leftarrow D$ (blocked: collider E with neither it nor a descendant observed); $C \rightarrow E \rightarrow H \leftarrow B \rightarrow D$ (blocked: collider H with neither it nor a descendant observed); $C \rightarrow F \leftarrow E \leftarrow D$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow F \leftarrow E \rightarrow H \leftarrow B \rightarrow D$ (blocked: collider F with neither it nor a descendant observed).

(D) $C \perp E | B$: **FALSE** — the path $C \leftarrow A \rightarrow D \rightarrow E$ is active given $\{B\}$.

Q.23 [MCQ, 2M · Sampling] Answer: D (D: $A=T, B=T, C=T, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.75$; $u = 0.7 < 0.75 \Rightarrow A = T$
- B: $P(B=T | A=T) = 0.8$; $u = 0.79 < 0.8 \Rightarrow B = T$
- C: $P(C=T | B=T) = 0.75$; $u = 0.59 < 0.75 \Rightarrow C = T$
- D: $P(D=T | A=T, C=T) = 0.2$; $u = 0.83 \geq 0.2 \Rightarrow D = F$

Sample: **$\{A=T, B=T, C=T, D=F\}$** .

Q.24 [NAT, 2M · Exact Inference] Answer: 0.641 to 0.645

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution} | \text{Smoker}) \cdot P(\text{Cancer} | \text{Smoker}) \cdot P(\text{XRay} | \text{Smoker}, \text{Cancer})$.

Inference by enumeration: $P(\text{XRay} | e) = \alpha \sum_{\text{hidden}} P(\text{XRay}, e, \text{hidden})$.

Hidden: Sm, Ca	XR	product of CPT entries (Sm · Po · Ca · XR)	value
T, T	T	$0.6 \times 0.95 \times 0.2 \times 0.5$	0.057
T, F	T	$0.6 \times 0.95 \times 0.8 \times 0.5$	0.228
F, T	T	$0.4 \times 0.75 \times 0.3 \times 0.95$	0.0855

Hidden: Sm, Ca	XR	product of CPT entries (Sm · Po · Ca · XR)	value
F, F	T	$0.4 \times 0.75 \times 0.7 \times 0.9$	0.189
T, T	F	$0.6 \times 0.95 \times 0.2 \times 0.5$	0.057
T, F	F	$0.6 \times 0.95 \times 0.8 \times 0.5$	0.228
F, T	F	$0.4 \times 0.75 \times 0.3 \times 0.05$	0.0045
F, F	F	$0.4 \times 0.75 \times 0.7 \times 0.1$	0.021

$P(\text{XRay}=T, e) = 0.5595$; $P(\text{XRay}=F, e) = 0.3105$; $P(e) = 0.87$.

$P(\text{XRay}=T \mid e) = 0.5595 / 0.87 = \mathbf{0.6431} \approx 0.643$.

Q.25 [MSQ, 2M · Probability for AI] **Answer: B, C**

The chain rule $P(x_1, \dots, x_n) = \prod_i P(x_i \mid x_1, \dots, x_{i-1})$ holds for any ordering of the variables. (B) is the chain rule for order X_1, X_2, X_3, X_4 ; (C) is the chain rule for the reverse order X_4, X_3, X_2, X_1 . (A) drops conditioning variables, which is valid only under Markov-chain independence assumptions. (D) replaces $P(x_3 \mid x_1, x_2)$ by $P(x_3 \mid x_2)$, which needs $X_3 \perp X_1 \mid X_2$, not generally true.

Mock Test 047

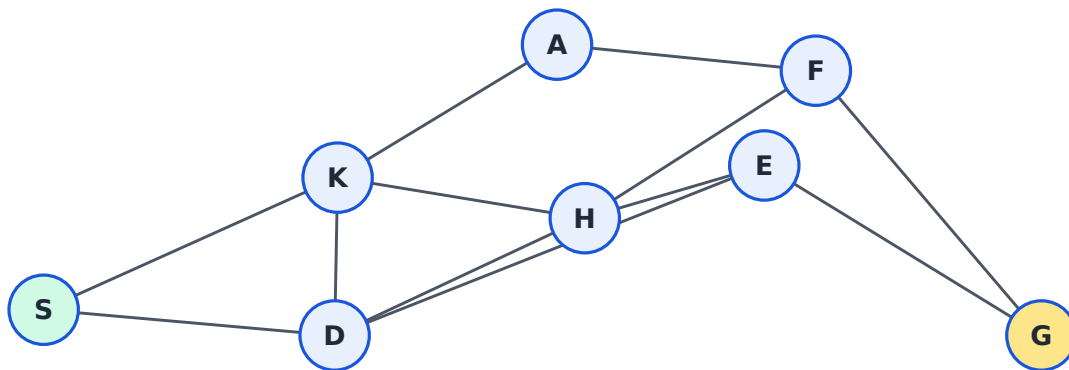
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 6 / 12	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

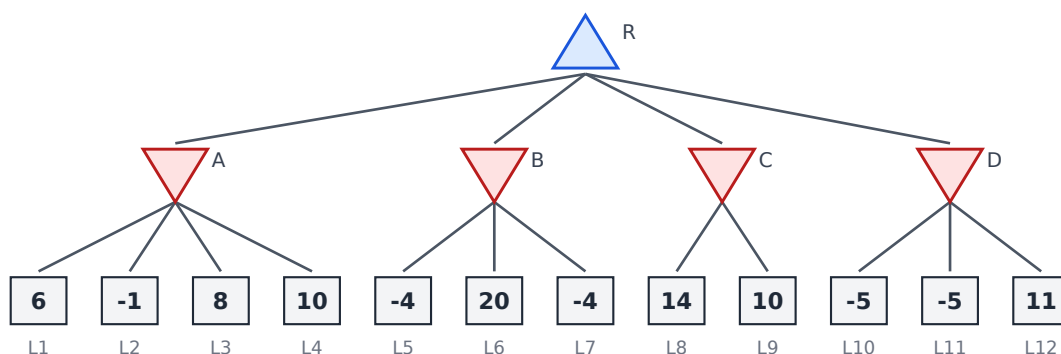
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, E, G
- ☐ (B) S, D, K, E, H, A, G
- ☐ (C) S, E, D, G
- ☐ (D) S, K, H, F, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 4$ (so $b^d = 81$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Informed Search

In route finding on a road map, why is the straight-line (Euclidean) distance to the destination an admissible heuristic?

- ☐ (A) Because it is easy to compute
- ☐ (B) Because the shortest route between two points can never be shorter than the straight line between them
- ☐ (C) Because roads are always straight
- ☐ (D) Because it equals the true road distance

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Over the finite domain $\{a, b\}$, the sentence $\forall x \exists y R(x, y)$ is equivalent to which ground formula?

- ☐ (A) $(R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$
- ☐ (B) $(R(a, a) \wedge R(a, b)) \vee (R(b, a) \wedge R(b, b))$
- ☐ (C) $(R(a, a) \vee R(b, a)) \wedge (R(a, b) \vee R(b, b))$
- ☐ (D) $R(a, a) \wedge R(b, b)$

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q) . For how many of them is $\neg \exists x (P(x) \wedge \neg Q(x)) \wedge \exists x P(x)$ true?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

For propositional sentences α and β , $\alpha \models \beta$ holds if and only if

- ☐ (A) $\alpha \wedge \neg \beta$ is unsatisfiable
- ☐ (B) $\alpha \wedge \beta$ is satisfiable
- ☐ (C) $\neg \alpha \vee \beta$ is satisfiable
- ☐ (D) $\alpha \vee \neg \beta$ is valid

Q.8 [NAT · 1 mark · no negative marking] Sampling

To estimate $P(\text{Rain} \mid \text{sprinkler})$ by rejection sampling, 500 samples are generated by prior sampling. 140 of them have $\text{Sprinkler} = \text{true}$, and of these 35 have $\text{Rain} = \text{true}$. What is the rejection-sampling estimate of $P(\text{rain} \mid \text{sprinkler})$? (Round to 2 decimal places.)

Answer (numerical): _____

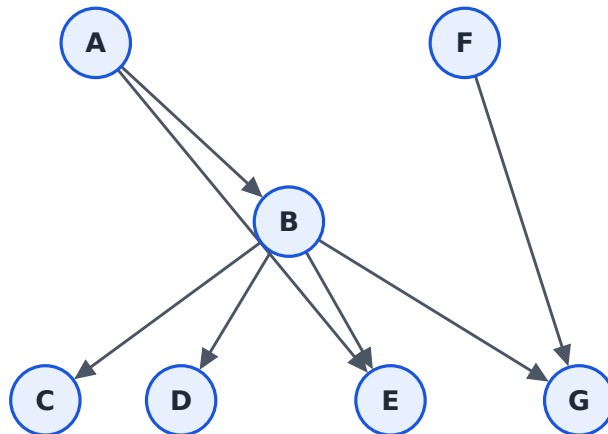
Q.9 [MCQ · 1 mark · negative: -1/3] Sampling

For sampling-based estimates of a probability, the standard deviation of the estimate decreases as $O(1/\sqrt{N})$ in the number of samples N . Roughly how many samples are needed to halve the error obtained with N samples?

- ☐ (A) $4N$
- ☐ (B) $2N$
- ☐ (C) $N\sqrt{2}$
- ☐ (D) N^2

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **B** in the Bayesian network below?

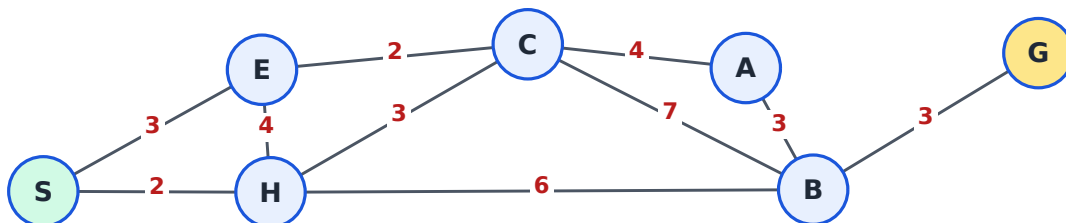


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

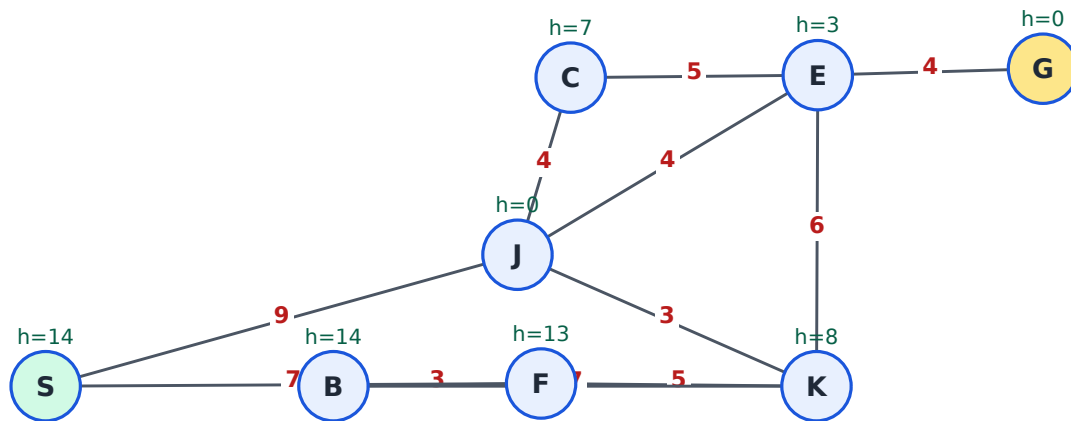
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

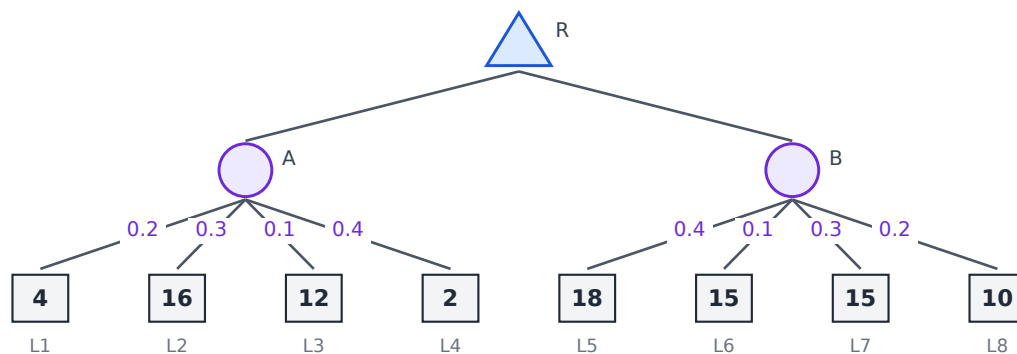
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
☐ (B) h is consistent (monotone).
☐ (C) $h(C)$ overestimates the true cost from C to G.
☐ (D) The edge C-J violates the consistency condition $h(C) \leq c(C, J) + h(J)$.

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



▲ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search

In the 8-puzzle, the goal is 1 2 3 / 8 _ 4 / 7 6 5 (rows top to bottom, _ = blank), and the current state is 2 8 3 / 1 6 4 / 7 _ 5. Let h_1 be the number of misplaced tiles and h_2 the total Manhattan distance. What is $h_1 + h_2$?

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (R \leftrightarrow \neg R) \rightarrow (\neg R \vee \neg P)$?

- ☐ (A) $\neg(\neg R \vee \neg P) \rightarrow \neg(R \leftrightarrow \neg R)$
☐ (B) $\neg(R \leftrightarrow \neg R) \vee (\neg R \vee \neg P)$
☐ (C) $(R \leftrightarrow \neg R) \vee \neg(\neg R \vee \neg P)$
☐ (D) $\neg(R \leftrightarrow \neg R) \wedge (\neg R \vee \neg P)$

Q.16 [MCQ · 2 marks · negative: -2/3] Predicate Logic

The negation of $\forall x \exists y (P(x, y) \rightarrow Q(y))$ is logically equivalent to

- ☐ (A) $\exists x \forall y (P(x, y) \wedge \neg Q(y))$
☐ (B) $\exists x \forall y (\neg P(x, y) \rightarrow Q(y))$
☐ (C) $\forall x \exists y (P(x, y) \wedge \neg Q(y))$
☐ (D) $\exists x \exists y (P(x, y) \wedge \neg Q(y))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

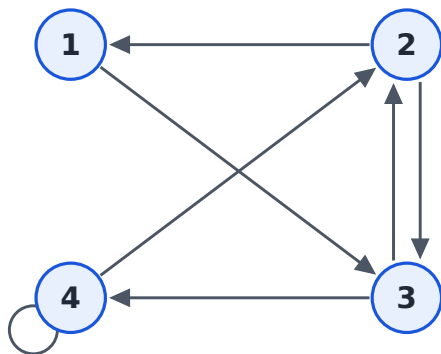
C1: $(\neg R)$, C2: $(\neg P \vee R \vee S)$, C3: $(\neg P \vee \neg R)$, C4: $(\neg Q \vee \neg R \vee \neg S)$, C5: $(P \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) (P)
☐ (B) $(\neg P \vee \neg Q \vee \neg R)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(\neg R \vee \neg S)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
☐ (B) $\forall x \exists y R(x, y)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

The formula $F = ((Q \rightarrow S) \rightarrow (S \rightarrow \neg P)) \rightarrow ((Q \rightarrow \neg Q) \rightarrow R)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R, S . How many maxterms (clauses) does it contain?

Answer (numerical): _____

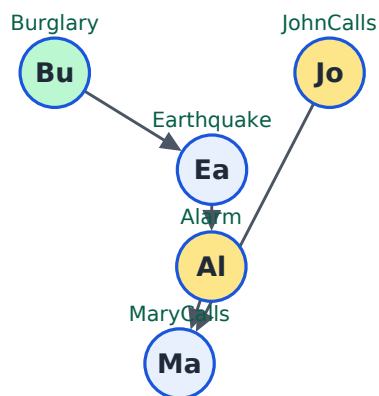
Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic

Consider the interpretation with domain $D = \{1, 2, 3\}$ in which $L(x, y)$ holds iff $x < y$. Which of the following sentences are TRUE in this interpretation?

- ☐ (A) $\forall x \exists y L(x, y)$
☐ (B) $\exists x \forall y \neg L(y, x)$
☐ (C) $\forall x \forall y (L(x, y) \rightarrow \neg L(y, x))$
☐ (D) $\forall x \forall y (L(x, y) \rightarrow \exists z (L(x, z) \wedge L(z, y)))$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls, Ma = MaryCalls.) Compute **$P(\text{Burglary} = T \mid \text{Alarm} = F, \text{JohnCalls} = T)$** (round off to 3 decimal places).



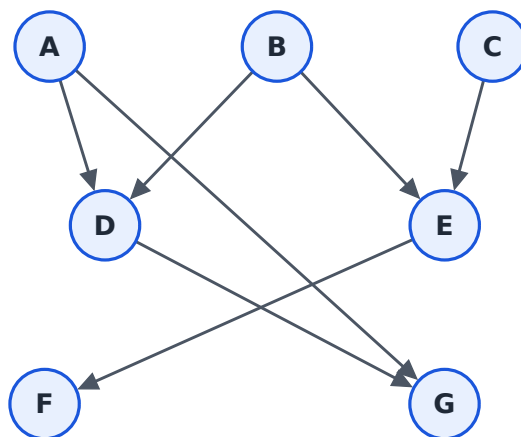
$P(\text{Bu}=T)$	Bu	$P(\text{Ea}=T \dots)$	Ea	$P(\text{Al}=T \dots)$
0.7	T	0.75	T	0.95
	F	0.95	F	0.6

$P(\text{Jo}=T)$	Al	Jo	$P(\text{Ma}=T \dots)$
0.4	T	T	0.75
	T	F	0.1
	F	T	0.95
	F	F	0.7

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp E \mid A$
☐ (B) $A \perp D \mid C, F$
☐ (C) $E \perp D \mid G$
☐ (D) $F \perp G \mid B$

Q.23 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

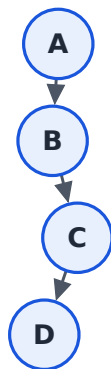
The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D, E. Using **rejection sampling**, estimate $P(E = T \mid B = T)$ (round off to 2 decimal places).

#	A	B	C	D	E
1	T	T	F	T	F
2	T	T	F	T	T
3	T	F	T	F	T
4	T	F	F	F	T
5	F	F	T	F	T
6	F	T	F	T	T
7	T	T	F	T	T
8	F	T	F	T	T
9	T	T	F	T	T
10	F	T	F	T	T
11	F	T	F	T	T
12	F	T	F	F	T
13	F	T	F	T	T
14	T	T	F	F	T
15	F	T	T	F	F
16	T	T	F	T	F
17	T	F	F	F	T
18	F	F	T	T	F

Answer (numerical): _____

Q.24 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle 0.57, 0.6, 0.39, 0.29 \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.8

A	P(B=T ...)	B	P(C=T ...)
T	0.5	T	0.1
F	0.1	F	0.5

C	P(D=T ...)
T	0.9
F	0.7

- ☐ (A) A=T, B=F, C=T, D=T
☐ (B) A=T, B=T, C=T, D=T
☐ (C) A=F, B=T, C=T, D=F
☐ (D) A=F, B=F, C=T, D=T

Q.25 [NAT · 2 marks · no negative marking] Probability for AI

$P(\text{cavity}) = 0.2$. Toothache and Catch are conditionally independent given Cavity with $P(\text{toothache} \mid \text{cavity}) = 0.6$, $P(\text{toothache} \mid \neg \text{cavity}) = 0.1$, $P(\text{catch} \mid \text{cavity}) = 0.9$, $P(\text{catch} \mid \neg \text{cavity}) = 0.2$. Compute $P(\text{cavity} \mid \text{toothache}, \text{catch})$. (Round to 3 decimal places.)

Answer (numerical): _____

Answer Key — Mock Test 047

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	NAT	2	Informed Search	9
2	NAT	1	Adversarial Search	10	15	MSQ	2	Propositional Logic	A, B
3	NAT	1	Adversarial Search	17	16	MCQ	2	Predicate Logic	A
4	MCQ	1	Informed Search	B	17	MSQ	2	Propositional Logic	B, D
5	MCQ	1	Predicate Logic	A	18	MSQ	2	Predicate Logic	B, C
6	NAT	1	Predicate Logic	19	19	NAT	2	Propositional Logic	3
7	MCQ	1	Propositional Logic	A	20	MSQ	2	Predicate Logic	B, C
8	NAT	1	Sampling	0.245 to 0.255	21	NAT	2	Exact Inference	0.824 to 0.828
9	MCQ	1	Sampling	A	22	MSQ	2	Conditional Independence	D
10	NAT	1	Conditional Independence	6	23	NAT	2	Sampling	0.76 to 0.78
11	NAT	2	Uninformed Search	11	24	MCQ	2	Sampling	A
12	MSQ	2	Informed Search	A, D	25	NAT	2	Probability for AI	0.87 to 0.872
13	NAT	2	Adversarial Search	15.19 to 15.21					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: A (A: S, D, E, G)

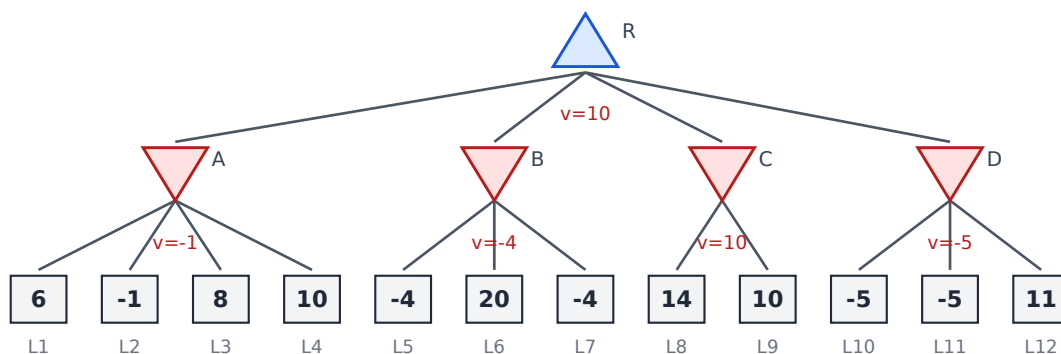
Adjacency lists (alphabetical): A: {F, K}; D: {E, H, K, S}; E: {D, G, H}; F: {A, G, H}; G: {E, F}; H: {D, E, F, K}; K: {A, D, H, S}; S: {D, K}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, E, G**; the path found is $S \rightarrow D \rightarrow E \rightarrow G$ (3 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] Answer: 10

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -1$, $B = -4$, $C = 10$, $D = -5$. Root value = **10**; MAX's optimal first move is towards C.

Q.3 [NAT, 1M · Adversarial Search] Answer: 17

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 3^2 + 3^2 - 1 = 9 + 9 - 1 = \mathbf{17}$, versus 81 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.4 [MCQ, 1M · Informed Search] Answer: B (B: Because the shortest route between two points can never be shorter than the straight line between them)

Admissibility needs $h(n) \leq h^*(n)$. Any road route is a path in the plane, and no path is **shorter than the straight line** between its endpoints, so the heuristic never overestimates. Being cheap to compute (A) is useful but irrelevant to admissibility. Roads are generally not straight (C), and the heuristic generally underestimates rather than equals the road distance (D).

Q.5 [MCQ, 1M · Predicate Logic] Answer: A ($A: (R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$)

Over a finite domain with every element named, $\forall$ becomes a conjunction over x and $\exists$ a disjunction over y . For $x = a$: $R(a, a) \vee R(a, b)$; for $x = b$: $R(b, a) \vee R(b, b)$. Their conjunction is option A. Option B is $\exists x \forall y R(x, y)$. Option C is $\forall y \exists x R(x, y)$ (every element has an R-predecessor). Option D is reflexivity.

Q.6 [NAT, 1M · Predicate Logic] Answer: 19

Interpret P and Q as subsets of the domain; $P \subseteq Q$ gives 3^n ; subtract the 2^n with P empty $\Rightarrow 3^n - 2^n$. With $n = 3$ this gives **19** (verified by enumerating all interpretations).

Q.7 [MCQ, 1M · Propositional Logic] Answer: A ($A: \alpha \wedge \neg\beta$ is unsatisfiable)

$\alpha \models \beta$ means every model of α is a model of β , i.e. no model makes α true and β false, i.e. $\alpha \wedge \neg\beta$ is unsatisfiable. Equivalently $\alpha \rightarrow \beta$ is valid. Satisfiability of $\neg\alpha \vee \beta$ or $\alpha \wedge \beta$ is too weak (one model is enough). $\alpha \vee \neg\beta$ being valid would mean $\beta \models \alpha$, the reverse direction.

Q.8 [NAT, 1M · Sampling] Answer: 0.245 to 0.255

Rejection sampling discards the $500 - 140 = 360$ samples inconsistent with the evidence. Among the 140 retained samples, the fraction with Rain = true is $35/140 = 0.25$. Dividing by 500 would be wrong because it estimates $P(\text{rain}, \text{sprinkler})$, not the conditional.

Q.9 [MCQ, 1M · Sampling] Answer: A ($A: 4N$)

Error $\propto 1/\sqrt{N}$, so halving the error requires $\sqrt{N'} = 2\sqrt{N}$, i.e. $N' = 4N$. $2N$ would reduce error only by a factor $\sqrt{2} \approx 1.41$; $N/2$ by $2^{1/4}$; N^2 is excessive and depends on N .

Q.10 [NAT, 1M · Conditional Independence] Answer: 6

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of B: {A}; children: {C, D, E, G}; co-parents: {F}.

$MB(B) = \{A, C, D, E, F, G\}$ — **6** nodes. Given its Markov blanket, B is conditionally independent of every other node in the network.

Q.11 [NAT, 2M · Uninformed Search] Answer: 11

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	H(2), E(3)
2	H (2)	E(3), C(5), B(8)
3	E (3)	C(5), B(8)
4	C (5)	B(8), A(9)
5	B (8)	A(9), G(11)
6	A (9)	G(11)
7	G (11)	← goal, stop

Path returned: **S** $\rightarrow$ **H** $\rightarrow$ **B** $\rightarrow$ **G**, cost = $2 + 6 + 3 = 11$. Nodes expanded (including S and G): **7** (S, H, E, C, B, A, G).

Q.12 [MSQ, 2M · Informed Search] Answer: A, D ($A: h$ is admissible.; $D: \text{The edge } C \rightarrow J \text{ violates the consistency condition } h(C) \leq c(C, J) + h(J).$)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	E	F	G	J	K	S
$h(n)$	14	7	3	13	0	0	8	14
$h^*(n)$	17	9	4	15	0	8	10	17
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u, v) + h(v)$ for every edge in both directions. Violating directed edges: $C \rightarrow J$ ($7 > 4 + 0$), $K \rightarrow J$ ($8 > 3 + 0$), $S \rightarrow J$ ($14 > 9 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Adversarial Search] Answer: 15.19 to 15.21

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

$$\bullet A = 0.2 \times 4 + 0.3 \times 16 + 0.1 \times 12 + 0.4 \times 2 = 7.6$$

$$\bullet B = 0.4 \times 18 + 0.1 \times 15 + 0.3 \times 15 + 0.2 \times 10 = 15.2$$

$$\text{Root} = \max(7.6, 15.2) = \mathbf{15.2}.$$

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.14 [NAT, 2M · Informed Search] Answer: 9

Misplaced tiles: 2 (at (0,0), belongs at (0,1)), 8 (at (0,1), belongs at (1,0)), 1 (at (1,0), belongs at (0,0)), and 6 (at (1,1), belongs at (2,1)). Tiles 3, 4, 7 and 5 are in place, so $h_1 = 4$. Manhattan distances: $2 \rightarrow 1$, $8 \rightarrow 1+1 = 2$, $1 \rightarrow 1$, $6 \rightarrow 1$, so $h_2 = 5$. $h_1 + h_2 = \mathbf{9}$. (The sum is not necessarily admissible. This question is purely arithmetic.)

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, B (A: $\neg(\neg R \vee \neg P) \rightarrow \neg(R \leftrightarrow \neg R)$; B: $\neg(R \leftrightarrow \neg R) \vee (\neg R \vee \neg P)$)

Compare truth tables with $F = (R \leftrightarrow \neg R) \rightarrow (\neg R \vee \neg P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg R \vee \neg P) \rightarrow \neg(R \leftrightarrow \neg R)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $\neg(R \leftrightarrow \neg R) \vee (\neg R \vee \neg P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $(R \leftrightarrow \neg R) \vee \neg(\neg R \vee \neg P)$ — differs at $P=T, R=F$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(D) $\neg(R \leftrightarrow \neg R) \wedge (\neg R \vee \neg P)$ — differs at $P=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

Q.16 [MCQ, 2M · Predicate Logic] Answer: A (A: $\exists x \forall y (P(x, y) \wedge \neg Q(y))$)

Push $\neg$ inward: $\neg \forall x \exists y \phi \equiv \exists x \neg \exists y \phi \equiv \exists x \forall y \neg \phi$. Then $\neg(P \rightarrow Q) \equiv P \wedge \neg Q$, giving $\exists x \forall y (P(x, y) \wedge \neg Q(y))$.

Option B uses $\neg P \rightarrow Q \equiv P \vee Q$, which is not $\neg(P \rightarrow Q)$. Option C does not flip the quantifiers. Option D flips only one quantifier correctly ($\exists y$ should become $\forall y$).

Q.17 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $(\neg P \vee \neg Q \vee \neg R)$; D: $(\neg R \vee \neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) (P) never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(\neg P \vee \neg Q \vee \neg R)$ is derivable:

1. resolve $(\neg R)$ and $(\neg P \vee R \vee S)$ on R $\Rightarrow (\neg P \vee S)$

2. resolve $(\neg Q \vee \neg R \vee \neg S)$ and $(\neg P \vee S)$ on S $\Rightarrow (\neg P \vee \neg Q \vee \neg R)$

(C) The resolution closure contains 9 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(D) $(\neg R \vee \neg S)$ is derivable:

1. resolve $(\neg P \vee \neg R)$ and $(P \vee \neg S)$ on P $\Rightarrow (\neg R \vee \neg S)$

Q.18 [MSQ, 2M · Predicate Logic] Answer: B, C (B: $\forall x \exists y R(x, y)$; C: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,3), (2,1), (2,3), (3,2), (3,4), (4,2), (4,4)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.

(B) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,4)$ nor $R(4,1)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [NAT, 2M · Propositional Logic] Answer: 3

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(Q \rightarrow S) \rightarrow (S \rightarrow \neg P)$	$(Q \rightarrow \neg Q) \rightarrow R$	$((Q \rightarrow S) \rightarrow (S \rightarrow \neg P)) \rightarrow ((Q \rightarrow \neg Q) \rightarrow R)$
T	T	T	T	F	T	T
T	T	T	F	T	T	T
T	T	F	T	F	T	T
T	T	F	F	T	T	T
T	F	T	T	F	T	T
T	F	T	F	T	T	T
T	F	F	T	F	F	T
T	F	F	F	T	F	F

P	Q	R	S	$(Q \rightarrow S) \rightarrow (S \rightarrow \neg P)$	$(Q \rightarrow \neg Q) \rightarrow R$	$((Q \rightarrow S) \rightarrow (S \rightarrow \neg P)) \rightarrow ((Q \rightarrow \neg Q) \rightarrow R)$
F	T	T	T	T	T	T
F	T	T	F	T	T	T
F	T	F	T	T	T	T
F	T	F	F	T	T	T
F	F	T	T	T	T	T
F	F	T	F	T	T	T
F	F	F	T	T	F	F
F	F	F	F	T	F	F

F has 13 models, so it is false in $16 - 13 = 3$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.20 [MSQ, 2M · Predicate Logic] **Answer: B, C** (B: $\exists x \forall y \neg L(y, x)$; C: $\forall x \forall y (L(x, y) \rightarrow \neg L(y, x))$)

(A) False: for $x = 3$ no y in D is larger. (B) True: $x = 1$ has no y with $y < 1$. (C) True: $<$ is asymmetric. (D) False: the order on $\{1, 2, 3\}$ is not dense; for $x = 1, y = 2$ there is no z with $1 < z < 2$. Evaluating all cases by enumeration confirms B and C.

Q.21 [NAT, 2M · Exact Inference] **Answer: 0.824 to 0.828**

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm} \mid \text{Earthquake}) \cdot P(\text{JohnCalls}) \cdot P(\text{MaryCalls} \mid \text{Alarm}, \text{JohnCalls})$.

Inference by enumeration: $P(\text{Burglary} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Burglary}, e, \text{hidden})$.

Hidden: Ea, Ma	Bu	product of CPT entries (Bu · Ea · Al · Jo · Ma)	value
T, T	T	$0.7 \times 0.75 \times 0.05 \times 0.4 \times 0.95$	0.00998
T, F	T	$0.7 \times 0.75 \times 0.05 \times 0.4 \times 0.05$	0.00053
F, T	T	$0.7 \times 0.25 \times 0.4 \times 0.4 \times 0.95$	0.0266
F, F	T	$0.7 \times 0.25 \times 0.4 \times 0.4 \times 0.05$	0.0014
T, T	F	$0.3 \times 0.95 \times 0.05 \times 0.4 \times 0.95$	0.00542
T, F	F	$0.3 \times 0.95 \times 0.05 \times 0.4 \times 0.05$	0.00029
F, T	F	$0.3 \times 0.05 \times 0.4 \times 0.4 \times 0.95$	0.00228
F, F	F	$0.3 \times 0.05 \times 0.4 \times 0.4 \times 0.05$	0.00012

$P(\text{Burglary}=T, e) = 0.0385$; $P(\text{Burglary}=F, e) = 0.0081$; $P(e) = 0.0466$.

$P(\text{Burglary}=T \mid e) = 0.0385 / 0.0466 = \mathbf{0.8262} \approx 0.826$.

Q.22 [MSQ, 2M · Conditional Independence] **Answer: D** ($D: F \perp G \mid B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp E \mid A$: **FALSE** — the path $G \leftarrow D \leftarrow B \rightarrow E$ is active given $\{A\}$.

(B) $A \perp D \mid C, F$: **FALSE** — the path $A \rightarrow D$ is active given $\{C, F\}$.

(C) $E \perp D \mid G$: **FALSE** — the path $E \leftarrow B \rightarrow D$ is active given $\{G\}$.

(D) $F \perp G \mid B$: **TRUE** — $F \leftarrow E \leftarrow B \rightarrow D \leftarrow A \rightarrow G$ (blocked: observed fork node B); $F \leftarrow E \leftarrow B \rightarrow D \rightarrow G$ (blocked: observed fork node B).

Q.23 [NAT, 2M · Sampling] **Answer: 0.76 to 0.78**

Rejection sampling keeps only samples consistent with the evidence $B = T$: samples 1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (13 accepted, 5 rejected).

Among the accepted samples, $E = T$ in 10. Estimate = $10/13 = \mathbf{0.7692}$.

(For this network the exact value is 0.585; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.24 [MCQ, 2M · Sampling] **Answer: A** (A: $A=T, B=F, C=T, D=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.8$; $u = 0.57 < 0.8 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.5$; $u = 0.6 \geq 0.5 \Rightarrow B = F$
- C: $P(C=T \mid B=F) = 0.5$; $u = 0.39 < 0.5 \Rightarrow C = T$
- D: $P(D=T \mid C=T) = 0.9$; $u = 0.29 < 0.9 \Rightarrow D = T$

Sample: **(A=T, B=F, C=T, D=T)**.

Q.25 [NAT, 2M · Probability for AI] **Answer: 0.87 to 0.872**

Using conditional independence, $P(\text{Cavity} \mid \text{toothache}, \text{catch}) = \alpha P(\text{toothache} \mid \text{Cavity}) P(\text{catch} \mid \text{Cavity}) P(\text{Cavity})$. For cavity: $0.6 \times 0.9 \times 0.2 = 0.108$. For $\neg\text{cavity}$: $0.1 \times 0.2 \times 0.8 = 0.016$. $\alpha = 1/(0.108 + 0.016) = 1/0.124$, so $P(\text{cavity} \mid \text{toothache}, \text{catch}) = 0.108/0.124 \approx 0.871$.

Mock Test 048

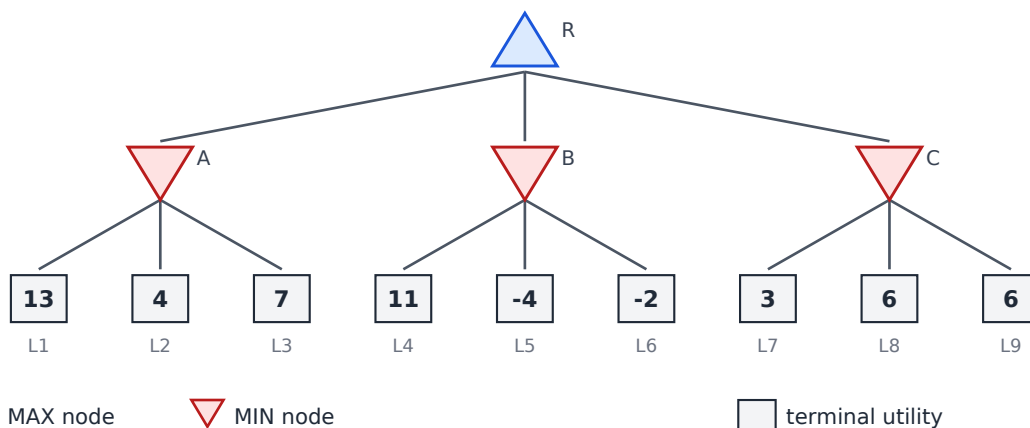
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 6 / 12	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

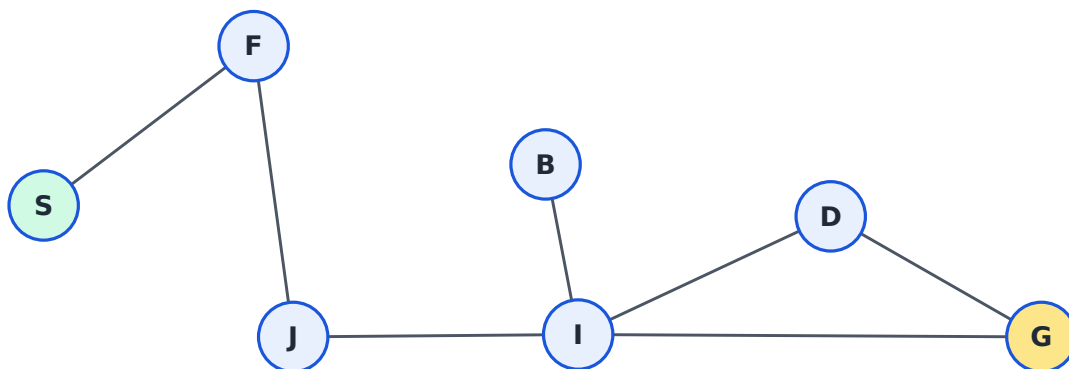
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, F, J, I, G
- ☐ (B) S, J, F, I, B, D, G
- ☐ (C) S, F, J, I, B, G
- ☐ (D) S, F, J, I, B, D, G

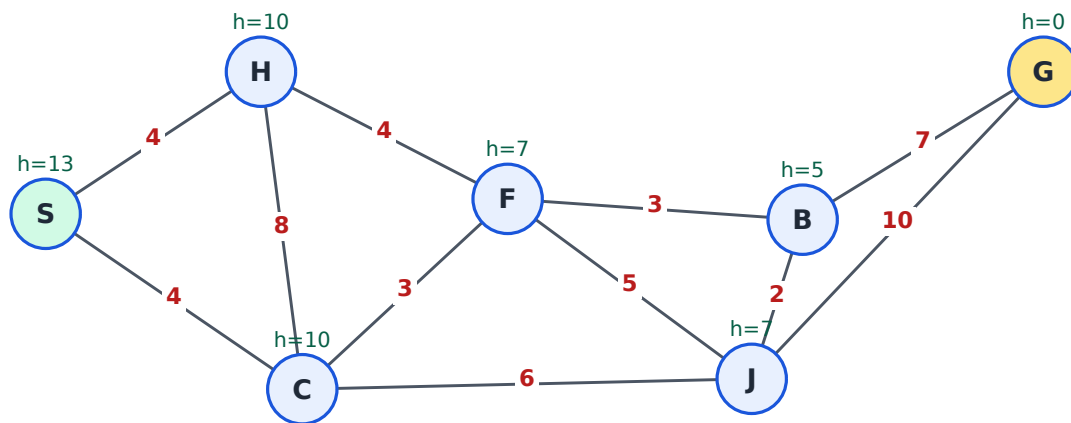
Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search

When successors are examined in **random** order, alpha-beta search is expected to examine roughly

- ☐ (A) $O(b^{m/2})$ nodes
- ☐ (B) $O(b^{3m/4})$ nodes
- ☐ (C) $O(b^m)$ nodes
- ☐ (D) $O(m \cdot b)$ nodes

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

What is the most general unifier (MGU) of $\text{Knows}(\text{John}, x)$ and $\text{Knows}(y, \text{Mother}(y))$?

- ☐ (A) $\{y/\text{John}, x/\text{Mother}(\text{John})\}$
- ☐ (B) $\{x/\text{John}, y/\text{Mother}(\text{John})\}$
- ☐ (C) $\{y/\text{John}, x/\text{Mother}(y)\}$
- ☐ (D) The two atoms are not unifiable

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((R \wedge Q) \vee (R \leftrightarrow \neg R)) \leftrightarrow ((R \wedge P) \rightarrow (P \leftrightarrow R)) ?$$

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic

The formula $(A \wedge B) \vee (C \wedge D) \vee (E \wedge F)$ is converted to CNF by repeatedly distributing $\vee$ over $\wedge$ (no simplification). How many clauses does the result have?

Answer (numerical): _____

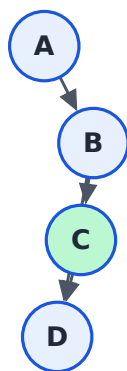
Q.9 [MCQ · 1 mark · negative: -1/3] Probability for AI

In AIMA's enumeration-style inference, $P(X \mid e)$ is computed as $\alpha P(X, e)$, where the sum over hidden variables gives $P(X, e)$. What is the value of α ?

- (A) $1/P(e)$
 ○ (B) $P(e)$
 ○ (C) $1/P(X)$
 ○ (D) $1/|\text{domain}(X)|$

Q.10 [NAT · 1 mark · no negative marking] Exact Inference - Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



P(A=T)
0.5

A	P(B=T ...)	B	P(C=T ...)
T	0.95	T	0.1
F	0.5	F	0.2

B	C	P(D=T ...)
T	T	0.95
T	F	0.6
F	T	0.7
F	F	0.5

Answer (numerical): _____

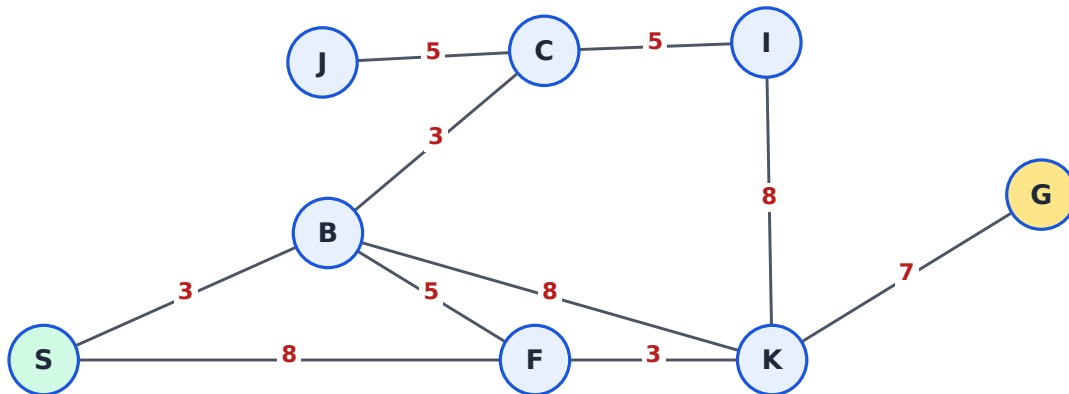
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Problem Formulation

In the missionaries-and-cannibals problem (3 of each, a boat with capacity 2), a state is (m, c, b) , where m and c are the numbers of missionaries and cannibals on the left bank (0 to 3 each) and $b \in \{\text{left}, \text{right}\}$ is the boat's side. A state is safe if, on each bank, the missionaries are either zero or at least as many as the cannibals. How many safe states are there, counting both boat positions and ignoring reachability?

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

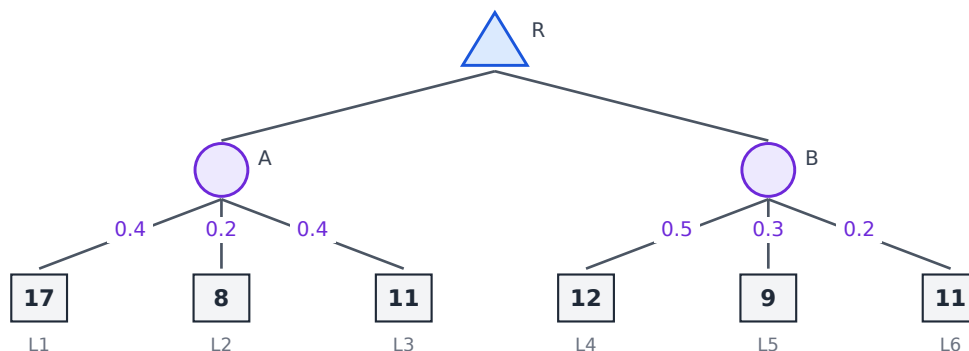
Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search

BFS runs on a problem with $b = 10$ and solution depth $d = 8$. It stores every generated node, and each node needs 1 KB. Approximately how much memory is required? (1 GB = 10^6 KB.)

- ☐ (A) About 100 MB
☐ (B) About 1 GB
☐ (C) About 100 GB
☐ (D) About 10 TB

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node

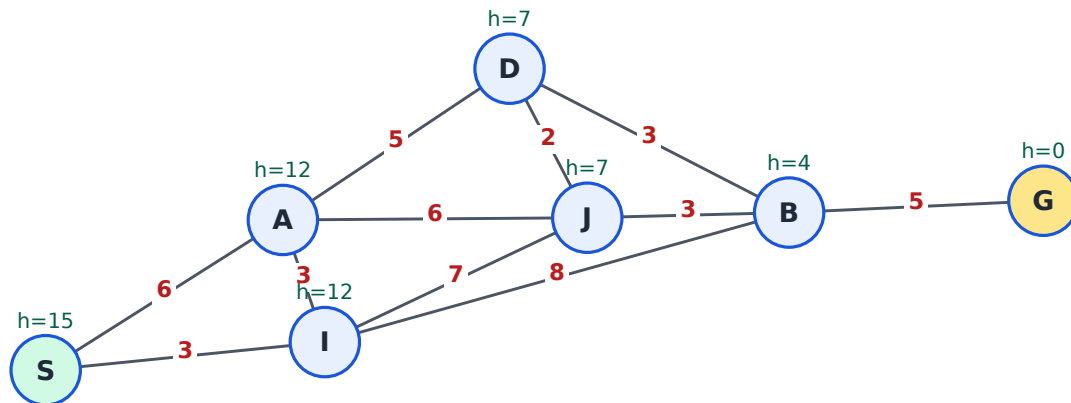
○ CHANCE node (edge = probability)

□ terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge S-I violates the consistency condition (in at least one direction).
- ☐ (B) h is admissible.
- ☐ (C) h is consistent (monotone).
- ☐ (D) $h(D)$ overestimates the true cost from D to G.

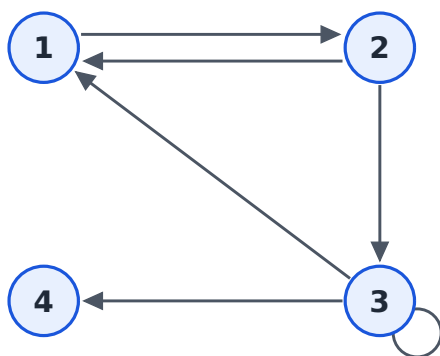
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following formulas are valid?

- ☐ (A) $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
- ☐ (B) $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$
- ☐ (C) $((p \rightarrow q) \rightarrow r) \leftrightarrow (p \rightarrow (q \rightarrow r))$
- ☐ (D) $(\neg p \rightarrow \perp) \rightarrow p$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
- ☐ (B) $\exists x R(x, x)$
- ☐ (C) $\exists x \forall y \neg R(x, y)$
- ☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(P \vee Q \vee S)$, C2: $(\neg S)$, C3: $(P \vee S)$, C4: (S) , C5: $(P \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) (Q)
☐ (B) $(\neg P \vee Q)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) (P)

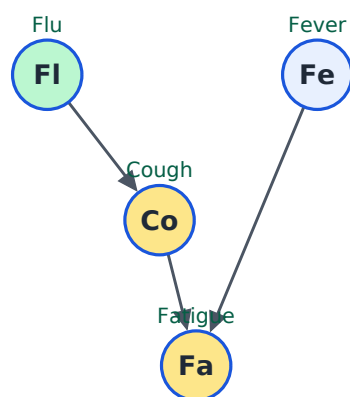
Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = Q \leftrightarrow (Q \rightarrow R)$?

- ☐ (A) $(Q \rightarrow (Q \rightarrow R)) \wedge ((Q \rightarrow R) \rightarrow Q)$
☐ (B) $(Q \wedge (Q \rightarrow R)) \vee (Q \wedge \neg(Q \rightarrow R))$
☐ (C) $Q \leftrightarrow \neg(Q \rightarrow R)$
☐ (D) $(Q \wedge (Q \rightarrow R)) \vee (\neg Q \wedge \neg(Q \rightarrow R))$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Flu} = T \mid \text{Cough} = F, \text{Fatigue} = T)$ (round off to 3 decimal places).



$P(\text{Fl}=T)$
0.5

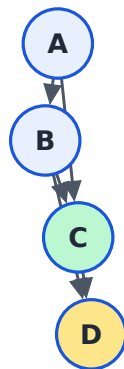
$P(\text{Fe}=T)$	Fl	$P(\text{Co}=T \dots)$
0.25	T	0.75
	F	0.05

Fe	Co	$P(\text{Fa}=T \dots)$
T	T	0.05
T	F	0.8
F	T	0.8
F	F	0.9

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{D=F}$. The current state of the chain is $\mathbf{(A=F, B=F, C=F, D=F)}$. The next step resamples $\mathbf{C}$. With what probability is $\mathbf{C}$ set to $\mathbf{T}$? (round off to 3 decimal places)



P(A=T)
0.9

A	P(B=T .)	A	B	P(C=T ...)
T	0.7	T	T	0.1
F	0.6	T	F	0.5
		F	T	0.6
		F	F	0.1

B	C	P(D=T ...)
T	T	0.9
T	F	0.6
F	T	0.05
F	F	0.5

Answer (numerical): _____

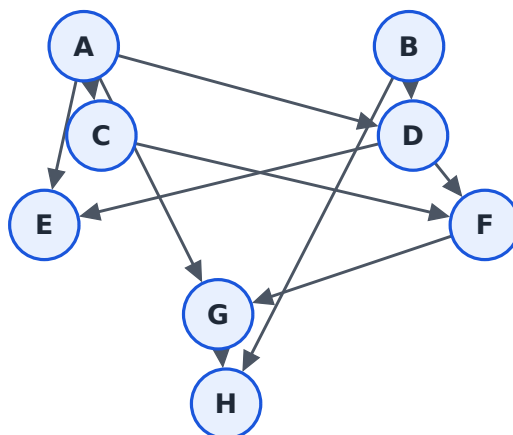
Q.22 [MCQ · 2 marks · negative: -2/3] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. For the query $P(j | b)$ (J given Burglary = true), which variables are irrelevant to the query and can be removed before running variable elimination?

- ☐ (A) {M}
☐ (B) {E, M}
☐ (C) {E}
☐ (D) No variable is irrelevant

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

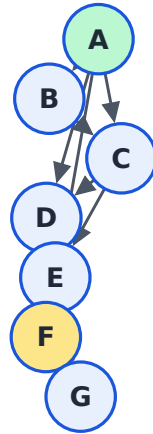
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp C | A$
☐ (B) $C \perp H | G$
☐ (C) $F \perp H | C$
☐ (D) $F \perp H | A, G$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

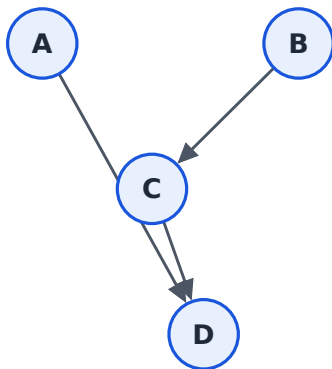
Variable elimination is used to compute $P(A \mid F = T)$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, C, E, G, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.98, 0.77, 0.01, 0.23)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.3

$P(B=T)$	B	$P(C=T \dots)$
0.6	T	0.1
	F	0.9

A	C	$P(D=T \dots)$
T	T	0.05
T	F	0.6
F	T	0.6
F	F	0.5

- ☐ (A) $A=T, B=F, C=T, D=T$
☐ (B) $A=F, B=F, C=T, D=T$
☐ (C) $A=F, B=T, C=T, D=T$
☐ (D) $A=T, B=T, C=F, D=F$

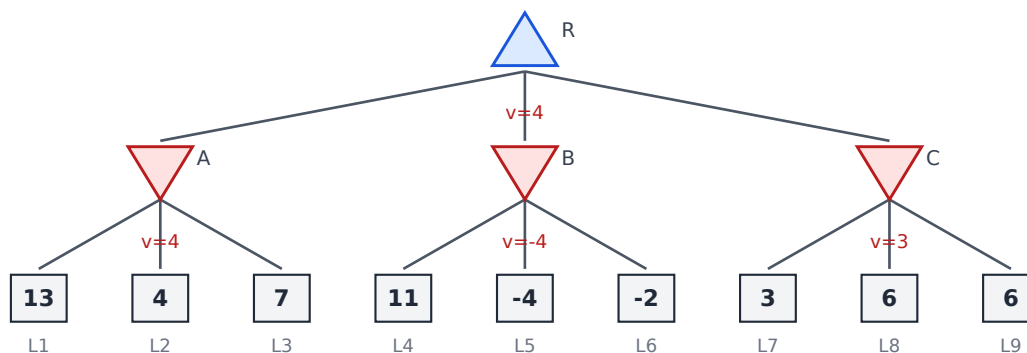
Answer Key — Mock Test 048

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	4	14	NAT	2	Adversarial Search	12.79 to 12.81
2	MCQ	1	Uninformed Search	D	15	MSQ	2	Informed Search	B, C
3	MCQ	1	Adversarial Search	B	16	MSQ	2	Propositional Logic	A, B, D
4	NAT	1	Informed Search	17	17	MSQ	2	Predicate Logic	B, C
5	MCQ	1	Predicate Logic	A	18	MSQ	2	Propositional Logic	C, D
6	NAT	1	Predicate Logic	65	19	MSQ	2	Propositional Logic	A, D
7	NAT	1	Propositional Logic	2	20	NAT	2	Exact Inference	0.206 to 0.21
8	NAT	1	Propositional Logic	8	21	NAT	2	Sampling	0.172 to 0.176
9	MCQ	1	Probability for AI	A	22	MCQ	2	Variable Elimination	A
10	NAT	1	Exact Inference	0.126 to 0.13	23	MSQ	2	Conditional Independence	A
11	NAT	2	Problem Formulation	20	24	NAT	2	Exact Inference	3
12	NAT	2	Uninformed Search	8	25	MCQ	2	Sampling	B
13	MCQ	2	Uninformed Search	C					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 4

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 4$, $B = -4$, $C = 3$. Root value = **4**; MAX's optimal first move is towards A.

Q.2 [MCQ, 1M · Uninformed Search] Answer: D (D: S, F, J, I, B, D, G)

Adjacency lists (alphabetical): B: {I}; D: {G, I}; F: {J, S}; G: {D, I}; I: {B, D, G, J}; J: {F, I}; S: {F}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[F]
2	F	[J]
3	J	[I]
4	I	[B, D, G]
5	B	[D, G]
6	D	[G]
7	G	[] $\leftarrow$ goal popped, stop

Expansion order: **S, F, J, I, B, D, G**. The path returned is $S \rightarrow F \rightarrow J \rightarrow I \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [MCQ, 1M · Adversarial Search] Answer: B ($B: O(b^{3m/4})$ nodes)

AIMA reports that random move ordering gives about **$O(b^{3m/4})$** for moderate b . $O(b^{m/2})$ (A) needs perfect ordering. $O(b^m)$ (C) is the worst case. $O(mb)$ (D) is a space bound.

Q.4 [NAT, 1M · Informed Search] Answer: 17

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (13)	C(10), H(10)
2	C (10)	F(7), J(7), H(10)
3	F (7)	B(5), J(7), H(10)
4	B (5)	G(0), J(7), H(10)
5	G (0)	J(7), H(10) ← goal, stop

Path returned: **S** → **C** → **F** → **B** → **G**, cost = 4 + 3 + 3 + 7 = **17**.

The optimal cost is $h^*(S) = 17$ — greedy happens to be optimal here, but in general it is not.

Q.5 [MCQ, 1M · Predicate Logic] **Answer: A** (A: {y/John, x/Mother(John)})

Unify arguments left to right. John vs y gives y/John. Then x vs Mother(y) becomes x vs Mother(John), giving x/Mother(John). Applying {y/John, x/Mother(John)} to both atoms yields Knows(John, Mother(John)). Option B binds the wrong variables. Option C leaves y in the binding of x, so it is not idempotent and does not make the atoms identical in one application.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 65**

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 4$ this gives **65** (verified by enumerating all interpretations).

Q.7 [NAT, 1M · Propositional Logic] **Answer: 2**

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \wedge Q) \vee (R \leftrightarrow \neg R)$	$(R \wedge P) \rightarrow (P \leftrightarrow R)$	$((R \wedge Q) \vee (R \leftrightarrow \neg R)) \leftrightarrow ((R \wedge P) \rightarrow (P \leftrightarrow R))$
T	T	T	T	T	T
T	T	F	F	T	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	F	T	F
F	F	T	F	T	F
F	F	F	F	T	F

Rows where F = T: **2**.

Q.8 [NAT, 1M · Propositional Logic] **Answer: 8**

Distribution produces one clause per way of choosing one literal from each conjunction: $\{A, B\} \times \{C, D\} \times \{E, F\}$, i.e. $2 \times 2 \times 2 = 8$ clauses such as $A \vee C \vee E$, $A \vee C \vee F$, ..., $B \vee D \vee F$. No clause is a tautology or a duplicate because all six symbols are distinct. This illustrates the exponential blow-up of naive CNF conversion.

Q.9 [MCQ, 1M · Probability for AI] **Answer: A** (A: $1/P(e)$)

By definition $P(X | e) = P(X, e)/P(e)$, so $\alpha = 1/P(e)$. In practice α is found by normalizing the vector $P(X, e)$ to sum to 1, because $\sum_x P(x, e) = P(e)$. The other options do not make the entries sum to 1 in general.

Q.10 [NAT, 1M · Exact Inference] **Answer: 0.126 to 0.13**

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T, D=T	0.04512
A=T, B=T, D=F	0.00238
A=T, B=F, D=T	0.0035
A=T, B=F, D=F	0.0015
A=F, B=T, D=T	0.02375
A=F, B=T, D=F	0.00125
A=F, B=F, D=T	0.035
A=F, B=F, D=F	0.015

Sum = **0.1275** ≈ 0.128 .

Q.11 [NAT, 2M · Problem Formulation] Answer: 20

Count the safe (m, c) pairs. m = 0: the right bank has 3 missionaries, so every c = 0..3 is safe, giving 4 pairs. m = 3: the left bank needs c ≤ 3 and the right bank has no missionaries, giving 4 pairs. m = 1: the left bank needs c ≤ 1 and the right bank (2 missionaries, 3−c cannibals) needs 3−c ≤ 2, so c = 1: 1 pair. m = 2: c ≤ 2 and 3−c ≤ 1, so c = 2: 1 pair. That gives 10 pairs, and 2 boat positions give **20** states. (Only 16 of them are reachable from (3,3,left).)

Q.12 [NAT, 2M · Uninformed Search] Answer: 8

UCS always expands the frontier node with the smallest path cost g(n); if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(3), F(8)
2	B (3)	C(6), F(8), K(11)
3	C (6)	F(8), I(11), J(11), K(11)
4	F (8)	I(11), J(11), K(11)
5	I (11)	J(11), K(11)
6	J (11)	K(11)
7	K (11)	G(18)
8	G (18)	← goal, stop

Path returned: **S → B → K → G**, cost = 3 + 8 + 7 = **18**. Nodes expanded (including S and G): **8** (S, B, C, F, I, J, K, G).

Q.13 [MCQ, 2M · Uninformed Search] Answer: C (C: About 100 GB)

The nodes generated are $b + \dots + b^8 \approx 1.11 \times 10^8$. At 1 KB each, that is about 1.11×10^8 KB $\approx$ **111 GB, i.e. about 100 GB**. This is AIMA's point that memory, not time, is the binding constraint for BFS. 100 MB (A) and 1 GB (B) are 2 to 3 orders of magnitude too small, and 10 TB (D) is 100 times too large.

Q.14 [NAT, 2M · Adversarial Search] Answer: 12.79 to 12.81

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.4 \times 17 + 0.2 \times 8 + 0.4 \times 11 = 12.8$
- $B = 0.5 \times 12 + 0.3 \times 9 + 0.2 \times 11 = 10.9$

Root = max(12.8, 10.9) = **12.8**.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [MSQ, 2M · Informed Search] Answer: B, C (B: h is admissible.; C: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	D	G	I	J	S
h(n)	12	4	7	0	12	7	15
h*(n)	13	5	8	0	13	8	16
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A, B, D (A: $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$; B: $(p \rightarrow (q \rightarrow r)) \Leftrightarrow ((p \wedge q) \rightarrow r)$; D: $(\neg p \rightarrow \perp) \rightarrow p$)

(A) Transitivity of implication; a truth table shows no falsifying row. (B) Exportation: both sides are equivalent to $\neg p \vee \neg q \vee r$. (C) Not valid: at $p=F, q=F, r=F$ the left side $(p \rightarrow q) \rightarrow r$ is $T \rightarrow F = F$, while the right side is T. Implication is not associative. (D) $\neg p \rightarrow \perp \equiv \neg \neg p \equiv p$, so the formula is $p \rightarrow p$, which is valid (this is the basis of proof by contradiction).

Q.17 [MSQ, 2M · Predicate Logic] **Answer: B, C** (B: $\exists x R(x, x)$; C: $\exists x \forall y \neg R(x, y)$)

$R = \{(1,2), (2,1), (2,3), (3,1), (3,3), (3,4)\}$.

(A) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(1,2)$ and $R(2,1)$ with $1 \neq 2$.

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 3.

(C) $\exists x \forall y \neg R(x, y)$: **TRUE** — 4 has no outgoing edge.

(D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 4 has no edge to a different element.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable); D: (P))

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) (Q) never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(\neg P \vee Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $\square$ is derivable:

1. resolve $(\neg S)$ and (S) on $S \Rightarrow \square$ (empty clause)

(D) (P) is derivable:

1. resolve $(\neg S)$ and $(P \vee S)$ on $S \Rightarrow (P)$

Q.19 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(Q \rightarrow (Q \rightarrow R)) \wedge ((Q \rightarrow R) \rightarrow Q)$; D: $(Q \wedge (Q \rightarrow R)) \vee (\neg Q \wedge \neg(Q \rightarrow R))$)

Compare truth tables with $F = Q \leftrightarrow (Q \rightarrow R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(Q \rightarrow (Q \rightarrow R)) \wedge ((Q \rightarrow R) \rightarrow Q)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $(Q \wedge (Q \rightarrow R)) \vee (Q \wedge \neg(Q \rightarrow R))$ — differs at $Q=T, R=F$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(C) $Q \leftrightarrow \neg(Q \rightarrow R)$ — differs at $Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(D) $(Q \wedge (Q \rightarrow R)) \vee (\neg Q \wedge \neg(Q \rightarrow R))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.20 [NAT, 2M · Exact Inference] **Answer: 0.206 to 0.21**

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever}) \cdot P(\text{Cough} | \text{Flu}) \cdot P(\text{Fatigue} | \text{Fever}, \text{Cough})$.

Inference by enumeration: $P(\text{Flu} | e) = \alpha \sum_{\text{hidden}} P(\text{Flu}, e, \text{hidden})$.

Hidden: Fe	FI	product of CPT entries (FI · Fe · Co · Fa)	value
T	T	$0.5 \times 0.25 \times 0.25 \times 0.8$	0.025
F	T	$0.5 \times 0.75 \times 0.25 \times 0.9$	0.08438
T	F	$0.5 \times 0.25 \times 0.95 \times 0.8$	0.095
F	F	$0.5 \times 0.75 \times 0.95 \times 0.9$	0.32062

$P(\text{Flu}=T, e) = 0.10938$; $P(\text{Flu}=F, e) = 0.41563$; $P(e) = 0.525$.

$P(\text{Flu}=T | e) = 0.10938 / 0.525 = \mathbf{0.2083} \approx 0.208$.

Q.21 [NAT, 2M · Sampling] **Answer: 0.172 to 0.176**

Gibbs sampling draws C from $P(C | \text{mb}(C))$, where $\text{mb}(C) = \{A, B, D\}$:

$P(C | \text{mb}) \propto P(C | \text{parents}(C)) \times \prod_{\text{children } Y} P(Y | \text{parents}(Y))$.

$C=T$: $P(C=T | A=F, B=F) \times P(D=F | B=F, C=T) = 0.1 \times 0.95 = 0.095$

$C=F$: $P(C=F | A=F, B=F) \times P(D=F | B=F, C=F) = 0.9 \times 0.5 = 0.45$

Normalise: $0.095 / (0.095 + 0.45) = \mathbf{0.1743}$.

Q.22 [MCQ, 2M · Variable Elimination] **Answer: A** (A: $\{M\}$)

AIMA: every variable that is not an ancestor of a query variable or an evidence variable is irrelevant.

Ancestors of J and B: $\{A, B, E\}$. M is neither, so it is irrelevant: $\sum_m P(m | a) = 1$. E is an ancestor of J ($E \rightarrow A \rightarrow J$), and $P(a | b) = \sum_e P(e) P(a | b, e)$ genuinely depends on E's CPT, so E cannot be dropped.

Q.23 [MSQ, 2M · Conditional Independence] Answer: A (A: $D \perp C \mid A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $D \perp C \mid A$: **TRUE** — $D \leftarrow A \rightarrow C$ (blocked: observed fork node A); $D \leftarrow A \rightarrow G \leftarrow F \leftarrow C$ (blocked: observed fork node A); $D \leftarrow B \rightarrow H \leftarrow G \leftarrow A \rightarrow C$ (blocked: collider H with neither it nor a descendant observed); $D \leftarrow B \rightarrow H \leftarrow G \leftarrow F \leftarrow C$ (blocked: collider H with neither it nor a descendant observed); $D \rightarrow E \leftarrow A \rightarrow C$ (blocked: collider E with neither it nor a descendant observed); $D \rightarrow E \leftarrow A \rightarrow G \leftarrow F \leftarrow C$ (blocked: collider E with neither it nor a descendant observed) (+2 more paths, all blocked).

(B) $C \perp H \mid G$: **FALSE** — the path $C \leftarrow A \rightarrow D \leftarrow B \rightarrow H$ is active given $\{G\}$.

(C) $F \perp H \mid C$: **FALSE** — the path $F \leftarrow D \leftarrow A \rightarrow G \rightarrow H$ is active given $\{C\}$.

(D) $F \perp H \mid A, G$: **FALSE** — the path $F \leftarrow D \leftarrow B \rightarrow H$ is active given $\{A, G\}$.

Q.24 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(A, C, D)$, $f(C, D, E)$, $f(A, E)$, $f(D, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B)$, $f(A, B, C)$	$\{A, B, C\}$	$f(A, C)$
C	$f(A, C, D)$, $f(C, D, E)$, $f(A, C)$	$\{A, C, D, E\}$	$f(A, D, E)$
E	$f(A, E)$, $f(A, D, E)$	$\{A, D, E\}$	$f(A, D)$
G	$f(D, G)$	$\{D, G\}$	$f(D)$
D	$f(A, D)$, $f(D)$	$\{A, D\}$	$f(A)$

The largest factor produced has **3** variable(s) (created when eliminating C); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over A and normalise.

Q.25 [MCQ, 2M · Sampling] Answer: B (B: $A=F$, $B=F$, $C=T$, $D=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.3$; $u = 0.98 \geq 0.3 \Rightarrow A = F$
- B: $P(B=T \mid \text{—}) = 0.6$; $u = 0.77 \geq 0.6 \Rightarrow B = F$
- C: $P(C=T \mid B=F) = 0.9$; $u = 0.01 < 0.9 \Rightarrow C = T$
- D: $P(D=T \mid A=F, C=T) = 0.6$; $u = 0.23 < 0.6 \Rightarrow D = T$

Sample: **$\{A=F, B=F, C=T, D=T\}$** .

Mock Test 049

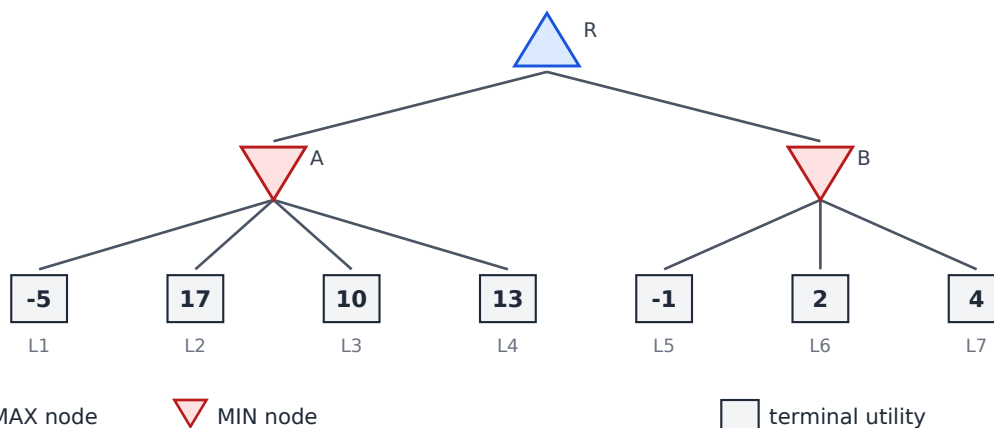
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 5 / 16	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 3$ (so $b^d = 8$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

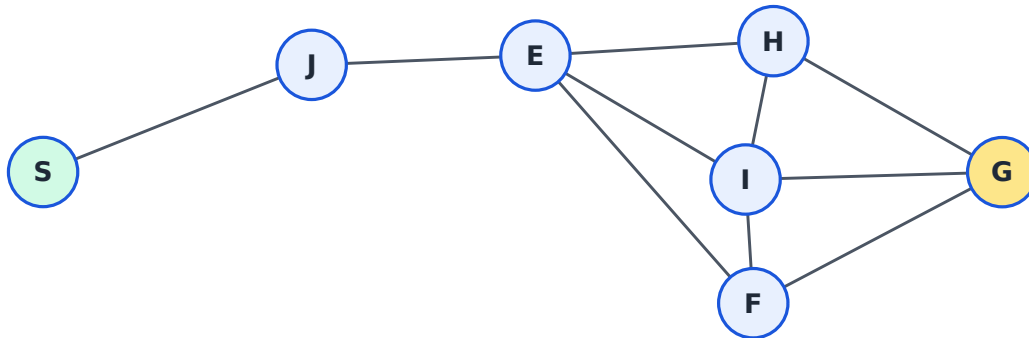
Q.3 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search

Breadth-first search (graph search) is guaranteed to return an optimal (least-cost) solution whenever a solution exists, provided that

- ☐ (A) all step costs are identical (e.g. every action costs 1)
- ☐ (B) an admissible heuristic is supplied
- ☐ (C) the branching factor is infinite
- ☐ (D) the state space contains no cycles

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, J, E, I, H, F, G
- ☐ (B) S, J, E, F, G
- ☐ (C) S, J, E, F, H, I, G
- ☐ (D) S, J, F, E, H, I, G

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following propositional formulas is a tautology?

- ☐ (A) $(p \rightarrow q) \vee (q \rightarrow p)$
- ☐ (B) $(p \rightarrow q) \rightarrow (q \rightarrow p)$
- ☐ (C) $(p \wedge \neg q) \rightarrow q$
- ☐ (D) $(p \vee q) \rightarrow (p \wedge q)$

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In the formula $\forall x (P(x, y) \rightarrow \exists y Q(x, y, z))$, which variables have free occurrences?

- ☐ (A) y and z
- ☐ (B) z only
- ☐ (C) x, y and z
- ☐ (D) x and z

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x R(x, x)$ true?

Answer (numerical): _____

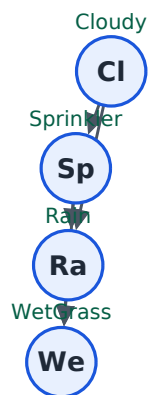
Q.8 [NAT · 1 mark · no negative marking] Variable Elimination

Factors over Boolean variables: $f(A, B)$ with $f(a, b) = 0.4$, $f(a, \neg b) = 0.6$, $f(\neg a, b) = 0.8$, $f(\neg a, \neg b) = 0.2$; $g(B, C)$ with $g(b, c) = 0.3$, $g(b, \neg c) = 0.7$, $g(\neg b, c) = 0.5$, $g(\neg b, \neg c) = 0.5$. Let $h(A, C) = \sum_B f(A, B) \times g(B, C)$. Compute $h(a, c)$. (Round to 2 decimal places.)

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample **(Cl=T, Sp=T, Ra=T, We=F)**? (round off to 4 decimal places)



P(Cl=T)
0.95

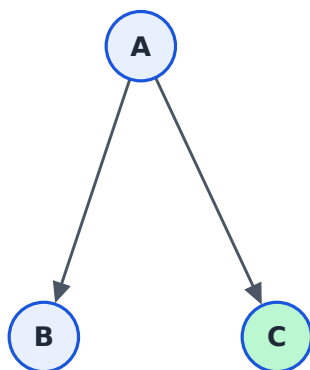
Cl	P(Sp=T Cl)	Cl	Sp	P(Ra=T ...)
T	0.75	T	T	0.05
F	0.95	T	F	0.8
		F	T	0.6
		F	F	0.75

Sp	Ra	P(We=T ...)
T	T	0.1
T	F	0.2
F	T	0.05
F	F	0.6

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability **$P(C = T)$** (round off to 3 decimal places).



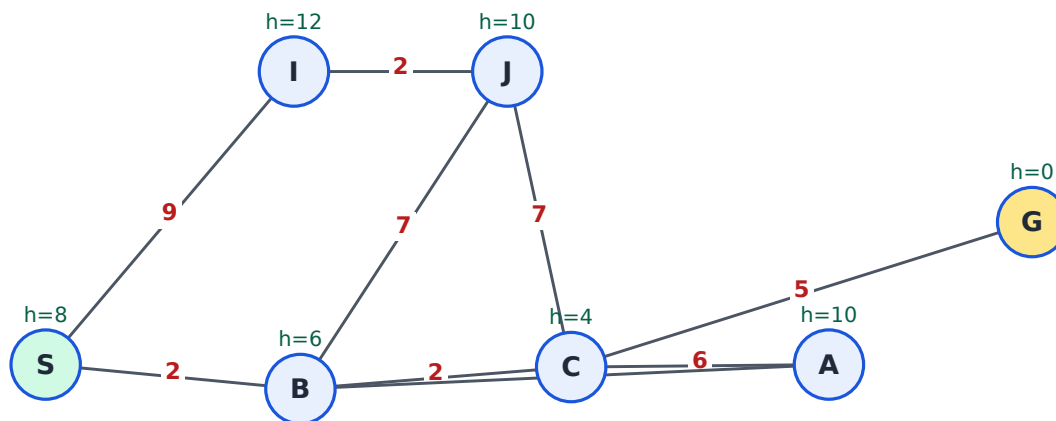
P(A=T)	A	P(B=T ...)	A	P(C=T ...)
0.7	T	0.8	T	0.95
	F	0.4	F	0.7

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

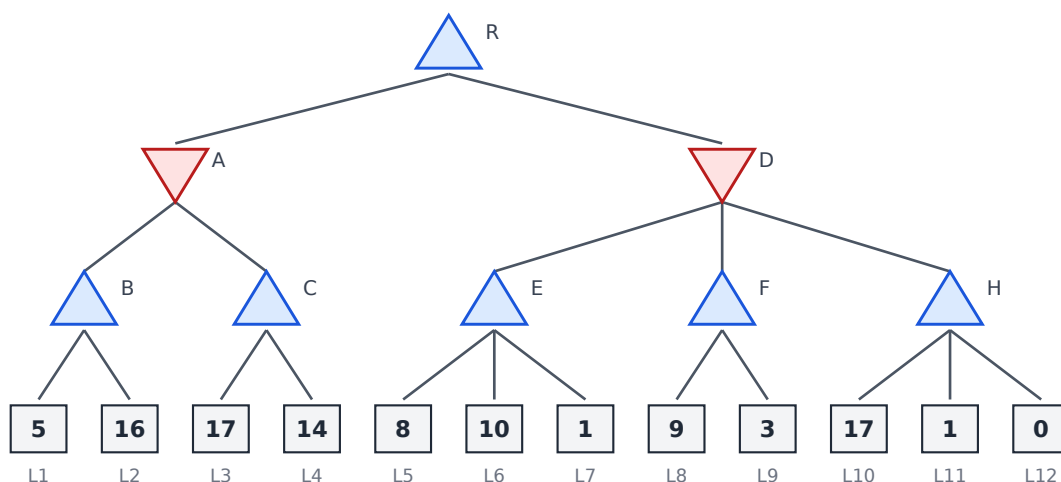
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

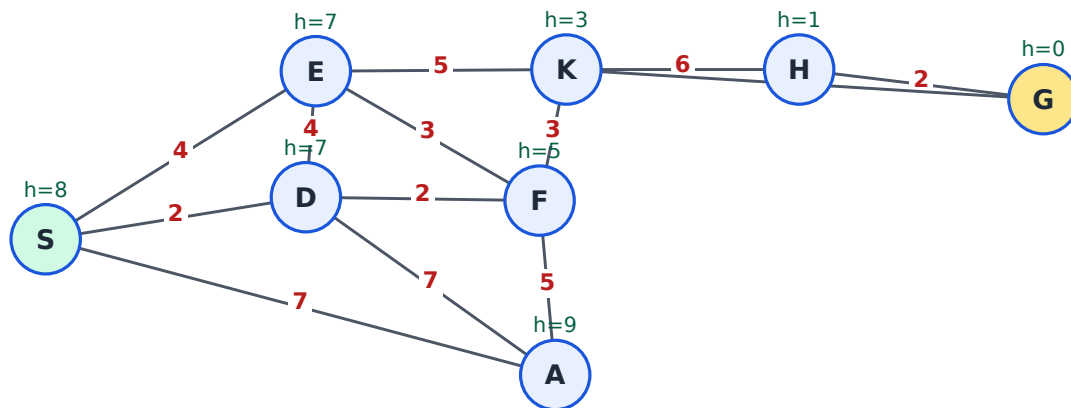
Q.13 [NAT · 2 marks · no negative marking] Adversarial Search

A two-ply game tree has a MAX root with four MIN children, whose leaves are listed left to right: MIN_1 : (10, 9, 14); MIN_2 : (11, 8, 20); MIN_3 : (2, 15, 30); MIN_4 : (12, 13, 7). Alpha-beta pruning visits children left to right. How many leaves are **pruned** (never evaluated)?

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

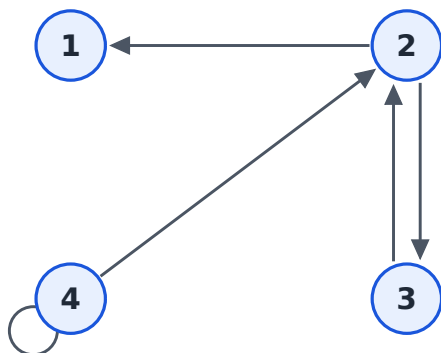
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall y \exists x R(x, y)$
☐ (B) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (C) $\exists x \forall y R(x, y)$
☐ (D) $\exists y \forall x R(x, y)$

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = ((P \vee \neg R) \rightarrow (S \wedge Q)) \leftrightarrow \neg(R \rightarrow Q) ?$$

Answer (numerical): _____

Q.17 [NAT · 2 marks · no negative marking] Predicate Logic

A first-order language has constants a, b and one binary function symbol g . Let $H_0 = \{a, b\}$ and $H_{i+1} = H_i \cup \{g(s, t) : s, t \in H_i\}$. How many ground terms are in H_2 ?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: (P), C2: (P $\vee$ $\neg$ S), C3: ($\neg$ R $\vee$ $\neg$ S), C4: (Q $\vee$ S)

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) (Q $\vee$ $\neg$ R)
- ☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
- ☐ (C) (P $\vee$ Q)
- ☐ (D) ($\neg$ P $\vee$ $\neg$ S)

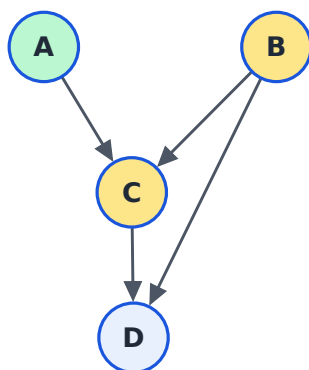
Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) ((P $\rightarrow$ R) $\rightarrow$ Q) $\leftrightarrow$ (P $\rightarrow$ (R $\rightarrow$ Q))
- ☐ (B) (Q $\vee$ R) $\rightarrow$ (Q $\wedge$ R)
- ☐ (C) ($\neg$ P $\rightarrow$ (R $\rightarrow$ Q)) $\leftrightarrow$ (($\neg$ P $\wedge$ R) $\rightarrow$ Q)
- ☐ (D) ((Q $\rightarrow$ R) $\wedge$ (R $\rightarrow$ P)) $\rightarrow$ (Q $\rightarrow$ P)

Q.20 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence **B = T, C = T**. The values of the non-evidence variables in the generated samples are listed in the table. Estimate **P(A = T | B = T, C = T)** (round off to 3 decimal places).



P(A=T)
0.6

P(B=T)	A	B	P(C=T ...)
0.6	T	T	0.2
	T	F	0.25
	F	T	0.7
	F	F	0.95

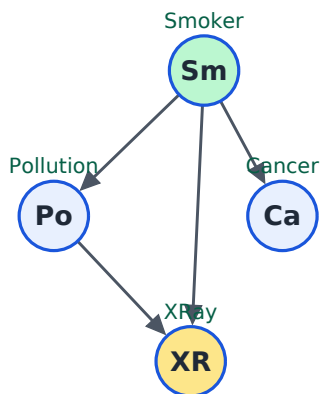
B	C	P(D=T ...)
T	T	0.1
T	F	0.3
F	T	0.25
F	F	0.2

#	A	D
1	F	T
2	T	F
3	T	T
4	T	F

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $P(\text{Smoker} = T \mid \text{XRay} = T)$ (round off to 3 decimal places).



$P(\text{Sm}=T)$
0.5

Sm	$P(\text{Po}=T \mid \dots)$	Sm	$P(\text{Ca}=T \mid \dots)$
T	0.8	T	0.5
F	0.6	F	0.2

Sm	Po	$P(\text{XR}=T \mid \dots)$
T	T	0.25
T	F	0.5
F	T	0.25
F	F	0.2

Answer (numerical): _____

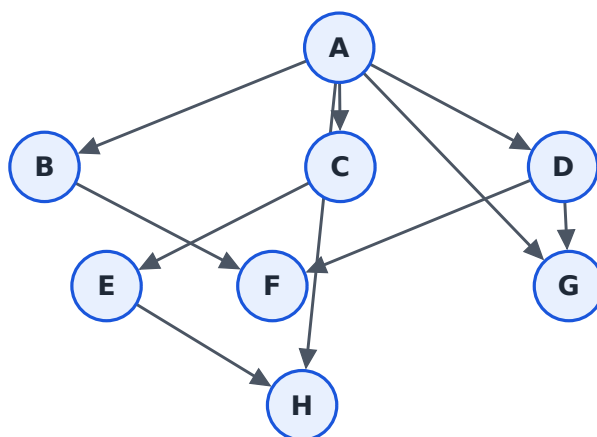
Q.22 [MSQ · 2 marks · no negative marking] Bayesian Networks

A Bayesian network is the chain $A \rightarrow B \rightarrow C$ over Boolean variables with $P(a) = 0.6$, $P(b \mid a) = 0.7$, $P(b \mid \neg a) = 0.2$, $P(c \mid b) = 0.9$, $P(c \mid \neg b) = 0.3$. Which of the following are correct?

- ☐ (A) $P(b) = 0.5$
☐ (B) $P(c) = 0.6$
☐ (C) $P(a \mid c) = 0.72$
☐ (D) $P(a \mid c) = P(a)$, since A and C are not adjacent

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

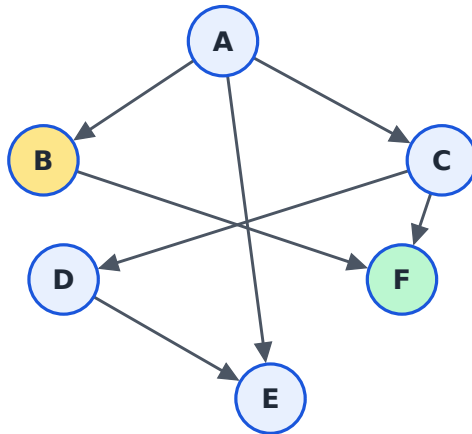
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp C \mid A$
☐ (B) $A \perp E \mid D, G$
☐ (C) $D \perp G \mid A$
☐ (D) $E \perp F \mid B, D$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{F} \mid \mathbf{B} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{B} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, E, C, A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. Compute $P(b \mid j, m)$. (Round to 3 decimal places.)

Answer (numerical): _____

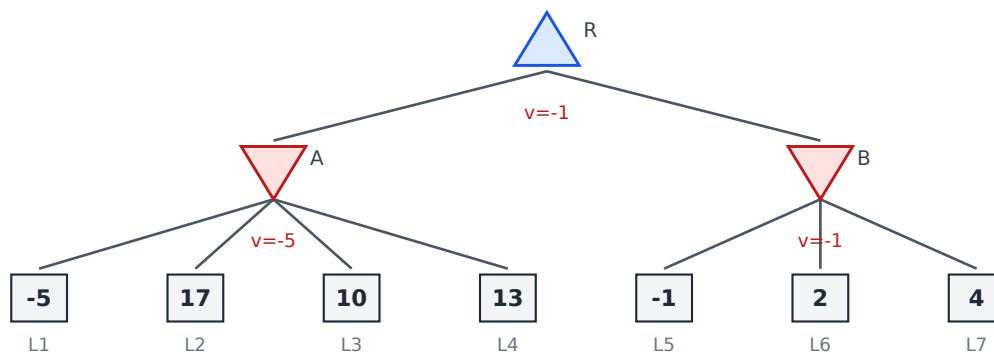
Answer Key — Mock Test 049

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	-1	14	NAT	2	Informed Search	6
2	NAT	1	Adversarial Search	5	15	MSQ	2	Predicate Logic	A
3	MCQ	1	Uninformed Search	A	16	NAT	2	Propositional Logic	9
4	MCQ	1	Uninformed Search	C	17	NAT	2	Predicate Logic	38
5	MCQ	1	Propositional Logic	A	18	MSQ	2	Propositional Logic	A, C
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Propositional Logic	C, D
7	NAT	1	Predicate Logic	64	20	NAT	2	Sampling	0.46 to 0.464
8	NAT	1	Variable Elimination	0.415 to 0.425	21	NAT	2	Exact Inference	0.564 to 0.568
9	NAT	1	Sampling	0.0319 to 0.0323	22	MSQ	2	Bayesian Networks	A, B, C
10	NAT	1	Exact Inference	0.873 to 0.877	23	MSQ	2	Conditional Independence	A, D
11	NAT	2	Informed Search	4	24	NAT	2	Exact Inference	3
12	NAT	2	Adversarial Search	6	25	NAT	2	Bayesian Networks	0.555 to 0.557
13	NAT	2	Adversarial Search	3					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] **Answer: -1**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = -5, B = -1. Root value = -1; MAX's optimal first move is towards B.

Q.2 [NAT, 1M · Adversarial Search] **Answer: 5**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 2^2 + 2^1 - 1 = 4 + 2 - 1 = 5$, versus 8 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [MCQ, 1M · Uninformed Search] **Answer: A** (A: all step costs are identical (e.g. every action costs 1))

BFS always returns the **shallowest** goal. The shallowest goal is the cheapest one only when path cost is a non-decreasing function of depth, the standard case being identical step costs. BFS uses no heuristic, so (B) has nothing to do with it. With an infinite branching factor BFS cannot even finish a single level, so (C) is wrong. Removing cycles does not help when step costs differ: a deep cheap path can still beat a shallow expensive one, so (D) is wrong.

Q.4 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, J, E, F, H, I, G)

Adjacency lists (alphabetical): E: {F, H, I, J}; F: {E, G, I}; G: {F, H, I}; H: {E, G, I}; I: {E, F, G, H}; J: {E, S}; S: {J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[J]
2	J	[E]
3	E	[F, H, I]
4	F	[H, I, G]
5	H	[I, G]
6	I	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, J, E, F, H, I, G**. The path returned is $S \rightarrow J \rightarrow E \rightarrow F \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [MCQ, 1M · Propositional Logic] **Answer: A** (A: $(p \rightarrow q) \vee (q \rightarrow p)$)

$(p \rightarrow q) \vee (q \rightarrow p) \equiv \neg p \vee q \vee \neg q \vee p$, which contains $q \vee \neg q$, so it is true in all 4 rows. $(p \rightarrow q) \rightarrow (q \rightarrow p)$ is false at $p=F, q=T$ (antecedent T, consequent F). $(p \wedge \neg q) \rightarrow q$ is false at $p=T, q=F$. $(p \vee q) \rightarrow (p \wedge q)$ is false whenever exactly one of p, q is true. Hence only the first is valid.

Q.6 [MCQ, 1M · Predicate Logic] **Answer: A** (A: y and z)

x is bound by $\forall x$ throughout. The occurrence of y in $P(x, y)$ is not in the scope of any $\exists y/\forall y$, so it is free. The y in $Q(x, y, z)$ is bound by $\exists y$. z is never quantified, so it is free. Hence the variables with free occurrences are y and z .

Q.7 [NAT, 1M · Predicate Logic] **Answer: 64**

View R as an 3×3 0/1 matrix. The sentence says R is reflexive: the n diagonal pairs are forced, the other $n^2 - n$ pairs are free $\Rightarrow 2^{(n^2 - n)}$. With $n = 3$: **64**.

Q.8 [NAT, 1M · Variable Elimination] **Answer: 0.415 to 0.425**

$h(a, c) = f(a, b) g(b, c) + f(a, \neg b) g(\neg b, c) = 0.4 \times 0.3 + 0.6 \times 0.5 = 0.12 + 0.30 = 0.42$. First the pointwise product over $\{A, B, C\}$ is formed, then B is summed out, leaving a factor over $\{A, C\}$.

Q.9 [NAT, 1M · Sampling] Answer: 0.0319 to 0.0323

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(CI=T) \times P(Sp=T|CI=T) \times P(Ra=T|CI=T, Sp=T) \times P(We=F|Sp=T, Ra=T) \\ = 0.95 \times 0.75 \times 0.05 \times 0.9 = \mathbf{0.032063} \approx 0.0321.$$

Q.10 [NAT, 1M · Exact Inference] Answer: 0.873 to 0.877

Only the ancestors of C (A) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T	0.532
A=T, B=F	0.133
A=F, B=T	0.084
A=F, B=F	0.126

Sum = **0.875** ≈ 0.875 .

Q.11 [NAT, 2M · Informed Search] Answer: 4

Heuristic values: $h(A)=10$, $h(B)=6$, $h(C)=4$, $h(G)=0$, $h(I)=12$, $h(J)=10$, $h(S)=8$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (8)	B(8), I(21)
2	B (8)	C(8), A(17), J(19), I(21)
3	C (8)	G(9), A(17), J(19), I(21)
4	G (9)	A(17), J(19), I(21) ← goal, stop

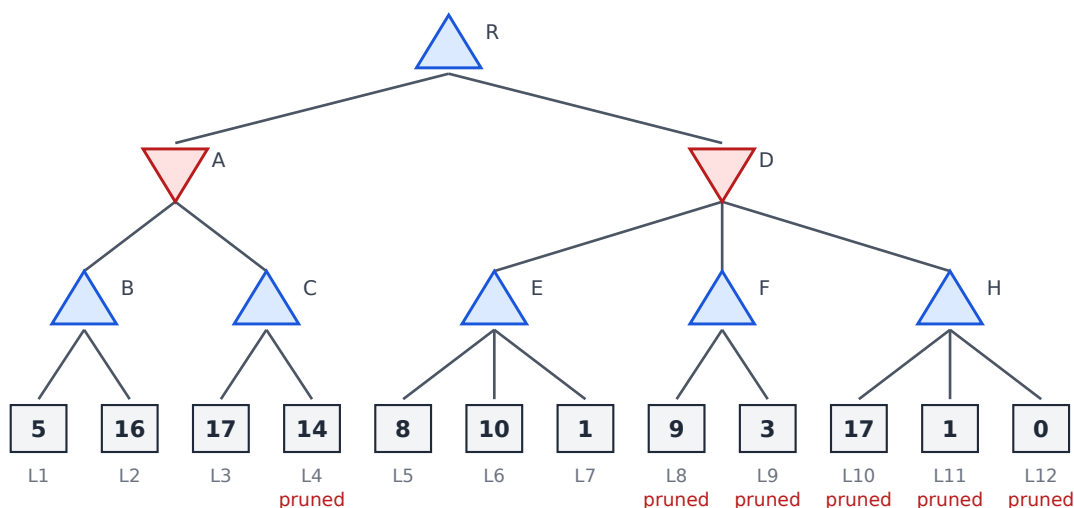
Path returned: **S** → **B** → **C** → **G** with cost **9**; nodes expanded = **4** (S, B, C, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 9$. For comparison UCS ($h = 0$) would expand 5 nodes.

Q.12 [NAT, 2M · Adversarial Search] Answer: 6

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value 17 $\geq \beta = 16 \Rightarrow$ prune remaining children (leaves L4).
- At MIN node D: value 10 $\leq \alpha = 16 \Rightarrow$ prune remaining children (leaves L8, L9, L10, L11, L12).



Leaves evaluated (6): L1, L2, L3, L5, L6, L7. Leaves pruned (6): L4, L8, L9, L10, L11, L12. Root value = 16 (same as plain minimax, 16 — pruning never changes the root value).

Q.13 [NAT, 2M · Adversarial Search] Answer: 3

MIN_1 : evaluate 10, 9, 14, giving 9, so $\alpha = 9$ at the root. MIN_2 : 11, then $8 \leq \alpha$, so prune 20. MIN_3 : $2 \leq \alpha$, so prune 15 and 30. MIN_4 : 12, 13, 7 are all evaluated, because the running minimum stays above 9 until the last leaf. Evaluated leaves: 9 of 12. Pruned: 20, 15, 30 = **3**.

Q.14 [NAT, 2M · Informed Search] **Answer: 6**

Heuristic values: $h(A)=9$, $h(D)=7$, $h(E)=7$, $h(F)=5$, $h(G)=0$, $h(H)=1$, $h(K)=3$, $h(S)=8$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (8)	D(9), E(11), A(16)
2	D (9)	F(9), E(11), A(16)
3	F (9)	K(10), E(11), A(16)
4	K (10)	E(11), G(12), H(14), A(16)
5	E (11)	G(12), H(14), A(16)
6	G (12)	H(14), A(16) ← goal, stop

Path returned: **S** → **D** → **F** → **K** → **G** with cost **12**; nodes expanded = **6** (S, D, F, K, E, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 12$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [MSQ, 2M · Predicate Logic] **Answer: A** (A: $\forall y \exists x R(x, y)$)

$R = \{(2,1), (2,3), (3,2), (4,2), (4,4)\}$.

(A) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(B) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 1 has no edge to a different element.

(C) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(D) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.16 [NAT, 2M · Propositional Logic] **Answer: 9**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(P \vee \neg R) \rightarrow (S \wedge Q)$	$\neg(R \rightarrow Q)$	$((P \vee \neg R) \rightarrow (S \wedge Q)) \leftrightarrow \neg(R \rightarrow Q)$
T	T	T	T	T	F	F
T	T	T	F	F	F	T
T	T	F	T	T	F	F
T	T	F	F	F	F	T
T	F	T	T	F	T	F
T	F	T	F	F	T	F
T	F	F	T	F	F	T
T	F	F	F	F	F	T
F	T	T	T	T	F	F
F	T	T	F	T	F	F
F	T	F	T	T	F	F
F	T	F	F	F	F	T
F	F	T	T	T	T	T
F	F	T	F	T	T	T
F	F	F	T	F	F	T
F	F	F	F	F	F	T

Rows where $F = T$: **9**.

Q.17 [NAT, 2M · Predicate Logic] **Answer: 38**

$|H_1| = 2 + 2^2 = 6$: $a, b, g(a,a), g(a,b), g(b,a), g(b,b)$. $H_2 = H_1 \cup \{g(s, t) : s, t \in H_1\}$. The 36 terms $g(s, t)$ are syntactically distinct, and they already include the 4 g -terms of H_1 (since $H_0 \subseteq H_1$); adding a and b gives $|H_2| = 2 + 6^2 = 38$.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, C ($A: (Q \vee \neg R)$; $C: (P \vee Q)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(Q \vee \neg R)$ is derivable:

1. resolve $(\neg R \vee \neg S)$ and $(Q \vee S)$ on $S \Rightarrow (Q \vee \neg R)$

(B) The resolution closure contains 6 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(P \vee Q)$ is derivable:

1. resolve $(P \vee \neg S)$ and $(Q \vee S)$ on $S \Rightarrow (P \vee Q)$

(D) $(\neg P \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.19 [MSQ, 2M · Propositional Logic] Answer: C, D ($C: (\neg P \rightarrow (R \rightarrow Q)) \leftrightarrow ((\neg P \wedge R) \rightarrow Q)$; $D: ((Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow (Q \rightarrow P)$)

(A) $((P \rightarrow R) \rightarrow Q) \leftrightarrow (P \rightarrow (R \rightarrow Q))$: **not valid** — counter-model $P=F, Q=F, R=T$.

(B) $(Q \vee R) \rightarrow (Q \wedge R)$: **not valid** — counter-model $Q=T, R=F$.

(C) $(\neg P \rightarrow (R \rightarrow Q)) \leftrightarrow ((\neg P \wedge R) \rightarrow Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow (Q \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.20 [NAT, 2M · Sampling] Answer: 0.46 to 0.464

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(B=T)=0.6 \times P(C=T A=F, B=T)=0.7$	0.42	F
2	$P(B=T)=0.6 \times P(C=T A=T, B=T)=0.2$	0.12	T
3	$P(B=T)=0.6 \times P(C=T A=T, B=T)=0.2$	0.12	T
4	$P(B=T)=0.6 \times P(C=T A=T, B=T)=0.2$	0.12	T

$\sum w (\text{all}) = 0.78$; $\sum w (A = T) = 0.36$.

Estimate $P(A=T \mid B = T, C = T) = 0.36 / 0.78 = \mathbf{0.4615}$.

Q.21 [NAT, 2M · Exact Inference] Answer: 0.564 to 0.568

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution} \mid \text{Smoker}) \cdot P(\text{Cancer} \mid \text{Smoker}) \cdot P(\text{XRay} \mid \text{Smoker}, \text{Pollution})$.

Inference by enumeration: $P(\text{Smoker} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Smoker}, e, \text{hidden})$.

Hidden: Po, Ca	Sm	product of CPT entries ($Sm \cdot Po \cdot Ca \cdot XR$)	value
T, T	T	$0.5 \times 0.8 \times 0.5 \times 0.25$	0.05
T, F	T	$0.5 \times 0.8 \times 0.5 \times 0.25$	0.05
F, T	T	$0.5 \times 0.2 \times 0.5 \times 0.5$	0.025
F, F	T	$0.5 \times 0.2 \times 0.5 \times 0.5$	0.025
T, T	F	$0.5 \times 0.6 \times 0.2 \times 0.25$	0.015
T, F	F	$0.5 \times 0.6 \times 0.8 \times 0.25$	0.06
F, T	F	$0.5 \times 0.4 \times 0.2 \times 0.2$	0.008
F, F	F	$0.5 \times 0.4 \times 0.8 \times 0.2$	0.032

$P(\text{Smoker}=T, e) = 0.15$; $P(\text{Smoker}=F, e) = 0.115$; $P(e) = 0.265$.

$P(\text{Smoker}=T \mid e) = 0.15 / 0.265 = \mathbf{0.566} \approx 0.566$.

Q.22 [MSQ, 2M · Bayesian Networks] Answer: A, B, C ($A: P(b) = 0.5$; $B: P(c) = 0.6$; $C: P(a \mid c) = 0.72$)

$P(b) = 0.6 \times 0.7 + 0.4 \times 0.2 = 0.42 + 0.08 = 0.5$ (A true). $P(c) = 0.5 \times 0.9 + 0.5 \times 0.3 = 0.45 + 0.15 = 0.6$ (B true). $P(a, c) = 0.6 \times (0.7 \times 0.9 + 0.3 \times 0.3) = 0.6 \times 0.72 = 0.432$, so $P(a \mid c) = 0.432 / 0.6 = 0.72$ (C true). (D) is false: A and C are connected by the active path $A \rightarrow B \rightarrow C$ (B unobserved), so observing c raises $P(a)$ from 0.6 to 0.72.

Q.23 [MSQ, 2M · Conditional Independence] Answer: A, D ($A: D \perp C \mid A; D: E \perp F \mid B, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $D \perp C \mid A$: **TRUE** — $D \leftarrow A \rightarrow C$ (blocked: observed fork node A); $D \leftarrow A \rightarrow H \leftarrow E \leftarrow C$ (blocked: observed fork node A); $D \rightarrow F \leftarrow B \leftarrow A \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $D \rightarrow F \leftarrow B \leftarrow A \rightarrow H \leftarrow E \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $D \rightarrow G \leftarrow A \rightarrow C$ (blocked: collider G with neither it nor a descendant observed); $D \rightarrow G \leftarrow A \rightarrow H \leftarrow E \leftarrow C$ (blocked: collider G with neither it nor a descendant observed).

(B) $A \perp E \mid D, G$: **FALSE** — the path $A \rightarrow C \rightarrow E$ is active given $\{D, G\}$.

(C) $D \perp G \mid A$: **FALSE** — the path $D \rightarrow G$ is active given $\{A\}$.

(D) $E \perp F \mid B, D$: **TRUE** — $E \leftarrow C \leftarrow A \rightarrow B \rightarrow F$ (blocked: observed chain node B); $E \leftarrow C \leftarrow A \rightarrow D \rightarrow F$ (blocked: observed chain node D); $E \leftarrow C \leftarrow A \rightarrow G \leftarrow D \rightarrow F$ (blocked: collider G with neither it nor a descendant observed); $E \rightarrow H \leftarrow A \rightarrow B \rightarrow F$ (blocked: collider H with neither it nor a descendant observed); $E \rightarrow H \leftarrow A \rightarrow D \rightarrow F$ (blocked: collider H with neither it nor a descendant observed); $E \rightarrow H \leftarrow A \rightarrow G \leftarrow D \rightarrow F$ (blocked: collider H with neither it nor a descendant observed).

Q.24 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A)$, $f(A, C)$, $f(C, D)$, $f(A, D, E)$, $f(C, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(C, D)$, $f(A, D, E)$	$\{A, C, D, E\}$	$f(A, C, E)$
E	$f(A, C, E)$	$\{A, C, E\}$	$f(A, C)$
C	$f(A, C)$, $f(C, F)$, $f(A, C)$	$\{A, C, F\}$	$f(A, F)$
A	$f(A)$, $f(A)$, $f(A, F)$	$\{A, F\}$	$f(F)$

The largest factor produced has **3** variable(s) (created when eliminating D); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over F and normalise.

Q.25 [NAT, 2M · Bayesian Networks] Answer: 0.555 to 0.557

$P(B \mid j, m) = \alpha P(B) \sum_e P(e) \sum_a P(a \mid B, e) P(j \mid a) P(m \mid a)$. Let $g(a) = P(j \mid a)P(m \mid a)$: $g(a) = 0.56$, $g(\neg a) = 0.001$. For b: e: $0.95 \times 0.56 + 0.05 \times 0.001 = 0.53205$; $\neg e$: $0.90 \times 0.56 + 0.10 \times 0.001 = 0.5041$; weighted: $0.01 \times 0.53205 + 0.99 \times 0.5041 = 0.504381$; times $P(b) = 0.02 \rightarrow 0.0100876$. For $\neg b$: e: $0.30 \times 0.56 + 0.70 \times 0.001 = 0.1687$; $\neg e$: $0.01 \times 0.56 + 0.99 \times 0.001 = 0.006589$; weighted: $0.01 \times 0.1687 + 0.99 \times 0.006589 = 0.0082101$; times $0.98 \rightarrow 0.0080459$. $P(b \mid j, m) = 0.0100876 / (0.0100876 + 0.0080459) \approx 0.556$.

Mock Test 050

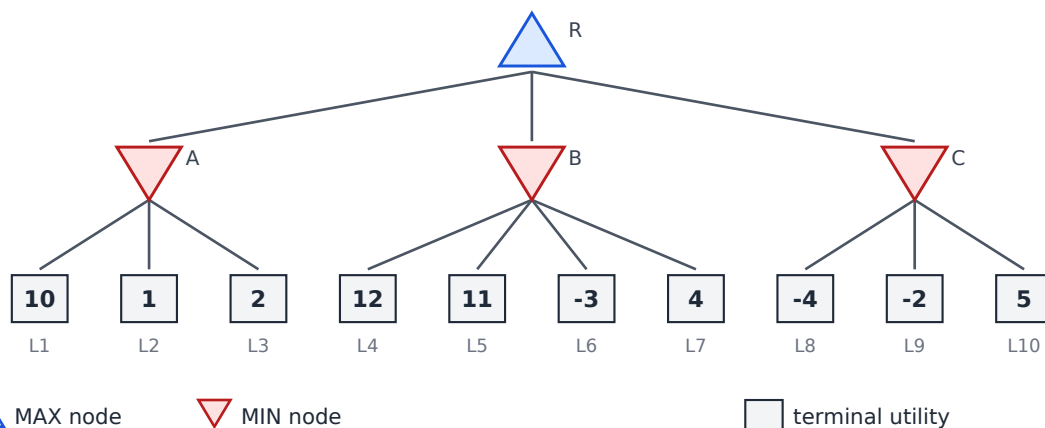
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 7 / 14	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

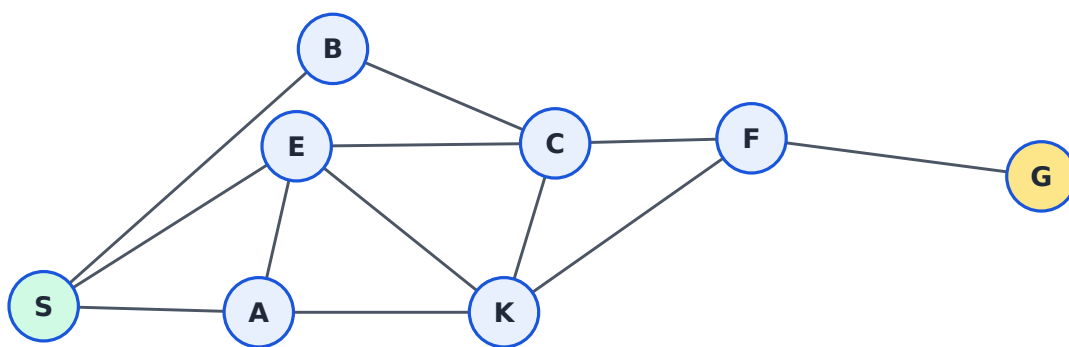
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, A, B, E, K, C, F, G
- ☐ (B) S, E, K, F, G
- ☐ (C) S, A, E, C, B, F, G
- ☐ (D) S, E, A, C, B, F, G

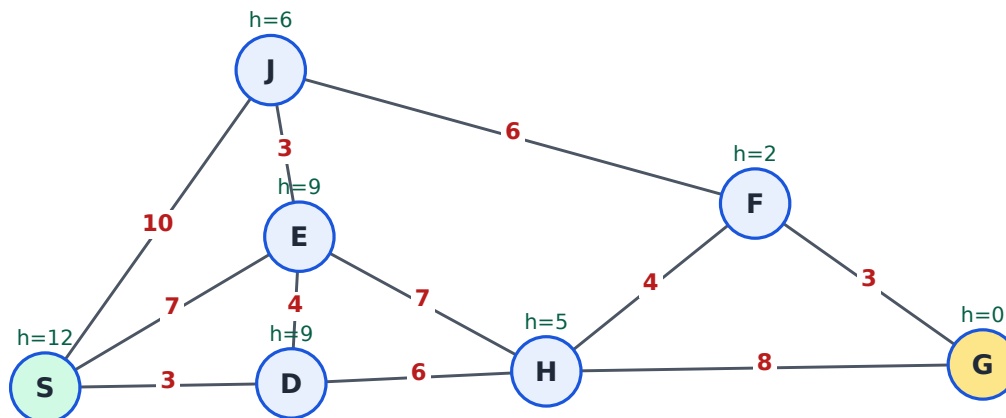
Q.3 [NAT · 1 mark · no negative marking] Local Search

In the 8-queens complete-state formulation (one queen per column), the successors of a state are all states obtained by moving a single queen to a different square in the same column. How many successors does each state have?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 2 elements, so there are $4^2 = 16$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \rightarrow Q(x))$ true?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following clauses are Horn clauses?

- ☐ (A) $\neg p \vee \neg q \vee r$
☐ (B) $p \vee q$
☐ (C) $\neg p \vee \neg q$
☐ (D) r

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (\neg Q \vee (\neg Q \vee \neg Q)) \rightarrow ((R \rightarrow \neg P) \leftrightarrow \neg \neg P) ?$$

Answer (numerical): _____

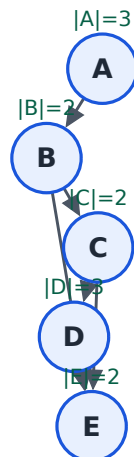
Q.8 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$ and let f be a unary function symbol. How many interpretations of f satisfy $\forall x f(f(x)) = x$?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

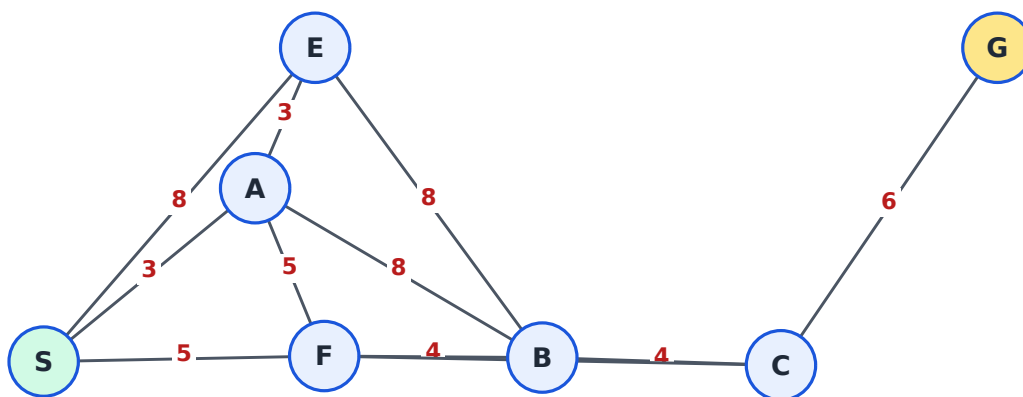
Q.10 [MCQ · 1 mark · negative: -1/3] Variable Elimination

In VE for a query with evidence $J = \text{true}$, the CPT $P(J | A)$ (A and J Boolean) is converted into which factor?

- ☐ (A) A factor over A with 2 entries: $P(j | a)$, $P(j | \neg a)$
- ☐ (B) A factor over $\{A, J\}$ with 4 entries
- ☐ (C) A factor over J with 2 entries
- ☐ (D) A scalar equal to $P(j)$

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search – UCS

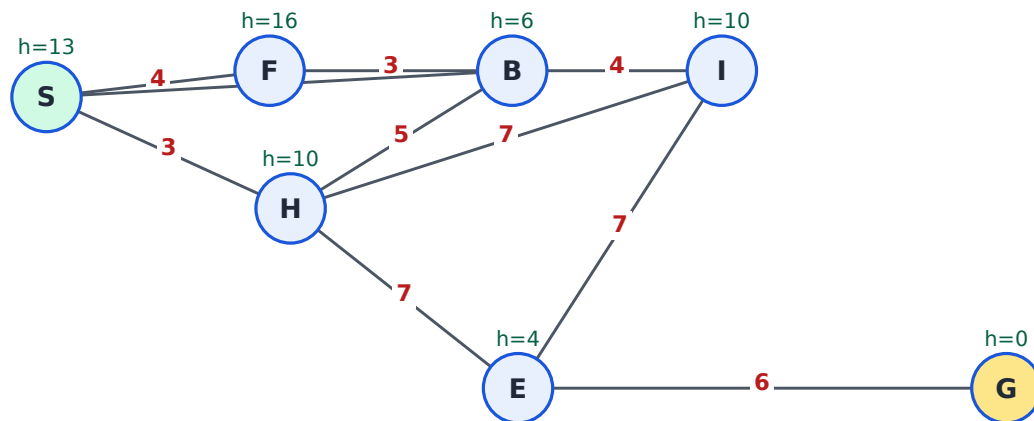
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
☐ (B) $h(B)$ overestimates the true cost from B to G.
☐ (C) h is consistent (monotone).
☐ (D) The edge S-B violates the consistency condition (in at least one direction).

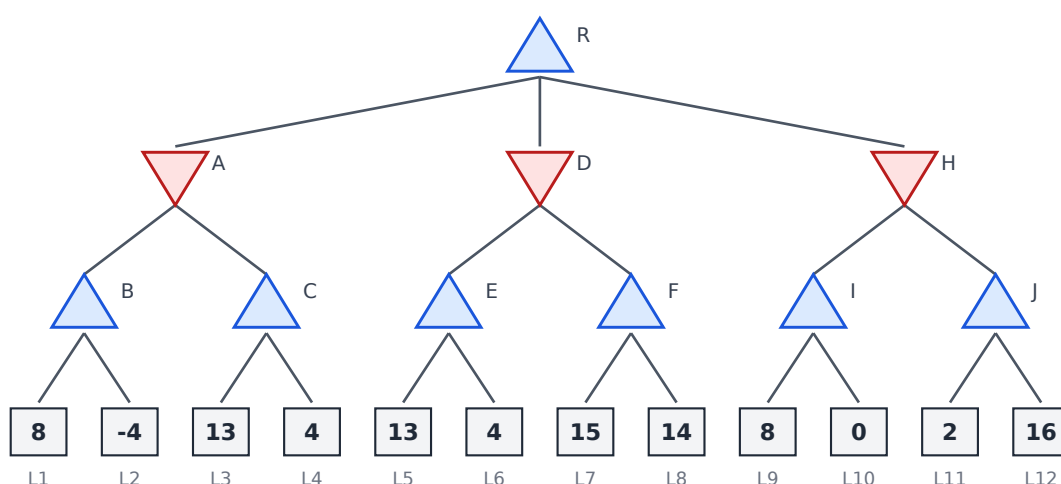
Q.13 [NAT · 2 marks · no negative marking] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Iterative deepening search uses depth limits 0, 1, 2, 3, ... in turn, expands left children before right children, and goal-tests a node when it is visited (popped). The goal is node 11. Over all iterations, counting repeated visits, how many node visits happen up to and including the visit at which node 11 is found?

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic

How many truth assignments to $x_1, \dots, x_5$ satisfy the 2-CNF $(x_1 \vee x_2) \wedge (x_2 \vee x_3) \wedge (x_3 \vee x_4) \wedge (x_4 \vee x_5)$?

Answer (numerical): _____

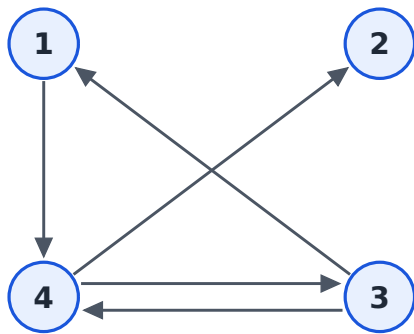
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (R \vee \neg R) \wedge Q$?

- ☐ (A) $\neg(\neg(R \vee \neg R) \wedge \neg Q)$
☐ (B) $Q \wedge (R \vee \neg R)$
☐ (C) $(R \vee \neg R) \vee Q$
☐ (D) $\neg(R \vee \neg R) \wedge Q$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\exists x \forall y \neg R(x, y)$
☐ (C) $\exists x \forall y \neg R(y, x)$
☐ (D) $\forall y \exists x R(x, y)$

Q.18 [NAT · 2 marks · no negative marking] Propositional Logic

How many of the 32 truth assignments to $x_1, \dots, x_5$ satisfy $(x_1 \rightarrow x_2) \wedge (x_2 \rightarrow x_3) \wedge (x_3 \rightarrow x_4) \wedge (x_4 \rightarrow x_5)$?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

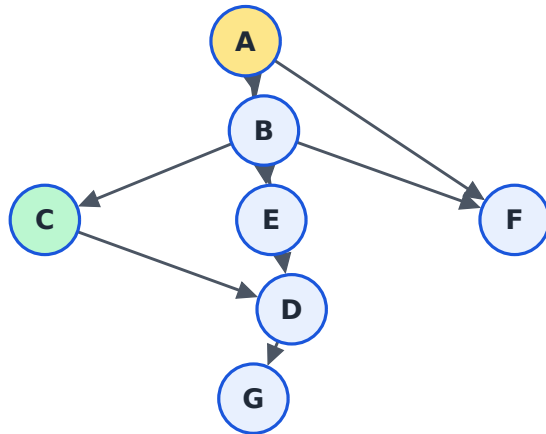
C1: $(\neg P \vee \neg R)$, C2: $(\neg Q)$, C3: $(\neg P \vee R \vee S)$, C4: $(\neg Q \vee R \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P \vee \neg S)$
☐ (B) $(\neg P \vee \neg Q)$
☐ (C) $(P \vee \neg R)$
☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

Q.20 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

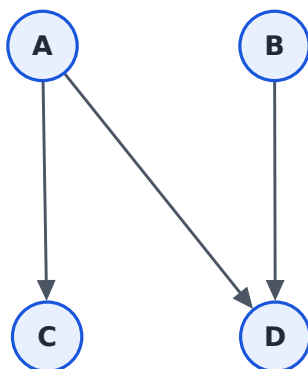
Variable elimination is used to compute $P(\mathbf{C} \mid \mathbf{A} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $A = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, F, B, E, G**. What is the scope of the new factor produced when **F** is summed out?



- ☐ (A) $f(B, F)$
☐ (B) $f(F)$
☐ (C) $f(B)$
☐ (D) $f(B, C)$

Q.21 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.58, 0.93, 0.64, 0.02)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.9

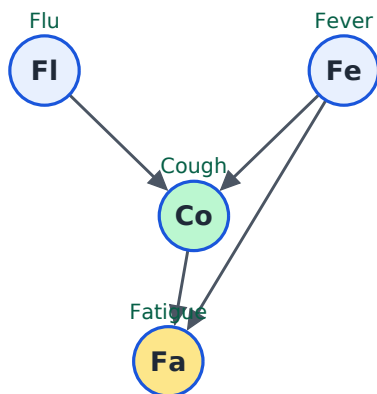
$P(B=T)$	A	$P(C=T \dots)$
0.7	T	0.75
	F	0.75

A	B	$P(D=T \dots)$
T	T	0.8
T	F	0.8
F	T	0.6
F	F	0.2

- ☐ (A) $A=F, B=T, C=F, D=F$
☐ (B) $A=F, B=F, C=T, D=T$
☐ (C) $A=T, B=F, C=T, D=T$
☐ (D) $A=T, B=T, C=T, D=T$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Cough} = T \mid \text{Fatigue} = T)$ (round off to 3 decimal places).



$P(FI=T)$
0.8

$P(Fe=T)$	FI	Fe	$P(Co=T \dots)$
0.25	T	T	0.5
	T	F	0.4
	F	T	0.9
	F	F	0.25

Fe	Co	$P(Fa=T \dots)$
T	T	0.3
T	F	0.1
F	T	0.05
F	F	0.5

Answer (numerical): _____

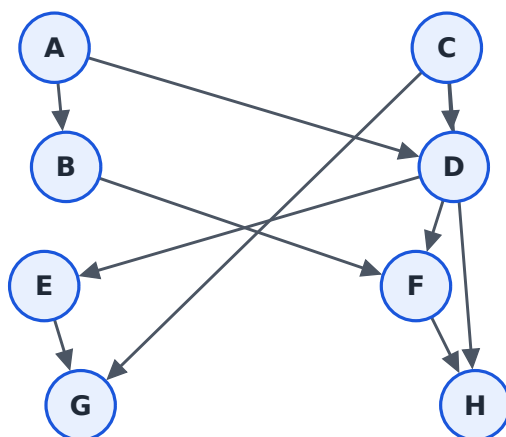
Q.23 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network (DAG) with edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$, $E \rightarrow D$, $D \rightarrow F$, $C \rightarrow G$, $H \rightarrow G$. Which of the following nodes belong to the Markov blanket of C?

- ☐ (A) B
☐ (B) F
☐ (C) H
☐ (D) A

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

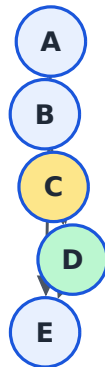
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $F \perp D$
☐ (B) $H \perp B \mid C$
☐ (C) $C \perp B$
☐ (D) $B \perp H \mid E$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{C = T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(D = T \mid C = T)}$ (round off to 3 decimal places).



P(A=T)
0.95

A	P(B=T ...)	B	P(C=T ...)
T	0.4	T	0.75
F	0.1	F	0.75

B	C	P(D=T ...)	A	D	P(E=T ...)
T	T	0.5	T	T	0.7
T	F	0.05	T	F	0.9
F	T	0.8	F	T	0.1
F	F	0.3	F	F	0.25

#	A	B	D	E
1	T	F	F	T
2	T	T	F	F
3	T	T	F	T
4	T	F	T	F

Answer (numerical): _____

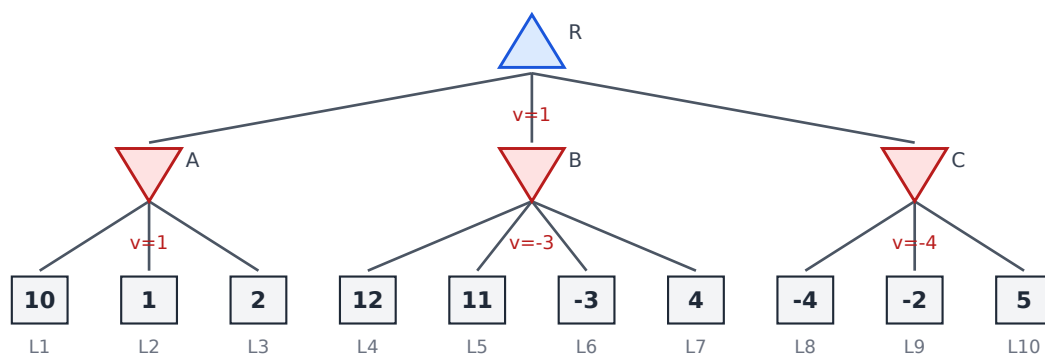
Answer Key — Mock Test 050

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	1	14	NAT	2	Adversarial Search	4
2	MCQ	1	Uninformed Search	C	15	NAT	2	Propositional Logic	13
3	NAT	1	Local Search	56	16	MSQ	2	Propositional Logic	B
4	NAT	1	Informed Search	19	17	MSQ	2	Predicate Logic	B, D
5	NAT	1	Predicate Logic	9	18	NAT	2	Propositional Logic	6
6	MSQ	1	Propositional Logic	A, C, D	19	MSQ	2	Propositional Logic	B
7	NAT	1	Propositional Logic	5	20	MCQ	2	Exact Inference	C
8	NAT	1	Predicate Logic	4	21	MCQ	2	Sampling	C
9	NAT	1	Bayesian Networks	23	22	NAT	2	Exact Inference	0.187 to 0.191
10	MCQ	1	Variable Elimination	A	23	MSQ	2	Bayesian Networks	A, C, D
11	NAT	2	Uninformed Search	7	24	MSQ	2	Conditional Independence	C
12	MSQ	2	Informed Search	A	25	NAT	2	Sampling	0.248 to 0.252
13	NAT	2	Uninformed Search	19					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 1

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 1$, $B = -3$, $C = -4$. Root value = **1**; MAX's optimal first move is towards A.

Q.2 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, A, E, C, B, F, G)

Adjacency lists (alphabetical): A: {E, K, S}; B: {C, S}; C: {B, E, F, K}; E: {A, C, K, S}; F: {C, G, K}; G: {F}; K: {A, C, E, F}; S: {A, B, E}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, A, E, C, B, F, G**; the path found is $S \rightarrow A \rightarrow E \rightarrow C \rightarrow F \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.3 [NAT, 1M · Local Search] **Answer: 56**

Each of the 8 queens can move to any of the 7 other squares in its column, giving $8 \times 7 = 56$ successors. Steepest-ascent hill climbing evaluates all 56 and picks the best.

Q.4 [NAT, 1M · Informed Search] **Answer: 19**

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (12)	J(6), D(9), E(9)
2	J (6)	F(2), D(9), E(9)
3	F (2)	G(0), H(5), D(9), E(9)
4	G (0)	H(5), D(9), E(9) ← goal, stop

Path returned: **S** → **J** → **F** → **G**, cost = $10 + 6 + 3 = 19$.

The optimal cost is $h^*(S) = 16$ — greedy search is not optimal here.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 9**

Interpret P and Q as subsets of the domain; each element independently has 3 allowed (P,Q) combinations (FF, FT, TT) $\Rightarrow 3^n$. With $n = 2$ this gives **9** (verified by enumerating all interpretations).

Q.6 [MSQ, 1M · Propositional Logic] **Answer: A, C, D** (A: $\neg p \vee \neg q \vee r$; C: $\neg p \vee \neg q$; D: r)

A Horn clause is a disjunction of literals with at most one positive literal. $\neg p \vee \neg q \vee r$ has one positive literal (a definite clause, i.e. $p \wedge q \rightarrow r$). $\neg p \vee \neg q$ has zero positive literals (a goal clause). r is a single positive literal (a fact). $p \vee q$ has two positive literals, so it is not Horn.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 5**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg Q \vee (\neg Q \vee \neg Q)$	$(R \rightarrow \neg P) \leftrightarrow \neg \neg P$	$(\neg Q \vee (\neg Q \vee \neg Q)) \rightarrow ((R \rightarrow \neg P) \leftrightarrow \neg \neg P)$
T	T	T	F	F	T
T	T	F	F	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	F	F	T
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	T	F	F

Rows where F = T: **5**.

Q.8 [NAT, 1M · Predicate Logic] Answer: 4

$f(f(x)) = x$ for all x means f is an involution: it is a bijection whose cycles have length 1 or 2. On 3 elements: the identity (1 way) or exactly one transposition with one fixed point (3 ways). Total = 4.

Q.9 [NAT, 1M · Bayesian Networks] Answer: 23

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (2-1) \times 3 + C: (2-1) \times 2 + D: (3-1) \times 2 + E: (2-1) \times 2 \times 2 \times 3 = 23$.

Full joint table: $\prod |X| - 1 = 72 - 1 = 71$; saving = 48.

Q.10 [MCQ, 1M · Variable Elimination] Answer: A (A: A factor over A with 2 entries: $P(j | a)$, $P(j | \neg a)$)

Evidence variables are instantiated, so only the column $J = \text{true}$ is kept, giving $f(A) = \langle P(j | a), P(j | \neg a) \rangle$. J no longer appears as a dimension. $P(j)$ would require summing over A weighted by $P(A)$, which VE does only later and only if needed.

Q.11 [NAT, 2M · Uninformed Search] Answer: 7

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	A(3), F(5), E(8)
2	A (3)	F(5), E(6), B(11)
3	F (5)	E(6), B(9), C(11)
4	E (6)	B(9), C(11)
5	B (9)	C(11)
6	C (11)	G(17)
7	G (17)	← goal, stop

Path returned: **S** → **F** → **C** → **G**, cost = 5 + 6 + 6 = **17**. Nodes expanded (including S and G): **7** (S, A, F, E, B, C, G).

Q.12 [MSQ, 2M · Informed Search] Answer: A (A: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	E	F	G	H	I	S
$h(n)$	6	4	16	0	10	10	13
$h^*(n)$	17	6	20	0	13	13	16
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $F \rightarrow B$ ($16 > 3 + 6$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

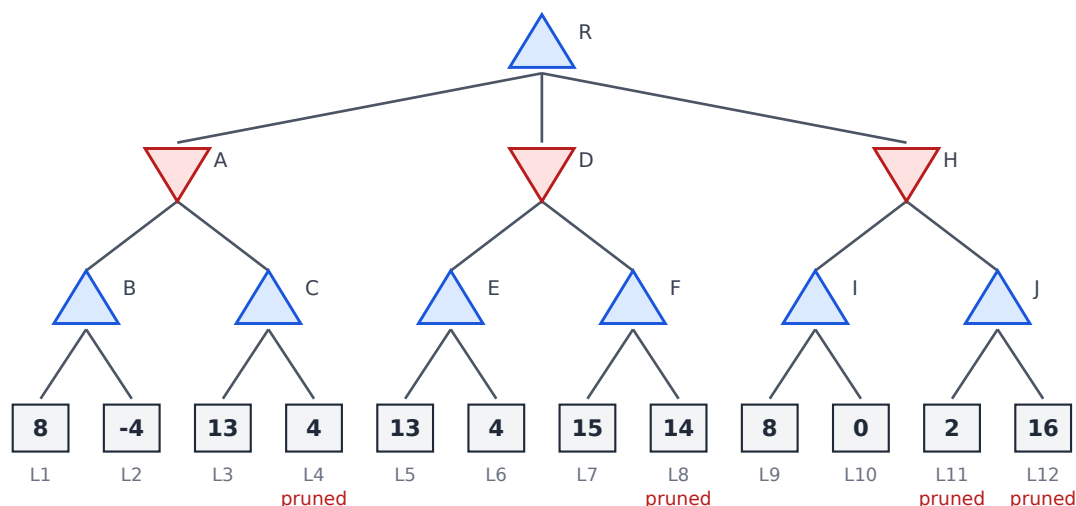
Q.13 [NAT, 2M · Uninformed Search] Answer: 19

Node 11 lies at depth 3 (path $1 \rightarrow 2 \rightarrow 5 \rightarrow 11$). Limit 0 visits $\{1\}$: 1 visit. Limit 1 visits $\{1,2,3\}$: 3 visits. Limit 2 visits the whole depth-2 tree $\{1,2,4,5,3,6,7\}$: 7 visits. Limit 3 visits 1,2,4,8,9,5,10,11 in preorder and stops at 11: 8 visits. Total = 1 + 3 + 7 + 8 = **19**.

Q.14 [NAT, 2M · Adversarial Search] Answer: 4

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $13 \geq \beta = 8 \Rightarrow$ prune remaining children (leaves L4).
- At MAX node F: value $15 \geq \beta = 13 \Rightarrow$ prune remaining children (leaves L8).
- At MIN node H: value $8 \leq \alpha = 13 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (8): L1, L2, L3, L5, L6, L7, L9, L10. Leaves pruned (4): L4, L8, L11, L12. Root value = 13 (same as plain minimax, 13 — pruning never changes the root value).

Q.15 [NAT, 2M · Propositional Logic] Answer: 13

The formula says no two consecutive variables are both F. Let a_n be the number of such strings of length n . A valid string either ends in T (preceded by any valid string of length $n-1$) or ends in F, in which case the previous symbol must be T (preceded by any valid string of length $n-2$). So $a_n = a_{n-1} + a_{n-2}$ with $a_1 = 2$, $a_2 = 3$. Then $a_3 = 5$, $a_4 = 8$, $a_5 = 13$. Enumeration of all 32 assignments confirms 13.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B (B: $Q \wedge (R \vee \neg R)$)

Compare truth tables with $F = (R \vee \neg R) \wedge Q$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg(R \vee \neg R) \wedge \neg Q)$ — differs at $Q=F, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(B) $Q \wedge (R \vee \neg R)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $(R \vee \neg R) \vee Q$ — differs at $Q=F, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(D) $\neg(R \vee \neg R) \wedge Q$ — differs at $Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

Q.17 [MSQ, 2M · Predicate Logic] Answer: B, D (B: $\exists x \forall y \neg R(x, y)$; D: $\forall y \exists x R(x, y)$)

$R = \{(1,4), (3,1), (3,4), (4,2), (4,3)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(3,4) \wedge R(4,2)$ but $\neg R(3,2)$.

(B) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.

(C) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.

(D) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [NAT, 2M · Propositional Logic] Answer: 6

The constraints say once some x_i is T, every later x_j ($j > i$) must be T. So a model is a string of F's followed by a string of T's, e.g. FTTT. The position where T's begin can be 1, 2, 3, 4, 5 or 'never' (all F), giving 6 models. Brute-force over all 32 assignments confirms 6.

Q.19 [MSQ, 2M · Propositional Logic] Answer: B (B: $(\neg P \vee \neg Q)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(\neg P \vee \neg Q)$ is derivable:

1. resolve $(\neg P \vee R \vee S)$ and $(\neg Q \vee R \vee \neg S)$ on $S \Rightarrow (\neg P \vee \neg Q \vee R)$

2. resolve $(\neg P \vee \neg R)$ and $(\neg P \vee \neg Q \vee R)$ on $R \Rightarrow (\neg P \vee \neg Q)$

(C) $(P \vee \neg R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) The resolution closure contains 8 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

Q.20 [MCQ, 2M · Exact Inference] Answer: C (C: $f(B)$)

Initial factors (one per CPT, evidence removed from scopes): $f()$, $f(B)$, $f(B, C)$, $f(C, D)$, $f(B, E)$, $f(B, F)$, $f(D, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(C, D), f(D, G)$	$\{C, D, G\}$	$f(C, G)$
F	$f(B, F)$	$\{B, F\}$	$f(B)$
B	$f(B), f(B, C), f(B, E), f(B)$	$\{B, C, E\}$	$f(C, E)$
E	$f(C, E)$	$\{C, E\}$	$f(C)$
G	$f(C, G)$	$\{C, G\}$	$f(C)$

The largest factor produced has **2** variable(s) (created when eliminating D); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over C and normalise.

Q.21 [MCQ, 2M · Sampling] **Answer: C** (C: A=T, B=F, C=T, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.9$; $u = 0.58 < 0.9 \Rightarrow A = T$
- B: $P(B=T \mid \text{—}) = 0.7$; $u = 0.93 \geq 0.7 \Rightarrow B = F$
- C: $P(C=T \mid A=T) = 0.75$; $u = 0.64 < 0.75 \Rightarrow C = T$
- D: $P(D=T \mid A=T, B=F) = 0.8$; $u = 0.02 < 0.8 \Rightarrow D = T$

Sample: **(A=T, B=F, C=T, D=T)**.

Q.22 [NAT, 2M · Exact Inference] **Answer: 0.187 to 0.191**

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever}) \cdot P(\text{Cough} \mid \text{Flu}, \text{Fever}) \cdot P(\text{Fatigue} \mid \text{Fever}, \text{Cough})$.

Inference by enumeration: $P(\text{Cough} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cough}, e, \text{hidden})$.

Hidden: Fl, Fe	Co	product of CPT entries (Fl · Fe · Co · Fa)	value
T, T	T	$0.8 \times 0.25 \times 0.5 \times 0.3$	0.03
T, F	T	$0.8 \times 0.75 \times 0.4 \times 0.05$	0.012
F, T	T	$0.2 \times 0.25 \times 0.9 \times 0.3$	0.0135
F, F	T	$0.2 \times 0.75 \times 0.25 \times 0.05$	0.00187
T, T	F	$0.8 \times 0.25 \times 0.5 \times 0.1$	0.01
T, F	F	$0.8 \times 0.75 \times 0.6 \times 0.5$	0.18
F, T	F	$0.2 \times 0.25 \times 0.1 \times 0.1$	0.0005
F, F	F	$0.2 \times 0.75 \times 0.75 \times 0.5$	0.05625

$P(\text{Cough}=T, e) = 0.05738$; $P(\text{Cough}=F, e) = 0.24675$; $P(e) = 0.30413$.

$P(\text{Cough}=T \mid e) = 0.05738 / 0.30413 = \mathbf{0.1887} \approx 0.189$.

Q.23 [MSQ, 2M · Bayesian Networks] **Answer: A, C, D** (A: B; C: H; D: A)

Parents of C: {A}. Children of C: {D, G}. Other parents of the children: D has parents B, C, E and G has parents C, H, giving {B, E, H}. Markov blanket of C = {A, B, D, E, G, H}. So A (parent), B (co-parent via D) and H (co-parent via G) belong to it. F is a child of D, i.e. a grandchild of C, and is not in the blanket: given D, F is independent of C.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: C** (C: $C \perp B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $F \perp D$: **FALSE** — the path $F \leftarrow B \leftarrow A \rightarrow D$ is active given $\{\}$.

(B) $H \perp B \mid C$: **FALSE** — the path $H \leftarrow F \leftarrow B$ is active given $\{C\}$.

(C) $C \perp B$: **TRUE** — $C \rightarrow D \leftarrow A \rightarrow B$ (blocked: collider D with neither it nor a descendant observed); $C \rightarrow D \rightarrow F \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow G \leftarrow E \leftarrow D \leftarrow A \rightarrow B$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow G \leftarrow E \leftarrow D \rightarrow F \leftarrow B$ (blocked: collider G with neither it nor a descendant observed); $C \rightarrow H \leftarrow F \leftarrow B$ (blocked: collider H with neither it nor a descendant observed); $C \rightarrow H \leftarrow F \leftarrow D \leftarrow A \rightarrow B$ (blocked: collider H with neither it nor a descendant observed).

(D) $B \perp H \mid E$: **FALSE** — the path $B \leftarrow A \rightarrow D \leftarrow C \rightarrow H$ is active given $\{E\}$.

Q.25 [NAT, 2M · Sampling] **Answer: 0.248 to 0.252**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	D
1	$P(C=T \mid B=F) = 0.75$	0.75	F
2	$P(C=T \mid B=T) = 0.75$	0.75	F

#	weight $w = \prod P(e \mid \text{parents})$	w	D
3	$P(C=T B=T)=0.75$	0.75	F
4	$P(C=T B=F)=0.75$	0.75	T

$\sum w \text{ (all)} = 3$; $\sum w \text{ (D = T)} = 0.75$.

Estimate $P(D=T \mid C = T) = 0.75 / 3 = \mathbf{0.25}$.

Mock Test 051

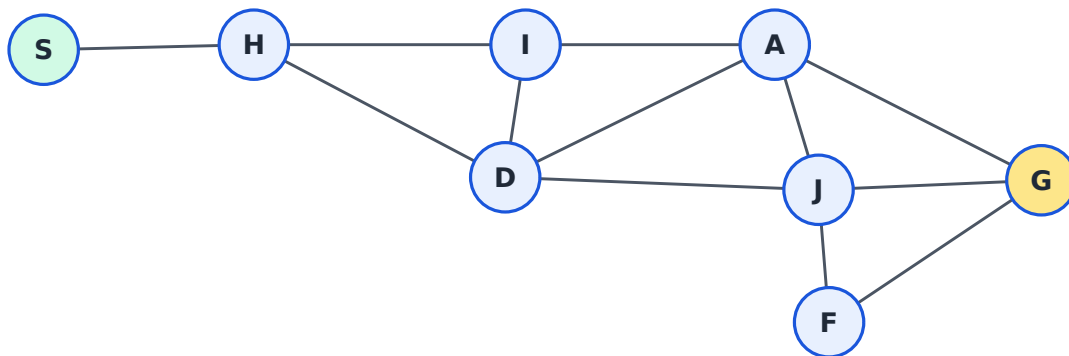
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 7 / 13	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, H, D, I, A, J, G
- ☐ (B) S, D, H, A, G
- ☐ (C) S, H, I, D, J, G
- ☐ (D) S, H, D, A, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 4$ (so $b^d = 16$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

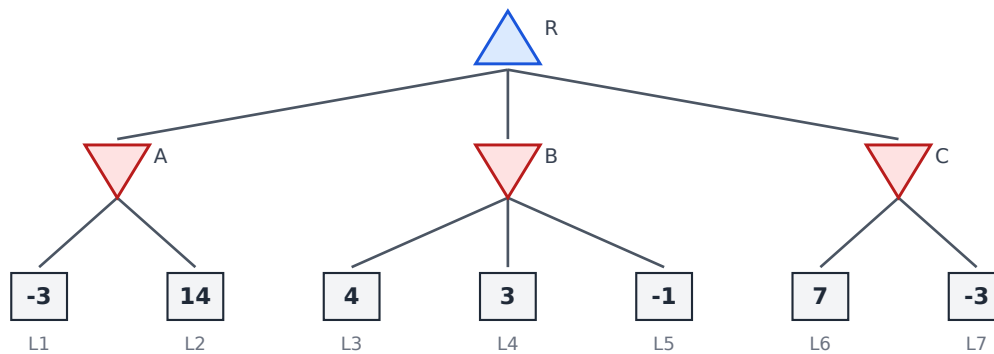
Q.3 [MCQ · 1 mark · negative: $-1/3$] Informed Search

A* search with $h(n) = 0$ for every node n is equivalent to

- ☐ (A) breadth-first search
- ☐ (B) depth-first search
- ☐ (C) uniform-cost search
- ☐ (D) greedy best-first search

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to B; root value 0
- ☐ (B) Move to A; root value -3
- ☐ (C) Move to B; root value -1
- ☐ (D) Move to C; root value -3

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((\neg Q \leftrightarrow P) \rightarrow \neg\neg P) \rightarrow ((\neg Q \vee R) \rightarrow (R \wedge \neg Q))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \rightarrow Q(x))$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over the symbols p, q, r does $(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)$ have?

Answer (numerical): _____

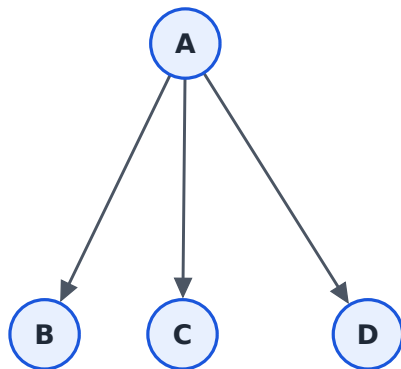
Q.8 [NAT · 1 mark · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. Compute the joint probability $P(j, m, a, \neg b, \neg e)$. (Round to 4 decimal places.)

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=T, B=F, C=T, D=F)**? (round off to 4 decimal places)



P(A=T)
0.8

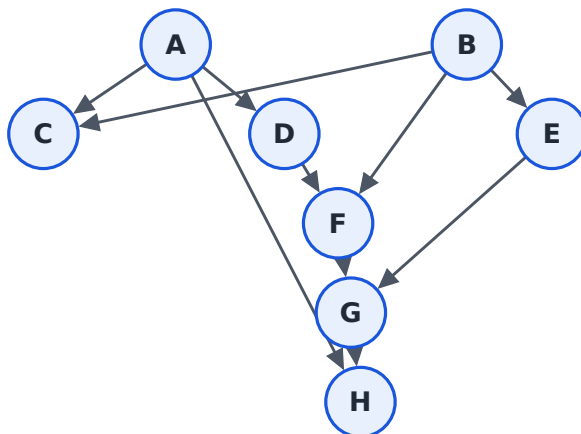
A	P(B=T ...)	A	P(C=T ...)
T	0.8	T	0.95
F	0.25	F	0.25

A	P(D=T ...)
T	0.2
F	0.1

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **F** in the Bayesian network below?

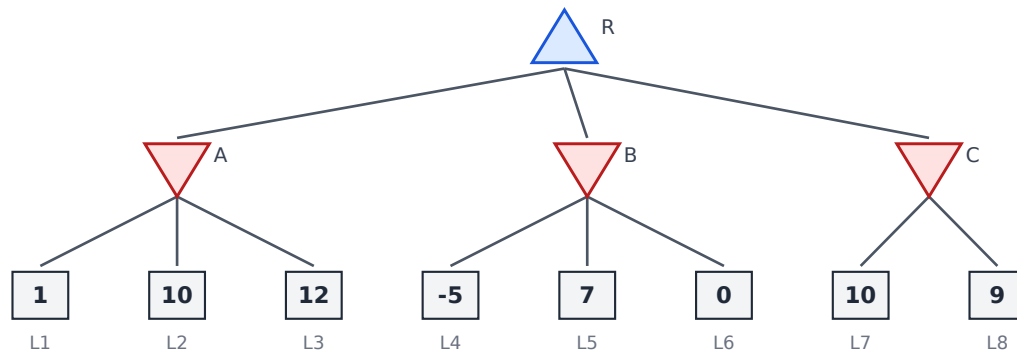


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



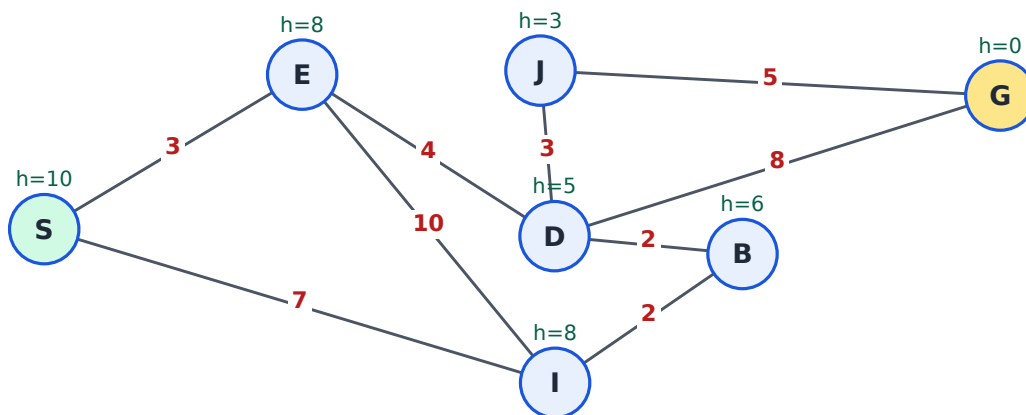
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L5 is pruned.
☐ (B) Leaf L6 is pruned.
☐ (C) Leaf L7 is pruned.
☐ (D) Leaf L8 is pruned.

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

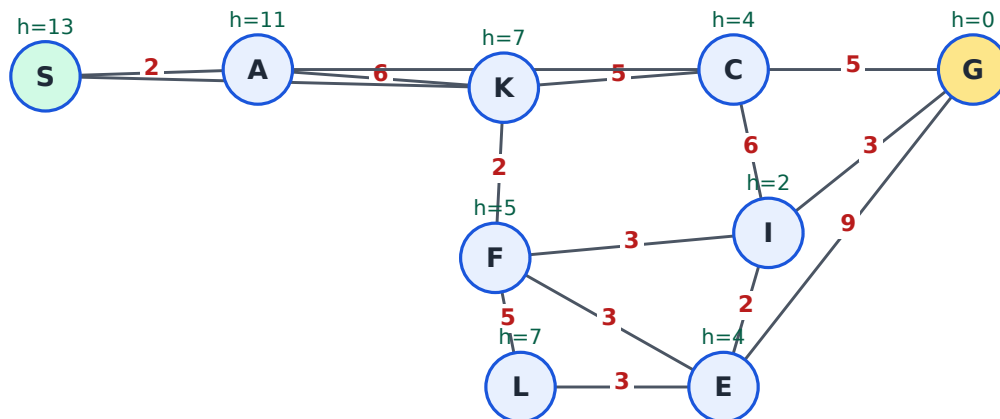
Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Consider the directed graph with start S and goal G and edges (with costs): S→A (1), S→B (4), A→B (2), A→C (5), B→C (1), B→G (6), C→G (3). Uniform-cost search (graph search, goal test when a node is selected for expansion) is run from S. Which path does it return?

- ☐ (A) S→B→G (cost 10)
☐ (B) S→A→C→G (cost 9)
☐ (C) S→B→C→G (cost 8)
☐ (D) S→A→B→C→G (cost 7)

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node L is expanded.
- ☐ (B) Node S is expanded first.
- ☐ (C) The path returned has cost 14.
- ☐ (D) Node K is expanded.

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

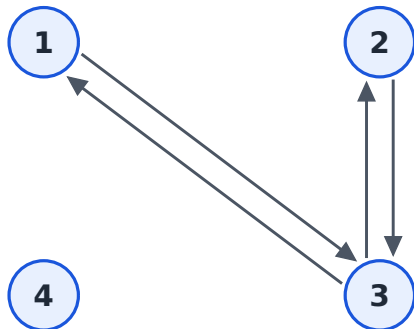
- (1) $(Q \leftrightarrow P) \wedge R$
- (2) $\neg R \rightarrow (P \leftrightarrow \neg S)$
- (3) $(\neg P \wedge \neg S) \rightarrow S$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $Q \rightarrow Q$
- ☐ (B) $Q \leftrightarrow Q$
- ☐ (C) $\neg \neg S$
- ☐ (D) $\neg Q \rightarrow Q$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\forall x R(x, x)$
☐ (D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $\mathbf{F} = \mathbf{Q} \rightarrow (\mathbf{Q} \wedge \neg \mathbf{R})$?

- ☐ (A) $\neg Q \wedge (Q \wedge \neg R)$
☐ (B) $\neg Q \vee (Q \wedge \neg R)$
☐ (C) $\neg(Q \wedge \neg R) \rightarrow \neg Q$
☐ (D) $\neg(Q \wedge \neg(Q \wedge \neg R))$

Q.18 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \exists y R(x, y) \wedge \exists y \forall x R(x, y)$?

Answer (numerical): _____

Q.19 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following formulas is in disjunctive normal form and is equivalent to $(p \vee q) \wedge \neg(p \wedge q \wedge r)$?

- ☐ (A) $(p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg r)$
☐ (B) $(p \wedge \neg q) \vee (\neg p \wedge q)$
☐ (C) $(p \wedge \neg r) \vee (q \wedge \neg r)$
☐ (D) $(p \vee q) \wedge (\neg p \vee \neg q \vee \neg r)$

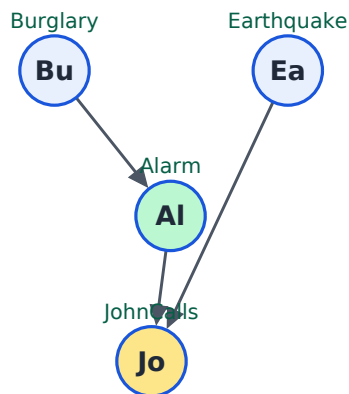
Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic

Here x and y are variables, and a, b are constants. Which of the following pairs of atoms are unifiable (with the occurs check)?

- ☐ (A) $P(x, f(x))$ and $P(f(y), y)$
☐ (B) $Q(a, x, f(x))$ and $Q(y, b, f(b))$
☐ (C) $R(x, g(x))$ and $R(g(y), g(g(a)))$
☐ (D) $S(f(a), x)$ and $S(x, f(b))$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute $P(\text{Alarm} = T \mid \text{JohnCalls} = T)$ (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.95

$P(\text{Ea}=T)$	Bu	$P(\text{Al}=T \dots)$
0.8	T	0.1
	F	0.2

Ea	Al	$P(\text{Jo}=T \dots)$
T	T	0.5
T	F	0.3
F	T	0.25
F	F	0.25

Answer (numerical): _____

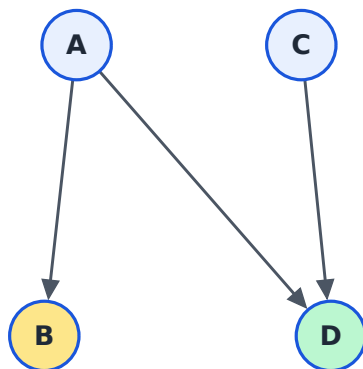
Q.22 [NAT · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $C \rightarrow A$ and $C \rightarrow B$ (Boolean variables) with $P(c) = 0.4$, $P(a \mid c) = 0.9$, $P(a \mid \neg c) = 0.2$, $P(b \mid c) = 0.7$, $P(b \mid \neg c) = 0.1$. Compute $P(a, b)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{B=T}$. The current state of the chain is $(\mathbf{A=T}, \mathbf{B=T}, \mathbf{C=F}, \mathbf{D=F})$. The next step resamples $\mathbf{D}$. With what probability is D set to T? (round off to 3 decimal places)



$P(A=T)$
0.4

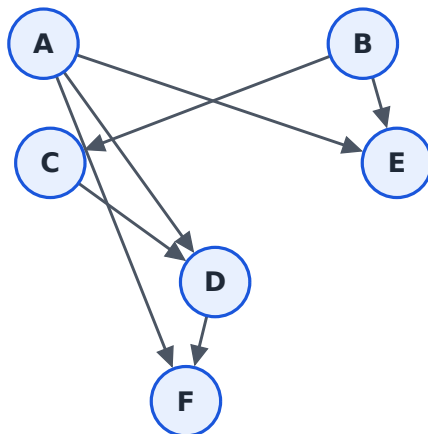
A	$P(B=T \dots)$	$P(C=T)$
T	0.75	0.5
F	0.1	

A	C	$P(D=T \dots)$
T	T	0.8
T	F	0.6
F	T	0.25
F	F	0.8

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

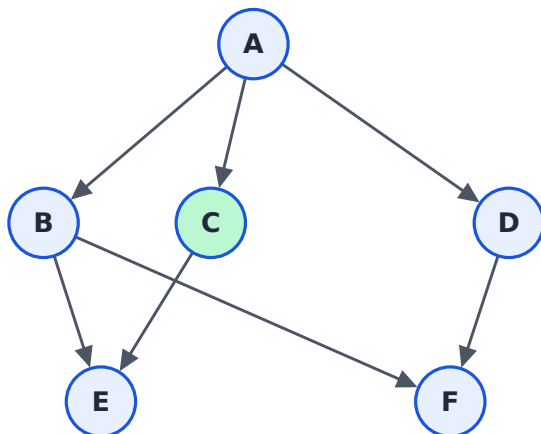
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp A$
☐ (B) $E \perp D$
☐ (C) $B \perp E \mid C, F$
☐ (D) $F \perp C \mid A, D$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(C)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E, F, A, D, B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 051

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	MSQ	2	Informed Search	B
2	NAT	1	Adversarial Search	7	15	MSQ	2	Propositional Logic	A, B
3	MCQ	1	Informed Search	C	16	MSQ	2	Predicate Logic	A, D
4	MCQ	1	Adversarial Search	C	17	MSQ	2	Propositional Logic	B, C, D
5	NAT	1	Propositional Logic	3	18	NAT	2	Predicate Logic	169
6	NAT	1	Predicate Logic	27	19	MCQ	2	Propositional Logic	A
7	NAT	1	Propositional Logic	2	20	MSQ	2	Predicate Logic	B, C
8	NAT	1	Bayesian Networks	0.0054 to 0.0055	21	NAT	2	Exact Inference	0.152 to 0.156
9	NAT	1	Sampling	0.1214 to 0.1218	22	NAT	2	Conditional Independence	0.263 to 0.265
10	NAT	1	Conditional Independence	4	23	NAT	2	Sampling	0.598 to 0.602
11	MSQ	2	Adversarial Search	A, B	24	MSQ	2	Conditional Independence	A, D
12	NAT	2	Informed Search	6	25	NAT	2	Exact Inference	3
13	MCQ	2	Uninformed Search	D					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: D (D: S, H, D, A, G)

Adjacency lists (alphabetical): A: {D, G, I, J}; D: {A, H, I, J}; F: {G, J}; G: {A, F, J}; H: {D, I, S}; I: {A, D, H}; J: {A, D, F, G}; S: {H}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, H, D, A, G**; the path found is $S \rightarrow H \rightarrow D \rightarrow A \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] Answer: 7

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

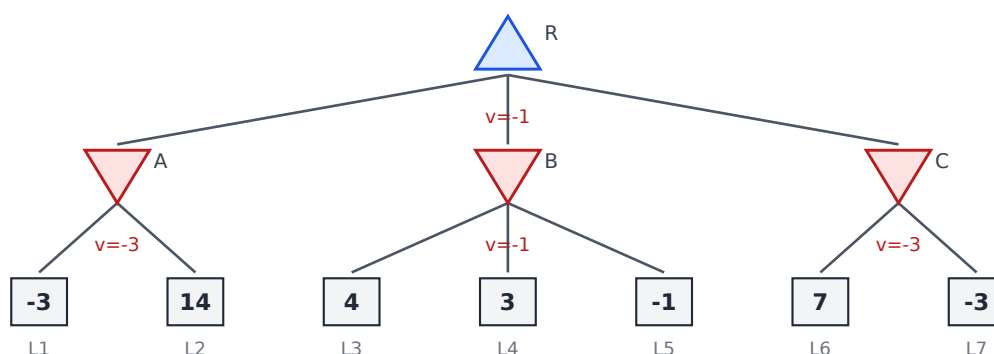
$= 2^2 + 2^2 - 1 = 4 + 4 - 1 = 7$, versus 16 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [MCQ, 1M · Informed Search] Answer: C (C: uniform-cost search)

With $h = 0$, $f(n) = g(n)$, so A* expands nodes in order of path cost, which is **uniform-cost search**. It equals BFS (A) only when all step costs are equal. DFS (D) uses a LIFO order. Greedy search with $h \equiv 0$ has every node tied, so its behaviour depends entirely on tie-breaking.

Q.4 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to B; root value -1)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -3$, $B = -1$, $C = -3$. Root value = **-1**; MAX's optimal first move is towards B.

Q.5 [NAT, 1M · Propositional Logic] Answer: 3

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg Q \leftrightarrow P) \rightarrow \neg\neg P$	$(\neg Q \vee R) \rightarrow (R \wedge \neg Q)$	$((\neg Q \leftrightarrow P) \rightarrow \neg\neg P) \rightarrow ((\neg Q \vee R) \rightarrow (R \wedge \neg Q))$
T	T	T	T	F	F
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	T
F	T	F	F	T	T
F	F	T	T	T	T
F	F	F	T	F	F

F has 5 models, so it is false in $8 - 5 = 3$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 27**

Interpret P and Q as subsets of the domain; each element independently has 3 allowed (P,Q) combinations (FF, FT, TT) $\Rightarrow 3^n$. With $n = 3$ this gives **27** (verified by enumerating all interpretations).

Q.7 [NAT, 1M · Propositional Logic] **Answer: 2**

The cycle of implications $p \rightarrow q \rightarrow r \rightarrow p$ forces p, q, r to have the same truth value: if any one is T, all following ones are T, and the cycle returns. So the only models are all-T and all-F, i.e. 2 models.

Q.8 [NAT, 1M · Bayesian Networks] **Answer: 0.0054 to 0.0055**

By the BN factorization, $P(j, m, a, \neg b, \neg e) = P(j | a) P(m | a) P(a | \neg b, \neg e) P(\neg b) P(\neg e) = 0.8 \times 0.7 \times 0.01 \times 0.98 \times 0.99 = 0.56 \times 0.01 \times 0.9702 = 0.005433 \approx 0.0054$.

Q.9 [NAT, 1M · Sampling] **Answer: 0.1214 to 0.1218**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(A=T) \times P(B=F|A=T) \times P(C=T|A=T) \times P(D=F|A=T)$
 $= 0.8 \times 0.2 \times 0.95 \times 0.8 = \mathbf{0.1216} \approx 0.1216$.

Q.10 [NAT, 1M · Conditional Independence] **Answer: 4**

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

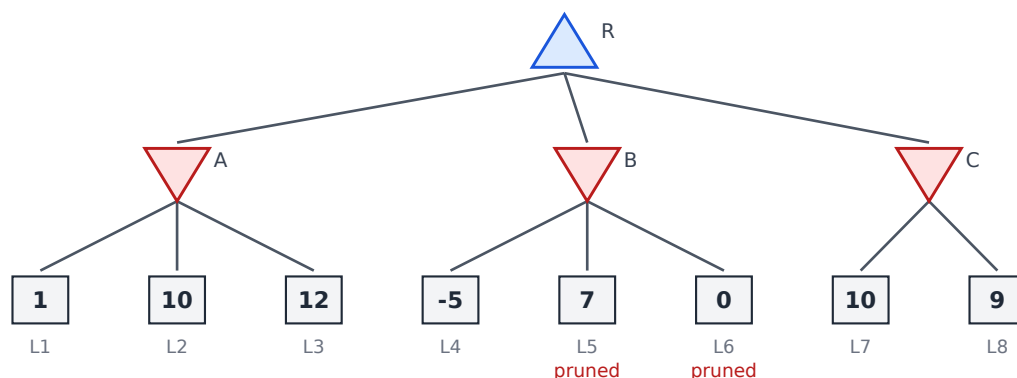
Parents of F: {B, D}; children: {G}; co-parents: {E}.

MB(F) = {B, D, E, G} — **4** nodes. Given its Markov blanket, F is conditionally independent of every other node in the network.

Q.11 [MSQ, 2M · Adversarial Search] **Answer: A, B** (A: Leaf L5 is pruned.; B: Leaf L6 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-5 \leq \alpha = 1 \Rightarrow$ prune remaining children (leaves L5, L6).



Leaves evaluated (6): L1, L2, L3, L4, L7, L8. Leaves pruned (2): L5, L6. Root value = 9 (same as plain minimax, 9 — pruning never changes the root value).

Q.12 [NAT, 2M · Informed Search] **Answer: 6**

Heuristic values: $h(B)=6$, $h(D)=5$, $h(E)=8$, $h(G)=0$, $h(I)=8$, $h(J)=3$, $h(S)=10$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded (f = g + h)	Frontier after expansion (sorted by priority)
1	S (10)	E(11), I(15)
2	E (11)	D(12), I(15)
3	D (12)	J(13), B(15), G(15), I(15)
4	J (13)	B(15), G(15), I(15)
5	B (15)	G(15), I(15)
6	G (15)	I(15) ← goal, stop

Path returned: **S** → **E** → **D** → **G** with cost **15**; nodes expanded = **6** (S, E, D, J, B, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.13 [MCQ, 2M · Uninformed Search] **Answer: D** (D: $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$ (cost 7))

UCS expands nodes in order of g . Pop S(0): push A(1), B(4). Pop A(1): push B(3), C(6). Pop B(3): push C(4), G(9). Pop C(4): push G(7). The stale entries B(4) and C(6) are skipped. Pop G(7) and return **S** → **A** → **B** → **C** → **G** with cost **7**. The other paths are real paths in the graph, but each costs more than 7. UCS returns the cheapest path because it tests the goal on expansion.

Q.14 [MSQ, 2M · Informed Search] **Answer: B** (B: Node S is expanded first.)

Heuristic values: $h(A)=11$, $h(C)=4$, $h(E)=4$, $h(F)=5$, $h(G)=0$, $h(I)=2$, $h(K)=7$, $h(L)=7$, $h(S)=13$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded (f = g + h)	Frontier after expansion (sorted by priority)
1	S (13)	A(13), K(16)
2	A (13)	C(14), K(15)
3	C (14)	G(15), K(15), I(18)
4	G (15)	K(15), I(18) ← goal, stop

Path returned: **S** → **A** → **C** → **G** with cost **15**; nodes expanded = **4** (S, A, C, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $Q \rightarrow Q$; B: $Q \leftrightarrow Q$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 3 of them:

P	Q	R	S	KB
T	T	T	T	T
T	T	T	F	T
F	F	T	T	T

(A) $Q \rightarrow Q$ is true in all models of KB ⇒ **entailed**.

(B) $Q \leftrightarrow Q$ is true in all models of KB ⇒ **entailed**.

(C) $\neg\neg S$ is false in the KB-model $P=T, Q=T, R=T, S=F$ ⇒ not entailed.

(D) $\neg Q \rightarrow Q$ is false in the KB-model $P=F, Q=F, R=T, S=T$ ⇒ not entailed.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: A, D** (A: $\exists x \forall y \neg R(y, x)$; D: $\forall x \forall y (R(x, y) \rightarrow R(y, x))$)

$R = \{(1,3), (2,3), (3,1), (3,2)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

(B) $\forall y \exists x R(x, y)$: **FALSE** — 4 has no incoming edge.

(C) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **TRUE** — every edge has its reverse.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: B, C, D** (B: $\neg Q \vee (Q \wedge \neg R)$; C: $\neg(Q \wedge \neg R) \rightarrow \neg Q$; D: $\neg(Q \wedge \neg(Q \wedge \neg R))$)

Compare truth tables with $F = Q \rightarrow (Q \wedge \neg R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg Q \wedge (Q \wedge \neg R)$ — differs at $Q=T, R=F$ ($F=T$, option = F) ⇒ not equivalent.

(B) $\neg Q \vee (Q \wedge \neg R)$ — agrees with F on all 4 rows ⇒ **equivalent**.

(C) $\neg(Q \wedge \neg R) \rightarrow \neg Q$ — agrees with F on all 4 rows ⇒ **equivalent**.

(D) $\neg(Q \wedge \neg(Q \wedge \neg R))$ — agrees with F on all 4 rows ⇒ **equivalent**.

Q.18 [NAT, 2M · Predicate Logic] Answer: 169

$\exists y \forall x R(x, y)$ says some column of the 3×3 matrix is all T. This already implies $\forall x \exists y R(x, y)$ (that column is a witness for every row), so we only count relations with at least one full column. Relations with no full column: each column has 7 non-full choices $\rightarrow 7^3 = 343$. Answer = $512 - 343 = 169$.

Q.19 [MCQ, 2M · Propositional Logic] Answer: A (A: $(p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg r)$)

Distribute: $(p \vee q) \wedge (\neg p \vee \neg q \vee \neg r) = (p \wedge \neg p) \vee (p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg q) \vee (q \wedge \neg r)$; dropping the contradictory terms gives option A. Option B ($p \oplus q$) misses the model $p=q=T, r=F$. Option C misses $p=T, q=F, r=T$. Option D is equivalent but is a CNF, not a DNF. Truth tables confirm A has exactly the 5 models of the original.

Q.20 [MSQ, 2M · Predicate Logic] Answer: B, C (B: $Q(a, x, f(x))$ and $Q(y, b, f(b))$; C: $R(x, g(x))$ and $R(g(y), g(g(a)))$)

(A) $x/f(y)$; then $f(f(y))$ must unify with y , which fails the occurs check. (B) a/y gives $y/a, x/b$, then $f(b) = f(b)$: MGU $\{y/a, x/b\}$. (C) $x/g(y)$, then $g(g(y))$ vs $g(g(a))$ gives y/a : MGU $\{x/g(a), y/a\}$. (D) $x/f(a)$, then $f(a)$ must equal $f(b)$, which fails since $a \neq b$ are distinct constants.

Q.21 [NAT, 2M · Exact Inference] Answer: 0.152 to 0.156

Factorisation: $P(\text{Burglary, Earthquake, Alarm, JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot P(\text{Alarm} \mid \text{Burglary}) \cdot P(\text{JohnCalls} \mid \text{Earthquake, Alarm})$.

Inference by enumeration: $P(\text{Alarm} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Alarm, e, hidden})$.

Hidden: Bu, Ea	Al	product of CPT entries (Bu · Ea · Al · Jo)	value
T, T	T	$0.95 \times 0.8 \times 0.1 \times 0.5$	0.038
T, F	T	$0.95 \times 0.2 \times 0.1 \times 0.25$	0.00475
F, T	T	$0.05 \times 0.8 \times 0.2 \times 0.5$	0.004
F, F	T	$0.05 \times 0.2 \times 0.2 \times 0.25$	0.0005
T, T	F	$0.95 \times 0.8 \times 0.9 \times 0.3$	0.2052
T, F	F	$0.95 \times 0.2 \times 0.9 \times 0.25$	0.04275
F, T	F	$0.05 \times 0.8 \times 0.8 \times 0.3$	0.0096
F, F	F	$0.05 \times 0.2 \times 0.8 \times 0.25$	0.002

$P(\text{Alarm}=T, e) = 0.04725$; $P(\text{Alarm}=F, e) = 0.25955$; $P(e) = 0.3068$.

$P(\text{Alarm}=T \mid e) = 0.04725 / 0.3068 = \mathbf{0.154} \approx 0.154$.

Q.22 [NAT, 2M · Conditional Independence] Answer: 0.263 to 0.265

$A \perp B \mid C$, so $P(a, b) = \sum_c P(c) P(a \mid c) P(b \mid c)$. c : $0.4 \times 0.9 \times 0.7 = 0.252$. $\neg c$: $0.6 \times 0.2 \times 0.1 = 0.012$. Sum = 0.264 . Note $P(a) = 0.36 + 0.12 = 0.48$ and $P(b) = 0.28 + 0.06 = 0.34$, so $P(a)P(b) = 0.1632 \neq 0.264$: A and B are marginally dependent.

Q.23 [NAT, 2M · Sampling] Answer: 0.598 to 0.602

Gibbs sampling draws D from $P(D \mid \text{mb}(D))$, where $\text{mb}(D) = \{A, C\}$:

$P(D \mid \text{mb}) \propto P(D \mid \text{parents}(D)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$D=T$: $P(D=T \mid A=T, C=F) = 0.6 = 0.6$

$D=F$: $P(D=F \mid A=T, C=F) = 0.4 = 0.4$

Normalise: $0.6 / (0.6 + 0.4) = \mathbf{0.6}$.

Q.24 [MSQ, 2M · Conditional Independence] Answer: A, D (A: $B \perp A$; D: $F \perp C \mid A, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp A$: **TRUE** — $B \rightarrow C \rightarrow D \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow C \rightarrow D \rightarrow F \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow E \leftarrow A$ (blocked: collider E with neither it nor a descendant observed).

(B) $E \perp D$: **FALSE** — the path $E \leftarrow A \rightarrow D$ is active given $\{\}$.

(C) $B \perp E \mid C, F$: **FALSE** — the path $B \rightarrow E$ is active given $\{C, F\}$.

(D) $F \perp C \mid A, D$: **TRUE** — $F \leftarrow A \rightarrow D \leftarrow C$ (blocked: observed fork node A); $F \leftarrow A \rightarrow E \leftarrow B \rightarrow C$ (blocked: observed fork node A); $F \leftarrow D \leftarrow A \rightarrow E \leftarrow B \rightarrow C$ (blocked: observed chain node D); $F \leftarrow D \leftarrow C$ (blocked: observed chain node D).

Q.25 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(A, C), f(A, D), f(B, C, E), f(B, D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(B, C, E)$	$\{B, C, E\}$	$f(B, C)$
F	$f(B, D, F)$	$\{B, D, F\}$	$f(B, D)$
A	$f(A), f(A, B), f(A, C), f(A, D)$	$\{A, B, C, D\}$	$f(B, C, D)$
D	$f(B, D), f(B, C, D)$	$\{B, C, D\}$	$f(B, C)$
B	$f(B, C), f(B, C)$	$\{B, C\}$	$f(C)$

The largest factor produced has **3** variable(s) (created when eliminating A); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over C and normalise.

Mock Test 052

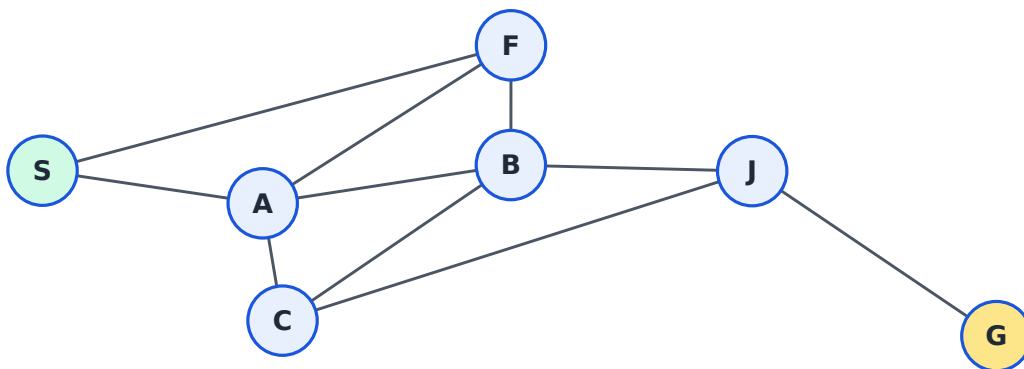
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	8 / 6 / 11	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

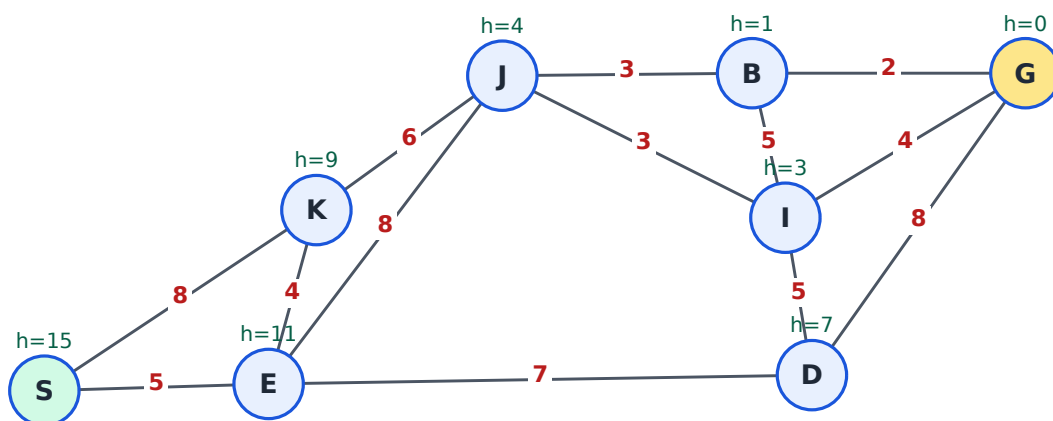
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, A, B, C, J, G
- ☐ (B) S, A, F, B, C, J, G
- ☐ (C) S, A, B, F, C, J, G
- ☐ (D) S, F, A, B, C, J, G

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

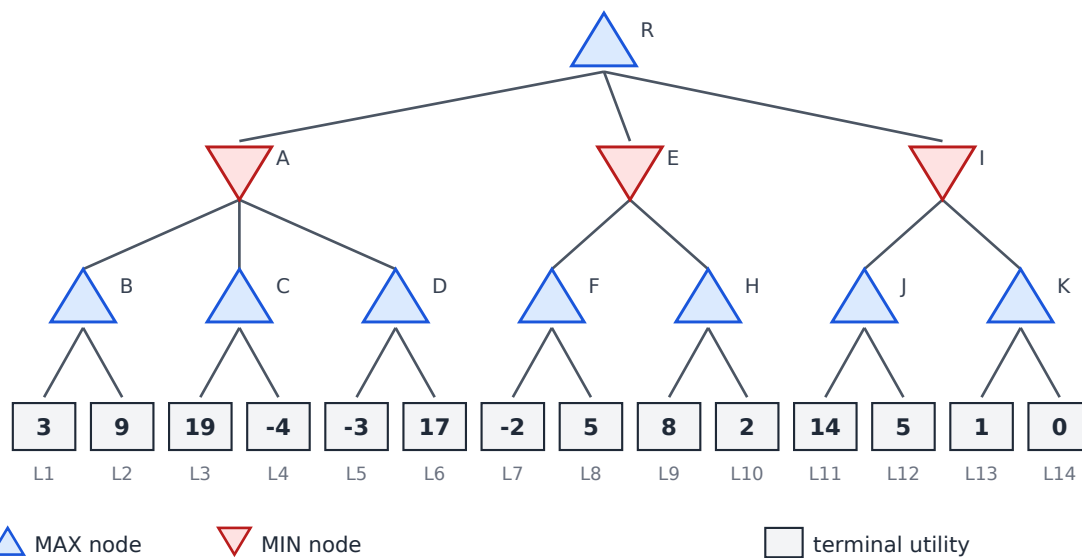
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Informed Search

h_1 and h_2 are both admissible, and $h_2(n) \geq h_1(n)$ for every n . Which statement is correct?

- ☐ (A) h_1 dominates h_2 , so A^* with h_1 is preferable
- ☐ (B) h_2 dominates h_1 , and A^* with h_2 never expands more nodes than A^* with h_1 (apart from ties at $f = C^*$)
- ☐ (C) A^* with h_2 may return a more expensive solution than A^* with h_1
- ☐ (D) Dominance requires $h_2(n) = h_1(n) + c$ for a constant c

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$ true?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In Existential Instantiation, the sentence $\exists x \text{Crown}(x) \wedge \text{OnHead}(x, \text{John})$ is replaced by $\text{Crown}(C_1) \wedge \text{OnHead}(C_1, \text{John})$. Which condition must C_1 satisfy?

- ☐ (A) C_1 must be a new constant symbol not appearing anywhere else in the KB
- ☐ (B) C_1 must be John
- ☐ (C) C_1 may be any ground term already in the KB
- ☐ (D) C_1 must be a variable

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Let $\text{KB} = \{P \rightarrow Q, Q \rightarrow R, \neg R\}$. Which one of the following is entailed by KB?

- ☐ (A) $\neg P \wedge \neg Q$
- ☐ (B) $P \vee Q$
- ☐ (C) $\neg P \wedge Q$
- ☐ (D) R

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((\neg R \vee P) \vee (R \vee P)) \rightarrow ((\neg Q \rightarrow \neg P) \vee \neg \neg Q) ?$$

Answer (numerical): _____

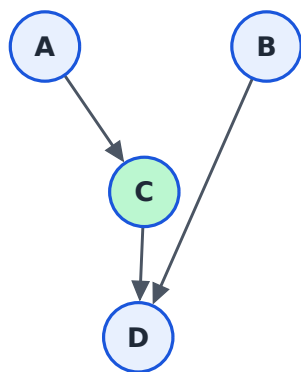
Q.9 [MCQ · 1 mark · negative: -1/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. When Gibbs sampling resamples Rain (R), the values of which variables must be consulted?

- ☐ (A) C, S and W
- ☐ (B) C only
- ☐ (C) W only
- ☐ (D) C and W only

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T | \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



P(A=T)
0.05

P(B=T)	A	P(C=T ...)
0.4	T	0.1
	F	0.9

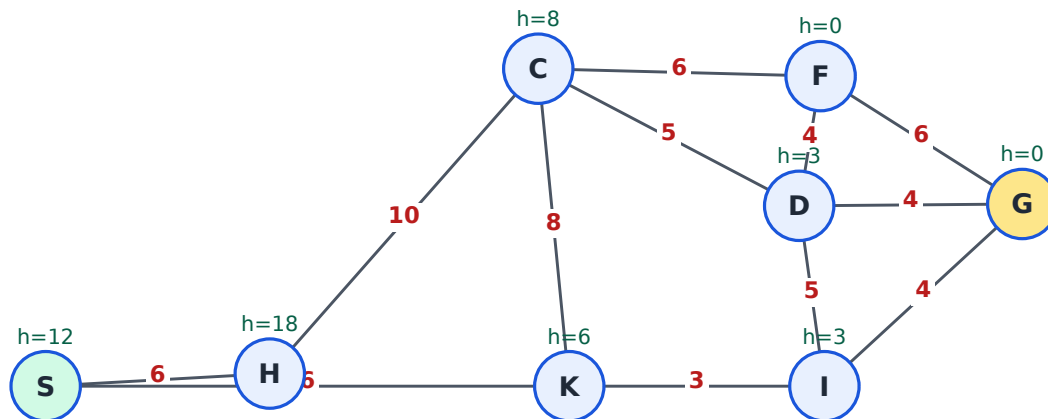
B	C	P(D=T ...)
T	T	0.5
T	F	0.75
F	T	0.4
F	F	0.7

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

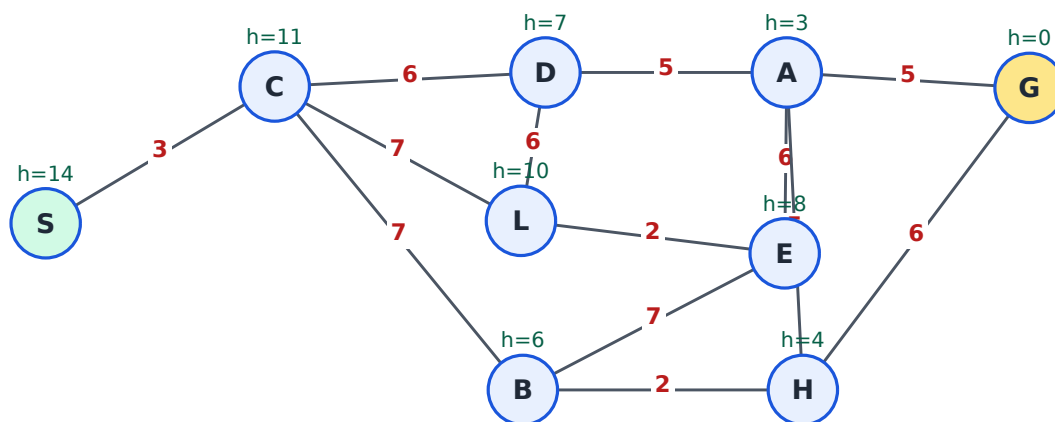
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(H)$ overestimates the true cost from H to G.
- ☐ (B) h is consistent (monotone).
- ☐ (C) h is admissible.
- ☐ (D) The edge C-F violates the consistency condition $h(C) \leq c(C,F) + h(F)$.

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – A*

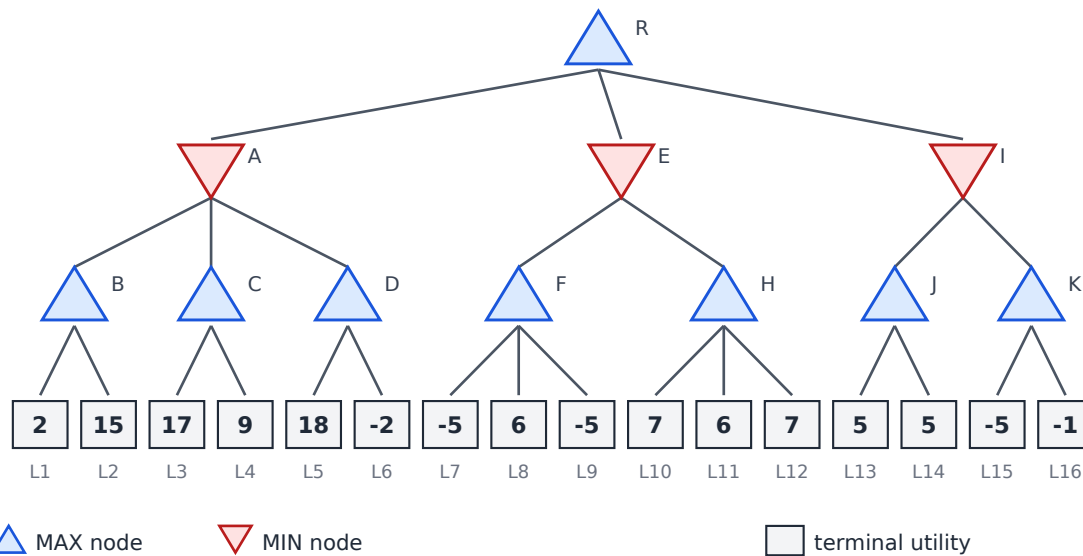
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node D is expanded.
- ☐ (B) The path returned has cost 18.
- ☐ (C) Node A is expanded.
- ☐ (D) Node C is expanded.

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search

A game tree has a MAX root with three MIN children. Each MIN child has two MAX children, and each of those has two leaves, for 12 leaves in total. The leaves, left to right, are 4, 8, 9, 3, 2, 1, 7, 6, 5, 10, 2, 11. Alpha-beta pruning processes children left to right. How many leaves are **evaluated**?

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg Q)$, C2: $(P \vee Q \vee \neg S)$, C3: (R) , C4: $(P \vee S)$, C5: $(\neg P \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(P \vee \neg S)$
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) $(\neg P \vee R)$
☐ (D) $(P \vee R)$

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic

To prove R from $KB = \{P \vee Q, P \rightarrow R, Q \rightarrow R\}$ by resolution refutation, the clause $\neg R$ is added to the clauses of KB . What is the minimum number of resolution steps needed to derive the empty clause? (Each step resolves two clauses on one complementary pair; duplicate literals are merged.)

Answer (numerical): _____

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(\neg P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg \neg P)$
☐ (B) $((Q \rightarrow R) \wedge (P \rightarrow R)) \rightarrow (Q \wedge P)$
☐ (C) $R \rightarrow (P \rightarrow R)$
☐ (D) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$

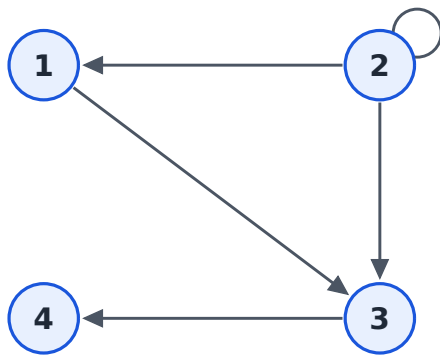
Q.18 [NAT · 2 marks · no negative marking] Propositional Logic

How many of the 16 truth assignments to p, q, r, s satisfy exactly two of the four clauses $p \vee q, \neg p \vee r, \neg q \vee s, \neg r \vee \neg s$? (Enter the number of assignments, not clauses.)

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

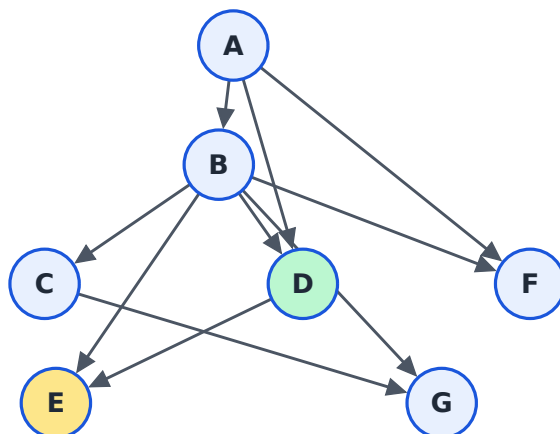
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
☐ (B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (C) $\exists x \forall y \neg R(x, y)$
☐ (D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$

Q.20 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

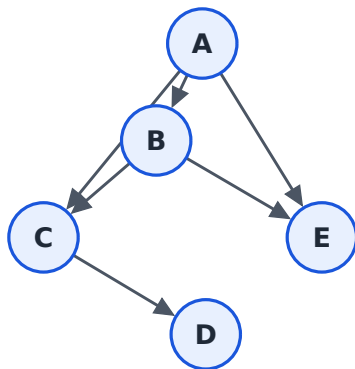
Variable elimination is used to compute $\mathbf{P(D \mid E = T)}$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, A, C, F, G**. What is the scope of the new factor produced when **A** is summed out?



- ☐ (A) $f(A, C, D, F, G)$
☐ (B) $f(C, F, G)$
☐ (C) $f()$
☐ (D) $f(C, D, F, G)$

Q.21 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.23, 0.98, 0.61, 0.45, 0.56)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.6	T	0.6	T	T	0.9
	F	0.75	T	F	0.95
			F	T	0.8
			F	F	0.6

C	P(D=T ...)	A	B	P(E=T ...)
T	0.2	T	T	0.75
F	0.1	T	F	0.5
		F	T	0.05
		F	F	0.9

- ☐ (A) A=F, B=T, C=F, D=T, E=T
☐ (B) A=T, B=F, C=T, D=F, E=F
☐ (C) A=F, B=F, C=T, D=F, E=F
☐ (D) A=T, B=T, C=T, D=F, E=F

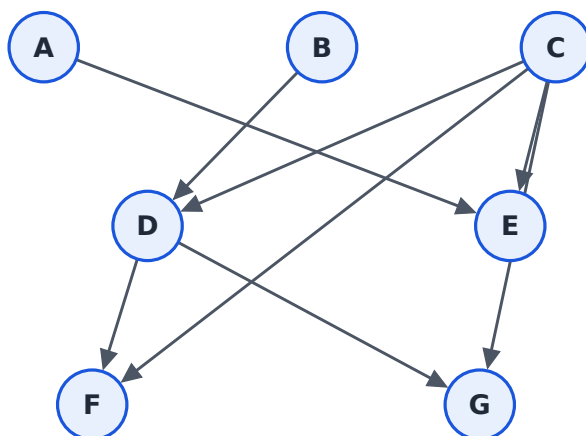
Q.22 [MCQ · 2 marks · negative: -2/3] Sampling

In likelihood weighting with evidence e and non-evidence variables z , let $S_{WS}(z, e)$ be the sampling probability and $w(z, e)$ the weight. Which statement is correct?

- ☐ (A) $S_{WS}(z, e) = \prod P(z_i \mid \text{parents}(Z_i))$, $w(z, e) = \prod P(e_j \mid \text{parents}(E_j))$, and $S_{WS}(z, e) \cdot w(z, e) = P(z, e)$
☐ (B) $S_{WS}(z, e) = P(z \mid e)$, so all weights equal 1
☐ (C) $S_{WS}(z, e) = P(z, e)$ and $w(z, e) = P(e)$
☐ (D) $S_{WS}(z, e) = \prod P(e_j \mid \text{parents}(E_j))$, $w(z, e) = \prod P(z_i \mid \text{parents}(Z_i))$

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

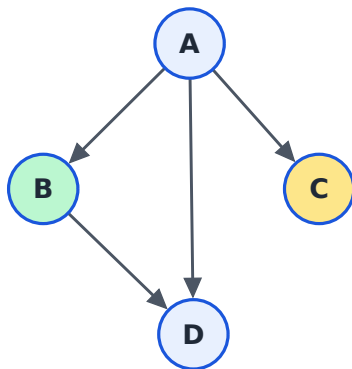
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp E \mid A$
☐ (B) $A \perp G \mid C, F$
☐ (C) $E \perp A \mid G$
☐ (D) $A \perp B \mid F$

Q.24 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{C=F}$. The current state of the chain is $(\mathbf{A=T, B=F, C=F, D=F})$. The next step resamples $\mathbf{B}$. With what probability is B set to T? (round off to 3 decimal places)



P(A=T)
0.5

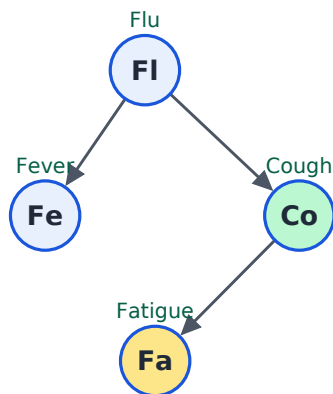
A	P(B=T ...)	A	P(C=T ...)
T	0.2	T	0.6
F	0.9	F	0.2

A	B	P(D=T ...)
T	T	0.25
T	F	0.9
F	T	0.2
F	F	0.75

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $\mathbf{P(\text{Cough} = T \mid \text{Fatigue} = F)}$ (round off to 3 decimal places).



P(FI=T)
0.8

FI	P(Fe=T ...)	FI	P(Co=T ...)
T	0.8	T	0.1
F	0.3	F	0.05

Co	P(Fa=T ...)
T	0.9
F	0.75

Answer (numerical): _____

Answer Key — Mock Test 052

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	B	14	NAT	2	Adversarial Search	9
2	NAT	1	Informed Search	19	15	MSQ	2	Propositional Logic	A
3	NAT	1	Adversarial Search	9	16	NAT	2	Propositional Logic	3
4	MCQ	1	Informed Search	B	17	MSQ	2	Propositional Logic	A, C
5	NAT	1	Predicate Logic	216	18	NAT	2	Propositional Logic	2
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Predicate Logic	C, D
7	MCQ	1	Propositional Logic	A	20	MCQ	2	Exact Inference	D
8	NAT	1	Propositional Logic	6	21	MCQ	2	Sampling	B
9	MCQ	1	Sampling	A	22	MCQ	2	Sampling	A
10	NAT	1	Exact Inference	0.858 to 0.862	23	MSQ	2	Conditional Independence	A, B, D
11	MSQ	2	Informed Search	C, D	24	NAT	2	Sampling	0.65 to 0.654
12	MSQ	2	Informed Search	A, B, C, D	25	NAT	2	Exact Inference	0.036 to 0.04
13	NAT	2	Adversarial Search	9					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: B** (B: S, A, F, B, C, J, G)

Adjacency lists (alphabetical): A: {B, C, F, S}; B: {A, C, F, J}; C: {A, B, J}; F: {A, B, S}; G: {J}; J: {B, C, G}; S: {A, F}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[A, F]
2	A	[F, B, C]
3	F	[B, C]
4	B	[C, J]
5	C	[J]
6	J	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, A, F, B, C, J, G**. The path returned is $S \rightarrow A \rightarrow B \rightarrow J \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Informed Search] **Answer: 19**

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

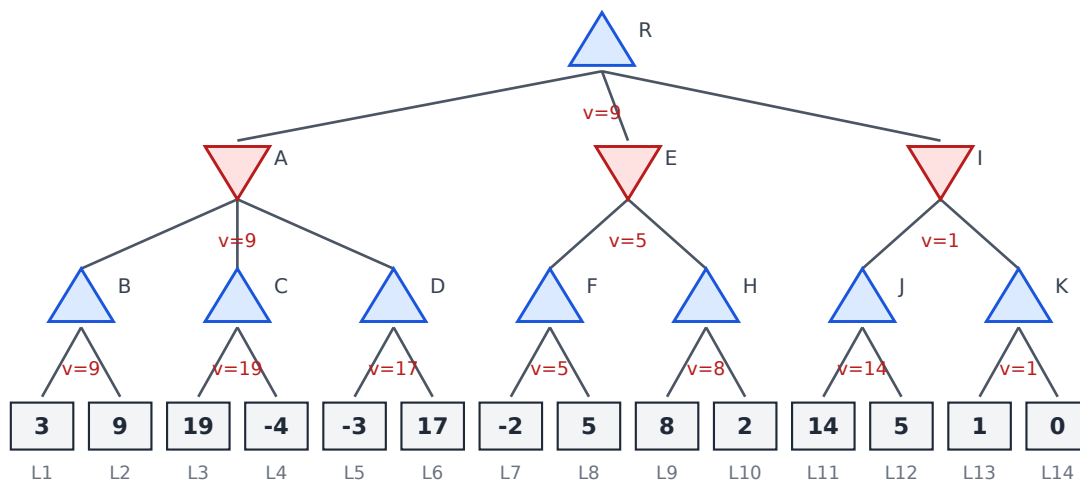
Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (15)	K(9), E(11)
2	K (9)	J(4), E(11)
3	J (4)	B(1), I(3), E(11)
4	B (1)	G(0), I(3), E(11)
5	G (0)	I(3), E(11) ← goal, stop

Path returned: **S** → **K** → **J** → **B** → **G**, cost = 8 + 6 + 3 + 2 = **19**.

The optimal cost is $h^*(S) = 18$ — greedy search is not optimal here.

Q.3 [NAT, 1M · Adversarial Search] **Answer: 9**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 9$, $C = 19$, $D = 17$, $A = 9$, $F = 5$, $H = 8$, $E = 5$, $J = 14$, $K = 1$, $I = 1$. Root value = **9**; MAX's optimal first move is towards A.

Q.4 [MCQ, 1M · Informed Search] Answer: B (B : h_2 dominates h_1 , and A^* with h_2 never expands more nodes than A^* with h_1 (apart from ties at $f = C^*$))

h_2 dominates h_1 . With consistent heuristics, A^* expands every node with $f(n) < C^*$, which means $h(n) < C^* - g(n)$. A larger h satisfies this inequality at fewer nodes, so A^* with h_2 expands a subset. (A) reverses the relation. (C) is wrong because both heuristics are admissible, so both runs are optimal. (D) is not the definition of dominance.

Q.5 [NAT, 1M · Predicate Logic] Answer: 216

View R as an 3×3 0/1 matrix. The sentence says R is antisymmetric: diagonal free (2^n); each unordered pair has 3 options (neither, one way, other way) $\Rightarrow 2^n \cdot 3^{(n-1)/2}$. With $n = 3$: **216**.

Q.6 [MCQ, 1M · Predicate Logic] Answer: A (A : C_1 must be a new constant symbol not appearing anywhere else in the KB)

Existential Instantiation introduces a fresh Skolem constant naming the (unknown) witness. Using an existing term such as John would assert, without justification, that John is the witness. The result is inferentially equivalent (satisfiable iff the original), not logically equivalent. A variable would make it a non-ground formula.

Q.7 [MCQ, 1M · Propositional Logic] Answer: A (A : $\neg P \wedge \neg Q$)

From $\neg R$ and $Q \rightarrow R$, modus tollens gives $\neg Q$; from $\neg Q$ and $P \rightarrow Q$ it gives $\neg P$. The only model of KB (over P, Q, R) is $P=F, Q=F, R=F$, in which $\neg P \wedge \neg Q$ is true. $P \vee Q, \neg P \wedge Q$ and R are all false in that model, so they are not entailed.

Q.8 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg R \vee P) \vee (R \vee P)$	$(\neg Q \rightarrow \neg P) \vee \neg \neg Q$	$((\neg R \vee P) \vee (R \vee P)) \rightarrow ((\neg Q \rightarrow \neg P) \vee \neg \neg Q)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

Rows where $F = T$: **6**.

Q.9 [MCQ, 1M · Sampling] Answer: A (A : C, S and W)

Gibbs samples R from $P(R \mid \text{mb}(R))$. $\text{mb}(R) = \text{parents}\{C\} \cup \text{children}\{W\} \cup \text{children's other parents}\{S\} = \{C, S, W\}$. S is needed because $P(w \mid S, R)$ depends on it (explaining away). The other options omit blanket members.

Q.10 [NAT, 1M · Exact Inference] Answer: 0.858 to 0.862

Only the ancestors of C (A) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T, D=T	0.001
A=T, B=T, D=F	0.001
A=T, B=F, D=T	0.0012
A=T, B=F, D=F	0.0018
A=F, B=T, D=T	0.171
A=F, B=T, D=F	0.171
A=F, B=F, D=T	0.2052
A=F, B=F, D=F	0.3078

Sum = **0.86** $\approx$ 0.860.

Q.11 [MSQ, 2M · Informed Search] Answer: C, D (C: h is admissible.; D: The edge C-F violates the consistency condition $h(C) \leq c(C,F) + h(F)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	D	F	G	H	I	K	S
$h(n)$	8	3	0	0	18	3	6	12
$h^*(n)$	9	4	6	0	19	4	7	13
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges:

$C \rightarrow F$ ($8 > 6 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [MSQ, 2M · Informed Search] Answer: A, B, C, D (A: Node D is expanded.; B: The path returned has cost 18.; C: Node A is expanded.; D: Node C is expanded.)

Heuristic values: $h(A)=3$, $h(B)=6$, $h(C)=11$, $h(D)=7$, $h(E)=8$, $h(G)=0$, $h(H)=4$, $h(L)=10$, $h(S)=14$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (14)	C(14)
2	C (14)	B(16), D(16), L(20)
3	B (16)	D(16), H(16), L(20), E(25)
4	D (16)	H(16), A(17), L(20), E(25)
5	H (16)	A(17), G(18), L(20), E(25)
6	A (17)	G(18), L(20), E(25)
7	G (18)	L(20), E(25) $\leftarrow$ goal, stop

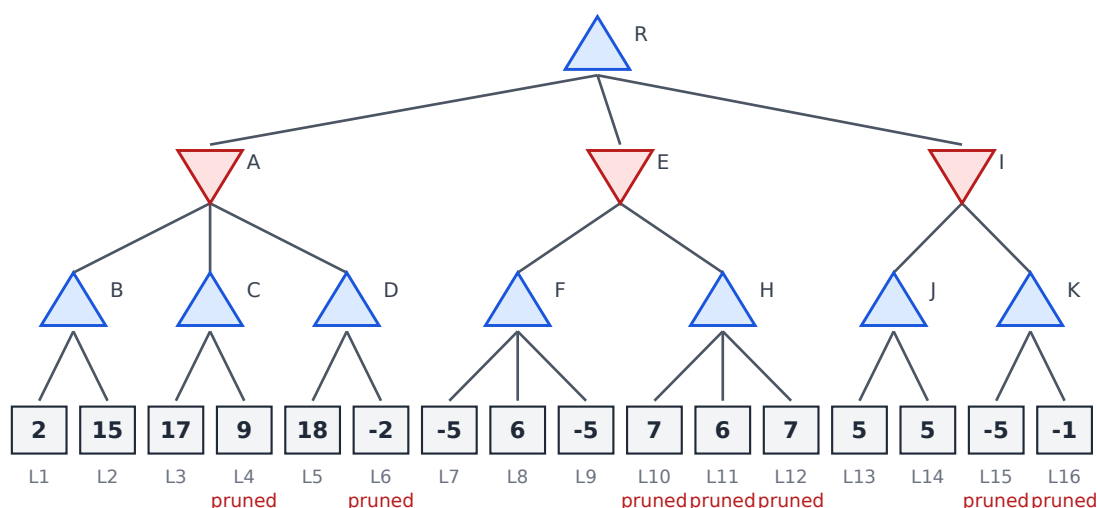
Path returned: **S $\rightarrow$ C $\rightarrow$ B $\rightarrow$ H $\rightarrow$ G** with cost **18**; nodes expanded = **7** (S, C, B, D, H, A, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 18$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.13 [NAT, 2M · Adversarial Search] Answer: 9

Run alpha-beta depth-first, left to right, passing (α , β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $17 \geq \beta = 15 \Rightarrow$ prune remaining children (leaves L4).
- At MAX node D: value $18 \geq \beta = 15 \Rightarrow$ prune remaining children (leaves L6).
- At MIN node E: value $6 \leq \alpha = 15 \Rightarrow$ prune remaining children (leaves L10, L11, L12).
- At MIN node I: value $5 \leq \alpha = 15 \Rightarrow$ prune remaining children (leaves L15, L16).



Leaves evaluated (9): L1, L2, L3, L5, L7, L8, L9, L13, L14. Leaves pruned (7): L4, L6, L10, L11, L12, L15, L16. Root value = 15 (same as plain minimax, 15 — pruning never changes the root value).

Q.14 [NAT, 2M · Adversarial Search] **Answer: 9**

MIN_1 : $\text{MAX}(4, 8) = 8$, so $\beta = 8$. The next MAX node sees $9 \geq 8$ and prunes leaf 3. $\text{MIN}_1 = 8$, so root $\alpha = 8$. MIN_2 : $\text{MAX}(2, 1) = 2 \leq \alpha$, so the MAX node with leaves (7, 6) is pruned. MIN_3 : $\text{MAX}(5, 10) = 10$, so $\beta = 10$. $\text{MAX}(2, 11)$: leaf 2 does not cut and $11 \geq 10$ cuts, but no siblings remain, so both are evaluated. $\text{MIN}_3 = 10$, and the root value is 10. Pruned: 3, 7, 6. Evaluated: 12 — 3 = **9**.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: A** ($A: (P \vee \neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(P \vee \neg S)$ is derivable:

1. resolve $(\neg Q)$ and $(P \vee Q \vee \neg S)$ on $Q \Rightarrow (P \vee \neg S)$

(B) The resolution closure contains 10 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(\neg P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.16 [NAT, 2M · Propositional Logic] **Answer: 3**

Clauses: $P \vee Q$, $\neg P \vee R$, $\neg Q \vee R$, $\neg R$. A 3-step refutation: $(P \vee Q, \neg P \vee R) \Rightarrow Q \vee R$; $(Q \vee R, \neg Q \vee R) \Rightarrow R$; $(R, \neg R) \Rightarrow \square$. 2 steps are impossible: the last step must resolve two complementary unit clauses, and the only unit initially is $\neg R$. So R (a unit) must be produced in a single step, but every one-step resolvent is $Q \vee R$, $P \vee R$, $\neg P$, $\neg Q$ or a tautology — never R . Hence the minimum is 3 (a breadth-first search over resolution derivations confirms this).

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, C** ($A: (\neg P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$; $C: R \rightarrow (P \rightarrow R)$)

(A) $(\neg P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $((Q \rightarrow R) \wedge (P \rightarrow R)) \rightarrow (Q \wedge P)$: **not valid** — counter-model $P=T, Q=F, R=T$.

(C) $R \rightarrow (P \rightarrow R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$: **not valid** — counter-model $P=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [NAT, 2M · Propositional Logic] **Answer: 2**

Each clause, being a disjunction, is false only in specific situations: C1 false iff $p=q=F$; C2 iff $p=T, r=F$; C3 iff $q=T, s=F$; C4 iff $r=s=T$. With 4 clauses, satisfying exactly two means falsifying exactly two. C1 is incompatible with C2 and C3 (they need $p=T$ or $q=T$). C1 & C4: $p=q=F, r=s=T$ — then C2, C3 hold $\rightarrow$ 1 assignment. C2 & C3: $p=T, r=F, q=T, s=F$ — C4 holds, C1 holds $\rightarrow$ 1 assignment. C2 & C4 needs $r=F$ and $r=T$; C3 & C4 needs $s=F$ and $s=T$ — impossible. Total = 2 (confirmed by enumeration).

Q.19 [MSQ, 2M · Predicate Logic] **Answer: C, D** (C: $\exists x \forall y \neg R(x, y)$; D: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$)

$R = \{(1,3), (2,1), (2,2), (2,3), (3,4)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.

(B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,1)$ holds but $R(1,2)$ does not.

(C) $\exists x \forall y \neg R(x, y)$: **TRUE** — 4 has no outgoing edge.

(D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MCQ, 2M · Exact Inference] **Answer: D** (D: $f(C, D, F, G)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(B, D)$, $f(A, B, F)$, $f(B, C, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(B, D)$, $f(A, B, F)$, $f(B, C, G)$	$\{A, B, C, D, F, G\}$	$f(A, C, D, F, G)$
A	$f(A)$, $f(A, C, D, F, G)$	$\{A, C, D, F, G\}$	$f(C, D, F, G)$
C	$f(C, D, F, G)$	$\{C, D, F, G\}$	$f(D, F, G)$
F	$f(D, F, G)$	$\{D, F, G\}$	$f(D, G)$
G	$f(D, G)$	$\{D, G\}$	$f(D)$

The largest factor produced has **5** variable(s) (created when eliminating B); with Boolean variables it has $2^5 = 32$ entries. Finally, multiply the remaining factors over D and normalise.

Q.21 [MCQ, 2M · Sampling] **Answer: B** (B: $A=T$, $B=F$, $C=T$, $D=F$, $E=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.6$; $u = 0.23 < 0.6 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.6$; $u = 0.98 \geq 0.6 \Rightarrow B = F$
- C: $P(C=T \mid A=T, B=F) = 0.95$; $u = 0.61 < 0.95 \Rightarrow C = T$
- D: $P(D=T \mid C=T) = 0.2$; $u = 0.45 \geq 0.2 \Rightarrow D = F$
- E: $P(E=T \mid A=T, B=F) = 0.5$; $u = 0.56 \geq 0.5 \Rightarrow E = F$

Sample: **$\{A=T, B=F, C=T, D=F, E=F\}$** .

Q.22 [MCQ, 2M · Sampling] **Answer: A**

Each non-evidence variable is sampled from its CPT given sampled/fixed parents, giving $S_{WS} = \prod P(z_i \mid \text{parents})$. The weight multiplies the evidence likelihoods. The product of these two is exactly the BN factorization of $P(z, e)$, which proves consistency. (B) is false: evidence at descendants is not taken into account while sampling, so $S_{WS} \neq P(z \mid e)$ in general. (C) and (D) misstate or swap the two products.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: A, B, D** (A: $B \perp E \mid A$; B: $A \perp G \mid C, F$; D: $A \perp B \mid F$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp E \mid A$: **TRUE** — $B \rightarrow D \leftarrow C \rightarrow E$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow D \rightarrow F \leftarrow C \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow D \rightarrow G \leftarrow C \rightarrow E$ (blocked: collider G with neither it nor a descendant observed).

(B) $A \perp G \mid C, F$: **TRUE** — $A \rightarrow E \leftarrow C \rightarrow D \rightarrow G$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow E \leftarrow C \rightarrow F \leftarrow D \rightarrow G$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow E \leftarrow C \rightarrow G$ (blocked: collider E with neither it nor a descendant observed).

(C) $E \perp A \mid G$: **FALSE** — the path $E \leftarrow A$ is active given $\{G\}$.

(D) $A \perp B \mid F$: **TRUE** — $A \rightarrow E \leftarrow C \rightarrow D \leftarrow B$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow E \leftarrow C \rightarrow F \leftarrow D \leftarrow B$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow E \leftarrow C \rightarrow G \leftarrow D \leftarrow B$ (blocked: collider E with neither it nor a descendant observed).

Q.24 [NAT, 2M · Sampling] **Answer: 0.65 to 0.654**

Gibbs sampling draws B from $P(B \mid \text{mb}(B))$, where $\text{mb}(B) = \{A, D\}$:

$P(B \mid \text{mb}) \propto P(B \mid \text{parents}(B)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$B=T$: $P(B=T \mid A=T) \times P(D=F \mid A=T, B=T) = 0.2 \times 0.75 = 0.15$

$B=F$: $P(B=F \mid A=T) \times P(D=F \mid A=T, B=F) = 0.8 \times 0.1 = 0.08$

Normalise: $0.15 / (0.15 + 0.08) = \mathbf{0.6522}$.

Q.25 [NAT, 2M · Exact Inference] **Answer: 0.036 to 0.04**

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever} \mid \text{Flu}) \cdot P(\text{Cough} \mid \text{Flu}) \cdot P(\text{Fatigue} \mid \text{Cough})$.

Inference by enumeration: $P(\text{Cough} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cough}, e, \text{hidden})$.

Hidden: Fl, Fe	Co	product of CPT entries (Fl · Fe · Co · Fa)	value
T, T	T	$0.8 \times 0.8 \times 0.1 \times 0.1$	0.0064
T, F	T	$0.8 \times 0.2 \times 0.1 \times 0.1$	0.0016
F, T	T	$0.2 \times 0.3 \times 0.05 \times 0.1$	0.0003
F, F	T	$0.2 \times 0.7 \times 0.05 \times 0.1$	0.0007
T, T	F	$0.8 \times 0.8 \times 0.9 \times 0.25$	0.144
T, F	F	$0.8 \times 0.2 \times 0.9 \times 0.25$	0.036
F, T	F	$0.2 \times 0.3 \times 0.95 \times 0.25$	0.01425
F, F	F	$0.2 \times 0.7 \times 0.95 \times 0.25$	0.03325

$P(\text{Cough}=\text{T}, e) = 0.009$; $P(\text{Cough}=\text{F}, e) = 0.2275$; $P(e) = 0.2365$.

$P(\text{Cough}=\text{T} \mid e) = 0.009 / 0.2365 = \mathbf{0.0381} \approx 0.038$.

Mock Test 053

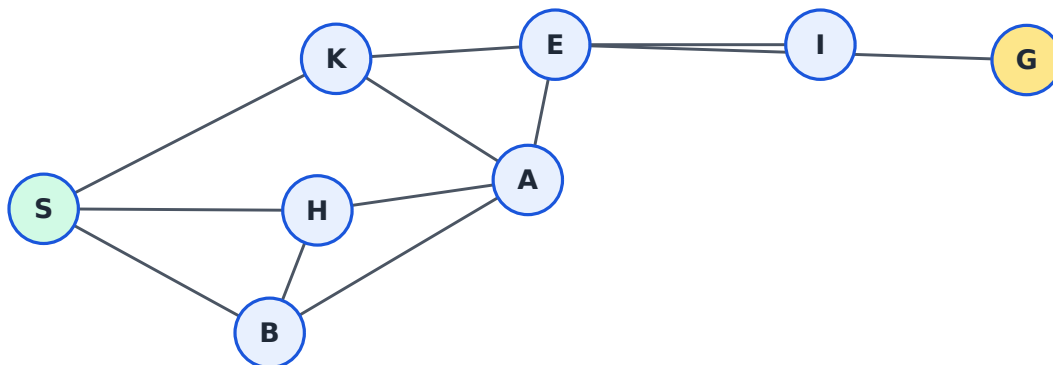
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 7 / 12	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, A, B, E, G
- ☐ (B) S, B, H, K, A, E, G
- ☐ (C) S, K, E, I, G
- ☐ (D) S, B, A, E, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 5$ (so $b^d = 1024$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

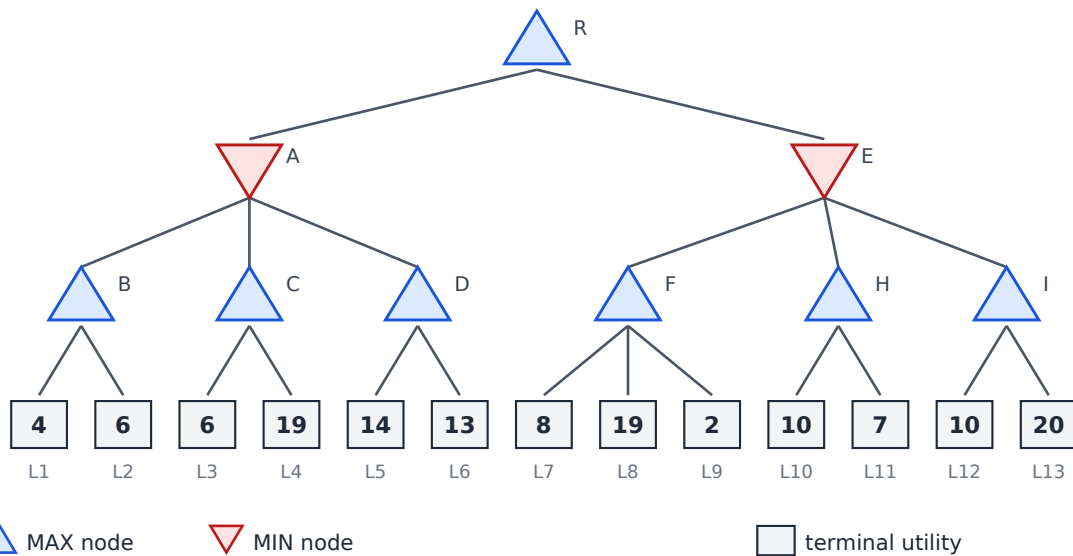
Q.3 [NAT · 1 mark · no negative marking] Adversarial Search

Tic-tac-toe has 9 squares. If early wins are ignored and every game continues until all 9 squares are filled, how many distinct move sequences (orderings of the 9 moves) are there?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following statements are TRUE?

- ☐ (A) Propositional resolution is refutation-complete
- ☐ (B) Modus ponens alone is complete for arbitrary propositional knowledge bases
- ☐ (C) Forward chaining is sound and complete for knowledge bases of definite clauses
- ☐ (D) A sound inference procedure derives only sentences that are entailed by the KB

Q.6 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following correctly express 'Not every graph is connected'?

- ☐ (A) $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$
- ☐ (B) $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$
- ☐ (C) $\forall x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$
- ☐ (D) $\exists x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$\mathbf{F = ((R \vee \neg R) \wedge (R \vee Q)) \wedge \neg P ?}$$

Answer (numerical): _____

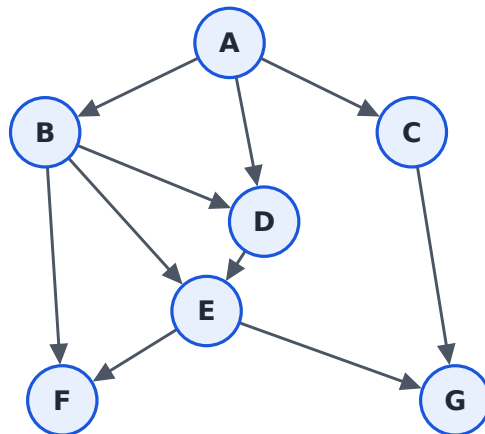
Q.8 [MCQ · 1 mark · negative: -1/3] Conditional Independence

For random variables A, B, C with all conditioning events of positive probability, $A \perp B \mid C$ holds if and only if

- ☐ (A) $P(A \mid B, C) = P(A \mid C)$
- ☐ (B) $P(A, B) = P(A) P(B)$
- ☐ (C) $P(A \mid C) = P(B \mid C)$
- ☐ (D) $P(A, B, C) = P(A) P(B) P(C)$

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **E** in the Bayesian network below?



Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Probability for AI

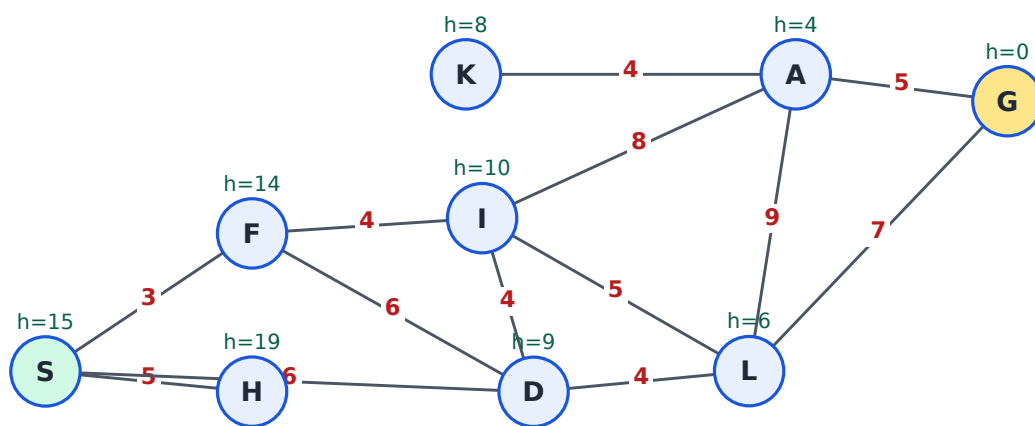
A domain has four Boolean random variables and one random variable with 3 possible values. How many independent numbers are needed to specify an arbitrary full joint distribution over these five variables?

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

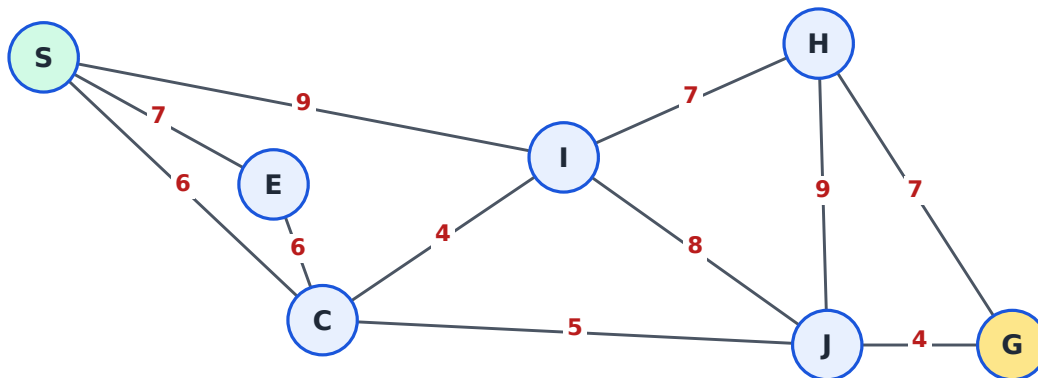
Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Suppose breadth-first search were changed to apply the goal test when a node is selected for expansion instead of when it is generated. The branching factor is b and the shallowest goal is at depth d . In the worst case, the number of nodes generated becomes

- ☐ (A) $O(b^d)$
- ☐ (B) $O(b^{d+1})$
- ☐ (C) $O(b^{d/2})$
- ☐ (D) $O(bd)$

Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

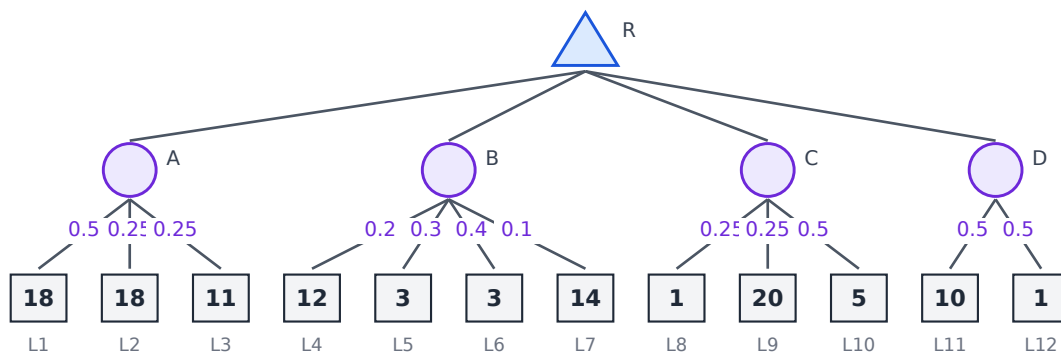
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow C \rightarrow I \rightarrow J \rightarrow G$
☐ (B) $S \rightarrow C \rightarrow J \rightarrow G$
☐ (C) $S \rightarrow I \rightarrow J \rightarrow G$
☐ (D) $S \rightarrow I \rightarrow C \rightarrow J \rightarrow G$

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



MAX node
 CHANCE node (edge = probability)
 terminal utility

Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic

Of the 16 binary connectives $\circ$, for how many is the formula $(p \circ q) \rightarrow p$ a tautology?

Answer (numerical): _____

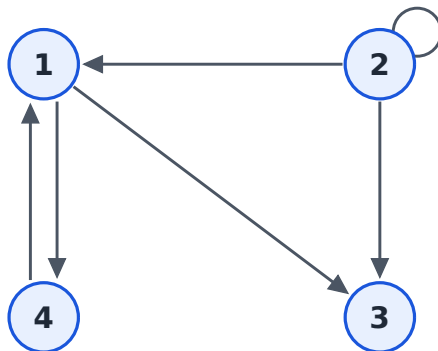
Q.16 [MCQ · 2 marks · negative: -2/3] Predicate Logic

What is the most general unifier of $P(f(x), g(y), y)$ and $P(f(g(z)), w, z)$, where x, y, z, w are variables?

- ☐ (A) $\{x/g(z), w/g(z), y/z\}$
☐ (B) $\{x/g(z), w/g(y)\}$
☐ (C) $\{x/g(a), w/g(a), y/a, z/a\}$
☐ (D) The atoms are not unifiable

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists y \forall x R(x, y)$
☐ (B) $\exists x R(x, x)$
☐ (C) $\forall y \exists x R(x, y)$
☐ (D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg S \leftrightarrow (S \vee R)$
 (2) $(\neg R \leftrightarrow P) \leftrightarrow (P \rightarrow R)$
 (3) $\neg S \wedge (S \vee \neg Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $S \wedge P$
☐ (B) $R \rightarrow \neg P$
☐ (C) $\neg P \leftrightarrow (R \vee R)$
☐ (D) $(\neg S \wedge S) \wedge Q$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $\neg(Q \wedge R) \leftrightarrow (\neg Q \wedge \neg R)$
☐ (B) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$
☐ (C) $(\neg Q \rightarrow R) \vee (R \rightarrow \neg Q)$
☐ (D) $((R \rightarrow Q) \wedge (P \rightarrow Q)) \rightarrow (R \wedge P)$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **D, B, C** and rules:

- R1: $B \rightarrow A$
 R2: $B \wedge F \rightarrow K$
 R3: $D \wedge C \rightarrow F$
 R4: $E \wedge K \rightarrow D$
 R5: $B \wedge J \wedge A \rightarrow D$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) E is entailed
☐ (B) F is entailed
☐ (C) H is entailed
☐ (D) J is entailed

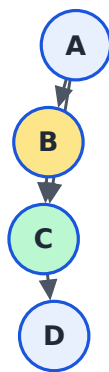
Q.21 [MCQ · 2 marks · negative: -2/3] Variable Elimination

For a Bayesian network with n Boolean variables, the depth-first ENUMERATION-ASK algorithm (inference by enumeration) has worst-case

- ☐ (A) $O(2^n)$ time and $O(n)$ space
☐ (B) $O(n)$ time and $O(2^n)$ space
☐ (C) $O(2^n)$ time and $O(2^n)$ space
☐ (D) $O(n^2)$ time and $O(n)$ space

Q.22 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence **B = T**. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $P(C = T \mid B = T)$ (round off to 3 decimal places).



P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.5	T	0.3	T	T	0.95
	F	0.4	T	F	0.75
			F	T	0.75
			F	F	0.9

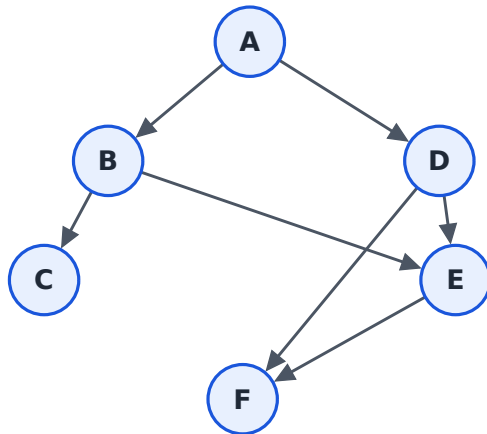
C	P(D=T ...)
T	0.4
F	0.1

#	A	C	D
1	T	T	F
2	T	T	T
3	F	F	T
4	T	T	F
5	T	T	F

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

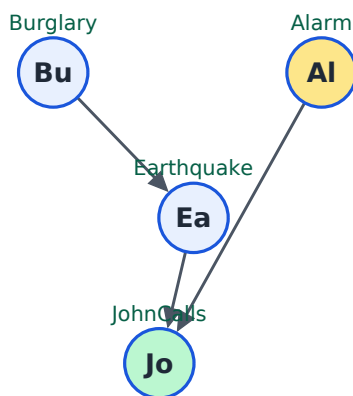
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp B$
☐ (B) $F \perp D \mid E$
☐ (C) $E \perp A \mid B, D$
☐ (D) $C \perp F \mid B, E$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute $\mathbf{P(\text{JohnCalls} = T \mid \text{Alarm} = F)}$ (round off to 3 decimal places).



P(Bu=T)
0.05

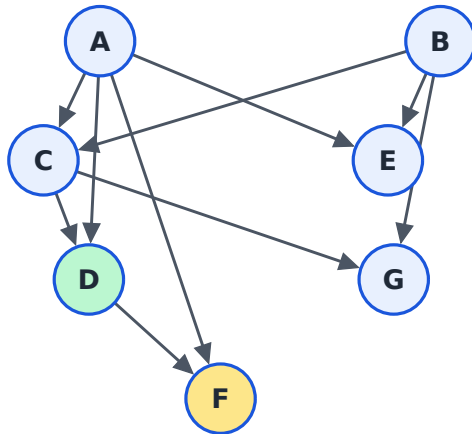
Bu	P(Ea=T ...)	P(Al=T)
T	0.95	0.3
F	0.3	

Ea	Al	P(Jo=T ...)
T	T	0.8
T	F	0.3
F	T	0.2
F	F	0.7

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(D \mid F = T)$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, C, G, E, A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Answer Key — Mock Test 053

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	NAT	2	Adversarial Search	16.24 to 16.26
2	NAT	1	Adversarial Search	79	15	NAT	2	Propositional Logic	4
3	NAT	1	Adversarial Search	362880	16	MCQ	2	Predicate Logic	A
4	NAT	1	Adversarial Search	10	17	MSQ	2	Predicate Logic	B, C
5	MSQ	1	Propositional Logic	A, C, D	18	MSQ	2	Propositional Logic	B, C
6	MSQ	1	Predicate Logic	A, B	19	MSQ	2	Propositional Logic	C
7	NAT	1	Propositional Logic	3	20	MSQ	2	Propositional Logic	B
8	MCQ	1	Conditional Independence	A	21	MCQ	2	Variable Elimination	A
9	NAT	1	Conditional Independence	5	22	NAT	2	Sampling	0.748 to 0.752
10	NAT	1	Probability for AI	47	23	MSQ	2	Conditional Independence	C, D
11	NAT	2	Informed Search	17	24	NAT	2	Exact Inference	0.565 to 0.569
12	MCQ	2	Uninformed Search	B	25	NAT	2	Exact Inference	4
13	MCQ	2	Uninformed Search	B					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: D** (D: S, B, A, E, G)

Adjacency lists (alphabetical): A: {B, E, H, K}; B: {A, H, S}; E: {A, G, I, K}; G: {E}; H: {A, B, S}; I: {E}; K: {A, E, S}; S: {B, H, K}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, B, A, E, G**; the path found is $S \rightarrow B \rightarrow A \rightarrow E \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] Answer: 79

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

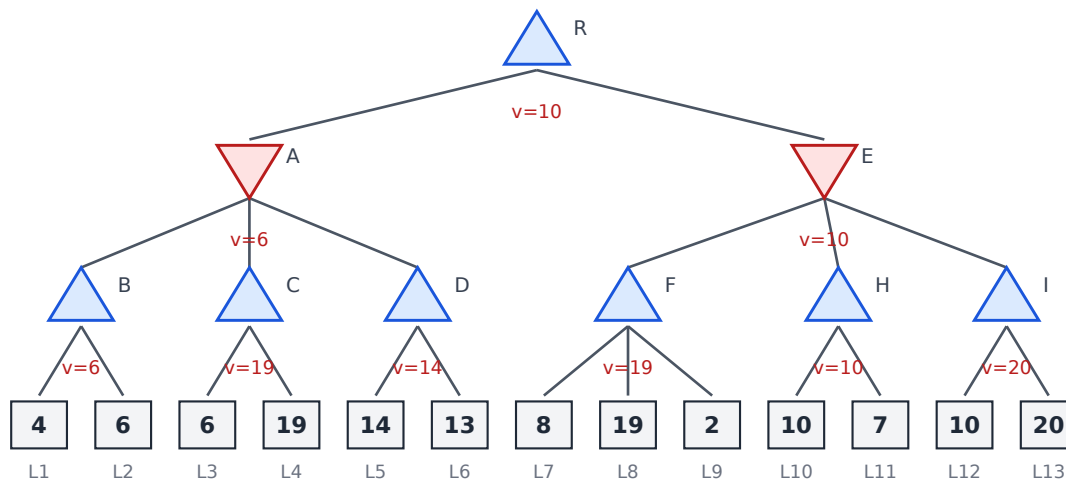
$= 4^3 + 4^2 - 1 = 64 + 16 - 1 = 79$, versus 1024 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [NAT, 1M · Adversarial Search] Answer: 362880

The first move has 9 choices, the second 8, and so on, giving $9! = 362880$ sequences. This is AIMA's upper bound on the size of the tic-tac-toe game tree. Games that end early make the actual tree smaller.

Q.4 [NAT, 1M · Adversarial Search] Answer: 10

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 6$, $C = 19$, $D = 14$, $A = 6$, $F = 19$, $H = 10$, $I = 20$, $E = 10$. Root value = **10**; MAX's optimal first move is towards E.

Q.5 [MSQ, 1M · Propositional Logic] Answer: A, C, D

(A) True: if $KB \wedge \neg\alpha$ is unsatisfiable, resolution derives the empty clause. (B) False: from $p \vee q$, $p \rightarrow r$, $q \rightarrow r$, modus ponens cannot derive r , although r is entailed. (C) True for definite clauses (Horn KBs with exactly one positive literal per clause). (D) This is the definition of soundness (truth-preserving).

Q.6 [MSQ, 1M · Predicate Logic] Answer: A, B (A: $\neg\forall x (Graph(x) \rightarrow Connected(x))$; B: $\exists x (Graph(x) \wedge \neg Connected(x))$)

Option A is the direct translation, and pushing the negation inward gives $\exists x \neg(Graph(x) \rightarrow Connected(x)) \equiv \exists x (Graph(x) \wedge \neg Connected(x))$ — option B. Option C says no graph is connected, which is stronger. Option D is true whenever any non-graph exists (vacuous implication), so it does not express the statement.

Q.7 [NAT, 1M · Propositional Logic] Answer: 3

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \vee \neg R) \wedge (R \vee Q)$	$\neg P$	$((R \vee \neg R) \wedge (R \vee Q)) \wedge \neg P$
T	T	T	T	F	F
T	T	F	T	F	F
T	F	T	T	F	F
T	F	F	F	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	F	T	F

Rows where F = T: **3**.

Q.8 [MCQ, 1M · Conditional Independence] Answer: A (A: $P(A | B, C) = P(A | C)$)

Conditional independence of A and B given C means $P(A, B | C) = P(A | C)P(B | C)$, equivalently $P(A | B, C) = P(A | C)$. (B) is marginal independence, which neither implies nor is implied by $A \perp B | C$. (C) compares two different variables' distributions and is meaningless here. (D) is mutual independence, a much stronger condition.

Q.9 [NAT, 1M · Conditional Independence] Answer: 5

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of E: {B, D}; children: {F, G}; co-parents: {C}.

MB(E) = {B, C, D, F, G} — **5** nodes. Given its Markov blanket, E is conditionally independent of every other node in the network.

Q.10 [NAT, 1M · Probability for AI] Answer: 47

The full joint table has $2^4 \times 3 = 48$ entries. Because the entries must sum to 1, one of them is determined by the rest, giving $48 - 1 = 47$ independent numbers. No independence assumptions are made, so no further reduction is possible.

Q.11 [NAT, 2M · Informed Search] Answer: 17

Heuristic values: $h(A)=4$, $h(D)=9$, $h(F)=14$, $h(G)=0$, $h(H)=19$, $h(I)=10$, $h(K)=8$, $h(L)=6$, $h(S)=15$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	D(15), F(17), H(24)
2	D (15)	L(16), F(17), I(20), H(24)
3	L (16)	F(17), G(17), I(20), A(23), H(24)
4	F (17)	G(17), I(17), A(23), H(24)
5	G (17)	I(17), A(23), H(24) ← goal, stop

Path returned: **S** → **D** → **L** → **G** with cost **17**; nodes expanded = **5** (S, D, L, F, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 17$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.12 [MCQ, 2M · Uninformed Search] Answer: B (B: $O(b^{d+1})$)

With a late goal test, BFS expands every node at depth d that precedes the goal and generates their children at depth $d+1$. That adds up to $b^{d+1} - b$ nodes, so the bound is **$O(b^{d+1})$** . Early testing (at generation) gives $O(b^d)$ (A). $O(b^{d/2})$ is the bidirectional bound (C). $O(bd)$ is a space bound for IDS (D).

Q.13 [MCQ, 2M · Uninformed Search] Answer: B (B: $S \rightarrow C \rightarrow J \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	C(6), E(7), I(9)
2	C (6)	E(7), I(9), J(11)
3	E (7)	I(9), J(11)
4	I (9)	J(11), H(16)
5	J (11)	G(15), H(16)
6	G (15)	H(16) ← goal, stop

Path returned: **S** → **C** → **J** → **G**, cost = $6 + 5 + 4 = \mathbf{15}$. Nodes expanded (including S and G): **6** (S, C, E, I, J, G).

Q.14 [NAT, 2M · Adversarial Search] Answer: 16.24 to 16.26

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.5 \times 18 + 0.25 \times 18 + 0.25 \times 11 = 16.25$
- $B = 0.2 \times 12 + 0.3 \times 3 + 0.4 \times 3 + 0.1 \times 14 = 5.9$
- $C = 0.25 \times 1 + 0.25 \times 20 + 0.5 \times 5 = 7.75$
- $D = 0.5 \times 10 + 0.5 \times 1 = 5.5$

Root = $\max(16.25, 5.9, 7.75, 5.5) = \mathbf{16.25}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [NAT, 2M · Propositional Logic] Answer: 4

$(p \circ q) \rightarrow p$ can fail only in rows where $p = F$, and it fails there iff $p \circ q = T$. So we need $F \circ T = F$ and $F \circ F = F$, while $T \circ T$ and $T \circ F$ are free. That gives $2^2 = 4$ connectives: constant F , $p \wedge q$, $p \wedge \neg q$, and the projection p .

Q.16 [MCQ, 2M · Predicate Logic] Answer: A ($A: \{x/g(z), w/g(z), y/z\}$)

$f(x)$ vs $f(g(z))$ gives $x/g(z)$. $g(y)$ vs w gives $w/g(y)$. y vs z gives y/z , and composing updates $w/g(y)$ to $w/g(z)$. Both atoms become $P(f(g(z)), g(z), z)$. Option B is not a unifier because y and z remain different. Option C is a unifier but not most general (it is an instance of A with z/a). No occurs-check failure arises, so D is wrong.

Q.17 [MSQ, 2M · Predicate Logic] Answer: B, C ($B: \exists x R(x, x)$; $C: \forall y \exists x R(x, y)$)

$R = \{(1,3), (1,4), (2,1), (2,2), (2,3), (4,1)\}$.

(A) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

(C) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(2,1) \wedge R(1,4)$ but $\neg R(2,4)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, C ($B: R \rightarrow \neg P$; $C: \neg P \Leftrightarrow (R \vee R)$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 16 assignments, KB is true in exactly 1 of them:

P	Q	R	S	KB
F	F	T	F	T

(A) $S \wedge P$ is false in the KB-model $P=F, Q=F, R=T, S=F \Rightarrow$ not entailed.

(B) $R \rightarrow \neg P$ is true in all models of $KB \Rightarrow$ **entailed**.

(C) $\neg P \Leftrightarrow (R \vee R)$ is true in all models of $KB \Rightarrow$ **entailed**.

(D) $(\neg S \wedge S) \wedge Q$ is false in the KB-model $P=F, Q=F, R=T, S=F \Rightarrow$ not entailed.

Q.19 [MSQ, 2M · Propositional Logic] Answer: C ($C: (\neg Q \rightarrow R) \vee (R \rightarrow \neg Q)$)

(A) $\neg(Q \wedge R) \Leftrightarrow (\neg Q \wedge \neg R)$: **not valid** — counter-model $Q=T, R=F$.

(B) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$: **not valid** — counter-model $Q=F, R=T$.

(C) $(\neg Q \rightarrow R) \vee (R \rightarrow \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((R \rightarrow Q) \wedge (P \rightarrow Q)) \rightarrow (R \wedge P)$: **not valid** — counter-model $P=T, Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.20 [MSQ, 2M · Propositional Logic] Answer: B ($B: F$ is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $B \rightarrow A, D \wedge C \rightarrow F$; add A, F .

Round 2: fire $B \wedge F \rightarrow K$; add K .

Fixed point: $\{A, B, C, D, F, K\}$ — 6 atoms are entailed (facts: B, C, D ; newly derived: A, F, K). Atoms never derived: E, H, J .

Q.21 [MCQ, 2M · Variable Elimination] Answer: A ($A: O(2^n)$ time and $O(n)$ space)

Enumeration evaluates a depth-first expression tree that branches over every hidden variable, giving $O(2^n)$ time. Being depth-first it stores only the current path, so space is $O(n)$. It never builds the full joint table, so it does not need $O(2^n)$ space; polynomial time is impossible in general since inference is NP-hard. Variable elimination improves time by caching repeated subexpressions as factors.

Q.22 [NAT, 2M · Sampling] Answer: 0.748 to 0.752

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	C
1	$P(B=T A=T)=0.3$	0.3	T
2	$P(B=T A=T)=0.3$	0.3	T
3	$P(B=T A=F)=0.4$	0.4	F
4	$P(B=T A=T)=0.3$	0.3	T
5	$P(B=T A=T)=0.3$	0.3	T

$$\sum w(\text{all}) = 1.6; \sum w(C = T) = 1.2.$$

$$\text{Estimate } P(C=T \mid B = T) = 1.2 / 1.6 = \mathbf{0.75}.$$

Q.23 [MSQ, 2M · Conditional Independence] **Answer: C, D** (C: $E \perp A \mid B, D$; D: $C \perp F \mid B, E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp B$: **FALSE** — the path $C \leftarrow B$ is active given $\{\}$.

(B) $F \perp D \mid E$: **FALSE** — the path $F \leftarrow D$ is active given $\{E\}$.

(C) $E \perp A \mid B, D$: **TRUE** — $E \leftarrow B \leftarrow A$ (blocked: observed chain node B); $E \leftarrow D \leftarrow A$ (blocked: observed chain node D); $E \rightarrow F \leftarrow D \leftarrow A$ (blocked: collider F with neither it nor a descendant observed).

(D) $C \perp F \mid B, E$: **TRUE** — $C \leftarrow B \leftarrow A \rightarrow D \rightarrow E \rightarrow F$ (blocked: observed chain node B); $C \leftarrow B \leftarrow A \rightarrow D \rightarrow F$ (blocked: observed chain node B); $C \leftarrow B \rightarrow E \leftarrow D \rightarrow F$ (blocked: observed fork node B); $C \leftarrow B \rightarrow E \rightarrow F$ (blocked: observed fork node B).

Q.24 [NAT, 2M · Exact Inference] **Answer: 0.565 to 0.569**

Factorisation: $P(\text{Burglary, Earthquake, Alarm, JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm}) \cdot P(\text{JohnCalls} \mid \text{Earthquake, Alarm})$.

Inference by enumeration: $P(\text{JohnCalls} \mid e) = \alpha \sum_{\text{hidden}} P(\text{JohnCalls}, e, \text{hidden})$.

Hidden: Bu, Ea	Jo	product of CPT entries (Bu · Ea · Al · Jo)	value
T, T	T	$0.05 \times 0.95 \times 0.7 \times 0.3$	0.00997
T, F	T	$0.05 \times 0.05 \times 0.7 \times 0.7$	0.00123
F, T	T	$0.95 \times 0.3 \times 0.7 \times 0.3$	0.05985
F, F	T	$0.95 \times 0.7 \times 0.7 \times 0.7$	0.32585
T, T	F	$0.05 \times 0.95 \times 0.7 \times 0.7$	0.02327
T, F	F	$0.05 \times 0.05 \times 0.7 \times 0.3$	0.00053
F, T	F	$0.95 \times 0.3 \times 0.7 \times 0.7$	0.13965
F, F	F	$0.95 \times 0.7 \times 0.7 \times 0.3$	0.13965

$$P(\text{JohnCalls}=T, e) = 0.3969; P(\text{JohnCalls}=F, e) = 0.3031; P(e) = 0.7.$$

$$P(\text{JohnCalls}=T \mid e) = 0.3969 / 0.7 = \mathbf{0.567} \approx 0.567.$$

Q.25 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, B, C)$, $f(A, C, D)$, $f(A, B, E)$, $f(A, D)$, $f(B, C, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(B)$, $f(A, B, C)$, $f(A, B, E)$, $f(B, C, G)$	$\{A, B, C, E, G\}$	$f(A, C, E, G)$
C	$f(A, C, D)$, $f(A, C, E, G)$	$\{A, C, D, E, G\}$	$f(A, D, E, G)$
G	$f(A, D, E, G)$	$\{A, D, E, G\}$	$f(A, D, E)$
E	$f(A, D, E)$	$\{A, D, E\}$	$f(A, D)$
A	$f(A)$, $f(A, D)$, $f(A, D)$	$\{A, D\}$	$f(D)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over D and normalise.

Mock Test 054

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 6 / 13	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

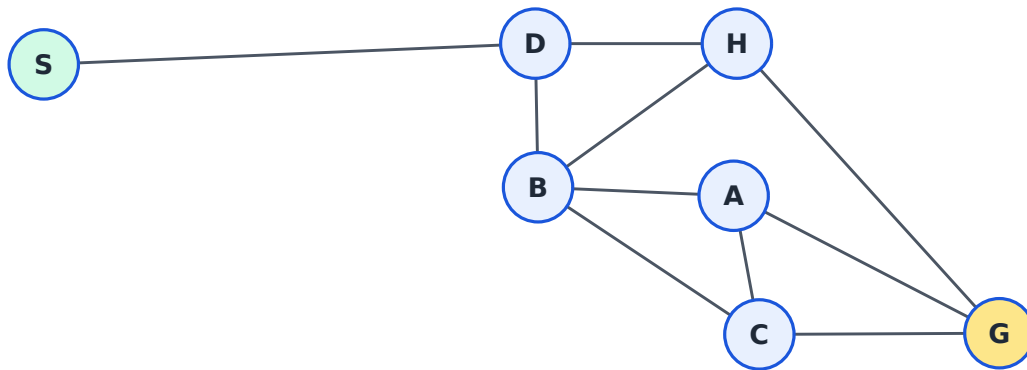
Q.1 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Depth-limited search with limit $l = 5$ is run on a tree with branching factor $b = 3$. No goal exists within depth 5. How many nodes (including the root) does it generate before returning cutoff?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

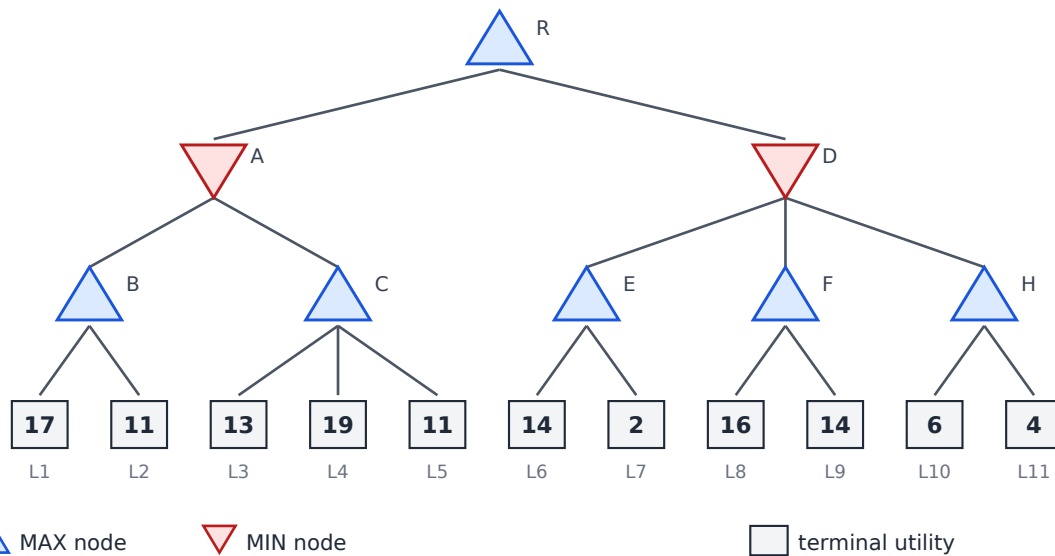
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, H, G
- ☐ (B) S, D, B, H, A, C, G
- ☐ (C) S, D, B, A, C, G
- ☐ (D) S, D, A, B, C, G

Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to A; root value 20
☐ (B) Move to A; root value 19
☐ (C) Move to A; root value 17
☐ (D) Move to D; root value 6

Q.4 [MCQ · 1 mark · negative: -1/3] Problem Formulation

In AIMA, a search problem is formally defined by

- ☐ (A) initial state, actions, transition model, goal test, and path (action) cost function
☐ (B) initial state and goal state only
☐ (C) a heuristic function and an evaluation function
☐ (D) a frontier and an explored set

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

The clauses $A \vee B \vee \neg C$ and $\neg A \vee B \vee D$ are resolved on the symbol A. How many literals are in the resolvent (after merging duplicate literals)?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, A** and rules:

- R1: $J \wedge B \rightarrow H$
 R2: $K \wedge C \rightarrow B$
 R3: $K \rightarrow E$
 R4: $A \rightarrow K$
 R5: $A \wedge B \wedge E \rightarrow F$
 R6: $H \rightarrow C$
 R7: $K \wedge F \wedge H \rightarrow B$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic

Interpret $\text{Div}(x, y)$ as 'x divides y' over the domain $D = \{1, 2, \dots, 12\}$. For how many $x \in D$ is the formula $\neg(x = 1) \wedge \forall y (\text{Div}(y, x) \rightarrow (y = 1 \vee y = x))$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Sampling

A continuous variable X has exponential density $f(x) = 2e^{-2x}$ for $x \geq 0$. Using inverse-CDF sampling with uniform random number $u = 0.75$, what value of X is generated? (Round to 3 decimal places.)

Answer (numerical): _____

Q.9 [MSQ · 1 mark · no negative marking] Conditional Independence

Which of the following statements about d-separation in three-node structures are correct?

- ☐ (A) In the chain $X \rightarrow Y \rightarrow Z$, X and Z are d-separated given Y
- ☐ (B) In the fork $X \leftarrow Y \rightarrow Z$, X and Z are d-separated given Y
- ☐ (C) In the collider $X \rightarrow Y \leftarrow Z$, X and Z are d-separated given Y
- ☐ (D) In the collider $X \rightarrow Y \leftarrow Z$, X and Z are d-separated given $\emptyset$

Q.10 [MCQ · 1 mark · negative: -1/3] Probability for AI

While answering a query $P(\text{Cavity} \mid \text{toothache})$, the unnormalized vector obtained for $\text{Cavity} = \langle \text{true}, \text{false} \rangle$ is $\langle 0.12, 0.08 \rangle$. Which option gives the normalization constant α and the resulting posterior distribution?

- ☐ (A) $\alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.6, 0.4 \rangle$
- ☐ (B) $\alpha = 0.2$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.024, 0.016 \rangle$
- ☐ (C) $\alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.4, 0.6 \rangle$
- ☐ (D) $\alpha = 2$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.24, 0.16 \rangle$

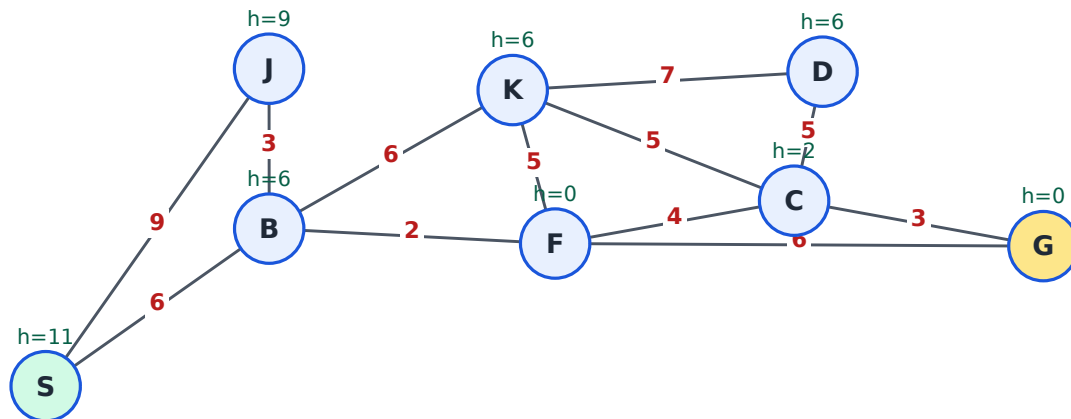
Section B — 2-mark questions**Q.11** [MCQ · 2 marks · negative: -2/3] Uninformed Search

Depth-limited search (DLS) with limit l is applied to a problem with unit step costs whose shallowest goal lies at depth d . Which statement is correct?

- ☐ (A) If $l < d$, DLS is incomplete. If $l > d$, it finds a solution but not necessarily the shallowest (optimal) one
- ☐ (B) If $l < d$, DLS still finds the goal because it backtracks
- ☐ (C) If $l > d$, DLS is always optimal because it searches deeper than needed
- ☐ (D) DLS is complete and optimal for every $l \geq 0$

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

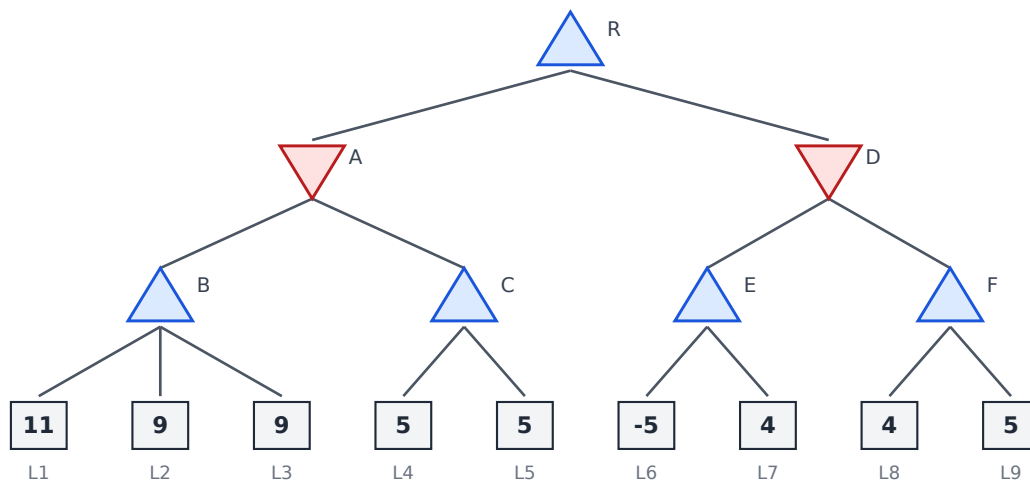
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(C)$ overestimates the true cost from C to G.
☐ (B) h is consistent (monotone).
☐ (C) h is admissible.
☐ (D) The edge B-F violates the consistency condition $h(B) \leq c(B,F) + h(F)$.

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



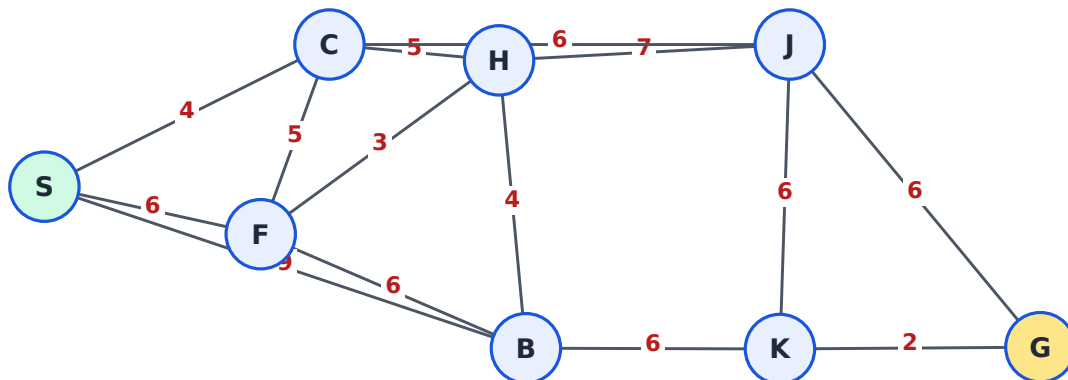
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = \neg P \leftrightarrow (\neg Q \vee \neg R)$?

- ☐ (A) $(\neg P \wedge (\neg Q \vee \neg R)) \vee (\neg P \wedge \neg(\neg Q \vee \neg R))$
- ☐ (B) $(\neg P \rightarrow (\neg Q \vee \neg R)) \wedge ((\neg Q \vee \neg R) \rightarrow \neg P)$
- ☐ (C) $(\neg P \rightarrow (\neg Q \vee \neg R)) \wedge (\neg(\neg Q \vee \neg R) \rightarrow \neg\neg P)$
- ☐ (D) $(\neg P \wedge (\neg Q \vee \neg R)) \vee (\neg\neg P \wedge \neg(\neg Q \vee \neg R))$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

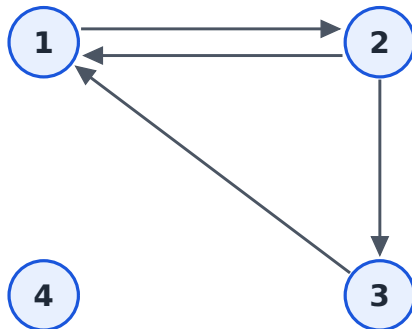
- (1) $(Q \leftrightarrow R) \wedge (\neg P \leftrightarrow Q)$
- (2) $R \rightarrow (R \vee \neg R)$
- (3) $\neg Q$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $Q \leftrightarrow Q$
- ☐ (B) $(Q \wedge P) \vee (R \rightarrow \neg P)$
- ☐ (C) $(\neg Q \vee \neg Q) \rightarrow (Q \vee R)$
- ☐ (D) $Q \vee Q$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
☐ (B) $\exists x \forall y \neg R(x, y)$
☐ (C) $\forall y \exists x R(x, y)$
☐ (D) $\exists x R(x, x)$

Q.18 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

The formula $F = ((S \rightarrow \neg Q) \rightarrow (P \vee P)) \leftrightarrow ((R \vee \neg Q) \vee P)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R, S . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following first-order sentences are valid?

- ☐ (A) $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$
☐ (B) $\forall y \exists x P(x, y) \rightarrow \exists x \forall y P(x, y)$
☐ (C) $\forall x (P(x) \vee Q(x)) \rightarrow (\forall x P(x) \vee \forall x Q(x))$
☐ (D) $\exists x (P(x) \wedge Q(x)) \rightarrow (\exists x P(x) \wedge \exists x Q(x))$

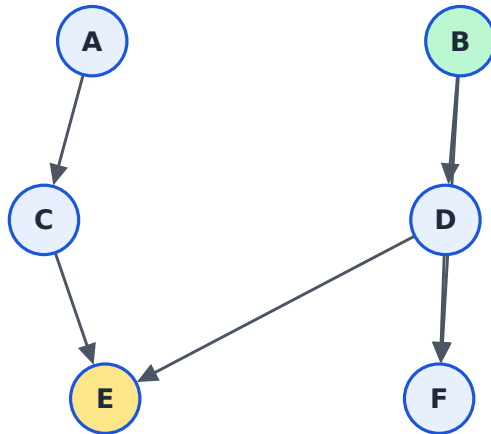
Q.20 [NAT · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. Suppose a new network is built for the same distribution by adding nodes in the order M, J, A, B, E , choosing for each node a minimal set of parents among the previously added nodes such that the node is conditionally independent of its other predecessors given its parents. How many independent CPT parameters does this new network have (all variables Boolean)? Assume no independences beyond those implied by the original structure.

Answer (numerical): _____

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{B} \mid \mathbf{E} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{E} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{F}$, $\mathbf{D}$, $\mathbf{A}$, $\mathbf{C}$. What is the scope of the new factor produced when $\mathbf{D}$ is summed out?



- ☐ (A) $f(\mathbf{B})$
☐ (B) $f(\mathbf{B}, \mathbf{C})$
☐ (C) $f(\mathbf{B}, \mathbf{C}, \mathbf{D})$
☐ (D) $f(\mathbf{C})$

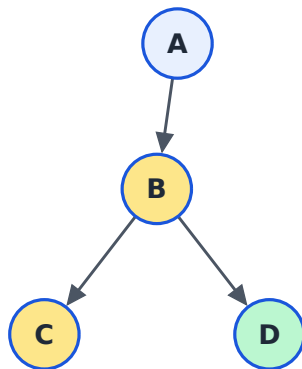
Q.22 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. Gibbs sampling with evidence $S = \text{true}$, $W = \text{true}$ starts in state $C = \text{true}$, $R = \text{false}$. One sweep first resamples C (given its Markov blanket), then resamples R (given its Markov blanket, using the new C). What is the probability that $R = \text{true}$ at the end of the sweep? (Round to 3 decimal places.)

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{C = T, B = F}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(D = T \mid C = T, B = F)}$ (round off to 3 decimal places).



P(A=T)
0.5

A	P(B=T ...)	B	P(C=T ...)
T	0.4	T	0.9
F	0.6	F	0.4

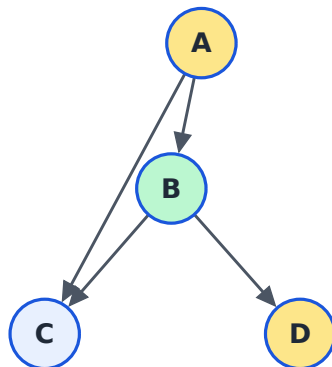
B	P(D=T ...)
T	0.05
F	0.6

#	A	D
1	F	T
2	F	T
3	F	T
4	T	T
5	F	F

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $\mathbf{P(B = T \mid A = F, D = T)}$ (round off to 3 decimal places).



P(A=T)
0.8

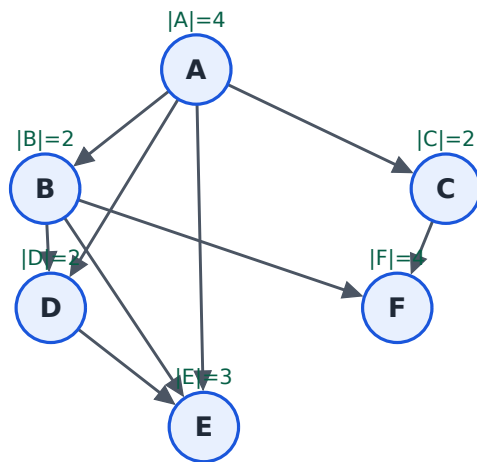
A	P(B=T ...)	A	B	P(C=T ...)
T	0.2	T	T	0.05
F	0.8	T	F	0.05
		F	T	0.95
		F	F	0.1

B	P(D=T ...)
T	0.9
F	0.7

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Answer Key — Mock Test 054

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	364	14	NAT	2	Uninformed Search	8
2	MCQ	1	Uninformed Search	C	15	MSQ	2	Propositional Logic	B, D
3	MCQ	1	Adversarial Search	C	16	MSQ	2	Propositional Logic	A, B
4	MCQ	1	Problem Formulation	A	17	MSQ	2	Predicate Logic	A, B
5	NAT	1	Propositional Logic	3	18	NAT	2	Propositional Logic	6
6	NAT	1	Propositional Logic	5	19	MSQ	2	Predicate Logic	A, D
7	NAT	1	Predicate Logic	5	20	NAT	2	Bayesian Networks	13
8	NAT	1	Sampling	0.692 to 0.694	21	MCQ	2	Exact Inference	B
9	MSQ	1	Conditional Independence	A, B, D	22	NAT	2	Sampling	0.244 to 0.245
10	MCQ	1	Probability for AI	A	23	NAT	2	Sampling	0.816 to 0.82
11	MCQ	2	Uninformed Search	A	24	NAT	2	Exact Inference	0.835 to 0.839
12	MSQ	2	Informed Search	C, D	25	NAT	2	Bayesian Networks	320
13	NAT	2	Adversarial Search	2					

Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] Answer: 364

DLS explores the complete tree down to depth l : $1 + b + \dots + b^l = 1 + 3 + 9 + 27 + 81 + 243 = 364 = (b^{(l+1)} - 1)/(b - 1)$.

Q.2 [MCQ, 1M · Uninformed Search] Answer: C (C: S, D, B, A, C, G)

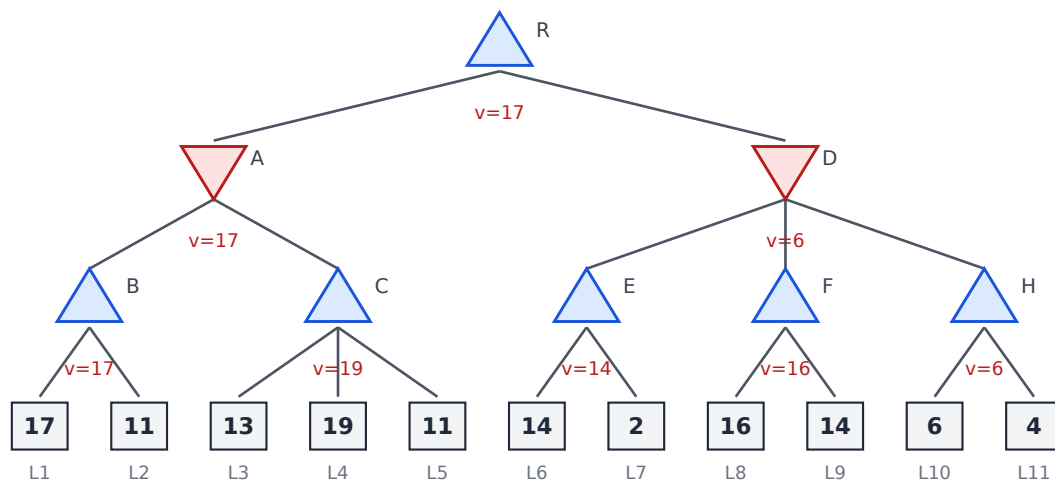
Adjacency lists (alphabetical): A: {B, C, G}; B: {A, C, D, H}; C: {A, B, G}; D: {B, H, S}; G: {A, C, H}; H: {B, D, G}; S: {D}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, B, A, C, G**; the path found is $S \rightarrow D \rightarrow B \rightarrow A \rightarrow C \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.3 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to A; root value 17)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 17$, $C = 19$, $A = 17$, $E = 14$, $F = 16$, $H = 6$, $D = 6$. Root value = **17**; MAX's optimal first move is towards A.

Q.4 [MCQ, 1M · Problem Formulation] **Answer: A** (A: initial state, actions, transition model, goal test, and path (action) cost function)

A problem consists of the **initial state**, the **actions available in each state**, the **transition model** **RESULT(s,a)**, the **goal test**, and the **action-cost / path-cost function**. Together these define the state space implicitly. (B) leaves out actions and costs. (C) belongs to particular algorithms, not to the problem. (D) is the data structure of a search algorithm.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 3**

Remove A and $\neg A$ and take the union of the remaining literals: $\{B, \neg C\} \cup \{B, D\} = \{B, \neg C, D\}$. The duplicate B is merged (factoring), so the resolvent $B \vee \neg C \vee D$ has 3 literals.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 5**

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $A \rightarrow K$; add K.

Round 2: fire $K \rightarrow E$; add E.

Round 3: fire $A \wedge B \wedge E \rightarrow F$; add F.

Fixed point: $\{A, B, E, F, K\}$ — 5 atoms are entailed (facts: A, B; newly derived: E, F, K). Atoms never derived: C, D, H, J.

Q.7 [NAT, 1M · Predicate Logic] **Answer: 5**

The formula says $x \neq 1$ and the only divisors of x in D are 1 and x itself — i.e. x is prime. (Every divisor of $x \leq 12$ lies in D, so the restriction to D loses nothing.) The primes in $\{1, \dots, 12\}$ are 2, 3, 5, 7, 11, so the answer is 5.

Q.8 [NAT, 1M · Sampling] **Answer: 0.692 to 0.694**

The CDF is $F(x) = 1 - e^{-2x}$. Solving $F(x) = u$ gives $x = -\ln(1 - u)/2$. With $u = 0.75$: $x = -\ln(0.25)/2 = \ln 4/2 = 1.3863/2 \approx 0.693$.

Q.9 [MSQ, 1M · Conditional Independence] **Answer: A, B, D**

An observed non-collider blocks a path, so (A) and (B) are true. A collider blocks the path when neither it nor any descendant is observed, so (D) is true. Observing the collider activates the path, so (C) is false.

Q.10 [MCQ, 1M · Probability for AI] **Answer: A** (A: $\alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \{0.6, 0.4\}$)

The entries must sum to 1 after scaling, so $\alpha = 1/(0.12 + 0.08) = 1/0.20 = 5$. Then $\alpha(0.12, 0.08) = \{0.6, 0.4\}$. $\alpha = 0.2$ is the sum itself (multiplying by it does not normalize), $\{0.4, 0.6\}$ swaps the order of the values, and $\alpha = 2$ gives entries summing to 0.4, not 1.

Q.11 [MCQ, 2M · Uninformed Search] Answer: A (A: If $l < d$, DLS is incomplete. If $l > d$, it finds a solution but not necessarily the shallowest (optimal) one)

DLS treats nodes at depth l as having no successors. If $l < d$, the goal can never be reached, so DLS is incomplete; backtracking does not help, so (B) is wrong. If $l > d$, a goal is reachable, but DLS explores depth-first and may meet a deeper goal (depth between d and l) before the shallowest one, so it is not optimal and (C) is wrong. (D) is wrong for both reasons.

Q.12 [MSQ, 2M · Informed Search] Answer: C, D (C: h is admissible.; D: The edge B-F violates the consistency condition $h(B) \leq c(B,F) + h(F)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	D	F	G	J	K	S
$h(n)$	6	2	6	0	0	9	6	11
$h^*(n)$	8	3	8	6	0	11	8	14
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges:

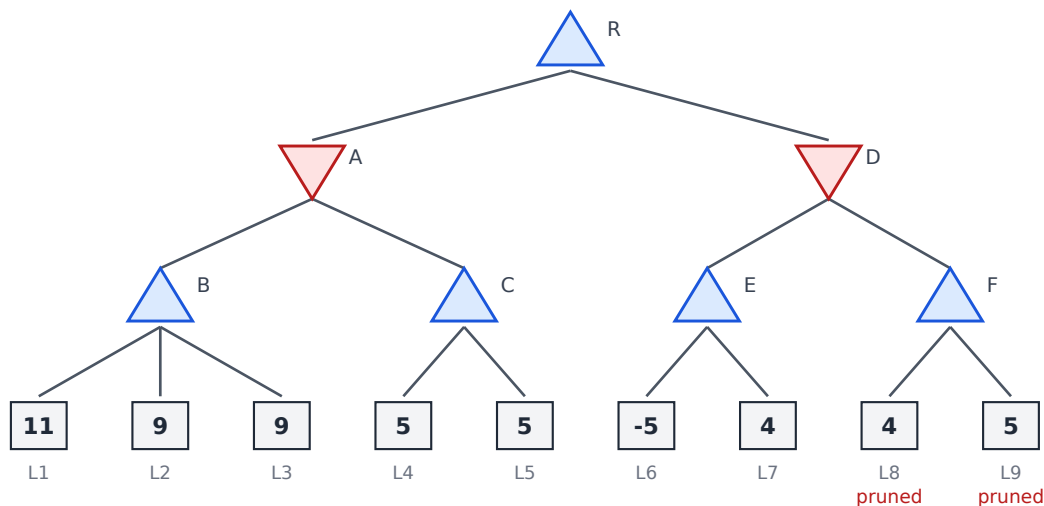
$K \rightarrow F$ ($6 > 5 + 0$), $B \rightarrow F$ ($6 > 2 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Adversarial Search] Answer: 2

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $4 \leq \alpha = 5 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (7): L1, L2, L3, L4, L5, L6, L7. Leaves pruned (2): L8, L9. Root value = 5 (same as plain minimax, 5 — pruning never changes the root value).

Q.14 [NAT, 2M · Uninformed Search] Answer: 8

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	C(4), F(6), B(9)
2	C (4)	F(6), B(9), H(9), J(10)
3	F (6)	B(9), H(9), J(10)
4	B (9)	H(9), J(10), K(15)
5	H (9)	J(10), K(15)
6	J (10)	K(15), G(16)
7	K (15)	G(16)
8	G (16)	← goal, stop

Path returned: **S** $\rightarrow$ **C** $\rightarrow$ **J** $\rightarrow$ **G**, cost = $4 + 6 + 6 = 16$. Nodes expanded (including S and G): **8** (S, C, F, B, H, J, K, G).

Q.15 [MSQ, 2M · Propositional Logic] **Answer: B, D** (B: $(\neg P \rightarrow (\neg Q \vee \neg R)) \wedge ((\neg Q \vee \neg R) \rightarrow \neg P)$; D: $(\neg P \wedge (\neg Q \vee \neg R)) \vee (\neg \neg P \wedge \neg(\neg Q \vee \neg R))$)

Compare truth tables with $F = \neg P \leftrightarrow (\neg Q \vee \neg R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(\neg P \wedge (\neg Q \vee \neg R)) \vee (\neg P \wedge \neg(\neg Q \vee \neg R))$ — differs at $P=T, Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(B) $(\neg P \rightarrow (\neg Q \vee \neg R)) \wedge ((\neg Q \vee \neg R) \rightarrow \neg P)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(C) $(\neg P \rightarrow (\neg Q \vee \neg R)) \wedge (\neg(\neg Q \vee \neg R) \rightarrow \neg \neg P)$ — differs at $P=T, Q=T, R=F$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(D) $(\neg P \wedge (\neg Q \vee \neg R)) \vee (\neg \neg P \wedge \neg(\neg Q \vee \neg R))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $Q \leftrightarrow Q$; B: $(Q \wedge P) \vee (R \rightarrow \neg P)$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
T	F	F	T

(A) $Q \leftrightarrow Q$ is true in all models of $KB \Rightarrow$ **entailed**.

(B) $(Q \wedge P) \vee (R \rightarrow \neg P)$ is true in all models of $KB \Rightarrow$ **entailed**.

(C) $(\neg Q \vee \neg Q) \rightarrow (Q \vee R)$ is false in the KB -model $P=T, Q=F, R=F \Rightarrow$ not entailed.

(D) $Q \vee Q$ is false in the KB -model $P=T, Q=F, R=F \Rightarrow$ not entailed.

Q.17 [MSQ, 2M · Predicate Logic] **Answer: A, B** (A: $\exists x \forall y \neg R(y, x)$; B: $\exists x \forall y \neg R(x, y)$)

$R = \{(1,2), (2,1), (2,3), (3,1)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

(B) $\exists x \forall y \neg R(x, y)$: **TRUE** — 4 has no outgoing edge.

(C) $\forall y \exists x R(x, y)$: **FALSE** — 4 has no incoming edge.

(D) $\exists x R(x, x)$: **FALSE** — there are no self-loops.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [NAT, 2M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(S \rightarrow \neg Q) \rightarrow (P \vee P)$	$(R \vee \neg Q) \vee P$	$((S \rightarrow \neg Q) \rightarrow (P \vee P)) \leftrightarrow ((R \vee \neg Q) \vee P)$
T	T	T	T	T	T	T
T	T	T	F	T	T	T
T	T	F	T	T	T	T
T	T	F	F	T	T	T
T	F	T	T	T	T	T
T	F	T	F	T	T	T
T	F	F	T	T	T	T
T	F	F	F	T	T	T
F	T	T	T	T	T	T
F	T	T	F	F	T	F
F	T	F	T	T	F	F
F	T	F	F	F	F	T
F	F	T	T	F	T	F
F	F	T	F	F	T	F
F	F	F	T	F	T	F
F	F	F	F	F	T	F

F has 10 models, so it is false in $16 - 10 = 6$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: A, D** (A: $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$; D: $\exists x (P(x) \wedge Q(x)) \rightarrow (\exists x P(x) \wedge \exists x Q(x))$)

(A) If some a satisfies $P(a, y)$ for all y , then for each y that same a is a witness — valid. (B) Fails on domain $\{1, 2\}$ with $P(x, y) \iff x = y$: each y has a witness, but no single x works for all y . (C) Fails on domain $\{1, 2\}$ with $P = \{1\}$, $Q = \{2\}$: every element is P or Q , but neither P nor Q holds everywhere. (D) A witness a for $P(a) \wedge Q(a)$ witnesses both $\exists x P(x)$ and $\exists x Q(x)$ — valid.

Q.20 [NAT, 2M · Bayesian Networks] Answer: 13

M: no predecessors $\rightarrow$ 1 parameter. J: not independent of M (both depend on A) $\rightarrow$ parent M $\rightarrow$ 2. A: depends on both M and J and neither screens the other $\rightarrow$ parents {M, J} $\rightarrow$ 4. B: given A, B is independent of M and J $\rightarrow$ parent {A} $\rightarrow$ 2. E: given A alone, E still depends on B (explaining away), but given {A, B} it is independent of M, J $\rightarrow$ parents {A, B} $\rightarrow$ 4. Total = 1 + 2 + 4 + 2 + 4 = 13, versus 10 for the causal ordering B, E, A, J, M.

Q.21 [MCQ, 2M · Exact Inference] Answer: B (B: f(B, C))

Initial factors (one per CPT, evidence removed from scopes): f(A), f(B), f(A, C), f(B, D), f(C, D), f(B, D, F).

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	f(B, D, F)	{B, D, F}	f(B, D)
D	f(B, D), f(C, D), f(B, D)	{B, C, D}	f(B, C)
A	f(A), f(A, C)	{A, C}	f(C)
C	f(B, C), f(C)	{B, C}	f(B)

The largest factor produced has **2** variable(s) (created when eliminating F); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over B and normalise.

Q.22 [NAT, 2M · Sampling] Answer: 0.244 to 0.245

Step 1: $P(c | s, \neg r) = \alpha P(c)P(s | c)P(\neg r | c)$: c : $0.5 \times 0.1 \times 0.2 = 0.01$; $\neg c$: $0.5 \times 0.5 \times 0.8 = 0.2 \rightarrow P(c) = 0.01/0.21 \approx 0.04762$. Step 2: $P(r | c, s, w) = 0.792/0.972 \approx 0.81481$; $P(r | \neg c, s, w) = (0.2 \times 0.99)/(0.2 \times 0.99 + 0.8 \times 0.90) = 0.198/0.918 \approx 0.21569$. Total: $0.04762 \times 0.81481 + 0.95238 \times 0.21569 \approx 0.03880 + 0.20542 = 0.2442 \approx 0.244$.

Q.23 [NAT, 2M · Sampling] Answer: 0.816 to 0.82

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	D
1	$P(B=F A=F)=0.4 \times P(C=T B=F)=0.4$	0.16	T
2	$P(B=F A=F)=0.4 \times P(C=T B=F)=0.4$	0.16	T
3	$P(B=F A=F)=0.4 \times P(C=T B=F)=0.4$	0.16	T
4	$P(B=F A=T)=0.6 \times P(C=T B=F)=0.4$	0.24	T
5	$P(B=F A=F)=0.4 \times P(C=T B=F)=0.4$	0.16	F

$\Sigma w(\text{all}) = 0.88$; $\Sigma w(D=T) = 0.72$.

Estimate $P(D=T | C=T, B=F) = 0.72 / 0.88 = \mathbf{0.8182}$.

Q.24 [NAT, 2M · Exact Inference] Answer: 0.835 to 0.839

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B | A) \cdot P(C | A, B) \cdot P(D | B)$.

Inference by enumeration: $P(B | e) = \alpha \sum_{\text{hidden}} P(B, e, \text{hidden})$.

Hidden: C	B	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T	T	$0.2 \times 0.8 \times 0.95 \times 0.9$	0.1368
F	T	$0.2 \times 0.8 \times 0.05 \times 0.9$	0.0072
T	F	$0.2 \times 0.2 \times 0.1 \times 0.7$	0.0028
F	F	$0.2 \times 0.2 \times 0.9 \times 0.7$	0.0252

$P(B=T, e) = 0.144$; $P(B=F, e) = 0.028$; $P(e) = 0.172$.

$P(B=T | e) = 0.144 / 0.172 = \mathbf{0.8372} \approx 0.837$.

Q.25 [NAT, 2M · Bayesian Networks] Answer: 320

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (4-1) + B: (2-1) \times 4 + C: (2-1) \times 4 + D: (2-1) \times 4 \times 2 + E: (3-1) \times 4 \times 2 \times 2 + F: (4-1) \times 2 \times 2 = \mathbf{63}$.

Full joint table: $\prod |X| - 1 = 384 - 1 = 383$; saving = 320.

Mock Test 055

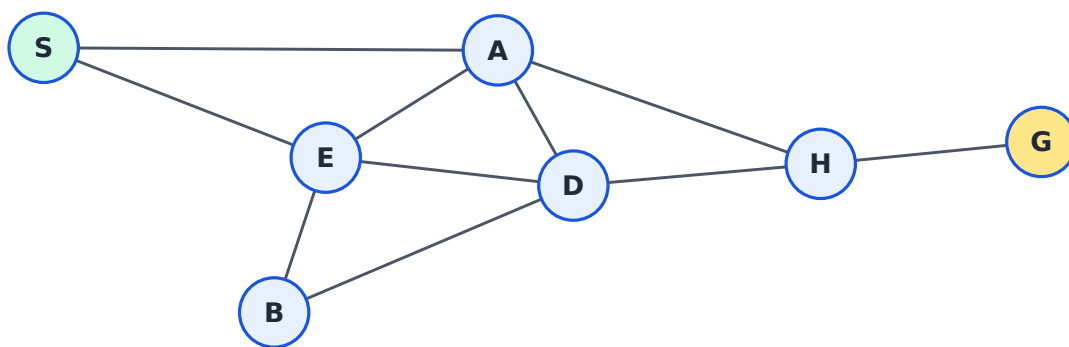
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 9 / 12	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

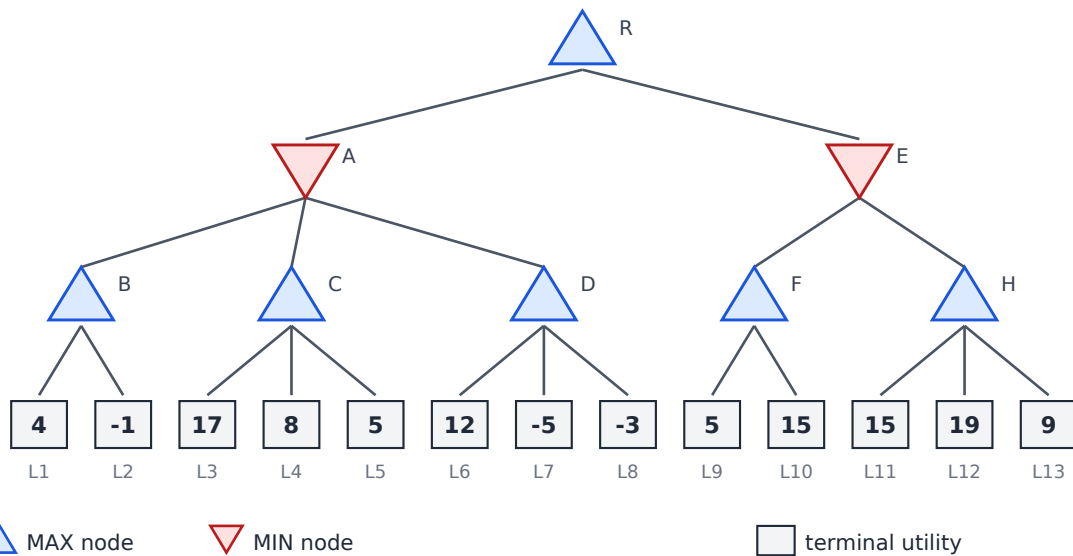
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, A, D, B, E, H, G
- ☐ (B) S, A, E, D, H, B, G
- ☐ (C) S, D, A, B, E, H, G
- ☐ (D) S, E, D, H, G

Q.2 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

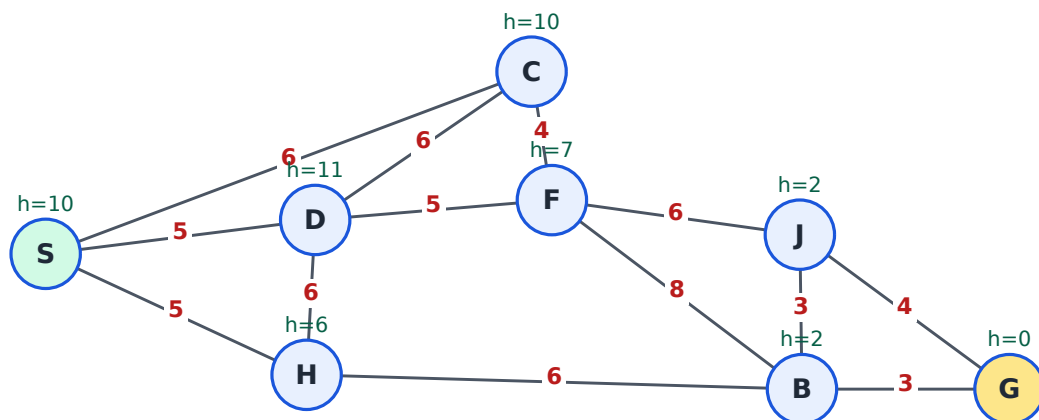
In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to E; root value 19
☐ (B) Move to A; root value 4
☐ (C) Move to E; root value 15
☐ (D) Move to E; root value 18

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Problem Formulation

The 8-puzzle has $9!/2$ states reachable from any given start state. Compute this number.

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following are sentences, i.e. formulas with no free variables (a is a constant)?

- ☐ (A) $(\forall x P(x)) \rightarrow Q(x)$
- ☐ (B) $\forall x (P(x) \rightarrow \exists y R(x, y))$
- ☐ (C) $\exists x R(x, y)$
- ☐ (D) $\forall x \exists y (R(x, y) \wedge P(a))$

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Backward chaining over a propositional definite-clause KB is used to prove the query Q. Which clause is used FIRST?

- ☐ (A) A clause whose head (conclusion) is Q
- ☐ (B) A clause in which Q appears in the body
- ☐ (C) Any fact in the KB, chosen arbitrarily
- ☐ (D) The negated query $\neg Q$ added to the KB

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

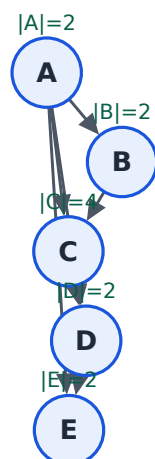
How many of the 8 truth assignments to P, Q, R satisfy the formula

$$\mathbf{F} = (\mathbf{Q} \vee (\mathbf{P} \rightarrow \mathbf{R})) \rightarrow (\mathbf{P} \rightarrow \mathbf{R}) ?$$

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

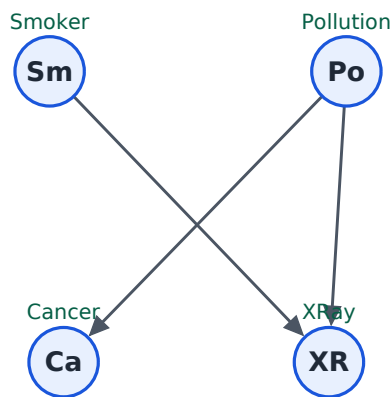
The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) What is the probability that a single run produces the sample **(Sm=T, Po=T, Ca=F, XR=F)**? (round off to 4 decimal places)



P(Sm=T)
0.3

P(Po=T)	Po	P(Ca=T ...)
0.8	T	0.2
	F	0.05

Sm	Po	P(XR=T ...)
T	T	0.05
T	F	0.4
F	T	0.2
F	F	0.25

Answer (numerical): _____

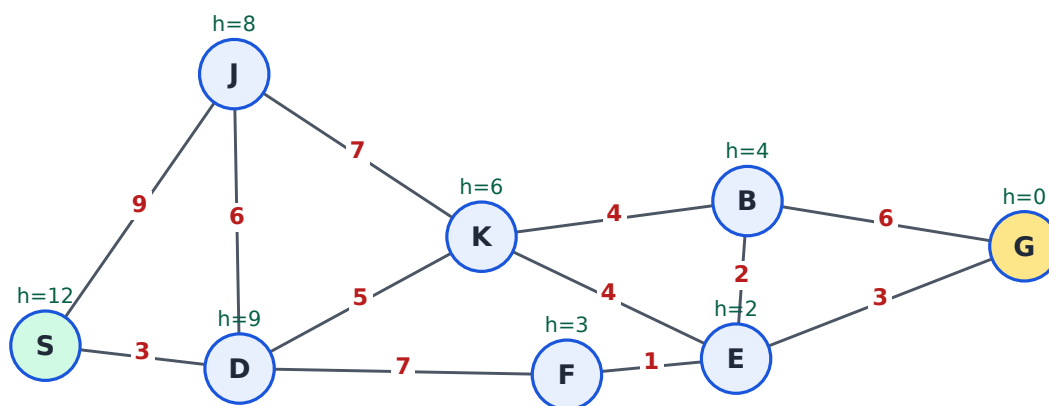
Section B – 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Informed Search

In the 8-puzzle, the goal configuration is 1 2 3 / 4 5 6 / 7 8 _ and the current state is 8 1 3 / 4 _ 2 / 7 6 5 (rows top to bottom, _ = blank). What is h_2 , the sum of the Manhattan distances of the tiles from their goal positions (excluding the blank)?

Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

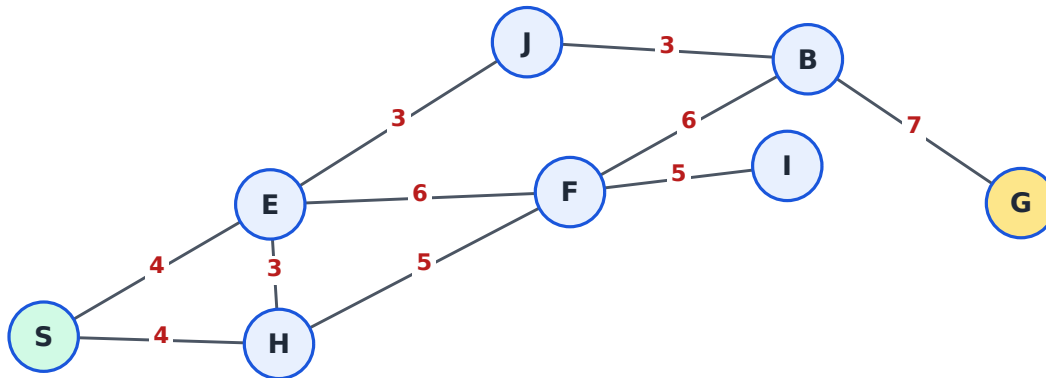
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge E-G violates the consistency condition (in at least one direction).
- ☐ (B) h is consistent (monotone).
- ☐ (C) h is admissible.
- ☐ (D) $h(E)$ overestimates the true cost from E to G.

Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow E \rightarrow F \rightarrow B \rightarrow G$
☐ (B) $S \rightarrow E \rightarrow J \rightarrow B \rightarrow G$
☐ (C) $S \rightarrow H \rightarrow F \rightarrow B \rightarrow G$
☐ (D) $S \rightarrow H \rightarrow E \rightarrow J \rightarrow B \rightarrow G$

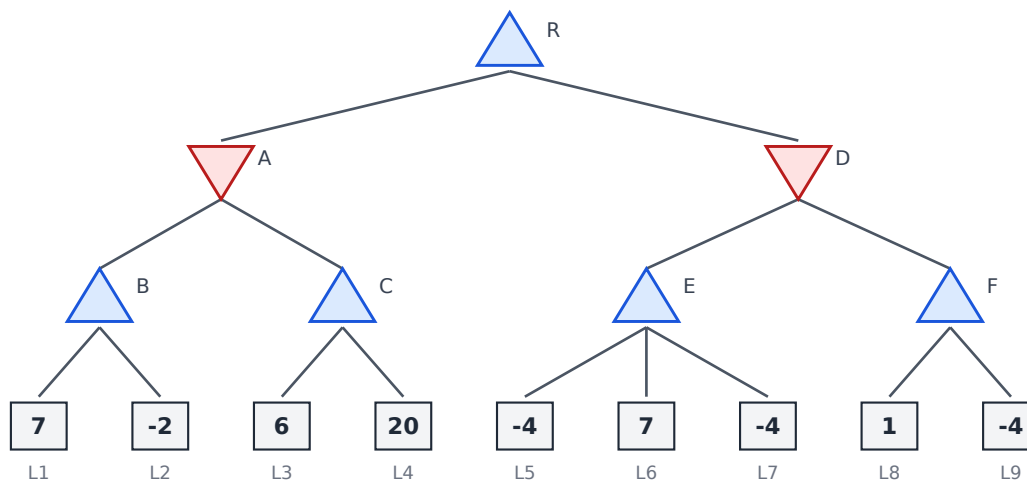
Q.14 [NAT · 2 marks · no negative marking] Informed Search

A search algorithm generates $N = 52$ nodes (not counting the root) to find a solution at depth $d = 5$. Using $N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^5$, find the effective branching factor b^* , correct to two decimal places.

Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = P \rightarrow (\neg R \wedge Q)$?

- ☐ (A) $\neg P \vee (\neg R \wedge Q)$
☐ (B) $\neg(P \wedge \neg(\neg R \wedge Q))$
☐ (C) $\neg P \rightarrow \neg(\neg R \wedge Q)$
☐ (D) $\neg(\neg R \wedge Q) \rightarrow \neg P$

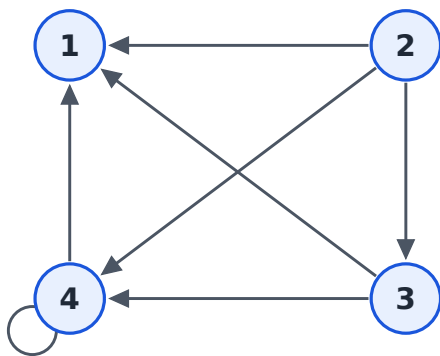
Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following sets of propositional formulas are unsatisfiable?

- ☐ (A) $\{p \vee q, \neg p \vee q, p \vee \neg q, \neg p \vee \neg q\}$
☐ (B) $\{p \rightarrow q, q \rightarrow r, r \rightarrow \neg p, p\}$
☐ (C) $\{p \leftrightarrow \neg q, q \leftrightarrow \neg r, r \leftrightarrow \neg p\}$
☐ (D) $\{p \vee q \vee r, \neg p \vee \neg q, \neg q \vee \neg r, \neg p \vee \neg r\}$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (C) $\exists y \forall x R(x, y)$
☐ (D) $\exists x R(x, x)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

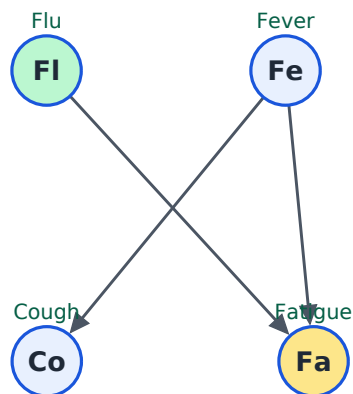
$C1: (\neg R), C2: (P \vee \neg R), C3: (R \vee S), C4: (\neg Q \vee \neg R \vee \neg S), C5: (P \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) $(P \vee Q)$
☐ (C) $(P \vee \neg Q \vee R)$
☐ (D) (R)

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Flu} = T \mid \text{Fatigue} = T)$ (round off to 3 decimal places).



$P(\text{Fl}=T)$
0.6

$P(\text{Fe}=T)$	Fe	$P(\text{Co}=T \dots)$
0.1	T	0.5
	F	0.4

Fl	Fe	$P(\text{Fa}=T \dots)$
T	T	0.1
T	F	0.95
F	T	0.05
F	F	0.8

Answer (numerical): _____

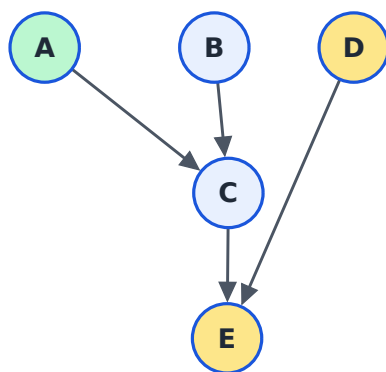
Q.21 [MSQ · 2 marks · no negative marking] Probability for AI

A hidden variable H takes values h_1, h_2, h_3 with prior probabilities 0.5, 0.3, 0.2. For an observation e , $P(e \mid h_1) = 0.1$, $P(e \mid h_2) = 0.4$, $P(e \mid h_3) = 0.7$. Which of the following are correct?

- ☐ (A) $P(e) = 0.31$
☐ (B) $P(h_3 \mid e) \approx 0.452$
☐ (C) $P(h_1 \mid e) \approx 0.161$
☐ (D) The most probable hypothesis after observing e (MAP) is h_2

Q.22 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $E = T$, $D = F$. The values of the non-evidence variables in the generated samples are listed in the table. What is the weight of sample #6? (round off to 4 decimal places)



$P(A=T)$	$P(B=T)$
0.4	0.25

A	B	$P(C=T \dots)$
T	T	0.1
T	F	0.75
F	T	0.9
F	F	0.7

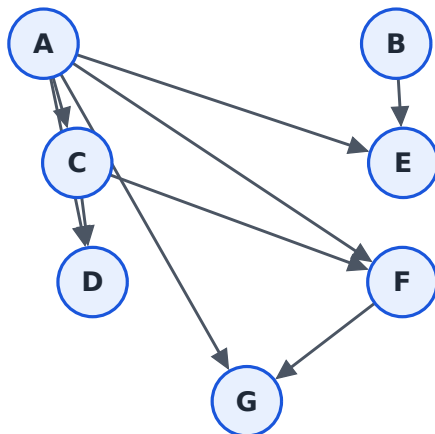
$P(D=T)$	C	D	$P(E=T \dots)$
0.75	T	T	0.6
	T	F	0.7
	F	T	0.7
	F	F	0.8

#	A	B	C
1	F	F	T
2	T	F	T
3	F	F	F
4	F	T	T
5	T	F	T
6	T	F	T

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

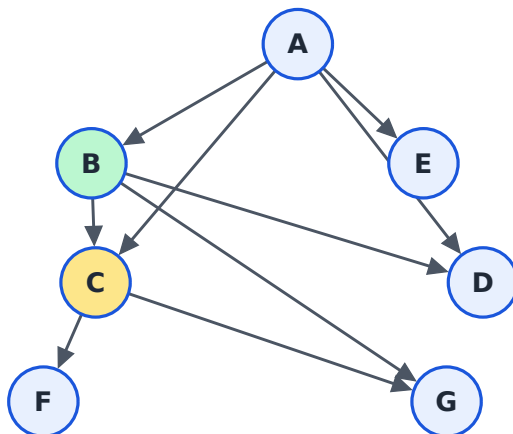
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp D \mid A, G$
☐ (B) $B \perp F$
☐ (C) $F \perp G \mid D$
☐ (D) $B \perp A$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{B} \mid \mathbf{C} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{C} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E, A, G, D, F**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Bayesian Networks

Which of the following statements about I-maps of a distribution P are correct? (G is an I-map of P if every independence implied by G by d-separation holds in P .)

- ☐ (A) A complete (fully connected) DAG over the variables is an I-map of every distribution
☐ (B) Adding nodes in a causal order (causes before effects) usually yields a sparser minimal I-map
☐ (C) Adding an arc (without creating a cycle) to an I-map of P can make it fail to be an I-map of P
☐ (D) The minimal I-map of P is the same DAG for every variable ordering

Answer Key — Mock Test 055

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	NAT	2	Informed Search	1.9 to 1.93
2	MCQ	1	Adversarial Search	C	15	NAT	2	Adversarial Search	7
3	NAT	1	Informed Search	14	16	MSQ	2	Propositional Logic	A, B, D
4	NAT	1	Problem Formulation	181440	17	MSQ	2	Propositional Logic	A, B, C
5	NAT	1	Predicate Logic	64	18	MSQ	2	Predicate Logic	A, D
6	MSQ	1	Predicate Logic	B, D	19	MSQ	2	Propositional Logic	C
7	MCQ	1	Propositional Logic	A	20	NAT	2	Exact Inference	0.64 to 0.644
8	NAT	1	Propositional Logic	7	21	MSQ	2	Probability for AI	A, B, C
9	NAT	1	Bayesian Networks	27	22	NAT	2	Sampling	0.1748 to 0.1752
10	NAT	1	Sampling	0.1822 to 0.1826	23	MSQ	2	Conditional Independence	A, B, D
11	NAT	2	Informed Search	10	24	NAT	2	Exact Inference	2
12	MSQ	2	Informed Search	B, C	25	MSQ	2	Bayesian Networks	A, B
13	MCQ	2	Uninformed Search	B					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: A (A: S, A, D, B, E, H, G)

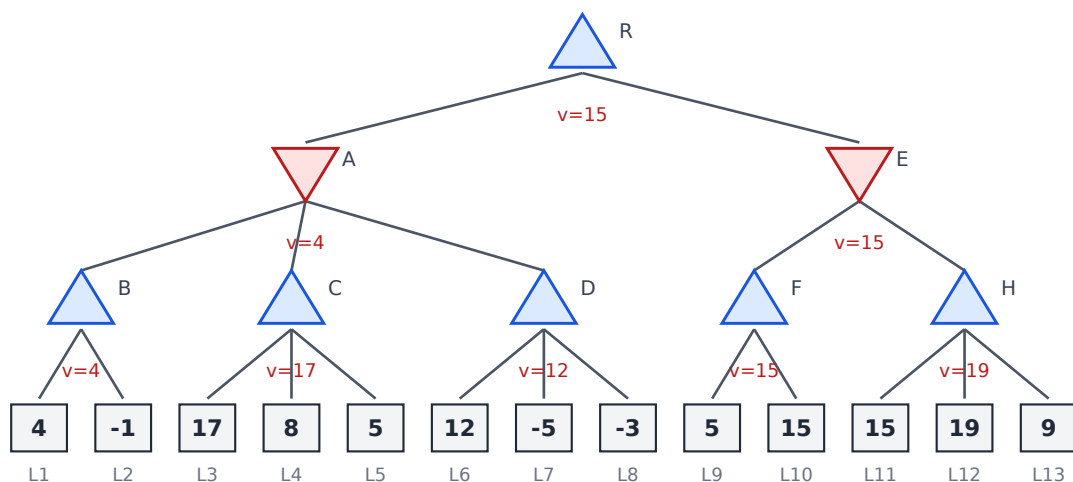
Adjacency lists (alphabetical): A: {D, E, H, S}; B: {D, E}; D: {A, B, E, H}; E: {A, B, D, S}; G: {H}; H: {A, D, G}; S: {A, E}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, A, D, B, E, H, G**; the path found is $S \rightarrow A \rightarrow D \rightarrow H \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.2 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to E; root value 15)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 4$, $C = 17$, $D = 12$, $A = 4$, $F = 15$, $H = 19$, $E = 15$. Root value = **15**; MAX's optimal first move is towards E.

Q.3 [NAT, 1M · Informed Search] Answer: 14

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (10)	H(6), C(10), D(11)
2	H (6)	B(2), C(10), D(11)
3	B (2)	G(0), J(2), F(7), C(10), D(11)
4	G (0)	J(2), F(7), C(10), D(11) ← goal, stop

Path returned: $S \rightarrow H \rightarrow B \rightarrow G$, cost = $5 + 6 + 3 = 14$.

The optimal cost is $h^*(S) = 14$ — greedy happens to be optimal here, but in general it is not.

Q.4 [NAT, 1M · Problem Formulation] Answer: 181440

$9! = 362880$. The parity of a permutation splits the states into two disjoint halves, and no sequence of moves crosses from one half to the other. So $362880/2 = \mathbf{181440}$ states are reachable.

Q.5 [NAT, 1M · Predicate Logic] Answer: 64

View R as an 3×3 0/1 matrix. The sentence says R is symmetric: choose freely the n diagonal pairs and one decision per unordered pair $\{x, y\} \Rightarrow 2^{(n(n+1)/2)}$. With $n = 3$: **64**.

Q.6 [MSQ, 1M · Predicate Logic] Answer: B, D (B: $\forall x (P(x) \rightarrow \exists y R(x, y))$; D: $\forall x \exists y (R(x, y) \wedge P(a))$)

(A) The scope of $\forall x$ ends before $\rightarrow$, so x in $Q(x)$ is free. (B) x and y are bound everywhere. (C) y is free. (D) x and y are bound; a is a constant, not a variable. Hence B and D are sentences.

Q.7 [MCQ, 1M · Propositional Logic] Answer: A (A: A clause whose head (conclusion) is Q)

Backward chaining works from the goal: it looks for rules (or facts) whose conclusion is Q and recursively tries to prove their premises. Using clauses with Q in the body is forward reasoning. Starting from facts is forward chaining. Adding $\neg Q$ is the resolution-refutation approach, not backward chaining.

Q.8 [NAT, 1M · Propositional Logic] Answer: 7

Build the truth table (sub-formulas shown first):

P	Q	R	$Q \vee (P \rightarrow R)$	$P \rightarrow R$	$(Q \vee (P \rightarrow R)) \rightarrow (P \rightarrow R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

Rows where F = T: **7**.

Q.9 [NAT, 1M · Bayesian Networks] Answer: 27

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (2-1) \times 2 + C: (4-1) \times 2 \times 2 + D: (2-1) \times 2 \times 4 + E: (2-1) \times 2 \times 2 = \mathbf{27}$.

Full joint table: $\prod |X| - 1 = 64 - 1 = 63$; saving = 36.

Q.10 [NAT, 1M · Sampling] Answer: 0.1822 to 0.1826

Prior sampling generates each complete event with exactly its joint probability $S_{\text{ps}}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Sm}=\text{T}) \times P(\text{Po}=\text{T}) \times P(\text{Ca}=\text{F} | \text{Po}=\text{T}) \times P(\text{XR}=\text{F} | \text{Sm}=\text{T}, \text{Po}=\text{T})$

$= 0.3 \times 0.8 \times 0.8 \times 0.95 = \mathbf{0.1824} \approx 0.1824$.

Q.11 [NAT, 2M · Informed Search] Answer: 10

Use (row, column) coordinates from (0,0). Tile 8 is at (0,0) and belongs at (2,1): distance 3. Tile 1 is at (0,1) and belongs at (0,0): 1. Tile 3: 0. Tile 4: 0. Tile 2 is at (1,2) and belongs at (0,1): 2. Tile 7: 0. Tile 6 is at (2,1) and belongs at (1,2): 2. Tile 5 is at (2,2) and belongs at (1,1): 2. Sum = $3+1+2+2+2 = \mathbf{10}$.

Q.12 [MSQ, 2M · Informed Search] Answer: B, C (B: h is consistent (monotone).; C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	D	E	F	G	J	K	S
$h(n)$	4	9	2	3	0	8	6	12
$h^*(n)$	5	11	3	4	0	14	7	14
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [MCQ, 2M · Uninformed Search] **Answer: B** ($B: S \rightarrow E \rightarrow J \rightarrow B \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	E(4), H(4)
2	E (4)	H(4), J(7), F(10)
3	H (4)	J(7), F(9)
4	J (7)	F(9), B(10)
5	F (9)	B(10), I(14)
6	B (10)	I(14), G(17)
7	I (14)	G(17)
8	G (17)	← goal, stop

Path returned: **S** → **E** → **J** → **B** → **G**, cost = 4 + 3 + 3 + 7 = **17**. Nodes expanded (including S and G): **8** (S, E, H, J, F, B, I, G).

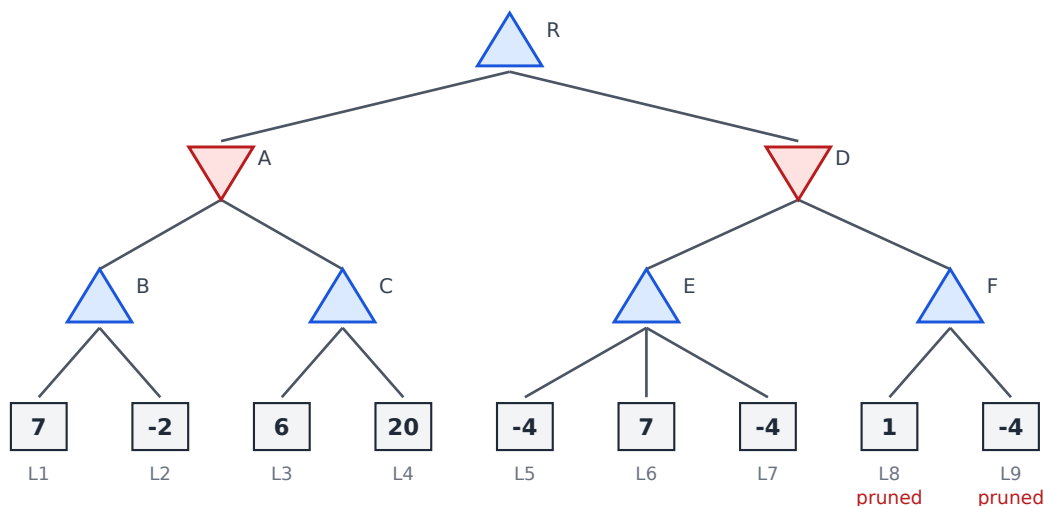
Q.14 [NAT, 2M · Informed Search] **Answer: 1.9 to 1.93**

Solve $b^* + b^{*2} + b^{*3} + b^{*4} + b^{*5} = 52$. For $b^*=1.9$: $1.9+3.61+6.859+13.03+24.76 \approx 50.2$, which is too small. For $b^*=1.95$: about 54.0, which is too large. Bisection gives $b^* \approx \mathbf{1.92}$, the AIMA textbook example. A b^* close to 1 indicates a very good heuristic.

Q.15 [NAT, 2M · Adversarial Search] **Answer: 7**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $7 \leq \alpha = 7 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (7): L1, L2, L3, L4, L5, L6, L7. Leaves pruned (2): L8, L9. Root value = 7 (same as plain minimax, 7 — pruning never changes the root value).

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A, B, D** ($A: \neg P \vee (\neg R \wedge Q)$; $B: \neg(P \wedge \neg(\neg R \wedge Q))$; $D: \neg(\neg R \wedge Q) \rightarrow \neg P$)

Compare truth tables with $F = P \rightarrow (\neg R \wedge Q)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

- (A) $\neg P \vee (\neg R \wedge Q)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.
- (B) $\neg(P \wedge \neg(\neg R \wedge Q))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.
- (C) $\neg P \rightarrow \neg(\neg R \wedge Q)$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.
- (D) $\neg(\neg R \wedge Q) \rightarrow \neg P$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, B, C** ($A: \{p \vee q, \neg p \vee q, p \vee \neg q, \neg p \vee \neg q\}$; $B: \{p \rightarrow q, q \rightarrow r, r \rightarrow \neg p, p\}$; $C: \{p \leftrightarrow \neg q, q \leftrightarrow \neg r, r \leftrightarrow \neg p\}$)

(A) The four clauses exclude all four (p, q) assignments, one each. (B) From p we get q , then r , then $\neg p$ — a contradiction. (C) $p = \neg q$ and $q = \neg r$ give $p = r$, but $r \leftrightarrow \neg p$ demands $r = \neg p$ — contradiction (an odd cycle of negations). (D) is satisfiable, e.g. $p=T, q=F, r=F$ (exactly one true).

Q.18 [MSQ, 2M · Predicate Logic] **Answer: A, D** (A: $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$; D: $\exists x R(x, x)$)

$R = \{(2,1), (2,3), (2,4), (3,1), (3,4), (4,1), (4,4)\}$.

(A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **TRUE** — every two distinct elements are related in some direction.

(B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,4)$ holds but $R(4,2)$ does not.

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\exists x R(x, x)$: **TRUE** — self-loop at 4.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: C** (C: $(P \vee \neg Q \vee R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 15 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(P \vee Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $(P \vee \neg Q \vee R)$ is derivable:

1. resolve $(\neg Q \vee \neg R \vee \neg S)$ and $(P \vee R)$ on $R \Rightarrow (P \vee \neg Q \vee \neg S)$

2. resolve $(R \vee S)$ and $(P \vee \neg Q \vee \neg S)$ on $S \Rightarrow (P \vee \neg Q \vee R)$

(D) (R) never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.20 [NAT, 2M · Exact Inference] **Answer: 0.64 to 0.644**

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever}) \cdot P(\text{Cough} \mid \text{Fever}) \cdot P(\text{Fatigue} \mid \text{Flu}, \text{Fever})$.

Inference by enumeration: $P(\text{Flu} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Flu}, e, \text{hidden})$.

Hidden: Fe, Co	Fl	product of CPT entries (Fl · Fe · Co · Fa)	value
T, T	T	$0.6 \times 0.1 \times 0.5 \times 0.1$	0.003
T, F	T	$0.6 \times 0.1 \times 0.5 \times 0.1$	0.003
F, T	T	$0.6 \times 0.9 \times 0.4 \times 0.95$	0.2052
F, F	T	$0.6 \times 0.9 \times 0.6 \times 0.95$	0.3078
T, T	F	$0.4 \times 0.1 \times 0.5 \times 0.05$	0.001
T, F	F	$0.4 \times 0.1 \times 0.5 \times 0.05$	0.001
F, T	F	$0.4 \times 0.9 \times 0.4 \times 0.8$	0.1152
F, F	F	$0.4 \times 0.9 \times 0.6 \times 0.8$	0.1728

$P(\text{Flu}=\text{T}, e) = 0.519$; $P(\text{Flu}=\text{F}, e) = 0.29$; $P(e) = 0.809$.

$P(\text{Flu}=\text{T} \mid e) = 0.519 / 0.809 = \mathbf{0.6415} \approx 0.642$.

Q.21 [MSQ, 2M · Probability for AI] **Answer: A, B, C** (A: $P(e) = 0.31$; B: $P(h_3 \mid e) \approx 0.452$; C: $P(h_1 \mid e) \approx 0.161$)

Unnormalized posteriors $P(e \mid h_i)P(h_i)$: $0.1 \times 0.5 = 0.05$, $0.4 \times 0.3 = 0.12$, $0.7 \times 0.2 = 0.14$. Marginalizing, $P(e) = 0.05 + 0.12 + 0.14 = 0.31$ (A true). $P(h_3 \mid e) = 0.14/0.31 \approx 0.452$ (B true); $P(h_1 \mid e) = 0.05/0.31 \approx 0.161$ (C true); $P(h_2 \mid e) = 0.12/0.31 \approx 0.387$. The MAP hypothesis is h_3 (largest posterior), even though it had the smallest prior, so (D) is false.

Q.22 [NAT, 2M · Sampling] **Answer: 0.1748 to 0.1752**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(D=F)=0.25 \times P(E=T \mid C=T, D=F)=0.7$	0.175	F
2	$P(D=F)=0.25 \times P(E=T \mid C=T, D=F)=0.7$	0.175	T
3	$P(D=F)=0.25 \times P(E=T \mid C=F, D=F)=0.8$	0.2	F
4	$P(D=F)=0.25 \times P(E=T \mid C=T, D=F)=0.7$	0.175	F
5	$P(D=F)=0.25 \times P(E=T \mid C=T, D=F)=0.7$	0.175	T
6	$P(D=F)=0.25 \times P(E=T \mid C=T, D=F)=0.7$	0.175	T

$\sum w (\text{all}) = 1.075$; $\sum w (A = T) = 0.525$.

Estimate $P(A=T \mid E = T, D = F) = 0.525 / 1.075 = \mathbf{0.4884}$.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: A, B, D** (A: $E \perp D \mid A, G$; B: $B \perp F$; D: $B \perp A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp D \mid A, G$: **TRUE** — $E \leftarrow A \rightarrow C \rightarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow F \leftarrow C \rightarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow G \leftarrow F \leftarrow C \rightarrow D$ (blocked: observed fork node A).

(B) $B \perp F$: **TRUE** — $B \rightarrow E \leftarrow A \rightarrow C \rightarrow F$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow E \leftarrow A \rightarrow D \leftarrow C \rightarrow F$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow E \leftarrow A \rightarrow F$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow E \leftarrow A \rightarrow G \leftarrow F$ (blocked: collider E with neither it nor a descendant observed).

(C) $F \perp G \mid D$: **FALSE** — the path $F \leftarrow A \rightarrow G$ is active given $\{D\}$.

(D) $B \perp A$: **TRUE** — $B \rightarrow E \leftarrow A$ (blocked: collider E with neither it nor a descendant observed).

Q.24 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B)$, $f(A, B, D)$, $f(A, E)$, $f(F)$, $f(B, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(A, E)$	$\{A, E\}$	$f(A)$
A	$f(A)$, $f(A, B)$, $f(A, B)$, $f(A, B, D)$, $f(A)$	$\{A, B, D\}$	$f(B, D)$
G	$f(B, G)$	$\{B, G\}$	$f(B)$
D	$f(B, D)$	$\{B, D\}$	$f(B)$
F	$f(F)$	$\{F\}$	$f()$

The largest factor produced has **2** variable(s) (created when eliminating A); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over B and normalise.

Q.25 [MSQ, 2M · Bayesian Networks] **Answer: A, B**

(A) A complete DAG implies no independences, so it is trivially an I-map. (B) AIMA shows causal orderings (e.g. B, E, A, J, M) give compact networks while diagnostic orderings (M, J, E, B, A) can be fully connected. (C) is false: adding an arc can only remove d-separation statements, so the set of implied independences shrinks and remains true in P. (D) is false: for the alarm example, order B, E, A, J, M gives 10 parameters, order M, J, A, B, E gives 13, and M, J, E, B, A gives 31.

Mock Test 056

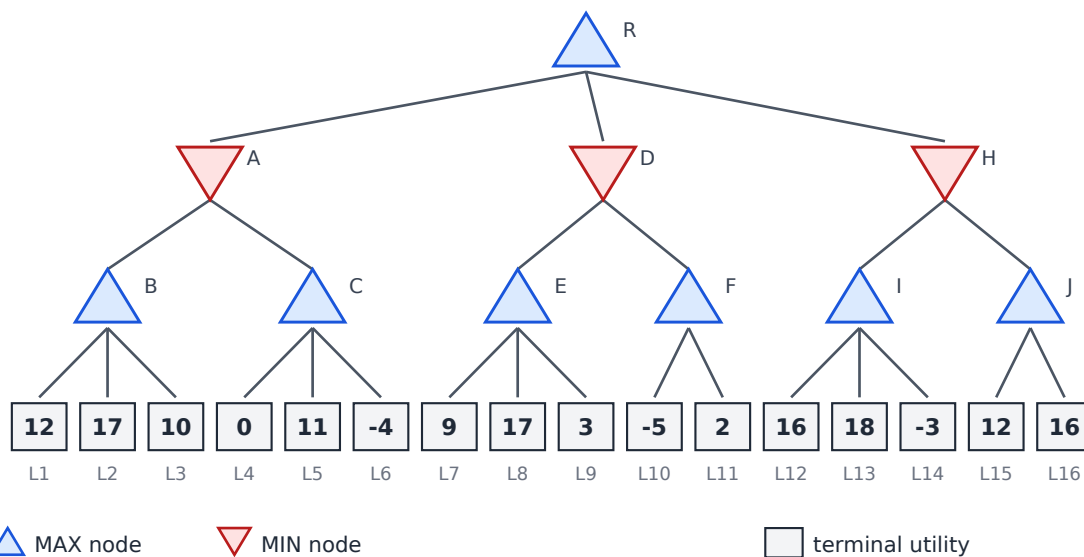
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 8 / 11	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

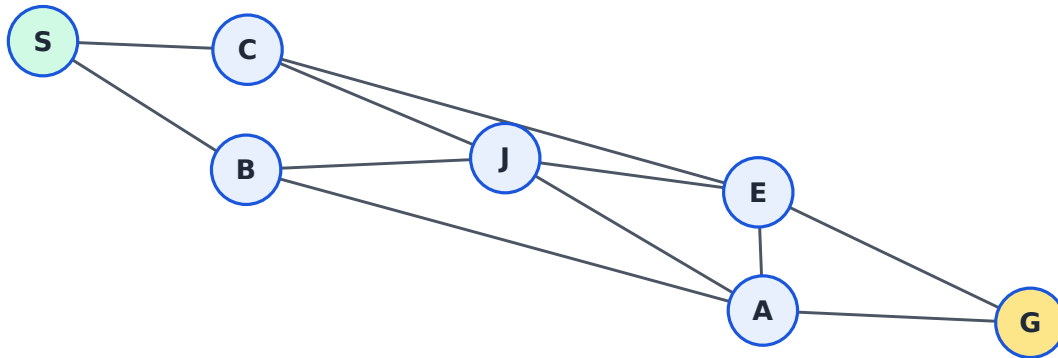
In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to D; root value 2
- ☐ (B) Move to A; root value 11
- ☐ (C) Move to H; root value 16
- ☐ (D) Move to H; root value 17

Q.2 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, C, A, J, E, G
☐ (B) S, B, A, E, C, J, G
☐ (C) S, C, B, J, E, A, G
☐ (D) S, B, C, J, A, E, G

Q.3 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Iterative deepening search is applied to a tree with branching factor $b = 2$ whose shallowest goal is at depth $d = 6$ (goal is the last node at that depth). Using the standard count $N(\text{IDS}) = d \cdot b + (d-1) \cdot b^2 + \dots + 1 \cdot b^d$ (root not counted), how many nodes are generated?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Problem Formulation

In the vacuum world with $n = 3$ squares in a row, a state specifies the agent's location and whether each square is dirty or clean. How many states are there?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

How many of the 8 truth assignments to p, q, r are models of $(p \vee q) \wedge (\neg p \vee r)$?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 2 elements, so there are $4^2 = 16$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following expresses 'None of my friends are perfect' ($F(x)$: x is my friend, $P(x)$: x is perfect)?

- ☐ (A) $\forall x (F(x) \rightarrow \neg P(x))$
☐ (B) $\exists x (F(x) \wedge \neg P(x))$
☐ (C) $\neg \forall x (F(x) \rightarrow P(x))$
☐ (D) $\forall x (\neg F(x) \rightarrow P(x))$

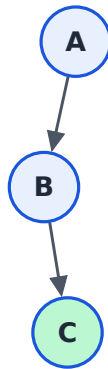
Q.8 [NAT · 1 mark · no negative marking] Bayesian Networks

A naive Bayes classifier has a class variable with 3 values and 10 Boolean features. How many independent parameters does the Bayesian network have?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $\mathbf{P(C = T)}$ (round off to 3 decimal places).

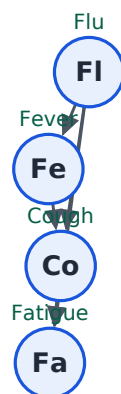


$P(A=T)$	A	$P(B=T \dots)$	B	$P(C=T \dots)$
0.75	T	0.4	T	0.2
	F	0.25	F	0.2

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Fl, Fe, Co, Fa. (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) What is the probability that a single run produces the sample **(Fl=F, Fe=F, Co=T, Fa=T)**? (round off to 4 decimal places)



$P(Fl=T)$
0.2

Fl	$P(Fe=T \dots)$	Fl	Fe	$P(Co=T \dots)$
T	0.75	T	T	0.7
F	0.05	T	F	0.1
		F	T	0.4
		F	F	0.9

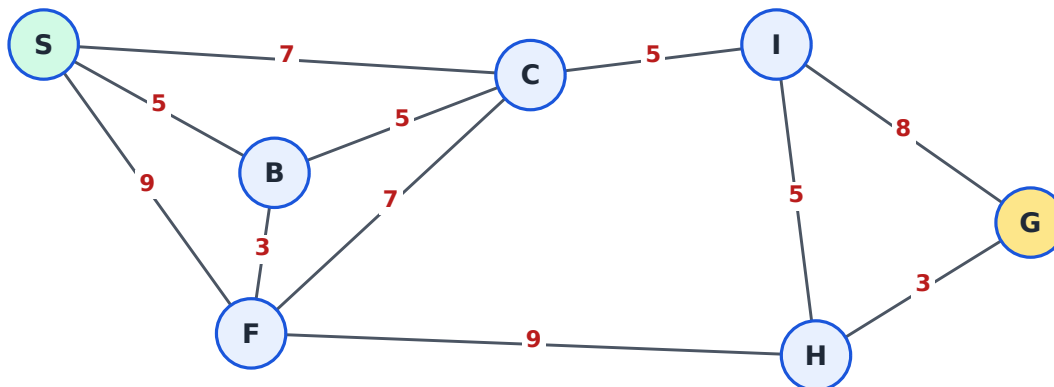
Fl	Co	$P(Fa=T \dots)$
T	T	0.5
T	F	0.7
F	T	0.4
F	F	0.9

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

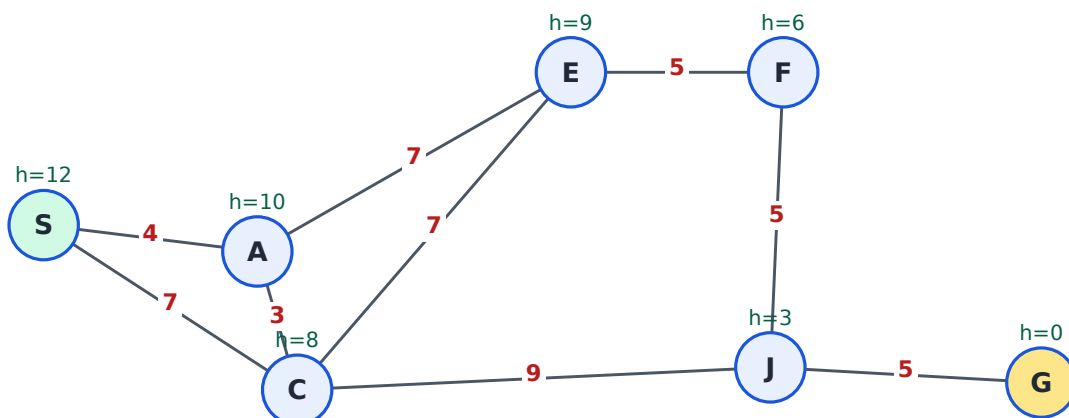
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow C \rightarrow I \rightarrow G$
☐ (B) $S \rightarrow B \rightarrow F \rightarrow H \rightarrow G$
☐ (C) $S \rightarrow F \rightarrow H \rightarrow G$
☐ (D) $S \rightarrow C \rightarrow I \rightarrow H \rightarrow G$

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) The path returned has cost 22.
☐ (B) Node F is expanded.
☐ (C) Node E is expanded.
☐ (D) Node J is expanded.

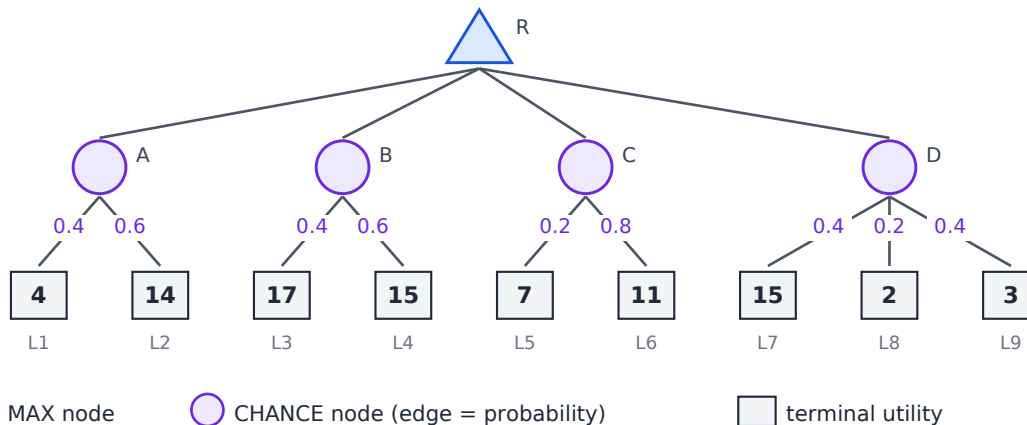
Q.13 [MSQ · 2 marks · no negative marking] Informed Search

Which of the following statements about greedy best-first search are TRUE?

- ☐ (A) The graph-search version is complete in finite state spaces
☐ (B) It is optimal whenever the heuristic is admissible
☐ (C) Its worst-case time and space complexity is $O(b^m)$, where m is the maximum depth
☐ (D) If $h(n) = h^*(n)$ exactly, greedy best-first search always returns an optimal solution

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Local Search

Which of the following statements about simulated annealing (maximisation) are TRUE?

- ☐ (A) If the temperature is lowered slowly enough, it finds a global optimum with probability approaching 1
- ☐ (B) It always accepts a move that improves the objective
- ☐ (C) At very high temperature, it behaves almost like a random walk
- ☐ (D) At $T = 0$, it accepts worsening moves with probability 1

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = ((S \rightarrow Q) \vee (\neg R \vee P)) \rightarrow ((P \wedge R) \leftrightarrow (P \rightarrow \neg Q)) ?$$

Answer (numerical): _____

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following statements about Horn clauses and Horn-form knowledge bases are TRUE?

- ☐ (A) Entailment of an atom from a Horn KB can be decided in time linear in the size of the KB
- ☐ (B) Every Horn clause has at most one positive literal
- ☐ (C) The clause $\neg p \vee q \vee r$ is a Horn clause
- ☐ (D) The resolvent of two Horn clauses is again a Horn clause

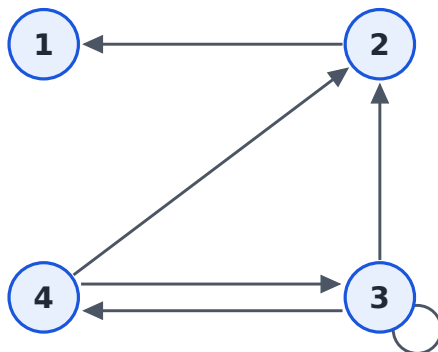
Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (P \rightarrow Q) \rightarrow (\neg P \rightarrow \neg R)$?

- ☐ (A) $(\neg P \rightarrow \neg R) \rightarrow (P \rightarrow Q)$
- ☐ (B) $\neg(\neg P \rightarrow \neg R) \rightarrow \neg(P \rightarrow Q)$
- ☐ (C) $(P \rightarrow Q) \vee \neg(\neg P \rightarrow \neg R)$
- ☐ (D) $\neg((P \rightarrow Q) \wedge \neg(\neg P \rightarrow \neg R))$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x R(x, x)$
☐ (B) $\exists x \forall y \neg R(x, y)$
☐ (C) $\forall y \exists x R(x, y)$
☐ (D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

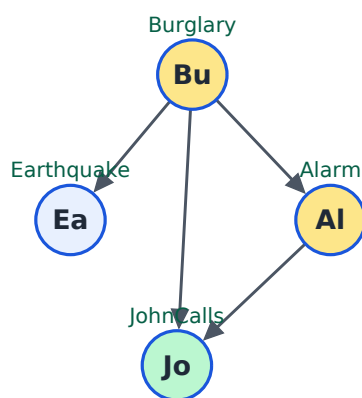
- (1) $(R \rightarrow Q) \leftrightarrow P$
 (2) $(Q \leftrightarrow Q) \leftrightarrow \neg R$
 (3) $(R \leftrightarrow \neg R) \rightarrow (\neg R \leftrightarrow P)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $(R \leftrightarrow P) \wedge (Q \rightarrow R)$
☐ (B) $\neg Q \leftrightarrow \neg R$
☐ (C) $(Q \vee P) \rightarrow (Q \rightarrow Q)$
☐ (D) $\neg R \wedge (R \vee R)$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute $P(\text{JohnCalls} = T \mid \text{Burglary} = T, \text{Alarm} = T)$ (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.9

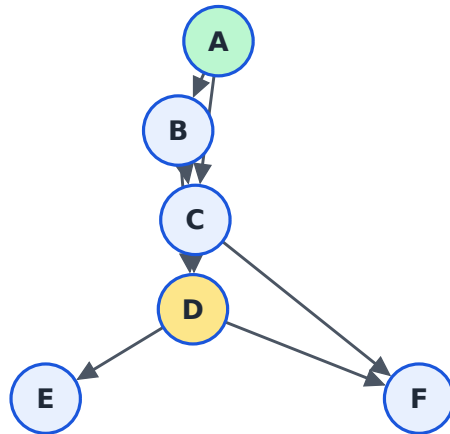
Bu	$P(\text{Ea}=T \mid \dots)$	Bu	$P(\text{Al}=T \mid \dots)$
T	0.4	T	0.9
F	0.3	F	0.05

Bu	Al	$P(\text{Jo}=T \mid \dots)$
T	T	0.9
T	F	0.1
F	T	0.4
F	F	0.5

Answer (numerical): _____

Q.22 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{A} \mid \mathbf{D} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{D} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B**, **C**, **F**, **E**. What is the scope of the new factor produced when **E** is summed out?



- ☐ (A) $f(E)$
- ☐ (B) $f(D)$
- ☐ (C) $f(A)$
- ☐ (D) $f()$ — a constant (scope $\emptyset$)

Q.23 [MCQ · 2 marks · negative: -2/3] Probability for AI

The prior odds of hypothesis h against $\neg h$ are 1 : 4. Evidence e has likelihood ratio $P(e \mid h)/P(e \mid \neg h) = 6$. What is $P(h \mid e)$?

- ☐ (A) 0.600
- ☐ (B) 0.857
- ☐ (C) 0.250
- ☐ (D) 0.200

Q.24 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

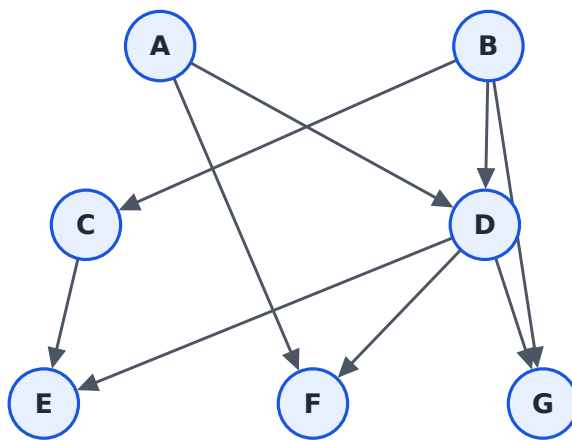
The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(C = T \mid A = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	F	T	F
2	F	F	T	F
3	T	F	T	F
4	F	F	T	F
5	F	F	T	F
6	F	T	T	T
7	F	F	T	T
8	F	F	T	T
9	T	T	T	F
10	T	F	T	F
11	F	F	F	F
12	F	F	T	F
13	F	F	T	T
14	F	F	T	T
15	T	F	F	T

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp A \mid C, F$
- ☐ (B) $A \perp C \mid B$
- ☐ (C) $C \perp E \mid B$
- ☐ (D) $A \perp C$

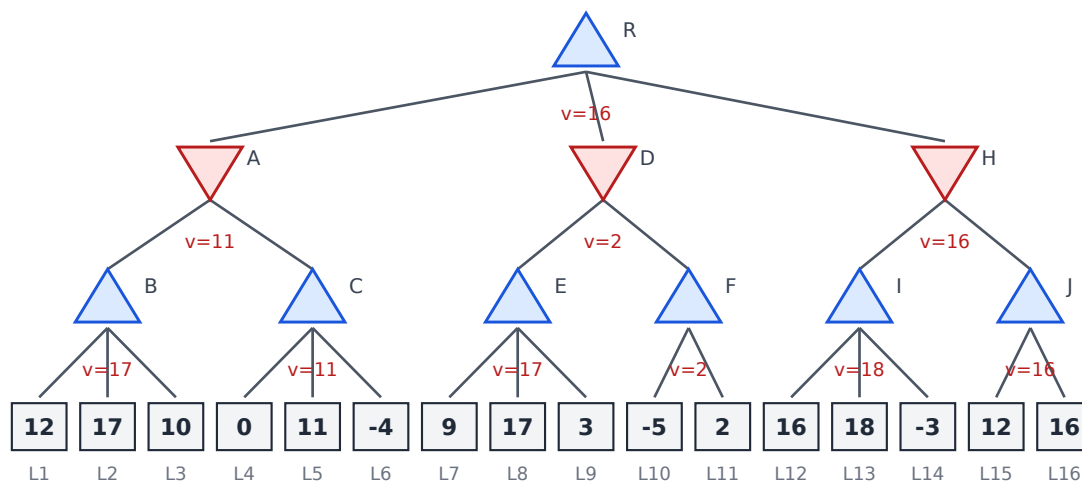
Answer Key — Mock Test 056

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	C	14	NAT	2	Adversarial Search	15.79 to 15.81
2	MCQ	1	Uninformed Search	A	15	MSQ	2	Local Search	A, B, C
3	NAT	1	Uninformed Search	240	16	NAT	2	Propositional Logic	5
4	NAT	1	Problem Formulation	24	17	MSQ	2	Propositional Logic	A, B, D
5	NAT	1	Propositional Logic	4	18	MSQ	2	Propositional Logic	B, D
6	NAT	1	Predicate Logic	4	19	MSQ	2	Predicate Logic	A, B, C
7	MCQ	1	Predicate Logic	A	20	MSQ	2	Propositional Logic	C
8	NAT	1	Bayesian Networks	32	21	NAT	2	Exact Inference	0.898 to 0.902
9	NAT	1	Exact Inference	0.198 to 0.202	22	MCQ	2	Exact Inference	D
10	NAT	1	Sampling	0.2734 to 0.2738	23	MCQ	2	Probability for AI	A
11	MCQ	2	Uninformed Search	A	24	NAT	2	Sampling	0.9 to 0.92
12	MSQ	2	Informed Search	C, D	25	MSQ	2	Conditional Independence	B, D
13	MSQ	2	Informed Search	A, C					

Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to H; root value 16)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 17, C = 11, A = 11, E = 17, F = 2, D = 2, I = 18, J = 16, H = 16. Root value = **16**; MAX's optimal first move is towards H.

Q.2 [MCQ, 1M · Uninformed Search] Answer: A (A: S, B, C, A, J, E, G)

Adjacency lists (alphabetical): A: {B, E, G, J}; B: {A, J, S}; C: {E, J, S}; E: {A, C, G, J}; G: {A, E}; J: {A, B, C, E}; S: {B, C}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B, C]
2	B	[C, A, J]
3	C	[A, J, E]
4	A	[J, E, G]
5	J	[E, G]
6	E	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, B, C, A, J, E, G**. The path returned is S → B → A → G with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [NAT, 1M · Uninformed Search] **Answer: 240**

Nodes at depth i are regenerated in every iteration with limit $\geq i$, i.e. $(d + 1 - i)$ times.

$$N = 6 \times 2 + 5 \times 4 + 4 \times 8 + 3 \times 16 + 2 \times 32 + 1 \times 64 = \mathbf{240}.$$

Compare BFS: 126; the overhead of IDS is small for $b \geq 2$.

Q.4 [NAT, 1M · Problem Formulation] **Answer: 24**

The agent can be in any of 3 squares, and each of the 3 squares is dirty or clean, giving 2^3 combinations.

$$\text{States} = n \cdot 2^n = 3 \times 8 = \mathbf{24}.$$
 The 2-square world has $2 \cdot 2^2 = 8$ states.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 4**

Case $p=T$: the first clause is satisfied and the second requires $r=T$; q is free, giving 2 models. Case $p=F$: the second clause is satisfied and the first requires $q=T$; r is free, giving 2 models. Total = $2 + 2 = 4$.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 4**

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 2$ this gives **4** (verified by enumerating all interpretations).

Q.7 [MCQ, 1M · Predicate Logic] **Answer: A** ($A: \forall x (F(x) \rightarrow \neg P(x))$)

'None of my friends are perfect' $\equiv \neg \exists x (F(x) \wedge P(x)) \equiv \forall x \neg (F(x) \wedge P(x)) \equiv \forall x (F(x) \rightarrow \neg P(x))$. Options B and C (which are equivalent to each other) only say some friend is not perfect. Option D speaks about non-friends.

Q.8 [NAT, 1M · Bayesian Networks] **Answer: 32**

$P(C)$ needs $3 - 1 = 2$ numbers. Each feature's CPT $P(F_i | C)$ has 3 rows with 1 free number each, i.e. 3 numbers. Total = $2 + 10 \times 3 = 32$ (versus $3 \times 2^{10} - 1 = 3071$ for the full joint).

Q.9 [NAT, 1M · Exact Inference] **Answer: 0.198 to 0.202**

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
$A=T, B=T$	0.06
$A=T, B=F$	0.09
$A=F, B=T$	0.0125
$A=F, B=F$	0.0375

Sum = **0.2** ≈ 0.200 .

Q.10 [NAT, 1M · Sampling] **Answer: 0.2734 to 0.2738**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(FI=F) \times P(Fe=F|FI=F) \times P(Co=T|FI=F, Fe=F) \times P(Fa=T|FI=F, Co=T)$$

$$= 0.8 \times 0.95 \times 0.9 \times 0.4 = \mathbf{0.2736} \approx 0.2736.$$

Q.11 [MCQ, 2M · Uninformed Search] **Answer: A** ($A: S \rightarrow C \rightarrow I \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(5), C(7), F(9)
2	B (5)	C(7), F(8)
3	C (7)	F(8), I(12)
4	F (8)	I(12), H(17)
5	I (12)	H(17), G(20)
6	H (17)	G(20)
7	G (20)	← goal, stop

Path returned: **S** $\rightarrow$ **C** $\rightarrow$ **I** $\rightarrow$ **G**, cost = $7 + 5 + 8 = \mathbf{20}$. Nodes expanded (including S and G): **7** (S, B, C, F, I, H, G).

Q.12 [MSQ, 2M · Informed Search] Answer: C, D (C: Node E is expanded.; D: Node J is expanded.)

Heuristic values: $h(A)=10$, $h(C)=8$, $h(E)=9$, $h(F)=6$, $h(G)=0$, $h(J)=3$, $h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	A(14), C(15)
2	A (14)	C(15), E(20)
3	C (15)	J(19), E(20)
4	J (19)	E(20), G(21), F(27)
5	E (20)	G(21), F(22)
6	G (21)	F(22) ← goal, stop

Path returned: **S** → **C** → **J** → **G** with cost **21**; nodes expanded = **6** (S, A, C, J, E, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 21$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.13 [MSQ, 2M · Informed Search] Answer: A, C (A: The graph-search version is complete in finite state spaces; C: Its worst-case time and space complexity is $O(b^m)$, where m is the maximum depth)

(A) is true: in a finite space with an explored set, it eventually expands every state. (B) is false: greedy search ignores g , so admissibility does not help. (C) is true, as stated in AIMA. (D) is false. Counterexample: $S \rightarrow A$ costs 100 and $A \rightarrow G$ costs 1, while $S \rightarrow B$ costs 1 and $B \rightarrow G$ costs 2. Then $h^*(A)=1 < h^*(B)=2$, so greedy expands A and returns cost 101 instead of 3. A perfect heuristic makes A* optimal, not greedy search.

Q.14 [NAT, 2M · Adversarial Search] Answer: 15.79 to 15.81

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.4 \times 4 + 0.6 \times 14 = 10$
- $B = 0.4 \times 17 + 0.6 \times 15 = 15.8$
- $C = 0.2 \times 7 + 0.8 \times 11 = 10.2$
- $D = 0.4 \times 15 + 0.2 \times 2 + 0.4 \times 3 = 7.6$

Root = $\max(10, 15.8, 10.2, 7.6) = \mathbf{15.8}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [MSQ, 2M · Local Search] Answer: A, B, C

(A) is the classical convergence result for sufficiently slow cooling. (B) is true: moves with $\Delta E > 0$ are always taken. (C) is true: $e^{\Delta E/T} \rightarrow 1$ as $T \rightarrow \infty$, so almost every move is accepted. (D) is false: as $T \rightarrow 0$, $e^{\Delta E/T} \rightarrow 0$ for $\Delta E < 0$, so worsening moves are rejected and the algorithm becomes hill climbing.

Q.16 [NAT, 2M · Propositional Logic] Answer: 5

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(S \rightarrow Q) \vee (\neg R \vee P)$	$(P \wedge R) \leftrightarrow (P \rightarrow \neg Q)$	$((S \rightarrow Q) \vee (\neg R \vee P)) \rightarrow ((P \wedge R) \leftrightarrow (P \rightarrow \neg Q))$
T	T	T	T	T	F	F
T	T	T	F	T	F	F
T	T	F	T	T	T	T
T	T	F	F	T	T	T
T	F	T	T	T	T	T
T	F	T	F	T	T	T
T	F	F	T	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	T	T	F	T	F	F
F	T	F	T	T	F	F
F	T	F	F	T	F	F
F	F	T	T	F	F	T
F	F	T	F	T	F	F
F	F	F	T	T	F	F
F	F	F	F	T	F	F

Rows where F = T: **5**.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A, B, D

(A) True: forward chaining with premise counters runs in linear time. (B) True: this is the definition. (C) False: q and r are both positive. (D) True: if the clashing positive literal comes from C_1 (≤ 1 positive) and the negative from C_2 , the resolvent contains C_1 's other literals (all negative) plus C_2 's remaining literals (≤ 1 positive), so it has at most one positive literal.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $\neg(\neg P \rightarrow \neg R) \rightarrow \neg(P \rightarrow Q)$; D: $\neg((P \rightarrow Q) \wedge \neg(\neg P \rightarrow \neg R))$)

Compare truth tables with $F = (P \rightarrow Q) \rightarrow (\neg P \rightarrow \neg R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(\neg P \rightarrow \neg R) \rightarrow (P \rightarrow Q)$ — differs at $P=T, Q=F, R=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(B) $\neg(\neg P \rightarrow \neg R) \rightarrow \neg(P \rightarrow Q)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(C) $(P \rightarrow Q) \vee \neg(\neg P \rightarrow \neg R)$ — differs at $P=T, Q=F, R=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(D) $\neg((P \rightarrow Q) \wedge \neg(\neg P \rightarrow \neg R))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

Q.19 [MSQ, 2M · Predicate Logic] Answer: A, B, C (A: $\exists x R(x, x)$; B: $\exists x \forall y \neg R(x, y)$; C: $\forall y \exists x R(x, y)$)

$R = \{(2,1), (3,2), (3,3), (3,4), (4,2), (4,3)\}$.

(A) $\exists x R(x, x)$: **TRUE** — self-loop at 3.

(B) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

(C) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(D) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,1)$ holds but $R(1,2)$ does not.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Propositional Logic] Answer: C (C: $(Q \vee P) \rightarrow (Q \rightarrow Q)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	T	F	T
T	F	F	T

(A) $(R \leftrightarrow P) \wedge (Q \rightarrow R)$ is false in the KB-model $P=T, Q=T, R=F \Rightarrow$ not entailed.

(B) $\neg Q \leftrightarrow \neg R$ is false in the KB-model $P=T, Q=T, R=F \Rightarrow$ not entailed.

(C) $(Q \vee P) \rightarrow (Q \rightarrow Q)$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $\neg R \wedge (R \vee R)$ is false in the KB-model $P=T, Q=T, R=F \Rightarrow$ not entailed.

Q.21 [NAT, 2M · Exact Inference] Answer: 0.898 to 0.902

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm} \mid \text{Burglary}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Alarm})$.

Inference by enumeration: $P(\text{JohnCalls} \mid e) = \alpha \sum_{\text{hidden}} P(\text{JohnCalls}, e, \text{hidden})$.

Hidden: Ea	Jo	product of CPT entries ($Bu \cdot Ea \cdot Al \cdot Jo$)	value
T	T	$0.9 \times 0.4 \times 0.9 \times 0.9$	0.2916
F	T	$0.9 \times 0.6 \times 0.9 \times 0.9$	0.4374
T	F	$0.9 \times 0.4 \times 0.9 \times 0.1$	0.0324
F	F	$0.9 \times 0.6 \times 0.9 \times 0.1$	0.0486

$P(\text{JohnCalls}=T, e) = 0.729$; $P(\text{JohnCalls}=F, e) = 0.081$; $P(e) = 0.81$.

$P(\text{JohnCalls}=T \mid e) = 0.729 / 0.81 = \mathbf{0.9} \approx 0.900$.

Q.22 [MCQ, 2M · Exact Inference] Answer: D (D: $f()$ — a constant (scope $\emptyset$))

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(B, C)$, $f(E)$, $f(C, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B)$, $f(A, B, C)$, $f(B, C)$	$\{A, B, C\}$	$f(A, C)$
C	$f(C, F)$, $f(A, C)$	$\{A, C, F\}$	$f(A, F)$
F	$f(A, F)$	$\{A, F\}$	$f(A)$
E	$f(E)$	$\{E\}$	$f()$

The largest factor produced has **2** variable(s) (created when eliminating B); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.23 [MCQ, 2M · Probability for AI] Answer: A (A: 0.600)

Bayes rule in odds form: posterior odds = likelihood ratio $\times$ prior odds = $6 \times (1/4) = 1.5$. Then $P(h | e) = 1.5/(1 + 1.5) = 0.6$. Check directly: $P(h) = 0.2$; with $P(e | h) = 6k$ and $P(e | \neg h) = k$, $P(h | e) = 1.2k/(1.2k + 0.8k) = 0.6$. $0.857 = 6/7$ ignores the prior, 0.250 reads the prior odds $1 : 4$ as a probability, and 0.200 is the prior $P(h) = 1/5$, ignoring the evidence.

Q.24 [NAT, 2M · Sampling] Answer: 0.9 to 0.92

Rejection sampling keeps only samples consistent with the evidence $A = F$: samples 1, 2, 4, 5, 6, 7, 8, 11, 12, 13, 14 (11 accepted, 4 rejected).

Among the accepted samples, $C = T$ in 10. Estimate = $10/11 = \mathbf{0.9091}$.

(For this network the exact value is 0.7725; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.25 [MSQ, 2M · Conditional Independence] Answer: B, D (B: $A \perp C | B$; D: $A \perp C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $B \perp A | C, F$: **FALSE** — the path $B \rightarrow D \leftarrow A$ is active given $\{C, F\}$.

(B) $A \perp C | B$: **TRUE** — $A \rightarrow D \leftarrow B \rightarrow C$ (blocked: collider D with neither it nor a descendant observed); $A \rightarrow D \rightarrow E \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow D \rightarrow G \leftarrow B \rightarrow C$ (blocked: collider G with neither it nor a descendant observed); $A \rightarrow F \leftarrow D \leftarrow B \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow F \leftarrow D \rightarrow E \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow F \leftarrow D \rightarrow G \leftarrow B \rightarrow C$ (blocked: collider F with neither it nor a descendant observed).

(C) $C \perp E | B$: **FALSE** — the path $C \rightarrow E$ is active given $\{B\}$.

(D) $A \perp C$: **TRUE** — $A \rightarrow D \leftarrow B \rightarrow C$ (blocked: collider D with neither it nor a descendant observed); $A \rightarrow D \rightarrow E \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow D \rightarrow G \leftarrow B \rightarrow C$ (blocked: collider G with neither it nor a descendant observed); $A \rightarrow F \leftarrow D \leftarrow B \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow F \leftarrow D \rightarrow E \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow F \leftarrow D \rightarrow G \leftarrow B \rightarrow C$ (blocked: collider F with neither it nor a descendant observed).

Mock Test 057

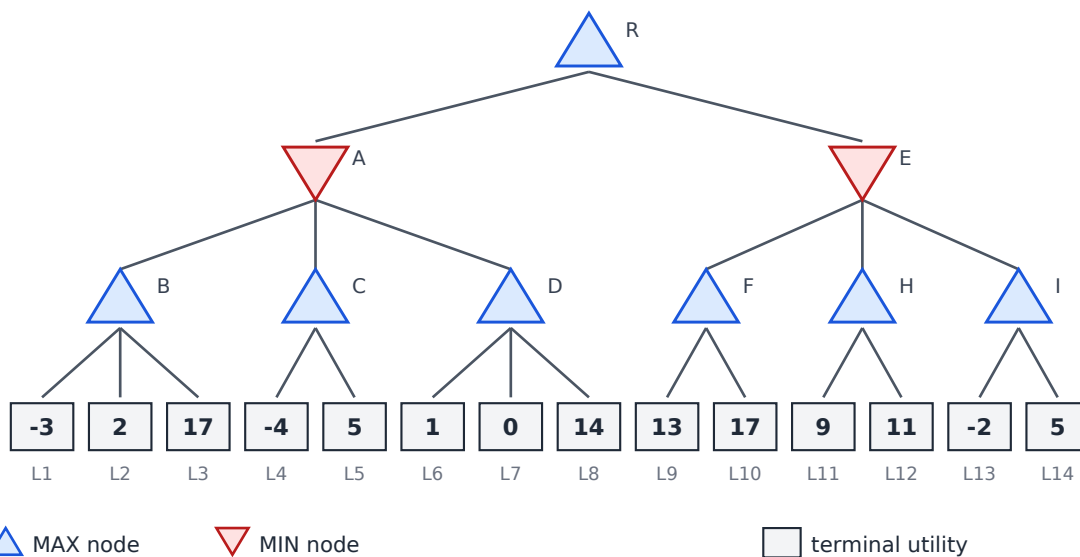
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 8 / 13	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

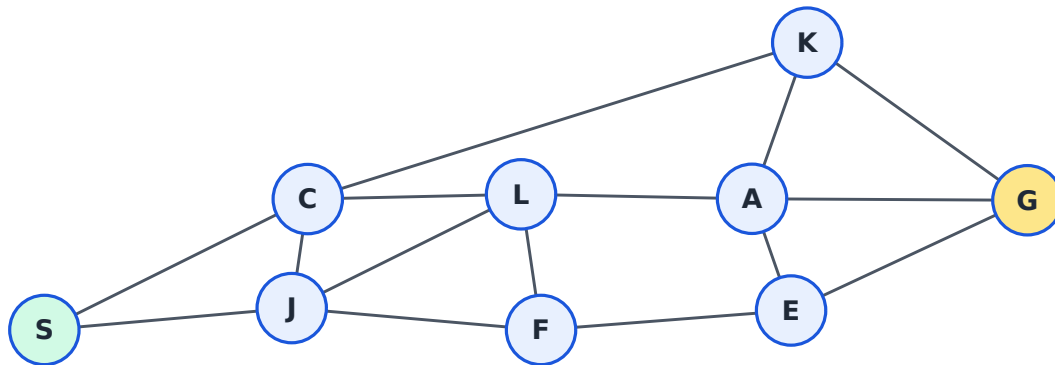
Q.2 [MSQ · 1 mark · no negative marking] Informed Search

h_1 and h_2 are admissible heuristics. Which of the following are **guaranteed** to be admissible?

- ☐ (A) $\max(h_1, h_2)$
- ☐ (B) $(h_1 + h_2) / 2$
- ☐ (C) $h_1 + h_2$
- ☐ (D) $0.5 \cdot h_1$

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, J, C, L, F, K, A, E, G
☐ (B) S, C, J, K, L, A, F, G
☐ (C) S, C, J, F, E, A, G
☐ (D) S, C, J, K, L, F, A, G

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 5$ (so $b^d = 243$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, D, A** and rules:

- R1: $J \wedge A \rightarrow K$
 R2: $K \wedge J \rightarrow B$
 R3: $B \wedge K \rightarrow D$
 R4: $H \rightarrow F$
 R5: $D \rightarrow H$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3, 4\}$. How many interpretations of a unary predicate P over D make $\exists x P(x) \wedge \neg \forall x P(x)$ true?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Skolemizing the sentence $\forall x \exists y P(x, y)$ yields

- ☐ (A) $\forall x P(x, F(x))$, where F is a new function symbol
☐ (B) $\forall x P(x, A)$, where A is a new constant
☐ (C) $\exists y P(y, y)$
☐ (D) $\forall x \forall y P(x, y)$

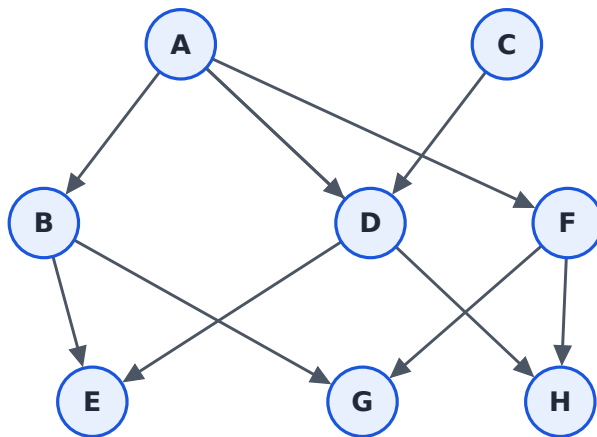
Q.8 [MSQ · 1 mark · no negative marking] Propositional Logic

In the DPLL algorithm, a pure symbol is one that appears with the same sign in every clause in which it occurs. For the clause set $\{A \vee \neg B, \neg B \vee \neg C, C \vee A\}$, which symbols are pure?

- ☐ (A) A
☐ (B) B
☐ (C) C
☐ (D) None of the symbols is pure

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **D** in the Bayesian network below?



Answer (numerical): _____

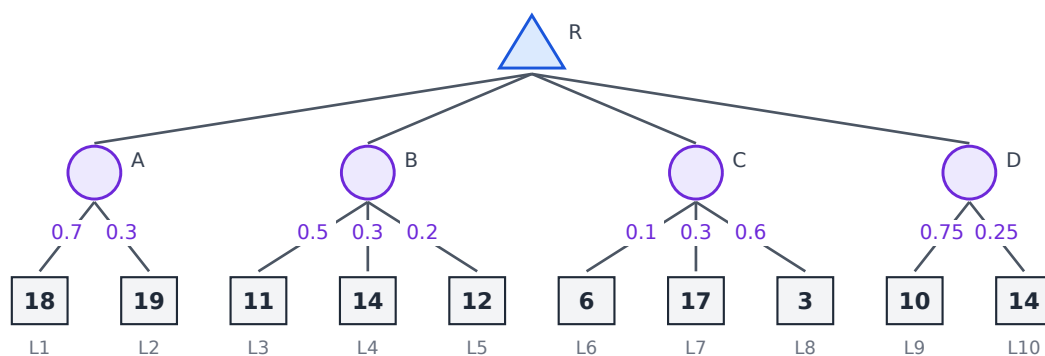
Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks

A Bayesian network over Boolean variables has edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $C \rightarrow E$. How many independent parameters do its CPTs contain?

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



MAX node



CHANCE node (edge = probability)

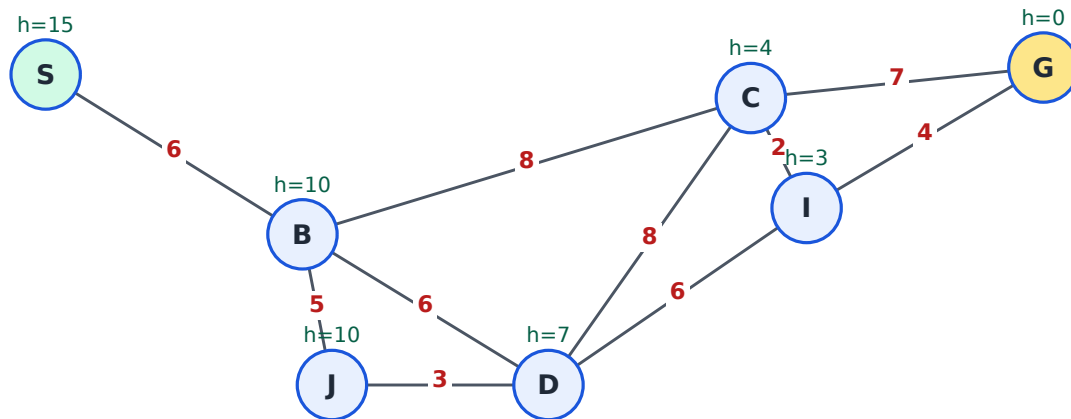


terminal utility

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

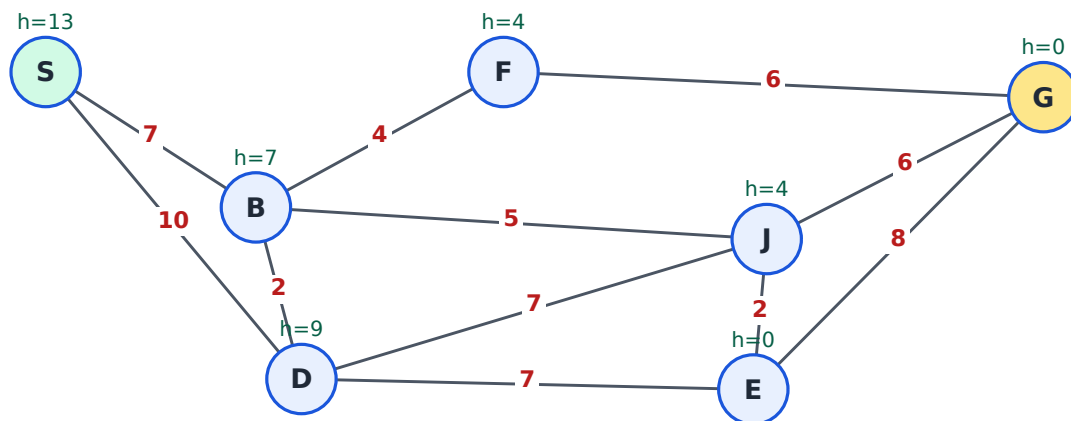
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(E)$ overestimates the true cost from E to G.
- ☐ (B) h is admissible.
- ☐ (C) The edge J-E violates the consistency condition $h(J) \leq c(J,E) + h(E)$.
- ☐ (D) h is consistent (monotone).

Q.14 [MCQ · 2 marks · negative: -2/3] Adversarial Search

MAX has two moves, each leading to a chance node with two equally likely outcomes. Move A has leaf values (0, 10) and move B has leaf values (4, 5). Every leaf value is then replaced by its square root. Which move does expectimax choose before and after the transformation?

- ☐ (A) A before, A after
- ☐ (B) A before, B after
- ☐ (C) B before, A after
- ☐ (D) B before, B after

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(R \rightarrow Q) \vee (Q \rightarrow R)$
☐ (B) $(Q \wedge (Q \rightarrow R)) \rightarrow R$
☐ (C) $((\neg P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (\neg P \rightarrow Q)$
☐ (D) $((Q \rightarrow P) \wedge P) \rightarrow Q$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic

On the domain $D = \{1, 2, 3\}$, let $R = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$. Which of the following sentences are TRUE in this interpretation?

- ☐ (A) $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$
☐ (B) $\forall x R(x, x)$
☐ (C) $\exists x \forall y R(x, y)$
☐ (D) $\forall y \exists x R(x, y)$

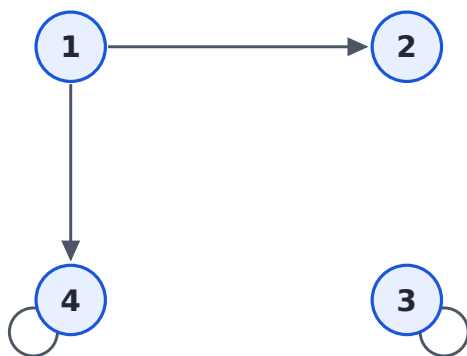
Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic

Let $\phi = \forall x \exists y (R(x, y) \wedge \neg R(y, x))$. Which of the following statements are TRUE?

- ☐ (A) ϕ has no model with a domain of size 1
☐ (B) ϕ has a model with domain $\{1, 2\}$
☐ (C) ϕ has a model with domain $\{1, 2, 3\}$
☐ (D) ϕ has a model with domain $\mathbb{N}$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x R(x, x)$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$

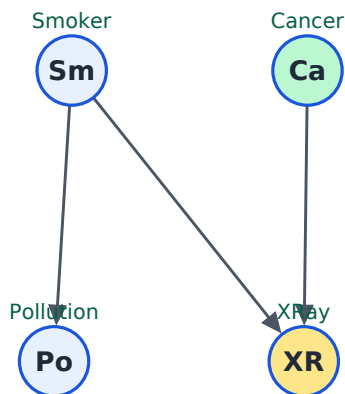
Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

The formula $F = ((P \wedge R) \vee S) \leftrightarrow ((R \vee Q) \rightarrow R)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R, S . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $P(\text{Cancer} = T \mid \text{XRay} = F)$ (round off to 3 decimal places).



$P(\text{Sm}=T)$
0.1

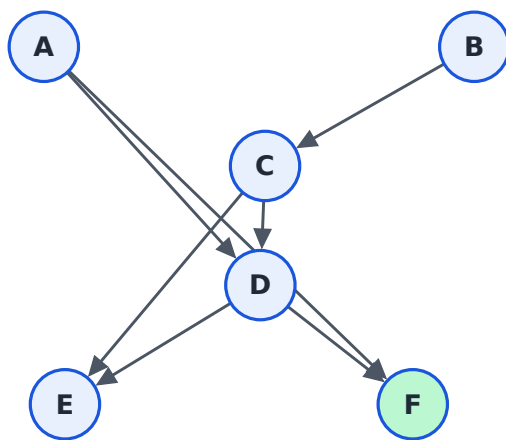
Sm	$P(\text{Po}=T \dots)$	$P(\text{Ca}=T)$
T	0.25	0.3
F	0.5	

Sm	Ca	$P(\text{XR}=T \dots)$
T	T	0.25
T	F	0.4
F	T	0.9
F	F	0.9

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

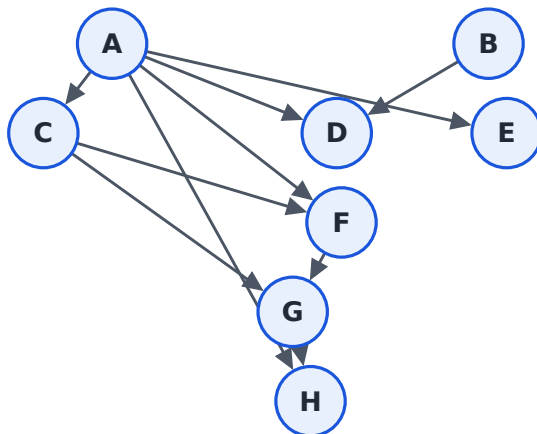
Variable elimination is used to compute $P(F)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, E, D, A, C**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp H \mid C, F$
☐ (B) $D \perp B \mid G$
☐ (C) $F \perp B \mid A, G$
☐ (D) $C \perp G$

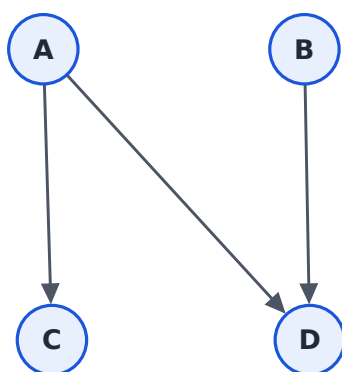
Q.23 [NAT · 2 marks · no negative marking] Variable Elimination

A Bayesian network is the chain $A \rightarrow B \rightarrow C \rightarrow D$ over Boolean variables with $P(a) = 0.2$; $P(b \mid a) = 0.9$, $P(b \mid \neg a) = 0.3$; $P(c \mid b) = 0.6$, $P(c \mid \neg b) = 0.1$; $P(d \mid c) = 0.7$, $P(d \mid \neg c) = 0.2$. Compute $P(d)$ by eliminating A, B, C in that order. (Round to 3 decimal places.)

Answer (numerical): _____

Q.24 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.62, 0.57, 0.86, 0.46)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.05

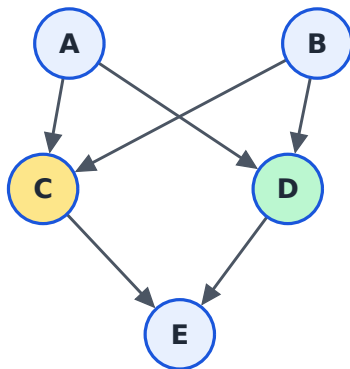
P(B=T)	A	P(C=T ...)
0.7	T	0.5
	F	0.7

A	B	P(D=T ...)
T	T	0.05
T	F	0.25
F	T	0.2
F	F	0.4

- ☐ (A) A=T, B=T, C=F, D=F
☐ (B) A=T, B=F, C=T, D=T
☐ (C) A=F, B=T, C=F, D=F
☐ (D) A=F, B=F, C=F, D=F

Q.25 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{C=F}$. The current state of the chain is $(\mathbf{A=F, B=T, C=F, D=T, E=T})$. The next step resamples $\mathbf{D}$. With what probability is $\mathbf{D}$ set to $\mathbf{T}$? (round off to 3 decimal places)



P(A=T)
0.25

P(B=T)
0.75

A	B	P(C=T ...)
T	T	0.3
T	F	0.9
F	T	0.95
F	F	0.6

A	B	P(D=T C, D)	P(E=T ...)
T	T	0.6	0.75
T	F	0.8	0.9
F	T	0.05	0.75
F	F	0.6	0.9

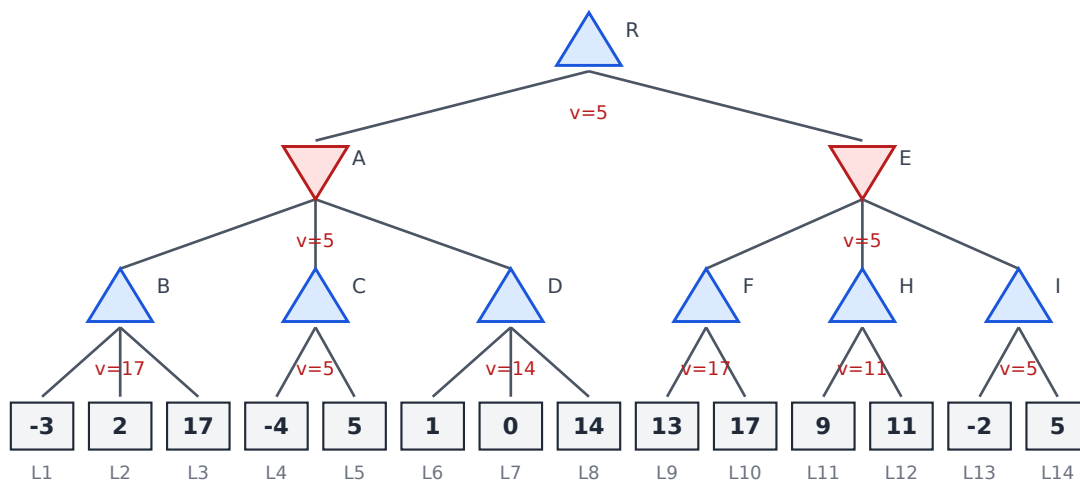
Answer (numerical): _____

Answer Key — Mock Test 057

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	5	14	MCQ	2	Adversarial Search	B
2	MSQ	1	Informed Search	A, B, D	15	MSQ	2	Propositional Logic	A, B, C
3	MCQ	1	Uninformed Search	D	16	MSQ	2	Predicate Logic	A, C, D
4	NAT	1	Adversarial Search	35	17	MSQ	2	Predicate Logic	A, C, D
5	NAT	1	Propositional Logic	5	18	MSQ	2	Predicate Logic	A, D
6	NAT	1	Predicate Logic	14	19	NAT	2	Propositional Logic	6
7	MCQ	1	Predicate Logic	A	20	NAT	2	Exact Inference	0.318 to 0.322
8	MSQ	1	Propositional Logic	A, B	21	NAT	2	Exact Inference	3
9	NAT	1	Conditional Independence	4	22	MSQ	2	Conditional Independence	C
10	NAT	1	Bayesian Networks	10	23	NAT	2	Variable Elimination	0.354 to 0.356
11	NAT	2	Adversarial Search	18.29 to 18.31	24	MCQ	2	Sampling	C
12	NAT	2	Informed Search	6	25	NAT	2	Sampling	0.04 to 0.044
13	MSQ	2	Informed Search	B, C					

Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 5**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 17$, $C = 5$, $D = 14$, $A = 5$, $F = 17$, $H = 11$, $I = 5$, $E = 5$. Root value = **5**; MAX's optimal first move is towards A, E.

Q.2 [MSQ, 1M · Informed Search] **Answer: A, B, D** (A: $\max(h_1, h_2)$; B: $(h_1 + h_2) / 2$; D: $0.5 \cdot h_1$)

Since $h_1, h_2 \leq h^*$, the max is $\leq h^*$ (A), the average is $\leq h^*$ (B), and any non-negative multiple ≤ 1 such as $0.5 \cdot h_1$ is $\leq h^*$ (D). The sum (C) can reach $2h^*$, for example when both heuristics equal h^* , so it is not guaranteed admissible.

Q.3 [MCQ, 1M · Uninformed Search] **Answer: D** (D: S, C, J, K, L, F, A, G)

Adjacency lists (alphabetical): A: {E, G, K, L}; C: {J, K, L, S}; E: {A, F, G}; F: {E, J, L}; G: {A, E, K}; J: {C, F, L, S}; K: {A, C, G}; L: {A, C, F, J}; S: {C, J}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[C, J]
2	C	[J, K, L]
3	J	[K, L, F]
4	K	[L, F, A, G]
5	L	[F, A, G]
6	F	[A, G, E]
7	A	[G, E]
8	G	[E] $\leftarrow$ goal popped, stop

Expansion order: **S, C, J, K, L, F, A, G**. The path returned is $S \rightarrow C \rightarrow K \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [NAT, 1M · Adversarial Search] **Answer: 35**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 3^3 + 3^2 - 1 = 27 + 9 - 1 = \mathbf{35}$, versus 243 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.5 [NAT, 1M · Propositional Logic] **Answer: 5**

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \rightarrow H$; add H.

Round 2: fire $H \rightarrow F$; add F.

Fixed point: {A, B, D, F, H} — 5 atoms are entailed (facts: A, B, D; newly derived: F, H). Atoms never derived: C, E, J, K.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 14**

An interpretation of P is a subset of D: $2^4 = 16$ choices. $\exists x P(x)$ excludes the empty set, $\neg \forall x P(x)$ excludes D itself. Hence $16 - 2 = 14$.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A ($A: \forall x P(x, F(x))$, where F is a new function symbol)

Since the existential $\exists y$ lies within the scope of $\forall x$, the chosen y may depend on x , so it is replaced by a Skolem function $F(x)$. A Skolem constant would wrongly assert one y works for all x . The result is satisfiable iff the original is (though not logically equivalent). $\forall x \forall y P(x, y)$ is much stronger.

Q.8 [MSQ, 1M · Propositional Logic] Answer: A, B ($A: A; B: B$)

A occurs only positively (in $A \vee \neg B$ and $C \vee A$), so it is pure; DPLL can set $A = T$. B occurs only negatively ($\neg B$ twice), so it is pure; DPLL can set $B = F$. C occurs negatively in $\neg B \vee \neg C$ and positively in $C \vee A$, so it is not pure.

Q.9 [NAT, 1M · Conditional Independence] Answer: 4

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of D : $\{A, C\}$; children: $\{E\}$; co-parents: $\{B\}$.

$MB(D) = \{A, B, C, E\}$ — 4 nodes. Given its Markov blanket, D is conditionally independent of every other node in the network.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 10

A Boolean node with k Boolean parents needs 2^k numbers. A : 1, B : 1, C (2 parents): 4, D (1 parent): 2, E (1 parent): 2. Total = 10, compared with $2^5 - 1 = 31$ for the full joint.

Q.11 [NAT, 2M · Adversarial Search] Answer: 18.29 to 18.31

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.7 \times 18 + 0.3 \times 19 = 18.3$
- $B = 0.5 \times 11 + 0.3 \times 14 + 0.2 \times 12 = 12.1$
- $C = 0.1 \times 6 + 0.3 \times 17 + 0.6 \times 3 = 7.5$
- $D = 0.75 \times 10 + 0.25 \times 14 = 11$

Root = $\max(18.3, 12.1, 7.5, 11) = 18.3$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.12 [NAT, 2M · Informed Search] Answer: 6

Heuristic values: $h(B)=10$, $h(C)=4$, $h(D)=7$, $h(G)=0$, $h(I)=3$, $h(J)=10$, $h(S)=15$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	B(16)
2	B (16)	C(18), D(19), J(21)
3	C (18)	D(19), I(19), G(21), J(21)
4	D (19)	I(19), G(21), J(21)
5	I (19)	G(20), J(21)
6	G (20)	J(21) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **B** $\rightarrow$ **C** $\rightarrow$ **I** $\rightarrow$ **G** with cost **20**; nodes expanded = **6** (S, B, C, D, I, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 20$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.13 [MSQ, 2M · Informed Search] Answer: B, C (B: h is admissible.; C: The edge $J \rightarrow E$ violates the consistency condition $h(J) \leq c(J, E) + h(E)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G :

n	B	D	E	F	G	J	S
$h(n)$	7	9	0	4	0	4	13
$h^*(n)$	10	12	8	6	0	6	17
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u, v) + h(v)$ for every edge in both directions. Violating directed edges: $J \rightarrow E$ ($4 > 2 + 0$), $D \rightarrow E$ ($9 > 7 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [MCQ, 2M · Adversarial Search] **Answer: B** (B: A before, B after)

Before: $E[A] = (0+10)/2 = 5$ and $E[B] = 4.5$, so A is chosen. After: $E[A] = (0 + 3.162)/2 \approx 1.58$ and $E[B] = (2 + 2.236)/2 \approx 2.12$, so B is chosen. Expectimax decisions are preserved only by positive linear transformations of the evaluation function. A merely order-preserving map such as $\sqrt{x}$ can change them. Minimax decisions are invariant under any order-preserving map.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: A, B, C** (A: $(R \rightarrow Q) \vee (Q \rightarrow R)$; B: $(Q \wedge (Q \rightarrow R)) \rightarrow R$; C: $((\neg P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (\neg P \rightarrow Q)$)

(A) $(R \rightarrow Q) \vee (Q \rightarrow R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $(Q \wedge (Q \rightarrow R)) \rightarrow R$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $((\neg P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (\neg P \rightarrow Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow P) \wedge P) \rightarrow Q$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: A, C, D** (A: $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$; C: $\exists x \forall y R(x, y)$; D: $\forall y \exists x R(x, y)$)

(A) The composable pairs are $(1,1)(1,y) \rightarrow (1,y) \in R$, and $(1,2)(2,3) \rightarrow (1,3) \in R$; so R is transitive. (B) False: $(2,2) \notin R$. (C) True with $x = 1$: $(1,1), (1,2), (1,3) \in R$. (D) True: $x = 1$ works for every y . Enumeration over all triples confirms A, C, D.

Q.17 [MSQ, 2M · Predicate Logic] **Answer: A, C, D** (A: $\emptyset$ has no model with a domain of size 1; C: $\emptyset$ has a model with domain $\{1, 2, 3\}$; D: $\emptyset$ has a model with domain $\mathbb{N}$)

On $\{1\}$: y must be 1, and $R(1, 1) \wedge \neg R(1, 1)$ is impossible, so no model. On $\{1, 2\}$: element 1 needs y with $R(1, y) \wedge \neg R(y, 1)$; $y = 1$ is impossible, so $y = 2$: $R(1,2), \neg R(2,1)$. Element 2 similarly needs $R(2,1) \wedge \neg R(1,2)$ — contradiction; brute force over all 16 relations finds no model. On $\{1, 2, 3\}$ the 3-cycle $R = \{(1,2), (2,3), (3,1)\}$ works, and on $\mathbb{N}$ $R(x, y) \iff x < y$ works.

Q.18 [MSQ, 2M · Predicate Logic] **Answer: A, D** (A: $\exists x R(x, x)$; D: $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$)

$R = \{(1,2), (1,4), (3,3), (4,4)\}$.

(A) $\exists x R(x, x)$: **TRUE** — self-loop at 3.

(B) $\forall y \exists x R(x, y)$: **FALSE** — 1 has no incoming edge.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 2 has no edge to a different element.

(D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **TRUE** — all two-step paths are short-cut.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [NAT, 2M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(P \wedge R) \vee S$	$(R \vee Q) \rightarrow R$	$((P \wedge R) \vee S) \leftrightarrow ((R \vee Q) \rightarrow R)$
T	T	T	T	T	T	T
T	T	T	F	T	T	T
T	T	F	T	T	F	F
T	T	F	F	F	F	T
T	F	T	T	T	T	T
T	F	T	F	T	T	T
T	F	F	T	T	T	T
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	T	F	F	T	F
F	T	F	T	T	F	F
F	T	F	F	F	F	T
F	F	T	T	T	T	T
F	F	T	F	F	T	F
F	F	F	T	T	T	T
F	F	F	F	F	T	F

F has 10 models, so it is false in $16 - 10 = 6$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.318 to 0.322

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution} | \text{Smoker}) \cdot P(\text{Cancer}) \cdot P(\text{XRay} | \text{Smoker}, \text{Cancer})$.

Inference by enumeration: $P(\text{Cancer} | e) = \alpha \sum_{\text{hidden}} P(\text{Cancer}, e, \text{hidden})$.

Hidden: Sm, Po	Ca	product of CPT entries (Sm · Po · Ca · XR)	value
T, T	T	$0.1 \times 0.25 \times 0.3 \times 0.75$	0.00562
T, F	T	$0.1 \times 0.75 \times 0.3 \times 0.75$	0.01688
F, T	T	$0.9 \times 0.5 \times 0.3 \times 0.1$	0.0135
F, F	T	$0.9 \times 0.5 \times 0.3 \times 0.1$	0.0135
T, T	F	$0.1 \times 0.25 \times 0.7 \times 0.6$	0.0105
T, F	F	$0.1 \times 0.75 \times 0.7 \times 0.6$	0.0315
F, T	F	$0.9 \times 0.5 \times 0.7 \times 0.1$	0.0315
F, F	F	$0.9 \times 0.5 \times 0.7 \times 0.1$	0.0315

$P(\text{Cancer}=\text{T}, e) = 0.0495$; $P(\text{Cancer}=\text{F}, e) = 0.105$; $P(e) = 0.1545$.

$P(\text{Cancer}=\text{T} | e) = 0.0495 / 0.1545 = \mathbf{0.3204} \approx 0.320$.

Q.21 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(B, C)$, $f(A, C, D)$, $f(C, D, E)$, $f(A, D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(B)$, $f(B, C)$	$\{B, C\}$	$f(C)$
E	$f(C, D, E)$	$\{C, D, E\}$	$f(C, D)$
D	$f(A, C, D)$, $f(A, D, F)$, $f(C, D)$	$\{A, C, D, F\}$	$f(A, C, F)$
A	$f(A)$, $f(A, C, F)$	$\{A, C, F\}$	$f(C, F)$
C	$f(C)$, $f(C, F)$	$\{C, F\}$	$f(F)$

The largest factor produced has **3** variable(s) (created when eliminating D); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over F and normalise.

Q.22 [MSQ, 2M · Conditional Independence] Answer: C (C: $F \perp B | A, G$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $A \perp H | C, F$: **FALSE** — the path $A \rightarrow H$ is active given $\{C, F\}$.

(B) $D \perp B | G$: **FALSE** — the path $D \leftarrow B$ is active given $\{G\}$.

(C) $F \perp B | A, G$: **TRUE** — $F \leftarrow A \rightarrow D \leftarrow B$ (blocked: observed fork node A); $F \leftarrow C \leftarrow A \rightarrow D \leftarrow B$ (blocked: observed fork node A); $F \leftarrow C \rightarrow G \rightarrow H \leftarrow A \rightarrow D \leftarrow B$ (blocked: observed chain node G); $F \rightarrow G \leftarrow C \leftarrow A \rightarrow D \leftarrow B$ (blocked: observed fork node A); $F \rightarrow G \rightarrow H \leftarrow A \rightarrow D \leftarrow B$ (blocked: observed chain node G).

(D) $C \perp G$: **FALSE** — the path $C \leftarrow A \rightarrow F \rightarrow G$ is active given $\{\}$.

Q.23 [NAT, 2M · Variable Elimination] Answer: 0.354 to 0.356

Eliminate A: $f_A(b) = 0.2 \times 0.9 + 0.8 \times 0.3 = 0.18 + 0.24 = 0.42$, $f_A(\neg b) = 0.58$. Eliminate B: $f_B(c) = 0.42 \times 0.6 + 0.58 \times 0.1 = 0.252 + 0.058 = 0.31$, $f_B(\neg c) = 0.69$. Eliminate C: $P(d) = 0.31 \times 0.7 + 0.69 \times 0.2 = 0.217 + 0.138 = 0.355$. Each step creates a factor over a single variable, so cost is linear in the chain length.

Q.24 [MCQ, 2M · Sampling] Answer: C (C: $A=F, B=T, C=F, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.05$; $u = 0.62 \geq 0.05 \Rightarrow A = F$
- B: $P(B=T | \text{—}) = 0.7$; $u = 0.57 < 0.7 \Rightarrow B = T$
- C: $P(C=T | A=F) = 0.7$; $u = 0.86 \geq 0.7 \Rightarrow C = F$
- D: $P(D=T | A=F, B=T) = 0.2$; $u = 0.46 \geq 0.2 \Rightarrow D = F$

Sample: **$\{A=F, B=T, C=F, D=F\}$** .

Q.25 [NAT, 2M · Sampling] Answer: 0.04 to 0.044

Gibbs sampling draws D from $P(D | \text{mb}(D))$, where $\text{mb}(D) = \{A, B, C, E\}$:

$P(D | \text{mb}) \propto P(D | \text{parents}(D)) \times \prod_{\text{children } Y} P(Y | \text{parents}(Y))$.

$D=T$: $P(D=T | A=F, B=T) \times P(E=T | C=F, D=T) = 0.05 \times 0.75 = 0.0375$

$D=F$: $P(D=F | A=F, B=T) \times P(E=T | C=F, D=F) = 0.95 \times 0.9 = 0.855$

Normalise: $0.0375 / (0.0375 + 0.855) = \mathbf{0.042}$.

Mock Test 058

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	8 / 7 / 10	11 / 16 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

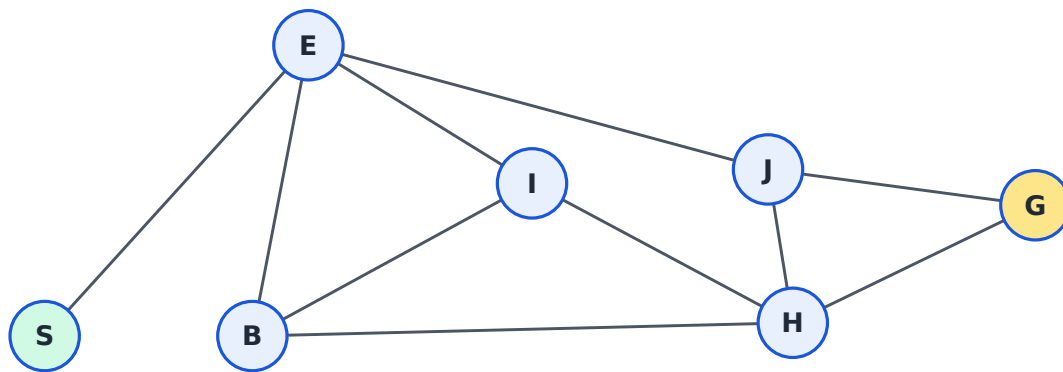
Q.1 [NAT · 1 mark · no negative marking] Informed Search

A* generates $N = 84$ nodes (not counting the root) to find a solution at depth $d = 3$. Using AIMA's definition $N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^d$, what is the effective branching factor b^* ?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, E, I, J, H, G
- ☐ (B) S, E, B, I, J, H, G
- ☐ (C) S, E, B, H, G
- ☐ (D) S, E, J, I, B, H, G

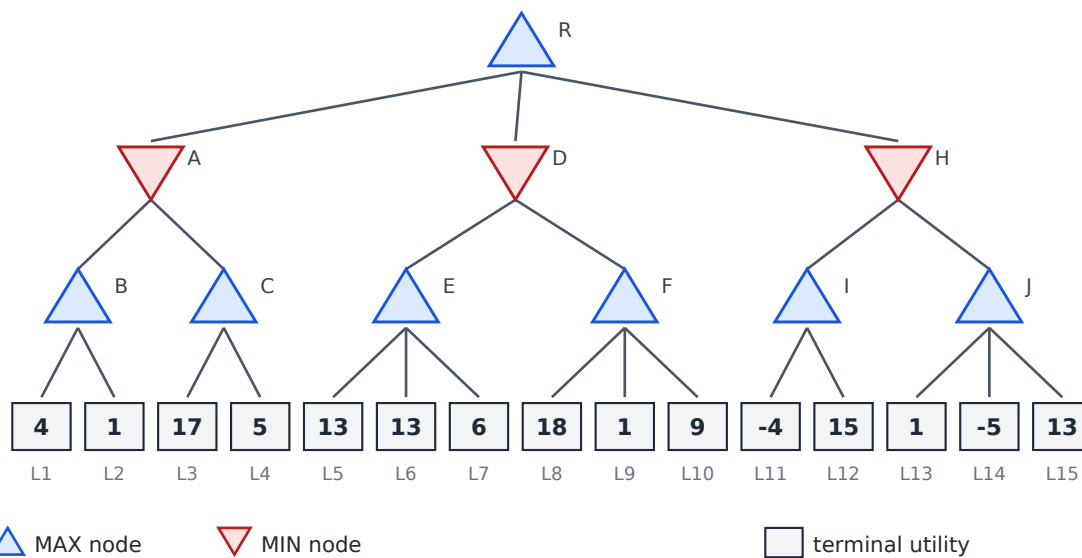
Q.3 [MCQ · 1 mark · negative: $-1/3$] Informed Search

A heuristic h is **consistent** (monotone) if, for every node n , every action a and its successor n' ,

- ☐ (A) $h(n) \leq h^*(n)$
- ☐ (B) $h(n) \leq c(n, a, n') + h(n')$
- ☐ (C) $h(n') \leq h(n)$
- ☐ (D) $h(n) + h(n') \leq c(n, a, n')$

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

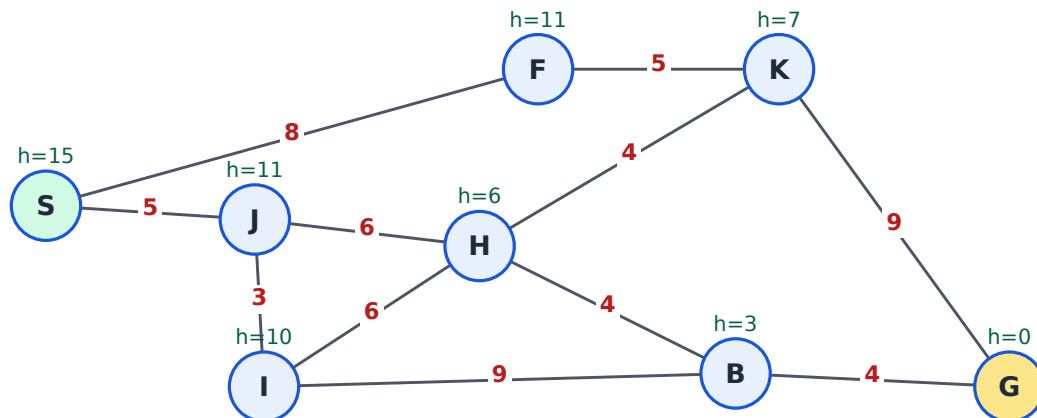
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following expresses 'None of my friends are perfect' ($F(x)$: x is my friend, $P(x)$: x is perfect)?

- ☐ (A) $\forall x (F(x) \rightarrow \neg P(x))$
☐ (B) $\exists x (F(x) \wedge \neg P(x))$
☐ (C) $\neg \forall x (F(x) \rightarrow P(x))$
☐ (D) $\forall x (\neg F(x) \rightarrow P(x))$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over the symbols p, q, r does $(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)$ have?

Answer (numerical): _____

Q.8 [MCQ · 1 mark · negative: -1/3] Conditional Independence

Burglary (B) and Earthquake (E) are independent causes of Alarm (A). Given that the alarm is ringing, learning that an earthquake occurred lowers the probability of a burglary. This phenomenon is called

- ☐ (A) explaining away (intercausal reasoning)
- ☐ (B) the local Markov property
- ☐ (C) the naive Bayes assumption
- ☐ (D) marginalization

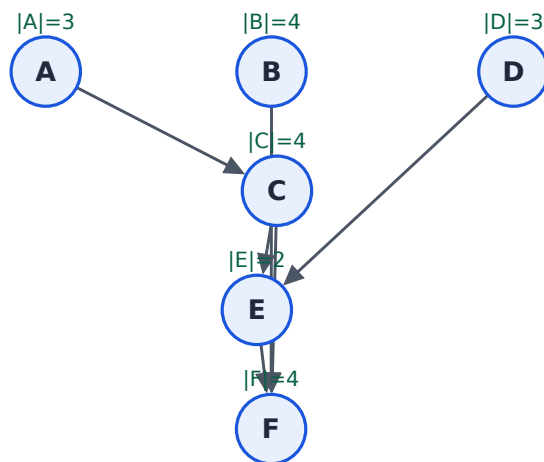
Q.9 [MCQ · 1 mark · negative: -1/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Prior (direct) sampling visits variables in the order C, S, R, W and sets a variable to true if and only if the next uniform random number u satisfies $u < P(\text{variable} = \text{true} | \text{sampled parents})$. The random numbers used are 0.42, 0.07, 0.85, 0.30 (in this order). What sample is generated?

- ☐ (A) (c, s, $\neg r$, w)
- ☐ (B) (c, $\neg s$, r, w)
- ☐ (C) (c, s, r, w)
- ☐ (D) ($\neg c$, s, $\neg r$, $\neg w$)

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?

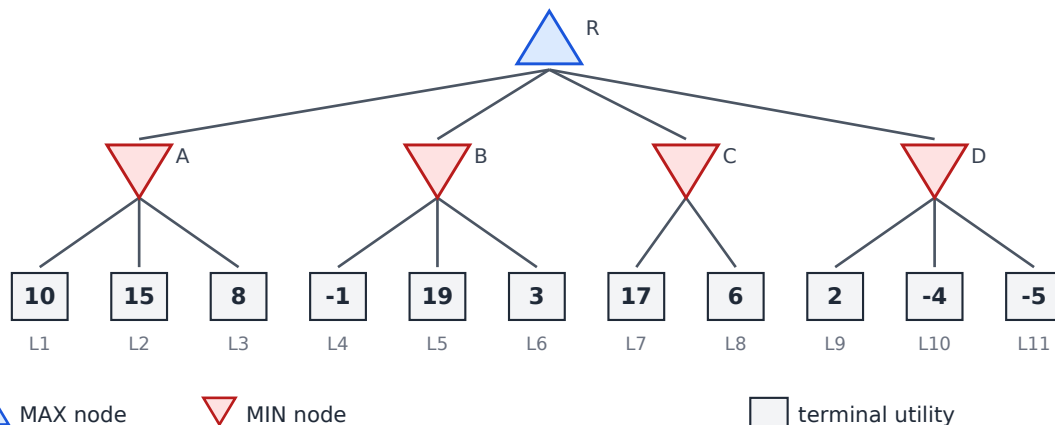


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

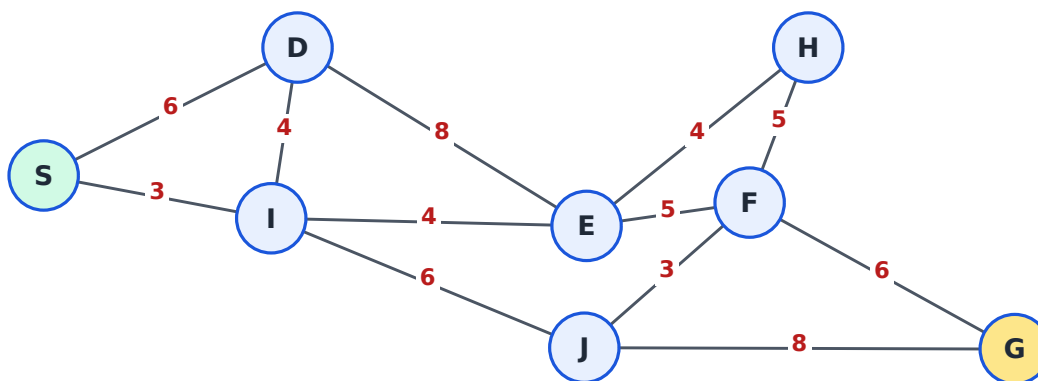
Q.12 [MCQ · 2 marks · negative: -2/3] Local Search

Steepest-ascent hill climbing solves a random 8-queens instance with probability $p \approx 0.14$. Random-restart hill climbing repeats independent trials until one succeeds. What is the expected number of trials?

- ☐ (A) About 1.14
☐ (B) About 3
☐ (C) About 7
☐ (D) About 14

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

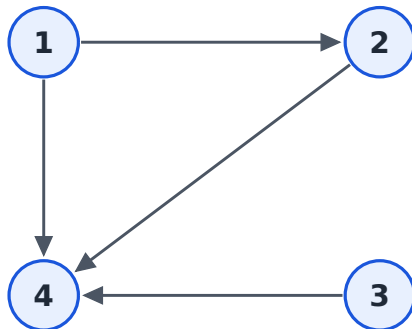
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (D) $\exists x \forall y R(x, y)$

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = ((\neg Q \leftrightarrow S) \leftrightarrow (P \vee S)) \rightarrow (\neg R \rightarrow (\neg P \vee Q)) ?$$

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **C, E** and rules:

- R1: $A \rightarrow C$
 R2: $A \wedge E \rightarrow B$
 R3: $C \wedge E \rightarrow H$
 R4: $H \wedge C \rightarrow F$
 R5: $D \wedge B \wedge K \rightarrow C$
 R6: $B \wedge J \rightarrow A$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) B is entailed
☐ (B) D is entailed
☐ (C) H is entailed
☐ (D) J is entailed

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following sentences (with equality) are true in exactly those interpretations where exactly one element satisfies P ?

- ☐ (A) $\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x))$
☐ (B) $\exists x \forall y (P(y) \leftrightarrow y = x)$
☐ (C) $\exists x \forall y ((P(x) \wedge P(y)) \rightarrow x = y)$
☐ (D) $\exists x (P(x) \wedge \forall y (y = x \rightarrow P(y)))$

Q.18 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$ and let f be a unary function symbol. How many interpretations of f (functions $D \rightarrow D$) satisfy $\forall x f(f(x)) = f(x)$?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \rightarrow P) \rightarrow Q) \rightarrow Q$
- ☐ (B) $(P \rightarrow R) \rightarrow (R \rightarrow P)$
- ☐ (C) $\neg(R \wedge P) \leftrightarrow (\neg R \vee \neg P)$
- ☐ (D) $((Q \rightarrow P) \wedge (R \rightarrow P)) \rightarrow (Q \wedge R)$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(P \wedge \neg S) \wedge \neg Q$
- (2) $(P \rightarrow Q) \vee \neg R$
- (3) $(P \leftrightarrow \neg P) \rightarrow (\neg P \vee Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $R \rightarrow \neg Q$
- ☐ (B) $(P \wedge \neg Q) \rightarrow R$
- ☐ (C) $(\neg S \vee P) \vee (P \wedge R)$
- ☐ (D) $\neg(P \wedge \neg R)$

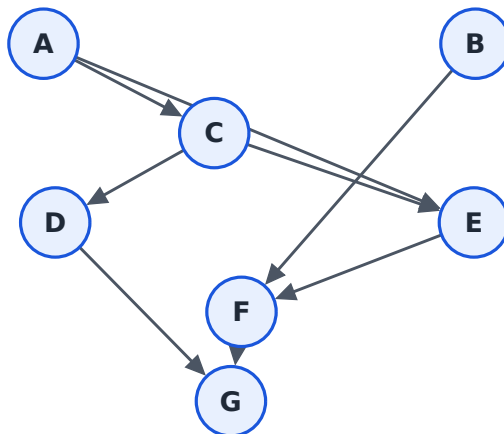
Q.21 [MSQ · 2 marks · no negative marking] Sampling

Which statements about Gibbs sampling (as an MCMC method in Bayesian networks) are correct?

- ☐ (A) Successive samples are generally correlated, not independent
- ☐ (B) With fixed evidence and all CPT entries strictly between 0 and 1, the chain's stationary distribution is $P(X \mid e)$
- ☐ (C) Each state of the Markov chain is a complete assignment to all variables, with evidence variables fixed
- ☐ (D) Each sample must be accepted or rejected depending on whether it agrees with the evidence

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

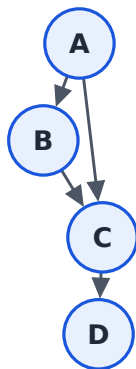
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp D \mid A, C$
☐ (B) $F \perp A \mid D, E$
☐ (C) $B \perp G \mid A$
☐ (D) $F \perp G$

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.56, 0.07, 0.05, 0.47)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.6

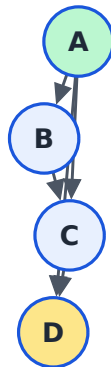
A	P(B=T ...)	A	B	P(C=T ...)
T	0.6	T	T	0.5
F	0.9	T	F	0.95
		F	T	0.8
		F	F	0.6

C	P(D=T ...)
T	0.2
F	0.05

- ☐ (A) A=T, B=T, C=T, D=F
☐ (B) A=T, B=F, C=T, D=F
☐ (C) A=F, B=F, C=F, D=T
☐ (D) A=F, B=T, C=T, D=F

Q.24 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{D} = \mathbf{T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(A = T \mid D = T)}$ (round off to 3 decimal places).



A	P(A=T)
T	0.9

A	P(B=T .)	A	B	P(C=T ...)
T	0.05	T	T	0.75
F	0.95	T	F	0.05
		F	T	0.9
		F	F	0.95

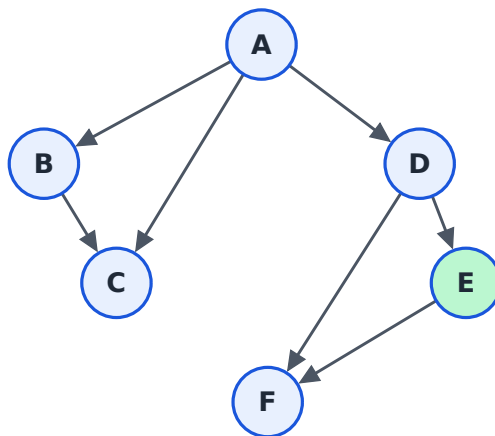
A	C	P(D=T ...)
T	T	0.4
T	F	0.7
F	T	0.05
F	F	0.7

#	A	B	C
1	F	T	T
2	T	F	F
3	T	F	F
4	T	F	F
5	T	F	F

Answer (numerical): _____

Q.25 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(E)}$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{F, C, A, D, B}$. What is the scope of the new factor produced when $\mathbf{D}$ is summed out?



- ☐ (A) $f(B, D, E)$
☐ (B) $f(E)$
☐ (C) $f(B, E)$
☐ (D) $f(B)$

Answer Key — Mock Test 058

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Informed Search	4	14	MSQ	2	Predicate Logic	A, C
2	MCQ	1	Uninformed Search	B	15	NAT	2	Propositional Logic	15
3	MCQ	1	Informed Search	B	16	MSQ	2	Propositional Logic	C
4	NAT	1	Adversarial Search	13	17	MSQ	2	Predicate Logic	A, B
5	NAT	1	Informed Search	22	18	NAT	2	Predicate Logic	10
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Propositional Logic	A, C
7	NAT	1	Propositional Logic	2	20	MSQ	2	Propositional Logic	A, C
8	MCQ	1	Conditional Independence	A	21	MSQ	2	Sampling	A, B, C
9	MCQ	1	Sampling	A	22	MSQ	2	Conditional Independence	A, B
10	NAT	1	Bayesian Networks	124	23	MCQ	2	Sampling	A
11	NAT	2	Adversarial Search	7	24	NAT	2	Sampling	0.98 to 0.984
12	MCQ	2	Local Search	C	25	MCQ	2	Exact Inference	C
13	NAT	2	Uninformed Search	8					

Detailed Solutions

Q.1 [NAT, 1M · Informed Search] Answer: 4

Solve $1 + b^* + b^{*2} + b^{*3} = 85$, which means $b^* + b^{*2} + b^{*3} = 84$. Try $b^* = 4$: $4 + 16 + 64 = 84$. So **$b^* = 4$** .

Q.2 [MCQ, 1M · Uninformed Search] Answer: B (B: S, E, B, I, J, H, G)

Adjacency lists (alphabetical): B: {E, H, I}; E: {B, I, J, S}; G: {H, J}; H: {B, G, I, J}; I: {B, E, H}; J: {E, G, H}; S: {E}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[E]
2	E	[B, I, J]
3	B	[I, J, H]
4	I	[J, H]
5	J	[H, G]
6	H	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, E, B, I, J, H, G**. The path returned is $S \rightarrow E \rightarrow J \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

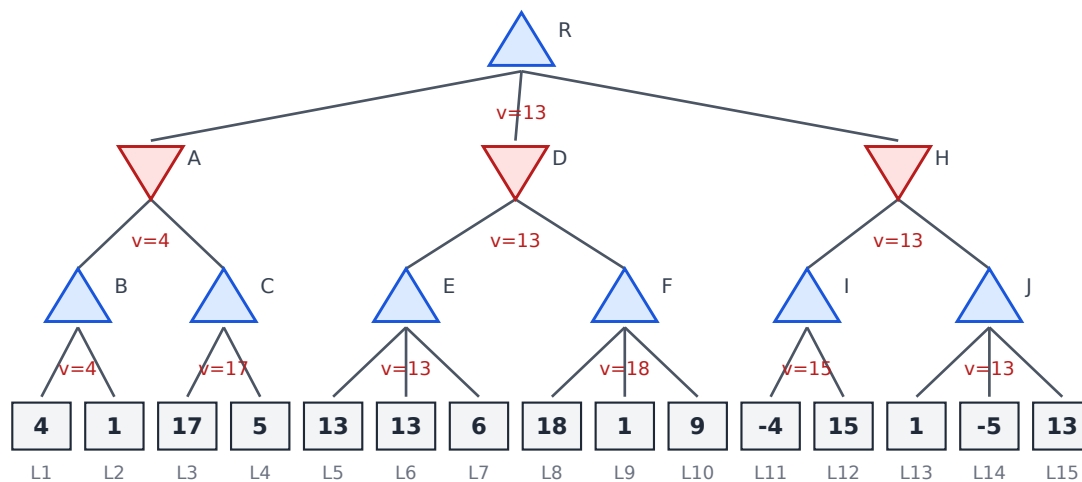
Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [MCQ, 1M · Informed Search] Answer: B (B: $h(n) \leq c(n, a, n') + h(n')$)

Consistency is the triangle inequality **$h(n) \leq c(n, a, n') + h(n')$** . (A) is admissibility. (C) is not required: a successor can have a larger h than its parent. (D) is meaningless as a condition and would force h to be tiny.

Q.4 [NAT, 1M · Adversarial Search] Answer: 13

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 4$, $C = 17$, $A = 4$, $E = 13$, $F = 18$, $D = 13$, $I = 15$, $J = 13$, $H = 13$. Root value = **13**; MAX's optimal first move is towards D, H.

Q.5 [NAT, 1M · Informed Search] Answer: 22

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (15)	F(11), J(11)
2	F (11)	K(7), J(11)
3	K (7)	G(0), H(6), J(11)
4	G (0)	H(6), J(11) ← goal, stop

Path returned: **S** → **F** → **K** → **G**, cost = $8 + 5 + 9 = 22$.

The optimal cost is $h^*(S) = 19$ — greedy search is not optimal here.

Q.6 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x (F(x) \rightarrow \neg P(x))$)

'None of my friends are perfect' $\equiv \neg \exists x (F(x) \wedge P(x)) \equiv \forall x \neg (F(x) \wedge P(x)) \equiv \forall x (F(x) \rightarrow \neg P(x))$. Options B and C (which are equivalent to each other) only say some friend is not perfect. Option D speaks about non-friends.

Q.7 [NAT, 1M · Propositional Logic] Answer: 2

The cycle of implications $p \rightarrow q \rightarrow r \rightarrow p$ forces p, q, r to have the same truth value: if any one is T, all following ones are T, and the cycle returns. So the only models are all-T and all-F, i.e. 2 models.

Q.8 [MCQ, 1M · Conditional Independence] Answer: A (A: explaining away (intercausal reasoning))

In the v-structure $B \rightarrow A \leftarrow E$ the causes are marginally independent but become dependent once the common effect is observed. One cause 'explains away' the effect and reduces the posterior of the other cause. The local Markov property concerns independence of a node from non-descendants given parents; naive Bayes is a class-feature model; marginalization is summing out variables.

Q.9 [MCQ, 1M · Sampling] Answer: A (A: $(c, s, \neg r, w)$)

C: $0.42 < P(c) = 0.5 \rightarrow c$. S: $P(s | c) = 0.1$, $0.07 < 0.1 \rightarrow s$. R: $P(r | c) = 0.8$, $0.85 \geq 0.8 \rightarrow \neg r$. W: $P(w | s, \neg r) = 0.90$, $0.30 < 0.90 \rightarrow w$. Sample = $(c, s, \neg r, w)$. (B) mis-samples S and R; (C) mis-samples R; (D) mis-samples C.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 124

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

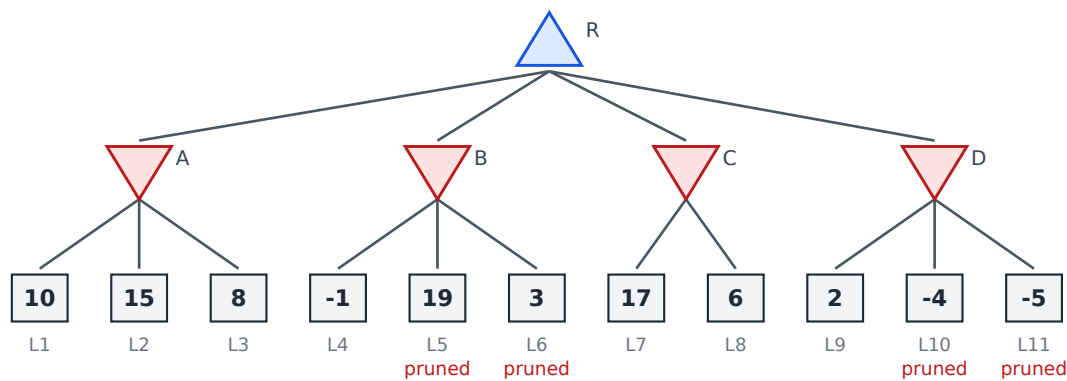
$\Sigma = A: (3-1) + B: (4-1) + C: (4-1) \times 3 + D: (3-1) + E: (2-1) \times 4 \times 3 + F: (4-1) \times 4 \times 4 \times 2 = 124$.

Full joint table: $\prod |X| - 1 = 1152 - 1 = 1151$; saving = 1027.

Q.11 [NAT, 2M · Adversarial Search] Answer: 7

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-1 \leq \alpha = 8 \Rightarrow$ prune remaining children (leaves L5, L6).
- At MIN node D: value $2 \leq \alpha = 8 \Rightarrow$ prune remaining children (leaves L10, L11).



Leaves evaluated (7): L1, L2, L3, L4, L7, L8, L9. Leaves pruned (4): L5, L6, L10, L11. Root value = 8 (same as plain minimax, 8 — pruning never changes the root value).

Q.12 [MCQ, 2M · Local Search] **Answer: C** (C: About 7)

The number of trials until the first success is geometrically distributed with mean $1/p = 1/0.14 \approx 7$. That means about 6 failures and 1 success, matching AIMA's analysis. (A) confuses $1/p$ with $1 + p$. (D) confuses $1/p$ with $100 \cdot p$.

Q.13 [NAT, 2M · Uninformed Search] **Answer: 8**

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	I(3), D(6)
2	I (3)	D(6), E(7), J(9)
3	D (6)	E(7), J(9)
4	E (7)	J(9), H(11), F(12)
5	J (9)	H(11), F(12), G(17)
6	H (11)	F(12), G(17)
7	F (12)	G(17)
8	G (17)	← goal, stop

Path returned: **S** → **I** → **J** → **G**, cost = 3 + 6 + 8 = **17**. Nodes expanded (including S and G): **8** (S, I, D, E, J, H, F, G).

Q.14 [MSQ, 2M · Predicate Logic] **Answer: A, C** (A: $\exists x \forall y \neg R(x, y)$; C: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$)

$R = \{(1,2), (1,4), (2,4), (3,4)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 4 has no outgoing edge.

(B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,3)$ nor $R(3,1)$.

(C) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(D) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.15 [NAT, 2M · Propositional Logic] **Answer: 15**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(\neg Q \leftrightarrow S) \leftrightarrow (P \vee S)$	$\neg R \rightarrow (\neg P \vee Q)$	$((\neg Q \leftrightarrow S) \leftrightarrow (P \vee S)) \rightarrow (\neg R \rightarrow (\neg P \vee Q))$
T	T	T	T	F	T	T
T	T	T	F	T	T	T
T	T	F	T	F	T	T
T	T	F	F	T	T	T
T	F	T	T	T	T	T
T	F	T	F	F	T	T
T	F	F	T	T	F	F
T	F	F	F	F	F	T
F	T	T	T	F	T	T
F	T	T	F	F	T	T
F	T	F	T	F	T	T
F	T	F	F	F	T	T

P	Q	R	S	$(\neg Q \leftrightarrow S) \leftrightarrow (P \vee S)$	$\neg R \rightarrow (\neg P \vee Q)$	$((\neg Q \leftrightarrow S) \leftrightarrow (P \vee S)) \rightarrow (\neg R \rightarrow (\neg P \vee Q))$
F	F	T	T	T	T	T
F	F	T	F	T	T	T
F	F	F	T	T	T	T
F	F	F	F	T	T	T

Rows where F = T: **15**.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: C** (C: H is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $C \wedge E \rightarrow H$; add H.

Round 2: fire $H \wedge C \rightarrow F$; add F.

Fixed point: $\{C, E, F, H\}$ — 4 atoms are entailed (facts: C, E; newly derived: F, H). Atoms never derived: A, B, D, J, K.

Q.17 [MSQ, 2M · Predicate Logic] **Answer: A, B** (A: $\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x))$; B: $\exists x \forall y (P(y) \leftrightarrow y = x)$)

(A) Standard 'exactly one': some x is P and every P is x. (B) For some x, P(y) holds iff $y = x$ — also exactly one; A and B are equivalent. (C) Is true when no element satisfies P ($P(x)$ false makes the implication vacuous), so it expresses 'at most one'. (D) The inner part is trivially implied by $P(x)$, so it is just $\exists x P(x)$ ('at least one'). Brute force on domains of size 1–4 confirms that only A and B match 'exactly one'.

Q.18 [NAT, 2M · Predicate Logic] **Answer: 10**

The condition says f is idempotent: f fixes every element of its image. Choose the image S (size k) — each element of S maps to itself — and map every element outside S anywhere into S: k^{3-k} ways. Sum over k: $k=1: 3 \times 1^2 = 3$; $k=2: 3 \times 2^1 = 6$; $k=3: 1 \times 3^0 = 1$. Total = 10.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: $((Q \rightarrow P) \rightarrow Q) \rightarrow Q$; C: $\neg(R \wedge P) \leftrightarrow (\neg R \vee \neg P)$)

(A) $((Q \rightarrow P) \rightarrow Q) \rightarrow Q$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $(P \rightarrow R) \rightarrow (R \rightarrow P)$: **not valid** — counter-model $P=F, R=T$.

(C) $\neg(R \wedge P) \leftrightarrow (\neg R \vee \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow P) \wedge (R \rightarrow P)) \rightarrow (Q \wedge R)$: **not valid** — counter-model $P=T, Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: $R \rightarrow \neg Q$; C: $(\neg S \vee P) \vee (P \wedge R)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 1 of them:

P	Q	R	S	KB
T	F	F	F	T

(A) $R \rightarrow \neg Q$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $(P \wedge \neg Q) \rightarrow R$ is false in the KB-model $P=T, Q=F, R=F, S=F \Rightarrow$ not entailed.

(C) $(\neg S \vee P) \vee (P \wedge R)$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $\neg(P \wedge \neg R)$ is false in the KB-model $P=T, Q=F, R=F, S=F \Rightarrow$ not entailed.

Q.21 [MSQ, 2M · Sampling] **Answer: A, B, C**

(A) Each state is obtained by modifying the previous one, so samples are autocorrelated. (B) Gibbs steps satisfy detailed balance with respect to $P(x | e)$ and the chain is ergodic when there are no deterministic (0/1) entries, so it converges to the posterior. (C) The state is a full assignment with evidence clamped. (D) describes rejection sampling; Gibbs never rejects since evidence is clamped.

Q.22 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $E \perp D \mid A, C$; B: $F \perp A \mid D, E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp D \mid A, C$: **TRUE** — $E \leftarrow A \rightarrow C \rightarrow D$ (blocked: observed fork node A); $E \leftarrow C \rightarrow D$ (blocked: observed fork node C); $E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed).

(B) $F \perp A \mid D, E$: **TRUE** — $F \leftarrow E \leftarrow A$ (blocked: observed chain node E); $F \leftarrow E \leftarrow C \leftarrow A$ (blocked: observed chain node E); $F \rightarrow G \leftarrow D \leftarrow C \leftarrow A$ (blocked: collider G with neither it nor a descendant observed); $F \rightarrow G \leftarrow D \leftarrow C \rightarrow E \leftarrow A$ (blocked: collider G with neither it nor a descendant observed).

(C) $B \perp G \mid A$: **FALSE** — the path $B \rightarrow F \rightarrow G$ is active given $\{A\}$.

(D) $F \perp G$: **FALSE** — the path $F \leftarrow E \leftarrow A \rightarrow C \rightarrow D \rightarrow G$ is active given $\{\}$.

Q.23 [MCQ, 2M · Sampling] **Answer: A** (A: $A=T, B=T, C=T, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.6$; $u = 0.56 < 0.6 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.6$; $u = 0.07 < 0.6 \Rightarrow B = T$
- C: $P(C=T \mid A=T, B=T) = 0.5$; $u = 0.05 < 0.5 \Rightarrow C = T$
- D: $P(D=T \mid C=T) = 0.2$; $u = 0.47 \geq 0.2 \Rightarrow D = F$

Sample: **(A=T, B=T, C=T, D=F)**.

Q.24 [NAT, 2M · Sampling] **Answer: 0.98 to 0.984**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(D=T \mid A=F, C=T) = 0.05$	0.05	F
2	$P(D=T \mid A=T, C=F) = 0.7$	0.7	T
3	$P(D=T \mid A=T, C=F) = 0.7$	0.7	T
4	$P(D=T \mid A=T, C=F) = 0.7$	0.7	T
5	$P(D=T \mid A=T, C=F) = 0.7$	0.7	T

$\sum w (\text{all}) = 2.85$; $\sum w (A = T) = 2.8$.

Estimate $P(A=T \mid D = T) = 2.8 / 2.85 = \mathbf{0.9825}$.

Q.25 [MCQ, 2M · Exact Inference] **Answer: C** (C: $f(B, E)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(A, D)$, $f(D, E)$, $f(D, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(D, E, F)$	$\{D, E, F\}$	$f(D, E)$
C	$f(A, B, C)$	$\{A, B, C\}$	$f(A, B)$
A	$f(A)$, $f(A, B)$, $f(A, D)$, $f(A, B)$	$\{A, B, D\}$	$f(B, D)$
D	$f(D, E)$, $f(D, E)$, $f(B, D)$	$\{B, D, E\}$	$f(B, E)$
B	$f(B, E)$	$\{B, E\}$	$f(E)$

The largest factor produced has **2** variable(s) (created when eliminating F); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over E and normalise.

Mock Test 059

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	1 / 10 / 14	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

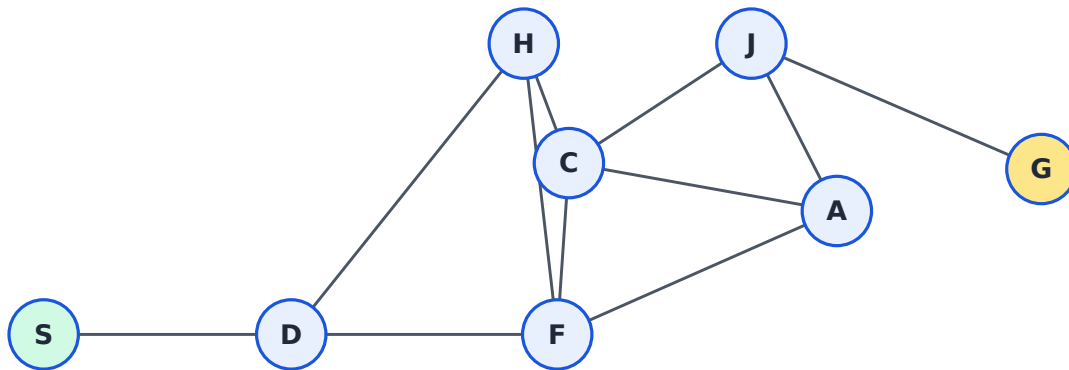
Q.1 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

How many nodes are there in a complete search tree of depth 5 (root at depth 0) with branching factor 5?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

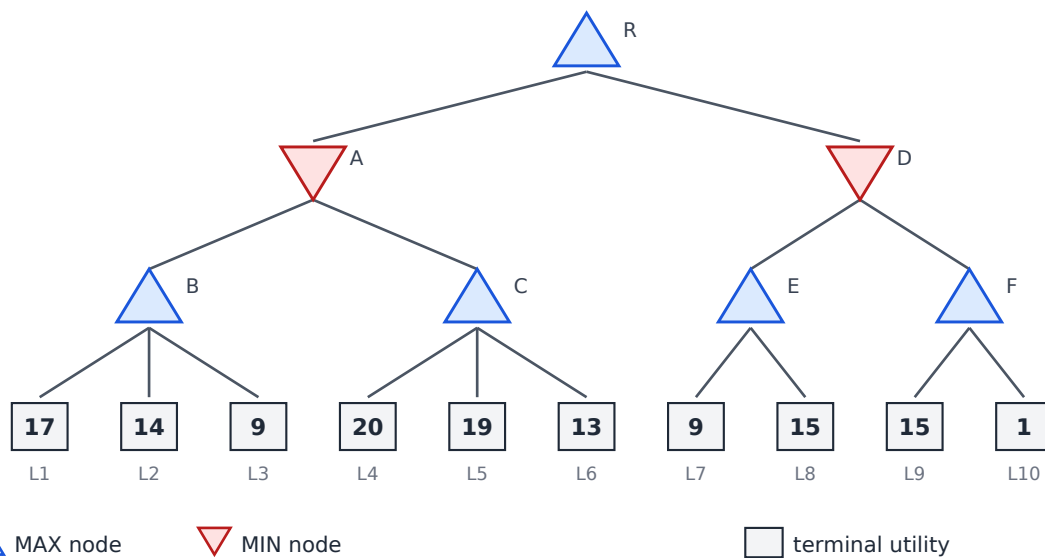
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, H, F, C, J, G
- ☐ (B) S, D, F, A, C, H, J, G
- ☐ (C) S, D, F, H, A, C, J, G
- ☐ (D) S, D, F, A, C, J, H, G

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [MSQ · 1 mark · no negative marking] Local Search

Which of the following statements about local search are TRUE?

- ☐ (A) Local beam search with k states passes useful information among the k parallel searches
- ☐ (B) Local beam search with $k = 1$ is equivalent to hill climbing
- ☐ (C) Hill climbing stores the whole search tree in memory
- ☐ (D) Typical local-search algorithms use only a constant amount of memory

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following correctly express 'Not every graph is connected'?

- ☐ (A) $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$
- ☐ (B) $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$
- ☐ (C) $\forall x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$
- ☐ (D) $\exists x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **E, A, B** and rules:

- R1: $H \wedge A \rightarrow J$
- R2: $F \rightarrow H$
- R3: $D \rightarrow A$
- R4: $J \rightarrow B$
- R5: $K \rightarrow F$
- R6: $E \rightarrow F$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

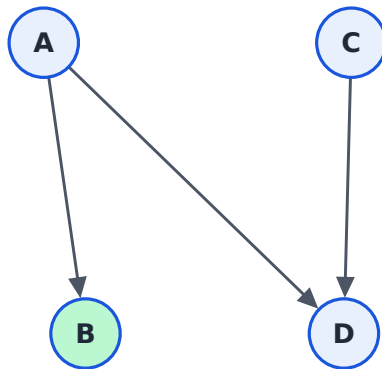
Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((R \wedge P) \wedge Q) \rightarrow (Q \wedge \neg P)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(\mathbf{B} = \mathbf{T})$ (round off to 3 decimal places).



$P(A=T)$
0.75

A	$P(B=T \dots)$	$P(C=T)$
T	0.2	0.3
F	0.5	

A	C	$P(D=T \dots)$
T	T	0.7
T	F	0.05
F	T	0.9
F	F	0.4

Answer (numerical): _____

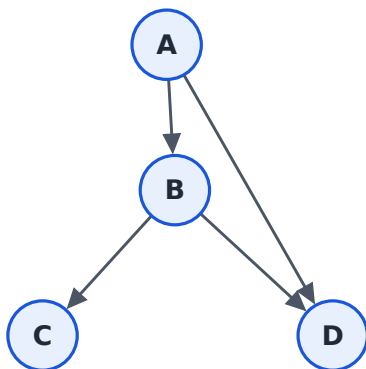
Q.9 [NAT · 1 mark · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. With prior sampling, what is the probability that a single generated sample is exactly (c, $\neg s$, r, w)? (Round to 3 decimal places.)

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample $\mathbf{A=F, B=T, C=T, D=T}$? (round off to 4 decimal places)



$P(A=T)$
0.1

A	$P(B=T \dots)$	B	$P(C=T \dots)$
T	0.05	T	0.7
F	0.9	F	0.3

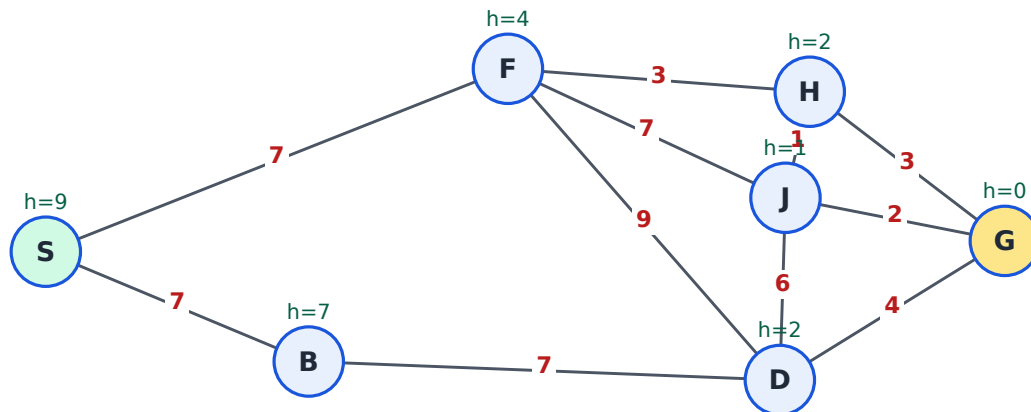
A	B	$P(D=T \dots)$
T	T	0.1
T	F	0.3
F	T	0.8
F	F	0.8

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

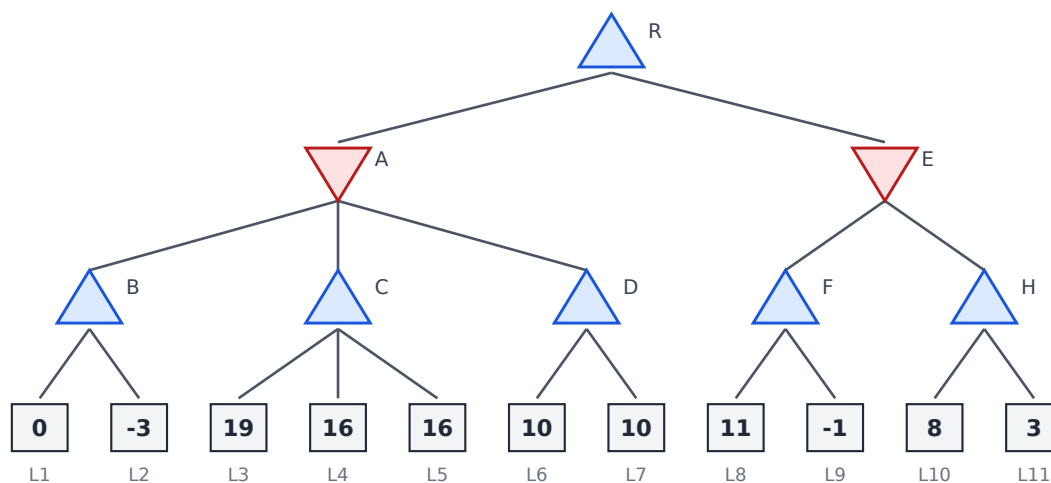
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



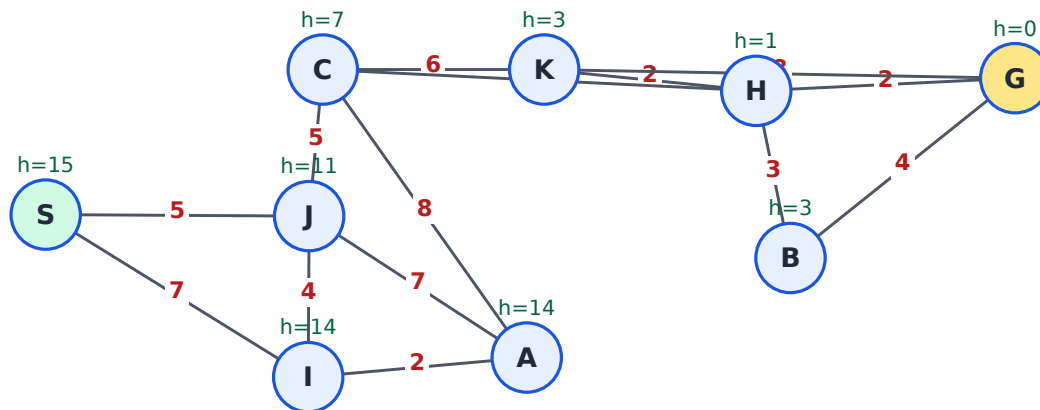
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L2 is pruned.
- ☐ (B) Leaf L5 is pruned.
- ☐ (C) Leaf L7 is pruned.
- ☐ (D) Leaf L10 is pruned.

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node A is expanded.
- ☐ (B) Node K is expanded.
- ☐ (C) The path returned has cost 19.
- ☐ (D) Node C is expanded.

Q.14 [MSQ · 2 marks · no negative marking] Problem Formulation

In a grid route-finding problem, the agent moves N, S, E or W at cost 1 per move. The start state is (0,0), the only goal is (3,2), and there are no obstacles in an unbounded grid. Which of the following are TRUE?

- ☐ (A) The optimal solution cost is 5
- ☐ (B) BFS with graph search is optimal for this problem
- ☐ (C) Manhattan distance is a consistent heuristic for this problem
- ☐ (D) The branching factor b is 8

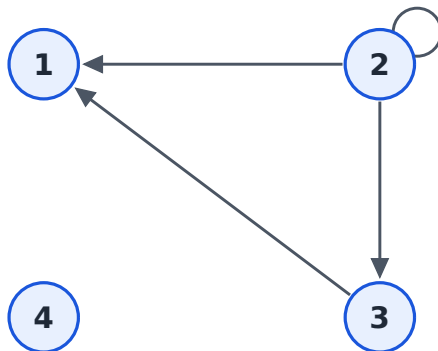
Q.15 [NAT · 2 marks · no negative marking] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Depth-first search pushes the right child before the left child, so the left child is explored first, and it goal-tests a node when the node is removed from the frontier. The goal is node 12. How many nodes are removed from the frontier, including node 12?

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (B) $\exists y \forall x R(x, y)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\exists x \forall y \neg R(x, y)$

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic

Consider all propositional formulas over the symbols p, q, r , identified up to logical equivalence. How many of these equivalence classes ϕ satisfy $(p \wedge q) \models \phi$?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(R \vee Q) \rightarrow (R \wedge Q)$
☐ (B) $\neg(Q \wedge R) \leftrightarrow (\neg Q \vee \neg R)$
☐ (C) $(R \rightarrow Q) \vee (Q \rightarrow R)$
☐ (D) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (R \vee \neg S), C2: (P \vee R \vee S), C3: (\neg P \vee \neg R), C4: (P)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) $(P \vee R)$
☐ (C) $(\neg P \vee \neg Q)$
☐ (D) $(\neg S)$

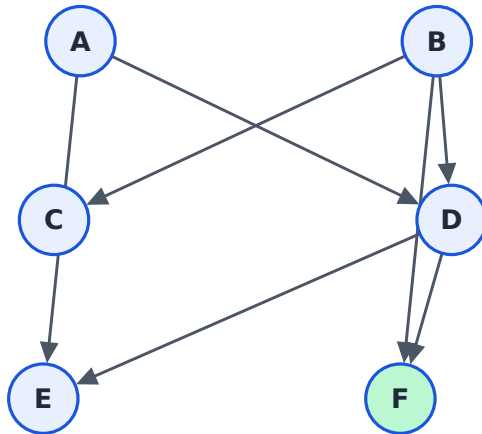
Q.20 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of two unary predicates P and Q over D (i.e. ordered pairs of subsets) satisfy $\forall x (P(x) \rightarrow Q(x)) \wedge \exists x (Q(x) \wedge \neg P(x))$?

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{F})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, B, D, C, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

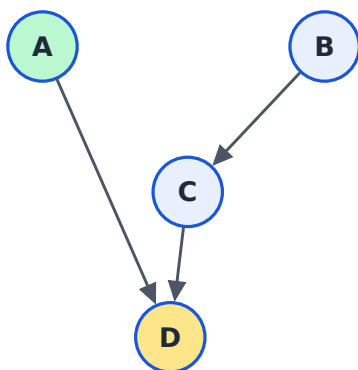
Q.22 [MSQ · 2 marks · no negative marking] Sampling

Which of the following statements about approximate inference in Bayesian networks are correct?

- ☐ (A) Likelihood weighting produces consistent estimates
- ☐ (B) Likelihood weighting performs poorly when evidence variables are downstream (e.g. leaves), since samples of upstream variables ignore that evidence
- ☐ (C) Gibbs sampling resamples each non-evidence variable from its distribution conditioned on the current values of its Markov blanket
- ☐ (D) In rejection sampling, the fraction of accepted samples grows as more evidence variables are added

Q.23 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{D}=\mathbf{F}$. The current state of the chain is $(\mathbf{A}=\mathbf{F}, \mathbf{B}=\mathbf{F}, \mathbf{C}=\mathbf{T}, \mathbf{D}=\mathbf{F})$. The next step resamples **A**. With what probability is A set to T? (round off to 3 decimal places)



P(A=T)
0.25

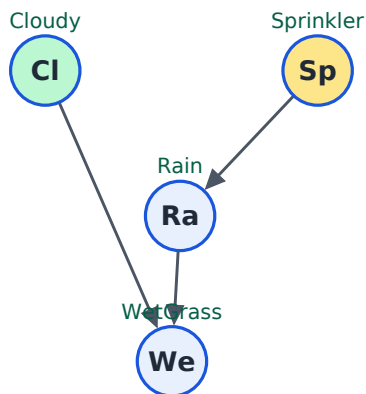
P(B=T)	B	P(C=T ...)
0.2	T	0.05
	F	0.9

A	C	P(D=T ...)
T	T	0.9
T	F	0.05
F	T	0.8
F	F	0.05

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\text{Cloudy} = T \mid \text{Sprinkler} = T)$ (round off to 3 decimal places).



$P(Cl=T)$
0.95

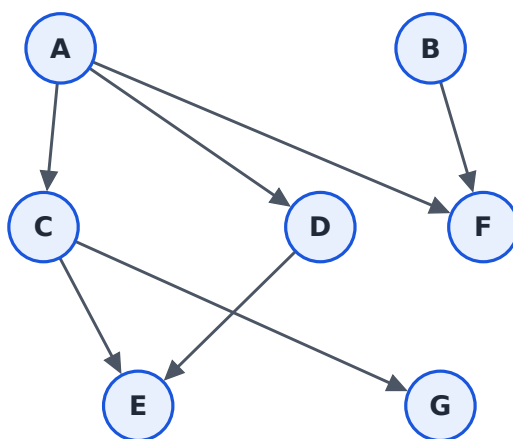
$P(Sp=T)$	Sp	$P(Ra=T \dots)$
0.05	T	0.5
	F	0.05

Cl	Ra	$P(We=T \dots)$
T	T	0.3
T	F	0.05
F	T	0.9
F	F	0.8

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp C \mid F$
☐ (B) $E \perp A \mid C, D$
☐ (C) $G \perp D$
☐ (D) $E \perp G$

Answer Key — Mock Test 059

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	3906	14	MSQ	2	Problem Formulation	A, B, C
2	MCQ	1	Uninformed Search	B	15	NAT	2	Uninformed Search	11
3	NAT	1	Adversarial Search	17	16	MSQ	2	Predicate Logic	D
4	MSQ	1	Local Search	A, B, D	17	NAT	2	Propositional Logic	64
5	MSQ	1	Predicate Logic	A, B	18	MSQ	2	Propositional Logic	B, C
6	NAT	1	Propositional Logic	6	19	MSQ	2	Propositional Logic	B, D
7	NAT	1	Propositional Logic	1	20	NAT	2	Predicate Logic	19
8	NAT	1	Exact Inference	0.273 to 0.277	21	NAT	2	Exact Inference	4
9	NAT	1	Sampling	0.323 to 0.325	22	MSQ	2	Sampling	A, B, C
10	NAT	1	Sampling	0.4534 to 0.4538	23	NAT	2	Sampling	0.141 to 0.145
11	NAT	2	Informed Search	5	24	NAT	2	Exact Inference	0.948 to 0.952
12	MSQ	2	Adversarial Search	B, C	25	MSQ	2	Conditional Independence	B
13	MSQ	2	Informed Search	C, D					

Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] **Answer: 3906**

$$\sum_{i=0..5} 5^i = (5^6 - 1)/(5 - 1) = \mathbf{3906}.$$

Q.2 [MCQ, 1M · Uninformed Search] **Answer: B** (B: S, D, F, A, C, H, J, G)

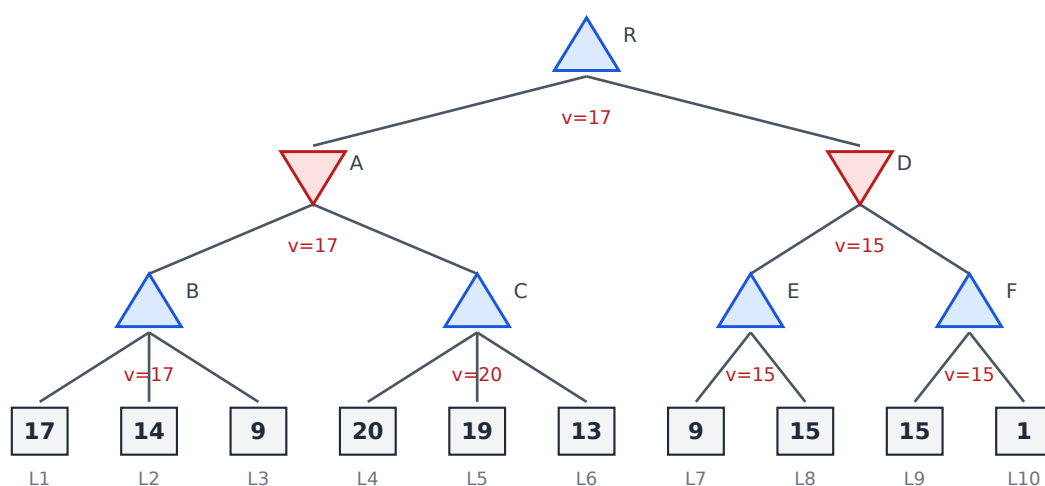
Adjacency lists (alphabetical): A: {C, F, J}; C: {A, F, H, J}; D: {F, H, S}; F: {A, C, D, H}; G: {J}; H: {C, D, F}; J: {A, C, G}; S: {D}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, F, A, C, H, J, G**; the path found is $S \rightarrow D \rightarrow F \rightarrow A \rightarrow C \rightarrow J \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.3 [NAT, 1M · Adversarial Search] **Answer: 17**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 17$, $C = 20$, $A = 17$, $E = 15$, $F = 15$, $D = 15$. Root value = **17**; MAX's optimal first move is towards A.

Q.4 [MSQ, 1M · Local Search] **Answer: A, B, D**

(A) is true: beam search selects the k best successors from the pooled successors of all k states. (B) is true: with one state, keeping the best successor is hill climbing. (C) is false: hill climbing keeps only the current state, not a tree or paths. (D) is true: local search keeps just the current state or states.

Q.5 [MSQ, 1M · Predicate Logic] Answer: A, B (A: $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$; B: $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$)

Option A is the direct translation, and pushing the negation inward gives $\exists x \neg(\text{Graph}(x) \rightarrow \text{Connected}(x)) \equiv \exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$ — option B. Option C says no graph is connected, which is stronger. Option D is true whenever any non-graph exists (vacuous implication), so it does not express the statement.

Q.6 [NAT, 1M · Propositional Logic] Answer: 6

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $E \rightarrow F$; add F.

Round 2: fire $F \rightarrow H$; add H.

Round 3: fire $H \wedge A \rightarrow J$; add J.

Fixed point: {A, B, E, F, H, J} — 6 atoms are entailed (facts: A, B, E; newly derived: F, H, J). Atoms never derived: C, D, K.

Q.7 [NAT, 1M · Propositional Logic] Answer: 1

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \wedge P) \wedge Q$	$Q \wedge \neg P$	$((R \wedge P) \wedge Q) \rightarrow (Q \wedge \neg P)$
T	T	T	T	F	F
T	T	F	F	F	T
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	F	T	T
F	T	F	F	T	T
F	F	T	F	F	T
F	F	F	F	F	T

F has 7 models, so it is false in $8 - 7 = 1$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.8 [NAT, 1M · Exact Inference] Answer: 0.273 to 0.277

Only the ancestors of B (A) matter: every other variable sums out to 1.

$P(B=T) = \sum \text{over its ancestors of } P(B=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, B=T)$
A=T, C=T, D=T	0.0315
A=T, C=T, D=F	0.0135
A=T, C=F, D=T	0.00525
A=T, C=F, D=F	0.09975
A=F, C=T, D=T	0.03375
A=F, C=T, D=F	0.00375
A=F, C=F, D=T	0.035
A=F, C=F, D=F	0.0525

Sum = **0.275** ≈ 0.275 .

Q.9 [NAT, 1M · Sampling] Answer: 0.323 to 0.325

Prior sampling generates each event with probability equal to its joint probability: $S_{PS}(x) = \prod P(x_i \mid \text{parents})$.
 $= P(c) P(\neg s \mid c) P(r \mid c) P(w \mid \neg s, r) = 0.5 \times 0.9 \times 0.8 \times 0.9 = 0.324$.

Q.10 [NAT, 1M · Sampling] Answer: 0.4534 to 0.4538

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(A=F) \times P(B=T \mid A=F) \times P(C=T \mid B=T) \times P(D=T \mid A=F, B=T)$
 $= 0.9 \times 0.9 \times 0.7 \times 0.8 = \mathbf{0.4536} \approx 0.4536$.

Q.11 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(B)=7$, $h(D)=2$, $h(F)=4$, $h(G)=0$, $h(H)=2$, $h(J)=1$, $h(S)=9$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	F(11), B(14)
2	F (11)	H(12), B(14), J(15), D(18)
3	H (12)	J(12), G(13), B(14), D(18)
4	J (12)	G(13), B(14), D(18)
5	G (13)	B(14), D(18) ← goal, stop

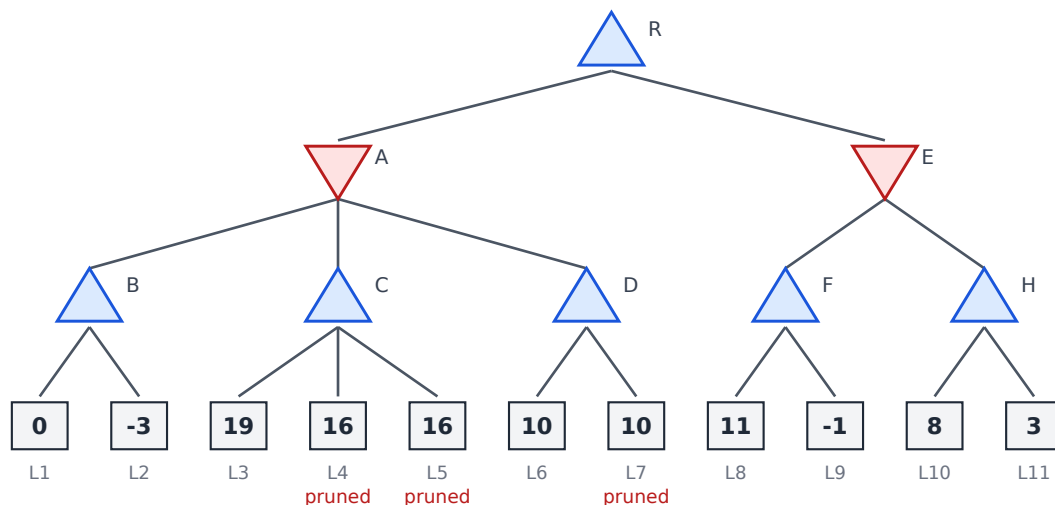
Path returned: **S** → **F** → **H** → **G** with cost **13**; nodes expanded = **5** (S, F, H, J, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 13$. For comparison UCS ($h = 0$) would expand 6 nodes.

Q.12 [MSQ, 2M · Adversarial Search] **Answer: B, C** (B: Leaf L5 is pruned.; C: Leaf L7 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $19 \geq \beta = 0 \Rightarrow$ prune remaining children (leaves L4, L5).
- At MAX node D: value $10 \geq \beta = 0 \Rightarrow$ prune remaining children (leaves L7).



Leaves evaluated (8): L1, L2, L3, L6, L8, L9, L10, L11. Leaves pruned (3): L4, L5, L7. Root value = 8 (same as plain minimax, 8 — pruning never changes the root value).

Q.13 [MSQ, 2M · Informed Search] **Answer: C, D** (C: The path returned has cost 19.; D: Node C is expanded.)

Heuristic values: $h(A)=14$, $h(B)=3$, $h(C)=7$, $h(G)=0$, $h(H)=1$, $h(I)=14$, $h(J)=11$, $h(K)=3$, $h(S)=15$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	J(16), I(21)
2	J (16)	C(17), I(21), A(26)
3	C (17)	H(18), K(19), I(21), A(26)
4	H (18)	G(19), K(19), I(21), B(23), A(26)
5	G (19)	K(19), I(21), B(23), A(26) ← goal, stop

Path returned: **S** → **J** → **C** → **H** → **G** with cost **19**; nodes expanded = **5** (S, J, C, H, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.14 [MSQ, 2M · Problem Formulation] **Answer: A, B, C** (A: The optimal solution cost is 5; B: BFS with graph search is optimal for this problem; C: Manhattan distance is a consistent heuristic for this problem)

(A) is true: the shortest path needs $|3| + |2| = 5$ unit moves. (B) is true: costs are uniform, so the shallowest goal is cheapest. (C) is true: one move changes the Manhattan distance by exactly 1 = the step cost, so $h(n) \leq 1 + h(n')$. (D) is false: there are 4 actions, so $b = 4$. $b = 8$ would need diagonal moves.

Q.15 [NAT, 2M · Uninformed Search] **Answer: 11**

DFS visits nodes in preorder: 1, 2, 4, 8, 9, 5, 10, 11, 3, 6, 12, ... Node 12 is the **11th** node removed. For comparison, BFS would remove nodes in label order and find node 12 at the 12th removal.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: D** ($D: \exists x \forall y \neg R(x, y)$)

$R = \{(2,1), (2,2), (2,3), (3,1)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,3)$ holds but $R(3,2)$ does not.

(B) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 1 has no edge to a different element.

(D) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [NAT, 2M · Propositional Logic] **Answer: 64**

Up to equivalence a formula is determined by its set of models, a subset of the 8 assignments. $(p \wedge q) \models \phi$ iff $M(p \wedge q) \subseteq M(\phi)$. Over p, q, r , $M(p \wedge q) = \{TTT, TTF\}$ has 2 elements. So $M(\phi)$ must contain these 2 assignments and may contain any subset of the other 6: $2^6 = 64$.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: B, C** ($B: \neg(Q \wedge R) \leftrightarrow (\neg Q \vee \neg R)$; $C: (R \rightarrow Q) \vee (Q \rightarrow R)$)

(A) $(R \vee Q) \rightarrow (R \wedge Q)$: **not valid** — counter-model $Q=T, R=F$.

(B) $\neg(Q \wedge R) \leftrightarrow (\neg Q \vee \neg R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $(R \rightarrow Q) \vee (Q \rightarrow R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$: **not valid** — counter-model $P=T, Q=F, R=T$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: B, D** ($B: (P \vee R)$; $D: (\neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 10 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(P \vee R)$ is derivable:

1. resolve $(R \vee \neg S)$ and $(P \vee R \vee S)$ on $S \Rightarrow (P \vee R)$

(C) $(\neg P \vee \neg Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg S)$ is derivable:

1. resolve $(\neg P \vee \neg R)$ and (P) on $P \Rightarrow (\neg R)$

2. resolve $(R \vee \neg S)$ and $(\neg R)$ on $R \Rightarrow (\neg S)$

Q.20 [NAT, 2M · Predicate Logic] **Answer: 19**

$\forall x (P(x) \rightarrow Q(x))$ leaves each element 3 choices of (P, Q) : $(F, F), (F, T), (T, T)$; so $3^3 = 27$ pairs. The second conjunct requires at least one element with (F, T) . Pairs with no (F, T) element: $2^3 = 8$. Answer = $27 - 8 = 19$ (P is a proper subset of Q).

Q.21 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(B), f(B, C), f(A, B, D), f(A, D, E), f(B, D, F)$.

Eliminating Z : multiply all factors whose scope contains Z , then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A), f(A, B, D), f(A, D, E)$	$\{A, B, D, E\}$	$f(B, D, E)$
B	$f(B), f(B, C), f(B, D, F), f(B, D, E)$	$\{B, C, D, E, F\}$	$f(C, D, E, F)$
D	$f(C, D, E, F)$	$\{C, D, E, F\}$	$f(C, E, F)$
C	$f(C, E, F)$	$\{C, E, F\}$	$f(E, F)$
E	$f(E, F)$	$\{E, F\}$	$f(F)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over F and normalise.

Q.22 [MSQ, 2M · Sampling] **Answer: A, B, C**

(A) AIMA shows $S_{WS}(z, e) w(z, e) = P(z, e)$, so weighted estimates are consistent. (B) Non-evidence variables are sampled from their parents only, so evidence at leaves does not influence them; many samples get tiny weights. (C) This is exactly the Gibbs step: $P(x_i | mb(X_i))$. (D) is false: each extra evidence variable generally reduces $P(e)$, so fewer samples are accepted.

Q.23 [NAT, 2M · Sampling] Answer: 0.141 to 0.145

Gibbs sampling draws A from $P(A \mid \text{mb}(A))$, where $\text{mb}(A) = \{C, D\}$:

$$P(A \mid \text{mb}) \propto P(A \mid \text{parents}(A)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y)).$$

$$A=T: P(A=T) \times P(D=F \mid A=T, C=T) = 0.25 \times 0.1 = 0.025$$

$$A=F: P(A=F) \times P(D=F \mid A=F, C=T) = 0.75 \times 0.2 = 0.15$$

$$\text{Normalise: } 0.025 / (0.025 + 0.15) = \mathbf{0.1429}.$$

Q.24 [NAT, 2M · Exact Inference] Answer: 0.948 to 0.952

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler}) \cdot P(\text{Rain} \mid \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Cloudy}, \text{Rain})$.

Inference by enumeration: $P(\text{Cloudy} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cloudy}, e, \text{hidden})$.

Hidden: Ra, We	Cl	product of CPT entries (Cl · Sp · Ra · We)	value
T, T	T	$0.95 \times 0.05 \times 0.5 \times 0.3$	0.00712
T, F	T	$0.95 \times 0.05 \times 0.5 \times 0.7$	0.01662
F, T	T	$0.95 \times 0.05 \times 0.5 \times 0.05$	0.00119
F, F	T	$0.95 \times 0.05 \times 0.5 \times 0.95$	0.02256
T, T	F	$0.05 \times 0.05 \times 0.5 \times 0.9$	0.00113
T, F	F	$0.05 \times 0.05 \times 0.5 \times 0.1$	0.00013
F, T	F	$0.05 \times 0.05 \times 0.5 \times 0.8$	0.001
F, F	F	$0.05 \times 0.05 \times 0.5 \times 0.2$	0.00025

$$P(\text{Cloudy}=T, e) = 0.0475; P(\text{Cloudy}=F, e) = 0.0025; P(e) = 0.05.$$

$$P(\text{Cloudy}=T \mid e) = 0.0475 / 0.05 = \mathbf{0.95} \approx 0.950.$$

Q.25 [MSQ, 2M · Conditional Independence] Answer: B (B: $E \perp A \mid C, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp C \mid F$: **FALSE** — the path $E \leftarrow C$ is active given $\{F\}$.

(B) $E \perp A \mid C, D$: **TRUE** — $E \leftarrow C \leftarrow A$ (blocked: observed chain node C); $E \leftarrow D \leftarrow A$ (blocked: observed chain node D).

(C) $G \perp D$: **FALSE** — the path $G \leftarrow C \leftarrow A \rightarrow D$ is active given $\{\}$.

(D) $E \perp G$: **FALSE** — the path $E \leftarrow C \rightarrow G$ is active given $\{\}$.

Mock Test 060

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 7 / 14	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search

Chess has an average branching factor of about $b = 36$. With perfect move ordering, alpha-beta reduces the effective branching factor from b to $\sqrt{b}$. What is the effective branching factor?

Answer (numerical): _____

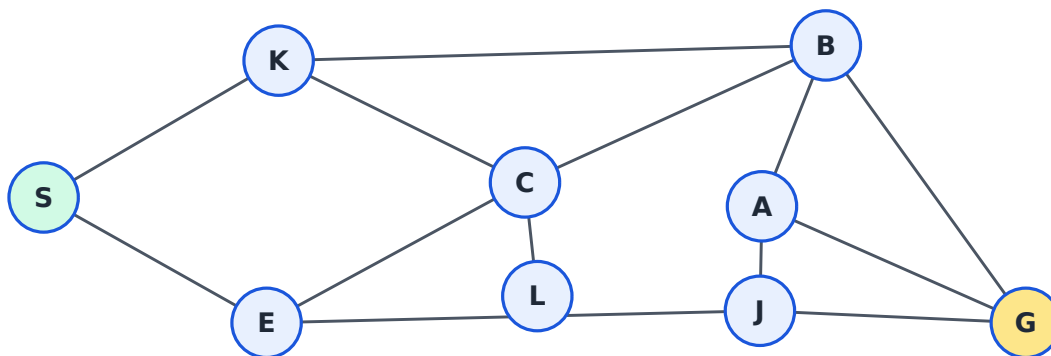
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 5$ and depth $d = 3$ (so $b^d = 125$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

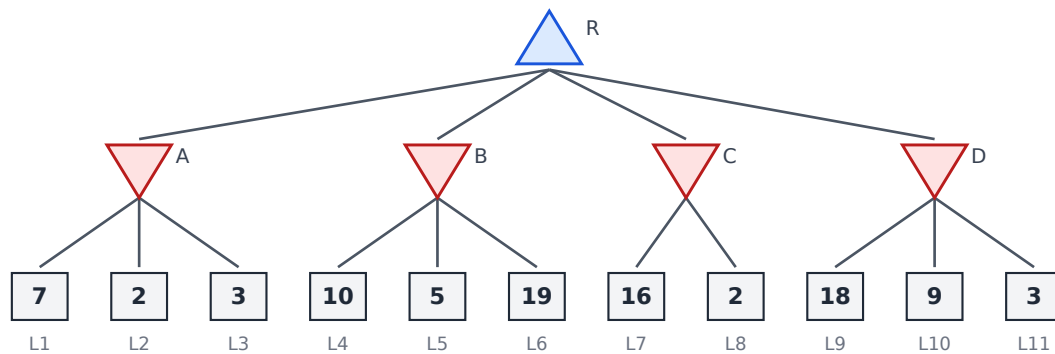
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, E, K, C, J, B, L, A, G
- ☐ (B) S, K, E, C, B, J, L, G
- ☐ (C) S, E, C, B, A, G
- ☐ (D) S, E, K, C, J, L, B, A, G

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



MAX node MIN node

terminal utility

- ☐ (A) Move to D; root value 3
- ☐ (B) Move to B; root value 5
- ☐ (C) Move to C; root value 2
- ☐ (D) Move to A; root value 2

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\forall x P(x) \vee \forall x Q(x)$ true?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

How many binary relations on {1, 2, 3} are both reflexive and symmetric?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (\neg Q \vee \neg R) \rightarrow ((P \vee P) \vee R) ?$$

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

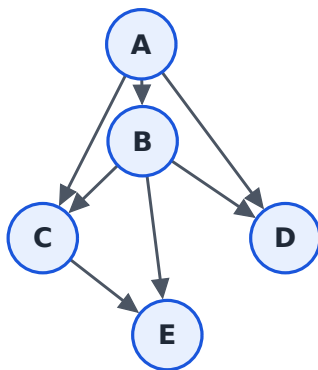
The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(A = T \mid C = T)$, how many of these samples are **rejected**?

#	A	B	C	D
1	T	T	T	T
2	T	T	F	T
3	T	T	F	F
4	T	F	T	F
5	T	F	T	T
6	T	F	T	F
7	T	F	T	T
8	T	F	T	F
9	T	F	T	T
10	F	F	T	F
11	T	T	T	F
12	T	T	F	T
13	T	F	T	T
14	F	F	T	F
15	T	T	F	F
16	T	T	T	T
17	T	F	T	T
18	T	T	T	T
19	T	F	T	F
20	F	F	T	F

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D, E. What is the probability that a single run produces the sample **(A=F, B=T, C=T, D=F, E=T)**? (round off to 4 decimal places)



P(A=T)
0.3

A	P(B=T ...)	A	B	P(C=T ...)
T	0.25	T	T	0.7
F	0.6	T	F	0.05
		F	T	0.5
		F	F	0.5

A	B	P(D=...)	B	C	P(E=T ...)
T	T	0.1	T	T	0.8
T	F	0.1	T	F	0.9
F	T	0.05	F	T	0.5
F	F	0.9	F	F	0.6

Answer (numerical): _____

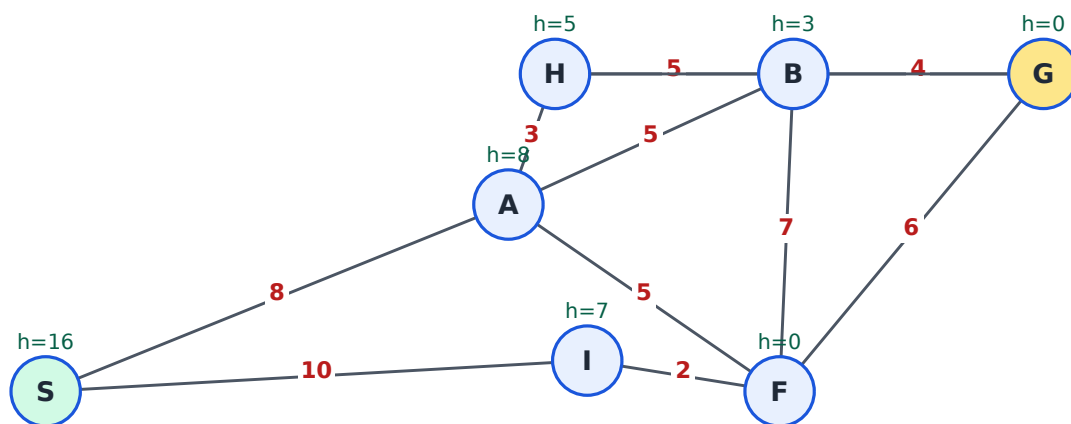
Q.10 [MCQ · 1 mark · negative: -1/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Likelihood weighting is used with evidence $C = \text{true}$ only (a root node), to estimate $P(R | c)$. What is the weight of every sample?

- ☐ (A) 0.5
☐ (B) 1.0
☐ (C) It depends on the sampled value of R
☐ (D) 0.8

Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
☐ (B) $h(H)$ overestimates the true cost from H to G.
☐ (C) h is admissible.
☐ (D) The edge I-F violates the consistency condition $h(I) \leq c(I, F) + h(F)$.

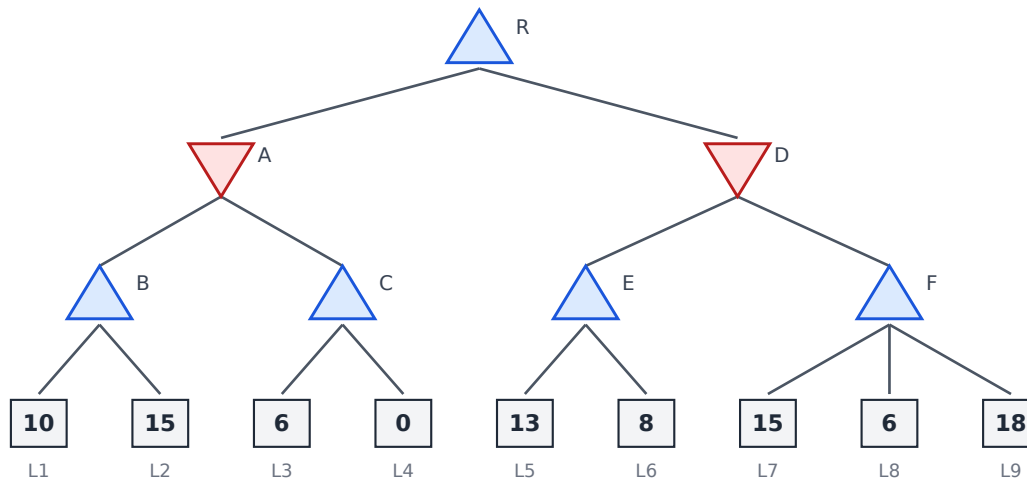
Q.12 [NAT · 2 marks · no negative marking] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (2), $S \rightarrow B$ (4), $A \rightarrow C$ (2), $A \rightarrow D$ (4), $B \rightarrow C$ (1), $C \rightarrow D$ (1), $C \rightarrow G$ (5), $D \rightarrow G$ (2). The heuristic is $h(S)=5$, $h(A)=4$, $h(B)=4$, $h(C)=3$, $h(D)=2$, $h(G)=0$. A* graph search is run from S. It goal-tests a node when the node is selected from the frontier, and it keeps the cheapest entry for each state. How many nodes are selected from the frontier for expansion, including the selection of G that ends the search?

Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



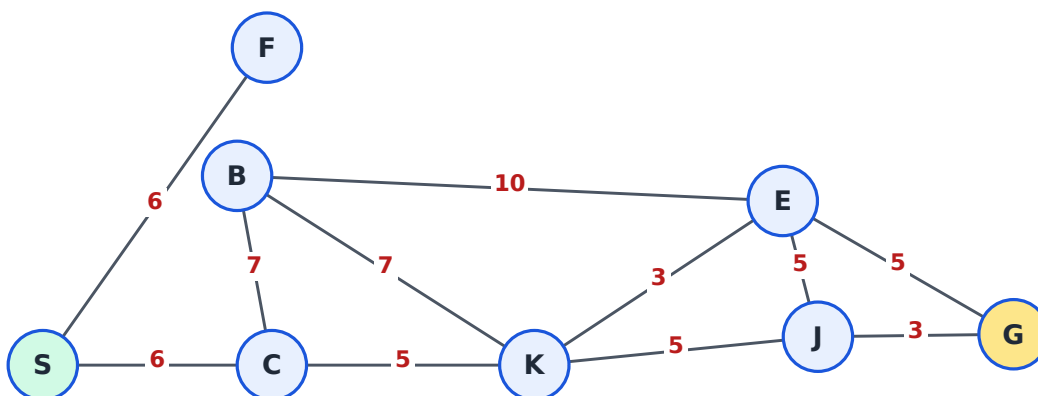
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L3 is pruned.
☐ (B) Leaf L5 is pruned.
☐ (C) Leaf L7 is pruned.
☐ (D) Leaf L8 is pruned.

Q.14 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(Q \rightarrow R) \rightarrow (R \rightarrow Q)$
☐ (B) $((R \rightarrow Q) \wedge \neg R) \rightarrow \neg Q$
☐ (C) $((P \rightarrow Q) \rightarrow P) \rightarrow P$
☐ (D) $((Q \rightarrow P) \wedge P) \rightarrow Q$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

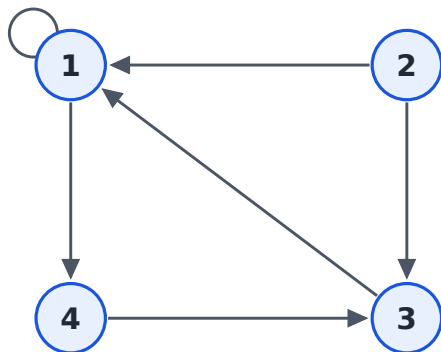
- (1) $\neg Q \wedge \neg \neg R$
- (2) $\neg P$
- (3) $\neg \neg P \vee R$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $S \leftrightarrow P$
- ☐ (B) $(Q \leftrightarrow \neg P) \vee \neg \neg P$
- ☐ (C) $\neg R \vee R$
- ☐ (D) $R \vee (Q \wedge \neg R)$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x R(x, x)$
- ☐ (B) $\forall x \exists y (x \neq y \wedge R(x, y))$
- ☐ (C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
- ☐ (D) $\exists x \forall y \neg R(x, y)$

Q.18 [NAT · 2 marks · no negative marking] Propositional Logic

A propositional Horn knowledge base contains the facts A and B and the definite clauses $A \wedge B \rightarrow C$, $C \rightarrow D$, $D \wedge E \rightarrow F$, $B \rightarrow E$, $F \wedge G \rightarrow H$, $H \rightarrow A$. Forward chaining is run to completion. How many distinct proposition symbols are inferred to be true (count the initial facts too)?

Answer (numerical): _____

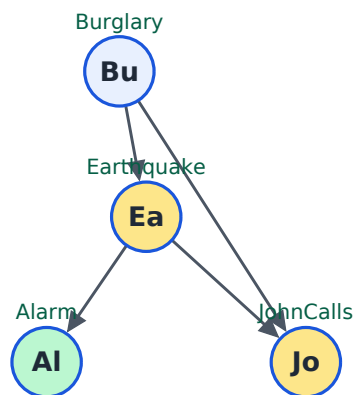
Q.19 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ are partial orders, i.e. satisfy reflexivity $\forall x R(x, x)$, antisymmetry $\forall x \forall y (R(x, y) \wedge R(y, x) \rightarrow x = y)$, and transitivity?

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute $P(\text{Alarm} = T \mid \text{Earthquake} = T, \text{JohnCalls} = T)$ (round off to 3 decimal places).



Bu	P(Bu=T)
T	0.05
F	

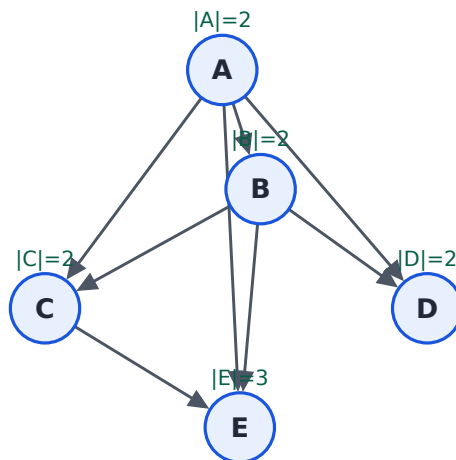
Bu	P(Ea=T ..)	Ea	P(Al=T ...)
T	0.6	T	0.2
F	0.4	F	0.4

Bu	Ea	P(Jo=T ...)
T	T	0.7
T	F	0.7
F	T	0.7
F	F	0.7

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $C \rightarrow A$ and $C \rightarrow B$ (Boolean variables) with $P(c) = 0.4$, $P(a \mid c) = 0.9$, $P(a \mid \neg c) = 0.2$, $P(b \mid c) = 0.7$, $P(b \mid \neg c) = 0.1$. Which of the following are true?

- ☐ (A) $P(a) = 0.48$
☐ (B) $A \perp B$ (marginally)
☐ (C) $P(a \mid b, c) = 0.9$
☐ (D) $P(a \mid b) \approx 0.776$

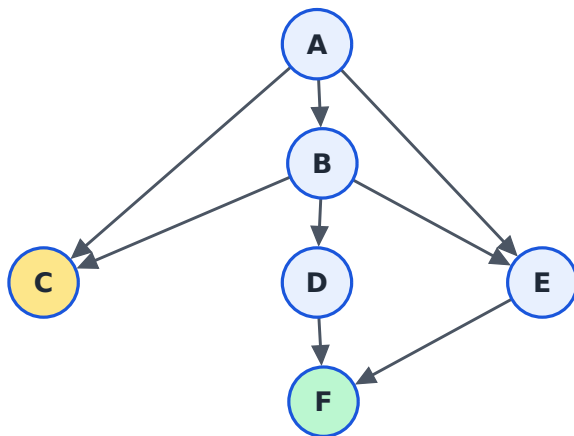
Q.23 [MCQ · 2 marks · negative: -2/3] Bayesian Networks

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. What is $P(r | w)$ (rounded to 3 decimal places)?

- (A) 0.708
 ○ (B) 0.647
 ○ (C) 0.430
 ○ (D) 0.800

Q.24 [NAT · 2 marks · no negative marking] Exact Inference - Variable elimination

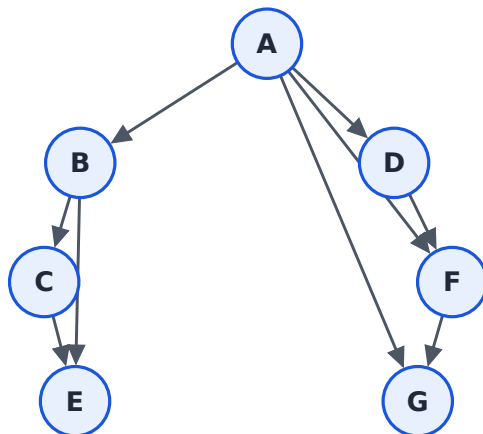
Variable elimination is used to compute $P(F | C = T)$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E**, **B**, **D**, **A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp D$
☐ (B) $A \perp E \mid B, C$
☐ (C) $D \perp G \mid B$
☐ (D) $C \perp D$

Answer Key — Mock Test 060

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	6	14	NAT	2	Uninformed Search	8
2	NAT	1	Adversarial Search	29	15	MSQ	2	Propositional Logic	C
3	MCQ	1	Uninformed Search	A	16	MSQ	2	Propositional Logic	C, D
4	MCQ	1	Adversarial Search	B	17	MSQ	2	Predicate Logic	A, B
5	NAT	1	Predicate Logic	15	18	NAT	2	Propositional Logic	6
6	NAT	1	Predicate Logic	8	19	NAT	2	Predicate Logic	19
7	NAT	1	Propositional Logic	6	20	NAT	2	Exact Inference	0.198 to 0.202
8	NAT	1	Sampling	4	21	NAT	2	Bayesian Networks	20
9	NAT	1	Sampling	0.1594 to 0.1598	22	MSQ	2	Conditional Independence	A, C, D
10	MCQ	1	Sampling	A	23	MCQ	2	Bayesian Networks	A
11	MSQ	2	Informed Search	C, D	24	NAT	2	Exact Inference	4
12	NAT	2	Informed Search	5	25	MSQ	2	Conditional Independence	B
13	MSQ	2	Adversarial Search	D					

Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 6**

Perfect ordering gives $O(b^{m/2}) = O((\sqrt{b})^m)$. The effective branching factor is $\sqrt{36} = 6$. Equivalently, alpha-beta can search roughly twice as deep as minimax in the same time.

Q.2 [NAT, 1M · Adversarial Search] **Answer: 29**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 5^2 + 5^1 - 1 = 25 + 5 - 1 = 29$, versus 125 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [MCQ, 1M · Uninformed Search] **Answer: A** (A: S, E, K, C, J, B, L, A, G)

Adjacency lists (alphabetical): A: {B, G, J}; B: {A, C, G, K}; C: {B, E, K, L}; E: {C, J, S}; G: {A, B, J}; J: {A, E, G}; K: {B, C, S}; L: {C}; S: {E, K}.

BFS trace (queue shown front → back):

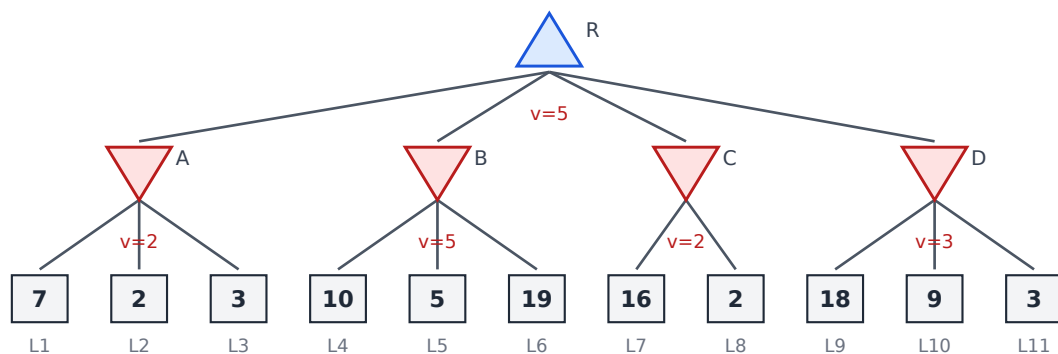
Step	Expanded	Frontier (FIFO) after expansion
1	S	[E, K]
2	E	[K, C, J]
3	K	[C, J, B]
4	C	[J, B, L]
5	J	[B, L, A, G]
6	B	[L, A, G]
7	L	[A, G]
8	A	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, E, K, C, J, B, L, A, G**. The path returned is $S \rightarrow E \rightarrow J \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [MCQ, 1M · Adversarial Search] **Answer: B** (B: Move to B; root value 5)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 2$, $B = 5$, $C = 2$, $D = 3$. Root value = **5**; MAX's optimal first move is towards B.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 15**

Interpret P and Q as subsets of the domain; inclusion-exclusion: $2^n + 2^n - 1$. With $n = 3$ this gives **15** (verified by enumerating all interpretations).

Q.6 [NAT, 1M · Predicate Logic] **Answer: 8**

Reflexivity fixes all 3 diagonal pairs to be present. Symmetry ties (x, y) and (y, x) together for $x \neq y$, leaving 3 independent unordered pairs, each in or out. Total = $2^3 = 8$.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg Q \vee \neg R$	$(P \vee P) \vee R$	$(\neg Q \vee \neg R) \rightarrow ((P \vee P) \vee R)$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	F	F

Rows where F = T: **6**.

Q.8 [NAT, 1M · Sampling] Answer: 4

Rejection sampling keeps only samples consistent with the evidence $C = T$: samples 1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 16, 17, 18, 19, 20 (16 accepted, 4 rejected).

Among the accepted samples, $A = T$ in 13. Estimate = $13/16 = \mathbf{0.8125}$.

(For this network the exact value is 0.7789; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.9 [NAT, 1M · Sampling] Answer: 0.1594 to 0.1598

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(A=F) \times P(B=T|A=F) \times P(C=T|A=F, B=T) \times P(D=F|A=F, B=T) \times P(E=T|B=T, C=T)$

$= 0.7 \times 0.6 \times 0.5 \times 0.95 \times 0.8 = \mathbf{0.1596} \approx 0.1598$.

Q.10 [MCQ, 1M · Sampling] Answer: A (A: 0.5)

The weight is the product of $P(e_i | \text{parents})$ over evidence variables; here only C , with no parents, so $w = P(c) = 0.5$ for every sample. Since all weights are equal, LW reduces to sampling R from $P(R | c)$, which is the exact posterior. The weight does not depend on R , and 0.8 is $P(r | c)$, a sampling probability, not a weight.

Q.11 [MSQ, 2M · Informed Search] Answer: C, D (C: h is admissible.; D: The edge $I \rightarrow F$ violates the consistency condition $h(I) \leq c(I, F) + h(F)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G :

n	A	B	F	G	H	I	S
$h(n)$	8	3	0	0	5	7	16
$h^*(n)$	9	4	6	0	6	8	17
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u, v) + h(v)$ for every edge in both directions. Violating directed edges: $I \rightarrow F$ ($7 > 2 + 0$), $A \rightarrow F$ ($8 > 5 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

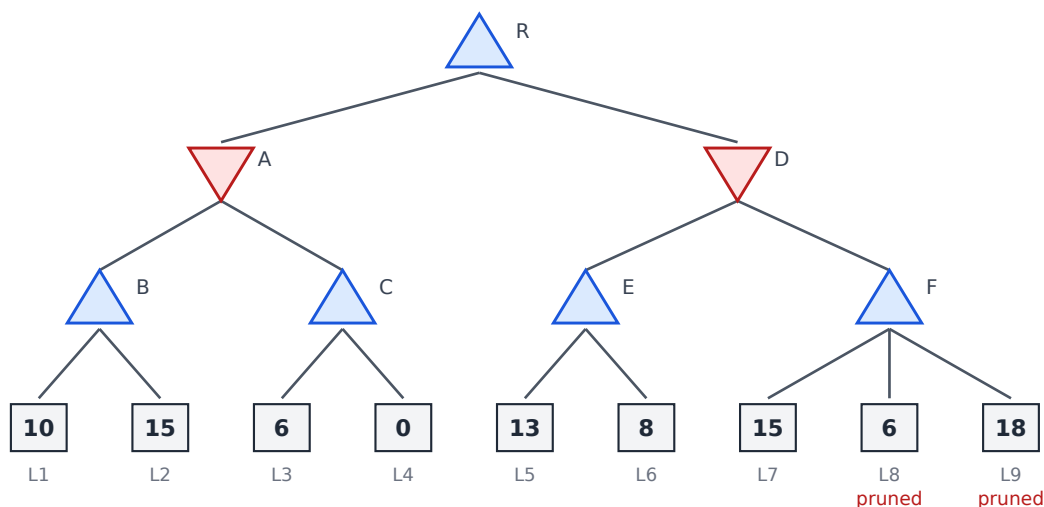
Q.12 [NAT, 2M · Informed Search] Answer: 5

$f(S)=5$. Expand S : $A(g=2, f=6)$, $B(g=4, f=8)$. Expand A ($f=6$): $C(g=4, f=7)$, $D(g=6, f=8)$. Expand C ($f=7$): D improves to $g=5, f=7$, and G is added with $g=9, f=9$. Expand D ($f=7$): G improves to $g=7, f=7$. Select G ($f=7$) and stop with cost 7 via $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$. Selections: $S, A, C, D, G = \mathbf{5}$. B ($f=8$) is never expanded.

Q.13 [MSQ, 2M · Adversarial Search] Answer: D (D: Leaf L8 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MAX node F : value $15 \geq \beta = 13 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (7): L1, L2, L3, L4, L5, L6, L7. Leaves pruned (2): L8, L9. Root value = 13 (same as plain minimax, 13 — pruning never changes the root value).

Q.14 [NAT, 2M · Uninformed Search] Answer: 8

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	C(6), F(6)
2	C (6)	F(6), K(11), B(13)
3	F (6)	K(11), B(13)
4	K (11)	B(13), E(14), J(16)
5	B (13)	E(14), J(16)
6	E (14)	J(16), G(19)
7	J (16)	G(19)
8	G (19)	← goal, stop

Path returned: **S** → **C** → **K** → **E** → **G**, cost = 6 + 5 + 3 + 5 = **19**. Nodes expanded (including S and G): **8** (S, C, F, K, B, E, J, G).

Q.15 [MSQ, 2M · Propositional Logic] Answer: C (C: $((P \rightarrow Q) \rightarrow P) \rightarrow P$)

(A) $(Q \rightarrow R) \rightarrow (R \rightarrow Q)$: **not valid** — counter-model $Q=F, R=T$.

(B) $((R \rightarrow Q) \wedge \neg R) \rightarrow \neg Q$: **not valid** — counter-model $Q=T, R=F$.

(C) $((P \rightarrow Q) \rightarrow P) \rightarrow P$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow P) \wedge P) \rightarrow Q$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.16 [MSQ, 2M · Propositional Logic] Answer: C, D (C: $\neg R \vee R$; D: $R \vee (Q \wedge \neg R)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
F	F	T	T	T
F	F	T	F	T

(A) $S \leftrightarrow P$ is false in the KB-model $P=F, Q=F, R=T, S=T \Rightarrow$ not entailed.

(B) $(Q \leftrightarrow \neg P) \vee \neg \neg P$ is false in the KB-model $P=F, Q=F, R=T, S=T \Rightarrow$ not entailed.

(C) $\neg R \vee R$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $R \vee (Q \wedge \neg R)$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, B (A: $\exists x R(x, x)$; B: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,1), (1,4), (2,1), (2,3), (3,1), (4,3)\}$.

(A) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

(B) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,1)$ holds but $R(1,2)$ does not.

(D) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [NAT, 2M · Propositional Logic] Answer: 6

Start with $\{A, B\}$. $A \wedge B \rightarrow C$ adds C; $C \rightarrow D$ adds D; $B \rightarrow E$ adds E; $D \wedge E \rightarrow F$ adds F. $F \wedge G \rightarrow H$ never fires because G is never derived, and $H \rightarrow A$ adds nothing new. The agenda empties with $\{A, B, C, D, E, F\}$, i.e. 6 symbols. Because forward chaining is sound and complete for Horn KBs, these are exactly the entailed atoms.

Q.19 [NAT, 2M · Predicate Logic] Answer: 19

Count labelled posets on 3 elements by shape: the antichain (1); one comparable pair plus an isolated point ($3 \times 2 = 6$); a chain ($3! = 6$); one element below two incomparable ones (3); one element above two incomparable ones (3). Total = 1 + 6 + 6 + 3 + 3 = 19, which matches brute-force enumeration of all 512 relations.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.198 to 0.202

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm} \mid \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Earthquake})$.

Inference by enumeration: $P(\text{Alarm} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Alarm}, e, \text{hidden})$.

Hidden: Bu	Al	product of CPT entries (Bu · Ea · Al · Jo)	value
T	T	$0.05 \times 0.6 \times 0.2 \times 0.7$	0.0042
F	T	$0.95 \times 0.4 \times 0.2 \times 0.7$	0.0532
T	F	$0.05 \times 0.6 \times 0.8 \times 0.7$	0.0168
F	F	$0.95 \times 0.4 \times 0.8 \times 0.7$	0.2128

$P(\text{Alarm}=T, e) = 0.0574$; $P(\text{Alarm}=F, e) = 0.2296$; $P(e) = 0.287$.

$P(\text{Alarm}=T \mid e) = 0.0574 / 0.287 = \mathbf{0.2} \approx 0.200$.

Q.21 [NAT, 2M · Bayesian Networks] Answer: 20

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (2-1) \times 2 + C: (2-1) \times 2 \times 2 + D: (2-1) \times 2 \times 2 + E: (3-1) \times 2 \times 2 \times 2 = \mathbf{27}$.

Full joint table: $\prod |X| - 1 = 48 - 1 = 47$; saving = 20.

Q.22 [MSQ, 2M · Conditional Independence] Answer: A, C, D (A: $P(a) = 0.48$; C: $P(a \mid b, c) = 0.9$; D: $P(a \mid b) \approx 0.776$)

$P(a) = 0.4 \times 0.9 + 0.6 \times 0.2 = 0.36 + 0.12 = 0.48$ (A true). $P(a, b) = 0.252 + 0.012 = 0.264$ while $P(a)P(b) = 0.48 \times 0.34 = 0.1632$, so A and B are dependent (B false). In the fork $C \rightarrow A, C \rightarrow B, A \perp B \mid C$, so $P(a \mid b, c) = P(a \mid c) = 0.9$ (C true). $P(a \mid b) = 0.264/0.34 \approx 0.7765$ (D true): observing b raises belief in c and hence in a.

Q.23 [MCQ, 2M · Bayesian Networks] Answer: A (A: 0.708)

Compute $P(r, w) = \sum_{c,s} P(c)P(s \mid c)P(r \mid c)P(w \mid s, r)$. With c: $P(r \mid c) = 0.8$ and $P(w \mid r)$ averaged over S:

$0.1 \times 0.99 + 0.9 \times 0.90 = 0.909 \rightarrow 0.5 \times 0.8 \times 0.909 = 0.3636$. With $\neg c$: $P(r \mid \neg c) = 0.2$ and $0.5 \times 0.99 + 0.5 \times 0.90 = 0.945 \rightarrow 0.5 \times 0.2 \times 0.945 = 0.0945$. So $P(r, w) = 0.4581$. $P(\neg r, w)$: c: $0.5 \times 0.2 \times (0.1 \times 0.9 + 0.9 \times 0) = 0.009$; $\neg c$: $0.5 \times 0.8 \times (0.5 \times 0.9) = 0.18$; total 0.189. $P(w) = 0.6471$. $P(r \mid w) = 0.4581/0.6471 \approx 0.708$. 0.647 is $P(w)$, 0.430 is $P(s \mid w)$, 0.800 is $P(r \mid c)$.

Q.24 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(A, B), f(B, D), f(A, B, E), f(D, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(A, B, E), f(D, E, F)$	$\{A, B, D, E, F\}$	$f(A, B, D, F)$
B	$f(A, B), f(A, B), f(B, D), f(A, B, D, F)$	$\{A, B, D, F\}$	$f(A, D, F)$
D	$f(A, D, F)$	$\{A, D, F\}$	$f(A, F)$
A	$f(A), f(A, F)$	$\{A, F\}$	$f(F)$

The largest factor produced has **4** variable(s) (created when eliminating E); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over F and normalise.

Q.25 [MSQ, 2M · Conditional Independence] Answer: B (B: $A \perp E \mid B, C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp D$: **FALSE** — the path $A \rightarrow D$ is active given $\{\}$.

(B) $A \perp E \mid B, C$: **TRUE** — $A \rightarrow B \rightarrow C \rightarrow E$ (blocked: observed chain node B); $A \rightarrow B \rightarrow E$ (blocked: observed chain node B).

(C) $D \perp G \mid B$: **FALSE** — the path $D \leftarrow A \rightarrow F \rightarrow G$ is active given $\{B\}$.

(D) $C \perp D$: **FALSE** — the path $C \leftarrow B \leftarrow A \rightarrow D$ is active given $\{\}$.

Mock Test 061

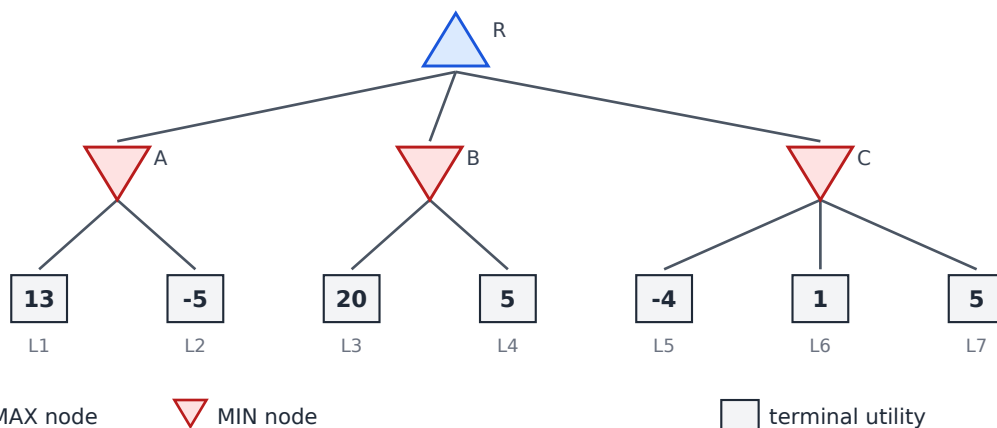
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 9 / 12	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

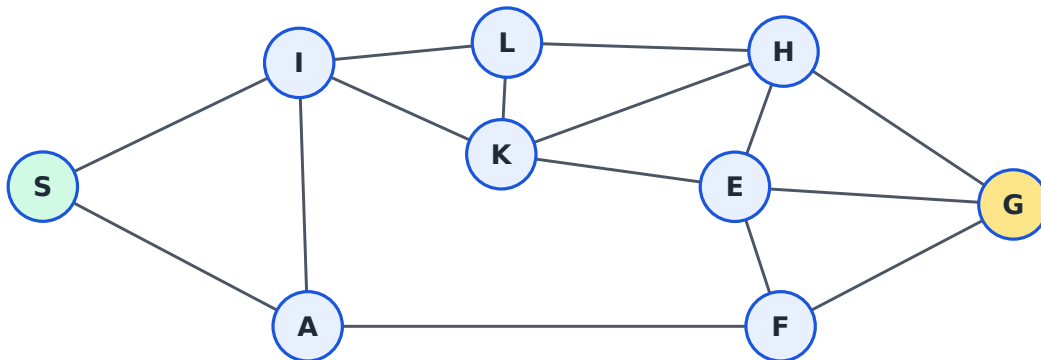
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search

A two-ply game tree has a MAX root with four MIN children. Their leaf values, left to right, are: MIN_1 : (10, 9, 14); MIN_2 : (11, 8, 20); MIN_3 : (2, 15, 30); MIN_4 : (12, 13, 7). What is the minimax value of the root?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

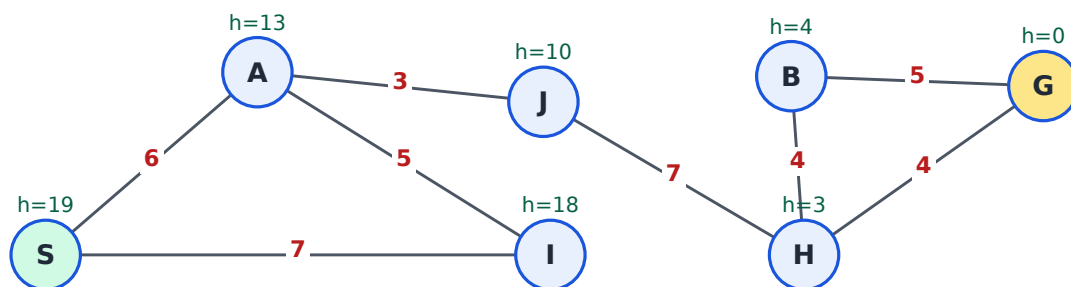
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- (A) S, I, L, K, H, G
 ○ (B) S, A, E, F, G
 ○ (C) S, A, F, E, G
 ○ (D) S, A, I, F, K, L, E, G

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((Q \rightarrow P) \leftrightarrow (P \leftrightarrow \neg Q)) \wedge ((R \leftrightarrow P) \wedge (R \leftrightarrow P)) ?$$

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q) . For how many of them is $\exists x (P(x) \wedge Q(x))$ true?

Answer (numerical): _____

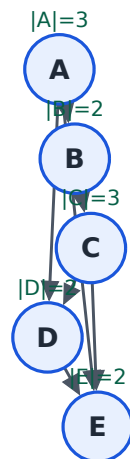
Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are in conjunctive normal form (CNF)?

- ☐ (A) $(p \vee \neg q) \wedge r$
☐ (B) $\neg(p \wedge q)$
☐ (C) $p \vee \neg q \vee r$
☐ (D) $(p \wedge q) \vee r$

Q.8 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

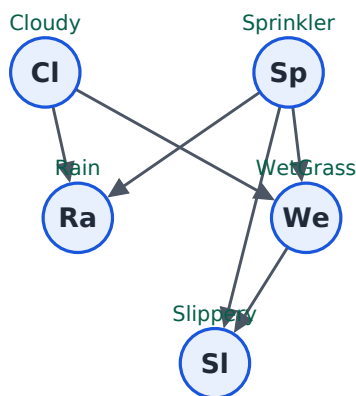
The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We, Sl. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, Sl = Slippery.) What is the probability that a single run produces the sample **(Cl=T, Sp=T, Ra=F, We=F, Sl=T)**? (round off to 4 decimal places)



P(Cl=T)
0.5

P(Sp=T)
0.9

Cl	Sp	P(Ra=T ...)
T	T	0.1
T	F	0.05
F	T	0.25
F	F	0.2

Cl	Sp	P(We=T Sp, We)	P(Sl=T ...)
T	T	0.2	0.2
T	F	0.1	0.75
F	T	0.05	0.4
F	F	0.4	0.4

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Conditional Independence

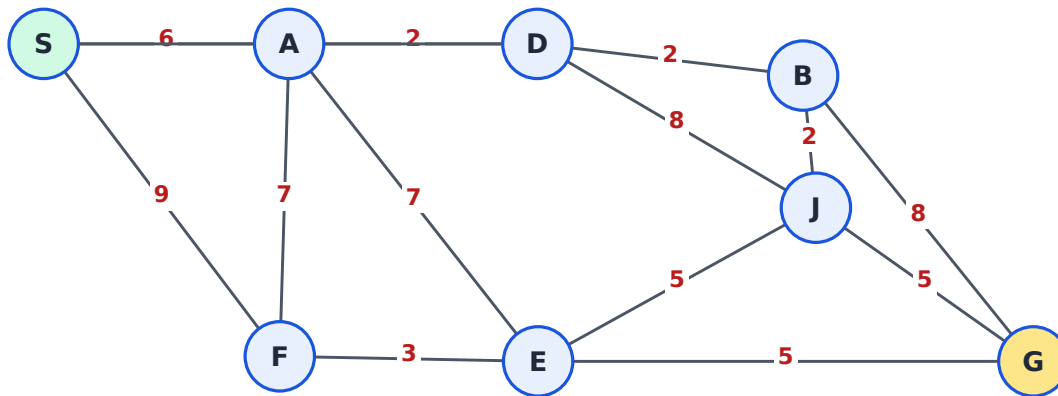
Which three-node structure is such that X and Z are d-separated by the empty set but **not** d-separated given Y?

- ☐ (A) $X \rightarrow Y \rightarrow Z$
☐ (B) $X \leftarrow Y \rightarrow Z$
☐ (C) $X \rightarrow Y \leftarrow Z$
☐ (D) $X \leftarrow Y \leftarrow Z$

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search

Which of the following statements about expectimax / expectiminimax are TRUE?

- ☐ (A) The value of a chance node is the probability-weighted average of its children's values
- ☐ (B) With n distinct chance outcomes per chance layer, the time complexity becomes $O(b^m n^m)$
- ☐ (C) Pruning at chance nodes is impossible even when leaf values are known to lie in a bounded range
- ☐ (D) Expectimax is appropriate when the opponent's or environment's moves are random rather than optimally adversarial

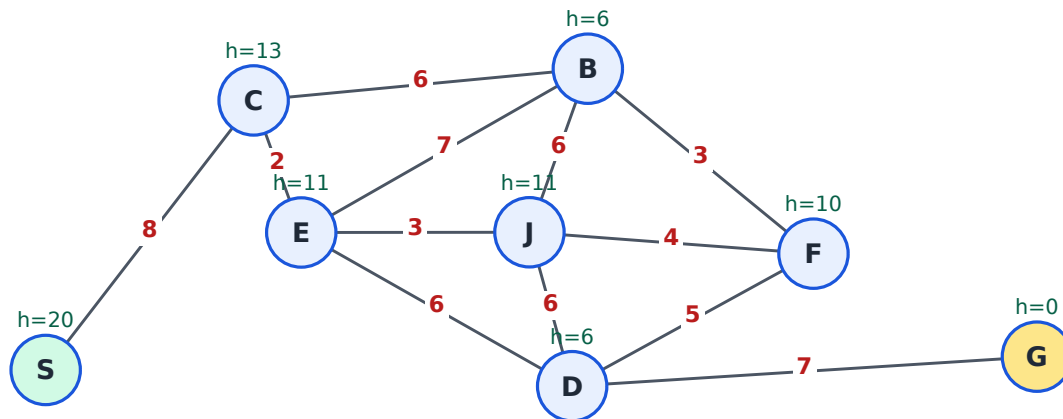
Q.13 [MCQ · 2 marks · negative: -2/3] Informed Search

A* tree search uses an admissible h , the optimal goal is G_1 with cost C^* , and G_2 is a suboptimal goal with $g(G_2) > C^*$. Why can G_2 never be selected for expansion before G_1 ?

- ☐ (A) Because $h(G_2) > h(G_1)$
- ☐ (B) Because some node n on the optimal path is always in the frontier with $f(n) \leq C^* < g(G_2) = f(G_2)$
- ☐ (C) Because A* expands nodes in order of depth
- ☐ (D) Because G_2 is never generated

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

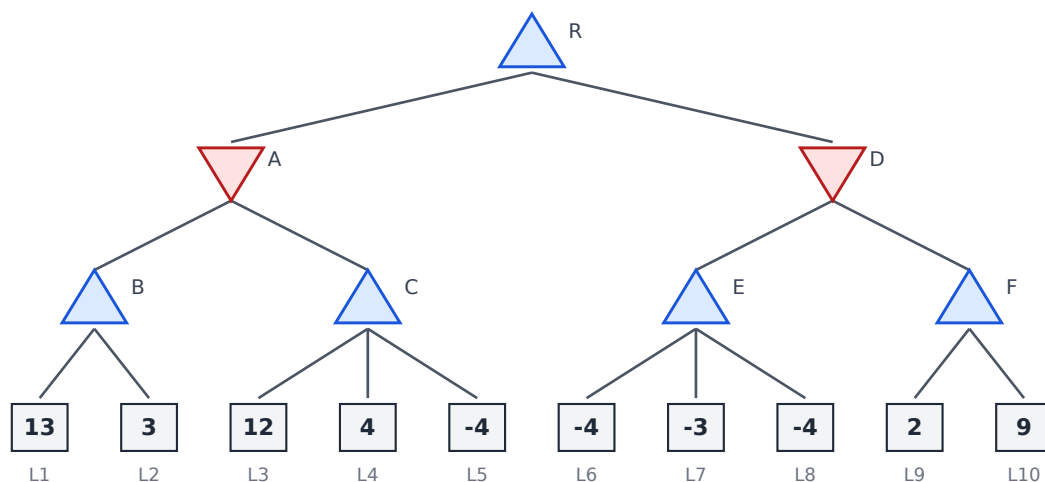
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge F-B violates the consistency condition $h(F) \leq c(F,B) + h(B)$.
☐ (B) h is consistent (monotone).
☐ (C) $h(C)$ overestimates the true cost from C to G.
☐ (D) h is admissible.

Q.15 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



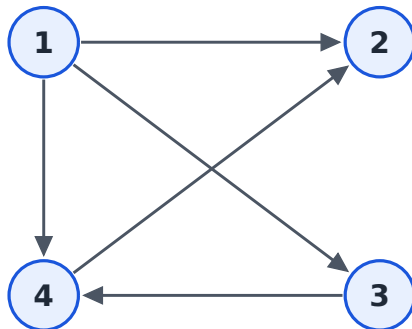
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall y \exists x R(x, y)$
☐ (B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (C) $\forall x R(x, x)$
☐ (D) $\exists x \forall y \neg R(x, y)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (\neg Q \vee \neg R)$, $C2: (\neg P \vee \neg Q \vee S)$, $C3: (\neg P \vee R)$, $C4: (\neg P \vee Q \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P \vee R \vee S)$
☐ (B) $(\neg Q \vee R)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(\neg P \vee \neg Q)$

Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Given the definite clauses $\text{King}(x) \wedge \text{Greedy}(x) \rightarrow \text{Evil}(x)$, $\text{King}(\text{John})$, and $\text{Greedy}(y)$, Generalized Modus Ponens derives $\text{Evil}(\text{John})$ using which substitution?

- ☐ (A) $\{x/\text{John}, y/\text{John}\}$
☐ (B) $\{x/y\}$
☐ (C) $\{x/\text{John}\}$
☐ (D) $\{y/x, x/\text{Evil}\}$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following statements about first-order logic are TRUE?

- ☐ (A) Entailment in first-order logic is semidecidable
☐ (B) Resolution (with factoring and unification) is refutation-complete for first-order logic
☐ (C) Validity of first-order sentences is decidable
☐ (D) Forward chaining on a Datalog KB (definite clauses without function symbols) always terminates

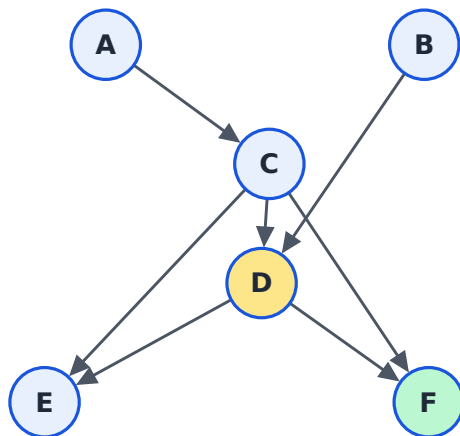
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = \neg Q \vee \neg R$?

- ☐ (A) $\neg(\neg\neg Q \wedge \neg\neg R)$
☐ (B) $\neg\neg Q \vee \neg R$
☐ (C) $\neg Q \wedge \neg R$
☐ (D) $\neg R \vee \neg Q$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(F \mid D = T)$ in the Bayesian network below (Boolean variables; evidence $D = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, B, E, A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

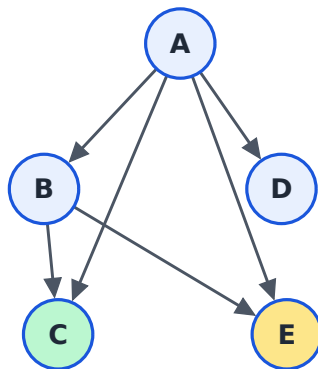
Q.22 [MSQ · 2 marks · no negative marking] Probability for AI

A full joint distribution over Boolean variables X, Y, Z is: $P(x, y, z) = 0.10$, $P(x, y, \neg z) = 0.05$, $P(x, \neg y, z) = 0.15$, $P(x, \neg y, \neg z) = 0.10$, $P(\neg x, y, z) = 0.05$, $P(\neg x, y, \neg z) = 0.20$, $P(\neg x, \neg y, z) = 0.15$, $P(\neg x, \neg y, \neg z) = 0.20$. Which of the following statements are true?

- ☐ (A) $P(x) = 0.4$
☐ (B) $X \perp Y$ (X and Y are marginally independent)
☐ (C) $X \perp Y \mid Z$
☐ (D) $P(y \mid \neg z) \approx 0.455$

Q.23 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{E=F}$. The current state of the chain is $(\mathbf{A=F, B=T, C=F, D=T, E=F})$. The next step resamples $\mathbf{C}$. With what probability is C set to T? (round off to 3 decimal places)



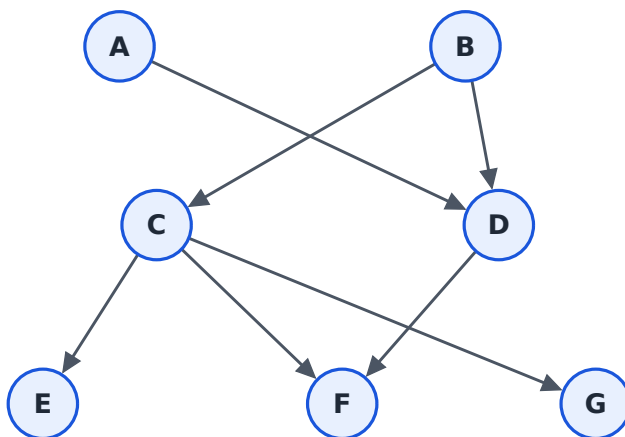
P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.4	T	0.4	T	T	0.9
	F	0.05	T	F	0.7
			F	T	0.2
			F	F	0.8

A	P(D=T ...)	A	B	P(E=T ...)
T	0.2	T	T	0.1
F	0.6	T	F	0.2
		F	T	0.8
		F	F	0.9

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

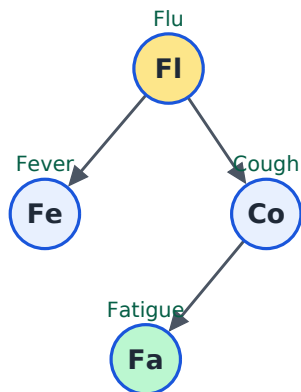
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp F$
☐ (B) $D \perp F \mid E$
☐ (C) $D \perp G \mid B$
☐ (D) $E \perp D \mid C$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Fatigue} = T \mid \text{Flu} = T)$ (round off to 3 decimal places).



FI	P(FI=T)
T	0.95
F	

FI	P(Fe=T ...)	FI	P(Co=T ...)
T	0.4	T	0.2
F	0.25	F	0.4

Co	P(Fa=T ...)
T	0.1
F	0.3

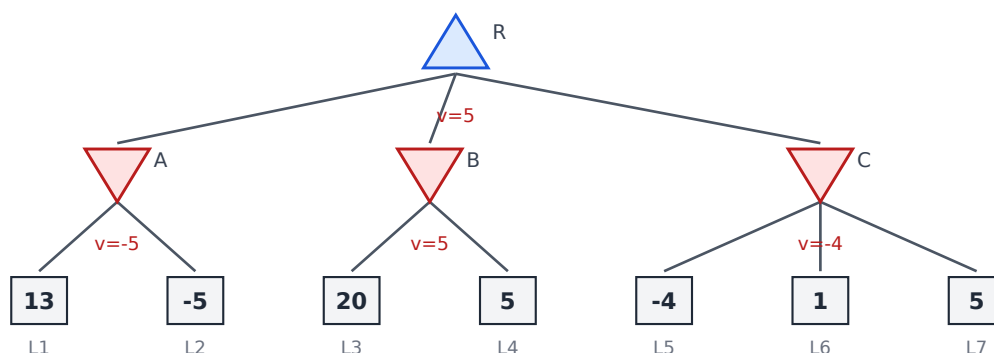
Answer (numerical): _____

Answer Key — Mock Test 061

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	5	14	MSQ	2	Informed Search	A, D
2	NAT	1	Adversarial Search	9	15	NAT	2	Adversarial Search	8
3	MCQ	1	Uninformed Search	C	16	MSQ	2	Predicate Logic	B, D
4	NAT	1	Informed Search	20	17	MSQ	2	Propositional Logic	A, D
5	NAT	1	Propositional Logic	1	18	MCQ	2	Predicate Logic	A
6	NAT	1	Predicate Logic	175	19	MSQ	2	Predicate Logic	A, B, D
7	MSQ	1	Propositional Logic	A, C	20	MSQ	2	Propositional Logic	A, D
8	NAT	1	Bayesian Networks	36	21	NAT	2	Exact Inference	4
9	NAT	1	Sampling	0.2428 to 0.2432	22	MSQ	2	Probability for AI	A, D
10	MCQ	1	Conditional Independence	C	23	NAT	2	Sampling	0.198 to 0.202
11	NAT	2	Uninformed Search	8	24	MSQ	2	Conditional Independence	C, D
12	MSQ	2	Adversarial Search	A, B, D	25	NAT	2	Exact Inference	0.258 to 0.262
13	MCQ	2	Informed Search	B					

Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 5**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = -5, B = 5, C = -4. Root value = **5**; MAX's optimal first move is towards B.

Q.2 [NAT, 1M · Adversarial Search] Answer: 9

Each MIN node takes the minimum of its leaves: $\text{MIN}_1 = 9$, $\text{MIN}_2 = 8$, $\text{MIN}_3 = 2$, $\text{MIN}_4 = 7$. The root takes the maximum: $\max(9, 8, 2, 7) = 9$, so MAX plays the first move.

Q.3 [MCQ, 1M · Uninformed Search] Answer: C (C: S, A, F, E, G)

Adjacency lists (alphabetical): A: {F, I, S}; E: {F, G, H, K}; F: {A, E, G}; G: {E, F, H}; H: {E, G, K, L}; I: {A, K, L, S}; K: {E, H, I, L}; L: {H, I, K}; S: {A, I}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, A, F, E, G**; the path found is $S \rightarrow A \rightarrow F \rightarrow E \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Informed Search] Answer: 20

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (19)	A(13), I(18)
2	A (13)	J(10), I(18)
3	J (10)	H(3), I(18)
4	H (3)	G(0), B(4), I(18)
5	G (0)	B(4), I(18) ← goal, stop

Path returned: **S → A → J → H → G**, cost = $6 + 3 + 7 + 4 = 20$.

The optimal cost is $h^*(S) = 20$ — greedy happens to be optimal here, but in general it is not.

Q.5 [NAT, 1M · Propositional Logic] Answer: 1

Build the truth table (sub-formulas shown first):

P	Q	R	$(Q \rightarrow P) \leftrightarrow (P \leftrightarrow \neg Q)$	$(R \leftrightarrow P) \wedge (R \leftrightarrow P)$	$((Q \rightarrow P) \leftrightarrow (P \leftrightarrow \neg Q)) \wedge ((R \leftrightarrow P) \wedge (R \leftrightarrow P))$
T	T	T	F	T	F
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	T	F	F	T	F
F	F	T	F	F	F
F	F	F	F	T	F

Rows where $F = T$: **1**.

Q.6 [NAT, 1M · Predicate Logic] Answer: 175

Interpret P and Q as subsets of the domain; complement: no element has both, 3 combos each $\Rightarrow 4^n - 3^n$. With $n = 4$ this gives **175** (verified by enumerating all interpretations).

Q.7 [MSQ, 1M · Propositional Logic] Answer: A, C (A: $(p \vee \neg q) \wedge r$; C: $p \vee \neg q \vee r$)

A CNF is a conjunction of clauses, each a disjunction of literals (negation only on atoms). $(p \vee \neg q) \wedge r$ is two clauses. $p \vee \neg q \vee r$ is a single clause, which is a (one-clause) CNF. $\neg(p \wedge q)$ has negation applied to a compound formula, so it is not in CNF (its CNF is $\neg p \vee \neg q$). $(p \wedge q) \vee r$ has a conjunction inside a disjunction; its CNF is $(p \vee r) \wedge (q \vee r)$.

Q.8 [NAT, 1M · Bayesian Networks] Answer: 36

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (2-1) \times 3 + C: (3-1) \times 2 + D: (2-1) \times 3 \times 3 + E: (2-1) \times 3 \times 3 \times 2 = 36$.

Full joint table: $\prod |X| - 1 = 72 - 1 = 71$; saving = 35.

Q.9 [NAT, 1M · Sampling] Answer: 0.2428 to 0.2432

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(\text{Cl}=T) \times P(\text{Sp}=T) \times P(\text{Ra}=F | \text{Cl}=T, \text{Sp}=T) \times P(\text{We}=F | \text{Cl}=T, \text{Sp}=T) \times P(\text{Sl}=T | \text{Sp}=T, \text{We}=F)$
 $= 0.5 \times 0.9 \times 0.9 \times 0.8 \times 0.75 = 0.243 \approx 0.2430$.

Q.10 [MCQ, 1M · Conditional Independence] Answer: C ($C: X \rightarrow Y \leftarrow Z$)

In a chain (A, D) or fork (B), the path $X \rightarrow Y \leftarrow Z$ is active when Y is unobserved and blocked when Y is observed. In the collider (v-structure) $X \rightarrow Y \leftarrow Z$ the path is blocked when neither Y nor a descendant is observed, so X and Z are d-separated by $\emptyset$, and it becomes active when Y is observed. Hence only (C) has the stated behaviour.

Q.11 [NAT, 2M · Uninformed Search] Answer: 8

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	A(6), D(9), F(9)
2	A (6)	D(8), F(9), E(13)
3	D (8)	F(9), B(10), E(13), J(16)
4	F (9)	B(10), E(12), J(16)
5	B (10)	E(12), J(12), G(18)
6	E (12)	J(12), G(17)
7	J (12)	G(17)
8	G (17)	← goal, stop

Path returned: **S** → **F** → **E** → **G**, cost = 9 + 3 + 5 = **17**. Nodes expanded (including S and G): **8** (S, A, D, F, B, E, J, G).

Q.12 [MSQ, 2M · Adversarial Search] Answer: A, B, D

(A) is the definition of a chance node's value. (B) is the AIMA bound for expectiminimax. (C) is false: if utilities are bounded, a chance node's average can be bounded before all of its children are seen, so some pruning is possible. (D) is true: expectimax models stochastic behaviour, while minimax models a worst-case adversary.

Q.13 [MCQ, 2M · Informed Search] Answer: B (B: Because some node n on the optimal path is always in the frontier with $f(n) \leq C^* < g(G_2) = f(G_2)$)

This is the standard optimality proof. Until G_1 is selected, some node n on the optimal path is still in the frontier, and admissibility gives $f(n) = g(n) + h(n) \leq C^*$. Also $f(G_2) = g(G_2) > C^*$ because $h(\text{goal}) = 0$. So n , and eventually G_1 , are selected before G_2 . (A) is wrong because $h = 0$ at every goal. (C) is wrong because A^* orders by f , not depth. (D) is wrong because G_2 may well be generated, just not selected.

Q.14 [MSQ, 2M · Informed Search] Answer: A, D (A: The edge $F \rightarrow B$ violates the consistency condition $h(F) \leq c(F, B) + h(B)$.; D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	D	E	F	G	J	S
$h(n)$	6	13	6	11	10	0	11	20
$h^*(n)$	15	15	7	13	12	0	13	23
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

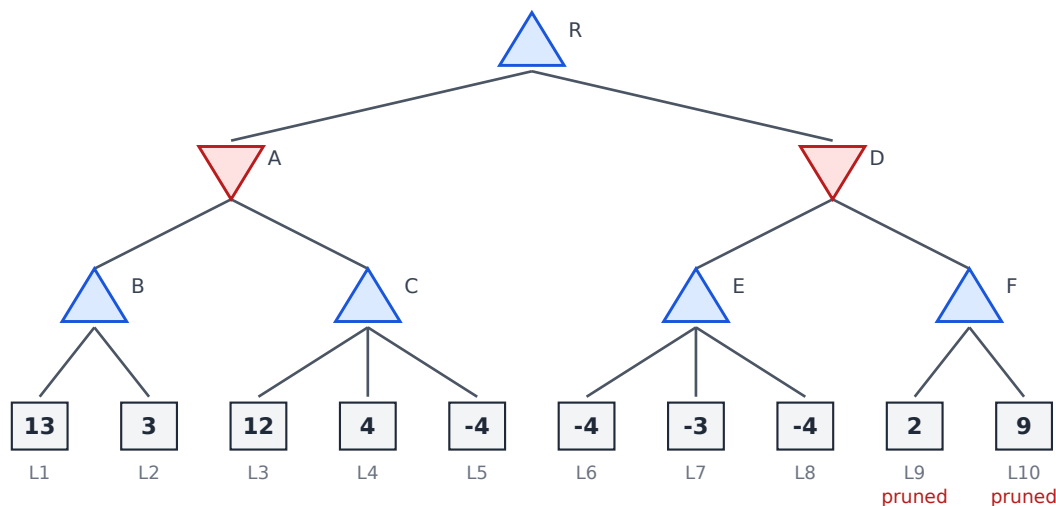
Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u, v) + h(v)$ for every edge in both directions. Violating directed edges: $C \rightarrow B$ ($13 > 6 + 6$), $F \rightarrow B$ ($10 > 3 + 6$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.15 [NAT, 2M · Adversarial Search] Answer: 8

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $-3 \leq \alpha = 12 \Rightarrow$ prune remaining children (leaves L9, L10).



Leaves evaluated (8): L1, L2, L3, L4, L5, L6, L7, L8. Leaves pruned (2): L9, L10. Root value = 12 (same as plain minimax, 12 — pruning never changes the root value).

Q.16 [MSQ, 2M · Predicate Logic] **Answer: B, D** (B: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$; D: $\exists x \forall y \neg R(x, y)$)

$R = \{(1,2), (1,3), (1,4), (3,4), (4,2)\}$.

(A) $\forall y \exists x R(x, y)$: **FALSE** — 1 has no incoming edge.

(B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(C) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(D) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(\neg P \vee R \vee S)$; D: $(\neg P \vee \neg Q)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee R \vee S)$ is derivable:

1. resolve $(\neg P \vee \neg Q \vee S)$ and $(\neg P \vee Q \vee R)$ on $Q \Rightarrow (\neg P \vee R \vee S)$

(B) $(\neg Q \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) The resolution closure contains 6 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(D) $(\neg P \vee \neg Q)$ is derivable:

1. resolve $(\neg Q \vee \neg R)$ and $(\neg P \vee R)$ on $R \Rightarrow (\neg P \vee \neg Q)$

Q.18 [MCQ, 2M · Predicate Logic] **Answer: A** (A: $\{x/\text{John}, y/\text{John}\}$)

GMP needs one θ with $\text{King}(x)\theta = \text{King}(\text{John})\theta$ and $\text{Greedy}(x)\theta = \text{Greedy}(y)\theta$. x/John matches the first premise; y/John then makes $\text{Greedy}(y)$ match $\text{Greedy}(\text{John})$. With only $\{x/\text{John}\}$, $\text{Greedy}(\text{John})$ and $\text{Greedy}(y)$ are not syntactically identical, so the GMP side condition fails; $\{x/y\}$ does not match $\text{King}(\text{John})$. The conclusion is $\text{Evil}(x)\theta = \text{Evil}(\text{John})$.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: A, B, D**

(A) True: an algorithm can confirm every entailed sentence but may run forever on non-entailed ones

(Turing/Church). (B) True: Robinson's resolution theorem. (C) False: FOL validity is undecidable. (D) True:

with no function symbols only finitely many ground atoms exist, so a fixed point is reached in finitely many steps.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $\neg(\neg\neg Q \wedge \neg\neg R)$; D: $\neg R \vee \neg Q$)

Compare truth tables with $F = \neg Q \vee \neg R$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg\neg Q \wedge \neg\neg R)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $\neg\neg Q \vee \neg R$ — differs at $Q=T, R=T$ ($F=F$, option $=T$) $\Rightarrow$ not equivalent.

(C) $\neg Q \wedge \neg R$ — differs at $Q=T, R=F$ ($F=T$, option $=F$) $\Rightarrow$ not equivalent.

(D) $\neg R \vee \neg Q$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.21 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, C)$, $f(B, C)$, $f(C, E)$, $f(C, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(A, C), f(B, C), f(C, E), f(C, F)$	$\{A, B, C, E, F\}$	$f(A, B, E, F)$
B	$f(B), f(A, B, E, F)$	$\{A, B, E, F\}$	$f(A, E, F)$
E	$f(A, E, F)$	$\{A, E, F\}$	$f(A, F)$
A	$f(A), f(A, F)$	$\{A, F\}$	$f(F)$

The largest factor produced has **4** variable(s) (created when eliminating C); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over F and normalise.

Q.22 [MSQ, 2M · Probability for AI] **Answer: A, D** (A: $P(x) = 0.4$; D: $P(y \mid \neg z) \approx 0.455$)

$P(x) = 0.10 + 0.05 + 0.15 + 0.10 = 0.40$, so (A) is true. $P(y) = 0.10 + 0.05 + 0.05 + 0.20 = 0.40$ and $P(x, y) = 0.15 \neq 0.4 \times 0.4 = 0.16$, so (B) is false. Given z : $P(z) = 0.45$, $P(x, y \mid z) = 0.10/0.45 \approx 0.222$, while $P(x \mid z)P(y \mid z) = (0.25/0.45)(0.15/0.45) \approx 0.556 \times 0.333 \approx 0.185$, so (C) is false. $P(\neg z) = 0.55$ and $P(y, \neg z) = 0.05 + 0.20 = 0.25$, so $P(y \mid \neg z) = 0.25/0.55 \approx 0.4545$, making (D) true.

Q.23 [NAT, 2M · Sampling] **Answer: 0.198 to 0.202**

Gibbs sampling draws C from $P(C \mid \text{mb}(C))$, where $\text{mb}(C) = \{A, B\}$:

$P(C \mid \text{mb}) \propto P(C \mid \text{parents}(C)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$C=T$: $P(C=T \mid A=F, B=T) = 0.2 = 0.2$

$C=F$: $P(C=F \mid A=F, B=T) = 0.8 = 0.8$

Normalise: $0.2 / (0.2 + 0.8) = \mathbf{0.2}$.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: C, D** (C: $D \perp G \mid B$; D: $E \perp D \mid C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp F$: **FALSE** — the path $C \leftarrow B \rightarrow D \rightarrow F$ is active given $\{\}$.

(B) $D \perp F \mid E$: **FALSE** — the path $D \leftarrow B \rightarrow C \rightarrow F$ is active given $\{E\}$.

(C) $D \perp G \mid B$: **TRUE** — $D \leftarrow B \rightarrow C \rightarrow G$ (blocked: observed fork node B); $D \rightarrow F \leftarrow C \rightarrow G$ (blocked: collider F with neither it nor a descendant observed).

(D) $E \perp D \mid C$: **TRUE** — $E \leftarrow C \leftarrow B \rightarrow D$ (blocked: observed chain node C); $E \leftarrow C \rightarrow F \leftarrow D$ (blocked: observed fork node C).

Q.25 [NAT, 2M · Exact Inference] **Answer: 0.258 to 0.262**

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever} \mid \text{Flu}) \cdot P(\text{Cough} \mid \text{Flu}) \cdot P(\text{Fatigue} \mid \text{Cough})$.

Inference by enumeration: $P(\text{Fatigue} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Fatigue}, e, \text{hidden})$.

Hidden: Fe, Co	Fa	product of CPT entries ($F_l \cdot F_e \cdot C_o \cdot F_a$)	value
T, T	T	$0.95 \times 0.4 \times 0.2 \times 0.1$	0.0076
T, F	T	$0.95 \times 0.4 \times 0.8 \times 0.3$	0.0912
F, T	T	$0.95 \times 0.6 \times 0.2 \times 0.1$	0.0114
F, F	T	$0.95 \times 0.6 \times 0.8 \times 0.3$	0.1368
T, T	F	$0.95 \times 0.4 \times 0.2 \times 0.9$	0.0684
T, F	F	$0.95 \times 0.4 \times 0.8 \times 0.7$	0.2128
F, T	F	$0.95 \times 0.6 \times 0.2 \times 0.9$	0.1026
F, F	F	$0.95 \times 0.6 \times 0.8 \times 0.7$	0.3192

$P(\text{Fatigue}=T, e) = 0.247$; $P(\text{Fatigue}=F, e) = 0.703$; $P(e) = 0.95$.

$P(\text{Fatigue}=T \mid e) = 0.247 / 0.95 = \mathbf{0.26} \approx 0.260$.

Mock Test 062

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	3 / 9 / 13	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MSQ · 1 mark · no negative marking] Uninformed Search

Every action has the same positive cost. Which of the following algorithms are guaranteed to return an optimal solution on a problem with a finite branching factor that has a solution?

- ☐ (A) Breadth-first search
- ☐ (B) Iterative deepening search
- ☐ (C) Uniform-cost search
- ☐ (D) Depth-first search (graph search)

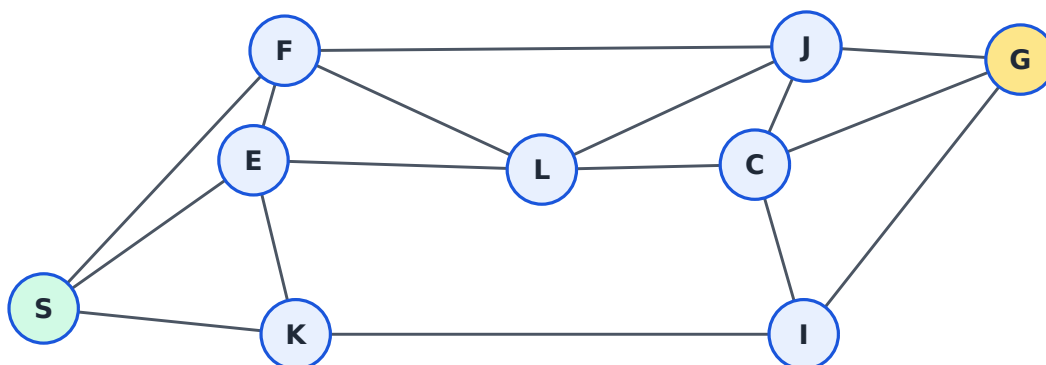
Q.2 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

A search tree has uniform branching factor $b = 3$; the shallowest goal is at depth $d = 5$ and happens to be the **last** node generated at that depth. BFS applies the goal test when a node is **generated**. How many nodes are generated in total (do **not** count the root)?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

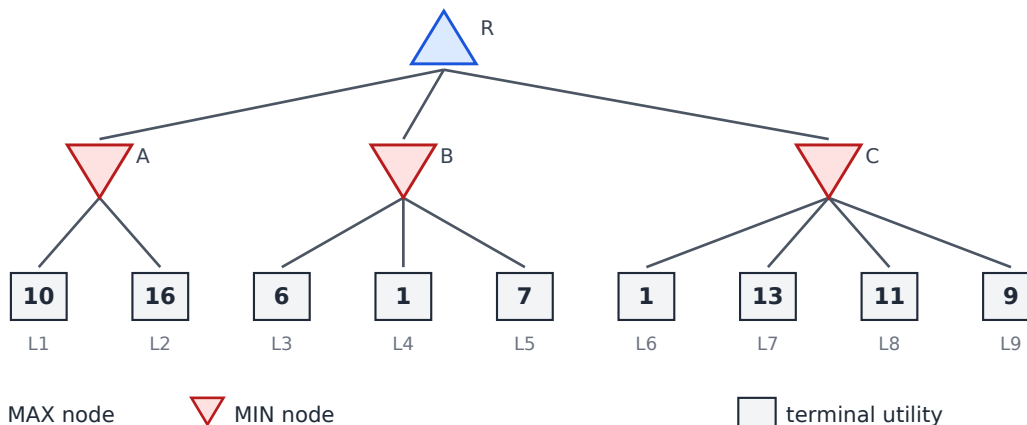
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, F, E, K, L, J, I, C, G
- ☐ (B) S, E, F, J, C, G
- ☐ (C) S, E, F, K, L, J, I, C, G
- ☐ (D) S, K, F, E, I, L, J, G

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

What is the most general unifier (MGU) of $\text{Knows}(\text{John}, x)$ and $\text{Knows}(y, \text{Mother}(y))$?

- ☐ (A) $\{y/\text{John}, x/\text{Mother}(\text{John})\}$
- ☐ (B) $\{x/\text{John}, y/\text{Mother}(\text{John})\}$
- ☐ (C) $\{y/\text{John}, x/\text{Mother}(y)\}$
- ☐ (D) The two atoms are not unifiable

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (R \rightarrow (P \wedge \neg Q)) \wedge (R \rightarrow \neg R) ?$$

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q) . For how many of them is $\exists x (P(x) \wedge Q(x))$ true?

Answer (numerical): _____

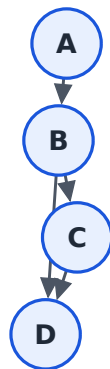
Q.8 [NAT · 1 mark · no negative marking] Predicate Logic

Interpret $\text{Div}(x, y)$ as 'x divides y' over the domain $D = \{1, 2, \dots, 12\}$. For how many $x \in D$ is the formula $\neg(x = 1) \wedge \forall y (\text{Div}(y, x) \rightarrow (y = 1 \vee y = x))$ true?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=F, B=F, C=T, D=T)**? (round off to 4 decimal places)



P(A=T)
0.2

A	P(B=T ...)	B	P(C=T ...)
T	0.7	T	0.5
F	0.05	F	0.95

B	C	P(D=T ...)
T	T	0.1
T	F	0.4
F	T	0.75
F	F	0.4

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(C = T \mid D = F)$, how many of these samples are **rejected**?

#	A	B	C	D
1	T	F	T	T
2	F	T	F	F
3	T	F	T	F
4	T	F	T	F
5	T	F	F	F
6	F	F	T	F
7	T	F	T	F
8	T	F	F	F
9	T	F	T	F
10	T	F	T	F
11	T	F	T	F
12	T	F	F	F
13	T	F	T	F
14	T	F	F	F
15	T	F	T	F
16	T	F	T	T
17	F	F	T	T
18	T	F	T	F
19	T	F	T	T
20	T	F	F	F

Answer (numerical): _____

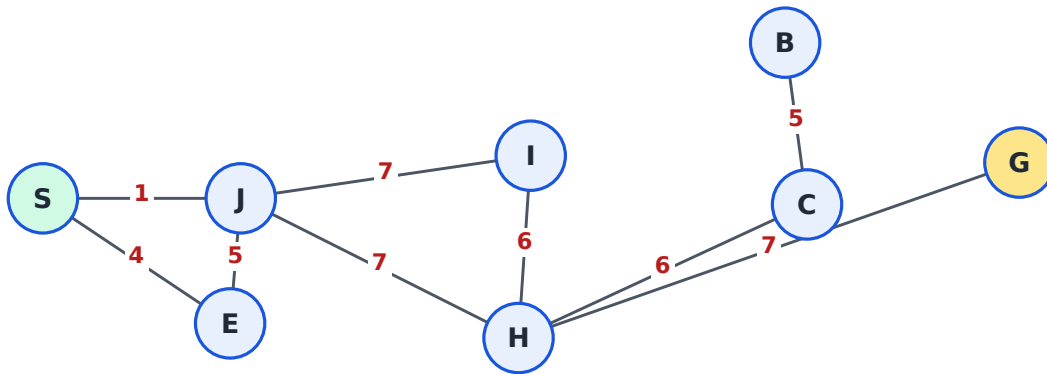
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search

Depth-first tree search generates all $b = 3$ children of a node when it expands that node. The tree has depth $m = 10$, and no goal is found along the leftmost path. Consider the moment just after DFS expands the leftmost node at depth 9, so that its 3 children at depth 10 have been added to the frontier. How many nodes are in the frontier at that moment?

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

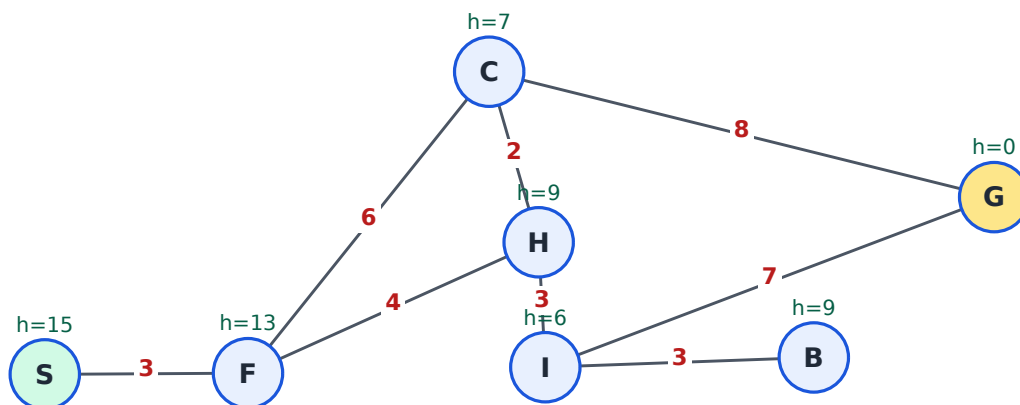
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

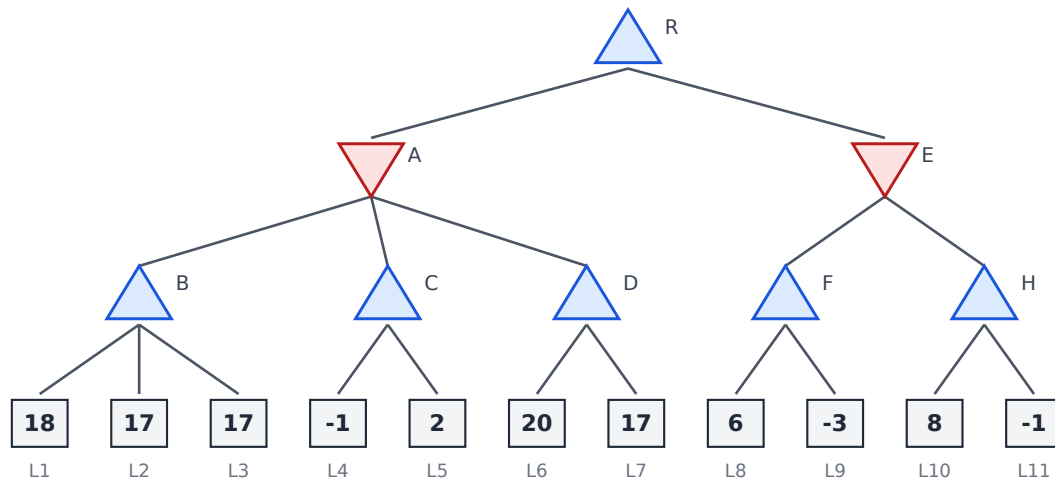
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(I)$ overestimates the true cost from I to G.
- ☐ (B) The edge H-C violates the consistency condition (in at least one direction).
- ☐ (C) h is admissible.
- ☐ (D) h is consistent (monotone).

Q.14 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L6 is pruned.
☐ (B) Leaf L7 is pruned.
☐ (C) Leaf L9 is pruned.
☐ (D) Leaf L11 is pruned.

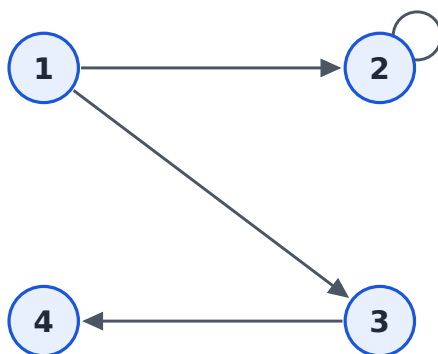
Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following formulas are contingent (satisfiable but not valid)?

- ☐ (A) $(p \rightarrow q) \wedge (p \rightarrow \neg q)$
☐ (B) $(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$
☐ (C) $(p \leftrightarrow q) \wedge (p \oplus q)$
☐ (D) $(p \vee q) \rightarrow (p \wedge \neg q)$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (C) $\exists x R(x, x)$
☐ (D) $\exists x \forall y R(x, y)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (P \vee \neg R) \vee (Q \wedge \neg P)$?

- ☐ (A) $(Q \wedge \neg P) \vee (P \vee \neg R)$
- ☐ (B) $((P \vee \neg R) \vee Q) \wedge ((P \vee \neg R) \vee \neg P)$
- ☐ (C) $\neg(P \vee \neg R) \vee (Q \wedge \neg P)$
- ☐ (D) $\neg(\neg(P \vee \neg R) \vee \neg(Q \wedge \neg P))$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg R)$, C2: $(R \vee S)$, C3: (S) , C4: $(P \vee Q \vee \neg R)$, C5: $(P \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) (P)
- ☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
- ☐ (C) $(P \vee Q \vee S)$
- ☐ (D) $(\neg Q \vee \neg R)$

Q.19 [NAT · 2 marks · no negative marking] Bayesian Networks

Fever has three possible causes Cold, Flu and Malaria, modeled with a noisy-OR CPT with no leak. The inhibition probabilities are $q_{\text{cold}} = 0.6$, $q_{\text{flu}} = 0.2$, $q_{\text{malaria}} = 0.1$ (q_i = probability that cause i , when present alone, fails to produce fever). Compute $P(\text{fever} \mid \text{cold, flu, } \neg \text{malaria})$. (Round to 2 decimal places.)

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $A \rightarrow C$ and $B \rightarrow C$ (Boolean variables) with $P(a) = 0.1$, $P(b) = 0.2$, $P(c \mid a, b) = 0.95$, $P(c \mid a, \neg b) = 0.90$, $P(c \mid \neg a, b) = 0.80$, $P(c \mid \neg a, \neg b) = 0.05$. Compute $P(a \mid c, b)$. (Round to 3 decimal places.)

Answer (numerical): _____

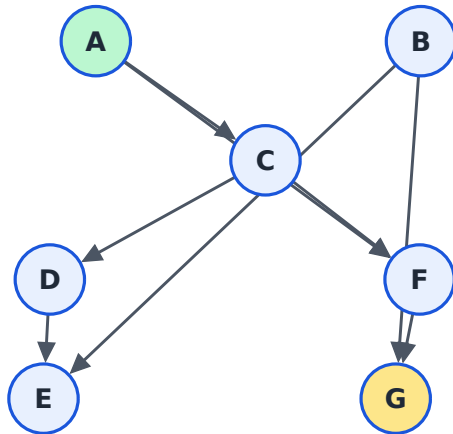
Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence

Consider the Bayesian network (DAG) with edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$, $B \rightarrow F$. Which of the following d-separation statements are true?

- ☐ (A) A and B are d-separated given $\emptyset$
- ☐ (B) A and F are d-separated given $\{B\}$
- ☐ (C) E and F are d-separated given $\{C\}$
- ☐ (D) E and F are d-separated given $\emptyset$

Q.22 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

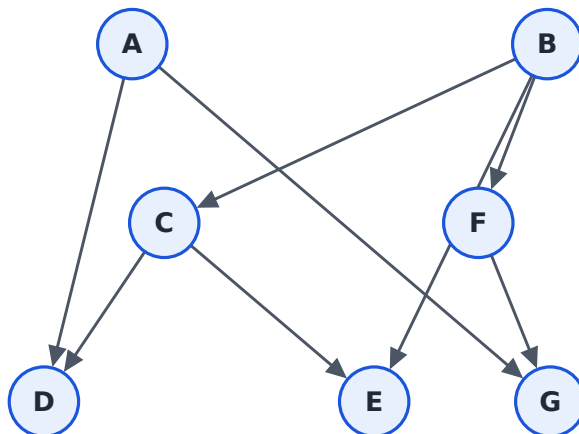
Variable elimination is used to compute $P(A \mid G = T)$ in the Bayesian network below (Boolean variables; evidence $G = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C**, **F**, **D**, **E**, **B**. What is the scope of the new factor produced when **C** is summed out?



- ☐ (A) $f(A, C, D, F)$
☐ (B) $f(D, F)$
☐ (C) $f(A, D, F)$
☐ (D) $f(A, F)$

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

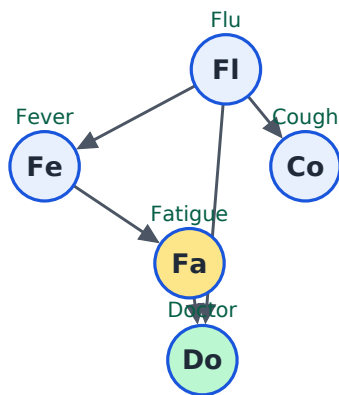
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp A$
☐ (B) $D \perp B \mid C$
☐ (C) $G \perp B$
☐ (D) $A \perp F \mid E$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) Compute $P(\text{Doctor} = T \mid \text{Fatigue} = T)$ (round off to 3 decimal places).



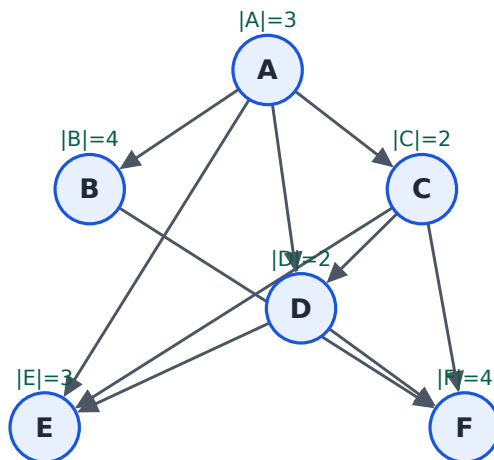
$P(\text{Fl}=T)$	Fl	$P(\text{Fe}=T \dots)$	Fl	$P(\text{Co}=T \dots)$
0.5	T	0.8	T	0.75
	F	0.1	F	0.05

Fe	$P(\text{Fa}=T \dots)$	Fl	Fa	$P(\text{Do}=T \dots)$
T	0.95	T	T	0.5
F	0.25	T	F	0.2
		F	T	0.7
		F	F	0.4

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Answer Key — Mock Test 062

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MSQ	1	Uninformed Search	A, B, C	14	MSQ	2	Adversarial Search	B, D
2	NAT	1	Uninformed Search	363	15	MSQ	2	Propositional Logic	A, D
3	MCQ	1	Uninformed Search	C	16	MSQ	2	Predicate Logic	A, C
4	NAT	1	Adversarial Search	10	17	MSQ	2	Propositional Logic	A, B
5	MCQ	1	Predicate Logic	A	18	MSQ	2	Propositional Logic	A, C
6	NAT	1	Propositional Logic	4	19	NAT	2	Bayesian Networks	0.875 to 0.885
7	NAT	1	Predicate Logic	37	20	NAT	2	Conditional Independence	0.116 to 0.117
8	NAT	1	Predicate Logic	5	21	MSQ	2	Conditional Independence	A, B, C
9	NAT	1	Sampling	0.5413 to 0.5417	22	MCQ	2	Exact Inference	C
10	NAT	1	Sampling	4	23	MSQ	2	Conditional Independence	B, D
11	NAT	2	Uninformed Search	21	24	NAT	2	Exact Inference	0.555 to 0.559
12	NAT	2	Uninformed Search	7	25	NAT	2	Bayesian Networks	483
13	MSQ	2	Informed Search	C, D					

Detailed Solutions

Q.1 [MSQ, 1M · Uninformed Search] Answer: A, B, C (A: Breadth-first search; B: Iterative deepening search; C: Uniform-cost search)

With equal step costs, the cheapest solution is the shallowest one. BFS (A) and IDS (B) find the shallowest goal first. UCS (C) is optimal for any positive step costs. DFS (D) can follow a deep branch and return a deep goal before discovering a shallower one, so it is not optimal.

Q.2 [NAT, 1M · Uninformed Search] Answer: 363

With goal test at generation, BFS generates every node at depths 1 ... d before (and including) the goal (the worst case). $N = b + b^2 + \dots + b^d = 3 + 9 + 27 + 81 + 243 = 363$. This is the $O(b^d)$ of AIMA.

Q.3 [MCQ, 1M · Uninformed Search] Answer: C (C: S, E, F, K, L, J, I, C, G)

Adjacency lists (alphabetical): C: {G, I, J, L}; E: {F, K, L, S}; F: {E, J, L, S}; G: {C, I, J}; I: {C, G, K}; J: {C, F, G, L}; K: {E, I, S}; L: {C, E, F, J}; S: {E, F, K}.

BFS trace (queue shown front → back):

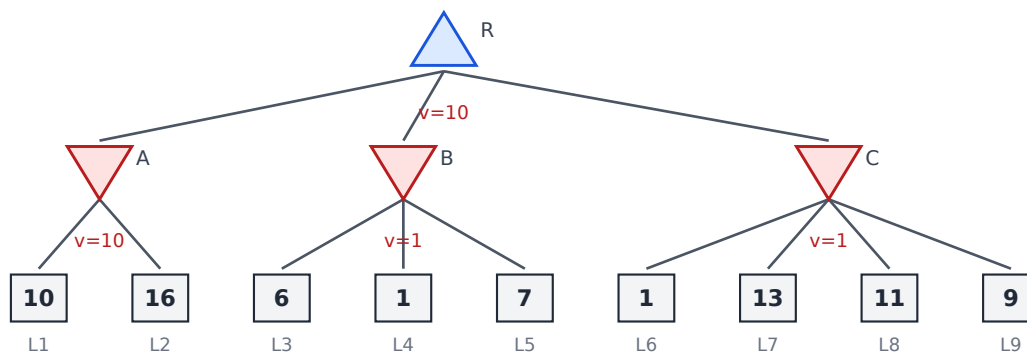
Step	Expanded	Frontier (FIFO) after expansion
1	S	[E, F, K]
2	E	[F, K, L]
3	F	[K, L, J]
4	K	[L, J, I]
5	L	[J, I, C]
6	J	[I, C, G]
7	I	[C, G]
8	C	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, E, F, K, L, J, I, C, G**. The path returned is $S \rightarrow F \rightarrow J \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [NAT, 1M · Adversarial Search] Answer: 10

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = 10, B = 1, C = 1. Root value = **10**; MAX's optimal first move is towards A.

Q.5 [MCQ, 1M · Predicate Logic] **Answer: A** (A: {y/John, x/Mother(John)})

Unify arguments left to right. John vs y gives y/John. Then x vs Mother(y) becomes x vs Mother(John), giving x/Mother(John). Applying {y/John, x/Mother(John)} to both atoms yields Knows(John, Mother(John)). Option B binds the wrong variables. Option C leaves y in the binding of x, so it is not idempotent and does not make the atoms identical in one application.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 4**

Build the truth table (sub-formulas shown first):

P	Q	R	$R \rightarrow (P \wedge \neg Q)$	$R \rightarrow \neg R$	$(R \rightarrow (P \wedge \neg Q)) \wedge (R \rightarrow \neg R)$
T	T	T	F	F	F
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	T	F	T	T	T
F	F	T	F	F	F
F	F	F	T	T	T

Rows where F = T: **4**.

Q.7 [NAT, 1M · Predicate Logic] **Answer: 37**

Interpret P and Q as subsets of the domain; complement: no element has both, 3 combos each $\Rightarrow 4^n - 3^n$. With $n = 3$ this gives **37** (verified by enumerating all interpretations).

Q.8 [NAT, 1M · Predicate Logic] **Answer: 5**

The formula says $x \neq 1$ and the only divisors of x in D are 1 and x itself — i.e. x is prime. (Every divisor of $x \leq 12$ lies in D, so the restriction to D loses nothing.) The primes in $\{1, \dots, 12\}$ are 2, 3, 5, 7, 11, so the answer is 5.

Q.9 [NAT, 1M · Sampling] **Answer: 0.5413 to 0.5417**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(A=F) \times P(B=F|A=F) \times P(C=T|B=F) \times P(D=T|B=F, C=T)$
 $= 0.8 \times 0.95 \times 0.95 \times 0.75 = \mathbf{0.5415} \approx 0.5415$.

Q.10 [NAT, 1M · Sampling] **Answer: 4**

Rejection sampling keeps only samples consistent with the evidence $D = F$: samples 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 18, 20 (16 accepted, 4 rejected).

Among the accepted samples, $C = T$ in 10. Estimate = $10/16 = \mathbf{0.625}$.

(For this network the exact value is 0.6625; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.11 [NAT, 2M · Uninformed Search] **Answer: 21**

Along the leftmost path, the nodes at depths 0 to 9 have all been expanded. At each depth from 1 to 9, the $b - 1 = 2$ unexpanded siblings of the path node are still in the frontier: $9 \times 2 = 18$ nodes. The 3 children at depth 10 have just been added. Frontier size = $18 + 3 = \mathbf{21} = (b-1)m + 1$, which illustrates DFS's $O(bm)$ space.

Q.12 [NAT, 2M · Uninformed Search] Answer: 7

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(1), E(4)
2	J (1)	E(4), H(8), I(8)
3	E (4)	H(8), I(8)
4	H (8)	I(8), C(14), G(15)
5	I (8)	C(14), G(15)
6	C (14)	G(15), B(19)
7	G (15)	B(19) ← goal, stop

Path returned: **S** → **J** → **H** → **G**, cost = 1 + 7 + 7 = **15**. Nodes expanded (including S and G): **7** (S, J, E, H, I, C, G).

Q.13 [MSQ, 2M · Informed Search] Answer: C, D (C: h is admissible.; D: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	F	G	H	I	S
$h(n)$	9	7	13	0	9	6	15
$h^*(n)$	10	8	14	0	10	7	17
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

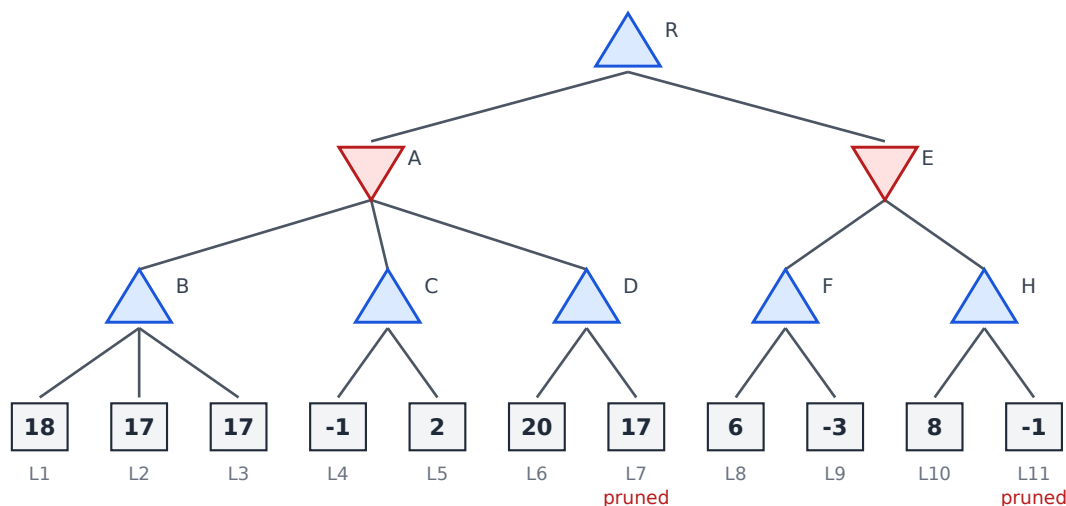
Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [MSQ, 2M · Adversarial Search] Answer: B, D (B: Leaf L7 is pruned.; D: Leaf L11 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node D: value 20 $\geq \beta = 2 \Rightarrow$ prune remaining children (leaves L7).
- At MAX node H: value 8 $\geq \beta = 6 \Rightarrow$ prune remaining children (leaves L11).



Leaves evaluated (9): L1, L2, L3, L4, L5, L6, L8, L9, L10. Leaves pruned (2): L7, L11. Root value = 6 (same as plain minimax, 6 — pruning never changes the root value).

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $(p \rightarrow q) \wedge (p \rightarrow \neg q)$; D: $(p \vee q) \rightarrow (p \wedge \neg q)$)

(A) $\equiv p \rightarrow (q \wedge \neg q) \equiv \neg p$: true at $p=F$, false at $p=T$, so contingent. (B) An implication and its contrapositive are equivalent, so this is valid. (C) $p \leftrightarrow q$ and $p \oplus q$ are negations of each other, so their conjunction is unsatisfiable. (D) True at $p=q=F$ (antecedent false), false at $p=F, q=T$, so contingent.

Q.16 [MSQ, 2M · Predicate Logic] Answer: A, C (A: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$; C: $\exists x R(x, x)$)

$R = \{(1,2), (1,3), (2,2), (3,4)\}$.

(A) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,4)$ nor $R(4,1)$.

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

(D) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A, B (A: $(Q \wedge \neg P) \vee (P \vee \neg R)$; B: $((P \vee \neg R) \vee Q) \wedge ((P \vee \neg R) \vee \neg P)$)

Compare truth tables with $F = (P \vee \neg R) \vee (Q \wedge \neg P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(Q \wedge \neg P) \vee (P \vee \neg R)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(B) $((P \vee \neg R) \vee Q) \wedge ((P \vee \neg R) \vee \neg P)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(C) $\neg(P \vee \neg R) \vee (Q \wedge \neg P)$ — differs at $P=T, Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

(D) $\neg(\neg(P \vee \neg R) \vee \neg(Q \wedge \neg P))$ — differs at $P=T, Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, C (A: (P); C: $(P \vee Q \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) (P) is derivable:

1. resolve $(\neg R)$ and $(P \vee R)$ on R $\Rightarrow$ (P)

(B) The resolution closure contains 8 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(P \vee Q \vee S)$ is derivable:

1. resolve $(R \vee S)$ and $(P \vee Q \vee \neg R)$ on R $\Rightarrow$ $(P \vee Q \vee S)$

(D) $(\neg Q \vee \neg R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.19 [NAT, 2M · Bayesian Networks] Answer: 0.875 to 0.885

In noisy-OR, $P(\neg \text{fever} \mid \text{causes}) = \text{product of } q_i \text{ over the causes that are present}$. Present causes: cold and flu, so $P(\neg \text{fever}) = 0.6 \times 0.2 = 0.12$. $P(\text{fever} \mid \text{cold, flu, } \neg \text{malaria}) = 1 - 0.12 = 0.88$. Malaria is absent and contributes no factor.

Q.20 [NAT, 2M · Conditional Independence] Answer: 0.116 to 0.117

Since $A \perp B$, $P(A \mid c, b) = \alpha P(A) P(c \mid A, b)$ ($P(b)$ cancels). For a: $0.1 \times 0.95 = 0.095$. For $\neg a$: $0.9 \times 0.80 = 0.72$. $P(a \mid c, b) = 0.095 / (0.095 + 0.72) = 0.095 / 0.815 \approx 0.1166 \approx 0.117$. B already explains C, so A remains near its prior 0.1.

Q.21 [MSQ, 2M · Conditional Independence] Answer: A, B, C (A: A and B are d-separated given $\emptyset$; B: A and F are d-separated given $\{B\}$; C: E and F are d-separated given $\{C\}$)

(A) The only path $A \rightarrow C \leftarrow B$ has collider C with no observed descendant: blocked, so true. (B) The only path $A \rightarrow C \leftarrow B \rightarrow F$ is blocked by the unobserved collider C (and also by the observed fork node B): true. (C) The only path $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ contains observed non-collider C: blocked, true. (D) With nothing observed, $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ has no colliders and is active, so false.

Q.22 [MCQ, 2M · Exact Inference] Answer: C (C: $f(A, D, F)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, C)$, $f(C, D)$, $f(B, D, E)$, $f(A, C, F)$, $f(B, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(A, C)$, $f(C, D)$, $f(A, C, F)$	$\{A, C, D, F\}$	$f(A, D, F)$
F	$f(B, F)$, $f(A, D, F)$	$\{A, B, D, F\}$	$f(A, B, D)$
D	$f(B, D, E)$, $f(A, B, D)$	$\{A, B, D, E\}$	$f(A, B, E)$
E	$f(A, B, E)$	$\{A, B, E\}$	$f(A, B)$
B	$f(B)$, $f(A, B)$	$\{A, B\}$	$f(A)$

The largest factor produced has **3** variable(s) (created when eliminating C); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over A and normalise.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: B, D** (B: $D \perp B \mid C$; D: $A \perp F \mid E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp A$: **FALSE** — the path $G \leftarrow A$ is active given $\{\}$.

(B) $D \perp B \mid C$: **TRUE** — $D \leftarrow A \rightarrow G \leftarrow F \leftarrow B$ (blocked: collider G with neither it nor a descendant observed); $D \leftarrow C \leftarrow B$ (blocked: observed chain node C); $D \leftarrow C \rightarrow E \leftarrow B$ (blocked: observed fork node C).

(C) $G \perp B$: **FALSE** — the path $G \leftarrow F \leftarrow B$ is active given $\{\}$.

(D) $A \perp F \mid E$: **TRUE** — $A \rightarrow D \leftarrow C \leftarrow B \rightarrow F$ (blocked: collider D with neither it nor a descendant observed); $A \rightarrow D \leftarrow C \rightarrow E \leftarrow B \rightarrow F$ (blocked: collider D with neither it nor a descendant observed); $A \rightarrow G \leftarrow F$ (blocked: collider G with neither it nor a descendant observed).

Q.24 [NAT, 2M · Exact Inference] **Answer: 0.555 to 0.559**

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}, \text{Doctor}) = P(\text{Flu}) \cdot P(\text{Fever} \mid \text{Flu}) \cdot P(\text{Cough} \mid \text{Flu}) \cdot P(\text{Fatigue} \mid \text{Fever}) \cdot P(\text{Doctor} \mid \text{Flu}, \text{Fatigue})$.

Inference by enumeration: $P(\text{Doctor} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Doctor}, e, \text{hidden})$.

Hidden: Fl, Fe, Co	Do	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, T, T	T	$0.5 \times 0.8 \times 0.75 \times 0.95 \times 0.5$	0.1425
T, T, F	T	$0.5 \times 0.8 \times 0.25 \times 0.95 \times 0.5$	0.0475
T, F, T	T	$0.5 \times 0.2 \times 0.75 \times 0.25 \times 0.5$	0.00937
T, F, F	T	$0.5 \times 0.2 \times 0.25 \times 0.25 \times 0.5$	0.00312
F, T, T	T	$0.5 \times 0.1 \times 0.05 \times 0.95 \times 0.7$	0.00166
F, T, F	T	$0.5 \times 0.1 \times 0.95 \times 0.95 \times 0.7$	0.03159
F, F, T	T	$0.5 \times 0.9 \times 0.05 \times 0.25 \times 0.7$	0.00394
F, F, F	T	$0.5 \times 0.9 \times 0.95 \times 0.25 \times 0.7$	0.07481
T, T, T	F	$0.5 \times 0.8 \times 0.75 \times 0.95 \times 0.5$	0.1425
T, T, F	F	$0.5 \times 0.8 \times 0.25 \times 0.95 \times 0.5$	0.0475
T, F, T	F	$0.5 \times 0.2 \times 0.75 \times 0.25 \times 0.5$	0.00937
T, F, F	F	$0.5 \times 0.2 \times 0.25 \times 0.25 \times 0.5$	0.00312
F, T, T	F	$0.5 \times 0.1 \times 0.05 \times 0.95 \times 0.3$	0.00071
F, T, F	F	$0.5 \times 0.1 \times 0.95 \times 0.95 \times 0.3$	0.01354
F, F, T	F	$0.5 \times 0.9 \times 0.05 \times 0.25 \times 0.3$	0.00169
F, F, F	F	$0.5 \times 0.9 \times 0.95 \times 0.25 \times 0.3$	0.03206

$P(\text{Doctor}=T, e) = 0.3145$; $P(\text{Doctor}=F, e) = 0.2505$; $P(e) = 0.565$.

$P(\text{Doctor}=T \mid e) = 0.3145 / 0.565 = \mathbf{0.5566} \approx 0.557$.

Q.25 [NAT, 2M · Bayesian Networks] **Answer: 483**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (4-1) \times 3 + C: (2-1) \times 3 + D: (2-1) \times 3 \times 2 + E: (3-1) \times 3 \times 2 \times 2 + F: (4-1) \times 4 \times 2 \times 2 = \mathbf{92}$.

Full joint table: $\prod |X| - 1 = 576 - 1 = 575$; saving = 483.

Mock Test 063

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	3 / 7 / 15	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

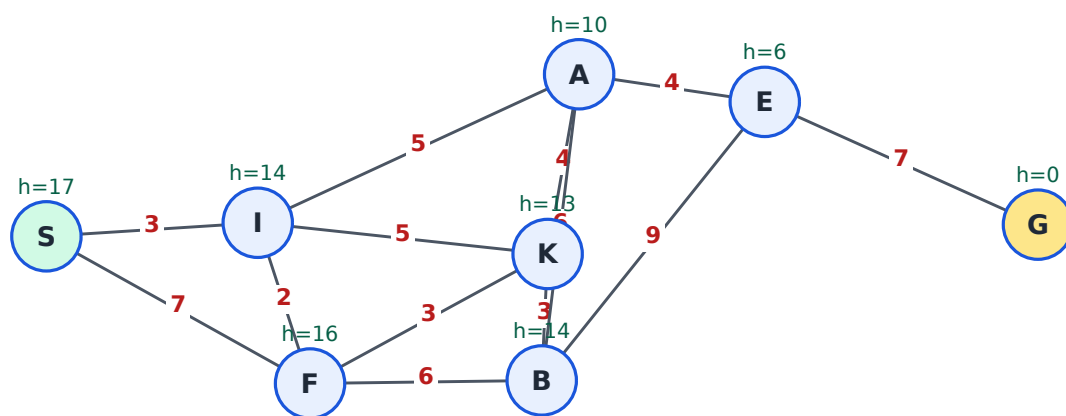
Q.1 [NAT · 1 mark · no negative marking] Informed Search

Weighted A* uses $f(n) = g(n) + W \cdot h(n)$ with $W = 1.5$ and an admissible heuristic h . The optimal solution cost is $C^* = 40$. What is the largest solution cost that weighted A* is guaranteed not to exceed?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

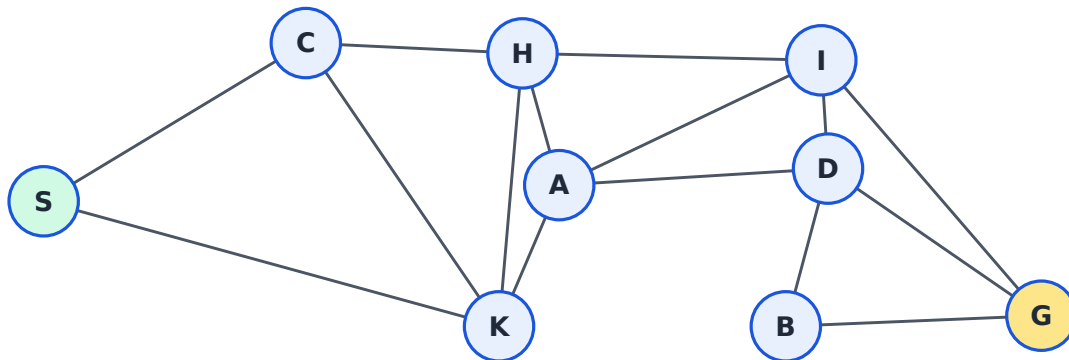
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

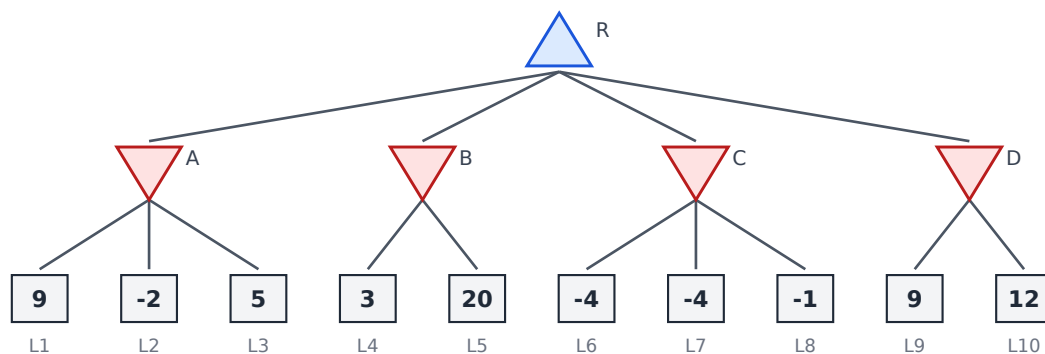
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, C, H, D, A, B, G
☐ (B) S, K, H, I, G
☐ (C) S, C, H, A, D, B, G
☐ (D) S, C, K, H, A, I, D, G

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (R \wedge \neg P) \leftrightarrow (\neg R \leftrightarrow (R \vee Q)) ?$$

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over a, b, c does the CNF $(a \vee b \vee c) \wedge (\neg a \vee \neg b) \wedge (\neg b \vee \neg c) \wedge (\neg a \vee \neg c)$ have?

Answer (numerical): _____

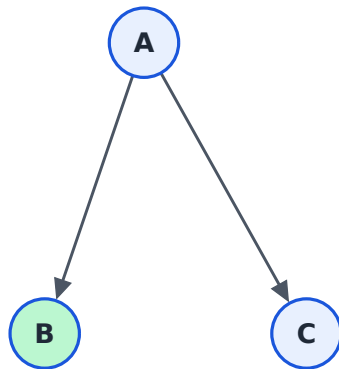
Q.7 [NAT · 1 mark · no negative marking] Predicate Logic

Let the domain be $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \forall y (R(x, y) \rightarrow R(y, x))$?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(B = T)$ (round off to 3 decimal places).



$P(A=T)$	A	$P(B=T \dots)$	A	$P(C=T \dots)$
0.2	T	0.2	T	0.4
	F	0.6	F	0.5

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

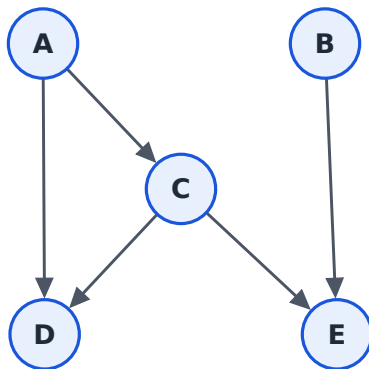
The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(C = T \mid D = T)$, how many of these samples are **rejected**?

#	A	B	C	D
1	F	F	T	F
2	T	F	F	T
3	T	F	F	F
4	T	F	F	F
5	T	F	T	T
6	T	F	F	F
7	T	F	F	T
8	T	F	F	F
9	T	F	F	F
10	F	F	T	F
11	T	F	F	F
12	T	T	F	F
13	T	F	F	F
14	T	F	F	F
15	T	F	F	F
16	T	F	T	F
17	T	F	F	F
18	T	F	F	F
19	T	T	F	T
20	T	F	F	F

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D, E. What is the probability that a single run produces the sample **(A=F, B=T, C=F, D=T, E=T)**? (round off to 4 decimal places)



P(A=T)
0.2

P(B=T)
0.8

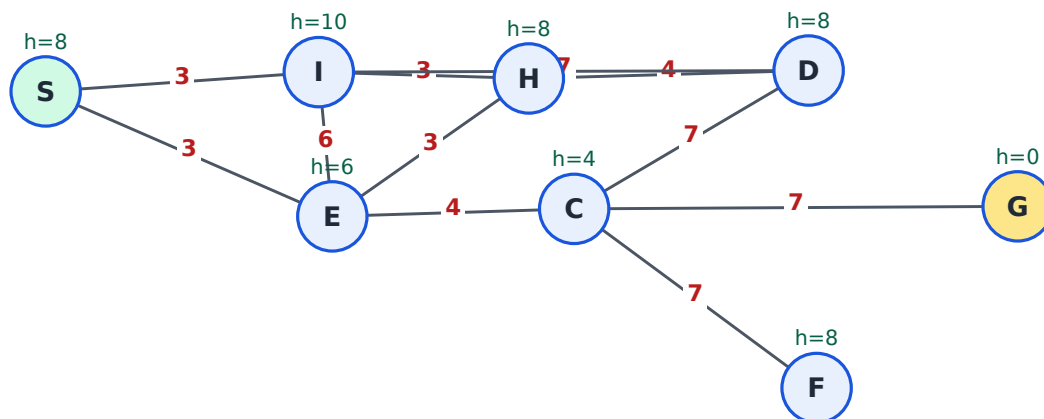
A	P(C=T ...)
T	0.2
F	0.1

A	C	P(D=T B, C)	B	C	P(E=T ...)
T	T	0.95	T	T	0.75
T	F	0.5	T	F	0.4
F	T	0.05	F	T	0.6
F	F	0.8	F	F	0.7

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

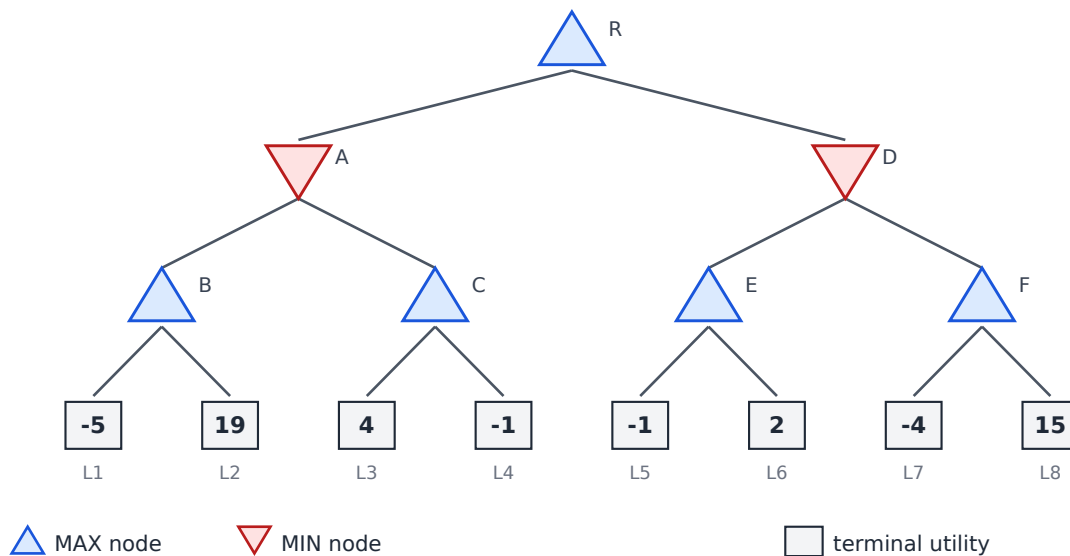
Q.12 [MCQ · 2 marks · negative: -2/3] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): S→A (2), S→B (4), A→C (2), A→D (4), B→C (1), C→D (1), C→G (5), D→G (2). The heuristic is $h(S)=5$, $h(A)=4$, $h(B)=4$, $h(C)=3$, $h(D)=2$, $h(G)=0$. What is the cost of the optimal path from S to G, and is h consistent?

- ☐ (A) Cost 7; h is consistent
☐ (B) Cost 7; h is not consistent
☐ (C) Cost 8; h is consistent
☐ (D) Cost 9; h is not consistent

Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

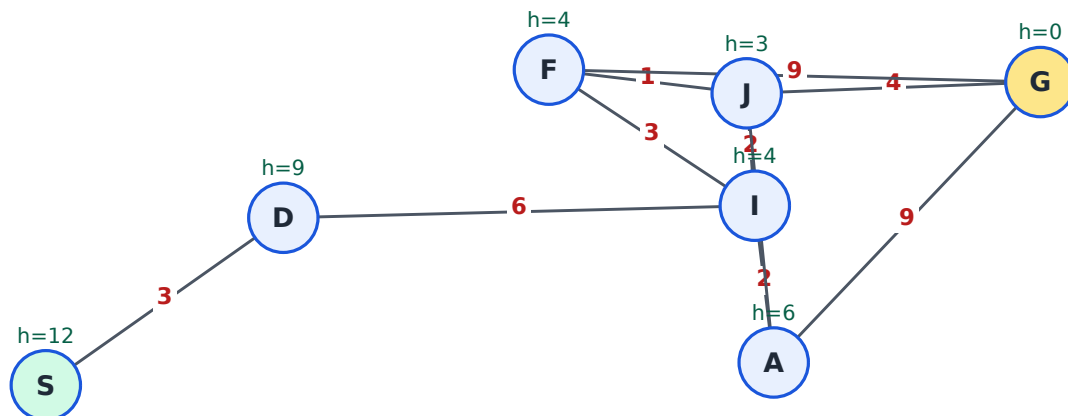
Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



- ☐ (A) Leaf L4 is pruned.
☐ (B) Leaf L6 is pruned.
☐ (C) Leaf L7 is pruned.
☐ (D) Leaf L8 is pruned.

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ satisfy all three of $\forall x R(x, x)$, $\forall x \forall y (R(x, y) \rightarrow R(y, x))$ and $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$?

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic

Let α and β be propositional sentences. Which of the following statements are ALWAYS true?

- ☐ (A) If α is satisfiable then $\neg\alpha$ is unsatisfiable
- ☐ (B) If α is valid then $\neg\alpha$ is unsatisfiable
- ☐ (C) $\alpha \vee \beta$ is satisfiable if and only if α is satisfiable or β is satisfiable
- ☐ (D) $\alpha \wedge \beta$ is satisfiable if and only if α is satisfiable and β is satisfiable

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (Q \wedge Q) \rightarrow (R \rightarrow Q)$?

- ☐ (A) $\neg(Q \wedge Q) \wedge (R \rightarrow Q)$
- ☐ (B) $\neg(Q \wedge Q) \vee (R \rightarrow Q)$
- ☐ (C) $(R \rightarrow Q) \rightarrow (Q \wedge Q)$
- ☐ (D) $\neg((Q \wedge Q) \wedge \neg(R \rightarrow Q))$

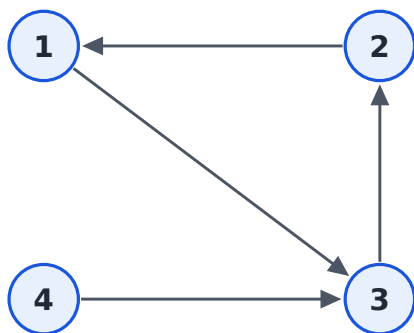
Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following are logically equivalent to $p \rightarrow (q \vee r)$?

- ☐ (A) $(p \wedge \neg q) \rightarrow r$
- ☐ (B) $(p \rightarrow q) \vee (p \rightarrow r)$
- ☐ (C) $\neg r \rightarrow (p \rightarrow q)$
- ☐ (D) $(p \rightarrow q) \wedge (p \rightarrow r)$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
- ☐ (B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
- ☐ (C) $\exists x R(x, x)$
- ☐ (D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg Q \wedge (P \vee \neg P)$
- (2) $\neg R \vee (\neg Q \rightarrow R)$
- (3) $\neg\neg R$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg R \rightarrow P$
- ☐ (B) $\neg\neg Q$
- ☐ (C) $P \vee Q$
- ☐ (D) $\neg Q$

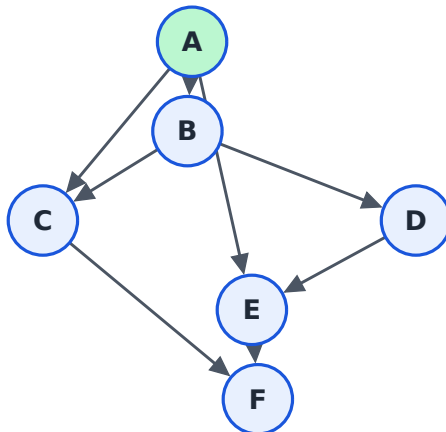
Q.21 [NAT · 2 marks · no negative marking] Variable Elimination

A Bayesian network over Boolean variables has edges $H \rightarrow X_1, H \rightarrow X_2, H \rightarrow X_3, H \rightarrow X_4$. To compute $P(X_1)$ by variable elimination, H is eliminated **first** (then X_2, X_3, X_4). How many entries does the new factor produced by summing out H contain?

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

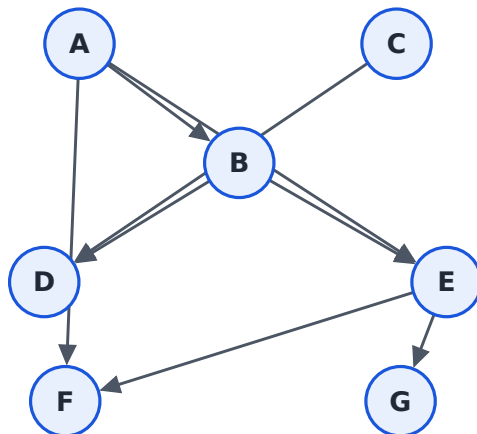
Variable elimination is used to compute **P(A)** in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, F, D, B, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp E \mid A, B$
☐ (B) $C \perp A \mid E$
☐ (C) $F \perp B \mid D, G$
☐ (D) $G \perp D \mid E$

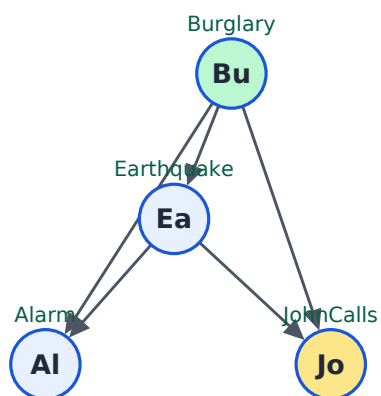
Q.24 [MCQ · 2 marks · negative: -2/3] Conditional Independence

A Bayesian network has edges $A \rightarrow C$ and $B \rightarrow C$ (Boolean variables) with $P(a) = 0.1$, $P(b) = 0.2$, $P(c \mid a, b) = 0.95$, $P(c \mid a, \neg b) = 0.90$, $P(c \mid \neg a, b) = 0.80$, $P(c \mid \neg a, \neg b) = 0.05$. Which ordering of the three quantities $P(a)$, $P(a \mid c)$ and $P(a \mid c, b)$ is correct?

- ☐ (A) $P(a) < P(a \mid c, b) < P(a \mid c)$
☐ (B) $P(a \mid c, b) < P(a) < P(a \mid c)$
☐ (C) $P(a) < P(a \mid c) < P(a \mid c, b)$
☐ (D) $P(a \mid c) < P(a) < P(a \mid c, b)$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute **$P(\text{Burglary} = T \mid \text{JohnCalls} = F)$** (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.7

Bu	$P(\text{Ea}=T \mid \dots)$	Bu	Ea	$P(\text{Al}=T \mid \dots)$
T	0.9	T	T	0.05
F	0.05	T	F	0.6
		F	T	0.7
		F	F	0.2

Bu	Ea	$P(\text{Jo}=T \mid \dots)$
T	T	0.75
T	F	0.7
F	T	0.1
F	F	0.8

Answer (numerical): _____

Answer Key — Mock Test 063

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Informed Search	60	14	NAT	2	Informed Search	5
2	NAT	1	Informed Search	19	15	NAT	2	Predicate Logic	5
3	MCQ	1	Uninformed Search	C	16	MSQ	2	Propositional Logic	B, C
4	NAT	1	Adversarial Search	9	17	MSQ	2	Propositional Logic	B, D
5	NAT	1	Propositional Logic	4	18	MSQ	2	Propositional Logic	A, B, C
6	NAT	1	Propositional Logic	3	19	MSQ	2	Predicate Logic	D
7	NAT	1	Predicate Logic	64	20	MSQ	2	Propositional Logic	A, D
8	NAT	1	Exact Inference	0.518 to 0.522	21	NAT	2	Variable Elimination	16
9	NAT	1	Sampling	16	22	NAT	2	Exact Inference	4
10	NAT	1	Sampling	0.1841 to 0.1845	23	MSQ	2	Conditional Independence	B, D
11	NAT	2	Informed Search	5	24	MCQ	2	Conditional Independence	A
12	MCQ	2	Informed Search	A	25	NAT	2	Exact Inference	0.715 to 0.719
13	MSQ	2	Adversarial Search	C, D					

Detailed Solutions

Q.1 [NAT, 1M · Informed Search] Answer: 60

With an admissible h , weighted A^* is W -admissible: the cost it returns is $\leq W \cdot C^* = 1.5 \times 40 = \mathbf{60}$. Inflating h makes the search greedier and faster, and the price is this bounded loss of optimality.

Q.2 [NAT, 1M · Informed Search] Answer: 19

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (17)	I(14), F(16)
2	I (14)	A(10), K(13), F(16)
3	A (10)	E(6), K(13), B(14), F(16)
4	E (6)	G(0), K(13), B(14), F(16)
5	G (0)	K(13), B(14), F(16) ← goal, stop

Path returned: **S** → **I** → **A** → **E** → **G**, cost = 3 + 5 + 4 + 7 = **19**.

The optimal cost is $h^*(S) = 19$ — greedy happens to be optimal here, but in general it is not.

Q.3 [MCQ, 1M · Uninformed Search] Answer: C (C: S, C, H, A, D, B, G)

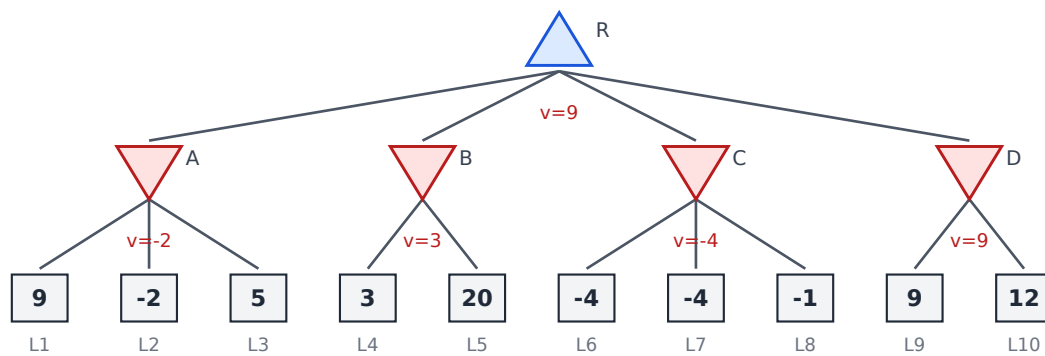
Adjacency lists (alphabetical): A: {D, H, I, K}; B: {D, G}; C: {H, K, S}; D: {A, B, G, I}; G: {B, D, I}; H: {A, C, I, K}; I: {A, D, G, H}; K: {A, C, H, S}; S: {C, K}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, C, H, A, D, B, G**; the path found is $S \rightarrow C \rightarrow H \rightarrow A \rightarrow D \rightarrow B \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Adversarial Search] Answer: 9

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = -2, B = 3, C = -4, D = 9. Root value = 9; MAX's optimal first move is towards D.

Q.5 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$R \wedge \neg P$	$\neg R \leftrightarrow (R \vee Q)$	$(R \wedge \neg P) \leftrightarrow (\neg R \leftrightarrow (R \vee Q))$
T	T	T	F	F	T
T	T	F	F	T	F
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	T	F	F
F	T	F	F	T	F
F	F	T	T	F	F
F	F	F	F	F	T

Rows where F = T: **4**.

Q.6 [NAT, 1M · Propositional Logic] Answer: 3

The first clause says at least one of a, b, c is true. The three binary clauses say no two are simultaneously true. Together they encode 'exactly one of a, b, c is true', which has 3 models: (T,F,F), (F,T,F), (F,F,T).

Q.7 [NAT, 1M · Predicate Logic] Answer: 64

An interpretation of R is a subset of $D \times D$ (9 pairs). The sentence says R is symmetric. The 3 diagonal pairs (x, x) can be chosen freely; the 3 unordered pairs {x, y} with $x \neq y$ must be included or excluded as a whole. That is $2^3 \times 2^3 = 64$ relations.

Q.8 [NAT, 1M · Exact Inference] Answer: 0.518 to 0.522

Only the ancestors of B (A) matter: every other variable sums out to 1.

$P(B=T) = \sum \text{over its ancestors of } P(B=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, B=T)$
A=T, C=T	0.016
A=T, C=F	0.024
A=F, C=T	0.24
A=F, C=F	0.24

Sum = **0.52** ≈ 0.520 .

Q.9 [NAT, 1M · Sampling] Answer: 16

Rejection sampling keeps only samples consistent with the evidence $D = T$: samples 2, 5, 7, 19 (4 accepted, 16 rejected).

Among the accepted samples, $C = T$ in 1. Estimate = $1/4 = \mathbf{0.25}$.

(For this network the exact value is 0.2869; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.10 [NAT, 1M · Sampling] Answer: 0.1841 to 0.1845

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(A=F) \times P(B=T) \times P(C=F \mid A=F) \times P(D=T \mid A=F, C=F) \times P(E=T \mid B=T, C=F)$

$= 0.8 \times 0.8 \times 0.9 \times 0.8 \times 0.4 = \mathbf{0.18432} \approx 0.1843$.

Q.11 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(C)=4$, $h(D)=8$, $h(E)=6$, $h(F)=8$, $h(G)=0$, $h(H)=8$, $h(I)=10$, $h(S)=8$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (8)	E(9), I(13)
2	E (9)	C(11), I(13), H(14)
3	C (11)	I(13), G(14), H(14), D(22), F(22)
4	I (13)	G(14), H(14), D(18), F(22)
5	G (14)	H(14), D(18), F(22) ← goal, stop

Path returned: **S** → **E** → **C** → **G** with cost **14**; nodes expanded = **5** (S, E, C, I, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 8 nodes.

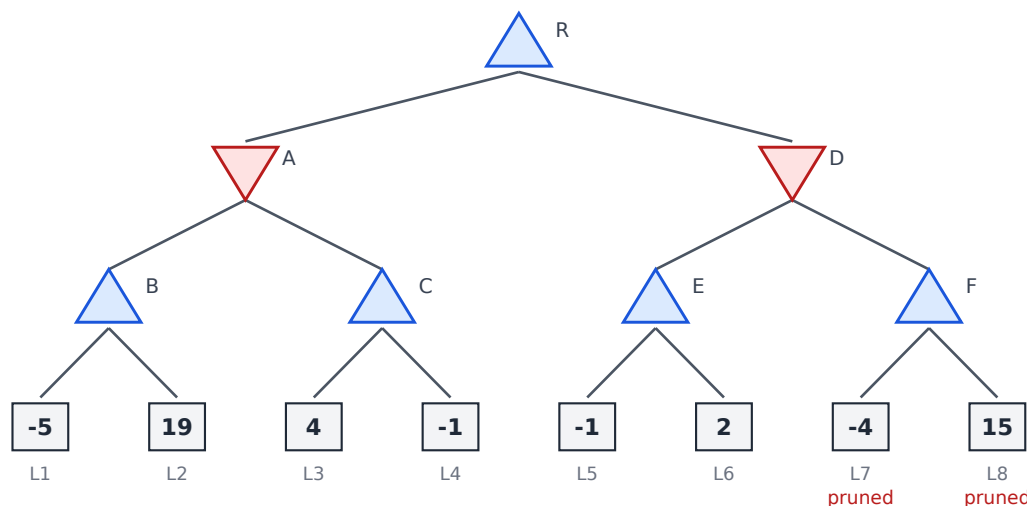
Q.12 [MCQ, 2M · Informed Search] Answer: A (A: Cost 7; h is consistent)

Path costs: $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 2+2+1+2 = 7$; $S \rightarrow A \rightarrow C \rightarrow G = 9$; $S \rightarrow A \rightarrow D \rightarrow G = 8$; $S \rightarrow B \rightarrow C \rightarrow D \rightarrow G = 8$; $S \rightarrow B \rightarrow C \rightarrow G = 10$. The optimum is **7**. Consistency, edge by edge: $S \rightarrow A$: $5 \leq 2+4$. $S \rightarrow B$: $5 \leq 4+4$. $A \rightarrow C$: $4 \leq 2+3$. $A \rightarrow D$: $4 \leq 4+2$. $B \rightarrow C$: $4 \leq 1+3$. $C \rightarrow D$: $3 \leq 1+2$. $C \rightarrow G$: $3 \leq 5+0$. $D \rightarrow G$: $2 \leq 2+0$. All hold, so **h is consistent**.

Q.13 [MSQ, 2M · Adversarial Search] Answer: C, D (C: Leaf L7 is pruned.; D: Leaf L8 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $2 \leq \alpha = 4 \Rightarrow$ prune remaining children (leaves L7, L8).



Leaves evaluated (6): L1, L2, L3, L4, L5, L6. Leaves pruned (2): L7, L8. Root value = 4 (same as plain minimax, 4 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(A)=6$, $h(D)=9$, $h(F)=4$, $h(G)=0$, $h(I)=4$, $h(J)=3$, $h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	D(12)
2	D (12)	I(13)
3	I (13)	J(14), F(16), A(17)
4	J (14)	G(15), F(16), A(17)
5	G (15)	F(16), A(17) ← goal, stop

Path returned: **S** → **D** → **I** → **J** → **G** with cost **15**; nodes expanded = **5** (S, D, I, J, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [NAT, 2M · Predicate Logic] Answer: 5

These are exactly the equivalence relations on a 3-element set, which correspond one-to-one with its partitions. The number of partitions of a 3-set is the Bell number $B_3 = 5$: $\{123\}$, $\{1|23\}$, $\{2|13\}$, $\{3|12\}$, $\{1|2|3\}$. Enumeration of all 512 relations confirms 5.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, C (B: If α is valid then $\neg\alpha$ is unsatisfiable; C: $\alpha \vee \beta$ is satisfiable if and only if α is satisfiable or β is satisfiable)

(A) False: p is satisfiable and so is $\neg p$. (B) True: valid means true in all models, so $\neg\alpha$ is true in none. (C) True: a model of $\alpha \vee \beta$ is a model of α or of β , and conversely any model of α (or of β) is a model of $\alpha \vee \beta$. (D) False: $\alpha = p$ and $\beta = \neg p$ are each satisfiable but $p \wedge \neg p$ is not (the 'if' direction fails).

Q.17 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $\neg(Q \wedge Q) \vee (R \rightarrow Q)$; D: $\neg((Q \wedge Q) \wedge \neg(R \rightarrow Q))$)

Compare truth tables with $F = (Q \wedge Q) \rightarrow (R \rightarrow Q)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(Q \wedge Q) \wedge (R \rightarrow Q)$ — differs at $Q=T, R=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(B) $\neg(Q \wedge Q) \vee (R \rightarrow Q)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $(R \rightarrow Q) \rightarrow (Q \wedge Q)$ — differs at $Q=F, R=F$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(D) $\neg((Q \wedge Q) \wedge \neg(R \rightarrow Q))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, B, C (A: $(p \wedge \neg q) \rightarrow r$; B: $(p \rightarrow q) \vee (p \rightarrow r)$; C: $\neg r \rightarrow (p \rightarrow q)$)

$p \rightarrow (q \vee r) \equiv \neg p \vee q \vee r$. (A) $(p \wedge \neg q) \rightarrow r \equiv \neg p \vee q \vee r$. (B) $(\neg p \vee q) \vee (\neg p \vee r) \equiv \neg p \vee q \vee r$. (C) $\neg r \rightarrow (\neg p \vee q) \equiv r \vee \neg p \vee q$. (D) $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$, which is false at $p=T, q=T, r=F$ whereas the original is true there.

Q.19 [MSQ, 2M · Predicate Logic] Answer: D (D: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$)

$R = \{(1,3), (2,1), (3,2), (4,3)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(3,2) \wedge R(2,1)$ but $\neg R(3,1)$.

(B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,4)$ nor $R(4,1)$.

(C) $\exists x R(x, x)$: **FALSE** — there are no self-loops.

(D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $\neg R \rightarrow P$; D: $\neg Q$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	F	T	T
F	F	T	T

(A) $\neg R \rightarrow P$ is true in all models of $KB \Rightarrow$ **entailed**.

(B) $\neg\neg Q$ is false in the KB -model $P=T, Q=F, R=T \Rightarrow$ not entailed.

(C) $P \vee Q$ is false in the KB -model $P=F, Q=F, R=T \Rightarrow$ not entailed.

(D) $\neg Q$ is true in all models of $KB \Rightarrow$ **entailed**.

Q.21 [NAT, 2M · Variable Elimination] Answer: 16

Eliminating H multiplies all factors that mention H : $P(H)$, $P(X_1 | H)$, ..., $P(X_4 | H)$. Summing out H leaves a factor over $\{X_1, X_2, X_3, X_4\}$, with $2^4 = 16$ entries. By contrast, eliminating X_2, X_3, X_4 first gives trivial factors over H (each $\sum_x P(x | H) = 1$) and the largest new factor has just 1 variable.

Q.22 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(B, D)$, $f(A, D, E)$, $f(C, E, F)$.

Eliminating Z : multiply all factors whose scope contains Z , then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(A, B, C)$, $f(C, E, F)$	$\{A, B, C, E, F\}$	$f(A, B, E, F)$
F	$f(A, B, E, F)$	$\{A, B, E, F\}$	$f(A, B, E)$
D	$f(B, D)$, $f(A, D, E)$	$\{A, B, D, E\}$	$f(A, B, E)$
B	$f(A, B)$, $f(A, B, E)$, $f(A, B, E)$	$\{A, B, E\}$	$f(A, E)$
E	$f(A, E)$	$\{A, E\}$	$f(A)$

The largest factor produced has **4** variable(s) (created when eliminating C); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over A and normalise.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: B, D** (B: $C \perp A \mid E$; D: $G \perp D \mid E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp E \mid A, B$: **FALSE** — the path $G \leftarrow E$ is active given $\{A, B\}$.

(B) $C \perp A \mid E$: **TRUE** — $C \rightarrow D \leftarrow B \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $C \rightarrow D \leftarrow B \rightarrow E \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $C \rightarrow D \leftarrow B \rightarrow E \rightarrow F \leftarrow A$ (blocked: collider D with neither it nor a descendant observed).

(C) $F \perp B \mid D, G$: **FALSE** — the path $F \leftarrow A \rightarrow B$ is active given $\{D, G\}$.

(D) $G \perp D \mid E$: **TRUE** — $G \leftarrow E \leftarrow A \rightarrow B \rightarrow D$ (blocked: observed chain node E); $G \leftarrow E \leftarrow B \rightarrow D$ (blocked: observed chain node E); $G \leftarrow E \rightarrow F \leftarrow A \rightarrow B \rightarrow D$ (blocked: observed fork node E).

Q.24 [MCQ, 2M · Conditional Independence] **Answer: A** (A: $P(a) < P(a \mid c, b) < P(a \mid c)$)

$P(c) = 0.1 \times 0.2 \times 0.95 + 0.1 \times 0.8 \times 0.90 + 0.9 \times 0.2 \times 0.80 + 0.9 \times 0.8 \times 0.05 = 0.019 + 0.072 + 0.144 + 0.036 = 0.271$, and $P(a, c) = 0.019 + 0.072 = 0.091$, so $P(a \mid c) = 0.091/0.271 \approx 0.336$. $P(a \mid c, b) = 0.095/(0.095 + 0.72) \approx 0.117$. $P(a) = 0.1$. Hence $0.1 < 0.117 < 0.336$: observing c raises P(a); additionally observing b explains c away and lowers it, but not below the prior here.

Q.25 [NAT, 2M · Exact Inference] **Answer: 0.715 to 0.719**

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm} \mid \text{Burglary}, \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Earthquake})$.

Inference by enumeration: $P(\text{Burglary} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Burglary}, e, \text{hidden})$.

Hidden: Ea, Al	Bu	product of CPT entries (Bu · Ea · Al · Jo)	value
T, T	T	$0.7 \times 0.9 \times 0.05 \times 0.25$	0.00788
T, F	T	$0.7 \times 0.9 \times 0.95 \times 0.25$	0.14962
F, T	T	$0.7 \times 0.1 \times 0.6 \times 0.3$	0.0126
F, F	T	$0.7 \times 0.1 \times 0.4 \times 0.3$	0.0084
T, T	F	$0.3 \times 0.05 \times 0.7 \times 0.9$	0.00945
T, F	F	$0.3 \times 0.05 \times 0.3 \times 0.9$	0.00405
F, T	F	$0.3 \times 0.95 \times 0.2 \times 0.2$	0.0114
F, F	F	$0.3 \times 0.95 \times 0.8 \times 0.2$	0.0456

$P(\text{Burglary}=T, e) = 0.1785$; $P(\text{Burglary}=F, e) = 0.0705$; $P(e) = 0.249$.

$P(\text{Burglary}=T \mid e) = 0.1785 / 0.249 = \mathbf{0.7169} \approx 0.717$.

Mock Test 064

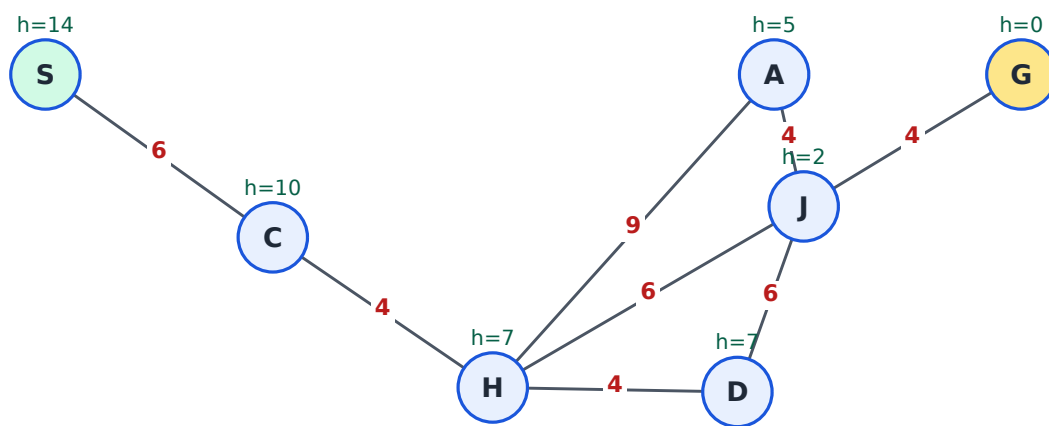
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 7 / 11	13 / 12 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

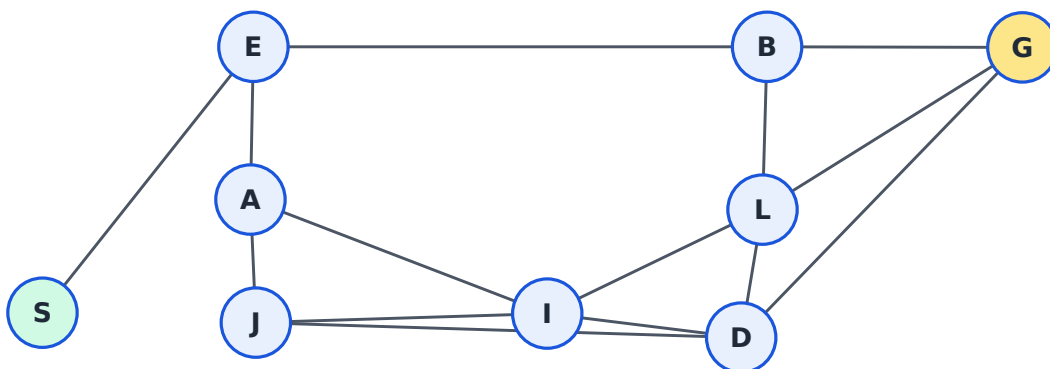
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

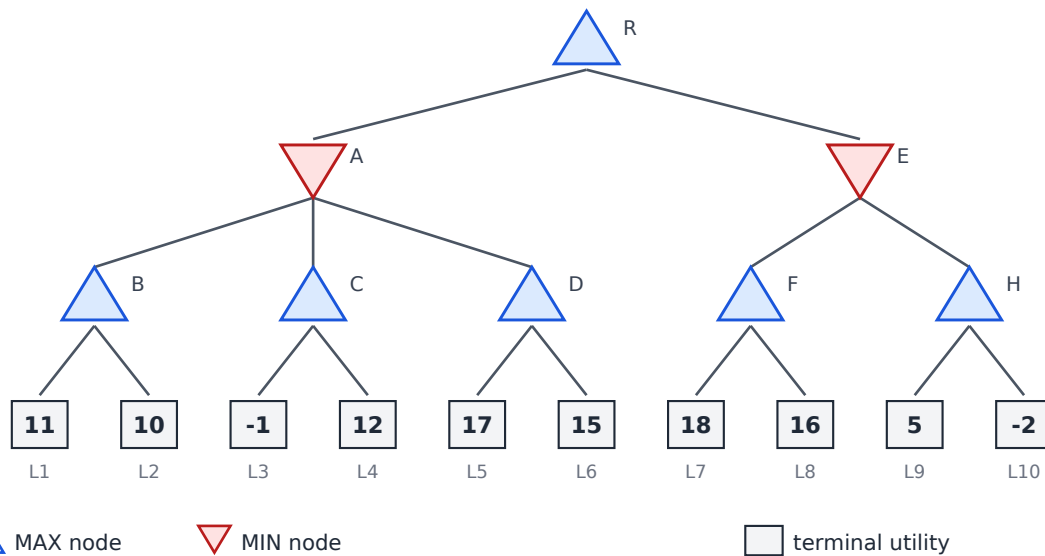
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, E, A, I, D, G
- ☐ (B) S, E, A, B, I, J, G
- ☐ (C) S, E, B, L, I, J, D, G
- ☐ (D) S, A, E, I, D, G

Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to A; root value 18
☐ (B) Move to E; root value 5
☐ (C) Move to A; root value 11
☐ (D) Move to A; root value 12

Q.4 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x R(x, x) \wedge \forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((Q \wedge R) \wedge (Q \leftrightarrow P)) \leftrightarrow \neg R ?$$

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

The KB contains the clauses $\neg \text{Man}(x) \vee \text{Mortal}(x)$ and $\text{Man}(\text{Socrates})$. To prove $\text{Mortal}(\text{Socrates})$ by resolution refutation, $\neg \text{Mortal}(\text{Socrates})$ is added. What is the minimum number of resolution steps needed to derive the empty clause?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Backward chaining over a propositional definite-clause KB is used to prove the query Q. Which clause is used FIRST?

- ☐ (A) A clause whose head (conclusion) is Q
☐ (B) A clause in which Q appears in the body
☐ (C) Any fact in the KB, chosen arbitrarily
☐ (D) The negated query $\neg Q$ added to the KB

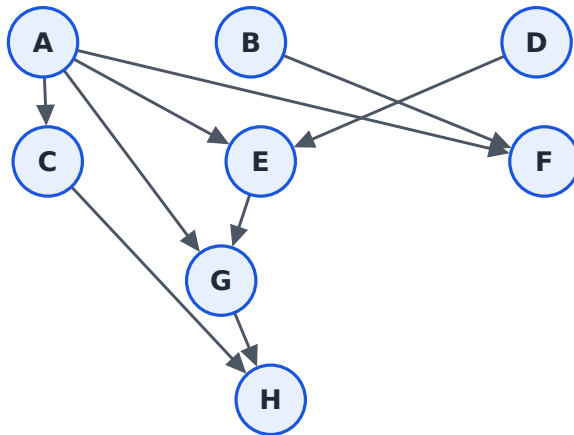
Q.8 [MCQ · 1 mark · negative: -1/3] Variable Elimination

For a polytree (singly connected) Bayesian network, the time and space complexity of exact inference by variable elimination is

- ☐ (A) linear in the size of the network (total CPT entries)
- ☐ (B) exponential in the number of variables in the worst case
- ☐ (C) quadratic in the number of variables, independent of CPT sizes
- ☐ (D) NP-hard

Q.9 [MCQ · 1 mark · negative: -1/3] Conditional Independence – Markov blanket

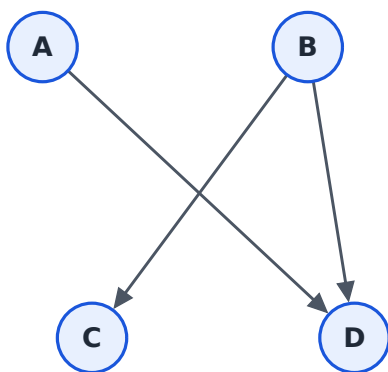
Which set is the Markov blanket of **E** in the Bayesian network below?



- ☐ (A) {G}
- ☐ (B) {A, D, G}
- ☐ (C) {A, D, F, G}
- ☐ (D) {A, D, G, H}

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=T, B=F, C=T, D=F)**? (round off to 4 decimal places)



P(A=T)
0.9

P(B=T)	B	P(C=T ...)
0.5	T	0.95
	F	0.9

A	B	P(D=T ...)
T	T	0.75
T	F	0.2
F	T	0.2
F	F	0.1

Answer (numerical): _____

Section B — 2-mark questions

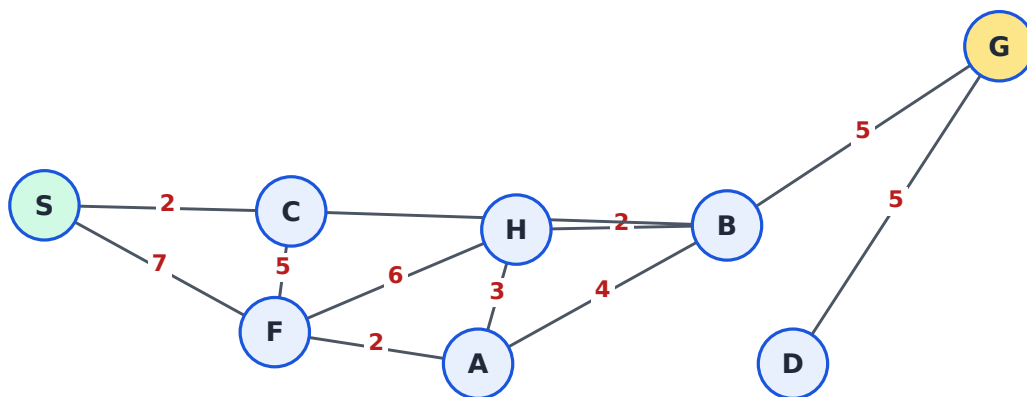
Q.11 [MSQ · 2 marks · no negative marking] Adversarial Search

Which of the following statements about alpha-beta pruning are TRUE?

- ☐ (A) It returns the same root value as minimax
- ☐ (B) A pruned subtree cannot change the root's minimax decision
- ☐ (C) Move ordering affects how much is pruned, but not the root value
- ☐ (D) It can be applied unchanged to expectimax trees containing chance nodes, with the same pruning conditions

Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow C \rightarrow B \rightarrow G$
- ☐ (B) $S \rightarrow C \rightarrow F \rightarrow A \rightarrow H \rightarrow B \rightarrow G$
- ☐ (C) $S \rightarrow C \rightarrow F \rightarrow A \rightarrow B \rightarrow G$
- ☐ (D) $S \rightarrow F \rightarrow A \rightarrow B \rightarrow G$

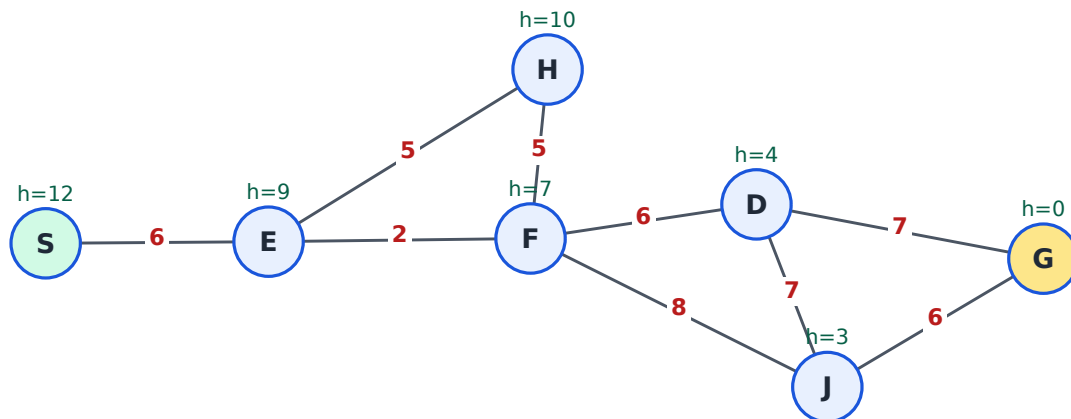
Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search

Which of the following statements about depth-limited (cutoff) game search with heuristic evaluation functions are TRUE?

- ☐ (A) The horizon effect occurs when an unavoidable damaging event is pushed beyond the search depth by delaying moves
- ☐ (B) Quiescence search extends the search at non-quiet positions, such as pending captures, before applying the evaluation function
- ☐ (C) A good evaluation function should be strongly correlated with the actual chance of winning
- ☐ (D) Increasing the cutoff depth by a fixed amount always removes the horizon effect

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – A*

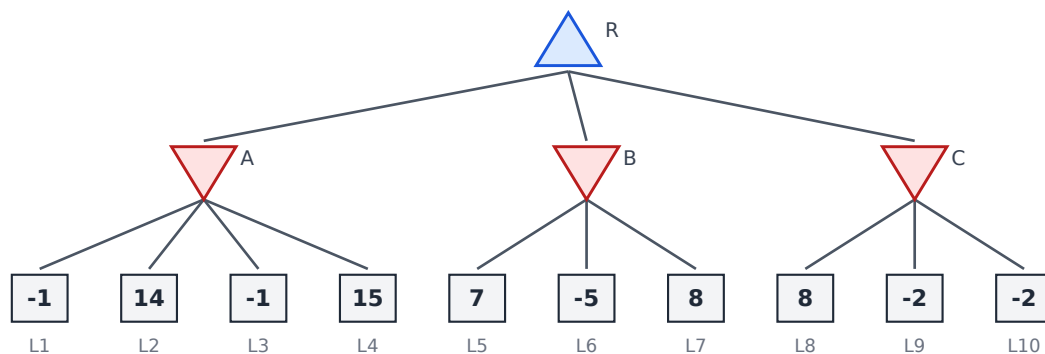
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node E is expanded.
☐ (B) The path returned has cost 22.
☐ (C) Node H is expanded.
☐ (D) Node J is expanded.

Q.15 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



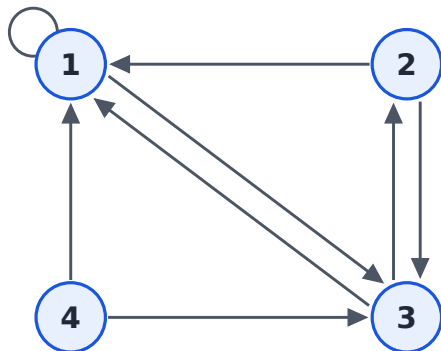
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
☐ (B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (C) $\exists x R(x, x)$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(\neg P \wedge (\neg P \rightarrow R)) \rightarrow R$
☐ (B) $R \rightarrow (Q \rightarrow R)$
☐ (C) $((Q \rightarrow R) \wedge (P \rightarrow R)) \rightarrow (Q \wedge P)$
☐ (D) $(Q \vee P) \rightarrow (Q \wedge P)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg\neg S \wedge (\neg Q \leftrightarrow R)$
 (2) $\neg S \leftrightarrow (\neg Q \vee R)$
 (3) $(S \vee S) \wedge (\neg Q \rightarrow S)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg P \leftrightarrow \neg R$
☐ (B) $\neg S$
☐ (C) $\neg P \rightarrow Q$
☐ (D) $P \wedge Q$

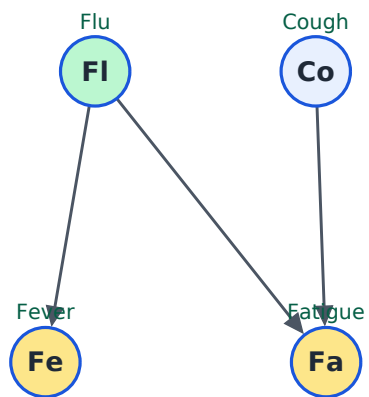
Q.19 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$ and let f be a unary function symbol. How many interpretations of f (functions $D \rightarrow D$) satisfy $\forall x f(f(x)) = f(x)$?

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Flu} = T \mid \text{Fatigue} = T, \text{Fever} = F)$ (round off to 3 decimal places).



FI	Co	P(Fa=T ...)
T	T	0.9
T	F	0.25
F	T	0.25
F	F	0.5

FI	Co	P(Fa=T ...)
T	T	0.9
T	F	0.25
F	T	0.25
F	F	0.5

Answer (numerical): _____

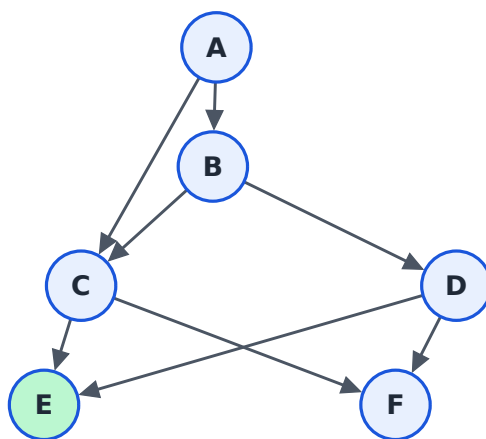
Q.21 [NAT · 2 marks · no negative marking] Bayesian Networks

A naive Bayes spam filter has $P(\text{spam}) = 0.3$ and three Boolean features with $P(f_1 \mid \text{spam}) = 0.8$, $P(f_1 \mid \text{ham}) = 0.1$; $P(f_2 \mid \text{spam}) = 0.6$, $P(f_2 \mid \text{ham}) = 0.3$; $P(f_3 \mid \text{spam}) = 0.2$, $P(f_3 \mid \text{ham}) = 0.5$. An email has $f_1 = \text{true}$, $f_2 = \text{false}$, $f_3 = \text{true}$. Compute $P(\text{spam} \mid f_1, \neg f_2, f_3)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

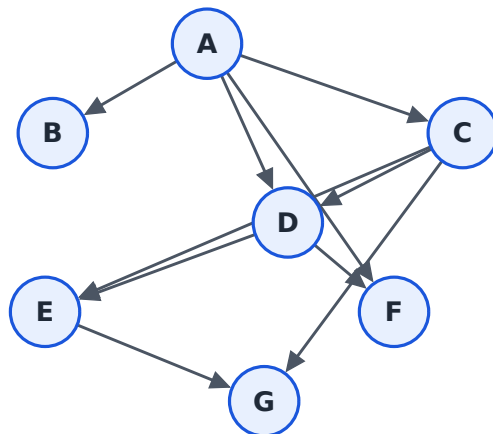
Variable elimination is used to compute $P(E)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F, D, A, B, C**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

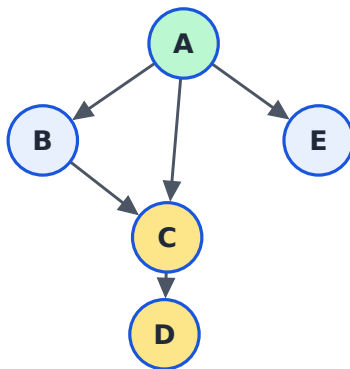
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp F \mid A$
☐ (B) $E \perp A \mid C, D$
☐ (C) $A \perp C \mid D$
☐ (D) $F \perp E \mid D$

Q.24 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $D = F$, $C = T$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $P(A = T \mid D = F, C = T)$ (round off to 3 decimal places).



P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.25	T	0.8	T	T	0.3
	F	0.7	T	F	0.25
			F	T	0.8
			F	F	0.4

C	P(D=T ...)	A	P(E=T ...)
T	0.3	T	0.3
F	0.4	F	0.3

#	A	B	E
1	T	F	T
2	T	F	F
3	T	T	F
4	F	T	T

Answer (numerical): _____

Q.25 [MCQ · 2 marks · negative: -2/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Gibbs sampling with evidence $S = \text{true}$, $W = \text{true}$ is in the state where $R = \text{true}$. What is the probability that C is resampled to true, $P(c | s, r)$?

- ☐ (A) 0.444
☐ (B) 0.040
☐ (C) 0.500
☐ (D) 0.878

Answer Key — Mock Test 064

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Informed Search	20	14	MSQ	2	Informed Search	A, D
2	MCQ	1	Uninformed Search	A	15	NAT	2	Adversarial Search	8
3	MCQ	1	Adversarial Search	C	16	MSQ	2	Predicate Logic	A, C, D
4	NAT	1	Predicate Logic	8	17	MSQ	2	Propositional Logic	A, B
5	NAT	1	Propositional Logic	3	18	MSQ	2	Propositional Logic	C
6	NAT	1	Predicate Logic	2	19	NAT	2	Predicate Logic	10
7	MCQ	1	Propositional Logic	A	20	NAT	2	Exact Inference	0.871 to 0.875
8	MCQ	1	Variable Elimination	A	21	NAT	2	Bayesian Networks	0.439 to 0.44
9	MCQ	1	Conditional Independence	B	22	NAT	2	Exact Inference	3
10	NAT	1	Sampling	0.3238 to 0.3242	23	MSQ	2	Conditional Independence	A, B
11	MSQ	2	Adversarial Search	A, B, C	24	NAT	2	Sampling	0.498 to 0.502
12	MCQ	2	Uninformed Search	A	25	MCQ	2	Sampling	A
13	MSQ	2	Adversarial Search	A, B, C					

Detailed Solutions

Q.1 [NAT, 1M · Informed Search] **Answer: 20**

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (14)	C(10)
2	C (10)	H(7)
3	H (7)	J(2), A(5), D(7)
4	J (2)	G(0), A(5), D(7)
5	G (0)	A(5), D(7) ← goal, stop

Path returned: **S** → **C** → **H** → **J** → **G**, cost = 6 + 4 + 6 + 4 = **20**.

The optimal cost is $h^*(S) = 20$ — greedy happens to be optimal here, but in general it is not.

Q.2 [MCQ, 1M · Uninformed Search] **Answer: A** (A: S, E, A, I, D, G)

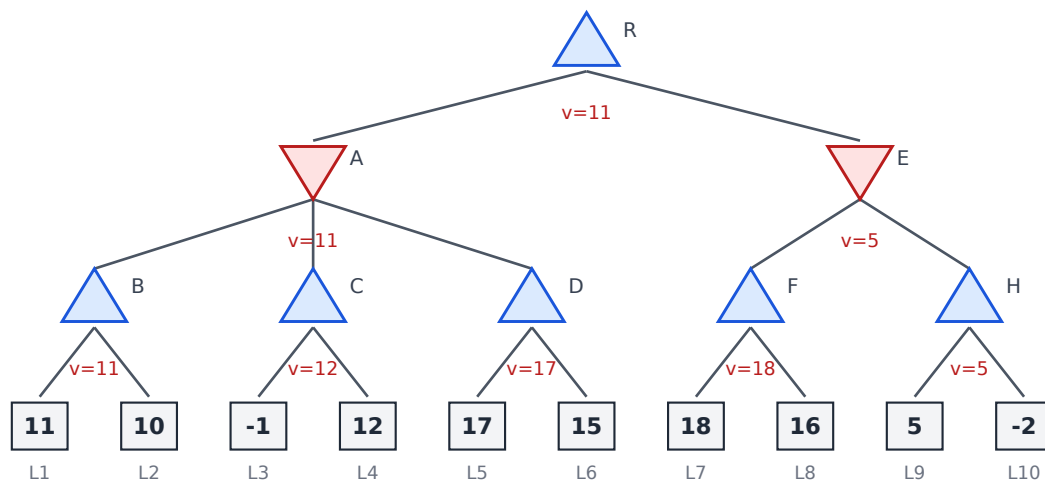
Adjacency lists (alphabetical): A: {E, I, J}; B: {E, G, L}; D: {G, I, J, L}; E: {A, B, S}; G: {B, D, L}; I: {A, D, J, L}; J: {A, D, I}; L: {B, D, G, I}; S: {E}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, E, A, I, D, G**; the path found is $S \rightarrow E \rightarrow A \rightarrow I \rightarrow D \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.3 [MCQ, 1M · Adversarial Search] **Answer: C** (C: Move to A; root value 11)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 11$, $C = 12$, $D = 17$, $A = 11$, $F = 18$, $H = 5$, $E = 5$. Root value = **11**; MAX's optimal first move is towards A.

Q.4 [NAT, 1M · Predicate Logic] **Answer: 8**

View R as an 3×3 0/1 matrix. The sentence says R is reflexive and symmetric: only the $n(n-1)/2$ unordered off-diagonal pairs are free $\Rightarrow 2^{n(n-1)/2}$. With $n = 3$: **8**.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 3**

Build the truth table (sub-formulas shown first):

P	Q	R	$(Q \wedge R) \wedge (Q \leftrightarrow P)$	$\neg R$	$((Q \wedge R) \wedge (Q \leftrightarrow P)) \leftrightarrow \neg R$
T	T	T	T	F	F
T	T	F	F	T	F
T	F	T	F	F	T
T	F	F	F	T	F
F	T	T	F	F	T
F	T	F	F	T	F
F	F	T	F	F	T
F	F	F	F	T	F

Rows where F = T: **3**.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 2**

Step 1: resolve $\neg \text{Mortal}(\text{Socrates})$ with $\neg \text{Man}(x) \vee \text{Mortal}(x)$ using $\{x/\text{Socrates}\}$, giving $\neg \text{Man}(\text{Socrates})$. Step 2: resolve $\neg \text{Man}(\text{Socrates})$ with $\text{Man}(\text{Socrates})$, giving $\square$. One step cannot suffice since no two initial clauses are complementary units. So the answer is 2.

Q.7 [MCQ, 1M · Propositional Logic] **Answer: A** (A: A clause whose head (conclusion) is Q)

Backward chaining works from the goal: it looks for rules (or facts) whose conclusion is Q and recursively tries to prove their premises. Using clauses with Q in the body is forward reasoning. Starting from facts is forward chaining. Adding $\neg Q$ is the resolution-refutation approach, not backward chaining.

Q.8 [MCQ, 1M · Variable Elimination] **Answer: A** (A: linear in the size of the network (total CPT entries))

AIMA: in polytrees there is at most one undirected path between any two nodes, and VE (with a suitable ordering) runs in time and space linear in the size of the network, measured by CPT entries. Exponential/NP-hard behaviour arises for multiply connected networks. Quadratic in variables ignores that CPT sizes grow with the number of parents.

Q.9 [MCQ, 1M · Conditional Independence] **Answer: B** (B: $\{A, D, G\}$)

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of E: $\{A, D\}$; children: $\{G\}$; co-parents: $\{\emptyset\}$.

$\text{MB}(E) = \{A, D, G\}$ — **3** nodes. Given its Markov blanket, E is conditionally independent of every other node in the network.

Q.10 [NAT, 1M · Sampling] Answer: 0.3238 to 0.3242

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(A=T) \times P(B=F) \times P(C=T|B=F) \times P(D=F|A=T, B=F) \\ = 0.9 \times 0.5 \times 0.9 \times 0.8 = \mathbf{0.324} \approx 0.3240.$$

Q.11 [MSQ, 2M · Adversarial Search] Answer: A, B, C

(A), (B) and (C) are the defining properties of alpha-beta: it prunes only subtrees that provably cannot affect the decision, so the result matches minimax for any ordering, and only the amount of pruning changes. (D) is false: the value of a chance node is an average, so one child does not bound it. Pruning there needs extra bounds on the leaf values and modified conditions.

Q.12 [MCQ, 2M · Uninformed Search] Answer: A (A: $S \rightarrow C \rightarrow B \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	C(2), F(7)
2	C (2)	F(7), B(10)
3	F (7)	A(9), B(10), H(13)
4	A (9)	B(10), H(12)
5	B (10)	H(12), G(15)
6	H (12)	G(15)
7	G (15)	← goal, stop

Path returned: **S** → **C** → **B** → **G**, cost = 2 + 8 + 5 = **15**. Nodes expanded (including S and G): **7** (S, C, F, A, B, H, G).

Q.13 [MSQ, 2M · Adversarial Search] Answer: A, B, C

(A) defines the horizon effect. (B) defines quiescence search, which applies the evaluation function only at quiet positions. (C) is the AIMA requirement for evaluation functions. (D) is false: at any fixed depth, a delaying tactic can push the bad event just beyond the new horizon.

Q.14 [MSQ, 2M · Informed Search] Answer: A, D (A: Node E is expanded.; D: Node J is expanded.)

Heuristic values: $h(D)=4$, $h(E)=9$, $h(F)=7$, $h(G)=0$, $h(H)=10$, $h(J)=3$, $h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	E(15)
2	E (15)	F(15), H(21)
3	F (15)	D(18), J(19), H(21)
4	D (18)	J(19), G(21), H(21)
5	J (19)	G(21), H(21)
6	G (21)	H(21) ← goal, stop

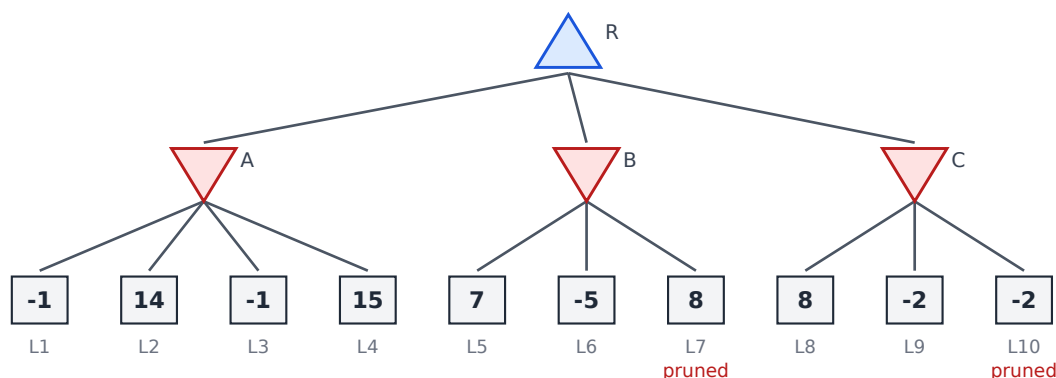
Path returned: **S** → **E** → **F** → **D** → **G** with cost **21**; nodes expanded = **6** (S, E, F, D, J, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 21$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [NAT, 2M · Adversarial Search] Answer: 8

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-5 \leq \alpha = -1 \Rightarrow$ prune remaining children (leaves L7).
- At MIN node C: value $-2 \leq \alpha = -1 \Rightarrow$ prune remaining children (leaves L10).



Leaves evaluated (8): L1, L2, L3, L4, L5, L6, L8, L9. Leaves pruned (2): L7, L10. Root value = -1 (same as plain minimax, -1 — pruning never changes the root value).

Q.16 [MSQ, 2M · Predicate Logic] Answer: A, C, D (A: $\forall x \exists y R(x, y)$; C: $\exists x R(x, x)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,1), (1,3), (2,1), (2,3), (3,1), (3,2), (4,1), (4,3)\}$.

(A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(3,1) \wedge R(1,3)$ but $\neg R(3,3)$.

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A, B (A: $(\neg P \wedge (\neg P \rightarrow R)) \rightarrow R$; B: $R \rightarrow (Q \rightarrow R)$)

(A) $(\neg P \wedge (\neg P \rightarrow R)) \rightarrow R$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $R \rightarrow (Q \rightarrow R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $((Q \rightarrow R) \wedge (P \rightarrow R)) \rightarrow (Q \wedge P)$: **not valid** — counter-model $P=T, Q=F, R=T$.

(D) $(Q \vee P) \rightarrow (Q \wedge P)$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Propositional Logic] Answer: C (C: $\neg P \rightarrow Q$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
T	T	F	T	T
F	T	F	T	T

(A) $\neg P \leftrightarrow \neg R$ is false in the KB-model $P=T, Q=T, R=F, S=T \Rightarrow$ not entailed.

(B) $\neg S$ is false in the KB-model $P=T, Q=T, R=F, S=T \Rightarrow$ not entailed.

(C) $\neg P \rightarrow Q$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $P \wedge Q$ is false in the KB-model $P=F, Q=T, R=F, S=T \Rightarrow$ not entailed.

Q.19 [NAT, 2M · Predicate Logic] Answer: 10

The condition says f is idempotent: f fixes every element of its image. Choose the image S (size k) — each element of S maps to itself — and map every element outside S anywhere into S : k^{3-k} ways. Sum over k : $k=1: 3 \times 1^2 = 3$; $k=2: 3 \times 2^1 = 6$; $k=3: 1 \times 3^0 = 1$. Total = 10.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.871 to 0.875

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever} | \text{Flu}) \cdot P(\text{Cough}) \cdot P(\text{Fatigue} | \text{Flu}, \text{Cough})$.

Inference by enumeration: $P(\text{Flu} | e) = \alpha \sum_{\text{hidden}} P(\text{Flu}, e, \text{hidden})$.

Hidden: Co	Fl	product of CPT entries (Fl · Fe · Co · Fa)	value
T	T	$0.75 \times 0.5 \times 0.6 \times 0.9$	0.2025
F	T	$0.75 \times 0.5 \times 0.4 \times 0.25$	0.0375
T	F	$0.25 \times 0.4 \times 0.6 \times 0.25$	0.015
F	F	$0.25 \times 0.4 \times 0.4 \times 0.5$	0.02

$P(\text{Flu}=T, e) = 0.24$; $P(\text{Flu}=F, e) = 0.035$; $P(e) = 0.275$.

$P(\text{Flu}=T | e) = 0.24 / 0.275 = \mathbf{0.8727} \approx 0.873$.

Q.21 [NAT, 2M · Bayesian Networks] Answer: 0.439 to 0.44

Spam: $0.3 \times 0.8 \times (1 - 0.6) \times 0.2 = 0.3 \times 0.8 \times 0.4 \times 0.2 = 0.0192$. Ham: $0.7 \times 0.1 \times (1 - 0.3) \times 0.5 = 0.7 \times 0.1 \times 0.7 \times 0.5 = 0.0245$. Normalize: $0.0192 / (0.0192 + 0.0245) = 0.0192 / 0.0437 \approx 0.4394 \approx 0.439$. Note $P(\neg f_2 | \text{class}) = 1 - P(f_2 | \text{class})$ must be used for the false feature.

Q.22 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(B, D)$, $f(C, D, E)$, $f(C, D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(C, D, F)$	$\{C, D, F\}$	$f(C, D)$
D	$f(B, D)$, $f(C, D, E)$, $f(C, D)$	$\{B, C, D, E\}$	$f(B, C, E)$
A	$f(A)$, $f(A, B)$, $f(A, B, C)$	$\{A, B, C\}$	$f(B, C)$
B	$f(B, C, E)$, $f(B, C)$	$\{B, C, E\}$	$f(C, E)$
C	$f(C, E)$	$\{C, E\}$	$f(E)$

The largest factor produced has **3** variable(s) (created when eliminating D); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over E and normalise.

Q.23 [MSQ, 2M · Conditional Independence] Answer: A, B (A: $B \perp F | A$; B: $E \perp A | C, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $B \perp F | A$: **TRUE** — $B \leftarrow A \rightarrow C \rightarrow D \rightarrow F$ (blocked: observed fork node A); $B \leftarrow A \rightarrow C \rightarrow E \leftarrow D \rightarrow F$ (blocked: observed fork node A); $B \leftarrow A \rightarrow C \rightarrow G \leftarrow E \leftarrow D \rightarrow F$ (blocked: observed fork node A); $B \leftarrow A \rightarrow D \rightarrow F$ (blocked: observed fork node A); $B \leftarrow A \rightarrow F$ (blocked: observed fork node A).

(B) $E \perp A | C, D$: **TRUE** — $E \leftarrow C \leftarrow A$ (blocked: observed chain node C); $E \leftarrow C \rightarrow D \leftarrow A$ (blocked: observed fork node C); $E \leftarrow C \rightarrow D \rightarrow F \leftarrow A$ (blocked: observed fork node C); $E \leftarrow D \leftarrow A$ (blocked: observed chain node D); $E \leftarrow D \leftarrow C \leftarrow A$ (blocked: observed chain node D); $E \leftarrow D \rightarrow F \leftarrow A$ (blocked: observed fork node D) (+3 more paths, all blocked).

(C) $A \perp C | D$: **FALSE** — the path $A \rightarrow C$ is active given $\{D\}$.

(D) $F \perp E | D$: **FALSE** — the path $F \leftarrow A \rightarrow C \rightarrow E$ is active given $\{D\}$.

Q.24 [NAT, 2M · Sampling] Answer: 0.498 to 0.502

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	A
1	$P(C=T A=T, B=F)=0.25 \times P(D=F C=T)=0.7$	0.175	T
2	$P(C=T A=T, B=F)=0.25 \times P(D=F C=T)=0.7$	0.175	T
3	$P(C=T A=T, B=T)=0.3 \times P(D=F C=T)=0.7$	0.21	T
4	$P(C=T A=F, B=T)=0.8 \times P(D=F C=T)=0.7$	0.56	F

$\Sigma w (\text{all}) = 1.12$; $\Sigma w (A = T) = 0.56$.

Estimate $P(A=T | D = F, C = T) = 0.56 / 1.12 = \mathbf{0.5}$.

Q.25 [MCQ, 2M · Sampling] Answer: A (A: 0.444)

$MB(C) = \{S, R\}$ (its children; they have no other parents). $P(C | s, r) = \alpha P(C) P(s | C) P(r | C)$. c: $0.5 \times 0.1 \times 0.8 = 0.04$. $\neg c$: $0.5 \times 0.5 \times 0.2 = 0.05$. $P(c | s, r) = 0.04 / 0.09 \approx 0.444$. 0.040 is the unnormalized value, 0.500 is the prior, and 0.878 is $P(c | \neg s, r)$.

Mock Test 065

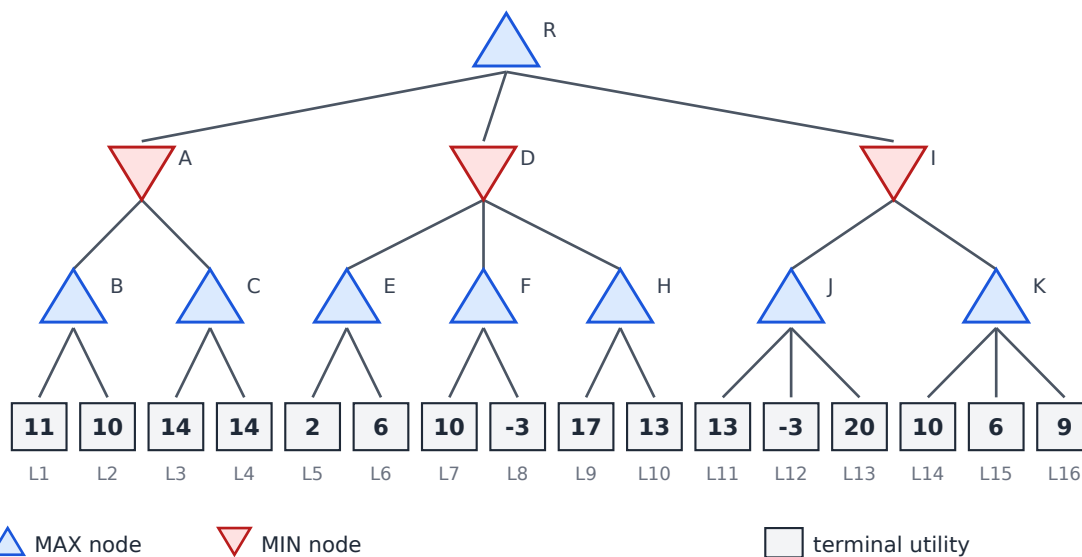
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	9 / 6 / 10	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

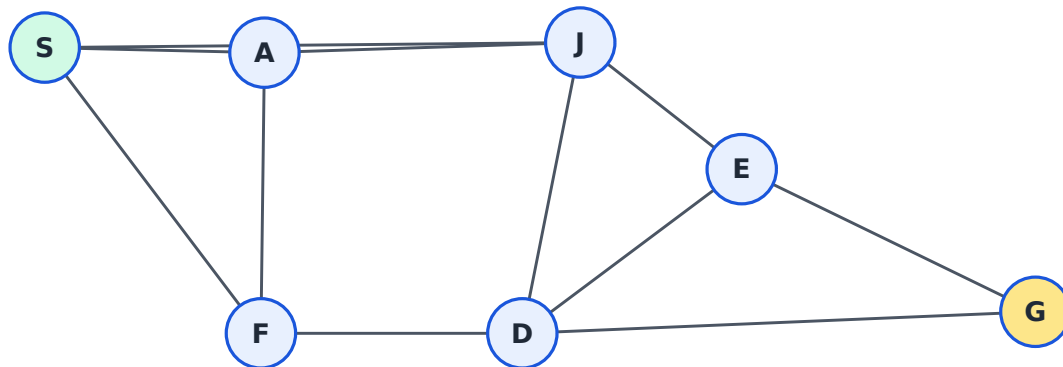
In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to A; root value 12
- ☐ (B) Move to I; root value 10
- ☐ (C) Move to D; root value 6
- ☐ (D) Move to A; root value 11

Q.2 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, A, F, J, D, E, G
- ☐ (B) S, J, E, G
- ☐ (C) S, A, D, F, E, G
- ☐ (D) S, A, F, D, E, G

Q.3 [MCQ · 1 mark · negative: -1/3] Informed Search

Heuristics are often derived from relaxed problems, in which some constraints on the actions are removed. The cost of an optimal solution to a relaxed problem is

- ☐ (A) an admissible (and in fact consistent) heuristic for the original problem
- ☐ (B) always an overestimate of the original problem's optimal cost
- ☐ (C) always equal to the original problem's optimal cost
- ☐ (D) inadmissible, but useful for greedy search only

Q.4 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

How many nodes are there in a complete search tree of depth 3 (root at depth 0) with branching factor 5?

Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following propositional formulas is a tautology?

- ☐ (A) $(p \rightarrow q) \vee (q \rightarrow p)$
- ☐ (B) $(p \rightarrow q) \rightarrow (q \rightarrow p)$
- ☐ (C) $(p \wedge \neg q) \rightarrow q$
- ☐ (D) $(p \vee q) \rightarrow (p \wedge q)$

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following is the correct first-order translation of 'Every student likes some course'?

- ☐ (A) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
- ☐ (B) $\forall x (\text{Student}(x) \wedge \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
- ☐ (C) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \rightarrow \text{Likes}(x, y)))$
- ☐ (D) $\exists y \forall x (\text{Student}(x) \wedge \text{Course}(y) \wedge \text{Likes}(x, y))$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = (\neg P \wedge (P \leftrightarrow R)) \rightarrow ((R \wedge \neg Q) \wedge (R \leftrightarrow P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

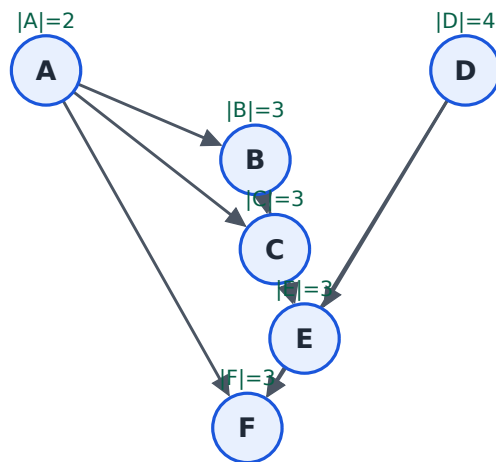
Q.8 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\exists x \forall y R(x, y)$ true?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

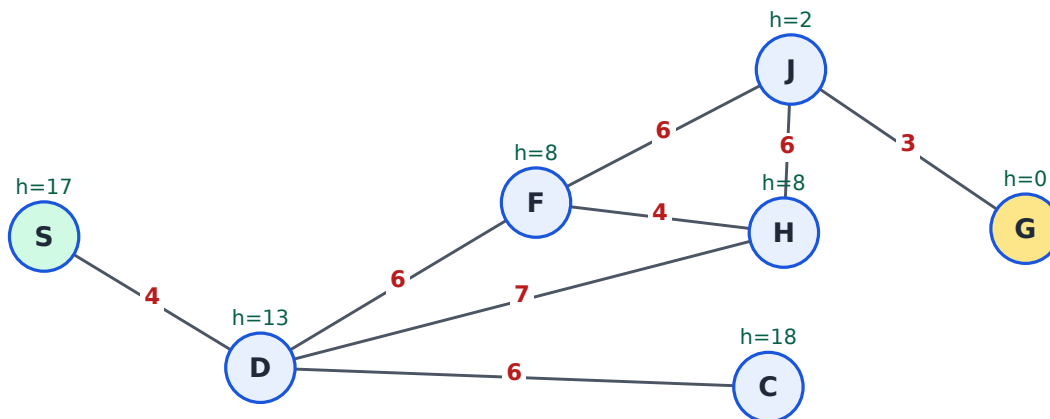
A naive Bayes model with class variable C and features $F_1, \dots, F_n$, viewed as a Bayesian network, has which structure and assumption?

- ☐ (A) Edges $C \rightarrow F_i$ for each i; features are conditionally independent given C
- ☐ (B) Edges $F_i \rightarrow C$ for each i; features are marginally independent
- ☐ (C) A chain $C \rightarrow F_1 \rightarrow F_2 \rightarrow \dots \rightarrow F_n$
- ☐ (D) Edges $C \rightarrow F_i$ for each i; features are marginally independent

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

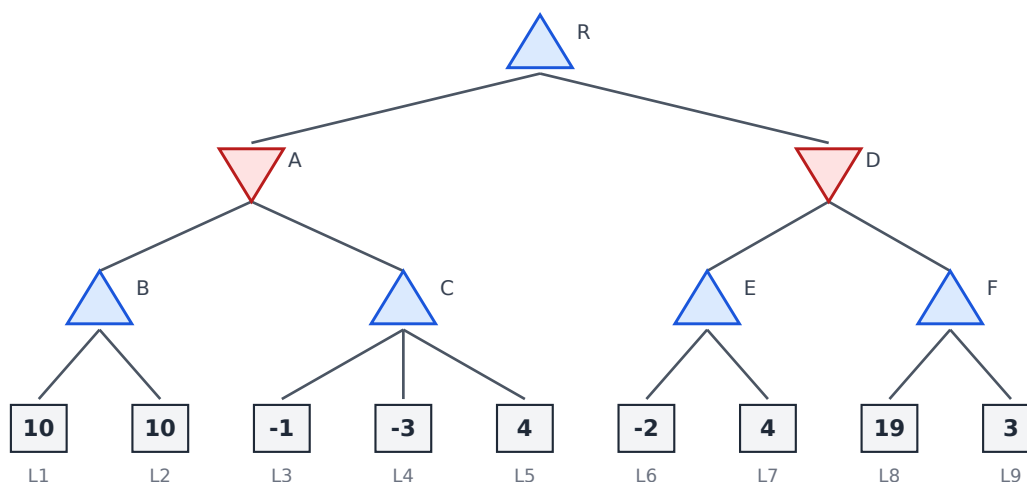
Q.12 [MSQ · 2 marks · no negative marking] Informed Search

The 8-puzzle rule is: a tile can move from square X to square Y if X is adjacent to Y and Y is blank. Which of the following statements about relaxations are TRUE?

- ☐ (A) Dropping both conditions (a tile can move anywhere in one step) gives the misplaced-tiles heuristic h_1
- ☐ (B) Dropping only the 'Y is blank' condition gives the Manhattan-distance heuristic h_2
- ☐ (C) Dropping only the adjacency condition gives the Manhattan-distance heuristic h_2
- ☐ (D) The heuristics obtained from these relaxations are admissible for the original 8-puzzle

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



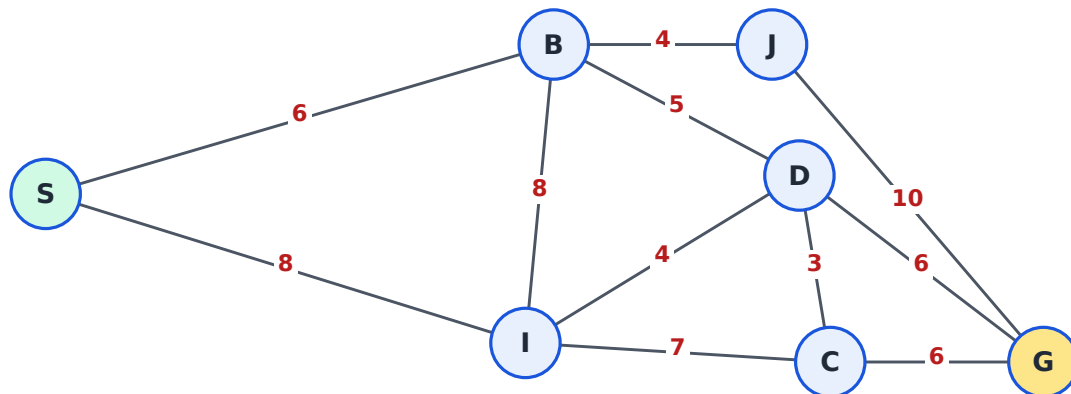
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.14 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow B \rightarrow D \rightarrow G$
- ☐ (B) $S \rightarrow I \rightarrow D \rightarrow G$
- ☐ (C) $S \rightarrow B \rightarrow D \rightarrow C \rightarrow G$
- ☐ (D) $S \rightarrow B \rightarrow J \rightarrow G$

Q.15 [MSQ · 2 marks · no negative marking] Uninformed Search

Uniform-cost search runs with every step cost at least some $\epsilon > 0$. Which of the following are TRUE?

- ☐ (A) UCS is optimal
- ☐ (B) If every step cost equals the same constant, UCS behaves essentially like BFS (expanding level by level), but because it goal-tests on expansion it may generate a whole extra level of nodes
- ☐ (C) UCS is complete when the branching factor is finite
- ☐ (D) UCS remains complete even when the graph contains a cycle of zero-cost actions

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(Q \rightarrow R) \rightarrow (R \rightarrow Q)$
- ☐ (B) $((R \rightarrow P) \wedge (Q \rightarrow P)) \rightarrow (R \wedge Q)$
- ☐ (C) $\neg P \rightarrow (R \rightarrow \neg P)$
- ☐ (D) $(R \rightarrow (Q \rightarrow P)) \leftrightarrow ((R \wedge Q) \rightarrow P)$

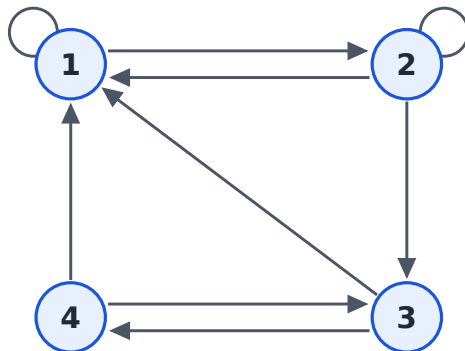
Q.17 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on the set $\{1, 2, 3\}$ are transitive, i.e. satisfy $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (C) $\forall y \exists x R(x, y)$
☐ (D) $\exists y \forall x R(x, y)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

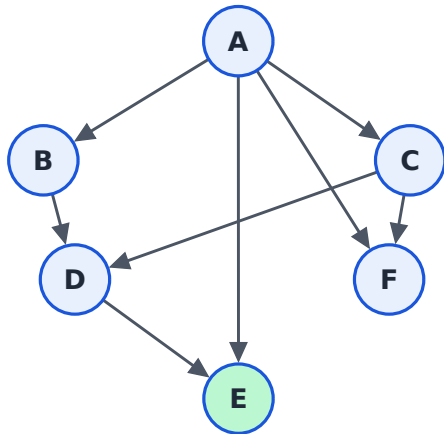
- (1) $\neg(Q \rightarrow R)$
- (2) $(P \vee \neg P) \rightarrow (R \rightarrow Q)$
- (3) $(Q \vee \neg P) \vee (Q \leftrightarrow \neg P)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $R \leftrightarrow (P \leftrightarrow R)$
☐ (B) $\neg R \rightarrow (Q \rightarrow Q)$
☐ (C) $\neg P \vee P$
☐ (D) $R \vee (R \wedge \neg P)$

Q.20 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

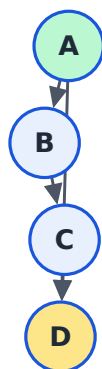
Variable elimination is used to compute $P(\mathbf{E})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, B, F, C, D**. What is the scope of the new factor produced when **F** is summed out?



- ☐ (A) $f(C, D, E)$
☐ (B) $f(C, D)$
☐ (C) $f(C, D, E, F)$
☐ (D) $f(C, E)$

Q.21 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $D = T$. The values of the non-evidence variables in the generated samples are listed in the table. What is the weight of sample #6? (round off to 4 decimal places)



$P(A=T)$
0.5

A	$P(B=T \dots)$	B	$P(C=T \dots)$
T	0.2	T	0.6
F	0.2	F	0.5

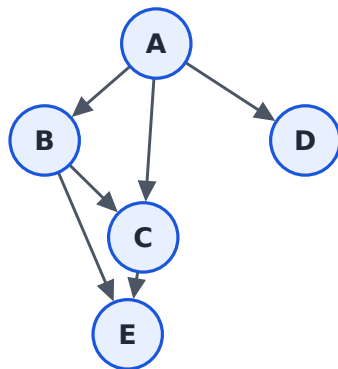
A	C	$P(D=T \dots)$
T	T	0.7
T	F	0.4
F	T	0.9
F	F	0.6

#	A	B	C
1	T	F	F
2	F	F	F
3	F	T	F
4	F	T	F
5	F	T	F
6	T	F	T

Answer (numerical): _____

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.18, 0.24, 0.31, 0.88, 0.49)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



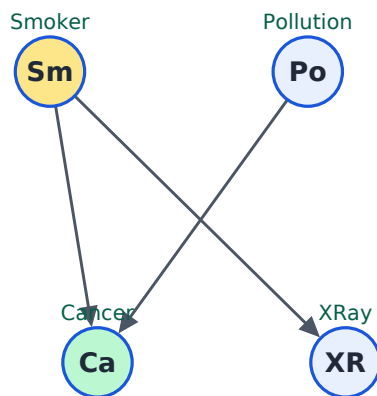
P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.6	T	0.25	T	T	0.2
	F	0.8	T	F	0.25
			F	T	0.1
			F	F	0.75

A	P(D=T ...)	B	C	P(E=T ...)
T	0.3	T	T	0.25
F	0.95	T	F	0.1
		F	T	0.6
		F	F	0.2

- ☐ (A) A=F, B=F, C=F, D=F, E=T
☐ (B) A=T, B=F, C=F, D=F, E=F
☐ (C) A=F, B=T, C=F, D=F, E=F
☐ (D) A=T, B=T, C=F, D=F, E=F

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $\mathbf{P(\text{Cancer} = T \mid \text{Smoker} = T)}$ (round off to 3 decimal places).



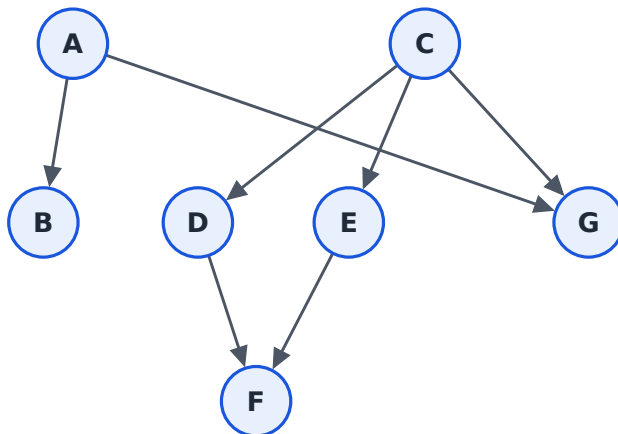
P(Sm=T)	P(Po=T)	Sm	Po	P(Ca=T ...)
0.05	0.05	T	T	0.8
		T	F	0.9
		F	T	0.1
		F	F	0.2

Sm	P(XR=T ...)
T	0.95
F	0.05

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp B \mid F$
☐ (B) $C \perp B$
☐ (C) $B \perp D \mid F, G$
☐ (D) $B \perp A \mid F, G$

Q.25 [NAT · 2 marks · no negative marking] Probability for AI

A disease D has prior $P(d) = 0.01$. A test has $P(+ \mid d) = 0.9$ and $P(+ \mid \neg d) = 0.05$. The test is performed twice; the two results are conditionally independent given D. Both tests come out positive. Compute $P(d \mid +, +)$. (Round to 3 decimal places.)

Answer (numerical): _____

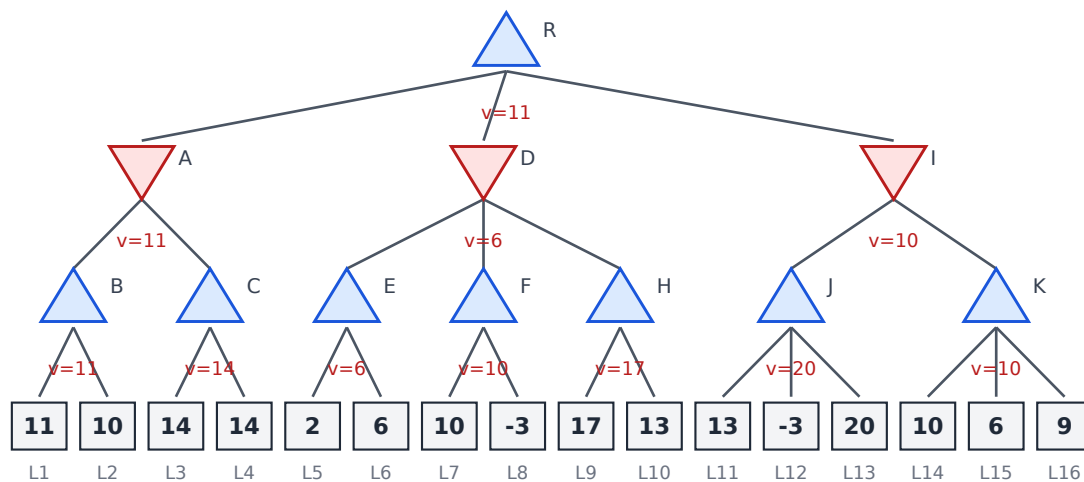
Answer Key — Mock Test 065

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	D	14	MCQ	2	Uninformed Search	A
2	MCQ	1	Uninformed Search	D	15	MSQ	2	Uninformed Search	A, B, C
3	MCQ	1	Informed Search	A	16	MSQ	2	Propositional Logic	C, D
4	NAT	1	Uninformed Search	156	17	NAT	2	Predicate Logic	171
5	MCQ	1	Propositional Logic	A	18	MSQ	2	Predicate Logic	C, D
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Propositional Logic	B, C
7	NAT	1	Propositional Logic	2	20	MCQ	2	Exact Inference	A
8	NAT	1	Predicate Logic	169	21	NAT	2	Sampling	0.6998 to 0.7002
9	NAT	1	Bayesian Networks	140	22	MCQ	2	Sampling	D
10	MCQ	1	Bayesian Networks	A	23	NAT	2	Exact Inference	0.893 to 0.897
11	NAT	2	Informed Search	19	24	MSQ	2	Conditional Independence	A, B
12	MSQ	2	Informed Search	A, B, D	25	NAT	2	Probability for AI	0.765 to 0.767
13	NAT	2	Adversarial Search	7					

Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] **Answer: D** (D: Move to A; root value 11)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 11$, $C = 14$, $A = 11$, $E = 6$, $F = 10$, $H = 17$, $D = 6$, $J = 20$, $K = 10$, $I = 10$. Root value = **11**; MAX's optimal first move is towards A.

Q.2 [MCQ, 1M · Uninformed Search] Answer: D (D: S, A, F, D, E, G)

Adjacency lists (alphabetical): A: {F, J, S}; D: {E, F, G, J}; E: {D, G, J}; F: {A, D, S}; G: {D, E}; J: {A, D, E, S}; S: {A, F, J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, A, F, D, E, G**; the path found is $S \rightarrow A \rightarrow F \rightarrow D \rightarrow E \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.3 [MCQ, 1M · Informed Search] Answer: A (A: an admissible (and in fact consistent) heuristic for the original problem)

The relaxed problem's state-space graph is a supergraph of the original, because it has extra edges. So every original solution is also a relaxed solution, and the relaxed optimum is $\leq$ **the true cost: admissible**. It is also an exact cost in a real problem, so it satisfies the triangle inequality and is consistent. Hence (B) and (D) are wrong. (C) holds only when the relaxation happens to be tight.

Q.4 [NAT, 1M · Uninformed Search] Answer: 156

$$\sum_{i=0..3} 5^i = (5^4 - 1)/(5 - 1) = \mathbf{156}.$$

Q.5 [MCQ, 1M · Propositional Logic] Answer: A (A: $(p \rightarrow q) \vee (q \rightarrow p)$)

$(p \rightarrow q) \vee (q \rightarrow p) \equiv \neg p \vee q \vee \neg q \vee p$, which contains $q \vee \neg q$, so it is true in all 4 rows. $(p \rightarrow q) \rightarrow (q \rightarrow p)$ is false at $p=F, q=T$ (antecedent T, consequent F). $(p \wedge \neg q) \rightarrow q$ is false at $p=T, q=F$. $(p \vee q) \rightarrow (p \wedge q)$ is false whenever exactly one of p, q is true. Hence only the first is valid.

Q.6 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$)

Universal quantifiers pair naturally with $\rightarrow$, existential with $\wedge$. Option B claims everything in the domain is a student. Option C is satisfied by any non-course y (making $\text{Course}(y) \rightarrow \dots$ vacuously true), so it does not require liking a course. Option D says there is a single y that is a course liked by everything, and everything is a student — far too strong.

Q.7 [NAT, 1M · Propositional Logic] Answer: 2

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg P \wedge (P \leftrightarrow R)$	$(R \wedge \neg Q) \wedge (R \leftrightarrow P)$	$(\neg P \wedge (P \leftrightarrow R)) \rightarrow ((R \wedge \neg Q) \wedge (R \leftrightarrow P))$
T	T	T	F	F	T
T	T	F	F	F	T
T	F	T	F	T	T
T	F	F	F	F	T
F	T	T	F	F	T
F	T	F	T	F	F
F	F	T	F	F	T
F	F	F	T	F	F

F has 6 models, so it is false in $8 - 6 = 2$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.8 [NAT, 1M · Predicate Logic] Answer: 169

View R as an 3×3 0/1 matrix. The sentence says R is complement of 'every row has at least one 0': $2^{(n^2)} - (2^n - 1)^n$. With $n = 3$: **169**.

Q.9 [NAT, 1M · Bayesian Networks] Answer: 140

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (3-1) \times 2 + C: (3-1) \times 2 \times 3 + D: (4-1) + E: (3-1) \times 3 \times 3 \times 4 + F: (3-1) \times 2 \times 4 \times 3 = 140$.

Full joint table: $\prod |X| - 1 = 648 - 1 = 647$; saving = 507.

Q.10 [MCQ, 1M · Bayesian Networks] Answer: A (A: Edges $C \rightarrow F_i$ for each i ; features are conditionally independent given C)

Naive Bayes uses $P(C, F_1, \dots, F_n) = P(C) \prod P(F_i | C)$, i.e. the class is the single parent of every feature. The DAG implies $F_i \perp F_j | C$. (D) is wrong because the features are generally dependent marginally (common cause C is unobserved). (B) reverses the edges (that would make features marginally independent but dependent given C); (C) is a different model.

Q.11 [NAT, 2M · Informed Search] Answer: 19

Heuristic values: $h(C)=18$, $h(D)=13$, $h(F)=8$, $h(G)=0$, $h(H)=8$, $h(J)=2$, $h(S)=17$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (17)	D(17)
2	D (17)	F(18), H(19), C(28)
3	F (18)	J(18), H(19), C(28)
4	J (18)	G(19), H(19), C(28)
5	G (19)	H(19), C(28) ← goal, stop

Path returned: **S** → **D** → **F** → **J** → **G** with cost **19**; nodes expanded = **5** (S, D, F, J, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 7 nodes.

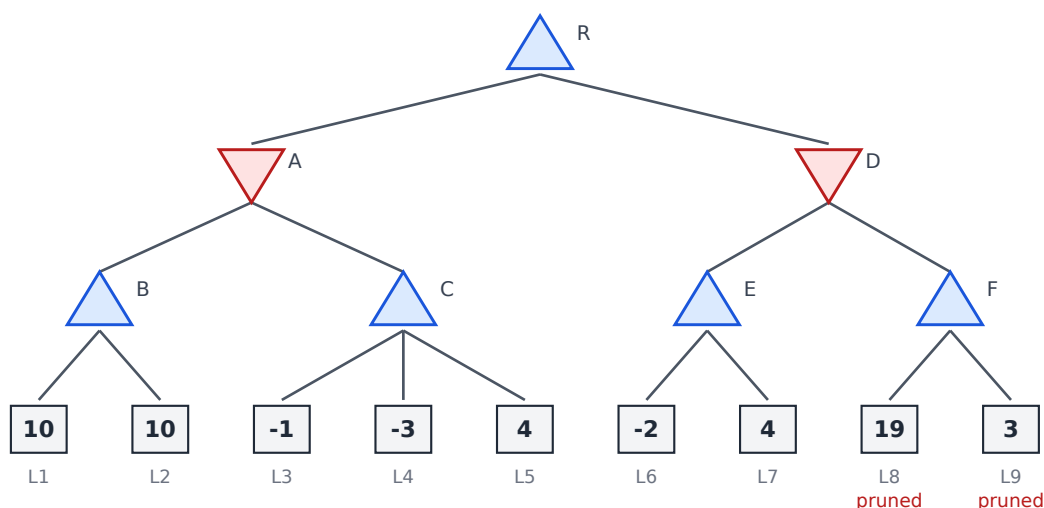
Q.12 [MSQ, 2M · Informed Search] Answer: A, B, D

(A) is true: if a tile can jump anywhere, each misplaced tile needs exactly 1 move. (B) is true: if tiles move to adjacent squares regardless of the blank, each tile needs exactly its Manhattan distance. (C) is false: dropping adjacency but keeping 'Y is blank' gives Gaschnig's heuristic, not Manhattan distance. (D) is true: optimal costs of relaxed problems are admissible.

Q.13 [NAT, 2M · Adversarial Search] Answer: 7

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node D: value $4 \leq \alpha = 4 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (7): L1, L2, L3, L4, L5, L6, L7. Leaves pruned (2): L8, L9. Root value = 4 (same as plain minimax, 4 — pruning never changes the root value).

Q.14 [MCQ, 2M · Uninformed Search] **Answer: A** (A: $S \rightarrow B \rightarrow D \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(6), I(8)
2	B (6)	I(8), J(10), D(11)
3	I (8)	J(10), D(11), C(15)
4	J (10)	D(11), C(15), G(20)
5	D (11)	C(14), G(17)
6	C (14)	G(17)
7	G (17)	← goal, stop

Path returned: **S** → **B** → **D** → **G**, cost = 6 + 5 + 6 = **17**. Nodes expanded (including S and G): **7** (S, B, I, J, D, C, G).

Q.15 [MSQ, 2M · Uninformed Search] **Answer: A, B, C**

(A) is true: UCS expands nodes in order of g , so the first goal it selects is the cheapest. (B) is true: equal costs make g proportional to depth, but the goal test at expansion means the children at depth $d+1$ may already have been generated. (C) is true: with costs $\geq \epsilon$ and finite b , only finitely many nodes have $g \leq C^*$. (D) is false: a zero-cost cycle lets g stay constant while the path grows forever, so UCS may never terminate. This is why the $\epsilon > 0$ condition is needed.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: C, D** (C: $\neg P \rightarrow (R \rightarrow \neg P)$; D: $(R \rightarrow (Q \rightarrow P)) \leftrightarrow ((R \wedge Q) \rightarrow P)$)

(A) $(Q \rightarrow R) \rightarrow (R \rightarrow Q)$: **not valid** — counter-model $Q=F, R=T$.

(B) $((R \rightarrow P) \wedge (Q \rightarrow P)) \rightarrow (R \wedge Q)$: **not valid** — counter-model $P=T, Q=T, R=F$.

(C) $\neg P \rightarrow (R \rightarrow \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $(R \rightarrow (Q \rightarrow P)) \leftrightarrow ((R \wedge Q) \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [NAT, 2M · Predicate Logic] **Answer: 171**

There is no simple closed formula; we enumerate all $2^9 = 512$ relations and test the sentence for all 27 triples (x, y, z) . The count of transitive relations on a 3-element set is 171 (OEIS A006905: 1, 2, 13, 171, 3994, ...). Sanity check: on a 2-element set the same enumeration gives 13 of 16; the 3 failures contain both $(1,2)$ and $(2,1)$ but miss $(1,1)$ or $(2,2)$.

Q.18 [MSQ, 2M · Predicate Logic] **Answer: C, D** (C: $\forall y \exists x R(x, y)$; D: $\exists y \forall x R(x, y)$)

$R = \{(1,1), (1,2), (2,1), (2,2), (2,3), (3,1), (3,4), (4,1), (4,3)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(3,1)$ holds but $R(1,3)$ does not.

(B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(2,4)$ nor $R(4,2)$.

(C) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(D) $\exists y \forall x R(x, y)$: **TRUE** — $y = 1$ receives an edge from every x (incl. itself).

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: B, C** (B: $\neg R \rightarrow (Q \rightarrow Q)$; C: $\neg P \vee P$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	T	F	T
F	T	F	T

(A) $R \leftrightarrow (P \leftrightarrow R)$ is false in the KB -model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(B) $\neg R \rightarrow (Q \rightarrow Q)$ is true in all models of $KB \Rightarrow$ **entailed**.

(C) $\neg P \vee P$ is true in all models of $KB \Rightarrow$ **entailed**.

(D) $R \vee (R \wedge \neg P)$ is false in the KB -model $P=T, Q=T, R=F \Rightarrow$ not entailed.

Q.20 [MCQ, 2M · Exact Inference] Answer: A (A: f(C, D, E))

Initial factors (one per CPT, evidence removed from scopes): f(A), f(A, B), f(A, C), f(B, C, D), f(A, D, E), f(A, C, F).

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	f(A), f(A, B), f(A, C), f(A, D, E), f(A, C, F)	{A, B, C, D, E, F}	f(B, C, D, E, F)
B	f(B, C, D), f(B, C, D, E, F)	{B, C, D, E, F}	f(C, D, E, F)
F	f(C, D, E, F)	{C, D, E, F}	f(C, D, E)
C	f(C, D, E)	{C, D, E}	f(D, E)
D	f(D, E)	{D, E}	f(E)

The largest factor produced has **5** variable(s) (created when eliminating A); with Boolean variables it has $2^5 = 32$ entries. Finally, multiply the remaining factors over E and normalise.

Q.21 [NAT, 2M · Sampling] Answer: 0.6998 to 0.7002

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(D=T \mid A=T, C=F)=0.4$	0.4	T
2	$P(D=T \mid A=F, C=F)=0.6$	0.6	F
3	$P(D=T \mid A=F, C=F)=0.6$	0.6	F
4	$P(D=T \mid A=F, C=F)=0.6$	0.6	F
5	$P(D=T \mid A=F, C=F)=0.6$	0.6	F
6	$P(D=T \mid A=T, C=T)=0.7$	0.7	T

$\Sigma w (\text{all}) = 3.5$; $\Sigma w (A = T) = 1.1$.

Estimate $P(A=T \mid D = T) = 1.1 / 3.5 = \mathbf{0.3143}$.

Q.22 [MCQ, 2M · Sampling] Answer: D (D: A=T, B=T, C=F, D=F, E=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.6$; $u = 0.18 < 0.6 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.25$; $u = 0.24 < 0.25 \Rightarrow B = T$
- C: $P(C=T \mid A=T, B=T) = 0.2$; $u = 0.31 \geq 0.2 \Rightarrow C = F$
- D: $P(D=T \mid A=T) = 0.3$; $u = 0.88 \geq 0.3 \Rightarrow D = F$
- E: $P(E=T \mid B=T, C=F) = 0.1$; $u = 0.49 \geq 0.1 \Rightarrow E = F$

Sample: **{A=T, B=T, C=F, D=F, E=F}**.

Q.23 [NAT, 2M · Exact Inference] Answer: 0.893 to 0.897

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution}) \cdot P(\text{Cancer} \mid \text{Smoker}, \text{Pollution}) \cdot P(\text{XRay} \mid \text{Smoker})$.

Inference by enumeration: $P(\text{Cancer} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cancer}, e, \text{hidden})$.

Hidden: Po, XR	Ca	product of CPT entries ($S_m \cdot P_o \cdot C_a \cdot X_R$)	value
T, T	T	$0.05 \times 0.05 \times 0.8 \times 0.95$	0.0019
T, F	T	$0.05 \times 0.05 \times 0.8 \times 0.05$	0.0001
F, T	T	$0.05 \times 0.95 \times 0.9 \times 0.95$	0.04061
F, F	T	$0.05 \times 0.95 \times 0.9 \times 0.05$	0.00214
T, T	F	$0.05 \times 0.05 \times 0.2 \times 0.95$	0.00047
T, F	F	$0.05 \times 0.05 \times 0.2 \times 0.05$	0.00003
F, T	F	$0.05 \times 0.95 \times 0.1 \times 0.95$	0.00451
F, F	F	$0.05 \times 0.95 \times 0.1 \times 0.05$	0.00024

$P(\text{Cancer}=T, e) = 0.04475$; $P(\text{Cancer}=F, e) = 0.00525$; $P(e) = 0.05$.

$P(\text{Cancer}=T \mid e) = 0.04475 / 0.05 = \mathbf{0.895} \approx 0.895$.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $C \perp B \mid F$; B: $C \perp B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp B \mid F$: **TRUE** — $C \rightarrow G \leftarrow A \rightarrow B$ (blocked: collider G with neither it nor a descendant observed).

(B) $C \perp B$: **TRUE** — $C \rightarrow G \leftarrow A \rightarrow B$ (blocked: collider G with neither it nor a descendant observed).

(C) $B \perp D \mid F, G$: **FALSE** — the path $B \leftarrow A \rightarrow G \leftarrow C \rightarrow D$ is active given $\{F, G\}$.

(D) $B \perp A \mid F, G$: **FALSE** — the path $B \leftarrow A$ is active given $\{F, G\}$.

Q.25 [NAT, 2M · Probability for AI] **Answer: 0.765 to 0.767**

By Bayes rule with conditional independence, $P(d \mid +, +) = \alpha P(+ \mid d)^2 P(d)$. Unnormalized for d: $0.9^2 \times 0.01 = 0.81 \times 0.01 = 0.0081$. For $\neg d$: $0.05^2 \times 0.99 = 0.0025 \times 0.99 = 0.002475$. Normalizing: $0.0081 / (0.0081 + 0.002475) = 0.0081 / 0.010575 \approx 0.766$. (For comparison, a single positive test gives only $0.009/0.0585 \approx 0.154$.)

Mock Test 066

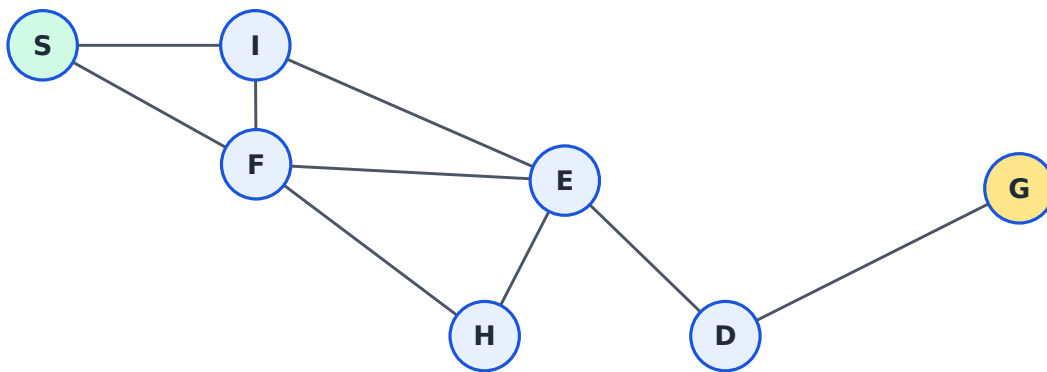
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	8 / 8 / 9	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

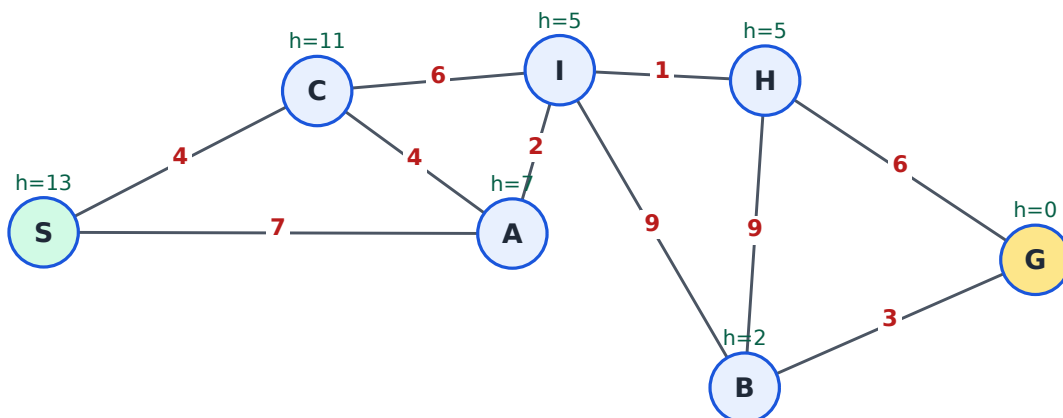
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, F, I, E, D, H, G
- ☐ (B) S, F, E, D, G
- ☐ (C) S, I, F, E, H, D, G
- ☐ (D) S, F, I, E, H, D, G

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

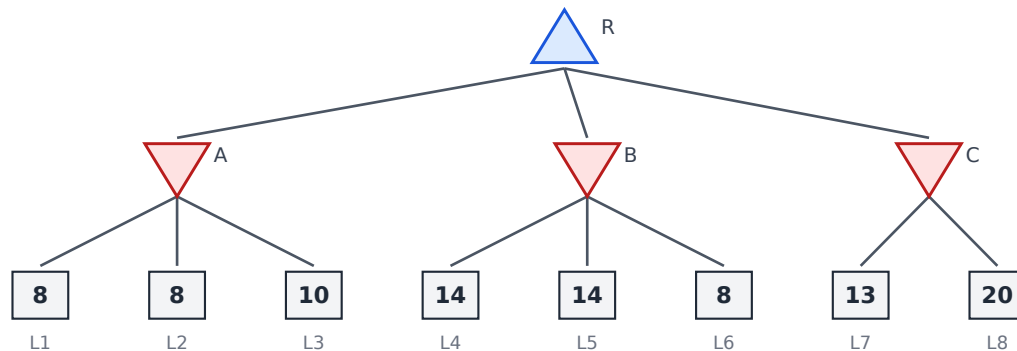
Q.3 [MSQ · 1 mark · no negative marking] Uninformed Search

Which of the following are requirements for applying bidirectional search?

- ☐ (A) Predecessors of a state can be computed, so the problem can be searched backward
- ☐ (B) The goal state is explicitly known (or there are only a few goal states), so the backward search has somewhere to start
- ☐ (C) An admissible heuristic must be available
- ☐ (D) The state space must be a tree

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



MAX node MIN node

terminal utility

- ☐ (A) Move to C; root value 14
- ☐ (B) Move to B; root value 8
- ☐ (C) Move to A; root value 8
- ☐ (D) Move to C; root value 13

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Over the finite domain $\{a, b\}$, the sentence $\forall x \exists y R(x, y)$ is equivalent to which ground formula?

- ☐ (A) $(R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$
- ☐ (B) $(R(a, a) \wedge R(a, b)) \vee (R(b, a) \wedge R(b, b))$
- ☐ (C) $(R(a, a) \vee R(b, a)) \wedge (R(a, b) \vee R(b, b))$
- ☐ (D) $R(a, a) \wedge R(b, b)$

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In Existential Instantiation, the sentence $\exists x \text{Crown}(x) \wedge \text{OnHead}(x, \text{John})$ is replaced by $\text{Crown}(C_1) \wedge \text{OnHead}(C_1, \text{John})$. Which condition must C_1 satisfy?

- ☐ (A) C_1 must be a new constant symbol not appearing anywhere else in the KB
- ☐ (B) C_1 must be John
- ☐ (C) C_1 may be any ground term already in the KB
- ☐ (D) C_1 must be a variable

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following formulas is satisfiable but NOT valid?

- ☐ (A) $p \rightarrow (q \rightarrow p)$
- ☐ (B) $p \wedge \neg p$
- ☐ (C) $(p \wedge q) \rightarrow (p \vee q)$
- ☐ (D) $(p \rightarrow q) \rightarrow p$

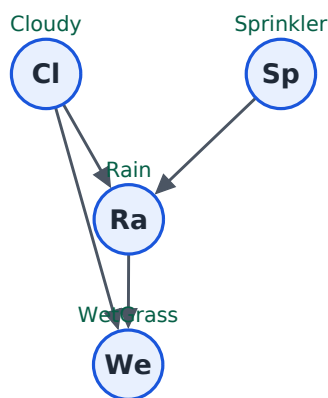
Q.8 [MCQ · 1 mark · negative: -1/3] Variable Elimination

A Bayesian network is called a polytree (singly connected) if

- ☐ (A) there is at most one undirected path between any two nodes
- ☐ (B) every node has at most one parent
- ☐ (C) there is at most one directed path between any two nodes
- ☐ (D) it has exactly one root node

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample **(Cl=T, Sp=F, Ra=F, We=T)**? (round off to 4 decimal places)



P(Cl=T)
0.25

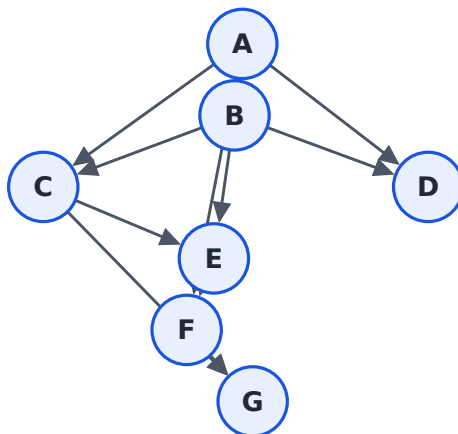
P(Sp=T)	Cl	Sp	P(Ra=T ...)
0.5	T	T	0.05
	T	F	0.7
	F	T	0.3
	F	F	0.6

Cl	Ra	P(We=T ...)
T	T	0.7
T	F	0.95
F	T	0.8
F	F	0.8

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **C** in the Bayesian network below?

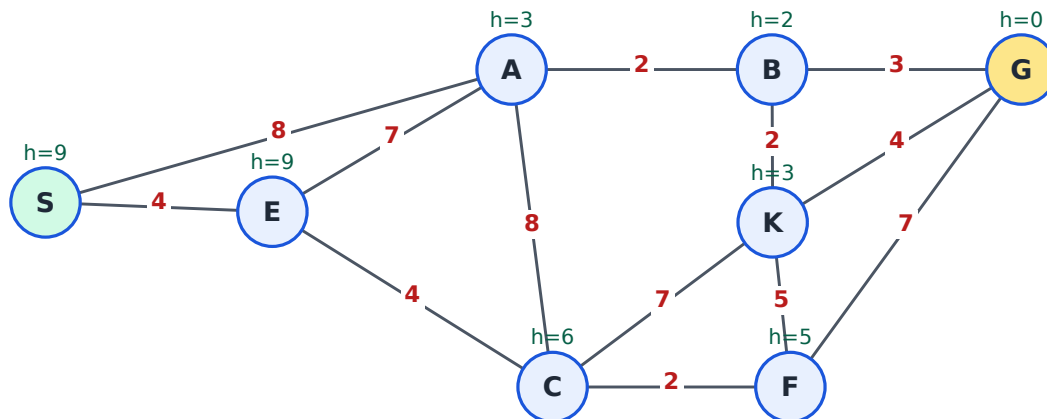


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search - A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

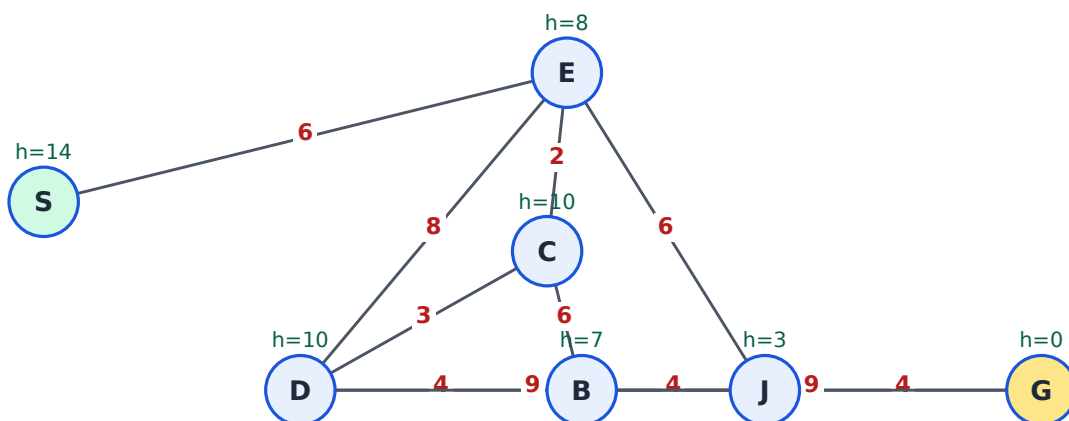
Q.12 [MSQ · 2 marks · no negative marking] Uninformed Search

Which of the following statements about tree search versus graph search are TRUE?

- ☐ (A) Graph search keeps an explored (reached) set so that no state is expanded more than once
- ☐ (B) On a finite state space with reversible actions, tree search can face an infinite search tree because of loopy paths
- ☐ (C) Depth-first graph search is complete on infinite state spaces
- ☐ (D) The memory used by depth-first graph search can grow in proportion to the number of reachable states

Q.13 [NAT · 2 marks · no negative marking] Informed Search - A*

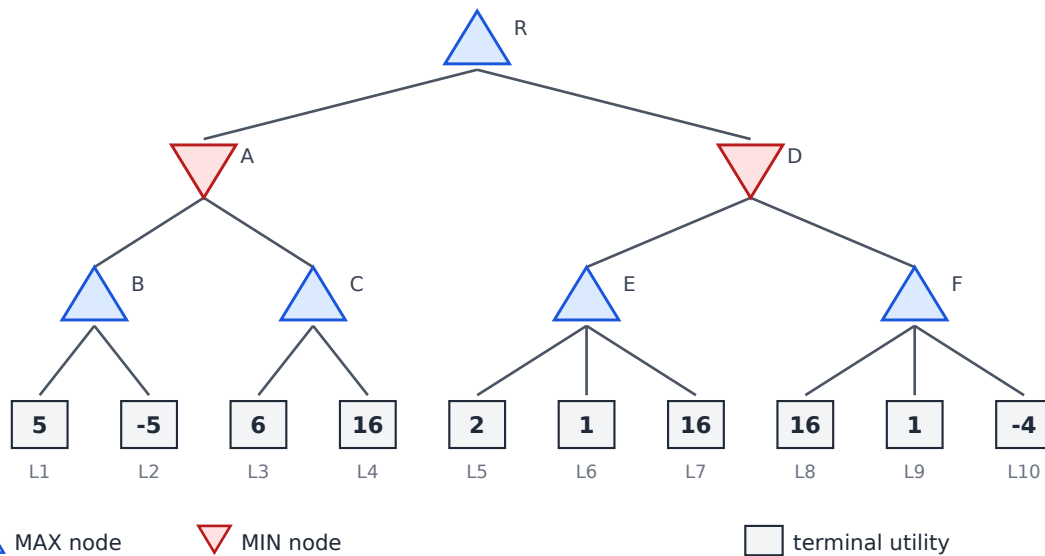
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Informed Search

In the 8-puzzle, h_1 is the number of misplaced tiles and h_2 is the total Manhattan distance. Which of the following are TRUE?

- ☐ (A) $h_2(n) \geq h_1(n)$ for every state n
- ☐ (B) Both h_1 and h_2 are consistent
- ☐ (C) $h_1 + h_2$ is admissible
- ☐ (D) The number of reachable states from any start state is $9!/2 = 181440$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg S \wedge \neg S$
- (2) $(\neg R \wedge \neg Q) \rightarrow R$
- (3) $(S \leftrightarrow Q) \leftrightarrow \neg R$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $Q \rightarrow Q$
- ☐ (B) $(P \rightarrow R) \wedge (S \vee S)$
- ☐ (C) $Q \wedge (\neg S \vee P)$
- ☐ (D) $(\neg Q \vee \neg R) \leftrightarrow (S \leftrightarrow P)$

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = (R \leftrightarrow (S \wedge \neg P)) \rightarrow ((\neg P \leftrightarrow \neg S) \leftrightarrow (Q \leftrightarrow \neg Q)) ?$$

Answer (numerical): _____

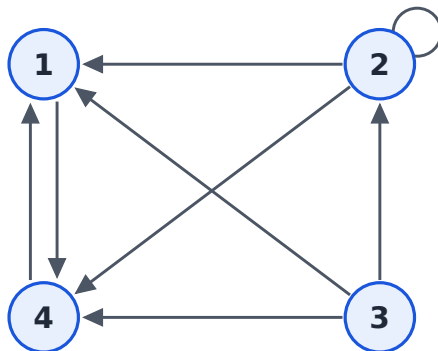
Q.18 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following statements about the computational complexity of propositional reasoning is correct (assuming $P \neq NP$)?

- ☐ (A) Satisfiability of 3-CNF formulas can be decided in polynomial time
- ☐ (B) Deciding whether a propositional formula is valid is co-NP-complete
- ☐ (C) Satisfiability of Horn formulas is NP-complete
- ☐ (D) Satisfiability of 2-CNF formulas is NP-complete

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (B) $\exists x \forall y R(x, y)$
☐ (C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **D, E, A** and rules:

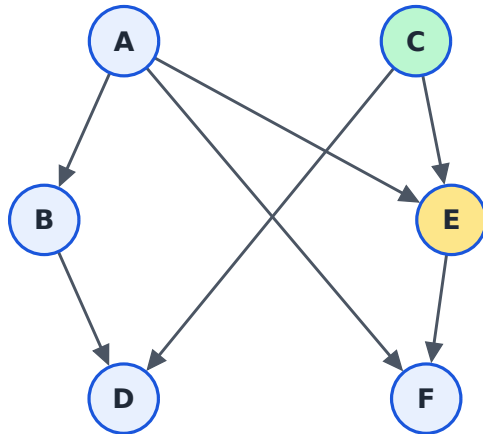
- R1: $A \wedge F \rightarrow K$
 R2: $C \wedge D \rightarrow H$
 R3: $A \wedge H \rightarrow K$
 R4: $C \rightarrow K$
 R5: $J \rightarrow D$
 R6: $D \rightarrow B$
 R7: $A \rightarrow F$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) C is entailed
☐ (B) H is entailed
☐ (C) J is entailed
☐ (D) K is entailed

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(C \mid E = T)$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, D, A, F**. What is the scope of the new factor produced when **D** is summed out?



- ☐ (A) $f(A)$
☐ (B) $f(C)$
☐ (C) $f(A, C)$
☐ (D) $f(A, C, D)$

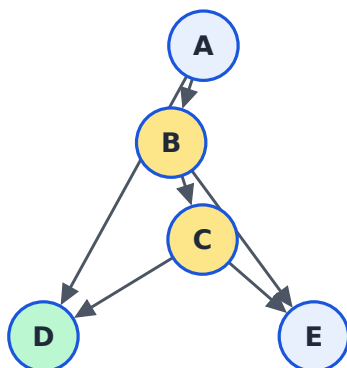
Q.22 [MSQ · 2 marks · no negative marking] Variable Elimination

Which statements about variable elimination (VE) are correct?

- ☐ (A) Every elimination ordering yields the same final answer
☐ (B) Finding an optimal elimination ordering is NP-hard in general
☐ (C) VE runs in time polynomial in the number of variables for every network and ordering
☐ (D) VE avoids the repeated computation of identical subexpressions that occurs in enumeration

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(D = T \mid C = F, B = T)$ (round off to 3 decimal places).



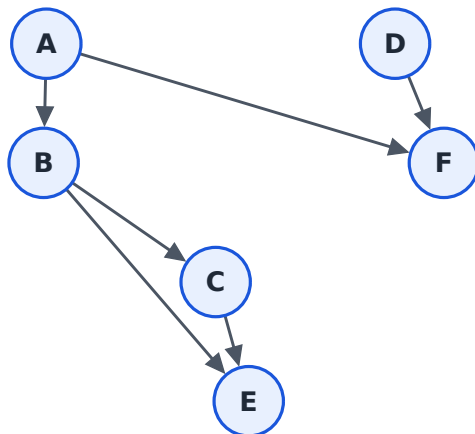
P(A=T)	A	P(B=T ...)	B	P(C=T ...)
0.25	T	0.7	T	0.5
	F	0.05	F	0.4

A	C	P(D=T ...)	B	C	P(E=T ...)
T	T	0.1	T	T	0.7
T	F	0.8	T	F	0.5
F	T	0.7	F	T	0.95
F	F	0.5	F	F	0.3

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp D$
☐ (B) $F \perp E \mid B$
☐ (C) $B \perp F \mid A$
☐ (D) $B \perp A$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(B = T \mid D = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	T	T	F
2	F	T	T	F
3	F	T	T	T
4	T	F	F	F
5	F	F	T	F
6	T	F	T	T
7	F	F	T	T
8	F	T	T	T
9	F	T	T	F
10	F	T	F	F
11	F	T	T	T
12	F	F	T	F
13	F	T	T	F
14	F	F	T	F
15	T	F	T	F
16	F	T	T	F
17	T	T	T	F
18	F	T	T	T
19	F	T	F	T
20	F	F	F	F

Answer (numerical): _____

Answer Key — Mock Test 066

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	NAT	2	Adversarial Search	3
2	NAT	1	Informed Search	21	15	MSQ	2	Informed Search	A, B, D
3	MSQ	1	Uninformed Search	A, B	16	MSQ	2	Propositional Logic	A, C
4	MCQ	1	Adversarial Search	D	17	NAT	2	Propositional Logic	12
5	MCQ	1	Predicate Logic	A	18	MCQ	2	Propositional Logic	B
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Predicate Logic	C, D
7	MCQ	1	Propositional Logic	D	20	MSQ	2	Propositional Logic	D
8	MCQ	1	Variable Elimination	A	21	MCQ	2	Exact Inference	C
9	NAT	1	Sampling	0.0354 to 0.0358	22	MSQ	2	Variable Elimination	A, B, D
10	NAT	1	Conditional Independence	5	23	NAT	2	Exact Inference	0.745 to 0.749
11	NAT	2	Informed Search	5	24	MSQ	2	Conditional Independence	A, B, C
12	MSQ	2	Uninformed Search	A, B, D	25	NAT	2	Sampling	0.53 to 0.55
13	NAT	2	Informed Search	16					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: D (D: S, F, I, E, H, D, G)

Adjacency lists (alphabetical): D: {E, G}; E: {D, F, H, I}; F: {E, H, I, S}; G: {D}; H: {E, F}; I: {E, F, S}; S: {F, I}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[F, I]
2	F	[I, E, H]
3	I	[E, H]
4	E	[H, D]
5	H	[D]
6	D	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, F, I, E, H, D, G**. The path returned is **S → F → E → D → G** with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Informed Search] Answer: 21

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (13)	A(7), C(11)
2	A (7)	I(5), C(11)
3	I (5)	B(2), H(5), C(11)
4	B (2)	G(0), H(5), C(11)
5	G (0)	H(5), C(11) ← goal, stop

Path returned: **S → A → I → B → G**, cost = 7 + 2 + 9 + 3 = **21**.

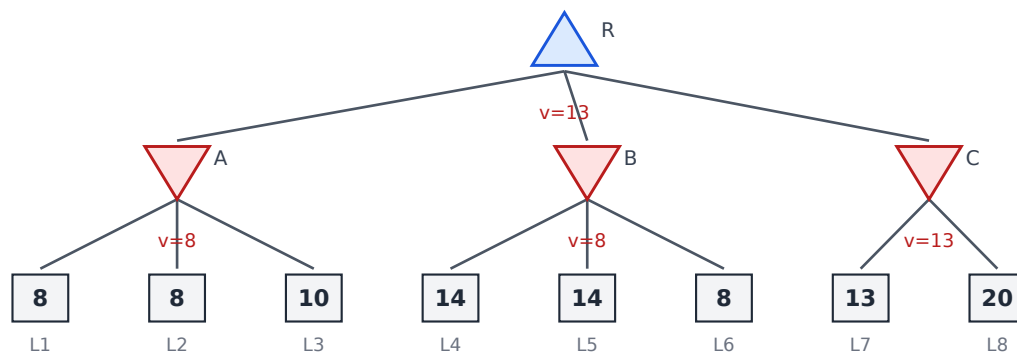
The optimal cost is $h^*(S) = 16$ — greedy search is not optimal here.

Q.3 [MSQ, 1M · Uninformed Search] Answer: A, B

The backward search expands states toward their predecessors, so (A) predecessors must be computable and (B) the goal must be explicitly known. An implicit goal test such as 'checkmate' gives backward search no starting state. A heuristic is not needed (C): bidirectional BFS and bidirectional UCS are uninformed. The state space may be any graph (D).

Q.4 [MCQ, 1M · Adversarial Search] Answer: D (D: Move to C; root value 13)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = 8, B = 8, C = 13. Root value = **13**; MAX's optimal first move is towards C.

Q.5 [MCQ, 1M · Predicate Logic] **Answer: A** (A: $(R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$)

Over a finite domain with every element named, $\forall$ becomes a conjunction over x and $\exists$ a disjunction over y . For $x = a$: $R(a, a) \vee R(a, b)$; for $x = b$: $R(b, a) \vee R(b, b)$. Their conjunction is option A. Option B is $\exists x \forall y R(x, y)$. Option C is $\forall y \exists x R(x, y)$ (every element has an R-predecessor). Option D is reflexivity.

Q.6 [MCQ, 1M · Predicate Logic] **Answer: A** (A: C_1 must be a new constant symbol not appearing anywhere else in the KB)

Existential Instantiation introduces a fresh Skolem constant naming the (unknown) witness. Using an existing term such as John would assert, without justification, that John is the witness. The result is inferentially equivalent (satisfiable iff the original), not logically equivalent. A variable would make it a non-ground formula.

Q.7 [MCQ, 1M · Propositional Logic] **Answer: D** (D: $(p \rightarrow q) \rightarrow p$)

$(p \rightarrow q) \rightarrow p$: when $p=T$ it is true (consequent T); when $p=F$, $p \rightarrow q$ is T and the consequent is F, so it is false. Hence it is satisfiable but not valid. $p \rightarrow (q \rightarrow p)$ is valid (if p is T the consequent is T). $p \wedge \neg p$ is unsatisfiable. $(p \wedge q) \rightarrow (p \vee q)$ is valid because $p \wedge q \models p \vee q$.

Q.8 [MCQ, 1M · Variable Elimination] **Answer: A** (A: there is at most one undirected path between any two nodes)

AIMA defines singly connected networks (polytrees) as those with at most one undirected path between any pair of nodes. Nodes may have several parents (e.g. the alarm network is a polytree although A has two parents). Requiring at most one directed path is weaker: $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow C$ has two directed paths A to C, but $A \rightarrow B$, $C \rightarrow B$, $A \rightarrow D$, $C \rightarrow D$ has unique directed paths yet the undirected cycle $A-B-C-D-A$. A polytree may also have several roots.

Q.9 [NAT, 1M · Sampling] **Answer: 0.0354 to 0.0358**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(x_i))$.

$P(CI=T) \times P(Sp=F) \times P(Ra=F|CI=T, Sp=F) \times P(We=T|CI=T, Ra=F)$
 $= 0.25 \times 0.5 \times 0.3 \times 0.95 = \mathbf{0.035625} \approx 0.0356$.

Q.10 [NAT, 1M · Conditional Independence] **Answer: 5**

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of C: {A, B}; children: {E, G}; co-parents: {F}.

$MB(C) = \{A, B, E, F, G\}$ — **5** nodes. Given its Markov blanket, C is conditionally independent of every other node in the network.

Q.11 [NAT, 2M · Informed Search] **Answer: 5**

Heuristic values: $h(A)=3$, $h(B)=2$, $h(C)=6$, $h(E)=9$, $h(F)=5$, $h(G)=0$, $h(K)=3$, $h(S)=9$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	A(11), E(13)
2	A (11)	B(12), E(13), C(22)
3	B (12)	E(13), G(13), K(15), C(22)
4	E (13)	G(13), C(14), K(15)
5	G (13)	C(14), K(15) ← goal, stop

Path returned: **S** → **A** → **B** → **G** with cost **13**; nodes expanded = **5** (S, A, B, E, G).

Check: $h(n) = \lceil c \cdot h^*(n) \rceil$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 13$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.12 [MSQ, 2M · Uninformed Search] Answer: A, B, D

(A) is true: the explored set is what defines graph search. (B) is true: with reversible actions, a path such as $A \rightarrow B \rightarrow A \rightarrow B \dots$ can be extended forever, so even a finite state space produces an infinite tree. (C) is false: in an infinite state space, DFS can follow an infinite branch forever. (D) is true: the explored set can hold every reachable state, which removes DFS's linear-space advantage.

Q.13 [NAT, 2M · Informed Search] Answer: 16

Heuristic values: $h(B)=7$, $h(C)=10$, $h(D)=10$, $h(E)=8$, $h(G)=0$, $h(J)=3$, $h(S)=14$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (14)	E(14)
2	E (14)	J(15), C(18), D(24)
3	J (15)	G(16), C(18), B(23), D(24)
4	G (16)	C(18), B(23), D(24) ← goal, stop

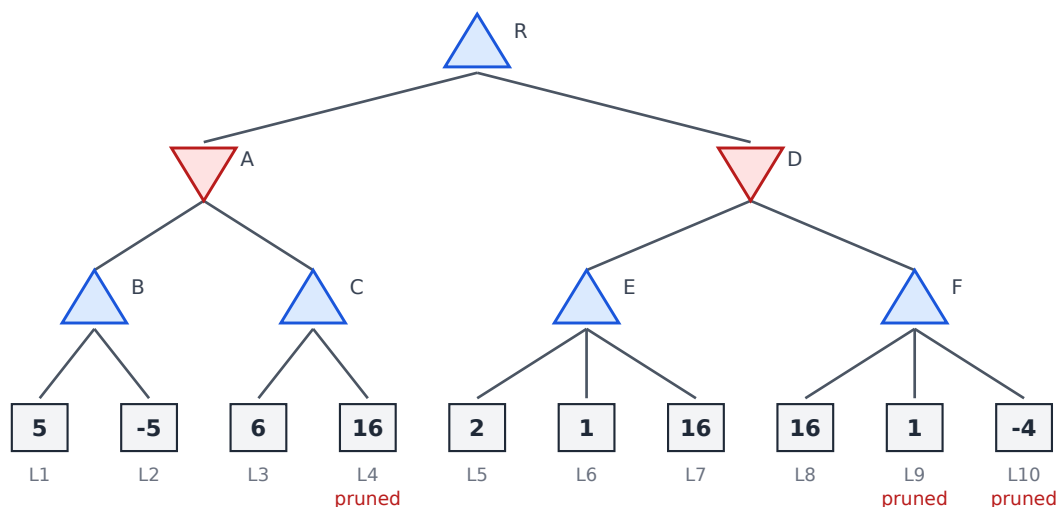
Path returned: **S** → **E** → **J** → **G** with cost **16**; nodes expanded = **4** (S, E, J, G).

Check: $h(n) = \lceil c \cdot h^*(n) \rceil$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.14 [NAT, 2M · Adversarial Search] Answer: 3

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $6 \geq \beta = 5 \Rightarrow$ prune remaining children (leaves L4).
- At MAX node F: value $16 \geq \beta = 16 \Rightarrow$ prune remaining children (leaves L9, L10).



Leaves evaluated (7): L1, L2, L3, L5, L6, L7, L8. Leaves pruned (3): L4, L9, L10. Root value = 16 (same as plain minimax, 16 — pruning never changes the root value).

Q.15 [MSQ, 2M · Informed Search] Answer: A, B, D

(A) is true: every misplaced tile is at Manhattan distance at least 1, so $h_2 \geq h_1$. (B) is true: one move changes h_1 by at most 1 and h_2 by exactly 1, while the step cost is 1. (C) is false: in a state one move from the goal, $h_1 = h_2 = 1$, so the sum is $2 > h^* = 1$. (D) is true: the parity invariant splits the $9!$ states into two halves that cannot reach each other.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $Q \rightarrow Q$; C: $Q \wedge (\neg S \vee P)$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
T	T	T	F	T
F	T	T	F	T

(A) $Q \rightarrow Q$ is true in all models of $KB \Rightarrow$ **entailed**.

(B) $(P \rightarrow R) \wedge (S \vee S)$ is false in the KB -model $P=T, Q=T, R=T, S=F \Rightarrow$ not entailed.

(C) $Q \wedge (\neg S \vee P)$ is true in all models of $KB \Rightarrow$ **entailed**.

(D) $(\neg Q \vee \neg R) \leftrightarrow (S \leftrightarrow P)$ is false in the KB-model $P=F, Q=T, R=T, S=F \Rightarrow$ not entailed.

Q.17 [NAT, 2M · Propositional Logic] **Answer: 12**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$R \leftrightarrow (S \wedge \neg P)$	$(\neg P \leftrightarrow \neg S) \leftrightarrow (Q \leftrightarrow \neg Q)$	$(R \leftrightarrow (S \wedge \neg P)) \rightarrow ((\neg P \leftrightarrow \neg S) \leftrightarrow (Q \leftrightarrow \neg Q))$
T	T	T	T	F	F	T
T	T	T	F	F	T	T
T	T	F	T	T	F	F
T	T	F	F	T	T	T
T	F	T	T	F	F	T
T	F	T	F	F	T	T
T	F	F	T	T	F	F
T	F	F	F	T	T	T
F	T	T	T	T	T	T
F	T	T	F	F	F	T
F	T	F	T	F	T	T
F	T	F	F	T	F	F
F	F	T	T	T	T	T
F	F	T	F	F	F	T
F	F	F	T	F	T	T
F	F	F	F	T	F	F

Rows where F = T: **12**.

Q.18 [MCQ, 2M · Propositional Logic] **Answer: B** (B: Deciding whether a propositional formula is valid is co-NP-complete)

α is valid iff $\neg\alpha$ is unsatisfiable, and SAT is NP-complete, so VALIDITY (TAUT) is co-NP-complete. 3-SAT is NP-complete, so not in P under $P \neq NP$. Horn-SAT is decidable in linear time (unit propagation / forward chaining). 2-SAT is decidable in linear time via the implication graph and strongly connected components.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: C, D** (C: $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$; D: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,4), (2,1), (2,2), (2,4), (3,1), (3,2), (3,4), (4,1)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,4)$ holds but $R(4,2)$ does not.

(B) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **TRUE** — every two distinct elements are related in some direction.

(D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: D** (D: K is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \rightarrow B, A \rightarrow F$; add B, F.

Round 2: fire $A \wedge F \rightarrow K$; add K.

Fixed point: $\{A, B, D, E, F, K\}$ — 6 atoms are entailed (facts: A, D, E; newly derived: B, F, K). Atoms never derived: C, H, J.

Q.21 [MCQ, 2M · Exact Inference] **Answer: C** (C: $f(A, C)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(C), f(B, C, D), f(A, C), f(A, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B), f(B, C, D)$	$\{A, B, C, D\}$	$f(A, C, D)$
D	$f(A, C, D)$	$\{A, C, D\}$	$f(A, C)$
A	$f(A), f(A, C), f(A, F), f(A, C)$	$\{A, C, F\}$	$f(C, F)$
F	$f(C, F)$	$\{C, F\}$	$f(C)$

The largest factor produced has **3** variable(s) (created when eliminating B); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over C and normalise.

Q.22 [MSQ, 2M · Variable Elimination] Answer: A, B, D

(A) Sums of products can be reordered (distributive law), so every ordering computes the same exact result; only cost differs. (B) AIMA notes finding the optimal ordering is intractable (NP-hard), so heuristics such as min-fill or min-degree are used. (D) VE stores intermediate results as factors so each is computed once. (C) is false: the cost is exponential in the size of the largest factor, which can be large for multiply connected networks or bad orderings.

Q.23 [NAT, 2M · Exact Inference] Answer: 0.745 to 0.749

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B | A) \cdot P(C | B) \cdot P(D | A, C) \cdot P(E | B, C)$.

Inference by enumeration: $P(D | e) = \alpha \sum_{\text{hidden}} P(D, e, \text{hidden})$.

Hidden: A, E	D	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T	T	$0.25 \times 0.7 \times 0.5 \times 0.8 \times 0.5$	0.035
T, F	T	$0.25 \times 0.7 \times 0.5 \times 0.8 \times 0.5$	0.035
F, T	T	$0.75 \times 0.05 \times 0.5 \times 0.5 \times 0.5$	0.00469
F, F	T	$0.75 \times 0.05 \times 0.5 \times 0.5 \times 0.5$	0.00469
T, T	F	$0.25 \times 0.7 \times 0.5 \times 0.2 \times 0.5$	0.00875
T, F	F	$0.25 \times 0.7 \times 0.5 \times 0.2 \times 0.5$	0.00875
F, T	F	$0.75 \times 0.05 \times 0.5 \times 0.5 \times 0.5$	0.00469
F, F	F	$0.75 \times 0.05 \times 0.5 \times 0.5 \times 0.5$	0.00469

$P(D=T, e) = 0.07937$; $P(D=F, e) = 0.02687$; $P(e) = 0.10625$.

$P(D=T | e) = 0.07937 / 0.10625 = \mathbf{0.7471} \approx 0.747$.

Q.24 [MSQ, 2M · Conditional Independence] Answer: A, B, C (A: $A \perp D$; B: $F \perp E | B$; C: $B \perp F | A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $A \perp D$: **TRUE** — $A \rightarrow F \leftarrow D$ (blocked: collider F with neither it nor a descendant observed).

(B) $F \perp E | B$: **TRUE** — $F \leftarrow A \rightarrow B \rightarrow C \rightarrow E$ (blocked: observed chain node B); $F \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed chain node B).

(C) $B \perp F | A$: **TRUE** — $B \leftarrow A \rightarrow F$ (blocked: observed fork node A).

(D) $B \perp A$: **FALSE** — the path $B \leftarrow A$ is active given $\{\}$.

Q.25 [NAT, 2M · Sampling] Answer: 0.53 to 0.55

Rejection sampling keeps only samples consistent with the evidence $D = F$: samples 1, 2, 4, 5, 9, 10, 12, 13, 14, 15, 16, 17, 20 (13 accepted, 7 rejected).

Among the accepted samples, $B = T$ in 7. Estimate = $7/13 = \mathbf{0.5385}$.

(For this network the exact value is 0.72; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Mock Test 067

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 6 / 12	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

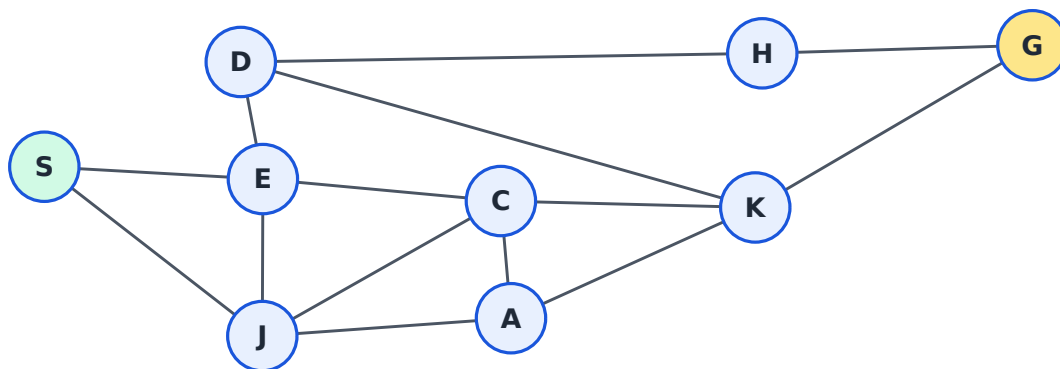
Q.1 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

How many nodes are there in a complete search tree of depth 4 (root at depth 0) with branching factor 2?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

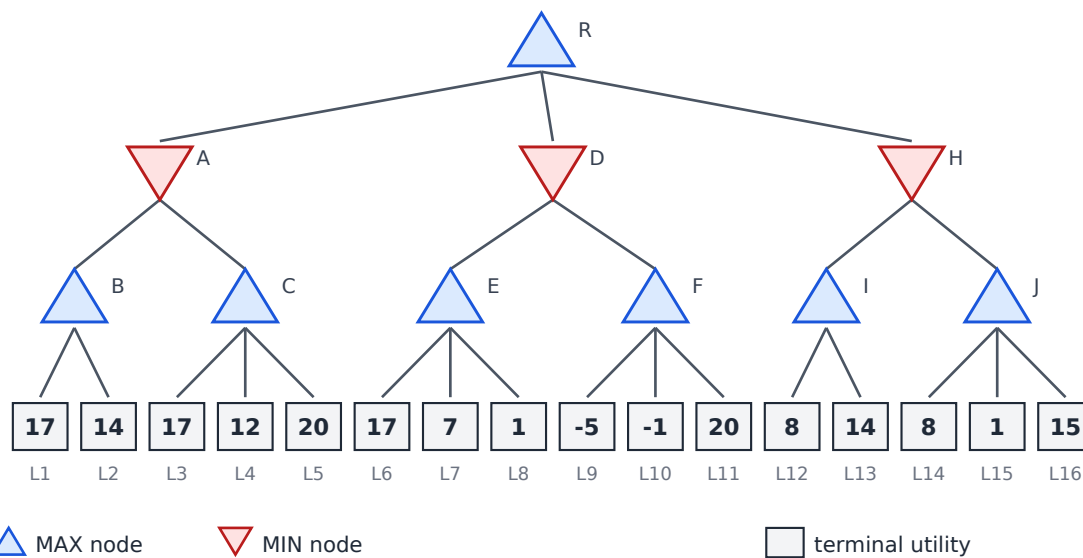
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, J, E, C, D, A, K, H, G
- ☐ (B) S, E, C, A, J, K, D, H, G
- ☐ (C) S, J, E, C, A, D, K, H, G
- ☐ (D) S, E, J, C, D, A, K, H, G

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search

For depth-first tree search with branching factor b and maximum depth m , the space complexity is

- ☐ (A) $O(b^m)$
☐ (B) $O(bm)$
☐ (C) $O(b^d)$
☐ (D) $O(m^b)$

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following statements about forward and backward chaining over Horn knowledge bases is correct?

- ☐ (A) Forward chaining is data-driven while backward chaining is goal-driven
☐ (B) Forward chaining is goal-driven while backward chaining is data-driven
☐ (C) Both are complete for arbitrary (non-Horn) propositional knowledge bases
☐ (D) Backward chaining can derive facts that forward chaining cannot

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Let $KB = \{P \rightarrow Q, Q \rightarrow R, \neg R\}$. Which one of the following is entailed by KB?

- ☐ (A) $\neg P \wedge \neg Q$
☐ (B) $P \vee Q$
☐ (C) $\neg P \wedge Q$
☐ (D) R

Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following clauses are Horn clauses?

- ☐ (A) $\neg p \vee \neg q \vee r$
☐ (B) $p \vee q$
☐ (C) $\neg p \vee \neg q$
☐ (D) r

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic

The clauses $A \vee B \vee \neg C$ and $\neg A \vee B \vee D$ are resolved on the symbol A. How many literals are in the resolvent (after merging duplicate literals)?

Answer (numerical): _____

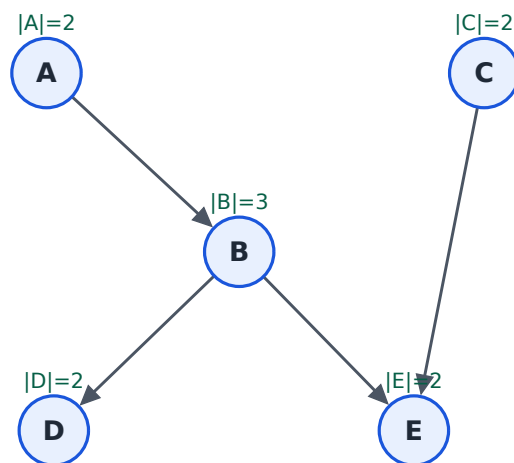
Q.9 [MCQ · 1 mark · negative: -1/3] Variable Elimination

What is the computational complexity of exact inference in general (multiply connected) Bayesian networks?

- ☐ (A) NP-hard (in fact #P-hard)
- ☐ (B) Polynomial time
- ☐ (C) Linear time
- ☐ (D) Undecidable

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?

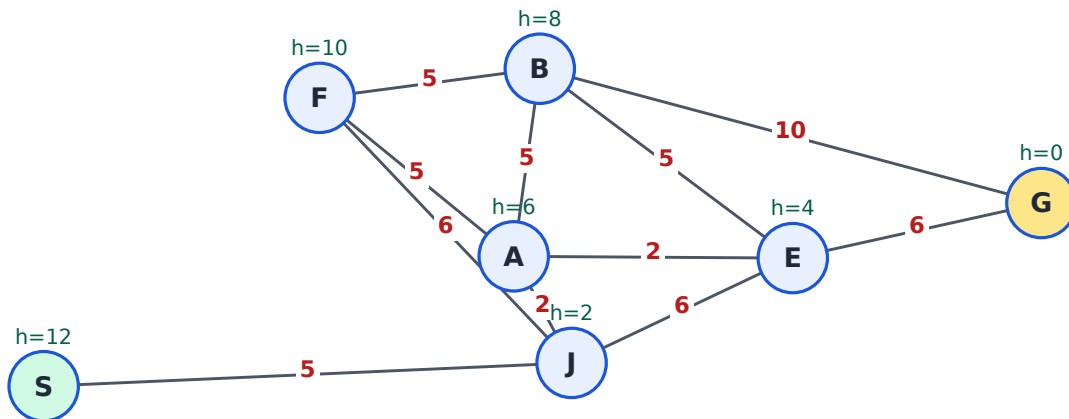


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
☐ (B) The edge A-J violates the consistency condition $h(A) \leq c(A,J) + h(J)$.
☐ (C) h is admissible.
☐ (D) $h(A)$ overestimates the true cost from A to G.

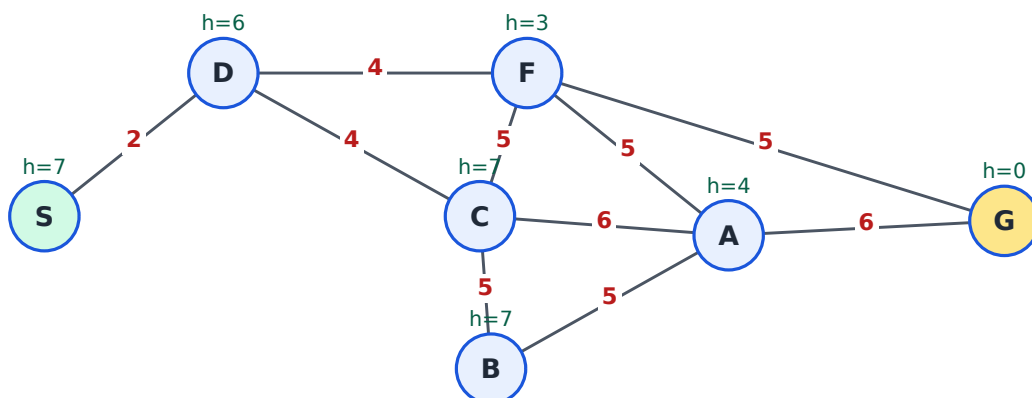
Q.12 [MCQ · 2 marks · negative: -2/3] Adversarial Search

To get the maximum pruning from alpha-beta search, how should children be ordered?

- ☐ (A) At MAX nodes in decreasing order of value, at MIN nodes in increasing order of value
☐ (B) At MAX nodes in increasing order of value, at MIN nodes in decreasing order of value
☐ (C) In increasing order at both node types
☐ (D) Ordering does not affect pruning

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

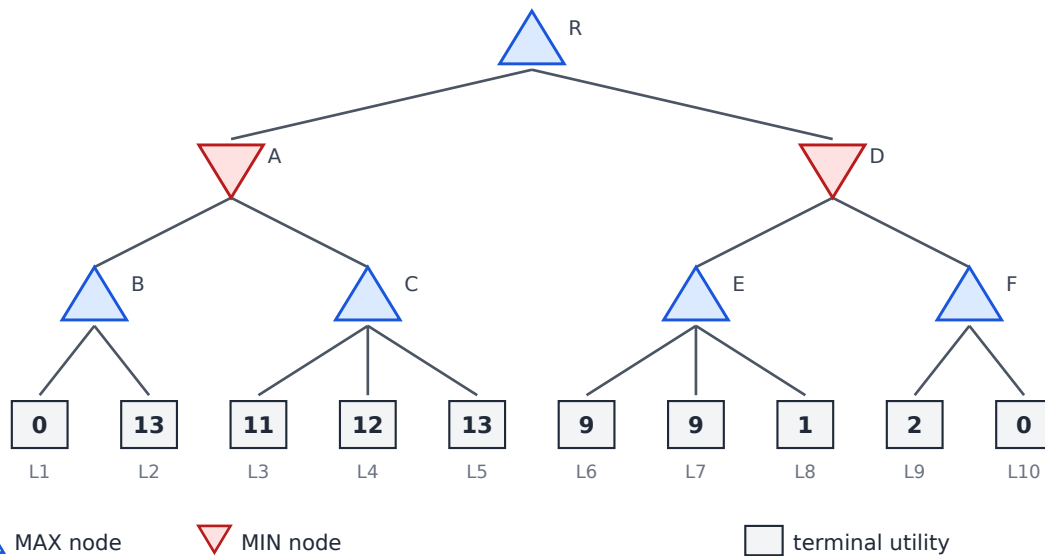
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

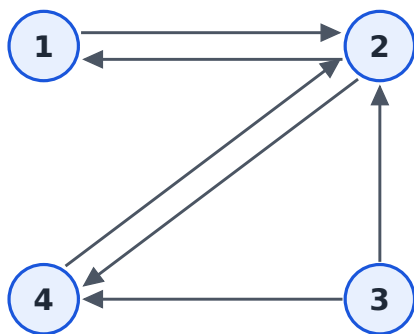
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = ((Q \vee P) \rightarrow \neg\neg R) \rightarrow ((\neg P \rightarrow S) \wedge S) ?$$

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (B) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (C) $\exists x \forall y R(x, y)$
☐ (D) $\forall x \exists y R(x, y)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, E, D** and rules:

- R1: $A \rightarrow B$
 R2: $E \wedge H \rightarrow J$
 R3: $K \wedge C \rightarrow J$
 R4: $E \wedge D \rightarrow A$
 R5: $H \wedge F \rightarrow E$
 R6: $H \wedge J \rightarrow D$
 R7: $A \wedge D \rightarrow F$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) A is entailed
☐ (B) C is entailed
☐ (C) J is entailed
☐ (D) K is entailed

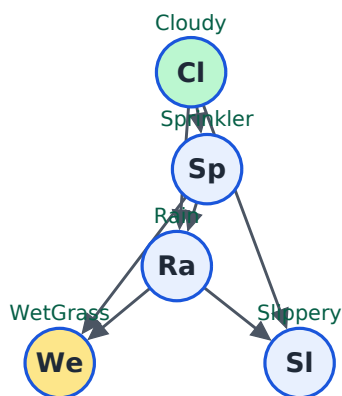
Q.18 [NAT · 2 marks · no negative marking] Propositional Logic

Consider all propositional formulas over the symbols p, q, r, identified up to logical equivalence. How many of these equivalence classes ϕ satisfy $(p \wedge q) \models \phi$?

Answer (numerical): _____

Q.19 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: CI = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, SI = Slippery.) Compute **$P(\text{Cloudy} = T \mid \text{WetGrass} = T)$** (round off to 3 decimal places).



P(CI=T)
0.1

CI	P(Sp=T ...)	CI	Sp	P(Ra=T ...)
T	0.1	T	T	0.4
F	0.95	T	F	0.2
		F	T	0.8
		F	F	0.95

Sp	Ra	P(We=T ...)	CI	Ra	P(SI=T ...)
T	T	0.8	T	T	0.7
T	F	0.75	T	F	0.05
F	T	0.95	F	T	0.8
F	F	0.7	F	F	0.6

Answer (numerical): _____

Q.20 [MSQ · 2 marks · no negative marking] Sampling

Which statements about Gibbs sampling (as an MCMC method in Bayesian networks) are correct?

- ☐ (A) Successive samples are generally correlated, not independent
☐ (B) With fixed evidence and all CPT entries strictly between 0 and 1, the chain's stationary distribution is $P(X \mid e)$
☐ (C) Each state of the Markov chain is a complete assignment to all variables, with evidence variables fixed
☐ (D) Each sample must be accepted or rejected depending on whether it agrees with the evidence

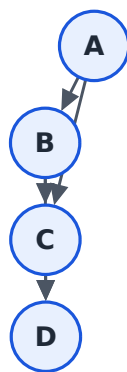
Q.21 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Rejection sampling is used with evidence $S = \text{true}$, $W = \text{true}$. What fraction of the prior samples is expected to be rejected? (Round to 3 decimal places.)

Answer (numerical): _____

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.84, 0.1, 0.75, 0.59)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T | \text{sampling parent values})$. Which sample is generated?



P(A=T)
0.6

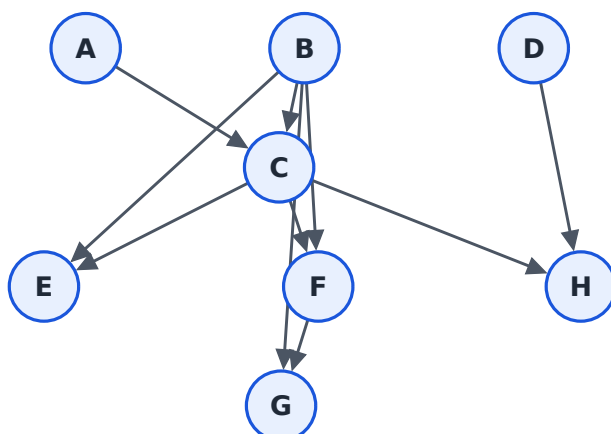
A	P(B=T .)	A	B	P(C=T ...)
T	0.9	T	T	0.25
F	0.7	T	F	0.05
		F	T	0.05
		F	F	0.4

B	C	P(D=T ...)
T	T	0.8
T	F	0.4
F	T	0.2
F	F	0.6

- ☐ (A) A=F, B=T, C=F, D=F
☐ (B) A=T, B=F, C=T, D=T
☐ (C) A=T, B=T, C=F, D=F
☐ (D) A=F, B=F, C=F, D=F

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

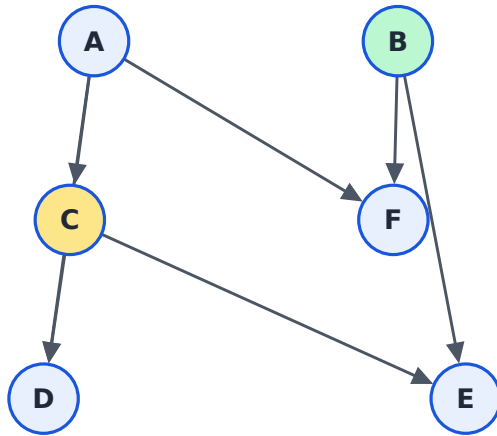
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp C$
☐ (B) $A \perp D \mid G$
☐ (C) $F \perp G$
☐ (D) $A \perp H \mid C$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

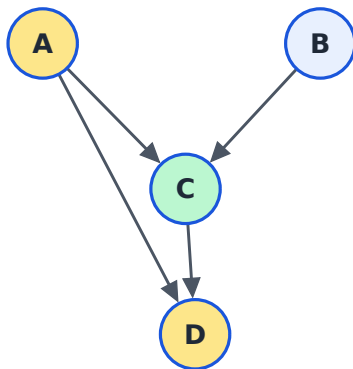
Variable elimination is used to compute $P(\mathbf{B} \mid \mathbf{C} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{C} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F**, **A**, **E**, **D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{A} = \mathbf{F}$, $\mathbf{D} = \mathbf{F}$. The values of the non-evidence variables in the generated samples are listed in the table. What is the weight of sample #3? (round off to 4 decimal places)



P(A=T)
0.3

P(B=T)	A	B	P(C=T ...)
0.6	T	T	0.95
	T	F	0.9
	F	T	0.75
	F	F	0.7

A	C	P(D=T ...)
T	T	0.5
T	F	0.75
F	T	0.2
F	F	0.4

#	B	C
1	T	T
2	T	T
3	T	F
4	F	T

Answer (numerical): _____

Answer Key — Mock Test 067

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	31	14	NAT	2	Adversarial Search	8
2	MCQ	1	Uninformed Search	D	15	NAT	2	Propositional Logic	11
3	NAT	1	Adversarial Search	17	16	MSQ	2	Predicate Logic	B, D
4	MCQ	1	Uninformed Search	B	17	MSQ	2	Propositional Logic	A
5	MCQ	1	Propositional Logic	A	18	NAT	2	Propositional Logic	64
6	MCQ	1	Propositional Logic	A	19	NAT	2	Exact Inference	0.093 to 0.097
7	MSQ	1	Propositional Logic	A, C, D	20	MSQ	2	Sampling	A, B, C
8	NAT	1	Propositional Logic	3	21	NAT	2	Sampling	0.721 to 0.723
9	MCQ	1	Variable Elimination	A	22	MCQ	2	Sampling	A
10	NAT	1	Bayesian Networks	15	23	MSQ	2	Conditional Independence	B, D
11	MSQ	2	Informed Search	B, C	24	NAT	2	Exact Inference	2
12	MCQ	2	Adversarial Search	A	25	NAT	2	Sampling	0.4198 to 0.4202
13	NAT	2	Informed Search	4					

Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] **Answer: 31**

$$\sum_{i=0..4} 2^i = (2^5 - 1)/(2 - 1) = \mathbf{31}.$$

Q.2 [MCQ, 1M · Uninformed Search] **Answer: D** (D: S, E, J, C, D, A, K, H, G)

Adjacency lists (alphabetical): A: {C, J, K}; C: {A, E, J, K}; D: {E, H, K}; E: {C, D, J, S}; G: {H, K}; H: {D, G}; J: {A, C, E, S}; K: {A, C, D, G}; S: {E, J}.

BFS trace (queue shown front → back):

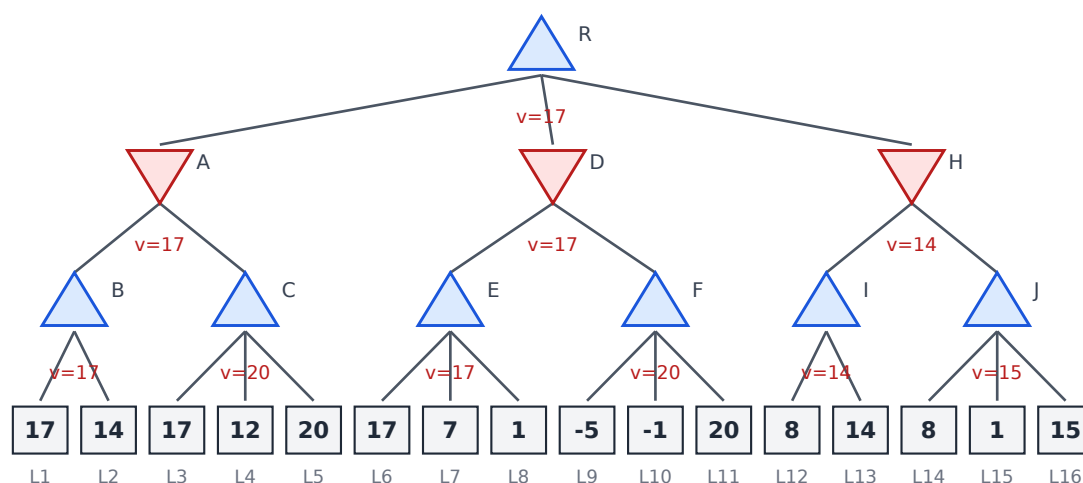
Step	Expanded	Frontier (FIFO) after expansion
1	S	[E, J]
2	E	[J, C, D]
3	J	[C, D, A]
4	C	[D, A, K]
5	D	[A, K, H]
6	A	[K, H]
7	K	[H, G]
8	H	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, E, J, C, D, A, K, H, G**. The path returned is $S \rightarrow E \rightarrow C \rightarrow K \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.3 [NAT, 1M · Adversarial Search] **Answer: 17**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 17$, $C = 20$, $A = 17$, $E = 17$, $F = 20$, $D = 17$, $I = 14$, $J = 15$, $H = 14$. Root value = **17**; MAX's optimal first move is towards A, D.

Q.4 [MCQ, 1M · Uninformed Search] **Answer: B** (B: $O(bm)$)

DFS tree search stores only the current path from the root plus the unexpanded siblings of each node on that path. That is at most about b nodes per level over m levels, so the space is **$O(bm)$** . $O(b^m)$ is its worst-case time, not its space. $O(b^d)$ is the space of BFS. $O(m^b)$ does not arise for any standard algorithm.

Q.5 [MCQ, 1M · Propositional Logic] **Answer: A** (A: Forward chaining is data-driven while backward chaining is goal-driven)

Forward chaining starts from known facts and fires rules whose premises are satisfied (data-driven). Backward chaining starts from the query and works back through rules whose conclusion matches (goal-driven). Both are complete only for Horn (definite clause) KBs, not arbitrary ones. For Horn KBs both derive exactly the entailed atoms, so neither derives anything the other cannot.

Q.6 [MCQ, 1M · Propositional Logic] **Answer: A** (A: $\neg P \wedge \neg Q$)

From $\neg R$ and $Q \rightarrow R$, modus tollens gives $\neg Q$; from $\neg Q$ and $P \rightarrow Q$ it gives $\neg P$. The only model of KB (over P, Q, R) is $P=F, Q=F, R=F$, in which $\neg P \wedge \neg Q$ is true. $P \vee Q, \neg P \wedge Q$ and R are all false in that model, so they are not entailed.

Q.7 [MSQ, 1M · Propositional Logic] **Answer: A, C, D** (A: $\neg p \vee \neg q \vee r$; C: $\neg p \vee \neg q$; D: r)

A Horn clause is a disjunction of literals with at most one positive literal. $\neg p \vee \neg q \vee r$ has one positive literal (a definite clause, i.e. $p \wedge q \rightarrow r$). $\neg p \vee \neg q$ has zero positive literals (a goal clause). r is a single positive literal (a fact). $p \vee q$ has two positive literals, so it is not Horn.

Q.8 [NAT, 1M · Propositional Logic] **Answer: 3**

Remove A and $\neg A$ and take the union of the remaining literals: $\{B, \neg C\} \cup \{B, D\} = \{B, \neg C, D\}$. The duplicate B is merged (factoring), so the resolvent $B \vee \neg C \vee D$ has 3 literals.

Q.9 [MCQ, 1M · Variable Elimination] **Answer: A** (A: NP-hard (in fact #P-hard))

AIMA shows that exact inference includes propositional satisfiability as a special case (3-SAT can be encoded), so it is NP-hard; computing the probability is #P-hard. It is linear only for polytrees and is certainly decidable (enumeration always terminates).

Q.10 [NAT, 1M · Bayesian Networks] **Answer: 15**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (3-1) \times 2 + C: (2-1) + D: (2-1) \times 3 + E: (2-1) \times 3 \times 2 = \mathbf{15}$.

Full joint table: $\prod |X| - 1 = 48 - 1 = 47$; saving = 32.

Q.11 [MSQ, 2M · Informed Search] **Answer: B, C** (B: The edge $A-J$ violates the consistency condition $h(A) \leq c(A, J) + h(J)$.; C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G :

n	A	B	E	F	G	J	S
$h(n)$	6	8	4	10	0	2	12

$h^*(n)$	8	10	6	13	0	10	15
$h \leq h^*$	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $A \rightarrow J$ ($6 > 2 + 2$), $F \rightarrow J$ ($10 > 6 + 2$), $S \rightarrow J$ ($12 > 5 + 2$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [MCQ, 2M · Adversarial Search] **Answer: A** (A: At MAX nodes in decreasing order of value, at MIN nodes in increasing order of value)

Pruning is greatest when the best move is examined first: the highest value for MAX and the lowest for MIN. That gives the tightest α (or β) as early as possible, so later siblings are cut sooner. (B) is the worst ordering, under which alpha-beta degenerates to $O(b^m)$. (C) is best for only one of the two players. (D) is false: ordering decides whether the cost is $O(b^{m/2})$ or $O(b^m)$.

Q.13 [NAT, 2M · Informed Search] **Answer: 4**

Heuristic values: $h(A)=4$, $h(B)=7$, $h(C)=7$, $h(D)=6$, $h(F)=3$, $h(G)=0$, $h(S)=7$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (7)	D(8)
2	D (8)	F(9), C(13)
3	F (9)	G(11), C(13), A(15)
4	G (11)	C(13), A(15) $\leftarrow$ goal, stop

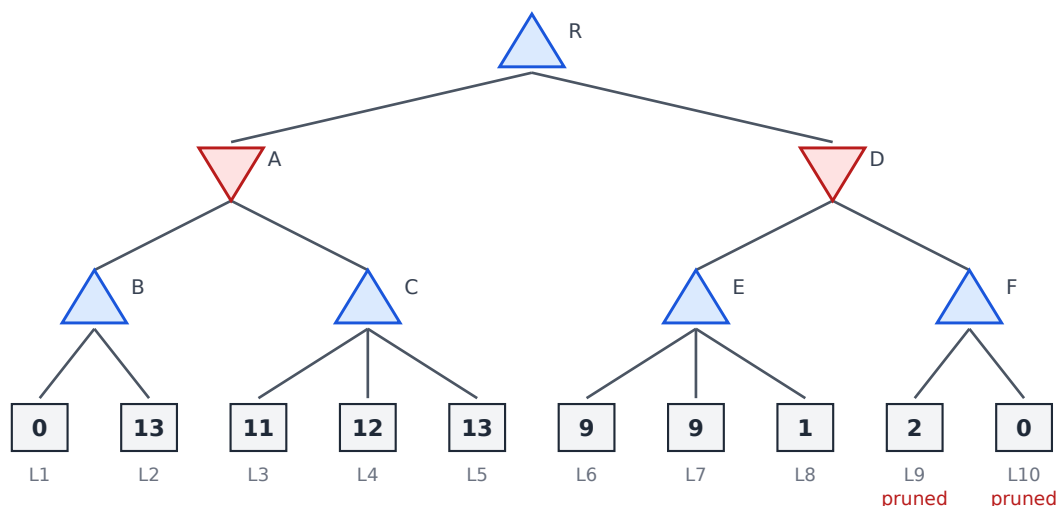
Path returned: **S $\rightarrow$ D $\rightarrow$ F $\rightarrow$ G** with cost **11**; nodes expanded = **4** (S, D, F, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 11$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.14 [NAT, 2M · Adversarial Search] **Answer: 8**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node D: value $9 \leq \alpha = 13 \Rightarrow$ prune remaining children (leaves L9, L10).



Leaves evaluated (8): L1, L2, L3, L4, L5, L6, L7, L8. Leaves pruned (2): L9, L10. Root value = 13 (same as plain minimax, 13 — pruning never changes the root value).

Q.15 [NAT, 2M · Propositional Logic] **Answer: 11**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(Q \vee P) \rightarrow \neg \neg R$	$(\neg P \rightarrow S) \wedge S$	$((Q \vee P) \rightarrow \neg \neg R) \rightarrow ((\neg P \rightarrow S) \wedge S)$
T	T	T	T	T	T	T
T	T	T	F	T	F	F
T	T	F	T	F	T	T
T	T	F	F	F	F	T
T	F	T	T	T	T	T
T	F	T	F	T	F	F

P	Q	R	S	$(Q \vee P) \rightarrow \neg \neg R$	$(\neg P \rightarrow S) \wedge S$	$((Q \vee P) \rightarrow \neg \neg R) \rightarrow ((\neg P \rightarrow S) \wedge S)$
T	F	F	T	F	T	T
T	F	F	F	F	F	T
F	T	T	T	T	T	T
F	T	T	F	T	F	F
F	T	F	T	F	T	T
F	T	F	F	F	F	T
F	F	T	T	T	T	T
F	F	T	F	T	F	F
F	F	F	T	T	T	T
F	F	F	F	T	F	F

Rows where F = T: **11**.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: B, D** (B: $\forall x \exists y (x \neq y \wedge R(x, y))$; D: $\forall x \exists y R(x, y)$)

$R = \{(1,2), (2,1), (2,4), (3,2), (3,4), (4,2)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(3,4)$ holds but $R(4,3)$ does not.

(B) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(C) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(D) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A** (A: A is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $E \wedge D \rightarrow A$; add A.

Round 2: fire $A \wedge D \rightarrow F$; add F.

Fixed point: {A, B, D, E, F} — 5 atoms are entailed (facts: B, D, E; newly derived: A, F). Atoms never derived: C, H, J, K.

Q.18 [NAT, 2M · Propositional Logic] **Answer: 64**

Up to equivalence a formula is determined by its set of models, a subset of the 8 assignments. $(p \wedge q) \models \phi$ iff $M(p \wedge q) \subseteq M(\phi)$. Over p, q, r, $M(p \wedge q) = \{TTT, TTF\}$ has 2 elements. So $M(\phi)$ must contain these 2 assignments and may contain any subset of the other 6: $2^6 = 64$.

Q.19 [NAT, 2M · Exact Inference] **Answer: 0.093 to 0.097**

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}, \text{Slippery}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Sprinkler}, \text{Rain}) \cdot P(\text{Slippery} \mid \text{Cloudy}, \text{Rain})$.

Inference by enumeration: $P(\text{Cloudy} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cloudy}, e, \text{hidden})$.

Hidden: Sp, Ra, SI	CI	product of CPT entries (CI · Sp · Ra · We · SI)	value
T, T, T	T	$0.1 \times 0.1 \times 0.4 \times 0.8 \times 0.7$	0.00224
T, T, F	T	$0.1 \times 0.1 \times 0.4 \times 0.8 \times 0.3$	0.00096
T, F, T	T	$0.1 \times 0.1 \times 0.6 \times 0.75 \times 0.05$	0.00023
T, F, F	T	$0.1 \times 0.1 \times 0.6 \times 0.75 \times 0.95$	0.00428
F, T, T	T	$0.1 \times 0.9 \times 0.2 \times 0.95 \times 0.7$	0.01197
F, T, F	T	$0.1 \times 0.9 \times 0.2 \times 0.95 \times 0.3$	0.00513
F, F, T	T	$0.1 \times 0.9 \times 0.8 \times 0.7 \times 0.05$	0.00252
F, F, F	T	$0.1 \times 0.9 \times 0.8 \times 0.7 \times 0.95$	0.04788
T, T, T	F	$0.9 \times 0.95 \times 0.8 \times 0.8 \times 0.8$	0.43776
T, T, F	F	$0.9 \times 0.95 \times 0.8 \times 0.8 \times 0.2$	0.10944
T, F, T	F	$0.9 \times 0.95 \times 0.2 \times 0.75 \times 0.6$	0.07695
T, F, F	F	$0.9 \times 0.95 \times 0.2 \times 0.75 \times 0.4$	0.0513
F, T, T	F	$0.9 \times 0.05 \times 0.95 \times 0.95 \times 0.8$	0.03249
F, T, F	F	$0.9 \times 0.05 \times 0.95 \times 0.95 \times 0.2$	0.00812
F, F, T	F	$0.9 \times 0.05 \times 0.05 \times 0.7 \times 0.6$	0.00095
F, F, F	F	$0.9 \times 0.05 \times 0.05 \times 0.7 \times 0.4$	0.00063

$P(\text{Cloudy}=T, e) = 0.0752$; $P(\text{Cloudy}=F, e) = 0.71764$; $P(e) = 0.79284$.

$P(\text{Cloudy}=T \mid e) = 0.0752 / 0.79284 = \mathbf{0.0948} \approx 0.095$.

Q.20 [MSQ, 2M · Sampling] Answer: A, B, C

(A) Each state is obtained by modifying the previous one, so samples are autocorrelated. (B) Gibbs steps satisfy detailed balance with respect to $P(x \mid e)$ and the chain is ergodic when there are no deterministic (0/1) entries, so it converges to the posterior. (C) The state is a full assignment with evidence clamped. (D) describes rejection sampling; Gibbs never rejects since evidence is clamped.

Q.21 [NAT, 2M · Sampling] Answer: 0.721 to 0.723

A sample is kept with probability $P(s, w)$. $P(s, w) = \sum_{c,r} P(c)P(s \mid c)P(r \mid c)P(w \mid s, r)$. $c: 0.5 \times 0.1 \times (0.8 \times 0.99 + 0.2 \times 0.90) = 0.05 \times 0.972 = 0.0486$. $\neg c: 0.5 \times 0.5 \times (0.2 \times 0.99 + 0.8 \times 0.90) = 0.25 \times 0.918 = 0.2295$. $P(s, w) = 0.2781$. Expected rejected fraction = $1 - 0.2781 = 0.7219 \approx 0.722$.

Q.22 [MCQ, 2M · Sampling] Answer: A (A: A=F, B=T, C=F, D=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.6$; $u = 0.84 \geq 0.6 \Rightarrow A = F$
- B: $P(B=T \mid A=F) = 0.7$; $u = 0.1 < 0.7 \Rightarrow B = T$
- C: $P(C=T \mid A=F, B=T) = 0.05$; $u = 0.75 \geq 0.05 \Rightarrow C = F$
- D: $P(D=T \mid B=T, C=F) = 0.4$; $u = 0.59 \geq 0.4 \Rightarrow D = F$

Sample: **(A=F, B=T, C=F, D=F)**.

Q.23 [MSQ, 2M · Conditional Independence] Answer: B, D (B: $A \perp D \mid G$; D: $A \perp H \mid C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp C$: **FALSE** — the path $G \leftarrow B \rightarrow C$ is active given $\{\}$.

(B) $A \perp D \mid G$: **TRUE** — $A \rightarrow C \rightarrow H \leftarrow D$ (blocked: collider H with neither it nor a descendant observed).

(C) $F \perp G$: **FALSE** — the path $F \leftarrow B \rightarrow G$ is active given $\{\}$.

(D) $A \perp H \mid C$: **TRUE** — $A \rightarrow C \rightarrow H$ (blocked: observed chain node C).

Q.24 [NAT, 2M · Exact Inference] Answer: 2

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A)$, $f(A, D)$, $f(B, E)$, $f(A, B, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(A, B, F)$	$\{A, B, F\}$	$f(A, B)$
A	$f(A)$, $f(A)$, $f(A, D)$, $f(A, B)$	$\{A, B, D\}$	$f(B, D)$
E	$f(B, E)$	$\{B, E\}$	$f(B)$
D	$f(B, D)$	$\{B, D\}$	$f(B)$

The largest factor produced has **2** variable(s) (created when eliminating F); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over B and normalise.

Q.25 [NAT, 2M · Sampling] Answer: 0.4198 to 0.4202

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	C
1	$P(A=F)=0.7 \times P(D=F A=F,C=T)=0.8$	0.56	T
2	$P(A=F)=0.7 \times P(D=F A=F,C=T)=0.8$	0.56	T
3	$P(A=F)=0.7 \times P(D=F A=F,C=F)=0.6$	0.42	F
4	$P(A=F)=0.7 \times P(D=F A=F,C=T)=0.8$	0.56	T

$\Sigma w (\text{all}) = 2.1$; $\Sigma w (C = T) = 1.68$.

Estimate $P(C=T \mid A = F, D = F) = 1.68 / 2.1 = \mathbf{0.8}$.

Mock Test 068

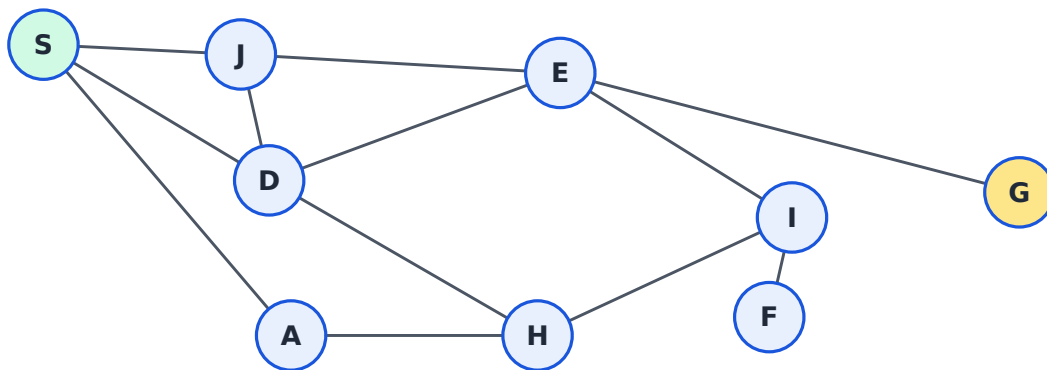
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 6 / 12	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

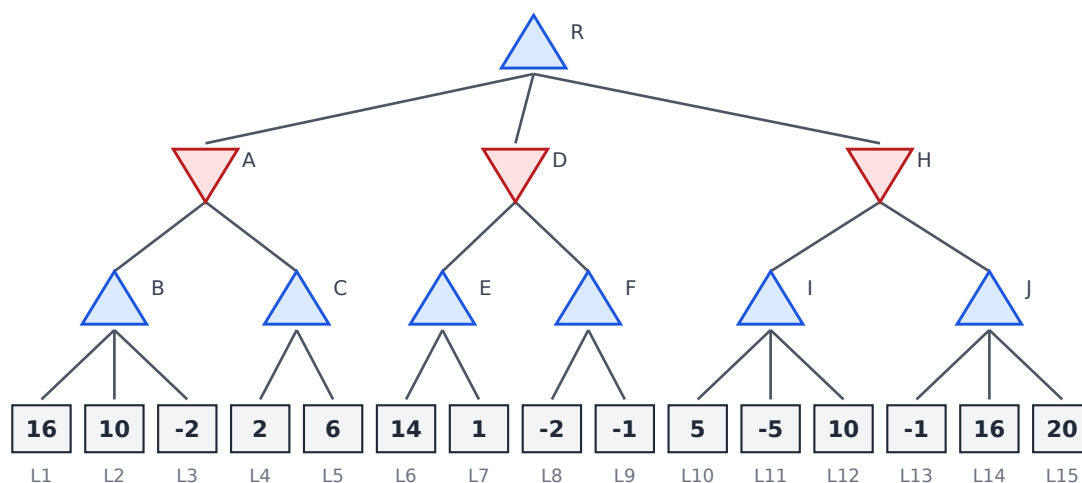
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, A, D, J, H, E, I, G
- ☐ (B) S, A, H, D, E, G
- ☐ (C) S, J, E, I, H, D, A, F, G
- ☐ (D) S, H, A, D, E, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Informed Search

A heuristic h is **admissible** if, for every node n (where $h^*(n)$ is the true cost of an optimal path from n to a goal),

- ☐ (A) $h(n) \geq h^*(n)$
☐ (B) $h(n) \leq h^*(n)$
☐ (C) $h(n) \leq c(n, a, n') + h(n')$ for every successor n'
☐ (D) $h(n) = 0$

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 5$ (so $b^d = 1024$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following are tautologies?

- ☐ (A) $p \vee \neg p$
☐ (B) $\neg(p \wedge \neg p)$
☐ (C) $p \rightarrow (p \wedge q)$
☐ (D) $(p \wedge (p \rightarrow q)) \rightarrow q$

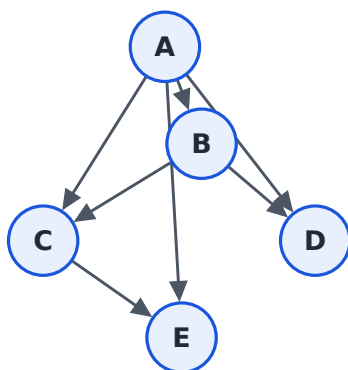
Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following is logically equivalent to $p \rightarrow q$?

- ☐ (A) $\neg q \rightarrow \neg p$
☐ (B) $q \rightarrow p$
☐ (C) $\neg p \rightarrow \neg q$
☐ (D) $p \wedge \neg q$

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D, E . What is the probability that a single run produces the sample **(A=F, B=F, C=F, D=F, E=F)**? (round off to 4 decimal places)



$P(A=T)$
0.4

A	$P(B=T \dots)$	A	B	$P(C=T \dots)$
T	0.5	T	T	0.3
F	0.25	T	F	0.25
		F	T	0.75
		F	F	0.3

A	B	$P(D=T \dots)$	A	C	$P(E=T \dots)$
T	T	0.05	T	T	0.2
T	F	0.8	T	F	0.05
F	T	0.05	F	T	0.05
F	F	0.1	F	F	0.1

Answer (numerical): _____

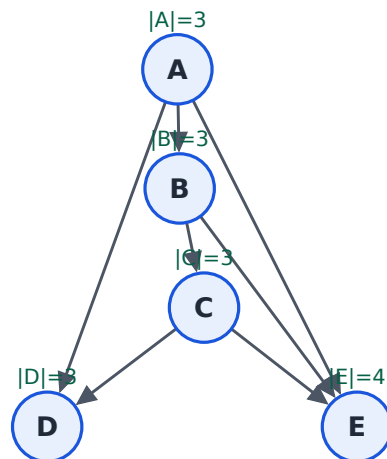
Q.9 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

Which statement is the local Markov property (local semantics) of a Bayesian network?

- ☐ (A) Each node is conditionally independent of its non-descendants given its parents
- ☐ (B) Each node is conditionally independent of its descendants given its parents
- ☐ (C) Each node is marginally independent of all nodes that are not its parents
- ☐ (D) Each node is conditionally independent of its parents given its children

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

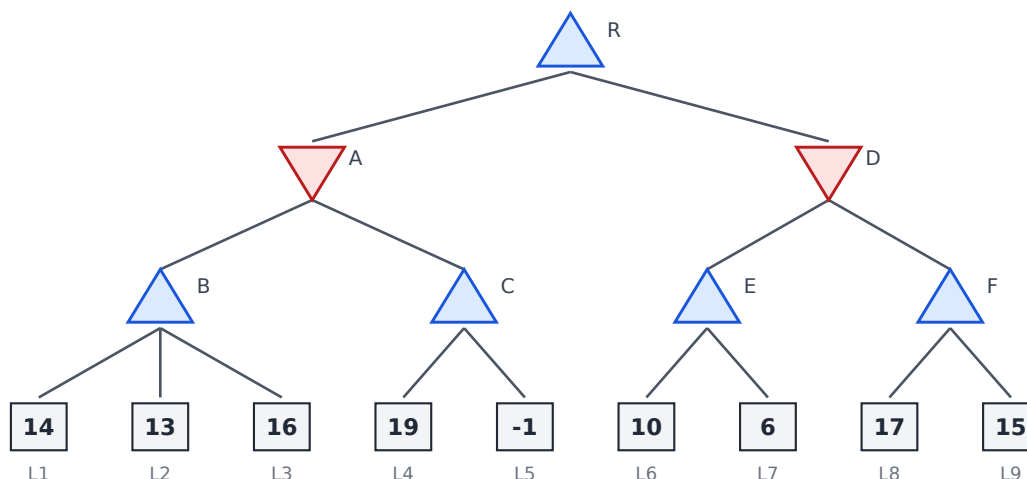
The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Section B – 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



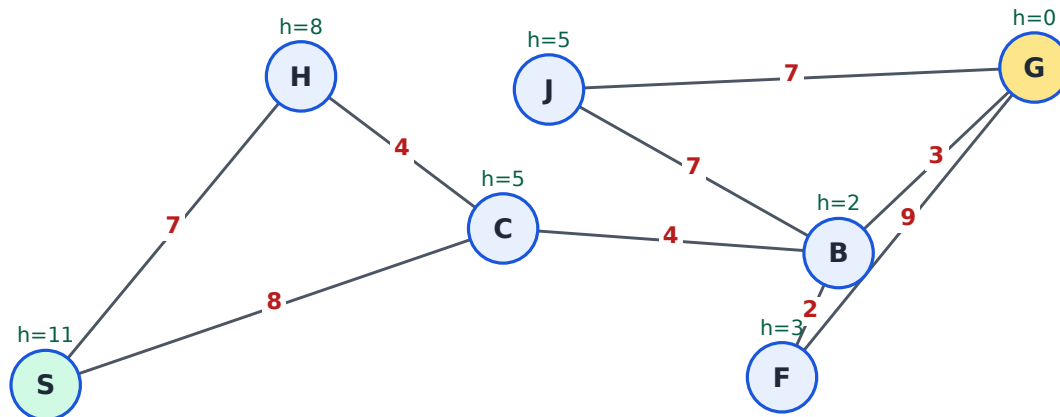
MAX node MIN node

terminal utility

- ☐ (A) Leaf L4 is pruned.
- ☐ (B) Leaf L7 is pruned.
- ☐ (C) Leaf L8 is pruned.
- ☐ (D) Leaf L9 is pruned.

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

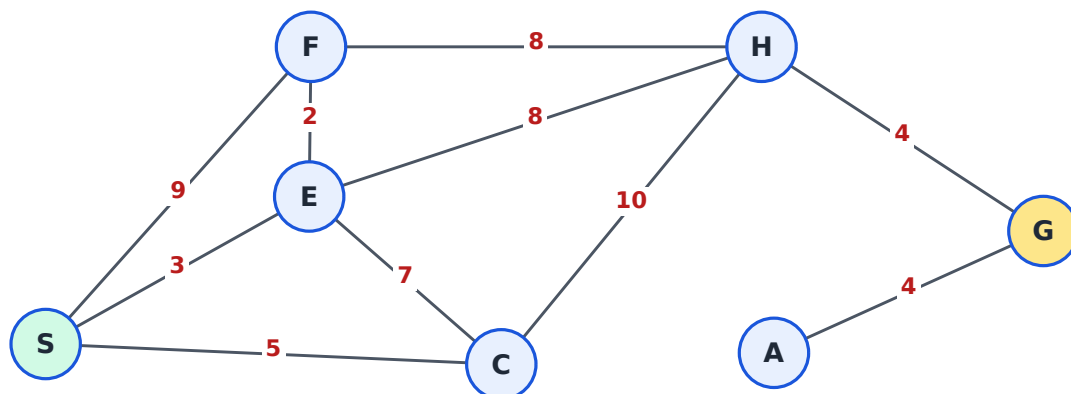
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow C \rightarrow H \rightarrow G$
☐ (B) $S \rightarrow F \rightarrow H \rightarrow G$
☐ (C) $S \rightarrow E \rightarrow H \rightarrow G$
☐ (D) $S \rightarrow E \rightarrow F \rightarrow H \rightarrow G$

Q.14 [MCQ · 2 marks · negative: -2/3] Informed Search

A* graph search runs with a consistent heuristic, and C^* is the optimal solution cost. Which statement is correct?

- ☐ (A) Every node reachable from the start with $f(n) < C^*$ is expanded
☐ (B) No node with $f(n) < C^*$ is ever expanded
☐ (C) Nodes with $f(n) > C^*$ may be expanded before the goal is selected
☐ (D) A* expands exactly the nodes on the optimal path and no others

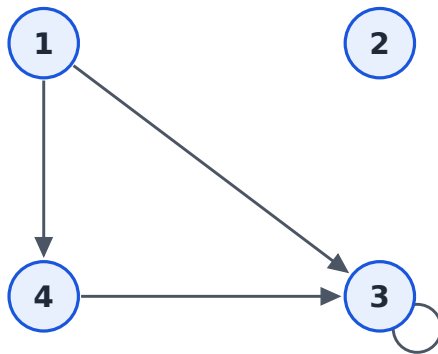
Q.15 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2\}$. How many interpretations of a binary predicate R over D make the sentence $\forall x \exists y R(x, y) \rightarrow \exists y \forall x R(x, y)$ true?

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists y \forall x R(x, y)$
☐ (B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (C) $\forall x \exists y R(x, y)$
☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (R \rightarrow Q) \rightarrow R$?

- ☐ (A) $(R \rightarrow Q) \vee \neg R$
☐ (B) $\neg(R \rightarrow Q) \vee R$
☐ (C) $R \rightarrow (R \rightarrow Q)$
☐ (D) $\neg((R \rightarrow Q) \wedge \neg R)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg\neg Q \leftrightarrow (P \wedge \neg R)$
 (2) $\neg P \vee (R \wedge R)$
 (3) $\neg(P \vee \neg R)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg Q \wedge (\neg P \vee \neg R)$
☐ (B) $(R \leftrightarrow P) \wedge (\neg Q \leftrightarrow Q)$
☐ (C) $P \leftrightarrow \neg R$
☐ (D) $(Q \vee R) \leftrightarrow (Q \leftrightarrow \neg Q)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

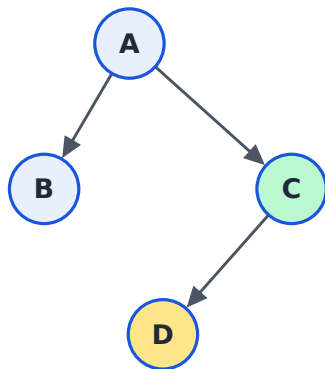
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$F = (\neg\neg P \wedge (Q \rightarrow \neg Q)) \leftrightarrow ((\neg S \leftrightarrow \neg Q) \leftrightarrow (S \leftrightarrow \neg R))$?

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{D} = \mathbf{F}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(C = T \mid D = F)}$ (round off to 3 decimal places).



P(A=T)
0.95

A	P(B=T ...)	A	P(C=T ...)
T	0.3	T	0.5
F	0.75	F	0.3

C	P(D=T ...)
T	0.95
F	0.3

#	A	B	C
1	T	F	F
2	T	F	F
3	T	T	T
4	T	F	T
5	T	T	F

Answer (numerical): _____

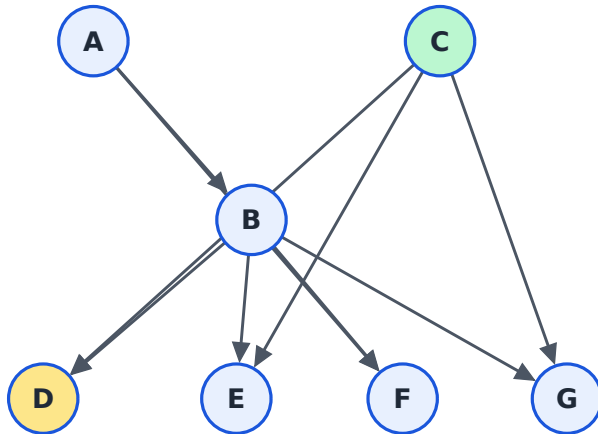
Q.21 [MCQ · 2 marks · negative: -2/3] Variable Elimination

For a Bayesian network with n Boolean variables, the depth-first ENUMERATION-ASK algorithm (inference by enumeration) has worst-case

- ☐ (A) $O(2^n)$ time and $O(n)$ space
- ☐ (B) $O(n)$ time and $O(2^n)$ space
- ☐ (C) $O(2^n)$ time and $O(2^n)$ space
- ☐ (D) $O(n^2)$ time and $O(n)$ space

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

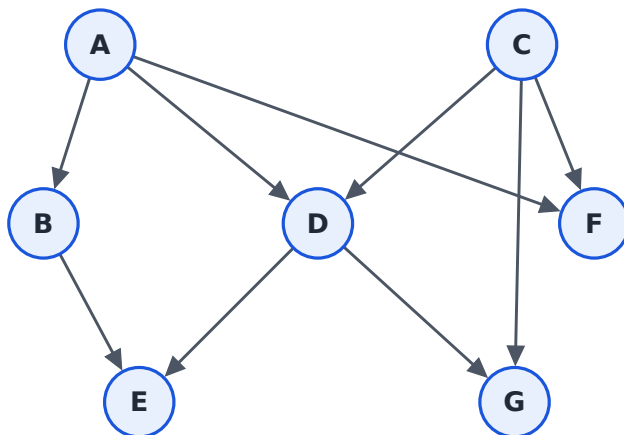
Variable elimination is used to compute $P(\mathbf{C} \mid \mathbf{D} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{D} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, G, F, E, B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp G \mid B, F$
☐ (B) $G \perp E \mid A, D$
☐ (C) $E \perp C \mid B, D$
☐ (D) $B \perp C$

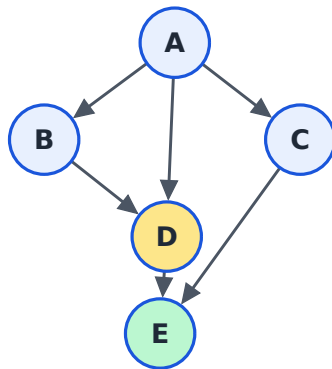
Q.24 [NAT · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $A \rightarrow C$ and $B \rightarrow C$ (Boolean variables) with $P(a) = 0.1$, $P(b) = 0.2$, $P(c \mid a, b) = 0.95$, $P(c \mid a, \neg b) = 0.90$, $P(c \mid \neg a, b) = 0.80$, $P(c \mid \neg a, \neg b) = 0.05$. Compute $P(a \mid c, b)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(E = T \mid D = T)$ (round off to 3 decimal places).



P(A=T)
0.25

A	P(B=T ...)	A	P(C=T ...)
T	0.2	T	0.4
F	0.9	F	0.5

A	B	P(D=T)	C	D	P(E=T ...)
T	T	0.3	T	T	0.4
T	F	0.3	T	F	0.5
F	T	0.9	F	T	0.25
F	F	0.05	F	F	0.05

Answer (numerical): _____

Answer Key — Mock Test 068

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	B	14	MCQ	2	Informed Search	A
2	NAT	1	Adversarial Search	10	15	NAT	2	Predicate Logic	14
3	MCQ	1	Informed Search	B	16	MSQ	2	Predicate Logic	B
4	NAT	1	Adversarial Search	79	17	MSQ	2	Propositional Logic	B, D
5	NAT	1	Predicate Logic	65	18	MSQ	2	Propositional Logic	A, C
6	MSQ	1	Propositional Logic	A, B, D	19	NAT	2	Propositional Logic	8
7	MCQ	1	Propositional Logic	A	20	NAT	2	Sampling	0.043 to 0.047
8	NAT	1	Sampling	0.255 to 0.2553	21	MCQ	2	Variable Elimination	A
9	MCQ	1	Bayesian Networks	A	22	NAT	2	Exact Inference	2
10	NAT	1	Bayesian Networks	113	23	MSQ	2	Conditional Independence	B, C, D
11	MSQ	2	Adversarial Search	C, D	24	NAT	2	Conditional Independence	0.116 to 0.117
12	NAT	2	Informed Search	4	25	NAT	2	Exact Inference	0.321 to 0.325
13	MCQ	2	Uninformed Search	C					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: B** (B: S, A, H, D, E, G)

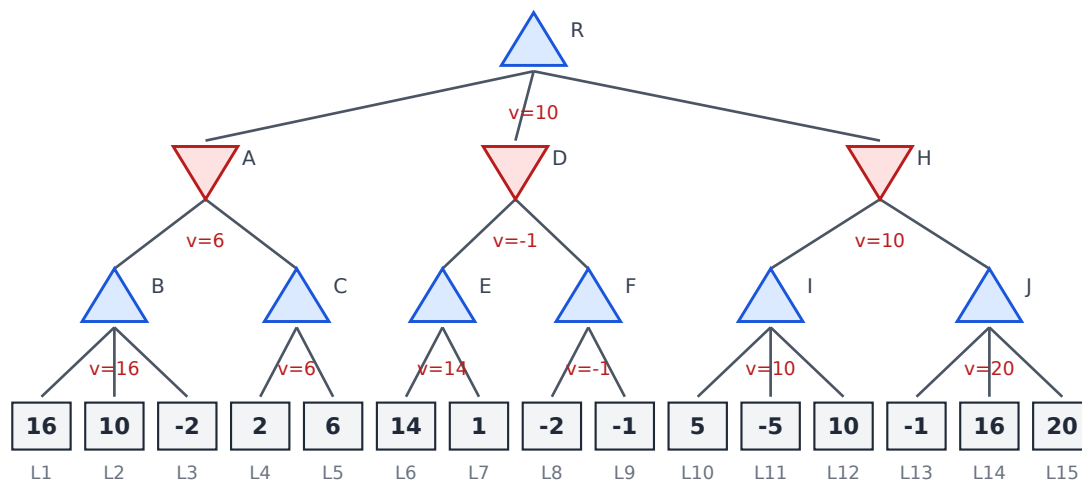
Adjacency lists (alphabetical): A: {H, S}; D: {E, H, J, S}; E: {D, G, I, J}; F: {I}; G: {E}; H: {A, D, I}; I: {E, F, H}; J: {D, E, S}; S: {A, D, J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, A, H, D, E, G**; the path found is $S \rightarrow A \rightarrow H \rightarrow D \rightarrow E \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] **Answer: 10**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 16$, $C = 6$, $A = 6$, $E = 14$, $F = -1$, $D = -1$, $I = 10$, $J = 20$, $H = 10$. Root value = **10**; MAX's optimal first move is towards H.

Q.3 [MCQ, 1M · Informed Search] Answer: B ($B: h(n) \leq h^*(n)$)

An admissible heuristic never overestimates the cost to reach a goal: $h(n) \leq h^*(n)$. (A) describes a pessimistic, inadmissible heuristic. (C) is the definition of consistency, which is a stronger condition. (D) is one particular admissible heuristic, not the definition.

Q.4 [NAT, 1M · Adversarial Search] Answer: 79

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 4^3 + 4^2 - 1 = 64 + 16 - 1 = \mathbf{79}$, versus 1024 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.5 [NAT, 1M · Predicate Logic] Answer: 65

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 4$ this gives **65** (verified by enumerating all interpretations).

Q.6 [MSQ, 1M · Propositional Logic] Answer: A, B, D ($A: p \vee \neg p$; $B: \neg(p \wedge \neg p)$; $D: (p \wedge (p \rightarrow q)) \rightarrow q$)

$p \vee \neg p$ (excluded middle) and $\neg(p \wedge \neg p)$ (non-contradiction) are valid. $(p \wedge (p \rightarrow q)) \rightarrow q$ is the modus ponens tautology. $p \rightarrow (p \wedge q)$ is false at $p=T, q=F$, so it is not valid.

Q.7 [MCQ, 1M · Propositional Logic] Answer: A ($A: \neg q \rightarrow \neg p$)

The contrapositive $\neg q \rightarrow \neg p \equiv q \vee \neg p \equiv p \rightarrow q$. The converse $q \rightarrow p$ and the inverse $\neg p \rightarrow \neg q$ are equivalent to each other but differ from $p \rightarrow q$ at $p=F, q=T$ ($p \rightarrow q$ is T, $q \rightarrow p$ is F). $p \wedge \neg q$ is the negation of $p \rightarrow q$.

Q.8 [NAT, 1M · Sampling] Answer: 0.255 to 0.2553

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(A=F) \times P(B=F|A=F) \times P(C=F|A=F, B=F) \times P(D=F|A=F, B=F) \times P(E=F|A=F, C=F)$
 $= 0.6 \times 0.75 \times 0.7 \times 0.9 \times 0.9 = \mathbf{0.25515} \approx 0.2551$.

Q.9 [MCQ, 1M · Bayesian Networks] Answer: A (A: Each node is conditionally independent of its non-descendants given its parents)

AIMA: a node is conditionally independent of its non-descendants given its parents; this is equivalent to the global factorization semantics. (B) is false because descendants (e.g. children) carry information about the node even given the parents. (C) ignores dependence through common ancestors, and (D) has no basis.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 113

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

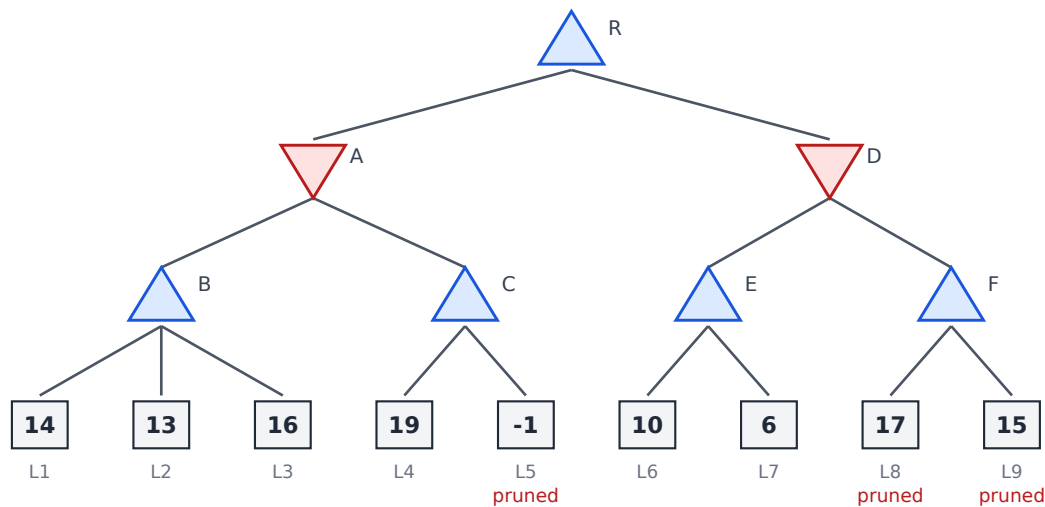
$\Sigma = A: (3-1) + B: (3-1) \times 3 + C: (3-1) \times 3 + D: (3-1) \times 3 \times 3 + E: (4-1) \times 3 \times 3 \times 3 = \mathbf{113}$.

Full joint table: $\prod |X| - 1 = 324 - 1 = 323$; saving = 210.

Q.11 [MSQ, 2M · Adversarial Search] Answer: C, D (C: Leaf L8 is pruned.; D: Leaf L9 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $19 \geq \beta = 16 \Rightarrow$ prune remaining children (leaves L5).
- At MIN node D: value $10 \leq \alpha = 16 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (6): L1, L2, L3, L4, L6, L7. Leaves pruned (3): L5, L8, L9. Root value = 16 (same as plain minimax, 16 — pruning never changes the root value).

Q.12 [NAT, 2M · Informed Search] Answer: 4

Heuristic values: $h(B)=2$, $h(C)=5$, $h(F)=3$, $h(G)=0$, $h(H)=8$, $h(J)=5$, $h(S)=11$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	C(13), H(15)
2	C (13)	B(14), H(15)
3	B (14)	G(15), H(15), F(17), J(24)
4	G (15)	H(15), F(17), J(24) ← goal, stop

Path returned: **S** → **C** → **B** → **G** with cost **15**; nodes expanded = **4** (S, C, B, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 6 nodes.

Q.13 [MCQ, 2M · Uninformed Search] Answer: C (C: $S \rightarrow E \rightarrow H \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	E(3), C(5), F(9)
2	E (3)	C(5), F(5), H(11)
3	C (5)	F(5), H(11)
4	F (5)	H(11)
5	H (11)	G(15)
6	G (15)	← goal, stop

Path returned: **S** → **E** → **H** → **G**, cost = $3 + 8 + 4 = 15$. Nodes expanded (including S and G): **6** (S, E, C, F, H, G).

Q.14 [MCQ, 2M · Informed Search] Answer: A (A: Every node reachable from the start with $f(n) < C^*$ is expanded)

With a consistent h , A* expands nodes in non-decreasing order of f . It therefore expands **all nodes with $f < C^*$** , some nodes with $f = C^*$, and none with $f > C^*$. (B) is the opposite. (C) contradicts that ordering, because the goal with $f = C^*$ would be selected first. (D) is true only for a perfect heuristic with favourable tie-breaking.

Q.15 [NAT, 2M · Predicate Logic] **Answer: 14**

There are $2^4 = 16$ relations. The sentence is false iff every row of the 2×2 matrix is non-empty but no column is full. A column is full if both rows contain that column; so each row must be non-empty and the rows must have no common column, forcing each row to be a single, different column: $R = \{(1,1), (2,2)\}$ or $R = \{(1,2), (2,1)\}$. So 2 interpretations falsify it and $16 - 2 = 14$ satisfy it.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: B** (B: $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$)

$R = \{(1,3), (1,4), (3,3), (4,3)\}$.

(A) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **TRUE** — all two-step paths are short-cut.

(C) $\forall x \exists y R(x, y)$: **FALSE** — 2 has no outgoing edge.

(D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 2 has no edge to a different element.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: B, D** (B: $\neg(R \rightarrow Q) \vee R$; D: $\neg((R \rightarrow Q) \wedge \neg R)$)

Compare truth tables with $F = (R \rightarrow Q) \rightarrow R$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(R \rightarrow Q) \vee \neg R$ — differs at $Q=T, R=F$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(B) $\neg(R \rightarrow Q) \vee R$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $R \rightarrow (R \rightarrow Q)$ — differs at $Q=T, R=F$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(D) $\neg((R \rightarrow Q) \wedge \neg R)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: $\neg Q \wedge (\neg P \vee \neg R)$; C: $P \leftrightarrow \neg R$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
F	F	T	T

(A) $\neg Q \wedge (\neg P \vee \neg R)$ is true in all models of $KB \Rightarrow$ **entailed**.

(B) $(R \leftrightarrow P) \wedge (\neg Q \leftrightarrow Q)$ is false in the KB -model $P=F, Q=F, R=T \Rightarrow$ not entailed.

(C) $P \leftrightarrow \neg R$ is true in all models of $KB \Rightarrow$ **entailed**.

(D) $(Q \vee R) \leftrightarrow (Q \leftrightarrow \neg Q)$ is false in the KB -model $P=F, Q=F, R=T \Rightarrow$ not entailed.

Q.19 [NAT, 2M · Propositional Logic] **Answer: 8**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$\neg\neg P \wedge (Q \rightarrow \neg Q)$	$(\neg S \leftrightarrow \neg Q) \leftrightarrow (S \leftrightarrow \neg R)$	$(\neg\neg P \wedge (Q \rightarrow \neg Q)) \leftrightarrow ((\neg S \leftrightarrow \neg Q) \leftrightarrow (S \leftrightarrow \neg R))$
T	T	T	T	F	F	T
T	T	T	F	F	F	T
T	T	F	T	F	T	F
T	T	F	F	F	T	F
T	F	T	T	T	T	T
T	F	T	F	T	T	T
T	F	F	T	T	F	F
T	F	F	F	T	F	F
F	T	T	T	F	F	T
F	T	T	F	F	F	T
F	T	F	T	F	T	F
F	T	F	F	F	T	F
F	F	T	T	F	T	F
F	F	T	F	F	T	F
F	F	F	T	F	F	T
F	F	F	F	F	F	T

Rows where $F = T$: **8**.

Q.20 [NAT, 2M · Sampling] **Answer: 0.043 to 0.047**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	C
1	$P(D=F C=F)=0.7$	0.7	F

#	weight $w = \prod P(e \mid \text{parents})$	w	C
2	$P(D=F C=F)=0.7$	0.7	F
3	$P(D=F C=T)=0.05$	0.05	T
4	$P(D=F C=T)=0.05$	0.05	T
5	$P(D=F C=F)=0.7$	0.7	F

$\sum w \text{ (all)} = 2.2$; $\sum w \text{ (C = T)} = 0.1$.

Estimate $P(C=T \mid D = F) = 0.1 / 2.2 = \mathbf{0.0455}$.

Q.21 [MCQ, 2M · Variable Elimination] **Answer: A** (A: $O(2^n)$ time and $O(n)$ space)

Enumeration evaluates a depth-first expression tree that branches over every hidden variable, giving $O(2^n)$ time. Being depth-first it stores only the current path, so space is $O(n)$. It never builds the full joint table, so it does not need $O(2^n)$ space; polynomial time is impossible in general since inference is NP-hard. Variable elimination improves time by caching repeated subexpressions as factors.

Q.22 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(C)$, $f(B, C)$, $f(B, C, E)$, $f(A, B, F)$, $f(B, C, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after $\sum$)
A	$f(A)$, $f(A, B)$, $f(A, B, F)$	$\{A, B, F\}$	$f(B, F)$
G	$f(B, C, G)$	$\{B, C, G\}$	$f(B, C)$
F	$f(B, F)$	$\{B, F\}$	$f(B)$
E	$f(B, C, E)$	$\{B, C, E\}$	$f(B, C)$
B	$f(B, C)$, $f(B, C)$, $f(B)$, $f(B, C)$	$\{B, C\}$	$f(C)$

The largest factor produced has **2** variable(s) (created when eliminating A); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over C and normalise.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: B, C, D** (B: $G \perp E \mid A, D$; C: $E \perp C \mid B, D$; D: $B \perp C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp G \mid B, F$: **FALSE** — the path $E \leftarrow D \leftarrow A \rightarrow F \leftarrow C \rightarrow G$ is active given $\{B, F\}$.

(B) $G \perp E \mid A, D$: **TRUE** — $G \leftarrow C \rightarrow D \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed fork node A); $G \leftarrow C \rightarrow D \rightarrow E$ (blocked: observed chain node D); $G \leftarrow C \rightarrow F \leftarrow A \rightarrow B \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow C \rightarrow F \leftarrow A \rightarrow D \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow D \leftarrow C \rightarrow F \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed chain node D) (+1 more paths, all blocked).

(C) $E \perp C \mid B, D$: **TRUE** — $E \leftarrow B \leftarrow A \rightarrow D \leftarrow C$ (blocked: observed chain node B); $E \leftarrow B \leftarrow A \rightarrow D \rightarrow G \leftarrow C$ (blocked: observed chain node B); $E \leftarrow B \leftarrow A \rightarrow F \leftarrow C$ (blocked: observed chain node B); $E \leftarrow D \leftarrow A \rightarrow F \leftarrow C$ (blocked: observed chain node D); $E \leftarrow D \leftarrow C$ (blocked: observed chain node D); $E \leftarrow D \rightarrow G \leftarrow C$ (blocked: observed fork node D).

(D) $B \perp C$: **TRUE** — $B \leftarrow A \rightarrow D \leftarrow C$ (blocked: collider D with neither it nor a descendant observed); $B \leftarrow A \rightarrow D \rightarrow G \leftarrow C$ (blocked: collider G with neither it nor a descendant observed); $B \leftarrow A \rightarrow F \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow E \leftarrow D \leftarrow A \rightarrow F \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow E \leftarrow D \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow E \leftarrow D \rightarrow G \leftarrow C$ (blocked: collider E with neither it nor a descendant observed).

Q.24 [NAT, 2M · Conditional Independence] **Answer: 0.116 to 0.117**

Since $A \perp B$, $P(A \mid c, b) = \alpha P(A) P(c \mid A, b)$ ($P(b)$ cancels). For a: $0.1 \times 0.95 = 0.095$. For $\neg a$: $0.9 \times 0.80 = 0.72$. $P(a \mid c, b) = 0.095 / (0.095 + 0.72) = 0.095 / 0.815 \approx 0.1166 \approx 0.117$. B already explains C, so A remains near its prior 0.1.

Q.25 [NAT, 2M · Exact Inference] **Answer: 0.321 to 0.325**

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B \mid A) \cdot P(C \mid A) \cdot P(D \mid A, B) \cdot P(E \mid C, D)$.

Inference by enumeration: $P(E \mid e) = \alpha \sum_{\text{hidden}} P(E, e, \text{hidden})$.

Hidden: A, B, C	E	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T, T	T	$0.25 \times 0.2 \times 0.4 \times 0.3 \times 0.4$	0.0024
T, T, F	T	$0.25 \times 0.2 \times 0.6 \times 0.3 \times 0.25$	0.00225
T, F, T	T	$0.25 \times 0.8 \times 0.4 \times 0.3 \times 0.4$	0.0096

Hidden: A, B, C	E	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, F, F	T	$0.25 \times 0.8 \times 0.6 \times 0.3 \times 0.25$	0.009
F, T, T	T	$0.75 \times 0.9 \times 0.5 \times 0.9 \times 0.4$	0.1215
F, T, F	T	$0.75 \times 0.9 \times 0.5 \times 0.9 \times 0.25$	0.07594
F, F, T	T	$0.75 \times 0.1 \times 0.5 \times 0.05 \times 0.4$	0.00075
F, F, F	T	$0.75 \times 0.1 \times 0.5 \times 0.05 \times 0.25$	0.00047
T, T, T	F	$0.25 \times 0.2 \times 0.4 \times 0.3 \times 0.6$	0.0036
T, T, F	F	$0.25 \times 0.2 \times 0.6 \times 0.3 \times 0.75$	0.00675
T, F, T	F	$0.25 \times 0.8 \times 0.4 \times 0.3 \times 0.6$	0.0144
T, F, F	F	$0.25 \times 0.8 \times 0.6 \times 0.3 \times 0.75$	0.027
F, T, T	F	$0.75 \times 0.9 \times 0.5 \times 0.9 \times 0.6$	0.18225
F, T, F	F	$0.75 \times 0.9 \times 0.5 \times 0.9 \times 0.75$	0.22781
F, F, T	F	$0.75 \times 0.1 \times 0.5 \times 0.05 \times 0.6$	0.00112
F, F, F	F	$0.75 \times 0.1 \times 0.5 \times 0.05 \times 0.75$	0.00141

$P(E=T, e) = 0.22191$; $P(E=F, e) = 0.46434$; $P(e) = 0.68625$.

$P(E=T | e) = 0.22191 / 0.68625 = \mathbf{0.3234} \approx 0.323$.

Mock Test 069

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 8 / 10	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

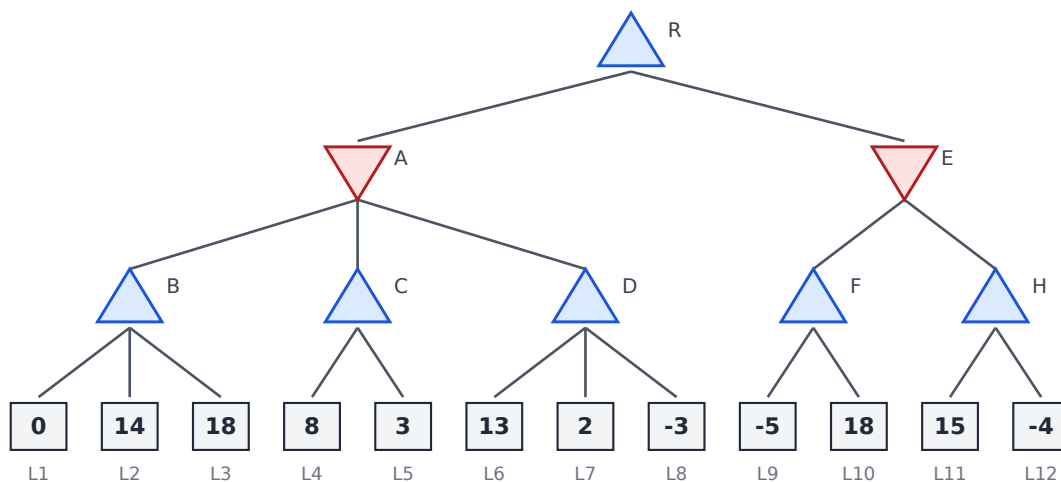
Q.1 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

A search tree has uniform branching factor $b = 3$; the shallowest goal is at depth $d = 6$ and happens to be the **last** node generated at that depth. BFS applies the goal test when a node is **generated**. How many nodes are generated in total (do **not** count the root)?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to E; root value 18
- ☐ (B) Move to E; root value 16
- ☐ (C) Move to E; root value 15
- ☐ (D) Move to A; root value 8

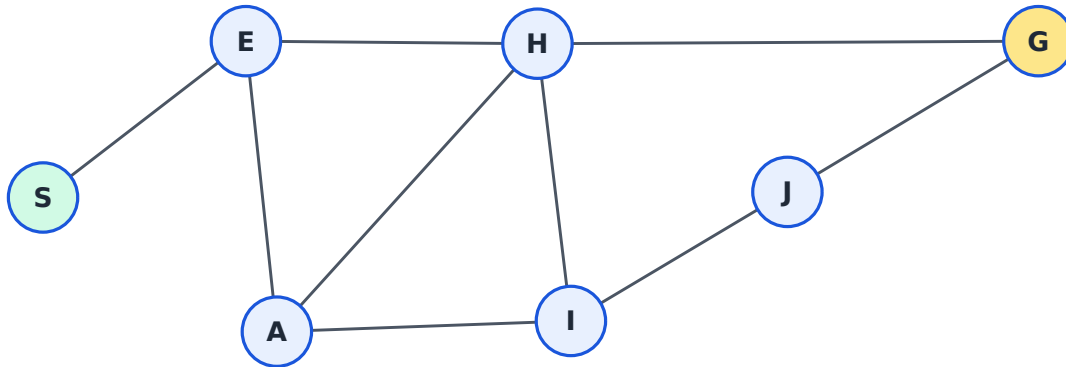
Q.3 [NAT · 1 mark · no negative marking] Uninformed Search

AIMA bounds the time and space of uniform-cost search by $O(b^{1+\lceil C^*/\epsilon \rceil})$, where C^* is the optimal solution cost and ϵ is the minimum step cost. For $C^* = 10$ and $\epsilon = 2$, what is the exponent of b in this bound?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, A, E, H, I, G
- ☐ (B) S, E, A, H, I, G
- ☐ (C) S, E, H, A, I, G
- ☐ (D) S, E, A, H, G

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\exists x P(x) \wedge \exists x Q(x)$ true?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following inference rules are sound in propositional logic?

- ☐ (A) Modus ponens: from $\alpha \rightarrow \beta$ and α infer β
- ☐ (B) And-elimination: from $\alpha \wedge \beta$ infer α
- ☐ (C) Abduction: from $\alpha \rightarrow \beta$ and β infer α
- ☐ (D) Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Skolemizing the sentence $\forall x \exists y P(x, y)$ yields

- ☐ (A) $\forall x P(x, F(x))$, where F is a new function symbol
- ☐ (B) $\forall x P(x, A)$, where A is a new constant
- ☐ (C) $\exists y P(y, y)$
- ☐ (D) $\forall x \forall y P(x, y)$

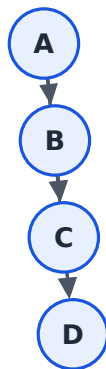
Q.8 [MCQ · 1 mark · negative: -1/3] Probability for AI

Given $P(b | a) = 0.8$, $P(a) = 0.25$ and $P(b) = 0.4$, what is $P(a | b)$?

- ☐ (A) 0.50
- ☐ (B) 0.32
- ☐ (C) 0.20
- ☐ (D) 0.80

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=F, B=T, C=F, D=T)**? (round off to 4 decimal places)



P(A=T)
0.1

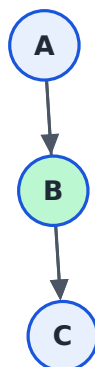
A	P(B=T .)	A	B	P(C=T ...)
T	0.8	T	T	0.25
F	0.9	T	F	0.9
		F	T	0.25
		F	F	0.8

A	C	P(D=T ...)
T	T	0.9
T	F	0.2
F	T	0.6
F	F	0.9

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability **$P(B = T)$** (round off to 3 decimal places).



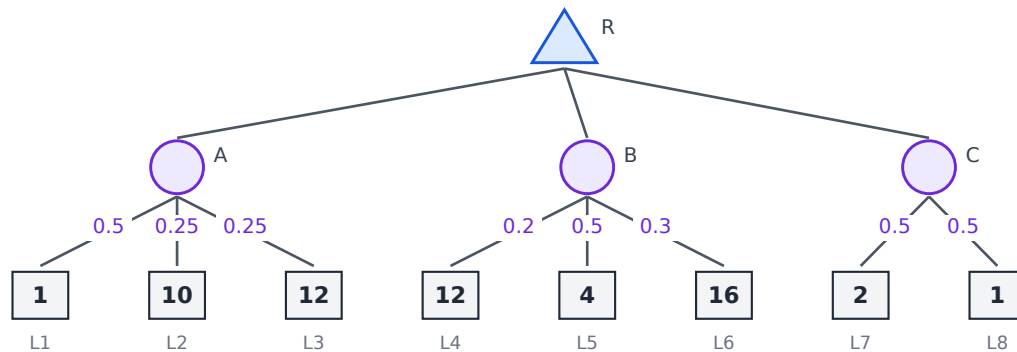
P(A=T)	A	P(B=T ...)	A	B	P(C=T ...)
0.2	T	0.95	T	T	0.8
	F	0.3	T	F	0.5
			F	T	0.9
			F	F	0.6

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)

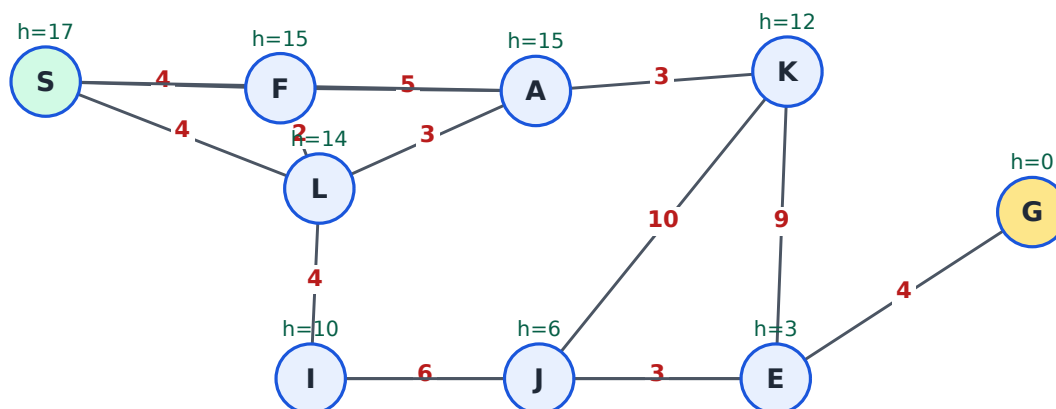


△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – A*

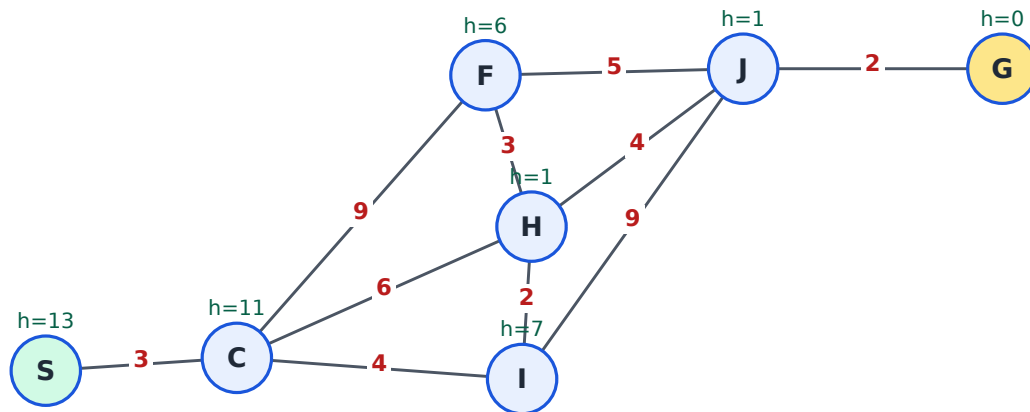
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node E is expanded.
- ☐ (B) Node F is expanded.
- ☐ (C) Node A is expanded.
- ☐ (D) The path returned has cost 21.

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(C)$ overestimates the true cost from C to G.
☐ (B) The edge J-H violates the consistency condition (in at least one direction).
☐ (C) h is consistent (monotone).
☐ (D) h is admissible.

Q.14 [MCQ · 2 marks · negative: -2/3] Informed Search

Weighted A* uses $f(n) = g(n) + W \cdot h(n)$. As the weight $W \rightarrow \infty$, the order in which it expands nodes approaches that of

- ☐ (A) uniform-cost search
☐ (B) A* search
☐ (C) greedy best-first search
☐ (D) breadth-first search

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic

Let KB be a propositional knowledge base and α a sentence, and let $M(\cdot)$ denote the set of models. Which of the following are equivalent to $KB \models \alpha$?

- ☐ (A) $M(KB) \subseteq M(\alpha)$
☐ (B) $KB \rightarrow \alpha$ is valid
☐ (C) $KB \wedge \neg\alpha$ is unsatisfiable
☐ (D) $KB \wedge \alpha$ is satisfiable

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg R \vee S)$, C2: $(Q \vee \neg R \vee S)$, C3: $(\neg Q \vee R \vee \neg S)$, C4: $(R \vee \neg S)$, C5: (R)

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) (S)
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) $(\neg Q)$
☐ (D) $(\neg P \vee \neg Q)$

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

F = $\neg(R \vee \neg P) \wedge ((S \leftrightarrow Q) \rightarrow (\neg P \leftrightarrow R))$?

Answer (numerical): _____

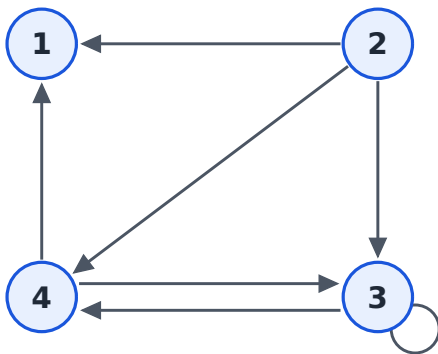
Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $R \rightarrow (P \rightarrow R)$
- ☐ (B) $((Q \rightarrow R) \wedge R) \rightarrow Q$
- ☐ (C) $((P \rightarrow Q) \wedge \neg P) \rightarrow \neg Q$
- ☐ (D) $(Q \rightarrow R) \rightarrow ((Q \vee P) \rightarrow R)$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

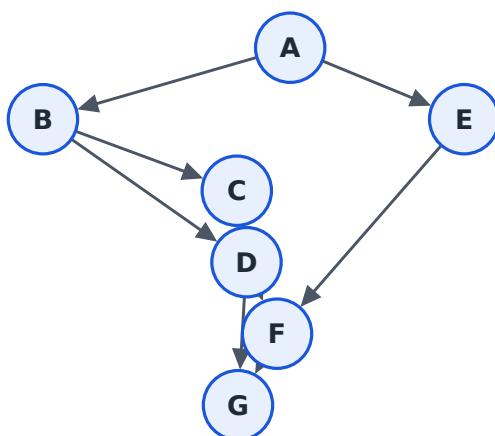
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
- ☐ (B) $\exists x \forall y \neg R(y, x)$
- ☐ (C) $\exists x \forall y \neg R(x, y)$
- ☐ (D) $\exists x R(x, x)$

Q.20 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp D \mid F$
- ☐ (B) $A \perp D \mid B$
- ☐ (C) $E \perp D \mid A$
- ☐ (D) $D \perp E$

Q.21 [NAT · 2 marks · no negative marking] Bayesian Networks

A naive Bayes spam filter has $P(\text{spam}) = 0.3$ and three Boolean features with $P(f_1 | \text{spam}) = 0.8$, $P(f_1 | \text{ham}) = 0.1$; $P(f_2 | \text{spam}) = 0.6$, $P(f_2 | \text{ham}) = 0.3$; $P(f_3 | \text{spam}) = 0.2$, $P(f_3 | \text{ham}) = 0.5$. An email has $f_1 = \text{true}$, $f_2 = \text{false}$, $f_3 = \text{true}$. Compute $P(\text{spam} | f_1, \neg f_2, f_3)$. (Round to 3 decimal places.)

Answer (numerical): _____

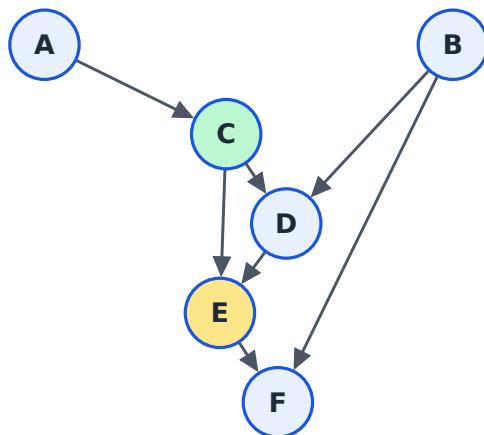
Q.22 [MCQ · 2 marks · negative: -2/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Gibbs sampling with evidence $S = \text{true}$, $W = \text{true}$ is in the state where $R = \text{true}$. What is the probability that C is resampled to true, $P(c | s, r)$?

- ☐ (A) 0.444
☐ (B) 0.040
☐ (C) 0.500
☐ (D) 0.878

Q.23 [MCQ · 2 marks · negative: -2/3] Exact Inference - Variable elimination

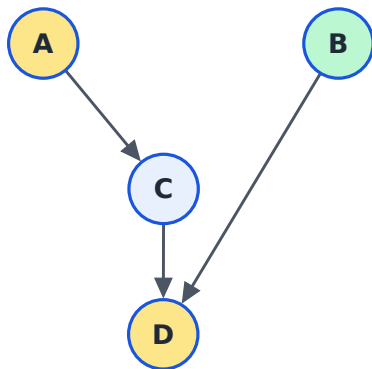
Variable elimination is used to compute $P(\mathbf{C} | \mathbf{E} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{E} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order $\mathbf{F}$, $\mathbf{D}$, $\mathbf{A}$, $\mathbf{B}$. What is the scope of the new factor produced when $\mathbf{A}$ is summed out?



- ☐ (A) $f(A, C)$
☐ (B) $f()$
☐ (C) $f(A)$
☐ (D) $f(C)$

Q.24 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $A = T$, $D = T$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $P(B = T \mid A = T, D = T)$ (round off to 3 decimal places).



$P(A=T)$
0.25

$P(B=T)$	A	$P(C=T \dots)$
0.9	T	0.5
	F	0.7

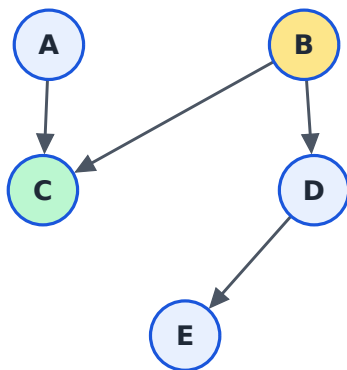
B	C	$P(D=T \dots)$
T	T	0.3
T	F	0.9
F	T	0.25
F	F	0.6

#	B	C
1	T	T
2	F	F
3	T	T
4	T	F
5	T	F
6	T	F

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(C = T \mid B = T)$ (round off to 3 decimal places).



$P(A=T)$
0.3

$P(B=T)$
0.4

A	B	$P(C=T \dots)$
T	T	0.2
T	F	0.9
F	T	0.75
F	F	0.05

B	$P(D=T \dots)$	D	$P(E=T \dots)$
T	0.4	T	0.75
F	0.9	F	0.4

Answer (numerical): _____

Answer Key — Mock Test 069

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	1092	14	MCQ	2	Informed Search	C
2	MCQ	1	Adversarial Search	C	15	MSQ	2	Propositional Logic	A, B, C
3	NAT	1	Uninformed Search	6	16	MSQ	2	Propositional Logic	A
4	MCQ	1	Uninformed Search	B	17	NAT	2	Propositional Logic	4
5	NAT	1	Predicate Logic	49	18	MSQ	2	Propositional Logic	A
6	MSQ	1	Propositional Logic	A, B, D	19	MSQ	2	Predicate Logic	B, C, D
7	MCQ	1	Predicate Logic	A	20	MSQ	2	Conditional Independence	B, C
8	MCQ	1	Probability for AI	A	21	NAT	2	Bayesian Networks	0.439 to 0.44
9	NAT	1	Sampling	0.5466 to 0.547	22	MCQ	2	Sampling	A
10	NAT	1	Exact Inference	0.428 to 0.432	23	MCQ	2	Exact Inference	D
11	NAT	2	Adversarial Search	9.19 to 9.21	24	NAT	2	Sampling	0.844 to 0.848
12	MSQ	2	Informed Search	A, B	25	NAT	2	Exact Inference	0.583 to 0.587
13	MSQ	2	Informed Search	D					

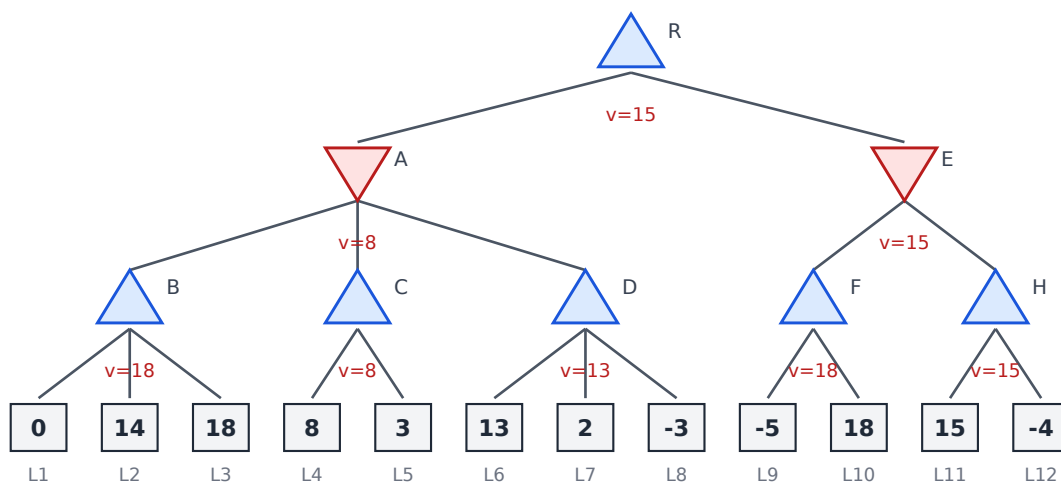
Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] **Answer: 1092**

With goal test at generation, BFS generates every node at depths 1 ... d before (and including) the goal (the worst case). $N = b + b^2 + \dots + b^d = 3 + 9 + 27 + 81 + 243 + 729 = 1092$. This is the $O(b^d)$ of AIMA.

Q.2 [MCQ, 1M · Adversarial Search] **Answer: C** (C: Move to E; root value 15)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 18$, $C = 8$, $D = 13$, $A = 8$, $F = 18$, $H = 15$, $E = 15$. Root value = **15**; MAX's optimal first move is towards E.

Q.3 [NAT, 1M · Uninformed Search] **Answer: 6**

The exponent is $1 + \lceil C^*/\epsilon \rceil = 1 + \lceil 10/2 \rceil = 1 + 5 = 6$. Each step costs at least ϵ , so an optimal path has at most $C^*/\epsilon = 5$ steps. UCS can then generate one further level when it expands nodes of cost about C^* .

Q.4 [MCQ, 1M · Uninformed Search] **Answer: B** (B: S, E, A, H, I, G)

Adjacency lists (alphabetical): A: {E, H, I}; E: {A, H, S}; G: {H, J}; H: {A, E, G, I}; I: {A, H, J}; J: {G, I}; S: {E}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[E]
2	E	[A, H]
3	A	[H, I]

Step	Expanded	Frontier (FIFO) after expansion
4	H	[I, G]
5	I	[G, J]
6	G	[J] ← goal popped, stop

Expansion order: **S, E, A, H, I, G**. The path returned is $S \rightarrow E \rightarrow H \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [NAT, 1M · Predicate Logic] Answer: 49

Interpret P and Q as subsets of the domain; each of P, Q must be non-empty $\Rightarrow (2^n - 1)^2$. With $n = 3$ this gives **49** (verified by enumerating all interpretations).

Q.6 [MSQ, 1M · Propositional Logic] Answer: A, B, D (A: Modus ponens: from $\alpha \rightarrow \beta$ and α infer β ; B: And-elimination: from $\alpha \wedge \beta$ infer α ; D: Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$)

Modus ponens, and-elimination and resolution are truth-preserving: whenever their premises are true, their conclusion is true. For resolution, if β is true then γ must be true, else α must be true. Abduction is unsound: $\alpha=F, \beta=T$ makes both premises true and the conclusion false.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x P(x, F(x))$, where F is a new function symbol)

Since the existential $\exists y$ lies within the scope of $\forall x$, the chosen y may depend on x , so it is replaced by a Skolem function $F(x)$. A Skolem constant would wrongly assert one y works for all x . The result is satisfiable iff the original is (though not logically equivalent). $\forall x \forall y P(x, y)$ is much stronger.

Q.8 [MCQ, 1M · Probability for AI] Answer: A (A: 0.50)

By Bayes rule, $P(a | b) = P(b | a)P(a)/P(b) = 0.8 \times 0.25 / 0.4 = 0.2 / 0.4 = 0.5$. $0.32 = 0.8 \times 0.4$ multiplies by $P(b)$ instead of dividing; $0.20 = P(a \wedge b)$ is the joint, not the conditional; 0.80 confuses $P(a | b)$ with $P(b | a)$.

Q.9 [NAT, 1M · Sampling] Answer: 0.5466 to 0.547

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(A=F) \times P(B=T|A=F) \times P(C=F|A=F, B=T) \times P(D=T|A=F, C=F)$
 $= 0.9 \times 0.9 \times 0.75 \times 0.9 = \mathbf{0.54675} \approx 0.5468$.

Q.10 [NAT, 1M · Exact Inference] Answer: 0.428 to 0.432

Only the ancestors of B (A) matter: every other variable sums out to 1.

$P(B=T) = \sum \text{over its ancestors of } P(B=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, B=T)$
A=T, C=T	0.152
A=T, C=F	0.038
A=F, C=T	0.216
A=F, C=F	0.024

Sum = **0.43** ≈ 0.430 .

Q.11 [NAT, 2M · Adversarial Search] Answer: 9.19 to 9.21

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.5 \times 1 + 0.25 \times 10 + 0.25 \times 12 = 6$
- $B = 0.2 \times 12 + 0.5 \times 4 + 0.3 \times 16 = 9.2$
- $C = 0.5 \times 2 + 0.5 \times 1 = 1.5$

Root = $\max(6, 9.2, 1.5) = \mathbf{9.2}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.12 [MSQ, 2M · Informed Search] Answer: A, B (A: Node E is expanded.; B: Node F is expanded.)

Heuristic values: $h(A)=15, h(E)=3, h(F)=15, h(G)=0, h(I)=10, h(J)=6, h(K)=12, h(L)=14, h(S)=17$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (17)	L(18), F(19), A(20)
2	L (18)	I(18), F(19), A(20)

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
3	I (18)	E(18), F(19), A(20), J(20)
4	E (18)	F(19), G(19), A(20), J(20), K(36)
5	F (19)	G(19), A(20), J(20), K(36)
6	G (19)	A(20), J(20), K(36) ← goal, stop

Path returned: **S** → **L** → **I** → **E** → **G** with cost **19**; nodes expanded = **6** (S, L, I, E, F, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.13 [MSQ, 2M · Informed Search] **Answer: D** (D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	F	G	H	I	J	S
$h(n)$	11	6	0	1	7	1	13
$h^*(n)$	12	7	0	6	8	2	15
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $I \rightarrow H$ ($7 > 2 + 1$), $F \rightarrow H$ ($6 > 3 + 1$), $C \rightarrow H$ ($11 > 6 + 1$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [MCQ, 2M · Informed Search] **Answer: C** (C: greedy best-first search)

For very large W , the term $W \cdot h$ dominates g , so nodes are effectively ordered by h alone, which is **greedy best-first search**. $W = 1$ gives A^* (B). $W = 0$ gives UCS (A). BFS (D) would need ordering by depth.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: A, B, C** (A: $M(KB) \subseteq M(\alpha)$; B: $KB \rightarrow \alpha$ is valid; C: $KB \wedge \neg \alpha$ is unsatisfiable)

(A) is the definition of entailment: α is true in every model of KB . (B) is the deduction theorem. (C) is the basis of proof by refutation (resolution). (D) is weaker: $KB = T$, $\alpha = p$ gives $KB \wedge \alpha$ satisfiable, but $KB \not\models \alpha$.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A** (A: (S))

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) (S) is derivable:

1. resolve $(\neg R \vee S)$ and (R) on $R \Rightarrow (S)$

(B) The resolution closure contains 11 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(\neg Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg P \vee \neg Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.17 [NAT, 2M · Propositional Logic] **Answer: 4**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$\neg(R \vee \neg P)$	$(S \leftrightarrow Q) \rightarrow (\neg P \leftrightarrow R)$	$\neg(R \vee \neg P) \wedge ((S \leftrightarrow Q) \rightarrow (\neg P \leftrightarrow R))$
T	T	T	T	F	F	F
T	T	T	F	F	T	F
T	T	F	T	T	T	T
T	T	F	F	T	T	T
T	F	T	T	F	T	F
T	F	T	F	F	F	F
T	F	F	T	T	T	T
T	F	F	F	T	T	T
F	T	T	T	F	T	F
F	T	T	F	F	T	F
F	T	F	T	F	F	F
F	T	F	F	F	T	F
F	F	T	T	F	T	F
F	F	T	F	F	T	F

P	Q	R	S	$\neg(R \vee \neg P)$	$(S \leftrightarrow Q) \rightarrow (\neg P \leftrightarrow R)$	$\neg(R \vee \neg P) \wedge ((S \leftrightarrow Q) \rightarrow (\neg P \leftrightarrow R))$
F	F	F	T	F	T	F
F	F	F	F	F	F	F

Rows where F = T: **4**.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A** (A: $R \rightarrow (P \rightarrow R)$)

(A) $R \rightarrow (P \rightarrow R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $((Q \rightarrow R) \wedge R) \rightarrow Q$: **not valid** — counter-model $Q=F, R=T$.

(C) $((P \rightarrow Q) \wedge \neg P) \rightarrow \neg Q$: **not valid** — counter-model $P=F, Q=T$.

(D) $(Q \rightarrow R) \rightarrow ((Q \vee P) \rightarrow R)$: **not valid** — counter-model $P=T, Q=F, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: B, C, D** (B: $\exists x \forall y \neg R(y, x)$; C: $\exists x \forall y \neg R(x, y)$; D: $\exists x R(x, x)$)

$R = \{(2,1), (2,3), (2,4), (3,3), (3,4), (4,1), (4,3)\}$.

(A) $\forall x \exists y R(x, y)$: **FALSE** — 1 has no outgoing edge.

(B) $\exists x \forall y \neg R(y, x)$: **TRUE** — 2 has no incoming edge.

(C) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

(D) $\exists x R(x, x)$: **TRUE** — self-loop at 3.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Conditional Independence] **Answer: B, C** (B: $A \perp D \mid B$; C: $E \perp D \mid A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp D \mid F$: **FALSE** — the path $B \leftarrow A \rightarrow E \rightarrow F \leftarrow D$ is active given $\{F\}$.

(B) $A \perp D \mid B$: **TRUE** — $A \rightarrow B \rightarrow C \rightarrow D$ (blocked: observed chain node B); $A \rightarrow B \rightarrow D$ (blocked: observed chain node B); $A \rightarrow E \rightarrow F \leftarrow D$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed).

(C) $E \perp D \mid A$: **TRUE** — $E \leftarrow A \rightarrow B \rightarrow C \rightarrow D$ (blocked: observed fork node A); $E \leftarrow A \rightarrow B \rightarrow D$ (blocked: observed fork node A); $E \rightarrow F \leftarrow D$ (blocked: collider F with neither it nor a descendant observed); $E \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed).

(D) $D \perp E$: **FALSE** — the path $D \leftarrow B \leftarrow A \rightarrow E$ is active given $\{\}$.

Q.21 [NAT, 2M · Bayesian Networks] **Answer: 0.439 to 0.44**

Spam: $0.3 \times 0.8 \times (1 - 0.6) \times 0.2 = 0.3 \times 0.8 \times 0.4 \times 0.2 = 0.0192$. Ham: $0.7 \times 0.1 \times (1 - 0.3) \times 0.5 = 0.7 \times 0.1 \times 0.7 \times 0.5 = 0.0245$. Normalize: $0.0192 / (0.0192 + 0.0245) = 0.0192 / 0.0437 \approx 0.4394 \approx 0.439$. Note $P(\neg f_2 \mid \text{class}) = 1 - P(f_2 \mid \text{class})$ must be used for the false feature.

Q.22 [MCQ, 2M · Sampling] **Answer: A** (A: 0.444)

$MB(C) = \{S, R\}$ (its children; they have no other parents). $P(C \mid s, r) = \alpha P(C) P(s \mid C) P(r \mid C)$. c: $0.5 \times 0.1 \times 0.8 = 0.04$. $\neg c$: $0.5 \times 0.5 \times 0.2 = 0.05$. $P(c \mid s, r) = 0.04 / 0.09 \approx 0.444$. 0.040 is the unnormalized value, 0.500 is the prior, and 0.878 is $P(c \mid \neg s, r)$.

Q.23 [MCQ, 2M · Exact Inference] **Answer: D** (D: $f(C)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, C)$, $f(B, C, D)$, $f(C, D)$, $f(B, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(B, F)$	$\{B, F\}$	$f(B)$
D	$f(B, C, D)$, $f(C, D)$	$\{B, C, D\}$	$f(B, C)$
A	$f(A)$, $f(A, C)$	$\{A, C\}$	$f(C)$
B	$f(B)$, $f(B, C)$	$\{B, C\}$	$f(C)$

The largest factor produced has **2** variable(s) (created when eliminating D); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over C and normalise.

Q.24 [NAT, 2M · Sampling] **Answer: 0.844 to 0.848**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	B
1	$P(A=T)=0.25 \times P(D=T B=T,C=T)=0.3$	0.075	T
2	$P(A=T)=0.25 \times P(D=T B=F,C=F)=0.6$	0.15	F
3	$P(A=T)=0.25 \times P(D=T B=T,C=T)=0.3$	0.075	T
4	$P(A=T)=0.25 \times P(D=T B=T,C=F)=0.9$	0.225	T
5	$P(A=T)=0.25 \times P(D=T B=T,C=F)=0.9$	0.225	T
6	$P(A=T)=0.25 \times P(D=T B=T,C=F)=0.9$	0.225	T

$\sum w (\text{all}) = 0.975$; $\sum w (B = T) = 0.825$.

Estimate $P(B=T \mid A = T, D = T) = 0.825 / 0.975 = \mathbf{0.8462}$.

Q.25 [NAT, 2M · Exact Inference] **Answer: 0.583 to 0.587**

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B) \cdot P(C \mid A, B) \cdot P(D \mid B) \cdot P(E \mid D)$.

Inference by enumeration: $P(C \mid e) = \alpha \sum_{\text{hidden}} P(C, e, \text{hidden})$.

Hidden: A, D, E	C	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T, T	T	$0.3 \times 0.4 \times 0.2 \times 0.4 \times 0.75$	0.0072
T, T, F	T	$0.3 \times 0.4 \times 0.2 \times 0.4 \times 0.25$	0.0024
T, F, T	T	$0.3 \times 0.4 \times 0.2 \times 0.6 \times 0.4$	0.00576
T, F, F	T	$0.3 \times 0.4 \times 0.2 \times 0.6 \times 0.6$	0.00864
F, T, T	T	$0.7 \times 0.4 \times 0.75 \times 0.4 \times 0.75$	0.063
F, T, F	T	$0.7 \times 0.4 \times 0.75 \times 0.4 \times 0.25$	0.021
F, F, T	T	$0.7 \times 0.4 \times 0.75 \times 0.6 \times 0.4$	0.0504
F, F, F	T	$0.7 \times 0.4 \times 0.75 \times 0.6 \times 0.6$	0.0756
T, T, T	F	$0.3 \times 0.4 \times 0.8 \times 0.4 \times 0.75$	0.0288
T, T, F	F	$0.3 \times 0.4 \times 0.8 \times 0.4 \times 0.25$	0.0096
T, F, T	F	$0.3 \times 0.4 \times 0.8 \times 0.6 \times 0.4$	0.02304
T, F, F	F	$0.3 \times 0.4 \times 0.8 \times 0.6 \times 0.6$	0.03456
F, T, T	F	$0.7 \times 0.4 \times 0.25 \times 0.4 \times 0.75$	0.021
F, T, F	F	$0.7 \times 0.4 \times 0.25 \times 0.4 \times 0.25$	0.007
F, F, T	F	$0.7 \times 0.4 \times 0.25 \times 0.6 \times 0.4$	0.0168
F, F, F	F	$0.7 \times 0.4 \times 0.25 \times 0.6 \times 0.6$	0.0252

$P(C=T, e) = 0.234$; $P(C=F, e) = 0.166$; $P(e) = 0.4$.

$P(C=T \mid e) = 0.234 / 0.4 = \mathbf{0.585} \approx 0.585$.

Mock Test 070

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 9 / 11	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

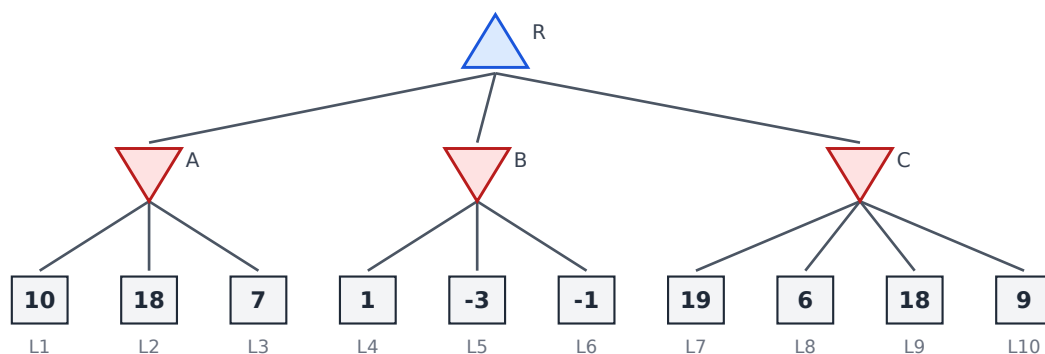
Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search

Chess has an average branching factor of about 35, and a game lasts about 100 plies. The full game tree therefore has about how many nodes?

- ☐ (A) 10^{40}
- ☐ (B) 10^{100}
- ☐ (C) 10^{154}
- ☐ (D) 10^{400}

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



 MAX node  MIN node

 terminal utility

Answer (numerical): _____

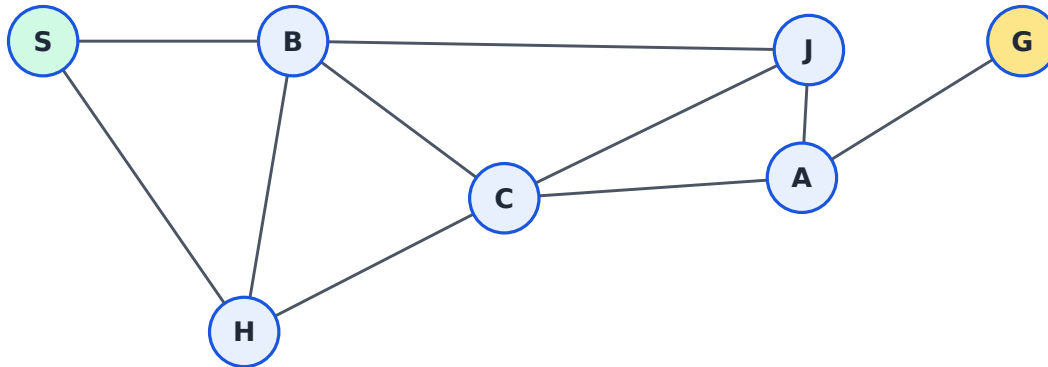
Q.3 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Depth-limited search with limit $l = 2$ is run on a tree with branching factor $b = 5$. No goal exists within depth 2. How many nodes (including the root) does it generate before returning cutoff?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, C, A, G
- ☐ (B) S, H, B, C, J, A, G
- ☐ (C) S, B, H, C, J, A, G
- ☐ (D) S, B, C, J, A, G

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Without standardizing variables apart, $\text{Unify}(\text{Knows}(\text{John}, x), q)$ succeeds for which of the following atoms q ?

- ☐ (A) $\text{Knows}(\text{John}, \text{Jane})$
- ☐ (B) $\text{Knows}(y, \text{Bill})$
- ☐ (C) $\text{Knows}(y, \text{Mother}(y))$
- ☐ (D) $\text{Knows}(x, \text{Elizabeth})$

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 2 elements, so there are $4^2 = 16$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

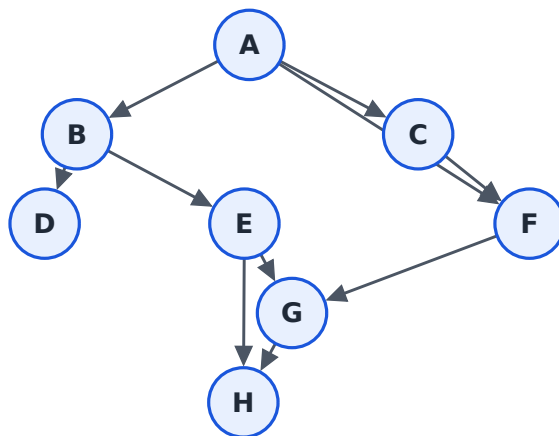
Q.7 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following are sentences, i.e. formulas with no free variables (a is a constant)?

- ☐ (A) $(\forall x P(x)) \rightarrow Q(x)$
- ☐ (B) $\forall x (P(x) \rightarrow \exists y R(x, y))$
- ☐ (C) $\exists x R(x, y)$
- ☐ (D) $\forall x \exists y (R(x, y) \wedge P(a))$

Q.8 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **G** in the Bayesian network below?



Answer (numerical): _____

Q.9 [MCQ · 1 mark · negative: -1/3] Sampling

What is the main drawback of rejection sampling for computing $P(X | e)$?

- ☐ (A) It rejects most samples when $P(e)$ is small, and the fraction accepted drops exponentially with the number of evidence variables
- ☐ (B) It is not consistent: estimates do not converge even with infinitely many samples
- ☐ (C) It requires the Markov blanket of every variable
- ☐ (D) It can only be used when the evidence variables are root nodes

Q.10 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(A = T | C = T)$, how many of these samples are **rejected**?

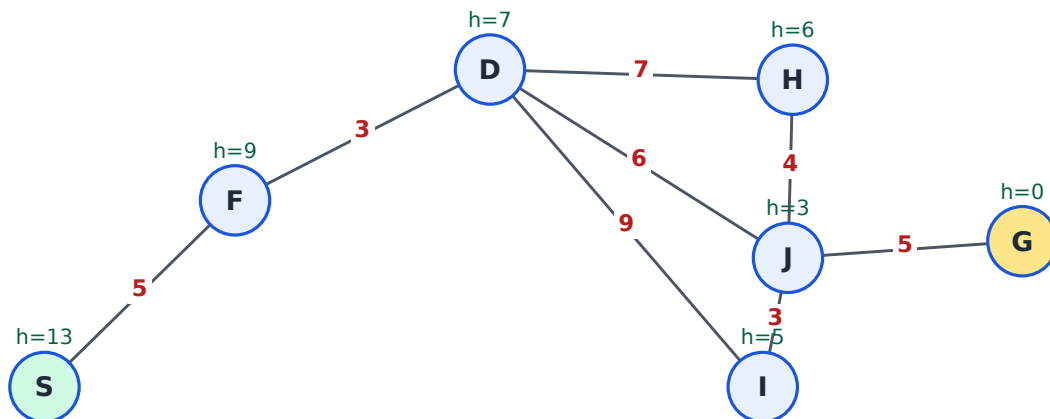
#	A	B	C	D
1	T	F	T	T
2	T	F	F	F
3	F	F	F	T
4	F	F	F	F
5	T	F	T	T
6	T	F	F	F
7	T	T	T	T
8	F	F	T	T
9	T	F	F	F
10	T	F	F	F
11	T	F	T	T
12	T	F	T	T
13	F	T	T	T
14	T	F	T	T
15	F	T	F	F

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

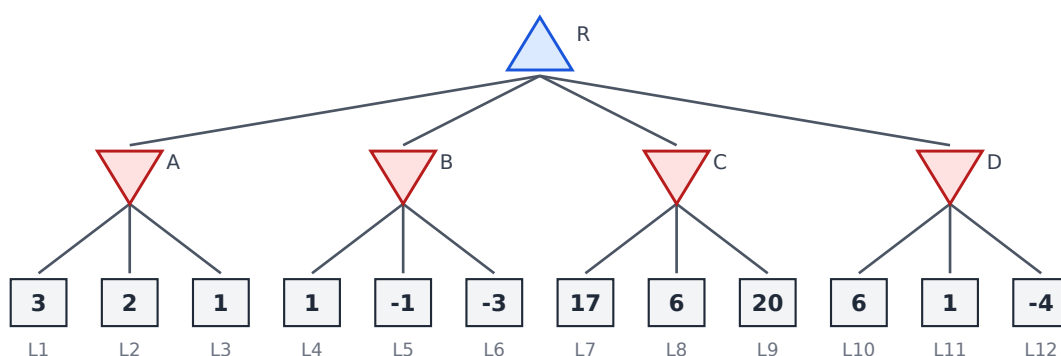
Q.12 [MCQ · 2 marks · negative: -2/3] Informed Search

Consider the directed graph with start **S** and goal **G** and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). The heuristic is $h(S)=7$, $h(A)=6$, $h(B)=2$, $h(C)=1$, $h(G)=0$. A* **graph search** is run from **S**. It keeps an explored set, never re-opens a state that has already been expanded, and goal-tests on expansion. What is the cost of the solution it returns?

- ☐ (A) 7
☐ (B) 8
☐ (C) 9
☐ (D) 10

Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



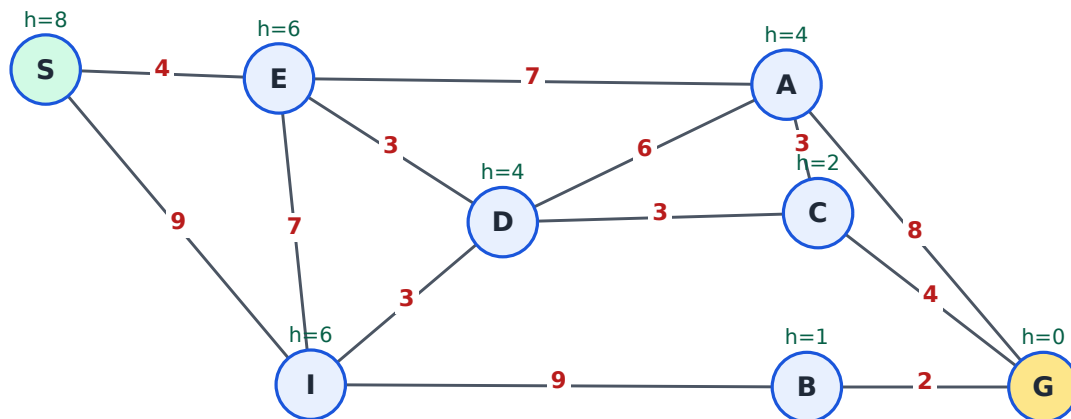
MAX node MIN node

terminal utility

- ☐ (A) Leaf L2 is pruned.
☐ (B) Leaf L3 is pruned.
☐ (C) Leaf L6 is pruned.
☐ (D) Leaf L10 is pruned.

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(\neg R \vee \neg R) \vee Q$
- (2) $\neg P \wedge (P \leftrightarrow P)$
- (3) $(\neg Q \leftrightarrow R) \wedge (Q \rightarrow \neg R)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg(\neg R \wedge Q)$
- ☐ (B) $(Q \rightarrow P) \wedge (R \wedge R)$
- ☐ (C) $(P \rightarrow R) \vee (\neg R \rightarrow \neg R)$
- ☐ (D) $(P \rightarrow R) \vee (\neg Q \vee Q)$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \vee P) \wedge (\neg Q \vee R)) \rightarrow (P \vee R)$
- ☐ (B) $(R \rightarrow Q) \rightarrow ((R \vee P) \rightarrow Q)$
- ☐ (C) $P \rightarrow (Q \rightarrow P)$
- ☐ (D) $\neg(Q \wedge P) \leftrightarrow (\neg Q \vee \neg P)$

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

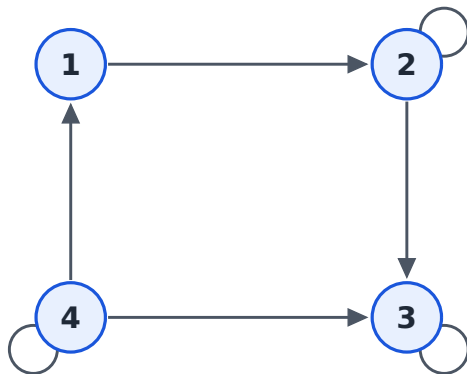
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = (\neg Q \leftrightarrow (\neg S \rightarrow P)) \vee (\neg \neg R \rightarrow (\neg S \rightarrow P)) ?$$

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x R(x, x)$
☐ (B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (D) $\exists y \forall x R(x, y)$

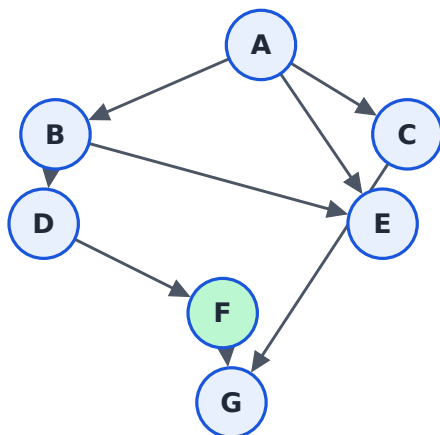
Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic

Consider the interpretation with domain $D = \{1, 2, 3\}$ in which $L(x, y)$ holds iff $x < y$. Which of the following sentences are TRUE in this interpretation?

- ☐ (A) $\forall x \exists y L(x, y)$
☐ (B) $\exists x \forall y \neg L(y, x)$
☐ (C) $\forall x \forall y (L(x, y) \rightarrow \neg L(y, x))$
☐ (D) $\forall x \forall y (L(x, y) \rightarrow \exists z (L(x, z) \wedge L(z, y)))$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

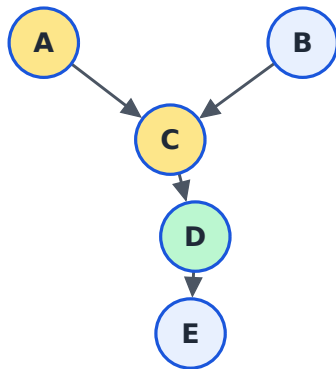
Variable elimination is used to compute $P(\mathbf{F})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, A, G, C, E, B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(D = T \mid C = F, A = F)$ (round off to 3 decimal places).



P(A=T)		P(B=T)		A	B	P(C=T ...)
0.25		0.95		T	T	0.9
				T	F	0.75
				F	T	0.7
				F	F	0.25

C	P(D=T ...)	D	P(E=T ...)
T	0.4	T	0.3
F	0.9	F	0.2

Answer (numerical): _____

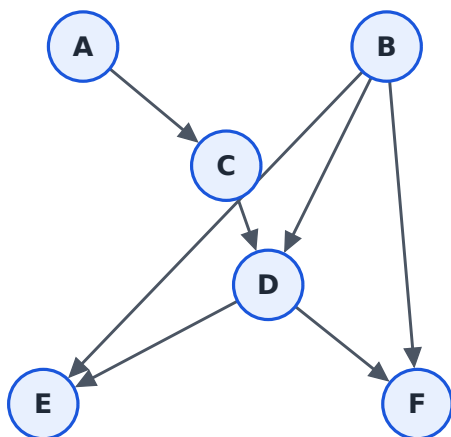
Q.22 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. Which of the following are correct (values rounded to 3 decimal places)?

- ☐ (A) $P(s \mid w, r) \approx 0.194$
☐ (B) $P(s \mid w) \approx 0.430$
☐ (C) $P(s \mid w, r) > P(s \mid w)$
☐ (D) $P(w) \approx 0.647$

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp A \mid C, F$
☐ (B) $D \perp A \mid C, E$
☐ (C) $B \perp A \mid C$
☐ (D) $B \perp F \mid D$

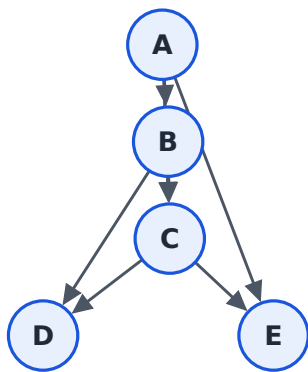
Q.24 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Gibbs sampling is run with evidence $S = \text{true}$, $W = \text{true}$. In the current state $C = \text{true}$. Compute the probability with which R is resampled to true, i.e. $P(r | c, s, w)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = \{0.03, 0.23, 0.25, 0.33, 0.64\}$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T | \text{sampling parent values})$. Which sample is generated?



P(A=T)					
0.2					
A	P(B=T ...)	A	B	P(C=T ...)	
T	0.7	T	T	0.05	
F	0.5	T	F	0.75	
		F	T	0.95	
		F	F	0.1	

B	C	P(D=T ...)	A	C	P(E=T ...)
T	T	0.75	T	T	0.75
T	F	0.1	T	F	0.05
F	T	0.9	F	T	0.5
F	F	0.95	F	F	0.95

- ☐ (A) A=F, B=T, C=F, D=F, E=F
☐ (B) A=T, B=F, C=F, D=F, E=F
☐ (C) A=T, B=T, C=F, D=F, E=F
☐ (D) A=F, B=F, C=T, D=F, E=T

Answer Key — Mock Test 070

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	C	14	NAT	2	Informed Search	5
2	NAT	1	Adversarial Search	7	15	MSQ	2	Propositional Logic	C, D
3	NAT	1	Uninformed Search	31	16	MSQ	2	Propositional Logic	A, C, D
4	MCQ	1	Uninformed Search	C	17	NAT	2	Propositional Logic	15
5	MSQ	1	Predicate Logic	A, B, C	18	MSQ	2	Predicate Logic	A, B
6	NAT	1	Predicate Logic	4	19	MSQ	2	Predicate Logic	B, C
7	MSQ	1	Predicate Logic	B, D	20	NAT	2	Exact Inference	3
8	NAT	1	Conditional Independence	3	21	NAT	2	Exact Inference	0.898 to 0.902
9	MCQ	1	Sampling	A	22	MSQ	2	Bayesian Networks	A, B, D
10	NAT	1	Sampling	7	23	MSQ	2	Conditional Independence	A, B, C
11	NAT	2	Informed Search	19	24	NAT	2	Sampling	0.814 to 0.815
12	MCQ	2	Informed Search	B	25	MCQ	2	Sampling	C
13	MSQ	2	Adversarial Search	C					

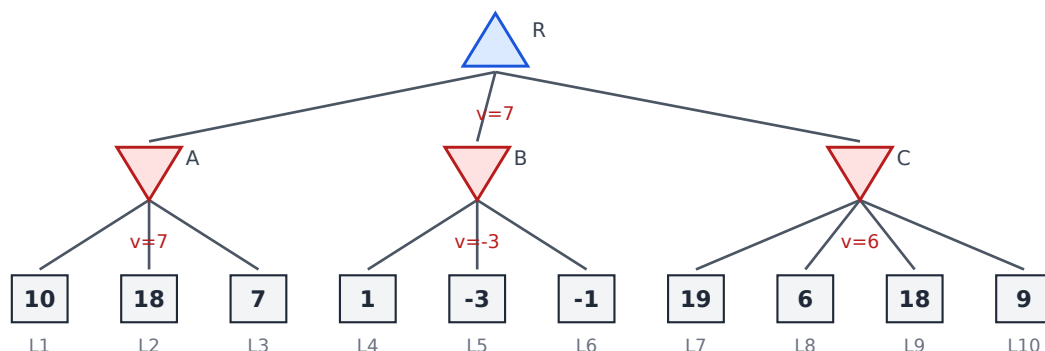
Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] Answer: C (C: 10^{154})

$35^{100} = 10^{100 \cdot \log_{10} 35} \approx 10^{100 \times 1.544} \approx 10^{154}$, the AIMA estimate. 10^{40} (A) is AIMA's estimate of the number of distinct chess positions, not of the tree. (B) and (D) do not match the calculation.

Q.2 [NAT, 1M · Adversarial Search] Answer: 7

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = 7, B = -3, C = 6. Root value = 7; MAX's optimal first move is towards A.

Q.3 [NAT, 1M · Uninformed Search] Answer: 31

DLS explores the complete tree down to depth ℓ : $1 + b + \dots + b^\ell = 1 + 5 + 25 = 31 = (b^{\ell+1} - 1)/(b - 1)$.

Q.4 [MCQ, 1M · Uninformed Search] Answer: C (C: S, B, H, C, J, A, G)

Adjacency lists (alphabetical): A: {C, G, J}; B: {C, H, J, S}; C: {A, B, H, J}; G: {A}; H: {B, C, S}; J: {A, B, C}; S: {B, H}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B, H]
2	B	[H, C, J]
3	H	[C, J]
4	C	[J, A]
5	J	[A]
6	A	[G]
7	G	[] $\leftarrow$ goal popped, stop

Expansion order: **S, B, H, C, J, A, G**. The path returned is $S \rightarrow B \rightarrow C \rightarrow A \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [MSQ, 1M · Predicate Logic] Answer: A, B, C (A: $\text{Knows}(\text{John}, \text{Jane})$; B: $\text{Knows}(y, \text{Bill})$; C: $\text{Knows}(y, \text{Mother}(y))$)

(A) $\{x/\text{Jane}\}$. (B) $\{x/\text{Bill}, y/\text{John}\}$. (C) $\{y/\text{John}, x/\text{Mother}(\text{John})\}$. (D) fails: x would have to be both John and Elizabeth. Renaming the second x (standardizing apart) would make it succeed, which is why AIMA insists on standardizing apart.

Q.6 [NAT, 1M · Predicate Logic] Answer: 4

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 2$ this gives **4** (verified by enumerating all interpretations).

Q.7 [MSQ, 1M · Predicate Logic] Answer: B, D (B: $\forall x (P(x) \rightarrow \exists y R(x, y))$; D: $\forall x \exists y (R(x, y) \wedge P(a))$)

(A) The scope of $\forall x$ ends before $\rightarrow$, so x in $Q(x)$ is free. (B) x and y are bound everywhere. (C) y is free. (D) x and y are bound; a is a constant, not a variable. Hence B and D are sentences.

Q.8 [NAT, 1M · Conditional Independence] Answer: 3

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of G: {E, F}; children: {H}; co-parents: $\{\emptyset\}$.

$\text{MB}(G) = \{E, F, H\}$ — **3** nodes. Given its Markov blanket, G is conditionally independent of every other node in the network.

Q.9 [MCQ, 1M · Sampling] Answer: A (A: It rejects most samples when $P(e)$ is small, and the fraction accepted drops exponentially with the number of evidence variables)

Rejection sampling keeps only samples consistent with e , so about $N \cdot P(e)$ samples are useful; $P(e)$ typically shrinks exponentially as evidence grows. It is consistent (B is false). Markov blankets are used by Gibbs sampling, not rejection sampling (C false). Evidence may be anywhere in the network (D false).

Q.10 [NAT, 1M · Sampling] Answer: 7

Rejection sampling keeps only samples consistent with the evidence $C = T$: samples 1, 5, 7, 8, 11, 12, 13, 14 (8 accepted, 7 rejected).

Among the accepted samples, $A = T$ in 6. Estimate = $6/8 = 0.75$.

(For this network the exact value is 0.3913; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.11 [NAT, 2M · Informed Search] Answer: 19

Heuristic values: $h(D)=7$, $h(F)=9$, $h(G)=0$, $h(H)=6$, $h(I)=5$, $h(J)=3$, $h(S)=13$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	F(14)
2	F (14)	D(15)
3	D (15)	J(17), H(21), I(22)
4	J (17)	G(19), H(21), I(22)
5	G (19)	H(21), I(22) ← goal, stop

Path returned: **S** → **F** → **D** → **J** → **G** with cost **19**; nodes expanded = **5** (S, F, D, J, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.12 [MCQ, 2M · Informed Search] Answer: B (B: 8)

Expand S: push A ($g=1$, $f=7$) and B ($g=4$, $f=6$). Expand B ($f=6$): push C ($g=5$, $f=6$) and G ($g=10$, $f=10$).

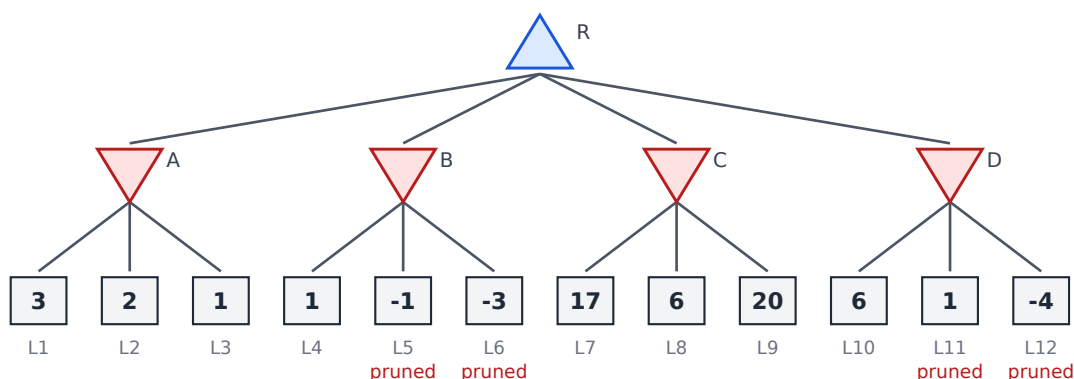
Expand C ($f=6$): push G ($g=8$, $f=8$). Expand A ($f=7$): its successors B ($g=3$) and C ($g=6$) are already explored and are discarded. Expand G ($f=8$) and return cost **8** ($S \rightarrow B \rightarrow C \rightarrow G$). The optimal cost is 7 ($S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$).

Optimality fails because h is admissible but not consistent ($h(A)=6 > c(A,B)+h(B)=4$), and the cheaper path to B is discovered only after B is closed.

Q.13 [MSQ, 2M · Adversarial Search] Answer: C (C: Leaf L6 is pruned.)

Run alpha-beta depth-first, left to right, passing (α , β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $1 \leq \alpha = 1 \Rightarrow$ prune remaining children (leaves L5, L6).
- At MIN node D: value $6 \leq \alpha = 6 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (8): L1, L2, L3, L4, L7, L8, L9, L10. Leaves pruned (4): L5, L6, L11, L12. Root value = 6 (same as plain minimax, 6 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(A)=4$, $h(B)=1$, $h(C)=2$, $h(D)=4$, $h(E)=6$, $h(G)=0$, $h(I)=6$, $h(S)=8$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (8)	E(10), I(15)
2	E (10)	D(11), A(15), I(15)

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
3	D (11)	C(12), A(15), I(15)
4	C (12)	G(14), A(15), I(15)
5	G (14)	A(15), I(15) ← goal, stop

Path returned: **S** → **E** → **D** → **C** → **G** with cost **14**; nodes expanded = **5** (S, E, D, C, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: C, D** ($C: (P \rightarrow R) \vee (\neg R \rightarrow \neg R)$; $D: (P \rightarrow R) \vee (\neg Q \vee Q)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
F	T	F	T

(A) $\neg(\neg R \wedge Q)$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(B) $(Q \rightarrow P) \wedge (R \wedge R)$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(C) $(P \rightarrow R) \vee (\neg R \rightarrow \neg R)$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $(P \rightarrow R) \vee (\neg Q \vee Q)$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A, C, D** ($A: ((Q \vee P) \wedge (\neg Q \vee R)) \rightarrow (P \vee R)$; $C: P \rightarrow (Q \rightarrow P)$; $D: \neg(Q \wedge P) \leftrightarrow (\neg Q \vee \neg P)$)

(A) $((Q \vee P) \wedge (\neg Q \vee R)) \rightarrow (P \vee R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(B) $(R \rightarrow Q) \rightarrow ((R \vee P) \rightarrow Q)$: **not valid** — counter-model $P=T, Q=F, R=F$.

(C) $P \rightarrow (Q \rightarrow P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $\neg(Q \wedge P) \leftrightarrow (\neg Q \vee \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [NAT, 2M · Propositional Logic] **Answer: 15**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$\neg Q \leftrightarrow (\neg S \rightarrow P)$	$\neg \neg R \rightarrow (\neg S \rightarrow P)$	$(\neg Q \leftrightarrow (\neg S \rightarrow P)) \vee (\neg \neg R \rightarrow (\neg S \rightarrow P))$
T	T	T	T	F	T	T
T	T	T	F	F	T	T
T	T	F	T	F	T	T
T	T	F	F	F	T	T
T	F	T	T	T	T	T
T	F	T	F	T	T	T
T	F	F	T	T	T	T
T	F	F	F	T	T	T
F	T	T	T	F	T	T
F	T	T	F	T	F	T
F	T	F	T	F	T	T
F	T	F	F	T	T	T
F	F	T	T	T	T	T
F	F	T	F	F	F	F
F	F	F	T	T	T	T
F	F	F	F	F	T	T

Rows where $F = T$: **15**.

Q.18 [MSQ, 2M · Predicate Logic] **Answer: A, B** ($A: \exists x R(x, x)$; $B: \forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$)

$R = \{(1,2), (2,2), (2,3), (3,3), (4,1), (4,3), (4,4)\}$.

(A) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

(B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,3)$ nor $R(3,1)$.

(D) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: B, C** (B: $\exists x \forall y \neg L(y, x)$; C: $\forall x \forall y (L(x, y) \rightarrow \neg L(y, x))$)

(A) False: for $x = 3$ no y in D is larger. (B) True: $x = 1$ has no y with $y < 1$. (C) True: $<$ is asymmetric. (D) False: the order on $\{1, 2, 3\}$ is not dense; for $x = 1, y = 2$ there is no z with $1 < z < 2$. Evaluating all cases by enumeration confirms B and C.

Q.20 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(A, B), f(A, C), f(B, D), f(A, B, E), f(D, F), f(C, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(B, D), f(D, F)$	$\{B, D, F\}$	$f(B, F)$
A	$f(A), f(A, B), f(A, C), f(A, B, E)$	$\{A, B, C, E\}$	$f(B, C, E)$
G	$f(C, F, G)$	$\{C, F, G\}$	$f(C, F)$
C	$f(B, C, E), f(C, F)$	$\{B, C, E, F\}$	$f(B, E, F)$
E	$f(B, E, F)$	$\{B, E, F\}$	$f(B, F)$
B	$f(B, F), f(B, F)$	$\{B, F\}$	$f(F)$

The largest factor produced has **3** variable(s) (created when eliminating A); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over F and normalise.

Q.21 [NAT, 2M · Exact Inference] **Answer: 0.898 to 0.902**

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B) \cdot P(C | A, B) \cdot P(D | C) \cdot P(E | D)$.

Inference by enumeration: $P(D | e) = \alpha \sum_{\text{hidden}} P(D, e, \text{hidden})$.

Hidden: B, E	D	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T	T	$0.75 \times 0.95 \times 0.3 \times 0.9 \times 0.3$	0.05771
T, F	T	$0.75 \times 0.95 \times 0.3 \times 0.9 \times 0.7$	0.13466
F, T	T	$0.75 \times 0.05 \times 0.75 \times 0.9 \times 0.3$	0.00759
F, F	T	$0.75 \times 0.05 \times 0.75 \times 0.9 \times 0.7$	0.01772
T, T	F	$0.75 \times 0.95 \times 0.3 \times 0.1 \times 0.2$	0.00427
T, F	F	$0.75 \times 0.95 \times 0.3 \times 0.1 \times 0.8$	0.0171
F, T	F	$0.75 \times 0.05 \times 0.75 \times 0.1 \times 0.2$	0.00056
F, F	F	$0.75 \times 0.05 \times 0.75 \times 0.1 \times 0.8$	0.00225

$P(D=T, e) = 0.21769$; $P(D=F, e) = 0.02419$; $P(e) = 0.24188$.

$P(D=T | e) = 0.21769 / 0.24188 = \mathbf{0.9} \approx 0.900$.

Q.22 [MSQ, 2M · Bayesian Networks] **Answer: A, B, D** (A: $P(s | w, r) \approx 0.194$; B: $P(s | w) \approx 0.430$; D: $P(w) \approx 0.647$)

$P(s, w) = c: 0.5 \times 0.1 \times (0.8 \times 0.99 + 0.2 \times 0.90) = 0.0486$; $\neg c: 0.5 \times 0.5 \times (0.2 \times 0.99 + 0.8 \times 0.90) = 0.2295$; total 0.2781. $P(\neg s, w) = c: 0.5 \times 0.9 \times 0.8 \times 0.90 = 0.324$; $\neg c: 0.5 \times 0.5 \times 0.2 \times 0.90 = 0.045$; total 0.369. $P(w) = 0.6471$ (D true) and $P(s | w) = 0.2781 / 0.6471 \approx 0.430$ (B true). $P(s, w, r) = (0.5 \times 0.1 \times 0.8 + 0.5 \times 0.5 \times 0.2) \times 0.99 = 0.0891$; $P(\neg s, w, r) = (0.36 + 0.05) \times 0.90 = 0.369$; $P(s | w, r) = 0.0891 / 0.4581 \approx 0.194$ (A true). Rain explains away the wet grass, so $P(s | w, r) < P(s | w)$: (C) is false.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: A, B, C** (A: $D \perp A | C, F$; B: $D \perp A | C, E$; C: $B \perp A | C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $D \perp A | C, F$: **TRUE** — $D \leftarrow C \leftarrow A$ (blocked: observed chain node C).

(B) $D \perp A | C, E$: **TRUE** — $D \leftarrow C \leftarrow A$ (blocked: observed chain node C).

(C) $B \perp A | C$: **TRUE** — $B \rightarrow D \leftarrow C \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow E \leftarrow D \leftarrow C \leftarrow A$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow F \leftarrow D \leftarrow C \leftarrow A$ (blocked: collider F with neither it nor a descendant observed).

(D) $B \perp F | D$: **FALSE** — the path $B \rightarrow F$ is active given $\{D\}$.

Q.24 [NAT, 2M · Sampling] **Answer: 0.814 to 0.815**

The Markov blanket of R is $\{C, S, W\}$. $P(R | \text{mb}) = \alpha P(R | c) P(w | s, R)$. For $r: 0.8 \times 0.99 = 0.792$. For $\neg r: 0.2 \times 0.90 = 0.18$. $P(r | c, s, w) = 0.792 / (0.792 + 0.18) = 0.792 / 0.972 \approx 0.8148 \approx 0.815$.

Q.25 [MCQ, 2M · Sampling] **Answer: C** (C: A=T, B=T, C=F, D=F, E=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.2$; $u = 0.03 < 0.2 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.7$; $u = 0.23 < 0.7 \Rightarrow B = T$
- C: $P(C=T \mid A=T, B=T) = 0.05$; $u = 0.25 \geq 0.05 \Rightarrow C = F$
- D: $P(D=T \mid B=T, C=F) = 0.1$; $u = 0.33 \geq 0.1 \Rightarrow D = F$
- E: $P(E=T \mid A=T, C=F) = 0.05$; $u = 0.64 \geq 0.05 \Rightarrow E = F$

Sample: (**A=T, B=T, C=F, D=F, E=F**).

Mock Test 071

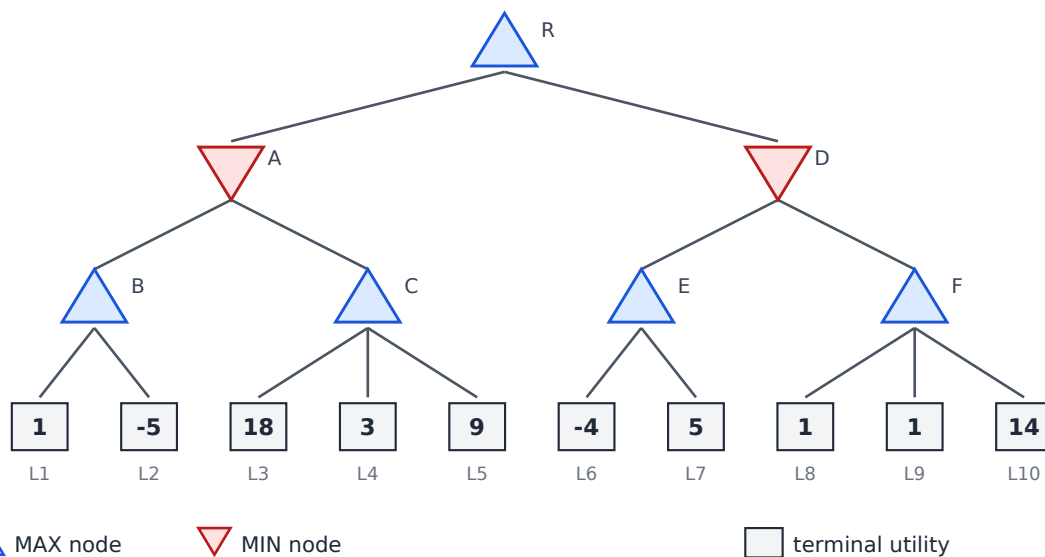
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 7 / 11	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

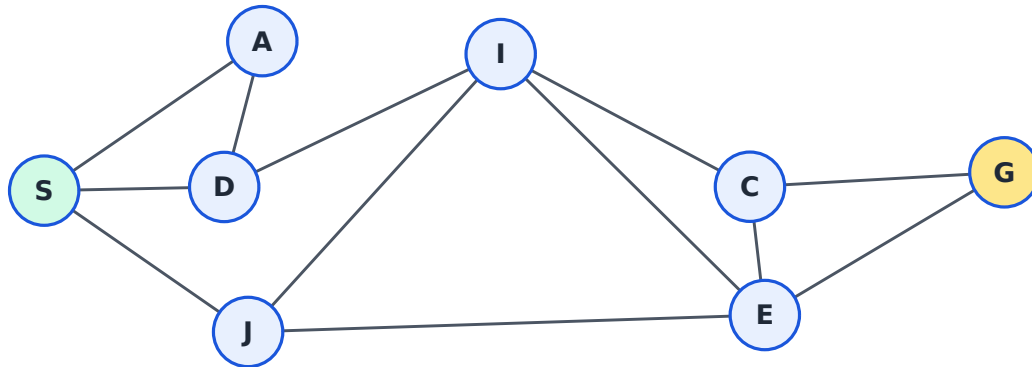
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 5$ and depth $d = 3$ (so $b^d = 125$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, A, D, J, I, C, E, G
☐ (B) S, A, D, I, C, E, G
☐ (C) S, J, D, A, I, E, C, G
☐ (D) S, A, D, J, I, E, C, G

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search

In alpha-beta search, while a MIN node is being evaluated with current bounds (α, β) , the remaining children of the MIN node can be pruned as soon as its running value v satisfies

- ☐ (A) $v \geq \beta$
☐ (B) $v \leq \alpha$
☐ (C) $v \geq \alpha$
☐ (D) $v \leq \beta$

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((P \rightarrow \neg R) \vee (R \leftrightarrow Q)) \vee ((P \vee \neg R) \rightarrow Q) ?$$

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

How many of the 8 truth assignments to p, q, r are models of $(p \vee q) \wedge (\neg p \vee r)$?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3, 4\}$. How many interpretations of a unary predicate P over D make $\exists x P(x) \wedge \neg \forall x P(x)$ true?

Answer (numerical): _____

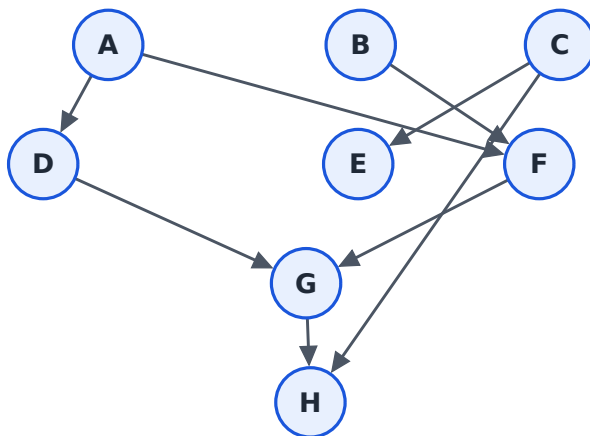
Q.8 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **F** in the Bayesian network below?



Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Conditional Independence

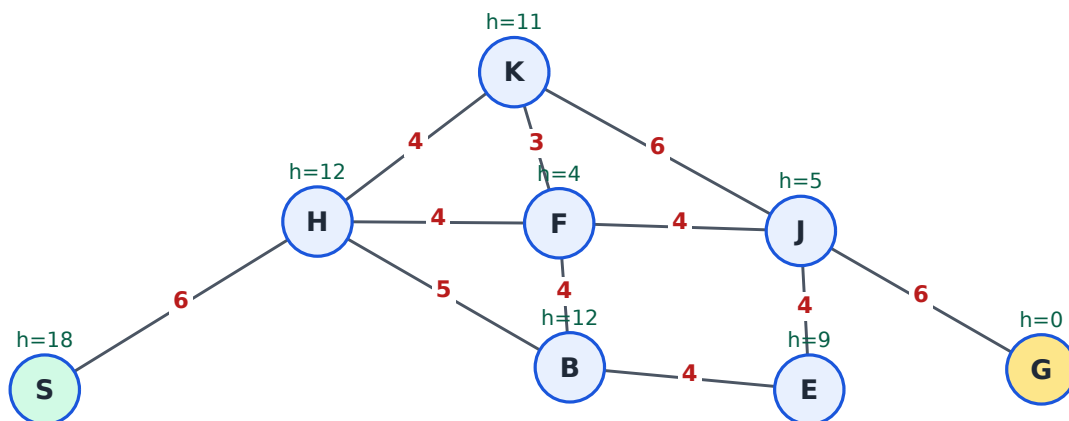
Boolean variables A, B, C satisfy $A \perp B \mid C$, and no other independence is known. What is the minimum number of independent parameters needed to specify $P(A, B, C)$?

- ☐ (A) 5
☐ (B) 7
☐ (C) 6
☐ (D) 4

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

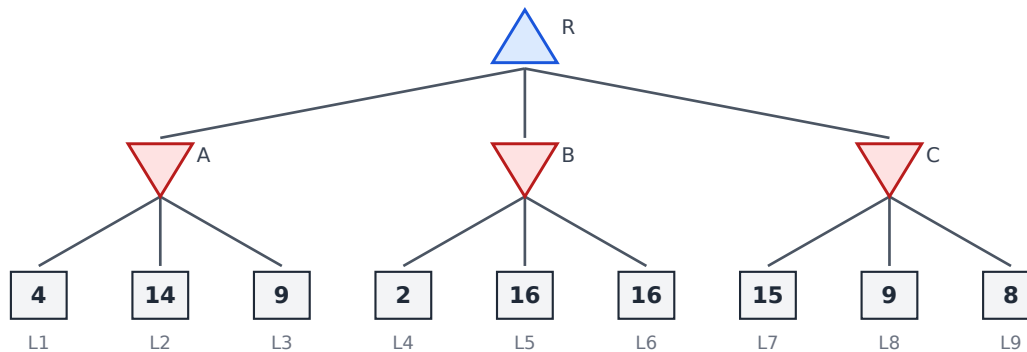
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(K)$ overestimates the true cost from K to G.
☐ (B) h is consistent (monotone).
☐ (C) h is admissible.
☐ (D) The edge H-B violates the consistency condition (in at least one direction).

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



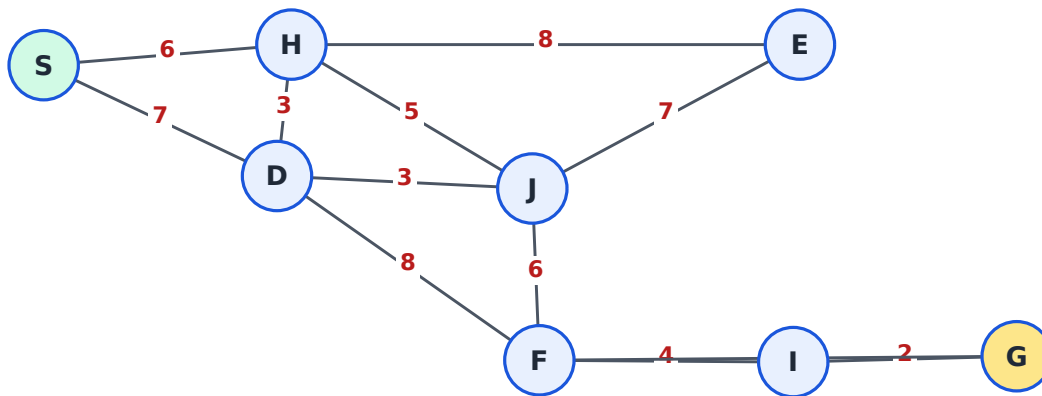
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L1 is pruned.
☐ (B) Leaf L5 is pruned.
☐ (C) Leaf L6 is pruned.
☐ (D) Leaf L8 is pruned.

Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow D \rightarrow F \rightarrow I \rightarrow G$
☐ (B) $S \rightarrow H \rightarrow J \rightarrow F \rightarrow I \rightarrow G$
☐ (C) $S \rightarrow D \rightarrow J \rightarrow F \rightarrow I \rightarrow G$
☐ (D) $S \rightarrow H \rightarrow D \rightarrow F \rightarrow I \rightarrow G$

Q.14 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Breadth-first search goal-tests a node when it is generated, generating the left child before the right, and the goal is node 13. Depth-first search goal-tests on removal from the frontier, exploring the left child first. Which is correct about the number of nodes each algorithm goal-tests up to and including node 13 (counting the root)?

- ☐ (A) BFS: 13, DFS: 12
☐ (B) BFS: 13, DFS: 11
☐ (C) BFS: 12, DFS: 12
☐ (D) BFS: 7, DFS: 12

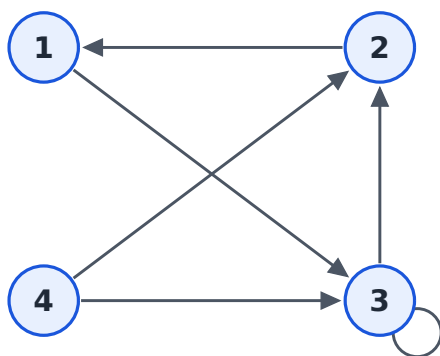
Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \rightarrow P) \rightarrow Q) \leftrightarrow (R \rightarrow (P \rightarrow Q))$
☐ (B) $\neg(P \wedge R) \leftrightarrow (\neg P \vee \neg R)$
☐ (C) $\neg(R \wedge P) \leftrightarrow (\neg R \wedge \neg P)$
☐ (D) $((P \rightarrow Q) \wedge \neg Q) \rightarrow \neg P$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\forall x \exists y R(x, y)$
☐ (C) $\exists x R(x, x)$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(P \vee \neg Q \vee \neg R)$, C2: $(\neg P \vee \neg R)$, C3: $(Q \vee S)$, C4: (R) , C5: $(\neg Q \vee \neg R \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(P \vee \neg S)$
☐ (B) $(P \vee \neg Q)$
☐ (C) $(R \vee \neg S)$
☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Skolemizing $\exists x \forall y \exists z R(x, y, z)$ gives (A is a new constant, F, G new function symbols)

- ☐ (A) $\forall y R(A, y, F(y))$
☐ (B) $\forall y R(A, y, F(A))$
☐ (C) $\forall y R(G(y), y, F(y))$
☐ (D) $\forall y R(A, y, B)$, with B a new constant

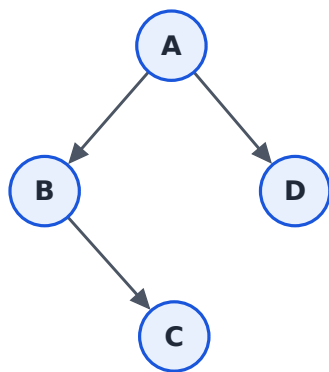
Q.19 [NAT · 2 marks · no negative marking] Bayesian Networks

Fever has three possible causes Cold, Flu and Malaria, modeled with a noisy-OR CPT with no leak. The inhibition probabilities are $q_{\text{cold}} = 0.6$, $q_{\text{flu}} = 0.2$, $q_{\text{malaria}} = 0.1$ (q_i = probability that cause i , when present alone, fails to produce fever). Compute $P(\text{fever} \mid \text{cold}, \text{flu}, \neg \text{malaria})$. (Round to 2 decimal places.)

Answer (numerical): _____

Q.20 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle 0.41, 0.32, 0.82, 0.74 \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampling parent values})$. Which sample is generated?



P(A=T)
0.75

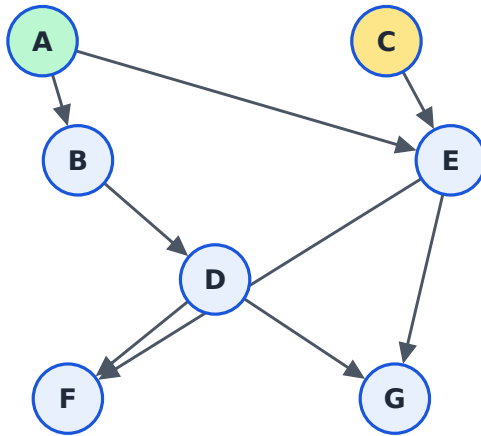
A	P(B=T ...)	B	P(C=T ...)
T	0.95	T	0.5
F	0.2	F	0.9

A	P(D=T ...)
T	0.7
F	0.7

- ☐ (A) A=F, B=T, C=F, D=F
☐ (B) A=F, B=T, C=T, D=T
☐ (C) A=T, B=T, C=F, D=F
☐ (D) A=T, B=F, C=F, D=F

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

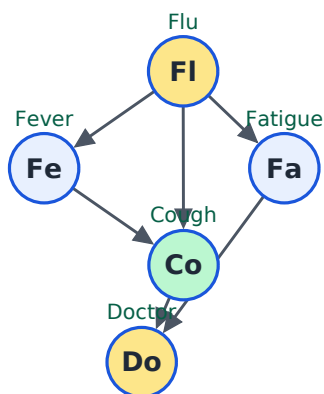
Variable elimination is used to compute $P(A \mid C = T)$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, E, G, D, F**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) Compute $P(\text{Cough} = T \mid \text{Flu} = F, \text{Doctor} = F)$ (round off to 3 decimal places).



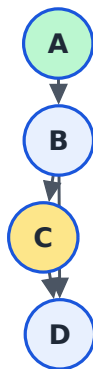
$P(\text{Fl}=T)$	Fl	$P(\text{Fe}=T \dots)$	Fl	Fe	$P(\text{Co}=T \dots)$
0.75	T	0.95	T	T	0.9
	F	0.4	T	F	0.9
			F	T	0.5
			F	F	0.3

Fl	$P(\text{Fa}=T)$	Co	Fa	$P(\text{Do}=T \dots)$
T	0.9	T	T	0.8
F	0.95	T	F	0.1
		F	T	0.7
		F	F	0.6

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{C} = \mathbf{F}$. The values of the non-evidence variables in the generated samples are listed in the table. What is the weight of sample #1? (round off to 4 decimal places)



P(A=T)
0.7

A	P(B=T ...)	B	P(C=T ...)
T	0.1	T	0.8
F	0.9	F	0.2

B	C	P(D=T ...)
T	T	0.1
T	F	0.25
F	T	0.5
F	F	0.4

#	A	B	D
1	T	F	F
2	F	T	F
3	T	F	T
4	F	T	F

Answer (numerical): _____

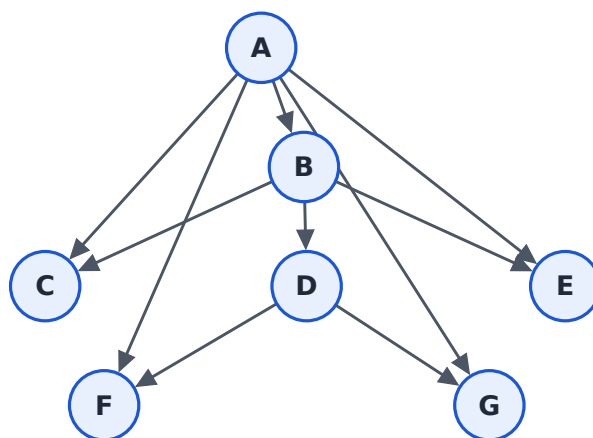
Q.24 [MSQ · 2 marks · no negative marking] Probability for AI

A hidden variable H takes values h_1, h_2, h_3 with prior probabilities 0.5, 0.3, 0.2. For an observation e , $P(e | h_1) = 0.1$, $P(e | h_2) = 0.4$, $P(e | h_3) = 0.7$. Which of the following are correct?

- ☐ (A) $P(e) = 0.31$
☐ (B) $P(h_3 | e) \approx 0.452$
☐ (C) $P(h_1 | e) \approx 0.161$
☐ (D) The most probable hypothesis after observing e (MAP) is h_2

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp B$
☐ (B) $F \perp D | G$
☐ (C) $C \perp D | B, E$
☐ (D) $A \perp D | B$

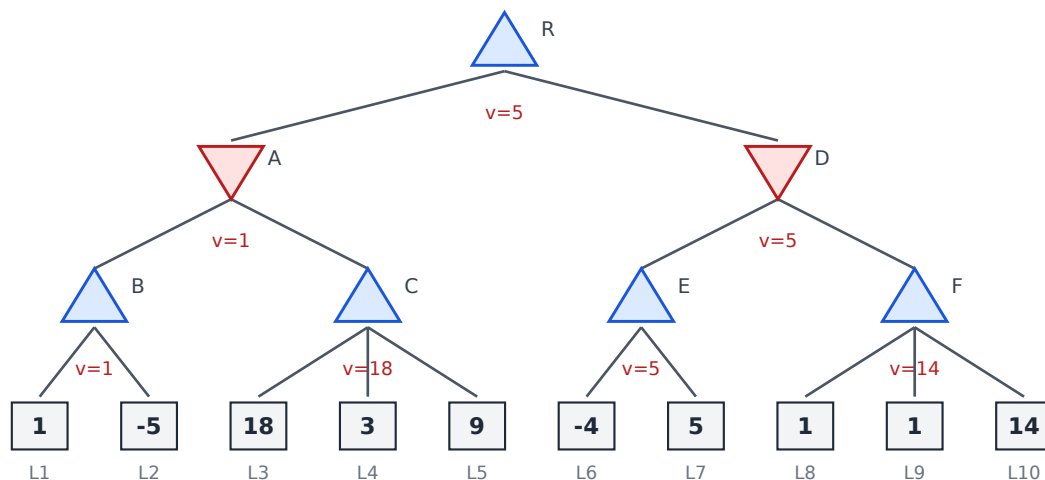
Answer Key — Mock Test 071

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	5	14	MCQ	2	Uninformed Search	A
2	NAT	1	Adversarial Search	29	15	MSQ	2	Propositional Logic	B, D
3	MCQ	1	Uninformed Search	D	16	MSQ	2	Predicate Logic	B, C, D
4	MCQ	1	Adversarial Search	B	17	MSQ	2	Propositional Logic	B
5	NAT	1	Propositional Logic	7	18	MCQ	2	Predicate Logic	A
6	NAT	1	Propositional Logic	4	19	NAT	2	Bayesian Networks	0.875 to 0.885
7	NAT	1	Predicate Logic	14	20	MCQ	2	Sampling	C
8	NAT	1	Predicate Logic	65	21	NAT	2	Exact Inference	4
9	NAT	1	Conditional Independence	4	22	NAT	2	Exact Inference	0.319 to 0.323
10	MCQ	1	Conditional Independence	A	23	NAT	2	Sampling	0.7998 to 0.8002
11	MSQ	2	Informed Search	C	24	MSQ	2	Probability for AI	A, B, C
12	MSQ	2	Adversarial Search	B, C	25	MSQ	2	Conditional Independence	C, D
13	MCQ	2	Uninformed Search	A					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 5

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 1$, $C = 18$, $A = 1$, $E = 5$, $F = 14$, $D = 5$. Root value = **5**; MAX's optimal first move is towards D.

Q.2 [NAT, 1M · Adversarial Search] Answer: 29

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 5^2 + 5^1 - 1 = 25 + 5 - 1 = \mathbf{29}$, versus 125 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.3 [MCQ, 1M · Uninformed Search] Answer: D (D: S, A, D, J, I, E, C, G)

Adjacency lists (alphabetical): A: {D, S}; C: {E, G, I}; D: {A, I, S}; E: {C, G, I, J}; G: {C, E}; I: {C, D, E, J}; J: {E, I, S}; S: {A, D, J}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[A, D, J]
2	A	[D, J]
3	D	[J, I]

Step	Expanded	Frontier (FIFO) after expansion
4	J	[I, E]
5	I	[E, C]
6	E	[C, G]
7	C	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, A, D, J, I, E, C, G**. The path returned is $S \rightarrow J \rightarrow E \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [MCQ, 1M · Adversarial Search] **Answer: B** ($B: v \leq \alpha$)

A MIN node's value can only decrease as more children are seen. Once $v \leq \alpha$, MAX already has an alternative worth α higher up the tree and will never let play reach this node. $v \geq \beta$ (A) is the cutoff test at MAX nodes. (C) and (D) do not justify pruning.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 7**

Build the truth table (sub-formulas shown first):

P	Q	R	$(P \rightarrow \neg R) \vee (R \leftrightarrow Q)$	$(P \vee \neg R) \rightarrow Q$	$((P \rightarrow \neg R) \vee (R \leftrightarrow Q)) \vee ((P \vee \neg R) \rightarrow Q)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	T	F	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	F	T

Rows where F = T: **7**.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 4**

Case $p=T$: the first clause is satisfied and the second requires $r=T$; q is free, giving 2 models. Case $p=F$: the second clause is satisfied and the first requires $q=T$; r is free, giving 2 models. Total = $2 + 2 = 4$.

Q.7 [NAT, 1M · Predicate Logic] **Answer: 14**

An interpretation of P is a subset of D : $2^4 = 16$ choices. $\exists x P(x)$ excludes the empty set, $\neg \forall x P(x)$ excludes D itself. Hence $16 - 2 = 14$.

Q.8 [NAT, 1M · Predicate Logic] **Answer: 65**

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 4$ this gives **65** (verified by enumerating all interpretations).

Q.9 [NAT, 1M · Conditional Independence] **Answer: 4**

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of F : $\{A, B\}$; children: $\{G\}$; co-parents: $\{D\}$.

$MB(F) = \{A, B, D, G\}$ — **4** nodes. Given its Markov blanket, F is conditionally independent of every other node in the network.

Q.10 [MCQ, 1M · Conditional Independence] **Answer: A** ($A: 5$)

Write $P(A, B, C) = P(C) P(A | C) P(B | C)$, using $A \perp B | C$. $P(C)$ needs 1 number, $P(A | C)$ needs 2 (one per value of C), $P(B | C)$ needs 2. Total = 5. $7 = 2^3 - 1$ is the unconstrained full joint; 6 and 4 do not correspond to this factorization.

Q.11 [MSQ, 2M · Informed Search] **Answer: C** ($C: h$ is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G :

n	B	E	F	G	H	J	K	S
$h(n)$	12	9	4	0	12	5	11	18
$h^*(n)$	14	10	10	0	14	6	12	20
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges:

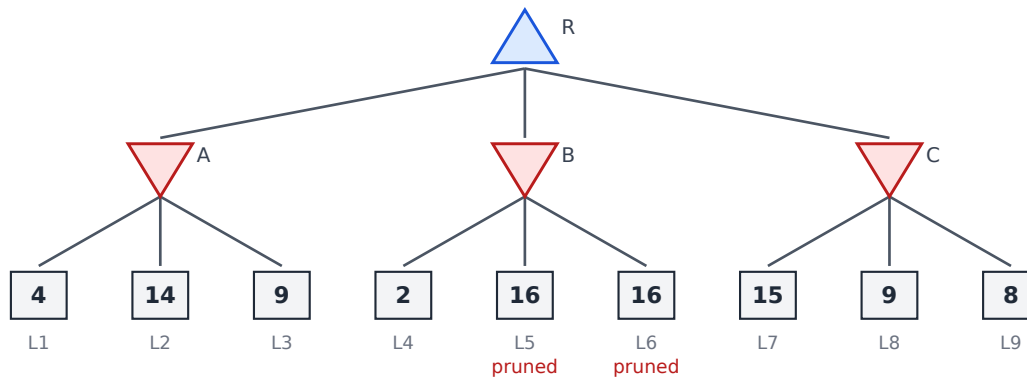
$B \rightarrow F$ ($12 > 4 + 4$), $K \rightarrow F$ ($11 > 3 + 4$), $H \rightarrow F$ ($12 > 4 + 4$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [MSQ, 2M · Adversarial Search] **Answer: B, C** (B: Leaf L5 is pruned.; C: Leaf L6 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node B: value $2 \leq \alpha = 4 \Rightarrow$ prune remaining children (leaves L5, L6).



Leaves evaluated (7): L1, L2, L3, L4, L7, L8, L9. Leaves pruned (2): L5, L6. Root value = 8 (same as plain minimax, 8 — pruning never changes the root value).

Q.13 [MCQ, 2M · Uninformed Search] **Answer: A** (A: $S \rightarrow D \rightarrow F \rightarrow I \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	H(6), D(7)
2	H (6)	D(7), J(11), E(14)
3	D (7)	J(10), E(14), F(15)
4	J (10)	E(14), F(15)
5	E (14)	F(15)
6	F (15)	I(19), G(23)
7	I (19)	G(21)
8	G (21)	← goal, stop

Path returned: **S** $\rightarrow$ **D** $\rightarrow$ **F** $\rightarrow$ **I** $\rightarrow$ **G**, cost = $7 + 8 + 4 + 2 = 21$. Nodes expanded (including S and G): **8** (S, H, D, J, E, F, I, G).

Q.14 [MCQ, 2M · Uninformed Search] **Answer: A** (A: BFS: 13, DFS: 12)

BFS generates nodes in label order (root, then 2, 3, 4, ...), so node 13 is the **13th** node tested. DFS preorder is 1, 2, 4, 8, 9, 5, 10, 11, 3, 6, 12, 13, ..., so node 13 is the **12th**. The other options miscount one or both orders.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: B, D** (B: $\neg(P \wedge R) \Leftrightarrow (\neg P \vee \neg R)$; D: $((P \rightarrow Q) \wedge \neg Q) \rightarrow \neg P$)

(A) $((R \rightarrow P) \rightarrow Q) \Leftrightarrow (R \rightarrow (P \rightarrow Q))$: **not valid** — counter-model $P=T, Q=F, R=F$.

(B) $\neg(P \wedge R) \Leftrightarrow (\neg P \vee \neg R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $\neg(R \wedge P) \Leftrightarrow (\neg R \wedge \neg P)$: **not valid** — counter-model $P=T, R=F$.

(D) $((P \rightarrow Q) \wedge \neg Q) \rightarrow \neg P$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: B, C, D** (B: $\forall x \exists y R(x, y)$; C: $\exists x R(x, x)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,3), (2,1), (3,2), (3,3), (4,2), (4,3)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(2,1) \wedge R(1,3)$ but $\neg R(2,3)$.

(B) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 3.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: B** (B: $(P \vee \neg Q)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(P \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(P \vee \neg Q)$ is derivable:

1. resolve $(P \vee \neg Q \vee \neg R)$ and (R) on R $\Rightarrow (P \vee \neg Q)$

(C) $(R \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) The resolution closure contains 14 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

Q.18 [MCQ, 2M · Predicate Logic] **Answer: A** (A: $\forall y R(A, y, F(y))$)

x is existential and not inside any universal quantifier, so it becomes a constant A. z is inside the scope of $\forall y$, so it becomes F(y) (arguments are the enclosing universally quantified variables). F(A) ignores the dependence on y and is an unwarranted strengthening; G(y) for x is wrong because x precedes $\forall y$; a constant for z also loses the dependence on y.

Q.19 [NAT, 2M · Bayesian Networks] **Answer: 0.875 to 0.885**

In noisy-OR, $P(\neg \text{fever} \mid \text{causes}) = \text{product of } q_i \text{ over the causes that are present}$. Present causes: cold and flu, so $P(\neg \text{fever}) = 0.6 \times 0.2 = 0.12$. $P(\text{fever} \mid \text{cold, flu, } \neg \text{malaria}) = 1 - 0.12 = 0.88$. Malaria is absent and contributes no factor.

Q.20 [MCQ, 2M · Sampling] **Answer: C** (C: A=T, B=T, C=F, D=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.75$; $u = 0.41 < 0.75 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.95$; $u = 0.32 < 0.95 \Rightarrow B = T$
- C: $P(C=T \mid B=T) = 0.5$; $u = 0.82 \geq 0.5 \Rightarrow C = F$
- D: $P(D=T \mid A=T) = 0.7$; $u = 0.74 \geq 0.7 \Rightarrow D = F$

Sample: **(A=T, B=T, C=F, D=F)**.

Q.21 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f()$, $f(B, D)$, $f(A, E)$, $f(D, E, F)$, $f(D, E, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B)$, $f(B, D)$	$\{A, B, D\}$	$f(A, D)$
E	$f(A, E)$, $f(D, E, F)$, $f(D, E, G)$	$\{A, D, E, F, G\}$	$f(A, D, F, G)$
G	$f(A, D, F, G)$	$\{A, D, F, G\}$	$f(A, D, F)$
D	$f(A, D)$, $f(A, D, F)$	$\{A, D, F\}$	$f(A, F)$
F	$f(A, F)$	$\{A, F\}$	$f(A)$

The largest factor produced has **4** variable(s) (created when eliminating E); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over A and normalise.

Q.22 [NAT, 2M · Exact Inference] **Answer: 0.319 to 0.323**

Factorisation: $P(\text{Flu, Fever, Cough, Fatigue, Doctor}) = P(\text{Flu}) \cdot P(\text{Fever} \mid \text{Flu}) \cdot P(\text{Cough} \mid \text{Flu, Fever}) \cdot P(\text{Fatigue} \mid \text{Flu}) \cdot P(\text{Doctor} \mid \text{Cough, Fatigue})$.

Inference by enumeration: $P(\text{Cough} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cough}, e, \text{hidden})$.

Hidden: Fe, Fa	Co	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, T	T	$0.25 \times 0.4 \times 0.5 \times 0.95 \times 0.2$	0.0095
T, F	T	$0.25 \times 0.4 \times 0.5 \times 0.05 \times 0.9$	0.00225
F, T	T	$0.25 \times 0.6 \times 0.3 \times 0.95 \times 0.2$	0.00855

Hidden: Fe, Fa	Co	product of CPT entries ($F_I \cdot F_e \cdot C_o \cdot F_a \cdot D_o$)	value
F, F	T	$0.25 \times 0.6 \times 0.3 \times 0.05 \times 0.9$	0.00203
T, T	F	$0.25 \times 0.4 \times 0.5 \times 0.95 \times 0.3$	0.01425
T, F	F	$0.25 \times 0.4 \times 0.5 \times 0.05 \times 0.4$	0.001
F, T	F	$0.25 \times 0.6 \times 0.7 \times 0.95 \times 0.3$	0.02993
F, F	F	$0.25 \times 0.6 \times 0.7 \times 0.05 \times 0.4$	0.0021

$P(\text{Cough}=T, e) = 0.02233$; $P(\text{Cough}=F, e) = 0.04728$; $P(e) = 0.0696$.

$P(\text{Cough}=T \mid e) = 0.02233 / 0.0696 = \mathbf{0.3208} \approx 0.321$.

Q.23 [NAT, 2M · Sampling] **Answer: 0.7998 to 0.8002**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(C=F \mid B=F)=0.8$	0.8	T
2	$P(C=F \mid B=T)=0.2$	0.2	F
3	$P(C=F \mid B=F)=0.8$	0.8	T
4	$P(C=F \mid B=T)=0.2$	0.2	F

$\sum w (\text{all}) = 2$; $\sum w (A = T) = 1.6$.

Estimate $P(A=T \mid C = F) = 1.6 / 2 = \mathbf{0.8}$.

Q.24 [MSQ, 2M · Probability for AI] **Answer: A, B, C** (A: $P(e) = 0.31$; B: $P(h_3 \mid e) \approx 0.452$; C: $P(h_1 \mid e) \approx 0.161$)

Unnormalized posteriors $P(e \mid h_i)P(h_i)$: $0.1 \times 0.5 = 0.05$, $0.4 \times 0.3 = 0.12$, $0.7 \times 0.2 = 0.14$. Marginalizing, $P(e) = 0.05 + 0.12 + 0.14 = 0.31$ (A true). $P(h_3 \mid e) = 0.14/0.31 \approx 0.452$ (B true); $P(h_1 \mid e) = 0.05/0.31 \approx 0.161$ (C true); $P(h_2 \mid e) = 0.12/0.31 \approx 0.387$. The MAP hypothesis is h_3 (largest posterior), even though it had the smallest prior, so (D) is false.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: C, D** (C: $C \perp D \mid B, E$; D: $A \perp D \mid B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp B$: **FALSE** — the path $E \leftarrow A \rightarrow B$ is active given $\{\}$.

(B) $F \perp D \mid G$: **FALSE** — the path $F \leftarrow A \rightarrow B \rightarrow D$ is active given $\{G\}$.

(C) $C \perp D \mid B, E$: **TRUE** — $C \leftarrow A \rightarrow B \rightarrow D$ (blocked: observed chain node B); $C \leftarrow A \rightarrow E \leftarrow B \rightarrow D$ (blocked: observed fork node B); $C \leftarrow A \rightarrow F \leftarrow D$ (blocked: collider F with neither it nor a descendant observed); $C \leftarrow A \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed); $C \leftarrow B \leftarrow A \rightarrow F \leftarrow D$ (blocked: observed chain node B); $C \leftarrow B \leftarrow A \rightarrow G \leftarrow D$ (blocked: observed chain node B) (+3 more paths, all blocked).

(D) $A \perp D \mid B$: **TRUE** — $A \rightarrow B \rightarrow D$ (blocked: observed chain node B); $A \rightarrow C \leftarrow B \rightarrow D$ (blocked: collider C with neither it nor a descendant observed); $A \rightarrow E \leftarrow B \rightarrow D$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow F \leftarrow D$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed).

Mock Test 072

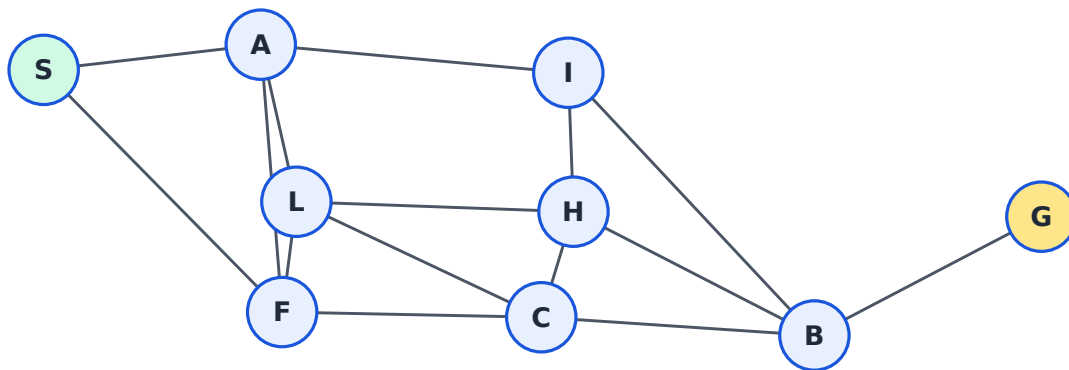
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 8 / 13	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, F, L, H, I, B, G
- ☐ (B) S, A, F, I, L, C, B, H, G
- ☐ (C) S, A, C, F, B, G
- ☐ (D) S, A, F, C, B, G

Q.2 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Depth-limited search with limit $l = 4$ is run on a tree with branching factor $b = 4$. No goal exists within depth 4. How many nodes (including the root) does it generate before returning cutoff?

Answer (numerical): _____

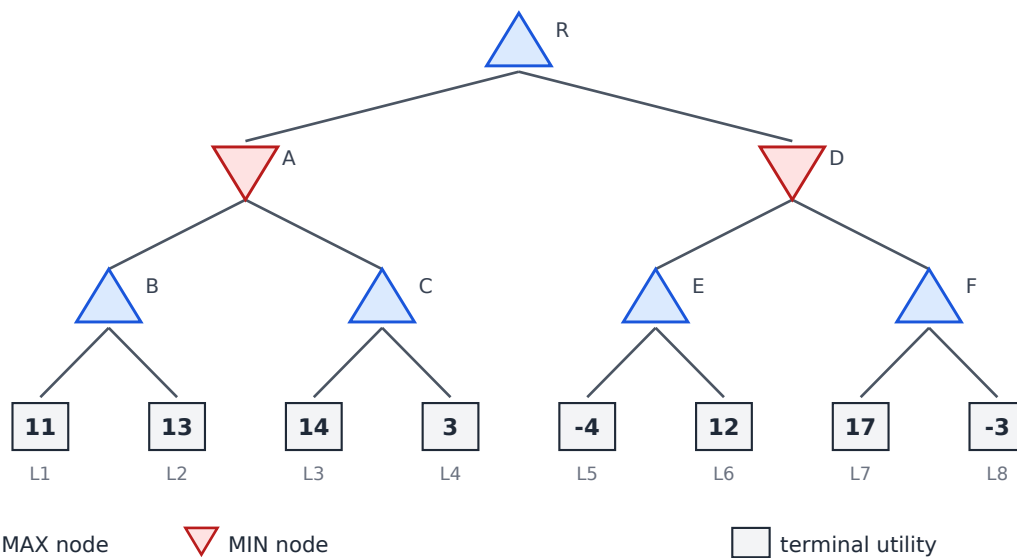
Q.3 [NAT · 1 mark · no negative marking] Adversarial Search

A chess evaluation function is the weighted linear function $\text{Eval} = 9 \cdot \Delta q + 5 \cdot \Delta r + 3 \cdot \Delta b + 1 \cdot \Delta p$, where each Δ is White's count minus Black's count of queens, rooks, bishops and pawns. In a position, the queens are equal, White has one extra rook, Black has one extra bishop, and White has two extra pawns. What is Eval?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

The formula $(A \wedge B) \vee (C \wedge D) \vee (E \wedge F)$ is converted to CNF by repeatedly distributing $\vee$ over $\wedge$ (no simplification). How many clauses does the result have?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((\neg P \leftrightarrow R) \wedge (P \wedge \neg P)) \leftrightarrow \neg(Q \leftrightarrow R) ?$$

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

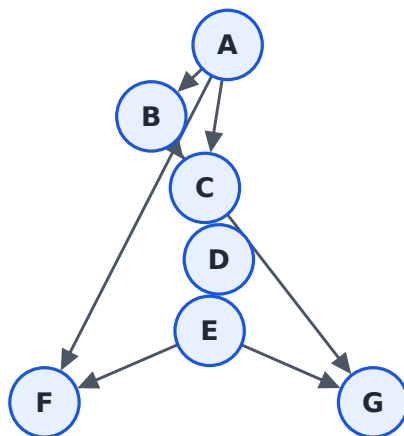
Q.8 [NAT · 1 mark · no negative marking] Propositional Logic

Of the 16 possible binary Boolean connectives $\circ$ (functions $\{T, F\}^2 \rightarrow \{T, F\}$), how many are commutative, i.e. satisfy $p \circ q \equiv q \circ p$?

Answer (numerical): _____

Q.9 [MCQ · 1 mark · negative: -1/3] Conditional Independence – Markov blanket

Which set is the Markov blanket of **C** in the Bayesian network below?



- ☐ (A) {A, B, D}
- ☐ (B) {A, B, D, E, F, G}
- ☐ (C) {D}
- ☐ (D) {A, B, D, E}

Q.10 [MCQ · 1 mark · negative: -1/3] Sampling

A discrete variable Weather has distribution (sunny: 0.2, cloudy: 0.5, rainy: 0.3) (in that order).

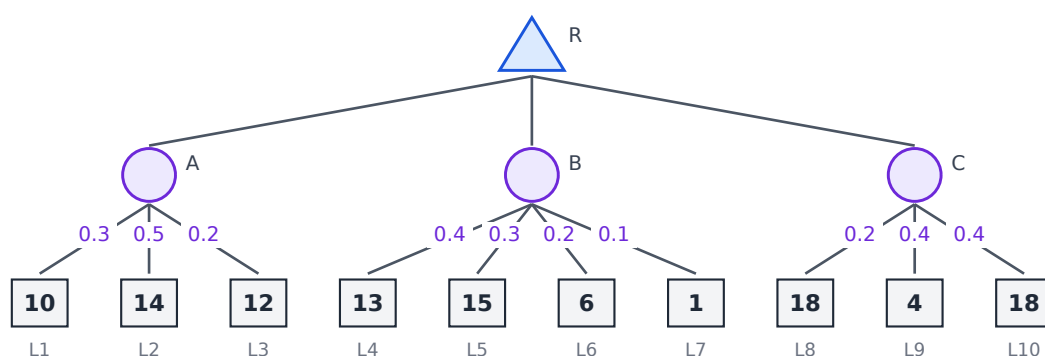
Using inverse-CDF sampling with cumulative intervals in the same order, which value is produced by the uniform random number $u = 0.65$?

- ☐ (A) cloudy
- ☐ (B) sunny
- ☐ (C) rainy
- ☐ (D) Cannot be determined from u alone

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node ○ CHANCE node (edge = probability)

□ terminal utility

Answer (numerical): _____

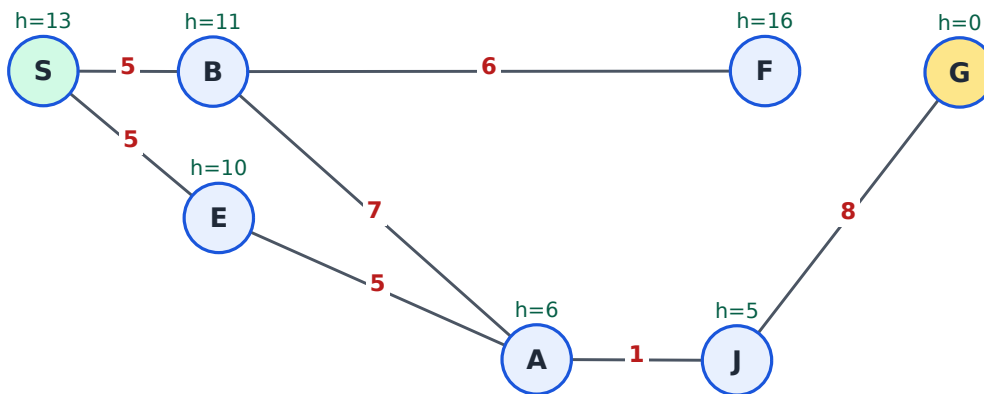
Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search

Minimax search (depth-first) runs on a finite game tree with branching factor b and maximum depth m . Which of the following are TRUE?

- ☐ (A) Its time complexity is $O(b^m)$
- ☐ (B) Its space complexity is $O(bm)$ when all successors are generated at once
- ☐ (C) Against an optimal opponent, the minimax strategy achieves the minimax value of the root
- ☐ (D) It needs a heuristic evaluation function even when every path can be searched to a terminal state

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

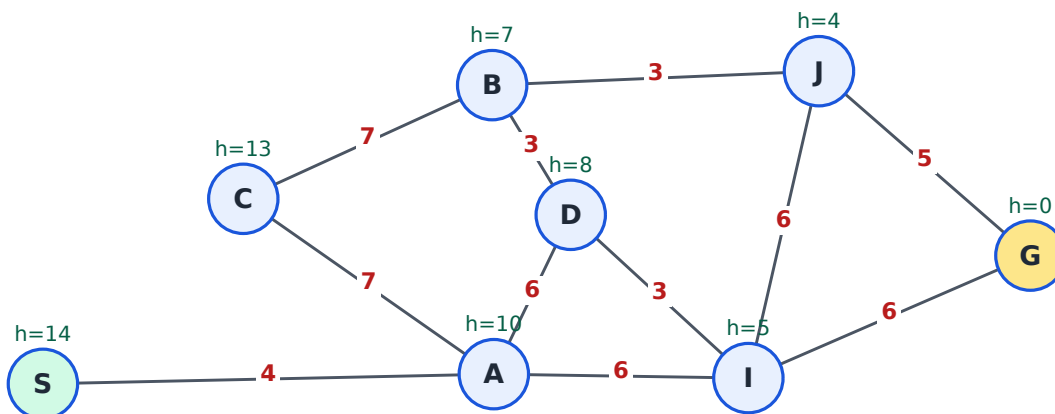
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic

On the domain $D = \{1, 2, 3\}$, let $R = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$. Which of the following sentences are TRUE in this interpretation?

- ☐ (A) $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$
☐ (B) $\forall x R(x, x)$
☐ (C) $\exists x \forall y R(x, y)$
☐ (D) $\forall y \exists x R(x, y)$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(\neg P \rightarrow R) \vee (R \rightarrow \neg P)$
☐ (B) $(P \vee R) \rightarrow (P \wedge R)$
☐ (C) $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$
☐ (D) $(P \rightarrow Q) \rightarrow ((P \vee R) \rightarrow Q)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

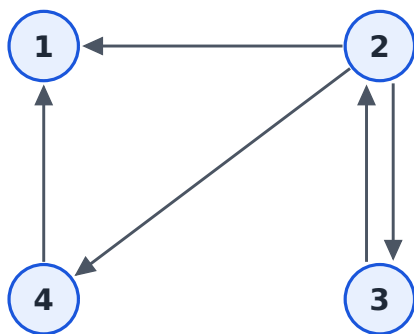
- (1) $(\neg R \wedge R) \vee \neg \neg R$
 (2) $(\neg R \leftrightarrow \neg R) \wedge \neg \neg P$
 (3) $(R \leftrightarrow \neg Q) \vee (R \vee Q)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg \neg \neg R$
☐ (B) $(R \vee Q) \wedge (R \leftrightarrow \neg Q)$
☐ (C) $\neg Q \vee P$
☐ (D) $\neg Q \vee \neg Q$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (D) $\forall x R(x, x)$

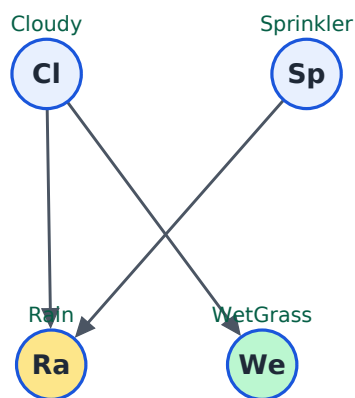
Q.19 [MSQ · 2 marks · no negative marking] Bayesian Networks

A Bayesian network is the chain $A \rightarrow B \rightarrow C$ over Boolean variables with $P(a) = 0.6$, $P(b | a) = 0.7$, $P(b | \neg a) = 0.2$, $P(c | b) = 0.9$, $P(c | \neg b) = 0.3$. Which of the following are correct?

- ☐ (A) $P(b) = 0.5$
☐ (B) $P(c) = 0.6$
☐ (C) $P(a | c) = 0.72$
☐ (D) $P(a | c) = P(a)$, since A and C are not adjacent

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T | \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute **$P(\text{WetGrass} = T | \text{Rain} = F)$** (round off to 3 decimal places).



$P(Cl=T)$
0.4

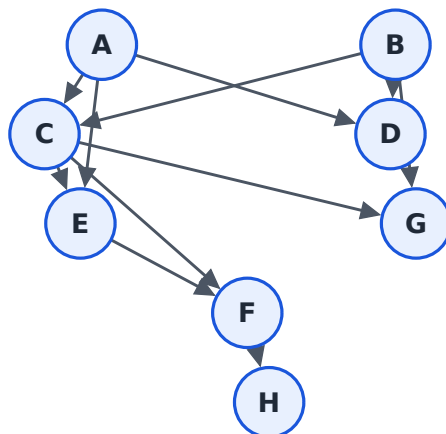
$P(Sp=T)$	Cl	Sp	$P(Ra=T ...)$
0.95	T	T	0.25
	T	F	0.1
	F	T	0.05
	F	F	0.9

Cl	$P(We=T ...)$
T	0.4
F	0.7

Answer (numerical): _____

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

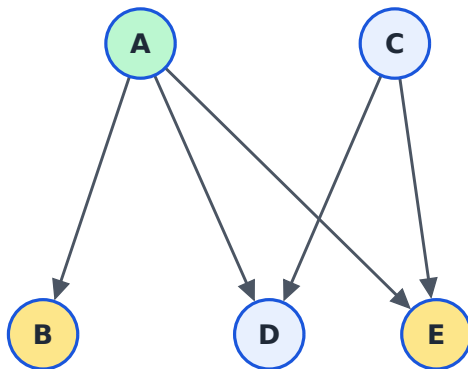
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp A$
☐ (B) $F \perp E | H$
☐ (C) $H \perp A | B, F$
☐ (D) $C \perp D | H$

Q.22 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $E = T$, $B = T$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $P(A = T \mid E = T, B = T)$ (round off to 3 decimal places).



P(A=T)
0.3

A	P(B=T ...)	P(C=T)
T	0.1	0.25
F	0.2	

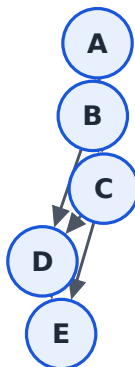
A	C	P(D=T ...)	A	C	P(E=T ...)
T	T	0.25	T	T	0.9
T	F	0.6	T	F	0.7
F	T	0.95	F	T	0.9
F	F	0.9	F	F	0.6

#	A	C	D
1	F	F	T
2	T	F	T
3	F	T	T
4	F	F	F

Answer (numerical): _____

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $u = \{0.48, 0.4, 0.6, 0.9, 0.67\}$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampling parent values})$. Which sample is generated?



P(A=T)
0.9

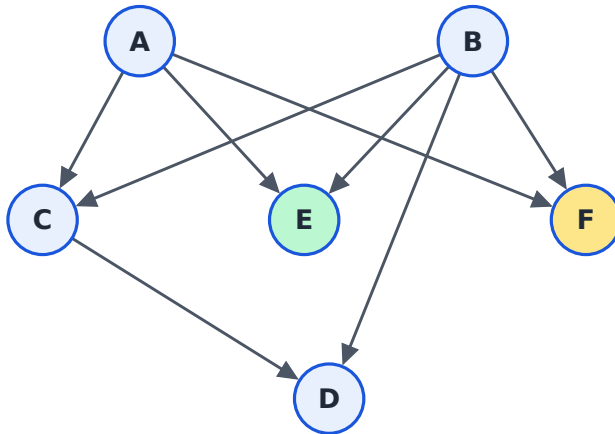
A	P(B=T ...)	A	B	P(C=T ...)
T	0.8	T	T	0.25
F	0.95	T	F	0.9
		F	T	0.9
		F	F	0.3

B	C	P(D=T ...)	C	D	P(E=T ...)
T	T	0.7	T	T	0.5
T	F	0.05	T	F	0.95
F	T	0.3	F	T	0.2
F	F	0.3	F	F	0.05

- ☐ (A) A=F, B=T, C=F, D=F, E=F
☐ (B) A=T, B=T, C=F, D=F, E=F
☐ (C) A=T, B=F, C=F, D=F, E=F
☐ (D) A=F, B=F, C=T, D=T, E=T

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(E \mid F = T)$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D, B, A, C**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network (DAG) with edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$, $E \rightarrow D$, $D \rightarrow F$, $C \rightarrow G$, $H \rightarrow G$. Which of the following nodes belong to the Markov blanket of C?

- ☐ (A) B
☐ (B) F
☐ (C) H
☐ (D) A

Answer Key — Mock Test 072

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	NAT	2	Informed Search	16
2	NAT	1	Uninformed Search	341	15	MSQ	2	Predicate Logic	A, C, D
3	NAT	1	Adversarial Search	4	16	MSQ	2	Propositional Logic	A, C
4	NAT	1	Adversarial Search	13	17	MSQ	2	Propositional Logic	C
5	NAT	1	Propositional Logic	8	18	MSQ	2	Predicate Logic	A, B
6	NAT	1	Propositional Logic	4	19	MSQ	2	Bayesian Networks	A, B, C
7	NAT	1	Predicate Logic	8	20	NAT	2	Exact Inference	0.591 to 0.595
8	NAT	1	Propositional Logic	8	21	MSQ	2	Conditional Independence	A, C
9	MCQ	1	Conditional Independence	A	22	NAT	2	Sampling	0.141 to 0.145
10	MCQ	1	Sampling	A	23	MCQ	2	Sampling	B
11	NAT	2	Adversarial Search	12.39 to 12.41	24	NAT	2	Exact Inference	3
12	MSQ	2	Adversarial Search	A, B, C	25	MSQ	2	Bayesian Networks	A, C, D
13	NAT	2	Informed Search	6					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: D (D: S, A, F, C, B, G)

Adjacency lists (alphabetical): A: {F, I, L, S}; B: {C, G, H, I}; C: {B, F, H, L}; F: {A, C, L, S}; G: {B}; H: {B, C, I, L}; I: {A, B, H}; L: {A, C, F, H}; S: {A, F}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, A, F, C, B, G**; the path found is $S \rightarrow A \rightarrow F \rightarrow C \rightarrow B \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Uninformed Search] Answer: 341

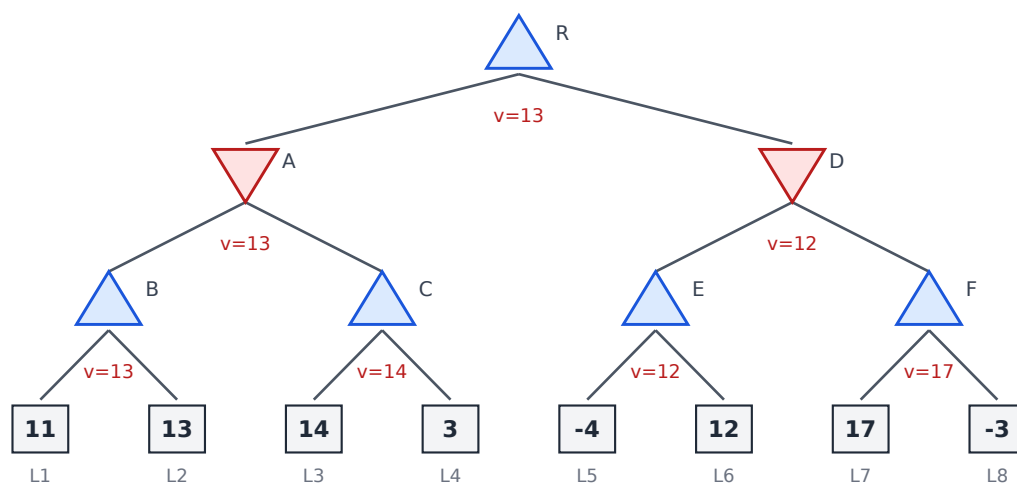
DLS explores the complete tree down to depth l : $1 + b + \dots + b^l = 1 + 4 + 16 + 64 + 256 = \mathbf{341} = (b^{(l+1)} - 1)/(b - 1)$.

Q.3 [NAT, 1M · Adversarial Search] Answer: 4

$\Delta q = 0$, $\Delta r = +1$, $\Delta b = -1$, $\Delta p = +2$. $\text{Eval} = 9 \cdot 0 + 5 \cdot 1 + 3 \cdot (-1) + 1 \cdot 2 = 0 + 5 - 3 + 2 = \mathbf{4}$. A weighted linear evaluation function treats features as independent contributions.

Q.4 [NAT, 1M · Adversarial Search] Answer: 13

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 13$, $C = 14$, $A = 13$, $E = 12$, $F = 17$, $D = 12$. Root value = **13**; MAX's optimal first move is towards A.

Q.5 [NAT, 1M · Propositional Logic] Answer: 8

Distribution produces one clause per way of choosing one literal from each conjunction: $\{A, B\} \times \{C, D\} \times \{E, F\}$, i.e. $2 \times 2 \times 2 = 8$ clauses such as $A \vee C \vee E$, $A \vee C \vee F$, ..., $B \vee D \vee F$. No clause is a tautology or a duplicate because all six symbols are distinct. This illustrates the exponential blow-up of naive CNF conversion.

Q.6 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg P \leftrightarrow R) \wedge (P \wedge \neg P)$	$\neg(Q \leftrightarrow R)$	$((\neg P \leftrightarrow R) \wedge (P \wedge \neg P)) \leftrightarrow \neg(Q \leftrightarrow R)$
T	T	T	F	F	T
T	T	F	F	T	F
T	F	T	F	T	F
T	F	F	F	F	T
F	T	T	F	F	T
F	T	F	F	T	F
F	F	T	F	T	F
F	F	F	F	F	T

Rows where $F = T$: **4**.

Q.7 [NAT, 1M · Predicate Logic] Answer: 8

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 3$ this gives **8** (verified by enumerating all interpretations).

Q.8 [NAT, 1M · Propositional Logic] Answer: 8

A binary connective is fixed by its outputs on TT, TF, FT, FF. Commutativity requires $\text{output}(TF) = \text{output}(FT)$. So we may freely choose the outputs on TT, FF and the common value on TF/FT: $2^3 = 8$. Examples: $\wedge$, $\vee$, $\leftrightarrow$, $\oplus$, NAND, NOR, constant T, constant F.

Q.9 [MCQ, 1M · Conditional Independence] Answer: A (A: {A, B, D})

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of C: {A, B}; children: {D}; co-parents: $\{\emptyset\}$.

$MB(C) = \{A, B, D\}$ — 3 nodes. Given its Markov blanket, C is conditionally independent of every other node in the network.

Q.10 [MCQ, 1M · Sampling] Answer: A (A: cloudy)

Cumulative distribution: sunny [0, 0.2), cloudy [0.2, 0.7), rainy [0.7, 1.0). $u = 0.65$ lies in [0.2, 0.7), so the sample is cloudy. Inverse-CDF sampling is deterministic given u and the ordering, so (D) is wrong.

Q.11 [NAT, 2M · Adversarial Search] Answer: 12.39 to 12.41

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.3 \times 10 + 0.5 \times 14 + 0.2 \times 12 = 12.4$
- $B = 0.4 \times 13 + 0.3 \times 15 + 0.2 \times 6 + 0.1 \times 1 = 11$
- $C = 0.2 \times 18 + 0.4 \times 4 + 0.4 \times 18 = 12.4$

Root = $\max(12.4, 11, 12.4) = 12.4$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.12 [MSQ, 2M · Adversarial Search] Answer: A, B, C

(A) and (B) are true: minimax is a complete depth-first exploration of the tree. (C) is true: minimax is optimal against an optimal adversary and guarantees the root's minimax value. (D) is false: when the search reaches terminal states, the exact utilities are used. An evaluation function is needed only when the search is cut off early.

Q.13 [NAT, 2M · Informed Search] Answer: 6

Heuristic values: $h(A)=6$, $h(B)=11$, $h(E)=10$, $h(F)=16$, $h(G)=0$, $h(J)=5$, $h(S)=13$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	E(15), B(16)
2	E (15)	A(16), B(16)
3	A (16)	B(16), J(16)
4	B (16)	J(16), F(27)
5	J (16)	G(19), F(27)
6	G (19)	F(27) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **E** $\rightarrow$ **A** $\rightarrow$ **J** $\rightarrow$ **G** with cost **19**; nodes expanded = **6** (S, E, A, B, J, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.14 [NAT, 2M · Informed Search] Answer: 16

Heuristic values: $h(A)=10$, $h(B)=7$, $h(C)=13$, $h(D)=8$, $h(G)=0$, $h(I)=5$, $h(J)=4$, $h(S)=14$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (14)	A(14)
2	A (14)	I(15), D(18), C(24)
3	I (15)	G(16), D(18), J(20), C(24)
4	G (16)	D(18), J(20), C(24) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **A** $\rightarrow$ **I** $\rightarrow$ **G** with cost **16**; nodes expanded = **4** (S, A, I, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [MSQ, 2M · Predicate Logic] Answer: A, C, D (A: $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$; C: $\exists x \forall y R(x, y)$; D: $\forall y \exists x R(x, y)$)

(A) The composable pairs are $(1,1)(1,y) \rightarrow (1,y) \in R$, and $(1,2)(2,3) \rightarrow (1,3) \in R$; so R is transitive. (B) False: $(2,2) \notin R$. (C) True with $x = 1$: $(1,1), (1,2), (1,3) \in R$. (D) True: $x = 1$ works for every y . Enumeration over all triples confirms A, C, D.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $(\neg P \rightarrow R) \vee (R \rightarrow \neg P)$; C: $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$)

(A) $(\neg P \rightarrow R) \vee (R \rightarrow \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $(P \vee R) \rightarrow (P \wedge R)$: **not valid** — counter-model $P=T, R=F$.

(C) $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $(P \rightarrow Q) \rightarrow ((P \vee R) \rightarrow Q)$: **not valid** — counter-model $P=F, Q=F, R=T$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [MSQ, 2M · Propositional Logic] Answer: C (C: $\neg Q \vee P$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	T	T	T
T	F	T	T

(A) $\neg\neg\neg R$ is false in the KB-model $P=T, Q=T, R=T \Rightarrow$ not entailed.

(B) $(R \vee Q) \wedge (R \leftrightarrow \neg Q)$ is false in the KB-model $P=T, Q=T, R=T \Rightarrow$ not entailed.

(C) $\neg Q \vee P$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $\neg Q \vee \neg Q$ is false in the KB-model $P=T, Q=T, R=T \Rightarrow$ not entailed.

Q.18 [MSQ, 2M · Predicate Logic] Answer: A, B (A: $\exists x \forall y \neg R(x, y)$; B: $\forall y \exists x R(x, y)$)

$R = \{(2,1), (2,3), (2,4), (3,2), (4,1)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

(B) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,4)$ holds but $R(4,2)$ does not.

(D) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Bayesian Networks] Answer: A, B, C (A: $P(b) = 0.5$; B: $P(c) = 0.6$; C: $P(a | c) = 0.72$)

$P(b) = 0.6 \times 0.7 + 0.4 \times 0.2 = 0.42 + 0.08 = 0.5$ (A true). $P(c) = 0.5 \times 0.9 + 0.5 \times 0.3 = 0.45 + 0.15 = 0.6$ (B true). $P(a, c) = 0.6 \times (0.7 \times 0.9 + 0.3 \times 0.3) = 0.6 \times 0.72 = 0.432$, so $P(a | c) = 0.432/0.6 = 0.72$ (C true). (D) is false: A and C are connected by the active path $A \rightarrow B \rightarrow C$ (B unobserved), so observing c raises P(a) from 0.6 to 0.72.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.591 to 0.595

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler}) \cdot P(\text{Rain} | \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{WetGrass} | \text{Cloudy})$.

Inference by enumeration: $P(\text{WetGrass} | e) = \alpha \sum_{\text{hidden}} P(\text{WetGrass}, e, \text{hidden})$.

Hidden: Cl, Sp	We	product of CPT entries (Cl · Sp · Ra · We)	value
T, T	T	$0.4 \times 0.95 \times 0.75 \times 0.4$	0.114
T, F	T	$0.4 \times 0.05 \times 0.9 \times 0.4$	0.0072
F, T	T	$0.6 \times 0.95 \times 0.95 \times 0.7$	0.37905
F, F	T	$0.6 \times 0.05 \times 0.1 \times 0.7$	0.0021
T, T	F	$0.4 \times 0.95 \times 0.75 \times 0.6$	0.171
T, F	F	$0.4 \times 0.05 \times 0.9 \times 0.6$	0.0108
F, T	F	$0.6 \times 0.95 \times 0.95 \times 0.3$	0.16245
F, F	F	$0.6 \times 0.05 \times 0.1 \times 0.3$	0.0009

$P(\text{WetGrass}=T, e) = 0.50235$; $P(\text{WetGrass}=F, e) = 0.34515$; $P(e) = 0.8475$.

$P(\text{WetGrass}=T | e) = 0.50235 / 0.8475 = \mathbf{0.5927} \approx 0.593$.

Q.21 [MSQ, 2M · Conditional Independence] Answer: A, C (A: $B \perp A$; C: $H \perp A \mid B, F$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp A$: **TRUE** — $B \rightarrow C \leftarrow A$ (blocked: collider C with neither it nor a descendant observed); $B \rightarrow C \rightarrow E \leftarrow A$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow C \rightarrow F \leftarrow E \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow D \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow G \leftarrow C \leftarrow A$ (blocked: collider G with neither it nor a descendant observed); $B \rightarrow G \leftarrow C \rightarrow E \leftarrow A$ (blocked: collider G with neither it nor a descendant observed) (+1 more paths, all blocked).

(B) $F \perp E \mid H$: **FALSE** — the path $F \leftarrow C \leftarrow A \rightarrow E$ is active given $\{H\}$.

(C) $H \perp A \mid B, F$: **TRUE** — $H \leftarrow F \leftarrow C \leftarrow A$ (blocked: observed chain node F); $H \leftarrow F \leftarrow C \rightarrow B \rightarrow D \leftarrow A$ (blocked: observed chain node F); $H \leftarrow F \leftarrow C \rightarrow E \leftarrow A$ (blocked: observed chain node F); $H \leftarrow F \leftarrow C \rightarrow G \rightarrow B \rightarrow D \leftarrow A$ (blocked: observed chain node F); $H \leftarrow F \leftarrow E \leftarrow A$ (blocked: observed chain node F); $H \leftarrow F \leftarrow E \leftarrow C \leftarrow A$ (blocked: observed chain node F) (+2 more paths, all blocked).

(D) $C \perp D \mid H$: **FALSE** — the path $C \leftarrow A \rightarrow D$ is active given $\{H\}$.

Q.22 [NAT, 2M · Sampling] Answer: 0.141 to 0.145

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(B=T A=F)=0.2 \times P(E=T A=F,C=F)=0.6$	0.12	F
2	$P(B=T A=T)=0.1 \times P(E=T A=T,C=F)=0.7$	0.07	T
3	$P(B=T A=F)=0.2 \times P(E=T A=F,C=T)=0.9$	0.18	F
4	$P(B=T A=F)=0.2 \times P(E=T A=F,C=F)=0.6$	0.12	F

$\sum w (\text{all}) = 0.49$; $\sum w (A = T) = 0.07$.

Estimate $P(A=T \mid E = T, B = T) = 0.07 / 0.49 = \mathbf{0.1429}$.

Q.23 [MCQ, 2M · Sampling] Answer: B (B: $A=T, B=T, C=F, D=F, E=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.9$; $u = 0.48 < 0.9 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.8$; $u = 0.4 < 0.8 \Rightarrow B = T$
- C: $P(C=T \mid A=T, B=T) = 0.25$; $u = 0.6 \geq 0.25 \Rightarrow C = F$
- D: $P(D=T \mid B=T, C=F) = 0.05$; $u = 0.9 \geq 0.05 \Rightarrow D = F$
- E: $P(E=T \mid C=F, D=F) = 0.05$; $u = 0.67 \geq 0.05 \Rightarrow E = F$

Sample: **$\{A=T, B=T, C=F, D=F, E=F\}$** .

Q.24 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, B, C)$, $f(B, C, D)$, $f(A, B, E)$, $f(A, B)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(B, C, D)$	$\{B, C, D\}$	$f(B, C)$
B	$f(B)$, $f(A, B, C)$, $f(A, B, E)$, $f(A, B)$, $f(B, C)$	$\{A, B, C, E\}$	$f(A, C, E)$
A	$f(A)$, $f(A, C, E)$	$\{A, C, E\}$	$f(C, E)$
C	$f(C, E)$	$\{C, E\}$	$f(E)$

The largest factor produced has **3** variable(s) (created when eliminating B); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over E and normalise.

Q.25 [MSQ, 2M · Bayesian Networks] Answer: A, C, D (A: B; C: H; D: A)

Parents of C: $\{A\}$. Children of C: $\{D, G\}$. Other parents of the children: D has parents B, C, E and G has parents C, H, giving $\{B, E, H\}$. Markov blanket of C = $\{A, B, D, E, G, H\}$. So A (parent), B (co-parent via D) and H (co-parent via G) belong to it. F is a child of D, i.e. a grandchild of C, and is not in the blanket: given D, F is independent of C.

Mock Test 073

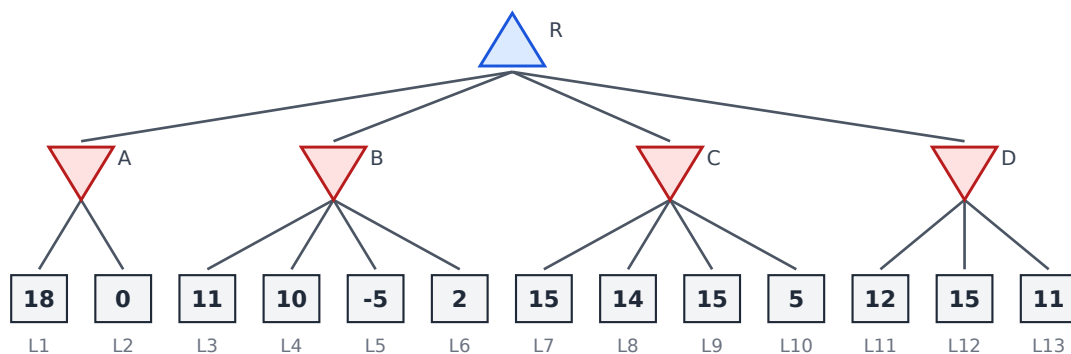
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	8 / 8 / 9	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



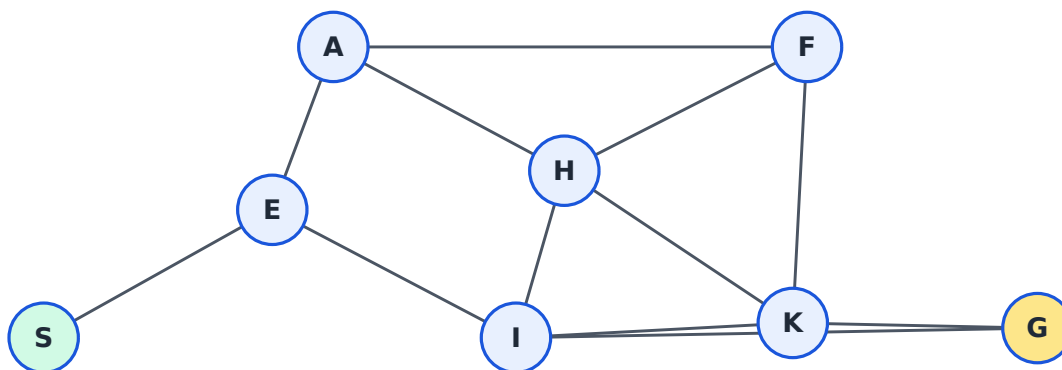
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to D; root value 11
- ☐ (B) Move to A; root value 0
- ☐ (C) Move to B; root value -5
- ☐ (D) Move to C; root value 5

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, E, I, K, H, F, A, G
- ☐ (B) S, E, A, I, F, H, G
- ☐ (C) S, E, A, F, H, I, G
- ☐ (D) S, A, E, F, H, I, G

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 2$ (so $b^d = 16$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Informed Search

Iterative deepening A* (IDA*) uses a cutoff on $f = g + h$. After an iteration fails, the cutoff for the next iteration is set to

- ☐ (A) the previous cutoff plus 1
- ☐ (B) the smallest f-value of any node that exceeded the cutoff in the previous iteration
- ☐ (C) the largest f-value seen in the previous iteration
- ☐ (D) twice the previous cutoff

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((\neg Q \vee R) \vee (R \wedge R)) \wedge ((\neg P \leftrightarrow P) \leftrightarrow \neg \neg P) ?$$

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

For propositional sentences α and β , $\alpha \models \beta$ holds if and only if

- ☐ (A) $\alpha \wedge \neg \beta$ is unsatisfiable
- ☐ (B) $\alpha \wedge \beta$ is satisfiable
- ☐ (C) $\neg \alpha \vee \beta$ is satisfiable
- ☐ (D) $\alpha \vee \neg \beta$ is valid

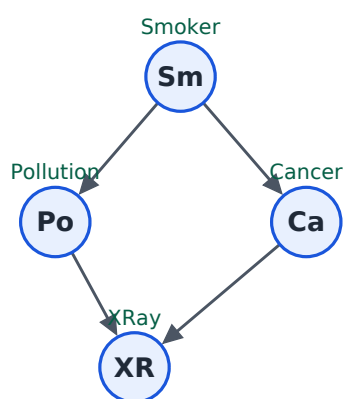
Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following statements are TRUE?

- ☐ (A) Propositional resolution is refutation-complete
- ☐ (B) Modus ponens alone is complete for arbitrary propositional knowledge bases
- ☐ (C) Forward chaining is sound and complete for knowledge bases of definite clauses
- ☐ (D) A sound inference procedure derives only sentences that are entailed by the KB

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Sm, Po, Ca, XR. (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) What is the probability that a single run produces the sample **(Sm=F, Po=F, Ca=T, XR=F)**? (round off to 4 decimal places)



P(Sm=T)
0.2

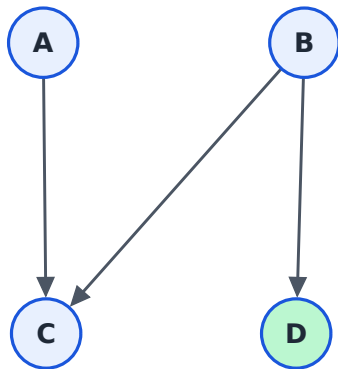
Sm	P(Po=T .. Sm)	Sm	P(Ca=T ...)
T	0.05	T	0.9
F	0.5	F	0.9

Po	Ca	P(XR=T ...)
T	T	0.25
T	F	0.7
F	T	0.25
F	F	0.1

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(D = T)$ (round off to 3 decimal places).



P(A=T)
0.05

P(B=T)	A	B	P(C=T ...)
0.4	T	T	0.3
	T	F	0.5
	F	T	0.3
	F	F	0.75

B	P(D=T ...)
T	0.1
F	0.5

Answer (numerical): _____

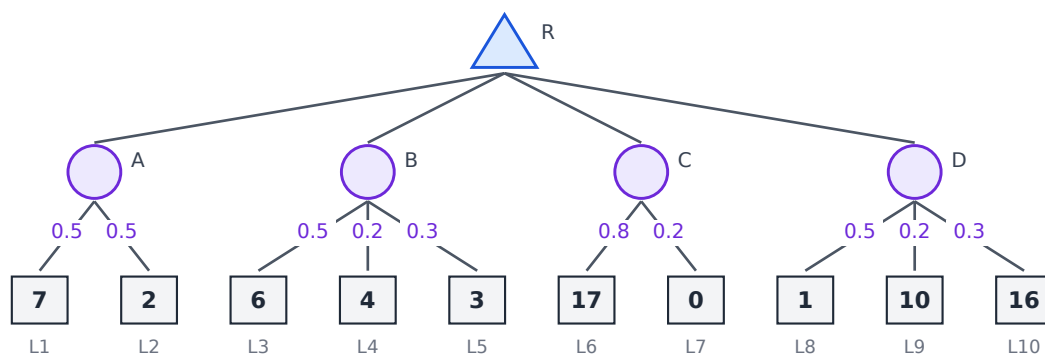
Q.10 [MSQ · 1 mark · no negative marking] Bayesian Networks

Which of the following are correct about Bayesian networks with Boolean variables?

- ☐ (A) The graph must be a directed acyclic graph
- ☐ (B) The CPT of a node with k Boolean parents has 2^k independent numbers
- ☐ (C) Each row of a CPT (one parent configuration) sums to 1 over the node's values
- ☐ (D) If there is no arc between X and Y , then $X \perp Y$ marginally

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node

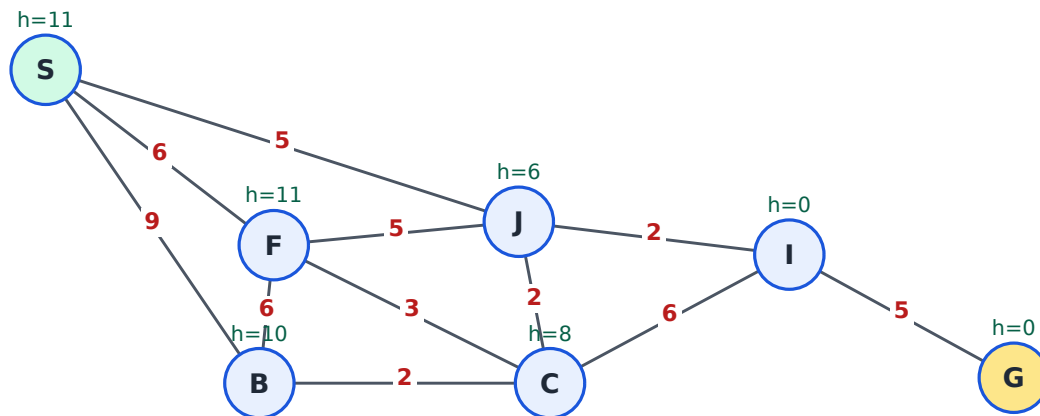
○ CHANCE node (edge = probability)

□ terminal utility

Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

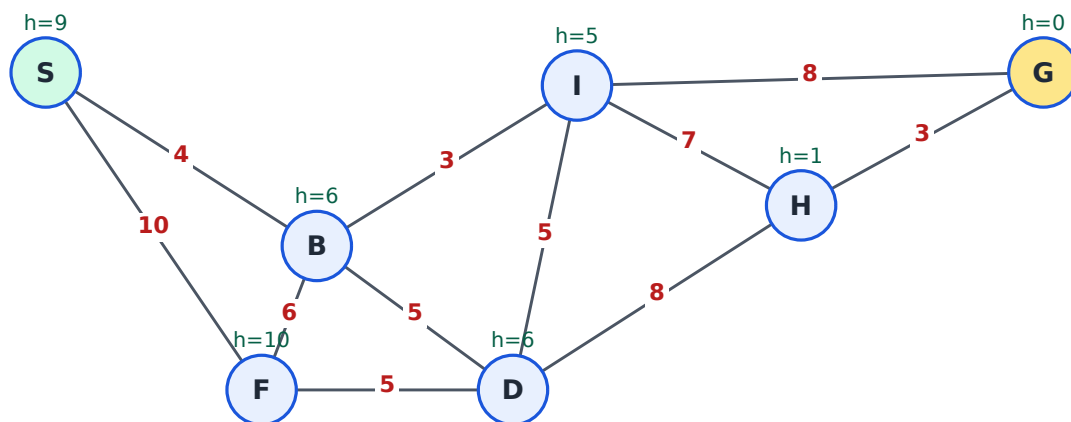
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
☐ (B) h is admissible.
☐ (C) The edge C-I violates the consistency condition $h(C) \leq c(C,I) + h(I)$.
☐ (D) $h(C)$ overestimates the true cost from C to G.

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Depth-first search pushes the right child before the left child, so the left child is explored first, and it goal-tests a node when the node is removed from the frontier. The goal is node 12. How many nodes are removed from the frontier, including node 12?

Answer (numerical): _____

Q.15 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following expresses 'No student likes every course' (S: student, C: course, L(x, y): x likes y)?

- ☐ (A) $\forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$
☐ (B) $\forall x (S(x) \rightarrow \exists y (C(y) \rightarrow \neg L(x, y)))$
☐ (C) $\forall x \forall y ((S(x) \wedge C(y)) \rightarrow \neg L(x, y))$
☐ (D) $\exists x (S(x) \wedge \exists y (C(y) \wedge \neg L(x, y)))$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(R \rightarrow Q) \vee (\neg P \wedge P)$
 (2) $\neg P \rightarrow (\neg R \vee R)$
 (3) $(Q \vee P) \rightarrow \neg P$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $P \rightarrow (R \rightarrow Q)$
☐ (B) $\neg \neg R \vee (Q \wedge R)$
☐ (C) $(\neg Q \wedge Q) \rightarrow (R \wedge Q)$
☐ (D) $P \wedge (P \rightarrow P)$

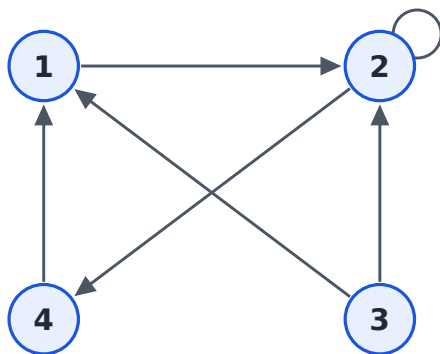
Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(P \wedge (P \rightarrow Q)) \rightarrow Q$
☐ (B) $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$
☐ (C) $((P \rightarrow Q) \wedge (R \rightarrow Q)) \rightarrow (P \wedge R)$
☐ (D) $P \rightarrow (Q \rightarrow P)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.19 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following is logically equivalent to $\neg \exists x (P(x) \wedge \forall y Q(x, y))$?

- ☐ (A) $\forall x (P(x) \rightarrow \exists y \neg Q(x, y))$
☐ (B) $\forall x (\neg P(x) \wedge \exists y \neg Q(x, y))$
☐ (C) $\exists x (\neg P(x) \vee \exists y \neg Q(x, y))$
☐ (D) $\forall x (P(x) \wedge \exists y \neg Q(x, y))$

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **A, D** and rules:

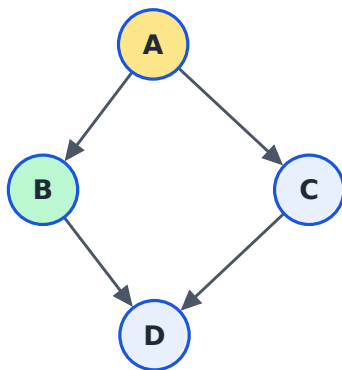
- R1: $J \wedge K \rightarrow H$
 R2: $A \rightarrow H$
 R3: $A \wedge H \wedge J \rightarrow E$
 R4: $B \wedge D \rightarrow C$
 R5: $H \wedge D \rightarrow B$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) C is entailed
☐ (B) E is entailed
☐ (C) F is entailed
☐ (D) K is entailed

Q.21 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence **A=T**. The current state of the chain is **(A=T, B=T, C=F, D=F)**. The next step resamples **B**. With what probability is B set to T? (round off to 3 decimal places)



P(A=T)
0.05

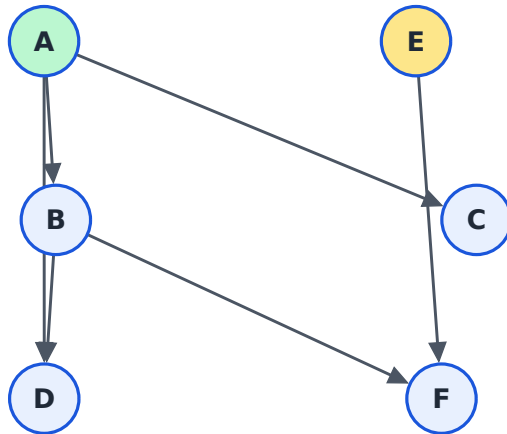
A	P(B=T ...)	A	P(C=T ...)
T	0.05	T	0.8
F	0.05	F	0.2

B	C	P(D=T ...)
T	T	0.7
T	F	0.9
F	T	0.8
F	F	0.3

Answer (numerical): _____

Q.22 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

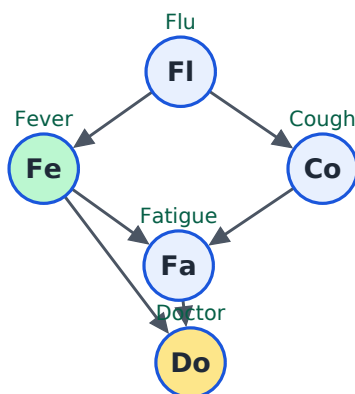
Variable elimination is used to compute $P(A \mid E = T)$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F**, **C**, **D**, **B**. What is the scope of the new factor produced when **F** is summed out?



- ☐ (A) $f(F)$
☐ (B) $f(B, F)$
☐ (C) $f(A, B)$
☐ (D) $f(B)$

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) Compute $P(\text{Fever} = T \mid \text{Doctor} = F)$ (round off to 3 decimal places).



P(FI=T)		P(Fe=T ...)		P(Co=T ...)	
FI		FI		FI	
T	0.95	T	0.6	T	0.6
F		F	0.3	F	0.75

Fe	Co	P(Fa=	Fe	Fa	P(Do=T ...)
T	T	0.05	T	T	0.7
T	F	0.1	T	F	0.9
F	T	0.2	F	T	0.1
F	F	0.05	F	F	0.6

Answer (numerical): _____

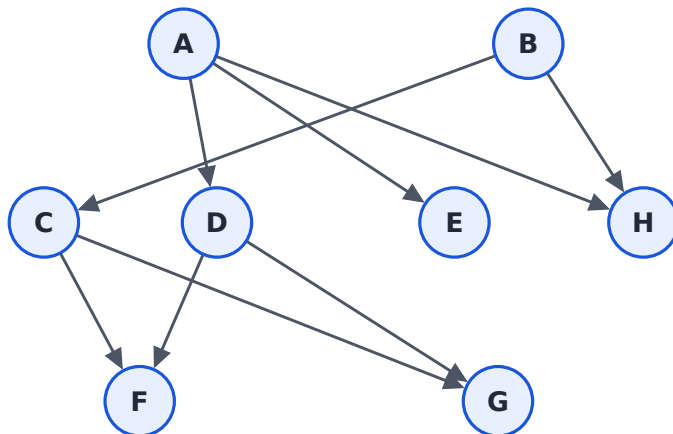
Q.24 [MCQ · 2 marks · negative: -2/3] Conditional Independence

Consider the Bayesian network (DAG) with edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$, $B \rightarrow F$. Which of the following pairs is d-separated by the given set?

- ☐ (A) A and B given {E}
☐ (B) A and F given {D}
☐ (C) A and E given {D}
☐ (D) E and F given $\emptyset$

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp B \mid E$
☐ (B) $D \perp C$
☐ (C) $E \perp A \mid H$
☐ (D) $F \perp H \mid C, E$

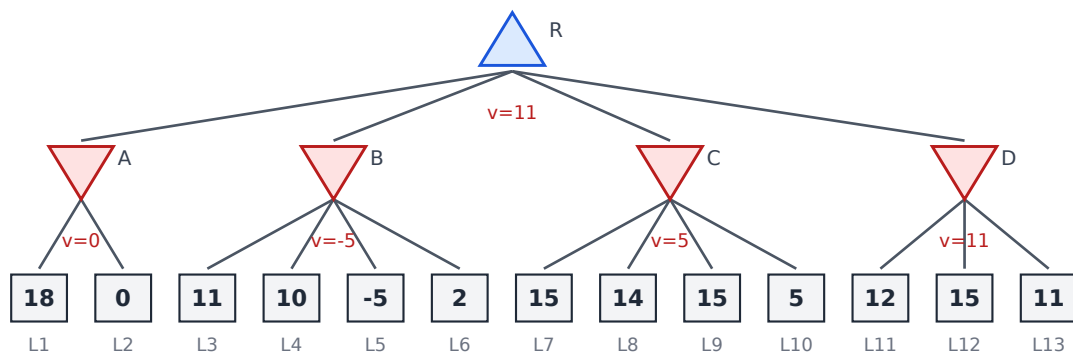
Answer Key — Mock Test 073

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	A	14	NAT	2	Uninformed Search	11
2	MCQ	1	Uninformed Search	C	15	MCQ	2	Predicate Logic	A
3	NAT	1	Adversarial Search	7	16	MSQ	2	Propositional Logic	A, C
4	MCQ	1	Informed Search	B	17	MSQ	2	Propositional Logic	A, B, D
5	NAT	1	Propositional Logic	3	18	MSQ	2	Predicate Logic	A, B, D
6	MCQ	1	Propositional Logic	A	19	MCQ	2	Predicate Logic	A
7	MSQ	1	Propositional Logic	A, C, D	20	MSQ	2	Propositional Logic	A
8	NAT	1	Sampling	0.2698 to 0.2702	21	NAT	2	Sampling	0.005 to 0.009
9	NAT	1	Exact Inference	0.338 to 0.342	22	MCQ	2	Exact Inference	D
10	MSQ	1	Bayesian Networks	A, B, C	23	NAT	2	Exact Inference	0.252 to 0.256
11	NAT	2	Adversarial Search	13.59 to 13.61	24	MCQ	2	Conditional Independence	C
12	MSQ	2	Informed Search	B, C	25	MSQ	2	Conditional Independence	B
13	NAT	2	Informed Search	5					

Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] **Answer: A** (A: Move to D; root value 11)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = 0, B = -5, C = 5, D = 11. Root value = **11**; MAX's optimal first move is towards D.

Q.2 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, E, A, F, H, I, G)

Adjacency lists (alphabetical): A: {E, F, H}; E: {A, I, S}; F: {A, H, K}; G: {I, K}; H: {A, F, I, K}; I: {E, G, H, K}; K: {F, G, H, I}; S: {E}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, E, A, F, H, I, G**; the path found is $S \rightarrow E \rightarrow A \rightarrow F \rightarrow H \rightarrow I \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.3 [NAT, 1M · Adversarial Search] **Answer: 7**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 4^{\lceil 1 \rceil} + 4^{\lfloor 1 \rfloor} - 1 = 4 + 4 - 1 = 7$, versus 16 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.4 [MCQ, 1M · Informed Search] **Answer: B** (B: the smallest f-value of any node that exceeded the cutoff in the previous iteration)

IDA* raises the threshold to the **smallest f-cost that exceeded the old threshold**. This keeps it optimal with an admissible h, because no solution with a smaller f is skipped. Increment-by-1 (A) assumes integer costs and can waste iterations. The largest f (C) and doubling (D) can jump past the optimal cost and return a suboptimal solution.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 3**

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg Q \vee R) \vee (R \wedge R)$	$(\neg P \leftrightarrow P) \leftrightarrow \neg \neg P$	$((\neg Q \vee R) \vee (R \wedge R)) \wedge ((\neg P \leftrightarrow P) \leftrightarrow \neg \neg P)$
T	T	T	T	F	F
T	T	F	F	F	F
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	F	T	F
F	F	T	T	T	T
F	F	F	T	T	T

Rows where F = T: **3**.

Q.6 [MCQ, 1M · Propositional Logic] **Answer: A** (A: $\alpha \wedge \neg \beta$ is unsatisfiable)

$\alpha \models \beta$ means every model of α is a model of β , i.e. no model makes α true and β false, i.e. $\alpha \wedge \neg \beta$ is unsatisfiable. Equivalently $\alpha \rightarrow \beta$ is valid. Satisfiability of $\neg \alpha \vee \beta$ or $\alpha \wedge \beta$ is too weak (one model is enough). $\alpha \vee \neg \beta$ being valid would mean $\beta \models \alpha$, the reverse direction.

Q.7 [MSQ, 1M · Propositional Logic] **Answer: A, C, D**

(A) True: if $KB \wedge \neg \alpha$ is unsatisfiable, resolution derives the empty clause. (B) False: from $p \vee q$, $p \rightarrow r$, $q \rightarrow r$, modus ponens cannot derive r , although r is entailed. (C) True for definite clauses (Horn KBs with exactly one positive literal per clause). (D) This is the definition of soundness (truth-preserving).

Q.8 [NAT, 1M · Sampling] Answer: 0.2698 to 0.2702

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(\text{Sm}=F) \times P(\text{Po}=F|\text{Sm}=F) \times P(\text{Ca}=T|\text{Sm}=F) \times P(\text{XR}=F|\text{Po}=F, \text{Ca}=T) \\ = 0.8 \times 0.5 \times 0.9 \times 0.75 = \mathbf{0.27} \approx 0.2700.$$

Q.9 [NAT, 1M · Exact Inference] Answer: 0.338 to 0.342

Only the ancestors of D (B) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
A=T, B=T, C=T	0.0006
A=T, B=T, C=F	0.0014
A=T, B=F, C=T	0.0075
A=T, B=F, C=F	0.0075
A=F, B=T, C=T	0.0114
A=F, B=T, C=F	0.0266
A=F, B=F, C=T	0.21375
A=F, B=F, C=F	0.07125

Sum = **0.34** ≈ 0.340 .

Q.10 [MSQ, 1M · Bayesian Networks] Answer: A, B, C

(A) BNs are DAGs by definition. (B) There are 2^k rows, each needing one number $P(X = \text{true} | \text{row})$. (C) Each row is a conditional distribution, so it sums to 1. (D) is false: in $A \rightarrow B \rightarrow C$ there is no arc between A and C, but they are dependent through B.

Q.11 [NAT, 2M · Adversarial Search] Answer: 13.59 to 13.61

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.5 \times 7 + 0.5 \times 2 = 4.5$
- $B = 0.5 \times 6 + 0.2 \times 4 + 0.3 \times 3 = 4.7$
- $C = 0.8 \times 17 + 0.2 \times 0 = 13.6$
- $D = 0.5 \times 1 + 0.2 \times 10 + 0.3 \times 16 = 7.3$

Root = $\max(4.5, 4.7, 13.6, 7.3) = \mathbf{13.6}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.12 [MSQ, 2M · Informed Search] Answer: B, C (B: h is admissible.; C: The edge C-I violates the consistency condition $h(C) \leq c(C,I) + h(I)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	F	G	I	J	S
$h(n)$	10	8	11	0	0	6	11
$h^*(n)$	11	9	12	0	5	7	12
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $J \rightarrow I$ ($6 > 2 + 0$), $C \rightarrow I$ ($8 > 6 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(B)=6$, $h(D)=6$, $h(F)=10$, $h(G)=0$, $h(H)=1$, $h(I)=5$, $h(S)=9$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (9)	B(10), F(20)
2	B (10)	I(12), D(15), F(20)
3	I (12)	D(15), G(15), H(15), F(20)
4	D (15)	G(15), H(15), F(20)
5	G (15)	H(15), F(20) $\leftarrow$ goal, stop

Path returned: **S $\rightarrow$ B $\rightarrow$ I $\rightarrow$ G** with cost **15**; nodes expanded = **5** (S, B, I, D, G).

Check: $h(n) = \lceil c \cdot h^*(n) \rceil$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.14 [NAT, 2M · Uninformed Search] Answer: 11

DFS visits nodes in preorder: 1, 2, 4, 8, 9, 5, 10, 11, 3, 6, 12, ... Node 12 is the **11th** node removed. For comparison, BFS would remove nodes in label order and find node 12 at the 12th removal.

Q.15 [MCQ, 2M · Predicate Logic] Answer: A (A: $\forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$)

'No student likes every course' $\equiv \neg \exists x (S(x) \wedge \forall y (C(y) \rightarrow L(x, y))) \equiv \forall x (S(x) \rightarrow \exists y (C(y) \wedge \neg L(x, y)))$. Option B is satisfied by any non-course y , so it is too weak. Option C says no student likes any course — too strong. Option D only says some student fails to like some course.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $P \rightarrow (R \rightarrow Q)$; C: $(\neg Q \wedge Q) \rightarrow (R \wedge Q)$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 3 of them:

P	Q	R	KB
F	T	T	T
F	T	F	T
F	F	F	T

(A) $P \rightarrow (R \rightarrow Q)$ is true in all models of $KB \Rightarrow$ **entailed**.

(B) $\neg \neg R \vee (Q \wedge R)$ is false in the KB -model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(C) $(\neg Q \wedge Q) \rightarrow (R \wedge Q)$ is true in all models of $KB \Rightarrow$ **entailed**.

(D) $P \wedge (P \rightarrow P)$ is false in the KB -model $P=F, Q=T, R=T \Rightarrow$ not entailed.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A, B, D (A: $(P \wedge (P \rightarrow Q)) \rightarrow Q$; B: $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$; D: $P \rightarrow (Q \rightarrow P)$)

(A) $(P \wedge (P \rightarrow Q)) \rightarrow Q$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $((R \rightarrow Q) \wedge (Q \rightarrow P)) \rightarrow (R \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $((P \rightarrow Q) \wedge (R \rightarrow Q)) \rightarrow (P \wedge R)$: **not valid** — counter-model $P=T, Q=T, R=F$.

(D) $P \rightarrow (Q \rightarrow P)$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Predicate Logic] Answer: A, B, D (A: $\forall x \exists y (x \neq y \wedge R(x, y))$; B: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,2), (2,2), (2,4), (3,1), (3,2), (4,1)\}$.

(A) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

(C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,4)$ holds but $R(4,2)$ does not.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 3 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MCQ, 2M · Predicate Logic] Answer: A (A: $\forall x (P(x) \rightarrow \exists y \neg Q(x, y))$)

$\neg \exists x (...) \equiv \forall x \neg (P(x) \wedge \forall y Q(x, y)) \equiv \forall x (\neg P(x) \vee \exists y \neg Q(x, y)) \equiv \forall x (P(x) \rightarrow \exists y \neg Q(x, y))$. Option B uses $\wedge$ instead of $\vee$ (De Morgan applied wrongly). Option C fails to flip $\exists x$ to $\forall x$. Option D requires every element to satisfy P .

Q.20 [MSQ, 2M · Propositional Logic] Answer: A (A: C is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $A \rightarrow H$; add H .

Round 2: fire $H \wedge D \rightarrow B$; add B .

Round 3: fire $B \wedge D \rightarrow C$; add C .

Fixed point: $\{A, B, C, D, H\}$ — 5 atoms are entailed (facts: A, D ; newly derived: B, C, H). Atoms never derived: E, F, J, K .

Q.21 [NAT, 2M · Sampling] Answer: 0.005 to 0.009

Gibbs sampling draws B from $P(B \mid \text{mb}(B))$, where $\text{mb}(B) = \{A, C, D\}$:

$$P(B \mid \text{mb}) \propto P(B \mid \text{parents}(B)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y)).$$

$$B=T: P(B=T \mid A=T) \times P(D=F \mid B=T, C=F) = 0.05 \times 0.1 = 0.005$$

$$B=F: P(B=F \mid A=T) \times P(D=F \mid B=F, C=F) = 0.95 \times 0.7 = 0.665$$

$$\text{Normalise: } 0.005 / (0.005 + 0.665) = \mathbf{0.0075}.$$

Q.22 [MCQ, 2M · Exact Inference] Answer: D (D: f(B))

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, C)$, $f(A, B, D)$, $f()$, $f(B, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(B, F)$	$\{B, F\}$	$f(B)$
C	$f(A, C)$	$\{A, C\}$	$f(A)$
D	$f(A, B, D)$	$\{A, B, D\}$	$f(A, B)$
B	$f(A, B), f(B), f(A, B)$	$\{A, B\}$	$f(A)$

The largest factor produced has **2** variable(s) (created when eliminating D); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.23 [NAT, 2M · Exact Inference] Answer: 0.252 to 0.256

Factorisation: $P(\text{Flu, Fever, Cough, Fatigue, Doctor}) = P(\text{Flu}) \cdot P(\text{Fever} \mid \text{Flu}) \cdot P(\text{Cough} \mid \text{Flu}) \cdot P(\text{Fatigue} \mid \text{Fever, Cough}) \cdot P(\text{Doctor} \mid \text{Fever, Fatigue})$.

Inference by enumeration: $P(\text{Fever} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Fever, e, hidden})$.

Hidden: Fl, Co, Fa	Fe	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, T, T	T	$0.95 \times 0.6 \times 0.6 \times 0.05 \times 0.3$	0.00513
T, T, F	T	$0.95 \times 0.6 \times 0.6 \times 0.95 \times 0.1$	0.03249
T, F, T	T	$0.95 \times 0.6 \times 0.4 \times 0.1 \times 0.3$	0.00684
T, F, F	T	$0.95 \times 0.6 \times 0.4 \times 0.9 \times 0.1$	0.02052
F, T, T	T	$0.05 \times 0.3 \times 0.75 \times 0.05 \times 0.3$	0.00017
F, T, F	T	$0.05 \times 0.3 \times 0.75 \times 0.95 \times 0.1$	0.00107
F, F, T	T	$0.05 \times 0.3 \times 0.25 \times 0.1 \times 0.3$	0.00011
F, F, F	T	$0.05 \times 0.3 \times 0.25 \times 0.9 \times 0.1$	0.00034
T, T, T	F	$0.95 \times 0.4 \times 0.6 \times 0.2 \times 0.9$	0.04104
T, T, F	F	$0.95 \times 0.4 \times 0.6 \times 0.8 \times 0.4$	0.07296
T, F, T	F	$0.95 \times 0.4 \times 0.4 \times 0.05 \times 0.9$	0.00684
T, F, F	F	$0.95 \times 0.4 \times 0.4 \times 0.95 \times 0.4$	0.05776
F, T, T	F	$0.05 \times 0.7 \times 0.75 \times 0.2 \times 0.9$	0.00473
F, T, F	F	$0.05 \times 0.7 \times 0.75 \times 0.8 \times 0.4$	0.0084
F, F, T	F	$0.05 \times 0.7 \times 0.25 \times 0.05 \times 0.9$	0.00039
F, F, F	F	$0.05 \times 0.7 \times 0.25 \times 0.95 \times 0.4$	0.00333

$$P(\text{Fever}=T, e) = 0.06667; P(\text{Fever}=F, e) = 0.19544; P(e) = 0.26211.$$

$$P(\text{Fever}=T \mid e) = 0.06667 / 0.26211 = \mathbf{0.2543} \approx 0.254.$$

Q.24 [MCQ, 2M · Conditional Independence] Answer: C (C: A and E given {D})

(C): every path from A to E goes $A \rightarrow C \rightarrow D \rightarrow E$; D is a non-collider on it and is observed, so it is blocked:

d-separated. (A): the path $A \rightarrow C \leftarrow B$ has collider C whose descendant E is observed, so it is active. (B): the path $A \rightarrow C \leftarrow B \rightarrow F$ has collider C with observed descendant D, and non-collider B unobserved, so it is active.

(D): $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ has no colliders and nothing observed, so it is active.

Q.25 [MSQ, 2M · Conditional Independence] Answer: B (B: $D \perp C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp B \mid E$: **FALSE** — the path $C \leftarrow B$ is active given {E}.

(B) $D \perp C$: **TRUE** — $D \leftarrow A \rightarrow H \leftarrow B \rightarrow C$ (blocked: collider H with neither it nor a descendant observed); $D \rightarrow F \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $D \rightarrow G \leftarrow C$ (blocked: collider G with neither it nor a descendant observed).

(C) $E \perp A \mid H$: **FALSE** — the path $E \leftarrow A$ is active given {H}.

(D) $F \perp H \mid C, E$: **FALSE** — the path $F \leftarrow D \leftarrow A \rightarrow H$ is active given {C, E}.

Mock Test 074

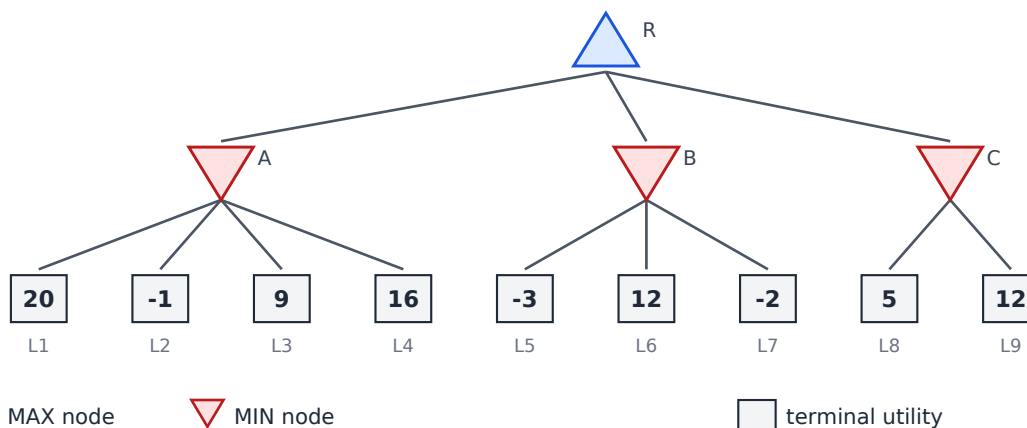
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	3 / 8 / 14	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

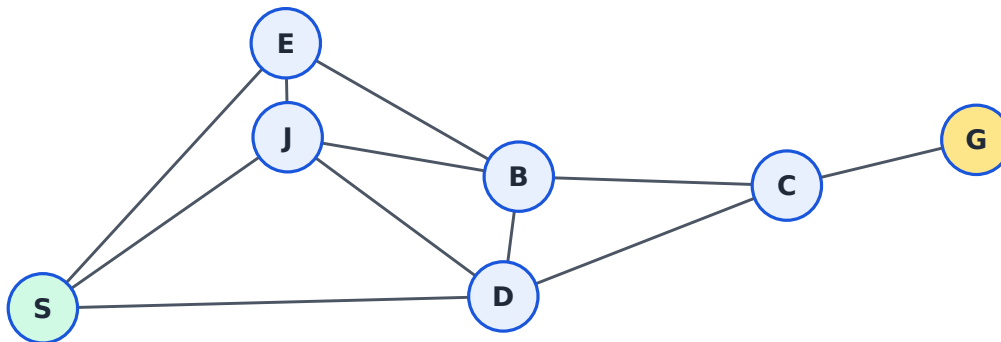
Q.2 [NAT · 1 mark · no negative marking] Uninformed Search

A uniform tree has branching factor $b = 10$, and the only goal is at depth $d = 5$. Breadth-first search applies the goal test when a node is generated. In the worst case the goal is the last node generated at depth 5. How many nodes are generated in total, not counting the root?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

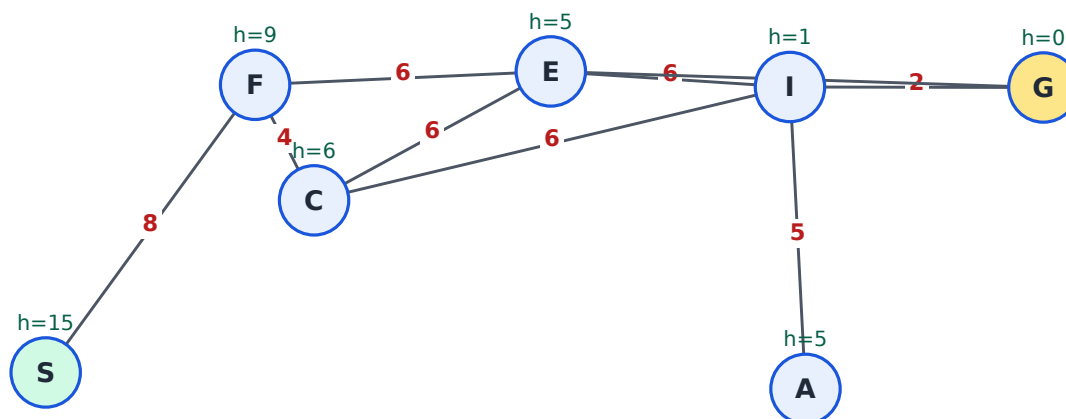
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, C, B, G
☐ (B) S, J, E, B, D, C, G
☐ (C) S, D, B, C, G
☐ (D) S, D, E, J, B, C, G

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are logically equivalent to p ?

- ☐ (A) $p \vee (p \wedge q)$
☐ (B) $p \wedge (p \vee q)$
☐ (C) $(p \vee q) \wedge (p \vee \neg q)$
☐ (D) $(p \rightarrow q) \rightarrow q$

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 4 elements and R a binary predicate symbol. In how many of the 2^{16} possible interpretations of R over D is the sentence $\forall x \forall y (R(x, y) \rightarrow \neg R(y, x))$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (\neg Q \leftrightarrow (R \vee Q)) \wedge ((Q \vee \neg P) \rightarrow (R \rightarrow \neg Q)) ?$$

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic

In how many rows of the truth table over p, q is $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow p)$ true?

Answer (numerical): _____

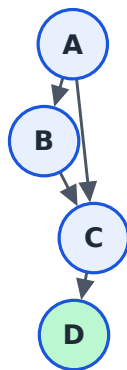
Q.9 [MCQ · 1 mark · negative: -1/3] Probability for AI

For random variables Y and Z, which expression is the conditioning rule for computing the marginal $P(Y)$?

- ☐ (A) $P(Y) = \sum_z P(Y | z) P(z)$
☐ (B) $P(Y) = \sum_z P(z | Y) P(Y)$
☐ (C) $P(Y) = \prod_z P(Y | z)$
☐ (D) $P(Y) = \sum_z P(Y | z)$

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T | \text{parents})$), compute the marginal probability $P(D = T)$ (round off to 3 decimal places).



P(A=T)
0.6

A	P(B=T ...)	A	B	P(C=T ...)
T	0.7	T	T	0.05
F	0.1	T	F	0.4
		F	T	0.6
		F	F	0.25

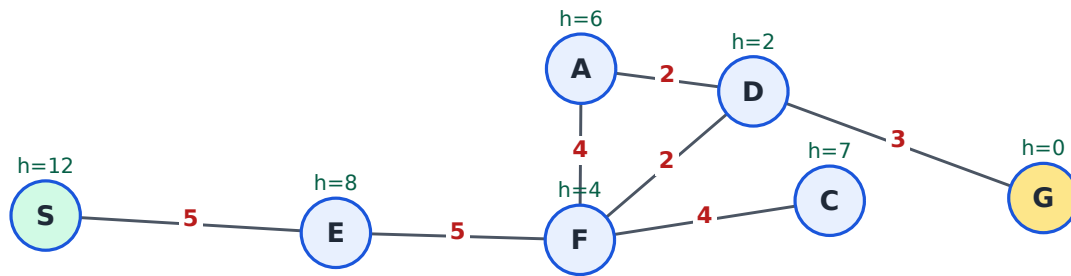
C	P(D=T ...)
T	0.8
F	0.05

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) h is admissible.
- ☐ (C) $h(C)$ overestimates the true cost from C to G.
- ☐ (D) The edge A-D violates the consistency condition $h(A) \leq c(A,D) + h(D)$.

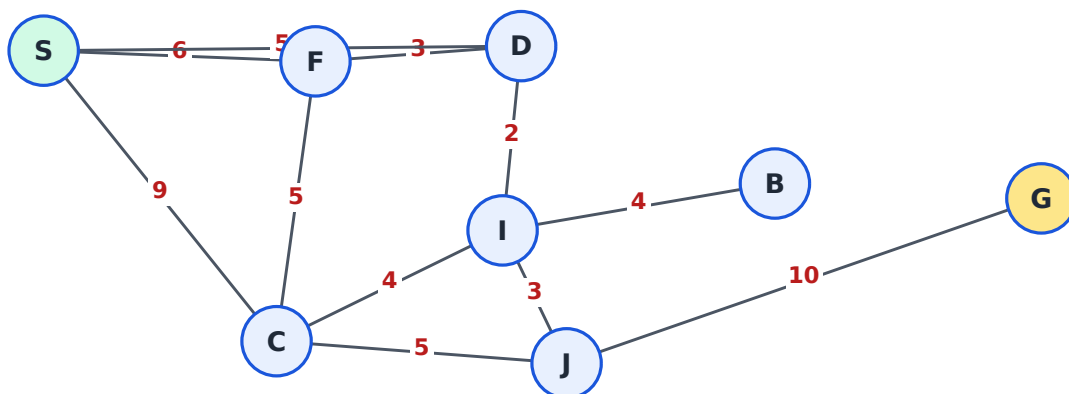
Q.12 [NAT · 2 marks · no negative marking] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (2), $S \rightarrow B$ (4), $A \rightarrow C$ (2), $A \rightarrow D$ (4), $B \rightarrow C$ (1), $C \rightarrow D$ (1), $C \rightarrow G$ (5), $D \rightarrow G$ (2). The heuristic is $h(S)=5$, $h(A)=4$, $h(B)=4$, $h(C)=3$, $h(D)=2$, $h(G)=0$. A* graph search is run from S. It goal-tests a node when the node is selected from the frontier, and it keeps the cheapest entry for each state. How many nodes are selected from the frontier for expansion, including the selection of G that ends the search?

Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

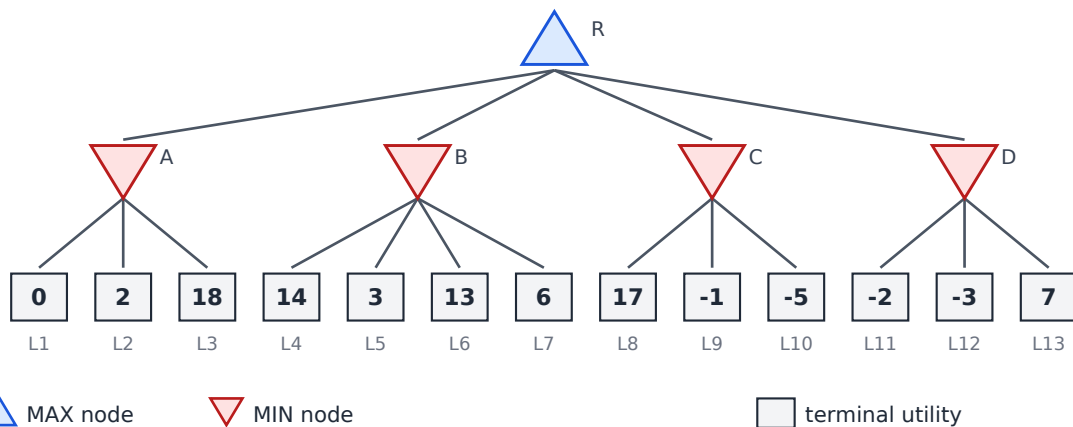
Q.14 [NAT · 2 marks · no negative marking] Informed Search

In the 8-puzzle, the goal configuration is 1 2 3 / 4 5 6 / 7 8 _ and the current state is 8 1 3 / 4 _ 2 / 7 6 5 (rows top to bottom, _ = blank). What is h_2 , the sum of the Manhattan distances of the tiles from their goal positions (excluding the blank)?

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



- ☐ (A) Leaf L6 is pruned.
- ☐ (B) Leaf L9 is pruned.
- ☐ (C) Leaf L12 is pruned.
- ☐ (D) Leaf L13 is pruned.

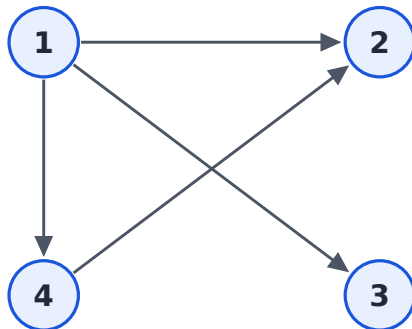
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(P \vee R) \rightarrow (P \wedge R)$
- ☐ (B) $(P \wedge (P \rightarrow Q)) \rightarrow Q$
- ☐ (C) $(P \rightarrow R) \rightarrow ((P \vee Q) \rightarrow R)$
- ☐ (D) $((P \rightarrow R) \wedge R) \rightarrow P$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x R(x, x)$
☐ (B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (C) $\exists x \forall y \neg R(x, y)$
☐ (D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following formulas are valid?

- ☐ (A) $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
☐ (B) $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$
☐ (C) $((p \rightarrow q) \rightarrow r) \leftrightarrow (p \rightarrow (q \rightarrow r))$
☐ (D) $(\neg p \rightarrow \perp) \rightarrow p$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

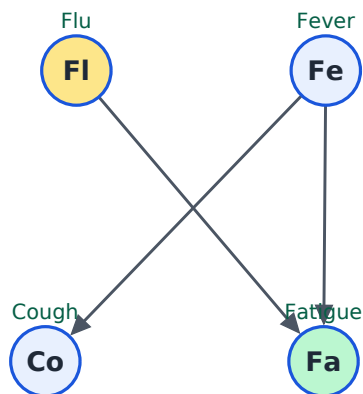
- (1) $R \rightarrow (R \leftrightarrow P)$
- (2) $\neg P \rightarrow (Q \rightarrow Q)$
- (3) $(Q \leftrightarrow Q) \rightarrow (P \leftrightarrow \neg R)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg Q \vee \neg R$
☐ (B) $\neg Q \wedge Q$
☐ (C) $R \wedge Q$
☐ (D) $P \wedge P$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Fatigue} = T \mid \text{Flu} = T)$ (round off to 3 decimal places).



$P(\text{Fl}=\text{T})$
0.75

$P(\text{Fe}=\text{T})$	Fe	$P(\text{Co}=\text{T} \dots)$
0.95	T	0.4
	F	0.8

Fl	Fe	$P(\text{Fa}=\text{T} \dots)$
T	T	0.3
T	F	0.05
F	T	0.5
F	F	0.1

Answer (numerical): _____

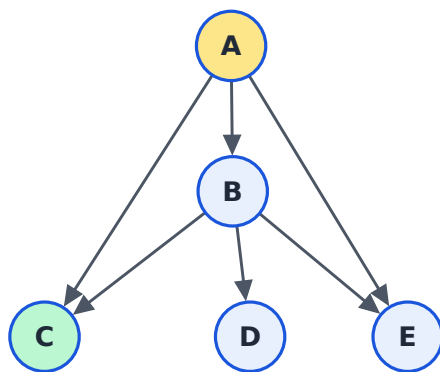
Q.21 [NAT · 2 marks · no negative marking] Variable Elimination

A Bayesian network over Boolean variables has edges $H \rightarrow X_1, H \rightarrow X_2, H \rightarrow X_3, H \rightarrow X_4$. To compute $P(X_1)$ by variable elimination, H is eliminated **first** (then X_2, X_3, X_4). How many entries does the new factor produced by summing out H contain?

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{A}=\mathbf{T}$. The current state of the chain is $(\mathbf{A}=\mathbf{T}, \mathbf{B}=\mathbf{T}, \mathbf{C}=\mathbf{T}, \mathbf{D}=\mathbf{F}, \mathbf{E}=\mathbf{T})$. The next step resamples $\mathbf{C}$. With what probability is $\mathbf{C}$ set to T? (round off to 3 decimal places)



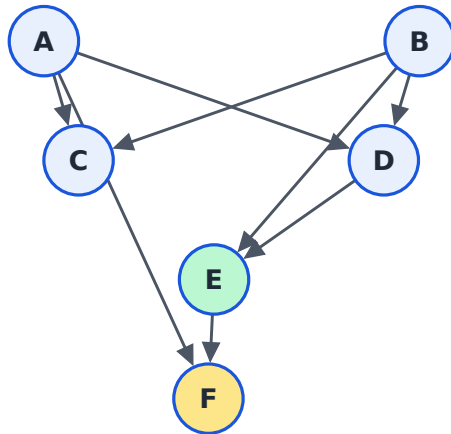
$P(\text{A}=\text{T})$	A	$P(\text{B}=\text{T} \dots)$	A	B	$P(\text{C}=\text{T} \dots)$
0.6	T	0.3	T	T	0.25
	F	0.1	T	F	0.3
			F	T	0.05
			F	F	0.05

B	$P(\text{D}=\text{T} \dots)$	A	B	$P(\text{E}=\text{T} \dots)$
T	0.6	T	T	0.25
F	0.95	T	F	0.75
		F	T	0.8
		F	F	0.6

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

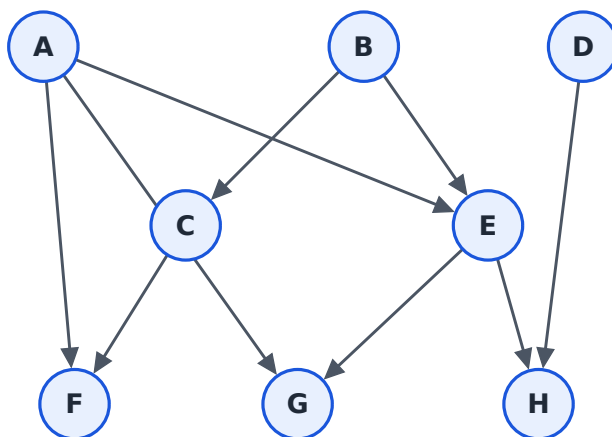
Variable elimination is used to compute $P(E \mid F = T)$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B**, **A**, **C**, **D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

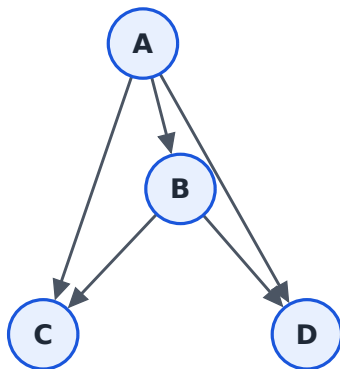
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp D \mid H$
- ☐ (B) $D \perp E \mid B$
- ☐ (C) $D \perp F \mid C$
- ☐ (D) $A \perp C$

Q.25 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.8}, \mathbf{0.52}, \mathbf{0.77}, \mathbf{0.31} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.7

A	P(B=T .)	A	B	P(C=T ...)
T	0.5	T	T	0.8
F	0.1	T	F	0.2
		F	T	0.4
		F	F	0.6

A	B	P(D=T ...)
T	T	0.6
T	F	0.95
F	T	0.05
F	F	0.9

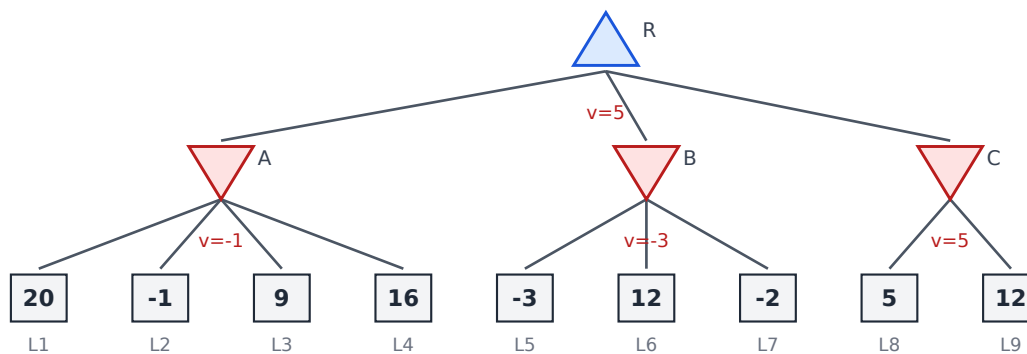
- ☐ (A) A=F, B=F, C=F, D=T
☐ (B) A=T, B=F, C=F, D=T
☐ (C) A=F, B=T, C=F, D=T
☐ (D) A=T, B=T, C=F, D=F

Answer Key — Mock Test 074

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	5	14	NAT	2	Informed Search	10
2	NAT	1	Uninformed Search	111110	15	MSQ	2	Adversarial Search	C, D
3	MCQ	1	Uninformed Search	C	16	MSQ	2	Propositional Logic	B
4	NAT	1	Informed Search	21	17	MSQ	2	Predicate Logic	B, C, D
5	MSQ	1	Propositional Logic	A, B, C	18	MSQ	2	Propositional Logic	A, B, D
6	NAT	1	Predicate Logic	729	19	MSQ	2	Propositional Logic	A, D
7	NAT	1	Propositional Logic	2	20	NAT	2	Exact Inference	0.285 to 0.289
8	NAT	1	Propositional Logic	2	21	NAT	2	Variable Elimination	16
9	MCQ	1	Probability for AI	A	22	NAT	2	Sampling	0.248 to 0.252
10	NAT	1	Exact Inference	0.203 to 0.207	23	NAT	2	Exact Inference	4
11	MSQ	2	Informed Search	D	24	MSQ	2	Conditional Independence	B, C, D
12	NAT	2	Informed Search	5	25	MCQ	2	Sampling	A
13	NAT	2	Uninformed Search	20					

Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 5**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -1$, $B = -3$, $C = 5$. Root value = **5**; MAX's optimal first move is towards C.

Q.2 [NAT, 1M · Uninformed Search] **Answer: 111110**

With the goal test at generation, BFS stops as soon as the goal is generated, so it generates at most every node down to depth d . Nodes generated = $b + b^2 + b^3 + b^4 + b^5 = 10 + 100 + 1000 + 10000 + 100000 = \mathbf{111110}$. This is the AIMA figure behind BFS's $O(b^d)$ time bound.

Q.3 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, D, B, C, G)

Adjacency lists (alphabetical): B: {C, D, E, J}; C: {B, D, G}; D: {B, C, J, S}; E: {B, J, S}; G: {C}; J: {B, D, E, S}; S: {D, E, J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, B, C, G**; the path found is $S \rightarrow D \rightarrow B \rightarrow C \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Informed Search] **Answer: 21**

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (15)	F(9)
2	F (9)	E(5), C(6)
3	E (5)	G(0), I(1), C(6)
4	G (0)	I(1), C(6) ← goal, stop

Path returned: **S** → **F** → **E** → **G**, cost = $8 + 6 + 7 = \mathbf{21}$.

The optimal cost is $h^*(S) = 20$ — greedy search is not optimal here.

Q.5 [MSQ, 1M · Propositional Logic] **Answer: A, B, C** (A: $p \vee (p \wedge q)$; B: $p \wedge (p \vee q)$; C: $(p \vee q) \wedge (p \vee \neg q)$)

(A) and (B) are the absorption laws. (C) By distribution, $(p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q) \equiv p$. (D) $(p \rightarrow q) \rightarrow q \equiv \neg(\neg p \vee q) \vee q \equiv (p \wedge \neg q) \vee q \equiv p \vee q$, which differs from p at $p=F, q=T$.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 729**

View R as an 4×4 0/1 matrix. The sentence says R is asymmetric: no self-loops; each unordered pair has 3 options $\Rightarrow 3^{(n(n-1)/2)}$. With $n = 4$: **729**.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 2**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg Q \leftrightarrow (R \vee Q)$	$(Q \vee \neg P) \rightarrow (R \rightarrow \neg Q)$	$(\neg Q \leftrightarrow (R \vee Q)) \wedge ((Q \vee \neg P) \rightarrow (R \rightarrow \neg Q))$
T	T	T	F	F	F
T	T	F	F	T	F
T	F	T	T	T	T
T	F	F	F	T	F
F	T	T	F	F	F
F	T	F	F	T	F
F	F	T	T	T	T
F	F	F	F	T	F

Rows where F = T: **2**.

Q.8 [NAT, 1M · Propositional Logic] Answer: 2

$p \rightarrow q \equiv \neg p \vee q$ and $\neg q \rightarrow p \equiv p \vee q$. Row TT: $T, T \rightarrow T$. Row TF: $F, T \rightarrow F$. Row FT: $T, T \rightarrow T$. Row FF: $T, F \rightarrow F$. So the biconditional is true in 2 rows.

Q.9 [MCQ, 1M · Probability for AI] Answer: A ($A: P(Y) = \sum_z P(Y | z) P(z)$)

Marginalization gives $P(Y) = \sum_z P(Y, z)$, and the product rule rewrites $P(Y, z) = P(Y | z)P(z)$; together this is the conditioning rule (A). (B) equals $P(Y) \sum_z P(z | Y) = P(Y)$, which is circular since it needs $P(Y)$. (C) multiplies probabilities and is not a marginal. (D) omits the weights $P(z)$ and need not even be ≤ 1 .

Q.10 [NAT, 1M · Exact Inference] Answer: 0.203 to 0.207

Only the ancestors of D (A, B, C) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
A=T, B=T, C=T	0.0168
A=T, B=T, C=F	0.01995
A=T, B=F, C=T	0.0576
A=T, B=F, C=F	0.0054
A=F, B=T, C=T	0.0192
A=F, B=T, C=F	0.0008
A=F, B=F, C=T	0.072
A=F, B=F, C=F	0.0135

Sum = **0.2053** ≈ 0.205 .

Q.11 [MSQ, 2M · Informed Search] Answer: D (D: The edge A-D violates the consistency condition $h(A) \leq c(A,D) + h(D)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	C	D	E	F	G	S
$h(n)$	6	7	2	8	4	0	12
$h^*(n)$	5	9	3	10	5	0	15
$h \leq h^*$?	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **no** (overestimates at A).

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges:

$A \rightarrow D$ ($6 > 2 + 2$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [NAT, 2M · Informed Search] Answer: 5

$f(S)=5$. Expand S: $A(g=2, f=6)$, $B(g=4, f=8)$. Expand A ($f=6$): $C(g=4, f=7)$, $D(g=6, f=8)$. Expand C ($f=7$): D improves to $g=5, f=7$, and G is added with $g=9, f=9$. Expand D ($f=7$): G improves to $g=7, f=7$. Select G ($f=7$) and stop with cost 7 via $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$. Selections: S, A, C, D, G = **5**. B ($f=8$) is never expanded.

Q.13 [NAT, 2M · Uninformed Search] Answer: 20

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	D(5), F(6), C(9)
2	D (5)	F(6), I(7), C(9)
3	F (6)	I(7), C(9)
4	I (7)	C(9), J(10), B(11)
5	C (9)	J(10), B(11)
6	J (10)	B(11), G(20)
7	B (11)	G(20)
8	G (20)	$\leftarrow$ goal, stop

Path returned: **S $\rightarrow$ D $\rightarrow$ I $\rightarrow$ J $\rightarrow$ G**, cost = $5 + 2 + 3 + 10 = 20$. Nodes expanded (including S and G): **8** (S, D, F, I, C, J, B, G).

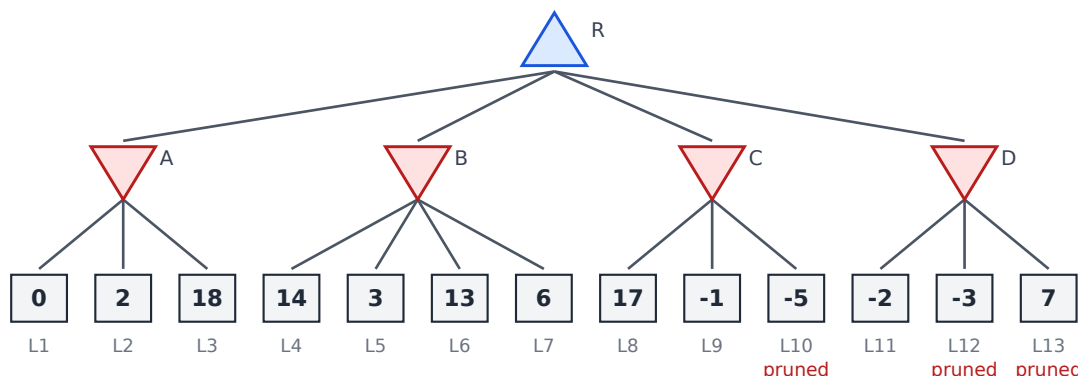
Q.14 [NAT, 2M · Informed Search] Answer: 10

Use (row, column) coordinates from (0,0). Tile 8 is at (0,0) and belongs at (2,1): distance 3. Tile 1 is at (0,1) and belongs at (0,0): 1. Tile 3: 0. Tile 4: 0. Tile 2 is at (1,2) and belongs at (0,1): 2. Tile 7: 0. Tile 6 is at (2,1) and belongs at (1,2): 2. Tile 5 is at (2,2) and belongs at (1,1): 2. Sum = 3+1+2+2+2 = **10**.

Q.15 [MSQ, 2M · Adversarial Search] Answer: C, D (C: Leaf L12 is pruned.; D: Leaf L13 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node C: value $-1 \leq \alpha = 3 \Rightarrow$ prune remaining children (leaves L10).
- At MIN node D: value $-2 \leq \alpha = 3 \Rightarrow$ prune remaining children (leaves L12, L13).



Leaves evaluated (10): L1, L2, L3, L4, L5, L6, L7, L8, L9, L11. Leaves pruned (3): L10, L12, L13. Root value = 3 (same as plain minimax, 3 — pruning never changes the root value).

Q.16 [MSQ, 2M · Propositional Logic] Answer: B (B: $(P \wedge (P \rightarrow Q)) \rightarrow Q$)

- (A) $(P \vee R) \rightarrow (P \wedge R)$: **not valid** — counter-model $P=T, R=F$.
 (B) $(P \wedge (P \rightarrow Q)) \rightarrow Q$: true in all 4 assignments $\Rightarrow$ **valid**.
 (C) $(P \rightarrow R) \rightarrow ((P \vee Q) \rightarrow R)$: **not valid** — counter-model $P=F, Q=T, R=F$.
 (D) $((P \rightarrow R) \wedge R) \rightarrow P$: **not valid** — counter-model $P=F, R=T$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [MSQ, 2M · Predicate Logic] Answer: B, C, D (B: $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$; C: $\exists x \forall y \neg R(x, y)$; D: $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$)

$R = \{(1,2), (1,3), (1,4), (4,2)\}$.

- (A) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.
 (B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **TRUE** — all two-step paths are short-cut.
 (C) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.
 (D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.
 Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, B, D (A: $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$; B: $(p \rightarrow (q \rightarrow r)) \Leftrightarrow ((p \wedge q) \rightarrow r)$; D: $(\neg p \rightarrow \perp) \rightarrow p$)

(A) Transitivity of implication; a truth table shows no falsifying row. (B) Exportation: both sides are equivalent to $\neg p \vee \neg q \vee r$. (C) Not valid: at $p=F, q=F, r=F$ the left side $(p \rightarrow q) \rightarrow r$ is $T \rightarrow F = F$, while the right side is T. Implication is not associative. (D) $\neg p \rightarrow \perp \equiv \neg \neg p \equiv p$, so the formula is $p \rightarrow p$, which is valid (this is the basis of proof by contradiction).

Q.19 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $\neg Q \vee \neg R$; D: $P \wedge P$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	T	F	T
T	F	F	T

- (A) $\neg Q \vee \neg R$ is true in all models of KB $\Rightarrow$ **entailed**.
 (B) $\neg Q \wedge Q$ is false in the KB-model $P=T, Q=T, R=F \Rightarrow$ not entailed.
 (C) $R \wedge Q$ is false in the KB-model $P=T, Q=T, R=F \Rightarrow$ not entailed.
 (D) $P \wedge P$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.285 to 0.289

Factorisation: $P(\text{Flu, Fever, Cough, Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever}) \cdot P(\text{Cough} \mid \text{Fever}) \cdot P(\text{Fatigue} \mid \text{Flu, Fever})$.

Inference by enumeration: $P(\text{Fatigue} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Fatigue, e, hidden})$.

Hidden: Fe, Co	Fa	product of CPT entries (Fl · Fe · Co · Fa)	value
T, T	T	$0.75 \times 0.95 \times 0.4 \times 0.3$	0.0855
T, F	T	$0.75 \times 0.95 \times 0.6 \times 0.3$	0.12825
F, T	T	$0.75 \times 0.05 \times 0.8 \times 0.05$	0.0015
F, F	T	$0.75 \times 0.05 \times 0.2 \times 0.05$	0.00038
T, T	F	$0.75 \times 0.95 \times 0.4 \times 0.7$	0.1995
T, F	F	$0.75 \times 0.95 \times 0.6 \times 0.7$	0.29925
F, T	F	$0.75 \times 0.05 \times 0.8 \times 0.95$	0.0285
F, F	F	$0.75 \times 0.05 \times 0.2 \times 0.95$	0.00713

$P(\text{Fatigue}=\text{T}, e) = 0.21562$; $P(\text{Fatigue}=\text{F}, e) = 0.53438$; $P(e) = 0.75$.

$P(\text{Fatigue}=\text{T} \mid e) = 0.21562 / 0.75 = \mathbf{0.2875} \approx 0.287$.

Q.21 [NAT, 2M · Variable Elimination] Answer: 16

Eliminating H multiplies all factors that mention H: $P(H)$, $P(X_1 \mid H)$, ..., $P(X_4 \mid H)$. Summing out H leaves a factor over $\{X_1, X_2, X_3, X_4\}$, with $2^4 = 16$ entries. By contrast, eliminating X_2, X_3, X_4 first gives trivial factors over H (each $\sum_x P(x \mid H) = 1$) and the largest new factor has just 1 variable.

Q.22 [NAT, 2M · Sampling] Answer: 0.248 to 0.252

Gibbs sampling draws C from $P(C \mid \text{mb}(C))$, where $\text{mb}(C) = \{A, B\}$:

$P(C \mid \text{mb}) \propto P(C \mid \text{parents}(C)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$C=\text{T}$: $P(C=\text{T} \mid A=\text{T}, B=\text{T}) = 0.25 = 0.25$

$C=\text{F}$: $P(C=\text{F} \mid A=\text{T}, B=\text{T}) = 0.75 = 0.75$

Normalise: $0.25 / (0.25 + 0.75) = \mathbf{0.25}$.

Q.23 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, B, C)$, $f(A, B, D)$, $f(B, D, E)$, $f(A, E)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(B)$, $f(A, B, C)$, $f(A, B, D)$, $f(B, D, E)$	$\{A, B, C, D, E\}$	$f(A, C, D, E)$
A	$f(A)$, $f(A, E)$, $f(A, C, D, E)$	$\{A, C, D, E\}$	$f(C, D, E)$
C	$f(C, D, E)$	$\{C, D, E\}$	$f(D, E)$
D	$f(D, E)$	$\{D, E\}$	$f(E)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over E and normalise.

Q.24 [MSQ, 2M · Conditional Independence] Answer: B, C, D (B: $D \perp E \mid B$; C: $D \perp F \mid C$; D: $A \perp C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp D \mid H$: **FALSE** — the path $A \rightarrow E \rightarrow H \leftarrow D$ is active given $\{H\}$.

(B) $D \perp E \mid B$: **TRUE** — $D \rightarrow H \leftarrow E$ (blocked: collider H with neither it nor a descendant observed).

(C) $D \perp F \mid C$: **TRUE** — $D \rightarrow H \leftarrow E \leftarrow A \rightarrow F$ (blocked: collider H with neither it nor a descendant observed); $D \rightarrow H \leftarrow E \rightarrow B \rightarrow C \rightarrow F$ (blocked: collider H with neither it nor a descendant observed); $D \rightarrow H \leftarrow E \rightarrow G \leftarrow A \rightarrow F$ (blocked: collider H with neither it nor a descendant observed).

(D) $A \perp C$: **TRUE** — $A \rightarrow E \leftarrow B \rightarrow C$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow F \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow G \leftarrow E \leftarrow B \rightarrow C$ (blocked: collider G with neither it nor a descendant observed).

Q.25 [MCQ, 2M · Sampling] Answer: A (A: $A=F$, B=F, C=F, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=\text{T} \mid \text{—}) = 0.7$; $u = 0.8 \geq 0.7 \Rightarrow A = \text{F}$
- B: $P(B=\text{T} \mid A=\text{F}) = 0.1$; $u = 0.52 \geq 0.1 \Rightarrow B = \text{F}$
- C: $P(C=\text{T} \mid A=\text{F}, B=\text{F}) = 0.6$; $u = 0.77 \geq 0.6 \Rightarrow C = \text{F}$
- D: $P(D=\text{T} \mid A=\text{F}, B=\text{F}) = 0.9$; $u = 0.31 < 0.9 \Rightarrow D = \text{T}$

Sample: **(A=F, B=F, C=F, D=T)**.

Mock Test 075

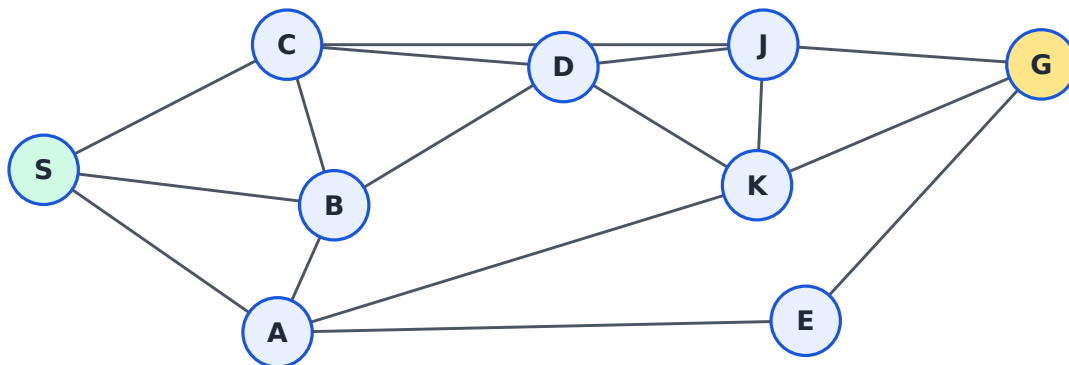
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 8 / 11	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, C, J, K, G
- ☐ (B) S, A, B, D, C, J, G
- ☐ (C) S, A, B, C, E, K, D, J, G
- ☐ (D) S, A, B, C, D, J, G

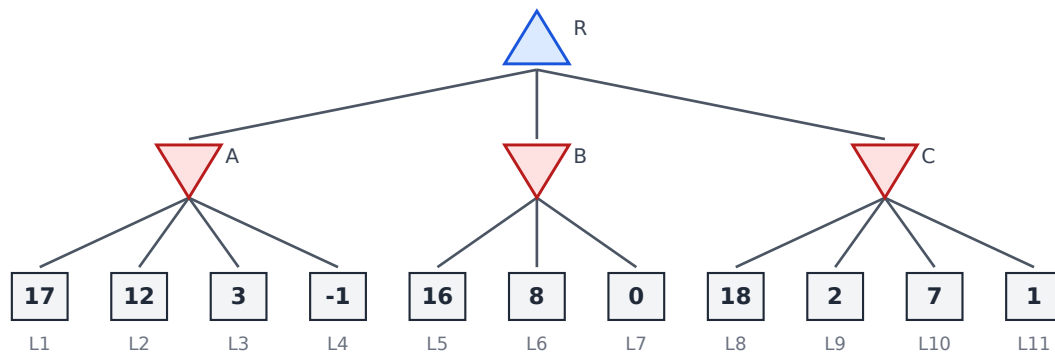
Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search

Which statement about depth-first search on a **finite** state space is correct?

- ☐ (A) The graph-search version is complete, but the tree-search version can loop forever on cycles
- ☐ (B) Both versions are complete and optimal
- ☐ (C) The tree-search version is complete, but the graph-search version is not
- ☐ (D) Neither version is complete

Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to C; root value 1
- ☐ (B) Move to A; root value -1
- ☐ (C) Move to C; root value 2
- ☐ (D) Move to B; root value 0

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 5$ (so $b^d = 1024$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **A, D** and rules:

- R1: $K \rightarrow A$
- R2: $C \rightarrow J$
- R3: $C \wedge B \rightarrow A$
- R4: $A \wedge H \rightarrow E$
- R5: $F \wedge K \rightarrow H$
- R6: $D \rightarrow H$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Resolving the first-order clauses $P(x) \vee Q(f(x))$ and $\neg P(g(y)) \vee R(y)$ (variables already standardized apart) on P gives

- ☐ (A) $Q(f(g(y))) \vee R(y)$
- ☐ (B) $Q(f(x)) \vee R(y)$
- ☐ (C) $Q(g(f(y))) \vee R(y)$
- ☐ (D) $Q(f(y)) \vee R(g(y))$

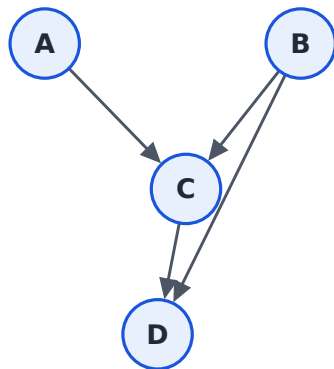
Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((\neg P \vee \neg Q) \wedge \neg P) \rightarrow (R \rightarrow (\neg P \wedge P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=F, B=F, C=F, D=T)**? (round off to 4 decimal places)



P(A=T)
0.6

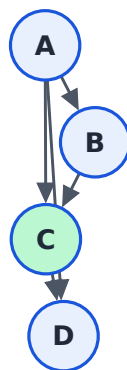
P(B=T)	A	B	P(C=T ...)
0.2	T	T	0.9
	T	F	0.7
	F	T	0.3
	F	F	0.05

B	C	P(D=T ...)
T	T	0.3
T	F	0.7
F	T	0.9
F	F	0.9

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability **$P(C = T)$** (round off to 3 decimal places).



P(A=T)
0.9

A	P(B=T .)	A	B	P(C=T ...)
T	0.3	T	T	0.6
F		T	F	0.5
	0.1	F	T	0.95
		F	F	0.5

A	C	P(D=T ...)
T	T	0.7
T	F	0.6
F	T	0.8
F	F	0.05

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling

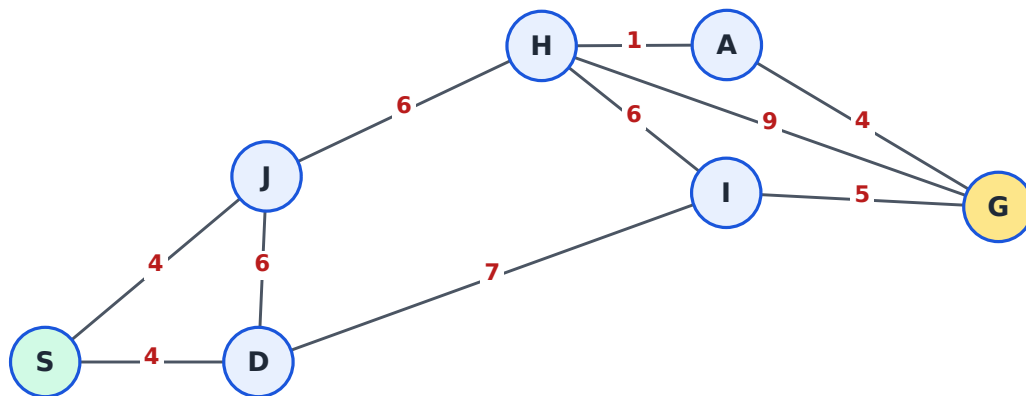
Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. Likelihood weighting is used with evidence $S = \text{true}$, $W = \text{true}$. A sample is generated with $C = \text{false}$ and $R = \text{false}$. What is the weight of this sample? (Round to 2 decimal places.)

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

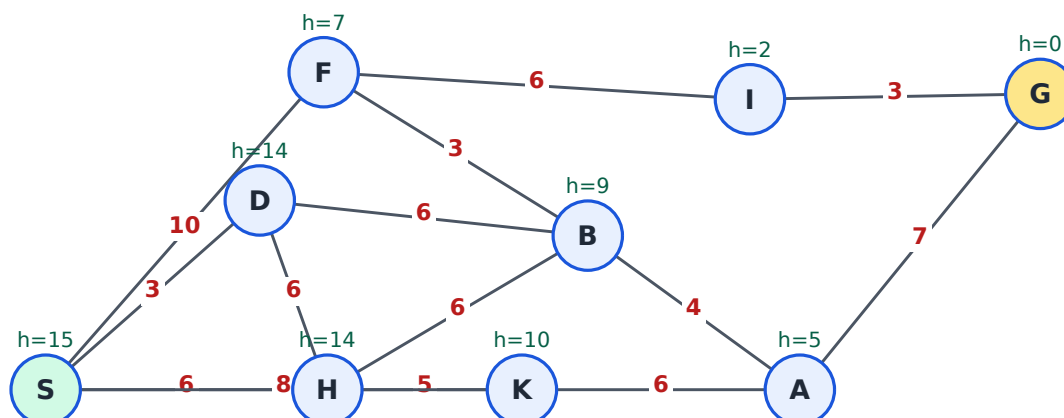
Q.12 [MSQ · 2 marks · no negative marking] Problem Formulation

Which of the following statements match standard (AIMA) definitions of the terms used to analyse search algorithms?

- ☐ (A) Completeness: the algorithm is guaranteed to find a solution when one exists
- ☐ (B) Cost-optimality: the algorithm finds a solution with the lowest path cost among all solutions
- ☐ (C) b , the branching factor, is the maximum number of successors of any node
- ☐ (D) d is the depth of the deepest node in the state space

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

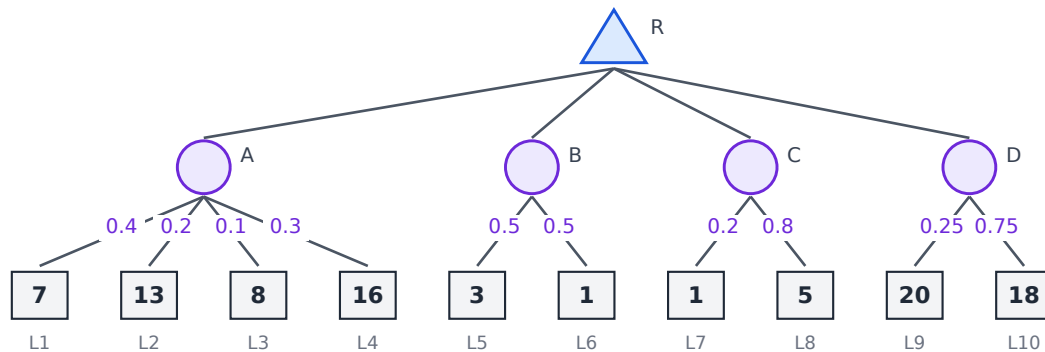
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following correctly express 'Every person who owns a dog is happy'?

- ☐ (A) $\forall x ((\text{Person}(x) \wedge \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y))) \rightarrow \text{Happy}(x))$
☐ (B) $\forall x \forall y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$
☐ (C) $\forall x \exists y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$
☐ (D) $\forall x (\text{Person}(x) \rightarrow \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y) \wedge \text{Happy}(x)))$

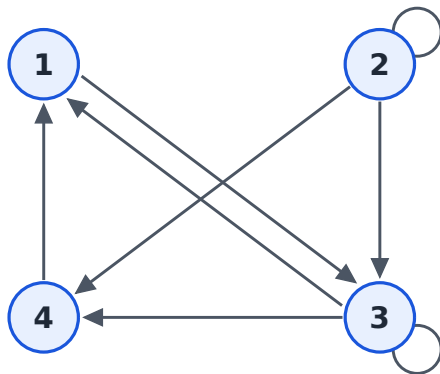
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $\mathbf{F} = \neg P \wedge (R \vee \neg Q)$?

- ☐ (A) $(R \vee \neg Q) \wedge \neg P$
☐ (B) $\neg P \vee (R \vee \neg Q)$
☐ (C) $\neg\neg P \wedge (R \vee \neg Q)$
☐ (D) $\neg(\neg\neg P \vee \neg(R \vee \neg Q))$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall y \exists x R(x, y)$
☐ (B) $\forall x \exists y R(x, y)$
☐ (C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (D) $\exists x R(x, x)$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $R \vee (P \vee \neg P)$
 (2) $\neg(S \rightarrow \neg R)$
 (3) $(P \rightarrow \neg S) \leftrightarrow (S \wedge S)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $(Q \rightarrow R) \vee (\neg R \wedge \neg P)$
☐ (B) $R \leftrightarrow (Q \wedge \neg S)$
☐ (C) $(P \leftrightarrow P) \leftrightarrow (\neg R \wedge S)$
☐ (D) $R \rightarrow R$

Q.19 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following formulas is in disjunctive normal form and is equivalent to $(p \vee q) \wedge \neg(p \wedge q \wedge r)$?

- ☐ (A) $(p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg r)$
☐ (B) $(p \wedge \neg q) \vee (\neg p \wedge q)$
☐ (C) $(p \wedge \neg r) \vee (q \wedge \neg r)$
☐ (D) $(p \vee q) \wedge (\neg p \vee \neg q \vee \neg r)$

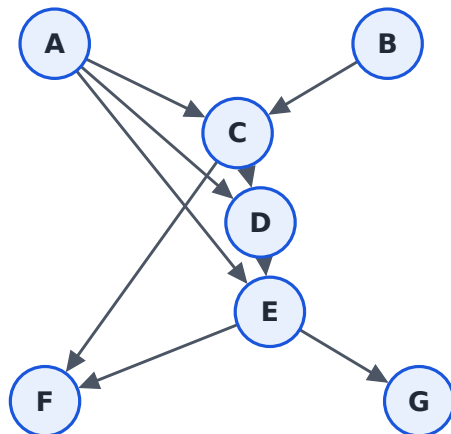
Q.20 [MSQ · 2 marks · no negative marking] Probability for AI

A full joint distribution over Boolean variables X, Y, Z is: $P(x, y, z) = 0.10$, $P(x, y, \neg z) = 0.05$, $P(x, \neg y, z) = 0.15$, $P(x, \neg y, \neg z) = 0.10$, $P(\neg x, y, z) = 0.05$, $P(\neg x, y, \neg z) = 0.20$, $P(\neg x, \neg y, z) = 0.15$, $P(\neg x, \neg y, \neg z) = 0.20$. Which of the following statements are true?

- ☐ (A) $P(x) = 0.4$
☐ (B) $X \perp Y$ (X and Y are marginally independent)
☐ (C) $X \perp Y \mid Z$
☐ (D) $P(y \mid \neg z) \approx 0.455$

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

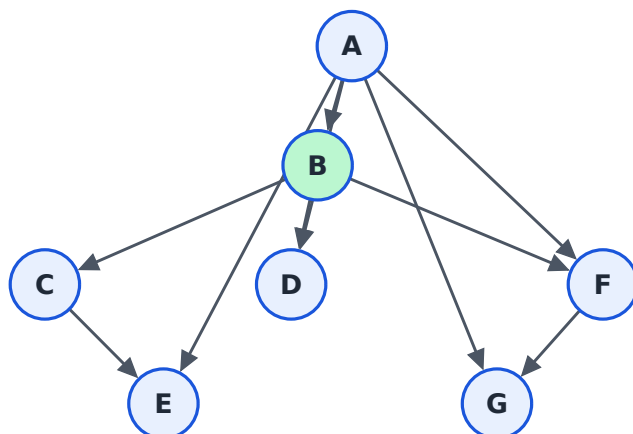
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp B \mid A, F$
☐ (B) $G \perp C \mid A$
☐ (C) $D \perp C \mid G$
☐ (D) $D \perp F \mid C, E$

Q.22 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{B})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, G, E, F, D, A**. What is the scope of the new factor produced when **A** is summed out?



- ☐ (A) $f()$
☐ (B) $f(B)$
☐ (C) $f(A, B)$
☐ (D) $f(A)$

Q.23 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid A = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	F	T	T
2	T	F	F	T
3	T	F	T	F
4	T	F	T	F
5	T	F	F	T
6	T	F	F	T
7	F	F	T	T
8	T	F	T	F
9	T	F	F	T
10	F	F	T	T
11	T	F	T	F
12	T	F	T	T
13	F	F	T	T
14	F	F	T	T
15	T	F	F	T
16	T	F	T	F
17	T	F	F	T
18	F	F	T	F
19	T	F	T	F
20	T	F	T	F

Answer (numerical): _____

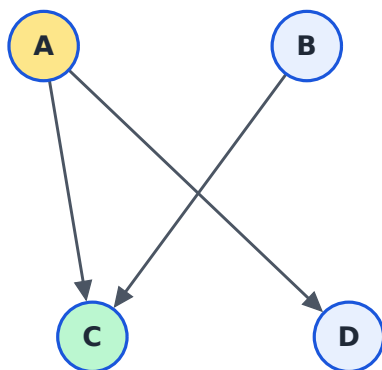
Q.24 [MSQ · 2 marks · no negative marking] Variable Elimination

Which of the following Bayesian networks (edges listed) are polytrees?

- ☐ (A) $A \rightarrow C, B \rightarrow C, C \rightarrow D, C \rightarrow E$
☐ (B) $A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D$
☐ (C) $A \rightarrow B, B \rightarrow C, D \rightarrow C, D \rightarrow E$
☐ (D) $A \rightarrow B, A \rightarrow C, B \rightarrow C$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(C = T \mid A = F)$ (round off to 3 decimal places).



P(A=T)
0.5

P(B=T)	A	B	P(C=T ...)
0.25	T	T	0.6
	T	F	0.3
	F	T	0.2
	F	F	0.2

A	P(D=T ...)
T	0.3
F	0.9

Answer (numerical): _____

Answer Key — Mock Test 075

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	NAT	2	Adversarial Search	18.49 to 18.51
2	MCQ	1	Uninformed Search	A	15	MSQ	2	Predicate Logic	A, B
3	MCQ	1	Adversarial Search	A	16	MSQ	2	Propositional Logic	A, D
4	NAT	1	Adversarial Search	79	17	MSQ	2	Predicate Logic	A, B, D
5	NAT	1	Propositional Logic	4	18	MSQ	2	Propositional Logic	A, D
6	MCQ	1	Predicate Logic	A	19	MCQ	2	Propositional Logic	A
7	NAT	1	Propositional Logic	2	20	MSQ	2	Probability for AI	A, D
8	NAT	1	Sampling	0.2734 to 0.2738	21	MSQ	2	Conditional Independence	D
9	NAT	1	Exact Inference	0.529 to 0.533	22	MCQ	2	Exact Inference	B
10	NAT	1	Sampling	0.445 to 0.455	23	NAT	2	Sampling	0.49 to 0.51
11	NAT	2	Uninformed Search	15	24	MSQ	2	Variable Elimination	A, C
12	MSQ	2	Problem Formulation	A, B, C	25	NAT	2	Exact Inference	0.198 to 0.202
13	NAT	2	Informed Search	8					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: D (D: S, A, B, C, D, J, G)

Adjacency lists (alphabetical): A: {B, E, K, S}; B: {A, C, D, S}; C: {B, D, J, S}; D: {B, C, J, K}; E: {A, G}; G: {E, J, K}; J: {C, D, G, K}; K: {A, D, G, J}; S: {A, B, C}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

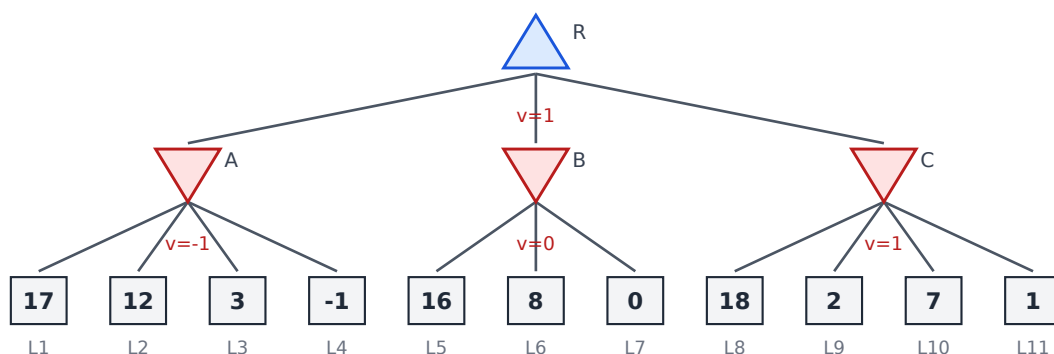
Visit order: **S, A, B, C, D, J, G**; the path found is $S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow J \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.2 [MCQ, 1M · Uninformed Search] Answer: A (A: The graph-search version is complete, but the tree-search version can loop forever on cycles)

Graph-search DFS never re-expands a state, so in a finite space it eventually expands every reachable state and is complete. Tree-search DFS keeps no explored set, so it can follow a cycle such as $A \rightarrow B \rightarrow A \rightarrow \dots$ forever and is incomplete. Neither version is optimal: DFS can return a deep, expensive goal before a shallow one. So (B), (C) and (D) are all wrong.

Q.3 [MCQ, 1M · Adversarial Search] Answer: A (A: Move to C; root value 1)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -1$, $B = 0$, $C = 1$. Root value = **1**; MAX's optimal first move is towards C.

Q.4 [NAT, 1M · Adversarial Search] Answer: 79

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 4^3 + 4^2 - 1 = 64 + 16 - 1 = 79$, versus 1024 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.5 [NAT, 1M · Propositional Logic] Answer: 4

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \rightarrow H$; add H.

Round 2: fire $A \wedge H \rightarrow E$; add E.

Fixed point: $\{A, D, E, H\}$ — 4 atoms are entailed (facts: A, D; newly derived: E, H). Atoms never derived: B, C, F, J, K.

Q.6 [MCQ, 1M · Predicate Logic] Answer: A ($A: Q(f(g(y))) \vee R(y)$)

Unify $P(x)$ with $P(g(y))$: MGU $\theta = \{x/g(y)\}$. The resolvent is $(Q(f(x)) \vee R(y))\theta = Q(f(g(y))) \vee R(y)$. Option B forgets to apply θ . Options C and D apply the substitution incorrectly.

Q.7 [NAT, 1M · Propositional Logic] Answer: 2

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg P \vee \neg Q) \wedge \neg P$	$R \rightarrow (\neg P \wedge P)$	$((\neg P \vee \neg Q) \wedge \neg P) \rightarrow (R \rightarrow (\neg P \wedge P))$
T	T	T	F	F	T
T	T	F	F	T	T
T	F	T	F	F	T
T	F	F	F	T	T
F	T	T	T	F	F
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	T	T	T

F has 6 models, so it is false in $8 - 6 = 2$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.8 [NAT, 1M · Sampling] Answer: 0.2734 to 0.2738

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(A=F) \times P(B=F) \times P(C=F|A=F, B=F) \times P(D=T|B=F, C=F) \\ = 0.4 \times 0.8 \times 0.95 \times 0.9 = \mathbf{0.2736} \approx 0.2736.$$

Q.9 [NAT, 1M · Exact Inference] Answer: 0.529 to 0.533

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T, D=T	0.1134
A=T, B=T, D=F	0.0486
A=T, B=F, D=T	0.2205
A=T, B=F, D=F	0.0945
A=F, B=T, D=T	0.0076
A=F, B=T, D=F	0.0019
A=F, B=F, D=T	0.036
A=F, B=F, D=F	0.009

Sum = **0.5315** ≈ 0.531 .

Q.10 [NAT, 1M · Sampling] Answer: 0.445 to 0.455

In likelihood weighting the weight is the product of the likelihoods of the evidence variables given their parents in the sample. $w = P(s | \neg c) \times P(w | s, \neg r) = 0.5 \times 0.90 = 0.45$. The probabilities of the sampled non-evidence values ($P(\neg c)$, $P(\neg r | \neg c)$) are not part of the weight.

Q.11 [NAT, 2M · Uninformed Search] Answer: 15

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	D(4), J(4)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
2	D (4)	J(4), I(11)
3	J (4)	H(10), I(11)
4	H (10)	A(11), I(11), G(19)
5	A (11)	I(11), G(15)
6	I (11)	G(15)
7	G (15)	← goal, stop

Path returned: **S** → **J** → **H** → **A** → **G**, cost = 4 + 6 + 1 + 4 = **15**. Nodes expanded (including S and G): **7** (S, D, J, H, A, I, G).

Q.12 [MSQ, 2M · Problem Formulation] **Answer: A, B, C**

(A), (B) and (C) are the textbook definitions. (D) is false: d is the depth of the **shallowest** goal node. The maximum length of any path in the state space is m.

Q.13 [NAT, 2M · Informed Search] **Answer: 8**

Heuristic values: h(A)=5, h(B)=9, h(D)=14, h(F)=7, h(G)=0, h(H)=14, h(I)=2, h(K)=10, h(S)=15. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded (f = g + h)	Frontier after expansion (sorted by priority)
1	S (15)	D(17), F(17), K(18), H(20)
2	D (17)	F(17), B(18), K(18), H(20)
3	F (17)	B(18), I(18), K(18), H(20)
4	B (18)	A(18), I(18), K(18), H(20)
5	A (18)	I(18), K(18), G(20), H(20)
6	I (18)	K(18), G(19), H(20)
7	K (18)	G(19), H(20)
8	G (19)	H(20) ← goal, stop

Path returned: **S** → **F** → **I** → **G** with cost **19**; nodes expanded = **8** (S, D, F, B, A, I, K, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.14 [NAT, 2M · Adversarial Search] **Answer: 18.49 to 18.51**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.4 \times 7 + 0.2 \times 13 + 0.1 \times 8 + 0.3 \times 16 = 11$
- $B = 0.5 \times 3 + 0.5 \times 1 = 2$
- $C = 0.2 \times 1 + 0.8 \times 5 = 4.2$
- $D = 0.25 \times 20 + 0.75 \times 18 = 18.5$

Root = $\max(11, 2, 4.2, 18.5) = \mathbf{18.5}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [MSQ, 2M · Predicate Logic] **Answer: A, B** (A: $\forall x ((\text{Person}(x) \wedge \exists y (\text{Dog}(y) \wedge \text{Owns}(x, y))) \rightarrow \text{Happy}(x))$; B: $\forall x \forall y ((\text{Person}(x) \wedge \text{Dog}(y) \wedge \text{Owns}(x, y)) \rightarrow \text{Happy}(x))$)

(A) is the direct reading. (B) is equivalent: $(\exists y \phi(y)) \rightarrow \psi \equiv \forall y (\phi(y) \rightarrow \psi)$ when y is not free in ψ . (C) $\exists y (\phi(y) \rightarrow \psi) \equiv (\forall y \phi(y)) \rightarrow \psi$, which is vacuously true for x as soon as some object is not a dog owned by x, so it imposes almost no constraint. (D) asserts every person owns a dog. Brute force on small domains confirms $A \equiv B$ and that C, D differ.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(R \vee \neg Q) \wedge \neg P$; D: $\neg(\neg\neg P \vee \neg(R \vee \neg Q))$)

Compare truth tables with $F = \neg P \wedge (R \vee \neg Q)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(R \vee \neg Q) \wedge \neg P$ — agrees with F on all 8 rows ⇒ **equivalent**.

(B) $\neg P \vee (R \vee \neg Q)$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) ⇒ not equivalent.

(C) $\neg\neg P \wedge (R \vee \neg Q)$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) ⇒ not equivalent.

(D) $\neg(\neg\neg P \vee \neg(R \vee \neg Q))$ — agrees with F on all 8 rows ⇒ **equivalent**.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, B, D (A: $\forall y \exists x R(x, y)$; B: $\forall x \exists y R(x, y)$; D: $\exists x R(x, x)$)

$R = \{(1,3), (2,2), (2,3), (2,4), (3,1), (3,3), (3,4), (4,1)\}$.

(A) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(B) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,2)$ nor $R(2,1)$.

(D) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $(Q \rightarrow R) \vee (\neg R \wedge \neg P)$; D: $R \rightarrow R$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
F	T	T	T	T
F	F	T	T	T

(A) $(Q \rightarrow R) \vee (\neg R \wedge \neg P)$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $R \leftrightarrow (Q \wedge \neg S)$ is false in the KB-model $P=F, Q=T, R=T, S=T \Rightarrow$ not entailed.

(C) $(P \leftrightarrow P) \leftrightarrow (\neg R \wedge S)$ is false in the KB-model $P=F, Q=T, R=T, S=T \Rightarrow$ not entailed.

(D) $R \rightarrow R$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.19 [MCQ, 2M · Propositional Logic] Answer: A (A: $(p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg r)$)

Distribute: $(p \vee q) \wedge (\neg p \vee \neg q \vee \neg r) = (p \wedge \neg p) \vee (p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg p) \vee (q \wedge \neg q) \vee (q \wedge \neg r)$;

dropping the contradictory terms gives option A. Option B ($p \oplus q$) misses the model $p=q=T, r=F$. Option C misses $p=T, q=F, r=T$. Option D is equivalent but is a CNF, not a DNF. Truth tables confirm A has exactly the 5 models of the original.

Q.20 [MSQ, 2M · Probability for AI] Answer: A, D (A: $P(x) = 0.4$; D: $P(y \mid \neg z) \approx 0.455$)

$P(x) = 0.10 + 0.05 + 0.15 + 0.10 = 0.40$, so (A) is true. $P(y) = 0.10 + 0.05 + 0.05 + 0.20 = 0.40$ and $P(x, y) = 0.15 \neq 0.4 \times 0.4 = 0.16$, so (B) is false. Given z : $P(z) = 0.45$, $P(x, y \mid z) = 0.10/0.45 \approx 0.222$, while $P(x \mid z)P(y \mid z) = (0.25/0.45)(0.15/0.45) \approx 0.556 \times 0.333 \approx 0.185$, so (C) is false. $P(\neg z) = 0.55$ and $P(y, \neg z) = 0.05 + 0.20 = 0.25$, so $P(y \mid \neg z) = 0.25/0.55 \approx 0.4545$, making (D) true.

Q.21 [MSQ, 2M · Conditional Independence] Answer: D (D: $D \perp F \mid C, E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp B \mid A, F$: **FALSE** — the path $E \leftarrow D \leftarrow C \leftarrow B$ is active given $\{A, F\}$.

(B) $G \perp C \mid A$: **FALSE** — the path $G \leftarrow E \leftarrow D \leftarrow C$ is active given $\{A\}$.

(C) $D \perp C \mid G$: **FALSE** — the path $D \leftarrow A \rightarrow C$ is active given $\{G\}$.

(D) $D \perp F \mid C, E$: **TRUE** — $D \leftarrow A \rightarrow C \rightarrow F$ (blocked: observed chain node C); $D \leftarrow A \rightarrow E \rightarrow F$ (blocked: observed chain node E); $D \leftarrow C \leftarrow A \rightarrow E \rightarrow F$ (blocked: observed chain node C); $D \leftarrow C \rightarrow F$ (blocked: observed fork node C); $D \rightarrow E \leftarrow A \rightarrow C \rightarrow F$ (blocked: observed chain node C); $D \rightarrow E \rightarrow F$ (blocked: observed chain node E).

Q.22 [MCQ, 2M · Exact Inference] Answer: B (B: $f(B)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(A, C, E)$, $f(A, B, F)$, $f(A, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C)$, $f(A, C, E)$	$\{A, B, C, E\}$	$f(A, B, E)$
G	$f(A, F, G)$	$\{A, F, G\}$	$f(A, F)$
E	$f(A, B, E)$	$\{A, B, E\}$	$f(A, B)$
F	$f(A, B, F)$, $f(A, F)$	$\{A, B, F\}$	$f(A, B)$
D	$f(A, B, D)$	$\{A, B, D\}$	$f(A, B)$
A	$f(A)$, $f(A, B)$, $f(A, B)$, $f(A, B)$, $f(A, B)$	$\{A, B\}$	$f(B)$

The largest factor produced has **3** variable(s) (created when eliminating C); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over B and normalise.

Q.23 [NAT, 2M · Sampling] Answer: 0.49 to 0.51

Rejection sampling keeps only samples consistent with the evidence $A = T$: samples 2, 3, 4, 5, 6, 8, 9, 11, 12, 15, 16, 17, 19, 20 (14 accepted, 6 rejected).

Among the accepted samples, $D = T$ in 7. Estimate = $7/14 = 0.5$.

(For this network the exact value is 0.6525; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.24 [MSQ, 2M · Variable Elimination] Answer: A, C (A: $A \rightarrow C, B \rightarrow C, C \rightarrow D, C \rightarrow E$; C: $A \rightarrow B, B \rightarrow C, D \rightarrow C, D \rightarrow E$)

A polytree has no undirected cycle. (A) Underlying undirected graph A-C, B-C, C-D, C-E is a tree: polytree.

(B) A-B-D-C-A is an undirected cycle: multiply connected. (C) A-B-C-D-E is a simple path, no cycle: polytree.

(C has two parents, which is allowed). (D) A-B-C-A forms a triangle: not a polytree.

Q.25 [NAT, 2M · Exact Inference] Answer: 0.198 to 0.202

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B) \cdot P(C | A, B) \cdot P(D | A)$.

Inference by enumeration: $P(C | e) = \alpha \sum_{\text{hidden}} P(C, e, \text{hidden})$.

Hidden: B, D	C	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.5 \times 0.25 \times 0.2 \times 0.9$	0.0225
T, F	T	$0.5 \times 0.25 \times 0.2 \times 0.1$	0.0025
F, T	T	$0.5 \times 0.75 \times 0.2 \times 0.9$	0.0675
F, F	T	$0.5 \times 0.75 \times 0.2 \times 0.1$	0.0075
T, T	F	$0.5 \times 0.25 \times 0.8 \times 0.9$	0.09
T, F	F	$0.5 \times 0.25 \times 0.8 \times 0.1$	0.01
F, T	F	$0.5 \times 0.75 \times 0.8 \times 0.9$	0.27
F, F	F	$0.5 \times 0.75 \times 0.8 \times 0.1$	0.03

$P(C=T, e) = 0.1$; $P(C=F, e) = 0.4$; $P(e) = 0.5$.

$P(C=T | e) = 0.1 / 0.5 = 0.2 \approx 0.200$.

Mock Test 076

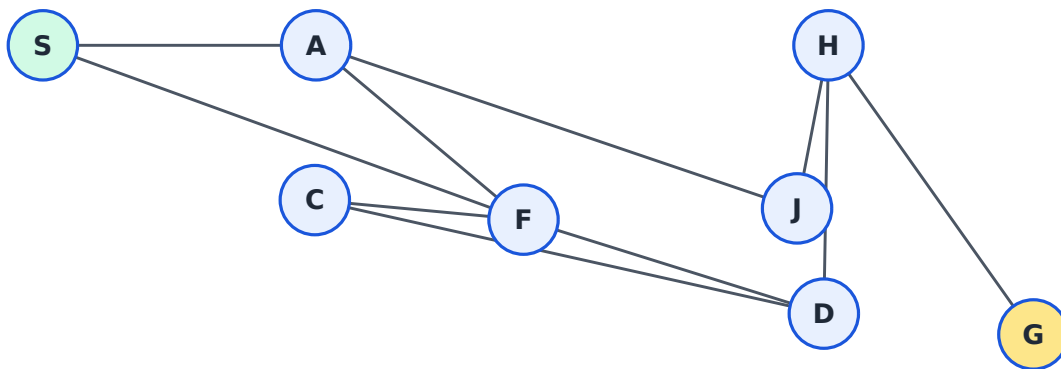
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 7 / 14	15 / 12 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

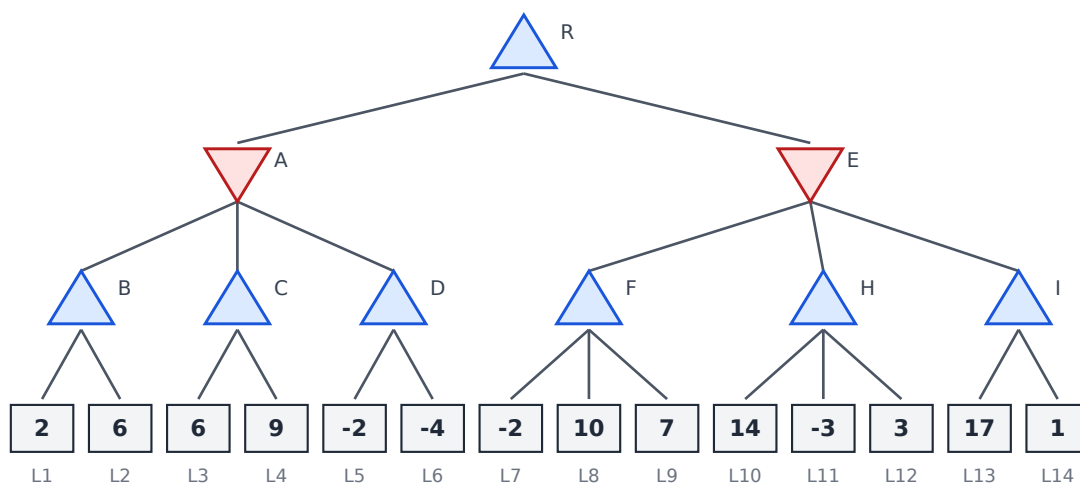
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, F, A, D, C, J, H, G
- ☐ (B) S, A, F, J, C, H, D, G
- ☐ (C) S, A, F, J, C, D, H, G
- ☐ (D) S, A, F, C, D, H, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



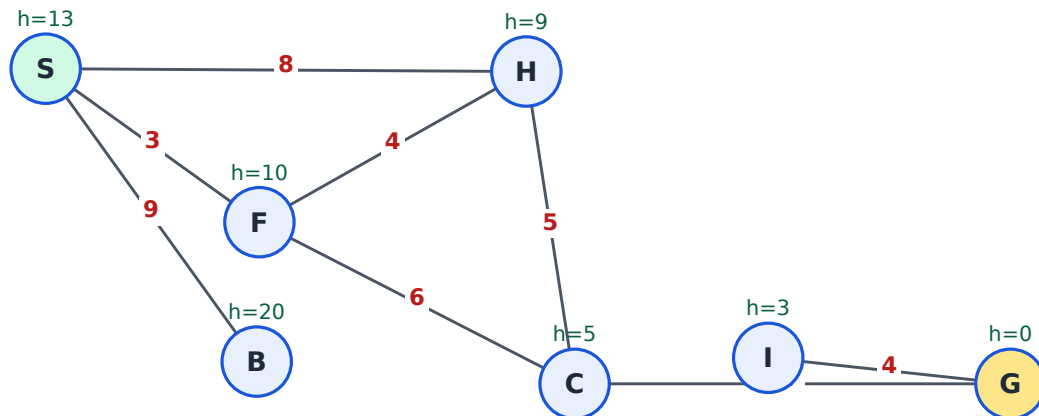
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Propositional Logic

$\neg(p \rightarrow q)$ is logically equivalent to

- ☐ (A) $p \wedge \neg q$
☐ (B) $\neg p \wedge q$
☐ (C) $\neg p \rightarrow \neg q$
☐ (D) $\neg p \vee q$

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$\mathbf{F} = \neg P \rightarrow (Q \vee (R \wedge Q)) ?$$

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following equivalences are valid in first-order logic?

- ☐ (A) $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
☐ (B) $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
☐ (C) $\forall x \exists y P(x, y) \equiv \exists y \forall x P(x, y)$
☐ (D) $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q) . For how many of them is $\forall x P(x) \vee \forall x Q(x)$ true?

Answer (numerical): _____

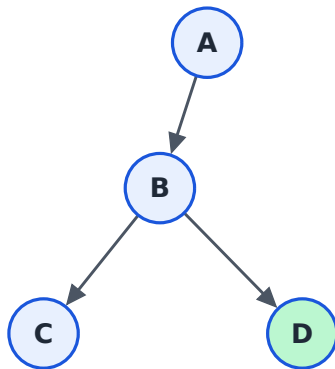
Q.8 [MCQ · 1 mark · negative: -1/3] Conditional Independence

In a Bayesian network, a node is conditionally independent of all other nodes in the network given its Markov blanket. The Markov blanket consists of the node's

- ☐ (A) parents, children, and children's other parents
☐ (B) parents and children only
☐ (C) parents and all ancestors
☐ (D) all non-descendants

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(D = T)$ (round off to 3 decimal places).



A	P(A=T)
T	0.25
F	

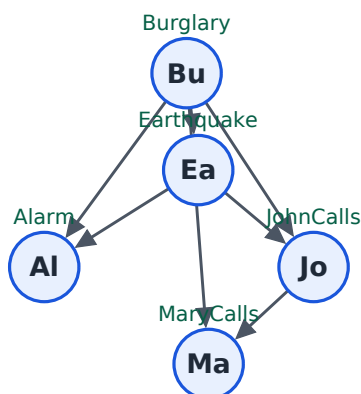
A	P(B=T ...)	B	P(C=T ...)
T	0.95	T	0.25
F	0.2	F	0.9

B	P(D=T ...)
T	0.7
F	0.1

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Bu, Ea, Al, Jo, Ma. (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls, Ma = MaryCalls.) What is the probability that a single run produces the sample **(Bu=T, Ea=T, Al=T, Jo=F, Ma=T)**? (round off to 4 decimal places)



Bu	P(Bu=T)
T	0.75
F	

Bu	P(Ea=T ...)	Bu	Ea	P(Al=T ...)
T	0.95	T	T	0.95
F	0.25	T	F	0.25
		F	T	0.75
		F	F	0.6

Bu	Ea	P(Jo=T ...)	Bu	Jo	P(Ma=T ...)
T	T	0.4	T	T	0.3
T	F	0.25	T	F	0.9
F	T	0.95	F	T	0.95
F	F	0.75	F	F	0.95

Answer (numerical): _____

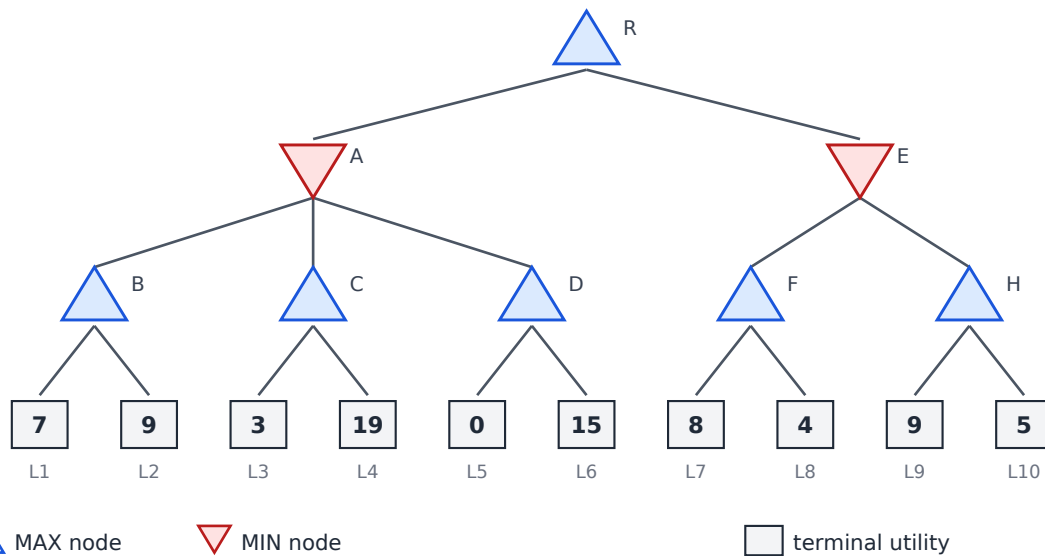
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search

Consider a complete binary tree of depth 3 whose 15 nodes are labelled 1 to 15 in level order: the root is 1 and the children of node i are $2i$ (left) and $2i+1$ (right). Iterative deepening search uses depth limits 0, 1, 2, 3, ... in turn, expands left children before right children, and goal-tests a node when it is visited (popped). The goal is node 11. Over all iterations, counting repeated visits, how many node visits happen up to and including the visit at which node 11 is found?

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

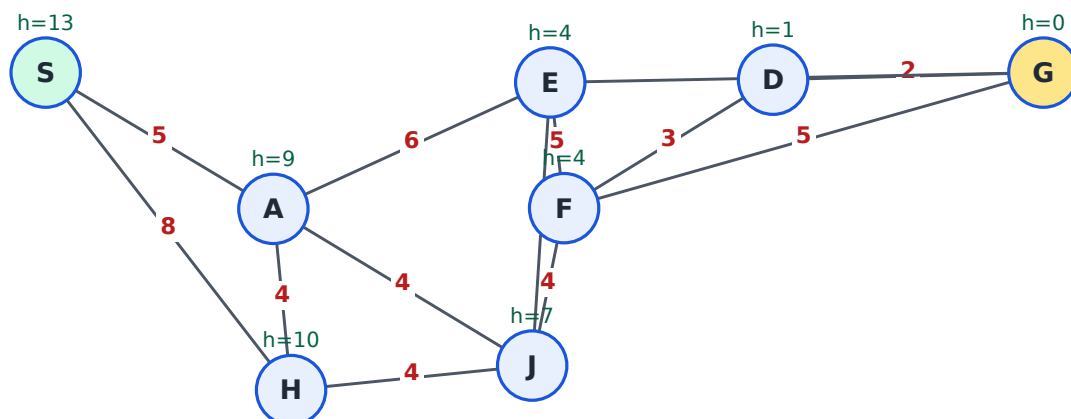
Q.13 [MSQ · 2 marks · no negative marking] Problem Formulation

In a grid route-finding problem, the agent moves N, S, E or W at cost 1 per move. The start state is (0,0), the only goal is (3,2), and there are no obstacles in an unbounded grid. Which of the following are TRUE?

- ☐ (A) The optimal solution cost is 5
- ☐ (B) BFS with graph search is optimal for this problem
- ☐ (C) Manhattan distance is a consistent heuristic for this problem
- ☐ (D) The branching factor b is 8

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

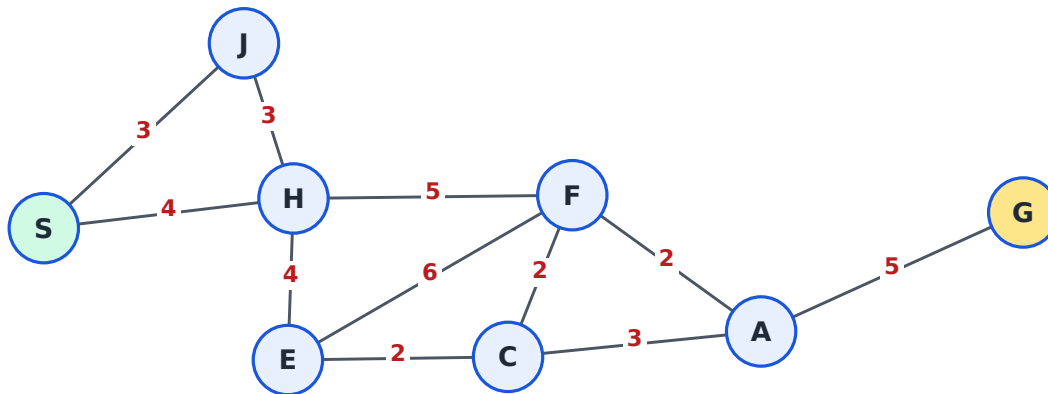
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

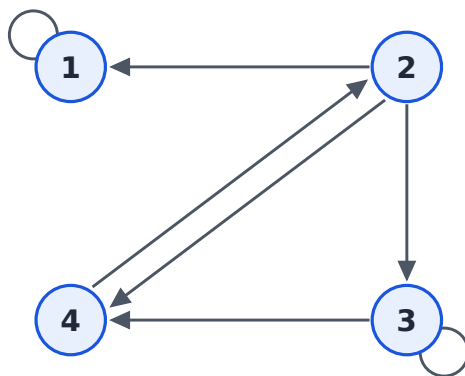
Q.16 [MSQ · 2 marks · no negative marking] Uninformed Search

Iterative deepening search (IDS) runs on a problem with finite branching factor b and shallowest goal depth d . Which of the following statements are TRUE?

- ☐ (A) IDS is complete
- ☐ (B) IDS is optimal when all step costs are equal
- ☐ (C) The space complexity of IDS is $O(bd)$
- ☐ (D) The time complexity of IDS is asymptotically worse than BFS by a factor of d , namely $O(d \cdot b^d)$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall y \exists x R(x, y)$
- ☐ (B) $\exists x R(x, x)$
- ☐ (C) $\forall x \exists y R(x, y)$
- ☐ (D) $\exists x \forall y \neg R(x, y)$

Q.18 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of two unary predicates P and Q over D (i.e. ordered pairs of subsets) satisfy $\forall x (P(x) \rightarrow Q(x)) \wedge \exists x (Q(x) \wedge \neg P(x))$?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $(Q \vee R) \rightarrow (Q \leftrightarrow R)$
- (2) $Q \leftrightarrow R$
- (3) $(R \vee P) \wedge R$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg P \rightarrow (\neg P \vee P)$
- ☐ (B) $\neg Q \leftrightarrow R$
- ☐ (C) $\neg \neg P$
- ☐ (D) $(R \leftrightarrow P) \vee P$

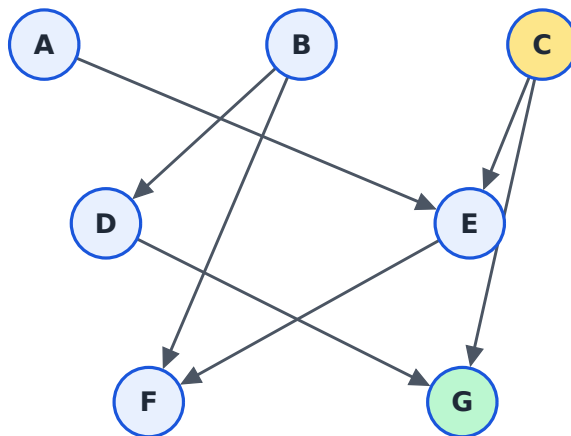
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (Q \rightarrow Q) \rightarrow P$?

- ☐ (A) $\neg P \rightarrow \neg(Q \rightarrow Q)$
- ☐ (B) $(Q \rightarrow Q) \vee \neg P$
- ☐ (C) $P \rightarrow (Q \rightarrow Q)$
- ☐ (D) $\neg((Q \rightarrow Q) \wedge \neg P)$

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

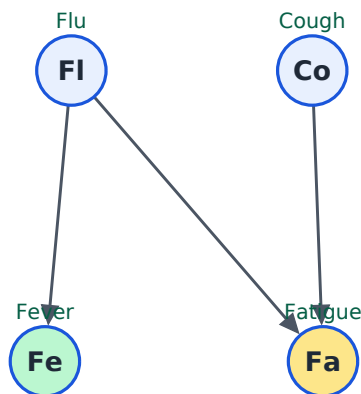
Variable elimination is used to compute $P(G \mid C = T)$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E**, **B**, **D**, **F**, **A**. What is the scope of the new factor produced when **F** is summed out?



- ☐ (A) $f(A)$
- ☐ (B) $f(A, G)$
- ☐ (C) $f(B, E)$
- ☐ (D) $f(A, F, G)$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Fever} = T \mid \text{Fatigue} = T)$ (round off to 3 decimal places).



FI	P(FI=T)
T	0.3

FI	Co	P(Fe=T ...)	P(Co=T)
T	T	0.3	0.6
F	T	0.2	

FI	Co	P(Fa=T ...)
T	T	0.3
T	F	0.05
F	T	0.05
F	F	0.75

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid C = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	T	F	F
2	F	T	T	T
3	F	T	F	F
4	T	T	F	T
5	F	T	T	T
6	F	T	T	T
7	F	T	T	T
8	T	F	F	T
9	F	T	T	T
10	F	T	F	F
11	F	F	F	F
12	F	T	T	T
13	F	T	T	F
14	T	F	F	T
15	F	T	T	T
16	F	T	T	F
17	F	T	T	T
18	F	T	T	T

Answer (numerical): _____

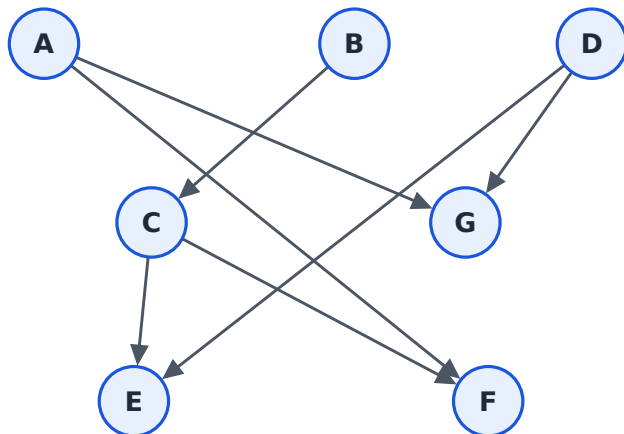
Q.24 [NAT · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. Compute $P(b \mid j, m)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp D \mid G$
☐ (B) $C \perp E \mid B$
☐ (C) $A \perp B \mid G$
☐ (D) $E \perp D \mid B$

Answer Key — Mock Test 076

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	C	14	NAT	2	Informed Search	7
2	NAT	1	Adversarial Search	10	15	NAT	2	Uninformed Search	8
3	NAT	1	Informed Search	20	16	MSQ	2	Uninformed Search	A, B, C
4	MCQ	1	Propositional Logic	A	17	MSQ	2	Predicate Logic	A, B, C
5	NAT	1	Propositional Logic	6	18	NAT	2	Predicate Logic	19
6	MSQ	1	Predicate Logic	A, B, D	19	MSQ	2	Propositional Logic	A
7	NAT	1	Predicate Logic	15	20	MSQ	2	Propositional Logic	A, D
8	MCQ	1	Conditional Independence	A	21	MCQ	2	Exact Inference	B
9	NAT	1	Exact Inference	0.331 to 0.335	22	NAT	2	Exact Inference	0.219 to 0.223
10	NAT	1	Sampling	0.3653 to 0.3657	23	NAT	2	Sampling	0.81 to 0.83
11	NAT	2	Uninformed Search	19	24	NAT	2	Bayesian Networks	0.555 to 0.557
12	NAT	2	Adversarial Search	8	25	MSQ	2	Conditional Independence	A, C
13	MSQ	2	Problem Formulation	A, B, C					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, A, F, J, C, D, H, G)

Adjacency lists (alphabetical): A: {F, J, S}; C: {D, F}; D: {C, F, H}; F: {A, C, D, S}; G: {H}; H: {D, G, J}; J: {A, H}; S: {A, F}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[A, F]
2	A	[F, J]
3	F	[J, C, D]
4	J	[C, D, H]
5	C	[D, H]

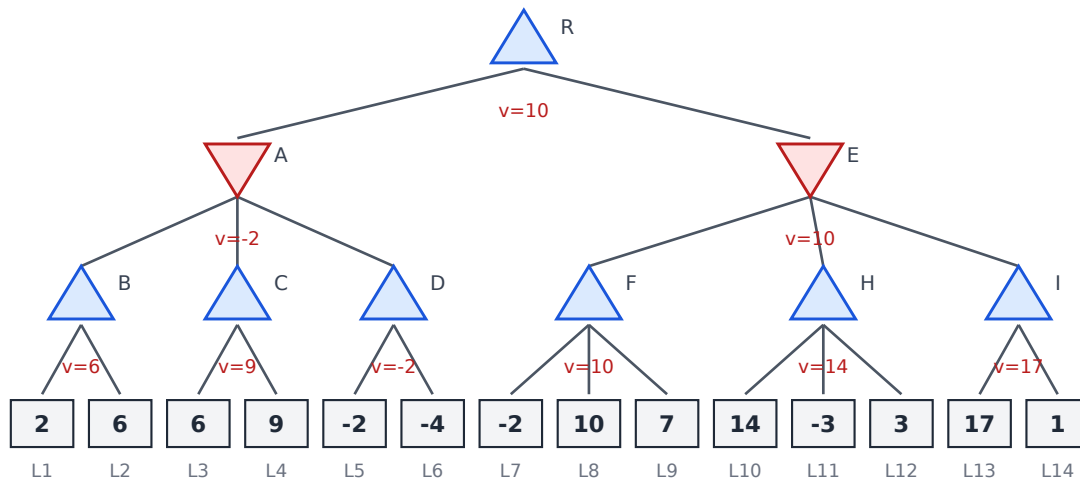
Step	Expanded	Frontier (FIFO) after expansion
6	D	[H]
7	H	[G]
8	G	[] ← goal popped, stop

Expansion order: **S, A, F, J, C, D, H, G**. The path returned is $S \rightarrow A \rightarrow J \rightarrow H \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Adversarial Search] Answer: 10

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 6$, $C = 9$, $D = -2$, $A = -2$, $F = 10$, $H = 14$, $I = 17$, $E = 10$. Root value = **10**; MAX's optimal first move is towards E.

Q.3 [NAT, 1M · Informed Search] Answer: 20

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (13)	H(9), F(10), B(20)
2	H (9)	C(5), F(10), B(20)
3	C (5)	G(0), F(10), B(20)
4	G (0)	F(10), B(20) ← goal, stop

Path returned: $S \rightarrow H \rightarrow C \rightarrow G$, cost = $8 + 5 + 7 = 20$.

The optimal cost is $h^*(S) = 16$ — greedy search is not optimal here.

Q.4 [MCQ, 1M · Propositional Logic] Answer: A ($A: p \wedge \neg q$)

$p \rightarrow q \equiv \neg p \vee q$, so $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q$ by De Morgan's law. It is true only in the row $p=T, q=F$, which is exactly the row where $p \rightarrow q$ is false. $\neg p \wedge q$ is true only at $p=F, q=T$; $\neg p \rightarrow \neg q$ is the inverse; $\neg p \vee q$ is $p \rightarrow q$ itself.

Q.5 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg P$	$Q \vee (R \wedge Q)$	$\neg P \rightarrow (Q \vee (R \wedge Q))$
T	T	T	F	T	T
T	T	F	F	T	T
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	T	F	F

Rows where F = T: **6**.

Q.6 [MSQ, 1M · Predicate Logic] Answer: A, B, D (A: $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$; B: $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$; D: $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$)

Quantifiers of the same kind commute (A, B). $\exists$ distributes over $\vee$ (D), just as $\forall$ distributes over $\wedge$. (C) Mixed quantifiers do not commute: with $P(x, y) \iff x = y$ on $\{1, 2\}$, $\forall x \exists y P$ is true but $\exists y \forall x P$ is false.

Q.7 [NAT, 1M · Predicate Logic] Answer: 15

Interpret P and Q as subsets of the domain; inclusion-exclusion: $2^n + 2^n - 1$. With $n = 3$ this gives **15** (verified by enumerating all interpretations).

Q.8 [MCQ, 1M · Conditional Independence] Answer: A (A: parents, children, and children's other parents)

AIMA defines the Markov blanket as parents, children and children's parents. Children's other parents are needed because of explaining away: given a child, a co-parent carries information about the node. (B) omits the co-parents; (C) and (D) describe sets relevant to the local Markov property (independence from non-descendants given parents), not the blanket.

Q.9 [NAT, 1M · Exact Inference] Answer: 0.331 to 0.335

Only the ancestors of D (A, B) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
A=T, B=T, C=T	0.04156
A=T, B=T, C=F	0.12469
A=T, B=F, C=T	0.00113
A=T, B=F, C=F	0.00013
A=F, B=T, C=T	0.02625
A=F, B=T, C=F	0.07875
A=F, B=F, C=T	0.054
A=F, B=F, C=F	0.006

Sum = **0.3325** $\approx$ 0.333.

Q.10 [NAT, 1M · Sampling] Answer: 0.3653 to 0.3657

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(\text{Bu}=T) \times P(\text{Ea}=T \mid \text{Bu}=T) \times P(\text{Al}=T \mid \text{Bu}=T, \text{Ea}=T) \times P(\text{Jo}=F \mid \text{Bu}=T, \text{Ea}=T) \times P(\text{Ma}=T \mid \text{Bu}=T, \text{Jo}=F)$
 $= 0.75 \times 0.95 \times 0.95 \times 0.6 \times 0.9 = \mathbf{0.365512} \approx 0.3655$.

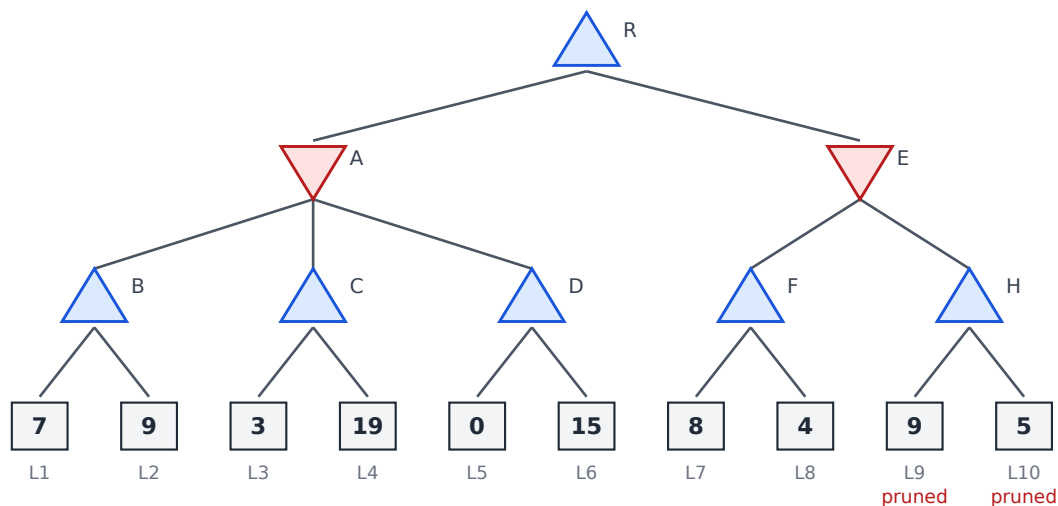
Q.11 [NAT, 2M · Uninformed Search] Answer: 19

Node 11 lies at depth 3 (path $1 \rightarrow 2 \rightarrow 5 \rightarrow 11$). Limit 0 visits $\{1\}$: 1 visit. Limit 1 visits $\{1, 2, 3\}$: 3 visits. Limit 2 visits the whole depth-2 tree $\{1, 2, 4, 5, 3, 6, 7\}$: 7 visits. Limit 3 visits 1, 2, 4, 8, 9, 5, 10, 11 in preorder and stops at 11: 8 visits. Total = $1 + 3 + 7 + 8 = \mathbf{19}$.

Q.12 [NAT, 2M · Adversarial Search] Answer: 8

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node E: value $8 \leq \alpha = 9 \Rightarrow$ prune remaining children (leaves L9, L10).



Leaves evaluated (8): L1, L2, L3, L4, L5, L6, L7, L8. Leaves pruned (2): L9, L10. Root value = 9 (same as plain minimax, 9 — pruning never changes the root value).

Q.13 [MSQ, 2M · Problem Formulation] Answer: A, B, C (A: The optimal solution cost is 5; B: BFS with graph search is optimal for this problem; C: Manhattan distance is a consistent heuristic for this problem)

(A) is true: the shortest path needs $|3| + |2| = 5$ unit moves. (B) is true: costs are uniform, so the shallowest goal is cheapest. (C) is true: one move changes the Manhattan distance by exactly 1 = the step cost, so $h(n) \leq 1 + h(n')$. (D) is false: there are 4 actions, so $b = 4$. $b = 8$ would need diagonal moves.

Q.14 [NAT, 2M · Informed Search] Answer: 7

Heuristic values: $h(A)=9$, $h(D)=1$, $h(E)=4$, $h(F)=4$, $h(G)=0$, $h(H)=10$, $h(J)=7$, $h(S)=13$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	A(14), H(18)
2	A (14)	E(15), J(16), H(18)
3	E (15)	J(16), G(17), H(18), F(20)
4	J (16)	F(17), G(17), H(18)
5	F (17)	D(17), G(17), H(18)
6	D (17)	G(17), H(18)
7	G (17)	H(18) ← goal, stop

Path returned: **S** → **A** → **E** → **G** with cost **17**; nodes expanded = **7** (S, A, E, J, F, D, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 17$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [NAT, 2M · Uninformed Search] Answer: 8

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(3), H(4)
2	J (3)	H(4)
3	H (4)	E(8), F(9)
4	E (8)	F(9), C(10)
5	F (9)	C(10), A(11)
6	C (10)	A(11)
7	A (11)	G(16)
8	G (16)	← goal, stop

Path returned: **S** → **H** → **F** → **A** → **G**, cost = $4 + 5 + 2 + 5 = 16$. Nodes expanded (including S and G): **8** (S, J, H, E, F, C, A, G).

Q.16 [MSQ, 2M · Uninformed Search] Answer: A, B, C (A: IDS is complete; B: IDS is optimal when all step costs are equal; C: The space complexity of IDS is $O(bd)$)

IDS combines DFS's memory use with BFS's level-by-level completeness. (A) is true: with finite b it eventually reaches limit d and finds the goal. (B) is true: like BFS, it finds the shallowest goal first, which is optimal for unit costs. (C) is true: each iteration is a DFS of depth at most d , so the space is $O(bd)$. (D) is false: the total $d \cdot b + (d-1)b^2 + \dots + b^d$ is still $O(b^d)$, because the deepest level dominates and the repeated shallow levels add only a constant factor of about $b/(b-1)$.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, B, C (A: $\forall y \exists x R(x, y)$; B: $\exists x R(x, x)$; C: $\forall x \exists y R(x, y)$)

$R = \{(1,1), (2,1), (2,3), (2,4), (3,3), (3,4), (4,2)\}$.

(A) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

(C) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(D) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [NAT, 2M · Predicate Logic] Answer: 19

$\forall x (P(x) \rightarrow Q(x))$ leaves each element 3 choices of (P, Q) : (F,F), (F,T), (T,T); so $3^3 = 27$ pairs. The second conjunct requires at least one element with (F,T). Pairs with no (F,T) element: $2^3 = 8$. Answer = $27 - 8 = 19$ (P is a proper subset of Q).

Q.19 [MSQ, 2M · Propositional Logic] Answer: A (A: $\neg P \rightarrow (\neg P \vee P)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
T	T	T	T
F	T	T	T

(A) $\neg P \rightarrow (\neg P \vee P)$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $\neg Q \leftrightarrow R$ is false in the KB-model $P=T, Q=T, R=T \Rightarrow$ not entailed.

(C) $\neg \neg P$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

(D) $(R \leftrightarrow P) \vee P$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

Q.20 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $\neg P \rightarrow \neg(Q \rightarrow Q)$; D: $\neg((Q \rightarrow Q) \wedge \neg P)$)

Compare truth tables with $F = (Q \rightarrow Q) \rightarrow P$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg P \rightarrow \neg(Q \rightarrow Q)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $(Q \rightarrow Q) \vee \neg P$ — differs at $P=F, Q=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(C) $P \rightarrow (Q \rightarrow Q)$ — differs at $P=F, Q=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(D) $\neg((Q \rightarrow Q) \wedge \neg P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.21 [MCQ, 2M · Exact Inference] Answer: B (B: $f(A, G)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(B), f(), f(B, D), f(A, E), f(B, E, F), f(D, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(A, E), f(B, E, F)$	$\{A, B, E, F\}$	$f(A, B, F)$
B	$f(B), f(B, D), f(A, B, F)$	$\{A, B, D, F\}$	$f(A, D, F)$
D	$f(D, G), f(A, D, F)$	$\{A, D, F, G\}$	$f(A, F, G)$
F	$f(A, F, G)$	$\{A, F, G\}$	$f(A, G)$
A	$f(A), f(A, G)$	$\{A, G\}$	$f(G)$

The largest factor produced has **3** variable(s) (created when eliminating E); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over G and normalise.

Q.22 [NAT, 2M · Exact Inference] Answer: 0.219 to 0.223

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever} | \text{Flu}) \cdot P(\text{Cough}) \cdot P(\text{Fatigue} | \text{Flu}, \text{Cough})$.

Inference by enumeration: $P(\text{Fever} | e) = \alpha \sum_{\text{hidden}} P(\text{Fever}, e, \text{hidden})$.

Hidden: Fl, Co	Fe	product of CPT entries (Fl · Fe · Co · Fa)	value
T, T	T	$0.3 \times 0.3 \times 0.6 \times 0.3$	0.0162
T, F	T	$0.3 \times 0.3 \times 0.4 \times 0.05$	0.0018
F, T	T	$0.7 \times 0.2 \times 0.6 \times 0.05$	0.0042

Hidden: Fl, Co	Fe	product of CPT entries (Fl · Fe · Co · Fa)	value
F, F	T	$0.7 \times 0.2 \times 0.4 \times 0.75$	0.042
T, T	F	$0.3 \times 0.7 \times 0.6 \times 0.3$	0.0378
T, F	F	$0.3 \times 0.7 \times 0.4 \times 0.05$	0.0042
F, T	F	$0.7 \times 0.8 \times 0.6 \times 0.05$	0.0168
F, F	F	$0.7 \times 0.8 \times 0.4 \times 0.75$	0.168

$P(\text{Fever}=T, e) = 0.0642$; $P(\text{Fever}=F, e) = 0.2268$; $P(e) = 0.291$.

$P(\text{Fever}=T \mid e) = 0.0642 / 0.291 = \mathbf{0.2206} \approx 0.221$.

Q.23 [NAT, 2M · Sampling] **Answer: 0.81 to 0.83**

Rejection sampling keeps only samples consistent with the evidence $C = T$: samples 2, 5, 6, 7, 9, 12, 13, 15, 16, 17, 18 (11 accepted, 7 rejected).

Among the accepted samples, $D = T$ in 9. Estimate = $9/11 = \mathbf{0.8182}$.

(For this network the exact value is 0.843; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.24 [NAT, 2M · Bayesian Networks] **Answer: 0.555 to 0.557**

$P(B \mid j, m) = \alpha P(B) \sum_e P(e) \sum_a P(a \mid B, e) P(j \mid a) P(m \mid a)$. Let $g(a) = P(j \mid a)P(m \mid a)$: $g(a) = 0.56$, $g(\neg a) = 0.001$. For b : e : $0.95 \times 0.56 + 0.05 \times 0.001 = 0.53205$; $\neg e$: $0.90 \times 0.56 + 0.10 \times 0.001 = 0.5041$; weighted: $0.01 \times 0.53205 + 0.99 \times 0.5041 = 0.504381$; times $P(b) = 0.02 \rightarrow 0.0100876$. For $\neg b$: e : $0.30 \times 0.56 + 0.70 \times 0.001 = 0.1687$; $\neg e$: $0.01 \times 0.56 + 0.99 \times 0.001 = 0.006589$; weighted: $0.01 \times 0.1687 + 0.99 \times 0.006589 = 0.0082101$; times $0.98 \rightarrow 0.0080459$. $P(b \mid j, m) = 0.0100876 / (0.0100876 + 0.0080459) \approx 0.556$.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: A, C** (A: $C \perp D \mid G$; C: $A \perp B \mid G$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp D \mid G$: **TRUE** — $C \rightarrow E \leftarrow D$ (blocked: collider E with neither it nor a descendant observed); $C \rightarrow F \leftarrow A \rightarrow G \leftarrow D$ (blocked: collider F with neither it nor a descendant observed).

(B) $C \perp E \mid B$: **FALSE** — the path $C \rightarrow E$ is active given $\{B\}$.

(C) $A \perp B \mid G$: **TRUE** — $A \rightarrow F \leftarrow C \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow G \leftarrow D \rightarrow E \leftarrow C \leftarrow B$ (blocked: collider E with neither it nor a descendant observed).

(D) $E \perp D \mid B$: **FALSE** — the path $E \leftarrow D$ is active given $\{B\}$.

Mock Test 077

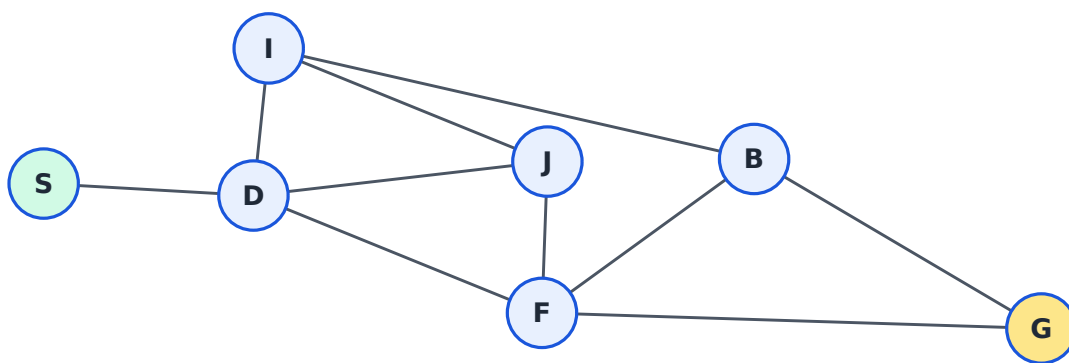
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	4 / 6 / 15	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

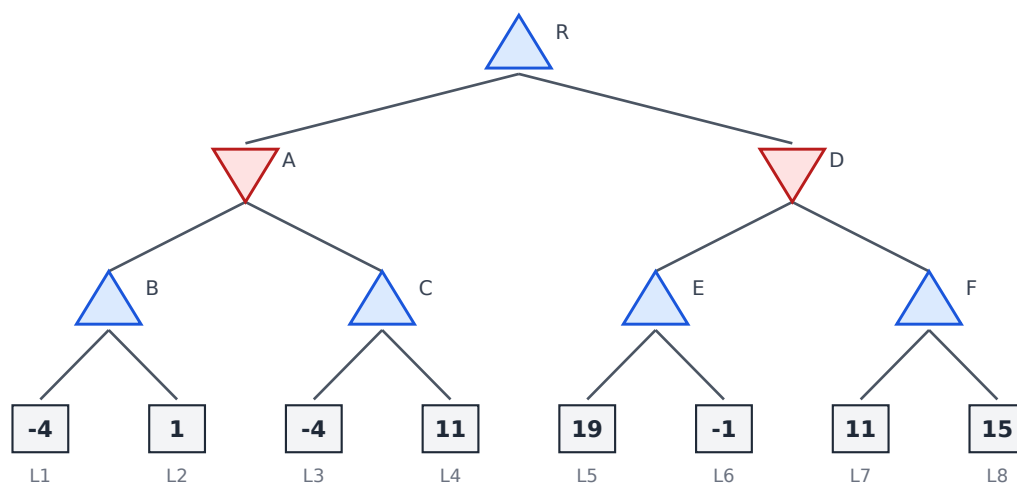
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, J, I, B, G
- ☐ (B) S, D, F, I, J, B, G
- ☐ (C) S, D, B, F, G
- ☐ (D) S, D, F, B, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



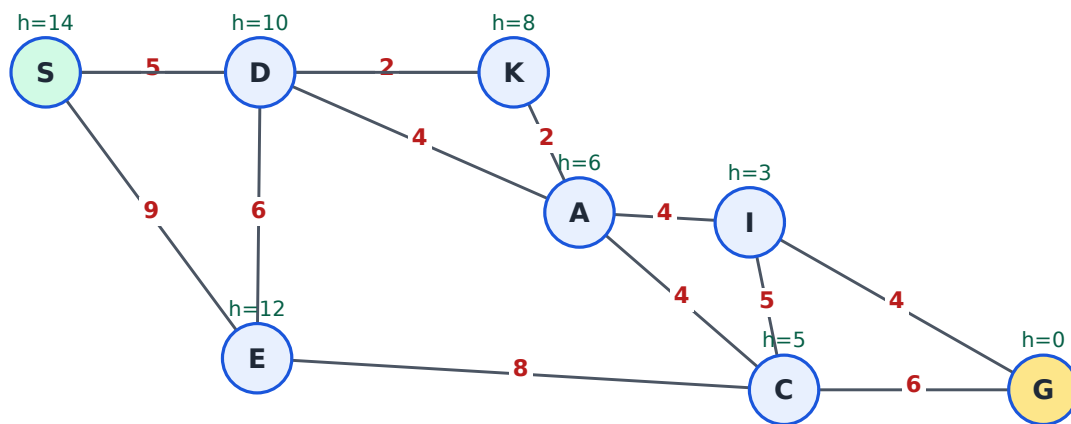
MAX node MIN node

terminal utility

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Uninformed Search

Bidirectional BFS runs on a problem with uniform branching factor $b = 10$ in both directions, and the solution has depth $d = 6$. Each search goes to depth $d/2 = 3$, and the two frontiers meet. Assume each direction generates every node down to depth 3, excluding its own root. How many nodes are generated in total by the two searches?

Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

In the DPLL algorithm, a pure symbol is one that appears with the same sign in every clause in which it occurs. For the clause set $\{A \vee \neg B, \neg B \vee \neg C, C \vee A\}$, which symbols are pure?

- ☐ (A) A
☐ (B) B
☐ (C) C
☐ (D) None of the symbols is pure

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$ true?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In the formula $\forall x (P(x, y) \rightarrow \exists y Q(x, y, z))$, which variables have free occurrences?

- ☐ (A) y and z
☐ (B) z only
☐ (C) x, y and z
☐ (D) x and z

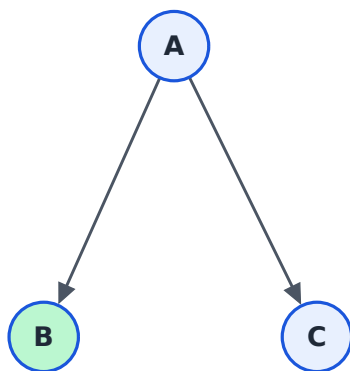
Q.8 [MCQ · 1 mark · negative: -1/3] Variable Elimination

In variable elimination, $f_1(A, B)$ and $f_2(B, C)$ are factors over Boolean variables. Their pointwise product $f_1 \times f_2$ is

- ☐ (A) a factor over $\{A, B, C\}$ with 8 entries
- ☐ (B) a factor over $\{A, C\}$ with 4 entries
- ☐ (C) a factor over $\{B\}$ with 2 entries
- ☐ (D) a factor over $\{A, B, B, C\}$ with 16 entries

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $\mathbf{P(B = T)}$ (round off to 3 decimal places).

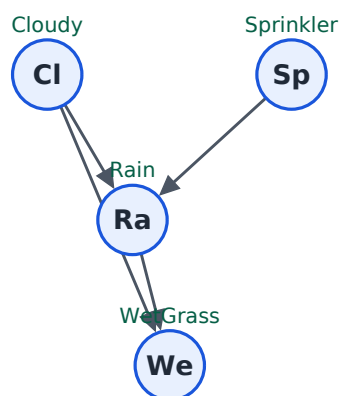


$P(A=T)$	A	$P(B=T \dots)$	A	$P(C=T \dots)$
0.2	T	0.95	T	0.2
	F	0.1	F	0.9

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample **(Cl=F, Sp=F, Ra=T, We=F)**? (round off to 4 decimal places)



$P(Cl=T)$
0.5

$P(Sp=T)$	Cl	Sp	$P(Ra=T \dots)$
0.4	T	T	0.25
	T	F	0.7
	F	T	0.2
	F	F	0.95

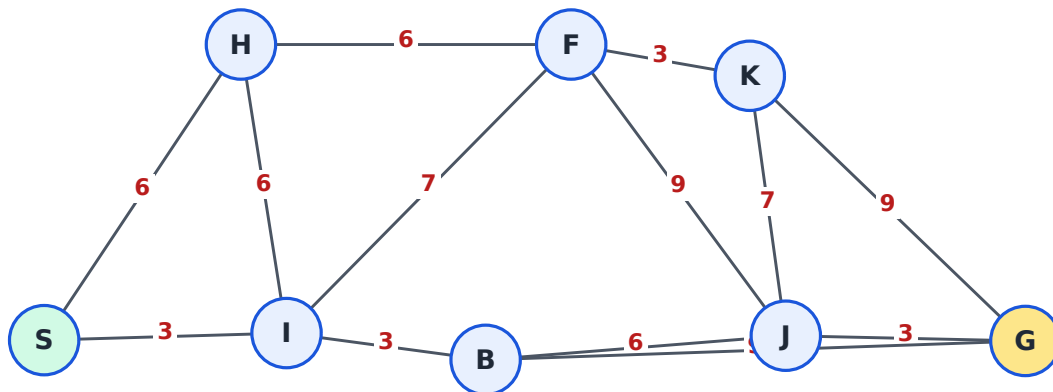
Cl	Ra	$P(We=T \dots)$
T	T	0.05
T	F	0.05
F	T	0.05
F	F	0.05

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

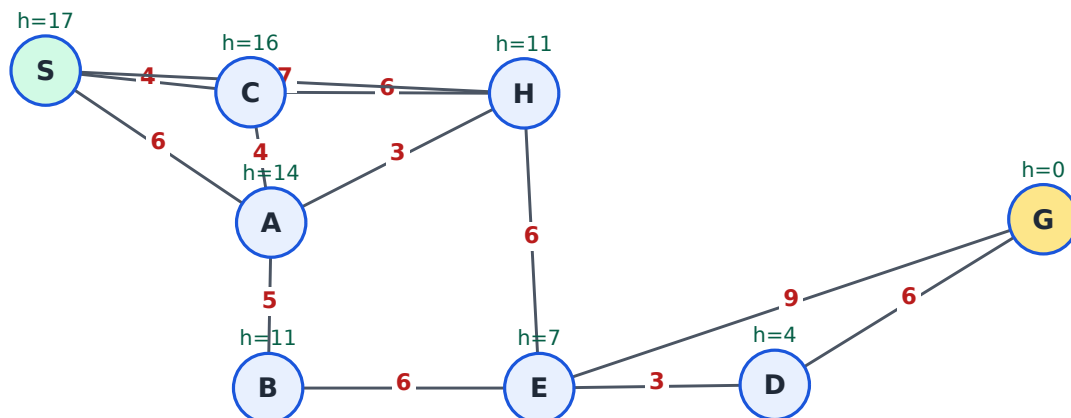
Q.12 [NAT · 2 marks · no negative marking] Uninformed Search

Depth-first tree search generates all $b = 3$ children of a node when it expands that node. The tree has depth $m = 10$, and no goal is found along the leftmost path. Consider the moment just after DFS expands the leftmost node at depth 9, so that its 3 children at depth 10 have been added to the frontier. How many nodes are in the frontier at that moment?

Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

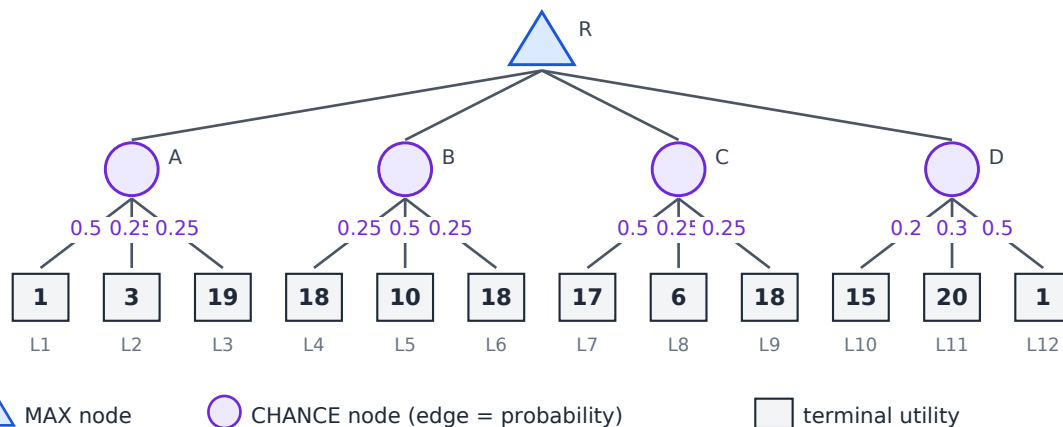
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
- ☐ (B) h is consistent (monotone).
- ☐ (C) The edge B-A violates the consistency condition (in at least one direction).
- ☐ (D) $h(H)$ overestimates the true cost from H to G.

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



Answer (numerical): _____

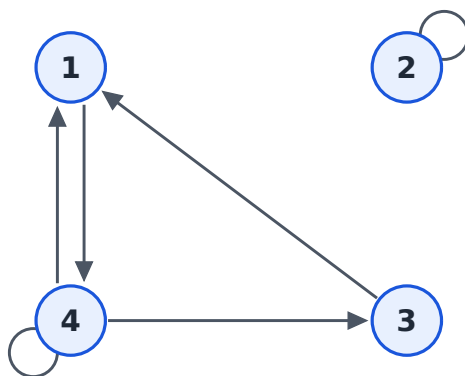
Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (Q \leftrightarrow \neg R) \leftrightarrow (\neg P \rightarrow R)$?

- ☐ (A) $\neg(Q \leftrightarrow \neg R) \leftrightarrow \neg(\neg P \rightarrow R)$
☐ (B) $((Q \leftrightarrow \neg R) \wedge (\neg P \rightarrow R)) \vee (\neg(Q \leftrightarrow \neg R) \wedge \neg(\neg P \rightarrow R))$
☐ (C) $((Q \leftrightarrow \neg R) \rightarrow (\neg P \rightarrow R)) \wedge ((\neg P \rightarrow R) \rightarrow (Q \leftrightarrow \neg R))$
☐ (D) $(Q \leftrightarrow \neg R) \leftrightarrow \neg(\neg P \rightarrow R)$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (C) $\exists x R(x, x)$
☐ (D) $\forall x \exists y R(x, y)$

Q.17 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following statements about resolution is correct?

- ☐ (A) Resolution is complete for deriving every entailed clause directly, without refutation
- ☐ (B) If a clause set is unsatisfiable, the resolution closure of the set contains the empty clause
- ☐ (C) If the resolution closure of a clause set does not contain the empty clause, the set may still be unsatisfiable
- ☐ (D) Resolution can derive the empty clause from a satisfiable set of clauses

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg R)$, C2: $(\neg Q \vee \neg R)$, C3: $(\neg P \vee \neg Q)$, C4: $(\neg Q \vee \neg S)$, C5: $(Q \vee \neg R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
- ☐ (B) $(\neg R \vee \neg S)$
- ☐ (C) $(Q \vee R)$
- ☐ (D) $(\neg P \vee \neg R)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

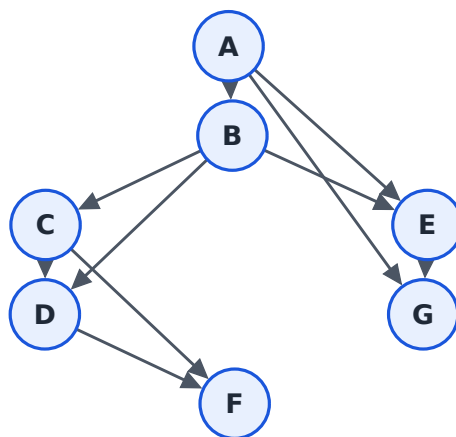
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = ((\neg R \wedge \neg P) \wedge \neg \neg P) \vee ((Q \rightarrow R) \vee S) ?$$

Answer (numerical): _____

Q.20 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp A \mid D$
- ☐ (B) $E \perp F \mid B$
- ☐ (C) $F \perp G \mid B$
- ☐ (D) $C \perp A \mid G$

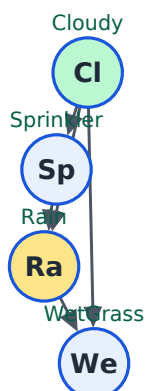
Q.21 [NAT · 2 marks · no negative marking] Probability for AI

A disease D has prior $P(d) = 0.01$. A test has $P(+ \mid d) = 0.9$ and $P(+ \mid \neg d) = 0.05$. The test is performed twice; the two results are conditionally independent given D. Both tests come out positive. Compute $P(d \mid +, +)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\text{Cloudy} = T \mid \text{Rain} = F)$ (round off to 3 decimal places).



Cl	P(Cl=T)
T	0.1

Cl	P(Sp=T Cl)	Cl	Sp	P(Ra=T ...)
T	0.5	T	T	0.3
F	0.6	T	F	0.8
		F	T	0.25
		F	F	0.6

Cl	Ra	P(We=T ...)
T	T	0.2
T	F	0.25
F	T	0.6
F	F	0.1

Answer (numerical): _____

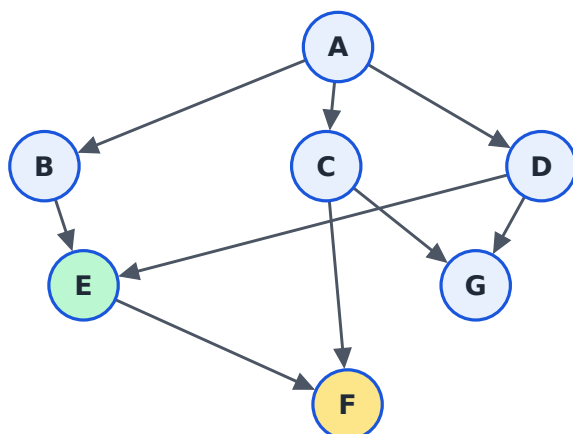
Q.23 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. Likelihood weighting with evidence $S = \text{true}$, $W = \text{true}$ produces exactly four samples of (C, R): (c, r), ($\neg c$, r), ($\neg c$, $\neg r$), (c, $\neg r$). Using the correct weights, compute the likelihood-weighting estimate of $P(r \mid s, w)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

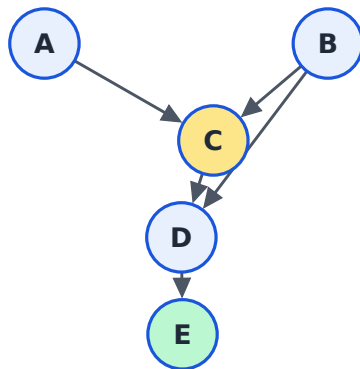
Variable elimination is used to compute $P(E \mid F = T)$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **G**, **D**, **C**, **A**, **B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{C} = \mathbf{F}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(E = T \mid C = F)}$ (round off to 3 decimal places).



P(A=T)
0.8

P(B=T)
0.95

A	B	P(C=T ...)
T	T	0.6
T	F	0.4
F	T	0.1
F	F	0.9

B	C	P(D=T D)	P(E=T ...)
T	T	0.25	0.3
T	F	0.7	0.7
F	T	0.95	
F	F	0.25	

#	A	B	D	E
1	T	T	T	T
2	T	T	T	T
3	T	T	T	T
4	T	T	F	F
5	T	T	T	T

Answer (numerical): _____

Answer Key — Mock Test 077

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	NAT	2	Adversarial Search	14.49 to 14.51
2	NAT	1	Adversarial Search	15	15	MSQ	2	Propositional Logic	A, B, C
3	NAT	1	Informed Search	18	16	MSQ	2	Predicate Logic	C, D
4	NAT	1	Uninformed Search	2220	17	MCQ	2	Propositional Logic	B
5	MSQ	1	Propositional Logic	A, B	18	MSQ	2	Propositional Logic	B, D
6	NAT	1	Predicate Logic	216	19	NAT	2	Propositional Logic	14
7	MCQ	1	Predicate Logic	A	20	MSQ	2	Conditional Independence	B, C
8	MCQ	1	Variable Elimination	A	21	NAT	2	Probability for AI	0.765 to 0.767
9	NAT	1	Exact Inference	0.268 to 0.272	22	NAT	2	Exact Inference	0.074 to 0.078
10	NAT	1	Sampling	0.2706 to 0.2709	23	NAT	2	Sampling	0.523 to 0.524
11	NAT	2	Uninformed Search	15	24	NAT	2	Exact Inference	4
12	NAT	2	Uninformed Search	21	25	NAT	2	Sampling	0.798 to 0.802
13	MSQ	2	Informed Search	A, B					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: D** (D: S, D, F, B, G)

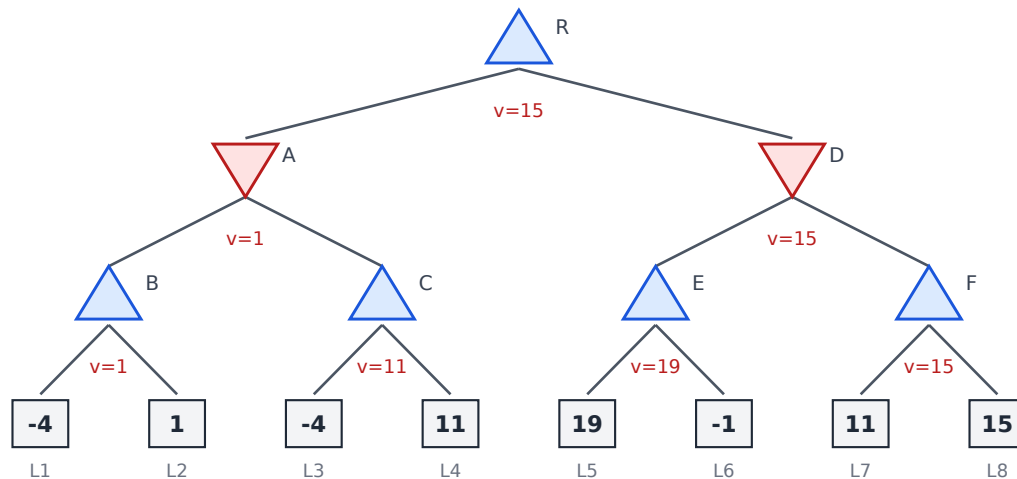
Adjacency lists (alphabetical): B: {F, G, I}; D: {F, I, J, S}; F: {B, D, G, J}; G: {B, F}; I: {B, D, J}; J: {D, F, I}; S: {D}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, F, B, G**; the path found is $S \rightarrow D \rightarrow F \rightarrow B \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.2 [NAT, 1M · Adversarial Search] Answer: 15

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 1$, $C = 11$, $A = 1$, $E = 19$, $F = 15$, $D = 15$. Root value = **15**; MAX's optimal first move is towards D.

Q.3 [NAT, 1M · Informed Search] Answer: 18

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (14)	K(8), D(10), E(12)
2	K (8)	A(6), D(10), E(12)
3	A (6)	I(3), C(5), D(10), E(12)
4	I (3)	G(0), C(5), D(10), E(12)
5	G (0)	C(5), D(10), E(12) ← goal, stop

Path returned: **S** → **K** → **A** → **I** → **G**, cost = $8 + 2 + 4 + 4 = 18$.

The optimal cost is $h^*(S) = 17$ — greedy search is not optimal here.

Q.4 [NAT, 1M · Uninformed Search] Answer: 2220

Each direction generates $b + b^2 + b^3 = 10 + 100 + 1000 = 1110$ nodes. Two directions give $2 \times 1110 = 2220$. A single BFS to depth 6 would generate about 1.11×10^6 nodes. That is the $O(b^{d/2})$ versus $O(b^d)$ advantage.

Q.5 [MSQ, 1M · Propositional Logic] Answer: A, B (A: A; B: B)

A occurs only positively (in $A \vee \neg B$ and $C \vee A$), so it is pure; DPLL can set $A = T$. B occurs only negatively ($\neg B$ twice), so it is pure; DPLL can set $B = F$. C occurs negatively in $\neg B \vee \neg C$ and positively in $C \vee A$, so it is not pure.

Q.6 [NAT, 1M · Predicate Logic] Answer: 216

View R as an 3×3 0/1 matrix. The sentence says R is antisymmetric: diagonal free (2^n); each unordered pair has 3 options (neither, one way, other way) $\Rightarrow 2^n \cdot 3^{(n(n-1)/2)}$. With $n = 3$: **216**.

Q.7 [MCQ, 1M · Predicate Logic] Answer: A (A: y and z)

x is bound by $\forall x$ throughout. The occurrence of y in $P(x, y)$ is not in the scope of any $\exists y/\forall y$, so it is free. The y in $Q(x, y, z)$ is bound by $\exists y$. z is never quantified, so it is free. Hence the variables with free occurrences are y and z.

Q.8 [MCQ, 1M · Variable Elimination] Answer: A (A: a factor over $\{A, B, C\}$ with 8 entries)

The pointwise product is defined on the union of the variables: $f(a, b, c) = f_1(a, b) \times f_2(b, c)$. The union $\{A, B, C\}$ has $2^3 = 8$ entries. $\{A, C\}$ would result only after summing out B; shared variable B appears once, not twice.

Q.9 [NAT, 1M · Exact Inference] Answer: 0.268 to 0.272

Only the ancestors of B (A) matter: every other variable sums out to 1.

$P(B=T) = \sum \text{over its ancestors of } P(B=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, B=T)$
$A=T, C=T$	0.038
$A=T, C=F$	0.152
$A=F, C=T$	0.072
$A=F, C=F$	0.008

Sum = **0.27** ≈ 0.270 .

Q.10 [NAT, 1M · Sampling] **Answer: 0.2706 to 0.2709**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(CI=F) \times P(Sp=F) \times P(Ra=T \mid CI=F, Sp=F) \times P(We=F \mid CI=F, Ra=T)$
 $= 0.5 \times 0.6 \times 0.95 \times 0.95 = \mathbf{0.27075} \approx 0.2707$.

Q.11 [NAT, 2M · Uninformed Search] **Answer: 15**

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	I(3), H(6)
2	I (3)	B(6), H(6), F(10)
3	B (6)	H(6), F(10), J(12), G(15)
4	H (6)	F(10), J(12), G(15)
5	F (10)	J(12), K(13), G(15)
6	J (12)	K(13), G(15)
7	K (13)	G(15)
8	G (15)	← goal, stop

Path returned: **S** → **I** → **B** → **G**, cost = 3 + 3 + 9 = **15**. Nodes expanded (including S and G): **8** (S, I, B, H, F, J, K, G).

Q.12 [NAT, 2M · Uninformed Search] **Answer: 21**

Along the leftmost path, the nodes at depths 0 to 9 have all been expanded. At each depth from 1 to 9, the $b - 1 = 2$ unexpanded siblings of the path node are still in the frontier: $9 \times 2 = 18$ nodes. The 3 children at depth 10 have just been added. Frontier size = $18 + 3 = \mathbf{21} = (b-1)m + 1$, which illustrates DFS's $O(bm)$ space.

Q.13 [MSQ, 2M · Informed Search] **Answer: A, B** (A: h is admissible.; B: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	C	D	E	G	H	S
$h(n)$	14	11	16	4	7	0	11	17
$h^*(n)$	18	15	21	6	9	0	15	22
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [NAT, 2M · Adversarial Search] **Answer: 14.49 to 14.51**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.5 \times 1 + 0.25 \times 3 + 0.25 \times 19 = 6$
- $B = 0.25 \times 18 + 0.5 \times 10 + 0.25 \times 18 = 14$
- $C = 0.5 \times 17 + 0.25 \times 6 + 0.25 \times 18 = 14.5$
- $D = 0.2 \times 15 + 0.3 \times 20 + 0.5 \times 1 = 9.5$

Root = $\max(6, 14, 14.5, 9.5) = \mathbf{14.5}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, B, C (A: $\neg(Q \leftrightarrow \neg R) \leftrightarrow \neg(\neg P \rightarrow R)$; B: $((Q \leftrightarrow \neg R) \wedge (\neg P \rightarrow R)) \vee (\neg(Q \leftrightarrow \neg R) \wedge \neg(\neg P \rightarrow R))$; C: $((Q \leftrightarrow \neg R) \rightarrow (\neg P \rightarrow R)) \wedge ((\neg P \rightarrow R) \rightarrow (Q \leftrightarrow \neg R))$)

Compare truth tables with $F = (Q \leftrightarrow \neg R) \leftrightarrow (\neg P \rightarrow R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(Q \leftrightarrow \neg R) \leftrightarrow \neg(\neg P \rightarrow R)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(B) $((Q \leftrightarrow \neg R) \wedge (\neg P \rightarrow R)) \vee (\neg(Q \leftrightarrow \neg R) \wedge \neg(\neg P \rightarrow R))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(C) $((Q \leftrightarrow \neg R) \rightarrow (\neg P \rightarrow R)) \wedge ((\neg P \rightarrow R) \rightarrow (Q \leftrightarrow \neg R))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(D) $(Q \leftrightarrow \neg R) \leftrightarrow \neg(\neg P \rightarrow R)$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

Q.16 [MSQ, 2M · Predicate Logic] Answer: C, D (C: $\exists x R(x, x)$; D: $\forall x \exists y R(x, y)$)

$R = \{(1,4), (2,2), (3,1), (4,1), (4,3), (4,4)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(4,3)$ holds but $R(3,4)$ does not.

(B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(1,4)$ and $R(4,1)$ with $1 \neq 4$.

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

(D) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MCQ, 2M · Propositional Logic] Answer: B (B: If a clause set is unsatisfiable, the resolution closure of the set contains the empty clause)

The ground resolution theorem: a clause set is unsatisfiable iff its resolution closure contains $\square$. So B is correct and C is false. D would contradict soundness. A is false: resolution is only refutation-complete; e.g. from $\{p\}$ it cannot derive the entailed clause $p \vee q$ directly.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $(\neg R \vee \neg S)$; D: $(\neg P \vee \neg R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(\neg R \vee \neg S)$ is derivable:

1. resolve $(\neg Q \vee \neg S)$ and $(Q \vee \neg R)$ on $Q \Rightarrow (\neg R \vee \neg S)$

(C) $(Q \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg P \vee \neg R)$ is derivable:

1. resolve $(\neg P \vee \neg Q)$ and $(Q \vee \neg R)$ on $Q \Rightarrow (\neg P \vee \neg R)$

Q.19 [NAT, 2M · Propositional Logic] Answer: 14

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(\neg R \wedge \neg P) \wedge \neg \neg P$	$(Q \rightarrow R) \vee S$	$((\neg R \wedge \neg P) \wedge \neg \neg P) \vee ((Q \rightarrow R) \vee S)$
T	T	T	T	F	T	T
T	T	T	F	F	T	T
T	T	F	T	F	T	T
T	T	F	F	F	F	F
T	F	T	T	F	T	T
T	F	T	F	F	T	T
T	F	F	T	F	T	T
T	F	F	F	F	T	T
F	T	T	T	F	T	T
F	T	T	F	F	T	T
F	T	F	T	F	T	T
F	T	F	F	F	F	F
F	F	T	T	F	T	T
F	F	T	F	F	T	T
F	F	F	T	F	T	T
F	F	F	F	F	T	T

Rows where $F = T$: **14**.

Q.20 [MSQ, 2M · Conditional Independence] Answer: B, C ($B: E \perp F \mid B$; $C: F \perp G \mid B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp A \mid D$: **FALSE** — the path $C \leftarrow B \leftarrow A$ is active given $\{D\}$.

(B) $E \perp F \mid B$: **TRUE** — $E \leftarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow F$ (blocked: observed chain node B); $E \leftarrow A \rightarrow B \rightarrow C \rightarrow F$ (blocked: observed chain node B); $E \leftarrow A \rightarrow B \rightarrow D \leftarrow C \rightarrow F$ (blocked: observed chain node B); $E \leftarrow A \rightarrow B \rightarrow D \rightarrow F$ (blocked: observed chain node B); $E \leftarrow B \rightarrow C \rightarrow D \rightarrow F$ (blocked: observed fork node B); $E \leftarrow B \rightarrow C \rightarrow F$ (blocked: observed fork node B) (+6 more paths, all blocked).

(C) $F \perp G \mid B$: **TRUE** — $F \leftarrow C \leftarrow B \leftarrow A \rightarrow E \rightarrow G$ (blocked: observed chain node B); $F \leftarrow C \leftarrow B \leftarrow A \rightarrow G$ (blocked: observed chain node B); $F \leftarrow C \leftarrow B \rightarrow E \leftarrow A \rightarrow G$ (blocked: observed fork node B); $F \leftarrow C \leftarrow B \rightarrow E \rightarrow G$ (blocked: observed fork node B); $F \leftarrow C \rightarrow D \leftarrow B \leftarrow A \rightarrow E \rightarrow G$ (blocked: collider D with neither it nor a descendant observed); $F \leftarrow C \rightarrow D \leftarrow B \leftarrow A \rightarrow G$ (blocked: collider D with neither it nor a descendant observed) (+10 more paths, all blocked).

(D) $C \perp A \mid G$: **FALSE** — the path $C \leftarrow B \leftarrow A$ is active given $\{G\}$.

Q.21 [NAT, 2M · Probability for AI] Answer: 0.765 to 0.767

By Bayes rule with conditional independence, $P(d \mid +, +) = \alpha P(+ \mid d)^2 P(d)$. Unnormalized for d: $0.9^2 \times 0.01 = 0.81 \times 0.01 = 0.0081$. For $\neg d$: $0.05^2 \times 0.99 = 0.0025 \times 0.99 = 0.002475$. Normalizing: $0.0081 / (0.0081 + 0.002475) = 0.0081 / 0.010575 \approx 0.766$. (For comparison, a single positive test gives only $0.009/0.0585 \approx 0.154$.)

Q.22 [NAT, 2M · Exact Inference] Answer: 0.074 to 0.078

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Cloudy}, \text{Rain})$.

Inference by enumeration: $P(\text{Cloudy} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cloudy}, e, \text{hidden})$.

Hidden: Sp, We	Cl	product of CPT entries ($Cl \cdot Sp \cdot Ra \cdot We$)	value
T, T	T	$0.1 \times 0.5 \times 0.7 \times 0.25$	0.00875
T, F	T	$0.1 \times 0.5 \times 0.7 \times 0.75$	0.02625
F, T	T	$0.1 \times 0.5 \times 0.2 \times 0.25$	0.0025
F, F	T	$0.1 \times 0.5 \times 0.2 \times 0.75$	0.0075
T, T	F	$0.9 \times 0.6 \times 0.75 \times 0.1$	0.0405
T, F	F	$0.9 \times 0.6 \times 0.75 \times 0.9$	0.3645
F, T	F	$0.9 \times 0.4 \times 0.4 \times 0.1$	0.0144
F, F	F	$0.9 \times 0.4 \times 0.4 \times 0.9$	0.1296

$P(\text{Cloudy}=T, e) = 0.045$; $P(\text{Cloudy}=F, e) = 0.549$; $P(e) = 0.594$.

$P(\text{Cloudy}=T \mid e) = 0.045 / 0.594 = \mathbf{0.0758} \approx 0.076$.

Q.23 [NAT, 2M · Sampling] Answer: 0.523 to 0.524

Weights $w = P(s \mid C) \times P(w \mid s, R)$: (c, r) : $0.1 \times 0.99 = 0.099$; $(\neg c, r)$: $0.5 \times 0.99 = 0.495$; $(\neg c, \neg r)$: $0.5 \times 0.90 = 0.45$; $(c, \neg r)$: $0.1 \times 0.90 = 0.09$. Total weight = $0.099 + 0.495 + 0.45 + 0.09 = 1.134$; weight with $r = 0.099 + 0.495 = 0.594$. Estimate = $0.594/1.134 \approx 0.5238 \approx 0.524$. (The exact $P(r \mid s, w) \approx 0.320$; with only four samples the estimate is far off, but LW is consistent as $N \rightarrow \infty$.)

Q.24 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, C)$, $f(A, D)$, $f(B, D, E)$, $f(C, E)$, $f(C, D, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(C, D, G)$	$\{C, D, G\}$	$f(C, D)$
D	$f(A, D)$, $f(B, D, E)$, $f(C, D)$	$\{A, B, C, D, E\}$	$f(A, B, C, E)$
C	$f(A, C)$, $f(C, E)$, $f(A, B, C, E)$	$\{A, B, C, E\}$	$f(A, B, E)$
A	$f(A)$, $f(A, B)$, $f(A, B, E)$	$\{A, B, E\}$	$f(B, E)$
B	$f(B, E)$	$\{B, E\}$	$f(E)$

The largest factor produced has **4** variable(s) (created when eliminating D); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over E and normalise.

Q.25 [NAT, 2M · Sampling] **Answer: 0.798 to 0.802**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	E
1	$P(C=F A=T,B=T)=0.4$	0.4	T
2	$P(C=F A=T,B=T)=0.4$	0.4	T
3	$P(C=F A=T,B=T)=0.4$	0.4	T
4	$P(C=F A=T,B=T)=0.4$	0.4	F
5	$P(C=F A=T,B=T)=0.4$	0.4	T

$\sum w (\text{all}) = 2$; $\sum w (E = T) = 1.6$.

Estimate $P(E=T \mid C = F) = 1.6 / 2 = \mathbf{0.8}$.

Mock Test 078

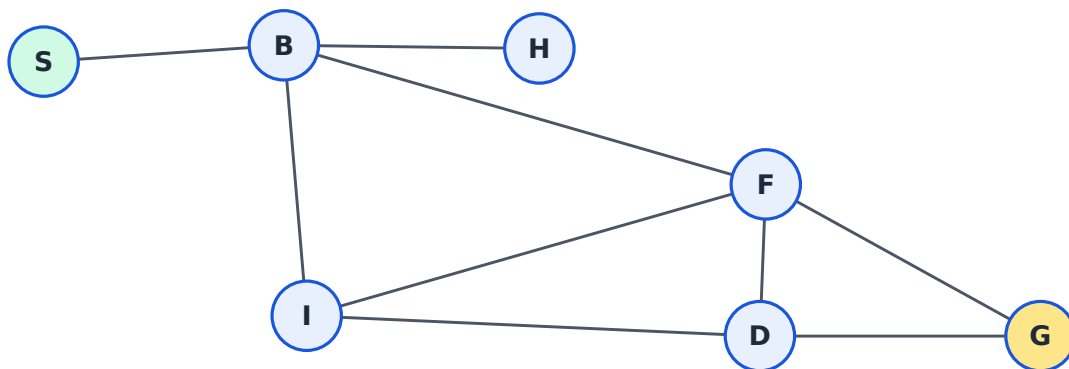
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 11 / 8	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, F, D, G
- ☐ (B) S, F, B, H, I, D, G
- ☐ (C) S, B, I, H, F, D, G
- ☐ (D) S, B, F, H, I, D, G

Q.2 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Iterative deepening search is applied to a tree with branching factor $b = 4$ whose shallowest goal is at depth $d = 5$ (goal is the last node at that depth). Using the standard count $N(IDS) = d \cdot b + (d-1) \cdot b^2 + \dots + 1 \cdot b^d$ (root not counted), how many nodes are generated?

Answer (numerical): _____

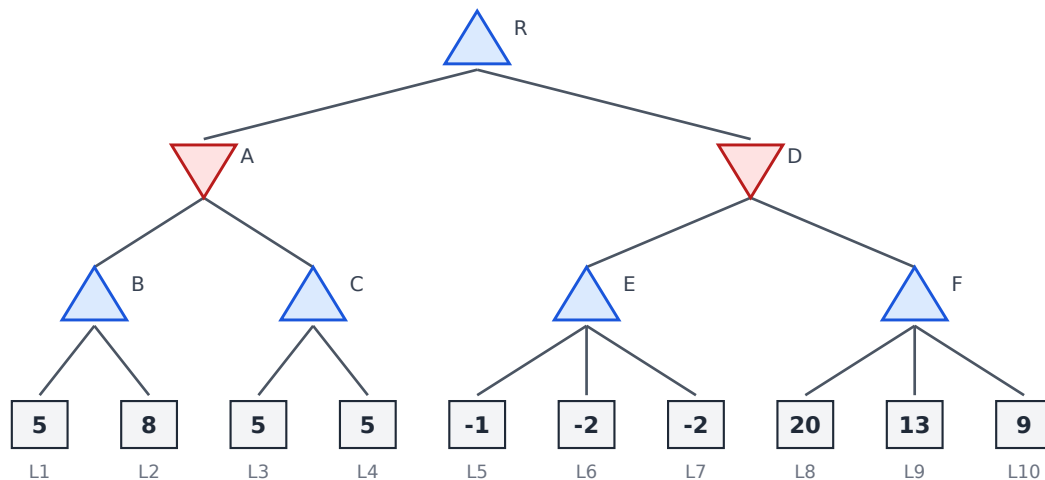
Q.3 [MSQ · 1 mark · no negative marking] Local Search

Steepest-ascent hill climbing (maximisation, moves only to a strictly better neighbour) can halt without reaching a global maximum because of which of the following?

- ☐ (A) A local maximum
- ☐ (B) A plateau or shoulder (a flat region where all neighbours have equal value)
- ☐ (C) A ridge (a sequence of local maxima that single moves cannot climb)
- ☐ (D) A state where the neighbour with the highest value is better than the current state

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to A; root value 5
- ☐ (B) Move to A; root value 20
- ☐ (C) Move to D; root value -1
- ☐ (D) Move to A; root value 7

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x R(x, x) \wedge \forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following is the correct first-order translation of 'Every student likes some course'?

- ☐ (A) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
- ☐ (B) $\forall x (\text{Student}(x) \wedge \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$
- ☐ (C) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \rightarrow \text{Likes}(x, y)))$
- ☐ (D) $\exists y \forall x (\text{Student}(x) \wedge \text{Course}(y) \wedge \text{Likes}(x, y))$

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$ and let f be a unary function symbol. How many interpretations of f satisfy $\forall x f(f(x)) = x$?

Answer (numerical): _____

Q.8 [MSQ · 1 mark · no negative marking] Conditional Independence

Which of the following hold for **every** probability distribution over variables A, B, C, D ?

- ☐ (A) $A \perp B \mid C$ implies $B \perp A \mid C$
- ☐ (B) $A \perp \{B, C\} \mid D$ implies $A \perp B \mid D$
- ☐ (C) $A \perp B$ implies $A \perp B \mid C$
- ☐ (D) $A \perp B \mid C$ implies $A \perp B$

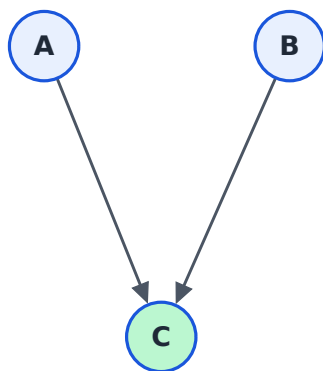
Q.9 [MSQ · 1 mark · no negative marking] Probability for AI

A and B are independent Boolean variables with $P(a) = 0.4$ and $P(b) = 0.5$. Which statements are true?

- ☐ (A) $P(a \vee b) = 0.7$
☐ (B) $P(a | b) = 0.4$
☐ (C) $P(\neg a \wedge \neg b) = 0.3$
☐ (D) $P(a \wedge \neg b) = 0.25$

Q.10 [NAT · 1 mark · no negative marking] Exact Inference - Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T | \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).

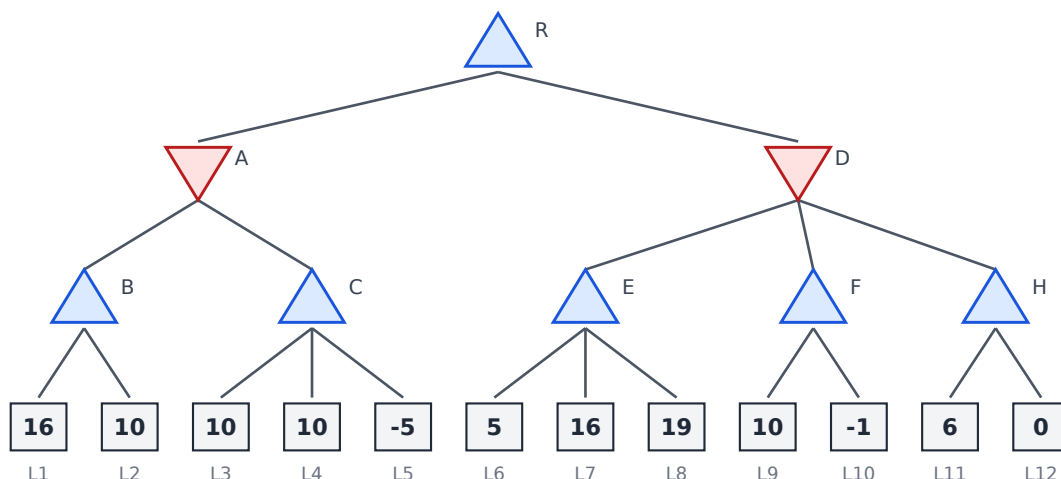


$P(A=T)$	$P(B=T)$	A	B	$P(C=T \dots)$
0.7	0.95	T	T	0.8
		T	F	0.8
		F	T	0.25
		F	F	0.25

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Adversarial Search - Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L6 is pruned.
☐ (B) Leaf L10 is pruned.
☐ (C) Leaf L11 is pruned.
☐ (D) Leaf L12 is pruned.

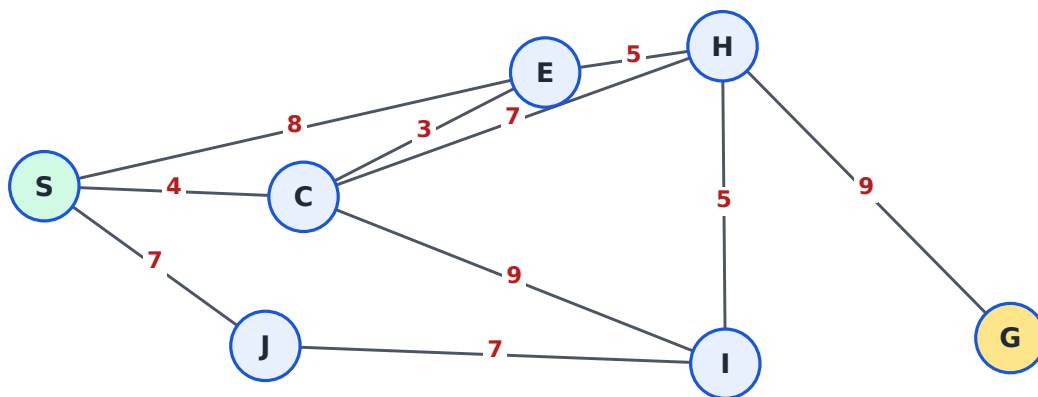
Q.12 [MSQ · 2 marks · no negative marking] Uninformed Search

Uniform-cost search runs with every step cost at least some $\epsilon > 0$. Which of the following are TRUE?

- ☐ (A) UCS is optimal
- ☐ (B) If every step cost equals the same constant, UCS behaves essentially like BFS (expanding level by level), but because it goal-tests on expansion it may generate a whole extra level of nodes
- ☐ (C) UCS is complete when the branching factor is finite
- ☐ (D) UCS remains complete even when the graph contains a cycle of zero-cost actions

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

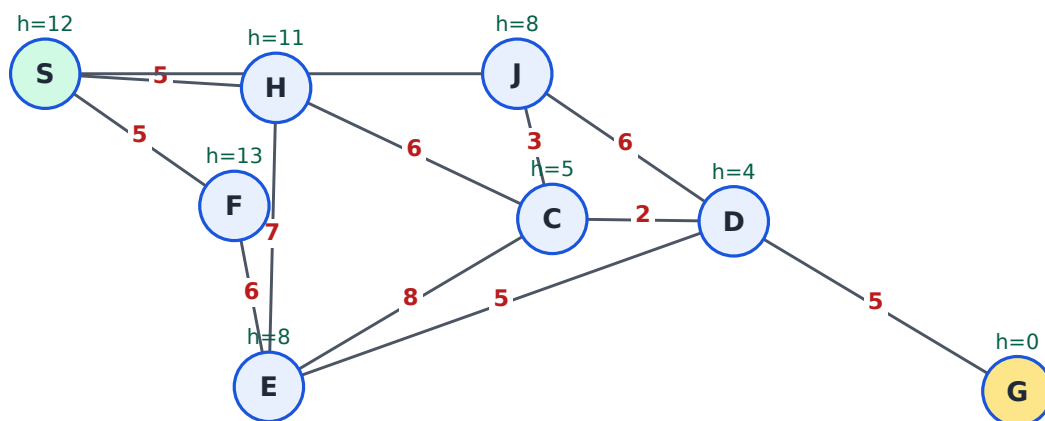
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(H)$ overestimates the true cost from H to G.
- ☐ (B) h is consistent (monotone).
- ☐ (C) The edge H-E violates the consistency condition (in at least one direction).
- ☐ (D) h is admissible.

Q.15 [MCQ · 2 marks · negative: -2/3] Informed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). The heuristic is $h(S)=7$, $h(A)=6$, $h(B)=2$, $h(C)=1$, $h(G)=0$. A* **graph search** is run from S. It keeps an explored set, never re-opens a state that has already been expanded, and goal-tests on expansion. What is the cost of the solution it returns?

- ☐ (A) 7
☐ (B) 8
☐ (C) 9
☐ (D) 10

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg R \vee S)$, C2: $(\neg P)$, C3: $(R \vee \neg S)$, C4: $(\neg P \vee S)$, C5: $(\neg Q \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) $(P \vee R)$
☐ (C) $(\neg P \vee R)$
☐ (D) $(P \vee S)$

Q.17 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ are partial orders, i.e. satisfy reflexivity $\forall x R(x, x)$, antisymmetry $\forall x \forall y (R(x, y) \wedge R(y, x) \rightarrow x = y)$, and transitivity?

Answer (numerical): _____

Q.18 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$F = (S \wedge (\neg P \vee R)) \vee ((P \wedge P) \wedge (R \leftrightarrow Q))$?

Answer (numerical): _____

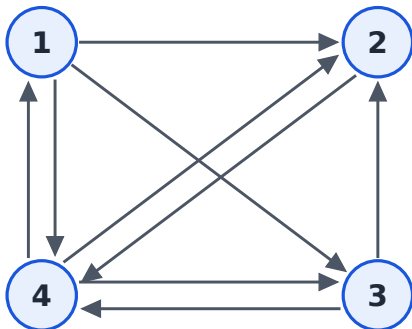
Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $\neg(P \wedge R) \leftrightarrow (\neg P \wedge \neg R)$
☐ (B) $(R \rightarrow P) \rightarrow ((R \vee Q) \rightarrow P)$
☐ (C) $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$
☐ (D) $((Q \rightarrow P) \wedge \neg Q) \rightarrow \neg P$

Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

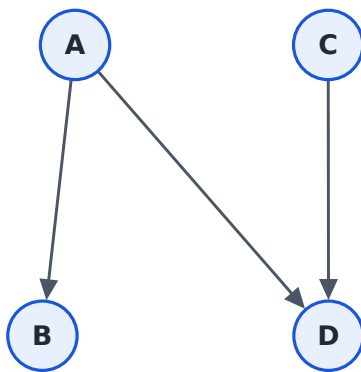
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (D) $\exists x \forall y R(x, y)$

Q.21 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.84, 0.76, 0.35, 0.58)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.2

A	$P(B=T \dots)$	$P(C=T)$
T	0.4	0.1
F	0.3	

A	C	$P(D=T \dots)$
T	T	0.3
T	F	0.8
F	T	0.9
F	F	0.6

- ☐ (A) $A=F, B=T, C=F, D=T$
☐ (B) $A=T, B=F, C=F, D=T$
☐ (C) $A=F, B=F, C=F, D=T$
☐ (D) $A=T, B=T, C=T, D=T$

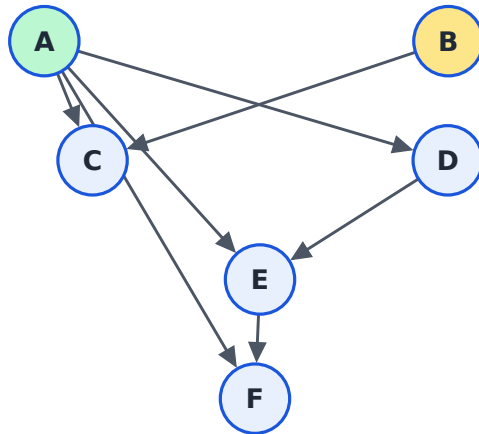
Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence

Consider the Bayesian network (DAG) with edges $P \rightarrow Q, P \rightarrow R, Q \rightarrow S, R \rightarrow S, S \rightarrow T, U \rightarrow R$. Which of the following independences are guaranteed by d-separation (i.e., hold in every distribution that factorizes over this DAG)?

- ☐ (A) $Q \perp R \mid \{P\}$
☐ (B) $Q \perp R \mid \{P, T\}$
☐ (C) $U \perp P$
☐ (D) $U \perp Q \mid \{P, S\}$

Q.23 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

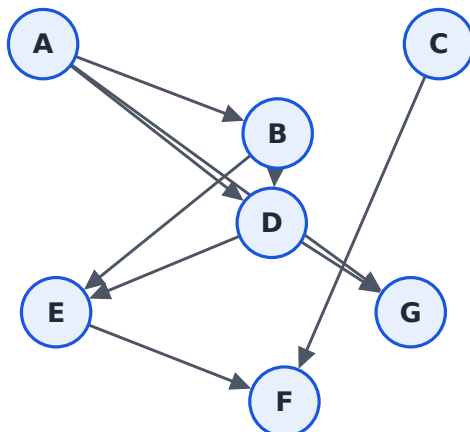
Variable elimination is used to compute $P(A \mid B = T)$ in the Bayesian network below (Boolean variables; evidence $B = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F**, **D**, **C**, **E**. What is the scope of the new factor produced when **F** is summed out?



- ☐ (A) $f(A, C, D, E, F)$
☐ (B) $f(A, E, F)$
☐ (C) $f(A, E)$
☐ (D) $f(E)$

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp D \mid G$
☐ (B) $F \perp A \mid E$
☐ (C) $B \perp A \mid D, E$
☐ (D) $E \perp D \mid G$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D, E. Using **rejection sampling**, estimate $P(E = T \mid B = F)$ (round off to 2 decimal places).

#	A	B	C	D	E
1	F	F	F	T	T
2	F	T	F	T	F
3	F	F	T	F	T
4	T	F	F	T	T
5	T	F	F	F	T
6	F	T	F	T	F
7	T	F	F	F	T
8	T	F	F	F	T
9	F	F	T	F	T
10	F	F	T	F	F
11	T	F	F	T	T
12	T	F	T	T	T
13	F	T	F	T	T
14	F	F	F	T	T
15	T	F	F	F	T
16	T	T	T	T	T
17	T	F	F	T	T
18	T	T	F	T	F
19	T	T	F	F	T
20	T	T	F	F	T

Answer (numerical): _____

Answer Key — Mock Test 078

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	MSQ	2	Informed Search	B, D
2	NAT	1	Uninformed Search	1812	15	MCQ	2	Informed Search	B
3	MSQ	1	Local Search	A, B, C	16	MSQ	2	Propositional Logic	C
4	MCQ	1	Adversarial Search	A	17	NAT	2	Predicate Logic	19
5	NAT	1	Predicate Logic	8	18	NAT	2	Propositional Logic	9
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Propositional Logic	C
7	NAT	1	Predicate Logic	4	20	MSQ	2	Predicate Logic	A, B
8	MSQ	1	Conditional Independence	A, B	21	MCQ	2	Sampling	C
9	MSQ	1	Probability for AI	A, B, C	22	MSQ	2	Conditional Independence	A, C
10	NAT	1	Exact Inference	0.633 to 0.637	23	MCQ	2	Exact Inference	C
11	MSQ	2	Adversarial Search	C, D	24	MSQ	2	Conditional Independence	B
12	MSQ	2	Uninformed Search	A, B, C	25	NAT	2	Sampling	0.91 to 0.93
13	NAT	2	Uninformed Search	7					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: D** (D: S, B, F, H, I, D, G)

Adjacency lists (alphabetical): B: {F, H, I, S}; D: {F, G, I}; F: {B, D, G, I}; G: {D, F}; H: {B}; I: {B, D, F}; S: {B}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B]

Step	Expanded	Frontier (FIFO) after expansion
2	B	[F, H, I]
3	F	[H, I, D, G]
4	H	[I, D, G]
5	I	[D, G]
6	D	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, B, F, H, I, D, G**. The path returned is $S \rightarrow B \rightarrow F \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Uninformed Search] Answer: 1812

Nodes at depth i are regenerated in every iteration with limit $\geq i$, i.e. $(d + 1 - i)$ times.

$$N = 5 \times 4 + 4 \times 16 + 3 \times 64 + 2 \times 256 + 1 \times 1024 = \mathbf{1812}.$$

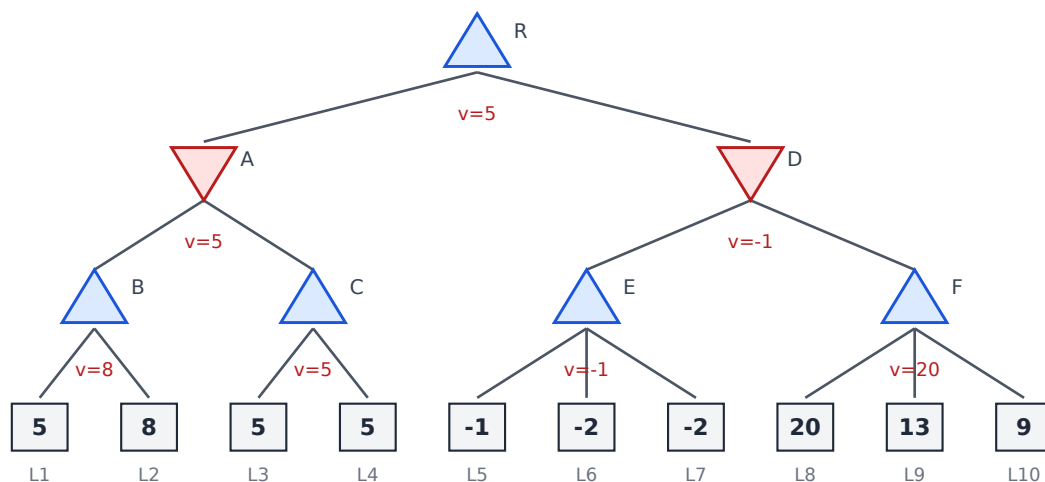
Compare BFS: 1364; the overhead of IDS is small for $b \geq 2$.

Q.3 [MSQ, 1M · Local Search] Answer: A, B, C (A: A local maximum; B: A plateau or shoulder (a flat region where all neighbours have equal value); C: A ridge (a sequence of local maxima that single moves cannot climb))

(A), (B) and (C) are the three classic traps listed in AIMA. At a local maximum no neighbour is better. On a plateau no neighbour is strictly better. On a ridge every single-move neighbour is worse even though the ridge rises. (D) is false: if the best neighbour is better, hill climbing moves there and does not halt.

Q.4 [MCQ, 1M · Adversarial Search] Answer: A (A: Move to A; root value 5)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 8$, $C = 5$, $A = 5$, $E = -1$, $F = 20$, $D = -1$. Root value = **5**; MAX's optimal first move is towards A.

Q.5 [NAT, 1M · Predicate Logic] Answer: 8

View R as an 3×3 0/1 matrix. The sentence says R is reflexive and symmetric: only the $n(n-1)/2$ unordered off-diagonal pairs are free $\Rightarrow 2^{n(n-1)/2}$. With $n = 3$: **8**.

Q.6 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Likes}(x, y)))$)

Universal quantifiers pair naturally with $\rightarrow$, existential with $\wedge$. Option B claims everything in the domain is a student. Option C is satisfied by any non-course y (making $\text{Course}(y) \rightarrow \dots$ vacuously true), so it does not require liking a course. Option D says there is a single y that is a course liked by everything, and everything is a student — far too strong.

Q.7 [NAT, 1M · Predicate Logic] Answer: 4

$f(f(x)) = x$ for all x means f is an involution: it is a bijection whose cycles have length 1 or 2. On 3 elements: the identity (1 way) or exactly one transposition with one fixed point (3 ways). Total = 4.

Q.8 [MSQ, 1M · Conditional Independence] Answer: A, B (A: $A \perp B \mid C$ implies $B \perp A \mid C$; B: $A \perp \{B, C\} \mid D$ implies $A \perp B \mid D$)

(A) Symmetry: $P(A, B \mid C) = P(A \mid C)P(B \mid C)$ is symmetric in A and B. (B) Decomposition: summing $P(A, B, C \mid D) = P(A \mid D)P(B, C \mid D)$ over C gives $P(A, B \mid D) = P(A \mid D)P(B \mid D)$. (C) is false: in a v-structure $A \rightarrow C \leftarrow B$, $A \perp B$ but conditioning on C makes them dependent (explaining away). (D) is false: in a fork $A \leftarrow C \rightarrow B$, $A \perp B \mid C$ but A and B are generally dependent.

Q.9 [MSQ, 1M · Probability for AI] Answer: A, B, C (A: $P(a \vee b) = 0.7$; B: $P(a \mid b) = 0.4$; C: $P(\neg a \wedge \neg b) = 0.3$)

Independence gives $P(a \wedge b) = 0.4 \times 0.5 = 0.2$, so $P(a \vee b) = 0.4 + 0.5 - 0.2 = 0.7$ (A true). Independence means $P(a \mid b) = P(a) = 0.4$ (B true). $P(\neg a \wedge \neg b) = 0.6 \times 0.5 = 0.3$ (C true). $P(a \wedge \neg b) = 0.4 \times 0.5 = 0.2$, not 0.25 (D false).

Q.10 [NAT, 1M · Exact Inference] Answer: 0.633 to 0.637

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

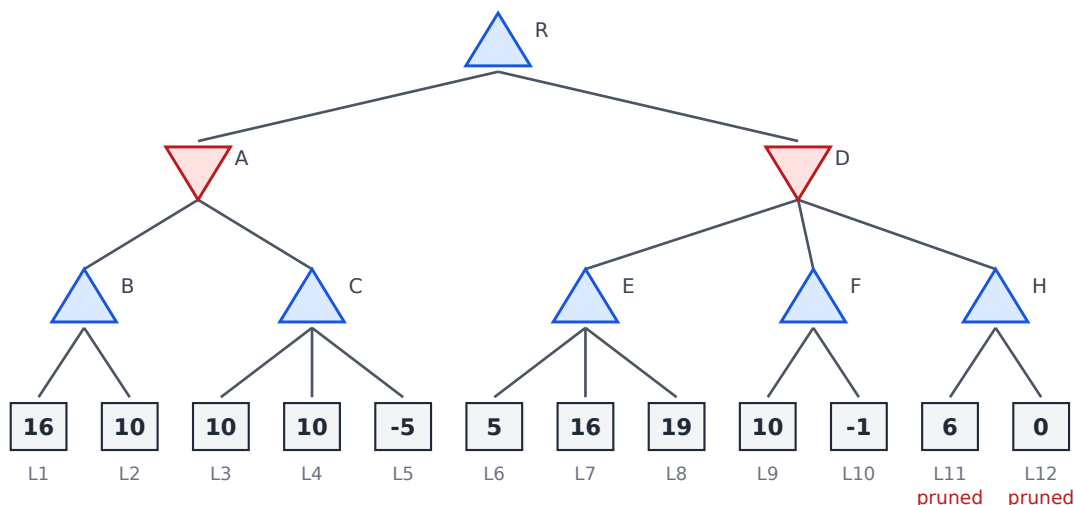
Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T	0.532
A=T, B=F	0.028
A=F, B=T	0.07125
A=F, B=F	0.00375

Sum = **0.635** ≈ 0.635 .

Q.11 [MSQ, 2M · Adversarial Search] Answer: C, D (C: Leaf L11 is pruned.; D: Leaf L12 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $10 \leq \alpha = 10 \Rightarrow$ prune remaining children (leaves L11, L12).



Leaves evaluated (10): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10. Leaves pruned (2): L11, L12. Root value = 10 (same as plain minimax, 10 — pruning never changes the root value).

Q.12 [MSQ, 2M · Uninformed Search] Answer: A, B, C

(A) is true: UCS expands nodes in order of g, so the first goal it selects is the cheapest. (B) is true: equal costs make g proportional to depth, but the goal test at expansion means the children at depth d+1 may already have been generated. (C) is true: with costs $\geq \epsilon$ and finite b, only finitely many nodes have $g \leq C^*$. (D) is false: a zero-cost cycle lets g stay constant while the path grows forever, so UCS may never terminate. This is why the $\epsilon > 0$ condition is needed.

Q.13 [NAT, 2M · Uninformed Search] Answer: 7

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	C(4), J(7), E(8)
2	C (4)	E(7), J(7), H(11), I(13)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
3	E (7)	J(7), H(11), I(13)
4	J (7)	H(11), I(13)
5	H (11)	I(13), G(20)
6	I (13)	G(20)
7	G (20)	← goal, stop

Path returned: **S** → **C** → **H** → **G**, cost = 4 + 7 + 9 = **20**. Nodes expanded (including S and G): **7** (S, C, E, J, H, I, G).

Q.14 [MSQ, 2M · Informed Search] **Answer: B, D** (B: h is consistent (monotone).; D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	D	E	F	G	H	J	S
$h(n)$	5	4	8	13	0	11	8	12
$h^*(n)$	7	5	10	16	0	13	10	15
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.15 [MCQ, 2M · Informed Search] **Answer: B** (B: 8)

Expand S: push A ($g=1, f=7$) and B ($g=4, f=6$). Expand B ($f=6$): push C ($g=5, f=6$) and G ($g=10, f=10$).

Expand C ($f=6$): push G ($g=8, f=8$). Expand A ($f=7$): its successors B ($g=3$) and C ($g=6$) are already explored and are discarded. Expand G ($f=8$) and return cost **8** ($S \rightarrow B \rightarrow C \rightarrow G$). The optimal cost is 7 ($S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$).

Optimality fails because h is admissible but not consistent ($h(A)=6 > c(A,B)+h(B)=4$), and the cheaper path to B is discovered only after B is closed.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: C** (C: $(\neg P \vee R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $(\neg P \vee R)$ is derivable:

1. resolve $(R \vee \neg S)$ and $(\neg P \vee S)$ on S $\Rightarrow (\neg P \vee R)$

(D) $(P \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.17 [NAT, 2M · Predicate Logic] **Answer: 19**

Count labelled posets on 3 elements by shape: the antichain (1); one comparable pair plus an isolated point ($3 \times 2 = 6$); a chain ($3! = 6$); one element below two incomparable ones (3); one element above two incomparable ones (3). Total = 1 + 6 + 6 + 3 + 3 = 19, which matches brute-force enumeration of all 512 relations.

Q.18 [NAT, 2M · Propositional Logic] **Answer: 9**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$S \wedge (\neg P \vee R)$	$(P \wedge P) \wedge (R \leftrightarrow Q)$	$(S \wedge (\neg P \vee R)) \vee ((P \wedge P) \wedge (R \leftrightarrow Q))$
T	T	T	T	T	T	T
T	T	T	F	F	T	T
T	T	F	T	F	F	F
T	T	F	F	F	F	F
T	F	T	T	T	F	T
T	F	T	F	F	F	F
T	F	F	T	F	T	T
T	F	F	F	F	T	T
F	T	T	T	T	F	T
F	T	T	F	F	F	F

P	Q	R	S	$S \wedge (\neg P \vee R)$	$(P \wedge P) \wedge (R \leftrightarrow Q)$	$(S \wedge (\neg P \vee R)) \vee ((P \wedge P) \wedge (R \leftrightarrow Q))$
F	T	F	T	T	F	T
F	T	F	F	F	F	F
F	F	T	T	T	F	T
F	F	T	F	F	F	F
F	F	F	T	T	F	T
F	F	F	F	F	F	F

Rows where F = T: **9**.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: C** ($C: ((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$)

(A) $\neg(P \wedge R) \leftrightarrow (\neg P \wedge \neg R)$: **not valid** — counter-model $P=T, R=F$.

(B) $(R \rightarrow P) \rightarrow ((R \vee Q) \rightarrow P)$: **not valid** — counter-model $P=F, Q=T, R=F$.

(C) $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow P) \wedge \neg Q) \rightarrow \neg P$: **not valid** — counter-model $P=T, Q=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.20 [MSQ, 2M · Predicate Logic] **Answer: A, B** ($A: \forall x \exists y (x \neq y \wedge R(x, y))$; $B: \forall y \exists x R(x, y)$)

$R = \{(1,2), (1,3), (1,4), (2,4), (3,2), (3,4), (4,1), (4,2), (4,3)\}$.

(A) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(B) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(1,2)$ holds but $R(2,1)$ does not.

(D) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.21 [MCQ, 2M · Sampling] **Answer: C** ($C: A=F, B=F, C=F, D=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.2$; $u = 0.84 \geq 0.2 \Rightarrow A = F$
- B: $P(B=T \mid A=F) = 0.3$; $u = 0.76 \geq 0.3 \Rightarrow B = F$
- C: $P(C=T \mid \text{—}) = 0.1$; $u = 0.35 \geq 0.1 \Rightarrow C = F$
- D: $P(D=T \mid A=F, C=F) = 0.6$; $u = 0.58 < 0.6 \Rightarrow D = T$

Sample: **(A=F, B=F, C=F, D=T)**.

Q.22 [MSQ, 2M · Conditional Independence] **Answer: A, C** ($A: Q \perp R \mid \{P\}$; $C: U \perp P$)

Paths between Q and R: $Q \leftarrow P \rightarrow R$ (fork at P) and $Q \rightarrow S \leftarrow R$ (collider S). (A) Given P, the fork is blocked and the collider S has no observed descendant: d-separated, true. (B) T is a descendant of S, so observing T activates $Q \rightarrow S \leftarrow R$: false. (C) Paths from U to P all pass through R as a collider ($U \rightarrow R \leftarrow P$, or $U \rightarrow R \rightarrow S \leftarrow P$ with collider S); with nothing observed they are blocked: true. (D) Given S, the path $U \rightarrow R \rightarrow S \leftarrow Q$ has observed collider S and unobserved non-collider R, so it is active: false.

Q.23 [MCQ, 2M · Exact Inference] **Answer: C** ($C: f(A, E)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(), f(A, C), f(A, D), f(A, D, E), f(A, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(A, E, F)$	$\{A, E, F\}$	$f(A, E)$
D	$f(A, D), f(A, D, E)$	$\{A, D, E\}$	$f(A, E)$
C	$f(A, C)$	$\{A, C\}$	$f(A)$
E	$f(A, E), f(A, E)$	$\{A, E\}$	$f(A)$

The largest factor produced has **2** variable(s) (created when eliminating F); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: B** ($B: F \perp A \mid E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp D \mid G$: **FALSE** — the path $A \rightarrow B \rightarrow D$ is active given $\{G\}$.

(B) $F \perp A \mid E$: **TRUE** — $F \leftarrow E \leftarrow B \leftarrow A$ (blocked: observed chain node E); $F \leftarrow E \leftarrow B \rightarrow D \leftarrow A$ (blocked: observed chain node E); $F \leftarrow E \leftarrow B \rightarrow D \rightarrow G \leftarrow A$ (blocked: observed chain node E); $F \leftarrow E \leftarrow D \leftarrow A$ (blocked: observed chain node E); $F \leftarrow E \leftarrow D \leftarrow B \leftarrow A$ (blocked: observed chain node E); $F \leftarrow E \leftarrow D \rightarrow G \leftarrow A$ (blocked: observed chain node E).

(C) $B \perp A \mid D, E$: **FALSE** — the path $B \leftarrow A$ is active given $\{D, E\}$.

(D) $E \perp D \mid G$: **FALSE** — the path $E \leftarrow B \leftarrow A \rightarrow D$ is active given $\{G\}$.

Q.25 [NAT, 2M · Sampling] **Answer: 0.91 to 0.93**

Rejection sampling keeps only samples consistent with the evidence $B = F$: samples 1, 3, 4, 5, 7, 8, 9, 10, 11, 12, 14, 15, 17 (13 accepted, 7 rejected).

Among the accepted samples, $E = T$ in 12. Estimate = $12/13 = \mathbf{0.9231}$.

(For this network the exact value is 0.864; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Mock Test 079

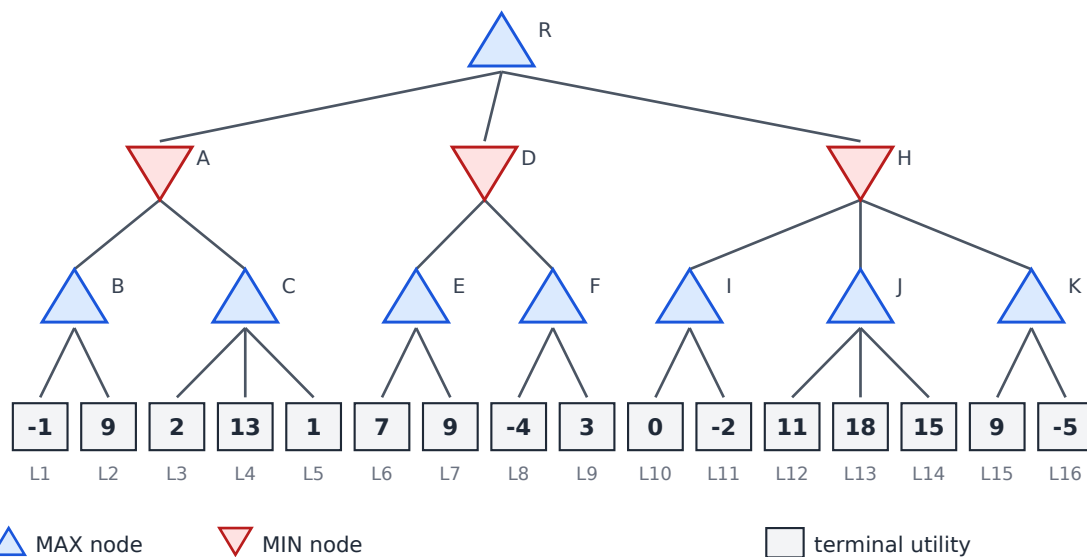
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 6 / 14	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

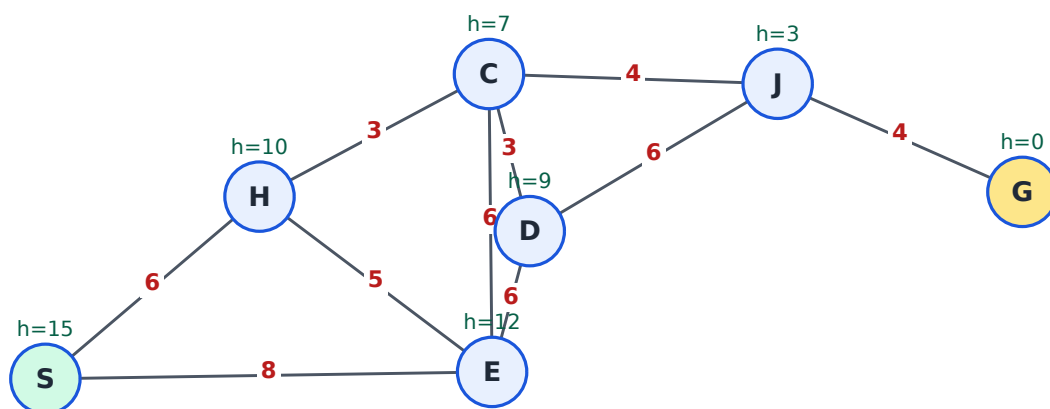
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

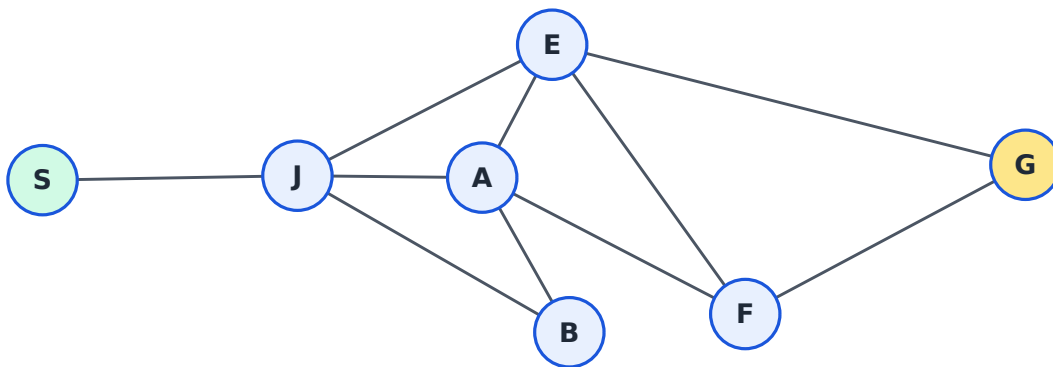
Q.3 [MCQ · 1 mark · negative: -1/3] Adversarial Search

In alpha-beta search, α denotes

- ☐ (A) the value of the best choice (highest value) found so far for MAX along the current path
- ☐ (B) the value of the best choice (lowest value) found so far for MIN along the current path
- ☐ (C) the depth limit
- ☐ (D) the number of pruned nodes

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, J, A, B, E, F, G
- ☐ (B) S, J, B, A, E, F, G
- ☐ (C) S, A, B, E, F, G
- ☐ (D) S, J, E, B, A, G

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In the formula $\forall x (P(x, y) \rightarrow \exists y Q(x, y, z))$, which variables have free occurrences?

- ☐ (A) y and z
- ☐ (B) z only
- ☐ (C) x, y and z
- ☐ (D) x and z

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

In Existential Instantiation, the sentence $\exists x \text{Crown}(x) \wedge \text{OnHead}(x, \text{John})$ is replaced by $\text{Crown}(C_1) \wedge \text{OnHead}(C_1, \text{John})$. Which condition must C_1 satisfy?

- ☐ (A) C_1 must be a new constant symbol not appearing anywhere else in the KB
- ☐ (B) C_1 must be John
- ☐ (C) C_1 may be any ground term already in the KB
- ☐ (D) C_1 must be a variable

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$ and let f be a unary function symbol. How many interpretations of f satisfy $\forall x f(f(x)) = x$?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (R \rightarrow (P \leftrightarrow Q)) \leftrightarrow ((Q \rightarrow R) \wedge (\neg Q \vee \neg P)) ?$$

Answer (numerical): _____

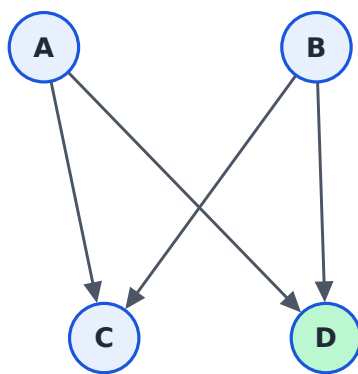
Q.9 [MSQ · 1 mark · no negative marking] Bayesian Networks

Which of the following are correct about Bayesian networks with Boolean variables?

- ☐ (A) The graph must be a directed acyclic graph
- ☐ (B) The CPT of a node with k Boolean parents has 2^k independent numbers
- ☐ (C) Each row of a CPT (one parent configuration) sums to 1 over the node's values
- ☐ (D) If there is no arc between X and Y, then $X \perp Y$ marginally

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(D = T)$ (round off to 3 decimal places).



P(A=T)
0.1

P(B=T)	A	B	P(C=T ...)
0.05	T	T	0.8
	T	F	0.4
	F	T	0.7
	F	F	0.2

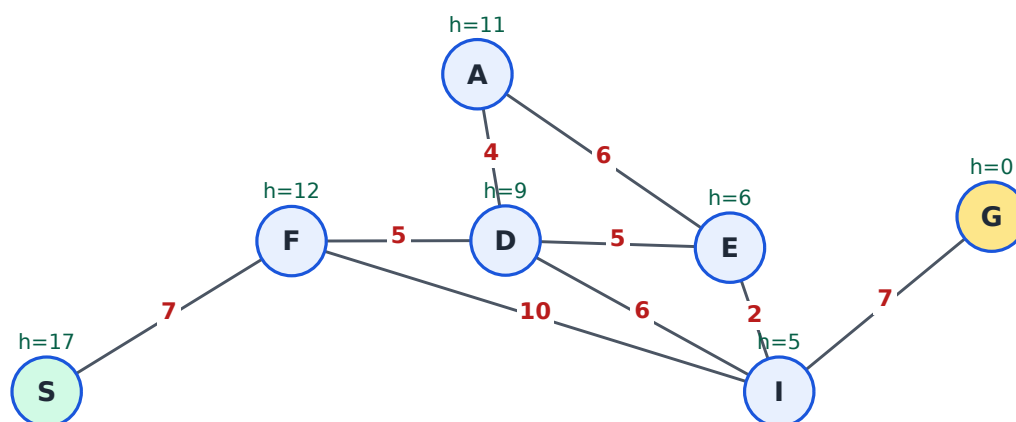
A	B	P(D=T ...)
T	T	0.4
T	F	0.2
F	T	0.9
F	F	0.25

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

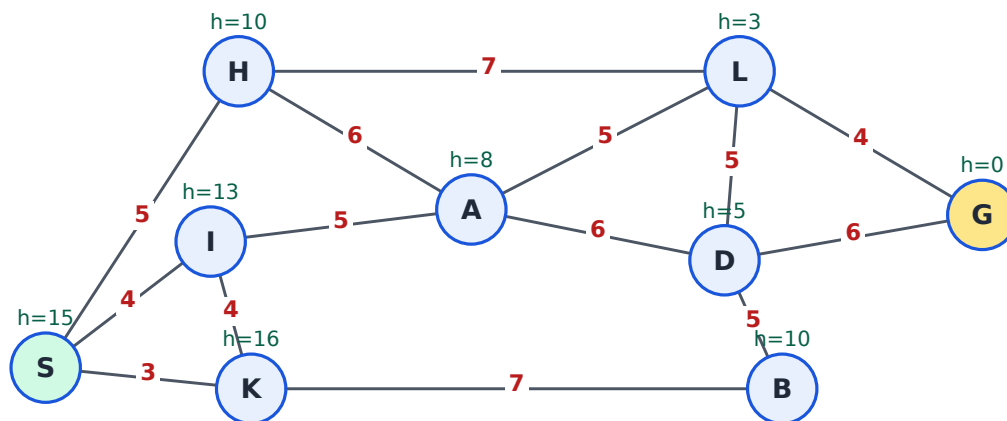
Q.12 [NAT · 2 marks · no negative marking] Problem Formulation

The 4-queens problem uses the improved incremental formulation. A state places k queens ($0 \leq k \leq 4$), one in each of the leftmost k columns, with no two queens attacking each other. Including the empty board ($k = 0$), how many states are in this state space?

Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

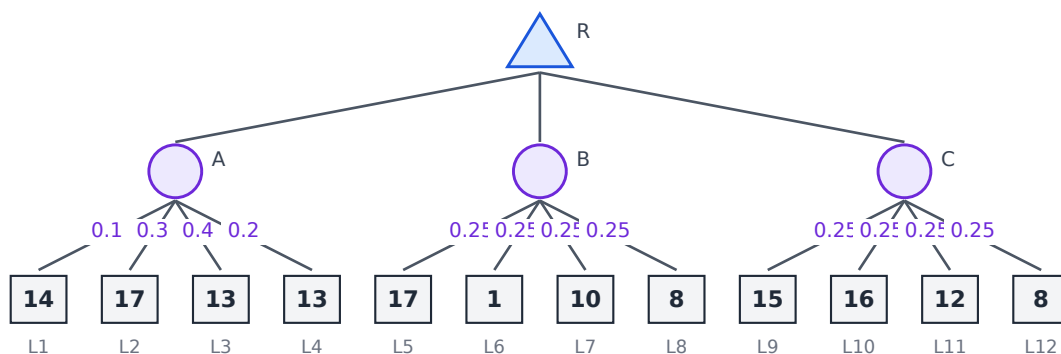
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both **S** and **G**?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



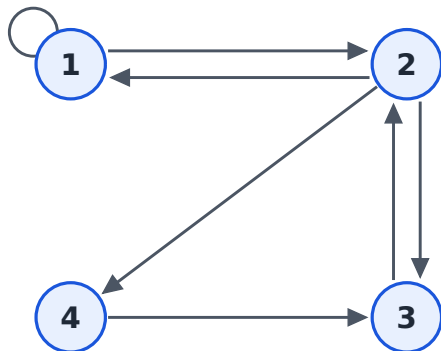
△ MAX node ○ CHANCE node (edge = probability)

□ terminal utility

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (C) $\forall x \exists y R(x, y)$
☐ (D) $\forall y \exists x R(x, y)$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following statements about first-order logic are TRUE?

- ☐ (A) Entailment in first-order logic is semidecidable
☐ (B) Resolution (with factoring and unification) is refutation-complete for first-order logic
☐ (C) Validity of first-order sentences is decidable
☐ (D) Forward chaining on a Datalog KB (definite clauses without function symbols) always terminates

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **B, C, A** and rules:

- R1: $A \rightarrow E$
 R2: $E \wedge F \rightarrow C$
 R3: $E \wedge K \rightarrow J$
 R4: $E \wedge D \rightarrow A$
 R5: $B \wedge E \rightarrow F$
 R6: $J \rightarrow H$
 R7: $K \rightarrow D$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) E is entailed
☐ (B) H is entailed
☐ (C) J is entailed
☐ (D) K is entailed

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

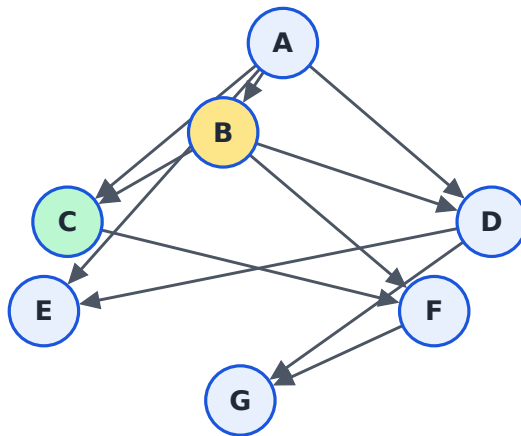
C1: (Q), C2: ($\neg Q$), C3: ($P \vee \neg Q \vee \neg S$), C4: (S), C5: ($P \vee Q \vee \neg R$)

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) ($P \vee \neg R \vee \neg S$)
☐ (B) ($\neg Q \vee S$)
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) ($P \vee \neg Q$)

Q.19 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{C} \mid \mathbf{B} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{B} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **G, F, A, E, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

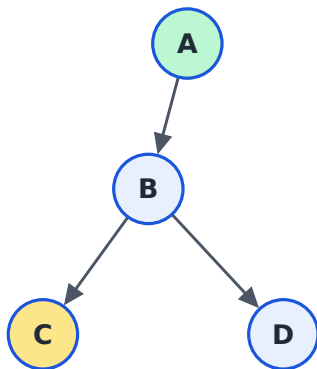
Q.20 [NAT · 2 marks · no negative marking] Bayesian Networks

A Bayesian network has: A with 3 values (no parents); B Boolean (no parents); C with 4 values and parents A, B; D Boolean with parent C; E with 3 values and parents C, D. How many independent parameters do the CPTs require?

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{C=F}$. The current state of the chain is $(\mathbf{A=T, B=F, C=F, D=F})$. The next step resamples $\mathbf{A}$. With what probability is A set to T? (round off to 3 decimal places)



P(A=T)
0.1

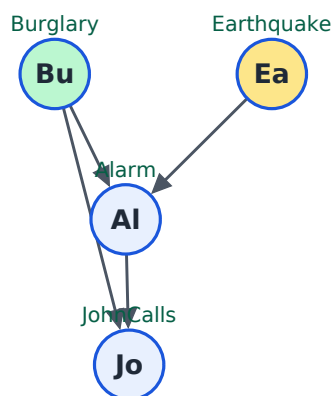
A	P(B=T ...)	B	P(C=T ...)
T	0.1	T	0.7
F	0.2	F	0.2

B	P(D=T ...)
T	0.9
F	0.25

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute $\mathbf{P(\text{Burglary} = T \mid \text{Earthquake} = T)}$ (round off to 3 decimal places).



P(Bu=T)
0.95

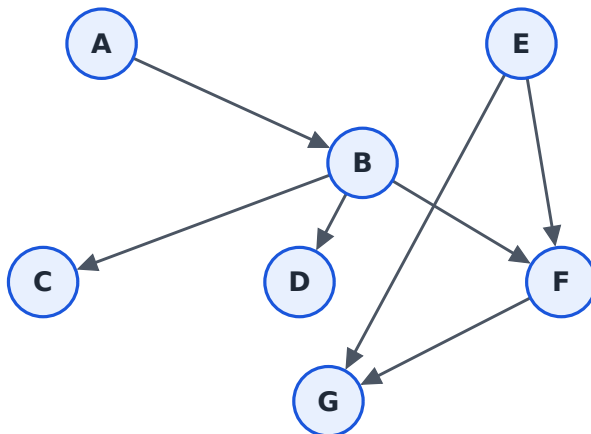
P(Ea=T)	Bu	Ea	P(Al=T ...)
0.4	T	T	0.1
	T	F	0.4
	F	T	0.5
	F	F	0.7

Bu	Al	P(Jo=T ...)
T	T	0.9
T	F	0.4
F	T	0.1
F	F	0.2

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

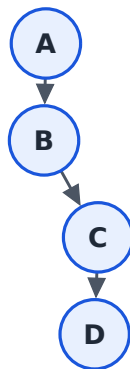
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp C \mid E$
☐ (B) $C \perp E \mid B$
☐ (C) $C \perp F \mid D$
☐ (D) $F \perp A$

Q.24 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.84}, \mathbf{0.46}, \mathbf{0.23}, \mathbf{0.74} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.6

A	P(B=T ...)	B	P(C=T ...)
T	0.05	T	0.75
F	0.3	F	0.4

C	P(D=T ...)
T	0.3
F	0.75

- ☐ (A) A=T, B=T, C=F, D=F
☐ (B) A=F, B=T, C=T, D=F
☐ (C) A=T, B=F, C=T, D=F
☐ (D) A=F, B=F, C=T, D=F

Q.25 [NAT · 2 marks · no negative marking] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. In computing $P(B \mid j, m)$ by variable elimination, A is summed out first, producing a factor $f(B, E) = \sum_a P(a \mid B, E) P(j \mid a) P(m \mid a)$. Compute $f(b, e)$. (Round to 3 decimal places.)

Answer (numerical): _____

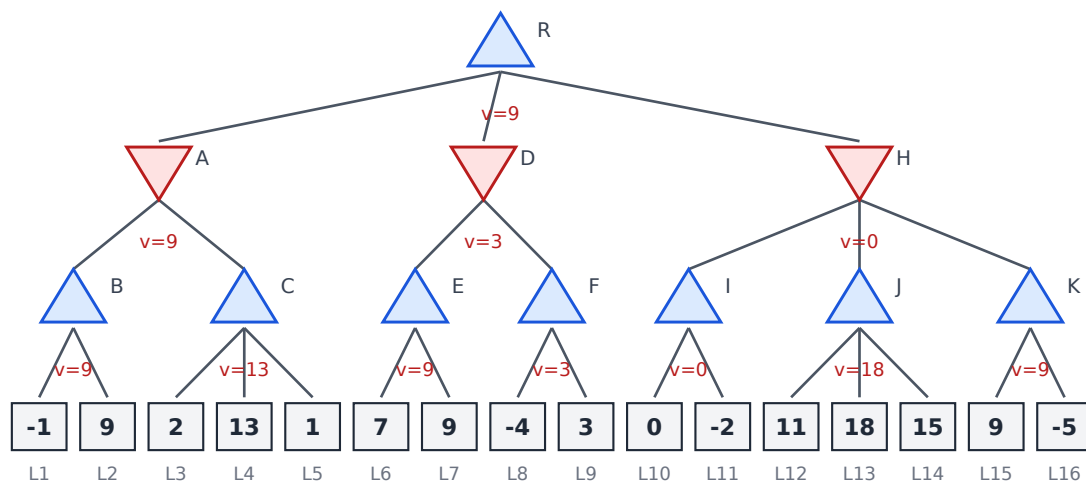
Answer Key — Mock Test 079

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	9	14	NAT	2	Adversarial Search	14.29 to 14.31
2	NAT	1	Informed Search	17	15	MSQ	2	Predicate Logic	C, D
3	MCQ	1	Adversarial Search	A	16	MSQ	2	Predicate Logic	A, B, D
4	MCQ	1	Uninformed Search	A	17	MSQ	2	Propositional Logic	A
5	MCQ	1	Predicate Logic	A	18	MSQ	2	Propositional Logic	A, C, D
6	MCQ	1	Predicate Logic	A	19	NAT	2	Exact Inference	3
7	NAT	1	Predicate Logic	4	20	NAT	2	Bayesian Networks	41
8	NAT	1	Propositional Logic	3	21	NAT	2	Sampling	0.109 to 0.113
9	MSQ	1	Bayesian Networks	A, B, C	22	NAT	2	Exact Inference	0.948 to 0.952
10	NAT	1	Exact Inference	0.273 to 0.277	23	MSQ	2	Conditional Independence	B
11	NAT	2	Informed Search	6	24	MCQ	2	Sampling	D
12	NAT	2	Problem Formulation	17	25	NAT	2	Variable Elimination	0.531 to 0.533
13	NAT	2	Informed Search	4					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 9

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 9$, $C = 13$, $A = 9$, $E = 9$, $F = 3$, $D = 3$, $I = 0$, $J = 18$, $K = 9$, $H = 0$. Root value = **9**; MAX's optimal first move is towards A.

Q.2 [NAT, 1M · Informed Search] Answer: 17

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (15)	H(10), E(12)
2	H (10)	C(7), E(12)
3	C (7)	J(3), D(9), E(12)
4	J (3)	G(0), D(9), E(12)
5	G (0)	D(9), E(12) ← goal, stop

Path returned: **S** → **H** → **C** → **J** → **G**, cost = $6 + 3 + 4 + 4 = \mathbf{17}$.

The optimal cost is $h^*(S) = 17$ — greedy happens to be optimal here, but in general it is not.

Q.3 [MCQ, 1M · Adversarial Search] Answer: A (A: the value of the best choice (highest value) found so far for MAX along the current path)

α is MAX's best guaranteed value so far, a lower bound, along the path to the root. β is MIN's best value so far, an upper bound (B). A branch is cut when the bounds cross, i.e. when $\alpha \geq \beta$. (C) and (D) are not what α means.

Q.4 [MCQ, 1M · Uninformed Search] Answer: A (A: S, J, A, B, E, F, G)

Adjacency lists (alphabetical): A: {B, E, F, J}; B: {A, J}; E: {A, F, G, J}; F: {A, E, G}; G: {E, F}; J: {A, B, E, S}; S: {J}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[J]
2	J	[A, B, E]
3	A	[B, E, F]
4	B	[E, F]
5	E	[F, G]
6	F	[G]
7	G	[] $\leftarrow$ goal popped, stop

Expansion order: **S, J, A, B, E, F, G**. The path returned is $S \rightarrow J \rightarrow E \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [MCQ, 1M · Predicate Logic] Answer: A (A: y and z)

x is bound by $\forall x$ throughout. The occurrence of y in $P(x, y)$ is not in the scope of any $\exists y/\forall y$, so it is free. The y in $Q(x, y, z)$ is bound by $\exists y$. z is never quantified, so it is free. Hence the variables with free occurrences are y and z.

Q.6 [MCQ, 1M · Predicate Logic] Answer: A (A: C_1 must be a new constant symbol not appearing anywhere else in the KB)

Existential Instantiation introduces a fresh Skolem constant naming the (unknown) witness. Using an existing term such as John would assert, without justification, that John is the witness. The result is inferentially equivalent (satisfiable iff the original), not logically equivalent. A variable would make it a non-ground formula.

Q.7 [NAT, 1M · Predicate Logic] Answer: 4

$f(f(x)) = x$ for all x means f is an involution: it is a bijection whose cycles have length 1 or 2. On 3 elements: the identity (1 way) or exactly one transposition with one fixed point (3 ways). Total = 4.

Q.8 [NAT, 1M · Propositional Logic] Answer: 3

Build the truth table (sub-formulas shown first):

P	Q	R	$R \rightarrow (P \leftrightarrow Q)$	$(Q \rightarrow R) \wedge (\neg Q \vee \neg P)$	$(R \rightarrow (P \leftrightarrow Q)) \leftrightarrow ((Q \rightarrow R) \wedge (\neg Q \vee \neg P))$
T	T	T	T	F	F
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

Rows where F = T: **3**.

Q.9 [MSQ, 1M · Bayesian Networks] Answer: A, B, C

(A) BNs are DAGs by definition. (B) There are 2^k rows, each needing one number $P(X = \text{true} \mid \text{row})$. (C) Each row is a conditional distribution, so it sums to 1. (D) is false: in $A \rightarrow B \rightarrow C$ there is no arc between A and C, but they are dependent through B.

Q.10 [NAT, 1M · Exact Inference] Answer: 0.273 to 0.277

Only the ancestors of D (A, B) matter: every other variable sums out to 1.

$P(D=T) = \sum \text{over its ancestors of } P(D=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, D=T)$
A=T, B=T, C=T	0.0016
A=T, B=T, C=F	0.0004
A=T, B=F, C=T	0.0076
A=T, B=F, C=F	0.0114
A=F, B=T, C=T	0.02835
A=F, B=T, C=F	0.01215
A=F, B=F, C=T	0.04275
A=F, B=F, C=F	0.171

Sum = **0.2752** ≈ 0.275 .

Q.11 [NAT, 2M · Informed Search] **Answer: 6**

Heuristic values: $h(A)=11$, $h(D)=9$, $h(E)=6$, $h(F)=12$, $h(G)=0$, $h(I)=5$, $h(S)=17$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (17)	F(19)
2	F (19)	D(21), I(22)
3	D (21)	I(22), E(23), A(27)
4	I (22)	E(23), G(24), A(27)
5	E (23)	G(24), A(27)
6	G (24)	A(27) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **F** $\rightarrow$ **I** $\rightarrow$ **G** with cost **24**; nodes expanded = **6** (S, F, D, I, E, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 24$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.12 [NAT, 2M · Problem Formulation] **Answer: 17**

$k = 0$: 1 state. $k = 1$: 4 states. $k = 2$: the pairs of rows (r_1, r_2) with $|r_1 - r_2| \geq 2$ are (1,3), (1,4), (2,4), (3,1), (4,1), (4,2), so 6 states. $k = 3$: extending these without attack gives (1,4,2), (2,4,1), (3,1,4), (4,1,3), so 4 states. $k = 4$: the two solutions (2,4,1,3) and (3,1,4,2). Total = $1+4+6+4+2 = 17$. For 8-queens, the same formulation gives AIMA's 2057 states.

Q.13 [NAT, 2M · Informed Search] **Answer: 4**

Heuristic values: $h(A)=8$, $h(B)=10$, $h(D)=5$, $h(G)=0$, $h(H)=10$, $h(I)=13$, $h(K)=16$, $h(L)=3$, $h(S)=15$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	H(15), I(17), K(19)
2	H (15)	L(15), I(17), A(19), K(19)
3	L (15)	G(16), I(17), A(19), K(19), D(22)
4	G (16)	I(17), A(19), K(19), D(22) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **H** $\rightarrow$ **L** $\rightarrow$ **G** with cost **16**; nodes expanded = **4** (S, H, L, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.14 [NAT, 2M · Adversarial Search] **Answer: 14.29 to 14.31**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.1 \times 14 + 0.3 \times 17 + 0.4 \times 13 + 0.2 \times 13 = 14.3$
- $B = 0.25 \times 17 + 0.25 \times 1 + 0.25 \times 10 + 0.25 \times 8 = 9$
- $C = 0.25 \times 15 + 0.25 \times 16 + 0.25 \times 12 + 0.25 \times 8 = 12.75$

Root = $\max(14.3, 9, 12.75) = 14.3$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [MSQ, 2M · Predicate Logic] **Answer: C, D** (C: $\forall x \exists y R(x, y)$; D: $\forall y \exists x R(x, y)$)

$R = \{(1,1), (1,2), (2,1), (2,3), (2,4), (3,2), (4,3)\}$.

(A) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

(B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(2,4)$ holds but $R(4,2)$ does not.

(C) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(D) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: A, B, D**

(A) True: an algorithm can confirm every entailed sentence but may run forever on non-entailed ones

(Turing/Church). (B) True: Robinson's resolution theorem. (C) False: FOL validity is undecidable. (D) True:

with no function symbols only finitely many ground atoms exist, so a fixed point is reached in finitely many steps.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A** (A: E is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $A \rightarrow E$; add E.

Round 2: fire $B \wedge E \rightarrow F$; add F.

Fixed point: $\{A, B, C, E, F\}$ — 5 atoms are entailed (facts: A, B, C; newly derived: E, F). Atoms never derived: D, H, J, K.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, C, D** (A: $(P \vee \neg R \vee \neg S)$; C: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable); D: $(P \vee \neg Q)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(P \vee \neg R \vee \neg S)$ is derivable:

1. resolve $(P \vee \neg Q \vee \neg S)$ and $(P \vee Q \vee \neg R)$ on $Q \Rightarrow (P \vee \neg R \vee \neg S)$

(B) $(\neg Q \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $\square$ is derivable:

1. resolve (Q) and $(\neg Q)$ on $Q \Rightarrow \square$ (empty clause)

(D) $(P \vee \neg Q)$ is derivable:

1. resolve $(P \vee \neg Q \vee \neg S)$ and (S) on $S \Rightarrow (P \vee \neg Q)$

Q.19 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A)$, $f(A, C)$, $f(A, D)$, $f(A, D, E)$, $f(C, F)$, $f(D, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	$f(D, F, G)$	$\{D, F, G\}$	$f(D, F)$
F	$f(C, F)$, $f(D, F)$	$\{C, D, F\}$	$f(C, D)$
A	$f(A)$, $f(A)$, $f(A, C)$, $f(A, D)$, $f(A, D, E)$	$\{A, C, D, E\}$	$f(C, D, E)$
E	$f(C, D, E)$	$\{C, D, E\}$	$f(C, D)$
D	$f(C, D)$, $f(C, D)$	$\{C, D\}$	$f(C)$

The largest factor produced has **3** variable(s) (created when eliminating A); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over C and normalise.

Q.20 [NAT, 2M · Bayesian Networks] **Answer: 41**

Each CPT needs (number of parent configurations) $\times$ (arity $- 1$). A: $1 \times 2 = 2$. B: $1 \times 1 = 1$. C: $(3 \times 2) \times 3 = 18$. D: $4 \times 1 = 4$. E: $(4 \times 2) \times 2 = 16$. Total = $2 + 1 + 18 + 4 + 16 = 41$, versus $3 \times 2 \times 4 \times 2 \times 3 - 1 = 143$ for the full joint.

Q.21 [NAT, 2M · Sampling] **Answer: 0.109 to 0.113**

Gibbs sampling draws A from $P(A \mid \text{mb}(A))$, where $\text{mb}(A) = \{B\}$:

$P(A \mid \text{mb}) \propto P(A \mid \text{parents}(A)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$A=T$: $P(A=T) \times P(B=F \mid A=T) = 0.1 \times 0.9 = 0.09$

$A=F$: $P(A=F) \times P(B=F \mid A=F) = 0.9 \times 0.8 = 0.72$

Normalise: $0.09 / (0.09 + 0.72) = \mathbf{0.1111}$.

Q.22 [NAT, 2M · Exact Inference] Answer: 0.948 to 0.952

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot P(\text{Alarm} \mid \text{Burglary}, \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Alarm})$.

Inference by enumeration: $P(\text{Burglary} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Burglary}, e, \text{hidden})$.

Hidden: Al, Jo	Bu	product of CPT entries (Bu · Ea · Al · Jo)	value
T, T	T	$0.95 \times 0.4 \times 0.1 \times 0.9$	0.0342
T, F	T	$0.95 \times 0.4 \times 0.1 \times 0.1$	0.0038
F, T	T	$0.95 \times 0.4 \times 0.9 \times 0.4$	0.1368
F, F	T	$0.95 \times 0.4 \times 0.9 \times 0.6$	0.2052
T, T	F	$0.05 \times 0.4 \times 0.5 \times 0.1$	0.001
T, F	F	$0.05 \times 0.4 \times 0.5 \times 0.9$	0.009
F, T	F	$0.05 \times 0.4 \times 0.5 \times 0.2$	0.002
F, F	F	$0.05 \times 0.4 \times 0.5 \times 0.8$	0.008

$P(\text{Burglary}=\text{T}, e) = 0.38$; $P(\text{Burglary}=\text{F}, e) = 0.02$; $P(e) = 0.4$.

$P(\text{Burglary}=\text{T} \mid e) = 0.38 / 0.4 = \mathbf{0.95} \approx 0.950$.

Q.23 [MSQ, 2M · Conditional Independence] Answer: B (B: $C \perp E \mid B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp C \mid E$: **FALSE** — the path $A \rightarrow B \rightarrow C$ is active given $\{E\}$.

(B) $C \perp E \mid B$: **TRUE** — $C \leftarrow B \rightarrow F \leftarrow E$ (blocked: observed fork node B); $C \leftarrow B \rightarrow F \rightarrow G \leftarrow E$ (blocked: observed fork node B).

(C) $C \perp F \mid D$: **FALSE** — the path $C \leftarrow B \rightarrow F$ is active given $\{D\}$.

(D) $F \perp A$: **FALSE** — the path $F \leftarrow B \leftarrow A$ is active given $\{\}$.

Q.24 [MCQ, 2M · Sampling] Answer: D (D: $A=F, B=F, C=T, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid _) = 0.6$; $u = 0.84 \geq 0.6 \Rightarrow A = F$
- B: $P(B=T \mid A=F) = 0.3$; $u = 0.46 \geq 0.3 \Rightarrow B = F$
- C: $P(C=T \mid B=F) = 0.4$; $u = 0.23 < 0.4 \Rightarrow C = T$
- D: $P(D=T \mid C=T) = 0.3$; $u = 0.74 \geq 0.3 \Rightarrow D = F$

Sample: **(A=F, B=F, C=T, D=F)**.

Q.25 [NAT, 2M · Variable Elimination] Answer: 0.531 to 0.533

$f(b, e) = P(a \mid b, e) P(j \mid a) P(m \mid a) + P(\neg a \mid b, e) P(j \mid \neg a) P(m \mid \neg a)$. $= 0.95 \times 0.8 \times 0.7 + 0.05 \times 0.05 \times 0.02$
 $= 0.532 + 0.00005 = 0.53205 \approx 0.532$. This is not a probability distribution over B, E; it is an intermediate factor.

Mock Test 080

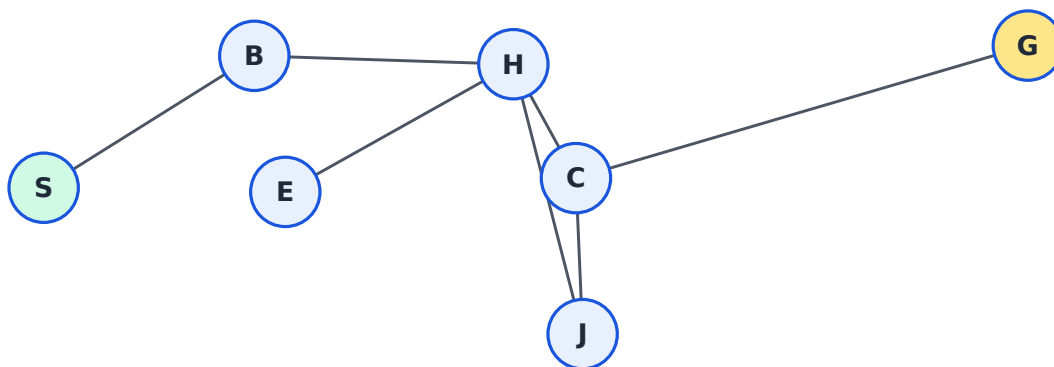
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 7 / 11	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, H, C, E, J, G
- ☐ (B) S, B, H, J, E, C, G
- ☐ (C) S, H, B, C, E, J, G
- ☐ (D) S, B, H, C, G

Q.2 [MCQ · 1 mark · negative: $-1/3$] Informed Search

Greedy best-first search orders its frontier by

- ☐ (A) $g(n)$
- ☐ (B) $h(n)$
- ☐ (C) $g(n) + h(n)$
- ☐ (D) $\text{depth}(n)$

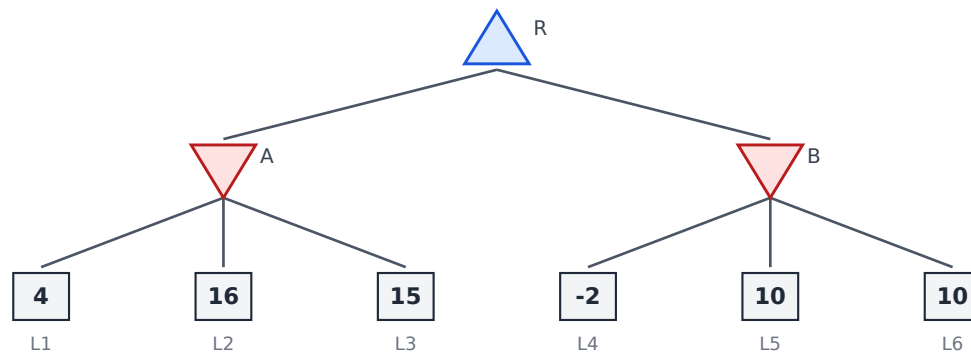
Q.3 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

A search tree has uniform branching factor $b = 4$; the shallowest goal is at depth $d = 3$ and happens to be the **last** node generated at that depth. BFS applies the goal test when a node is **generated**. How many nodes are generated in total (do **not** count the root)?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



MAX node MIN node

terminal utility

- ☐ (A) Move to A; root value 16
- ☐ (B) Move to A; root value 7
- ☐ (C) Move to B; root value -2
- ☐ (D) Move to A; root value 4

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following are sentences, i.e. formulas with no free variables (a is a constant)?

- ☐ (A) $(\forall x P(x)) \rightarrow Q(x)$
- ☐ (B) $\forall x (P(x) \rightarrow \exists y R(x, y))$
- ☐ (C) $\exists x R(x, y)$
- ☐ (D) $\forall x \exists y (R(x, y) \wedge P(a))$

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$\mathbf{F} = \neg \mathbf{R} \wedge (\neg \mathbf{P} \rightarrow (\neg \mathbf{R} \rightarrow \mathbf{Q})) ?$$

Answer (numerical): _____

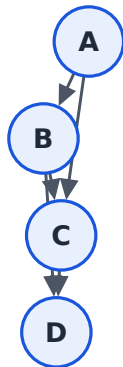
Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over the symbols p, q, r does $(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)$ have?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=T, B=F, C=F, D=T)**? (round off to 4 decimal places)



P(A=T)
0.95

A	P(B=T .)	A	B	P(C=T ...)
T	0.3	T	T	0.95
F	0.25	T	F	0.2
		F	T	0.1
		F	F	0.95

B	C	P(D=T ...)
T	T	0.95
T	F	0.75
F	T	0.05
F	F	0.75

Answer (numerical): _____

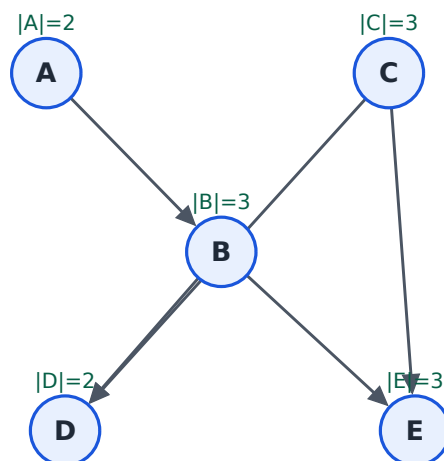
Q.9 [NAT · 1 mark · no negative marking] Probability for AI

A domain has four Boolean random variables and one random variable with 3 possible values. How many independent numbers are needed to specify an arbitrary full joint distribution over these five variables?

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?

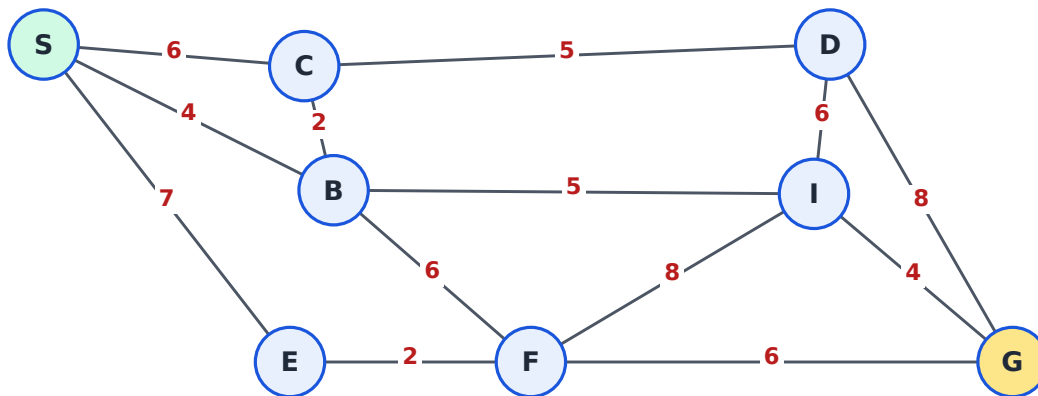


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

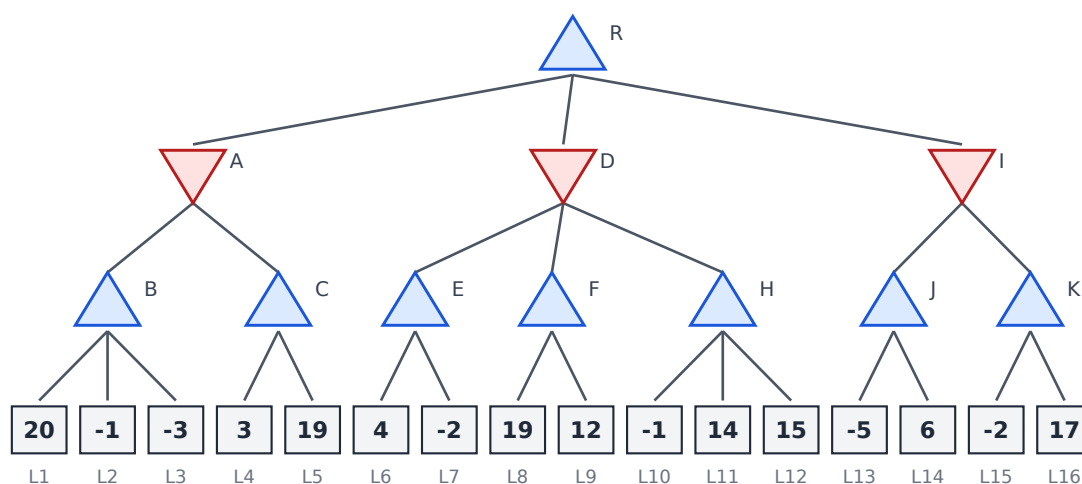
Q.12 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Suppose breadth-first search were changed to apply the goal test when a node is selected for expansion instead of when it is generated. The branching factor is b and the shallowest goal is at depth d . In the worst case, the number of nodes generated becomes

- ☐ (A) $O(b^d)$
☐ (B) $O(b^{d+1})$
☐ (C) $O(b^{d/2})$
☐ (D) $O(bd)$

Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



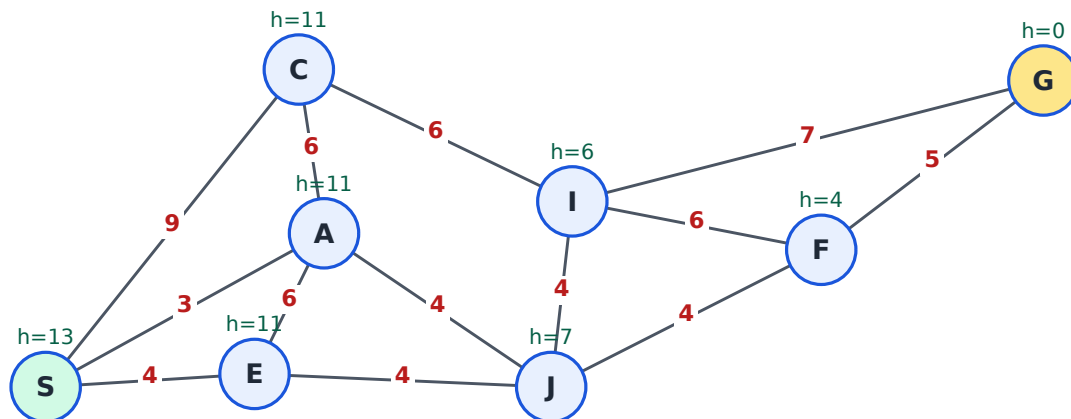
MAX node
 MIN node

terminal utility

- ☐ (A) Leaf L1 is pruned.
☐ (B) Leaf L3 is pruned.
☐ (C) Leaf L5 is pruned.
☐ (D) Leaf L16 is pruned.

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

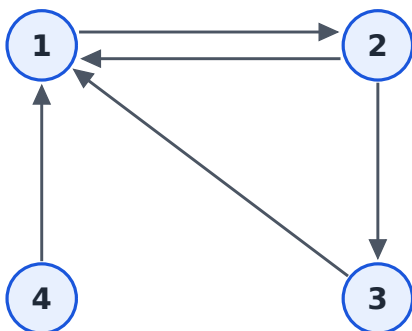
- (1) $\neg P \vee \neg Q$
- (2) $(P \leftrightarrow Q) \rightarrow Q$
- (3) $\neg R \rightarrow Q$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $(Q \vee P) \rightarrow \neg Q$
- ☐ (B) $(R \vee Q) \leftrightarrow Q$
- ☐ (C) $R \rightarrow (P \leftrightarrow \neg Q)$
- ☐ (D) $(R \rightarrow Q) \rightarrow (\neg R \leftrightarrow R)$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
- ☐ (B) $\exists x \forall y \neg R(y, x)$
- ☐ (C) $\exists x \forall y \neg R(x, y)$
- ☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \rightarrow Q) \wedge \neg R) \rightarrow \neg Q$
☐ (B) $(R \rightarrow P) \rightarrow ((R \vee Q) \rightarrow P)$
☐ (C) $((Q \rightarrow P) \rightarrow R) \leftrightarrow (Q \rightarrow (P \rightarrow R))$
☐ (D) $(Q \rightarrow (P \rightarrow R)) \leftrightarrow ((Q \wedge P) \rightarrow R)$

Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Which one of the following is a prenex normal form equivalent to $(\forall x P(x)) \rightarrow (\exists y Q(y))$?

- ☐ (A) $\exists x \exists y (P(x) \rightarrow Q(y))$
☐ (B) $\forall x \exists y (P(x) \rightarrow Q(y))$
☐ (C) $\forall x \forall y (P(x) \rightarrow Q(y))$
☐ (D) $\exists y \forall x (P(x) \rightarrow Q(y))$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **C, D** and rules:

- R1: $D \rightarrow F$
 R2: $C \wedge E \rightarrow H$
 R3: $K \wedge A \rightarrow E$
 R4: $F \wedge D \rightarrow E$
 R5: $J \rightarrow D$
 R6: $A \wedge F \wedge J \rightarrow D$
 R7: $K \wedge C \wedge H \rightarrow D$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) B is entailed
☐ (B) F is entailed
☐ (C) H is entailed
☐ (D) J is entailed

Q.20 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following formulas is in conjunctive normal form AND is logically equivalent to $p \leftrightarrow q$?

- ☐ (A) $(\neg p \vee q) \wedge (p \vee \neg q)$
☐ (B) $(p \wedge q) \vee (\neg p \wedge \neg q)$
☐ (C) $(p \vee q) \wedge (\neg p \vee \neg q)$
☐ (D) $(\neg p \vee q) \wedge (p \vee \neg q) \wedge (p \vee q)$

Q.21 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

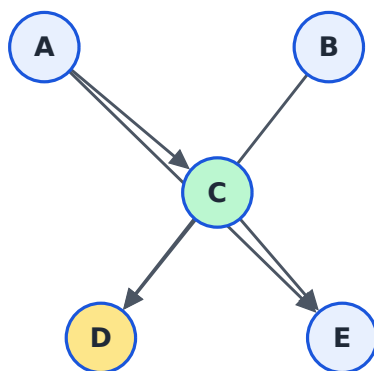
The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(B = T \mid D = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	T	T	F
2	F	T	F	F
3	F	T	T	F
4	F	T	F	F
5	F	T	F	T
6	F	F	T	F
7	F	T	F	F
8	T	T	T	F
9	F	T	T	T
10	T	T	T	F
11	T	T	T	T
12	F	F	T	F
13	F	T	F	T
14	F	F	T	F
15	F	T	T	F
16	F	T	F	F
17	T	F	F	T
18	F	T	T	F

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(C = T \mid D = T)$ (round off to 3 decimal places).



P(A=T)
0.9

P(B=T)
0.25

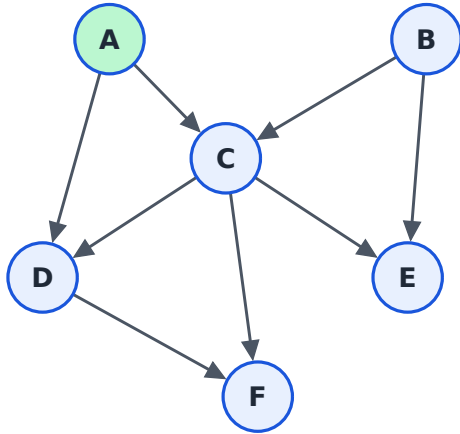
A	P(C=T ...)
T	0.2
F	0.05

B	C	P(D=T ...)	A	C	P(E=T ...)
T	T	0.05	T	T	0.75
T	F	0.9	T	F	0.25
F	T	0.3	F	T	0.8
F	F	0.7	F	F	0.5

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

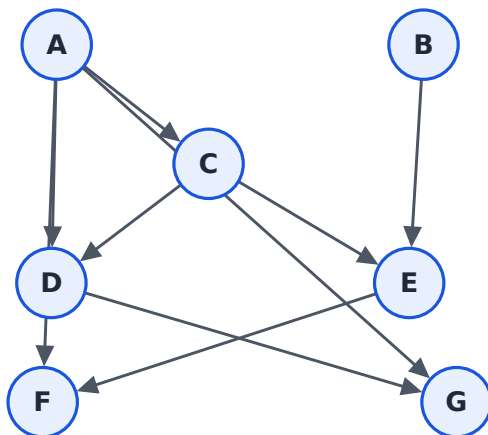
Variable elimination is used to compute $P(\mathbf{A})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F, B, E, C, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp B \mid G$
☐ (B) $F \perp G \mid D$
☐ (C) $E \perp A \mid G$
☐ (D) $A \perp B \mid D$

Q.25 [MCQ · 2 marks · negative: -2/3] Sampling

In likelihood weighting with evidence e and non-evidence variables z , let $S_{WS}(z, e)$ be the sampling probability and $w(z, e)$ the weight. Which statement is correct?

- ☐ (A) $S_{WS}(z, e) = \prod P(z_i \mid \text{parents}(Z_i))$, $w(z, e) = \prod P(e_j \mid \text{parents}(E_j))$, and $S_{WS}(z, e) \cdot w(z, e) = P(z, e)$
☐ (B) $S_{WS}(z, e) = P(z \mid e)$, so all weights equal 1
☐ (C) $S_{WS}(z, e) = P(z, e)$ and $w(z, e) = P(e)$
☐ (D) $S_{WS}(z, e) = \prod P(e_j \mid \text{parents}(E_j))$, $w(z, e) = \prod P(z_i \mid \text{parents}(Z_i))$

Answer Key — Mock Test 080

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	A	14	NAT	2	Informed Search	6
2	MCQ	1	Informed Search	B	15	MSQ	2	Propositional Logic	C
3	NAT	1	Uninformed Search	84	16	MSQ	2	Predicate Logic	A, B, D
4	MCQ	1	Adversarial Search	D	17	MSQ	2	Propositional Logic	D
5	MSQ	1	Predicate Logic	B, D	18	MCQ	2	Predicate Logic	A
6	NAT	1	Propositional Logic	3	19	MSQ	2	Propositional Logic	B, C
7	NAT	1	Propositional Logic	2	20	MCQ	2	Propositional Logic	A
8	NAT	1	Sampling	0.3988 to 0.3992	21	NAT	2	Sampling	0.76 to 0.78
9	NAT	1	Probability for AI	47	22	NAT	2	Exact Inference	0.065 to 0.069
10	NAT	1	Bayesian Networks	34	23	NAT	2	Exact Inference	3
11	NAT	2	Uninformed Search	8	24	MSQ	2	Conditional Independence	A, D
12	MCQ	2	Uninformed Search	B	25	MCQ	2	Sampling	A
13	MSQ	2	Adversarial Search	D					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: A** (A: S, B, H, C, E, J, G)

Adjacency lists (alphabetical): B: {H, S}; C: {G, H, J}; E: {H}; G: {C}; H: {B, C, E, J}; J: {C, H}; S: {B}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B]
2	B	[H]
3	H	[C, E, J]
4	C	[E, J, G]
5	E	[J, G]
6	J	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, B, H, C, E, J, G**. The path returned is $S \rightarrow B \rightarrow H \rightarrow C \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [MCQ, 1M · Informed Search] **Answer: B** (B: $h(n)$)

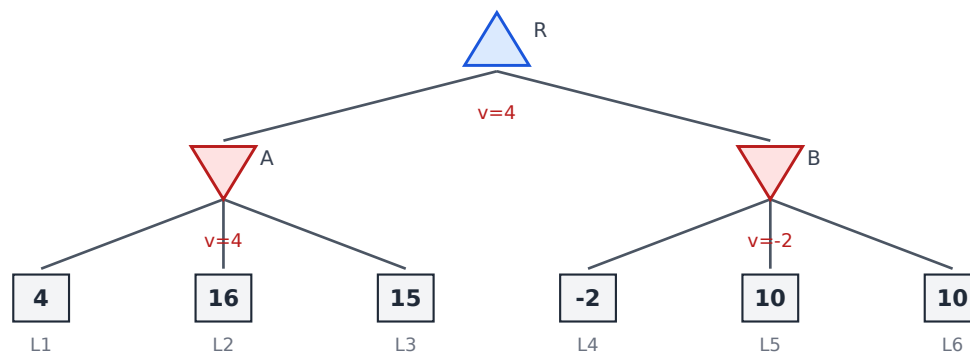
Greedy best-first search always expands the node that looks closest to the goal, so it uses $f(n) = h(n)$. $g(n)$ gives UCS (A). $g + h$ gives A* (C). Depth gives BFS when the shallowest node is expanded first, or DFS when the deepest is (D).

Q.3 [NAT, 1M · Uninformed Search] **Answer: 84**

With goal test at generation, BFS generates every node at depths 1 ... d before (and including) the goal (the worst case). $N = b + b^2 + \dots + b^d = 4 + 16 + 64 = \mathbf{84}$. This is the $O(b^d)$ of AIMA.

Q.4 [MCQ, 1M · Adversarial Search] **Answer: D** (D: Move to A; root value 4)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = 4, B = -2. Root value = 4; MAX's optimal first move is towards A.

Q.5 [MSQ, 1M · Predicate Logic] **Answer: B, D** (B: $\forall x (P(x) \rightarrow \exists y R(x, y))$; D: $\forall x \exists y (R(x, y) \wedge P(a))$)

(A) The scope of $\forall x$ ends before $\rightarrow$, so x in $Q(x)$ is free. (B) x and y are bound everywhere. (C) y is free. (D) x and y are bound; a is a constant, not a variable. Hence B and D are sentences.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 3**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg R$	$\neg P \rightarrow (\neg R \rightarrow Q)$	$\neg R \wedge (\neg P \rightarrow (\neg R \rightarrow Q))$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	T	F
F	F	F	T	F	F

Rows where F = T: **3**.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 2**

The cycle of implications $p \rightarrow q \rightarrow r \rightarrow p$ forces p, q, r to have the same truth value: if any one is T, all following ones are T, and the cycle returns. So the only models are all-T and all-F, i.e. 2 models.

Q.8 [NAT, 1M · Sampling] **Answer: 0.3988 to 0.3992**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(A=T) \times P(B=F|A=T) \times P(C=F|A=T, B=F) \times P(D=T|B=F, C=F) \\ = 0.95 \times 0.7 \times 0.8 \times 0.75 = \mathbf{0.399} \approx 0.3990.$$

Q.9 [NAT, 1M · Probability for AI] **Answer: 47**

The full joint table has $2^4 \times 3 = 48$ entries. Because the entries must sum to 1, one of them is determined by the rest, giving $48 - 1 = 47$ independent numbers. No independence assumptions are made, so no further reduction is possible.

Q.10 [NAT, 1M · Bayesian Networks] **Answer: 34**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$$\Sigma = A: (2-1) + B: (3-1) \times 2 + C: (3-1) + D: (2-1) \times 3 \times 3 + E: (3-1) \times 3 \times 3 = \mathbf{34}.$$

Full joint table: $\prod |X| - 1 = 108 - 1 = 107$; saving = 73.

Q.11 [NAT, 2M · Uninformed Search] **Answer: 8**

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(4), C(6), E(7)
2	B (4)	C(6), E(7), I(9), F(10)
3	C (6)	E(7), I(9), F(10), D(11)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
4	E (7)	F(9), I(9), D(11)
5	F (9)	I(9), D(11), G(15)
6	I (9)	D(11), G(13)
7	D (11)	G(13)
8	G (13)	← goal, stop

Path returned: **S → B → I → G**, cost = 4 + 5 + 4 = **13**. Nodes expanded (including S and G): **8** (S, B, C, E, F, I, D, G).

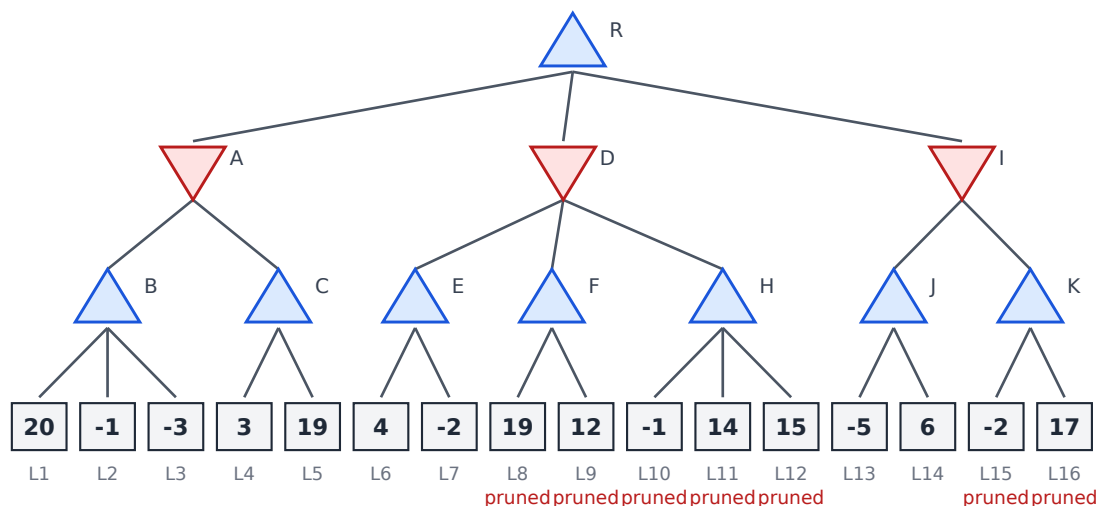
Q.12 [MCQ, 2M · Uninformed Search] **Answer: B** (B: $O(b^{d+1})$)

With a late goal test, BFS expands every node at depth d that precedes the goal and generates their children at depth $d+1$. That adds up to $b^{d+1} - b$ nodes, so the bound is **$O(b^{d+1})$** . Early testing (at generation) gives $O(b^d)$ (A). $O(b^{d/2})$ is the bidirectional bound (C). $O(bd)$ is a space bound for IDS (D).

Q.13 [MSQ, 2M · Adversarial Search] **Answer: D** (D: Leaf L16 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $4 \leq \alpha = 19 \Rightarrow$ prune remaining children (leaves L8, L9, L10, L11, L12).
- At MIN node I: value $6 \leq \alpha = 19 \Rightarrow$ prune remaining children (leaves L15, L16).



Leaves evaluated (9): L1, L2, L3, L4, L5, L6, L7, L13, L14. Leaves pruned (7): L8, L9, L10, L11, L12, L15, L16. Root value = 19 (same as plain minimax, 19 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] **Answer: 6**

Heuristic values: $h(A)=11$, $h(C)=11$, $h(E)=11$, $h(F)=4$, $h(G)=0$, $h(I)=6$, $h(J)=7$, $h(S)=13$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	A(14), E(15), C(20)
2	A (14)	J(14), E(15), C(20)
3	J (14)	E(15), F(15), I(17), C(20)
4	E (15)	F(15), I(17), C(20)
5	F (15)	G(16), I(17), C(20)
6	G (16)	I(17), C(20) ← goal, stop

Path returned: **S → A → J → F → G** with cost **16**; nodes expanded = **6** (S, A, J, E, F, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: C** (C: $R \rightarrow (P \leftrightarrow \neg Q)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 3 of them:

P	Q	R	KB
T	F	T	T
F	T	T	T
F	T	F	T

- (A) $(Q \vee P) \rightarrow \neg Q$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.
 (B) $(R \vee Q) \leftrightarrow Q$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.
 (C) $R \rightarrow (P \leftrightarrow \neg Q)$ is true in all models of KB $\Rightarrow$ **entailed**.
 (D) $(R \rightarrow Q) \rightarrow (\neg R \leftrightarrow R)$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: A, B, D** (A: $\forall x \exists y R(x, y)$; B: $\exists x \forall y \neg R(y, x)$; D: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,2), (2,1), (2,3), (3,1), (4,1)\}$.

- (A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.
 (B) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.
 (C) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.
 (D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.
 Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: D** (D: $(Q \rightarrow (P \rightarrow R)) \leftrightarrow ((Q \wedge P) \rightarrow R)$)

- (A) $((R \rightarrow Q) \wedge \neg R) \rightarrow \neg Q$: **not valid** — counter-model $Q=T, R=F$.
 (B) $(R \rightarrow P) \rightarrow ((R \vee Q) \rightarrow P)$: **not valid** — counter-model $P=F, Q=T, R=F$.
 (C) $((Q \rightarrow P) \rightarrow R) \leftrightarrow (Q \rightarrow (P \rightarrow R))$: **not valid** — counter-model $P=T, Q=F, R=F$.
 (D) $(Q \rightarrow (P \rightarrow R)) \leftrightarrow ((Q \wedge P) \rightarrow R)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MCQ, 2M · Predicate Logic] **Answer: A** (A: $\exists x \exists y (P(x) \rightarrow Q(y))$)

A quantifier pulled out of the antecedent of $\rightarrow$ flips: $(\forall x P(x)) \rightarrow \psi \equiv \exists x (P(x) \rightarrow \psi)$. From the consequent it does not flip: $\phi \rightarrow \exists y Q(y) \equiv \exists y (\phi \rightarrow Q(y))$. So we get $\exists x \exists y (P(x) \rightarrow Q(y))$. Check option B on domain $\{1, 2\}$ with $P = \{1\}$, $Q = \emptyset$: the original is true ($\forall x P(x)$ is false), but at $x=1$ no y makes $P(1) \rightarrow Q(y)$ true, so B is false. C and D are false on the same interpretation.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: B, C** (B: F is entailed; C: H is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \rightarrow F$; add F.

Round 2: fire $F \wedge D \rightarrow E$; add E.

Round 3: fire $C \wedge E \rightarrow H$; add H.

Fixed point: $\{C, D, E, F, H\}$ — 5 atoms are entailed (facts: C, D; newly derived: E, F, H). Atoms never derived: A, B, J, K.

Q.20 [MCQ, 2M · Propositional Logic] **Answer: A** (A: $(\neg p \vee q) \wedge (p \vee \neg q)$)

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv (\neg p \vee q) \wedge (p \vee \neg q)$, which is a CNF. $(p \wedge q) \vee (\neg p \wedge \neg q)$ is equivalent but is in DNF, not CNF. $(p \vee q) \wedge (\neg p \vee \neg q)$ is true exactly when one of p, q is true, i.e. $p \oplus q$. The last option adds the clause $p \vee q$, which removes the model $p=q=F$, leaving only $p \wedge q$.

Q.21 [NAT, 2M · Sampling] **Answer: 0.76 to 0.78**

Rejection sampling keeps only samples consistent with the evidence $D = F$: samples 1, 2, 3, 4, 6, 7, 8, 10, 12, 14, 15, 16, 18 (13 accepted, 5 rejected).

Among the accepted samples, $B = T$ in 10. Estimate = $10/13 = \mathbf{0.7692}$.

(For this network the exact value is 0.7542; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.22 [NAT, 2M · Exact Inference] **Answer: 0.065 to 0.069**

Factorisation: $P(A, B, C, D, E) = P(A) \cdot P(B) \cdot P(C | A) \cdot P(D | B, C) \cdot P(E | A, C)$.

Inference by enumeration: $P(C | e) = \alpha \sum_{\text{hidden}} P(C, e, \text{hidden})$.

Hidden: A, B, E	C	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
T, T, T	T	$0.9 \times 0.25 \times 0.2 \times 0.05 \times 0.75$	0.00169
T, T, F	T	$0.9 \times 0.25 \times 0.2 \times 0.05 \times 0.25$	0.00056
T, F, T	T	$0.9 \times 0.75 \times 0.2 \times 0.3 \times 0.75$	0.03037
T, F, F	T	$0.9 \times 0.75 \times 0.2 \times 0.3 \times 0.25$	0.01013
F, T, T	T	$0.1 \times 0.25 \times 0.05 \times 0.05 \times 0.8$	0.00005
F, T, F	T	$0.1 \times 0.25 \times 0.05 \times 0.05 \times 0.2$	0.00001
F, F, T	T	$0.1 \times 0.75 \times 0.05 \times 0.3 \times 0.8$	0.0009

Hidden: A, B, E	C	product of CPT entries ($A \cdot B \cdot C \cdot D \cdot E$)	value
F, F, F	T	$0.1 \times 0.75 \times 0.05 \times 0.3 \times 0.2$	0.00022
T, T, T	F	$0.9 \times 0.25 \times 0.8 \times 0.9 \times 0.25$	0.0405
T, T, F	F	$0.9 \times 0.25 \times 0.8 \times 0.9 \times 0.75$	0.1215
T, F, T	F	$0.9 \times 0.75 \times 0.8 \times 0.7 \times 0.25$	0.0945
T, F, F	F	$0.9 \times 0.75 \times 0.8 \times 0.7 \times 0.75$	0.2835
F, T, T	F	$0.1 \times 0.25 \times 0.95 \times 0.9 \times 0.5$	0.01069
F, T, F	F	$0.1 \times 0.25 \times 0.95 \times 0.9 \times 0.5$	0.01069
F, F, T	F	$0.1 \times 0.75 \times 0.95 \times 0.7 \times 0.5$	0.02494
F, F, F	F	$0.1 \times 0.75 \times 0.95 \times 0.7 \times 0.5$	0.02494

$P(C=T, e) = 0.04394$; $P(C=F, e) = 0.61125$; $P(e) = 0.65519$.

$P(C=T | e) = 0.04394 / 0.65519 = \mathbf{0.0671} \approx 0.067$.

Q.23 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, B, C)$, $f(A, C, D)$, $f(B, C, E)$, $f(C, D, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(C, D, F)$	$\{C, D, F\}$	$f(C, D)$
B	$f(B)$, $f(A, B, C)$, $f(B, C, E)$	$\{A, B, C, E\}$	$f(A, C, E)$
E	$f(A, C, E)$	$\{A, C, E\}$	$f(A, C)$
C	$f(A, C, D)$, $f(C, D)$, $f(A, C)$	$\{A, C, D\}$	$f(A, D)$
D	$f(A, D)$	$\{A, D\}$	$f(A)$

The largest factor produced has **3** variable(s) (created when eliminating B); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over A and normalise.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: A, D** ($A: C \perp B | G$; $D: A \perp B | D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $C \perp B | G$: **TRUE** — $C \leftarrow A \rightarrow F \leftarrow E \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow D \leftarrow A \rightarrow F \leftarrow E \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow D \rightarrow G \leftarrow A \rightarrow F \leftarrow E \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $C \rightarrow E \leftarrow B$ (blocked: collider E with neither it nor a descendant observed).

(B) $F \perp G | D$: **FALSE** — the path $F \leftarrow A \rightarrow G$ is active given $\{D\}$.

(C) $E \perp A | G$: **FALSE** — the path $E \leftarrow C \leftarrow A$ is active given $\{G\}$.

(D) $A \perp B | D$: **TRUE** — $A \rightarrow C \rightarrow E \leftarrow B$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow D \leftarrow C \rightarrow E \leftarrow B$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow F \leftarrow E \leftarrow B$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow G \leftarrow D \leftarrow C \rightarrow E \leftarrow B$ (blocked: collider G with neither it nor a descendant observed).

Q.25 [MCQ, 2M · Sampling] **Answer: A**

Each non-evidence variable is sampled from its CPT given sampled/fixed parents, giving $S_{WS} = \prod P(z_i | \text{parents})$. The weight multiplies the evidence likelihoods. The product of these two is exactly the BN factorization of $P(z, e)$, which proves consistency. (B) is false: evidence at descendants is not taken into account while sampling, so $S_{WS} \neq P(z | e)$ in general. (C) and (D) misstate or swap the two products.

Mock Test 081

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 6 / 13	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

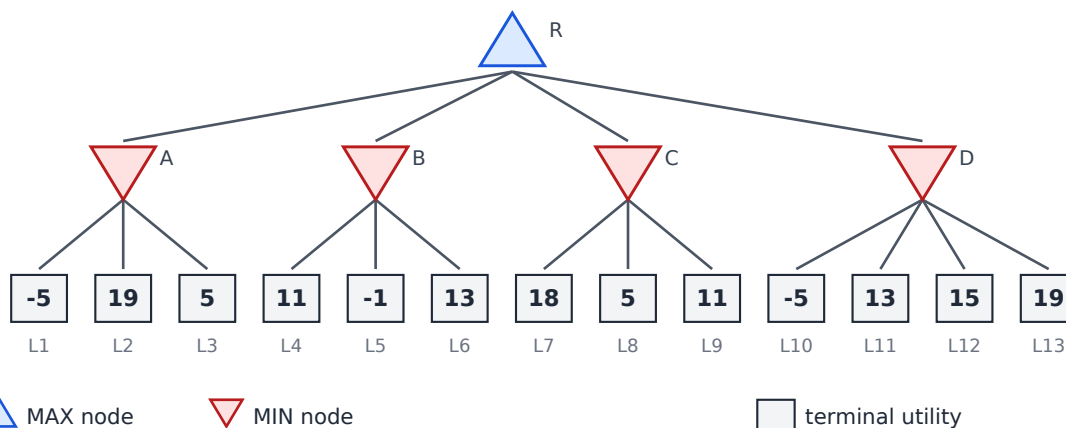
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 3$ and depth $d = 5$ (so $b^d = 243$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

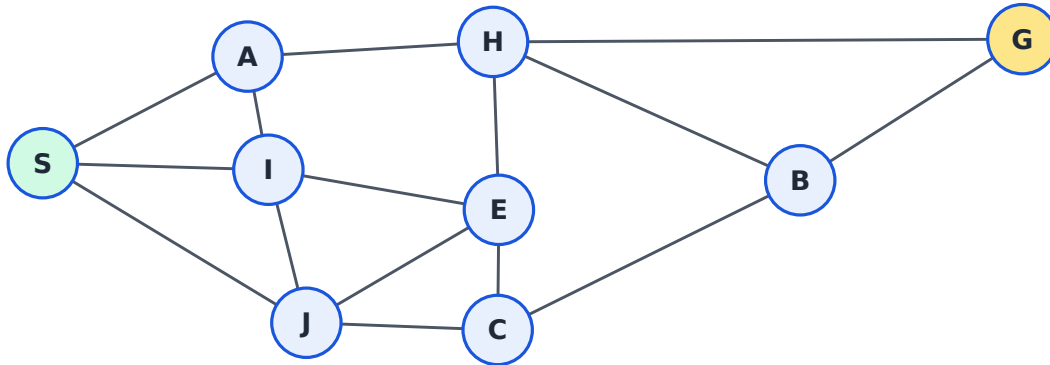
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, A, I, J, H, E, C, B, G
- ☐ (B) S, J, I, A, E, C, H, B, G
- ☐ (C) S, A, H, B, C, E, I, J, G
- ☐ (D) S, A, I, J, H, E, B, C, G

Q.4 [NAT · 1 mark · no negative marking] Problem Formulation

The Tower of Hanoi has 3 pegs and 4 disks of distinct sizes, and a larger disk may never sit on a smaller one. How many distinct legal states are there?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \exists y R(x, y)$ true?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are logically equivalent to p?

- ☐ (A) $p \vee (p \wedge q)$
- ☐ (B) $p \wedge (p \vee q)$
- ☐ (C) $(p \vee q) \wedge (p \vee \neg q)$
- ☐ (D) $(p \rightarrow q) \rightarrow q$

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following propositional formulas is a tautology?

- ☐ (A) $(p \rightarrow q) \vee (q \rightarrow p)$
- ☐ (B) $(p \rightarrow q) \rightarrow (q \rightarrow p)$
- ☐ (C) $(p \wedge \neg q) \rightarrow q$
- ☐ (D) $(p \vee q) \rightarrow (p \wedge q)$

Q.8 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

A Bayesian network has edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$. Which is the joint distribution it represents?

- ☐ (A) $P(A) P(B | A) P(C | A) P(D | B, C)$
☐ (B) $P(A) P(B | A) P(C | B) P(D | C)$
☐ (C) $P(A) P(B) P(C) P(D | B, C)$
☐ (D) $P(D) P(B | D) P(C | D) P(A | B, C)$

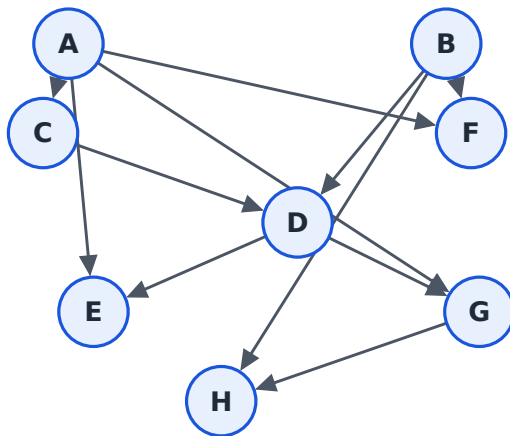
Q.9 [NAT · 1 mark · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. With prior sampling, what is the probability that a single generated sample is exactly (c, $\neg s$, r, w)? (Round to 3 decimal places.)

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **C** in the Bayesian network below?

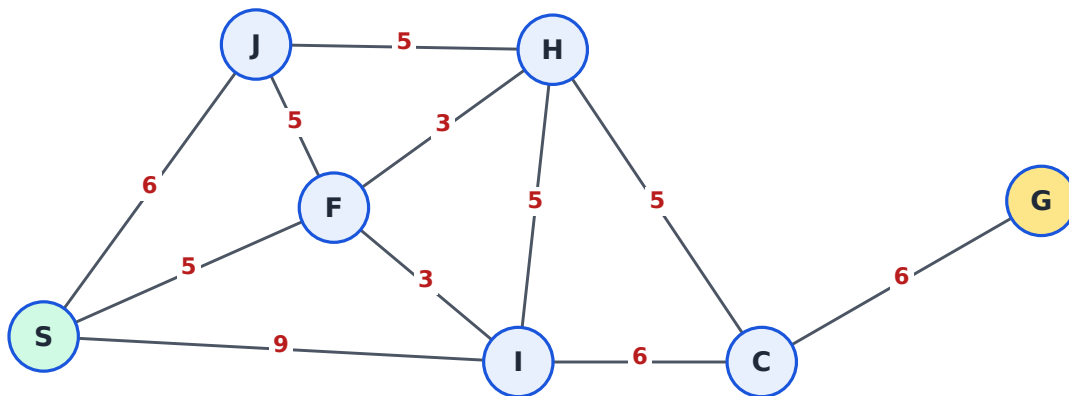


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

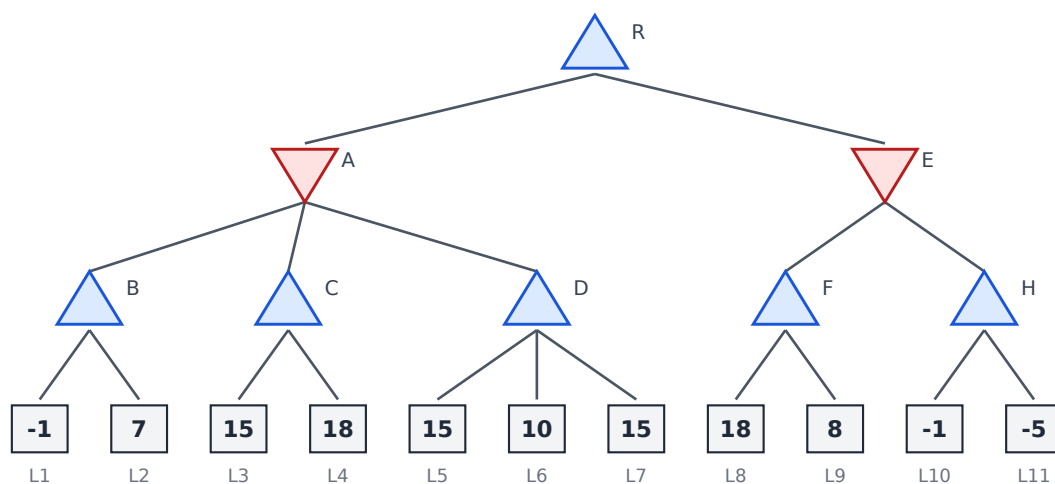
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



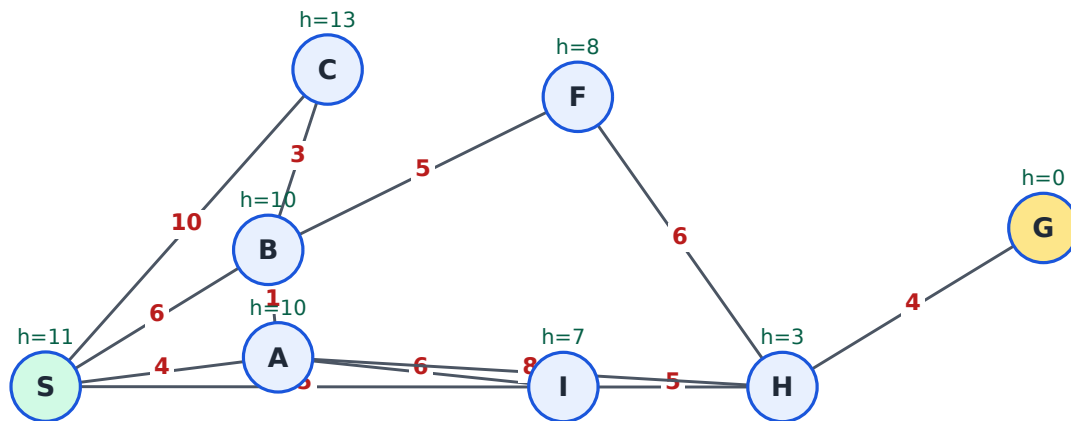
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Adversarial Search

Which of the following statements about expectimax / expectiminimax are TRUE?

- ☐ (A) The value of a chance node is the probability-weighted average of its children's values
- ☐ (B) With n distinct chance outcomes per chance layer, the time complexity becomes $O(b^m n^m)$
- ☐ (C) Pruning at chance nodes is impossible even when leaf values are known to lie in a bounded range
- ☐ (D) Expectimax is appropriate when the opponent's or environment's moves are random rather than optimally adversarial

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = \neg(P \vee R) \vee ((\neg S \leftrightarrow S) \leftrightarrow (P \leftrightarrow Q)) ?$$

Answer (numerical): _____

Q.16 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \exists y R(x, y) \wedge \exists y \forall x R(x, y)$?

Answer (numerical): _____

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg(\neg P \rightarrow R)$
- (2) $\neg R$
- (3) $\neg R \vee \neg\neg Q$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg\neg R \vee (\neg P \vee P)$
- ☐ (B) $R \wedge (\neg P \wedge Q)$
- ☐ (C) $Q \rightarrow \neg Q$
- ☐ (D) $(Q \vee \neg R) \vee (R \rightarrow P)$

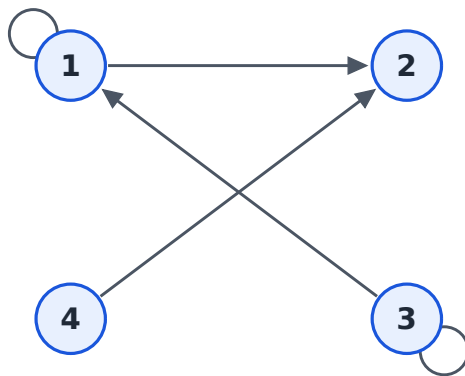
Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Given the definite clauses $\text{King}(x) \wedge \text{Greedy}(x) \rightarrow \text{Evil}(x)$, $\text{King}(\text{John})$, and $\text{Greedy}(y)$, Generalized Modus Ponens derives $\text{Evil}(\text{John})$ using which substitution?

- ☐ (A) $\{x/\text{John}, y/\text{John}\}$
- ☐ (B) $\{x/y\}$
- ☐ (C) $\{x/\text{John}\}$
- ☐ (D) $\{y/x, x/\text{Evil}\}$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
- ☐ (B) $\exists x R(x, x)$
- ☐ (C) $\exists x \forall y R(x, y)$
- ☐ (D) $\exists x \forall y \neg R(x, y)$

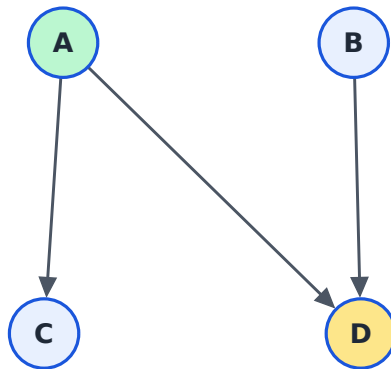
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $\mathbf{F} = (\mathbf{P} \vee \neg \mathbf{P}) \leftrightarrow \mathbf{R}$?

- ☐ (A) $((P \vee \neg P) \rightarrow R) \wedge (R \rightarrow (P \vee \neg P))$
- ☐ (B) $((P \vee \neg P) \wedge R) \vee (\neg(P \vee \neg P) \wedge \neg R)$
- ☐ (C) $(P \vee \neg P) \leftrightarrow \neg R$
- ☐ (D) $((P \vee \neg P) \wedge R) \vee ((P \vee \neg P) \wedge \neg R)$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(A = T \mid D = T)$ (round off to 3 decimal places).



$P(A=T)$
0.7

$P(B=T)$	A	$P(C=T \dots)$
0.4	T	0.5
	F	0.1

A	B	$P(D=T \dots)$
T	T	0.7
T	F	0.3
F	T	0.95
F	F	0.7

Answer (numerical): _____

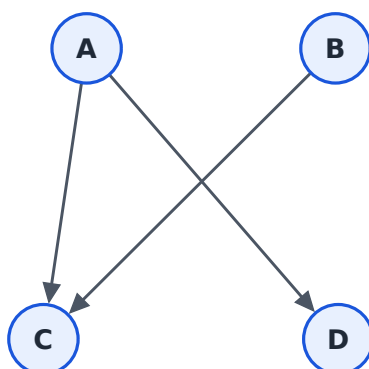
Q.22 [MCQ · 2 marks · negative: -2/3] Bayesian Networks

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. What is $P(r \mid w)$ (rounded to 3 decimal places)?

- ☐ (A) 0.708
☐ (B) 0.647
☐ (C) 0.430
☐ (D) 0.800

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle 0.98, 0.67, 0.93, 0.57 \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.6

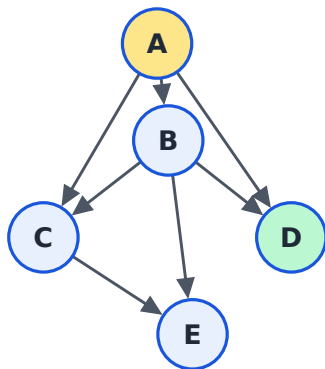
$P(B=T)$	A	B	$P(C=T \dots)$
0.25	T	T	0.3
	T	F	0.95
	F	T	0.2
	F	F	0.5

A	$P(D=T \dots)$
T	0.1
F	0.7

- ☐ (A) A=T, B=T, C=T, D=T
☐ (B) A=F, B=T, C=F, D=T
☐ (C) A=T, B=F, C=F, D=T
☐ (D) A=F, B=F, C=F, D=T

Q.24 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{A=T}$. The current state of the chain is $(\mathbf{A=T, B=F, C=F, D=F, E=T})$. The next step resamples $\mathbf{D}$. With what probability is D set to T? (round off to 3 decimal places)



P(A=T)
0.5

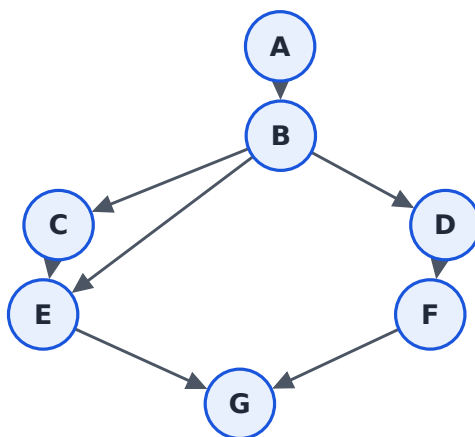
A	P(B=T ...)	A	B	P(C=T ...)
T	0.9	T	T	0.1
F	0.3	T	F	0.8
		F	T	0.7
		F	F	0.05

A	B	P(D=T B, C)	P(E=T ...)
T	T	0.4	0.95
T	F	0.8	0.75
F	T	0.8	0.5
F	F	0.6	0.4

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp E \mid B$
☐ (B) $C \perp B$
☐ (C) $E \perp F \mid B$
☐ (D) $E \perp F \mid D$

Answer Key — Mock Test 081

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	35	14	MSQ	2	Adversarial Search	A, B, D
2	NAT	1	Adversarial Search	5	15	NAT	2	Propositional Logic	10
3	MCQ	1	Uninformed Search	A	16	NAT	2	Predicate Logic	169
4	NAT	1	Problem Formulation	81	17	MSQ	2	Propositional Logic	A, D
5	NAT	1	Predicate Logic	343	18	MCQ	2	Predicate Logic	A
6	MSQ	1	Propositional Logic	A, B, C	19	MSQ	2	Predicate Logic	B, D
7	MCQ	1	Propositional Logic	A	20	MSQ	2	Propositional Logic	A, B
8	MCQ	1	Bayesian Networks	A	21	NAT	2	Exact Inference	0.571 to 0.575
9	NAT	1	Sampling	0.323 to 0.325	22	MCQ	2	Bayesian Networks	A
10	NAT	1	Conditional Independence	3	23	MCQ	2	Sampling	D
11	NAT	2	Uninformed Search	19	24	NAT	2	Sampling	0.798 to 0.802
12	NAT	2	Adversarial Search	8	25	MSQ	2	Conditional Independence	A, C, D
13	NAT	2	Informed Search	5					

Detailed Solutions

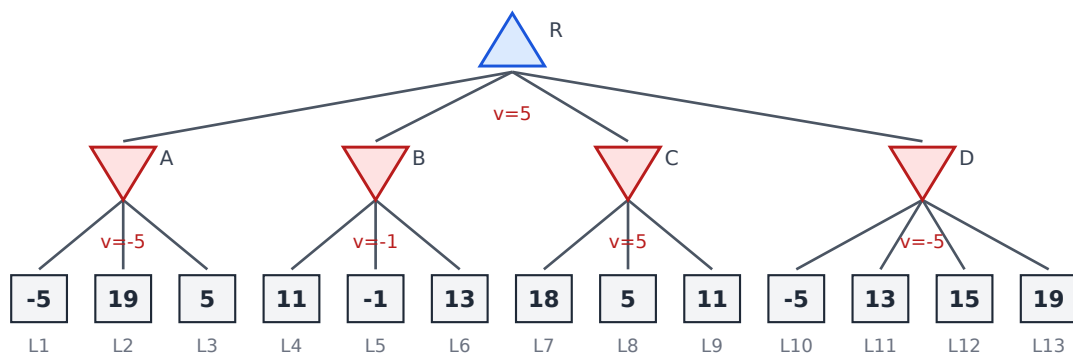
Q.1 [NAT, 1M · Adversarial Search] Answer: 35

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 3^{\lceil 3 \rceil} + 3^{\lfloor 3 \rfloor} - 1 = 27 + 9 - 1 = \mathbf{35}$, versus 243 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.2 [NAT, 1M · Adversarial Search] Answer: 5

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: A = -5, B = -1, C = 5, D = -5. Root value = **5**; MAX's optimal first move is towards C.

Q.3 [MCQ, 1M · Uninformed Search] Answer: A (A: S, A, I, J, H, E, C, B, G)

Adjacency lists (alphabetical): A: {H, I, S}; B: {C, G, H}; C: {B, E, J}; E: {C, H, I, J}; G: {B, H}; H: {A, B, E, G}; I: {A, E, J, S}; J: {C, E, I, S}; S: {A, I, J}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[A, I, J]
2	A	[I, J, H]
3	I	[J, H, E]
4	J	[H, E, C]
5	H	[E, C, B, G]
6	E	[C, B, G]
7	C	[B, G]

Step	Expanded	Frontier (FIFO) after expansion
8	B	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, A, I, J, H, E, C, B, G**. The path returned is $S \rightarrow A \rightarrow H \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [NAT, 1M · Problem Formulation] Answer: 81

Each disk can be on any of the 3 pegs. For a given assignment of disks to pegs, the size rule allows exactly one legal stacking order on each peg. So the number of states is $3^4 = 81$.

Q.5 [NAT, 1M · Predicate Logic] Answer: 343

View R as an 3×3 0/1 matrix. The sentence says R is every row of the relation matrix must be non-empty: $(2^n - 1)$ choices per row $\Rightarrow (2^n - 1)^n$. With $n = 3$: **343**.

Q.6 [MSQ, 1M · Propositional Logic] Answer: A, B, C (A: $p \vee (p \wedge q)$; B: $p \wedge (p \vee q)$; C: $(p \vee q) \wedge (p \vee \neg q)$)

(A) and (B) are the absorption laws. (C) By distribution, $(p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q) \equiv p$. (D) $(p \rightarrow q) \rightarrow q \equiv \neg(\neg p \vee q) \vee q \equiv (p \wedge \neg q) \vee q \equiv p \vee q$, which differs from p at $p=F, q=T$.

Q.7 [MCQ, 1M · Propositional Logic] Answer: A (A: $(p \rightarrow q) \vee (q \rightarrow p)$)

$(p \rightarrow q) \vee (q \rightarrow p) \equiv \neg p \vee q \vee \neg q \vee p$, which contains $q \vee \neg q$, so it is true in all 4 rows. $(p \rightarrow q) \rightarrow (q \rightarrow p)$ is false at $p=F, q=T$ (antecedent T, consequent F). $(p \wedge \neg q) \rightarrow q$ is false at $p=T, q=F$. $(p \vee q) \rightarrow (p \wedge q)$ is false whenever exactly one of p, q is true. Hence only the first is valid.

Q.8 [MCQ, 1M · Bayesian Networks] Answer: A (A: $P(A) P(B | A) P(C | A) P(D | B, C)$)

A BN represents $\prod_i P(X_i | \text{Parents}(X_i))$. Parents: A: none; B: A; C: A; D: B, C. This gives (A). (B) corresponds to a chain $A \rightarrow B \rightarrow C \rightarrow D$; (C) drops edges from A; (D) reverses every edge.

Q.9 [NAT, 1M · Sampling] Answer: 0.323 to 0.325

Prior sampling generates each event with probability equal to its joint probability: $S_{PS}(x) = \prod P(x_i | \text{parents})$. $= P(c) P(\neg s | c) P(r | c) P(w | \neg s, r) = 0.5 \times 0.9 \times 0.8 \times 0.9 = 0.324$.

Q.10 [NAT, 1M · Conditional Independence] Answer: 3

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of C: {A}; children: {D}; co-parents: {B}.

MB(C) = {A, B, D} — **3** nodes. Given its Markov blanket, C is conditionally independent of every other node in the network.

Q.11 [NAT, 2M · Uninformed Search] Answer: 19

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

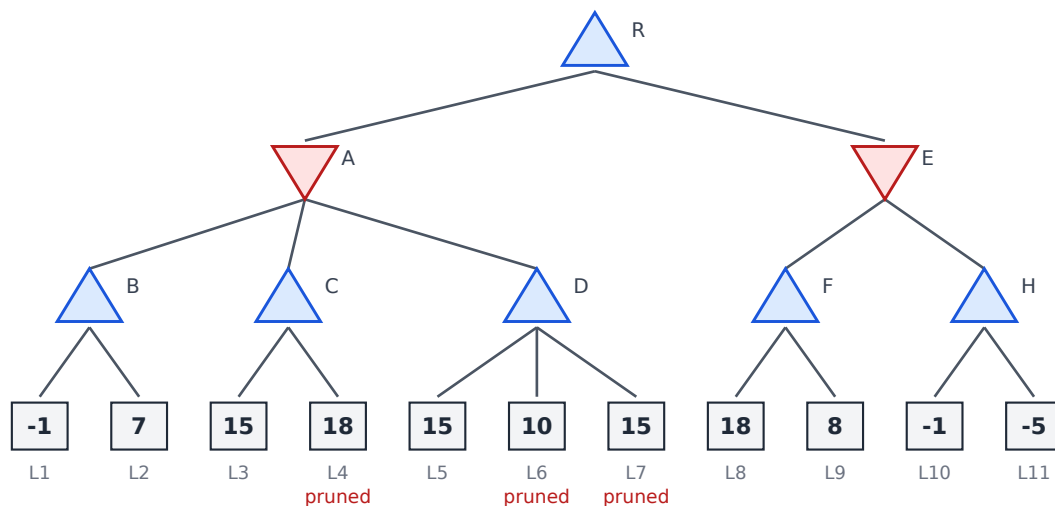
Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	F(5), J(6), I(9)
2	F (5)	J(6), H(8), I(8)
3	J (6)	H(8), I(8)
4	H (8)	I(8), C(13)
5	I (8)	C(13)
6	C (13)	G(19)
7	G (19)	← goal, stop

Path returned: **S** $\rightarrow$ **F** $\rightarrow$ **H** $\rightarrow$ **C** $\rightarrow$ **G**, cost = $5 + 3 + 5 + 6 = 19$. Nodes expanded (including S and G): **7** (S, F, J, H, I, C, G).

Q.12 [NAT, 2M · Adversarial Search] Answer: 8

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $15 \geq \beta = 7 \Rightarrow$ prune remaining children (leaves L4).
- At MAX node D: value $15 \geq \beta = 7 \Rightarrow$ prune remaining children (leaves L6, L7).



Leaves evaluated (8): L1, L2, L3, L5, L8, L9, L10, L11. Leaves pruned (3): L4, L6, L7. Root value = 7 (same as plain minimax, 7 — pruning never changes the root value).

Q.13 [NAT, 2M · Informed Search] **Answer: 5**

Heuristic values: $h(A)=10$, $h(B)=10$, $h(C)=13$, $h(F)=8$, $h(G)=0$, $h(H)=3$, $h(I)=7$, $h(S)=11$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	I(12), A(14), B(16), C(23)
2	I (12)	H(13), A(14), B(16), C(23)
3	H (13)	A(14), G(14), B(16), C(23), F(24)
4	A (14)	G(14), B(15), C(23), F(24)
5	G (14)	B(15), C(23), F(24) ← goal, stop

Path returned: **S** → **I** → **H** → **G** with cost **14**; nodes expanded = **5** (S, I, H, A, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 14$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.14 [MSQ, 2M · Adversarial Search] **Answer: A, B, D**

(A) is the definition of a chance node's value. (B) is the AIMA bound for expectiminimax. (C) is false: if utilities are bounded, a chance node's average can be bounded before all of its children are seen, so some pruning is possible. (D) is true: expectimax models stochastic behaviour, while minimax models a worst-case adversary.

Q.15 [NAT, 2M · Propositional Logic] **Answer: 10**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$\neg(P \vee R)$	$(\neg S \leftrightarrow S) \leftrightarrow (P \leftrightarrow Q)$	$\neg(P \vee R) \vee ((\neg S \leftrightarrow S) \leftrightarrow (P \leftrightarrow Q))$
T	T	T	T	F	F	F
T	T	T	F	F	F	F
T	T	F	T	F	F	F
T	T	F	F	F	F	F
T	F	T	T	F	T	T
T	F	T	F	F	T	T
T	F	F	T	F	T	T
T	F	F	F	F	T	T
F	T	T	T	F	T	T
F	T	T	F	F	T	T
F	T	F	T	T	T	T
F	T	F	F	T	T	T
F	F	T	T	F	F	F
F	F	T	F	F	F	F
F	F	F	T	T	F	T
F	F	F	F	T	F	T

Rows where $F = T$: **10**.

Q.16 [NAT, 2M · Predicate Logic] **Answer: 169**

$\exists y \forall x R(x, y)$ says some column of the 3×3 matrix is all T. This already implies $\forall x \exists y R(x, y)$ (that column is a witness for every row), so we only count relations with at least one full column. Relations with no full column: each column has 7 non-full choices $\rightarrow 7^3 = 343$. Answer = $512 - 343 = 169$.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $\neg\neg R \vee (\neg P \vee P)$; D: $(Q \vee \neg R) \vee (R \rightarrow P)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 2 of them:

P	Q	R	KB
F	T	F	T
F	F	F	T

(A) $\neg\neg R \vee (\neg P \vee P)$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $R \wedge (\neg P \wedge Q)$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(C) $Q \rightarrow \neg Q$ is false in the KB-model $P=F, Q=T, R=F \Rightarrow$ not entailed.

(D) $(Q \vee \neg R) \vee (R \rightarrow P)$ is true in all models of KB $\Rightarrow$ **entailed**.

Q.18 [MCQ, 2M · Predicate Logic] **Answer: A** (A: $\{x/\text{John}, y/\text{John}\}$)

GMP needs one θ with $\text{King}(x)\theta = \text{King}(\text{John})\theta$ and $\text{Greedy}(x)\theta = \text{Greedy}(y)\theta$. x/John matches the first premise; y/John then makes $\text{Greedy}(y)$ match $\text{Greedy}(\text{John})$. With only $\{x/\text{John}\}$, $\text{Greedy}(\text{John})$ and $\text{Greedy}(y)$ are not syntactically identical, so the GMP side condition fails; $\{x/y\}$ does not match $\text{King}(\text{John})$. The conclusion is $\text{Evil}(x)\theta = \text{Evil}(\text{John})$.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: B, D** (B: $\exists x R(x, x)$; D: $\exists x \forall y \neg R(x, y)$)

$R = \{(1,1), (1,2), (3,1), (3,3), (4,2)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(1,2)$ holds but $R(2,1)$ does not.

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

(C) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(D) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $((P \vee \neg P) \rightarrow R) \wedge (R \rightarrow (P \vee \neg P))$; B: $((P \vee \neg P) \wedge R) \vee (\neg(P \vee \neg P) \wedge \neg R)$)

Compare truth tables with $F = (P \vee \neg P) \leftrightarrow R$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $((P \vee \neg P) \rightarrow R) \wedge (R \rightarrow (P \vee \neg P))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $((P \vee \neg P) \wedge R) \vee (\neg(P \vee \neg P) \wedge \neg R)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $(P \vee \neg P) \leftrightarrow \neg R$ — differs at $P=T, R=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(D) $((P \vee \neg P) \wedge R) \vee ((P \vee \neg P) \wedge \neg R)$ — differs at $P=T, R=F$ ($F=F$, option = T) $\Rightarrow$ not equivalent.

Q.21 [NAT, 2M · Exact Inference] **Answer: 0.571 to 0.575**

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B) \cdot P(C | A) \cdot P(D | A, B)$.

Inference by enumeration: $P(A | e) = \alpha \sum_{\text{hidden}} P(A, e, \text{hidden})$.

Hidden: B, C	A	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.7 \times 0.4 \times 0.5 \times 0.7$	0.098
T, F	T	$0.7 \times 0.4 \times 0.5 \times 0.7$	0.098
F, T	T	$0.7 \times 0.6 \times 0.5 \times 0.3$	0.063
F, F	T	$0.7 \times 0.6 \times 0.5 \times 0.3$	0.063
T, T	F	$0.3 \times 0.4 \times 0.1 \times 0.95$	0.0114
T, F	F	$0.3 \times 0.4 \times 0.9 \times 0.95$	0.1026
F, T	F	$0.3 \times 0.6 \times 0.1 \times 0.7$	0.0126
F, F	F	$0.3 \times 0.6 \times 0.9 \times 0.7$	0.1134

$P(A=T, e) = 0.322$; $P(A=F, e) = 0.24$; $P(e) = 0.562$.

$P(A=T | e) = 0.322 / 0.562 = \mathbf{0.573} \approx 0.573$.

Q.22 [MCQ, 2M · Bayesian Networks] Answer: A (A: 0.708)

Compute $P(r, w) = \sum_{c,s} P(c)P(s | c)P(r | c)P(w | s, r)$. With c : $P(r | c) = 0.8$ and $P(w | r)$ averaged over S : $0.1 \times 0.99 + 0.9 \times 0.90 = 0.909 \rightarrow 0.5 \times 0.8 \times 0.909 = 0.3636$. With $\neg c$: $P(r | \neg c) = 0.2$ and $0.5 \times 0.99 + 0.5 \times 0.90 = 0.945 \rightarrow 0.5 \times 0.2 \times 0.945 = 0.0945$. So $P(r, w) = 0.4581$. $P(\neg r, w)$: c : $0.5 \times 0.2 \times (0.1 \times 0.9 + 0.9 \times 0) = 0.009$; $\neg c$: $0.5 \times 0.8 \times (0.5 \times 0.9) = 0.18$; total 0.189 . $P(w) = 0.6471$. $P(r | w) = 0.4581 / 0.6471 \approx 0.708$. 0.647 is $P(w)$, 0.430 is $P(s | w)$, 0.800 is $P(r | c)$.

Q.23 [MCQ, 2M · Sampling] Answer: D (D: A=F, B=F, C=F, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \neg) = 0.6$; $u = 0.98 \geq 0.6 \Rightarrow A = F$
- B: $P(B=T | \neg) = 0.25$; $u = 0.67 \geq 0.25 \Rightarrow B = F$
- C: $P(C=T | A=F, B=F) = 0.5$; $u = 0.93 \geq 0.5 \Rightarrow C = F$
- D: $P(D=T | A=F) = 0.7$; $u = 0.57 < 0.7 \Rightarrow D = T$

Sample: **(A=F, B=F, C=F, D=T)**.

Q.24 [NAT, 2M · Sampling] Answer: 0.798 to 0.802

Gibbs sampling draws D from $P(D | mb(D))$, where $mb(D) = \{A, B\}$:

$P(D | mb) \propto P(D | \text{parents}(D)) \times \prod_{\text{children } Y} P(y | \text{parents}(Y))$.

$D=T$: $P(D=T | A=T, B=F) = 0.8 = 0.8$

$D=F$: $P(D=F | A=T, B=F) = 0.2 = 0.2$

Normalise: $0.8 / (0.8 + 0.2) = \mathbf{0.8}$.

Q.25 [MSQ, 2M · Conditional Independence] Answer: A, C, D (A: $A \perp E | B$; C: $E \perp F | B$; D: $E \perp F | D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $A \perp E | B$: **TRUE** — $A \rightarrow B \rightarrow C \rightarrow E$ (blocked: observed chain node B); $A \rightarrow B \rightarrow D \rightarrow F \rightarrow G \leftarrow E$ (blocked: observed chain node B); $A \rightarrow B \rightarrow E$ (blocked: observed chain node B).

(B) $C \perp B$: **FALSE** — the path $C \leftarrow B$ is active given $\{\}$.

(C) $E \perp F | B$: **TRUE** — $E \leftarrow B \rightarrow D \rightarrow F$ (blocked: observed fork node B); $E \leftarrow C \leftarrow B \rightarrow D \rightarrow F$ (blocked: observed fork node B); $E \rightarrow G \leftarrow F$ (blocked: collider G with neither it nor a descendant observed).

(D) $E \perp F | D$: **TRUE** — $E \leftarrow B \rightarrow D \rightarrow F$ (blocked: observed chain node D); $E \leftarrow C \leftarrow B \rightarrow D \rightarrow F$ (blocked: observed chain node D); $E \rightarrow G \leftarrow F$ (blocked: collider G with neither it nor a descendant observed).

Mock Test 082

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 8 / 12	13 / 15 / 12

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

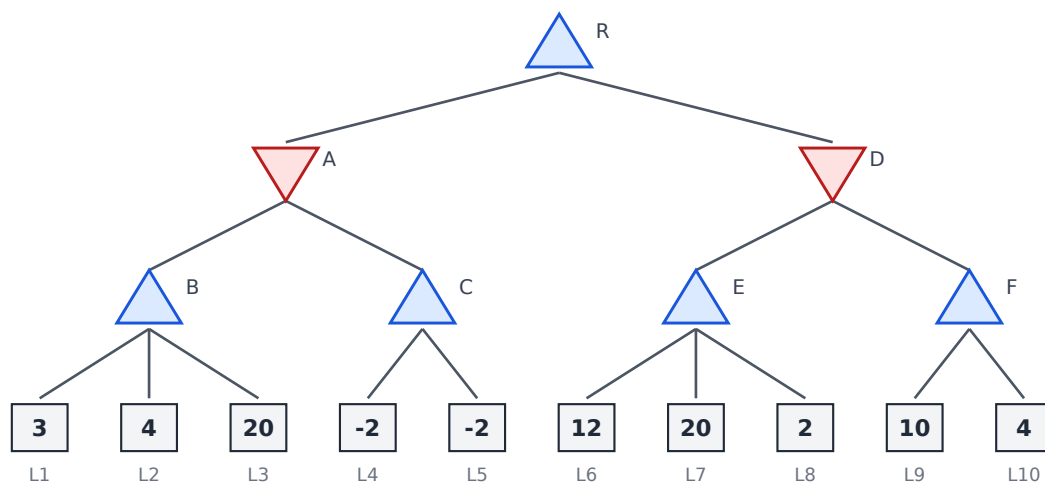
Q.1 [NAT · 1 mark · no negative marking] Informed Search – Effective branching factor

A* finds a solution at depth $d = 3$ after generating $N = 18$ nodes (root excluded). The effective branching factor b^* satisfies $N = b^* + (b^*)^2 + \dots + (b^*)^d$. Find b^* (round off to 2 decimal places).

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



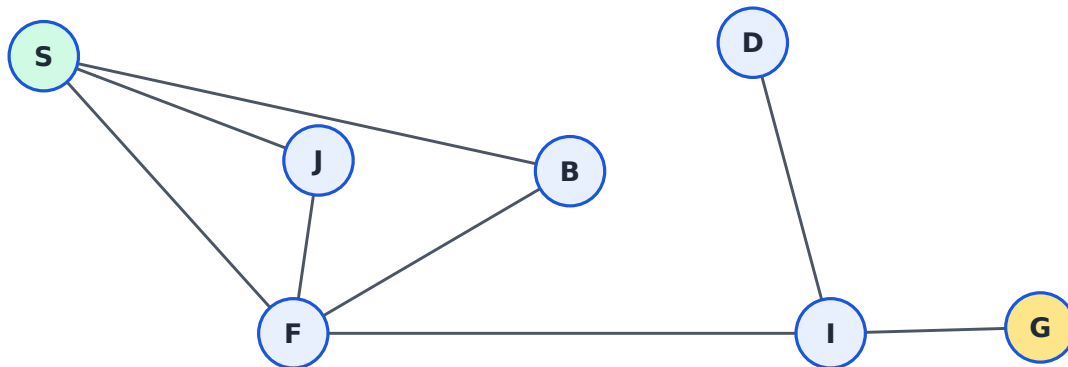
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to D; root value 10
- ☐ (B) Move to D; root value 11
- ☐ (C) Move to D; root value 20
- ☐ (D) Move to A; root value -2

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, B, I, F, D, G
☐ (B) S, J, F, I, G
☐ (C) S, B, F, I, D, G
☐ (D) S, B, F, J, I, D, G

Q.4 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (\neg Q \vee (Q \leftrightarrow Q)) \wedge \neg(R \rightarrow \neg P) ?$$

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ are both reflexive and symmetric?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 3 elements and R a binary predicate symbol. In how many of the 2^9 possible interpretations of R over D is the sentence $\forall x \exists y R(x, y)$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(B = T \mid A = T)$, how many of these samples are **rejected**?

#	A	B	C	D
1	T	F	F	T
2	F	F	F	F
3	T	T	T	T
4	T	T	F	T
5	F	F	F	F
6	T	T	F	T
7	F	F	F	F
8	F	F	F	F
9	F	F	F	T
10	F	F	F	F
11	F	F	T	T
12	T	F	F	T
13	F	F	F	F
14	F	T	F	F
15	T	T	F	F
16	F	F	F	F
17	T	T	F	T
18	T	T	T	T

Answer (numerical): _____

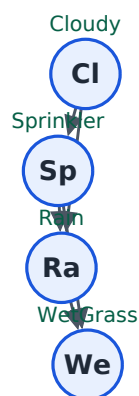
Q.8 [MCQ · 1 mark · negative: -1/3] Conditional Independence

Burglary (B) and Earthquake (E) are independent causes of Alarm (A). Given that the alarm is ringing, learning that an earthquake occurred lowers the probability of a burglary. This phenomenon is called

- ☐ (A) explaining away (intercausal reasoning)
☐ (B) the local Markov property
☐ (C) the naive Bayes assumption
☐ (D) marginalization

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample **(Cl=T, Sp=F, Ra=T, We=T)**? (round off to 4 decimal places)



P(Cl=T)
0.75

Cl	P(Sp=T Cl)	Cl	Sp	P(Ra=T ...)
T	0.1	T	T	0.9
F	0.1	T	F	0.5
		F	T	0.3
		F	F	0.8

Sp	Ra	P(We=T ...)
T	T	0.95
T	F	0.25
F	T	0.25
F	F	0.9

Answer (numerical): _____

Q.10 [MSQ · 1 mark · no negative marking] Sampling

Which statements correctly describe how sampling algorithms treat the evidence e ?

- ☐ (A) Likelihood weighting fixes evidence variables to their observed values and samples only non-evidence variables
- ☐ (B) Prior sampling ignores evidence while generating samples
- ☐ (C) Gibbs sampling changes the values of evidence variables during the Markov chain
- ☐ (D) Rejection sampling discards samples that are inconsistent with e

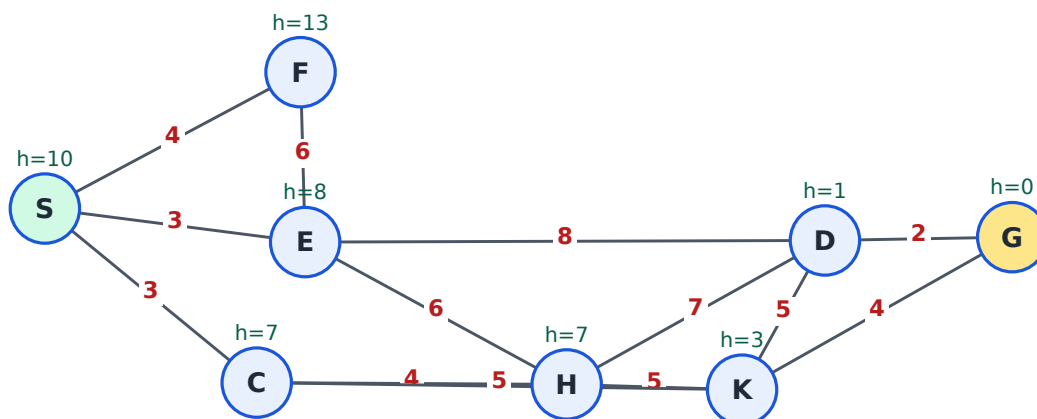
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Uninformed Search

Consider the directed graph with start S and goal G and edges (with costs): $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $B \rightarrow C$ (1), $B \rightarrow G$ (6), $C \rightarrow G$ (3). Uniform-cost search (graph search with an explored set, goal test when a node is selected for expansion) is run from S . How many nodes are expanded before the goal G is selected? Count S , and do not count G .

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

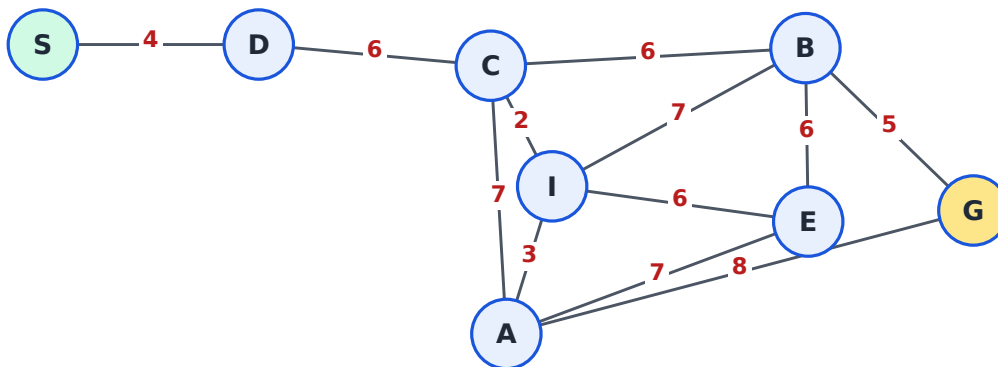
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from S to G on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G ?



Answer (numerical): _____

Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

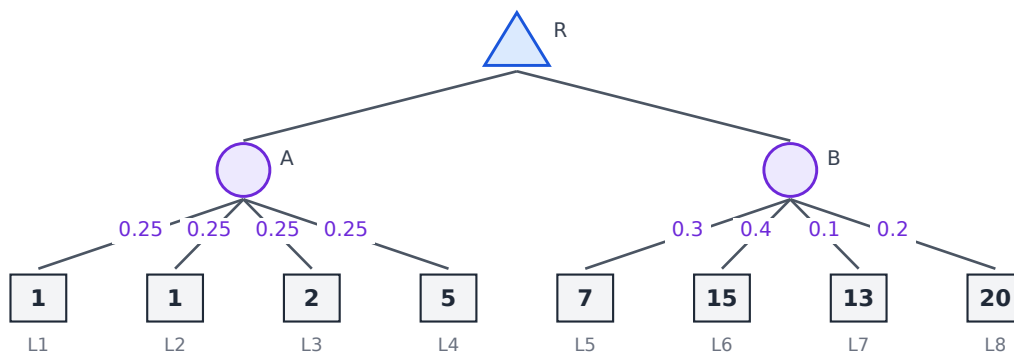
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow D \rightarrow C \rightarrow I \rightarrow B \rightarrow G$
☐ (B) $S \rightarrow D \rightarrow C \rightarrow I \rightarrow A \rightarrow G$
☐ (C) $S \rightarrow D \rightarrow C \rightarrow A \rightarrow G$
☐ (D) $S \rightarrow D \rightarrow C \rightarrow B \rightarrow G$

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



MAX node CHANCE node (edge = probability) terminal utility

Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Adversarial Search

A game tree has a MAX root with three MIN children. Each MIN child has two MAX children, and each of those has two leaves, for 12 leaves in total. The leaves, left to right, are 4, 8, 9, 3, 2, 1, 7, 6, 5, 10, 2, 11. Alpha-beta pruning processes children left to right. How many leaves are **evaluated**?

Answer (numerical): _____

Q.16 [MCQ · 2 marks · negative: -2/3] Predicate Logic

The negation of $\forall x \exists y (P(x, y) \rightarrow Q(y))$ is logically equivalent to

- ☐ (A) $\exists x \forall y (P(x, y) \wedge \neg Q(y))$
☐ (B) $\exists x \forall y (\neg P(x, y) \rightarrow Q(y))$
☐ (C) $\forall x \exists y (P(x, y) \wedge \neg Q(y))$
☐ (D) $\exists x \exists y (P(x, y) \wedge \neg Q(y))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following statements about Horn clauses and Horn-form knowledge bases are TRUE?

- ☐ (A) Entailment of an atom from a Horn KB can be decided in time linear in the size of the KB
- ☐ (B) Every Horn clause has at most one positive literal
- ☐ (C) The clause $\neg p \vee q \vee r$ is a Horn clause
- ☐ (D) The resolvent of two Horn clauses is again a Horn clause

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $\mathbf{F} = \mathbf{Q} \vee (\neg \mathbf{Q} \rightarrow \mathbf{P})$?

- ☐ (A) $(\neg Q \rightarrow P) \vee Q$
- ☐ (B) $\neg(\neg Q \wedge \neg(\neg Q \rightarrow P))$
- ☐ (C) $\neg Q \vee (\neg Q \rightarrow P)$
- ☐ (D) $\neg(\neg Q \vee \neg(\neg Q \rightarrow P))$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

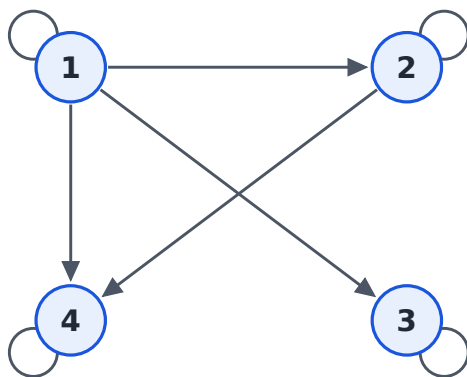
- (1) $\neg Q$
- (2) $(P \wedge \neg Q) \leftrightarrow \neg Q$
- (3) $\neg(R \rightarrow Q)$

Which of the following sentences is/are entailed by KB ($\text{KB} \models \alpha$)?

- ☐ (A) $(\neg R \wedge \neg Q) \leftrightarrow (R \leftrightarrow Q)$
- ☐ (B) $\neg R \vee \neg P$
- ☐ (C) $\neg Q \vee (\neg R \wedge R)$
- ☐ (D) $(Q \leftrightarrow P) \vee (Q \wedge Q)$

Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
- ☐ (B) $\exists x \forall y \neg R(x, y)$
- ☐ (C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
- ☐ (D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$

Q.21 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following are logically equivalent to $p \rightarrow (q \vee r)$?

- ☐ (A) $(p \wedge \neg q) \rightarrow r$
☐ (B) $(p \rightarrow q) \vee (p \rightarrow r)$
☐ (C) $\neg r \rightarrow (p \rightarrow q)$
☐ (D) $(p \rightarrow q) \wedge (p \rightarrow r)$

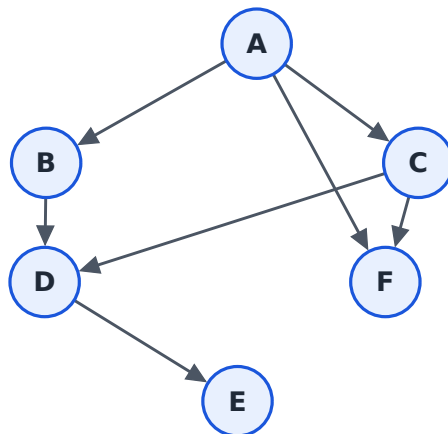
Q.22 [MSQ · 2 marks · no negative marking] Probability for AI

For arbitrary random variables X_1, X_2, X_3, X_4 (no independence assumed), which of the following expressions are always equal to $P(x_1, x_2, x_3, x_4)$?

- ☐ (A) $P(x_4 | x_3) P(x_3 | x_2) P(x_2 | x_1) P(x_1)$
☐ (B) $P(x_4 | x_1, x_2, x_3) P(x_3 | x_1, x_2) P(x_2 | x_1) P(x_1)$
☐ (C) $P(x_1 | x_2, x_3, x_4) P(x_2 | x_3, x_4) P(x_3 | x_4) P(x_4)$
☐ (D) $P(x_4 | x_1, x_2, x_3) P(x_3 | x_2) P(x_2 | x_1) P(x_1)$

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

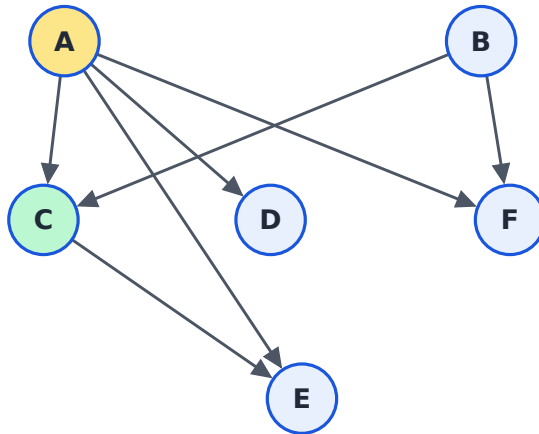
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp A \mid D$
☐ (B) $A \perp E \mid D$
☐ (C) $F \perp B \mid A$
☐ (D) $A \perp E$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

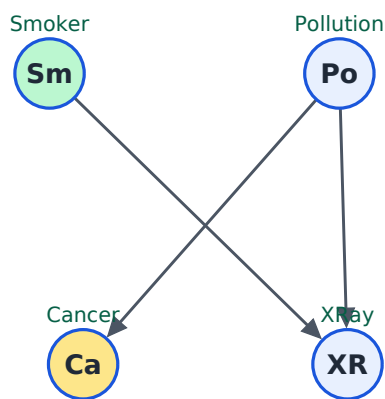
Variable elimination is used to compute $P(\mathbf{C} \mid \mathbf{A} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{A} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E**, **B**, **F**, **D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $P(\mathbf{Smoker} = \mathbf{T} \mid \mathbf{Cancer} = \mathbf{T})$ (round off to 3 decimal places).



$P(\text{Sm}=\text{T})$
0.95

$P(\text{Po}=\text{T})$	Po	$P(\text{Ca}=\text{T} \dots)$
0.1	T	0.9
	F	0.9

Sm	Po	$P(\text{XR}=\text{T} \dots)$
T	T	0.6
T	F	0.9
F	T	0.6
F	F	0.25

Answer (numerical): _____

Answer Key — Mock Test 082

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Informed Search	2.2 to 2.24	14	NAT	2	Adversarial Search	13.39 to 13.41
2	MCQ	1	Adversarial Search	A	15	NAT	2	Adversarial Search	9
3	MCQ	1	Uninformed Search	C	16	MCQ	2	Predicate Logic	A
4	NAT	1	Propositional Logic	2	17	MSQ	2	Propositional Logic	A, B, D
5	NAT	1	Predicate Logic	8	18	MSQ	2	Propositional Logic	A, B
6	NAT	1	Predicate Logic	343	19	MSQ	2	Propositional Logic	A, C
7	NAT	1	Sampling	10	20	MSQ	2	Predicate Logic	D
8	MCQ	1	Conditional Independence	A	21	MSQ	2	Propositional Logic	A, B, C
9	NAT	1	Sampling	0.0842 to 0.0846	22	MSQ	2	Probability for AI	B, C
10	MSQ	1	Sampling	A, B, D	23	MSQ	2	Conditional Independence	A, B, C
11	NAT	2	Uninformed Search	4	24	NAT	2	Exact Inference	2
12	NAT	2	Informed Search	6	25	NAT	2	Exact Inference	0.948 to 0.952
13	MCQ	2	Uninformed Search	D					

Detailed Solutions

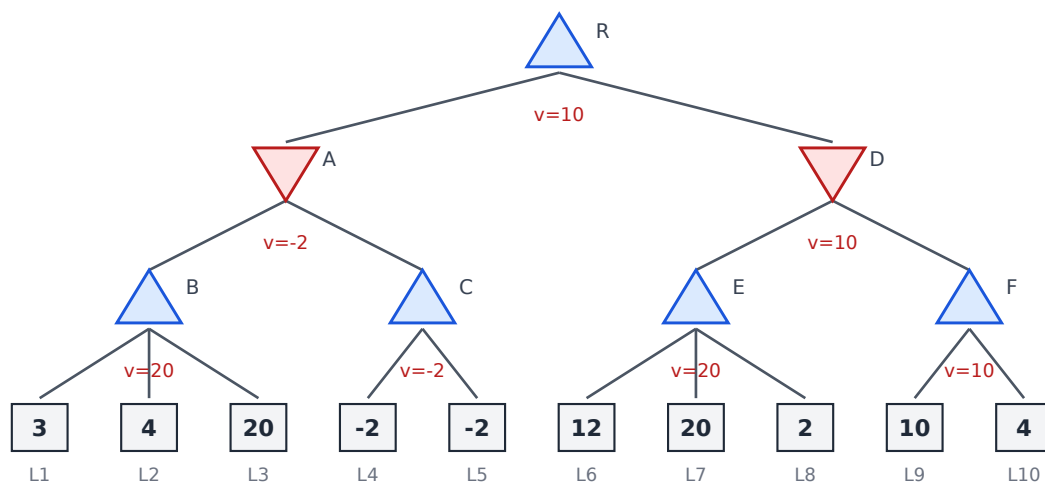
Q.1 [NAT, 1M · Informed Search] **Answer: 2.2 to 2.24**

Solve $b + b^2 + \dots + b^3 = 18$ numerically (bisection): $b^* \approx 2.22$.

Check: $2.22^1 + 2.22^2 + 2.22^3 \approx 18.09 \approx 18$.

Q.2 [MCQ, 1M · Adversarial Search] **Answer: A** (A: Move to D; root value 10)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 20$, $C = -2$, $A = -2$, $E = 20$, $F = 10$, $D = 10$. Root value = **10**; MAX's optimal first move is towards D.

Q.3 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, B, F, I, D, G)

Adjacency lists (alphabetical): B: {F, S}; D: {I}; F: {B, I, J, S}; G: {I}; I: {D, F, G}; J: {F, S}; S: {B, F, J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, B, F, I, D, G**; the path found is $S \rightarrow B \rightarrow F \rightarrow I \rightarrow G$ (4 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Propositional Logic] **Answer: 2**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg Q \vee (Q \leftrightarrow Q)$	$\neg(R \rightarrow \neg P)$	$(\neg Q \vee (Q \leftrightarrow Q)) \wedge \neg(R \rightarrow \neg P)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	T	F	F
F	F	F	T	F	F

Rows where F = T: **2**.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 8**

Reflexivity fixes all 3 diagonal pairs to be present. Symmetry ties (x, y) and (y, x) together for $x \neq y$, leaving 3 independent unordered pairs, each in or out. Total = $2^3 = 8$.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 343**

View R as an 3×3 0/1 matrix. The sentence says R is every row of the relation matrix must be non-empty: $(2^n - 1)$ choices per row $\Rightarrow (2^n - 1)^n$. With $n = 3$: **343**.

Q.7 [NAT, 1M · Sampling] **Answer: 10**

Rejection sampling keeps only samples consistent with the evidence $A = T$: samples 1, 3, 4, 6, 12, 15, 17, 18 (8 accepted, 10 rejected).

Among the accepted samples, $B = T$ in 6. Estimate = $6/8 = 0.75$.

(For this network the exact value is 0.6; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.8 [MCQ, 1M · Conditional Independence] **Answer: A** (A: explaining away (intercausal reasoning))

In the v-structure $B \rightarrow A \leftarrow E$ the causes are marginally independent but become dependent once the common effect is observed. One cause 'explains away' the effect and reduces the posterior of the other cause. The local Markov property concerns independence of a node from non-descendants given parents; naive Bayes is a class-feature model; marginalization is summing out variables.

Q.9 [NAT, 1M · Sampling] **Answer: 0.0842 to 0.0846**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(CI=T) \times P(Sp=F|CI=T) \times P(Ra=T|CI=T, Sp=F) \times P(We=T|Sp=F, Ra=T) \\ = 0.75 \times 0.9 \times 0.5 \times 0.25 = \mathbf{0.084375} \approx 0.0844.$$

Q.10 [MSQ, 1M · Sampling] **Answer: A, B, D**

(A) LW clamps evidence and weights each sample by $\prod P(e_i | \text{parents})$. (B) Prior sampling generates from the prior; rejection sampling then filters these samples (D). (C) is false: in Gibbs sampling evidence variables stay fixed at their observed values; only non-evidence variables are resampled.

Q.11 [NAT, 2M · Uninformed Search] **Answer: 4**

The expansion order by g-value is S(0), A(1), B(3) (reached via A), C(4) (reached via B). After that, the cheapest entry in the frontier is G(7), which is selected and terminates the search. The stale entries B(4) and C(6) are never expanded. Nodes expanded = {S, A, B, C} = **4**.

Q.12 [NAT, 2M · Informed Search] **Answer: 6**

Heuristic values: $h(C)=7$, $h(D)=1$, $h(E)=8$, $h(F)=13$, $h(G)=0$, $h(H)=7$, $h(K)=3$, $h(S)=10$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (10)	C(10), E(11), F(17)
2	C (10)	E(11), K(11), H(14), F(17)
3	E (11)	K(11), D(12), H(14), F(17)
4	K (11)	D(12), G(12), H(14), F(17)
5	D (12)	G(12), H(14), F(17)
6	G (12)	H(14), F(17) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **C** $\rightarrow$ **K** $\rightarrow$ **G** with cost **12**; nodes expanded = **6** (S, C, E, K, D, G).

Check: $h(n) = \lceil c \cdot h^*(n) \rceil$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 12$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.13 [MCQ, 2M · Uninformed Search] **Answer: D** (D: $S \rightarrow D \rightarrow C \rightarrow B \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	D(4)
2	D (4)	C(10)
3	C (10)	I(12), B(16), A(17)
4	I (12)	A(15), B(16), E(18)
5	A (15)	B(16), E(18), G(23)
6	B (16)	E(18), G(21)
7	E (18)	G(21)
8	G (21)	← goal, stop

Path returned: **S** → **D** → **C** → **B** → **G**, cost = 4 + 6 + 6 + 5 = **21**. Nodes expanded (including S and G): **8** (S, D, C, I, A, B, E, G).

Q.14 [NAT, 2M · Adversarial Search] **Answer: 13.39 to 13.41**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

$$\bullet A = 0.25 \times 1 + 0.25 \times 1 + 0.25 \times 2 + 0.25 \times 5 = 2.25$$

$$\bullet B = 0.3 \times 7 + 0.4 \times 15 + 0.1 \times 13 + 0.2 \times 20 = 13.4$$

$$\text{Root} = \max(2.25, 13.4) = \mathbf{13.4}.$$

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [NAT, 2M · Adversarial Search] **Answer: 9**

MIN_1 : $\text{MAX}(4, 8) = 8$, so $\beta = 8$. The next MAX node sees $9 \geq 8$ and prunes leaf 3. $\text{MIN}_1 = 8$, so root $\alpha = 8$.

MIN_2 : $\text{MAX}(2, 1) = 2 \leq \alpha$, so the MAX node with leaves (7, 6) is pruned. MIN_3 : $\text{MAX}(5, 10) = 10$, so $\beta = 10$.

$\text{MAX}(2, 11)$: leaf 2 does not cut and $11 \geq 10$ cuts, but no siblings remain, so both are evaluated. $\text{MIN}_3 = 10$, and the root value is 10. Pruned: 3, 7, 6. Evaluated: $12 - 3 = \mathbf{9}$.

Q.16 [MCQ, 2M · Predicate Logic] **Answer: A** ($A: \exists x \forall y (P(x, y) \wedge \neg Q(y))$)

Push $\neg$ inward: $\neg \forall x \exists y \phi \equiv \exists x \neg \exists y \phi \equiv \exists x \forall y \neg \phi$. Then $\neg(P \rightarrow Q) \equiv P \wedge \neg Q$, giving $\exists x \forall y (P(x, y) \wedge \neg Q(y))$.

Option B uses $\neg P \rightarrow Q \equiv P \vee Q$, which is not $\neg(P \rightarrow Q)$. Option C does not flip the quantifiers. Option D flips only one quantifier correctly ($\exists y$ should become $\forall y$).

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, B, D**

(A) True: forward chaining with premise counters runs in linear time. (B) True: this is the definition. (C) False: q and r are both positive. (D) True: if the clashing positive literal comes from C_1 (≤ 1 positive) and the negative from C_2 , the resolvent contains C_1 's other literals (all negative) plus C_2 's remaining literals (≤ 1 positive), so it has at most one positive literal.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A, B** ($A: (\neg Q \rightarrow P) \vee Q$; $B: \neg(\neg Q \wedge \neg(\neg Q \rightarrow P))$)

Compare truth tables with $F = Q \vee (\neg Q \rightarrow P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(\neg Q \rightarrow P) \vee Q$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $\neg(\neg Q \wedge \neg(\neg Q \rightarrow P))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $\neg Q \vee (\neg Q \rightarrow P)$ — differs at $P=F, Q=F$ ($F=F$, option = T) $\Rightarrow$ not equivalent.

(D) $\neg(\neg Q \vee \neg(\neg Q \rightarrow P))$ — differs at $P=T, Q=F$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: A, C** ($A: (\neg R \wedge \neg Q) \leftrightarrow (R \leftrightarrow Q)$; $C: \neg Q \vee (\neg R \wedge R)$)

$\text{KB} \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
T	F	T	T

(A) $(\neg R \wedge \neg Q) \leftrightarrow (R \leftrightarrow Q)$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $\neg R \vee \neg P$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.

(C) $\neg Q \vee (\neg R \wedge R)$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $(Q \leftrightarrow P) \vee (Q \wedge Q)$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.

Q.20 [MSQ, 2M · Predicate Logic] Answer: D ($D: \forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$)

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.

(B) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

(C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(2,3)$ nor $R(3,2)$.

(D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **TRUE** — no pair of distinct elements is related both ways.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.21 [MSQ, 2M · Propositional Logic] Answer: A, B, C ($A: (p \wedge \neg q) \rightarrow r$; $B: (p \rightarrow q) \vee (p \rightarrow r)$; $C: \neg r \rightarrow (p \rightarrow q)$)

$p \rightarrow (q \vee r) \equiv \neg p \vee q \vee r$. (A) $(p \wedge \neg q) \rightarrow r \equiv \neg p \vee q \vee r$. (B) $(\neg p \vee q) \vee (\neg p \vee r) \equiv \neg p \vee q \vee r$. (C) $\neg r \rightarrow (\neg p \vee q) \equiv r \vee \neg p \vee q$. (D) $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$, which is false at $p=T, q=T, r=F$ whereas the original is true there.

Q.22 [MSQ, 2M · Probability for AI] Answer: B, C

The chain rule $P(x_1, \dots, x_n) = \prod_i P(x_i | x_1, \dots, x_{i-1})$ holds for any ordering of the variables. (B) is the chain rule for order X_1, X_2, X_3, X_4 ; (C) is the chain rule for the reverse order X_4, X_3, X_2, X_1 . (A) drops conditioning variables, which is valid only under Markov-chain independence assumptions. (D) replaces $P(x_3 | x_1, x_2)$ by $P(x_3 | x_2)$, which needs $X_3 \perp X_1 | X_2$, not generally true.

Q.23 [MSQ, 2M · Conditional Independence] Answer: A, B, C ($A: E \perp A | D$; $B: A \perp E | D$; $C: F \perp B | A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $E \perp A | D$: **TRUE** — $E \leftarrow D \leftarrow B \leftarrow A$ (blocked: observed chain node D); $E \leftarrow D \leftarrow C \leftarrow A$ (blocked: observed chain node D); $E \leftarrow D \leftarrow C \rightarrow F \leftarrow A$ (blocked: observed chain node D).

(B) $A \perp E | D$: **TRUE** — $A \rightarrow B \rightarrow D \rightarrow E$ (blocked: observed chain node D); $A \rightarrow C \rightarrow D \rightarrow E$ (blocked: observed chain node D); $A \rightarrow F \leftarrow C \rightarrow D \rightarrow E$ (blocked: collider F with neither it nor a descendant observed).

(C) $F \perp B | A$: **TRUE** — $F \leftarrow A \rightarrow B$ (blocked: observed fork node A); $F \leftarrow A \rightarrow C \rightarrow D \leftarrow B$ (blocked: observed fork node A); $F \leftarrow C \leftarrow A \rightarrow B$ (blocked: observed fork node A); $F \leftarrow C \rightarrow D \leftarrow B$ (blocked: collider D with neither it nor a descendant observed).

(D) $A \perp E$: **FALSE** — the path $A \rightarrow B \rightarrow D \rightarrow E$ is active given $\{\}$.

Q.24 [NAT, 2M · Exact Inference] Answer: 2

Initial factors (one per CPT, evidence removed from scopes): $f()$, $f(B)$, $f(B, C)$, $f(D)$, $f(C, E)$, $f(B, F)$.

Eliminating Z : multiply all factors whose scope contains Z , then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(C, E)$	$\{C, E\}$	$f(C)$
B	$f(B)$, $f(B, C)$, $f(B, F)$	$\{B, C, F\}$	$f(C, F)$
F	$f(C, F)$	$\{C, F\}$	$f(C)$
D	$f(D)$	$\{D\}$	$f()$

The largest factor produced has **2** variable(s) (created when eliminating B); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over C and normalise.

Q.25 [NAT, 2M · Exact Inference] Answer: 0.948 to 0.952

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution}) \cdot P(\text{Cancer} | \text{Pollution}) \cdot P(\text{XRay} | \text{Smoker}, \text{Pollution})$.

Inference by enumeration: $P(\text{Smoker} | e) = \alpha \sum_{\text{hidden}} P(\text{Smoker}, e, \text{hidden})$.

Hidden: Po, XR	Sm	product of CPT entries ($Sm \cdot Po \cdot Ca \cdot XR$)	value
T, T	T	$0.95 \times 0.1 \times 0.9 \times 0.6$	0.0513
T, F	T	$0.95 \times 0.1 \times 0.9 \times 0.4$	0.0342
F, T	T	$0.95 \times 0.9 \times 0.9 \times 0.9$	0.69255
F, F	T	$0.95 \times 0.9 \times 0.9 \times 0.1$	0.07695
T, T	F	$0.05 \times 0.1 \times 0.9 \times 0.6$	0.0027
T, F	F	$0.05 \times 0.1 \times 0.9 \times 0.4$	0.0018
F, T	F	$0.05 \times 0.9 \times 0.9 \times 0.25$	0.01013
F, F	F	$0.05 \times 0.9 \times 0.9 \times 0.75$	0.03038

$P(\text{Smoker}=T, e) = 0.855$; $P(\text{Smoker}=F, e) = 0.045$; $P(e) = 0.9$.

$P(\text{Smoker}=T | e) = 0.855 / 0.9 = \mathbf{0.95} \approx 0.950$.

Mock Test 083

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 8 / 12	13 / 12 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

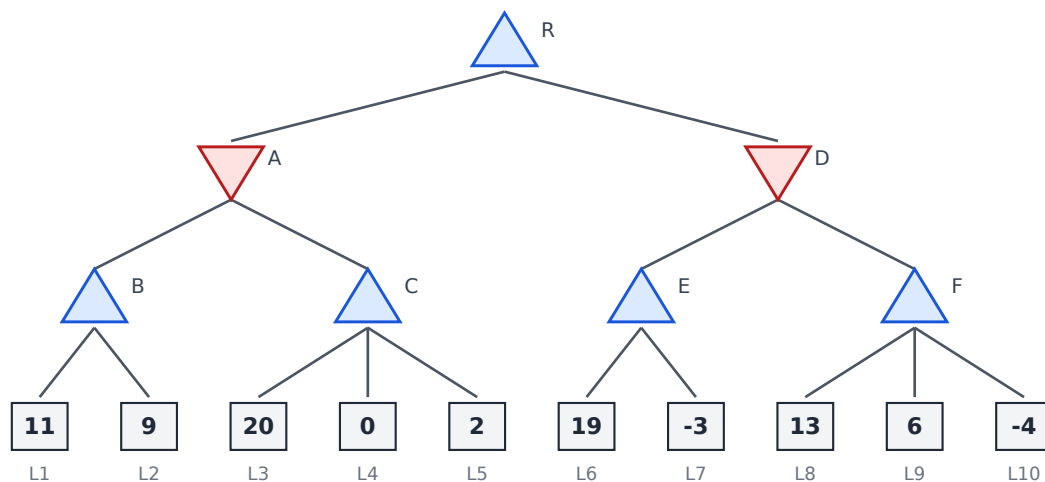
Q.1 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

A search tree has uniform branching factor $b = 5$; the shallowest goal is at depth $d = 5$ and happens to be the **last** node generated at that depth. BFS applies the goal test when a node is **generated**. How many nodes are generated in total (do **not** count the root)?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



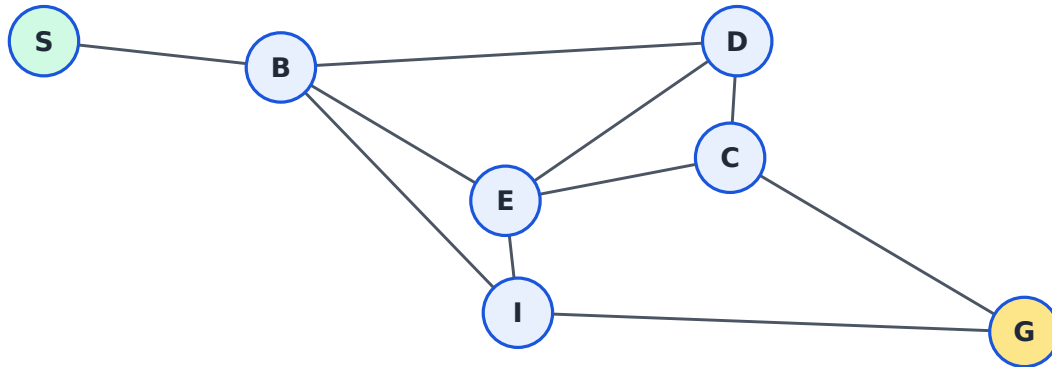
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, I, E, D, G
☐ (B) S, B, D, E, I, C, G
☐ (C) S, B, D, C, E, I, G
☐ (D) S, B, D, E, C, I, G

Q.4 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = (R \wedge (P \rightarrow \neg Q)) \vee (P \vee R) ?$$

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic

Interpret $\text{Div}(x, y)$ as 'x divides y' over the domain $D = \{1, 2, \dots, 12\}$. For how many $x \in D$ is the formula $\neg(x = 1) \wedge \forall y (\text{Div}(y, x) \rightarrow (y = 1 \vee y = x))$ true?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\exists x (P(x) \wedge Q(x))$ true?

Answer (numerical): _____

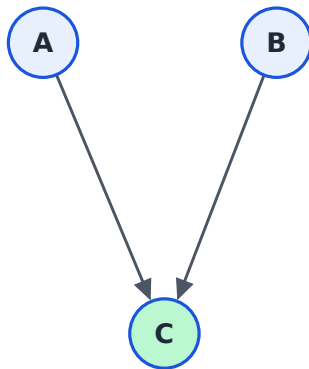
Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

How many models over a, b, c does the CNF $(a \vee b \vee c) \wedge (\neg a \vee \neg b) \wedge (\neg b \vee \neg c) \wedge (\neg a \vee \neg c)$ have?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(\mathbf{C} = \mathbf{T})$ (round off to 3 decimal places).



$P(A=T)$	$P(B=T)$	A	B	$P(C=T \dots)$
0.75	0.05	T	T	0.05
		T	F	0.75
		F	T	0.95
		F	F	0.5

Answer (numerical): _____

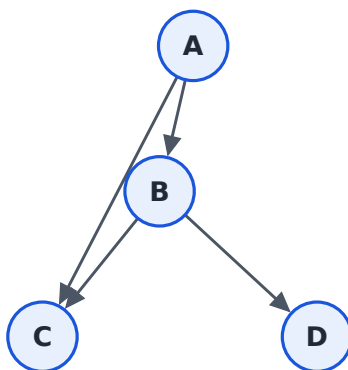
Q.9 [MSQ · 1 mark · no negative marking] Conditional Independence

Which of the following statements about d-separation in three-node structures are correct?

- ☐ (A) In the chain $X \rightarrow Y \rightarrow Z$, X and Z are d-separated given Y
☐ (B) In the fork $X \leftarrow Y \rightarrow Z$, X and Z are d-separated given Y
☐ (C) In the collider $X \rightarrow Y \leftarrow Z$, X and Z are d-separated given Y
☐ (D) In the collider $X \rightarrow Y \leftarrow Z$, X and Z are d-separated given $\emptyset$

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample $\mathbf{A=F, B=F, C=T, D=F}$? (round off to 4 decimal places)



$P(A=T)$	A	$P(B=T \dots)$	A	B	$P(C=T \dots)$
0.25	T	0.8	T	T	0.5
	F	0.1	T	F	0.5
			F	T	0.05
			F	F	0.9

B	$P(D=T \dots)$
T	0.3
F	0.05

Answer (numerical): _____

Section B — 2-mark questions

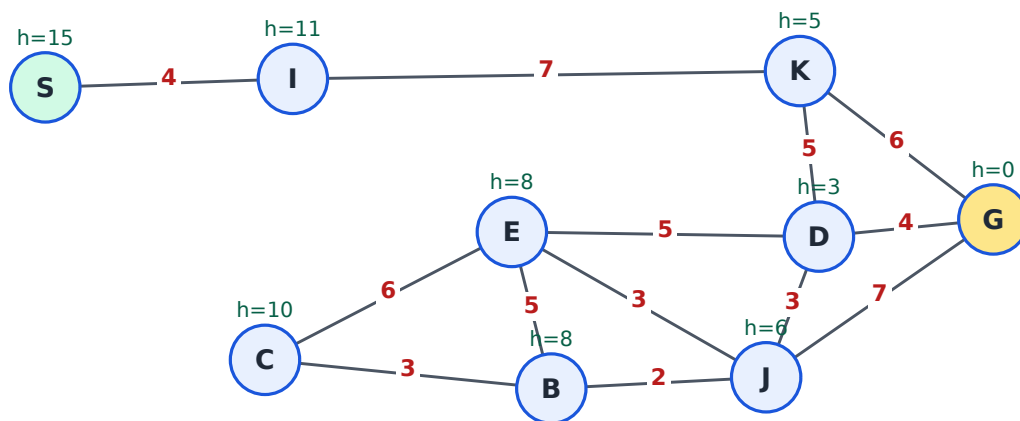
Q.11 [MSQ · 2 marks · no negative marking] Informed Search

The 8-puzzle rule is: a tile can move from square X to square Y if X is adjacent to Y and Y is blank. Which of the following statements about relaxations are TRUE?

- ☐ (A) Dropping both conditions (a tile can move anywhere in one step) gives the misplaced-tiles heuristic h_1
- ☐ (B) Dropping only the 'Y is blank' condition gives the Manhattan-distance heuristic h_2
- ☐ (C) Dropping only the adjacency condition gives the Manhattan-distance heuristic h_2
- ☐ (D) The heuristics obtained from these relaxations are admissible for the original 8-puzzle

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – A*

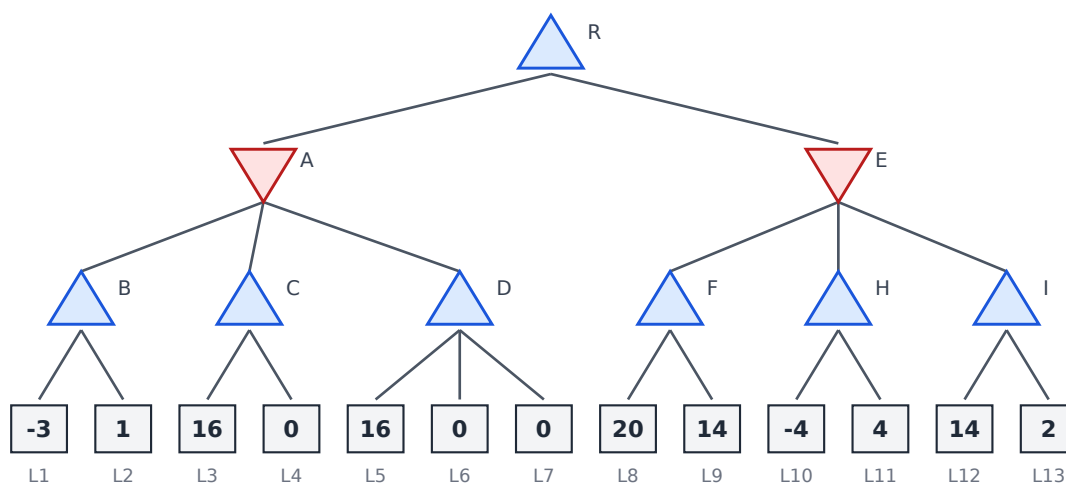
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node J is expanded.
- ☐ (B) Node S is expanded first.
- ☐ (C) The path returned has cost 16.
- ☐ (D) Node C is expanded.

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



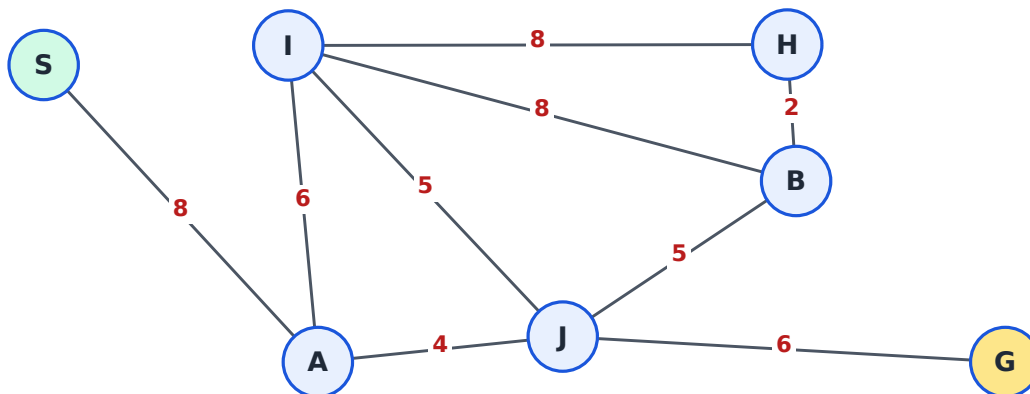
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

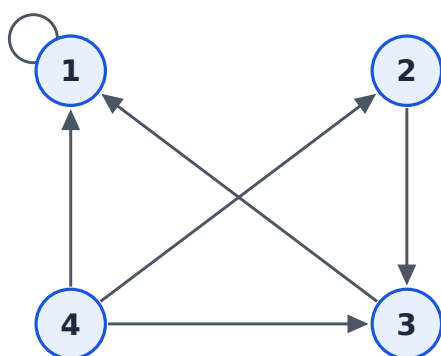
Q.15 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Consider the directed graph with start S and goal G and edges (with costs): S→A (1), S→B (4), A→B (2), A→C (5), B→C (1), B→G (6), C→G (3). Uniform-cost search (graph search, goal test when a node is selected for expansion) is run from S. Which path does it return?

- ☐ (A) S→B→G (cost 10)
☐ (B) S→A→C→G (cost 9)
☐ (C) S→B→C→G (cost 8)
☐ (D) S→A→B→C→G (cost 7)

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (C) $\exists x R(x, x)$
☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic

Here x and y are variables, and a, b are constants. Which of the following pairs of atoms are unifiable (with the occurs check)?

- ☐ (A) $P(x, f(x))$ and $P(f(y), y)$
- ☐ (B) $Q(a, x, f(x))$ and $Q(y, b, f(b))$
- ☐ (C) $R(x, g(x))$ and $R(g(y), g(g(a)))$
- ☐ (D) $S(f(a), x)$ and $S(x, f(b))$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (Q \leftrightarrow Q) \leftrightarrow P$?

- ☐ (A) $((Q \leftrightarrow Q) \rightarrow P) \wedge (\neg P \rightarrow \neg(Q \leftrightarrow Q))$
- ☐ (B) $(Q \leftrightarrow Q) \leftrightarrow \neg P$
- ☐ (C) $((Q \leftrightarrow Q) \wedge P) \vee ((Q \leftrightarrow Q) \wedge \neg P)$
- ☐ (D) $((Q \leftrightarrow Q) \rightarrow P) \wedge (P \rightarrow (Q \leftrightarrow Q))$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (\neg Q \vee \neg R \vee S)$, $C2: (Q \vee R)$, $C3: (P \vee Q \vee S)$, $C4: (R \vee S)$, $C5: (\neg P \vee Q)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
- ☐ (B) $(\neg P \vee S)$
- ☐ (C) $(P \vee \neg R \vee S)$
- ☐ (D) $(\neg S)$

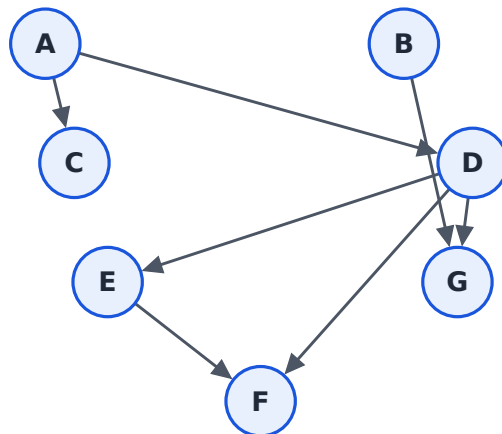
Q.20 [MCQ · 2 marks · negative: -2/3] Conditional Independence

Consider the Bayesian network (DAG) with edges $P \rightarrow Q$, $P \rightarrow R$, $Q \rightarrow S$, $R \rightarrow S$, $S \rightarrow T$, $U \rightarrow R$. Which pair of variables is **NOT** d-separated by the given set?

- ☐ (A) T and P given $\{Q, R\}$
- ☐ (B) U and Q given $\{P\}$
- ☐ (C) U and P given $\{T\}$
- ☐ (D) Q and R given $\{P\}$

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

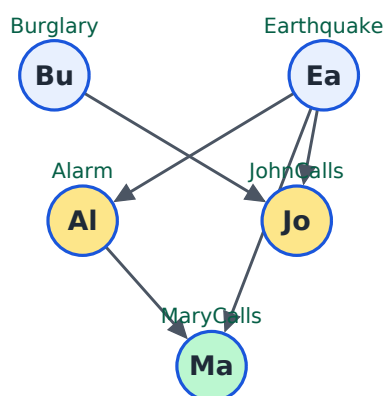
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp B \mid C$
☐ (B) $A \perp C$
☐ (C) $B \perp E \mid C, G$
☐ (D) $F \perp D$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls, Ma = MaryCalls.) Compute $P(\text{MaryCalls} = T \mid \text{JohnCalls} = T, \text{Alarm} = T)$ (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.5

$P(\text{Ea}=T)$
0.2

Ea	$P(\text{Al}=T \dots)$
T	0.3
F	0.6

Bu	Ea	$P(\text{Jo}=T \mid \text{Bu}, \text{Ea})$	Al	$P(\text{Ma}=T \mid \text{Al}, \text{Jo})$
T	T	0.75	T	0.7
T	F	0.9	F	0.3
F	T	0.5	F	0.25
F	F	0.75	F	0.2

Answer (numerical): _____

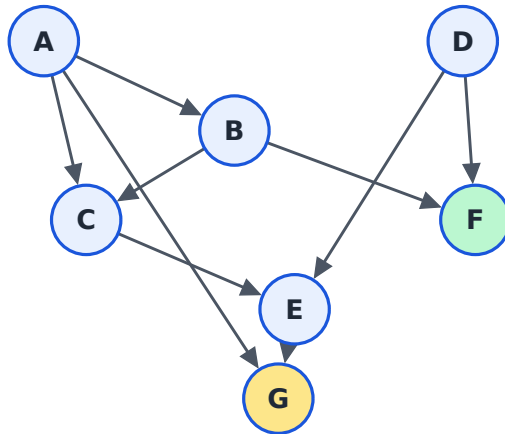
Q.23 [MCQ · 2 marks · negative: -2/3] Variable Elimination

A Bayesian network over Boolean variables has edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$, $D \rightarrow E$. $P(E)$ is computed by variable elimination with no evidence. For which elimination ordering does **every** new factor created by summing out a variable contain at most two variables?

- ☐ (A) A, B, C, D
☐ (B) D, C, B, A
☐ (C) B, C, A, D
☐ (D) C, B, A, D

Q.24 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(F \mid G = T)$ in the Bayesian network below (Boolean variables; evidence $G = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, B, C, D, E**. What is the scope of the new factor produced when **D** is summed out?



- ☐ (A) $f()$
☐ (B) $f(E)$
☐ (C) $f(E, F)$
☐ (D) $f(D, E, F)$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(C = T \mid B = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	T	F	T	F
2	F	F	F	F
3	T	F	T	F
4	T	F	T	F
5	T	F	F	F
6	T	T	T	T
7	T	T	F	T
8	F	F	T	F
9	F	F	T	F
10	F	F	F	F
11	T	T	T	T
12	F	F	T	F
13	T	F	F	F
14	F	T	T	F
15	T	F	F	F
16	F	T	T	T
17	T	F	F	F
18	T	F	F	F
19	F	F	F	F
20	T	T	T	T

Answer (numerical): _____

Answer Key — Mock Test 083

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	3905	14	NAT	2	Uninformed Search	18
2	NAT	1	Adversarial Search	13	15	MCQ	2	Uninformed Search	D
3	MCQ	1	Uninformed Search	B	16	MSQ	2	Predicate Logic	C
4	NAT	1	Propositional Logic	6	17	MSQ	2	Predicate Logic	B, C
5	NAT	1	Predicate Logic	5	18	MSQ	2	Propositional Logic	A, D
6	NAT	1	Predicate Logic	37	19	MSQ	2	Propositional Logic	B, C
7	NAT	1	Propositional Logic	3	20	MCQ	2	Conditional Independence	C
8	NAT	1	Exact Inference	0.665 to 0.669	21	MSQ	2	Conditional Independence	A
9	MSQ	1	Conditional Independence	A, B, D	22	NAT	2	Exact Inference	0.287 to 0.291
10	NAT	1	Sampling	0.5769 to 0.5773	23	MCQ	2	Variable Elimination	A
11	MSQ	2	Informed Search	A, B, D	24	MCQ	2	Exact Inference	C
12	MSQ	2	Informed Search	B	25	NAT	2	Sampling	0.82 to 0.84
13	NAT	2	Adversarial Search	4					

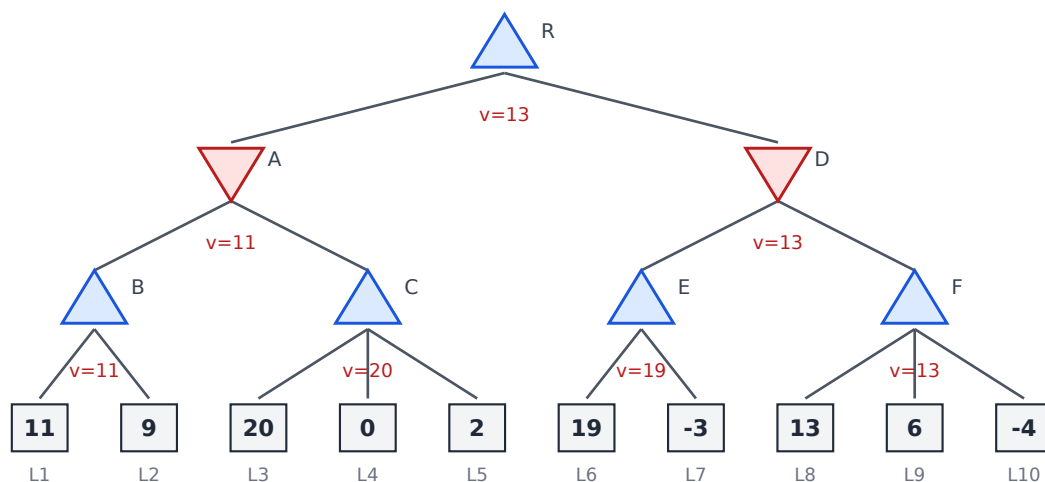
Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] Answer: 3905

With goal test at generation, BFS generates every node at depths 1 ... d before (and including) the goal (the worst case). $N = b + b^2 + \dots + b^d = 5 + 25 + 125 + 625 + 3125 = \mathbf{3905}$. This is the $O(b^d)$ of AIMA.

Q.2 [NAT, 1M · Adversarial Search] Answer: 13

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 11$, $C = 20$, $A = 11$, $E = 19$, $F = 13$, $D = 13$. Root value = **13**; MAX's optimal first move is towards D.

Q.3 [MCQ, 1M · Uninformed Search] Answer: B (B: S, B, D, E, I, C, G)

Adjacency lists (alphabetical): B: {D, E, I, S}; C: {D, E, G}; D: {B, C, E}; E: {B, C, D, I}; G: {C, I}; I: {B, E, G}; S: {B}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B]
2	B	[D, E, I]
3	D	[E, I, C]
4	E	[I, C]
5	I	[C, G]
6	C	[G]

Step	Expanded	Frontier (FIFO) after expansion
7	G	[] ← goal popped, stop

Expansion order: **S, B, D, E, I, C, G**. The path returned is $S \rightarrow B \rightarrow I \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [NAT, 1M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	$R \wedge (P \rightarrow \neg Q)$	$P \vee R$	$(R \wedge (P \rightarrow \neg Q)) \vee (P \vee R)$
T	T	T	F	T	T
T	T	F	F	T	T
T	F	T	T	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	F	F	F
F	F	T	T	T	T
F	F	F	F	F	F

Rows where F = T: **6**.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 5**

The formula says $x \neq 1$ and the only divisors of x in D are 1 and x itself — i.e. x is prime. (Every divisor of $x \leq 12$ lies in D , so the restriction to D loses nothing.) The primes in $\{1, \dots, 12\}$ are 2, 3, 5, 7, 11, so the answer is 5.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 37**

Interpret P and Q as subsets of the domain; complement: no element has both, 3 combos each $\Rightarrow 4^n - 3^n$. With $n = 3$ this gives **37** (verified by enumerating all interpretations).

Q.7 [NAT, 1M · Propositional Logic] **Answer: 3**

The first clause says at least one of a, b, c is true. The three binary clauses say no two are simultaneously true. Together they encode 'exactly one of a, b, c is true', which has 3 models: (T,F,F), (F,T,F), (F,F,T).

Q.8 [NAT, 1M · Exact Inference] **Answer: 0.665 to 0.669**

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
$A=T, B=T$	0.00188
$A=T, B=F$	0.53437
$A=F, B=T$	0.01188
$A=F, B=F$	0.11875

Sum = **0.6669** $\approx$ 0.667.

Q.9 [MSQ, 1M · Conditional Independence] **Answer: A, B, D**

An observed non-collider blocks a path, so (A) and (B) are true. A collider blocks the path when neither it nor any descendant is observed, so (D) is true. Observing the collider activates the path, so (C) is false.

Q.10 [NAT, 1M · Sampling] **Answer: 0.5769 to 0.5773**

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(A=F) \times P(B=F \mid A=F) \times P(C=T \mid A=F, B=F) \times P(D=F \mid B=F)$
 $= 0.75 \times 0.9 \times 0.9 \times 0.95 = \mathbf{0.577125} \approx 0.5771$.

Q.11 [MSQ, 2M · Informed Search] **Answer: A, B, D**

(A) is true: if a tile can jump anywhere, each misplaced tile needs exactly 1 move. (B) is true: if tiles move to adjacent squares regardless of the blank, each tile needs exactly its Manhattan distance. (C) is false: dropping adjacency but keeping 'Y is blank' gives Gaschnig's heuristic, not Manhattan distance. (D) is true: optimal costs of relaxed problems are admissible.

Q.12 [MSQ, 2M · Informed Search] **Answer: B** (B: Node S is expanded first.)

Heuristic values: $h(B)=8$, $h(C)=10$, $h(D)=3$, $h(E)=8$, $h(G)=0$, $h(I)=11$, $h(J)=6$, $h(K)=5$, $h(S)=15$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	I(15)
2	I (15)	K(16)
3	K (16)	G(17), D(19)
4	G (17)	D(19) ← goal, stop

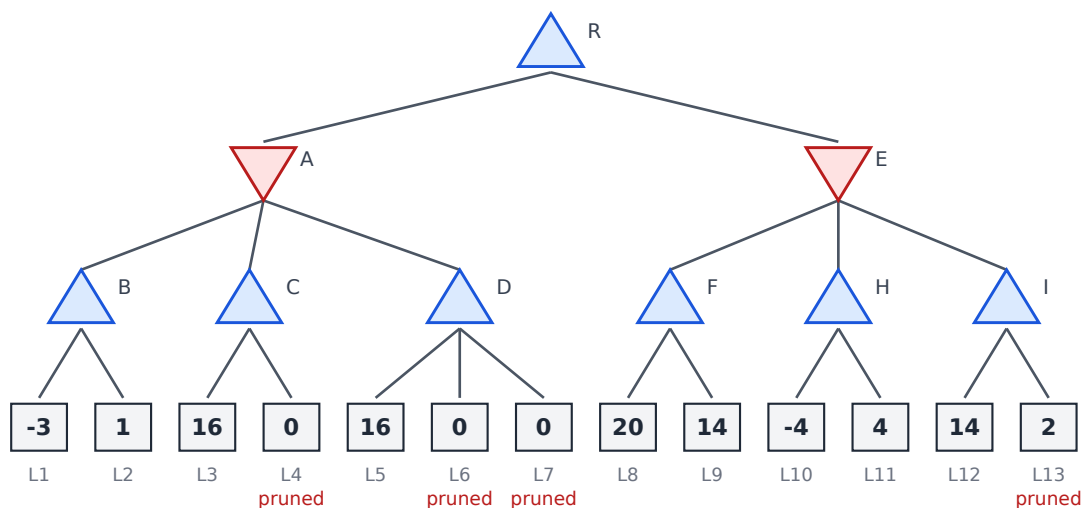
Path returned: **S → I → K → G** with cost **17**; nodes expanded = **4** (S, I, K, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 17$. For comparison UCS ($h = 0$) would expand 5 nodes.

Q.13 [NAT, 2M · Adversarial Search] **Answer: 4**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $16 \geq \beta = 1 \Rightarrow$ prune remaining children (leaves L4).
- At MAX node D: value $16 \geq \beta = 1 \Rightarrow$ prune remaining children (leaves L6, L7).
- At MAX node I: value $14 \geq \beta = 4 \Rightarrow$ prune remaining children (leaves L13).



Leaves evaluated (9): L1, L2, L3, L5, L8, L9, L10, L11, L12. Leaves pruned (4): L4, L6, L7, L13. Root value = 4 (same as plain minimax, 4 — pruning never changes the root value).

Q.14 [NAT, 2M · Uninformed Search] **Answer: 18**

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	A(8)
2	A (8)	J(12), I(14)
3	J (12)	I(14), B(17), G(18)
4	I (14)	B(17), G(18), H(22)
5	B (17)	G(18), H(19)
6	G (18)	H(19) ← goal, stop

Path returned: **S → A → J → G**, cost = $8 + 4 + 6 = \mathbf{18}$. Nodes expanded (including S and G): **6** (S, A, J, I, B, G).

Q.15 [MCQ, 2M · Uninformed Search] **Answer: D** (D: $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$ (cost 7))

UCS expands nodes in order of g . Pop S(0): push A(1), B(4). Pop A(1): push B(3), C(6). Pop B(3): push C(4), G(9). Pop C(4): push G(7). The stale entries B(4) and C(6) are skipped. Pop G(7) and return **S → A → B → C → G with cost 7**. The other paths are real paths in the graph, but each costs more than 7. UCS returns the cheapest path because it tests the goal on expansion.

Q.16 [MSQ, 2M · Predicate Logic] Answer: C (C: $\exists x R(x, x)$)

$R = \{(1,1), (2,3), (3,1), (4,1), (4,2), (4,3)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(2,3) \wedge R(3,1)$ but $\neg R(2,1)$.

(B) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(4,3)$ holds but $R(3,4)$ does not.

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

(D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **FALSE** — 1 has no edge to a different element.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Predicate Logic] Answer: B, C (B: $Q(a, x, f(x))$ and $Q(y, b, f(b))$; C: $R(x, g(x))$ and $R(g(y), g(g(a)))$)

(A) $x/f(y)$; then $f(f(y))$ must unify with y , which fails the occurs check. (B) a/y gives y/a , x/b , then $f(b) = f(b)$: MGU $\{y/a, x/b\}$. (C) $x/g(y)$, then $g(g(y))$ vs $g(g(a))$ gives y/a : MGU $\{x/g(a), y/a\}$. (D) $x/f(a)$, then $f(a)$ must equal $f(b)$, which fails since $a \neq b$ are distinct constants.

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, D (A: $((Q \leftrightarrow Q) \rightarrow P) \wedge (\neg P \rightarrow \neg(Q \leftrightarrow Q))$; D: $((Q \leftrightarrow Q) \rightarrow P) \wedge (P \rightarrow (Q \leftrightarrow Q))$)

Compare truth tables with $F = (Q \leftrightarrow Q) \leftrightarrow P$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $((Q \leftrightarrow Q) \rightarrow P) \wedge (\neg P \rightarrow \neg(Q \leftrightarrow Q))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(B) $(Q \leftrightarrow Q) \leftrightarrow \neg P$ — differs at $P=T, Q=T$ ($F=T$, option $=F$) $\Rightarrow$ not equivalent.

(C) $((Q \leftrightarrow Q) \wedge P) \vee ((Q \leftrightarrow Q) \wedge \neg P)$ — differs at $P=F, Q=T$ ($F=F$, option $=T$) $\Rightarrow$ not equivalent.

(D) $((Q \leftrightarrow Q) \rightarrow P) \wedge (P \rightarrow (Q \leftrightarrow Q))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.19 [MSQ, 2M · Propositional Logic] Answer: B, C (B: $(\neg P \vee S)$; C: $(P \vee \neg R \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 15 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(\neg P \vee S)$ is derivable:

1. resolve $(\neg Q \vee \neg R \vee S)$ and $(\neg P \vee Q)$ on $Q \Rightarrow (\neg P \vee \neg R \vee S)$
2. resolve $(R \vee S)$ and $(\neg P \vee \neg R \vee S)$ on $R \Rightarrow (\neg P \vee S)$

(C) $(P \vee \neg R \vee S)$ is derivable:

1. resolve $(\neg Q \vee \neg R \vee S)$ and $(P \vee Q \vee S)$ on $Q \Rightarrow (P \vee \neg R \vee S)$

(D) $(\neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.20 [MCQ, 2M · Conditional Independence] Answer: C (C: U and P given $\{T\}$)

(C): T is a descendant of collider R on the path $U \rightarrow R \leftarrow P$ ($R \rightarrow S \rightarrow T$), so observing T activates it: not

d-separated. (A): every path from T to P goes through Q or R as non-colliders ($T \leftarrow S \leftarrow Q \leftarrow P$, $T \leftarrow S \leftarrow R \leftarrow P$), both observed: blocked. (B): $U \rightarrow R \leftarrow P \rightarrow Q$ is blocked at observed P and at unobserved collider R ; $U \rightarrow R \rightarrow S \leftarrow Q$ is blocked at unobserved collider S . (D): fork P is observed and collider S is unobserved with no observed descendants: blocked.

Q.21 [MSQ, 2M · Conditional Independence] Answer: A (A: $A \perp B \mid C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp B \mid C$: **TRUE** — $A \rightarrow D \rightarrow G \leftarrow B$ (blocked: collider G with neither it nor a descendant observed).

(B) $A \perp C$: **FALSE** — the path $A \rightarrow C$ is active given $\{ \}$.

(C) $B \perp E \mid C, G$: **FALSE** — the path $B \rightarrow G \leftarrow D \rightarrow E$ is active given $\{C, G\}$.

(D) $F \perp D$: **FALSE** — the path $F \leftarrow D$ is active given $\{ \}$.

Q.22 [NAT, 2M · Exact Inference] Answer: 0.287 to 0.291

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot P(\text{Alarm} \mid \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Earthquake}) \cdot P(\text{MaryCalls} \mid \text{Earthquake}, \text{Alarm})$.

Inference by enumeration: $P(\text{MaryCalls} \mid e) = \alpha \sum_{\text{hidden}} P(\text{MaryCalls}, e, \text{hidden})$.

Hidden: Bu, Ea	Ma	product of CPT entries (Bu · Ea · Al · Jo · Ma)	value
T, T	T	$0.5 \times 0.2 \times 0.3 \times 0.75 \times 0.7$	0.01575
T, F	T	$0.5 \times 0.8 \times 0.6 \times 0.9 \times 0.25$	0.054
F, T	T	$0.5 \times 0.2 \times 0.3 \times 0.5 \times 0.7$	0.0105
F, F	T	$0.5 \times 0.8 \times 0.6 \times 0.75 \times 0.25$	0.045
T, T	F	$0.5 \times 0.2 \times 0.3 \times 0.75 \times 0.3$	0.00675

Hidden: Bu, Ea	Ma	product of CPT entries (Bu · Ea · Al · Jo · Ma)	value
T, F	F	$0.5 \times 0.8 \times 0.6 \times 0.9 \times 0.75$	0.162
F, T	F	$0.5 \times 0.2 \times 0.3 \times 0.5 \times 0.3$	0.0045
F, F	F	$0.5 \times 0.8 \times 0.6 \times 0.75 \times 0.75$	0.135

$P(\text{MaryCalls}=T, e) = 0.12525$; $P(\text{MaryCalls}=F, e) = 0.30825$; $P(e) = 0.4335$.

$P(\text{MaryCalls}=T | e) = 0.12525 / 0.4335 = \mathbf{0.2889} \approx 0.289$.

Q.23 [MCQ, 2M · Variable Elimination] Answer: A (A: A, B, C, D)

Order A, B, C, D: eliminating A combines $P(A)$, $P(B | A)$, $P(C | A) \rightarrow f(B, C)$; B combines $f(B, C)$, $P(D | B, C) \rightarrow f(C, D)$; C $\rightarrow f(D)$; D with $P(E | D) \rightarrow f(E)$. Max 2 variables. D first: $P(D | B, C)$ and $P(E | D) \rightarrow f(B, C, E)$ (3 variables). B first: $P(B | A)$, $P(D | B, C) \rightarrow f(A, C, D)$ (3 variables); C first similarly gives $f(A, B, D)$. So only (A) keeps all factors at two variables or fewer.

Q.24 [MCQ, 2M · Exact Inference] Answer: C (C: $f(E, F)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(D)$, $f(C, D, E)$, $f(B, D, F)$, $f(A, E)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, B)$, $f(A, B, C)$, $f(A, E)$	$\{A, B, C, E\}$	$f(B, C, E)$
B	$f(B, D, F)$, $f(B, C, E)$	$\{B, C, D, E, F\}$	$f(C, D, E, F)$
C	$f(C, D, E)$, $f(C, D, E, F)$	$\{C, D, E, F\}$	$f(D, E, F)$
D	$f(D)$, $f(D, E, F)$	$\{D, E, F\}$	$f(E, F)$
E	$f(E, F)$	$\{E, F\}$	$f(F)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over F and normalise.

Q.25 [NAT, 2M · Sampling] Answer: 0.82 to 0.84

Rejection sampling keeps only samples consistent with the evidence $B = T$: samples 6, 7, 11, 14, 16, 20 (6 accepted, 14 rejected).

Among the accepted samples, $C = T$ in 5. Estimate = $5/6 = \mathbf{0.8333}$.

(For this network the exact value is 0.6; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Mock Test 084

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 6 / 12	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 5$ and depth $d = 2$ (so $b^d = 25$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

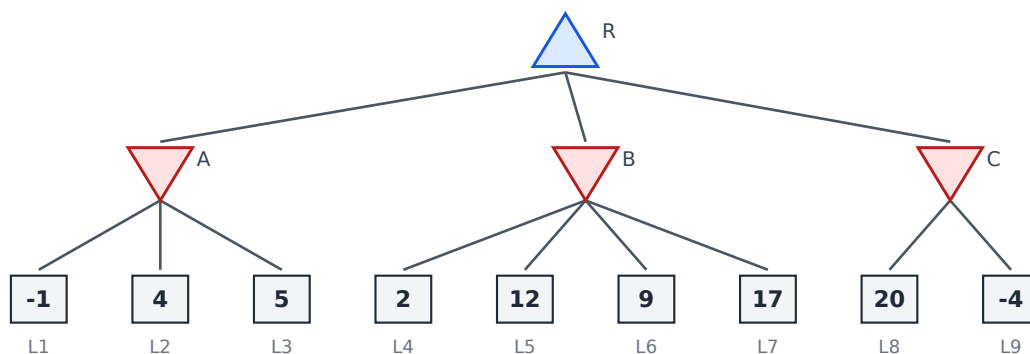
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search

A uniform game tree has branching factor $b = 3$ and depth $d = 4$ plies. Alpha-beta search examines the moves in perfect order. Using the best-case formula $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$, how many leaves are evaluated?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



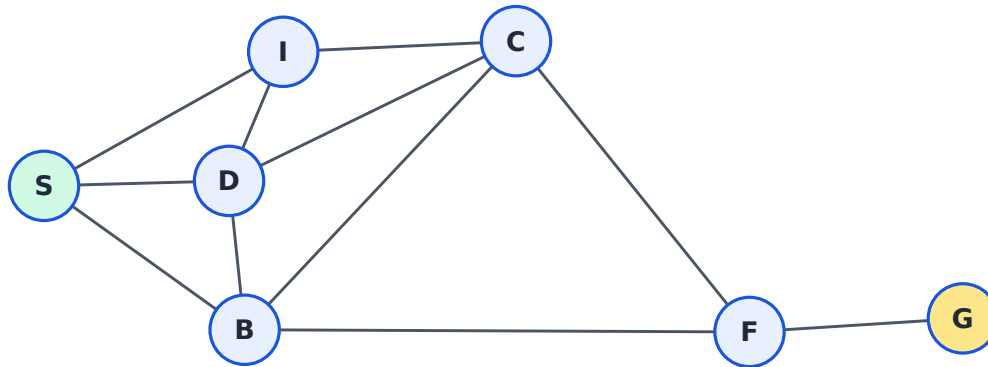
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to B; root value 4
- ☐ (B) Move to A; root value -1
- ☐ (C) Move to B; root value 2
- ☐ (D) Move to C; root value -4

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, D, C, I, F, G
- ☐ (B) S, B, D, I, C, F, G
- ☐ (C) S, I, D, B, C, F, G
- ☐ (D) S, B, C, D, I, F, G

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **E, D, A** and rules:

- R1: $F \wedge J \rightarrow D$
- R2: $E \wedge H \wedge K \rightarrow D$
- R3: $A \rightarrow B$
- R4: $K \wedge A \rightarrow D$
- R5: $F \wedge H \rightarrow A$
- R6: $B \wedge D \rightarrow K$
- R7: $J \wedge B \rightarrow E$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following formulas is satisfiable but NOT valid?

- ☐ (A) $p \rightarrow (q \rightarrow p)$
- ☐ (B) $p \wedge \neg p$
- ☐ (C) $(p \wedge q) \rightarrow (p \vee q)$
- ☐ (D) $(p \rightarrow q) \rightarrow p$

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((\neg Q \rightarrow \neg Q) \rightarrow (R \wedge R)) \leftrightarrow \neg \neg P$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.8 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Let $KB = \{P \rightarrow Q, Q \rightarrow R, \neg R\}$. Which one of the following is entailed by KB?

- ☐ (A) $\neg P \wedge \neg Q$
- ☐ (B) $P \vee Q$
- ☐ (C) $\neg P \wedge Q$
- ☐ (D) R

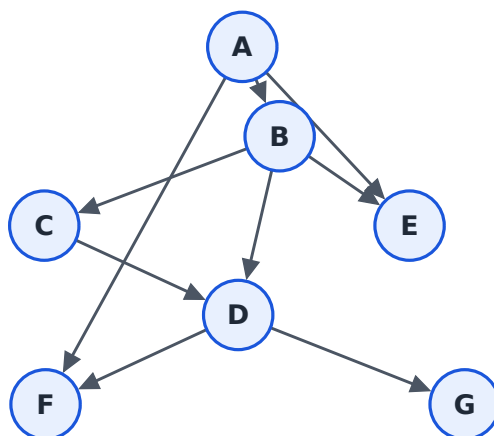
Q.9 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

A Boolean node has k Boolean parents. Its CPT is specified using a noisy-OR model (no leak). How many parameters are needed, versus a general CPT?

- ☐ (A) k for noisy-OR versus 2^k for a general CPT
- ☐ (B) 2^k for both
- ☐ (C) k^2 for noisy-OR versus 2^k for a general CPT
- ☐ (D) $2k$ for noisy-OR versus k^2 for a general CPT

Q.10 [MCQ · 1 mark · negative: -1/3] Conditional Independence – Markov blanket

Which set is the Markov blanket of **C** in the Bayesian network below?

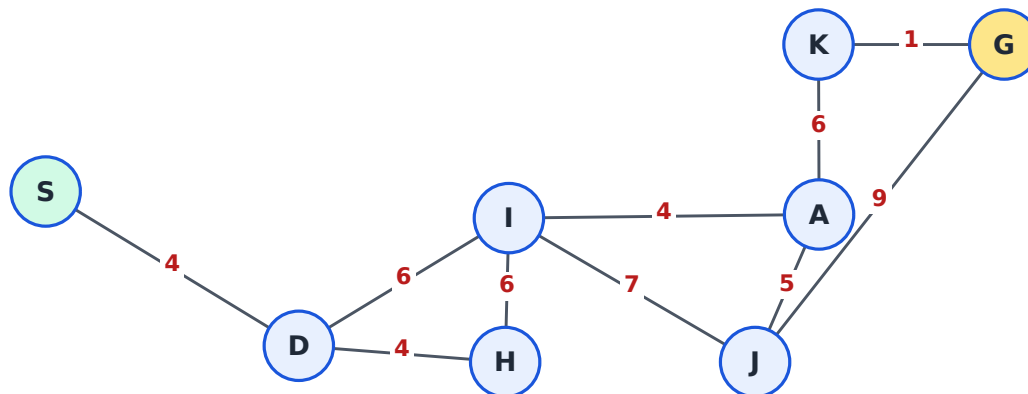


- ☐ (A) $\{B, D, E\}$
- ☐ (B) $\{B, D\}$
- ☐ (C) $\{B, D, F, G\}$
- ☐ (D) $\{D\}$

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

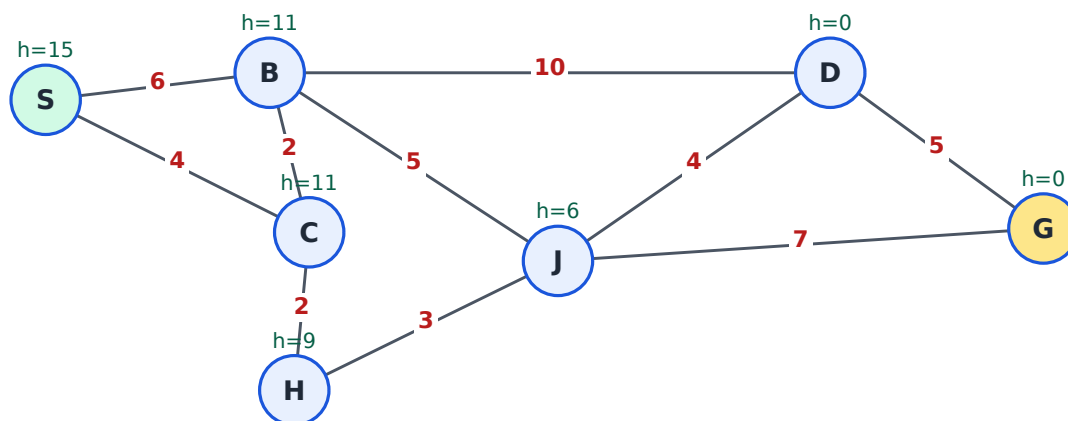
Q.12 [NAT · 2 marks · no negative marking] Uninformed Search

Iterative deepening search runs on a complete binary tree of depth 3 (15 nodes) that contains **no** goal. It uses limits 0, 1, 2, 3 and then stops. A node visit is one goal test on one node, and repeated visits across iterations count separately. What is the total number of node visits?

Answer (numerical): _____

Q.13 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

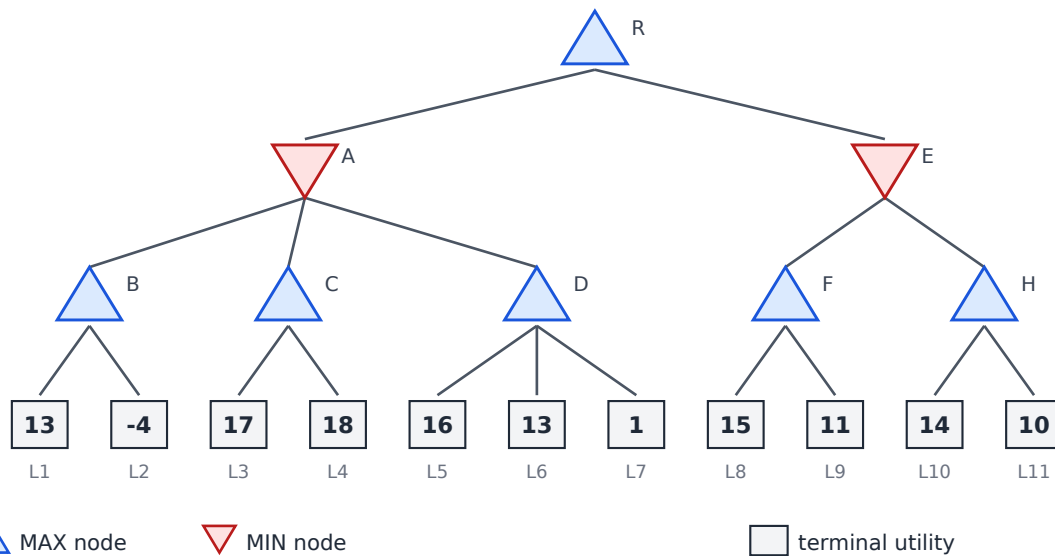
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge J-D violates the consistency condition $h(J) \leq c(J,D) + h(D)$.
- ☐ (B) h is admissible.
- ☐ (C) h is consistent (monotone).
- ☐ (D) $h(H)$ overestimates the true cost from H to G.

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic

A propositional Horn knowledge base contains the facts A and B and the definite clauses $A \wedge B \rightarrow C$, $C \rightarrow D$, $D \wedge E \rightarrow F$, $B \rightarrow E$, $F \wedge G \rightarrow H$, $H \rightarrow A$. Forward chaining is run to completion. How many distinct proposition symbols are inferred to be true (count the initial facts too)?

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \rightarrow P) \wedge (R \rightarrow P)) \rightarrow (Q \wedge R)$
☐ (B) $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$
☐ (C) $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$
☐ (D) $((R \rightarrow Q) \wedge Q) \rightarrow R$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

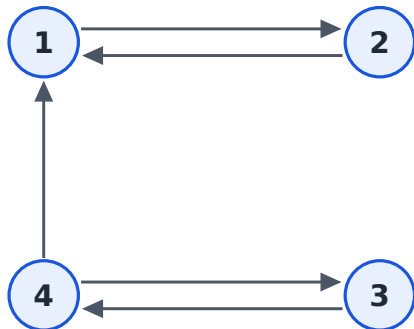
C1: $(\neg R \vee S)$, C2: (Q) , C3: $(\neg P \vee Q)$, C4: $(\neg P \vee \neg Q \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P \vee S)$
☐ (B) (P)
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) (R)

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

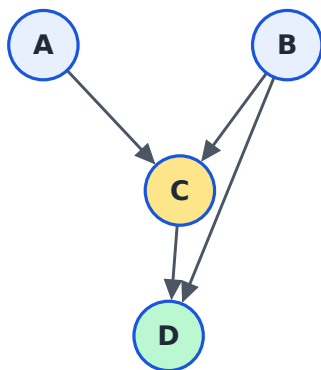
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (B) $\exists x \forall y \neg R(x, y)$
☐ (C) $\forall y \exists x R(x, y)$
☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.19 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(D = T \mid C = T)$ (round off to 3 decimal places).



$P(A=T)$
0.4

$P(B=T)$	A	B	$P(C=T \dots)$
0.95	T	T	0.9
	T	F	0.4
	F	T	0.75
	F	F	0.25

B	C	$P(D=T \dots)$
T	T	0.4
T	F	0.6
F	T	0.7
F	F	0.6

Answer (numerical): _____

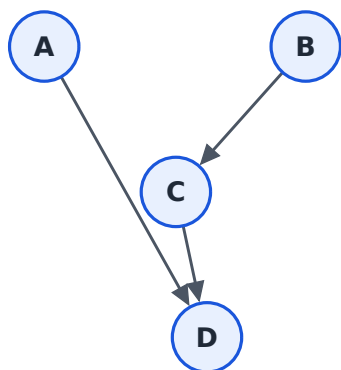
Q.20 [MSQ · 2 marks · no negative marking] Sampling

Which of the following statements about approximate inference in Bayesian networks are correct?

- ☐ (A) Likelihood weighting produces consistent estimates
☐ (B) Likelihood weighting performs poorly when evidence variables are downstream (e.g. leaves), since samples of upstream variables ignore that evidence
☐ (C) Gibbs sampling resamples each non-evidence variable from its distribution conditioned on the current values of its Markov blanket
☐ (D) In rejection sampling, the fraction of accepted samples grows as more evidence variables are added

Q.21 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.04}, \mathbf{0.04}, \mathbf{0.31}, \mathbf{0.96} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.8

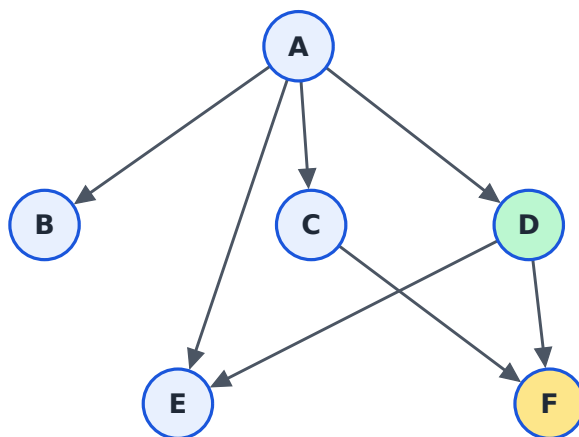
$P(B=T)$	B	$P(C=T \dots)$
0.3	T	0.4
	F	0.95

A	C	$P(D=T \dots)$
T	T	0.1
T	F	0.75
F	T	0.1
F	F	0.4

- ☐ (A) $A=F, B=F, C=F, D=T$
☐ (B) $A=T, B=T, C=T, D=F$
☐ (C) $A=T, B=F, C=T, D=F$
☐ (D) $A=F, B=T, C=T, D=F$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(D \mid F = T)}$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, A, C, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

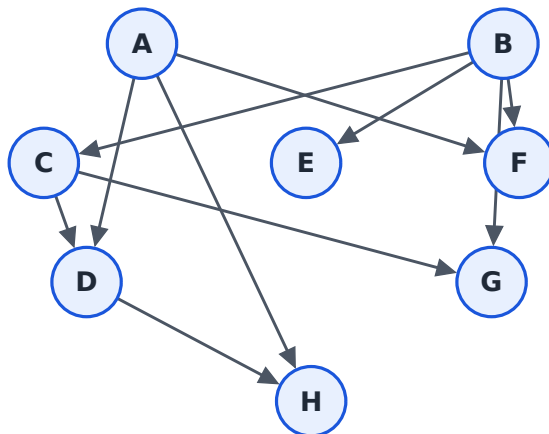
Q.23 [NAT · 2 marks · no negative marking] Probability for AI

$P(\text{cavity}) = 0.2$. Toothache and Catch are conditionally independent given Cavity with $P(\text{toothache} \mid \text{cavity}) = 0.6$, $P(\text{toothache} \mid \neg \text{cavity}) = 0.1$, $P(\text{catch} \mid \text{cavity}) = 0.9$, $P(\text{catch} \mid \neg \text{cavity}) = 0.2$. Compute $P(\text{cavity} \mid \text{toothache}, \text{catch})$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

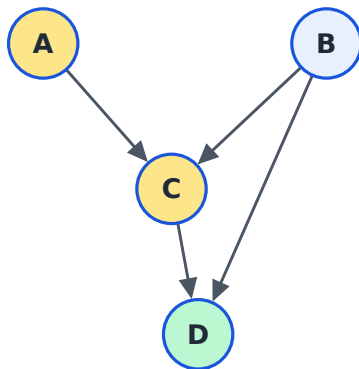
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp A$
☐ (B) $E \perp B \mid A$
☐ (C) $B \perp A$
☐ (D) $H \perp E \mid B$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $C = F$, $A = T$. The values of the non-evidence variables in the generated samples are listed in the table. What is the weight of sample #4? (round off to 4 decimal places)



P(A=T)
0.7

P(B=T)	A	B	P(C=T ...)
0.5	T	T	0.4
	T	F	0.9
	F	T	0.8
	F	F	0.3

B	C	P(D=T ...)
T	T	0.4
T	F	0.8
F	T	0.4
F	F	0.3

#	B	D
1	T	T
2	F	F
3	F	F
4	F	T
5	T	F

Answer (numerical): _____

Answer Key — Mock Test 084

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	9	14	NAT	2	Adversarial Search	3
2	NAT	1	Adversarial Search	17	15	NAT	2	Propositional Logic	6
3	MCQ	1	Adversarial Search	C	16	MSQ	2	Propositional Logic	B, C
4	MCQ	1	Uninformed Search	B	17	MSQ	2	Propositional Logic	A
5	NAT	1	Propositional Logic	5	18	MSQ	2	Predicate Logic	C, D
6	MCQ	1	Propositional Logic	D	19	NAT	2	Exact Inference	0.404 to 0.408
7	NAT	1	Propositional Logic	4	20	MSQ	2	Sampling	A, B, C
8	MCQ	1	Propositional Logic	A	21	MCQ	2	Sampling	B
9	MCQ	1	Bayesian Networks	A	22	NAT	2	Exact Inference	3
10	MCQ	1	Conditional Independence	B	23	NAT	2	Probability for AI	0.87 to 0.872
11	NAT	2	Uninformed Search	8	24	MSQ	2	Conditional Independence	A, C, D
12	NAT	2	Uninformed Search	26	25	NAT	2	Sampling	0.0698 to 0.0702
13	MSQ	2	Informed Search	A, B					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 9

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

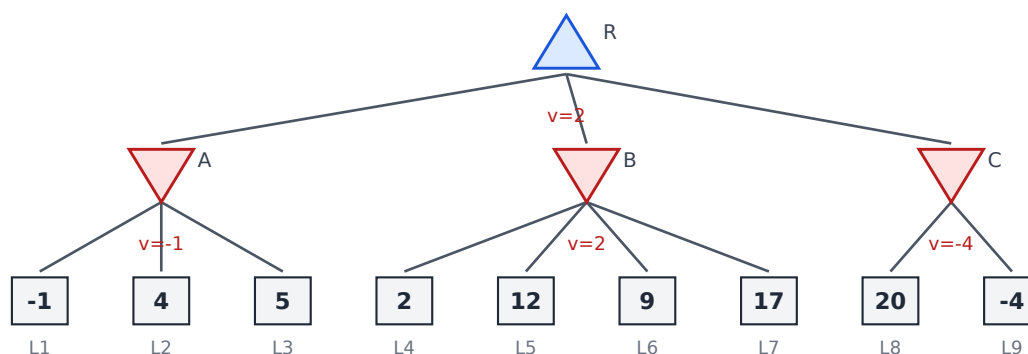
$= 5^{\lceil 1 \rceil} + 5^{\lfloor 1 \rfloor} - 1 = 5 + 5 - 1 = 9$, versus 25 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.2 [NAT, 1M · Adversarial Search] Answer: 17

$3^2 + 3^2 - 1 = 9 + 9 - 1 = 17$. Plain minimax evaluates $3^4 = 81$ leaves, so perfect ordering saves 64 leaf evaluations.

Q.3 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to B; root value 2)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -1$, $B = 2$, $C = -4$. Root value = **2**; MAX's optimal first move is towards B.

Q.4 [MCQ, 1M · Uninformed Search] Answer: B (B: S, B, D, I, C, F, G)

Adjacency lists (alphabetical): B: {C, D, F, S}; C: {B, D, F, I}; D: {B, C, I, S}; F: {B, C, G}; G: {F}; I: {C, D, S}; S: {B, D, I}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[B, D, I]
2	B	[D, I, C, F]
3	D	[I, C, F]
4	I	[C, F]
5	C	[F]

Step	Expanded	Frontier (FIFO) after expansion
6	F	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, B, D, I, C, F, G**. The path returned is $S \rightarrow B \rightarrow F \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [NAT, 1M · Propositional Logic] Answer: 5

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $A \rightarrow B$; add B.

Round 2: fire $B \wedge D \rightarrow K$; add K.

Fixed point: {A, B, D, E, K} — 5 atoms are entailed (facts: A, D, E; newly derived: B, K). Atoms never derived: C, F, H, J.

Q.6 [MCQ, 1M · Propositional Logic] Answer: D ($D: (p \rightarrow q) \rightarrow p$)

$(p \rightarrow q) \rightarrow p$: when $p=T$ it is true (consequent T); when $p=F$, $p \rightarrow q$ is T and the consequent is F, so it is false. Hence it is satisfiable but not valid. $p \rightarrow (q \rightarrow p)$ is valid (if p is T the consequent is T). $p \wedge \neg p$ is unsatisfiable. $(p \wedge q) \rightarrow (p \vee q)$ is valid because $p \wedge q \models p \vee q$.

Q.7 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg Q \rightarrow \neg Q) \rightarrow (R \wedge R)$	$\neg \neg P$	$((\neg Q \rightarrow \neg Q) \rightarrow (R \wedge R)) \leftrightarrow \neg \neg P$
T	T	T	T	T	T
T	T	F	F	T	F
T	F	T	T	T	T
T	F	F	F	T	F
F	T	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	F	F	T

F has 4 models, so it is false in $8 - 4 = 4$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.8 [MCQ, 1M · Propositional Logic] Answer: A ($A: \neg P \wedge \neg Q$)

From $\neg R$ and $Q \rightarrow R$, modus tollens gives $\neg Q$; from $\neg Q$ and $P \rightarrow Q$ it gives $\neg P$. The only model of KB (over P, Q, R) is $P=F, Q=F, R=F$, in which $\neg P \wedge \neg Q$ is true. $P \vee Q, \neg P \wedge Q$ and R are all false in that model, so they are not entailed.

Q.9 [MCQ, 1M · Bayesian Networks] Answer: A ($A: k$ for noisy-OR versus 2^k for a general CPT)

Noisy-OR needs one inhibition probability per parent, i.e. k numbers, and the full table is derived as $P(\neg x \mid \text{parents}) = \prod q_i$ over true parents. A general Boolean CPT has one free number for each of the 2^k parent configurations. This linear-vs-exponential saving is why canonical distributions are used.

Q.10 [MCQ, 1M · Conditional Independence] Answer: B ($B: \{B, D\}$)

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of C: {B}; children: {D}; co-parents: $\{\emptyset\}$.

$MB(C) = \{B, D\}$ — 2 nodes. Given its Markov blanket, C is conditionally independent of every other node in the network.

Q.11 [NAT, 2M · Uninformed Search] Answer: 8

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	D(4)
2	D (4)	H(8), I(10)
3	H (8)	I(10)
4	I (10)	A(14), J(17)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
5	A (14)	J(17), K(20)
6	J (17)	K(20), G(26)
7	K (20)	G(21)
8	G (21)	← goal, stop

Path returned: **S** → **D** → **I** → **A** → **K** → **G**, cost = 4 + 6 + 4 + 6 + 1 = **21**. Nodes expanded (including S and G): **8** (S, D, H, I, A, J, K, G).

Q.12 [NAT, 2M · Uninformed Search] **Answer: 26**

The iteration with limit L visits the whole tree of depth L, which has $2^{L+1} - 1$ nodes. Limit 0: 1. Limit 1: 3. Limit 2: 7. Limit 3: 15. Total = 1 + 3 + 7 + 15 = **26**. For b = 2 the repetition overhead is largest, here 26 versus 15 nodes, which is still less than double.

Q.13 [MSQ, 2M · Informed Search] **Answer: A, B** (A: The edge J-D violates the consistency condition $h(J) \leq c(J,D) + h(D)$.; B: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	B	C	D	G	H	J	S
$h(n)$	11	11	0	0	9	6	15
$h^*(n)$	12	12	5	0	10	7	16
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

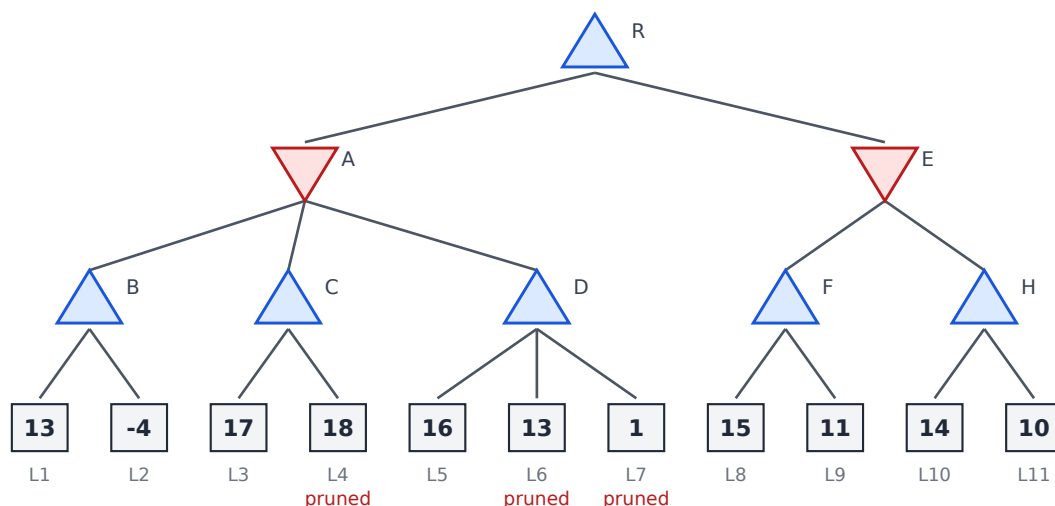
Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: J→D ($6 > 4 + 0$), B→D ($11 > 10 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.14 [NAT, 2M · Adversarial Search] **Answer: 3**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value 17 $\geq \beta = 13 \Rightarrow$ prune remaining children (leaves L4).
- At MAX node D: value 16 $\geq \beta = 13 \Rightarrow$ prune remaining children (leaves L6, L7).



Leaves evaluated (8): L1, L2, L3, L5, L8, L9, L10, L11. Leaves pruned (3): L4, L6, L7. Root value = 14 (same as plain minimax, 14 — pruning never changes the root value).

Q.15 [NAT, 2M · Propositional Logic] **Answer: 6**

Start with {A, B}. $A \wedge B \rightarrow C$ adds C; $C \rightarrow D$ adds D; $B \rightarrow E$ adds E; $D \wedge E \rightarrow F$ adds F. $F \wedge G \rightarrow H$ never fires because G is never derived, and $H \rightarrow A$ adds nothing new. The agenda empties with {A, B, C, D, E, F}, i.e. 6 symbols. Because forward chaining is sound and complete for Horn KBs, these are exactly the entailed atoms.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, C (B: $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$; C: $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$)

(A) $((Q \rightarrow P) \wedge (R \rightarrow P)) \rightarrow (Q \wedge R)$: **not valid** — counter-model $P=T, Q=T, R=F$.

(B) $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((R \rightarrow Q) \wedge Q) \rightarrow R$: **not valid** — counter-model $Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A (A: $(\neg P \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee S)$ is derivable:

1. resolve (Q) and $(\neg P \vee \neg Q \vee R)$ on Q $\Rightarrow (\neg P \vee R)$

2. resolve $(\neg R \vee S)$ and $(\neg P \vee R)$ on R $\Rightarrow (\neg P \vee S)$

(B) (P) never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(D) (R) never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.18 [MSQ, 2M · Predicate Logic] Answer: C, D (C: $\forall y \exists x R(x, y)$; D: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,2), (2,1), (3,4), (4,1), (4,3)\}$.

(A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,3)$ nor $R(3,1)$.

(B) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

(C) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [NAT, 2M · Exact Inference] Answer: 0.404 to 0.408

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B) \cdot P(C | A, B) \cdot P(D | B, C)$.

Inference by enumeration: $P(D | e) = \alpha \sum_{\text{hidden}} P(D, e, \text{hidden})$.

Hidden: A, B	D	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T, T	T	$0.4 \times 0.95 \times 0.9 \times 0.4$	0.1368
T, F	T	$0.4 \times 0.05 \times 0.4 \times 0.7$	0.0056
F, T	T	$0.6 \times 0.95 \times 0.75 \times 0.4$	0.171
F, F	T	$0.6 \times 0.05 \times 0.25 \times 0.7$	0.00525
T, T	F	$0.4 \times 0.95 \times 0.9 \times 0.6$	0.2052
T, F	F	$0.4 \times 0.05 \times 0.4 \times 0.3$	0.0024
F, T	F	$0.6 \times 0.95 \times 0.75 \times 0.6$	0.2565
F, F	F	$0.6 \times 0.05 \times 0.25 \times 0.3$	0.00225

$P(D=T, e) = 0.31865$; $P(D=F, e) = 0.46635$; $P(e) = 0.785$.

$P(D=T | e) = 0.31865 / 0.785 = \mathbf{0.4059} \approx 0.406$.

Q.20 [MSQ, 2M · Sampling] Answer: A, B, C

(A) AIMA shows $S_{WS}(z, e) w(z, e) = P(z, e)$, so weighted estimates are consistent. (B) Non-evidence variables are sampled from their parents only, so evidence at leaves does not influence them; many samples get tiny weights. (C) This is exactly the Gibbs step: $P(x_i | mb(X_i))$. (D) is false: each extra evidence variable generally reduces $P(e)$, so fewer samples are accepted.

Q.21 [MCQ, 2M · Sampling] Answer: B (B: $A=T, B=T, C=T, D=F$)

Sample each variable after its parents, using the parent values already drawn:

• A: $P(A=T | \text{—}) = 0.8$; $u = 0.04 < 0.8 \Rightarrow A = T$

• B: $P(B=T | \text{—}) = 0.3$; $u = 0.04 < 0.3 \Rightarrow B = T$

• C: $P(C=T | B=T) = 0.4$; $u = 0.31 < 0.4 \Rightarrow C = T$

• D: $P(D=T | A=T, C=T) = 0.1$; $u = 0.96 \geq 0.1 \Rightarrow D = F$

Sample: **(A=T, B=T, C=T, D=F)**.

Q.22 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, C)$, $f(A, D)$, $f(A, D, E)$, $f(C, D)$.
Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B)$	$\{A, B\}$	$f(A)$
A	$f(A)$, $f(A, C)$, $f(A, D)$, $f(A, D, E)$, $f(A)$	$\{A, C, D, E\}$	$f(C, D, E)$
C	$f(C, D)$, $f(C, D, E)$	$\{C, D, E\}$	$f(D, E)$
E	$f(D, E)$	$\{D, E\}$	$f(D)$

The largest factor produced has **3** variable(s) (created when eliminating A); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over D and normalise.

Q.23 [NAT, 2M · Probability for AI] Answer: 0.87 to 0.872

Using conditional independence, $P(\text{Cavity} \mid \text{toothache}, \text{catch}) = \alpha P(\text{toothache} \mid \text{Cavity}) P(\text{catch} \mid \text{Cavity}) P(\text{Cavity})$. For cavity: $0.6 \times 0.9 \times 0.2 = 0.108$. For $\neg$ cavity: $0.1 \times 0.2 \times 0.8 = 0.016$. $\alpha = 1/(0.108 + 0.016) = 1/0.124$, so $P(\text{cavity} \mid \text{toothache}, \text{catch}) = 0.108/0.124 \approx 0.871$.

Q.24 [MSQ, 2M · Conditional Independence] Answer: A, C, D (A: $G \perp A$; C: $B \perp A$; D: $H \perp E \mid B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp A$: **TRUE** — $G \leftarrow B \rightarrow C \rightarrow D \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $G \leftarrow B \rightarrow C \rightarrow D \rightarrow H \leftarrow A$ (blocked: collider H with neither it nor a descendant observed); $G \leftarrow B \rightarrow F \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow C \leftarrow B \rightarrow F \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow C \rightarrow D \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $G \leftarrow C \rightarrow D \rightarrow H \leftarrow A$ (blocked: collider H with neither it nor a descendant observed).

(B) $E \perp B \mid A$: **FALSE** — the path $E \leftarrow B$ is active given $\{A\}$.

(C) $B \perp A$: **TRUE** — $B \rightarrow C \rightarrow D \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow C \rightarrow D \rightarrow H \leftarrow A$ (blocked: collider H with neither it nor a descendant observed); $B \rightarrow F \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow G \leftarrow C \rightarrow D \leftarrow A$ (blocked: collider G with neither it nor a descendant observed); $B \rightarrow G \leftarrow C \rightarrow D \rightarrow H \leftarrow A$ (blocked: collider G with neither it nor a descendant observed).

(D) $H \perp E \mid B$: **TRUE** — $H \leftarrow A \rightarrow D \leftarrow C \leftarrow B \rightarrow E$ (blocked: collider D with neither it nor a descendant observed); $H \leftarrow A \rightarrow D \leftarrow C \rightarrow G \leftarrow B \rightarrow E$ (blocked: collider D with neither it nor a descendant observed); $H \leftarrow A \rightarrow F \leftarrow B \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $H \leftarrow D \leftarrow A \rightarrow F \leftarrow B \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $H \leftarrow D \leftarrow C \leftarrow B \rightarrow E$ (blocked: observed fork node B); $H \leftarrow D \leftarrow C \rightarrow G \leftarrow B \rightarrow E$ (blocked: collider G with neither it nor a descendant observed).

Q.25 [NAT, 2M · Sampling] Answer: 0.0698 to 0.0702

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	D
1	$P(A=T)=0.7 \times P(C=F A=T, B=T)=0.6$	0.42	T
2	$P(A=T)=0.7 \times P(C=F A=T, B=F)=0.1$	0.07	F
3	$P(A=T)=0.7 \times P(C=F A=T, B=F)=0.1$	0.07	F
4	$P(A=T)=0.7 \times P(C=F A=T, B=F)=0.1$	0.07	T
5	$P(A=T)=0.7 \times P(C=F A=T, B=T)=0.6$	0.42	F

Σw (all) = 1.05; Σw (D = T) = 0.49.

Estimate $P(D=T \mid C = F, A = T) = 0.49 / 1.05 = \mathbf{0.4667}$.

Mock Test 085

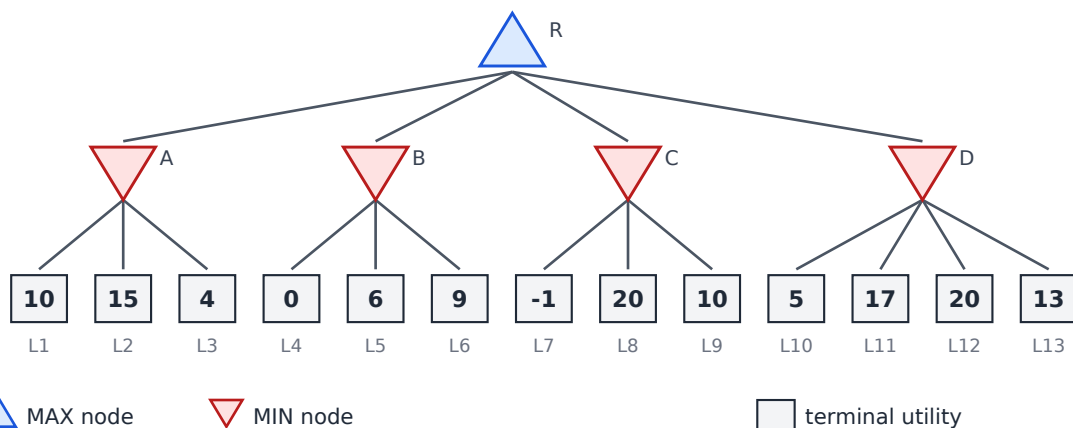
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 8 / 11	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Depth-limited search with limit $l = 3$ is run on a tree with branching factor $b = 3$. No goal exists within depth 3. How many nodes (including the root) does it generate before returning cutoff?

Answer (numerical): _____

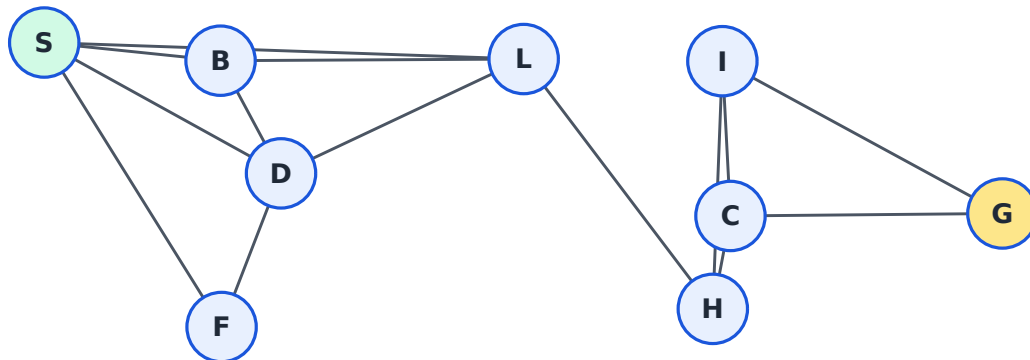
Q.3 [MSQ · 1 mark · no negative marking] Adversarial Search

Which of the following statements about two-player zero-sum games are TRUE?

- ☐ (A) The players' utilities sum to the same constant in every terminal state
- ☐ (B) Chess, with outcomes $+1/0/-1$ or $1/1/2/0$, can be modelled as a zero-sum (constant-sum) game
- ☐ (C) Minimax assumes that the opponent plays optimally
- ☐ (D) In a zero-sum game, both players can strictly gain from the same outcome change

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, L, H, I, G
- ☐ (B) S, B, D, F, L, H, C, I, G
- ☐ (C) S, B, D, L, F, H, C, G
- ☐ (D) S, B, D, F, L, H, C, G

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((\neg P \leftrightarrow Q) \leftrightarrow (\neg Q \vee P)) \rightarrow ((R \wedge Q) \vee \neg R) ?$$

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Backward chaining over a propositional definite-clause KB is used to prove the query Q. Which clause is used FIRST?

- ☐ (A) A clause whose head (conclusion) is Q
- ☐ (B) A clause in which Q appears in the body
- ☐ (C) Any fact in the KB, chosen arbitrarily
- ☐ (D) The negated query $\neg Q$ added to the KB

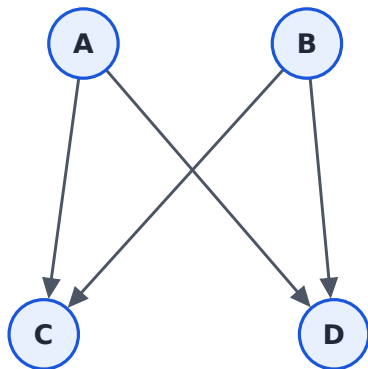
Q.8 [MCQ · 1 mark · negative: -1/3] Propositional Logic

$\neg(p \rightarrow q)$ is logically equivalent to

- ☐ (A) $p \wedge \neg q$
- ☐ (B) $\neg p \wedge q$
- ☐ (C) $\neg p \rightarrow \neg q$
- ☐ (D) $\neg p \vee q$

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=F, B=F, C=F, D=F)**? (round off to 4 decimal places)



P(A=T)
0.75

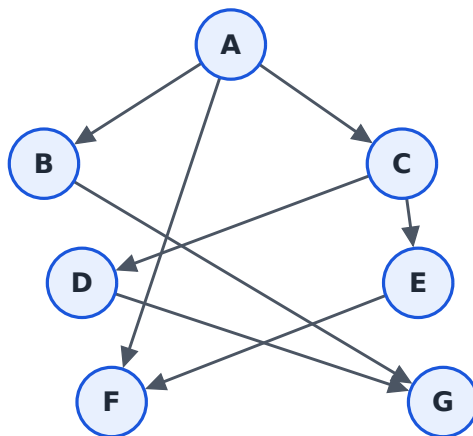
P(B=T)	A	B	P(C=T ...)
0.75	T	T	0.3
	T	F	0.25
	F	T	0.2
	F	F	0.9

A	B	P(D=T ...)
T	T	0.8
T	F	0.4
F	T	0.75
F	F	0.2

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **D** in the Bayesian network below?

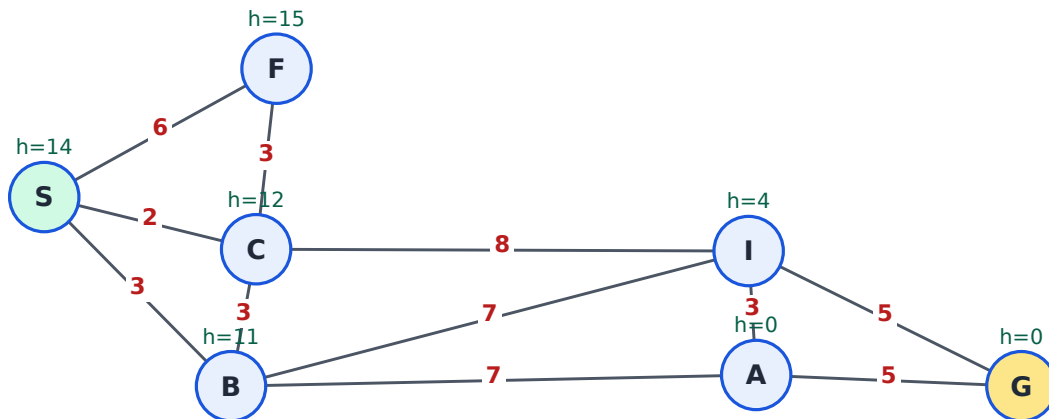


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

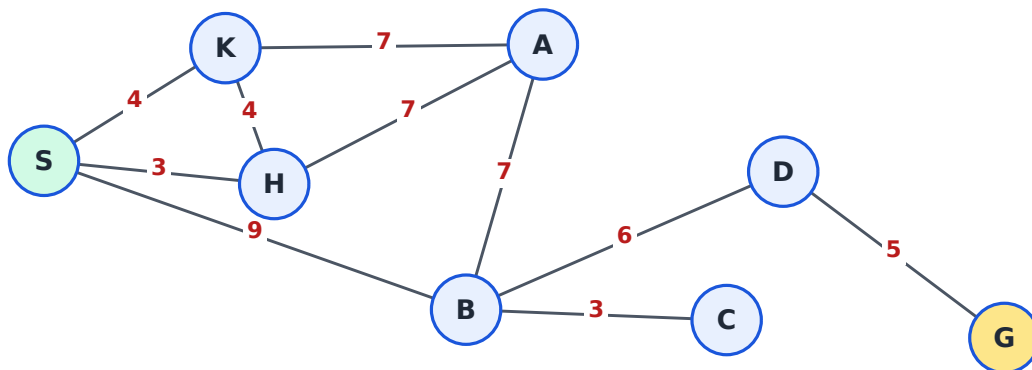
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
☐ (B) h is consistent (monotone).
☐ (C) The edge I-A violates the consistency condition $h(I) \leq c(I,A) + h(A)$.
☐ (D) $h(B)$ overestimates the true cost from B to G.

Q.12 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

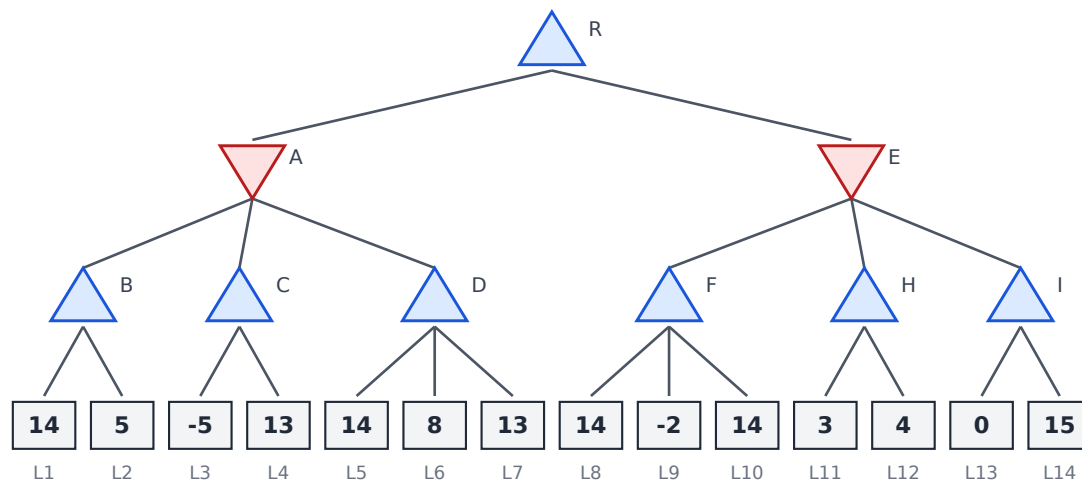
Q.13 [NAT · 2 marks · no negative marking] Local Search

In simulated annealing, a randomly chosen move worsens the objective (which is being maximised) by $\Delta E = -2$ at temperature $T = 4$. With what probability is this move accepted? Give the answer to two decimal places.

Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



- ☐ (A) Leaf L2 is pruned.
☐ (B) Leaf L4 is pruned.
☐ (C) Leaf L7 is pruned.
☐ (D) Leaf L11 is pruned.

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $R \wedge (\neg R \leftrightarrow S)$
- (2) $R \leftrightarrow (R \leftrightarrow \neg P)$
- (3) $(S \rightarrow \neg P) \wedge (\neg R \leftrightarrow S)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $P \rightarrow \neg R$
☐ (B) $(P \vee \neg Q) \vee (\neg S \leftrightarrow \neg P)$
☐ (C) $Q \rightarrow S$
☐ (D) $\neg S \wedge Q$

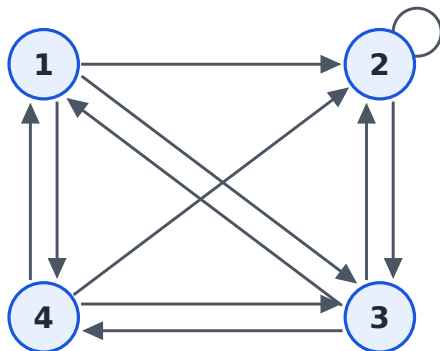
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(Q \wedge (Q \rightarrow R)) \rightarrow R$
☐ (B) $(Q \vee P) \rightarrow (Q \wedge P)$
☐ (C) $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$
☐ (D) $((Q \rightarrow R) \wedge R) \rightarrow Q$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
☐ (B) $\exists y \forall x R(x, y)$
☐ (C) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (D) $\exists x \forall y \neg R(x, y)$

Q.18 [NAT · 2 marks · no negative marking] Predicate Logic

Let $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \forall y (R(x, y) \rightarrow \neg R(y, x))$?

Answer (numerical): _____

Q.19 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D . Using **rejection sampling**, estimate $P(B = T \mid D = T)$ (round off to 2 decimal places).

#	A	B	C	D
1	T	T	F	F
2	T	F	F	T
3	T	F	T	F
4	T	T	T	T
5	T	F	T	T
6	T	F	F	T
7	T	T	T	T
8	T	F	T	T
9	T	T	F	F
10	T	F	F	T
11	T	T	F	F
12	F	T	F	T
13	T	F	T	T
14	T	T	F	F
15	T	T	T	T
16	T	F	F	T
17	T	T	T	T
18	F	T	T	T
19	T	T	T	T
20	T	F	F	F

Answer (numerical): _____

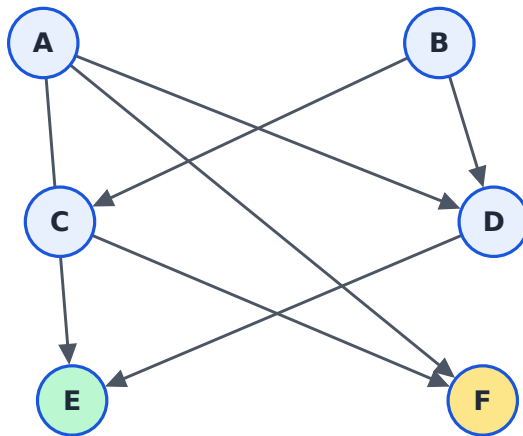
Q.20 [MCQ · 2 marks · negative: -2/3] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. For $P(B | j, m)$, the elimination order A then E is used. After summing out A, which factors remain to be multiplied when E is eliminated?

- ☐ (A) $P(E)$ and $f_A(B, E)$
- ☐ (B) $P(E)$, $P(B)$ and $f_A(B, E)$
- ☐ (C) $P(E)$ and $f_A(B, E, J, M)$
- ☐ (D) $P(E)$, $P(A | B, E)$ and $f_A(B)$

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(E | F = T)$ in the Bayesian network below (Boolean variables; evidence $F = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **D**, **C**, **A**, **B**. What is the scope of the new factor produced when **B** is summed out?



- ☐ (A) $f(E)$
- ☐ (B) $f(B, E)$
- ☐ (C) $f()$
- ☐ (D) $f(B)$

Q.22 [MCQ · 2 marks · negative: -2/3] Probability for AI

The prior odds of hypothesis h against $\neg h$ are 1 : 4. Evidence e has likelihood ratio $P(e | h)/P(e | \neg h) = 6$. What is $P(h | e)$?

- ☐ (A) 0.600
- ☐ (B) 0.857
- ☐ (C) 0.250
- ☐ (D) 0.200

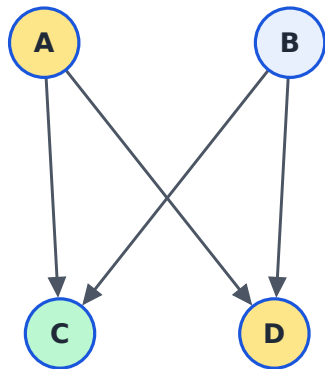
Q.23 [MSQ · 2 marks · no negative marking] Bayesian Networks

Which of the following statements about I-maps of a distribution P are correct? (G is an I-map of P if every independence implied by G by d-separation holds in P .)

- ☐ (A) A complete (fully connected) DAG over the variables is an I-map of every distribution
- ☐ (B) Adding nodes in a causal order (causes before effects) usually yields a sparser minimal I-map
- ☐ (C) Adding an arc (without creating a cycle) to an I-map of P can make it fail to be an I-map of P
- ☐ (D) The minimal I-map of P is the same DAG for every variable ordering

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(C = T \mid D = T, A = T)$ (round off to 3 decimal places).



$P(A=T)$
0.75

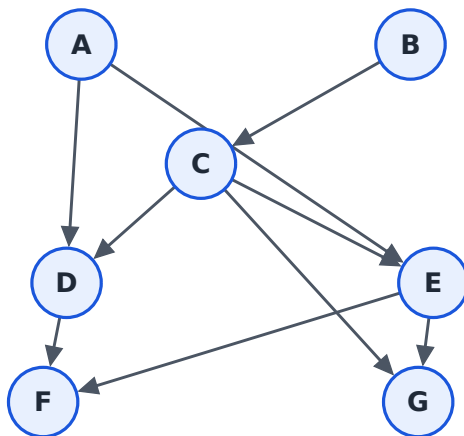
$P(B=T)$	A	B	$P(C=T \dots)$
0.5	T	T	0.6
	T	F	0.7
	F	T	0.5
	F	F	0.75

A	B	$P(D=T \dots)$
T	T	0.25
T	F	0.75
F	T	0.95
F	F	0.1

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp B \mid A$
- ☐ (B) $B \perp A \mid C$
- ☐ (C) $B \perp A \mid F$
- ☐ (D) $B \perp A$

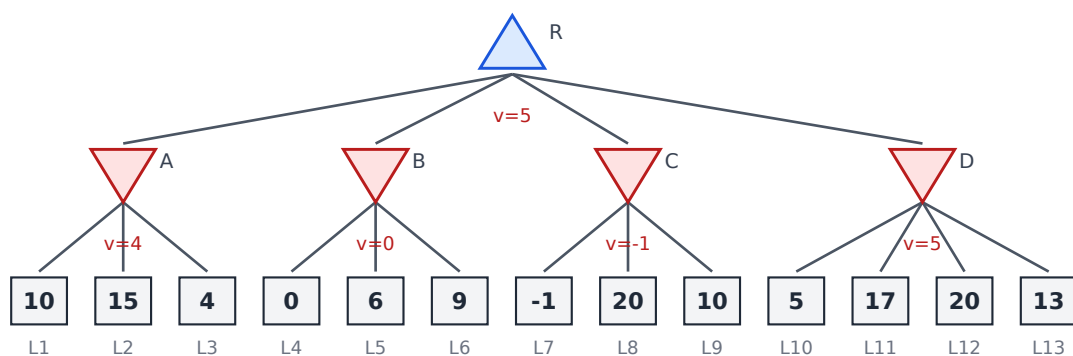
Answer Key — Mock Test 085

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	5	14	MSQ	2	Adversarial Search	C
2	NAT	1	Uninformed Search	40	15	MSQ	2	Propositional Logic	A, B
3	MSQ	1	Adversarial Search	A, B, C	16	MSQ	2	Propositional Logic	A, C
4	MCQ	1	Uninformed Search	D	17	MSQ	2	Predicate Logic	B
5	NAT	1	Predicate Logic	19	18	NAT	2	Predicate Logic	27
6	NAT	1	Propositional Logic	7	19	NAT	2	Sampling	0.49 to 0.51
7	MCQ	1	Propositional Logic	A	20	MCQ	2	Variable Elimination	A
8	MCQ	1	Propositional Logic	A	21	MCQ	2	Exact Inference	A
9	NAT	1	Sampling	0.0048 to 0.0052	22	MCQ	2	Probability for AI	A
10	NAT	1	Conditional Independence	3	23	MSQ	2	Bayesian Networks	A, B
11	MSQ	2	Informed Search	A, C	24	NAT	2	Exact Inference	0.673 to 0.677
12	NAT	2	Uninformed Search	20	25	MSQ	2	Conditional Independence	B, D
13	NAT	2	Local Search	0.6 to 0.61					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 5

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 4$, $B = 0$, $C = -1$, $D = 5$. Root value = **5**; MAX's optimal first move is towards D.

Q.2 [NAT, 1M · Uninformed Search] Answer: 40

DLS explores the complete tree down to depth ℓ : $1 + b + \dots + b^{\ell} = 1 + 3 + 9 + 27 = \mathbf{40} = (b^{(\ell+1)} - 1)/(b - 1)$.

Q.3 [MSQ, 1M · Adversarial Search] Answer: A, B, C

(A) is the definition of zero-sum (constant-sum). (B) is true: AIMA notes that chess's payoffs always total a constant. (C) is true: minimax computes the best play against a perfect, adversarial opponent. (D) is false: since the sum is constant, one player's gain is exactly the other's loss.

Q.4 [MCQ, 1M · Uninformed Search] Answer: D (D: S, B, D, F, L, H, C, G)

Adjacency lists (alphabetical): B: {D, L, S}; C: {G, H, I}; D: {B, F, L, S}; F: {D, S}; G: {C, I}; H: {C, I, L}; I: {C, G, H}; L: {B, D, H, S}; S: {B, D, F, L}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, B, D, F, L, H, C, G**; the path found is $S \rightarrow B \rightarrow D \rightarrow L \rightarrow H \rightarrow C \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.5 [NAT, 1M · Predicate Logic] Answer: 19

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 3$ this gives **19** (verified by enumerating all interpretations).

Q.6 [NAT, 1M · Propositional Logic] Answer: 7

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg P \leftrightarrow Q) \leftrightarrow (\neg Q \vee P)$	$(R \wedge Q) \vee \neg R$	$((\neg P \leftrightarrow Q) \leftrightarrow (\neg Q \vee P)) \rightarrow ((R \wedge Q) \vee \neg R)$
T	T	T	F	T	T
T	T	F	F	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	T	T
F	F	T	F	F	T
F	F	F	F	T	T

Rows where F = T: **7**.

Q.7 [MCQ, 1M · Propositional Logic] Answer: A (A: A clause whose head (conclusion) is Q)

Backward chaining works from the goal: it looks for rules (or facts) whose conclusion is Q and recursively tries to prove their premises. Using clauses with Q in the body is forward reasoning. Starting from facts is forward chaining. Adding $\neg Q$ is the resolution-refutation approach, not backward chaining.

Q.8 [MCQ, 1M · Propositional Logic] Answer: A (A: $p \wedge \neg q$)

$p \rightarrow q \equiv \neg p \vee q$, so $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q$ by De Morgan's law. It is true only in the row $p=T, q=F$, which is exactly the row where $p \rightarrow q$ is false. $\neg p \wedge q$ is true only at $p=F, q=T$; $\neg p \rightarrow \neg q$ is the inverse; $\neg p \vee q$ is $p \rightarrow q$ itself.

Q.9 [NAT, 1M · Sampling] Answer: 0.0048 to 0.0052

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(A=F) \times P(B=F) \times P(C=F|A=F, B=F) \times P(D=F|A=F, B=F) \\ = 0.25 \times 0.25 \times 0.1 \times 0.8 = \mathbf{0.005} \approx 0.0050.$$

Q.10 [NAT, 1M · Conditional Independence] Answer: 3

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of D: {C}; children: {G}; co-parents: {B}.

MB(D) = {B, C, G} — **3** nodes. Given its Markov blanket, D is conditionally independent of every other node in the network.

Q.11 [MSQ, 2M · Informed Search] Answer: A, C (A: h is admissible.; C: The edge I→A violates the consistency condition $h(I) \leq c(I, A) + h(A)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	C	F	G	I	S
$h(n)$	0	11	12	15	0	4	14
$h^*(n)$	5	12	13	16	0	5	15
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u, v) + h(v)$ for every edge in both directions. Violating directed edges: I→A ($4 > 3 + 0$), B→A ($11 > 7 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [NAT, 2M · Uninformed Search] Answer: 20

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	H(3), K(4), B(9)
2	H (3)	K(4), B(9), A(10)
3	K (4)	B(9), A(10)
4	B (9)	A(10), C(12), D(15)
5	A (10)	C(12), D(15)
6	C (12)	D(15)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
7	D (15)	G(20)
8	G (20)	← goal, stop

Path returned: **S** → **B** → **D** → **G**, cost = 9 + 6 + 5 = **20**. Nodes expanded (including S and G): **8** (S, H, K, B, A, C, D, G).

Q.13 [NAT, 2M · Local Search] **Answer: 0.6 to 0.61**

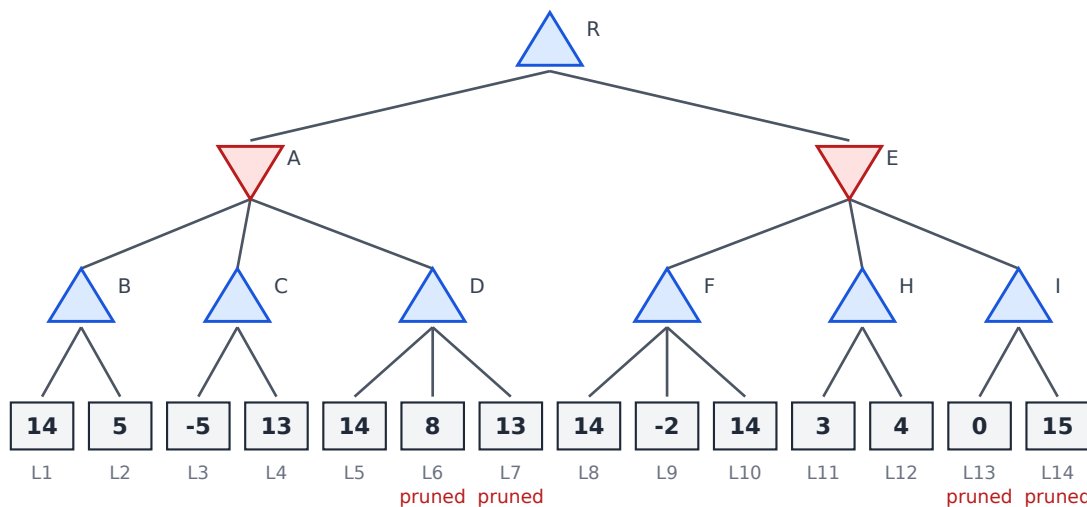
A worsening move is accepted with probability $e^{\Delta E/T} = e^{-2/4} = e^{-0.5} \approx \mathbf{0.6065}$, which rounds to 0.61. Lower temperatures or larger losses make acceptance less likely. Improving moves ($\Delta E > 0$) are always accepted.

Q.14 [MSQ, 2M · Adversarial Search] **Answer: C** (C: Leaf L7 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MAX node D: value 14 $\geq \beta = 13 \Rightarrow$ prune remaining children (leaves L6, L7).

• At MIN node E: value 4 $\leq \alpha = 13 \Rightarrow$ prune remaining children (leaves L13, L14).



Leaves evaluated (10): L1, L2, L3, L4, L5, L8, L9, L10, L11, L12. Leaves pruned (4): L6, L7, L13, L14. Root value = 13 (same as plain minimax, 13 — pruning never changes the root value).

Q.15 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: $P \rightarrow \neg R$; B: $(P \vee \neg Q) \vee (\neg S \leftrightarrow \neg P)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
F	T	T	F	T
F	F	T	F	T

(A) $P \rightarrow \neg R$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $(P \vee \neg Q) \vee (\neg S \leftrightarrow \neg P)$ is true in all models of KB $\Rightarrow$ **entailed**.

(C) $Q \rightarrow S$ is false in the KB-model $P=F, Q=T, R=T, S=F \Rightarrow$ not entailed.

(D) $\neg S \wedge Q$ is false in the KB-model $P=F, Q=F, R=T, S=F \Rightarrow$ not entailed.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: $(Q \wedge (Q \rightarrow R)) \rightarrow R$; C: $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$)

(A) $(Q \wedge (Q \rightarrow R)) \rightarrow R$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $(Q \vee P) \rightarrow (Q \wedge P)$: **not valid** — counter-model $P=T, Q=F$.

(C) $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow R) \wedge R) \rightarrow Q$: **not valid** — counter-model $Q=F, R=T$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [MSQ, 2M · Predicate Logic] Answer: B ($B: \exists y \forall x R(x, y)$)

$R = \{(1,2), (1,3), (1,4), (2,2), (2,3), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.

(B) $\exists y \forall x R(x, y)$: **TRUE** — $y = 2$ receives an edge from every x (incl. itself).

(C) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(3,4) \wedge R(4,3)$ but $\neg R(3,3)$.

(D) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [NAT, 2M · Predicate Logic] Answer: 27

Taking $y = x$ gives $R(x, x) \rightarrow \neg R(x, x)$, so no diagonal pair may be in R (R is irreflexive). For each of the 3 unordered pairs $\{x, y\}$ with $x \neq y$, at most one of (x, y) , (y, x) is in R : 3 choices (neither, one, the other).

Total = $1 \times 3^3 = 27$ (asymmetric relations).

Q.19 [NAT, 2M · Sampling] Answer: 0.49 to 0.51

Rejection sampling keeps only samples consistent with the evidence $D = T$: samples 2, 4, 5, 6, 7, 8, 10, 12, 13, 15, 16, 17, 18, 19 (14 accepted, 6 rejected).

Among the accepted samples, $B = T$ in 7. Estimate = $7/14 = 0.5$.

(For this network the exact value is 0.6874; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.20 [MCQ, 2M · Variable Elimination] Answer: A ($A: P(E)$ and $f_A(B, E)$)

Initially the factors are $P(B)$, $P(E)$, $P(A | B, E)$, $f(A) = P(j | A)$, $g(A) = P(m | A)$. Summing out A multiplies the three factors containing A and yields $f_A(B, E)$. Eliminating E then multiplies only the factors mentioning E : $P(E)$ and $f_A(B, E)$, giving $f_E(B)$. $P(B)$ does not mention E so it is left aside until the end; J and M are evidence, not dimensions; $P(A | B, E)$ was already consumed.

Q.21 [MCQ, 2M · Exact Inference] Answer: A ($A: f(E)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(B, C)$, $f(A, B, D)$, $f(A, D, E)$, $f(A, C)$.

Eliminating Z : multiply all factors whose scope contains Z , then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(A, B, D)$, $f(A, D, E)$	$\{A, B, D, E\}$	$f(A, B, E)$
C	$f(B, C)$, $f(A, C)$	$\{A, B, C\}$	$f(A, B)$
A	$f(A)$, $f(A, B, E)$, $f(A, B)$	$\{A, B, E\}$	$f(B, E)$
B	$f(B)$, $f(B, E)$	$\{B, E\}$	$f(E)$

The largest factor produced has **3** variable(s) (created when eliminating D); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over E and normalise.

Q.22 [MCQ, 2M · Probability for AI] Answer: A ($A: 0.600$)

Bayes rule in odds form: posterior odds = likelihood ratio $\times$ prior odds = $6 \times (1/4) = 1.5$. Then $P(h | e) = 1.5/(1 + 1.5) = 0.6$. Check directly: $P(h) = 0.2$; with $P(e | h) = 6k$ and $P(e | \neg h) = k$, $P(h | e) = 1.2k/(1.2k + 0.8k) = 0.6$. $0.857 = 6/7$ ignores the prior, 0.250 reads the prior odds $1 : 4$ as a probability, and 0.200 is the prior $P(h) = 1/5$, ignoring the evidence.

Q.23 [MSQ, 2M · Bayesian Networks] Answer: A, B

(A) A complete DAG implies no independences, so it is trivially an I-map. (B) AIMA shows causal orderings (e.g. B, E, A, J, M) give compact networks while diagnostic orderings (M, J, E, B, A) can be fully connected. (C) is false: adding an arc can only remove d-separation statements, so the set of implied independences shrinks and remains true in P . (D) is false: for the alarm example, order B, E, A, J, M gives 10 parameters, order M, J, A, B, E gives 13, and M, J, E, B, A gives 31.

Q.24 [NAT, 2M · Exact Inference] Answer: 0.673 to 0.677

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B) \cdot P(C | A, B) \cdot P(D | A, B)$.

Inference by enumeration: $P(C | e) = \alpha \sum_{\text{hidden}} P(C, e, \text{hidden})$.

Hidden: B	C	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T	T	$0.75 \times 0.5 \times 0.6 \times 0.25$	0.05625
F	T	$0.75 \times 0.5 \times 0.7 \times 0.75$	0.19687
T	F	$0.75 \times 0.5 \times 0.4 \times 0.25$	0.0375
F	F	$0.75 \times 0.5 \times 0.3 \times 0.75$	0.08438

$P(C=T, e) = 0.25312$; $P(C=F, e) = 0.12188$; $P(e) = 0.375$.

$$P(C=T | e) = 0.25312 / 0.375 = \mathbf{0.675} \approx 0.675.$$

Q.25 [MSQ, 2M · Conditional Independence] **Answer: B, D** (B: $B \perp A | C$; D: $B \perp A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $E \perp B | A$: **FALSE** — the path $E \leftarrow C \leftarrow B$ is active given $\{A\}$.

(B) $B \perp A | C$: **TRUE** — $B \rightarrow C \rightarrow D \leftarrow A$ (blocked: observed chain node C); $B \rightarrow C \rightarrow D \rightarrow F \leftarrow E \leftarrow A$ (blocked: observed chain node C); $B \rightarrow C \rightarrow E \leftarrow A$ (blocked: observed chain node C); $B \rightarrow C \rightarrow E \rightarrow F \leftarrow D \leftarrow A$ (blocked: observed chain node C); $B \rightarrow C \rightarrow G \leftarrow E \leftarrow A$ (blocked: observed chain node C); $B \rightarrow C \rightarrow G \leftarrow E \rightarrow F \leftarrow D \leftarrow A$ (blocked: observed chain node C).

(C) $B \perp A | F$: **FALSE** — the path $B \rightarrow C \rightarrow D \leftarrow A$ is active given $\{F\}$.

(D) $B \perp A$: **TRUE** — $B \rightarrow C \rightarrow D \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow C \rightarrow D \rightarrow F \leftarrow E \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow C \rightarrow E \leftarrow A$ (blocked: collider E with neither it nor a descendant observed); $B \rightarrow C \rightarrow E \rightarrow F \leftarrow D \leftarrow A$ (blocked: collider F with neither it nor a descendant observed); $B \rightarrow C \rightarrow G \leftarrow E \leftarrow A$ (blocked: collider G with neither it nor a descendant observed); $B \rightarrow C \rightarrow G \leftarrow E \rightarrow F \leftarrow D \leftarrow A$ (blocked: collider G with neither it nor a descendant observed).

Mock Test 086

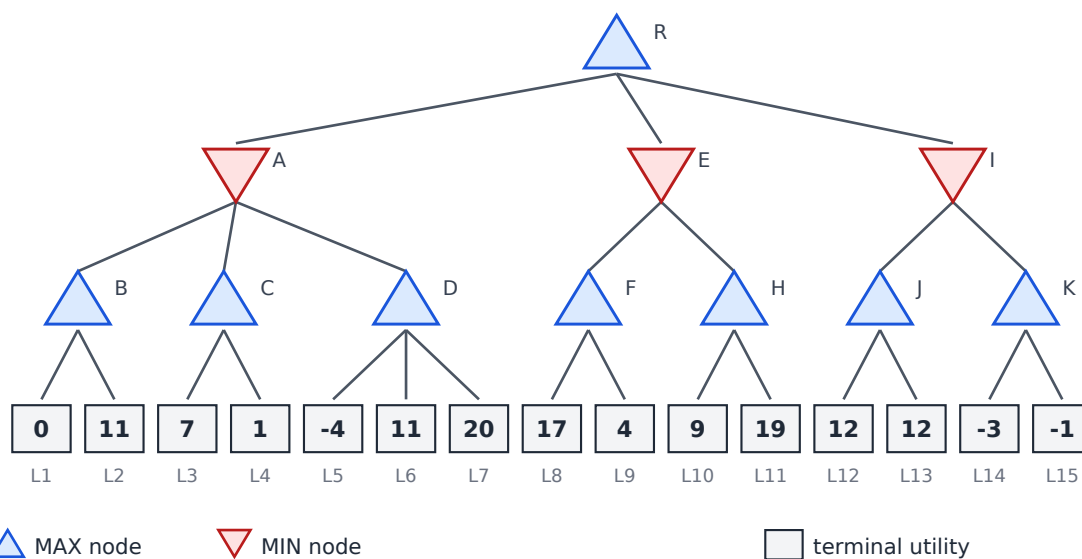
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	2 / 9 / 14	15 / 12 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

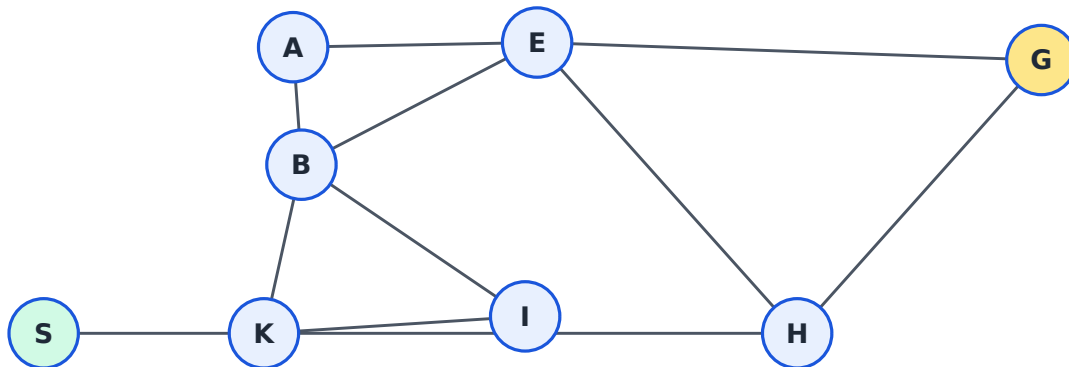
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 4$ and depth $d = 4$ (so $b^d = 256$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, K, I, B, E, H, G
☐ (B) S, B, K, A, E, G
☐ (C) S, K, B, H, I, A, E, G
☐ (D) S, K, B, A, E, G

Q.4 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following equivalences are valid in first-order logic?

- ☐ (A) $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
☐ (B) $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
☐ (C) $\forall x \exists y P(x, y) \equiv \exists y \forall x P(x, y)$
☐ (D) $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = (R \rightarrow (P \leftrightarrow Q)) \leftrightarrow ((R \leftrightarrow Q) \rightarrow \neg P)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.6 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following are tautologies?

- ☐ (A) $p \vee \neg p$
☐ (B) $\neg(p \wedge \neg p)$
☐ (C) $p \rightarrow (p \wedge q)$
☐ (D) $(p \wedge (p \rightarrow q)) \rightarrow q$

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

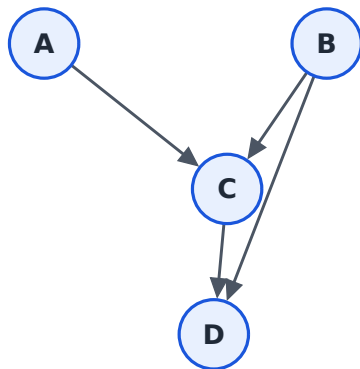
Q.8 [NAT · 1 mark · no negative marking] Bayesian Networks

A naive Bayes classifier has a class variable with 3 values and 10 Boolean features. How many independent parameters does the Bayesian network have?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order A, B, C, D. What is the probability that a single run produces the sample **(A=T, B=T, C=T, D=T)**? (round off to 4 decimal places)



P(A=T)
0.3

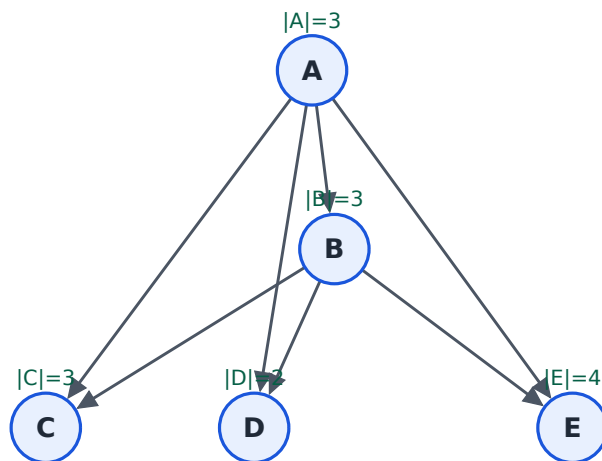
P(B=T)	A	B	P(C=T ...)
0.8	T	T	0.8
	T	F	0.05
	F	T	0.25
	F	F	0.6

B	C	P(D=T ...)
T	T	0.95
T	F	0.75
F	T	0.2
F	F	0.3

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search

Which of the following statements about greedy best-first search are TRUE?

- ☐ (A) The graph-search version is complete in finite state spaces
- ☐ (B) It is optimal whenever the heuristic is admissible
- ☐ (C) Its worst-case time and space complexity is $O(b^m)$, where m is the maximum depth
- ☐ (D) If $h(n) = h^*(n)$ exactly, greedy best-first search always returns an optimal solution

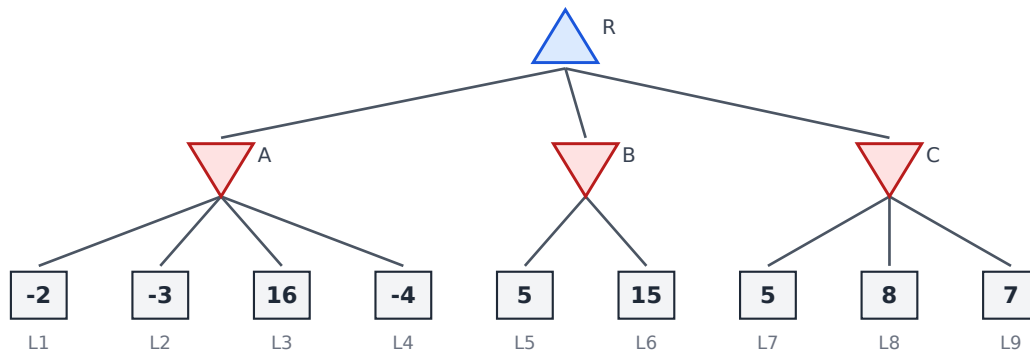
Q.12 [MSQ · 2 marks · no negative marking] Local Search

Which of the following statements about simulated annealing (maximisation) are TRUE?

- ☐ (A) If the temperature is lowered slowly enough, it finds a global optimum with probability approaching 1
- ☐ (B) It always accepts a move that improves the objective
- ☐ (C) At very high temperature, it behaves almost like a random walk
- ☐ (D) At $T = 0$, it accepts worsening moves with probability 1

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



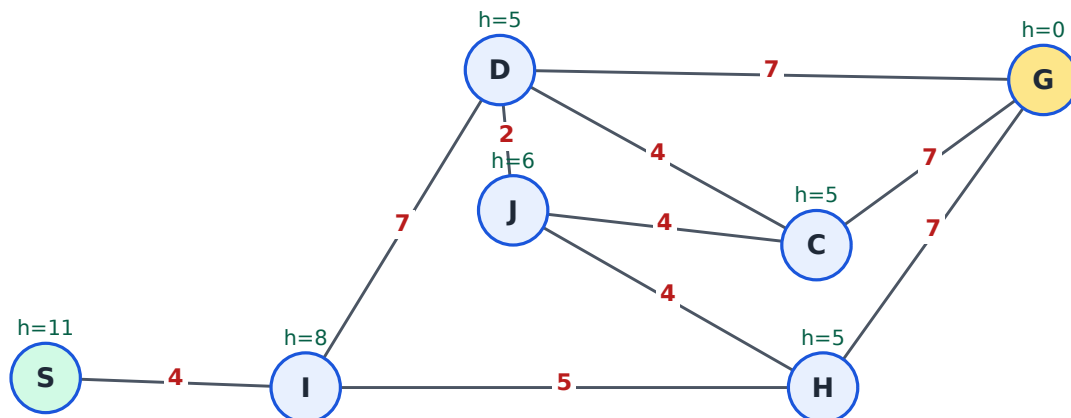
△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

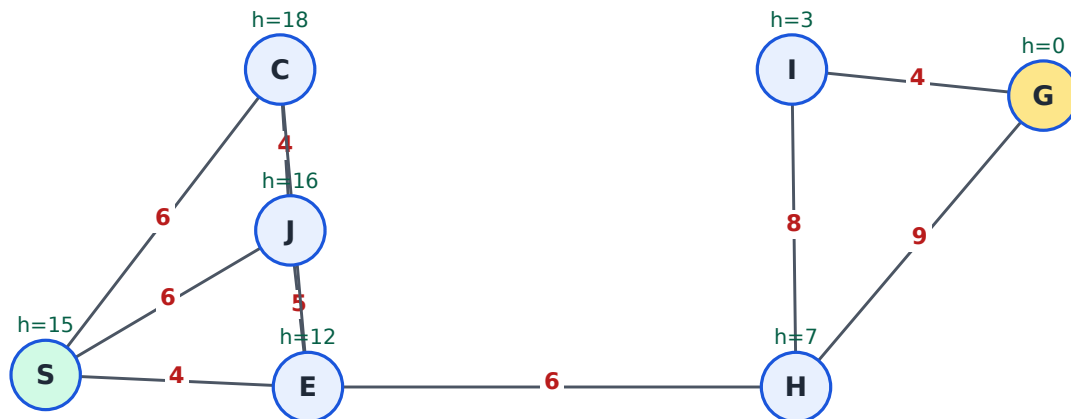
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.16 [MCQ · 2 marks · negative: -2/3] Adversarial Search

MAX has two moves, each leading to a chance node with two equally likely outcomes. Move A has leaf values (0, 10) and move B has leaf values (4, 5). Every leaf value is then replaced by its square root. Which move does expectimax choose before and after the transformation?

- ☐ (A) A before, A after
☐ (B) A before, B after
☐ (C) B before, A after
☐ (D) B before, B after

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic

To prove R from $KB = \{P \vee Q, P \rightarrow R, Q \rightarrow R\}$ by resolution refutation, the clause $\neg R$ is added to the clauses of KB . What is the minimum number of resolution steps needed to derive the empty clause? (Each step resolves two clauses on one complementary pair; duplicate literals are merged.)

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (P \rightarrow \neg Q) \wedge (Q \vee R)$?

- ☐ (A) $(Q \vee R) \wedge (P \rightarrow \neg Q)$
☐ (B) $\neg(\neg(P \rightarrow \neg Q) \vee \neg(Q \vee R))$
☐ (C) $\neg(\neg(P \rightarrow \neg Q) \wedge \neg(Q \vee R))$
☐ (D) $((P \rightarrow \neg Q) \wedge Q) \vee ((P \rightarrow \neg Q) \wedge R)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

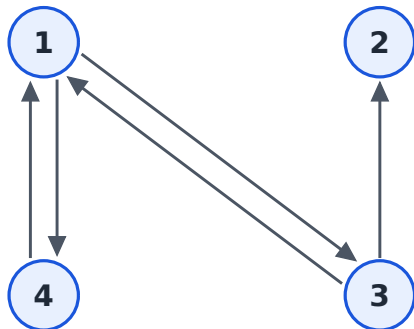
- (1) $(Q \wedge R) \leftrightarrow (P \leftrightarrow P)$
- (2) $(Q \wedge R) \wedge R$
- (3) $(\neg P \leftrightarrow P) \vee \neg P$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg R$
☐ (B) $R \wedge R$
☐ (C) $P \leftrightarrow Q$
☐ (D) $\neg R \wedge R$

Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

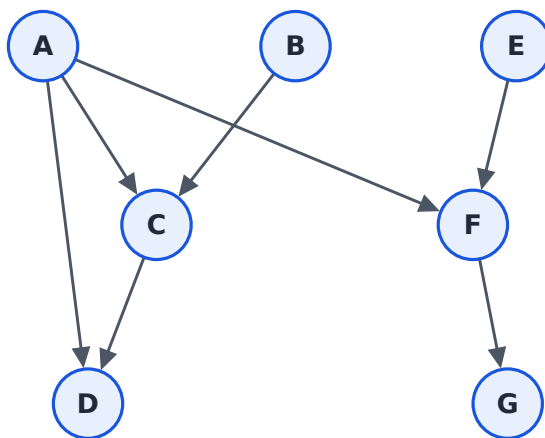
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(x, y)$
☐ (B) $\exists x R(x, x)$
☐ (C) $\exists x \forall y \neg R(y, x)$
☐ (D) $\forall y \exists x R(x, y)$

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp C$
☐ (B) $F \perp C \mid B, D$
☐ (C) $B \perp A$
☐ (D) $G \perp A \mid D, F$

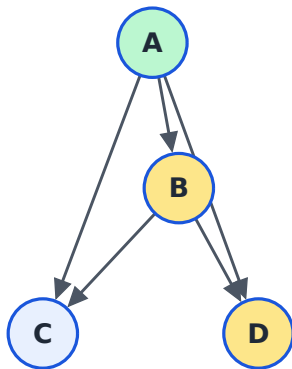
Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence

A Bayesian network over A, B, C, D encodes the factorization $P(A) P(B \mid A) P(C \mid A) P(D \mid B, C)$. Which independences are guaranteed to hold in every distribution with this factorization?

- ☐ (A) $B \perp C \mid \{A\}$
☐ (B) $B \perp C \mid \{A, D\}$
☐ (C) $A \perp D \mid \{B, C\}$
☐ (D) $A \perp D$

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). Compute $P(A = T \mid D = T, B = T)$ (round off to 3 decimal places).



A	P(A=T)
T	0.9

A	P(B=T .)	A	B	P(C=T ...)
T	0.05	T	T	0.05
F	0.05	T	F	0.8
		F	T	0.25
		F	F	0.5

A	B	P(D=T ...)
T	T	0.8
T	F	0.8
F	T	0.5
F	F	0.3

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

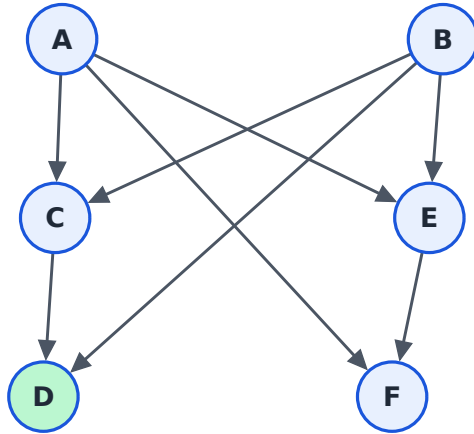
The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid C = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	T	T	F
2	F	T	T	F
3	F	T	T	F
4	T	T	T	T
5	F	T	F	T
6	T	F	F	T
7	T	T	F	T
8	T	T	T	T
9	F	T	F	F
10	T	F	F	T
11	T	T	T	T
12	T	T	T	T
13	T	T	F	T
14	T	T	T	T
15	F	T	T	T

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute **P(D)** in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, C, E, F, A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

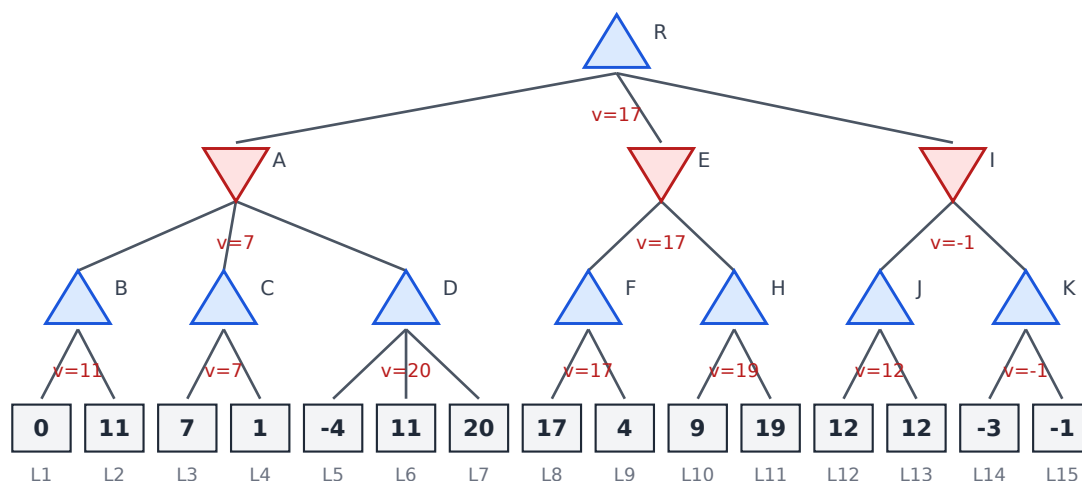
Answer Key — Mock Test 086

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	17	14	NAT	2	Informed Search	5
2	NAT	1	Adversarial Search	31	15	NAT	2	Informed Search	4
3	MCQ	1	Uninformed Search	D	16	MCQ	2	Adversarial Search	B
4	MSQ	1	Predicate Logic	A, B, D	17	NAT	2	Propositional Logic	3
5	NAT	1	Propositional Logic	4	18	MSQ	2	Propositional Logic	A, B, D
6	MSQ	1	Propositional Logic	A, B, D	19	MSQ	2	Propositional Logic	B
7	NAT	1	Predicate Logic	8	20	MSQ	2	Predicate Logic	A, D
8	NAT	1	Bayesian Networks	32	21	MSQ	2	Conditional Independence	C, D
9	NAT	1	Sampling	0.1822 to 0.1826	22	MSQ	2	Conditional Independence	A, C
10	NAT	1	Bayesian Networks	62	23	NAT	2	Exact Inference	0.933 to 0.937
11	MSQ	2	Informed Search	A, C	24	NAT	2	Sampling	0.82 to 0.84
12	MSQ	2	Local Search	A, B, C	25	NAT	2	Exact Inference	4
13	NAT	2	Adversarial Search	7					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] Answer: 17

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 11$, $C = 7$, $D = 20$, $A = 7$, $F = 17$, $H = 19$, $E = 17$, $J = 12$, $K = -1$, $I = -1$. Root value = **17**; MAX's optimal first move is towards E.

Q.2 [NAT, 1M · Adversarial Search] Answer: 31

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 4^2 + 4^2 - 1 = 16 + 16 - 1 = \mathbf{31}$, versus 256 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.3 [MCQ, 1M · Uninformed Search] Answer: D (D: S, K, B, A, E, G)

Adjacency lists (alphabetical): A: {B, E}; B: {A, E, I, K}; E: {A, B, G, H}; G: {E, H}; H: {E, G, K}; I: {B, K}; K: {B, H, I, S}; S: {K}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, K, B, A, E, G**; the path found is $S \rightarrow K \rightarrow B \rightarrow A \rightarrow E \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.4 [MSQ, 1M · Predicate Logic] Answer: A, B, D (A: $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$; B: $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$; D: $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$)

Quantifiers of the same kind commute (A, B). $\exists$ distributes over $\vee$ (D), just as $\forall$ distributes over $\wedge$. (C) Mixed quantifiers do not commute: with $P(x, y) \iff x = y$ on $\{1, 2\}$, $\forall x \exists y P$ is true but $\exists y \forall x P$ is false.

Q.5 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$R \rightarrow (P \leftrightarrow Q)$	$(R \leftrightarrow Q) \rightarrow \neg P$	$(R \rightarrow (P \leftrightarrow Q)) \leftrightarrow ((R \leftrightarrow Q) \rightarrow \neg P)$
T	T	T	T	F	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	F	F
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

F has 4 models, so it is false in $8 - 4 = \mathbf{4}$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.6 [MSQ, 1M · Propositional Logic] Answer: A, B, D (A: $p \vee \neg p$; B: $\neg(p \wedge \neg p)$; D: $(p \wedge (p \rightarrow q)) \rightarrow q$)

$p \vee \neg p$ (excluded middle) and $\neg(p \wedge \neg p)$ (non-contradiction) are valid. $(p \wedge (p \rightarrow q)) \rightarrow q$ is the modus ponens tautology. $p \rightarrow (p \wedge q)$ is false at $p=T, q=F$, so it is not valid.

Q.7 [NAT, 1M · Predicate Logic] Answer: 8

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 3$ this gives **8** (verified by enumerating all interpretations).

Q.8 [NAT, 1M · Bayesian Networks] Answer: 32

$P(C)$ needs $3 - 1 = 2$ numbers. Each feature's CPT $P(F_i | C)$ has 3 rows with 1 free number each, i.e. 3 numbers. Total = $2 + 10 \times 3 = 32$ (versus $3 \times 2^{10} - 1 = 3071$ for the full joint).

Q.9 [NAT, 1M · Sampling] Answer: 0.1822 to 0.1826

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(A=T) \times P(B=T) \times P(C=T|A=T, B=T) \times P(D=T|B=T, C=T) \\ = 0.3 \times 0.8 \times 0.8 \times 0.95 = \mathbf{0.1824} \approx 0.1824.$$

Q.10 [NAT, 1M · Bayesian Networks] Answer: 62

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$$\Sigma = A: (3-1) + B: (3-1) \times 3 + C: (3-1) \times 3 \times 3 + D: (2-1) \times 3 \times 3 + E: (4-1) \times 3 \times 3 = \mathbf{62}.$$

Full joint table: $\prod |X| - 1 = 216 - 1 = 215$; saving = 153.

Q.11 [MSQ, 2M · Informed Search] Answer: A, C (A: The graph-search version is complete in finite state spaces; C: Its worst-case time and space complexity is $O(b^m)$, where m is the maximum depth)

(A) is true: in a finite space with an explored set, it eventually expands every state. (B) is false: greedy search ignores g , so admissibility does not help. (C) is true, as stated in AIMA. (D) is false. Counterexample: $S \rightarrow A$ costs 100 and $A \rightarrow G$ costs 1, while $S \rightarrow B$ costs 1 and $B \rightarrow G$ costs 2. Then $h^*(A)=1 < h^*(B)=2$, so greedy expands A and returns cost 101 instead of 3. A perfect heuristic makes A^* optimal, not greedy search.

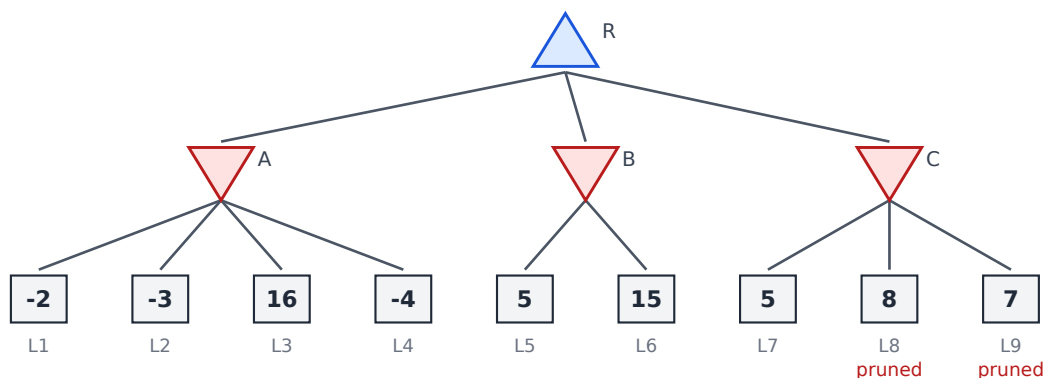
Q.12 [MSQ, 2M · Local Search] Answer: A, B, C

(A) is the classical convergence result for sufficiently slow cooling. (B) is true: moves with $\Delta E > 0$ are always taken. (C) is true: $e^{\Delta E/T} \rightarrow 1$ as $T \rightarrow \infty$, so almost every move is accepted. (D) is false: as $T \rightarrow 0$, $e^{\Delta E/T} \rightarrow 0$ for $\Delta E < 0$, so worsening moves are rejected and the algorithm becomes hill climbing.

Q.13 [NAT, 2M · Adversarial Search] Answer: 7

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node C: value $5 \leq \alpha = 5 \Rightarrow$ prune remaining children (leaves L8, L9).



Leaves evaluated (7): L1, L2, L3, L4, L5, L6, L7. Leaves pruned (2): L8, L9. Root value = 5 (same as plain minimax, 5 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(C)=5$, $h(D)=5$, $h(G)=0$, $h(H)=5$, $h(I)=8$, $h(J)=6$, $h(S)=11$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	I(12)
2	I (12)	H(14), D(16)
3	H (14)	D(16), G(16), J(19)
4	D (16)	G(16), J(19), C(20)
5	G (16)	J(19), C(20) ← goal, stop

Path returned: **S** → **I** → **H** → **G** with cost **16**; nodes expanded = **5** (S, I, H, D, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [NAT, 2M · Informed Search] Answer: 4

Heuristic values: $h(C)=18$, $h(E)=12$, $h(G)=0$, $h(H)=7$, $h(I)=3$, $h(J)=16$, $h(S)=15$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	E(16), J(22), C(24)
2	E (16)	H(17), J(22), C(24)
3	H (17)	G(19), I(21), J(22), C(24)
4	G (19)	I(21), J(22), C(24) ← goal, stop

Path returned: **S** → **E** → **H** → **G** with cost **19**; nodes expanded = **4** (S, E, H, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.16 [MCQ, 2M · Adversarial Search] Answer: B (B: A before, B after)

Before: $E[A] = (0+10)/2 = 5$ and $E[B] = 4.5$, so A is chosen. After: $E[A] = (0 + 3.162)/2 \approx 1.58$ and $E[B] = (2 + 2.236)/2 \approx 2.12$, so B is chosen. Expectimax decisions are preserved only by positive linear transformations of the evaluation function. A merely order-preserving map such as $\sqrt{x}$ can change them. Minimax decisions are invariant under any order-preserving map.

Q.17 [NAT, 2M · Propositional Logic] Answer: 3

Clauses: $P \vee Q$, $\neg P \vee R$, $\neg Q \vee R$, $\neg R$. A 3-step refutation: $(P \vee Q, \neg P \vee R) \Rightarrow Q \vee R$; $(Q \vee R, \neg Q \vee R) \Rightarrow R$; $(R, \neg R) \Rightarrow \square$. 2 steps are impossible: the last step must resolve two complementary unit clauses, and the only unit initially is $\neg R$. So R (a unit) must be produced in a single step, but every one-step resolvent is $Q \vee R$, $P \vee R$, $\neg P$, $\neg Q$ or a tautology — never R . Hence the minimum is 3 (a breadth-first search over resolution derivations confirms this).

Q.18 [MSQ, 2M · Propositional Logic] Answer: A, B, D (A: $(Q \vee R) \wedge (P \rightarrow \neg Q)$; B: $\neg(\neg(P \rightarrow \neg Q) \vee \neg(Q \vee R))$; D: $((P \rightarrow \neg Q) \wedge Q) \vee ((P \rightarrow \neg Q) \wedge R)$)

Compare truth tables with $F = (P \rightarrow \neg Q) \wedge (Q \vee R)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

- (A) $(Q \vee R) \wedge (P \rightarrow \neg Q)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.
 (B) $\neg(\neg(P \rightarrow \neg Q) \vee \neg(Q \vee R))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.
 (C) $\neg(\neg(P \rightarrow \neg Q) \wedge \neg(Q \vee R))$ — differs at $P=T, Q=T, R=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.
 (D) $((P \rightarrow \neg Q) \wedge Q) \vee ((P \rightarrow \neg Q) \wedge R)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

Q.19 [MSQ, 2M · Propositional Logic] Answer: B (B: $R \wedge R$)

$KB \models \alpha$ iff α is true in every model of KB . Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
F	T	T	T

- (A) $\neg R$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.
 (B) $R \wedge R$ is true in all models of $KB \Rightarrow$ **entailed**.
 (C) $P \leftrightarrow Q$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.
 (D) $\neg R \wedge R$ is false in the KB-model $P=F, Q=T, R=T \Rightarrow$ not entailed.

Q.20 [MSQ, 2M · Predicate Logic] Answer: A, D (A: $\exists x \forall y \neg R(x, y)$; D: $\forall y \exists x R(x, y)$)

$R = \{(1,3), (1,4), (3,1), (3,2), (4,1)\}$.

- (A) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.
 (B) $\exists x R(x, x)$: **FALSE** — there are no self-loops.
 (C) $\exists x \forall y \neg R(y, x)$: **FALSE** — every element has an incoming edge.
 (D) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.21 [MSQ, 2M · Conditional Independence] Answer: C, D ($C: B \perp A$; $D: G \perp A \mid D, F$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp C$: **FALSE** — the path $A \rightarrow C$ is active given $\{\}$.

(B) $F \perp C \mid B, D$: **FALSE** — the path $F \leftarrow A \rightarrow C$ is active given $\{B, D\}$.

(C) $B \perp A$: **TRUE** — $B \rightarrow C \leftarrow A$ (blocked: collider C with neither it nor a descendant observed); $B \rightarrow C \rightarrow D \leftarrow A$ (blocked: collider D with neither it nor a descendant observed).

(D) $G \perp A \mid D, F$: **TRUE** — $G \leftarrow F \leftarrow A$ (blocked: observed chain node F).

Q.22 [MSQ, 2M · Conditional Independence] Answer: A, C ($A: B \perp C \mid \{A\}$; $C: A \perp D \mid \{B, C\}$)

The DAG is $A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D$. (A) B-C paths: $B \leftarrow A \rightarrow C$ (fork, A observed: blocked) and $B \rightarrow D \leftarrow C$ (collider D unobserved, no descendants: blocked): d-separated, true. (B) Observing D activates the collider path $B \rightarrow D \leftarrow C$: false. (C) Both A-D paths $A \rightarrow B \rightarrow D$ and $A \rightarrow C \rightarrow D$ pass through observed non-colliders B and C: d-separated, true. (D) With nothing observed, $A \rightarrow B \rightarrow D$ is active: false.

Q.23 [NAT, 2M · Exact Inference] Answer: 0.933 to 0.937

Factorisation: $P(A, B, C, D) = P(A) \cdot P(B \mid A) \cdot P(C \mid A, B) \cdot P(D \mid A, B)$.

Inference by enumeration: $P(A \mid e) = \alpha \sum_{\text{hidden}} P(A, e, \text{hidden})$.

Hidden: C	A	product of CPT entries ($A \cdot B \cdot C \cdot D$)	value
T	T	$0.9 \times 0.05 \times 0.05 \times 0.8$	0.0018
F	T	$0.9 \times 0.05 \times 0.95 \times 0.8$	0.0342
T	F	$0.1 \times 0.05 \times 0.25 \times 0.5$	0.00062
F	F	$0.1 \times 0.05 \times 0.75 \times 0.5$	0.00187

$P(A=T, e) = 0.036$; $P(A=F, e) = 0.0025$; $P(e) = 0.0385$.

$P(A=T \mid e) = 0.036 / 0.0385 = \mathbf{0.9351} \approx 0.935$.

Q.24 [NAT, 2M · Sampling] Answer: 0.82 to 0.84

Rejection sampling keeps only samples consistent with the evidence $C = F$: samples 5, 6, 7, 9, 10, 13 (6 accepted, 9 rejected).

Among the accepted samples, $D = T$ in 5. Estimate = $5/6 = \mathbf{0.8333}$.

(For this network the exact value is 0.8368; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.25 [NAT, 2M · Exact Inference] Answer: 4

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(B)$, $f(A, B, C)$, $f(B, C, D)$, $f(A, B, E)$, $f(A, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(B)$, $f(A, B, C)$, $f(B, C, D)$, $f(A, B, E)$	$\{A, B, C, D, E\}$	$f(A, C, D, E)$
C	$f(A, C, D, E)$	$\{A, C, D, E\}$	$f(A, D, E)$
E	$f(A, E, F)$, $f(A, D, E)$	$\{A, D, E, F\}$	$f(A, D, F)$
F	$f(A, D, F)$	$\{A, D, F\}$	$f(A, D)$
A	$f(A)$, $f(A, D)$	$\{A, D\}$	$f(D)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over D and normalise.

Mock Test 087

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 8 / 11	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Depth-limited search with limit $l = 4$ is run on a tree with branching factor $b = 4$. No goal exists within depth 4. How many nodes (including the root) does it generate before returning cutoff?

Answer (numerical): _____

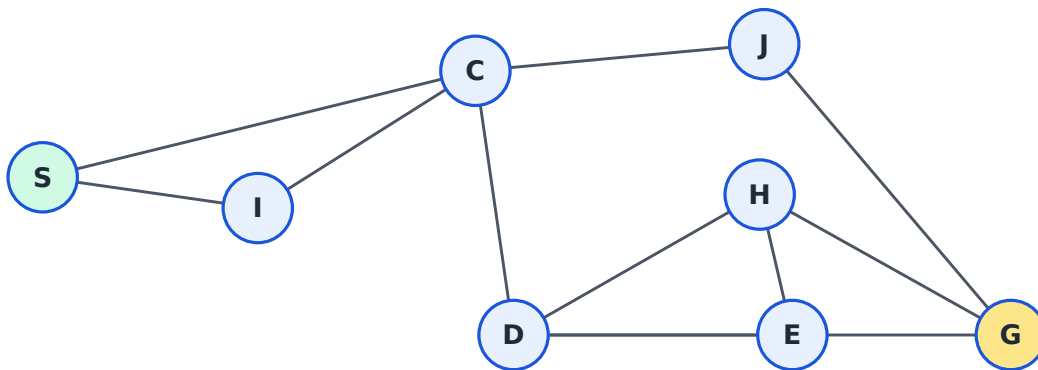
Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search

The frontier of depth-first search is implemented as a

- ☐ (A) FIFO queue
- ☐ (B) LIFO stack
- ☐ (C) priority queue ordered by $g(n)$
- ☐ (D) priority queue ordered by $h(n)$

Q.3 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

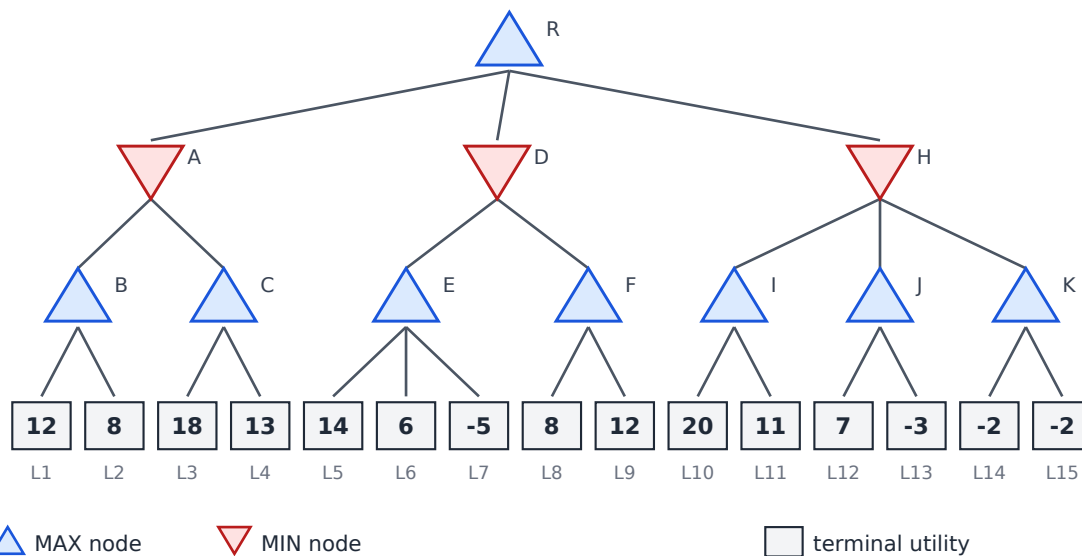
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, C, I, D, J, E, G
- ☐ (B) S, C, D, E, H, G
- ☐ (C) S, I, C, J, G
- ☐ (D) S, C, E, D, H, G

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

For the game tree below, what is the minimax value of the root?



Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic

Let $D = \{1, 2, 3, 4\}$. How many interpretations of a unary predicate P over D make $\exists x P(x) \wedge \neg \forall x P(x)$ true?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = (\neg R \leftrightarrow (\neg Q \rightarrow P)) \vee ((Q \rightarrow \neg Q) \rightarrow (P \wedge P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **D, B, C** and rules:

- R1: $H \rightarrow C$
- R2: $D \wedge H \rightarrow C$
- R3: $E \wedge C \rightarrow H$
- R4: $D \wedge C \rightarrow E$
- R5: $B \rightarrow C$
- R6: $D \wedge B \rightarrow F$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.8 [MSQ · 1 mark · no negative marking] Predicate Logic

Without standardizing variables apart, $\text{Unify}(\text{Knows}(\text{John}, x), q)$ succeeds for which of the following atoms q ?

- ☐ (A) $\text{Knows}(\text{John}, \text{Jane})$
- ☐ (B) $\text{Knows}(y, \text{Bill})$
- ☐ (C) $\text{Knows}(y, \text{Mother}(y))$
- ☐ (D) $\text{Knows}(x, \text{Elizabeth})$

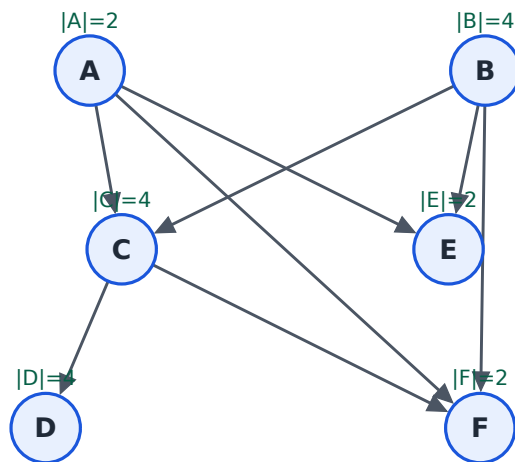
Q.9 [MCQ · 1 mark · negative: -1/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Likelihood weighting is used with evidence $C = \text{true}$ only (a root node), to estimate $P(R | c)$. What is the weight of every sample?

- ☐ (A) 0.5
☐ (B) 1.0
☐ (C) It depends on the sampled value of R
☐ (D) 0.8

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks - Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

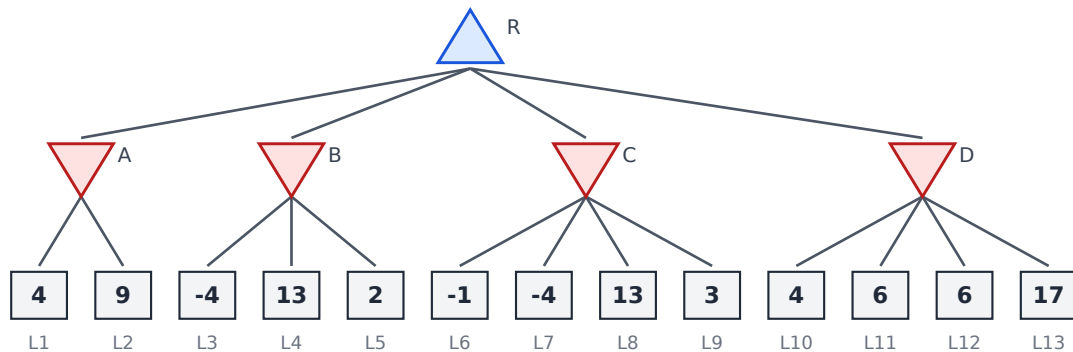
Section B — 2-mark questions**Q.11** [MCQ · 2 marks · negative: -2/3] Uninformed Search

BFS runs on a problem with $b = 10$ and solution depth $d = 8$. It stores every generated node, and each node needs 1 KB. Approximately how much memory is required? (1 GB = 10^6 KB.)

- ☐ (A) About 100 MB
☐ (B) About 1 GB
☐ (C) About 100 GB
☐ (D) About 10 TB

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



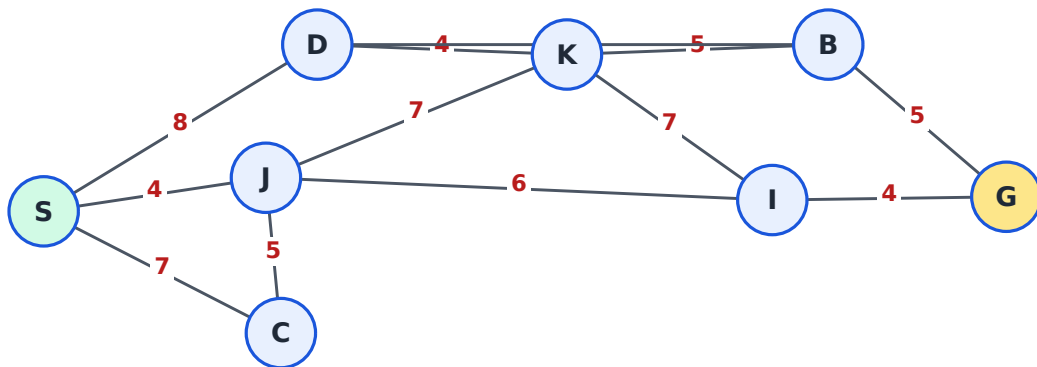
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L3 is pruned.
☐ (B) Leaf L4 is pruned.
☐ (C) Leaf L6 is pruned.
☐ (D) Leaf L10 is pruned.

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

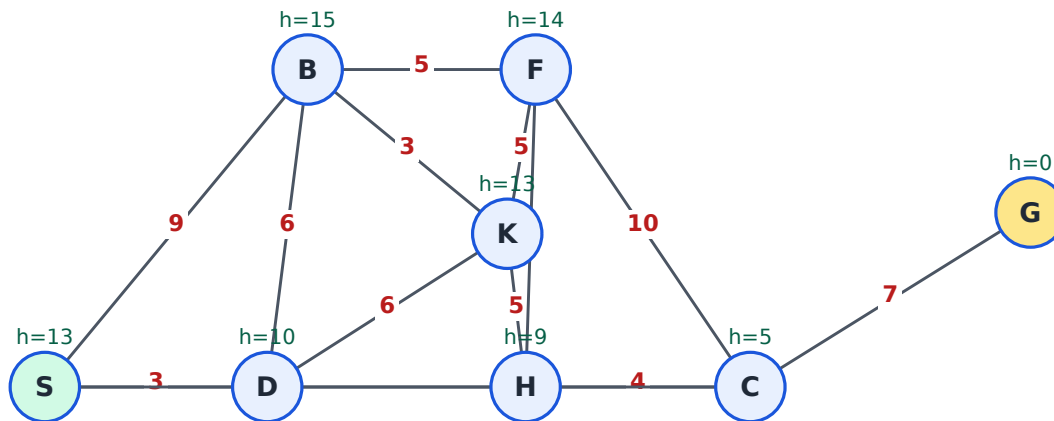
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – A*

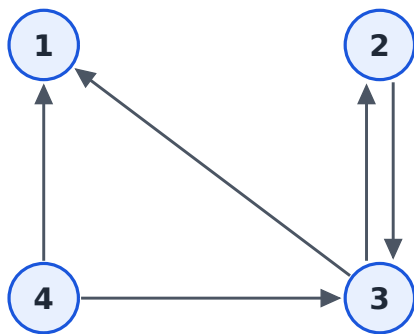
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node B is expanded.
☐ (B) The path returned has cost 16.
☐ (C) Node C is expanded.
☐ (D) Node K is expanded.

Q.15 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\forall x R(x, x)$
☐ (C) $\exists x \forall y \neg R(y, x)$
☐ (D) $\exists x \forall y \neg R(x, y)$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following sentences (with equality) are true in exactly those interpretations where exactly one element satisfies P ?

- ☐ (A) $\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x))$
☐ (B) $\exists x \forall y (P(y) \leftrightarrow y = x)$
☐ (C) $\exists x \forall y ((P(x) \wedge P(y)) \rightarrow x = y)$
☐ (D) $\exists x (P(x) \wedge \forall y (y = x \rightarrow P(y)))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$
☐ (B) $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$
☐ (C) $\neg(R \wedge Q) \leftrightarrow (\neg R \wedge \neg Q)$
☐ (D) $((R \rightarrow Q) \rightarrow R) \rightarrow R$

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

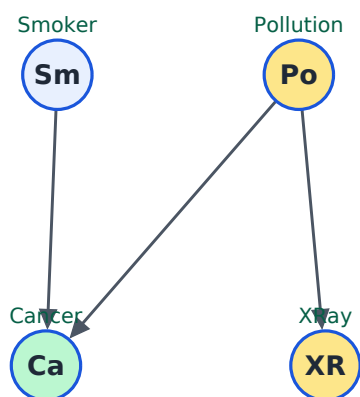
C1: $(P \vee R \vee S)$, C2: $(P \vee \neg R)$, C3: $(\neg P \vee Q \vee S)$, C4: $(P \vee Q \vee \neg S)$, C5: $(Q \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P)$
☐ (B) $(P \vee Q)$
☐ (C) $(P \vee Q \vee R)$
☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

Q.19 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute **$P(\text{Cancer} = T \mid \text{Pollution} = T, \text{XRay} = F)$** (round off to 3 decimal places).



P(Sm=T)
0.6

P(Po=T)	Sm	Po	P(Ca=T ...)
0.6	T	T	0.4
	T	F	0.6
	F	T	0.2
	F	F	0.4

Po	P(XR=T ...)
T	0.8
F	0.95

Answer (numerical): _____

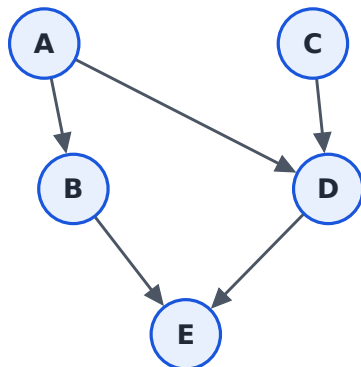
Q.20 [NAT · 2 marks · no negative marking] Probability for AI

A full joint distribution over Boolean variables X, Y, Z is: $P(x, y, z) = 0.10$, $P(x, y, \neg z) = 0.05$, $P(x, \neg y, z) = 0.15$, $P(x, \neg y, \neg z) = 0.10$, $P(\neg x, y, z) = 0.05$, $P(\neg x, y, \neg z) = 0.20$, $P(\neg x, \neg y, z) = 0.15$, $P(\neg x, \neg y, \neg z) = 0.20$. Compute $P(x \mid y \vee z)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.21 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = (0.62, 0.17, 0.99, 0.61, 0.42)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.7

A	P(B=T ...)
T	0.5
F	0.3

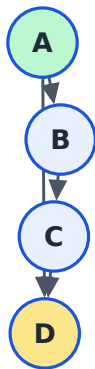
P(C=T)
0.05

A	C	P(D=T ...)	B	D	P(E=T ...)
T	T	0.4	T	T	0.95
T	F	0.4	T	F	0.7
F	T	0.95	F	T	0.7
F	F	0.05	F	F	0.4

- ☐ (A) A=T, B=F, C=F, D=F, E=T
☐ (B) A=F, B=T, C=F, D=F, E=T
☐ (C) A=F, B=F, C=T, D=F, E=T
☐ (D) A=T, B=T, C=F, D=F, E=T

Q.22 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{D=F}$. The current state of the chain is $(\mathbf{A=T}, \mathbf{B=F}, \mathbf{C=T}, \mathbf{D=F})$. The next step resamples **A**. With what probability is A set to T? (round off to 3 decimal places)



P(A=T)
0.9

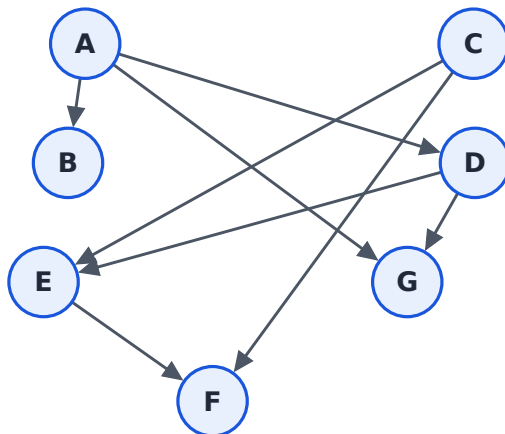
A	P(B=T ...)	B	P(C=T ...)
T	0.5	T	0.6
F	0.75	F	0.05

A	C	P(D=T ...)
T	T	0.6
T	F	0.75
F	T	0.05
F	F	0.5

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp C$
☐ (B) $A \perp E \mid C, F$
☐ (C) $A \perp F \mid C, D$
☐ (D) $C \perp A \mid E$

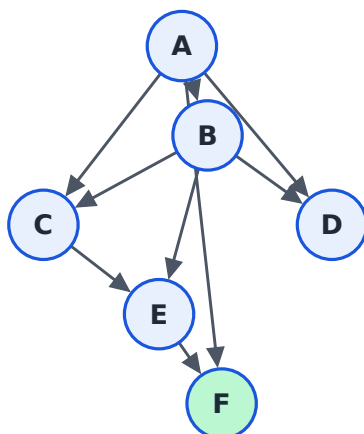
Q.24 [NAT · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $C \rightarrow A$ and $C \rightarrow B$ (Boolean variables) with $P(c) = 0.4$, $P(a \mid c) = 0.9$, $P(a \mid \neg c) = 0.2$, $P(b \mid c) = 0.7$, $P(b \mid \neg c) = 0.1$. Compute $P(a, b)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(\mathbf{F})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, D, E, A, B**. What is the scope of the new factor produced when **A** is summed out?



- ☐ (A) $f(B)$
☐ (B) $f()$
☐ (C) $f(B, F)$
☐ (D) $f(A, B, F)$

Answer Key — Mock Test 087

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	341	14	MSQ	2	Informed Search	B, C
2	MCQ	1	Uninformed Search	B	15	MSQ	2	Predicate Logic	C, D
3	MCQ	1	Uninformed Search	B	16	MSQ	2	Predicate Logic	A, B
4	NAT	1	Adversarial Search	12	17	MSQ	2	Propositional Logic	B, D
5	NAT	1	Predicate Logic	14	18	MSQ	2	Propositional Logic	B, C
6	NAT	1	Propositional Logic	1	19	NAT	2	Exact Inference	0.318 to 0.322
7	NAT	1	Propositional Logic	6	20	NAT	2	Probability for AI	0.428 to 0.429
8	MSQ	1	Predicate Logic	A, B, C	21	MCQ	2	Sampling	D
9	MCQ	1	Sampling	A	22	NAT	2	Sampling	0.881 to 0.885
10	NAT	1	Bayesian Networks	80	23	MSQ	2	Conditional Independence	A, C
11	MCQ	2	Uninformed Search	C	24	NAT	2	Conditional Independence	0.263 to 0.265
12	MSQ	2	Adversarial Search	B	25	MCQ	2	Exact Inference	C
13	NAT	2	Uninformed Search	14					

Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] Answer: 341

DLS explores the complete tree down to depth l : $1 + b + \dots + b^l = 1 + 4 + 16 + 64 + 256 = 341 = (b^{(l+1)} - 1)/(b - 1)$.

Q.2 [MCQ, 1M · Uninformed Search] Answer: B (B: LIFO stack)

DFS always expands the most recently generated (deepest) node, which is **LIFO** behaviour. A FIFO queue gives BFS (A). A priority queue on $g(n)$ gives UCS (C). A priority queue on $h(n)$ gives greedy best-first search (D).

Q.3 [MCQ, 1M · Uninformed Search] Answer: B (B: S, C, D, E, H, G)

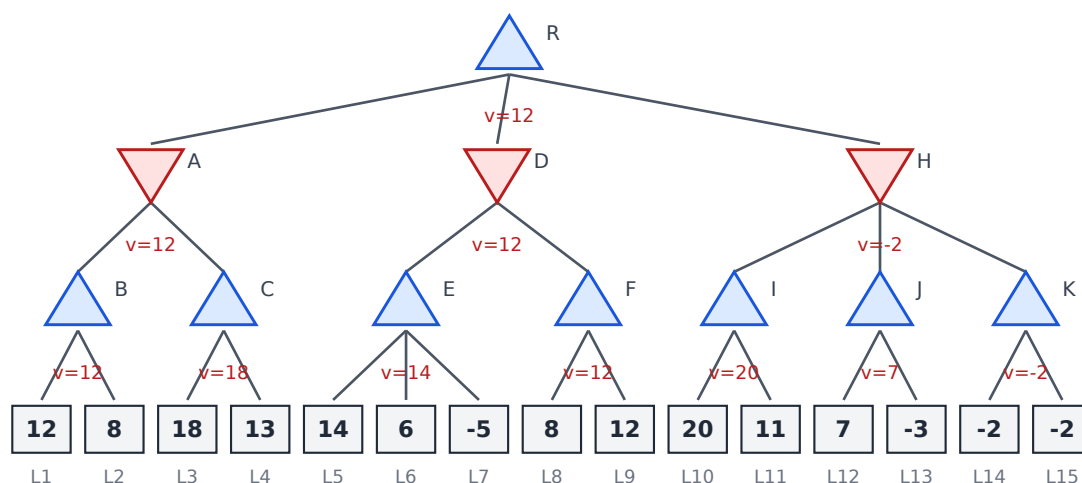
Adjacency lists (alphabetical): C: {D, I, J, S}; D: {C, E, G, H}; E: {D, H}; G: {D, H, J}; H: {D, E, G}; I: {C, S}; J: {C, G}; S: {C, I}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, C, D, E, H, G**; the path found is $S \rightarrow C \rightarrow D \rightarrow E \rightarrow H \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Adversarial Search] Answer: 12

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 12$, $C = 18$, $A = 12$, $E = 14$, $F = 12$, $D = 12$, $I = 20$, $J = 7$, $K = -2$, $H = -2$. Root value = **12**; MAX's optimal first move is towards A, D.

Q.5 [NAT, 1M · Predicate Logic] **Answer: 14**

An interpretation of P is a subset of D: $2^4 = 16$ choices. $\exists x P(x)$ excludes the empty set, $\neg \forall x P(x)$ excludes D itself. Hence $16 - 2 = 14$.

Q.6 [NAT, 1M · Propositional Logic] **Answer: 1**

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg R \leftrightarrow (\neg Q \rightarrow P)$	$(Q \rightarrow \neg Q) \rightarrow (P \wedge P)$	$(\neg R \leftrightarrow (\neg Q \rightarrow P)) \vee ((Q \rightarrow \neg Q) \rightarrow (P \wedge P))$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	T	T	T
F	F	T	T	F	T
F	F	F	F	F	F

F has 7 models, so it is false in $8 - 7 = 1$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.7 [NAT, 1M · Propositional Logic] **Answer: 6**

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $D \wedge C \rightarrow E$, $D \wedge B \rightarrow F$; add E, F.

Round 2: fire $E \wedge C \rightarrow H$; add H.

Fixed point: {B, C, D, E, F, H} — 6 atoms are entailed (facts: B, C, D; newly derived: E, F, H). Atoms never derived: A, J, K.

Q.8 [MSQ, 1M · Predicate Logic] **Answer: A, B, C** (A: Knows(John, Jane); B: Knows(y, Bill); C: Knows(y, Mother(y)))

(A) {x/Jane}. (B) {x/Bill, y/John}. (C) {y/John, x/Mother(John)}. (D) fails: x would have to be both John and Elizabeth. Renaming the second x (standardizing apart) would make it succeed, which is why AIMA insists on standardizing apart.

Q.9 [MCQ, 1M · Sampling] **Answer: A** (A: 0.5)

The weight is the product of $P(e_i | \text{parents})$ over evidence variables; here only C, with no parents, so $w = P(c) = 0.5$ for every sample. Since all weights are equal, LW reduces to sampling R from $P(R | c)$, which is the exact posterior. The weight does not depend on R, and 0.8 is $P(r | c)$, a sampling probability, not a weight.

Q.10 [NAT, 1M · Bayesian Networks] **Answer: 80**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (4-1) + C: (4-1) \times 2 \times 4 + D: (4-1) \times 4 + E: (2-1) \times 2 \times 4 + F: (2-1) \times 2 \times 4 \times 4 = 80$.

Full joint table: $\prod |X| - 1 = 512 - 1 = 511$; saving = 431.

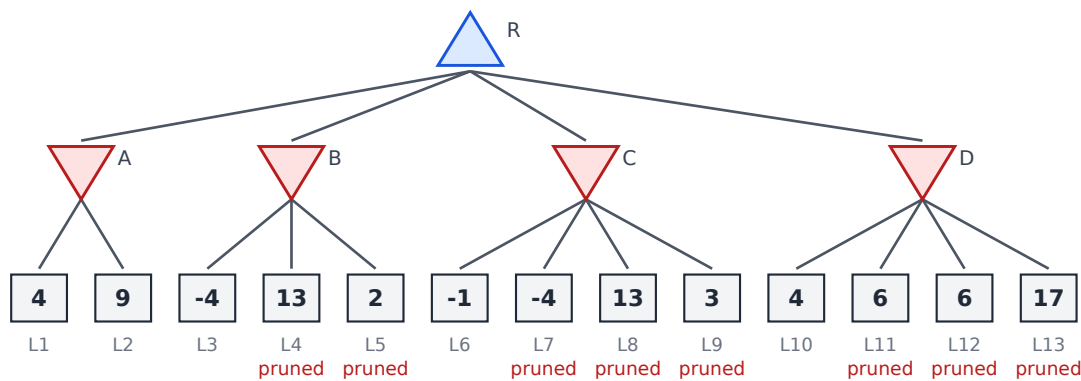
Q.11 [MCQ, 2M · Uninformed Search] **Answer: C** (C: About 100 GB)

The nodes generated are $b + \dots + b^8 \approx 1.11 \times 10^8$. At 1 KB each, that is about $1.11 \times 10^8 \text{ KB} \approx 111 \text{ GB}$, i.e. **about 100 GB**. This is AIMA's point that memory, not time, is the binding constraint for BFS. 100 MB (A) and 1 GB (B) are 2 to 3 orders of magnitude too small, and 10 TB (D) is 100 times too large.

Q.12 [MSQ, 2M · Adversarial Search] **Answer: B** (B: Leaf L4 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value $-4 \leq \alpha = 4 \Rightarrow$ prune remaining children (leaves L4, L5).
- At MIN node C: value $-1 \leq \alpha = 4 \Rightarrow$ prune remaining children (leaves L7, L8, L9).
- At MIN node D: value $4 \leq \alpha = 4 \Rightarrow$ prune remaining children (leaves L11, L12, L13).



Leaves evaluated (5): L1, L2, L3, L6, L10. Leaves pruned (8): L4, L5, L7, L8, L9, L11, L12, L13. Root value = 4 (same as plain minimax, 4 — pruning never changes the root value).

Q.13 [NAT, 2M · Uninformed Search] Answer: 14

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(4), C(7), D(8)
2	J (4)	C(7), D(8), I(10), K(11)
3	C (7)	D(8), I(10), K(11)
4	D (8)	I(10), K(11), B(14)
5	I (10)	K(11), B(14), G(14)
6	K (11)	B(14), G(14)
7	B (14)	G(14)
8	G (14)	← goal, stop

Path returned: **S** → **J** → **I** → **G**, cost = 4 + 6 + 4 = **14**. Nodes expanded (including S and G): **8** (S, J, C, D, I, K, B, G).

Q.14 [MSQ, 2M · Informed Search] Answer: B, C (B: The path returned has cost 16.; C: Node C is expanded.)

Heuristic values: $h(B)=15$, $h(C)=5$, $h(D)=10$, $h(F)=14$, $h(G)=0$, $h(H)=9$, $h(K)=13$, $h(S)=13$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	D(13), H(14), B(24)
2	D (13)	C(14), H(14), K(22), B(24)
3	C (14)	H(14), G(16), K(22), B(24), F(33)
4	H (14)	G(16), K(22), B(24), F(25)
5	G (16)	K(22), B(24), F(25) ← goal, stop

Path returned: **S** → **D** → **C** → **G** with cost **16**; nodes expanded = **5** (S, D, C, H, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 16$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [MSQ, 2M · Predicate Logic] Answer: C, D (C: $\exists x \forall y \neg R(y, x)$; D: $\exists x \forall y \neg R(x, y)$)

$R = \{(2,3), (3,1), (3,2), (4,1), (4,3)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(4,3) \wedge R(3,2)$ but $\neg R(4,2)$.

(B) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(C) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

(D) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.16 [MSQ, 2M · Predicate Logic] Answer: A, B (A: $\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x))$; B: $\exists x \forall y (P(y) \leftrightarrow y = x)$)

(A) Standard 'exactly one': some x is P and every P is x . (B) For some x , $P(y)$ holds iff $y = x$ — also exactly one; A and B are equivalent. (C) Is true when no element satisfies P ($P(x)$ false makes the implication vacuous), so it expresses 'at most one'. (D) The inner part is trivially implied by $P(x)$, so it is just $\exists x P(x)$ ('at least one'). Brute force on domains of size 1–4 confirms that only A and B match 'exactly one'.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: B, D** (B: $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$; D: $((R \rightarrow Q) \rightarrow R) \rightarrow R$)

(A) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$: **not valid** — counter-model $Q=F, R=T$.

(B) $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $\neg(R \wedge Q) \leftrightarrow (\neg R \wedge \neg Q)$: **not valid** — counter-model $Q=T, R=F$.

(D) $((R \rightarrow Q) \rightarrow R) \rightarrow R$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: B, C** (B: $(P \vee Q)$; C: $(P \vee Q \vee R)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(P \vee Q)$ is derivable:

1. resolve $(P \vee R \vee S)$ and $(P \vee Q \vee \neg S)$ on $S \Rightarrow (P \vee Q \vee R)$

2. resolve $(P \vee \neg R)$ and $(P \vee Q \vee R)$ on $R \Rightarrow (P \vee Q)$

(C) $(P \vee Q \vee R)$ is derivable:

1. resolve $(P \vee R \vee S)$ and $(P \vee Q \vee \neg S)$ on $S \Rightarrow (P \vee Q \vee R)$

(D) The resolution closure contains 17 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

Q.19 [NAT, 2M · Exact Inference] **Answer: 0.318 to 0.322**

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution}) \cdot P(\text{Cancer} \mid \text{Smoker}, \text{Pollution}) \cdot P(\text{XRay} \mid \text{Pollution})$.

Inference by enumeration: $P(\text{Cancer} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cancer}, e, \text{hidden})$.

Hidden: Sm	Ca	product of CPT entries (Sm · Po · Ca · XR)	value
T	T	$0.6 \times 0.6 \times 0.4 \times 0.2$	0.0288
F	T	$0.4 \times 0.6 \times 0.2 \times 0.2$	0.0096
T	F	$0.6 \times 0.6 \times 0.6 \times 0.2$	0.0432
F	F	$0.4 \times 0.6 \times 0.8 \times 0.2$	0.0384

$P(\text{Cancer}=T, e) = 0.0384$; $P(\text{Cancer}=F, e) = 0.0816$; $P(e) = 0.12$.

$P(\text{Cancer}=T \mid e) = 0.0384 / 0.12 = \mathbf{0.32} \approx 0.320$.

Q.20 [NAT, 2M · Probability for AI] **Answer: 0.428 to 0.429**

$P(y \vee z) = 1 - P(\neg y \wedge \neg z) = 1 - [P(x, \neg y, \neg z) + P(\neg x, \neg y, \neg z)] = 1 - (0.10 + 0.20) = 0.70$. $P(x \wedge (y \vee z)) = P(x, y, z) + P(x, y, \neg z) + P(x, \neg y, z) = 0.10 + 0.05 + 0.15 = 0.30$. Therefore $P(x \mid y \vee z) = 0.30 / 0.70 \approx 0.4286 \approx 0.429$.

Q.21 [MCQ, 2M · Sampling] **Answer: D** (D: $A=T, B=T, C=F, D=F, E=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \neg) = 0.7$; $u = 0.62 < 0.7 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.5$; $u = 0.17 < 0.5 \Rightarrow B = T$
- C: $P(C=T \mid \neg) = 0.05$; $u = 0.99 \geq 0.05 \Rightarrow C = F$
- D: $P(D=T \mid A=T, C=F) = 0.4$; $u = 0.61 \geq 0.4 \Rightarrow D = F$
- E: $P(E=T \mid B=T, D=F) = 0.7$; $u = 0.42 < 0.7 \Rightarrow E = T$

Sample: **(A=T, B=T, C=F, D=F, E=T)**.

Q.22 [NAT, 2M · Sampling] **Answer: 0.881 to 0.885**

Gibbs sampling draws A from $P(A \mid \text{mb}(A))$, where $\text{mb}(A) = \{B, C, D\}$:

$P(A \mid \text{mb}) \propto P(A \mid \text{parents}(A)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$A=T$: $P(A=T) \times P(B=F \mid A=T) \times P(D=F \mid A=T, C=T) = 0.9 \times 0.5 \times 0.4 = 0.18$

$A=F$: $P(A=F) \times P(B=F \mid A=F) \times P(D=F \mid A=F, C=T) = 0.1 \times 0.25 \times 0.95 = 0.02375$

Normalise: $0.18 / (0.18 + 0.02375) = \mathbf{0.8834}$.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: A, C** (A: $G \perp C$; C: $A \perp F \mid C, D$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp C$: **TRUE** — $G \leftarrow A \rightarrow D \rightarrow E \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $G \leftarrow A \rightarrow D \rightarrow E \rightarrow F \leftarrow C$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow D \rightarrow E \leftarrow C$ (blocked: collider E with neither it nor a descendant observed); $G \leftarrow D \rightarrow E \rightarrow F \leftarrow C$ (blocked: collider F with neither it nor a descendant observed).

(B) $A \perp E \mid C, F$: **FALSE** — the path $A \rightarrow D \rightarrow E$ is active given $\{C, F\}$.

(C) $A \perp F \mid C, D$: **TRUE** — $A \rightarrow D \rightarrow E \leftarrow C \rightarrow F$ (blocked: observed chain node D); $A \rightarrow D \rightarrow E \rightarrow F$ (blocked: observed chain node D); $A \rightarrow G \leftarrow D \rightarrow E \leftarrow C \rightarrow F$ (blocked: collider G with neither it nor a descendant observed); $A \rightarrow G \leftarrow D \rightarrow E \rightarrow F$ (blocked: collider G with neither it nor a descendant observed).

(D) $C \perp A \mid E$: **FALSE** — the path $C \rightarrow E \leftarrow D \leftarrow A$ is active given $\{E\}$.

Q.24 [NAT, 2M · Conditional Independence] **Answer: 0.263 to 0.265**

$A \perp B \mid C$, so $P(a, b) = \sum_c P(c) P(a \mid c) P(b \mid c)$. c : $0.4 \times 0.9 \times 0.7 = 0.252$. $\neg c$: $0.6 \times 0.2 \times 0.1 = 0.012$. Sum = 0.264. Note $P(a) = 0.36 + 0.12 = 0.48$ and $P(b) = 0.28 + 0.06 = 0.34$, so $P(a)P(b) = 0.1632 \neq 0.264$: A and B are marginally dependent.

Q.25 [MCQ, 2M · Exact Inference] **Answer: C** (C: $f(B, F)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(A, B, D)$, $f(B, C, E)$, $f(A, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(A, B, C)$, $f(B, C, E)$	$\{A, B, C, E\}$	$f(A, B, E)$
D	$f(A, B, D)$	$\{A, B, D\}$	$f(A, B)$
E	$f(A, E, F)$, $f(A, B, E)$	$\{A, B, E, F\}$	$f(A, B, F)$
A	$f(A)$, $f(A, B)$, $f(A, B)$, $f(A, B, F)$	$\{A, B, F\}$	$f(B, F)$
B	$f(B, F)$	$\{B, F\}$	$f(F)$

The largest factor produced has **3** variable(s) (created when eliminating C); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over F and normalise.

Mock Test 088

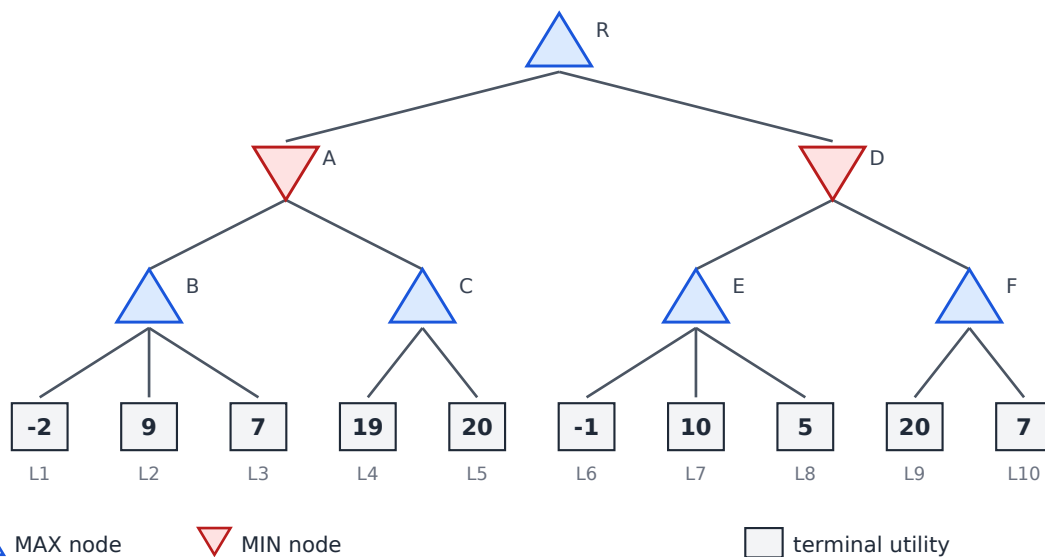
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	7 / 6 / 12	14 / 12 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to A; root value 9
- ☐ (B) Move to D; root value 10
- ☐ (C) Move to D; root value 11
- ☐ (D) Move to D; root value 20

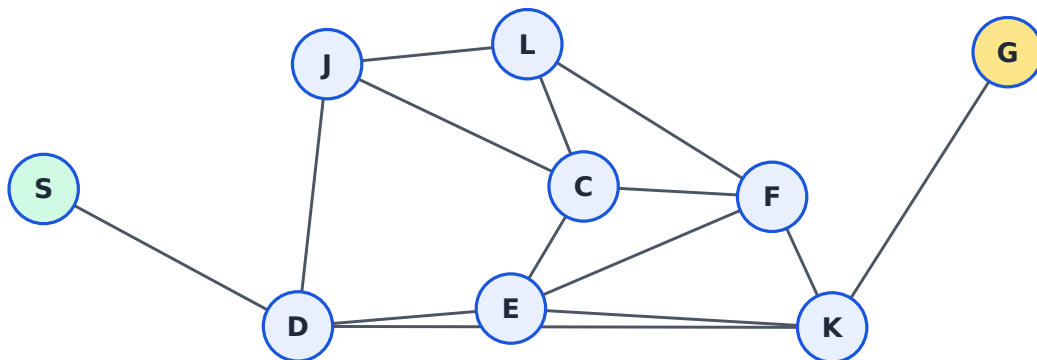
Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search

In alpha-beta search, while a MIN node is being evaluated with current bounds (α , β), the remaining children of the MIN node can be pruned as soon as its running value v satisfies

- ☐ (A) $v \geq \beta$
- ☐ (B) $v \leq \alpha$
- ☐ (C) $v \geq \alpha$
- ☐ (D) $v \leq \beta$

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, E, C, F, K, G
☐ (B) S, D, K, G
☐ (C) S, E, D, C, F, K, G
☐ (D) S, D, E, J, K, C, F, L, G

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 5$ (so $b^d = 32$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$\mathbf{F = ((R \leftrightarrow \neg Q) \wedge (R \rightarrow R)) \leftrightarrow ((\neg R \leftrightarrow R) \vee \neg P) \text{ ?}}$$

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \rightarrow Q(x))$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

The clauses $A \vee B \vee \neg C$ and $\neg A \vee B \vee D$ are resolved on the symbol A. How many literals are in the resolvent (after merging duplicate literals)?

Answer (numerical): _____

Q.8 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following clauses are Horn clauses?

- ☐ (A) $\neg p \vee \neg q \vee r$
☐ (B) $p \vee q$
☐ (C) $\neg p \vee \neg q$
☐ (D) r

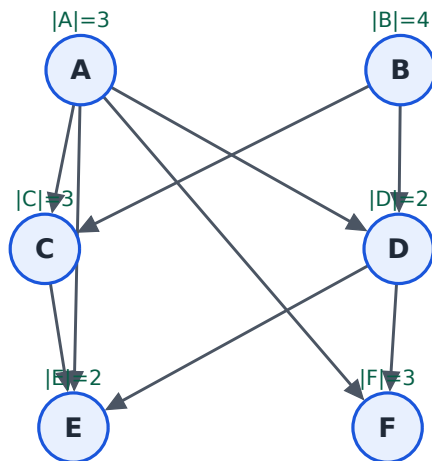
Q.9 [MCQ · 1 mark · negative: -1/3] Conditional Independence

For random variables A, B, C with all conditioning events of positive probability, $A \perp B \mid C$ holds if and only if

- ☐ (A) $P(A \mid B, C) = P(A \mid C)$
- ☐ (B) $P(A, B) = P(A) P(B)$
- ☐ (C) $P(A \mid C) = P(B \mid C)$
- ☐ (D) $P(A, B, C) = P(A) P(B) P(C)$

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks – Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?



Answer (numerical): _____

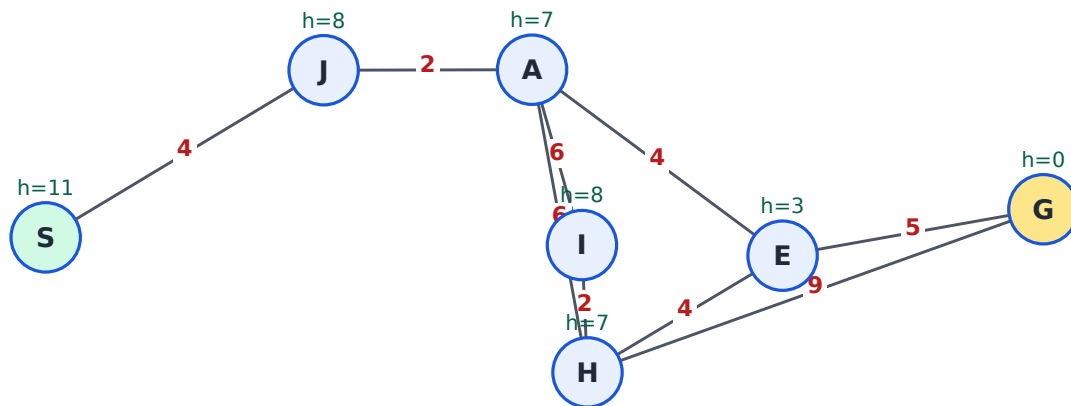
Section B — 2-mark questions**Q.11** [MCQ · 2 marks · negative: -2/3] Informed Search

$h_1, \dots, h_m$ are admissible heuristics for a problem. Which composite heuristic is guaranteed to be admissible and to dominate each h_i ?

- ☐ (A) $h(n) = h_1(n) + \dots + h_m(n)$
- ☐ (B) $h(n) = \max(h_1(n), \dots, h_m(n))$
- ☐ (C) $h(n) = \min(h_1(n), \dots, h_m(n))$
- ☐ (D) $h(n) = m \cdot h_1(n)$

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

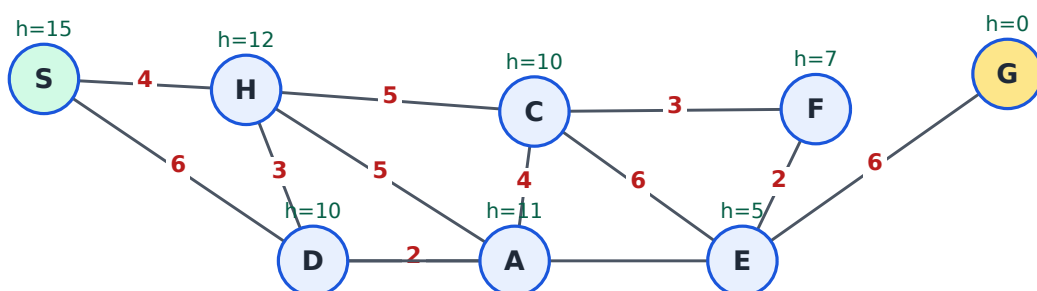
Q.13 [MCQ · 2 marks · negative: -2/3] Uninformed Search

Depth-limited search (DLS) with limit l is applied to a problem with unit step costs whose shallowest goal lies at depth d . Which statement is correct?

- ☐ (A) If $l < d$, DLS is incomplete. If $l > d$, it finds a solution but not necessarily the shallowest (optimal) one
- ☐ (B) If $l < d$, DLS still finds the goal because it backtracks
- ☐ (C) If $l > d$, DLS is always optimal because it searches deeper than needed
- ☐ (D) DLS is complete and optimal for every $l \geq 0$

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

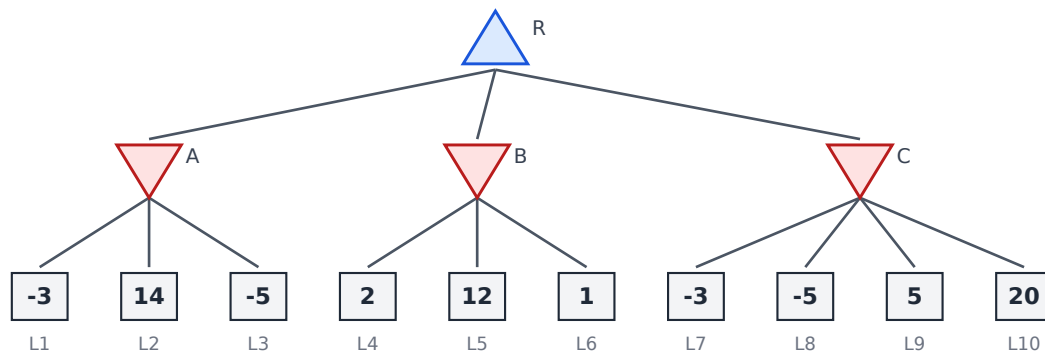
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



MAX node



MIN node

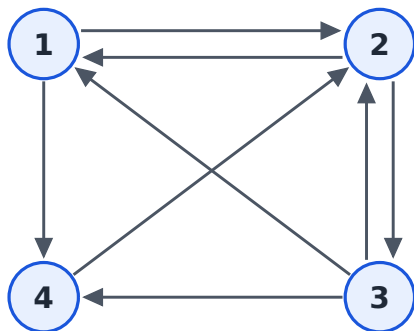


terminal utility

Answer (numerical): _____

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\exists x \forall y \neg R(x, y)$
☐ (D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(\neg P \vee R)$, C2: $(\neg Q)$, C3: $(\neg P \vee Q \vee S)$, C4: $(\neg P)$, C5: $(\neg P \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) $(R \vee S)$
☐ (C) $(\neg P \vee Q)$
☐ (D) $(Q \vee \neg R)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic

Let $\phi = \forall x \exists y (R(x, y) \wedge \neg R(y, x))$. Which of the following statements are TRUE?

- ☐ (A) ϕ has no model with a domain of size 1
- ☐ (B) ϕ has a model with domain $\{1, 2\}$
- ☐ (C) ϕ has a model with domain $\{1, 2, 3\}$
- ☐ (D) ϕ has a model with domain $\mathbb{N}$

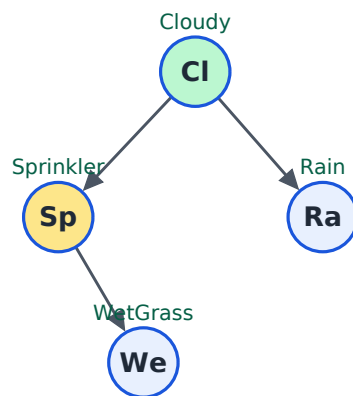
Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (P \rightarrow Q)$
- ☐ (B) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$
- ☐ (C) $(\neg Q \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg \neg Q)$
- ☐ (D) $\neg(P \wedge R) \leftrightarrow (\neg P \wedge \neg R)$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: CI = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute **$P(\text{Cloudy} = T \mid \text{Sprinkler} = T)$** (round off to 3 decimal places).



P(CI=T)
0.95

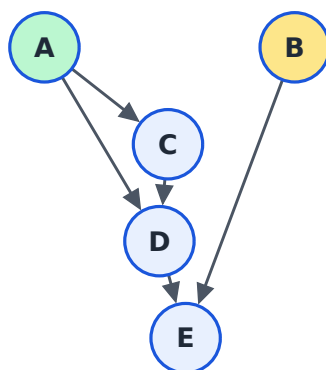
CI	P(Sp=T ...)	CI	P(Ra=T ...)
T	0.5	T	0.9
F	0.6	F	0.7

Sp	P(We=T ...)
T	0.8
F	0.1

Answer (numerical): _____

Q.21 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence **$B=F$** . The current state of the chain is **$A=T, B=F, C=T, D=F, E=F$** . The next step resamples **A**. With what probability is A set to T? (round off to 3 decimal places)



P(A=T)
0.1

P(B=T)
0.2

A	P(C=T ...)
T	0.95
F	0.75

A	C	P(D=T ...)	B	D	P(E=T ...)
T	T	0.25	T	T	0.1
T	F	0.2	T	F	0.5
F	T	0.95	F	T	0.4
F	F	0.05	F	F	0.2

Answer (numerical): _____

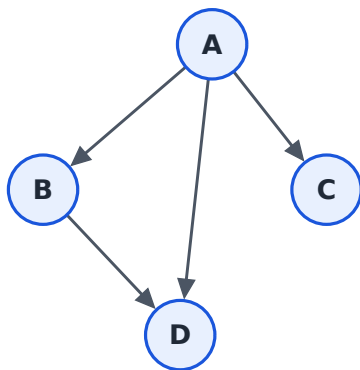
Q.22 [NAT · 2 marks · no negative marking] Variable Elimination

A Bayesian network is the chain $A \rightarrow B \rightarrow C \rightarrow D$ over Boolean variables with $P(a) = 0.2$; $P(b | a) = 0.9$, $P(b | \neg a) = 0.3$; $P(c | b) = 0.6$, $P(c | \neg b) = 0.1$; $P(d | c) = 0.7$, $P(d | \neg c) = 0.2$. Compute $P(d)$ by eliminating A, B, C in that order. (Round to 3 decimal places.)

Answer (numerical): _____

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.86, 0.36, 0.31, 0.7)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T | \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.5

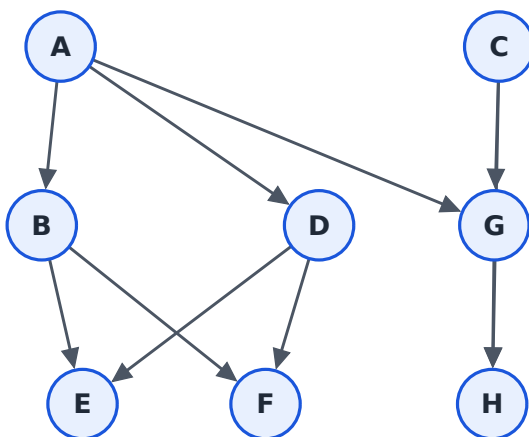
A	P(B=T ...)	A	P(C=T ...)
T	0.95	T	0.25
F	0.25	F	0.5

A	B	P(D=T ...)
T	T	0.8
T	F	0.5
F	T	0.05
F	F	0.6

- ☐ (A) A=T, B=F, C=T, D=T
☐ (B) A=F, B=F, C=T, D=F
☐ (C) A=F, B=T, C=T, D=F
☐ (D) A=T, B=F, C=T, D=F

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

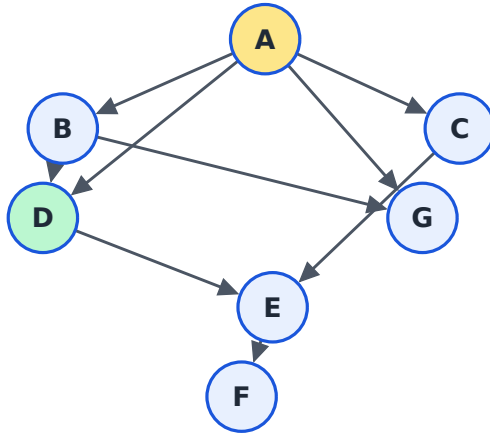
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $D \perp E$
☐ (B) $F \perp G \mid A, B$
☐ (C) $A \perp D \mid C, F$
☐ (D) $F \perp H \mid B$

Q.25 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(D \mid A = T)$ in the Bayesian network below (Boolean variables; evidence $A = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E**, **F**, **G**, **C**, **B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

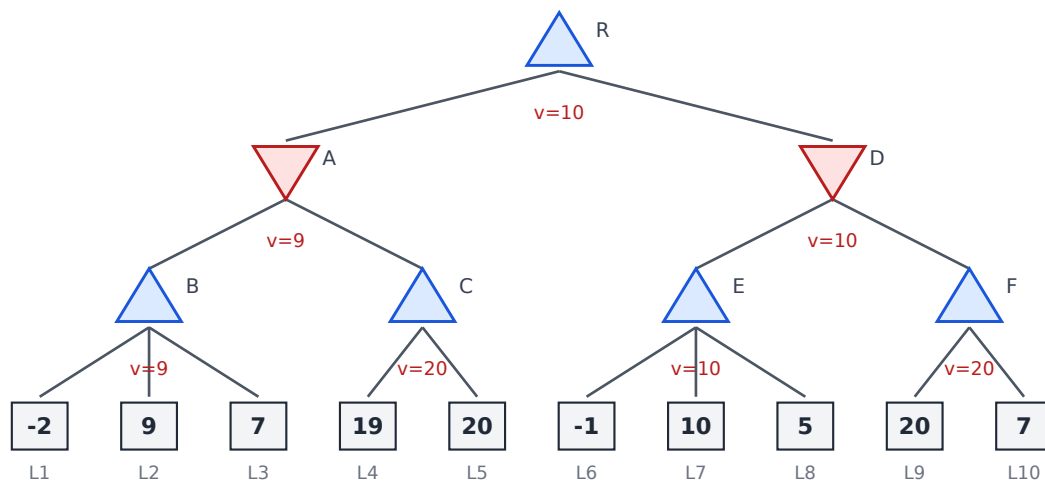
Answer Key — Mock Test 088

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Adversarial Search	B	14	NAT	2	Informed Search	17
2	MCQ	1	Adversarial Search	B	15	NAT	2	Adversarial Search	7
3	MCQ	1	Uninformed Search	A	16	MSQ	2	Predicate Logic	B, D
4	NAT	1	Adversarial Search	11	17	MSQ	2	Propositional Logic	C
5	NAT	1	Propositional Logic	4	18	MSQ	2	Predicate Logic	A, C, D
6	NAT	1	Predicate Logic	27	19	MSQ	2	Propositional Logic	A, C
7	NAT	1	Propositional Logic	3	20	NAT	2	Exact Inference	0.939 to 0.943
8	MSQ	1	Propositional Logic	A, C, D	21	NAT	2	Sampling	0.677 to 0.681
9	MCQ	1	Conditional Independence	A	22	NAT	2	Variable Elimination	0.354 to 0.356
10	NAT	1	Bayesian Networks	71	23	MCQ	2	Sampling	B
11	MCQ	2	Informed Search	B	24	MSQ	2	Conditional Independence	B
12	NAT	2	Informed Search	5	25	NAT	2	Exact Inference	3
13	MCQ	2	Uninformed Search	A					

Detailed Solutions

Q.1 [MCQ, 1M · Adversarial Search] **Answer: B** (B: Move to D; root value 10)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 9$, $C = 20$, $A = 9$, $E = 10$, $F = 20$, $D = 10$. Root value = **10**; MAX's optimal first move is towards D.

Q.2 [MCQ, 1M · Adversarial Search] **Answer: B** ($B: v \leq \alpha$)

A MIN node's value can only decrease as more children are seen. Once $v \leq \alpha$, MAX already has an alternative worth α higher up the tree and will never let play reach this node. $v \geq \beta$ (A) is the cutoff test at MAX nodes. (C) and (D) do not justify pruning.

Q.3 [MCQ, 1M · Uninformed Search] **Answer: A** (A: S, D, E, C, F, K, G)

Adjacency lists (alphabetical): C: {E, F, J, L}; D: {E, J, K, S}; E: {C, D, F, K}; F: {C, E, K, L}; G: {K}; J: {C, D, L}; K: {D, E, F, G}; L: {C, F, J}; S: {D}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, E, C, F, K, G**; the path found is $S \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow K \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Adversarial Search] **Answer: 11**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 2^3 + 2^2 - 1 = 8 + 4 - 1 = \mathbf{11}$, versus 32 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.5 [NAT, 1M · Propositional Logic] **Answer: 4**

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \leftrightarrow \neg Q) \wedge (R \rightarrow R)$	$(\neg R \leftrightarrow R) \vee \neg P$	$((R \leftrightarrow \neg Q) \wedge (R \rightarrow R)) \leftrightarrow ((\neg R \leftrightarrow R) \vee \neg P)$
T	T	T	F	F	T
T	T	F	T	F	F
T	F	T	T	F	F
T	F	F	F	F	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	F	T	F

Rows where $F = T$: **4**.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 27**

Interpret P and Q as subsets of the domain; each element independently has 3 allowed (P,Q) combinations (FF, FT, TT) $\Rightarrow 3^n$. With $n = 3$ this gives **27** (verified by enumerating all interpretations).

Q.7 [NAT, 1M · Propositional Logic] **Answer: 3**

Remove A and $\neg A$ and take the union of the remaining literals: $\{B, \neg C\} \cup \{B, D\} = \{B, \neg C, D\}$. The duplicate B is merged (factoring), so the resolvent $B \vee \neg C \vee D$ has 3 literals.

Q.8 [MSQ, 1M · Propositional Logic] Answer: A, C, D (A: $\neg p \vee \neg q \vee r$; C: $\neg p \vee \neg q$; D: r)

A Horn clause is a disjunction of literals with at most one positive literal. $\neg p \vee \neg q \vee r$ has one positive literal (a definite clause, i.e. $p \wedge q \rightarrow r$). $\neg p \vee \neg q$ has zero positive literals (a goal clause). r is a single positive literal (a fact). $p \vee q$ has two positive literals, so it is not Horn.

Q.9 [MCQ, 1M · Conditional Independence] Answer: A (A: $P(A \mid B, C) = P(A \mid C)$)

Conditional independence of A and B given C means $P(A, B \mid C) = P(A \mid C)P(B \mid C)$, equivalently $P(A \mid B, C) = P(A \mid C)$. (B) is marginal independence, which neither implies nor is implied by $A \perp B \mid C$. (C) compares two different variables' distributions and is meaningless here. (D) is mutual independence, a much stronger condition.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 71

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (3-1) + B: (4-1) + C: (3-1) \times 3 \times 4 + D: (2-1) \times 3 \times 4 + E: (2-1) \times 3 \times 3 \times 2 + F: (3-1) \times 3 \times 2 = 71$.

Full joint table: $\prod |X| - 1 = 432 - 1 = 431$; saving = 360.

Q.11 [MCQ, 2M · Informed Search] Answer: B (B: $h(n) = \max(h_1(n), \dots, h_m(n))$)

Each $h_i(n) \leq h^*(n)$, so their maximum is also $\leq h^*(n)$, and it is $\geq$ each component by construction. So the **max** is admissible and dominates every h_i . The sum (A) and the multiple (D) can exceed h^* . The min (C) is admissible but is dominated by every h_i , not dominating.

Q.12 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(A)=7$, $h(E)=3$, $h(G)=0$, $h(H)=7$, $h(I)=8$, $h(J)=8$, $h(S)=11$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	J(12)
2	J (12)	A(13)
3	A (13)	E(13), H(19), I(20)
4	E (13)	G(15), H(19), I(20)
5	G (15)	H(19), I(20) ← goal, stop

Path returned: **S** → **J** → **A** → **E** → **G** with cost **15**; nodes expanded = **5** (S, J, A, E, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.13 [MCQ, 2M · Uninformed Search] Answer: A (A: If $l < d$, DLS is incomplete. If $l > d$, it finds a solution but not necessarily the shallowest (optimal) one)

DLS treats nodes at depth l as having no successors. If $l < d$, the goal can never be reached, so DLS is incomplete; backtracking does not help, so (B) is wrong. If $l > d$, a goal is reachable, but DLS explores depth-first and may meet a deeper goal (depth between d and l) before the shallowest one, so it is not optimal and (C) is wrong. (D) is wrong for both reasons.

Q.14 [NAT, 2M · Informed Search] Answer: 17

Heuristic values: $h(A)=11$, $h(C)=10$, $h(D)=10$, $h(E)=5$, $h(F)=7$, $h(G)=0$, $h(H)=12$, $h(S)=15$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (15)	D(16), H(16)
2	D (16)	E(16), H(16), A(19)
3	E (16)	H(16), G(17), A(19), F(20), C(27)
4	H (16)	G(17), A(19), C(19), F(20)
5	G (17)	A(19), C(19), F(20) ← goal, stop

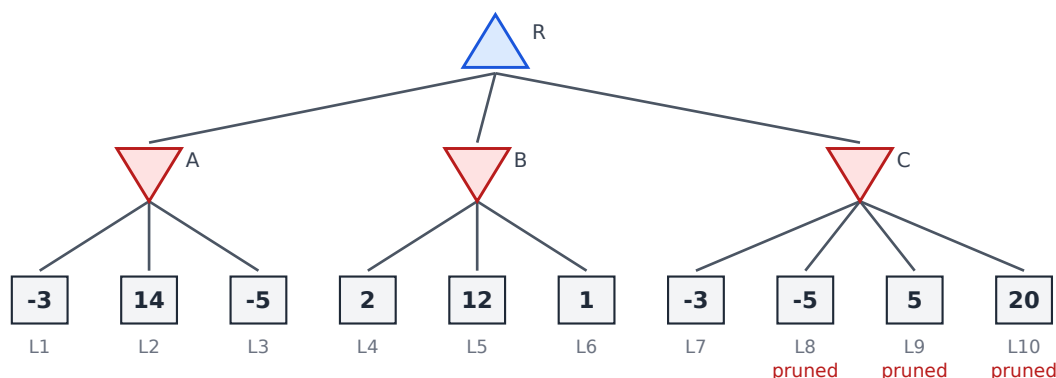
Path returned: **S** → **D** → **E** → **G** with cost **17**; nodes expanded = **5** (S, D, E, H, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 17$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [NAT, 2M · Adversarial Search] Answer: 7

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node C: value $-3 \leq \alpha = 1 \Rightarrow$ prune remaining children (leaves L8, L9, L10).



Leaves evaluated (7): L1, L2, L3, L4, L5, L6, L7. Leaves pruned (3): L8, L9, L10. Root value = 1 (same as plain minimax, 1 — pruning never changes the root value).

Q.16 [MSQ, 2M · Predicate Logic] Answer: B, D (B: $\forall y \exists x R(x, y)$; D: $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$)

$R = \{(1,2), (1,4), (2,1), (2,3), (3,1), (3,2), (3,4), (4,2)\}$.

(A) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(3,4)$ holds but $R(4,3)$ does not.

(B) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(C) $\exists x \forall y \neg R(x, y)$: **FALSE** — every element has an outgoing edge.

(D) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **TRUE** — every two distinct elements are related in some direction.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] Answer: C (C: $(\neg P \vee Q)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(R \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(C) $(\neg P \vee Q)$ is derivable:

1. resolve $(\neg P \vee Q \vee S)$ and $(\neg P \vee \neg S)$ on $S \Rightarrow (\neg P \vee Q)$

(D) $(Q \vee \neg R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.18 [MSQ, 2M · Predicate Logic] Answer: A, C, D (A: $\emptyset$ has no model with a domain of size 1; C: $\emptyset$ has a model with domain $\{1, 2, 3\}$; D: $\emptyset$ has a model with domain $\mathbb{N}$)

On $\{1\}$: y must be 1, and $R(1, 1) \wedge \neg R(1, 1)$ is impossible, so no model. On $\{1, 2\}$: element 1 needs y with $R(1, y) \wedge \neg R(y, 1)$; $y = 1$ is impossible, so $y = 2$: $R(1, 2), \neg R(2, 1)$. Element 2 similarly needs $R(2, 1) \wedge \neg R(1, 2)$ — contradiction; brute force over all 16 relations finds no model. On $\{1, 2, 3\}$ the 3-cycle $R = \{(1, 2), (2, 3), (3, 1)\}$ works, and on $\mathbb{N}$ $R(x, y) \Leftrightarrow x < y$ works.

Q.19 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $((P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (P \rightarrow Q)$; C: $(\neg Q \rightarrow P) \Leftrightarrow (\neg P \rightarrow \neg \neg Q)$)

(A) $((P \rightarrow R) \wedge (R \rightarrow Q)) \rightarrow (P \rightarrow Q)$: true in all 8 assignments $\Rightarrow$ **valid**.

(B) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$: **not valid** — counter-model $P=T, Q=F, R=T$.

(C) $(\neg Q \rightarrow P) \Leftrightarrow (\neg P \rightarrow \neg \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $\neg(P \wedge R) \Leftrightarrow (\neg P \wedge \neg R)$: **not valid** — counter-model $P=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.939 to 0.943

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Cloudy}) \cdot P(\text{WetGrass} \mid \text{Sprinkler})$.

Inference by enumeration: $P(\text{Cloudy} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cloudy}, e, \text{hidden})$.

Hidden: Ra, We	Cl	product of CPT entries (Cl · Sp · Ra · We)	value
T, T	T	$0.95 \times 0.5 \times 0.9 \times 0.8$	0.342
T, F	T	$0.95 \times 0.5 \times 0.9 \times 0.2$	0.0855
F, T	T	$0.95 \times 0.5 \times 0.1 \times 0.8$	0.038
F, F	T	$0.95 \times 0.5 \times 0.1 \times 0.2$	0.0095

Hidden: Ra, We	CI	product of CPT entries (CI · Sp · Ra · We)	value
T, T	F	$0.05 \times 0.6 \times 0.7 \times 0.8$	0.0168
T, F	F	$0.05 \times 0.6 \times 0.7 \times 0.2$	0.0042
F, T	F	$0.05 \times 0.6 \times 0.3 \times 0.8$	0.0072
F, F	F	$0.05 \times 0.6 \times 0.3 \times 0.2$	0.0018

$P(\text{Cloudy}=T, e) = 0.475$; $P(\text{Cloudy}=F, e) = 0.03$; $P(e) = 0.505$.

$P(\text{Cloudy}=T | e) = 0.475 / 0.505 = \mathbf{0.9406} \approx 0.941$.

Q.21 [NAT, 2M · Sampling] **Answer: 0.677 to 0.681**

Gibbs sampling draws A from $P(A | \text{mb}(A))$, where $\text{mb}(A) = \{C, D\}$:

$P(A | \text{mb}) \propto P(A | \text{parents}(A)) \times \prod_{\text{children } Y} P(Y | \text{parents}(Y))$.

$A=T$: $P(A=T) \times P(C=T|A=T) \times P(D=F|A=T, C=T) = 0.1 \times 0.95 \times 0.75 = 0.07125$

$A=F$: $P(A=F) \times P(C=T|A=F) \times P(D=F|A=F, C=T) = 0.9 \times 0.75 \times 0.05 = 0.03375$

Normalise: $0.07125 / (0.07125 + 0.03375) = \mathbf{0.6786}$.

Q.22 [NAT, 2M · Variable Elimination] **Answer: 0.354 to 0.356**

Eliminate A: $f_A(b) = 0.2 \times 0.9 + 0.8 \times 0.3 = 0.18 + 0.24 = 0.42$, $f_A(\neg b) = 0.58$. Eliminate B: $f_B(c) = 0.42 \times 0.6 + 0.58 \times 0.1 = 0.252 + 0.058 = 0.31$, $f_B(\neg c) = 0.69$. Eliminate C: $P(d) = 0.31 \times 0.7 + 0.69 \times 0.2 = 0.217 + 0.138 = 0.355$. Each step creates a factor over a single variable, so cost is linear in the chain length.

Q.23 [MCQ, 2M · Sampling] **Answer: B** (B: A=F, B=F, C=T, D=F)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.5$; $u = 0.86 \geq 0.5 \Rightarrow A = F$
- B: $P(B=T | A=F) = 0.25$; $u = 0.36 \geq 0.25 \Rightarrow B = F$
- C: $P(C=T | A=F) = 0.5$; $u = 0.31 < 0.5 \Rightarrow C = T$
- D: $P(D=T | A=F, B=F) = 0.6$; $u = 0.7 \geq 0.6 \Rightarrow D = F$

Sample: **(A=F, B=F, C=T, D=F)**.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: B** (B: $F \perp G | A, B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $D \perp E$: **FALSE** — the path $D \leftarrow A \rightarrow B \rightarrow E$ is active given $\{\}$.

(B) $F \perp G | A, B$: **TRUE** — $F \leftarrow B \leftarrow A \rightarrow G$ (blocked: observed chain node B); $F \leftarrow B \rightarrow E \leftarrow D \leftarrow A \rightarrow G$ (blocked: observed fork node B); $F \leftarrow D \leftarrow A \rightarrow G$ (blocked: observed fork node A); $F \leftarrow D \rightarrow E \leftarrow B \leftarrow A \rightarrow G$ (blocked: collider E with neither it nor a descendant observed).

(C) $A \perp D | C, F$: **FALSE** — the path $A \rightarrow B \rightarrow F \leftarrow D$ is active given $\{C, F\}$.

(D) $F \perp H | B$: **FALSE** — the path $F \leftarrow D \leftarrow A \rightarrow G \rightarrow H$ is active given $\{B\}$.

Q.25 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f()$, $f(B)$, $f(C)$, $f(B, D)$, $f(C, D, E)$, $f(E, F)$, $f(B, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(C, D, E), f(E, F)$	$\{C, D, E, F\}$	$f(C, D, F)$
F	$f(C, D, F)$	$\{C, D, F\}$	$f(C, D)$
G	$f(B, G)$	$\{B, G\}$	$f(B)$
C	$f(C), f(C, D)$	$\{C, D\}$	$f(D)$
B	$f(B), f(B, D), f(B)$	$\{B, D\}$	$f(D)$

The largest factor produced has **3** variable(s) (created when eliminating E); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over D and normalise.

Mock Test 089

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 8 / 12	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

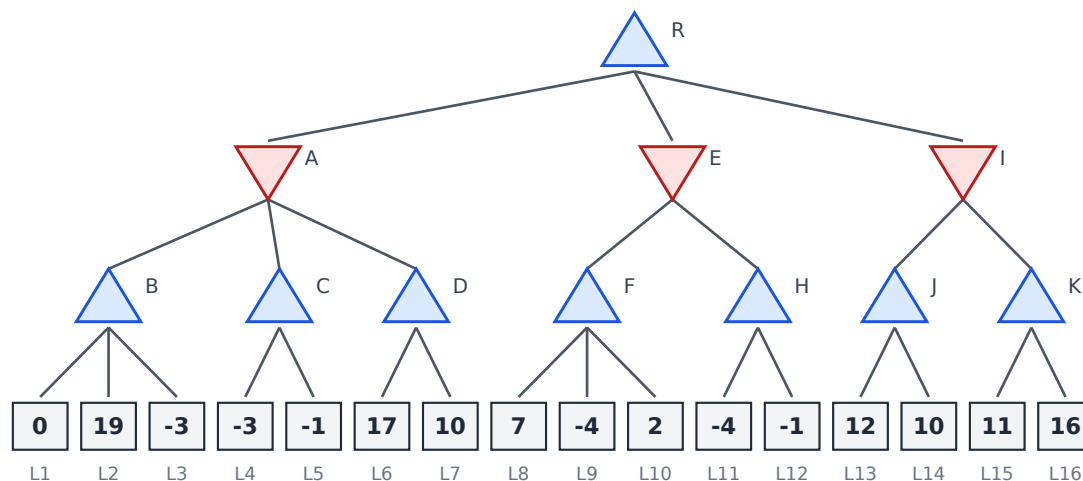
Q.1 [MCQ · 1 mark · negative: $-1/3$] Problem Formulation

In AIMA, a search problem is formally defined by

- ☐ (A) initial state, actions, transition model, goal test, and path (action) cost function
- ☐ (B) initial state and goal state only
- ☐ (C) a heuristic function and an evaluation function
- ☐ (D) a frontier and an explored set

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



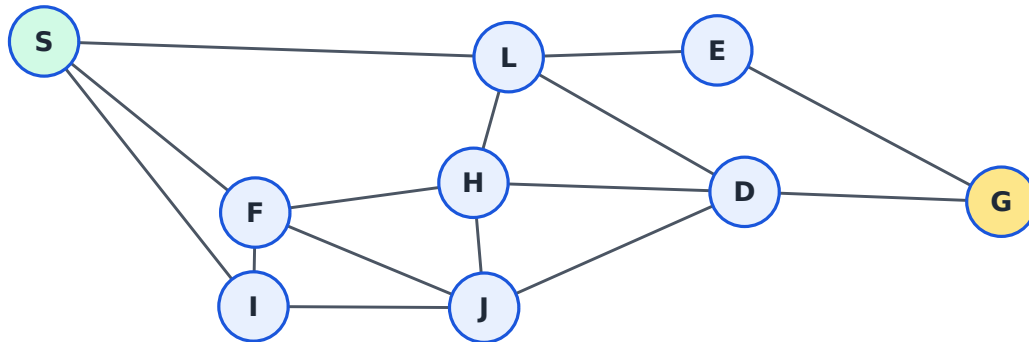
△ MAX node ▽ MIN node

 terminal utility

Answer (numerical): _____

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

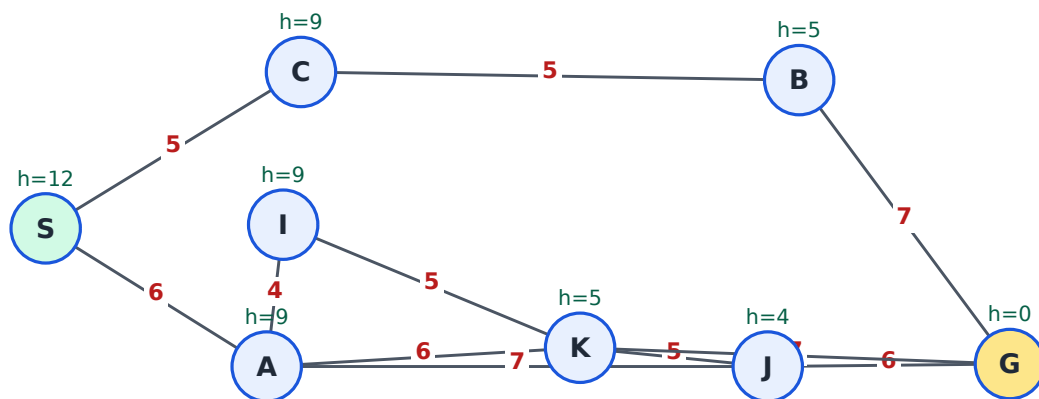
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, F, L, I, H, J, D, E, G
☐ (B) S, L, I, F, H, E, D, J, G
☐ (C) S, F, I, L, H, J, D, E, G
☐ (D) S, F, H, D, G

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following statements are TRUE?

- ☐ (A) Propositional resolution is refutation-complete
☐ (B) Modus ponens alone is complete for arbitrary propositional knowledge bases
☐ (C) Forward chaining is sound and complete for knowledge bases of definite clauses
☐ (D) A sound inference procedure derives only sentences that are entailed by the KB

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q). For how many of them is $\forall x (P(x) \rightarrow Q(x))$ true?

Answer (numerical): _____

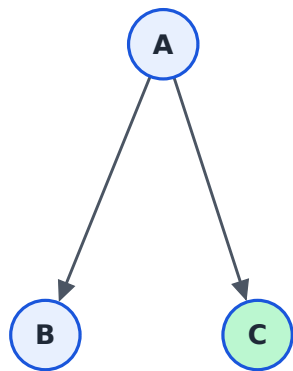
Q.7 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following inference rules are sound in propositional logic?

- ☐ (A) Modus ponens: from $\alpha \rightarrow \beta$ and α infer β
- ☐ (B) And-elimination: from $\alpha \wedge \beta$ infer α
- ☐ (C) Abduction: from $\alpha \rightarrow \beta$ and β infer α
- ☐ (D) Resolution: from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$ infer $\alpha \vee \gamma$

Q.8 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



$P(A=T)$	A	$P(B=T \dots)$	A	$P(C=T \dots)$
0.25	T	0.7	T	0.8
	F	0.9	F	0.75

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Rejection sampling

The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. If rejection sampling is used to estimate $P(A = T \mid C = F)$, how many of these samples are **rejected**?

#	A	B	C	D
1	T	T	T	F
2	T	F	T	F
3	F	F	F	F
4	F	T	T	F
5	F	F	F	F
6	T	T	T	F
7	T	F	T	T
8	T	T	T	F
9	T	F	F	T
10	F	F	F	T
11	F	F	F	T
12	F	T	F	T
13	F	F	F	F
14	T	F	F	F
15	F	F	F	F

Answer (numerical): _____

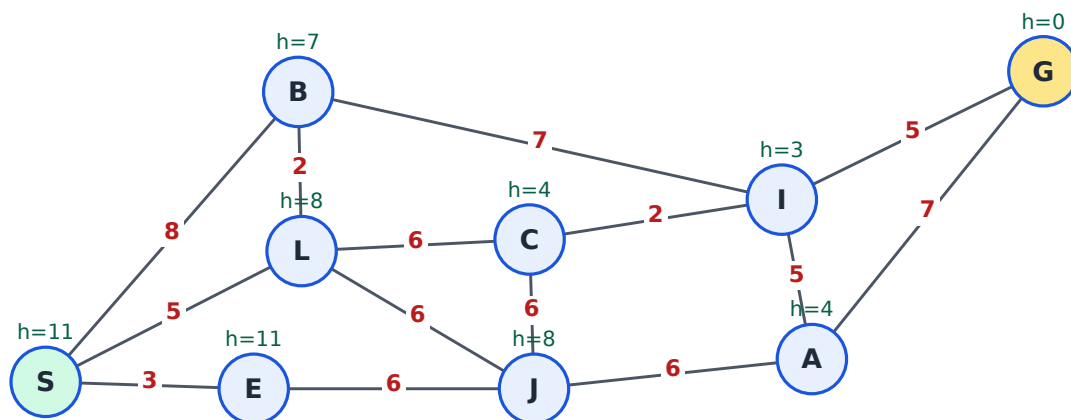
Q.10 [MCQ · 1 mark · negative: -1/3] Variable Elimination

In VE for a query with evidence $J = \text{true}$, the CPT $P(J | A)$ (A and J Boolean) is converted into which factor?

- ☐ (A) A factor over A with 2 entries: $P(j | a)$, $P(j | \neg a)$
- ☐ (B) A factor over $\{A, J\}$ with 4 entries
- ☐ (C) A factor over J with 2 entries
- ☐ (D) A scalar equal to $P(j)$

Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

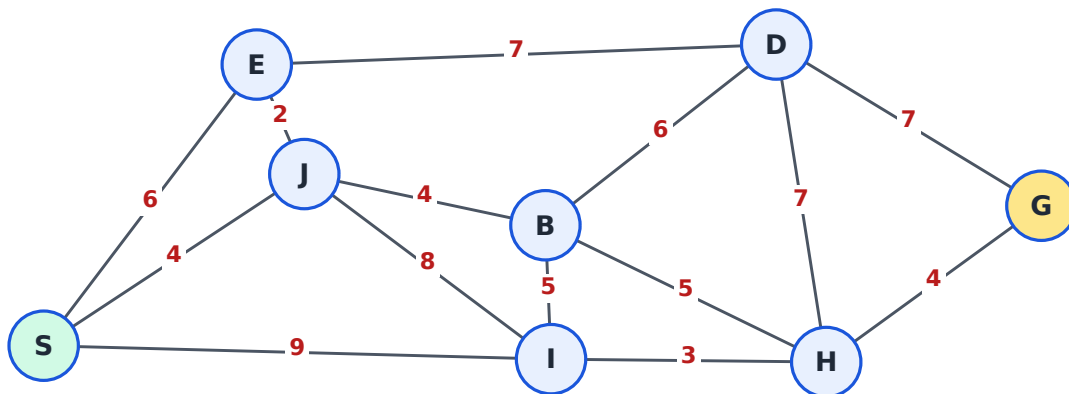
Q.12 [MCQ · 2 marks · negative: -2/3] Informed Search

A* tree search uses an admissible h , the optimal goal is G_1 with cost C^* , and G_2 is a suboptimal goal with $g(G_2) > C^*$. Why can G_2 never be selected for expansion before G_1 ?

- ☐ (A) Because $h(G_2) > h(G_1)$
- ☐ (B) Because some node n on the optimal path is always in the frontier with $f(n) \leq C^* < g(G_2) = f(G_2)$
- ☐ (C) Because A* expands nodes in order of depth
- ☐ (D) Because G_2 is never generated

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

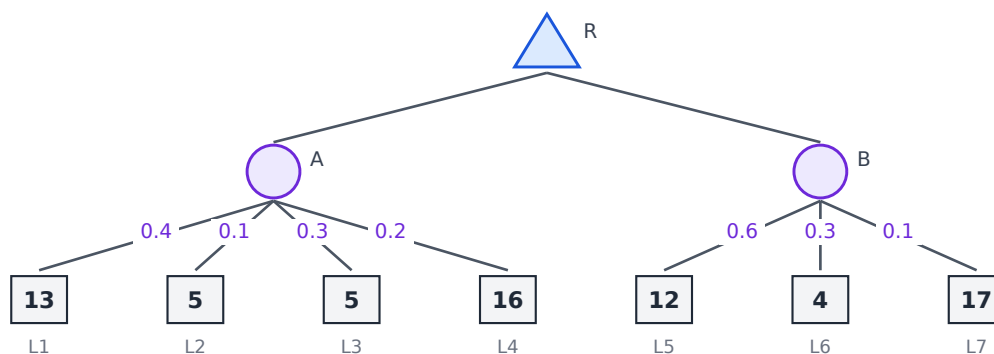
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

The formula $F = (Q \wedge (S \rightarrow S)) \wedge ((P \vee P) \rightarrow (\neg R \wedge Q))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R, S. How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.16 [NAT · 2 marks · no negative marking] Propositional Logic

How many of the 16 truth assignments to p, q, r, s satisfy exactly two of the four clauses $p \vee q$, $\neg p \vee r$, $\neg q \vee s$, $\neg r \vee \neg s$? (Enter the number of assignments, not clauses.)

Answer (numerical): _____

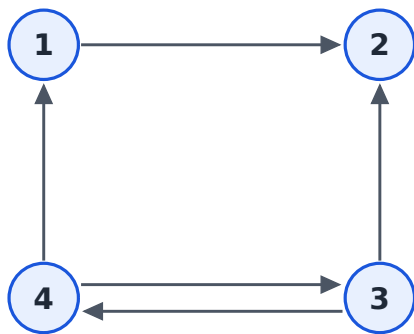
Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(Q \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg Q)$
☐ (B) $((R \rightarrow P) \rightarrow Q) \leftrightarrow (R \rightarrow (P \rightarrow Q))$
☐ (C) $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$
☐ (D) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$
☐ (C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (D) $\forall y \exists x R(x, y)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(R \vee \neg S)$, C2: $(R \vee S)$, C3: (S) , C4: $(\neg P \vee \neg S)$, C5: (P)

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(P \vee R)$
☐ (B) $(\neg S)$
☐ (C) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (D) $(P \vee Q)$

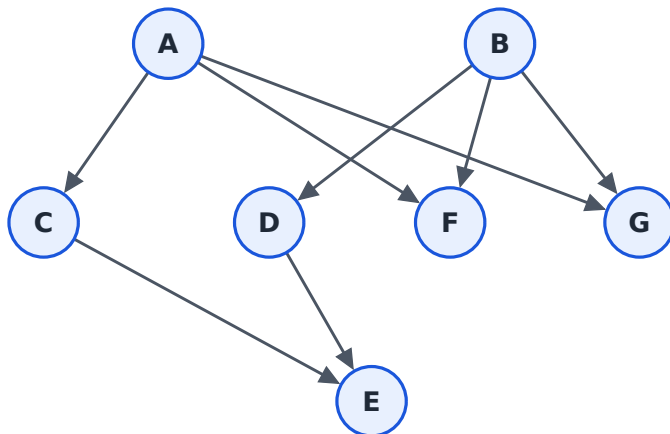
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following sets of propositional formulas are unsatisfiable?

- ☐ (A) $\{p \vee q, \neg p \vee q, p \vee \neg q, \neg p \vee \neg q\}$
☐ (B) $\{p \rightarrow q, q \rightarrow r, r \rightarrow \neg p, p\}$
☐ (C) $\{p \leftrightarrow \neg q, q \leftrightarrow \neg r, r \leftrightarrow \neg p\}$
☐ (D) $\{p \vee q \vee r, \neg p \vee \neg q, \neg q \vee \neg r, \neg p \vee \neg r\}$

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp A \mid C, D$
☐ (B) $E \perp G \mid C$
☐ (C) $D \perp A \mid B, G$
☐ (D) $F \perp B \mid C$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\text{WetGrass} = T \mid \text{Cloudy} = T)$ (round off to 3 decimal places).



Cl	P(Cl=T)
T	0.9

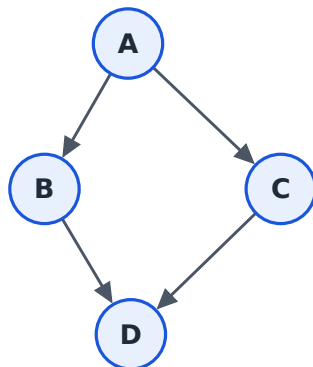
Cl	P(Sp=T ...)	Sp	P(Ra=T ...)
T	0.75	T	0.5
F	0.25	F	0.9

Ra	P(We=T ...)
T	0.7
F	0.8

Answer (numerical): _____

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.04, 0.03, 0.4, 0.95)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.6

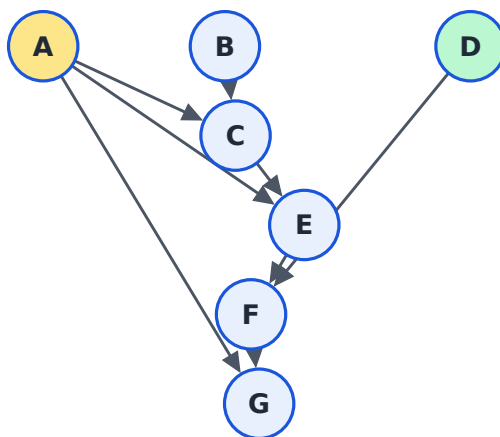
A	P(B=T ...)	A	P(C=T ...)
T	0.3	T	0.2
F	0.7	F	0.8

B	C	P(D=T ...)
T	T	0.05
T	F	0.7
F	T	0.3
F	F	0.1

- ☐ (A) A=F, B=T, C=F, D=F
☐ (B) A=T, B=T, C=F, D=F
☐ (C) A=F, B=F, C=F, D=T
☐ (D) A=T, B=F, C=F, D=F

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $\mathbf{P(D \mid A = T)}$ in the Bayesian network below (Boolean variables; evidence $A = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E, F, C, G, B**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence

X and Y are independent fair coin flips (Boolean) and $Z = X \text{ XOR } Y$. Which statements are true?

- ☐ (A) $X \perp Z$
☐ (B) $Y \perp Z$
☐ (C) $X \perp Y \mid Z$
☐ (D) $X \perp \{Y, Z\}$

Answer Key — Mock Test 089

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Problem Formulation	A	14	NAT	2	Adversarial Search	10.39 to 10.41
2	NAT	1	Adversarial Search	12	15	NAT	2	Propositional Logic	10
3	MCQ	1	Uninformed Search	C	16	NAT	2	Propositional Logic	2
4	NAT	1	Informed Search	19	17	MSQ	2	Propositional Logic	A, C
5	MSQ	1	Propositional Logic	A, C, D	18	MSQ	2	Predicate Logic	D
6	NAT	1	Predicate Logic	81	19	MSQ	2	Propositional Logic	B, C
7	MSQ	1	Propositional Logic	A, B, D	20	MSQ	2	Propositional Logic	A, B, C
8	NAT	1	Exact Inference	0.761 to 0.765	21	MSQ	2	Conditional Independence	A, C
9	NAT	1	Sampling	6	22	NAT	2	Exact Inference	0.738 to 0.742
10	MCQ	1	Variable Elimination	A	23	MCQ	2	Sampling	B
11	NAT	2	Informed Search	18	24	NAT	2	Exact Inference	3
12	MCQ	2	Informed Search	B	25	MSQ	2	Conditional Independence	A, B
13	NAT	2	Uninformed Search	16					

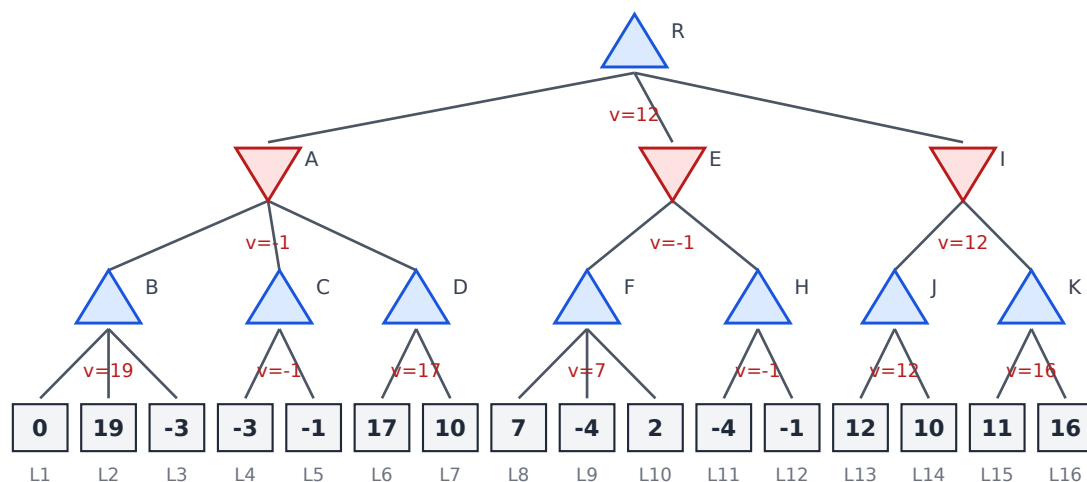
Detailed Solutions

Q.1 [MCQ, 1M · Problem Formulation] Answer: A (A: initial state, actions, transition model, goal test, and path (action) cost function)

A problem consists of the **initial state**, the **actions available in each state**, the **transition model** **RESULT(s,a)**, the **goal test**, and the **action-cost / path-cost function**. Together these define the state space implicitly. (B) leaves out actions and costs. (C) belongs to particular algorithms, not to the problem. (D) is the data structure of a search algorithm.

Q.2 [NAT, 1M · Adversarial Search] Answer: 12

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 19$, $C = -1$, $D = 17$, $A = -1$, $F = 7$, $H = -1$, $E = -1$, $J = 12$, $K = 16$, $I = 12$. Root value = **12**; MAX's optimal first move is towards I.

Q.3 [MCQ, 1M · Uninformed Search] Answer: C (C: S, F, I, L, H, J, D, E, G)

Adjacency lists (alphabetical): D: {G, H, J, L}; E: {G, L}; F: {H, I, J, S}; G: {D, E}; H: {D, F, J, L}; I: {F, J, S}; J: {D, F, H, I}; L: {D, E, H, S}; S: {F, I, L}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[F, I, L]
2	F	[I, L, H, J]
3	I	[L, H, J]

Step	Expanded	Frontier (FIFO) after expansion
4	L	[H, J, D, E]
5	H	[J, D, E]
6	J	[D, E]
7	D	[E, G]
8	E	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, F, I, L, H, J, D, E, G**. The path returned is $S \rightarrow L \rightarrow D \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.4 [NAT, 1M · Informed Search] **Answer: 19**

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (12)	A(9), C(9)
2	A (9)	J(4), K(5), C(9), I(9)
3	J (4)	G(0), K(5), C(9), I(9)
4	G (0)	K(5), C(9), I(9) ← goal, stop

Path returned: **S** → **A** → **J** → **G**, cost = 6 + 7 + 6 = **19**.

The optimal cost is $h^*(S) = 17$ — greedy search is not optimal here.

Q.5 [MSQ, 1M · Propositional Logic] **Answer: A, C, D**

(A) True: if $KB \wedge \neg \alpha$ is unsatisfiable, resolution derives the empty clause. (B) False: from $p \vee q$, $p \rightarrow r$, $q \rightarrow r$, modus ponens cannot derive r , although r is entailed. (C) True for definite clauses (Horn KBs with exactly one positive literal per clause). (D) This is the definition of soundness (truth-preserving).

Q.6 [NAT, 1M · Predicate Logic] **Answer: 81**

Interpret P and Q as subsets of the domain; each element independently has 3 allowed (P, Q) combinations $(FF, FT, TT) \Rightarrow 3^n$. With $n = 4$ this gives **81** (verified by enumerating all interpretations).

Q.7 [MSQ, 1M · Propositional Logic] **Answer: A, B, D** (A: Modus ponens: from $\alpha \rightarrow \beta$ and α infer β ; B: And-elimination: from $\alpha \wedge \beta$ infer α ; D: Resolution: from $\alpha \vee \beta$ and $\neg \beta \vee \gamma$ infer $\alpha \vee \gamma$)

Modus ponens, and-elimination and resolution are truth-preserving: whenever their premises are true, their conclusion is true. For resolution, if β is true then γ must be true, else α must be true. Abduction is unsound: $\alpha=F$, $\beta=T$ makes both premises true and the conclusion false.

Q.8 [NAT, 1M · Exact Inference] **Answer: 0.761 to 0.765**

Only the ancestors of C (A) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T	0.14
A=T, B=F	0.06
A=F, B=T	0.50625
A=F, B=F	0.05625

Sum = **0.7625** $\approx$ 0.763.

Q.9 [NAT, 1M · Sampling] **Answer: 6**

Rejection sampling keeps only samples consistent with the evidence $C = F$: samples 3, 5, 9, 10, 11, 12, 13, 14, 15 (9 accepted, 6 rejected).

Among the accepted samples, $A = T$ in 2. Estimate = $2/9 =$ **0.2222**.

(For this network the exact value is 0.3; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.10 [MCQ, 1M · Variable Elimination] **Answer: A** (A: A factor over A with 2 entries: $P(j \mid a)$, $P(j \mid \neg a)$)

Evidence variables are instantiated, so only the column $J = \text{true}$ is kept, giving $f(A) = \langle P(j \mid a), P(j \mid \neg a) \rangle$. J no longer appears as a dimension. $P(j)$ would require summing over A weighted by $P(A)$, which VE does only later and only if needed.

Q.11 [NAT, 2M · Informed Search] Answer: 18

Heuristic values: $h(A)=4$, $h(B)=7$, $h(C)=4$, $h(E)=11$, $h(G)=0$, $h(I)=3$, $h(J)=8$, $h(L)=8$, $h(S)=11$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	L(13), E(14), B(15)
2	L (13)	B(14), E(14), C(15), J(19)
3	B (14)	E(14), C(15), I(17), J(19)
4	E (14)	C(15), I(17), J(17)
5	C (15)	I(16), J(17)
6	I (16)	J(17), G(18), A(22)
7	J (17)	G(18), A(19)
8	G (18)	A(19) ← goal, stop

Path returned: **S** → **L** → **C** → **I** → **G** with cost **18**; nodes expanded = **8** (S, L, B, E, C, I, J, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 18$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.12 [MCQ, 2M · Informed Search] Answer: B (B: Because some node n on the optimal path is always in the frontier with $f(n) \leq C^* < g(G_2) = f(G_2)$)

This is the standard optimality proof. Until G_1 is selected, some node n on the optimal path is still in the frontier, and admissibility gives $f(n) = g(n) + h(n) \leq C^*$. Also $f(G_2) = g(G_2) > C^*$ because $h(\text{goal}) = 0$. So n , and eventually G_1 , are selected before G_2 . (A) is wrong because $h = 0$ at every goal. (C) is wrong because A^* orders by f , not depth. (D) is wrong because G_2 may well be generated, just not selected.

Q.13 [NAT, 2M · Uninformed Search] Answer: 16

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(4), E(6), I(9)
2	J (4)	E(6), B(8), I(9)
3	E (6)	B(8), I(9), D(13)
4	B (8)	I(9), D(13), H(13)
5	I (9)	H(12), D(13)
6	H (12)	D(13), G(16)
7	D (13)	G(16)
8	G (16)	← goal, stop

Path returned: **S** → **I** → **H** → **G**, cost = $9 + 3 + 4 = \mathbf{16}$. Nodes expanded (including S and G): **8** (S, J, E, B, I, H, D, G).

Q.14 [NAT, 2M · Adversarial Search] Answer: 10.39 to 10.41

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

• $A = 0.4 \times 13 + 0.1 \times 5 + 0.3 \times 5 + 0.2 \times 16 = 10.4$

• $B = 0.6 \times 12 + 0.3 \times 4 + 0.1 \times 17 = 10.1$

Root = $\max(10.4, 10.1) = \mathbf{10.4}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [NAT, 2M · Propositional Logic] Answer: 10

Build the truth table (sub-formulas shown first):

P	Q	R	S	$Q \wedge (S \rightarrow S)$	$(P \vee P) \rightarrow (\neg R \wedge Q)$	$(Q \wedge (S \rightarrow S)) \wedge ((P \vee P) \rightarrow (\neg R \wedge Q))$
T	T	T	T	T	F	F
T	T	T	F	T	F	F
T	T	F	T	T	T	T
T	T	F	F	T	T	T
T	F	T	T	F	F	F
T	F	T	F	F	F	F
T	F	F	T	F	F	F
T	F	F	F	F	F	F

P	Q	R	S	$Q \wedge (S \rightarrow S)$	$(P \vee P) \rightarrow (\neg R \wedge Q)$	$(Q \wedge (S \rightarrow S)) \wedge ((P \vee P) \rightarrow (\neg R \wedge Q))$
F	T	T	T	T	T	T
F	T	T	F	T	T	T
F	T	F	T	T	T	T
F	T	F	F	T	T	T
F	F	T	T	F	T	F
F	F	T	F	F	T	F
F	F	F	T	F	T	F
F	F	F	F	F	T	F

F has 6 models, so it is false in $16 - 6 = 10$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.16 [NAT, 2M · Propositional Logic] **Answer: 2**

Each clause, being a disjunction, is false only in specific situations: C1 false iff $p=q=F$; C2 iff $p=T, r=F$; C3 iff $q=T, s=F$; C4 iff $r=s=T$. With 4 clauses, satisfying exactly two means falsifying exactly two. C1 is incompatible with C2 and C3 (they need $p=T$ or $q=T$). C1 & C4: $p=q=F, r=s=T$ — then C2, C3 hold $\rightarrow$ 1 assignment. C2 & C3: $p=T, r=F, q=T, s=F$ — C4 holds, C1 holds $\rightarrow$ 1 assignment. C2 & C4 needs $r=F$ and $r=T$; C3 & C4 needs $s=F$ and $s=T$ — impossible. Total = 2 (confirmed by enumeration).

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: $(Q \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg Q)$; C: $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$)

(A) $(Q \rightarrow P) \leftrightarrow (\neg P \rightarrow \neg Q)$: true in all 4 assignments $\Rightarrow$ **valid**.

(B) $((R \rightarrow P) \rightarrow Q) \leftrightarrow (R \rightarrow (P \rightarrow Q))$: **not valid** — counter-model $P=T, Q=F, R=F$.

(C) $((P \vee Q) \wedge (\neg P \vee R)) \rightarrow (Q \vee R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $((P \rightarrow R) \wedge (Q \rightarrow R)) \rightarrow (P \wedge Q)$: **not valid** — counter-model $P=T, Q=F, R=T$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.18 [MSQ, 2M · Predicate Logic] **Answer: D** (D: $\forall y \exists x R(x, y)$)

$R = \{(1,2), (3,2), (3,4), (4,1), (4,3)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(3,4) \wedge R(4,1)$ but $\neg R(3,1)$.

(B) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(3,4)$ and $R(4,3)$ with $3 \neq 4$.

(C) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,3)$ nor $R(3,1)$.

(D) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: B, C** (B: $(\neg S)$; C: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable))

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(\neg S)$ is derivable:

1. resolve $(\neg P \vee \neg S)$ and (P) on $P \Rightarrow (\neg S)$

(C) $\square$ is derivable:

1. resolve $(\neg P \vee \neg S)$ and (P) on $P \Rightarrow (\neg S)$
2. resolve (S) and $(\neg S)$ on $S \Rightarrow \square$ (empty clause)

(D) $(P \vee Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.20 [MSQ, 2M · Propositional Logic] **Answer: A, B, C** (A: $\{p \vee q, \neg p \vee q, p \vee \neg q, \neg p \vee \neg q\}$; B: $\{p \rightarrow q, q \rightarrow r, r \rightarrow \neg p, p\}$; C: $\{p \leftrightarrow \neg q, q \leftrightarrow \neg r, r \leftrightarrow \neg p\}$)

(A) The four clauses exclude all four (p, q) assignments, one each. (B) From p we get q , then r , then $\neg p$ — a contradiction. (C) $p = \neg q$ and $q = \neg r$ give $p = r$, but $r \leftrightarrow \neg p$ demands $r = \neg p$ — contradiction (an odd cycle of negations). (D) is satisfiable, e.g. $p=T, q=F, r=F$ (exactly one true).

Q.21 [MSQ, 2M · Conditional Independence] **Answer: A, C** (A: $E \perp A \mid C, D$; C: $D \perp A \mid B, G$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp A \mid C, D$: **TRUE** — $E \leftarrow C \leftarrow A$ (blocked: observed chain node C); $E \leftarrow D \leftarrow B \rightarrow F \leftarrow A$ (blocked: observed chain node D); $E \leftarrow D \leftarrow B \rightarrow G \leftarrow A$ (blocked: observed chain node D).

(B) $E \perp G \mid C$: **FALSE** — the path $E \leftarrow D \leftarrow B \rightarrow G$ is active given $\{C\}$.

(C) $D \perp A \mid B, G$: **TRUE** — $D \leftarrow B \rightarrow F \leftarrow A$ (blocked: observed fork node B); $D \leftarrow B \rightarrow G \leftarrow A$ (blocked: observed fork node B); $D \rightarrow E \leftarrow C \leftarrow A$ (blocked: collider E with neither it nor a descendant observed).

(D) $F \perp B \mid C$: **FALSE** — the path $F \leftarrow B$ is active given $\{C\}$.

Q.22 [NAT, 2M · Exact Inference] **Answer: 0.738 to 0.742**

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Rain})$.

Inference by enumeration: $P(\text{WetGrass} \mid e) = \alpha \sum_{\text{hidden}} P(\text{WetGrass}, e, \text{hidden})$.

Hidden: Sp, Ra	We	product of CPT entries ($Cl \cdot Sp \cdot Ra \cdot We$)	value
T, T	T	$0.9 \times 0.75 \times 0.5 \times 0.7$	0.23625
T, F	T	$0.9 \times 0.75 \times 0.5 \times 0.8$	0.27
F, T	T	$0.9 \times 0.25 \times 0.9 \times 0.7$	0.14175
F, F	T	$0.9 \times 0.25 \times 0.1 \times 0.8$	0.018
T, T	F	$0.9 \times 0.75 \times 0.5 \times 0.3$	0.10125
T, F	F	$0.9 \times 0.75 \times 0.5 \times 0.2$	0.0675
F, T	F	$0.9 \times 0.25 \times 0.9 \times 0.3$	0.06075
F, F	F	$0.9 \times 0.25 \times 0.1 \times 0.2$	0.0045

$P(\text{WetGrass}=T, e) = 0.666$; $P(\text{WetGrass}=F, e) = 0.234$; $P(e) = 0.9$.

$P(\text{WetGrass}=T \mid e) = 0.666 / 0.9 = \mathbf{0.74} \approx 0.740$.

Q.23 [MCQ, 2M · Sampling] **Answer: B** (B: $A=T, B=T, C=F, D=F$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.6$; $u = 0.04 < 0.6 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.3$; $u = 0.03 < 0.3 \Rightarrow B = T$
- C: $P(C=T \mid A=T) = 0.2$; $u = 0.4 \geq 0.2 \Rightarrow C = F$
- D: $P(D=T \mid B=T, C=F) = 0.7$; $u = 0.95 \geq 0.7 \Rightarrow D = F$

Sample: **(A=T, B=T, C=F, D=F)**.

Q.24 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f()$, $f(B)$, $f(B, C)$, $f(D)$, $f(C, E)$, $f(D, E, F)$, $f(F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(C, E)$, $f(D, E, F)$	$\{C, D, E, F\}$	$f(C, D, F)$
F	$f(F, G)$, $f(C, D, F)$	$\{C, D, F, G\}$	$f(C, D, G)$
C	$f(B, C)$, $f(C, D, G)$	$\{B, C, D, G\}$	$f(B, D, G)$
G	$f(B, D, G)$	$\{B, D, G\}$	$f(B, D)$
B	$f(B)$, $f(B, D)$	$\{B, D\}$	$f(D)$

The largest factor produced has **3** variable(s) (created when eliminating E); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over D and normalise.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: A, B** (A: $X \perp Z$; B: $Y \perp Z$)

$P(z) = 0.5$ and $P(x, z) = P(x, \neg y) = 0.25 = P(x)P(z)$; every entry factorizes, so $X \perp Z$ (A true) and by symmetry $Y \perp Z$ (B true). Given Z, X determines Y: $P(x \mid y, z) = 0$ while $P(x \mid z) = 0.5$, so X and Y are dependent given Z (C false). Since $P(x \mid y, z) = 0 \neq P(x) = 0.5$, X is not independent of the pair $\{Y, Z\}$ (D false). Thus pairwise independence does not imply joint independence.

Mock Test 090

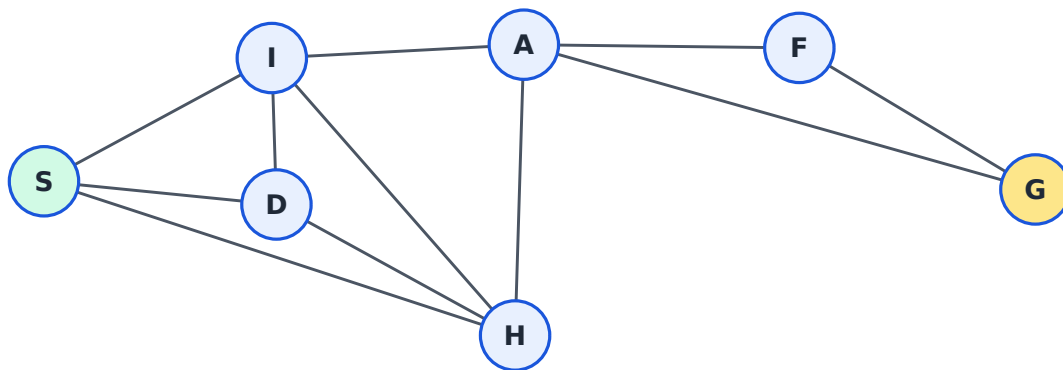
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 7 / 12	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

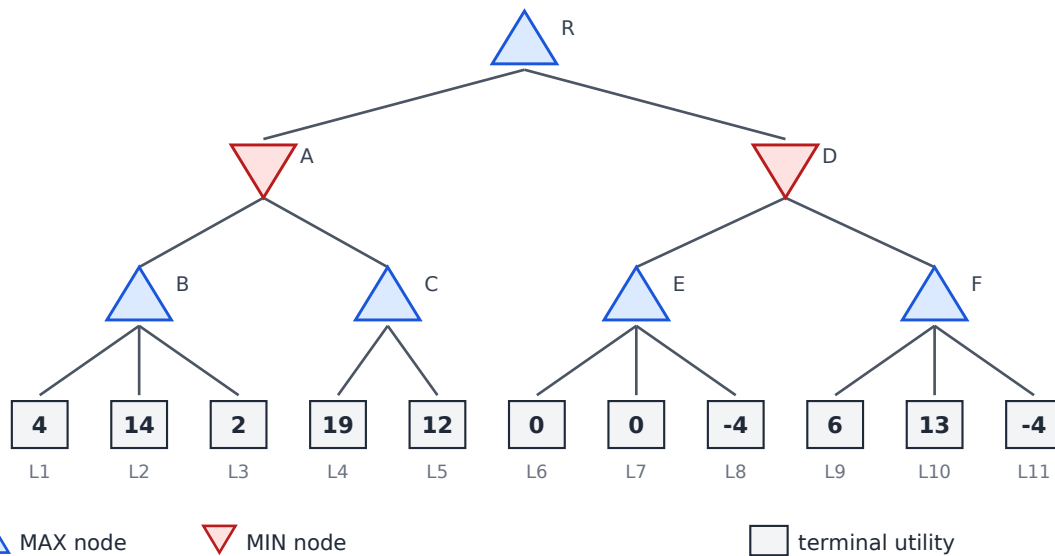
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, D, H, F, A, G
- ☐ (B) S, D, H, I, A, F, G
- ☐ (C) S, D, H, A, F, G
- ☐ (D) S, I, H, D, A, G

Q.2 [MCQ · 1 mark · negative: -1/3] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



- ☐ (A) Move to D; root value 0
☐ (B) Move to A; root value 19
☐ (C) Move to A; root value 14
☐ (D) Move to A; root value 15

Q.3 [NAT · 1 mark · no negative marking] Problem Formulation

In the vacuum world with $n = 3$ squares in a row, a state specifies the agent's location and whether each square is dirty or clean. How many states are there?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 2$ (so $b^d = 4$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

What is the most general unifier (MGU) of $\text{Knows}(\text{John}, x)$ and $\text{Knows}(y, \text{Mother}(y))$?

- ☐ (A) $\{y/\text{John}, x/\text{Mother}(\text{John})\}$
☐ (B) $\{x/\text{John}, y/\text{Mother}(\text{John})\}$
☐ (C) $\{y/\text{John}, x/\text{Mother}(y)\}$
☐ (D) The two atoms are not unifiable

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$\mathbf{F = (\neg P \rightarrow \neg \neg Q) \rightarrow ((\neg R \leftrightarrow \neg R) \wedge \neg P) ?}$$

Answer (numerical): _____

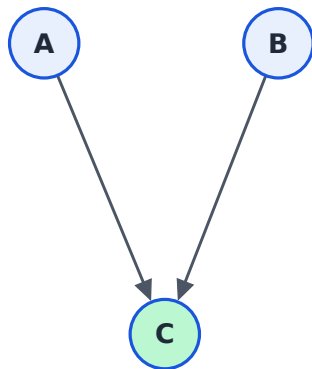
Q.7 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following correctly express 'Not every graph is connected'?

- ☐ (A) $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$
☐ (B) $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$
☐ (C) $\forall x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$
☐ (D) $\exists x (\text{Graph}(x) \rightarrow \neg \text{Connected}(x))$

Q.8 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).

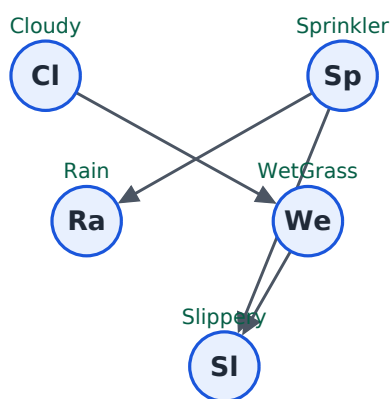


$P(A=T)$	$P(B=T)$	A	B	$P(C=T \dots)$
0.95	0.8	T	T	0.8
		T	F	0.2
		F	T	0.8
		F	F	0.6

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We, Sl. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, Sl = Slippery.) What is the probability that a single run produces the sample **(Cl=F, Sp=T, Ra=F, We=F, Sl=F)**? (round off to 4 decimal places)



$P(Cl=T)$	$P(Sp=T)$	Sp	$P(Ra=T \dots)$
0.5	0.9	T	0.9
		F	0.75

Cl	$P(We=T \dots)$	Sp	We	$P(Sl=T \dots)$
T	0.7	T	T	0.2
F	0.4	T	F	0.25
		F	T	0.95
		F	F	0.9

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

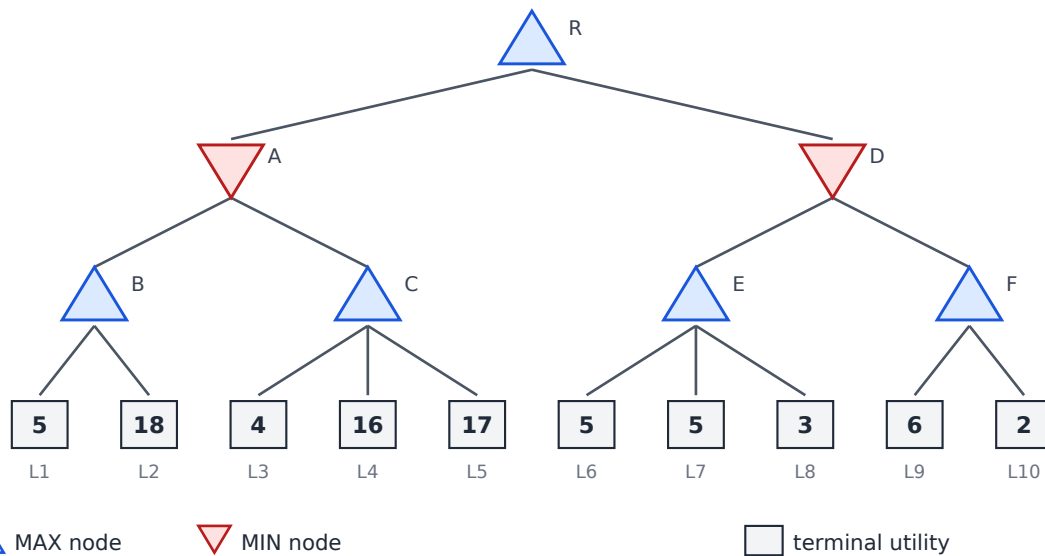
Which statement is the local Markov property (local semantics) of a Bayesian network?

- ☐ (A) Each node is conditionally independent of its non-descendants given its parents
☐ (B) Each node is conditionally independent of its descendants given its parents
☐ (C) Each node is marginally independent of all nodes that are not its parents
☐ (D) Each node is conditionally independent of its parents given its children

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



Answer (numerical): _____

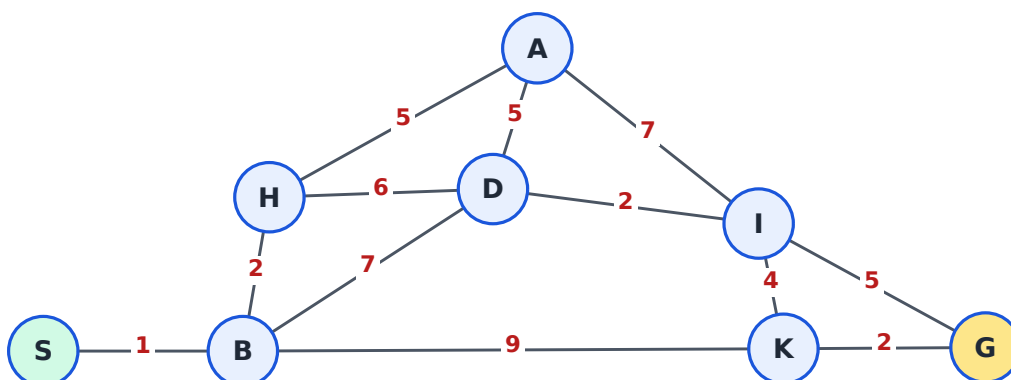
Q.12 [MCQ · 2 marks · negative: -2/3] Local Search

Steepest-ascent hill climbing solves a random 8-queens instance with probability $p \approx 0.14$. Random-restart hill climbing repeats independent trials until one succeeds. What is the expected number of trials?

- ☐ (A) About 1.14
☐ (B) About 3
☐ (C) About 7
☐ (D) About 14

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

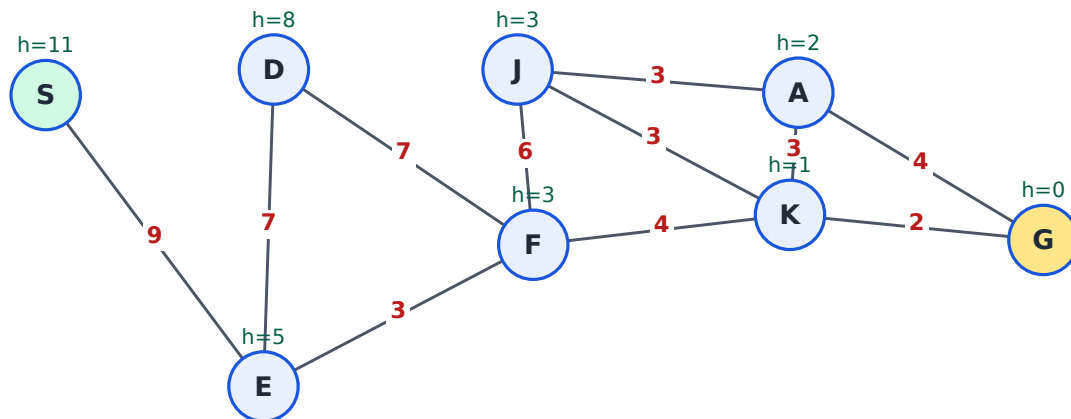
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **E, C, B** and rules:

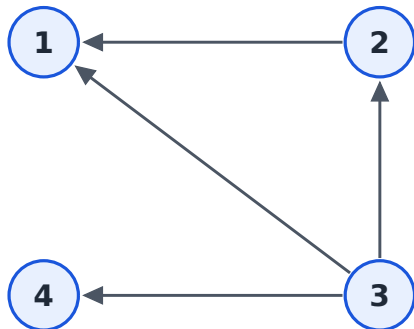
- R1: $C \wedge E \rightarrow A$
- R2: $C \wedge J \wedge D \rightarrow H$
- R3: $F \wedge D \rightarrow H$
- R4: $B \wedge C \rightarrow F$
- R5: $E \rightarrow D$
- R6: $J \wedge H \wedge F \rightarrow K$
- R7: $B \wedge D \rightarrow H$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) A is entailed
- ☐ (B) F is entailed
- ☐ (C) J is entailed
- ☐ (D) K is entailed

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y R(x, y)$
☐ (B) $\exists x \forall y \neg R(x, y)$
☐ (C) $\exists x R(x, x)$
☐ (D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $S \leftrightarrow \neg R$
 (2) $(S \wedge \neg Q) \leftrightarrow (\neg R \vee Q)$
 (3) $R \leftrightarrow (P \leftrightarrow S)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg Q$
☐ (B) $\neg S \rightarrow (R \rightarrow Q)$
☐ (C) $\neg P$
☐ (D) $Q \vee (\neg P \wedge Q)$

Q.18 [NAT · 2 marks · no negative marking] Propositional Logic

The canonical CNF (product of maxterms, each clause containing all variables) of $p \oplus q \oplus r$ has how many clauses?

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following entailments hold?

- ☐ (A) $\{p \rightarrow q, q \rightarrow r\} \models p \rightarrow r$
☐ (B) $\{p \vee q, \neg p \vee r\} \models q \vee r$
☐ (C) $\{p \rightarrow q, q\} \models p$
☐ (D) $\{p \rightarrow q, \neg q\} \models \neg p$

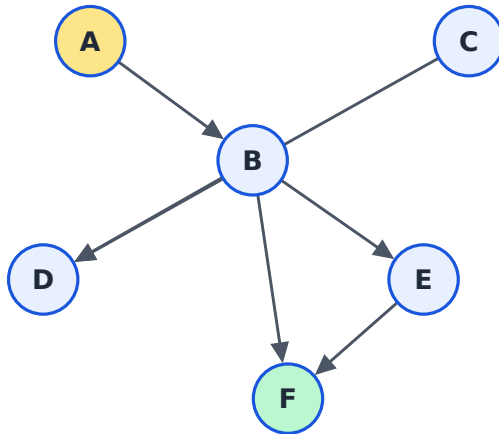
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \vee Q) \wedge (\neg R \vee P)) \rightarrow (Q \vee P)$
☐ (B) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$
☐ (C) $(R \rightarrow (Q \rightarrow P)) \leftrightarrow ((R \wedge Q) \rightarrow P)$
☐ (D) $((Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow (Q \rightarrow P)$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(F \mid A = T)$ in the Bayesian network below (Boolean variables; evidence $A = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, D, C, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

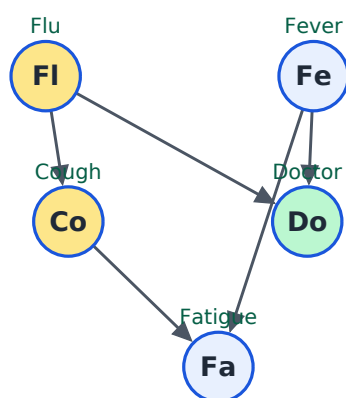
Q.22 [MCQ · 2 marks · negative: -2/3] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. For the query $P(J \mid b)$ (J given Burglary = true), which variables are irrelevant to the query and can be removed before running variable elimination?

- ☐ (A) {M}
☐ (B) {E, M}
☐ (C) {E}
☐ (D) No variable is irrelevant

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) Compute $P(\text{Doctor} = T \mid \text{Flu} = F, \text{Cough} = T)$ (round off to 3 decimal places).



$P(FI=T)$
0.4

$P(Fe=T)$
0.05

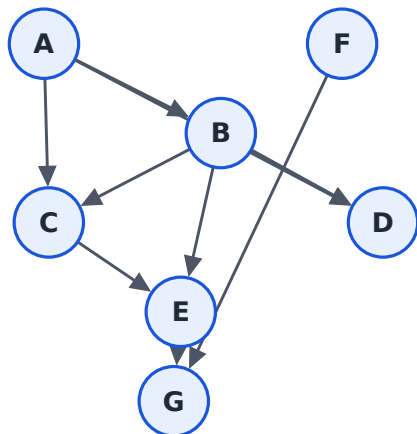
FI	$P(Co=T \dots)$
T	0.4
F	0.9

Fe	Co	$P(Fa=T \dots)$	FI	Fe	$P(Do=T \dots)$
T	T	0.9	T	T	0.05
T	F	0.9	T	F	0.6
F	T	0.1	F	T	0.7
F	F	0.6	F	F	0.5

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

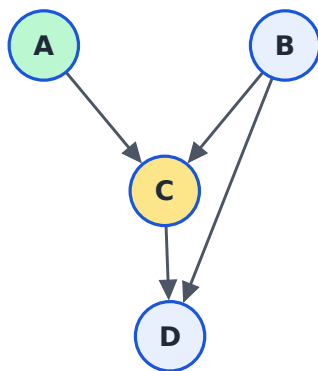
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp G \mid F$
☐ (B) $D \perp F \mid B, C$
☐ (C) $C \perp G \mid A, D$
☐ (D) $C \perp B$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $C = T$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $P(A = T \mid C = T)$ (round off to 3 decimal places).



P(A=T)
0.4

P(B=T)	A	B	P(C=T ...)
0.4	T	T	0.2
	T	F	0.9
	F	T	0.9
	F	F	0.4

B	C	P(D=T ...)
T	T	0.1
T	F	0.7
F	T	0.9
F	F	0.75

#	A	B	D
1	F	F	F
2	F	F	T
3	T	T	F
4	T	F	T

Answer (numerical): _____

Answer Key — Mock Test 090

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	C	14	NAT	2	Informed Search	5
2	MCQ	1	Adversarial Search	C	15	MSQ	2	Propositional Logic	A, B
3	NAT	1	Problem Formulation	24	16	MSQ	2	Predicate Logic	B, D
4	NAT	1	Adversarial Search	3	17	MSQ	2	Propositional Logic	A, C
5	MCQ	1	Predicate Logic	A	18	NAT	2	Propositional Logic	4
6	NAT	1	Propositional Logic	4	19	MSQ	2	Propositional Logic	A, B, D
7	MSQ	1	Predicate Logic	A, B	20	MSQ	2	Propositional Logic	A, C, D
8	NAT	1	Exact Inference	0.682 to 0.686	21	NAT	2	Exact Inference	4
9	NAT	1	Sampling	0.02 to 0.0204	22	MCQ	2	Variable Elimination	A
10	MCQ	1	Bayesian Networks	A	23	NAT	2	Exact Inference	0.508 to 0.512
11	NAT	2	Adversarial Search	2	24	MSQ	2	Conditional Independence	B
12	MCQ	2	Local Search	C	25	NAT	2	Sampling	0.577 to 0.581
13	NAT	2	Uninformed Search	8					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: C** (C: S, D, H, A, F, G)

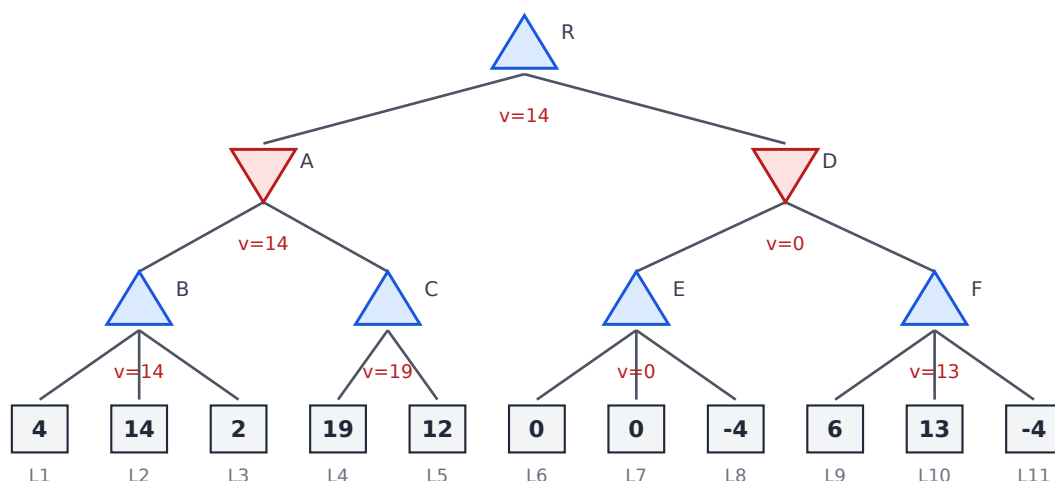
Adjacency lists (alphabetical): A: {F, G, H, I}; D: {H, I, S}; F: {A, G}; G: {A, F}; H: {A, D, I, S}; I: {A, D, H, S}; S: {D, H, I}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, D, H, A, F, G**; the path found is $S \rightarrow D \rightarrow H \rightarrow A \rightarrow F \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.2 [MCQ, 1M · Adversarial Search] **Answer: C** (C: Move to A; root value 14)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 14, C = 19, A = 14, E = 0, F = 13, D = 0. Root value = **14**; MAX's optimal first move is towards A.

Q.3 [NAT, 1M · Problem Formulation] **Answer: 24**

The agent can be in any of 3 squares, and each of the 3 squares is dirty or clean, giving 2^3 combinations. States = $n \cdot 2^n = 3 \times 8 = \mathbf{24}$. The 2-square world has $2 \cdot 2^2 = 8$ states.

Q.4 [NAT, 1M · Adversarial Search] Answer: 3

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 2^{\lceil 1 \rceil} + 2^{\lfloor 1 \rfloor} - 1 = 2 + 2 - 1 = 3$, versus 4 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.5 [MCQ, 1M · Predicate Logic] Answer: A (A: $\{y/\text{John}, x/\text{Mother}(\text{John})\}$)

Unify arguments left to right. John vs y gives y/John. Then x vs Mother(y) becomes x vs Mother(John), giving x/Mother(John). Applying $\{y/\text{John}, x/\text{Mother}(\text{John})\}$ to both atoms yields Knows(John, Mother(John)). Option B binds the wrong variables. Option C leaves y in the binding of x, so it is not idempotent and does not make the atoms identical in one application.

Q.6 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg P \rightarrow \neg \neg Q$	$(\neg R \leftrightarrow \neg R) \wedge \neg P$	$(\neg P \rightarrow \neg \neg Q) \rightarrow ((\neg R \leftrightarrow \neg R) \wedge \neg P)$
T	T	T	T	F	F
T	T	F	T	F	F
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	F	T	T
F	F	F	F	T	T

Rows where F = T: **4**.

Q.7 [MSQ, 1M · Predicate Logic] Answer: A, B (A: $\neg \forall x (\text{Graph}(x) \rightarrow \text{Connected}(x))$; B: $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$)

Option A is the direct translation, and pushing the negation inward gives $\exists x \neg(\text{Graph}(x) \rightarrow \text{Connected}(x)) \equiv \exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$ — option B. Option C says no graph is connected, which is stronger. Option D is true whenever any non-graph exists (vacuous implication), so it does not express the statement.

Q.8 [NAT, 1M · Exact Inference] Answer: 0.682 to 0.686

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T	0.608
A=T, B=F	0.038
A=F, B=T	0.032
A=F, B=F	0.006

Sum = **0.684** ≈ 0.684 .

Q.9 [NAT, 1M · Sampling] Answer: 0.02 to 0.0204

Prior sampling generates each complete event with exactly its joint probability $S_{p_5}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(CI=F) \times P(Sp=T) \times P(Ra=F|Sp=T) \times P(We=F|CI=F) \times P(SI=F|Sp=T, We=F)$
 $= 0.5 \times 0.9 \times 0.1 \times 0.6 \times 0.75 = \mathbf{0.02025} \approx 0.0202$.

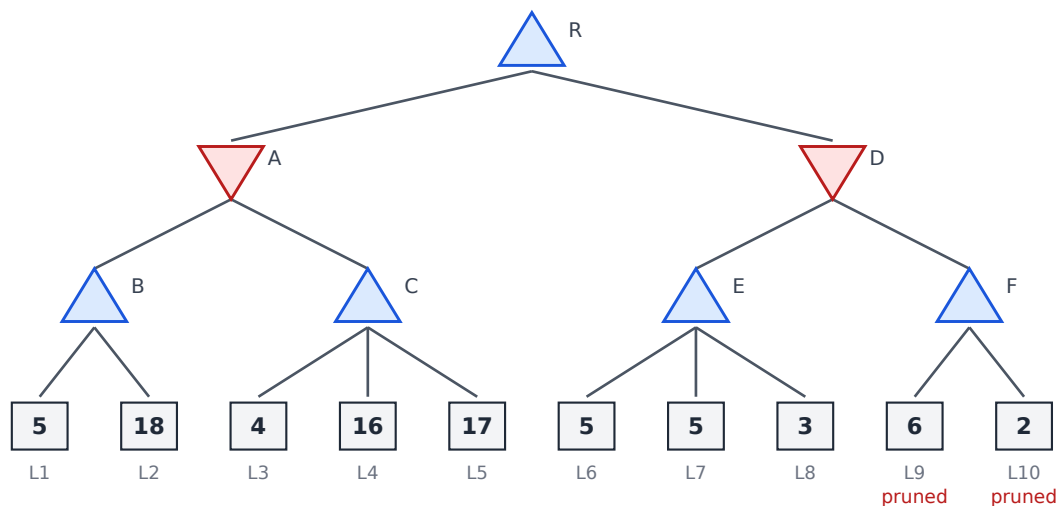
Q.10 [MCQ, 1M · Bayesian Networks] Answer: A (A: Each node is conditionally independent of its non-descendants given its parents)

AIMA: a node is conditionally independent of its non-descendants given its parents; this is equivalent to the global factorization semantics. (B) is false because descendants (e.g. children) carry information about the node even given the parents. (C) ignores dependence through common ancestors, and (D) has no basis.

Q.11 [NAT, 2M · Adversarial Search] Answer: 2

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

• At MIN node D: value $5 \leq \alpha = 17 \Rightarrow$ prune remaining children (leaves L9, L10).



Leaves evaluated (8): L1, L2, L3, L4, L5, L6, L7, L8. Leaves pruned (2): L9, L10. Root value = 17 (same as plain minimax, 17 — pruning never changes the root value).

Q.12 [MCQ, 2M · Local Search] **Answer: C** (C: About 7)

The number of trials until the first success is geometrically distributed with mean $1/p = 1/0.14 \approx 7$. That means about 6 failures and 1 success, matching AIMA's analysis. (A) confuses $1/p$ with $1 + p$. (D) confuses $1/p$ with $100 \cdot p$.

Q.13 [NAT, 2M · Uninformed Search] **Answer: 8**

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(1)
2	B (1)	H(3), D(8), K(10)
3	H (3)	A(8), D(8), K(10)
4	A (8)	D(8), K(10), I(15)
5	D (8)	I(10), K(10)
6	I (10)	K(10), G(15)
7	K (10)	G(12)
8	G (12)	← goal, stop

Path returned: **S** → **B** → **K** → **G**, cost = 1 + 9 + 2 = **12**. Nodes expanded (including S and G): **8** (S, B, H, A, D, I, K, G).

Q.14 [NAT, 2M · Informed Search] **Answer: 5**

Heuristic values: $h(A)=2$, $h(D)=8$, $h(E)=5$, $h(F)=3$, $h(G)=0$, $h(J)=3$, $h(K)=1$, $h(S)=11$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (11)	E(14)
2	E (14)	F(15), D(24)
3	F (15)	K(17), J(21), D(24)
4	K (17)	G(18), A(21), J(21), D(24)
5	G (18)	A(21), J(21), D(24) ← goal, stop

Path returned: **S** → **E** → **F** → **K** → **G** with cost **18**; nodes expanded = **5** (S, E, F, K, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 18$. For comparison UCS ($h = 0$) would expand 6 nodes.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: A, B** (A: A is entailed; B: F is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $C \wedge E \rightarrow A$, $B \wedge C \rightarrow F$, $E \rightarrow D$; add A, D, F.

Round 2: fire $F \wedge D \rightarrow H$, $B \wedge D \rightarrow H$; add H.

Fixed point: {A, B, C, D, E, F, H} — 7 atoms are entailed (facts: B, C, E; newly derived: A, D, F, H). Atoms never derived: J, K.

Q.16 [MSQ, 2M · Predicate Logic] **Answer: B, D** (B: $\exists x \forall y \neg R(x, y)$; D: $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$)

$R = \{(2,1), (3,1), (3,2), (3,4)\}$.

(A) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(B) $\exists x \forall y \neg R(x, y)$: **TRUE** — 1 has no outgoing edge.

(C) $\exists x R(x, x)$: **FALSE** — there are no self-loops.

(D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **TRUE** — all two-step paths are short-cut.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: $\neg Q$; C: $\neg P$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
F	F	T	F	T
F	F	F	T	T

(A) $\neg Q$ is true in all models of KB $\Rightarrow$ **entailed**.

(B) $\neg S \rightarrow (R \rightarrow Q)$ is false in the KB-model $P=F, Q=F, R=T, S=F \Rightarrow$ not entailed.

(C) $\neg P$ is true in all models of KB $\Rightarrow$ **entailed**.

(D) $Q \vee (\neg P \wedge Q)$ is false in the KB-model $P=F, Q=F, R=T, S=F \Rightarrow$ not entailed.

Q.18 [NAT, 2M · Propositional Logic] **Answer: 4**

In the canonical CNF there is one maxterm (clause) for every falsifying assignment. $p \oplus q \oplus r$ is false exactly when an even number of variables is true: FFF, TTF, TFT, FTT. That gives 4 falsifying rows and hence 4 clauses. It is also known that no shorter CNF exists for parity, since every clause must contain all 3 variables.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: A, B, D** (A: $\{p \rightarrow q, q \rightarrow r\} \models p \rightarrow r$; B: $\{p \vee q, \neg p \vee r\} \models q \vee r$; D: $\{p \rightarrow q, \neg q\} \models \neg p$)

(A) Hypothetical syllogism: every model of both premises with $p=T$ has $q=T$ and hence $r=T$. (B) This is exactly the resolution rule on p : if $p=T$ then r must be T; if $p=F$ then q must be T. (C) Fails: $p=F, q=T$ satisfies both premises but falsifies p (affirming the consequent). (D) Modus tollens: if p were T then q would be T, contradicting $\neg q$. Truth tables confirm A, B, D.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: A, C, D** (A: $((R \vee Q) \wedge (\neg R \vee P)) \rightarrow (Q \vee P)$; C: $(R \rightarrow (Q \rightarrow P)) \leftrightarrow ((R \wedge Q) \rightarrow P)$; D: $((Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow (Q \rightarrow P)$)

(A) $((R \vee Q) \wedge (\neg R \vee P)) \rightarrow (Q \vee P)$: true in all 8 assignments $\Rightarrow$ **valid**.

(B) $((Q \rightarrow R) \wedge \neg Q) \rightarrow \neg R$: **not valid** — counter-model $Q=F, R=T$.

(C) $(R \rightarrow (Q \rightarrow P)) \leftrightarrow ((R \wedge Q) \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

(D) $((Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow (Q \rightarrow P)$: true in all 8 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.21 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f()$, $f(B)$, $f(C)$, $f(B, C, D)$, $f(B, E)$, $f(B, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(B), f(B, C, D), f(B, E), f(B, E, F)$	$\{B, C, D, E, F\}$	$f(C, D, E, F)$
D	$f(C, D, E, F)$	$\{C, D, E, F\}$	$f(C, E, F)$
C	$f(C), f(C, E, F)$	$\{C, E, F\}$	$f(E, F)$
E	$f(E, F)$	$\{E, F\}$	$f(F)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over F and normalise.

Q.22 [MCQ, 2M · Variable Elimination] **Answer: A** (A: $\{M\}$)

AIMA: every variable that is not an ancestor of a query variable or an evidence variable is irrelevant.

Ancestors of J and B: $\{A, B, E\}$. M is neither, so it is irrelevant: $\sum_m P(m | a) = 1$. E is an ancestor of J ($E \rightarrow A \rightarrow J$), and $P(a | b) = \sum_e P(e) P(a | b, e)$ genuinely depends on E's CPT, so E cannot be dropped.

Q.23 [NAT, 2M · Exact Inference] **Answer: 0.508 to 0.512**

Factorisation: $P(\text{Flu}, \text{Fever}, \text{Cough}, \text{Fatigue}, \text{Doctor}) = P(\text{Flu}) \cdot P(\text{Fever}) \cdot P(\text{Cough} | \text{Flu}) \cdot P(\text{Fatigue} | \text{Fever}, \text{Cough}) \cdot P(\text{Doctor} | \text{Flu}, \text{Fever})$.

Inference by enumeration: $P(\text{Doctor} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Doctor}, e, \text{hidden})$.

Hidden: Fe, Fa	Do	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, T	T	$0.6 \times 0.05 \times 0.9 \times 0.9 \times 0.7$	0.01701
T, F	T	$0.6 \times 0.05 \times 0.9 \times 0.1 \times 0.7$	0.00189
F, T	T	$0.6 \times 0.95 \times 0.9 \times 0.1 \times 0.5$	0.02565
F, F	T	$0.6 \times 0.95 \times 0.9 \times 0.9 \times 0.5$	0.23085
T, T	F	$0.6 \times 0.05 \times 0.9 \times 0.9 \times 0.3$	0.00729
T, F	F	$0.6 \times 0.05 \times 0.9 \times 0.1 \times 0.3$	0.00081
F, T	F	$0.6 \times 0.95 \times 0.9 \times 0.1 \times 0.5$	0.02565
F, F	F	$0.6 \times 0.95 \times 0.9 \times 0.9 \times 0.5$	0.23085

$P(\text{Doctor}=T, e) = 0.2754$; $P(\text{Doctor}=F, e) = 0.2646$; $P(e) = 0.54$.

$P(\text{Doctor}=T \mid e) = 0.2754 / 0.54 = \mathbf{0.51} \approx 0.510$.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: B** (B: $D \perp F \mid B, C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp G \mid F$: **FALSE** — the path $A \rightarrow B \rightarrow C \rightarrow E \rightarrow G$ is active given $\{F\}$.

(B) $D \perp F \mid B, C$: **TRUE** — $D \leftarrow A \rightarrow B \rightarrow C \rightarrow E \rightarrow G \leftarrow F$ (blocked: observed chain node B); $D \leftarrow A \rightarrow B \rightarrow E \rightarrow G \leftarrow F$ (blocked: observed chain node B); $D \leftarrow A \rightarrow C \leftarrow B \rightarrow E \rightarrow G \leftarrow F$ (blocked: observed fork node B); $D \leftarrow A \rightarrow C \rightarrow E \rightarrow G \leftarrow F$ (blocked: observed chain node C); $D \leftarrow B \leftarrow A \rightarrow C \rightarrow E \rightarrow G \leftarrow F$ (blocked: observed chain node B); $D \leftarrow B \rightarrow C \rightarrow E \rightarrow G \leftarrow F$ (blocked: observed fork node B) (+1 more paths, all blocked).

(C) $C \perp G \mid A, D$: **FALSE** — the path $C \leftarrow B \rightarrow E \rightarrow G$ is active given $\{A, D\}$.

(D) $C \perp B$: **FALSE** — the path $C \leftarrow A \rightarrow B$ is active given $\{\}$.

Q.25 [NAT, 2M · Sampling] **Answer: 0.577 to 0.581**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	A
1	$P(C=T \mid A=F, B=F)=0.4$	0.4	F
2	$P(C=T \mid A=F, B=F)=0.4$	0.4	F
3	$P(C=T \mid A=T, B=T)=0.2$	0.2	T
4	$P(C=T \mid A=T, B=F)=0.9$	0.9	T

$\sum w (\text{all}) = 1.9$; $\sum w (A = T) = 1.1$.

Estimate $P(A=T \mid C = T) = 1.1 / 1.9 = \mathbf{0.5789}$.

Mock Test 091

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	3 / 7 / 15	12 / 13 / 15

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

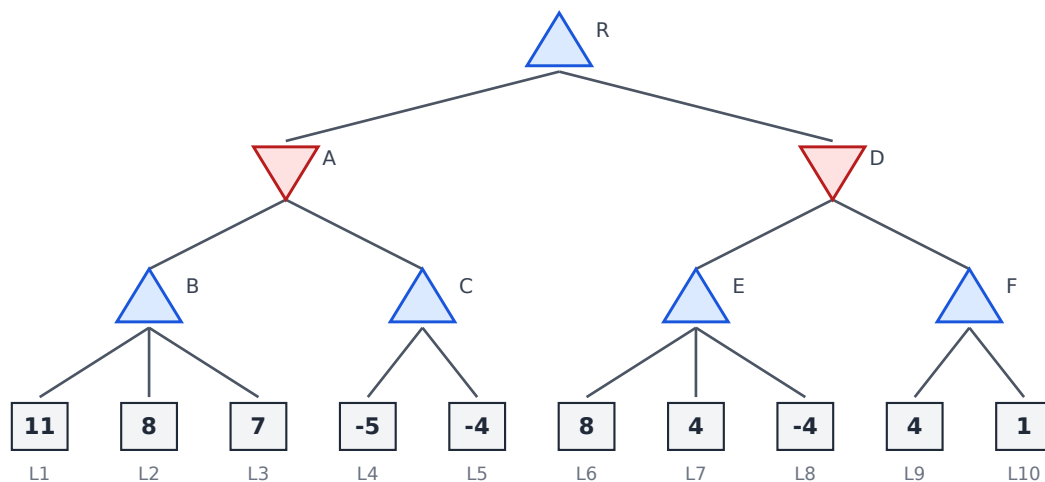
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 5$ and depth $d = 2$ (so $b^d = 25$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



△ MAX node ▽ MIN node

□ terminal utility

Answer (numerical): _____

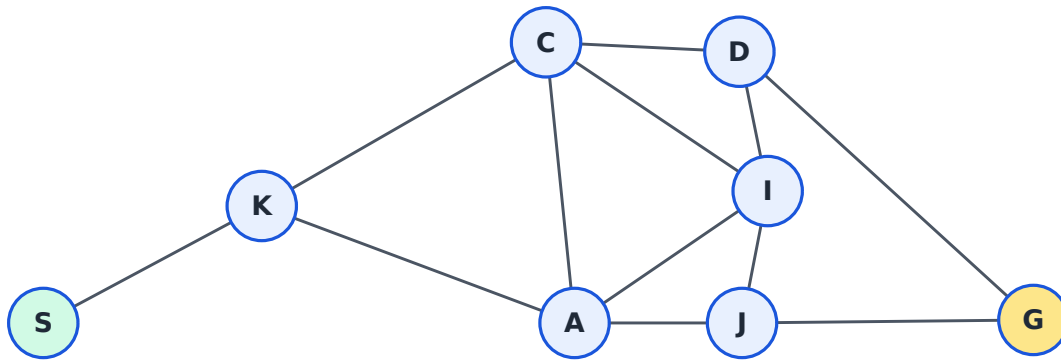
Q.3 [NAT · 1 mark · no negative marking] Adversarial Search

A uniform game tree has branching factor $b = 3$ and depth $d = 4$ plies. Alpha-beta search examines the moves in perfect order. Using the best-case formula $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$, how many leaves are evaluated?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, K, A, C, I, J, D, G
☐ (B) S, K, A, I, C, J, D, G
☐ (C) S, K, C, A, I, D, J, G
☐ (D) S, K, A, C, D, G

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = ((P \leftrightarrow R) \leftrightarrow \neg R) \rightarrow ((\neg P \leftrightarrow Q) \vee P) ?$$

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Resolving the first-order clauses $P(x) \vee Q(f(x))$ and $\neg P(g(y)) \vee R(y)$ (variables already standardized apart) on P gives

- ☐ (A) $Q(f(g(y))) \vee R(y)$
☐ (B) $Q(f(x)) \vee R(y)$
☐ (C) $Q(g(f(y))) \vee R(y)$
☐ (D) $Q(f(y)) \vee R(g(y))$

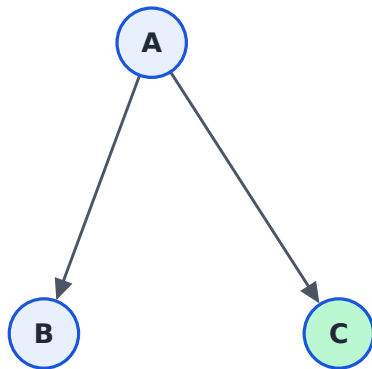
Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

In how many rows of the truth table over p, q is $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow p)$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).

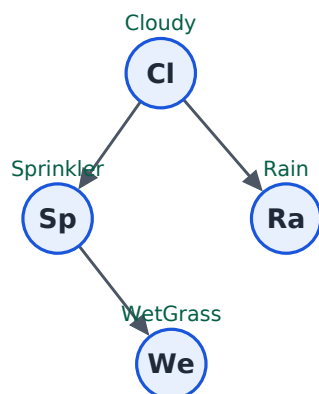


$P(A=T)$	A	$P(B=T \dots)$	A	$P(C=T \dots)$
0.25	T	0.4	T	0.05
	F	0.6	F	0.1

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample **(Cl=T, Sp=F, Ra=F, We=T)**? (round off to 4 decimal places)



$P(Cl=T)$
0.8

Cl	$P(Sp=T \dots)$	Cl	$P(Ra=T \dots)$
T	0.4	T	0.7
F	0.4	F	0.1

Sp	$P(We=T \dots)$
T	0.4
F	0.95

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling

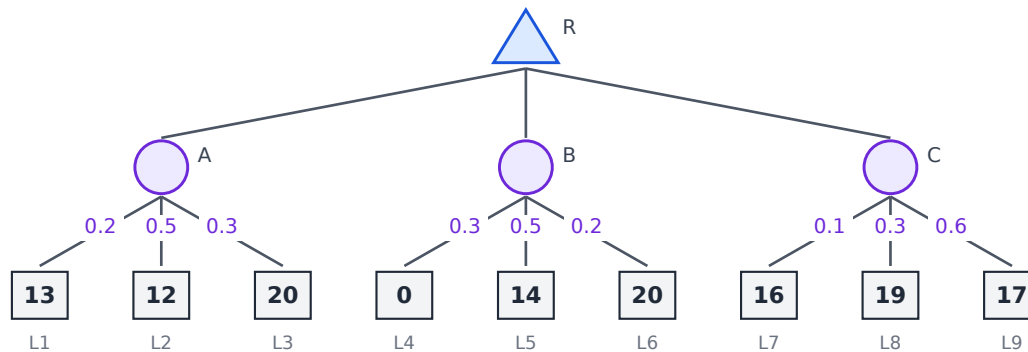
Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$; $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$; $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = 0.90$, $P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0.0$. Likelihood weighting is used with evidence $S = \text{true}$, $W = \text{true}$. A sample is generated with $C = \text{false}$ and $R = \text{false}$. What is the weight of this sample? (Round to 2 decimal places.)

Answer (numerical): _____

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)

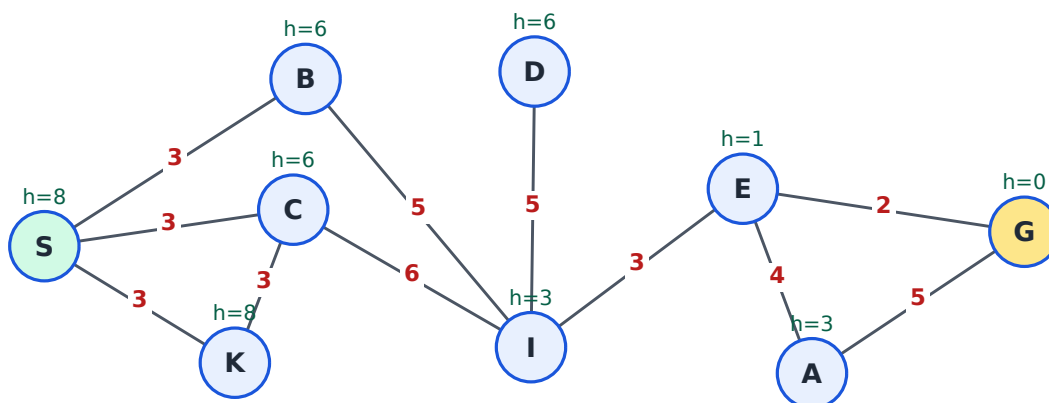


△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

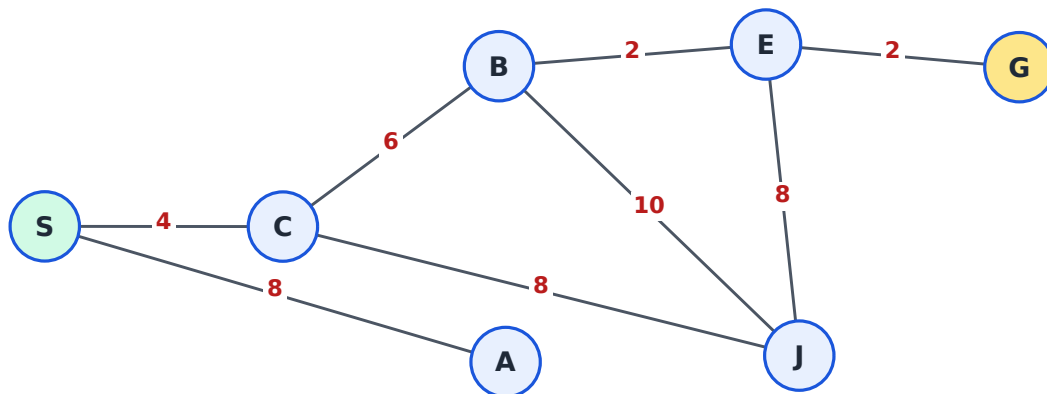
Q.13 [MCQ · 2 marks · negative: -2/3] Informed Search

Weighted A* uses $f(n) = g(n) + W \cdot h(n)$. As the weight $W \rightarrow \infty$, the order in which it expands nodes approaches that of

- ☐ (A) uniform-cost search
- ☐ (B) A* search
- ☐ (C) greedy best-first search
- ☐ (D) breadth-first search

Q.14 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **B, A** and rules:

- R1: $F \rightarrow K$
- R2: $A \rightarrow D$
- R3: $F \wedge C \rightarrow B$
- R4: $A \wedge D \rightarrow F$
- R5: $B \wedge H \rightarrow E$
- R6: $B \wedge F \rightarrow J$
- R7: $J \wedge C \wedge F \rightarrow E$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) E is entailed
- ☐ (B) F is entailed
- ☐ (C) J is entailed
- ☐ (D) K is entailed

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(P \rightarrow Q) \rightarrow ((P \vee R) \rightarrow Q)$
- ☐ (B) $(P \rightarrow R) \leftrightarrow (\neg R \rightarrow \neg P)$
- ☐ (C) $((P \rightarrow R) \wedge R) \rightarrow P$
- ☐ (D) $(R \rightarrow Q) \rightarrow (Q \rightarrow R)$

Q.17 [NAT · 2 marks · no negative marking] Predicate Logic

A first-order language has constants a, b and one binary function symbol g. Let $H_0 = \{a, b\}$ and $H_{i+1} = H_i \cup \{g(s, t) : s, t \in H_i\}$. How many ground terms are in H_2 ?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

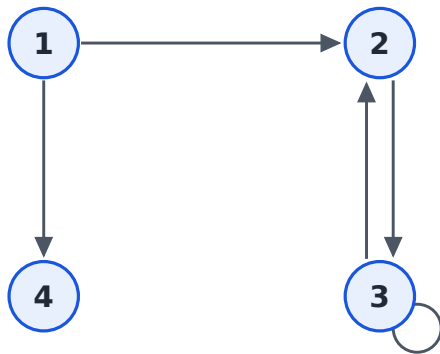
C1: $(\neg P \vee R)$, C2: $(\neg P)$, C3: $(P \vee R \vee \neg S)$, C4: $(\neg P \vee Q \vee R)$, C5: $(\neg P \vee \neg Q)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(R \vee \neg S)$
- ☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
- ☐ (C) $(P \vee \neg Q)$
- ☐ (D) $(P \vee S)$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

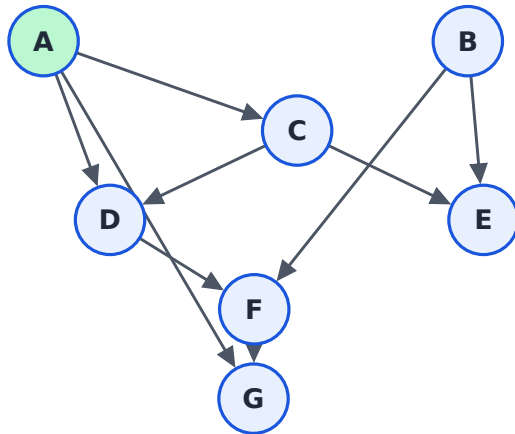
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y \neg R(y, x)$
- ☐ (B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
- ☐ (C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$
- ☐ (D) $\forall x R(x, x)$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

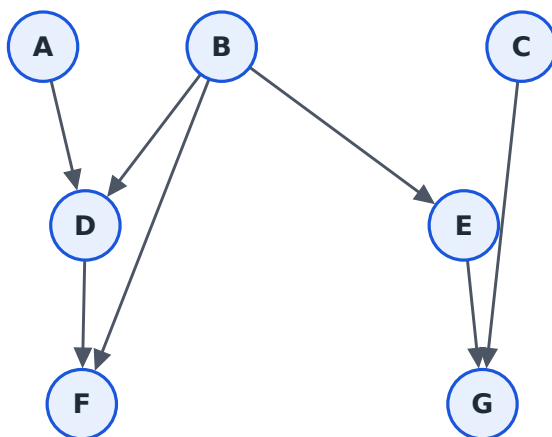
Variable elimination is used to compute $P(\mathbf{A})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **G, C, D, E, B, F**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp F \mid C, G$
- ☐ (B) $B \perp A$
- ☐ (C) $E \perp G \mid A$
- ☐ (D) $C \perp A \mid B, G$

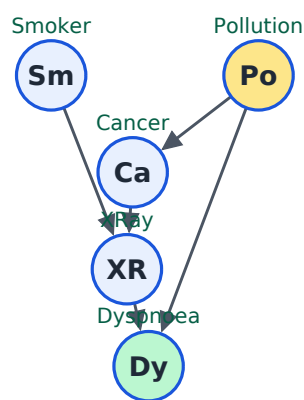
Q.22 [MSQ · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. Which of the following are correct (values rounded)?

- ☐ (A) $P(a) \approx 0.0307$
☐ (B) $P(b | a) \approx 0.588$
☐ (C) $P(b | a, e) > P(b | a)$
☐ (D) $P(j | b) \approx 0.725$

Q.23 [NAT · 2 marks · no negative marking] Exact Inference - Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T | \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay, Dy = Dyspnoea.) Compute **$P(\text{Dyspnoea} = T | \text{Pollution} = F)$** (round off to 3 decimal places).



$P(\text{Sm}=T)$
0.25

$P(\text{Po}=T)$
0.6

Po	$P(\text{Ca}=T ...)$
T	0.3
F	0.2

Sm	Ca	$P(\text{XR}=T ...)$	Po	XR	$P(\text{Dy}=T ...)$
T	T	0.5	T	T	0.3
T	F	0.4	T	F	0.9
F	T	0.25	F	T	0.9
F	F	0.6	F	F	0.7

Answer (numerical): _____

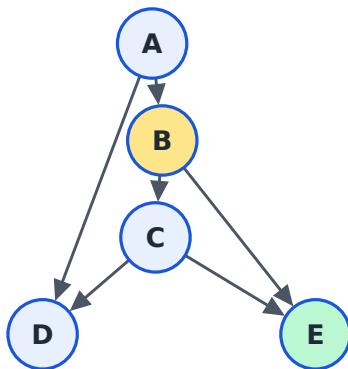
Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence

Consider the Bayesian network (DAG) with edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$, $B \rightarrow F$. Which of the following d-separation statements are true?

- ☐ (A) A and B are d-separated given $\emptyset$
☐ (B) A and F are d-separated given $\{B\}$
☐ (C) E and F are d-separated given $\{C\}$
☐ (D) E and F are d-separated given $\emptyset$

Q.25 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence **B=T**. The current state of the chain is (**A=F, B=T, C=T, D=T, E=F**). The next step resamples **E**. With what probability is E set to T? (round off to 3 decimal places)



P(A=T)
0.6

A	P(B=T ...)	B	P(C=T ...)
T	0.95	T	0.25
F	0.95	F	0.2

A	C	P(D=T ...)	B	C	P(E=T ...)
T	T	0.3	T	T	0.5
T	F	0.9	T	F	0.3
F	T	0.7	F	T	0.6
F	F	0.25	F	F	0.5

Answer (numerical): _____

Answer Key — Mock Test 091

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	9	14	NAT	2	Uninformed Search	7
2	NAT	1	Adversarial Search	4	15	MSQ	2	Propositional Logic	B, C, D
3	NAT	1	Adversarial Search	17	16	MSQ	2	Propositional Logic	B
4	MCQ	1	Uninformed Search	A	17	NAT	2	Predicate Logic	38
5	NAT	1	Propositional Logic	6	18	MSQ	2	Propositional Logic	A
6	MCQ	1	Predicate Logic	A	19	MSQ	2	Predicate Logic	A
7	NAT	1	Propositional Logic	2	20	NAT	2	Exact Inference	4
8	NAT	1	Exact Inference	0.085 to 0.089	21	MSQ	2	Conditional Independence	B, D
9	NAT	1	Sampling	0.1366 to 0.137	22	MSQ	2	Bayesian Networks	A, B, D
10	NAT	1	Sampling	0.445 to 0.455	23	NAT	2	Exact Inference	0.798 to 0.802
11	NAT	2	Adversarial Search	17.49 to 17.51	24	MSQ	2	Conditional Independence	A, B, C
12	NAT	2	Informed Search	13	25	NAT	2	Sampling	0.498 to 0.502
13	MCQ	2	Informed Search	C					

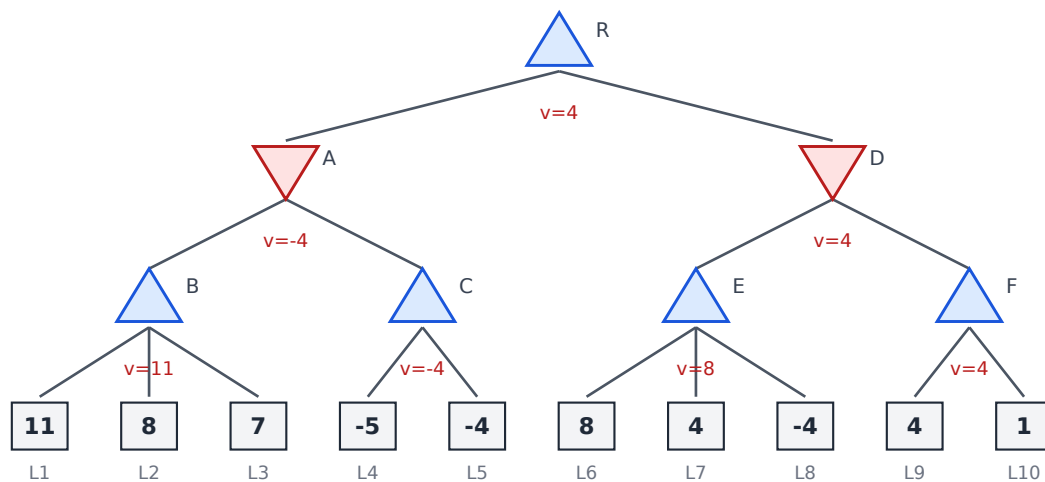
Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 9**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 5^{\lceil 1 \rceil} + 5^{\lfloor 1 \rfloor} - 1 = 5 + 5 - 1 = 9$, versus 25 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.2 [NAT, 1M · Adversarial Search] **Answer: 4**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 11$, $C = -4$, $A = -4$, $E = 8$, $F = 4$, $D = 4$. Root value = **4**; MAX's optimal first move is towards D.

Q.3 [NAT, 1M · Adversarial Search] **Answer: 17**

$3^2 + 3^2 - 1 = 9 + 9 - 1 = \mathbf{17}$. Plain minimax evaluates $3^4 = 81$ leaves, so perfect ordering saves 64 leaf evaluations.

Q.4 [MCQ, 1M · Uninformed Search] **Answer: A** (A: S, K, A, C, I, J, D, G)

Adjacency lists (alphabetical): A: {C, I, J, K}; C: {A, D, I, K}; D: {C, G, I}; G: {D, J}; I: {A, C, D, J}; J: {A, G, I}; K: {A, C, S}; S: {K}.

BFS trace (queue shown front $\rightarrow$ back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[K]
2	K	[A, C]
3	A	[C, I, J]
4	C	[I, J, D]
5	I	[J, D]
6	J	[D, G]
7	D	[G]
8	G	[] $\leftarrow$ goal popped, stop

Expansion order: **S, K, A, C, I, J, D, G**. The path returned is $S \rightarrow K \rightarrow A \rightarrow J \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [NAT, 1M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	$(P \leftrightarrow R) \leftrightarrow \neg R$	$(\neg P \leftrightarrow Q) \vee P$	$((P \leftrightarrow R) \leftrightarrow \neg R) \rightarrow ((\neg P \leftrightarrow Q) \vee P)$
T	T	T	F	T	T
T	T	F	F	T	T
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	T	F	F

Rows where $F = T$: **6**.

Q.6 [MCQ, 1M · Predicate Logic] **Answer: A** (A: $Q(f(g(y))) \vee R(y)$)

Unify $P(x)$ with $P(g(y))$: MGU $\theta = \{x/g(y)\}$. The resolvent is $(Q(f(x)) \vee R(y))\theta = (Q(f(g(y))) \vee R(y))$. Option B forgets to apply θ . Options C and D apply the substitution incorrectly.

Q.7 [NAT, 1M · Propositional Logic] Answer: 2

$p \rightarrow q \equiv \neg p \vee q$ and $\neg q \rightarrow p \equiv p \vee q$. Row TT: $T, T \rightarrow T$. Row TF: $F, T \rightarrow F$. Row FT: $T, T \rightarrow T$. Row FF: $T, F \rightarrow F$. So the biconditional is true in 2 rows.

Q.8 [NAT, 1M · Exact Inference] Answer: 0.085 to 0.089

Only the ancestors of C (A) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T	0.005
A=T, B=F	0.0075
A=F, B=T	0.045
A=F, B=F	0.03

Sum = **0.0875** ≈ 0.087 .

Q.9 [NAT, 1M · Sampling] Answer: 0.1366 to 0.137

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(CI=T) \times P(Sp=F|CI=T) \times P(Ra=F|CI=T) \times P(We=T|Sp=F)$
 $= 0.8 \times 0.6 \times 0.3 \times 0.95 = \mathbf{0.1368} \approx 0.1368$.

Q.10 [NAT, 1M · Sampling] Answer: 0.445 to 0.455

In likelihood weighting the weight is the product of the likelihoods of the evidence variables given their parents in the sample. $w = P(s \mid \neg c) \times P(w \mid s, \neg r) = 0.5 \times 0.90 = 0.45$. The probabilities of the sampled non-evidence values ($P(\neg c)$, $P(\neg r \mid \neg c)$) are not part of the weight.

Q.11 [NAT, 2M · Adversarial Search] Answer: 17.49 to 17.51

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.2 \times 13 + 0.5 \times 12 + 0.3 \times 20 = 14.6$
- $B = 0.3 \times 0 + 0.5 \times 14 + 0.2 \times 20 = 11$
- $C = 0.1 \times 16 + 0.3 \times 19 + 0.6 \times 17 = 17.5$

Root = $\max(14.6, 11, 17.5) = \mathbf{17.5}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.12 [NAT, 2M · Informed Search] Answer: 13

Heuristic values: $h(A)=3, h(B)=6, h(C)=6, h(D)=6, h(E)=1, h(G)=0, h(I)=3, h(K)=8, h(S)=8$. A^* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (8)	B(9), C(9), K(11)
2	B (9)	C(9), I(11), K(11)
3	C (9)	I(11), K(11)
4	I (11)	K(11), E(12), D(19)
5	K (11)	E(12), D(19)
6	E (12)	G(13), A(18), D(19)
7	G (13)	A(18), D(19) $\leftarrow$ goal, stop

Path returned: **S** $\rightarrow$ **B** $\rightarrow$ **I** $\rightarrow$ **E** $\rightarrow$ **G** with cost **13**; nodes expanded = **7** (S, B, C, I, K, E, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 13$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.13 [MCQ, 2M · Informed Search] Answer: C (C: greedy best-first search)

For very large W , the term $W \cdot h$ dominates g , so nodes are effectively ordered by h alone, which is **greedy best-first search**. $W = 1$ gives A^* (B). $W = 0$ gives UCS (A). BFS (D) would need ordering by depth.

Q.14 [NAT, 2M · Uninformed Search] Answer: 7

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	C(4), A(8)

Step	Expanded (g)	Frontier after expansion (sorted by priority)
2	C (4)	A(8), B(10), J(12)
3	A (8)	B(10), J(12)
4	B (10)	E(12), J(12)
5	E (12)	J(12), G(14)
6	J (12)	G(14)
7	G (14)	← goal, stop

Path returned: **S** → **C** → **B** → **E** → **G**, cost = 4 + 6 + 2 + 2 = **14**. Nodes expanded (including S and G): **7** (S, C, A, B, E, J, G).

Q.15 [MSQ, 2M · Propositional Logic] **Answer: B, C, D** (B: F is entailed; C: J is entailed; D: K is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $A \rightarrow D$; add D.

Round 2: fire $A \wedge D \rightarrow F$; add F.

Round 3: fire $F \rightarrow K$, $B \wedge F \rightarrow J$; add J, K.

Fixed point: {A, B, D, F, J, K} — 6 atoms are entailed (facts: A, B; newly derived: D, F, J, K). Atoms never derived: C, E, H.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: B** (B: $(P \rightarrow R) \leftrightarrow (\neg R \rightarrow \neg P)$)

(A) $(P \rightarrow Q) \rightarrow ((P \vee R) \rightarrow Q)$: **not valid** — counter-model $P=F, Q=F, R=T$.

(B) $(P \rightarrow R) \leftrightarrow (\neg R \rightarrow \neg P)$: true in all 4 assignments $\Rightarrow$ **valid**.

(C) $((P \rightarrow R) \wedge R) \rightarrow P$: **not valid** — counter-model $P=F, R=T$.

(D) $(R \rightarrow Q) \rightarrow (Q \rightarrow R)$: **not valid** — counter-model $Q=T, R=F$.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [NAT, 2M · Predicate Logic] **Answer: 38**

$|H_1| = 2 + 2^2 = 6$: a, b, g(a,a), g(a,b), g(b,a), g(b,b). $H_2 = H_1 \cup \{g(s, t) : s, t \in H_1\}$. The 36 terms g(s, t) are syntactically distinct, and they already include the 4 g-terms of H_1 (since $H_0 \subseteq H_1$); adding a and b gives $|H_2| = 2 + 6^2 = 38$.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: A** (A: $(R \vee \neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(R \vee \neg S)$ is derivable:

1. resolve $(\neg P \vee R)$ and $(P \vee R \vee \neg S)$ on $P \Rightarrow (R \vee \neg S)$

(B) The resolution closure contains 9 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(P \vee \neg Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(P \vee S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.19 [MSQ, 2M · Predicate Logic] **Answer: A** (A: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,2), (1,4), (2,3), (3,2), (3,3)\}$.

(A) $\exists x \forall y \neg R(y, x)$: **TRUE** — 1 has no incoming edge.

(B) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(1,3)$ nor $R(3,1)$.

(C) $\forall x \forall y (R(x, y) \rightarrow R(y, x))$: **FALSE** — $R(1,2)$ holds but $R(2,1)$ does not.

(D) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): f(A), f(B), f(A, C), f(A, C, D), f(B, C, E), f(B, D, F), f(A, F, G).

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
G	f(A, F, G)	{A, F, G}	f(A, F)
C	f(A, C), f(A, C, D), f(B, C, E)	{A, B, C, D, E}	f(A, B, D, E)

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
D	$f(B, D, F), f(A, B, D, E)$	$\{A, B, D, E, F\}$	$f(A, B, E, F)$
E	$f(A, B, E, F)$	$\{A, B, E, F\}$	$f(A, B, F)$
B	$f(B), f(A, B, F)$	$\{A, B, F\}$	$f(A, F)$
F	$f(A, F), f(A, F)$	$\{A, F\}$	$f(A)$

The largest factor produced has **4** variable(s) (created when eliminating C); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over A and normalise.

Q.21 [MSQ, 2M · Conditional Independence] **Answer: B, D** (B: $B \perp A$; D: $C \perp A \mid B, G$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $A \perp F \mid C, G$: **FALSE** — the path $A \rightarrow D \rightarrow F$ is active given $\{C, G\}$.

(B) $B \perp A$: **TRUE** — $B \rightarrow D \leftarrow A$ (blocked: collider D with neither it nor a descendant observed); $B \rightarrow F \leftarrow D \leftarrow A$ (blocked: collider F with neither it nor a descendant observed).

(C) $E \perp G \mid A$: **FALSE** — the path $E \rightarrow G$ is active given $\{A\}$.

(D) $C \perp A \mid B, G$: **TRUE** — $C \rightarrow G \leftarrow E \leftarrow B \rightarrow D \leftarrow A$ (blocked: observed fork node B); $C \rightarrow G \leftarrow E \leftarrow B \rightarrow F \leftarrow D \leftarrow A$ (blocked: observed fork node B).

Q.22 [MSQ, 2M · Bayesian Networks] **Answer: A, B, D** (A: $P(a) \approx 0.0307$; B: $P(b \mid a) \approx 0.588$; D: $P(j \mid b) \approx 0.725$)

$P(a) = 0.02 \times 0.01 \times 0.95 + 0.02 \times 0.99 \times 0.90 + 0.98 \times 0.01 \times 0.30 + 0.98 \times 0.99 \times 0.01 = 0.00019 + 0.01782 + 0.00294 + 0.009702 = 0.030652$ (A true). $P(b, a) = 0.00019 + 0.01782 = 0.01801$, so $P(b \mid a) = 0.01801 / 0.030652 \approx 0.588$ (B true). $P(b \mid a, e) = 0.019 / 0.313 \approx 0.061 < 0.588$ (explaining away), so (C) is false. $P(a \mid b) = 0.01 \times 0.95 + 0.99 \times 0.90 = 0.9005$, so $P(j \mid b) = 0.9005 \times 0.8 + 0.0995 \times 0.05 = 0.7204 + 0.004975 \approx 0.725$ (D true).

Q.23 [NAT, 2M · Exact Inference] **Answer: 0.798 to 0.802**

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}, \text{Dyspnoea}) = P(\text{Smoker}) \cdot P(\text{Pollution}) \cdot P(\text{Cancer} \mid \text{Pollution}) \cdot P(\text{XRay} \mid \text{Smoker}, \text{Cancer}) \cdot P(\text{Dyspnoea} \mid \text{Pollution}, \text{XRay})$.

Inference by enumeration: $P(\text{Dyspnoea} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Dyspnoea}, e, \text{hidden})$.

Hidden: Sm, Ca, XR	Dy	product of CPT entries (Sm · Po · Ca · XR · Dy)	value
T, T, T	T	$0.25 \times 0.4 \times 0.2 \times 0.5 \times 0.9$	0.009
T, T, F	T	$0.25 \times 0.4 \times 0.2 \times 0.5 \times 0.7$	0.007
T, F, T	T	$0.25 \times 0.4 \times 0.8 \times 0.4 \times 0.9$	0.0288
T, F, F	T	$0.25 \times 0.4 \times 0.8 \times 0.6 \times 0.7$	0.0336
F, T, T	T	$0.75 \times 0.4 \times 0.2 \times 0.25 \times 0.9$	0.0135
F, T, F	T	$0.75 \times 0.4 \times 0.2 \times 0.75 \times 0.7$	0.0315
F, F, T	T	$0.75 \times 0.4 \times 0.8 \times 0.6 \times 0.9$	0.1296
F, F, F	T	$0.75 \times 0.4 \times 0.8 \times 0.4 \times 0.7$	0.0672
T, T, T	F	$0.25 \times 0.4 \times 0.2 \times 0.5 \times 0.1$	0.001
T, T, F	F	$0.25 \times 0.4 \times 0.2 \times 0.5 \times 0.3$	0.003
T, F, T	F	$0.25 \times 0.4 \times 0.8 \times 0.4 \times 0.1$	0.0032
T, F, F	F	$0.25 \times 0.4 \times 0.8 \times 0.6 \times 0.3$	0.0144
F, T, T	F	$0.75 \times 0.4 \times 0.2 \times 0.25 \times 0.1$	0.0015
F, T, F	F	$0.75 \times 0.4 \times 0.2 \times 0.75 \times 0.3$	0.0135
F, F, T	F	$0.75 \times 0.4 \times 0.8 \times 0.6 \times 0.1$	0.0144
F, F, F	F	$0.75 \times 0.4 \times 0.8 \times 0.4 \times 0.3$	0.0288

$P(\text{Dyspnoea}=T, e) = 0.3202$; $P(\text{Dyspnoea}=F, e) = 0.0798$; $P(e) = 0.4$.

$P(\text{Dyspnoea}=T \mid e) = 0.3202 / 0.4 = \mathbf{0.8005} \approx 0.800$.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: A, B, C** (A: A and B are d-separated given $\emptyset$; B: A and F are d-separated given $\{B\}$; C: E and F are d-separated given $\{C\}$)

(A) The only path $A \rightarrow C \leftarrow B$ has collider C with no observed descendant: blocked, so true. (B) The only path $A \rightarrow C \leftarrow B \rightarrow F$ is blocked by the unobserved collider C (and also by the observed fork node B): true. (C) The only path $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ contains observed non-collider C: blocked, true. (D) With nothing observed, $E \leftarrow D \leftarrow C \leftarrow B \rightarrow F$ has no colliders and is active, so false.

Q.25 [NAT, 2M · Sampling] **Answer: 0.498 to 0.502**

Gibbs sampling draws E from $P(E \mid \text{mb}(E))$, where $\text{mb}(E) = \{B, C\}$:

$$P(E \mid \text{mb}) \propto P(E \mid \text{parents}(E)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y)).$$

$$E=T: P(E=T \mid B=T, C=T) = 0.5 = 0.5$$

$$E=F: P(E=F \mid B=T, C=T) = 0.5 = 0.5$$

$$\text{Normalise: } 0.5 / (0.5 + 0.5) = \mathbf{0.5}.$$

Mock Test 092

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 9 / 11	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

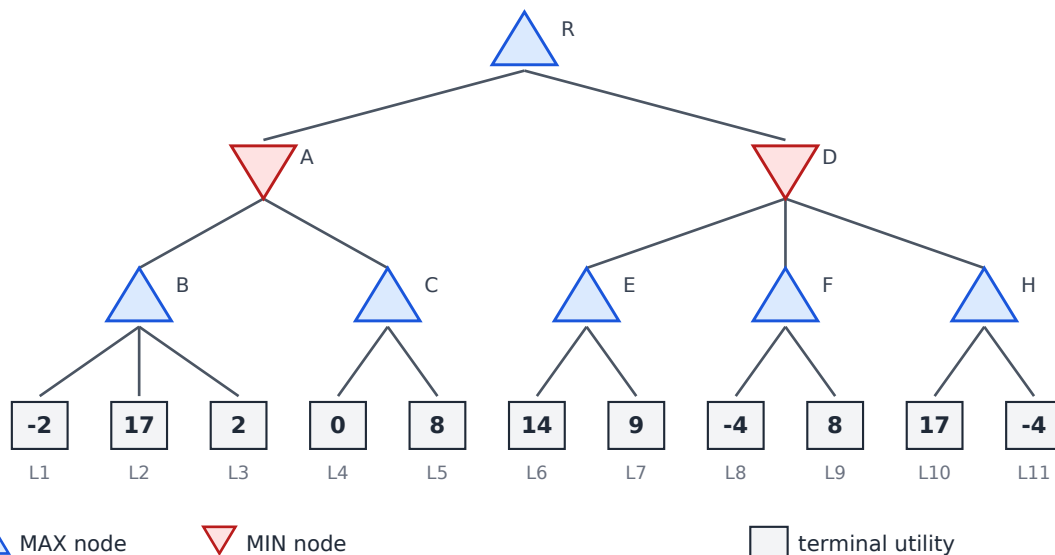
Q.1 [MSQ · 1 mark · no negative marking] Adversarial Search

Which of the following statements about two-player zero-sum games are TRUE?

- ☐ (A) The players' utilities sum to the same constant in every terminal state
- ☐ (B) Chess, with outcomes $+1/0/-1$ or $1/1/2/0$, can be modelled as a zero-sum (constant-sum) game
- ☐ (C) Minimax assumes that the opponent plays optimally
- ☐ (D) In a zero-sum game, both players can strictly gain from the same outcome change

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

For the game tree below, what is the minimax value of the root?



Answer (numerical): _____

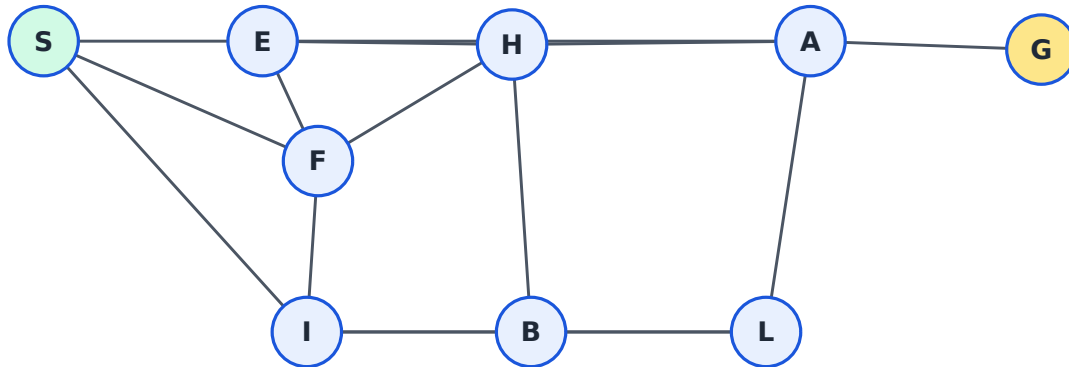
Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 5$ and depth $d = 4$ (so $b^d = 625$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, E, F, I, A, B, H, L, G
- ☐ (B) S, I, F, E, L, B, H, A, G
- ☐ (C) S, E, F, I, A, H, B, L, G
- ☐ (D) S, E, A, G

Q.5 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Which one of the following expresses 'None of my friends are perfect' ($F(x)$: x is my friend, $P(x)$: x is perfect)?

- ☐ (A) $\forall x (F(x) \rightarrow \neg P(x))$
- ☐ (B) $\exists x (F(x) \wedge \neg P(x))$
- ☐ (C) $\neg \forall x (F(x) \rightarrow P(x))$
- ☐ (D) $\forall x (\neg F(x) \rightarrow P(x))$

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

For propositional sentences α and β , $\alpha \models \beta$ holds if and only if

- ☐ (A) $\alpha \wedge \neg \beta$ is unsatisfiable
- ☐ (B) $\alpha \wedge \beta$ is satisfiable
- ☐ (C) $\neg \alpha \vee \beta$ is satisfiable
- ☐ (D) $\alpha \vee \neg \beta$ is valid

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 2 elements, so there are $4^2 = 16$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \leftrightarrow Q(x))$ true?

Answer (numerical): _____

Q.8 [MCQ · 1 mark · negative: -1/3] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Prior (direct) sampling visits variables in the order C, S, R, W and sets a variable to true if and only if the next uniform random number u satisfies $u < P(\text{variable} = \text{true} | \text{sampled parents})$. The random numbers used are 0.42, 0.07, 0.85, 0.30 (in this order). What sample is generated?

- ☐ (A) (c, s, $\neg r$, w)
☐ (B) (c, $\neg s$, r, w)
☐ (C) (c, s, r, w)
☐ (D) ($\neg c$, s, $\neg r$, $\neg w$)

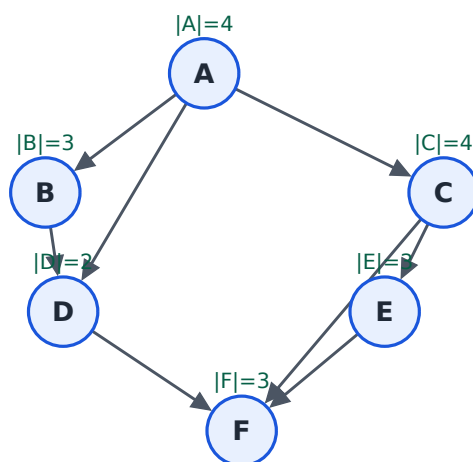
Q.9 [NAT · 1 mark · no negative marking] Bayesian Networks

A Bayesian network over Boolean variables has edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $C \rightarrow E$. How many independent parameters do its CPTs contain?

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Bayesian Networks - Parameter count

The Bayesian network below has variables whose domain sizes are written above each node. How many independent parameters are required to specify all of its CPTs?

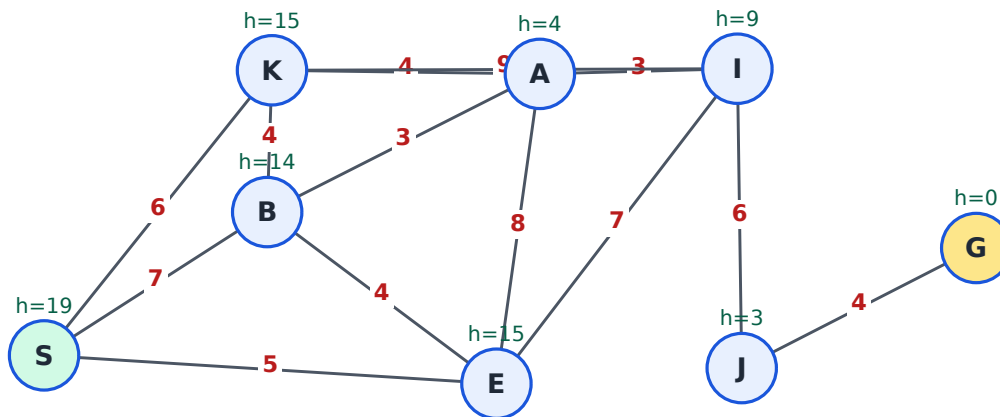


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

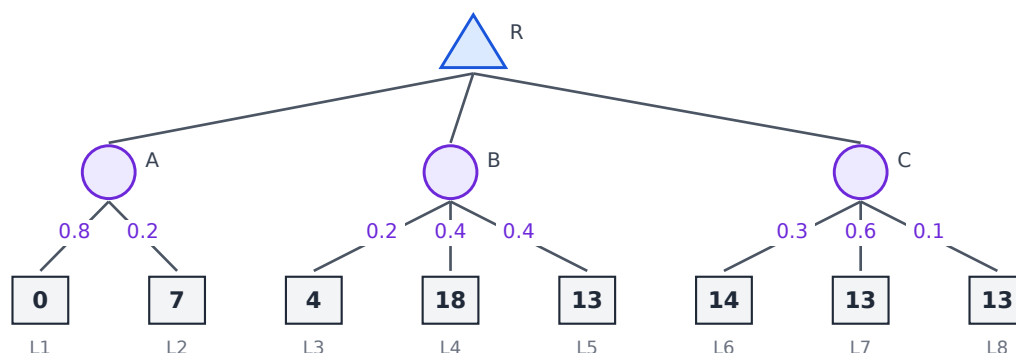
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) The edge K-A violates the consistency condition $h(K) \leq c(K,A) + h(A)$.
☐ (B) h is consistent (monotone).
☐ (C) h is admissible.
☐ (D) $h(J)$ overestimates the true cost from J to G.

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

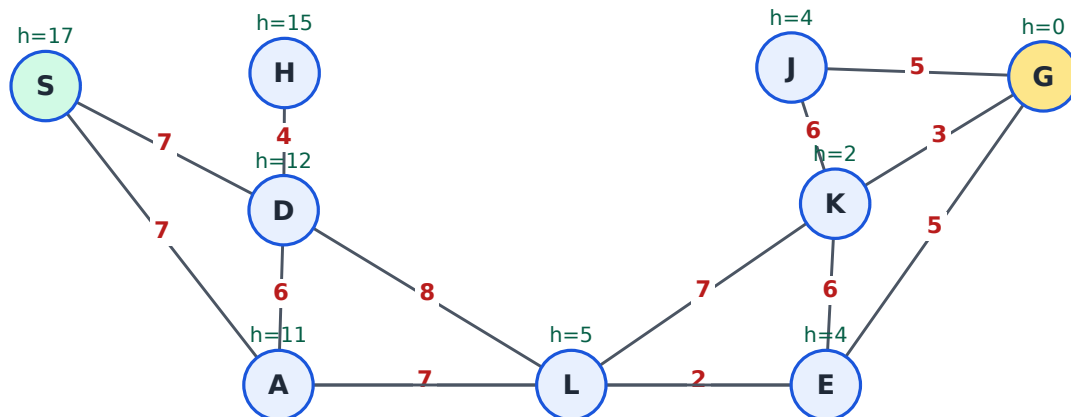
Q.13 [MSQ · 2 marks · no negative marking] Informed Search

Which of the following statements about IDA* are TRUE?

- ☐ (A) Its memory requirement is linear in the depth of the solution
☐ (B) With an admissible heuristic, it returns an optimal solution
☐ (C) It keeps every previously visited state in memory to avoid repeated states
☐ (D) When step costs are real-valued and almost all distinct, each iteration may add only one new node, which leads to many iterations

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Adversarial Search

Which of the following statements about depth-limited (cutoff) game search with heuristic evaluation functions are TRUE?

- ☐ (A) The horizon effect occurs when an unavoidable damaging event is pushed beyond the search depth by delaying moves
- ☐ (B) Quiescence search extends the search at non-quiet positions, such as pending captures, before applying the evaluation function
- ☐ (C) A good evaluation function should be strongly correlated with the actual chance of winning
- ☐ (D) Increasing the cutoff depth by a fixed amount always removes the horizon effect

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg(\neg R \wedge \neg Q)$
- (2) $(Q \leftrightarrow \neg R) \leftrightarrow (\neg Q \leftrightarrow P)$
- (3) $\neg(P \wedge \neg R)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $Q \wedge R$
- ☐ (B) $(\neg R \vee R) \vee P$
- ☐ (C) $(R \vee P) \leftrightarrow \neg R$
- ☐ (D) $\neg P \leftrightarrow (P \vee Q)$

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic

How many truth assignments to $x_1, \dots, x_5$ satisfy the 2-CNF $(x_1 \vee x_2) \wedge (x_2 \vee x_3) \wedge (x_3 \vee x_4) \wedge (x_4 \vee x_5)$?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = \neg Q \rightarrow R$?

- ☐ (A) $\neg Q \vee \neg R$
- ☐ (B) $\neg R \rightarrow \neg \neg Q$
- ☐ (C) $\neg \neg Q \rightarrow \neg R$
- ☐ (D) $\neg(\neg Q \wedge \neg R)$

Q.19 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

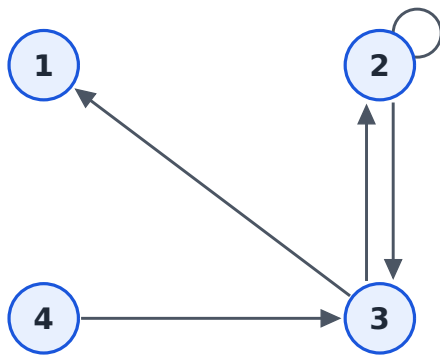
How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = (\neg R \leftrightarrow (P \wedge S)) \vee ((\neg Q \wedge P) \vee (P \vee R)) ?$$

Answer (numerical): _____

Q.20 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

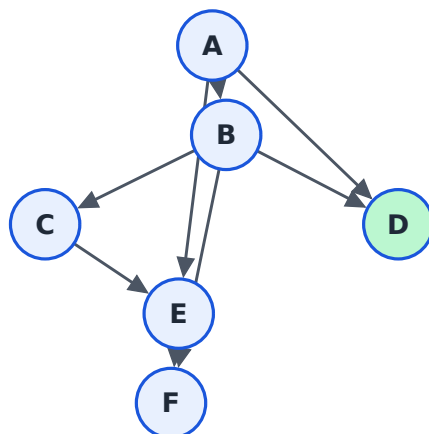
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists x \forall y R(x, y)$
- ☐ (B) $\exists x \forall y \neg R(y, x)$
- ☐ (C) $\exists y \forall x R(x, y)$
- ☐ (D) $\exists x R(x, x)$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

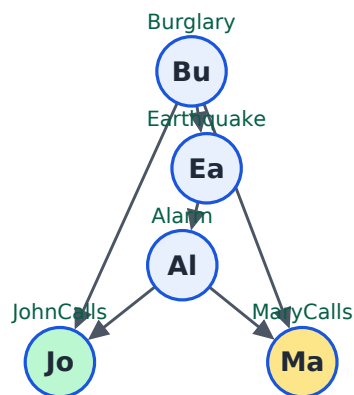
Variable elimination is used to compute $P(D)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **A, F, B, C, E**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls, Ma = MaryCalls.) Compute $P(\text{JohnCalls} = T \mid \text{MaryCalls} = T)$ (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.7

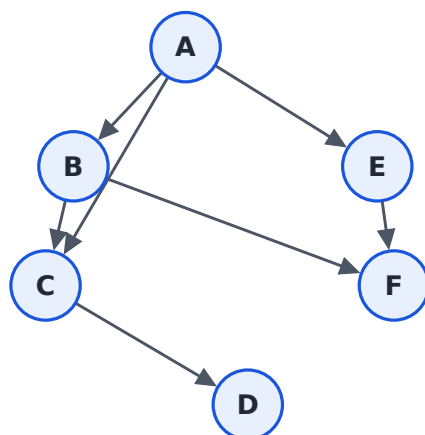
Bu	$P(\text{Ea}=T \dots)$	Ea	$P(\text{Al}=T \dots)$
T	0.1	T	0.2
F	0.6	F	0.7

Bu	Al	$P(\text{Jo}=T \dots)$	Bu	Al	$P(\text{Ma}=T \dots)$
T	T	0.4	T	T	0.95
T	F	0.95	T	F	0.5
F	T	0.8	F	T	0.05
F	F	0.4	F	F	0.8

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

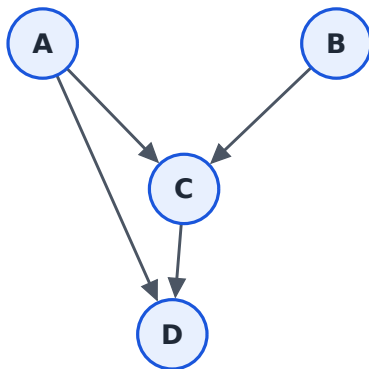
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp E$
☐ (B) $E \perp D \mid B, C$
☐ (C) $A \perp D \mid C$
☐ (D) $B \perp A \mid E$

Q.24 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.16}, \mathbf{0.68}, \mathbf{0.75}, \mathbf{0.44} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.95

P(B=T)	A	B	P(C=T ...)
0.8	T	T	0.2
	T	F	0.7
	F	T	0.5
	F	F	0.8

A	C	P(D=T ...)
T	T	0.6
T	F	0.8
F	T	0.8
F	F	0.5

- ☐ (A) A=T, B=T, C=F, D=T
☐ (B) A=T, B=F, C=F, D=T
☐ (C) A=F, B=T, C=F, D=T
☐ (D) A=F, B=F, C=F, D=F

Q.25 [MSQ · 2 marks · no negative marking] Variable Elimination

A Bayesian network has edges $A \rightarrow B$ and $A \rightarrow C$ over Boolean variables with $P(a) = 0.3$, $P(b \mid a) = 0.8$, $P(b \mid \neg a) = 0.4$, $P(c \mid a) = 0.5$, $P(c \mid \neg a) = 0.1$. $P(A \mid b, c)$ is computed by variable elimination, multiplying the evidence-instantiated factors $f_1(A) = P(A)$, $f_2(A) = P(b \mid A)$, $f_3(A) = P(c \mid A)$ into an unnormalized factor $f(A)$. Which are correct?

- ☐ (A) $f(a) = 0.12$
☐ (B) $f(\neg a) = 0.028$
☐ (C) $P(a \mid b, c) \approx 0.811$
☐ (D) $P(b, c) = 0.12$

Answer Key — Mock Test 092

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MSQ	1	Adversarial Search	A, B, C	14	NAT	2	Informed Search	21
2	NAT	1	Adversarial Search	8	15	MSQ	2	Adversarial Search	A, B, C
3	NAT	1	Adversarial Search	49	16	MSQ	2	Propositional Logic	B
4	MCQ	1	Uninformed Search	C	17	NAT	2	Propositional Logic	13
5	MCQ	1	Predicate Logic	A	18	MSQ	2	Propositional Logic	B, D
6	MCQ	1	Propositional Logic	A	19	NAT	2	Propositional Logic	12
7	NAT	1	Predicate Logic	4	20	MSQ	2	Predicate Logic	B, D
8	MCQ	1	Sampling	A	21	NAT	2	Exact Inference	4
9	NAT	1	Bayesian Networks	10	22	NAT	2	Exact Inference	0.497 to 0.501
10	NAT	1	Bayesian Networks	91	23	MSQ	2	Conditional Independence	B, C
11	MSQ	2	Informed Search	A, C	24	MCQ	2	Sampling	A
12	NAT	2	Adversarial Search	13.29 to 13.31	25	MSQ	2	Variable Elimination	A, B, C
13	MSQ	2	Informed Search	A, B, D					

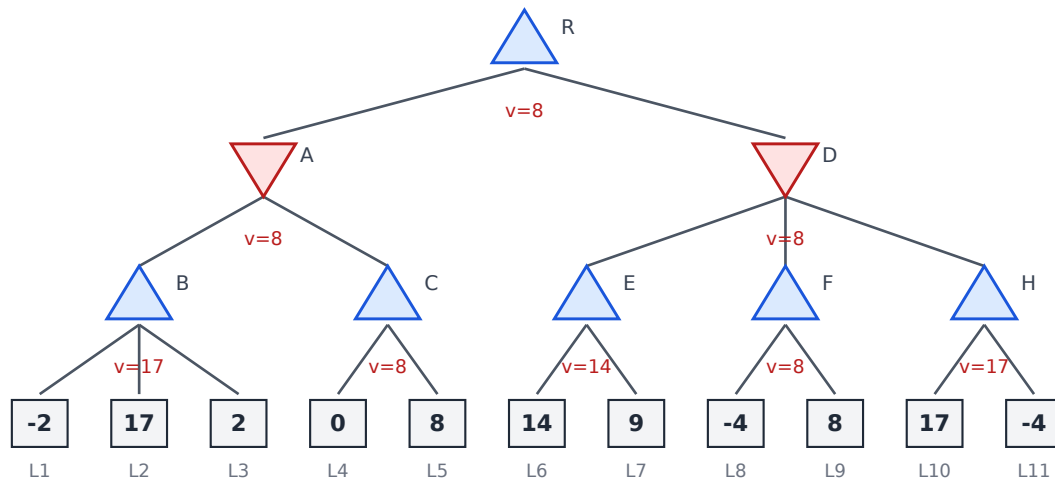
Detailed Solutions

Q.1 [MSQ, 1M · Adversarial Search] Answer: A, B, C

(A) is the definition of zero-sum (constant-sum). (B) is true: AIMA notes that chess's payoffs always total a constant. (C) is true: minimax computes the best play against a perfect, adversarial opponent. (D) is false: since the sum is constant, one player's gain is exactly the other's loss.

Q.2 [NAT, 1M · Adversarial Search] Answer: 8

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 17, C = 8, A = 8, E = 14, F = 8, H = 17, D = 8. Root value = **8**; MAX's optimal first move is towards A, D.

Q.3 [NAT, 1M · Adversarial Search] Answer: 49

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 5^2 + 5^2 - 1 = 25 + 25 - 1 = 49$, versus 625 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.4 [MCQ, 1M · Uninformed Search] Answer: C (C: S, E, F, I, A, H, B, L, G)

Adjacency lists (alphabetical): A: {E, G, H, L}; B: {H, I, L}; E: {A, F, H, S}; F: {E, H, I, S}; G: {A}; H: {A, B, E, F}; I: {B, F, L, S}; L: {A, B, I}; S: {E, F, I}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[E, F, I]
2	E	[F, I, A, H]
3	F	[I, A, H]
4	I	[A, H, B, L]
5	A	[H, B, L, G]
6	H	[B, L, G]
7	B	[L, G]
8	L	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, E, F, I, A, H, B, L, G**. The path returned is $S \rightarrow E \rightarrow A \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x (F(x) \rightarrow \neg P(x))$)

'None of my friends are perfect' $\equiv \neg \exists x (F(x) \wedge P(x)) \equiv \forall x \neg (F(x) \wedge P(x)) \equiv \forall x (F(x) \rightarrow \neg P(x))$. Options B and C (which are equivalent to each other) only say some friend is not perfect. Option D speaks about non-friends.

Q.6 [MCQ, 1M · Propositional Logic] Answer: A (A: $\alpha \wedge \neg \beta$ is unsatisfiable)

$\alpha \models \beta$ means every model of α is a model of β , i.e. no model makes α true and β false, i.e. $\alpha \wedge \neg \beta$ is unsatisfiable. Equivalently $\alpha \rightarrow \beta$ is valid. Satisfiability of $\neg \alpha \vee \beta$ or $\alpha \wedge \beta$ is too weak (one model is enough). $\alpha \vee \neg \beta$ being valid would mean $\beta \models \alpha$, the reverse direction.

Q.7 [NAT, 1M · Predicate Logic] Answer: 4

Interpret P and Q as subsets of the domain; P and Q must coincide $\Rightarrow 2^n$. With $n = 2$ this gives **4** (verified by enumerating all interpretations).

Q.8 [MCQ, 1M · Sampling] Answer: A (A: (c, s, $\neg r$, w))

C: $0.42 < P(c) = 0.5 \rightarrow c$. S: $P(s | c) = 0.1$, $0.07 < 0.1 \rightarrow s$. R: $P(r | c) = 0.8$, $0.85 \geq 0.8 \rightarrow \neg r$. W: $P(w | s, \neg r) = 0.90$, $0.30 < 0.90 \rightarrow w$. Sample = (c, s, $\neg r$, w). (B) mis-samples S and R; (C) mis-samples R; (D) mis-samples C.

Q.9 [NAT, 1M · Bayesian Networks] Answer: 10

A Boolean node with k Boolean parents needs 2^k numbers. A: 1, B: 1, C (2 parents): 4, D (1 parent): 2, E (1 parent): 2. Total = 10, compared with $2^5 - 1 = 31$ for the full joint.

Q.10 [NAT, 1M · Bayesian Networks] Answer: 91

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (4-1) + B: (3-1) \times 4 + C: (4-1) \times 4 + D: (2-1) \times 4 \times 3 + E: (3-1) \times 4 + F: (3-1) \times 4 \times 2 \times 3 = 91$.

Full joint table: $\prod |X| - 1 = 864 - 1 = 863$; saving = 772.

Q.11 [MSQ, 2M · Informed Search] Answer: A, C (A: The edge K-A violates the consistency condition $h(K) \leq c(K,A) + h(A)$.; C: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	E	G	I	J	K	S
$h(n)$	4	14	15	0	9	3	15	19
$h^*(n)$	13	16	17	0	10	4	17	22
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $I \rightarrow A$ ($9 > 3 + 4$), $K \rightarrow A$ ($15 > 4 + 4$), $B \rightarrow A$ ($14 > 3 + 4$), $E \rightarrow A$ ($15 > 8 + 4$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [NAT, 2M · Adversarial Search] Answer: 13.29 to 13.31

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.8 \times 0 + 0.2 \times 7 = 1.4$
- $B = 0.2 \times 4 + 0.4 \times 18 + 0.4 \times 13 = 13.2$
- $C = 0.3 \times 14 + 0.6 \times 13 + 0.1 \times 13 = 13.3$

Root = $\max(1.4, 13.2, 13.3) = 13.3$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.13 [MSQ, 2M · Informed Search] Answer: A, B, D

(A) is true: each iteration is a depth-first search bounded by the f-cutoff. (B) is true: thresholds rise to the smallest f that exceeded the old one, so the first goal found is optimal. (C) is false: IDA* does not keep an explored set, which is exactly why it uses little memory and why it can re-expand states. (D) is true: this is the well-known weakness, which motivates RBFS and SMA*.

Q.14 [NAT, 2M · Informed Search] Answer: 21

Heuristic values: $h(A)=11$, $h(D)=12$, $h(E)=4$, $h(G)=0$, $h(H)=15$, $h(J)=4$, $h(K)=2$, $h(L)=5$, $h(S)=17$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (17)	A(18), D(19)
2	A (18)	D(19), L(19), E(20)
3	D (19)	L(19), E(20), H(26)
4	L (19)	E(20), K(23), H(26)
5	E (20)	G(21), K(23), H(26)
6	G (21)	K(23), H(26) $\leftarrow$ goal, stop

Path returned: **S $\rightarrow$ A $\rightarrow$ E $\rightarrow$ G** with cost **21**; nodes expanded = **6** (S, A, D, L, E, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 21$. For comparison UCS ($h = 0$) would expand 7 nodes.

Q.15 [MSQ, 2M · Adversarial Search] Answer: A, B, C

(A) defines the horizon effect. (B) defines quiescence search, which applies the evaluation function only at quiet positions. (C) is the AIMA requirement for evaluation functions. (D) is false: at any fixed depth, a delaying tactic can push the bad event just beyond the new horizon.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B (B: $(\neg R \vee R) \vee P$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 3 of them:

P	Q	R	KB
T	T	T	T
T	F	T	T
F	T	F	T

- (A) $Q \wedge R$ is false in the KB-model $P=T, Q=F, R=T \Rightarrow$ not entailed.
 (B) $(\neg R \vee R) \vee P$ is true in all models of KB $\Rightarrow$ **entailed**.
 (C) $(R \vee P) \leftrightarrow \neg R$ is false in the KB-model $P=T, Q=T, R=T \Rightarrow$ not entailed.
 (D) $\neg P \leftrightarrow (P \vee Q)$ is false in the KB-model $P=T, Q=T, R=T \Rightarrow$ not entailed.

Q.17 [NAT, 2M · Propositional Logic] Answer: 13

The formula says no two consecutive variables are both F. Let a_n be the number of such strings of length n . A valid string either ends in T (preceded by any valid string of length $n-1$) or ends in F, in which case the previous symbol must be T (preceded by any valid string of length $n-2$). So $a_n = a_{n-1} + a_{n-2}$ with $a_1 = 2, a_2 = 3$. Then $a_3 = 5, a_4 = 8, a_5 = 13$. Enumeration of all 32 assignments confirms 13.

Q.18 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $\neg R \rightarrow \neg\neg Q$; D: $\neg(\neg Q \wedge \neg R)$)

Compare truth tables with $F = \neg Q \rightarrow R$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

- (A) $\neg Q \vee \neg R$ — differs at $Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.
 (B) $\neg R \rightarrow \neg\neg Q$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.
 (C) $\neg\neg Q \rightarrow \neg R$ — differs at $Q=T, R=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.
 (D) $\neg(\neg Q \wedge \neg R)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.19 [NAT, 2M · Propositional Logic] Answer: 12

Build the truth table (sub-formulas shown first):

P	Q	R	S	$\neg R \leftrightarrow (P \wedge S)$	$(\neg Q \wedge P) \vee (P \vee R)$	$(\neg R \leftrightarrow (P \wedge S)) \vee ((\neg Q \wedge P) \vee (P \vee R))$
T	T	T	T	F	T	T
T	T	T	F	T	T	T
T	T	F	T	T	T	T
T	T	F	F	F	T	T
T	F	T	T	F	T	T
T	F	T	F	T	T	T
T	F	F	T	T	T	T
T	F	F	F	F	T	T
F	T	T	T	T	T	T
F	T	T	F	T	T	T
F	T	F	T	F	F	F
F	T	F	F	F	F	F
F	F	T	T	T	T	T
F	F	T	F	T	T	T
F	F	F	T	F	F	F
F	F	F	F	F	F	F

Rows where F = T: **12**.

Q.20 [MSQ, 2M · Predicate Logic] **Answer: B, D** (B: $\exists x \forall y \neg R(y, x)$; D: $\exists x R(x, x)$)

$R = \{(2,2), (2,3), (3,1), (3,2), (4,3)\}$.

(A) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).

(B) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.21 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(A, C, E)$, $f(B, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
A	$f(A)$, $f(A, B)$, $f(A, B, D)$, $f(A, C, E)$	$\{A, B, C, D, E\}$	$f(B, C, D, E)$
F	$f(B, E, F)$	$\{B, E, F\}$	$f(B, E)$
B	$f(B, C)$, $f(B, C, D, E)$, $f(B, E)$	$\{B, C, D, E\}$	$f(C, D, E)$
C	$f(C, D, E)$	$\{C, D, E\}$	$f(D, E)$
E	$f(D, E)$	$\{D, E\}$	$f(D)$

The largest factor produced has **4** variable(s) (created when eliminating A); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over D and normalise.

Q.22 [NAT, 2M · Exact Inference] **Answer: 0.497 to 0.501**

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm} \mid \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Alarm}) \cdot P(\text{MaryCalls} \mid \text{Burglary}, \text{Alarm})$.

Inference by enumeration: $P(\text{JohnCalls} \mid e) = \alpha \sum_{\text{hidden}} P(\text{JohnCalls}, e, \text{hidden})$.

Hidden: Bu, Ea, Al	Jo	product of CPT entries (Bu · Ea · Al · Jo · Ma)	value
T, T, T	T	$0.7 \times 0.1 \times 0.2 \times 0.4 \times 0.95$	0.00532
T, T, F	T	$0.7 \times 0.1 \times 0.8 \times 0.95 \times 0.5$	0.0266
T, F, T	T	$0.7 \times 0.9 \times 0.7 \times 0.4 \times 0.95$	0.16758
T, F, F	T	$0.7 \times 0.9 \times 0.3 \times 0.95 \times 0.5$	0.08978
F, T, T	T	$0.3 \times 0.6 \times 0.2 \times 0.8 \times 0.05$	0.00144
F, T, F	T	$0.3 \times 0.6 \times 0.8 \times 0.4 \times 0.8$	0.04608
F, F, T	T	$0.3 \times 0.4 \times 0.7 \times 0.8 \times 0.05$	0.00336
F, F, F	T	$0.3 \times 0.4 \times 0.3 \times 0.4 \times 0.8$	0.01152
T, T, T	F	$0.7 \times 0.1 \times 0.2 \times 0.6 \times 0.95$	0.00798
T, T, F	F	$0.7 \times 0.1 \times 0.8 \times 0.05 \times 0.5$	0.0014
T, F, T	F	$0.7 \times 0.9 \times 0.7 \times 0.6 \times 0.95$	0.25137
T, F, F	F	$0.7 \times 0.9 \times 0.3 \times 0.05 \times 0.5$	0.00473
F, T, T	F	$0.3 \times 0.6 \times 0.2 \times 0.2 \times 0.05$	0.00036
F, T, F	F	$0.3 \times 0.6 \times 0.8 \times 0.6 \times 0.8$	0.06912
F, F, T	F	$0.3 \times 0.4 \times 0.7 \times 0.2 \times 0.05$	0.00084
F, F, F	F	$0.3 \times 0.4 \times 0.3 \times 0.6 \times 0.8$	0.01728

$P(\text{JohnCalls}=T, e) = 0.35168$; $P(\text{JohnCalls}=F, e) = 0.35307$; $P(e) = 0.70475$.

$P(\text{JohnCalls}=T \mid e) = 0.35168 / 0.70475 = \mathbf{0.499} \approx 0.499$.

Q.23 [MSQ, 2M · Conditional Independence] **Answer: B, C** (B: $E \perp D \mid B, C$; C: $A \perp D \mid C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp E$: **FALSE** — the path $B \leftarrow A \rightarrow E$ is active given $\{\}$.

(B) $E \perp D \mid B, C$: **TRUE** — $E \leftarrow A \rightarrow B \rightarrow C \rightarrow D$ (blocked: observed chain node B); $E \leftarrow A \rightarrow C \rightarrow D$ (blocked: observed chain node C); $E \rightarrow F \leftarrow B \leftarrow A \rightarrow C \rightarrow D$ (blocked: collider F with neither it nor a descendant observed); $E \rightarrow F \leftarrow B \rightarrow C \rightarrow D$ (blocked: collider F with neither it nor a descendant observed).

(C) $A \perp D \mid C$: **TRUE** — $A \rightarrow B \rightarrow C \rightarrow D$ (blocked: observed chain node C); $A \rightarrow C \rightarrow D$ (blocked: observed chain node C); $A \rightarrow E \rightarrow F \leftarrow B \rightarrow C \rightarrow D$ (blocked: collider F with neither it nor a descendant observed).

(D) $B \perp A \mid E$: **FALSE** — the path $B \leftarrow A$ is active given $\{E\}$.

Q.24 [MCQ, 2M · Sampling] **Answer: A** (A: A=T, B=T, C=F, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.95$; $u = 0.16 < 0.95 \Rightarrow A = T$
- B: $P(B=T \mid \text{—}) = 0.8$; $u = 0.68 < 0.8 \Rightarrow B = T$
- C: $P(C=T \mid A=T, B=T) = 0.2$; $u = 0.75 \geq 0.2 \Rightarrow C = F$
- D: $P(D=T \mid A=T, C=F) = 0.8$; $u = 0.44 < 0.8 \Rightarrow D = T$

Sample: **(A=T, B=T, C=F, D=T)**.

Q.25 [MSQ, 2M · Variable Elimination] **Answer: A, B, C** (A: $f(a) = 0.12$; B: $f(\neg a) = 0.028$; C: $P(a \mid b, c) \approx 0.811$)

$f(a) = 0.3 \times 0.8 \times 0.5 = 0.12$ (A true). $f(\neg a) = 0.7 \times 0.4 \times 0.1 = 0.028$ (B true). Normalizing, $P(a \mid b, c) = 0.12 / (0.12 + 0.028) = 0.12 / 0.148 \approx 0.811$ (C true). The normalizer is $P(b, c) = f(a) + f(\neg a) = 0.148$, not 0.12 (which is $P(a, b, c)$), so (D) is false.

Mock Test 093

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 4 / 16	11 / 16 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Depth-limited search with limit $l = 5$ is run on a tree with branching factor $b = 3$. No goal exists within depth 5. How many nodes (including the root) does it generate before returning cutoff?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Problem Formulation

In a search problem, the state space is defined implicitly by

- ☐ (A) the initial state, the actions, and the transition model
- ☐ (B) the goal test only
- ☐ (C) the heuristic function
- ☐ (D) the path cost function only

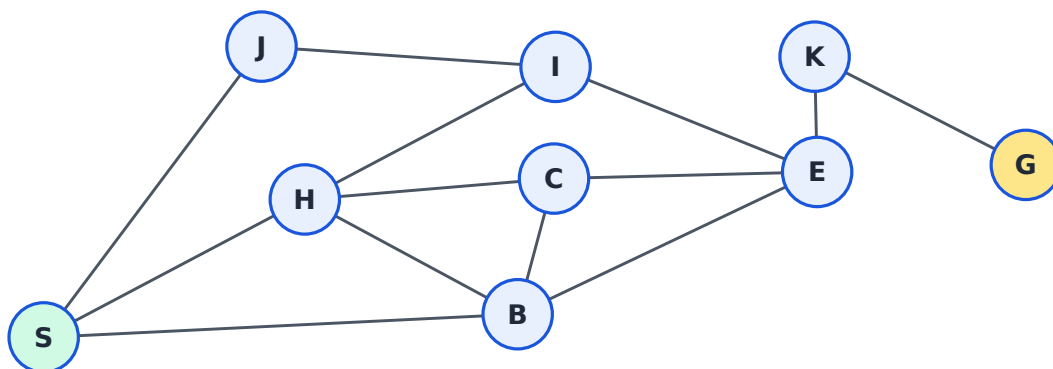
Q.3 [NAT · 1 mark · no negative marking] Adversarial Search

A two-ply game tree has a MAX root with four MIN children. Their leaf values, left to right, are: MIN_1 : (10, 9, 14); MIN_2 : (11, 8, 20); MIN_3 : (2, 15, 30); MIN_4 : (12, 13, 7). What is the minimax value of the root?

Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

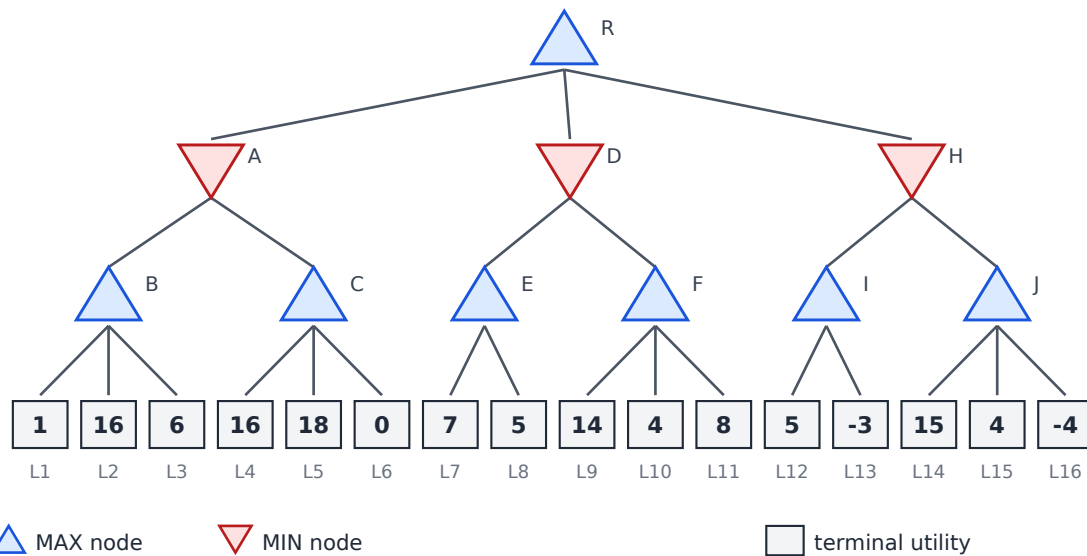
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, B, H, J, E, C, I, K, G
- ☐ (B) S, B, H, J, C, E, I, K, G
- ☐ (C) S, J, H, B, I, C, E, K, G
- ☐ (D) S, B, C, E, I, H, J, K, G

Q.5 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic

Let the domain be $D = \{1, 2, 3\}$. How many interpretations of a binary predicate R over D satisfy $\forall x \forall y (R(x, y) \rightarrow R(y, x))$?

Answer (numerical): _____

Q.7 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Over the finite domain $\{a, b\}$, the sentence $\forall x \exists y R(x, y)$ is equivalent to which ground formula?

- ☐ (A) $(R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$
☐ (B) $(R(a, a) \wedge R(a, b)) \vee (R(b, a) \wedge R(b, b))$
☐ (C) $(R(a, a) \vee R(b, a)) \wedge (R(a, b) \vee R(b, b))$
☐ (D) $R(a, a) \wedge R(b, b)$

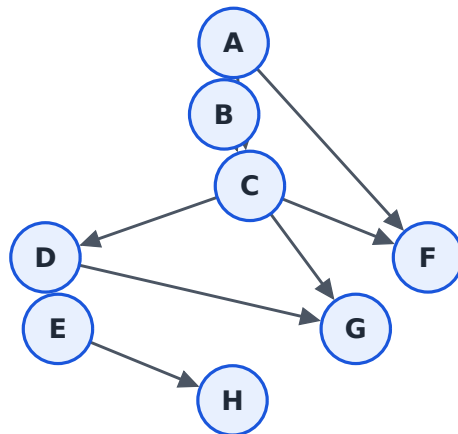
Q.8 [NAT · 1 mark · no negative marking] Sampling

To estimate $P(\text{Rain} \mid \text{sprinkler})$ by rejection sampling, 500 samples are generated by prior sampling. 140 of them have Sprinkler = true, and of these 35 have Rain = true. What is the rejection-sampling estimate of $P(\text{rain} \mid \text{sprinkler})$? (Round to 2 decimal places.)

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **B** in the Bayesian network below?



Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Probability for AI

While answering a query $P(\text{Cavity} \mid \text{toothache})$, the unnormalized vector obtained for $\text{Cavity} = \langle \text{true}, \text{false} \rangle$ is $\langle 0.12, 0.08 \rangle$. Which option gives the normalization constant α and the resulting posterior distribution?

- ☐ (A) $\alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.6, 0.4 \rangle$
- ☐ (B) $\alpha = 0.2$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.024, 0.016 \rangle$
- ☐ (C) $\alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.4, 0.6 \rangle$
- ☐ (D) $\alpha = 2$, $P(\text{Cavity} \mid \text{toothache}) = \langle 0.24, 0.16 \rangle$

Section B — 2-mark questions

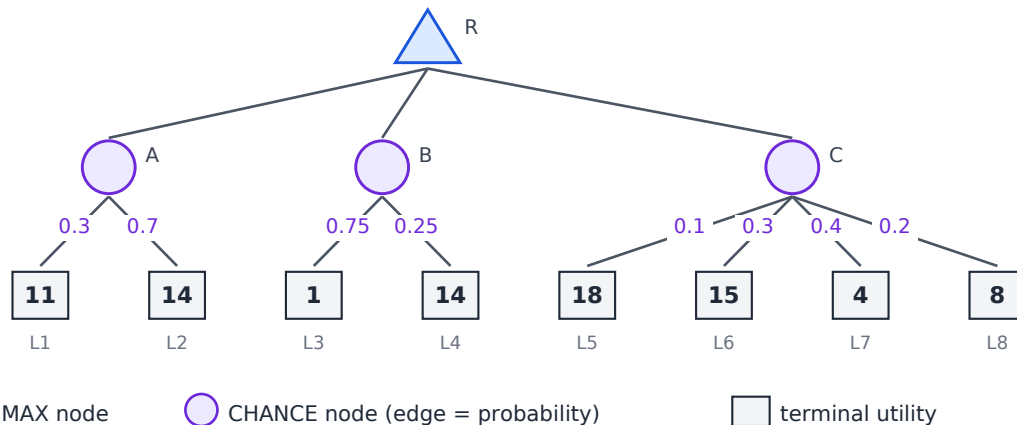
Q.11 [NAT · 2 marks · no negative marking] Uninformed Search

Iterative deepening search (IDS) runs on a uniform tree with $b = 10$ and the shallowest goal at $d = 5$. Assume the worst case: the final iteration (limit 5) generates every node at depth 5. Each time a non-root node is generated, it counts once, so a node regenerated in a later iteration counts again. The root is not counted. What is the total number of node generations?

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)

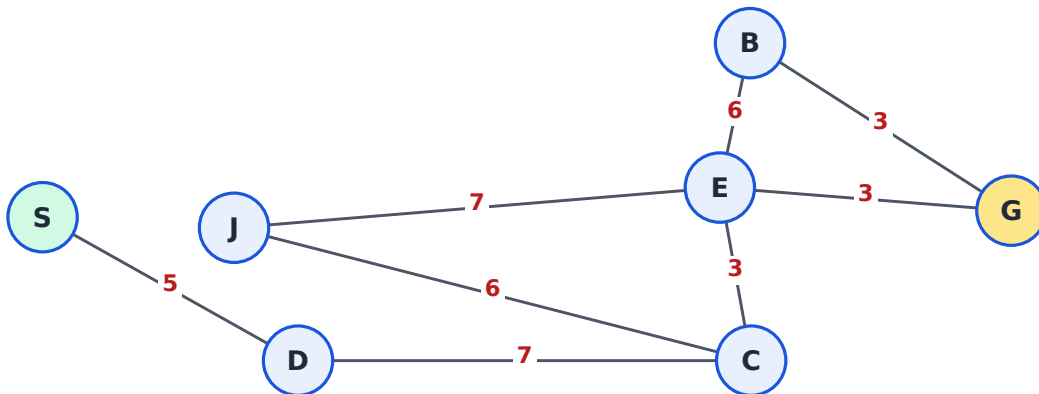


△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. What is the cost of the path returned?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

C1: $(R \vee S)$, C2: $(P \vee S)$, C3: $(Q \vee \neg S)$, C4: $(Q \vee S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg P \vee R)$
- ☐ (B) $(Q \vee R)$
- ☐ (C) $(P \vee Q)$
- ☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic

Of the 16 binary connectives $\circ$, for how many is the formula $(p \circ q) \rightarrow p$ a tautology?

Answer (numerical): _____

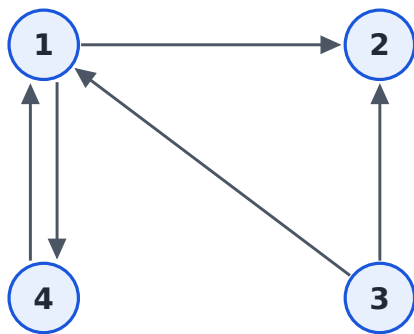
Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = Q \leftrightarrow (\neg R \leftrightarrow \neg P)$?

- ☐ (A) $(Q \wedge (\neg R \leftrightarrow \neg P)) \vee (Q \wedge \neg(\neg R \leftrightarrow \neg P))$
☐ (B) $(Q \wedge (\neg R \leftrightarrow \neg P)) \vee (\neg Q \wedge \neg(\neg R \leftrightarrow \neg P))$
☐ (C) $\neg Q \leftrightarrow \neg(\neg R \leftrightarrow \neg P)$
☐ (D) $(Q \rightarrow (\neg R \leftrightarrow \neg P)) \wedge ((\neg R \leftrightarrow \neg P) \rightarrow Q)$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\exists x R(x, x)$
☐ (C) $\exists x \forall y \neg R(x, y)$
☐ (D) $\forall y \exists x R(x, y)$

Q.18 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = (R \wedge (Q \leftrightarrow \neg P)) \wedge (S \vee (R \rightarrow \neg R)) ?$$

Answer (numerical): _____

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms $A, B, C, D, E, F, H, J, K$ has facts **E, B, A** and rules:

- R1: $F \wedge A \rightarrow D$
 R2: $B \wedge A \rightarrow F$
 R3: $H \wedge B \wedge D \rightarrow K$
 R4: $J \rightarrow B$
 R5: $F \rightarrow H$
 R6: $H \wedge F \wedge B \rightarrow E$

Using forward chaining, which of the following is/are TRUE?

- ☐ (A) C is entailed
☐ (B) D is entailed
☐ (C) H is entailed
☐ (D) K is entailed

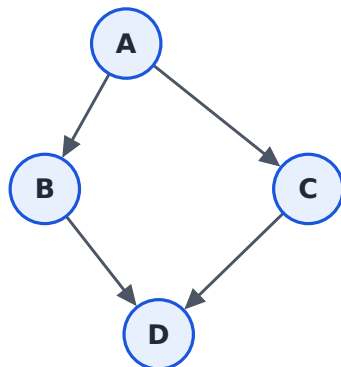
Q.20 [NAT · 2 marks · no negative marking] Predicate Logic

How many binary relations on $\{1, 2, 3\}$ satisfy all three of $\forall x R(x, x)$, $\forall x \forall y (R(x, y) \rightarrow R(y, x))$ and $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$?

Answer (numerical): _____

Q.21 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.91, 0.29, 0.93, 0.2)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.3

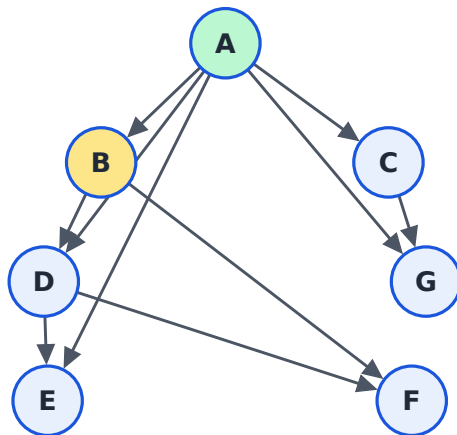
A	P(B=T ...)	A	P(C=T ...)
T	0.25	T	0.25
F	0.1	F	0.8

B	C	P(D=T ...)
T	T	0.25
T	F	0.5
F	T	0.8
F	F	0.75

- ☐ (A) A=T, B=F, C=F, D=T
☐ (B) A=T, B=T, C=T, D=F
☐ (C) A=F, B=T, C=F, D=T
☐ (D) A=F, B=F, C=F, D=T

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

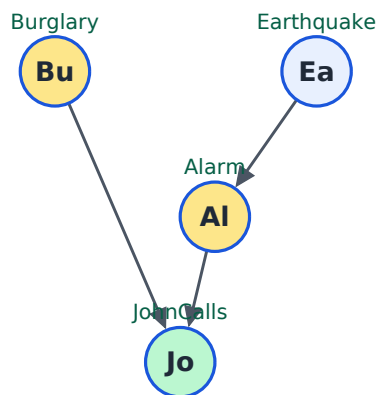
Variable elimination is used to compute $\mathbf{P(A \mid B = T)}$ in the Bayesian network below (Boolean variables; evidence $B = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F, D, E, C, G**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Bu = Burglary, Ea = Earthquake, Al = Alarm, Jo = JohnCalls.) Compute $P(\text{JohnCalls} = T \mid \text{Alarm} = T, \text{Burglary} = T)$ (round off to 3 decimal places).



$P(\text{Bu}=T)$
0.2

$P(\text{Ea}=T)$	Ea	$P(\text{Al}=T ...)$
0.7	T	0.2
	F	0.25

Bu	Al	$P(\text{Jo}=T ...)$
T	T	0.9
T	F	0.75
F	T	0.8
F	F	0.95

Answer (numerical): _____

Q.24 [NAT · 2 marks · no negative marking] Bayesian Networks

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a \mid b, e) = 0.95$, $P(a \mid b, \neg e) = 0.90$, $P(a \mid \neg b, e) = 0.30$, $P(a \mid \neg b, \neg e) = 0.01$; $P(j \mid a) = 0.8$, $P(j \mid \neg a) = 0.05$; $P(m \mid a) = 0.7$, $P(m \mid \neg a) = 0.02$. Suppose a new network is built for the same distribution by adding nodes in the order M, J, A, B, E, choosing for each node a minimal set of parents among the previously added nodes such that the node is conditionally independent of its other predecessors given its parents. How many independent CPT parameters does this new network have (all variables Boolean)? Assume no independences beyond those implied by the original structure.

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 15 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D, E. Using **rejection sampling**, estimate $P(C = T \mid E = F)$ (round off to 2 decimal places).

#	A	B	C	D	E
1	T	T	T	T	F
2	T	F	F	T	T
3	T	T	T	F	T
4	F	F	F	F	F
5	F	F	T	F	T
6	T	T	F	F	T
7	T	T	T	F	F
8	F	F	T	F	T
9	T	T	T	F	T
10	F	T	T	F	F
11	T	T	F	T	T
12	T	T	F	F	T
13	T	F	T	F	F
14	T	F	F	T	F
15	T	F	F	T	T

Answer (numerical): _____

Answer Key — Mock Test 093

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Uninformed Search	364	14	MSQ	2	Propositional Logic	B, C
2	MCQ	1	Problem Formulation	A	15	NAT	2	Propositional Logic	4
3	NAT	1	Adversarial Search	9	16	MSQ	2	Propositional Logic	B, C, D
4	MCQ	1	Uninformed Search	B	17	MSQ	2	Predicate Logic	C
5	NAT	1	Adversarial Search	16	18	NAT	2	Propositional Logic	2
6	NAT	1	Predicate Logic	64	19	MSQ	2	Propositional Logic	B, C, D
7	MCQ	1	Predicate Logic	A	20	NAT	2	Predicate Logic	5
8	NAT	1	Sampling	0.245 to 0.255	21	MCQ	2	Sampling	D
9	NAT	1	Conditional Independence	2	22	NAT	2	Exact Inference	2
10	MCQ	1	Probability for AI	A	23	NAT	2	Exact Inference	0.898 to 0.902
11	NAT	2	Uninformed Search	123450	24	NAT	2	Bayesian Networks	13
12	NAT	2	Adversarial Search	13.09 to 13.11	25	NAT	2	Sampling	0.66 to 0.68
13	NAT	2	Uninformed Search	18					

Detailed Solutions

Q.1 [NAT, 1M · Uninformed Search] Answer: 364

DLS explores the complete tree down to depth l : $1 + b + \dots + b^l = 1 + 3 + 9 + 27 + 81 + 243 = 364 = (b^{(l+1)} - 1)/(b - 1)$.

Q.2 [MCQ, 1M · Problem Formulation] Answer: A (A: the initial state, the actions, and the transition model)

Starting from the **initial state** and applying the **actions** through the **transition model** generates every reachable state and the edges between them, which is the state-space graph. The goal test (B) only identifies goal states. The heuristic (C) guides search but does not define states. The path cost (D) only labels edges.

Q.3 [NAT, 1M · Adversarial Search] Answer: 9

Each MIN node takes the minimum of its leaves: $\text{MIN}_1 = 9$, $\text{MIN}_2 = 8$, $\text{MIN}_3 = 2$, $\text{MIN}_4 = 7$. The root takes the maximum: $\max(9, 8, 2, 7) = 9$, so MAX plays the first move.

Q.4 [MCQ, 1M · Uninformed Search] Answer: B (B: S, B, H, J, C, E, I, K, G)

Adjacency lists (alphabetical): B: {C, E, H, S}; C: {B, E, H}; E: {B, C, I, K}; G: {K}; H: {B, C, I, S}; I: {E, H, J}; J: {I, S}; K: {E, G}; S: {B, H, J}.

BFS trace (queue shown front $\rightarrow$ back):

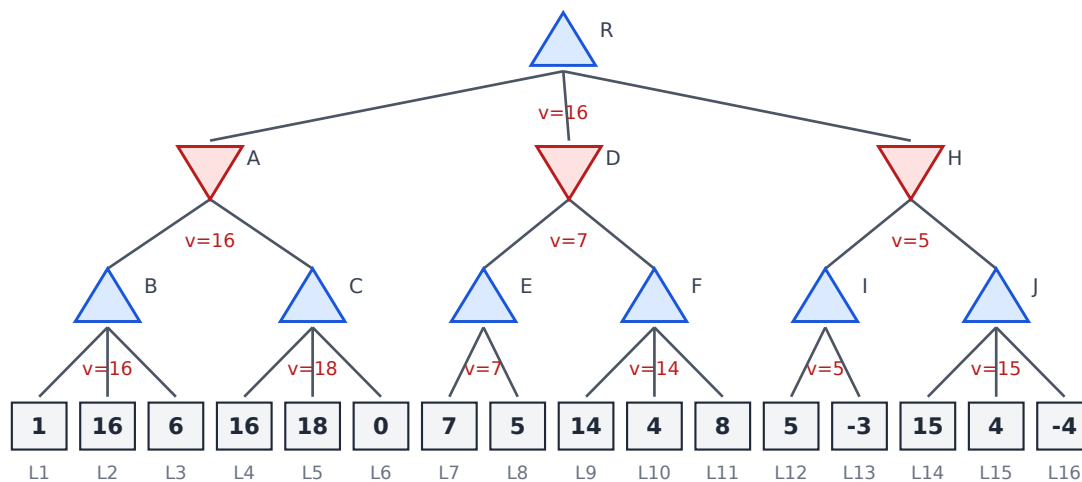
Step	Expanded	Frontier (FIFO) after expansion
1	S	[B, H, J]
2	B	[H, J, C, E]
3	H	[J, C, E, I]
4	J	[C, E, I]
5	C	[E, I]
6	E	[I, K]
7	I	[K]
8	K	[G]
9	G	[] $\leftarrow$ goal popped, stop

Expansion order: **S, B, H, J, C, E, I, K, G**. The path returned is $S \rightarrow B \rightarrow E \rightarrow K \rightarrow G$ with 4 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.5 [NAT, 1M · Adversarial Search] Answer: 16

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 16$, $C = 18$, $A = 16$, $E = 7$, $F = 14$, $D = 7$, $I = 5$, $J = 15$, $H = 5$. Root value = **16**; MAX's optimal first move is towards A.

Q.6 [NAT, 1M · Predicate Logic] **Answer: 64**

An interpretation of R is a subset of $D \times D$ (9 pairs). The sentence says R is symmetric. The 3 diagonal pairs (x, x) can be chosen freely; the 3 unordered pairs $\{x, y\}$ with $x \neq y$ must be included or excluded as a whole. That is $2^3 \times 2^3 = 64$ relations.

Q.7 [MCQ, 1M · Predicate Logic] **Answer: A** ($A: (R(a, a) \vee R(a, b)) \wedge (R(b, a) \vee R(b, b))$)

Over a finite domain with every element named, $\forall$ becomes a conjunction over x and $\exists$ a disjunction over y . For $x = a$: $R(a, a) \vee R(a, b)$; for $x = b$: $R(b, a) \vee R(b, b)$. Their conjunction is option A. Option B is $\exists x \forall y R(x, y)$. Option C is $\forall y \exists x R(x, y)$ (every element has an R -predecessor). Option D is reflexivity.

Q.8 [NAT, 1M · Sampling] **Answer: 0.245 to 0.255**

Rejection sampling discards the $500 - 140 = 360$ samples inconsistent with the evidence. Among the 140 retained samples, the fraction with $\text{Rain} = \text{true}$ is $35/140 = 0.25$. Dividing by 500 would be wrong because it estimates $P(\text{rain}, \text{sprinkler})$, not the conditional.

Q.9 [NAT, 1M · Conditional Independence] **Answer: 2**

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of B: $\{A\}$; children: $\{C\}$; co-parents: $\{\emptyset\}$.

$MB(B) = \{A, C\}$ — **2** nodes. Given its Markov blanket, B is conditionally independent of every other node in the network.

Q.10 [MCQ, 1M · Probability for AI] **Answer: A** ($A: \alpha = 5$, $P(\text{Cavity} \mid \text{toothache}) = \{0.6, 0.4\}$)

The entries must sum to 1 after scaling, so $\alpha = 1/(0.12 + 0.08) = 1/0.20 = 5$. Then $\alpha(0.12, 0.08) = \{0.6, 0.4\}$. $\alpha = 0.2$ is the sum itself (multiplying by it does not normalize), $\{0.4, 0.6\}$ swaps the order of the values, and $\alpha = 2$ gives entries summing to 0.4, not 1.

Q.11 [NAT, 2M · Uninformed Search] **Answer: 123450**

Nodes at depth i are generated once in each iteration with $\text{limit} \geq i$, so they are generated $(d - i + 1)$ times. $N(\text{IDS}) = d \cdot b + (d-1) \cdot b^2 + \dots + 1 \cdot b^d = 5 \cdot 10 + 4 \cdot 100 + 3 \cdot 1000 + 2 \cdot 10000 + 1 \cdot 100000 = 50 + 400 + 3000 + 20000 + 100000 = \mathbf{123450}$. BFS generates 111110 nodes on the same problem, so the overhead of repeating the shallow levels is only about 11%.

Q.12 [NAT, 2M · Adversarial Search] **Answer: 13.09 to 13.11**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.3 \times 11 + 0.7 \times 14 = 13.1$
- $B = 0.75 \times 1 + 0.25 \times 14 = 4.25$
- $C = 0.1 \times 18 + 0.3 \times 15 + 0.4 \times 4 + 0.2 \times 8 = 9.5$

Root = $\max(13.1, 4.25, 9.5) = \mathbf{13.1}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.13 [NAT, 2M · Uninformed Search] Answer: 18

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	D(5)
2	D (5)	C(12)
3	C (12)	E(15), J(18)
4	E (15)	G(18), J(18), B(21)
5	G (18)	J(18), B(21) ← goal, stop

Path returned: **S** → **D** → **C** → **E** → **G**, cost = 5 + 7 + 3 + 3 = **18**. Nodes expanded (including S and G): **5** (S, D, C, E, G).

Q.14 [MSQ, 2M · Propositional Logic] Answer: B, C (B: $(Q \vee R)$; C: $(P \vee Q)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee R)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(Q \vee R)$ is derivable:

1. resolve $(R \vee S)$ and $(Q \vee \neg S)$ on $S \Rightarrow (Q \vee R)$

(C) $(P \vee Q)$ is derivable:

1. resolve $(P \vee S)$ and $(Q \vee \neg S)$ on $S \Rightarrow (P \vee Q)$

(D) The resolution closure contains 7 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

Q.15 [NAT, 2M · Propositional Logic] Answer: 4

$(p \circ q) \rightarrow p$ can fail only in rows where $p = F$, and it fails there iff $p \circ q = T$. So we need $F \circ T = F$ and $F \circ F = F$, while $T \circ T$ and $T \circ F$ are free. That gives $2^2 = 4$ connectives: constant F , $p \wedge q$, $p \wedge \neg q$, and the projection p .

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, C, D (B: $(Q \wedge (\neg R \leftrightarrow \neg P)) \vee (\neg Q \wedge \neg(\neg R \leftrightarrow \neg P))$; C: $\neg Q \leftrightarrow \neg(\neg R \leftrightarrow \neg P)$; D: $(Q \rightarrow (\neg R \leftrightarrow \neg P)) \wedge ((\neg R \leftrightarrow \neg P) \rightarrow Q)$)

Compare truth tables with $F = Q \leftrightarrow (\neg R \leftrightarrow \neg P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $(Q \wedge (\neg R \leftrightarrow \neg P)) \vee (Q \wedge \neg(\neg R \leftrightarrow \neg P))$ — differs at $P=T, Q=T, R=F$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

(B) $(Q \wedge (\neg R \leftrightarrow \neg P)) \vee (\neg Q \wedge \neg(\neg R \leftrightarrow \neg P))$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(C) $\neg Q \leftrightarrow \neg(\neg R \leftrightarrow \neg P)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

(D) $(Q \rightarrow (\neg R \leftrightarrow \neg P)) \wedge ((\neg R \leftrightarrow \neg P) \rightarrow Q)$ — agrees with F on all 8 rows $\Rightarrow$ **equivalent**.

Q.17 [MSQ, 2M · Predicate Logic] Answer: C (C: $\exists x \forall y \neg R(x, y)$)

$R = \{(1,2), (1,4), (3,1), (3,2), (4,1)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(3,1) \wedge R(1,4)$ but $\neg R(3,4)$.

(B) $\exists x R(x, x)$: **FALSE** — there are no self-loops.

(C) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.

(D) $\forall y \exists x R(x, y)$: **FALSE** — 3 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [NAT, 2M · Propositional Logic] Answer: 2

Build the truth table (sub-formulas shown first):

P	Q	R	S	$R \wedge (Q \leftrightarrow \neg P)$	$S \vee (R \rightarrow \neg R)$	$(R \wedge (Q \leftrightarrow \neg P)) \wedge (S \vee (R \rightarrow \neg R))$
T	T	T	T	F	T	F
T	T	T	F	F	F	F
T	T	F	T	F	T	F
T	T	F	F	F	T	F
T	F	T	T	T	T	T
T	F	T	F	T	F	F
T	F	F	T	F	T	F
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	T	F	T	F	F

P	Q	R	S	$R \wedge (Q \leftrightarrow \neg P)$	$S \vee (R \rightarrow \neg R)$	$(R \wedge (Q \leftrightarrow \neg P)) \wedge (S \vee (R \rightarrow \neg R))$
F	T	F	T	F	T	F
F	T	F	F	F	T	F
F	F	T	T	F	T	F
F	F	T	F	F	F	F
F	F	F	T	F	T	F
F	F	F	F	F	T	F

Rows where F = T: **2**.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: B, C, D** (B: D is entailed; C: H is entailed; D: K is entailed)

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $B \wedge A \rightarrow F$; add F.

Round 2: fire $F \wedge A \rightarrow D$, $F \rightarrow H$; add D, H.

Round 3: fire $H \wedge B \wedge D \rightarrow K$; add K.

Fixed point: {A, B, D, E, F, H, K} — 7 atoms are entailed (facts: A, B, E; newly derived: D, F, H, K). Atoms never derived: C, J.

Q.20 [NAT, 2M · Predicate Logic] **Answer: 5**

These are exactly the equivalence relations on a 3-element set, which correspond one-to-one with its partitions. The number of partitions of a 3-set is the Bell number $B_3 = 5$: {123}, {1|23}, {2|13}, {3|12}, {1|2|3}. Enumeration of all 512 relations confirms 5.

Q.21 [MCQ, 2M · Sampling] **Answer: D** (D: A=F, B=F, C=F, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \neg) = 0.3$; $u = 0.91 \geq 0.3 \Rightarrow A = F$
- B: $P(B=T \mid A=F) = 0.1$; $u = 0.29 \geq 0.1 \Rightarrow B = F$
- C: $P(C=T \mid A=F) = 0.8$; $u = 0.93 \geq 0.8 \Rightarrow C = F$
- D: $P(D=T \mid B=F, C=F) = 0.75$; $u = 0.2 < 0.75 \Rightarrow D = T$

Sample: **{A=F, B=F, C=F, D=T}**.

Q.22 [NAT, 2M · Exact Inference] **Answer: 2**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A)$, $f(A, C)$, $f(A, D)$, $f(A, D, E)$, $f(D, F)$, $f(A, C, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(D, F)$	{D, F}	$f(D)$
D	$f(A, D)$, $f(A, D, E)$, $f(D)$	{A, D, E}	$f(A, E)$
E	$f(A, E)$	{A, E}	$f(A)$
C	$f(A, C)$, $f(A, C, G)$	{A, C, G}	$f(A, G)$
G	$f(A, G)$	{A, G}	$f(A)$

The largest factor produced has **2** variable(s) (created when eliminating D); with Boolean variables it has $2^2 = 4$ entries. Finally, multiply the remaining factors over A and normalise.

Q.23 [NAT, 2M · Exact Inference] **Answer: 0.898 to 0.902**

Factorisation: $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot P(\text{Alarm} \mid \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Burglary}, \text{Alarm})$.

Inference by enumeration: $P(\text{JohnCalls} \mid e) = \alpha \sum_{\text{hidden}} P(\text{JohnCalls}, e, \text{hidden})$.

Hidden: Ea	Jo	product of CPT entries (Bu · Ea · Al · Jo)	value
T	T	$0.2 \times 0.7 \times 0.2 \times 0.9$	0.0252
F	T	$0.2 \times 0.3 \times 0.25 \times 0.9$	0.0135
T	F	$0.2 \times 0.7 \times 0.2 \times 0.1$	0.0028
F	F	$0.2 \times 0.3 \times 0.25 \times 0.1$	0.0015

$P(\text{JohnCalls}=T, e) = 0.0387$; $P(\text{JohnCalls}=F, e) = 0.0043$; $P(e) = 0.043$.

$P(\text{JohnCalls}=T \mid e) = 0.0387 / 0.043 = \mathbf{0.9} \approx 0.900$.

Q.24 [NAT, 2M · Bayesian Networks] **Answer: 13**

M: no predecessors $\rightarrow$ 1 parameter. J: not independent of M (both depend on A) $\rightarrow$ parent M $\rightarrow$ 2. A: depends on both M and J and neither screens the other $\rightarrow$ parents {M, J} $\rightarrow$ 4. B: given A, B is independent of M and J $\rightarrow$ parent {A} $\rightarrow$ 2. E: given A alone, E still depends on B (explaining away), but given {A, B} it is independent of M, J $\rightarrow$ parents {A, B} $\rightarrow$ 4. Total = 1 + 2 + 4 + 2 + 4 = 13, versus 10 for the causal ordering B, E, A, J, M.

Q.25 [NAT, 2M · Sampling] **Answer: 0.66 to 0.68**

Rejection sampling keeps only samples consistent with the evidence $E = F$: samples 1, 4, 7, 10, 13, 14 (6 accepted, 9 rejected).

Among the accepted samples, $C = T$ in 4. Estimate = $4/6 = \mathbf{0.6667}$.

(For this network the exact value is 0.3391; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Mock Test 094

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	8 / 6 / 11	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

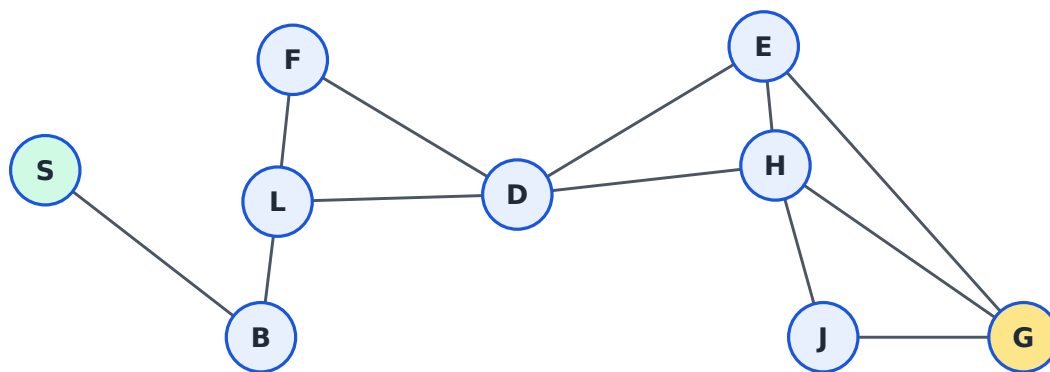
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 2$ and depth $d = 5$ (so $b^d = 32$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – DFS

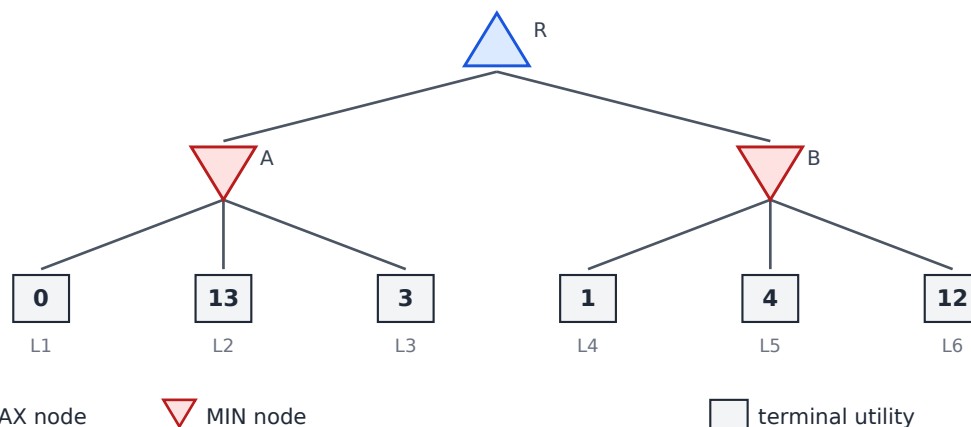
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, B, D, L, E, G
- ☐ (B) S, B, L, F, D, H, J, G
- ☐ (C) S, B, L, D, F, E, H, G
- ☐ (D) S, B, L, D, E, G

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Adversarial Search

Chess has an average branching factor of about 35, and a game lasts about 100 plies. The full game tree therefore has about how many nodes?

- ☐ (A) 10^{40}
☐ (B) 10^{100}
☐ (C) 10^{154}
☐ (D) 10^{400}

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((\neg R \vee Q) \leftrightarrow (R \wedge \neg P)) \rightarrow \neg(R \leftrightarrow \neg Q)$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.6 [MCQ · 1 mark · negative: -1/3] Predicate Logic

Skolemizing the sentence $\forall x \exists y P(x, y)$ yields

- ☐ (A) $\forall x P(x, F(x))$, where F is a new function symbol
☐ (B) $\forall x P(x, A)$, where A is a new constant
☐ (C) $\exists y P(y, y)$
☐ (D) $\forall x \forall y P(x, y)$

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 4 elements and R a binary predicate symbol. In how many of the 2^{16} possible interpretations of R over D is the sentence $\forall x \forall y (R(x, y) \rightarrow R(y, x))$ true?

Answer (numerical): _____

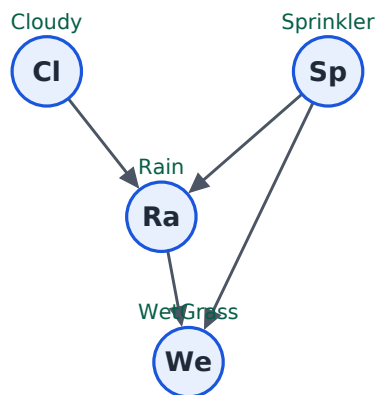
Q.8 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following is logically equivalent to $p \rightarrow q$?

- ☐ (A) $\neg q \rightarrow \neg p$
☐ (B) $q \rightarrow p$
☐ (C) $\neg p \rightarrow \neg q$
☐ (D) $p \wedge \neg q$

Q.9 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample **(Cl=T, Sp=F, Ra=F, We=F)**? (round off to 4 decimal places)



P(Cl=T)
0.75

P(Sp=T)	Cl	Sp	P(Ra=T ...)
0.5	T	T	0.5
	T	F	0.9
	F	T	0.1
	F	F	0.2

Sp	Ra	P(We=T ...)
T	T	0.3
T	F	0.1
F	T	0.95
F	F	0.9

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Probability for AI

If n Boolean random variables are mutually (absolutely) independent, how many independent numbers suffice to specify their full joint distribution?

- ☐ (A) n
☐ (B) $2^n - 1$
☐ (C) $2n$
☐ (D) n^2

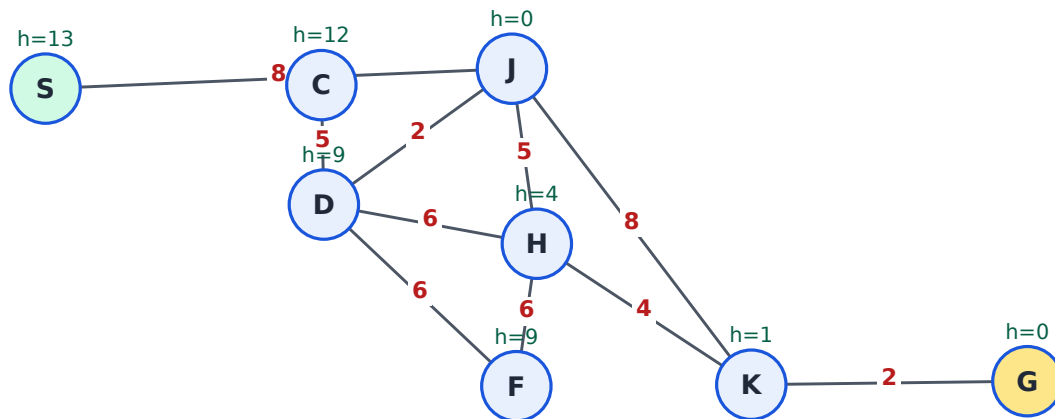
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search

A MAX root has two chance-node children. Chance node C_1 leads with probability 0.4 to a MIN node with leaves (6, 10) and with probability 0.6 to a MIN node with leaves (3, 9). Chance node C_2 leads with probability 0.5 to a leaf of value 8 and with probability 0.5 to a leaf of value 1. What is the expectiminimax value of the root?

Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

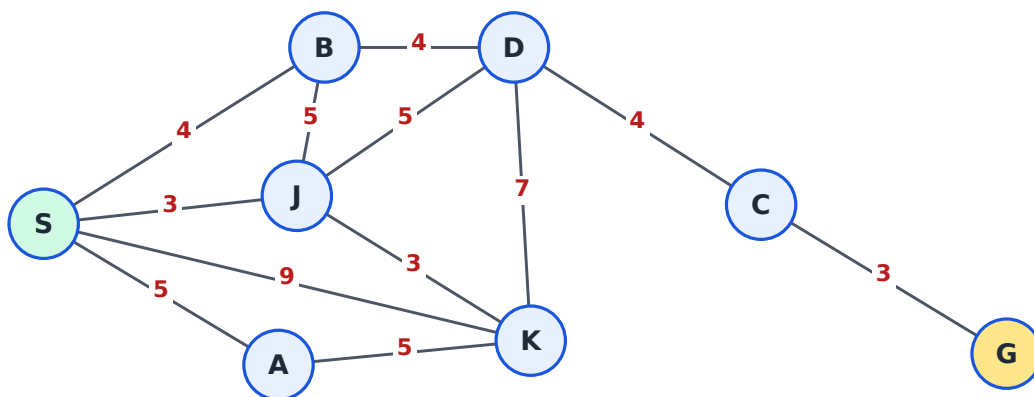
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
☐ (B) The edge D-J violates the consistency condition $h(D) \leq c(D,J) + h(J)$.
☐ (C) $h(F)$ overestimates the true cost from F to G.
☐ (D) h is consistent (monotone).

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

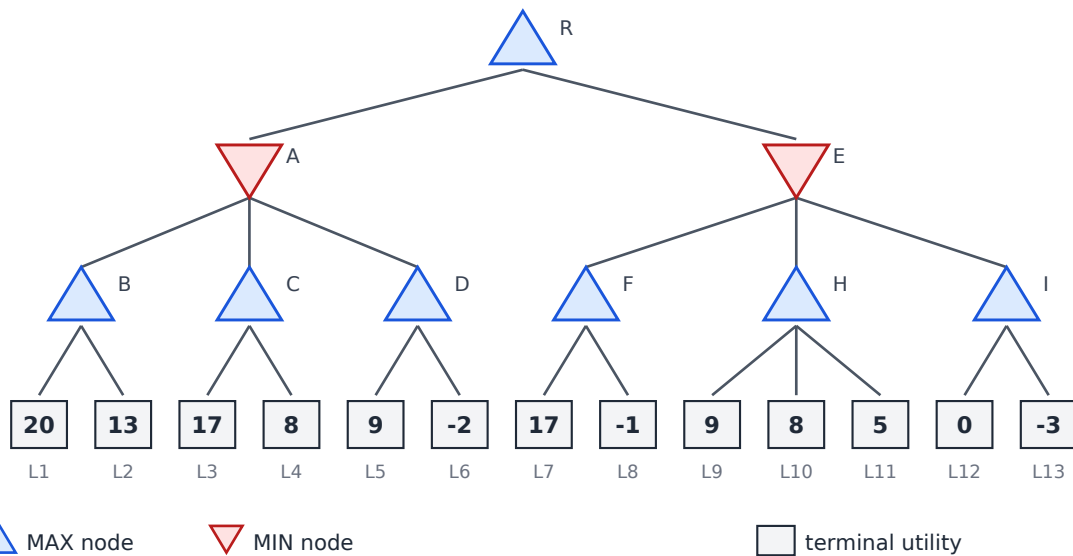
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are evaluated?



Answer (numerical): _____

Q.15 [MCQ · 2 marks · negative: -2/3] Predicate Logic

A KB consists of the definite clauses $\text{Nat}(0)$ and $\text{Nat}(x) \rightarrow \text{Nat}(S(x))$. Forward chaining is used to answer the query $\text{Nat}(\text{Foo})$, where Foo is a constant different from 0. Which one of the following is correct?

- ☐ (A) Forward chaining never terminates, since it keeps adding $\text{Nat}(S(\dots S(0)\dots))$
- ☐ (B) Forward chaining terminates and answers 'not entailed'
- ☐ (C) Forward chaining derives $\text{Nat}(\text{Foo})$
- ☐ (D) Forward chaining fails immediately since the KB is not in Datalog form and therefore cannot be processed

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = \neg Q \vee \neg P$?

- ☐ (A) $\neg Q \wedge \neg P$
- ☐ (B) $\neg(\neg\neg Q \wedge \neg\neg P)$
- ☐ (C) $\neg(\neg\neg Q \vee \neg\neg P)$
- ☐ (D) $\neg P \vee \neg Q$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

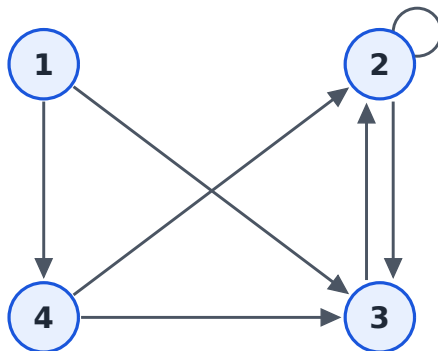
C1: (P), C2: (P $\vee$ Q), C3: ($\neg$ P), C4: (S), C5: ($\neg$ P $\vee$ $\neg$ R $\vee$ $\neg$ S)

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) ($\neg$ P $\vee$ $\neg$ R)
- ☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
- ☐ (C) ($\neg$ P $\vee$ Q)
- ☐ (D) ($\neg$ R $\vee$ $\neg$ S)

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
☐ (B) $\exists x R(x, x)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$

Q.19 [MCQ · 2 marks · negative: -2/3] Conditional Independence

A Bayesian network has edges $A \rightarrow C$ and $B \rightarrow C$ (Boolean variables) with $P(a) = 0.1$, $P(b) = 0.2$, $P(c | a, b) = 0.95$, $P(c | a, \neg b) = 0.90$, $P(c | \neg a, b) = 0.80$, $P(c | \neg a, \neg b) = 0.05$. Which ordering of the three quantities $P(a)$, $P(a | c)$ and $P(a | c, b)$ is correct?

- ☐ (A) $P(a) < P(a | c, b) < P(a | c)$
☐ (B) $P(a | c, b) < P(a) < P(a | c)$
☐ (C) $P(a) < P(a | c) < P(a | c, b)$
☐ (D) $P(a | c) < P(a) < P(a | c, b)$

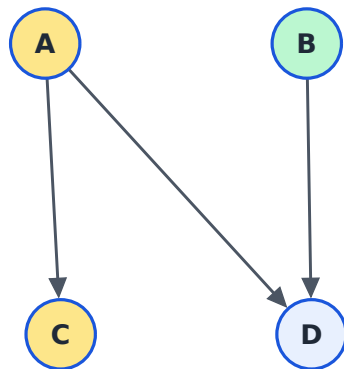
Q.20 [MSQ · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $C \rightarrow A$ and $C \rightarrow B$ (Boolean variables) with $P(c) = 0.4$, $P(a | c) = 0.9$, $P(a | \neg c) = 0.2$, $P(b | c) = 0.7$, $P(b | \neg c) = 0.1$. Which of the following are true?

- ☐ (A) $P(a) = 0.48$
☐ (B) $A \perp B$ (marginally)
☐ (C) $P(a | b, c) = 0.9$
☐ (D) $P(a | b) \approx 0.776$

Q.21 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{A = T, C = T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P(B = T | A = T, C = T)}$ (round off to 3 decimal places).



P(A=T)
0.2

P(B=T)	A	P(C=T ...)
0.5	T	0.75
	F	0.9

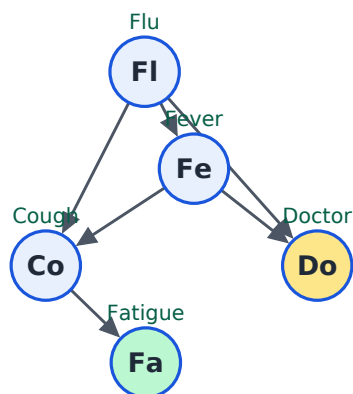
A	B	P(D=T ...)
T	T	0.95
T	F	0.95
F	T	0.5
F	F	0.5

#	B	D
1	T	T
2	T	T
3	F	T
4	T	F
5	F	T

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T | \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) Compute $\mathbf{P(\text{Fatigue} = T | \text{Doctor} = T)}$ (round off to 3 decimal places).



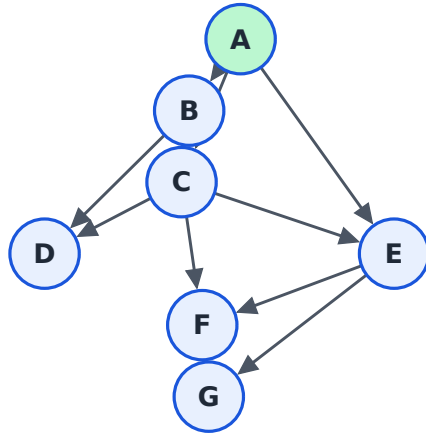
P(FI=T)	FI	P(Fe=T ...)	FI	Fe	P(Co=T ...)
0.25	T	0.95	T	T	0.5
	F	0.95	T	F	0.1
			F	T	0.8
			F	F	0.5

Co	P(Fa=T ...)	FI	Fe	P(Do=T ...)
T	0.95	T	T	0.05
F	0.6	T	F	0.4
		F	T	0.3
		F	F	0.6

Answer (numerical): _____

Q.23 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

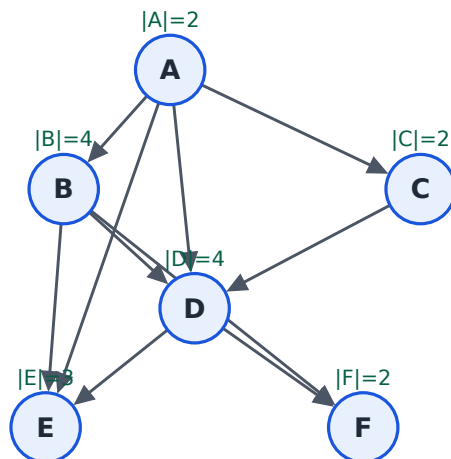
Variable elimination is used to compute $P(\mathbf{A})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F, E, G, D, C, B**. What is the scope of the new factor produced when **C** is summed out?



- ☐ (A) $f(A)$
☐ (B) $f(B)$
☐ (C) $f(A, B, C)$
☐ (D) $f(A, B)$

Q.24 [NAT · 2 marks · no negative marking] Bayesian Networks – Parameter count

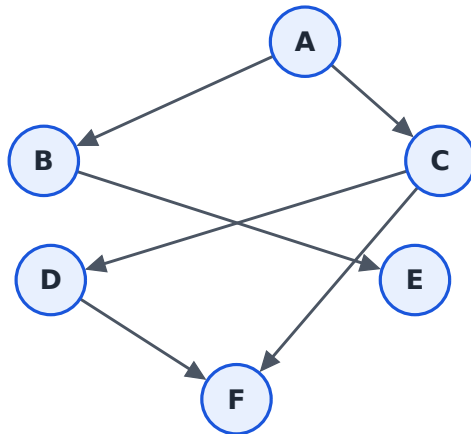
The domain sizes of the variables of the Bayesian network below are written above the nodes. By how many does the number of independent parameters of the full joint distribution exceed the number of independent parameters of the network's CPTs?



Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $E \perp F \mid A, B$
☐ (B) $A \perp D \mid B, F$
☐ (C) $F \perp E \mid D$
☐ (D) $F \perp B \mid A, E$

Answer Key — Mock Test 094

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	11	14	NAT	2	Adversarial Search	11
2	MCQ	1	Uninformed Search	D	15	MCQ	2	Predicate Logic	A
3	NAT	1	Adversarial Search	1	16	MSQ	2	Propositional Logic	B, D
4	MCQ	1	Adversarial Search	C	17	MSQ	2	Propositional Logic	A, B, D
5	NAT	1	Propositional Logic	1	18	MSQ	2	Predicate Logic	A, B, C
6	MCQ	1	Predicate Logic	A	19	MCQ	2	Conditional Independence	A
7	NAT	1	Predicate Logic	1024	20	MSQ	2	Conditional Independence	A, C, D
8	MCQ	1	Propositional Logic	A	21	NAT	2	Sampling	0.598 to 0.602
9	NAT	1	Sampling	0.0035 to 0.0039	22	NAT	2	Exact Inference	0.859 to 0.863
10	MCQ	1	Probability for AI	A	23	MCQ	2	Exact Inference	D
11	NAT	2	Adversarial Search	4.5	24	NAT	2	Bayesian Networks	246
12	MSQ	2	Informed Search	A, B	25	MSQ	2	Conditional Independence	A, D
13	NAT	2	Uninformed Search	8					

Detailed Solutions**Q.1** [NAT, 1M · Adversarial Search] **Answer: 11**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 2^3 + 2^2 - 1 = 8 + 4 - 1 = \mathbf{11}$, versus 32 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{(d/2)})$).

Q.2 [MCQ, 1M · Uninformed Search] Answer: D (D: S, B, L, D, E, G)

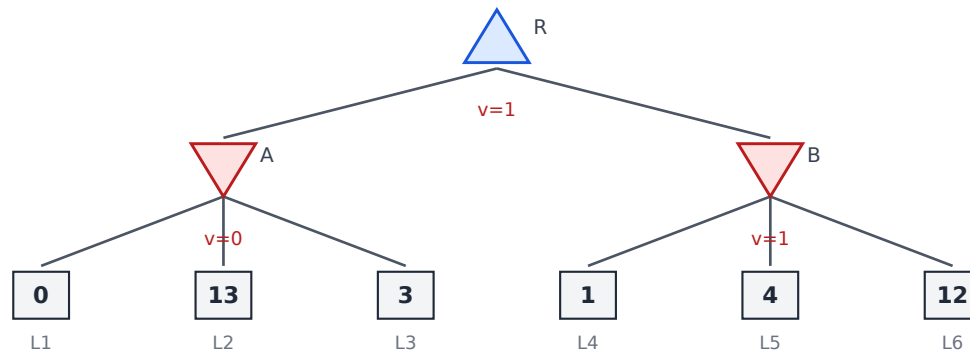
Adjacency lists (alphabetical): B: {L, S}; D: {E, F, H, L}; E: {D, G, H}; F: {D, L}; G: {E, H, J}; H: {D, E, G, J}; J: {G, H}; L: {B, D, F}; S: {B}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, B, L, D, E, G**; the path found is $S \rightarrow B \rightarrow L \rightarrow D \rightarrow E \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.3 [NAT, 1M · Adversarial Search] Answer: 1

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = 0$, $B = 1$. Root value = **1**; MAX's optimal first move is towards B.

Q.4 [MCQ, 1M · Adversarial Search] Answer: C (C: 10^{154})

$35^{100} = 10^{100 \cdot \log_{10} 35} \approx 10^{100 \times 1.544} \approx \mathbf{10^{154}}$, the AIMA estimate. 10^{40} (A) is AIMA's estimate of the number of distinct chess positions, not of the tree. (B) and (D) do not match the calculation.

Q.5 [NAT, 1M · Propositional Logic] Answer: 1

Build the truth table (sub-formulas shown first):

P	Q	R	$(\neg R \vee Q) \leftrightarrow (R \wedge \neg P)$	$\neg(R \leftrightarrow \neg Q)$	$((\neg R \vee Q) \leftrightarrow (R \wedge \neg P)) \rightarrow \neg(R \leftrightarrow \neg Q)$
T	T	T	F	T	T
T	T	F	F	F	T
T	F	T	T	F	F
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	F	F	T
F	F	T	F	F	T
F	F	F	F	T	T

F has 7 models, so it is false in $8 - 7 = \mathbf{1}$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.6 [MCQ, 1M · Predicate Logic] Answer: A (A: $\forall x P(x, F(x))$, where F is a new function symbol)

Since the existential $\exists y$ lies within the scope of $\forall x$, the chosen y may depend on x, so it is replaced by a Skolem function $F(x)$. A Skolem constant would wrongly assert one y works for all x. The result is satisfiable iff the original is (though not logically equivalent). $\forall x \forall y P(x, y)$ is much stronger.

Q.7 [NAT, 1M · Predicate Logic] Answer: 1024

View R as an 4×4 0/1 matrix. The sentence says R is symmetric: choose freely the n diagonal pairs and one decision per unordered pair $\{x, y\} \Rightarrow 2^{(n(n+1)/2)}$. With $n = 4$: **1024**.

Q.8 [MCQ, 1M · Propositional Logic] Answer: A (A: $\neg q \rightarrow \neg p$)

The contrapositive $\neg q \rightarrow \neg p \equiv q \vee \neg p \equiv p \rightarrow q$. The converse $q \rightarrow p$ and the inverse $\neg p \rightarrow \neg q$ are equivalent to each other but differ from $p \rightarrow q$ at $p=F, q=T$ ($p \rightarrow q$ is T, $q \rightarrow p$ is F). $p \wedge \neg q$ is the negation of $p \rightarrow q$.

Q.9 [NAT, 1M · Sampling] Answer: 0.0035 to 0.0039

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(Cl=T) \times P(Sp=F) \times P(Ra=F|Cl=T, Sp=F) \times P(We=F|Sp=F, Ra=F) \\ = 0.75 \times 0.5 \times 0.1 \times 0.1 = \mathbf{0.00375} \approx 0.0037.$$

Q.10 [MCQ, 1M · Probability for AI] Answer: A (A: n)

Under mutual independence $P(x_1, \dots, x_n) = \prod P(x_i)$, and each Boolean marginal needs one number $P(x_i = \text{true})$. Hence n numbers suffice. $2^n - 1$ is the count with no independence; $2n$ double-counts $P(\neg x_i) = 1 - P(x_i)$; n^2 has no basis.

Q.11 [NAT, 2M · Adversarial Search] Answer: 4.5

The MIN nodes under C_1 have values $\min(6, 10) = 6$ and $\min(3, 9) = 3$. $C_1 = 0.4 \cdot 6 + 0.6 \cdot 3 = 2.4 + 1.8 = 4.2$. $C_2 = 0.5 \cdot 8 + 0.5 \cdot 1 = 4.5$. Root = $\max(4.2, 4.5) = \mathbf{4.5}$, so MAX chooses C_2 .

Q.12 [MSQ, 2M · Informed Search] Answer: A, B (A: h is admissible.; B: The edge D-J violates the consistency condition $h(D) \leq c(D, J) + h(J)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	D	F	G	H	J	K	S
$h(n)$	12	9	9	0	4	0	1	13
$h^*(n)$	17	12	12	0	6	10	2	18
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u, v) + h(v)$ for every edge in both directions. Violating directed edges: $D \rightarrow J$ ($9 > 2 + 0$), $S \rightarrow J$ ($13 > 8 + 0$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [NAT, 2M · Uninformed Search] Answer: 8

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

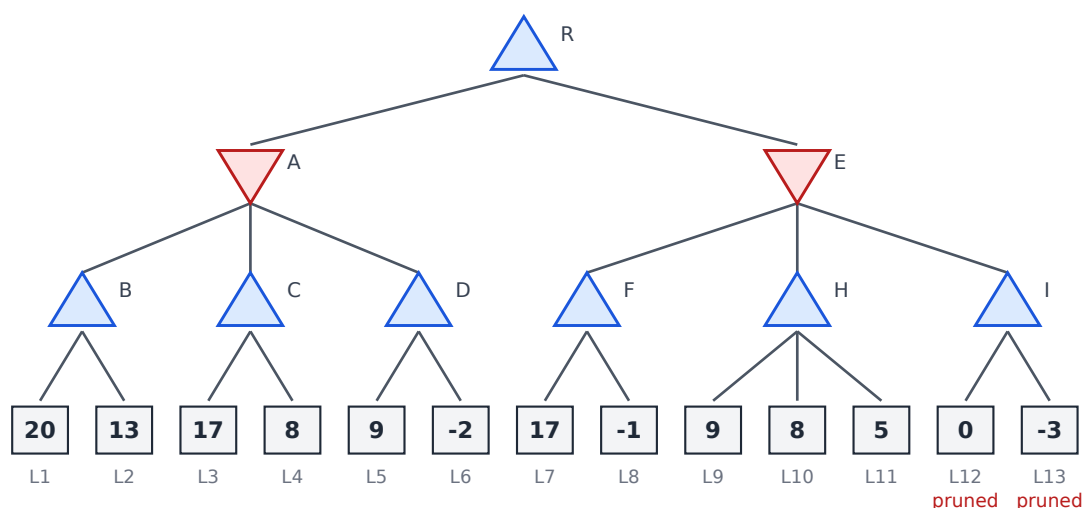
Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(3), B(4), A(5), K(9)
2	J (3)	B(4), A(5), K(6), D(8)
3	B (4)	A(5), K(6), D(8)
4	A (5)	K(6), D(8)
5	K (6)	D(8)
6	D (8)	C(12)
7	C (12)	G(15)
8	G (15)	← goal, stop

Path returned: **S** $\rightarrow$ **J** $\rightarrow$ **D** $\rightarrow$ **C** $\rightarrow$ **G**, cost = 3 + 5 + 4 + 3 = **15**. Nodes expanded (including S and G): **8** (S, J, B, A, K, D, C, G).

Q.14 [NAT, 2M · Adversarial Search] Answer: 11

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node E: value $9 \leq \alpha = 9 \Rightarrow$ prune remaining children (leaves L12, L13).



Leaves evaluated (11): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10, L11. Leaves pruned (2): L12, L13. Root value = 9 (same as plain minimax, 9 — pruning never changes the root value).

Q.15 [MCQ, 2M · Predicate Logic] Answer: A (A: Forward chaining never terminates, since it keeps adding $\text{Nat}(S(\dots S(0)\dots))$)

$\text{Nat}(\text{Foo})$ is not entailed (there is a model where Foo is not a natural number). Forward chaining derives $\text{Nat}(S(0))$, $\text{Nat}(S(S(0)))$, ... forever, because the function symbol S yields infinitely many ground terms. This illustrates that FOL entailment with definite clauses is only semidecidable: entailed queries are eventually found, non-entailed ones may run forever. Forward chaining still applies to non-Datalog KBs; it just lacks a termination guarantee.

Q.16 [MSQ, 2M · Propositional Logic] Answer: B, D (B: $\neg(\neg\neg Q \wedge \neg\neg P)$; D: $\neg P \vee \neg Q$)

Compare truth tables with $F = \neg Q \vee \neg P$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg Q \wedge \neg P$ — differs at $P=T, Q=F$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(B) $\neg(\neg\neg Q \wedge \neg\neg P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $\neg(\neg\neg Q \vee \neg\neg P)$ — differs at $P=T, Q=F$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(D) $\neg P \vee \neg Q$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.17 [MSQ, 2M · Propositional Logic] Answer: A, B, D (A: $(\neg P \vee \neg R)$; B: The empty clause $\square$ (i.e. the clause set is shown unsatisfiable); D: $(\neg R \vee \neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg P \vee \neg R)$ is derivable:

1. resolve (S) and $(\neg P \vee \neg R \vee \neg S)$ on S $\Rightarrow (\neg P \vee \neg R)$

(B) $\square$ is derivable:

1. resolve (P) and $(\neg P)$ on P $\Rightarrow \square$ (empty clause)

(C) $(\neg P \vee Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg R \vee \neg S)$ is derivable:

1. resolve (P) and $(\neg P \vee \neg R \vee \neg S)$ on P $\Rightarrow (\neg R \vee \neg S)$

Q.18 [MSQ, 2M · Predicate Logic] Answer: A, B, C (A: $\forall x \exists y R(x, y)$; B: $\exists x R(x, x)$; C: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,3), (1,4), (2,2), (2,3), (3,2), (4,2), (4,3)\}$.

(A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(B) $\exists x R(x, x)$: **TRUE** — self-loop at 2.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(D) $\forall x \forall y ((R(x, y) \wedge R(y, x)) \rightarrow x = y)$: **FALSE** — $R(2,3)$ and $R(3,2)$ with $2 \neq 3$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MCQ, 2M · Conditional Independence] Answer: A (A: $P(a) < P(a | c, b) < P(a | c)$)

$P(c) = 0.1 \times 0.2 \times 0.95 + 0.1 \times 0.8 \times 0.90 + 0.9 \times 0.2 \times 0.80 + 0.9 \times 0.8 \times 0.05 = 0.019 + 0.072 + 0.144 + 0.036 = 0.271$, and $P(a, c) = 0.019 + 0.072 = 0.091$, so $P(a | c) = 0.091/0.271 \approx 0.336$. $P(a | c, b) = 0.095/(0.095 + 0.072) \approx 0.117$. $P(a) = 0.1$. Hence $0.1 < 0.117 < 0.336$: observing c raises P(a); additionally observing b explains c away and lowers it, but not below the prior here.

Q.20 [MSQ, 2M · Conditional Independence] Answer: A, C, D (A: $P(a) = 0.48$; C: $P(a | b, c) = 0.9$; D: $P(a | b) \approx 0.776$)

$P(a) = 0.4 \times 0.9 + 0.6 \times 0.2 = 0.36 + 0.12 = 0.48$ (A true). $P(a, b) = 0.252 + 0.012 = 0.264$ while $P(a)P(b) = 0.48 \times 0.34 = 0.1632$, so A and B are dependent (B false). In the fork $C \rightarrow A$, $C \rightarrow B$, $A \perp B | C$, so $P(a | b, c) = P(a | c) = 0.9$ (C true). $P(a | b) = 0.264/0.34 \approx 0.7765$ (D true): observing b raises belief in c and hence in a.

Q.21 [NAT, 2M · Sampling] Answer: 0.598 to 0.602

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \text{parents})$	w	B
1	$P(A=T)=0.2 \times P(C=T A=T)=0.75$	0.15	T
2	$P(A=T)=0.2 \times P(C=T A=T)=0.75$	0.15	T
3	$P(A=T)=0.2 \times P(C=T A=T)=0.75$	0.15	F
4	$P(A=T)=0.2 \times P(C=T A=T)=0.75$	0.15	T
5	$P(A=T)=0.2 \times P(C=T A=T)=0.75$	0.15	F

$\sum w (\text{all}) = 0.75$; $\sum w (B = T) = 0.45$.

Estimate $P(B=T | A = T, C = T) = 0.45 / 0.75 = \mathbf{0.6}$.

Q.22 [NAT, 2M · Exact Inference] Answer: 0.859 to 0.863

Factorisation: $P(\text{Flu, Fever, Cough, Fatigue, Doctor}) = P(\text{Flu}) \cdot P(\text{Fever} | \text{Flu}) \cdot P(\text{Cough} | \text{Flu, Fever}) \cdot P(\text{Fatigue} | \text{Cough}) \cdot P(\text{Doctor} | \text{Flu, Fever})$.

Inference by enumeration: $P(\text{Fatigue} | e) = \alpha \sum_{\text{hidden}} P(\text{Fatigue, e, hidden})$.

Hidden: Fl, Fe, Co	Fa	product of CPT entries (Fl · Fe · Co · Fa · Do)	value
T, T, T	T	$0.25 \times 0.95 \times 0.5 \times 0.95 \times 0.05$	0.00564
T, T, F	T	$0.25 \times 0.95 \times 0.5 \times 0.6 \times 0.05$	0.00356
T, F, T	T	$0.25 \times 0.05 \times 0.1 \times 0.95 \times 0.4$	0.00048
T, F, F	T	$0.25 \times 0.05 \times 0.9 \times 0.6 \times 0.4$	0.0027
F, T, T	T	$0.75 \times 0.95 \times 0.8 \times 0.95 \times 0.3$	0.16245
F, T, F	T	$0.75 \times 0.95 \times 0.2 \times 0.6 \times 0.3$	0.02565
F, F, T	T	$0.75 \times 0.05 \times 0.5 \times 0.95 \times 0.6$	0.01069
F, F, F	T	$0.75 \times 0.05 \times 0.5 \times 0.6 \times 0.6$	0.00675
T, T, T	F	$0.25 \times 0.95 \times 0.5 \times 0.05 \times 0.05$	0.0003
T, T, F	F	$0.25 \times 0.95 \times 0.5 \times 0.4 \times 0.05$	0.00238
T, F, T	F	$0.25 \times 0.05 \times 0.1 \times 0.05 \times 0.4$	0.00003
T, F, F	F	$0.25 \times 0.05 \times 0.9 \times 0.4 \times 0.4$	0.0018
F, T, T	F	$0.75 \times 0.95 \times 0.8 \times 0.05 \times 0.3$	0.00855
F, T, F	F	$0.75 \times 0.95 \times 0.2 \times 0.4 \times 0.3$	0.0171
F, F, T	F	$0.75 \times 0.05 \times 0.5 \times 0.05 \times 0.6$	0.00056
F, F, F	F	$0.75 \times 0.05 \times 0.5 \times 0.4 \times 0.6$	0.0045

$P(\text{Fatigue}=T, e) = 0.21792$; $P(\text{Fatigue}=F, e) = 0.03521$; $P(e) = 0.25312$.

$P(\text{Fatigue}=T | e) = 0.21792 / 0.25312 = \mathbf{0.8609} \approx 0.861$.

Q.23 [MCQ, 2M · Exact Inference] Answer: D (D: $f(A, B)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(B, C, D)$, $f(A, C, E)$, $f(C, E, F)$, $f(E, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(C, E, F)$, $f(E, F, G)$	$\{C, E, F, G\}$	$f(C, E, G)$
E	$f(A, C, E)$, $f(C, E, G)$	$\{A, C, E, G\}$	$f(A, C, G)$
G	$f(A, C, G)$	$\{A, C, G\}$	$f(A, C)$
D	$f(B, C, D)$	$\{B, C, D\}$	$f(B, C)$
C	$f(A, B, C)$, $f(A, C)$, $f(B, C)$	$\{A, B, C\}$	$f(A, B)$
B	$f(A, B)$, $f(A, B)$	$\{A, B\}$	$f(A)$

The largest factor produced has **3** variable(s) (created when eliminating F); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over A and normalise.

Q.24 [NAT, 2M · Bayesian Networks] **Answer: 246**

A CPT for X with parents $\text{Pa}(X)$ needs $(|X| - 1) \times \prod_{U \in \text{Pa}(X)} |U|$ independent parameters (the last entry of each row is 1 minus the others).

$\Sigma = A: (2-1) + B: (4-1) \times 2 + C: (2-1) \times 2 + D: (4-1) \times 2 \times 4 \times 2 + E: (3-1) \times 2 \times 4 \times 4 + F: (2-1) \times 4 \times 4 = 137$.

Full joint table: $\prod |X| - 1 = 384 - 1 = 383$; saving = 246.

Q.25 [MSQ, 2M · Conditional Independence] **Answer: A, D** (A: $E \perp F \mid A, B$; D: $F \perp B \mid A, E$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \rightarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $E \perp F \mid A, B$: **TRUE** — $E \leftarrow B \leftarrow A \rightarrow C \rightarrow D \rightarrow F$ (blocked: observed chain node B); $E \leftarrow B \leftarrow A \rightarrow C \rightarrow F$ (blocked: observed chain node B).

(B) $A \perp D \mid B, F$: **FALSE** — the path $A \rightarrow C \rightarrow D$ is active given $\{B, F\}$.

(C) $F \perp E \mid D$: **FALSE** — the path $F \leftarrow C \leftarrow A \rightarrow B \rightarrow E$ is active given $\{D\}$.

(D) $F \perp B \mid A, E$: **TRUE** — $F \leftarrow C \leftarrow A \rightarrow B$ (blocked: observed fork node A); $F \leftarrow D \leftarrow C \leftarrow A \rightarrow B$ (blocked: observed fork node A).

Mock Test 095

Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 8 / 11	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

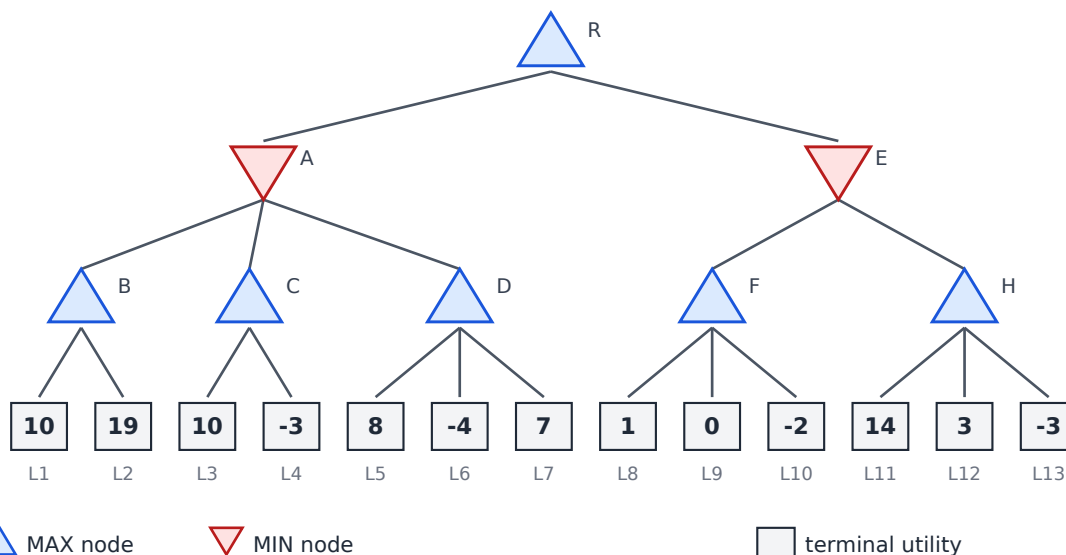
Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Alpha-beta complexity

A uniform game tree has branching factor $b = 5$ and depth $d = 4$ (so $b^d = 625$ leaves). With **perfect move ordering**, what is the minimum number of leaf nodes that alpha-beta pruning must evaluate?

Answer (numerical): _____

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

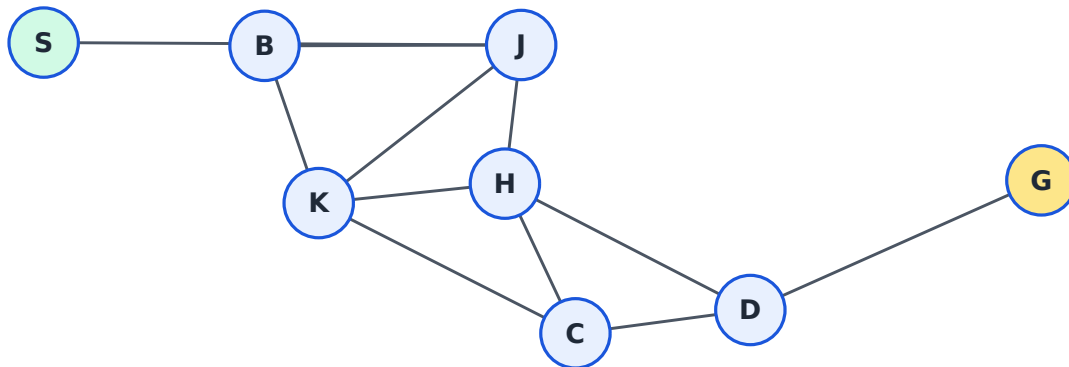
Q.3 [MCQ · 1 mark · negative: $-1/3$] Informed Search

Iterative deepening A* (IDA*) uses a cutoff on $f = g + h$. After an iteration fails, the cutoff for the next iteration is set to

- ☐ (A) the previous cutoff plus 1
- ☐ (B) the smallest f-value of any node that exceeded the cutoff in the previous iteration
- ☐ (C) the largest f-value seen in the previous iteration
- ☐ (D) twice the previous cutoff

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, B, J, K, C, D, G
☐ (B) S, J, B, K, C, D, G
☐ (C) S, J, K, H, D, G
☐ (D) S, J, B, H, K, C, D, G

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

How many of the 8 truth assignments to p, q, r are models of $(p \vee q) \wedge (\neg p \vee r)$?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 2 elements, so there are $4^2 = 16$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

How many of the 8 truth assignments to P, Q, R satisfy the formula

$$F = \neg(P \wedge \neg P) \wedge (\neg Q \rightarrow (P \leftrightarrow R)) ?$$

Answer (numerical): _____

Q.8 [MSQ · 1 mark · no negative marking] Propositional Logic

In the DPLL algorithm, a pure symbol is one that appears with the same sign in every clause in which it occurs. For the clause set $\{A \vee \neg B, \neg B \vee \neg C, C \vee A\}$, which symbols are pure?

- ☐ (A) A
☐ (B) B
☐ (C) C
☐ (D) None of the symbols is pure

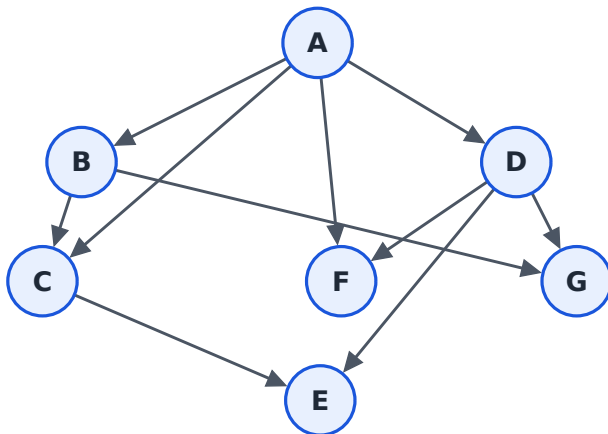
Q.9 [MSQ · 1 mark · no negative marking] Probability for AI

Let A and B be Boolean random variables with $P(b) > 0$ and $P(\neg b) > 0$. Which of the following identities hold for **every** such distribution?

- ☐ (A) $P(a \wedge b) = P(a \mid b) P(b)$
☐ (B) $P(a \mid b) = P(b \mid a) P(a) / P(b)$
☐ (C) $P(a \mid b) + P(a \mid \neg b) = 1$
☐ (D) $P(a) = P(a \mid b) P(b) + P(a \mid \neg b) P(\neg b)$

Q.10 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **B** in the Bayesian network below?

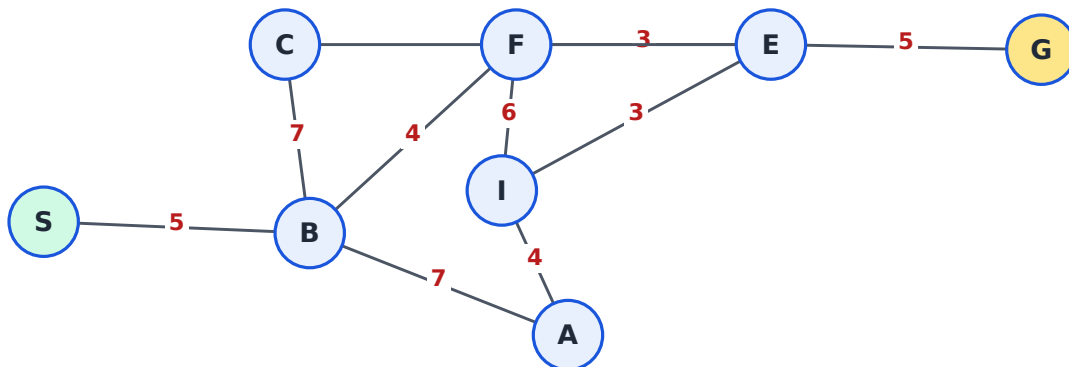


Answer (numerical): _____

Section B — 2-mark questions

Q.11 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

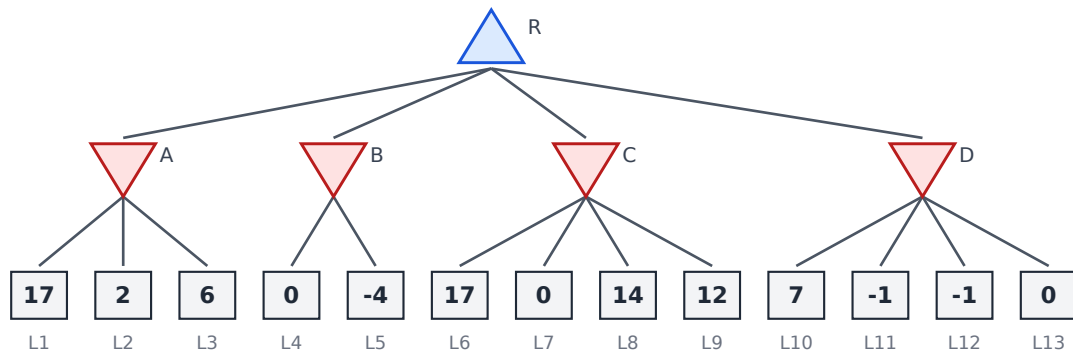
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow B \rightarrow A \rightarrow I \rightarrow E \rightarrow G$
- ☐ (B) $S \rightarrow B \rightarrow F \rightarrow E \rightarrow G$
- ☐ (C) $S \rightarrow B \rightarrow F \rightarrow I \rightarrow E \rightarrow G$
- ☐ (D) $S \rightarrow B \rightarrow C \rightarrow E \rightarrow G$

Q.12 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L3 is pruned.
☐ (B) Leaf L4 is pruned.
☐ (C) Leaf L6 is pruned.
☐ (D) Leaf L12 is pruned.

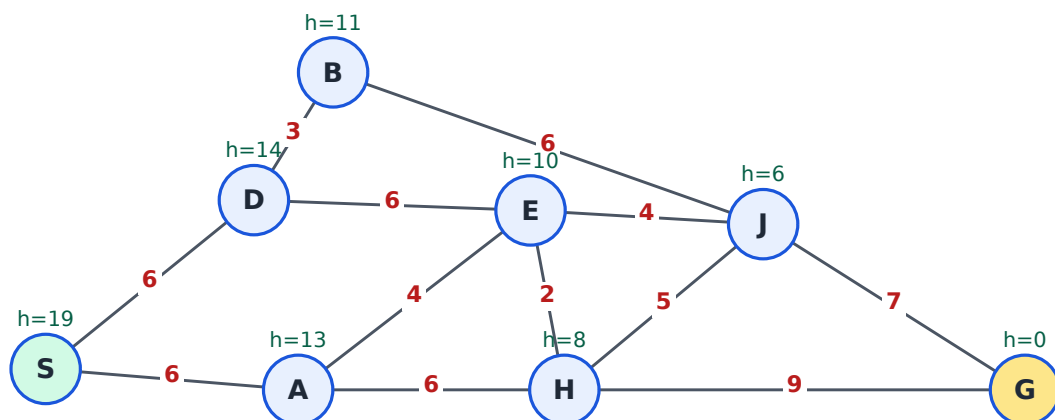
Q.13 [NAT · 2 marks · no negative marking] Informed Search

In the 8-puzzle, the goal is 1 2 3 / 8 _ 4 / 7 6 5 (rows top to bottom, _ = blank), and the current state is 2 8 3 / 1 6 4 / 7 _ 5. Let h_1 be the number of misplaced tiles and h_2 the total Manhattan distance. What is $h_1 + h_2$?

Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg Q \leftrightarrow S$
- (2) $(R \vee \neg S) \leftrightarrow (\neg Q \vee P)$
- (3) $\neg P \wedge \neg P$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $P \rightarrow Q$
- ☐ (B) $(P \leftrightarrow P) \leftrightarrow (R \rightarrow \neg R)$
- ☐ (C) $(R \wedge \neg R) \rightarrow \neg R$
- ☐ (D) $\neg Q \rightarrow \neg R$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $(Q \rightarrow (R \rightarrow P)) \leftrightarrow ((Q \wedge R) \rightarrow P)$
- ☐ (B) $((R \rightarrow P) \rightarrow Q) \leftrightarrow (R \rightarrow (P \rightarrow Q))$
- ☐ (C) $(Q \rightarrow R) \vee (R \rightarrow Q)$
- ☐ (D) $(\neg Q \wedge (\neg Q \rightarrow R)) \rightarrow R$

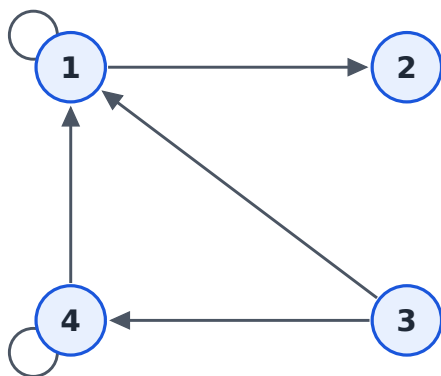
Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic

Which of the following first-order sentences are valid?

- ☐ (A) $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$
- ☐ (B) $\forall y \exists x P(x, y) \rightarrow \exists x \forall y P(x, y)$
- ☐ (C) $\forall x (P(x) \vee Q(x)) \rightarrow (\forall x P(x) \vee \forall x Q(x))$
- ☐ (D) $\exists x (P(x) \wedge Q(x)) \rightarrow (\exists x P(x) \wedge \exists x Q(x))$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
- ☐ (B) $\exists x \forall y \neg R(x, y)$
- ☐ (C) $\exists y \forall x R(x, y)$
- ☐ (D) $\exists x \forall y \neg R(y, x)$

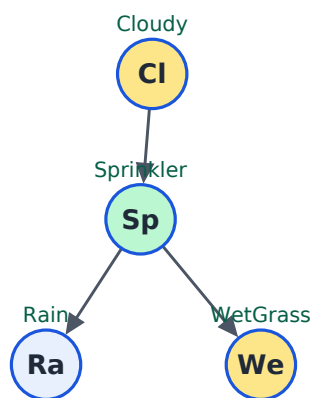
Q.19 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Which one of the following sets of connectives is NOT functionally complete?

- (A) $\{\neg, \wedge\}$
 ○ (B) {NAND}
 ○ (C) $\{\rightarrow, \perp\}$
 ○ (D) $\{\wedge, \vee, \leftrightarrow\}$

Q.20 [NAT · 2 marks · no negative marking] Exact Inference - Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\text{Sprinkler} = T \mid \text{WetGrass} = F, \text{Cloudy} = T)$ (round off to 3 decimal places).



P(Cl=T)
0.2

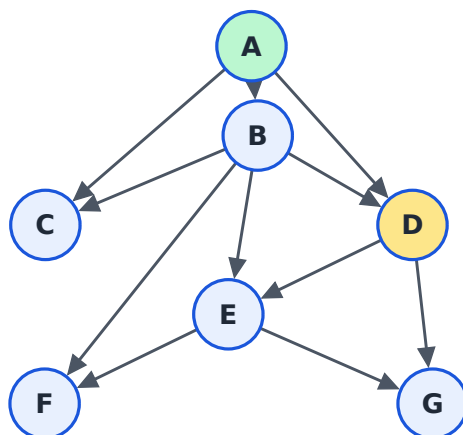
Cl	P(Sp=T ...)	Sp	P(Ra=T ...)
T	0.7	T	0.75
F	0.2	F	0.75

Sp	P(We=T ...)
T	0.25
F	0.75

Answer (numerical): _____

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference - Variable elimination

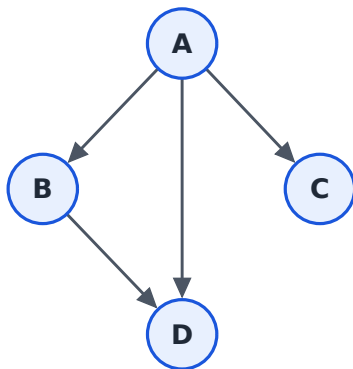
Variable elimination is used to compute $P(A \mid D = T)$ in the Bayesian network below (Boolean variables; evidence $D = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **E**, **B**, **G**, **F**, **C**. What is the scope of the new factor produced when **C** is summed out?



- (A) $f()$
 ○ (B) $f(A, C)$
 ○ (C) $f(C)$
 ○ (D) $f(A)$

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = (0.84, 0.23, 0.89, 0.07)$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.9

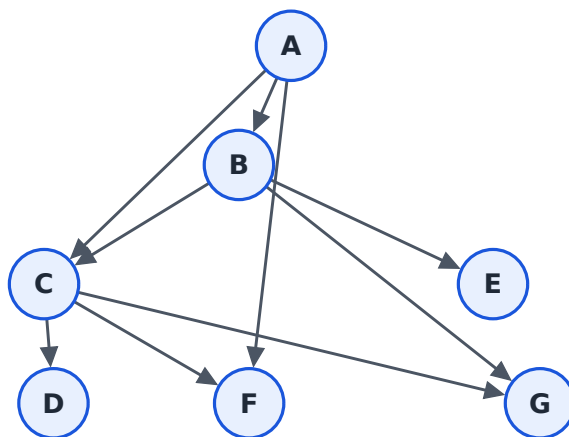
A	P(B=T ...)	A	P(C=T ...)
T	0.3	T	0.95
F	0.75	F	0.8

A	B	P(D=T ...)
T	T	0.9
T	F	0.1
F	T	0.05
F	F	0.4

- ☐ (A) A=T, B=F, C=T, D=T
☐ (B) A=F, B=F, C=T, D=F
☐ (C) A=F, B=T, C=T, D=T
☐ (D) A=T, B=T, C=T, D=T

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

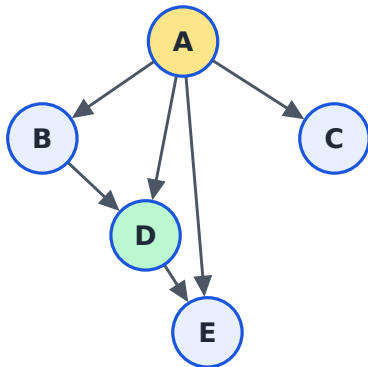
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $G \perp E \mid B, C$
☐ (B) $B \perp G \mid D$
☐ (C) $F \perp B$
☐ (D) $D \perp A$

Q.24 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence **A=F**. The current state of the chain is **(A=F, B=F, C=F, D=T, E=T)**. The next step resamples **D**. With what probability is D set to T? (round off to 3 decimal places)



P(A=T)
0.05

A	P(B=T ...)	A	P(C=T ...)
T	0.1	T	0.4
F	0.7	F	0.3

A	B	P(D=T)	A	D	P(E=T ...)
T	T	0.4	T	T	0.9
T	F	0.3	T	F	0.3
F	T	0.3	F	T	0.5
F	F	0.25	F	F	0.7

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Sampling

Consider the Bayesian network over Boolean variables Cloudy (C), Sprinkler (S), Rain (R) and WetGrass (W) with edges $C \rightarrow S$, $C \rightarrow R$, $S \rightarrow W$, $R \rightarrow W$ and CPTs: $P(c) = 0.5$; $P(s | c) = 0.1$, $P(s | \neg c) = 0.5$; $P(r | c) = 0.8$, $P(r | \neg c) = 0.2$; $P(w | s, r) = 0.99$, $P(w | s, \neg r) = 0.90$, $P(w | \neg s, r) = 0.90$, $P(w | \neg s, \neg r) = 0.0$. Gibbs sampling with evidence $S = \text{true}$, $W = \text{true}$ starts in state $C = \text{true}$, $R = \text{false}$. One sweep first resamples C (given its Markov blanket), then resamples R (given its Markov blanket, using the new C). What is the probability that $R = \text{true}$ at the end of the sweep? (Round to 3 decimal places.)

Answer (numerical): _____

Answer Key — Mock Test 095

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	49	14	NAT	2	Informed Search	21
2	NAT	1	Adversarial Search	8	15	MSQ	2	Propositional Logic	A, C
3	MCQ	1	Informed Search	B	16	MSQ	2	Propositional Logic	A, C, D
4	MCQ	1	Uninformed Search	B	17	MSQ	2	Predicate Logic	A, D
5	NAT	1	Propositional Logic	4	18	MSQ	2	Predicate Logic	B, D
6	NAT	1	Predicate Logic	5	19	MCQ	2	Propositional Logic	D
7	NAT	1	Propositional Logic	6	20	NAT	2	Exact Inference	0.873 to 0.877
8	MSQ	1	Propositional Logic	A, B	21	MCQ	2	Exact Inference	D
9	MSQ	1	Probability for AI	A, B, D	22	MCQ	2	Sampling	D
10	NAT	1	Conditional Independence	4	23	MSQ	2	Conditional Independence	A
11	MCQ	2	Uninformed Search	B	24	NAT	2	Sampling	0.19 to 0.194
12	MSQ	2	Adversarial Search	D	25	NAT	2	Sampling	0.244 to 0.245
13	NAT	2	Informed Search	9					

Detailed Solutions

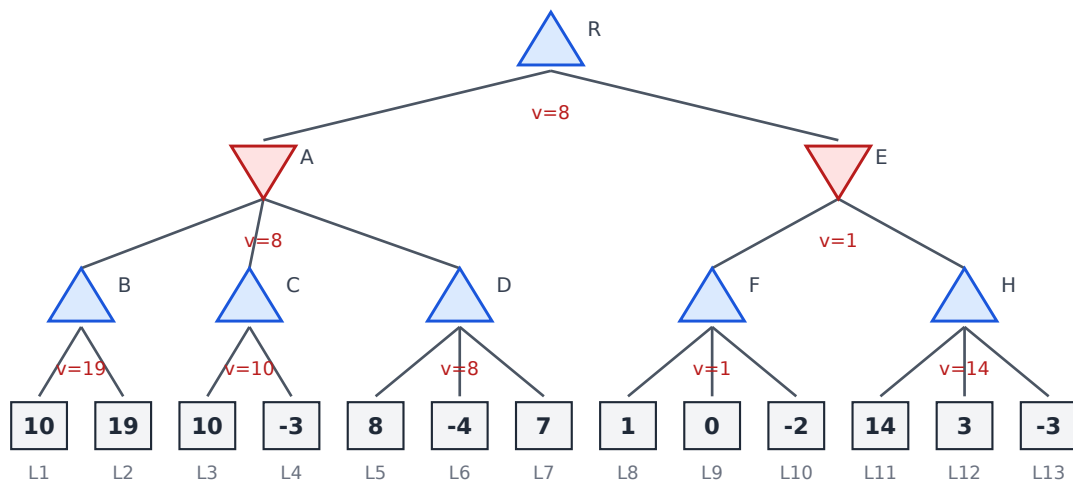
Q.1 [NAT, 1M · Adversarial Search] **Answer: 49**

Knuth & Moore (1975): with perfect ordering the minimum number of leaves examined is $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$.

$= 5^{\lceil 2 \rceil} + 5^{\lfloor 2 \rfloor} - 1 = 25 + 25 - 1 = 49$, versus 625 for plain minimax. This is why alpha-beta can search roughly twice as deep in the same time ($O(b^{\lceil d/2 \rceil})$).

Q.2 [NAT, 1M · Adversarial Search] Answer: 8

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 19, C = 10, D = 8, A = 8, F = 1, H = 14, E = 1. Root value = 8; MAX's optimal first move is towards A.

Q.3 [MCQ, 1M · Informed Search] Answer: B (B: the smallest f-value of any node that exceeded the cutoff in the previous iteration)

IDA* raises the threshold to the **smallest f-cost that exceeded the old threshold**. This keeps it optimal with an admissible h, because no solution with a smaller f is skipped. Increment-by-1 (A) assumes integer costs and can waste iterations. The largest f (C) and doubling (D) can jump past the optimal cost and return a suboptimal solution.

Q.4 [MCQ, 1M · Uninformed Search] Answer: B (B: S, J, B, K, C, D, G)

Adjacency lists (alphabetical): B: {J, K}; C: {D, H, K}; D: {C, G, H}; G: {D}; H: {C, D, J, K}; J: {B, H, K, S}; K: {B, C, H, J}; S: {J}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, J, B, K, C, D, G**; the path found is S → J → B → K → C → D → G (6 edges). Note that DFS need not find the shortest path.

Q.5 [NAT, 1M · Propositional Logic] Answer: 4

Case p=T: the first clause is satisfied and the second requires r=T; q is free, giving 2 models. Case p=F: the second clause is satisfied and the first requires q=T; r is free, giving 2 models. Total = 2 + 2 = 4.

Q.6 [NAT, 1M · Predicate Logic] Answer: 5

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With n = 2 this gives **5** (verified by enumerating all interpretations).

Q.7 [NAT, 1M · Propositional Logic] Answer: 6

Build the truth table (sub-formulas shown first):

P	Q	R	$\neg(P \wedge \neg P)$	$\neg Q \rightarrow (P \leftrightarrow R)$	$\neg(P \wedge \neg P) \wedge (\neg Q \rightarrow (P \leftrightarrow R))$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	T	T	T

Rows where F = T: **6**.

Q.8 [MSQ, 1M · Propositional Logic] Answer: A, B (A: A; B: B)

A occurs only positively (in $A \vee \neg B$ and $C \vee A$), so it is pure; DPLL can set $A = T$. B occurs only negatively ($\neg B$ twice), so it is pure; DPLL can set $B = F$. C occurs negatively in $\neg B \vee \neg C$ and positively in $C \vee A$, so it is not pure.

Q.9 [MSQ, 1M · Probability for AI] Answer: A, B, D (A: $P(a \wedge b) = P(a | b) P(b)$; B: $P(a | b) = P(b | a) P(a) / P(b)$; D: $P(a) = P(a | b) P(b) + P(a | \neg b) P(\neg b)$)

(A) is the product rule. (B) is Bayes rule, derived from writing $P(a \wedge b)$ in two ways via the product rule. (D) is the conditioning rule (marginalization of $P(a, B)$ with the product rule). (C) is false: conditional distributions sum to 1 over the values of the conditioned variable ($P(a | b) + P(\neg a | b) = 1$), not over the conditioning variable; e.g. if $A \perp B$ with $P(a) = 0.9$, then $P(a | b) + P(a | \neg b) = 0.9 + 0.9 = 1.8$.

Q.10 [NAT, 1M · Conditional Independence] Answer: 4

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of B: {A}; children: {C, G}; co-parents: {D}.

MB(B) = {A, C, D, G} — 4 nodes. Given its Markov blanket, B is conditionally independent of every other node in the network.

Q.11 [MCQ, 2M · Uninformed Search] Answer: B (B: $S \rightarrow B \rightarrow F \rightarrow E \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

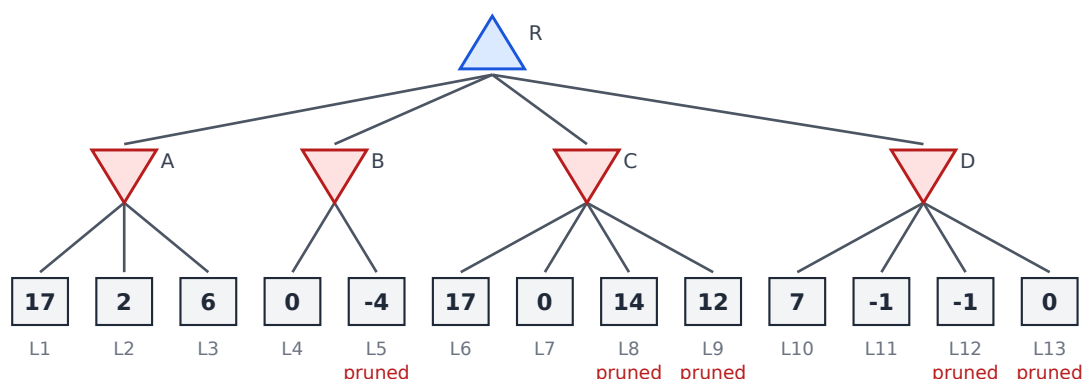
Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	B(5)
2	B (5)	F(9), A(12), C(12)
3	F (9)	A(12), C(12), E(12), I(15)
4	A (12)	C(12), E(12), I(15)
5	C (12)	E(12), I(15)
6	E (12)	I(15), G(17)
7	I (15)	G(17)
8	G (17)	← goal, stop

Path returned: **S** $\rightarrow$ **B** $\rightarrow$ **F** $\rightarrow$ **E** $\rightarrow$ **G**, cost = 5 + 4 + 3 + 5 = **17**. Nodes expanded (including S and G): **8** (S, B, F, A, C, E, I, G).

Q.12 [MSQ, 2M · Adversarial Search] Answer: D (D: Leaf L12 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node B: value 0 $\leq \alpha = 2 \Rightarrow$ prune remaining children (leaves L5).
- At MIN node C: value 0 $\leq \alpha = 2 \Rightarrow$ prune remaining children (leaves L8, L9).
- At MIN node D: value -1 $\leq \alpha = 2 \Rightarrow$ prune remaining children (leaves L12, L13).



Leaves evaluated (8): L1, L2, L3, L4, L6, L7, L10, L11. Leaves pruned (5): L5, L8, L9, L12, L13. Root value = 2 (same as plain minimax, 2 — pruning never changes the root value).

Q.13 [NAT, 2M · Informed Search] Answer: 9

Misplaced tiles: 2 (at (0,0), belongs at (0,1)), 8 (at (0,1), belongs at (1,0)), 1 (at (1,0), belongs at (0,0)), and 6 (at (1,1), belongs at (2,1)). Tiles 3, 4, 7 and 5 are in place, so $h_1 = 4$. Manhattan distances: $2 \rightarrow 1$, $8 \rightarrow 1+1 = 2$, $1 \rightarrow 1$, $6 \rightarrow 1$, so $h_2 = 5$. $h_1 + h_2 = 9$. (The sum is not necessarily admissible. This question is purely arithmetic.)

Q.14 [NAT, 2M · Informed Search] Answer: 21

Heuristic values: $h(A)=13$, $h(B)=11$, $h(D)=14$, $h(E)=10$, $h(G)=0$, $h(H)=8$, $h(J)=6$, $h(S)=19$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (19)	A(19), D(20)
2	A (19)	D(20), E(20), H(20)
3	D (20)	B(20), E(20), H(20)
4	B (20)	E(20), H(20), J(21)
5	E (20)	H(20), J(20)
6	H (20)	J(20), G(21)
7	J (20)	G(21)
8	G (21)	← goal, stop

Path returned: **S** → **A** → **H** → **G** with cost **21**; nodes expanded = **8** (S, A, D, B, E, H, J, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 21$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [MSQ, 2M · Propositional Logic] Answer: A, C (A: $P \rightarrow Q$; C: $(R \wedge \neg R) \rightarrow \neg R$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 1 of them:

P	Q	R	S	KB
F	F	T	T	T

(A) $P \rightarrow Q$ is true in all models of KB ⇒ **entailed**.

(B) $(P \leftrightarrow P) \leftrightarrow (R \rightarrow \neg R)$ is false in the KB-model $P=F, Q=F, R=T, S=T$ ⇒ not entailed.

(C) $(R \wedge \neg R) \rightarrow \neg R$ is true in all models of KB ⇒ **entailed**.

(D) $\neg Q \rightarrow \neg R$ is false in the KB-model $P=F, Q=F, R=T, S=T$ ⇒ not entailed.

Q.16 [MSQ, 2M · Propositional Logic] Answer: A, C, D (A: $(Q \rightarrow (R \rightarrow P)) \leftrightarrow ((Q \wedge R) \rightarrow P)$; C: $(Q \rightarrow R) \vee (R \rightarrow Q)$; D: $(\neg Q \wedge (\neg Q \rightarrow R)) \rightarrow R$)

(A) $(Q \rightarrow (R \rightarrow P)) \leftrightarrow ((Q \wedge R) \rightarrow P)$: true in all 8 assignments ⇒ **valid**.

(B) $((R \rightarrow P) \rightarrow Q) \leftrightarrow (R \rightarrow (P \rightarrow Q))$: **not valid** — counter-model $P=T, Q=F, R=F$.

(C) $(Q \rightarrow R) \vee (R \rightarrow Q)$: true in all 4 assignments ⇒ **valid**.

(D) $(\neg Q \wedge (\neg Q \rightarrow R)) \rightarrow R$: true in all 4 assignments ⇒ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.17 [MSQ, 2M · Predicate Logic] Answer: A, D (A: $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$; D: $\exists x (P(x) \wedge Q(x)) \rightarrow (\exists x P(x) \wedge \exists x Q(x))$)

(A) If some a satisfies $P(a, y)$ for all y , then for each y that same a is a witness — valid. (B) Fails on domain $\{1, 2\}$ with $P(x, y) \iff x = y$: each y has a witness, but no single x works for all y . (C) Fails on domain $\{1, 2\}$ with $P = \{1\}$, $Q = \{2\}$: every element is P or Q , but neither P nor Q holds everywhere. (D) A witness a for $P(a) \wedge Q(a)$ witnesses both $\exists x P(x)$ and $\exists x Q(x)$ — valid.

Q.18 [MSQ, 2M · Predicate Logic] Answer: B, D (B: $\exists x \forall y \neg R(x, y)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,1), (1,2), (3,1), (3,4), (4,1), (4,4)\}$.

(A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(2,3)$ nor $R(3,2)$.

(B) $\exists x \forall y \neg R(x, y)$: **TRUE** — 2 has no outgoing edge.

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 3 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MCQ, 2M · Propositional Logic] Answer: D (D: $\{\wedge, \vee, \leftrightarrow\}$)

$\wedge$, $\vee$ and $\leftrightarrow$ all output T when every input is T ($T \wedge T = T$, $T \vee T = T$, $T \leftrightarrow T = T$). Hence every formula built from them is T under the all-true assignment, so $\neg p$ cannot be expressed. $\{\neg, \wedge\}$ is complete by De Morgan. NAND alone is complete ($\neg p = p \text{ NAND } p$). With $\rightarrow$ and $\perp$ we get $\neg p \equiv p \rightarrow \perp$ and $p \vee q \equiv \neg p \rightarrow q$, so $\{\rightarrow, \perp\}$ is complete.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.873 to 0.877

Factorisation: $P(\text{Cloudy, Sprinkler, Rain, WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Sprinkler})$.

Inference by enumeration: $P(\text{Sprinkler} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Sprinkler, e, hidden})$.

Hidden: Ra	Sp	product of CPT entries (Cl · Sp · Ra · We)	value
T	T	$0.2 \times 0.7 \times 0.75 \times 0.75$	0.07875
F	T	$0.2 \times 0.7 \times 0.25 \times 0.75$	0.02625
T	F	$0.2 \times 0.3 \times 0.75 \times 0.25$	0.01125
F	F	$0.2 \times 0.3 \times 0.25 \times 0.25$	0.00375

$P(\text{Sprinkler}=T, e) = 0.105$; $P(\text{Sprinkler}=F, e) = 0.015$; $P(e) = 0.12$.

$P(\text{Sprinkler}=T \mid e) = 0.105 / 0.12 = \mathbf{0.875} \approx 0.875$.

Q.21 [MCQ, 2M · Exact Inference] Answer: D (D: f(A))

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B, C)$, $f(A, B)$, $f(B, E)$, $f(B, E, F)$, $f(E, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
E	$f(B, E)$, $f(B, E, F)$, $f(E, G)$	$\{B, E, F, G\}$	$f(B, F, G)$
B	$f(A, B)$, $f(A, B, C)$, $f(A, B)$, $f(B, F, G)$	$\{A, B, C, F, G\}$	$f(A, C, F, G)$
G	$f(A, C, F, G)$	$\{A, C, F, G\}$	$f(A, C, F)$
F	$f(A, C, F)$	$\{A, C, F\}$	$f(A, C)$
C	$f(A, C)$	$\{A, C\}$	$f(A)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over A and normalise.

Q.22 [MCQ, 2M · Sampling] Answer: D (D: A=T, B=T, C=T, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T \mid \text{—}) = 0.9$; $u = 0.84 < 0.9 \Rightarrow A = T$
- B: $P(B=T \mid A=T) = 0.3$; $u = 0.23 < 0.3 \Rightarrow B = T$
- C: $P(C=T \mid A=T) = 0.95$; $u = 0.89 < 0.95 \Rightarrow C = T$
- D: $P(D=T \mid A=T, B=T) = 0.9$; $u = 0.07 < 0.9 \Rightarrow D = T$

Sample: **{A=T, B=T, C=T, D=T}**.

Q.23 [MSQ, 2M · Conditional Independence] Answer: A (A: $G \perp E \mid B, C$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $G \perp E \mid B, C$: **TRUE** — $G \leftarrow B \rightarrow E$ (blocked: observed fork node B); $G \leftarrow C \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed chain node C); $G \leftarrow C \leftarrow B \rightarrow E$ (blocked: observed chain node C); $G \leftarrow C \rightarrow F \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed fork node C).

(B) $B \perp G \mid D$: **FALSE** — the path $B \leftarrow A \rightarrow C \rightarrow G$ is active given $\{D\}$.

(C) $F \perp B$: **FALSE** — the path $F \leftarrow A \rightarrow B$ is active given $\{\}$.

(D) $D \perp A$: **FALSE** — the path $D \leftarrow C \leftarrow A$ is active given $\{\}$.

Q.24 [NAT, 2M · Sampling] Answer: 0.19 to 0.194

Gibbs sampling draws D from $P(D \mid \text{mb}(D))$, where $\text{mb}(D) = \{A, B, E\}$:

$P(D \mid \text{mb}) \propto P(D \mid \text{parents}(D)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

$D=T$: $P(D=T \mid A=F, B=F) \times P(E=T \mid A=F, D=T) = 0.25 \times 0.5 = 0.125$

$D=F$: $P(D=F \mid A=F, B=F) \times P(E=T \mid A=F, D=F) = 0.75 \times 0.7 = 0.525$

Normalise: $0.125 / (0.125 + 0.525) = \mathbf{0.1923}$.

Q.25 [NAT, 2M · Sampling] Answer: 0.244 to 0.245

Step 1: $P(c \mid s, \neg r) = \alpha P(c)P(s \mid c)P(\neg r \mid c)$: c : $0.5 \times 0.1 \times 0.2 = 0.01$; $\neg c$: $0.5 \times 0.5 \times 0.8 = 0.2 \rightarrow P(c) = 0.01/0.21 \approx 0.04762$. Step 2: $P(r \mid c, s, w) = 0.792/0.972 \approx 0.81481$; $P(r \mid \neg c, s, w) = (0.2 \times 0.99)/(0.2 \times 0.99 + 0.8 \times 0.90) = 0.198/0.918 \approx 0.21569$. Total: $0.04762 \times 0.81481 + 0.95238 \times 0.21569 \approx 0.03880 + 0.20542 = 0.2442 \approx 0.244$.

Mock Test 096

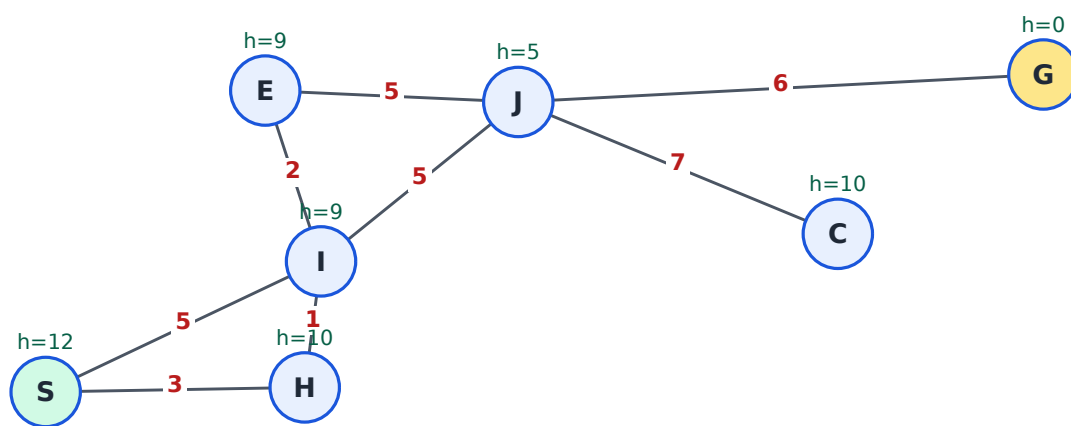
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 9 / 10	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

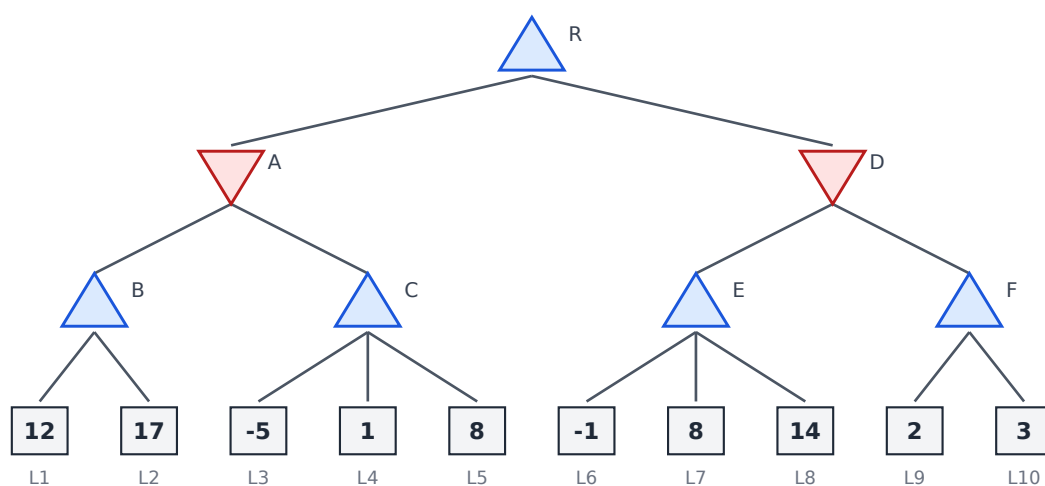
Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.2 [MCQ · 1 mark · negative: $-1/3$] Adversarial Search – Minimax

In the game tree below MAX is to move at the root. Assuming both players play optimally, which option gives MAX's best move and the minimax value of the root?



△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Move to D; root value 3
- ☐ (B) Move to A; root value 17
- ☐ (C) Move to A; root value 8
- ☐ (D) Move to A; root value 10

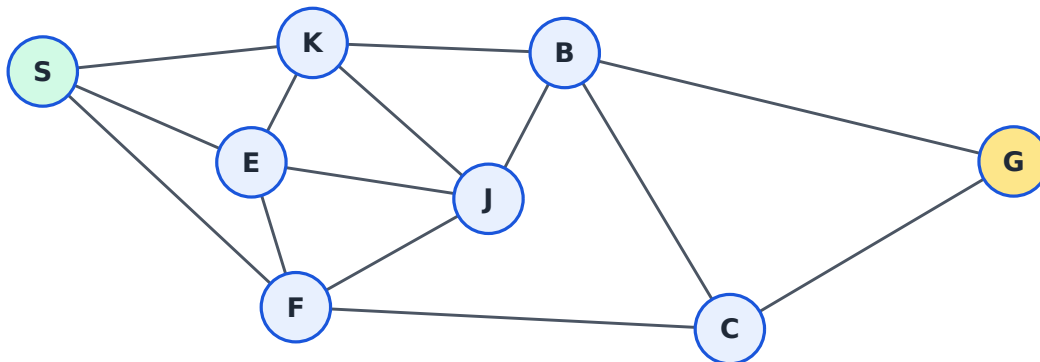
Q.3 [MCQ · 1 mark · negative: -1/3] Informed Search

A heuristic h is **admissible** if, for every node n (where $h^*(n)$ is the true cost of an optimal path from n to a goal),

- ☐ (A) $h(n) \geq h^*(n)$
- ☐ (B) $h(n) \leq h^*(n)$
- ☐ (C) $h(n) \leq c(n, a, n') + h(n')$ for every successor n'
- ☐ (D) $h(n) = 0$

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, F, E, C, B, G
- ☐ (B) S, E, F, C, B, G
- ☐ (C) S, K, J, F, E, C, G
- ☐ (D) S, E, F, K, J, C, B, G

Q.5 [MSQ · 1 mark · no negative marking] Propositional Logic

Which of the following formulas are in conjunctive normal form (CNF)?

- ☐ (A) $(p \vee \neg q) \wedge r$
- ☐ (B) $\neg(p \wedge q)$
- ☐ (C) $p \vee \neg q \vee r$
- ☐ (D) $(p \wedge q) \vee r$

Q.6 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Which one of the following statements about forward and backward chaining over Horn knowledge bases is correct?

- ☐ (A) Forward chaining is data-driven while backward chaining is goal-driven
- ☐ (B) Forward chaining is goal-driven while backward chaining is data-driven
- ☐ (C) Both are complete for arbitrary (non-Horn) propositional knowledge bases
- ☐ (D) Backward chaining can derive facts that forward chaining cannot

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Forward chaining

A propositional definite-clause knowledge base over atoms A, B, C, D, E, F, H, J, K has facts **E, D, B** and rules:

R1: $H \rightarrow K$

R2: $A \rightarrow K$

R3: $H \wedge C \rightarrow A$

R4: $E \rightarrow A$

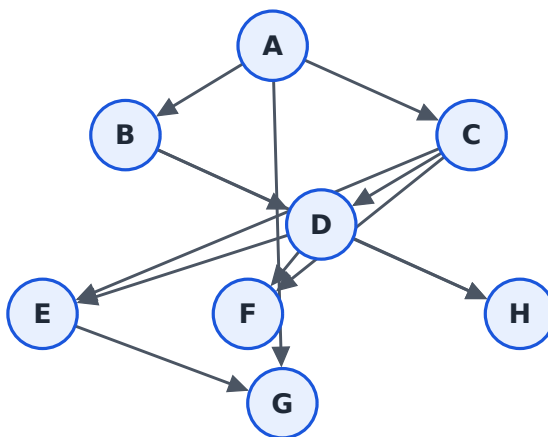
R5: $C \rightarrow H$

How many atoms (including the given facts) are entailed by the KB?

Answer (numerical): _____

Q.8 [MCQ · 1 mark · negative: -1/3] Conditional Independence – Markov blanket

Which set is the Markov blanket of **D** in the Bayesian network below?



- ☐ (A) {A, B, C, E, F, H}
- ☐ (B) {B, C, E, F, H}
- ☐ (C) {B, C, E, F, G, H}
- ☐ (D) {E, F, H}

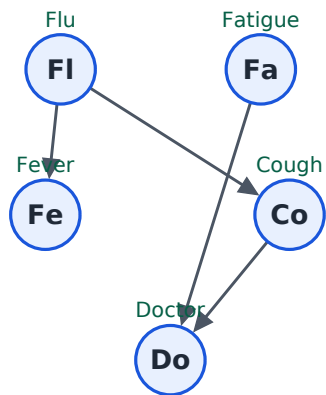
Q.9 [MCQ · 1 mark · negative: -1/3] Probability for AI

In AIMA's enumeration-style inference, $P(X | e)$ is computed as $\alpha \sum P(X, e)$, where the sum over hidden variables gives $P(X, e)$. What is the value of α ?

- ☐ (A) $1/P(e)$
- ☐ (B) $P(e)$
- ☐ (C) $1/P(X)$
- ☐ (D) $1/|\text{domain}(X)|$

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Fl, Fe, Co, Fa, Do. (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue, Do = Doctor.) What is the probability that a single run produces the sample **(Fl=F, Fe=F, Co=F, Fa=T, Do=T)**? (round off to 4 decimal places)



P(Fl=T)	Fl	P(Fe=T ...)	Fl	P(Co=T ...)
0.6	T	0.75	T	0.05
	F	0.4	F	0.3

P(Fa=T)	Co	Fa	P(Do=T ...)
0.75	T	T	0.9
	T	F	0.7
	F	T	0.3
	F	F	0.7

Answer (numerical): _____

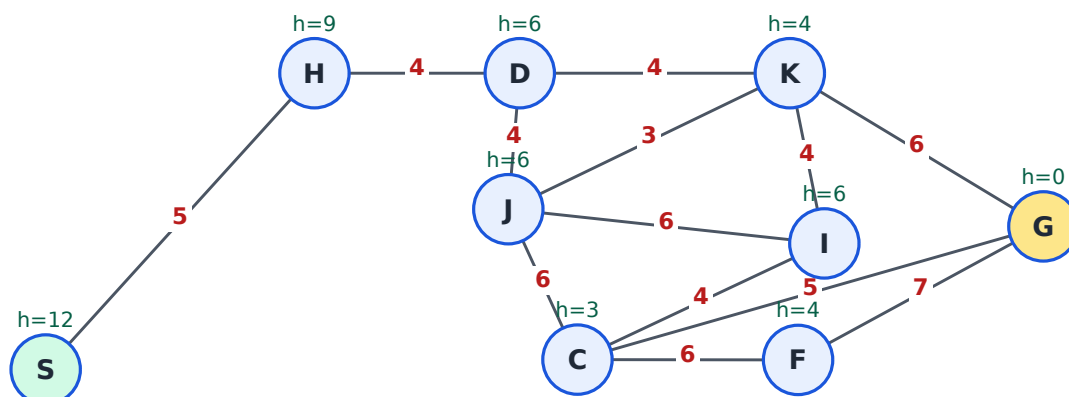
Section B – 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Uninformed Search

Which of the following statements about tree search versus graph search are TRUE?

- ☐ (A) Graph search keeps an explored (reached) set so that no state is expanded more than once
- ☐ (B) On a finite state space with reversible actions, tree search can face an infinite search tree because of loopy paths
- ☐ (C) Depth-first graph search is complete on infinite state spaces
- ☐ (D) The memory used by depth-first graph search can grow in proportion to the number of reachable states

Q.12 [NAT · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

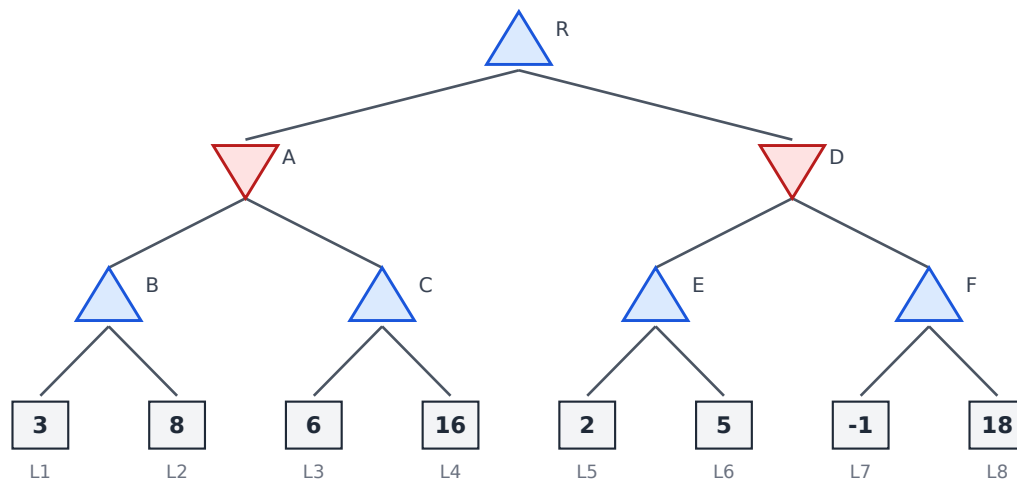
Q.13 [NAT · 2 marks · no negative marking] Adversarial Search

A three-ply binary game tree has levels MAX (root), MIN, MAX, and then leaves. The eight leaf values, left to right, are 5, 6, 7, 4, 5, 3, 6, 7. Each MAX node at the third level owns two consecutive leaves. Alpha-beta pruning processes children left to right. How many leaves are **evaluated**?

Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



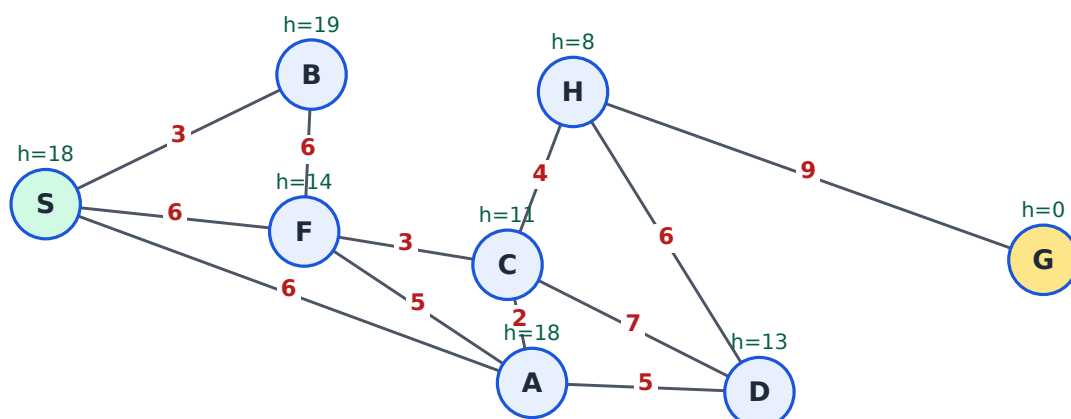
△ MAX node ▽ MIN node

□ terminal utility

- ☐ (A) Leaf L3 is pruned.
- ☐ (B) Leaf L4 is pruned.
- ☐ (C) Leaf L6 is pruned.
- ☐ (D) Leaf L7 is pruned.

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) $h(G) = 0$.
- ☐ (B) The edge F-A violates the consistency condition (in at least one direction).
- ☐ (C) h is admissible.
- ☐ (D) h is consistent (monotone).

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

- (1) $\neg Q \leftrightarrow S$
- (2) $\neg Q \wedge R$
- (3) $(\neg R \rightarrow S) \wedge (S \wedge R)$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $S \leftrightarrow Q$
- ☐ (B) $R \rightarrow (R \wedge S)$
- ☐ (C) $P \leftrightarrow R$
- ☐ (D) $\neg P \vee \neg S$

Q.17 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

How many of the 16 truth assignments to P, Q, R, S satisfy the formula

$$F = ((S \vee R) \vee (\neg P \leftrightarrow P)) \wedge (Q \leftrightarrow (P \rightarrow R)) ?$$

Answer (numerical): _____

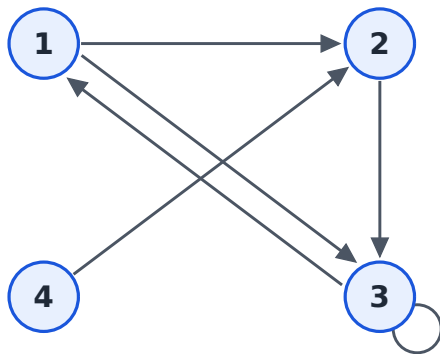
Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = (Q \wedge P) \leftrightarrow Q$?

- ☐ (A) $((Q \wedge P) \wedge Q) \vee ((Q \wedge P) \wedge \neg Q)$
- ☐ (B) $(Q \wedge P) \leftrightarrow \neg Q$
- ☐ (C) $((Q \wedge P) \rightarrow Q) \wedge (Q \rightarrow (Q \wedge P))$
- ☐ (D) $((Q \wedge P) \rightarrow Q) \wedge (\neg Q \rightarrow \neg(Q \wedge P))$

Q.19 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\exists y \forall x R(x, y)$
- ☐ (B) $\exists x R(x, x)$
- ☐ (C) $\exists x \forall y R(x, y)$
- ☐ (D) $\exists x \forall y \neg R(y, x)$

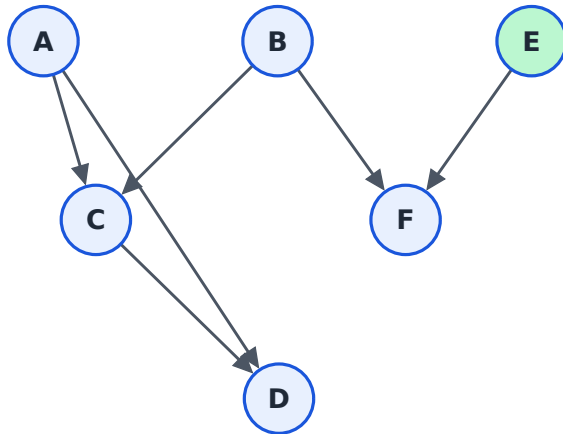
Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following formulas are contingent (satisfiable but not valid)?

- ☐ (A) $(p \rightarrow q) \wedge (p \rightarrow \neg q)$
- ☐ (B) $(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$
- ☐ (C) $(p \leftrightarrow q) \wedge (p \oplus q)$
- ☐ (D) $(p \vee q) \rightarrow (p \wedge \neg q)$

Q.21 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(E)$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, F, C, A, D**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

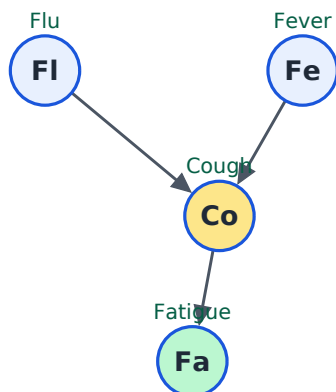
The following 20 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D, E. Using **rejection sampling**, estimate $P(B = T \mid C = F)$ (round off to 2 decimal places).

#	A	B	C	D	E
1	T	T	T	F	T
2	T	F	T	F	F
3	T	F	F	F	F
4	T	T	F	T	T
5	T	T	T	F	T
6	T	F	T	T	T
7	T	T	F	T	T
8	T	T	T	F	T
9	T	T	T	F	T
10	T	T	F	T	F
11	T	F	T	F	F
12	T	F	T	F	T
13	T	T	T	T	F
14	T	F	T	F	F
15	T	F	F	T	F
16	T	T	F	T	T
17	T	T	F	F	T
18	T	F	T	F	F
19	T	T	F	T	T
20	T	F	T	T	F

Answer (numerical): _____

Q.23 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Fl = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Fatigue} = T \mid \text{Cough} = T)$ (round off to 3 decimal places).



Fl	P(Fl=T)
	0.4

Fl	Fe	P(Co=T ...)
T	T	0.5
T	F	0.7
F	T	0.2
F	F	0.8

Co	P(Fa=T ...)
T	0.75
F	0.6

Answer (numerical): _____

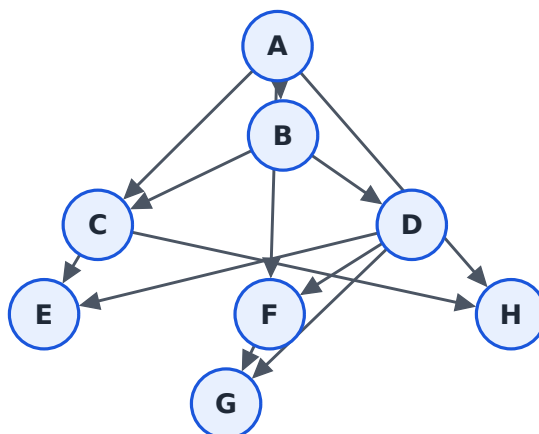
Q.24 [NAT · 2 marks · no negative marking] Probability for AI

A full joint distribution over Boolean variables X, Y, Z is: $P(x, y, z) = 0.10$, $P(x, y, \neg z) = 0.05$, $P(x, \neg y, z) = 0.15$, $P(x, \neg y, \neg z) = 0.10$, $P(\neg x, y, z) = 0.05$, $P(\neg x, y, \neg z) = 0.20$, $P(\neg x, \neg y, z) = 0.15$, $P(\neg x, \neg y, \neg z) = 0.20$. Compute $P(x \mid y \vee z)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp A$
☐ (B) $A \perp F$
☐ (C) $F \perp E \mid C, D$
☐ (D) $A \perp D \mid B$

Answer Key — Mock Test 096

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Informed Search	16	14	MSQ	2	Adversarial Search	D
2	MCQ	1	Adversarial Search	C	15	MSQ	2	Informed Search	A
3	MCQ	1	Informed Search	B	16	MSQ	2	Propositional Logic	B
4	MCQ	1	Uninformed Search	B	17	NAT	2	Propositional Logic	6
5	MSQ	1	Propositional Logic	A, C	18	MSQ	2	Propositional Logic	C
6	MCQ	1	Propositional Logic	A	19	MSQ	2	Predicate Logic	B, D
7	NAT	1	Propositional Logic	5	20	MSQ	2	Propositional Logic	A, D
8	MCQ	1	Conditional Independence	B	21	NAT	2	Exact Inference	4
9	MCQ	1	Probability for AI	A	22	NAT	2	Sampling	0.74 to 0.76
10	NAT	1	Sampling	0.0376 to 0.038	23	NAT	2	Exact Inference	0.748 to 0.752
11	MSQ	2	Uninformed Search	A, B, D	24	NAT	2	Probability for AI	0.428 to 0.429
12	NAT	2	Informed Search	5	25	MSQ	2	Conditional Independence	C, D
13	NAT	2	Adversarial Search	5					

Detailed Solutions

Q.1 [NAT, 1M · Informed Search] Answer: 16

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

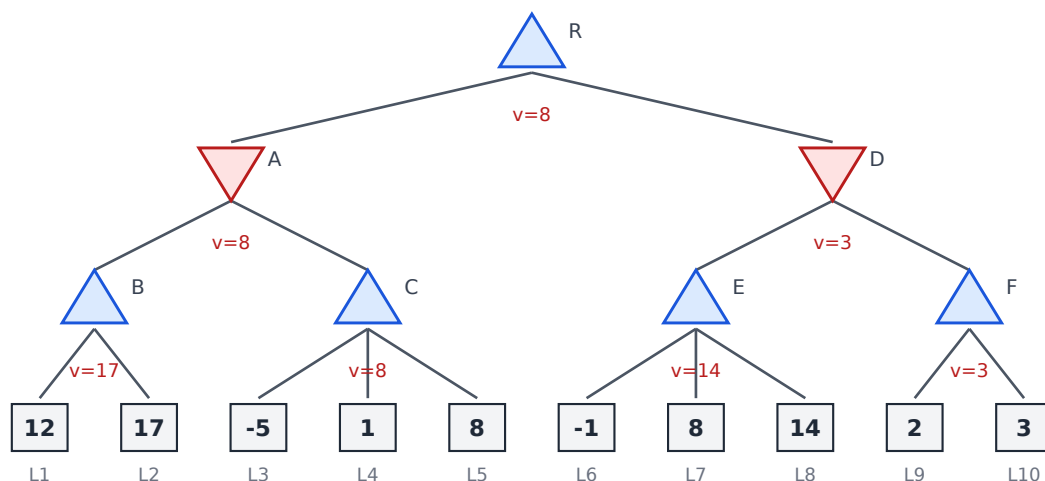
Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (12)	I(9), H(10)
2	I (9)	J(5), E(9), H(10)
3	J (5)	G(0), E(9), C(10), H(10)
4	G (0)	E(9), C(10), H(10) ← goal, stop

Path returned: **S** → **I** → **J** → **G**, cost = 5 + 5 + 6 = **16**.

The optimal cost is $h^*(S) = 15$ — greedy search is not optimal here.

Q.2 [MCQ, 1M · Adversarial Search] Answer: C (C: Move to A; root value 8)

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: B = 17, C = 8, A = 8, E = 14, F = 3, D = 3. Root value = **8**; MAX's optimal first move is towards A.

Q.3 [MCQ, 1M · Informed Search] Answer: B ($B: h(n) \leq h^*(n)$)

An admissible heuristic never overestimates the cost to reach a goal: **$h(n) \leq h^*(n)$** . (A) describes a pessimistic, inadmissible heuristic. (C) is the definition of consistency, which is a stronger condition. (D) is one particular admissible heuristic, not the definition.

Q.4 [MCQ, 1M · Uninformed Search] Answer: B (B: S, E, F, C, B, G)

Adjacency lists (alphabetical): B: {C, G, J, K}; C: {B, F, G}; E: {F, J, K, S}; F: {C, E, J, S}; G: {B, C}; J: {B, E, F, K}; K: {B, E, J, S}; S: {E, F, K}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, E, F, C, B, G**; the path found is $S \rightarrow E \rightarrow F \rightarrow C \rightarrow B \rightarrow G$ (5 edges). Note that DFS need not find the shortest path.

Q.5 [MSQ, 1M · Propositional Logic] Answer: A, C (A: $(p \vee \neg q) \wedge r$; C: $p \vee \neg q \vee r$)

A CNF is a conjunction of clauses, each a disjunction of literals (negation only on atoms). $(p \vee \neg q) \wedge r$ is two clauses. $p \vee \neg q \vee r$ is a single clause, which is a (one-clause) CNF. $\neg(p \wedge q)$ has negation applied to a compound formula, so it is not in CNF (its CNF is $\neg p \vee \neg q$). $(p \wedge q) \vee r$ has a conjunction inside a disjunction; its CNF is $(p \vee r) \wedge (q \vee r)$.

Q.6 [MCQ, 1M · Propositional Logic] Answer: A (A: Forward chaining is data-driven while backward chaining is goal-driven)

Forward chaining starts from known facts and fires rules whose premises are satisfied (data-driven).

Backward chaining starts from the query and works back through rules whose conclusion matches (goal-driven). Both are complete only for Horn (definite clause) KBs, not arbitrary ones. For Horn KBs both derive exactly the entailed atoms, so neither derives anything the other cannot.

Q.7 [NAT, 1M · Propositional Logic] Answer: 5

Forward chaining fires every rule whose premises are all known, adds the conclusion, and repeats until no new atom is added (sound and complete for Horn/definite clauses).

Round 1: fire $E \rightarrow A$; add A.

Round 2: fire $A \rightarrow K$; add K.

Fixed point: {A, B, D, E, K} — 5 atoms are entailed (facts: B, D, E; newly derived: A, K). Atoms never derived: C, F, H, J.

Q.8 [MCQ, 1M · Conditional Independence] Answer: B (B: {B, C, E, F, H})

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of D: {B, C}; children: {E, F, H}; co-parents: $\{\emptyset\}$.

$MB(D) = \{B, C, E, F, H\}$ — 5 nodes. Given its Markov blanket, D is conditionally independent of every other node in the network.

Q.9 [MCQ, 1M · Probability for AI] Answer: A (A: $1/P(e)$)

By definition $P(X | e) = P(X, e)/P(e)$, so $\alpha = 1/P(e)$. In practice α is found by normalizing the vector $P(X, e)$ to sum to 1, because $\sum_x P(x, e) = P(e)$. The other options do not make the entries sum to 1 in general.

Q.10 [NAT, 1M · Sampling] Answer: 0.0376 to 0.038

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$P(FI=F) \times P(Fe=F|FI=F) \times P(Co=F|FI=F) \times P(Fa=T) \times P(Do=T|Co=F, Fa=T)$

$= 0.4 \times 0.6 \times 0.7 \times 0.75 \times 0.3 = \mathbf{0.0378} \approx 0.0378$.

Q.11 [MSQ, 2M · Uninformed Search] Answer: A, B, D

(A) is true: the explored set is what defines graph search. (B) is true: with reversible actions, a path such as $A \rightarrow B \rightarrow A \rightarrow B \dots$ can be extended forever, so even a finite state space produces an infinite tree. (C) is false: in an infinite state space, DFS can follow an infinite branch forever. (D) is true: the explored set can hold every reachable state, which removes DFS's linear-space advantage.

Q.12 [NAT, 2M · Informed Search] Answer: 5

Heuristic values: $h(C)=3$, $h(D)=6$, $h(F)=4$, $h(G)=0$, $h(H)=9$, $h(I)=6$, $h(J)=6$, $h(K)=4$, $h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	H(14)
2	H (14)	D(15)
3	D (15)	K(17), J(19)
4	K (17)	G(19), J(19), I(23)
5	G (19)	J(19), I(23) $\leftarrow$ goal, stop

Path returned: **S → H → D → K → G** with cost **19**; nodes expanded = **5** (S, H, D, K, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 19$. For comparison UCS ($h = 0$) would expand 8 nodes.

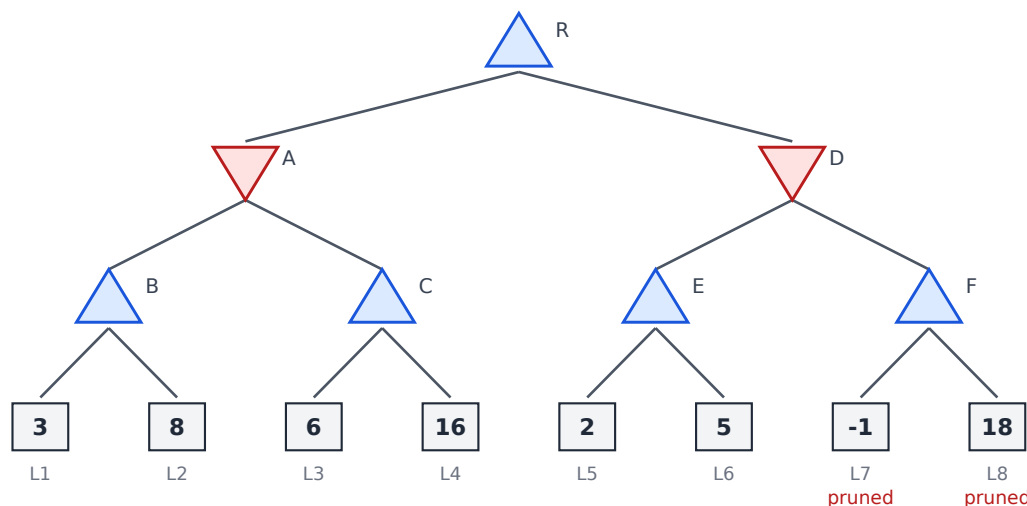
Q.13 [NAT, 2M · Adversarial Search] **Answer: 5**

The leftmost MAX node sees (5, 6) and returns 6, so its MIN parent has $\beta = 6$. The next MAX node sees $7 \geq \beta$ and prunes leaf 4. The left MIN node therefore has value 6, and the root gets $\alpha = 6$. In the right MIN subtree, the first MAX node sees 5 and then 3 and returns 5. The MIN node's value is now $\leq 5 \leq \alpha$, so the whole MAX subtree with leaves (6, 7) is pruned. Evaluated leaves: 5, 6, 7, 5, 3 = **5**. The root value is 6.

Q.14 [MSQ, 2M · Adversarial Search] **Answer: D** (D: Leaf L7 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node D: value $5 \leq \alpha = 8 \Rightarrow$ prune remaining children (leaves L7, L8).



Leaves evaluated (6): L1, L2, L3, L4, L5, L6. Leaves pruned (2): L7, L8. Root value = 8 (same as plain minimax, 8 — pruning never changes the root value).

Q.15 [MSQ, 2M · Informed Search] **Answer: A** (A: $h(G) = 0$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	C	D	F	G	H	S
$h(n)$	18	19	11	13	14	0	8	18
$h^*(n)$	15	22	13	15	16	0	9	21
$h \leq h^*$?	X	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n: **no** (overestimates at A).

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges:

$A \rightarrow C$ ($18 > 2 + 11$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: B** (B: $R \rightarrow (R \wedge S)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 16 assignments, KB is true in exactly 2 of them:

P	Q	R	S	KB
T	F	T	T	T
F	F	T	T	T

(A) $S \leftrightarrow Q$ is false in the KB-model $P=T, Q=F, R=T, S=T \Rightarrow$ not entailed.

(B) $R \rightarrow (R \wedge S)$ is true in all models of KB $\Rightarrow$ **entailed**.

(C) $P \leftrightarrow R$ is false in the KB-model $P=F, Q=F, R=T, S=T \Rightarrow$ not entailed.

(D) $\neg P \vee \neg S$ is false in the KB-model $P=T, Q=F, R=T, S=T \Rightarrow$ not entailed.

Q.17 [NAT, 2M · Propositional Logic] **Answer: 6**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(S \vee R) \vee (\neg P \leftrightarrow P)$	$Q \leftrightarrow (P \rightarrow R)$	$((S \vee R) \vee (\neg P \leftrightarrow P)) \wedge (Q \leftrightarrow (P \rightarrow R))$
T	T	T	T	T	T	T

P	Q	R	S	$(S \vee R) \vee (\neg P \leftrightarrow P)$	$Q \leftrightarrow (P \rightarrow R)$	$((S \vee R) \vee (\neg P \leftrightarrow P)) \wedge (Q \leftrightarrow (P \rightarrow R))$
T	T	T	F	T	T	T
T	T	F	T	T	F	F
T	T	F	F	F	F	F
T	F	T	T	T	F	F
T	F	T	F	T	F	F
T	F	F	T	T	T	T
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	T	F	T	T	T
F	T	F	T	T	T	T
F	T	F	F	F	T	F
F	F	T	T	T	F	F
F	F	T	F	T	F	F
F	F	F	T	T	F	F
F	F	F	F	F	F	F

Rows where F = T: **6**.

Q.18 [MSQ, 2M · Propositional Logic] **Answer: C** (C: $((Q \wedge P) \rightarrow Q) \wedge (Q \rightarrow (Q \wedge P))$)

Compare truth tables with $F = (Q \wedge P) \leftrightarrow Q$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

- (A) $((Q \wedge P) \wedge Q) \vee ((Q \wedge P) \wedge \neg Q)$ — differs at $P=T, Q=F$ ($F = T$, option = F) $\Rightarrow$ not equivalent.
 (B) $(Q \wedge P) \leftrightarrow \neg Q$ — differs at $P=T, Q=T$ ($F = T$, option = F) $\Rightarrow$ not equivalent.
 (C) $((Q \wedge P) \rightarrow Q) \wedge (Q \rightarrow (Q \wedge P))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.
 (D) $((Q \wedge P) \rightarrow Q) \wedge (\neg Q \rightarrow \neg(Q \wedge P))$ — differs at $P=F, Q=T$ ($F = F$, option = T) $\Rightarrow$ not equivalent.

Q.19 [MSQ, 2M · Predicate Logic] **Answer: B, D** (B: $\exists x R(x, x)$; D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,2), (1,3), (2,3), (3,1), (3,3), (4,2)\}$.

- (A) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).
 (B) $\exists x R(x, x)$: **TRUE** — self-loop at 3.
 (C) $\exists x \forall y R(x, y)$: **FALSE** — no element has edges to all elements (incl. itself).
 (D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 4 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(p \rightarrow q) \wedge (p \rightarrow \neg q)$; D: $(p \vee q) \rightarrow (p \wedge \neg q)$)

(A) $\equiv p \rightarrow (q \wedge \neg q) \equiv \neg p$: true at $p=F$, false at $p=T$, so contingent. (B) An implication and its contrapositive are equivalent, so this is valid. (C) $p \leftrightarrow q$ and $p \oplus q$ are negations of each other, so their conjunction is unsatisfiable. (D) True at $p=q=F$ (antecedent false), false at $p=F, q=T$, so contingent.

Q.21 [NAT, 2M · Exact Inference] **Answer: 4**

Initial factors (one per CPT, evidence removed from scopes): $f(A), f(B), f(A, B, C), f(A, C, D), f(E), f(B, E, F)$.
 Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(B), f(A, B, C), f(B, E, F)$	$\{A, B, C, E, F\}$	$f(A, C, E, F)$
F	$f(A, C, E, F)$	$\{A, C, E, F\}$	$f(A, C, E)$
C	$f(A, C, D), f(A, C, E)$	$\{A, C, D, E\}$	$f(A, D, E)$
A	$f(A), f(A, D, E)$	$\{A, D, E\}$	$f(D, E)$
D	$f(D, E)$	$\{D, E\}$	$f(E)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over E and normalise.

Q.22 [NAT, 2M · Sampling] **Answer: 0.74 to 0.76**

Rejection sampling keeps only samples consistent with the evidence $C = F$: samples 3, 4, 7, 10, 15, 16, 17, 19 (8 accepted, 12 rejected).

Among the accepted samples, $B = T$ in 6. Estimate = $6/8 = \mathbf{0.75}$.

(For this network the exact value is 0.8571; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Q.23 [NAT, 2M · Exact Inference] Answer: 0.748 to 0.752

Factorisation: $P(\text{Flu, Fever, Cough, Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever}) \cdot P(\text{Cough} \mid \text{Flu, Fever}) \cdot P(\text{Fatigue} \mid \text{Cough})$.

Inference by enumeration: $P(\text{Fatigue} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Fatigue, e, hidden})$.

Hidden: Fl, Fe	Fa	product of CPT entries (Fl · Fe · Co · Fa)	value
T, T	T	$0.4 \times 0.9 \times 0.5 \times 0.75$	0.135
T, F	T	$0.4 \times 0.1 \times 0.7 \times 0.75$	0.021
F, T	T	$0.6 \times 0.9 \times 0.2 \times 0.75$	0.081
F, F	T	$0.6 \times 0.1 \times 0.8 \times 0.75$	0.036
T, T	F	$0.4 \times 0.9 \times 0.5 \times 0.25$	0.045
T, F	F	$0.4 \times 0.1 \times 0.7 \times 0.25$	0.007
F, T	F	$0.6 \times 0.9 \times 0.2 \times 0.25$	0.027
F, F	F	$0.6 \times 0.1 \times 0.8 \times 0.25$	0.012

$P(\text{Fatigue}=\text{T}, e) = 0.273$; $P(\text{Fatigue}=\text{F}, e) = 0.091$; $P(e) = 0.364$.

$P(\text{Fatigue}=\text{T} \mid e) = 0.273 / 0.364 = \mathbf{0.75} \approx 0.750$.

Q.24 [NAT, 2M · Probability for AI] Answer: 0.428 to 0.429

$P(y \vee z) = 1 - P(\neg y \wedge \neg z) = 1 - [P(x, \neg y, \neg z) + P(\neg x, \neg y, \neg z)] = 1 - (0.10 + 0.20) = 0.70$. $P(x \wedge (y \vee z)) = P(x, y, z) + P(x, y, \neg z) + P(x, \neg y, z) = 0.10 + 0.05 + 0.15 = 0.30$. Therefore $P(x \mid y \vee z) = 0.30 / 0.70 \approx 0.4286 \approx 0.429$.

Q.25 [MSQ, 2M · Conditional Independence] Answer: C, D (C: $F \perp E \mid C, D$; D: $A \perp D \mid B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp A$: **FALSE** — the path $B \leftarrow A$ is active given $\{\}$.

(B) $A \perp F$: **FALSE** — the path $A \rightarrow B \rightarrow D \rightarrow F$ is active given $\{\}$.

(C) $F \perp E \mid C, D$: **TRUE** — $F \leftarrow A \rightarrow B \rightarrow C \rightarrow E$ (blocked: observed chain node C); $F \leftarrow A \rightarrow B \rightarrow D \rightarrow E$ (blocked: observed chain node D); $F \leftarrow A \rightarrow C \leftarrow B \rightarrow D \rightarrow E$ (blocked: observed chain node D); $F \leftarrow A \rightarrow C \rightarrow E$ (blocked: observed chain node C); $F \leftarrow A \rightarrow H \leftarrow C \leftarrow B \rightarrow D \rightarrow E$ (blocked: collider H with neither it nor a descendant observed); $F \leftarrow A \rightarrow H \leftarrow C \rightarrow E$ (blocked: collider H with neither it nor a descendant observed) (+8 more paths, all blocked).

(D) $A \perp D \mid B$: **TRUE** — $A \rightarrow B \rightarrow C \rightarrow E \leftarrow D$ (blocked: observed chain node B); $A \rightarrow B \rightarrow D$ (blocked: observed chain node B); $A \rightarrow C \leftarrow B \rightarrow D$ (blocked: collider C with neither it nor a descendant observed); $A \rightarrow C \rightarrow E \leftarrow D$ (blocked: collider E with neither it nor a descendant observed); $A \rightarrow F \leftarrow D$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow F \rightarrow G \leftarrow D$ (blocked: collider G with neither it nor a descendant observed) (+2 more paths, all blocked).

Mock Test 097

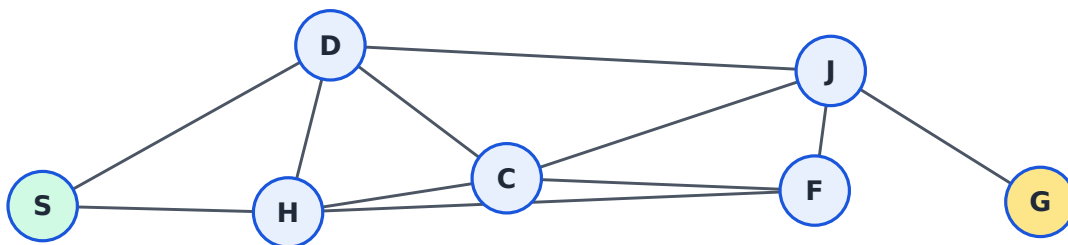
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 5 / 14	12 / 15 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A – 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

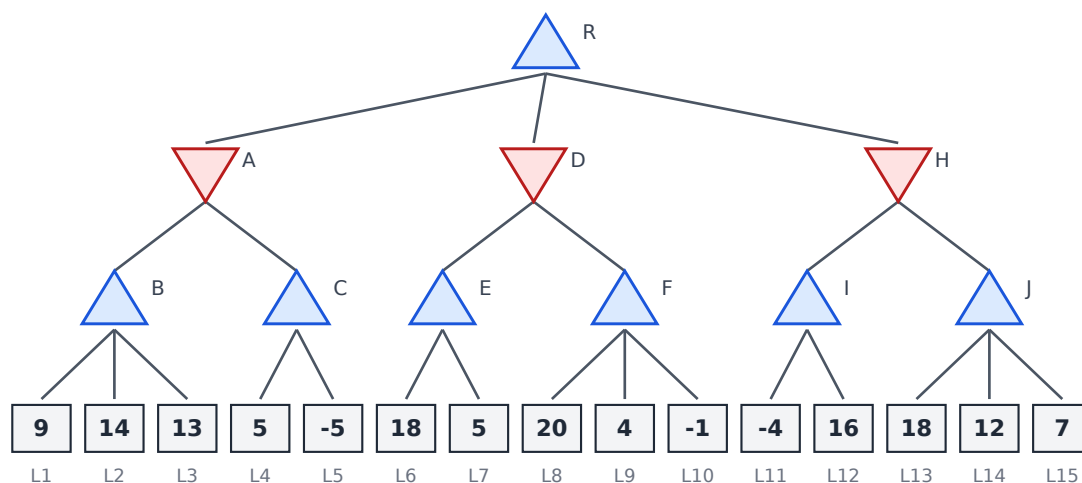
Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, H, D, F, C, J, G
- ☐ (B) S, D, C, F, H, J, G
- ☐ (C) S, D, C, H, J, F, G
- ☐ (D) S, D, H, C, J, F, G

Q.2 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



MAX node MIN node

terminal utility

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Uninformed Search

AIMA bounds the time and space of uniform-cost search by $O(b^{1+\lceil C^*/\epsilon \rceil})$, where C^* is the optimal solution cost and ϵ is the minimum step cost. For $C^* = 10$ and $\epsilon = 2$, what is the exponent of b in this bound?

Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

A search tree has uniform branching factor $b = 5$; the shallowest goal is at depth $d = 5$ and happens to be the **last** node generated at that depth. BFS applies the goal test when a node is **generated**. How many nodes are generated in total (do **not** count the root)?

Answer (numerical): _____

Q.5 [NAT · 1 mark · no negative marking] Propositional Logic

The formula $(A \wedge B) \vee (C \wedge D) \vee (E \wedge F)$ is converted to CNF by repeatedly distributing $\vee$ over $\wedge$ (no simplification). How many clauses does the result have?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 4 elements, so there are $4^4 = 256$ ways to interpret the pair (P, Q) . For how many of them is $\forall x (P(x) \vee Q(x)) \wedge \exists x \neg P(x)$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = (R \rightarrow (Q \wedge Q)) \leftrightarrow (P \wedge (\neg Q \rightarrow Q))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

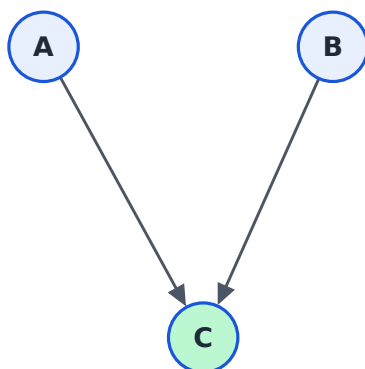
Q.8 [MCQ · 1 mark · negative: -1/3] Sampling

What is the main drawback of rejection sampling for computing $P(X | e)$?

- ☐ (A) It rejects most samples when $P(e)$ is small, and the fraction accepted drops exponentially with the number of evidence variables
- ☐ (B) It is not consistent: estimates do not converge even with infinitely many samples
- ☐ (C) It requires the Markov blanket of every variable
- ☐ (D) It can only be used when the evidence variables are root nodes

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T | \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).

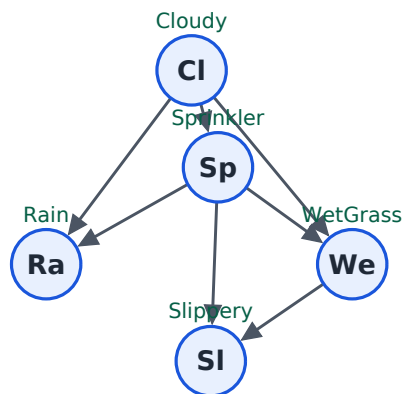


$P(A=T)$	$P(B=T)$	A	B	$P(C=T ...)$
0.1	0.95	T	T	0.25
		T	F	0.25
		F	T	0.25
		F	F	0.1

Answer (numerical): _____

Q.10 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We, Sl. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, Sl = Slippery.) What is the probability that a single run produces the sample **Cl=T, Sp=T, Ra=T, We=F, Sl=T**? (round off to 4 decimal places)



P(Cl=T)
0.8

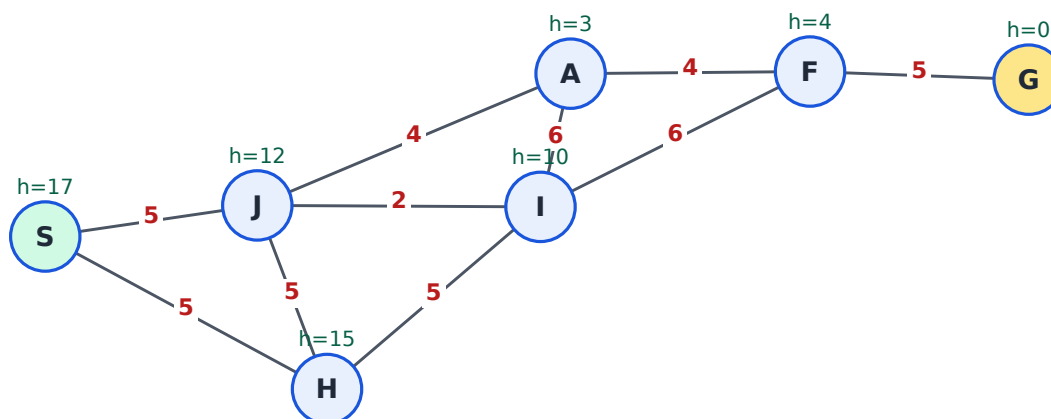
Cl	P(Sp=T ...)	Cl	Sp	P(Ra=T ...)
T	0.9	T	T	0.8
F	0.1	T	F	0.1
		F	T	0.4
		F	F	0.05

Cl	Sp	P(We=T ...)	Sp	We	P(Sl=T ...)
T	T	0.3	T	T	0.7
T	F	0.25	T	F	0.8
F	T	0.2	F	T	0.95
F	F	0.95	F	F	0.9

Answer (numerical): _____

Section B — 2-mark questions**Q.11** [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
☐ (B) $h(J)$ overestimates the true cost from J to G.
☐ (C) h is admissible.
☐ (D) The edge J-A violates the consistency condition $h(J) \leq c(J,A) + h(A)$.

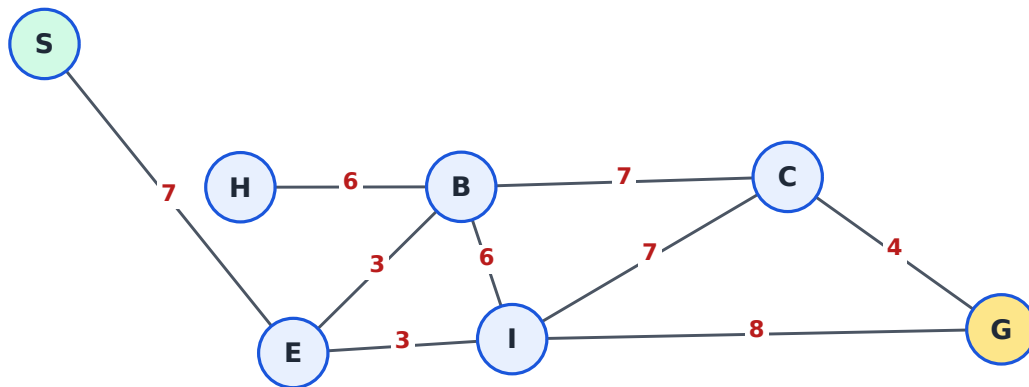
Q.12 [MCQ · 2 marks · negative: -2/3] Adversarial Search

To get the maximum pruning from alpha-beta search, how should children be ordered?

- ☐ (A) At MAX nodes in decreasing order of value, at MIN nodes in increasing order of value
☐ (B) At MAX nodes in increasing order of value, at MIN nodes in decreasing order of value
☐ (C) In increasing order at both node types
☐ (D) Ordering does not affect pruning

Q.13 [NAT · 2 marks · no negative marking] Uninformed Search – UCS

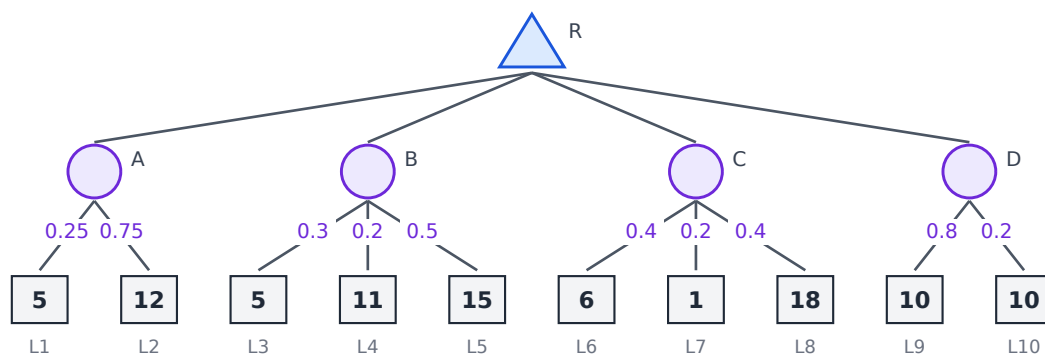
Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. How many nodes are **expanded** (count S and G)?



Answer (numerical): _____

Q.14 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



△ MAX node ○ CHANCE node (edge = probability) □ terminal utility

Answer (numerical): _____

Q.15 [NAT · 2 marks · no negative marking] Propositional Logic

How many of the 32 truth assignments to $x_1, \dots, x_5$ satisfy $(x_1 \rightarrow x_2) \wedge (x_2 \rightarrow x_3) \wedge (x_3 \rightarrow x_4) \wedge (x_4 \rightarrow x_5)$?

Answer (numerical): _____

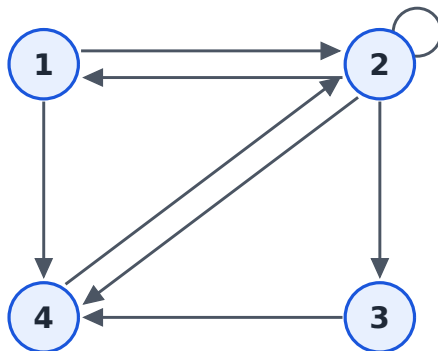
Q.16 [MCQ · 2 marks · negative: -2/3] Predicate Logic

Converting $\forall x ((\exists y P(x, y)) \rightarrow (\exists z Q(x, z)))$ to clausal form (with F a new Skolem function) gives

- ☐ (A) $\neg P(x, y) \vee Q(x, F(x))$
- ☐ (B) $\neg P(x, G(x)) \vee Q(x, F(x))$
- ☐ (C) $\neg P(x, y) \vee Q(x, C)$, with C a new constant
- ☐ (D) $\neg P(x, y) \wedge Q(x, F(x))$

Q.17 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
☐ (B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (C) $\exists y \forall x R(x, y)$
☐ (D) $\forall x \exists y (x \neq y \wedge R(x, y))$

Q.18 [MCQ · 2 marks · negative: -2/3] Predicate Logic

What is the most general unifier of $P(f(x), g(y), y)$ and $P(f(g(z)), w, z)$, where x, y, z, w are variables?

- ☐ (A) $\{x/g(z), w/g(z), y/z\}$
☐ (B) $\{x/g(z), w/g(y)\}$
☐ (C) $\{x/g(a), w/g(a), y/a, z/a\}$
☐ (D) The atoms are not unifiable

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C1: (\neg R), C2: (P \vee \neg S), C3: (\neg P \vee S), C4: (\neg P \vee \neg S)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg Q \vee \neg S)$
☐ (B) $(\neg S)$
☐ (C) $(\neg R \vee \neg S)$
☐ (D) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)

Q.20 [MSQ · 2 marks · no negative marking] Propositional Logic – Validity

Which of the following formulas is/are **valid** (tautologies)?

- ☐ (A) $((R \rightarrow P) \wedge \neg R) \rightarrow \neg P$
☐ (B) $((\neg Q \vee P) \wedge (\neg \neg Q \vee R)) \rightarrow (P \vee R)$
☐ (C) $\neg R \rightarrow (Q \rightarrow \neg R)$
☐ (D) $((\neg R \rightarrow Q) \rightarrow \neg R) \rightarrow \neg R$

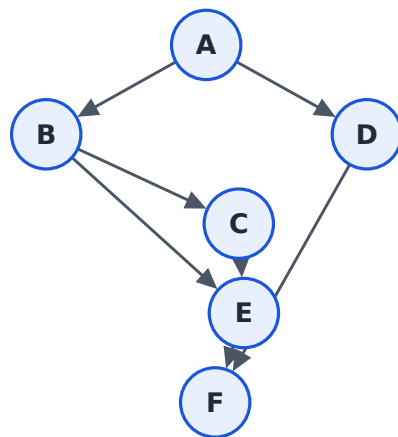
Q.21 [NAT · 2 marks · no negative marking] Variable Elimination

Consider the Bayesian network over Boolean variables Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M) with edges $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow J$, $A \rightarrow M$ and CPTs: $P(b) = 0.02$, $P(e) = 0.01$; $P(a | b, e) = 0.95$, $P(a | b, \neg e) = 0.90$, $P(a | \neg b, e) = 0.30$, $P(a | \neg b, \neg e) = 0.01$; $P(j | a) = 0.8$, $P(j | \neg a) = 0.05$; $P(m | a) = 0.7$, $P(m | \neg a) = 0.02$. In computing $P(B | j, m)$ by variable elimination, A is summed out first, producing a factor $f(B, E) = \sum_a P(a | B, E) P(j | a) P(m | a)$. Compute $f(b, e)$. (Round to 3 decimal places.)

Answer (numerical): _____

Q.22 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

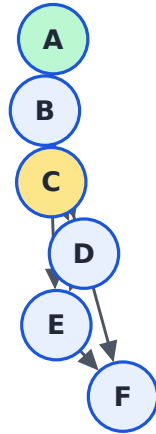
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp C | B$
- ☐ (B) $D \perp A | B, C$
- ☐ (C) $A \perp E$
- ☐ (D) $D \perp E | A$

Q.23 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

Variable elimination is used to compute $P(A \mid C = T)$ in the Bayesian network below (Boolean variables; evidence $C = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B**, **D**, **E**, **F**. What is the scope of the new factor produced when **D** is summed out?



- ☐ (A) $f(A, E)$
☐ (B) $f(A, D, E, F)$
☐ (C) $f(E, F)$
☐ (D) $f(A, E, F)$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: FI = Flu, Fe = Fever, Co = Cough, Fa = Fatigue.) Compute $P(\text{Fatigue} = T \mid \text{Fever} = F)$ (round off to 3 decimal places).



FI	Co	P(Fa=T ...)
T	T	0.75
T	F	0.1
F	T	0.1
F	F	0.4

FI	P(Fe=T FI)	FI	Fe	P(Co=T ...)
T	0.4	T	T	0.1
F	0.3	T	F	0.6
		F	T	0.6
		F	F	0.05

FI	Co	P(Fa=T ...)
T	T	0.75
T	F	0.1
F	T	0.1
F	F	0.4

Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Sampling – Rejection sampling

The following 18 samples were generated by prior sampling from a Bayesian network over Boolean variables A, B, C, D. Using **rejection sampling**, estimate $P(D = T \mid A = F)$ (round off to 2 decimal places).

#	A	B	C	D
1	F	T	F	T
2	F	T	F	T
3	F	F	F	F
4	F	T	T	T
5	T	T	T	T
6	F	F	T	F
7	F	T	F	T
8	F	T	F	T
9	F	T	F	T
10	F	F	F	F
11	F	T	F	T
12	F	T	T	T
13	F	T	T	T
14	F	F	F	T
15	F	T	F	T
16	T	T	T	T
17	T	T	T	T
18	T	T	T	T

Answer (numerical): _____

Answer Key — Mock Test 097

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	NAT	2	Adversarial Search	11.19 to 11.21
2	NAT	1	Adversarial Search	18	15	NAT	2	Propositional Logic	6
3	NAT	1	Uninformed Search	6	16	MCQ	2	Predicate Logic	A
4	NAT	1	Uninformed Search	3905	17	MSQ	2	Predicate Logic	A, D
5	NAT	1	Propositional Logic	8	18	MCQ	2	Predicate Logic	A
6	NAT	1	Predicate Logic	65	19	MSQ	2	Propositional Logic	B
7	NAT	1	Propositional Logic	4	20	MSQ	2	Propositional Logic	B, C, D
8	MCQ	1	Sampling	A	21	NAT	2	Variable Elimination	0.531 to 0.533
9	NAT	1	Exact Inference	0.241 to 0.245	22	MSQ	2	Conditional Independence	A, D
10	NAT	1	Sampling	0.3224 to 0.3228	23	MCQ	2	Exact Inference	D
11	MSQ	2	Informed Search	C, D	24	NAT	2	Exact Inference	0.482 to 0.486
12	MCQ	2	Adversarial Search	A	25	NAT	2	Sampling	0.78 to 0.8
13	NAT	2	Uninformed Search	7					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: D** (D: S, D, H, C, J, F, G)

Adjacency lists (alphabetical): C: {D, F, H, J}; D: {C, H, J, S}; F: {C, H, J}; G: {J}; H: {C, D, F, S}; J: {C, D, F, G}; S: {D, H}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[D, H]
2	D	[H, C, J]
3	H	[C, J, F]

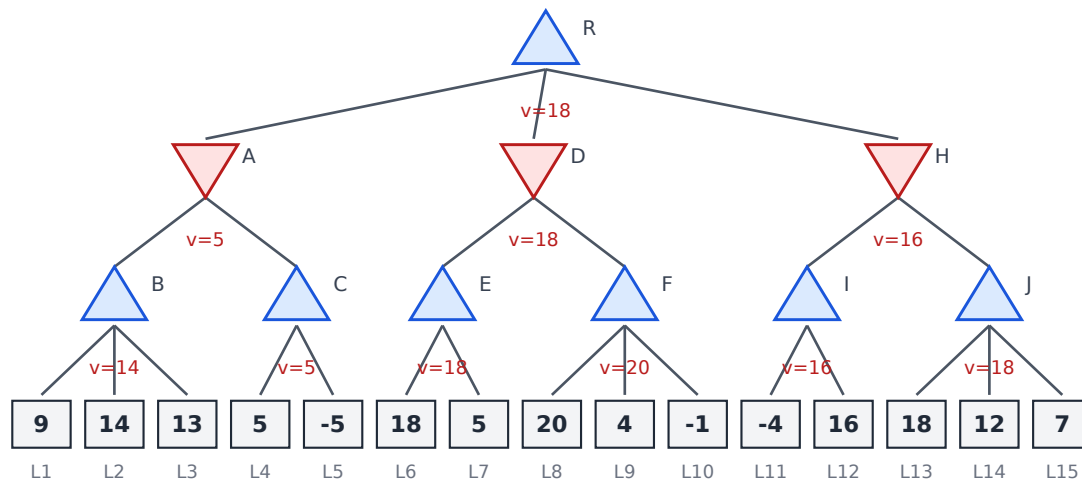
Step	Expanded	Frontier (FIFO) after expansion
4	C	[J, F]
5	J	[F, G]
6	F	[G]
7	G	[] ← goal popped, stop

Expansion order: **S, D, H, C, J, F, G**. The path returned is $S \rightarrow D \rightarrow J \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Adversarial Search] Answer: 18

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 14$, $C = 5$, $A = 5$, $E = 18$, $F = 20$, $D = 18$, $I = 16$, $J = 18$, $H = 16$. Root value = **18**; MAX's optimal first move is towards D.

Q.3 [NAT, 1M · Uninformed Search] Answer: 6

The exponent is $1 + \lceil C^*/\epsilon \rceil = 1 + \lceil 10/2 \rceil = 1 + 5 = \mathbf{6}$. Each step costs at least ϵ , so an optimal path has at most $C^*/\epsilon = 5$ steps. UCS can then generate one further level when it expands nodes of cost about C^* .

Q.4 [NAT, 1M · Uninformed Search] Answer: 3905

With goal test at generation, BFS generates every node at depths $1 \dots d$ before (and including) the goal (the worst case). $N = b + b^2 + \dots + b^d = 5 + 25 + 125 + 625 + 3125 = \mathbf{3905}$. This is the $O(b^d)$ of AIMA.

Q.5 [NAT, 1M · Propositional Logic] Answer: 8

Distribution produces one clause per way of choosing one literal from each conjunction: $\{A, B\} \times \{C, D\} \times \{E, F\}$, i.e. $2 \times 2 \times 2 = 8$ clauses such as $A \vee C \vee E$, $A \vee C \vee F$, ..., $B \vee D \vee F$. No clause is a tautology or a duplicate because all six symbols are distinct. This illustrates the exponential blow-up of naive CNF conversion.

Q.6 [NAT, 1M · Predicate Logic] Answer: 65

Interpret P and Q as subsets of the domain; 3^n interpretations satisfy the $\forall$ -part (each element: not both false); remove the 2^n of them with P true everywhere (Q then arbitrary) $\Rightarrow 3^n - 2^n$. With $n = 4$ this gives **65** (verified by enumerating all interpretations).

Q.7 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$R \rightarrow (Q \wedge Q)$	$P \wedge (\neg Q \rightarrow Q)$	$(R \rightarrow (Q \wedge Q)) \leftrightarrow (P \wedge (\neg Q \rightarrow Q))$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	F	T
T	F	F	T	F	F
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	F	F	T

P	Q	R	$R \rightarrow (Q \wedge Q)$	$P \wedge (\neg Q \rightarrow Q)$	$(R \rightarrow (Q \wedge Q)) \leftrightarrow (P \wedge (\neg Q \rightarrow Q))$
F	F	F	T	F	F

F has 4 models, so it is false in $8 - 4 = 4$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.8 [MCQ, 1M · Sampling] Answer: A (A: It rejects most samples when $P(e)$ is small, and the fraction accepted drops exponentially with the number of evidence variables)

Rejection sampling keeps only samples consistent with e , so about $N \cdot P(e)$ samples are useful; $P(e)$ typically shrinks exponentially as evidence grows. It is consistent (B is false). Markov blankets are used by Gibbs sampling, not rejection sampling (C false). Evidence may be anywhere in the network (D false).

Q.9 [NAT, 1M · Exact Inference] Answer: 0.241 to 0.245

Only the ancestors of C (A, B) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
A=T, B=T	0.02375
A=T, B=F	0.00125
A=F, B=T	0.21375
A=F, B=F	0.0045

Sum = **0.2432** ≈ 0.243 .

Q.10 [NAT, 1M · Sampling] Answer: 0.3224 to 0.3228

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i \mid \text{parents}(X_i))$.

$P(CI=T) \times P(Sp=T \mid CI=T) \times P(Ra=T \mid CI=T, Sp=T) \times P(We=F \mid CI=T, Sp=T) \times P(SI=T \mid Sp=T, We=F)$
 $= 0.8 \times 0.9 \times 0.8 \times 0.7 \times 0.8 = \mathbf{0.32256} \approx 0.3226$.

Q.11 [MSQ, 2M · Informed Search] Answer: C, D (C: h is admissible.; D: The edge J-A violates the consistency condition $h(J) \leq c(J,A) + h(A)$.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	F	G	H	I	J	S
$h(n)$	3	4	0	15	10	12	17
$h^*(n)$	9	5	0	16	11	13	18
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: $I \rightarrow A$ ($10 > 6 + 3$), $J \rightarrow A$ ($12 > 4 + 3$). Hence consistent: **no**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.12 [MCQ, 2M · Adversarial Search] Answer: A (A: At MAX nodes in decreasing order of value, at MIN nodes in increasing order of value)

Pruning is greatest when the best move is examined first: the highest value for MAX and the lowest for MIN. That gives the tightest α (or β) as early as possible, so later siblings are cut sooner. (B) is the worst ordering, under which alpha-beta degenerates to $O(b^m)$. (C) is best for only one of the two players. (D) is false: ordering decides whether the cost is $O(b^{m/2})$ or $O(b^m)$.

Q.13 [NAT, 2M · Uninformed Search] Answer: 7

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	E(7)
2	E (7)	B(10), I(10)
3	B (10)	I(10), H(16), C(17)
4	I (10)	H(16), C(17), G(18)
5	H (16)	C(17), G(18)
6	C (17)	G(18)
7	G (18)	← goal, stop

Path returned: **S** → **E** → **I** → **G**, cost = 7 + 3 + 8 = **18**. Nodes expanded (including S and G): **7** (S, E, B, I, H, C, G).

Q.14 [NAT, 2M · Adversarial Search] **Answer: 11.19 to 11.21**

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.25 \times 5 + 0.75 \times 12 = 10.25$
- $B = 0.3 \times 5 + 0.2 \times 11 + 0.5 \times 15 = 11.2$
- $C = 0.4 \times 6 + 0.2 \times 1 + 0.4 \times 18 = 9.8$
- $D = 0.8 \times 10 + 0.2 \times 10 = 10$

Root = $\max(10.25, 11.2, 9.8, 10) = \mathbf{11.2}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.15 [NAT, 2M · Propositional Logic] **Answer: 6**

The constraints say once some x_i is T, every later x_j ($j > i$) must be T. So a model is a string of F's followed by a string of T's, e.g. FTTT. The position where T's begin can be 1, 2, 3, 4, 5 or 'never' (all F), giving 6 models. Brute-force over all 32 assignments confirms 6.

Q.16 [MCQ, 2M · Predicate Logic] **Answer: A** ($A: \neg P(x, y) \vee Q(x, F(x))$)

Eliminate $\rightarrow$: $\forall x (\neg \exists y P(x, y) \vee \exists z Q(x, z)) \equiv \forall x (\forall y \neg P(x, y) \vee \exists z Q(x, z))$. y is now universal, so it stays a variable. z lies only within the scope of $\forall x$, so it becomes $F(x)$. Dropping universals gives the clause $\neg P(x, y) \vee Q(x, F(x))$. Option B wrongly Skolemizes the universal y ; option C ignores the dependence of z on x ; option D is not a clause and changes the meaning.

Q.17 [MSQ, 2M · Predicate Logic] **Answer: A, D** ($A: \forall x \exists y R(x, y)$; $D: \forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,2), (1,4), (2,1), (2,2), (2,3), (2,4), (3,4), (4,2)\}$.

(A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(B) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(1,2) \wedge R(2,1)$ but $\neg R(1,1)$.

(C) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(D) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.18 [MCQ, 2M · Predicate Logic] **Answer: A** ($A: \{x/g(z), w/g(z), y/z\}$)

$f(x)$ vs $f(g(z))$ gives $x/g(z)$. $g(y)$ vs w gives $w/g(y)$. y vs z gives y/z , and composing updates $w/g(y)$ to $w/g(z)$.

Both atoms become $P(f(g(z)), g(z), z)$. Option B is not a unifier because y and z remain different. Option C is a unifier but not most general (it is an instance of A with z/a). No occurs-check failure arises, so D is wrong.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: B** ($B: (\neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg Q \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(B) $(\neg S)$ is derivable:

1. resolve $(P \vee \neg S)$ and $(\neg P \vee \neg S)$ on $P \Rightarrow (\neg S)$

(C) $(\neg R \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) The resolution closure contains 6 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

Q.20 [MSQ, 2M · Propositional Logic] **Answer: B, C, D** ($B: ((\neg Q \vee P) \wedge (\neg \neg Q \vee R)) \rightarrow (P \vee R)$; $C: \neg R \rightarrow (Q \rightarrow \neg R)$; $D: ((\neg R \rightarrow Q) \rightarrow \neg R) \rightarrow \neg R$)

(A) $((R \rightarrow P) \wedge \neg R) \rightarrow \neg P$: **not valid** — counter-model $P=T, R=F$.

(B) $((\neg Q \vee P) \wedge (\neg \neg Q \vee R)) \rightarrow (P \vee R)$: true in all 8 assignments $\Rightarrow$ **valid**.

(C) $\neg R \rightarrow (Q \rightarrow \neg R)$: true in all 4 assignments $\Rightarrow$ **valid**.

(D) $((\neg R \rightarrow Q) \rightarrow \neg R) \rightarrow \neg R$: true in all 4 assignments $\Rightarrow$ **valid**.

Useful facts: contraposition, hypothetical syllogism, modus ponens/tollens, Peirce's law, De Morgan, resolution and exportation are tautologies; affirming the consequent, denying the antecedent and the converse are fallacies.

Q.21 [NAT, 2M · Variable Elimination] Answer: 0.531 to 0.533

$f(b, e) = P(a | b, e) P(j | a) P(m | a) + P(\neg a | b, e) P(j | \neg a) P(m | \neg a) = 0.95 \times 0.8 \times 0.7 + 0.05 \times 0.05 \times 0.02 = 0.532 + 0.00005 = 0.53205 \approx 0.532$. This is not a probability distribution over B, E; it is an intermediate factor.

Q.22 [MSQ, 2M · Conditional Independence] Answer: A, D (A: $A \perp C | B$; D: $D \perp E | A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $A \perp C | B$: **TRUE** — $A \rightarrow B \rightarrow C$ (blocked: observed chain node B); $A \rightarrow B \rightarrow E \leftarrow C$ (blocked: observed chain node B); $A \rightarrow D \rightarrow F \leftarrow E \leftarrow B \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $A \rightarrow D \rightarrow F \leftarrow E \leftarrow C$ (blocked: collider F with neither it nor a descendant observed).

(B) $D \perp A | B, C$: **FALSE** — the path $D \leftarrow A$ is active given $\{B, C\}$.

(C) $A \perp E$: **FALSE** — the path $A \rightarrow B \rightarrow C \rightarrow E$ is active given $\{\}$.

(D) $D \perp E | A$: **TRUE** — $D \leftarrow A \rightarrow B \rightarrow C \rightarrow E$ (blocked: observed fork node A); $D \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed fork node A); $D \rightarrow F \leftarrow E$ (blocked: collider F with neither it nor a descendant observed).

Q.23 [MCQ, 2M · Exact Inference] Answer: D (D: $f(A, E, F)$)

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(A, B)$, $f(B, D)$, $f(D, E)$, $f(B, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B)$, $f(A, B)$, $f(B, D)$, $f(B, E, F)$	$\{A, B, D, E, F\}$	$f(A, D, E, F)$
D	$f(D, E)$, $f(A, D, E, F)$	$\{A, D, E, F\}$	$f(A, E, F)$
E	$f(A, E, F)$	$\{A, E, F\}$	$f(A, F)$
F	$f(A, F)$	$\{A, F\}$	$f(A)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over A and normalise.

Q.24 [NAT, 2M · Exact Inference] Answer: 0.482 to 0.486

Factorisation: $P(\text{Flu, Fever, Cough, Fatigue}) = P(\text{Flu}) \cdot P(\text{Fever} | \text{Flu}) \cdot P(\text{Cough} | \text{Flu, Fever}) \cdot P(\text{Fatigue} | \text{Flu, Cough})$.

Inference by enumeration: $P(\text{Fatigue} | e) = \alpha \sum_{\text{hidden}} P(\text{Fatigue, e, hidden})$.

Hidden: Fl, Co	Fa	product of CPT entries (Fl · Fe · Co · Fa)	value
T, T	T	$0.95 \times 0.6 \times 0.6 \times 0.75$	0.2565
T, F	T	$0.95 \times 0.6 \times 0.4 \times 0.1$	0.0228
F, T	T	$0.05 \times 0.7 \times 0.05 \times 0.1$	0.00018
F, F	T	$0.05 \times 0.7 \times 0.95 \times 0.4$	0.0133
T, T	F	$0.95 \times 0.6 \times 0.6 \times 0.25$	0.0855
T, F	F	$0.95 \times 0.6 \times 0.4 \times 0.9$	0.2052
F, T	F	$0.05 \times 0.7 \times 0.05 \times 0.9$	0.00158
F, F	F	$0.05 \times 0.7 \times 0.95 \times 0.6$	0.01995

$P(\text{Fatigue}=\text{T}, e) = 0.29277$; $P(\text{Fatigue}=\text{F}, e) = 0.31222$; $P(e) = 0.605$.

$P(\text{Fatigue}=\text{T} | e) = 0.29277 / 0.605 = \mathbf{0.4839} \approx 0.484$.

Q.25 [NAT, 2M · Sampling] Answer: 0.78 to 0.8

Rejection sampling keeps only samples consistent with the evidence $A = F$: samples 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 (14 accepted, 4 rejected).

Among the accepted samples, $D = T$ in 11. Estimate = $11/14 = \mathbf{0.7857}$.

(For this network the exact value is 0.84; the estimate converges to it as $N \rightarrow \infty$, but with so few accepted samples the variance is high — the main weakness of rejection sampling when $P(e)$ is small.)

Mock Test 098

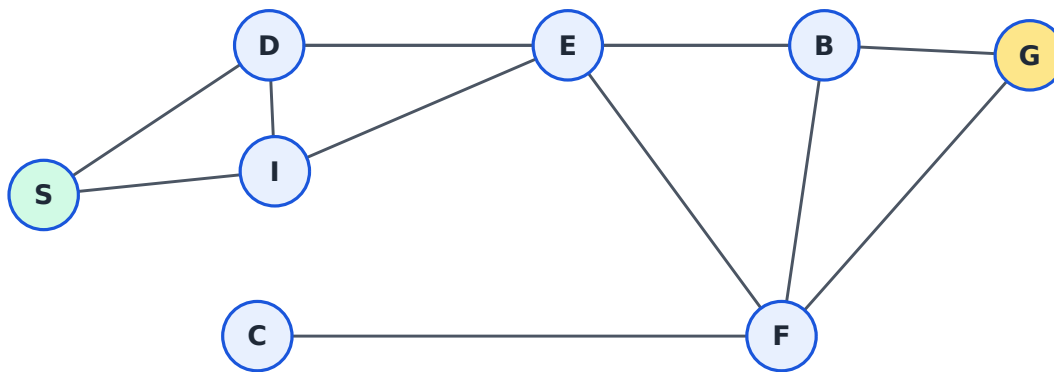
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	6 / 5 / 14	14 / 13 / 13

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, D, B, E, F, C, G
- ☐ (B) S, D, B, I, E, F, G
- ☐ (C) S, I, D, E, B, F, G
- ☐ (D) S, D, I, B, E, F, G

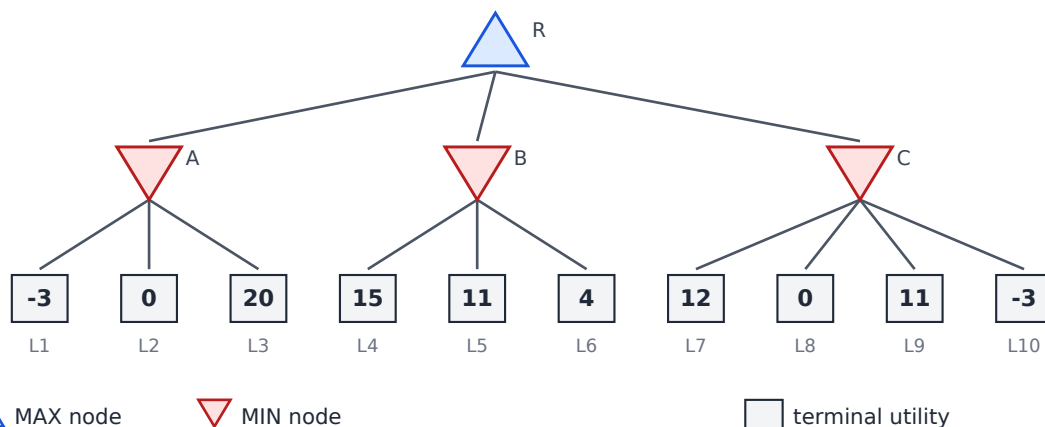
Q.2 [NAT · 1 mark · no negative marking] Uninformed Search – Complexity

Depth-limited search with limit $l = 4$ is run on a tree with branching factor $b = 4$. No goal exists within depth 4. How many nodes (including the root) does it generate before returning cutoff?

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [MCQ · 1 mark · negative: -1/3] Uninformed Search

The frontier of depth-first search is implemented as a

- ☐ (A) FIFO queue
☐ (B) LIFO stack
☐ (C) priority queue ordered by $g(n)$
☐ (D) priority queue ordered by $h(n)$

Q.5 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

Let D be a domain with exactly 4 elements and R a binary predicate symbol. In how many of the 2^{16} possible interpretations of R over D is the sentence $\forall x \forall y (R(x, y) \rightarrow \neg R(y, x))$ true?

Answer (numerical): _____

Q.6 [NAT · 1 mark · no negative marking] Propositional Logic

Of the 16 possible binary Boolean connectives $\circ$ (functions $\{T, F\}^2 \rightarrow \{T, F\}$), how many are commutative, i.e. satisfy $p \circ q \equiv q \circ p$?

Answer (numerical): _____

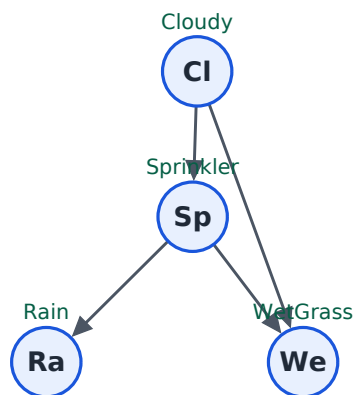
Q.7 [NAT · 1 mark · no negative marking] Predicate Logic

The KB contains the clauses $\neg \text{Man}(x) \vee \text{Mortal}(x)$ and $\text{Man}(\text{Socrates})$. To prove $\text{Mortal}(\text{Socrates})$ by resolution refutation, $\neg \text{Mortal}(\text{Socrates})$ is added. What is the minimum number of resolution steps needed to derive the empty clause?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Sampling – Prior sampling

Prior (direct) sampling is run on the Bayesian network below, sampling variables in topological order Cl, Sp, Ra, We. (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) What is the probability that a single run produces the sample **(Cl=F, Sp=T, Ra=T, We=T)**? (round off to 4 decimal places)



P(Cl=T)
0.3

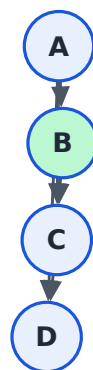
Cl	P(Sp=T ..)	Sp	P(Ra=T ...)
T	0.95	T	0.8
F	0.9	F	0.95

Cl	Sp	P(We=T ...)
T	T	0.8
T	F	0.5
F	T	0.8
F	F	0.2

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability **$P(B = T)$** (round off to 3 decimal places).



P(A=T)
0.5

A	P(B=T .)	A	B	P(C=T ...)
T	0.5	T	T	0.9
F	0.25	T	F	0.9
		F	T	0.95
		F	F	0.4

A	C	P(D=T ...)
T	T	0.95
T	F	0.2
F	T	0.3
F	F	0.7

Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Bayesian Networks

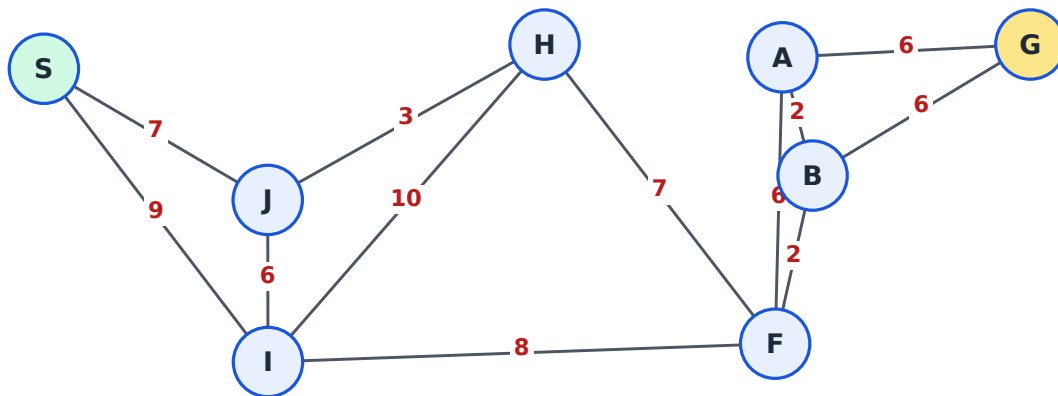
A naive Bayes model with class variable C and features $F_1, \dots, F_n$, viewed as a Bayesian network, has which structure and assumption?

- ☐ (A) Edges $C \rightarrow F_i$ for each i ; features are conditionally independent given C
- ☐ (B) Edges $F_i \rightarrow C$ for each i ; features are marginally independent
- ☐ (C) A chain $C \rightarrow F_1 \rightarrow F_2 \rightarrow \dots \rightarrow F_n$
- ☐ (D) Edges $C \rightarrow F_i$ for each i ; features are marginally independent

Section B — 2-mark questions

Q.11 [MCQ · 2 marks · negative: -2/3] Uninformed Search – UCS

Uniform-cost search (graph search with an explored set, goal test when a node is **selected for expansion**, ties broken alphabetically) is run on the weighted undirected graph below from **S** to **G**. Edge labels are step costs. Which path is returned?



- ☐ (A) $S \rightarrow J \rightarrow H \rightarrow F \rightarrow B \rightarrow G$
☐ (B) $S \rightarrow I \rightarrow F \rightarrow B \rightarrow A \rightarrow G$
☐ (C) $S \rightarrow J \rightarrow H \rightarrow F \rightarrow B \rightarrow A \rightarrow G$
☐ (D) $S \rightarrow I \rightarrow F \rightarrow B \rightarrow G$

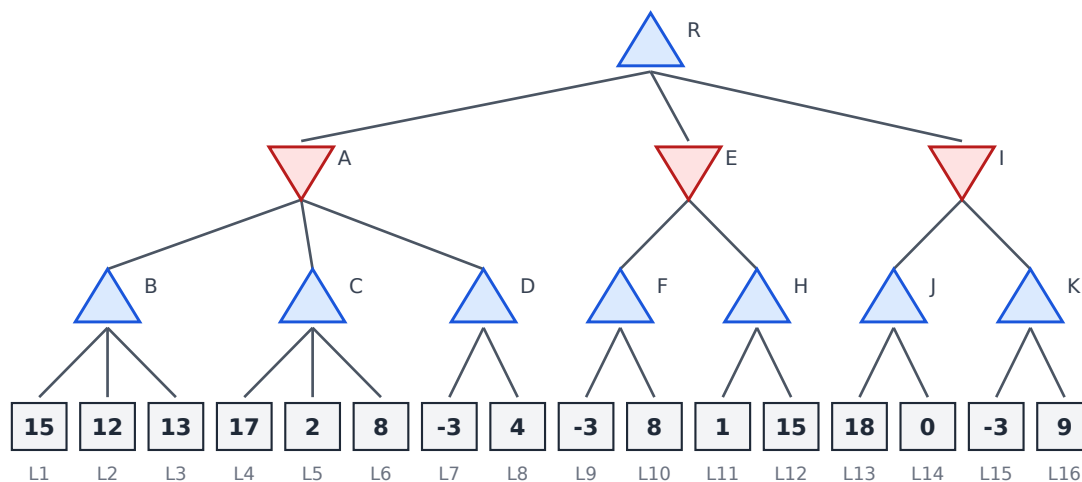
Q.12 [MCQ · 2 marks · negative: -2/3] Adversarial Search

A three-ply binary game tree has levels MAX (root), MIN, MAX, and then leaves. The leaves, left to right, are 3, 4, 2, 9, 7, 1, 8, 6. Each bottom MAX node owns two consecutive leaves. Alpha-beta processes children left to right. Which statement is correct?

- ☐ (A) Root value 7; only the last leaf (6) is pruned
☐ (B) Root value 4; leaves 2 and 9 are pruned
☐ (C) Root value 7; leaves 1 and 6 are pruned
☐ (D) Root value 8; no leaf is pruned

Q.13 [NAT · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). How many leaf nodes are **not** evaluated (pruned)?



MAX node MIN node

terminal utility

Answer (numerical): _____

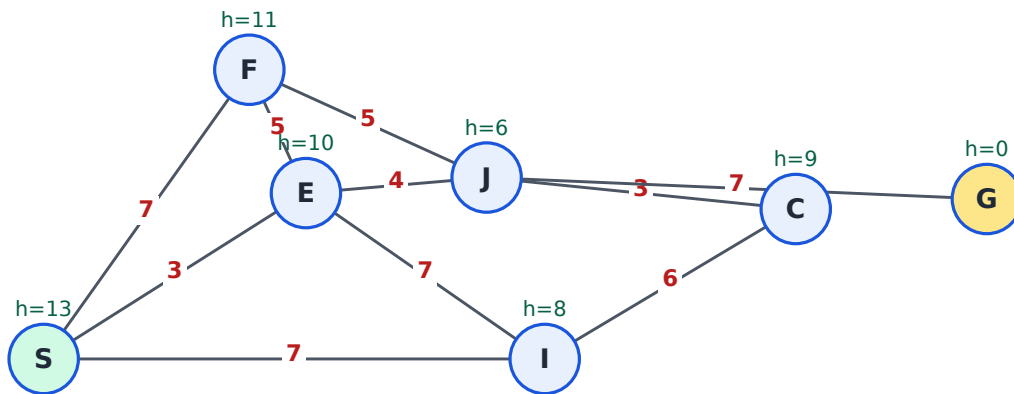
Q.14 [NAT · 2 marks · no negative marking] Informed Search

A search algorithm generates $N = 52$ nodes (not counting the root) to find a solution at depth $d = 5$. Using $N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^5$, find the effective branching factor b^* , correct to two decimal places.

Answer (numerical): _____

Q.15 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is consistent (monotone).
- ☐ (B) $h(E)$ overestimates the true cost from E to G .
- ☐ (C) The edge S - F violates the consistency condition (in at least one direction).
- ☐ (D) h is admissible.

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = Q \rightarrow (\neg Q \leftrightarrow P)$?

- ☐ (A) $\neg Q \rightarrow \neg(\neg Q \leftrightarrow P)$
- ☐ (B) $\neg Q \vee (\neg Q \leftrightarrow P)$
- ☐ (C) $(\neg Q \leftrightarrow P) \rightarrow Q$
- ☐ (D) $\neg(Q \wedge \neg(\neg Q \leftrightarrow P))$

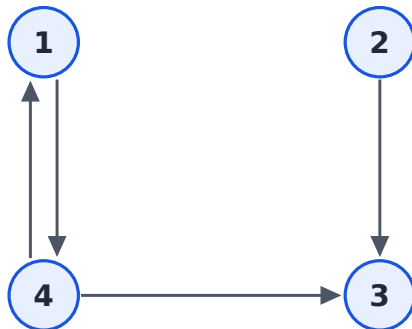
Q.17 [NAT · 2 marks · no negative marking] Propositional Logic – Model counting

The formula $F = ((R \vee Q) \leftrightarrow (\neg P \rightarrow R)) \vee (S \leftrightarrow (\neg S \wedge S))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R, S . How many maxterms (clauses) does it contain?

Answer (numerical): _____

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
☐ (B) $\exists y \forall x R(x, y)$
☐ (C) $\forall x R(x, x)$
☐ (D) $\exists x \forall y \neg R(y, x)$

Q.19 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

$C_1: (\neg Q \vee R \vee S)$, $C_2: (S)$, $C_3: (P \vee \neg R)$, $C_4: (\neg Q \vee R)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (B) $(P \vee \neg Q)$
☐ (C) $(P \vee \neg Q \vee S)$
☐ (D) $(\neg P)$

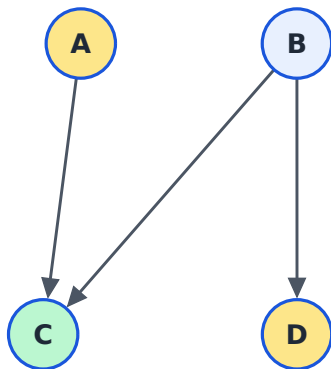
Q.20 [MCQ · 2 marks · negative: -2/3] Propositional Logic

Consider the clauses $C_1 = p \vee q \vee \neg r$ and $C_2 = \neg p \vee r \vee s$. Which one of the following statements is correct?

- ☐ (A) $q \vee s$ is a valid resolvent of C_1 and C_2
☐ (B) Every resolvent of C_1 and C_2 is a tautology
☐ (C) C_1 and C_2 have no resolvent
☐ (D) The empty clause can be derived from C_1 and C_2

Q.21 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{D} = \mathbf{T}$, $\mathbf{A} = \mathbf{T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P}(\mathbf{C} = \mathbf{T} \mid \mathbf{D} = \mathbf{T}, \mathbf{A} = \mathbf{T})$ (round off to 3 decimal places).



P(A=T)
0.8

P(B=T)	A	B	P(C=T ...)
0.1	T	T	0.05
	T	F	0.7
	F	T	0.9
	F	F	0.9

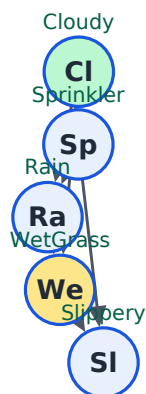
B	P(D=T ...)
T	0.2
F	0.1

#	B	C
1	T	F
2	F	T
3	F	T
4	F	F
5	F	T

Answer (numerical): _____

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Cl = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass, Sl = Slippery.) Compute $\mathbf{P}(\mathbf{Cloudy} = \mathbf{T} \mid \mathbf{WetGrass} = \mathbf{F})$ (round off to 3 decimal places).



P(Cl=T)
0.6

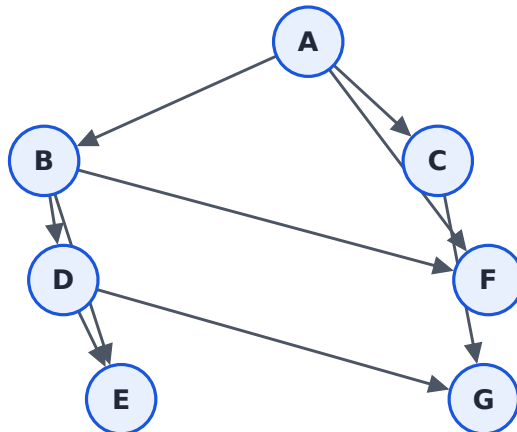
Cl	P(Sp=T ...)	Cl	Sp	P(Ra=T ...)
T	0.9	T	T	0.1
F	0.9	T	F	0.05
		F	T	0.5
		F	F	0.05

Cl	Ra	P(We=T ...)	Sp	We	P(Sl=T ...)
T	T	0.7	T	T	0.3
T	F	0.4	T	F	0.25
F	T	0.75	F	T	0.5
F	F	0.95	F	F	0.5

Answer (numerical): _____

Q.23 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

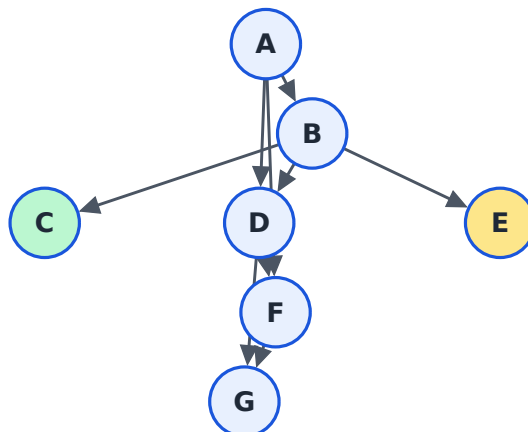
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $B \perp F$
☐ (B) $G \perp E \mid B, D$
☐ (C) $D \perp C \mid A$
☐ (D) $F \perp D \mid A, B$

Q.24 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

Variable elimination is used to compute $P(C \mid E = T)$ in the Bayesian network below (Boolean variables; evidence $E = T$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **F**, **B**, **D**, **G**, **A**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.25 [NAT · 2 marks · no negative marking] Bayesian Networks

A Bayesian network has: A with 3 values (no parents); B Boolean (no parents); C with 4 values and parents A, B; D Boolean with parent C; E with 3 values and parents C, D. How many independent parameters do the CPTs require?

Answer (numerical): _____

Answer Key — Mock Test 098

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	NAT	2	Informed Search	1.9 to 1.93
2	NAT	1	Uninformed Search	341	15	MSQ	2	Informed Search	A, D
3	NAT	1	Adversarial Search	4	16	MSQ	2	Propositional Logic	B, D
4	MCQ	1	Uninformed Search	B	17	NAT	2	Propositional Logic	2
5	NAT	1	Predicate Logic	729	18	MSQ	2	Predicate Logic	D
6	NAT	1	Propositional Logic	8	19	MSQ	2	Propositional Logic	B, C
7	NAT	1	Predicate Logic	2	20	MCQ	2	Propositional Logic	B
8	NAT	1	Sampling	0.403 to 0.4034	21	NAT	2	Sampling	0.498 to 0.502
9	NAT	1	Exact Inference	0.373 to 0.377	22	NAT	2	Exact Inference	0.857 to 0.861
10	MCQ	1	Bayesian Networks	A	23	MSQ	2	Conditional Independence	B, C, D
11	MCQ	2	Uninformed Search	D	24	NAT	2	Exact Inference	3
12	MCQ	2	Adversarial Search	A	25	NAT	2	Bayesian Networks	41
13	NAT	2	Adversarial Search	2					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] Answer: D (D: S, D, I, B, E, F, G)

Adjacency lists (alphabetical): B: {D, E, F, G}; C: {F}; D: {B, E, I, S}; E: {B, D, F, I}; F: {B, C, E, G}; G: {B, F}; I: {D, E, S}; S: {D, I}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[D, I]
2	D	[I, B, E]
3	I	[B, E]
4	B	[E, F, G]
5	E	[F, G]
6	F	[G, C]
7	G	[C] ← goal popped, stop

Expansion order: **S, D, I, B, E, F, G**. The path returned is $S \rightarrow D \rightarrow B \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

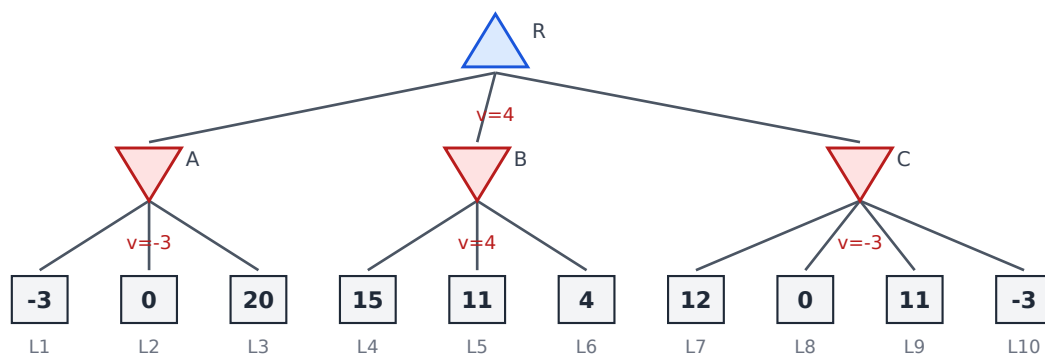
Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Uninformed Search] Answer: 341

DLS explores the complete tree down to depth ℓ : $1 + b + \dots + b^\ell = 1 + 4 + 16 + 64 + 256 = \mathbf{341} = (b^{\ell+1} - 1)/(b - 1)$.

Q.3 [NAT, 1M · Adversarial Search] Answer: 4

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $A = -3$, $B = 4$, $C = -3$. Root value = **4**; MAX's optimal first move is towards B.

Q.4 [MCQ, 1M · Uninformed Search] Answer: B (B: LIFO stack)

DFS always expands the most recently generated (deepest) node, which is **LIFO** behaviour. A FIFO queue gives BFS (A). A priority queue on $g(n)$ gives UCS (C). A priority queue on $h(n)$ gives greedy best-first search (D).

Q.5 [NAT, 1M · Predicate Logic] Answer: 729

View R as an 4×4 0/1 matrix. The sentence says R is asymmetric: no self-loops; each unordered pair has 3 options $\Rightarrow 3^{(n(n-1)/2)}$. With $n = 4$: **729**.

Q.6 [NAT, 1M · Propositional Logic] Answer: 8

A binary connective is fixed by its outputs on TT, TF, FT, FF. Commutativity requires $\text{output}(TF) = \text{output}(FT)$. So we may freely choose the outputs on TT, FF and the common value on TF/FT: $2^3 = 8$. Examples: $\wedge$, $\vee$, $\leftrightarrow$, $\oplus$, NAND, NOR, constant T, constant F.

Q.7 [NAT, 1M · Predicate Logic] Answer: 2

Step 1: resolve $\neg \text{Mortal}(\text{Socrates})$ with $\neg \text{Man}(x) \vee \text{Mortal}(x)$ using $\{x/\text{Socrates}\}$, giving $\neg \text{Man}(\text{Socrates})$. Step 2: resolve $\neg \text{Man}(\text{Socrates})$ with $\text{Man}(\text{Socrates})$, giving $\square$. One step cannot suffice since no two initial clauses are complementary units. So the answer is 2.

Q.8 [NAT, 1M · Sampling] Answer: 0.403 to 0.4034

Prior sampling generates each complete event with exactly its joint probability $S_{PS}(x_1, \dots, x_n) = \prod P(x_i | \text{parents}(X_i))$.

$$P(CI=F) \times P(Sp=T|CI=F) \times P(Ra=T|Sp=T) \times P(We=T|CI=F, Sp=T) \\ = 0.7 \times 0.9 \times 0.8 \times 0.8 = \mathbf{0.4032} \approx 0.4032.$$

Q.9 [NAT, 1M · Exact Inference] Answer: 0.373 to 0.377

Only the ancestors of B (A) matter: every other variable sums out to 1.

$P(B=T) = \sum \text{over its ancestors of } P(B=T | \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, B=T)$
A=T, C=T, D=T	0.21375
A=T, C=T, D=F	0.01125
A=T, C=F, D=T	0.005
A=T, C=F, D=F	0.02
A=F, C=T, D=T	0.03562
A=F, C=T, D=F	0.08312
A=F, C=F, D=T	0.00438
A=F, C=F, D=F	0.00188

Sum = **0.375** ≈ 0.375 .

Q.10 [MCQ, 1M · Bayesian Networks] Answer: A (A: Edges $C \rightarrow F_i$ for each i ; features are conditionally independent given C)

Naive Bayes uses $P(C, F_1, \dots, F_n) = P(C) \prod P(F_i | C)$, i.e. the class is the single parent of every feature. The DAG implies $F_i \perp F_j | C$. (D) is wrong because the features are generally dependent marginally (common cause C is unobserved). (B) reverses the edges (that would make features marginally independent but dependent given C); (C) is a different model.

Q.11 [MCQ, 2M · Uninformed Search] Answer: D (D: $S \rightarrow I \rightarrow F \rightarrow B \rightarrow G$)

UCS always expands the frontier node with the smallest path cost $g(n)$; if a cheaper path to a frontier node is found its entry is replaced.

Step	Expanded (g)	Frontier after expansion (sorted by priority)
1	S (0)	J(7), I(9)
2	J (7)	I(9), H(10)
3	I (9)	H(10), F(17)
4	H (10)	F(17)
5	F (17)	B(19), A(23)
6	B (19)	A(21), G(25)
7	A (21)	G(25)
8	G (25)	← goal, stop

Path returned: **S** → **I** → **F** → **B** → **G**, cost = 9 + 8 + 2 + 6 = **25**. Nodes expanded (including S and G): **8** (S, J, I, H, F, B, A, G).

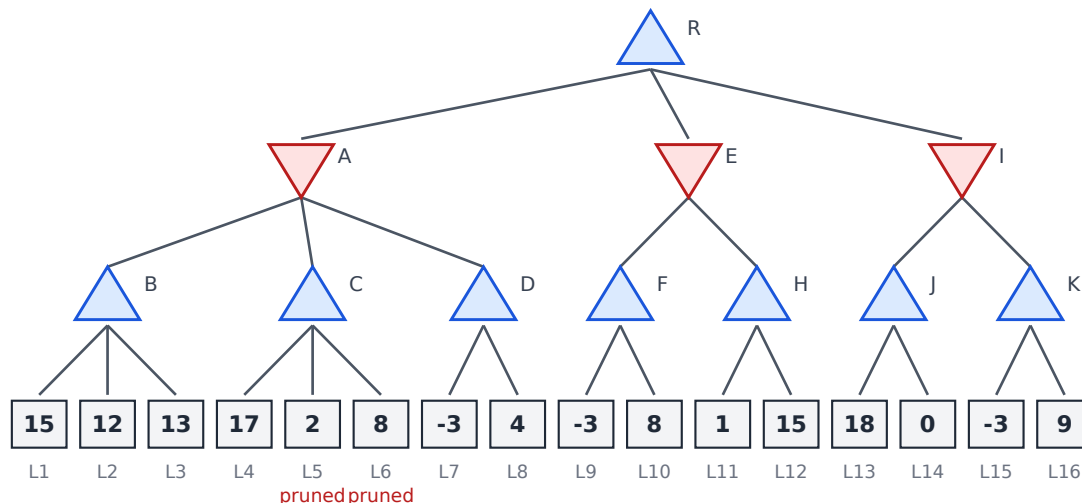
Q.12 [MCQ, 2M · Adversarial Search] **Answer: A** (A: Root value 7; only the last leaf (6) is pruned)

Left MIN: $\text{MAX}(3, 4) = 4$, so $\beta = 4$. $\text{MAX}(2, 9)$: leaf 2 does not cut, and $9 \geq 4$ cuts, but it is the last child, so both leaves are evaluated. Left MIN = 4 and root $\alpha = 4$. Right MIN: $\text{MAX}(7, 1) = 7$ ($7 > \alpha$, so no cut), so $\beta = 7$. $\text{MAX}(8, 6)$: $8 \geq \beta = 7$, so leaf 6 is pruned. Right MIN = 7, and the root is $\max(4, 7) = 7$. Only leaf 6 is pruned, so (A) is correct. Leaf 1 must be evaluated because $7 > \alpha$ at that point.

Q.13 [NAT, 2M · Adversarial Search] **Answer: 2**

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MAX node C: value $17 \geq \beta = 15 \Rightarrow$ prune remaining children (leaves L5, L6).



Leaves evaluated (14): L1, L2, L3, L4, L7, L8, L9, L10, L11, L12, L13, L14, L15, L16. Leaves pruned (2): L5, L6. Root value = 9 (same as plain minimax, 9 — pruning never changes the root value).

Q.14 [NAT, 2M · Informed Search] **Answer: 1.9 to 1.93**

Solve $b^* + b^{*2} + b^{*3} + b^{*4} + b^{*5} = 52$. For $b^*=1.9$: $1.9 + 3.61 + 6.859 + 13.03 + 24.76 \approx 50.2$, which is too small. For $b^*=1.95$: about 54.0, which is too large. Bisection gives $b^* \approx 1.92$, the AIMA textbook example. A b^* close to 1 indicates a very good heuristic.

Q.15 [MSQ, 2M · Informed Search] **Answer: A, D** (A: h is consistent (monotone).; D: h is admissible.)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	C	E	F	G	I	J	S
$h(n)$	9	10	11	0	8	6	13
$h^*(n)$	10	11	12	0	16	7	14
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: B, D** (B: $\neg Q \vee (\neg Q \leftrightarrow P)$; D: $\neg(Q \wedge \neg(\neg Q \leftrightarrow P))$)

Compare truth tables with $F = Q \rightarrow (\neg Q \leftrightarrow P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg Q \rightarrow \neg(\neg Q \leftrightarrow P)$ — differs at $P=T, Q=T$ ($F=F$, option = T) $\Rightarrow$ not equivalent.

(B) $\neg Q \vee (\neg Q \leftrightarrow P)$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

(C) $(\neg Q \leftrightarrow P) \rightarrow Q$ — differs at $P=T, Q=T$ ($F=F$, option = T) $\Rightarrow$ not equivalent.

(D) $\neg(Q \wedge \neg(\neg Q \leftrightarrow P))$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.17 [NAT, 2M · Propositional Logic] **Answer: 2**

Build the truth table (sub-formulas shown first):

P	Q	R	S	$(R \vee Q) \leftrightarrow (\neg P \rightarrow R)$	$S \leftrightarrow (\neg S \wedge S)$	$((R \vee Q) \leftrightarrow (\neg P \rightarrow R)) \vee (S \leftrightarrow (\neg S \wedge S))$
T	T	T	T	T	F	T
T	T	T	F	T	T	T
T	T	F	T	T	F	T
T	T	F	F	T	T	T
T	F	T	T	T	F	T
T	F	T	F	T	T	T
T	F	F	T	F	F	F
T	F	F	F	F	T	T
F	T	T	T	T	F	T
F	T	T	F	T	T	T
F	T	F	T	F	F	F
F	T	F	F	F	T	T
F	F	T	T	T	F	T
F	F	T	F	T	T	T
F	F	F	T	T	F	T
F	F	F	F	T	T	T

F has 14 models, so it is false in $16 - 14 = 2$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.18 [MSQ, 2M · Predicate Logic] **Answer: D** (D: $\exists x \forall y \neg R(y, x)$)

$R = \{(1,4), (2,3), (4,1), (4,3)\}$.

(A) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(4,1) \wedge R(1,4)$ but $\neg R(4,4)$.

(B) $\exists y \forall x R(x, y)$: **FALSE** — no element receives edges from all elements (including a self-loop).

(C) $\forall x R(x, x)$: **FALSE** — no self-loop at 1.

(D) $\exists x \forall y \neg R(y, x)$: **TRUE** — 2 has no incoming edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [MSQ, 2M · Propositional Logic] **Answer: B, C** (B: $(P \vee \neg Q)$; C: $(P \vee \neg Q \vee S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) The resolution closure contains 6 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(B) $(P \vee \neg Q)$ is derivable:

1. resolve $(P \vee \neg R)$ and $(\neg Q \vee R)$ on $R \Rightarrow (P \vee \neg Q)$

(C) $(P \vee \neg Q \vee S)$ is derivable:

1. resolve $(\neg Q \vee R \vee S)$ and $(P \vee \neg R)$ on $R \Rightarrow (P \vee \neg Q \vee S)$

(D) $(\neg P)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.20 [MCQ, 2M · Propositional Logic] **Answer: B** (B: Every resolvent of C_1 and C_2 is a tautology)

C_1 and C_2 clash on two pairs, $p/\neg p$ and $\neg r/r$. Resolution removes exactly one complementary pair at a time. Resolving on p gives $q \vee \neg r \vee r \vee s$; resolving on r gives $p \vee \neg p \vee q \vee s$. Both contain a complementary pair, so both are tautologies. Removing both pairs at once to get $q \vee s$ is unsound: $p=T, q=F, r=T, s=F$ satisfies C_1 and C_2 but falsifies $q \vee s$. The pair is satisfiable, so $\square$ cannot be derived.

Q.21 [NAT, 2M · Sampling] **Answer: 0.498 to 0.502**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	C
1	$P(A=T)=0.8 \times P(D=T B=T)=0.2$	0.16	F
2	$P(A=T)=0.8 \times P(D=T B=F)=0.1$	0.08	T
3	$P(A=T)=0.8 \times P(D=T B=F)=0.1$	0.08	T
4	$P(A=T)=0.8 \times P(D=T B=F)=0.1$	0.08	F
5	$P(A=T)=0.8 \times P(D=T B=F)=0.1$	0.08	T

$\sum w (\text{all}) = 0.48$; $\sum w (C = T) = 0.24$.

Estimate $P(C=T \mid D = T, A = T) = 0.24 / 0.48 = 0.5$.

Q.22 [NAT, 2M · Exact Inference] Answer: 0.857 to 0.861

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}, \text{Slippery}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Cloudy}, \text{Rain}) \cdot P(\text{Slippery} \mid \text{Sprinkler}, \text{WetGrass})$.

Inference by enumeration: $P(\text{Cloudy} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cloudy}, e, \text{hidden})$.

Hidden: Sp, Ra, SI	CI	product of CPT entries (CI · Sp · Ra · We · SI)	value
T, T, T	T	$0.6 \times 0.9 \times 0.1 \times 0.3 \times 0.25$	0.00405
T, T, F	T	$0.6 \times 0.9 \times 0.1 \times 0.3 \times 0.75$	0.01215
T, F, T	T	$0.6 \times 0.9 \times 0.9 \times 0.6 \times 0.25$	0.0729
T, F, F	T	$0.6 \times 0.9 \times 0.9 \times 0.6 \times 0.75$	0.2187
F, T, T	T	$0.6 \times 0.1 \times 0.05 \times 0.3 \times 0.5$	0.00045
F, T, F	T	$0.6 \times 0.1 \times 0.05 \times 0.3 \times 0.5$	0.00045
F, F, T	T	$0.6 \times 0.1 \times 0.95 \times 0.6 \times 0.5$	0.0171
F, F, F	T	$0.6 \times 0.1 \times 0.95 \times 0.6 \times 0.5$	0.0171
T, T, T	F	$0.4 \times 0.9 \times 0.5 \times 0.25 \times 0.25$	0.01125
T, T, F	F	$0.4 \times 0.9 \times 0.5 \times 0.25 \times 0.75$	0.03375
T, F, T	F	$0.4 \times 0.9 \times 0.5 \times 0.05 \times 0.25$	0.00225
T, F, F	F	$0.4 \times 0.9 \times 0.5 \times 0.05 \times 0.75$	0.00675
F, T, T	F	$0.4 \times 0.1 \times 0.05 \times 0.25 \times 0.5$	0.00025
F, T, F	F	$0.4 \times 0.1 \times 0.05 \times 0.25 \times 0.5$	0.00025
F, F, T	F	$0.4 \times 0.1 \times 0.95 \times 0.05 \times 0.5$	0.00095
F, F, F	F	$0.4 \times 0.1 \times 0.95 \times 0.05 \times 0.5$	0.00095

$P(\text{Cloudy}=\text{T}, e) = 0.3429$; $P(\text{Cloudy}=\text{F}, e) = 0.0564$; $P(e) = 0.3993$.

$P(\text{Cloudy}=\text{T} \mid e) = 0.3429 / 0.3993 = \mathbf{0.8588} \approx 0.859$.

Q.23 [MSQ, 2M · Conditional Independence] Answer: B, C, D (B: $G \perp E \mid B, D$; C: $D \perp C \mid A$; D: $F \perp D \mid A, B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $B \perp F$: **FALSE** — the path $B \leftarrow A \rightarrow F$ is active given $\{\}$.

(B) $G \perp E \mid B, D$: **TRUE** — $G \leftarrow C \leftarrow A \rightarrow B \rightarrow D \rightarrow E$ (blocked: observed chain node B); $G \leftarrow C \leftarrow A \rightarrow B \rightarrow E$ (blocked: observed chain node B); $G \leftarrow C \leftarrow A \rightarrow F \leftarrow B \rightarrow D \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow C \leftarrow A \rightarrow F \leftarrow B \rightarrow E$ (blocked: collider F with neither it nor a descendant observed); $G \leftarrow D \leftarrow B \rightarrow E$ (blocked: observed chain node D); $G \leftarrow D \rightarrow E$ (blocked: observed fork node D).

(C) $D \perp C \mid A$: **TRUE** — $D \leftarrow B \leftarrow A \rightarrow C$ (blocked: observed fork node A); $D \leftarrow B \rightarrow F \leftarrow A \rightarrow C$ (blocked: collider F with neither it nor a descendant observed); $D \rightarrow E \leftarrow B \leftarrow A \rightarrow C$ (blocked: collider E with neither it nor a descendant observed); $D \rightarrow E \leftarrow B \rightarrow F \leftarrow A \rightarrow C$ (blocked: collider E with neither it nor a descendant observed); $D \rightarrow G \leftarrow C$ (blocked: collider G with neither it nor a descendant observed).

(D) $F \perp D \mid A, B$: **TRUE** — $F \leftarrow A \rightarrow B \rightarrow D$ (blocked: observed fork node A); $F \leftarrow A \rightarrow B \rightarrow E \leftarrow D$ (blocked: observed fork node A); $F \leftarrow A \rightarrow C \rightarrow G \leftarrow D$ (blocked: observed fork node A); $F \leftarrow B \leftarrow A \rightarrow C \rightarrow G \leftarrow D$ (blocked: observed chain node B); $F \leftarrow B \rightarrow D$ (blocked: observed fork node B); $F \leftarrow B \rightarrow E \leftarrow D$ (blocked: observed fork node B).

Q.24 [NAT, 2M · Exact Inference] Answer: 3

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(B)$, $f(A, D, F)$, $f(D, F, G)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
F	$f(A, D, F)$, $f(D, F, G)$	$\{A, D, F, G\}$	$f(A, D, G)$
B	$f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(B)$	$\{A, B, C, D\}$	$f(A, C, D)$
D	$f(A, D, G)$, $f(A, C, D)$	$\{A, C, D, G\}$	$f(A, C, G)$
G	$f(A, C, G)$	$\{A, C, G\}$	$f(A, C)$
A	$f(A)$, $f(A, C)$	$\{A, C\}$	$f(C)$

The largest factor produced has **3** variable(s) (created when eliminating F); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over C and normalise.

Q.25 [NAT, 2M · Bayesian Networks] **Answer: 41**

Each CPT needs (number of parent configurations) $\times$ (arity $- 1$). A: $1 \times 2 = 2$. B: $1 \times 1 = 1$. C: $(3 \times 2) \times 3 = 18$. D: $4 \times 1 = 4$. E: $(4 \times 2) \times 2 = 16$. Total = $2 + 1 + 18 + 4 + 16 = 41$, versus $3 \times 2 \times 4 \times 2 \times 3 - 1 = 143$ for the full joint.

Mock Test 099

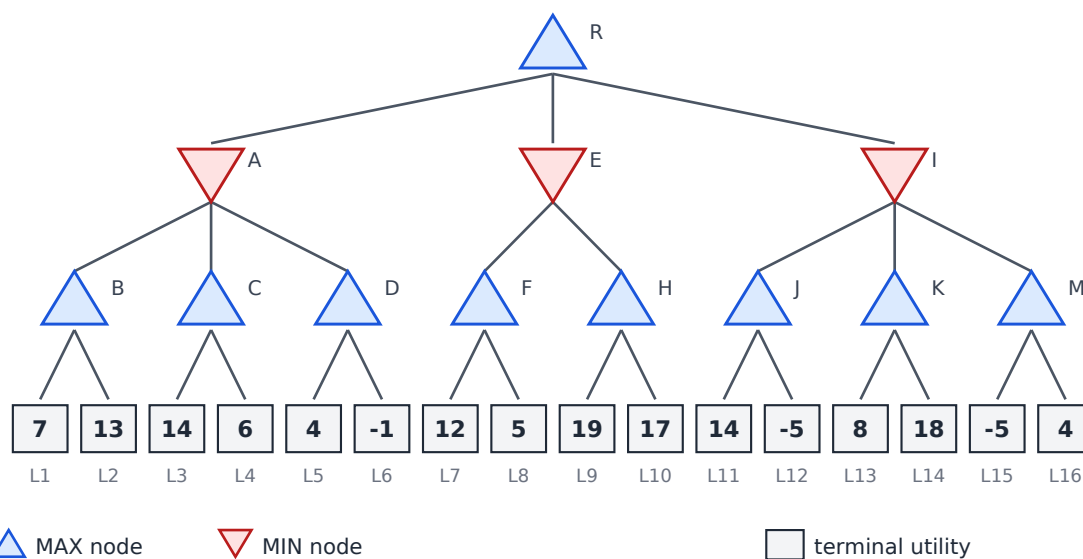
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 10 / 10	12 / 14 / 14

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

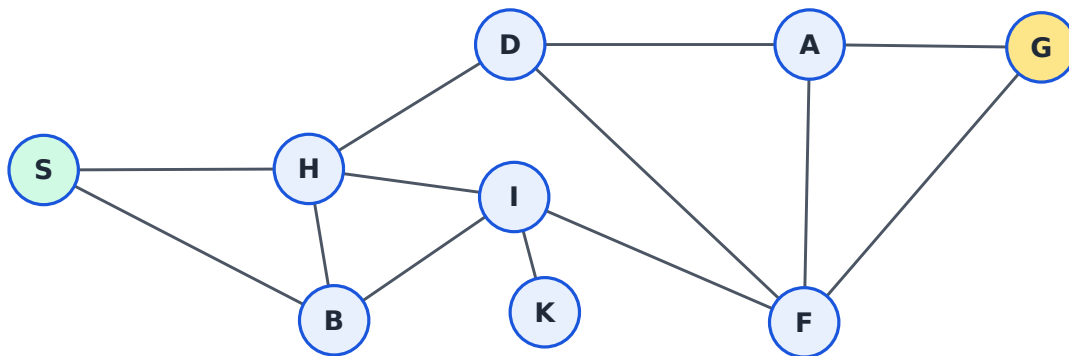
Q.2 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search

Which statement about depth-first search on a **finite** state space is correct?

- ☐ (A) The graph-search version is complete, but the tree-search version can loop forever on cycles
- ☐ (B) Both versions are complete and optimal
- ☐ (C) The tree-search version is complete, but the graph-search version is not
- ☐ (D) Neither version is complete

Q.3 [MCQ · 1 mark · negative: -1/3] Uninformed Search – DFS

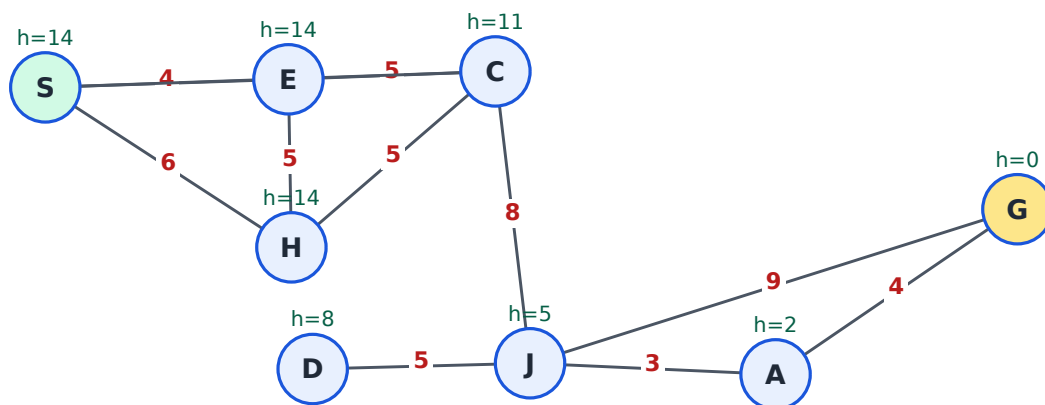
For the undirected graph shown, a **recursive depth-first search** is started at **S**. From every state, unvisited neighbours are explored in **alphabetical order**; a state already visited is never visited again; the search stops as soon as **G** is visited. In what order are the states visited?



- ☐ (A) S, B, H, I, D, F, K, A, G
☐ (B) S, H, B, D, A, F, G
☐ (C) S, H, I, K, F, G
☐ (D) S, B, H, D, A, F, G

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [MSQ · 1 mark · no negative marking] Predicate Logic

Which of the following equivalences are valid in first-order logic?

- ☐ (A) $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
☐ (B) $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
☐ (C) $\forall x \exists y P(x, y) \equiv \exists y \forall x P(x, y)$
☐ (D) $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$

Q.6 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q) . For how many of them is $\neg \exists x (P(x) \wedge \neg Q(x)) \wedge \exists x P(x)$ true?

Answer (numerical): _____

Q.7 [NAT · 1 mark · no negative marking] Propositional Logic

In how many rows of the truth table over p, q is $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow p)$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

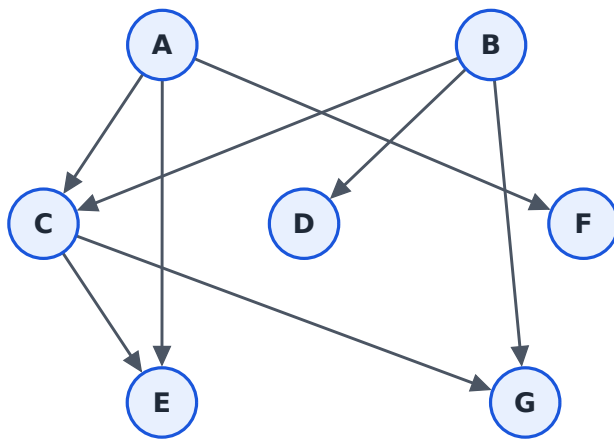
How many of the 8 truth assignments to P, Q, R satisfy the formula

$F = ((R \rightarrow R) \rightarrow (P \wedge \neg R)) \leftrightarrow (\neg \neg R \leftrightarrow (\neg Q \leftrightarrow P))$?

Answer (numerical): _____

Q.9 [NAT · 1 mark · no negative marking] Conditional Independence – Markov blanket

How many nodes are in the Markov blanket of **C** in the Bayesian network below?



Answer (numerical): _____

Q.10 [MCQ · 1 mark · negative: -1/3] Variable Elimination

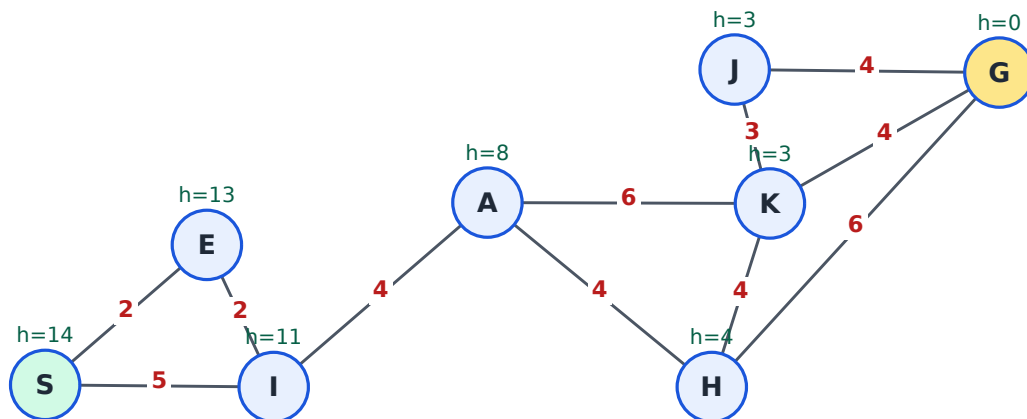
For a polytree (singly connected) Bayesian network, the time and space complexity of exact inference by variable elimination is

- ☐ (A) linear in the size of the network (total CPT entries)
- ☐ (B) exponential in the number of variables in the worst case
- ☐ (C) quadratic in the number of variables, independent of CPT sizes
- ☐ (D) NP-hard

Section B — 2-mark questions

Q.11 [NAT · 2 marks · no negative marking] Informed Search – A*

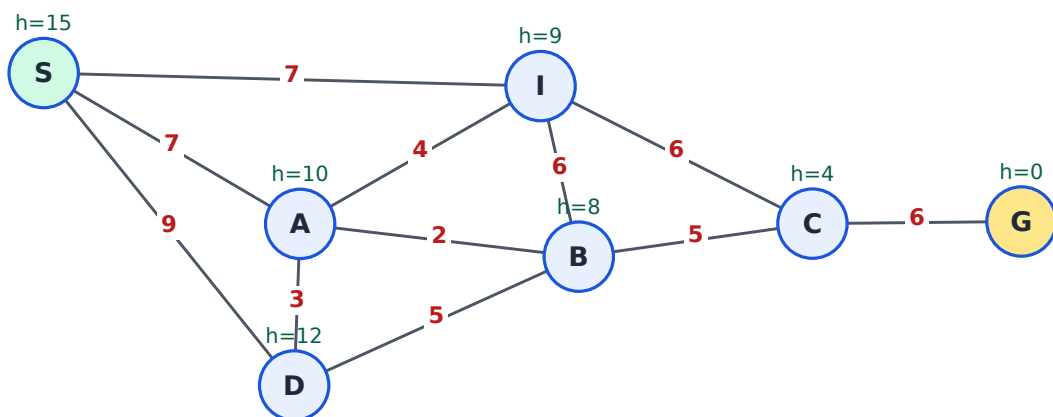
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. How many nodes are expanded, counting both S and G?



Answer (numerical): _____

Q.12 [MSQ · 2 marks · no negative marking] Informed Search – Heuristics

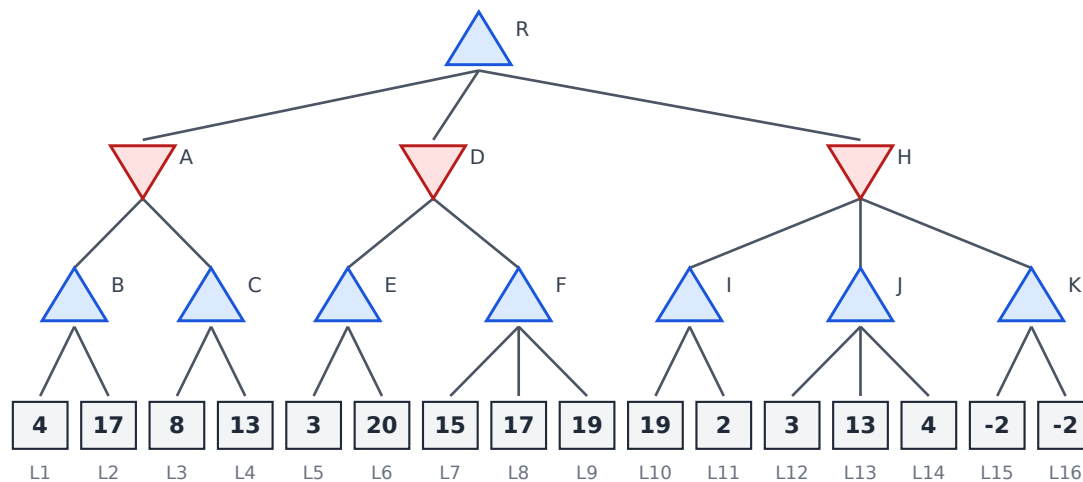
The heuristic values $h(n)$ for the graph below are printed above the nodes; edge labels are step costs and **G** is the only goal. Which of the following statements is/are TRUE?



- ☐ (A) h is admissible.
- ☐ (B) $h(C)$ overestimates the true cost from C to G.
- ☐ (C) The edge A-I violates the consistency condition (in at least one direction).
- ☐ (D) h is consistent (monotone).

Q.13 [MSQ · 2 marks · no negative marking] Adversarial Search – Alpha-beta

Alpha-beta pruning is applied to the game tree below, exploring children strictly from **left to right** (leaves are labelled L1, L2, ... under the boxes). Which of the following leaves is/are pruned (never evaluated)?



- ☐ (A) Leaf L3 is pruned.
☐ (B) Leaf L6 is pruned.
☐ (C) Leaf L10 is pruned.
☐ (D) Leaf L16 is pruned.

Q.14 [MSQ · 2 marks · no negative marking] Informed Search

Assume $h(\text{goal}) = 0$ and all step costs are positive. Which of the following are TRUE?

- ☐ (A) Every consistent heuristic is admissible
☐ (B) Every admissible heuristic is consistent
☐ (C) If h is consistent, the values of $f(n) = g(n) + h(n)$ along any path are non-decreasing
☐ (D) A* tree search with an admissible heuristic is optimal

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Resolution

Consider the clause set

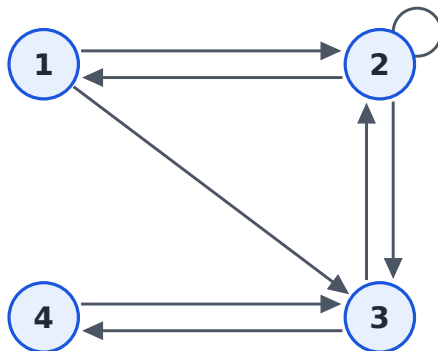
C1: $(R \vee \neg S)$, C2: $(P \vee \neg S)$, C3: $(\neg P \vee \neg Q \vee R)$, C4: $(\neg P)$

Which of the following can be derived by repeatedly applying the propositional **resolution** rule (each step resolves exactly one complementary pair of literals; tautological resolvents are discarded; derived clauses may be reused)?

- ☐ (A) $(\neg S)$
☐ (B) The empty clause $\square$ (i.e. the clause set is shown unsatisfiable)
☐ (C) $(\neg P \vee Q)$
☐ (D) $(\neg Q \vee \neg S)$

Q.16 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \exists y R(x, y)$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\forall x \exists y (x \neq y \wedge R(x, y))$
☐ (D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic

Let α and β be propositional sentences. Which of the following statements are ALWAYS true?

- ☐ (A) If α is satisfiable then $\neg\alpha$ is unsatisfiable
☐ (B) If α is valid then $\neg\alpha$ is unsatisfiable
☐ (C) $\alpha \vee \beta$ is satisfiable if and only if α is satisfiable or β is satisfiable
☐ (D) $\alpha \wedge \beta$ is satisfiable if and only if α is satisfiable and β is satisfiable

Q.18 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $\mathbf{F} = \neg R \rightarrow (P \wedge P)$?

- ☐ (A) $\neg\neg R \wedge (P \wedge P)$
☐ (B) $(P \wedge P) \rightarrow \neg R$
☐ (C) $\neg\neg R \rightarrow \neg(P \wedge P)$
☐ (D) $\neg(P \wedge P) \rightarrow \neg\neg R$

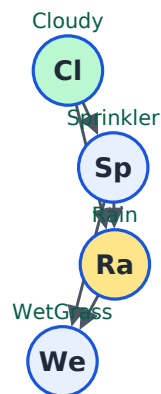
Q.19 [NAT · 2 marks · no negative marking] Propositional Logic

The canonical CNF (product of maxterms, each clause containing all variables) of $p \oplus q \oplus r$ has how many clauses?

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: CI = Cloudy, Sp = Sprinkler, Ra = Rain, We = WetGrass.) Compute $P(\text{Cloudy} = T \mid \text{Rain} = T)$ (round off to 3 decimal places).



$P(\text{CI}=T)$
0.6

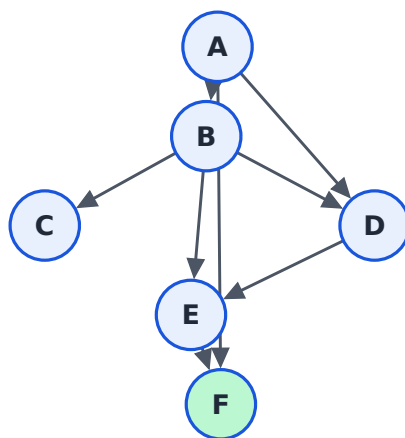
CI	$P(\text{Sp}=T \mid \text{CI})$	CI	Sp	$P(\text{Ra}=T \mid \dots)$
T	0.1	T	T	0.2
F	0.9	T	F	0.6
		F	T	0.9
		F	F	0.05

Sp	Ra	$P(\text{We}=T \mid \dots)$
T	T	0.95
T	F	0.05
F	T	0.8
F	F	0.25

Answer (numerical): _____

Q.21 [MCQ · 2 marks · negative: -2/3] Exact Inference – Variable elimination

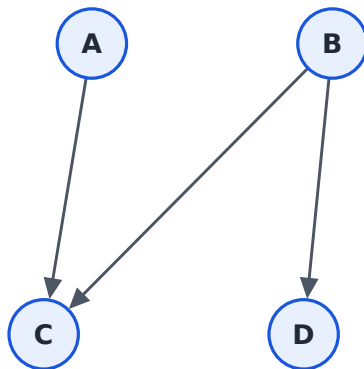
Variable elimination is used to compute $P(\mathbf{F})$ in the Bayesian network below (Boolean variables; there is no evidence; the query node is green). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **B, C, A, D, E**. What is the scope of the new factor produced when **C** is summed out?



- ☐ (A) $f(A, D, E)$
☐ (B) $f(A, C, D, E)$
☐ (C) $f(A, E)$
☐ (D) $f(A, D, E, F)$

Q.22 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, using the uniform random numbers $\mathbf{u} = \langle \mathbf{0.16}, \mathbf{0.21}, \mathbf{0.6}, \mathbf{0.41} \rangle$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



P(A=T)
0.05

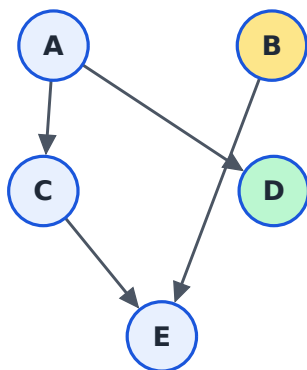
P(B=T)	A	B	P(C=T ...)
0.5	T	T	0.05
	T	F	0.7
	F	T	0.8
	F	F	0.6

B	P(D=T ...)
T	0.75
F	0.5

- ☐ (A) A=T, B=T, C=T, D=T
☐ (B) A=T, B=F, C=F, D=F
☐ (C) A=F, B=T, C=T, D=T
☐ (D) A=F, B=F, C=T, D=T

Q.23 [NAT · 2 marks · no negative marking] Sampling – Gibbs sampling

Gibbs sampling (MCMC) is run on the Bayesian network below with evidence $\mathbf{B=F}$. The current state of the chain is $\langle \mathbf{A=F}, \mathbf{B=F}, \mathbf{C=F}, \mathbf{D=F}, \mathbf{E=T} \rangle$. The next step resamples $\mathbf{D}$. With what probability is D set to T? (round off to 3 decimal places)



P(A=T)
0.2

P(B=T)
0.5

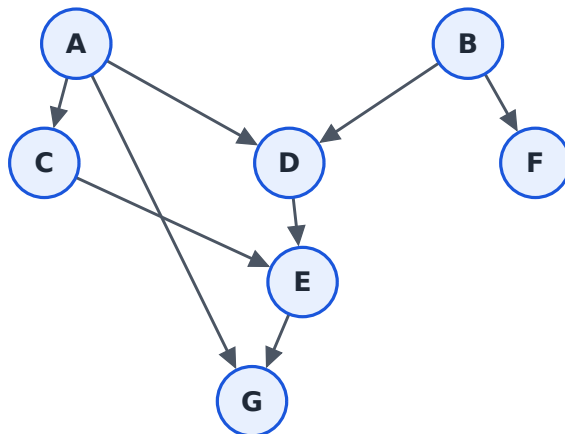
A	P(C=T ...)
T	0.2
F	0.1

A	P(D=T ...)	B	C	P(E=T ...)
T	0.9	T	T	0.9
F	0.25	T	F	0.1
		F	T	0.75
		F	F	0.2

Answer (numerical): _____

Q.24 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $C \perp D \mid F$
☐ (B) $F \perp B \mid E$
☐ (C) $G \perp F \mid B$
☐ (D) $F \perp E \mid A, B$

Q.25 [MSQ · 2 marks · no negative marking] Conditional Independence

A Bayesian network has edges $C \rightarrow A$ and $C \rightarrow B$ (Boolean variables) with $P(c) = 0.4$, $P(a \mid c) = 0.9$, $P(a \mid \neg c) = 0.2$, $P(b \mid c) = 0.7$, $P(b \mid \neg c) = 0.1$. Which of the following are true?

- ☐ (A) $P(a) = 0.48$
☐ (B) $A \perp B$ (marginally)
☐ (C) $P(a \mid b, c) = 0.9$
☐ (D) $P(a \mid b) \approx 0.776$

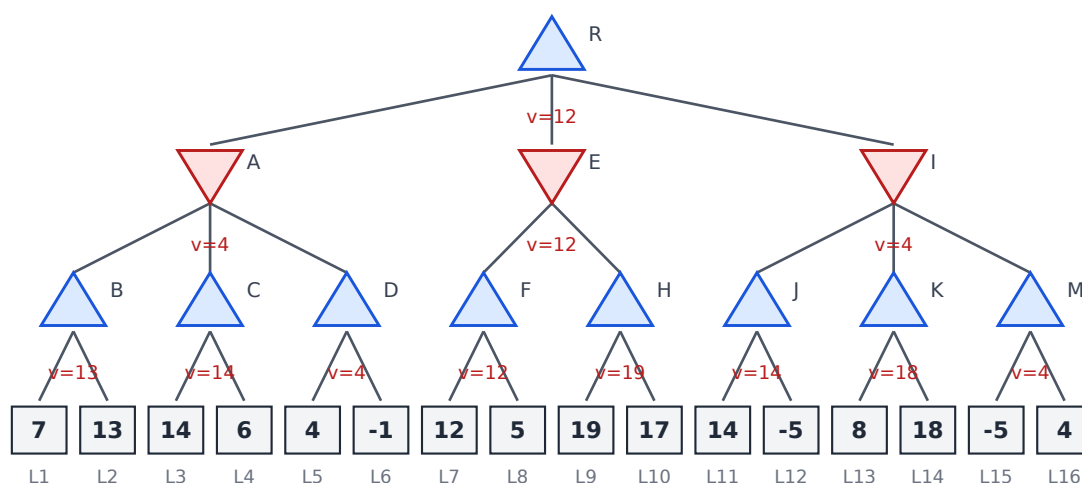
Answer Key — Mock Test 099

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	NAT	1	Adversarial Search	12	14	MSQ	2	Informed Search	A, C, D
2	MCQ	1	Uninformed Search	A	15	MSQ	2	Propositional Logic	A
3	MCQ	1	Uninformed Search	D	16	MSQ	2	Predicate Logic	A, B, C
4	NAT	1	Informed Search	21	17	MSQ	2	Propositional Logic	B, C
5	MSQ	1	Predicate Logic	A, B, D	18	MSQ	2	Propositional Logic	D
6	NAT	1	Predicate Logic	19	19	NAT	2	Propositional Logic	4
7	NAT	1	Propositional Logic	2	20	NAT	2	Exact Inference	0.506 to 0.51
8	NAT	1	Propositional Logic	4	21	MCQ	2	Exact Inference	A
9	NAT	1	Conditional Independence	4	22	MCQ	2	Sampling	C
10	MCQ	1	Variable Elimination	A	23	NAT	2	Sampling	0.248 to 0.252
11	NAT	2	Informed Search	7	24	MSQ	2	Conditional Independence	C, D
12	MSQ	2	Informed Search	A, D	25	MSQ	2	Conditional Independence	A, C, D
13	MSQ	2	Adversarial Search	D					

Detailed Solutions

Q.1 [NAT, 1M · Adversarial Search] **Answer: 12**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 13$, $C = 14$, $D = 4$, $A = 4$, $F = 12$, $H = 19$, $E = 12$, $J = 14$, $K = 18$, $M = 4$, $I = 4$. Root value = **12**; MAX's optimal first move is towards E.

Q.2 [MCQ, 1M · Uninformed Search] Answer: A (A: The graph-search version is complete, but the tree-search version can loop forever on cycles)

Graph-search DFS never re-expands a state, so in a finite space it eventually expands every reachable state and is complete. Tree-search DFS keeps no explored set, so it can follow a cycle such as $A \rightarrow B \rightarrow A \rightarrow \dots$ forever and is incomplete. Neither version is optimal: DFS can return a deep, expensive goal before a shallow one. So (B), (C) and (D) are all wrong.

Q.3 [MCQ, 1M · Uninformed Search] Answer: D (D: S, B, H, D, A, F, G)

Adjacency lists (alphabetical): A: {D, F, G}; B: {H, I, S}; D: {A, F, H}; F: {A, D, G, I}; G: {A, F}; H: {B, D, I, S}; I: {B, F, H, K}; K: {I}; S: {B, H}.

DFS always goes deeper into the alphabetically first unvisited neighbour and backtracks only when a state has no unvisited neighbours.

Visit order: **S, B, H, D, A, F, G**; the path found is $S \rightarrow B \rightarrow H \rightarrow D \rightarrow A \rightarrow F \rightarrow G$ (6 edges). Note that DFS need not find the shortest path.

Q.4 [NAT, 1M · Informed Search] Answer: 21

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (14)	C(11), E(14), H(14)
2	C (11)	J(5), E(14), H(14)
3	J (5)	G(0), A(2), D(8), E(14), H(14)
4	G (0)	A(2), D(8), E(14), H(14) ← goal, stop

Path returned: **S** → **C** → **J** → **G**, cost = $4 + 8 + 9 = 21$.

The optimal cost is $h^*(S) = 19$ — greedy search is not optimal here.

Q.5 [MSQ, 1M · Predicate Logic] Answer: A, B, D (A: $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$; B: $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$; D: $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$)

Quantifiers of the same kind commute (A, B). $\exists$ distributes over $\vee$ (D), just as $\forall$ distributes over $\wedge$. (C) Mixed quantifiers do not commute: with $P(x, y) \iff x = y$ on $\{1, 2\}$, $\forall x \exists y P$ is true but $\exists y \forall x P$ is false.

Q.6 [NAT, 1M · Predicate Logic] Answer: 19

Interpret P and Q as subsets of the domain; $P \subseteq Q$ gives 3^n ; subtract the 2^n with P empty $\Rightarrow 3^n - 2^n$. With $n = 3$ this gives **19** (verified by enumerating all interpretations).

Q.7 [NAT, 1M · Propositional Logic] Answer: 2

$p \rightarrow q \equiv \neg p \vee q$ and $\neg q \rightarrow p \equiv p \vee q$. Row TT: T, $T \rightarrow T$. Row TF: F, $T \rightarrow F$. Row FT: T, $T \rightarrow T$. Row FF: T, $F \rightarrow F$. So the biconditional is true in 2 rows.

Q.8 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$(R \rightarrow R) \rightarrow (P \wedge \neg R)$	$\neg \neg R \leftrightarrow (\neg Q \leftrightarrow P)$	$((R \rightarrow R) \rightarrow (P \wedge \neg R)) \leftrightarrow (\neg \neg R \leftrightarrow (\neg Q \leftrightarrow P))$
T	T	T	F	F	T
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	F	F
F	T	T	F	T	F
F	T	F	F	F	T
F	F	T	F	F	T
F	F	F	F	T	F

Rows where F = T: **4**.

Q.9 [NAT, 1M · Conditional Independence] **Answer: 4**

Markov blanket = parents $\cup$ children $\cup$ children's other parents.

Parents of C: {A, B}; children: {E, G}; co-parents: $\{\emptyset\}$.

MB(C) = {A, B, E, G} — **4** nodes. Given its Markov blanket, C is conditionally independent of every other node in the network.

Q.10 [MCQ, 1M · Variable Elimination] **Answer: A** (A: linear in the size of the network (total CPT entries))

AIMA: in polytrees there is at most one undirected path between any two nodes, and VE (with a suitable ordering) runs in time and space linear in the size of the network, measured by CPT entries.

Exponential/NP-hard behaviour arises for multiply connected networks. Quadratic in variables ignores that CPT sizes grow with the number of parents.

Q.11 [NAT, 2M · Informed Search] **Answer: 7**

Heuristic values: $h(A)=8$, $h(E)=13$, $h(G)=0$, $h(H)=4$, $h(I)=11$, $h(J)=3$, $h(K)=3$, $h(S)=14$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (14)	E(15), I(16)
2	E (15)	I(15)
3	I (15)	A(16)
4	A (16)	H(16), K(17)
5	H (16)	K(17), G(18)
6	K (17)	G(18), J(20)
7	G (18)	J(20) ← goal, stop

Path returned: **S** → **E** → **I** → **A** → **H** → **G** with cost **18**; nodes expanded = **7** (S, E, I, A, H, K, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 18$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.12 [MSQ, 2M · Informed Search] **Answer: A, D** (A: h is admissible.; D: h is consistent (monotone).)

First compute the true cost-to-goal $h^*(n)$ by running Dijkstra/UCS backwards from G:

n	A	B	C	D	G	I	S
$h(n)$	10	8	4	12	0	9	15
$h^*(n)$	13	11	6	16	0	12	19
$h \leq h^*$?	✓	✓	✓	✓	✓	✓	✓

Admissible $\Leftrightarrow h(n) \leq h^*(n)$ for every n : **yes**.

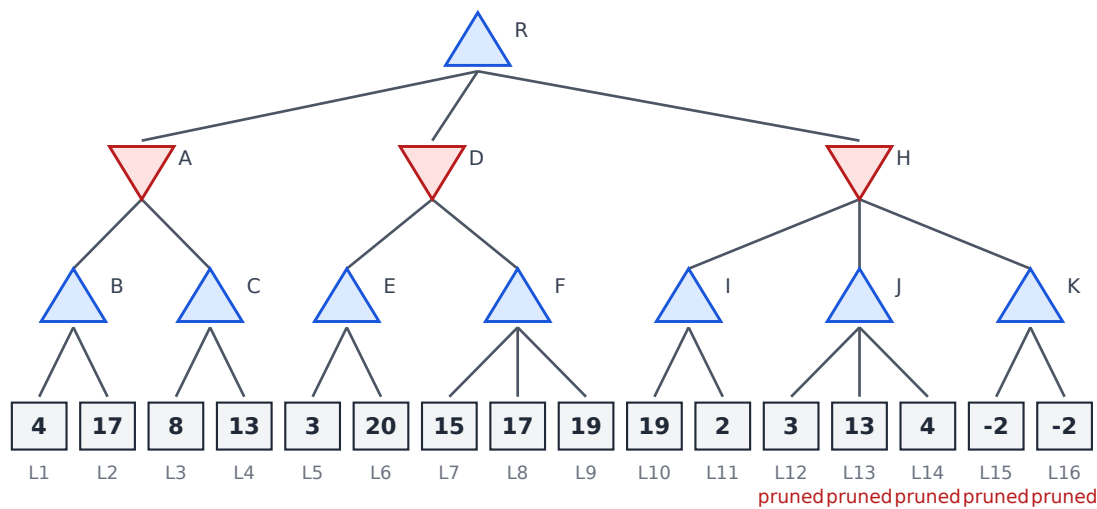
Consistent $\Leftrightarrow h(G)=0$ and $h(u) \leq c(u,v) + h(v)$ for every edge in both directions. Violating directed edges: none. Hence consistent: **yes**.

Recall: consistency $\Rightarrow$ admissibility, but not conversely.

Q.13 [MSQ, 2M · Adversarial Search] **Answer: D** (D: Leaf L16 is pruned.)

Run alpha-beta depth-first, left to right, passing (α, β) down. A MIN node stops as soon as its value $\leq \alpha$; a MAX node stops as soon as its value $\geq \beta$.

- At MIN node H: value $19 \leq \alpha = 19 \Rightarrow$ prune remaining children (leaves L12, L13, L14, L15, L16).



Leaves evaluated (11): L1, L2, L3, L4, L5, L6, L7, L8, L9, L10, L11. Leaves pruned (5): L12, L13, L14, L15, L16. Root value = 19 (same as plain minimax, 19 — pruning never changes the root value).

Q.14 [MSQ, 2M · Informed Search] Answer: A, C, D

(A) is true: induct backward along an optimal path from the goal, using $h(n) \leq c + h(n') \leq c + h^*(n') = h^*(n)$. (B) is false: admissible but inconsistent heuristics exist, for example $h(A)=6$ and $h(B)=2$ with $c(A,B)=2$. (C) is true: $f(n') = g(n) + c + h(n') \geq g(n) + h(n) = f(n)$. (D) is true: for tree search, admissibility is enough for optimality. Consistency is needed for graph search without reopening.

Q.15 [MSQ, 2M · Propositional Logic] Answer: A (A: $(\neg S)$)

Saturate the clause set under resolution (only the steps needed for the options are shown).

(A) $(\neg S)$ is derivable:

1. resolve $(P \vee \neg S)$ and $(\neg P)$ on $P \Rightarrow (\neg S)$

(B) The resolution closure contains 6 clauses and no $\square$, so the set is satisfiable (resolution is refutation-complete) $\Rightarrow$ not derivable.

(C) $(\neg P \vee Q)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

(D) $(\neg Q \vee \neg S)$ never appears in the resolution closure $\Rightarrow$ not derivable (note: a clause can be entailed yet not derivable, e.g. a weakening of a derived clause; resolution is complete only for refutation).

Q.16 [MSQ, 2M · Predicate Logic] Answer: A, B, C (A: $\forall x \exists y R(x, y)$; B: $\forall y \exists x R(x, y)$; C: $\forall x \exists y (x \neq y \wedge R(x, y))$)

$R = \{(1,2), (1,3), (2,1), (2,2), (2,3), (3,2), (3,4), (4,3)\}$.

(A) $\forall x \exists y R(x, y)$: **TRUE** — every element has an outgoing edge.

(B) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(C) $\forall x \exists y (x \neq y \wedge R(x, y))$: **TRUE** — every element has an edge to a different element.

(D) $\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$: **FALSE** — $R(1,2) \wedge R(2,1)$ but $\neg R(1,1)$.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.17 [MSQ, 2M · Propositional Logic] Answer: B, C (B: If α is valid then $\neg \alpha$ is unsatisfiable; C: $\alpha \vee \beta$ is satisfiable if and only if α is satisfiable or β is satisfiable)

(A) False: p is satisfiable and so is $\neg p$. (B) True: valid means true in all models, so $\neg \alpha$ is true in none. (C)

True: a model of $\alpha \vee \beta$ is a model of α or of β , and conversely any model of α (or of β) is a model of $\alpha \vee \beta$. (D)

False: $\alpha = p$ and $\beta = \neg p$ are each satisfiable but $p \wedge \neg p$ is not (the 'if' direction fails).

Q.18 [MSQ, 2M · Propositional Logic] Answer: D (D: $\neg(P \wedge P) \rightarrow \neg \neg R$)

Compare truth tables with $F = \neg R \rightarrow (P \wedge P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg \neg R \wedge (P \wedge P)$ — differs at $P=T, R=F$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(B) $(P \wedge P) \rightarrow \neg R$ — differs at $P=T, R=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(C) $\neg \neg R \rightarrow \neg(P \wedge P)$ — differs at $P=T, R=T$ ($F=T$, option = F) $\Rightarrow$ not equivalent.

(D) $\neg(P \wedge P) \rightarrow \neg \neg R$ — agrees with F on all 4 rows $\Rightarrow$ **equivalent**.

Q.19 [NAT, 2M · Propositional Logic] Answer: 4

In the canonical CNF there is one maxterm (clause) for every falsifying assignment. $p \oplus q \oplus r$ is false exactly when an even number of variables is true: FFF, TTF, TFT, FTT. That gives 4 falsifying rows and hence 4 clauses. It is also known that no shorter CNF exists for parity, since every clause must contain all 3 variables.

Q.20 [NAT, 2M · Exact Inference] Answer: 0.506 to 0.51

Factorisation: $P(\text{Cloudy}, \text{Sprinkler}, \text{Rain}, \text{WetGrass}) = P(\text{Cloudy}) \cdot P(\text{Sprinkler} \mid \text{Cloudy}) \cdot P(\text{Rain} \mid \text{Cloudy}, \text{Sprinkler}) \cdot P(\text{WetGrass} \mid \text{Sprinkler}, \text{Rain})$.

Inference by enumeration: $P(\text{Cloudy} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cloudy}, e, \text{hidden})$.

Hidden: Sp, We	Cl	product of CPT entries (Cl · Sp · Ra · We)	value
T, T	T	$0.6 \times 0.1 \times 0.2 \times 0.95$	0.0114
T, F	T	$0.6 \times 0.1 \times 0.2 \times 0.05$	0.0006
F, T	T	$0.6 \times 0.9 \times 0.6 \times 0.8$	0.2592
F, F	T	$0.6 \times 0.9 \times 0.6 \times 0.2$	0.0648
T, T	F	$0.4 \times 0.9 \times 0.9 \times 0.95$	0.3078
T, F	F	$0.4 \times 0.9 \times 0.9 \times 0.05$	0.0162
F, T	F	$0.4 \times 0.1 \times 0.05 \times 0.8$	0.0016
F, F	F	$0.4 \times 0.1 \times 0.05 \times 0.2$	0.0004

$P(\text{Cloudy}=\text{T}, e) = 0.336$; $P(\text{Cloudy}=\text{F}, e) = 0.326$; $P(e) = 0.662$.

$P(\text{Cloudy}=\text{T} \mid e) = 0.336 / 0.662 = \mathbf{0.5076} \approx 0.508$.

Q.21 [MCQ, 2M · Exact Inference] Answer: A (A: f(A, D, E))

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(B, D, E)$, $f(A, E, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
B	$f(A, B)$, $f(B, C)$, $f(A, B, D)$, $f(B, D, E)$	$\{A, B, C, D, E\}$	$f(A, C, D, E)$
C	$f(A, C, D, E)$	$\{A, C, D, E\}$	$f(A, D, E)$
A	$f(A)$, $f(A, E, F)$, $f(A, D, E)$	$\{A, D, E, F\}$	$f(D, E, F)$
D	$f(D, E, F)$	$\{D, E, F\}$	$f(E, F)$
E	$f(E, F)$	$\{E, F\}$	$f(F)$

The largest factor produced has **4** variable(s) (created when eliminating B); with Boolean variables it has $2^4 = 16$ entries. Finally, multiply the remaining factors over F and normalise.

Q.22 [MCQ, 2M · Sampling] Answer: C (C: A=F, B=T, C=T, D=T)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=\text{T} \mid \text{—}) = 0.05$; $u = 0.16 \geq 0.05 \Rightarrow A = \text{F}$
- B: $P(B=\text{T} \mid \text{—}) = 0.5$; $u = 0.21 < 0.5 \Rightarrow B = \text{T}$
- C: $P(C=\text{T} \mid A=\text{F}, B=\text{T}) = 0.8$; $u = 0.6 < 0.8 \Rightarrow C = \text{T}$
- D: $P(D=\text{T} \mid B=\text{T}) = 0.75$; $u = 0.41 < 0.75 \Rightarrow D = \text{T}$

Sample: **(A=F, B=T, C=T, D=T)**.

Q.23 [NAT, 2M · Sampling] Answer: 0.248 to 0.252

Gibbs sampling draws D from $P(D \mid \text{mb}(D))$, where $\text{mb}(D) = \{A\}$:

$P(D \mid \text{mb}) \propto P(D \mid \text{parents}(D)) \times \prod_{\text{children } Y} P(Y \mid \text{parents}(Y))$.

D=T: $P(D=\text{T} \mid A=\text{F}) = 0.25 = 0.25$

D=F: $P(D=\text{F} \mid A=\text{F}) = 0.75 = 0.75$

Normalise: $0.25 / (0.25 + 0.75) = \mathbf{0.25}$.

Q.24 [MSQ, 2M · Conditional Independence] **Answer: C, D** (C: $G \perp F \mid B$; D: $F \perp E \mid A, B$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y \mid Z$ holds iff every undirected path is blocked.

(A) $C \perp D \mid F$: **FALSE** — the path $C \leftarrow A \rightarrow D$ is active given $\{F\}$.

(B) $F \perp B \mid E$: **FALSE** — the path $F \leftarrow B$ is active given $\{E\}$.

(C) $G \perp F \mid B$: **TRUE** — $G \leftarrow A \rightarrow C \rightarrow E \leftarrow D \leftarrow B \rightarrow F$ (blocked: collider E with neither it nor a descendant observed); $G \leftarrow A \rightarrow D \leftarrow B \rightarrow F$ (blocked: collider D with neither it nor a descendant observed); $G \leftarrow E \leftarrow C \leftarrow A \rightarrow D \leftarrow B \rightarrow F$ (blocked: collider D with neither it nor a descendant observed); $G \leftarrow E \leftarrow D \leftarrow B \rightarrow F$ (blocked: observed fork node B).

(D) $F \perp E \mid A, B$: **TRUE** — $F \leftarrow B \rightarrow D \leftarrow A \rightarrow C \rightarrow E$ (blocked: observed fork node B); $F \leftarrow B \rightarrow D \leftarrow A \rightarrow G \leftarrow E$ (blocked: observed fork node B); $F \leftarrow B \rightarrow D \rightarrow E$ (blocked: observed fork node B).

Q.25 [MSQ, 2M · Conditional Independence] **Answer: A, C, D** (A: $P(a) = 0.48$; C: $P(a \mid b, c) = 0.9$; D: $P(a \mid b) \approx 0.776$)

$P(a) = 0.4 \times 0.9 + 0.6 \times 0.2 = 0.36 + 0.12 = 0.48$ (A true). $P(a, b) = 0.252 + 0.012 = 0.264$ while $P(a)P(b) = 0.48 \times 0.34 = 0.1632$, so A and B are dependent (B false). In the fork $C \rightarrow A$, $C \rightarrow B$, $A \perp B \mid C$, so $P(a \mid b, c) = P(a \mid c) = 0.9$ (C true). $P(a \mid b) = 0.264/0.34 \approx 0.7765$ (D true): observing b raises belief in c and hence in a.

Mock Test 100

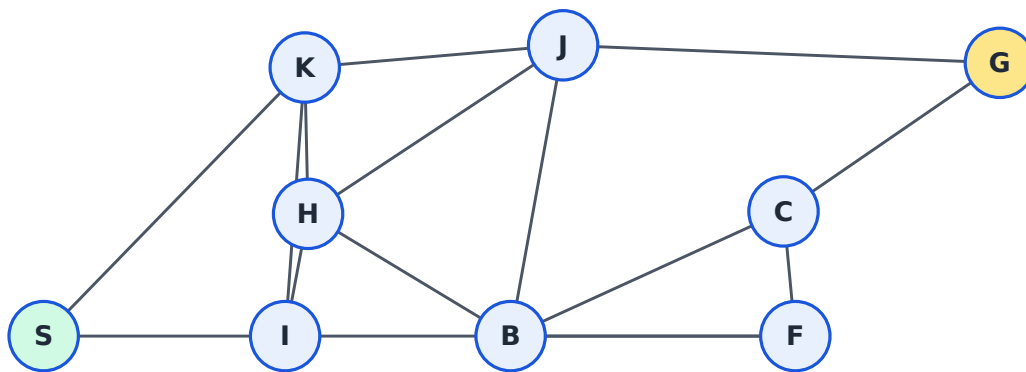
Questions	Total marks	Time	MCQ / MSQ / NAT	Search / Logic / Uncertainty (marks)
25	40	75 minutes	5 / 8 / 12	12 / 12 / 16

Q.1 – Q.10 carry 1 mark each; Q.11 – Q.25 carry 2 marks each. MCQ: one correct option, wrong answer $-1/3$ (1-mark) or $-2/3$ (2-mark). MSQ: one or more correct options, full credit only if exactly all correct options are chosen, no negative marking. NAT: enter a number, no negative marking. Unless stated otherwise, ties are broken alphabetically and all random variables are Boolean.

Section A — 1-mark questions

Q.1 [MCQ · 1 mark · negative: $-1/3$] Uninformed Search – BFS

Consider the undirected graph shown below. **Breadth-first search (graph-search version)** is run from start state **S** to find goal **G**. Successors are generated in **alphabetical order**, a state is added to the FIFO frontier only if it has not been reached before, and the goal test is applied when a state is **selected for expansion**. Which of the following is the sequence of states expanded (including S and G)?



- ☐ (A) S, K, I, F, H, J, B, C, G
- ☐ (B) S, I, F, B, C, G
- ☐ (C) S, K, I, J, H, F, G
- ☐ (D) S, I, K, F, H, J, B, C, G

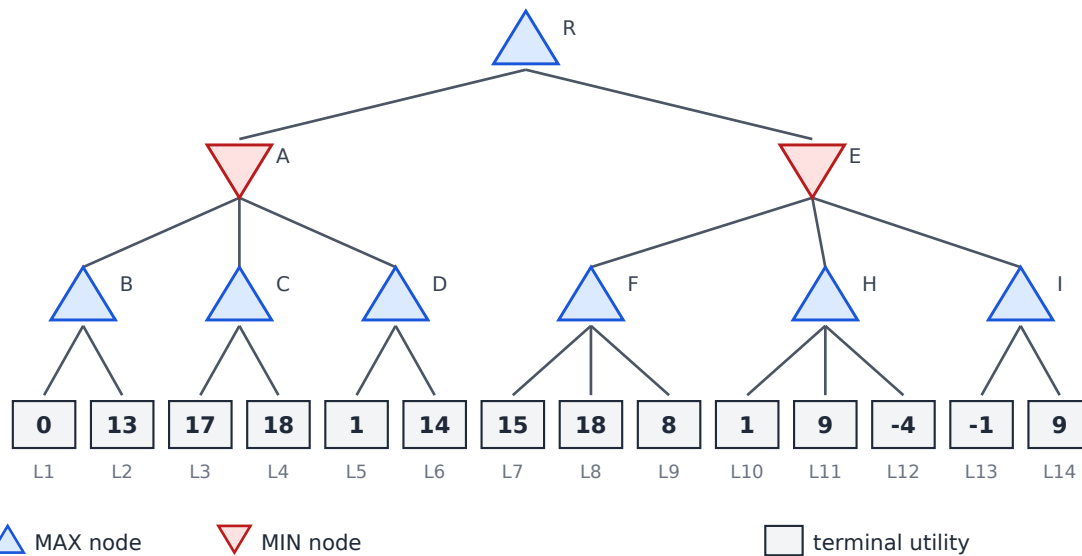
Q.2 [NAT · 1 mark · no negative marking] Adversarial Search

Chess has an average branching factor of about $b = 36$. With perfect move ordering, alpha-beta reduces the effective branching factor from b to $\sqrt{b}$. What is the effective branching factor?

Answer (numerical): _____

Q.3 [NAT · 1 mark · no negative marking] Adversarial Search – Minimax

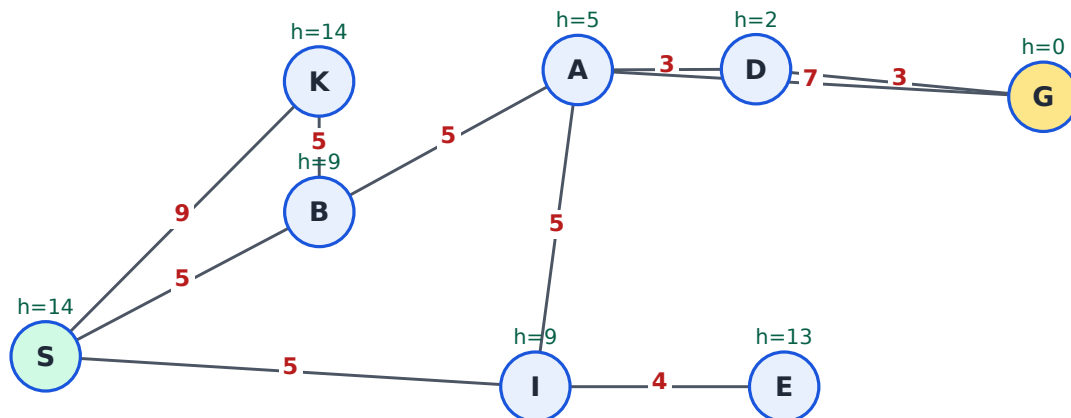
Consider the two-player zero-sum game tree below (MAX moves at the root, levels alternate). What is the minimax value of the root?



Answer (numerical): _____

Q.4 [NAT · 1 mark · no negative marking] Informed Search – Greedy best-first

Greedy best-first search (graph search; a state is added to the frontier only if it has not been reached before; goal test on expansion; ties broken alphabetically) uses the heuristic h shown above each node to search from **S** to **G**. Edge labels are step costs. What is the cost of the path returned by greedy best-first search?



Answer (numerical): _____

Q.5 [MCQ · 1 mark · negative: -1/3] Propositional Logic

Let $KB = \{P \rightarrow Q, Q \rightarrow R, \neg R\}$. Which one of the following is entailed by KB ?

- ☐ (A) $\neg P \wedge \neg Q$
- ☐ (B) $P \vee Q$
- ☐ (C) $\neg P \wedge Q$
- ☐ (D) R

Q.6 [MSQ · 1 mark · no negative marking] Predicate Logic

Without standardizing variables apart, $\text{Unify}(\text{Knows}(\text{John}, x), q)$ succeeds for which of the following atoms q ?

- ☐ (A) $\text{Knows}(\text{John}, \text{Jane})$
☐ (B) $\text{Knows}(y, \text{Bill})$
☐ (C) $\text{Knows}(y, \text{Mother}(y))$
☐ (D) $\text{Knows}(x, \text{Elizabeth})$

Q.7 [NAT · 1 mark · no negative marking] Predicate Logic – Counting interpretations

P and Q are unary predicate symbols and the domain has exactly 3 elements, so there are $4^3 = 64$ ways to interpret the pair (P, Q) . For how many of them is $\forall x P(x) \vee \forall x Q(x)$ true?

Answer (numerical): _____

Q.8 [NAT · 1 mark · no negative marking] Propositional Logic – Model counting

The formula $F = ((Q \vee R) \wedge (\neg P \vee P)) \leftrightarrow (\neg P \wedge (Q \vee \neg P))$ is written in canonical (full) conjunctive normal form, i.e. as a conjunction of maxterms over P, Q, R . How many maxterms (clauses) does it contain?

Answer (numerical): _____

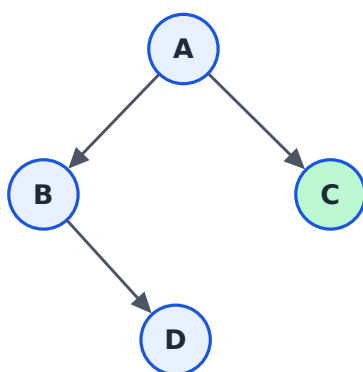
Q.9 [MCQ · 1 mark · negative: -1/3] Sampling

For sampling-based estimates of a probability, the standard deviation of the estimate decreases as $O(1/\sqrt{N})$ in the number of samples N . Roughly how many samples are needed to halve the error obtained with N samples?

- ☐ (A) $4N$
☐ (B) $2N$
☐ (C) $N\sqrt{2}$
☐ (D) N^2

Q.10 [NAT · 1 mark · no negative marking] Exact Inference – Marginal

For the Bayesian network below (Boolean variables; CPTs list $P(X = T \mid \text{parents})$), compute the marginal probability $P(C = T)$ (round off to 3 decimal places).



P(A=T)
0.75

A	P(B=T ...)	A	P(C=T ...)
T	0.3	T	0.7
F	0.3	F	0.3

B	P(D=T ...)
T	0.8
F	0.2

Answer (numerical): _____

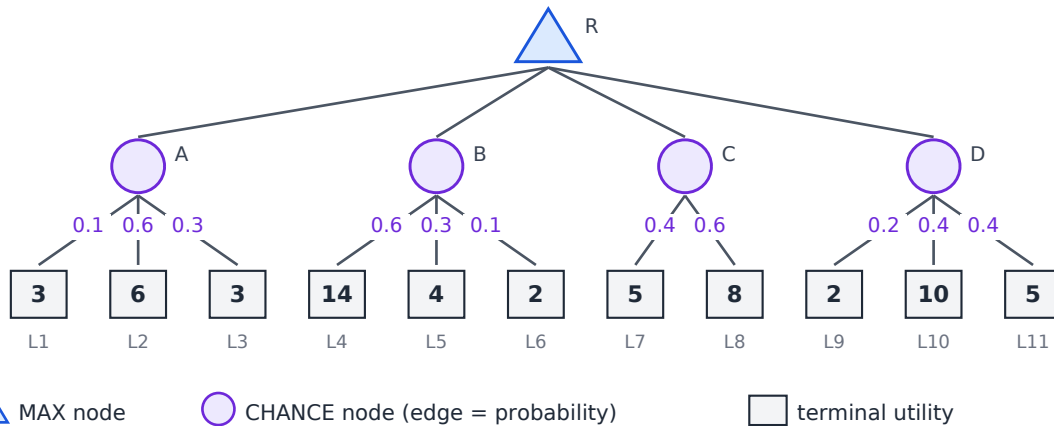
Section B — 2-mark questions**Q.11** [NAT · 2 marks · no negative marking] Adversarial Search

A two-ply game tree has a MAX root with four MIN children, whose leaves are listed left to right: MIN_1 : (10, 9, 14); MIN_2 : (11, 8, 20); MIN_3 : (2, 15, 30); MIN_4 : (12, 13, 7). Alpha-beta pruning visits children left to right. How many leaves are **pruned** (never evaluated)?

Answer (numerical): _____

Q.12 [NAT · 2 marks · no negative marking] Adversarial Search – Expectimax

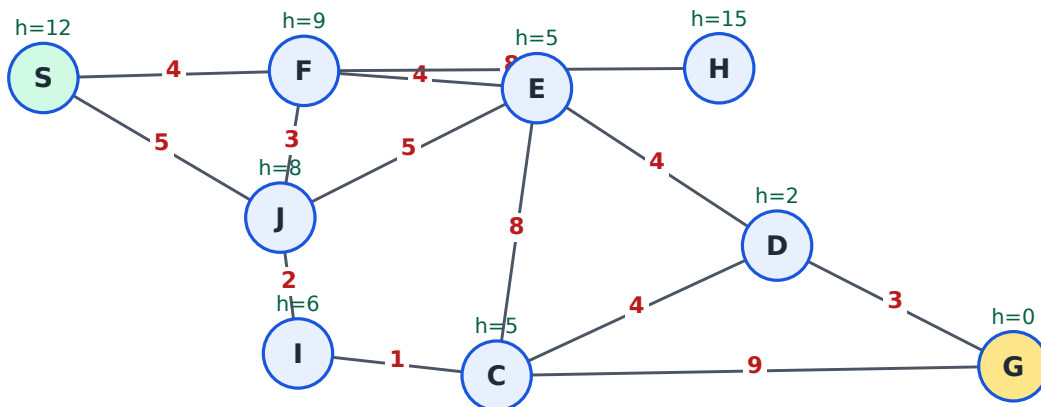
In the expectimax tree below, the root is a MAX node and each child of the root is a CHANCE node; probabilities are written on the edges. What is the expectimax value of the root? (round off to 2 decimal places)



Answer (numerical): _____

Q.13 [NAT · 2 marks · no negative marking] Informed Search – A*

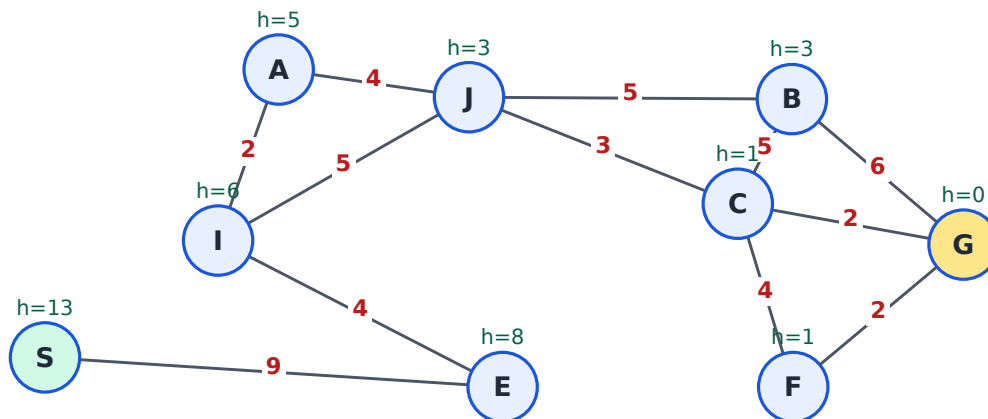
A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. What is the cost of the solution returned by A*?



Answer (numerical): _____

Q.14 [MSQ · 2 marks · no negative marking] Informed Search – A*

A* search (graph search with explored set, goal test on expansion, ties broken alphabetically) is run from **S** to **G** on the graph below. Edge labels are step costs and the heuristic value $h(n)$ is printed above each node. Which of the following statements is/are TRUE?



- ☐ (A) Node B is expanded.
- ☐ (B) The path returned has cost 24.
- ☐ (C) Node I is expanded.
- ☐ (D) Node A is expanded.

Q.15 [MSQ · 2 marks · no negative marking] Propositional Logic – Equivalence

Which of the following formulas is/are logically equivalent to $F = Q \vee (R \rightarrow P)$?

- ☐ (A) $\neg(\neg Q \wedge \neg(R \rightarrow P))$
- ☐ (B) $Q \wedge (R \rightarrow P)$
- ☐ (C) $(R \rightarrow P) \vee Q$
- ☐ (D) $\neg(\neg Q \vee \neg(R \rightarrow P))$

Q.16 [MSQ · 2 marks · no negative marking] Propositional Logic

Which of the following formulas are contingent (satisfiable but not valid)?

- ☐ (A) $(p \rightarrow q) \wedge (p \rightarrow \neg q)$
- ☐ (B) $(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$
- ☐ (C) $(p \leftrightarrow q) \wedge (p \oplus q)$
- ☐ (D) $(p \vee q) \rightarrow (p \wedge \neg q)$

Q.17 [MSQ · 2 marks · no negative marking] Propositional Logic – Entailment

A knowledge base KB contains exactly the following three sentences:

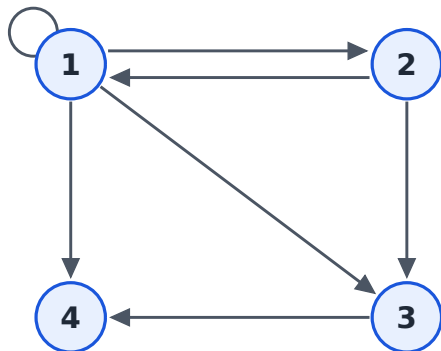
- (1) $(\neg P \wedge Q) \vee (Q \vee P)$
- (2) $(\neg P \rightarrow \neg Q) \wedge Q$
- (3) $R \leftrightarrow P$

Which of the following sentences is/are entailed by KB ($KB \models \alpha$)?

- ☐ (A) $\neg Q \wedge P$
- ☐ (B) $Q \rightarrow R$
- ☐ (C) $(Q \leftrightarrow \neg Q) \wedge P$
- ☐ (D) $\neg(Q \rightarrow \neg P)$

Q.18 [MSQ · 2 marks · no negative marking] Predicate Logic – FOL semantics

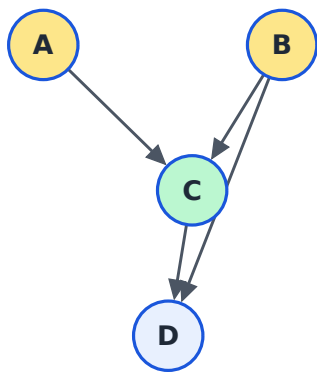
Let the domain be $D = \{1, 2, 3, 4\}$ and let the binary predicate R be interpreted as the relation drawn below (an arrow $x \rightarrow y$ means $R(x, y)$ is true; a small loop at a node means $R(x, x)$ is true). Which of the following sentences is/are TRUE in this interpretation?



- ☐ (A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$
☐ (B) $\forall y \exists x R(x, y)$
☐ (C) $\exists x R(x, x)$
☐ (D) $\forall x \exists y R(x, y)$

Q.19 [NAT · 2 marks · no negative marking] Sampling – Likelihood weighting

Likelihood weighting is used on the Bayesian network below with evidence $\mathbf{B} = \mathbf{F}$, $\mathbf{A} = \mathbf{T}$. The values of the non-evidence variables in the generated samples are listed in the table. Estimate $\mathbf{P}(\mathbf{C} = \mathbf{T} \mid \mathbf{B} = \mathbf{F}, \mathbf{A} = \mathbf{T})$ (round off to 3 decimal places).



$P(A=T)$
0.9

$P(B=T)$	A	B	$P(C=T \dots)$
0.25	T	T	0.4
	T	F	0.5
	F	T	0.3
	F	F	0.25

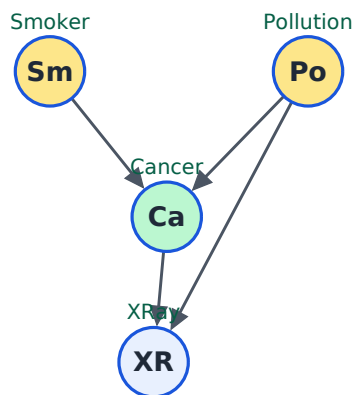
B	C	$P(D=T \dots)$
T	T	0.8
T	F	0.95
F	T	0.05
F	F	0.75

#	C	D
1	T	F
2	F	T
3	F	F
4	F	T
5	T	F

Answer (numerical): _____

Q.20 [NAT · 2 marks · no negative marking] Exact Inference – Exact inference

Consider the Bayesian network below over Boolean variables with the given CPTs (only $P(X = T \mid \text{parents})$ is listed). (Nodes are abbreviated by their first two letters: Sm = Smoker, Po = Pollution, Ca = Cancer, XR = XRay.) Compute $P(\text{Cancer} = T \mid \text{Pollution} = T, \text{Smoker} = F)$ (round off to 3 decimal places).



$P(\text{Sm}=T)$
0.8

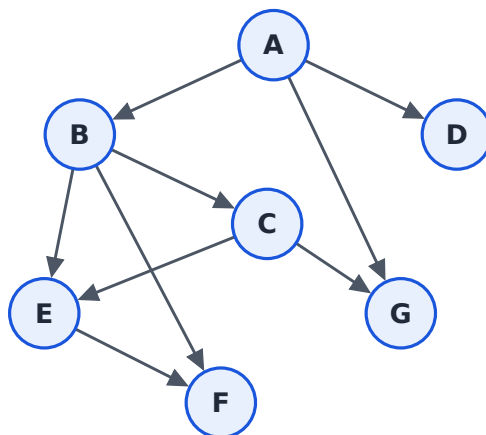
$P(\text{Po}=T)$	Sm	Po	$P(\text{Ca}=T \dots)$
0.4	T	T	0.95
	T	F	0.3
	F	T	0.9
	F	F	0.6

Po	Ca	$P(\text{XR}=T \dots)$
T	T	0.4
T	F	0.1
F	T	0.4
F	F	0.75

Answer (numerical): _____

Q.21 [MSQ · 2 marks · no negative marking] Conditional Independence – d-separation

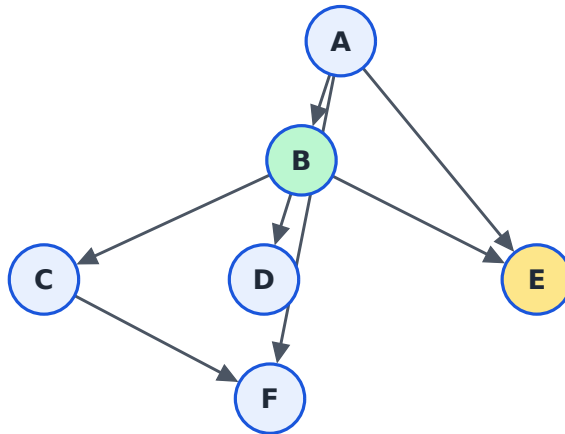
For the Bayesian network structure shown below, which of the following conditional independence statements is/are guaranteed by d-separation?



- ☐ (A) $A \perp E \mid B, D$
☐ (B) $B \perp G \mid E$
☐ (C) $E \perp D \mid A$
☐ (D) $D \perp C \mid E, F$

Q.22 [NAT · 2 marks · no negative marking] Exact Inference – Variable elimination

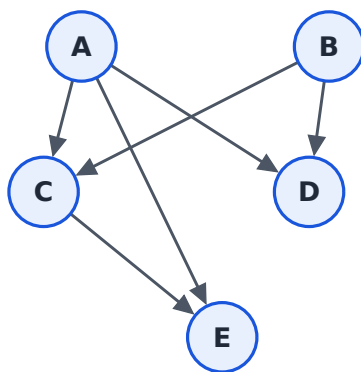
Variable elimination is used to compute $P(\mathbf{B} \mid \mathbf{E} = \mathbf{T})$ in the Bayesian network below (Boolean variables; evidence $\mathbf{E} = \mathbf{T}$ is observed; the query node is green, evidence yellow). Every CPT becomes an initial factor (no pruning of irrelevant variables), evidence variables are instantiated and so do not appear in any factor's scope, and the hidden variables are eliminated in the order **C, A, D, F**. What is the number of variables in the largest factor **generated** by a summation step?



Answer (numerical): _____

Q.23 [MCQ · 2 marks · negative: -2/3] Sampling – Prior sampling

Prior sampling is applied to the network below in the topological order A, B, C, D, E, using the uniform random numbers $\mathbf{u} = \{0.15, 0.54, 0.47, 0.59, 0.33\}$ (one per variable, in that order). A variable X is set to T iff $u < P(X = T \mid \text{sampled parent values})$. Which sample is generated?



$P(A=T)$
0.95

$P(B=T)$
0.7

A	B	$P(C=T \dots)$
T	T	0.3
T	F	0.95
F	T	0.8
F	F	0.25

A	B	$P(D=$	A	C	$P(E=T \dots)$
T	T	0.4	T	T	0.4
T	F	0.9	T	F	0.8
F	T	0.25	F	T	0.3
F	F	0.95	F	F	0.95

- ☐ (A) $A=F, B=T, C=F, D=F, E=T$
☐ (B) $A=T, B=F, C=F, D=F, E=T$
☐ (C) $A=T, B=T, C=F, D=F, E=T$
☐ (D) $A=F, B=F, C=T, D=F, E=T$

Q.24 [MCQ · 2 marks · negative: -2/3] Variable Elimination

For a Bayesian network with n Boolean variables, the depth-first ENUMERATION-ASK algorithm (inference by enumeration) has worst-case

- ☐ (A) $O(2^n)$ time and $O(n)$ space
☐ (B) $O(n)$ time and $O(2^n)$ space
☐ (C) $O(2^n)$ time and $O(2^n)$ space
☐ (D) $O(n^2)$ time and $O(n)$ space

Q.25 [MSQ · 2 marks · no negative marking] Sampling

Which of the following statements about approximate inference in Bayesian networks are correct?

- ☐ (A) Likelihood weighting produces consistent estimates
- ☐ (B) Likelihood weighting performs poorly when evidence variables are downstream (e.g. leaves), since samples of upstream variables ignore that evidence
- ☐ (C) Gibbs sampling resamples each non-evidence variable from its distribution conditioned on the current values of its Markov blanket
- ☐ (D) In rejection sampling, the fraction of accepted samples grows as more evidence variables are added

Answer Key — Mock Test 100

Q	Type	Marks	Topic	Key	Q	Type	Marks	Topic	Key
1	MCQ	1	Uninformed Search	D	14	MSQ	2	Informed Search	C, D
2	NAT	1	Adversarial Search	6	15	MSQ	2	Propositional Logic	A, C
3	NAT	1	Adversarial Search	13	16	MSQ	2	Propositional Logic	A, D
4	NAT	1	Informed Search	17	17	MSQ	2	Propositional Logic	B, D
5	MCQ	1	Propositional Logic	A	18	MSQ	2	Predicate Logic	B, C
6	MSQ	1	Predicate Logic	A, B, C	19	NAT	2	Sampling	0.398 to 0.402
7	NAT	1	Predicate Logic	15	20	NAT	2	Exact Inference	0.898 to 0.902
8	NAT	1	Propositional Logic	4	21	MSQ	2	Conditional Independence	A, C
9	MCQ	1	Sampling	A	22	NAT	2	Exact Inference	3
10	NAT	1	Exact Inference	0.598 to 0.602	23	MCQ	2	Sampling	C
11	NAT	2	Adversarial Search	3	24	MCQ	2	Variable Elimination	A
12	NAT	2	Adversarial Search	9.79 to 9.81	25	MSQ	2	Sampling	A, B, C
13	NAT	2	Informed Search	15					

Detailed Solutions

Q.1 [MCQ, 1M · Uninformed Search] **Answer: D** (D: S, I, K, F, H, J, B, C, G)

Adjacency lists (alphabetical): B: {C, F, H, J}; C: {B, F, G}; F: {B, C, I}; G: {C, J}; H: {B, I, J, K}; I: {F, H, K, S}; J: {B, G, H, K}; K: {H, I, J, S}; S: {I, K}.

BFS trace (queue shown front → back):

Step	Expanded	Frontier (FIFO) after expansion
1	S	[I, K]
2	I	[K, F, H]
3	K	[F, H, J]
4	F	[H, J, B, C]
5	H	[J, B, C]
6	J	[B, C, G]
7	B	[C, G]
8	C	[G]
9	G	[] ← goal popped, stop

Expansion order: **S, I, K, F, H, J, B, C, G**. The path returned is $S \rightarrow K \rightarrow J \rightarrow G$ with 3 edges, which is the path with the fewest edges (BFS is optimal for unit step costs).

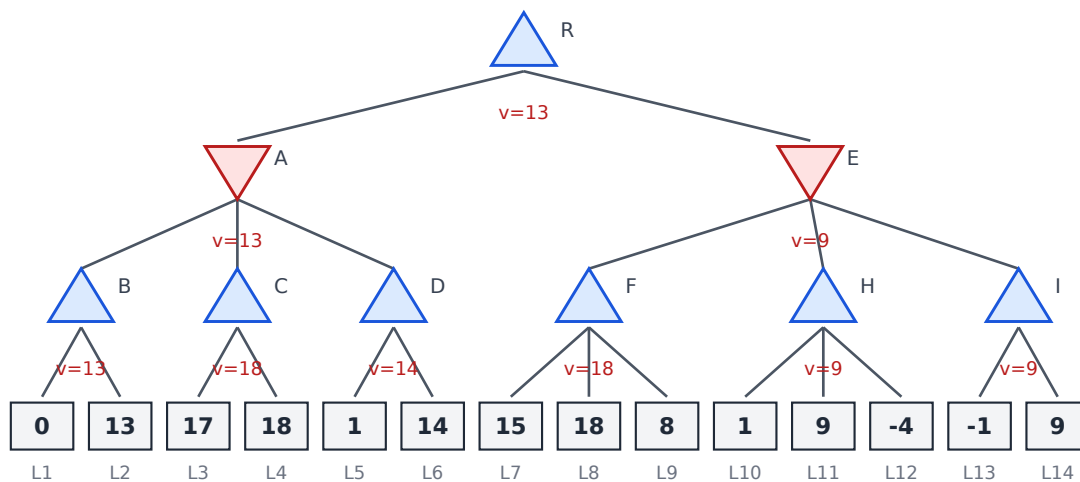
Distractors include the DFS order and BFS with reverse-alphabetical tie-breaking.

Q.2 [NAT, 1M · Adversarial Search] **Answer: 6**

Perfect ordering gives $O(b^{m/2}) = O((\sqrt{b})^m)$. The effective branching factor is $\sqrt{36} = 6$. Equivalently, alpha-beta can search roughly twice as deep as minimax in the same time.

Q.3 [NAT, 1M · Adversarial Search] **Answer: 13**

Back up values bottom-up: a MIN node takes the minimum of its children, a MAX node the maximum.



Values of internal nodes: $B = 13$, $C = 18$, $D = 14$, $A = 13$, $F = 18$, $H = 9$, $I = 9$, $E = 9$. Root value = **13**; MAX's optimal first move is towards A.

Q.4 [NAT, 1M · Informed Search] Answer: 17

Greedy best-first search orders the frontier by $h(n)$ only, ignoring the cost already incurred.

Step	Expanded (h)	Frontier after expansion (sorted by priority)
1	S (14)	B(9), I(9), K(14)
2	B (9)	A(5), I(9), K(14)
3	A (5)	G(0), D(2), I(9), K(14)
4	G (0)	D(2), I(9), K(14) ← goal, stop

Path returned: **S** → **B** → **A** → **G**, cost = $5 + 5 + 7 = 17$.

The optimal cost is $h^*(S) = 16$ — greedy search is not optimal here.

Q.5 [MCQ, 1M · Propositional Logic] Answer: A ($A: \neg P \wedge \neg Q$)

From $\neg R$ and $Q \rightarrow R$, modus tollens gives $\neg Q$; from $\neg Q$ and $P \rightarrow Q$ it gives $\neg P$. The only model of KB (over P, Q, R) is $P=F$, $Q=F$, $R=F$, in which $\neg P \wedge \neg Q$ is true. $P \vee Q$, $\neg P \wedge Q$ and R are all false in that model, so they are not entailed.

Q.6 [MSQ, 1M · Predicate Logic] Answer: A, B, C ($A: \text{Knows}(\text{John}, \text{Jane})$; $B: \text{Knows}(y, \text{Bill})$; $C: \text{Knows}(y, \text{Mother}(y))$)

(A) $\{x/\text{Jane}\}$. (B) $\{x/\text{Bill}, y/\text{John}\}$. (C) $\{y/\text{John}, x/\text{Mother}(\text{John})\}$. (D) fails: x would have to be both John and Elizabeth. Renaming the second x (standardizing apart) would make it succeed, which is why AIMA insists on standardizing apart.

Q.7 [NAT, 1M · Predicate Logic] Answer: 15

Interpret P and Q as subsets of the domain; inclusion-exclusion: $2^n + 2^n - 1$. With $n = 3$ this gives **15** (verified by enumerating all interpretations).

Q.8 [NAT, 1M · Propositional Logic] Answer: 4

Build the truth table (sub-formulas shown first):

P	Q	R	$(Q \vee R) \wedge (\neg P \vee P)$	$\neg P \wedge (Q \vee \neg P)$	$((Q \vee R) \wedge (\neg P \vee P)) \leftrightarrow (\neg P \wedge (Q \vee \neg P))$
T	T	T	T	F	F
T	T	F	T	F	F
T	F	T	T	F	F
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	F	T	F

F has 4 models, so it is false in $8 - 4 = 4$ rows; each falsifying row contributes exactly one maxterm to the canonical CNF.

Q.9 [MCQ, 1M · Sampling] Answer: A (A: 4N)

Error $\propto 1/\sqrt{N}$, so halving the error requires $\sqrt{N'} = 2\sqrt{N}$, i.e. $N' = 4N$. $2N$ would reduce error only by a factor $\sqrt{2} \approx 1.41$; $N/2$ by $2^{1/4}$; N^2 is excessive and depends on N .

Q.10 [NAT, 1M · Exact Inference] Answer: 0.598 to 0.602

Only the ancestors of C (A) matter: every other variable sums out to 1.

$P(C=T) = \sum \text{over its ancestors of } P(C=T \mid \text{parents}) \times P(\text{parents' configuration})$. Summing the joint over all other variables gives:

Assignment to other variables	joint $P(\dots, C=T)$
$A=T, B=T, D=T$	0.126
$A=T, B=T, D=F$	0.0315
$A=T, B=F, D=T$	0.0735
$A=T, B=F, D=F$	0.294
$A=F, B=T, D=T$	0.018
$A=F, B=T, D=F$	0.0045
$A=F, B=F, D=T$	0.0105
$A=F, B=F, D=F$	0.042

Sum = **0.6** ≈ 0.600 .

Q.11 [NAT, 2M · Adversarial Search] Answer: 3

MIN_1 : evaluate 10, 9, 14, giving 9, so $\alpha = 9$ at the root. MIN_2 : 11, then $8 \leq \alpha$, so prune 20. MIN_3 : $2 \leq \alpha$, so prune 15 and 30. MIN_4 : 12, 13, 7 are all evaluated, because the running minimum stays above 9 until the last leaf. Evaluated leaves: 9 of 12. Pruned: 20, 15, 30 = **3**.

Q.12 [NAT, 2M · Adversarial Search] Answer: 9.79 to 9.81

A chance node's value is the probability-weighted average of its children; MAX takes the maximum.

- $A = 0.1 \times 3 + 0.6 \times 6 + 0.3 \times 3 = 4.8$
- $B = 0.6 \times 14 + 0.3 \times 4 + 0.1 \times 2 = 9.8$
- $C = 0.4 \times 5 + 0.6 \times 8 = 6.8$
- $D = 0.2 \times 2 + 0.4 \times 10 + 0.4 \times 5 = 6.4$

Root = $\max(4.8, 9.8, 6.8, 6.4) = \mathbf{9.8}$.

Note: unlike minimax, expectimax values are sensitive to positive affine vs. non-linear transformations of the utilities — only positive linear transformations preserve the optimal decision.

Q.13 [NAT, 2M · Informed Search] Answer: 15

Heuristic values: $h(C)=5, h(D)=2, h(E)=5, h(F)=9, h(G)=0, h(H)=15, h(I)=6, h(J)=8, h(S)=12$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (12)	F(13), J(13)
2	F (13)	E(13), J(13), H(27)
3	E (13)	J(13), D(14), C(21), H(27)
4	J (13)	I(13), D(14), C(21), H(27)
5	I (13)	C(13), D(14), H(27)
6	C (13)	D(14), G(17), H(27)
7	D (14)	G(15), H(27)
8	G (15)	H(27) ← goal, stop

Path returned: **S** → **F** → **E** → **D** → **G** with cost **15**; nodes expanded = **8** (S, F, E, J, I, C, D, G).

Check: $h(n) = [c \cdot h^*(n)]$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 15$. For comparison UCS ($h = 0$) would expand 9 nodes.

Q.14 [MSQ, 2M · Informed Search] Answer: C, D (C: Node I is expanded.; D: Node A is expanded.)

Heuristic values: $h(A)=5, h(B)=3, h(C)=1, h(E)=8, h(F)=1, h(G)=0, h(I)=6, h(J)=3, h(S)=13$. A* expands the frontier node with the smallest $f(n) = g(n) + h(n)$.

Step	Expanded ($f = g + h$)	Frontier after expansion (sorted by priority)
1	S (13)	E(17)
2	E (17)	I(19)
3	I (19)	A(20), J(21)
4	A (20)	J(21)

Step	Expanded (f = g + h)	Frontier after expansion (sorted by priority)
5	J (21)	C(22), B(26)
6	C (22)	G(23), B(26), F(26)
7	G (23)	B(26), F(26) ← goal, stop

Path returned: **S** → **E** → **I** → **J** → **C** → **G** with cost **23**; nodes expanded = **7** (S, E, I, A, J, C, G).

Check: $h(n) = \lfloor c \cdot h^*(n) \rfloor$ for a constant $c < 1$, so h is consistent (and admissible) and the path is optimal — the true optimal cost $h^*(S) = 23$. For comparison UCS ($h = 0$) would expand 8 nodes.

Q.15 [MSQ, 2M · Propositional Logic] **Answer: A, C** (A: $\neg(\neg Q \wedge \neg(R \rightarrow P))$; C: $(R \rightarrow P) \vee Q$)

Compare truth tables with $F = Q \vee (R \rightarrow P)$ (or use the standard laws: implication elimination $\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$, contraposition, De Morgan, biconditional elimination, distributivity).

(A) $\neg(\neg Q \wedge \neg(R \rightarrow P))$ — agrees with F on all 8 rows ⇒ **equivalent**.

(B) $Q \wedge (R \rightarrow P)$ — differs at $P=T, Q=F, R=T$ ($F = T$, option = F) ⇒ not equivalent.

(C) $(R \rightarrow P) \vee Q$ — agrees with F on all 8 rows ⇒ **equivalent**.

(D) $\neg(\neg Q \vee \neg(R \rightarrow P))$ — differs at $P=T, Q=F, R=T$ ($F = T$, option = F) ⇒ not equivalent.

Q.16 [MSQ, 2M · Propositional Logic] **Answer: A, D** (A: $(p \rightarrow q) \wedge (p \rightarrow \neg q)$; D: $(p \vee q) \rightarrow (p \wedge \neg q)$)

(A) $\equiv p \rightarrow (q \wedge \neg q) \equiv \neg p$: true at $p=F$, false at $p=T$, so contingent. (B) An implication and its contrapositive are equivalent, so this is valid. (C) $p \leftrightarrow q$ and $p \oplus q$ are negations of each other, so their conjunction is unsatisfiable. (D) True at $p=q=F$ (antecedent false), false at $p=F, q=T$, so contingent.

Q.17 [MSQ, 2M · Propositional Logic] **Answer: B, D** (B: $Q \rightarrow R$; D: $\neg(Q \rightarrow \neg P)$)

$KB \models \alpha$ iff α is true in every model of KB. Enumerating all 8 assignments, KB is true in exactly 1 of them:

P	Q	R	KB
T	T	T	T

(A) $\neg Q \wedge P$ is false in the KB-model $P=T, Q=T, R=T$ ⇒ not entailed.

(B) $Q \rightarrow R$ is true in all models of KB ⇒ **entailed**.

(C) $(Q \leftrightarrow \neg Q) \wedge P$ is false in the KB-model $P=T, Q=T, R=T$ ⇒ not entailed.

(D) $\neg(Q \rightarrow \neg P)$ is true in all models of KB ⇒ **entailed**.

Q.18 [MSQ, 2M · Predicate Logic] **Answer: B, C** (B: $\forall y \exists x R(x, y)$; C: $\exists x R(x, x)$)

$R = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,3), (3,4)\}$.

(A) $\forall x \forall y (x \neq y \rightarrow (R(x, y) \vee R(y, x)))$: **FALSE** — neither $R(2,4)$ nor $R(4,2)$.

(B) $\forall y \exists x R(x, y)$: **TRUE** — every element has an incoming edge.

(C) $\exists x R(x, x)$: **TRUE** — self-loop at 1.

(D) $\forall x \exists y R(x, y)$: **FALSE** — 4 has no outgoing edge.

Remember: $\exists y \forall x R(x, y) \Rightarrow \forall x \exists y R(x, y)$ but not conversely; quantifier order matters.

Q.19 [NAT, 2M · Sampling] **Answer: 0.398 to 0.402**

In likelihood weighting the evidence variables are fixed and each sample is weighted by the product of the probabilities of the evidence values given their (sampled) parents.

#	weight $w = \prod P(e \mid \text{parents})$	w	C
1	$P(A=T)=0.9 \times P(B=F)=0.75$	0.675	T
2	$P(A=T)=0.9 \times P(B=F)=0.75$	0.675	F
3	$P(A=T)=0.9 \times P(B=F)=0.75$	0.675	F
4	$P(A=T)=0.9 \times P(B=F)=0.75$	0.675	F
5	$P(A=T)=0.9 \times P(B=F)=0.75$	0.675	T

$\sum w (\text{all}) = 3.375$; $\sum w (C = T) = 1.35$.

Estimate $P(C=T \mid B = F, A = T) = 1.35 / 3.375 = \mathbf{0.4}$.

Q.20 [NAT, 2M · Exact Inference] **Answer: 0.898 to 0.902**

Factorisation: $P(\text{Smoker}, \text{Pollution}, \text{Cancer}, \text{XRay}) = P(\text{Smoker}) \cdot P(\text{Pollution}) \cdot P(\text{Cancer} \mid \text{Smoker}, \text{Pollution}) \cdot P(\text{XRay} \mid \text{Pollution}, \text{Cancer})$.

Inference by enumeration: $P(\text{Cancer} \mid e) = \alpha \sum_{\text{hidden}} P(\text{Cancer}, e, \text{hidden})$.

Hidden: XR	Ca	product of CPT entries ($Sm \cdot Po \cdot Ca \cdot XR$)	value
T	T	$0.2 \times 0.4 \times 0.9 \times 0.4$	0.0288
F	T	$0.2 \times 0.4 \times 0.9 \times 0.6$	0.0432
T	F	$0.2 \times 0.4 \times 0.1 \times 0.1$	0.0008

Hidden: XR	Ca	product of CPT entries ($S_m \cdot P_o \cdot C_a \cdot X_R$)	value
F	F	$0.2 \times 0.4 \times 0.1 \times 0.9$	0.0072

$P(\text{Cancer}=T, e) = 0.072$; $P(\text{Cancer}=F, e) = 0.008$; $P(e) = 0.08$.

$P(\text{Cancer}=T | e) = 0.072 / 0.08 = \mathbf{0.9} \approx 0.900$.

Q.21 [MSQ, 2M · Conditional Independence] **Answer: A, C** (A: $A \perp E | B, D$; C: $E \perp D | A$)

Rules: a chain ($\rightarrow B \rightarrow$) or fork ($\leftarrow B \leftarrow$) is blocked iff B is observed; a collider ($\rightarrow B \leftarrow$) is blocked iff neither B nor any descendant of B is observed. $X \perp Y | Z$ holds iff every undirected path is blocked.

(A) $A \perp E | B, D$: **TRUE** — $A \rightarrow B \rightarrow C \rightarrow E$ (blocked: observed chain node B); $A \rightarrow B \rightarrow E$ (blocked: observed chain node B); $A \rightarrow B \rightarrow F \leftarrow E$ (blocked: observed chain node B); $A \rightarrow G \leftarrow C \leftarrow B \rightarrow E$ (blocked: collider G with neither it nor a descendant observed); $A \rightarrow G \leftarrow C \leftarrow B \rightarrow F \leftarrow E$ (blocked: collider G with neither it nor a descendant observed); $A \rightarrow G \leftarrow C \rightarrow E$ (blocked: collider G with neither it nor a descendant observed).

(B) $B \perp G | E$: **FALSE** — the path $B \leftarrow A \rightarrow G$ is active given $\{E\}$.

(C) $E \perp D | A$: **TRUE** — $E \leftarrow B \leftarrow A \rightarrow D$ (blocked: observed fork node A); $E \leftarrow B \rightarrow C \rightarrow G \leftarrow A \rightarrow D$ (blocked: collider G with neither it nor a descendant observed); $E \leftarrow C \leftarrow B \leftarrow A \rightarrow D$ (blocked: observed fork node A); $E \leftarrow C \rightarrow G \leftarrow A \rightarrow D$ (blocked: collider G with neither it nor a descendant observed); $E \rightarrow F \leftarrow B \leftarrow A \rightarrow D$ (blocked: collider F with neither it nor a descendant observed); $E \rightarrow F \leftarrow B \rightarrow C \rightarrow G \leftarrow A \rightarrow D$ (blocked: collider F with neither it nor a descendant observed).

(D) $D \perp C | E, F$: **FALSE** — the path $D \leftarrow A \rightarrow B \rightarrow C$ is active given $\{E, F\}$.

Q.22 [NAT, 2M · Exact Inference] **Answer: 3**

Initial factors (one per CPT, evidence removed from scopes): $f(A)$, $f(A, B)$, $f(B, C)$, $f(B, D)$, $f(A, B)$, $f(A, C, F)$.

Eliminating Z: multiply all factors whose scope contains Z, then sum Z out.

Eliminate	Factors multiplied	Product scope	New factor (after Σ)
C	$f(B, C)$, $f(A, C, F)$	$\{A, B, C, F\}$	$f(A, B, F)$
A	$f(A)$, $f(A, B)$, $f(A, B)$, $f(A, B, F)$	$\{A, B, F\}$	$f(B, F)$
D	$f(B, D)$	$\{B, D\}$	$f(B)$
F	$f(B, F)$	$\{B, F\}$	$f(B)$

The largest factor produced has **3** variable(s) (created when eliminating C); with Boolean variables it has $2^3 = 8$ entries. Finally, multiply the remaining factors over B and normalise.

Q.23 [MCQ, 2M · Sampling] **Answer: C** (C: $A=T$, $B=T$, $C=F$, $D=F$, $E=T$)

Sample each variable after its parents, using the parent values already drawn:

- A: $P(A=T | \text{—}) = 0.95$; $u = 0.15 < 0.95 \Rightarrow A = T$
- B: $P(B=T | \text{—}) = 0.7$; $u = 0.54 < 0.7 \Rightarrow B = T$
- C: $P(C=T | A=T, B=T) = 0.3$; $u = 0.47 \geq 0.3 \Rightarrow C = F$
- D: $P(D=T | A=T, B=T) = 0.4$; $u = 0.59 \geq 0.4 \Rightarrow D = F$
- E: $P(E=T | A=T, C=F) = 0.8$; $u = 0.33 < 0.8 \Rightarrow E = T$

Sample: **$\{A=T, B=T, C=F, D=F, E=T\}$** .

Q.24 [MCQ, 2M · Variable Elimination] **Answer: A** (A: $O(2^n)$ time and $O(n)$ space)

Enumeration evaluates a depth-first expression tree that branches over every hidden variable, giving $O(2^n)$ time. Being depth-first it stores only the current path, so space is $O(n)$. It never builds the full joint table, so it does not need $O(2^n)$ space; polynomial time is impossible in general since inference is NP-hard. Variable elimination improves time by caching repeated subexpressions as factors.

Q.25 [MSQ, 2M · Sampling] **Answer: A, B, C**

(A) AIMA shows $S_{WS}(z, e) w(z, e) = P(z, e)$, so weighted estimates are consistent. (B) Non-evidence variables are sampled from their parents only, so evidence at leaves does not influence them; many samples get tiny weights. (C) This is exactly the Gibbs step: $P(x_i | mb(X_i))$. (D) is false: each extra evidence variable generally reduces $P(e)$, so fewer samples are accepted.